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INTERMEDIATE DESIGN SPECIFICATIONS

Project Title:	**Army Family Housing Renovate Tower B743** **Camp Zama, Japan**
Project Number:	**PN100964**
Program Year:	**FY27**
Installation Name:	**Camp Zama, Japan**
Date:	**January 29th, 2026**
Prepared by:	**Design Partners Incorporated**

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SECTION 00 10 00

SOLICITATION, CONTRACT LINE ITEM NUMBER (CLIN) SCHEDULE

PART 1 GENERAL

1.1 ITEMS

ITEM NO.	DESCRIPTION	QUANTITY	UNIT	UNIT PRICE	AMOUNT
BASE ITEMS					
1	Demolition and construction/renovation of Building 743, including 68 family housing units, common areas, multi-purpose rooms, community activities center, and all associated supporting facilities work, except as specified in Item Nos. 0002 and 0003.	1	JB	-	¥________
TOTAL BASE CLIN SCHEDULE					¥________
OPTION ITEMS					
2	Demolish Building S-726 and construct new parking area and playground.(OPTION 1)	1	JB	-	¥________
3	Upgrade hydronic distribution from a two-pipe system to a four-Pipe system.(OPTION 2)	1	JB	-	¥________
TOTAL (BASE + OPTION ITEMS) CLIN SCHEDULE					¥________

NOTES:

1. Failure of the offeror to propose prices for all items in the CLIN Schedule may be cause for rejection of the proposal as unacceptable.

2. See Section 01 20 00, for additional line item description.

3. Any reference to "bid", bidder, or "bidding schedule" shall be construed to mean "offeror" and "CLIN schedule"

4. The Government reserves the right to exercise option item(s) by written notice to the Contractor either singularly or in any combination for up to ONE-HUNDRED EIGHTY (180) calendar days after award of the Base Schedule at the prices stated in the CLIN Schedule. Completion of the added option(s) shall continue at the same schedule as the Base Schedule unless otherwise noted in Section 00 70 00, 52.211-10, COMMENCEMENT, PROSECUTION AND COMPLETION OF WORK.

5. LEGEND: EA = Each, JB = Job

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION 01 11 00.00 10

GENERAL CONTRACT REQUIREMENTS

PART 1 GENERAL

1.1 DEFINITIONS

The term "Government" refers to "United States Government" whenever the term "Government" appears in this Contract, unless otherwise indicated within the Contract.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 189.1	(2017) Standard for the Design of High-Performance Green Buildings Except Low-Rise Residential Buildings
ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASTM INTERNATIONAL (ASTM)

ASTM D6245	~~(2021) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation~~(2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation
ASTM D6345	~~(2010) Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air~~(2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	~~(2022) Standard for Portable Fire Extinguishers~~(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008	(2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	~~(2014) Safety and Health Requirements Manual~~ Safety -- Safety and Health Requirements Manual

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

NAVFAC P-307	(2016) Weight Handling Program Management

U.S. ARMY GARRISON JAPAN (USAG-J)

Regulation 190-13	(2013) Installation Access and Control Procedures

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information onlysubmittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Initial Site Conditions Photos
Key Personnel Qualifications; G
~~[~~ ~~Conformed Drawings Submittal]~~
~~[~~ INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN; G~~]~~

SD-07 Certificates

Monthly Progress Photos;

SD-11 Closeout Submittals

~~[~~ Construction Completion Photos;~~]~~

1.4 ACCEPTABLE ALTERNATIVE JAPANESE STANDARDS

Where indicated in the technical specifications REFERENCES paragraphs, acceptable Japanese standards are listed which are considered acceptable alternatives to the US Standards within the Contract Specifications. Contract specifications authorize the use of specified acceptable alternative Japanese standards. A limited set of modified UFGS incorporating previously identified and accepted Japanese standards and methods (also known as Japan Edited Specifications, or JES) can be found at https://www.poj.usace.army.mil/Business-With-Us/References/. Other portions of the specifications also authorize the use of specified acceptable alternative Japanese standards. The use of products not meeting the applicable US standards of the Contract or alternative Japanese standards specifically authorized by the Contract is prohibited unless authorized by the Contracting Officer. Japanese standards not identified in these specifications as being acceptable alternatives to US standards may be submitted to the Contracting Officer for consideration as acceptable alternatives after award using the variation request process set forth in Section 01 33 00 SUBMITTAL PROCEDURES. Prior to the use of such proposed additional alternative Japanese standards, written approval by the Contracting Officer is required. Incorporation of methods, materials, and equipment that will promote cost-effective and timely maintenance, and that are otherwise authorized by the contract, is highly encouraged.

~~[1.5 GOVERNMENT FURNISHED CONTRACTOR INSTALLED ITEMS~~

~~Attachment 01 11 00.00 10-A "Government Furnished Property" provides the items that shall be Government Furnished, Contractor Installed (GFCI).~~

~~]~~1.5 MANDATORY U.S. TESTED PRODUCTS

The following items shall meet U.S. testing methods (e.g. UL listed, FM approved, ASTM, etc.) and shall be labeled as required. These items shall not be substituted with Japanese testing methods (e.g. JIS, JASS, etc.) for certification:

a. Fire suppression systems, including valves, alarm valves, sprinklers ~~(pipes and fittings of Japanese manufacturer may meet the function requirements)~~. Fire suppression systems includes wet chemical, dry chemical, mist, deluge, pre-action, foam, and clean agent, in addition to standard sprinkler systems.

b. Fire pumps, including motors, controllers, drives, and valves.

c. Fire alarm and mass notification systems, including panels, initiating devices, notification appliances, smoke alarms~~ (conductors and conduits of Japanese manufacturer may meet the functional requirements)~~.

d. Engineering technician requirements for development of fire suppression systems and alarm system shop drawings, calculations, and material submittals.

e. Fire doors, smoke doors, and frames.

f. Fire dampers and smoke dampers.

g. Interior finishes with flame spread and smoke development ratings required for installation of rated assemblies.

h. Insulation with flame spread and smoke development ratings.

i. Fire proofing and fire stopping materials.

j. Plenum rated cables.

k. Domestic water supply piping in the building and plumbing fixtures that directly dispense potable drinking water (NSF stamped and/or labeled). ~~Japanese plumbing fixtures~~Plumbing fixtures conforming to Japanese standards that do not directly dispense potable drinking water are allowed, such as urinals and water closets, bath and shower faucets, utility/janitor faucets, lavatories and sinks.

1.6 KEY PERSONNEL QUALIFICATIONS

The Contractor's Project Key personnel shall not be assigned duties to any other Contracts ~~(excluding the project manager)~~ without approval from the Contracting Officer. ~~[~~The Project Superintendent, CQC System Manager, and Site Safety Health Officer shall all be separate persons, and shall not have other duties assigned.~~]~~ ~~[One person may be assigned the duties of Project Superintendent and CQC System Manager provided they meet all qualification requirements and maintain presence on site as required for the various duties. The Site Safety Health Officer shall not be assigned other duties.]~~ Within five (5) working days after receipt of the Notice to Proceed, the Contractor shall submit in writing to the Contracting Officer an organizational chart, the qualifications and background history of the proposed Key Personnel for approval. The Contracting Officer shall have the explicit right to determine acceptability (or rejection) of the proposed individuals. In addition, the Contractor shall be responsible to replace said individuals upon notification by the Government should performance become inadequate during the Contract period. Key Personnel shall attend the Preconstruction conference.

~~[~~1.6.1 Program Manager

The Program Manager is responsible for the overall management of the Contract. The Program Manager shall have a minimum of ~~[~~five (5)~~]~~ years experience in the administration of construction Contracts on construction projects similar in size and scope to this Contract, and shall have a thorough knowledge of the duties of key management personnel assigned to this Contract.

~~]~~1.6.2 Project Manager

The Project Manager shall have a minimum of ~~[~~five (5)~~]~~ years experience as a Project Manager on construction projects ~~[containing [___] components]~~ similar in size and scope to this Contract. And also, Construction Management and experience in building construction and renovation, ideally housing (multi-unit residential housing). The Project Manager shall maintain oversight of Contract proposals prepared by the Contractor staff and be authorized to negotiate Contract terms and sign Contract documents on behalf of the Contractor. The project manager does not need to be present on the site daily, but shall attend weekly progress meetings and be available on site within 24 hours upon request. Project Manager ~~[~~shall not have other duties assigned~~]~~~~[Project Manager may have other duties assigned]~~.

1.6.3 Project Superintendent (Supervisor)

The Project Superintendent shall be on the work site when on-site work is being performed and shall be available to the Contracting Officer or his representatives upon request. The Project Superintendent shall have overall responsibility for all operations at the job site and be authorized to make decisions, negotiate Contract terms and sign Contract documents on behalf of the Contractor. The Project Superintendent shall have a minimum of ~~[~~five (5)~~]~~ years experience as a superintendent on construction projects ~~[containing [___] components]~~similar in size and scope to this Contract~~[~~.~~][~~ And also, Construction Management and experience in building construction and renovation, ideally housing (multi-unit residential housing), and have at least one the following qualifications:~~]~~

~~[~~ a. U.S. Registered Structural Engineer or U.S. Registered Architect or 1 Kyu Kenchikushi (1st Class Qualified Architect).

~~][~~b. U.S. Registered Civil Engineer or 1 Kyu Doboku Sekou Kanrigishi (1st Class Civil Engineering Works Management Engineer).

~~][~~c. U.S. Registered Mechanical or Electrical Engineer or 1 Kyu Kankouji Sekou Kanrigishi (1st Class Building Mechanical and Electrical Engineer).

~~][~~d. U.S. Registered Electrical Engineer or 1 Kyu Denkikouji Sekou Kanrigishi (1st Class Electric Construction Management Engineer).

~~][~~e. U.S. Bachelors Degree in Construction Management or 1 Kyu Kenchiku Sekou Kanrigishi (1st Class Building Construction Management Engineer)

~~][~~f. U.S. Bachelors Degree in Construction Management or AI*DD Sougoushu (AI*DD Construction Engineer).

~~][g. U.S. Registered Fire Protection Engineer.~~

~~]~~1.6.4 English Speaking Representative (Interpreter)

At all times during the Contract period the Contractor shall have an employee capable of fluent bilingual speech in the Japanese and English languages at the job site. The bilingual interpreter shall have the capability to receive and issue concise and technical explanation and instructions between the Government representative(s) and Contractor supervisory personnel concerning all aspects of Contract administration and construction. Within fifteen (15) days after receipt of the Notice to Proceed, the Contractor shall submit in writing to the Contracting Officer ~~'s Representative~~ the qualifications and background history of the proposed interpreter for approval. The Contracting Officer shall have the explicit right to determine acceptability or rejection of the proposed individual. In addition, the Contractor shall be responsible to replace said individual upon notification by the Government should performance become inadequate during the Contract period. The interpreter shall attend the preconstruction conference. Interpreter shall not have other duties assigned.

1.6.5 Contractor Quality Control

1.6.5.1 Contractor Quality Control System Manager (CQCSM)

The CQC System Manager is required to be a graduate of an accredited college with an engineering, architecture, or construction management degree, with a minimum of ~~[~~5~~]~~ years construction experience on construction of similar size and scope to this Contract. Refer to Section 01 45 00.00 10 QUALITY CONTROL, Paragraph "CQC System Manager" for additional requirements.

1.6.5.2 Contractor Quality Control Personnel

The following Contractor Quality Control (CQC) Personnel are required:~~[~~ Civil, ~~][~~Mechanical, ~~][~~Electrical, ~~][~~Structural, ~~][~~Architectural, ~~][~~Fire Protection, ~~][~~Communications, ~~][~~Environmental, ~~][~~Submittals, ~~][Occupied Family Housing,][~~Concrete, Pavements, and Soils, ~~][~~Testing, Adjusting, and Balancing~~,][Design Quality Control Manager]~~. All other specialties in section 01 45 00.00 10 not identified in this paragraph does not apply to this project.

See Section 01 45 00.00 10 QUALITY CONTROL for qualification requirements for CQC personnel. These individuals or specialized technical companies ~~[~~ must be directly employed by the prime Contractor and can not be employed by a supplier or subcontractor on this project~~][may be employed by the prime contractor or a subcontractor on this project]~~. These individuals ~~[~~ shall have no other duties other than quality control~~] [can perform other duties but need to be allowed sufficient time to perform the specialized personnel's assigned quality control duties as described in the Quality Control Plan]~~. ~~[~~A single person can cover more than one area provided that the single person is qualified to perform quality control activities in each designated and that workload allows.~~][A single person cannot cover more than one area].~~

1.6.6 Site Safety and Health Officer (SSHO)

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

~~[1.6.7 [SUSTAINABILITY REPRESENTATIVE][TPC SUSTAINABILITY PROFESSIONAL]~~

~~See Section 01 33 29 SUSTAINABILITY REPORTING.~~

~~]1.6.8 Pass Coordinator~~

~~A person on the contractor's staff who is familiar with the requirements of USAG-J Regulation 190-13, Individual and Vehicle Access Pass Procedures, and USAG-J vehicle security inspection procedures. The Pass Coordinator must be able to obtain a DBIDS access pass with escort privileges and may be required to escort other contract personnel. Although a TOEIC score is not required, the Pass Coordinator must be able to understand pass requirements and complete pass application forms in the English language.~~

1.7 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER

a. This paragraph specifies the procedure for determination of time extensions for unusually severe weather in accordance with the Contract clause entitled "DEFAULT (FIXED-PRICE CONSTRUCTION)". In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:

   (1) The weather experienced at the project site during the Contract period must be found to be unusually severe, that is, more severe than the adverse weather anticipated for the project location during any given month.

   (2) The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the Contractor.

b. The listing below defines the monthly anticipated adverse weather days for the Contract period and is based upon 20th Weather Squadron (MAC), U.S. Air Force; Iwakuni U.S. Marine Corps Air Station, Weather Service; Sasebo Naval Pacific Meteorology and Oceanography Detachment; JASDF (Japan Air Self Defense Force), or similar data for the geographic location of the project. (More specific information may be obtained as specified under Special Contract Requirements Clause entitled "Physical Data.")

MONTHLY ANTICIPATED ADVERSE WEATHER DELAYS
WORK DAYS BASED ON (5) DAY WORK WEEK

	Misawa-area	Kanto Plain	Iwakuni	Okinawa	Sasebo	Osaka
JAN	8	1	2	5	7	4
FEB	9	3	4	7	6	4
MAR	6	6	7	8	7	5
APR	3	6	6	6	7	4
MAY	4	4	6	6	6	4
JUN	5	9	8	6	12	5
JUL	6	6	6	6	9	4
AUG	6	6	5	9	9	3
SEP	6	7	6	7	9	4
OCT	4	5	4	4	8	4
NOV	6	4	4	4	5	3
DEC	7	1	3	5	5	3

The above schedule of anticipated adverse weather days shall constitute the base line for monthly (or portion thereof) weather time evaluations.

c. Upon acknowledgment of the Notice-to-Proceed (NTP) and continuing throughout the Contract, the Contractor shall record on the daily CQC report, the occurrence of adverse weather and resultant impact to normally scheduled work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the Contractor's scheduled work day.

d. The number of actual adverse weather delay days shall include days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated in paragraph (b), above, the Contracting Officer shall convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the Contract clause entitled "DEFAULT (FIXED PRICE CONSTRUCTION)".

e. For all work under this Contract, adverse weather is defined as:

(1) Rainfall - Number of occurrences of precipitation greater than or equal to ~~0.10 inches (~~2.54mm~~)~~.

(2) Snowfall - Number of occurrences of precipitation greater than or equal to ~~1.00 inches (~~25.4mm~~)~~. ~~(Not applicable to Okinawa Area)~~

(3) Cold Temperature - Number of occurrences when daily maximum temperature does not exceed the monthly mean low temperature or ~~32 degrees Fahrenheit (~~0 degrees Celsius~~)~~, whichever is lower. ~~(Not applicable to Okinawa Area)~~

(4) Concurrence between snowfall and cold temperature is 80 percent, i.e. 80 percent of the time snow falls, the temperature is "cold".

(5) Wind - Number of occurrences when the wind is gusting ~~30 knots (~~56 kilometers/hour~~)~~ or greater.

f. The Contractor's schedule must reflect the above anticipated adverse weather delays on all weather-dependent activities.

1.8 PERMITS AND RESPONSIBILITIES

The Contractor shall, without additional expense to the Government, be responsible for obtaining any necessary licenses and permits, and for complying with any laws, codes, and regulations (including the requirements of material, prefectural, and local Government of Japan, and associated military installation) applicable to the performance of work. The Contractor shall also be responsible for all damages to persons or property that occur as a result of the Contractor's fault or negligence. These damages shall be repaired or replaced by the Contractor at no cost to the Government. The Contractor shall take proper safety and health precautions to protect all work and workers.

Submit a schedule of planned road closures to the Contracting Officer with the initial project schedule. Notification of specific road closures shall be in writing to the Contracting Officer not less than thirty (30) calendar days in advance of the intended closure. The road closure request shall include the planned traffic control measures as well as the general information about the closure. All road closures shall be coordinated with base officials and are subject to base requirements. No road closure shall be permitted until the Contractor receives written approval from the Contracting Officer. Full road closures is generally not permitted (at least one way traffic shall always be provided).

~~1.8.1 Naha Port Authority (NPA) Work Permit~~

~~The Contractor shall obtain a Harbor Facility Usage Permission Application (Form Number 5), also known as the "Kowan Shisetsu Shiyou Kyoka Shinseisho" permit form addressed to the Naha Port Authority Administrator.~~

~~The submission shall include basic information regarding the Contractor's point of contact information, project title, description of work, and usage area information, as well as a work plan (to include scheduling), emergency contact plan, and protection plans. Application shall be hand delivered to:~~

~~Naha Port Authority~~
~~General Affairs Division~~
~~2-1 Tondocho (2nd Floor)~~
~~Naha City, Okinawa Prefecture~~
~~900-0035~~

~~The NPA will require scheduling coordination and due consideration to all Naha Commercial Port end users and coordination with other stakeholders, such as the Local Fisheries Association, to ensure that stakeholder operations are not negatively impacted by the proposed activity and ensure that appropriate coordination is performed by the construction activity proponent. The final Harbor Facility Usage Permit Application will be authorized by the Okinawa Prefectural Governor's office.~~

~~1.8.2 Work Application Package~~

~~In accordance with Article 31 of the GOJ Act on Port Regulations "Kou Sokuhou", the Contractor shall submit a Work Application Package known as a "Koji Sagyou Kyoka Shinseisho" for review and approval by the Maritime Traffic Division Office of the Japan Coast Guard (JCG) at Naha. A~~

~~physical printed out copy shall be hand delivered to the following address:~~

~~Japan Coast Guard~~
~~Naha Coast Guard Office~~
~~Maritime Traffic Division~~
~~2-11-1 Minatomachi~~
~~Naha City, Okinawa Prefecture~~
~~900-8548~~

~~This application shall contain project safety information, work planning and scheduling, emergency management information as well as copies of all pertinent certifications and licenses (e.g. boat licenses, commercial diver's certifications, etc.) The minimum application review period is 30 days, although in the case of complicated work sites, review of work permit application packages by the JCG may take longer.~~

~~The Naha Coast Guard Office normally does not accept a work permit package from a Contractor until approval of work is formalized first with the NPA. Therefore, ample time must be secured for the dredging Contractor to clear both NPA and JCG Naha Coast Guard Office work permitting processes during the pre-construction phase of the dredging project.~~

~~1.8.3 Safety Panel and Committee (SPC)~~

~~As part of the work permitting process, the Contractor shall also participate in a Safety Panel and Committee (SPC). This SPC is required by the Japan Coast Guard, and shall be facilitated by a consultant hired by the Okinawa Defense Bureau. Intent of the SPC is to provide a forum under which all stakeholder and port operator concerns can be identified and addressed. The Contractor shall attend and present proposed safety, environmental, and operational plans at the forum.~~

1.9 SPECIAL CONTRACT REQUIREMENTS

1.9.1 Meetings

a. ~~Predesign Conference. A predesign conference per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.~~

a. ~~Preconstruction Conference. A Preconstruction Conference per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.~~

a. Post Award Orientation. A post award orientation (often referred to as a Preconstruction Conference) per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.

b. Weekly Progress Meetings. A weekly progress meeting shall be conducted, with the day, time, and location to be determined at the Preconstruction Conference. The Contractor's Project Manager, Superintendent, Interpreter, and Quality Control Manager shall attend. Representatives from the Government may include, but are not limited to, Director of Public Works/Facilities Engineering personnel, Project's Resident Office personnel, and Project Manager. The Contractor shall be prepared to discuss work completed during the previous week, work currently in progress, and work forecasted for the following week, as well as the status of any construction issues and open action items. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the progress meeting for review.

c. Red Zone Meeting. A Red Zone meeting shall be conducted for Contracts with a value in excess of 55,000,000 JPY or those deemed sufficiently complex by the Contracting Officer to warrant one. The purpose of the Red Zone meeting is to discuss closeout requirements for the contract and to establish a timeline to get those items completed (see Section 01 78 00 CLOSEOUT SUBMITTALS for typical closeout requirements). The meeting is a good time for the Contractor to gain approval on format for items such as Operations and Maintenance manuals, the equipment-in-place list, the warranty plan, and any other items. The Initial Red Zone is typically scheduled when project reached 75 percent completion milestone. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the Red Zone meeting.

d. Safety and Quality Control Meetings. Safety and quality control meetings shall be held as needed and determined by the Contractor. See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and Section 01 45 00.00 10 USACE QUALITY CONTROL for specific requirements.

e. Construction Waste Management Meetings. See Section 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT for requirements.

f. Monthly Design Progress Meetings. A monthly design progress meeting shall be conducted, with the day, time, and location to be determined at the project kickoff meeting. The Contractor's Design team, Interpreter, and Design Quality Control manager shall attend. Representatives from the Government may include, but are not limited to, Client, Project Manager, and POJ Engineering personnel. The Contractor shall be prepared to discuss work completed during the previous week, work currently in progress, and work forecasted for the following week, as well as the status of any design issues and open action items. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the progress meeting for review.

~~[1.9.2 Conformed Drawings Submittal~~

~~Contractor shall submit one (1) hard copy Japanese-style folded and bound set of conformed A3 size drawings. Submittal shall be provided to the Resident Office at the Pre-Construction meeting. Conformed drawings are defined as the original advertised drawings with amended drawing sheets replaced in-kind with the corresponding original sheet number. If applicable, add new sheets and remove deleted sheets in accordance with amendment instructions. If multiple amendments are issued, supersede any previously amended drawing so that only the highest numbered amendment drawing is included.~~

~~]~~1.9.2 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 days after Notice to Proceed and not less than 10 days before the preconstruction meeting. The Plan shall include IAQ management practices implemented during construction and preoccupancy phases and describe how each requirement in ANSI/SMACNA 008 ~~[and TPC requirements]~~will be met, addressed, and managed on the job site. The plan shall cover the following additional items:

a. Identify the key players and person(s) responsible for implementing the plan.

b. Specify procedures for protecting stored and installed absorptive materials from moisture damage.

c. Prohibit the use of tobacco products inside the building and within 7.5 meters~~ (50 feet)~~ of the building entrance at all times during construction. Consider prohibiting smoking on the entire job site. If smoking will be allowed, locate on a site plan the designated locations and signage. Comply with the tobacco use policy of the local jurisdiction.

d. Indicate whether air handlers will be operated during construction, and specify compliant filtration procedures for permanent equipment that will be used.

e. Take photographs of each IAQ measure for documentation. Annotate photographs to indicate each IAQ measure depicted and its general location. Provide photographs of the methods employed to protect stored and installed absorptive materials from moisture damage during construction and preoccupancy.

Revise and resubmit the Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.

1.9.2.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.9.2.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008, Chapter 3, to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.

b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building.

Designate receiving/storage areas for incoming material that minimize IAQ impacts.

c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.

d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.

e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.9.2.1.2 Moisture Contamination

a. Remove accumulated water and keep work dry.

b. Use dehumidification to remove moist, humid air from a work area.

c. Do not use combustion heaters or generators inside the building.

d. Protect porous materials from exposure to moisture.

e. Remove and replace items which remain damp for more than a few hours.

1.9.2.2 Requirements after Construction

After construction ends and prior to occupancy, conduct IAQ testing in accordance with ASHRAE 189.1 Section 10.3.1.4.

In the event that IAQ testing fails, conduct a building flush-out in accordance with ASHRAE 189.1 Section 10.3.1.3~~[and TPC requirements]~~. The space shall be ventilated at a minimum rate of 1.5 Liters per second per square meter of outdoor air while maintaining an internal temperature of at least 15 degrees C ~~(60 degrees F)~~ and no higher than 27 degrees C ~~(80 degrees F)~~ and relative humidity no higher than 60%. Total air volume of outdoor air will be determined by ASHRAE 189.1 Section 10.3.1.4 Equation 10-1. Total volume of outdoor air shall be approved by the Contracting Office prior to building flush-out

Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Ambient Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.10 PROGRESS AND COMPLETION PHOTOGRAPHS

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600 by 1200 by 24 bit true color 300 DPI minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of ~~[20]~~50 digital photographs each week throughout the entire project~~[~~ and ~~[20]~~50 digital photographs at completion of project~~]~~. Submit the Initial Site Conditions Photos, Monthly Progress Photos~~[~~, and Construction Completion Photos~~]~~. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Include a date designator in file names. Cross reference submittals in the appropriate daily report. Photographs provided are for unrestricted use by the Government.

1.11 BASE REGULATIONS

The Contractor and Subcontractor(s) shall become familiar with and obey all base regulations, including fire, traffic, safety, and security regulations. All Contractors shall keep within the limits of the work (and avenues of ingress and egress), and shall not enter any restricted areas unless required to do so and are cleared for such entry. The Contractor's equipment shall be conspicuously marked for identification.

1.11.1 Request for Contractor's Employee Passes and Vehicle Passes

No employee or representative of the Contractor shall be admitted to the work site without a Contracting Officer's furnished authorized admittance. Refer to USAG-J Regulation 190-13. Prior to the start of on-site work, submit applications for base passes to the Contracting Officer for key employees (project manager, site supervisor, interpreter, etc.) for long term DBIDS passes. The Contractor (through the employment of a Pass Coordinator), with assistance from the Resident Office, is responsible for securing sufficient passes for other workers, including subcontractor's employees and vehicles, required to access the base for the project duration. Additional personnel data shall also be furnished.

~~The Contractor shall use the USAG-J Form 1529 "EZ Pass" as the primary means of obtaining passes for all personnel who do not qualify for DBIDS passes. The USAG-J Form 1529 "EZ Pass" One Time/Multiple Access Roster must be complete and submitted to the Provost Marshall Pass and Vehicle Registration Office a minimum of three (3) U.S. workdays prior to access. The use of the USAG-J "one day" pass (AJ Form 47) should be avoided, and used only for urgent situations. Applicants for a "one day" pass may be required to wait for several hours for their pass to be issued. Contractor personnel applying for more than one (1) "one-day" pass within any 15-day period will not be issued multiple one-day passes, and will instead be required to apply for access via the USAG-J Form 1529, and wait three or more days for approval.~~

Provide all information required for background checks to meet base access requirements to be accomplished by Base Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must comply with all personal identity verification requirements as directed by DOD, HQDA and/or local policy. The Government reserves the right to make changes to the Contractor security requirements or processes due to a change in the Force Protection Condition (FPCON).

In addition to security requirements, cranes, delivery "unic" trucks, forklifts, backhoes, and other equipment capable of vertical lift operations must comply with NAVFAC rules for access to the base. Refer to NAVFAC P-307. The operator must submit a P-1 Pass request and submit the equipment, its records, and operator records for inspection by Public Works prior to the initial entry onto base property. Vehicle passes for vertical lift equipment must be renewed monthly with Public Works as long as the equipment remains on base property.

Upon completion of this Contract, return all employee and vehicle passes to the base, and obtain a certification of receipt. Final payment shall be withheld until all passes have been returned.

1.11.2 No Smoking Policy

Smoking is prohibited on installations except in designated smoking areas. This applies to existing buildings, buildings under construction, and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer shall identify designated smoking areas.

1.11.3 Munitions and Explosives of Concern (MEC)

Munitions and Explosives of Concern (MEC): Unexploded Ordnance (UXO), Material Presenting a Potential Explosive Hazard (MPPEH), Chemical Agents (CA), or Discarded Military Munitions (DMM) on jobsites shall be treated as extremely dangerous and must be reported immediately. Follow the 3Rs: RECOGNIZE, RETREAT, and REPORT. In the event MEC are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer's Representative. Contractor shall provide dispatch and Contracting Officer's Representative with specific location of the item. Contractor personnel shall stop work in the immediate vicinity of the discovery and maintain a minimum distance of 300 meters from the item. Contractor shall maintain flexibility in redirecting personnel and work effort in the event that items possessing an explosive hazard are discovered and construction personnel are excluded from areas during the destruction/removal process.

~~1.11.3.1 Additional Requirements for Camp Foster~~

~~All workers on site involved in the earthwork, in any manner, must attend a UXO brief on Marine Camp Foster. Upon completion of the brief, the worker will receive a laminated card verifying that they attended the brief. The card does not expire and is valid for all USACE construction contracts on Okinawa, but is non-transferrable between workers. The USACE team will help coordinate the training and assist with processing base access requests for Marine Camp Foster as needed. The brief is free of charge, but the contractor is responsible for all other costs associated with transport, labor, and all other costs related to the training. The expected duration of this training is two hours.~~

~~1.11.4 Unexploded Ordnance (UXO) Procedures on Kadena Air Base~~

~~Kadena Air Base 718 CES/CEN assesses the area as UXO unlikely with a low probability of uncovering UXO's on the site. However, since there is always a possibility of encountering UXOs anywhere on Okinawa, albeit unlikely at this location, excavations and other construction activities related earth work on the project site, require a measure of caution. The Ground Penetrating Radar (GPR) survey specified in 01 35 26 paragraph "Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces" for the identification of underground utilities will identify anomalies. Best practices of conducting GPR surveys are to repeat them in areas greater than the used method's level of detection (LOD), down to a depth of 2 meters, to ensure no utility lines or other suspicious objects are present. If suspected UXOs are identified, the construction team needs to follow the standard UXO notification procedures per the Kadena policies identified in attachment 01 11 00.00 10-[B] "UXO Emergency Mitigation Steps and Survey Policies." The contact number for~~

~~the Emergency Dispatch Center for Kadena Air Base is 098-934-5911. See the following website for additional information: www.kadena.af.mil/About-Us/Emergency-Actions/.~~

1.12 ORDER OF WORK

1.12.1 Schedule

Schedule all work to cause the least amount of interference with activity operations. Permission to interrupt any activity or roads shall be requested in writing a minimum of 30 calendar days prior to the desired date of interruption. Interruptions of activities, roads, and utility services shall be allowed only when they will not cause interference with the operations of the activity. The Contractor shall remove and dispose off Government property all Contractor generated debris at the end of each shift, or more frequently if required, to keep the space usable. Upon award of this Contract, the Contractor shall begin and complete all required work; ready for use and including cleanup, within the time period specified on this Contract. All work scheduled in occupied areas shall be accomplished in such a manner as to cause the least possible inconvenience to the occupants.

1.12.2 Working Hours

Normal Working Hours and Days are Monday through Friday, 0800 hours through 1700 hours. Saturday work is allowed on this contract and does not require additional approvals. Notify the Contracting Officer at least three days in advance when scheduling Saturday work.

1.12.3 Noise and Vibration

Noise greater than 70 decibels and vibration producing work in occupied buildings shall be conducted outside core school/business hours (0700 - 1430) unless written permission is given by the Contracting Officer.

1.12.4 Work Outside of Regular Hours

If the Contractor desires to carry on work outside the regular hours, on Sundays, or holidays, a written application shall be submitted to the Contracting Officer or his representative for approval. The Contractor shall allow three working days notice to enable satisfactory arrangements to be made by the Government for inspecting the work in progress. If work is to be accomplished after daylight hours, the Contractor shall illuminate the area in a manner approved by the Contracting Officer or their representative and in accordance with EM 385-1-1. Unless directed by a Contracting Officer, work accomplished outside regular working hours shall be at no additional cost to the Government.

~~1.13 NUCLEAR DENSOMETER TESTING~~

~~Nuclear densometer testing and all forms of radiography are not permitted on U.S. Navy installations in Japan because such testing may interfere with Government of Japan monitoring activities.~~

1.13 FIRE PREVENTION DURING CONSTRUCTION

1.13.1 General

Comply with all pertinent fire prevention provisions of the US Army Corps of Engineers Manual EM 385-1-1, NFPA 241 and shall follow the Installation fire regulations. Prior to commencement of welding or other hot work operations, obtain approval from the Installation Fire Chief.

1.13.2 Supply

No more than one day's supply of paint, paint materials, or compounds shall be allowed within the area of the building, and shall be removed from the job site after each working day. No gasoline or similar low flash point flammable liquid shall be allowed within the building area. After proper coordination with the Facility's Emergency Services (to confirm the maximum allowable quantities and location of storage), storage of additional product may be authorized.

1.13.3 Fire Extinguishers

Provide, as a minimum, the number, size, and type of fire extinguishers in accordance with the latest NFPA 10. The Contractor shall comply with the Installation Fire Chief's policies if they are more stringent than NFPA 10. Fire extinguishers shall remain the property of the Contractor and shall be removed upon completion of the project.

1.13.4 Housekeeping

Accumulations of combustible material shall be removed from the building area on a daily basis.

1.13.5 Handling of Gasoline

Gasoline shall be stored in industry standard approved safety containers. Adequate ventilation shall be provided to safely dispose of flammable vapors where flammable liquids are utilized. Gasoline powered equipment shall be refueled a minimum 6 meters away from the building area.

1.13.6 Notification of Fire

Be familiar with methods for notifying the Installation Fire Department. The Installation's fire poster shall be posted in conspicuous locations and at telephones in construction shacks.

1.14 CONTRACTOR FURNISHED MATERIAL AND WORKMANSHIP

The Contractor shall furnish all materials necessary for performance of the work of this contract unless otherwise specified. Materials procured shall be new and shall meet any specifications and standards listed in these specifications. If no specification for a material needed to perform this contract are stated, the material shall be new, of acceptable industrial grade and quality, equal to or better than the manufacturer's original equipment for equipment being repaired or replaced, and will be compatible with existing materials and systems. All materials provided under this contract shall be free of asbestos, lead in paint, and PCB. The Contracting Officer reserves the right to request submittal of any material being provided under this contract. The Contracting Officer shall make the final determination of the acceptability of any material used on this contract.

1.15 CORRESPONDENCE

All correspondence addressed to the Government shall be made through serialized letters furnished with one original and two copies. Serialized letters shall begin with the number S-0001 and shall be continuous without a break in numbering. Serialized letters shall include the contract title and number, date, subject, and shall be signed by an authorized representative of the Contractor.

1.16 CYBERSECURITY DURING CONSTRUCTION

In addition to the control system cybersecurity requirements indicated in any other section, meet the following requirements throughout the construction process.

1.16.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.16.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.16.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must be scanned using the installed software at least once per day.

1.16.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.16.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.16.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.16.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Each Statement must be signed by a cybersecurity representative for the relevant company. See Attachment 01 11 00.00.10-A.

1.16.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.16.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than those specifically provided or furnished for this project. Any and all access to the network from outside the project site is prohibited.

1.16.3 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

PART 1 GENERAL

1.1 GENERAL

Payment items for the work of this Contract for which payments shall be made are listed in the CONTRACT LINE ITEM NUMBER (CLIN) SCHEDULE and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, are included in the listed job item most closely associated with the work involved. The price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, topographic surveys, cost of bond premiums, and for performing all work required by drawings and specifications for which separate payment is not otherwise provided. Work paid for under one item shall not be paid for under any other item. No separate payment shall be made for the work, services, or operations required by the Contractor, as specified in DIVISION 01, GENERAL REQUIREMENTS, to complete the project in accordance with these specifications; all costs thereof shall be considered as incidental to the work.

1.2 Submittals

Government approval is required for submittals with a "G" designation; DESIGN-BID-BUILD:submittals not haveing a "G" designation are for information only; DESIGN-BUILD: submittals not having a "G" designation are for Contractor Quality Control or Designer of Record appproval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

Bill Of Lading; G

1.3 LINE ITEMS

1.3.1 CLIN 0001, ~~Construction of Primary Facility~~Demolition and construction/renovation of Building 743, including 68 family housing units, common areas, multi-purpose rooms, community activities center, and all associated supporting facilities work

1.3.1.1 Payment

~~Payment shall be made at the Contract job price for CLIN 0001 "Construction of Primary Facility" for work completed including, but not limited to []. Payment for work completed shall constitute full compensation of CLIN 0001, complete.~~Payment shall be made at the Contract job price for CLIN 0001 "Demolition and construction/renovation of Building 743" for work completed including,
but not limited to update of existing dwelling unit layouts, repair and replacement of unit and common area finishes, cabinetry, and counter tops, new fire pump room, balcony floor repair, refinish roof areas, replacement

of roof drains, downspouts, and splash blocks, replace aluminum windows, replace doors, new HVAC system, new plumbing system including water, sanitary, vent, and drain piping, new plumbing fixtures, new electrical wiring for new HVAC and water heater systems, new main service switchboard, new pad-mounted transformer, new standby generator, interior power and lighting for dwelling units, common area, and utility spaces, new LED exit signs, telecommunications infrastructure upgrades, new cabling, equipment, pathways, and telecommunications room, replace all fire sprinklers, replace all smoke/CO detectors, replace existing fire alarm strobes and speaker strobes, cybersecurity, commissioning, mold, asbestos, lead/cadmium/chromium, PCB, and low level radioactive material abatement. Payment for all work completed shall constitute full compensation of CLIN 0001, complete.

1.3.1.2 Unit of Measure

Unit of measure: job.

1.3.2 CLIN 0002, ~~Construction of Supporting Facilities, 1.5 meters outside of Buildings~~Demolish Building S-726 and construct new parking area and playground.(OPTION 1)

1.3.2.1 Payment

~~Payment shall be made at the Contract job price for CLIN 0002 "Construction of Supporting Facilities, 1.5 meters outside of buildings" for work completed including, but not limited to, construction of Supporting Facilities including special construction features [Pile Foundations], canopies, electrical utilities, communication utilities, water/sewer/utilities, site preparation, roads, sidewalks and parking, site improvements, AT/FP, demolition, low impact development, and environmental mitigation as shown on the drawings and specifications 1.5 meter outside of buildings, complete. Payment for work completed shall constitute full compensation of CLIN 0002, complete.~~Payment shall be made at the Contract job price for CLIN 0002 "Demolish Building S-726 and construct new parking area and playground." for work completed. Payment for all work completed shall constitute full
compensation of CLIN 0002, complete.

1.3.2.2 Unit of Measure

Unit of measure: job.

1.3.3 CLIN 0003, ~~Excavation~~Upgrade hydronic distribution from a two-pipe system to a four-Pipe system.(OPTION 2)

1.3.3.1 Payment

~~Payment will be made for costs associated with excavation for the channel and for the structure, which includes performing required excavation and other operations incidental thereto, Contractor-furnished disposal area(s) and disposition of excess excavated material and unsuitable and frozen materials.~~Payment shall be made at the Contract job price for CLIN 0003 "Upgrade hydronic distribution from a two-pipe system to a four-Pipe system." for work completed. Payment for all work completed shall constitute full
compensation of CLIN 0003, complete.

~~1.3.3.2 Measurement~~

~~The total quantity of excavated material for which payment will be made will be the theoretical quantity between the ground surface as determined by a survey and the grade and slope of the theoretical cross sections indicated. No allowance will be made for overdepth excavation or for the removal of any material outside the required slope lines unless authorized.~~

1.3.3.2 Unit of Measure

Unit of measure: ~~cubic yard (CY)~~job.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PROGRESS PAYMENT INVOICE

Requests for payment shall be submitted in accordance with Federal Acquisition Regulations (FAR) Subpart 32.9, entitled "PROMPT PAYMENT", and Paragraphs 52.232-5 and 52.232-27, entitled "Payments Under Fixed-Price Construction Contracts", and "Prompt Payment for Construction Contracts", respectively. In addition each request shall be submitted in the number of copies and to the designated billing office as shown in the Contract.

When submitting payment requests, the Contractor shall complete Blocks 1 through 12 of the "PROGRESS PAYMENT INVOICE" Form as directed by the Contracting Officer (provided in RMS CM). The completed form shall then become the cover document to which all other support data shall be attached.

One additional copy of the entire request for payment, to include the "PROGRESS PAYMENT INVOICE" cover document, shall be forwarded to a separate address as designated by the Contracting Officer.

The Contractor shall submit with each pay request, a list of subcontractors that have worked during that pay period. The listing shall be broken down into weeks, identifying each subcontractor that has worked during a particular week, and indicate the total number of employees that have worked on site for each subcontractor for each week. The prime Contractor shall also indicate the total number of employees for its on site staff for each week.

Formal invoices shall be submitted every month between the first and the tenth of the month. Any invoice received after the tenth will be considered as not meeting the contract requirements and rejected. It may be resubmitted on the first of the following month.

3.2 TRANSPORTATION OF SUPPLIES BY SEA, BILL OF LADING

The Contractor shall, within 30 days after each shipment covered by Defense Federal Acquisition Regulation (DFAR) Subpart 252.247-7023, provide the Contracting Officer and the Maritime Administration (MARAD) one copy of the rated on board vessel operating carrier's ocean Bill of Lading. See DFAR 252.247-7023 for information required in the bill of lading. MARAD contact information: usarmy.scott.sddc.mbx.g3-ffw-team@mail.mil and msc.n101.ffw@navy.mil.

The Contractor shall use U.S.-flag vessels when transporting any supplies by sea under this contract. If U.S.-flag vessels are not available, the Contractor may submit a request for use of foreign-flag vessels in writing to the Contracting Officer at least 45 days prior to the sailing date necessary to meet its delivery schedules. Refer to DFAR 252.247-7023 for required documentation when making this request. The Contracting Officer will process requests submitted as expeditiously as possible, but the Contracting Officer's failure to grant approvals to meet the shipper's sailing date will not of itself constitute a compensable delay. Failure to get approval prior to the use of foreign-flag vessels may result in monetary penalty.

3.3 CONTRACT COST BREAKDOWN

The Contractor must furnish within 30 days after the date of Notice to Proceed, and prior to the submission of its first partial payment estimate, a breakdown of its single job pay item or items which will be reviewed by the Contracting Officer as to propriety of distribution of the total cost to the various accounts. Any unbalanced items as between early and late payment items or other discrepancies will be revised by the Contracting Officer to agree with a reasonable cost of the work included in the various items. This contract cost breakdown will then be utilized as the basis for progress payments to the Contractor.

3.4 CONTRACT ADMINISTRATION

3.4.1 Contracting Officer (KO)

This Contract shall be administered by a Contracting Officer (KO) assigned to the U.S. Army Corps of Engineers, Japan Engineer District, APO AP 96338-5010.

3.4.2 Administrative Contracting Officer (ACO)

In accordance with the USACE Acquisition Instruction (UAI) 5101.603-3-100, Appointment, Administrative Contracting Officer(s) (ACO's) may be delegated a part of the Contracting Officer's authority to modify this Contract and perform Contract administration functions under Federal Acquisition Regulation (FAR) Section 42.302, subject to the authority and limitations set forth in the appointment and delegation letter.

3.4.3 Contracting Officer's Representative (COR)

The Contracting Officer may designate one or more individuals as his/her authorized representatives in administering this Contract. Refer to the clause in Section 00 70 00 CONDITIONS OF THE CONTRACT entitled CONTRACTING OFFICER'S REPRESENTATIVE (DFARS 252.201-7000) for the definition and limited authority of CORs.

3.5 DESIGNATED BILLING OFFICE

~~The designated billing office for this Contract is:~~

~~Resident Engineer, Zukeran Resident Office~~
~~U.S. Army Corps of Engineers, Japan Engineer District~~
~~Unit 35132~~
~~APO AP 96376-5132~~
~~The designated billing office for this Contract is:~~

~~Resident Engineer, Yokosuka Resident Office~~
~~U.S. Army Corps of Engineers, Japan Engineer District~~
~~PSC 473, Box 1571~~
~~FPO AP 96349-0016~~

~~USAED-J~~
~~Box 1571, Bldg. 4328, 3F~~
~~Yokosuka Naval Base~~
~~2381 Tomari-Cho~~
~~Yokosuka-City, Kanagawa~~
~~238-0001 JAPAN~~
~~The designated billing office for this Contract is:~~
~~Iwakuni Resident Office~~
~~U.S. Army Corps of Engineers, Japan Engineer District~~
~~PSC 561, Box 1860~~
~~FPO AP 96310-0019~~

~~USAED-J~~
~~Bldg. 53~~
~~MCAS Iwakuni~~
~~Misumi-Cho~~
~~Iwakuni-City, Yamaguchi~~
~~740-0025 JAPAN~~
The designated billing office for this Contract is:
Kanagawa Resident Office
U.S. Army Corps of Engineers, Japan Engineer District
Unit 45010
APO AP 96343-5010

USAED-J
Bldg. 250
Camp Zama
Zama-City, Kanagawa
252-8511 JAPAN
~~The designated billing office for this Contract is:~~
~~Misawa Resident Office~~
~~U.S. Army Corps of Engineers, Japan Engineer District~~
~~Unit 5254~~
~~APO AP 96319-5254~~

~~USAED-J~~
~~Bldg. 793~~
~~Misawa Air Base~~
~~64 Hirahata~~
~~Misawa-City, Aomori~~
~~033-0012 JAPAN~~
~~The designated billing office for this Contract is:~~
~~Sasebo Resident Office~~
~~U.S. Army Corps of Engineers, Japan Engineer District~~
~~PSC 476, Box 83~~
~~FPO AP 96322-0001~~

~~USAED-J~~
~~Bldg. 200, Rm. 212~~
~~Sasebo Naval Base~~
~~Hirase-Cho~~
~~Sasebo City, Nagasaki~~
~~857-0056 JAPAN~~
~~The designated billing office for this Contract is:~~

~~Yokota Resident Office~~
~~U.S. Army Corps of Engineers, Japan Engineer District~~
~~Unit 5101~~
~~APO AP 96328-5101~~

~~USAED-J~~
~~Unit 5101, Bldg 445~~
~~Yokota Air Base~~
~~Fussa-City, Tokyo~~
~~197-0001 JAPAN~~

-- End of Section --

SECTION 01 32 01.00 10

PROJECT SCHEDULE
08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

AACE 29R-03	(2011) Forensic Schedule Analysis
AACE 52R-06	(2006) Time Impact Analysis - As Applied in Construction
AACE 84R-13	(2015) Planning and Accounting for Adverse Weather

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 67-17	(2017) Schedule Delay Analysis

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1-1-11	(2017) Administration -- Project Schedules

1.2 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S"~~ classification. Submittals not having a "G" ~~or "S"~~ classification are ~~[for Contractor Quality Control approval.][~~for information only. ~~When used, a code following the "G" classification identifies the office that will review the submittal for the Government.]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Project Scheduler Qualifications; G~~[, [_____]]~~

Preliminary Project Schedule; G~~[, [_____]]~~

Initial Project Schedule; G~~[, [_____]]~~

Periodic Schedule Update; G~~[, [_____]]~~

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules for projects of similar size and complexity to this contract, using Primavera P6.~~.~~ Representative must have a comprehensive knowledge of CPM scheduling principles and application.

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

2.1.1 Government Default Software

The Government uses Primavera P6. Ensure exported schedule files are compatible with the version of P6 used by the Government.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required sdef file must be created and supported by the software manufacturer.

2.1.2.1 Primavera

If Primavera P6 is selected for use, provide the "xer" export file in a version of P6 importable by the Government system. Verify at the SEKO meeting which version of P6 is in use by the Government. Export the schedule in a version of P6 no newer than that used by the Government.

2.1.2.2 Other Than Primavera

If the Contractor chooses software other than Primavera P6, that is compliant with this specification, provide for the Government's use two licenses, two computers, and training for two Government employees in the use of the software. These computers will be stand-alone and not connected to Government network. Computers and licenses will be returned at project completion.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. Show in the schedule the proposed sequence to perform the work and dates contemplated for starting and completing all schedule activities. The scheduling of the entire project is required. The scheduling of design and construction is the responsibility of the Contractor. Contractor management personnel must actively participate in its development. Designers, Subcontractors, and suppliers working on the project must also contribute in developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project monitoring tool. Use the Critical Path Method (CPM) of network calculation to generate all Project Schedules. Prepare each Project Schedule using the Precedence Diagram Method (PDM).

3.1.1 Scheduling Expectations Kickoff Meeting(SEKO)

In conjunction with the post award conference or pre construction conference the Government and the Contractor will hold a separate and distinct Scheduling Expectations Kickoff Meeting(SEKO). The intent of this meeting is for the Government to review with the contractor the requirements of the scheduling specification, communicate the Government’s expectations regarding schedule development and management, and answer any questions the Contractor may have regarding the schedule requirements.

3.2 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Match the progress payment "Pay Period Thru" date in RMS to the schedule data date.

3.2.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.

3.2.2 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial, or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, the Contracting Officer may withhold 10 percent of pay request amount from each payment period until such revisions to the project schedule have been made.

3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.3.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution. Failure to develop the Project Schedule to an appropriate level of detail will result in its disapproval. The Contracting Officer will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail:

3.3.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Non-procurement activities Original Durations (OD) are not to exceed 20 workdays or 30 calendar days.

3.3.3 Design and Permit Activities

Include design and permit activities with the necessary conferences and follow-up actions and design package submission dates. Include the design schedule in the project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific contract period. Provide a detailed level of scheduling sufficient to identify all major design tasks, including those that control the flow of work. Also include review and correction periods associated with each item.

3.3.4 Procurement Activities

Include in the schedule activities associated with the critical submittals and their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement activities are those with an anticipated procurement sequence of over 90 calendar days.

3.3.5 Mandatory Activities

Include the following activities in the initial project schedule and all updates.

a. Submission, review and acceptance of SD-01 Preconstruction Submittals (individual activity for each).

b. Submission, review and acceptance of features requiring design completion Submission, review and acceptance of design packages.

c. Submission of mechanical/electrical/information systems layout drawings.

d. Submission and approval of O & M manuals.

e. Submission and approval of as-built drawings.

f. Submission and approval of DD1354 data and installed equipment lists.

g. Submission and approval of testing and air balance (TAB).

h. Submission of TAB specialist design review report.

i. Submission and approval of fire protection specialist.

j. Submission and approval of Building Commissioning Plan, Controls testing plan, test data, and reports: Develop the schedule logic associated with testing and commissioning of mechanical systems to a level of detail consistent with the contract commissioning requirements. All tasks associated with building testing and commissioning will be completed prior to submission of building commissioning report and subsequent contract completion.

   These activities may include but are not limited to: air and water balancing, building commissioning -functional performance testing, controls testing, and performance verification testing. Include other systems testing, if required. Include additional commissioning requirements noted elsewhere in the Contract as required in the schedule and discuss during the SEKO meeting, noted in paragraph SCHEDULING EXPECTATIONS KICKOFF MEETING (SEKO).

k. Contractor's pre-final inspection.

m. Correction of punch list from Contractor's pre-final inspection.

n. Government's pre-final inspection.

o. Correction of punch list from Government's pre-final inspection.

p. Final inspection.

3.3.6 Government Activities

Show in schedule, Government and other agency activities that could impact progress. These activities include, but are not limited to approvals, acceptance, design reviews, environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE) and Notice to Proceed (NTP) for phasing requirements.

3.3.7 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in ER 1-1-11. This exact structure is mandatory. Develop and assign all Activity Codes to activities as detailed herein.

The SDEF format is as follows:

Field	Activity Code	Length	Description
1	WRKP	3	Workers per day
2	RESP	4	Responsible party
3	AREA	4	Area of work
4	MODF	6	Modification Number
5	BIDI	6	Bid Item (CLIN)
6	PHAS	2	Phase of work
7	CATW	1	Category of work
8	FOW	20	Feature of work*
*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements.			

3.3.7.1 Workers Per Day (WRKP)

Assign Workers per Day for all field construction or direct work activities, if directed by the Contracting Officer. Workers per day is based on the average number of workers expected each day to perform a task for the duration of that activity.

3.3.7.2 Responsible Party Coding (RESP)

Assign responsibility code for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the activity.

a. Activities coded with a Government Responsibility code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements.

b. Activities cannot have more than one Responsibility Code. Examples of acceptable activity code values are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE).

3.3.7.3 Area of Work Coding (AREA)

Assign Work Area code to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space. Examples of Work Area Coding include different areas within a floor of a building, different floors within a building, and different buildings within a complex of buildings. Activities cannot have more than one Work Area Code.

3.3.7.4 Modification Number (MODF)

Assign a Modification Number Code to any activity or sequence of activities added to the schedule as a result of a Contract Modification, when approved by Contracting Officer. Key all Code values to the Government's modification numbering system. An activity can have only one Modification Number Code.

3.3.7.5 Bid Item Coding (BIDI)

Assign a Bid Item Code to all activities using the Contract Line Item Number (CLIN) to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI Code.

3.3.7.6 Phase of Work Coding (PHAS)

Assign Phase of Work Code to all activities. Examples of phase of work are design phase, procurement phase and construction phase. Each activity can have only one Phase of Work code.

a. Code proposed fast track design and construction phases proposed to allow filtering and organizing the schedule by fast track design and construction packages.

b. If the contract specifies phasing with separately defined performance periods, identify a Phase Code to allow filtering and organizing the schedule accordingly.

3.3.7.7 Category of Work Coding (CATW)

Assign a Category of Work Code to all activities. Category of Work Codes include, but are not limited to design, design submittal, design reviews, review conferences, permits, construction submittal, procurement, fabrication, weather sensitive installation, non-weather sensitive installation, start-up, and testing activities. Each activity can have no more than one Category of Work Code.

3.3.7.8 Feature of Work Coding (FOW)

Assign a Feature of Work Code to appropriate activities based on the Definable Feature of Work to which the activity belongs based on the approved QC plan.

Definable Feature of Work is defined in Section 01 45 00.00 10 QUALITY CONTROL. An activity can have only one Feature of Work Code.

3.3.8 Contract Milestones and Constraints

Milestone activities are to be used for significant project events including, but not limited to, project phasing, project start and end activities, or interim completion dates. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

Mandatory constraints that ignore or effect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case by case basis.

3.3.8.1 Project Start Date Milestone and Constraint

The first activity in the project schedule must be a start milestone titled "NTP Acknowledged," which must have a "Start On" constraint date equal to the date that the NTP is acknowledged.

3.3.8.2 End Project Finish Milestone and Constraint

The last activity in the schedule must be a finish milestone titled "End Project."

Constrain the project schedule to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path.

3.3.8.3 Interim Completion Dates and Constraints

Constrain contractually specified interim completion dates to show negative float when the calculated late finish date of the last activity in that phase is later than the specified interim completion date.

3.3.8.3.1 Start Phase

Use a start milestone as the first activity for a project phase. Call the start milestone "Start Phase X" where "X" refers to the phase of work.

3.3.8.3.2 End Phase

Use a finish milestone as the last activity for a project phase. Call the finish milestone "End Phase X" where "X" refers to the phase of work.

3.3.9 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project Schedule. AACE 84R-13 provides methodologies for techniques in planning for adverse weather. The preferred methodology is the use of a weather calendar. If a method other than a weather calendar is proposed the discuss the reason during the SEKO meeting.

3.3.10 Calendars

Schedule activities on a Calendar to which the activity logically belongs. Develop calendars to accommodate any contract defined work period such as a 7-day calendar for Government Acceptance activities, concrete cure times, etc. Develop the default Calendar to match the physical work plan with non-work periods identified including weekends and holidays. Develop Seasonal Calendar(s) and assign to seasonally affected activities as applicable.

If a weather calendar is used its implementation and execution is to be in accordance with AACE 84R-13

3.3.11 Open Ended Logic

Only two open ended activities are allowed: the first activity "NTP Acknowledged" is to have no predecessor logic, and the last activity -"End Project" is to have no successor logic. Except as noted above all activities are to have at least a start-to-start or finish-to-start relationship with its predecessor(s) and a finish-to-finish or finish-to-start relationship with its successor(s).

Predecessor open ended logic may be allowed in a time impact analyses upon the Contracting Officer's approval.

3.3.12 Default Progress Data Disallowed

Actual Start and Finish dates must not automatically update with default mechanisms included in the scheduling software. Updating of the percent complete and the remaining duration of any activity must be independent functions. Disable program features that calculate one of these parameters from the other. Activity Actual Start (AS) and Actual Finish (AF) dates assigned during the updating process must match those dates provided in the Contractor Quality Control Reports. Failure to document the AS and AF dates in the Daily Quality Control report will result in disapproval of the Contractor's schedule.

3.3.13 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the Contracting Officer. Propose logic corrections to eliminate out of sequence progress or justify not changing the sequencing for approval prior to submitting an updated project schedule. Address out of sequence progress or logic changes in the Narrative Report and in the periodic schedule update meetings.

3.3.14 Added and Deleted Activities

Do not delete activities from the project schedule or add new activities to the schedule without approval from the Contracting Officer. Activity ID and description changes are considered new activities and cannot be changed without Contracting Officer approval.

3.3.15 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

3.3.16 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government and not used in place of realistic original durations, must not be in place to artificially absorb or create float, or to replace proper schedule logic.

a. Leads (negative lags) are prohibited.

b. Start to Finish (SF) relationships are prohibited.

3.3.17 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be allowed.

3.3.18 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value. Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management. Budgeted cost of activity is to be reduced by the amount needed to correct deficiency. The activity referenced in paragraph COST LOADING of CLOSEOUT ACTIVITIES must have its budgeted cost increased by this amount.

3.3.19 Remaining Duration

Update the remaining duration for each activity based on the number of estimated workdays it will take to complete the activity. Remaining duration may not mathematically correlate with percentage found under paragraph entitled Percent Complete. The remaining duration for unstarted activities are not to be less than its orginal duration.

3.3.20 Cost Loading of Closeout Activities

Cost load the "Correction of punch list from Government pre-final inspection" activity(ies) not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

3.3.20.1 As-Built Drawings

If there is no separate contract line item (CLIN) for as-built drawings, cost load the "Submission and approval of as-built drawings" activity not less than $35,000 or 1 percent of the present contract value, which ever is greater, up to $200,000. Activity will be declared 100 percent complete upon the Government's approval.

3.3.20.2 O & M Manuals

Cost load the "Submission and approval of O & M manuals" activity not less than $20,000. Activity will be declared 100 percent complete upon the Government's approval of all O & M manuals.

3.3.21 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that indicates all scope of the required contract work will be completed before the contractually required completion date.

a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and manhours) and the Government agreeing that the schedule is reasonable and achievable.

b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.

3.4 PROJECT SCHEDULE SUBMISSIONS

Provide the submissions as described below. The data reports and network diagrams required for each submission are contained in paragraph SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.

Review comments made by the Government on the schedule(s) do not relieve the Contractor from compliance with requirements of the Contract Documents.

3.4.1 Preliminary Project Schedule Submission

Within 15 calendar days after the NTP is acknowledged, submit the Preliminary Project Schedule defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. Completely cost load the Preliminary Project Schedule to balance the contract award CLINS shown on the Price Schedule. The Preliminary Project Schedule may be summary in nature for the remaining performance period. It must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as design activities, planned submissions of all early design packages, permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. Government acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. Activity code any activities that are summary in nature after the first 90 calendar days with Bid Item (CLIN) code (BIDI), Responsibility Code (RESP) and Feature of Work code (FOW).

3.4.2 Initial Project Schedule Submission

Submit the Initial Project Schedule for approval within 42 calendar days after notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the entire contract performance period. Include in the design-build schedule detailed design and permitting activities, including but not limited to identification of individual design packages, design submission, reviews and conferences; permit submissions and any required Government actions; and long lead item acquisition prior to design completion. Also cover in the initial design-build schedule the entire construction effort with as much detail as is known at the time but, as a minimum, include all construction start and completion milestones, and detailed construction activities through the dry-in, including all activity coding and cost loading. Include the remaining construction, including cost loading, but it may be scheduled summary in nature. As the design proceeds and design packages are developed, fully detail the remaining construction activities concurrent with the monthly schedule updating process. Constrain construction activities by Government acceptance of associated designs. When the design is complete, incorporate into the then approved schedule update all remaining detailed construction activities that are planned to occur after the dry-in milestone. No payment will be made for work items not fully detailed in the Project Schedule.

3.4.2.1 Design Package Schedule Submission

With each design package submitted to the Government, submit a fragnet schedule extracted from the then current Preliminary, Initial or Updated schedule which covers the activities associated with that Design Package including construction, procurement and permitting activities.

3.4.3 Periodic Schedule Updates

Update the Project Schedule on a regular basis, monthly at a minimum. Provide a draft Periodic Schedule Update for review at the schedule update meetings as prescribed in the paragraph PERIODIC SCHEDULE UPDATE MEETINGS. These updates will enable the Government to assess Contractor's progress. Update the schedule to include detailed construction activities as the design progresses, but not later than the submission of the final un-reviewed design submission for each separate design package. The Contracting Officer may require submission of detailed schedule activities for any distinct construction that is started prior to submission of a final design submission if such activity is authorized.

a. Update information including Actual Start Dates (AS), Actual Finish Dates (AF), Remaining Durations (RD), and Percent Complete. Updated information is subject to the approval of the Government at the meeting.

b. AS and AF dates must match the date(s) reported on the Contractor's

Quality Control Report for an activity start or finish.

3.5 SUBMISSION REQUIREMENTS

Submit the following items for the Preliminary Schedule, Initial Schedule, and every Periodic Schedule Update throughout the life of the project:

3.5.1 Electronic Scheduling Data

Provide electronic scheduling data containing the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Preliminary, Initial, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings. The file naming convention is to be determined at SEKO meeting.

3.5.2 Narrative Report

Provide a Narrative Report with each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Indicate who is responsible for the schedule update and narrative report. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for problems, either current or potential, which are revealed through that analysis. Include the following information at a minimum in the Narrative Report:

a. Identify and discuss the work scheduled to start in the next update period.

b. Describe activities along the critical path in addition to activities on the next float path after the critical path.

c. Describe current and anticipated problem areas, delaying factors and their impact, and an explanation of corrective actions taken or required to be taken. Identify the party responsible for delay, if applicable.

d. Identify and explain why activities, based on their calculated late dates, should have either started or finished during the update period but did not.

e. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, resource changes, and actual start and finish date changes.

f. Identify and discuss out-of-sequence work.

g. If the longest path changes from a monthly update to the next, provide an explanation of the factors driving the change.

h. Respond to USACE Schedule Review Comments from the previous update(s).

3.5.3 Schedule Reports

The format, filtering, organizing, and sorting for each schedule report will be as directed by the Contracting Officer. Typically, reports contain Activity Numbers, Activity Description, Original Duration, Remaining Duration, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Total Float, Actual Start Date, Actual Finish Date, and Percent Complete. Provide the reports electronically in .pdf format. Provide ~~[_____]~~2 set(s) of hardcopy reports. The following lists typical reports that will be requested:

3.5.3.1 Activity Report

List of all activities sorted according to activity number.

3.5.3.2 Logic Report

List of detailed predecessor and successor activities for every activity in ascending order by activity number.

3.5.3.3 Total Float Report

A list of all incomplete activities sorted in ascending order of total float. List activities which have the same amount of total float in ascending order of Early Start Dates. Do not show completed activities on this report.

3.5.3.4 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date, which reflects the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments. Group activities by CLIN number and sort by activity number. Provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. The printed report must contain the following for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings this period, Total Quantity, Quantity to Date, and Percent Complete (based on cost).

3.5.3.5 Scheduling and Leveling Report

Provide a Scheduling/Leveling Report generated from the current project schedule being submitted.

3.5.4 Network Diagram

The Network Diagram is required for the Preliminary, Initial and Periodic Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.4.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram.

3.5.4.2 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.4.3 Critical Path

Show all activities on the critical path. The critical path is defined as the longest path.

3.5.4.4 Grouping and Sorting

Group and sort activities as directed to assist in the understanding of the activity sequence. Typically, this flow will group activities by major elements of work, category of work, work area and/or responsibility.

3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram

With each schedule submission, provide a SVC diagram showing 1) Cash Flow S-Curves indicating planned project cost based on projected early and late activity finish dates, and 2) Earned Value to-date.

3.6 PERIODIC SCHEDULE UPDATE

3.6.1 Periodic Schedule Update Meetings

Conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly within five days of the proposed schedule data date. Provide a computer with the scheduling software loaded and a projector which allows all meeting participants to view the proposed schedule during the meeting. The Contractor's authorized scheduler must organize, group, sort, filter, perform schedule revisions as needed and review functions as requested by the Contractor and/or Government. The Contractor scheduler must attend in person unless virtual attendance is approved by Contracting Officer. The meeting is a working interactive exchange which allows the Government and Contractor the opportunity to review the updated schedule on a real time and interactive basis. The meeting will last no longer than 8 hours. Submit schedule file and narrative to the Government a minimum of two workdays in advance of the meeting. The Contractor's Project Manager must attend the meeting with the authorized representative of the Contracting Officer. Superintendents, foremen and major subcontractors must attend the meeting as required to discuss the project schedule and work. Following the schedule update meeting, make corrections to the schedule and resubmit, following the required schedule naming convention. Include only those changes approved by the Government in the schedule review meeting. Only approved schedules may be used to generate invoices for payment.

3.6.2 Update Submission Following Progress Meeting

Submit the complete Periodic Schedule Update of the Project Schedule containing all approved progress, revisions, and adjustments, pursuant to paragraph SUBMISSION REQUIREMENTS not later than 4 workdays after the periodic schedule update meeting.

3.7 WEEKLY PROGRESS MEETINGS

Conduct a weekly meeting with the Government (or as otherwise mutually agreed to) between the meetings described in paragraph entitled PERIODIC SCHEDULE UPDATE MEETINGS for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming two weeks. Use the current approved schedule update for the purposes of this meeting and for the production and review of reports. At the weekly progress meeting, address the status of RFIs, RFPs and Submittals.

3.8 REQUESTS FOR TIME EXTENSIONS

ASCE 67-17 provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in ASCE 67-17, the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occuring. Also prepare a time impact analysis for each Government request for proposal (RFP).

3.8.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date, or interim completion date(s). Evaluate multiple impacts chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.8.2 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the Project schedule.

3.8.3 Time Impact Analysis (Prospective Analysis)

Prepare a time impact analysis for approval by the Contracting Officer based on industry standard AACE 52R-06. Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

3.8.4 Forensic Schedule Analysis (Retrospective Analysis)

Prepare an analysis for approval by the Contracting Officer based on methodologies identified in industry standard AACE 29R-03. Identify which methodology was chosen and explain why it is more appropriate than other methodologies available.

3.8.5 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis consisting of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the project schedule.

3.8.6 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay, are not a cause for an extension to the performance period, completion date, or any interim milestone date.

3.8.7 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless the Contractor actually performed work in accordance with that Early Completion Schedule. The Contractor must show that an early completion was achievable had it not been for the impact.

3.9 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the Contracting Officer may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra workdays, etc.

3.9.1 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying, or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.9.2 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.9.3 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to FAR 52.236-15 Schedules for Construction Contracts.

3.10 OWNERSHIP OF FLOAT

Except for the provision given in the paragraph IMPACT TO EARLY COMPLETION SCHEDULE, float available in the schedule, at any time, belongs to the Project and is available for Contractor and Government use. This includes activity and project float. Activity float is the number of workdays that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the "End Project" finish.

3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM

Ensure schedule data is uploaded to RMS. This data is additional supporting data in a form and detail required by the Contracting Officer pursuant to FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The receipt of a proper payment request pursuant to FAR 52.232-27 Prompt Payment for Construction Contracts is contingent upon the Government receiving both acceptable and approvable hard copies and matching electronic versions of the application for progress payment.

3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

Ensure Primavera P6 settings provide a schedule capable of fulfilling the requirements of the contract. The following settings are mandatory and required in all schedule submissions to the Government:

a. Activity Codes must be Project Level, not Global or EPS level.

b. Calendars must be Project Level, not Global or Resource level.

c. Activity Duration Types must be set to "Fixed Duration & Units".

d. Percent Complete Types must be set to "Physical".

e. Time Period Admin Preferences must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days.

f. Set Schedule Option for defining Critical Activities to "Longest Path".

g. Set Schedule Option for defining progressed activities to "Retained Logic".

h. Set up cost loading using a single lump sum non-labor resource. The Price/Unit must be $1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" ~~un-~~selected.

i. Activity ID's must not exceed 10 characters.

j. Activity Names must have a verb-noun structure and contain the most defining and detailed description within the first 30 characters.

k. The daily ending hour for all work calendars must be the same.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor's Quality Control (CQC) System Manager are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

The Contractor and the Designer of Record (DOR), if applicable, are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

~~[For Government approved division 01 preconstruction submittals that are required prior to or commencing with the start of work shall be submitted within 30 calendar days of contract award unless specified elsewhere in the specifications. For contractor approved division 01 submittals that are required prior to or commencing with the start of work shall be submitted within 45 calendar days of contract award unless specified elsewhere in the specifications.]~~

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan Health And Safety Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan ~~[Explosive Safety Submission ESS Work Plan]~~
~~SD-02~~

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

Design submittals, design substantiation submittals and extensions of design submittals.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS)concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S"~~ classification. Submittals not having a "G" ~~or "S"~~ classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

~~1.4.2 Design-Build Submittal Classifications~~

~~1.4.2.1 Designer of Record Approved (DA)~~

~~Designer of Record (DOR) approval is required for extensions of design; critical materials; any variations from the Solicitation, the Accepted Proposal, or the completed design; equipment whose compatibility with the entire system must be checked; and other items as designated by the Contracting Officer. Provide the Government with the number of copies designated hereinafter of all DOR approved submittals. The Government may review any or all Designer of Record approved submittals for conformance~~

~~with the Solicitation, the Accepted Proposal, and the completed design. The Government will review all submittals designated as varying from the Solicitation or Accepted Proposal, as described below. Provide design submittals in accordance with Section 01 33 16.00 10 DESIGN DATA. Generally, list design submittals under SD-05 Design Data.~~

~~1.4.2.2 Government Conformance Review of Design (CR)~~

~~The Government will review all intermediate and final design submittals for conformance with the technical requirements of the Solicitation. Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD) covers the design submittal and review process in detail. Review will be only for conformance with the applicable codes, standards, and contract requirements. Design data includes the design documents described in Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD).~~

~~1.4.2.3 Designer of Record Approved/Government Conformance Review (DA/CR)~~

~~1.4.2.3.1 Variations from the Accepted Design~~

~~DOR approval and the Government's concurrence are required for any proposed variation from the accepted design that still complies with the contract before the Contractor is authorized to proceed with material acquisition or installation. If necessary to facilitate the project schedule, before official submission to the Government, the Contractor and the DOR may discuss with the Contracting Officer's Representative a submittal proposing a variation. However, the Government reserves the right to review the submittal before providing an opinion. In any case, the Government will not formally agree to or provide a preliminary opinion on any variation without the DOR's approval or recommended approval. The Government reserves the right to reject any design, variation that may affect furniture, furnishings, equipment selections, or operational decisions that were made, based on the reviewed and concurred design.~~

~~1.4.2.3.2 Substitutions~~

~~Unless prohibited or otherwise provided for elsewhere in the contract, where the Accepted Proposal named products, systems, materials or equipment by manufacturer, brand name, model number, or other specific identification, and the Contractor desires to substitute a manufacturer or model after award, submit a requested substitution for Government concurrence. Include substantiation, through identifying information and the DOR's approval, that the substitute meets the contract requirements and that it is equal in function, performance, quality, and salient features to that in the accepted contract proposal. If the contract otherwise prohibits substitutions of equal named products, systems, materials or equipment by manufacturer, brand name, model number or other specific identification, the request is considered a "variation" to the contract. Variations are discussed below in paragraphs: "DESIGNER OF RECORD APPROVED/GOVERNMENT APPROVED" and VARIATIONS.~~

~~1.4.2.4 Designer of Record Approved/Government Approved (DA/GA)~~

~~In addition to the above-stated requirements for proposed variations to the accepted design, both DOR and Government Approval and, where applicable, a contract modification are required before the Contractor is authorized to proceed with material acquisition or installation for any proposed variation to the contract (the Solicitation or the Accepted Proposal), that constitutes a change to the contract terms. The Government~~

~~reserves the right to accept or reject any such proposed variation.~~

1.4.2 For Information Only

Submittals not requiring Government approval will be for information only. ~~For Design-build construction all submittals not requiring DOR or Government approval will be for information only.~~ Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

~~1.4.3 Sustainability Reporting Submittals (S)~~

~~Submittals for Guiding Principle Validation (GPV) or Third Party Certification (TPC) are indicated with an "S" designation. These submittals are for information only and for use as specified in Section 01 33 29 SUSTAINABILITY REPORTING.~~

~~Schedule submittals for these items throughout the course of construction as provided; do not wait until closeout.~~

~~1.5 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL~~

~~As soon as practicable after award of contract, and before procurement or fabrication, forward to the [Commander, NAVFAC [_____], Code CI4[_____], [_____]] [Architect-Engineer: [_____],] submittals required in the technical sections of this specification, including shop drawings, product data and samples. In addition, forward a copy of the submittals to the Contracting Officer.~~

~~1.5.1 O&M Data~~

~~Submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.~~

~~In the event the Contractor fails to deliver O&M data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the items to which such O&M data apply.~~

~~[1.5.2 Submittals Reserved for NAVFAC [_____] Approval~~

~~As an exception to the standard submittal procedure for Government Approval, submit the following to the Commander, NAVFAC [_____], Code CI4[_____], [_____]:~~

~~[a. Section [_____] [_____]: Pile driving records~~

~~][b. Section [_____] [_____]: All fire protection system submittals~~

~~][c. Section [_____] [_____]: All fire alarm system submittals~~

~~][d. Section [_____] [_____]: All elevator submittals~~

~~][e. Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING: SD-06 Commissioning Plan, Certificate of Readiness, and Commissioning Report submittals~~

~~][f. Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC: SD-06 field test report submittals~~

~~][g Section 23 09 53.00 20 SPACE TEMPERATURE CONTROL SYSTEMS: SD-06 field test report submittals~~

~~][h. Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC: All submittals~~

~~][i. Section 23 08 01.00 20 TESTING INDUSTRIAL VENTILATION SYSTEMS: All submittals~~

~~][j. Section 26 12 19.10 THREE-PHASE PAD-MOUNTED TRANSFORMERS: All submittals~~

~~][k. Section 26 12 21 SINGLE-PHASE PAD-MOUNTED TRANSFORMERS: All submittals~~

~~][l. Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION: Transformer submittals~~

~~][m. Section 26 11 16 SECONDARY UNIT SUBSTATIONS: Transformer submittals~~

~~][n. Section 26 11 13.00 20 PRIMARY UNIT SUBSTATION: Transformer submittals~~

~~]][1.5.3 Overseas Shop Drawing Submittals~~

~~Send submittals via overnight express mail service. All costs associated with the overnight express mail service are borne by the Contractor. Costs associated with the overnight express mail of submittals related to proposed submittal variances of resubmittals necessary as a result of noncompliant or incomplete Contractor submittals are the responsibility of the Contractor.~~

~~]~~1.5 PREPARATION

1.5.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels to the office of the approving authority using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms ~~[will be furnished to the Contractor][~~are included in the RMS CM software that the Contractor is required to use for this contract~~]~~. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Identifying Submittals

The Contractor's ~~[~~Quality Control Manager~~] [approving authority]~~ must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

a. Project title and location

b. Construction contract number

c. Dates of the drawings and revisions

d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.

e. Section number of the specification by which submittal is required

f. Submittal description (SD) number of each component of submittal

g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission

h. Product identification and location in project.

1.5.3 Submittal Format

1.5.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 210 by 297 mm nor more than 1189 by 841 mm, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.

b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Present 210 by 297 mm shop drawings sized as part of the bound volume for submittals. Present larger drawings in sets. Submit an electronic copy of drawings in PDF format.

1.5.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than ~~[_____]~~50 millimeter on the right-hand side of each sheet for the Government disposition stamp.

1.5.3.3 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in metric dimensions. Where product data are included in preprinted catalogs with English units only, submit metric dimensions on separate sheet.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will ~~[~~not~~]~~ be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.3.4 Format of SD-04 Samples

1.5.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

a. Sample of Equipment or Device: Full size.

b. Sample of Materials Less Than 50 by 75 mm: Built up to 210 by 297 mm.

c. Sample of Materials Exceeding 210 by 297 mm: Cut down to 210 by 297 mm and adequate to indicate color, texture, and material variations.

d. Sample of Linear Devices or Materials: 250 mm length or length to be supplied, if less than 250 mm. Examples of linear devices or materials are conduit and handrails.

e. Sample Volume of Nonsolid Materials: 750 ml. Examples of nonsolid materials are sand and paint.

f. Color Selection Samples: 50 by 100 mm. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.

g. Sample Panel: 1200 by 1200 mm.

h. Sample Installation: 10 square meters.

1.5.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.3.5 Format of SD-05 Design Data

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.6 Format of SD-06 Test Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.7 Format of SD-07 Certificates

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section as a complete, bound volume. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.9 Format of SD-09 Manufacturer's Field Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

Provide all dimensions in administrative submittals in metric. Where data are included in preprinted material with English units only, submit metric dimensions on separate sheet.

1.5.4 Source Drawings for Shop Drawings

1.5.4.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.4.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. ~~[~~In addition to the electronic submittal, ~~provide [three] [_____] hard copies of the submittals~~see paragraph titled QUANTITY OF SUBMITTALS for number of hard copies required. ~~]~~ Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the AMRDEC SAFE Web Application located at the following website: https://safe.amrdec.army.mil/safe/.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit ~~[~~two~~][three]~~ sets of administrative submittals.

1.6.2 Number of SD-02 Shop Drawing Copies

Submit ~~[six][_____]~~two copies of submittals of shop drawings requiring review and approval by a QC organization. Submit ~~[seven][_____]~~two copies of shop drawings requiring review and approval by the Contracting Officer.

1.6.3 Number of SD-03 Product Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.4 Number of SD-04 Samples

a. Submit ~~[~~two~~] [_____]~~ samples, or ~~[~~two~~] [_____]~~ sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.

b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.

c. Submit one sample installation, where directed.

d. Submit one sample of nonsolid materials.

1.6.5 Number of SD-05 Design Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.6 Number of SD-06 Test Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings, other than field test results that will be submitted with QC reports.

1.6.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC reports.

1.6.10 Number of SD-10 Operation and Maintenance Data Copies

Submit ~~[~~five~~][three][_____]~~ copies of O&M data to the Contracting Officer for review and approval.

1.6.11 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit ~~[~~two~~][three]~~ sets of administrative submittals.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.~~ For Design-Build construction, the Government will retain [_____] copies of information-only submittals.~~

1.8 PROJECT SUBMITTAL REGISTER AND DATABASE

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Use an electronic submittal register program furnished by the Government. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and

(i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM).~~[~~ The Government will provide the initial submittal register ~~][~~in electronic format~~][~~ with the following fields completed, to the extent that will be required by the Government during subsequent usage.~~]~~

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Column (f): Lists the approving authority for each submittal.

The database and submittal management program will be furnished to the Contractor on a writable compact disk (CD-R), for operation on a Windows-based personal computer.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

~~1.8.2 Design-Build Submittal Register~~

~~The Designer of Record develops a complete list of submittals during design and identify required submittals in the specifications, and use the list to prepare the Submittal Register. The list may not be all inclusive and additional submittals may be required by other parts of the contract. Complete the submittal register and submit it to the Contracting Officer for approval within 30 calendar days after Notice to Proceed. The approved submittal register will serve as a scheduling document for submittals and will be used to control submittal actions throughout the contract period. Coordinate the submit dates and need dates with dates in the Contractor prepared progress schedule. Submit monthly or until all submittals have been satisfactorily completed, updates to the submittal register showing the Contractor action codes and actual dates with Government action codes. Revise the submittal register when the progress schedule is revised and submit both for approval.~~

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register as an electronic database, using the submittal management program furnished to Contractor. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register database submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

Entries for columns (j) and (o) are to be used as follows (others may be prescribed by the Transmittal Form):

1.8.5.1 Government Review Action Codes

"A" - "Approved as submitted"; "Completed"

"B" - "Approved, except as noted on drawings"; "Completed"

"C" - "Approved, except as noted on drawings; resubmission required"; "Resubmit"

"D" - "Returned by separate correspondence"; "Completed"

"E" - "Disapproved (See attached)"; "Resubmit"

"F" - "Receipt acknowledged"; "Completed"

"G" - "Other (Specify)"; "Resubmit"

"X" - "Receipt acknowledged, does not comply with contract requirements"; "Resubmit"

1.8.5.2 Government Review Action Codes

"A" - "Approved as submitted"

"AN" - "Approved as noted"

"RR" - "Disapproved as submitted"; "Completed"

"NR" - "Not Reviewed"

"RA" - "Receipt Acknowledged"

1.8.5.3 Contractor Action Codes

DESIGN BID BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS – The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA
BLANK	Submittal is For Information Only (FIO)	BLANK	FIO
S	Submittal is for documentation of Sustainable requirements	S	S/FIO

1.8.5.4 Contractor Action Codes

DESIGN BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS – The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA
BLANK	Submittal is For Information Only(FIO)	BLANK	FIO
DA	Submittal requires Designer of Record Approval	D	DA
CR	Submittal requires Government Conformance Review	C	CR
DA/CR	Submittal requires Designer of Record Approval and Government Conformance Review	R	DA/CR
DA/GA	Submittal requires Designer of Record Approval and Government Approval	A	DA/GA

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request, unless a paper copy is requested by the Contracting Officer. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission ~~[of a variation submittal]~~ will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a ~~[~~ transmittal letter~~][variation submittal]~~. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

~~[~~When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.~~]~~

~~[~~The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk.~~]~~"

Specifically point out variations from contract requirements in a ~~[~~ transmittal letter~~][variation submittal]~~. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of ~~[~~14~~] [_____]~~ calendar working days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. ~~[Allow an additional [_____] calendar working days for review and approval of submittals for [food service equipment] [and] [refrigeration and HVAC control systems]].~~

a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within ~~[_____]~~14 calendar working days after the date of submission.

d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least ~~[15] [_____]~~ 10 working days for submittals for QC manager approval and ~~[20] [_____]~~ 30 working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.

~~e. For submittals requiring review by a Government fire protection engineer, allow a review period, beginning when the Government receives the submittal from the QC organization, of [30][_____] working days for return of the submittal to the Contractor.~~

~~[Within [30][15] calendar days of Notice To Proceed][At the Preconstruction conference], provide the following schedule of submittals for approval by the Contracting Officer:~~

~~d. A schedule of shop drawings and technical submittals required by the specifications and drawings. Indicate the specification or drawing~~

~~reference requiring the submittal; the material, item, or process for which the submittal is required; the "SD" number and identifying title of the submittal; the anticipated submission date, and the approval need date.~~

~~e. A separate schedule of other submittals required under the contract but not listed in the specifications or drawings. Indicate the contract requirement reference, the type or title of the submittal, the anticipated submission date, and the approval need date (if approval is required).~~

~~1.10.1 Reviewing, Certifying, and Approving Authority~~

~~The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification sections, a notation "G" following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government Approving authority~~

~~1.10.2 Constraints~~

~~Conform to provisions of this section, unless explicitly stated otherwise for submittals listed or specified in this contract.~~

~~Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.~~

~~When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.~~

~~Approval of a separate material, product, or component does not imply approval of the assembly in which the item functions.~~

~~1.10.3 QC Organization Responsibilities~~

~~a. Review submittals for conformance with project design concepts and compliance with contract documents.~~

~~b. Process submittals based on the approving authority indicated in the submittal register.~~

~~(1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.~~

~~(2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.~~

~~c. Ensure that material is clearly legible.~~

~~d. Stamp each sheet of each submittal with a QC certifying statement or~~

~~an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.~~

~~(1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to the Contracting Officer with the following certifying statement:~~

~~"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number [_____] is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.~~

~~Certified by Submittal Reviewer _____________________, Date _______ (Signature when applicable)~~

~~Certified by QC Manager _____________________________, Date ______" (Signature)~~

~~(2) When approving authority is the QC manager, the QC manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."~~

~~"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number [_____] is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is approved for use.~~

~~Certified by Submittal Reviewer ______________________, Date ______ (Signature when applicable)~~

~~Approved by QC Manager _______________________________, Date _____" (Signature)~~

~~e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.~~

~~f. Update the submittal register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.~~

~~g. Retain a copy of approved submittals and approved samples at the project site.~~

~~h. For "S" submittals, provide a copy of the approved submittal to the Government Approving authority.~~

~~1.10.4 Government Reviewed Design~~

~~The Government will review design submittals for conformance with the technical requirements of the Solicitation. Section 01 33 16.00 10 DESIGN DATA (DESIGN AFTER AWARD) covers the design submittal and review process in detail. Government review is required for variations from the completed design. Review will be only for conformance with the contract requirements. Included are only those construction submittals for which~~

~~the DOR's design documents do not include enough detail to ascertain contract compliance. The Government may, but is not required to, review extensions of design such as structural steel or reinforcement shop drawings.~~

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

a. Note the date on which the submittal was received from the QC manager.

b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.

c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. ~~[_____]~~Two copies of the submittal will be retained by the Contracting Officer and ~~[_____]~~one copies of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.

b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.

c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.

d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory. the design, general method of construction, materials, detailing, and other information appear to meet the Solicitation and Accepted Proposal.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained with in each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

______NAME OF CONTRACTOR ______ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 35 13

SPECIAL PROJECT PROCEDURES
11/20, CHG 1: 02/22

PART 1 GENERAL

~~1.1 DEFINITIONS~~
~~1.1.1 Landing Areas~~

~~"Landing Areas" means:~~

~~a. The primary surfaces, comprising the surface of the runway, runway shoulders, and lateral safety zones. The length of each primary surface is the same as the runway length. The width of each primary surface is 609.6 meters (304.8 meters on each side of the runway centerline).[Exceptions: Some airfields are based on a primary width of 457.2 meters (228.6 meters on each side of the runway centerline). In such instances, substitute the proper width in the applicable statements.]~~

~~b. The "clear zone" beyond the ends of each runway is the extension of the primary surface for a distance of 914.4 meters in length for fixed wing aircraft and 121.92 meters in length for helicopter only runways beyond each end of each runway.~~

~~c. All taxiways, plus the lateral clearance zones along each side for the length of the taxiways (the outer edge of each lateral clearance zone is laterally 76.2 meters from the far or opposite edge of the taxiway (example: a 22.86 meter taxiway must have a combined width and lateral clearance zone of 129.54 meters.)~~

~~d. All aircraft parking aprons, plus the area 38.1 meters in width extending beyond each edge around the aprons.~~

~~1.1.2 Safety Precaution Areas~~

~~"Safety Precaution Areas" means those portions of approach-departure clearance zones and transitional zones where placement of objects incident to Contract performance might result in vertical projections at or above the approach-departure clearance, or the transitional surface.~~

~~a. The "approach-departure clearance surface" is an extension of the primary surface and the clear zone at each end of each runway, for a distance of 15240 meters, first along an inclined (glide angle) and then along a horizontal plane, both flaring symmetrically about the runway centerline extended.~~

~~(1) The inclined plane (glide angle) begins in the clear zone 61 meters past the end of the runway (and primary surface) at the same elevation as the end of the runway. It continues upward at a slope of 50:1 (0.305 meters vertically for each 15.24 meters horizontally) to an elevation of 152.4 meters above the established airfield elevation. At that point the plane becomes horizontal, continuing at that same uniform elevation to a point 15240 meters longitudinally from the beginning of the inclined plane (glide angle) and ending there.~~

~~(2) The width of the surface at the beginning of the inclined plane (glide angle) is the same as the width of the clear zone. It then flares uniformly, reaching the maximum width of 4876.8 meters at the end.~~

~~b. The "approach-departure clearance zone" is the ground area under the approach-departure clearance surface.~~

~~c. The "transitional surface" is a sideways extension of all primary surfaces, clear zones, and approach-departure clearance surfaces along inclined planes.~~

~~(1) The inclined plane in each case begins at the edge of the surface.~~

~~(2) The slope of the incline plane is 7:1 (.305 meters vertically for each 2.13 meters horizontally). It continues to the point of intersection with the:~~

~~(a) Inner horizontal surface (which is the horizontal plane 45.72 meters above the established airfield elevation); or~~

~~(b) Outer horizontal surface (which is the horizontal plane 152.4 meters above the established airfield elevation), whichever is applicable.~~

~~d. The "transitional zone" is the ground area under the transitional surface. (It adjoins the primary surface, clear zone, and approach-departure clearance zone.)~~

~~1.1.3 Federal Aviation Administration (FAA) Notice of Proposed Construction or Alteration~~

~~a. FAA Notice of Proposed Construction or Alteration may be required in accordance with~~

~~49 USC 44718 and~~

~~14 CFR 77, depending on height of construction equipment on site, height of temporary structures, proximity to an airport or heliport, and specific location of equipment or temporary structure. For the purpose of notifying the FAA, proximity shall be defined as within 5 nautical miles of a Government or civilian airfield, including landing areas, taxiways, runways and helicopter pads.~~

~~b. In order to determine if a FAA Notice of Proposed Construction or Alteration is required, refer to~~

~~14 CFR 77 Subpart B. Alternately, utilize the FAA's Notice Criteria Tool located at: https://oeaaa.faa.gov/oeaaa/external/portal.jsp. The FAA will determine if the equipment or temporary structure exceeds obstruction standards and may pose a hazard to air navigation.~~

~~c. Failure to comply with the provisions of~~

~~14 CFR 77 are subject to Civil Penalty under Section 902 of the Federal Aviation Act of 1958, as amended and pursuant to~~

~~49 USC 46301 Subpart (a).~~

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[for Contractor Quality Control approval.]~~~~[~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

~~SD-01 Preconstruction Submittals~~

~~[~~ ~~Heavy Equipment and Vehicle List~~

~~][~~ ~~Existing Conditions Survey~~

~~]~~ ~~FAA Form 7460-1~~

~~FAA Form 7460-2~~

~~Construction Operations Plan~~

~~[~~ ~~Watercraft List~~

~~][~~ SD-04 Samples

Model Unit

~~]~~PART 2 PRODUCTS

NOT USED

~~2.1 AIRFIELD OBSTRUCTION LIGHTS~~

~~Airfield obstruction lights must be in accordance with FAA AC 70/7460-1 and have red or white lenses.~~

PART 3 EXECUTION

~~[3.1 HAZARDS TO [AIRFIELD][HELIPORT] OPERATION~~

~~In addition to DFARS 252.236-7005 Airfield Safety Precautions, the following paragraphs apply.~~

~~3.1.1 Work in Proximity to [Landing Areas] [Landing Strips] [Landing Pad(s)]~~

~~Place nothing upon the landing area or applicable portions of safety precaution areas without authority of the Contracting Officer.~~

~~Use of [landing areas] [landing strips] [landing pads] for purposes other than aircraft operation, is prohibited without permission of the Contracting Officer, and the [landing area] [landing strip] [landing pad] is closed by order of the Contracting Officer and marked as indicated herein.~~

~~Accomplish all construction work on [the runways, taxiways, and parking aprons and in the end zones of the runways and 23 m to each side of the runways and taxiways][the landing strip, 23 m to each side thereof, and on the taxiways and parking aprons][the landing pad(s)]with extreme care~~

~~regarding the operation of aircraft. Cooperate closely, and coordinate with [the Operations Officer and]the Contracting Officer. Park equipment in an area designated by the Contracting Officer. Parking of equipment, vehicles, or any storage features overnight or for any extended period of time in the proximity of the [landing areas or taxiways][landing strip][landing pad]is strictly prohibited. Leave no material in areas where extreme care is to be taken regarding the operation of aircraft.~~

~~During periods of active performance of work on the airfield by the Contractor, govern all operations of mobile equipment in accordance with the safety provisions.~~

~~3.1.2 Contractor FAA Notification~~

~~When required in accordance with 49 USC 44718 and 14 CFR 77, submit FAA Form 7460-1 and attachments directly to the FAA a minimum of 60 calendar days prior to the start date of the operations that may affect air traffic. Submit supplemental notification FAA Form 7460-2 to the FAA within 48 hours prior to start of the construction. Simultaneous with submission to the FAA, submit both forms to the Contracting Officer for information. It is the Contractor's responsibility to notify the FAA when required.~~

~~3.1.3 Schedule of Work/Aircraft Operating Schedules~~

~~Schedule work to conform to aircraft operating schedules. The Government will exert every effort to schedule aircraft operations so as to permit the maximum amount of time for the Contractor's activities; however, in the event of emergency, intense operational demands, adverse wind conditions, and other such unforeseen difficulties, the Contractor must cease operations at the specified locations in the aircraft operational area for the safety of the Contractor and military personnel and Government property. Submit a schedule of the work to the Contracting Officer [for transmittal to the Operations Officer]describing the work to be accomplished; the location of the work, noting distances from the ends of [landing areas, taxiways][landing strips][landing pads]and buildings and other structures as necessary; and dates and hours during which the work is to be accomplished. Keep the approved schedule of work current, and notify the Contracting Officer of changes prior to beginning each day's work.~~

~~Where flying is controlled, obtain permission from the control tower operator to enter a landing area unless such area is marked as hazardous to aircraft.~~

~~3.1.3.1 Construction Operations Plan~~

~~Submit a Construction Operations Plan prior to the start of work that includes a description of the airfield work to be accomplished; the exact location of the work, noting distances from the ends of [landing areas, taxiways][landing strips][landing pads]and buildings and other structures as necessary; and dates and hours during which the work is to be accomplished. Keep the approved schedule of work current and notify the Contracting Officer of changes prior to beginning each day's work.~~

~~[3.1.3.2 Existing Conditions Survey~~

~~Perform an Existing Conditions Survey and submit results to the Government prior to the start of work. As part of this survey, inspect the work site~~

~~to identify existing conditions located within the vicinity of the contract work. Utilize original as-builts if available showing possible utilities, type of pavement, thickness, airfield lighting, equipment, vehicles, structures, and parked aircraft. Unless already specified in the Contract documents, utilize the RFI process to clarify responsibility for relocating appurtenances that will interfere with work. If the site is opened up (in/under construction), update the existing survey condition to include the actual condition of the underlying conditions such as sub base, base, and pavement thickness.~~

~~]3.1.4 Daytime Markings~~

~~During daylight, mark stationary and mobile equipment with international orange and white checkered flags, mark the material, and work with yellow flags.~~

~~Submit a Heavy Equipment and Vehicle List identifying all stationary and mobile equipment and vehicles that will be operating on the airfield. All equipment and vehicles must be identified by means of a flag on a staff attached to and flying above the vehicle. Flag size must be not less than one meter square and consist of a checkered pattern of international orange and white squares not less than 300 millimeter on each side. Flags varying in any dimension by not more than 10 percent of the specified dimensions are considered to comply with the stated requirements.~~

~~3.1.5 Nighttime Markings~~

~~During nighttime, which begins 2-hours before sundown and ends 2-hours after sunrise, mark stationary and mobile equipment and material, and work with [red lanterns][battery-operated, low-intensity, red flasher lights]. Where the [Operations Officer][Contracting Officer]determines that the [red lanterns][red flasher lights]may confuse pilots approaching for landings, the [Operations Officer][Contracting Officer]may direct that the [red lanterns][red flasher lights]be left off or that the color of the [globes][lights]be changed.~~

~~Provide lighting in accordance with FAA AC 70/7460-1. Provide a minimum of two aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 30 meter above ground level. Lights must be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.~~

~~No separate payments will be made for lighting and protection necessitated by the safety provisions.~~

~~[3.1.6 Excavation~~

~~Open only those trenches for which material is on hand and ready for placing therein. As soon as possible after the material has been placed and work approved, backfill and compact the trenches as specified.~~

~~Maintain [landing areas][landings strips][landing pads]at all times free from hazards, holes, material piles, or projecting shoulders that might damage tires or landing gear. Keep paved surfaces clean at all times and free from small stones or other objects which could cause damage to propellers, craft, and personnel.~~

~~][3.1.7 Contractor Safety Precautions~~

~~The Contractor is advised that aircraft operations will produce extremely high noise levels and will induce vibrations in pavements, structures, and equipment in the vicinity, and may result in high velocity flying debris in the area. These anticipated hazards must be appropriately addressed in the Activity Hazard Analysis (AHA) associated with airfield work - the Activity Hazard Analysis must be in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. The Contractor is responsible for providing personal protective equipment (PPE) and other safety devices required to ensure protection of contractor personnel and equipment. Schedule the work to eliminate hazards to personnel and equipment and to prevent damage to work performed.~~

~~Boundary areas for hazardous work locations and restrictions are defined in FAA AC 150/5300-13. Construction activity within the limits of the boundary areas without approval of the Contracting Officer is prohibited.~~

~~]3.1.8 Base Civil Engineering (BCE) Work Clearance Request~~

~~Obtain an approved BCE Work Clearance Request, AF Form 103, prior to the start of excavation, digging work, or work that disrupts aircraft or vehicular traffic flow, base utility services, fire and intrusion alarm system, or routine activities of the Activity.~~

~~3.1.9 Radio Contact~~

~~Provide necessary battery powered portable radios, including one radio for the tower. During work within the landing area, have an operator (who speaks fluent English) available for radio contact with the tower at all times. Obtain approval of radio frequency from the tower.~~

~~][3.2 HARBOR WATERWAYS~~

~~In addition to DFARS 252.236-7002 Obstruction of Navigable Waterways, obtain from the [Operations Officer at the Naval Base, Pearl Harbor via the] [_____] Contracting Officer, permission to use [Pearl Harbor] [_____] waterways and the regulations and directives governing such usage. Submit a watercraft list with a description of crafts, including sizes, types, numbers and boat crew for approval.~~

~~[3.2.1 Hazards to Navigation~~

~~Maintain complete control of the movement of floating equipment and material. Loose floating equipment and material are not permitted. Keep in readiness at all times a powered craft capable of moving, securing and disposing of floating equipment which may get loose and become a hazard to navigation.~~

~~][3.2.2 Submarine Cables or Underwater Utilities~~

~~Physically locate submarine cables or underwater utilities as indicated prior to performance of work.~~

~~][3.2.3 RIMPAC Operational Exercises~~

~~Due to the RIMPAC exercises conducted every even numbered years, the Contractor must anticipate delays and disruption in construction activities for a period of 60 consecutive calendar days during the~~

~~Contract. Disruption may range from complete shutdown of construction activity for a prolonged period of time or periodic short term shutdown of less than a full day, but in no event will such shutdowns in the aggregate exceed 60 calendar days per calendar year. This affected period is anticipated between 1 April and 31 July of every even numbered year. Bidders must take into consideration such delays when preparing their bids.~~

~~][3.2.4 December 7, 1941 Commemorative Ceremonies~~

~~To minimize disturbances during Pearl Harbor's December 7, 1941 Commemorative Ceremonies, no construction activities will be permitted on the day that ceremonies are held.~~

~~][3.2.5 Annual Pearl Harbor "Hydrofest"~~

~~The annual Pearl Harbor "Hydrofest" activities will prevent access to the waterways for a 2-week period during October or November. Bidders must take into consideration such delays when preparing their bid.~~

~~]][~~3.1 MODEL UNIT

Prior to placement of material orders for components of the ~~[BEQ living suites] [BOQ living suites] [~~family housing units~~] [_____]~~ and the installation of those components throughout the project, the Contractor must complete the construction of the components of the model unit, and gain the approval of the Contracting Officer for the materials and workmanship therein.

3.1.1 Model Unit Description

Provide a model unit made up of ~~enough [living suites consisting of two bedrooms and an interconnecting bathroom] [single family housing units] [_____]~~one Type A, and one Type B to show all of the different levels of completion required below.~~[~~ The model unit must have enough ~~[suites] [housing units]~~ to show all level of completion simultaneously~~[, except for foundation and floor slab levels which are approved by the Contracting Officer and covered by following levels]~~.~~]~~ The model unit must be ~~[living suites] [family housing units] [_____]~~ constructed at the location agreed upon with the Contracting Officer~~ and [a part of the permanent structure as indicated] [a temporary structure constructed to full scale at the location indicated, and demolished at the completion of the Contract]~~.

3.1.2 Model Unit Requirements

Adhere to the following requirements for model construction:

a. Provide materials in model unit identical to those actually approved in submittals. The QC Manager and Organization must approve and process submittals in accordance with Section 01 33 00 SUBMITTAL PROCEDURES. Do not procure, fabricate, or install materials prior to approval by the Contracting Officer.

b. The Contracting Officer retains the right to disapprove and reject materials or items which are not in compliance with the project drawings and specifications.

c. After approval of initial product submittals, provide full size samples of these components in the required color, and in sufficient quantity to completely outfit the model unit. Provide full size samples as dictated by the construction schedule for installation of those components in the model unit. Do not order or approve production of materials beyond those required for the model unit, until the full size samples have been installed in the model unit and approved by the Contracting Officer.

d. Proceed immediately with construction of the model unit after the Contracting Officer acknowledges the required product submittals are approved.

e. Provide temporary utilities and climate control necessary to construct, inspect, approve, and maintain the model unit.

f. Provide and maintain the following throughout the construction period of the entire project: Temporary parking at the model unit; signage from the temporary parking to the model unit; the visual approach to the unit including trash removal, lawn ~~[~~and landscaping ~~]~~maintenance.

g. Locate the model unit in an easily accessible location for public tours. The Government retains the right to use the model unit for supervised tours and public relations activities following each level of completion. Tours will take place during normal working hours.

~~[~~ h. Prior to completion and approval of the model unit, and with written permission to start from the Contracting Officer, the Contractor may perform general sitework and run major utility lines for the rest of the project.

~~]~~
~~[~~ i. As each model unit's level of completion is approved by the Contracting Officer, construction of that level's components throughout the remainder of the project may proceed.

~~]~~3.1.3 Model Unit Levels of Completion

~~[~~ The completed model unit must be exhibited for approval. Upon completion, the model unit must be available to the Government for a period of 15 working days for inspection after notification is received by Contracting Officer.

~~][Exhibit and secure approval of model unit at the following levels of completion:~~

~~a. Level 1 (Pre-Finished Level) - Model unit construction within a completed building exterior must exhibit plumbing, ductwork, electrical locations, exposed framing, and windows. Leave walls and floors unfinished but ready for application of final finish. Upon completion of this level, model unit must be available to the Government for a period of five working days for inspection.~~

~~b. Level 2 (Finished Level) - Model unit construction must exhibit all materials, completely finished as specified, including fixtures, wall and floor finishes, cabinets, and shelving. Upon completion, the model unit must be available to the Government for a period of 15 working days for inspection.~~

~~][~~Model unit must be exhibited and approved at the following ~~[7] [_____]~~3 levels of completion:

Level 1, model must be available for Government inspection for 28 calendar days.

Levels 2 through 3~~7~~, model unit must be available for Government inspection for 10 calendar days per level.

~~a. Level 1 (Foundation & Floor Slab Level) - Clearly demonstrate treatment of earth excavation, testing of underslab mechanical piping, earth compaction, assembly and support of concrete reinforcing, technique of concrete movement and placement, framing of foundation steps, placement of insulation, final exposed concrete finished surface treatment, location outlines of all walls, all slab penetrations, and structural support points.~~

~~b. Level 2 (Framing Level) - Clearly exhibit all load bearing and non-load bearing framing, sheathing, anchorage for exterior shell materials, and pre-trim frame-out.~~

~~c. Level 3 (Exterior Shell Level) - Exhibit all exterior finishes, windows and doors with temporary hardware, caulking, sealant, sheathing paper, wall ties, locations of mechanical equipment, roofing, louvers, vents and trim.~~

~~d~~a. Level ~~4~~1 (Plumbing, Mechanical and Electrical rough-in Level) - Exhibit electrical conduit and box location for switches, outlets fixtures, cable and data collection, ductwork, built-in plumbing fixtures, plumbing piping, and water pressure test of all plumbing pipe.

~~e~~b. Level ~~5~~2 (Interior Finishes-rough Level) - Exhibit non-veneer plaster, unspackled and uncaulked wall and ceiling finishes, insulation, interior doors, final locations of all outlets and fixtures.

~~f~~c. Level ~~6~~3 (Interior Finishes & Fixtures-final Level) - Exhibit final interior finishes, veneer plaster, painting, caulking, built-in shelving, cabinetwork, cabinet supported sinks, final door hardware, mechanical louvers, final electrical equipment connections, electrical fixtures, freestanding bathroom fixtures, properly functioning appliances, and operation of mechanical and electrical systems.

~~[~~ ~~g. Level 7 (Site Finishes Level) - Exhibit final driveway, walks, exterior stairs, final concrete and asphalt finished surface treatment, topsoil quality and application, sodding/seeding of lawn, tree and shrub planting, landscaping beds, playground fall areas and playground equipment. Contracting Officer retains the right to inspect all preparation and installation of pavings. Provide adequate notice to the Contracting Officer prior to commencing paving work.~~

~~]]]~~ -- End of Section --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ANSI/ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ANSI/ASSP A10.8	(2019) Scaffolding Safety Requirements

ASTM INTERNATIONAL (ASTM)

ASTM D6245	(2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation
ASTM D6345	(2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air
ASTM F855	(2020) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1	(2014; R 2019) American National Standard for Industrial Head Protection

Japanese Standards Association (JSA)

JIS A 8951	(1995) Tubular Steel Scaffolds

MINISTRY OF HEALTH, LABOUR AND WELFARE, GOVERNMENT OF JAPAN (MHLW)

MHLW	(1972) Amendment No. 57 - 2006 Industrial Safety and Health Law
ORDINANCE NO. 34	(1972) The Safety Ordinance for Cranes
ORDINANCE NO. 47	(2007) Ordinance on Industrial Safety and Health

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023; ERTA 7 2023; TIA 23-15) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 306	(2024) Standard for the Control of Gas Hazards on Vessels

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008	(2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	~~(2024) Safety -- Safety and Health Requirements Manual~~ Safety -- Safety and Health Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.1000	Air Contaminants
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1926	Safety and Health Regulations for Construction
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR

1910.146

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, Chapter 2 applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. In conjunction with the local Safety and Occupational Health Office (SOHO), the Contracting Officer will evaluate proposed Japanese equivalent training for applicability to the contract scope of work being performed.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. Submittals not having a "G" classification are for Contractor Quality Control or Designer of Record approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

~~Dive Operations Plan; G~~

~~[~~ Final IAQ Management Plan; S

~~]~~ Indoor Air Quality (IAQ) Management Plan; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Crane Operators/Riggers

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Hot Work Permit

Standard Lift Plan; G

Certifications

Licenses

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by installation emergency management requirement.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher. Monthly exposure reports are to be tracked and generated through RMS.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, MHLW Lawsand follow host nation laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1, Chapter 2 and this Contract. In conjunction with the local Safety and Occupational Health Office (SOHO), the Contracting Officer will evaluate proposed Japanese equivalent training for applicability to the contract scope of work being performed.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1, Chapter 2. All SSHOs will be designated on an ENG Form 6282, as approved by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1, Chapter 2 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed. For features of work with work with multiple high or extremely high RAC codes, a Level One SSHO or Alternate Level One SSHO presence is required on site at all times the work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

(1) Are designated on an ENG Form 6282.

(2) Meet minimum training and experience requirements identified in EM 385-1-1, Chapter 2.

(3) Execute roles and responsibilities identified in EM 385-1-1, Chapter 2.

1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1, Chapter 2 for this project.

1.7.4 Competent Person for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of EM 385-1-1, Chapter 34 and herein. The CP for Confined Space Entry must supervise the entry into each confined space.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, 29 CFR 1915, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable in 29 CFR 1910 for general industry, 29 CFR 1926 for construction.

1.7.5 Crane Operators/Riggers

Provide Operators meeting the requirements in EM 385-1-1, Chapter 15 for Riggers and Chapter 16 for Crane Operators. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 22,680 kg or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.7.5.1 Load Handling Equipment Operator Accepted Alterative

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan (GOJ) Ordinance to achieve the required protections for crane operator licensing and practical training requirements employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Load Handling Equipment, Qualifications, dated 8 January 2020.

Provisions within the JISHA (ORDINANCE NO. 34) Articles 223-228 provide sufficient confirmation of an operator's certification/qualification and practical training to achieve the intent of EM 385-1-1, Chapter 16-3.

JISHA (ORDINANCE NO. 34) Articles 223 establishes the Director of the Prefectural Labour Bureau as the licensing authority. Based upon this Article the Ministry of Labour is the recognized auditor, established for local accreditation of certification and training programs.

1.7.6 Preconstruction Conference

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures, and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.

c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Design Accident Prevention Plan (APP)

Provide a site-specific Accident Prevention Plan (APP), including Activity Hazard Analyses (AHA), in accordance with EM 385-1-1, ENG Form 6293, for the design team to follow during site visits and investigations. For subsequent visits, update the plan if there are changes in the personnel who will be attending, or the tasks to be performed. Submit the APP for review and acceptance by the Government at least 15 calendar days prior to the start of the design field work after being given Notice to Proceed. Field work must not begin until the design APP is accepted by the Contracting Officer. Prior to the start of construction incorporate the Design APP into the Construction APP so that one site specific APP exists for the project and submit to the Contracting Officer for acceptance.

If the design scope includes borings or other subsurface investigations, include in the APP the type of field investigation and verification techniques, such as visual, local utility locating service scanning and third party subcontractor scanning, potholing, or hand digging within two feet of a known utility that will be required. Mark underground utilities before starting any ground-disturbing actions. Notify the Contracting Officer 15 days prior to the start of soil borings or sub-surface investigations.

1.8.2 Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer as attachment 01 35 26-A "ACCIDENT PREVENTION PLAN (APP) WORKSHEET; ENG Form 6293" 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

Copies of the accepted APP shall be maintained at the resident engineer's office and at the job site. Continuously review and amend the APP, as necessary, throughout the life of the contract. Incorporate unusual or high-hazard activities not identified in the original APP as discovered.

1.8.3 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.8.4 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including the following:

1.8.4.1 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

1.8.4.2 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

~~1.8.4.3 Site Safety and Health Plan~~

~~Identify the safety and health aspects, and prepare in accordance with Section 01 35 29.13 HEALTH, SAFETY, AND EMERGENCY RESPONSE PROCEDURES FOR CONTAMINATED SITES.~~

1.8.4.3 Polychlorinated Biphenyls (PCB) Plan

Identify the safety and health aspects of Polychlorinated Biphenyls work, and prepare in accordance with Sections 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS~~ (PCBs) and 02 61 23 REMOVAL AND DISPOSAL OF PCB CONTAMINATED SOILS~~.

1.8.4.4 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION and referenced sources.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting

Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Chapter 2 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	**Notify KO or COR**	**Complete Final Accident Report on ENG 3394 and provide to KO or COR**

Table
Accident Reporting Required Timeline

Fatality, in-patient hospitalization, amputation, eye loss, or property damage over $600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1 to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.

b. Near Misses: Report all "Near Misses" to the Contracting Officer, using local accident reporting procedures, within 24 hours. The Contracting Officer will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1, Chapter 16-10, paragraph b (16-10.b). Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Chapter 16-10, paragraph a (16-10.a) and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Installation Fire Department. Contractors are required to meet all criteria before a permit is issued. Provide at least two (2) 9kg 4A:20 BC rated extinguishers for normal "Hot Work". Extinguishers shall be current inspection tagged, and have approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch shall be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Base Fire Department phone number. Report any fire, no matter how small, to the responsible Base Fire Department immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Chapter 6.

1.14 RADIATION SAFETY REQUIREMENTS

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall request approval in writing to the Contracting Officer not less than thirty (30) calendar days in advance of any request for exception. Following Contracting Officer approval of any radiation related exception, the Contractor shall coordinate with the Contracting Officer's Representative at least fifteen (15) calendar days in advance to notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with EM 385-1-1, Chapter 34, 29 CFR 1926, 29 CFR 1910, Host Nation, local regulatory requirements and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used. Obtain written permission from the Contracting Officer prior to confined space entry by personnel.

1.15.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.16 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 calendar days after design Contract award and not less than 10 calendar days before the preconstruction conference. Revise and resubmit Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants. Submit Final IAQ Management Plan.

1.16.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.16.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008 to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.

b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.

c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.

d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.

e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.16.1.2 Moisture Contamination

a. Remove accumulated water and keep work dry.

b. Use dehumidification to remove moist, humid air from a work area.

c. Do not use combustion heaters or generators inside the building.

d. Protect porous materials from exposure to moisture.

e. Remove and replace items which remain damp for more than a few hours.

1.16.2 Requirements After Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum 2 weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.17 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

a. Secure outside equipment and materials and place materials that could be damaged in protected areas.

b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.

c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

a. Head Protection that meets ANSI/ISEA Z89.1

b. Long Pants

c. Appropriate Safety Footwear

d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall request approval in writing to the Contracting Officer not less than thirty (30) calendar days in advance of any request for exception. Following Contracting Officer approval of any radiation related exception, the Contractor shall coordinate with the Contracting Officer's Representative at least fifteen (15) calendar days in advance to notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations. Stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least sixty (60) days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Chapter 12. In accordance with EM 385-1-1, Chapter 12 where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.2.1 Utility Outages

Outages (power, water, sewer, and communications) for the following listed dates, times, and locations shall not be permitted during the performance of this Contract, unless approved by the authorized agent of the Installation Commander and Contracting Officer's Representative:

a. DoDEA Schools: No outages from 01 September through 31 July, Monday through Friday between the hours of 0700 and 1800.

b. Family Housing Areas: No outages will be allowed on Saturdays when DoDEA schools are not in session. No more than two (2) outages for the same utility service shall be allowed in a one-month period (30 calendar days), and no more than one (1) outage in a two-week period (14 calendar days).

c. U.S. Government Holidays: No outages during the following periods:

(1) New Year's Day - to include seven (7) calendar days before and one (1) calendar day after this date (contiguous with the Christmas Day outage requirement).

(2) Martin Luther King's Birthday - to include the following Tuesday.

(3) President's Day - to include the following Tuesday.

(4) Memorial Day - to include the following Tuesday.

(5) Juneteenth - to include one (1) calendar day before and after this date.

(6) Independence Day - to include one (1) calendar day before and after this date.

(7) Labor Day - to include the following Tuesday.

(8) Columbus Day - to include the preceding Friday.

(9) Veteran's Day - to include two (2) calendar days before and after this date.

(10)Thanksgiving Day - to include one (1) calendar day before and after this date.

(11)Christmas Day - to include seven (7) calendar days before and after this date (contiguous with the New Year's Day outage requirement).

d. Child Development Centers (CDC): Outages allowed only when the CDC is closed or on Sundays.

3.2.2 Reimbursement for Public Works Support of Electrical Outages

The Contractor shall reimburse the NAF Atsugi Public Works Department for its costs to prepare for, support, and recover from Contractor requested utility outages associated with this work. Public Works efforts include setting up generators, securing elevators and securing utilities before the outage, and restoring these systems after the outage. The work includes resetting steam systems, air conditioning systems, hot water systems and electrical systems. Outages during the heating and cooling season cost $1,369 to support during normal working hours, and $3,200 during overtime hours. Outages in the spring and fall, between heating and cooling seasons, cost $1,150 to support during normal working hours, and $2,696 during overtime hours.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Chapter 12. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1, Chapter 12. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, Chapter 12 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Chapter 12. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Chapter 12. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Chapter 12.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Chapter 12, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1, Chapter 21.

3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Chapter 21-4.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Chapter 21-8.t. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 1633 kg in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 1.8 m, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Chapter 21-8.t.

3.6 SCAFFOLDING ACCEPTED ALTERNATIVE

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan Ordinance to achieve the required protections for supported metal scaffolding employed as part of contracted activities provided for U.S. Army Corps of Engineers (USACE) within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Scaffolding Standards Review, dated 24 September 2019.

This determination only includes supported metal scaffolding structures and does not include other forms of scaffolding methods included in EM 385-1-1, Chapter 22.

JISHA requirements for steel scaffolding structures to conform to the Japanese Industrial Standard JIS A 8951 are comparable to those of ANSI/ASSP A10.8, providing sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 22.

3.7 RIGGING ACCEPTED ALTERNATIVE

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan (GOJ) Ordinance, which warrant a partial acceptance for rigging procedures and rigging equipment, employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Rigging Requirements and Procedures, dated 15 January 2020.

This acceptance excludes Multiple Lift Rigging (MLR) procedures as well as standards for all hooks used for lifting or load handling purposes. MLR and hooks used for load handling purposes will adhere to the respective sections of the EM 385-1-1, Chapter 15 without exception.

Accepted standards include, JISHA ORDINANCE NO. 34, "The Safety Ordinance for Cranes", Chapter VIII Sling Work, Section 1 Slinging Equipment Articles 213-220. Also referenced, is the JISHA's "Guideline for sling work safety (2000)".

3.8 EXCAVATIONS

Soil classification shall be performed by a competent person in accordance with EM 385-1-1, Chapter 25.

3.8.1 Excavations and Trenching Accepted Alterative

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan Ordinance to achieve the required protections for excavation and trenching systems including shoring, shielding, and engineered designed systems employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Excavation and Trenching Review, dated 7 November 2019.

This determination only includes excavation and trenching systems including shoring, shielding, and engineered designed systems where the manufactures or a registered professional engineer's tabulated data has been employed.

Provisions within JISHA ORDINANCE NO. 47, Articles 368-371 provide sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 25-2, paragraph j (25-2.j). JISHA ORDINANCE NO. 47 instructions included in Articles 368-371 conform with the provisions for manufactured shoring/shielding and engineered designed support systems including material serviceability, structural design, and erection/installation. These provisions provided sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 25-2, paragraph j (25-2.j).

3.8.2 Utility Locations

The Contractor shall positively identify underground utilities in the work area and coordinate with the installation utility department. The Contractor shall obtain digging permits prior to start of excavation by contacting the Contracting Officer at least fifteen (15) calendar days in advance.

3.8.3 Utility Location Verification

The Contractor shall scan the construction site with electromagnetic or sonic equipment, and mark the surface of the ground where existing underground utilities are discovered. Verify the elevations of existing piping, utilities, and any type of underground or encased obstruction not indicated to be specified or removed, but indicated or discovered during scanning in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made. Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within one meter of the underground system.

3.8.4 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with installation utility departments in addition to location and depth verification by the Contractor. The Contractor shall locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.8.5 Munitions and Explosives of Concern (MEC)

Munitions and Explosives of Concern (MEC): Unexploded Ordinance (UXO), Material Presenting a Potential Explosive Hazard (MPPEH), Chemical Agents (CA), or Discarded Military Munitions (DMM) on jobsites shall be treated as extremely dangerous and must be reported immediately. Follow the 3Rs: RECOGNIZE, RETREAT, and REPORT.

3.8.5.1 Identification and Notification

If you encounter or suspect you have encountered MEC, the on-site supervisor shall immediately suspend all operations put at risk due to the suspected MEC identified. STOP WORK, and DO NOT TOUCH, mark the location, keep people out of the area, and report to the Government Designated Authority. Project personnel will withdraw along cleared path upwind 300 meters from the discovery or as determined by emergency responders.

The on-site supervisor shall contact the installation emergency services who will intern contact the appropriate Explosives Ordinance Disposal (EOD) unit. EOD will determine the threat and mitigate any immediate explosives hazard(s).

Where suspected MEC are encountered during construction operations and other activities, notify the Contracting Officer's Representative/Contracting Officer immediately. If no hazard is present or poses no danger, the Government will direct the Contractor to proceed without change after given the all clear from the emergency responders. If MEC are found the condition or probability of encountering MEC may also change. The responsible government authority will conduct a new probability assessment, this determination will be used to plan the level of support required and determine further requirements. Pursuant to FAR 52.243-4, "Changes" the Government may issue a modification.

3.8.6 Training

All workers on site involved in the earthwork, in any manner, must attend a UXO brief. Coordinate with Contracting Officer for scheduling of this Briefing. Upon completion of the brief, the worker will receive a laminated card verifying that they attended the brief. The card does not expire and is valid for all USACE construction contracts on Okinawa, but is non-transferrable between workers. The USACE team will help coordinate the training and assist with processing base access requests as needed. The brief is free of charge, but the contractor is responsible for all other costs associated with transport to Camp Foster, labor, and all other costs related to the training. The expected duration of this training is two hours.

3.9 EQUIPMENT

3.9.1 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.10 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Chapters 11 and 12.

3.10.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910and 29 CFR 1926.

3.10.1.1 Energized Electrical Work Permit (EEWP)

All equipment and circuits must be placed into an Establishing an Electrically Safe Working Condition (ESWC) must be de-energized before work is started unless energized work can be justified. A QP must complete and verify the following to establish electrically safe working condition in accordance with EM 385-1-1, paragraph 11-8.d unless the work meets the EEWP exemption located under EM 385-1-1, paragraph 11-8.e(5) or the host employer or system owner can show that there are additional hazards or an increased risk from de-energizing.

Hazardous Energy: Any energy including but not limited to mechanical (for example, power transmission apparatus, counterbalances, springs, pressure, gravity), pneumatic, hydraulic, electrical, stored electrical energy exists such as Uninterrupted Power Systems (UPS), chemical, nuclear, and thermal (for example, high or low temperature) energies, which could cause injury to employees.

Once it has been determined that equipment must be worked on in an energized condition, an EEWP must be submitted to the USACE supervisor/KO or COR and AHJ for acceptance. Do not perform energized work without prior authorization. All permits must be prepared, signed, and authorized in advance of performing any electrical work. A non-mandatory ENG Form 6277 (Energized Electrical Work Permit) is provided in EM 385-1-1, Chapter 11-10.

3.10.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with Host Nation requirements applicable to where work is being performed.

3.10.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.10.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.10.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AACE INTERNATIONAL (AACE)
1265 Suncrest Towne Centre Drive
Morgantown, WV 26505-1876 USA
Ph: 304-296-8444
Fax: 304-291-5728
Internet: https://web.aacei.org/

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)
30 West University Drive
Arlington Heights, IL 60004 U.S.A.
Ph: +1 847-394-0150
E-mail: communications@amca.org
Internet: https://www.amca.org/

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2111 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: http://www.ahrinet.org

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)
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Washington, DC 20001
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Internet: https://www.transportation.org/

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130 East Randolph, Suite 2000
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Internet: https://www.aisc.org/

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Internet: https://www.asce.org/

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)
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Internet: https://www.ashrae.org/

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E-mail: customerservice@assp.org
Internet: https://www.assp.org/

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Fax: 708-479-6139
Internet: http://www.asse-plumbing.org

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Fax: 303-347-0804
Internet: https://www.awwa.org/

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Internet: https://www.aws.org/

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EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)
CEN-CENELEC Management Centre
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B - 1040 Brussels, Belgium
Ph: 32-2-550-08-11
Fax: 32-2-550-08-19
Internet: https://www.cencenelec.eu/

ELECTRICAL SAFETY INSPECTION ASSOCIATIONS
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Internet: https://denkihoan.org/en/

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)
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Internet: https://theicpa.com/

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E-mail: info@iec.ch
Internet: https://www.iec.ch/

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Internet: https://www.iso.org

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Internet: https://www.mhlw.go.jp/index.html

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Fax: 630-790-3095
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Greenbelt, MD 20770
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Ph: 901-377-1818
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E-mail: washington.contact@oecd.org

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1215 Geneva - Switzerland
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SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)
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Fax: 703-803-3732
Internet: https://www.smacna.org/

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)
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Internet: https://www.sae.org/

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Washington, D.C. 20585
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HUD User
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Washington, DC 20026-3268
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E-mail: helpdesk@huduser.gov
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U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)
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Internet: https://www.gsaelibrary.gsa.gov/ElibMain/home.do
Obtain documents from:
Acquisition Streamlining and Standardization Information System
(ASSIST)

Internet: https://assist.dla.mil/online/start/; account registration required

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)
8601 Adelphi Road
College Park, MD 20740-6001
Ph: 866-272-6272
Internet: https://www.archives.gov/
Order documents from:
Superintendent of Documents
U.S. Government Publishing Office (GPO)
732 N. Capitol Street, NW
Washington, DC 20401
Ph: 202-512-1800 or 866-512-1800
Bookstore: 202-512-0132
Internet: https://www.gpo.gov/

U.S. NAVAL SEA SYSTEMS COMMAND (NAVSEA)
Commander Naval Sea Systems Command
1333 Isaac Hull Ave., SE
Washington Navy Yard, DC 20376-1080
Ph: 202-781-0000
E-mail: navsea_publicqueries@navy.mil
Internet: http://www.navsea.navy.mil/PublicInquiries.aspx

WOOD MOULDING AND MILLWORK PRODUCERS ASSOCIATION (WMMPA)
1282 Stabler Lane, STE 630 #146
Yuba City, CA95993-2625
Ph: 530-661-9591
Fax: 530-661-9586
E-mail: info@wmmpa.com
Internet: https://www.wmmpa.com/

WOODWORK INSTITUTE (WI)
1455 Response Road, Suite 110
Sacremento, CA 95815
Ph: 916-372-9943
Fax: 916-372-9950
E-mail: info@woodinst.com
Internet: https://woodworkinstitute.com

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)
1500 SW First Ave., Suite 870
Portland, OR 97201
Ph: 503-224-3930
E-mail: info@wwpa.org
Internet: http://www.wwpa.org

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION 01 42 15

METRIC MEASUREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM SI10	(201~~0~~6) American National Standard for Use of the International System of Units (SI): The Modern Metric System

1.2 GENERAL

This project includes metric units of measurements. The metric units used are the International System of Units (SI) developed and maintained by the General Conference on Weights and Measures (CGPM); the name International System of Units and the international abbreviation SI were adopted by the 11th CGPM in 1960. The following criteria allow both metric and English inch-pound (I-P) measurements to be shown for a product:

a. To subsitute products that are manufactured to an industry recognized rounded metric (hard metric) dimension with I-P products.

b. To indicate industry and/or Government standards, test values or other controlling factors (such as the code requirements where I-P values are needed for clarity).

c. To trace back to the referenced standard(s), test value(s) or code(s).

1.3 USE OF MEASUREMENTS IN DRAWINGS AND SPECIFICATIONS

Measurements in drawings and specifications shall be in SI units, with I-P units in parenthesis in cases as indicated in Paragraph GENERAL and as otherwise authorized by the Contracting Officer. Procure the product in SI units unless otherwise authorized by the Contracting Officer. The Contractor is responsible for all associated labor and materials when authorized to substitute one system of units for another and for the final assembly and performance of the specified work and/or products.

1.3.1 Hard Metric

Hard metric measurements are often used for field data such as distance from one point to another or distance above the floor. Products are considered to be hard metric when they are manufactured to metric dimensions or have an industry recognized metric designation.

1.3.2 Soft Metric

a. A soft metric measurement is a non-mathematical, industry related conversion. Soft metric measurements are used for measurements pertaining to products, test values, and other situations where the I-P units are the standard for manufacture, verification, or other controlling factor.

b. A soft metric measurement is also indicated for products that are manufactured in industry designated metric dimensions but are required by law to allow substitute I-P products.

1.3.3 Neutral

A neutral measurement is indicated by an identifier which has no expressed relation to either an SI or an I-P value (e.g., American Wire Gage (AWG) which indicates thickness but in itself is neither SI nor I-P).

1.4 COORDINATION

Bring discrepancies, such as mismatches or product unavailability, arising from use of both metric and non-metric measurements and discrepancies between the measurements in the specifications and the measurements in the drawings to the attention of the Contracting Officer for resolution.

1.5 RELATIONSHIP TO SUBMITTALS

Submittals for Government approval or for information only covers the SI products actually being furnished for the project. Submit the required drawings and calculations in the same units used in the Contract documents describing the product or requirement unless otherwise instructed or approved. Use ASTM SI10 as the basis for establishing metric measurements required to be used in submittals.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION 01 45 00.00 10

QUALITY CONTROL
11/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-12	(2006; Change 1) Engineering and Design -- Quality Management

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program, all associated costs shall be included in the applicable CLIN Schedule unit or job prices.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information onlysubmittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval.Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

Design Quality Control (DQC) Plan

SD-05 Design Data

Discipline-Specific Checklists

Design Quality Control

SD-06 Test Reports

Verification Statement

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC consist of plans, procedures, and organization necessary to produce an end product which complies with the Contract requirements. The QC system covers all design and/or construction operations, both onsite and offsite, and be keyed to the proposed design and construction sequence. The project superintendent will be held responsible for the quality of work and is subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. In this context the highest level manager responsible for the overall construction activities at the site, including quality and production is the project superintendent. The project superintendent maintains a physical presence at the site at all times and is responsible for all construction and related activities at the site, except as otherwise acceptable to the Contracting Officer.

3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN

Submit no later than 15 days after receipt of notice to proceed, the Contractor Quality Control (CQC) Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Include the submittal register with CQC Plan submissions. The Government will consider an interim plan for the first 14 days of operation. ConstructionDesign and/or construction will be permitted to begin only after acceptance of the CQC Plan or acceptance of an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of the~~a~~ CQC Plan~~ or another interim plan containing the additional work~~.

3.2.1 Content of the CQC Plan

Include, as a minimum, the following to cover all design and construction-operations, both onsite and offsite, including work by subcontractors designers of record, consultants, architect/engineers (AE), fabricators, suppliers and purchasing agents:

a. A description of the quality control organization, including a chart showing lines of authority and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified. ~~Include a CQC System Manager that reports to the project superintendent if the CQC System Manager and project superintendent are separate persons~~The CQC System Manager shall report to the assigned Project Manager.

b. The name, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function.

c. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authorities to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the Contract. Letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities will be issued by the CQC System Manager. Furnish copies of these letters to the Contracting Officer.

d. Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, designers of record, consultants, architect engineers (AE), offsite fabricators, suppliers, and purchasing agents. These procedures must be in accordance with

Section 01 33 00 SUBMITTAL PROCEDURES.

e. Control, verification, and acceptance testing procedures for each specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test frequency, and person responsible for each test. (Laboratory facilities approved by the Contracting Officer are required to be used.)

f. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.

g. Procedures for tracking design and construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected.

h. Reporting procedures, including proposed reporting formats.

i. A list of the definable features of work. A definable feature of work is a task which is separate and distinct from other tasks, has separate control requirements, and is identified by different trades or disciplines, or it is work by the same trade in a different environment. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. This list will be agreed upon during the coordination meeting.

j. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections. Where the applicable Code issue by the International Code Council (ICC) calls for inspections by the Building Official, the Contractor must include the inspections in the Quality Control Plan and must perform the inspections required by the applicable ICC. The Contractor must perform these inspections using independent qualified inspectors. Include the Special Inspection Plan requirements in the QC Plan.

3.2.2 Additional Requirements for Design Quality Control (DQC) Plan

The following additional requirements apply to the Design Quality Control (DQC) plan:

a. Submit and maintain a Design Quality Control (DQC) Plan as an effective quality control program which assures that all services required by this contract are performed and provided in a manner that meets professional architectural and engineering quality standards. As a minimum, all documents must be technically reviewed by competent, independent reviewers identified in the DQC Plan. The same element that produced the product may not perform the independent technical review (ITR). Correct errors and deficiencies in the design documents prior to submitting them to the Government.

b. Include the design schedule in the master project schedule, showing the sequence of events involved in carrying out the project design tasks within the specific Contract period. This should be at a detailed level of scheduling sufficient to identify all major design tasks, including those that control the flow of work. Include review and correction periods associated with each item. This should be a forward planning as well as a project monitoring tool. The schedule reflects calendar days and not dates for each activity. If the schedule is changed, submit a revised schedule reflecting the change within 7 calendar days. Include in the DQC Plan the discipline-specific checklists to be used during the design and quality control of each submittal. Submit at each design phase as part of the project documentation these completed discipline-specific checklists. ER 1110-1-12 provides some useful information in developing checklists.

c. Implement the DQC Plan by a Design Quality Control Manager who has the responsibility of being cognizant of and assuring that all documents on the project have been coordinated. This individual must be a person who has verifiable engineering or architectural design experience and is a registered professional engineer or architect. Notify the Contracting Officer, in writing, of the name of the individual, and the name of an alternate person assigned to the position.

The Contracting Officer will notify the Contractor in writing of the acceptance of the DQC Plan. After acceptance, any changes proposed by the Contractor are subject to the acceptance of the Contracting Officer.

3.2.3 Acceptance of Plan

Acceptance of the Contractor's plan is required prior to the start of design and construction. Acceptance is conditional and will be predicated on satisfactory performance during the design and construction. The Government reserves the right to require the Contractor to make changes in the Contractor Quality Control(CQC) Plan and operations including removal of personnel, as necessary, to obtain the quality specified.

3.2.4 Notification of Changes

After acceptance of the CQC Plan, notify the Contracting Officer in writing a minimum of seven (7) days prior of any proposed change. Proposed changes are subject to acceptance by the Contracting Officer.

3.3 COORDINATION MEETING

After the Preconstruction Conference,Postaward Conference, before start of design or construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. Submit the CQC Plan a minimum of 15 calendar days prior to the Coordination Meeting. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, design activities, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the Government, signed by both the Contractor and the Contracting Officer and will become a part of the contract file. There can be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

3.4 QUALITY CONTROL ORGANIZATION

3.4.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), CQC System Manager, a Design Quality Manager, and sufficient number of additional qualified personnel to ensure safety and Contract compliance. The SSHO reports directly to a senior project (or corporate) official independent from the CQC System Manager. The SSHO will also serve as a member of the CQC Staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the CQC organization. The Contractor's CQC staff maintains a presence at the site at all times during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules and all other project documentation to the CQC organization. The CQC organization is responsible to maintain these documents and records at the site at all times, except as otherwise acceptable to the Contracting Officer.

3.4.2 CQC System Manager

Identify as CQC System Manager an individual within the onsite work organization that is responsible for overall management of CQC and has the authority to act in all CQC matters for the Contractor. This CQC System Manager is on the site at all times during construction and is employed by the prime Contractor. Identify in the plan an alternate to serve in the event of the CQC System Manager's absence. The requirements for the alternate are the same as the CQC System Manager. The CQC System Manager shall report to the assigned Project Manager. Refer to Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, Paragraph "Contractor Quality Control System Manager" for experience requirements.

3.4.3 CQC Personnel

In addition to CQC personnel specified elsewhere in the contract, provide as part of the CQC organization specialized personnel to assist the CQC System Manager for the required areas listed in Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, Paragraph "Contractor Quality Control Personnel". All personnel listed in this table may not be required on this Contract. These individuals or specialized technical companies must be responsible to the CQC System Manager; must be physically present at the construction site during work on the specialized personnel's areas of responsibility; must have the necessary education or experience in accordance with the experience matrix listed herein. The table below defines only the qualification requirements. Refer to the Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, Paragraph "Contractor Quality Control Personnel" for actual required personnel.

Experience Matrix	
Area	Qualifications
Civil	Graduate of an accredited university with an Civil Engineering or Construction Manager degree with 3 years experience in the type of work being performed on this project or technician with 5 yrs related experience. Or Non-US degree plus 1 Kyu Doboku Sekou Kanrigishi (1st Class Civil Engineering Works Management Engineer).
Mechanical	Graduate of an accredited university with an Mechanical Engineer degree with 3 years experience or person with 5 years of experience supervising mechanical features of work in the field with a construction company. Or Non-US degree plus 1 Kyu Kankouji Sekou Kanrigishi (1st Class Building Mechanical and Electrical Engineer).
Electrical	Graduate of an accredited university with an Electrical Engineering degree with 3 years related experience or person 5 years of experience supervising electrical features of work in the field with a construction company. Or Non-US degree plus 1 Kyu Kankouji Sekou Kanrigishi (1st Class Building Mechanical and Electrical Engineer) or 1 Kyu Denkikouji Sekou Kanrigishi (1st Class Electric Construction Management Engineer).
Structural	Graduate of an accredited university with an Civil Engineering degree (with Structural Track or Focus) or Construction Management degree with 3 years experience or technician 5 years of experience supervising structural features of work in the field with a construction company. Or Non-US degree plus 1 Kyu Kenchikushi (1st Class Qualified Architect) or 1 Kyu Kenchiku Sekou Kanrigishi (1st Class Building Construstion Managemnet Engineer).
Architectural	Graduate of an accredited university with an Architecture degree with 3 years experience or person with 5 years related experience. Or Non-US degree plus 1 Kyu Kenchikushi (1st Class Qualified Architect) or 1 Kyu Kenchiku Sekou Kanrigishi (1st Class Building Construction Management Engineer).

Experience Matrix	
Area	Qualifications
Fire Protection	3 to 5 years of documented experience of similar projects.~~Registered U.S. Professional Engineer who has passed the fire protection engineering written examination administered by the National Council of Examiners and Surveying (NCEES) and has a minimum of four (4) years of relevant fire protection engineering experience.~~
Communications	Graduate of an accredited university with an Electrical Engineering degree with 3 yrs related experience or technician with 5 years of experience supervising communications features of work in the field with a construction company. Non-US degree plus AI*DD Sougoushu(AI*DD Construction Engineer).
Environmental	Refer to Section ~~[01 57 19][~~01 57 19.01~~]~~ ENVIRONMENTAL MANAGER.
Submittals	Submittal Clerk with 1 year experience
Occupied Family Housing	Person, customer relations type, coordinator experience
Concrete, Pavements and Soils	Materials Technician with 2 years experience for the appropriate area
Testing, Adjusting and Balancing (TAB) Personnel	Specialist must be a member of AABC or an experienced technician of the firm certified by the NEBB
Design Quality Control Manager	Registered Architect or Professional Engineer or Japanese equivalent (i.e. 1st Class License in any specific discpline, etc.) Minimum of 5 years experience as a Design Quality Control Manager.

3.4.4 Site Safety and Health Officer (SSHO)

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMNTS.

3.4.5 Additional Requirement

In addition to the above experience and education requirements, the Contractor Quality Control(CQC) System Manager and Alternate CQC System Manager are required to have completed the Construction Quality Management (CQM) for Contractors course. If the CQC System Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the CQC System Manager's certificate has expired, retake the course to remain current.

3.4.6 Organizational Changes

Maintain the CQC staff at full strength at all times. When it is necessary to make changes to the CQC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

3.5 SUBMITTALS AND DELIVERABLES

Submittals, if needed, have to comply with the requirements in Section 01 33 00 SUBMITTAL PROCEDURES. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the contract requirements. When Section 01 91 00.15 10 TOTAL BUILDING COMMISSIONING is included the submittals required by those sections must be coordinated with Section 01 33 00 SUBMITTAL PROCEDURES to ensure adequate time is allowed for each type of submittal required.

3.6 CONTROL

CQC is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the contract. At least three phases of control are required to be conducted by the CQC System Manager for each definable feature of the construction work as follows:

3.6.1 Preparatory Phase

This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase includes:

a. A review of each paragraph of applicable specifications, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

b. Review of the Contract drawings.

c. Check to assure that all materials and equipment have been tested, submitted, and approved.

d. Review of provisions that have been made to provide required control inspection and testing.

e. Review Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

f. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the Contract.

g. Examination of required materials, equipment, and sample work to assure that they are on hand, conform to approved shop drawings or submitted data, and are properly stored.

h. Review of the appropriate activity hazard analysis to assure safety requirements are met.

i. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.

j. Check to ensure that the portion of the plan for the work to be performed has been accepted by the Contracting Officer.

k. Discussion of the initial control phase.

l. The Government needs to be notified at least 72 hours in advance of beginning the preparatory control phase. Include a meeting conducted by the CQC System Manager and attended by the superintendent, other CQC personnel (as applicable), and the foreman responsible for the definable feature. Document the results of the preparatory phase actions by separate minutes prepared by the CQC System Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required in order to meet contract specifications.

3.6.2 Initial Phase

This phase is accomplished at the beginning of a definable feature of work. Accomplish the following:

a. Check work to ensure that it is in full compliance with contract requirements. Review minutes of the preparatory meeting.

b. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing are in compliance with the contract.

c. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

d. Resolve all differences.

e. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.

f. The Government needs to be notified at least 24 hours in advance of beginning the initial phase for definable feature of work. Prepare separate minutes of this phase by the CQC System Manager and attach to the daily CQC report. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases.

g. The initial phase for each definable feature of work is repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.

h. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

3.6.3 Follow-up Phase

Perform daily checks to assure control activities, including control testing, are providing continued compliance with contract requirements, until completion of the particular feature of work. Record the checks in the CQC documentation. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work. Do not build upon nor conceal non-conforming work. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

3.6.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if other problems develop.

3.7 TESTS

3.7.1 Testing Procedure

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of a Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site. Perform the following activities and record and provide the following data:

a. Verify that testing procedures comply with contract requirements.

b. Verify that facilities and testing equipment are available and comply with testing standards.

c. Check test instrument calibration data against certified standards.

d. Verify that recording forms and test identification control number system, including all of the test documentation requirements, have been prepared.

e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

3.7.2 Onsite Laboratory

The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

3.8 COMPLETION INSPECTION

3.8.1 Punch-Out Inspection

Conduct an inspection of the work by the CQC System Manager near the end of the work, or any increment of the work established by a time stated in FAR 52.211-10 Commencement, Prosecution, and Completion of Work. Prepare and include in the CQC documentation a punch list of items which do not conform to the approved drawings and specifications, as required by paragraph DOCUMENTATION. Include within the list of deficiencies the estimated date by which the deficiencies will be corrected. Make a second inspection the CQC System Manager or staff to ascertain that all deficiencies have been corrected. Once this is accomplished, notify the Government that the facility is ready for the Government Pre-Final inspection.

3.8.2 Pre-Final Inspection

The Government will perform the pre-final inspection to verify that the facility is complete and ready to be occupied. A Government Pre-Final Punch List may be developed as a result of this inspection. Ensure that all items on this list have been corrected before notifying the Government, so that a Final inspection with the customer can be scheduled. Correct any items noted on the Pre-Final inspection in a timely manner. These inspections and any deficiency corrections required by this paragraph need to be accomplished within the time slated for completion of the entire work or any particular increment of the work if the project is divided into increments by separate completion dates.

3.8.3 Final Acceptance Inspection

The Contractor's Quality Control Inspection personnel, plus the superintendent or other primary management person, and the Contracting Officer's Representative is required to be in attendance at the final acceptance inspection. Additional Government personnel including, but not limited to, those from Base/Post Civil Facility Engineer user groups, and major commands can also be in attendance. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the Pre-Final inspection. Notify the Contracting Officer at least 14 days prior to the final acceptance inspection and include the Contractor's assurance that all specific items previously identified to the Contractor as being unacceptable, along with all remaining work performed under the Contract, will be complete and acceptable by the date scheduled for the final acceptance inspection. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance FAR 52.246-12 Inspection of Construction.

3.9 DOCUMENTATION

3.9.1 Quality Control Activities

Maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

a. The name and area of responsibility of the Contractor/Subcontractor.

b. Operating plant/equipment with hours worked, idle, or down for repair.

c. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.

d. Test and control activities performed with results and references to specifications/drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action.

e. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications/drawings requirements.

f. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.

g. Offsite surveillance activities, including actions taken.

h. Job safety evaluations stating what was checked, results, and instructions or corrective actions.

i. Instructions given/received and conflicts in plans and specifications.

j. Provide documentation of design quality control activities. For independent design reviews, provide, as a minimum, identification of the Independent Technical Review (ITR) team, the ITR review comments, responses and the record of resolution of the comments.

3.9.2 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract. Furnish the original and one copy of these records in report form to the Government daily within 24 hours after the date covered by the report, except that reports need not be submitted for days on which no work is performed. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the Contractor Quality Control(CQC) System Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the CQC System Manager Report.

3.10 SAMPLE FORMS

Sample forms can be obtained from the Resident Office or found through Contractor link of the RMS website ~~http://rms.usace.army.mil.~~ https://rms.usace.army.mil/.

3.11 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor of any detected noncompliance with the foregoing requirements. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, will be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer can issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

-- End of Section --

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	~~(2024) Safety -- Safety and Health Requirements Manual~~ Safety -- Safety and Health Requirements Manual

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

Administration
Finances
Quality Control
Submittal Monitoring
Scheduling
Closeout
Import/Export of Data

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some correspondence, including pay requests and payrolls, are also to be provided in paper format with original signatures. Paper documents will govern, in the event of discrepancy with the electronic version.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Priced Construction Contracts; ~~[~~Section 01 32 01.00 10 PROJECT SCHEDULE~~][Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES]~~; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 00.00 10 QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a Windows-based program that can be run on a Windows-based PC meeting the requirements as specified in paragraph SYSTEM REQUIREMENTS. Download, install and be able to utilize the latest version of the RMS software within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (https://rms.usace.army.mil). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 SYSTEM REQUIREMENTS

The following is the recommended system configuration to run the Contractor Mode RMS for full utilization of all features for all types and sizes of contracts. Smaller, less complicated, projects may not require the configuration levels described below. Required configuration also noted below.

Recommended RMS System Requirements	
Hardware	
Windows-based PC	1.7 GHz i3; AMD A6 3650 GHz or higher processor (REQUIRED)
RAM	8 GB
Hard drive disk	100 GB space for sole use by RMS system
Monitor	Screen resolution 1366 x 768
Mouse or other pointing device	
Windows compatible printer	Laser printer must have 4 MB+ of RAM
Connection to the Internet	minimum 4 Mbs per user
Software	
MS Windows	Windows 10 x 64 bit (RMS requires 64 bit O/S) or newer (REQUIRED)
Word Processing software	Viewer for MS Word 2013, MS Excel 2013 or newer (REQUIRED)

Recommended RMS System Requirements	
E-mail	MAPI compatible (REQUIRED)
Virus protection software	Regularly upgraded with all issued Manufacturer's updates and is able to detect most zero day viruses (REQUIRED)

1.6 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.7 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.7.1 Administration

1.7.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.7.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within 7 calendar days of receiving access to the contract in RMS or within 7 calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.7.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.7.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates.

1.7.1.5 Reports

Track the status of the project utilizing the reports available in RMS. The value of these reports is reflective of the quality of the data input. These reports include the Progress Payment Request worksheet, Quality Control (QC) comments, Submittal Register Status, and Three-Phase Control worksheets.

1.7.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response.

1.7.2 Finances

1.7.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule. The sum of pay activities equals the total contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.7.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data either electronically or by hard copy. Unless waived by the Contracting Officer, a signed paper copy of the approved payment certification and request is also required and will govern in the event of discrepancy with the electronic version.

1.7.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Insure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments of same within 7 calendar days of Government acceptance of the CQC Plan.

1.7.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.7.3.2 Deficiency Tracking.

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.7.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS. Worksheets for the three-phase control meetings are generated within RMS.

1.7.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.7.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.7.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.7.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager.

1.7.4 Submittal Management

Enter all current submittal register data and information into RMS within 7 calendar days of receiving access to the contract in RMS. The information shown on the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES will already be entered into the RMS database when access is granted. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the contract. Submit hard copies of all submittals unless waived by the Contracting Officer.

1.7.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in ~~[~~Section 01 32 01.00 10 PROJECT SCHEDULE~~][Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES]~~.

1.7.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.8 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.9 MONTHLY COORDINATION MEETING

Update the RMS CM database each workday. At least monthly, generate and submit one schedule update. At least one week prior to submittal, meet with the Government representative to review the planned progress payment data submission for errors and omissions.

Make required corrections prior to Government acceptance of the export file and progress payment request. Payment requests accompanied by incomplete or incorrect data submittals will not be accepted. The Government will not process progress payments until all required corrections are processed.

1.10 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within 7 calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 45 35

SPECIAL INSPECTIONS
02/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7	(2017) Minimum Design Loads for Buildings and Other Structures

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2018) International Building Code

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS Z2305	(2013) Non-destructive Testing - Qualification and Certification of Personnel

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections and Chapter 17 of ICC IBC. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the prime contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.~~[~~

Structural observations will be performed by the Government. The contractor must provide notification to the Contracting Officer 14 days prior to the following points of construction:

~~a. [_____]~~
~~b. [_____]~~
~~c. [_____]]~~a. Concrete rehabilitation: repair work for concrete cracks, concrete strengthening and concrete wall infill

b. Utility building: reinforcement and concrete placement for foundations, walls and roof slab

1.3 DEFINITIONS

1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.

1.3.2 Periodic Special Inspections

Periodic Special Inspections is Special Inspections by the special inspector who is intermittently present where the work to be inspected has been or is being performed.

1.3.3 Perform

Perform these Special Inspections tasks for each welded joint or member, bolted or fastener connection, and required verification.

1.3.4 Observe

Observe these Special Inspections items on a random daily basis. Operations need not be delayed pending these inspections.

1.3.5 Special Inspector (SI)

A qualified person retained by the contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

1.3.6 Associate Special Inspector (ASI)

A qualified person who assists the SI in performing Special Inspections but must perform inspection under the direct supervision of the SI and cannot perform inspections without the SI on site.

1.3.7 Third Party

A third party inspector must not be company employee of the Contractor or any Sub-Contractor performing the work to be inspected.

~~[~~1.3.8 Special Inspector of Record (SIOR)

A licensed engineer in responsible charge of supervision all special inspectors for the project and approved by the Contracting officer. The SIOR must be an independent third party hired directly by the Prime Contractor.

~~]~~1.3.9 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).

1.3.10 Contractor's Quality Control (QC) Manager

An individual retained by the prime contractor and qualified in accordance with the Section ~~[~~01 45 00.00 10 QUALITY CONTROL~~]~~ having the overall responsibility for the contractor's QC organization.

1.3.11 Designer of Record (DOR)

A registered design professional ~~[employed by the Government] [~~contracted by the Government as an A/E~~]~~ responsible for the overall design and review of submittal documents prepared by others. The DOR is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in state in which the design professional works. The DOR is also referred to as the Engineer of Record (EOR) in design code documents.

1.3.12 Statement of Special Inspections (SSI)

A document developed by the DOR identifying the material, systems, components and work required to have Special Inspections.

1.3.13 Schedule of Special Inspections

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspections is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17.

~~[~~1.3.14 Designated Seismic System

Those nonstructural components that require design in accordance with ASCE 7 Chapter 13 and for which the component importance factor, Ip, is greater than 1.0. This designation applies to systems that are required to be operational following the Design Earthquake for RC I - IV structures~~ [and following the MCER for RC V structures. All systems in RC V facilities designated as MC-1 in accordance with UFC 03-301-01 are considered part of the Designated Seismic Systems]~~. ~~[~~Designated Seismic Systems will be identified by Owner and will have an Importance Factor Ip = 1.5~~]~~.

~~][~~1.3.15 Component Certification and O&M Manual

For any electrical or mechanical component required by ASCE 7 Section 13.2.2 to be certified, evidence demonstrating compliance with the requirement shall be maintained in a file identified as "Equipment Certification Documentation." This file shall be a part of the Operations & Maintenance Manual that is turned over to the Contracting Officer. The project specifications shall require the Operations & Maintenance Manual state that replaced or modified components need to be certified per the original certification criteria.

~~][~~1.3.16 Component Identification Nameplate

Any electrical or mechanical component required by ASCE 7 ~~S~~Section 13.2.2 to be certified shall bear permanent marking or nameplates constructed of a durable heat and water resistant material. Nameplates shall be mechanically attached to such nonstructural components and placed on each component for clear identification. The nameplate shall not be less than 125 x 180 with red letters 25 in height on a white background stating "Certified Equipment." The following statement shall be on the nameplate: "This equipment/component is certified. No modifications are allowed unless authorized in advance and documented in the Equipment Certification Documentation file." The nameplate shall also contain the component identification number in accordance with the drawings/specifications and the O&M manuals.

~~]~~1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.][~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

SIOR Letter of Acceptance; G~~[, [_____]]~~

Special Inspections Project Manual; G~~[, [_____]]~~

Special Inspections Agency's Written Practices

NDT Procedures and Equipment Calibration Records

SD-06 Test Reports

Special Inspections Daily Reports

Special Inspections Biweekly Reports

SD-07 Certificates

Fabrication Plant

Certificate of Compliance

Special Inspector of Record Qualifications; G~~[, [_____]]~~

Special Inspector Qualifications; G~~[, [_____]]~~

Qualification Records for NDT technicians

SD-11 Closeout Submittals

Interim Final Report of Special Inspections

Comprehensive Final Report of Special Inspections; G~~[, [_____]]~~

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector ~~[~~and the special inspector of record~~]~~.

Certifying Associations	
FM	Factory Mutual
ICC	International Code Council

Certifying Associations	
JCI	Japan Concrete Institute
JSFA	Japan Steel Fabricators Association
JWES	Japan Welding Engineering Society
UL	Underwriters Laboratories
Architect Under Bldg Standard Law of Japan	First Class Architect (Kenchikushi) Registered Under Standard Law of Japan

1.5.1 Steel Construction and High Strength Bolting

1.5.1.1 Special Inspector

a. Structural Steel Work Manager (Tekkotsu Koji Kanri Sekininsha), or

b. Japan Welding Society Certified Inspector (Nihon Yosetsu Kyokai Kensa-in), or

c. Non-Destructive Test Technician (Hihakai Kensa Gijutsusha), or

d. Architectural or Civil Construction Managing Engineer (Kenchiku/ Doboku Koji Sekou Kanrigishi), or

e. JSFA Certified Architectural High Strength Bolt Joint Management Engineer

1.5.2 Welding Structural Steel

1.5.2.1 Special Inspector

a. JWES Certified Welding Management Engineer

b. JSFA Certified Architectural Structural Steel Products Inspection Engineer

1.5.3 Nondestructive Testing of Welds

1.5.3.1 Special Inspector

JIS Z2305 NDT Level III Certificate.

1.5.3.2 Associate Special Inspector

JIS Z2305 NDT Level II Certificate plus one year of related experience.

~~1.5.4 Cold Formed Steel Framing~~

~~1.5.4.1 Special Inspector~~

~~a. JSFA Certified Architectural Structural Steel Products Inspection Engineer, or~~

~~b. Registered Professional Engineer with related experience.~~

1.5.4 Concrete Construction

1.5.4.1 Special Inspector

a. Japan Concrete Institute Concrete Engineer (JCI Concrete Gishi), or

b. Registered Professional Engineer with related experience, or

c. First Class Kenchikushi with four years of related experience.

d. Architectural or Civil Construction Managing Engineer (Kenchiku/ Doboku Sekou Kanrigishi)

~~1.5.5 Prestressed Concrete Construction~~

~~1.5.5.1 Special Inspector~~

~~a. Registered Professional Engineer with related experience, or~~

~~b. First Class Kenchikushi with four years of related experience~~

~~1.5.6 Post-tensioned Concrete Construction~~

~~1.5.6.1 Special Inspector~~

~~a. Registered Professional Engineer with related experience, or~~

~~b. First Class Kenchikushi with four years of related experience.~~

~~1.5.7 Masonry Construction~~

~~1.5.7.1 Special Inspector~~

~~a. Registered Professional Engineer with related experience, or~~

~~b. First Class Kenchikushi with four years of related experience.~~

~~1.5.8 Wood~~

~~1.5.8.1 Special Inspector~~

~~a. Registered Professional Engineer with related experience~~

~~b. First Class Kenchikushi with four years of related experience~~

1.5.5 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements

1.5.5.1 Special Inspector

a. First Class Kenchikushi, or

b. First Class Doboku Sekou Kanrigishi (1st class Civil Engineering Works Management Engineer), or

c. Geotechnical Survey Technician (Chishitsu Chosa Gishi), or

d. Soil Investigation Technician (Jiban Kensa Gishi)

~~1.5.6 Deep Foundations~~

~~1.5.6.1 Special Inspector~~

~~Chishitsu Chosa Gishi (Professional Geotechnical Engineer)~~

~~1.5.7 Sprayed Fire Resistant Material~~

~~1.5.7.1 Special Inspector~~

~~a. ICC Spray-applied Fireproofing Special Inspector Certificate, or~~

~~b. ICC Fire Inspector I Certificate with one year of related experience, or~~

~~c. Registered Professional Engineer with related experience, or~~

~~d. First Class Kenchikushi with four years related experience.~~

1.5.6 Mastic and Intumescent Fire Resistant Coatings

1.5.6.1 Special Inspector

a. ICC Spray-applied Fireproofing Special Inspector Certificate, or

b. ICC Fire Inspector I Certificate with one year of related experience, or

c. Registered Professional Engineer with related experience, or

d. First Class Kenchikushi with four years related experience

~~1.5.7 Exterior Insulation and Finish System (EIFS)~~

~~1.5.7.1 Special Inspector~~

~~a. Registered Professional Engineer with related experience, or~~

~~b. First Class Kenchikushi with four years related experience.~~

1.5.7 Fire-Resistant Penetrations and Joints

1.5.7.1 Special Inspector

a. Passed the UL Firestop Exam with one year of related experience, or

b. Passed the FM Firestop Exam with one year of related experience, or

c. Registered Professional Engineer with related experience, or

d. First Class Kenchikushi with four years related experience.

~~1.5.8 Smoke Control~~

~~1.5.8.1 Special Inspector~~

~~a. AABC Technician Certification with one year of related experience, or~~

~~b. Registered Professional Engineer with related experience, or~~

~~c. First Class Kenchikushi with four years related experience~~

1.5.8 Architectural Components

1.5.8.1 Special Inspector

a. Registered Professional Architect with related experience, or

b. Registered Professional Engineer with related experience, or

c. First Class Kenchikushi with four years related experience.

1.5.9 Mechanical, Electrical, and Plumbing Designated Seismic Systems

1.5.9.1 Special Inspector

a. Registered Professional Engineer with related experience, or

b. First Class Kenchikushi with four years related experience.

1.5.10 Verification of Electrical Systems

a. Chief Electrical Engineer (Denki Shunin Gijutusha), or

b. Electrical Construction Managing Engineer (Denki Sekou Kanrigishi), or

c. Japanese Licensed Electrician (Denki Koujishi)

~~[~~1.5.11 Special Inspector of Record (SIOR)

Registered Professional Engineer or First Class Kenchikushi with four years of related experience.

~~]~~PART 2 PRODUCTS

2.1 FABRICATOR SPECIAL INSPECTIONS

Special Inspections of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following certification ~~[certifications]~~ to the Contracting Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

Minister of Land, Infrastructure and Transportation (MLIT) Certified Fabrication Plant, Category as specified

At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special Inspections, stating that the materials supplied and work performed by the fabricator are in accordance the construction documents.

PART 3 EXECUTION

3.1 RESPONSIBILITIES

3.1.1 Special Inspector of Record

a. Supervise all Special Inspectors required by the contract documents and the IBC.

b. Submit a SIOR Letter of Acceptance to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR.

c. Verify the qualifications of all of the Special Inspectors.

d. Verify the qualifications of fabricators.

e. Submit Special Inspections agency's written practices for the monitoring and control of the agency's operations to include the following:

(1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.

(2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.

f. Submit qualification records for nondestructive testing (NDT) technicians designated for the project.

g. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.

h. Prepare a Special Inspections Project Manual, which will cover the following:

(1) Roles and responsibilities of the following individuals during Special Inspections: SIOR, SI, General Contractor, Subcontractors, QC Manager, and DOR.

(2) Organizational chart and/or communication plan, indicating lines of communication.

(3) Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, who to contact for inspection requests, and availability of alternate inspectors.

(4) Indicate the government reporting procedures.

(5) Propose forms or templates to be used by SI and SIOR to document inspections.

(6) Indicate procedures for tracking nonconforming work and verification that corrective work is complete.

(7) Indicate how the SIOR and/or SI will participate in weekly QC meetings.

(8) Indicate how Special Inspections of shop fabricated items will be handled when the fabricator's shop is not certified per paragraph FABRICATOR SPECIAL INSPECTIONS.

(9) Include a section in the manual that covers each specific item requiring Special Inspections that is indicated on the Schedule of Special Inspections. Provide names and qualifications of each special inspector who will be performing the Special Inspections for each specific item. Provide detail on how the Special Inspections are to be carried out for each item so that the expectations are clear for the General Contractor and the Subcontractor performing the work.

Make a copy of the Special Inspections Project Manual available on the job site during construction. Submit a copy of the Special Inspections Project Manual for approval.

i. Attend coordination and mutual understanding meeting where the information in the Special Inspections Project Manual will be reviewed to verify that all parties have a clear understanding of the Special Inspections provisions and the individual duties and responsibilities of each party.

j. Maintain a 3- ring binder for the Special Inspector's daily and biweekly reports and the Special Inspections Project Manual. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the DOR.

k. Submit a copy of the Special Inspector's daily reports to the QC Manager.

l. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report.

m. Submit a biweekly Special Inspections report until all work requiring Special Inspections is complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:

(1) A brief summary of the work performed during the reporting time frame.

(2) Changes and/or discrepancies with the drawings, specifications and mechanical or electrical component certification, that were observed during the reporting period.

(3) Discrepancies which were resolved or corrected.

(4) A list of nonconforming items requiring resolution.

(5) All applicable test results including nondestructive testing reports.

n. At the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an interim final report of Special Inspections that documents the Special Inspections completed for that

DFOW and corrections of all discrepancies noted in the daily reports. The interim final report of Special Inspections must be signed, dated and bear the seal of the SIOR.]

o. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and bear the seal of the SIOR.

]3.1.2 Quality Control Manager

[a. Supervise all Special Inspectors required by the contract documents and the IBC.

b. Verify the qualifications of all of the Special Inspectors.

c. Verify the qualifications of fabricators.

d. Maintain a 3- ring binder for the Special Inspector's daily and biweekly reports. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the DOR.

] ~~[a.]~~[e.] Maintain a rework items list that includes discrepancies noted on the Special Inspectors daily report.

3.1.3 Special Inspectors

a. Inspect all elements of the project for which the special inspector is qualified to inspect and are identified in the Schedule of Special Inspections.

b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) for which the special inspector is qualified to inspect.

[c. Submit Special Inspections agency's written practices for the monitoring and control of the agency's operations to include the following:

(1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.

(2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.

d. Submit qualification records for nondestructive testing (NDT) technicians designated for the project.

e. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.]

f. Submit a copy of the daily reports to the QC Manager.

g. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report.

h. Submit a biweekly Special Inspection Report until all inspections are complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:

(1) A brief summary of the work performed during the reporting time frame.

(2) Changes and/or discrepancies with the drawings, specifications [and mechanical or electrical component certification,] that were observed during the reporting period.

(3) Discrepancies which were resolved or corrected.

(4) A list of nonconforming items requiring resolution.

5) All applicable test result including nondestructive testing reports.

[i. At the completion of each DFOW requiring Special Inspections, submit an interim final report of Special Inspections that documents the Special Inspections completed for that DFOW. Identify the inspector responsible for each item inspected and corrections of all discrepancies noted in the daily reports. The interim final report of Special Inspections must be signed, dated and indicate the certification of the special inspector qualifying them to conduct the inspection.]

j. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project and corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and indicate the certification of the special inspector qualifying them to conduct the inspection.

[k. Submit daily reports to the SIOR.

]3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept such work.

3.3 DESIGNATED SEISMIC SYSTEMS

Designated seismic systems shall require a final walk-down inspection by the Designer of Record or Registered Design Professional in Responsible Charge as defined in ICC IBC Chapter 2. The final review shall be documented in a report. The final report prepared by the Designer of Record or Registered Design Professional in Responsible Charge shall include the following:

a. Record/observations of final site visit.

b. Documentation that all required inspections were performed in accordance with the Statement of Special Inspections.

c. Documentation, including certificates of compliance, that the Designated Seismic Systems were installed in accordance with the construction documents and the requirements of ICC IBC Chapter 17, and as modified by this section.

-- End of Section --

Project: RENOVATE TOWER B743
Location: CAMP ZAMA, JAPAN
Project #: PN100964
Date: 1/29/2026

STATEMENT OF SPECIAL INSPECTIONS

Project Seismic Design Category: E
Project Risk Category: II
Project Design Wind Speed (km/h): 248
Number of Stories: 9
Structure Height Above Grade (M): 31
Hazardous Occupancy or attached to such? No Group H Occupancies

Special Inspector of Record (SIOR)

A Special Inspector of Record (SIOR) **IS** required (per UFGS 01 45 35, Section 1.3.7)

SIOR Name (Registered Professional): Contractor to provide prior to construction start
Professional Registration Number: Contractor to provide prior to construction start
Consulting Firm Name (if any): Contractor to provide prior to construction start
SIOR Office AND Mobile Phone Number: Contractor to provide prior to construction start

2024 IBC 1704.3.2 and 1704.3.3

Following is a listing of critical main wind/seismic force resisting systems for this structure. Carefully inspect these elements as part of the roles and responsibilities of the Special Inspector (reference the Schedule of Special Inspections for inspection checklists).

Vertical LFRS Elements	Notes
	Added Shearwalls and Column Strengthening at Tower
	Shearwalls at Utility Building

Horizontal LFRS Elements	Notes
	Roof slab at Utility Building

Project: RENOVATE TOWER B743
Location: CAMP ZAMA, JAPAN
Project #: PN100964
Date: 1/29/2026

Designated Seismic Systems (DSS)

(2024 IBC 1705.13.3.4) (ASCE 7-22, 13.2.3, C13.2.2) (UFC 3-301-1, 2-5.3)

Non-structural 'Designated Seismic Systems' (DSS) have an I_p greater than 1.0, and must be certified by the manufacturer to remain operable after a design earthquake. The Prime Contractor shall collect the manufacturers Certificates of Compliance for incorporation into the Operations and Maintenance manual per Section 01 78 23 paragraph 1.3. The SIOR shall review the Certificates of Compliance as part of the special inspection. Additionally, the below listed Designated Seismic Systems must be carefully inspected by the Special Inspector according to the requirements noted in the Schedule of Special Inspections.

ELECTRICAL Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	DSS LIghting that supports the emergency light fixtures
2.	DSS Conduits and Cable Tray
3.	DSS Electrical System (including conduits) and Panels Serving Fire Pump Controller and Fire Pump: Fire and Jocky Pump Controller, 3P200A Fire Pump Closed Circuit Breaker, Main Switchboard "MSB"
4.	DSS Elevator Power Supply Disconnect Switch
5.	Emergency Distribution Panel "2EDP"

If additional space is required, append an additional sheet listing the remaining DSS

MECHANICAL/PLUMBING Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	None
2.	

If additional space is required, append an additional sheet listing the remaining DSS

OTHER Designated Seismic Systems (DSS) Requiring a Certificate of Compliance	
1.	DSS Building fire sprinkler systems
2.	DSS Elevator System, see specification section 14 21 23
3.	
4.	

Final Walk Down Inspection and Report

(UFC 3-301-01 SECTION 2-5.4)

Designated Seismic Systems shall receive a final walk-down inspection by the Registered Design Professional in Responsible Charge.

Final Walk Down Report, Prepared by the Registered Design Professional in Responsible Charge, Must Include:

1. Record observations of Final Walk Down Inspection
2. Document that Inspections were performed in accordance with the Schedule of Special Inspections
3. Document that all Designated Seismic Systems are installed according to construction/manufacturer document requirements, and that Compliance Certificates have been collected (UFC 3- 301-01, 01, 2-13.2.2.1).

SCHEDULE OF SPECIAL INSPECTIONS

Reference UFGS 01 45 35 for all requirements not noted as part of this schedule.

INSPECTION DEFINITIONS:

PERFORM: Perform these tasks for each weld, fastener or bolted connection, and required verification.

OBSERVE: Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Operations need not be delayed pending these inspections at contractor's risk.

DOCUMENT: Document, with a report, that the work has been performed in accordance with the contract documents. This is in addition to any other reports required in the Special Inspections guide specification.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

PERIODIC: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

The Seismic Design Category for this project is: ☐ A, ☐ B, ☐ C, ☐ D, ☒ E, ☐ F (check appropriate box)

DESIGNER NOTES (delete this box after reviewing):

1. This schedule contains minimum requirements. Do not delete applicable inspection tasks unless notes in blue indicate it is acceptable to do so.
2. Blue text = designers notes. The designer must review and edit all blue text in this schedule prior to inserting this schedule into the special inspections spec (UFGS 01 45 35).
3. Check section boxes with ANY inspection tasks applicable to your project. You may choose to delete unchecked sections or leave them in the schedule unchecked.
4. Individual rows/tasks that are not applicable to the project may be left in the section, as the inspector can determine whether they occur/apply (e.g. metal trusses in the light gauge framing section for example).
5. Design discipline sections are color coded for easier reference by designers. This schedule does NOT need to be printed in color.
6. When finished editing, delete this note box and save this schedule as a PDF and insert into the project specifications (special inspections section).

A. STRUCTURAL STEEL – WELDING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

STEEL INSPECTION PRIOR TO WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-1 and Table C-N5.4-1		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify that the welding procedures specification (WPS) is available	**PERFORM**	
2. Verify manufacturer certifications for welding consumables are available	**PERFORM**	
3. Verify material identification	**PERFORM**	Type and grade.
4. Verify use of qualified welders	**PERFORM**	Welding by welders, welding operators, and tack welders who are qualified in conformance with requirements.
5. Welder Identification System	OBSERVE	✓ Fabricator or erector, as applicable, shall maintain a system by which a welder who has welded a joint or member can be identified. ✓ Die stamping of members subjected to fatigue shall be prohibited unless approved by the engineer of record.
6. Fit-up of groove welds (including joint geometry)	OBSERVE	✓ Joint preparation ✓ Dimensions (alignment, root opening, root face, bevel) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location) ✓ Backing type and fit (if applicable)
7. Fit-up of CJP groove welds of HSS T-, Y-, and K-connections without backing (including joint geometry	**PERFORM**	✓ Joint preparations ✓ Dimensions (alignment, root opening, root face, bevel) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location)
8. Configuration and finish of access holes	OBSERVE	
9. Fit-up of fillet welds	OBSERVE	✓ Dimensions (alignment, gaps at root) ✓ Cleanliness (condition of steel surfaces) ✓ Tacking (tack weld quality and location)
STEEL INSPECTION DURING WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-2 and Table C-N5.4-2		
TASK	INSPECTION TYPE	DESCRIPTION
10. Control and handling of welding consumables	OBSERVE	✓ Packaging ✓ Electrode atmospheric exposure control
11. No welding over cracked tack welds	OBSERVE	
12. Environmental conditions	OBSERVE	✓ Wind speed within limits ✓ Precipitation and temperature

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

STEEL INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-2 and Table C-N5.4-2		
TASK	INSPECTION TYPE	DESCRIPTION
13. Welding Procedures Specification followed	OBSERVE	✓ Settings on welding equipment ✓ Travel speed ✓ Selected welding materials ✓ Shielding gas type/flow rate ✓ Preheat applied ✓ Interpass temperature maintained (min./max.) ✓ Proper position (F, V, H, OH) ✓ Intermix of filler metals avoided
14. Welding techniques	OBSERVE	✓ Interpass and final cleaning ✓ Each pass within profile limitations ✓ Each pass meets quality requirements
STEEL INSPECTION <u>AFTER</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-3 and Table C-N5.4-3		
TASK	INSPECTION TYPE[1]	DESCRIPTION
15. Welds cleaned	OBSERVE	✓
16. Verify size, length, and location of all welds	**PERFORM**	✓ Size, length, and location of all welds conform to the requirements of the construction drawings.
17. Verify welds meet visual acceptance criteria	**PERFORM AND DOCUMENT**	✓ Crack prohibition ✓ Weld/base-metal fusion ✓ Crater cross section ✓ Weld profiles ✓ Weld size ✓ Undercut ✓ Porosity
18. Check for arc strikes	**PERFORM**	
19. Visually inspect *k*-area	**PERFORM**	When welding of doubler plates, continuity plates or stiffeners has been performed in the k-area, visually inspect the web k-area for cracks within 3 in. (75 mm) of the weld
20. Verify backing removed, weld tabs removed and finished, and fillet welds added where required	**PERFORM**	
21. Repair activities	**PERFORM AND DOCUMENT**	
22. Acceptance or rejection of welded joint or member	**PERFORM AND DOCUMENT**	
23. No prohibited welds have been added without the approval of the engineer of record	OBSERVE	

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

B. STRUCTURAL STEEL – BOLTING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

STEEL INSPECTION TASKS PRIOR TO BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.6-1 and Table C-N5.6-1		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify manufacture's certifications available for fastener materials	**PERFORM**	
2. Fasteners marked in accordance with ASTM requirements	OBSERVE	
3. Proper fasteners selected for joint detail (grade, type, bolt length if threads are to be excluded from shear plane)	OBSERVE	
4. Proper bolting procedure selected for joint detail	OBSERVE	
5. Connecting elements, including appropriate faying surface condition and hole preparation, if specified, meet applicable requirements	OBSERVE	
6. Pre-installation verification testing for each fastener assembly performed by installation personnel, carefully observing and documenting the installation methods.	OBSERVE	Check if the fastener assemblies and the installation methods are capable of achieving 105%, or more, of the minimum required bolt pretension.
7. Proper storage provided for bolts, nuts, washers, and other fastener components	OBSERVE	
STEEL INSPECTION TASKS DURING BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.6-2 and Table C-N5.6-2		
TASK	INSPECTION TYPE	DESCRIPTION
8. Fastener assemblies of suitable condition, placed in all holes and washers (if required) are positioned as required	OBSERVE	
9. Joint brought to the snug-tight condition prior to pretensioning operation	OBSERVE	
10. Fastener component not turned by the wrench prevented from rotating	OBSERVE	
11. Bolts are pretensioned in accordance with RCSC *Specification*, progressing systematically from the most rigid point toward the free edges	OBSERVE	
STEEL INSPECTION TASKS AFTER BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.6.3 and Table C-N5.6-3		
TASK	INSPECTION TYPE	DESCRIPTION
12. Acceptance or rejection of all bolted connections	**DOCUMENT**	

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

C. STRUCTURAL STEEL - NONDESTRUCTIVE TESTING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Section N5.5		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify use of qualified nondestructive testing personnel	**PERFORM**	Visual weld inspection and nondestructive testing (NDT) shall be conducted by personnel qualified in accordance with AWS D1.8 Clause 7.2
2. CJP groove welds	OBSERVE	**[NOTE: DOR must delete this row if Section D (SEISMIC PROVISIONS SECTION) is checked]** Dye penetrant testing (DT) and ultrasonic testing (UT) shall be performed on 20% of CJP groove welds for materials greater than 5/16" (8 mm) thick. Testing rate must be increased to 100% if greater than 5% of welds tested have unacceptable defects.
3. Welded joints subject to fatigue	OBSERVE	Dye penetrant testing (DT) and Ultrasonic testing (UT) shall be performed on 100% of welded joints identified on contract drawings as being subject to fatigue. Table A-3.1 in AISC 360-22 provides Fatigue Design Parameters for welded joints that must be checked.

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

D. STRUCTURAL STEEL – AISC 341 REQUIREMENTS (SEISMIC PROVISIONS)

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 341-22: Section J7.2		
TASK	INSPECTION TYPE[1]	DESCRIPTION
[NOTE: DOR may uncheck this section for projects NOT designed in accordance with AISC 341 (Seismic Provisions) or for projects designed according to AISC 341, but using an R value equal to 3]		
4. CJP groove welds	OBSERVE	Ultrasonic testing (UT) shall be performed on 100% of complete-joint-penetration (CJP) groove welds in materials 5/16 in. (8 mm) thick or greater. UT in materials less than 5/16 in. (8 mm) thick is not required. Welds shall be inspected by UT in compliance with AWS D1.8/D1.8M. Magnetic particle testing (MT) shall be performed on 25% of all beam-to-column CJP groove welds. Welds shall be inspected by MT in compliance with AWS D1.8/ D1.8M. For ordinary moment frames in structures in risk categories I or II, UT and MT of CJP groove welds shall be required only for demand critical welds.
5. Beam cope and access hole.	OBSERVE	At welded splices and connections, thermally cut surfaces of beam copes and access holes shall be tested using magnetic particle testing (MT) or dye penetrant testing (DT), when the flange thickness exceeds 1 1/2 in. for rolled shapes, or when the web thickness exceeds 1 1/2 in. for built-up shapes.
6. *k*-area NDT (AISC 341 Table J7.1 and AWS D1.8 Clause 7.4)	**PERFORM AND DOCUMENT**	Where welding of doubler plates, continuity plates or stiffeners has been performed in the *k*-area, the web shall be tested for cracks using magnetic particle testing (MT). The MT inspection area shall include the *k*-area base metal within 3 in. of the weld. The MT shall be performed no sooner than 48 hours following completion of the welding.
7. Placement of reinforcing or contouring fillet welds (Table J7.1)	**PERFORM AND DOCUMENT**	
8. Weld tab removal sites	OBSERVE	At the end of welds where weld tabs have been removed, magnetic particle testing shall be performed on the same beam-to-column joints receiving UT.

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports

E. STRUCTURAL STEEL - COMPOSITE CONSTRUCTION[1]

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

COMPOSITE CONSTRUCTION PRIOR TO PLACING CONCRETE – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-2, AISC 341-22: Table J10.1		
TASK	INSPECTION TYPE	DESCRIPTION
1. Placement and installation of steel headed stud anchors	**PERFORM**	
2. Material identification of reinforcing steel (Type/Grade)	OBSERVE	
3. Determination of carbon equivalent for reinforcing steel other than ASTM A706	OBSERVE	
4. Proper reinforcing steel size, spacing, clearances, support, and orientation	OBSERVE	
5. Reinforcing steel has not been rebent in the field	OBSERVE	
6. Required reinforcing steel clearances have been provided	OBSERVE	
7. Reinforcing steel has been tied and supported as required	OBSERVE	
8. Composite member has required size	OBSERVE	

END SECTION

F. STRUCTURAL STEEL - OTHER INSPECTIONS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

OTHER STEEL INSPECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22 Section N5.8, AISC 341-22: Tables J9.1 & J11.1		
TASK	INSPECTION TYPE [2]	DESCRIPTION
1. Verify anchor rods and other embedments supporting structural steel	**PERFORM**	Verify the diameter, grade, type, and length of the anchor rod or embedded item, and the extent or depth of embedment prior to placement of concrete.
2. Fabricated steel or erected steel frame	OBSERVE	Verify compliance with the details shown on the construction documents, such as braces, stiffeners, member locations and proper application of joint details at each connection.
3. Check reduced beam sections (RBS) where/if they occur	**PERFORM AND DOCUMENT**	✓ Contour and finish ✓ Dimensional tolerances
4. Check protected zones	**PERFORM AND DOCUMENT**	No holes or unapproved attachments made by fabricator or erector
5. Check H-piles where/if they occur	**PERFORM AND DOCUMENT**	No holes or unapproved attachments made by the responsible contractor

END SECTION

G. STRUCTURAL COLD-FORMED METAL DECK PLACEMENT

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[1] See Concrete Construction Section for all concrete related inspection of composite steel construction.

[2] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

METAL DECK INSPECTION PRIOR TO DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.1		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Verify compliance of materials (deck and all deck accessories) with construction documents, including profiles, material properties, and base metal thickness	**PERFORM**	
2. Acceptance or rejection of deck and deck accessories	**DOCUMENT**	
METAL DECK INSPECTION AFTER DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.2		
TASK	INSPECTION TYPE	DESCRIPTION
3. Verify compliance of deck and all deck accessory installation with construction documents	**PERFORM**	
4. Verify deck materials are represented by the mill certifications that comply with the construction documents	**PERFORM**	
5. Acceptance or rejection of installation of deck and deck accessories	**DOCUMENT**	
METAL DECK INSPECTION PRIOR TO WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.3		
TASK	INSPECTION TYPE	DESCRIPTION
6. Verify welding procedure specification (WPS) is available	**PERFORM**	
7. Verify manufacturer's certifications for welding consumables are available	**PERFORM**	
8. Material identification (type/grade)	OBSERVE	
9. Check welding equipment	**PERFORM**	

END SECTION

[1]**PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

H. STRUCTURAL COLD-FORMED METAL DECK – WELDING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

METAL DECK INSPECTION DURING WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.4		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Use of qualified welders	OBSERVE	
2. Control and handling of welding consumables	OBSERVE	
3. Environmental conditions (wind speed, moisture, temperature)	OBSERVE	
4. WPS followed	OBSERVE	
METAL DECK INSPECTION AFTER WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.5		
TASK	INSPECTION TYPE	DESCRIPTION
5. Verify size and location of welds, including support, sidelap, and perimeter welds.	**PERFORM**	
6. Verify welds meet visual acceptance criteria	**PERFORM**	
7. Verify repair activities	**PERFORM**	
8. Acceptance or rejection of welds	**DOCUMENT**	

END SECTION

[1]**PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

I. STRUCTURAL COLD-FORMED METAL DECK – FASTENING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

METAL DECK INSPECTION BEFORE MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.6		
TASK	INSPECTION TYPE	DESCRIPTION
1. Manufacturer installation instructions available for mechanical fasteners	OBSERVE	
2. Proper tools available for fastener installation	OBSERVE	
3. Proper storage for mechanical fasteners	OBSERVE	✓ Fasteners are located away from moisture and in a protected location, such as off the ground and away from excessive handling. ✓ Uncoated fasteners are stored in airtight containers to reduce corrosion and rust.
METAL DECK INSPECTION DURING MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.7		
TASK	INSPECTION TYPE	DESCRIPTION
4. Fasteners are positioned as required	OBSERVE	
5. Fasteners are installed in accordance with manufacturer's instructions	OBSERVE	
METAL DECK INSPECTION AFTER MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.8		
TASK	INSPECTION TYPE[1]	DESCRIPTION
6. Check spacing, type, and installation of support fasteners	**PERFORM**	
7. Check spacing, type, and installation of side-lap fasteners	**PERFORM**	
8. Check spacing, type, and installation of perimeter fasteners	**PERFORM**	
9. Verify repair activities	**PERFORM**	
10. Acceptance or rejection of mechanical fasteners	**DOCUMENT**	

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.

J. STRUCTURAL COLD-FORMED STEEL FRAMING AND/OR COLD-FORMED TRUSSES

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

COLD-FORMED STEEL CONSTRUCTION AND CONNECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.5, 1705.12.2, 1705.12.3, 1705.13.3, 1705.13.9, UFC 4 023 03		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify compliance of trusses spanning 60-feet or greater where/if applicable	**PERFORM**	Verify that temporary and permanent truss restraint/bracing is installed in accordance with approved truss submittal package.
2. Welded connections (seismic and/or wind resisting system)	OBSERVE	Visually inspect all welds that are part of the main wind or seismic force resisting system, including shear walls, braces, collectors (drag struts), and hold-downs. **[NOTE: DOR must identify critical wind and/or seismic force resisting welds in the contract drawings so that the special inspector can confirm compliance.]**
3. Connections (seismic and/or wind resisting system)	OBSERVE	Visually inspect all screw attachment, bolting, anchoring and other fastening of components within the main wind or seismic force resisting system, including roof deck, roof framing, exterior wall covering, wall to roof/floor connections, braces, collectors (drag struts) and hold-downs. **[NOTE: DOR must identify critical wind and/or seismic force resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]**
4. Cold-formed steel (progressive collapse resisting system where/if applicable)	OBSERVE	Verify proper welding operations, screw attachment, bolting, anchoring and other fastening of components within the progressive collapse resisting system, including horizontal tie force elements, vertical tie force elements and bridging elements (UFC 4 023 03). **[NOTE: DOR must identify critical progressive collapse resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]**

END SECTION

K. STRUCTURAL OPEN-WEB STEEL JOISTS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

OPEN-WEB STEEL JOISTS AND JOIST GIRDERS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.2.4, 2024 IBC 2207.1, SJI 100 OR 200		
	INSPECTION TYPE [1]	DESCRIPTION
1. Installation of open-web steel joists and joist girders	OBSERVE	✓ End connections – welded or bolted for compliance with SJI 100 or 200 ✓ Bridging – horizontal and diagonal for compliance with SJI 100 or 200

END SECTION

[1] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.
OBSERVE: Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

L. STRUCTURAL CONCRETE CONSTRUCTION

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Inspect reinforcement, including prestressing tendons, and verify placement.	PERIODIC	Verify prior to placing concrete that reinforcement is of specified type, grade and size; that it is free of oil, dirt and unacceptable rust; that it is located and spaced properly; that specified cover has been provided; that hooks, bends, ties, stirrups and supplemental reinforcement are placed correctly; that specified lap lengths, stagger and offsets are provided; and that all mechanical connections are installed per the manufacturer's instructions and/or evaluation report.
2. Inspect reinforcing bar welding	PERIODIC	✓ Verify weldability of reinforcing bars other than ASTM A 706 ✓ Inspect single-pass fillet welds, maximum 5/16" in accordance with AWS D1.4
	CONTINUOUS	✓ Inspect welding of reinforcement for special moment frame, boundary elements of special shear walls, and coupling beams. ✓ Inspect welded reinforcement splices. ✓ Inspect welding of primary tension reinforcement in corbels.
	PERIODIC	✓ Inspect all other welds in accordance with AWS D1.4
3. Inspect cast-in-place anchors, post-installed drilled anchors, and adhesive anchors other than those in item 4 below	PERIODIC	✓ Verify prior to placing concrete that cast in place anchors and post installed drilled anchors have proper embedment, spacing and edge distance. ✓ Inspection requirements for adhesive anchors are different from those for other post-installed anchors and require assessment and qualification under the provisions of ACI 355.4 which may require proof loading.
4. Inspect post-installed adhesive anchors in horizontal or upward inclined orientations and	CONTINUOUS AND DOCUMENT	✓ Inspect as required per approved ICC-ES report

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.
DOCUMENT: Document in a report that the work has been performed as required. This is in addition to all other required reports.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE [1]	DESCRIPTION
subject to sustained tension loading.		✓ Verify that installer is certified for installation of horizontal and overhead installation applications ✓ Verify proof loading as required by the contract documents
5. Verify use of required mix design	PERIODIC	Verify that all mixes used comply with the approved construction documents
6. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete	CONTINUOUS	At the time fresh concrete is sampled to fabricate specimens for strength test, verify these tests are performed by qualified technicians.
7. Inspect concrete and/or shotcrete placement for proper application techniques	CONTINUOUS	Verify proper application techniques are used during concrete conveyance and that depositing avoids segregation or contamination. Verify that concrete is properly consolidated.
8. Verify maintenance of specified curing temperature and technique	PERIODIC	Inspect curing and cold weather, or hot weather protection procedures.
9. Inspect prestressed concrete	CONTINUOUS	Verify application of prestressing forces and grouting of bonded prestressing tendons.
10. Inspect erection of precast concrete members	PERIODIC	✓ Verify dimensional tolerances for precast members and interfacing members. ✓ Verify details of lifting devices, embedments, and related reinforcement required to resist temporary loads from handling, storage, transportation, and erection, if designed by the licensed design professional.
11. Inspect precast concrete diaphragm connections or reinforcement at joints classified as moderate or high deformability element (MDE or HDE) in structures assigned to Seismic Design Category C, D, E, or F	**CONTINUOUS**	Inspect such connections and reinforcement in the field for: ✓ Installation of the embedded parts ✓ Completion of the continuity of reinforcement across joints. ✓ Completion of connections in the field
12. Inspect installation tolerances of precast concrete diaphragm connections for compliance with ACI 550.5	PERIODIC	

CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)		
TASK	INSPECTION TYPE [1]	DESCRIPTION
13. Verify in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs.	PERIODIC	See ACI 318 26.11.2
14. Inspect formwork for shape, location and dimensions of the concrete member being formed.	PERIODIC	Formwork fabrication and installation shall result in final structure that conforms to shapes, lines, and dimensions of the members as required by the construction documents.

END SECTION

M. STRUCTURAL - MASONRY CONSTRUCTION (ALL RISK CATEGORIES)

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH AT START OF CONSTRUCTION 2024 IBC 1705.4 (TMS 402/602-22 TABLE 3 & 4)		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Verify compliance with approved submittals prior to start	PERIODIC	
2. Verify proportions of site-mixed mortar.	PERIODIC	
3. Verify grade, type and size of reinforcement, connectors, and anchor bolts	PERIODIC	
4. Verify sample panel construction	PERIODIC **CONTINUOUS**	**[NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]**
MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH PRIOR TO GROUTING IBC 1705.4 (TMS 402/602-22 TABLE 4)		
TASK	INSPECTION TYPE	DESCRIPTION
5. Verify grout space	PERIODIC **CONTINUOUS**	**[NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]**
6. Verify placement of reinforcement, connectors, and anchor bolts	PERIODIC **CONTINUOUS**	**[NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]**
7. Verify proportions of site-prepared grout	PERIODIC	
MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH DURING CONSTRUCTION IBC 1705.4 (ACI 530-16 TABLE 4)		
TASK	INSPECTION TYPE	DESCRIPTION
8. Verify compliance of materials and procedures with the approved submittals	PERIODIC	
9. Verify placement of masonry units and mortar joint construction	PERIODIC	
10. Verify size and location of structural elements	PERIODIC	
11. Verify type, size and placement of reinforcement, connectors, anchor bolts, and anchorages, including details of anchorage of masonry to structural members, frames, or other construction	PERIODIC **CONTINUOUS**	**[NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/III]**
12. Verify type, size and location of veneer ties and movement joints	PERIODIC	Periodic inspection of veneers is required when the height of the veneer exceeds 60 ft (18.3 m) above grade plane. See Commentary in Appendix A.

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH DURING CONSTRUCTION IBC 1705.4 (ACI 530-16 TABLE 4)		
TASK	INSPECTION TYPE	DESCRIPTION
13. Verify installation of adhered veneer	PERIODIC	Periodic inspection of veneers is required when the height of the veneer exceeds 60 ft (18.3 m) above grade plane
14. Verify welding of reinforcement	**CONTINUOUS**	
15. Verify preparation, construction, and protection of masonry during cold weather (temperature below 40°F (4.4°c) or hot weather (temp above 90°F (32.2°C))	PERIODIC	
16. Verify placement of grout	**CONTINUOUS**	
17. Observe preparation of grout specimens, mortar specimens, and/or prisms	PERIODIC **CONTINUOUS**	**[NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/III]**

END SECTION

N. STRUCTURAL – WOOD CONSTRUCTION – SPECIALTY ITEMS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

WOOD CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.5		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. High-load diaphragms where applicable	PERIODIC	Verify thickness and grade of sheathing, size of framing members at panel edges, nail diameters and length, and the number of fastener lines and that fastener spacing is per approved contract documents.
2. Metal-plate connected wood trusses spanning 60 feet or greater	PERIODIC	Verify that the temporary installation restraint/bracing and the permanent individual truss member restraint/bracing are installed in accordance with the approved truss submittal package
Mass timber elements in Type IV-A, IV-B, and IV-C construction. Refer to 2024 IBC 602.4 for details about Type IV construction		
3. Inspect anchorage and connections of mass timber construction to timber deep foundation systems	PERIODIC	
4. Inspect erection of mass timber construction	PERIODIC	
5. Inspect connections where installation methods are required to meet design loads		
a) Threaded fasteners	PERIODIC	✓ Verify use of proper installation equipment ✓ Verify use of pre-drilled holes where required ✓ Inspect screws, including diameter, length, head type, spacing, installation angle and depth
b) Adhesive anchors connections installed in horizontal or upwardly inclined orientation to resist sustained tension loads.	CONTINUOUS	
c) Adhesive anchors not defined in preceding cell	PERIODIC	
d) Bolted connections	PERIODIC	
e) Concealed connections	PERIODIC	

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

O. STRUCTURAL WOOD CONSTRUCTION - SEISMIC & WIND

THIS SECTION IS APPLICABLE IF BOX IS CHECKED: ☐

WOOD CONSTRUCTION SEISMIC AND WIND – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.12 & 1705.13.2		
TASK	INSPECTION TYPE[1]	DESCRIPTION
[NOTE: DOR may uncheck this section where the lateral resistance is provided by structural sheathing, the sheathing nailing/fasteners (shear walls, shear panels and diaphragms) are consistently greater than 4" on center, or if the design wind speed (ASD) is less than 140 mph (62.6 meters/sec) AND the seismic design category is A or B]		
1. Nailing, bolting, anchoring and other fastening of elements of the main wind/seismic force-resisting system	PERIODIC	Includes connectors for: shear wall sheathing, roof/floor sheathing, drag struts/collectors (double top plates), braces, hold downs, roof connections to exterior walls.
2. Field gluing operations of elements of the main wind/seismic force-resisting system	**CONTINUOUS**	

END SECTION

P. STRUCTURAL ISOLATION AND ENERGY DISSIPATION SYSTEMS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

ISOLATION AND ENERGY DISSIPATION SYSTEMS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.13.8, 1705.14.4		
[NOTE: This section is not applicable to Seismic Design Category A. Uncheck this section if this category applies]		
TASK	INSPECTION TYPE[2]	DESCRIPTION
1. Verify fabrication and installation of isolator units and energy dissipation devices	PERFORM	Verify that fabrication and installation of isolator units and energy dissipation devices conform to manufacturer's recommendations and approved construction documents
2. Test of seismic isolation systems in seismically isolated structures	PERFORM	Seismic isolation systems in seismically isolated structures shall be tested in accordance with ASCE 7, Section 17.8

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

[2] **PERFORM**: Perform these tasks for each weld, fastener, or bolted connection, and required verification.

Q. GEOTECHNICAL - SOILS INSPECTION

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

SOILS INSPECTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.6 and TABLE 1705.6		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify materials below shallow foundations are adequate to achieve the design bearing capacity.	PERIODIC	
2. Verify excavations are extended to proper depth and have reached proper material	PERIODIC	
3. Perform classification and testing of compacted fill materials	PERIODIC	
4. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly	PERIODIC	
5. During fill placement, verify use of proper materials and procedures in accordance with the provisions of the approved geotechnical report. Verify densities and lift thicknesses during placement and compaction of compacted fill	**CONTINUOUS**	

END SECTION

R. GEOTECHNICAL - DRIVEN DEEP FOUNDATION ELEMENTS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

DRIVEN DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.7 and TABLE 1705.7		
TASK	INSPECTION TYPE	DESCRIPTION
1. Verify element materials, sizes and lengths comply with requirements	**CONTINUOUS**	
2. Inspect driving operations and maintain complete and accurate records for each element	**CONTINUOUS**	
3. Verify placement locations and plumbness, confirm type and size of hammer, record number of blows per foot of penetration, determine required penetrations to achieve design capacity, record tip and butt elevations and document any damage to foundation element	**CONTINUOUS**	
4. Determine capacities of test elements and conduct additional load tests if required.	**CONTINUOUS**	
5. For steel, concrete, concrete-filled, perform additional special inspections in accordance with the steel and concrete sections in this schedule		
6. For specialty elements, perform additional inspections as determined by the licensed design professional in responsible charge		In accordance with Statement of Special Inspections

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.
CONTINUOUS: Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

S. GEOTECHNICAL - HELICAL PILE FOUNDATIONS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

HELICAL PILE FOUNDATIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.9		
TASK	INSPECTION TYPE	DESCRIPTION
1. Record installation equipment used, pile dimensions, tip elevations, final depth, final installation torque and other pertinent installation data as required. The approved geotechnical report and the contract documents prepared by the licensed design professional in responsible charge shall be used to determine compliance	**CONTINUOUS**	
2. Determine capacities of test elements and conduct additional load tests if required.	**CONTINUOUS**	

END SECTION

T. GEOTECHNICAL – CAST-IN-PLACE CONCRETE DEEP FOUNDATION ELEMENTS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

CAST-IN-PLACE CONCRETE DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.8		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Inspect drilling operations and maintain complete and accurate records for each element	**CONTINUOUS**	
2. Verify placement locations and plumbness, confirm element diameters, bell diameters (if applicable), lengths, embedment into bedrock (if applicable) and adequate end-bearing strata capacity. Record concrete or grout volumes	**CONTINUOUS**	
3. For concrete elements, perform tests and additional special inspections in accordance with Concrete section of this schedule	**CONTINUOUS**	
4. Determine capacities of test elements and conduct additional load tests if required.	**CONTINUOUS**	

END SECTION

[1] **CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

U. FIRE PROTECTION - SPRAYED FIRE-RESISTANT MATERIALS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

SPRAYED FIRE RESISTANT MATERIALS (SFRM) – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.15		
TASK	INSPECTION TYPE [1]	DESCRIPTION
1. Check substrate condition	PERIODIC	Prior to application, confirm that surfaces have been prepared according to the approved fire-resistant design and manufacturer's instructions. See 2024 IBC 1705.15.2 and 1705.15.3 for more details.
2. Verify material thickness	PERIODIC	Verify SFRM thickness according to 2024 IBC 1705.15.4
3. Verify material density	PERIODIC	Verify SFRM density according to 2024 IBC 1705.15.5
4. Verify bond strength	PERIODIC	Verify bond strength of cured SFRM according to 2024 IBC 1705.15.6
5. Inspect the condition of finished application	PERIODIC	

END SECTION

V. FIRE PROTECTION - INTUMESCENT FIRE-RESISTIVE MATERIALS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

INTUMESCENT FIRE-RESISTIVE MATERIALS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.16		
TASK	INSPECTION TYPE	DESCRIPTION
1. Inspect according to AWCI 12-B and the contract documents	PERIODIC	Inspections shall be performed in accordance with AWCI 12-B, Standard Practice for the Testing and Inspection of Field Applied Thin Film Intumescent Fire-Resistive Materials. Inspections shall be performed during construction. Additional visual inspection shall be performed after rough installation and where applicable, prior to the concealment of electrical, automatic sprinkler, mechanical and plumbing systems.

END SECTION

W. FIRE PROTECTION – FIRE-RESISTANT PENETRATIONS AND JOINTS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

FIRE-RESISTANT PENETRATIONS AND JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.18		
TASK	INSPECTION TYPE	DESCRIPTION
1. Inspect of penetration firestop systems conducted in accordance with ASTM E2174.	PERIODIC	**[NOTE: This section applies to Risk Category III, IV, & V structures, or in fire areas containing Group R occupancies with an occupant load greater than 250. DOR may choose to uncheck this section where project is assigned to Risk Category I or II. Confirm Risk Category with licensed design professional in responsible charge]**
2. Inspecti of fire-resistant joint systems conducted in accordance with ASTM E2393	PERIODIC	

END SECTION

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[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

X. FIRE PROTECTION – SMOKE CONTROL

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

SMOKE CONTROL – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.19		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Verify device locations and perform leakage testing	PERIODIC	Perform during erection of ductwork and prior to concealment
2. Perform pressure difference testing, flow measurements and detection and control verification	PERIODIC	Perform prior to occupancy and after sufficient completion

END SECTION

Y. SEALANT – SEALING OF MASS TIMBER

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

SEALING OF MASS TIMBER – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.20		
TASK	INSPECTION TYPE	DESCRIPTION
1. Inspect sealants or adhesives	PERIODIC	Inspect sealants or adhesives where sealant and adhesive are required by 2024 IBC 703.7

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

Z. ARCHITECTURAL - EXTERIOR INSULATION AND FINISH SYSTEMS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

EXTERIOR INSULATION AND FINISH SYSTEMS (EIFS) – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.17		
TASK	INSPECTION TYPE[1]	DESCRIPTION
1. Water-resistive barrier coating applied over a sheathing substrate.	PERIODIC	Verify that water-resistive barrier coating complies with ASTM E2570. **[NOTE: Not required for EIFS applications installed over:** **a) a water-resistive barrier with a means of draining moisture to the exterior.** **b) masonry or concrete walls** **Uncheck this section in those cases]**

END SECTION

AA. ARCHITECTURAL – ARCHITECTURAL COMPONENTS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

ARCHITECTURAL COMPONENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.13.5, 1705.13.7 and TABLE 1705.13.7		
TASK	INSPECTION TYPE	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A, B, & C. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with the licensed design professional in responsible charge.]		
1. Erection and fastening of exterior cladding and interior and exterior veneer.	PERIODIC	Verify appropriate materials, fasteners and attachment at commencement of work and at completion. **[NOTE: Inspection not required if height is less than 30 feet or weight is less than 5psf]**
2. Interior and exterior non-load-bearing walls	PERIODIC	Verify appropriate materials, fasteners and attachment at commencement of work and at completion. **[NOTE: Inspection not required if interior non-load-bearing walls weigh less than 15psf]**
3. Access floors	PERIODIC	Verify that anchorage complies with approved construction documents.
4. Storage racks	PERIODIC	Steel storage racks and steel cantilevered storage racks that are 8 feet in height or greater shall be inspected for: ✓ Materials used, to verify compliance with one or more of the material test reports in accordance with the approved construction documents ✓ Fabricated storage rack elements ✓ Storage rack anchorage installation ✓ Completion of storage rack system Inspection of post-installed anchors shall comply with approved ICC-ES report. **[NOTE: Not required for racks less than 8 feet in height]**

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

BB. PLUMBING/MECHANICAL/ELECTRICAL - DESIGNATED SEISMIC SYSTEMS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☒

PLUMBING, MECHANICAL AND ELECTRICAL – DESIGNATED SEISMIC SYSTEMS 2024 IBC 1705.13.6		
TASK	INSPECTION TYPE[1]	DESCRIPTION
[NOTE: This section is not applicable to Seismic Design Categories A or B. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with licensed design professional in responsible charge.]		
1. Check anchorage of electrical equipment for emergency and standby power systems	PERIODIC	✓ Check for general conformance
2. Check anchorage of all other electrical equipment in Seismic Design Categories E and F only (See first page of this schedule for Seismic Design Category)	PERIODIC	✓ Check for general conformance
3. Check installation and anchorage of piping designed to carry hazardous materials and their associated mechanical units.	PERIODIC	✓ Check for general conformance
4. Check installation and anchorage of vibration isolation systems where the construction documents require a nominal clearance of ¼” or less between support framing and restraint.	PERIODIC	✓ Check for general conformance
5. Verify clearance between fire sprinkler piping and surrounding mechanical and electrical equipment, including ductwork, piping and their structural supports.	PERIODIC	✓ Check for minimum clearances noted in ASCE 7 13.2.3 or a nominal clearance of not less than 3 in.

END SECTION

[1] **PERIODIC**: Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

APPPENDIX A

Commentary on M12

Table 4 of TMS 602-22 does not have any Level 3 inspection requirements (for Risk Category IV structures). The leadership of the standard committee was contacted. They indicated that there are no requirements because veneers are covered under Part 4: Prescriptive Design Methods, and for prescriptive design there is never any Level 3 inspection requirement according to Table 3.1 of TMS 402-22. Although veneer is covered in Part 4, Chapter 13 of TMS 402-22, in Section 13.2.3, there is Engineered Design of Anchored Masonry Veneer and in Section 13.3.3, there is Engineered Design of Adhered Masonry Veneer. Engineered Design is covered in Part 3 of the standard, and for Part 3, Engineered Design, there are Level 3 inspection requirements in Table 3.1 of TMS 402-22. Upon further discussion, the leadership of the standard committee suggested to keep inspection Periodic for all Risk Category structures.

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511	(2007) Standard for Reduced-Pressure Principle Backflow Prevention Assembly

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH (FCCCHR)

FCCCHR List	(continuously updated) List of Approved Backflow Prevention Assemblies
FCCCHR Manual	(10th Edition) Manual of Cross-Connection Control

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 70	~~(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code~~(2023; ERTA 7 2023; TIA 23-15) National Electrical Code

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	~~(2024) Safety -- Safety and Health Requirements Manual~~ Safety -- Safety and Health Requirements Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

PL 93-523	(1974; A 1999) Safe Drinking Water Act

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1	(2016; Rev L; Change 2) Obstruction Marking and Lighting

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only~~submittals not having a "G" designation are for Contractor Quality Control or~~

~~Designer of Record approval~~. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G
Traffic Control Plan; G
Temporary Utility Connections Plan; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 BACKFLOW PREVENTERS CERTIFICATE

Certificate of Full Approval from FCCCHR List, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval will not be acceptable.

1.4.1 Backflow Tester Certificate

Certification issued by the local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with any company participating in any other phase of this Contract.

1.4.2 Backflow Prevention Training Certificate

Certification recognized by the local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current. Training and certification can be obtained from the following sources:

a. Kyuusuikouji Gijutsu Koshin Zaidan; https://www.kyuukou.or.jp/

b. Zenkoku Kensetsu Kenshu Center; https://www.jctc.jp/

c. Tester Training Course, Foundation for Cross-Connection Control and Hydraulic Research, University of Southern California; https://fccchr.usc.edu/tester.html

1.5 TYPHOON AND STORM PREPARATIONS

In order to minimize damage to public properties and prevent personal injury, the following actions shall be taken upon declaration of the typhoon conditions described below. Ensure that the construction sites are well-prepared for protection from the damages of heavy rain and strong winds. Develop and establish the necessary procedures to be taken for storm preparation. In Part 1 of the Accident Prevention Plan, provide emergency phone numbers and addresses where at least three Contractor officials may be reached and notified in the event that an immediate typhoon alert is declared.

1.5.1 Tropical Cyclone Conditions of Readiness (TCCOR)

a. TCCOR STORM WATCH: The winds are not forecast to exceed 50 knots (58 mph/26 m/s) sustained (the criteria for "destructive winds") but there still exists a probability of high winds due to the proximity of the storm. High winds may include gusts exceeding 50 knots and/or sustained winds meeting TCCOR 1 Caution criteria. The storm is also close enough to the area that a heightened alert status is necessary in order to rapidly establish elevated TCCOR conditions should the storm deviate from the forecast track. Personnel should follow Standard Operating Procedures for TCCOR Storm Watch and stay alert for any changes to TCCOR status.

When wind gusts of 30 knots (35 mph/16 m/s) or greater are forecast, the Contractor shall be required to perform general area cleanup and monitor TCCOR levels.

When a local storm warning has been issued, the Contractor competent person and/or SSHO shall determine whether it is necessary to implement manufacturer recommendations for securing the equipment to include Weight Handling Equipment (WHE).

Contractor reviews project site storm preparation plans and discuss plan of action with Contracting Officer's Representative.

b. TCCOR 4: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 72 hours.

When wind gusts of 30 knots (35 mph/16 m/s) or greater are forecast, the Contractor will be required to perform general area cleanup and monitor TCCOR levels.

Cranes shall not be operated when wind speeds at the site attain the maximum wind velocity based on the surface/load ratio recommendations of the manufacturer.

At winds greater than 20 mph (17 knots/9 m/s), the operator, rigger, lift supervisor, and SSHO shall cease all crane operations, evaluate conditions and determine if the lift can proceed with Contracting Officer's Representative acceptance.

All elevated work above 6 ft (1.8 m) shall be risk assessed by the Contractor's competent person and/or SSHO and discussed with the Contracting Officer's Representative for acceptance.

c. TCCOR 3: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 48 hours.

(1) Clean and remove all loose debris and waste including scrap wood and metal, empty barrels, and construction form materials no longer required on the job site, to a safe area for proper disposition.

(2) Secure, tie down, and anchor construction field office and storage facilities, scaffolding, concrete forms and supports, doors,

windows, opening covers, stored lumber and other materials, mechanized construction equipment, temporary power lines and supports, and other items that may be blown away or that might cause injury or damage.

(3) Inspect all excavation and trenching work in process, and provide necessary temporary drainage and proper protection and shoring for excavation sides and openings, in order to prevent damage to public roads and facilities by slides or flooding. Accumulation of water in the excavation of structure foundation work shall be controlled and minimized.

(4) Inspect scaffolding or work platforms for loose materials, planking, etc. that could become airborne projectile hazards and secure scaffold netting, tarps, etc. from wind loads.

d. TCCOR 2: Winds of 50 knots (58 mph/ 26 m/s) sustained or greater are possible within 24 hours.

Work required at remote areas such as off-shore facilities or high elevations shall cease immediately and the workers shall be evacuated to a safe area. During TCCOR 2, the Contractor shall continue the actions described in TCCOR 3 above and the construction site shall be inspected for storm preparation by the Contracting Officer's Representative. The Contractor shall request an inspection by calling the Contracting Officer's Representative at the appropriate Japan Engineer District field office.

e. TCCOR 1: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 12 hours.

All work shall cease immediately and the Contractor's representative shall insure that all necessary storm preparations, including the items listed below, are completed.

(1) All electrical circuits and equipment including temporary power lines are cut off and secured against unauthorized use.

(2) Gas cylinders, hot work equipment, and flammable materials properly stored at a safe area.

(3) No igniting source is present.

(4) All workers have been evacuated from the construction site. When TCCOR 1 is declared without the normal progression through TCCOR 3 and/or 2, the Contractor shall take the actions listed in TCCOR 3 above, and also follow the procedures described herein.

f. TCCOR 1 Caution: Winds of 35 to 49 knots (39/17 to 56/25 mph/ m/s) sustained are occurring.

Construction Sites Secured.

g. TCCOR 1 Emergency: Winds of 50 knots (58 mph/26 m/s) sustained or greater are occurring.

Construction Sites Secured.

h. TCCOR 1 Recovery: Winds of 50 knots (58 mph/26 m/s) sustained or greater are no longer forecast to occur. Strong winds may still exist.

Installation personnel begin storm damage assessments and clean up.

i. TCCOR Storm Clear: The storm is over and not forecast to return. Used to inform personnel that the threat of the storm is over, but the storm dama~~m~~ge could still present a danger.

Contractor may resume normal activities. The construction site shall be investigated for all damage caused by the typhoon or high winds, and the result of the investigation shall be furnished in verbal or written form to the Contracting Officer's Representative as soon as practicable.

Complete and submit PODWP 134 Typhoon Damage Report for storm damages or negative report of damage to Emergency Management, Construction Division, and Safety Office.

j. TCCOR All Clear: The storm is over and not forecast to return, and recovery efforts are complete.

Resume normal activities.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Immediately upon beginning of work, provide a weatherproof glass-covered bulletin board not less than 915 by 1220 mm in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, and other information approved by the Contracting Officer. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

2.1.2 Project and Safety Signs

Provide ~~[~~one (1)~~]~~ project sign and ~~[~~one (1)~~]~~ safety sign fabricated to size and design as shown in Attachment 01 50 00-A. The signs shall be rigidly formed and erected at location designated by the Contracting Officer prior to commencement of work. A blue-line drawing of different letter sizes and style shall be made available by the Contracting Officer upon request. Prior to painting the sign, submit for approval a sketch, similar to diagram shown in Attachment 01 50 00-A, indicating the actual information for the project. The sketch shall indicate lettering dimensions and locations. The Corps of Engineers castle logo shall be painted red per Attachment 01 50 00-A. Correct the data required by the safety sign daily, with light-colored metallic or non-metallic numerals. No separate payment shall be made for the sign, and all costs in connection therewith shall be included in the Contract price for the project. The sign shall be subject to the approval of the Contracting Officer. Erect signs within 15 days after receipt of the Notice to Proceed. Correct the data required by the safety sign daily, with light-colored metallic or non-metallic numerals. Upon completion of work under this Contract, the signs shall be removed from the job site and shall remain the property of the Contractor.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract. Construct with suitable grades and widths; sharp curves, blind corners, and dangerous cross traffic are to be avoided. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and hauling roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic barricades will be required. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.2.3 Temporary Wiring

Provide temporary wiring in accordance with NFPA 241 and NFPA 70. Include frequent inspection of all equipment and apparatus.

2.2.4 Backflow Preventers

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 65 kg flanged, mounted gate valve and strainer, stainless steel or bronze, internal parts. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Contractor employees will park privately owned vehicles in an area designated by the Contracting Officer. This area shall be within reasonable walking distance of the construction site. Contractor employee parking must not interfere with existing and established parking requirements of the government installation.

3.2 TEMPORARY BULLETIN BOARD

Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

3.3 AVAILABILITY AND USE OF UTILITY SERVICES

3.3.1 Payment for Utility Services

The Government shall make all reasonably required utilities available to the Contractor~~[~~, without charge,~~]~~ from existing outlets and supplies, as specified in the Contract. Carefully conserve any utilities.

3.3.2 Meters and Temporary Connections

At the Contractor's expense and in a manner satisfactory to the Contracting Officer, provide and maintain necessary temporary connections, distribution lines, and meter bases required to measure the amount of each utility used. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards. Submit a Temporary Utility Connections Plan for approval to the Contracting Officer, fifteen (15) working days before making temporary utility connections.

~~[~~

~~The Government shall~~

~~[Furnish the water meter]~~
~~[Furnish the electric meter]~~
~~[Furnish the backflow preventer]~~
~~[Install the water meter]~~
~~[Install the electric meter]~~
~~[Install the backflow preventer]~~
~~][~~

The Contractor shall

~~[~~ Furnish the water meter~~]~~
~~[~~ Furnish the electric meter~~]~~
~~[~~ Furnish the backflow preventer~~]~~
~~[~~ Install the water meter~~]~~
~~[~~ Install the electric meter~~]~~
~~[~~ Install the backflow preventer~~]~~
~~]~~

~~[3.3.3 Advance Deposit~~

~~An advance deposit for utilities consisting of an estimated month's usage or a minimum of [6,000] JPY shall be required. Adjustments to the monthly bills shall be settled at the end of the fiscal year ending on 30 September. Services to be rendered for the next fiscal year, beginning 1 October, shall require a new deposit. Notification of the due date for the new deposit shall be mailed to the Contractor prior to the end of the previous fiscal year.~~

~~]~~3.3.3 Final Meter Reading

Before completion of the work and final acceptance of the work by the Government, notify the Contracting Officer, in writing, five working days before termination is desired. A final reading shall be taken by the Government. Afterwards, remove all temporary distribution lines, meter bases, meters, and associated appurtenances.~~[Pay all outstanding utility bills before final acceptance of the work by the Government.]~~

3.3.4 Sanitation

Provide and properly maintain temporary toilet facility for use of Contractor personnel. Government toilet facilities are not available to Contractor personnel. Toilet facility shall be constructed as required in EM 385-1-1 and by local installation regulations, where directed. Include provisions for pest control and elimination of odors. Upon completion of the Contract, dispose facility outside the limits of Government-controlled land and at Contractor's expense.

3.3.5 Electricity

The Contractor shall be responsible for all temporary connections required, as well as the removal of these temporary connections upon completion of the project. The Contractor's plan for temporary connections shall be submitted to the Contracting Officer Representative for approval prior to making any temporary connections.

Electrical power system frequencies for different installations are as follows:

a. 50 Hertz (Hz): Camp Zama, Sagami Depot, Sagamihara Housing Area, Yokota Air Base, Commander Fleet Activities Yokosuka, Naval Air Facility Atsugi, Misawa Air Base, Camp Fuji, and Naval Support Facility Kamiseya.

~~b. 60 Hertz (Hz): Marine Corps Air Station Iwakuni, Commander Fleet Activities Sasebo, and all of Okinawa.~~

3.3.6 Telephone and Data

Make arrangements and pay all costs for telephone facilities and data connections.

3.3.7 Obstruction Lighting of Cranes

Provide a minimum of 2 aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 30 meter above ground level. Light construction and installation shall comply with FAA AC 70/7460-1. Lights shall be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

3.3.8 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

a. Conduct operations in a manner that will not close any thoroughfare or interfere in any way with traffic except with written permission of the Contracting Officer at least 30 calendar days prior to the proposed modification date, and provide a Traffic Control Plan detailing the proposed controls to traffic movement for approval. The plan must be in accordance with installation regulations. Coordinate with installation officials. Contractor may move oversized and slow-moving vehicles to the worksite once the requirements of the highway authority have been met. Include a schedule of planned road closures in the
Initial Project Schedule.

b. ~~Conduct work so [there is no obstruction of traffic, and maintain traffic on the full roadway width at all times][as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times].~~ Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.

c. At the Contractor's expense, provide, erect, and maintain: lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage overhead protection authority having jurisdiction.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment, the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of any damage to roads caused by construction operations.

3.4.3 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Treat dust abatement on access roads with applications of calcium chloride, water sprinklers, or similar methods or treatment.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

3.5.1 Safety

Protect the integrity of any installed safety systems or personnel safety devices. If entrance into systems serving safety devices is required, the Contractor must obtain prior approval from the Contracting Officer. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.2 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

3.5.3 Temporary Project Safety and Storage Area Fencing

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety and storage area fencing to control access by unauthorized people. Fencing shall be able to restrain a force of at least 114 kg. Safety and storage area fencing shall be ~~[1.8 m high wire mesh fence with lockable gates and visibility screening (similar to tennis court screening)][~~2.5 to 3.0m high sheet metal fence (similar to construction fencing used throughout Japan)~~] [located around the perimeter of project work areas and the perimeter of Contractor lay-down areas]~~.~~[Use plastic panels or perforated steel sheets to improve visibility for vehicular traffic when erecting fencing within 3 meters of street corners.]~~ The intent is to block (screen) public view of the construction, and prevent unauthorized access to the site. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Maintain the safety and storage area fencing during the life of the Contract and, upon completion and acceptance of the work, shall become the property of the Contractor and be removed from the work site.

3.5.4 Storage Area

Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on any given day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

3.5.5 Supplemental Storage Area

Upon Contractor's request, the Contracting Officer will designate another or supplemental area for the Contractor's use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Fencing of materials or equipment will be required at this site. The Contractor is responsible for cleanliness and orderliness of the area used and for the security of any material or equipment stored in this area. Utilities will not be provided to this area by the Government.

3.5.6 Appearance of Trailers

Trailers utilized by the Contractor for administrative or material storage purposes must present a clean and neat exterior appearance and be in a state of good repair. Trailers which, in the opinion of the Contracting Officer, require exterior painting or maintenance will not be allowed on installation property.

3.5.7 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Grassed or unpaved areas, which are not established roadways, will be covered with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways, should the Contractor elect to traverse them with construction equipment or other vehicles; gravel gradation will be at the Contractor's discretion. Mow and maintain grass to a maximum height of 100 mm located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers will be edged or trimmed neatly.

3.5.8 Security Provisions

Provide adequate outside security lighting at the Contractor's temporary facilities. The Contractor will be responsible for the security of its own equipment.

3.5.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 GOVERNMENT FIELD OFFICE

The Contractor shall be responsible for providing and maintaining, in good condition, an office for the sole use of Government personnel, in a structure physically separated from the Contractor's site offices. Facility shall be furnished at a location designated and approved by the Contracting Officer, and available for use concurrent with availability of Contractor's site office or within 15 calendar days of Notice to Proceed, whichever occurs first. The office facility, including all furniture, equipment, and connections, shall remain the property of the Contractor and, upon completion of work under this Contract, shall be removed from the site by the Contractor and the site returned as nearly as practical to the original conditions.

Contractor shall provide two designated USACE parking spaces as close to the Government Field Office as practicable. The spaces shall not exceed 50 meters from the field office without written approval from the Contracting Officer.

3.6.1 Facility Requirements

Provide the Government an office with a minimum of ~~[~~67 square meters ~~(720 square feet)]~~ in floor area, windows, a door with lockset individually keyed with two keys, and a battery operated smoke detector in each room. Provide batteries for the smoke detectors, and replace batteries in each smoke detector at least twice annually.

The walls and ceiling of the office shall be insulated. All windows shall be screened. Windows shall be openable and be securely fastened from the inside. Glass panels in windows shall be protected by bars or heavy mesh screens to prevent easy access to the facility through these panels. The office shall contain a minimum of eight (8) standard electrical outlets meeting local requirements and additionally provide for delivery of a minimum 3,000 Watts.

The office shall contain an integral toilet facility with western-type (with a push-type flush button) toilet fixture (bowl), which shall hold water in it at all times, and a wash basin.

The office facility shall contain a kitchenette type facility with a countertop/work area with a minimum surface area of 55 cm (W) by 150 cm (L). A minimum of two electrical outlets (local current) shall be located adjacent to the countertop.

Hot water for all faucets shall be provided.

3.6.2 Furniture and Equipment

Furnish the office with the following furniture and equipment. All furnishing shall be new and manufactured by a recognized supplier of office equipment. All items shall be assembled, installed, and maintained by the Contractor.

a. ~~Two~~Three desks (minimum 80 cm by 150 cm) double pedestal with lockable drawers.

b. ~~Two~~Three desk chairs with arms: swivel tilt, hydraulic height-adjustable, medium back, fabric covered seat cushion, castored.

c. One conference table (minimum 120 cm by 250 cm), with a minimum of six new arm-chairs, castored.

d. Three file cabinets, capable of holding US letter-size documents, with a minimum five drawers each.

e. Two bookcases with a minimum of five shelves each. Minimum dimensions are 120 cm (W) by 200 cm (H) by 30 cm (D).

f. One clock: wall hung, quartz, 12 dial, battery powered.

g. Plan racks capable of holding two full-size set of Contract plans.

h. One dry-erase board with a minimum dimension of 1.5 m by 1.0 m. Dry erase markers (minimum four different colors) shall be provided and replaced as depleted.

i. One microwave (minimum 0.028 cubic meter).

j. One refrigerator (minimum 0.70 cubic meter).

3.6.3 Lighting Requirements

The office facility shall include suitable electric lighting to meet the minimum lighting requirements (luminance) of administrative areas as indicated in EM 385-1-1.

3.6.4 Heating/Cooling Requirements

Provide heating and cooling capable of maintaining an interior temperature of between 16 degrees Celsius and 25 degrees Celsius in all seasons. If applicable, use a fluorocarbon gas refrigerant with an Ozone Depletion Potential (ODP) of less than or equal to 0.05.

3.6.5 Drinking Water

Provide potable drinking water for the facility that meets or exceeds PL 93-523.

3.6.6 Data Connection

Provide network connectivity (ADSL, Cable or Fiber), one USACE-dedicated router and one switching hub, eight-5 m network cables, and eight power strips.

3.6.7 Maintenance

Provide maintenance and janitorial services, to include:

a. Emptying of trash at least twice a week.

b. Cleaning of the office, including the toilet facilities, weekly.

c. Replenishment of liquid soap, toilet paper, and paper towels for daily use. Adequate quantity of these supplies shall be provided on-site and replaced as needed.

3.6.8 Security Requirements

The physical security of the Government personnel, property, and field office shall be the responsibility of the Contractor throughout the duration of the Contract.

If the project site is outside of the controlled area of an existing military installation, the Government field office shall have 30 meters of standoff from a fence enclosing the office and all parking or roads. Additional fencing, earth berms, concrete barriers, controlled access points, and similar security features shall be provided to limit access to the site and provide additional building standoff or security.

3.7 PLANT COMMUNICATION

Whenever the Contractor has the individual elements of its plant so located that operation by normal voice between these elements is not satisfactory, the Contractor must install a satisfactory means of communication, such as telephone or other suitable devices and made available for use by Government personnel.

3.8 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store any salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.9 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and any other temporary products from the site. All tools, equipment, and materials not the property of the Government shall be removed from the premises and properly disposed of off Government property. After removal of trailers, materials, and equipment from within the fenced area, remove the temporary project safety and storage area fencing. Restore areas used by the Contractor for the storage of equipment or material, or other use to the original or better condition.

Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary. Upon completion of the work, notify the Contracting Officer for final inspection.

-- End of Section --

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS

01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS	(Apr 202~~2~~4) Japan Environmental Governing Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 261	Identification and Listing of Hazardous Waste

1.2 DEFINITIONS

In some cases, definitions are given only for illustrative purposes. Prevailing JEGS definitions shall be used for environmental compliance requirements.

1.2.1 Aboveground Storage Tank

A portable or fixed POL aboveground storage container as defined in the JEGS with a capacity greater than 55 gallons.

1.2.2 Asbestos Containing Material (ACM)

Any material containing more than one tenth of one percent (0.1%) asbestos by weight.

1.2.3 Bulky Waste

Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid waste collection, processing, or disposal methods.

1.2.4 Chemical Wastes

Salts, acids, alkalis, herbicides, pesticides, organic chemicals, and spent products, which serve no purpose.

1.2.5 Class I and II Ozone Depleting Substance (ODS)

Class I and II ODS are listed in the JEGS.

1.2.6 Construction and Demolition Waste

The waste building materials, packaging, and rubble resulting from construction, remodeling, repair and demolition operations on pavements, houses, commercial buildings, and other structures.

1.2.7 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.8 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.9 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.10 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.11 Food Waste

Organic residues generated by the handling, storage, sale, preparation, cooking, and serving of foods (commonly called garbage).

1.2.12 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with the JEGS. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with the JEGS.

1.2.13 Hazardous Materials

Hazardous material is any material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in the JEGS, "Typical Hazardous Materials

Characteristics," or the material is listed in the "List of Hazardous Waste/Substances/Materials". Munitions are excluded.

1.2.14 Hazardous Substances

Any substance having the potential to do serious harm to human health or the environment if spilled or released in reportable quantity. A list of these substances and the corresponding reportable quantities is contained in the JEGS, "Characteristics of Hazardous Waste and Lists of Hazardous Waste and Hazardous Material."

1.2.15 Hazardous Waste

Hazardous Waste is discarded material that may be solid, semi-solid, liquid, or contained gas, and either exhibits a characteristic of a hazardous waste as defined in the JEGS. Excluded from this definition are domestic sewage sludge, household wastes, and medical wastes.

1.2.16 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan.

1.2.17 Installation Pest Management Consultant

Installation Pest Management Consultant (IPMC) is the professional DoD pest management personnel located at component headquarters, field operating agencies, major commands, facilities engineering field divisions or activities, or area support activities who provide technical and management guidance for the conduct of installation pest management operations. Some pest management consultants may be designated by their component as certifying officials.

1.2.18 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, or discharge into defined drainage areas (includes drainage ditches, streams, rivers, ocean, etc.) must occur. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.19 Lead-based Paint (LBP)

Paint or other surface coatings that contain lead greater than or equal to 1.0 milligram per square centimeter, or 0.5 percent by weight, or 5,000 ppm by weight.

1.2.20 Oily Wastes

Oily wastes are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from those POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements..

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.21 Pesticide

Pesticide is any substance or mixture of substances, including biological control agents that may prevent, destroy, repel, or mitigate pests.

1.2.22 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.23 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.24 Project Pesticide Coordinator

The Project Pesticide Coordinator (PPC) is an individual who resides at a Civil Works Project office and who is responsible overseeing of pesticide application on project grounds.

1.2.25 Petroleum, Oil, and Lubricants

Refined petroleum, oils, and lubricants, including, but not limited to, petroleum, fuel, lubricant oils, synthetic oils, mineral oils, animal fats, vegetable oil, sludge, and POL mixed with wastes other than dredged spoil.

1.2.26 Polychlorinated Biphenyl (PCB)

Any PCB article, PCB article container, PCB container, or PCB equipment that deliberately or unintentionally contains or has as a part of it any detectable concentration of PCB.

1.2.27 Regulated Waste

Regulated waste are solid wastes that have specific additional Federal, GOJ national, and prefectural controls for handling, storage, or disposal.

1.2.28 Rubbish

A general term for solid waste, excluding food wastes and ashes, taken from residences, commercial establishments, and institutions.

1.2.29 Sanitary Waste

a. Sewage: Wastes characterized as domestic sanitary sewage.

b. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.30 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.31 Solid Waste

Solid waste is garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community activities. It does not include solids or dissolved material in domestic sewage or their significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants. Types of solid waste typically generated at construction sites may include:

1.2.31.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.31.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.31.3 Material Not Regulated as Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges;

regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.31.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, characteristic of a hazardous waste as defined in the JEGS or is listed as a hazardous waste in the JEGS. Excluded from this definition is medical wastes.

1.2.31.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable if sold to a scrap metal company.

1.2.31.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.31.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.31.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.32 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks, or waters of Japan. Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.33 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant.

1.2.33.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.34 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.3 PROHIBITED PRODUCTS

The following items are forbidden for use by the JEGS or other criteria. Details of each are included in the text of each chapter of the Environmental Protection Plan (EPP) described in this section.

a. Asbestos Containing Materials (ACM).

b. Lead-Containing Paint. Paint containing greater than 0.009 percent lead by weight.

c. Polychlorinated Biphenyls (PCBs). Materials containing PCBs greater than 0.5mg/kg shall not be used.

d. Class I Ozone Depleting Substances (ODS). Class 1 ODS listed in the JEGS shall not be used.

e. Lead Drinking-Water Pipes, Solders, Flux, and Fittings.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Environmental Protection Plan; G

Environmental Manager Qualifications; G

Written Assessment Of Friable Asbestos Disturbance

SD-06 Test Reports

Solid Waste Management Report; G

Nonhazardous Solid Waste Diversion Report

SD-07 Certificates

Asbestos Certification

Lead Certification

SD-11 Closeout Submittals

Disposal Documentation for Hazardous and Regulated Waste; G

Solid Waste Management Report; G

Hazardous Waste/Debris Management; G

Environmental Records Binder

1.5 PAYMENT

No separate payment shall be made for work covered under this section. Payment of fees associated with environmental permits, application, and/or notices obtained by the Contractor, and payment of all fines/fees for violation or non-compliance with GOJ, Federal, and local laws and regulations are the Contractor's responsibility. All costs associated with this section shall be included in the Contract price.

1.6 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with applicable Environmental Laws may be required. Laboratory analyses necessary to implement the JEGS shall be conducted in a laboratory certified by a U.S. or GOJ regulatory authority for the applicable test method, and where required by law, the laboratories shall be certified. In the absence of a certified laboratory, contact the Contracting Officer for further guidance.

Contractor shall be responsible to ensure that subcontractors comply with all environmental protection requirements of this section.

The Contractor shall record any problems in complying with laws, regulations, permit requirements, ordinances, and corrective actions taken. The Contractor shall immediately inform the Contracting Officer of any environmental problems.

1.6.1 Conformance with the Environmental Management System

Perform work under this Contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the Contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.7 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the special environmental requirements listed here and attached at the end of this section.

1.7.1 Asbestos Prohibition and Certification

a. Materials or products containing more than one-tenth of one percent (0.1 percent) by total weight, of the material or product, of asbestos shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain asbestos in excess of one-tenth of one percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if asbestos (exceeding 0.1 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform asbestos abatement and restoration work, as required, at the Contractor's sole cost. Asbestos abatement work (removal and disposal of asbestos-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

   Exception: Where suitable asbestos-free substitutes do not exist for a material or product, the Contractor may use a material or product containing asbestos in the excess of 0.1 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of asbestos present in the material or product. When available, laboratory analysis of the material or product for asbestos content shall be included with the submittal.

b. The Government may conduct asbestos testing on suspected asbestos-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of asbestos exceeding 0.1 percent by weight. In the event test results indicate 0.1 percent or less asbestos content or complete absence of asbestos, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.

c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are asbestos-free and do not contain asbestos in excess of 0.1 percent by weight, as applicable. However, when presence of asbestos is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Asbestos certification shall be required for the items applicable to this project only.

(1) Vinyl sheet/vinyl tile flooring, including accessories and adhesives.

(2) Insulation materials, including facing.

(3) Gaskets for piping and duct work.

(4) Acoustical tiles.

(5) Firestopping materials.

(6) Fireproofing materials.

(7) Special coating, including factory applied coatings, on sheet metal roofing and siding.

(8) Wallboard for all interior and exterior applications, including joint compounds.

(9) Adhesives (other than item 1) used in the project.

(10) Tape materials used in the project.

(11) Roofing and siding, nonmetallic.

(12) Felt materials and cushion materials.

(13) Pre-mixed mortars, grouts, leveling compounds, fillers, and other cementitious materials.

(14) Caulking and sealing materials.

d. All submittals shall be accompanied by a certification from the manufacturer of the material or product that the material or product is asbestos-free; or shall set forth, in specific detail, the amount of asbestos present in the material or product. Documentary evidence of laboratory analysis of the material or product for asbestos content, conducted by a qualified independent testing laboratory, shall be included with the submittal.

1.7.2 Lead Prohibition and Certification

a. Paint or product coating containing more than 0.009 percent) by total weight of lead shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain lead in excess of 0.009 percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if lead (exceeding 0.009 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform lead abatement and restoration work, as required, at the Contractor's sole cost. Lead abatement work (removal and disposal of lead-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable lead-free substitutes do not exist for a paint or product coating, the Contractor may use a material or product containing lead in the excess of 0.009 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of lead present in the material or product. When available, laboratory analysis of the material or product for lead content shall be included with the submittal.

b. The Government may conduct lead testing on suspected lead-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of lead exceeding 0.009 percent by weight. In the event test results indicate 0.009 percent or less lead content or complete absence of lead, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.

c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are lead-free and do not contain lead in excess of 0.009 percent by weight, as applicable. However, when presence of lead is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Lead certification shall be required for the items applicable to this project only.

1. Paints and Coatings

2. Any product or material with a factory applied coating

1.7.3 Class I and Class II Ozone Depleting Chemicals (ODC) or Substances (ODS)

Class I and II Ozone Depleting Substances listed in the JEGS are prohibited from being used. Contactor must provide certifications that materials utilized do not contain Class I and Class II ODC/ODS.

1.7.4 Polychlorinated Biphenyls (PCB)

Materials (ballasts, capacitors, transformers, dielectric fluid, switches, etc.) that contain PCBs are prohibited. Contractor must provide certifications that materials utilized do not contain PCBs, in accordance with the JEGS.

1.7.5 Hazardous Material Survey

The Contractor shall review the Hazardous Materials Inspection report - Army Family Housing Renovate Tower B743 Camp Zama, Japan Survey by ESC dated January 2025, attached at the end of this section, to familiarize themselves with the materials that have been sampled and tested for this project. The Contractor shall utilize the information contained within the report to develop their work and compliance plans with regards to Hazardous Materials. During construction, should potentially hazardous material be discovered, which has not been previously tested, the Contractor shall take action to assure that the untested material is not disturbed and contact the Contracting Officer. The quantities provided in the report represents a rough order of magnitude estimate of materials and shall not be used for bidding purposes as conditions may have changed from the date of the survey and the present existence of the material(s).

1.8 QUALITY ASSURANCE

1.8.1 Regulatory Notifications

Provide regulatory notification requirements in accordance with GOJ national and prefectural laws and regulations and installation requirements.. Include copies of regulatory notifications in the approved EPP prior to commencement of work activities translated to English.

1.8.2 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP). Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

Permits, licenses, or other forms of official approvals are not required by DoD activities and installations. Permits, licenses, or other forms of official approvals may, however, be required under GOJ law for certain contracted activities. When required, all such permits, licenses and other forms of official approval shall be obtained by the Contractor from the appropriate GOJ authorities and included in the Environmental Protection Plan. DoD Components shall assist Contractors when they are applying for a required permit, license or other form of official approval by providing necessary information only.

1.8.3 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for Environmental Quality Control and coordinating Contractor compliance with the JEGS, Federal, GOJ national and prefectural laws and regulations, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in JEGS requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Insert Environmental Manager Qualifications into the Environmental Records Binder to indicate the training and past experience which meets the requirements of this position as described in this section.

1.8.4 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the Contract meeting applicable JEGS requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations and installation requirements.

Train personnel to meet the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations, and Installation requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of Japan, and endangered species and their habitat that are known to be in the area.

1.8.5 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the JEGS, applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the Contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.9 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 15 days after receiving notice to proceed. Commencement of work will not begin until the Environmental Protection Plan has been approved.

Revise the EPP throughout the project to include any reporting requirements, changes in site conditions, or Contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of the JEGS, and applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version of the EPP onsite.

The EPP includes, but is not limited to, the following elements as they apply to this project:

1.9.1 General Overview and Purpose

1.9.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as stormwater pollution prevention plan, solid waste management plan, traffic control plan Non-Hazardous Solid Waste Disposal Plan.

Include a list of applicable Federal, GOJ, JEGS, prefectural laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the Contractor's proposed operations and requirements imposed by those laws, regulations, and permits. Whenever there is conflict between Federal, GOJ, JEGS, or prefectural laws, regulations, and permit requirements, the strictest applicable rule applies.

1.9.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

Provide the name, telephone number, and address of a Primary and Alternate Environmental Representative.

1.9.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

Provide schedule of digging or trenching actions to include date/time, location, purpose, method and depth.

1.9.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.9.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.9.2 General Site Information

1.9.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.9.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

1.9.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this Contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.9.3 Management of Natural Resources

a. Land resources

b. Tree protection

c. Replacement of damaged landscape features

d. Temporary construction

e. Stream crossings

f. Fish and wildlife resources

g. Wetland areas

h. Waters of Japan

1.9.4 Protection of Historical and Archaeological Resources

a. Objectives

b. Methods

1.9.5 Stormwater Management and Control

a. Ground cover

b. Erodible soils

c. Temporary measures

(1) Structural Practices

(2) Temporary and permanent stabilization

d. Effective selection, implementation and maintenance of Best Management Practices (BMPs).

e. Erosion Control Plan. Prepared in accordance with the JEGS.

f. Storm Water Pollution Prevention Plan. Prepared in accordance with paragraph Stormwater Pollution Prevention Plan and the JEGS (if required)

1.9.6 Hazardous Material List

Prepared in accordance with paragraph Hazardous Material and the JEGS (if required). Safety Data Sheets (SDS): SDS in English for all Hazardous materials to be used in accordance with the JEGS (if required).

List shall include all pesticides and herbicides that will be brought onto the Installation.

1.9.7 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste. Control and disposal of hazardous waste.

Include a Hazardous Waste Management and Disposal Plan prepared in accordance with paragraph Control and Management of Hazardous Waste and the JEGS (if required). This item consists of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan. As a minimum, include the following:

a. List of the types of hazardous wastes expected to be generated

b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated

c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications

d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)

e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)

f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions, applicable Federal, GOJ and prefectural laws and regulations, JEGS, and Installation Hazardous Waste Management Plan. .

g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar

h. Used oil management procedures in accordance with the JEGS; Hazardous waste minimization procedures, and Installation Hazardous Waste Management Plan.

i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained.

j. Procedures to be employed to ensure required employee training records are maintained.

k. Hazardous Waste Disposal local permits or licenses for hazardous waste disposal in accordance with the JEGS (if required).

l. Hazardous Waste Disposal Statement of Agreement. From a treatment, storage, or disposal (TSD) facility that will accept the waste from the Contractor and also a statement from a certified hazardous waste transporter who will transport the waste to the TSD facility in accordance with the JEGS (if required).

1.9.8 Solid Waste Management Plan

Prepared in accordance with the Waste Management Plan requirements in paragraph Section 01 74 19 Solid Waste Management Plan and the JEGS.

1.9.9 Site Specific Spill Prevention Plan

Prepared in accordance with paragraph SPILLS and the JEGS (if required).

1.9.10 Prevention of Releases to the Environment

a. Procedures to prevent releases to the environment

b. Notifications in the event of a release to the environment

1.9.11 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.9.12 Air Quality Compliance

1.9.12.1 Haul Route

Include the truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways as a part of the EPP. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.9.12.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may generate air emissions and require compliance with the JEGS or GOJ national or prefectural laws and regulations. Determine requirements based on any current JEGS and installation requirements and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager) in the EPP. Air emissions must comply with the JEGS.

1.9.12.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced in the EPP. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation.

1.9.12.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants and ozone depleting substances comply with the JEGS and installation requirements. Technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Japanese equivalent. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.9.12.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.9.12.6 Monitoring

For the protection of public health, monitor and control contaminant emissions to the air from Hazardous, Toxic, and Radioactive Waste remedial action area sources to minimize short-term risks that might be posed to the community during implementation of the remedial alternative in accordance with the following.

a. Perimeter Air Contaminant of Concern Asbestos.

b. Time Averaged Perimeter Action Levels 0.1f/cc.

Concentration	
Time	

c. Perimeter Sampling/Monitoring Locationnegative air machice exhaust.

d. Monitoring Instruments/Sampling and Analysis Methods PCM.

e. Staffing CP.

1.9.12.7 Compliant Materials

Provide the Government a list of and SDSs for all hazardous materials proposed for use on site. Materials must be compliant with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.10 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7.

A copy of all Licenses and Permits shall be included in the Environmental Protection Plan.

1.11 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.12 SOLID WASTE MANAGEMENT PERMIT

Include as a part of the Environmental Protection Plan (EPP) written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.12.1 Solid Waste Management Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

PART 2 PRODUCTS

2.1 GREEN PROCUREMENT

The Contractor is required to buy items using recovered, or bio-based, materials listed at the following website: http://www.epa.gov/epp/pubs/products/index.htm provided, however, the cost of buying and transporting these materials does not exceed the cost of buying the listed recovered, or bio-based, materials, and providing that these materials are competitively available and available within reasonable period. A reasonable period is the time required to buy, ship, receive, and use the materials that does not exceed the time allowed for completion of a delivery order. Recovered and bio-based materials include items listed by the EPA in the following categories:

a. Bio-based.

b. Construction Products.

c. Non-Paper Office Products.

d. Miscellaneous Products.

e. Transportation Products.

f. Vehicular Products.

g. Park and Recreation Products.

h. Landscape Products.

i. Paper and Paper Products.

The Contractor shall provide a Recovered Materials Certification and estimate of Percentage of Recovered Material Content for EPA designated items. 2.1.1 Recycled Paper The Contractor shall use, as a minimum, paper that is Processed Chlorine Free (PCF) and contains at least 30 percent recycled content. All printed or copied products shall be double-sided and on recycled paper.

2.1.2 Energy Star and Energy Efficient Products

All applicable products delivered, used in the performance of the work, furnished for Government use, or specified in the design of a building or work shall be Energy Star and Energy Efficient Products.

2.1.3 Electronic Product Environmental Assessment Tool (EPEAT)

Purchased computers, notebook and monitors shall, as a minimum, meet the Electronic Product Environmental Assessment Tool (EPEAT) Bronze level of certification.

2.1.4 Leadership in Energy and Environmental Design (LEED) All new construction Contracts or those performing major renovations shall incorporate the Leadership in Energy and Environmental Design (LEED) System guidelines with Silver being the goal, or shall use Comprehensive Assessment System for Building Environmental Efficiency (CASBEE).

PART 3 EXECUTION

3.1 PROTECTION OF NATURAL RESOURCES

Minimize interference with, disturbance to, and damage to fish, wildlife, and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Office, regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Office or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

The Contractor shall take precautions to preserve all such resources as they existed at the time they were first pointed out. The Contractor shall provide and install protection for these resources and be responsible for their preservation during the life of the Contract. Environmental protection shall be conducted as follows:

a. Except in areas indicated on the drawing or specified to be cleared, the Contractor shall not remove, cut, deface, injure, or destroy land resources, including trees, shrubs, vines, grasses, topsoil, and land forms without the Contracting Officer's permission. Vegetation resources, land forms and other landscape features indicated and defined on the drawings to be preserved shall be clearly identified by marking, fencing, or wrapping with boards, or any other approved techniques. Any anticipated vegetation disturbance needs to be coordinated with the Government before it occurs.

3.1.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified and permitted.

3.1.2 Vegetation

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor is responsible for any resultant damage.

Protect existing trees that are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. Coordinate with the Contracting Officer and Installation Environmental Office to determine appropriate action for trees and other landscape features scarred or damaged by equipment operations.

3.1.3 GOJ-Protected Species

GOJ-Protected species are typically found in undeveloped and unmaintained portions of the base. Projects that may affect protected species shall include mitigation measures to eliminate or minimize effects.

3.1.4 Streams

Stream crossings must allow movement of materials or equipment without violating water pollution control standards of the JEGS and applicable, Federal, and GOJ national and prefectural laws and regulations, and installation requirements. Construction of stream crossing structures must be in compliance with the JEGS, applicable Federal and GOJ national and prefectural laws and regulations, and installation requirements, including the Storm Water Pollution Prevention Plan.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to ford live streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer. A copy of all secured permits shall be included in the EPP.

3.1.5 Endangered Species

At any time an endangered or threatened species, flora or fauna, to include sea turtle nests are encountered, all activities shall stop and the Contracting Officer shall be notified.

3.1.6 Indigenous/Native Flora and Fauna

The Contractor shall place emphasis on the protection of habitats favorable to the reproduction and survival of indigenous flora and fauna.

The Contractor shall use indigenous flora for planting/sodding.

3.1.7 Invasive Species

Invasive Species are prohibited to be raised, planted, stored or possessed on DoD installations. The Contractor shall not bring in any invasive species to DoD installations.

3.2 STORMWATER

Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

3.2.1 Stormwater Pollution Prevention Plan

If applicable, submit a project-specific Stormwater Pollution Prevention Plan (SWPPP) to the Contracting Officer as a part of the EPP, for approval, prior to the commencement of work. The Contractor shall maintain an approved copy of the SWPPP at the construction on-site office, and continually update as regulations require, to reflect current site conditions. The Contractor may be required to install, inspect, maintain best management practices (BMPs), and submit storm water BMPs inspection reports and storm water pollution prevention plan inspection reports.

The SWPPP must meet the requirements of the JEGS and the Installation's SWPPP.

Include the following:

a. Identify potential sources of pollution which may be reasonably expected to affect the quality of storm water discharge from the site.

b. Comply with terms of the Installation SWPPP for stormwater discharges from construction activities. Prepare SWPPP in accordance with the JEGS and installation requirements.

c. Select applicable BMPs from the JEGS and/or the Installation SWPPP. Applicable Best Management Practices in the JEGS shall be incorporated into the site-specific SWPPP and implemented.

d. The SWPPP shall also address erosion and sediment control measures and stormwater management and control including, but not limited to ground cover, erodible soils, temporary measures - structural practices, temporary and permanent stabilization.

3.2.1.1 Inspection Reports

Insert "Inspection Reports" into the Environmental Records Binder in accordance with the JEGS and applicable GOJ or local laws and regulations.

3.2.2 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with the JEGS and applicable GOJ or local laws and regulations. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

3.2.2.1 Erosion Control

Prevent erosion by Geotextiles,. Stabilize slopes by sodding, or such combination of these methods necessary for effective erosion control. Use of hay bales is prohibited.

Any disturbed area with exposed soil that is not being worked on must be seeded or sodded no later than two weeks from the last disturbance regardless of the installation of any other erosion control measures in place. Remove any temporary measures after the area has been stabilized.

3.2.3 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.2.4 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2.5 Protection of Excavated Soil

During the temporary storing of excavated soil until off base disposal or backfill, the excavated soil/rocks shall be completely segregated with materials from other locations, and covered with vinyl sheets to prevent the soil from run-off by rain, and label indicates "Project Name", "Contract Number", "Contractor's Name", "POC phone number" shall be posted on the vinyl sheet.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing GOJ, Prefectural, installation water quality standards and designated uses of the surface water body. Comply with water quality standards and anti-degradation provisions. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Office. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is specifically permitted. Obtain authorization for any contaminated groundwater release in advance from the Installation Environmental Officer and the GOJ, Prefectural, installation, or local authority, as applicable. Discharge of hazardous substances will not be permitted under any circumstances.

3.3.2 Waters of Japan

Do not enter, disturb, destroy, or allow discharge of contaminants into waters of Japan.

3.3.3 Water Protection

Contractor shall prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

3.3.4 Sewage

Contractor is not authorized sewage holding tanks on base and must procure a porta-potty service Contract. The porta-potty Contract must include the correct removal of sewage and maintenance of the facilities.

3.3.5 Wastewater

If the project generates wastewater from rinsing tanks, dewatering sites, etc., contact the Contracting Officer on proper disposal.

3.4 PROTECTION OF CULTURAL RESOURCES

3.4.1 Archaeological Resources

If, during excavation or other construction activities, any previously unidentified or unanticipated historical, archaeological, and cultural resources are discovered or found, activities that may damage or alter such resources will be suspended. Resources covered by this paragraph include, but are not limited to: any human skeletal remains or burials; artifacts; shell, midden, bone, charcoal, or other deposits; rock or coral alignments, pavings, wall, or other constructed features; and any indication of agricultural or other human activities. Upon such discovery or find, immediately notify the Installation Cultural Resources Manager via the Contracting Officer so that the appropriate authorities may be notified and a determination made as to their significance and what, if any, special disposition of the finds should be made. Cease all activities that may result in impact to or the destruction of these resources. Secure the area and prevent employees or other persons from trespassing on, removing, or otherwise disturbing such resources. The Government retains ownership and control over archaeological resources.

3.5 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with the JEGS and all applicable Federal, GOJ national or prefectural air emission and performance laws and regulations.

3.5.1 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Product data shall be provided prior to equipment installation.

3.5.2 Burning

Burning is prohibited on the Government premises.

3.5.3 Class I and II ODS Prohibition

Class I and II ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the required recovery cylinders and the appropriate location for turn in of all reclaimed refrigerant.Class I and II ODS as defined and identified herein shall not be used in the performance of this Contract.

The Contractor shall coordinate with the Refrigerant, AC and Heating Shop, on the quantity of ODS recovered from system for the purpose of recycle, reuse or dilution. In the event the ODS can be recycled, reused or diluted, the entire quantity will be turned over to the Refrigerant, AC and Heating shop.

In cases where the ODS is sufficiently contaminated and cannot be recycled, reused and/or diluted, the Contractor shall dispose of the ODS according to the JEGS and local regulations.

In the event of a release of ODS, base personnel and/or Contractor shall inform Fire and Emergency's Service Branch and Environmental Division.

3.5.4 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Comply with the JEGS.

3.5.5 Training/Certification Requirements

Heating and air conditioning technicians must be trained to meet requirements in the JEGS. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by GOJ national or prefectural laws and regulations.

3.5.6 Dust Control

Keep dust down at all times, including during nonworking periods. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

3.5.6.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed the JEGS and applicable US Federal, GOJ national or prefectural air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

3.5.6.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

3.5.7 Odors

Control odors from construction activities. The odors must be in compliance with applicable GOJ national or prefectural laws and regulations and may not constitute a health hazard.

3.6 WATER RESOURCES

Keep construction activities under surveillance, management, and control, and monitor all water areas affected by construction activities in order to prevent pollution of surface and ground waters. Toxic or hazardous chemicals shall not be applied to soil or vegetation when such application may cause contamination. Prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

3.6.1 Lead-Free Drinking Water Pipe, Solders, Flux and Fittings

The maximum allowable lead content for pipes, fittings, and fixtures intended to convey or dispense water for human consumption and cooking shall be a weighted average of 0.25 percent with respect to the wetted surfaces of pipes and pipe fittings, plumbing fittings, and fixtures in accordance with NSF 61.

3.7 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.7.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the GOJ, Prefectural, installation or local permit (cover) or license for recycling.

All solid wastes and materials that have been separated for the purpose of recycling shall be stored in such a manner that they do not constitute a fire, health or safety hazard, or provide food or harborage for vectors, and shall be contained or bundled so as to not result in spillage. Containers must be: leak proof, water proof, and vermin proof including sides, seams, tops and bottoms, durable enough to withstand anticipated usage, and stored on a firm, level, well-drained surface).

3.7.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Installation Environmental Office the Contracting Officer on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	(_____) kilograms or metric tons as appropriate
C&D Debris Recycled	(_____) kilograms or metric tons as appropriate
Total C&D Debris Generated	(_____) kilograms or metric tons as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	(_____) kilograms or metric tons as appropriate

3.8 WASTE MANAGEMENT AND DISPOSAL

3.8.1 Waste Determination Documentation

Incorporate the Waste Determination Documentation for Contractor-derived wastes to be generated into the Environmental Records Binder. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g. scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination Documentation. As a minimum, provide Waste Determination Documentation for the following waste (this listing is not inclusive): oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.8.1.1 Sampling and Analysis of Waste

3.8.1.1.1 Waste Sampling

Sample waste in accordance with the JEGS, Installation Hazardous Waste Management Plan, and appropriate Japanese or US EPA testing protocols that meet the purpose of the testing. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.

Excess excavated soil shall become the property of the Contractor and shall be disposed of off base at a properly licensed facility in accordance with the latest version of the JEGS and applicable GOJ federal and local laws and regulations. Any excess soil shall be sampled and tested according to the applicable GOJ federal and local laws and regulations and disposal facility requirements prior to disposal.

3.8.1.1.2 Laboratory Analysis

Follow the analytical procedure and methods in accordance with the JEGS and all applicable GOJ national or prefectural laws and regulations and the Installation Hazardous Waste Management Plan. Provide analytical results and reports performed in the Environmental Records Binder.

3.8.1.1.3 Analysis Type

Identify hazardous waste by analyzing for the following characteristics: ignitability, corrosivity, reactivity, toxicity based on TCLP results,.

3.8.2 Solid Waste Management

3.8.2.1 Solid Waste Management Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, Contractor Certification and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation must include the receiver's tax identification number and business, GOJ or prefectural registration number, along with the receiver's delivery and business addresses and telephone numbers.

3.8.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off

Government property and dispose of it in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for solid waste disposal. Solid waste disposal offsite must comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Include a copy of all licenses and permits secured in the Environmental Protection Plan

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.8.3 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

3.8.3.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements

Manage hazardous waste in accordance with the approved Hazardous Waste Management Section of the EPP. Store hazardous wastes in approved containers in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in the JEGS. For hazardous wastes spills, verbally notify the Contracting Officer immediately.

3.8.3.2 Hazardous Waste Records

The Contractor shall maintain sampling, analysis, and turn-in records for all hazardous waste generated during the project. These records shall include, but not be limited to: logs of sample locations or container identification data (including time and date of sample collection), analytical results, and quality control data provided by the analytical lab pertaining to the samples analyzed. Copies of this data shall be submitted to teh installation Environmental Division via the COR after the work is completed.

3.8.3.3 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such as computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.8.3.4 Disposal Documentation for Hazardous and Regulated Waste

Submit a copy of the applicable local permit(s), manifest(s), or license(s) for transportation, treatment, storage, and disposal of hazardous and regulated waste by permitted facilities. Hazardous or toxic waste manifests must be reviewed, signed, and approved by the Contracting Officer before the Contractor may ship waste. To obtain specific disposal instructions, coordinate with the Installation Environmental Office.

3.8.4 Releases/Spills of Oil and Hazardous Substances

3.8.4.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with the JEGS and Installation plans. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Installation Fire Department dial 911 on base, the Installation Command Duty Officer, the Installation Environmental Office, the Contracting Officer. The Contractor shall immediately report all POL or Hazardous Substances spills to the Contracting Officer. Submit verbal and written notifications as required by the JEGS and installation and service component instructions and plans, and local regulations. Provide copies of the written notification and documentation that a verbal notification was made within the timeframes required by the JEGS and installation instructions. Spill response must be in accordance with the JEGS and installation requirements. Contain and clean up these spills without cost to the Government.

3.8.4.2 Clean Up

Clean up hazardous and non-hazardous waste spills in accordance with the JEGS, local regulations and the installation's spill response procedures. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor-responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.8.5 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.8.6 Wastewater

3.8.6.1 Disposal of wastewater must be as specified below.

3.8.6.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction- related waste water off-Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.8.6.1.2 Surface Discharge

For discharge of ground water, Surface discharge shall be done in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.

3.8.6.1.3 Land Application

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing must be discharged into the sanitary sewer with prior approval and notification to the owner of the sanitary sewer system.

3.9 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this Contract. Submit an Safety Data Sheet in both English and Japanese and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent.

Storage and handling of hazardous materials will adhere to the DoD Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction (DLAI) 4145.11, Army Technical Manual (TM) 38-410, Naval Supply Publication (NAVSUP PUB) 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order (MCO) 4450.12A, "Storage and Handling of Hazardous Materials," January 13, 1999 provide additional guidance on the storage and handling of hazardous materials. The International Maritime Dangerous Goods (IMDG) Code and appropriate DoD and Component instructions provide requirements for international maritime transport of hazardous materials originating from DoD installations. International air shipments of hazardous materials originating from DoD installations are subject to International Civil Aviation Organization Technical Instructions or DoD Component guidance, including Air Force Manual 24-204, (Interservice) TM

38-250, NAVSUP PUB 505, MCO P4030.19J, and DLAI 4145.3, DCMAD1, Ch3.4 (HM24), "Preparing Hazardous Materials for Military Air Shipments," 3 December 2012. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with the 40 CFR 261, JEGS, and installation requirements.

3.10 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds.

3.11 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos- containing waste in accordance with the JEGS and applicable Federal, GOJ national or prefectural laws and regulations. Refer to Section 02 82 00 ASBESTOS REMEDIATION. Manifest asbestos-containing waste and provide the manifest to the Contracting Officer. Notifications to the Contracting Officer and Installation Air Program Manager are required before starting any asbestos work.

Removal of mastics, necessary to achieve the Contract's objectives, shall be performed in a manner as if the material contains asbestos.

When applicable, a minimum of 14 calendar days prior to the demolition or renovation of a facility that involves removing or disturbing friable ACM, the Contractor shall prepare a Written Assessment Of Friable Asbestos Disturbance and submit to the COR, who will, in turn submit to the Installation Commander in accordance with the JEGS.

For any previously untested material suspected to contain asbestos and located in areas impacted by the work, notify the Contracting Officer (CO) who will order up to 20 bulk samples to be obtained and analyzed at the Contractor's expense. Samples shall be collected in sufficient amounts such that a follow-on test utilizing TEM or 1000 point count analysis can be performed, should it be necessary. The Contractor shall deliver the sample(s) to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)" and analyzed by PLM. The laboratory shall have a working definition of “Trace” amounts of asbestos, and the laboratory shall report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantification of 1% as a “Trace” concentration. If PLM does not detect the presence of asbestos (e.g. “non-detect”), the material shall be considered <0.1% asbestos. If PLM analysis detects asbestos in any discernible amount (to include “trace” or “less than 1%”), the material shall be considered >0.1% asbestos unless proven to be non-ACM by the use of quantification methods capable of achieving an analytical sensitivity of less than 0.1%, such as Transmission Electron Microscopy (TEM) or 1000 point counting.

The Contracting Officer (CO) will order, testing by TEM or 1000 point counting, for up to 10 samples, to be obtained at the Contractor's expense and delivered to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)", for those samples with results of < 1% ACM as analyzed by PLM.

Any additional components identified as ACM that have been approved by the CO for removal shall be removed and will be paid for by an equitable adjustment to the Contract price under the CONTRACT CLAUSE titled "changes". Sampling shall be conducted by personnel who have successfully completed the EPA Model Accreditation Plan (MAP) "Building Inspector" training course and is AHERA certified as a "Building Inspector".

3.12 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements. Refer to Section 02 83 00 LEAD REMEDIATION. The Contractor shall assume all paints and coatings that will be disturbed during the course of meeting the contractual requirements, contain detectable levels of lead, requiring the Contractor to incorporate OSHA regulations for worker and occupant protection. The Contractor is to assume that all existing paint encountered in this Contract contain detectable levels of lead, and the OSHA regulations referenced in these specification applies. Any attached lead material survey information is provided for disposal purposes only. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer.

3.13 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)

Manage and dispose of PCB-contaminated waste in accordance with the JEGS, installation requirements, and Section 02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCB). Purchase of electrical equipment and transformers containing PCBs is prohibited.

3.13.1 Removal Or Installation

Work on PCB equipment is forbidden using this supplement alone. If the project involves the disturbance, relocation, installation, or removal of products that have historically contained PCBs, follow the instructions in the supplement to the specification that pertains to that product's installation or removal (Transformer Replacement, Lighting Replacement, etc.).

3.14 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with the JEGS and installation requirements.

3.15 CONTROL AND MANAGEMENT OF ARSENIC/CADMIUM CONTAINING GYPSUM BOARD

Inspect all gypsum boards removed during construction for manufacturers serial and lot numbers know to contain elevated levels or arsenic or cadmium. Manage and dispose of target boards in accordance with JEGS, and Government of Japan regulations pertaining to such waste.

3.16 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with the JEGS and installation requirements and applicable GOJ national or prefectural laws and regulations. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or navigable waters of Japan. Describe in the EPP (see paragraph ENVIRONMENTAL PROTECTION PLAN) how POL tanks and containers must be stored, managed, and inspected and what protections must be provided. Storage of fuel on the project site must be in accordance with the JEGS and installation requirements, and local laws and regulations and paragraph OIL STORAGE INCLUDING FUEL TANKS.

3.16.1 Used Oil Management

"Used oil," means any oil or other waste petroleum, oil, or lubricant (POL) product that has been refined from crude oil, or is synthetic oil, has been used and as a result of such use, is contaminated by physical or chemical impurities, or is off specification and cannot be used as intended. Although used oil may exhibit the characteristics of reactivity, toxicity, ignitability, or corrosivity, it is still considered used oil, unless it has been mixed with hazardous waste. Manage used oil generated on site in accordance with the JEGS and installation requirements. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL.

3.16.2 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 12 centimeters freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 5000 liter will be used onsite (only containers with a capacity of 208 liter or greater are counted), provide and implement a SPCC plan meeting the requirements of 40 CFR 112. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Monitor and remove any rainwater that accumulates in open containment dikes or berms. Inspect the accumulated rainwater prior to draining from a containment dike to the environment, to determine there is no oil sheen present.

3.17 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.18 PEST MANAGEMENT

In order to minimize impacts to existing fauna and flora, coordinate with the Installation Pest Management Coordinator (IPMC) or the Natural Resources Manager in the DPW Environmental Division, through the Contracting Officer, at the earliest possible time prior to pesticide application. Discuss integrated pest management strategies with the IPMC and receive concurrence from the IPMC through the Contracting Officer prior to the application of any pesticide associated with these specifications. Provide Installation Project Office Pest Management personnel the opportunity to be present at meetings concerning treatment measures for pest or disease control and during application of the pesticide. The use and management of pesticides are regulated under the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.

3.18.1 Application

Apply pesticides using a DoD-certified or equivalent Japanese-certified pesticide applicator in accordance with guidance given on the pesticide label. The certified applicator must wear clothing and personal protective equipment as specified on the pesticide label. If local nationals will be using the pesticides, the precautionary messages and use instructions shall be in English and Japanese. The Contracting Officer will designate locations for water used in formulating. Do not allow the equipment to overflow. Inspect equipment for leaks, clogging, wear, or damage and repair prior to application of pesticide.

3.18.2 Pesticide Treatment Plan

Include and update a pesticide treatment plan, as information becomes available. Include in the plan the sequence of treatment, dates, times, locations, pesticide trade name, EPA registration numbers if applicable, authorized uses, chemical composition, formulation, original and applied concentration, application rates of active ingredient (that is, pounds of active ingredient applied), equipment used for application and calibration of equipment. Comply with the JEGS or any applicable Federal GOJ national or prefectural and installation pest management record-keeping and reporting requirements as well as any additional Installation Project Office specific requirements in conformance with DA AR 200-1 Chapter 5, Pest Management, Section 5-4 "Program requirements" for data required to be reported to the Installation.

3.19 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

-- End of Section --

SECTION 01 74 19

CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS	(Ap~~ril~~r 202~~2~~4) Japan Environmental Governing Standards

1.2 GOVERNMENT POLICY

Government policy is to apply sound environmental principles in the design, construction and use of facilities. As part of the implementation of that policy, practice efficient waste management when sizing, cutting, and installing products and materials, and use all reasonable means to divert construction and demolition waste from landfills and incinerators and to facilitate their recycling or reuse. Divert a minimum of 60 percent by weight of total project solid waste from the landfill.~~[Comply with TPC requirements as specified in Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING.]~~

1.3 MANAGEMENT

Develop and implement a waste management program. Take a pro-active, responsible role in the management of construction and demolition waste and require all subcontractors, vendors, and suppliers to participate in the effort. The Environmental Manager, as specified in Section ~~[~~01 57 19~~][01 57 19.01]~~ TEMPORARY ENVIRONMENTAL CONTROLS, is responsible for instructing workers and overseeing and documenting results of the Waste Management Plan for the project. Construction and demolition waste includes products of demolition or removal, excess or unusable construction materials, packaging materials for construction products, and other materials generated during the construction process but not incorporated into the work. In the management of waste, consider the availability of viable markets, the condition of the material, the ability to provide the material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates. Implement any special programs involving rebates or similar incentives related to recycling of waste. Revenues or other savings obtained for salvage, or recyclable materials not claimed by the Government will accrue to the Contractor. Appropriately permit firms and facilities used for recycling, reuse, and disposal for the intended use to the extent required by federal, state, and local regulations. Also, provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the project.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; ~~submittals not having a "G" designation are for information only~~submittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Waste Management Plan; G

SD-11 Closeout Submittals

Records; G

1.5 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Waste Management Plan and to develop a mutual understanding relative to the details of waste management. The requirements for this meeting may be fulfilled during the coordination and mutual understanding meeting outlined in Section 01 45 00.00 10 QUALITY CONTROL. At a minimum, discuss environmental and waste management goals and issues at the following additional meetings:

a. Pre-bid meeting.

b. Preconstruction meeting.

c. Regular QC meetings.

d. Work safety meetings.

1.6 WASTE MANAGEMENT PLAN

Submit a Waste Management Plan within 15 days after Notice to Proceed. The plan demonstrates how to meet ~~the~~ the project waste diversion goal. Also, include the following in the plan:

a. Name of individuals on the Contractor's staff responsible for waste prevention and management.

b. Actions that will be taken to reduce solid waste generation, including coordination with Subcontractors to ensure awareness and participation.

c. Description of the regular meetings to be held to address waste management.

d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of wastes.

e. Characterization, including estimated types and quantities, of the waste to be generated.

f. Name of landfill and/or incinerator to be used and the estimated costs for use, assuming that there would be no salvage or recycling on the project.

g. Identification of local and regional reuse programs, including non-profit organizations such as schools, local housing agencies, and organizations that accept used materials such as materials exchange networks and Habitat for Humanity. Include the name, location, and phone number for each reuse facility to be used, and provide a copy of the permit or license for each facility.

h. List of specific waste materials that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, location, and phone number, including a copy of the permit or license for each facility.

i. Identification of materials that cannot be recycled/reused with an explanation or justification, to be approved by the Contracting Officer.

j. Description of the means by which any waste materials identified in item (h) above will be protected from contamination.

k. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).

l. Anticipated net cost savings determined by subtracting Contractor program management costs and the cost of disposal from the revenue generated by sale of the materials and the incineration and/or landfill cost avoidance.

Revise and resubmit Plan as required by the Contracting Officer. Approval of Contractor's Plan shall not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Distribute copies of the Waste Management Plan to each Subcontractor, the Quality Control Manager, and the Contracting Officer.

1.7 RECORDS

Maintain records to document the quantity of waste generated; the quantity of waste diverted through sale, reuse, or recycling; and the quantity of waste disposed by landfill or incineration. Quantities may be measured by weight or by volume, but must be consistent throughout. List each type of waste separately noting the disposal or diversion date. Identify the landfill, recycling center, waste processor, or other organization used to process or receive the solid waste. Provide explanations for any waste not recycled or reused. With each application for payment, submit updated documentation for solid waste disposal and diversion, and submit manifests, weight tickets, receipts, and invoices specifically identifying the project and waste material. Make the records available to the Contracting Officer during construction, and deliver to the Contracting Officer upon completion of the construction a copy of the records.

1.8 REPORTS

Provide quarterly reports and a final report to Contracting Officer.

Include project name, information for waste generated this quarter, and cumulative totals for the project in quarterly and final reports. Also include in each report, supporting documentation to include manifests, weight tickets, receipts, and invoices specifically identifying the project and waste material. Include timber harvest and demolition information, if any. See Section ~~[~~01 57 19~~]~~ ~~[01 57 19.01]~~ TEMPORARY ENVIRONMENTAL CONTROLS for Nonhazardous Solid Waste Diversion Report requirements.

1.9 COLLECTION

Separate, store, protect, and handle at the site identified recyclable and salvageable waste products in a manner that maximizes recyclability and salvagability of identified materials. Provide the necessary containers, bins and storage areas to facilitate effective waste management and clearly and appropriately identify them. Provide materials for barriers and enclosures around recyclable material storage areas which are nonhazardous and recyclable or reusable. Locate out of the way of construction traffic. Provide adequate space for pick-up and delivery and convenience to Subcontractors. Recycling and waste bin areas are to be kept neat and clean, and handle recyclable materials to prevent contamination of materials from incompatible products and materials. Clean contaminated materials prior to placing in collection containers. Use cleaning materials that are nonhazardous and biodegradable. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section ~~[~~01 57 19~~][01 57 19.01]~~ TEMPORARY ENVIRONMENTAL CONTROLS. Separate materials by one of the following methods:

1.9.1 Source Separated Method

Separate waste products and materials that are recyclable from trash and sorted as described below into appropriately marked separate containers. Transport materials to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the following category types as appropriate to the project waste and to the available recycling and reuse programs in the project area:

a. Land clearing debris.

b. Asphalt.

c. Concrete and masonry.

d. Metal (e.g. banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, lead brass, bronze).

(1) Ferrous.

(2) Non-ferrous.

e. Wood (nails and staples allowed).

f. Debris.

g. Glass (colored glass allowed).

h. Paper.

(1) Bond.

(2) Newsprint.

(3) Cardboard and paper packaging materials.

i. Plastic.

(1) Type 1 - Polyethylene Terephthalate (PET, PETE).

(2) Type 2 - High Density Polyethylene (HDPE).

(3) Type 3 - Vinyl (Polyvinyl Chloride or PVC).

(4) Type 4 - Low Density Polyethylene (LDPE).

(5) Type 5 - Polypropylene (PP).

(6) Type 6 - Polystyrene (PS).

(7) Type 7 - Other. Use of this code indicates that the package in question is made with a resin other than the six listed above, or is made of more than one resin listed above, and used in a multi-layer combination.

j. Gypsum.

k. Non-hazardous paint and paint cans.

l. Carpet.

m. Ceiling tiles.

n. Insulation.

o. Beverage containers.

1.9.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to a recycling facility where the recyclable materials are sorted and processed.

1.9.3 Other Methods

Other proposed methods may be used when approved by the Contracting Officer.

1.10 DISPOSAL

Control accumulation of waste materials and trash. Where materials are to be turned over to the Contractor for disposal, every effort shall be made to obtain credit from the disposal to reduce the cost of the Contract. Recycle or dispose of collected materials off of Government property at intervals approved by the Contracting Officer and in compliance with waste management procedures of JEGS and local laws and regulations.~~[For MCAS Iwakuni, metal Items to be demolished shall be sent to the Recycle Center, Building 7725.]~~ Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1. Reuse. Give first consideration to salvage for reuse since little or no re-processing is necessary for this method, and less pollution is created when items are reused in their original form. Consider sale or donation of waste suitable for reuse.

2. Recycle. Recycle waste materials not suitable for reuse, but having value as being recyclable. Recycle all fluorescent lamps, HID lamps, and mercury-containing thermostats removed from the site. Arrange for timely pickups from the site or deliveries to recycling facilities in order to prevent contamination of recyclable materials.

3. Compost. Consider composting on-site if a reasonable amount of compostable material will be available. Compostable materials include plant material, sawdust, and certain food scraps.

4. Waste. Dispose of materials with no practical use or economic benefit to waste-to-energy plants where available. As the last choice, dispose of materials at a landfill or incinerator.

5. Return. Set aside and protect misdelivered and substandard products and materials and return to supplier for credit.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION 01 78 00

CLOSEOUT SUBMITTALS
05/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2005; R 2011) Stewardship for the Cleaning of Commercial and Institutional Buildings

GREEN SEAL (GS)

GS-37	(2017) Cleaning Products for Industrial and Institutional Use

~~SPATIAL DATA STANDARDS FOR FACILITIES, INFRASTRUCTURE, AND ENVIRONMENT (SDSFIE)~~SPATIAL DATA STANDARD FOR FACILITIES, INFRASTRUCTURE, AND ENVIRONMENT (SDSFIE)

SDSFIE 3.1.0.1	(2015) Air Force Data Model

U.S. ARMY CORPS OF ENGINEERS (USACE)

ERDC/ITL TR-12-1	(2015) A/E/C Graphics Standard, Release 2.0
ERDC/ITL TR-12-6	(2015) A/E/C CAD Standard - Release 6.0

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-300-08	(2009, with Change 2, 2011) Criteria for Transfer and Acceptance of DoD Real Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site and/or red-lined PDF files. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.2.3 USACE CAD/BIM Technology Center

The USACE CAD/BIM Technology Center hosts all standard content for USACE. This content can be accessed through the CAD/BIM Technology Center website, https://cadbimcenter.erdc.dren.mil/.

1.3 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction drawings and data for the referenced project. Any other use or reuse shall be at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction drawings anddata related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information onlysubmittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan; G

Warranty Tags

Final Cleaning

~~Spare Parts Data~~

Spare Parts and Spare Parts Data

SD-08 Manufacturer's Instructions

Posted Instructions

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

As-Built Record of Equipment and Materials; G

Final Approved Shop Drawings; G

Construction Contract Specifications; G

Interim DD FORM 1354; G

Checklist for DD FORM 1354; G

1.5 SPARE PARTS AND SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

a. Supply the necessary quantity of items of each part for spare parts required by this contract, identified in the technical specification herein. Indicate manufacturer's name, part number, nomenclature, and stock level required for maintenance and repair. List those items that may be standard to the normal maintenance of the system.

1.6 QUALITY CONTROL

Additions and corrections to the contract drawings must be equal in quality and detail to that of the originals. Line colors, line weights, lettering, layering conventions, and symbols must be the same as the original line colors, line weights, lettering, layering conventions, and symbols.

1.7 WARRANTY MANAGEMENT

1.7.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

b. For each warranty, the name, address, telephone number, and email of each of the guarantor's representatives nearest to the project location.

c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.

d. As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:

(1) Name of item.
(2) Model and serial numbers.
(3) Location where installed.
(4) Name and phone numbers of manufacturers or suppliers.
(5) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have extended warranties must be indicated with separate warranty expiration dates.
(6) Cross-reference to warranty certificates as applicable.
(7) Starting point and duration of warranty period.
(8) Summary of maintenance procedures required to continue the warranty in force.
(9) Cross-reference to specific pertinent Operation and Maintenance manuals.
(10) Organization, names and phone numbers of persons to call for warranty service.
(11) Typical response time and repair time expected for various warranted equipment.

e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.

f. Procedure and status of tagging of equipment covered by extended warranties.

g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.7.2 Performance Bond

The Performance Bond must remain effective throughout the construction period.

a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.

b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.7.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.7.4 Contractor's Response to Construction Warranty Service Requirements

Following oral or written notification by the Contracting Officer, respond to construction warranty service requirements in accordance with the "Construction Warranty Service Priority List" and the three categories of priorities listed below. Submit a report on any warranty item that has been repaired during the warranty period. Include within the report the cause of the problem, date reported, corrective action taken, and when the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and back charge the construction warranty payment item established.

a. First Priority Code 1. Perform onsite inspection to evaluate situation, and determine course of action within 4 hours, initiate work within 6 hours and work continuously to completion or relief.

b. Second Priority Code 2. Perform onsite inspection to evaluate situation, and determine course of action within 8 hours, initiate work within 24 hours and work continuously to completion or relief.

c. Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.

d. The "Construction Warranty Service Priority List" is as follows:

Code 1-Life Safety Systems
(1) Fire suppression systems.
(2) Fire alarm system(s) in place in the building.

Code 1-Air Conditioning Systems
(1) Recreational support.
(2) Air conditioning leak in part of building, if causing damage.
(3) Air conditioning system not cooling properly.

Code 1-Doors
(1) Overhead doors not operational, causing a security, fire, or safety problem.
(2) Interior, exterior personnel doors or hardware, not functioning properly, causing a security, fire, or safety problem.

Code 3-Doors
(1) Overhead doors not operational.
(2) Interior/exterior personnel doors or hardware not functioning properly.

Code 1-Electrical
(1) Power failure (entire area or any building operational after 1600 hours).
(2) Security lights
(3) Smoke detectors

Code 2-Electrical
(1) Power failure (no power to a room or part of building).
(2) Receptacle and lights (in a room or part of building).

Code 3-Electrical
Street lights.

Code 1-Gas
(1) Leaks and breaks.
(2) No gas to family housing unit or cantonment area.

Code 1-Heat
(1) Area power failure affecting heat.
(2) Heater in unit not working.

Code 2-Kitchen Equipment
(1) Dishwasher not operating properly.
(2) All other equipment hampering preparation of a meal.

Code 1-Plumbing
(1) Hot water heater failure.

(2) Leaking water supply pipes.

Code 2-Plumbing
(1) Flush valves not operating properly.
(2) Fixture drain, supply line to commode, or any water pipe leaking.
(3) Commode leaking at base.

Code 3 -Plumbing
Leaky faucets.

Code 3-Interior
(1) Floors damaged.
(2) Paint chipping or peeling.
(3) Casework.

Code 1-Roof Leaks
Temporary repairs will be made where major damage to property is occurring.

Code 2-Roof Leaks
Where major damage to property is not occurring, check for location of leak during rain and complete repairs on a Code 2 basis.

Code 2-Water (Exterior)
No water to facility.

Code 2-Water (Hot)
No hot water in portion of building listed.

Code 3-All other work not listed above.

1.7.5 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	

Construction Contractor	
Address	
Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

PART 2 PRODUCTS

2.1 RECORD DRAWINGS

Prepare the CAD drawing files in AutoCAD Release 2013 format and Bentley Microstation V8i format compatible with a Windows 7 or higher operating system.

2.1.1 Additional Drawings

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings.

2.1.1.1 Sheet Numbers and File Names

If a sheet needs to be added between two sequential sheets, append a Supplemental Drawing Designator in accordance with ERDC/ITL TR-12-6 Adding a drawing sheet, and ERDC/ITL TR-12-1.

2.2 PDF AS-BUILT FILES

Provide electronic PDF "plots" of all contract drawings sheets associated with the as-built drawing submittal. Compile and organize the PDF set to match the contract drawings.

2.3 REDLINES AND MARKUPS

Provide PDFs of the current working redlines and/or markups complying with the as-builts drawing and markup requirements contained in this specification.

2.4 AS-BUILT RE-SUBMISSION REQUIREMENTS

If elements of an as-built submittal or advanced modeling package are rejected, provide the following for each re-submission, in addition to any information required in Section 01 33 00 SUBMITTAL PROCEDURES:

a. Re-submit all components required under paragraph As-Builts or Advanced Modeling Package, including a new Advanced Modeling Submittal Checklist and updated content in response to Government comments.

b. Provide a copy of all Government review comments.

c. Provide a disposition/response to each Government review comment for a back-check of the re-submission deliverable.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site and/or red-lined PDF files. Submit As-Built Drawings 30 days prior to Beneficial Occupance Date (BOD).

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

a. Use base colors of red, green, and blue. Color code for changes as follows:

(1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.

(2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.

(3) Additions (Green) - Added items, lettering in notes and leaders.

b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.

c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.

d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.

e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.

f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.

g. For deletions, cross out all features, data and captions that relate to that revision.

h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.

i. Indicate one of the following when attaching a print or sketch to a markup print:

1) Add an entire drawing to contract drawings

2) Change the contract drawing to show

3) Provided for reference only to further detail the initial design.

j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content

Revise As-Built Drawings in accordance with ERDC/ITL TR-12-1 and ERDC/ITL TR-12-6. Keep these working as-built markup drawings current on a weekly basis and at least one set available on the jobsite at all times. Changes from the contract drawings which are made during construction or additional information which might be uncovered in the course of construction must be accurately and neatly recorded as they occur by means of details and notes. Submit the working as-built markup drawings for approval prior to submission of each monthly pay estimate. For failure to maintain the working and final record drawings as specified herein, the Contracting Officer will withhold 10 percent of the monthly progress payment until approval of updated drawings. Show on the as-built drawings, but not limited to, the following information:

a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.

b. The location and dimensions of any changes within the building structure.

c. Layout and schematic drawings of electrical circuits and piping.

d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.

e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.

f. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.

g. Changes or Revisions which result from the final inspection.

h. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.

i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.

j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.

k. Changes in location of equipment and architectural features.

j. Modifications (include within change order price the cost to change working as-built markup drawings to reflect modifications).

l. Actual location of anchors, construction and control joints, etc., in concrete.

m. Unusual or uncharted obstructions that are encountered in the contract work area during construction.

n. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.1.3 GIS Data

Provide GIS data with as-builts. GIS data shall be in a format accepted by ESRI software. Data shall comply with SDSFIE 3.1.0.1. The Air Force adaptation data dictionary can be obtained from the Kadena GeoBase Office (718ces.geobase@us.af.mil or DSN 634-9322). All GIS data shall be delivered with the following requirements:

a. Projection: UTM Zone 52N

b. Datum: WGS 1984

c. Geoid: EGM 96

d. Units: Meters

3.1.4 GIS Data

Provide GIS data with as-builts. GIS data shall be in a format accepted by ESRI software. Data shall comply with SDSFIE 3.1.0.1. All GIS data shall be delivered with the following requirements:

a. Projection: JGD 2000 Zone 15

b. Datum: WGS 1984

c. Geoid: EGM 96

d. Units: Meters

3.2 RECORD DRAWING FILES

If additional drawings are required, prepare them using the specified electronic file format applying the same graphic standards specified for original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used on the contract drawings. Accomplish additions and corrections to the contract drawings using CAD files. Provide all program files and hardware necessary to prepare final PDF record drawings. The Contracting Officer will review final PDF record drawings for accuracy and return them to the Contractor for required corrections, changes, additions, and deletions.

3.2.1 Rename the CAD Drawing files

Rename the CAD Drawing files using the ~~[ACES-PM Project Number as the Project Code field, (e.g., LXEZ11745C-101.DWG)][~~Contract number as the Project Code field, (e.g., W912HV-15-C-0001-A-101.DWG)~~]~~ as instructed in the Pre-Construction conference. Use only those renamed files for the Marked-up changes. Make all changes on the layer/level as the original item.

a. For AutoCAD files (DWG), enter all as-built delta changes and notations on the AS-BUILT layer.

c. When final revisions have been completed, show the wording "RECORD DRAWING AS-BUILTS" followed by the name of the Contractor in letters at least 5 mm high on the cover sheet drawing. Date RECORD DRAWING AS-BUILTS" drawing revisions in the revision block.

d. Within 20 days after Government approval of all of the working record drawings for a phase of work, prepare the final CAD record drawings for that phase of work and submit PDF drawing files and two sets of prints for review and approval. The Government will promptly return one set of prints annotated with any necessary corrections. Within 10 days revise the CAD files accordingly at no additional cost and submit one set of final prints for the completed phase of work to the Government. Within 20 days of substantial completion of all phases of work, submit the final record drawing package for the entire project. Submit one set of electronic CAD files, and one set of the approved working record PDF files on an optical disc with two sets of prints. The CAD files must be complete in all details and identical in form and function to the CAD drawing files supplied by the Government. Prepare AutoCAD files for transmittal using e-Transmit. Make any transactions or adjustments necessary to accomplish this. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CAD system. Paper prints, drawing files and storage media submitted will become the property of the Government upon final approval. Failure to submit final record PDF drawing files, CAD files and marked prints as specified will be cause for withholding any payment due under this contract. Approval and acceptance of final record drawings must be accomplished before final payment is made.

3.3 RECORD DRAWINGS

Prepare final record drawings after the completion of each definable feature of work as listed in the Contractor Quality Control Plan (such as

Foundations, Utilities, or Structural Steel as appropriate for the project). Transfer the changes from the approved working as-built markup drawings to the original electronic CAD drawing files. Modify the as-built CAD drawing files to correctly show the features of the project as-built by bringing the working CAD drawing set into agreement with approved working as-built markup drawings, and adding such additional drawings as may be necessary. Refer to ERDC/ITL TR-12-1. Jointly review the working as-built markup drawings with printouts from working as-built CAD drawing PDF files for accuracy and completeness. Monthly review of working as-built CAD drawing PDF file printouts must cover all sheets revised since the previous review. These PDF drawing files are part of the permanent records of this project. Any drawings damaged or lost must be satisfactorily replaced at no expense to the Government.

Drawing revisions (include within change order price the cost to change working and final record drawings to reflect revisions) and compliance with the following procedures.

a. Follow directions in the revision for posting descriptive changes.

b. The revision delta size must be 8 mm unless the area where the delta is to be placed is crowded. Use a smaller size delta for crowded areas.

c. Place a revision delta at the location of each deletion.

d. For new details or sections which are added to a drawing, place a revision delta by the detail or section title.

e. For minor changes, place a revision delta by the area changed on the drawing (each location).

f. For major changes to a drawing, place a revision delta by the title of the affected plan, section, or detail at each location.

g. For changes to schedules or drawings, place a revision delta either by the schedule heading or by the change in the schedule.

3.3.1 Final Record Drawing Package

Submit the final record PDF and CAD drawings package for the entire project within 20 days of substantial completion of all phases of work. Submit one set of A1 size PDF and CAD files on optical disc, read-only memory (ROM), two sets of A1 size prints and one set of the approved working record drawings. The package must be complete in all details and identical in form and function to the contract drawing files supplied by the Government.

3.4 FINAL APPROVED SHOP DRAWINGS

Submit final approved project shop drawings 30 days after transfer of the completed facility.

3.5 CONSTRUCTION CONTRACT SPECIFICATIONS

Submit final PDF file record construction contract specifications, including revisions thereto, 30 days after transfer of the completed facility.

3.6 AS-BUILT RECORD OF EQUIPMENT AND MATERIALS

Furnish one copy of preliminary record of equipment and materials used on the project 15 days prior to final inspection. This preliminary submittal will be reviewed and returned 2 days after final inspection with Government comments. Submit Two sets of final record of equipment and materials 10 days after final inspection. Key the designations to the related area depicted on the contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

3.7 OPERATION AND MAINTENANCE MANUALS

Provide project operation and maintenance manuals as specified in Section 01 78 23 OPERATION AND MAINTENANCE MANUALS DATA. Provide one electronic copy of the Operation and Maintenance Manual files and three hard copies of the Operation and Maintenance Manuals. Submit to the Contracting Officer for approval within 30 calendar days of the Beneficial Occupancy Date (BOD). Update and resubmit files for final approval at BOD.

3.8 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit two copies of the listing of completed final clean-up items. Leave premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Clean filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan as required. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS, and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

3.9 REAL PROPERTY RECORD

Near the completion of Project, but a minimum of 60 days prior to final acceptance of the work, complete and submit an accounting of all installed property with Interim DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354. Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Refer to Attachment 01 78 00-A for the Draft DD FORM 1354.~~Attach the Real~~

~~Property receiving Component's completed High Performance and Sustainable Building (HPSB) Checklist for each applicable building to the completed DD 1354, in accordance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ For convenience, a blank fillable PDF DD FORM 1354 may be obtained at the following link: www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf

Submit the completed Checklist for DD FORM 1354 of Installed Building Equipment items. Attach this list to the updated DD FORM 1354.

-- End of Section --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2005; R 2011) Stewardship for the Cleaning of Commercial and Institutional Buildings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information onlysubmittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

O&M Database; G
Training Plan; G
Training Outline; G
Training Content; G

SD-11 Closeout Submittals

Training Video Recording; G
Validation of Training Completion; G

1.3 OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data for the provided equipment, product, or system, defining the importance of system interactions, troubleshooting, and long-term preventive operation and maintenance. Compile, prepare, and aggregate O&M data to include clarifying and updating the original sequences of operation to as-built conditions. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this section and Section 01 33 00 SUBMITTAL PROCEDURES.

1.3.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.3.2 Package Content

Provide data package content in both English and Japanese, and in accordance with paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES. Comply with the data package requirements specified in the individual technical sections, including the content of the packages and addressing each product, component, and system designated for data package submission, except as follows. Use Data Package 5 for commissioned items without a specified data package requirement in the individual technical sections. Provide a Data Package 5 instead of Data Package 1 or 2, as specified in the individual technical section, for items that are commissioned.

1.3.3 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

1.3.4 Commissioning Authority Review and Approval

Submit the commissioned systems and equipment submittals to the Commissioning Authority (CxA) to review for completeness and applicability. Obtain validation from the CxA that the systems and equipment provided meet the requirements of the Contract documents and design intent, particularly as they relate to functionality, energy performance, water performance, maintainability, sustainability, system cost, indoor environmental quality, and local environmental impacts. The CxA communicates deficiencies to the Contracting Officer. Submit the O&M manuals to the Contracting Officer upon a successful review of the corrections, and with the CxA recommendation for approval and acceptance of these O&M manuals. This work is in addition to the normal review procedures for O&M data.

1.4 O&M DATABASE

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program. As a minimum, provide list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL FILE FORMAT

Assemble data packages into electronic Operation and Maintenance Manuals. Assemble each manual into a composite electronically indexed file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete electronically linked operation and maintenance directory.

1.5.1 Organization

Bookmark Product and Drawing Information documents using the current version of CSI Masterformat numbering system, and arrange submittals using the specification sections as a structure. Use CSI Masterformat and UFGS numbers along with descriptive bookmarked titles that explain the content of the information that is being bookmarked.

1.5.2 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

a. Building Number.

b. Project Title.

c. Activity and Location.

d. Construction Contract Number.

e. Prepared For: (Contracting Agency).

f. Prepared By: (Name, title, phone number and email address).

g. Include the disk content on the disk label.

h. Date.

i. Virus scanning program used.

1.6 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

The following are a detailed description of the data package items listed in paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES.

1.6.1 Operating Instructions

Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.1.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard.

1.6.1.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.1.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.1.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal

Operating Procedures.

1.6.1.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.1.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.1.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.1.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.6.1.9 Additional Requirements for HVAC Control Systems

Provide Data Package 5 and the following for control systems:

a. Narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.

b. Full as-built sequence of operations.

c. Copies of checkout tests and calibrations performed by the Contractor (not Cx tests).

d. Full points list. Provide a listing of rooms with the following information for each room:

(1) Floor.

(2) Room number.

(3) Room name.

(4) Air handler unit ID.

(5) Reference drawing number.

(6) Air terminal unit tag ID.

(7) Heating or cooling valve tag ID.

(8) Minimum cfm.

(9) Maximum cfm.

e. Full print out of all schedules and set points after testing and acceptance of the system.

f. Full as-built print out of software program.

g. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.2 Preventive Maintenance

Provide the following information for preventive and scheduled maintenance to minimize repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.6.2.1 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

a. A table showing recommended lubricants for specific temperature ranges and applications.

b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities.

c. A Lubrication Schedule showing service interval frequency.

1.6.2.2 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems shall be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

a. Define the anticipated time required to perform each of each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.

b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.2.3 Cleaning Recommendations

Provide environmentally preferable cleaning recommendations in accordance with ASTM E1971.

1.6.3 Repair

Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.6.3.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.3.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.3.3 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards.

1.6.3.4 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.3.5 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.3.6 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.4 Real Property Equipment

Provide a list of installed equipment furnished under this Contract. Include all information usually listed on manufacturer's name plate. In the "EQUIPMENT-IN-PLACE LIST" include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Submit the final list 30 days after transfer of the completed facility.

Key the designations to the related area depicted on the Contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA

Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

1.6.5 Appendices

Provide information required below and information not specified in the preceding paragraphs but pertinent to the maintenance or operation of the product or equipment. Include the following:

1.6.5.1 Product Submittal Data

Provide a copy of SD-03 Product Data submittals documented with the required approval.

1.6.5.2 Manufacturer's Instructions

Provide a copy of SD-08 Manufacturer's Instructions submittals documented with the required approval.

1.6.5.3 O&M Submittal Data

Provide a copy of SD-10 Operation and Maintenance Data submittals documented with the required approval.

1.6.5.4 Parts Identification

Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that shall cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master parts catalog.

1.6.5.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or Contract documents in order to keep warranties in force. Include warranty information for primary componentsof the system. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.5.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.5.7 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.5.8 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.5.9 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5.10 Field Test Reports

Provide a copy of Field Test Reports (SD-06) submittals documented with the required approval.

1.6.5.11 Contractor Information

Provide a list that includes the name, address, and telephone number of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. The information required in each type of data package follows:

1.7.1 Data Package 1

a. Safety precautions and hazards.

b. Cleaning recommendations.

c. Maintenance and repair procedures.

d. Warranty information.

e. Extended warranty information.

f. Contractor information.

g. Spare parts and supply list.

1.7.2 Data Package 2

a. Safety precautions and hazards.

b. Normal operations.

c. Environmental conditions.

d. Lubrication data.

e. Preventive maintenance plan, schedule, and procedures.

f. Cleaning recommendations.

g. Maintenance and repair procedures.

h. Removal and replacement instructions.

i. Spare parts and supply list.

j. Parts identification.

k. Warranty information.

l. Extended warranty information.

m. Contractor information.

1.7.3 Data Package 3

a. Safety precautions and hazards.

b. Operator prestart.

c. Startup, shutdown, and post-shutdown procedures.

d. Normal operations.

e. Emergency operations.

f. Environmental conditions.

g. Operating log.

h. Lubrication data.

i. Preventive maintenance plan, schedule, and procedures.

j. Cleaning recommendations.

k. Troubleshooting guides and diagnostic techniques.

l. Wiring diagrams and control diagrams.

m. Maintenance and repair procedures.

n. Removal and replacement instructions.

o. Spare parts and supply list.

p. Product submittal data.

q. O&M submittal data.

r. Parts identification.

s. Warranty information.

t. Extended warranty information.

u. Testing equipment and special tool information.

v. Testing and performance data.

w. Contractor information.

x. Field test reports.

1.7.4 Data Package 4

a. Safety precautions and hazards.

b. Operator prestart.

c. Startup, shutdown, and post-shutdown procedures.

d. Normal operations.

e. Emergency operations.

f. Operator service requirements.

g. Environmental conditions.

h. Operating log.

i. Lubrication data.

j. Preventive maintenance plan, schedule, and procedures.

k. Cleaning recommendations.

l. Troubleshooting guides and diagnostic techniques.

m. Wiring diagrams and control diagrams.

n. Repair procedures.

o. Removal and replacement instructions.

p. Spare parts and supply list.

q. Repair work-hours.

r. Product submittal data.

s. O&M submittal data.

t. Parts identification.

u. Warranty information.

v. Extended warranty information.

w. Personnel training requirements.

x. Testing equipment and special tool information.

y. Testing and performance data.

z. Contractor information.

aa. Field test reports.

1.7.5 Data Package 5

a. Safety precautions and hazards.

b. Operator prestart.

c. Start-up, shutdown, and post-shutdown procedures.

d. Normal operations.

e. Environmental conditions.

f. Preventive maintenance plan, schedule, and procedures.

g. Troubleshooting guides and diagnostic techniques.

h. Wiring and control diagrams.

i. Maintenance and repair procedures.

j. Removal and replacement instructions.

k. Spare parts and supply list.

l. Product submittal data.

m. Manufacturer's instructions.

n. O&M submittal data.

o. Parts identification.

p. Testing equipment and special tool information.

q. Warranty information.

r. Extended warranty information.

s. Testing and performance data.

t. Contractor information.

u. Field test reports.

v. Additional requirements for HVAC control systems.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training conducted in English for the systems and equipment specified in the technical specifications. The training must be targeted for the Facilities Management Specialist, building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS. Training must include classroom or field lectures based on the system operating requirements. Notify the Contracting Officer in writing seven (7) calendar days prior of the scheduled instructional services. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training.~~[~~ Training plan must be approved by the Commissioning Authority (CxA) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and CxA.~~]~~ Include within the plan the following elements:

a. Equipment included in training.

b. Intended audience.

c. Location of training.

d. Dates of training.

e. Objectives.

f. Outline of the information to be presented and subjects covered including description.

g. Start and finish times and duration of training on each subject.

h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts).

i. Instructor names and instructor qualifications for each subject.

j. List of texts and other materials to be furnished by the Contractor that are required to support training.

k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information.~~[~~ The CxA is responsible for overseeing and approving the content and adequacy of the training.~~]~~ Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.

b. Relevant health and safety issues.

c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

d. Design intent.

e. Use of O&M Manual Files.

f. Review of control drawings and schematics.

g. Interactions with other systems.

h. Special maintenance and replacement sources.

i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with~~[~~ the CxA in accordance with~~]~~ Section 01 45 00.00 10 QUALITY CONTROL.

-- End of Section --

SECTION 01 91 00.15 10

TOTAL BUILDING COMMISSIONING
05/16

PART 1 GENERAL

1.1 SUMMARY

Commission the building systems listed herein. Employ the services of an independent Commissioning Firm. The Commissioning Firm must be a 1st tier subcontractor of the General or Prime Contractor and must be financially and corporately independent of all other subcontractors. The Commissioning Firm must employ a Lead Commissioning Specialist for the Contractor (CxC) that coordinates all aspects of the commissioning process. Perform Commissioning in accordance with the requirements of the contract specifications and the standard under which the Commissioning Firm's qualifications are approved. In the case of a conflict, the most stringent requirement will prevail.

The Commissioning Specialist for the Government (CxG) is a Government employee. In general, the CxC coordinates the commissioning activities and reports to the Contracting Officer's Representative/CxG, copying the General or Prime Contractor on all results in accordance with Third Party Certification (TPC) requirements. The CxC's responsibilities, along with all other contractors' commissioning responsibilities are detailed in the specifications. All members work together to fulfill their contracted responsibilities and meet the objectives of the Contract Documents.

1.2 SYSTEMS TO BE COMMISSIONED

Commission the following systems:

~~[~~ Heating, Ventilating, Air Conditioning, and Refrigeration Systems (HVAC)
~~][~~ Building Automation System
~~][~~ Utility Monitoring and Control System
~~][~~ Lighting Systems
~~][~~ Power Distribution Systems
~~][~~ Power Generation Systems
~~][~~ ~~Renewable Energy Systems~~
~~][~~ Service Water Heating Systems
~~][~~ Plumbing Systems
~~][~~ ~~Natural Gas and Propane Systems~~
~~][~~ Water Pumping and Mixing Systems
~~][~~ ~~Irrigation Systems~~
~~][~~ ~~Water Harvesting/Reclaim Systems~~
~~][~~ ~~Compressed Air and Vacuum Systems~~
~~][~~ Energy and Water Utility Metering Systems and Sub-Meters
~~]~~ ~~Building Envelope: moisture and thermal integrity and air tightness~~
~~[~~ ~~Fenestration Control Systems~~
~~]~~

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 180 (2012) Standard Practice for Inspection and Maintenance of Commercial Building HVAC Systems

1.4 COMMUNICATION WITH THE GOVERNMENT

The Lead Commissioning Specialist (CxC) must submit all plans, schedules, reports, and documentation directly to the Contracting Officer's Representative concurrent with submission to the ~~CQC System Manager~~ Contractor Quality Control (CQC) System Manager. The Lead Commissioning Specialist must have direct communication with the Contracting Officer's Representative regarding all elements of the commissioning process; however, the Government has no direct contract authority with the Lead Commissioning Specialist.

1.5 SEQUENCING AND SCHEDULING

1.5.1 Sequencing

Complete the following prior to starting Functional Performance Tests of mechanical systems:

a. All equipment and systems have been completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications.

b. Performance Verification Tests of the controls systems have been completed and the Performance Verification Test Report has been submitted and approved in accordance with Specifications.

c. Testing, Adjusting, and Balancing has been completed and the Testing, Adjusting, and Balancing Report,~~-~~ has been submitted and approved in accordance with Specification Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~d. The building envelope is enclosed according to contract documents with final construction completed, the Air Barrier Pressure Tests have been completed and the Air Leakage Test Reports and Diagnostic Test Reports have been submitted and approved in accordance with Specification Section 07 05 23 PRESSURE TESTING AN AIR BARRIER SYSTEM FOR AIR TIGHTNESS.~~

~~e~~d. The Pre-Functional Checklists have been submitted and approved.

~~f~~e. The Certificate of Readiness for mechanical systems has been submitted and approved.

Complete the following prior to starting Functional Performance Tests of the electrical systems:

a. All electrical, power generation, and lighting equipment and systems have been completed, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications.

b. The building envelope is enclosed according to contract documents with final construction completed.

c. Ceiling tiles, floor coverings, and window coverings are in place.

d. The Certificate of Readiness for electrical systems has been submitted and approved.

e. Lamps have completed a minimum 100 hour burn-in period.

Refer to the following technical sections for additional commissioning requirements for respective systems:

Section 23 08 00 COMMISSIONING OF MECHANICAL AND PLUMBING SYSTEMS

Section 26 08 00 APPARATUS INSPECTION AND TESTING

1.5.2 Project Schedule

Include the following tasks in the project schedule required by Section ~~[~~ 01 32 01.00 10 PROJECT SCHEDULE~~][01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES]~~. Ensure sufficient time is scheduled to accommodate the requirements of this specification section. The order of items listed below is not intended to imply a specified sequence:

a. Submission and approval of the Commissioning Firm and Commissioning Specialist.

b. Submission and approval of the Testing, Adjusting, and Balancing (TAB) Firm and TAB Specialist specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

c. Submission of the Design Review Report specified herein.

d. Submission of the Design Review Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

e. Submission and approval of the Construction Phase Commissioning Plan.

f. Installation of permanent utilities (~~gas,~~ water, electric).

~~g. Building Envelope Construction~~

~~h. Submission and approval of the Building Envelope Inspection Checklists~~

~~i. Air Barrier Pressure Tests specified in Section 07 05 23 PRESSURE TESTING AN AIR BARRIER SYSTEM FOR AIR TIGHTNESS~~

~~j~~g. Drainage and Vent, Building Sewers, Water Supply Systems and Backflow Prevention Assembly Tests specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

~~k~~h. Factory Acceptance Testing for each of the systems to be commissioned as required by technical specifications.

~~l~~i. Manufacturer's Equipment Start-Up for each of the systems to be commissioned.

~~m~~j. Potable Water System Flushing specified in Section 22 00 00 PLUMBING,

GENERAL PURPOSE.

~~n~~k. Operational Tests of the plumbing system specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

~~o~~l. Potable Water System Disinfection specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

~~p~~m. Submission and approval of the TAB Schematic Drawings, Report Forms, and Procedures specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~q~~n. Submission and approval of Duct Air Leakage Test Procedures specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~r~~o. Duct Air Leakage Test Execution specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~s~~p. Submission and approval of the Final Duct Air Leakage Test Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~t~~q. Testing, Adjusting, and Balancing (TAB) Field Work required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~u~~r. Submission and approval of the TAB Report specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~v~~s. TAB Field Acceptance Testing required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

~~w~~t. Submission and approval of the Start-Up Testing Report.

~~x~~u. Submission and approval of the Performance Verification Test Procedures.

~~y~~v. Performance Verification Tests.

~~z~~w. Performance Verification Test Report.

~~aa~~x. Pre-Functional Checklist Submittal.

~~bb~~y. Functional Performance Testing for each system to be commissioned.

~~cc. Integrated Systems Tests~~

~~dd~~z. Post-Test Deficiency Correction for each system to be commissioned.

~~ee~~aa. Re-Testing.

~~ff~~bb. Endurance Tests.

~~gg~~cc. Training for each of the systems to be commissioned.

~~hh~~dd. Systems Manual/Computerized Maintenance Management System Manual, Maintenance Plan, and Service Life Plan submission and approval.

~~ii~~ee. Submission and approval of the Commissioning Report.

~~jj~~ff. ~~-~~Seasonal Testing.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Commissioning Firm; G

Lead Commissioning Specialist

Technical Commissioning Specialists

Commissioning Firm's Contract

SD-06 Test Reports

~~Design Review Report; G~~

Interim Construction Phase Commissioning Plan; G

Final Construction Phase Commissioning Plan; G

~~Building Envelope Inspection Checklists; G~~

Pre-Functional Checklists; G

Issues Log

Commissioning Report; G

Post-Construction Trend Log Report; G

SD-07 Certificates

Certificate of Readiness; G

SD-10 Operation and Maintenance Data

Training Plan; G

Training Attendance Rosters

Systems Manual; G

Maintenance and Service Life Plans

SD-11 Closeout Submittals

Construction Phase Commissioning Plan; G

Final Commissioning Report; G

1.7 COMMISSIONING FIRM

Provide a Commissioning Firm that is certified in commissioning by one of the following: the AABC Commissioning Group (ACG); the National Environmental Balancing Bureau (NEBB); the International Certification Board/Testing, Adjusting, and Balancing Bureau (ICB/TABB), the Building Commissioning Association (BCA); the Association of Energy Engineers (AEE). The Commissioning Firm must be certified in all systems to be commissioned to the extent such certifications are available from the certifying body. Describe any lapses in certification or disciplinary action taken by the certifying body against the proposed Commissioning Firm or Lead Commissioning Specialist in detail. Any firm or commissioning professional that has been the subject of disciplinary action by the certifying body within the five years preceding contract award is not eligible to perform any duties related to commissioning.

a. Submit the Commissioning Firm's certification of qualifications including the name of the firm and certifications no later than 60 calendar days after Notice to Proceed. Submit four hard copies and an electronic copy.

b. The Commissioning Firm's and Commissioning Specialists' certifications must be maintained for the entire duration of the duties specified herein. If, for any reason, the firm or a specialist loses a certification during this period, immediately notify the Contracting Officer's Representative and submit another Commissioning Firm or Commissioning Specialist for approval. All work specified in this specification section performed by the Commissioning Firm or associated Commissioning Specialists is invalid if the Commissioning Firm or Commissioning Specialist loses its certification prior to contract completion and must be performed by an approved successor.

c. The Commissioning Firm must oversee and assist the General or Prime Contractor with the work specified herein. Submit the Commissioning Firm's Contract including the Scope of Work associated with the paragraph POST-CONSTRUCTION SUPPORT no later than 30 calendar days after approval of the Commissioning Firm. Submit four hard copies and an electronic copy.

~~d. The Commissioning Firm may act as the Pressure Test Agency required by Section 07 05 23 PRESSURE TESTING AN AIR BARRIER SYSTEM FOR AIR TIGHTNESS provided that all qualification requirements of that specification section are met.~~

1.7.1 Lead Commissioning Specialist

The Commissioning Firm must provide a Lead Commissioning Specialist (CxC) that has a minimum of five years of commissioning experience, including two projects of similar size and complexity, and that is one of the following: a NEBB qualified Systems Commissioning Administrator (SCA); ACG Certified Commissioning Authority (CxA); ICB/TABB Certified Commissioning Supervisor; BCA Certified Commissioning Professional (CCP); AEE Certified Building Commissioning Professional (CBCP); University of Wisconsin-Madison Qualified Commissioning Process Provider (QCxP); ASHRAE Building Commissioning Professional (BCxP).

a. Submit the Lead Commissioning Specialist's certification of qualifications including the name of the specialist and firm; certifications; years of experience; and a listing of representative projects of similar size and complexity no later than 60 calendar days after Notice to Proceed. Submit four hard copies and an electronic copy.

b. The Lead Commissioning Specialists certifications must be maintained for the entire duration of the duties specified herein. If, for any reason, the specialist loses a certification during this period, immediately notify the Contracting Officer's Representative and submit another Lead Commissioning Specialist for approval. All work specified in this specification section to be performed by the Lead Commissioning Specialist is invalid if the Lead Commissioning Specialist loses its certification prior to contract completion and must be performed by an approved successor.

c. The Lead Commissioning Specialist must lead and oversee the commissioning work specified herein and be the primary point of contact for the Government regarding the commissioning work.

1.7.2 Technical Commissioning Specialists

Technical Commissioning Specialists, employed by the Commissioning Firm or financially and corporately independent subcontractor hired by the Commissioning Firm, and that have the following qualifications, must perform the technical work specified herein associated with each system to be commissioned:

a. The technical work associated with mechanical systems including Heating, Ventilating, Air Conditioning, and Refrigeration Systems; Building Automation System; Utility Monitoring and Control System; Service Water Heating Systems; Plumbing Systems; Water Pumping and Mixing Systems; ~~[Irrigation Systems]; [Compressed Air and Vacuum Systems];~~ Energy and Water Utility Metering Systems must be performed by a Commissioning Specialist certified by NEBB, ACG, ICB/TABB, or BCA in the commissioning of HVAC systems with five years of experience in the commissioning of HVAC systems.

b. The technical work associated with electrical systems including Lighting Systems; Power Distribution Systems; Power Generation Systems; Renewable Energy Systems must be performed by an engineering technician certified by the InterNational Electrical Testing Association (NETA) or the National Institute for Certification in Engineering Technologies (NICET) with five years of experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices.

~~c. The technical work associated with the Building Envelope system must be performed by a registered architect with five years of building envelope design or construction experience. The Commissioning Firm team member with the required experience related to the building envelope may act as the Air Barrier Inspector required by specification section 07 27 10.00 10 BUILDING AIR BARRIER SYSTEM provided that all qualification requirements of that specification section are met. The Commissioning Firm team member with the required experience related to the building envelope may act as the thermographer required by specification section 07 05 23 PRESSURE TESTING AN AIR BARRIER SYSTEM FOR AIR TIGHTNESS provided that all of the qualification requirements of that specification section are met.~~

~~d~~c. Submit the Technical Commissioning Specialist's certification of qualifications including the name of the specialist and firm; certifications; years of experience; and a listing of representative projects of similar size and complexity no later than 60 calendar days after Notice to Proceed. Submit four hard copies and an electronic copy.

1.7.3 Commissioning Standard

Comply with the requirements of the commissioning standard under which the Commissioning Firm and Specialists qualifications are approved.~~ When the firm and specialists are certified by BCA, AEE, ASHRAE, or the University of Wisconsin-Madison, comply with the requirements of one of the acceptable standards unless otherwise stated herein. The acceptable standards are ACG Commissioning Guideline, NEBB Commissioning Standard, SMACNA 1429, or ASHRAE 202. Comply with applicable NETA and NICET testing standards for electrical systems.~~

a. Implement all recommendations and suggested practices contained in the Commissioning Standard and electrical test standards.

b. Use the Commissioning Standard for all aspects of Commissioning, including calibration of instruments.

c. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the Commissioning Standard, adhere to the manufacturer calibration recommendations.

d. All quality assurance provisions of the Commissioning Standard such as performance guarantees are part of this contract.

e. The Commissioning Specialists must develop commissioning procedures for any systems or system components not covered in the Commissioning Standard.

f. Use any new requirements, recommendations, and procedures published or adopted prior to contract solicitation by the body responsible for the Commissioning Standard.

~~[1.8 SUSTAINABILITY THIRD PARTY CERTIFICATION (TPC)~~

~~The Commissioning Specialists must execute and document the commissioning activities required of the Commissioning Authority for the purposes of complying with the Third Party Certification (TPC) requirements for the project in accordance with Section 01 33 29 SUSTAINABILITY REPORTING. Provide all commissioning documentation required to meet the TPC requirements.~~

~~]~~1.8 ISSUES LOG

The Lead Commissioning Specialist must develop and maintain an Issues Log for tracking and resolution of all deficiencies discovered through commissioning review, inspection, and testing. Include the date of final resolution of issues as confirmed by the Commissioning Specialist. Submit the Issues Log on a monthly basis at a minimum. At any point during construction, any commissioning team member finding deficiencies may communicate those deficiencies in writing to the Commissioning Specialist for inclusion into the Issues Log.

Track construction deficiencies identified in the Issues Log using QCS as specified in Specification Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM).

1.9 CERTIFICATE OF READINESS

Prior to scheduling Functional Performance Tests for each system, issue a Certificate of Readiness for the system certifying that the system is ready for Functional Performance Testing. The Certificate of Readiness must include, for each system to be commissioned, all equipment and system start-up reports; Performance Verification Test Reports; ~~completed Building Envelope Inspection Checklists;~~ completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; HVAC Controls Start-Up Reports; and the Air Leakage Test Reports and Diagnostic Test Reports to the extent applicable to the system. The Contractor; the Lead Commissioning Specialist; the Contractor's Quality Control Representative; the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness. Submit the Certificate of Readiness for each system no later than 14 calendar days prior to Functional Performance Tests of that system. Submit four hard copies and an electronic copy. Do not schedule Functional Performance Tests for a system until the Certificate of Readiness for that system receives approval by the Government.

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

3.1 CONSTRUCTION PHASE

3.1.1 Construction Commissioning Coordination Meeting

The Lead Commissioning Specialist must lead a Construction Commissioning Coordination Meeting no later than 14 days after approval of the Commissioning Firm and Commissioning Specialists to discuss the commissioning process including contract requirements, lines of communication, roles and responsibilities, schedules, documentation requirements, inspection and test procedures, and logistics as specified in this specification section. The Contractor's Superintendent or Project Manager, the Contractor's Quality Control Representative, and the Government must attend this meeting. Invite the User and a Directorate of Public Works Representative, Reserve Support Command Representative, or a Base Civil Engineer Office Representative to attend this meeting.

3.1.2 Construction Phase Commissioning Plan

3.1.2.1 Interim Construction Phase Commissioning Plan

The Lead Commissioning Specialist (CxC) must prepare the Interim Construction Phase Commissioning Plan. Submit the Interim Construction Phase Commissioning Plan no later than 30 calendar days after the Construction Commissioning Coordination Meeting and no later than 14 days prior to the start of construction of the building envelope. Submit four hard copies and an electronic copy.

Identify the commissioning and testing standards and outline the overall commissioning process, the commissioning schedule, the commissioning team members and responsibilities, lines of communication, and documentation requirements for the construction phase of the project~~, and Template Building Envelope Inspection Checklists~~ in the Interim Construction Phase Commissioning Plan. A Template Construction Phase Commissioning Plan is attached to this specification and is available in word format upon request. Please note that the Template Construction Phase Commissioning Plan is provided as a supplement and contains additional detailed requirements not specifically addressed within this specification section. The CxC is required to develop a commissioning plan that is comprehensive in scope and conforms to the requirements of the contract documents.

3.1.2.1.1 Checklists

~~Download example~~Develop ~~Building Envelope Inspection Checklists,~~ Pre-Functional Checklists, Functional Performance Test Checklists, and Integrated Systems Test Checklists~~ for specification section 01 91 00.15 TOTAL BUILDING COMMISSIONING at the following location: http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphic The checklists submitted in the Interim and Final Construction Phase Commissioning Plans must contain the same level of detail shown in the examples. The submitted checklists are not required to match the format of the examples~~. Refer to sections below for description of checklists.

3.1.2.1.2 Contents

In addition to the requirements listed above and those included in the Template Commissioning Plan, include the following in the Interim Construction Phase Commissioning Plan:

a. Plan purpose

b. Commissioning scope

c. Systems to be commissioned

d. Examples and description of development of pre-functional, integrated systems test, and functional performance test checklists

e. Building information

f. Contact information for the Commissioning Specialists, the Contracting Officer's Technical Representative, and the Commissioning Team listed in paragraph Commissioning Team

g. Roles and responsibilities

h. Management plan

i. Owner's Project Requirements

j. Description of design reviews by the Commissioning Specialists

k. Description and templates for site observation reports and the issues log

l. Listing and description of required meetings

m. Identification and sequence of commissioning and acceptance tasks for incorporation into the Project Schedule

n. Listing of required submittals to Government and Commissioning Specialists

o. Description of execution of ~~building envelope inspection,~~ pre-functional checks, integrated systems tests, and functional performance tests

p. Description of Endurance Tests

q. Acceptance testing of critical systems as identified in contract documents

r. Operation and maintenance manual requirements

s. Description of training requirements

t. Description of required Systems Manual

u. Description of the Commissioning Report

~~3.1.2.1.3 Template Building Envelope Inspection Checklists~~

~~The Building Envelope Technical Commissioning Specialist must develop the Template Building Envelope Inspection Checklists. Include items that verify the building materials and construction maintain the required thermal and moisture integrity and air tightness of the building envelope system in the Building Envelope Inspection Checklists.~~

3.1.2.2 Final Construction Phase Commissioning Plan

The Lead Commissioning Specialist (CxC) must prepare the Final Construction Phase Commissioning Plan. Submit the Final Construction Phase Commissioning Plan no later than 30 calendar days prior to the start of Pre-Functional Checks. Submit four hard copies and an electronic copy.

Include the information provided in the Interim Construction Phase Commissioning Plan. In addition, the Technical Commissioning Specialist must develop the Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists for each building, for each system required to be commissioned, and for each component for inclusion in the Final Construction Phase Commissioning Plan.

3.1.2.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. See paragraph Pre-Functional Checks for more information.

3.1.2.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. See paragraph Functional Performance and Integrated Systems Tests for more information. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

a. Notable system features including information about controls to facilitate understanding of system operation.

b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements. Recommendation must clearly indicate that the system should or should not be accepted by the Government.

c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions.

d. Attendees~~—~~.

e. Identification of the equipment involved in the test.

f. Control system feature identification~~—~~.

g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station.

h. Actuator operation observations demonstrating actuator responses to commands from the control system.

i. As-found condition of the system operation.

j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.

k. Space for comments for each test item.

3.1.2.2.3 Integrated Systems Test Checklists

Integrated Systems Test Checklists must include test procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. See paragraph Functional Performance and Integrated Systems Tests for more information. Include the following sections in the Integrated Systems Test Checklists:

a. Notable features of the interconnected systems organized by discipline including information to facilitate understanding of system operation.

b. Conclusions and recommendations. Conclusions must clearly indicate if the systems do or do not perform in accordance with contract requirements. Recommendation must clearly indicate that the systems should or should not be accepted by the Government.

c. Test conditions including date and beginning and ending time.

d. Attendees~~—~~.

e. Identification of the equipment and systems involved in the test.

f. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.

g. Space for comments for each test item.

~~3.1.3 Design Review~~

~~The Lead Commissioning Specialist and Technical Commissioning Specialists must review the construction contract plans and specifications.~~

~~a. Advise the Contracting Officer's Representative of any deficiencies that would prevent the building systems and features from operating or performing effectively and from being adequately maintainable.~~

~~b. The Commissioning Specialists must provide a Design Review Report individually listing each deficiency and the corresponding proposed corrective action necessary for proper system operation or performance. Submit four hard copies and an electronic copy of the report to the Contracting Officer's Representative no later than 30 days after approval of the Commissioning Specialists.~~

~~c. The Lead Commissioning Specialist must participate in a meeting to discuss any items contained in the report no later than 14 calendar days after submission of the report.~~

3.1.3 Construction Submittals

Provide all submittals associated with the systems to be commissioned, including shop drawings; equipment submittals; test plans, procedures, and reports; and resubmittal's to the Commissioning Specialists. The Technical Commissioning Specialist must review the submittals to the extent necessary to verify that the equipment and system installation will comply with the contract requirements.

3.1.4 Inspection and Testing

Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents and the Owner's Project Requirements. Requirements in related specification sections are independent from the requirements of this section and do not satisfy any of the requirements specified in this specification section. Provide all materials, services, and labor required to perform the Pre-Functional Checks~~, Building Envelope Inspection~~, Integrated Systems Tests, and Functional Performance Tests.

3.1.4.1 Commissioning Team

Provide a commissioning representative for each sub-contractor associated with the systems to be commissioned. Each commissioning representative is responsible for coordination of their respective sub-contractor's execution of the commissioning activities and participation in the inspection and testing required by this specification section. The designers listed below are the designers of record for their respective systems. Substitutes must be approved by the Contracting Officer's Representative. The General or Prime Contractor shall provide a detailed schedule of all Functional Performance and Integrated System Testing activities to be accomplished at least 14 calendar days prior to the start date of testing.

~~3.1.4.1.1 Building Envelope Inspections Team~~

~~The following team members must participate in building envelope inspections:~~

~~Designation~~	~~Function~~
~~CxB~~	~~Building Envelope Technical Commissioning Specialist~~
~~QAR~~	~~Contracting Officer's Quality Assurance Representative~~
~~CQC~~	~~Contractor's Quality Control Personnel~~
~~BEC~~	~~Contractor's Building Envelope Commissioning Representative~~
~~[AD]~~	~~[Architectural Designer]~~

3.1.4.1.1 Mechanical System Pre-Functional Checks Team

The following team members must participate in Pre-Functional checks of mechanical systems:

Designation	Function
CxM	Mechanical System Technical Commissioning Specialist
QAR	Contracting Officer's Quality Assurance Representative
CQC	Contractor's Quality Control Personnel
MC	Contractor's Mechanical Commissioning Representative
EC	Contractor's Electrical Commissioning Representative
CC	Contractor's Controls Commissioning Representative
TABC	Contractor's TAB Commissioning Representative
PC	Contractor's Plumbing Commissioning Representative
IC	Contractor's Irrigation Commissioning Representative

3.1.4.1.2 Electrical System Pre-Functional Checks Team

The following team members must participate in Pre-Functional checks of electrical systems:

Designation	Function
CxE	Electrical System Technical Commissioning Specialist

Designation	Function
QAR	Contracting Officer's Quality Assurance Representative
CQC	Contractor's Quality Control Personnel
EC	Contractor's Electrical Commissioning Representative

3.1.4.1.3 Mechanical Systems Test Team

The following team members must participate in Functional Performance, Seasonal,and Integrated Systems Testing of mechanical systems:

Designation	Function
CxM	Mechanical System Technical Commissioning Specialist
QAR	Contracting Officer's Quality Assurance Representative
CQC	Contractor's Quality Control Personnel
MC	Contractor's Mechanical Commissioning Representative
EC	Contractor's Electrical Commissioning Representative
CC	Contractor's Controls Commissioning Representative
TABC	Contractor's TAB Commissioning Representative
PC	Contractor's Plumbing Commissioning Representative
IC	Contractor's Irrigation Commissioning Representative

3.1.4.1.4 Electrical Systems Test Team

The following team members must participate in Functional Performance and Integrated Systems Testing of electrical systems:

Designation	Function
CxE	Electrical System Technical Commissioning Specialist
QAR	Contracting Officer's Quality Assurance Representative
CQC	Contractor's Quality Control Personnel
EC	Contractor's Electrical Commissioning Representative

3.1.4.1.5 Other Pre-Functional and Functional Performance Participants

The following may participate as team members during Pre-Functional Checks and Functional Performance Testing:

Designation	Function
User	Using Agent's Representative

~~3.1.4.2 Building Envelope Inspection~~

~~Document building envelope inspection by the commissioning team using the approved Template Building Envelope Inspection Checklists. Indicate commissioning team member inspection and acceptance of each Building Envelope Inspection Checklist item by initials at the time they are inspected and found to be in conformance with contract requirements. Inspect checklist items before they become hidden as construction progresses.~~

~~a. Submit the completed and initialed Building Envelope Inspection Checklists no later than 7 calendar days after completion of inspection of all checklists items. Submit four hard copies and an electronic copy.~~

~~b. The Building Envelope Technical Commissioning Specialist must make at least two site visits to the site to observe construction of the building envelope in-progress. On each visit, the Building Envelope Commissioning Specialist must review the Contractor's in-progress checklists to ensure that the commissioning team is inspecting the building envelope as required.~~

~~c. The Building Envelope Technical Commissioning Specialist must witness the building envelope pressure tests and diagnostic tests specified in Specification Section 07 05 23 PRESSURE TESTING AN AIR BARRIER SYSTEM FOR AIR TIGHTNESS. The Building Envelope Technical Commissioning Specialist must review the resulting reports and provide recommendations for correction of any deficiencies or further testing.~~

3.1.4.2 Pre-Functional Checks

Pre-Functional Checklists from the approved Final Construction Phase Commissioning Plan must be completed by the commissioning team. Complete one Pre-Functional Checklist for each individual item of equipment or for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures (lighting and plumbing), and controls. Indicate commissioning team member inspection and acceptance of each Pre-Functional Checklist item by initials. Acceptance of each Pre-Functional Checklist item by each team member indicates that item conforms to the construction contract requirements in their area of responsibility. Technical Commissioning Specialist acceptance of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents and the Owner's Project Requirements. Submit the completed and initialed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklists items for each system. Submit four hard copies and an electronic copy. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists.

3.1.4.3 Testing, Adjusting, and Balancing (TAB) Report and Field Acceptance Testing

The Mechanical System Technical Commissioning Specialist must review the pre-final TAB Report required by Specification Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. Resolve all deficiencies prior to TAB Field Acceptance Testing.

The Mechanical System Technical Commissioning Specialist must witness the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Include a certification by the Mechanical Technical Specialist that no outstanding deficiencies exist in the systems relative to Testing, Adjusting, and Balancing with the final TAB Report submittal.

3.1.4.4 HVAC Controls Test Reports

The Mechanical System Technical Commissioning Specialist must review the Performance Verification Testing Plan, Checklists, and Report. Include a certification by the Mechanical System Technical Commissioning Specialist that the submittals contain no deficiencies or that the submittals do not indicate any deficiencies in the HVAC systems or HVAC control systems with each of these submittals.

3.1.4.5 Tests

3.1.4.5.1 Functional Performance and Integrated Systems Tests

Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests. Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

a. Functional Performance Tests and Integrated Systems Tests must be performed with the Contracting Officer's Quality Assurance Representative present.

b. Abort Functional Performance Tests or Integrated Systems Tests when any system deficiency prevents the successful completion of the test.

c. Technical Commissioning Specialists must lead and document all Functional Performance Tests and Integrated Systems Tests for the systems to be commissioned with the Contractor and appropriate sub-contractors performing the Functional Performance Tests and Integrated Systems Tests. The representatives listed in the paragraph Commissioning Team must attend the tests. Abort Functional Performance Tests or Integrated Systems Tests when any required commissioning team member is not present for the test.

3.1.4.5.1.1 Checklist

Use the Functional Performance Test and Integrated Systems Test Checklists from the approved Final Construction Phase Commissioning Plan to guide the

Functional Performance Tests and Integrated Systems Tests. Functional Performance Tests must be performed for each item of equipment and each system required to be commissioned and verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, lighting levels, and all other performance requirements comply with construction contract regardless of the specific items listed within the Functional Performance Test and Integrated Systems Test Checklists provided. Testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Integrated Systems Tests must be performed for the interactive operation between systems such as HVAC systems, fire protection systems, back-up electrical supply, energy generation systems, and other systems, and verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation, correct operation of uninterruptible power supplies or energy generators and connected systems, or lighting system operation during power outage or emergency system activation. The order of components and systems to be tested must be determined by the Technical Commissioning Specialists.

3.1.4.5.1.2 Acceptance

Indicate acceptance of each item of equipment and systems tested by signature of each commissioning team member for each Functional Performance Test or Integrated Systems Test Checklist. The Contractor's Quality Control Representative and the Technical Commissioning Specialists must indicate acceptance after the equipment and systems are free of deficiencies.

3.1.4.5.2 HVAC Test Methods

Perform Functional Performance Tests in accordance with the following:

3.1.4.5.2.1 Prior to Testing

Prior to testing operating modes, sequences of operation, interlocks, and safeties, complete control point-to-point observations, test sensor calibrations, and test actuator commands.

3.1.4.5.2.2 Simulating Conditions

Over-writing control input values through the controls system is not acceptable, unless approved by the Contracting Officer's Representative. Identify proposed exceptions in a protocol submitted to the Contracting Officer's Representative for approval. Before simulating conditions, overwriting values (if approved), or changing set-points, calibrate all sensors, transducers and devices. Below are several examples of exceptions that would be considered acceptable:

a. When varying static pressures inside ductwork can not be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).

b. Dirty filter pressure drops can be simulated using sheets of cardboard at filter face.

c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.

d. High outside air temperatures can be simulated with a hair blower.

e. High entering cooling coil temperatures can be used to simulate entering cooling coil conditions.

f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer's Representative , as noted above, for special cases.

g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 13 degrees C, when the outside air temperature is above 13 degrees C, temporarily change the lockout set point to be minus 18 degrees C above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point such that damage to the components, systems, or the building structure and/or contents will occur.

h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition in ductwork.

i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer's Representative~~-~~.

j. Test dehumidifiers in accordance with the manufacturer's recommendations. Humid conditions can be simulated through an external humidifier or humidifying agent present during testing.

3.1.4.5.2.3 Setup

Perform each test under conditions that simulate actual conditions as close as is practically possible. Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

3.1.4.5.3 Sample Strategy

Perform Functional Performance Tests and Integrated System Tests for all equipment and systems. Prepare and complete a Functional Performance Test checklist for each piece of equipment or system. Prepare and complete an Integrated Systems Test Checklist for all systems and equipment having interactive operation. Test 100 percent of all HVAC Central Plant equipment and primary air handling units. ~~[Test [100] percent of renewable energy systems/equipment].~~ Sample testing of ~~10 units or [~~twenty~~]~~ percent, ~~whichever is greater,~~ is allowed for HVAC equipment with identical controllers such as air terminal units and fan coil units. ~~[Sample testing of [twenty] percent is allowed for other equipment and systems.]~~ Test 100 percent of DOAS (Dedicated outside air units, and all central exhaust fans.

3.1.4.5.4 Seasonal Tests

3.1.4.5.4.1 Initial Functional Performance Tests

Perform Initial Functional Performance Tests as soon as all contract work is completed, regardless of the season. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads.

3.1.4.5.4.2 Full-Load Conditions

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems under full-load conditions during peak heating and cooling seasons during outdoor air condition design extremes. Test cooling equipment and systems with the building fully occupied when performing the Functional Performance Tests during peak cooling season.

Schedule Seasonal Functional Performance Tests in coordination with the Government.

3.1.4.5.4.3 System Acceptance

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract that can be tested during initial Functional Performance Tests. All Functional Performance Test procedures must be completed prior to full systems acceptance.

3.1.4.5.5 Aborted Tests and Re-Testing

Abort Functional Performance Tests, Integrated Systems Tests, or Seasonal Tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. ~~reimburse~~Reimburse the Government for all costs associated with effort lost due to re-testing due to test failures and aborted tests. These costs must include salary, travel costs, and per diem for Government commissioning team members. Re-test only after all deficiencies identified during the original tests have been corrected.

3.1.4.5.5.1 100 Percent Sample

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.1.4.5.5.2 Less than 100 Percent Sample

For systems tests with a sample size less than 100 percent, ~~–~~ if one or more of the test procedures for an item of equipment or a system results in discovery of a deficiency, regardless of whether the deficiency is corrected during the sample tests, the item of equipment or system fails the test.

a. If the system failure rate is 5 percent or less, meaning that 5 percent or less of the equipment or systems tested had at least one deficiency, re-test only on the items which experienced the initial failures.

b. If the system failure rate is greater than 5 percent, meaning that more than 5 percent of equipment or systems tested had at least one deficiency, re-test the items which experienced the initial failures to the extent necessary to confirm that the deficiencies have been corrected. In addition, test another random sample of the same size as the initial sample for the first time. If the second random sample set has any failures, re-test those failed items and all remaining equipment and systems to complete 100 percent testing of that system type.

3.1.5 Training Plan

Develop a training plan which identifies all training required by specification sections associated with commissioned systems. Include a matrix listing each training requirement, content of the training, the trainer name, trainer contact information, and schedule and location of training. Submit four hard copies and an electronic copy of the Training Plan to the Commissioning Specialists and the Government no later than 30 calendar days prior to the associated training.

Document training attendance using training attendance rosters and provide completed attendance rosters to the Commissioning Specialists and the Government no later than 7 calendar days following the completion of training for each system to be commissioned. Submit four hard copies and an electronic copy.

3.1.6 Systems Manual

Prepare and submit a Systems Manual including, for all commissioned systems, the Owner's Project Requirements, system single line diagrams, as-built sequences of operation and controls drawings, as-built control setpoints, recommended schedule for sensor and actuator calibration, recommended schedule of maintenance when not in the O&M manuals, recommended re-testing schedule with proposed testing forms, and full equipment warranty information. Update and resubmit the Systems Manual based on any corrective action taken during the warranty period. Include a signed certification or letter from the Lead Commissioning Specialist with the submittal stating that the Systems Manual is complete, clear, and accurate.

Submit Systems Manual no later than 30 calendar days following completion of Functional Performance Tests. Submit four hard copies and an electronic copy.

3.1.7 Maintenance and Service Life Plans

3.1.7.1 Maintenance Plan

Prepare and submit a Maintenance Plan for the project mechanical, electrical, plumbing, and fire protection systems. Prepare the HVAC and refrigeration sections of the Maintenance Plan in accordance with ASHRAE 180. Develop required inspection and maintenance tasks similar to

Section 5 of ASHRAE 180 for the other commissioned systems and fire protection systems. Ensure Maintenance Plan is coordinated with the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA.

Submit the Maintenance Plan no later than 30 calendar days following the completion of Functional Performance tests. Submit four hard copies and an electronic copy.

3.1.7.2 Service Life Plan

Prepare and submit a Service Life Plan for the building envelope, structural systems, and site hardscape that includes the following for each assembly or component:

a. A description of each including the materials or products.

b. The estimated service life, in years.

c. The estimated maintenance frequency and description of maintenance tasks.

d. The point of maintenance access for the components with estimated service life less than service life of the building.

Submit the Service Life Plan no later than 30 calendar days following the completion of Functional Performance tests. Submit four hard copies and an electronic copy.

3.2 COMMISSIONING REPORT

Following the completion of Functional Performance Tests and Integrated Systems Tests, with the exception of Seasonal Tests, the Lead Commissioning Specialist must prepare a Commissioning Report.

a. Include an executive summary describing the overall commissioning process, the results of the commissioning process, any outstanding deficiencies and recommended resolutions, and any seasonal testing that must be scheduled for a later date. Indicate, in the executive summary, whether the systems meet the requirements of the construction contract and the Owner's Project Requirements.

b. Detail any deficiencies discovered during the commissioning process and the corrective actions taken in the report. Include the completed ~~Building Envelope Inspection Checklists,~~ Pre-Functional Checklists, Functional Performance Test Checklists, Integrated Systems Test Checklists, the Commissioning Plans, the Issues Log, control sequences, meeting minutes, progress and site visit reports, Performance Verification Test Reports, Training Attendance Rosters, the Design Review Report, the final TAB Report.

c. Submit the Commissioning Report no later than 14 calendar days following commissioning team acceptance of all Functional Performance Tests and Integrated Systems Tests with the exception of Seasonal Tests. Submit four hard copies and an electronic copy.

d. Following any Seasonal Tests or Post-Construction Activities, update the Final Commissioning Report to reflect any changes and resubmit.

3.3 POST-CONSTRUCTION SUPPORT

3.3.1 Post-Construction Endurance Test

Perform a one-week Endurance Test once during the peak heating season and once during the peak cooling season during outdoor air condition extremes using the building control system to trend all points necessary to evaluate the installed equipment and system performance or as shown as requiring a trend on the project schedules. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost. Poll all points shown in the project schedules with an alarm condition at 5 minute intervals. Poll all points shown in the project schedules required for trending, overrides, or graphical displays at 15 minute intervals.

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation and operate as required by the construction contract and the Owner's Project Requirements. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation and includes a graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit four hard copies and one electronic copy of the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist.

3.3.2 Post-Construction Site Visit

The Commissioning Specialists must visit the building site concurrent with the 9 month warranty inspection to inspect building system equipment and review building operation with the building operating/maintenance staff. The Commissioning Specialists must identify any deficiency of the building systems to operate in accordance with the contract requirements and the Owner's Project Requirements. The Commissioning Specialists must advise the Contracting Officer's Representative of any identified deficiencies and the proposed corrective action. Submit an updated commissioning report and systems manual documenting the results of the post-construction inspection.

-- End of Section --

SECTION 02 82 00

ASBESTOS REMEDIATION
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~AMERICAN SOCIETY OF SAFETY ENGINEERS (ASSE/SAFE)~~AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSE/SAFE Z9.2	(2012) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM C732	(2006; R 2012) Aging Effects of Artificial Weathering on Latex Sealants
ASTM D2794	(1993; R 2010) Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM D4397	(2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications
ASTM D522/D522M	(2014) Mandrel Bend Test of Attached Organic Coatings
ASTM E119	(20~~18~~20) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E1368	(2014) Visual Inspection of Asbestos Abatement Projects
ASTM E736/E736M	(2017) Standard Test Method for Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members
ASTM E84	~~(2018a) Standard Test Method for Surface Burning Characteristics of Building Materials~~(2023) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(20~~16~~21) Standard Test Methods for Water Vapor Transmission of Materials

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7	(2014) Compressed Air for Human Respiration; 6th Edition

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z87.1	(20~~15~~20) Occupational and Educational Personal Eye and Face Protection Devices

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701	(2015) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH NMAM	(2016; 5th Ed) NIOSH Manual of Analytical Methods

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	~~(2014) Safety and Health Requirements Manual~~ Safety -- Safety and Health Requirements Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS	(Apr 202~~2~~4) Japan Environmental Governing Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1101	Asbestos
29 CFR 1926.200	Accident Prevention Signs and Tags
29 CFR 1926.51	Sanitation
29 CFR 1926.59	Hazard Communication
40 CFR 61-SUBPART A	General Provisions
40 CFR 61-SUBPART M	National Emission Standard for Asbestos
40 CFR 763	Asbestos
42 CFR 84	Approval of Respiratory Protective Devices
49 CFR 107	Hazardous Materials Program Procedures

49 CFR 171	General Information, Regulations, and Definitions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 173	Shippers - General Requirements for Shipments and Packagings

UNDERWRITERS LABORATORIES (UL)

UL 586	(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

1.2 DEFINITIONS

1.2.1 ACM

Asbestos Containing Materials.

1.2.2 Amended Water

Water containing a wetting agent or surfactant with a maximum surface tension of 2.9 Pa.

1.2.3 Area Sampling

Sampling of asbestos fiber concentrations which approximates the concentrations of asbestos in the theoretical breathing zone but is not actually collected in the breathing zone of an employee.

1.2.4 Asbestos

The term asbestos includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, and actinolite asbestos and any of these minerals that has been chemically treated or altered. Materials are considered to contain asbestos if the asbestos content of the material is determined to be at least one tenth of one percent.

1.2.4.1 Asbestos-Containing Material (ACM)

Any materials containing more than one tenth of one percent (0.1%) asbestos.

1.2.5 Asbestos Control Area

That area where asbestos removal operations are performed which is isolated by physical boundaries which assist in the prevention of the uncontrolled release of asbestos dust, fibers, or debris.

1.2.6 Asbestos Fibers

Those fibers having an aspect ratio of at least 3:1 and longer than 5 micrometers as determined by National Institute for Occupational Safety and Health (NIOSH) Method 7400.

1.2.7 Asbestos Permissible Exposure Limit

0.1 fibers per cubic centimeter of air as an 8-hour time weighted average measured in the breathing zone as defined by 29 CFR 1926.1101 or other Federal legislation having legal jurisdiction for the protection of workers health.

1.2.8 Authorized Person

Any person authorized by the Contractor and required by work duties to be present in the regulated areas.

1.2.9 Background

The ambient airborne asbestos concentration in an uncontaminated area as measured prior to any asbestos hazard abatement efforts. Background concentrations for other (contaminated) areas are measured in similar but asbestos free locations.

1.2.10 Building Inspector

Individual who inspects buildings for asbestos and has EPA Model Accreditation Plan (MAP) "Building Inspector" training; accreditation required by 40 CFR 763, Subpart E, Appendix C, has EPA/State certification/license as a "Building Inspector".

1.2.11 Competent Person (CP)

A person meeting the requirements for competent person as specified in 29 CFR 1926.1101 including a person capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, and is specifically trained in a training course which meet the criteria of EPA's Model Accreditation Plan (40 CFR 763) for contractor/supervisor, and is AHERA certified as a contractor/supervisor.

1.2.12 Contractor

The Contractor is that individual, or entity under contract to perform the herein listed work.

1.2.13 Class I Asbestos Work

Activities defined by OSHA involving the removal of thermal system insulation (TSI) and surfacing ACM.

1.2.14 Class II Asbestos Work

Activities defined by OSHA involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos - containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastic. Certain "incidental" roofing materials such as mastic, flashing and cements when they are still intact are excluded from Class II asbestos work. Removal of small amounts of these materials which would fit into a glovebag may be classified as a Class III job.

1.2.15 Class III Asbestos Work

Activities defined by OSHA that involve repair and maintenance operations, where ACM, including TSI and surfacing ACM, is likely to be disturbed. Operations may include drilling, abrading, cutting a hole, cable pulling, crawling through tunnels or attics and spaces above the ceiling, where asbestos is actively disturbed or asbestos-containing debris is actively disturbed.

1.2.16 Class IV Asbestos Work

Maintenance and custodial construction activities during which employees contact but do not disturb ACM and activities to clean-up dust, waste and debris resulting from Class I, II, and III activities. This may include dusting surfaces where ACM waste and debris and accompanying dust exists and cleaning up loose ACM debris from TSI or surfacing ACM following construction

1.2.17 Contractor/Supervisor

Individual who supervises asbestos abatement work and has EPA Model Accreditation Plan "Contractor/Supervisor" training; and is AHERA certified as a "Contractor/Supervisor".

1.2.18 Disposal Bag

A 0.15 mm thick, leak-tight plastic bag, pre-labeled in accordance with 29 CFR 1926.1101, used for transporting asbestos waste from containment to disposal site.

1.2.19 Disturbance

Activities that disrupt the matrix of ACM, crumble or pulverize ACM, or generate visible debris from ACM. Disturbance includes cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glovebag or waste bag, not larger than 1.5 m in length and width in order to access a building component.

1.2.20 Encapsulation

The abatement of an asbestos hazard through the appropriate use of chemical encapsulants.

1.2.21 Encapsulants

Specific materials in various forms used to chemically or physically entrap asbestos fibers in various configurations to prevent these fibers from becoming airborne. There are four types of encapsulants as follows which must comply with performance requirements as specified herein.

a. Removal Encapsulant (can be used as a wetting agent)

b. Bridging Encapsulant (used to provide a tough, durable surface coating to asbestos containing material)

c. Penetrating Encapsulant (used to penetrate the asbestos containing material encapsulating all asbestos fibers and preventing fiber release due to routine mechanical damage)

d. Lock-Down Encapsulant (used to seal off or "lock-down" minute asbestos fibers left on surfaces from which asbestos containing material has been removed).

1.2.22 Friable Asbestos Material

A term defined in the JEGS meaning any material which contains more than 0.1 percent asbestos, as determined using the method specified in the JEGS, that when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

1.2.23 Glovebag Technique

Those asbestos removal and control techniques put forth in 29 CFR 1926.1101 utilizing a bag not more than 1.5 by 1.5 m impervious plastic bag-like enclosure affixed around an asbestos-containing material, with glove-like appendages through which material and tools may be handled.

1.2.24 Government Consultant (GC)

That qualified person employed directly by the Government to monitor, sample, inspect the work or in some other way advise the Contracting Officer. The GC is normally a private consultant, but can be an employee of the Government.

1.2.25 HEPA Filter Equipment

High efficiency particulate air (HEPA) filtered vacuum and exhaust ventilation equipment with a filter system capable of collecting and retaining asbestos fibers. Filters must retain 99.97 percent of particles 0.3 microns or larger as indicated in UL 586.

1.2.26 Industrial Hygienist (IH)/Private Qualified Person (PQP)/ Qualified Person (QP)

A registered Architect, Professional Engineer (licensed), or U.S. Certified Industrial Hygienist who has successfully completed training in, and is accredited under a legitimate Model Accreditation Plan, as described in 40 CFR 763, as a certified Building Inspector, Contractor/Supervisor, and Project Designer. The IH/PQP shall have working knowledge of applicable U.S. Federal and Japanese occupational safety and health regulations for asbestos in construction.

The IH/PQP shall be considered a competent person as defined in 29 CFR 1926.1101. The IH/PQP shall be a third party Contractor and have no employee/employer relationship or financial relationship with the abatement Contractor, which could constitute a conflict of interest.

1.2.27 Industrial Hygiene Technician

The Industrial Hygiene Technician (IHT) shall be an employee of the IH/PQP or the testing laboratory, have a minimum of 2 years experience in the industrial hygiene field working under the direction of the IH/PQP, and have completed the following courses: "Contractor Supervisor Abatement Worker" and "Asbestos Project Designer".

1.2.28 Model Accreditation Plan (MAP)

USEPA training accreditation requirements for persons who work with asbestos as specified in 40 CFR 763.

1.2.29 Negative Initial Exposure Assessment

A demonstration by the Contractor to show that employee exposure during an operation is expected to be consistently below the OSHA Permissible Exposure Limits (PELs).

1.2.30 Negative Pressure Enclosure (NPE)

That engineering control technique described as a negative pressure enclosure in 29 CFR 1926.1101.

1.2.31 NESHAP

National Emission Standards for Hazardous Air Pollutants. The USEPA NESHAP regulation for asbestos is at 40 CFR 61-SUBPART M.

1.2.32 Nonfriable Asbestos Material

A term defined in the JEGS meaning a material that contains asbestos in which the fibers have been immobilized by a bonding agent, coating, binder, or other material so that the asbestos is well bound and will not normally release asbestos fibers during any appropriate use, handling, storage or transportation. It is understood that asbestos fibers may be released under other conditions such as demolition, removal, or mishap.

1.2.33 Permissible Exposure Limits (PELs)

1.2.33.1 PEL-Time Weighted Average(TWA)

Concentration of asbestos not in excess of 0.1 fibers per cubic centimeter of air (f/cc) as an 8-hour time weighted average (TWA).

1.2.33.2 PEL-Excursion Limit

An airborne concentration of asbestos not in excess of 1.0 f/cc of air as averaged over a sampling period of 30 minutes.

1.2.34 Personal Sampling

Air sampling which is performed to determine asbestos fiber concentrations within the breathing zone of a specific employee, as performed in accordance with 29 CFR 1926.1101.

1.2.35 Private Qualified Person (PQP)

See 1.2.26.

1.2.36 Qualified Person (QP)

See 1.2.26.

1.2.37 Regulated Asbestos Containing Material

Friable asbestos material; Category I nonfriable ACM that has become friable; Category I nonfriable ACM that will be or has been subjected to sanding grinding, cutting, or abrading; or Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations.

1.2.38 TEM

Refers to Transmission Electron Microscopy.

1.2.39 Time Weighted Average (TWA)

The TWA is an 8-hour time weighted average airborne concentration of asbestos fibers.

1.2.40 Transite

A generic name for asbestos cement wallboard and pipe.

1.2.41 Wetting Agent

A chemical added to water to reduce the water's surface tension thereby increasing the water's ability to soak into the material to which it is applied. An equivalent wetting agent must have a surface tension of at most 2.9 Pa.

1.2.42 Worker

Individual (not designated as the Competent Person or a supervisor) who performs asbestos work and has completed asbestos worker training required by 29 CFR 1926.1101, to include EPA Model Accreditation Plan (MAP) "Worker" training; accreditation, if required by the OSHA Class of work to be performed.

1.3 REQUIREMENTS

1.3.1 Description of Work

The work covered by this section includes the handling and control of asbestos containing materials and describes some of the resultant procedures and equipment required to protect workers, the environment and occupants of the building or area, or both, from contact with airborne asbestos fibers. The work also includes the disposal of any asbestos containing materials generated by the work. More specific operational procedures must be outlined in the Asbestos Hazard Abatement Plan called for elsewhere in this specification. The asbestos work includes the ~~[~~ demolition and removal~~][encapsulation]~~ of ~~[_____]~~mortar, mastics, puttys and other sealants located ~~[_____]~~on, but not limited to, walls, flooring and covebase material ~~[~~which is governed by 40 CFR 763.~~[~~ Under normal conditions non-friable or chemically bound materials containing asbestos would not be considered hazardous; however, this material may release airborne asbestos fibers during demolition and removal and therefore must be handled in accordance with the removal and disposal procedures as specified herein and/or developed by the IH/PQP and pre-approved by the Government.~~]~~ Provide techniques as outlined in this specification. The ~~[building][~~work area~~]~~ will be evacuated during the asbestos abatement work. A competent person must supervise asbestos removal work as specified herein.

1.3.1.1 Wallboard/Joint Compound

~~[~~ Discrete samples of the components (wallboard and joint compound) have been tested and results are shown in the attached Hazardous Material Survey.

~~][Discrete samples of the wallboard were tested and found to contain [less than one tenth of one percent asbestos] [_____]. Discrete samples of the joint compound were tested and found to contain [greater than one tenth of one percent asbestos] [_____].~~

~~]~~1.3.2 Unexpected Discovery of Asbestos

The Contractor shall be responsible to review the available asbestos survey reports for this project, including, but not limited to:Hazardous Materials Inspection Report - Army Family Housing Renovate Tower B743, by ESC, January 2025.

Additionally, the contractor shall conduct supplemental destructive sampling and inspection prior to mobilization and demolition activities. This inspection should target any suspect materials identified in the original inspection report, as well as identify any additional asbestos containing material in the construction area. It is the contractors responsibility to identify all additional/not previously identified and tested materials during this survey. Except as otherwise determined by the Contracting officer, additional materials found during demolition or other construction activities post-destructive sampling, shall not be considered an unforeseen site condition, as such materials should have been identified during the destructive inspection by the contractor.~~[insert title of report, Company who developed the report and the date of the report]~~

The Contractor's IH/PQP shall review the existing asbestos survey report(s) to develop the appropriate work response to ACM which may be disturbed as part of this project. Should the contract require the Contractor to perform any hazardous material survey, the Contractor shall provide a copy of all surveys/sampling activities and analytical results regarding asbestos to the Contracting Officer. The Asbestos Abatement Contractor shall also submit an Asbestos Hazard Abatement Plan, reviewed and certified by the IH/PQP, to the Contracting Officer for review before asbestos abatement work begins.

The Contractor shall review the existing Hazardous Material Reports and familiarize themselves on the specific homogeneous materials that were tested and all materials that were determined to be ACM. The Contractor shall be responsible for visually confirming the presence of these materials. The Contractor shall remove ACM which would otherwise be disturbed during normal work practices to execute the work required by this project:

The Contractor shall notify the Contracting Officer if any previously untested building components suspected to contain asbestos are discovered and will be impacted by the work. The material(s) in question shall not be disturbed until a response is received by the Contracting Officer. For any previously untested building components suspected to contain asbestos and located in areas impacted by the work, notify the Contracting Officer (CO) who will order up to ~~[_____]~~75 bulk samples to be obtained and delivered to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)" and analyzed by PLM. Samples shall be collected in sufficient amounts such that a follow-on test utilizing TEM or 1000 point count analysis can be performed, should it be necessary.

This cost shall be included in the Contractor's bid and will be performed at no additional cost to the Government. The laboratory shall have a working definition of “Trace” amounts of asbestos, and the laboratory shall report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantification of 1% as a “Trace” concentration. If PLM does not detect the presence of asbestos (e.g. “non-detect”), the material shall be considered <0.1% asbestos. If PLM analysis detects asbestos in any discernible amount (to include “trace” or “less than 1%”), the material shall be considered >0.1% asbestos unless proven to be non-ACM by the use of quantification methods capable of achieving an analytical sensitivity of less than 0.1%, such as Transmission Electron Microscopy (TEM) or 1000 point counting. The CO will order up to ~~[______]~~10 samples to be analyzed further by TEM or 1000 point counting counting at no additional cost to the Government. Any additional components identified as ACM (not identified as ACM at the time of bidding) that will be disturbed will be paid for by an equitable adjustment to the contract price under the CONTRACT CLAUSE titled "changes". Sampling shall be conducted by personnel who have successfully completed the EPA Model Accreditation Plan (MAP) "Building Inspector" training course and is EPA/State certified/licensed as a "Building Inspector".

1.3.3 Medical Requirements

Provide medical requirements including but not limited to medical surveillance and medical record keeping as listed in 29 CFR 1926.1101.

1.3.3.1 Medical Examinations

Before exposure to airborne asbestos fibers, provide workers with a comprehensive medical examination as required by 29 CFR 1926.1101 or other pertinent local, prefectural or Government of Japan (GOJ) directives. This requirement must have been satisfied within the 12 months prior to the start of work on this contract. The same medical examination must be given on an annual basis to employees engaged in an occupation involving asbestos and within 30 calendar days before or after the termination of employment in such occupation. Specifically identify x-ray films of asbestos workers to the consulting radiologist and mark medical record jackets with the word "ASBESTOS."

1.3.3.2 Medical Records

Maintain complete and accurate records of employees' medical examinations, medical records, and exposure data for a period of as required by Government of Japan law~~[30 years][indefinite time]~~ after termination of employment and make records of the required medical examinations and exposure data available for inspection and copying to: The Assistant Secretary of Labor for Occupational Safety and Health (OSHA), or authorized representatives of them, and an employee's physician upon the request of the employee or former employee.

1.3.4 Employee Training

Submit certificates, prior to the start of work but after the main abatement submittal, signed by each employee indicating that the employee has received training in the proper handling of materials and wastes that contain asbestos in accordance with 40 CFR 763; understands the health implications and risks involved, including the illnesses possible from exposure to airborne asbestos fibers; understands the use and limits of the respiratory equipment to be used; and understands the results of monitoring of airborne quantities of asbestos as related to health and respiratory equipment as indicated in 29 CFR 1926.1101 on an initial and annual basis. Organize certificates by individual worker, not grouped by type of certification.~~[Post appropriate evidence of compliance with the training requirements of 40 CFR 763.]~~ Train personnel involved in the asbestos control work in accordance with United States Environmental Protection Agency (USEPA) Asbestos Hazard Emergency Response Act (AHERA) training criteria. Document the training by providing: dates of training, training entity, course outline, names of instructors, and qualifications of instructors upon request by the Contracting Officer. Furnish each employee with respirator training and fit testing administered by the PQP as required by 29 CFR 1926.1101 and 29 CFR 1926.103. Fully cover engineering and other hazard control techniques and procedures.

1.3.5 Permits~~[, Licenses,]~~ and Notifications

Prior to the start of work, obtain necessary permits~~[and licenses]~~ in conjunction with asbestos removal, encapsulation, hauling, and disposition, and furnish notification of such actions required by local, prefectural and GOG regulations. Notify the Contracting Officer in writing ~~[10]~~ ~~[~~20~~]~~ ~~[_____]~~ working days prior to commencement of work in accordance with 40 CFR 61-SUBPART M and applicable laws, ordinances, criteria, rules, and regulations~~[_____]~~. In accordance with the JEGS, prior to demolition or renovation of a facility that involves removing or disturbing friable ACM, a written assessment of the action will be prepared and submitted to the installation commander via the Contracting Officer.

The Asbestos assessment shall include the following information:

1. Type of Operation: demolition or renovation
2. Description of the facility or affected part of the facility, including the age, size, and present and past use of the facility.
3. Procedure including analytical methods, use to detect the presence of regulated ACM and category I and category II nonfriable ACM.
4. Estimated amount of a)regulated ACM to be removed from the facility and b)Category I and II nonfriable ACM in the affected part of the facility that will not be removed before demolition
5. Location, street address, and building number of the facility being demolished or renovated.
6. Scheduled starting and completion dates of asbestos removal work, or any other activity such as site preparation, that would break up, dislodge or similarly disturb asbestos material.
7. Scheduled staring and completion dates of demolition or renovation.
8. Description of planned demolition or renovation work and method(s) to be used, including a description of affected facility components.
9. Description of work practices and engineering controls, including asbestos removal and waste-handling emission control procedures.
10. Name and location of the waste disposal site where the asbestos-containing waste material will be deposited.
11.Description of procedures to follow if unexpected regulated ACM is found or Category II nonfriable ACM becomes crumbled, pulverized, or reduced to powder.

Submit copies of all Notifications to the Contracting Officer who will need to provide to the Installation Asbestos Manager for their records in accordance with the JEGS.~~[Notify the local fire department 3 days prior to removing fire-proofing material from the building including notice that~~

~~the material contains asbestos.]~~

1.3.6 Environment, Safety and Health Compliance

In addition to detailed requirements of this specification, comply with those applicable laws, ordinances, criteria, rules, and regulations of local prefectural and GOJ regulations regarding handling, storing, transporting, and disposing of asbestos waste materials. Comply with the applicable requirements of the current issue of EM 385-1-1, 29 CFR 1926.1101, 40 CFR 61-SUBPART A, 40 CFR 61-SUBPART M, 40 CFR 763~~, and ND OPNAVINST 5100.23~~.~~.~~ Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting the work. Where the requirements of this specification, applicable laws, rules, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirement as defined by the Government apply. The following laws, ordinances, criteria, rules and regulations regarding removal, handling, storing, transporting and disposing of asbestos materials apply:

a. ~~[_____]~~JEGS

~~b. [_____]~~

~~c. [_____].~~

1.3.7 Respiratory Protection Program

Establish and implement a respirator program as required by 29 CFR 1926.1101, and 29 CFR 1926.103. Submit a written description of the program to the Contracting Officer. Submit a written program manual or operating procedure including methods of compliance with regulatory statutes.

1.3.7.1 Respirator Program Records

Submit records of the respirator program as required by 29 CFR 1926.103, and 29 CFR 1926.1101.

1.3.7.2 Respirator Fit Testing

The Contractor's PQP must conduct a qualitative or quantitative fit test conforming to 29 CFR 1926.103 for each worker required to wear a respirator, and any authorized visitors who enter a regulated area where respirators are required to be worn. A respirator fit test must be performed prior to initially wearing a respirator and every 12 months thereafter. If physical changes develop that will affect the fit, a new fit test must be performed. Functional fit checks must be performed each time a respirator is put on and in accordance with the manufacturer's recommendation.

1.3.7.3 Respirator Selection and Use Requirements

Provide respirators, and ensure that they are used as required by 29 CFR 1926.1101 and in accordance with CGA G-7 and the manufacturer's recommendations. Respirators must be approved by the National Institute for Occupational Safety and Health NIOSH, under the provisions of 42 CFR 84, for use in environments containing airborne asbestos fibers. For air-purifying respirators, the particulate filter must be high-efficiency particulate air (HEPA)/(N-,R-,P-100). The initial respirator selection and the decisions regarding the upgrading or downgrading of respirator type must be made by the Contractor's Designated IH based on the measured or anticipated airborne asbestos fiber concentrations to be encountered.

1.3.8 Asbestos Hazard Control Supervisor

The Contractor must be represented on site by a supervisor, trained using the model Contractor accreditation plan as indicated in the Federal statutes for all portions of the herein listed work.

1.3.9 Hazard Communication

Adhere to all parts of 29 CFR 1926.59 and provide the Contracting Officer with a copy of the Safety Data Sheets (SDS) for all materials brought to the site.

1.3.10 Asbestos Hazard Abatement Plan

Submit a detailed plan of the safety precautions such as lockout, tagout, tryout, fall protection, and confined space entry procedures and equipment and work procedures to be used in the ~~[encapsulation] [~~removal~~]~~ ~~[~~and demolition~~]~~ of materials containing asbestos. The plan, not to be combined with other hazard abatement plans, must be prepared, signed, and sealed by the PQP. Provide a Table of Contents for each abatement submittal, which follows the sequence of requirements in the contract. The plan must include but not be limited to the precise personal protective equipment to be used including, but not limited to, respiratory protection, type of whole-body protection~~[and if reusable coveralls are to be employed decontamination methods (operations and quality control plan)]~~, the location of asbestos control areas including clean and dirty areas, buffer zones, showers, storage areas, change rooms, ~~[~~removal~~]~~ ~~[encapsulation]~~ method, interface of trades involved in the construction, sequencing of asbestos related work, disposal plan, type of wetting agent and asbestos sealer to be used, locations of local exhaust equipment, planned air monitoring strategies, and a detailed description of the method to be employed in order to control environmental pollution. The plan must also include (both fire and medical emergency) response plans and an Activity Hazard Analyses (AHAs) in accordance with EM 385-1-1. The Asbestos Hazard Abatement Plan must be approved in writing prior to starting any asbestos work. The Contractor, Asbestos Hazard Control Supervisor,, CP and PQP must meet with the Contracting Officer prior to beginning work, to discuss in detail the Asbestos Hazard Abatement Plan, including work procedures and safety precautions. Once approved by the Contracting Officer, the plan will be enforced as if an addition to the specification. Any changes required in the specification as a result of the plan must be identified specifically in the plan to allow for free discussion and approval by the Contracting Officer prior to starting work.

1.3.11 Testing Laboratory

Submit the name, address, and telephone number of each testing laboratory selected for the~~[sampling,]~~ analysis, and reporting of airborne concentrations of asbestos fibers along with~~[~~ evidence that each laboratory selected holds the appropriate State license and permits and~~]~~ certification that each laboratory is American Industrial Hygiene Association (AIHA) accredited and that persons counting the samples have been judged proficient by current inclusion on the AIHA Asbestos Analysis Registry (AAR) and successful participation of the laboratory in the Proficiency Analytical Testing (PAT) Program. Where analysis to determine asbestos content in bulk materials or transmission electron microscopy is required, submit evidence that the laboratory is accredited by the National Institute of Science and Technology (NIST) under National Voluntary Laboratory Accreditation Program (NVLAP) for asbestos analysis. The testing laboratory firm must be independent of the asbestos contractor and must have no employee or employer relationship which could constitute a conflict of interest.

1.3.12 Landfill Approval

Submit written evidence that the landfill is approved for asbestos disposal by local prefectural and GOJ regulatory agencies. Within three working days after delivery, submit detailed delivery tickets, prepared, signed, and dated by an agent of the landfill, certifying the amount of asbestos materials delivered to the landfill. Submit a copy of the waste shipment records within one day of the shipment leaving the project site.

1.3.13 Transporter Certification

Submit written evidence that the transporter is approved to transport asbestos waste in accordance with the DOT requirements of 49 CFR 171, 49 CFR 172 and 49 CFR 173 as well as registration requirements of 49 CFR 107 and all other GOJ and local prefectural regulatory agency requirements.

1.3.14 Medical Certification

Provide a written certification for each worker and supervisor, signed by a licensed physician indicating that the worker and supervisor has met or exceeded all of the medical prerequisites listed herein and in 29 CFR 1926.1101 and 29 CFR 1926.103 as prescribed by law. Submit certificates prior to the start of work but after the main abatement submittal.

1.4 SUBMITTALS

The Contractor's PQP shall, at a minimum, review and certify all submittals required under this section prior to submission to the Government. Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government~~]~~. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Amended Water;

Safety Data Sheets (SDS) for All Materials;

Encapsulants;

Respirators;

Local Exhaust Equipment;

Pressure Differential Automatic Recording Instrument;

Vacuums;

[Glovebags;

] SD-06 Test Reports

Air Sampling Results;G

Pressure Differential Recordings for Local Exhaust System;

[~~Encapsulation Test Patches;~~

] Clearance Sampling;

Asbestos Disposal Quantity Report;G

SD-07 Certificates

Employee Training;G

Notifications;

Respiratory Protection Program;

Asbestos Hazard Abatement Plan;G

Testing Laboratory; G

Landfill Approval; G

Delivery Tickets;

Waste Shipment Records;

Transporter Certification;

Medical Certification;

Private Qualified Person Documentation;G

Competent Person; G

Worker's License; G

Contractor's License; G

Encapsulants;

Equipment Used to Contain Airborne Asbestos Fibers;

Water Filtration Equipment;

Vacuums;

Ventilation Systems;

SD-11 Closeout Submittals

Permits~~[and Licenses]~~;

Notifications;

Respirator Program Records;

Rental Equipment;

1.5 QUALITY ASSURANCE

1.5.1 Additional Requirements

In addition to detailed requirements of this specification, work performed under this contract shall comply with the JEGS, EM 385-1-1, applicable federal, GOJ, and local laws, ordinances, criteria, rules and regulations regarding handling, storing, transporting, and disposing of asbestos waste materials. Matters of interpretation of standards shall be submitted to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements shall apply. The following GOJ and local laws, rules and regulations regarding demolition, removal, encapsulation, construction alteration, repair, maintenance, renovation, spill/emergency cleanup, housekeeping, handling, storing, transporting and disposing of asbestos material apply:

1. Japan Environmental Governing Standards (JEGS), most recent version.

1.5.2 Private Qualified Person Documentation

Submit the name, address, telephone number, and resume of the Private Qualified Person (PQP) selected to prepare the Asbestos Hazard Abatement Plan, direct monitoring and training, direct air monitoring and assist the Contractor's Competent Person in implementing and ensuring that safety and health requirements are complied with during the performance of all required work, and documented evidence that the PQP has successfully completed training in and is accredited and where required is certified as, a Building Inspector and Contractor/Supervisor Abatement Worker. as described by 40 CFR 763. Submit certification to show that the PQP is a Registered Architect, Professional Engineer (licensed) or US Certified Industrial Hygienist determined and documented by the American Board of Industrial Hygiene (ABIH). A copy of the PQP's current valid ABIH certification and/or documentation that the PQP is a registered Architect or Professional Engineer shall be provided. The PQP and the asbestos contractor must not have an employee/employer relationship or financial relationship which could constitute a conflict of interest. The PQP must be a first tier subcontractor. The Designated IH/PQP shall ~~[~~be onsite at all times~~]~~ ~~[visit the site at least [_____] per [month] [week]]~~ for the duration of asbestos activities and shall be available for emergencies. In addition, submit resumes of additional IH/PQP's and industrial hygiene technicians (IHT) who will be assisting the Designated IH/PQP in performing onsite tasks. IH/PQPs and IHTs supporting the Designated IH/PQP shall have a minimum of ~~[~~2 years~~]~~ ~~[_____]~~ of practical onsite asbestos abatement experience. Indicate the formal reporting relationship between the Designated IH/PQP and the support IH/PQPs and IHTs, the Designated Competent Person, and the Contractor.

1.5.3 Competent Person Documentation

The Competent Person must be experienced in the administration and supervision of asbestos abatement projects including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, notification of other employees onsite, ~~[______]~~. The Competent Person must be on-site at all times when asbestos abatement activities are underway. Submit training certification and a current EPA MAP "Contractor/Supervisor" training certificates.. Submit evidence that the Competent Person has a minimum of 2 years of comprehensive experience in planning and overseeing asbestos abatement activities and is experienced in the administration and supervision of asbestos abatement projects relevant to OSHA competent person requirements, including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, notification of other employees onsite, etc. The Designated Competent Person shall be responsible for compliance with the JEGS, applicable U.S. federal, GOJ, and local requirements, the Contractor's Accident Prevention Plan (APP) and Asbestos Hazard Abatement Plan (AHAP).

1.5.4 Worker's License

Submit documentation that workers meet the requirements of 29 CFR 1926.1101, 40 CFR 61-SUBPART M and other applicable federal, GOJ, and local requirements. Worker training documentation shall be provided as required on the "Certificate of Workers Acknowledgment". Training documentation is required for each employee who will perform OSHA Class I, Class II, Class III, or Class IV asbestos abatement operations. Such documentation shall be submitted on a Contractor generated form titled "Certificate of Workers Acknowledgment", to be completed for each employee in the same format. Training course completion certificates (initial and most recent update refresher) required by the information checked on the form shall be attached..

1.5.5 Contractor's License

Submit a copy of the asbestos contractor's AHERA Asbestos Worker(s) certification. Submit the following certification along with the license: "I certify that the personnel I am responsible for during the course of this project fully understand the contents of 29 CFR 1926.1101, 40 CFR 61-SUBPART M, EM 385-1-1, and the Federal, JEGS, GOJ and local prefecture requirement for those asbestos abatement activities that they will be involved in." This certification statement must be signed by the Company's President or Chief Executive.

1.5.6 Air Sampling Results

Complete fiber counting and provide results to the PQP for review within 16 hours of the "time off" of the sample pump. Notify the Contracting Officer immediately of any airborne levels of asbestos fibers in excess of the acceptable limits. Submit sampling results to the Contracting Officer and the affected Contractor employees where required by law within three working days, signed by the testing laboratory employee performing air sampling, the employee that analyzed the sample, and the PQP and GC. Notify the Contractor and the Contracting Officer immediately of any results in excess of 0.01 fibers per cubic centimeter or background whichever is higher. In no circumstance must levels exceed 0.1 fibers per cubic centimeter.

1.5.7 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.51 mm of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the asbestos control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer within 24-hours from the end of each work day. Notify the Contractor and the Contracting Officer immediately of any variance in the pressure differential which could cause adjacent unsealed areas to have asbestos fiber concentrations in excess of 0.01 fibers per cubic centimeter or background whichever is higher.

~~[1.5.8 Protective Clothing Decontamination Quality Control Records~~

~~Keep all records that document quality control for the decontamination of reusable outer protective clothing in the Environmental Records Binder, on the project site.~~

~~][1.5.9 Protective Clothing Decontamination Facility Notification~~

~~Submit written evidence that persons who decontaminate, store, or transport asbestos contaminated clothing used in the performance of this contract were duly notified in accordance with 29 CFR 1926.1101.~~

~~]~~1.5.8 GOJ, Prefectural or Local Citations on Previous Projects

Keep a statement, signed by an officer of the company, containing a record of any citations issued by GOJ, Prefectural or local regulatory agencies relating to asbestos activities within the last 5 years (including projects, dates, and resolutions); a list of penalties incurred through non-compliance with asbestos project specifications, including liquidated damages, overruns in scheduled time limitations and resolutions; and situations in which an asbestos-related contract has been terminated (including projects, dates, and reasons for terminations) in the Environmental Records Binder on the project site. If there are none, a negative declaration signed by an officer of the company must be placed in the binder.

1.5.9 Preconstruction Conference

Conduct a safety preconstruction conference to discuss the details of the Asbestos Hazard Abatement Plan reviewed and certified by the Contractor's PQP, Accident Prevention Plan (APP) including the AHAs required in specification Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS ~~[_____]~~. The safety preconstruction conference must include the Contractor and their Designated Competent Person, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted. ~~[_____]~~

1.6 SECURITY

All work areas shall be secured to keep unauthorized personnel out of the area. This may be by locking of doors, installation of barricades, or other means. However, the contractor shall ensure that the security measures are adequate to prevent access 24 hours a day.~~[[Twenty-four hour security guard] [Fenced and locked security area] [[_____] shall be provided for each regulated area.] A log book must be kept documenting entry into and out of the regulated area. Entry into regulated areas must only be by personnel authorized by the Contractor and the Contracting Officer. Personnel authorized to enter regulated areas must be trained, medically evaluated, and wear the required personal protective equipment.]~~

1.7 EQUIPMENT

1.7.1 Rental Equipment

Provide a copy of the written notification to the rental company concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

PART 2 PRODUCTS

2.1 ENCAPSULANTS

Encapsulants must conform to current USEPA requirements, contain no toxic or hazardous substances as defined in 29 CFR 1926.59, and conform to the following performance requirements. Submit certificates stating that encapsulants meet the applicable specified performance requirements.

2.1.1 Removal Encapsulants

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3 hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 245.5 mm/N	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.2 Bridging Encapsulant

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 245.5 mm/N	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.3 Penetrating Encapsulant

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Cohesion/Adhesion Test - 729.5 N of force/meter	ASTM E119
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Classified by UL for use over fibrous and cementitious sprayed fireproofing)	ASTM E119
Impact Resistance - Minimum 245.5 mm/N	ASTM D2794 Gardner Impact Test
Flexibility - no rupture or cracking	ASTM D522/D522M Mandrel Bend Test

2.1.4 Lock-down Encapsulant

Requirement	Test Standard
Flame Spread - 25, Smoke Emission - 50	ASTM E84
Life Expectancy - 20 years	ASTM C732 Accelerated Aging Test
Permeability - Minimum 0.4 perms	ASTM E96/E96M
Fire Resistance - Negligible affect on fire resistance rating over 3-hour test (Tested with fireproofing over encapsulant applied directly to steel member)	ASTM E119
Bond Strength: 1459 N of force/meter	ASTM E736/E736M
(Tests compatibility with cementitious and fibrous fireproofing)	

~~[2.2 ENCASEMENT PRODUCTS~~

~~Encasement must consist of primary cellular polymer coat, polymer finish coat, and any other finish coat as approved by the Contracting Officer.~~

~~]~~2.2 DUCT TAPE

Industrial grade duct tape of appropriate widths suitable for bonding sheet plastic and disposal container.

2.3 DISPOSAL CONTAINERS

Leak-tight (defined as solids, liquids, or dust that cannot escape or spill out) disposal containers must be provided for ACM wastes as required by 29 CFR 1926.1101. Disposal containers can be in the form of:

a. Disposal Bags

b. Fiberboard Drums

c. Cardboard Boxes

2.4 SHEET PLASTIC

Sheet plastic must be polyethylene of 0.15 mm minimum thickness and must be provided in the largest sheet size necessary to minimize seams~~[, as indicated on the project drawings]~~. Film must be ~~[~~clear~~][~~frosted~~][~~ or ~~][~~black~~]~~ and conform to ASTM D4397, except as specified below

2.4.1 Flame Resistant

Where a potential for fire exists, flame-resistant sheets must be provided. Film must be ~~[~~frosted~~][~~ or ~~][~~black~~]~~ and must conform to the requirements of NFPA 701.

2.4.2 Reinforced

Reinforced sheets must be provided where high skin strength is required, such as where it constitutes the only barrier between the regulated area and the outdoor environment. The sheet stock must consist of translucent, nylon-reinforced or woven-polyethylene thread laminated between 2 layers of polyethylene film. Film must meet flame resistant standards of NFPA 701.

2.5 MASTIC REMOVING SOLVENT

Mastic removing solvent must be nonflammable and must not contain methylene chloride, glycol ether, or halogenated hydrocarbons. Solvents used onsite must have a flash point greater than 60 degrees C.

2.6 LEAK-TIGHT WRAPPING

Two layers of 0.15 mm minimum thick polyethylene sheet stock must be used for the containment of removed asbestos-containing components or materials such as large tanks, boilers, insulated pipe segments and other materials. Upon placement of the ACM component or material, each layer must be individually leak-tight sealed with duct tape.

2.7 VIEWING INSPECTION WINDOW

Where feasible, a minimum of one clear, 3 mm thick, acrylic sheet,450 by 610 mm, must be installed as a viewing inspection window at eye level on a wall in each containment enclosure. The windows must be sealed leak-tight with industrial grade duct tape.

2.8 WETTING AGENTS

Removal encapsulant (a penetrating encapsulant) must be provided when conducting removal abatement activities that require a longer removal time or are subject to rapid evaporation of amended water. The removal encapsulant must be capable of wetting the ACM and retarding fiber release during disturbance of the ACM greater than or equal to that provided by amended water. Performance requirements for penetrating encapsulants are specified in paragraph ENCAPSULANTS above.

PART 3 EXECUTION

3.1 EQUIPMENT

Provide the Contracting Officer or the Contracting Officer's Representative, with at least ~~[~~two~~]~~ ~~[_____]~~ complete sets of personal protective equipment ~~[including decontaminating reusable coveralls]~~as required for entry to and inspection of the asbestos control area. Provide equivalent training to the Contracting Officer or a designated representative as provided to Contractor employees in the use of the required personal protective equipment. Provide manufacturer's certificate of compliance for all equipment used to contain airborne asbestos fibers.

3.1.1 Air Monitoring Equipment

The Contractor's PQP must approve air monitoring equipment. The equipment must include, but must not be limited to:

a. High-volume sampling pumps that can be calibrated and operated at a constant airflow up to 16 liters per minute.

b. Low-volume, battery powered, body-attachable, portable personal pumps that can be calibrated to a constant airflow up to approximately 3.5 liters per minute, and a self-contained rechargeable power pack capable of sustaining the calibrated flow rate for a minimum of 10 hours. The pumps must also be equipped with an automatic flow control unit which must maintain a constant flow, even as filter resistance increases due to accumulation of fiber and debris on the filter surface.

c. Single use standard 25 mm diameter cassette, open face, 0.8 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive extension cowl, and shrink bands for personal air sampling.

~~[~~ d. Single use standard 25 mm diameter cassette, open face, 0.45 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive cowl, and shrink bands when conducting environmental area sampling using NIOSH NMAM Methods 7400 and 7402, (and the transmission electric microscopy method specified at 40 CFR 763 if required).

~~]~~ e. A flow calibrator capable of calibration to within plus or minus 2 percent of reading over a temperature range of minus 20 to plus 60 degrees C and traceable to a NIST primary standard.

3.1.2 Respirators

Select respirators from those approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services.

3.1.2.1 Respirators for Handling Asbestos

Provide personnel engaged in pre-cleaning, cleanup, handling, ~~[encapsulation]~~~~[~~removal~~][~~ and ~~][~~ or ~~][~~demolition~~]~~ of asbestos materials with respiratory protection as indicated in 29 CFR 1926.1101 and 29 CFR 1926.103. Breathing air must comply with CGA G-7.

3.1.3 Exterior Whole Body Protection

3.1.3.1 Outer Protective Clothing

Provide personnel exposed to asbestos with disposable "non-breathable,"~~[or reusable "non-breathable"]~~ whole body outer protective clothing, head coverings, gloves, and foot coverings. Provide disposable plastic or rubber gloves to protect hands. Cloth gloves may be worn inside the plastic or rubber gloves for comfort, but must not be used alone. Make sleeves secure at the wrists, make foot coverings secure at the ankles, and make clothing secure at the neck by the use of tape.~~[Reusable whole body outer protective clothing must be either disposed of as asbestos contaminated waste upon exiting from the asbestos regulated work area or be properly decontaminated.]~~

3.1.3.2 Work Clothing

Provide cloth work clothes for wear under the outer protective clothing and foot coverings and either dispose of or properly decontaminate them as recommended by the ~~[GC][~~PQP~~]~~ after each use.

3.1.3.3 Personal Decontamination Unit

Provide a temporary, negative pressure unit with a separate decontamination locker room and clean locker room with a shower that complies with 29 CFR 1926.51(f)(4)(ii) through (V) in between for personnel required to wear whole body protective clothing. Provide two separate lockers for each asbestos worker, one in each locker room. Keep street clothing and street shoes in the clean locker. HEPA vacuum and remove asbestos contaminated disposable protective clothing while still wearing respirators at the boundary of the asbestos work area and seal in impermeable bags or containers for disposal.~~[HEPA vacuum and remove asbestos contaminated reusable protective clothing while still wearing respirators at the boundary of the asbestos work area, seal in two impermeable bags, label outer bag as asbestos contaminated waste, and transport for decontamination.]~~ Do not wear work clothing between home and work. Locate showers between the decontamination locker room and the clean locker room and require that all employees shower before changing into street clothes. Collect used shower water and filter with approved water filtration equipment to remove asbestos contamination. Wastewater filters must be installed in series with the first stage pore size ~~[~~20~~]~~ ~~[_____]~~ microns and the second stage pore size of ~~[~~5~~]~~ ~~[_____]~~ microns. Dispose of filters and residue as asbestos waste. Discharge clean water to the sanitary system. Dispose of asbestos contaminated work clothing as asbestos contaminated waste~~[or properly decontaminate as specified in the Contractor's Asbestos Hazard Abatement Plan]~~. Keep the floor of the decontamination unit's clean room dry and clean at all times. Proper housekeeping and hygiene requirements must be maintained. Provide soap and towels for showering, washing and drying. Cloth towels provided must be disposed of as ACM waste or must be laundered in accordance with 29 CFR 1926.1101. Physically attach the decontamination units to the asbestos control area. Construct both a personnel decontamination unit and an equipment decontamination unit onto and integral with each asbestos control area.

~~[3.1.3.4 Decontamination of Reusable Outer Protective Clothing~~

~~When reusable outer protective clothing is used, transport the double bagged clothing to a previously notified commercial/industrial decontamination facility for decontamination. Perform non-destructive testing to determine the effectiveness of asbestos decontamination. If representative sampling is used, ensure the statistical validity of the sampling results. If representative sampling is used, reject any entire batch in which any of the pieces exceed 40 fibers per square millimeter. Inspect reusable protective clothing prior to use to ensure that it will provide adequate protection and is not or is not about to become ripped, torn, deteriorated, or damaged, and that it is not visibly contaminated. Notify, in writing, all personnel involved in the decontamination of reusable outer protective clothing as indicated in 29 CFR 1926.1101.~~

~~]~~3.1.3.4 Eye Protection

Provide eye protection that complies with ANSI/ISEA Z87.1 when operations present a potential eye injury hazard. Provide goggles to personnel engaged in asbestos abatement operations when the use of a full face respirator is not required.

3.1.4 Regulated Areas

All Class I, II, and III asbestos work must be conducted within regulated areas. The regulated area must be demarcated to minimize the number of persons within the area and to protect persons outside the area from exposure to airborne asbestos. Control access to regulated areas, ensure that only authorized personnel enter, and verify that Contractor required medical surveillance, training and respiratory protection program requirements are met prior to allowing entrance.

3.1.5 Load-out Unit

Provide a temporary load-out unit that is adjacent and connected to the regulated area~~[~~ and ~~][~~access tunnel~~]~~. Attach the load-out unit in a leak-tight manner to each regulated area.

3.1.6 Warning Signs and Labels

Provide bilingual warning signs printed in English and Japanese at all approaches to asbestos control areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scrap, waste, debris, and other products contaminated with asbestos. Containers with preprinted warning labels conforming to the requirements are acceptable

3.1.6.1 Warning Sign

Provide vertical format conforming to 29 CFR 1926.200, and 29 CFR 1926.1101 minimum 500 by 355 mm displaying the following legend in the lower panel:

Legend	Notation
DANGER	25 mm Sans Serif Gothic or Block
ASBESTOS	25 mm Sans Serif Gothic or Block
MAY CAUSE CANCER	25 mm Sans Serif Gothic or Block
CAUSES DAMAGE TO LUNGS	6 mm Sans Serif Gothic or Block
AUTHORIZED PERSONNEL ONLY	6 mm Sans Serif Gothic or Block
~~[~~WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA~~]~~	6 mm Sans Serif Gothic or Block

Spacing between lines must be at least equal to the height of the upper of any two lines.

3.1.6.2 Warning Labels

Provide labels conforming to 29 CFR 1926.1101 of sufficient size to be clearly legible, displaying the following legend:

DANGER
CONTAINS ASBESTOS FIBERS
MAY CAUSE CANCER
CAUSES DAMAGE TO LUNGS
DO NOT BREATHE DUST AVOID CREATING DUST

3.1.7 Local Exhaust System

Provide a local exhaust system in the asbestos control area in accordance with ASSE/SAFE Z9.2 and 29 CFR 1926.1101 that will provide at least four air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the asbestos control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the control area of minus 0.51 mm of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the asbestos control area. Filters on exhaust equipment must conform to ASSE/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.8 Tools

Vacuums must be leak proof to the filter and equipped with HEPA filters. Filters on vacuums must conform to ASSE/SAFE Z9.2 and UL 586. Do not use power tools to remove asbestos containing materials unless the tool is equipped with effective, integral HEPA filtered exhaust ventilation systems. Remove all residual asbestos from reusable tools prior to storage or reuse. Reusable tools must be thoroughly decontaminated prior to being removed from the regulated areas.

3.1.9 Rental Equipment

If rental equipment is to be used, furnish written notification to the rental agency concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

3.1.10 Glovebags

Submit written manufacturers proof that glovebags will not break down under expected temperatures and conditions. The glovebag assembly shall be 0.15 mm thick plastic, prefabricated and seamless at the bottom with preprinted OSHA warning label.

~~]~~3.1.11 Single Stage Decontamination Area

A decontamination area (equipment room/area) must be provided for Class I work involving less than 7.5 m or 0.9 square meters of TSI or surfacing ACM, and for Class II and Class III asbestos work operations where exposures exceed the PELs or where there is no negative exposure assessment. The equipment room or area must be adjacent to the regulated area for the decontamination of employees, material, and their equipment which could be contaminated with asbestos. The area must be covered by an impermeable drop cloth on the floor or horizontal working surface. The area must be of sufficient size to accommodate cleaning of equipment and removing personal protective equipment without spreading contamination beyond the area.

3.1.12 Decontamination Area Exit Procedures

Ensure that the following procedures are followed:

a. Before leaving the regulated area, remove all gross contamination and debris from work clothing using a HEPA vacuum.

b. Employees must remove their protective clothing in the equipment room and deposit the clothing in labeled impermeable bags or containers for disposal or laundering.

c. Employees must not remove their respirators until showering.

d. Employees must shower prior to entering the clean room. If a shower has not been located between the equipment room and the clean room or the work is performed outdoors, ensure that employees engaged in Class I asbestos jobs: a) Remove asbestos contamination from their work suits in the equipment room or decontamination area using a HEPA vacuum before proceeding to a shower that is not adjacent to the work area; or b) Remove their contaminated work suits in the equipment room, without cleaning worksuits, and proceed to a shower that is not adjacent to the work area.

3.2 WORK PROCEDURE

Perform asbestos related work in accordance with the JEGS, 29 CFR 1926.1101, 40 CFR 61-SUBPART M, and as specified herein. Use ~~[[~~wet~~][or][if given prior EPA approval, dry]~~ removal procedures~~][appropriate encapsulation procedures as listed in the asbestos hazard abatement plan]~~ and ~~[~~negative pressure enclosure~~] [_____] ~~techniques. Wear and utilize protective clothing and equipment as specified herein. No eating, smoking, drinking, chewing gum, tobacco, or applying cosmetics is permitted in the asbestos work or control areas. Personnel of other trades not engaged in the ~~[encapsulation][~~removal and demolition~~]~~ of asbestos containing material must not be exposed at any time to airborne concentrations of asbestos unless all the personnel protection and training provisions of this specification are complied with by the trade personnel.~~[~~ Seal all roof top penetrations, except plumbing vents, prior to asbestos roofing work.~~]~~ Shut down the building heating, ventilating, and air conditioning system, cap the openings to the system, ~~[and provide temporary [heating,][and][ventilation,][and][air conditioning]]~~ prior to the commencement of asbestos work. Power to the regulated area must be locked-out and tagged in accordance with 29 CFR 1910.147.~~[~~ Disconnect electrical service when ~~[encapsulation][~~wet removal~~]~~ is performed and provide temporary electrical service with verifiable ground fault circuit interrupter (GFCI) protection prior to the use of any ~~[~~water~~][encapsulant]~~.~~]~~ All electrical work must be performed by a licensed electrician. Stop abatement work in the regulated area immediately when the airborne total fiber concentration: (1) equals or exceeds 0.01 f/cc, or the pre-abatement concentration, whichever is greater, outside the regulated area; or (2) equals or exceeds 1.0 f/cc inside the regulated area. Correct the condition to the satisfaction of the PQP , including visual inspection and air sampling. Work must resume only upon notification by the Contracting Officer once the PQP has certified that the condition has been corrected. Corrective actions must be documented. If an asbestos fiber release or spill occurs~~[~~ outside of the asbestos control area~~]~~, stop work immediately, correct the condition to the satisfaction of the PQP. Submit the documents certified by the PQP to the Contracting Officer including clearance sampling,and obtain the Contracting Officer's approval prior to resumption of work.

3.2.1 Building Ventilation System and Critical Barriers

Building ventilation system supply and return air ducts in a regulated area must be~~[shut down and isolated by lockable switch or other positive means in accordance with 29 CFR 1910.147.][~~ isolated by airtight seals to prevent the spread of contamination throughout the system.~~]~~ The airtight seals must consist of ~~[air-tight rigid covers for building ventilation supply and exhaust grills where the ventilation system is required to remain in service during abatement][~~2 layers of polyethylene~~]~~. Edges to wall, ceiling and floor surfaces must be sealed with industrial grade duct tape.

a. A Competent Person must supervise the work.

b. For indoor work, critical barriers must be placed over all openings to the regulated area.

c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.2 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that asbestos may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection and air sampling results are obtained from the ~~[~~PQP~~][GC]~~ work may proceed at the discretion of the Contracting Officer.

3.2.3 Furnishings

~~[~~ Furniture ~~[,(______)]~~ and equipment will be removed from the area of work by the Government before asbestos work begins.

~~][Furniture [,(______)] and equipment will remain in the building. Cover and seal furnishings with 0.15 mm plastic sheet or remove from the work area and store in a location on site approved by the Contracting Officer.~~

~~][Furnishings listed below and located in the work area are considered to be contaminated with asbestos fibers. Transfer these items to an area on~~

~~site approved by the Contracting Officer, decontaminate (wet methods where possible), and then store until the room from which they came is declared clean and safe for entry.[Carpets, draperies, and other items with porous, non-solid surfaces can not be suitably cleaned and must be properly disposed of as contaminated waste.] At the conclusion of the asbestos removal work and cleanup operations, transfer all objects so removed and cleaned back to the area from which they came and re-install them. Base bids on decontaminating:~~

~~a. [_____] Desks~~

~~b. [_____] Filing cabinets~~

~~c. [_____] Linear meters of shelving~~

~~d. [_____] Cubic meters of books, papers, files, [_____].~~

~~e. [_____].~~

~~]~~3.2.4 Precleaning

Wet wipe and HEPA vacuum all surfaces potentially contaminated with asbestos prior to establishment of an enclosure.

3.2.5 Asbestos Control Area Requirements

3.2.5.1 Negative Pressure Enclosure

Removal of asbestos conatining materials~~[asbestos contaminated acoustical ceiling tiles,] [spray applied fireproofing,] [thermal system insulation,] [gypsum wallboard/joint compound] [_____]~~ require the use of a negative pressure enclosure. Block and seal openings in areas where the release of airborne asbestos fibers can be expected. Establish an asbestos negative pressure enclosure with the use of curtains, portable partitions, or other enclosures in order to prevent the escape of asbestos fibers from the contaminated asbestos work area. Negative pressure enclosure development must include protective covering of uncontaminated walls, and ceilings with a continuous membrane of two layers of minimum 0.15 mm plastic sheet sealed with tape to prevent water or other damage. Provide two layers of 0.15 mm plastic sheet over floors and extend a minimum of 300 mm up walls. Seal all joints with tape. Provide local exhaust system in the asbestos control area. Openings will be allowed in enclosures of asbestos control areas for personnel and equipment entry and exit, the supply and exhaust of air for the local exhaust system and the removal of properly containerized asbestos containing materials. Replace local exhaust system filters as required to maintain the efficiency of the system.

3.2.5.2 Glovebag

If the construction of a negative pressure enclosure is infeasible for the ~~[~~removal~~] [encapsulation]~~ of ACM~~[_____]~~ located in the work area~~[_____]~~. Use alternate techniques as indicated in 29 CFR 1926.1101. Establish designated limits for the asbestos regulated area with the use of rope or other continuous barriers, and maintain all other requirements for asbestos control areas. The PQP must conduct personal samples of each worker engaged in asbestos handling (removal, disposal, transport and other associated work) throughout the duration of the project. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers at any time exceeds background or 0.01 fibers per cubic centimeter whichever is greater, stop work, evacuate personnel in adjacent areas or provide personnel with approved protective equipment at the discretion of the Contracting Officer. This sampling may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those obtained by the Contractor, the Government will determine which results predominate. If adjacent areas are contaminated as determined by the Contracting Officer, clean the contaminated areas, monitor, and visually inspect the area as specified herein.

3.2.5.3 Regulated Area for Class II Removal

Removal of ~~[~~asbestos containing floor tile/mastic,~~]~~ ~~[~~carpet/mastic,~~]~~ ~~[sealants,] [_____]~~ are Class II removal activities. Establish designated limits for the asbestos regulated work area with the use of red barrier tape; install critical barriers, splash guards and signs, and maintain all other requirements for asbestos control area except local exhaust. Place impermeable dropcloths on surfaces beneath removal activity extending out 3 feet in all directions. A detached decontamination system may be used. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If workers the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.6 Removal Procedures

Wet asbestos material with a fine spray of ~~[~~amended water~~][a specific wetting agent such as light oil]~~ during removal, cutting, or other handling so as to reduce the emission of airborne fibers. Remove material and immediately place in 0.15 mm plastic disposal bags. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. Where unusual circumstances prohibit the use of 0.15 mm plastic bags, submit an alternate proposal for containment of asbestos fibers to the Contracting Officer for approval. For example, in the case where both piping and insulation are to be removed, the Contractor may elect to wet the insulation, wrap the pipes and insulation in plastic and remove the pipe by sections. Containerize asbestos containing material while wet. Do not allow asbestos material to accumulate or become dry. Lower and otherwise handle asbestos containing material as indicated in 40 CFR 61-SUBPART M.

3.2.6.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as venetian blinds, full-height partitions, carpeting, duct work, pipes and fittings, radiators, light fixtures, conduit, panels, and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. These items need not be vacuumed. The asbestos lock-down encapsulant must be tinted a contrasting color and spray-applied by airless method. Thoroughness of sealing operation must be visually gauged by the extent of colored coating on exposed surfaces. Lock-down encapsulants must comply with the performance requirements specified herein.

3.2.6.2 Exposed Pipe Insulation Edges

Contain edges of asbestos insulation to remain that are exposed by a removal operation. Wet and cut the rough ends true and square with sharp tools and then encapsulate the edges with a 6 mm thick layer of non-asbestos containing insulating cement troweled to a smooth hard finish. When cement is dry, lag the end with a layer of non-asbestos lagging cloth, overlapping the existing ends by at least 100 mm. When insulating cement and cloth is an impractical method of sealing a raw edge of asbestos, take appropriate steps to seal the raw edges as approved by the Contracting Officer.

3.2.7 Methods of Compliance

3.2.7.1 Mandated Practices

The specific abatement techniques and items identified must be detailed in the Contractor's AHAP. Use the following engineering controls and work practices in all operations, regardless of the levels of exposure:

a. Vacuum cleaners equipped with HEPA filters.

b. Wet methods or wetting agents except where it can be demonstrated that the use of wet methods is unfeasible due to the creation of electrical hazards, equipment malfunction, and in roofing.

c. Prompt clean-up and disposal.

d. Inspection and repair of polyethylene.

e. Cleaning of equipment and surfaces of containers prior to removing them from the equipment room or area.

3.2.7.2 Control Methods

Use the following control methods:

a. Local exhaust ventilation equipped with HEPA filter;

b. Enclosure or isolation of processes producing asbestos dust;

c. Where the feasible engineering and work practice controls are not sufficient to reduce employee exposure to or below the PELs, use them to reduce employee exposure to the lowest levels attainable and must supplement them by the use of respiratory protection.

3.2.7.3 Unacceptable Practices

The following work practices must not be used:

a. High-speed abrasive disc saws that are not equipped with point of cut ventilator or enclosures with HEPA filtered exhaust air.

b. Compressed air used to remove asbestos containing materials, unless the compressed air is used in conjunction with an enclosed ventilation system designed to capture the dust cloud created by the compressed air.

c. Dry sweeping, shoveling, or other dry clean up.

d. Employee rotation as a means of reducing employee exposure to asbestos.

3.2.8 Class I Work Procedures

In addition to requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

a. A Competent Person must supervise the installation and operation of the control methods.

b. For jobs involving the removal of more than 7.5 m or 0.9 square m of TSI or surfacing material, place critical barriers over all openings to the regulated area.

c. HVAC systems must be isolated in the regulated area by sealing with a double layer of plastic or air-tight rigid covers.

d. Impermeable dropcloths (0.15 mm or greater thickness) must be placed on surfaces beneath all removal activity.

e. Where a negative exposure assessment has not been provided or where exposure monitoring shows the PEL was exceeded, the regulated area must be ventilated with a HEPA unit and employees must use PPE.

3.2.9 Specific Control Methods for Class I Work

Use Class I work procedures, control methods and removal methods for the following ACM:

a. Spray Applied Fireproofing

b. Gypsum Wallboard and Joint Compound

c. Thermal System Insulation and Mudded Pipe Fittings

d. Plaster and Textured Ceilings and Walls

e. Vermiculite

3.2.9.1 Negative Pressure Enclosure (NPE) System

The system must provide at least four air changes per hour inside the containment. The local exhaust unit equipment must be operated 24-hours per day until the containment is removed. The NPE must be smoke tested for leaks at the beginning of each shift and be sufficient to maintain a minimum pressure differential of minus 0.5 mm of water column relative to adjacent, unsealed areas. Pressure differential must be monitored continuously, 24-hours per day, with an automatic manometric recording instrument and Records must be provided daily on the same day collected to the Contracting Officer. The Contracting Officer must be notified immediately if the pressure differential falls below the prescribed minimum. The building ventilation system must not be used as the local exhaust system for the regulated area. The NPE must terminate outdoors unless an alternate arrangement is allowed by the Contracting Officer. All filters used must be new at the beginning of the project and must be periodically changed as necessary and disposed of as ACM waste.

3.2.9.2 Glovebag Systems

Glovebags must be used without modification, smoke-tested for leaks, and completely cover the circumference of pipe or other structures where the work is to be done. Glovebags must be used only once and must not be moved. Glovebags must not be used on surfaces that have temperatures exceeding 66 degrees C. Prior to disposal, glovebags must be collapsed using a HEPA vacuum. Before beginning the operation, loose and friable material adjacent to the glovebag operation must be wrapped and sealed in 2 layers of plastic or otherwise rendered intact. At least two persons must perform glovebag removal. Asbestos regulated work areas must be established for glovebag abatement. Designated boundary limits for the asbestos work must be established with rope or other continuous barriers and all other requirements for asbestos control areas must be maintained, including area signage and boundary warning tape.

a. Attach HEPA vacuum systems to the bag to prevent collapse during removal of ACM.

b. The negative pressure glove boxes must be fitted with gloved apertures and a bagging outlet and constructed with rigid sides from metal or other material which can withstand the weight of the ACM and water used during removal. A negative pressure must be created in the system using a HEPA filtration system. The box must be smoke tested for leaks prior to each use.

3.2.9.3 Mini-Enclosure

[Single bulkhead containment][Double bulkhead containment][or][Mini-containment (small walk-in enclosure)] to accommodate no more than two persons, may be used if the disturbance or removal can be completely contained by the enclosure. The mini-enclosure must be inspected for leaks and smoke tested before each use. Air movement must be directed away from the employee's breathing zone within the mini-enclosure.

3.2.9.4 Wrap and Cut Operation

Prior to cutting pipe, the asbestos-containing insulation must be wrapped with polyethylene and securely sealed with duct tape to prevent asbestos becoming airborne as a result of the cutting process. The following steps must be taken: install glovebag, strip back sections to be cut 150 mm from point of cut, and cut pipe into manageable sections.

3.2.9.5 Class I Removal Method

Class I ACM must be removed using a control method described above. Prepare work area as previously specified. Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area. Spread one layer of 0.15 mm seamless plastic sheeting on the floor below the work area. Remove asbestos containing spray applied fireproofing using a scraper and wet methods and immediately place into 0.15 mm thickness disposal bag. After removal of the material use a wire brush to clean the exposed substrate to remove residual material. Continue wet cleaning until surfaces are free of visible debris. Cut manageable sections of gypsum wallboard and joint compound and immediately place into a 0.15 mm minimum thickness disposal bag or other approved container. Make every effort to keep the material from falling to the floor of the work area. Use a wire brush and wet clean to remove residual material from studs. Continue wet cleaning until the surface is clean of visible material and encapsulate stud walls. Remove ACM thermal system insulation and mudded pipe fittings using mechanical means and wet methods and immediately place into 0.15 mm thickness disposal bag. Continue wet cleaning until surfaces are free of visible debris. Remove ACM plaster ceilings or walls using mechanical means and adequately wet methods and immediately place into 0.15 mm thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris. Remove ACM textured ceiling finish using a scraper and wet methods and immediately place into 0.15 mm thickness disposal bag. Floors are considered contaminated from fallen textured ceiling finish. Clean up debris on floor and dispose of carpet as asbestos contaminated material. After removal of the material use a wire brush to clean the exposed concrete ceiling to remove residual material. Continue wet cleaning until surfaces are free of visible debris. Remove ACM vermiculite using mechanical means and adequately wet methods and immediately place into 0.15 mm thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris. Bag all asbestos debris which has fallen to the floor as asbestos-containing debris. Place all debris in plastic disposal bags of 0.15 mm minimum thickness. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers or the designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work, and immediately correct the situation.

3.2.10 Class II Work Procedures

In addition to the requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

a. A Competent Person must supervise the work.

b. For indoor work, critical barriers must be placed over all openings to the regulated area.

c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

3.2.11 Specific Control Methods for Class II Work

3.2.11.1 ~~[~~Vinyl and Asphaltic Flooring Materials~~]~~ ~~[~~Carpet and Mastic~~]~~

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. When removing ~~[~~vinyl floor tile and mastic~~][~~carpet and mastic~~]~~ which contains ACM, use the following practices. Remove ~~[~~floor tile and mastic~~][~~carpet and mastic~~]~~ using adequately wet methods. Remove ~~[~~floor tiles~~][~~carpet and mastic~~]~~ intact (if possible).~~[Wetting is not required when floor tiles are heated and removed intact.]~~ Do not sand flooring or its backing. Scrape residual adhesive and backing using wet methods. Mechanical chipping is prohibited unless performed in a negative pressure enclosure. Dry sweeping is prohibited. Use vacuums equipped with HEPA filter, disposable dust bag, and metal floor tool (no brush) to clean floors. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If workers the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.2 Sealants and Mastic

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers and signs, and maintain all other requirements for asbestos control area except local exhaust. Spread 0.15 mm plastic sheeting on the ground around the perimeter of the work area extending out in all directions. Using adequately wet methods, carefully remove the ACM sealants and mastics using a scraper of knife blade. As it is removed place the material into a disposal bag. Make every effort to keep the asbestos material from falling to the ground or work area floor below. Dry sweeping is prohibited. Use vacuums equipped with HEPA filter and disposable dust bag. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or at designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.3 Suspect Fire Doors

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. Spread 0.15 mm plastic sheeting on the ground beneath the work area and around the perimeter of the work area extending out in all directions. Remove door intact from hinges and wrap with 6-mil plastic sheeting. Inspect the interior areas of the door to determine if ACM is present. If ACM is not present the door may be disposed of as general construction debris. If ACM is present place whole door in enclosed container for disposal. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.4 Roofing Materials

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing roofing materials which contain ACM as described in 29 CFR 1926.1101(g)(8)(ii), use the following practices. Roofing material must be removed in an intact state. Wet methods must be used to remove roofing materials that are not intact, or that will be rendered not intact during removal, unless such wet methods are not feasible or will create safety hazards. When removing built-up roofs, with asbestos-containing roofing felts and an aggregate surface, using a power roof cutter, all dust resulting from the cutting operations must be collected by a HEPA dust collector, or must be HEPA vacuumed by vacuuming along the cut line. Asbestos-containing roofing material must not be dropped or thrown to the ground, but must be lowered to the ground via covered, dust-tight chute, crane, hoist or other method approved by the Contracting Officer. Any ACM that is not intact must be lowered to the ground as soon as practicable, but not later than the end of the work shift. While the material remains on the roof it must be kept wet or placed in an impermeable waste bag or wrapped in plastic sheeting. Intact ACM must be lowered to the ground as soon as practicable, but not later than the end of the work shift. Unwrapped material must be transferred to a closed receptacle. Critical barriers must be placed over roof level heating and ventilation air intakes. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.5 Cementitious Siding and Shingles or Transite Panels

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing cementitious asbestos-containing siding, shingles or Transite panels use the following work practices. Intentionally cutting, abrading or breaking is prohibited. Each panel or shingle must be sprayed with amended water prior to removal. Nails must be cut with flat, sharp instruments. Unwrapped or unbagged panels or shingles must be immediately lowered to the ground via covered dust-tight chute, crane or hoist, or placed in an impervious waste bag or wrapped in plastic sheeting and lowered to the ground no later than the end of the work shift. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.11.6 Gaskets

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Gaskets must be thoroughly wetted with amended water prior to removal and immediately placed in a disposal container. If a gasket is visibly deteriorated and unlikely to be removed intact, removal must be undertaken within a glovebag. Any scraping to remove residue must be performed wet. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

~~[3.2.12 Encapsulation Procedures~~

~~3.2.12.1 Preparation of Test Patches~~

~~Install [three] [_____] test patches of encapsulant in [_____], as indicated. Use airless spray at the lowest pressure and as recommended by the encapsulant manufacturer. Follow exactly the manufacturer's instructions for thinning recommendations, application procedures and rates. Curing time must be not less than five days or that recommended by the manufacturer, whichever is more. A test patch must be 0.8 square meter in size.~~

~~3.2.12.2 Field Testing~~

~~Field test the encapsulation test patches in accordance with ASTM E1494, paragraph "Required Field Test," in the presence of the Contracting Officer. Keep a written record of the testing procedures and test results. Upon successful testing of the encapsulant, submit a signed statement to the Contracting Officer certifying that the encapsulant is suitable for installation on the particular asbestos containing material.~~

~~3.2.12.3 Large-Scale Application~~

~~Apply encapsulant using the same equipment and procedures as employed for the test patches. Keep the encapsulant material stirred to prevent settling. Keep a clean work area. Change pre-filters in the ventilation equipment as soon as they appear clogged by encapsulant aerosol or~~

~~pressure differential drops below 0.02 Hg.~~

~~]~~3.2.12 Abatement of Asbestos Contaminated Soil

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Asbestos contaminated soil must be removed from areas to a minimum depth of 50 mm. Soil must be thoroughly dampened with amended water and then removed by manual shoveling into labeled containers. Place debris into a 0.15 mm minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 0.15 mm minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.01 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

3.2.13 Air Sampling

Perform sampling of airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101, the Contractor's air monitoring plan and as specified herein. Sampling performed in accordance with 29 CFR 1926.1101 must be performed by the PQP.~~[Sampling performed for environmental and quality control reasons must be performed by the [PQP][GC].]~~ Unless otherwise specified, use NIOSH Method 7400 for sampling and analysis. Monitoring may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those results obtained by the Contractor, the Government will determine which results predominate. Results of breathing zone samples must be posted at the job site and made available to the Contracting Officer. Submit all documentation regarding initial exposure assessments, negative exposure assessments, and air-monitoring results.

3.2.13.1 Sampling Prior to Asbestos Work

Provide area air sampling and establish the baseline one day prior to the masking and sealing operations for each ~~[~~demolition~~]~~ ~~[~~removal~~]~~ ~~[encapsulation]~~ site. Establish the background by performing area sampling in similar but uncontaminated sites in the building.

3.2.13.2 Sampling During Asbestos Work

~~[The PQP must provide personal and area sampling as indicated in 29 CFR 1926.1101 and governing environmental regulations. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours (breathing zone samples), and [_____] hours (environmental/clearance monitoring) after completion of a sampling period. In addition, provided the same type of work is being performed, provide area sampling at least once every work shift close to the work inside the enclosure, outside the clean room~~

~~entrance to the enclosure, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.01 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer immediately.[Where alternate methods are used, perform personal and area air sampling at locations and frequencies that will accurately characterize the evolving airborne asbestos levels.] The written results must be signed by testing laboratory analyst, testing laboratory principal and the [Contractor's PQP][GC]. The air sampling results must be documented on a Contractor's daily air monitoring log.~~

~~][~~The PQP must provide personal sampling as indicated in 29 CFR 1926.1101. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours (breathing zone samples), and 72~~[_____]~~ hours (environmental/clearance monitoring) after completion of a sampling period. At the same time the GC will provide area sampling close to the work inside the enclosure, outside the clean room entrance to the enclosure, and at the exhaust opening of the local exhaust system. In addition, provided the same type of work is being performed, the GC will provide area sampling once every work shift close to the work inside the enclosure, outside the clean room entrance to the enclosure, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.01 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer immediately.~~[~~ Where alternate methods are used, perform personal and area air sampling at locations and frequencies that will accurately characterize the evolving airborne asbestos levels.~~]~~ The written results must be signed by testing laboratory analyst, testing laboratory principal and the ~~[~~Contractor's PQP~~][GC]~~. The air sampling results must be documented on a Contractor's daily air monitoring log.

~~]~~3.2.13.3 Final Clearance Requirements, NIOSH PCM Method

For PCM sampling and analysis using NIOSH NMAM Method 7400, the fiber concentration inside the abated regulated area, for each airborne sample, must be less than 0.01 f/cc. The abatement inside the regulated area is considered complete when every PCM final clearance sample is below the clearance limit. If any sample result is greater than 0.01 total f/cc, the asbestos fiber concentration (asbestos f/cc) must be confirmed from that same filter using NIOSH NMAM Method 7402 (TEM) at Contractor's expense. If any confirmation sample result is greater than 0.01 asbestos f/cc, abatement is incomplete and cleaning must be repeated at the Contractor's expense. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.13.4 Final Clearance Requirements, EPA TEM Method

For EPA TEM sampling and analysis, using the EPA Method specified in 40 CFR 763, abatement inside the regulated area is considered complete when the arithmetic mean asbestos concentration of the five inside samples is less than or equal to 70 structures per square millimeter (70 S/mm). When the arithmetic mean is greater than 70 S/mm, the three blank samples must be analyzed. If the three blank samples are greater than 70 S/mm, resampling must be done. If less than 70 S/mm, the five outside samples must be analyzed and a Z-test analysis performed. When the Z-test results are less than 1.65, the decontamination must be considered complete. If the Z-test results are more than 1.65, the abatement is incomplete and cleaning must be repeated. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

3.2.13.5 Sampling After Final Clean-Up (Clearance Sampling)

The Contractor must perform any Clearance Sampling utilizing a third-party independent person who has no ties to the Contractor (except as being a subcontractor), the Asbestos Abatement Subcontractor, and the IH/PQP for the project. Provide area sampling of asbestos fibers~~[using aggressive air sampling techniques as defined in the EPA 560/5-85-024]~~ and establish an airborne asbestos concentration of less than 0.01 fibers per cubic centimeter after final clean-up but before removal of the enclosure or the asbestos work control area. After final cleanup and the asbestos control area is dry but prior to clearance sampling, the ~~[~~PQP~~][~~ and ~~][~~GC~~]~~ must perform a visual inspection in accordance with ASTM E1368 to ensure that the asbestos control and work area is free of any accumulations of dirt, dust, or debris.~~[~~ Prepare a written report signed and dated by the PQP documenting that the asbestos control area is free of dust, dirt, and debris and all waste has been removed.~~][Perform at least [_____] samples.][Use transmission electron microscopy (TEM) to analyze clearance samples and report the results in accordance with current NIOSH criteria.]~~ The asbestos fiber counts from these samples must be less than 0.01 fibers per cubic centimeter or be not greater than the background, whichever is greater. Should any of the final samples indicate a higher value take appropriate actions to re-clean the area and repeat the sampling and ~~[TEM]~~ analysis at the Contractor's expense.

3.2.13.6 Air Clearance Failure

If clearance sampling results fail to meet the final clearance requirements, pay all costs associated with the required recleaning, resampling, and analysis, until final clearance requirements are met.

3.2.14 Lock-Down

Prior to removal of plastic barriers and after pre-clearance clean up of gross contamination, the ~~[~~PQP~~][GC]~~ must conduct a visual inspection of all areas affected by the ~~[~~removal~~][encapsulation]~~ in accordance with ASTM E1368. Inspect for any visible fibers~~[~~, and to ensure that encapsulants were applied evenly and appropriately~~]~~.~~[~~ Spray apply a post removal (lock-down) encapsulant to ceiling, walls, floors and other areas exposed in the removal area. The exposed areaincludes but is not limited to plastic barriers, furnishings and articles to be discarded as well as dirty change room, air locks for bag removal and decontamination chambers.~~]~~

3.2.15 Site Inspection

While performing asbestos engineering control work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

3.3 CLEAN-UP AND DISPOSAL

3.3.1 Housekeeping

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete, all asbestos waste is removed from the work-site, and final clean-up is completed, the Contracting Officer will attest that the area is safe before the signs can be removed. After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the building HVAC system and provide new pre-filters. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order. The Contracting Officer will visually inspect all surfaces within the enclosure for residual material or accumulated dust or debris. The Contractor must re-clean all areas showing dust or residual materials. If re-cleaning is required, air sample and establish an acceptable asbestos airborne concentration after re-cleaning. The Contracting Officer must agree that the area is safe in writing before unrestricted entry will be permitted. The Government must have the option to perform monitoring to determine if the areas are safe before entry is permitted.

3.3.2 Title to Materials

All waste materials, except as specified otherwise, become the property of the Contractor and must be disposed of as specified in accordance with the JEGS, applicable U.S. Federal, GOJ, and local regulations.

3.3.3 Disposal of Asbestos

3.3.3.1 Procedure for Disposal

Coordinate all waste disposal manifests with the Contracting Officer and NAVFAC EV. Collect asbestos waste, contaminated waste water filters, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 0.15 mm thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with 40 CFR 61-SUBPART M. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 0.15 mm thick bags with the approved warnings and DOT labeling preprinted on the bag. Clearly indicate on the outside of each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete. Dispose of waste asbestos material at an approved asbestos landfill off Government property. For temporary storage, store sealed impermeable bags in asbestos waste drums or skids. An area for interim storage of asbestos waste-containing drums or skids will be assigned by the Contracting Officer or his authorized representative. Comply with 40 CFR 61-SUBPART M, GOJ, regional, and local standards for hauling and disposal. Sealed plastic bags may be dumped from drums into the burial site unless the bags have been broken or damaged. Damaged bags must remain in the drum and the entire contaminated drum must be buried. Uncontaminated drums may be recycled. Workers unloading the sealed drums must wear appropriate respirators and personal protective equipment when handling asbestos materials at the disposal site.

3.3.3.2 Asbestos Disposal Quantity Report

~~[Direct the PQP to record and report, to the Contracting Officer, the amount of asbestos containing material removed and released for disposal. Deliver the report for the previous day at the beginning of each day shift with amounts of material removed during the previous day reported in linear meters or square meters as described initially in this specification and in cubic meters for the amount of asbestos containing material released for disposal.~~

~~][~~Allow the GC to inspect, record and report the amount of asbestos containing material removed and released for disposal on a daily basis.~~]~~

-- End of Section --

SECTION 02 83 00

LEAD REMEDIATION

01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~AMERICAN SOCIETY OF SAFETY ENGINEERS (ASSE/SAFE)~~AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSE/SAFE Z9.2	(2012) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701	(2015) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	~~(2014) Safety and Health Requirements Manual~~ Safety -- Safety and Health Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS	(Apr 202~~2~~4) Japan Environmental Governing Standards

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780	(1995; Errata Aug 1996;Rev Ch. 7 - 1997) Guidelines for the Evaluation and Control of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.103	Respiratory Protection
29 CFR 1926.1126	Chromium
29 CFR 1926.1127	Cadmium
29 CFR 1926.21	Safety Training and Education
29 CFR 1926.33	Access to Employee Exposure and Medical Records
29 CFR 1926.55	Gases, Vapors, Fumes, Dusts, and Mists
29 CFR 1926.59	Hazard Communication

29 CFR 1926.62	Lead
29 CFR 1926.65	Hazardous Waste Operations and Emergency Response
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste
40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 268	Land Disposal Restrictions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 178	Specifications for Packagings

UNDERWRITERS LABORATORIES (UL)

UL 586	(2009; Reprint Dec 2017) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

1.2 DEFINITIONS

1.2.1 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

1.2.2 Area Sampling

Sampling of lead, cadmium, chromium concentrations within the lead, cadmium, chromium control area and inside the physical boundaries which is representative of the airborne lead, cadmium, chromium concentrations but is not collected in the breathing zone of personnel (approximately 1.5 to 1.8 meters above the floor).

1.2.3 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

PEL (micrograms/cubic meter of air) = 40/No. hrs worked per day

1.2.4 Competent Person (CP)

As used in this section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium hazards in accordance with current U.S. federal, GOJ national, prefectural, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene or a Certified Safety Professional (CSP) certified by the Board of Certified Safety Professionals or a licensed lead-based paint abatement Supervisor/Project Designer.

1.2.5 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

1.2.6 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

1.2.7 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium or to plan such activities in commercial buildings, bridges or other structures.

1.2.8 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, chromium to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.2.9 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, chromium contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

1.2.10 Industrial Hygienist/Private Qualified Person (IH/PQP)

The IH/PQP shall have demonstratable experience in lead air monitoring techniques and in the establishment of a respiratory protection program for employees. The IH/PQP shall have working knowledge of applicable Federal and Japanese regulations occupational safety and health regulations for lead in construction. The IH/PQP shall have attended and completed Lead Abatement Supervision and Monitoring training and shall be considered a competent person as defined in 29 CFR 1926.62. The IH/PQP shall be a registered Architect, Professional Engineer, or Certified Industrial Hygienist, who has successfully completed training and is therefore accredited under a legitimate Model Accreditation Plan. The IH/PQP must be qualified to perform visual inspections.

1.2.11 Industrial Hygiene Technician (IHT)

The Industrial Hygiene Technician (IHT) shall be an employee of the IH/PQP or the testing laboratory and shall have a minimum of 2 years experience in the industrial hygiene field working under the direction of the IH/PQP and has completed the following course: Lead Abatement Supervision and Monitoring - Covering practices and procedures in lead abatement, lead air sampling and abatement monitoring.

1.2.12 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. Excludes other forms of organic lead compounds. The use of the term Lead in this section also refers to paints which contain detectable concentrations of Cadmium and Chromium. For the purposes of the section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium.

1.2.13 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

1.2.14 Lead, Cadmium, Chromium Control Area

A system of control methods to prevent the spread of lead, cadmium, chromium dust, paint chips or debris to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

1.2.15 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

PEL (micrograms/cubic meter of air) = 400/No. hrs worked per day

1.2.16 Material Containing Lead/Paint with Lead (MCL/PWL)

Any material, including paint, which contains lead as determined by the testing laboratory using a valid test method. The requirements of this section does not apply if no detectable levels of lead are found using a quantitative method for analyzing paint or MCL using laboratory instruments with specified limits of detection (usually 0.01 percent). An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

1.2.17 Personal Sampling

Sampling of airborne lead, cadmium, chromium concentrations within the breathing zone of an employee to determine the 8-hour time weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 150 to 225 mm and centered at the nose or mouth of an employee.

1.2.18 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, chromium control area to limit unauthorized entry of personnel.

1.3 DESCRIPTION

Construction activities impacting PWL or material containing lead, cadmium, chromium which are covered by this specification include the d~~emolition or removal~~isturbance of material containing lead, cadmium, chromium in in tact~~[_____]~~ condition~~, located [_____] and as indicated on the drawings~~. ~~[_____]~~ The work covered by this section includes work tasks and the precautions specified in this section for the protection of building occupants and the environment during and after the performance of the hazard abatement activities.

The Contractor shall assume that all coatings will contain some level of lead, cadmium, and Chromium. The Contractor shall review existing Lead Paint survey data~~, if available~~.

1.3.1 Protection of Existing Areas To Remain

Project work including, but not limited to, lead, cadmium, chromium hazard abatement work, storage, transportation, and disposal must be performed without damaging or contaminating adjacent work and areas. Where such work or areas are damaged or contaminated, restore work and areas to the original condition.

1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, Chromium Compliance Plan and describe how the Contractor will prevent lead, cadmium and chromium exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium activities.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only.~~When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.]~~Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Occupational and Environmental Assessment Data Report

Medical Examinations

[~~Initial Sample Results;~~

] Lead, Cadmium, Chromium Compliance Plan; G

SD-03 Product Data

Respirators

Vacuum Filters

Negative Air Pressure System

Materials and Equipment

Expendable Supplies

Local Exhaust Equipment

Pressure Differential Automatic Recording Instrument

Pressure Differential Log

SD-06 Test Reports

Occupational and Environmental Assessment Data Report

Pressure Differential Recordings For Local Exhaust System

Sampling Results; G

SD-07 Certificates

Notification of the Commencement of LBP Hazard Abatement

SD-11 Closeout Submittals

Turn-In Documents or Weight Tickets

Written Evidence Of TSD Approval

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Include in the Lead, Cadmium, Chromium Compliance plan, the name, address, and telephone number of the CP selected to perform responsibilities specified in paragraph COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium standard (29 CFR 1926.1126), Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, chromium; experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 3 years experience implementing OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and certified in accordance with U.S.

federal, GOJ national, prefectural and local laws.

1.5.1.2 Training Certification

Include in the Lead, Cadmium, Chromium Compliance plan, a certificate for each worker and supervisor, signed and dated by the training provider, stating that the employee has received the required lead, cadmium and chromium training specified in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.5.1.3 Testing Laboratory

Include in the Lead, Cadmium, Chromium Compliance plan, the name, address, and telephone number of the testing laboratory selected to perform the air and wipe analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium. Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved.

1.5.1.4 Third Party Consultant Qualifications

Include in the Lead, Cadmium, Chromium Compliance plan, the name, address and telephone number of the third party consultant selected to perform the wipe sampling for determining concentrations of lead, cadmium and chromium in dust. Submit proper documentation that the consultant is trained and certified as an inspector technician or inspector/risk assessor by the USEPA authorized certification and accreditation program.

1.5.2 Requirements

1.5.2.1 Competent Person (CP) Responsibilities

a. Verify training meets all U.S. federal, GOJ national, prefectural, and local requirements.

b. Review and approve Lead, Cadmium, Chromium Compliance Plan for conformance to the applicable referenced standards.

c. Continuously inspect LBP/PWL or MCL work for conformance with the approved plan.

d. Perform (or oversee performance of) air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.

e. Ensure work is performed in strict accordance with specifications at all times.

f. Control work to prevent hazardous exposure to human beings and to the environment at all times.

g. Supervise final cleaning of the lead, cadmium, chromium control area, take clearance wipe samples if necessary; review clearance sample results and make recommendations for further cleaning.

h. Certify the conditions of the work as called for elsewhere in this specification.

1.5.2.2 Lead, Cadmium, Chromium Compliance Plan

The purpose of the plan for this project is to verify that the Contractor understands that they will encounter paint with lead (PWL), cadmium and Chromium. The Contractor shall assume that all coatings will contain some level of lead, cadmium and chromium. The Contractor's LCCCP shall identify the Contractor's process to comply with all applicable regulations depending on the method of work the Contractor choose when encountering paints and coatings.

The Lead, Cadmium, Chromium Compliance Plan shall be reviewed and approved (with signature, date of approval, and certification number) by the IH/PQP.

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium, LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, chromium control areas, critical barriers, physical boundaries, location and details of decontamination facilities, viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity from which lead, cadmium, chromium is emitted. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, chromium related work, collected waste water and dust containing lead, cadmium, chromium and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, chromium is not released outside of the lead, cadmium, chromium control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the air sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksites to inform affected employees and to clarify responsibilities to control exposures.

The plan shall also include the requirements of paragraphs 1.5.1.1 through 1.5.1.5.

The plan must be developed and signed by the IH/PQP . The plan must include the name, certificate, and certification number of the person signing the plan.

In occupied buildings, the plan must also include an occupant protection program that describes the measures that will be taken during the work to protect the building occupants.

1.5.2.3 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational and environmental sampling results to the Contracting Officer within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 the Contractor must provide documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62,29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, Chromium Compliance Plan.

a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to estimate worker exposures per 29 CFR 1926.62,29 CFR 1926.1126, 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, chromium for stated work.

b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 with a complete process description. This includes manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium containing coatings are present.

c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, chromium compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

1.5.2.4 Medical Examinations

Submit pre-work blood lead levels and post-work blood lead levels for all workers performing lead, cadmium, chromium activities during the execution of the work. Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, chromium at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, chromium as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

1.5.2.5 Training

Train each employee performing work that disturbs lead, cadmium, chromium, who performs LBP/MCL/PWL disposal, and air sampling operations prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and GOJ and local regulations where appropriate.

1.5.2.6 Respiratory Protection Program

a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.55.

1.5.2.7 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

1.5.2.8 Lead, Cadmium, Chromium Waste Management

The Lead, Cadmium, Chromium Waste Management Plan must comply with applicable requirements of the JEGS. U.S federal, GOJ National, prefectural, and local hazardous waste regulations and address:

a. Identification and classification of wastes associated with the work.

b. Estimated quantities of wastes to be generated and disposed of.

c. Names and qualifications of each contractor that will be transporting, storing, treating, and disposing of the wastes. Include the facility location and a 24-hour point of contact. Furnish two copies of local hazardous waste permits.

d. Names and qualifications (experience and training) of personnel who will be working on-site with hazardous wastes.

e. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.

f. Spill prevention, containment, and cleanup contingency measures including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.

g. Work plan and schedule for waste containment, removal and disposal. Proper containment of the waste includes using acceptable waste containers (e.g., 55-gallon drums) as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.

h. Include any process that may alter or treat waste rendering a hazardous waste non hazardous.

i. Unit cost for hazardous waste disposal according to this plan.

1.5.2.9 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of the JEGS, U.S. federal, GOJ national, prefectural, and local authorities regarding lead, cadmium and chromium. Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and EM 385-1-1. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply. The following local laws, ordinances, criteria, rules and regulations regarding removing, handling, storing, transporting, and disposing of lead, cadmium and chromium-contaminated materials apply:

a. Waste Disposal and Public Cleaning Law - licensing and certification in accordance with Japanese law

1.5.3 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.51 mm of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, chromium control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer within 24-hours from the end of each work day.

1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits and notifications have been obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the contract price.

1.5.5 Occupant Protection Plan

The certified project designer must develop and implement an Occupant Protection Plan describing the measures and management procedures to be taken during lead, cadmium and chromium hazard abatement activities to protect the building occupants/building facilities and the outside environment from exposure to any lead, cadmium and chromium contamination while lead, cadmium and chromium hazard abatement activities are performed.

1.5.6 Pre-Construction Conference

Along with the CP, meet with the Contracting Officer to discuss in detail the Lead, Cadmium, Chromium Waste Management Plan and the Lead, Cadmium, Chromium Compliance Plan, including procedures and precautions for the work.

1.6 EQUIPMENT

1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and any other GOJ and local prefectural requirements on respirators.

1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, chromium contaminated dust must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

1.6.3 Rental Equipment Notification

If rental equipment is to be used during PWL or MCL handling and disposal, notify the rental agency in writing concerning the intended use of the equipment.

1.6.4 Vacuum Filters

UL 586 labeled HEPA filters.

1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of the lead, cadmium and chromium removal work within the lead, cadmium and chromium controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

1.6.6 Abrasive Removal Equipment

The use of powered machine for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters.

1.6.7 Negative Air Pressure System

1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of ASSE/SAFE Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of minus 0.005 kPa relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.

b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.

c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 0.625 kPa static pressure differential on a magnehelic gauge.

d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.005 kPa and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.

e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.

f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.

g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.

h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 590 degrees C. Heat blower must be (grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

1.7 PROJECT/SITE CONDITIONS

1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent areas. Where existing work is damaged or contaminated, restore work to its original condition or better as determined by the Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 0.15 mm thick (0.1 mm/ thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust cannot escape or spill out.

2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, chromium contaminated debris must be polyethylene plastic that is a minimum of 0.15 mm thick or any other thick plastic material shown to demonstrate at least equivalent performance.

2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 0.15 mm, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out of the ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 50 mm wide, industrial strength.

2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT, GOJ, local prefecture, and OSHA standards.

2.1.1.6 Chemical Paint Strippers

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

2.1.1.7 Chemical Paint Stripper Neutralizer

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

2.1.1.8 Detergents and Cleaners

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium control work using cleaning techniques specified by HUD 6780 guidelines.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Protection

3.1.1.1 Notification

a. Notify the Contracting Officer 20 days prior to the start of any lead, cadmium and chromium work.

c. Notification of the Commencement of LBP Hazard Abatement

3.1.1.2 Lead, Cadmium, Chromium Control Area

a. Physical Boundary - Provide physical boundaries around the lead, cadmium, chromium control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium will not escape outside of the lead, cadmium and chromium control area. Prohibit the general public from accessing the lead, cadmium, chromium control areas.

b. Warning Signs - Provide bilingual warning signs in English and Japanese at approaches to lead, cadmium, chromium control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

3.1.1.3 Furnishings

The Government will remove furniture and equipment from the building before lead, cadmium and chromium work begins.

3.1.1.4 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, chromium control areas. Seal intake and exhaust vents in the lead, cadmium, chromium control area with 0.15 mm plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, chromium control area.

3.1.1.5 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, chromium control area in accordance with ASSE/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the lead, cadmium, chromium control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, chromium control area of minus 0.51 mm of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, chromium control area. Filters on exhaust equipment must conform to ASSE/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

3.1.1.6 Negative Air Pressure System Containment

a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium contaminated waste.

b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during abatement activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Dismantling of the negative air pressure systems must conform to written decontamination procedures be as presented in the Lead, Cadmium, Chromium Compliance Plan. Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination.

3.1.1.7 Decontamination Shower Facility

Provide clean and contaminated change rooms and shower facilities in accordance with this specification and 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

3.1.1.8 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

3.1.1.9 Mechanical Ventilation System

a. Use adequate ventilation to control personnel exposure to lead, cadmium and chromium in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127. To the extent feasible, use local exhaust ventilation or other collection systems, approved by the CP. Evaluate and maintain local exhaust ventilation systems in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127.

b. Vent local exhaust outside the building and away from building ventilation intakes or ensure system is connected to HEPA filters.

c. Use locally exhausted, power actuated tools or manual hand tools.

3.1.1.10 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, chromium control area. No one will be permitted in the lead, cadmium, chromium control area unless they have been appropriately trained and provided with protective equipment.

3.2 ERECTION

3.2.1 Lead, Cadmium, Chromium Control Area Requirements

Establish a lead, cadmium, chromium control area by completely establishing barriers and physical boundaries around the area or structure where PWL or MCL removal operations will be performed.

3.3 APPLICATION

3.3.1 Lead, Cadmium, Chromium Work

Perform lead, cadmium, chromium work in accordance with approved Lead, Cadmium, Chromium Compliance Plan. Use procedures and equipment required to limit occupational exposure and environmental contamination with lead, cadmium, chromium when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with the JEGS. U.S. federal, GOJ national, prefectural laws and regulations, and local requirements.

3.3.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium Removal

Manual or power sanding or grinding of lead, cadmium, chromium surfaces or materials is not permitted unless tools are equipped with HEPA attachments or wet methods. The dry sanding or grinding of surfaces that contain lead, cadmium, chromium is prohibited. Provide methodology for removing lead, cadmium, chromium in the Lead, Cadmium, Chromium Compliance Plan. Select lead, cadmium, chromium removal processes to minimize contamination of work areas outside the control area with lead, cadmium, chromium contaminated dust or other lead, cadmium, chromium contaminated debris or waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, chromium. Describe this removal process in the Lead, Cadmium, Chromium Compliance Plan. Provide methodology for lead, cadmium and chromium, LBP/PWL control and processes to minimize contamination of work areas outside the control area with lead, cadmium, chromium contaminated dust or other lead, cadmium, chromium contaminated debris/waste and to ensure that unprotected personnel are not exposed to hazardous concentrations of lead, cadmium, chromium. Describe this lead,, cadmium and chromium, LBP/PWL removal/control process in the Lead, Cadmium, Chromium Compliance Plan.

3.3.2.1 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium - Indoor Removal

Perform manual removal in the lead, cadmium, chromium control areas using enclosures, barriers or containments. Collect residue and debris for disposal in accordance with the JEGS, U.S. federal, GOJ national, prefectural, and local requirements.

3.3.2.2 Paint with Lead, Cadmium, Chromium or Material Containing Lead, Cadmium, Chromium - Outdoor Removal

Perform outdoor removal as indicated in the JEGS, U.S. federal, GOJ national, prefectural, and local regulations and in the Lead, Cadmium, Chromium Compliance Plan. The worksite preparation (barriers or containments) must be job dependent and presented in the Lead, Cadmium, Chromium Compliance Plan.

3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, chromium controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

a. Vacuum all clothing before entering the contaminated change room.

b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.

d. Change to clean clothes prior to leaving the clean clothes storage area.

3.4 FIELD QUALITY CONTROL

3.4.1 Tests

3.4.1.1 Air and Wipe Sampling

Conduct sampling for lead, cadmium, chromium in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and as specified herein. Air and wipe sampling must be directed or performed by the CP.

a. The CP must be on the job site directing the air and wipe sampling and inspecting the PWL or MCL removal work to ensure that the requirements of the contract have been satisfied during the entire PWL or MCL operation.

b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.

c. Submit results of air samples, signed by the CP, within 72-hours after the air samples are taken.

d. Conduct area air sampling daily, on each shift in which lead, cadmium and chromium and lead-based paint removal operations are performed, in areas immediately adjacent to the lead, cadmium and chromium control area. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium per cubic meter of air. If 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium per cubic meter of air is reached or exceeded, stop work, correct the conditions(s) causing the increased levels. Notify the Contracting Officer immediately. Determine if condition(s) require any further change in work methods. Resume removal work only after the CP and the Contracting Officer give approval.

3.4.1.2 Sampling After Removal

After the visual inspection, collect wipe samples according to the HUD protocol contained in HUD 6780 to determine the lead, cadmium and chromium content of settled dust in micrograms per square meter foot of surface area.

3.5 CLEANING AND DISPOSAL

3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, chromium control area free of accumulations of dust and debris. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each shift and when the lead, cadmium, chromium operation has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area as indicated by the Lead, Cadmium, Chromium Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become contaminated at any time during the work, clean, visually inspect, and then wipe sample all contaminated areas. The CP must then certify in writing that the area has been cleaned of lead, cadmium and chromium contamination before clearance testing.

3.5.1.1 Clearance Certification

The CP must certify in writing that air samples collected outside the lead, cadmium, chromium control area during paint removal operations are less than 30 micrograms of lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; and that there were no visible accumulations of material and dust containing lead, cadmium, chromium left in the work site. Do not remove the lead, cadmium, chromium control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt of the CP certification.

The third party consultant must certify surface wipe sample results collected inside and outside the work area are less than 40 micrograms of lead per 0.1 square meter on floors, less than 250 micrograms of lead per 0.1 square meter on interior window sills and less than 400 micrograms of lead per 0.1 square meter on window troughs.

3.5.2 Disposal

a. Dispose of material, whether hazardous or non-hazardous in accordance with all laws and provisions and all JEGS, U.S. federal, GOJ national, prefectural laws and regulations or local regulations. Ensure all waste is properly characterized. The result of each waste characterization (TCLP for RCRA materials) will dictate disposal requirements.

b. Contractor is responsible for segregation of waste. Collect lead, cadmium, chromium contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, chromium contaminated clothing that may produce airborne concentrations of lead, cadmium, chromium particles. Label the containers in accordance with the JEGS, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.

c. Dispose of lead, cadmium, chromium contaminated material classified as hazardous waste at a locally approved hazardous waste treatment, storage, or disposal facility off Government property.

d. Accumulate waste materials in U.S. Department of Transportation (49 CFR 178) approved 55 gallon drums or appropriately sized container for smaller volumes. Properly label each drum to identify the type of hazardous material (49 CFR 172). For hazardous waste, the collection container requires marking/labeling in accordance with the JEGS, 40 CFR 262 and corresponding GOJ national and local prefectural regulations during the accumulation/collection timeframe. The

Contracting Officer or an authorized representative will assign an area for accumulation of waste containers. Coordinate authorized accumulation volumes and time limits with the host installation environmental function.

e. Handle, store, transport, and dispose lead, cadmium, chromium or lead, cadmium, chromium contaminated waste in accordance with the JEGS, 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265. Comply with land disposal restriction notification requirements as required by 40 CFR 268.

f. All lead, cadmium, and chromium waste generation, management,and disposal will be coordinated with the host installation environmental function.

3.5.2.1 Disposal Documentation

Coordinate all disposal or off-site shipments of lead, cadmium, and chromium waste with the host installation environmental function. Submit written evidence of TSD approval to demonstrate the hazardous waste treatment, storage, or disposal facility (TSD) is approved for lead, cadmium, chromium disposal by the EPA, State or local regulatory agencies. Submit one copy of the completed hazardous waste manifest, signed and dated by the initial transporter in accordance with 40 CFR 262, JEGS, GOJ National, prefectural laws and local regulations. Provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

3.5.2.2 Payment for Hazardous Waste

Payment for disposal of hazardous and non-hazardous waste will not be made until a signed copy of the manifest from the treatment or disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of lead, cadmium, chromium containing materials or non-hazardous waste delivered to the treatment or disposal facility.

-- End of Section --

SECTION 02 84 33

REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs)
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPAN MINISTRY OF THE ENVIRONMENT(MOE)~~MINISTRY OF THE ENVIRONMENT GOVERNMENT OF JAPAN (MOE)

Act No. 137	(1970, Amended 1991 and 2006) Waste Management and Public Cleansing Law

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS	(Apr 202~~2~~4) ~~-~~Japan Environmental Governing Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.145	Specifications for Accident Prevention Signs and Tags
40 CFR 761	Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions

1.2 REQUIREMENTS

The work includes the management of PCB containing ~~[~~ballasts~~][~~ and ~~][~~ transformers~~]~~. The Contractor shall inspect, confirm and identify all PCB-containing items prior to commencement of work. Perform work in accordance with 40 CFR 761, JEGS, and the requirements specified herein.

1.3 DEFINITIONS

1.3.1 Industrial Hygienist (IH)/Private Qualified Person (PQP)

An industrial hygienist/private qualified person (IH/PQP) hired by the Contractor shall be a registered Architect, Professional Engineer (licensed), or US Certified Industrial Hygienist and having demonstrable experience in hazardous materials management (i.e. lighting ballasts and lamps containing PCBs or mercury), who is trained in the recognition and control of hazardous chemical related hazards, and has the authority to take prompt corrective actions to control the hazard.

1.3.2 Leak

Leak or leaking means any instance in which a PCB Article, PCB Container, or PCB Equipment has any PCBs on any portion of its external surface.

~~[~~1.3.3 Lamps

Lamp is defined as the bulb or tube portion of an electric lighting device. A lamp is specifically designed to produce radiant energy, most often in the ultraviolet, visible, and infra-red regions of the electromagnetic
spectrum. Examples of common universal waste electric lamps include, but are not limited to, fluorescent, high intensity discharge, neon, mercury vapor, high pressure sodium, and metal halide lamps.

~~]~~1.3.4 PCBs

PCBs as used in this specification shall mean the same as PCBs, PCB Article, PCB Article Container, PCB Container, PCB Equipment, PCB Item, PCB Transformer, PCB-Contaminated Electrical Equipment, as defined in the JEGS.

1.3.5 Spill

Spill means both intentional and unintentional spills, leaks, and other uncontrolled discharges when the release results in any quantity of PCBs running off or about to run off the external surface of the equipment or other PCB source, as well as the contamination resulting from those releases.

1.4 QUALITY ASSURANCE

1.4.1 PCB and Lamp Management Plan

Prior to handling any PCB items, the Contractor shall submit a PCB and Lamp Management and Disposal Plan. The submitted plan shall include the PCB and Lamp removal and disposal procedures in accordance with the JEGS, GoJ, and prefectural laws, qualifications for the IH/PQP, Installation specific requirements and plans, a designated Installation point of contact (POC),and a Japan Industrial Waste Collection and Transport Permit.

Include within the PCB and Lamp Management and Disposal Plan, a job-specific plan within ~~[~~20~~]~~ calendar days after ~~[~~receipt of Notice to Proceed~~]~~ explaining the work procedures to be used in the removal, packaging, and storage of PCB-containing lighting ballasts and associated hazardous material-containing lamps.

Submit a detailed job-specific plan of the work procedures to be used in the removal of PCB-containing materials, not to be combined with other hazardous abatement plans. Provide a Table of Contents for each abatement submittal which shall follow the sequence of requirements in the contract. The plan shall include a sketch showing the location, size, and details of PCB control areas~~[, location and details of decontamination rooms, change rooms, shower facilities, and mechanical ventilation system]~~. Include in the plan, eating, drinking, smoking and restroom procedures, interface of trades, sequencing of PCB related work, PCB disposal plan, respirators, protective equipment, and a detailed description of the method of containment of the operation to ensure that PCB contamination is not spread or carried outside of the control area. ~~[Include provisions to ensure that airborne PCB concentrations of 0.50 milligrams per cubic meter of air are not exceeded outside of the PCB control area. Include air sampling, training and strategy, sampling methodology, frequency, duration of sampling, and qualifications of air monitoring personnel in the air~~

~~sampling portion of the plan.]~~ Obtain approval of the plan prior to the start of PCB removal work. Include in the plan: Requirements for Personal Protective Equipment (PPE), spill cleanup procedures and equipment, eating, smoking and restroom procedures.

The Plan shall comply with applicable requirements of Federal, JEGS, GOJ and prefectural PCB and hazardous material-related regulations. The plan shall be approved and signed by the IH/PQP. Obtain approval of the plan by the Contracting Officer prior to the start of PCB and/or lamp removal work.

1.4.1.1 PCB and Lamp Disposal Plan

Include within the PCB and Lamp Management Plan, a PCB and lamp Disposal Plan within ~~[~~45~~]~~ ~~[_____]~~ calendar days after ~~[receipt of Notice to Proceed]~~ ~~[~~prior to removal work~~]~~. The PCB and Lamp Disposal Plan shall comply with applicable requirements of the JEGS and applicable U.S. federal, GOJ, and local prefecture and JEGS PCB regulations and address:

a. Identification of PCB wastes associated with the work.

b. Estimated quantities of wastes to be generated, disposed of, and recycled.

c. Names and qualifications of each Contractor personnel who will be working on-site with PCB and hazardous material-containing lamp wastes and that will be transporting, storing, treating, and disposing of the wastes. Include the facility location. Include a 24-hour point of contact. Furnish two copies of PCB and hazardous material-containing lamp waste permit applications and pertinent identification numbers, as required.

d. Spill prevention, containment, and cleanup contingency measures to be implemented.

e. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.

f. Work plan and schedule for PCB and hazardous material-containing lamp waste removal, containment, storage, transportation, disposal and or recycling. Wastes shall be cleaned up and containerized daily.

1.4.2 Regulatory Requirements

Perform PCB related work in accordance with the JEGS, Act No. 137, and other applicable US Federal, GOJ, and local laws and regulations. Perform hazardous material-containing lamps storage and transport in accordance with the JEGS, Act No. 137, and other applicable US Federal, GOJ, and local laws and regulations.

1.4.2.1 No Smoking

Smoking is not permitted within 15 m of the PCB control area. Provide "No Smoking" signs as directed by the Contracting Officer.

1.4.2.2 Work Operations

Ensure that work operations or processes involving PCB or PCB-contaminated materials are conducted in accordance with JEGS and the applicable requirements of this section, including but not limited to:

a. Obtaining advance approval of PCB storage sites.

b. Notifying Contracting Officer prior to commencing the operation.

c. Reporting leaks and spills to the Contracting Officer.

d. Cleaning up spills.

e. Maintaining an access log of employees working in a PCB control area and providing a copy to the Contracting Officer upon completion of the operation.

f. Inspecting PCB and PCB-contaminated items and waste containers for leaks and forwarding copies of inspection reports to the Contracting Officer.

g. Maintaining a spill kit as specified in paragraph entitled "PCB Spill Kit."

h. Maintaining inspection, inventory and spill records.

1.4.3 Training

Industrial Hygienist/Private Qualified Person (IH/PQP) shall instruct and certify the training of all persons involved in the removal of PCB containing materials/equipment and lamps containing hazardous substances. The instruction shall include: The dangers of PCB and hazardous material exposure, decontamination, safe work practices, and 29 CFR 1910, JEGS, GOJ, and prefectural regulations.

The IH/PQP shall review and approve the PCB and Lamp Management Plan and temporary on-site Storage Plans.

1.4.4 Regulation Documents

Maintain at all times one copy each at the office and one copy each in view at the job site of pertinent JEGS and a copy of the Contractor PCB and Lamp Management Plan.

1.4.5 Qualifications of IH/PQP

An industrial hygienist/private qualified person (IH/PQP) hired by the Contractor shall be a registered Architect, Professional Engineer (licensed), or US Certified Industrial Hygienist and having demonstrable experience in hazardous materials management (i.e. lighting ballasts and lamps containing PCBs or mercury), who is trained in the recognition and control of hazardous chemical related hazards, and has the authority to take prompt corrective actions to control the hazard. An IH/PQP must have working knowledge of applicable GOJ, Federal, local prefecture, and JEGS regulations as well as occupational safety and health regulations and shall be capable of recognizing chemical hazards associated with lighting ballasts, lamps, and other similar building equipment or materials.

1.4.6 Training Certification

Submit certificates, prior to the start of work but after the main abatement submittals, signed and dated by the CIH and by each employee stating that the employee has received training. Certificates shall be organized by individual worker, not grouped by type of certificates.

1.4.7 Notification

Notify the Contracting Officer 20 days prior to the start of PCB removal work.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

PCB and Lamp Management Plan; G

SD-07 Certificates

*Qualifications of IH/PQP; G

Training certification

Copy E of the Japanese Hazardous Waste Manifest

Notification

Certification Of Decontamination; G

Post Cleanup Sampling; G

1.6 EQUIPMENT

1.6.1 Special Clothing

Work clothes shall consist of PPE as required by OSHA regulations, including, but not limited to the following:

a. Disposable coveralls

b. Gloves (Disposable rubber gloves may be worn under these)

c. Disposable foot covers (polyethylene)

d. Chemical safety goggles

e. Half mask cartridge respirator.

1.6.2 Special Clothing for Government Personnel

Provide PPE specified in paragraph entitled "Special Clothing" to the

Contracting Officer as required for inspection of the work.

1.6.3 PCB Spill Kit

Assemble a spill kit to include the following items:

ITEM	MINIMUM QUANTITY
1. Disposable gloves (polyethylene)	6 prs
2. Gloves with a high degree of impermeability to PCB	6 prs
3. Disposable coveralls with permeation resistance to PCB	4 ea
4. Chemical safety goggles	2 ea
5. Disposable foot covers (polyethylene)	6 prs
6. PCB Caution Sign: "PCB Spill--Authorized Personnel Only"	2 ea
7. Banner guard or equivalent banner material	30 m
8. Absorbent material	
9. Blue polyethylene waste bags	5 bags
10. Cloth backed tape	5 ea
11. Area access logs, blank	1 roll
12. Brattice cloth, 2 m x 2 m	10 ea
13. Rags	1 piece
14. Ball point pens	20 ea
15. Herculite, 1.5 m x 1.5 m and 3 m x 3 m	2 ea and 1 ea
16. Blank metal signs and grease pencils	
17. Waste containers 208 liters drum, may be used as container for kit	2 ea ~~[1]~~ ~~[____]~~ ea

1.7 SCHEDULING

Notify the Contracting Officer 20 days prior to the start of PCB and hazardous material-containing lamp removal work.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 WORK PROCEDURE

Furnish labor, materials, services, and equipment necessary for the removal of PCB containing lighting ballasts, associated mercury-containing fluorescent lamps, and high intensity discharge (HID) lamps in accordance with the JEGS, applicable GOJ and local, prefectural regulations. Do not expose PCBs to open flames or other high temperature sources since toxic decomposition by-products may be produced. Do not break mercury containing fluorescent lamps or high intensity discharge lamps.

3.1.1 Protection

a. Provide material and labor for construction of a decontamination room, a clean room, and shower facilities. Provide rooms with doors and attach to the exit ways of PCB work areas. Rooms shall be of sufficient size to accommodate the Contractor's operation within. ~~[~~ Existing facilities with water closets, urinals, wash basins and showers may be used if available to the Contractor.~~]~~ ~~[Provide portable toilet and shower facilities. Locate shower facilities between the clean room and decontamination room.]~~ Provide separate clothing lockers or containers in each room to prevent contamination of street and work clothes.

b. Remove PCB-contaminated PPE in the decontamination room. Workers shall then proceed to showers. Workers shall shower before lunch and at the end of each day's work. Hot water, towels, soap, and hygienic conditions are the responsibility of the Contractor.

3.1.2 PCB Control Area

Isolate PCB control area by physical boundaries to prevent unauthorized entry of personnel. Food, drink and smoking materials shall not be permitted in areas where PCBs are handled or PCB items are stored.

3.1.3 Personnel Protection

Workers shall wear and use PPE, as recommended by the Industrial Hygienist, upon entering a PCB control area. If PPE is not required per the CIH, specify in the PCB and Lamp Management Plan.

3.1.4 Footwear

Work footwear shall remain inside work area until completion of the job.

3.1.5 Permissible Exposure Limits (PEL)

PEL for PCBs is 0.5 mg/m3 on an 8-hour time weighted average basis.

3.1.6 Special Hazards

a. PCBs shall not be exposed to open flames or other high temperature sources since toxic decomposition by-products may be produced.

b. PCBs shall not be heated to temperatures of 55 degrees C or higher without Contracting Officer's concurrence.

3.1.7 PCB Caution Label

Affix labels to PCB waste containers and other PCB-contaminated items. Provide label with sufficient print size to be clearly legible, with bold print on a contrasting background, displaying the following: CAUTION: Contains PCBs (Polychlorinated Biphenyls). Installations shall label PCB waste as a hazardous waste per JEGS Appendix 16B.

3.1.8 PCB Caution Sign

29 CFR 1910.145. Provide signs at approaches to PCB control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area.

3.2 PCB TRANSFORMERS

3.2.1 Draining of Transformer Liquid

Perform work in accordance with the JEGS and approved PCB and Lamp Management Plan. Drain the transformer, switches, and regulators of free flowing liquid prior to transportation. The drums shall not contain more than 190 liters of oil. If the equipment cannot be drained, then place it in DOT Spec 17C drums.

3.2.2 Markings

Provide drums and drained PCB-contaminated electrical equipment with caution label markings as specified in paragraph entitled "PCB Caution Label" in both Japanese and English.

3.2.3 Laboratory Analysis

All transformers shall have a laboratory analysis for turn-in. DLA-DS prefers a gas chromatograph test. The only two exceptions to this rule are:

a. The transformer is hermetically sealed (solder sealed or fusion sealed. No access ports or openings).

b. The name plate states that the transformer contains pyranol, interteen, etc.

Attach a copy of the lab analysis to both the DD 1348-1 and the transformer itself.

3.2.4 Markings

3.2.4.1 Transformers, Less Than 0.5ppm

Add absorbent material to absorb residue oil remaining after draining. Write the date drained on the transformer. Turn in transformers to the DLA-DS Scrapyard. Telephone 471-3636 to schedule appointment for turn-in.

3.2.4.2 Transformers, Greater Than 0.5 ppm

Stencil date drained on the transformer. ~~Turn in transformer to [include~~

~~specific installation information]~~Transformers shall be turned in in accordance the the direction provided by the local Environmental Division with jurisdition.

3.2.4.3 Drums

Stencil on DOT-approved 208 liter drums containing PCB liquid the following (in English and Japanese):

a. ppm

b. Date drum filled

c. Serial number of transformer liquid came from

d. National Stock Number

(1) "9999-00-OIL" for <50 ppm

(2) "9999-00-CONPCB" for 50-499 ppm

(3) "9999-00-PCBOIL" for >500 ppm

Do not mix different ppms in the same drum. Drums must have a 50 mm ullage space from the top of the drum.

3.3 PCB REMOVAL

Select PCB removal procedure to minimize contamination of work areas with PCB or other PCB-contaminated debris/waste. Handle PCBs such that no skin contact occurs. PCB removal process should be described in the PCB and Lamp Management Plan.

3.3.1 Confined Spaces

As feasible, do not carry out PCB handling operations in confined spaces as defined in 29 CFR 1910.146.

3.3.2 Control Area

Establish a PCB control area around the PCB item as specified in paragraph entitled "PCB Control Area." Only personnel briefed on the elements in the paragraph entitled "Training" and on the handling precautions shall be allowed into the area.

3.3.3 Exhaust Ventilation

If used, exhaust ventilation for PCB operations shall discharge to the outside and away from personnel.

3.3.4 Temperatures

As feasible, handle PCBs at ambient temperatures and not at elevated temperatures.

3.3.5 Solvent Cleaning

Clean contaminated tools, containers, etc., after use by rinsing three times with an appropriate solvent or by wiping down three times with a solvent wetted rag. Suggested solvents are stoddard solvent or hexane.

3.3.6 Drip Pans

Drip pans are required under portable PCB transformers and rectifiers in use or stored for use. The pans shall have a containment volume of at least one and one-half times the internal volume of PCBs in the item.

3.3.7 Evacuation Procedures

Procedures shall be written for evacuation of injured workers. Aid for a seriously injured worker shall not be delayed for reasons of decontamination.

3.4 PCB SPILL CLEANUP REQUIREMENTS

3.4.1 PCB Spills

Immediately report to Installation ENV Office, Installation Emergency Response POC, and the Contracting Officer any PCB spills on the ground or in the water, PCB spills in drip pans, or PCB leaks.

3.4.2 PCB Spill Control Area

Rope off an area around the edges of a PCB leak or spill and post a "PCB Spill Authorized Personnel Only" caution sign. Immediately transfer leaking items to a drip pan or other container.

3.4.3 PCB Spill Cleanup

Perform in accordance with the JEGS and approved PCB and Lamp Management Plan. Initiate cleanup of spills as soon as possible, but no later than 48 hours of its discovery. ~~[To clean up spills, personnel shall wear the PPE prescribed in paragraph entitled "Special Clothing" of this section.]~~ If misting, elevated temperatures or open flames are present, or if the spill is situated in a confined space, notify the Installation ENV Office, Installation Emergency Response POC, and Contracting Officer. Mop up the liquid with rags or other conventional absorbent. The spent absorbent shall be properly contained and disposed of as solid PCB waste. The spent absorbent shall be properly contained and turned over as solid PCB waste.

3.4.4 Records and Certification

Document the cleanup with records of decontamination in accordance with the JEGS and applicable US Federal, GOJ and prefectural requirements for PCB Spill Cleanup. Provide certification of decontamination.

3.4.5 Sampling Requirements

Perform post cleanup sampling as required by JEGS Chapter 7.3. Do not remove boundaries of the PCB control area until site is determined satisfactorily clean by the Contracting Officer.

3.5 STORAGE FOR DISPOSAL

3.5.1 Storage Containers for PCBs

Store liquid PCBs in accordance with UN specification packaging requirements: PGII standards. Store nonliquid PCB mixtures, articles, or equipment in accordance with DLA Disposition Services requirements..

3.5.2 Waste Containers

Label with the following:

a. "Solid (or Liquid) Waste Polychlorinated Biphenyls"

b. The PCB Caution Label, paragraph entitled "PCB Caution Label"

c. The date the item was placed in storage and the name of the cognizant activity/building.

3.5.3 PCB Articles and PCB-Contaminated Items

Label with items b. through c. above.

3.5.4 Approval of Storage Site

Obtain in advance Contracting Officer approval using the following criteria without exception.

a. Adequate roof and walls to prevent rainwater from reaching the stored PCBs.

b. An adequate floor which has continuous curbing with a minimum 150 mm high curb. Such floor and curbing shall provide a containment volume equal to at least two times the internal volume of the largest PCB article or PCB container stored therein or 25 percent of the total internal volume of all PCB equipment or containers stored therein, whichever is greater.

c. No drain valves, floor drains, expansion joints, sewer lines, or other openings that would permit liquids to flow from the curbed area.

d. Floors and curbing constructed of continuous smooth and impervious materials such as portland cement, concrete or steel to prevent or minimize penetrations of PCBs.

e. Not located at a site which is below the 100-year flood water elevation.

f. Each storage site shall be posted with the PCB Caution Sign, paragraph entitled "PCB Caution Sign."

3.6 CLEANUP

Maintain surfaces of the PCB control area free of accumulations of PCBs. Restrict the spread of dust and debris; keep waste from being distributed over work area.

Do not remove the PCB control area and warning signs prior to the Contracting Officer's approval. Reclean areas showing residual PCBs.

3.7 DISPOSAL/TURNOVER

~~[~~Turn in hazardous material-containing lamps, ballasts, and other PCB or hazardous material-containing materials to the Installation Environmental Division. Contact Installation Environmental Division ~~at [phone number]~~ at least [5] working days in advance to make arrangements for delivery of these materials to the storage site.

Do not dispose of mercury or PCB related materials at an off Government facility. Do not consider US manufactured light ballasts with "No PCBs" labeling to contain PCBs less than Japan regulatory limits and always acquire PCB content related information from the manufacturer. All ballasts, regardless of PCB or non-PCB containing types, shall be turned into the Installation Environmental Division per the turnover procedures as follows.

3.8 Ballasts

[For removed ballasts that are not leaking, up to 300 ballasts may be stored on wooden or plastic pallets on-site for up to 30 days prior to delivery.

Ballasts that are leaking must be immediately stored in leak-proof drip pans, buckets, or double bagged in nitrile bags and stored in a secure area that is not exposed to rainfall. Mark the storage area in English and Japanese in accordance with the JEGS and Japan national and prefectural regulation.

Any PCB oil that has contaminated the ballast mount or other fixtures must be cleaned or removed and turned in with the ballasts. Prior to transport for disposal, the contractor shall perform the following:

a. Segregate ballasts by manufacturer.

b. Visit the websites of the manufacturers to verify if the ballasts contain PCB or not by product model number, product color, etc. Certificates shall be obtained stating that the ballast does not contain PCB.

c. Sort the ballast in cardboard cases by Non-PCB, PCB containing, unknown for each manufacturer/country of origin.

d. Bring ballasts to the Installation Environmental Division with certificates and supporting documents.

All non-Japanese ballasts, including those with a "NO PCB" label shall be assumed to contain as high as 49 ppm and shall be handled, stored, transported, and disposed of as PCB ballasts. The contractor shall deliver all non-Japanese ballasts to the Installation ENV Office.

For Japanese-made ballasts, the Contractor shall segregate regulated PCB ballasts from non-regulated ballasts according to GOJ and prefectural regulations. Information to discriminate between regulated and non-regulated ballasts can be obtained from:
https://www.env.go.jp/recycle/poly/
https://www.env.go.jp/recycle/poly/law/no_14091618.pdf

Regulated PCB ballasts shall be delivered to the Installation ENV Office. Non-regulated Japanese ballasts become property of the contractor and shall be disposed of in accordance with GOJ and prefectural regulations. In the submitted PCB and Lamp Management Plan, clearly state how non-regulated Japanese ballasts will be disposed.

~~]~~

~~[~~ Light ballasts may still contain PCBs even where a "No PCB" label exists on the ballast due to regulatory differences between U.S. and GOJ definitions for "PCB Free". ~~[Regardless of PCB concentration, manufacturer, country of origin, all light ballasts shall be removed, containerized, and turned into the Installation Environmental Division.]~~ PCB abatement or handling shall be in accordance with the JEGS .

The Contractor shall identify all project ballasts that contain and do not contain PCBs or are suspect to contain PCBs. The Contractor shall submit the PCB ballast list (to include suspect PCB ballasts) to the Installation ENV Office. PCB containing ballasts and suspect PCB containing ballasts shall be segregated by country of manufacture.

The following procedures for US manufactured and Japan manufactured ballasts shall be used in the identification of PCB ballasts and described in this section.

3.8.1 For US Manufactured Ballasts

US manufactured ballasts shall not be mixed with Japan manufactured ballasts. The Contractor shall segregate and palletize US manufactured ballasts separately from Japan manufactured ballasts.

3.8.2 For Japan Manufactured Ballasts

The Contractor shall carefully examine the ballast to identify the manufacturer name and year made. Consult with the manufacturer to identify any presence of PCBs. If absence of PCB cannot be confirmed, assume and treat the ballast as a PCB item.

~~[~~ Japan manufactured ballasts shall not be mixed with US manufactured ballasts. The Contractor shall segregate and palletize Japan manufactured ballasts separately from US manufactured ballasts.~~]~~

3.8.3 For Unmarked/Unlabeled Ballasts

The Contractor shall treat all unmarked/unlabeled ballasts as suspect PCB-containing materials. Segregate unmarked/unlabeled ballasts from other US or Japanese manufactured items.

3.8.4 ~~[~~Fluorescent Light Tubes (Bulbs)~~]~~

~~[The Contractor shall carefully remove and store fluorescent light bulbs in a manner such that they remain intact. Fluorescent light tubes suspect for containing mercury shall be handled and disposed of by the Contractor in accordance to all GOJ, Federal, local prefectural laws and regulations and the JEGS. In the event of a lighting tube/lamp breaking, sweep and place waste in double plastic taped bags and dispose of as appropriate to GOJ, Federal, local prefecture laws and regulations and the JEGS.]~~

3.9 Japan Specially Controlled Hazardous Waste Disposal Permit

3.10 Japan Hazardous Waste Manifest (JHWM)

A minimum of 5-days prior to the scheduled transportation of hazardous material containing lamps or other PCB containing ballasts from the installation, contact the Installation ENV Office to schedule a pre-transportation inspection. A government representative will inspect, annotate the (JHWM), and authorize the shipment. No toxic wastes shall be transported off the installation without the approval of a US government representative.

Within 60-days after the date of transportation of hazardous waste from the installation, submit Copy E of the Japanese Hazardous Waste Manifest bearing the signature/stamp of the final hazardous material waste disposal/recycling facility as an SD-07 closeout submittal.

For SCIW shipments to disposal facilities outside the prefecture, the manifest must be the "tsumikae" type manifest to allow multiple Chain of Custody entries.

Contract final payment will not be approved by the COR until the Copy E is submitted.

Contractors failing to submit Copy E of the JHWM may be reported to United States Forces Japan (USFJ) and the Ministry of the Environment.

-- End of Section --

SECTION 03 01 00

REHABILITATION OF CONCRETE
02/18

PART 1 GENERAL

1.1 SCOPE

This specification governs the rehabilitation of structural concrete.

1.2 DEFINITIONS

1.2.1 Bracing

Temporary supplemental members used to avoid local or global instability during construction, evaluation, or repair that are intended to be removed after completion of construction.

1.2.2 Delamination

A planar separation in a material that is roughly parallel to the surface of the material.

1.2.3 Rehabilitation

Repairing or modifying an existing structure to a desired useful condition.

1.2.4 Repair

The reconstruction or renewal of concrete parts of an existing structure for its maintenance or to correct deterioration, damage, or faulty construction of members or systems of a structure.

1.2.5 Shoring

Props or posts of timber or other material in compression used for the temporary support of excavations, formwork, or unsafe structures; the process of erecting shores.

1.2.6 Termination Joint

The interface where a placement of repair material meets existing concrete, the edge of an expansion joint, or other existing surfaces.

1.2.7 Unsound Concrete

Concrete that is fractured, delaminated, spalled, deteriorated, defective, contaminated or otherwise damaged.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 503.2-503.4	(2010, R 2003) Three Epoxy Specifications
ACI 503.7	(2007) Specification for Crack Repair by Epoxy Injection
ACI 548.4	(2011) Standard Specification for Latex-Modified Concrete (LMC) Overlays
ACI 548.8	(2007) Specification for Type EM (Epoxy Multi-Layer) Polymer Overlay for Bridge and Parking Garage Decks
ACI 548.9	(2008) Specification for Type ES (Epoxy Slurry) Polymer Overlay for Bridge and Parking Garage Decks
ACI 548.10	(2010) Specification for Type MMS (Methyl Methacrylate Slurry) Polymer Overlays for Bridge and Parking Garage Decks
ACI 548.12	(2012) Specification for Bonding Hardened Concrete and Steel to Hardened Concrete with an Epoxy Adhesive

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE/SEI 37	(2015) Design Loads on Structures During Construction

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M	(2018) Standard Specification for Concrete Aggregates
ASTM C387/C387M	(2017) Standard Specification for Packaged, Dry, Combined Materials for Concrete and High Strength Mortar
ASTM C881/C881M	(2020a) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete
ASTM C882/C882M	(2020) Bond Strength of Epoxy-Resin Systems Used with Concrete by Slant Shear
ASTM C928/C928M	(2020a) Standard Specification for Packaged, Dry, Rapid-Hardening Cementitious Materials for Concrete Repairs
ASTM C1059/C1059M	(2021) Standard Specification for Latex Agents for Bonding Fresh to Hardened Concrete
ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation

ASTM C1438	(2013; R 2017) Standard Specification for Latex and Powder Polymer Modifiers for use in Hydraulic Cement Concrete and Mortar
ASTM C1600/C1600M	(2017) Standard Specification for Rapid Hardening Hydraulic Cement
ASTM C1602/C1602M	(2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete
ASTM D93	(2019) Standard Test Methods for Flash-Point by Pensky-Martens Closed Cup Tester
ASTM D323	(2015a) Vapor Pressure of Petroleum Products (Reid Method)
ASTM D542	(2014) Index of Refraction of Transparent Organic Plastics
ASTM D1078	(2011) Standard Test Method for Distillation Range of Volatile Organic Liquids
ASTM D3418	(2015) Transition Temperatures of Polymers by Differential Scanning Calorimetry
ASTM D4016	(2014) Viscosity of Chemical Grouts by Brook field Viscometer (Laboratory Method)
ASTM D4580/D4580M	(2012) Standard Practice for Measuring Delaminations in Concrete Bridge Decks by Sounding
ASTM E329	(2021) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

JAPANESE ARCHITECTURAL STANDARD SPECIFICAITONS (JASS)

JASS 5	(2015) Reinforced Concrete Work

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete
JIS A 5308	(20149) Ready-Mixed Concrete

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM,

MLIT-SS Chapter 6	MLIT Architectural Standard

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are

~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G~~[, [_____]]~~

Work Plan; G~~[, [_____]]~~

Quality Control Plan; G~~[, [_____]]~~

SD-03 Product Data

Conventional Concrete

Polymers

Miscellaneous Materials And Equipment

~~SD-04 Samples~~

~~Reinforcement And Reinforcement Supports~~

~~[Polymers]~~

~~[Miscellaneous Materials And Equipment]~~

SD-05 Design Data

Formwork And Shoring; G~~[, [_____]]~~

Repair Procedures; G~~[, [_____]]~~

Mixture Proportioning; G~~[, [_____]]~~

SD-06 Test Reports

Mixture Proportioning

Quality Control

~~[Tolerence Report]~~

~~[Reinforcement And Reinforcement Supports]~~

~~[~~Conventional Concrete~~]~~

~~[Polymers]~~

~~[Miscellaneous Materials And Equipment]~~

SD-07 Certificates

Qualifications

Reinforcement And Reinforcement Supports

~~[~~Conventional Concrete~~]~~

~~[Polymers]~~

~~SD-08 Manufacturer's Instructions~~

~~[Equipment For Concrete Preparation]~~

~~[Conventional Concrete]~~

~~[Polymers]~~

~~[Miscellaneous Materials And Equipment]~~

1.5 QUALITY ASSURANCE

1.5.1 General Requirements

a. Follow the requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE for Work involving portland cement concrete.

b. To protect personnel from overexposure to toxic materials, conform to the applicable manufacturer's Safety data sheets or local regulations. Submit manufacturer's Safety Data Sheets for all polymers as well as other potentially hazardous materials.

c. Submit the repair procedures for executing the work as well as the test data and documentation on materials used for repair. Submittal must include component materials, mixture proportions, and supplier's quality control program.

d. Inspection and testing of surface preparation as well as placement of reinforcing steel must be in accordance with provisions included herein and the Contract Document.

e. Sampling and testing of materials, as well as inspection and testing of work, must be in accordance with established procedures, manufacturer's instructions, specific instructions from the Contracting Officer if given, or recommended practices as referenced herein and the Contract Documents.

f. Trial batches and testing requirements for various repair materials specified are the responsibility of the Contractor.

g. The testing agency must inspect, sample, and test repair materials and concrete production as required. When it appears that material furnished or work performed by Contractor fails to conform to Contract Documents the testing agency will immediately report such deficiency.

1.5.2 Quality Control Plan

Submit a quality control plan as specified in Sections 01 45 00.00 10 QUALITY CONTROL ~~[~~and~~]~~ ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~]~~.

1.5.3 Qualifications

The submittals must where applicable, identify agencies and individuals who will be working on this contract and their relevant experience. Do not make changes in approved agencies or personnel without prior approval of the Contracting Officer.

1.5.3.1 Testing Agencies

In addition to the requirements of Section 01 45 00.00 10 QUALITY CONTROL, agencies that test concrete materials must meet the requirements of ASTM C1077. Testing agencies that test or inspect placement of reinforcing steel must meet the requirement of ASTM E329. Submit data on qualifications of Contractor's proposed testing agency for acceptance.

1.5.3.2 Quality Control Personnel

Field tests of repair materials required must be made by an ICRI Concrete Surface Repair Technician Tier 2 or Japan Concrete Institute Concrete Engineer. Submit ~~resumes,~~ pertinent information, past experience, training and education of all operators of specialized demolition equipment if needed for this and the three paragraphs above.

1.5.3.3 Contractor Qualifications

The contractor performing the repair work must have been involved in a minimum of ~~[three][_____]~~2 concrete repair projects similar in size and scope to this project for at least ~~[five][_____]~~3 years. Submit information, including name, dollar value, date, and point-of-contact for similar projects which demonstrates the required experience and/or training.

1.5.3.4 Worker Qualifications

a. Each worker engaged in the use of specialized removal or application equipment, including ~~[~~saw operators~~]~~, ~~[~~milling machine operators,~~]~~ ~~[hydromilling equipment operators,] [~~epoxy injection~~] [_____]~~, must have satisfactorily completed an instruction program and three years of experience in the operation of the equipment. ~~[~~The worker must have active experience with the equipment within five years of the project.~~]~~

b. Workers installing adhesive anchors must be ACI Adhesive Anchor Installer certified or equivalent.

1.5.3.5 Regulatory Requirements

Perform all work in accordance with applicable Federal, State, and local safety, health, and environmental requirements. The Contractor is responsible for obtaining all permits required by Federal, State, and local agencies for the performance of the work.

1.5.4 ~~[~~Pre-Construction Conference~~]~~

~~[~~Conduct a pre-construction conference to discuss repair materials performance requirements, control provisions, and roles and responsibilities for the Work to ensure that the Contractor's personnel understand all aspects of the repair material, its properties and application procedures. The conference must include the Contracting Officer or authorized representative, the Contractor's field superintendent and foreman, and a competent Technical Representative of the material manufacturer, and other involved trades or supplier representatives. The Technical Representative must be fully qualified to perform the work.~~]~~

1.5.5 Work Plan

Prepare a work plan describing the methods of concrete removal and repair, including methods, equipment and materials to be used for each feature. Submit the work plan for approval at least 30 days prior to the start of the work. The plan must include, but not be limited to, repair materials to be used with specific information on products and/or constituents, and requirements for handling, storage, etc., equipment to be used, surface preparation, and requirements for placement, finishing, curing and protection specific to the materials used. Include a description of field demonstrations in the work plan. Do not commence work until the work plan and field demonstration representative of the type of work are approved.

1.6 ACCEPTANCE OF REHABILITATION WORK

1.6.1 General Requirements

a. Completed concrete rehabilitation work must conform to applicable requirements of Contract Document and this specification. The Contractor is responsible to bring Work into compliance with requirements of Contract Documents if the Concrete repair work fails to meet one or more requirements of Contract Documents.

b. Correct rejected repair work by removing and replacing or by strengthening with additional construction acceptable to the Contracting Officer. Use repair methods that meet applicable requirements for function, durability, dimensional tolerances, and appearance.

c. Submit proposed work plan, repair methods, materials, and modifications to the Work needed to correct rejected repair work to meet the requirements of Contract Documents.

1.6.2 Tolerances

a. Construction tolerances for repairs must conform to ~~[ACI 117][_____]~~ JASS 5 and MLIT-SS Chapter 6. Where existing conditions do not allow tolerances to conform to ~~[ACI 117][_____]~~JASS 5 and MLIT-SS Chapter 6, use the details and materials for such conditions as indicated in the Contract Documents. For conditions not shown or that are different than indicated in the Contract Documents, notify the Contracting Officer before proceeding with the work at those locations.~~[Provide a tolerence report as required by Section 03 30 00 CAST-IN-PLACE CONCRETE.]~~

b. Inaccurately formed concrete surfaces resulting in concrete members with dimensions that exceed ~~[ACI 117][_____]~~JASS 5 and MLIT-SS Chapter 6 tolerances are subject to rejection.

1.6.3 Appearance

Concrete surfaces not meeting the requirements of the Contract Documents must be brought into compliance.

1.7 PROTECTION OF COMPLETED REHABILITATION WORK

a. Do not allow construction loads to exceed the loads that a structural member or structure is safely capable of supporting without damage. Provide supplemental support if construction loads are expected to exceed safe load capacity.

b. Protect repaired and adjacent areas from damage by construction traffic, equipment, and materials. During the curing period, protect repair materials from damage by mechanical disturbances, including load-induced stresses, shock, and vibration.

c. Protect repair materials from environmental damage by weather events during the length of the curing period.

PART 2 PRODUCTS

Products or materials used must conform to the requirements included herein as well as the Contract Documents. The usage of other products or materials not covered by this requirement or specified in the Contract Documents are permitted upon approval by the Contracting Officer. Additional information and submittals for products and materials not included in this document including product data, samples, design data, test reports, certificates, manufacturer's instructions, and field reports must be submitted as requested by the Contracting Officer.

2.1 MATERIALS FOR SHORING AND BRACING

2.1.1 Shoring and Bracing Systems

Use commercially manufactured and engineered shoring and bracing systems and components, except where custom built assemblies of lumber or other suitable materials are permitted by the Contracting Officer.

2.1.2 Design Requirements

The design of the bracing and shoring must be based on ASCE/SEI 37.

a. Non-manufactured shoring and bracing systems must have calculations signed and sealed by a Licensed Design Professional.

b. Members of non-manufactured shoring systems, must be designed in accordance with the provisions of the governing building code for the specific material of the member.

c. Members of manufactured shoring systems, consisting of pre-engineered components designed and produced specifically for structural shoring, must be used in accordance with the manufacturer's recommendations.

2.2 EQUIPMENT FOR CONCRETE PREPARATION

Means and methods used for concrete removal and surface preparation must be selected and used such as to minimize damage to the structure and to the concrete substrate that remains.

2.2.1 Equipment for Concrete Removal

Removal equipment and techniques must be suitable to produce concrete surface profiles and level of cleanliness in designated areas as required by this specification and the contract Documents.

2.2.1.1 ~~[~~Cutting Equipment~~]~~

a. The following cutting equipment are permitted: ~~[High-pressure water~~

~~jet without abrasives][~~Saw cutting~~][~~, Diamond wire cutting~~][Mechanical shearing][Stitch drilling][_____]~~.

b. Cutting, lifting, and transporting equipment must be adequate to cut, support, and transport concrete sections without incurring any damage to the existing structure.

2.2.1.2 Concrete Breakers

a. Provide sharp tips on breaker equipment to minimize microcracking damage in partial depth removal.

b. The use of the following impact equipment and methods is permitted: ~~[~~ Hand-held breakers~~][Boom-mounted breakers][~~, Scabblers~~][~~, Needle scalers~~][~~, Scarifiers~~][~~, Milling methods~~][_____]~~.

~~c. [The maximum breaker size is [_____]][_____].~~

~~2.2.1.3 [Hydromilling Equipment]~~

~~a. Hydromilling equipment must include a trailer-mounted water tank, pumps, high-pressure hose, wand with safety release cutoff control, nozzle, and auxiliary water re-supply equipment. The water tank and auxiliary re-supply equipment must be of sufficient capacity to permit continuous operations.~~

~~b. Hydrodemolition for concrete removal is permitted in the following locations: [_____]~~

~~c. Use protective covers and barriers to protect adjacent surfaces not intended to be repaired from water blasting and over-spray.~~

~~d. Use equipment capable of delivering pressures of 35 MPa to 275 MPa at 7.5 liters/min to 190 liters/min for concrete removal and surface preparation.~~

~~e. [Noise resulting from hydrodemolition operations must be at a noise level of less than [90 decibels][_____] at a distance of [15 m][_____].]~~

2.2.2 Surface preparation and cleaning equipment

2.2.2.1 ~~[~~Abrasive Blasting~~]~~

a. Use ~~[dry][wet][~~dry or wet~~]~~ oil-free abrasive blasting capable of removing loose micro-fractured (bruised) or otherwise damaged or pulverized concrete surfaces, and rust from exposed steel reinforcement, and providing a surface profile in compliance with the Contract Documents.

~~b. [Use the following abrasive blasting methods: [Sandblasting] [Shotblasting] [_____]]~~

2.2.2.2 ~~[~~Low Pressure Water Cleaning~~]~~

Use equipment capable of delivering 7 MPa to 35 MPa to at 7.5 liters/min to 38 liters/min for cleaning loose material from repair areas.

2.2.2.3 Other Cleaning Equipment

Use equipment that delivers oil free air capable of cleaning loose material and debris from repair areas. If necessary to dry the concrete surface, ~~[gas-fired torches or]~~ clean, dry, compressed air may be used. Also, use vacuums capable of removing loose material and debris.

2.3 MATERIALS FOR FORMWORK AND EMBEDDED ITEMS

a. Formwork and embedded items must meet the requirements specified in Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

b. Install and remove formwork without damaging or staining the existing structure or repair material.

c. Forms used for polymer concrete/mortars must be tight enough to hold the material that is used without leaking. All surfaces where bond is not desired, but which are exposed to the monomer or resin, must be treated with a form release agent.

2.4 REINFORCEMENT AND REINFORCEMENT SUPPORTS

2.4.1 Steel Bars, Wires, and Fiber-reinforced Concrete

a. Reinforcement and reinforcement support must meet the requirements specified in Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

~~b. Repair coating damage incurred during shipment, storage, handling, and placing of reinforcing bars in accordance with [the appropriate ASTM standard practices for repair of damaged reinforcement][ASTM A780/A780M][ASTM A775/A775M][ASTM A934/A934M][_____]. Damaged areas must not exceed 2 percent of surface area in each linear foot of each bar.~~

~~c. Mechanical splices for coated reinforcement must have compatible coatings, in accordance with manufacturer's instructions. Splices for galvanized reinforcement must be galvanized or coated with dielectric material. Splices used with epoxy-coated or dual-coated reinforcement must be coated with dielectric material.~~

~~d~~b. Submit mill certificates and shop drawings as requirement by Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

~~2.4.2 Fiber-Reinforced Polymers~~

~~a. Fiber-Reinforced Polymers (FRP) bars used as internal reinforcement in concrete and their supports must meet the product requirements of ACI 440.5 and conform to [ACI 440.6][_____].~~

~~b. Submit test reports and certificates for FRP bars as required by ACI 440.5 and the Contract Documents.~~

~~c. Fiber-Reinforced Polymer (FRP) laminate materials externally bonded to concrete made by wet layup must meet the requirements of ACI 440.8 and the Contract Documents. Submit product data sheets for materials used for FRP layup systems as described in ACI 440.8.~~

~~d. The use of externally bonded FRP systems other than wet layup systems are permitted upon approval by the Contracting Officer. Submit product and materials data, design data, test reports, certificates, manufacturer's instructions, and field reports for those systems as requested by the Contracting Officer and required by Contract Documents.~~

2.5 CONVENTIONAL CONCRETE

a. Portland cement concrete materials must meet the requirements specified in Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

~~b. Materials for shotcrete must meet the requirements of Section 03 37 13 SHOTCRETE.~~

~~c~~b. ~~[~~For cement based bonding systems use neat portland cement or a blend of portland cement and an ASTM C33/C33M fine aggregate filler proportioned one to one by mass. The water-to-cement ratio of the bonding mixture must be ~~[~~equal to the water-to-cement ratio of concrete used as a repair or overlay material~~][_____]~~. Water used must meet ASTM C1602/C1602M requirements.~~]~~

~~d~~c. ~~[Use cementitious materials indicated in the Contract Documents.][~~Use cementitious materials of the same brand and type from the same manufacturing plant as the cementitious materials used in the concrete represented by the submitted field test records or used in trial mixtures.~~][_____]~~

~~e~~d. Aggregates used in concrete must be obtained from the same sources ~~[~~,be of the same type~~][and have the same size range]~~ as aggregates used in the concrete represented by submitted historical data or used in trial mixtures.

~~f~~e. Refer to Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][03 37 13 SHOTCRETE][_____]~~ for details on submittals involving conventional concrete.

2.6 POLYMERS

a. The requirements for the properties of polymers and aggregates used in polymers must meet the requirements specified in this paragraph as well as the properties specified in the referenced specifications and the Contract Documents.

b. Polymers used must be compatible with other polymers and materials used on the project. Unless repair materials are specified in the contract documents, the Contractor is responsible for verifying material compatibilities.

c. Submit product data, manufacturer's Safety Data Sheets,samples, design data, test reports, certificates, manufacturer's instructions, and field reports for materials as required by this document as well as the referenced specifications and the Contract Documents.

2.6.1 Epoxies

a. Epoxy mortars and epoxy compounds must conform to ASTM C881/C881M~~[~~.

~~][, Type [_____]; Class [_____]; Grade [_____]; [_____].]~~

b. Epoxy mortars used for repairing defects in hardened portland cement concrete must meet the requirements of ACI 503.2-503.4.

c. Epoxy used for crack repair must meet the requirements of ACI 503.7.

~~d. Epoxy used to produce a skid-resistant surface on hardened concrete must meet the requirements of ACI 503.3.~~

d~~e~~. Epoxy used for overlays must meet the requirements of ~~[~~ACI 548.8~~]~~ for multi-layer overlays and ~~[~~ACI 548.9~~]~~ for slurry overlays.

~~f~~e. Epoxy used for bonding freshly mixed concrete and hardened concrete must meet the requirements of ASTM C881/C881M, Type ~~[II][~~V~~], Grade [2][3], Class [A][B][C]~~.

f. Cementitious epoxy resin compensated 3-component, solvent-free, coating material with corrosion inhibitor, used as a bonding primer and reinforcement corrosion protection.

~~g~~g. Epoxy used for bonding hardened concrete and steel to hardened concrete must meet the requirements of ACI 548.12.

2.6.2 Latexes

a. Latex used in polymer modified portland cement concrete/mortar must meet the requirements of ASTM C1438.

b. Latex used in polymer modified portland cement concrete overlays must meet the requirements of ACI 548.4.

c. Latex used for bonding freshly mixed concrete and hardened concrete must meet the requirements of ASTM C1059/C1059M, Type II.

2.6.3 Methacrylates

a. Methyl methacrylate slurry (MMS) used for overlays must meet the requirements of ACI 548.10.

b. High molecular weight methacrylate (HMWM) must be a 2-component, rapid curing, and solvent-free system.

c. HMWM monomers must be a high molecular weight or substituted methacrylate that conforms the following properties:

Physical Properties of HMWM Monomer		
Property	Test Method	Criteria
Vapor Pressure Flash Point Density	ASTM D323 ASTM D93	Less than 133 Pa at 25 degrees C Greater than 93 degrees C Greater than 1.0 g per cubic cm at 25 degrees C

Physical Properties of HMWM Monomer		
Viscosity Index of Refraction Boiling point @ 133 Pa Shrinkage on cure	ASTM D4016 ASTM D542 ASTM D1078	0.012 ± 0.004 Pas at 23 degrees C; 1.470 ± 0.002 70 degrees C Less than 11 percent
Glass Transition Temperature (DSC)	ASTM D3418	57.2 degrees C
Curing Time (100 g mass)	ASTM D3418	Greater than 40 minutes at 25 degrees C, with 4 percent cuemene hydroperoxide
Bond Strength	ASTM C882/C882M	Greater than 10.3 mPa

d. The initiator/promoter system for HMWM must be capable of providing a surface cure time of not less than 40 minutes nor more than 3 hours at the surface temperature of the concrete during application. The initiator/promoter system must be such that the gel time may be adjusted to compensate for changes in temperature that may occur throughout the treatment application.

d. The initiator/promoter system for HMWM must meet the following criteria:

Initiator Cuemene Hydroperoxide	78 percent
Promoter Cobalt Napthenate	6 percent

~~2.6.4 [Other Polymers]~~

~~The use of [urethanes][silicones][acrylics][_____] is permitted.~~

~~Submit [product data], [samples], [design data], [test reports], [certificates], [manufacturer's instructions] for acceptance by the Contracting Officer.~~

2.6.4 Aggregate

a. Unless otherwise specified or recommended by the polymer material manufacturer, aggregate used with polymers must meet ~~ASTM C33/C33M~~ JIS A 5005 requirements.

b. Aggregate properties and proportions used with polymers must meet the requirements of the polymer material manufacturer, the requirements of the referenced polymer standard, and the Contract Documents.

c. Aggregate used with polymers must be dry and free of dirt, asphalt, and other organic materials. Aggregate moisture content must be less than ~~[0.2][~~1~~][_____]~~ percent by weight.

d. For patch repairs, the maximum-sized aggregate must not be greater than one third the depth of the patch area.

2.7 MISCELLANEOUS MATERIALS AND EQUIPMENT

2.7.1 Packaged and proprietary materials

The required properties for the materials listed in this paragraph must meet the properties specified in the Contract Documents. Submit ~~[~~Product data~~]~~, ~~[samples,] [~~design data,~~]~~ ~~[~~test reports,~~]~~ ~~[~~certificates,~~]~~ ~~[~~ manufacturer's instructions~~]~~, ~~[~~and field reports~~]~~ as required by the Contracting Officer and the Contract Documents.

a. Packaged, rapid hardening concrete repair materials must conform to ASTM C928/C928M.

b. Packaged, mortar and concrete must conform ASTM C387/C387M.

c. Rapid hardening cement must conform to ASTM C1600/C1600M.

Water used with packaged and proprietary materials must meet ~~ASTM C1602/C1602M~~JIS A 5308 requirements. Aggregates must meet the repair material manufacturer's requirements if available and ~~ASTM C33/C33M~~ JIS A 5005 if such requirements are not specified.

~~2.7.2 Bond Breakers~~

~~a. Bond breaker materials must meet the requirements of [ASTM D2822/D2822M], [ASTM D4869/D4869M], [ASTM D226/D226M, Type I], ASTM D2103, and must have a minimum thickness of [0.25 mm][_____], [AASHTO M 288, Erosion Control, Class B], [ASTM D450/D450M, Type II].~~

~~b. Bond breaker materials used must not have detrimental effects on portland cement concrete and reinforcement.~~

2.7.2 Structural steel

Structural steel used for repairs must meet the requirements of ~~05 12 00 STRUCTURAL STEEL~~05 50 13 MISCELLANEOUS METAL FABRICATIONS.

2.7.3 Concrete Accessories

All concrete accessories not included in this document must meet the requirements specified in Section 03 30 00 CAST-IN-PLACE CONCRETE~~ [and 03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE]~~.

2.7.4 Miscellaneous Equipment

a. Equipment designed specifically for the application of repair materials must be used as required by the repair material manufacturer and the referenced specification.

b. Equipment not listed in this specification but referenced or used for repairs must be clean and in good operating condition.

c. All supplies and equipment must be available in sufficient quantities to allow continuity in the installation project and quality assurance.

2.8 MIXTURE PROPORTIONING

a. Portland cement-based concrete mixtures must be in accordance with the requirements of Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][03 37 13 SHOTCRETE]~~.

b. Polymer concrete/mortar/resin/monomer proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must conform to the requirements of the referenced material specifications and the repair material manufacturer's directions.

c. Polymer-modified portland cement concrete proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must conform to the requirements provided by the repair material manufacturer as well as ACI 548.4 when such materials are used for overlays.

d. Proportioning and mixing materials not specified above must follow the requirements provided by the repair material manufacturer.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

3.1.1 Examination

Locate area of unsound concrete or delamination using hammer sounding or chain drag sound methods in accordance to ASTM D4580/D4580M. Denote and mark perimeter boundaries and notify the Contracting Officer to approve the unsound concrete layout boundaries.

3.1.2 Protection

Protect pedestrians, motorized traffic, mechanical, electrical, and plumbing equipment, surrounding construction, project site, landscaping, and surrounding buildings from damage or injury resulting from concrete rehabilitation work.

a. Construct dust and debris barriers surrounding repair work perimeter to control dust and to protect and control construction traffic.

b. Dispose of runoff from wet demolition or surface preparation operations in accordance with all local ordinances. Disposal methods must avoid soil erosion, avoid undermining pavements and foundations, damage to landscaping and vegetation, and minimize water penetration through other parts of buildings.

c. Collect and neutralize alkaline wastes and acid wastes and dispose in accordance with local, state, and federal regulations.

d. Comply with local noise ordinances during demolition operations.

e. Perform demolition work and surface preparation work in a manner that minimizes disturbances of operations. Coordinate work with the Contracting Officer.

f. Submit a proposed protection plan for approval by owner representative and Licensed Design Professional.

3.1.3 Formwork and Shoring

Execution of formwork and shoring must meet the requirements specified in Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

3.1.3.1 Formwork

a. Construct forms to sizes, shapes, lines, and dimensions to match existing adjacent surfaces and textures. Provide forms that match openings, offsets, chamfers, anchorages, inserts and other features as described on Contract Documents. Construct forms to accommodate installation of products by other trades. Provide forms for easy removal to minimize damage to concrete surfaces and adjacent surfaces. Apply form release coating over formwork surfaces prior to each concrete placement. Form release agents must not be applied to or come in contact with the repair area concrete substrate or reinforcement.

b. Do not damage repair material during removal of formwork for columns, walls, sides of beams, and other parts not supporting weight of concrete or repair material. Perform needed repair and treatment required on vertical surfaces at once and follow immediately with specified curing. Remove all formwork anchors embedded in existing concrete. Fill anchor holes and repair all damage to existing concrete at anchor holes.

3.1.3.2 Shoring

a. Provide shoring in accordance with the shoring drawings prior to performing work to brace the substrate structure temporarily while repair work is proceeding. Shoring must be designed, documented, and stamped by a Licensed Design Professional. Shoring designs must be submitted to and approved by the Contracting Officer prior to work commencing.

b. Leave formwork and shoring in place to support existing loads, construction loads and weight of repair material in beams, slabs, and other structural members until in-place strength of repair material determined in accordance with the Contract Documents. ~~For post-tensioned construction, leave formwork and shoring in place until stressing is complete.~~ When shores and other supports are arranged to allow removal of form-facing material without allowing structural slab or member to deflect, form-facing material and its horizontal supporting members may be removed at an earlier age.

3.1.4 Concrete ~~p~~Preparation

a. Remove concrete as needed per the removal requirements of this section. Limits on removal equipment are specified in the paragraph titled EQUIPMENT FOR CONCRETE PREPARATION.

b. Remove foreign material, such as dirt, oil, grease, or other chemicals, from the cracks before injection using compressed air, low-pressure water, or vacuuming. Allow wet surfaces to dry at least 24 hours.

c. Immediately before placing the repair material or installing formwork, make the repair area available for inspection by the Contracting Officer. Obtain acceptance by the Contracting Officer of surface preparation before proceeding with Work. If the Work is rejected, perform additional operations to the satisfaction of Contracting Officer.

d. ~~[Perform tensile pull-off tests in accordance with ASTM C1583/C1583M and guidance at [location]. Pull-off strength must meet or exceed [1.7 MPa][_____]. [Test a minimum of 3 specimens at locations no greater than 420 square meters of prepared surface.]]~~

e. ~~[The prepared surface must have a concrete surface profile equivalent to CSP [_____] as defined by ICRI 310.2R].~~

3.1.5 Quality Control

3.1.5.1 Quality ~~c~~Control of ~~s~~Surface ~~p~~Preparation

Evaluation of prepared substrate must be continuously monitored to assure that the prepared substrate surface meets project requirements.

3.1.5.2 Quality ~~c~~Control of ~~r~~Repair ~~o~~Overlays

All components of overlay PPCC materials must be certified by the material manufacturer or aggregate supplier to meet all project testing requirements. During the PPCC overlay, take mixed samples and check that the materials are mixed properly. Confirm that the right PC overlay thickness was applied by recording the volume of PC overlay materials and the substrate surface area covered by the overlay.

3.1.6 Curing

a. For portland cement concrete Work, follow the requirements indicated in ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~][_____]~~.

b. For polymer concrete/mortar Work, follow ~~[~~manufacturer's requirements for curing~~][_____]~~.

c. For polymer modified portland cement concrete Work follow ~~[~~ manufacturer's requirements for curing~~][_____]~~.

3.1.7 Clean up

a. Clean and remove all spills and leaks of injection adhesive and stains caused by the injection adhesives.

b. Dispose wastewater used for cutting and cleaning without staining or damaging the existing surfaces of the structure or the environment of the project area. The method of disposal must meet all the requirements of Section ~~01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS~~ 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

3.1.8 Safety

a. Provide Material Safety Data Sheets (MSDS) for products on site reviewing them before work begins.

b. Provide safety guards, maintenance, and warnings for all machinery and equipment.

c. Have personal protection equipment practice in place - eye protection and face guards.

d. Have all workers in contact with wet cementitious material wear protective gloves and clothing.

e. Provide eyewash facilities on-site with location signage.

f. Provide dust masks for workers operating mixers.

g. Have confined space procedures in place including adequate ventilation in closed spaces before operating equipment or using products that emit potentially dangerous or toxic exhaust, fumes, or dust.

h. Provide secured storage available for all hazardous or flammable materials.

i. Conduct safety meetings prior to beginning repair operations.

3.2 CRACK REPAIR

3.2.1 Preparation

3.2.1.1 General Requirements

a. Clean all cracks in accordance with the paragraph titled Concrete Preparation.

b. Do not repair cracks when the temperature of the concrete is below freezing and moisture conditions indicate the possibility of ice on the internal surfaces of the crack.

c. Do not apply adhesive if the temperature of the concrete is not within the range of application temperatures recommended by the manufacturer of the adhesive.

3.2.1.2 Crack ~~r~~Routing

Inspect surfaces adjacent to crack to receive repair material. If deteriorated, route a ~~V~~U-groove section at the crack face until sound concrete is reached.

3.2.1.3 Sealing

a. For epoxy injection, apply a surface seal over all exterior faces of the crack that can be reached to contain the injection adhesive in the crack.

b. For gravity fill repairs, apply a surface seal along the bottom surface of the element that can be reached to contain the repair material in the crack.

3.2.2 Application

3.2.2.1 Epoxy Injection

a. Install the injection entry and venting ports using flush mounted or drilled fittings per proprietary manufacturer's instructions.

b. Space the ports at ~~[a distance equal to the thickness of the member] [~~

200 mm~~]~~.

c. Inject the epoxy using material manufacturer's recommended equipment.

d. Apply recommended manufacturer's injection pressure.

e. For vertical or inclined cracks, apply injection by pumping epoxy into entry ports at the lowest elevation, cap, and move upward.

f. For horizontal cracks, apply injection by proceeding from one end of the crack to the other until the crack is fully sealed.

g. ~~[~~After 10 min., repeat injection procedure until all ports refuse injection.~~]~~

h. ~~[~~Remove ports and remove the surface seal by ~~[heat,]~~ chipping, or grinding or other acceptable means after the injected epoxy has cured.~~]~~

3.2.2.2 Gravity fill

a. Mix resin or monomer per material manufacturer's instructions.

~~b. Pre-fill cracks at least 3 mm wide with aggregate.~~

~~c~~b. Pour resin or monomer onto the surface, over the cracks and spread with brooms, rollers, or squeegees.

~~d~~c. Work material back and forth over the cracks to maximize fill in crack.

~~e~~d. Allow at least ~~[~~20 minutes~~][_____]~~ for material to penetrate cracks.

~~f~~e. Remove excess material once cracks have been filled to refusal.

~~g. [Broadcast [0.5 to 1.0 kg per square meter][_____] of sand.]~~

~~h~~f. Allow material to cure per material manufacturer's recommendations.

~~i~~g. ~~[~~Remove sealant and grind smooth.~~]~~

~~3.2.3 Quality Control~~

~~a. [Conduct quality [and control] tests for metering accuracy and mixing effectiveness of the continuous mixing pump in accordance with ACI 503.7.]~~

~~b. Qualify the test injection procedures in accordance with ACI 503.7.~~

~~3.2.4 Acceptance Criteria~~

~~3.2.4.1 Core Sampling~~

~~a. Obtain core samples in accordance with ASTM C42/C42M.~~

~~b. Allow 24 hours after injection before coring.~~

~~c. Obtain cores in a manner that includes as much of the bond line of the repaired concrete as possible. Replace cores that do not intersect the crack for at least 75 percent of the length of the core.~~

d. ~~Obtain three diameter core from first 30 m and one core for each 30 m thereafter.~~

e. ~~If cores would sever reinforcing steel or other embedded items, do not core, and notify the Contracting Officer so that an alternative location can be chosen.~~

f. ~~Obtain cores at least 50 mm in diameter for visual inspections and at least 100 mm in diameter for the splitting tensile test. Perform a splitting tensile test on one core from the first 30 m and one core for each 75 m thereafter.~~

g. ~~Fill core holes with [non-shrink grout][_____].~~

~~3.2.4.2 Core Testing~~

a. ~~Test a portion of the core samples for the splitting tensile strength in accordance with ASTM C496/C496M.~~

b. ~~Allow 72 hours after injection before beginning splitting tensile tests~~

c. ~~Prepare core sample per ASTM C42/C42M.~~

d. ~~Align the core so that the crack is in a plane as close to vertical as possible.~~

~~3.2.4.3 Acceptance~~

~~Work is acceptable if at least 90 percent of the depth of the crack in each core is filled with adhesive [and a or b is met][_____].~~

a. ~~[The splitting tensile strength of the core is at least 90 percent of the splitting tensile strength of a core taken from an uncracked area within 300 mm of the repaired crack.]~~

b. ~~[A splitting tensile test of the core indicates that no more than 10 percent of the bonded area of the crack in each core exhibits combined areas of separation of the adhesive from the concrete or cohesive failure within the adhesive.]~~

3.3 CORROSION AND SURFACE REPAIR

3.3.1 Preparation

3.3.1.1 Identification of Extent of Concrete Removal

a. Configure geometry of removal area to maximize the use of right-angle geometry, avoiding reentrant corners, and to obtain uniformity of depth. Determine the depth, location, and size of reinforcing bars prior to removal of concrete.

b. ~~[~~Perform visual inspection and hammer tapping, chain drag sounding, or other methods acceptable by the Contracting Officer to identify cracked, delaminated, spalled, disintegrated, and otherwise unsound concrete for removal. Mark boundaries of repair area before concrete removal.~~] [_____]~~

c. Inspect the marked boundaries with the Contracting Officer prior to commencing with the concrete removal. Revise the repair area boundaries as instructed by the Contracting Officer.

3.3.1.2 Shoring and Formwork

a. Provide shoring and formwork per the paragraph titled Formwork and Shoring.

b. For post-tensioned concrete, detension strands and wires as required by Contract Documents prior to repair.

3.3.1.3 Concrete Removal

a. Remove concrete from repair areas to indicated depth and profile. Notify Contracting Officer if additional delaminated, fractured, or unsound concrete is present.

b. Do not damage embedded reinforcing and adjacent concrete. The removal methods must produce minimal microcracking (bruising) of the prepared substrate surfaces. Avoid directly striking reinforcing steel with impact tools used for concrete removal.

c. Provide perpendicular edges at perimeter of repair area. The perimeter of the repair areas must be saw cut to a depth of 15 to 20 mm. ~~[~~For vertical or overhead surfaces, provide 45-degree slope at repair boundaries to facilitate air and rebound escape.~~]~~ Do not cut or damage embedded reinforcement or other embedded items. If embedded reinforcing steel or other embedded items are too close to the surface to provide the perpendicular edge cut, notify the Contracting Officer for direction before proceeding.

d. Extend concrete removal along the corroded reinforcing steel to a point where there is no further delamination, concrete cracking, or reinforcing steel corrosion, and where the reinforcement is bonded to the surrounding concrete.

e. Remove concrete around the exposed layer of reinforcement to a uniform depth beyond within the repair areas to provide a minimum clearance between exposed reinforcing steel and surrounding concrete of ~~[~~ 20 mm ~~][_____]~~, or at least 5 mm larger than the maximum nominal size of the coarse aggregate in the repair material.

~~f. [Do not remove concrete behind vertical reinforcing bars in columns.]~~

3.3.1.4 Preparation of Concrete Substrate Surface

a. Confirm perpendicular edges at repair area perimeter, and reinstate if damaged by concrete removal process. Remove loosely bonded concrete, bruised or fractured concrete, and bond-inhibiting materials such as dirt, concrete slurry, or any other detrimental materials from the concrete substrate using approved methods. Where concrete has been removed by impact methods, abrasive blasting must be used to prepare the surface and remove bruised concrete.

b. Provide substrate surface profiles as specified in the Contract Documents.

c. Visually inspect and sound substrate surface to confirm that no further delaminations or otherwise unsound concrete remains. If encountered, notify the Contracting Officer.

d. Clean the substrate per the paragraph titled Concrete preparation.

3.3.2 Application

3.3.2.1 ~~[~~Existing Reinforcement Preparation~~]~~

a. Clean existing reinforcement that will remain. Remove corrosion and/or other laitance and notify the Contracting Officer if section loss is greater than ~~[~~20%~~][_____]~~.

~~b. [Replace coating on reinforcement per [ASTM A780/A780M][ASTM A775/A775M][ASTM A934/A934M][_____]. Exposed areas must not exceed 2 percent of surface area in each linear foot of each bar.]~~

b. Apply cementitious epoxy resin compensated 3-component, solvent-free, coating material with corrosion inhibitor for corrosion protection of existing reinforcement. Proportioning, handling, and mixing procedures as well as equipment used for mixing these materials must follow the requirements provided by the manufacturer.

~~c~~c. ~~[~~Permit evaluation of existing reinforcement and placement of new reinforcement by the Contracting Officer.~~]~~

3.3.2.2 ~~[~~Placement of New Reinforcement~~]~~

Placement of new reinforcement

a. Placement of new reinforcement to replace or strengthen existing reinforcement is like new construction. Placement, splicing, and handling of new reinforcement must meet the requirements specified in Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

b. Reinforcement must be free of materials deleterious to bond. New reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM or JIS specification requirements.

3.3.2.3 Placement of Concrete

a. ~~[~~If portland cement concrete is used as the repair material, follow the requirements indicated in ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~][_____]~~ as well the Contract Document for proportioning, mixing, and placing concrete. For all other materials, follow material manufacturer's recommendations.~~][_____]~~

~~b. [For vertical and overhead applications of portland cement concrete, use shotcrete. Follow the requirements indicated in [03 37 13 SHOTCRETE][_____]].~~

~~c~~b. ~~[~~A bonding agent or cementitious epoxy resin compensated 3-component bonding primer with corrosion inhibitor ~~[~~must be used~~][must not be used][_____]~~.~~]~~

~~d. [Apply [corrosion inhibitors][sacrificial anodes][_____] as designated~~

~~by the Contract Documents.]~~

~~e. [Bristle broom a thin coat of the repair material into the saturated surface dry substrate filling roughened surface pores before placing the repair material in the repair area. Do not allow thin coat to dry before placing repair material.][_____]~~

~~f~~c. ~~[~~Consolidate the repair material after placement with a vibrating screed or internal vibrator.~~][_____]~~

~~g~~d. Finish the surface to match surface finish and texture requirements indicated in the Contract Document.~~ [Screed, float and trowel the repair material or broom the surface for non-slip texture. Follow the requirements of 03 30 00 CAST-IN-PLACE CONCRETE][For shotcrete, apply finishing techniques and requirements indicated in 03 37 13 SHOTCRETE][_____].~~

3.3.2.4 Placement of Other Repair Materials

a. Equilibrate repair material(s) and substrate to the temperature, cleanliness of substrate and reinforcement, and moisture requirements of the repair material manufacturer's requirements.

b. Comply with the repair material manufacturer's requirements for batching, mixing, placing and curing repair materials.

c. Review consistency of the mixed repair material(s) relative to the parameters documented in the repair material manufacturer product data sheet. If non-conforming, adjust consistency in compliance with the repair material manufacturer's requirements.

d. Apply or install repair material(s) within the application time frame (pot life) requirements of the repair material manufacturer's requirements, and place and consolidate to provide well-compacted repair.

e. Finish and tool repair materials, finished in accordance with the repair material manufacturer's written instructions and as indicated in Contract Documents.

f. Protect installed repair material(s) from damage, exposure to environmental conditions that are detrimental to the uncured or cured properties of the material. Cure in accordance with the requirements of the repair material manufacturer's requirements.

3.3.3 Quality Control

a. Protect concrete surfaces, beyond limits of surfaces receiving bonding agent adhesive, against spillage. Immediately remove any bonding agent adhesive that has spilled beyond desired area. Perform cleanup with material designated by bonding agent adhesive manufacturer. Avoid contamination of work area.

~~b. [The bond strength between the existing concrete and the repair material must be a minimum of [1.7 MPa][_____] per [ASTM C1583/C1583M][_____]. [Test a minimum of 3 specimens at locations no greater than 420 square meters of prepared surface.]]~~

~~3.4 OVERLAYS~~

~~3.4.1 Preparation~~

~~3.4.1.1 [Bonded Overlays]~~

~~a. Provide surface preparation as required in this Section.~~

~~b. [Repair cracks and patch deteriorated concrete prior to final surface preparation.]~~

~~c. Apply additional preparation requirements specified by the overlay material manufacturer~~

~~3.4.1.2 [Unbonded Overlays]~~

~~a. Repair distresses that cause a major loss of structural integrity when present[.][, including [_____]]~~

~~b. Clean the existing slab and remove any loose materials.~~

~~c. Install the separator layer as required by the Contract Documents and recommended by the material manufacturer.~~

~~3.4.2 Application~~

~~3.4.2.1 Portland Cement Concrete~~

~~a. [Apply the specified bonding agent. Follow the requirements of 3.4.2.4.]~~

~~b. Follow the requirements of Section [03 30 00 CAST-IN-PLACE CONCRETE] [_____] and the Contract Documents for installing forms, placing reinforcement, placing and consolidating concrete, and finishing concrete.~~

~~3.4.2.2 Polymer-modified Portland Cement Concrete~~

~~For polymer modified portland cement concrete overlays follow [ACI 548.4 requirements][manufacturer's requirements][_____] for placing and finishing the overlay.~~

~~3.4.2.3 Polymer Concrete/Mortar~~

~~For polymer concrete overlays, follow [manufacturer's requirements][ACI 548.8 requirements][ACI 548.9 requirements][ACI 548.10 requirements][_____] for placing and finishing the overlay.~~

~~3.4.2.4 [Bonding Agents]~~

~~a. Use a [cement slurry][epoxy bonding agent][latex bonding agent][_____] to improve the bonding between the overlay and the existing concrete.~~

~~b. [Follow material manufacturer's instructions for mixing, preparing, and applying bonding agent. Do not exceed the manufacturer's thickness recommendations][_____].~~

~~c. [Condition materials and the existing concrete surface to a~~

~~temperature consistent with manufacturer's recommendations at the time of installation.]~~

~~d. [Do not allow bond agents to dry before placement of repair material.]~~

~~3.4.3 Quality Control~~

~~a. [Concrete overlays must meet all the strength and durability requirements of 03 30 00 CAST-IN-PLACE CONCRETE][Material properties must meet the requirements defined in PRODUCTS.][_____].~~

~~b. [The bond strength between the existing concrete and the overlay must be a minimum of [1.8 MPa][_____] per [ASTM C1583/C1583M][_____]. [Test a minimum of 3 specimens at locations no greater than 420 square meters of prepared surface.]]~~

~~3.4.4 Joints~~

~~a. [Place joints as indicated in [03 30 00 CAST-IN-PLACE CONCRETE] [and as shown on the drawings] [_____].]~~

~~b. [Construct expansion and contraction joints in concrete overlay at the locations shown. Maintain alignment of control joints within 6 mm, to either side, of the required joint alignment.]~~

~~c. [Construct expansion and contraction joints at the locations shown and in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.]~~

~~d. [Construct expansion joints in the overlay at existing joint locations in the base slab while maintaining joint width and type[.][, and extending the full depth of the overlay.]]~~

~~e. [Construct control joints by tooling the plastic concrete, then sawcutting at the appropriate time. Saw control joints to a minimum [depth of [_____] mm] [of 25 percent of the thickness of the slab]. Maintain an ample supply of saw blades on the job before concrete placement is started, and have at least one standby sawing unit in good working order available at the jobsite at all times during the sawing operations. Begin sawcutting as soon as it is possible to saw the concrete without damaging adjacent concrete.]~~

~~f. [Inspect the faces of joints during sawcutting for undercutting or washing of the concrete due to early sawing. Complete sawcutting within 16 hours of concrete placement. Continue sawcutting regardless of weather conditions. Delay sawing if undercutting is sufficiently deep to cause structural weakness or excessive roughness in the joint or chipping, tearing, or spalling of the concrete occurs at the surface. Discontinue sawing when a crack develops ahead of the saw cut.]~~

~~g. [Immediately after the joint is sawed, flush the saw cut and adjacent concrete surface thoroughly with water until all residue from sawing is removed from the joint. Control and dispose of waste water from sawcutting and cleanup in accordance with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.]~~

3.4 CONCRETE STRENGTHENING

~~a. For enlargement of slabs using overlays see the paragraph titled~~

~~OVERLAYS.~~

~~b~~a. For ~~all other types of~~concrete strengthening follow the requirements contained in this paragraph.

3.4.1 Preparation

a. ~~[~~Use equipment and methods specified in the paragraph titled EQUIPMENT FOR CONCRETE PREPARATION and the Contract Documents to produce a sound, rough, open-pore surface at locations where bonding between existing and new concrete is required.~~]~~

~~b. [Round members of existing concrete with corners to minimum 13 mm radius. Roughened corners must be smoothed with putty]~~

~~c~~b. Clean all surfaces from contaminant and remove unsound concrete using the prescribed cleaning equipment and methods in the paragraphs titled PRODUCTS. All laitance, dust, dirt, oil, curing compound, existing coatings, and any other matter that could interfere with bonding concrete to the repair material must be removed.

~~d~~c. Follow the procedures of the paragraphs titled CRACK REPAIR and CORROSION AND SURFACE REPAIR. The concrete surface must be in good condition and all cracking, surface repair, and corrosion related problems must be adequately addressed prior to proceeding with concrete strengthening procedures.

~~e~~d. Insure that materials used for repairs are compatible with materials used for strengthening. Consult with the repair material manufacturers for information concerning material compatibility.

~~f~~e. Surfaces not intended to be strengthened must be covered as needed to protect against contamination and spills.

~~g~~f. Surfaces intended to be strengthened must be protected before application so that no materials that can interfere with bond are redeposited on the surface.

3.4.2 Application

3.4.2.1 Section enlargement

a. Install dowel reinforcement as required by the Contract Documents. Follow the ~~[~~adhesive~~][mechanical anchor]~~ manufacturer's procedures for installing dowels.

b. Install formwork and shoring following the requirements of this section.

c. Install reinforcement and reinforcement supports. Follow the requirements specified in Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~] [03 30 53 MISCELLANEOUS CAST-IN-PLACE CONCRETE][_____]~~.

d. Follow the requirements of Section ~~[~~03 30 00 CAST-IN-PLACE CONCRETE~~][03 37 13 SHOTCRETE][_____]~~ to place, consolidate, and finish concrete.

~~3.4.2.2 Externally bonded systems~~

~~3.4.2.2.1 Steel Plates~~

~~a. Bond steel plates to concrete using the methods and materials specified in the Contract Documents.~~

~~b. For bonding steel plates to concrete using an epoxy resin follow the requirements and procedures of ACI 548.12.~~

~~c. For bonding steel plates to concrete using mechanical or adhesive anchors, follow the procedures provided by the material manufacturer.~~

~~3.4.2.2.2 Fiber-reinforced Polymer Laminates~~

~~The following procedures are general procedures used for the installation of FRP laminates. If the FRP system used requires conflicting procedures, consult with the Contracting Officer before proceeding.~~

~~a. Insure that all surfaces that will receive FRP are clean, dry, and free of contaminants.~~

~~b. Insure that the workplace is well ventilated and that the repair material is applied at a time when the air temperature, concrete surface temperature, and the relative humidity are as required by the repair material manufacturer.~~

~~c. Temporary protection of the Work area is required during installation and until the resins have cured. If temporary shoring is required, the FRP system must be fully cured before removing the shoring and allowing the structural member to carry the design loads.~~

~~d. [If a primer is required, the primer must be applied uniformly to all areas on the concrete surface where the FRP system is to be placed at the manufacturer's specified rate of coverage. Protect the primer from dust, moisture, and other contaminants before applying the FRP system]~~

~~e. [Putty must be used in an appropriate thickness and sequence with the primer as recommended by the FRP manufacturer. The system-compatible putty must be used only to fill voids and smooth surface discontinuities before the application of other materials. Rough edges or trowel lines of cured putty must be ground smooth before continuing the installation. Allow the putty to cure as specified by the FRP system manufacturer before proceeding.]~~

~~f. Proportion, mix, and apply resins components in accordance with the FRP system manufacturer's recommended procedures.~~

~~g. Install and cure the FRP system per the manufacturer's recommendations.~~

~~h. [During installation of wet layup FRP systems, entrapped air between layers must be released or rolled out before the resin sets. Sufficient saturating resin must be applied to achieve full saturation of the fibers. Furthermore, successive layers of saturating resin and fiber materials must be placed before the complete cure of the previous layer of resin. If previous layers are cured, interlayer surface preparation, such as light sanding or solvent application as recommended by the system manufacturer, is required.]~~

~~i. Follow the FRP material manufacturer's recommendations for the application of protective coatings. Do not clean the installed FRP with a solvent before a protective coating is installed.~~

~~3.4.3 Quality Control~~

~~The cured FRP system must be evaluated for delaminations or air voids between multiple plies or between the FRP system and the concrete. Methods such as acoustic sounding (hammer sounding), ultrasonics, and thermography can be used to detect delaminations. The following requirements apply to wet layup systems:~~

~~a. Small delaminations less than 1300 square millimeter each are permissible as long as the delaminated area is less than 5 percent of the total laminate area and there are no more than 10 such delaminations per square meter.~~

~~b. Large delaminations, greater than 16,000 square millimeter, can affect the performance of the installed FRP and must be repaired by selectively cutting away the affected sheet and applying an overlapping sheet patch of equivalent plies.~~

~~c. Delaminations less than 16,000 square millimeter must be repaired by resin injection or ply replacement.~~

~~For other FRP systems, delamination must be evaluated and repaired in accordance with the material manufacturer direction. Upon completion of the Work, the laminate must be reinspected to verify that the repair was properly accomplished.~~

-- End of Section --

SECTION 03 30 00

CAST-IN-PLACE CONCRETE
02/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 301	(2016) Specifications for Structural Concrete
ACI 302.1R	(2015) Guide for Concrete Floor and Slab Construction
ACI 305R	(2020) Guide to Hot Weather Concreting
ACI SP-2	(2007; Abstract: 10th Edition) ACI Manual of Concrete Inspection

ASTM INTERNATIONAL (ASTM)

ASTM A970/A970M	(2018) Standard Specification for Headed Steel Bars for Concrete Reinforcement
ASTM E1155	(2020) Standard Test Method for Determining Floor Flatness and Floor Levelness Numbers

JAPANESE ARCHITECTURAL STANDARD SPECIFICAITONS (JASS)

JASS 5	(2015) Reinforced Concrete Work
JASS 6	(2015) Structural Steelwork Specification for Building Construction

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1101	(2014) Method of Test for Slump of Concrete
JIS A 1107	(2014) Method of Sampling and Testing for Compressive Strength of Drilled Cores of Concrete
JIS A 1108	(2018) Method of Test for Compressive Strength of Concrete
JIS A 1115	(2014) Method of Sampling Fresh Concrete

JIS A 1116	(2005) Method of Test for Unit Mass and Air Content of Fresh Concrete by Mass Method
JIS A 1118	(2017) Method of Test for Air Content of Fresh Concrete by Volumetric Method
JIS A 1128	(2014) Method of Test for Air Content of Fresh Concrete by Pressure Method
JIS A 1132	(2014) Method of Making and Curing Concrete Specimens
JIS A 1146	(2017) Method of Test for Alkali-Silica Reactivity of Aggregates by Mortar-bar Method
JIS A 1154	(2012) Method of Test for Chloride Ion Content in Hardened Concrete
JIS A 1804	(2009) Methods of Test for Production Control of Concrete - Method of Rapid Test for Identification of Alkali-Silica Reactivity of Aggregate
JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete
JIS A 5308	(20149) Ready-Mixed Concrete
JIS A 5758	(2016) Sealants for Sealing and Glazing in Buildings
JIS A 6201	(2015) Fly Ash for Use in Concrete
JIS A 6204	(2011) Chemical Admixtures for Concrete
JIS A 8652	(1995) Metal Panels for Concrete Form
JIS G 3112	(2010) Steel Bars for Concrete Reinforcement
JIS G 3444	(2016) Carbon Steel Tubes for General Structure
JIS G 3551	(2005) Welded Wire Mesh and Rebar Grid
JIS H 8641	(2007) Hot Dip Galvanized Coatings
JIS K 6251	(2017) Rubber, Vulcanized or Thermoplastic-Determination of Tensile Stress-Strain Properties
JIS K 6253	(2012) Rubber, Vulcanized or Thermoplastic - Determination of Hardness
JIS K 7124-1	(1999) Plastics Film and Sheeting - Determination of Impact Resistance by the

	Free-falling Dart Method - Part 1: Staircase Methods
JIS K 7127	(1999) Plastics - Determination of Tensile Properties - Part 3: Test Conditions for Films and Sheets
JIS K 7129-2	(2019) Plastics Film and Sheeting - Determination of Water Vapor Transmission Rate - Part 2: Infrared Detection Sensor Method
JIS Q 1011	(2009) Conformity Assessment – Conformity Assessment for Japanese Industrial Standards – Guidance on Third-party Certification System for Ready-mixed Concrete Products
JIS Q 17011	(2005) Conformity Assessment - General Requirements for Accreditation Bodies Accrediting Conformity Assessment Bodies
JIS Q 17025	(2005) General Requirements for the Competence of Testing and Calibration Laboratories
JIS R 5210	(2009) Portland Cement
JIS R 5211	(2019) Portland Blast-Furnace Slag Cement
JIS R 5212	(2019) Portland Pozzolan Cement
JIS Z 3881	(2014) Standard Qualification Procedure for Gas Pressure Welding Technique of Steel Bars for Concrete Reinforcement

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM,

MLIT-SS Chapter 6	MLIT Architectural Standard

U.S. GREEN BUILDING COUNCIL (USGBC)

LEED NC	(2009) Leadership in Energy and Environmental Design(tm) New Construction Rating System

1.2 DEFINITIONS

a. "Cementitious material" as used herein must include all portland cement, pozzolan, fly ash, and slag cement.

b. "Exposed to public view" means situated so that it can be seen from eye level from a public location after completion of the building. A public location is accessible to persons not responsible for operation or maintenance of the building.

c. "Chemical admixtures" are materials in the form of powder or fluids that are added to the concrete to give it certain characteristics not obtainable with plain concrete mixes.

d. "Supplementary cementing materials" (SCM) include coal fly ash, slag cement, natural or calcined pozzolans, and ultra-fine coal ash when used in such proportions to replace the portland cement that result in improvement to sustainability and durability and reduced cost.

e. "Design strength" (f'c) is the specified compressive strength of concrete at time(s) specified in this section to meet structural design criteria.

f. "Mass Concrete" is any concrete system that approaches a maximum temperature of 70 degrees C within the first 72 hours of placement. In addition, it includes all concrete elements with a section thickness of 1 meter or more regardless of temperature.

g. "Mixture proportioning" is the process of designing concrete mixture proportions to enable it to meet the strength, service life and constructability requirements of the project while minimizing the initial and life-cycle cost.

h. "Mixture proportions" are the masses or volumes of individual ingredients used to make a unit measure (cubic meter or cubic yard) of concrete.

i. "Pozzolan" is a siliceous or siliceous and aluminous material, which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.

j. "Workability (or consistence)" is the ability of a fresh (plastic) concrete mix to fill the form/mould properly with the desired work (vibration) and without reducing the concrete's quality. Workability depends on water content, chemical admixtures, aggregate (shape and size distribution), cementitious content and age (level of hydration).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

~~[~~ Concrete Curing Plan

~~]~~ Quality Control Plan; G~~[, [_____]]~~

Laboratory Accreditation; G~~[, [_____]]~~

SD-02 Shop Drawings

~~[~~ ~~Formwork~~

~~]~~ Reinforcing Steel; G~~[, [_____]]~~

SD-03 Product Data

Formwork Materials

Reinforcement; (LEED NC)

Liquid Chemical Floor Hardeners and Sealers

Mechanical Reinforcing Bar Connectors

~~[~~ Pumping Concrete

~~][~~ Finishing Plan

~~]~~ ~~SD-04 Samples~~

~~[~~ ~~Slab Finish Sample~~

~~][~~ ~~Surface Finish Samples~~

~~]~~ SD-05 Design Data

Concrete Mix Design; G~~[, [_____]]~~

SD-06 Test Reports

Concrete Mix Design; G~~[, [_____]]~~

Fly Ash

Pozzolan

Slag Cement

Aggregates

~~[~~ Tolerance Report

~~]~~ Compressive Strength Tests; G~~[, [_____]]~~

~~[~~ Unit Weight of Structural Concrete

~~][~~ Chloride Ion Concentration
~~]~~
~~[~~ Air Content

~~]~~ Slump Tests

Water

SD-07 Certificates

Reinforcing Bars

Welder Qualifications

~~[~~ VOC Content for Form Release Agents, Curing Compounds, and Concrete Penetrating Sealers

~~]~~ Safety Data Sheets

Field Testing Technician and Testing Agency

SD-08 Manufacturer's Instructions

Liquid Chemical Floor Hardeners and Sealers

Joint Sealants; (LEED NC)

~~[~~ Curing Compound

~~]~~1.4 MODIFICATION OF REFERENCES

Accomplish work in accordance with JASS 5 and MLIT-SS Chapter 6 or ACI publications except as modified herein. Consider the advisory or recommended provisions to be mandatory. Interpret reference to the "Building Official," the "Structural Engineer," and the "Architect/Engineer" to mean the Contracting Officer.

1.5 DELIVERY, STORAGE, AND HANDLING

Follow JASS 5 and MLIT-SS Chapter 6, requirements and recommendations. Do not deliver concrete until ~~vapor retarder, [~~vapor barrier,~~]~~ forms, reinforcement, embedded items, and chamfer strips are in place and ready for concrete placement. Do not store concrete curing compounds or sealers with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions~~, including [_____]~~. Do not store concrete curing compounds or sealers in occupied spaces.

1.5.1 Reinforcement

Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid excessive rusting. Protect from contaminants such as grease, oil, and dirt. Ensure bar sizes can be accurately identified after bundles are broken and tags removed.

1.6 QUALITY ASSURANCE

1.6.1 Design Data

1.6.1.1 Concrete Mix Design

Sixty days minimum prior to concrete placement, submit a mix design for each strength and type of concrete. Submit a complete list of materials including type; brand; source and amount of cement, supplementary cementitious materials, ~~[fibers],~~ and admixtures; and applicable reference specifications. Submit mill test and all other test for cement, supplementary cementitious materials, aggregates, and admixtures. Provide documentation of maximum nominal aggregate size, gradation analysis, percentage retained and passing sieve, and a graph of percentage retained verses sieve size. Provide mix proportion data using at least three different water-cementitious material ratios for each type of mixture, which produce a range of strength encompassing those required for each type of concrete required. If source material changes, resubmit mix proportion data using revised source material. Provide only materials that have been proven by trial mix studies to meet the requirements of this specification, unless otherwise approved in writing by the Contracting Officer. Indicate clearly in the submittal where each mix design is used when more than one mix design is submitted. Resubmit data on concrete components if the qualities or source of components changes. For previously approved concrete mix designs used within the past twelve months, the previous mix design may be re-submitted without further trial batch testing if accompanied by material test data conducted within the last six months. Obtain mix design approval from the contracting officer prior to concrete placement.

1.6.2 Shop Drawings

1.6.2.1 Reinforcing Steel

Indicate bending diagrams, assembly diagrams, splicing and laps of bars, shapes, dimensions, and details of bar reinforcing, accessories, and concrete cover. Do not scale dimensions from structural drawings to determine lengths of reinforcing bars. Reproductions of contract drawings are unacceptable.

1.6.3 Control Submittals

~~[~~1.6.3.1 Concrete Curing Plan

Submit proposed materials, methods and duration for curing concrete elements in accordance with JASS 5 and MLIT-SS Chapter 6.

~~][~~1.6.3.2 Pumping Concrete

Submit proposed materials and methods for pumping concrete. Submittal must include mix designs, pumping equipment including type of pump and size and material for pipe, and maximum length and height concrete is to be pumped.

~~][~~1.6.3.3 Finishing Plan

Submit proposed material and procedures to be used in obtaining the finish for ~~the [_____] floors~~floor and roof slabs. Include qualification of person to be used for obtaining ~~floor~~ tolerance measurement, description of measuring equipment to be used, and a sketch showing lines and locations the measuring equipment will follow.

~~][~~1.6.3.4 VOC Content for ~~form release agents, curing compounds, and concrete penetrating sealers~~Form Release Agents, Curing Compounds, and Concrete Penetrating Sealers

Submit certification for the form release agent, curing compounds, and concrete penetrating sealers that indicate the VOC content of each product.

~~]~~1.6.3.5 Safety Data Sheets

Submit Safety Data Sheets (SDS) for all materials that are regulated for hazardous health effects. SDS must be readily accessible during each work shift to employees when they are at the construction site.

1.6.4 Test Reports

1.6.4.1 Fly Ash and Pozzolan

Submit test results in accordance with JIS A 6201 for fly ash and pozzolan. Submit test results performed within 6 months of submittal date.

1.6.4.2 Slag Cement

Submit test results in accordance with JIS R 5211 for slag cement. Submit test results performed within 6 months of submittal date.

1.6.4.3 Aggregates

Submit test results in accordance with JIS A 5005, JIS A 1146 and JIS A 1804 as required in the paragraph titled ALKALI-AGGREGATE REACTION.

~~1.6.5 Field Samples~~

~~[1.6.5.1 Slab Finish Sample~~

~~Install minimum of 3000 mm by 3000 mm slab. Slab finish sample must not be part of the final project. Finish as required by specification.~~

~~][1.6.5.2 Surface Finish Samples~~

~~Provide a minimum of three sample concrete panels for each finish for each mix design, 1000 mm by 1000 mm, 75 mm thick. Use the approved concrete mix design(s). Provide sample panels on-site at locations directed. Once approved, each set of panels must be representative of each of the finishes specified and of the workmanship and finish(es) required. Do not remove or destroy samples until directed by the Contracting Officer.~~

~~]~~1.6.5 Quality Control Plan

Develop and submit for approval a concrete quality control program in accordance with the guidelines of JASS 5 and MLIT-SS Chapter 6 and as specified herein. The plan must include approved laboratories. Provide direct oversight for the concrete qualification program inclusive of associated sampling and testing. All quality control reports must be provided to the Contracting Officer, Quality Manager and Concrete Supplier.

1.6.6 Quality Control Personnel Certifications

The Contractor must submit for approval the responsibilities of the various quality control personnel, including the names and qualifications of the individuals in those positions and a quality control organizational chart defining the quality control hierarchy and the responsibility of the various positions. Quality control personnel must be employed by the Contractor.

Submit Japan Concrete Institute (JCI) or American Concrete Institute (ACI) certification for the following:

a. CQC personnel responsible for inspection of concrete operations.

b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.

c. Field Testing Technicians: ACI Concrete Field Testing Technician, Grade I or equivalent Japan Concrete Institute (JCI) Concrete Field Testing Technician.

1.6.6.1 Quality Manager Qualifications

The quality manager must hold a current license as a First Class Kenchikushi architect registered under standard law of Japan with experience on at least five similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

1.6.6.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

a. Work on concrete under this contract must be performed by an ACI Concrete Field Testing Technician Grade 1 qualified in accordance with ACI SP-2 or equivalent JCI Concrete Field Testing Technician. Equivalent certification programs must include requirements for written and performance examinations as stipulated in ACI SP-2.

b. Testing agencies that perform testing services on concrete materials including reinforcing steel must meet the requirements of JASS 5, JIS Q 1011, JIS Q 17011 and JIS Q 17025.

1.6.7 Laboratory Qualifications for Concrete Qualification Testing

The concrete testing laboratory must have the necessary equipment and experience to accomplish required testing. The laboratory must meet the requirements of JASS 5, JIS Q 1011, JIS Q 17011 and JIS Q 17025.

1.6.8 Laboratory Accreditation

Laboratory and testing facilities must be provided by and at the expense of the Contractor. The laboratories performing the tests must be accredited in accordance with JIS Q 17011 and JIS Q 17025. The accreditation must be current and must include the required test methods, as specified. Furthermore, the testing must comply with the following requirements:

a. Aggregate Testing and Mix Proportioning: Aggregate testing and mixture proportioning studies must be performed by an accredited laboratory and under the direction of a Chief Concrete Engineer or Concrete Engineer authorized by the Japan Concrete Institute or First Class Kenchikushi who is competent in concrete materials and must sign all reports and designs.

b. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by JIS A 1132.

c. Contractor Quality Control: All sampling and testing must be performed by an approved, onsite, independent, accredited laboratory.

1.7 ENVIRONMENTAL REQUIREMENTS

Provide space ventilation according to material manufacturer recommendations, at a minimum, during and following installation of concrete curing compound and sealer. Maintain one of the following ventilation conditions during the curing period or for 72 hours after installation:

a. Supply 100 percent outside air 24 hours a day.

b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 13 degrees C and 29 degrees C and humidity is between 30 percent and 60 percent.

c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

1.7.1 Submittals for Environmental Performance

a. Provide SDS product information data showing that form release agents meet any environmental performance goals such as using vegetable and soy based products.

b. Provide SDS product information data showing that concrete adhesives meet any environmental performance goals including low emitting, low volatile organic compound products.

1.8 QUALIFICATIONS FOR WELDING WORK

Welding procedures must be in accordance with JASS 6.

Verify that Welder qualifications are in accordance with JIS Z 3881 for welding of reinforcement or under an equivalent qualification test approved in advance. Welders are permitted to do only the type of welding for which each is specifically qualified.

PART 2 PRODUCTS

2.1 FORMWORK MATERIALS

a. Form-facing material in contact with concrete must be ~~[lumber,] [~~ plywood~~,]~~ ~~[tempered concrete-form-grade hardboard,] [~~or metal~~,]~~ ~~[plastic,] or [treated paper~~ that creates specified appearance and texture of concrete surface~~]~~. Submit product information on proposed form-facing materials if different from that specified herein.

b. Design formwork, shores, reshores, and backshores to support all vertical and lateral loads transmitted to them and to comply with applicable building code requirements.

c. Design formwork and shoring for load redistribution resulting from stressing of post-tensioned reinforcement. Ensure that formwork allows movement resulting from application of prestressing force.

d. Design formwork to withstand pressure resulting from placement and vibration of concrete and to maintain specified tolerances.

e. Design formwork to accommodate waterstop materials in joints at locations indicated in Contract Documents.

f. Provide temporary openings in formwork if needed to facilitate cleaning and inspection.

g. Design formwork joints to inhibit leakage of mortar.

h. Limit deflection of facing materials for concrete surfaces exposed to view to ~~[1/240][~~1/400~~][_____]~~ of center-to-center spacing of facing supports.

~~[i. Do not use earth cuts as forms for vertical or sloping surfaces.~~

~~]~~ ~~j~~i. Submit product information on proposed form-facing materials if different from that specified herein.

~~[~~ ~~k~~j. Submit shop drawings for formwork construction joints, shoring, reshoring, and backshoring. Shop drawings must be signed and sealed by a licensed design engineer or First Class Kenchikushi.

~~]~~ ~~l~~k. Submit procedure for reshoring and backshoring, including drawings signed and sealed by a First Class Kenchikushi. Include on shop drawings the formwork removal procedure and magnitude of construction loads used for design of reshoring or backshoring system. Indicate in procedure the magnitude of live and dead loads assumed for required capacity of the structure at time of reshoring or backshoring.

~~m~~l. Submit manufacturer's product data on form liner proposed for use with each formed surface.

2.1.1 Wood Forms

Provide lumber that is square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface defects. Provide plywood that complies with Japan Agricultural Standard concrete form panels or better or hardboard for smooth form lining.~~[Submit data verifying that composite wood products contain no urea formaldehyde resins.]~~

2.1.1.1 Concrete Form Plywood (Standard Rough)

Provide plywood that conforms to Japan Agricultural Standard concrete form, not less than 1~~6~~2 mm thick.

2.1.1.2 Overlaid Concrete Form Plywood (Standard Smooth)

Provide plywood that conforms to Japan Agricultural Standard high density form overlay, not less than 1~~6~~2 mm thick.

2.1.2 Steel Forms

Provide JIS A 8652 steel form surfaces that do not contain irregularities, dents, or sags.

2.2 FORMWORK ACCESSORIES

a. Use commercially manufactured formwork accessories, including ties and hangers.

b. Form ties and accessories must not reduce the effective cover of the reinforcement.

2.2.1 Form Ties

a. Use form ties with ends or end fasteners that can be removed without damage to concrete.

b. Where indicated in Contract Documents, use form ties with integral water barrier plates or other acceptable positive water barriers in walls.

c. The breakback distance for ferrous ties must be at least ~~[5~~40 mm~~] [19 mm] [_______]~~ for Surface Finish-2.0 or Surface Finish-3.0, as defined in ACI 301.

~~[d. If the breakback distance is less than 19 mm, use coated or corrosion-resistant ties.~~

~~]~~ ~~e~~d. Submit manufacturer's data sheet on form ties.

2.2.2 Waterstops

Submit manufacturer's data sheet on waterstop materials and splices.

~~2.2.2.1 PVC Waterstop~~

~~Polyvinylchloride waterstops must conform to JIS K 6773.~~

~~2.2.2.2 Rubber Waterstop~~

~~Rubber waterstops must conform to JIS K 6773.~~

~~2.2.2.3 Thermoplastic Elastomeric Rubber Waterstop~~

~~Thermoplastic elastomeric rubber waterstops must conform to JIS K 6258.~~

2.2.2.1 Hydrophilic Waterstop

Swellable strip type compound of polymer modified chloroprene rubber that swells upon contact with water must conform to the following requirements when tested in accordance to JIS K 6251: Tensile strength 2.9 MPa minimum; ultimate elongation 600 percent minimum. Hardness must be 50~~:[40]~~ minimum on the type A durometer when tested in accordance with JIS K 6253 and the volumetric expansion ratio in distilled water at 20 degrees C must be 3 to 1 minimum.

2.2.3 Biodegradable Form Release Agent

a. Provide form release agent that is colorless, biodegradable, and ~~[rapeseed oil-based] [soy oil-based] [water-based],~~ with a ~~[~~low ~~(maximum of 55 grams/liter (g/l))] [zero]~~ VOC content.~~[A minimum of [85][_______] percent of the total product must be biobased material.]~~

b. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces.

c. Provide form release agent that reduces formwork moisture absorption, and does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene. Submit documentation indicating type of biobased material in product and biobased content. Indicate relative dollar value of biobased content products to total dollar value of products included in project.

d. Submit manufacturer's product data on formwork release agent for use on each form-facing material. Provide products meeting VOC content requirements to attain F 4-Star or 4VOC rating.

2.2.4 Chamfer Materials

Use lumber materials with dimensions of 20 x 20 mm.

2.2.5 Construction and movement joints

a. Submit details and locations of construction joints in accordance with the requirements herein.

b. Locate construction joints within middle one-third of spans of slabs, beams, and girders. If a beam intersects a girder within the middle one-third of girder span, the distance between the construction joint in the girder and the edge of the beam must be at least twice the width of the larger member.

c. For members with post-tensioning tendons, locate construction joints where tendons pass through centroid of concrete section.

d. Locate construction joints in walls and columns at underside of slabs, beams, or girders and at tops of footings or slabs.

e. Make construction joints perpendicular to main reinforcement.

f. Provide movement joints where indicated in Contract Documents or in accepted alternate locations.

g. Submit location and detail of movement joints if different from those indicated in Contract Documents.

h. Submit manufacturer's data sheet on expansion joint materials.

i. Provide keyways where indicated in Contract Documents.~~[Longitudinal keyways indicated in Contract Documents must be at least 40 mm deep, measured perpendicular to the plane of the joint.]~~

~~2.2.6 Perimeter Insulation~~

~~Perimeter insulation must be expanded polystyrene conforming to JIS A 9521 meeting the following performance requirements:~~

~~a. Density: 300 kg/3^3 minimum.~~

~~b. Compressive Strength at yield or 10 percent deformation: 104 kPa, min.~~

~~c. Thermal Resistance of 25 mm thickness: 0.70 K·m^2/W~~

~~d. Flexural Strength: 240 kPa, min.~~

~~e. Water Absorption by total immersion, volume percent: 3 percent maximum.~~

~~f. Dimensional Stability (change in dimensions): 2 percent maximum.~~

~~g. Water Vapor Permeance of 25 mm thickness: 200 ng/Pa·s·m^2, maximum.~~

~~Comply with EPA requirements in accordance with Section 01 33 29 SUSTAINABILITY REPORTING~~

2.2.6 Other Embedded items

Use sleeves, inserts, anchors, and other embedded items of material and design indicated in Contract Documents.

2.3 CONCRETE MATERIALS

2.3.1 Cementitious Materials

2.3.1.1 Portland Cement

a. Unless otherwise specified, provide cement that conforms to JIS R 5210, ~~Ordinary~~Moderate Sulfate Resistant Portland Cement ~~[~~and meets ~~[~~low alkali content requirements~~][_____]~~.~~]~~

b. Use one brand and type of cement for formed concrete having exposed-to-view finished surfaces.

c. Submit information along with evidence demonstrating compliance with referenced standards. Submittals must include types of cementitious materials, manufacturing locations, shipping locations, and certificates showing compliance.

d. Cementitious materials must be stored and kept dry and free from contaminants.

~~[2.3.1.2 Blended Cements~~

~~Blended cements must conform to [JIS R 5211, Type [A] [B] [C]], [JIS R 5212, Type [A] [B] [C]], [JIS R 5213, Type [A] [B] [C]].~~

~~]~~2.3.2 Water

a. Water or ice must comply with the requirements of JIS A 5308, Annex C.

b. Minimize the amount of water in the mix. Improve workability by adjusting the grading of the aggregate and using admixture rather than by adding water.

c. Water must be ~~[~~potable~~] [from rainwater collection] [from graywater] [from recycled water]~~; free from injurious amounts of oils, acids, alkalis, salts, organic materials, or other substances deleterious to concrete.

d. Protect mixing water and ice from contamination during storage and delivery.

e. Submit test report showing water complies with JIS A 5308, Annex C.

~~[f. When nonpotable source is proposed for use, submit documentation on effects of water on strength and setting time in compliance with JIS A 5308, Annex C.~~

~~]~~2.3.3 Aggregate

2.3.3.1 Normal-Weight Aggregate

a. Aggregates must conform to JIS A 5005~~ [unless otherwise specified in the Contract Documents or approved by the contracting officer][_____]~~.

b. Aggregates used in concrete must be obtained from the same sources and have the same size range as aggregates used in concrete represented by submitted field test records or used in trial mixtures.

c. Store and handle aggregate in a manner that will avoid segregation and prevents contamination by other materials or other sizes of aggregates. Store aggregates in locations that will permit them to drain freely. Do not use aggregates that contain frozen lumps.

d. Submit types, pit or quarry locations, producers' names, aggregate supplier statement of compliance with JIS A 5005 and JIS A 1804 expansion data not more than 18 months old.

2.3.4 Admixtures

a. Chemical admixtures must conform to JIS A 6204.

b. Air-entraining admixtures must conform to JIS A 6204.

c. Chemical admixtures for use in producing flowing concrete must conform to JIS A 6204.

d. Do not use calcium chloride admixtures~~[~~.~~][unless approved by the contracting officer.]~~

e. Admixtures used in concrete must be the same as those used in the concrete represented by submitted field test records or used in trial mixtures.

f. Protect stored admixtures against contamination, evaporation, or damage.

g. To ensure uniform distribution of constituents, provide agitating equipment for admixtures used in the form of suspensions or unstable solutions. Protect liquid admixtures from freezing and from temperature changes that would adversely affect their characteristics.

h. Submit types, brand names, producers' names, manufacturer's technical data sheets, and certificates showing compliance with standards required herein.

2.4 MISCELLANEOUS MATERIALS

2.4.1 Concrete Curing Materials

Provide concrete curing material in accordance with JASS 5 and MLIT-SS Chapter 6. Submit product data for concrete curing compounds. Submit manufactures instructions for placement of curing compound. Provide products meeting VOC content requirements to attain F 4-Star or 4VOC rating.

2.4.2 Nonshrink Grout

Packaged dry, hydraulic cement non-shrink grout, that is non-metallic, non-corrosive, non-bleed, with the following performance requirements when prepared using the highest water-to-solids ratio, maximum flow, or most fluid consistency at 23.0 plus or minus 2.0 degrees C:

a. Minimum compressive strengths: 7.0 MPa at 1 day; 17.0 MPa at 3 day; 24.0 MPa at 7 day; and 34.0 MPa at 28 day.

b. Early height change (maximum percent at time of final setting): plus 4.0 percent.

c. Height change of moist cured hardened grout: 0.0 to plus or minus 0.3 percent at 1-day, 3-day, 14-day and 28-day.

2.4.3 Floor Finish Materials

2.4.3.1 Liquid Chemical Floor Hardeners and Sealers

a. Hardener must be a colorless aqueous solution containing a blend of inorganic silicate or siliconate material and proprietary components combined with a wetting agent; that penetrates, hardens, and densifies concrete surfaces. Submit manufactures instructions for placement of liquid chemical floor hardener.

b. Use concrete penetrating sealers with a low (maximum 100 grams/liter, less water and less exempt compounds) VOC content. Submit manufactures instructions for placement of sealers.

c. Provide products meeting VOC content requirements to attain F 4-Star or 4VOC rating.

2.4.4 Expansion/Contraction Joint Filler

Preformed, bituminous joint fiber filler for concrete paving and structural concrete construction. Provide joint filler meeting the following performance requirements:

a. Density: 300 kg/m^3 minimum.

b. Compression: The load required to compress the test specimen to 50 percent of its thickness before test shall not be less than 690 kPa and not more than 5100 kPa. If the nominal thickness of the specimen is less than 13 mm a maximum load of 8600 kPa is permitted. The sample after compression shall show a loss of not more than 3% of its original weight.

c. Extrusion/Protrusion: Test specimen shall be compressed to 50 percent of its thickness before test with three edges restrained. The amount of extrusion/protrusion of the free edge shall not exceed 6 mm.

d. Recovery: The test specimen shall be compressed to 50 percent of its thickness before test. The load shall be released immediately after application. At the end of 10 minutes after release of the load, the specimen shall have recovered to at least 70 percent of its thickness before test.

e. Water Absorption: The test specimen when submerged under 25 mm of water shall absorb not more than 15 percent by volume in 24 hours for 10 mm thickness and over.

Material must be 13 mm thick[, unless otherwise indicated]~~.~~ [and of a width applicable for the joint formed]. [Backer material shall be closed cell, polyethylene foam material with a density between 20-45 kg/m^3; Greater than 95 percent compression recovery; compression deflection 38 kPa; Greater than 160kPa tensile strength; and 200 degrees C heat resistance for hot applied sealants].

2.4.5 Joint Sealants and Seals

[Submit manufacturer's product data, indicating VOC content.] [Joint sealants conforming to the requirements of Section 07 92 00 JOINT SEALANTS].

2.4.5.1 Horizontal Surfaces, 3 Percent Slope, Maximum

JIS A 5758, multi-component sealant, Class 25, traffic~~ [non-traffic]. [Hot-applied joint sealant for sealing cracks in concrete and asphalt pavements shall be in conformance with the applicable requirements of the JRA PDCG - Pavement Design and Construction Guidelines~~.]

2.4.5.2 Vertical Surfaces Greater Than 3 Percent Slope

JIS A 5758, multi-component sealant, non-sag, Class 25, traffic ~~[non-traffic]~~.

2.4.6 ~~Vapor Retarder [and ~~Vapor Barrier]

~~Preformed, flexible polyethylene sheeting to be used as vapor retarder in contact with soil or granular fill under concrete slabs, minimum 0.25 mm [0.38 mm] thickness or other equivalent material meeting the following performance requirements:~~

~~a. Water Vapor Permeance, JIS K 7129-2 or equivalent test procedure: 2.28 ng/(m^2·s ·Pa), max.~~

~~b. Water Vapor Permeance after wetting, drying and soaking [, after heat conditioning] [, after low temperature conditioning] [, after soil organism conditioning], JIS K 7129-2 or equivalent test procedure: 2.28 ng/(m^2·s ·Pa), max.~~

~~c. Tensile Strength, JIS K 7127: 2.4 kN/m [5.3 kN/m] [7.9 kN/m], min.~~

~~d. Puncture Resistance, JIS K 7124-1: 475 g [1700 g] [2200 g], min.~~

[Preformed, flexible polyethylene sheeting or multi-layer plastic extruded sheeting manufactured with high grade, virgin polyolefin to be used as vapor barrier in contact with soil or granular fill under concrete slabs, minimum 0.38 mm thickness or other equivalent material with the following performance requirements:

a. Water Vapor Permeance, JIS K 7129-2 or equivalent test procedure: 0.57 ng/(m^2·s ·Pa), max.

b. Water Vapor Permeance after wetting, drying and soaking~~ [, after heat conditioning] [, after low temperature conditioning] [, after soil~~

~~organism conditioning]~~, JIS K 7129-2 or equivalent test procedure: 0.57 ng/(m^2·s ·Pa), max.

c. Tensile Strength, JIS K 7127: 7.9 kN/m, min.

d. Puncture Resistance, JIS K 7124-1: 2200 g, min.~~]~~

~~Consider plastic vapor retarders and vapor barriers and adhesives with a high recycled content, low toxicity low VOC (Volatile Organic Compounds) levels.~~

2.5 CONCRETE MIX DESIGN

2.5.1 Properties and Requirements

a. Use materials and material combinations listed in this section and the contract documents.

b. Cementitious material content must be adequate for concrete to satisfy the specified requirements for strength, w/cm, durability, and finishability described in this section and the contract documents.

~~[The minimum cementitious material content for concrete used in floors must meet the following requirements:~~

~~Nominal maximum size of aggregate, mm~~	~~Minimum cementitious material content, kg per cubic meter~~
~~37.5~~	~~280~~
~~25~~	~~310~~
~~19~~	~~320~~
~~9.5~~	~~360~~

~~]~~

c. Selected target slump must meet the requirements this section, the contract documents, and must not exceed 230 mm. Concrete must not show visible signs of segregation.

d. The target slump must be enforced for the duration of the project. Determine the slump by JIS A 1101. Slump tolerances must meet the requirements of JASS 5 and MLIT-SS Chapter 6.

e. The nominal maximum size of coarse aggregate for a mixture must not exceed three-fourths of the minimum clear spacing between reinforcement, one-fifth of the narrowest dimension between sides of forms, or one-third of the thickness of slabs or toppings.

f. Concrete must be air entrained for members assigned to Exposure Class F1, F2, or F3. The total air content must be in accordance with the requirements of the paragraph titled DURABILITY.

g. Measure air content at the point of delivery in accordance with JIS A 1118 or JIS A 1128.

h. Concrete for slabs to receive a hard-troweled finish must not contain an air-entraining admixture or have a total air content greater than 3 percent.

i. Concrete properties and requirements for each portion of the structure are specified in the table below. Refer to the paragraph titled DURABILITY for more details on exposure categories and their requirements.

	Minimum *f'c* MPa	**Exposure Categories^**	**Miscellaneous Requirements**
Footings	30 at 28 days	S0; C0	Nominal maximum aggregate size must be 20 mm Maximum w/cm ratio: 0.50
Columns, walls, and suspended slabs	30 at 28 days	S0; C0	Nominal maximum aggregate size must be 20 mm Maximum w/cm ratio: 0.50
Slabs-on-ground and grade beams	33 at 28 days	S0; C0	Nominal maximum aggregate size must be 20 mm Maximum w/cm ratio: 0.45

2.5.2 Durability

2.5.2.1 Alkali-Aggregate Reaction

Do not use any aggregate susceptible to alkali-carbonate reaction (ACR) to reduce the potential of alkali-silica reaction (ASR). For each aggregate used in qualifying concrete mixtures, the expansion result of the aggregate and cementitious materials combination determined in accordance with JIS A 1146 must not exceed 0.10 percent at an age of 16 days.

~~2.5.2.2 Freezing and Thawing Resistance~~

~~a. Provide concrete meeting the following requirements based on exposure class assigned to members for freezing-and-thawing exposure in Contract Documents:~~

~~**Exposure class**~~	~~**Maximum *w/cm****~~	~~**Minimum *f'c*, MPa**~~	~~**Air content**~~	~~**Additional Requirements**~~
~~F0~~	~~N/A~~	~~18~~	~~N/A~~	
~~F1~~	~~0.55~~	~~24~~	~~Depends on aggregate size~~	~~N/A~~

~~**Exposure class**~~	~~**Maximum *w/cm****~~	~~**Minimum *f'c*,**~~ **MPa**	~~**Air content**~~	~~**Additional Requirements**~~
~~F2~~	~~0.45~~	~~30~~	~~Depends on aggregate size~~	~~See limits on maximum cementitious material by mass~~
~~F3~~	~~0.40~~	~~36~~	~~Depends on aggregate size~~	~~See limits on maximum cementitious material by mass~~
~~F3 plain concrete~~	~~0.45~~	~~30~~	~~Depends on aggregate size~~	~~See limits on maximum cementitious material by mass~~

~~*The maximum *w/cm* limits do not apply to lightweight concrete.~~

~~b. Concrete must be air entrained for members assigned to Exposure Class F1, F2, or F3. The total air content must meet the requirements of the following table:~~

~~**Nominal maximum aggregate size, mm**~~	~~**Total air content, percent*^**~~	
	~~**Exposure Class F2 and F3**~~	~~**Exposure Class F1**~~
~~9.5~~	~~7.5~~	~~6.0~~
~~12.5~~	~~7.0~~	~~5.5~~
~~19.0~~	~~6.0~~	~~5.0~~
~~25.0~~	~~6.0~~	~~4.5~~
~~37.5~~	~~5.5~~	~~4.5~~
~~50~~	~~5.0~~	~~4.0~~
~~75~~	~~5.5~~	~~3.5~~

~~*Tolerance on air content as delivered must be plus/minus 1.5 percent.~~
~~^For f'c greater than 36 MPa psi, reducing air content by 1.0 percentage point is acceptable.~~

~~c. Submit documentation verifying compliance with specified requirements.~~

~~d. For sections of the structure that are assigned Exposure Class F3, submit certification on cement composition verifying that concrete mixture meets the requirements of the following table:~~

~~**Cementitious material**~~	~~**Maximum percent of total cementitious material by mass***~~
~~Fly ash or other pozzolans conforming to JIS A 6201~~	~~25~~

~~**Cementitious material**~~	~~**Maximum percent of total cementitious material by mass***~~
~~Slag cement conforming to JIS A 5011~~	~~50~~

~~*Total cementitious material also includes JIS R 5210, fly ash, other pozzolans, and slag cement. The maximum percentages above must include:~~

~~i. Fly ash or other pozzolans present in Type IP JIS R 5212 blended cement.~~

~~ii. Slag cement present in Type IS JIS R 5211 blended cement.~~

~~^Fly ash or other pozzolans must constitute no more than 25 percent of the total mass of the cementitious materials.~~

2.5.2.2 Corrosion and Chloride Content

a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members requiring protection against reinforcement corrosion in Contract Documents.

b. Submit documentation verifying compliance with specified requirements.

c. Water-soluble chloride ion content contributed from constituents including water, aggregates, cementitious materials, and admixtures must be determined for the concrete mixture by JIS A 1154 at age between 28 and 42 days.

d. The maximum water-soluble chloride ion (Cl-) content in concrete, percent by mass of cement is as follows:

Exposure class	**Maximum *w/cm******	**Minimum f'c, MPa**	**Maximum water-soluble chloride ion (CL-) content in concrete, percent by mass of cement**
Reinforced Concrete			
C0	0.5	30	0.30

*The maximum w/cm limits do not apply to lightweight concrete.

2.5.2.3 Sulfate Resistance

a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members for sulfate exposure.

Exposure class	**Maximum *w/cm****	Minimum f'c MPa	**Required cementitious materials types**		**Calcium chloride admixture**
			JIS R 5210	JIS R 5211, JIS R 5212	
S0	0.50	30	II (moderate sulfate resistant)	IP, portland pozzolan cement (moderate sulfate resistant); IS, portland blast furnace slag cement (<70) (moderate sulfate resistant)	Not permitted

~~* For seawater exposure, other types of portland cements with tricalcium aluminate (C3A) contents up to 10 percent are acceptable if the w/cm does not exceed 0.40.~~

~~** The amount of the specific source of the pozzolan or slag cement to be used shall be at least the amount determined by test or service record to improve sulfate resistance when used in concrete containing Type V cement. Alternatively, the amount of the specific source of the pozzolan or slag used shall not be less than the amount tested in accordance with ASTM C1012/C1012M and meeting the requirements maximum expansion requirements listed herein.~~

~~^ Other available types of cement, such as Type III or Type I, are acceptable in exposure classes S1 or S2 if the C3A contents are less than 8 or 5 percent, respectively.~~

b. The maximum w/cm limits for sulfate exposure do not apply to lightweight concrete.

~~c. Alternative combinations of cementitious materials of those listed in this paragraph are acceptable if they meet the maximum expansion requirements listed in the following table:~~

~~**Exposure class**~~	~~**Maximum expansion when tested using ASTM C1012/C1012M**~~		
	~~**At 6 months**~~	~~**At 6 months**~~	~~**At 18 months**~~
~~S1~~	~~0.10 percent~~	~~N/A~~	~~N/A~~
~~S2~~	~~0.05 percent~~	~~0.10 percent^~~	~~N/A~~
~~S3~~	~~N/A~~	~~N/A~~	~~0.10 percent~~

~~^The 12-month expansion limit applies only when the measured expansion exceeds the 6-month maximum expansion limit.~~

2.5.2.4 Concrete Temperature

The temperature of concrete as delivered must not exceed ~~[~~35 degrees C~~]~~

~~[______]~~.

~~2.5.2.5 Concrete Permeability~~

~~a. Provide concrete meeting the requirements of the following table based on exposure class assigned to members requiring low permeability in the Contract Documents.~~

~~Exposure class~~	~~Maximum w/cm*~~	~~Minimum f'c, MPa~~	~~Additional minimum requirements~~
~~W0~~	~~N/A~~	~~18~~	~~None~~
~~W1~~	~~0.5~~	~~30~~	~~None~~

~~*The maximum w/cm limits do not apply to lightweight concrete.~~

~~b. Submit documentation verifying compliance with specified requirements.~~

2.5.3 Trial Mixtures

Trial mixtures must be in accordance to JASS 5 and MLIT-SS Chapter 6.

2.5.4 Ready-Mix Concrete

Provide concrete that meets the requirements of JIS A 5308.

Ready-mixed concrete manufacturer must provide duplicate delivery tickets with each load of concrete delivered. Provide delivery tickets with the following information in addition to that required by JIS A 5308:

a. Type and brand cement

b. Cement and supplementary cementitious materials content in 43-kilogram bags per cubic meter of concrete

c. Maximum size of aggregate

d. Amount and brand name of admixtures

e. Total water content expressed by water cementitious material ratio

2.6 REINFORCEMENT

a. Bend reinforcement cold. Fabricate reinforcement in accordance with fabricating tolerances of JASS 5 and MLIT-SS Chapter 6.

b. When handling and storing coated reinforcement, use equipment and methods that do not damage the coating. If stored outdoors for more than 2 months, cover coated reinforcement with opaque protective material.

c. Submit manufacturer's certified test report for reinforcement.

d. Submit placing drawings showing fabrication dimensions and placement locations of reinforcement and reinforcement supports. Placing drawings must indicate locations of splices, lengths of lap splices, and details of mechanical and welded splices.

e. Submit request with locations and details of splices not indicated in Contract Documents.

f. Submit request to place column dowels without using templates.

~~[~~ g. Submit request and procedure to field-bend or straighten reinforcing bars partially embedded in concrete at locations not indicated in Contract Documents.~~ Field bending or straightening of reinforcing bars is permitted [where indicated in the Contract Documents][in the following locations: [_____]]~~

~~]~~ h. Submit request for field cutting, including location and type of bar to be cut and reason field cutting is required.

2.6.1 Reinforcing Bars

a. Reinforcing bars must be deformed, except spirals, load-transfer dowels, and welded wire reinforcement, which may be plain.

b. JIS G 3112, grades and sizes as indicated. ~~[~~Cold drawn wire used for spiral reinforcement must conform to JIS G 3551.~~]~~ Reinforcing bars actual yield strength shall not exceed specified yield strength by more than 124 MPa and actual tensile strength to actual yield strength ratio is not less than 1.25.

~~[Provide reinforcing bars that contain a minimum of [100][_____] percent recycled content.][See Section 01 33 29 SUSTAINABILITY REPORTING for cumulative total recycled content requirements.]~~

~~c. [Reinforcing bars may contain post-consumer or post-industrial recycled content.] [Submit documentation indicating percentage of post-industrial and post-consumer recycled content per unit of product. Indicate relative dollar value of recycled content products to total dollar value of products included in project.]~~

~~d~~c. Submit mill certificates for reinforcing bars.

2.6.1.1 Headed Reinforcing Bars

Headed reinforcing bars must conform to ASTM A970/A970M including Annex A1, and other specified requirements.

2.6.1.2 Bar Mats

a. Bar mats must conform to JIS G 3551.

2.6.2 Mechanical Reinforcing Bar Connectors

a. Provide 125 percent minimum yield strength of the reinforcement bar.

b. Mechanical splices for galvanized reinforcing bars must be galvanized or coated with dielectric material.

c. Mechanical splices used with epoxy-coated or dual-coated reinforcing bars must be coated with dielectric material.

d. Submit data on mechanical splices demonstrating compliance with this paragraph.

2.6.3 Wire

~~a. [Provide wire reinforcement that contains a minimum of [100] [_____] percent recycled content.][See Section 01 33 29 SUSTAINABILITY REPORTING for cumulative total recycled content requirements. Wire reinforcement may contain post-consumer or post-industrial recycled content.]Provide flat sheets of welded wire reinforcement for slabs and toppings.~~

~~b.~~ Plain or deformed steel wire must conform to JIS G 3551.

2.6.4 Welded Wire Reinforcement

a. Use welded wire reinforcement specified in Contract Documents and conforming to one or more of the specifications given herein.

b. Plain welded wire reinforcement must conform to JIS G 3551, with welded intersections spaced no greater than 300 mm apart in direction of principal reinforcement.

c. Deformed welded wire reinforcement must conform to JIS G 3551, with welded intersections spaced no greater than 400 mm apart in direction of principal reinforcement.

d. Zinc-coated (galvanized) welded wire reinforcement must conform to JIS H 8641. Coating damage incurred during shipment, storage, handling, and placing of zinc-coated (galvanized) welded wire reinforcement must be repaired in accordance with JASS 6. If damaged area exceeds 2 percent of surface area in each linear foot of each wire or welded wire reinforcement, the sheet containing the damaged area must not be used. The 2 percent limit on damaged coating area shall include repaired areas damaged before shipment.

2.6.5 Reinforcing Bar Supports

a. Provide reinforcement support types within structure as required by Contract Documents. Reinforcement supports must conform to JASS 5 and MLIT-SS Chapter 6.

b. Legs of supports in contact with formwork must be hot-dip galvanized, or plastic coated after fabrication, or stainless-steel bar supports.

~~c. [Minimum [5][10][_____] percent post-consumer recycled content, or minimum [20][40][_____] percent post-industrial recycled content.] [See Section 01 33 29 SUSTAINABILITY REPORTING for cumulative total recycled content requirements. Plastic and steel may contain post-consumer or post-industrial recycled content.]~~

2.6.6 Dowels for Load Transfer in Floors

Provide greased dowels for load transfer in floors of the type, design, weight, and dimensions indicated. Provide dowel bars that are plain-billet steel conforming to JIS G 3112, Grade SD295. Provide dowel pipe that is steel conforming to JIS G 3444 STK400.

~~[Plate dowels must conform to JIS G 3101 SS400, and must be of size and spacing indicated. Plate dowel system must minimize shrinkage restraint~~

~~by [using a tapered shape] [or] [formed void] [or] [by having compressible material on the vertical faces with a thin bond breaker on the top and bottom dowel surfaces.]~~

~~]~~2.6.7 Welding

a. Provide weldable reinforcing bars that conform to JIS G 3112, grades and sizes as indicated. The maximum carbon content shall not exceed 0.55 percent with carbon not exceeding 0.30 percent and manganese not exceeding 1.5 percent.

b. Comply with JIS Z 3881 unless otherwise specified. Do not tack weld reinforcing bars.

c. Welded assemblies of steel reinforcement produced under factory conditions, such as welded wire reinforcement, bar mats, and deformed bar anchors, are allowed.

PART 3 EXECUTION

3.1 EXAMINATION

a. Do not begin installation until substrates have been properly constructed; verify that substrates are level.

b. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before processing.

c. Check field dimensions before beginning installation. If dimensions vary too much from design dimensions for proper installation, notify Contracting Officer and wait for instructions before beginning installation.

3.2 PREPARATION

Determine quantity of concrete needed and minimize the production of excess concrete. Designate locations or uses for potential excess concrete before the concrete is poured.

3.2.1 General

a. Surfaces against which concrete is to be placed must be free of debris, loose material, standing water, snow, ice, and other deleterious substances before start of concrete placing.

b. Remove standing water without washing over freshly deposited concrete. Divert flow of water through side drains provided for such purpose.

3.2.2 Subgrade Under Foundations and Footings

a. When subgrade material is semi-porous and dry, sprinkle subgrade surface with water as required to eliminate suction at the time concrete is deposited, or seal subgrade surface by covering surface with specified vapor ~~retarder~~barrier.

b. When subgrade material is porous, seal subgrade surface by covering surface with specified vapor ~~retarder~~barrier.

3.2.3 Subgrade Under Slabs on Ground

a. Before construction of slabs on ground, have underground work on pipes and conduits completed and approved.

b. Previously constructed subgrade or fill must be cleaned of foreign materials

c. Finish surface of capillary water barrier under interior slabs on ground must not show deviation in excess of 6 mm when tested with a 3000 mm straightedge parallel with and at right angles to building lines.

d. Finished surface of subgrade or fill under exterior slabs on ground must not be more than 6 mm above or 30 mm below elevation indicated.

3.2.4 Edge Forms and Screed Strips for Slabs

a. Set edge forms or bulkheads and intermediate screed strips for slabs to obtain indicated elevations and contours in finished slab surface and must be strong enough to support vibrating bridge screeds or roller pipe screeds if nature of specified slab finish requires use of such equipment.

b. Align concrete surface to elevation of screed strips by use of strike-off templates or approved compacting-type screeds.

3.2.5 Reinforcement and Other Embedded Items

a. Secure reinforcement, joint materials, and other embedded materials in position, inspected, and approved before start of concrete placing.

b. When concrete is placed, reinforcement must be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM specification requirements.

3.3 FORMS

a. Provide forms, shoring, and scaffolding for concrete placement. Set forms mortar-tight and true to line and grade.

b. Chamfer above grade exposed joints, edges, and external corners of concrete ~~[~~20 mm~~]~~. Place chamfer strips in corners of formwork to produce beveled edges on permanently exposed surfaces.~~[~~ Do not bevel reentrant corners or edges of formed joints of concrete.~~]~~

c. Provide formwork with clean-out openings to permit inspection and removal of debris.

d. Inspect formwork and remove foreign material before concrete is placed.

e. At construction joints, lap form-facing materials over the concrete of previous placement. Ensure formwork is placed against hardened concrete so offsets at construction joints conform to specified tolerances.

f. Provide positive means of adjustment (such as wedges or jacks) of shores and struts. Do not make adjustments in formwork after concrete has reached initial setting. Brace formwork to resist lateral deflection and lateral instability.

g. Fasten form wedges in place after final adjustment of forms and before concrete placement.

h. Provide anchoring and bracing to control upward and lateral movement of formwork system.

i. Construct formwork for openings to facilitate removal and to produce opening dimensions as specified and within tolerances.

j. Provide runways for moving equipment. Support runways directly on formwork or structural members. Do not support runways on reinforcement. Loading applied by runways must not exceed capacity of formwork or structural members.

k. Position and support expansion joint materials, waterstops, and other embedded items to prevent displacement. Fill voids in sleeves, inserts, and anchor slots temporarily with removable material to prevent concrete entry into voids.

l. Clean surfaces of formwork and embedded materials of mortar, grout, and foreign materials before concrete placement.

3.3.1 Coating

a. Cover formwork surfaces with an acceptable material that inhibits bond with concrete.

b. If formwork release agent is used, apply to formwork surfaces in accordance with manufacturer's recommendations before placing reinforcement. Remove excess release agent on formwork prior to concrete placement.

c. Do not allow formwork release agent to contact reinforcement or hardened concrete against which fresh concrete is to be placed.

3.3.2 Reshoring

a. Do not allow structural members to be loaded with combined dead and construction loads in excess of loads indicated in the accepted procedure.

b. Install and remove reshores or backshores in accordance with accepted procedure.

c. For floors supporting shores under newly placed concrete, either leave original supporting shores in place, or install reshores or backshores. Shoring system and supporting slabs must resist anticipated loads. Locate reshores and backshores directly under a shore position or as indicated on formwork shop drawings.

d. In multistory buildings, place reshoring or backshoring over a sufficient number of stories to distribute weight of newly placed concrete, forms, and construction live loads.

3.3.3 Reuse

a. Reuse forms providing the structural integrity of concrete and the aesthetics of exposed concrete are not compromised.

b. Wood forms must not be clogged with paste and must be capable of absorbing high water-cementitious material ratio paste.

c. Remove leaked mortar from formwork joints before reuse.

3.3.4 Forms for Standard Rough Form Finish

Provide formwork in accordance with JASS 5 and MLIT-SS Chapter 6 with a surface finish, ACI 301 SF-1.0, for formed surfaces ~~that are to be concealed by other construction~~below grade.

3.3.5 Forms for Standard Smooth Form Finish

Provide formwork in accordance with JASS 5 and MLIT-SS Chapter 6 with a surface finish, ACI 301 SF-3.0, for all formed surfaces~~ that are exposed to view.[Do not provide mockup of concrete surface appearance and texture~~. ~~]~~

3.3.6 Form Ties

a. For post-tensioned structures, do not remove formwork supports until stressing records have been accepted by the Contracting Officer.

b. After ends or end fasteners of form ties have been removed, repair tie holes in accordance with JASS 5 and MLIT-SS Chapter 6.

3.3.7 Tolerances for Form Construction

a. Construct formwork so concrete surfaces conform to tolerances in JASS 5 and MLIT-SS Chapter 6.

b. Position and secure sleeves, inserts, anchors, and other embedded items such that embedded items are positioned within JASS 5 and MLIT-SS Chapter 6 tolerances.

c. To maintain specified elevation and thickness within tolerances, install formwork to compensate for deflection and anticipated settlement in formwork during concrete placement. Set formwork and intermediate screed strips for slabs to produce designated elevation, camber, and contour of finished surface before formwork removal. If specified finish requires use of vibrating screeds or roller pipe screeds, ensure that edge forms and screed strips are strong enough to support such equipment.

3.3.8 Removal of Forms and Supports

a. If vertical formed surfaces require finishing, remove forms as soon as removal operations will not damage concrete.

b. Remove top forms on sloping surfaces of concrete as soon as removal will not allow concrete to sag. Perform repairs and finishing operations required. If forms are removed before end of specified curing period, provide curing and protection.

c. Do not damage concrete during removal of vertical formwork for columns, walls, and sides of beams. Perform needed repair and finishing operations required on vertical surfaces. If forms are removed before end of specified curing period, provide curing and protection.

~~[~~ d. Leave formwork and shoring in place to support construction loads and weight of concrete in beams, slabs, and other structural members until in-place required strength of concrete is reached.

~~]~~ e. Form-facing material and horizontal facing support members may be removed before in-place concrete reaches specified compressive strength if shores and other supports are designed to allow facing removal without deflection of supported slab or member.

3.3.9 Strength of Concrete Required for Removal of Formwork

If removal of formwork, reshoring, or backshoring is based on concrete reaching a specified in-place strength, mold and field-cure cylinders in accordance with JIS A 1132. Test cylinders in accordance with JIS A 1108.

~~3.4 WATERSTOP INSTALLATION AND SPLICES~~

~~a. Provide waterstops in construction joints as indicated.~~

~~b. Install formwork to accommodate waterstop materials. Locate waterstops in joints where indicated in Contract Documents. Minimize number of splices in waterstop. Splice waterstops in accordance with manufacturer's written instructions. Install factory-manufactured premolded mitered corners.~~

~~c. Install waterstops to form a continuous diaphragm in each joint. Make adequate provisions to support and protect waterstops during progress of work. Protect waterstops protruding from joints from damage.~~

~~3.4.1 PVC Waterstop~~

~~Make splices by heat sealing the adjacent waterstop edges together using a thermoplastic splicing iron utilizing a non-stick surface specifically designed for waterstop welding. Reform waterstops at splices with a remolding iron with ribs or corrugations to match the pattern of the waterstop. The spliced area, when cooled, must show no signs of separation, holes, or other imperfections when bent by hand in as sharp an angle as possible.~~

~~3.4.2 Rubber Waterstop~~

~~Rubber waterstops must be spliced using cold bond adhesive as recommended by the manufacturer.~~

~~3.4.3 Thermoplastic Elastomeric Rubber Waterstop~~

~~Fittings must be shop made using a machine specifically designed to mechanically weld the waterstop. A portable power saw must be used to miter or straight cut the ends to be joined to ensure good alignment and contact between joined surfaces. Maintain continuity of the characteristic features of the cross section of the waterstop (for example ribs, tabular center axis, and protrusions) across the splice.~~

~~3.4.4 Hydrophilic Waterstop~~

~~Miter cut ends to be joined with sharp knife or shears. The ends must be adhered with adhesive.~~

3.4 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS

a. Unless otherwise specified, placing reinforcement and miscellaneous materials must be in accordance to JASS 5 and MLIT-SS Chapter 6. Provide bars, welded wire reinforcement, wire ties, supports, and other devices necessary to install and secure reinforcement.

b. Reinforcement must not have rust, scale, oil, grease, clay, or foreign substances that would reduce the bond. Rusting of reinforcement is a basis of rejection if the effective cross-sectional area or the nominal weight per unit length has been reduced. Remove loose rust prior to placing steel. Tack welding is prohibited.

c. Nonprestressed cast-in-place concrete members must have concrete cover for reinforcement given in the following table:

Concrete Exposure	**Member**	**Reinforcement**	**Specified cover, mm**
Cast against and permanently in contact with ground	All	All	75
Exposed to weather or in contact with ground	All	D19 through D51 bars	50
		D16, 16 mm diameter smooth, or 16 mm diameter deformed wire, and smaller	40
Not exposed to weather or in contact with ground	Slabs, joists, and walls	D38, D41 and D51 bars	40
		D35 bar and smaller	~~20~~25
	Beams, columns, pedestals, and tension ties	Primary reinforcement, stirrups, ties, spirals, and hoops	40

~~d. Cast-in-place prestressed concrete members must have concrete cover for reinforcement, ducts, and end fittings given in the following table:~~

~~Concrete Exposure~~	~~Member~~	~~Reinforcement~~	~~Specified cover, mm~~
~~Cast against and permanently in contact with ground~~	~~All~~	~~All~~	~~75~~
~~Exposed to weather or in contact with ground~~	~~Slabs, joists, and walls~~	~~All~~	~~25~~
	~~All other~~	~~All~~	~~40~~
~~Not exposed to weather or in contact with ground~~	~~Slabs, joists, and walls~~	~~All~~	~~20~~
	~~Beams, columns, and tension ties~~	~~Primary reinforcement~~	~~40~~
		~~Stirrups, ties, spirals, and hoops~~	~~25~~

~~e~~d. Precast nonprestressed or prestressed concrete members manufactured under plant conditions must have concrete cover for reinforcement, ducts, and end fittings given in the following table:

Concrete	**Member**	**Reinforcement**	**Specified**
Exposed to weather or in contact with ground	Walls	D38, D41 and D51 bars; tendons larger than 40 mm diameter	40
		D35 bars and smaller; 16 mm diameter smooth, or 16 mm diameter deformed wire, and smaller; tendons and strands 40 mm and smaller	20
	All other	D38, D41 and D51 bars; tendons and strands larger than 40 mm diameter	50
		D19 through D35 bars; tendons and strands larger than 16 mm diameter through 40 mm	40
		D16 bar, 16 mm diameter smooth, or 16 mm diameter deformed wire, and smaller; tendons and strands 16 mm diameter and smaller	32

Concrete	**Member**	**Reinforcement**	**Specified**
Not exposed to weather or in contact with ground	Slabs, joists, and walls	D38, D41 and D51 bars; tendons larger than 40 mm diameter	32
		Tendons and strands 40 mm diameter and smaller	20
		D35, 16 mm diameter smooth, or 16 mm diameter deformed wire, and smaller	16
	Beams, columns, pedestals, and tension ties	Primary reinforcement	Greater of bar diameter and 16 and need not exceed 40
		Stirrups, ties, spirals, and hoops	10

3.4.1 General

Provide details of reinforcement that are in accordance with the Contract Documents.

3.4.2 ~~Vapor Retarder [and ~~Vapor Barrier~~]~~

a. Level and compact base material and install ~~vapor retarder [~~vapor barrier~~]~~ with the longest dimension parallel with the direction of concrete pour and face laps away from direction of pour whenever possible. Extend ~~vapor retarder [~~vapor barrier~~]~~ over footings, and seal to foundation wall, grade beam, or slab at an elevation consistent with the top of the slab or terminate at impediments such as dowels or water stops. Seal around all penetrations such as utilities and columns with ~~vapor retarder [~~vapor barrier~~]~~ material and seal tape. Use the greatest widths and lengths practicable to eliminate joints wherever possible. Lap joints a minimum of 300 mm and tape. ~~[~~Extend ~~vapor retarder [~~vapor barrier~~]~~ over the tops of pile caps and grade beams to a distance acceptable to the contracting officer and terminate as recommended by the manufacturer.~~]~~

b. Protect ~~vapor retarder [~~vapor barrier~~]~~ from damage during installation of reinforcing steel, utilities and concrete. Provide reinforcing bar supports with base section that minimize the potential for puncture of ~~vapor retarder [~~vapor barrier~~]~~. Avoid use of stakes driven through the ~~vapor retarder [~~vapor barrier~~]~~.

c. Inspect installation of ~~vapor retarder [~~vapor barrier~~]~~ including sealing of joints and penetrations and mark all areas of damage and insufficient installation in advance of concrete placement such that deficiencies are corrected before concrete is placed. Remove torn, punctured or damaged ~~vapor retarder [~~vapor barrier~~]~~ and repair damaged areas prior to concrete placement with ~~vapor retarder [~~vapor barrier~~]~~ material lapped and sealed a minimum of 150 mm beyond damaged area or as instructed by the manufacturer.

~~[d. Place a 50 mm layer of clean concrete sand on vapor retarder [vapor barrier] before placing concrete.]~~

~~3.4.3 Perimeter Insulation~~

~~Install perimeter insulation at locations indicated. Adhesive must be used where insulation is applied to the interior surface of foundation walls and may be used for exterior application.~~

3.4.3 Reinforcement Supports

Provide reinforcement support in accordance with JASS 5 and MLIT-SS Chapter 6. Supports for coated or galvanized bars must also be coated with electrically compatible material for a distance of at least 50 mm beyond the point of contact with the bars.

3.4.4 Splicing

Lap slice lengths and locations as indicated in the Contract Documents. For splice locations not indicated follow JASS 5 and MLIT-SS Chapter 6, at no additional cost to the Government and subject to approval. Splicing must be by lapping, by gas pressure welding, or by mechanical or welded connection, except that lap splices must not be used for bars larger than D35. Do not splice at points of maximum stress and stagger splices a minimum of ~~[600][~~1200~~][_____]~~ mm or as otherwise indicated so no more than half of the bars are spliced at any one section.

Overlap welded wire reinforcement the spacing of the cross wires, plus 50 mm.

Approve welded splices prior to use.

3.4.5 Future Bonding

Plug exposed, threaded, mechanical reinforcement bar connectors with a greased bolt. Provide bolt threads that match the connector. Countersink the connector in the concrete. Caulk the depression after the bolt is installed.

3.4.6 Setting Miscellaneous Material

Place and secure anchors and bolts, pipe sleeves, conduits, and other such items in position before concrete placement and support against displacement. Plumb anchor bolts and check location and elevation. Temporarily fill voids in sleeves with readily removable material to prevent the entry of concrete.

3.4.7 Fabrication

Shop fabricate reinforcing bars to conform to shapes and dimensions indicated for reinforcement, and as follows:

a. Provide fabrication tolerances that are in accordance with JASS 5 and

MLIT-SS Chapter 6.

b. Provide hooks and bends that are in accordance with the Contract Documents.

Reinforcement must be bent cold to shapes as indicated. Bending must be done in the shop. Rebending of a reinforcing bar that has been bent incorrectly is not be permitted. Bending must be in accordance with standard approved practice and by approved machine methods.

Deliver reinforcing bars bundled, tagged, and marked. Tags must be metal with bar size, length, mark, and other information pressed in by machine. Marks must correspond with those used on the placing drawings.

Do not use reinforcement that has any of the following defects:

a. Bar lengths, depths, and bends beyond specified fabrication tolerances

b. Bends or kinks not indicated on drawings or approved shop drawings

c. Bars with reduced cross-section due to rusting or other cause

Replace defective reinforcement with new reinforcement having required shape, form, and cross-section area.

3.4.8 Placing Reinforcement

Place reinforcement in accordance with JASS 5 and MLIT-SS Chapter 6.

For slabs on grade (over earth or over capillary water barrier) and for footing reinforcement, support bars or welded wire reinforcement on precast concrete blocks, spaced at intervals required by size of reinforcement, to keep reinforcement the minimum height specified above the underside of slab or footing.

For slabs other than on grade, supports for which any portion is less than 25 mm from concrete surfaces that are exposed to view or to be painted must be of precast concrete units, plastic-coated steel, or stainless steel protected bar supports. Precast concrete units must be wedge shaped, not larger than 90 by 90 mm, and of thickness equal to that indicated for concrete protection of reinforcement. Provide precast units that have cast-in galvanized tie wire hooked for anchorage and blend with concrete surfaces after finishing is completed.

Provide reinforcement that is supported and secured together to prevent displacement by construction loads or by placing of wet concrete, and as follows:

a. Provide supports for reinforcing bars that are sufficient in number and have sufficient strength to carry the reinforcement they support, and in accordance with JASS 5 and MLIT-SS Chapter 6. Do not use supports to support runways for concrete conveying equipment and similar construction loads.

b. Equip supports on ground and similar surfaces with sand-plates.

c. Support welded wire reinforcement as required for reinforcing bars.

d. Secure reinforcements to supports by means of tie wire. Wire must be black, soft iron wire, not less than 1.6 mm.

e. Reinforcement must be accurately placed, securely tied at intersections, and held in position during placing of concrete by spacers, chairs, or other approved supports. Point wire-tie ends away from the form. Unless otherwise indicated, numbers, type, and spacing of supports must conform to the Contract Documents.

f. Bending of reinforcing bars partially embedded in concrete is permitted only as specified in the Contract Documents.

3.4.9 Spacing of Reinforcing Bars

a. Spacing must be as indicated in the Contract Documents.

b. Reinforcing bars may be relocated to avoid interference with other reinforcement, or with conduit, pipe, or other embedded items. If any reinforcing bar is moved a distance exceeding one bar diameter or specified placing tolerance, resulting rearrangement of reinforcement is subject to preapproval by the Contracting Officer.

3.4.10 Concrete Protection for Reinforcement

Additional concrete protection must be in accordance with the Contract Documents.

3.4.11 Welding

Welding must be in accordance with JASS 6. Welders shall be certified in accordance with JIS Z 3881 for gas pressure welding. Welded joint connections shall develop 125 percent of the specified yield strength of the reinforcing bar and 100 percent of the specified tensile strength of the reinforcing bar.

3.5 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE

In accordance with JIS A 5308, JASS 5 and MLIT-SS Chapter 6, except as modified herein. Batching equipment must be such that the concrete ingredients are consistently measured within the following tolerances: 1 percent for cement and water, 2 percent for aggregate, and 3 percent for admixtures. Furnish mandatory batch ticket information for each load of ready mix concrete.

3.5.1 Measuring

Make measurements at intervals as specified in paragraphs SAMPLING and TESTING.

3.5.2 Mixing

a. Mix concrete in accordance with JIS A 5308, JASS 5 and MLIT-SS Chapter 6.

b. Machine mix concrete. Begin mixing within 30 minutes after the cement has been added to the aggregates. Place concrete within 90 minutes of either addition of mixing water to cement and aggregates or addition of cement to aggregates if the air temperature is less than 29 degrees C.

c. Reduce mixing time and place concrete within 60 minutes if the air temperature is greater than 29 degrees C except as follows: if set retarding admixture is used and slump requirements can be met, limit for placing concrete may remain at 90 minutes. Additional water may be added, provided that both the specified maximum slump and submitted water-cementitious material ratio are not exceeded and the required concrete strength is still met. When additional water is added, an additional 30 revolutions of the mixer at mixing speed is required.

d. ~~[If the entrained air content falls below the specified limit, add a sufficient quantity of admixture to bring the entrained air content within the specified limits.]~~Dissolve admixtures in the mixing water and mix in the drum to uniformly distribute the admixture throughout the batch. Do not reconstitute concrete that has begun to solidify.

3.5.3 Transporting

Transport concrete from the mixer to the forms as rapidly as practicable. Prevent segregation or loss of ingredients. Clean transporting equipment thoroughly before each batch. Do not use aluminum pipe or chutes. Remove concrete which has segregated in transporting and dispose of as directed.

3.6 PLACING CONCRETE

Place concrete in accordance with JASS 5 and MLIT-SS Chapter 6.

~~[~~3.6.1 Footing Placement

Concrete for footings may be placed in excavations without forms upon inspection and approval by the Contracting Officer. Excavation width must be a minimum of 100 mm greater than indicated.

~~][~~3.6.2 Pumping

JASS 5 and MLIT-SS Chapter 6. Pumping must not result in separation or loss of materials nor cause interruptions sufficient to permit loss of plasticity between successive increments. Loss of slump in pumping equipment must not exceed 50 mm at discharge/placement. Do not convey concrete through pipe made of aluminum or aluminum alloy. Avoid rapid changes in pipe sizes. Limit maximum size of course aggregate to 33 percent of the diameter of the pipe. Limit maximum size of well-rounded aggregate to 40 percent of the pipe diameter. Take samples for testing at both the point of delivery to the pump and at the discharge end.

~~]~~3.6.3 Cold Weather

Cold weather concrete must meet the requirements of JASS 5 and MLIT-SS Chapter 6 unless otherwise specified. Do not allow concrete temperature to decrease below 10 degrees C. Obtain approval prior to placing concrete when the ambient temperature is below 4 degrees C or when concrete is likely to be subjected to freezing temperatures within 24 hours. Cover concrete and provide sufficient heat to maintain 10 degrees C minimum adjacent to both the formwork and the structure while curing. Limit the rate of cooling to 3 degrees C in any 1 hour and 10 degrees C per 24 hours after heat application.

3.6.4 Hot Weather

~~[~~Hot weather concrete must meet the requirements of JASS 5 and

MLIT-SS Chapter 6 unless otherwise specified. ~~]~~Maintain required concrete temperature using Figure 4.2 in ACI 305R to prevent the evaporation rate from exceeding 1 kg per square meter of exposed concrete per hour. Cool ingredients before mixing or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Shade the fresh concrete as soon as possible after placing. Start curing when the surface of the fresh concrete is sufficiently hard to permit curing without damage. Provide water hoses, pipes, spraying equipment, and water hauling equipment, where job site is remote to water source, to maintain a moist concrete surface throughout the curing period. Provide burlap cover or other suitable, permeable material with fog spray or continuous wetting of the concrete when weather conditions prevent the use of either liquid membrane curing compound or impervious sheets. For vertical surfaces, protect forms from direct sunlight and add water to top of structure once concrete is set.

3.6.5 Bonding

Surfaces of set concrete at joints, must be roughened and cleaned of laitance, coatings, loose particles, and foreign matter. Roughen surfaces in a manner that exposes the aggregate uniformly and does not leave laitance, loosened particles of aggregate, nor damaged concrete at the surface.

Obtain bonding of fresh concrete that has set as follows:

a. At joints between footings and walls or columns, between walls or columns and the beams or slabs they support, and elsewhere unless otherwise specified; roughened and cleaned surface of set concrete must be dampened, but not saturated, immediately prior to placing of fresh concrete.

b. At joints in exposed-to-view work; at vertical joints in walls; at joints near midpoint of span in girders, beams, supported slabs, other structural members; in work designed to contain liquids; the roughened and cleaned surface of set concrete must be dampened but not saturated and covered with a cement grout coating.

c. Provide cement grout that consists of equal parts of portland cement and fine aggregate by weight with not more than 22.5 liters of water per sack of cement. Apply cement grout with a stiff broom or brush to a minimum thickness of 1.6 mm. Deposit fresh concrete before cement grout has attained its initial set.

3.7 WASTE MANAGEMENT

Provide as specified in the Waste Management Plan and as follows.

3.7.1 Mixing Equipment

Before concrete pours, designate~~[~~ Contractor-owned site meeting environmental standards~~][~~ or approved on-site area to be paved later in project~~]~~ for cleaning out concrete mixing trucks. Minimize water used to wash equipment.

~~3.7.2 Hardened, Cured Waste Concrete~~

~~[Crush and reuse hardened, cured waste concrete as fill or as a base course for pavement.][Use hardened, cured waste concrete as aggregate in~~

~~concrete mix if approved by Contracting Officer.]~~

3.7.2 Reinforcing Steel

Collect reinforcing steel and place in designated area for recycling.

3.7.3 Other Waste

Identify concrete manufacturer's or supplier's policy for collection or return of construction waste, unused material, deconstruction waste, and/or packaging material.~~[Return excess cement to supplier.][~~ Institute deconstruction and construction waste separation and recycling for use in manufacturer's programs. When such a program is not available, seek local recyclers to reclaim the materials.~~]~~

3.8 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES

3.8.1 Defects

Repair surface defects in accordance with JASS 5 and MLIT-SS Chapter 6.

3.8.2 Not Against Forms (Top of Walls)

Surfaces not otherwise specified must be finished with wood floats to even surfaces. Finish must match adjacent finishes.

3.8.3 Formed Surfaces

3.8.3.1 Tolerances

Tolerances in accordance with JASS 5 and MLIT-SS Chapter 6 and as indicated.

3.8.3.2 As-Cast Rough Form

Provide for surfaces ~~not exposed to public view~~below grade a surface finish SF-1.0. Patch holes and defects in accordance with ACI 301 or JASS 5 and MLIT-SS Chapter 6.

3.8.3.3 Standard Smooth Finish

Provide for all concrete surfaces ~~exposed to public view~~ a surface finish SF-3.0. Patch holes and defects in accordance with ACI 301 or JASS 5 and MLIT-SS Chapter 6.

3.8.4 ~~[Smooth-Rubbed][~~Grout-Cleaned Rubbed~~][Cork-Floated][Exposed Aggregate]~~ Finish

~~[Provide a smooth-rubbed finish per ACI 301 Section 5 in the locations indicated.][~~Provide a grout-cleaned rubbed finish per ACI 301 Section 5 for surfaces exposed to public view and in the locations indicated. ~~][Provide a cork-floated finish per ACI 301 Section 5 in the locations indicated.][Provide an exposed aggregate finish per ACI 301 Section 5 in the locations indicated.]~~

3.9 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION

In accordance with JASS 5 and MLIT-SS Chapter 6, unless otherwise specified. Slope floors uniformly to drains where drains are provided~~.[~~, unless otherwise indicated. Depress the concrete base slab where quarry tile~~,~~ or ceramic tile~~, [or] [_____]~~ are indicated.~~][~~ Steel trowel and fine-broom finish concrete slabs that are to receive quarry tile or~~,~~ ceramic tile~~, or paver tile [_____]~~.~~]~~ Where straightedge measurements are specified, Contractor must provide straightedge.

3.9.1 Finish

Place, consolidate, and immediately strike off concrete to obtain proper contour, grade, and elevation before bleedwater appears. Permit concrete to attain a set sufficient for floating and supporting the weight of the finisher and equipment. If bleedwater is present prior to floating the surface, drag the excess water off or remove by absorption with porous materials. Do not use dry cement to absorb bleedwater.

3.9.1.1 Scratched

Use for surfaces intended to receive bonded applied cementitious applications. Finish concrete in accordance with ACI 301 Section 5 for a scratched finish.

3.9.1.2 Floated

Use for ~~[~~surfaces to receive ~~[~~roofing,~~]~~ ~~[~~waterproofing membranes~~,] [sand bed terrazzo,]] [_____] [and] [exterior slabs where not otherwise specified~~. ~~]~~ Finish concrete in accordance with ACI 301 Section 5 for a floated finish.

3.9.1.3 Steel Troweled

Use for floors intended as walking surfaces~~[,]~~ ~~[~~and~~]~~ for reception of floor coverings~~[, and] [_____]~~. Finish concrete in accordance with ACI 301 Section 5 for a steel troweled finish.

~~[~~3.9.1.4 Nonslip Finish

Use on surfaces of exterior platforms, steps, and landings; and on exterior and interior pedestrian ramps. Finish concrete in accordance with ACI 301 Section 5 for a dry-shake finish. After the selected material has been embedded by the two floatings, complete the operation with a ~~[~~broomed~~] [floated] [troweled]~~ finish.

~~]~~3.9.1.5 Broomed

Use on surfaces of exterior walks, platforms, ~~patios,~~ and ramps, unless otherwise indicated. Finish concrete in accordance with ACI 301 Section 5 for a broomed finish.

3.9.1.6 Pavement

Screed the concrete with a template advanced with a combined longitudinal and crosswise motion. Maintain a slight surplus of concrete ahead of the template. After screeding, float the concrete longitudinally. Use a straightedge to check slope and flatness; correct and refloat as necessary. Obtain final finish by ~~[~~belting. Lay belt flat on the concrete surface and advance with a sawing motion; continue until a uniform but gritty nonslip surface is obtained.~~] [a burlap drag. Drag a strip of clean, wet burlap from 900 to 3000 mm wide and 600 mm longer than the pavement width across the slab. Produce a fine, granular, sandy~~

~~textured surface without disfiguring marks.]~~ Round edges and joints with an edger having a radius of 3 mm.

3.9.1.7 Chemical-Hardener Treatment

~~[~~ Apply liquid-chemical floor hardener where indicated after curing and drying concrete surface. Dilute liquid hardener with water and apply in three coats. First coat must be one-third strength, second coat one-half strength, and third coat two-thirds strength. Apply each coat evenly and allow to dry 24 hours between coats.

Approved proprietary chemical hardeners must be applied in accordance with manufacturer's printed directions.

~~]~~3.9.2 Flat Floor Finishes

ACI 302.1R. Construct in accordance with one of the methods recommended in Table 7.15.3, "Typical Composite Ff/FL Values for Various Construction Methods." ACI 117 for tolerance tested by ASTM E1155.

a. Specified Conventional Value:

Floor Flatness (Ff) ~~[~~20~~] [______] [~~____13~~] [______]~~ minimum
Floor Levelness (FL) ~~[~~15~~] [______] [~~___10~~] [______]~~ minimum

~~b. Specified Industrial:~~

~~Floor Flatness (Ff) [30] [______] [15] [______] minimum~~
~~Floor Levelness (FL) [20] [______] [10] [______] minimum~~

3.9.2.1 Measurement of Floor Tolerances

Test slab within 24 hours of the final troweling. Provide tests to Contracting Officer within 12 hours after collecting the data. Floor flatness inspector is required to provide a tolerance report which must include:

a. Key plan showing location of data collected.

b. Results required by ASTM E1155.

3.9.2.2 Remedies for Out of Tolerance Work

Contractor is required to repair and retest any floors not meeting specified tolerances. Prior to repair, Contractor must submit and receive approval for the proposed repair, including product data from any materials proposed. Repairs must not result in damage to structural integrity of the floor. For floors exposed to public view, repairs must prevent any uneven or unusual coloring of the surface.

3.9.3 Concrete Walks

Provide 100 mm thick minimum unless otherwise indicated. Provide contraction joints spaced every 1500 lineal mm unless otherwise indicated. Cut contraction joints 25 mm deep, or one fourth the slab thickness whichever is deeper, with a jointing tool after the surface has been finished. Provide 13 mm thick transverse expansion joints at changes in direction where sidewalk abuts curb, steps, rigid pavement, or other similar structures; space expansion joints every 15 m maximum. Give walks a broomed finish. Unless indicated otherwise, provide a transverse slope of 1/48. Limit variation in cross section to 6 mm in 1500 mm.

3.9.4 Pits and Trenches

Place bottoms and walls monolithically or provide waterstops and keys.

3.9.5 Curbs~~[~~ and Gutters~~]~~

Provide contraction joints spaced every 3 m maximum unless otherwise indicated. Cut contraction joints 20 mm deep with a jointing tool after the surface has been finished. Provide expansion joints 13 mm thick and spaced every 30 m maximum unless otherwise indicated. Perform pavement finish.

~~[~~3.9.6 Splash Blocks

Provide at outlets of downspouts emptying at grade. Splash blocks may be precast concrete, and must be 600 mm long, 300 mm wide and 100 mm thick, unless otherwise indicated, with smooth-finished countersunk dishes sloped to drain away from the building.

~~]~~3.10 JOINTS

3.10.1 Construction Joints

Make and locate joints not indicated so as not to impair strength and appearance of the structure, as approved. Joints must be perpendicular to main reinforcement. Reinforcement must be continued and developed across construction joints. Locate construction joints as follows:

3.10.1.1 Maximum Allowable Construction Joint Spacing

a. In walls at not more than ~~18.0~~18.3 meter in any horizontal direction.

b. In slabs on ground, so as to divide slab into areas not in excess of 110 square meter.

3.10.1.2 Construction Joints for Constructability Purposes

a. In walls, at top of footing; at top of slabs on ground; at top and bottom of door and window openings or where required to conform to architectural details; and at underside of deepest beam or girder framing into wall.

b. In columns or piers, at top of footing; at top of slabs on ground; and at underside of deepest beam or girder framing into column or pier.

c. Near midpoint of spans for supported slabs, beams, and girders unless a beam intersects a girder at the center, in which case construction joints in girder must offset a distance equal to twice the width of the beam. Make transfer of shear through construction joint by use of inclined reinforcement.

Provide keyways at least 40 mm deep in construction joints in walls and slabs and between walls and footings; approved bulkheads may be used for slabs.

3.10.2 Isolation Joints in Slabs on Ground

a. Provide joints at points of contact between slabs on ground and vertical surfaces, such as column pedestals, foundation walls, grade beams, and elsewhere as indicated.

b. Fill joints with premolded joint filler strips 13 mm thick, extending full slab depth. Install filler strips at proper level below finish floor elevation with a slightly tapered, dress-and-oiled wood strip temporarily secured to top of filler strip to form a groove not less than 20 mm in depth where joint is sealed with sealing compound and not less than 6 mm in depth where joint sealing is not required. Remove wood strip after concrete has set. Contractor must clean groove of foreign matter and loose particles after surface has dried.

3.10.3 Contraction Joints in Slabs on Ground

a. Provide joints to form panels as indicated.

~~b. Under and on exact line of each control joint, cut 50 percent of welded wire reinforcement before placing concrete.~~

~~c~~b. Sawcut contraction joints into slab on ground in accordance with ACI 301 Section 5.

~~[~~ ~~d~~c. Joints must be 4 mm wide by 1/5 to 1/4 of slab depth and formed by inserting hand-pressed fiberboard strip into fresh concrete until top surface of strip is flush with slab surface. After concrete has cured for at least 7 days, the Contractor must remove inserts and clean groove of foreign matter and loose particles.
~~]~~
~~[~~ ~~e~~d. Sawcutting will be limited to within 12 hours after set and at 1/4 slab depth.

~~]~~3.10.4 Sealing Joints in Slabs on Ground

a. Contraction and control joints which are to receive finish flooring material must be sealed with joint sealing compound after concrete curing period. Slightly underfill groove with joint sealing compound to prevent extrusion of compound. Remove excess material as soon after sealing as possible.

b. Sealed groove must be left ready to receive filling material that is provided as part of finish floor covering work.

3.11 CONCRETE FLOOR TOPPING

3.11.1 Standard Floor Topping

Provide topping for floor and roof slabs, treads and platforms of metal steel stairs, and elsewhere as indicated.

3.11.1.1 Preparations Prior to Placing

a. When topping is placed on a green concrete base slab, screed surface of base slab to a level not more than 40 mm nor less than 25 mm below required finish surface. Remove water and laitance from surface of base slab before placing topping mixture. As soon as water ceases to rise to surface of base slab, place topping.

b. When topping is placed on a hardened concrete base slab, remove dirt, loose material, oil, grease, asphalt, paint, and other contaminants from base slab surface, leaving a clean surface. Prior to placing topping mixture, 64 mm minimum, slab surface must be dampened and left free of standing water. Immediately before topping mixture is placed, broom a coat of neat cement grout onto surface of slab. Do not allow cement grout to set or dry before topping is placed.

c. When topping is placed on a metal surface, such as metal pans for steel stairs, remove dirt, loose material, oil, grease, asphalt, paint, and other contaminants from metal surface.

3.11.1.2 Placing

Spread standard topping mixture evenly on previously prepared base slab or metal surface, brought to correct level with a straightedge, and struck off. Topping must be consolidated, floated, checked for trueness of surface, and refloated as specified for float finish.

3.11.1.3 Finishing

Give trowel finish standard floor topping surfaces.

Give other finishes standard floor topping surfaces as indicated.

3.12 CURING AND PROTECTION

Curing and protection in accordance with JASS 5 and MLIT-SS Chapter 6, unless otherwise specified. Begin curing immediately following form removal. Avoid damage to concrete from vibration created by blasting, pile driving, movement of equipment in the vicinity, disturbance of formwork or protruding reinforcement, and any other activity resulting in ground vibrations. Protect concrete from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks, and oil stains. Do not allow concrete to dry out from time of placement until the expiration of the specified curing period. Do not use membrane-forming compound on surfaces where appearance would be objectionable, on any surface to be painted, where coverings are to be bonded to the concrete, or on concrete to which other concrete is to be bonded. If forms are removed prior to the expiration of the curing period, provide another curing procedure specified herein for the remaining portion of the curing period. Provide moist curing for those areas receiving liquid chemical sealer, hardener, or epoxy coating. Allow curing compound/sealer installations to cure prior to the installation of materials that adsorb VOCs~~, including [_____]~~.

3.12.1 Curing Periods

JASS 5 and MLIT-SS Chapter 6, except 10 days for retaining walls~~,~~ or pavement~~ or chimneys~~. Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing are subject to approval by the Contracting Officer.

3.12.2 Curing Formed Surfaces

Accomplish curing of formed surfaces, including undersurfaces of girders, beams, supported slabs, and other similar surfaces by moist curing with forms in place for full curing period or until forms are removed. If forms are removed before end of curing period, accomplish final curing of formed surfaces by any of the curing methods specified above, as applicable.

3.12.3 Curing Unformed Surfaces

a. Accomplish initial curing of unformed surfaces, such as monolithic slabs, floor topping, and other flat surfaces, by membrane curing.

b. Accomplish final curing of unformed surfaces by any of curing methods specified, as applicable.

c. Accomplish final curing of concrete surfaces to receive liquid floor hardener of finish flooring by moisture-retaining cover curing.

3.12.4 Temperature of Concrete During Curing

When temperature of atmosphere is 5 degrees C and below, maintain temperature of concrete at not less than 13 degrees C throughout concrete curing period or 7 degrees C when the curing period is measured by maturity. When necessary, make arrangements before start of concrete placing for heating, covering, insulation, or housing as required to maintain specified temperature and moisture conditions for concrete during curing period.

When the temperature of atmosphere is 27 degrees C and above or during other climatic conditions which cause too rapid drying of concrete, make arrangements before start of concrete placing for installation of wind breaks, of shading, and for fog spraying, wet sprinkling, or moisture-retaining covering of light color as required to protect concrete during curing period.

Changes in temperature of concrete must be uniform and not exceed 3 degrees C in any 1 hour nor 27 degrees C in any 24-hour period.

3.12.5 Protection from Mechanical Injury

During curing period, protect concrete from damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration and from damage caused by rain or running water.

3.12.6 Protection After Curing

Protect finished concrete surfaces from damage by construction operations.

3.13 FIELD QUALITY CONTROL

3.13.1 Sampling

JIS A 1115. Collect samples of fresh concrete to perform tests specified. JIS A 1132 for making test specimens.

3.13.2 Testing

3.13.2.1 Slump Tests

JIS A 1101. Take concrete samples during concrete placement/discharge.

The maximum slump may be increased as specified with the addition of an approved admixture provided that the water-cementitious material ratio is not exceeded. Perform tests at commencement of concrete placement, when test cylinders are made, and for each batch (minimum) or every 16 cubic meters (maximum) of concrete.

3.13.2.2 Temperature Tests

Test the concrete delivered and the concrete in the forms. Perform tests in hot or cold weather conditions (below 10 degrees C and above 27 degrees C) for each batch (minimum) or every 16 cubic meters (maximum) of concrete, until the specified temperature is obtained, and whenever test cylinders and slump tests are made.

3.13.2.3 Compressive Strength Tests

JIS A 1108. Make ~~[~~six~~]~~ ~~[eight]~~ 150 mm by 300 mm ~~[100 mm by 200 mm]~~ test cylinders for each set of tests in accordance with JIS A 1132, JIS A 1115 and applicable requirements of JASS 5 and MLIT-SS Chapter 6. Take precautions to prevent evaporation and loss of water from the specimen. Test two cylinders at 7 days, two cylinders at 28 days, ~~[two cylinders at 56 days] [two cylinders at 90 days] [______]~~ and hold two cylinder in reserve. Take samples for strength tests of each ~~[~~mix design of~~] [and for] [______]~~ concrete placed each day not less than once a day, nor less than once for each 75 cubic meters of concrete for the first 380 cubic meters, then every 380 cubic meters thereafter, nor less than once for each 500 square meters of surface area for slabs or walls. For the entire project, take no less than five sets of samples and perform strength tests for each mix design of concrete placed. Each strength test result must be the average of two cylinders from the same concrete sample tested at 28 days~~[56 days] [90 days] [______]~~. Concrete compressive tests must meet the requirements of this section, the Contract Document, and JASS 5 and MLIT-SS Chapter 6. Retest locations represented by erratic core strengths. Where retest does not meet concrete compressive strength requirements submit a mitigation or remediation plan for review and approval by the contracting officer. Repair core holes with nonshrink grout. Match color and finish of adjacent concrete.

~~[~~3.13.2.4 Air Content

JIS A 1118 or JIS A 1128 for normal weight concrete. Test air-entrained concrete for air content at the same frequency as specified for slump tests.

~~][~~3.13.2.5 Unit Weight of Structural Concrete

JIS A 1116. Determine unit weight of normal weight concrete. Perform test for every 15 cubic meters maximum.

~~][~~3.13.2.6 Chloride Ion Concentration

Chloride ion concentration must meet the requirements of the paragraph titled CORROSION AND CHLORIDE CONTENT. Determine water soluble ion concentration in accordance with JIS A 1154. Perform test once for each mix design.

~~]~~3.13.2.7 Strength of Concrete Structure

The strength of the concrete structure will be considered to be deficient if any of the following conditions are identified:

a. Failure to meet compressive strength tests as evaluated.

b. Reinforcement not conforming to requirements specified.

c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.

d. Concrete curing and protection of concrete against extremes of temperature during curing, not conforming to requirements specified.

e. Concrete subjected to damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration.

f. Poor workmanship likely to result in deficient strength.

Where the strength of the concrete structure is considered deficient submit a mitigation or remediation plan for review and approval by the contracting officer.

3.13.2.8 Non-Conforming Materials

Factors that indicate that there are non-conforming materials include (but not limited to) excessive compressive strength, inadequate compressive strength, excessive slump, excessive voids and honeycombing, concrete delivery records that indicate excessive time between mixing and placement, or excessive water was added to the mixture during delivery and placement. Any of these indicators alone are sufficient reason for the Contracting Officer to request additional sampling and testing.

Investigations into non-conforming materials must be conducted at the Contractor's expense. The Contractor must be responsible for the investigation and must make written recommendations to adequately mitigate or remediate the non-conforming material. The Contracting Officer may accept, accept with reduced payment, require mitigation, or require removal and replacement of non-conforming material at no additional cost to the Government.

3.13.2.9 Testing Concrete Structure for Strength

When there is evidence that strength of concrete structure in place does not meet specification requirements or there are non-conforming materials, make cores drilled from hardened concrete for compressive strength determination in accordance with JIS A 1107, and as follows:

a. Take at least three representative cores from each member or area of concrete-in-place that is considered potentially deficient. Location of cores will be determined by the Contracting Officer.

b. Test cores after moisture conditioning in accordance with JIS A 1107 if concrete they represent is more than superficially wet under service.

c. Air dry cores, (16 to 27 degrees C with relative humidity less than 60 percent) for 7 days before test and test dry if concrete they represent is dry under service conditions.

d. Strength of cores from each member or area are considered satisfactory if their average is equal to or greater than 85 percent of the 28-day design compressive strength of the class of concrete.

~~[e. Core specimens will be taken and tested by the Government. If the results of core-boring tests indicate that the concrete as placed does not conform to the drawings and specification, the cost of such tests and restoration required must be borne by the Contractor.~~

~~]~~

Fill core holes solid with patching mortar and finished to match adjacent concrete surfaces.

Correct concrete work that is found inadequate by core tests in a manner approved by the Contracting Officer.

3.14 REPAIR, REHABILITATION AND REMOVAL

Before the Contracting Officer accepts the structure the Contractor must inspect the structure for cracks, damage and substandard concrete placements that may adversely affect the service life of the structure. A report documenting these defects must be prepared which includes recommendations for repair, removal or remediation must be submitted to the Contracting Officer for approval before any corrective work is accomplished.

~~[~~3.14.1 Crack Repair

Prior to final acceptance, all cracks in excess of 0.50 mm wide must be documented and repaired. The proposed method and materials to repair the cracks must be submitted to the Contracting Officer for approval. The proposal must address the amount of movement expected in the crack due to temperature changes and loading.

~~]~~3.14.2 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than 6 mm thick must be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than 6 mm thick must be removed and replaced or mitigated in a manner acceptable to the Contracting Officer.

3.14.3 Failure of Quality Assurance Test Results

Proposed mitigation efforts by the Contractor must be approved by the Contracting Officer prior to proceeding.

-- End of Section --

SECTION 05 12 00

STRUCTURAL STEEL
08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303 (2016) Code of Standard Practice for Steel Buildings and Bridges

AISC 325 (2017) Steel Construction Manual

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

AWS D1.8/D1.8M (2016) Structural Welding Code—Seismic Supplement

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 (2009) Surface Texture, Surface Roughness, Waviness and Lay

ASTM INTERNATIONAL (ASTM)

ASTM A143/A143M (2007; R 2014) Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement

JAPANESE ARCHITECTURAL STANDARD SPECIFICATIONS (JASS)

JASS 6 (2015) Structural Steelwork Specification for Building Construction

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1048 (2020) Fasteners - Hot Dip Galvanized Coatings

JIS B 1180 (2021) Hexagon Head Bolts and Hexagon Head

	Screws
JIS B 1181	(2014) Hexagon Nuts and Hexagon Thin Nuts
JIS B 1186	(2020) Sets of High Strength Hexagon Bolt, Hexagon Nut and Plain Washers for Friction Grip Joints
JIS B 1198	(2021) Headed Studs
JIS B 1220	(2021) Set of Anchor Bolts for Structures
JIS G 3106	(2020) Rolled Steels for Welded Structure (Amendment 1)
JIS B 1256	(2008) Plain Washers
JIS G 3136	(2012) Rolled Steel for Building Structures
JIS G 3192	(2014) Dimensions, Mass and Permissible Variations of Hot Rolled Steel Sections
JIS G 3193	(2008) Dimensions, Mass and Permissible Variations of Hot Rolled Steel Plates, Sheets, and Strips
JIS G 3444	(2022) Carbon Steel Tubes for General Structure
JIS G 3466	(2022) Carbon Steel Square and Rectangular Tubes for General Structure
JIS Z2305	(2013) Non-destructive Testing - Qualification and Certification of Personnel
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS K 5674	(2021) Lead-Free, Chromium-Free Anticorrosive Paints
JIS Z 3021	(2016) Welding and Allied Processes - Symbolic Representation
JIS Z 3060	(2015) Method of Ultrasonic Testing for Welds of Ferritic Steel

JAPANESE SOCIETY OF STEEL CONSTRUCTION (JSS)

JSS II 09	(2015) Sets of Torshear Type High Strength Bolt, Hexagonal Nut and Plain Washer for Structural Joints

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT-SS Chapter 7	(2019) Public Building Construction Standard Specifications - Ch.7 Steel Frame Work

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC SP 3	(1982; E 2004) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR Part 1926, Subpart R	Steel Erection

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Erection and Erection Bracing Drawings; G

SD-02 Shop Drawings

Fabrication Drawings Including Details of Connections; G

SD-03 Product Data

Shop Primer

Welding Electrodes and Rods

Direct Tension Indicator Washers

Non-Shrink Grout

Tension Control Bolts

SD-06 Test Reports

Weld Inspection Reports

Bolt Testing Reports

Embrittlement Test Reports

SD-07 Certificates

MLIT Structural Steel Fabricator Quality Certification

MLIT Structural Steel Erector Quality Certification

Welding Procedure Specifications (WPS)

1.3 MLIT QUALITY CERTIFICATION

Work must be fabricated by MLIT Structural Steel Fabricator, Category M. Submit MLIT Structural Steel Fabricator quality certification.

Work must be erected by MLIT Structural Steel Certified Erector. Submit MLIT Structural Steel erector quality certification.

1.4 QUALITY ASSURANCE

1.4.1 Preconstruction Submittals

1.4.1.1 Erection and Erection Bracing Drawings

Submit for record purposes. Indicate the sequence of erection, temporary shoring and bracing. The erection drawings must conform to JASS 6 and MLIT-SS Chapter 7.

1.4.2 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with JASS 6 and MLIT-SS Chapter 7. Fabrication drawings must not be reproductions of contract drawings. Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 or JIS Z 3021 standard welding symbols. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

1.4.3 Certifications

1.4.3.1 Welding Procedures and Qualifications

Prior to welding, submit certification for each welder stating the type of welding and positions qualified for, the code and procedure qualified under, date qualified, and the firm and individual certifying the qualification tests. If the qualification date of the welder or welding operator is more than 6 months old, the welding operator's qualification certificate must be accompanied by a current certificate by the welder attesting to the fact that he has been engaged in welding since the date of certification, with no break in welding service greater than 6 months.

Conform to all requirements specified in JASS 6, MLIT-SS Chapter 7, and AWS D1.8/D1.8M.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide the structural steel system, including shop primer and galvanizing, complete and ready for use. Provide structural steel systems including materials, installation, workmanship, fabrication, assembly, erection, inspection, quality control, and testing in accordance with JASS 6 and MLIT-SS Chapter 7, except as modified in this contract.

2.2 STEEL

2.2.1 Structural Steel

Wide flange and WT shapes, JIS G 3136, SN490B, 325 MPa. Angles and Channels JIS G 3106, SM400, 235 MPa. Plates, JIS G 3136, SN400B, 235 MPa and SN490B, 325 MPa, unless otherwise indicated on contract drawings.

2.2.2 Structural Steel Tubing

JIS G 3466, STKR400, 245MPa.

2.2.3 Steel Pipe

JIS G 3444, STK 400 and 235 MPa.

2.3 BOLTS, NUTS, AND WASHERS

Submit the certified manufacturer's mill reports which clearly show the applicable JIS mechanical and chemical requirements together with the actual test results for the supplied fasteners.

2.3.1 Common Grade Bolts

2.3.1.1 Bolts

JIS B 1180 with 420 MPa minimum tensile strength, hot dipped zinc coating in accordance with JIS B 1048. The bolt heads and the nuts of the supplied fasteners must be marked with the manufacturer's identification mark, the strength grade and type specified by JIS or JASS 6 specifications.

2.3.1.2 Nuts

Heavy hex style JIS B 1181, hot dipped zinc coating in accordance with JIS B 1048.

2.3.1.3 Washers

JIS B 1256, hot dipped zinc coating in accordance with JIS B 1048.

2.3.2 High-Strength Bolts

High strength bolts and nuts must be shipped together in the same shipping container. Fasteners indicated to be galvanized shall be tested by the supplier to show that the galvanized nut with the supplied lubricant provided may be rotated from the snug tight condition well in excess of the rotation required for pretentioned installation without stripping. The supplier shall supply nuts that have been lubricated and tested with the supplied bolts.

2.3.2.1 Bolts

JIS B 1186, Type F10T.

JIS B 1186, Type F8T hot dipped zinc coating at all exterior or exposed connections.

2.3.2.2 Nuts

JIS B 1186, F10. Provide hot dipped zinc coating with F8T bolts.

2.3.2.3 Washers

JIS B 1186, F35, plain carbon steel. Provide hot dipped zinc coating with F8T bolts.

2.3.3 Tension Control Bolts

MLIT approved JSS II 09, JIS B 1186 S10T twistoff style assemblies consisting of steel structural bolts with splined ends, heavy-hex carbon steel nuts, and hardened carbon steel washers. Assembly finish must be plain. Submit product data for tension control bolts.

2.3.4 Foundation Anchorage

2.3.4.1 Anchor Rods

JIS B 1220 ABR 400 hot dipped galvanized.

2.3.4.2 Anchor Nuts

Hexagon nuts JIS B 1220 ABR 400 hot dipped galvanized.

2.3.4.3 Anchor Washers

JIS B 1220 ABR 400 hot dipped galvanized.

2.3.4.4 Anchor Plate Washers

JIS B 1220 ABR 400 hot dipped galvanized.

2.4 STRUCTURAL STEEL ACCESSORIES

2.4.1 Welding Electrodes and Rods

AWS D1.1/D1.1M or JASS 6 and AWS D1.8/D1.8M. Submit product data for welding electrodes and rods.

2.4.2 Non-Shrink Grout

Packaged dry, hydraulic cement non-shrink grout, that is non-metallic, non-corrosive, non-bleed, with the following performance requirements when prepared using the highest water-to-solids ratio, maximum flow, or most fluid consistency at 23.0 plus/minus 2.0 degrees C:

a. Minimum compressive strengths: 7.0 MPa at 1 day; 17.0 MPa at 3 day; 24.0 MPa at 7 day; and 34.0 MPa at 28 day.

b. Early height change (maximum percent at time of final setting): + 4.0 percent.

c. Height change of moist cured hardened grout: 0.0 to + 0.3 percent at 1-day, 3-day, 14-day and 28-day.

2.4.3 Welded Shear Stud Connectors

JIS B 1198, 450MPa minimum ultimate tensile strength and 350MPa minimum yield strength.

2.5 GALVANIZING

JIS B 1048 for threaded parts or JIS H 8641 for structural steel members, as applicable, unless specified otherwise galvanize after fabrication where practicable.

2.6 FABRICATION

Fabrication must be in accordance with the applicable provisions of AISC 325and JASS 6. Fabrication and assembly must be done in the shop to the greatest extent possible. Punch, subpunch and ream, or drill bolt holes perpendicular to the surface of the member.

Compression joints depending on contact bearing must have a surface roughness not in excess of 13 micrometer as determined by ASME B46.1, and ends must be square within the tolerances for milled ends specified in JIS G 3192, JIS G 3193, and JASS 6.

Shop splices of members between field splices will be permitted only where indicated on the Contract Drawings. Splices not indicated require the approval of the Contracting Officer.

Do not splice truss top and bottom chords except as shown on the contract drawings or approved by the Structural Engineer of Record. Provide chord splices at panel joints at approximately the third point of the span. The center of gravity lines of truss members must intersect at panel points unless otherwise approved by the Contracting Officer. When the center of gravity lines do not intersect at a panel point, make provisions for the stresses due to eccentricity. Camber of trusses must be 3 mm in 3.0 meters unless otherwise indicated.

2.6.1 Markings

Prior to erection, identify members by a painted erection mark. Connecting parts assembled in the shop for reaming holes in field connections must be match marked with scratch and notch marks. Do not locate erection markings on areas to be welded. Do not locate match markings in areas that will decrease member strength or cause stress concentrations. Affix embossed tags to hot-dipped galvanized members.

2.6.2 Shop Primer

Shop prime structural steel, JIS K 5674 lead Free, chromium Free, anticorrosive paint, except as modified herein, in accordance with SSPC PA 1. Do not prime steel surfaces embedded in concrete, galvanized surfaces, surfaces to receive sprayed-on fireproofing, surfaces to receive epoxy coatings, surfaces designed as part of a composite steel concrete section, slip critical surfaces of high strength bolted connections, or surfaces within 13 mm of the toe of the welds prior to welding (except surfaces on which metal decking and shear studs are to be welded). If flash rusting occurs, re-clean the surface prior to application of primer. Apply primer to a minimum dry film thickness of 0.05 mm. Submit shop primer product data.

Prior to assembly, prime surfaces which will be concealed or inaccessible after assembly. Do not apply primer in foggy or rainy weather; when the ambient temperature is below 7 degrees C or over 35 degrees C; or when the primer may be exposed to temperatures below 4 degrees C within 48 hours after application, unless approved otherwise by the Contracting Officer. Repair damaged primed surfaces with an additional coat of primer.

2.6.2.1 Cleaning

SSPC SP 6/NACE No.3, except steel exposed in spaces above ceilings, attic spaces, furred spaces, and chases that will be hidden to view in finished construction may be cleaned to SSPC SP 3 when recommended by the shop primer manufacturer. Maintain steel surfaces free from rust, dirt, oil, grease, and other contaminants through final assembly.

2.6.3 ~~Fireproofing and~~ Epoxy Coated Surfaces

Clean and prepare surfaces to receive ~~sprayed-on fireproofing or~~ epoxy coatings in accordance with the manufacturer's recommendations~~, and as specified in Section 07 81 00 SPRAY-APPLIED FIREPROOFING~~.

2.7 DRAINAGE HOLES

Drill adequate drainage holes to eliminate water traps. Hole diameter must be 13 mm and location indicated on the detail drawings. Hole size and locations must not affect the structural integrity.

PART 3 EXECUTION

3.1 ERECTION

a. Erection of structural steel must be in accordance with the applicable provisions of AISC 325, AISC 303 and 29 CFR Part 1926, Subpart R or MLIT-SS Chapter 7 and JASS 6.

After final positioning of steel members, provide full bearing under base plates and bearing plates using nonshrink grout. Place nonshrink grout in accordance with the manufacturer's instructions or MLIT-SS Chapter 7 and JASS 6.

3.1.1 STORAGE

Store the material out of contact with the ground in such manner and location as to minimize deterioration.

3.2 CONNECTIONS

Except as modified in this section, design connections indicated in accordance with AISC 360. Build connections into existing work. Do not tighten anchor bolts set in concrete with impact torque wrenches. Holes must not be cut or enlarged by burning. Bolts, nuts, and washers must be clean of dirt and rust, and lubricated immediately prior to installation.

3.2.1 Common Grade Bolts

Tighten JIS B 1180 bolts to a "snug tight" fit. "Snug tight" is the tightness that exists when plies in a joint are in firm contact. If firm contact of joint plies cannot be obtained with a few impacts of an impact wrench, or the full effort of a man using a spud wrench, contact the

Contracting Officer for further instructions.

3.2.2 High-Strength Bolts

Provide direct tension indicator washers in all JIS B 1186 bolted connections. Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

Fastener components shall be protected from dirt and moisture in closed containers at the site of the installation. Fastener components that are not incorporated into the work shall be returned to protected storage at the end of the work shift.

3.2.3 Tension Control Bolts

Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

3.3 GAS CUTTING

Use of gas-cutting torch in the field for correcting fabrication errors is not permitted on any major member in the structural framing. Use of a gas cutting torch will be permitted on minor members not under stress only after approval has been obtained from the Contracting Officer.

3.4 WELDING

Welding must be in accordance with JASS 6 and AWS D1.8/D1.8M. Grind exposed welds smooth as indicated. Provide JASS 6 qualified welders, welding operators, and tackers.

Develop and submit the Welding Procedure Specifications (WPS) for all welding, including welding done using prequalified procedures. Submit for approval all WPS, whether prequalified or qualified by testing.

3.4.1 Removal of Temporary Welds, Run-Off Plates, and Backing Strips

Remove backing strips from bottom flange of moment connections, backgouge the root pass to sound weld metal and reinforce with a 8 mm fillet weld minimum.

3.5 SHOP PRIMER REPAIR

Repair shop primer in accordance with the paint manufacturer's recommendation for surfaces damaged by handling, transporting, cutting, welding, or bolting.

3.5.1 Field Priming

Field prime steel exposed to the weather, or located in building areas without HVAC for control of relative humidity. After erection, the field bolt heads and nuts, field welds, and any abrasions in the shop coat must be cleaned and primed with paint of the same quality as that used for the shop coat.

3.6 GALVANIZING REPAIR

Repair damage to galvanized coatings using JASS 6 zinc rich paint for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces to which repair paint has been applied.

3.7 FIELD QUALITY CONTROL

Perform field tests, and provide labor, equipment, and incidentals required for testing, except that electric power for field tests will be furnished as set forth in Division 1. Notify the Contracting Officer in writing of defective welds, bolts, nuts, and washers within 7 working days of the date of the inspection.

3.7.1 Welds

3.7.1.1 Visual Inspection

Perform in accordance with JASS 6. Furnish the services of certified welding inspectors for fabrication and erection inspection and testing and verification inspections. A Certified Welding Inspector must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Submit certificates indicating that certified welding inspectors meet the requirements of Japanese Welding Engineering Society (JWES) and JASS 6.

Inspect proper preparation, size, gaging location, and acceptability of all welds; identification marking; operation and current characteristics of welding sets in use.

3.7.1.2 Nondestructive Testing

Nondestructive testing must be in accordance with JASS 6 and JIS Z 3060 and AWS D1.8/D1.8M. Ultrasonic testing must be performed in accordance with Table or 6.3 of AWS D1.1/D1.1M. Test locations must be selected by the Contracting Officer. All personnel performing NDT must be certified in accordance with JIS Z2305 in the method of testing being performed. Submit certificates showing compliance with JIS Z2305 for all NDT technicians. If more than 20 percent of welds made by a welder contain defects identified by testing, then all groove welds made by that welder must be tested by ultrasonic testing, and all fillet welds made by that welder must be inspected by magnetic particle testing (MT) or dye penetrant testing (PT) as approved by the Contracting Officer. When groove welds made by an individual welder are required to be tested, magnetic particle or dye penetrant testing may be used only in areas inaccessible to ultrasonic testing. Retest all repaired areas. Submit weld inspection reports.

Testing frequency: Provide the following types and number of tests:

Test Type	Number of Tests
Ultrasonic	100 percent of CJP Welds
Magnetic Particle	50 percent of PJP and Fillet Welds

Test Type	Number of Tests
Dye Penetrant	50 percent of PJP and Fillet Welds

3.7.2 High-Strength Bolts

3.7.2.1 Testing Bolt, Nut, and Washer Assemblies

Test a minimum of 3 bolt, nut, and washer assemblies from each mill certificate batch in a tension measuring device at the job site prior to the beginning of bolting start-up. Demonstrate that the bolts and nuts, when used together, can develop tension not less than the provisions specified in AISC 360 or applicable JIS, depending on bolt size and grade. The bolt tension must be developed by tightening the nut. A representative of the manufacturer or supplier must be present to ensure that the fasteners are properly used, and to demonstrate that the fastener assemblies supplied satisfy the specified requirements. Submit bolt testing reports.

3.7.2.2 Inspection

Inspection procedures must be in accordance with JASS 6 and MLIT-SS Chapter 7. As a minimum, high-strength bolting inspection tasks shall be in accordance with Section 01 45 35 SPECIAL INSPECTION. Confirm and report to the Contracting Officer that the materials meet the project specification and that they are properly stored. Confirm that the faying surfaces have been properly prepared before the connections are assembled. Observe the specified job site testing and calibration, and confirm that the procedure to be used provides the required tension. Monitor the work to ensure the testing procedures are routinely followed on joints that are specified to be fully tensioned.

Inspect calibration of torque wrenches for high-strength bolts.

3.7.2.3 Testing

The Government has the option to perform nondestructive tests on 5 percent of the installed bolts to verify compliance with pre-load bolt tension requirements. Provide the required access for the Government to perform the tests. The nondestructive testing will be done in-place using an ultrasonic measuring device or any other device capable of determining in-place pre-load bolt tension. The test locations must be selected by the Contracting Officer. If more than 10 percent of the bolts tested contain defects identified by testing, then all bolts used from the batch from which the tested bolts were taken, must be tested at the Contractor's expense. Retest new bolts after installation at the Contractor's expense.

3.7.3 Testing for Embrittlement

ASTM A143/A143M for steel products hot-dip galvanized after fabrication. Submit embrittlement test reports.

3.7.4 Inspection and Testing of Steel Stud Welding

Perform verification inspection and testing of steel stud welding conforming to the requirements of JASS 6, Stud Welding Clause. The Contracting Officer will serve as the verification inspector. Bend test studs that do not show a full 360 degree weld flash or have been repaired by welding as required by JASS 6, Stud Welding Clause. Studs that crack under testing in the weld, base metal or shank will be rejected and replaced by the Contractor at no additional cost.

-- End of Section --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)

JAPANESE ARCHITECTURAL STANDARD SPECIFICAITONS (JASS)

JASS 5	(2015) Reinforced Concrete Work
JASS 6	(2015) Structural Steelwork Specification for Building Construction

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1111	(2017) Cross Recessed Machine Screws
JIS B 1112	(1995) Cross Recessed Head Wood Screws
JIS B 1180	(2014) Hexagon Head Bolts and Hexagon Head Screws
JIS B 1186	(2013) Sets of High Strength Hexagon Bolt, Hexagon Nut and Plain Washers for Friction Grip Joints
JIS B 1198	(2011) Headed Studs
JIS B 1220	(2015) Set of Anchor Bolt for Structures
JIS B 1250	(2008) Plain Washers for General Bolts, Machine Screws and Nuts - Overall System
JIS B 1251	(2018) Spring Lock Washers
JIS G 3101	(2017) Rolled Steels for General Structure (Amendment 1)
JIS G 3302	(2019) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3312	(2019) Prepainted Hot-Dip Zinc-Coated Steel Sheet and Strip
JIS G 3317	(2019) Hot-Dip Zinc-5 Percent Aluminum Allot-Coated Steel Sheet and Strip

JIS G 3323	(2019) Hot-dip Zinc-Aluminium-Magnesium Alloy-Coated Steel Sheet and Strip
JIS G 3444	(2016) Carbon Steel Tubes for General Structure
JIS G 3466	(2015) Carbon Steel Square and Rectangular Tubes for General Structure
JIS G 3475	(2014) Carbon Steel Tubes for Building Structure
JIS G 5702	(1998) Blackheart Malleable Iron Castings
JIS H 8641	(2007) Hot Dip Galvanized Coatings
JIS K 5553	(2006) Thick Film Zinc Rich Paint
JIS Z 0310	(2016) Abrasive Blast Cleaning Methods for Surface Preparation
JIS Z 3801	(2018) Standard Qualification Test and Acceptance Requirements for Manual Welding Technique
JIS Z 3821	(2018) Standard Qualification Test and Acceptance Requirements for Welding Technique of Stainless Steel
JIS Z 3841	(2018) Test Method and Criteria in Semi-Automatic Welding Technology Certification

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT-SS Chapter 7	(2019) Public Building Construction Standard Specifications - Ch.7 Steel Frame Work

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety~~ -- Safety~~ and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

~~Structural Steel Door Frames, Fabrication Drawings; G[, [_____]]~~

Cover Plates and Frames, Installation Drawings; G~~[, [_____]]~~

~~Expansion Joint Covers, Installation Drawings; G[, [_____]]~~

Floor Gratings, Installation Drawings;

~~Roof Walkways, Installation Drawings;~~

Bollards/Pipe Guards; G~~[, [_____]]~~

~~Wheel Guards, Installation Drawings; G[, [_____]]~~

~~Window[and Door] Guards, Installation Drawings;~~

Embedded Angles and Plates, Installation Drawings; G~~[, [_____]]~~

~~Roof Hatches, Installation Drawings; G[, [_____]]~~

SD-03 Product Data

Corner Guards

Cover Plates and Frames;

~~Expansion Joint Covers~~

Floor Gratings;

~~Roof Walkways;~~

~~Structural Steel Door Frames;~~

~~Wheel Guards~~

~~Window[and Door] Guards;~~

~~Roof Hatches;~~

~~SD-04 Samples~~

~~Expansion Joint Covers~~

~~SD-07 Certificates~~

~~[Certified Mill Test Reports for Chemistry and Mechanical Properties; G[, [_____]]~~
~~]~~

1.3 QUALIFICATION OF WELDERS

Qualify welders in accordance with JIS Z 3801, JIS Z 3821, or JIS Z 3841. Use procedures, materials, and equipment of the type required for the work.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in MLIT-SS Chapter 7 and/or JASS 6.

1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the building construction.

PART 2 PRODUCTS

2.1 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied.~~[Submit the manufacturer's certified mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied materials.]~~

2.1.1 Structural Carbon Steel

Provide in accordance with JIS G 3101, SS 400, 235 MPa.

2.1.2 Structural Tubing

Provide in accordance with JIS G 3466, SKTR 400, 245 MPa.

2.1.3 Steel Pipe

Provide in accordance with JIS G 3444 STK 400 and 235 MPa or JIS G 3475, STKN400B and 235 MPa.

2.1.4 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with JIS G 5702.

2.1.5 Gratings

~~a. Provide gray cast iron in accordance with JIS G 5501, FC300.~~

~~b~~a. Provide metal plank grating, non-slip requirement, ~~[aluminum in accordance with KS651-T6 or KD610-T6[DM1]] [~~steel in accordance with JIS G 3302 Z27~~]~~.

2.1.6 Floor Plates, Patterned

Provide steel plate not less than 1.9 mm.

2.1.7 Anchor Bolts

Provide in accordance with JIS B 1220. Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

2.1.7.1 ~~[~~Expansion Anchors~~] [Sleeve Anchors] [Adhesive Anchors]~~

Provide ~~[_____]~~16 mm diameter ~~[~~expansion anchors~~] [sleeve anchors] [adhesive anchors]~~. Minimum ~~[concrete] [~~masonry~~]~~ embedment of ~~[_____]~~100 mm. Design values listed are as tested in accordance with JASS 5.

a. Provide minimum ~~[ultimate] [allowable]~~ pullout value of ~~[_____]~~18 kN. Calculate pullout capacity according to JASS 5.

b. Provide minimum ~~[ultimate] [allowable]~~ shear value of ~~[_____]~~18 kN. Calculate shear capacity according to JASS 5.

2.1.7.2 Lag Screws and Bolts

Provide in accordance with JIS lag screws, type and grade best suited for the purpose.

2.1.7.3 Toggle Bolts

Provide in accordance with ASME B18.2.1.

2.1.7.4 Bolts, Nuts, Studs and Rivets

Provide in accordance with JIS B 1180, JIS B 1186, or JIS B 1220.

2.1.7.5 Screws

Provide in accordance with JIS B 1111 and JIS B 1112.

2.1.7.6 Washers

Provide plain washers in accordance with JIS B 1250. Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with JIS B 1251.

2.1.7.7 Welded Headed Shear Studs

Provide in accordance with JIS B 1198.

~~2.1.8 Aluminum Alloy Products~~

~~Provide in accordance with JIS H 4040 for extrusions and JIS H 5202 or JIS H 5202 for castings. Provide aluminum extrusions at least 3 mm thick and aluminum plate or sheet at least 1.3 mm thick.~~

2.2 FABRICATION FINISHES

2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with JIS G 3302, JIS G 3312, JIS G 3317, JIS G 3323 and/or JIS H 8641 Z27.

2.2.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

~~[~~2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint in accordance with JASS 6 12.4 and JIS K 5553 or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat, with a torch, surfaces to which stick or paste material will be applied. Heat to a temperature sufficient to melt the metals in the stick or paste. Spread molten material uniformly over surfaces to be coated and wipe off excess material.

~~]~~2.2.4 Shop Cleaning and Painting

2.2.4.1 Surface Preparation

Blast clean surfaces in accordance with JIS Z 0310. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned by means of power tools. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

2.2.4.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions. ~~[~~On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of 0.03 mm. Tint additional prime coat with a small amount of tinting pigment.~~]~~

2.2.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

~~2.2.6 Aluminum Surfaces~~

~~2.2.6.1 Surface Condition~~

~~Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.~~

~~2.2.6.2 Aluminum Finishes~~

~~Unexposed sheet, plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, JIS H 8601, JIS H 8602 or JIS H 4001. Unless otherwise specified, provide all other aluminum items with a[standard mill finish][hand sanded or machine finish to a 240 grit][anodized finish]. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or architectural Class I type finish, min. 0.7 mil for~~

~~items used in exterior locations. Provide in accordance with JIS H 8601, JIS H 8602 or JIS H 4001. Provide a polished satin finish on items to be anodized.~~

2.3 CORNER GUARDS

For jambs and sills of openings and edges of platforms provide steel shapes and plates anchored in masonry or concrete with welded steel straps or end-weld stud anchors. Form corner guards for use with glazed or ceramic tile finish on walls with 1.6 mm thick corrosion-resisting steel with~~[polished][or][~~ satin~~]~~ finish, extend 1.5 m above the top of cove base or to the top of the wainscot, whichever is less, and securely anchor to the supporting wall. Provide Hot-dipped ~~[~~galvanized~~][_____]~~ corner guards on exterior.~~[Provide interior corner guards as indicated in Section 10 26 00 WALL AND DOOR PROTECTION.]~~

2.4 COVER PLATES AND FRAMES

Fabricate cover plates of ~~[6][_____]~~5 mm thick rolled steel weighing not more than 45 kg per plate with a ~~[~~selected raised pattern nonslip top surface~~][slip-resistant, carbon steel in accordance with JIS G 3101. Provide aluminum oxide or silicon carbide on wearing surfaces]~~. Provide ~~[~~ galvanized~~][shop painted]~~ plate. Reinforce to sustain a live load of ~~[_____]~~as indicated on the drawings MPa. Provide structural steel shapes and plates for frames, ~~[with bent steel bars or headed anchors welded to frame for anchoring to concrete][~~securely fastened to the structure as indicated~~]~~. Miter and weld all corners. Butt joint straight runs. Allow for expansion on straight runs over 4500 mm.~~[~~ Provide holes for lifting tools.~~][Provide flush drop handles for removal where indicated; form from 6 mm round stock.][Provide holes and openings with 13 mm clearance for pipes and equipment.]~~ Remove sharp edges and burrs from cover plates and exposed edges of frames. Weld all connections and grind top surface smooth. Weld bar stops every 152 mm. Provide 3 mm clearance at edges and between cover plates.

~~2.5 EXPANSION JOINT COVERS~~

~~Provide expansion joint covers constructed of extruded aluminum with anodized satin aluminum finish for walls and ceilings and standard mill finish for floor covers and exterior covers. Furnish plates, backup angles, expansion filler strips and anchors as indicated.[Provide a [_____]-hour fire-rating for expansion joints.]~~

2.5 FLOOR GRATINGS~~ AND ROOF WALKWAYS~~

Design ~~[~~steel~~] [aluminum]~~ grating in accordance with manufacturer's charts for plank grating. ~~[~~Galvanize steel floor gratings.~~]~~

a. Design floor gratings to support a stress live load of ~~[_____]~~as indicated on the drawings MPa for the spans indicated, with maximum deflection of L/240.

~~[~~ b. In accordance with the manufacturer's standard for trim ~~[~~unless otherwise indicated~~]~~. Design tops of bearing bars, cross or intermediate bars to be in the same plane and to match grating finish.

~~][~~b. Band ends of gratings with bars of the same or greater thickness than the metal used for grating. Weld banding bars to bearing bars or channels at least every fourth bar or channel and in every corner.

Tack weld intervening bars or channels. Band diagonal or round cuts by welding bars of the same or greater thickness as the grating and in accordance with the manufacturer's standard for trim ~~[~~unless otherwise indicated~~]~~.

~~][c. [Attach gratings to structural members with welded-on anchors.][Anchor gratings to structural members with bolts, toggle bolts, or expansion shields and bolts.][Attach grating in accordance with manufacturer's roof attachment system.]~~

~~]~~

~~d~~c. Provide slip resistant surface finishes.

~~[e. Rooftop walkway: Minimum 600 mm wide, 1.8 mm, JIS G 3302, Z27 steel with slip resistant surface. Furnish all brackets, connectors and other accessories. Support at minimum 1500 mm intervals on hard rubber pads in accordance with manufacturer's instructions.~~

~~]~~2.6 BOLLARDS/PIPE GUARDS

Provide ~~[_____] mm [~~galvanized~~] [prime coated] [~~standard~~] [extra strong]~~ weight steel pipe in accordance with JIS G 3444, STK 400 and 235 MPa or JIS G 3475, STKN400B and 235 MPa. Anchor posts in concrete ~~[~~as indicated~~]~~ and fill solidly with concrete with minimum compressive strength of 17 MPa.

~~2.7 DOWNSPOUT TERMINATIONS~~

~~Provide [102 x 102 mm], [102 x 152 mm] [and] [or] [152 x 152 mm] [_____] aluminum downspout tile adapter with [mill] [manufacturer's standard powder coated] finish. Units shall have all seams welded.~~

~~Provide [nickel bronze] [polished bronze] [chrome plated] cast downspout nozzle and flange.~~

~~Provide [100 x 76 mm], [125 x 100 mm] [and] [or] [100 mm diameter] [_____] [cast iron] [galvanized cast iron] downspout boot with cleanout access and manufacturer's standard cast iron strap.~~

2.7 MISCELLANEOUS PLATES AND SHAPES

Provide items that do not form a part of the structural steel framework, such as lintels, sill angles, ~~[~~support framing for ceiling-mounted toilet partitions,~~]~~ miscellaneous mountings and frames. Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions ~~[as indicated and]~~ as required to support wall loads over openings. Provide with connections and ~~[~~fasteners~~] [welds]~~. Construct to have at least ~~[[_____] mm] [~~200 mm~~]~~ bearing on masonry at each end.

Provide angles and plates in accordance with JIS G 3101 SS400, 235 MPa, for embedment as indicated. Galvanize embedded items exposed to the elements in accordance with JIS H 8641.

~~2.8 SAFETY CHAINS~~

~~Construct safety chains of galvanized steel, straight link type, minimum 5 mm diameter, with a minimum of twelve links per 300 mm, and snap hooks on each end. Provide boat type snap hooks. Provide galvanized 10 mm bolt with 20 mm eye diameter for attachment of chain, anchored as indicated. Supply two chains, 100 mm longer than the anchorage spacing, for each guarded area.~~

~~2.9 SECURITY GRILLES~~

~~Fabricate of channel frames with not less than two masonry anchors at each jamb and 12 mm hardened steel bars spaced not over 100 mm both ways and welded to frame. Provide 18 by 16 mesh screen and two layers of 6 mm hardware cloth clamped to frame.~~

~~2.10 STEEL PLATE WAINSCOTS FOR CONCRETE OR MASONRY COLUMNS~~

~~Shop bend to radius for round columns and at right angles for square and rectangular columns with slight 6 mm radius on corners, with no horizontal joints and not more than 2 vertical joints single strapped and butt welded with a thickness of [_____].~~

~~2.11 STRUCTURAL STEEL DOOR FRAMES~~

~~[a. Provide frames as indicated. Unless otherwise indicated, construct frames of structural shapes, or shape and plate composite, to form a full depth channel shape with at least 40 mm outstanding legs. For single swing doors, provide continuous 16 by 40 mm bar stock stops at head and jambs. For freight elevator hoistway entrance, include a non-skid metal sill. Provide extruded metal frames as required by the elevator manufacturer.~~

~~] b. Provide support where track, guides, hoods, hangers, operators, and other accessories are required.~~

~~c. Provide jamb anchors near top, bottom, and at not more than 600 mm intervals. Provide the bottom of each jamb member with a clip angle welded in place with two 12 mm diameter floor bolts for adjustment.~~

~~[d. Provide spreaders between bottoms of floor jamb members. When floor construction permits, spreaders may be left in place and concealed in the floor.~~
~~]~~
~~[Provide frames of rolled shapes as indicated. Miter and weld heads to jambs, or provide riveted clip angle connections concealed in the finished work. Provide frames for swinging doors with 16 by 40 mm solid bar stops secured to the frame by welding or by 6 mm diameter countersunk machine screws spaced not more than 300 mm on centers. Stiffen head openings greater than 900 mm as necessary to limit deflection to not more than 2 mm. Secure frames to masonry with zinc-coated metal anchors spaced not more than 750 mm on centers. Where necessary to engage the threads of machine screws for fastening hardware, back frames on inside faces with steel plates of suitable thickness. Tap frames and reinforcing plates as necessary for the installation of hardware and other work. Countersink rivets and screw heads where they will be exposed in the finished work. Grind welds smooth.~~

~~]2.12 WHEEL GUARDS~~

~~Provide wheel guards of hollow, heavy-duty type cast iron with shaped, [rounded][half round][three quarters round]top, at least 450 mm high, and designed to provide a minimum of 150 mm of protection.~~

~~[2.13 ROOF HATCHES (SCUTTLES)~~

~~Provide [aluminum][zinc-coated steel] sheets not less than 1.9 mm with 75~~

~~mm beaded flange, welded and ground at corners. Provide a minimum clear opening of 760 by 900 mm. Insulate cover and curb with 25 mm thick rigid fiberboard insulation, covered and protected by [aluminum sheet][zinc-coated steel liner] of not less than 0.45 mm. Provide with 300 mm high curb, formed with 75 mm mounting flanges with holes for securing to the roof deck.~~

~~]2.14 WINDOW[AND DOOR] GUARDS, DIAMOND-MESH TYPE~~

~~Provide diamond-mesh window[and door] guards constructed of woven steel wire [or expanded metal]framed with hot-rolled or cold-formed structural steel shapes. Provide woven wire panels of 3.3 mm, 40 mm mesh secured through weaving bar to 25 by 12 by 3 mm thick channel frame.[Provide expanded metal panels in accordance with ASTM F1267.] Miter and weld corners of frames.[Mount window[and door] guards on interior of window[and door] frame with not less than two tamperproof hinged butts mounted on wood jambs with 6 mm lag bolts, to masonry jamb with toggle bolts, or welded to metal jambs.][Mount window[and door] guards on exterior of window frame with not less than two tamperproof hinged butts mounted on 25 by 12 by 3 mm jamb channel attached as indicated to 50 by 6 mm plate anchored to wood jamb with 6 mm lag bolts; to masonry jamb with toggle bolts, or to concrete jambs and solid masonry jambs with expansion shields and bolts.] Provide one additional butt for each 900 mm internal length of guard over 1500 mm. Provide one tamperproof hasp and padlock, with access from the interior, for each butt used and installed on the jamb opposite to that hinged.[Provide galvanized guards and accessories.]~~

~~2.15 WINDOW[AND DOOR] GUARDS~~

~~Provide woven wire window[and door] guards of size as necessary to completely fill opening. Construct guards with 10 mm round rod frame and 40 mm diamond-mesh of No. 10 U.S. Gage 3.4 mm wire. Provide all materials with zinc coating. Provide a minimum of three hinge side clips on one side and two lock ring hasps on the opposite side.~~

~~2.16 CLEANOUT DOORS~~

~~Provide [galvanized][cast iron]cleanout doors with frames, sized to match flues unless otherwise indicated. Provide continuous flange and anchors for securing frames to masonry. Provide smokeproof, hinged doors with[lockable] fastening devices to hold doors closed[and secured].~~

~~2.17 COAL HOPPER DOORS~~

~~Provide coal hopper doors of [galvanized][_____] steel plates and shapes. Provide complete assemblies including frames, stops, wall boxes, hinges, and hasp or lock-type latches. Weld joints and attachments.~~

~~2.18 GUY CABLES~~

~~Provide guy cables as pre-stretched, galvanized wire rope of sizes indicated. Provide wire rope high strength grade. Guys must have a factory attached clevis top-end fitting, a factory attached open-bridge strand socket bottom-end fitting, and must be complete with oval eye, threaded anchor rods. Provide hot-dip galvanized fittings and accessories.~~

~~2.19 WINDOW SUB-SILL~~

~~Provide window sub-sill of extruded aluminum alloy, standard mill finish,~~

~~of size(s) and design(s) indicated. Provide a minimum of two anchors per window section for securing to mortar joints of masonry sill course. Provide sills with protective coating for shipment, of two coats of a clear, colorless, methacrylate lacquer applied to all surfaces of the sills.~~

~~2.20 WINDOW WELLS~~

~~Provide window wells in a minimum 1.5 mm, corrugated sheet steel, hot-dip galvanized after fabrication, with top edge of window well walls with a 19 mm bead or rolled top. Provide windows wells with radiused corners and of sizes that overlap each window by a minimum of 75 mm on each side. Provide removable covers, hot-dipped galvanized after fabrication, consisting of steel bar grate, with bars spaced at not more than 50 mm centers and welded to 25 by 6 mm frame. Frames must fit into, and rest on top edge of, window wells.~~

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact except where tack welding is permitted. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersuck rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), through bolts for masonry, JIS B 1198 welded headed shear studs,machine and carriage bolts for steel, through bolts, lag bolts, and screws for wood. Do not use wood plugs. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise.~~[~~ Test a minimum of 2 bolt, nut, and washer assemblies from each certified mill batch in a tension measuring device at the job site prior to the beginning of bolting start-up.~~]~~

3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete or masonry. Provide built-in metal work in ample time for securing in place as the work progresses.

3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with JASS 6. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation. Provide welded headed shear studs in accordance with JASS 6, Clause 7, except as otherwise specified. Provide in accordance with the safety requirements of EM 385-1-1.

3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect with asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

3.7 PREPARATION

3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 15 degrees C above the dew point of the surrounding air, or when surface temperature is below 7 degrees C or over 35 degrees C, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

3.8 EXPANSION JOINT COVERS

Provide in accordance with manufacturer's written instructions~~[~~ and with seismic requirements indicated~~]~~. Verify installation allows specified movement prior to completion of work

3.9 COVER PLATES AND FRAMES

Provide tops of cover plates and frames flush with finished surface. Test for trip hazards and adjust for any encountered lippage.

~~3.10 WHEEL GUARDS~~

~~Anchor guards to concrete or masonry in accordance with manufacturer's instructions. Fill hollow cores solid with concrete with minimum compressive strength of 17 MPa.~~

~~[3.11 ROOF HATCH (SCUTTLES)~~

~~Construction and accessories as follows:~~

~~a. Provide insulated cover and curb with mounting flanges for securing to roof deck. Provide curbs with integral metal cap flashing of the same~~

~~gage and metal as the curb, fully welded and ground at corners for weather tightness.~~

~~b. Provide hatches completely assembled, with pintle hinges, compression spring operators enclosed in telescopic tubes, positive snap latches with turn handles on inside and outside, and neoprene draft seals. Provide fasteners for padlocking from the inside. Provide covers with automatic hold-open arms complete with grip handle to permit one hand release. Cover action must be smooth through its entire range of motion with an operating pressure of approximately 130 N.~~

~~]3.12 DOOR GUARD FRAME~~

~~Mount door guard frames over glazed openings using 6 mm lag bolts on the interiors of wood doors or tamperproof through bolts on the interiors of metal doors.~~

3.10 INSTALLATION OF BOLLARDS/PIPE GUARDS

Set bollards/pipe guards vertically in concrete piers. Fill hollow cores with concrete having a minimum compressive strength of 21 MPa.

~~3.11 INSTALLATION OF DOWNSPOUT TERMINATIONS~~

~~Secure downspouts terminations to downspouts and substrate per manufacturer's instructions.~~

~~3.12 MOUNTING OF SAFETY CHAINS~~

~~Provide safety chains where indicated. Mount the top chain 1050 mm[_____] above the [floor][ground] and mount the lower chain 600 mm [_____] above the [floor][ground].~~

~~3.13 STRUCTURAL STEEL DOOR FRAMES~~

~~Secure door frames to the floor slab by means of angle clips and expansion bolts. Provide any necessary reinforcements and drill and tap frames as required for hardware. Clean metal shavings from finished surfaces at the end of each work day.~~

~~For freight elevator hoistway entrances, include a non-skid metal sill installed in accordance with the elevator manufacturer's written installation instructions.~~

~~3.14 INSTALLATION OF WHEEL GUARDS~~

~~Fill wheel guards with concrete and anchor to slab in accordance with manufacturer's recommendations.~~

~~3.15 BAR-GRILLE WINDOW GUARDS~~

~~Securely anchor bar-grille window guards to masonry with 13 mm diameter prison-type screws or bolts and expansion shields, or other type of fastenings if the ends of such fastenings are welded to the adjoining metal grilles or otherwise made tamperproof in manner as approved by the Contracting Officer. Spanner-head screws or bolts are not considered prison-type fasteners.~~

~~3.16 DIAMOND MESH WINDOW [AND DOOR]GUARDS~~

~~Provide diamond mesh window guards on [interior window frames with not less than two tamperproof hinged butts mounted on wood jambs.][exterior of window frames with not less than two tamperproof hinged butts mounted on 25 by 300 by 3 mm jamb channel attached to 50 by 6 mm plate anchored][to wood jambs with 6 mm lag bolt,] to masonry jamb with toggle bolts[, or to concrete jambs and solid masonry jambs with expansion shields and bolts]. Provide one additional butt for each 900 mm internal length of guard over 1500 mm. Install hasp and padlock jamb opposite the hinged side.~~

~~3.17 INSTALLATION OF WINDOW WELLS~~

~~Provide window wells with walls securely anchored to foundation surface. Excavate the area within the well to the bottom of the well and cover with a 100 mm thick layer of coarse gravel or crushed rock.~~

3.11 INSTALLATION MISCELLANEOUS PLATES AND SHAPES

Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions~~[~~ as indicated and~~]~~ as required to support wall loads over openings. Provide with connections and ~~[~~fasteners~~][welds]~~. Construct to have at least 200 mm bearing on masonry at each end.

-- End of Section --

SECTION 05 52 00

METAL RAILINGS
02/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 898-1	(2013) Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel — Part 1: Bolts, Screws and Studs with Specified Property Classes — Coarse Thread and Fine Pitch Thread

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1111	(2017) Cross Recessed Machine Screws
JIS B 1112	(1995) Cross Recessed Head Wood Screws
JIS B 1251	(2018) Spring Lock Washers
JIS G 3459	(2017) Stainless Steel Pipes (Amendment 1)
JIS G 4304	(2012) Hot-Rolled Stainless Steel Plate, Sheet and Strip
JIS G 4305	(2008) Cold-rolled Stainless Steel Plate, Sheet and Strip
JIS H 4100	(2015) Aluminum and Aluminum Alloy Extruded Profiles
JIS H 5202	(2010) Aluminum Alloy Castings
JIS H 8641	(2007) Hot Dip Galvanized Coatings
JIS Z 3420	(2003) Specification and Approval of Welding Procedures for Metallic Materials - General Rules
JIS Z 3801	(2018) Standard Qualification Test and Acceptance Requirements for Manual Welding Technique
JIS Z 3841	(2018) Test Method and Criteria in Semi-Automatic Welding Technology Certification

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS B 1048	(2020) Fasteners - Hot Dip Galvanized Coatings
JIS B 1189	(2015) Hexagon Bolt with Flange
JIS B 1256	(2008) Plain Washers
JIS Z 3410	(2013) Welding Coordination - Tasks and Responsibilities
JIS Z 3604	(2016) Inert Gas Arc Welding Standard for Aluminum

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT Chapter 14	Metal Work

1.2 ADMINISTRATIVE REQUIREMENTS

1.2.1 Preinstallation Meetings

Within ~~[~~30~~]~~ ~~[_____]~~ days of contract award, submit fabrication drawings ~~[~~ to the Contracting Officer~~]~~ for the following items:

~~[a. Iron and steel hardware~~

~~][b. Steel shapes, plates, bars and strips~~

~~][c. Steel railings and handrails~~

~~][d~~a. Stainless Steel railings and handrails

~~][e~~b. Aluminum railings and handrails

~~]~~ ~~f~~c. Anchorage and fastening systems

Submit manufacturer's catalog data, including two copies of manufacturers specifications, load tables, dimension diagrams, and anchor details for the following items:

~~[a. Structural-steel plates, shapes, and bars~~

~~][b. Structural-steel tubing~~

~~][c. Cold-finished steel bars~~

~~][d. Hot-rolled carbon steel bars~~

~~][e. Cold-drawn steel tubing~~

~~][f~~a. Concrete inserts

~~][g. Masonry anchorage devices~~

~~][h. Protective coating~~

~~][i. Steel railings and handrails~~

~~][j~~b. Stainless Steel railings and handrails

~~][k~~c. Aluminum railings and handrails

~~]~~ ~~l~~d. Anchorage and fastening systems

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings; G~~[, [_____]]~~

~~Iron and Steel Hardware~~

~~Steel Shapes, Plates, Bars and Strips~~

SD-07 Certificates

Welding Procedures; G~~[, [_____]]~~

Welder Qualification; G~~[, [_____]]~~

SD-08 Manufacturer's Instructions

Installation Instructions

1.4 QUALITY CONTROL

1.4.1 Welding Procedures

~~[Section 05 05 23.16 STRUCTURAL WELDING applies to work specified in this section.~~

~~]~~ Submit results of welding procedures testing in accordance with JIS Z 3801 or JIS Z 3841 made in the presence of the Contracting Officer and by an approved testing laboratory at the Contractor's expense.

1.4.2 Welder Qualification

Submit certified welder qualification by tests in accordance with JIS Z 3801 or JIS Z 3841, or under an equivalent approved qualification test. In addition, perform tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, conduct an immediate retest of two test welds and ensure that each test weld passes. Failure in the immediate retest will require that the welder be retested after further practice or training and make a complete set of test welds.

PART 2 PRODUCTS

2.1 DESIGN

Design all handrails and guards to resist a concentrated load of~~[~~ 890 N~~][ ] [______]~~ in any direction at any point of the top of the rail or~~[~~ 730 N/m ~~][ ] [______]~~ applied horizontally to the top of the rail, whichever is more severe. Intermediate rails, balusters and panel fillers shall be design to resist a concentrated load of~~[~~ 220 N~~][ ]~~. MLIT Chapter 14, provide the same size rail and post. Provide pipe collars of the same material and finish as the handrail and posts.~~ [Provide series 300 stainless-steel pipe collars.]~~

~~In addition to the above loads exterior railings shall be also designed to withstand a wind load of [ ]N/m.~~

2.2 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning, treating, and applying surface finishes, including zinc coatings.

Provide railing and handrail detail plans and elevations at not less than 1 to 10 scale. Provide details of sections and connections at not less than 1 to 5 scale. Also detail setting drawings, diagrams, templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchors.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce adequate strength and durability in the finished product for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and straight sharp edges. Ensure that all exposed edges are eased to a radius of approximately 0.8 millimeter. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Weld corners and seams continuously and in accordance with the recommendations of JIS Z 3420 and JIS Z 3604. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form the exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use countersunk Phillips flathead screws or bolts.

Provide anchorage of the type indicated and coordinated with the supporting structure. Fabricate anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified to be fabricated from cold-finished or cold-rolled stock.

2.2.1 Aluminum Railings and Handrails

Fabrication: Provide fabrication jointing by one of the following methods:

a. Use flush-type rail fittings, welded and ground smooth with splice locks secured with 10 mm recessed-head set screws.

b. Ensure that mitered and welded joints made by fitting; post to top rail; intermediate rail to post; and corners, are groove welded and ground smooth. Where allowed by the Contracting Officer, provide butt splices reinforced by a tight-fitting dowel or sleeve not less than 150 mm in length. Tack-weld or epoxy-cement the dowel or sleeve to one side of the splice.

c. Assemble railings using slip-on aluminum-magnesium alloy fittings for joints. Fasten fittings to pipe or tube with 6 or 10 mm stainless-steel recessed-head setscrews. Provide assembled railings with fittings only at vertical supports or at rail terminations attached to walls. Provide expansion joints at the midpoint of panels. Provide a setscrew in only one side of the slip-on sleeve. Provide alloy fittings to conform to JIS H 5202.

~~[Provide removable railing sections as indicated. [Provide toe-boards and brackets where indicated, using flange castings as appropriate.]~~

~~]2.2.2 Steel Railings and Handrails~~

~~Fabricate joint posts, rail, and corners by one of the following methods:~~

~~a. Flush-type rail fittings of commercial standard, welded and ground smooth, with railing splice locks secured with 10 mm hexagonal-recessed-head setscrews.~~

~~b. Mitered and welded joints made by fitting post to top rail and intermediate rail to post, mitering corners, groove-welding joints, and grinding smooth. Butt railing splices and reinforce them by a tight-fitting interior sleeve not less than 150 mm long.~~

~~c. Railings may be bent at corners in lieu of jointing, provided that bends are made in suitable jigs and the pipe is not crushed.~~

~~[Provide removable sections as indicated.~~

~~][~~2.2.2 Stainless Steel Railings and Handrails

Provide stainless steel tubing, welded or seamless, conforming to JIS G 4304, JIS G 4305 or JIS G 3459 unless otherwise noted.

~~][~~2.2.3 Protective Coating

~~[Shop-prime the steelwork as indicated in accordance with Section 09 90 00 PAINTS AND COATINGS except the following:~~

~~a. steel surfaces encased in concrete~~

~~b. steel surfaces for welding~~

~~c. high-strength bolt-connected contact surfaces~~

~~d. crane rail surfaces~~

~~][~~Provide hot-dipped galvanized steelwork as indicated in accordance with JIS H 8641. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

~~]]~~2.3 COMPONENTS

~~[2.3.1 Structural Steel Plates, Shapes And Bars~~

~~Provide structural-size shapes and plates, except plates to be bent or cold-formed, conforming to JIS G 3101, unless otherwise noted.~~

~~Provide steel plates, to be bent or cold-formed, conforming to JIS G 3101, SS 400.~~

~~Provide steel bars and bar-size shapes conforming to JIS G 3101, unless otherwise noted.~~

~~][2.3.2 Structural-Steel Tubing~~

~~Provide structural-steel tubing, hot-formed, welded or seamless, conforming to JIS G 3466 unless otherwise noted.~~

~~][2.3.3 Hot-Rolled Carbon Steel Bars~~

~~Provide bars and bar-size shapes conforming to JIS G 3138, grade as selected by the fabricator.~~

~~][2.3.4 Cold-Finished Steel Bars~~

~~Provide cold-finished steel bars conforming to JIS G 4051, grade as selected by the fabricator.~~

~~][2.3.5 Cold-Drawn Steel Tubing~~

~~Provide tubing conforming to JIS G 3444, sunk-drawn, butt-welded, cold-finished, and stress-relieved.~~

~~][2.3.6 Steel Pipe~~

~~Provide pipe conforming to JIS G 3466 STK 400 and 235 MPa or JIS G 3475, STKN400B and 235 MPa; primed finish, unless galvanizing is required; standard weight (Schedule 40).~~

~~][2.3.7 Concrete Inserts~~

~~[Provide threaded-type concrete inserts consisting of galvanized ferrous castings, internally threaded to receive M20 diameter machine bolts; either malleable iron conforming to JIS G 5705 or cast steel conforming to JIS G 5101, SC 42, SC 46 or SC 49, hot-dip galvanized in accordance with JIS B 1048.~~

~~][Provide wedge-type concrete inserts consisting of galvanized box-type ferrous castings designed to accept M20 diameter bolts having special wedge-shaped heads, made of either malleable iron conforming to JIS G 5705 or cast steel conforming to JIS G 5101 and hot-dip galvanized in accordance with JIS B 1048.~~

~~][Provide carbon steel bolts having special wedge-shaped heads, nuts, washers, and shims, galvanized in accordance with JIS B 1048. Provide slotted-type concrete inserts consisting of a galvanized 3 millimeter thick pressed-steel plate conforming to JIS G 3101 SS41 or SB35, made of box-type welded construction with a slot designed to receive M20 diameter square-head bolt with knockout cover; and hot-dip galvanized in accordance with JIS H 8641.~~

~~]][~~2.3.1 Fasteners

Provide galvanized zinc-coated fasteners in accordance with JIS B 1048 used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of steel stair items.

~~[~~ Provide standard hexagon-head bolts, conforming to ISO 898-1.

~~][~~Provide square-head lag bolts conforming to JIS B 1189.

~~][~~Provide cadmium-plated steel machine screws conforming to JIS B 1111.

~~][~~Provide flat-head carbon steel wood screws conforming to JIS B 1112.

~~][~~Provide plain round, general-assembly-grade, carbon steel washers conforming toJIS B 1256.

~~][~~Provide helical spring, carbon steel lockwashers conforming to JIS B 1251.

~~]][2.3.2 Steel Railings and Handrails~~

~~2.3.2.1 Steel Handrails~~

~~Provide steel handrails, including inserts in concrete, [steel pipe conforming to JIS G 3466] [or] [structural tubing conforming to JIS G 3466]. Provide steel railings of [40] [50] mm nominal size,[hot-dip galvanized] [and] [shop-painted].~~

~~Provide kickplates between railing posts where indicated and consisting of 4 millimeter steel flat bars not less than 150 millimeter high. Secure kickplates as indicated.~~

~~[Galvanize exterior railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components. Provide black steel pipe for interior railings.~~

~~][Provide galvanized exterior and interior railings where indicated, including pipe, fittings, brackets, fasteners, and other ferrous metal components. Provide black steel pipe for interior railings not indicated as galvanized.~~

~~][Provide galvanized railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components.~~

~~]][~~2.3.2 Aluminum Railings and Handrails

Provide railings and handrails consisting of~~[[40] [50] ~~mm nominal schedule 40 pipe JIS H 4100~~]~~, ~~45 mm aluminum semi hollow tube with rounded corners JIS H 4040~~bottom rail, and extruded top rail as indicated. Provide ~~[mill-finish] [~~anodized~~]~~ aluminum ~~[_____] ~~color railings. Ensure that all fasteners are Series 300 stainless steel.

~~]2.3.3 Safety Chains [and Guardrails]~~

~~Provide safety chains of galvanized steel, straight-link type, 5 mm diameter, with at least 12 links per 300 mm, and with snap hooks on each end. Test safety chain in accordance with JIS F 2106. Provide snap hooks of boat type. Provide galvanized 10 mm bolt with 20 mm eye diameter for attachment of chain, anchored as indicated. Supply two chains, 100 mm longer than the anchorage spacing, for each guarded area. Locate [guardrails] safety chain where indicated. Mount the top chain [rail] 1050 mm [_____] above the [floor] [ground] and mount the lower chain [rail] 600 mm [_____] above the [floor] [ground].~~

PART 3 EXECUTION

3.1 PREPARATION

Adjust stair railings and handrails before securing in place in order to ensure proper matching at butting joints and correct alignment throughout their length. Space posts ~~not more than [2440 millimeter] [_____] on center~~as indicated, but not exceeding manufacturer's recommendations. Plumb posts in each direction. Secure posts and rail ends to building construction as follows:

~~[~~ a. Anchor posts in concrete by means of pipe sleeves set and anchored into concrete. Provide sleeves of galvanized, standard-weight, steel pipe, not less than 150 millimeter long, and having an inside diameter not less than 13 millimeter greater than the outside diameter of the inserted pipe post. Provide steel plate closure secured to the bottom of the sleeve, with closure width and length not less than 25 millimeter greater than the outside diameter of the sleeve. After posts have been inserted into sleeves, fill the annular space between the post and sleeve with non-shrink grout or a quick-setting hydraulic cement. Cover anchorage joint with a round steel flange welded to the post.

~~][b. Anchor posts to steel with oval steel flanges, angle type or floor type as required by conditions, welded to posts and bolted to the steel supporting members.~~

~~][c~~b. Anchor rail ends into concrete ~~and masonry ~~with round steel flanges welded to rail ends and anchored into the wall construction with lead expansion shields and bolts.

~~][d~~c. Anchor rail ends to steel with oval or round steel flanges welded to tail ends and bolted to the structural-steel members.

~~]~~ Secure handrails to walls by means of wall brackets and wall return fitting at handrail ends. Provide brackets of malleable iron castings, with not less than 75 millimeter projection from the finished wall surface to the center of the pipe, drilled to receive one M10 bolt. Locate brackets not more than 1525 millimeter on center. Provide wall return fittings of cast iron castings, flush type, with the same projection as that specified for wall brackets. Secure wall brackets and wall return fittings to building construction as follows:

~~[~~ a. For concrete ~~and solid masonry~~ anchorage, use bolt anchor expansion shields and lag bolts.

~~][~~b. For hollow masonry and stud partition anchorage, use toggle bolts having square heads.

~~]~~ Install toe boards and brackets where indicated. Make splices, where required, at expansion joints. Install removable sections as indicated.

3.2 INSTALLATION

Submit manufacturer's installation instructions for the following products to be used in the fabrication of ~~[~~steel~~] [_____]~~, aluminum ~~[stair~~ railing~~]~~s ~~[~~and~~]~~ ~~[~~hand rail work~~]~~:

~~[a. Structural-steel plates, shapes, and bars~~

~~][b. Structural-steel tubing~~

~~][c. Cold-finished steel bars~~

~~][d. Hot-rolled carbon steel bars~~

~~][e. Cold-drawn steel tubing~~

~~][f. Protective coating~~

~~][g. Masonry anchorage devices~~

~~][h~~a. Steel railings and handrails

~~][i~~b. Aluminum railings and handrails

~~][j~~c. Anchorage and fastening systems

~~]~~ Provide complete, detailed fabrication and installation drawings for all ~~iron and steel~~ hardware, and for all ~~steel~~ shapes, plates, bars, and strips used in accordance with the design specifications cited in this section.

~~[~~3.2.1 Steel and Aluminum Railings and Handrail

~~Install handrail [in pipe sleeves embedded in concrete and filled with non-shrink grout or quick-setting anchoring cement with anchorage covered with standard pipe collar pinned to post.][by means of pipe sleeves secured to wood with screws.][by means of masonry with expansion shields and bolts or toggle bolts.][by means of base plates bolted to stringers or structural-steel frame work.] Secure rail ends by steel pipe flanges [anchored by expansion shields and bolts.] [through-bolted to a back plate or by 6 mm lag bolts to studs or solid backing.]~~Install as recommended by the manufacturer.

~~][3.2.2 Aluminum Handrail~~

~~Affix to base structure by [flanges anchored to concrete or other existing masonry by expansion shields] [base plates or flanges bolted to stringers or structural-steel framework] [flanges through-bolted to a backing plate on the other side of a wall] [flanges lag-bolted to studs or other structural timbers]. Provide Series 300 stainless-steel bolts to anchor aluminum alloy flanges, of a size appropriate to the standard product of the manufacturer. Where aluminum or alloy fittings or extrusions are to be in contact with dissimilar metals or concrete, coat the contact surface with a heavy coating of bituminous paint.~~

~~][~~3.2.2 Touchup Painting

Immediately after installation, clean field welds, bolted connections, abraded areas of the shop paint, and exposed areas painted with the paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 0.051 millimeter.

~~]~~3.3 FIELD QUALITY CONTROL

3.3.1 Field Welding

Ensure that procedures of manual shielded metal arc welding, appearance and quality of welds made, and methods used in correcting welding work comply with JIS Z 3410, JIS Z 3420, or JIS Z 3604.

-- End of Section --

SECTION 06 20 00

FINISH CARPENTRY
~~08/16, CHG 2: 11/18~~

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

FOREST STEWARDSHIP COUNCIL (FSC)

FSC STD 01 001 (2015) Principles and Criteria for Forest Stewardship

JAPANESE ADHESIVE INDUSTRY ASSOCIATION (JAIA)

JAIA 4VOC (2010) Voluntary VOC Regulating Rule for Indoor Air Pollution Control

JAPANESE AGRICULTURAL STANDARDS (JAS)

JAS Notification No. 233 (2003) Japanese Agricultural Standards for Plywood

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 5508 (2009) Nails

JIS B 1112 (1995) Cross Recessed Head Wood Screws

JIS B 1180 (2014) Hexagon Head Bolts and Hexagon Head Screws

JIS B 1186 (2013) Sets of High Strength Hexagon Bolt, Hexagon Nut and Plain Washers for Friction Grip Joints

JIS B 1220 (2015) Set of Anchor Bolt for Structures

JIS K 1570 (2018) Wood Preservatives

JIS K 1571 (2020) Wood Preservatives - Performance Requirements and Their Test Methods for Determining. Effectiveness

PROGRAMME FOR ENDORSEMENT OF FOREST CERTIFICATION (PEFC)

PEFC ST 2002:2013 (2015) PEFC International Standard Chain of Custody of Forest Based Products Requirements

1.2 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S"~~

classification. Submittals not having a "G" ~~or "S"~~ classification are ~~[for Contractor Quality Control approval.]~~~~[~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings Indicating All Wood Assemblies; G~~[, [_____]]~~

SD-03 Product Data

Wood Products~~; G[, [_____]]~~

~~Countertops; G[, [_____]]~~

~~Engineered Wood Products; G[, [_____]]~~

Treated Wood Products~~; G[, [_____]]~~

~~Soffits; G[, [_____]]~~

~~Fascias and Trim; G[, [_____]]~~

Hardware and Accessories~~; G[, [_____]]~~

~~[~~ ~~VOC Content for Siding; S~~

~~][~~ ~~Recycled Content for MDF/Particleboard; S~~

~~]~~ SD-04 Samples

Samples; G~~[, [_____]]~~

SD-07 Certificates

Certificates of Grade; G~~[, [_____]]~~

~~[~~ ~~Certified Sustainably Harvested Wood for Trim and Frames; S~~

~~][~~ ~~Certified Sustainably Harvested Softwood Plywood; S~~

~~][~~ ~~Certified Sustainably Harvested Hardwood Plywood; S~~

~~][~~ ~~Certified Sustainably Harvested Hardboard; S~~

~~][~~ ~~Certified Sustainably Harvested Siding; S~~

~~][~~ ~~Indoor Air Quality for Hardwood Plywood; S~~

~~][~~ ~~Indoor Air Quality for MDF and Particleboard; S~~

~~][~~ ~~Indoor Air Quality for Non-aerosol Adhesives; S~~

~~][~~ ~~Indoor Air Quality for Aerosol Adhesives; S~~

~~]~~1.3 DETAIL DRAWINGS

Submit detail drawings indicating all wood assemblies proposed for use in the project. Indicate materials, species, grade, density, grain, finish details of construction, location of use in the project, finishes, types, method and arrangement of fasteners, and installation details. This includes all fabricated assemblies.

1.4 PRODUCT DATA

Submit Manufacturers printed data including proposed species, grade, density grain, and finish as applicable; sufficient to demonstrate compliance with this specification for each type of wood product specified. For treated wood products also provide documentation of environmentally safe preservatives for each type of wood product specified.

Provide Manufacturers printed data for hardware and all wood accessories including but not limited to edge banding, adhesives, and sealers.

1.5 SAMPLES

Samples indicating proposed species, grade, density grain, and finish for each type of wood product specified. Provide samples of sufficient size to show pattern and color ranges of proposed products.

1.6 DELIVERY, STORAGE, AND HANDLING

Deliver wood products to the jobsite in an undamaged condition. Stack materials to ensure ventilation and drainage. Protect against dampness before and after delivery. Store materials under cover in a well ventilated enclosure and protect against extreme changes in temperature and humidity. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Do not store products in building until wet trade materials are dry and humidity of the space is within wood manufacturer's tolerance limits for storage.

1.7 QUALITY ASSURANCE

1.7.1 Certifications

1.7.1.1 Certified Wood Grades

Provide certificates of grade from the grading agency on graded but unmarked lumber or plywood attesting that materials meet the grade requirements specified herein.

~~[~~1.7.1.2 Certified Sustainably Harvested Wood

Provide wood certified as sustainably harvested by FSC STD 01 001~~[, ATFS STANDARDS, CSA Z809-08, SFI 2015-2019~~ SGEC Certification Japan, or other third party program certified by PEFC ST 2002:2013~~]~~. Provide a letter of Certification of Sustainably Harvested Wood signed by the wood supplier. Identify certifying organization and their third party program name and indicate compliance with chain-of-custody program requirements. Submit sustainable wood certification data; identify each certified product on a line item basis. Submit copies of invoices bearing certification numbers.

~~][~~1.7.1.3 Indoor Air Quality Certifications

1.7.1.3.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements ~~by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section~~to attain F 4-Star or JAIA 4VOC rating. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

~~1.7.1.3.2 Composite Wood Products~~

~~For purposes of this specification, composite wood products include hardwood plywood, particleboard, medium density fiberboard (MDF), panel substrates, and door cores. Provide products certified to meet requirements of both 40 CFR 770 and CARB 93120. Provide current product certification documentation from certification body.~~

~~]~~1.7.2 Lumber

Identify each piece or each bundle of lumber, millwork, and trim by the grade mark of a recognized association or independent inspection agency ~~certified by the Board of Review of the ALSC to grade the species~~in accordance with JAS Notification No. 233.

1.7.3 Plywood

Provide each sheet of plywood with the mark of a recognized association or independent inspection agency in accordance with JAS Notification No. 233 that maintains continuing control over the quality of the plywood. Marks must identify plywood by species group or span rating, exposure durability classification, grade, and compliance with ~~APA L870~~ JAS Notification No. 233.

~~1.7.4 Hardboard [and Particleboard]~~

~~Provide materials marks or written documentation identifying the producer and the applicable standard.~~

1.7.4 Pressure Treated Lumber and Plywood

Inspect each treated piece in accordance with ~~AWPA U1~~JIS K 1571.

1.7.5 Non-Pressure Treated Woodwork and Millwork

Mark, stamp, or label to indicate compliance with ~~WDMA I.S.4~~JIS K 1570.

1.7.6 Fire-Retardant Treated Lumber

Each piece must bear an Underwriters Laboratories fire resistance label or comparable label of another nationally recognized independent fire retardant materials testing laboratory.

PART 2 PRODUCTS

2.1 WOOD PRODUCTS

2.1.1 Sizes and Patterns of Wood Products

Provide yard and board lumber sizes in accordance with ~~ALSC PS 20~~ JAS Notification No. 233. Provide shaped lumber and millwork in the patterns indicated and in standard patterns of the association covering the species. Size references, unless otherwise specified, are nominal sizes. Provide actual sizes within manufacturing tolerances allowed by the applicable standard.

2.1.2 Species and Grades

Provide in accordance with ~~AWPA U1~~JAS Notification No. 233 ~~Use Category System Tables~~ unless otherwise specified herein.

~~2.1.3 Trim, Finish, and Frames~~

~~Provide species and grades listed in the table below for wood materials that must be painted. For materials that must be stained, have a natural, or a transparent finish, provide materials one grade higher than those listed in the table below. Provide trim, except window stools and aprons with hollow backs.[Provide certified sustainably harvested wood for trim and frames.]~~

~~TABLE OF GRADES FOR WOOD TO RECEIVE PAINT FINISH~~

~~Grading Rules~~	~~Species~~	~~Exterior and Interior Trim, Finish, and Frames~~
~~WWPA G-5 standard grading rules~~	~~Aspen, Douglas Fir-Larch, Douglas Fir South, Engelmann Spruce-Lodgepole Pine, Engelmann Spruce, Hem-Fir, Idaho White Pine, Lodgepole Pine, Mountain Hemlock, Mountain Hemlock-Hem-Fir, Ponderosa Pine-Sugar Pine, (Ponderosa Pine-Lodgepole Pine,) White Woods, (Western Woods,) Western Cedars, Western Hemlock~~	~~All Species: C & BTR. Select (Choice & BTR Idaho White Pine) or Superior Finish. Western Red Cedar may be graded C & BTR. Select or A & BTR in accordance with Special Western Red Cedar Rules.~~
~~WCLIB 17 standard grading rules~~	~~Douglas Fir-Larch, Hem-Fir, Mountain Hemlock, Sitka Spruce, Western Cedars, Western Hemlock~~	~~All Species: C & BTR VG, except A for Western Red Cedar~~

~~TABLE OF GRADES FOR WOOD TO RECEIVE PAINT FINISH~~		
~~Grading Rules~~	~~Species~~	~~Exterior and Interior Trim, Finish, and Frames~~
~~SPIB 1003 standard grading rules~~	~~Southern Pine~~	~~C & BTR~~
~~NHLA Rules~~	~~Cypress~~	~~C Select~~
~~NELMA Grading Rules standard grading rules **~~	~~Balsam Fir, Eastern Hemlock-Tamarack, Eastern Spruce, Eastern White Pine, Northern Pine, Northern Pine, Northern White Cedar~~	~~All Species: C-Select except C & BTR for Eastern White Pine and Norway Pine~~
~~RIS Grade Use standard specifications~~	~~Redwood~~	~~Clear, Clear All Heart~~
~~NHLA Rules~~	~~Cypress~~	~~B Finish~~
	~~Red Gum, Soft Elm, Birch~~	~~Select or BTR (for interior use only)~~

~~Note: **~~
~~http://www.nelma.org/library/2013-standard-grading-rules-for-northeastern-lumber/~~

~~2.1.4 Utility Shelving~~

~~Provide utility shelving in a suitable species equal to or exceeding the requirements of No. 3 common white fir under WWPA G-5, 25 mm thick; or plywood, interior type, Grade A-B, 13 mm thick, any species group.~~

~~2.1.5 Softwood Plywood~~

~~Provide in accordance with APA L870.[Provide certified sustainably harvested softwood plywood.][When located on the interior of buildings, provide products with no added urea-formaldehyde resins.]~~

~~a. Plywood for Soffits: Exterior type, B-B medium density overlay.~~

~~b. Plywood for Shelving: Interior type, [A-B] [B-B] Grade, any species group.~~

~~c. Plywood for Countertops: Exterior type, A-C Grade.~~

~~2.1.6 Hardwood Plywood~~

~~HPVA HP-1, Type [Technical (Exterior)] [I (Exterior)] [II (Interior)] [III (Interior)], [Premium (A)] [Good (1)] [Sound (2)] [Utility (3)] [Backing (4)] [Specialty (SP)] Grade, [hardwood veneer core construction,] [lumber core construction,] face veneers of [_____], of thickness indicated.[Provide certified sustainably harvested hardwood plywood.][When located~~

~~on the interior of buildings, provide products with no added urea-formaldehyde resins. For products located on the interior of the building (inside of the weatherproofing system), provide certification of indoor air quality for hardwood plywood.]~~

~~2.1.7 Hardboard~~

~~AHA A135.4, [standard] [tempered] [service] type, [3] [6] mm thick.[Provide certified sustainably harvested hardboard.]~~

~~[2.1.8 Medium Density Fiberboard (MDF) and Particleboard~~

~~CPA A208.1, Grade 1-M-2 or 2-M-2 or better.[For products located on the interior of the building (inside of the weatherproofing system), provide certification of indoor air quality for MDF and particleboard.]~~

~~Provide products with 80 percent total recovered materials content. Provide data identifying percentage of recycled content for MDF/particleboard.~~

~~]2.1.9 Stairs~~

~~Treads 32 mm thickness, clear red or white oak. Risers 19 mm nominal finish lumber.~~

~~2.1.10 Shoe Mould~~

~~Clear red or white oak, 13 by 16 mm unless otherwise indicated.~~

~~2.1.11 Wood Seats~~

~~Clear maple, oak, or other suitable hardwood, not less than 40 mm thick, with rounded edges. Provide stainless steel stanchions or brackets.~~

~~2.1.12 Wood Bumpers~~

~~Clear oak[, maple] [, birch] [or] [_____], dressed to size indicated and with outer edges beveled.~~

~~2.1.13 Catwalks~~

~~Boards, 19 by 140 mm, species and grade equal to or exceeding 3 Common Hem-Fir under WWPA G-5.~~

~~2.1.14 Siding~~

~~Provide hardboard, plywood, or wood for horizontal siding. Provide hardboard or plywood for panel siding.[Provide certified sustainably harvested siding.][When located on the interior of buildings, provide products with no added urea-formaldehyde resins. Provide data identifying VOC content for siding.]~~

~~2.1.14.1 Horizontal Hardboard Siding~~

~~AHA A135.6, factory primed face and longitudinal edges, factory sealed back, lap type, [200] [225] [250] [300] mm wide, maximum practicable lengths, 9.5 or 11 mm thick, [smooth] [embossed] [textured] face.~~

~~2.1.14.2 Panel Hardboard Siding~~

~~AHA A135.6, factory primed face and longitudinal edges, factory sealed back, 1220 mm wide, maximum practicable lengths, 9.5 or 11 mm thick, [smooth] [embossed] face [, and grooved as selected from manufacturer's standard patterns].~~

~~2.1.14.3 Horizontal Plywood Siding~~

~~APA L870, exterior, [medium density overlay] lap type, [150] [200] [300] mm wide, maximum practicable lengths, [9.5] [11] [12] [13] mm thick, [smooth] [embossed] [rough sawn texture] [embossed] face.~~

~~2.1.14.4 Panel Plywood Siding~~

~~APA L870, exterior, [medium density overlay,] 1220 mm wide, maximum practicable lengths, span rating of [400] [600] mm on center, [smooth] [embossed] [rough sawn texture] [striated] face, [and grooved] as selected from manufacturer's standard patterns.~~

~~2.1.14.5 Horizontal Rated Siding~~

~~Qualified under APA E445, exterior type [medium density overlay], lap types, [150] [200] [250] [300] mm wide, maximum practicable lengths, [11] [12] [13] mm thick, [smooth] [embossed] [rough sawn texture] face.~~

~~2.1.14.6 Panel Rated Siding~~

~~Qualified under APA E445, exterior type, [medium density overlay] 1220 mm wide, maximum practicable lengths, [span rated at 400 mm] [span rated at 600 mm,] [smooth] [embossed] [striated] face [, and grooves] as selected from manufacturer's standard patterns.~~

~~2.1.14.7 Wood Siding~~

~~Provide [horizontal bevel type, minimum 5 mm thin edge by minimum 11 mm thick edge,] [horizontal plain lap type] [horizontal drop type] [vertical board, tongue and groove or shiplap on long edges,] [vertical board and batten type,] 25 mm thick, [150] [200] [250] mm wide, maximum practicable lengths, [smooth] [rough sawn texture].~~

~~2.1.14.8 Engineered Wood Structural Panels~~

~~Comply with ICC IBC, Chapter 23 Wood, and with APA S350, exterior, exposure [1] [2], [single-faced] [double-faced], 1200 mm wide, maximum practicable lengths, selected from manufacturer's standard patterns to satisfy the wind load for the specified span.~~

~~2.1.14.9 Epoxy Coated Wood Panels~~

~~Provide prefinished epoxy coated wood panels consisting of an asbestos-free cement board base sheet with a factory applied surface of epoxy resin and decorative natural stone chips. Provide factory applied finish in a minimum 0.5 mm thickness consisting of 100 percent solids, with a two component epoxy resin based coating followed by an application of inert aggregate. Provide stone color(s) and accessory colors as selected by Contracting Officer's Representative from manufacturer's full color and pattern ranges. Provide cement board base sheets in a minimum thickness of 6 mm thick. Dimensionally stable finished panels are~~

~~required. Water absorption on the surfaced side of panels cannot exceed 0.20 percent after 24 hours of submergence in water. Provide accessories in manufacturer's standard extruded aluminum. Provide mouldings for meeting strips, end caps, inside corners, outside corners. Provide non-corrosive, self-tapping screw type fasteners finished to match the color of the panel surface. Provide caulking compounds that are color compatible, low modulus silicone or urethane types.~~

~~2.2 SOFFITS~~

~~2.2.1 Hardboard and Plywood~~

~~Provide hardboard and plywood soffits in siding grade hardboard, 10 or 11 mm thick; plywood, APA L870, exterior type, [Grade A-C] [plywood panel siding] [rated siding],[9 mm thick for 600 mm on center] [12 mm thick for 800 mm on center] [15 mm thick for 1200 mm on center] maximum span with all edges supported.~~

~~2.3 FASCIAS AND TRIM~~

~~2.3.1 Wood~~

~~Provide species and grades for all fascia and trim, including exterior door and window casings, in accordance with AWPA U1 Use Category System Tables. Provide sizes indicated. Metal corners may be provided in lieu of wood corner boards for horizontal siding. If metal corners are used, provide galvanized steel or aluminum, completely coated with primer compatible for the specific metal substrate.~~

~~2.4 COUNTERTOPS~~

~~2.4.1 Laminated Plastic-faced Countertops~~

~~ANSI/NEMA LD 3.~~

~~2.4.1.1 Countertop Finishes~~

~~High pressure plastic laminate, Grade GP 50 or PF 42, satin or textured finish. Provide color and pattern [_____] [as selected by Contracting Officer's Representative from manufacturer's full color and pattern ranges].~~

~~2.4.1.2 Backing Sheet~~

~~Heavy gauge, BK 20.~~

~~2.4.2 Solid Surface~~

~~For solid surface countertops refer to Section 06 61 16, SOLID POLYMER (SOLID SURFACING) FABRICATIONS.~~

2.2 MOISTURE CONTENT OF WOOD PRODUCTS

Air dry or kiln dry lumber. Kiln dry treated lumber after treatment. Maximum moisture content of wood products at time of delivery to the jobsite, and when installed, must be as follows:

~~a. Interior Paneling: [6] [12] percent.~~

~~b~~a. Interior Finish Lumber~~,~~ and Trim~~, and Millwork~~: 25 mm or Less in Thickness: ~~[~~6~~]~~ ~~[12]~~ percent on 85 percent of the pieces and ~~[~~8~~]~~ ~~[15]~~ percent on remainder.

~~c. Exterior Treated and Untreated Finish Lumber and Trim: 89 mm or Less in Thickness: 19 percent.~~

~~d. Exterior Wood Siding: 15 percent.~~

~~e~~b. Provide moisture content of other materials in accordance with the applicable standards.

~~2.3 PRESERVATIVE TREATMENT OF WOOD PRODUCTS~~

~~2.3.1 Non-Pressure Treatment~~

~~Treat woodwork and millwork, such as [cabinets,]exterior trim, door trim, and window trim, in accordance with WDMA I.S.4, with either 2 percent copper napthenate, 3 percent zinc napthenate, or 1.8 percent copper-8-quinolinolate. Provide a liberal brush coat of preservative treatment to field cuts and holes.~~

~~2.3.2 Pressure Treatment~~

~~Treat lumber and plywood used on the exterior of buildings [or in contact with masonry or concrete] with a waterborne preservative listed in AWPA U1 (P series is included therein by reference) as applicable, and inspected in accordance with AWPA U1. Identify treatment on each piece of material by the quality mark of an agency accredited by the Board of Review of the American Lumber Standards Committee. Provide treated plywood to a reflection level as follows:~~

~~Preservative treat exterior wood moulding and millwork that will be within 455 mm of soil or in contact with water or concrete in accordance with WMMPA WM 6. Provide a field treatment in accordance with AWPA M4 of exposed areas of treated wood that have been cut or drilled. Items of all-heart material of cedar, cypress, or redwood do not require preservative treatment except when in direct contact with soil.~~

~~2.4 FIRE-RETARDANT TREATMENT~~

~~2.4.1 Wood Products~~

~~Pressure treat fire-retardant treated lumber and plywood in accordance with AWPA U1. Comply with material use as defined in AWPA U1 for Interior Type [A] [and] [B] and Exterior Type. Treatment and performance inspection must be conducted by a qualified independent testing agency that establishes performance ratings. Each piece or bundle of treated material must bear identification of the testing agency to indicate performance with such rating. Subject treated materials that will be exposed to rain wetting to an accelerated weathering technique in accordance with ASTM D2898, Method A, prior to being tested for compliance with AWPA U1.~~

~~Treat the following items:~~

~~[_____].~~

2.3 HARDWARE AND ACCESSORIES

Provide sizes, types, and spacings of hardware and accessories as recommended in writing by the wood product manufacturer, except as otherwise specified.

2.3.1 Wood Screws

~~ASME B18.6.1~~JIS B 1112.

2.3.2 Bolts, Nuts, Lag Screws, and Studs

~~ASME B18.2.1 and ASME B18.2.2~~JIS B 1180, JIS B 1186, JIS B 1220.

2.3.3 Nails

Use nails of a size and type best suited for each application and in accordance with ~~ASTM F547~~JIS A 5508. ~~Use hot-dipped galvanized or aluminum nails for exterior applications. For siding, provide nails of sufficient length to extend 40 mm into supports, including wood sheathing over framing.~~ Where nailing is impractical, provide screws of a size and type best suited for each application.

~~2.3.4 Adjustable Shelf Standards~~

~~ANSI/BHMA A156.9, Type [_____], with shelf rests Type [_____].~~

~~2.3.5 Vertical Slotted Shelf Standards~~

~~ANSI/BHMA A156.9, Type [_____], with shelf brackets Type [_____].~~

~~2.3.6 Closet Hanger Rods~~

~~Chromium plated steel rods, not less than 25 mm diameter by 1.3 mm thick. Rods may be adjustable with integral mounting brackets if smaller tube is 25 mm by 1.3 mm thick. Provide intermediate support brackets for rods more than 1200 mm long.~~

~~2.4 FABRICATION~~

~~2.4.1 Quality Standards (QS)~~

~~2.4.1.1 Grades~~

~~The terms "Premium," "Custom," and "Economy" refer to the quality grades defined in NAAWS 3.1. Provide items not otherwise specified in a specific grade as "Custom" grade.~~

~~2.4.1.2 Adhesives~~

~~Select adhesives for durability and permanent bonding. Address factors such as materials that must be bonded, expansion and contraction, bond strength, fire rating, moisture resistance, and manufacturer's recommendations.~~

~~[Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements~~

~~of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for non-aerosol adhesives applied on the interior of the building (inside of the weatherproofing system). Provide certification or validation of indoor air quality for aerosol adhesives used on the interior of the building.~~

~~]2.4.2 Countertops~~

~~Fabricate with lumber and a core of [exterior plywood] [or] [particleboard], glued and screwed to form an integral unit. Bond laminated plastic under pressure to exposed surfaces, using adhesive as recommended by the plastic manufacturer [, and bond a backing sheet under pressure to underside of countertop]. Provide countertop units as post-formed type, no-drip nose, cove mouldings, Style A backsplash, and surfaced with ANSI/NEMA LD 3, Grade PF 42 plastic. Provide backsplashes not less than 90 mm nor more than 115 mm high.~~

~~2.4.3 Cabinets~~

~~Unless specified otherwise, provide wall and base cabinets of the same construction, materials, and finishes as countertops. Fabricate cabinets with solid ends and frame fronts, or with frames all around. Provide frames of solid hardwood not less than 19 by 38 mm. Provide ends, bottoms, backs, partitions, and doors as hardwood plywood. Mortise and tenon, dovetail, or dowel and glue joints to produce a rigid unit. Cover exposed edges of plywood with hardwood strips. Provide cabinet doors, frames, and solid exposed ends 19 mm thick minimum. Provide cabinet bottoms, partitions, and framed ends to be 13 mm minimum. Provide shelves to be 16 mm thick minimum. Provide cabinet backs 6 mm thick minimum.~~

~~2.4.3.1 Cabinet Hardware~~

~~ANSI/BHMA A156.9. Provide cabinet hardware including two self, closing hinges for each door, two side mounted metal drawer slides for each drawer, and pulls for all doors and drawers as follows. Provide hardware exposed to view [as bright chromium plated][_____]. Comply with the following requirements for all cabinet hardware:~~

~~a. Provide frameless concealed European style, back mounted hinges with 165 degree opening and a self closing feature when at less than 90 degrees open.~~

~~b. Provide drawer slides having a static rating capacity of [444 N][_____]. Slides to have a self closing/stay closed action, zinc or epoxy coated steel finish, ball bearing rollers, and positive stop with lift out design.~~

~~c. Provide drawer pulls as [wire type pulls with center-to-center dimension of not less than 89 mm and a cross sectional diameter of 8 mm]. Provide handle projections not less than[33 mm]. [_____].~~

~~d. Provide heavy duty magnetic drawer catches.~~

~~2.4.3.2 Finish~~

~~Provide a clear factory finish on wood surfaces after fabrication. Provide~~

~~fabricator's standard natural finish equivalent to one coat of sealer, one coat of varnish on all surfaces and a second coat of varnish on surfaces exposed to view. Provide spar varnish in exterior or wet area applications. Sand lightly and wipe clean between coats.~~

~~2.4.4 Workbenches~~

~~Dovetail and glue drawer corners. Fasten frames with suitable wood screws or bolts. Sand exposed surfaces smooth, and ease exposed edges. Provide two side mounted, metal, ball bearing drawer slides [ANSI/BHMA A156.9, Type [_____],] for each drawer, and at least two surface mounted hinges[, Type [_____],] and a magnetic catch[, Type [_____],] for each door.~~

~~2.4.5 Casework with Transparent Finish (CTF)~~

~~2.4.5.1 AWI Quality Grade~~

~~[Premium] [Custom] [Economy] grade.~~

~~2.4.5.2 Construction~~

~~Provide [reveal overlay] [flush overlay] [exposed face frame] design details.~~

~~2.4.5.3 Exposed Parts~~

~~[_____] specie, [_____] cut.~~

~~2.4.5.4 Semi-Exposed Parts~~

~~As specified in the NAAWS 3.1 for the grade selected.~~

~~2.4.6 Casework with High Pressure Laminate Finish~~

~~2.4.6.1 AWI Quality Grade~~

~~[Premium] [Custom] grade.~~

~~2.4.6.2 Construction~~

~~Provide [reveal overlay] [flush overlay] [exposed face frame] design details.~~

~~2.4.6.3 Exposed Surfaces~~

~~High pressure plastic laminate, color and pattern [_____][as selected by Contracting Officer's Representative from manufacturer's full range].~~

~~2.4.6.4 Semi-Exposed Surfaces~~

~~As specified in the NAAWS 3.1 for the grade selected.~~

~~[2.4.6.5 Edge Banding~~

~~Provide edge banding for casework doors and drawer fronts in PVC vinyl [0.5 mm] [3 mm] [_____] thick. Provide width [23.8 mm] [_____].[Match color and pattern to exposed door and drawer front laminate pattern and color.][Provide color and pattern [_____].]~~

PART 3 EXECUTION

Do not install building construction materials that show visual evidence of biological growth.

3.1 FINISH WORK

Apply primer to finish work before installing. Where practicable, shop assemble and finish millwork items. Construct joints tight and in a manner to conceal shrinkage but to avoid cupping, twisting and warping after installation. Miter trim and mouldings at exterior angles; cope at interior angles and at returns. Provide millwork and trim in maximum practical lengths. Fasten finish work with finish nails. Provide blind nailing where practicable. Set face nails for putty stopping.

~~3.1.1 Exterior Finish Work~~

~~Machine sand exposed flat members and square edges. Machine finish semi-exposed surfaces. Construct joints to exclude water. In addition to nailing, glue joints with waterproof glue as necessary for weather resistant construction. Evenly distribute end joints in built-up members. Provide shoulder joints in flat work. Reinforce backs of wide-faced miters with metal rings and waterproof glue. Unless otherwise indicated, provide fascia and other flat members 19 mm thick minimum. Provide door and window trim in single lengths. Provide braced, blocked, and rigidly anchored cornices for support and protection of vertical joints. Provide soffits in largest practical size. Align joints of plywood over centerlines of supports. Fasten soffits with aluminum or stainless steel nails. Back prime all concealed surfaces of exterior trim.~~

3.1.1 Interior Finish Work

After installation, sand exposed surfaces smooth. Provide window and door trim in single lengths.

~~3.1.2 Door Frames~~

~~Set plumb and square. Provide solid blocking at not more than 400 mm on center for each jamb. Position blocking to occur behind hinges and lock strikes. Double wedge frames and fasten with finish nails. Set nails for putty stopping.~~

~~3.1.3 Thresholds~~

~~Unless otherwise indicated, provide thresholds [16 mm thick by 70 mm wide with beveled sides] and cut to fit at jambs. Fasten thresholds with casing nails. Set nails for putty stopping.~~

~~3.1.4 Window Stools and Aprons~~

~~Provide stools with rabbets over window sills. Provide aprons with returns cut accurately to profile of member.~~

~~3.1.5 Bases~~

~~Provide flat member with a moulded top [and oak shoe mould]. Fasten base to framing or to grounds.[Nail shoe mould to base.] Set [shoe mould] [one-piece wood base] after finish flooring is in place.~~

~~3.1.6 Finish Stair Work~~

~~Fit, nail, screw, bolt, and glue stair work together to form a strong, rigid structure without squeaks or vibrations. Anchor newels and posts securely to stair framing. Cut newels, posts, and drops accurately around floor construction to make a tight fit. Embed balusters into treads and landings and secure with glue. Provide railings with straight runs that follow the slope of the stairs and have smooth curved turns. Return railing profile at ends and secure joints with bolts and nuts in accordance with structural load requirements for railings. Secure railing to posts and newels with concealed anchors. Support wall rails on metal brackets spaced near ends and at not more than 1500 mm on center.~~

~~3.2 SHELVING~~

~~Support 25 mm nominal thick wood shelf material or 19 or 20 mm thick plywood shelf material with end and intermediate supports arranged to prevent buckling and sagging.[Provide hook strips25 mm by 100 mm nominal and cleats25 mm by 50 mm nominal.] Provide cleats except where hook strips are specified or indicated.[Where adjustable shelving is indicated, provide standards and brackets or shelf rests for each shelf.][Anchor standards to wall at not more than 600 mm on center.]~~

~~3.2.1 Linen Closets~~

~~Unless indicated otherwise, provide linen closets with a counter shelf 500 mm wide located 900 mm above the floor, a lower shelf approximately 450 mm wide and 450 mm above the floor, and three upper shelves 285 mm wide located 350 mm above the counter shelf and 350 mm apart.~~

~~3.2.2 Storage Rooms~~

~~Unless otherwise indicated, provide storage rooms with shelves 285 mm wide, bottom shelf 450 mm above the floor, top shelf 450 mm below the ceiling, and intermediate shelves approximately 450 mm apart.~~

~~3.2.3 Room Closets~~

~~Provide two shelves 285 mm wide. Support lower shelf by hook strips at back and ends, and provide full length wood or metal clothes hanger rods unless indicated otherwise.~~

~~3.2.4 Cleaning Gear Closets~~

~~Provide [shelves of size and arrangement indicated] [two shelves 350 mm wide].~~

~~3.3 CLOTHES HANGER RODS~~

~~Provide clothes hanger rods where indicated and in closets having hook strips. Set rods parallel with front edges of shelves and support by sockets at each end and intermediate brackets spaced not more than 1200 mm on center.~~

~~3.4 MISCELLANEOUS~~

~~3.4.1 Countertops~~

~~Conceal fastenings where practicable. Fit counters tight to adjoining~~

~~surfaces and scribe where necessary. Provide scribed joints neat and flush. Provide counter sections in longest lengths practicable with a minimum number of joints. Where joints are necessary, provide tight joints drawn up with concealed type heavy pull-up bolts. Glue joints with water resistant glue and make rigid with screws, bolts, or other approved fastenings.~~

~~3.4.2 Cabinets~~

~~Provide cabinets level, plumb, true, and tight to adjacent walls. Secure cabinets to walls with concealed toggle bolts. Secure top to cabinet with concealed screws.[Make cutouts for fixtures from templates supplied by fixture manufacturer. Locate cutouts for pipes so that edges of holes are covered by escutcheons after installation.]~~

~~3.4.3 Workbenches~~

~~Provide level, plumb, and tight to adjacent construction. Fasten workbenches to walls with screws or toggle bolts and to floors with expansion bolts.~~

~~3.4.4 Wood Seats~~

~~Support seats [on brackets spiked to the studs] [on stanchions]. Secure seats to supports with [screws] [bolts] as required; countersink heads of [screws] [bolts] and fill holes with hardwood filler, finished flush with tops of seats.~~

~~3.4.5 Wood Bumpers~~

~~Bore, countersink, and bolt wood bumpers in place where indicated.~~

~~3.4.6 Catwalks in Attic Spaces~~

~~Lay boards with 25 mm spaces between. Stagger end joints, with each joint on a support.~~

~~3.5 SIDING~~

~~3.5.1 Installation of Siding~~

~~Fit and position siding without springing or otherwise forcing panels into place.[For siding to have a stain finish, set nails and stop with nonstaining putty to match finished siding.][For siding to have a paint finish, drive nails flush.]~~

~~3.5.2 Horizontal Siding~~

~~Locate end joints over framing members and alternate such that there are a minimum of two boards between joints on the same support. Evenly distribute shorter pieces throughout the installation. Provide starter strips to establish proper cant for siding. Predrill ends of siding as necessary to prevent splitting when nailed.[Horizontal bevel or plain lap siding: Overlap and nail into each support in accordance with recommendations of siding manufacturer.][Horizontal drop siding: Work each course into top edge of previous course. Nail into each support with [two nails, one near lower edge to clear top of previous course, and one just above mid-height of course.] [one nail just above mid-height of course.]]~~

~~3.5.3 Vertical Board Siding~~

~~Apply siding with horizontal joints only at locations indicated. Work each board into edge of previous course. Nail into supports at 600 mm on center with [two nails, one blind if possible at or near joint with previous board, and one just outside board centerline.] [one nail just outside board center line.]~~

~~3.5.4 Vertical Board and Batten Siding~~

~~Apply with horizontal joints only at locations indicated. Install each board with 13 mm of space between boards. Nail at center of board and into supports at 600 mm on center. Center battens over space between boards and nail down center at 400 mm on center.~~

~~3.5.5 Panel Siding~~

~~Apply panels with edges at joints spaced in accordance with manufacturer's recommendations. Provide shiplapped edges or square edges that will be covered by battens in a [primed for paint finish,] [sealed for stain finish]. Back all edges with framing members. Nail panels at edges 150 mm on center and at intermediate supports 300 mm on center. Locate edge nailing 10 mm from edges. For shiplap joints, nail 10 mm from visible joint and at a location to penetrate lap with previous panel. When panel siding is part of an engineered shear wall or used as wall bracing, nail shiplap joints to supports with double rows of nails. Space battens at [300] [400] mm on center and nail down center at 600 mm on center.~~

~~3.5.6 Epoxy Coated Panels~~

~~Provide panels where indicated and install in accordance with panel manufacturer's written installation instructions.~~

~~3.6 SOFFITS~~

~~3.6.1 Wood~~

~~Provide panels with edges at joints spaced in accordance with manufacturer's written instructions and with all edges backed by framing members. Nail panels 10 mm from edges at 150 mm on center and at intermediate supports at 300 mm on center. Provide panels in maximum practicable lengths.~~

~~3.7 FASCIA AND EXTERIOR TRIM~~

~~Construct, caulk, and machine sand exposed surfaces and edges to exclude water. In addition to nailing, glue joints as necessary for weather resistance. Evenly distribute end joints in built-up members. Shoulder joints in flat work. Reinforce backs of wide-faced miters with metal rings and glue. Provide fascia and other flat members in maximum practicable lengths. Braced, block, and rigidly anchor cornices for support and protection of vertical joints.~~

3.2 MOULDING AND INTERIOR TRIM

Install mouldings and interior trim straight, plumb, level and with closely fitted joints. Provide exposed surfaces machine sanded at the shop. Cope returns and interior angles at moulded items and miter external corners. Shoulder intersections of flatwork to ease any inherent changes in plane. Provide window and door trim in single lengths. Blind nail to the extent practicable. Set and stop face nailing with a nonstaining putty to match the applied finish. Use screws for attachment to metal; set and stop screws in accordance with the same quality requirements for nails.

-- End of Section --

SECTION 06 41 16.00 10

PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS
08/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~ASTM INTERNATIONAL (ASTM)~~

~~ASTM E84~~	~~(2018a) Standard Test Method for Surface Burning Characteristics of Building Materials~~

JAPANESE ADHESIVE INDUSTRY ASSOCIATION (JAIA)

JAIA 4VOC	(2010) Voluntary VOC Regulating Rule for Indoor Air Pollution Control

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1099	~~(2019) Construction Standard Specification~~ (2021) Fastening Parts-General Requirements for Bolts, Machine Screws, Implanted Bolts and Nuts
JIS K 6903	(20~~08~~22) Laminated Thermosetting High-Pressure Decorative Sheets

~~MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORTATION AND TOURISM (MLIT)~~
MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 12	(2019) Public Building Construction Standard Specification Chapter 12: Wood Work

1.2 SYSTEM DESCRIPTION

Work in this section includes laminate clad custom casework ~~[~~cabinets~~]~~ ~~[vanities] [______]~~ as shown on the drawings and as described in this specification. This Section includes high-pressure laminate surfacing and cabinet hardware. Comply with ~~EPA~~JAIA requirements~~ in accordance with Section 01 33 29 SUSTAINABILITY REPORTING~~. All exposed and semi-exposed surfaces, whose finish is not otherwise noted on the drawings or finish schedule, shall be sanded smooth and shall receive a clear finish of polyurethane.~~ Wood finish may be shop finished or field applied in accordance with Section 09 90 00 PAINTS AND COATINGS.~~

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation;

submittals not having a "G" designation are for ~~[Contractor Quality Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings

Installation

SD-03 Product Data

Finish Schedule

SD-04 Samples

Plastic Laminates

Cabinet Hardware

SD-07 Certificates

1.4 QUALITY ASSURANCE

1.4.1 General Requirements

Unless otherwise noted on the drawings, all materials, construction methods, and fabrication shall conform to and comply with the ~~[premium] [~~ custom~~]~~ grade quality standards as outlined in this specification. Contractors and their personnel engaged in the work shall be able to demonstrate successful experience with work of comparable extent, complexity and quality to that shown and specified. Submit a quality control statement which illustrates compliance with and understanding of project requirements. The quality control statement shall also certify a minimum of ten years Contractor's experience in laminate clad casework fabrication and construction. The quality control statement shall provide a list of a minimum of five successfully completed projects of a similar scope, size, and complexity.

1.5 DELIVERY, STORAGE, AND HANDLING

Casework may be delivered knockdown or fully assembled. Deliver all units to the site in undamaged condition, stored off the ground in fully enclosed areas, and protected from damage. The storage area shall be well ventilated and not subject to extreme changes in temperature or humidity.

1.6 SEQUENCING AND SCHEDULING

Coordinate work with other trades. Units shall not be installed in any room or space until painting, and ceiling installation are complete within the room where the units are located. Floor cabinets shall be installed before finished flooring materials are installed.

PART 2 PRODUCTS

2.1 WOOD MATERIALS

2.1.1 Lumber

a. All framing lumber shall be kiln-dried with moisture content 6 to 8 percent, to dimensions as shown on the drawings. Frame front, where indicated on the drawings, shall be nominal 15 mm to 19 mm hardwood.

~~b. Standing or running trim casework components, which are specified to receive a transparent finish, shall be [_____] hardwood species, plain sawn. Grade shall be [premium] [custom]. Location, shape, and dimensions shall be as indicated on the drawings.~~

2.1.2 Panel Products

2.1.2.1 Plywood

All plywood panels used for framing purposes shall be veneer core hardwood plywood. Nominal thickness of plywood panels shall be as indicated in this specification and on the drawings.

~~2.1.2.2 Particleboard~~

~~All particleboard shall be industrial grade, medium density (640 to 800 kg per cubic meter), 15 mm to 19 mm thick. A moisture-resistant particleboard shall be used as the substrate for plastic laminate covered [countertops] [backsplashes] [_____] [components as located on the drawings] and other areas subjected to moisture. Particleboard shall meet the minimum standards listed in JIS A 5908.~~

~~2.1.2.3 Medium Density Fiberboard~~

~~Medium density fiberboard (MDF) shall be an acceptable panel substrate where noted on the drawings. Medium density fiberboard shall meet the minimum standards listed in JIS A 5905.~~

2.2 SOLID POLYMER MATERIAL

Solid surfacing casework components shall conform to the requirements of Section 06 61 16 SOLID SURFACING FABRICATIONS.

2.3 HIGH PRESSURE DECORATIVE LAMINATE (HPDL)

All plastic laminates shall meet the requirements of JIS K 6903 for high-pressure decorative laminates. Design, colors, surface finish and texture, and locations shall be as indicated on ~~[~~the drawings~~]~~. Submit two samples of each plastic laminate pattern and color. Samples shall be a minimum of 120 by 170 mm in size. Plastic laminate types and nominal minimum thicknesses for casework components shall be as indicated in the following paragraphs.

2.3.1 Horizontal General Purpose Standard (HGS) Grade

Horizontal general purpose standard grade plastic laminate shall be 0.95 mm to 1.2 mm in thickness. This laminate grade is intended for horizontal surfaces where postforming is not required.

2.3.2 Vertical General Purpose Standard (VGS) Grade

Vertical general purpose standard grade plastic laminate shall be 0.55 mm to 0.7 mm in thickness. This laminate grade is intended for exposed exterior vertical surfaces of casework components where postforming is not required.

~~2.3.3 Horizontal General Purpose Postformable (HGP) Grade~~

~~Horizontal general purpose postformable grade plastic laminate shall be 0.95 mm to 1.2 mm in thickness. This laminate grade is intended for horizontal surfaces where post forming is required.~~

~~2.3.4 Vertical General Purpose Postformable (VGP) Grade~~

~~Vertical general purpose postformable grade plastic laminate shall be 0.7 mmin thickness. This laminate grade is intended for exposed exterior vertical surfaces of components where postforming is required for curved surfaces.~~

~~2.3.5 Horizontal General Purpose Fire Rated (HGF) Grade (U.S. Product only)~~

~~Horizontal general purpose fire rated grade plastic laminate shall be 1.22 mm (plus or minus 0.127 mm) in thickness. Laminate grade shall have a class 1, class A fire rating in accordance with ASTM E84.~~

~~2.3.6 Vertical General Purpose Fire Rated (VGF) Grade~~

~~Vertical general purpose fire rated grade plastic laminate shall be 0.71 mm (plus or minus 0.012 mm) in thickness. This laminate grade shall have a class 1, class A fire rating in accordance with ASTM E84.~~

2.3.3 Cabinet Liner Standard (CLS) Grade

Cabinet liner standard grade plastic laminate shall be 0.55 mm in thickness. This laminate grade is intended for light duty semi-exposed interior surfaces of casework components.

2.3.4 Backing Sheet (BK) Grade

Undecorated backing sheet grade laminate is formulated specifically to be used on the backside of plastic laminated panel substrates to enhance dimensional stability of the substrate. Backing sheet thickness shall be 0.51 mm. Backing sheets shall be provided for all laminated casework components where plastic laminate finish is applied to only one surface of the component substrate.

2.4 THERMOSET DECORATIVE OVERLAYS (MELAMINE)

Thermoset decorative overlays (melamine panels) shall be used for ~~[~~ casework cabinet interior~~]~~, ~~[~~drawer interior~~]~~, and ~~[~~all semi-exposed~~]~~ ~~[_____]~~ surfaces.

2.5 EDGE BANDING

Edge banding for casework doors and drawer fronts shall be ~~PVC vinyl~~HDPL plastic laminate and shall be ~~[0.5 mm] [~~3 mm~~] [_____]~~ thick. Material width shall be ~~[as indicated on the drawings] [_____]~~as required. Color and pattern shall ~~[~~match exposed door and drawer front laminate pattern and color~~] [be as indicated on the drawings] [_____]~~.

2.6 CABINET HARDWARE

Submit one sample of each cabinet hardware item specified to include ~~[~~hinges~~]~~, ~~[~~pulls~~]~~, ~~[~~drawer glides~~]~~, and ~~[_____]~~locks. All hardware shall follow the requirements to MLIT SS Chapter 12, unless otherwise noted, and shall consist of the following components:

2.6.1 Door Hinges

~~[_____]~~Self-closing type.

2.6.2 Cabinet Pulls

~~[_____]~~Wire type. Satin stainless steel.

2.6.3 Drawer Slide

Side mounted ~~[_____]~~and closing type, with ~~[~~full~~] [_____]~~ extension and a minimum ~~[34 kg] [~~45 kg~~] [_____]~~ load capacity. Slides shall include an ~~[integral] [~~positive~~]~~ stop to avoid accidental drawer removal.

2.6.4 Adjustable Shelf Support System

~~[~~Recessed (mortised) metal standards, finish: ~~[_____]~~satin nickel plated, clear coated, steel, 652. Support clips for the standards shall be ~~[~~open type,~~] [closed type,]~~ finish: ~~[_____]] [Multiple holes with [metal] [plastic] [wood] pin supports]~~satin nickel plated, clear coated, steel, 652.

2.7 FASTENERS

Nails, screws, and other suitable fasteners shall be the size and type best suited for the purpose and shall conform to JIS B 1099 where applicable.

2.8 ADHESIVES, CAULKS, AND SEALANTS

2.8.1 Adhesives

Adhesives shall be of a formula and type recommended by manufacturer. Adhesives shall be selected for their ability to provide a durable, permanent bond and shall take into consideration such factors as materials to be bonded, expansion and contraction, bond strength, fire rating, and moisture resistance. ~~Adhesives shall meet local regulations and JIS A 1901 regarding VOC emissions and off-gassing.~~ Provide products certified to meet indoor air quality requirements to attain F 4-Star or JAIA 4VOC rating.

2.8.1.1 Wood Joinery

Adhesives used to bond wood members shall be for interior use ~~[~~ urea-formaldehyde resin formula~~]~~ or ~~[~~polyvinyl acetate resin emulsion~~] [_____]~~. Adhesives shall withstand a bond test as described in MLIT SS Chapter 12.

2.8.1.2 Laminate Adhesive

Adhesive used to join high-pressure decorative laminate to wood shall be

~~[a water-based contact adhesive] [_____] [~~adhesive consistent with ~~AWI and~~ laminate manufacturer's recommendations~~]~~.~~ PVC edgebanding shall be adhered using a polymer-based hot melt glue.~~

2.8.2 Caulk

Caulk used to fill voids and joints between laminated components and between laminated components and adjacent surfaces shall be clear, 100 percent silicone.

2.8.3 Sealant

Sealant shall be of a type and composition recommended by the substrate manufacturer to provide a moisture barrier at sink cutouts and all other locations where unfinished substrate edges may be subjected to moisture.

~~2.9 WOOD FINISHES~~

~~Paint, stain, varnish and their applications required for laminate clad casework components shall be [_____] [as indicated in Section 09 90 00 PAINTS AND COATINGS] [as indicated in Section 09 06 00 SCHEDULES FOR FINISHES]. Color and location shall be as indicated on the drawings.~~

~~2.10 ACCESSORIES~~

~~2.10.1 Glass and Glazing~~

~~Glass required in laminated casework shall be referenced by type in accordance with Section 08 81 00 GLAZING. Glass shall be one of the following:~~

~~a. Type [A] [_____].~~

~~b. [Float] [Patterned] glass: [Clear] [pattern] quality.~~

~~c. Safety glass: [Clear] [_____]; [heat strengthened] [fully tempered] [laminated] [_____]; [_____] mm thick minimum.~~

~~d. Wire Glass: [Clear] [_____], polished [both sides] [one side]; [square] [diagonal] [_____] mesh woven stainless steel wire of grid [_____] mm size; [_____] mm thick.~~

~~2.10.2 Grommets~~

~~Grommets shall be [plastic] [metal] [rubber] [_____] material for cutouts with a diameter of [_____] mm. Locations shall be as indicated on the drawings.~~

2.9 FABRICATION

Verify field measurements as indicated in the shop drawings before fabrication. Fabrication and assembly of components shall be accomplished at the shop site to the maximum extent possible. Construction and fabrication of cabinets and their components shall meet or exceed the requirements for ~~AWI [premium] [~~custom~~]~~ grade unless otherwise indicated in this specification. Cabinet style, shall be ~~[~~flush overlay~~] [reveal overlay] [flush inset without face frame] [flush inset with face frame] [as indicated on the drawings]~~.

2.9.1 Base and Wall Cabinet Case Body

2.9.1.1 Cabinet Components

Frame members shall be glued-together, kiln-dried hardwood lumber. Top corners, bottom corners, and cabinet bottoms shall be braced with either hardwood blocks or water-resistant glue and nailed in place metal or plastic corner braces. Cabinet components shall be constructed from the following materials and thicknesses:

2.9.1.1.1 Body Members (Ends, Divisions, Bottoms, and Tops)

~~15 mm to 20~~19 mm ~~[veneered particleboard per JIS A 5908] [medium density fiberboard (MDF)] [~~veneer core plywood~~]~~ panel product

2.9.1.1.2 Face Frames and Rails

~~15 mm to 20~~19 mm ~~[~~hardwood lumber~~] [panel product]~~

2.9.1.1.3 Shelving

~~15 mm to 20~~19 mm ~~[veneered particleboard per JIS A 5908 or ISO 16893] [medium density fiberboard (MDF)] [~~veneer core plywood~~]~~ panel product

2.9.1.1.4 Cabinet Backs

~~5~~6 mm ~~[veneered particleboard per JIS A 5908 or ISO 16893] [medium density fiberboard (MDF)] [~~veneer core plywood~~]~~ panel product

2.9.1.1.5 Drawer Sides, Backs, and Subfronts

13 mm ~~[~~hardwood lumber~~] [panel product]~~

2.9.1.1.6 Drawer Bottoms

~~5~~6 mm ~~[veneered particleboard per JIS A 5908 or ISO 16893] [medium density fiberboard (MDF)] [~~veneer core plywood~~]~~ panel product

2.9.1.1.7 Door and Drawer Fronts

~~15 mm to 20 mm [veneered particleboard] [medium density fiberboard (MDF)] panel product~~19 mm veneer core plywood

2.9.1.2 Joinery Method for Case Body Members

2.9.1.2.1 Tops, Exposed Ends, and Bottoms

a. Steel "European" assembly screws (37 mm from end, 128 mm on center, fasteners will not be visible on exposed parts).

b. Doweled, glued under pressure (approx. 4 dowels per 300 mm of joint).

c. Stop dado, glued under pressure, and either nailed, stapled or screwed (fasteners will not be visible on exposed parts).

d. Spline or biscuit, glued under pressure.

2.9.1.2.2 Exposed End Corner and Face Frame Attachment

2.9.1.2.2.1 Mitered Joint

Lock miter or spline or biscuit, glued under pressure (no visible fasteners)

2.9.1.2.2.2 Non-Mitered Joint (90 degree)

Butt joint glued under pressure (no visible fasteners)

2.9.1.2.2.3 Butt Joint

Glued and nailed

2.9.1.2.3 Cabinet Backs (Wall Hung Cabinets)

Wall hung cabinet backs must not be relied upon to support the full weight of the cabinet and its anticipated load for hanging/mounting purposes. Method of back joinery and hanging/mounting mechanisms should transfer the load to case body members. Fabrication method shall be:

2.9.1.2.3.1 Full Bound

Full bound, captured in grooves on cabinet sides, top, and bottom. Cabinet backs for floor standing cabinets shall be side bound, captured in grooves; glued and fastened to top and bottom.

2.9.1.2.3.2 Full Overlay

Full overlay, plant-on backs with minimum back thickness of 13 mm and minimum No. 12 plated (no case hardened) screws spaced a minimum 80 mm on center. Edge of back shall not be exposed on finished sides. Anchor strips are not required when so attached.

2.9.1.2.3.3 Side Bound

Side bound, captured in groove or rabbetts; glued and fastened.

2.9.1.2.4 Cabinet Backs (Floor Standing Cabinets)

2.9.1.2.4.1 Side Bound

Side bound, captured in grooves; glued and fastened to top and bottom.

2.9.1.2.4.2 Full Overlay

Full overlay, plant-on backs with minimum back thickness of 13 mm and minimum No. 12 plated (no case hardened) screws spaced a minimum 80 mm on center. Edge of back shall not be exposed on finished sides. Anchor strips are not required when so attached.

2.9.1.2.4.3 Side Bound with Rabbetts

Side bound, placed in rabbetts; glued and fastened in rabbetts.

2.9.1.2.5 Wall Anchor Strips

Wall Anchor Strips shall be required for all cabinets with backs less than

13 mm thick. Strips shall consist of minimum 13 mm thick lumber, minimum 60 mm width; securely attached to wall side of cabinet back - top and bottom for wall hung cabinets, top only for floor standing cabinets.

2.9.2 Cabinet Floor Base

Floor cabinets shall be mounted on a base constructed of ~~[~~nominal 50 mm thick lumber~~] [19 mm particleboard] [19 mm fiberboard] [19 mm veneer core exterior plywood]~~. Base assembly components shall be ~~[~~treated lumber~~] [a moisture-resistant panel product]~~. Finished height for each cabinet base shall be ~~[not less than the full height of the installed, specified wall base] [~~as indicated on the drawings~~]~~. Bottom edge of the cabinet door or drawer face shall ~~[~~be flush with top of base~~] [extend below the top of the base as indicated on the drawings]~~.

2.9.3 Cabinet Door and Drawer Fronts

Door and drawer fronts shall be fabricated from ~~[15 mm medium density particleboard] [15 mm medium density fiberboard (MDF]~~19 mm veneer core plywood. All door and drawer front edges shall be surfaced with ~~[~~high pressure plastic laminate~~] [PVC edgebanding], color and pattern [to match exterior face laminate] [as indicated on the drawings] [as indicated in Section 09 06 00 SCHEDULES FOR FINISHES]~~.

2.9.4 Drawer Assembly

2.9.4.1 Drawer Components

Drawer components shall consist of a removable drawer front, sides, backs, and bottom. Drawer components shall be constructed of the following materials and thicknesses:

~~2.9.4.1.1 Drawer Sides and Backs For Transparent Finish~~

~~13 mm thick [solid hardwood lumber] [7-ply hardwood veneer core plywood (no voids), any species]~~

2.9.4.1.1 Drawer Sides and Backs For Laminate Finish

13 mm thick 7-ply hardwood veneer core substrate

2.9.4.1.2 Drawer Sides and Back For Thermoset Decorative Overlay (Melamine) Finish

13 mm thick ~~medium density particleboard or MDF fiberboard substrate~~7 ply hardwood veneer core substrate

2.9.4.1.3 Drawer Bottom

6 mm thick ~~[veneer core panel product for transparent or plastic laminate finish] [~~thermoset decorative overlay melamine panel product~~]~~

2.9.4.2 Drawer Assembly Joinery Method

a. Multiple dovetail (all corners) or French dovetail front/dadoed back, glued under pressure.

b. Doweled, glued under pressure.

c. Lock shoulder, glued and pin nailed.

d. Bottoms shall be set into sides, front, and back, 6 mm deep groove with a minimum 9 mm standing shoulder.

2.9.5 Shelving

2.9.5.1 General Requirements

Shelving shall be fabricated from ~~[15~~19 mm ~~medium density particleboard] [15 mm medium density fiberboard (MDF] [15 mm~~ veneer core plywood~~]~~. All shelving top and bottom surfaces shall be finished with ~~[~~HPDL plastic laminate~~] [thermoset decorative overlay (melamine)]. Shelf edges shall be finished in a [HPDL plastic laminate] [thermoset decorative overlay (melamine)] [PVC edgebanding]~~.

2.9.5.2 Shelf Support System

The shelf support system shall be:

2.9.5.2.1 Recessed (Mortised) Metal Shelf Standards

Mortise standards flush with the finishes surface of the cabinet interior side walls, two per side. Position and space standards on the side walls to provide a stable shelf surface that eliminates tipping when shelf front is weighted. Install and adjust standards vertically to provide a level, stable shelf surface when clips are in place.

~~2.9.5.2.2 Pin Hole Method~~

~~Drill holes on the interior surface of the cabinet side walls. Evenly space holes in two vertical columns Space the holes in each column at [25 mm] [_____] increments starting [150 mm] [_____] from the cabinet interior bottom and extending to within [150 mm] [_____] of the top interior surface of the cabinet. Drill holes to provide a level, stable surface when the shelf is resting on the shelf pins. Coordinate hole diameter with pin insert size to provide a firm, tight fit.~~

2.9.6 Edge Style

Front ~~[~~and exposed side~~]~~ countertop edges shall be in shapes and to dimensions as shown on the drawings.~~ The countertop edge material shall be:~~

~~2.9.6.1 Post Formed Plastic Laminate~~

~~Laminate edge shall be integral with countertop surface. Shape and profile shall be [bullnose] [waterfall] [as indicated] [_____] and to dimensions as indicated.~~

~~2.9.6.2 Hardwood~~

~~Species, finish, profile, shape, and dimensions shall be as indicated on the drawings. Hardwood edge shall overlap the exposed countertop laminate edge and shall be installed flush with the countertop laminate surface.~~

~~2.9.6.3 Vinyl~~

~~Vinyl tee-mould edge shall be in shape, thickness, and color as indicated~~

~~on the drawings. Tee mould edge shall overlap the exposed countertop laminate edge and shall be installed flush with the countertop laminate surface.~~

~~2.9.6.4 Plastic Laminate Self Edge~~

~~Flat, 90 degree "self " edge. Edge must be applied before top. Laminate edge shall overlap countertop laminate and shall be eased to eliminate sharp corners.~~

~~2.9.7 Laminate Clad Splashes~~

~~Countertop splash substrate shall be 20 mm [particleboard] [MDF fiberboard] [veneer core plywood]. Laminate clad backsplash shall be [integral with countertop, coved to radius and to dimensions as indicated on the drawings] [loose, to be installed at the time of countertop installation]. Side splashes shall be straight profile and provided loose, to be installed at the time of countertop installation. Back and side splash laminate pattern and color shall match the adjacent countertop laminate.~~

2.9.7 Laminate Application

Laminate application to substrates shall follow the recommended procedures and instructions of the laminate manufacturer and JIS K 6903, using tools and devices specifically designed for laminate fabrication and application. Provide a balanced backer sheet ~~(Grade BK)~~ wherever only one surface of the component substrate requires a plastic laminate finish. Apply required grade of laminate in full uninterrupted sheets consistent with manufactured sizes using one piece for full length only, using adhesives specified herein or as recommended by the manufacturer. Fit corners and joints hairline. All laminate edges shall be machined flush, filed, sanded, or buffed to remove machine marks and eased (sharp corners removed). Clean up at easing shall be such that no overlap of the member eased is visible. Fabrication shall conform to laminate types and grades for component surfaces and shall be as follows unless otherwise indicated on the drawings:

2.9.7.1 Base/Wall Cabinet Case Body

a. Exterior (exposed) surfaces to include exposed and semi-exposed face frame surfaces: HPDL Grade ~~[~~VGS~~] [VGP]~~.

b. Interior (semi-exposed) surfaces to include interior back wall, bottom, and side walls: ~~[HPDL Grade CLS] [~~Thermoset Decorative Overlay (melamine)~~]~~.

2.9.7.2 Adjustable Shelving

2.9.7.2.1 Top and Bottom Surfaces

~~[~~HPDL Grade HGS~~] [Thermoset Decorative Overlay (melamine)]~~

2.9.7.2.2 All Edges

~~[~~HPDL Grade VGS~~] [Thermoset Decorative Overlay (melamine)][PVC edgebanding]~~

2.9.7.3 Fixed Shelving

2.9.7.3.1 Top and Bottom Surfaces

~~[~~HPDL Grade HGS~~] [Thermoset Decorative Overlay (melamine)]~~

2.9.7.3.2 Exposed Edges

~~[~~HPDL Grade VGS~~] [Thermoset Decorative Overlay (melamine)][PVC edgebanding]~~

2.9.7.4 Door, Drawer Fronts, Access Panels

2.9.7.4.1 Exterior (Exposed) and Interior (Semi-Exposed) Faces

HPDL Grade ~~[~~VGS~~] [VGP]~~

2.9.7.4.2 Edges

~~[~~HPDL Grade VGS~~] [PVC edgebanding]~~

2.9.7.5 Drawer Assembly

All interior and exterior surfaces: ~~[~~HPDL Grade CLS~~] [Thermoset Decorative Overlay (melamine)]~~.

~~2.9.7.6 Countertops and Splashes~~

~~All exposed and semi-exposed surfaces: HPDL Grade HGS~~

2.9.7.6 Tolerances

Flushness, flatness, and joint tolerances of laminated surfaces shall meet the ~~[premium] [~~custom~~]~~ grade requirements.

2.9.8 Finishing

2.9.8.1 Filling

No fasteners shall be exposed on laminated surfaces. All nails, screws, and other fasteners in non-laminated cabinet components shall be countersunk and the holes filled with wood filler consistent in color with the wood species.

~~2.9.8.2 Sanding~~

~~All surfaces requiring coatings shall be prepared by sanding with a grit and in a manner that scratches will not show in the final system.~~

~~2.9.8.3 Coatings~~

~~Types, method of application and location of casework finishes shall be in accordance with the finish schedule, drawings and Section 09 90 00 PAINTS AND COATINGS. All cabinet reveals shall be painted. Submit descriptive data which provides narrative written verification of all types of construction materials and finishes, methods of construction, etc. not clearly illustrated on the submitted shop drawings. Data shall provide written verification of conformance for the quality indicated to include materials, tolerances, and types of construction. Both the manufacturer of materials and the fabricator shall submit available literature which~~

~~describes re-cycled product content, operations and processes in place that support efficient use of natural resources, energy efficiency, emissions of ozone depleting chemicals, management of water and operational waste, indoor environmental quality, and other production techniques supporting sustainable design and products.~~

PART 3 EXECUTION

3.1 INSTALLATION

Installation shall comply with applicable requirements for ~~[premium] [~~ custom~~]~~ quality standards. Countertops and fabricated assemblies shall be installed level, plumb, and true to line, in locations shown on the drawings. Cabinets and other laminate clad casework assemblies shall be attached and anchored securely to the floor and walls with mechanical fasteners that are appropriate for the wall and floor construction.

3.1.1 Anchoring Systems

3.1.1.1 Floor

~~[~~Base cabinets~~]~~ ~~[______]~~ shall utilize a floor anchoring system~~ [as detailed on the drawings]~~. Anchoring and mechanical fasteners shall not be visible from the finished side of the casework assembly. ~~[~~Cabinet~~]~~ ~~[______]~~and other laminate clad casework assemblies shall be attached to anchored bases without visible fasteners~~ [as indicated in the drawings]~~. Where assembly abuts a wall surface, anchoring shall include a minimum 13 mm thick lumber or panel product hanging strip, minimum 60 mm width; securely attached to the top of the wall side of the cabinet back.

3.1.1.2 Wall

~~[~~Cabinet~~]~~ ~~[vanity] [______]~~ to be wall mounted shall utilize minimum 13 mm thick lumber or panel product hanging strips, minimum 60 mm width; securely attached to the wall side of the cabinet back, both top and bottom.

~~3.1.2 Countertops~~

~~Countertops shall be installed in locations as indicated on the drawings. Countertops shall be fastened to supporting casework structure with mechanical fasteners, hidden from view. All joints formed by the countertop or countertop splash and adjacent wall surfaces shall be filled with a clear silicone caulk. Loose [back] [side] splashes shall be adhered to both the countertop surface perimeter and the adjacent wall surface with adhesives appropriate for the type of materials to be adhered. Joints between the countertop surface and splash shall be filled with clear silicone caulk in a smooth consistent concave bead. Bead size shall be the minimum necessary to fill the joint and any surrounding voids or cracks.~~

3.1.2 Hardware

Casework hardware shall be installed in types and locations as indicated on the drawings.~~ Where fully concealed European-style hinges are specified to be used with particleboard or fiberboard doors, the use of plastic or synthetic insertion dowels shall be used to receive 5 mm "Euroscrews". The use of wood screws without insertion dowels is prohibited.~~

3.1.3 Doors, Drawers and Removable Panels

The fitting of doors, drawers and removable panels shall be accomplished within target fitting tolerances for gaps and flushness in accordance with ~~[~~premium~~]~~ ~~[custom]~~ grade requirements.

3.1.4 Plumbing Fixtures

Install sinks, sink hardware, and other plumbing fixtures in locations as indicated on the drawings and in accordance with ~~[~~Section 22 00 00 PLUMBING, GENERAL PURPOSE~~] [_____]~~.

~~3.1.5 Glass~~

~~Install glass and glazing in the casework using methods and materials specified in Section 08 81 00 GLAZING in locations as indicated on the drawings.~~

-- End of Section --

SECTION 06 61 16

SOLID SURFACING FABRICATIONS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D570	(1998; E 2010; R 2010) Standard Test Method for Water Absorption of Plastics
ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D696	(2016) Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30 degrees C and 30 degrees C With a Vitreous Silica Dilatometer
ASTM D790	(2017) Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D2583	(2013a) Indentation Hardness of Rigid Plastics by Means of a Barcol Impressor
ASTM E84	(2020) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM G21	(2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

INTERNATIONAL CAST POLYMER ASSOCIATION (ICPA)

ICPA SS-1	(2001) Performance Standard for Solid Surface Materials

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS K 6902	(2022) Testing Method for Laminated Thermosetting High-Pressure Decorative Sheets
JIS K 7060	(1995) Testing Method for Barcol Hardness of Glass Fiber Reinforced Plastics

JIS K 7161-1	(2014) Plastics - Determination of Tensile Properties - Part 1: General Principles
JIS K 7161-2	(2014) Plastics - Determination of Tensile Properties - Part 2: Test Conditions for Moulding and Extrusion Plastics
JIS K 7209	(2000) Plastics - Determination of Water Absorption
JIS Z 2801	(2012) Antibacterial Processed Products - Antibacterial Test Method / Antibacterial Effect

JAPANESE ADHESIVE INDUSTRY ASSOCIATION (JAIA)

JAIA 4VOC	(2010) Voluntary VOC Regulating Rule for Indoor Air Pollution Control

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3	(2005) Standard for High-Pressure Decorative Laminates

NSF INTERNATIONAL (NSF)

NSF/ANSI 51	(2012) Food Equipment Materials

1.2 SYSTEM DESCRIPTION

a. Work under this section includes ~~[_____]~~countertops and other items utilizing solid surfacing material fabrications as indicated on the drawings and as described in this specification. Do not change source of supply for materials after work has started, if the appearance of finished work would be affected.

b. In most instances, installation of solid surfacing material fabricated components and assemblies requires strong correctly located structural support provided by other trades. To provide a stable, sound, secure installation, close coordination is required between the solid surfacing material fabricator/installer and other trades to ensure that necessary structural wall support, cabinet counter top structural support, proper clearances, and other supporting components are provided for the installation of ~~wall panels,~~ counter tops, ~~shelving,~~ and all other solid surfacing material fabrications to the degree and extent recommended by the solid surfacing material manufacturer.

c. Provide appropriate staging areas for solid surfacing material fabrications. Allow variation in component size and location of openings of plus or minus 3 mm.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S"~~ classification. Submittals not having a "G" ~~or "S"~~ classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Fabrication Drawings; G~~[, [_____]]~~

Installation; G~~[, [_____]]~~

SD-03 Product Data

Solid Polymer; G~~[, [_____]]~~

Indoor air quality for solid surface seam and sealant products~~; S [, [_____]]~~

~~[Quartz Agglomerate Material; G[, [_____]]~~

~~]~~ SD-04 Samples

Material; G~~[, [_____]]~~

Counter Tops; G~~[, [_____]]~~

SD-06 Test Reports

Test Report Results

SD-07 Certificates

Qualifications

Indoor Air Quality for solid surface fabrication products~~; S[, [_____]]~~

SD-10 Operation and Maintenance Data

Solid Polymer, Data Package 1; G~~[, [_____]]~~

~~[Quartz Agglomerate Material, Data Package 1; G[, [_____]]~~

~~]~~1.4 QUALITY ASSURANCE

1.4.1 Qualifications

To ensure warranty coverage, provide manufacturer certified solid surfacing fabricators to fabricate the solid surfacing material being utilized. Mark all fabrications with the fabricator's certification label affixed in an inconspicuous location. Minimum of 5 years of experience working with solid surfacing materials is required of fabricators. Submit solid surfacing material manufacturer's certification attesting to fabricator qualification approval.

1.4.2 Mock-ups

Submit Detail Fabrication Drawings indicating locations, dimensions, component sizes, fabrication and joint details, attachment provisions, installation details, and coordination requirements with adjacent work. Prior to final approval of shop drawings, provide a full-size mock-up of a typical ~~[~~counter top~~] [shelving] [_____]~~ where multiple units are required. Include all solid surfacing material components required to provide a completed unit. Utilize finishes in patterns and colors~~[as specified in Section 09 06 00 SCHEDULES FOR FINISHES.][~~ as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.~~]~~ in the mock-up. Should the mock-up not be approved, re-work or remake it until approval is secured. Remove rejected units from the jobsite. Approved mock-up may remain as part of the finished work.

1.5 DELIVERY, STORAGE, AND HANDLING

Do not deliver materials to project site until areas are ready for installation. Deliver components and materials to the site undamaged, in containers clearly marked and labeled with manufacturer's name. Store materials indoors and take adequate precautions to prevent damage to finished surfaces. Provide protective coverings to prevent physical damage or staining following installation, for duration of project.

1.6 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials~~[~~, excluding damages caused by physical or chemical abuse or excessive heat,~~]~~ and workmanship for a period of ~~[~~10~~][______]~~ years from date of final acceptance of the work.

PART 2 PRODUCTS

2.1 MATERIAL

Submit detail fabrication drawings and installation drawings of each solid surfacing fabrication indicated. Include elevations, dimensions, clearances, details of construction and anchorage, and details of joints and connections.

Submit manufacturers' descriptive product data for ~~[each type of]~~ solid polymer fabrication ~~[and quartz agglomerate fabrication]~~ indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for ~~[each type of]~~ solid polymer fabrication ~~[and quartz agglomerate material fabrication]~~ in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

2.1.1 Solid Surfacing Material

Provide solid polymer~~[and][quartz agglomerate material]~~ that is a homogeneous filled solid polymer; not coated, laminated or of a composite construction, complying with ICPA SS-1~~[and ICPA SS-1 for quartz agglomerate, except for composition]~~. Provide material that meets or exceeds the minimum physical and performance properties specified. Superficial damage to a depth of 0.25 mm must be repairable by sanding or polishing. Material thickness is as~~[indicated below][~~ indicated on the drawings~~]~~; required minimum thickness is 6 mm. Submit a minimum ~~102 by 102 mm~~100 mm by 100 mm sample of each color and pattern for approval; include full range of color and pattern variation. Retain approved samples as a standard for this work. Submit test report results from an independent testing laboratory attesting that the submitted solid surfacing materials meet or exceed each of the specified performance requirements.

a. Horizontal Surfaces:~~[13 mm thick material][~~ not less than~~ ~~ 19 mm thick material~~][_____]~~

b. Vertical Surfaces:~~[6 mm thick material][~~ not less than~~ ~~ 13 mm thick material~~][_____]~~

c. Provide materials ~~that meet the emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type)~~certified to meet indoor air quality requirements to attain F 4-Star rating. Provide certification or validation of indoor air quality for solid surface fabrication products.

2.1.2 Cast, 100 Percent Acrylic Polymer Solid Surfacing Material

Cast, 100 percent acrylic solid polymer material composed of acrylic polymer, mineral fillers, and pigments. Provide acrylic polymer that meets or exceeds the following minimum performance requirements:

~~PROPERTY~~	~~REQUIREMENT (min. or max.)~~	~~TEST PROCEDURE~~
~~Tensile Strength~~	~~291 kg/cm2 (max.)~~	~~ASTM D638~~
~~Hardness~~	~~55-Barcol Impressor (min.)~~	~~ASTM D2583~~
~~Thermal Expansion~~	~~.0000386cm/cm/deg C (max.)~~	~~ASTM D696~~
~~Boiling Water Surface Resistance~~	~~No Change~~	~~ANSI/NEMA LD 3-3.05~~
~~High Temperature Resistance~~	~~No Change~~	~~ANSI/NEMA LD 3-3.06~~
~~Impact Resistance (Ball drop)~~		~~ANSI/NEMA LD 3-303~~
~~19 mm sheet~~	~~5080 mm, 227 g ball, no failure~~	
~~Mold & Mildew Growth~~	~~No growth~~	~~ASTM G21~~
~~Bacteria Growth~~	~~No growth~~	~~ASTM G21~~
~~Liquid Absorption (Weight in 24 hrs.)~~	~~0.1 percent max.~~	~~ASTM D570~~
~~Flammability~~		~~ASTM E84~~
~~Flame Spread~~	~~25 max.~~	
~~Smoke Developed~~	~~30 max.~~	
~~Sanitation~~	~~"Food Contact" approval~~	~~NSF/ANSI 51~~
~~Flexural Strength~~	~~10,400 psi (min.)~~	~~ASTM D790~~

PROPERTY	REQUIREMENT (min. or max.)	TEST PROCEDURE
Tensile Strength	291 kg/cm^2 (max.)	ASTM D638 or JIS K 7161-1 /JIS K 7161-2
Hardness	55-Barcol Impressor (min.)	ASTM D2583 or JIS K 7060
Thermal Expansion	.0000386cm/cm/deg C (max.)	ASTM D696
Boiling Water Surface Resistance	No Change	ANSI/NEMA LD 3-3.05 or JIS K 6902
High Temperature Resistance	No Change	ANSI/NEMA LD 3-3.06 or JIS K 6902
Impact Resistance (Ball drop)		ANSI/NEMA LD 3-303 or JIS K 6902
6 mm sheet	914 mm, 227 g ball, no failure	
13 mm sheet	3556 mm, 227 g ball, no failure	
19 mm sheet	5080 mm, 227 g ball, no failure	
Mold & Mildew Growth	No growth	ASTM G21
Bacteria Growth	No growth	ASTM G21
Liquid Absorption (Weight in 24 hrs.)	0.1 percent max.	ASTM D570 or JIS K 7209
Flammability		ASTM E84
Flame Spread	25 max.	
Smoke Developed	30 max.	
Sanitation	"Food Contact" approval	NSF/ANSI 51 or JIS Z 2801
Flexural Strength	10,400 psi (min.)	ASTM D790

~~2.1.3 Acrylic-modified Polymer Solid Surfacing Material~~

~~Cast, solid polymer material composed of a formulation containing acrylic and polyester polymers, mineral fillers, and pigments. Provide acrylic polymer content not less than 5 percent and not more than 10 percent in order to meet the following minimum performance requirements:~~

~~PROPERTY~~	~~REQUIREMENT (min. or max.)~~	~~TEST PROCEDURE~~
~~Tensile Strength~~	~~288 kg/cm^2 (max.)~~	~~ASTM D638~~
~~Hardness~~	~~50-Barcol Impressor (min.)~~	~~ASTM D2583~~
~~Thermal Expansion~~	~~.0000386cm/cm/deg C (max.)~~	~~ASTM D696~~
~~Boiling Water Surface Resistance~~	~~No Change~~	~~ANSI/NEMA LD 3-3.05~~
~~High Temperature Resistance~~	~~No Change~~	~~ANSI/NEMA LD 3-3.06~~
~~Impact Resistance (Ball drop)~~		~~ANSI/NEMA LD 3-303~~
~~6 mm sheet~~	~~914 mm, 227 g ball, no failure~~	
~~13 mm sheet~~	~~3556 mm, 227 g ball, no failure~~	
~~19 mm sheet~~	~~5080 mm, 227 g ball, no failure~~	
~~Mold & Mildew Growth~~	~~No growth~~	~~ASTM G21~~
~~Bacteria Growth~~	~~No growth~~	~~ASTM G21~~
~~Liquid Absorption (Weight in 24 hrs.)~~	~~0.6 percent max.~~	~~ASTM D570~~
~~Flammability~~		~~ASTM E84~~
~~Flame Spread~~	~~25 max.~~	
~~Smoke Developed~~	~~100 max.~~	
~~Sanitation~~	~~"Food Contact" approval~~	~~NSF/ANSI 51~~
~~Flexural Strength~~	~~[6,800][10,400] psi (min.)~~	~~ASTM D790~~

~~[2.1.4 Quartz Agglomerate (or "Engineered Quartz") Solid Surfacing Material~~

~~Solid sheets consisting of quartz aggregates in an acrylic or polyester, or a combination of the two, resin binder (or matrix) that is solid and nonporous with integral color.~~

~~]~~2.1.3 Material Patterns and Colors

Provide pattern and color for all solid surfacing material components and fabrications ~~[as specified in Section 09 96 00 SCHEDULES FOR FINISHES.][~~as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.~~]~~ Provide products with consistent patterned color throughout thickness of the product.

2.1.4 Surface Finish

Provide a uniform appearance on exposed finished surfaces and edges. Exposed surface finish is ~~[~~matte; gloss rating of 5-20~~] [semigloss; gloss rating of 25-50] [polished; gloss rating of 55-80] [as indicated]~~.

2.2 ACCESSORY PRODUCTS

Provide accessory products, as specified below, as manufactured by the solid surfacing material manufacturer or as approved by the solid surfacing material manufacturer for use with the solid surfacing materials being specified.

2.2.1 Adhesives

Provide a two-part seam adhesive kit to create permanent, inconspicuous, non-porous, hard seams and joints by chemical bond between solid surfacing materials and components to create a monolithic appearance of the fabrication. Provide adhesive approved by the solid surfacing material manufacturer. Color-match adhesive to the surfaces being bonded where solid-colored, solid surfacing materials are being bonded together. Provide clear or color matched seam adhesive where particulate patterned, solid surfacing materials are being bonded together.

2.2.2 Seam and Sealant Emissions

Provide seam and other accessory materials that ~~meet the emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type)~~are certified to meet indoor air quality requirements to attain F 4-Star or JAIA 4VOC rating. Provide validation of indoor air quality for solid surface seam and sealant products.

2.2.3 Silicone Sealant

Provide silicone sealant, mildew-resistant, single-component, nonsag, plus 25 percent and minus 25 percent movement capability, acid-curing; ~~ASTM C920, Type S, Grade NS, Class 25, Use NT;~~ clear formulation; approved for use by the solid surfacing material manufacturer.

2.2.4 Conductive Tape

Provide manufacturer's standard conductive foil tape, 0.1 mm thick, applied around the edges of cut outs containing hot or cold appliances.

~~2.2.5 Insulating Tape~~

~~Provide manufacturer's standard insulating tape for use with drop-in food wells used in commercial food service applications to insulate solid surfacing material from hot or cold appliances.~~

2.2.5 Heat Reflective Tape

Provide heat reflective tape as recommended by the solid surfacing material manufacturer for use with cutouts for heat sources.

2.2.6 Mounting Hardware

Provide mounting hardware, including sink/bowl clips, inserts and fasteners for attachment of undermount sinks and lavatories.

2.3 FABRICATIONS

Provide factory or shop fabricate components to sizes and shapes indicated, to the greatest extent practical, in accordance with approved Shop Drawings and manufacturer's requirements. Provide factory cutouts for sinks, lavatories, and plumbing fixtures where indicated on the drawings. Contours and radii must be routed to template, with edges smooth. Defective and inaccurate work will be rejected. Submit product data indicating product description, fabrication information, and compliance with specified performance requirements for solid surfacing material, joint adhesive, sealants, and heat reflective tape.~~[Both the manufacturer of materials and the fabricator are required to submit a detailed description of operations and processes in place that support efficient use of natural resources, energy efficiency, emissions of ozone depleting chemicals, management of water and operational waste, indoor environmental quality, and other production techniques supporting sustainable design and products.]~~

2.3.1 Joints and Seams

Form joints and seams between solid surfacing material components using manufacturer's approved seam adhesive. Provide inconspicuous joints in appearance without voids to create a monolithic appearance.

2.3.2 Edge Finishing

Rout and finish component edges to a smooth, uniform appearance and finish. Provide edge shapes and treatments, including any inserts, as detailed on the drawings. Rout all cutouts, then sand all edges smooth. Repair or reject defective or inaccurate work.

2.3.3 Counter Top Splashes

Fabricate backsplashes and end splashes from ~~[~~13 mm~~]~~ ~~[_____]~~ thick solid surfacing material to be ~~[[102~~100 mm~~]~~ ~~[_____]~~ high~~]~~ ~~[~~in conformance with dimensions and shapes as indicated~~]~~. Provide backsplashes and end splashes ~~[~~for all counter tops~~]~~ and ~~[~~at locations indicated~~]~~. Shop fabricate backsplashes and provide ~~[~~permanently ~~attached] [~~loose, to be field attached~~]~~.

~~2.3.3.1 Permanently Attached Backsplash~~

~~Provide permanently attached backsplashes [straight with seam adhesive to form a 90 degree transition] [with seam adhesive and to form a radiused coved transition from counter top to backsplash].~~

2.3.3.1 End Splashes

Provide end splashes loose for installation at the jobsite after horizontal surfaces to which they are to be attached have been installed.

~~2.3.4 Shelving~~

~~Fabricate shelving [and wall support brackets] from [13 mm] [_____] thick solid surfacing material; dimensions, edge shape, and other details as indicated.~~

~~2.3.5 Window Stools~~

~~Fabricate window stools from [13 mm] [_____] thick solid surfacing material; dimensions, edge shape, and other details [as indicated] [as selected from manufacturer's available pre-fabricated standards] [equal to the width of the window opening by a 13 mm overhang of the window sill depth] [_____]. Provide [square][bullnose][_____] edge profile.~~

2.3.4 Counter Tops

Fabricate all solid surfacing material, counter top components from~~[13 mm][~~ 19 mm ~~[_____]~~ thick material. Indicate details, dimensions, locations, and quantities on the drawings. Provide counter tops with~~[~~ ~~102~~100 mm~~] [_____]~~ high ~~[~~loose~~] [permanently attached, 90 degrees transition] [permanently attached with coved transition backsplash and loose endsplashes]~~ ~~[~~at all locations~~] as indicated]~~. Attach 51 mm wide reinforcing strip of solid surfacing material under each horizontal counter top seam. Submit a minimum ~~305~~300 mm wide by ~~152~~100 mm deep, full size sample for each type of counter top shown on the project drawings; include the edge profile and backsplash as detailed on the drawings and at least one seam. Retain approved sample as standard for this work. Provide ~~[~~square~~][bullnose][_____]~~ edge profile.

2.3.4.1 Counter Tops with Sinks

~~[~~ a. Provide stainless steel or vitreous china sink; include cutouts to template for counter tops with sinks as furnished by the sink manufacturer. Provide manufacturer's standard sink mounting hardware for ~~[stainless steel] [vitreous china] [rimless] [_____]~~ installation. Seal between sink and counter top with specified silicone sealant. Provide sink, faucet, and plumbing requirements in accordance with Section 22 00 00 PLUMBING,GENERAL PURPOSE.~~ [_____]~~

~~][b. Provide manufacturer's standard solid polymer sinks, pre-molded product specifically designed for attachment to solid surfacing material counter tops. See paragraph SOLID POLYMER SINKS for additional requirements.~~

~~][]2.3.5 Toilet Partition System~~

~~Refer to Section 10 21 13 TOILET COMPARTMENTS.~~

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Components

Install all components and fabricated units plumb, level, and rigid. Make field joints between solid surfacing material components using solid surfacing material manufacturer's approved seam adhesives, to provide a monolithic appearance with joints inconspicuous in the finished work. Attach metal or vitreous china sinks and lavatory bowls to counter tops using solid surfacing material manufacturer's recommended clear silicone sealant and mounting hardware. Install solid polymer sinks and bowls using a color-matched seam adhesive.

3.1.1.1 Loose Counter Top Splashes

Mount loose splashes in the locations noted on the drawings. Adhere loose splashes to the counter top with a color matched silicone sealant when the solid surfacing material components are solid colors. Use a clear silicone sealant to provide adhesion of particulate patterned solid surfacing material splashes to counter tops.

3.1.2 Silicone Sealant

Use specified silicone sealant to seal all expansion joints between solid surfacing material components and all joints between solid surfacing material components and other adjacent surfaces such as walls, floors, ceiling, and plumbing fixtures. Provide sealant bead smooth and uniform in appearance and minimum size necessary to bridge any gaps between the solid surfacing material and the adjacent surface. Provide continuous bead and run the entire length of the joint being sealed.

3.1.3 Plumbing

Make plumbing connections to sinks and lavatories in accordance with Section ~~[~~22 00 00 PLUMBING, GENERAL PURPOSE~~] [_____]~~.

3.2 CLEAN-UP

Components must be cleaned after installation and covered to protect against damage during completion of the remaining project items. Damaged components must be repaired or replaced at the Contractor's sole expense.

-- End of Section --

SECTION 07 21 13

BOARD AND BLOCK INSULATION
02/16, CHG 2: 08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C930	(2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories
ASTM D3833/D3833M	(1996; R 2011) Water Vapor Transmission of Pressure-Sensitive Tapes
ASTM E84	(2020) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(2016) Standard Test Methods for Water Vapor Transmission of Materials
ASTM E119	(2020) Standard Test Methods for Fire Tests of Building Construction and Materials

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2018) International Building Code

JAPANESE ADHESIVE INDUSTRY ASSOCIATION (JAIA)

JAIA 4VOC	(2010) Voluntary VOC Regulating Rule for Indoor Air Pollution Control

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 255	(2006; Errata 2006) Standard Method of Test of Surface Burning Characteristics of Building Materials

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS	SCS Global Services (SCS) Indoor Advantage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134	Respiratory Protection

UNDERWRITERS LABORATORIES (UL)

UL 723	(2018) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials
UL 2818	(2013) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Standard Details; G~~[, [_____]]~~

Block or Board Insulation; G~~[, [_____]]~~

~~Vapor Retarder; G[, [_____]]~~

Pressure Sensitive Tape; G~~[, [_____]]~~

~~Protection Board or Coatings; G[, [_____]]~~

Accessories including sealants; G~~[, [_____]]~~

Recycled Content for Block or Board Insulation; S

SD-07 Certificates

Block or Board Insulation; G~~[, [_____]]~~

~~Vapor Retarder; G[, [_____]]~~

~~Protection Board or Coating; G[, [_____]]~~

Draft Special Warranties; G~~[, [_____]]~~

Final Special Warranties; G~~[, [_____]]~~

Indoor Air Quality For Block Or Board Insulation; S

SD-08 Manufacturer's Instructions

Block or Board Insulation

Adhesive

1.3 PERFORMANCE REQUIREMENTS

Japanese manufactured products will be accepted if the insulation meets the inherent properties as indicated in this specification section and drawings. Inherent properties include, but are not limited to, dimensions; thermal resistance; surface burning characteristics; critical radiant flux; water vapor permanence; water vapor sorption; odor emission; corrosiveness; and fungi resistance and sustainability requirements. The Contractor shall bear responsibility of all components as the property of thermal resistance for a specific thickness of blanket is only part of the total thermal performance of a building element such as a wall, ceiling, and floor. The Contractor shall bear responsibility for all testing required to ensure requirements set forth in this specification are met.

1.3.1 Thermal and Acoustical

Thermal and acoustical insulation with a flame spread (FS) rating not higher than 75, and a smoke developed (SD) rating not higher than 150 when tested in accordance with ASTM E84, NFPA 255 or UL 723.

1.3.2 Exposed Insulation

Exposed insulation in concealed spaces of sprinklered buildings must have flame spread of 25 or less and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84, NFPA 255 or UL 723. Acceptable types of insulation blankets are Type I, Type II (Class A only) and Type III (Class A only).

1.3.3 Flame Spread - No Smoke Developed Rating Limitation

Compliance with the SD rating limitation is not required, and a FS rating up to 100 is permitted for insulation, including insulating sheathing installed within wall assemblies. In such assemblies, conform to the requirements for interior finish with a minimum fire resistance rating of 15 minutes when tested in accordance with ASTM E119.

1.3.4 No Flame Spread or Smoke Limitation

Compliance with FS and SD limitations are not required for the following applications.

a. Insulation installed above poured concrete or poured gypsum roof decks, nominal 50 mm thick tongue-and-grooved wood plank roof decks or precast roof panels or plants that are approved by an approved testing laboratory, as noncombustible roof deck construction.

b. Insulation installed above roof decks where the entire roof construction assembly, including the insulation, is UL-listed as Fire Classified, or FM-approved for Class I roof deck construction or equal listing or classification by an approved testing laboratory.

c. Insulation contained entirely within panels where the entire panel assembly used in the construction application meets the cited FS and SD limitations.

d. Insulation isolated from the interior of the building by masonry walls, masonry cavity walls, insulation encased in masonry cores, or concrete floors.

e. Insulation installed over concrete slabs and completely covered by wood tongue-and-groove flooring without creating air spaces within the flooring system.

f. Insulation completely enclosed in hollow metal doors.

g. Insulation installed between new exterior siding materials and existing siding or wood board, plywood, fiberboard, or gypsum exterior wall sheathing.

1.4 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

1.5 PRODUCT DATA

Include data for material descriptions, recommendations for product shelf life, ~~requirements for protection board or coatings,~~ and precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

~~[~~1.6 CERTIFICATIONS

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Japanese manufactured products must be certified to meet indoor air quality requirements to attain F 4-Star of JAIA 4VOC rating. Provide current product certification documentation from certification body.

~~]~~1.7 DELIVERY, STORAGE, AND HANDLING

1.7.1 Delivery

Deliver materials to the site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

1.7.2 Storage

Inspect materials delivered to the site for damage and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Comply with manufacturer's recommendations for handling, storage, and protection of materials before and during installation.

1.8 SAFETY PRECAUTIONS

~~[~~1.8.1 Respirators

Provide installers with dust/mist respirators, training in their use, and protective clothing, all approved by the National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) and in accordance with 29 CFR 1910.134.

~~]~~1.8.2 Other Safety Considerations

Comply with the safety requirements of ASTM C930.

1.9 SPECIAL WARRANTIES

1.9.1 Guarantee

Guarantee insulation installation against failure due to ultraviolet light exposure for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft and final guarantees in accordance with Sections ~~01 78 00 CLOSEOUT SUBMITTALS~~01 77 00.00 20 CLOSEOUT PROCEDURES ~~[~~and 01 78 23 OPERATION AND MAINTENANCE DATA~~]~~.

1.9.2 Warranty

Provide manufacturer's material warranty for all system components for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft and final warranties in accordance with Sections ~~01 78 00 CLOSEOUT SUBMITTALS~~01 77 00.00 20 CLOSEOUT PROCEDURES ~~[~~and 01 78 23 OPERATION AND MAINTENANCE DATA~~]~~.

PART 2 PRODUCTS

2.1 BLOCK OR BOARD INSULATION

Provide thermal insulating materials as recommended by manufacturer for each type of application indicated. Provide insulation with the following physical properties and in accordance with the following standards:

~~[a. Cellular Glass: ASTM C552~~

~~][b. Extruded Preformed Cellular Polystyrene: ASTM C578 REV A~~

~~][c~~a. Mineral Fiber Block and Board Exterior Wall Insulation: ASTM C612

~~][d. Unfaced Preformed Rigid Polyurethane and Polyisocyanurate Board: ASTM C591~~

~~][e. Faced Rigid Cellular Polyisocyanurate and Polyurethane Insulation: ASTM C1289 REV A~~

~~][(1) Type I Aluminum Foil on both major surfaces. [Class 1 - Non-reinforced core foam.] [Class 2 - Glass fiber reinforced core.]~~

~~][(2) Type II Fibrous felt or glass fiber mat membrane on both major surfaces of the core foam.~~

~~][(3) Type III Perlite insulation board on one major surface of the core foam and a fibrous felt or glass fiber mat membrane on the other major surface of the core foam.~~

~~][(4) Type IV Cellulosic fiber insulating board on the one major surface of the core foam and fibrous felt or glass fiber mat membrane on the other major surface of the core foam.~~

~~][(5) Type V Oriented strand board or water board on one major surface of the core foam and fibrous felt or glass fiber mat membrane or aluminum foil on the other major surface of the core foam.~~

~~][(6) Type VI Perlite insulation board on both major surfaces of the core foam.~~

~~]~~2.1.1 Thermal Resistance

~~[Unless otherwise indicated, Ceiling R-[_____] Wall R-[_____] Floor [R-_____].~~Unless otherwise indicated:

Wall: Minimum R-5.7 (IP) continuous; R-1.00377 (SI) continuous

Roof: Minimum R-25 (IP) continuous; R-4.4025 (SI) continuous

~~]~~2.1.2 Fire Protection Requirements

a. Flame spread index of 75 or less when tested in accordance with ASTM E84.

b. Smoke developed index of ~~[450] [200] [~~150~~]~~ ~~[_____]~~ or less when tested in accordance with ASTM E84.

c. Exposed insulation in concealed spaces of sprinklered buildings must be specified to have a flame spread of 25 or less and a smoke developed rating of 50 or less.

~~c~~d. Provide insulated assemblies in accordance ICC IBC Chapter Fire and Smoke Protection Features.

2.1.3 Other Material Properties

Provide thermal insulating materials with the following properties:

~~[a. Rigid cellular plastics: Compressive Resistance at Yield: Not less than [170] [_____] kilopascals (kPa) when measured according to ASTM D1621.~~

~~][b. Mineral fiber board: Compressive strength: Minimum load required to produce a reduction in thickness of 10 percent kilograms per square meter (kg/m2): [120] [4900] when tested according to ASTM C165.~~

~~][c. Block-type insulation: Block-type insulation: Flexural strength: Not less than [275] [_____] kPa when measured according to ASTM C203 REV A.~~

~~][d~~a. Water Vapor Permeance: Not more than ~~[6.3 by 10-8] [_____]~~1.15 by 10 g/Pa.s.m2 or less when measured according to ASTM E96/E96M, desiccant method, in the thickness required to provide the specified thermal resistance, including facings, if any.

~~]~~
~~[e. Water Absorption: Not more than [2] [_____] percent by total immersion, by volume, when measured according to ASTM C272/C272M.~~

~~][f. Water Adsorption: Not more than [1] [_____] percent by volume when measured in accordance with paragraph 14 of ASTM C553.~~

~~]2.1.4 Premolded Concrete Masonry Insert~~

~~Provide in accordance with ASTM C578 REV A. Provide inserts in concrete masonry units that are installed at the masonry unit manufacturing plant. Provide insert with thickness of not less than 32 mm.~~

2.1.4 Recycled Materials

Provide thermal insulation containing recycled materials to the extent practicable, provided that the material meets all other requirements of this section.~~ The minimum required recycled material contents (by weight, not volume) are:~~

~~Polyisocyanurate/Polyurethane:~~	~~9 percent~~
~~Phenolic Rigid Foam:~~	~~5 percent~~
~~Perlite Board:~~	~~75 percent post consumer paper~~

Provide data identifying percentage of recycled content for block or board insulation.

2.1.5 Indoor Air Quality

Provide certification of indoor air quality for block or board insulation.

2.1.6 Prohibited Materials

Do not provide materials containing asbestos.

~~[2.2 VAPOR RETARDER AND DAMPPROOFING~~

~~2.2.1 Vapor Retarder in Framed Walls and Roofs~~

~~[a. 0.15 mm thick polyethylene sheeting conforming to ASTM D4397 and having a water vapor permeance of 5.72 by 10-8g/Pa.s.m2 or less when tested in accordance with ASTM E96/E96M.~~

~~][b. Membrane with the following properties:~~

~~(1) Water Vapor Permeance: ASTM E96/E96M: 5.72 by 10-8 g/Pa.s.m2~~

~~(2) Maximum Flame Spread: ASTM E84: [25] [50] [_____]~~

~~(3) Combustion Characteristics: Passing ASTM E136~~

~~(4) Puncture Resistance: TAPPI T803 OM: [15] [25] [50]~~

~~][2.2.2 Dampproofing for Masonry Cavity Walls~~

~~[Bituminous material is specified in Section 07 11 13 BITUMINOUS~~

~~DAMPPROOFING.] [Parging material is specified in Section 04 20 00 MASONRY.]~~

~~][2.2.3 Vapor Retarder under Floor Slab~~

~~a. Water vapor permeance: 1.14 by 10-8 g/Pa.s.m2 or less when tested in accordance with ASTM E96/E96M.~~

~~b. Puncture resistance: Maximum load no less than 18 kilograms when tested according to ASTM E154/E154M REV A.~~

~~]]~~2.2 PRESSURE SENSITIVE TAPE

As recommended by manufacturer of vapor retarder(s). Match water vapor permeance rating for each vapor retarder specified. Provide tape in accordance with ASTM D3833/D3833M.

~~2.3 PROTECTION BOARD OR COATING~~

~~As recommended by insulation manufacturer.~~

2.3 ACCESSORIES

2.3.1 Adhesive

As recommended by insulation manufacturer.

2.3.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

Prior to installation, ensure all areas that are in contact with the insulation are dry and free of projections that could cause voids, compressed insulation, or punctured vapor retarders. For foundation perimeter or under slab applications, check that subsurface fill is flat, smooth, dry, and well tamped. Do not proceed with installation if moisture or other conditions are present, and notify the Contracting Officer of such conditions. Do not proceed with the work until conditions have been corrected and verified to be dry.

~~3.2 PREPARATION~~

~~3.2.1 Blocking Around Heat Producing Devices~~

~~Provide noncombustible blocking at all spaces between heat producing devices and the floors, ceilings and roofs through which they pass. Provide in accordance with ICC IBC Section 2111.12 Fireplace Blocking and with the following clearances:~~

~~a. Recessed lighting fixtures, including wiring compartments, ballasts, and other heat producing devices, unless certified for installation surrounded by insulation: 75 mm from outside face of fixtures and devices or as required by NFPA 70 and, if insulation is placed above fixture or device, 600 mm above fixture.~~

~~b. Masonry chimneys or masonry enclosing a flue: 50 mm from outside face of masonry. Masonry chimneys for medium and high heat operating appliances: Minimum clearances required by NFPA 211.~~

~~c. Vents and vent connectors used for venting products of combustion, flues, and chimneys other than masonry chimneys: Minimum clearances as required by NFPA 211.~~

~~d. Gas Fired Appliances: Clearances as required in NFPA 54.~~

~~e. Oil Fired Appliances: Clearances as required in NFPA 31.~~

~~Blocking is not required if chimneys or flues are certified in writing by the chimney or flue manufacturer for use in contact with specific insulating materials.~~

3.2 INSTALLATION

3.2.1 Installation and Handling

Provide insulation in accordance with the manufacturer's printed installation instructions. Keep material dry and free of extraneous materials.

3.2.2 Electrical Wiring

Do not install insulation in a manner that would enclose electrical wiring between two layers of insulation.

~~[3.2.3 Cold Climate Requirement~~

~~Place insulation on the outside of pipes.~~

~~]~~3.2.3 Continuity of Insulation

Butt tightly against adjoining boards, studs, ~~rafters, joists, sill plates, headers~~ and obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joint, roof, and floor. Avoid creating thermal bridges and voids. Provide and verify continuity of insulative barrier throughout the building enclosure.

3.2.4 Coordination

Verify final installed insulation thicknesses comply with thicknesses indicated, R-values specified herein, and with the approved insulation submittal(s).

3.3 INSTALLATION ON WALLS

~~3.3.1 Installation using Furring Strips~~

~~Install insulation [between] [on] members as recommended by insulation manufacturer.~~

~~3.3.2 Installation on Masonry Walls~~

~~[Apply board directly to masonry with adhesive or fasteners as recommended by the insulation manufacturer. Fit between obstructions without impaling~~

~~board on ties or anchors. Apply in parallel courses with joints breaking midway over course below. Place boards in moderate contact with adjoining insulation without forcing and without gaps. Cut and shape as required to fit around wall penetrations, projections or openings to accommodate conduit or other utilities. Seal around cutouts with sealant. Install insulation in wall cavities so that it leaves at least a nominal 25 mm air space outside of the insulation to allow for cavity drainage.~~

~~][Insert premolded or board insulation into masonry unit hollow cores as recommended by the insulation manufacturer.~~

~~]~~3.3.1 Adhesive Attachment to Concrete ~~and Masonry~~ Walls

Apply adhesive to wall and completely cover wall with insulation.

~~[~~ a. Full back bed method ~~[~~or~~]~~

~~][~~b. Spot method: Provide at least six spots having diameter of approximately 100 mm, located at each corner and mid points of each of the longer sides of each board.

~~][~~c. As recommended by the insulation manufacturer.

~~]~~ d. Use only full back method for pieces of 0.1 square meter or less.

e. Butt all edges of insulation and seal edges with tape.

3.3.2 Mechanical Attachment on Concrete ~~and Masonry Walls~~

Cut insulation to cover walls. Apply adhesive to wall and set clip or other mechanical fastener in adhesive as recommended by manufacturer. After curing of adhesive, install insulation over fasteners and bend split prongs to provide a flush condition with the insulation. Butt all edges of insulation and seal with tape.

~~[3.3.3 Protection Board or Coating~~

~~Install protection board or coating in accordance with manufacturer's printed instructions. Install protection over all exterior exposed insulation and to 300 mm below grade.~~

~~]3.4 INSTALLATION ON UNDERSIDE OF CONCRETE FLOOR SLAB~~

~~[3.4.1 Mechanically Fastened Systems~~

~~Size insulation to cover underside of slab. Apply adhesive to slab and set fasteners in adhesive as recommended by manufacturer. After curing of adhesive, install insulation over fasteners and bend split prongs to provide a flush condition with the insulation. Butt all edges of insulation and seal with tape.~~

~~][3.4.2 Adhesively Bonded Systems~~

~~Apply adhesive to underside of slab and completely cover wall with insulation.~~

~~[a. Full back bed method [or]~~

~~][b. Spot method: Provide at least six spots having a diameter of~~

~~approximately 100 mm, located at each corner and mid-point of each of the longer sides.~~

~~][c. As recommended by insulation manufacturer.~~

~~] d. Use full back method for insulation pieces 0.1 square meter or less.~~

~~e. Butt all edges of insulation and seal with tape.~~

~~]~~3.4 PERIMETER AND UNDER SLAB INSULATION

~~Install perimeter thermal insulation where heated spaces are adjacent to exterior walls, slab edges in slab-on-grade, or floating slab construction.~~

3.4.1 Manufacturer's Instructions

Layout insulation, tape edges, ~~provide vapor retarder~~ and other required accessories to protection against vermin, insects, and damage in accordance with manufacturer's printed instructions.

~~[~~3.4.2 Insulation on Vertical Surfaces

~~Provide thermal insulation [on exterior of foundation walls] [on grade beams] [partially] [below grade] [and] [on edges of slabs-on-grade.]~~ Fasten insulation with ~~[~~adhesive~~]~~ ~~[~~or~~]~~ ~~[~~mechanical fasteners~~]~~.

~~][3.4.3 Insulation Under Slab~~

~~Provide insulation horizontally under [entire] slab on grade [for a distance of [_____] mm from the edge of slab]. [Turn insulation up at slab edge, and extend full height of slab.] Install insulation on top of vapor retarder and turn retarder up over the outside edge of insulation to top of slab.~~

~~][~~3.4.3 Protection of Insulation

Protect insulation from damage during construction and back filling by application of protection board or a coating. Do not leave installed vertical insulation unprotected overnight. Protect installed insulation from weather, including rain and ultraviolet light, from mechanical abuse, compression, and dislocation. ~~[Install protection over entire exposed exterior insulation board.] [Extend protection at least 300 mm below grade.]~~

~~][3.5 VAPOR RETARDER~~

~~Apply vapor retarder continuous across all surfaces. Overlap all joints at least 150 mm and seal with pressure sensitive tape. Seal at sills, header, windows, doors and utility penetrations. Repair punctures or tears with pressure sensitive tape.~~

~~][3.6 ACCESS PANELS AND DOORS~~

~~Attach insulation to all access panels greater than 0.1 square meter and all access doors in insulated floors and ceilings. Use insulation with same R-Value as that for the floor or ceiling in which each panel occurs.~~

~~]~~ -- End of Section --

SECTION 07 21 16

MINERAL FIBER BLANKET INSULATION
11/11, CHG 4: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C665	(2017) Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing
ASTM C930	(2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories
ASTM D5359	(2015) Standard Specification for Glass Cullet Recovered from Waste for Use in Manufacture of Glass Fiber
ASTM E84	(2020) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2020) Standard Test Methods for Fire Tests of Building Construction and Materials

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350	(2010; Version 1.1) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

GREEN SEAL (GS)

GS-36	(2013) Adhesives for Commercial Use

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 9504	(2017) Man Made Mineral Fibre Thermal Insulation Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 255	(2006; Errata 2006) Standard Method of Test of Surface Burning Characteristics of Building Materials

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168	(2017) Adhesive and Sealant Applications

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134	Respiratory Protection

UNDERWRITERS LABORATORIES (UL)

UL 723	(2018) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

1.2 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S"~~ classification. Submittals not having a "G" ~~or "S"~~ classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Blanket Insulation

~~Recycled Content for Insulation Materials; S~~

~~[~~ ~~Sill Sealer Insulation~~

~~][~~ ~~Vapor Retarder~~

~~]~~ ~~Pressure Sensitive Tape~~

Accessories

SD-07 Certificates

~~Indoor Air Quality for Insulation Materials; S~~

Indoor Air Quality for Adhesives~~; S~~

SD-08 Manufacturer's Instructions

Insulation

~~1.3 CERTIFICATIONS~~

~~Submit required indoor air quality certifications and validations in one submittal package.~~

~~1.3.1 Insulation Products~~

~~Provide product certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Provide current product certification from certification body.~~

~~1.3.2 Adhesives and Sealants~~

~~Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.~~

1.3 PERFORMANCE REQUIREMENTS

Japanese manufactured products will be accepted if the insulation meets the inherent properties as indicated in this specification section and drawings. Inherent properties include, but are not limited to, dimensions; thermal resistance; surface burning characteristics; critical radiant flux; water vapor permanence; water vapor sorption; odor emission; corrosiveness; and fungi resistance and sustainability requirements. The Contractor shall bear responsibility of all components as the property of thermal resistance for a specific thickness of blanket is only part of the total thermal performance of a building element such as a wall, ceiling, and floor. The Contractor shall bear responsibility for all testing required to ensure requirements set forth in this specification are met.

1.3.1 Thermal and Acoustical

Thermal and acoustical insulation with a flame spread (FS) rating not higher than 75, and a smoke developed (SD) rating not higher than 150 when tested in accordance with ASTM E84, NFPA 255 or UL 723.

1.3.2 Exposed Insulation

Exposed insulation in concealed spaces of sprinklered buildings must have flame spread of 25 or less and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84, NFPA 255 or UL 723. Acceptable types of insulation blankets are Type I, Type II (Class A only) and Type III (Class A only).

1.3.3 Flame Spread - No Smoke Developed Rating Limitation

Compliance with the SD rating limitation is not required, and a FS rating up to 100 is permitted for insulation, including insulating sheathing installed within wall assemblies. In such assemblies, conform to the requirements for interior finish with a minimum fire resistance rating of 15 minutes when tested in accordance with ASTM E119.

1.3.4 No Flame Spread or Smoke Limitation

Compliance with FS and SD limitations are not required for the following applications.

a. Insulation installed above poured concrete or poured gypsum roof decks, nominal 50 mm thick tongue-and-grooved wood plank roof decks or precast roof panels or plants that are approved by an approved testing laboratory, as noncombustible roof deck construction.

b. Insulation installed above roof decks where the entire roof construction assembly, including the insulation, is UL-listed as Fire

Classified, or FM-approved for Class I roof deck construction or equal listing or classification by an approved testing laboratory.

c. Insulation contained entirely within panels where the entire panel assembly used in the construction application meets the cited FS and SD limitations.

d. Insulation isolated from the interior of the building by masonry walls, masonry cavity walls, insulation encased in masonry cores, or concrete floors.

e. Insulation installed over concrete slabs and completely covered by wood tongue-and-groove flooring without creating air spaces within the flooring system.

f. Insulation completely enclosed in hollow metal doors.

g. Insulation installed between new exterior siding materials and existing siding or wood board, plywood, fiberboard, or gypsum exterior wall sheathing.

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery

Deliver materials to site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

1.4.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling.

1.5 SAFETY PRECAUTIONS

1.5.1 Respirators

Provide installers with dust/mist respirators, training in their use, and protective clothing, all approved by National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) in accordance with 29 CFR 1910.134.

1.5.2 Other Safety Concerns

Consider other safety concerns and measures as outlined in ASTM C930.

PART 2 PRODUCTS

2.1 BLANKET INSULATION

ASTM C665 or JIS A 9504, Type ~~[~~I, blankets without membrane coverings~~]~~ ~~[~~and ~~]~~ ~~[~~II, blankets with non-reflecting coverings~~] [and] [III, blankets with~~

~~reflective coverings]; Class [A, membrane-faced surface with a flame spread of 25 or less] [B, membrane-faced surface with a flame propagation resistance; critical radiant flux of 0.12 W/m2 or greater], except a flame spread rating of~~ ~~[~~25~~] [75] [100]~~ or less ~~[~~and a smoke developed rating of 150 or less~~]~~ when tested in accordance with ASTM E84.

2.1.1 Thermal Resistance Value (R-VALUE)

The R-Value must be as indicated on drawings.

2.1.2 Recycled Materials

Provide insulation materials containing the following minimum percentage of recycled material content by weight:

Fiberglass: 20 percent glass cullet complying with ASTM D5359

~~Provide data identifying percentage of recycled content for insulation materials.~~

2.1.3 Prohibited Materials

Do not provide asbestos-containing materials.

~~[2.1.4 Reduced Volatile Organic Compounds (VOC) for Insulation Materials~~

~~Provide certification of indoor air quality for insulation materials.~~

~~][2.2 SILL SEALER INSULATION~~

~~Provide polyethylene foam sill sealer [89][139][190][241] millimeters in width with the following characteristics:.~~

~~Physical Properties~~	~~Test Method~~	~~Measurement~~
~~Nominal Thickness~~	~~ASTM D3575~~	~~4.76 mm~~
~~Compressive Strength~~	~~ASTM D3575~~	~~8.27 kPa~~
~~- Vertical Direction~~	~~Suffix D~~	
~~Tensile Strength~~	~~ASTM D3575~~	~~220 kPa~~
	~~Suffix T~~	

~~]2.3 BLOCKING~~

~~Wood, metal, unfaced mineral fiber blankets in accordance with ASTM C665 or JIS A 9504, Type I, or other approved materials. Use only non-combustible materials meeting the requirements of ASTM E136 for blocking around chimneys and heat producing devices.~~

~~[2.4 VAPOR RETARDER~~

~~[a. 0.15 mm thick polyethylene sheeting conforming to ASTM D4397 and having a water vapor permeance of 57.2 ng/(Pa * s * m2) or less when tested in accordance with ASTM E96/E96M.~~

~~][b. Membrane with the following properties:~~

~~[Water Vapor Permeance: ASTM E96/E96M: 57.2 ng/(Pa * s * m2)~~
~~][Maximum Flame Spread: ASTM E84: [25] [50] [_____]~~
~~][Combustion Characteristics: Passing ASTM E136~~
~~][Puncture Resistance: TAPPI T803 OM: [15] [25] [50]~~

~~]]]2.5 PRESSURE SENSITIVE TAPE~~

~~As recommended by the vapor retarder manufacturer and having a water vapor permeance rating of 57.2 ng/(Pa * s * m2) or less when tested in accordance with ASTM D3833/D3833M.~~

2.2 ACCESSORIES

2.2.1 Adhesive

As recommended by the insulation manufacturer. Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Japanese manufactured products must be certified to meet indoor air quality requirements to attain F 4-Star of JAIA 4VOC rating. Provide certification or validation of indoor air quality for adhesives.

2.2.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

2.2.3 Wire Mesh

Corrosion resistant and as recommended by the insulation manufacturer.

PART 3 EXECUTION

3.1 EXISTING CONDITIONS

Before installing insulation, ensure that areas that will be in contact with the insulation are dry and free of projections which could cause voids, compressed insulation, or punctured vapor retarders. If moisture or other conditions are found that do not allow the workmanlike installation of the insulation, do not proceed but notify Contracting Officer of such conditions.

3.2 PREPARATION

3.2.1 Blocking at Attic Vents and Access Doors

Prior to installation of insulation, install permanent blocking to prevent insulation from slipping over, clogging, or restricting air flow through soffit vents at eaves.~~[Install permanent blocking around attic trap doors.][~~ Install permanent blocking to maintain accessibility to equipment or controls that require maintenance or adjustment.~~]~~

~~3.2.2 Blocking Around Heat Producing Devices~~

~~Install non-combustible blocking around heat producing devices to provide the following clearances:~~

~~a. Recessed lighting fixtures, including wiring compartments, ballasts, and other heat producing devices, unless these are certified by the manufacturer for installation surrounded by insulation: 75 mm from outside face of fixtures and devices or as required by NFPA 70 and, if insulation is to be placed above fixture or device, 600 mm above fixture.~~

~~b. Masonry chimneys or masonry enclosing a flue: 50 mm from outside face of masonry. Masonry chimneys for medium and high heat operating appliances: Minimum clearances required by NFPA 211.~~

~~c. Vents and vent connectors used for venting the products of combustion, flues, and chimneys other than masonry chimneys: Minimum clearances as required by NFPA 211.~~

~~d. Gas Fired Appliances: Clearances as required in NFPA 54.~~

~~e. Oil Fired Appliances: Clearances as required in NFPA 31.~~

~~Blocking around flues and chimneys is not required when insulation blanket, including any attached vapor retarder, passed ASTM E136, in addition to meeting all other requirements stipulated in Part 2. Blocking is also not required if the chimneys are certified by the manufacturer for use in contact with insulating materials.~~

3.3 INSTALLATION

3.3.1 Insulation

Install and handle insulation in accordance with manufacturer's instructions. Keep material dry and free of extraneous materials. Any materials that show visual evidence of biological growth due to presence of moisture must not be installed on the building project. Ensure personal protective clothing and respiratory equipment is used as required. Observe safe work practices.

3.3.1.1 Electrical wiring

Do not install insulation in a manner that would sandwich electrical wiring between two layers of insulation.

3.3.1.2 Continuity of Insulation

Install blanket insulation to butt tightly against adjoining blankets and to studs, ~~rafters, joists,~~ sill plates, headers and any obstructions. ~~[Where insulation required is thicker than depth of joist, provide full width blankets to cover across top of joists.]~~ Provide continuity and integrity of insulation at corners, wall to ceiling joints, roof, and floor. Avoid creating thermal bridges.

3.3.1.3 Installation at Bridging and Cross Bracing

Insulate at bridging and cross bracing by splitting blanket vertically at center and packing one half into each opening. Butt insulation at bridging and cross bracing; fill in bridged area with loose or scrap insulation.

~~[~~3.3.1.4 Cold Climate Requirement

Place insulation to the outside of pipes.

~~][3.3.1.5 Insulation Blanket with Affixed Vapor Retarder~~

~~Locate vapor retarder as indicated. Do not install blankets with affixed vapor retarders unless so specified. Unless the insulation manufacturer's instructions specifically recommend not to staple the flanges of the vapor retarder facing, staple flanges of vapor retarder at 150 mm intervals flush with face or set in the side of truss, joist, or stud. Avoid gaps and bulges in insulation and "fishmouth" in vapor retarders. Overlap both flanges when using face method. Seal joints and edges of vapor retarder with pressure sensitive tape. Stuff pieces of insulation into small cracks between trusses, joists, studs and other framing, such as at attic access doors, door and window heads, jambs, and sills, band joists, and headers. Cover these insulated cracks with vapor retarder material and tape all joints with pressure sensitive tape to provide air and vapor tightness.~~

~~][~~3.3.1.5 Insulation without Affixed Vapor Retarder

Provide snug friction fit to hold insulation in place. Stuff pieces of insulation into cracks between trusses, joists, studs and other framing, such as at attic access doors, door and window heads, jambs, and sills, band joists, and headers.

~~]~~3.3.1.6 Sizing of Blankets

Provide only full width blankets when insulating between trusses, joists, or studs. Size width of blankets for a snug fit where trusses, joists or studs are irregularly spaced.

~~[~~3.3.1.7 Special Requirements for Ceilings

Place insulation under electrical wiring occurring across joists. Pack insulation into narrowly spaced framing. Do not block flow of air through soffit vents. ~~[~~Attach insulation to attic door by adhesive or staples.~~]~~

~~][~~3.3.1.8 Installation of Sill Sealer

Size sill sealer insulation and place insulation over top of masonry or concrete perimeter walls or concrete perimeter floor slab on grade. Fasten sill plate over insulation.

~~][3.3.1.9 Special Requirements for Floors~~

~~Hold insulation in place with corrosion resistant wire mesh, wire fasteners, or wire lacing.~~

~~][~~3.3.1.9 Access Panels and Doors

Affix blanket insulation to access panels greater than one square foot and access doors in insulated floors and ceilings. Use insulation with same

R-Value as that for floor or ceiling.

~~][3.3.2 Installation of Separate Vapor Retarder~~

~~Apply continuous vapor retarder as indicated. Overlap joints at least 150 mm and seal with pressure sensitive tape. Seal at sill, header, windows, doors and utility penetrations. Repair punctures or tears with pressure sensitive tape.~~

~~]~~ -- End of Section --

SECTION 07 56 00

~~FLUID-APPLIED WATERPROOFING~~FLUID-APPLIED ROOFING AND PEDESTRIAN TRAFFIC COATING SYSTEM
02/12

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE ARCHITECTURAL STANDARD SPECIFICATIONS (JASS)

JASS 8	Waterproofing Work Standard Specification

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 6013	(2014) Polymer-Modified Bitumen Roofing Sheets
JIS A 6021	(2011) Liquid-applied Compounds for Waterproofing Membrane Coatings of Buildings

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 9	(2019) Building Construction Standard Specifications - Chapter 9 Waterproofing Construction

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Fluid-Applied Membrane

Membrane Primer

Bond Breaker

Submit material description and physical properties, application details, and recommendations regarding shelf life, application procedures, and precautions on flammability and toxicity.

SD-07 Certificates

Qualifications of Installer

A certificate certifying that the installer is approved by the coating manufacturer to apply the coating products, and have verifiable minimum application experience. Certificates of compliance attesting that the waterproof coating material meet the specified requirements. Certificate from product manufacturers indicating that materials in contact with one another are compatible; i.e. roofing and caulking materials.

SD-11 Closeout Submittals

Warranty

Information Card

Instructions To Government Personnel

Include copies of Safety Data Sheets for maintenance/repair materials.

1.3 GENERAL REQUIREMENTS

1.3.1 Coordination

Fluid-applied roofing operations shall be coordinated with work of other trades to ensure that components are installed as required to permit continuous self-flashing of the roofing system. The installed roofing system shall be protected from damage. Damaged areas shall be repaired.

1.3.2 Preparation

Surfaces to receive liquid-applied roofing shall be dry and free of loose coatings, surface curing agents, dust, wax or other contaminants. Surface dryness shall be tested in terms of percentage of moisture content of concrete substrate, and assure that the surface dryness is within the range that coating manufacturer recommends for coating, prior to start of coating activity. They dryness testing shall be conducted using Delmhorst moisture meter or similar tester.

1.3.3 Protection of Adjacent Surfaces

Surfaces near roofing operations shall be protected from over spray of roofing materials.

1.4 PREWATERPROOFING CONFERENCE

Prior to starting application of waterproofing system, arrange and attend a prewaterproofing conference to ensure a clear understanding of drawings and specifications. Give the Contracting Officer 7 days advance written notice of the time and place of meeting. Ensure that the mechanical and electrical subcontractor, flashing and sheetmetal subcontractor, and other trades that may perform other types of work on or over the membrane after installation, attend this conference.

1.5 DELIVERY, STORAGE, AND HANDLING

Deliver waterproofing materials in manufacturer's original, unopened containers, with labels intact and legible. Containers of materials covered by a referenced specification number shall bear the specification number, type, and class of the contents. Deliver materials in sufficient quantity to continue work without interruption. Store and protect materials in accordance with manufacturer's instructions, and use within their indicated shelf life. When hazardous materials are involved, adhere to special precautions of the manufacturer, unless precautions conflict with local, state, and federal regulations. Promptly remove from the site materials or incomplete work adversely affected by exposure to moisture or freezing. Store materials on pallets and cover from top to bottom with canvas tarpaulins.

1.6 ENVIRONMENTAL ~~CONDITIONS~~REQUIREMENTS

Apply materials when ambient temperature is 4 degrees C or above for a period of 24 hours prior to the application and when there is no ice, frost, surface moisture, or visible dampness on the substrate surface. Apply materials when air temperature is expected to remain above 4 degrees C during the cure period recommended by the manufacturer. Moisture test for substrate is specified under paragraph entitled "Moisture Test." Work may be performed within heated enclosures, provided the surface temperature of the substrate is maintained at a minimum of 4 degrees C for 24 hours prior to the application of the waterproofing, and remains above that temperature during the cure period recommended by the manufacturer.

1.6.1 Personnel Protection During Coating Applications

Personnel engaged in painting of roof coating systems shall wear an appropriate personnel protective equipment to prevent skin or eye contact or inhalation. Smoking will not be permitted in the area where coating is being applied. Substrate and air temperature at the time of coating application and curing shall be in accordance with the manufacturer's instructions

1.6.2 Coating

Apply per manufacturers recommendations and warranty requirements.

1.7 WARRANTY

Provide roof system and pedestrian traffic coating material and workmanship warranties meeting specified requirements. Provide revisions or amendment to standard membrane and pedestrian traffic coating manufacturer warranty to comply with the specified requirements. Minimum manufacturer warranty shall have no dollar limit, cover full system water-tightness, and shall have a minimum duration of 10 years.

1.7.1 Roof Membrane and Pedestrian Traffic Coating Systems Manufacturer Warranty

Furnish the roof membrane and pedestrian traffic coating systems manufacturer's 10-year no dollar limit roof system materials and installation workmanship warranty, including flashing, insulation, and accessories necessary for a watertight roof system construction. Write the warranty directly to the Government commencing at time of Government's acceptance of the roof work. Provide the the following statements for such warranty:

a. If within the warranty period the roof system and pedestrian traffic coating systems, as installed for its intended use in the normal climatic and environmental conditions of the facility, becomes non-watertight, shows evidence of moisture intrusion within the assembly, blisters, splits, tears, cracks, delaminates, separates at the seams, or shows evidence of excessive weathering due to defective materials or installation workmanship, the repair or replacement of the defective and damaged materials of the roof system and pedestrian traffic coating systems assembly and correction of defective workmanship are the responsibility of the roof membrane and pedestrian traffic coating systems manufacturer. All cost associated with the repair or replacement work are the responsibility of the roof membrane and pedestrian traffic coating systems manufacturer.

b. The warranty must remain in full force and effect, including emergency temporary repairs performed by others, when the manufacturer or his approved applicator fail to perform the repairs within 72 hours of notification.

1.7.2 Roofing System and Pedestrian Traffic Coating Systems Installer Warranty

The roof system and pedestrian traffic coating systems installer must warrant for a minimum period of two years that the roof and pedestrian traffic coating system, as installed, is free from defects in installation workmanship, to include the roof membrane, flashing, insulation, accessories, attachments, and sheet metal installation integral to a complete watertight roof system and pedestrian traffic coating systems assembly. Write the warranty directly to the Government. The roof system installer is responsible for correction of defective workmanship and replacement of damaged or affected materials. The roof and pedestrian traffic coating system installer is responsible for all costs associated with the repair or replacement work.

1.7.3 Continuance of Warranty

Approve repair or replacement work that becomes necessary within the warranty period and accomplished in a manner so as to restore the integrity of the roof system assembly and validity of the roof membrane manufacturer warranty for the remainder of the manufacturer warranty period.

1.8 WIND UPLIFT

Installed roof coating system and accessories shall withstand design wind load and design pressure, as indicated in the drawings.

1.9 QUALIFICATIONS OF INSTALLER

Installation shall be by an applicator approved by the manufacturer of the roofing system. Furnish a written statement from the manufacturer indicating that the installer is acceptable.

1.10 CONFORMANCE AND COMPATIBILITY

Work not specifically addressed and any deviation from specified requirements must be in general accordance with recommendations of the MLIT SS Chapter 9, membrane manufacturer published recommendations and details, and compatible with surrounding components and construction.

PART 2 PRODUCTS

2.1 FLUID-APPLIED MEMBRANE

JIS A 6021 ~~and comply~~and comply with JASS 8.

2.2 PEDESTRIAN TRAFFIC COATING

Specialty coating used for purposes of pedestrian traffic, resists wear and remains flexible to absorb stresses. Used in conjunction with fluid-applied roofing system which shall meet or exceed the properties below. Installed system shall have a minimum solar reflectance of 0.29 and emittance of 0.85.

2.3 MEMBRANE PRIMER

As recommended by the fluid-applied membrane manufacturer unless specifically prohibited by the manufacturer of the fluid-applied membrane.

2.4 SEALANT

As specified in Section 07 92 00 JOINT SEALANTS.

2.5 SEALANT PRIMER

As specified in Section 07 92 00 JOINT SEALANTS.

2.6 BACKING MATERIAL

Premolded, closed-cell, polyethylene, or polyurethane foam rod having a diameter 25 percent larger than joint width before being compressed into joint. Provide bond breaker of polyethylene film or other suitable material between backing material and sealant.

2.7 ~~[~~JOINT FILLER

As specified in ~~[~~Section 03 30 00 CAST-IN-PLACE CONCRETE.~~]]~~

2.8 BOND BREAKER

As recommended by the fluid-applied membrane manufacturer. Bond breaker shall not interfere with the curing process or other performance properties of the fluid-applied membrane.

2.9 ELASTOMERIC SHEET

Preformed; as recommended by the fluid-applied membrane manufacturer. Bond strength between the fluid-applied membrane and the preformed elastomeric sheet shall be a minimum of 7 kPa when tested in accordance with JIS A 6013.

2.10 ELASTOMERIC SHEET ADHESIVE

As recommended by the elastomeric sheet manufacturer.

~~2.11 PROTECTION BOARD~~

~~Premolded bitumen composition board, 3 mm minimum thickness or other composition board compatible with the fluid-applied membrane.~~

~~2.12 DRAINAGE COURSE AGGREGATE~~

~~JIS A 5005, size 9.5 mm to 2.5 mm.~~

~~2.13 INSULATION~~

~~Polystyrene foam conforming to JIS A 9521, thickness as [indicated] [required by indicated R-value].~~

PART 3 EXECUTION

3.1 PREPARATION

Coordinate work with that of other trades to ensure that components to be incorporated into the waterproofing and pedestrian traffic coating systems are available when needed. Inspect and approve surfaces immediately before application of waterproofing and pedestrian traffic coating materials. Verify that structural concrete substrate surfaces are smooth, and not detrimental to full contact bond of materials. Remove laitance, loose aggregate, sharp projections, grease, oil, dirt, curing compounds, and other contaminants which could adversely affect the complete bonding of the fluid-applied membrane and pedestrian traffic coating to the concrete surface.

3.1.1 Existing Roof Coating Removal

Remove existing roofing to bare concrete as recommended by the manufacturer. Make repairs where required.

3.1.2 Flashings

Make penetrations through sleeves in concrete slab watertight before application of waterproofing. After flashing is completed, cover elastomeric sheet with fluid-applied waterproofing during waterproofing application.

3.1.2.1 Drains

Make drain flanges flush with surface of structural slab. Apply a full elastomeric sheet around the drain, with edges fully adhered to drain flange and to structural slab. Do not adhere elastomeric sheet over joint between drain and concrete slab. Do not plug drainage or weep holes. Cover elastomeric sheet with fluid-applied waterproofing during waterproofing application. Lap elastomeric sheet a minimum of 100 mm onto concrete slab.

3.1.2.2 Penetrations and Projections

Flash penetrations and projections through structural slab with an elastomeric sheet adhered to the concrete slab and the penetration. Leave elastomeric sheet unadhered for 25 mm over joint between penetration and concrete slab. Adhere elastomeric sheet a minimum of 100 mm onto horizontal deck.

3.1.2.3 Walls and Vertical Surfaces

Flash wall intersections which are not of monolithic pour or constructed with reinforced concrete joints with an elastomeric sheet adhered to both vertical wall surfaces and concrete slab. Flash intersections which are monolithically poured or constructed with reinforced concrete joints with either an elastomeric sheet or a vertical grade of fluid-applied waterproofing adhered to vertical wall surfaces and concrete slab. Leave sheet unadhered for a distance of 25 mm from the corner on both vertical and horizontal surfaces.

3.1.3 Cracks and Joints

Prepare visible cracks and joints in substrate to receive fluid-applied waterproofing membrane and pedestrian traffic coating by placing a bond breaker and an elastomeric slip sheet between membrane and substrate. Cracks that show movement shall receive a 50 mm bond breaker followed by an elastomeric sheet adhered to the deck. Nonmoving cracks shall be double coated with fluid-applied waterproofing.

3.1.4 Priming

Prime surfaces to receive fluid-applied waterproofing membrane and pedestrian traffic coating. Apply primer as required by membrane and pedestrian traffic coating manufacturer's printed instructions.

3.2 SPECIAL PRECAUTIONS

Protect waterproofing and pedestrian traffic coating materials during transport and application. Do not dilute primers and other materials, unless specifically recommended by materials manufacturer. Keep containers closed except when removing contents. Do not mix remains of unlike materials. Thoroughly remove residual materials before using application equipment for mixing and transporting materials. Do not permit equipment on the project site that has residue of materials used on previous projects. Use cleaners only for cleaning, not for thinning primers or membrane materials. Ensure that workers and others who walk on cured membrane wear clean, soft-soled shoes to avoid damaging the waterproofing materials.

3.3 APPLICATION

Over primed surfaces, provide a uniform, wet, monolithic coating of fluid-applied membrane, 1.5 mm thick, plus or minus 0.125 mm by following manufacturer's printed instructions. Apply material by trowel, squeegee, roller, brush, spray apparatus, or other method recommended by membrane manufacturer. Check wet film thickness as specified in paragraph entitled "Film Thickness" and adjust application rate as necessary to provide a uniform coating of the thickness specified. Where possible, mark off surface to be coated in equal units to facilitate proper coverage. At expansion joints, control joints, prepared cracks, flashing, and terminations, carry membrane over preformed elastomeric sheet in a uniform 1.5 mm thick, plus or minus 0.125 mm, wet thickness to provide a monolithic coating. If membrane cures before next application, wipe previously applied membrane with a solvent to remove dirt and dust that could inhibit adhesion of overlapping membrane coat. Use solvent recommended by the membrane manufacturer, as approved.

3.3.1 Work Sequence

Perform work so that protection board is installed prior to using the waterproofed surface. Do not permanently install protection board until the membrane has passed the flood test specified under paragraph entitled

"Flood Test." Move material storage areas as work progresses to prevent abuse of membrane and overloading of structural deck.

~~3.3.2 Protection Board~~

~~Protect fluid-applied membrane by placing protection board over membrane at a time recommended by the membrane manufacturer. Protect membrane application when protection board is not placed immediately. Butt protection boards together and do not overlap.~~

~~3.3.3 Drainage Course~~

~~Place drainage course where shown after flood tests are completed and concrete protection slab or wearing course is ready to be installed.~~

~~3.3.4 Insulation~~

~~Place insulation of thickness indicated, on top of drainage course just prior to placement of concrete protection slab.~~

3.4 FIELD QUALITY CONTROL

3.4.1 Moisture Test

Prior to application of fluid-applied waterproofing, measure moisture content of substrate with a moisture meter in the presence of the Contracting Officer. Do not begin application until meter reading indicates "dry" range.

3.4.2 Film Thickness

Measure wet film thickness every 10 square meters during application by placing flat metal plates on the substrate or using a mil-thickness gage especially manufactured for the purpose.

3.4.3 Flood Test

After application and curing is complete, plug drains and fill waterproofed area with water to a depth of 50 mm. A minimum 48 hour cure time, or longer cure time if recommended by the membrane manufacturer, shall be required prior to flood testing. Allow water to stand 24 hours. Test watertightness by measuring water level at beginning and end of the 24 hour period. If water level falls, drain water, allow installation to dry, and inspect. Make repairs or replace as required and repeat the test. Work shall not proceed before approval of repairs or replacement.

3.5 INSTRUCTIONS TO GOVERNMENT PERSONNEL

Furnish written and verbal instructions on proper maintenance procedures to designated Government personnel. Furnish instructions by a competent representative of the roof membrane manufacturer and include a minimum of 4 hours on maintenance and emergency repair of the membrane. Include a demonstration of membrane repair, and give sources of required special tools. Furnish information on safety requirements during maintenance and emergency repair operations.

3.6 INFORMATION CARD

For each roof application, furnish a minimum 215 mm information card for facility records and a card laminated in plastic and framed for interior display at roof access point, or a photoengraved 1 mm thick aluminum card for exterior display. Identify facility name and number; location; contract number; approximate roof area; detailed roof system description, including deck type, membrane, number of plies, method of application, manufacturer, insulation and cover board system and thickness; presence of tapered insulation for primary drainage, presence of vapor retarder; date of completion; installing contractor identification and contract information; membrane manufacturer warranty expiration, warranty reference number, and contact information. Install card at roof top or access location as directed by the Contracting Officer and provide a paper copy to the Contracting Officer.

~~FORM 1~~
~~FLUID-APPLIED WATERPROOFING SYSTEM COMPONENTS~~
~~1. Contract Number~~
~~2. Date Work Completed~~
~~3. Project Specification Designation~~
~~4. Substrate Material~~
~~5. Slope of Substrate~~
~~6. Drains Type/Manufacturer~~
~~7. Waterproofing~~
~~a. Membrane~~
~~b. Sealant~~
~~c. Elastomeric Sheet~~
~~d. Materials Manufacturer(s)~~
~~8. Protection Board~~
~~a. Type~~
~~b. Thickness~~
~~c. Manufacturer's Name~~
~~9. Drainage Course Material Graduation~~
~~10. Insulation~~
~~a. Type~~
~~b. Thickness~~
~~c. Manufacturer's Name~~
~~11. Protection Slab~~
~~a. Material~~
~~b. Thickness~~
~~c. Support~~
~~d. Joint System~~

~~FORM 1~~	
~~FLUID-APPLIED WATERPROOFING SYSTEM COMPONENTS~~	
~~12. Wearing Course~~	
~~a. Type~~	
~~b. Slope~~	
~~c. Joint System~~	
~~d. Sealant/Gasket Type~~	
~~13. Wearing Surface Type~~	
~~Manufacturer's Name~~	
~~14. Warranty~~	
~~a. Manufacturer warranty expiration~~	
~~b. Warranty reference number~~	
~~15. Statement of Compliance or Exception~~	
~~Contractor's Signature~~	~~Date Signed~~
~~Inspector's Signature~~	~~Date Signed~~

FORM 1	
FLUID-APPLIED WATERPROOFING SYSTEM COMPONENTS	
1. Contract Number	
2. Date Work Completed	
3. Project Specification Designation	
4. Substrate Material	
5. Slope of Substrate	
6. Drains Type/Manufacturer	
7. Waterproofing	
a. Membrane	
b. Sealant	
c. Elastomeric Sheet	
d. Materials Manufacturer(s)	
8. Warranty	
a. Manufacturer warranty expiration	
b. Warranty reference number	
9. Statement of Compliance or Exception	
Contractor's Signature	Date Signed
Inspector's Signature	Date Signed

-- End of Section --

SECTION 07 60 00

FLASHING AND SHEET METAL
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 6021	(2011) Liquid-applied Compounds for Waterproofing Membrane Coatings of Buildings
JIS G 4305	(2008) Cold-rolled Stainless Steel Plate, Sheet and Strip
JIS K 2208	(2009) Asphalt Emulsion
JIS Z 3282	(2017) Solder-Chemical Composition and Shape

1.2 GENERAL REQUIREMENTS

Finished sheet metal assemblies must form a weathertight enclosure without waves, warps, buckles, fastening stresses or distortion, while allowing for expansion and contraction without damage to the system. The sheet metal installer is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal modifications required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous, uninterrupted roofing operations.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

~~Exposed Sheet Metal Coverings; G[, [_____]]~~

~~Gutters; G[, [_____]]~~

Downspouts; G~~[, [_____]]~~

~~Expansion Joints; G[, [_____]]~~

~~Gravel Stops and fascia; G[, [_____]]~~

~~Splash Pans; G[, [_____]]~~

~~Flashing for Roof Drains; G[, [_____]]~~

Base Flashing; G~~[, [_____]]~~

Counterflashing; G~~[, [_____]]~~

~~Flashing at Roof Penetrations and Equipment Supports; G[, [_____]]~~

Scuppers; G~~[, [_____]]~~

~~Copings; G[, [_____]]~~

~~Drip Edges; G[, [_____]]~~

~~Conductor Heads; G[, [_____]]~~

~~Open Valley Flashing; G[, [_____]]~~

~~Eave Flashing; G[, [_____]]~~

SD-08 Manufacturer's Instructions

~~Instructions for Installation; G[, [_____]]~~

Quality Control Plan; G~~[, [_____]]~~

~~SD-10 Operation and Maintenance Data~~

~~Cleaning and Maintenance; G[, [_____]]~~

1.4 MISCELLANEOUS REQUIREMENTS

1.4.1 Product Data

Indicate thicknesses, dimensions, fastenings, anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction. Scaled manufacturer's catalog data may be submitted for factory fabricated items.

1.4.2 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

~~1.4.3 Operation and Maintenance Data~~

~~Submit detailed instructions for installation and quality control during installation, cleaning and maintenance, for each type of assembly indicated.~~

1.5 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until installation.

PART 2 PRODUCTS

2.1 RECYCLED CONTENT

Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.

2.2 MATERIALS

Do not use lead, lead-coated metal, or galvanized steel. Different items need not be of the same metal, except that~~[~~ if copper is selected for any exposed item, all exposed items must be copper, and that~~]~~ contact between dissimilar metals must be avoided.

Furnish sheet metal items in 2400 to 3000 mm lengths. Single pieces less than 2400 mm long may be used to connect to factory-fabricated inside and outside corners, and at ends of runs. Factory fabricate corner pieces with minimum 300 mm legs. Provide accessories and other items essential to complete the sheet metal installation. Provide accessories made of the same or compatible materials as the items to which they are applied. Fabricate sheet metal items of the materials specified below and to the gage, thickness, or weight shown in Table I at the end of this section. Provide sheet metal items with mill finish unless specified otherwise. Where more than one material is listed for a particular item in Table I, each is acceptable and may be used, except as follows:

2.2.1 Exposed Sheet Metal Items

Must be of the same material. Consider the following as exposed sheet metal: gutters, including hangers; downspouts; gravel stops and fascia; cap, valley, steeped, base, and eave flashings and related accessories.

2.2.2 Drainage

Do not use copper for an exposed item if drainage from that item will pass over exposed masonry, stonework or other metal surfaces. In addition to the metals listed in Table I, lead-coated copper may be used for such items.

~~2.2.3 Copper, Sheet and Strip~~

~~Provide in accordance with JIS H 3100, cold-rolled temper (standard).~~

~~2.2.4 Lead Sheet~~

~~Provide in a minimum weight of 19.6 kilograms per square meter.~~

~~2.2.5 Steel Sheet, Zinc-Coated (Galvanized)~~

~~Provide in accordance with JIS G 3302.~~

~~2.2.6 Zinc Sheet and Strip~~

~~Provide a minimum of 0.61 mm thick.~~

2.2.3 Stainless Steel

Provide in accordance with JIS G 4305, SUS302, 304 or SUS304, 2D Finish, fully annealed, dead-soft temper.

~~2.2.4 Terne-Coated Steel~~

~~Provide a minimum of 350 by 500 mm with minimum of 18 kilogram coating per double base box.~~

~~2.2.5 Aluminum Alloy Sheet and Plate~~

~~Provide in accordance with JIS H 4000 [anodized [clear] [color [_____][_____]]] form alloy, and temper appropriate for use. Provide material not less than[0.813 mm][1.651 mm] in thickness.~~

~~[2.2.5.1 Alclad~~

~~When fabricated of aluminum, fabricate the following items with Alclad 3003, Alclad 3004, or Alclad 3005, clad on [one side][both sides] unless otherwise indicated.~~

~~a. Gutters, downspouts, and hangers~~

~~b. Gravel stops and fascia~~

~~c. Flashing~~

~~]2.2.6 Finishes~~

~~Provide exposed exterior sheet metal and aluminum with a baked on, factory applied color coating of polyvinylidene fluoride (PVF2) or approved equal fluorocarbon coating. Dry film thickness of coatings must be 0.020 to 0.033 mm. Color to be selected from [manufacturer's full range of "cool roof" color choices][manufacturer's standard range of color choices][manufacturer's full range of color choices][as indicated on the Drawings]. Field applications of color coatings are prohibited and will be rejected.~~

~~2.2.7 Cool Roof Finishes~~

~~Provide cool roof finish coatings and colors in accordance with one of the following methods of analysis:~~

~~2.2.7.1 JIS K 5602 and JIS K 5675~~

~~Provide roof finishes having minimum initial solar reflectance of [40][50] when tested in accordance with JIS K 5602 and JIS K 5675.~~

~~2.2.8 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes~~

~~JIS H 4040.~~

2.2.4 Solder

Provide in accordance with JIS Z 3282, 95-5 tin-antimony.

~~2.2.5 Reglets~~

~~2.2.5.1 Polyvinyl Chloride Reglets~~

~~Provide in accordance with JIS K 6720, 1.9 mm minimum thickness.~~

~~2.2.5.2 Metal Reglets~~

~~Provide factory fabricated caulked type or friction type reglets with a minimum opening of 6 mm and a depth of 30 mm, as approved.~~

~~2.2.5.2.1 Caulked Reglets~~

~~Provide with rounded edges, temporary reinforcing cores, and accessories as required for securing to adjacent construction. Provide built-up mitered corner pieces for inside and outside corners.~~

~~2.2.5.2.2 Friction Reglets~~

~~Provide with flashing receiving slots not less than 16 mm deep, 25 mm jointing tongues, and upper and lower anchoring flanges installed at 600 mm maximum snap-lock type receiver.~~

2.2.5 Scuppers

Line interiors of scupper openings with sheet metal. Provide a drip edge at bottom edges with returns of not less than 25 mm against the face of the outside wall at the top and sides. Provide the perimeter of the lining approximately 13 mm less than the perimeter of the scupper.

~~2.2.6 Conductor Heads~~

~~Provide conductor heads and screens in the same material as downspouts. Provide outlet tubes not less than 100 mm long.~~

~~2.2.7 Splash Pans~~

~~Provide splash pans where downspouts discharge onto roof surfaces and at locations indicated. Unless otherwise indicated, provide pans not less than 600 mm long by 450 mm wide with metal ribs across bottoms of pans. Provide sides of pans with vertical baffles not less than 25 mm high in the front, and 100 mm high in the back.~~

2.2.6 Copings

Unless otherwise indicated, provide copings in copper sheets, 2400 or 3000 mm long, joined by a 20 mm locked and soldered seam.

2.2.7 Bituminous Plastic Cement

Provide in accordance with JIS A 6021.

~~2.2.8 Roofing Felt~~

~~Provide in accordance with A6005 [Asphalt Felt] or [Asphalt Roofing].~~

2.2.8 Asphalt Primer

Provide in accordance with JIS K 2208.

2.2.9 Fasteners

Use the same metal as, or a metal compatible with the item fastened.~~[Use stainless steel fasteners to fasten.]~~ Confirm compatibility of fasteners and items to be fastened to avoid galvanic corrosion due to dissimilar materials.

PART 3 EXECUTION

3.1 INSTALLATION

~~3.1.1 Metal Roofing~~

~~[3.1.1.1 [Flat Copper,] [Zinc,] [Terne-coated Steel] Roofing~~

~~Before applying roofing, cover deck with rosin-sized roofing felt. Lap 50 mm at joints and secure in place with roofing nails. Using solder of equal parts tin and lead, solder slowly with well-heated irons to thoroughly heat sheet and completely sweat solder through full width of seam.[Tin edges of copper to be soldered at least 20 mm before sheets are locked.][Use stainless nails in terne-coated steel]; [in copper, use solid copper or bronze roofing nails][in zinc, use zinc-coated roofing nails.] Where roof decks abut vertical surfaces, turn metal roofing up vertical surfaces about 200 mm where practicable; where vertical surfaces are covered with applied materials, turn up roofing behind applied materials. Use standing-seam method for roofs having rise of more than one in four, and use flat-seam method when rise is one in four or less. Walking not permitted directly on metal roofs; provide approved walkways.~~

~~][3.1.1.2 Standing-seam Method~~

~~Make standing seams parallel with slope of roof. Fabricate sheets into long lengths at shop by locking short dimensions together and thoroughly soldering joints thus formed. In applying metal, turn up one edge of course at each side seam at right angles 40 mm. Then install 50 by 75 mm cleats spaced 300 mm apart by fastening one end of each cleat to roof with two 25 mm long nails and folding roof end back over nail heads. Turn end adjoining turned-up side seam up over upstanding edge of course. Turn up adjoining edge of next course 45 mm and abutting upstanding edges locked, turned over, and flattened against one side of standing seam. Make standing seams straight, rounded neatly at the top edges, and stand about 25 mm above roof deck. All sheets must be same length, except as required to complete run or maintain pattern. Locate transverse joints of each panel half way between joints in adjacent sheets. Align joints of alternate sheets horizontally to produce uniform pattern.~~

~~][3.1.1.3 Flat-seam Method~~

~~Lay metal so short dimension is parallel to gutter or eave lines and so water will flow over and not into seams. Make seams by turning edges of sheet 20 mm and lock and solder together. If sheets are laid one at a time, secure to roof deck with cleats, using three cleats to each sheet, two on long side and one on short side. Use cleats 50 mm wide, hooked over 20 mm upturned edges of sheets, and nail to roof deck with two 25 mm long nails. Turn back roof end of cleat over nail heads before next sheet is applied. If desired, sheets may be made into long lengths at shop by locking short dimensions together and soldering seams thus formed. Turn long lengths 20 mm, and secure each length to roof deck by cleats spaced~~

~~300 mm apart. Mallet and solder seams after pans are in place. All sheets to be same length, except as required to complete run or maintain pattern. Locate transverse joints of each panel half way between joints in adjacent sheets. Align joints of alternate sheets horizontally to produce uniform pattern.~~

]3.1.1 Workmanship

Make lines and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 13 mm hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight. Join sheet metal items together as shown in Table II.

3.1.2 Nailing

Confine nailing of sheet metal generally to sheet metal having a maximum width of 450 mm. Confine nailing of flashing to one edge only. Space nails evenly not over 75 mm on center and approximately 13 mm from edge unless otherwise specified or indicated. Face nailing will not be permitted. Where sheet metal is applied to other than wood surfaces, include in shop drawings, the locations for sleepers and nailing strips required to secure the work.[Secure flashing at one-half the normal interval to ensure a wind-resistant installation.]

3.1.3 Cleats

Provide cleats for sheet metal 450 mm and over in width. Space cleats evenly not over 300 mm on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 50 mm wide by 75 mm long and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam. [Where the fastening is to be made to concrete or masonry, use screws and drive in expansion shields set in concrete or masonry.]Pre-tin cleats for soldered seams.

3.1.4 Bolts, Rivets, and Screws

Install bolts, rivets, and screws where indicated or required. Provide compatible washers where required to protect surface of sheet metal and to provide a watertight connection. Provide mechanically formed joints in aluminum sheets 1.0 mm or less in thickness.

3.1.5 Seams

Straight and uniform in width and height with no solder showing on the face.

3.1.5.1 Flat-lock Seams

Finish not less than 20 mm wide.

3.1.5.2 Lap Seams

Finish soldered seams not less than 25 mm wide. Overlap seams not soldered, not less than 75 mm.

3.1.5.3 Loose-Lock Expansion Seams

Not less than 75 mm wide; provide minimum 25 mm movement within the joint. Completely fill the joints with the specified sealant, applied at not less than 3 mm thick bed.

3.1.5.4 Standing Seams

Not less than 25 mm high, double locked without solder.

3.1.5.5 Flat Seams

Make seams in the direction of the flow.

3.1.6 Soldering

Where soldering is specified, apply to copper, terne-coated stainless steel, zinc-coated steel, and stainless steel items. Pre-tin edges of sheet metal before soldering is begun. Seal the joints in aluminum sheets of 1.0 mm or less in thickness with specified sealants. Do not solder aluminum.

3.1.6.1 Edges

Scrape or wire-brush the edges of lead-coated material to be soldered to produce a bright surface. Flux brush the seams in before soldering. Treat with soldering acid flux the edges of stainless steel to be pre-tinned. Seal the joints in aluminum sheets of 1.0 mm or less in thickness with specified sealants. Do not solder aluminum.

~~3.1.7 Welding and Mechanical Fastening~~

~~Use welding for aluminum of thickness greater than 1.0 mm. Aluminum 1.0 mm or less in thickness must be butted and the space backed with formed flashing plate; or lock joined, mechanically fastened, and filled with sealant as recommended by the aluminum manufacturer.~~

~~3.1.7.1 Welding of Aluminum~~

~~Use welding of the inert gas, shield-arc type. For procedures, appearance and quality of welds, and the methods used in correcting welding work, conform to JIS Z 3604.~~

~~3.1.7.2 Mechanical Fastening of Aluminum~~

~~Use No. 12, aluminum alloy, sheet metal screws or other suitable aluminum alloy or stainless steel fasteners. Drive fasteners in holes made with a No. 26 drill in securing side laps, end laps, and flashings. Space fasteners 300 mm maximum on center. Where end lap fasteners are required to improve closure, locate the end lap fasteners not more than 50 mm from the end of the overlapping sheet.~~

~~3.1.8 Protection from Contact with Dissimilar Materials~~

~~3.1.8.1 Copper or Copper-bearing Alloys~~

~~Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.~~

~~3.1.8.2 Aluminum~~

~~Do not allow aluminum surfaces in direct contact with other metals except stainless steel, zinc, or zinc coating. Where aluminum contacts another metal, paint the dissimilar metal with a primer followed by two coats of aluminum paint. Where drainage from a dissimilar metal passes over aluminum, paint the dissimilar metal with a non-lead pigmented paint.[Aluminum may be used over concrete construction, provided that required reglets are of stainless steel and aluminum surface in contact with concrete or masonry is coated with bituminous paint or zinc chromate primer.]~~

~~3.1.8.3 Metal Surfaces~~

~~Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.~~

~~3.1.8.4 Wood or Other Absorptive Materials~~

~~Paint surfaces that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.~~

~~3.1.9 Expansion and Contraction~~

~~Provide expansion and contraction joints at not more than 9750 mm intervals for aluminum and at not more than 12 meter intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval. Space joints evenly. Join extruded aluminum gravel stops and fascia by expansion and contraction joints spaced not more than 3600 mm apart.~~

3.1.7 Base Flashing

~~[Lay the base flashings with each course of the roof covering, shingle fashion, where practicable, where sloped roofs abut chimneys, curbs, walls, or other vertical surfaces.]~~Extend up vertical surfaces of the flashing not less than 200 mm and not less than 100 mm under the roof covering. Where finish wall coverings form a counterflashing, extend the vertical leg of the flashing up behind the applied wall covering not less than 150 mm. Overlap the flashing strips ~~[or shingles]~~ with the previously laid flashing not less than 75 mm. Fasten the strips ~~[or shingles]~~ at their upper edge to the deck. Horizontal flashing at vertical surfaces must extend vertically above the roof surface and fastened at their upper edge to the deck a minimum of 150 mm on center with ~~[~~large headed aluminum roofing nails~~]~~ ~~[hex headed, galvanized shielded screws]~~ a minimum of 150 mm lap of any surface. Solder end laps and provide for expansion and contraction. Extend the metal flashing over crickets at the up-slope side of ~~[chimneys,]~~ ~~[~~curbs,~~]~~ ~~[~~and similar~~]~~

vertical surfaces extending through sloping roofs, the metal flashings. Extend the metal flashings onto the roof covering not less than 115 mm at the lower side of ~~[dormer~~ walls,~~]~~ ~~[~~chimneys,~~]~~ ~~[~~and similar~~]~~ vertical surfaces extending through the roof decks. Install and fit the flashings so as to be completely weathertight. Provide factory-fabricated base flashing for interior and exterior corners. Do not use metal base flashing on built-up roofing.

3.1.8 Counterflashing

Except where indicated or specified otherwise, insert counterflashing in reglets located from 230 to 250 mm above roof decks, extend down vertical surfaces over upturned vertical leg of base flashings not less than 75 mm. Fold the exposed edges of counterflashings 13 mm. Where stepped counterflashings are required, they may be installed in short lengths a minimum ~~[200 mm by 200 mm]~~~~[~~200 mm by 250 mm~~]~~ or may be of the preformed single piece type. Provide end laps in counterflashings not less than 75 mm and make it weathertight with plastic cement. Do not make lengths of metal counterflashings exceed 3000 mm. Form flashings to the required shapes before installation. Factory form corners not less than 300 mm from the angle. Secure the flashings in the reglets with lead wedges and space not more than 450 mm apart; on~~[chimneys and]~~~~[~~ stair/elevator towers~~]~~ short runs, place wedges closer together. Fill caulked-type reglets or raked joints which receive counterflashing with caulking compound. Turn up the concealed edge of counterflashings built into masonry or concrete walls not less than 6 mm and extend not less than 50 mm into the walls. Install counterflashing to provide a spring action against base flashing. ~~[Where bituminous base flashings are provided, extend down the counter flashing as close as practicable to the top of the cant strip. Factory form counter flashing to provide spring action against the base flashing.]~~

~~3.1.9 Metal Reglets~~

~~Keep temporary cores in place during installation. Ensure factory fabricated caulked type or friction type, reglets have a minimum opening of 6 mm and a minimum depth of 30 mm, when installed.~~

~~3.1.9.1 Caulked Reglets~~

~~Wedge flashing in reglets with lead wedges every 450 mm, caulked full and solid with an approved compound.~~

~~3.1.9.2 Friction Reglets~~

~~Install flashing snap lock receivers at 600 mm by 600 mm on center maximum. When flashing has been inserted the full depth of the slot, caulk the slot, lock [with wedges], and fill with sealant.~~

~~3.1.10 Polyvinyl Chloride Reglets for Temporary Construction~~

~~Rigid polyvinyl chloride reglets may be provided in lieu of metal reglets for temporary construction.~~

~~3.1.11 Gravel Stops and fascia~~

~~Prefabricate in the shapes and sizes indicated and in lengths not less than 2400 mm. Extend flange at least 100 mm onto roofing. Provide prefabricated, mitered corners internal and external corners. Install gravel stops and fascia after all plies of the roofing membrane have been~~

~~applied, but before the flood coat of bitumen is applied. Prime roof flange of gravel stops and fascia on both sides with an asphalt primer. After primer has dried, set flange on roofing membrane and strip-in. Nail flange securely to wood nailer with large-head, barbed-shank roofing nails 38 mm long spaced not more than 75 mm on center, in two staggered rows.~~

~~3.1.11.1 Edge Strip~~

~~Hook the lower edge of fascia at least 20 mm over a continuous strip of the same material bent outward at an angle not more than 45 degrees to form a drip. Nail hook strip to a wood nailer at 150 mm maximum on center. Where fastening is made to concrete or masonry, use screws spaced 300 mm on center driven in expansion shields set in the concrete or masonry. Where horizontal wood nailers are slotted to provide for insulation venting, install strips to prevent obstruction of vent slots. Where necessary, install strips over 2 mm thick compatible spacer or washers.~~

~~3.1.11.2 Joints~~

~~Leave open the section ends of gravel stops and fascia 6 mm and backed with a formed flashing plate, mechanically fastened in place and lapping each section end a minimum of 100 mm set laps in plastic cement. Face nailing will not be permitted. Install prefabricated aluminum gravel stops and fascia in accordance with the manufacturer's printed instructions and details.~~

~~3.1.12 Metal Drip Edges~~

~~Provide a metal drip edge, designed to allow water run-off to drip free of underlying construction, at eaves and rakes prior to the application of roofing shingles. Apply directly on the wood deck at the eaves and over the underlay along the rakes. Extend back from the edge of the deck not more than 75 mm and secure with compatible nails spaced not more than 250 mm on center along upper edge.~~

~~3.1.13 Gutters~~

~~The hung type of shape indicated and supported on underside by brackets that permit free thermal movement of the gutter. Provide gutters in sizes indicated complete with mitered corners, end caps, outlets, brackets, and other accessories necessary for installation. Bead with hemmed edge or reinforce the outer edge of gutter with a stiffening bar not less than 20 by 5 mm of material compatible with gutter. Fabricate gutters in sections not less than 2400 mm. Lap the sections a minimum of 25 mm in the direction of flow or provide with concealed splice plate 150 mm minimum. Join the gutters, other than aluminum, by riveted and soldered joints. Join aluminum gutters with riveted sealed joints. Provide expansion-type slip joints midway between outlets. Install gutters below slope line of the roof so that snow and ice can slide clear. Support gutters on[adjustable hangers spaced not more than 750 mm on center][as indicated][by continuous cleats][and][or][by cleats spaced not less than 900 mm apart]. Adjust gutters to slope uniformly to outlets, with high points occurring midway between outlets. Fabricate hangers and fastenings from compatible metals.~~

3.1.9 Downspouts

Space supports for downspouts according to the manufacturer's recommendation for the ~~[wood][masonry] or [steel]~~ substrate. Types, shapes and sizes are indicated. Provide complete including elbows and offsets. Provide downspouts in approximately 3000 mm lengths. Provide end joints to telescope not less than 13 mm and lock longitudinal joints. Provide gutter outlets with wire ball strainers for each outlet. Provide strainers to fit tightly into outlets and be of the same material used for gutters. Keep downspouts not less than 25 mm away from walls. Fasten to the walls at top, bottom, and at an intermediate point not to exceed 1500 mm on center with leader straps or concealed rack-and-pin type fasteners. Form straps and fasteners of metal compatible with the downspouts.

3.1.9.1 Terminations

Neatly fit into the drainage connection the downspouts terminating in drainage lines and fill the joints with a portland cement mortar cap sloped away from the downspout. Provide downspouts terminating in splash blocks with elbow-type fittings. Provide splash pans as specified.

~~3.1.10 Flashing for Roof Drains~~

~~Provide a 750 mm square sheet indicated. Taper insulation to drain from 600 mm out. Set flashing on finished felts in a full bed of asphalt roof cement. Heavily coat the drain flashing ring with asphalt roof cement. Clamp the roof membrane, flashing sheet, and stripping felt in the drain clamping ring. Secure clamps so that felts and drain flashing are free of wrinkles and folds.~~

3.1.10 Scuppers

Extend the scupper liner through and project outside of, the wall it penetrates to form a bottom drip edge against the face of the wall. Fold outside edges under 13 mm on all sides. Join the top and sides of the lining on the roof deck side to a closure flange by a locked and soldered joint. Join the bottom edge by a locked and soldered joint to the closure flange, where required, form with a ridge to act as a gravel stop around the scupper inlet. Provide surfaces to receive the scupper lining and coat with bituminous plastic cement.

~~3.1.11 Conductor Heads~~

~~Set the depth of the top opening equal to two-thirds of the width or the conductor head. Flat-lock solder seams. Where conductor heads are used in conjunction with scuppers, set the conductor a minimum of 50 mm wider than the scupper. Attach conductor heads to the wall with masonry fasteners. Securely fasten screens to heads.~~

~~3.1.12 Splash Pans~~

~~Install splash pans lapped with horizontal roof flanges not less than 100 mm wide to form a continuous surface. Bend the rear flange of the pan to contour of can't strip and extend up 150 mm under the side wall covering or to height of base flashing under counterflashing. Bed the pans and roof flanges in plastic bituminous cement and strip-flash as specified.~~

~~3.1.13 Open Valley Flashing~~

~~Provide valley flashing free of longitudinal seams, of width sufficient to extend not less than 150 mm under the roof covering on each side. Provide a 13 mm fold on each side of the valley flashing. Lap the sheets not less~~

~~than 150 mm in the direction of flow and secure to roofing construction with cleats attached to the fold on each side. Nail the tops of sheets to roof sheathing. Space the cleats not more than 300 mm on center. Provide exposed flashing not less than 100 mm in width at the top and increase 25 mm in width for each additional 2400 mm in length. Where the slope of the valley is one in 2.67 or less, or the intersecting roofs are on different slopes, provide an inverted V-joint, 25 mm high, along the centerline of the valley; and extend the edge of the valley sheets 200 mm under the roof covering on each side.~~

~~Valley flashing for asphalt shingle roofs is specified in Section 07 31 13 ASPHALT SHINGLES.~~

~~3.1.14 Eave Flashing~~

~~One piece in width, applied in 2400 to 3000 mm lengths with expansion joints spaced as specified in paragraph EXPANSION AND CONTRACTION. Provide a 20 mm continuous fold in the upper edge of the sheet to engage cleats spaced not more than 250 mm on center. Locate the upper edge of flashing not less than 450 mm from the outside face of the building, measured along the roof slope. Fold lower edge of the flashing over and loose-lock into a continuous edge strip on the fascia. Where eave flashing intersects metal valley flashing, secure with 25 mm flat locked joints with cleats that are 250 mm on center.~~

~~3.1.15 Sheet Metal Covering on Flat, Sloped, or Curved Surfaces~~

~~Except as specified or indicated otherwise, cover and flash all minor flat, sloped, or curved surfaces such as crickets, bulkheads, dormers and small decks with metal sheets of the material used for flashing; maximum size of sheets, 375 by 455 mm. Fasten sheets to sheathing with metal cleats. Lock seams and solder. Lock aluminum seams as recommended by aluminum manufacturer. Provide an underlayment of roofing felt for all sheet metal covering.~~

~~3.1.16 Expansion Joints~~

~~Provide expansion joints for roofs, walls, and floors as [specified] [indicated]. Provide [expansion joints in continuous sheet metal at [12 meter intervals for copper and stainless steel] [and at 9750 mm intervals for aluminum], [aluminum gravel stops and fascia which must have expansion joints at not more than 3600 mm spacing]. Provide evenly spaced joints. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval spacing]. Conform to the requirements of Table I.~~

~~3.1.16.1 Roof Expansion Joints~~

~~Consist of curb with wood nailing members on each side of joint, bituminous base flashing, metal counterflashing, and metal joint cover. Bituminous base flashing is specified in Roofing Section. Provide counterflashing as specified in paragraph COUNTERFLASHING, except as follows: Provide counterflashing with vertical leg of suitable depth to enable forming into a horizontal continuous cleat. Secure the inner edge to the nailing member. Make the outer edge projection not less than 25 mm for flashing on one side of the expansion joint and be less than the width of the expansion joint plus 25 mm for flashing on the other side of the joint. Hook the expansion joint cover over the projecting outer edges of counterflashing. Provide roof joint with a joint cover of the width~~

~~indicated. Hook and lock one edge of the joint cover over the shorter projecting flange of the continuous cleat, and the other edge hooked over and loose locked with the longer projecting flange. Joints are specified in Table II.~~

~~3.1.16.2 Floor and Wall Expansion Joints~~

~~Provide U-shape with extended flanges for expansion joints in concrete and masonry walls and in floor slabs.~~

~~3.1.17 Flashing at Roof Penetrations and Equipment Supports~~

~~Provide metal flashing for all pipes, ducts, and conduits projecting through the roof surface and for equipment supports, guy wire anchors, and similar items supported by or attached to the roof deck. Goose-necks, rain hoods, power roof ventilators, and [_____] are specified in [_____].~~

3.1.11 Single Pipe Vents

~~See Table I, footnote (d).~~ Set flange of sleeve in bituminous plastic cement and nail 75 mm on center. Bend the top of sleeve over and extend down into the vent pipe a minimum of 50 mm.~~ For long runs or long rises above the deck, where it is impractical to cover the vent pipe with lead, use a two-piece formed metal housing. Set metal housing with a metal sleeve having a 100 mm roof flange in bituminous plastic cement and nailed 75 mm on center. Extend sleeve a minimum of 200 mm above the roof deck and lapped a minimum of 75 mm by a metal hood secured to the vent pipe by a draw band. Seal the area of hood in contact with vent pipe with an approved sealant.~~

~~3.1.12 Stepped Flashing~~

~~Provide stepped flashing where sloping roofs surfaced with shingles abut vertical surfaces. Place separate pieces of base flashing in alternate shingle courses.~~

3.1.12 Copings

Provide coping with locked and soldered seam. Terminate outer edges in edge strips. Install with sealed ~~[~~lap joints~~][~~, cover plate joints~~][~~, or standing seam joints~~] as indicated~~.

~~3.2 PAINTING~~

~~Touch ups in the field may be applied only after metal substrates have been cleaned and pretreated in accordance with manufacturer's written instructions and products.~~

~~Field-paint sheet metal for separation of dissimilar materials.~~

~~[3.2.1 Aluminum Surfaces~~

~~Clean with solvent and apply one coat of zinc-molybdate primer and one coat of aluminum paint.~~

~~]~~

3.2 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

3.3 REPAIRS TO FINISH

Scratches, abrasions, and minor surface defects of finish may be repaired in accordance with the manufacturer's printed instructions and as approved. Repair damaged surfaces caused by scratches, blemishes, and variations of color and surface texture. Replace items which cannot be repaired.

3.4 FIELD QUALITY CONTROL

Establish and maintain a Quality Control Plan for sheet metal used in conjunction with roofing to assure compliance of the installed sheet metalwork with the contract requirements. Remove work that is not in compliance with the contract and replace or correct. Include quality control, but not be limited to, the following:

a. Observation of environmental conditions; number and skill level of sheet metal workers; condition of substrate.

b. Verification that specified material is provided and installed.

c. Inspection of sheet metalwork, for proper size(s) and thickness(es), fastening and joining, and proper installation.

3.4.1 Procedure

Submit for approval prior to start of roofing work. Include a checklist of points to be observed. Document the actual quality control observations and inspections. Furnish a copy of the documentation to the Contracting Officer at the end of each day.

TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES

Sheet Metal Items	Stainless Steel, mm
Downspouts and leaders	0.38
Downspout clips and anchors	-
Downspout straps, 50 mm	1.27
Scupper lining	0.38
Pipe vent sleave (d)	

(d) 12.25 kilogram minimum lead sleeve with 100 mm flange. Where lead sleeve is impractical, refer to paragraph SINGLE PIPE VENTS for optional material.

~~TABLE II. SHEET METAL JOINTS~~			
~~TYPE OF JOINT~~			
~~Item Designation~~	~~Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel~~	~~Aluminum~~	~~Remarks~~
~~Joint cap for building expansion seam, cleated joint at roof~~	~~30 mm single lock, standing seam, cleated~~	~~30 mm single lock, standing~~	~~--~~
~~Flashings~~			
~~Base~~	~~25 mm 75 mm lap for expansion joint~~	~~25 mm flat locked, soldered; sealed; 75 mm lap for expansion joint~~	~~Aluminum manufacturer's recommended hard setting sealant for locked aluminum joints. Fill each metal expansion joint with a joint sealing compound.~~
~~Cap-in reglet~~	~~75 mm lap~~	~~75 mm lap~~	~~Seal groove with joint sealing compound.~~
~~Reglets~~	~~Butt joint~~	~~--~~	~~Seal reglet groove with joint sealing compound.~~
~~Eave~~	~~25 mm flat locked, cleated. 25 mm loose locked, sealed expansion joint, cleated.~~	~~25 mm flat locked, locked, cleated 25 mm loose locked, sealed expansion joints, cleated~~	~~Same as base flashing.~~
~~Stepped~~	~~75 mm lap~~	~~75 mm lap~~	~~--~~
~~Valley~~	~~150 mm lap cleated~~	~~150 mm lap cleated~~	~~--~~
~~Edge strip~~	~~Butt~~	~~Butt~~	~~--~~
~~Gravel stops:~~			
~~Extrusions~~	~~--~~	~~Butt with 13 mm space~~	~~Use sheet flashing beneath and a cover plate~~

~~TABLE II. SHEET METAL JOINTS~~			
~~TYPE OF JOINT~~			
~~Item Designation~~	~~Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel~~	~~Aluminum~~	~~Remarks~~
~~Sheet, smooth~~	~~Butt with 6 mm space~~	~~Butt with 6 mm space~~	~~Use sheet flashing backup plate.~~
~~Sheet, corrugated~~	~~Butt with 6 mm space~~	~~Butt with 6 mm space~~	~~Use sheet flashing beneath and a cover plate or a combination unit~~
~~Gutters~~	~~40 mm lap, riveted and soldered~~	~~25 mm flat locked riveted and sealed~~	~~Aluminum producers recommended hard setting sealant for locked aluminum joints.~~
~~(a) Provide a 75 mm lap elastomeric flashing with manufacturer's recommended sealant.~~			
~~(b) Seal Polyvinyl chloride reglet with manufacturer's recommended sealant.~~			

~~]~~ -- End of Section --

SECTION 07 84 00

FIRESTOPPING
05/10, CHG 1: 08/13

PART 1 GENERAL

1.1 SUMMARY

Furnish and install tested and listed firestopping systems, combination of materials, or devices to form an effective barrier against the spread of flame, smoke and gases, and maintain the integrity of fire resistance rated walls, partitions, floors, and ceiling-floor assemblies, including through-penetrations and construction joints and gaps.

a. Through-penetrations include the annular space around pipes, tubes, conduit, wires, cables and vents.

b. Construction joints include those used to accommodate expansion, contraction, wind, or seismic movement; do not allow firestopping material to interfere with the required movement of the joint.

c. Gaps requiring firestopping include gaps between the curtain wall and the floor slab and between the top of the fire-rated walls and the roof or floor deck above and at the intersection of shaft assemblies and adjoining fire resistance rated assemblies.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E699	(2009) Standard Practice for Evaluation of Agencies Involved in Testing, Quality Assurance, and Evaluating of Building Components
ASTM E814	(2024) Standard Test Method for Fire Tests of Penetration Firestop Systems
ASTM E1399/E1399M	(1997; R 2022) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems
ASTM E1966	(2024) Standard Test Method for Fire-Resistive Joint Systems

ASTM E2174	(2024) Standard Practice for On-Site Inspection of Installed Firestop Systems
ASTM E2307	(2025) Standard Test Method for Determining Fire Resistance of Perimeter Fire Barrier Systems Using Intermediate-Scale, Multi-story Test Apparatus
ASTM E2393	(2024) Standard Practice for On-Site Inspection of Installed Fire Resistive Joint Systems and Perimeter Fire Barriers

FM GLOBAL (FM)

FM 4991	(2013) Approval of Firestop Contractors
FM APP GUIDE	(updated on-line) Approval Guide https://www.approvalguide.com/

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

UL SOLUTIONS (UL)

UL 723	(2018; Reprint Jun 2025) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials
UL 1479	(2015; Reprint Apr 2025) UL Standard for Safety Fire Tests of Penetration Firestops
UL 2079	(2015; Reprint Jun 2025) Tests for Fire Resistance of Building Joint Systems
UL Fire Resistance	(2014) Fire Resistance Directory

1.3 SEQUENCING

Coordinate the specified work with other trades. Apply firestopping materials, at penetrations of pipes and ducts, prior to insulating, unless insulation meets requirements specified for firestopping. Apply firestopping materials. at building joints and construction gaps, prior to completion of enclosing walls or assemblies. Locate cast-in-place firestop devices and install in place before concrete placement. Install pipe, conduit or cable bundles through cast-in-place device after concrete placement but before area is concealed or made inaccessible. Firestop material must be inspected and approved prior to final completion and enclosing of any assemblies that may conceal installed firestop.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Firestopping System; G~~, [_____]~~

SD-03 Product Data

Firestopping Materials; G~~, [_____]~~

SD-06 Test Reports

Inspection; G~~, [_____]~~

SD-07 Certificates

Inspector Qualifications

Firestopping Materials

Installer Qualifications; G~~, [_____]~~

1.5 QUALITY ASSURANCE

1.5.1 Installer

Engage an experienced Installer who is:

a. FM Research approved in accordance with FM 4991, operating as a UL Certified Firestop Contractor, or

b. Certified, licensed, or otherwise qualified by the firestopping manufacturer as having the necessary staff, training, and a minimum of 3 years experience in the installation of manufacturer's products in accordance with specified requirements. Submit documentation of this experience. A manufacturer's willingness to sell its firestopping products to the Contractor or to an installer engaged by the Contractor does not in itself confer installer qualifications on the buyer. The Installer must be a trained representative of the manufacturer (not distributor or agent) in the proper selection and installation procedures. Obtain and submit installer's written certification of training, and retain proof of certification for duration of firestop installation.

1.5.2 Inspector Qualifications

The inspector must~~[~~ meet the criteria contained in ASTM E699 for agencies involved in quality assurance and must~~]~~ have a minimum of two years experience in construction field inspections of firestopping systems, products, and assemblies. The inspector must be completely independent of, and divested from, the installer, the manufacturer, and the supplier of any material or item being inspected. The inspector must not be a competitor of the installer, the contractor, the manufacturer, or supplier of any material or item being inspected. Include in the qualifications submittal a notarized statement assuring compliance with the requirements stated herein.

1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the original unopened packages or containers showing name of the manufacturer and the brand name. Store materials off the ground, protected from damage and exposure to elements and temperatures in accordance with manufacturer requirements. Remove damaged or deteriorated materials from the site. Use materials within their indicated shelf life.

PART 2 PRODUCTS

2.1 FIRESTOPPING SYSTEM

Submit detail drawings including manufacturer's descriptive data, typical details conforming to UL Fire Resistance or other details certified by another nationally recognized testing laboratory, installation instructions or UL listing details for a firestopping assembly in lieu of fire-test data or report. For those firestop applications for which no UL tested system is available through a manufacturer, submit a manufacturer's engineering judgment, derived from similar UL system designs or other tests for review and approval prior to installation. Submittal must indicate the firestopping material to be provided for each type of application. When more than a total of 5 penetrations and/or construction joints are to receive firestopping, provide drawings that indicate location, "F" "T" and "L" ratings, and type of application.

Also, submit a written report indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL list printed numbers.

2.2 FIRESTOPPING MATERIALS

Provide firestopping materials, supplied from a single domestic manufacturer, consisting of commercially manufactured, asbestos-free, nontoxic products FM APP GUIDE approved, or UL listed, for use with applicable construction and penetrating items, complying with the following minimum requirements:

2.2.1 Fire Hazard Classification

Provide material that has a flame spread of 25 or less, and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84 or UL 723. Provide an approved firestopping material as listed in UL Fire Resistance or by a nationally recognized testing laboratory.

2.2.2 Toxicity

Provide material that is nontoxic and carcinogen free to humans at all stages of application or during fire conditions and does not contain hazardous chemicals or require harmful chemicals to clean material or equipment.

2.2.3 Fire Resistance Rating

Firestop systems must be UL Fire Resistance listed or FM APP GUIDE approved with "F" rating at least equal to fire-rating of fire wall or floor in which penetrated openings are to be protected. Where required, firestop systems must also have "T" rating at least equal to the fire-rated floor in which the openings are to be protected.

2.2.3.1 Through-Penetrations

Firestopping materials for through-penetrations, as described in paragraph SUMMARY, must provide "F", "T" and "L" fire resistance ratings in accordance with ASTM E814 or UL 1479. Provide fire resistance ratings as follows:

2.2.3.1.1 Penetrations of Fire Resistance Rated Walls and Partitions

F Rating = ~~[[______] hour] [~~Rating of wall or partition being penetrated~~]~~.

2.2.3.1.2 Penetrations of Fire Resistance Rated Floors, Floor-Ceiling Assemblies and the Ceiling Membrane of Roof-Ceiling Assemblies

F Rating = ~~[______]~~2 hour, T Rating = ~~[______]~~2 hour. Where the penetrating item is outside of a wall cavity the F rating must be equal to the fire resistance rating of the floor penetrated, and the T rating must be in accordance with the requirements of ICC IBC.

2.2.3.1.3 Penetrations of Fire and Smoke Resistance Rated Walls, Floors, Floor-Ceiling Assemblies, and the ceiling membrane of Roof-Ceiling Assemblies

F Rating = ~~[______]~~2 hour, T Rating = ~~[______]~~2 hour and L Rating = ~~[[~~<10~~]~~ cfm/sf~~]~~ ~~[~~Where L rating is required~~]~~.

2.2.3.2 Construction Joints and Gaps

Fire resistance ratings of construction joints, as described in paragraph SUMMARY, and gaps such as those between floor slabs and curtain walls must be ~~[~~the same as the construction in which they occur.~~] [as follows: construction joints in walls, [______] hour; construction joints in floors, [______] hour; gaps between floor slabs and curtain walls, [______] hour; gaps between top of the walls and the bottom of roof and floor decks, [______] hour, and provide L rating of <5 cfm/lf where required.]~~ Provide construction joints and gaps with firestopping materials and systems that have been tested in accordance with ASTM E119, ASTM E1966 or UL 2079 to meet the required fire resistance rating. Provide curtain wall joints with firestopping materials and systems that have been tested in accordance with ASTM E2307 to meet the required fire resistance rating. Systems installed at construction joints must meet the cycling requirements of ASTM E1399/E1399M or UL 2079. Provide a minimum class II movement capability for all joints at the intersection of the top of a fire resistance rated wall and the underside of a fire-rated floor, floor ceiling, or roof ceiling assembly.

2.2.4 Material Certification

Submit certificates attesting that firestopping material complies with the specified requirements. Provide certification of compliance with UL 1479 for all intumescent firestop materials used in through penetration systems.

PART 3 EXECUTION

3.1 PREPARATION

Areas to receive firestopping must be free of dirt, grease, oil, or loose materials which may affect the fitting or fire resistance of the firestopping system. For cast-in-place firestop devices, formwork or metal deck to receive device prior to concrete placement must be sound and capable of supporting device. Prepare surfaces as recommended by the manufacturer.

3.2 INSTALLATION

Completely fill void spaces with firestopping material regardless of geometric configuration, subject to tolerance established by the manufacturer. Firestopping systems for filling floor voids 100 mm or more in any direction must be capable of supporting the same load as the floor is designed to support or be protected by a permanent barrier to prevent loading or traffic in the firestopped area. Install firestopping in accordance with manufacturer's written instructions. Provide tested and listed firestop systems in the following locations, except in floor slabs on grade:

a. Penetrations of duct, conduit, tubing, cable and pipe through floors and through fire-resistance rated walls, partitions, and ceiling-floor assemblies.

b. Penetrations of vertical shafts such as pipe chases, elevator shafts, and utility chutes.

c. Gaps at the intersection of floor slabs and curtain walls, including inside of hollow curtain walls at the floor slab.

d. Gaps at perimeter of fire-resistance rated walls and partitions, such as between the top of the walls and the bottom of roof decks.

e. Construction joints in floors and fire rated walls and partitions.

f. Other locations where required to maintain fire resistance rating of the construction.

3.2.1 Insulated Pipes and Ducts

Cut and remove thermal insulation where pipes or ducts pass through firestopping, unless insulation meets requirements specified for firestopping. Replace thermal insulation with a material having equal thermal insulating and firestopping characteristics.

3.2.2 Fire Dampers

Install and firestop fire dampers in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION. Firestop installed with fire damper must be tested and approved for use in fire damper system. Firestop installed with fire damper must be tested and approved for use in fire damper system.

3.2.3 Data and Communication Cabling

Seal cabling for data and communication applications with re-enterable firestopping ~~[products] [devices] [~~products and devices as indicated~~]~~.

3.2.3.1 Re-Enterable Devices

Provide firestopping devices that are pre-manufactured modular devices, containing built-in self-sealing intumescent inserts. Allow for cable moves, additions or changes without the need to remove or replace any firestop materials. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants; while maintaining "L" rating of <10 cfm/sf ~~[~~ measured at ambient temperature and 205 degrees C~~]~~ at 0 percent to 100 percent visual fill.

3.2.3.2 Re-Sealable Products

Provide firestopping pre-manufactured modular products, containing self-sealing intumescent inserts. Allow for cable moves, additions or changes. Provide devices capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants.

3.3 INSPECTION

~~[For Navy projects, install one of each type of penetration and have it inspected and accepted by the [_____] Division, Naval Facilities Engineering Systems Command, Fire Protection Engineer prior to the installation of the remainder of the penetrations. At this inspection, the manufacturer's technical representative of the firestopping material must be present.]~~ For all projects, do not cover or enclose~~[the remainder of][~~ the firestopped areas~~]~~ until inspection is complete and approved by the Contracting Officer. ~~[~~The inspector must inspect~~] [Inspect]~~ the applications initially to ensure adequate preparations (clean surfaces suitable for application, etc.) and periodically during the work to assure that the completed work has been accomplished according to the manufacturer's written instructions and the specified requirements. Submit written reports indicating locations of and types of penetrations and types of firestopping used at each location; record type by UL listed printed numbers.

3.3.1 Inspection Standards

Inspect all firestopping in accordance with ASTM E2393 and ASTM E2174 for firestop inspection, and document inspection results to be submitted.

3.3.2 Inspection Reports

Submit inspection report stating that firestopping work has been inspected and found to be applied according to the manufacturer's recommendations and the specified requirements.

-- End of Section --

SECTION 07 92 00

JOINT SEALANTS
08/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C919	(2012; R 2017) Standard Practice for Use of Sealants in Acoustical Applications
ASTM E84	(2020) Standard Test Method for Surface Burning Characteristics of Building Materials

JAPANESE ADHESIVE INDUSTRY ASSOCIATION (JAIA)

JAIA 4VOC	(2010) Voluntary VOC Regulating Rule for Indoor Air Pollution Control

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1901	(2015) Determination of the Emission of Volatile Organic Compounds and Aldehydes by Building Products - Small Chamber Test Method
JIS A 5758	(2016) Sealants for Sealing and Glazing in Buildings

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 9	(2019) Building Construction Standard Specifications - Chapter 9 Waterproofing Construction

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

1.3 PRODUCT DATA

Include storage requirements, shelf life, curing time, instructions for mixing and application, and accessories. Provide manufacturer's Safety Data Sheets (SDS) for each solvent, primer and sealant material proposed.

1.4 CERTIFICATIONS

1.4.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

1.4.1.1 Adhesives and Sealants

Adhesives and sealants shall be of a formula and type recommended by the manufacturer and be selected for their ability to provide a durable, permanent bond. Adhesives and sealants shall take into consideration such factors as materials to be bonded, expansion and contraction, bond strength, fire rating, and moisture resistance. Adhesives and sealants shall comply with F 4-Star and JAIA 4VOC, JIS A 1901, and contain no asbestos.

1.5 ENVIRONMENTAL CONDITIONS

Apply sealant when the ambient temperature is between 4 and 32 degrees C.

1.6 DELIVERY AND STORAGE

Deliver materials to the jobsite in unopened manufacturers' sealed shipping containers, with brand name, date of manufacture, ~~[~~color,~~]~~ and material designation clearly marked thereon. Label elastomeric sealant containers to identify type, class, grade, and use. Handle and store materials in accordance with manufacturer's printed instructions. Prevent exposure to foreign materials or subjection to sustained temperatures exceeding 32 degrees C or lower than 4 degrees C. Keep materials and containers closed and separated from absorptive materials such as wood and insulation.

1.7 QUALITY ASSURANCE

1.7.1 Compatibility with Substrate

Verify that each sealant is compatible for use with each joint substrate in accordance with sealant manufacturer's printed recommendations for each application.

1.7.2 Joint Tolerance

Provide joint tolerances in accordance with manufacturer's printed instructions.

1.7.3 Mock-Up

Provide a mock-up of each type of sealant using materials, colors, and techniques approved for use on the project. Approved mock-ups may be incorporated into the Work.

1.7.4 Adhesion

Provide in accordance with JIS A 5758.

PART 2 PRODUCTS

2.1 SEALANTS

Provide sealant products that have been tested, found suitable, and documented as such by the manufacturer for the particular substrates to which they will be applied, and comply with MLIT SS Chapter 9.

In areas with ambient temperatures that exceed 43.33 degrees C, do not use polybutene, bituminous, acrylic-latex, polyvinyl acetate latex sealants, polychloroprene (neoprene), and polyurethane foams, and neoprene, and styrene butadiene rubber extruded seals and closure strips due to these materials having maximum recommended surface temperature ranges from 54.44 degrees C to 82.22 degrees C.

2.1.1 Interior Sealants

Provide sealant products per JIS A 5758, used on the interior of the building (defined as inside of the weatherproofing system) meeting VOC content requirements to attain F4-Star, and meet Japan Sealants Industry Association requirement. Provide certification or validation of indoor air quality for interior sealants. Location(s) and color(s) of sealant for the following. Note, color "as selected" refers to manufacturer's full range of color options

LOCATION	COLOR
a. Small voids between walls or partitions and adjacent lockers, casework, shelving, door frames, built-in or surface mounted equipment and fixtures, and similar items.	As selected
b. Perimeter of frames at doors, windows, and access panels which adjoin exposed interior concrete and masonry surfaces.	As selected
c. Joints of interior masonry walls and partitions which adjoin columns, pilasters, concrete walls, and exterior walls unless otherwise detailed.	As selected
d. Joints between edge members for acoustical tile and adjoining vertical surfaces.	As selected
e. Interior locations, not otherwise indicated or specified, where small voids exist between materials specified to be painted.	As selected
f. Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where non-planar tile surfaces meet.	As selected
g. Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.	As selected
h. Behind escutcheon plates at valve pipe penetrations and showerheads in showers.	As selected

2.1.2 Exterior Sealants

For joints in vertical surfaces and horizontal surfaces, provide JIS A 5758. Provide location(s) and color(s) of sealant as follows. Note, color "as selected" refers to manufacturer's full range of color options:

LOCATION	COLOR
a. Joints and recesses formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Use sealant at both exterior and interior surfaces of exterior wall penetrations.	Match adjacent surface color
b. Joints between new and existing exterior masonry walls.	Match adjacent surface color
c. Masonry joints where shelf angles occur.	Match adjacent surface color
d. Joints in wash surfaces of stonework.	Match adjacent surface color
e. Expansion and control joints.	Match adjacent surface color
f. Interior face of expansion joints in exterior concrete or masonry walls where metal expansion joint covers are not required.	Match adjacent surface color
g. Voids where items pass through exterior walls.	Match adjacent surface color
h. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.	Match adjacent surface color
i. Metal-to-metal joints where sealant is indicated or specified.	Match adjacent surface color
j. Joints between ends of gravel stops, fascia, copings, and adjacent walls.	Match adjacent surface color

2.1.3 Floor Joint Sealants

Provide sealant products per JIS A 5758 used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of UL 2818(Greenguard) Gold, SCS Global Services Indoor Advantage Gold, CDPH Section 01350 (limit requirements of either office or classroom spaces regardless of space type), VOC content requirements of SCAQMD Rule 1168, or F4-star, and meet Japan Sealant Industry Association requirement. Provide location(s) and color(s) of sealant as follows. Note, color "as selected" refers to manufacturer's full range of color options:

LOCATION	COLOR
a. Seats of metal thresholds for exterior doors.	As selected

LOCATION	COLOR
b. Control and expansion joints in floors, slabs, ceramic tile, and walkways.	As selected

2.1.4 Acoustical Sealants

~~[_____]~~ Rubber or polymer based acoustical sealant in accordance to ASTM C919, to have a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E84. Provide non-staining acoustical sealant with a consistency of 250 to 310. Acoustical sealant must remain flexible and adhesive after 500 hours of accelerated weathering. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) to have low or no pollutant emissions.

2.1.5 Preformed Sealants

Provide preformed sealants of polybutylene or isoprene-butylene based pressure sensitive weather resistant tape or bead sealants capable of sealing out moisture, air and dust when installed as recommended by the manufacturer. At temperatures from minus 34 to plus 71 degrees C, sealants must be non-bleeding and have no loss of adhesion.

2.1.5.1 Tape

~~[_____]~~ Tape sealant: Provide cross section dimensions of ~~[_____]~~19 mm.

2.1.5.2 Bead

~~[_____]~~ Bead sealant: Provide cross section dimensions of ~~[_____]~~19 mm.

2.1.5.3 Foam Strip

Provide ~~[_____]~~19 mm foam strip of polyurethane foam with cross section dimensions of ~~[_____]~~19 mm. Provide foam strip capable of sealing out moisture, air, and dust when installed and compressed in accordance with manufacturer's printed instructions. Service temperature must be minus 40 to plus 135 degrees C. Furnish untreated strips with adhesive to hold them in place. Do not allow adhesive to stain or bleed onto adjacent finishes. Saturate treated strips with butylene waterproofing or impregnate with asphalt.

2.2 PRIMERS

Non-staining, quick drying type and consistency as recommended by the sealant manufacturer for the particular application. Provide primers for interior applications that meet the indoor air quality requirements of the paragraph SEALANTS above.

2.3 BOND BREAKERS

Type and consistency as recommended by the sealant manufacturer to prevent adhesion of the sealant to the backing or to the bottom of the joint. Provide bond breakers for interior applications that meet the indoor air quality requirements of the paragraph SEALANTS above.

2.4 BACKSTOPS

Provide glass fiber roving, neoprene, butyl, polyurethane, or polyethylene foams free from oil or other staining elements as recommended by sealant manufacturer. Provide 25 to 33 percent oversized backing for closed cell and 40 to 50 percent oversized backing for open cell material, unless otherwise indicated. Provide backstop material that is compatible with sealant. Do not use oakum~~[, [_____]]~~ or other types of absorptive materials as backstops.

2.4.1 Rubber

Provide in accordance per manufacturer requirement ~~[round] [_____]~~ cross section for ~~[_____]~~ cellular rubber sponge backing.

2.4.2 Silicone Rubber Base

Provide in accordance with JIS A 5758. Color ~~[~~as selected from manufacturer's full range of color choices~~] [_____]~~.

2.5 CAULKING

For interior use and only where there is little or no anticipated joint movement. Provide products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350(limit requirements of either office or classroom spaces regardless of space type), VOC content requirements of SCAQMD Rule 1168, or F-4 Star. Provide certification or validation of indoor air quality for interior caulking.

2.6 CLEANING SOLVENTS

Provide type(s) recommended by the sealant manufacturer and in accordance with environmental requirements herein. ~~[~~Protect adjacent aluminum and bronze surfaces from solvents~~]~~. Provide solvents for interior applications that meet the indoor air quality requirements of the paragraph SEALANTS above.

PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL

Perform a field adhesion test in accordance with manufacturer's instructions. Remove sealants that fail adhesion testing; clean substrates, reapply sealants, and re-test. Test sealants adjacent to failed sealants. Submit field adhesion test report indicating tests, locations, dates, results, and remedial actions taken.

3.2 SURFACE PREPARATION

Prepare surfaces according to manufacturer's printed installation instructions. Clean surfaces from dirt, frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter that would destroy or impair adhesion. Remove oil and grease with solvent; thoroughly remove solvents prior to sealant installation. Wipe surfaces dry with clean cloths. When resealing an existing joint, remove existing caulk or sealant prior to applying new sealant. For surface types not listed below, provide in accordance with sealant manufacturer's printed instructions for each specific surface.

3.2.1 Steel Surfaces

Remove loose mill scale by sandblasting or, if sandblasting is impractical or would damage finished work, scraping and wire brushing. Remove protective coatings by sandblasting or using a residue free solvent. Remove resulting debris and solvent residue prior to sealant installation.

3.2.2 Aluminum or Bronze Surfaces

Remove temporary protective coatings from surfaces that will be in contact with sealant. When masking tape is used as a protective coating, remove tape and any residual adhesive prior to sealant application. For removing protective coatings and final cleaning, use non-staining solvents recommended by the manufacturer of the item(s) containing aluminum or bronze surfaces.

3.2.3 Concrete and Masonry Surfaces

Where surfaces have been treated with curing compounds, oil, or other such materials, remove materials by sandblasting or wire brushing. Remove laitance, efflorescence and loose mortar from the joint cavity. Remove resulting debris prior to sealant installation.

3.2.4 Wood Surfaces

Ensure wood surfaces that will be in contact with sealants are free of splinters, sawdust and other loose particles.

~~[~~3.2.5 Removing Existing Hazardous Sealants

For sealants applied prior to 1979, or that have been tested and found to contain polychlorinated biphenyls (PCBs), remove and dispose of these sealants in accordance with Section ~~02 84 33 REMOVAL AND DISPOSAL OF POLYCHLORINATED BIPHENYLS (PCBs)~~02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS CONTAINING PCBS AND MERCURY.

~~]~~3.3 SEALANT PREPARATION

Do not add liquids, solvents, or powders to sealants. Mix multicomponent elastomeric sealants in accordance with manufacturer's printed instructions.

3.4 APPLICATION

3.4.1 Joint Width-To-Depth Ratios

Acceptable Ratios:

JOINT WIDTH	JOINT DEPTH	
	Minimum	Maximum
For metal, glass, or other nonporous surfaces:		
6 mm (minimum)	6 mm	6 mm

JOINT WIDTH	JOINT DEPTH	
	Minimum	Maximum
over 6 mm	1/2 of width	Equal to width
For wood, concrete, masonry, stone:		
6 mm (minimum)	6 mm	6 mm
over 6 mm to 13 mm	6 mm	Equal to width
over 13 mm to 25 mm	50 mm	16 mm
Over 25 mm	prohibited	

Unacceptable Ratios: Where joints of acceptable width-to-depth ratios have not been provided, clean out joints to acceptable depths and grind or cut to acceptable widths without damage to the adjoining work. Grinding is prohibited at metal surfaces.

3.4.2 Unacceptable Sealant Use

Do not install sealants in lieu of other required building enclosure weatherproofing components such as flashing, drainage components, and joint closure accessories, or to close gaps between walls, floors, roofs, windows, and doors, that exceed acceptable installation tolerances. Remove sealants that have been used in an unacceptable manner and correct building enclosure deficiencies to comply with contract documents requirements.

3.4.3 Masking Tape

Place masking tape on the finished surface on one or both sides of joint cavities to protect adjacent finished surfaces from primer or sealant smears. Remove masking tape within 10 minutes of joint filling and tooling.

3.4.4 Backstops

Provide backstops dry and free of tears or holes. Tightly pack the back or bottom of joint cavities with backstop material to provide joints in specified depths. Provide backstops where indicated and where backstops are not indicated but joint cavities exceed the acceptable maximum depths specified in JOINT WIDTH-TO-DEPTH RATIOS Table.

3.4.5 Primer

Clean out loose particles from joints immediately prior to application of. Apply primer to joints in concrete masonry units, wood, and other porous surfaces in accordance with sealant manufacturer's printed instructions. Do not apply primer to exposed finished surfaces.

3.4.6 Bond Breaker

Provide bond breakers to surfaces not intended to bond in accordance with, sealant manufacturer's printed instructions for each type of surface and sealant combination specified.

3.4.7 Sealants

Provide sealants compatible with the material(s) to which they are applied. Do not use a sealant that has exceeded its shelf life or has jelled and cannot be discharged in a continuous flow from the sealant gun. Apply sealants in accordance with the manufacturer's printed instructions with a gun having a nozzle that fits the joint width. Work sealant into joints so as to fill the joints solidly without air pockets. Tool sealant after application to ensure adhesion. Apply sealant uniformly smooth and free of wrinkles. Upon completion of sealant application, roughen partially filled or unfilled joints, apply additional sealant, and tool smooth as specified. Apply sealer over sealants in accordance with the sealant manufacturer's printed instructions.

3.5 PROTECTION AND CLEANING

3.5.1 Protection

Protect areas adjacent to joints from sealant smears. Masking tape may be used for this purpose if removed 5 to 10 minutes after the joint is filled and no residual tape marks remain.

3.5.2 Final Cleaning

Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean and neat condition.

a. Masonry and Other Porous Surfaces: Immediately remove fresh sealant that has been smeared on adjacent masonry, rub clean with a solvent, and remove solvent residue, in accordance with sealant manufacturer's printed instructions. Allow excess sealant to cure for 24 hour then remove by wire brushing or sanding. Remove resulting debris.

b. Metal and Other Non-Porous Surfaces: Remove excess sealant with a solvent moistened cloth. Remove solvent residue in accordance with solvent manufacturer's printed instructions.

-- End of Section --

SECTION 08 11 13

STEEL DOORS AND FRAMES
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 4702	(20~~15~~21) Doorsets
JIS G 3302	(2019) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3317	(2019) Hot-Dip Zinc-5 Percent Aluminum Allot-Coated Steel Sheet and Strip
JIS Z 3420	(2003) Specification and Approval of Welding Procedures for Metallic Materials - General Rules

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 16	(2019) Building Construction Standard Specifications - Chapter 16 Opening Construction

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2019) Standard for Fire Doors and Other Opening Protectives
NFPA 105	(2019) Standard for Smoke Door Assemblies and Other Opening Protectives
NFPA 252	(2017) Standard Methods of Fire Tests of Door Assemblies

UNDERWRITERS LABORATORIES (UL)

UL 10C	(2016; Reprint May 2021) UL Standard for Safety Positive Pressure Fire Tests of Door Assemblies

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion~~

~~in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Doors; G~~[, [_____]]~~

Frames; G~~[, [_____]]~~

Show elevations, construction details, metal gages, hardware provisions, method of glazing, and installation details.

Schedule of Doors; G~~[, [_____]]~~

Schedule of Frames; G~~[, [_____]]~~

Submit door and frame locations.

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver doors, frames, and accessories undamaged and with protective wrappings or packaging.~~[~~ Strap knock-down frames in bundles.~~][~~ Provide temporary steel spreaders securely fastened to the bottom of each welded frame.~~]~~ Store doors and frames on platforms under cover in clean, dry, ventilated, and accessible locations, with 6 mm airspace between doors. Remove damp or wet packaging immediately and wipe affected surfaces dry. Replace damaged materials with new.

PART 2 PRODUCTS

2.1 STANDARD STEEL DOORS

JIS A 4702, except as specified otherwise. Prepare doors to receive door hardware as specified in Section 08 71 00 DOOR HARDWARE. Undercut where indicated. ~~Provide exterior doors with top edge closed flush and sealed to prevent water intrusion.~~ Provide doors at 44.5 mm thick, unless otherwise indicated. Provide door material that uses a minimum of 25 percent recycled content.~~ [Provide exterior glazing in accordance with project load resistance requirement.]~~

2.1.1 Classification - Level, Performance, Model

~~2.1.1.1 Standard Duty Doors (Level 1)~~

~~35 mm to 45 mm door thickness, 0.8 mm door faces steel thickness, 1.0 mm frames steel thickness [full flush] of size(s) and design(s) indicated and core construction as required by the manufacturer. Provide door classification label per JIS A 4702.~~

2.1.1.1 Heavy Duty Doors (Level 2)

45 mm door thickness, 1.0 mm door faces steel thickness, 1.3 mm frames steel thickness ~~[~~full flush~~] [seamless]~~, with core construction as required by the manufacturer ~~[~~for interior doors~~][and][for exterior doors]~~, of size(s) and design(s) indicated. ~~[~~Where vertical stiffener cores are required, the space between the stiffeners must be filled with mineral board insulation.~~][~~ Provide door classification label per JIS A 4702.~~]~~

2.1.1.2 Extra Heavy Duty Doors (Level 3)

45 mm door thickness, 1.3mm door faces and frames steel thickness, ~~[~~full flush~~] [seamless] [stile and rail]~~ with core construction as required by the manufacturer ~~[~~for interior doors~~][and][for indicated exterior doors]~~, of size(s) and design(s) indicated.~~[~~ Where vertical stiffener cores are required, the space between the stiffeners must be filled with mineral board insulation.~~][~~ Provide door classification label per JIS A 4702.~~]~~

~~2.1.1.3 Maximum Duty Doors (Level 4)~~

~~45 mm door thickness, 1.7mm door faces and frames steel thickness, [full flush] [seamless] with core construction as required by the manufacturer [for interior doors][and][for indicated exterior doors], of size(s) and design(s) indicated.[Where vertical stiffener cores are required, the space between the stiffeners must be filled with mineral board insulation.][Provide door classification label per JIS A 4702.]~~

2.2 CUSTOM HOLLOW METAL DOORS

Provide custom hollow metal doors where nonstandard steel doors are indicated. Provide custom steel doors in the door size(s), design(s), materials, construction, gages, and finish as specified for custom steel doors and complying with the requirements. Fill all spaces in doors with insulation. Close top and bottom edges with steel channels not lighter than 1.5 mm thick.~~[Close tops of exterior doors flush with an additional channel and seal to prevent water intrusion.]~~ Prepare doors to receive hardware specified in Section 08 71 00 DOOR HARDWARE.~~[~~ Undercut doors where indicated.~~]~~ Provide doors at 45 mm thick, unless otherwise indicated.

~~2.3 INSULATED STEEL DOOR SYSTEMS~~

~~[At the option of the Contractor, insulated steel doors and frames may be provided in lieu of Level 1 standard steel doors and frames. Provide insulated steel doors in the door size(s), design, and material as specified for standard steel doors.]Provide insulated steel doors with a core of polyurethane foam; face sheets, edges, and frames of galvanized steel not lighter than 0.7 mm thick, 1.5 mm thick, and 1.5 mm respectively; magnetic weatherstripping; nonremovable-pin hinges; thermal-break aluminum threshold; and vinyl door bottom. Provide to doors and frames a phosphate treatment, rust-inhibitive primer, and baked acrylic enamel finish. Door shall meet JIS A 4702 and tested for 500,000 cycles. Prepare doors to receive specified hardware. Provide doors 44.5 mm thick.[Provide insulated steel doors and frames [at entrances to dwelling units][where shown][_____].]~~

~~[~~2.3 SOUND RATED STEEL DOORS

Provide sound rated doors with a Transmission Loss (TL) ~~[of [_____]][~~as indicated on the drawings~~]~~.

~~]2.4 ACCESSORIES~~

~~2.4.1 Shelves for Dutch Doors~~

~~Fabricate shelves of steel not lighter than 1.5 mm thick, [[_____] mm wide][of the size indicated]. Provide brackets of stock type fabricated~~

~~of the same metal used to fabricate shelves.~~

~~2.4.2 Louvers~~

~~2.4.2.1 Interior Louvers~~

~~Where indicated, provide louvers of stationary [sightproof][and][lightproof] type[where scheduled].[Louvers for lightproof must not transmit light.] Detachable moldings on room or non security side of door; on security side of door, moldings to be integral part of louver. Form louver frames of 1 mm thick steel and louver blades of a minimum 0.6 mm.[Louvers for lightproof doors must have minimum of 20 percent net-free opening.][Sightproof louvers to be inverted ["V" blade design with minimum 55][and][inverted ["Y"] blade design with minimum 40] percent net-free opening.]~~

~~2.4.2.2 Exterior Louvers~~

~~Provide louvers of the inverted ["Y"]["V"]["Z"] type with minimum of [30][55][35] percent net-free opening. Weld or tenon louver blades to continuous channel frame and weld assembly to door to form watertight assembly. Form louvers of hot-dip galvanized steel of same gage as door facings. At louvers provide steel-framed [insect][bird] screens secured to room side and readily removable. Provide [aluminum wire cloth, 7 by 7 per 10 mm or 7 by 6 per 10 mm mesh, for insect screens][galvanized steel, 13 by 13 mm mesh hardware cloth, for bird screens]. Net-free louver area to be before screening.~~

~~2.4.3 Astragals~~

~~For pairs of exterior steel doors which will not have aluminum astragals or removable mullions, as specified in Section 08 71 00 DOOR HARDWARE provide overlapping steel astragals with the doors.~~

~~2.4.4 Moldings~~

~~Provide moldings around glass of interior and exterior doors and louvers of interior doors. Provide nonremovable moldings on outside of exterior doors and on corridor side of interior doors. Other moldings may be stationary or removable. Secure inside moldings to stationary moldings, or provide snap-on moldings.~~

~~2.5 INSULATION CORES~~

~~Provide insulating cores of the type specified, and provide an apparent U-factor per insulation requirements in accordance with JIS A4702, [H-_____]~~

2.4 STANDARD STEEL FRAMES

Provide hardware reinforcing thickness ~~per ANSI/SDI A250.8 requirement~~in accordance with JIS A 4702. Form frames to sizes and shapes indicated, with ~~[~~welded corners~~][~~ or ~~][~~knock-down field-assembled corners~~]~~. Provide steel frames for doors, ~~[transoms,] [sidelights,] [mullions,] [cased openings,][and][interior glazed panels,]~~ unless otherwise indicated. Provide frame product that uses a minimum of 25 percent recycled content. Provide data indicating percentage of recycled content for steel frame product.

2.4.1 Welded Frames

Continuously weld frame faces at corner joints. Mechanically interlock or continuously weld stops and rabbets. Grind welds smooth.

Weld frames in accordance with the recommended practice of Welding Code per JIS Z 3420 and in accordance with the practice specified by the producer of the metal being welded.

2.4.2 Knock-Down Frames

Design corners for simple field assembly by concealed tenons, splice plates, or interlocking joints that produce square, rigid corners and a tight fit and maintain the alignment of adjoining members. Provide locknuts for bolted connections.

~~2.4.3 Mullions and Transom Bars~~

~~Provide mullions and transom bars of closed or tubular construction with heads and jambs butt-welded together[or knock-down for field assembly]. Bottom of door mullions must have adjustable floor anchors and spreader connections.~~

2.4.3 Stops and Beads

Provide form and loose stops and beads from 0.9 mm thick steel. Provide for glazed and other openings in standard steel frames. Secure beads to frames with oval-head, countersunk Phillips self-tapping sheet metal screws or concealed clips and fasteners. Space fasteners approximately 300 to 400 mm on center. Miter molded shapes at corners. Butt or miter square or rectangular beads at corners.

~~2.4.4 Terminated Stops~~

~~Where indicated, terminate interior door frame stops 150 mm above floor.[Do not terminate stops of frames for [lightproof,] [soundproof,] [or lead-lined] doors.]~~

~~2.4.5 Cased Openings~~

~~Fabricate frames for cased openings of same material, gage, and assembly as specified for metal door frames, except omit door stops and preparation for hardware.~~

2.4.4 Anchors

Provide anchors to secure the frame to adjoining construction. Provide steel anchors, zinc-coated not lighter than 1.2 mm thick.

2.4.4.1 Wall Anchors

Provide at least three anchors for each jamb. For frames which are more than 2285 mm in height, provide one additional anchor for each jamb for each additional 760 mm or fraction thereof.

a. Masonry: Provide anchors of corrugated or perforated steel straps or 5 mm diameter steel wire, adjustable or T-shaped;

b. Stud partitions: Weld or otherwise securely fasten anchors to backs of frames. Design anchors to be fastened ~~[to wood studs with nails,]~~ ~~[~~to closed steel studs with sheet metal screws~~, and to open steel studs by wiring or welding]~~;

c. Completed openings: Secure frames to previously placed concrete or masonry with expansion bolts in accordance with JIS A 4702~~; and~~.

~~d. Solid plaster partitions: Secure anchors solidly to back of frames and tie into the lath. Provide adjustable top strut anchors on each side of frame for fastening to structural members or ceiling construction above. Provide size and type of strut anchors as recommended by the frame manufacturer.~~

2.4.4.2 Floor Anchors

Provide floor anchors drilled for 10 mm anchor bolts at bottom of each jamb member. ~~[~~Where floor fill occurs, terminate bottom of frames at the indicated finished floor levels and support by adjustable extension clips resting on and anchored to the structural slabs.~~]~~

2.5 FIRE ~~[~~AND~~]~~ ~~[~~SMOKE~~]~~ DOORS AND FRAMES

Provide fire~~[~~ and smoke~~]~~ doors and frames in accordance with NFPA 80~~[~~ and ~~][~~NFPA 105~~]~~ and this specification.~~[Include insulated core materials in fire doors where indicated in the door schedule.]~~

2.5.1 Labels

Provide fire doors and frames bearing the label of Underwriters Laboratories (UL), Factory Mutual Engineering and Research (FM), or Warnock Hersey International (WHI) attesting to the rating required. Testing must be in accordance with NFPA 252 or UL 10C. Provide labels that are metal with raised letters, bearing the name or file number of the door and frame manufacturer. Labels must be permanently affixed at the factory to frames and to the hinge edge of the door. Do not paint door and labels. Use of Japanese tested and labeled fire doors, up to 20 minutes, is permitted. NFPA 252 paragraph 6.2.2 requires hose stream test for fire rated doors greater than 20 minutes. Japanese manufactured fire rated doors are not subjected to hose stream tests and are not deemed equivalent to US products or requirements.

Japanese manufactured fire doors and frames, greater than 20 minutes, with UL documentation that the door has successfully passed ANSI/UL testing, will be accepted. Provide labels that are metal with raised letters, bearing the name or file number of the door and frame manufacturer. Labels must be permanently affixed at the factory to frames and to the hinge edge of the door. Do not paint door and labels.

2.5.2 Oversized Doors

For fire doors and frames which exceed the size for which testing and labeling are available, furnish certificates stating that the doors and frames are identical in design, materials, and construction to a door which has been tested and meets the requirements for the class indicated.

2.5.3 Astragal on Fire ~~[~~and Smoke~~]~~ Doors

On pairs of labeled fire doors, conform to NFPA 80 and UL requirements.~~[~~ On smoke control doors, conform to NFPA 105.~~]~~

2.6 EXTERIOR FRAMES

Provide thermal insulation in all exterior frames. Provide frames of a minimum Level 4, with frames of a minimum thickness of 1.7 mm, 14 gage.

2.7 WEATHERSTRIPPING

As specified in Section 08 71 00 DOOR HARDWARE.

~~[2.7.1 Integral Gasket~~

~~Black synthetic rubber gasket with tabs for factory fitting into factory slotted frames, or extruded neoprene foam gasket made to fit into a continuous groove formed in the frame, may be provided in lieu of head and jamb seals specified in Section 08 71 00 DOOR HARDWARE. Insert gasket in groove after frame is finish painted. Provide doors where air leakage of weatherstripped doors does not exceed [2.19 by 10-5] [5.48 by 10-5] cubic meters per second of air per square meter of door area when tested in accordance with JIS A 1516.~~

~~]~~2.8 HARDWARE PREPARATION

Provide minimum hardware reinforcing gages as specified in JIS A 4702. Drill and tap doors and frames to receive finish hardware. Prepare doors and frames for hardware in accordance with the applicable requirements of JIS A 4702. Drill and tap for surface-applied hardware at the project site. Build additional reinforcing for surface-applied hardware into the door at the factory. Locate hardware in accordance with the requirements of JIS A 4702, as applicable. Punch door frames~~ [~~, with the exception of frames that will have weatherstripping ~~[or] [lightproof] [or~~~~]~~ ~~[~~soundproof~~]~~ gasketing,~~]~~ to receive a minimum of two rubber or vinyl door silencers on lock side of single doors and one silencer for each leaf at heads of double doors. Set lock strikes out to provide clearance for silencers.

2.9 FINISHES

~~[2.9.1 Factory-Primed Finish~~

~~Thoroughly clean all surfaces of doors and frames then chemically treat and factory prime with a rust inhibiting coating as specified in JIS A 4702.[, or paintable A25 galvannealed steel without primer. Where coating is removed by welding, apply touchup of factory primer.]~~

~~][~~2.9.1 Hot-Dip Zinc-Coated and Factory-Primed Finish

Fabricate ~~[exterior][interior][scheduled]~~ doors and frames from hot dipped zinc coated steel, alloyed type, that complies with JIS G 3302 and JIS G 3317. The coating weight must meet or exceed the minimum requirements for coatings having zinc-5 percent. Repair damaged zinc-coated surfaces by the application of zinc dust paint. Thoroughly clean and chemically treat to insure maximum paint adhesion. Factory prime as specified in JIS A 4702.~~[Provide for [exterior doors][and][interior doors][door openings No. [_____]]].~~

~~]2.9.2 Electrolytic Zinc-Coated Anchors and Accessories~~

~~Provide electrolytically deposited zinc-coated steel in accordance with JIS G 3313, Commercial Quality, Coating Class A. Phosphate treat and~~

~~factory prime zinc-coated surfaces as specified in JIS A 4702.~~

~~[2.9.3 Factory-Applied Enamel Finish~~

~~Provide coatings that meet test procedures and acceptance criteria in accordance with JIS A 4702. After factory priming, apply [one coat][two coats] of [low-gloss][medium-gloss] enamel to exposed surfaces. Separately bake or oven dry each coat. Drying time and temperature requirements must be in accordance with the coating manufacturer's recommendations. Provide finish coat color(s) [as indicated][_____] to match approved color sample(s).~~

~~]~~2.10 FABRICATION AND WORKMANSHIP

Provide finished doors and frames that are strong and rigid, neat in appearance, and free from defects, waves, scratches, cuts, dents, ridges, holes, warp, and buckle. Provide molded members that are clean cut, straight, and true, with joints coped or mitered, well formed, and in true alignment. Dress exposed welded and soldered joints smooth. Design door frame sections for use with the wall construction indicated. Corner joints must be well formed and in true alignment. Conceal fastenings where practicable.~~[Frames for use in solid plaster partitions must be welded construction.][~~ On wraparound frames for masonry partitions, provide a throat opening 3 mm larger than the actual masonry thickness.~~][~~ Design~~[~~ other~~]~~ frames in exposed masonry walls or partitions to allow sufficient space between the inside back of trim and masonry to receive caulking compound.~~]~~

~~2.10.1 Grouted Frames~~

~~For frames to be installed in exterior walls and to be filled with mortar or grout, fill the stops with strips of rigid insulation to keep the grout out of the stops and to facilitate installation of stop-applied head and jamb seals.~~

~~2.11 PROVISIONS FOR GLAZING~~

~~Materials are specified in Section 08 81 00, GLAZING.~~

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Frames

Set frames in accordance with JIS A 4702 and MLIT SS Chapter 16. Plumb, align, and brace securely until permanent anchors are set. Anchor bottoms of frames with expansion bolts or powder-actuated fasteners. Build in or secure wall anchors to adjoining construction. ~~[~~Where frames require ceiling struts or overhead bracing, anchor frames to the struts or bracing. ~~]~~

3.1.2 Doors

Hang doors in accordance with clearances specified in JIS A 4702 and MLIT SS Chapter 16. After erection and glazing, clean and adjust hardware.

3.1.3 Fire ~~[~~and Smoke~~]~~ Doors and Frames

Install fire doors and frames, including hardware, in accordance with NFPA 80.~~[~~ Install~~[~~ fire rated~~]~~ smoke doors and frames in accordance with ~~[~~ NFPA 80~~][~~ and ~~][~~NFPA 105~~]~~.~~]~~

3.2 PROTECTION

Protect doors and frames from damage. Repair damaged doors and frames prior to completion and acceptance of the project or replace with new, as directed. Wire brush rusted frames until rust is removed. Clean thoroughly. Apply an all-over coat of rust-inhibitive paint of the same type used for shop coat.

3.3 CLEANING

Upon completion, clean exposed surfaces of doors and frames thoroughly. Remove mastic smears and other unsightly marks.

-- End of Section --

SECTION 08 11 16

ALUMINUM DOORS~~ AND FRAMES~~, FRAMES, ENTRANCES AND WINDOWS
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16	(2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM F2006	Standard Safety Specification for Window Fall Prevention Devices for Nonemergency Escape (Egress) and Rescue (Ingress) Windows
ASTM F2090	Standard Specification for Window Fall Prevention Devices With Emergency Escape (Egress) Release Mechanisms

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1516	(1998) Windows and Doorsets - Air Permeability Test
JIS A 1517	(~~1996~~2020) Windows and Doorsets-Watertightness Test Under Dynamic
JIS A 4702	(20~~15~~21) Doorsets
JIS H 4000	(2017) Aluminium and Aluminium Alloy Sheets, Strips and Plates (Amendment 1)
JIS H 4040	(20~~15~~20) Aluminum and Aluminum Alloy Bars and Wires
JIS H 4100	(2015) Aluminum and Aluminum Alloy Extruded Profiles
JIS H 8602	(2010) Combined Coatings of Anodic Oxide and Organic Coatings on Aluminum and Aluminum Alloys

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 16	(2019) Building Construction Standard Specifications - Chapter 16 Opening

Construction

1.2 PERFORMANCE REQUIREMENTS

Aluminum doors, frames, entrances, and windows must meet the following performance requirements. Perform testing requirements by a testing laboratory or agency.

1.2.1 Structural Calculations

~~Commentary: Add wind pressure requirements in accordance with ASCE 7.~~
Design and test exterior aluminum doors and frames including frames, hardware, anchors, and glazing to withstand wind-loading design pressures as indicated in the structural drawings and ASCE 7-16.

Entrances and Window framing members for each individual light of glass must not deflect to the extent that deflection perpendicular to the glass light exceeds L/175 of the glass edge length when subjected to uniform loads at specified design pressures. Provide structural calculations to substantiate compliance with deflection requirements.

Structural test pressures on exterior doors, frames, entrances, and windows must be for positive load (inward) and negative load (outward). After testing, there will be no glass breakage, permanent damage to fasteners and hardware parts, including support arms or actuating mechanisms or any other damage which could cause doors to be inoperable.

1.2.1.1 Minimum ~~Antiterrorism~~Structural Performance

Provide ~~doors~~doors, frames, entrances and windows meeting the minimum ~~antiterrorism~~ performance as specified in the paragraphs below.

Aluminum ~~door, window frame and window wall~~doors, frames, entrances and windows shall be designed to support
the wind pressures ~~specified~~and Wind Borne Debris Impact Resistance as identified in the Structural Drawings, but not less than the minimum section
properties indicated. The anchorages shall be designed to support the wind load reactions calculated, but not less than the support loads indicated on the contract documents.

~~If the minimum section properties and anchorages were not indicated, one of the following methods can be used as noted in the following:~~

~~[a. Static Equivalent Load Design~~

~~][1.2.2 Wind Borne Debris~~

~~Provide impact resistant door [_____] assemblies meeting the Windborne-Debris-Impact Resistant Performance requirements of JIS R 3109 as follows:~~

~~(1) Pass missile-impact tests when tested according to JIS R 3109 for missiles A and D in Table 2 or ISO 16932 (Missile C).~~

~~]~~1.2.2 Air Infiltration

When tested in accordance with JIS A 1516, air infiltration per door leaf cannot exceed 2.83 by 10-4 cms per square meter of fixed area at a test pressure of 0.30 kPa.

1.2.3 Water Penetration

When tested in accordance with JIS A 1517, there can be no water penetration at a pressure of 0.14 kPa of fixed area.

1.2.4 Thermal Transmittance, Solar Heat Gain, Visible Light Transmittance

1.2.4.1 U-Factor

Provide exterior glazed assemblies, ~~including aluminum entrances doors with greater than 50 percent glazed area with U-Factor [_____]~~with maximum U-Factors indicated below:~~.~~

Fixed windows - 0.46
Operable windows - 0.60
Entrances - 0.65

1.2.4.2 Solar Heat Gain Coefficient (SHGC)

Provide exterior glazed assemblies, including aluminum entrances doors with greater than 50 percent glazed area, certified by the National Fenestration Rating Council with a whole window maximum SHGC of ~~[_____]~~0.25.

1.2.4.3 Visible ~~Light~~ Transmittance (V~~L~~T)

Provide exterior glazed assemblies, ~~including aluminum entrances doors with greater than 50 percent glazed area with VLT [_____]~~with VT/SHGC Ratio of 1.10.

~~[1.2.4.4 Doors with Less than 50 Percent Glazed Area~~

~~For exterior aluminum entrances doors with less than 50 percent glazed area, the glazed area is considered the fenestration area with a whole window U-Factor, SHGC and VLT as required above.~~

~~]~~1.2.4.4 Fall Protection

1. All operable windows with an opening large enough that a 100mm sphere can pass through and greater than 1.8M above the exterior adjacent surface must be provided with window fall prevention devices that comply with ASTM F2090(Standard Specification for Window Fall Prevention Devices with Emergency (Egress) Release Mechanisms for ASTM F2006, as applicable.

2. ASTM F2090 applies to all operable windows meeting the above description, are installed below 23m above exterior adjacent surface, and meet the size requirements for emergency escape and rescue. Minimum oening size of 0.53M2, 610mm high, and 508mm wide.

3. ASTM F2006 applies to windows installed at heights of more than 23m above exterior adjacent surfaces and windows that do not meet the size requirements for emergency escape and rescue.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality~~

~~Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

For Each Type ~~of Door and Frame~~Door, Frame, Entrance and Window Assembly; G~~[, [_____]]~~

SD-04 Samples

Finish Samples; G~~[, [_____]]~~

SD-05 Design Data

Structural Calculations for Deflection; G~~[, [_____]]~~

SD-06 Test Reports

Air Infiltration; G~~[, [_____]]~~

Water Penetration; G~~[, [_____]]~~

SD-10 Operation and Maintenance Data

Adjustments, Cleaning, and Maintenance; G~~[, [_____]]~~

1.4 DELIVERY, STORAGE, AND HANDLING

Inspect materials delivered to the site for damage. Unload and store with minimum handling. Provide storage space in dry location with adequate ventilation, free from dust or water, and easily accessible for inspection and handling. Stack materials on non-absorptive strips or wood platforms. Do not cover doors and frames with tarps, polyethylene film, or similar coverings. Protect finished surfaces during shipping and handling using manufacturer's standard method. Do not apply coatings or lacquers to surfaces to which caulking and glazing compounds must adhere.

1.5 QUALITY CONTROL

1.5.1 Qualifications

1.5.1.1 Installer Qualifications

Provide documentation of the installer's experience in performing the work specified in this section. Ensure that the installers are specialized in work similar to that required for this project, and that they are acceptable to product manufacturer.

1.5.1.2 Manufacturer Qualifications

Ensure that manufacturers meet the requirements specified in this section and project drawings. Ensure that the manufacturer is capable of providing field service representation during construction, approving acceptable installers and approving application methods.

1.5.2 Single-Source Responsibility

Provide aluminum doors, windows, and window wall products from a single-source manufacturer. Use a single source manufacturer with sole responsibility for providing design, structural engineering, and custom fabrication for door portal systems and for supplying components, materials, and products. Do not use products provided from numerous sources for assembly at the site.

1.5.3 Shop Drawing

Indicate elevations and sections for each type of ~~door and frame~~door, frame, entrance, and window assembly. Show sizes and details of each assembly, frame construction,~~[subframe attachment,]~~ thickness and gages of metal, details of door and frame construction, proposed method(s) of anchorage, glazing details, provisions for an location of hardware, ~~[~~ mullion details,~~]~~ method and materials for flashing and weatherstripping, miscellaneous trim, installation details, and other related items necessary for a complete representation of all components.~~ A qualified blast engineer must perform testing or calculations for door system design resistance to specified blast loads.~~

1.5.4 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

1.5.5 Operation and Maintenance Data

Submit detailed instructions for installation, adjustments, cleaning, and maintenance of each type of assembly indicated.

1.6 PROJECT / SITE CONDITIONS

1.6.1 Field Measurements

Verify actual measurements or openings by taking field measurements before fabrication; record these measurements on shop drawings. To avoid construction delays, coordinate field measurements, and fabrication schedule with construction progress.

1.7 WARRANTY

Provide a written manufacturer's warranty, executed by a company official, warranting against defects in materials and products for 2 years from the date of shipment. Warrant that the door corner construction is for the life of the project. Provide a written installer's warranty, warranting work to be watertight and free from defective materials, defective workmanship, and glass breakage as a result of defective design, and agreeing to replace components that fail within 2 years. The warranty states the following:

a. Watertight and airtight system installation is completed within specified tolerances.

b. The completed installation remains free of rattles, wind whistles and noise caused by thermal movement and wind pressure.

c. System is structurally sound and free from distortion.

d. Glass and glazing gaskets will not break or "pop" from frames as a result of design, wind load pressure, movement caused by expansion or contraction, or structural loading.

e. Glazing sealants and gaskets remain free of abnormal deterioration or dislocation as a result of sunlight, weather, or oxidation.

PART 2 PRODUCTS

2.1 DOORS AND FRAMES ALUMINUM DOORS, FRAMES, ENTRANCES AND WINDOWS

Provide swing-type aluminum doors and frames of size, design, and location indicated. Provide doors complete with frames, framing members~~[, subframes]~~[, transoms~~][, adjoining side lites]~~ , trim, and accessories.~~[Coordinate side lites, window walls, adjacent curtainwall with Section 08 41 13 ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS][and Section 08 44 00 CURTAIN WALL AND GLAZED ASSEMBLIES.]~~

2.2 MATERIALS

2.2.1 Anchors

316 or 304 Stainless steel~~ [or steel with hot-dipped galvanized finish]~~.

2.2.2 Weatherstripping

Continuous wool pile, silicone treated, or type recommended by door manufacturer, and per JIS A 4702.

2.2.3 Aluminum Alloy for Doors and Frames

JIS H 4040, Alloy 6063-T5 for extrusions. JIS H 4000, alloy and temper best suited for aluminum sheets and strips.

~~2.2.4 Fasteners~~

~~Hard aluminum or stainless steel.~~

2.2.4 Structural Steel

JIS H 4100.

~~2.2.5 Aluminum Paint~~

~~Aluminum door manufacturer's standard aluminum paint.~~

2.2.5 Sealants

Refer to Section 07 92 00 JOINT SEALANTS.

2.2.6 Glass

Refer to Section 08 81 00 GLAZING.

2.3 ACCESSORIES

2.3.1 Fasteners

Provide stainless steel fasteners in areas where the fasteners are exposed. Use non-corrosive and compatible fasteners with components being fastened. Do not use exposed fasteners, except where unavoidable for application of hardware. In areas where fasteners are not exposed, use aluminum, 304 or 316 stainless steel, or other materials warranted by the manufacturer. For exposed locations, provide countersunk Phillips head screws when items with a matching finish are fastened. For concealed locations, provide the manufacturer's standard fasteners. Provide nuts or washers that have been designed with a means to prevent disengagement; do not deform fastener threads.

2.3.2 Perimeter Anchors

When stainless steel anchors are used, provide insulation between steel material and aluminum material to prevent galvanic action.

2.3.2.1 Inserts and Anchorage Devices

Provide manufacturer's standard formed or fabricated assemblies, steel or aluminum, of shapes, plates, bars, or tubes. Shop-coat steel assemblies after fabrication with an alkyd zinc chromate primer.

2.4 FABRICATION

2.4.1 Aluminum Frames

Extruded aluminum shapes with contours approximately as indicated. Provide removable glass stops and glazing beads for frames accommodating fixed glass. Use countersunk stainless steel Phillips screws for exposed fastenings, and space not more than 300 mm on center. Mill joints in frame members to a hairline fit, reinforce, and secure mechanically.

2.4.2 Aluminum Doors

Of type, size, and design indicated and minimum 45 mm thick. minimum wall thickness, 3 mm, except beads and trim, 1.25 mm. Door sizes shown are nominal; include standard clearances as follows: 2.5 mm at hinge and lock stiles, 3 mm between meeting stiles, 3 mm at top rails, 5 mm between bottom and threshold, and 17 mm between bottom and floor.~~[~~ Provide bevel single-acting doors 2 or 3 mm at lock, hinge, and meeting stile edges.~~][~~ Provide double-acting doors rounded edges at hinge stile, lock stile, and meeting stile edges.~~]~~

~~2.4.2.1 Full Glazed Stile and Rail Doors~~

~~Provide doors with [narrow][medium][wide] stiles and rails as indicated. Fabricate from extruded aluminum hollow seamless tubes or from a combination of open-shaped members interlocked or welded together. Fasten top and bottom rail together by means of welding or by 10 or 13 mm diameter cadmium-plated tensioned steel tie rods. Provide an adjustable mechanism of jack screws or other methods in the top rail to allow for minor clearance adjustments after installation.~~

2.4.2.1 Aluminum Entrances

Provide aluminum entrances, with glass and glazing, door hardware, and components. Aluminum entrances include impact resistance entrances; wide stiles and rails as indicated, interior structural silicone glaze, for high-traffic/impact-resistant applications. Swinging door surfaces within 255 mm of the finish floor or ground measured vertically shall have a smooth surface on the push side extending the full width of the door. Provide door corner construction consisting of mechanical clip fastening, deep penetration plug welds and 2.8575 cm long fillet welds inside and outside all four corners. Provide a hook-in type exterior glazing stop with EPDM glazing gaskets reinforced with non-stretchable cord. Provide an interior glazing stop that is mechanically fastened to the door member and that incorporates a silicone-compatible spacer used with silicone sealant. Accurately fit and secure joints and corners. Make joints hairline in appearance. Remove burrs and smooth edges. Prepare components with internal reinforcement for door hardware. Arrange fasteners and attachments so that they are concealed from view. Separate dissimilar metals with protective coating or pre-formed separators to prevent contact and corrosion.

2.4.2.1.1 Shop Assembly

Fabricate and assemble units with joints only at the intersection of aluminum members with hairline joints; rigidly secure these units, and seal them in accordance with the manufacturer's recommendations.

2.4.2.1.2 Welding

Conceal welds on aluminum members in accordance with AWS recommendations or methods recommended by the manufacturer. Members showing welding bloom or discoloration on finish or material distortion will be rejected by the Contacting Officer.

2.4.2.2 Flush Doors

Use facing sheets with~~[a vertical ribbed][an embossed][or][~~ a plain smooth~~]~~ surface. Use one of the following constructions:

a. A phenolic resin-impregnated kraft paper honeycomb core, surrounded at edges and around glass and louvered areas with extruded aluminum shapes. Provide cores with a minimum impregnation of 18 percent resin content. Provide sheet aluminum door facings minimum 0.8 mm thick laminated to a 2.5 mm thick tempered hardboard backing, with the backing bonded to the honeycomb core. Bond facing sheets to cores under heat and pressure with thermosetting adhesive and mechanically lock to extruded edge members.

b. A phenolic resin-impregnated kraft paper honeycomb core. Use aluminum facing sheets minimum 1.25 mm thick and form into two pans to eliminate seams on faces. Bond honeycomb core to face sheets using epoxy resin or contact cement-type adhesive.

c. A solid fibrous core, surrounded at edges and around glass and louvered areas and cross braced at intermediate points with extruded aluminum shapes. Use aluminum facing sheets of minimum 1.25 mm thickness. Bond facing sheets to core under heat and pressure with a thermosetting adhesive, and mechanically lock to the extruded edge members.

d. Form from extruded tubular stiles and rails mitered at corners, reinforce, and continuously weld at miters. Provide facing sheets of minimum 0.8 mm thick sheet aluminum internally reinforced with aluminum channels or Z-bars placed horizontally not more than 400 mm apart and extending the full width of panels. Fit spaces between reinforcing with sound-deadening insulation. Weld facing sheets to reinforcing bars or channels and to stiles and rails. Finish facing sheets flush with faces of stiles and rails.

e. Form from an internal grid composed of extruded aluminum tubular sections. Provide tubular sections at all sides and perimeter of louver and glass openings. Provide three extruded aluminum tubular sections at top and bottom of each door. Provide wall thickness of tubular sections minimum 2.25 mm except at lock rails which must be minimum 3 mm thick, hinge lock rails which must be minimum 3 mm thick, and hinge rail edges which must be minimum 5 mm thick. Fill spaces in door with mineral insulation. Provide facing sheets of aluminum minimum 2.25 mm thick.

f. Form from extruded aluminum members at top and bottom, both sides, and at perimeters of louver and glass openings. Provide wall sections of extruded aluminum members minimum 2.25 mm thick and reinforce for application of hardware. Cover framing members on both sides with aluminum facing sheets minimum 2 mm thick. Fill door panels with ~~[~~172 kPa density polystyrene~~] [40 kg per cubic meter density, chlorofluorocarbon (CFC) free, foamed urethane]~~ with a flame spread rating of no more than 25.

2.4.3 Welding and Fastening

Where possible, locate welds on unexposed surfaces. Dress welds on exposed surfaces smoothly. Select welding rods, filler wire, and flux to produce a uniform texture and color in finished work. Remove flux and spatter from surfaces immediately after welding. Exposed screws or bolts will be permitted only in inconspicuous locations, and must have countersunk heads. Weld concealed reinforcements for hardware in place.

2.4.4 Weatherstripping

Provide on stiles and rails of exterior doors and entrances. Fit into slots which are integral with doors or frames. Weatherstripping must be replaceable without special tools, and adjustable at meeting rails of pairs of doors. During installation, verify doors swing freely and close positively. Refer to paragraph AIR INFILTRATION for air leakage requirements and testing.

2.4.5 Anchors

On the backs of subframes, provide anchors of the sizes and shapes indicated for securing subframes to adjacent construction. Anchor transom bars at ends and mullions at head and sill.~~[~~ Where indicated, reinforce vertical mullions with structural steel members of sufficient length to extend up to the overhead structural slab or framing and secure thereto.~~][~~ Reinforce and anchor freestanding door frames to floor construction as indicated on approved shop drawings and in accordance with manufacturer's recommendation.~~]~~ Place anchors ~~[as indicated][~~near top and bottom of each jamb and at intermediate points not more than 635 mm apart~~]~~.

2.4.6 Provisions for Hardware

Coordinate with Section 08 71 00 DOOR HARDWARE. Deliver hardware templates and hardware (except field-applied hardware) to the door manufacturer for use in fabrication of aluminum doors and frames. Cut, reinforce, drill, and tap doors and frames at the factory to receive template hardware. Provide doors to receive surface-applied hardware, except push plates, kick plates, and mop plates, with reinforcing only; drill and tap in the field. Provide hardware reinforcements of stainless steel or steel with hot-dipped galvanized finish, and secure with stainless steel screws.~~[~~ Provide reinforcement in core of flush doors as required to receive locks, door closers, and other hardware.~~]~~

2.4.7 Provisions for Glazing

~~[~~Provide extruded aluminum snap-in glazing beads on interior side of doors. ~~][Provide extruded aluminum, theft-proof, snap-in glazing beads or fixed glazing beads on exterior or security side of doors.][Provide glazing beads with vinyl insert glazing gaskets.][~~ Design glazing beads to receive thickness indicated for each glazed assembly.~~]~~ Coordinate requirements with Section 08 81 00 GLAZING.

2.4.8 Finishes

Provide exposed aluminum surfaces with ~~[mill finish] [~~factory finish of anodic coating or organic coating~~]~~.

2.4.8.1 Anodic Coating

Clean exposed aluminum surfaces and provide an anodized finish conforming to JIS H 8602. Provide ~~[as selected from manufacturer's [standard][complete] range of color options]~~clear (natural) anodized finish.

~~2.4.8.2 Organic Coating~~

~~Clean and prime exposed aluminum surfaces. Provide [a baked enamel finish in accordance with JIS K 5906 with total dry film thickness minimum 0.02 mm]. Finish color to be [_____] [as indicated] [as selected from manufacturer's [standard][complete] range of color options].~~

2.4.9 Fabrication Tolerance

Fabricate and assemble units with joints only at intersection of aluminum members with hairline joints; rigidly secure these units, and seal them in accordance with the manufacturer's recommendations. Fabricate aluminum entrances in accordance with the entrance manufacturer's prescribed tolerances.

2.4.10 Material Cuts

Square to 0.8 mm off square, over largest dimension; proportionate amount of 0.8 mm on the two dimensions.

2.4.11 Maximum Offset at Consecutive Members

0.4 mm in alignment between two consecutive members in line, end to end.

2.4.12 Maximum Offset at Glazing Pocket Corners

0.4 mm between framing members at glazing pocket corners.

2.4.13 Joints

Between adjacent members in same assembly: Joints are hairline and square to the adjacent member.

2.4.14 Variation

In squaring diagonals for doors and fabricated assemblies: 1.6 mm.

2.4.15 Flatness

For doors and fabricated assemblies: 1.6 mm plus/minus of neutral plane.

PART 3 EXECUTION

3.1 INSTALLATION

Plumb, square, level, and align frames and framing members to receive doors ~~[~~, transoms~~][~~, adjoining side lites~~][, and][, adjoining window walls]~~ per MLIT SS Chapter 16. Anchor frames to adjacent construction as indicated and in accordance with manufacturer's printed instructions and the approved shop drawings. Install anchorage that complies with applicable structural requirements. Anchor bottom of each frame to rough floor construction with 2.4 mm thick minimum stainless steel angle clips secured to back of each jamb and to floor construction; use stainless steel bolts and expansion rivets for fastening clip anchors. Hang doors to produce clearances specified in paragraph ALUMINUM DOORS. After erection and glazing, adjust doors and hardware to operate properly.

3.2 PROTECTION FROM DISSIMILAR MATERIALS

3.2.1 Dissimilar Metals

Where aluminum surfaces come in contact with metals other than stainless steel, zinc, or small areas of white bronze, protect from direct contact to dissimilar metals.

3.2.1.1 Protection

Provide one of the following systems to protect surfaces in contact with dissimilar metals:

a. Paint the dissimilar metal with one coat of heavy-bodied bituminous paint.

b. Apply elastomeric sealant between aluminum and dissimilar metals in accordance with Section 07 92 00 JOINT SEALANTS.

c. Paint dissimilar metals with one coat of primer and one coat of aluminum paint.

d. Use a non-absorptive tape or gasket in permanently dry locations.

3.2.2 Drainage from Dissimilar Metals

In locations where drainage from dissimilar metals has direct contact with aluminum, provide protective paint to prevent aluminum discoloration.

3.2.3 ~~Masonry and~~ Concrete

Provide aluminum surfaces in contact with mortar, concrete, or other masonry materials with one coat of heavy-bodied bituminous paint.

~~3.2.4 Wood or Other Absorptive Materials~~

~~Provide aluminum surfaces in contact with absorptive materials subject to frequent moisture, and aluminum surfaces in contact with treated wood, with two coats of aluminum paint or one coat of heavy-bodied bituminous paint. In lieu of painting aluminum, paint the wood or other absorptive surface with two coats of aluminum paint and seal joints with elastomeric sealant.~~

3.3 SEALING AROUND ASSEMBLIES

Seal ~~all penetrations of the air barrier by sealing~~ around ~~door~~all openings as necessary to ~~achieve compliance with~~prevent air leakage requirements indicated~~ in [the air barrier sections of the specifications][, the requirements of Section 07 27 10.00 10 BUILDING AIR BARRIER SYSTEM][, and Section 07 05 23 PRESSURE TESTING AN AIR BARRIER SYSTEM FOR AIR TIGHTNESS]~~. Flash all doors with corrosion resistant flashing to prevent water intrusion.

3.4 CLEANING

Upon completion of installation, clean ~~door and frame~~door, frame, entrance and window surfaces in accordance with door manufacturer's written recommended procedure. Do not use abrasive, caustic, or acid cleaning agents.

3.5 PROTECTION

Protect ~~doors and frames~~doors, frames, entrances and windows from damage and from contamination by other materials such as cement mortar. Prior to completion and acceptance of the work, restore damaged doors and frames to original condition, or replace with new ones.

-- End of Section --

SECTION 08 14 00

WOOD DOORS
08/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

FOREST STEWARDSHIP COUNCIL (FSC)

FSC STD 01 001	(2015) Principles and Criteria for Forest Stewardship

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 10140	(2016) Acoustics - Laboratory Measurement of Sound Insulation of Building Elements

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1416	(2000) Acoustics - Measuring Method of Air Sound Insulation Performance of Building Materials in Laboratory
JIS A 1460	(2015) Determination of the Emission of Formaldehyde from Building Boards - Desiccator Method

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT-SS Ch 16, Sec 7	(2019) Public Building Construction Standard Specification: Chapter 16, Section 7 Wood Door

PROGRAMME FOR ENDORSEMENT OF FOREST CERTIFICATION (PEFC)

PEFC ST 2002:2013	(2015) PEFC International Standard Chain of Custody of Forest Based Products Requirements

SUSTAINABLE GREEN ECOSYSTEM COUNCIL (SGEC)

SGEC Certification Japan	Sustainable Green Ecosystem Council - Documents 1 - 4

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Doors; G~~[, [_____]]~~

Submit drawings or catalog data showing each type of door unit~~[; include descriptive data of head and jamb weatherstripping with installation instructions]~~. Indicate within drawings and data the door types and construction, sizes, thickness, ~~[~~methods of assembly,~~]~~ ~~[~~door louvers~~,] and [glazing,]~~.

SD-04 Samples

Doors

Prior to the delivery of wood doors, submit a sample section of each type of door which shows the stile, rail, veneer, finish, and core construction.

Door Finish Colors

Submit a minimum of three color selection samples~~[~~, minimum 76 by 127 mm in size representing ~~wood stain]~~plastic laminates ~~[~~for selection by the Contracting Officer~~]~~.

SD-11 Closeout Submittals

Warranty

1.3 CERTIFICATIONS

1.3.1 Certified Wood Grades

Provide certificates of grade from the grading agency on ~~[~~acoustical doors~~], and [fire doors]~~.

~~[~~1.3.2 Certified Sustainably Harvested Wood

Provide wood certified as sustainably harvested by FSC STD 01 001, SGEC Certification Japan~~[~~, or other third party program certified by PEFC ST 2002:2013~~]~~. Provide a letter of Certification of Sustainably Harvested Wood signed by the wood supplier. Identify certifying organization and their third-party program name and indicate compliance with chain-of-custody program requirements. Submit sustainable wood certification data; identify each certified product on a line item basis. Submit copies of invoices bearing certification numbers.

~~][~~1.3.3 Indoor Air Quality Certification

~~[~~1.3.3.1 Composite Wood, Wood Structural Panel and Agrifiber Products

For purposes of this specification, composite wood and agrifiber products include particleboard, medium density fiberboard (MDF), wheatboard, strawboard, panel substrates, and door cores. Provide products certified to meet F 4-star and contents of ethylbenzene <1% 20 ppm, and xylene <5%

ppm, and methanol <1% 200 ppm.

~~]]~~1.4 DELIVERY, STORAGE, AND HANDLING

Deliver doors to the site in an undamaged condition and protect against damage and dampness. Stack doors flat under cover. Support on blocking, a minimum of 100 mm thick, located at each end and at the midpoint of the door. Store doors in a well-ventilated building so that they will not be exposed to excessive moisture, heat, dryness, direct sunlight, or extreme changes of temperature and humidity.~~[~~ Do not store in a building under construction until concrete, masonry work, and plaster are dry.~~]~~ Replace defective or damaged doors with new ones.

1.5 WARRANTY

Warrant doors free of defects as set forth in the door manufacturer's standard door warranty.

PART 2 PRODUCTS

2.1 DOORS

Provide doors of the types, sizes, and designs ~~[indicated] [~~specified~~]~~ free of urea-formaldehyde resins. Provide products certified to meet F 4-star requirements of JIS A 1460.

~~2.1.1 Stile and Rail Doors~~

~~Stile and rail doors conforming to MLIT-SS Ch 16, Sec 7. Furnish laminate panels in not less than three ply thickness. Provide flat panels with a minimum finished panel thickness of [_____] mm and [_____] thickness for raised panels.[Provide certified sustainably harvested stile and rail wood doors.]~~

2.1.1 Flush Doors

Conform to MLIT-SS Ch 16, Sec 7 for flush doors. Provide hollow core doors with lock blocks and 25 mm minimum thickness hinge stile. Hardwood stile edge bands of doors receives a natural finish, compatible with face veneer. Provide mill option for stile edge of doors scheduled to be painted. No visible finger joints will be accepted in stile edge bands. When used, locate finger-joints under hardware.~~[~~ Provide certified sustainably harvested flush wood doors.~~]~~

2.1.1.1 Interior Flush Doors

Provide ~~[~~staved lumber~~]~~ ~~[particleboard] [agrifiber] [hollow]~~or structural composite lumber engineered core, ~~Type II~~ flush doors conforming to MLIT-SS Ch 16, Sec 7 with faces of ~~[~~sound grade hardwood or hardboard for painted finish~~] [premium] [good] grade [natural birch] [select [premium white] [red] birch] [[premium] [good] grade [red] [white] oak] [[premium] [good] grade walnut] [plastic laminate]. [Hardwood veneers must be [[rotary cut] [plain sliced] [quarter sliced]] [[random] [slip] [book] matched]]. [Finish plastic laminate faced doors on both vertical edges with [wood] [laminated plastic] of color matching faces~~.~~]~~ ~~[~~Products must contain no added urea-formaldehyde resins.~~]~~

2.1.2 Bi-Fold Closet Doors

Provide ~~[~~hardboard grade flush doors conforming to MLIT-SS Ch 16, Sec 7.~~]~~ ~~[paneled] [louvered]~~ Paneled doors ~~[premium or select] [~~standard~~]~~ grade, conforming to MLIT-SS Ch 16, Sec 7 with ~~[_____] [_____] mm~~ thickness as indicated in the drawings. Equip doors with the manufacturer's standard hardware, including tracks, hinges, guides, and pulls.

2.1.3 Sliding Closet Doors

Provide flush wood doors to conform to MLIT-SS Ch 16, Sec 7. Provide ~~[~~ paneled~~]~~ ~~[and] [louvered]~~ doors to conform to MLIT-SS Ch 16, Sec 7 ~~[~~ premium ~~or select] [standard]~~ grade ~~with [_____] mm thickness~~. Equip doors with the manufacturer's standard hardware.

2.1.4 Acoustical Doors

MLIT-SS Ch 16, Sec 7, solid core, constructed to provide minimum Weighted Sound Reduction Index (Rw) rating of ~~[35] [_____]~~42 when tested in accordance with JIS A 1416 or ISO 10140.

~~2.1.5 [Composite-Type] Fire Doors~~

~~Provide doors specified or indicated to have a fire resistance rating conforming to the requirements of UL 10B, ASTM E2226, or NFPA 252 for the class of door indicated. Affix a permanent metal label with raised or incised markings indicating testing agency's name and approved hourly fire rating to hinge edge of each door.~~

~~2.1.6 Prehung Doors~~

~~Frames for prehung interior doors to be for [painted] [clear] finish, with [3 piece adjustable jamb units] [3 piece adjustable jamb units with pins]. Provide doors complete with frame, hinges, and prepared to receive finish hardware.~~

2.2 ACCESSORIES

2.2.1 Door Louvers

Fabricate from wood and of sizes indicated. Provide louvers with a minimum of 35 percent free air. Equip louvers with ~~[~~slat~~] [sightproof inverted vee slat]~~ type. ~~[Block hollow core doors to provide solid anchorage for the louvers.]~~ Mount louvers in the door with ~~[~~flush wood moldings.~~] [wood lip moldings.]~~

~~2.2.2 Door Light Openings~~

~~Provide glazed openings with the manufacturer's standard wood moldings. [Provide moldings for doors to receive natural finish of the same wood species and color as the wood face veneers.] Provide moldings on the exterior doors with sloped surfaces. [Lip type moldings for flush doors.]~~

~~2.2.3 Weatherstripping~~

~~Provide weatherstripping that is a standard cataloged product of a manufacturer regularly engaged in the manufacture of this specialized item. Provide weatherstripping [tempered spring bronze] [or] [looped neoprene or vinyl held in an extruded non-ferrous metal housing]. Install~~

~~[bronze weatherstripping with a minimum thickness of 0.23 mm for sills, and a minimum thickness of 0.16 mm elsewhere.] Air leakage of weatherstripped doors not to exceed [0.0025] [0.0031] cubic meter per second of air per square meter of door area when tested in accordance with JIS A 1516 or ISO 8272.~~

~~2.2.4 Additional Hardware Reinforcement~~

~~Provide the minimum lock blocks to secure the specified hardware. The measurement of top, bottom, and intermediate rail blocks is a minimum 125 mm by full core width. Comply with the manufacturer's labeling requirements for reinforcement blocking, but not mineral material similar to the core.~~

2.3 FABRICATION

2.3.1 Marking

Stamp each door with a brand, stamp, or other identifying mark indicating quality and construction of the door.

2.3.2 Quality and Construction

Identify the standard on which the construction of the door was based~~ [~~, identify the standard under which preservative treatment was made,~~]~~ and identify doors having a glue bond.

2.3.3 Preservative Treatment

Treat doors scheduled for restrooms, janitor closets and other possible wet locations including exterior doors with a water-repellent preservative treatment and so marketed at the manufacturer's plant.

2.3.4 Adhesives and Bonds

MLIT-SS Ch 16, Sec 7. Use bond as recommended by manufacturer for interior and exterior doors. Provide a nonstaining adhesive on doors with a natural finish.

2.3.5 Prefitting

Provide factory ~~[~~prefinished~~]~~, ~~[~~finished~~]~~, ~~[~~and~~]~~ factory prefitted doors for the specified hardware, door frame and door-swing indicated. Machine and size doors at the factory by the door manufacturer in accordance with the standards under which the doors are produced and manufactured. The work includes sizing, beveling edges, mortising, and drilling for hardware and providing necessary beaded openings for glass and louvers. Provide the door manufacturer with the necessary hardware samples, and frame and hardware schedules to coordinate the work.

2.3.6 Finishes

2.3.6.1 Field Painting

Factory prime or seal doors, and field paint.

~~2.3.6.2 Factory Finish~~

~~Provide doors finished at the factory by the door manufacturer. Use stain~~

~~when required to produce the finish specified for color. Seal edges, cutouts, trim, and wood accessories, and apply two coats of finish compatible with the door face finish. Touch-up finishes that are scratched or marred, or where exposed fastener holes are filled, in accordance with the door manufacturer's instructions. Match color and sheen of factory finish using materials compatible for field application.~~

~~2.3.6.3 Plastic Laminate Finish~~

~~Factory applied, JIS K 6903, General or Specific purpose type, 1.25 mm minimum thickness. Glue laminated plastic for hollow core doors to wood veneer, plywood, or hardboard backing to form door panel. Provide a combined thickness of laminate sheet and backing of 2.5 mm minimum.~~

2.3.6.2 Color

Provide door finish colors ~~in accordance with Section 09 06 00 SCHEDULES FOR FINISHES~~as indicated.

2.3.7 Water-Resistant Sealer

Provide manufacturer's standard water-resistant sealer compatible with the specified finish~~[es]~~.

~~2.4 SOURCE QUALITY CONTROL~~

~~Meet or exceed the following minimum performance criteria of stiles fire doors utilizing standard mortise leaf hinges:~~

~~a. Cycle-slam: [Standard Duty Doors: no loose hinge screws or other visible signs of failure when tested in accordance with the requirements of JIS A 1530] [Heavy Duty Doors: no loose hinge screws or other visible signs of failure when tested in accordance with the requirements of JIS A 1530] [Extra Heavy Duty Doors: no loose hinge screws or other visible signs of failure when tested in accordance with the requirements of JIS A 1530].~~

~~b. Hinge loading resistance: Averages of ten test samples not less than [Standard Duty doors: 1780 Newton force] [Heavy Duty doors: 2110 Newton force] [Extra Heavy Duty doors: 2440 Newton force] when tested for direct screw withdrawal in accordance with JIS A 1530. Do not use a steel plate to reinforce screw area.~~

PART 3 EXECUTION

3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Before installation, seal top and bottom edges of doors with the approved water-resistant sealer. Seal cuts made on the job immediately after cutting using approved water-resistant sealer. Fit, trim, and hang doors with manufacturer required maximum clearance at sides and top, and a clearance over thresholds per manufacturer recommendation. Provide 10 mm minimum, 11 mm maximum clearance at bottom where no threshold occurs. Bevel edges of doors at the rate of 3 mm in 50 mm. Door warp must not exceed 6 mm when measured in accordance with MLIT-SS Ch 16, Sec 7.

~~3.1.1 Fire[and Smoke] Doors~~

~~Install fire doors in accordance with NFPA 80. [Install smoke doors in accordance with NFPA 105.]Do not paint over labels.~~

~~3.1.2 Prehung Doors~~

~~Install doors in accordance with the manufacturer's instructions and details. Provide fasteners for [stops] [and] [casing trim] within 75 mm of each end and spaced 279 mm on center maximum. Provide side and head jambs joined together with a dado or notch of 5 mm minimum depth.~~

~~[3.1.3 Weatherstripping~~

~~Install doors in strict accordance with the door manufacturer's printed installation instructions and details. Weatherstrip exterior swing-type doors at sills, heads and jambs to provide weathertight installation. Apply weatherstripping at sills to bottom rails of doors and hold in place with a brass or bronze plate. Apply weatherstripping to door frames at jambs and head. Shape weatherstripping at sills to suit the threshold. [Meeting stiles of exterior double-doors must be made weathertight by means of [a looped vinyl or neoprene strip in an extruded nonferrous metal housing applied to the edge of one door leaf] [a neoprene, vinyl or spring-bronze weatherstripped astragal secured to the inactive door leaf].]~~

~~]~~ -- End of Section --

SECTION 08 31 00

ACCESS DOORS AND PANELS
05/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A653/A653M (2020) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A666 (2015) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar

ASTM A1008/A1008M (2021) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable

ASTM E90 (2009; R2016) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements

ASTM E119 (2020) Standard Test Methods for Fire Tests of Building Construction and Materials

ASTM E413 (2016) Classification for Rating Sound Insulation

ASTM E1332 (2016) Standard Classification for Rating Outdoor-Indoor Sound Attenuation

JAPANESE ARCHITECTURAL STANDARD SPECIFICAITONS (JASS)

JASS 6 (2015) Structural Steelwork Specification for Building Construction

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 4702	(20~~15~~21) Doorsets
JIS G 3136	(2008) Rolled Steels for Building Structure
JIS G 3302	(2019) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 4305	(2008) Cold-rolled Stainless Steel Plate, Sheet and Strip
JIS Z 3420	(2003) Specification and Approval of Welding Procedures for Metallic Materials - General Rules

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 18	(2019) Public Building Construction Standard Specification: Chapter 18 Painting Work

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2019) Standard for Fire Doors and Other Opening Protectives
NFPA 252	(2017) Standard Methods of Fire Tests of Door Assemblies
NFPA 288	(2017) Standard Methods of Fire Tests of Horizontal Fire Door Assemblies Installed in Horizontal Fire Resistance-Rated Assemblies

UNDERWRITERS LABORATORIES (UL)

UL 10B	(2008; Reprint May 2020) Fire Tests of Door Assemblies
UL 263	(2011; Reprint Aug 2021) UL Standard for Safety Fire Tests of Building Construction and Materials

1.2 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S"~~ classification. Submittals not having a "G" ~~or "S"~~ classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Access Doors And Panels; G~~[, [_____]]~~

SD-03 Product Data

Access Doors And Panels; G~~[, [_____]]~~

Hardware Including Locks and Keys; G~~[, [_____]]~~

Accessories; G~~[, [_____]]~~

~~[~~ ~~Power Transfer Components; G[, [_____]]~~

~~]~~ ~~Recycled Content; S~~

SD-04 Samples

Finishes; G~~[, [_____]]~~

~~[~~ SD-06 Test Reports

Fire-rating(s) of Assemblies; G~~[, [_____]]~~

Acoustical Ratings of Assemblies; G~~[, [_____]]~~

~~]~~1.3 MISCELLANEOUS REQUIREMENTS

For access doors and panels provide the following:

1.3.1 Shop Drawings

For field assembled access doors and panels, provide plans, elevations, sections, and details for each type of access door and panel assembly. Indicate frame, surface and edge construction, materials, and accessories. Indicate types of finished surfaces and details for panel edge conditions. Provide a door schedule with a unique number for each access door and panel, specific location in the project, location of hinges and hardware for each door.~~[~~ Indicate ~~[~~acoustical ratings of assemblies as sound transmission class (STC) ratings~~][,][~~ and~~][~~ fire-rating(s) of assemblies~~][and][locations and power transfer components for electrified locks and alarms]~~.~~]~~

1.3.2 Product Data

For shop assembled access doors and panels, provide literature indicating sizes, types, frame and edge construction, finishes, hardware, accessories such as gaskets, seals and weatherstripping, and location of each door and panel in the project. Indicate~~[~~ acoustical ratings of assemblies,~~][~~ fire-ratings of assemblies~~,][and][locations and power transfer components for electrified locks and alarms.]~~. Provide details of adjoining work for each condition indicated.

1.3.3 Finish Samples

Submit two color charts from manufacturer's standard color and finish options for each type of frame and panel assembly finish indicated.

~~[~~1.3.4 Test Reports

~~[~~Provide test reports for acoustical assemblies when tested in accordance with ASTM E90 and classified in accordance with ASTM E413 and ASTM E1332.~~][~~ Provide test reports for fire-rated assemblies when tested in accordance with NFPA 252 or UL 10B for fire-rated access door assemblies installed vertically and NFPA 288 for fire-rated access door assemblies installed horizontally.~~]~~

~~]~~1.4 PERFORMANCE REQUIREMENTS

1.4.1 Structural Requirements

Provide floor access assemblies to support live loads indicated for floors. Deflection must not exceed 1/180 of span.

~~[~~1.4.2 Acoustical Requirements

Provide access panels with a minimum sound transmission class (STC) of ~~[_____]~~~~[~~as indicated on the Drawings~~]~~. Provide gasketing in accordance with manufacturer's written recommendations.

~~][~~1.4.3 Fire-Rating Requirements

Provide access panels with a minimum fire-rating of ~~[[_____]-Hour][~~as indicated on the Drawing~~s]~~.

~~][~~1.4.4 Insulated Access Panels

Provide panels in a thickness as necessary to achieve a minimum R-value of ~~[_____]~~~~[~~as indicated on the Drawings~~]~~. Provide gasketing as necessary for an airtight installation.

~~][~~1.4.5 Access Panels for Wet Areas

Provide panel assemblies that will be located in wet areas with corrosion resistant finishes and hardware and water resistant gasketing.

~~]~~1.5 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

PART 2 PRODUCTS

~~2.1 RECYCLED CONTENT~~

~~Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.~~

2.1 MATERIALS

2.1.1 Steel Plates, Shapes, and Bars

Provide in accordance with ASTM A36/A36M or JIS A 4702.

2.1.2 Sheet Steel

Provide cold rolled steel sheet substrate in accordance with ASTM A1008/A1008M or JIS G 3136, Commercial Steel (CS), exposed.

2.1.3 Stainless Steel

Provide in accordance with ASTM A666 or JIS G 4305, type 302 or 304.

2.1.4 Metallic Coated Steel Sheet

Provide in accordance with ASTM A653/A653M or JIS G 3302, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

2.1.5 Hardware

Provide automatic closing devices. Provide latch releases operable from insides of doors.~~[~~ Provide anchors in accordance with applicable fire test parameters.~~]~~

2.1.6 Hinges

Provide concealed spring hinges, 175 degrees of opening, with ~~[non-]~~ removable hinge pins~~[~~ to allow removal of door panel from frame~~]~~. Provide hinges of same steel as door and frame or in accordance with manufacturer's written recommendations. If providing non-continuous hinges, provide in numbers required to maintain alignment of door panel with frame. Provide coatings as necessary to permanently protect dissimilar metals from contact with one another; see Part 3 herein for more information.

2.1.7 Locks

Unless otherwise indicated, provide flush ~~[~~screwdriver operated cam lock. Provide plastic sleeve or stainless steel bushings to protect holes in surface finishes for screwdriver to access lock~~.][keyed lock][~~ tamper proof screws (spanner head locks) for access panels in locations requiring such security.~~][Lock cylinders are specified in Section 08 71 00 DOOR HARDWARE.]~~

2.1.8 Accessories

Provide anchors in size, number and location on four sides to secure access door to substrate. Provide anchors in types as recommended by manufacturer's written installation instructions for each substrate indicated. Provide shims, bushings, clips, gaskets, and other devices as necessary for a complete installation.

2.2 FABRICATION

2.2.1 Thickness, Size, Edges

Fabricate frames for access doors of steel not lighter than 16 gage with welded joints and anchorage for securing to adjacent construction. Provide ~~doors a minimum of 600 by 600 mm~~various sized doors as required for sufficient access and of not lighter than 16 gage steel, with stiffened edges and welded attachments. Provide with eased (lightly rounded) edges, without burrs, snags or sharpness and exposed welds ground smooth.

2.2.2 Welding

Provide in accordance with AWS D1.1/D1.1M or JASS 6 and JIS Z 3420.

2.3 ACCESS ASSEMBLY TYPES

Unless indicated otherwise, provide flush-face steel access doors and panels with steel frames and flanges.

~~[2.3.1 Recessed Doors~~

~~Provide recessed access doors[with gypsum wallboard bead flanges]. Depth of door panel recess must accommodate the installed thickness of the finish material of the wall assembly for a flush finished condition of the wall and the access panel face. Reinforce panel and frame to prevent sagging.~~

~~][~~2.3.1 Fire-rated Doors

2.3.1.1 Door Construction

Provide ceiling access door construction in accordance with ASTM E119 or UL 263. Provide wall access doors in accordance with NFPA 252 or UL 10B.

2.3.1.2 Labels

Provide class B opening according to UL 10B or test by another nationally recognized laboratory, approved by the Contracting Officer. Provide fire-rating as indicated herein, with a maximum temperature rise of 120 degrees C.

2.3.1.3 Door Panel and Frame

~~[~~Steel~~][Stainless steel]~~ sheet, with mineral fiber insulation core, insulated sandwich type construction.

~~][~~2.3.2 Acoustical Doors

Manufacturer's standard assembly rated in accordance with STC requirements indicated herein. Acoustical insulating materials must have a flame spread rating of no more than 25.

~~][~~2.3.3 Insulated Doors

Provide access door panels with ~~[~~172 kPa density polystyrene~~][80 kg per cubic meter density, chlorofluorocarbon (CFC) free, foamed urethane]~~ with a flame spread rating of no more than 25.

~~[~~Provide ceiling access panels for terminal air blenders as indicated. Provide pin-tumbler cylinder locks with appropriate cams in lieu of screwdriver-operated latches.~~]~~

~~]~~2.4 FINISHES

~~[~~Provide steel frames and panel surfaces with a ~~[baked enamel][~~powder coated finish. ~~]~~Provide manufacturer's standard two coat finish system consisting of one coat primer and one thermoset topcoat. Provide dry film thickness in 0.05 mm minimum~~.][Provide steel frame and panel surfaces~~ with a shop applied prime coat. ~~[Field paint frames and panels to match wall and ceiling surfaces in which they occur.]][~~ Provide stainless steel frames and panels.~~][Provide brushed aluminum frames and panels.]~~ Provide exposed fastenings that approximately match the color and finish of the each material to which fastenings are applied.

PART 3 EXECUTION

3.1 PREPARATION

Field verify all measurements prior to fabrication. Verify access door locations and sizes provide required maintenance access to installed building services components. Protect existing construction and completed work from damage during installation.

3.2 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated, in accordance with manufacturer's written instructions. Include materials and parts as necessary for a complete installation of each item. Conceal fastenings where practicable. Poor matching of holes to fasteners is cause for rejection of the work.

3.3 ACCESS LOCATIONS

Install removable access panels directly below each valve, flow indicator, damper, air splitter or other utility requiring access that is located above ceilings, other than at acoustical panel ceilings, and that would otherwise not be accessible. Install access doors and panels permitting access to service valves, traps, dampers, cleanouts, and other mechanical, electrical and conveyor control items concealed in walls and partitions.

3.4 ACCESS LOCATIONS IN WET AREAS

When possible, avoid locating access panels in wet areas. When such locations cannot be avoided, provide moisture resistant assemblies as indicated in Part I herein.

~~[~~3.5 RECESSED ACCESS DOORS

Install fire-rated access doors in fire-rated partitions and ceilings in accordance with NFPA 80.

~~]~~3.6 FIELD PAINTING

Field painting primed access doors in accordance with the requirements of Section 09 90 00 PAINTS AND COATINGS.

3.7 DISSIMILAR MATERIALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with ~~MPI 79~~MLIT SS Chapter 18 to prevent galvanic or corrosive action.

3.8 ADJUSTMENT

Adjust hardware so that door panel opens freely. Adjust door when closed center door panel in frame.

3.9 ENVIRONMENTAL CONDITIONS

Do not paint surfaces when damp or exposed to weather, when surface temperature is below 7 degrees C or over 35 degrees C, unless approved by the Contracting Officer.

-- End of Section --

SECTION 08 33 23

OVERHEAD COILING DOORS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B29.400	(2001; (R 2008) (R 2013) (R 2018)) Combination, "H" Type Mill Chains, and Sprockets

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A653/A653M	(2020) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A924/A924M	(2020) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM F568M	(2007) Standard Specification for Carbon and Alloy Steel Externally Threaded Metric Fasteners

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1	(2000; R 2015) Standard for Industrial Control and Systems: General Requirements
NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA MG 1	(2018) Motors and Generators
NEMA ST 1	(1988; R 1994; R 1997) Specialty Transformers (Except General Purpose Type)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
NFPA 72	(20~~19; TIA 19-1; ERTA 1 2019; TIA 21-1; ERTA 1 2021~~22) National Fire Alarm and Signaling Code
NFPA 105	(2019) Standard for Smoke Door Assemblies and Other Opening Protectives

UNDERWRITERS LABORATORIES (UL)

UL 325	(2017; Reprint Feb 2020) UL Standard for Safety Door, Drapery, Gate, Louver, and Window Operators and Systems

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Overhead Coiling Doors~~[~~; G~~[, [_____]]]~~

Counterbalancing Mechanism~~[~~; G~~[, [_____]]]~~

~~Manual Door Operators[; G[, [_____]]]~~

Electric Door Operators~~[~~; G~~[, [_____]]]~~

Bottom Bars~~[~~; G~~[, [_____]]]~~

Guides~~[~~; G~~[, [_____]]]~~

Mounting Brackets~~[~~; G~~[, [____]]]~~

Hood~~[~~; G~~[, [____]]]~~

Installation Drawings~~[~~; G~~[, [____]]]~~

SD-03 Product Data

Overhead Coiling Doors~~[~~; G~~[, [____]]]~~

Hardware~~[~~; G~~[, [____]]]~~

Counterbalancing Mechanism~~[~~; G~~[, [____]]]~~

~~Manual Door Operators[; G[, [____]]]~~

Electric Door Operators~~[~~; G~~[, [____]]]~~

~~Fire~~Smoke-Rated Door Assembly~~[~~; G~~[, [____]]]~~

Recycled content for steel curtain slats; S

~~Recycled content for stainless steel curtain slats; S~~

SD-05 Design Data

Overhead Coiling Doors~~[~~; G~~[, [____]]]~~

Hardware~~[~~; G~~[, [____]]]~~

Counterbalancing Mechanism~~[~~; G~~[, [____]]]~~

~~Manual Door Operators[; G[, [____]]]~~

Electric Door Operators~~[~~; G~~[, [____]]]~~

~~Fire~~Smoke-Rated Door~~[~~; G~~[, [____]]]~~

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals, Data Package 2~~[~~; G~~[, [____]]]~~

SD-11 Closeout Submittals

Warranty~~[~~; G~~[, [____]]]~~

1.3 QUALITY CONTROL

Provide ~~fire~~smoke-rated door assemblies bearing the Underwriters Laboratories~~, Warnock Hersey, Factory Mutual or other nationally recognized testing laboratory~~ label for ~~[Class [_____] rating.] [the rating listed on the drawings.]~~"Leakage Rated Assembly" of "S" label demonstrating product tested to UL 1784. Provide a permanent label for each door showing the manufacturer's name and address, and the model/serial number of the door.

Provide oversized ~~fire~~smoke-rated door assemblies with a listing agency oversize label, or a certificate signed by an official of the manufacturing company certifying that the door and operator are designed to meet the specified requirements.

1.3.1 Warranty

Furnish a written guarantee that the helical spring and counterbalance mechanism are free from defects in material and workmanship for not less than ~~[~~two~~] [_____]~~ years after completion and acceptance of the project.

Warrant that upon notification by the Government, any defects in material, workmanship, and door operation are immediately correct within the same time period covered by the guarantee, at no cost to the Government.

1.3.2 Operation And Maintenance Submittals

Submit ~~[~~6~~] [_____]~~ copies of the operation and maintenance manuals 30 calendar days prior to testing the Overhead Coiling Door Assemblies. Update and resubmit data for final approval no later than 30 calendar days prior to cContract completion.

Submit Operation and Maintenance Manuals for Overhead Coiling Door Assemblies, including the following items:

~~[~~ ~~Manual Door Operators~~

~~]~~ Electric Door Operators

~~[~~ Hood

~~]~~ Counterbalancing Mechanism

Painting

Provide operation and maintenance manuals which are consistent with manufacturer's standard brochures, schematics, printed instructions, operating procedures, and safety precautions.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver doors to the jobsite wrapped in a protective covering with the brands and names clearly marked thereon. Store doors in an adequately ventilated dry location that is free from dirt and dust, water, or other contaminants. Store in a manner that permits easy access for inspection and handling. Handle doors carefully to prevent damage. Remove damaged items that cannot be restored to like-new condition and provide new items.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide overhead coiling doors with interlocking slats, complete with anchoring and door hardware, guides, hood, and operating mechanisms, and designed for use on openings as indicated. Doors must be spring counterbalanced, rolling type, and designed for use on ~~[exterior] [or] [~~ interior~~]~~ openings, as indicated. Doors must be operated ~~[by means of lifting handles] [by hand chain with gear or sprocket reduction] [by hand crank with gear or sprocket reduction] [~~by electric-power with auxiliary hand chain operation~~]~~. Doors to be surface-mounted type with guides at jambs set back a sufficient distance to provide a clear opening when door is in open position.~~ [Mount exterior doors [as indicated] [on interior~~

~~face of walls].] [Where doors are indicated to be chain- or crank-operated, the door design and construction must allow for future installation of electric-power operation.]~~

2.1.1 Design Requirements

2.1.1.1 Door Detail Shop Drawings

Provide installation drawings for door assemblies which show: elevations of each door type, shape and thickness of materials, finishes, details of joints and connections, details of guides and fittings, rough opening dimensions, location and description of hardware, anchorage locations, and counterbalancing mechanism and door operator details. Show locations of ~~replaceable fusible links on wiring diagrams for power, signal and controls.[~~smoke detectors installed in accordance with the requirements for smoke detectors for door release service in NFPA 72. For motor-operated doors include supporting brackets for motors, location, type, and ratings of motors, and safety devices.~~]~~ Include a schedule showing the location of each door with the drawings.

2.1.2 Performance Requirements

~~2.1.2.1 Wind Loading~~

~~Design and fabricate door assembly to withstand the wind loading pressure of at least [_____] kilopascal in accordace with ANSI/DASMA 108. Provide test data showing compliance with ASTM E330/E330M. Sound engineering principles may be used to interpolate or extrapolate test results to door sizes not specifically tested. Ensure that the complete assembly meets or exceeds the requirements of ASCE 7-16.~~

2.1.2.1 ~~Fire~~Smoke-Rated Doors, Frames, and Hardware

Provide ~~fire~~smoke-rated doors, frames, and hardware that are tested, rated, and labeled in accordance with Underwriters Laboratories~~, Factory Mutual or Warnock Hersey~~. ~~Fire~~Smoke doors must be complete with hardware, accessories, and automatic closing device as required by NFPA ~~80~~105. ~~Indicate on the labels the rating in hours, per NFPA 80, of fire exposure duration. Additionally, ensure a letter follows the hourly rating to designate the location for which the assembly is designed and the temperature rise on the unexposed door face at the end of 30 minutes of fire exposure is required.~~Smoke-rated door to pass UL test procedure 1784. The construction details necessary for labeled doors take precedence over details indicated or specified for non-labeled doors.

Provide and attach metal UL labels to the bottom bar.

2.1.2.2 Oversized Coiling Fire-rated Door Assemblies

Where ~~fire~~smoke-rated doors and frames exceed the size for which testing and labeling services are offered, furnish certificates of inspection from either UL~~,~~ or Factory Mutual~~ or Warnock Hersey~~. State within certificates that except for size; doors, frames, and hardware are identical in design, materials, and construction to a door that has been tested and rated.

2.1.2.3 Operational Cycle Life

Design all portions of the door, hardware and operating mechanism that are subject to movement, wear, or stress fatigue to operate through a minimum number of ~~[~~10~~] [_____]~~ cycles per ~~[~~day~~] [hour]~~. One complete cycle of door operation is defined as when the door is in the closed position, moves to the fully open position, and returns to the closed position.

2.2 COMPONENTS

2.2.1 Overhead Coiling Doors

2.2.1.1 Curtain Materials and Construction

~~[~~ Provide curtain slats fabricated from Grade A steel sheets conforming to ASTM A653/A653M, with the additional requirement of a minimum yield point of 228 Megapascal. Provide ~~[22][20][~~18~~]~~ gauge sheets, Grade 40 steel with galvanized steel zinc coating in conformance with ASTM A653/A653M and ASTM A924/A924M. Provide steel curtain slats containing a minimum of ~~[~~20 ~~][_____]~~ percent recycled content. Submit data identifying percentage of recycled content for steel curtain slats.

~~][Provide curtain slats fabricated from Type 304 stainless steel sheets conforming to ASTM A666; sheet thickness as required by the size of the door to meet the required windload. Provide stainless steel curtain slats containing a minimum of [60][_____] percent recycled content. Submit data identifying percentage of recycled content for stainless steel curtain slats.~~

~~][Provide curtain slats fabricated from aluminum sheets conforming to ASTM B209M , or ASTM B221M extrusions, alloy and tempering standard from the manufacturer for type of use and finish indicated; with a thickness as required by the size of the door to meet the required windload.~~

~~] Fabricate doors from interlocking cold-rolled slats, with section profiles as specified, designed to withstand the specified wind loading. Ensure the provided slats are continuous without splices for the width of the door.~~

~~Provide slats filled with manufacturer's standard thermal insulation, complying with the maximum flame-spread and smoke-developed indexes of 75 and 450, respectively, according to ASTM E84. Enclose the insulation completely within the slat faces on the interior surface of the slats.~~

~~2.2.1.2 Non-Insulated Curtains~~

~~Form curtains from the manufacturer's standard shapes of interlocking slats.~~

~~2.2.1.3 Insulated Curtains~~

~~Form curtains from manufacturer's standard shapes of interlocking slats. Supply a slat system with a minimum R-value of [4] [_____] when calculated in accordance with ASHRAE FUN IP. Slats to consist of a [urethane] [polystyrene] core not less than 17 mm thick, completely enclosed within metal facings. Slat steel thickness as required by the size of the door to meet specified performance requirements. The insulated slat assembly requires a flame spread rating of not more than 25 and a smoke development factor of not more than 50 when tested in accordance with ASTM E84.~~

2.2.1.2 Curtain Bottom Bar

Install curtain bottom bars as pairs of angles or using extrusions from the manufacturer's standard steel,~~ stainless and aluminum~~ extrusions not less than 50 by 50 millimeter by 4.8 millimeter.~~ Do not use aluminum on doors more than 1877 mm wide. Ensure steel extrusions conform to ASTM A36/A36M. Stainless steel extrusions conforming to ASTM A666, Type 304. Aluminum extrusions conforming to ASTM B221M.~~ Galvanize angles and fasteners in accordance with ASTM A653/A653M and ASTM A924/A924M. Coat welds and abrasions with paint conforming to ASTM A780/A780M.

~~[~~ Provide two minimum 50 mm by 50 mm by 3.2 mm structural steel angles.

~~][2.2.1.3 Vision Lites~~

~~Provide complete manufacturer's standard vision panels assembly consisting of clear acrylic glazing panels or fire-rated glass as required for the type door.~~

~~]2.2.1.4 Endlocks (and Windlocks)~~

~~Provide endlocks of Grade B cast steel conforming to ASTM A47/A47M, galvanized in accordance with ASTM A153/A153M. Secure locks at every other curtain slat. [In addition to endlocks, exterior doors which are more than 4877 mm wide or which have a design wind load of more than 0.96 kilopascals, must have windlocks of manufacturer's standard design. Windlocks must prevent curtain from leaving guide because of deflection from wind pressure or other forces.]~~

~~2.2.1.5 Weather Stripping~~

~~Provide a hood baffle inside the hood that is a minimum 1.6 millimeter thick sheet of vinyl, neoprene rubber or equivalent. Provide guide weather stripping that is a minimum 1.6 millimeter thick sheet of vinyl, neoprene rubber, or equivalent.~~

~~Provide bottom bar weather-stripping that is a minimum 1.6 millimeter thick sheet of vinyl, neoprene rubber, or equivalent.~~

~~2.2.1.6 Locking Devices~~

~~Ensure that the slide bolt engages through slots in tracks for locking by padlock, located on both left and right jamb sides, operable from coil side.~~

~~Provide a locking device assembly which includes cylinder lock, operating handle, cam plate, and adjustable locking bars to engage through slots in tracks.~~

~~[Provide a chain lock keeper suitable for a standard padlock.~~

~~]2.2.1.7 Safety Interlock~~

~~Equip power-operated doors with a safety interlock switch to disengage power supply when the door is locked, or provide an operator with an internal lock sensing device to prevent the door opening when the door is locked.~~

2.2.1.3 Smoke Seals

Provide UL-listed brush-type smoke seals "S" label.

2.2.2 Hardware

Ensure that all hardware conforms to ASTM A153/A153M, ASTM A307, and ASTM F568M.

2.2.2.1 Guides

Fabricate curtain jamb guides from the manufacturer's standard angles or channels of same material and finish as curtain slats unless otherwise indicated. Provide guides with sufficient depth or incorporate a steel locking bar to retain the curtain in place~~ under the wind pressure specified~~. Ensure curtain operates smoothly. Slot bolt holes for track adjustment. Securely attach guides to adjoining construction with not less than 10 mm diameter bolts, spaced near each end and not over 762 mm apart.

~~[Ensure guides are roll-formed steel channel bolted to angle or structural grade, three angle assembly of [steel][stainless steel][aluminum] to form a slot of sufficient depth to retain curtains in guides to achieve 13.8 kilopascals windload standard. Guides may be provided with integral windlock bars and removable bottom bar stops.~~

~~][Fabricate with [structural steel][stainless steel][aluminum] angles. Provide windlock bars of same material when windlocks are required to meet specified wind load. Flare the top of inner and outer guide angles outwards to form bellmouth for smooth entry of curtain into guides. Provide removable guide stoppers to prevent over travel of curtain and bottom bar.~~

~~]~~2.2.2.2 Hood

Provide a hood with a minimum~~[24-gauge][aluminum 18-gauge B&S][~~ galvanized~~][stainless steel]~~ sheet metal, flanged at top for attachment to header and flanged at bottom to provide longitudinal stiffness. The hood encloses the curtain coil and counterbalance mechanism.

~~[~~ Hoods for openings more than 3658 mm in width must have intermediate support brackets to prevent excessive sag.~~] [Provide a weather baffle at the lintel or inside the hood of each exterior door.~~

~~]~~2.2.3 Counterbalancing Mechanism

Counterbalance doors by means of manufacturer's standard mechanism with an adjustable-tension, steel helical torsion spring mounted, around a steel shaft and contained in a spring barrel connected to top of curtain with barrel rings. Use grease-sealed or self-lubricating bearings for rotating members.

2.2.3.1 Brackets

Provide the manufacturer's standard mounting brackets with one located at each end of the counterbalance barrel conforming to ASTM A36/A36M. Provide brackets of hot-rolled steel.

~~[~~ Brackets will be of~~[5 mm][~~ 6.35 mm~~]~~ minimum thick steel plates, with permanently sealed ball bearings. Designed to enclose ends of coil and provide support of counterbalance pipe at each end.

~~]~~2.2.3.2 Counterbalance Barrels

Curtain must roll up on a barrel supported at head of opening on brackets and be balanced by a torsion spring system in the barrel. Fabricate spring barrel of manufacturer's standard hot-formed, structural-quality, welded or seamless carbon-steel pipe, conforming to ASTM A53/A53M or equivalent. Ensure the barrel is of sufficient diameter and wall thickness to support rolled-up curtain without distortion of slats. Limit barrel deflection to not more than 2.5 mm per meter of span under full load.

a. Barrel

Provide steel pipe capable of supporting curtain load with maximum deflection of 2.5 mm per meter of width.

b. Spring Balance

Provide an oil-tempered, heat-treated steel helical torsion spring assembly designed for proper balance of door. Ensure that effort to operate manually operated units does not exceed 110 N. At least 80 percent of the door weight must be counterbalanced at any position. Provide wheel for applying and adjusting spring torque.

~~2.2.3.3 Spring Balance~~

~~Install one or more oil-tempered, heat-treated steel helical torsion springs within the barrel, capable of producing sufficient torque to assure easy operation of the door curtain. Provide and size springs to counterbalance weight of curtain, with uniform adjustment accessible from outside barrel. Secure ends of springs to barrel and shaft with cast-steel barrel plugs.~~

~~2.2.3.4 Torsion Rod for Counter Balance~~

~~Fabricate rod from the manufacturer's standard cold-rolled steel, sized to hold fixed spring ends and carry torsional load.~~

~~[2.2.3.5 Counterbalance Shaft Assembly~~

~~[a. Barrel~~

~~Provide steel pipe capable of supporting the curtain load with maximum deflection of 2.5 mm per meter of width.~~

~~][b. Spring Balance~~

~~Provide an oil-tempered, heat-treated steel helical torsion spring assembly designed for proper balance of door. Ensure that maximum effort to operate does not exceed 110 Newtons. Provide wheel for applying and adjusting spring torque.~~

~~]]2.2.4 Manual Door Operators~~

~~[2.2.4.1 Manual Push-Up Door Operators~~

~~Provide lifting handles on both sides of door and counterbalance in a manner to provide easy operation while raising or lowering the curtain by hand. Adjust counterbalance mechanisms so that the required lift or pull for operation does not exceed 11 kilogram unless another type of door operator is indicated. Provide pull-down straps or pole hooks on bottom rail of doors over 2134 mm high.~~

~~][2.2.4.2 Manual Chain-Hoist Door Operators~~

~~Provide door operators which consist of an endless steel hand chain, chain-pocket wheel, guard, and a geared reduction unit [of at least a 3 to 1 ratio] [with a maximum lifting force of[111 N][133 N]]. Required pull for operation cannot exceed 16 kilogram. Chain must extend to within 914 mm of floor.~~

~~Provide chain hoists with a mechanism allowing the curtain to be stopped at any point in its upward or downward travel and to remain in that position until moved to the fully open or closed position. Provide hand chains of galvanized steel. Ensure that the yield point of the chain is at least three times the required hand-chain pull.~~

~~Provide chain sprocket wheels of cast iron conforming to ASTM A48/A48M.~~

~~][2.2.4.3 Manual Crank-Hoist Door Operators~~

~~Provide door operators which consist of crank and crank gearbox, steel crank drive shaft, and gear-reduction unit with a maximum[111 N][133 N] force to turn crank. Fabricate gearbox to be oil tight and to completely enclose operating mechanism. Gears must be high-grade gray iron, cast from machine-cut patterns.~~

~~]~~2.2.4 Electric Door Operators

Provide electrical wiring and door operating controls conforming to the applicable requirements of NFPA 70 and UL 325. The door manufacturer must furnish automatic control and safety devices, including extra flexible type SO cable and spring-loaded automatic takeup reel or equivalent device, as required for proper operation of the doors. Conduit, wiring, and mounting of controls are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

~~[Electrical materials, equipment, and devices for installation in hazardous locations, as defined by NFPA 70, must be specifically approved by Underwriters Laboratories or an independent testing agency using equivalent standards, for the particular chemical group and the class and division of hazardous location involved.~~

~~]~~ Electric door-operator assemblies need to be the sizes and capacities recommended and provided by the door manufacturer for specified doors. Furnish complete assemblies with electric motors and factory-prewired motor controls, starter, gear reduction units, solenoid-operated brakes, clutch, remote-control stations, manual or automatic control devices, and accessories as required for proper operation of the doors.

Design the operators so that motors may be removed without disturbing the limit-switch adjustment and affecting the emergency auxiliary operators.

Provide a manual operator of crank-gear or chain-gear mechanisms with a release clutch to permit manual operation of doors in case of power failure. Arrange the emergency manual operator so that it may be put into and out of operation from floor level, and its use does not affect the adjustment of the limit switches. Provide an electrical or mechanical device that automatically disconnects the motor from the operating mechanism when the emergency manual operating mechanism is engaged.

2.2.4.1 Door-Operator Types

~~[Provide an operator mounted to the right or left door head plate with the operator on top of the door-hood assembly and connected to the door drive shaft with drive chain and sprockets. Headroom is required for this type of mounting.~~

~~][Provide an operator mounted to the right or left door head plate with the operator on coil side of the door-hood assembly and connected to the door drive shaft with drive chain and sprockets. Front clearance is required for this type of mounting.~~

~~][~~Provide an operator mounted to the inside front wall on the left or right side of door and connected to door drive shaft with drive chain and sprockets. Side room is required for this type of mounting. Wall mounted operator can also be mounted above or below shaft; if above shaft, headroom is required.

~~][Provide a bench mounted operator mounted to the right or left door head plate and connected to the door drive shaft with drive chain and sprockets. Side room is required for this type of mounting.~~

~~][Provide a through-wall operator which is mounted on other side of wall from coil side of door.~~

~~]~~2.2.4.2 Electric Motors

Provide motors which are the high-starting-torque, reversible, constant-duty electrical type with overload protection of sufficient torque and wattage to move the door in either direction from any position. Ensure they produce a door-travel speed of not less than 0.2 nor more than 0.3 meter per second without exceeding the wattage rating.

Provide motors which conform to NEMA MG 1 designation, temperature rating, service factor, enclosure type, and efficiency to the requirements specified. Motors must be suitable for operation on current of the characteristics indicated.~~[~~ Single-phase motors must not have commutation or more than one starting contact.~~][~~ Motor enclosures must be the drip-proof type or NEMA TEFC and TENV type.~~]~~ Install motors in approved locations.

~~[Certify and label explosion-proof motors to indicate conformance to the following:~~

~~[UL 674, Class I, Groups C and D]~~

~~[UL 674, Class II, Groups F and G]~~

~~]~~2.2.4.3 Motor Bearings

Select bearings with bronze-sleeve or heavy-duty ball or roller antifriction type with full provisions for the type of thrust imposed by the specific duty load.

Pre-lubricate and factory seal bearings in motors less than 375 watts.

Equip motors coupled to worm-gear reduction units with either ball or roller bearings.

Equip bearings in motors 375 watts or larger with lubrication service fittings. Fit lubrication fittings with color-coded plastic or metal dust caps.

In any motor, bearings that are lubricated at the factory for extended duty periods do not need to be lubricated for a given number of operating hours. Display this information on an appropriate tag or label on the motor with instructions for lubrication cycle maintenance.

2.2.4.4   Motor Starters, Controls, and Enclosures

Provide each door motor with: a factory-wired, unfused, disconnect switch; a reversing, across-the-line magnetic starter with thermal overload protection; 24-volt operating coils with a control transformer limit switch; and a safety interlock assembled in a NEMA ICS 6 type enclosure as specified herein. Ensure control equipment conforms to NEMA ICS 1 and NEMA ICS 2.

Provide adjustable switches, electrically interlocked with the motor controls and set to stop the door automatically at the fully open and fully closed position.

2.2.4.5   Control Enclosures

Provide control enclosures that conform to NEMA ICS 6 for ~~[NEMA Type 4][~~ NEMA Type 4X~~][general purpose NEMA Type 1]. [oil-tight and dust-tight NEMA Type 12.] [explosion-proof, NEMA Type 7, group as indicated.] [explosion-proof NEMA Type 9, group as indicated.]~~.

2.2.4.6   Transformer

Provide starters with 230/460 to 115 volt control transformers with one secondary fuse when required to reduce the voltage on control circuits to 24volts or less. Provide a transformer conforming to NEMA ST 1.

2.2.4.7   Sensing-Edge Device

Provide each door with a ~~pneumatic or electric~~photoelectric sensing device that meets UL 325, extends the full width of the door, and is located within a U-section neoprene or rubber astragal, mounted on the bottom rail of the bottom door section. Device needs to immediately stop and reverse the door upon contact with an obstruction in the door opening or upon failure of the device or any component of the control system and cause the door to return to its user-defined open position. Any momentary door-closing circuit must be automatically locked out and the door must be operable manually or with constant pressure controls until the failure or damage has been corrected. A sensing device is not a substitute for a limit switch.

Connect sensing device to the control circuit through a retracting cord and reel.

~~2.2.4.8 Remote-Control Stations~~

~~[Remote control stations must be at least 1500 mm above the floor line, and all switches must be located so that the operator will have complete visibility of the door at all times. Provide interior remote control stations that are full-guarded, momentary-contact three-button, heavy-duty, surface-mounted NEMA ICS 6 type enclosures as specified. Mark buttons "OPEN," "CLOSE," and "STOP." The "OPEN" and "STOP" buttons must be of the type requiring only momentary pressure to operate. The "CLOSE" button must be of the type either requiring constant pressure to maintain the closing motion of the door or momentary pressure when installed with a monitored entrapment detection device which, upon failure of the device or any component of the control system, cause the door to return to its full open position. When the door is in motion and the "STOP" button is pressed, ensure the door stops instantly and remains in the stopped position. From the stopped position, the door may then be operated in either direction by the "OPEN" or "CLOSE" buttons. When the door is in motion, and the "CLOSE" button of the constant pressure type is released, the door must stop and remain in the stop position or reverse to the user set up position; from the stop position, the door may then be operated in either direction by the "OPEN" or "CLOSE" buttons. Controls must be adjustable to automatically stop the doors at their fully open and closed positions. Open and closed positions must be readily adjustable.~~

~~][Provide exterior control stations that are full-guarded, momentary-contact three-button standard-duty, surface-mounted, weatherproof type, NEMA ICS 6, Type 4 enclosures, key-operated, with the same operating functions as specified herein for interior remote-control stations.~~

~~]~~2.2.4.8 Speed-Reduction Units

Provide speed-reduction units consisting of hardened-steel worm and bronze worm gear assemblies or planetary gear reducers running in oil or grease and inside a sealed casing, coupled to the motor through a flexible coupling. Drive shafts need to rotate on ball- or roller-bearing assemblies that are integral with the unit.

Provide minimum ratings of speed reduction units in accordance with AGMA provisions for class of service.

Ground worm gears to provide accurate thread form; machine teeth for all other types of gearing. Surface harden all gears.

Provide antifriction type bearings equipped with oil seals.

2.2.4.9 Chain Drives

Provide roller chains that are a power-transmission series steel roller type conforming to ASME B29.400, with a minimum safety factor of 10 times the design load.

Heat-treat or otherwise harden roller-chain side bars, rollers, pins, and bushings.

Provide high-carbon steel chain sprockets with machine-cut hardened teeth, finished bore and keyseat, and hollow-head setscrews.

2.2.4.10 Brakes

Provide 360-degree shoe brakes or shoe and drum brakes. Ensure the brakes are solenoid-operated and electrically interlocked to the control circuit to set automatically when power is interrupted.

2.2.4.11 Clutches

Ensure clutches are friction type or adjustable centrifugal type.

2.2.4.12 ~~Weather/~~Smoke Seal Sensing Edge

Provide automatic stop control by an automatic sensing switch within neoprene astragal extending the full width of door bottom bar.

Provide an electric sensing edge device. Ensure the door immediately stops downward travel when contact occurs before door fully closes. Provide a self-monitoring sensing edge connection to the motor operator.

2.2.5 ~~Fire~~Smoke-Rated Door Assembly

Provide ~~fire~~smoke-rated door assemblies with the dimensions,~~ fire rating,~~ and operating type indicated with electric operators and assemblies~~ that do not interfere with manufacturer's standard interconnecting fusible links~~. Equip ~~fire~~smoke doors with an automatic closing mechanism. Doors must be forced into a closed position at a rate of descent of not more than 0.61 meters per second and not less than 0.15 meters per second without impact. The curtain must be held against the sill until the release mechanism has been reset. The automatic closing mechanism must not interfere with normal operation of the door.

~~[Provide the door manufacturer's standard interconnecting fusible links for door assemblies on both sides of the wall opening.~~

~~]2.2.5.1 Fire Ratings~~

~~Provide fire-rated door assemblies complying with NFPA 80 Standard for Fire Doors and Other Opening Protectives. Fire doors must be complete with hardware, accessories, and automatic closing device as required by NFPA 80.~~

2.2.6 Surface Finishing

Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes. Noticeable variations in the same metal component are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved samples and are assembled or installed to minimize contrast.

2.2.6.1 Galvanized and Shop-Primed Finish

Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating must conform to ASTM A653/A653M, coating designation Z275 (G90), for steel sheets~~[~~, except that hoods located on interior of the building may be Z180 (G60)~~]~~, and ASTM A123/A123M for iron and steel products. The weight of coatings for products must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied to slats and hoods before forming, in lieu of prime coat specified.

2.2.6.2 ~~Baked-Enamel or~~ Powder-Coat Finish

Manufacturer's standard baked-on powder coat finish consisting of prime coat and thermosetting topcoat. Comply with the coating manufacturer's written instructions for cleaning, pretreatment, application, and minimum dry film thickness for corrosive environment.

PART 3 EXECUTION

3.1 INSTALLATION

Install overhead coiling door assembly, anchors and inserts for guides, brackets, motors, switches, hardware, and other accessories in accordance with approved detail drawings and manufacturer's written instructions. Upon completion of installation, ensure doors are free from all distortion.

~~Install overhead coiling doors, motors, hoods, and operators at the mounting locations as indicated for each door in the Contract Documents and as required by the manufacturer.~~

~~Install overhead coiling doors, switches, and controls along accessible routes in compliance with regulatory requirements for accessibility and as required by the manufacturer.~~

3.1.1 Field Painted Finish

Ensure field painted steel doors and frames are in accordance with Section 09 90 00 PAINTS AND COATINGS and the manufacturer's written instructions. Protect the weather stripping from paint. Ensure that the finishes are free of scratches or other blemishes.

3.2 ADJUSTING AND CLEANING

3.2.1 Acceptance Provisions

After installation, adjust the hardware and moving parts. Lubricate bearings and sliding parts as recommended by manufacturer to provide smooth operating functions for ease movement, free of warping, twisting, or distortion of the door assembly.

Adjust seals to provide a weather-tight fit around entire perimeter.

Engage a factory-authorized service representative to perform startup service and checks according to the manufacturer's written instructions.

Test the door opening and closing operation when activated by controls~~[or alarm-connected fire-release]~~ and in accordance with UL 1784 for safety of air leakage tests~~ system~~. Adjust controls and safeties. Replace damaged and malfunctioning controls and equipment. Reset the door-closing mechanism after a successful test.

Test and make final adjustment of new doors at no additional cost to the

Government.

3.2.1.1 Maintenance and Adjustment

Not more than 90 calendar days after completion and acceptance of the project, examine, lubricate, test, and re-adjust doors as required for proper operation.

3.2.1.2 Cleaning

Clean doors in accordance with manufacturer's approved instructions.

-- End of Section --

SECTION 08 71 00

DOOR HARDWARE
02/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1	(2017) Standard And Commentary Accessible and Usable Buildings and Facilities

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1510-2	(2019) Test Method for Door Fittings - Part 2: Fittings for Doors
JIS A 1510-3	(2001) Test Methods for Door Fittings - Part 3: Floor Concealed Door Closers, Door Closers and Hinge Closers
JIS A 1516	(1998) Windows and Doorsets - Air Permeability Test
JIS A 1525	(1996) Doorsets - Repeated and Opening and Closing Test
JIS A 1541-1	(2016) Building Hardware - Locks and Latches - Part 1: Test Methods for Locks and Latches
JIS A 1541-2	(2016) Building Hardware - Locks and Latches - Part 2: Methods of the Presentation and Grade of Criteria for Practical Performance Item
JIS A 4702	(20~~15~~21) Doorsets
JIS A 4722	(2017) Power Operated Pedestrian Door Sets - Safety
JIS A 5756	(2013) Preformed Gaskets Used in Buildings - Classification, Specifications and Test Methods

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code

NFPA 72	(20~~19; TIA 19-1; ERTA 1 2019; TIA 21-1; ERTA 1 2021~~22) National Fire Alarm and Signaling Code
NFPA 80	(2019) Standard for Fire Doors and Other Opening Protectives
NFPA 101	(2021) Life Safety Code
NFPA 252	(2017) Standard Methods of Fire Tests of Door Assemblies

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191	Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

UNDERWRITERS LABORATORIES (UL)

UL 305	(2012) UL Standard for Safety Panic Hardware
UL 228	(2006; Reprint Nov 2008) Door Closers-Holders, With or Without Integral Smoke Detectors
UL Bld Mat Dir	(updated continuously online) Building Materials Directory

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Manufacturer's Detail Drawings; G~~[, [_____]]~~

Verification of Existing Conditions; G~~[, [_____]]~~

Hardware Schedule; G~~[, [_____]]~~

Keying System; G~~[, [_____]]~~

SD-08 Manufacturer's Instructions

Installation

SD-10 Operation and Maintenance Data

Hardware Schedule Items, Data Package 1; G~~[, [_____]]~~

SD-11 Closeout Submittals

Key Bitting

1.3 SHOP DRAWINGS

Submit manufacturer's detail drawings indicating all hardware assembly components and interface with adjacent construction.~~[Indicate power components and wiring coordination for electrified hardware.]~~ Base shop drawings on verified field measurements and include verification of existing conditions.

Submit detailed system wiring diagrams for power, signaling, monitoring, communication, and control of the access control system electrified hardware. Differentiate between manufacturer-installed and field-installed wiring. Diagrams to include the following:

a. Elevation diagram of each unique access controlled opening showing location and interconnection of major system components with respect to their placement in the respective door openings.

b. Complete (risers, point-to-point) access control system block wiring diagrams.

c. Wiring instructions for each electronic component scheduled herein.

d. Coordinate with related sections the voltages and wiring details required at electrically controlled and operated hardware openings.

1.4 PRODUCT DATA

Indicate fire-ratings at applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

1.5 HARDWARE SCHEDULE

Prepare and submit hardware schedule in the following form:

Hardware Item	Quantity	Size	Reference Publi-cation Type No.	Finish	Mfr Name and Catalog No.	Key Control Symbols	UL Mark (If fire-rated and listed)	BHMA Finish Desig-nation

In addition, submit hardware schedule data package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6 KEY BITTING CHART REQUIREMENTS

1.6.1 Requirements

Submit key bitting charts to the Contracting Officer prior to completion of the work. Include:

a. Complete listing of all keys.

b. Complete listing of all key cuts.

c. Tabulation showing which key fits which door.

d. Copy of floor plan showing doors and door numbers.

e. Listing of 20 percent more key cuts than are presently required in each master system.

1.7 QUALITY ASSURANCE

1.7.1 Hardware Manufacturers and Modifications

Provide, as far as feasible, locks, hinges,~~[pivots,]~~ and closers of one lock, hinge,~~[pivot,]~~ or closer manufacturer's make. Modify hardware as necessary to provide features indicated or specified.

1.7.2 Key Shop Drawings Coordination Meeting

Prior to the submission of the key shop drawing, the Contracting Officer, Contractor, Door Hardware Subcontractor, using Activity and Base Locksmith must meet to discuss and coordinate key requirements for the facility.

1.8 DELIVERY, STORAGE, AND HANDLING

Deliver hardware in original individual containers, complete with necessary appurtenances including fasteners and instructions. Mark each individual container with item number as shown on hardware schedule.~~[~~ Deliver permanent keys~~[~~ and removable cores~~]~~ to the Contracting Officer, either directly or by certified mail. Deliver construction master keys with the locks.~~]~~

PART 2 PRODUCTS

2.1 TEMPLATE HARDWARE

Hardware applied to metal ~~[~~or to prefinished ~~]~~doors must be manufactured using a template. Provide templates to door and frame manufacturers in accordance with JIS A 1510-2 for template hinges. Coordinate hardware items to prevent interference with other hardware.

2.2 HARDWARE FOR FIRE DOORS AND EXIT DOORS

Provide all hardware necessary to meet the requirements of NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, NFPA 252 for fire tests of door assemblies, ABA/ADA accessibility requirements, and all other requirements indicated, even if such hardware is not specifically mentioned in paragraph HARDWARE SCHEDULE.~~[Provide swinging hardware for tin-clad fire doors in accordance with UL 14C.]~~ Provide Underwriters Laboratories, Inc. labels for such hardware in accordance with UL Bld Mat Dir or equivalent labels in accordance with another testing laboratory approved in writing by the Contracting Officer.

2.3 HARDWARE ITEMS

Clearly and permanently mark with the manufacturer's name or trademark, hinges, pivots, locks, latches, exit devices, bolts and closers where the identifying mark is visible after the item is installed. For closers with covers, the name or trademark may be beneath the cover. Coordinate electrified door hardware components with corresponding components specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

2.3.1 Hinges

Provide in accordance with JIS A 1510-3. Provide hinges that are 114 by 114 mm unless otherwise indicated. Construct loose pin hinges for interior doors and reverse-bevel exterior doors so that pins are non-removable when door is closed. Other anti-friction bearing hinges may be provided in lieu of ball bearing hinges.

~~[2.3.1.1 Protection Devices~~

~~Provide full height hand and finger protection device at the hinge-side area opening of doors and gates. Provide hinge-side protection devices on both sides of doors and gates, covering hinges and space between door and frame when doors are in the open position. The installed device must push hand and fingers out of the opening and away from a crushing hazard.~~

~~]~~2.3.2 Continuous Hinges

Where continuous hinges are required, provide in accordance with JIS A 1510-2.

~~2.3.3 Pivots~~

~~Provide in accordance with JIS A 1510-3.~~

~~2.3.4 Spring Hinges~~

~~Provide in accordance with JIS A 1510-3.~~

2.3.3 Locks and Latches

~~[~~ a. At exterior locations provide locksets of full stainless steel type 302 or 304 construction including fronts, strike, escutcheons, knobs, bolts and all interior working parts. Marine Grade I, fully non-ferrous.

b. In non-air-conditioned interior environments or humid interior environments, provide interior locksets on the same Marine Grade I, fully non-ferrous as exterior locksets.

~~]~~2.3.3.1 Mortise Locks and Latches

Provide in accordance with JIS A 1541-1.~~[Provide factory installed lead lining in locks for lead shielded doors.][~~ Provide mortise locks with escutcheons not less than 178 by 57 mm with a bushing at least 6 mm long. Cut escutcheons to fit cylinders and provide trim items with straight, beveled, or smoothly rounded sides, corners, and edges.~~]~~ Provide knobs and roses of mortise locks with screwless shanks and no exposed screws.

~~2.3.3.2 Bored Locks and Latches~~

~~Provide in accordance with JIS A 4702.[Provide factory installed lead lining in locks for lead -shielded doors.]~~

~~2.3.3.3 Residential Bored Locks and Latches~~

~~Provide in accordance with JIS A 4702. Install locks for exterior doors with threaded roses or concealed machine screws.~~

~~[2.3.3.4 Interconnected Locks and Latches~~

~~Provide in accordance with JIS A 1541-2.~~

~~]2.3.3.5 Hospital Latches~~

~~Push-pull latch set as required to meet operational requirements, 13 mm throw, [70 mm] [127 mm] backset, to fit 161 cutout. Cover approximately 64 by 140 mm, handle approximately 38 by 114 mm, projection approximately 64 mm, covers and handles of stainless steel finish, engraved "PUSH" and "PULL" on handles, push handle pointing up, pull handle pointing down.~~

~~2.3.3.6 Auxiliary Locks~~

~~Provide lock having a [latch bolt] or [dead bolt] operated by a [key], [paddle [and] [and/or] turn], which is used in addition to a primary lock or latching device to meet operational requirements.~~

~~2.3.3.7 Combination Locks~~

~~[Key pharmacy door locks separately from building master key system.]Heavy-duty, mechanical combination lockset with five push buttons, standard sized knobs, 20 mm deadlocking latch, 70 mm backset. Locks to operate by pressing two or more of the buttons in unison or individually in the proper sequence. Inside knob operates the latch. Provide a keyed cylinder on the interior to permit setting the combination.[Provide a keyed [removable core]cylinder on the exterior to permit bypassing the combination.][Provide a thumb turn on the interior to activate passage set function so that outside knob operates latch without using the combination.]~~

2.3.4 Exit Devices

Provide adjustable strikes for rim type and vertical rod devices. Provide open back strikes for pairs of doors with mortise and vertical rod devices. Provide ~~[touch bars in lieu of conventional crossbars and arms~~at doors not requiring a fire rating, provide devices complying with NFPA 101 and listed and labeled for 'Panic Hardware' in accordance with UL 305. Provide proper fasteners as required. Where exit devices are required on fire rated doors, provide devices complying with NFPA 80 and with UL labeling indicating 'Fire Exit Hardware'. Provide devices with proper fasteners for installation as tested and listed by UL. Electromechanical exit devices to comply with identical compliance standards and requirements as mechanical exit devices and shall be of type and design as indicated.~~][~~ Provide escutcheons not less than 178 by 57 mm.~~]~~

~~[Use stainless steel or bronze base metal with plated finishes. Also include stainless steel fasteners and screws.~~

~~]2.3.5 Exit Locks With Alarm~~

~~Provide [with full width horizontal actuating bar] for single doors; Type E0431 [with actuating bar] or E0471 [with actuating bar and top and bottom~~

~~bolts, both leaves active] for pairs of doors, unless otherwise specified.[Provide terminals for connection to remote indicating panel.][Provide outside control key.] Provide door alarms integrated with the fire alarm system in accordance with NFPA 72.~~

2.3.5 Cylinders and Cores

~~[~~Provide cylinders and cores for new locks, including locks provided under other sections of this specification. ~~]~~Provide cylinders and cores with ~~[~~ six~~]~~ ~~[seven]~~ pin tumblers. Provide cylinders from the products of one manufacturer, and provide cores from the products of one manufacturer.~~[Rim cylinders, mortise cylinders, and knobs of bored locksets have interchangeable cores which are removable by special control keys. Stamp each interchangeable core with a key control symbol in a concealed place on the core.]~~Provide threaded mortise cylinders with rings and cams to suit hardware application. Rim cylinders to be provided with back plate, flat-type vertical or horizontal tailpiece and raised trim ring. Mortise and rim cylinder collars to be solid and recessed to allow cylinder face to be flush and be free spinning with matching finishes.

~~[Provide cylinders for new locks, including locks provided under other sections of this specification. Provide fully compatible cylinders of Grade 1 products from products of one manufacturer with interchangeable cores that are removable by a special control key. Factory set the cores with [six] [seven] pin tumblers. Submit a core code sheet with the cores. Provide master keyed cores in one system for this project. Provide construction interchangeable cores.~~

~~][For medical projects, key pharmacy door locks separately from building master key system.~~

~~]2.3.5.1 High Security Cylinders~~

~~Provide high security cylinder with locking technology that limits the duplication of unauthorized keys or unauthorized electronic credentials that would operate the locks. High security cylinder must be able to stand up to forcing, drilling, sawing, prying, pulling, plug driving, picking and have corrosion resistance. [High security cylinder to comply with UL 437].~~

2.3.6 Push Button Mechanisms

Provide in accordance with JIS A 1541-1.

2.3.7 Electrified Hardware

Comply with the requirements of NFPA 70 for wiring of electrified hardware.

2.3.7.1 Electric Strikes and Frame Mounted Actuators

Provide electric strikes and actuators as required to meet operational requirements. Provide electric strikes that ~~[~~release automatically~~]~~ ~~[remain secure] [remain maintained]~~ during power failure.~~[~~ Provide a separate power supply for electric strikes, other locking devices and ancillary parts.~~][Provide battery backup for continued operation during power failure.]~~ Provide strikes and actuators with a minimum opening force of 101 kilonewtons (kN).

Provide facility interface devices that use direct current (dc) power to energize the solenoids. Provide electric strikes and actuators that incorporate end-of-line resistors to facilitate line supervision by the system. If not incorporated into the electric strike or local controller, provide metal oxide resistors (MOVs) to protect the controller from reverse current surges.

2.3.7.1.1 Solenoid

Provide actuating solenoid for strikes and actuators that are rated for continuous duty, cannot dissipate more than 12 Watts and must operate on 12 or 24 Volts dc. Inrush current cannot exceed 1 ampere and the holding current cannot be greater than 500 milliamperes. Actuating solenoid must move from fully secure to fully open positions in less than 500 milliseconds.

2.3.7.1.2 Signal Switches

Provide strikes and actuators with signal switches to indicate to the system when the bolt is not engaged or the strike mechanism is unlocked. Signal switches must report a forced entry to the system.

2.3.7.1.3 Tamper Resistance

~~[~~ Provide strike guards that prevent tampering with the latch bolt of the locking hardware or the latch bolt keeper of the electric strike. Strike guards to bolt through the door using tamper resistant screws. Provide strike guards made of 3 mm thick brass and that are 286 mm high by 41 mm, with a minimum 4 mm wide offset.

~~]~~2.3.7.1.4 Coordination

Provide electric strikes and actuators of a size, weight and profile compatible with each specified door frame. Field verify installation clearances prior to procurement.

2.3.7.1.5 Mounting Method

Provide electric strikes and actuators suitable for use with single and double doors, with mortise or rim type hardware specified, and for right or left hand mounting as specified. In double door installations, locate the lock in the active leaf and monitor the fixed leaf.

~~2.3.7.2 Electrified Mortise Locks~~

~~Provide electrified mortise locks that [release automatically] [remain secure] [remain maintained] during power failure. Provide facility interface devices that use dc power to energize solenoids. Provide solenoids, resisters, and signal switches in accordance with paragraph ELECTRIC STRIKES AND FRAME MOUNTED ACTUATORS.~~

~~2.3.7.2.1 Power Transfer Hinges~~

~~Provide power transfer hinges with each electrified lock that route power and monitoring signals from the lockset to the door frame. Coordinate power transfer hinges with door frames.~~

2.3.7.2 Card Readers and Keypad Access Control Hardware

Provide devices that are tamper alarmed, tamper and vandal resistant, solid state, and do not contain electronics which could compromise the access control subsystem should the subsystem be attacked. Provide surface, semi-flush, pedestal, or weatherproof mountable devices as specified for each individual location.~~[Each device to contain a visual display, either mounted on the face, or on an integral part of the device, to indicate access or exit request processing, request approval, and request denial.]~~ Provide ~~[~~proximity~~] [insertion] [swipe through]~~ type card readers capable of reading ~~[magnetic stripe] [high coercivity magnetic stripe] [Wiegand] [Hollerith] [~~proximity~~] [Transmissive Infrared] [Keypad] [[_____]/Keypad] [Smart Card] [Biometric] [_____]~~ type access control cards. ~~Provide keypads that contain an integral 12-digit tactile keyboard with digits [arranged in numerical order]. Provide keypads that are [a standalone device] [or] [integrated into the card reader].~~All doors with access control system to have a lockset with key overried capability. Coordinate access control hardware with corresponding devices and systems specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS). Coordinate with ISE?

~~2.3.7.3 Power Operated Pedestrian Door Hardware~~

~~Provide in accordance with JIS A 4722.~~

2.3.7.3 Release Devices

In accordance with JIS A 1510-3.

2.3.7.3.1 Closer Holders

Provide ~~[floor] [~~door~~] [header]~~ mounted closer holder devices connected by ~~[separate releasing] [~~integral releasing~~]~~ to ~~[~~fire~~]~~ and ~~[~~smoke~~]~~ detecting devices. Closer arms to be of heavy duty, forged steel unless otherwise indicated. Closers to comply with ICC A117.1.

2.3.7.3.2 Release Devices

Provide ~~[wall] [floor] [~~door~~]~~ mounted ~~[Electromagnetic] [~~electromechanical~~] [free swinging]~~ release devices connected to ~~[~~fire~~]~~ and ~~[~~smoke~~]~~ detecting devices.

2.3.7.4 Power Assist and Low Energy Power Operated Doors

Provide in accordance with JIS A 4722.

2.3.7.5 Electromagnetic Locks

Provide electromagnetic locks that do not contain any moving parts and depend solely upon electromagnetism to secure a portal by generating at least 5.3 kN of holding force. The lock must interface with the local processors without external, internal or functional alteration of the local processor. The electromagnetic lock must incorporate an end of line resistor to facilitate line supervision by the system. Provide metal-oxide resistors (MOVs) to protect controllers from reverse current surges, if not incorporated into the electromagnetic lock or local controller. Where indicated, provide electromagnetic locks with outside door lock/unlock trim control, latchbolt and lock/unlock status monitoring, deadbolt monitoring, and request-to-exit signaling. Support end-of-line resistors contained within the lock case.

2.3.7.5.1 Electromagnetic Locks

Magnalocks shall operate on 24VDC input and shall not consume more than 3 watts of power (125 mA at 24 VDC). The magnalock shall be capable of providing a pull-apart or tensile holding force of at least 272 kg. The strike plate shall be mounted using a steel sex bolt and roll pin to provide a "floating" movement to assure automatic self-alignment with the lock. Anti-tamper caps shall be provided for the exposed holes. The lock and strike shall be plated to provide corrosion proofing. The lock shall be full sealed in resin to make it tamper and weatherproof. The lock shall contain a suppression circuit to prevent residual magnetism and inductive kickback. The circuit also shall provide accelerated field collapse and radiation suppression. Three meters of jacketed stranded conductor shall be provided for electrical connection.

2.3.7.5.2 Electromagnetic Door Hold-Open Devices

Devices shall be attached to the walls unless otherwise indicated. Devices shall comply with the appropriate requirements of UL 228. Devices shall operate on 24 Volt dc power. Compatible magnetic component shall be attached to the door. Under normal conditions, the magnets shall attract and hold the doors open. When magnets are de-energized, they shall release the doors. Magnets shall have a holding force of 111.2 N. Devices shall be UL or FM approved. Housing for devices shall be brushed aluminum or stainless steel. Operation shall be fail secure with no moving parts. Electromagnetic door hold-open devices shall not be required to be held.

2.3.7.5.3 Armature

Provide electromagnetic locks with internal circuitry to eliminate residual magnetism and inductive kickback. Provide atuating armature that operates on 12 or 24 Volts dc and cannot dissipate more than 12 Watts. Holding current must be less than 500 milliamperes. Actuating armature must take less than 300 milliseconds to change the status of the lock from fully secure to fully open or fully open to fully secure.

2.3.7.5.4 Tamper Resistance

Provide lock mechanism encased in hardened guard barriers to deter forced entry.

2.3.7.5.5 Mounting Method

Provide electromagnetic lock suitable for use with single and double door with mortise or rim type hardware and compatible with right or left hand mounting.

~~2.3.7.6 Delayed Egress Locking System~~

~~Provide delayed egress product capable of allowing the door to be opened by actuating the lock which is equipped with a 15-second maximum delayed feature including a zero-to-three second pre-delay. The door shall be allowed to close by action of the door closer. Electrically re-lock the system so that the time delay is operative. A force, not to exceed 67 N, shall be continuously applied on the door or release device allowing the door to be opened after not more than 15 seconds.~~

~~2.3.7.7 Power and Manual Operated Revolving Pedestrian Doors~~

~~Provide in accordance with JIS A 1551 for powered revolving pedestrian doors and JIS A 4721 for manual operated revolving pedestrian doors.~~

2.3.8 Keying System

Provide~~[~~ a ~~[~~great~~][~~ grand~~]~~ master keying system~~][an extension of the~~ existing keying system. Existing locks were manufactured by ~~[_____]~~Miwa Lock System and ~~[~~do not~~]~~ have interchangeable cores.~~][~~ Provide~~[a construction master keying system][~~ construction interchangeable cores~~]~~. ~~][Provide key cabinet as specified.]~~

2.3.9 Lock Trim

Provide cast, forged, or heavy wrought construction and commercial plain design for lock trim.

2.3.9.1 Knobs and Roses

Provide in accordance with JIS A 4702 and JIS A 1541-1 for knobs, roses, and escutcheons. For unreinforced knobs, roses, and escutcheons, provide a 1.25 mm thickness. For reinforced knobs, roses, and escutcheons, provide an outer shell thickness of 0.89 mm and a combined total thickness of 1.78 mm, except at knob shanks. Provide knob shanks 1.52 mm thick.

2.3.9.2 Lever Handles

Provide lever handles ~~[~~where indicated in the Hardware Schedule~~]~~. Provide lever handle locks with a breakaway feature (such as a weakened spindle or a shear key) to prevent irreparable damage to the lock when force in excess of that specified in JIS A 1541-1 is applied to the lever handle. Provide lever handles return to within 13 mm of the door face.

~~2.3.9.3 Texture~~

~~Provide knurled or abrasive coated knobs or lever handles for doors which are accessible to blind persons and which lead to dangerous areas.~~

2.3.10 Keys

~~[Furnish][~~Provide~~]~~ one file key, one duplicate key, and one working key for each key change ~~[~~and for each master ~~[~~and grand master~~]~~ keying system~~]~~. ~~[Furnish][~~Provide~~]~~ one additional working key for each lock of each keyed-alike group.~~[[Furnish][Provide] two additional keys for each sleeping room.][[Furnish][~~ Provide~~]~~ ~~[[_____] great grand master keys,]~~ ~~[[_____] ~~construction master keys,~~]~~ ~~[~~and ~~[_____] ~~control keys for removable cores~~]~~.~~][~~ ~~[Furnish][~~Provide~~]~~ a quantity of key blanks equal to 20 percent of the total number of file keys.~~]~~ Stamp each key with appropriate key control symbol and "U.S. property - do not duplicate." Do not place room number on keys.

2.3.11 Door Bolts

Provide in accordance with JIS A 4702. Provide dustproof strikes for bottom bolts, except at doors having metal thresholds. Provide flush bolts with top rod of sufficient length to allow bolt retraction device location approximately 1830 mm from floor. Surface bolts to be minimum 205 mm in length for UL listed and labeled fire doors and UL listed windstorm components. Provide automatic latching flush bolts to meet operational requirements.

2.3.12 Closers

Provide in accordance with JIS A 1510-3. Provide with brackets, arms, mounting devices, fasteners, ~~[~~full size covers, except at storefront mounting,~~] [pivots,] [cement cases,]~~ and other features necessary for the particular application. Size closers in accordance with manufacturer's printed recommendations, or provide multi-size closers, Sizes 1 through 6, and list sizes in the Hardware Schedule. Provide manufacturer's 10 year warranty.

~~[~~ Use stainless steel inside bracketed or door mounted closers on exterior doors. Non-ferrous closers, such as aluminum or cast bronze, are permissible where door utilization is minimal. On interior doors use closers of 302 or 304 stainless steel or non-ferrous materials. On surface-mounted closers use or apply rust inhibiting finish on all ferrous parts. Also apply this finish on concealed closers.

~~]~~2.3.12.1 Identification Marking

Engrave each closer with manufacturer's name or trademark, date of manufacture, and manufacturer's size designation in locations that will be visible after installation.

2.3.13 Overhead Holders

Provide in accordance with JIS A 1510-3.

2.3.14 Door Protection Plates

Provide in accordance with JIS A 4702.

2.3.14.1 Sizes of ~~[Armor] [Mop] [and]~~ Kick Plates

50 mm less than door width for single doors; 25 mm less than door width for pairs of doors. ~~Provide [[200] [1200] mm kick plates for flush doors] [and] [125 mm less than height of bottom rail for panel doors]. Provide a minimum [900] [1200] [_____] mm armor plates for flush doors [and] completely cover lower panels of panel doors, except 400 mm high armor plates on fire doors. Provide [100] [150] mm mop plates.~~

~~2.3.14.2 Edge Guards~~

~~Stainless steel, of same height as armor plates. Apply to [hinge stile] [lock stile] [meeting stiles].~~

2.3.15 Door Stops and Silencers

Provide in accordance with JIS A 4702. Provide three silencers for each single door, two for each pair.

~~2.3.16 Padlocks~~

~~Provide padlock of [solid extruded brass] [stainless steel]. [Shackle to be cut-resistant]. Provide lock functions consisting of [key retained], [non-key retained], [frangible shackle], [double lockout], [weather cover], [car seal slot].~~

2.3.16 Thresholds

Use vinyl or silicone rubber insert in face of stop, for exterior doors opening out, unless specified otherwise.

2.3.17 Weatherstripping Gasketing

Provide in accordance with JIS A 5756. Provide the type and function designation where specified in paragraph HARDWARE SCHEDULE. Provide a set to include head and jamb seals~~[~~, sweep strips,~~]~~ ~~[~~and, for pairs of doors, astragals~~]~~. Air leakage of weatherstripped doors not to exceed ~~[2.19 by 10-5]~~ ~~[~~5.48 by 10-5~~]~~ cms per minute of air per square meter of door area when tested in accordance with JIS A 1516. Provide weatherstripping with one of the following:

~~2.3.17.1 Extruded Aluminum Retainers~~

~~Extruded aluminum retainers not less than 1.25 mm wall thickness with vinyl, neoprene, silicone rubber, or polyurethane inserts. Provide [clear (natural)] [bronze] anodized aluminum.~~

~~2.3.17.2 Interlocking Type~~

~~Zinc or bronze not less than 0.45 mm thick.~~

~~2.3.17.3 Spring Tension Type~~

~~Spring bronze or stainless steel not less than 0.20 mm thick.~~

~~2.3.18 [Lightproofing] [and] [Soundproofing] Gasketing~~

~~Provide in accordance with JIS A 5756. Provide adjustable doorstops at heads,jambs and automatic door bottoms in accordance with the hardware set, of extruded aluminum, [clear (natural)] [bronze] anodized, surface applied, with vinyl fin seals between plunger and housing. Provide doorstops with solid neoprene tube, silicone rubber, or closed cell sponge gasket. Provide door bottoms with adjustable operating rod and silicone rubber or closed cell sponge neoprene gasket. Provide doorstops that are mitered at corners. Provide type and function designation where specified in paragraph HARDWARE SETS.~~

~~2.3.19 Rain Drips~~

~~Provide extruded aluminum rain drips, not less than 2.03 mm thick, [clear anodized] [bronze anodized] [factory painted] [factory primed] finish. Provide the manufacturer's full range of color choices to the Contracting Officer for color selection.[Provide rain drips with a 102 mm overlap on each side of each exterior door that is not protected by an awning, roof, eave or other horizontal projection.] Set drips in sealant and fasten with stainless steel screws.~~

~~2.3.19.1 Door Rain Drips~~

~~Approximately 38 mm high by 16 mm projection. Align bottom with bottom edge of door.~~

~~2.3.19.2 Overhead Rain Drips~~

~~Approximately 38 mm high by 64 mm projection. Align bottom with door frame rabbet.~~

2.3.18 Auxiliary Hardware (Other than locks)

Provide in accordance with JIS A 4702.

2.3.19 Sliding and Folding Door Hardware

Provide in accordance with JIS A 1525. Finishes to match other hardware specified herein.

2.3.20 Special Tools

Provide special tools, such as spanner and socket wrenches and dogging keys, as required to service and adjust hardware items.

2.4 FASTENERS

Provide fasteners of type, quality, size, and quantity appropriate to the specific application. Fastener finish to match hardware. Provide stainless steel or nonferrous metal fasteners in locations exposed to weather. Verify metals in contact with one another are compatible and will avoid galvanic corrosion when exposed to weather.

2.5 FINISHES

~~[~~ Provide in accordance with JIS A 1541-2. Provide hardware in satin stainless steel, unless specified otherwise. Provide items not manufactured in stainless steel in satin chromium plated over brass or bronze, except ~~[~~aluminum paint~~] [prime coat]~~ finish for surface door closers~~, and except satin chromium plated [primed for painting] for steel hinges~~. Provide hinges for exterior doors in stainless steel finish~~ [or chromium plated brass or bronze finish]~~. Furnish exit devices in satin chrome finish in lieu of stainless steel finish ~~[~~except where specified under paragraph HARDWARE SETS~~]~~. Match exposed parts of concealed closers to lock and door trim. Match hardware finish for aluminum doors to the doors.

~~][Provide in accordance with JIS A 1541-2. Provide hardware in satin bronze, unless specified otherwise. Finish surface door closers [bronze paint] [prime coat] finish. Provide steel hinges in [satin bronze plated] [primed for painting]. Provide exposed parts of concealed closers finish to match lock and door trim. Match hardware finish for aluminum doors to match the doors. Provide hardware showing on interior of [bathrooms] [shower rooms] [toilet rooms] [washrooms] [laundry rooms] [and kitchens] in bright stainless steel or bright chromium plated.~~

~~]2.6 KEY CABINET AND CONTROL SYSTEM~~

~~Provide in accordance with project requirements, [[(25 hooks)] [(125 hooks)] [(150 hooks)] [(600 hooks)] [(700 hooks)].] [Type required to yield a capacity (number of hooks) 50 percent greater than the number of key changes used for door locks.]~~

PART 3 EXECUTION

3.1 INSTALLATION

Provide hardware in accordance with manufacturers' printed installation instructions. Fasten hardware to wood surfaces with full-threaded wood screws or sheet metal screws. Provide machine screws set in expansion shields for fastening hardware to solid concrete and masonry surfaces. Provide toggle bolts where required for fastening to hollow core construction. Provide through bolts where necessary for satisfactory installation.

3.1.1 Weatherstripping Installation

Provide full contact, weathertight seals that allow operation of doors without binding the weatherstripping.

3.1.1.1 Stop Applied Weatherstripping

Fasten in place with color matched sheet metal screws not more than 225 mm on center after doors and frames have been finish painted.

3.1.1.2 Interlocking Type Weatherstripping

Provide interlocking, self adjusting type on heads and jambs and flexible hook type at sills. Nail weatherstripping to door 25 mm on center and to heads and jambs at 100 mm on center.

3.1.1.3 Spring Tension Type Weatherstripping

Provide spring tension type on heads and jambs. Provide bronze nails with bronze. Provide stainless steel nails with stainless steel. Space nails not more than 38 mm on center.

3.1.2 ~~[Lightproofing] [and] [~~Soundproofing~~]~~ Installation

Provide as specified for stop applied weatherstripping.

3.1.3 Threshold Installation

Extend thresholds the full width of the opening and notch end for jamb stops. Set thresholds in a full bed of sealant and anchor to floor with cadmium-plated, countersunk, steel screws~~[~~ in expansion sleeves~~]~~. For aluminum thresholds placed on top of concrete surfaces, coat the underside surfaces that are in contact with the concrete with fluid applied waterproofing as a separation measure prior to placement.

3.2 FIRE DOORS AND EXIT DOORS

Provide hardware in accordance with NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, and NFPA 252 for fire tests of door assemblies.~~ [Provide tin-clad fire doors in accordance with UL 14C].~~

3.3 HARDWARE LOCATIONS

Provide as indicated or specified otherwise.

a. Kick ~~and Armor~~ Plates: Push side of single-acting doors. Both sides of double-acting doors.

~~b. Mop Plates: Bottom flush with bottom of door.~~

~~3.4 KEY CABINET AND CONTROL SYSTEM~~

~~Locate where [directed][indicated]. Tag one set of file keys and one set of duplicate keys. Place other keys in appropriately marked envelopes, or tag each key. Provide complete instructions for setup and use of key control system. On tags and envelopes, indicate door and room numbers or master or grand master key.~~

3.4 FIELD QUALITY CONTROL

After installation, protect hardware from paint, stains, blemishes, and other damage until acceptance of work. Submit notice of testing 15 days before scheduled, so that testing can be witnessed by the Contracting Officer. Adjust hinges, locks, latches, bolts, holders, closers, and other items to operate properly. Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer. Correct, repair, and finish, errors in cutting and fitting and damage to adjoining work.

3.5 HARDWARE SETS

Provide ~~[~~hardware for aluminum doors under this section. Deliver Hardware templates and hardware, except field applied hardware, to the aluminum door and frame manufacturer for use in fabricating doors and frames.~~]~~

Door Hardware Sets

Yokota Tower 3001

HW-1	Unit Entry Doors – Interior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 1000 F20, Grade 1
1 Each	Cylinder	E09211
1 Each	Door Closer	C02021
1 Each	Door Stop	L02251
1 Each	Threshold	J32100 127
1 Set	Smoke Seals	ROE154
HW-2	Barn Style Sliding Door - Interior	
1 Each	Track with Stop	
1 Set	Hangers	
1 Each	Flush Pull	
HW-3		
6 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F86, Grade 1
1 Each	Cylinder	E09231
1 Each	Door Closer	C02011
1 Each	Threshold	As Detailed
1 Set	Smoke Seals	ROE154
2 Each	Concealed Flush Bolt	L04081 (Top & Bottom at Inactive Door)

HW-4	Sliding Glass Door, 3-Panel	
All Hardware by Door Manufacturer		
HW-5	Unit Balcony Storage Double Door - Exterior	
6 Each	Hinges	A8133 (114mm x 114mm)
2 Each	Half Dummy Trim	Grade 2
2 Each	Magnetic Catch	E09051
2 Each	Door Stop	L02101
HW-6	Bedroom and Bathrooms - Interior	
3 Each	Hinges	A8133 (114mm x 114mm)
1 Each	Lockset	Series 4000, F76, Grade 2
1 Each	Door Stop	L02141
1 Each	Threshold	As Detailed
HW-7	Closet Bypass Door - Interior	
1 Each	Track with Stop	D8751
2 Sets	Hangers	D8731
2 Each	Flush Pulls	D0781
HW-8	Sliding Glass Door, 2-Panel	
All Hardware by Door Manufacturer		
HW-9	Stair Door - From Balcony (Fire Rated) - Exterior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Cylinder	E09231
1 Each	Door Closer	C02021
1 Each	Door Stop	L02161
1 Each	Threshold	As detailed
1 Set	Smoke Seals	ROE154
1 Each	Door Drip	R3A534
HW-10	Exterior Stair Door - 1st Floor	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Exit Device	Type 1, 08, Grade 1
1 Each	Cylinder	E09221
1 Each	Door Closer	C02041
1 Each	Door Stop	L02161
1 Each	Threshold	As Detailed
1 Each	Door Shoe with Drip	R3D515
1 set	Weatherstripping	By Door Manufacturer
HW-11	Stair Door - At Core (1st Floor) (Fire Rated) - Interior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Exit Device	Type 1, 08, Grade 1
1 Each	Door Closer	C02041
1 Each	Threshold	As Detailed
1 Set	Smoke Seals	ROE154
HW-11A	Stair Door - At Core (Basement) (Fire Rated) - Interior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F75, Grade 1
1 Each	Door Closer	C02041
1 Each	Door Stop	L02161

1 Each	Threshold	As Detailed
1 Set	Smoke Seals	ROE154
HW-12	Stair Door - At Core (Fire Rated) - Interior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Exit Device	Type 1, 14, Grade 1
1 Each	Door Closer	C02021
1 Each	Door Stop	L02161
1 Each	Threshold	As Detailed
1 Set	Smoke Seals	ROE154
HW-13	Lobby Doors (1st Floor)	
3 Each	Cylinder	E09211
Balance of Hardware by Door Manufacturer		
HW-14	Doors at Core and Roof	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F86, Grade 1
1 Each	Cylinder	E09231
1 Each	Door Closer	C02021
1 Each	Threshold	As Detailed
1 Set	Smoke Seals	ROE154
HW-15	Doors at Basement - Exterior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F86, Grade 1
1 Each	Cylinder	E09231
1 Each	Door Closer	C02021
1 Each	Threshold	As Detailed
1 Set	Weatherstripping	By Door Manufacturer
HW-16	Double Doors at Basement - Exterior	
6 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F86, Grade 1
1 Each	Cylinder	E09231
1 Each	Door Closer	C02011
1 Each	Threshold	As Detailed
1 Set	Smoke Seals	ROE154
2 Each	Concealed Flush Bolt	L04081 (Top & Bottom at Inactive Door)
HW-17	Storefront Doors at Multipurpose - Exterior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Exit Device	Type 1, 08, Grade 1
1 Each	Cylinder	E09221
1 Each	Door Closer	C02021
1 Each	Threshold	As Detailed
1 Set	Weatherstripping	By Door Manufacturer
HW-18	Double Storefront Doors at Multipurpose & Basement Overhead Coiling Doors - Exterior	
1 Each	Cylinder	E09211
Balance of Hardware by Door Manufacturer		
HW-19	Unisex Toilets - Interior	

3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Deadlock	E2141
1 Each	Pull	J405
1 Each	Push Plate	J301
1 Each	Kick Plate	J102
1 Each	Door Stop with Keeper	L01371
HW-20	Generator/Pump Room/ Exterior Double Doors	
6 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F86, Grade 1
1 Each	Threshold	As Detailed
2 Each	Concealed Flush Bolt	L04081 (Top & Bottom at Inactive Door)
2 Each	Door Shoe with Drip	R3D515
1 Set	Weatherstripping	By Door Manufacturer
HW-21	Door at Roof - Exterior	
3 Each	Hinges	A5112, NRP (114mm x 114mm)
1 Each	Lockset	Series 4000, F86, Grade 1
1 Each	Cylinder	E09231
1 Each	Door Closer	C02041
1 Each	Door Stop	L02161
1 Each	Threshold	As Detailed
1 Each	Door Shoe with Drip	R3D515
1 Set	Weatherstripping	By Door Manufacturer
HW-22		
1 Each	Magnetic Door holder	
Balance of Hardware by Door Manufacturer		
HW-23	Chain Link Gate at Core Storage - Interior	
1 Each	LFIC CORE 6PIN GMK	US15
Balance of Hardware by Door Manufacturer		
HW-24	Picket Gate, Pair - Exterior	
1 Each	LFIC CORE 6PIN GMK	US15
Balance of Hardware by Door Manufacturer		

-- End of Section --

SECTION 08 81 00

GLAZING
05/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7 (2017) Minimum Design Loads for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM E1996 (2017) Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2018) International Building Code

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 15099 (2003) Thermal Performance of Windows, Doors and Shading Devices - Detailed Calculations

ISO 16932 (2007) Glass in Buildings - Destructive Windstorm-Resistant Security Glazing - Test and Clarification

ISO 28278 (2011) Glass in Building - Glass Products for Structural Sealant Glazing

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1439 (2016) Testing Methods of Sealants for Sealing and Glazing in Buildings

JIS A 5756 (2013) Preformed Gaskets Used in Buildings - Classification, Specifications and Test Methods

JIS A 5758 (2016) Sealants for Sealing and Glazing in Buildings

JIS K 6262 (2013) Rubber, Vulcanized or Thermoplastic - Determination of Compression Set at Ambient, Elevated or Low Temperatures

JIS K 7366	(1999) Plastic-plasticized polyvinyl chloride (PVC-P) Molding and extrusion materials
JIS R 3109	(2018) Glass in Building - Destructive-Windstorm-Resistant Security Glazing - Test Method
JIS R 3202	(2011) Float Glass and Polished Plate Glass
JIS R 3205	(2005) Laminated Glass
JIS R 3206	(2014) Tempered Glass
JIS R 3220	(2011) Glass in Building - Silvered, Flat-Glass Mirror
JIS R 3222	(2003) Heat-Strengthened Glass

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT-SS Ch 16, Sec 14	(2019) Building Construction Standard Specifications - Chapter 16 Openings Construction, Section 14 Glazing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

16 CFR 1201	Safety Standard for Architectural Glazing Materials

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Insulating Glass

Sealants

Joint Backer

SD-08 Manufacturer's Instructions

Setting and Sealing Materials

Glass Setting

SD-11 Closeout Submittals

Warranty for Insulated Glass Units

~~[~~ ~~Warranty for Polycarbonate Sheet~~

~~][~~ ~~Warranty for Monolithic Reflective Glass~~

~~][~~ ~~Warranty for Monolithic Opacified Spandrel~~
~~]~~

~~[~~1.3 SYSTEM DESCRIPTION

Fabricate and install watertight and airtight glazing systems to withstand thermal movement and wind loading without glass breakage, gasket failure, deterioration of glazing accessories, or defects in the work. Glazed panels must comply with the safety standards, in accordance with JIS R 3205 for Laminated Glass or JIS R 3206 for Tempered Glass, comply with indicated wind/snow loading in accordance with ISO 28278, and relative displacement requirements in accordance with ASCE 7, section 13.5.9.1. Provide insulated laminated heat-strengthened exterior glazing with a minimum interlayer thickness as indicated on the drawings or as required to meet performance requirements.~~[Sloped glazing must comply with MLIT-SS Ch 16, Sec 14.]~~

1.3.1 Wind Pressure Requirements

Exterior glazing to withstand an allowable wind-loading design ~~pressure of [_____] kPa in Zone [_____] and [_____] kPa in Zone [_____]~~pressures as indicated in the drawings.~~ Zones [_____] and [_____] are defined by ASCE 7-10.~~

1.3.2 Windborne Debris Requirement

~~Exterior glazing shall be tested and certified for impact resistance with the window or door as applicable under JIS R 3109 to comply with minimum Missile Type [_____] and ICC IBC Section 1609, Wind Loads or tested and certified for impact resistance under ISO 16932 to comply with minimum Missile Type [_____] and Wind Zone [_____].~~Exterior glazing shall be tested and certified for impact resistance with the window or door as applicable under ISO 16932 or JIS R 3109, to comply with ICC IBC Section 1609, Wind Loads.

a. Glazed openings located within 9.1m of grade must meet the Basic Level of Protection requirements of Level 3 Protection requirements of lumber missile test of ISO 16932 and JIS R 3109, Wind Zone 3. Type D lumber missile shall apply for ISO 16932 and JIS R 3109.

b. Glazed openings located more than 9.1m above grade must meet the Basic Level of Protection requirements of the small missile test of ASTM E1996, Wind Zone 4 or Level 3 Protection requirements of small-ball missile test of ISO 16932 and JIS R 3109, Wind Zone 3. Type A steel ball missile shall apply for ISO 16932 and JIS R 3109.

~~]~~1.4 QUALITY CONTROL

Submit two 203 by 254 mm samples of each ~~of the following: tinted glass, patterned glass, heat-absorbing glass, [_____] and~~ insulating glass units.

Submit three samples of each other material.~~ Samples of plastic sheets must be minimum 125 by 175 mm.~~

1.5 DELIVERY, STORAGE, AND HANDLING

Deliver products to the site in unopened containers, labeled plainly with manufacturers' names and brands. Store glass and setting materials in safe, enclosed dry locations and do not unpack until needed for installation. Handle and install materials in a manner that will protect them from damage.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not start glazing work until the outdoor temperature is above 4 degrees C and rising, unless procedures recommended by the glass manufacturer and approved by the Contracting Officer are made to warm the glass and rabbet surfaces. Provide ventilation to prevent condensation of moisture on glazing work during installation. Do not perform glazing work during damp or rainy weather.

1.7 WARRANTY

1.7.1 Warranty for Insulated Glass Units

Warranty insulating glass units against development of material obstruction to vision (such as dust, fogging, or film formation on the inner glass surfaces) caused by failure of the hermetic seal, other than through glass breakage, for a 10-year period following acceptance of the work. Provide new units for any units failing to comply with terms of this warranty within 45 working days after receipt of notice from the Government.

~~1.7.2 Warranty for Polycarbonate Sheet~~

~~For a 5-year period following acceptance of the work:~~

~~a. Warranty Type I, Class A (UV stabilized) sheets against breakage;~~

~~b. Warranty Type III (coated, mar-resistant) sheets against breakage and against coating delamination;~~

~~c. Warranty Type IV (coated sheet) against breakage and against yellowing;~~

~~d. Warranty extruded polycarbonate profile sheet against breakage.~~

~~For a 10-year period following acceptance of the work, warranty Type IV against yellowing and loss of light transmission.~~

~~[1.7.3 Monolithic Reflective Glass~~

~~Manufacturer must warrant the monolithic reflective glass to be free of peeling or deteriorating of coating for a period of 10 years after Date of Substantial Completion. Warranty must be signed by manufacturer.~~

~~][1.7.4 Monolithic Opacified Spandrel~~

~~Manufacturer must warrant the opacifier film on the spandrel to be free of peeling for a period of five years after Date of Substantial Completion. Warranty must be signed by manufacturer.~~

~~]~~PART 2 PRODUCTS

2.1 PRODUCT SUSTAINABILITY CRITERIA

~~[2.1.1 Energy Efficient Equipment for Residential Windows~~

~~Provide energy efficient residential windows in accordance with Section 01 33 29 SUSTAINABILITY REPORTING paragraph ENERGY EFFICIENT PRODUCTS.~~

~~]~~2.2 GLASS

JIS R 3202, unless specified otherwise. In doors and sidelights, provide safety glazing material conforming to 16 CFR 1201. Glazing shall comply with the test criteria for Category II.

a. All fixed glazing in doors.

b. Glazing adjacent to doors where the nearest vertical edge of the glazing is within a 610 mm arc of either the vertical edge of the door in a closed position and where the bottom exposed edge of the glazing is less than 1524 mm above the walking surface.

c. Glazing in windows where the exposed area of any individual panel is greater than 0.84 m2; bottom edge of glazing is less than 457 mm above the floor; and one or more walking surface(s) are within 914 mm, measured horizontally and in a straight line, of the plane of the glazing.

d. Glazing where the bottom exposed edge of the glazing is less than 1524 mm above the plane of the adjacent walking surface of stairways, landings between flights of stairs and ramps.

e. Glazing adjacent to the landing at the bottom of a stairway where the glazing is less than 914 mm above the landing and within 1524 mm horizontally of the bottom tread.

~~2.2.1 Clear Glass~~

~~[For interior glazing (i.e., pass and observation windows), 6 mm thick glass should be used.~~

~~] Provide for glazing openings not indicated or specified otherwise. Use double-strength sheet glass or 3 mm float glass for openings up to and including 1.39 square meters, 4.5 mm for glazing openings over 1.39 square meters but not over 2.79 square meters, and 6 mm for glazing openings over 2.79 square meters but not over 4.18 square meters.~~

~~2.2.2 Annealed Glass~~

~~Annealed glass must be Type I transparent flat type, [Class 1 - clear,] Quality q3 - glazing select, [_____] percent light transmittance, [_____] percent shading coefficient, conforming to JIS R 3202.~~

~~2.2.3 Heat-Absorbing Glass~~

~~[Tinted], [_____] mm thick, [blue][green] in color, [_____] percent light transmittance, [_____] percent shading coefficient, conforming to JIS R 3202.~~

~~2.2.4 Reflective Coating Vision Glass~~

~~JIS R 3221.~~

~~2.2.5 Wired Glass~~

~~Provide UL listed glass for fire-rated windows rated for [45] [20] minutes when tested in accordance with ASTM E2226. Wired glass must be Type II flat type, Class [1 - translucent] [2 - tinted, heat-absorbing] [3 - tinted, light-reducing], Quality [q7 - decorative] [q8 - glazing], Form [1 - wired and polished both sides] [2 - patterned and wired], [_____] percent light transmittance, [_____] percent shading coefficient, conforming to JIS R 3202. Wire mesh must be polished stainless steel Mesh [1 - diamond] [2 - square] [3 - parallel]. Wired glass for fire-rated windows must bear an identifying UL label or the label of a nationally recognized testing agency, and be rated for [20] [45] minutes when tested in accordance with NFPA 257. Wired glass for fire-rated doors must be tested as part of a door assembly in accordance with NFPA 252.~~

~~2.2.6 Patterned Glass~~

~~[Translucent], [patterned], [decorative], [patterned one side], [patterned two sides], [linear], [geometric], [random], [special], [[_____] percent light transmittance, [_____] percent shading coefficient.] [3] [6] mm thick.[Provide [_____].]~~

2.2.1 Laminated Glass

~~[JIS R 3205, Laminated glass fabricated from two nominal [3] [_____] mm pieces of flat annealed [ultraclear]; [clear] [_____] glass conforming to JIS R 3202.] [~~JIS R 3205, Laminated glass fabricated from two minimum nominal ~~[3] [_____] mm~~ pieces of ~~[HS] [~~FT~~]~~, flat ~~[heat strengthened] [~~fully tempered~~]~~ ~~[~~clear~~]~~ ~~[_____]~~ glass conforming to JIS R 3222.~~]~~ Flat glass to be laminated together with a minimum of 0.75 mm ~~[_____] mm~~ thick, clear ~~[~~polyvinyl butyral~~] [ionoplast] [cast-in-place liquid resin]~~ laminate, conforming to requirements of JIS R 3205. The total minimum thickness of nominally 6 ~~[_____]~~ mm. Color to be ~~[~~clear~~] [gray] [bronze] [_____]~~.~~ The total thickness of nominally [_____] mm.~~

~~[Design window glazing using a dynamic analysis[testing from airblast loading in accordance with ISO 16933, or ISO 16934 by an independent testing agency regularly engaged in blast testing] to prove the glazing will provide performance equivalent to or better than a [low] [very low] [_____] hazard rating in accordance with ISO 16933 for the peak positive pressure of [_____] kilopascals (kPa) and peak positive phase impulse of [_____] kilopascal-millisecond (kPa-msec).~~

~~]2.2.2 Bullet-Resisting Glass~~

~~Fabricated from Type I, Class 1, Quality q3 glass with polyvinyl butyral plastic interlayers between the layers of glass and listed by UL MEAPD as bullet resisting, with a rating Level of [Level 1] [Level 2] [Level 3] [Level 4] [Level 5] [_____] in accordance with UL 752. Provide [_____] [where indicated].~~

~~[~~2.2.2 Mirrors

2.2.2.1 Glass Mirrors

Glass for mirrors must be transparent flat type, clear, 6 mm thick conforming to JIS R 3220. Glass must be coated on one surface with silver coating, copper protective coating, and mirror backing paint. Silver coating must be highly adhesive pure silver coating of a thickness which must provide reflectivity of 83 percent or more of incident light when viewed through 6 mm thick glass, and must be free of pinholes or other defects. Copper protective coating must be pure bright reflective copper, homogeneous without sludge, pinholes or other defects, and must be of proper thickness to prevent "adhesion pull" by mirror backing paint. Mirror backing paint must consist of two coats of special scratch and abrasion-resistant paint , and must be baked in uniform thickness to provide a protection for silver and copper coatings which will permit normal cutting and edge fabrication.

~~]2.2.3 One-Way Vision Glass (Transparent Mirrors)~~

~~6 mm thick, coated on one face with a hard, adherent film of chromium or other approved coating of equal durability. Glass must transmit not less than 5 percent or more than 11 percent of total incident visible light and must reflect from the front surface of the coating not less than 45 percent of the total incident visible light.[Provide [_____].]~~

2.2.3 Tempered Glass

Per fully tempered, uncoated, transparent per JIS R 3222 ~~[2 tinted heat absorbing], [_____]~~6 mm thick, ~~[_____] percent light transmittance, [_____] percent shading coefficient conforming~~ to JIS R 3206 or MLIT-SS Ch 16, Sec 14. Color must be ~~[[~~clear~~] [bronze] [gray] [_____]]~~. ~~[~~ Provide ~~[_____]] [~~and wherever safety glazing material is indicated or specified~~]~~.

2.2.4 Heat-Strengthened Glass

HS (heat strengthened), uncoated, ~~[~~clear JIS R 3222~~]~~ ~~[tinted heat absorbing], [_____]~~6 mm thick.~~ [Provide [_____].]~~

~~2.2.5 Spandrel Glass~~

~~2.2.5.1 Ceramic-Opacified Spandrel Glass~~

~~Ceramic-opacified spandrel glass must be Kind HS heat-strengthened transparent flat type, coated with a colored ceramic material on No. 2 surface, [_____] mm thick, conforming to JIS R 3222. Glass performance must be K-Value/Winter Nighttime [_____], shading coefficient [_____]. Color must be [_____].~~

~~2.2.5.2 Film-Opacified Spandrel Glass~~

~~Film-opacified spandrel glass must be Kind HS heat-strengthened transparent flat type glass with a polyester or polyethylene film 0.025 mm to 0.127 mm thick attached to No. 2 surface of a sputtered solar-reflective film, conforming to JIS R 3222. Film opacification must be compatible to and specifically developed for application to solar reflective films. Glass performance must be K-Value/Winter Nighttime [_____], shading coefficient [_____]. Color must be [_____].~~

~~2.2.5.3 Spandrel Glass With Adhered Backing~~

~~Kind HS or FT, ceramic coated, JIS R 3222, [_____] mm thick and must pass the fallout resistance test specified in JIS R 3222. [Provide [_____].]~~

~~[2.2.6 Fire/Safety Rated Glass~~

~~[2.2.6.1 Fire Protection Rated Glass~~

~~Clear tempered and meet 16 CFR 1201 Category I (under 0.836 sqm) or II (over 0.836 sqm) impact safety standard. Glass to make [20] [45] minute rating when tested in accordance with NFPA 257 and NFPA 252. Glass to be permanently labeled with appropriate markings.~~

~~][2.2.6.2 Fire Resistive Rated Glazing~~

~~Fire resistive glass must be laminated, with intumescent interlayer, Type I transparent flat type, Class 1-clear and meet 16 CFR 1201 Category I (under 0.836 sqm) or II (over 0.836 sqm). Glass must have a [60] [90] [120] minute rating when tested in accordance with JIS A 1304. Glass must be permanently labeled with appropriate markings.~~

~~]]~~2.3 INSULATING GLASS UNITS

~~[~~Two~~][Three]~~ panes of glass separated by a dehydrated airspace~~[, filled with argon gas][, filled with krypton gas,][, filled with aerogel]~~ and hermetically sealed, conforming to ISO 28278. Submit performance and compliance documentation for each type of insulating glass.

~~[~~ Insulated glass units must have a Solar Heat Gain Coefficient (SHGC) ~~maximum of [_____] determined according to ISO 15099 and~~ a U-factor~~ ~~ as identified in 08 11 16 ALUMINUM DOORS, FRAMES, ENTRANCES AND WINDOWS and maximum ~~of [_____] W per square m by K~~ in accordance with ISO 15099.

~~] [See section[s][_____] for energy performance requirements for glazed systems (glazing and frames).] [Glazed panels must be rated for not less than [26] [30] [35] [_____] Sound Transmission Class (STC) or equivalent Weighted Sound Reduction Index (Rw) when tested for laboratory sound transmission loss according to JIS A 1418 and determined by JIS A 1419.]~~

Spacer must be black, roll-formed, ~~[thin-gauge, C-section steel] [steel-reinforced butyl rubber] [thermally broken aluminum] [polyurethane and silicon foams],~~ with bent or tightly welded or keyed and sealed joints to completely seal the spacer periphery and eliminate moisture and hydrocarbon vapor transmission into airspace through the corners. Primary seal must be compressed polyisobutylene and the secondary seal must be a specially formulated silicone.

The inner light must be clear ~~annealed~~heat strengthened, laminated flat glass JIS R 3205, conforming with the performance requirements as indicated ~~JIS R 3202 [_____] mm thick, fully tempered, uncoated, transparent, JIS R 3222 [_____] mm thick. The intermediate light must be clear annealed flat glass, JIS R 3202, fully tempered, uncoated, transparent, JIS R 3222 [_____] mm thick~~. The outer light must be heat strengthened transparent, JIS R 3202, ~~[2 (tinted heat absorbing)], [~~2 (solar-reflective) ~~]~~, ~~[_____]~~6 mm thick, ~~fully tempered, uncoated, clear JIS R 3202 [2 (tinted heat absorbing)][solar-reflective], [_____] mm thick~~conforming to the performance requirements as indicated.

2.3.1 Low Emissivity Coatings

~~Interior and e~~Exterior glass panes for Low-E insulating units must be annealed heat strengthend flat glass, ~~Class [1-clear] [2-tinted]~~ with anti-reflective low-emissivity coating or heat-strengthened ~~or fully tempered~~ glass ~~complying with JIS R 3222, Condition C~~ on ~~[~~No. 2 surface (inside surface of exterior pane) JIS R 3222~~][No. 3 surface (inside surface of interior pane)], conforming to JIS R 3202. Glass performance must be U value maximum of [_____] [W/m2-K], Solar Heat Gain Coefficient (SHGC) maximum of [_____]~~. Color must be ~~[green] [gray] [bronze] [blue] [_____]~~ as indicated.

~~2.4 PLASTIC GLAZING~~

~~Plastic glazing must have a U-factor maximum of [_____] W per square m by K. [Plastic glazing must include a [16][32][_____] mm layer of aerogel between panels.]~~

~~Certificates stating that the plastic glazing meets the specified requirements. Labels or manufacturers marking affixed to the glass will be accepted in lieu of certificates.~~

~~2.4.1 Acrylic Sheet~~

~~JIS K 6718, [regular] [heat resistant,] [clear and smooth on both sides] [translucent, textured on both sides,] [gray tint,] [bronze tint,] ultraviolet stabilized, [scratch resistant,] [_____] [6] [_____] mm thick.~~

~~2.4.2 Polycarbonate Sheet~~

~~[Clear and smooth both sides] [Translucent, textured both sides] [Gray tint] [Bronze tint] [mar-resistant] [high abrasion resistant], ultraviolet stabilized, [_____] mm thick and listed in UL MEAPD as burglar resisting.~~

~~2.4.3 Extruded Polycarbonate Profiled Sheet~~

~~Provide [double] [triple] walled, surface treated for improved UV resistance, offering thermal efficiency and impact strength.~~

~~2.4.4 Bullet-Resistant Plastic Sheet~~

~~Cast acrylic sheet or mar-resistant polycarbonate sheet laminated with a special interlayer, and listed in UL 752 as bullet resisting, Class [I] [II] [III], [clear] [_____] in color.[Provide [_____].]~~

2.4 SETTING AND SEALING MATERIALS

Provide as specified in the MLIT-SS Ch 16, Sec 14, and manufacturer's recommendations, unless specified otherwise herein. Do not use metal sash putty, nonskinning compounds, nonresilient preformed sealers, or impregnated preformed gaskets. Materials exposed to view and unpainted must be gray or neutral color. Sealant testing must be performed by a qualified testing agency.

Submit glass manufacturer's recommendations for setting and sealing materials and for installation of each type of glazing material specified. ~~[Include cleaning instructions for plastic sheets.]~~

2.4.1 Putty and Glazing Compound

Provide glazing compound as recommended by manufacturer for face-glazing metal sash. Putty must be linseed oil type. Do not use putty and glazing compounds with insulating glass or laminated glass.

2.4.2 Glazing Compound

Use for face glazing metal sash. Do not use with insulating glass units or laminated glass.

2.4.3 Sealants

Provide elastomeric ~~[~~and structural~~]~~ sealants.

2.4.3.1 Elastomeric Sealant

JIS A 5758. Use for channel or stop glazing ~~[wood] [and] [~~metal~~]~~ sash. Sealants must be chemically compatible with setting blocks, edge blocks, and sealing tapes~~[~~, with sealants used in manufacture of insulating glass units~~] [, and with plastic sheet]~~. Color of sealant must be white.

2.4.3.2 Structural Sealant

JIS A 5758.

2.4.4 Joint Backer

Joint backer must have a diameter size at least 25 percent larger than joint width; type and material as recommended in writing by glass and sealant manufacturer.

2.4.5 Glazing Tapes

2.4.5.1 Back-Bedding Mastic Glazing Tapes

Preformed, butyl-based, 100 percent solids elastomeric tape; nonstaining and nonmigrating in contact with nonporous surfaces; with or without spacer rod as recommended in writing by tape and glass manufacturers for application indicated; and complying with JIS A 5758.

2.4.5.2 Expanded Cellular Glazing Tapes

Closed-cell, PVC foam tapes; factory coated with adhesive on both surfaces; and complying with JIS A 5758 as recommended by tape and glass manufacturers.

2.4.6 Sealing Tapes

Preformed, semisolid, PVC-based material of proper size and compressibility for the particular condition, complying with JIS K 7366. Use only where glazing rabbet is designed for tape and tape is recommended by the glass or sealant manufacturer. Provide spacer shims for use with compressible tapes. Tapes must be chemically compatible with the product being set.

2.4.7 Setting Blocks and Edge Blocks

Closed-cell neoprene setting blocks must be dense extruded type conforming to JIS A 5756 and JIS K 6262. Edge blocking as recommended by glazing manufacturer. Provide silicone setting blocks when blocks are in contact with silicone sealant. Profiles, lengths and locations must be as required and recommended in writing by glass manufacturer. Block color must be ~~[~~black~~][_____]~~.

2.4.8 Glazing Gaskets

Glazing gaskets must be extruded with continuous integral locking projection designed to engage into metal glass holding members to provide a watertight seal during dynamic loading, building movements and thermal movements. Glazing gaskets for a single glazed opening must be continuous one-piece units with factory-fabricated injection-molded corners free of flashing and burrs. Glazing gaskets must be in lengths or units recommended by manufacturer to ensure against pull-back at corners. Provide glazing gasket profiles as recommended by the manufacturer for the intended application.

2.4.8.1 Fixed Glazing Gaskets

Fixed glazing gaskets must be closed-cell (sponge) smooth extruded compression gaskets of cured elastomeric virgin neoprene compounds conforming to JIS A 5756.

2.4.8.2 Wedge Glazing Gaskets

Wedge glazing gaskets must be high-quality extrusions of cured elastomeric virgin neoprene compounds, ozone resistant, conforming to JIS A 5756.

2.4.8.3 Aluminum Framing Glazing Gaskets

Glazing gaskets for aluminum framing must be permanent, elastic, non-shrinking, non-migrating, watertight and weathertight.

2.4.9 Accessories

Provide as required for a complete installation, including glazing points, clips, shims, angles, beads, and spacer strips. Provide noncorroding metal accessories. Provide primer-sealers and cleaners as recommended by the glass and sealant manufacturers. Use JIS A 1439 to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of glazing sealants to surface.

~~[~~2.5 MIRROR ACCESSORIES

2.5.1 Mastic

Mastic for setting mirrors must be a ~~[~~polymer~~] [_____]~~ type mirror mastic resistant to water, shock, cracking, vibration and thermal expansion. Provide mastic compatible with mirror backing paint, and as approved by mirror manufacturer.

2.5.2 Mirror Frames

Provide mirrors with mirror frames (J-mold channels) fabricated of one-piece roll-formed Type 304 stainless steel with No. 4 brushed satin finish and concealed fasteners which will keep mirrors snug to wall. Frames must be 32 by 6 by 6 mm continuous at top and bottom of mirrors. Concealed fasteners of type to suit wall construction material must be provided with mirror frames.

2.5.3 Mirror Clips

Provide clips with concealed fasteners of type to suit wall construction material.

PART 3 EXECUTION

Any materials that show visual evidence of biological growth due to the presence of moisture must not be installed on the building project.

3.1 PREPARATION

Preparation, unless otherwise specified or approved, must conform to applicable recommendations in MLIT-SS Ch 16, Sec 14 and manufacturer's recommendations. Determine the sizes to provide the required edge clearances by measuring the actual opening to receive the glass. Grind smooth in the shop glass edges that will be exposed in finish work. Leave labels in place until the installation is approved, except remove applied labels on heat-absorbing glass and on insulating glass units as soon as glass is installed. Securely fix movable items or keep in a closed and locked position until glazing compound has thoroughly set.

3.2 GLASS SETTING

Shop glaze or field glaze items to be glazed using glass of the quality and thickness specified or indicated. Glazing, unless otherwise specified or approved, must conform to applicable recommendations in MLIT-SS Ch 16, Sec 14 and manufacturer's recommendations. Aluminum windows, wood doors, and wood windows may be glazed in conformance with one of the glazing methods described in the standards under which they are produced, except that face puttying with no bedding will not be permitted. Handle and install glazing materials in accordance with manufacturer's instructions. Use beads or stops which are furnished with items to be glazed to secure the glass in place. Verify products are properly installed, connected, and adjusted.

3.2.1 Sheet Glass

Cut and set with the visible lines or waves horizontal.

~~3.2.2 Patterned Glass~~

~~Set glass with one patterned surface with smooth surface on the weather side. When used for interior partitions, place the patterned surface in same direction in all openings.~~

3.2.2 Insulating Glass Units

Do not grind, nip, or cut edges or corners of units after the units have left the factory. Springing, forcing, or twisting of units during setting will not be permitted. Handle units so as not to strike frames or other objects. Installation must conform to applicable recommendations of MLIT-SS Ch 16, Sec 14.

~~3.2.3 Installation of Wire Glass~~

~~Install glass for fire doors in accordance with installation requirements~~

~~of NFPA 80.~~

~~3.2.4 Installation of Heat-Absorbing Glass~~

~~Provide glass with clean-cut, factory-fabricated edges. Field cutting will not be permitted.~~

3.2.3 Installation of Laminated Glass

Sashes which are to receive laminated glass must be weeped to the outside to allow water drainage into the channel.

~~3.2.4 Plastic Sheet~~

~~Conform to manufacturer's recommendations for edge clearance, type of sealant and tape, and method of installation.~~

3.3 CLEANING

Clean glass surfaces and remove labels, paint spots, putty, and other defacement as required to prevent staining. Glass must be clean at the time the work is accepted.~~[Clean plastic sheet in accordance with manufacturer's instructions.]~~

3.4 PROTECTION

Protect glass work immediately after installation. Identify glazed openings with suitable warning tapes, cloth or paper flags, attached with non-staining adhesives. Protect reflective glass with a protective material to eliminate any contamination of the reflective coating. Place protective material far enough away from the coated glass to allow air to circulate to reduce heat buildup and moisture accumulation on the glass. Upon removal, separate protective materials for reuse or recycling. Remove and replace glass units which are broken, chipped, cracked, abraded, or otherwise damaged during construction activities with new units.

-- End of Section --

SECTION 08 91 00

METAL ~~[~~WALL~~] [AND] [DOOR]~~ LOUVERS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-L	(2015) Laboratory Methods of Testing Louvers for Rating
AMCA 511	(2010; R 2016) Certified Ratings Program for Air Control Devices

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS G 3141	(2017) Cold-reduced carbon steel sheet and strip
JIS H 4000	(2017) Aluminium and Aluminium Alloy Sheets, Strips and Plates (Amendment 1)
JIS H 4100	(2015) Aluminum and Aluminum Alloy Extruded Profiles
JIS H 8602	(2010) Combined Coatings of Anodic Oxide and Organic Coatings on Aluminum and Aluminum Alloys

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Wall Louvers

SD-03 Product Data

Metal Wall Louvers

~~Door Louvers~~

SD-04 Samples

Wall Louver Samples; G~~[, [_____]]~~

~~Door Louver Samples; G[, [_____]]~~

1.3 DELIVERY, STORAGE, AND PROTECTION

Deliver materials to the site in an undamaged condition. Carefully store materials off the ground to provide proper ventilation, drainage, and protection against dampness. Louvers must be free from nicks, scratches, and blemishes. Replace defective or damaged materials with new.

1.4 DETAIL DRAWINGS

Show all information necessary for fabrication and installation of wall louvers. Indicate materials, sizes, thicknesses, fastenings, and profiles.

1.5 COLOR SAMPLES

Colors of finishes for wall louver samples ~~and door louver samples~~ must closely approximate colors indicated. Where color is not indicated, submit the manufacturer's standard colors to the Contracting Officer for selection.

PART 2 PRODUCTS

2.1 MATERIALS

~~2.1.1 Galvanized Steel Sheet~~

~~ASTM A653/A653M, coating designation Z275.~~

2.1.1 Aluminum Sheet

~~ASTM B209M or~~ JIS H 4000, alloy 3003 or 5005 with temper as required for forming.

2.1.2 Extruded Aluminum

~~ASTM B221M or~~ JIS H 4100, alloy 6063-T5 or -T52.

~~[2.1.3 Stainless Steel~~

~~Type 302 or 304, with 2B finish.~~

~~]~~2.1.3 Cold Rolled Steel Sheet

~~ASTM A1008/A1008M, Class 1, with matte finish. Use for interior louvers only~~As indicated in the drawings.

2.2 METAL WALL LOUVERS

~~[~~Weather~~][Wind driven rain]~~ resistant type, with bird screens and made to withstand a wind load ~~of not less than [1.44] [_____] kilopascals~~JIS G 3141, with matte finish, use for interior louvers only. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L and AMCA 511. The rating must show a water penetration of 0.06 kilograms or less per square meter of free area at a free velocity of 244 meters per minute.

2.2.1 Extruded Aluminum Louvers

Fabricated of extruded 6063-T5 or -T52 aluminum with a wall thickness of not less than 2 mm.

~~2.2.2 Formed Metal Louvers~~

~~Formed of [zinc-coated] [stainless] steel sheet not thinner than 16 U.S. gage, or aluminum sheet not less than 2 mm thick.~~

2.2.2 Mullions and Mullion Covers

Same material and finish as louvers. Provide mullions ~~[where indicated] [~~ for all louvers more than 1500 mm in width at not more than 1500 mm on centers~~]~~. Provide mullion covers on both faces of joints between louvers.

2.2.3 Screens and Frames

For aluminum louvers, provide 12.5 mm square mesh, 1.8 or 1.5 mm aluminum or 6 mm square mesh, 1.5 mm aluminum bird screening.~~ For steel louvers, provide 12.5 mm square mesh, 2.5 or 1.5 mm zinc-coated steel; 12.5 mm square mesh, 1.5 mm copper; or 6 mm square mesh, 1.5 mm thick zinc-coated steel or copper bird screening. Mount screens in removable, rewirable frames of same material and finish as the louvers.~~

~~2.3 DOOR LOUVERS~~

~~[Inverted "Y"][or][Inverted "V"] sightproof type not less than 25 mm thick with matching metal trim. Louvers for exterior doors must be weather resistant type.~~

~~2.3.1 Extruded Aluminum Door Louvers~~

~~Fabricate of 6063-T5 or -T52 aluminum alloy with a wall thickness of not less than 1.25 mm thick. Frames and trim must be clamp-in "L" type.~~

~~2.3.2 Formed Metal Door Louvers~~

~~Fabricate of [0.9 mm thick steel sheet][or][sheet aluminum not less than 1.25 mm thick]. Trim must be beveled "Z" molding both sides.~~

~~2.3.3 Screens and Frames~~

~~For exterior doors, provide aluminum insect screens, 18 by 16 or 18 by 14 mesh. Mount screens in removable, rewirable frames of same material and finish as the louvers.~~

2.3 FASTENERS AND ACCESSORIES

Provide stainless steel screws and fasteners for aluminum louvers and zinc-coated or stainless steel screws and fasteners for steel louvers. Provide other accessories as required for complete and proper installation.

2.4 FINISHES

2.4.1 Aluminum

Exposed aluminum surfaces must be factory finished with an ~~[~~anodic coating ~~][or][organic coating]~~.~~[~~ Color must be ~~[_____][~~as indicated~~]~~.~~]~~ Louvers

~~[for each building]~~ must have the same finish.

2.4.1.1 Anodic Coating

Clean exposed aluminum surfaces and provide an anodized finish conforming to ~~AA DAF45 and AAMA 611~~JIS H 8602. Finish must be:

~~[~~ a. ~~Architectural Class II (0.01 to 0.0175 mm), designation AA-M10-C22-[A31, clear (natural)] [A32, integral color] [A34, electrolytically deposited color] anodized.~~Anodic coating 0.0175 mm or thicker, clear (natural) anodozed.

~~][b. Architectural Class I (0.0175 mm or thicker), designation AA-M10-C22-[A41, clear (natural)] [A42, integral color] [A44, electrolytically deposited color] anodized.~~

~~]2.4.1.2 Organic Coating~~

~~Clean and prime exposed aluminum surfaces. Provide a [baked enamel finish conforming to AAMA 2603, with total dry film thickness not less than 0.02 mm] [superior performance finish in accordance with AAMA 2605 with total dry film thickness of not less than 0.03 mm], color [_____].~~

~~2.4.2 Steel~~

~~Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating must conform to ASTM A653/A653M, coating designation Z275 (G90)[, except that louvers located in conditioned spaces on interior of the building may be Z180 (G60)]. The weight of zinc coatings must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied before forming, in lieu of prime coat specified.~~

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Wall Louvers

Install using stops or moldings, flanges, strap anchors, or jamb fasteners as appropriate for the wall construction and in accordance with manufacturer's recommendations.

~~3.1.2 Door Louvers~~

~~Install louvers in wood doors by using metal "Z" or "L" moldings. Fasten moldings to door with screws.~~

3.1.2 Screens and Frames

Attach frames to louvers with screws or bolts.

3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS

~~3.2.1 Copper or Copper-Bearing Alloys~~

~~Paint copper or copper-bearing alloys in contact with dissimilar metal with heavy-bodied bituminous paint or separate with inert membrane.~~

3.2.1 Aluminum

Where aluminum contacts metal other than zinc, paint the dissimilar metal with a primer and two coats of aluminum paint.

3.2.2 Metal

Paint metal in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

~~3.2.3 Wood~~

~~Paint wood or other absorptive materials that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.~~

-- End of Section --

SECTION 09 22 00

SUPPORTS FOR PLASTER AND GYPSUM BOARD
02/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

BUILDING STANDARD LAW OF JAPAN

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 5505	(2014) Metal Laths
JIS A 6517	(2010) Steel Furrings for Wall and Ceiling in Buildings
JIS G 3302	(2019) Hot Dip Zinc Coated Steel Sheet and Strip
JIS G 3314	(2010) Hot-Dip Aluminum-Coated Steel Sheet and Strip
JIS G 3321	(2012) Molten 55 Percent Aluminum Zinc Alloy Plated Steel Sheet and Strip
JIS G 3505	(2017) Mild Steel Wire Rod

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Metal Support Systems; G~~[, [_____]]~~

Submit for the erection of metal~~[~~ framing,~~][~~ furring,~~][~~ and~~][~~ ceiling suspension systems~~]~~. Indicate materials, sizes, thicknesses, and fastenings.

SD-03 Product Data

Metal Support Systems

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the job site and store in ventilated dry locations permitting easy access for inspection and handling. If materials are stored outdoors, stack materials off the ground, supported on a level platform, and fully protected from the weather. Handle materials carefully to prevent damage. Remove damaged items and provide new items.

PART 2 PRODUCTS

2.1 MATERIALS

Provide steel materials for metal support systems with galvanized coating JIS G 3302, Z18; aluminum coating JIS G 3314, with a plating adhesion amount of 80, or JIS G 3321. Provide support systems, bracing and attachments per Building Standard Law of Japan.

Provide metal support systems containing a minimum of 20 percent recycled content.

2.1.1 Materials for Attachment of Lath

2.1.1.1 Suspended and Furred Ceiling Systems and Wall Furring

JIS A 5505.

2.1.1.2 Non-loadbearing Wall Framing and Furring

JIS A 6517.

2.1.2 Materials for Attachment of Gypsum Wallboard

2.1.2.1 Suspended and Furred Ceiling Systems

JIS A 6517. JIS G 3505 for hanger rod and nut.

2.1.2.2 Nonload-Bearing Wall Framing and Furring

JIS A 6517, but not thinner than~~[~~ 0.45 mm thickness, with 0.85 mm minimum thickness supporting wall hung items such as cabinetwork, equipment and fixtures~~][0.85 mm thickness regardless of the ASTM certified third party testing statement for equivalent thicknesses]~~ and partitions with ceramic tile.

~~2.1.2.3 Furring Structural Steel Columns~~

~~JIS A 6517.~~

~~2.1.2.4 Z-Furring Channels with Wall Insulation~~

~~Not lighter than 0.5 mm thick galvanized steel, Z-shaped, with 32 mm and 19 mm flanges and [[25] [38] [50] [75] mm furring depth] [depth as required by the insulation thickness provided].~~

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Systems for Attachment of Gypsum Board or Lath

Follow manufacturer's instructions.

3.2 ERECTION TOLERANCES

Provide framing members which will be covered by finish materials such as wallboard, plaster, or ceramic tile set in a mortar setting bed, within the following limits:

a. Layout of walls and partitions: 6 mm from intended position;

b. Plates and runners: 5 mm in 1.9 meters from a straight line;

c. Studs: 5 mm in 1.9 meters out of plumb, not cumulative; and

d. Face of framing members: 5 mm in 1.9 meters from a true plane.

Provide framing members which will be covered by ceramic tile set in dry-set mortar, latex-portland cement mortar, or organic adhesive within the following limits:

a. Layout of walls and partitions: 6 mm from intended position;

b. Plates and runners: 5 mm in 3.8 meters from a straight line;

c. Studs: 5 mm in 3.8 meters out of plumb, not cumulative; and

d. Face of framing members: 5 mm in 3.8 meters from a true plane.

-- End of Section --

SECTION 09 29 00

GYPSUM BOARD
08/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C475/C475M	(2017) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board
ASTM C514	(2004; R 2020) Standard Specification for Nails for the Application of Gypsum Board
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.84 mm to 2.84 mm in Thickness
ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM C1396/C1396M	(2017) Standard Specification for Gypsum Board
ASTM D3273	(2016) Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber

GYPSUM BOARD ASSOCIATION OF JAPAN (GRAJ)

GRAM	Gypsum Board Application Manual

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1408	(2017) Test Methods of Bending and Impact for Building Boards
JIS A 5430	(2018) Fiber Reinforced Cement Boards
JIS A 5508	(2009) Nails
JIS A 6901	(2014) Gypsum Boards
JIS A 6914	(2008) Jointing Materials for Gypsum

	Plasterboards
JIS B 1125	(2015) Self Drilling Tapping Screws
JIS Z 2911	(2018) Methods of Test for Fungus Resistance

UNDERWRITERS LABORATORIES (UL)

UL Fire Resistance	(2014) Fire Resistance Directory

1.2 SUBMITTALS

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SD-08 Manufacturer's Instructions

Safety Data Sheets

SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

1.3.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board and cementitious backer units outside, store off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Store per manufacturer's recommendations for allowable temperature and humidity range. Do not store gypsum wallboard with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants~~, including [_____]~~. Do not store panels near materials that may off gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth.

1.3.3 Handling

Neatly stack gypsum board and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces.

1.4 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of ~~[~~3~~]~~ ~~[_____]~~ years of documented successful experience.

1.5 SCHEDULING

~~[~~The gypsum wallboard must be taped, finished and primed before the installation of the highly emitting materials~~, including [_____]~~.~~]~~ ~~[The gypsum wallboard must be installed after the installation and ventilation period of the highly emitting materials, including [_____].]~~

Commence application only after the area scheduled for gypsum board work is completely weathertight. The heating, ventilating, and air-conditioning systems must be complete and in operation prior to application of the gypsum board. If the mechanical system cannot be activated before gypsum board is begun, the gypsum board work may proceed in accordance with an approved plan to maintain the environmental conditions specified below. Apply gypsum board prior to the installation of finish flooring and acoustic ceiling.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 10 degrees C and not more than 27 degrees C for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of air across the plastered surfaces. Provide ventilation to exhaust moist air to the outside during gypsum board application, set, and until gypsum board jointing is dry. In glazed areas, keep windows open top and bottom or side to side 75 to 100 mm. Reduce openings in cold weather to prevent freezing of joint compound when applied. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. In unglazed areas subjected to hot, dry winds or temperature differentials from day to night of 10 degrees C or more, screen openings with cheesecloth or similar materials. Avoid rapid drying. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

~~[~~1.7 FIRE RESISTIVE CONSTRUCTION

Comply with specified fire-rated assemblies for design numbers indicated.

~~]~~PART 2 PRODUCTS

2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, cementitious backing units, and joint treating materials manufactured from asbestos free materials only. Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

2.1.1 Gypsum Board

JIS A 6901.

2.1.1.1 Regular

900 mm wide, ~~[~~12.5~~]~~ ~~[15]~~ mm thick, ~~[~~tapered~~][, tapered and featured] edges. [Provide tapered and featured edge gypsum board [in Rooms [_____]] [as indicated]~~.~~]~~

~~2.1.1.2 Foil-Backed~~

~~900 mm wide,[12.5] [15] mm thick, square, [tapered] [tapered and featured] edges.~~

~~2.1.1.3 Type X or Type Z (Special Fire-Resistant)~~

~~900 mm wide, [12.5] [15] mm thick, [tapered] [tapered and featured] edges.~~

2.1.1.2 Mold Resistant / Anti-Microbial Gypsum

ASTM D3273 or JIS Z 2911. 900 mm wide, ~~[12.5] [~~15~~]~~ mm thick, ~~square,~~ tapered ~~or beveled~~ edges.

2.1.2 Gypsum Backing Board

ASTM C1396/C1396M or JIS A 6901, gypsum backing board must be used as a base in a multilayer system.

2.1.2.1 Regular

900 mm wide, ~~[~~12.5~~]~~ ~~[15]~~ mm thick, square, tapered or beveled edges.

2.1.2.2 Type Z or Type X (Special Fire-Resistant)

900 mm or 1200 mm wide, ~~[12.5] [~~15~~]~~ mm thick, square or tapered edges.

2.1.3 Regular Water-Resistant Gypsum Backing Board

2.1.3.1 Regular

900 mm wide, ~~[~~12.5~~] [15]~~ mm thick, tapered or beveled edges.

~~2.1.3.2 Type X or Type Z (Special Fire-Resistant)~~

~~900 mm wide, [12.5] [15] mm thick, tapered or beveled edges.~~

~~2.1.4 Glass Mat Water-Resistant Gypsum Tile Backing Board~~

~~Water absorption rate of less than 5 percent.~~

~~2.1.4.1 Regular~~

~~1200 mm wide, [12.5] [15] mm thick, square, tapered or beveled edges.~~

~~2.1.4.2 Type Z (Special Fire-Resistant)~~

~~900 mm wide, [12.5] [15] mm thick, square, tapered or beveled edges.~~

2.1.4 Abuse Resistant Gypsum Board

900 mm wide, ~~[15]~~12.5 mm thick, tapered edges. Reinforced gypsum panel with imbedded fiber mesh or lexan backing tested in accordance with the

following tests. Hard body impact test must attain a maximum indentation depth of 2.5 mm in accordance with JIS A 1408. Provide fasteners that meet manufacturer requirements and specifications stated within this section.~~ Abuse resistant gypsum board, when tested in accordance with ASTM E84, have [a flame spread rating of 25 or less and a smoke developed rating of 50 or less for [_____]] [and] [a flame spread rating of 75 or less and a smoke developed rating of 100 or less for [_____]].~~

~~2.1.4.1 Hard Body Impact Test~~

~~Comply with hard body impact test in accordance with JIS A 1408 with a maximum indentation depth of 2.5 mm.~~

~~2.1.4.2 Surface Abrasion Test~~

~~Comply with test surface abrasion test in accordance with JIS A 1453.~~

~~2.1.4.3 Indentation Test~~

~~JIS A 1408 for indentation resistance.~~

2.1.5 Cementitious Backer Units

In accordance with JIS A 5430 or Tile Council of America (TCA) Handbook.

2.1.6 Joint Treatment Materials

ASTM C475/C475M or JIS A 6914. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Provide data identifying VOC content of joint compound. ~~[~~Use all-purpose joint and texturing compound containing inert fillers and natural binders, including lime compound. Pre-mixed compounds must be free of antifreeze, vinyl adhesives, preservatives, biocides and other slow releasing compounds.~~]~~

2.1.6.1 Embedding Compound

Specifically formulated and manufactured for use in embedding tape at gypsum board joints and compatible with tape, substrate and fasteners.

2.1.6.2 Finishing or Topping Compound

Specifically formulated and manufactured for use as a finishing compound.

2.1.6.3 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate and fasteners.

2.1.6.4 Setting or Hardening Type Compound

Specifically formulated and manufactured for use with fiber glass mesh tape.

2.1.6.5 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

2.1.7 Fasteners

2.1.7.1 Nails

JIS A 5508 or ASTM C514.

2.1.7.2 Screws

ASTM C1002 or JIS B 1125 steel drill screws for fastening gypsum board to gypsum board, wood framing members and steel framing members less than 0.84 mm thick. ASTM C954 or JIS B 1125 steel drill screws for fastening gypsum board to steel framing members 0.84 to 2.84 mm thick. Provide cementitious backer unit screws with a polymer coating.

2.1.7.3 Staples

1.5 mm thick flattened galvanized wire staples with 11.1 mm wide crown outside measurement and divergent point for base ply of two-ply gypsum board application. Use as follows:

Length of Legs	Thickness of Gypsum Board
28.6 mm	12.5 mm
31.8 mm	15 mm

~~2.1.8 Adhesives~~

~~Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of UL 2818(Greenguard) Gold, SCS Global SErvices Indoor ADvantage Gold, or F 4-Star and JAIA 4VOC . Provide aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of JIS A 1901 (limit requirements for either office or classroom spaces regardless of space type. Provide certification or validation of indoor air quality for non-aerosol adhesives applied on the interior of the building (inside of the weatherproofing system). Provide certification or validation of indoor air quality for aerosol adhesives used on the interior of the building (inside of the weatherproofing system).~~

~~2.1.8.1 Adhesive for Laminating~~

~~[Not permitted.][Adhesive attachment is not permitted for multi-layer gypsum boards. For laminating gypsum studs to face panels, provide adhesive recommended by gypsum board manufacturer.]~~

~~2.1.9 Gypsum Studs~~

~~Provide 25 mm minimum thickness and 150 mm minimum width. Studs may be of 25 mm thick gypsum board or multilayers fastened to required thickness. Conform to JIS A 6901 for material and Gypsum Board Application Manual (GRAM) for installation.~~

2.1.8 Shaftwall Liner Panel

Conform to the UL Fire Resistance or MLIT for the Design Numbers(s) indicated for shaftwall liner panels. Manufacture liner panel for cavity shaftwall system, with water-resistant paper faces, bevel edges, single lengths to fit required conditions, as indicated.

2.1.9 Accessories

Fabricate from ~~[corrosion protected steel][or][~~plastic~~]~~ designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

~~2.1.10 Asphalt Impregnated Building Felt~~

~~Provide a 6.7 kg asphalt moisture barrier over glass mat covered or reinforced gypsum sheathing. Conforming to JIS A 6005 for asphalt impregnated building felt.~~

2.1.10 Water

Provide clean, fresh, and potable water.

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Framing and Furring

Verify that framing and furring are securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing and furring are acceptable for application of gypsum board and cementitious backer units.

~~3.1.2 [Gypsum Board] [and] [Framing]~~

~~Verify that surfaces of [gypsum board] [and] [framing] to be bonded with an adhesive are free of dust, dirt, grease, and any other foreign matter. Do not proceed with work until surfaces are acceptable for application of gypsum board with adhesive.~~

~~3.1.3 [Masonry] [and] [Concrete] Walls~~

~~Verify that surfaces of [masonry] [and] [concrete] walls to receive gypsum board applied with adhesive are dry, free of dust, oil, form release agents, protrusions and voids, and any other foreign matter. Do not proceed with work until surfaces are acceptable for application of gypsum board with adhesive.~~

~~3.1.4 Building Construction Materials~~

~~Do not install building construction materials that show visual evidence of biological growth.~~

3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing and furring members in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturers instructions and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may ~~[not]~~be bonded together with an adhesive~~[~~, except where prohibited by fire rating(s)~~]~~. Treat edges of cutouts for plumbing pipes, screwheads, and joints with water-resistant compound as recommended by the gypsum board manufacturer. ~~Minimize framing by floating corners with single studs and drywall clips. [Install [16 mm][_____] gypsum or [13 mm][_____] ceiling board over framing at [610 mm][_____] on center.]~~ Provide type of gypsum board for use in each system specified herein as indicated.

~~3.2.1 Application of Single-Ply Gypsum Board to Wood Framing~~

~~Apply in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.~~

~~3.2.2 Application of Two-Ply Gypsum Board to Wood Framing~~

~~Apply in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.~~

~~3.2.3 Adhesive Application to Interior Masonry or Concrete Walls~~

~~Apply in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.~~

3.2.1 Application of Gypsum Board to Steel Framing and Furring

Apply in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.

~~3.2.2 Arches and Bending Radii~~

~~Apply in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.~~

3.2.2 Gypsum Board for Wall Tile or Tile Base Applied with Adhesive

In dry areas (areas other than tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply glass mat water-resistant gypsum tile backing board ~~[~~or water-resistant gypsum backing board~~]~~ in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.

~~3.2.3 Exterior Application~~

~~Apply exterior gypsum board (such as at soffits) in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.~~

~~3.2.4 Floating Interior Angles~~

~~Minimize framing by floating corners with single studs and drywall clips.~~

~~Locate the attachment fasteners adjacent to ceiling and wall intersections in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions, for [single-ply] [and] [two-ply] applications of gypsum board to wood framing.~~

3.2.3 Control Joints

Install expansion and contraction joints in ceilings and walls in accordance with Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions. Fill control joints between studs in fire-rated construction with firesafing insulation to match the fire-rating of construction.

3.2.4 Application of Abuse Resistant Gypsum Board

Apply in accordance with applicable system of Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions. Follow manufacturers written instructions on how to cut, drill and attach board.

3.3 APPLICATION OF CEMENTITIOUS BACKER UNITS

3.3.1 Application

In wet areas (tubs, shower enclosures, saunas, steam rooms, gang shower rooms), apply cementitious backer units. Place a 7.6 kg asphalt impregnated, continuous felt paper membrane behind cementitious backer units, between backer units and studs or base layer of gypsum board. Place membrane with a minimum 150 mm overlap of sheets laid shingle style.

3.3.2 Joint Treatment

Gypsum Board Application Manual (GRAM), Gypsum Board Association of Japan or Manufacturer's Instructions.

3.4 FINISHING OF GYPSUM BOARD

Finish plenum areas above ceilings to Level 1. Finish water resistant gypsum backing board to receive ceramic tile to Level 2. Finish walls and ceilings to receive a heavy-grade wall covering or heave textured finish before painting to Level 3. Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4. Unless otherwise specified, finish all gypsum board walls, partitions and ceilings to Level 5. Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for water-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

3.4.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to Level 5. Apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

3.4.2 Gypsum Board Finish Levels

3.4.2.1 Level 1

All joints and interior angles shall have tape set in joint compound. Surface shall be free of excess joint compound. Tool marks and ridges are acceptable.

3.4.2.2 Level 2

All joints and interior angles shall have tape embedded in joint compound and wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. Fastener heads and accessories shall be covered with a coat of joint compound. Surface shall be free of excess joint compound. Tool marks and ridges are acceptable. Joint compound applied over the body of the tape at the time of tape embedment shall be considered a separate coat of joint compound and shall satisfy the conditions of this level.

3.4.2.3 Level 3

All joints and interior angles shall have tape embedded in joint compound and shall be immediately wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. One additional coat of joint compound shall be applied over all joints and interior angles. Fastener heads and accessories shall be covered with two separate coats of joint compound. All joint compound shall be smooth and free of tool marks and ridges. Note: It is recommended that the prepared surface be coated with a drywall primer prior to the application of final finishes.

3.4.2.4 Level 4

All joints and interior angles shall have tape embedded in joint compound and shall be immediately wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. Two separate coats of joint compound shall be applied overall flat joints and one separate coat of joint compound shall be applied over interior angles. Fastener heads and accessories shall be covered with three separate coats of joint compound. All joint compound shall be smooth and free of tool marks and ridges. Note: It is recommended that the prepared surface be coated with a drywall primer prior to the application of final finishes.

3.4.2.5 Level 5

All joints and interior angles shall have tape embedded in joint compound and shall be immediately wiped with a joint knife leaving a thin coating of joint compound over all joints and interior angles. Two separate coats of joint compound shall be applied over all flat joints and one separate coat of joint compound shall be applied over interior angles. Fastener heads and accessories shall be covered with three separate coats of joint compound. A thin skim coat of joint compound trowel applied, or a material manufactured especially for this purpose and applied in accordance with manufacturer's recommendations, applied to the entire surface. The surface shall be smooth and free of tool marks and ridges.

Note: It is recommended that the prepared surface be coated with a drywall primer prior to the application of finish paint.

3.5 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified in Section 07 92 00 JOINT SEALANTS. Apply material with exposed surface flush with gypsum board or cementitious backer units.

~~[~~3.5.1 Sealing for Glass Mat or Reinforced Gypsum Board Sheathing

Apply silicone sealant in a 9.5 mm bead to all joints and trowel flat. Apply enough of the same sealant to all fasteners penetrating through the glass mat gypsum board surface to completely cover the penetration when troweled flat.~~ [Do not place construction and materials behind sheathing until a visual inspection of sealed joints during daylight hours has been completed by Contracting Officer.]~~

~~]~~3.6 FIRE-RESISTANT ASSEMBLIES

Wherever fire-rated construction is indicated, provide materials and application methods, including types and spacing of fasteners,~~[~~ wall and ceiling framing~~]~~ in accordance with the specifications contained in ~~[~~UL Fire Resistance for the Design Number(s) indicated~~]~~ or ~~[~~GA 600 for the File Number(s) indicated~~]~~. Seal penetrations through rated partitions and ceilings tight in accordance with tested systems.

3.7 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

3.8 SHAFTWALL FRAMING

Install the shaftwall system in accordance with the system manufacturer's published instructions. Coordinate bucks, anchors, blocking and other items placed in or behind shaftwall framing with electrical and mechanical work. Patch or replace fireproofing materials which are damaged or removed during shaftwall construction.

-- End of Section --

SECTION 09 30 10

CERAMIC~~, QUARRY, AND GLASS ~~TILING
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1509-3	(2014) Test Method for Ceramic Tiles - Part 3: Determination of Water Absorption, Apparent Porosity and Bulk Density
JIS A 1509-4	(2008) Test Methods for Ceramic Tiles - Part 4: Determination of Modulus of Rupture and Breaking strength
JIS A 1509-8	(2014) Test Methods for Ceramic Tiles - Part 8: Determination for crazing resistance for glazed tiles
JIS A 1509-9	(2014) Test Methods for Ceramic Tiles - Part 9: Determination for frost resistance
JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete
JIS A 5209	(2010) Ceramic Tiles
JIS A 5505	(2014) Metal Laths
JIS A 6022	(2010) Stretchy Asphalt Roofing Felts (Synthetic Fiber Base)
JIS A 6902	(1995) Plastering Lime
JIS G 3551	(2005) Welded Wire Mesh and Rebar Grid
JIS R 5210	(2009) Portland Cement

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT-SS Chapter 11	(2019) Public Building Construction Standard Specifications - Ch.11 Tile Work and Wood Work

1.2 SUBMITTALS

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~~Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G

SD-10 Operation and Maintenance Data

Installation; G

1.3 CERTIFICATIONS

1.3.1 Indoor Air Quality Certifications

1.3.1.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by providing the certification or validation by other third-party program that products meet the requirements of this Section. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

1.4 QUALITY ASSURANCE

Provide installers having a minimum of two years experience with a company specializing in performing the type of work described. Each type and color of tile to be provided from a single source. Each type and color of mortar, adhesive, and grout to be provided from the same source.

1.5 DELIVERY, STORAGE, AND HANDLING

Ship tiles in sealed packages and clearly marked with the grade, type of tile, producer identification, and country of origin. Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and hallmarks intact. Protect materials from weather, and store them under cover in accordance with manufacturer's printed instructions.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not perform ceramic tile work unless the substrate and ambient temperature is at least 10 degrees C and rising. Maintain temperature above 10 degrees C while the work is being performed and for at least 7 days after completion of the work. When temporary heaters are used, ventilate the area to the outside to avoid carbon dioxide damage to new tilework.

1.7 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a 1-year period.

1.8 EXTRA MATERIALS

Supply an extra 2 percent of each type tile used in clean and marked cartons.

PART 2 PRODUCTS

2.1 TILE

Provide tiles that comply with JIS A 5209 and are standard grade tiles. Provide a minimum breaking strength for floor tile in accordance with JIS A 1509-4. Provide exterior building tile for cold climate projects that is approved by the manufacturer for exterior use in accordance with JIS A 1509-9 and JIS A 5209. Provide floor tiles with a wet dynamic coefficient of friction value per JIS A 5209 requirements. Provide glazed floor tile with commercial rating by the manufacturer per JIS A 1509-8, for visible abrasion resistance as related to foot traffic. For materials like tile, accessories, and transition strips submit samples of sufficient size to show color range, pattern, type and joints. Submit manufacturer's catalog data. Tiles and tile works shall comply with MLIT-SS Chapter 11.

~~2.1.1 Porcelain Tile~~

~~Provide [[unglazed] [or] [glazed],] [rectified] porcelain tile, [cove] [bullnose] base and trim pieces[with color extending uniformly through the body of the tile]. [Provide tile with a [V0] [V1] [V2] [V3] [V4] aesthetic classification. Blend tiles in factory and in a packages to have same color range and continuous blend for installation.] Provide nominal tile size(s) of [150 by 150] [300 by 300] [450 by 450] [300 by 450] [_____] mm and [8] [10] [_____] mm thick. Provide a 0.50 percent maximum water absorption in accordance with JIS A 1509-3.~~

~~Provide Porcelain Tiling Materials that contain a minimum of 10 percent recycled content.~~

~~2.1.2 Quarry Tile~~

~~Furnish an unglazed quarry tile, [cove] [bullnose] base and trim pieces. Provide tile with [smooth] [abrasive] surface. Provide nominal tile size(s) of [150 by 150] [_____] mm and 13 mm thick. Provide maximum water absorption in accordance with JIS A 5209 or JIS A 1509-3.~~

~~Provide Quarry Tiling Materials that contain a minimum of 10 percent recycled content.~~

~~2.1.3 Mosaic Tile~~

~~Furnish [unglazed] [glazed], mosaic tile[, [cove] [bullnose] [base] and trim composed of [natural clay] [porcelain]. Blend tiles in factory and in a packages to have same color range and continuous blend for installation.] Provide [nominal tile size(s) of [25 by 25] [25 by 50] [50 by 50] [_____] mm][a mixture of standard sizes in a stock pattern]. [Provide porcelain mosaics with a water absorption up to 0.50 percent] [Provide natural clay mosaics with a water absorption up to [3.0] [_____] percent] per JIS A 1509-3.~~

~~Provide Mosaic Tiling Materials that contain a minimum of 3 percent recycled content.~~

~~2.1.4 Glass Tile~~

~~Furnish glass mosaic tile that requires thermal shock resistance and can withstand the range of temperatures outdoors. Provide nominal tile size(s) of [25 by 25] [_____] mm.~~

~~2.1.5 Glazed Wall Tile~~

~~Furnish glazed wall tile that has cushioned edges and trim with lead-free [bright] [matte] finish. Provide nominal tile size(s) of [106 by 106] [106 by 150] [150 by 150] mm.~~

~~2.1.6 Accessories~~

~~Provide built-in type accessories of the same materials and finish as the wall tile. Provide accessories as follows:~~

	~~Quantity~~	~~Location~~
~~Recessed soap holders~~	~~[_____]~~	~~[_____]~~
~~Tumbler holders~~	~~[_____]~~	~~[_____]~~
~~Combination tumbler and toothbrush holders~~	~~[_____]~~	~~[_____]~~
~~Towel bars, [stainless steel][ceramic] [600] [750] mm long, two towel posts~~	~~[_____]~~	~~[_____]~~
~~Robe hooks~~	~~[_____]~~	~~[_____]~~
~~Roll paper holder~~	~~[_____]~~	~~[_____]~~
~~Recessed soap holder and hand hold combination~~	~~[_____]~~	~~[_____]~~

2.1.1 Ceramic Floor Tile

Provide unglazed interior slip-resistant floor tile, cove base and trim pieces with color extending uniformly through the body of the tile. Blend tiles in factory and in packages to have some color range and continuous blend for installation. Provide nominal tile size(s) of 600 by 600 mm and thicknesses as indicated on drawings. Provide a 0.50 percent maximum water absorption in accordance with JIS A 1509-3.

Provide ceramic tiling materials that contain a minimum of 10 percent recycled content.

2.2 SETTING-BED

Submit manufacturer's catalog data. Compose the setting-bed of the following materials:

2.2.1 Aggregate for Concrete Fill

Conform to JIS A 5005, for aggregate fill. Do not exceed one-half the thickness of concrete fill for maximum size of coarse aggregate.

2.2.2 Portland Cement

Conform to JIS R 5210, for cement, Type I, white for wall mortar and gray for other uses.

2.2.3 Sand

Conform to JIS A 5005, for sand.

2.2.4 Hydrated Lime

Conform to JIS A 6902, for hydrated lime.

2.2.5 Metal Lath

Conform to JIS A 5505, for flat expanded type metal lath, and weighing a minimum 1.4 kg/square meter.

2.2.6 Reinforcing Wire Fabric

Conform to JIS G 3551 for wire fabric. Provide ~~[~~50 by 50 mm mesh, 16/16 wire~~] [or] [38 by 50 mm mesh, 16/13 wire]~~.

2.3 WATER

Provide potable water.

2.4 MORTAR, GROUT, AND ADHESIVE

~~[~~Provide ~~non-aerosol adhesive~~ products ~~used on the interior of the building (defined as inside of the weatherproofing system) meeting formaldehyde emission rate below 0.005 mg/m2h or~~ meeting requirements of F 4-Star and 4VOC~~-~~.~~]~~

2.4.1 Dry-Set Portland Cement Mortar

MLIT-SS Chapter 11 Tile Work.

2.4.2 Latex-Portland Cement Mortar

MLIT-SS Chapter 11 Tile Work.

2.4.3 Ceramic Tile Grout

MLIT-SS Chapter 11 Tile Work; petroleum-free and plastic-free ~~[sand portland cement grout] [dry-set grout] [~~latex-portland cement grout~~] [commercial portland cement grout]~~.

~~2.4.4 Organic Adhesive~~

~~MLIT-SS Chapter 11 Tile Work. Comply with JIS A 5557 for exterior tile and JIS A 5548 for interior tile.~~

2.4.4 Sealants

Comply with applicable regulations regarding toxic and hazardous materials and as specified. Grout sealant must not change the color or alter the appearance of the grout. Refer to Section 07 92 00 JOINT SEALANTS.

[Provide~~ sealants used on the interior of the building (defined as inside of the weatherproofing system) meeting formaldehyde emissions rate below 0.005 mg/m2h or~~ products that meet the requirements of F 4-Star and 4VOC.

2.5 SUBSTRATES

2.5.1 Cementitious Backer Board

Provide cementitious backer units, for use as tile substrate over wood sub-floors, in accordance with MLIT-SS Chapter 11 Tile Work. Furnish ~~[6.35] [~~12.7~~]~~ mm thick cementitious backer units.

~~2.5.2 Glass Mat Gypsum Backer Panel~~

~~Provide glass mat water-resistant gypsum backer board, for use as tile substrate over wood subfloors, in accordance with JIS A 5403. Provide [6.35][12.7] mm thick glass mat gypsum backer board.~~

~~2.6 TRANSITION STRIPS~~

~~Provide [[clear] [_____] anodized aluminum transitions between tile and carpet or resilient flooring. Provide types as recommended by flooring manufacturer for both edges and transitions of flooring materials specified] [marble transitions appropriate for conditions].~~

2.6 MEMBRANE MATERIALS

Conform to JIS A 6022, Type 1 for 33 kg waterproofing membrane, asphalt-saturated building felt.

2.7 COLOR, TEXTURE, AND PATTERN

Provide color, pattern and texture ~~in accordance with [Section 09 06 00 SCHEDULES FOR FINISHES][~~as indicated~~][[_____]~~ in the drawings. Color listed is not intended to limit the selection of equal colors from other manufacturers~~]~~.~~[~~ Provide floor patterns as specified on the drawings.~~]~~

PART 3 EXECUTION

3.1 PREPARATORY WORK AND WORKMANSHIP

Inspect surface to receive tile in conformance to the requirements of MLIT-SS Chapter 11 Tile Work for surface conditions for the type setting bed specified and for workmanship. Provide variations of tiled surfaces that fall within maximum values shown below:

TYPE	WALLS	FLOORS
Polymer-Fortified Portland Cement	3 mm in 2.4 meter	3.0 mm in 3 meter
Epoxy	3 mm in 2.4 meter	3.0 mm in 3 meter

3.2 GENERAL INSTALLATION REQUIREMENTS

Do not start tile work until roughing in for mechanical and electrical work has been completed and tested, and built-in items requiring membrane waterproofing have been installed and tested. Close space, in which tile is being set, to traffic and other work. Keep closed until tile is firmly set. Do not start floor tile installation in spaces requiring wall tile until after wall tile has been installed. Apply tile in colors and patterns indicated in the area shown on the drawings. Install tile with the respective surfaces in true even planes to the elevations and grades shown. Provide special shapes as required for sills, jambs, recesses, offsets, external corners, and other conditions to provide a complete and neatly finished installation. Solidly back tile bases and coves with mortar. Do not walk or work on newly tiled floors without using kneeling boards or equivalent protection of the tiled surface. Keep traffic off horizontal portland cement mortar installations for at least 72 hours. Keep all traffic off epoxy installed floors for at least 40 hours after grouting, and heavy traffic off for at least 7 days, unless otherwise specifically authorized by manufacturer. ~~Dimension and draw detail drawings at a minimum scale of 6 mm = 300 mm. Include drawings of pattern at inside corners, outside corners, termination points and location of all equipment items such as thermostats, switch plates, mirrors and toilet accessories mounted on surface.~~ Submit detail drawings showing ceramic tile pattern ~~[~~elevations~~][~~ and ~~][~~floor plans~~]~~. Submit manufacturer's preprinted installation instructions.

Do not install building construction materials that show visual evidence of biological growth.

3.3 INSTALLATION OF WALL TILE

Install wall tile in accordance with the MLIT-SS Chapter 11 Tile Work, method ~~[_____]~~ and with grout joints ~~[[~~as recommended by the manufacturer for the type of tile~~][of [_____] mm]~~. ~~[~~Install thinner wall tile flush with thicker wall tile applied on same wall and provide installation materials as recommended by the tile and setting materials manufacturer's to achieve flush installation.~~]]~~

3.3.1 Workable or Cured Mortar Bed

Install tile over workable mortar bed or a cured mortar bed at the option of the Contractor. Install a 0.102 mm polyethylene membrane, metal lath, and scratch coat.

~~3.3.2 Dry-Set Mortar and Latex-Portland Cement Mortar~~

~~Use [Dry-set] [or] [Latex-Portland Cement] to install tile in accordance with MLIT-SS Chapter 11 Tile Work. Use Latex Portland Cement when installing porcelain ceramic tile.~~

~~3.3.3 Organic Adhesive~~

~~Conform to JIS A 5557 for exterior tile and JIS A 5548 for interior tile for the organic adhesive installation of ceramic tile.~~

3.3.2 Polymer-Fortified Portland Cement Mortar

Use Polymer-Fortified Partland Mortar to install tile in accordance with MLIT-SS Chapter 11 Tile Work. Use Polymer-Fortified Portland Mortar when installing ceramic tile.

3.3.3 Ceramic Tile Grout

Prepare and install ceramic tile grout in accordance with MLIT-SS Chapter 11 Tile Work. ~~[~~Provide and apply manufacturer's standard ~~[_____]~~ product for sealing grout joints in accordance with manufacturer's recommendations.~~]~~

3.4 INSTALLATION OF FLOOR TILE

Install floor tile in accordance with MLIT-SS Chapter 11 Tile Work method ~~[_____]~~ and with grout joints ~~[~~as recommended by the manufacturer for the type of tile~~][of [_____] mm]~~. Install shower receptors in accordance with manufacturer recommendations.

3.4.1 Workable or Cured Mortar Bed

Install floor tile over a workable mortar bed or a cured mortar bed at the option of the Contractor. Install in accordance with MLIT-SS Chapter 11 Tile Work. Provide minimum 6.35 mm to maximum 9.53 mm.

~~3.4.2 Dry-Set and Latex-Portland Cement~~

~~Use [dry-set] [or] [Latex-Portland cement] mortar to install tile directly over properly cured, plane, clean concrete slabs in accordance with Install in accordance with MLIT-SS Chapter 11 Tile Work. Use Latex Portland cement when installing porcelain ceramic tile.~~

3.4.2 Polymer-Fortified Portland Cement

Use Polymer-Fortified Portland mortar to install tile directly over properly cured, plane, clean concrete slabs in accordance with Install in accordance with MLIT-SS Chapter 11 Tile Work. Use Polymer-Fortified Portland Mortar cement when installing ceramic tile.

3.4.3 Resinous Grout

When resinous grout is indicated, grout quarry tile with either furan or epoxy resin grout. Rake and clean joints to the full depth of the tile and neutralize when recommended by the resin manufacturer. Install epoxy resin grout in conformance with MLIT-SS Chapter 11 Tile Work. Install resin grout in accordance with manufacturer's printed installation instructions. Provide a coating of wax applied from the manufacturer on all tile installed and furan resin. Follow manufacturer's printed installation instructions of installed resin grout for proportioning, mixing, installing, and curing. Maintain the recommended temperature in the area and on the surface to be grouted. Protect finished grout of grout stain.

3.4.4 Ceramic Tile Grout

Prepare and install ceramic tile grout in accordance with MLIT-SS Chapter 11 Tile Work. Provide and apply manufacturer's standard product for sealing grout joints in accordance with manufacturer's recommendations.

3.4.5 Waterproofing

Shower pans are specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE. ~~Conform to the requirements of Section 07 12 00 BUILT-UP BITUMINOUS~~

~~WATERPROOFING for waterproofing under concrete fill.~~

3.4.6 Concrete Fill

Provide a 24.1 MPa concrete fill mix to dry as consistency as practicable. ~~[~~Compose concrete fill by volume of 1 part Portland cement to 3 parts fine aggregate to 4 parts coarse aggregate, and mix with water to as dry a consistency as practicable.~~]~~ Spread, tamp, and screed concrete fill to a true plane, and pitch to drains or levels as shown. Thoroughly damp concrete fill before applying setting-bed material. Reinforce concrete fill with one layer of reinforcement, with the uncut edges lapped the width of one mesh and the cut ends and edges lapped a minimum 50 mm. Tie laps together with 1.3 mm wire every 250 mm along the finished edges and every 150 mm along the cut ends and edges. Provide reinforcement with support and secure in the centers of concrete fills. Provide a continuous mesh; except where expansion joints occur, cut mesh and discontinue across such joints. Provide reinforced concrete fill under the setting-bed where the distance between the under-floor surface and the finished tiles floor surface is a minimum of 50 mm, and of the same thickness that the mortar setting-bed over the concrete fill with the thickness required.

~~3.5 INSTALLATION OF TRANSITION STRIPS~~

~~Install transition strips where indicated, in a manner similar to that of the ceramic tile floor and as recommended by the manufacturer. Provide thresholds full width of the opening. Install head joints at ends not exceeding 6 mm in width and grouted full.~~

3.5 EXPANSION JOINTS

Form and seal joints as specified in Section 07 92 00 JOINT SEALANTS.

3.5.1 Walls

Provide expansion joints at control joints in backing material. Wherever backing material changes, install an expansion joint to separate the different materials.

3.5.2 Floors

Provide expansion joints over construction joints, control joints, and expansion joints in concrete slabs. Provide expansion joints where tile abuts restraining surfaces such as perimeter walls, curbs and columns and at intervals of 7 to 11 m each way in large interior floor areas and 3 to 5 m each way in large exterior areas or areas exposed to direct sunlight or moisture. Extend expansion joints through setting-beds and fill.

3.6 CLEANING AND PROTECTING

Upon completion, thoroughly clean tile surfaces in accordance with manufacturer's approved cleaning instructions. Do not use acid for cleaning glazed tile. Clean floor tile with resinous grout or with factory mixed grout in accordance with printed instructions of the grout manufacturer. After the grout has set, provide a protective coat of a noncorrosive soap or other approved method of protection for tile wall surfaces. Cover tiled floor areas with building paper before foot traffic is permitted over the finished tile floors. Provide board walkways on tiled floors that are to be continuously used as passageways by workmen.

Replace damaged or defective tiles. Submit copy of manufacturer's printed maintenance instructions.

-- End of Section --

SECTION 09 51 00

ACOUSTICAL CEILINGS
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E580/E580M	(2020) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions

JAPANESE ARCHITECTURAL STANDARD SPECIFICAITONS(JASS)

JASS 26	(2006) Interior Work

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1409	(1998) Method for Measurement of Sound Absorption Coefficients in a Reverberation Room
JIS A 1445	(2007) Testing Methods of Materials for Ceiling Suspension System
JIS A 5758	(2016) Sealants for Sealing and Glazing in Buildings
JIS A 6301	(2007) Sound Absorbing Materials
JIS A 6517	(2010) Steel Furrings for Wall and Ceiling in Buildings
JIS B 1168	(1994) Eyebolts
JIS G 4309	(2013) Stainless Steel Wires
JIS G 5111	(1991) High Tensile Strength Carbon Steel castings and low alloy steel castings for structural purposes
JIS H 8610	(1999) Electroplated-Coatings of Zinc on Iron or Steel

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT Notification No. 771	(2013) Establishment of specified ceilings and a construction method that is

	effective for structural resistance of specified ceilings

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-310-04	(2013; with Change 1, 2016) Seismic Design of Buildings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Approved Detail Drawings; G~~[, [_____]]~~

~~[~~1.3 ADHESIVES AND SEALANTS

Provide products certified to meet indoor air quality requirements by providing the certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

~~]~~1.4 DELIVERY, STORAGE. AND HANDLING

Deliver materials to the site in the manufacturer's original unopened containers with brand name and type clearly marked. Carefully handle and store materials in dry, watertight enclosures. Immediately before installation, store acoustical units for not less than 24 hours at the same temperature and relative humidity as the space where they will be installed in order to assure proper temperature and moisture acclimation.

1.5 ENVIRONMENTAL REQUIREMENTS

Maintain a uniform temperature of not less than 16 degrees C nor more than 29 degrees C and a relative humidity of not more than 70 percent for 24 hours before, during, and 24 hours after installation of acoustical units.

1.6 SCHEDULING

Complete and dry interior finish work such as plastering, concrete and terrazzo work before ceiling installation. Complete mechanical, electrical, and other work above the ceiling line; install and start operating heating, ventilating, and air conditioning systems in order to maintain temperature and humidity requirements.

1.7 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a one year period. Include an agreement to repair or replace acoustical panels that fail within the warranty period in the standard performance guarantee or warranty. Failures include, but are not limited to, sagging and warping of panels; rusting and manufacturers defects of grid system.

1.8 EXTRA MATERIALS

Furnish spare tiles, from the same lot as those installed, of each color at the rate of 5 tiles for each 1000 tiles installed.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide sound controlling units mechanically mounted on a ceiling suspension system for acoustical treatment. The unit size, texture, finish, and color must be as specified. Coordinate the whole ceiling system with other details, like the location of access panels and ceiling penetrations, etc., shown on the drawings. The Contractor is responsible for all associated labor and materials and for the final assembly and performance of the specified work and products are used. The location and extent of acoustical treatment must be as shown on the approved detail drawings. Submit drawings showing suspension system, method of anchoring and fastening, details, and reflected ceiling plan. Coordinate with paragraph RECLAMATION PROCEDURES for reclamation of mineral fiber acoustical ceiling panels to be removed from the job site.

~~2.1.1 Fire Resistive Ceilings~~

~~Rate acoustical ceiling systems, indicated as fire resistant, for fire endurance as specified when tested in accordance with ASTM E119. Test suspended ceiling with a specimen [roof][floor] assembly representative of the indicated construction, including mechanical and electrical work within ceiling space openings for light fixtures, and air outlets, and access panels. Provide ceiling assembly rating for [[1][1-1/2][2][3][4] hour [concealed grid system][exposed grid system]][as shown on drawings]. Provide acoustical units with a flame spread of 25 or less and smoke development of 50 or less when tested in accordance with ASTM E84.~~

2.1.1 Ceiling Attenuation Class and Test

Provide a ceiling system with an attenuation class (CAC) ~~of [36] for [_____] [and _____ for _____]~~as indicated. Provide fixture attenuators over light fixtures and other ceiling penetrations, and provide acoustical blanket insulation adjacent to partitions, as required to achieve the specified CAC. Provide test ceiling continuous at the partition and assembled in the suspension system in the same manner that the ceiling will be installed on the project.

2.1.2 Ceiling Sound Absorption

Determine the sound reduction rate in accordance with JIS A 1409.

2.1.3 Light Reflectance

Light reflectance to be 0.75 or above.

2.2 ACOUSTICAL UNITS

Submit two samples of each type of acoustical unit and each type of suspension grid tee section showing texture, finish, and color. Conform acoustical units to JIS A 6301 and the following requirements:

2.2.1 Units for Exposed-Grid System~~ [A] [_____]~~

2.2.1.1 Type

~~[~~III (non-asbestos mineral fiber with painted finish).~~ Provide Type III Acoustical Ceiling Tiles containing a minimum of 30 percent recycled content.]~~

~~[IV (non-asbestos mineral fiber with membrane-faced overlay). Provide Type IV Acoustical Ceiling Tiles containing a minimum of 60 percent recycled content.]~~

~~[IX (mineral fiber with scrubbable finish). [Provide Type IX Acoustical Ceiling Tiles containing a minimum [50] [_____] percent recycled content.]]~~

~~[X (mineral composition with plastic membrane).]~~

~~[XI (mineral fiber with fabric faced overlay).]~~

~~[XII (fiberglass base with membrane-faced overlay). [Provide Type XII Acoustical Ceiling Tiles containing a minimum of [25] [_____] percent recycled content.]]~~

2.2.1.2 Flame Spread

Class A, 25 or less

2.2.1.3 Pattern

~~[A] [B] [C] [D] [E] [F] [G] [I] [J] [K] [_____]~~Smooth finish

2.2.1.4 Minimum NRC

~~[~~0.7~~5~~0~~] [_____] in open office areas; [0.60] [_____] in conference rooms, executive offices, teleconferencing rooms, and other rooms as designated; [0.50] [_____] in all other rooms and areas~~.

2.2.1.5 Minimum Light Reflectance Coefficient
~~[LR-1, ~~0.75 or greater~~] [_____]~~

2.2.1.6 Nominal Size

~~[~~600 by 1200~~]~~ ~~[_____]~~ mm

2.2.1.7 Edge Detail

~~[Square]~~Beveled ~~[~~Reveal~~] [Trimmed and butt] [_____]~~

2.2.1.8 Finish

Factory-applied ~~[~~standard paint finish~~] [color finish]~~.

2.2.1.9 Minimum CAC

~~[40] [~~36~~]~~

~~2.2.2 Units for Concealed-Grid System [A] [_____]~~

~~2.2.2.1 Type~~

~~[III (non-asbestos mineral fiber with painted finish). Provide Type III Acoustical Ceiling Tiles containing a minimum of 30 percent recycled content.]~~

~~[IV (non-asbestos mineral fiber with membrane-faced overlay). Provide Type IV Acoustical Ceiling Tiles containing a minimum of 60 percent recycled content.]~~

~~[IX (mineral fiber with scrubbable finish). [Provide Type IX Acoustical Ceiling Tiles containing a minimum of [50][_____] percent recycled content.]]~~

~~[X (mineral composition with plastic membrane).]~~

~~[XI (mineral fiber with fabric faced overlay).]~~

~~[XII (fiberglass base with membrane-faced overlay). [Provide Type XII Acoustical Ceiling Tiles containing a minimum of [25][_____] percent recycled content.]]~~

~~2.2.2.2 Flame Spread~~

~~Class A, 25 or less~~

~~2.2.2.3 Pattern~~

~~[A] [B] [C] [D] [E] [F] [G] [I] [J] [K] [_____]~~

~~2.2.2.4 Minimum NRC~~

~~[0.50] [_____]~~

~~2.2.2.5 Minimum Light Reflectance Coefficient~~

~~[LR-1, 0.75 or greater] [_____]~~

~~2.2.2.6 Nominal Size~~

~~[300 by 300] [_____] mm~~

~~2.2.2.7 Edge Detail~~

~~[Beveled] [Square]~~

~~2.2.2.8 Joint Detail~~

~~[kerfed and rabbeted] [tongue and grooved]~~

~~2.2.2.9 Finish~~

~~Factory-applied [standard finish] [color finish]~~

~~2.2.2.10 Minimum CAC~~

~~[40] [_____]~~

~~2.2.3 Metal Pans [A] [_____]~~

~~2.2.3.1 Type~~

~~[V, steel.]~~

~~[VI, stainless steel.]~~

~~[VII, aluminum perforated pans with acoustical, non-asbestos, insulation backing.]~~

~~2.2.3.2 Flame Spread~~

~~Class: A, 25 or less~~

~~2.2.3.3 Pattern~~

~~[A] [C] [I] [_____]~~

~~2.2.3.4 Minimum NRC~~

~~[0.75] [_____] in open office areas; [0.60] [_____] in conference rooms, executive offices, teleconferencing rooms, and other rooms as designated; [0.50] [_____] in all other rooms.~~

~~2.2.3.5 Minimum Light Reflectance Coefficient~~

~~[LR-1, 0.75 or greater] [_____]~~

~~2.2.3.6 Nominal Size~~

~~[600 by 600] [_____] mm~~

~~2.2.3.7 Edge Detail~~

~~Manufacturer's standard.~~

~~2.2.3.8 Joint Detail~~

~~[Beveled] [_____]~~

~~2.2.3.9 Finish~~

~~Factory-applied standard finish~~

~~2.2.3.10 Pads~~

~~[Completely enclosed, of material and thickness required for acoustical and fire test ratings] [_____].~~

~~2.2.4 Impact/Abrasion Resistant Units~~

~~2.2.4.1 Type~~

~~Non-asbestos mineral composition with a hardened mineral surface and factory applied white paint finish. Provide a surface resistant to impact and abrasion.~~

~~2.2.4.2 Flame Spread~~

~~Class A, 25 or less~~

~~2.2.4.3 Pattern~~

~~[_____]~~

~~2.2.4.4 Minimum NRC~~

~~[0.50] [_____].~~

~~2.2.4.5 Minimum Light Reflectance Coefficient~~

~~LR-1, 0.75 or greater~~

~~2.2.4.6 Nominal Size~~

~~[300 by 300] [600 by 600] [600 by 1200] mm~~

~~2.2.4.7 Edge Detail~~

~~[Square] [Beveled]~~

~~2.2.4.8 Joint Detail~~

~~[Trimmed and butted] [Kerfed and rabbeted]~~

~~2.2.5 Humidity Resistant Composition Units~~

~~2.2.5.1 Type~~

~~Non-asbestos mineral or glass fibers bonded with ceramic, moisture resistant thermo-setting resin, or other moisture resistant material and having a factory applied white paint finish. Provide panels that do not sag or warp under conditions of heat, high humidity or chemical fumes.~~

~~2.2.5.2 Flame Spread~~

~~Class: A, 25 or less~~

~~2.2.5.3 Pattern~~

~~[_____]~~

~~2.2.5.4 Minimum NRC~~

~~Minimum [0.50] [_____].~~

~~2.2.5.5 Minimum Light Reflectance Coefficient~~

~~LR-1, 0.75 or greater~~

~~2.2.5.6 Nominal Size~~

~~[600 by 1200] [_____] mm~~

~~2.2.5.7 Edge Detail~~

~~Square~~

~~2.2.6 Metal Faced Composition Units~~

~~2.2.6.1 Type~~

~~[Type V (Steel facings with non-asbestos mineral composition absorbent backing).]~~

~~[Type VI (Stainless steel facings with non-asbestos mineral composition absorbent backing)]~~

~~[Type VII (Aluminum facings with non-asbestos mineral composition absorbent backing) with [anodized] [baked enamel] [acrylic] finish color [white] [_____].]~~

~~2.2.6.2 Flame Spread~~

~~Class: A, flame spread 25 or less~~

~~2.2.6.3 Pattern~~

~~[_____]~~

~~2.2.6.4 Minimum (NRC)~~

~~[0.75] [_____] in open office areas. [0.60] [_____] in conference rooms, executive offices, teleconferencing rooms, and other rooms as designated. [0.50] [_____] in all other rooms and areas.~~

~~2.2.6.5 Minimum Light Reflectance Coefficient~~

~~LR-1, 0.75 or greater~~

~~2.2.6.6 Nominal Size~~

~~600 by [600] [1200] mm~~

~~2.2.6.7 Edge Detail~~

~~Square~~

~~2.2.6.8 Joint Detail~~

~~Trimmed and butted~~

~~2.2.7 Unit Acoustical Absorbers~~

~~Absorbers must be individually mounted sound absorbing plaques composed of~~

~~glass fibers or non-asbestos mineral fibers and having a NRC range of not less than 0.60 - 0.70 when tested in accordance with JIS A 1409 and reported as a 4 frequency average.~~

2.2.2 Acoustical Ceiling

2.2.2.1 Type

Rock wool acoustical ceiling.

2.2.2.2 Flame Resistance

Non-flammable.

2.2.2.3 Pattern

Flat travertine.

2.2.2.4 Minimum NRC

0.51.

2.2.2.5 Nominal Size

303 by 606 mm, 12 mm thick

2.2.2.6 Edge Details

Concealed spline.

2.2.2.7 Finish

Factory-applied standard white.

2.3 SUSPENSION SYSTEM

Provide ~~[[~~standard~~] [fire-resistive] [snap-in metal pan]] [[~~ exposed-grid~~] [indirect hung concealed H and T or Zee] [direct hung, concealed, downward access] [direct hung, concealed, upward access]] [[standard width flange] [narrow width flange] [narrow width slotted flange]] [as shown on drawings]~~ suspension system conforming to JIS A 1445~~ [for intermediate-duty systems] [for heavy-duty systems]~~. Provide surfaces exposed to view of ~~[aluminum or [~~galvanized ~~]~~steel with a factory-applied ~~[~~ white~~] [black] [color]~~ baked-enamel finish~~] [aluminum with a clear anodized finish] [aluminum with colored factory-applied vinyl paint finish]~~. Provide wall molding having a flange of not less than ~~[~~23 mm~~] [_____]~~. ~~Provide [inside and outside corner caps] [[standard] [overlapped] [mitered] corners].~~ Suspended ceiling framing system must have the capability to support the finished ceiling, light fixtures, air diffusers, and accessories, as shown. Provide a suspension system with a maximum deflection of 1/360 of the span length. Conform seismic details to the ~~[~~ guidance in UFC 3-310-04 and ASTM E580/E580M or JIS A 1445~~] [contract drawings]~~, and in accordance with "Practical Guide on the Technical Standards concerning Measures to Prevent the Fall of Buildings" based on MLIT Notification No. 771 of the Ministry of Land, Infrastructure, Transport and Tourism.

For gypsum board ceilings greater than 5994mm AFF, with an area greater than 200 SM, and weight of 2 kg/SM, refer to the detail in the Ministry of

Land, Infrastructure, Transport and Tourism, MLIT Notification No. 771, and JIS A 1445 and JIS A 6517 for testing method. For gypsum board ceilings less than or equal to 5994mm AFF, refer to detail in JASS 26 Section 4, and JIS A 6517 for testing method. For lay-in acoustical ceilings refer to detail in JASS 26, Section 4.4 and JIS A 1445 for testing method.

~~Provide Suspension System containing a minimum of 15 percent recycled content.~~

2.4 HANGERS

Provide hangers and attachment capable of supporting a minimum 1330 N ultimate vertical load without failure of supporting material or attachment.

2.4.1 Wires

~~Conform wires to JIS G 3537, [~~JIS G 4309 condition annealed stainless steel, ~~[~~2.0~~]~~ ~~[_____]~~ mm in diameter.~~]~~

2.4.2 Straps

Provide straps of 25 by 5 mm galvanized steel with a light commercial zinc coating or JIS G 5111 with an electrodeposited zinc coating conforming to JIS H 8610.

2.4.3 Rods

Provide 5 mm diameter threaded steel rods, zinc or cadmium coated.

2.4.4 Eyebolts

Provide eyebolts of weldless, forged-carbon-steel, with a straight-shank in accordance with JIS B 1168. Eyebolt size must be a minimum ~~[_____] [~~7~~]~~ mm, ~~[~~zinc coated~~][cadmium plated]~~.

2.4.5 Anchorage Devices

Comply with JIS A 1445 for anchorage devices for ~~[~~eyebolts~~] [machine screws] [wood screws]~~. Where aluminum is in contact with concrete, coat aluminum with bituminous paint or where exposed, with a chromatic primer and 2-coats of enamel paint.

2.5 ADHESIVE

Use adhesive as recommended by tile manufacturer.

2.6 ACCESS PANELS

Provide access panels that match adjacent acoustical units, designed and equipped with suitable framing and fastenings for removal and replacement without damage. Size panel to be not less than 300 by 300 mm or more than 300 by 600 mm.

a. Attach an identification plate of 0.8 mm thick aluminum, 19 mm in diameter, stamped with the letters "AP" and finished the same as the unit, near one corner on the face of each access panel.

b. Identify ceiling access panel by a number utilizing white identification plates or plastic buttons with contrasting numerals. Provide plates or buttons of minimum 25 mm diameter and securely attached to one corner of each access unit. Provide a typewritten card framed under glass listing the code identification numbers and corresponding system descriptions listed above. Mount the framed card where directed and furnish a duplicate card to the Contracting Officer. Code identification system is as follows:

1 Fire detection/alarm system

2 Air conditioning controls

3 Plumbing system

4 Heating and steam systems

5 Air conditioning duct system

6 Sprinkler system

7 Intercommunication system

~~8 Nurse's call system~~

~~9 Pneumatic tube system~~

~~10 Medical piping system~~

~~11 Program entertainment~~

~~12~~8 Telephone junction boxes

~~13 Detector X-ray~~

~~14 [_____]~~

2.7 ADHESIVE

Use adhesive as recommended by tile manufacturer. Meet emissions requirements of F 4-STar and JAIA 4VOC.

2.8 FINISHES

Use manufacturer's standard textures, patterns and finishes as specified for acoustical units and suspension system members. Treat ceiling suspension system components to inhibit corrosion.

2.9 COLORS AND PATTERNS

Use colors and patterns for acoustical units and suspension system components as indicated.

2.10 ACOUSTICAL SEALANT

Conform acoustical sealant to JIS A 5758, nonstaining.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Examine surfaces to receive directly attached acoustical units for unevenness, irregularities, and dampness that would affect quality and execution of the work. Rid areas, where acoustical units will be cemented, of oils, form residue, or other materials that reduce bonding capabilities of the adhesive. Complete and dry interior finish work such as plastering, concrete, and terrazzo work before installation. Complete and approve mechanical, electrical, and other work above the ceiling line prior to the start of acoustical ceiling installation. Provide acoustical work complete with necessary fastenings, clips, and other accessories required for a complete installation. Do not expose mechanical fastenings in the finished work. Lay out hangers for each individual room or space. Provide hangers to support framing around beams, ducts, columns, grilles, and other penetrations through ceilings. Keep main runners and carrying channels clear of abutting walls and partitions. Provide at least two main runners for each ceiling span. Wherever required to bypass an object with the hanger wires, install a subsuspension system so that all hanger wires will be plumb.

3.1.1 Suspension System

Install suspension system in accordance with JIS A 1445 and in accordance with "Practical Guide on the Technical Standards concerning Measures to Prevent the Fall of Buildings" based on MLIT Notification No. 771 of the Ministry of Land, Infrastructure, Transport and Tourism and as specified herein. The suspension system shall conform to allowable stress performance per the static pressurization test designed by the Japan Ministry of Construction. Do not suspend hanger wires or other loads from underside of steel decking.

3.1.1.1 Plumb Hangers

Install hangers plumb and not pressing against insulation covering ducts and pipes. Where lighting fixtures are supported from the suspended ceiling system, provide hangers at a minimum of four hangers per fixture and located not more than 150 mm from each corner of each fixture.

3.1.1.2 Splayed Hangers

Where hangers must be splayed (sloped or slanted) around obstructions, offset the resulting horizontal force by bracing, countersplaying, or other acceptable means.

3.1.2 Wall Molding

Provide wall molding where ceilings abut vertical surfaces. Miter corners where wall moldings intersect or install corner caps. Secure wall molding not more than 75 mm from ends of each length and not more than 400 mm on centers between end fastenings. Provide wall molding springs at each acoustical unit in semi-exposed or concealed systems.

3.1.3 Acoustical Units

Install acoustical units in accordance with the approved installation instructions of the manufacturer. Ensure that edges of acoustical units are in close contact with metal supports, with each other, and in true alignment. Arrange acoustical units so that units less than one-half width are minimized. Hold units in exposed-grid system in place with manufacturer's standard hold-down clips, if units weigh less than 5 kg/square meter or if required for fire resistance rating.

3.1.4 Caulking

Seal all joints around pipes, ducts or electrical outlets penetrating the ceiling. Apply a continuous ribbon of acoustical sealant on vertical web of wall or edge moldings.

3.1.5 Adhesive Application

Wipe back of tile to remove accumulated dust. Daub acoustical units on back side with four equal daubs of adhesive. Apply daubs near corners of tiles. Ensure that contact area of each daub is at least 50 mm diameter in final position. Press units into place, aligning joints and abutting units tight and uniform without differences in joint widths.

3.2 CEILING ACCESS PANELS

Locate ceiling access panels directly under the items which require access.

3.3 CLEANING

Following installation, clean dirty or discolored surfaces of acoustical units and leave them free from defects. Remove units that are damaged or improperly installed and provide new units as directed.

3.4 RECLAMATION PROCEDURES

Neatly stack ceiling tile, designated for recycling by the Contracting Officer, on 1220 by 1220 mm pallets not higher than 1220 mm. Panels must be completely dry. Shrink wrap and symmetrically stack pallets on top of each other without falling over.

-- End of Section --

SECTION 09 65 00

RESILIENT FLOORING
08/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E648 (2019a) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 5536 (2015) Adhesives for Resilient Textile or Laminate Floor Coverings

JIS A 5705 (2010) Polyvinyl Chloride Floorcoverings

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 19 (2019) Public Building Construction Standard Specifications Chapter 19: Interior Finishing Works

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Resilient Flooring and Accessories; G~~[, [_____]]~~

SD-10 Operation and Maintenance Data

Resilient Flooring and Accessories; G~~[, [_____]]~~

1.3 JAPAN PUBLIC BUILDING CONSTRUCTION STANDARDS

Follow MLIT SS Chapter 19 for interior finishing works.

1.4 INDOOR AIR QUALITY

1.4.1 Floor Covering Materials

Provide ~~[~~Vinyl Composition Tile~~][Sheet Vinyl Flooring][Rubber Tile][Rubber Sheet Flooring][Luxury Vinyl Tile][Solid Vinyl Tile][Sheet Linoleum][Linoleum Tile][Cork Flooring]~~, and wall base products certified to meet indoor air quality requirements of third-party programs.

1.4.1.1 Adhesives, Caulking and Sealants

Provide products certified to meet emission requirement of F 4-Star and JAIA 4VOC.

1.5 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the building site in original unopened containers bearing the manufacturer's name, style name, pattern color name and number, production run, project identification, and handling instructions. Store materials in a clean, dry, secure, and well-ventilated area free from strong contaminant sources and residues with ambient air temperature maintained above 20 degrees C and below 30 degrees C, stacked according to manufacturer's recommendations. Remove resilient flooring products from packaging to allow ventilation prior to installation. Protect materials from the direct flow of heat from hot-air registers, radiators and other heating fixtures and appliances. Observe ventilation and safety procedures specified in the MSDS. Do not store rubber surface products with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions~~, including [_____]~~. Do not store exposed rubber surface materials in occupied spaces. ~~[~~Do not store ~~[_____]~~ near materials that may off gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives.~~]~~

1.6 ENVIRONMENTAL REQUIREMENTS

Maintain areas to receive resilient flooring at a temperature above 20 degrees C and below 30 degrees C for 3 days before application, during application and 2 days after application, unless otherwise directed by the flooring manufacturer for the flooring being installed. Maintain a minimum temperature of 13 degrees C thereafter. Provide adequate ventilation to remove moisture from area and to comply with regulations limiting concentrations of hazardous vapors.

1.7 SCHEDULING

Schedule resilient flooring application after the completion of other work which would damage the finished surface of the flooring.

1.8 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a one year period.

1.9 EXTRA MATERIALS

Provide extra flooring material of each color and pattern at the rate of 5 tiles for each 1000 tiles. Provide extra wall base material composed of 6 m of each type, color and pattern. Package all extra materials in original properly marked containers bearing the manufacturer's name, brand name, pattern color name and number, production run, and handling instructions. Provide extra materials from the same lot as those installed. Leave extra stock at the site in location assigned by Contracting Officer.

PART 2 PRODUCTS

2.1 VINYL COMPOSITION TILE~~ [TYPE [A][_____]]~~

Conform to JIS A 5705, asbestos-free, ~~[~~300~~]~~ ~~[_____] ~~mm square and ~~[2.4] [~~ 3.2~~]~~ mm thick. Provide color and pattern uniformly distributed throughout the thickness of the tile.

~~[Provide Vinyl Composition Tile containing a minimum of 10 percent recycled content.]~~

~~2.2 SHEET VINYL FLOORING [TYPE [A] [_____]]~~

~~Conform to JIS A 5705 and a minimum [1800 mm] [3660 mm] wide. Extend color and pattern through the total thickness of the material. As required, provide welding rods as recommended by the manufacturer for heat welding of joints.~~

2.2 LUXURY VINYL TILE~~ [TYPE [A][_____]]~~

Conform to JIS A 5705 printed film with a minimum wear layer thickness ~~[~~ 0.50 mm (20 mil)~~][0.70 mm (30 mil)][_____]~~ and minimum overall thickness ~~[[~~2.5 mm~~] [or] [3 mm]] [5 mm[~~ with non slip/skid backing~~]]~~. Provide ~~[[300 by 600] [_____] mm][[300] [400] [450] [600] [900] [_____] mm square][_____]~~ tile size as indicated in the drawings~~ tile~~.~~[~~ Provide tile with a factory protective finish that enhances cleanability and durability. ~~]~~

~~2.3 SOLID VINYL TILE [TYPE [A][_____]]~~

~~Conform to JIS A 5705 Class I monolithic (minimum wear layer thickness 3 mm and minimum overall thickness 3 mm, Type [A (smooth)] [B (embossed)]. Provide [300] [400] [450] [600] [900] [_____] mm square tile.~~

2.3 WALL BASE

Provide ~~[~~100~~]~~ ~~[150] ~~mm high and a minimum 3.175 mm thick wall base. Provide ~~[~~preformed~~] [job formed]~~ corners in matching height, shape, and color.

~~2.4 INTEGRAL COVE BASE~~

~~Extend integral coved base for [[sheet vinyl] [and] [sheet linoleum] flooring up the wall [100] [150] mm]. Provide a [vinyl] [or] [rubber] [clear anodized aluminum], [square] [round] cap strip and vinyl, rubber, or wood fillet strip with a minimum radius of 19 mm for integral coved bases [at perimeter and fixed vertical interruptions to flooring] [as shown]. Provide integral cove of the same material as flooring. [Provide inside and outside corner protectors of [[_____]-colored anodized aluminum] [clear anodized aluminum] [or] [plastic] approved by flooring manufacturer.]~~

~~2.5 STAIR TREADS, RISERS AND STRINGERS~~

~~Provide either a one piece nosing/tread/riser or a two piece nosing/tread design with a matching coved riser.~~

2.4 MOULDING

Provide tapered mouldings of ~~[[~~vinyl~~]~~ ~~[~~or~~]~~ ~~[~~rubber~~]] [[_____]-colored anodized aluminum] [clear anodized aluminum]~~ and types as recommended by flooring manufacturer for both edges and transitions of flooring materials specified. Provide vertical lip on moulding of maximum 6 mm. Provide bevel change in level between 6 and 13 mm with a slope no greater than 1:2.

2.5 ADHESIVES

Provide adhesives for flooring, base and accessories as recommended by the manufacturer or per JIS A 5536 and comply with local indoor air quality standards. Submit manufacturer's descriptive data, documentation stating physical characteristics, and mildew and germicidal characteristics.

2.6 SURFACE PREPARATION MATERIALS

Provide surface preparation materials, such as panel type underlayment, lining felt, and floor crack fillers as recommended by the flooring manufacturer for the subfloor conditions.~~ Use one of the following substrates:~~ Use one of the following substrates:

~~[a. Particleboard: As specified in Section 06 10 00 ROUGH CARPENTRY.]~~

~~[b. Fiberboard: As specified in Section 06 10 00 ROUGH CARPENTRY.]~~

~~[c. Cork: As specified in Section 06 10 00 ROUGH CARPENTRY.]~~

~~[d. Cement-fiber board: As specified in Section 09 29 00 GYPSUM BOARD.]~~

~~[e. Plywood: As specified in Section 06 10 00 ROUGH CARPENTRY.]~~

~~[f. Concrete.]~~

a. Rubber/cork type acoustic underlayment: 3mm thick sheet or panel type.

b. PVC form type acoustic underlayment: 3mm thick sheet or panel type.

2.7 POLISH/FINISH

Provide polish finish as recommended by the manufacturer.

2.8 CAULKING AND SEALANTS

Provide caulking and sealants in accordance with Section 07 92 00 JOINT SEALANTS.

2.9 MANUFACTURER'S COLOR, PATTERN AND TEXTURE

Provide color, pattern and texture for resilient flooring and accessories ~~[~~ selected from manufacturer's standard colors~~] [[_____]~~. Color listed is not intended to limit the selection of equal colors from other manufacturers~~]~~.~~[~~ Provide floor patterns as specified on the ~~[~~drawings~~ Sheet No. [_____]] [_____]~~.~~]~~ Provide flooring in any one continuous area or replacement of damaged flooring in continuous area from same production run with same shade and pattern. Submit scaled drawings indicating patterns (including location of patterns and colors) and dimensions. Submit manufacturer's descriptive data and ~~[three] [_____]~~ samples of each indicated color and type of flooring, base, mouldings, and accessories sized a minimum 60 by 100 mm. Submit Data Package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

2.10 FIRE RESISTANCE TESTING REQUIREMENTS

Provide a minimum average critical radiant flux of ~~[0.22][~~0.45~~]~~ watts per square centimeter for flooring in corridors and exits when tested in accordance with ASTM E648.

PART 3 EXECUTION

3.1 EXAMINATION

Examine and verify that site conditions are in agreement with the design package. Report all conditions that will prevent a proper installation. Do not take any corrective action without written permission from the Government. Work will proceed only when conditions have been corrected and accepted by the installer. Submit manufacturer's printed installation instructions for all flooring materials and accessories, including preparation of substrate, seaming techniques, and recommended adhesives.

3.2 SURFACE PREPARATION

Provide a smooth, true, level plane for surface preparation of the flooring, except where indicated as sloped. Floor to be flat to within 4.75 in 3048 mm. Prepare subfloor in accordance with flooring manufacturer's recommended instructions. Prepare the surfaces of lightweight concrete slabs (as defined by the flooring manufacturer) as recommended by the flooring manufacturer. Floor fills or toppings may be required as recommended by the flooring manufacturer. Install underlayments, when required by the flooring manufacturer, in accordance with manufacturer's recommended printed installation instructions. Before any work under this section is begun, correct all defects such as rough or scaling concrete, chalk and dust, cracks, low spots, high spots, and uneven surfaces. Repair all damaged portions of concrete slabs as recommended by the flooring manufacturer. Remove concrete curing and sealer compounds from the slabs, other than the type that does not adversely affect adhesion. Remove paint, varnish, oils, release agents, sealers, waxes, and adhesives, as required by the flooring product in accordance with manufacturer's printed installation instructions.

3.3 MOISTURE, ALKALINITY AND BOND TESTS

Determine the suitability of the concrete subfloor for receiving the resilient flooring with regard to moisture content and pH level by moisture and alkalinity tests. Conduct moisture testing recommended by the flooring manufacturer. Conduct alkalinity testing as recommended by the flooring manufacturer. Determine the compatibility of the resilient flooring adhesives to the concrete floors by a bond test in accordance with the flooring manufacturer's recommendations. Submit copy of test reports for moisture and alkalinity content of concrete slab, and bond test stating date of test, person conducting the test, and the area tested.

3.4 GENERAL INSTALLATION

Do not install building construction materials that show visual evidence of biological growth. Installation shall comply with MLIT SS Chapter 19.

3.5 PLACING VINYL COMPOSITION~~, LINOLEUM AND SOLID VINYL~~ TILES

Install tile flooring and accessories in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's directions. Keep tile lines and joints square, symmetrical, tight, and even. Keep each floor in true, level plane, except where slope is indicated. Vary edge width as necessary to maintain full-size tiles in the field, no edge tile to be less than one-half the field tile size, except where irregular shaped rooms make it impossible. Cut flooring to fit around all permanent fixtures, built-in furniture and cabinets, pipes, and outlets. Cut, fit, and scribe edge tile to walls and partitions after field flooring has been applied.

3.6 PLACING LUXURY VINYL TILES

~~[~~Install luxury vinyl tile flooring ~~using [glue down] [loose lay (room perimter adhesive only)] installation.]Install flooring and accessories~~ in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's directions for installation method specified. Keep tile lines and joints square, symmetrical, tight, and even. Keep each floor in true, level plane, except where slope is indicated. Vary edge width as necessary to maintain full-size tiles in the field, no edge tile to be less than one-half the field tile size, except where irregular shaped rooms make it impossible. Cut flooring to fit around all permanent fixtures, built-in furniture and cabinets, pipes, and outlets. Cut, fit, and scribe edge tile to walls and partitions after field flooring has been applied.

~~3.7 PLACING SHEET VINYL FLOORING~~

~~Install sheet vinyl flooring and accessories in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Provide square, symmetrical, tight, and even flooring lines and joints. Keep each floor in true, level plane, except where slope is indicated. Cut flooring to fit around all permanent fixtures, built-in furniture and cabinets, pipes, and outlets. Lay out sheets to minimize waste. Cut, fit, and scribe flooring to walls and partitions after field flooring has been applied.[Provide [chemically bonded] [or] [heat welded] seams and edges [in rooms [_____]] [shown on the drawings] in accordance with the manufacturer's written installation instructions. Finish joints flush, free from voids, recesses, and raised areas.] [Install flooring with an integral coved base.]~~

~~3.8 PLACING SHEET LINOLEUM FLOORING~~

~~Install sheet linoleum flooring and accessories in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Provide square, symmetrical, tight, and even flooring lines and joints. Keep each floor in true, level plane, except where slope is indicated. Cut flooring to fit around all permanent fixtures, built-in furniture and cabinets, pipes, and outlets. Lay out sheets to minimize waste. Cut, fit, and scribe flooring to walls and partitions after field flooring has been~~

~~applied. Cut seams by overlapping or underscribing as recommended by the manufacturer. [Provide heat welded seams [in rooms [_____]] [as shown on the drawings] in accordance with there manufacturer's written installation instructions.] Finish joints flush, free from voids, recesses, and raised areas. [Install flooring with an integral coved base.]~~

~~3.9 PLACING CORK FLOORING~~

~~Install cork [tile] [plank flooring] and accessories in accordance with manufacturer's installation instructions. Prepare and apply adhesives in accordance with manufacturer's directions. Provide square, symmetrical, tight, and even flooring lines and joints except where slope is indicated. Keep each floor in true, level plane, except where slope is indicated. [Vary width of edge tiles as necessary to maintain full-size tiles in field, while keeping edge tiles larger than one-half full size, except where irregular-shaped rooms make it impossible.] Cut and fit flooring around all permanent fixtures, built-in furniture and cabinets, pipes, and outlets. Cut, fit and scribe flooring to walls and partitions after field flooring has been applied.~~

~~3.10 PLACING FEATURE STRIPS~~

~~Install feature strips in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions.~~

3.7 PLACING MOULDING

Provide moulding where flooring termination is higher than the adjacent finished flooring and at transitions between different flooring materials. When required, locate moulding under door centerline. Moulding is not required at doorways where thresholds are provided. ~~[~~ Secure moulding with adhesive as recommended by the manufacturer. Prepare and apply adhesives in accordance with manufacturer's printed directions.~~]~~ ~~[Anchor aluminum moulding to floor surfaces as recommended by the manufacturer.]~~

3.8 PLACING WALL BASE

Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent resilient flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

~~3.9 PLACING STAIR TREADS, RISERS, AND STRINGERS~~

~~Secure and install stair treads, risers, and stringers in accordance with manufacturer's printed installation instructions. Cover the surface of treads and risers[the full width of the stairs][within 150 mm to the stair edges]. Provide equal length pieces butted together to cover the treads and risers for stairs wider than manufacturer's standard lengths. [Provide stringer angles on both the wall and banister sides of the stairs, and landing trim.]~~

~~3.10 PLACING INTEGRAL COVED BASE~~

~~Install integral cove base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Shape integral coved base by extending the flooring material [100] [150] [_____] mm onto the wall surface. Support cove by a filler. Provide a cap strip at the top of the base. Fill voids along the top edge of base at masonry walls with caulk.~~

3.9 CLEANING

Immediately upon completion of installation of flooring in a room or an area, dry and clean the flooring and adjacent surfaces to remove all surplus adhesive. Clean flooring as recommended in accordance with manufacturer's printed maintenance instructions and within the recommended time frame. As required by the manufacturer, apply the recommended number of coats and type of polish and finish in accordance with manufacturer's written instructions.

3.10 PROTECTION

From the time of installation until acceptance, protect flooring from damage as recommended by the flooring manufacturer. Remove and replace flooring which becomes damaged, loose, broken, or curled and wall base which is not tight to wall or securely adhered.

-- End of Section --

SECTION 09 90 00

PAINTS AND COATINGS
05/11

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 5758	(2016) Sealants for Sealing and Glazing in Buildings
JIS A 6909	(2014) Coating Materials for Textured Finishes of Buildings
JIS K5600	(1999) Testing Methods for Paints
JIS K5670	(2008) Non-Aqueous Dispersion Acrylic Paint

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT SS Chapter 18	(2019) Public Building Construction Standard Specification: Chapter 18 Painting Work

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety ~~-- Safety~~ and Health Requirements Manual

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

AMS-STD-595	(2017) Colors Used in Government Procurement

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~

Samples of specified materials may be taken and tested for compliance with specification requirements.

SD-02 Shop Drawings

Piping Identification

SD-08 Manufacturer's Instructions

Application Instructions

Mixing

SD-10 Operation and Maintenance Data

Coatings

1.3 CERTIFICATES

1.3.1 Indoor Air Quality

Submit required indoor air quality certifications in one submittal package.

1.3.1.1 Paints and Coatings

Provide paint and coating products certified to meet formaldehyde emission class ~~F****~~F 4-Star.

1.4 APPLICATOR'S QUALIFICATIONS

~~[~~1.4.1 Contractor Qualification

Submit the name, address, telephone number, FAX number, and e-mail address of the contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on ~~[_____] on~~ a minimum of three similar projects within the past three years. List information by individual and include the following:

a. Name of individual and proposed position for this work.

b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address, telephone number, and telex number (if non-US) of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

~~]~~1.5 QUALITY ASSURANCE

1.5.1 Field Samples and Tests

The Contracting Officer may choose up to two coatings that have been delivered to the site to be tested at no cost to the Government. Take samples of each chosen product as specified in the paragraph SAMPLING PROCEDURES. Test each chosen product as specified in the paragraph TESTING PROCEDURE. Remove products from the job site which do not conform, and replace with new products that conform to the referenced specification. Test replacement products that failed initial testing at no cost to the Government.

1.5.1.1 Sampling Procedure

The Contracting Officer will select paint at random from the products that have been delivered to the job site for sample testing. The Contractor will provide one liter samples of the selected paint materials. Take samples in the presence of the Contracting Officer, and label, and identify each sample. Provide labels in accordance with the paragraph PACKAGING, LABELING, AND STORAGE of this specification.

1.5.1.2 Testing Procedure

Qualification testing of coated surfaces per MLIT SS Chapter 18 Painting Work, Table 18.1.1 and Section 18.1.7 and JIS K5600.

Testing of film thickness per JIS K5600.

1.6 REGULATORY REQUIREMENTS

1.6.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, indoor coating materials to conform to formaldehyde emission class ~~F****~~F 4-Star. Notify Contracting Officer of any paint specified herein which fails to conform.

1.6.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

1.6.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

1.6.4 Asbestos Content

Provide asbestos-free materials.

1.6.5 Mercury Content

Provide materials free of mercury or mercury compounds.

1.6.6 Silica

Provide abrasive blast media containing no free crystalline silica.

1.6.7 Human Carcinogens

Provide materials that do not contain confirmed human carcinogens or suspected human carcinogens.

1.7 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the contract specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 20 liters. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 4 to 35 degrees C. Do not store paint, polyurethane, varnish, or wood stain products with materials that have a high capacity to adsorb VOC emissions~~, [including [_____]]~~. Do not store paint, polyurethane, varnish, or wood stain products in occupied spaces.

1.8 SAFETY AND HEALTH

Apply coating materials using safety methods and equipment in accordance with the following:

Comply with applicable local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS and in Appendix A of EM 385-1-1. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

1.8.1 Safety Methods Used During Coating Application

Comply with the requirements of MLIT SS Chapter 18.

1.8.2 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.

~~[~~ b. Removal and disposal of coatings which contain lead is specified in Section 02 83 00 LEAD REMEDIATION~~[_____]~~. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

~~]~~
~~[~~ c. Removal and disposal of coatings which contain asbestos materials is specified in Section 02 82 00 ASBESTOS REMEDIATION. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

~~]~~ Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials.

1.9 ENVIRONMENTAL CONDITIONS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation.~~[~~ Isolate area of application from rest of building when applying high-emission paints or coatings.~~]~~

1.9.1 Coatings

Do not apply coating when air or substrate conditions are:

a. Unsuitable for drying, including when the air temperature at a location for coating is below 5 degrees C, the humidity is 85 percent or more and condensation occurs due to inadequate ventilation. Where it is not possible to avoid coating, curing measures, such as warming and ventilation, shall be performed.

b. External coatings shall generally not be performed if rain is likely to occur or during strong winds.

1.9.2 Post-Application

Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

a. Supply 100 percent outside air 24 hours a day.

b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 13 degrees C and 29 degrees C and humidity is between 30 percent and 60 percent.

c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

1.10 SCHEDULING

Allow paint, polyurethane, varnish, and wood stain installations to cure prior to the installation of materials that adsorb VOCs~~, including [_____]~~.

1.11 COLOR SELECTION

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors approximate colors indicated and the product conforms to specified requirements.

Tint each coat progressively darker to enable confirmation of the number of coats.

Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes.

1.12 LOCATION AND SURFACE TYPE TO BE PAINTED

1.12.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.

b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.

c. Existing coated surfaces that are damaged during performance of the work.

1.12.1.1 Exterior Painting

Includes new surfaces~~[~~, existing coated surfaces,~~]~~ ~~[~~and~~]~~ ~~[~~existing uncoated surfaces,~~]~~ of the building~~[s]~~ and appurtenances. Also included are existing coated surfaces made bare by cleaning operations.

1.12.1.2 Interior Painting

Includes new surfaces~~[~~, existing uncoated surfaces,~~]~~ ~~[~~and~~]~~ ~~[~~existing coated surfaces~~]~~ of the building~~[s]~~ and appurtenances as indicated and existing coated surfaces made bare by cleaning operations. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

a. Exposed columns, girders, beams, joists, and metal deck; and

b. Other contiguous surfaces.

1.12.2 Painting Excluded

Do not paint the following unless indicated otherwise.

a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.

b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.

c. Steel to be embedded in concrete.

d. Copper, stainless steel, aluminum, brass, and lead except existing coated surfaces.

e. Hardware, fittings, and other factory finished items.

~~[f. Do not paint surfaces in the following areas: [_____].~~

~~]~~1.12.3 Mechanical and Electrical Painting

Includes field coating of ~~[~~interior~~]~~ ~~[~~and~~]~~ ~~[~~exterior~~]~~ new ~~[~~and existing~~]~~ surfaces.

a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.

(1) Exposed piping, conduit, and ductwork;

(2) Supports, hangers, air grilles, and registers;

(3) Miscellaneous metalwork and insulation coverings.

~~[~~ b. Do not paint the following, unless indicated otherwise:

~~[~~ (1) New zinc-coated, aluminum, and copper surfaces under insulation

~~][~~ (2) New aluminum jacket on piping

~~][~~ (3) New interior ferrous piping under insulation.

~~]][~~1.12.3.1 Fire Extinguishing Sprinkler Systems

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. Apply coatings to clean, dry surfaces, using clean brushes. Clean the surfaces to remove dust, dirt, rust, and loose mill scale. Immediately after cleaning, provide the metal surfaces with one coat primer per schedules. Shield sprinkler heads with protective covering while painting is in progress. Upon completion of painting, remove protective covering from sprinkler heads. Remove sprinkler heads which have been painted and replace with new sprinkler heads. Provide primed surfaces with the following:

a. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel applied to a minimum dry film thickness of 0.025 mm in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceiling are not painted or not constructed of a prefinished material.~~[In lieu of red enamel finish coat, provide piping with 50 mm wide red enamel bands or self-adhering red plastic bands spaced at maximum of 6 meters intervals.]~~

b. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel applied to a minimum dry film thickness of 0.025 mm. Provide piping with 50 mm wide red enamel bands or self-adhering red plastic bands spaced at maximum of 6 meters intervals throughout the piping systems.

~~][1.12.4 Exterior Painting of Site Work Items~~

~~Field coat the following items:~~

~~New Surfaces Existing Surfaces~~

~~a. [_____] [_____]~~

~~b. [_____] [_____]~~

~~c. [_____] [_____]~~

~~]~~1.12.4 MISCELLANEOUS PAINTING

Lettering ~~[~~Building ~~][Room~~ Number(s)~~]~~

Provide lettering ~~[as scheduled on the drawings] [~~block~~] [Gothic]~~ type, ~~[~~ black enamel~~] [water-type decalcomania, finished with a protective coating of spar varnish]~~. Samples must be approved before application.

1.12.5 Definitions and Abbreviations

1.12.5.1 Coating

A film or thin layer applied to a base material called a substrate. A coating may be a metal, alloy, paint, or solid/liquid suspensions on various substrates (such as metals, plastics, wood, paper, leather, cloth). They may be applied by electrolysis, vapor deposition, vacuum, or mechanical means such as brushing, spraying, calendaring, and roller coating. A coating may be applied for aesthetic or protective purposes or both. The term "coating" as used herein includes emulsions, enamels, stains, varnishes, sealers, epoxies, and other coatings, whether used as primer, intermediate, or finish coat. The terms paint and coating are used interchangeably.

1.12.5.2 Dry Film Thickness

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

1.12.5.3 micron / microns

The metric measurement for 0.001 mm or one/one-thousandth of a millimeter.

1.12.5.4 mm

The metric measurement for millimeter, 0.001 meter or one/one-thousandth of a meter.

1.12.5.5 Gloss Levels

Gloss Level	Description	Units at 60 degrees	Units at 85 degrees
Gloss 30	Semi-Gloss	15	30
Gloss 50	Semi-Gloss	35	50
Gloss 70	Gloss	60	70
Glossy	High-Gloss	70+	80+

Gloss is tested in accordance with JIS K5600. Historically, the Government has used Flat(Matte), Gloss 30(Eggshell), Gloss 50 (Semi-Gloss), and Gloss 70 (Gloss).

1.12.5.6 Paint

See Coating definition.

PART 2 PRODUCTS

2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit product data sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems.

Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, potlife, and curing and drying times between coats.

Provide certification of Indoor Air Quality for paints and primers.

[Provide certification of Indoor Air Quality for consolidated latex paints.

]PART 3 EXECUTION

3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

[3.2 REPUTTYING AND REGLAZING

Remove cracked, loose, and defective putty or glazing compound on glazed sash and provide new putty or glazing compound. Where defective putty or glazing compound constitutes 30 percent or more of the putty at any one light, remove the glass and putty or glazing compound and reset the glass. Remove putty or glazing compound without damaging sash or glass. Clean rabbets to bare wood or metal and prime prior to reglazing. Provide linseed oil putty for wood sash. Patch surfaces to provide smooth transition between existing and new surfaces. Finish putty or glazing compound to a neat and true bead. Allow glazing compound time to cure, in accordance with manufacturer's recommendation, prior to coating application. Allow putty to set one week prior to coating application.

][3.3 RESEALING OF EXISTING EXTERIOR JOINTS

3.3.1 Surface Condition

Begin with surfaces that are clean, dry to the touch, and free from frost and moisture; remove grease, oil, wax, lacquer, paint, defective backstop, or other foreign matter that would prevent or impair adhesion. Where adequate grooves have not been provided, clean out to a depth of 13 mm and grind to a minimum width of 6 mm without damage to adjoining work. Grinding is not required on metal surfaces.

3.3.2 Backstops

In joints more than 13 mm deep, install glass fiber roving or neoprene, butyl, polyurethane, or polyethylene foams free of oil or other staining elements as recommended by sealant manufacturer. Provide backstop material compatible with sealant. Do not use oakum and other types of absorptive materials as backstops.

3.3.3 Primer and Bond Breaker

Install the type recommended by the sealant manufacturer.

3.3.4 Ambient Temperature

Between 4 degrees C and 35 degrees C when applying sealant.

3.3.5 Exterior Sealant

For joints, provide JIS A 5758. Color(s) will be selected by the Contracting Officer. Apply the sealant in accordance with the manufacturer's printed instructions. Force sealant into joints with sufficient pressure to fill the joints solidly. Apply sealant uniformly smooth and free of wrinkles.

3.3.6 Cleaning

Immediately remove fresh sealant from adjacent areas using a solvent recommended by the sealant manufacturer. Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean condition. Allow sealant time to cure, in accordance with manufacturer's recommendations, prior to coating.

]3.4 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, [disintegrated coatings,] and other foreign matter and substances deleterious to coating performance per MLIT SS Chapter 18 for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas.

[3.4.1 Additional Requirements for Preparation of Surfaces With Existing Coatings

Before application of coatings, perform the following on surfaces covered by soundly-adhered coatings, defined as those which cannot be removed with a putty knife:

a. Test existing finishes for lead before sanding, scraping, or removing. If lead is present, refer to paragraph Toxic Materials.

b. Wipe previously painted surfaces to receive solvent-based coatings, except stucco and similarly rough surfaces clean with a clean, dry cloth saturated with mineral spirits or per paint manufacturer's requirements. Allow surface to dry. Wipe immediately preceding the application of the first coat of any coating, unless specified otherwise.

c. Sand existing glossy surfaces to be painted to reduce gloss. Brush, and wipe clean with a damp cloth to remove dust.

d. The requirements specified are minimum. Comply also with the application instructions of MLIT SS Chapter 18 and the paint manufacturer.

e. Thoroughly clean previously painted surfaces [specified to be repainted] [damaged during construction] of all grease, dirt, dust or other foreign matter.

f. Remove blistering, cracking, flaking and peeling or otherwise deteriorated coatings.

g. Roughen slick surfaces. Repair damaged areas such as, but not limited to, nail holes, cracks, chips, and spalls with suitable material to match adjacent undamaged areas.

h. Feather and sand smooth edges of chipped paint.

i. Clean rusty metal surfaces per MLIT SS Chapter 18 and per paint manufacturer's instructions. Use solvent, mechanical, or chemical cleaning methods to provide surfaces suitable for painting.

j. Provide new, proposed coatings that are compatible with existing coatings.

][3.4.2 Existing Coated Surfaces with Minor Defects

[Sand, spackle, and treat minor defects to render them smooth. Minor defects are defined as scratches, nicks, cracks, gouges, spalls, alligatoring, chalking, and irregularities due to partial peeling of previous coatings.][Remove chalking by sanding [or blasting].]

][3.4.3 Removal of Existing Coatings

Remove existing coatings from the following surfaces:

a. Surfaces containing large areas of minor defects;

b. Surfaces containing more than 20 percent peeling area; and

c. Surfaces designated by the Contracting Officer, such as surfaces where rust shows through existing coatings.

][3.4.4 Substrate Repair

a. Repair substrate surface damaged during coating removal;

b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and

c. Clean and prime the substrate as specified.

]3.5 PREPARATION OF METAL SURFACES

3.5.1 Existing and New Ferrous Surfaces

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

3.6 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

3.6.1 Concrete and Masonry

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.

b. Surface Cleaning: Remove the following deleterious substances.

(1) Dirt,~~[~~ Chalking,~~]~~ Grease, and Oil: Wash new~~[~~ and existing uncoated ~~]~~ surfaces with a solution per paint manufacturer's recommendations. Then rinse thoroughly with fresh water.~~[Wash existing coated surfaces with a suitable detergent and rinse thoroughly.]~~ For large areas, water blasting may be used.

(2) Fungus and Mold: Wash ~~[~~new~~][~~, existing coated,~~]~~ ~~[~~and existing uncoated~~]~~ surfaces with a solution per paint manufacturer's recommendations. Rinse thoroughly with fresh water.

(3) Paint and Loose Particles: Remove by wire brushing.

(4) Efflorescence: Remove by scraping or wire brushing followed by washing with a solution per paint manufacturer's recommendations.

~~[~~ (5) Removal of Existing Coatings: For surfaces to receive textured coating, remove existing coatings including soundly adhered coatings if recommended by textured coating manufacturer.

~~]~~ c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.

d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical or horizontal surfaces as determined manufacturer's recommendations. In all cases follow manufacturers recommendations. Allow surfaces to cure a minimum of 30 days before painting.

3.6.2 Gypsum Board

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

a. Surface Cleaning: Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint will be water-based.

b. Repair of Minor Defects: Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

c. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by the paint manufacturer. Verify that new plaster to be coated does not exceed maximum moisture content per manufacturer's recommendations. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before preparation for painting or as recommended by paint manufacturer.

3.6.3 Existing Asbestos Cement Surfaces

Remove oily stains by solvent cleaning with mineral spirits, per manufacturer's recommendations. Remove loose dirt, dust, and other deleterious substances by brushing with a soft brush or rubbing with a dry cloth prior to application of the first coat material. Do not wire brush or clean using other abrasive methods. Verify surfaces are dry and clean prior to application of the coating.

3.7 PREPARATION OF WOOD AND PLYWOOD SURFACES

3.7.1 New ~~[, Existing Uncoated,] [and] [Existing Coated]~~ Plywood and Wood Surfaces, Except Floors:

Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.

a. Clean wood surfaces of foreign matter.

Surface Cleaning: Verify that surfaces are free from dust and other deleterious substances and in a condition approved by the Contracting Officer prior to receiving paint or other finish. Do not use water to clean uncoated wood.~~[~~ Scrape to remove loose coatings. Lightly sand to roughen the entire area of previously enamel-coated wood surfaces.~~]~~

~~[~~ b. Removal of Fungus and Mold: Wash existing coated surfaces with a solution as recommended by paint manufacturer. Rinse thoroughly with fresh water.

~~]~~ c. Do not exceed 12 percent moisture content of the wood as measured by a moisture meter in accordance with paint manufacturer's recommendations.

d. Prime or touch up wood surfaces adjacent to surfaces to receive water-thinned paints before applying water-thinned paints.

e. Cracks and Nailheads: Set and putty stop nailheads and putty cracks after the prime coat has dried.

f. Cosmetic Repair of Minor Defects:

(1) Knots and Resinous Wood~~ [and Fire, Smoke, Water, and Color Marker Stained Existing Coated Surface]~~: Prior to application of coating, cover knots and stains with two or more coats of 1.3-kg-cut shellac varnish, plasticized with 0.14 liters of castor oil per liter or as recommended by coating manufacturer. . Scrape away existing coatings from knotty areas, and sand before treating. Prime before applying any putty over shellacked area.

(2) Open Joints and Other Openings: Fill with whiting putty, linseed oil putty. Sand smooth after putty has dried.

(3) Checking: Where checking of the wood is present, sand the surface, wipe and apply a coat of pigmented orange shellac. Allow to dry before paint is applied.

g. Prime Coat For New Exterior Surfaces: Prime coat ~~[wood doors,] [windows,] [frames,] [and] [~~trim~~]~~ before wood becomes dirty, warped, ~~[~~ or weathered~~]~~.

~~3.7.2 Wood Floor Surfaces, Natural Finish~~

~~Base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.~~

~~a. Initial Surface Cleaning: As specified in paragraph entitled "Surface Preparation."~~

~~[b. Existing Loose Boards and Shoe Molding: Before sanding, renail loose boards. Countersink nails and fill with an approved wood filler. Remove shoe molding before sanding and reinstall after completing other work. At Contractor's option, new shoe molding may be provided in lieu of reinstalling old. Provide new wood molding of the same size, wood species, and finish as the existing.~~

~~] c. Sanding and Scraping: Sanding of wood floors is specified in Section [09 64 29 WOOD STRIP AND PLANK FLOORING] [09 64 23 WOOD PARQUET FLOORING] [09 64 66 WOOD ATHLETIC FLOORING] [09 64 00 PORTABLE (DEMOUNTABLE) WOOD FLOORING]. Fill floors of oak or similar open-grain wood with wood filler recommended by the finish manufacturer and the excess filler removed.~~

~~d. Final Cleaning: After sanding, sweep and vacuum floors clean. Do not walk on floors thereafter until specified sealer has been applied and is dry.~~

~~3.7.3 Interior Wood Surfaces, Stain Finish~~

~~Sand interior wood surfaces to receive stain. Fill oak and other open-grain wood to receive stain with a coat of wood filler not less than 8 hours before the application of stain; remove excess filler and sand the surface smooth. Ensuing base surface preparation per MLIT SS Chapter 18 or per paint manufacturer's recommendations.~~

~~3.7.4 Water Blasting of Existing Coated Wood Surfaces:~~

~~Provide water blasting for the following surfaces: [_____].~~

~~a. Sample Panel: Prior to the initial surface cleaning, water blast a representative surface designated by the Contracting Officer. Provide surface cleaning of the remaining work to match the sample panel approved by the Contracting Officer.~~

~~b. Initial Surface Cleaning: Water blast surfaces to receive paint with a high pressure spray, to remove loose paint, dirt, and other foreign or deleterious materials. Do not flood vents or damage windows and floors. If the pressure specified will cause damage to existing wood, advise the Contracting Officer. Direct the wash nozzle at the surface at an angle of approximately 75 degrees with the surface and at a distance not greater than 1500 mm to apply water pressure required to remove loose paint, dirt, chalking, and other foreign matter.~~

~~c. Final Surface Cleaning: After allowing the surfaces to dry for a minimum of 24 hours, remove remaining dirt, splinters, loose particles, disintegrated and loose paint, grease, oil, and other foreign matter from the surface.~~

3.8 APPLICATION

3.8.1 Coating Application

Application of paint per MLIT SS Chapter 18. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.

Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.

Only apply paints, except water-thinned types to surfaces that are completely free of moisture as determined by sight or touch.

Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.

Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.

Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.

Apply paint to new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metal work, and accessories. Shield sprinkler heads with protective coverings while painting is in progress. Remove sprinkler heads which have been painted and replace with new sprinkler heads. For piping in unfinished spaces, provide primed surfaces with one coat of red alkyd gloss enamel to a minimum dry film thickness of 0.025 mm. Unfinished spaces include attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and space where walls or ceiling are not painted or not constructed of a prefinished material. For piping in finished areas, provide prime surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating accessories with one coat of red alkyd gloss enamel. Upon completion of painting, remove protective covering from sprinkler heads.

a. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.

b. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.

c. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.

d. Thermosetting Paints: Topcoats over thermosetting paints (epoxies and urethanes) should be applied within the overcoating window recommended by the manufacturer.

e. Floors: ~~[~~For nonslip surfacing on level floors, as the intermediate coat is applied, cover wet surface completely with almandite garnet, Grit No. 36, with maximum passing .420 mm less than 0.5 percent. When the coating is dry, use a soft bristle broom to sweep up excess grit, which may be reused, and vacuum up remaining residue before application of the topcoat.~~]~~ ~~[~~For nonslip surfacing on ramps, provide ~~MPI 77~~solvent based, gloss, two component, epoxy coating with non-skid additive, applied by roller in accordance with manufacturer's instructions.~~]~~

3.8.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

3.8.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.

3.8.4 Coating Systems

a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in MLIT SS Chapter 18.

b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness as specified in MLIT SS Chapter 18. Coating thickness where specified, refers to the minimum dry film thickness.

c. Coatings for Surfaces Not Specified Otherwise: Coat surfaces which have not been specified, the same as surfaces having similar conditions of exposure.

d. Existing Surfaces Damaged During Performance of the Work, Including New Patches In Existing Surfaces: Coat surfaces with the following:

(1) One coat of primer.

(2) One coat of undercoat or intermediate coat.

(3) One topcoat to match adjacent surfaces.

e. Existing Coated Surfaces To Be Painted: Apply coatings conforming to the respective specifications listed in the Tables herein, except that pretreatments, sealers and fillers need not be provided on surfaces where existing coatings are soundly adhered and in good condition. Do not omit undercoats or primers.

3.9 COATING SYSTEMS FOR METAL

Apply coatings per MLIT SS Chapter 18 for Exterior and Interior.

a. Apply specified ferrous metal primer on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.

b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.

c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.

d. Surface Previously Coated with Epoxy or Urethane: Apply ~~MPI 101~~ solvent based, two component, epoxy, anti-corrosive primer, 0.038 mm DFT immediately prior to application of epoxy or urethane coatings.

e. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.

f. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal per paint manufacturer's recommendations.

3.10 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings per MLIT SS Chapter 18 for Exterior and Interior.

3.11 COATING SYSTEMS FOR WOOD AND PLYWOOD

a. Apply coatings per MLIT SS Chapter 18 for Exterior and Interior.

b. Apply stains in accordance with manufacturer's printed instructions.

~~[c. Wood Floors to Receive Natural Finish: Thin first coat 2 to 1 using thinner recommended by coating manufacturer. Apply second coat not less than 2 hours and not over 24 hours after first coat has been applied. Apply with applicators as recommended by coating manufacturer. Buff or lightly sand between intermediate coats as recommended by coating manufacturer's printed instructions.~~

~~]~~3.12 PIPING IDENTIFICATION

Piping Identification, Including Surfaces In Concealed Spaces: Provide per the following.

a. Flammable Materials: Defined as all materials known ordinarily as flammables or combustibles. AMS-STD-595, Yellow, No. 13655.

b. Toxic and Poisonous Materials: Defined as all materials extremely hazardous to life or health under normal conditions as toxics or poisons. AMS-STD-595, Brown, No. 10080.

c. Anesthetics and Harmful Materials: Defined as all materials productive of anesthetic vapors and all liquid chemicals and compound hazardous to life and property but not normally productive of dangerous quantities of fumes or vapors. AMS-STD-595, Blue, No. 15102.

d. Oxidizing Materials: Defined as all materials which readily furnish oxygen for combustion and fire producers which react explosively or with the evolution of heat in contact with many other materials. AMS-STD-595, Green, No. 14187.

e. Physically Dangerous Materials: Defined as all materials, not dangerous in themselves, which are asphyxiating in confined areas or which are generally handled in a dangerous physical state of pressure or temperature. AMS-STD-595, Gray, No. 16187.

f. Fire Protection Materials: Defined as all materials provided in piping systems or in compressed gas cylinders exclusively for use in fire protection. AMS-STD-595, Red, No. 11105.

g. Water: Piping system containing water suitable for human consumption and installed for this purpose. AMS-STD-595, White, No. 1787 or painted to match surroundings when not in conflict with other color designations.

Place stenciling in clearly visible locations. On piping not covered by the aforementioned stencil approved names or code letters, in letters a minimum of 13 mm high for piping and a minimum of 50 mm high elsewhere. Stencil arrow-shaped markings on piping to indicate direction of flow using black stencil paint.

3.13 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

3.14 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers.

3.15 PAINT TABLES

3.15.1 Exterior Paint Tables

DIVISION 3: EXTERIOR CONCRETE PAINT TABLE

A. ~~{~~New and uncoated existing~~}~~ ~~{~~and Existing, previously painted~~}~~ concrete; vertical surfaces, including undersides of balconies, soffits, roofs, columns and beams but excluding tops of slabs:

1. Paint system shall be formaldehyde emission class ~~F****~~F 4-Star minimum, Weatherproof Class 3 Type. Coatings shall comply with JIS A 6909, Multi-Layer, Type E:

   Spray Tile Finish System (4-Layer System):

(1)	Base Coat 1:	Not less than 0.1 kg/sm
(2)	Texture Base Coat 1:	Not less than 0.7 kg/sm
(3)	Texture Top Coat:	Not less than 0.8 kg/sm
(4)	Top Coat, Acrylic, Resin, Emulsion Type, High Gloss Finish, 2 times:	Not less than 0.25 kg/sm

   Primer as recommended by manufacturer. Coating system shall be applied by spray application in accordance with manufacturer's instructions.

B. ~~{~~New and uncoated existing~~}~~ ~~{~~and Existing, previously painted~~}~~ concrete, textured system; vertical surfaces, including undersides of balconies and soffits but excluding tops of slabs:

1. MLIT SS Chapter 18

   Texture - ~~{~~Fine~~} [Medium] [Coarse]~~. Surface preparation and number of coats in accordance with manufacturer's instructions.
   Topcoat: Coating to match adjacent surfaces.

C. ~~{~~New and uncoated existing~~}~~ ~~{~~and Existing, previously painted~~}~~ concrete, elastomeric System; vertical surfaces, including undersides of balconies and soffits but excluding tops of slabs:

1. MLIT SS Chapter 18

   Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces. Surface preparation and number of coats in accordance with manufacturer's instructions.

D. ~~{~~New and uncoated existing~~}~~ ~~{~~and Existing, previously painted~~}~~ concrete: walls and bottom of swimming pools.

1. MLIT SS Chapter 18

E. ~~{~~New~~}~~ ~~{~~and Existing~~}~~ Cementitious composition board (including Asbestos cement board):

1. MLIT SS Chapter 18

   Topcoat: Coating to match adjacent surfaces.

~~DIVISION 4: EXTERIOR CONCRETE MASONRY UNITS PAINT TABLE~~

~~A. [New] [and Existing] concrete masonry on uncoated surface:~~

~~1. MLIT SS Chapter 18~~

~~Topcoat: Coating to match adjacent surfaces.~~

~~B. [New] [and Existing] concrete masonry, textured system; on uncoated surface:~~

~~1. MLIT SS Chapter 18~~

~~Texture - [Fine] [Medium] [Coarse]. Surface preparation and number of coats in accordance with manufacturer's instructions. Topcoat: Coating to match adjacent surfaces.~~

~~C. [New] [and Existing] concrete masonry, elastomeric system; on uncoated surface:~~

~~1. MLIT SS Chapter 18~~

~~Primer as recommended by manufacturer. Topcoat: Coating to match adjacent surfaces. Surface preparation and number of coats in accordance with manufacturer's instructions.~~

DIVISION 5: EXTERIOR METAL, FERROUS AND NON-FERROUS PAINT TABLE

STEEL / FERROUS SURFACES

A. New Steel that has been hand or power tool cleaned or per MLIT SS Chapter 18.

1. MLIT SS Chapter 18

B. New Steel that has been blast-cleaned:

2. MLIT SS Chapter 18

C. Existing steel that has been spot-blasted:

1. MLIT SS Chapter 18

2. ~~[~~Surface previously coated with epoxy:
MLIT SS Chapter 18~~]~~

D. New ~~[~~and existing~~]~~ steel blast cleaned:

1. MLIT SS Chapter 18

2. ~~[~~Pigmented Polyurethane
MLIT SS Chapter 18~~]~~

~~E. Metal floors (non-shop-primed surfaces or non-slip deck surfaces) with non-skid additive (NSA), load at manufacturer's recommendations.:~~

~~1. MLIT SS Chapter 18~~

EXTERIOR GALVANIZED SURFACES

F. New Galvanized surfaces:

~~1. MLIT SS Chapter 18~~

~~2. [Waterborne Primer / Latex~~
~~MLIT SS Chapter 18]~~

~~3. [Waterborne Primer / Waterborne Light Industrial Coating~~
~~MLIT SS Chapter 18~~
~~System DFT: 112 microns]~~

~~4. [Epoxy Primer / Waterborne Light Industrial Coating~~
~~MLIT SS Chapter 18]~~

~~5~~1. ~~[~~Pigmented Polyurethane
MLIT SS Chapter 18~~]~~

G. Galvanized surfaces with slight coating deterioration; little or no rusting:

1. MLIT SS Chapter 18

2. ~~[~~Pigmented Polyurethane
MLIT SS Chapter 18~~]~~

H. Galvanized surfaces with severely deteriorated coating or rusting:

1. MLIT SS Chapter 18

2. ~~[~~Pigmented Polyurethane
MLIT SS Chapter 18~~]~~

EXTERIOR SURFACES, OTHER METALS (NON-FERROUS)

I. Aluminum, aluminum alloy and other miscellaneous non-ferrous metal items not otherwise specified except hot metal surfaces, roof surfaces, and new prefinished equipment. Match surrounding finish:

~~1. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~2~~1. ~~[~~Waterborne Light Industrial Coating
MLIT SS Chapter 18~~]~~

J. Existing roof surfaces previously coated:

1. MLIT SS Chapter 18

2. ~~[~~Aluminum Paint
MLIT SS Chapter 18~~]~~

K. Surfaces adjacent to painted surfaces; ~~[~~Mechanical,~~]~~ ~~[~~Electrical,~~]~~ ~~[~~Fire extinguishing sprinkler systems including valves, conduit, hangers, supports,~~][~~ exposed copper piping,~~]~~ ~~[~~and miscellaneous metal items~~]~~ not otherwise specified except floors, hot metal surfaces, and new prefinished equipment. Match surrounding finish:

~~1. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~2~~1. ~~[~~Waterborne Light Industrial Coating
MLIT SS Chapter 18~~]~~

3.15.2 Interior Paint Tables

DIVISION 3: INTERIOR CONCRETE PAINT TABLE

A. ~~[~~New and uncoated existing~~]~~ ~~[~~and Existing, previously painted~~]~~ Concrete, vertical surfaces, not specified otherwise:

~~1. MLIT SS Chapter 18~~

~~2. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18]~~

~~3~~1. ~~[~~Institutional Low Odor / Low VOC Latex
MLIT SS Chapter 18~~]~~

~~B. Concrete ceilings, uncoated:~~

~~1. [Latex Aggregate~~
~~MLIT SS Chapter 18]~~

~~Texture - [Fine] [Medium] [Coarse]. Surface preparation, number of coats, and primer in accordance with manufacturer's instructions. Topcoat: Coating to match adjacent surfaces.~~

~~C~~B. ~~[~~New and uncoated existing~~][~~ and ~~][~~Existing, previously painted~~]~~ Concrete in ~~[~~toilets,~~] [food-preparation,] [food-serving,]~~ ~~[~~restrooms,~~]~~ ~~[~~laundry areas,~~]~~ ~~[~~shower areas,~~] [areas requiring a high degree of sanitation,]~~ ~~[_____]~~ ~~[~~and other high-humidity areas~~]~~ not otherwise specified except floors:

1. Coating shall conform to JIS K5670, 0.10 kg/sm for base coat and JIS K5670, 0.10 kg/sm for top coat.

DIVISION 4: INTERIOR CONCRETE MASONRY UNITS PAINT TABLE

~~A. New [and uncoated Existing] Concrete masonry:~~

~~1. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18~~
~~Fill all holes in masonry surface]~~

~~2. [Institutional Low Odor / Low VOC Latex~~
~~MLIT SS Chapter 18]~~

~~B~~A. Existing, previously painted Concrete masonry:

~~1. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18]~~

~~2~~1. ~~[~~Institutional Low Odor / Low VOC Latex
MLIT SS Chapter 18~~]~~

~~C. New [and uncoated Existing] Concrete masonry units in [toilets,] [food-preparation,] [food-serving,] [restrooms,] [laundry areas,] [shower areas,] [areas requiring a high degree of sanitation,] [_____,] [and other high humidity areas] unless otherwise specified:~~

~~1. [Waterborne Light Industrial Coating~~
~~MLIT SS Chapter 18~~
~~Fill all holes in masonry surface]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18~~
~~Fill all holes in masonry surface]~~

~~3. [Epoxy~~
~~MLIT SS Chapter 18~~
~~Fill all holes in masonry surface]~~

~~D. Existing, previously painted, concrete masonry units in [toilets,] [food-preparation,] [food-serving,] [restrooms,] [laundry areas,] [shower areas,] [areas requiring a high degree of sanitation,] [_____,] [and other high humidity areas] unless otherwise specified:~~

~~1. [Waterborne Light Industrial Coating~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~3. [Epoxy~~
~~MLIT SS Chapter 18]~~

DIVISION 5: INTERIOR METAL, FERROUS AND NON-FERROUS PAINT TABLE

INTERIOR STEEL / FERROUS SURFACES

A. Metal, ~~[~~Mechanical,~~]~~ ~~[~~Electrical,~~]~~ ~~[~~Fire extinguishing sprinkler systems including valves, conduit, hangers, supports~~]~~. ~~[~~Surfaces adjacent to painted surfaces (Match surrounding finish),~~]~~ ~~[~~exposed copper piping,~~]~~ ~~[~~and miscellaneous metal items~~]~~ not otherwise specified except floors, hot metal surfaces, and new prefinished equipment:

~~1. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~B. Metal floors (non-shop-primed surfaces or non-slip deck surfaces) with non-skid additive (NSA), load at manufacturer's recommendations.:~~

~~1. [Alkyd Floor Paint~~
~~MLIT SS Chapter 18]~~

~~2~~1. ~~[~~Epoxy
MLIT SS Chapter 18~~]~~

~~C. Metal in [toilets,] [food-preparation,] [food-serving,] [restrooms,] [laundry areas,] [shower areas,] [areas requiring a high degree of sanitation,] [_____,] [and other high-humidity areas] not otherwise specified except floors, hot metal surfaces, and new prefinished equipment:~~

~~1. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~D. Miscellaneous non-ferrous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment. Match surrounding finish:~~

~~1. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

DIVISION 6: INTERIOR WOOD PAINT TABLE

A. New ~~[~~and Existing, uncoated~~]~~ Wood and plywood not otherwise specified:

~~1. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~3~~1. ~~[~~Institutional Low Odor / Low VOC Latex
New; MPI INT 6.3V-~~G2 (Flat)~~G5 (Semigloss)
MLIT SS Chapter 18~~]~~

B. Existing, previously painted Wood and plywood not otherwise specified:

~~1. [High Performance Architectural Latex~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~3~~1. ~~[~~Institutional Low Odor / Low VOC Latex
MLIT SS Chapter 18~~]~~

C. New ~~[~~and Existing, previously finished or stained~~]~~ Wood and Plywood, except floors; natural finish or stained:

~~1. [Natural finish, oil-modified polyurethane~~
~~MLIT SS Chapter 18]~~

~~2~~1. ~~[~~Stained, oil-modified polyurethane
MLIT SS Chapter 18~~]~~

~~3. [Stained, Moisture Cured Urethane~~

~~MLIT SS Chapter 18]~~

~~D. New [and Existing, previously finished or stained] Wood Floors; Natural finish or stained:~~

~~1. [Natural finish, oil-modified polyurethane~~
~~MLIT SS Chapter 18]~~

~~2. [Natural finish, Moisture Cured Polyurethane~~
~~MLIT SS Chapter 18]~~

~~3. [Stained, oil-modified polyurethane~~
~~MLIT SS Chapter 18]~~

~~4. [Stained, Moisture Cured Polyurethane~~
~~MLIT SS Chapter 18]~~

~~E. New [and Existing, previously coated] Wood floors; pigmented finish:~~

~~1. [Latex Floor Paint~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd Floor Paint~~
~~MLIT SS Chapter 18]~~

~~F. New [and Existing, uncoated] wood surfaces in [toilets,] [food-preparation,] [food-serving,] [restrooms,] [laundry areas,] [shower areas,] [areas requiring a high degree of sanitation,] [_____] [and other high humidity areas] not otherwise specified.:~~

~~1. [As specified in Section 09 96 59 HIGH-BUILD GLAZE COATINGS.]~~

~~2. [MLIT SS Chapter 18]~~

~~3. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~G. Existing, previously painted wood surfaces in [toilets,] [food-preparation,] [food-serving,] [restrooms,] [laundry areas,] [shower areas,] [areas requiring a high degree of sanitation,] [_____] [and other high humidity areas] not otherwise specified:~~

~~1. [As specified in Section 09 96 59 HIGH-BUILD GLAZE COATINGS.]~~

~~2. [MLIT SS Chapter 18]~~

~~3. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~H. New [and Existing, previously finished or stained] Wood Doors; Natural Finish or Stained:~~

~~1. [MLIT SS Chapter 18]~~
~~Note: Sand between all coats per manufacturers recommendations.~~

~~2. [Stained, oil-modified polyurethane~~
~~MLIT SS Chapter 18]~~
~~Note: Sand between all coats per manufacturers recommendations.~~

~~3. [Stained, Moisture Cured Urethane~~
~~MLIT SS Chapter 18]~~
~~Note: Sand between all coats per manufacturers recommendations.~~

~~I. New [and Existing, uncoated] Wood Doors; Pigmented finish:~~

~~1. [Alkyd~~
~~MLIT SS Chapter 18]~~
~~Note: Sand between all coats per manufacturers recommendations.~~

~~2. [Pigmented Polyurethane~~
~~MLIT SS Chapter 18]~~
~~Note: Sand between all coats per manufacturers recommendations.~~

~~J. Existing, previously painted Wood Doors; Pigmented finish:~~

~~1. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~Note: Sand between all coats per manufacturers recommendations.~~

DIVISION 9: INTERIOR ~~PLASTER,~~ GYPSUM BOARD~~, TEXTURED SURFACES~~ PAINT TABLE

A. New ~~[~~and Existing, previously painted~~] [Plaster] [and] [~~Wallboard~~]~~ not otherwise specified:

~~1. [Latex~~
~~MLIT SS Chapter 18]~~

~~2. [High Performance Architectural Latex - High Traffic Areas~~
~~MLIT SS Chapter 18]~~

~~3~~1. ~~[~~Institutional Low Odor / Low VOC Latex
MLIT SS Chapter 18~~]~~

B. New ~~[~~and Existing, previously painted~~] [Plaster] [~~and~~] [~~Wallboard~~]~~ in ~~[~~toilets,~~] [food-preparation,] [food-serving,] [~~restrooms,~~] [~~laundry areas,~~] [~~shower areas,~~] [areas requiring a high degree of sanitation,] [______] [~~and other high humidity areas~~]~~ not otherwise specified.:

~~1. [Waterborne Light Industrial Coating~~
~~MLIT SS Chapter 18]~~

~~2. [Alkyd~~
~~MLIT SS Chapter 18]~~

~~3~~1. ~~[~~Epoxy
MLIT SS Chapter 18~~]~~

-- End of Section --

SECTION 10 14 00.20

INTERIOR SIGNAGE
08/17

PART 1 GENERAL

~~1.1 [Enter Appropriate Subpart Title Here][Enter Appropriate Subpart Title Here]REFERENCES~~

~~The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.~~

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1	(2015) Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191	Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~]~~ ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G~~[, [_____]]~~

SD-10 Operation and Maintenance Data

Approved Manufacturer's Instructions; G~~[, [_____]]~~

Protection and Cleaning; G~~[, [_____]]~~

1.3 EXTRA MATERIALS

~~Provide [_____] extra frames and extra stock of the following: [[_____]~~

~~blank plates of each color and size for sign types [_____].] [[_____] changeable message strips for sign type [_____].]~~ Provide ~~[[_____] paper inserts and [~~one~~][_____]~~ copy of the software for user produced signs and inserts after project completion~~]~~ ~~[~~and equipment necessary for removal of signage parts and pieces.~~]~~

1.4 QUALITY ASSURANCE

1.4.1 Samples

Submit interior signage samples of each of the following sign types showing typical quality, workmanship and color: Directional sign, Standard Room sign, Changeable message strip sign~~, [Facility Recognition Plaque] [_____]~~. The samples may be installed in the work, provided each sample is identified and location recorded.

1.4.2 Detail Drawings

Submit detail drawings showing elevations of each type of sign, dimensions, details and methods of mounting or anchoring, mounting height, shape and thickness of materials, and details of construction. Include a schedule showing the location, each sign type, and message.

1.5 DELIVERY, STORAGE, AND HANDLING

Materials must be packaged to prevent damage and deterioration during shipment, handling, storage and installation. Product must be delivered to the jobsite in manufacturer's original packaging and stored in a clean, dry area in accordance with manufacturer's instructions.

1.6 WARRANTY

Warrant the interior signage for a period of ~~[~~2~~][_____]~~ year~~[~~s~~]~~ against defective workmanship and material. Warranties must be signed by the authorized representative of the manufacturer. Submit warranty accompanied by the document authenticating the signer as an authorized representative of the guarantor. Guarantee that the signage products and the installation are free from any defects in material and workmanship from the date of delivery.

PART 2 PRODUCTS

2.1 STANDARD PRODUCTS

Signs, ~~plaques, directories,~~ and dimensional letters must be the standard product of a manufacturer regularly engaged in the manufacture of such products that essentially duplicate signs that have been in satisfactory use at least 2 years prior to bid opening. Obtain signage from a single manufacturer with edges and corners of finished letterforms and graphics true and clean.

2.2 ROOM IDENTIFICATION/DIRECTIONAL SIGNAGE SYSTEM

2.2.1 Standard Room Signs

Signs must consist of ~~[~~acrylic plastic 2 mm thickness minimum conforming to ~~] [laminated thermosetting Type MP plastic (three-ply melamine plastic laminate with phenolic core)] [6063-T5 extruded aluminum in accordance with] [_____]~~ANSI Z97.1 and must conform to the following:

~~a. Frames must be [[aluminum] [wood] [molded acrylic]], [[flat][radius]] [[3 mm] [6 mm] [_____] thick.]~~

~~b. End caps must be [aluminum] [wood] [molded acrylic] with [round] [square] [_____] style corners.~~

~~c~~a. Units must be frameless. Corners of signs must be ~~[squared] [rounded to [10] [13] [19] [_____]~~ mm radius~~]~~.

~~2.2.2 Changeable Message Strip Signs~~

~~Changeable message strip signs must be of same construction as standard room signs to include a clear sleeve that will accept a paper or plastic insert identifying changeable text. The insert must be prepared [die-cut vinyl letters applied to 0.38 mm rigid vinyl film] [typeset message mounted on paper card stock][typewritten message] [_____].[Provide[paper and] software for creating text and symbols for computers identified by owner for Owner production of paper inserts after project completion.][Furnish one [suction][_____] device to assist in removing face sheet.][Sliding inserts or slide knobs that slide horizontally exposing different graphic information must be provided as identified in the signage placement schedule and [drawings][attachments.]]~~

2.2.2 Type of Mounting For Signs

~~Provide extruded aluminum brackets for hanging, projecting, and double-sided signs. Mounting for framed, hanging, and projecting signs must be by mechanical fasteners. Surface mounted signs must be mounted with [countersunk mounting holes in plaques and mounting screws][1.6 mm thick closed cell vinyl foam with adhesive backing. Adhesive must be transparent, long aging, high tech formulation on two sides of the vinyl foam.][magnetic tape [silicone adhesive]][[hook and loop tape consisting of hooked part on sign back and looped side on mounting surface] [pin mount] for textile surfaces] [_____] fabricated from materials that are not corrosive to sign material and mounting surface.~~Surface mounted signs must be mounted with countersunk mounting holes in plaques and mounting screws.

2.2.3 Graphics

Signage graphics for modular signs must conform to the following:

~~[2.2.3.1 Subsurface Copy~~

~~Copy is transferred to the back face of clear acrylic sheeting forming the panel face to produce precisely formed opaque image. This method bonds all sign elements (color, graphics, lettering, Braille and substrate) into a single unit.~~

~~][2.2.3.2 First Surface Copy Direct Print or Silkscreened (Non-Tactile)~~

~~Message may be applied to panel using the silkscreen process. Silkscreened images must be executed with photo screens prepared from original art. Handcut screens will not be accepted. Original art is defined as artwork that is a first generation reproduction of the specified art. Edges and corners must be clean.~~

~~][2.2.3.3 Surface Applied Photopolymer~~

~~Integral graphics and Braille achieved by photomechanical stratification processes. Photopolymer used for ADA compliant graphics must be of the type that has a minimum durometer reading of 90. Tactile graphics must be raised 0.8 mm from the first surface of plaque by photomechanical stratification process.~~

~~][2.2.3.4 Engraved Copy~~

~~Machine engrave letters, numbers, symbols, and other graphics into panel sign on face to produce precisely formed copy and sharp images, incised to uniform depth. Melamine plastic engraving stock used for ADA compliant graphic must be three-ply lamination contrasting color core meeting ASTM D635.~~

~~][~~2.2.3.1 Graphic Blast Raised Copy

Background is sandblasted to a uniform depth of 0.8 mm leaving raised text and Braille. Background must be painted with polyurethane paint.

~~][2.2.3.2 Embossed~~

~~Methods other than sandblasting such as vacuum formed to create ADA compliant projected graphics.~~

~~][2.2.3.3 [Cast] [Fabricated] [Solid] Aluminum Letters~~

~~Provide [3] [6] [_____] mm thick and fasten to the message panel with concealed fasteners.~~

~~]~~2.2.4 Character Proportions and Heights

Letters and numbers on signs conform to 36 CFR 1191.

~~2.2.5 Tactile Letters, Symbols and Braille~~

~~Raised letters and numbers on signs must conform to 36 CFR 1191.~~

2.2.5 Raised and Braille Characters and Pictorial Symbol Signs (Pictograms)

Raised letters and numbers on signs shall conform to 36 CFR 1191.

~~2.3 STAIR SIGNAGE~~

~~Provide signs on stairs serving three or more stories with special signage within the enclosure at each floor landing conforming to NFPA 101. Indicate the floor level, the terminus of the top and bottom of the stair enclosure, and the identification of the stair enclosure. Also, state the floor level of, and the direction to, exit discharge. Locate the signage inside the enclosure in a position that is visible when the door is in the open or closed position and install in conformance with 36 CFR 1191. The floor level designation must also be tactile in accordance with ICC A117.1 COMM.~~

~~2.4 BUILDING DIRECTORIES~~

~~Building directories must be lobby directories or floor directories, and must be provided with a changeable directory listing consisting of the~~

~~areas, offices and personnel located within the facility. Dimensions, details, and materials of sign and message content must be as shown on the [drawings][attachments][signage placement schedule].~~

~~2.4.1 Header Panel~~

~~Header panel must [have background metal to match frame] [be acrylic with raised acrylic letters][be ES/MP plastic with raised letters] [_____].~~

~~2.4.2 Doors~~

~~2.4.2.1 Door Glazing~~

~~Door glazing must be [in accordance with JIS R 3202, minimum 3 mm thick][clear acrylic sheet 5 mm thick conforming to [_____]][clear polycarbonate sheet 5 mm thick][_____].~~

~~2.4.2.2 Door Construction~~

~~Extruded aluminum door frame must be of same finish as surrounding frame. Corners must be mitered [, reinforced] [, welded], and assembled with concealed fasteners. Hinges must be standard with the manufacturer, in finish to match frames and trim. Glazing must be set in frame with resilient glazing channels.~~

~~2.4.2.3 Door Locks~~

~~Door locks must be manufacturer's standard, and must be keyed alike. Provide two sets of keys.~~

~~2.4.3 Fabrication~~

~~Extruded aluminum frames and trim must be assembled with corners [reinforced] [welded] and mitered to a hairline fit, with no exposed fasteners.~~

~~2.4.4 Illuminated Units~~

~~Illuminated directory units must have concealed internal [top] [back] lighting with [LED] [rapid start fluorescent tube lamp] [_____], internal wiring, and lead at wire for connection. Electrical work must comply with NFPA 70 and must be UL or FM listed. Directory must consist of backlit photo negative directory strips and a black background. Unit must have a tinted [tempered safety solar glass][_____] door.~~

~~2.4.4.1 Construction~~

~~The directory must be [50][100][150] mm deep frame constructed of an [aluminum with [[satin [black][painted][dark bronze][_____] anodized finish]][[red oak][walnut][_____] with [natural][stained] finish]]. Unit must be [[semi][fully] recessed][surface][_____] mounted. Unit must have a [75][_____] mm high header lettering as shown. Unit must have a [10][_____]mm face door frame with concealed hinges and locking system or other secure method. Door frame must [match directory material and finish][_____].~~

~~2.4.4.2 Message Strips~~

~~Message strips must be photo negative type updatable by user. Message~~

~~strips must be [as shown on the drawings] [_____].~~

~~2.4.5 Non-Illuminated Unit~~

~~Directory must consist of a non-illuminated unit with [machine or laser engraved copy in interchangeable acrylic, metal, or high-pressure plastic laminate strips] [screen printed or vinyl copy applied to acrylic, metal, or high-pressure plastic laminate strips] [vinyl or screen printed lettering on plastic film held in interchangeable plastic carriers] [screen printed or vinyl copy laminated to magnetic tape]. Design of unit must be as shown in the drawings.~~

~~2.4.5.1 Construction~~

~~The directory must be constructed of an aluminum [50][100][150] mm deep frame with [satin [black][painted][dark bronze][_____] anodized finish][[red oak][walnut][_____] with [natural][stained] finish]. Unit must be [[semi][fully] recessed][surface][_____] mounted. Unit must have a [75][_____] mm high header lettering as shown. Unit must have a [9.3][_____] mm face door frame with concealed hinges and locking system or other secure method. Door frame must [match directory material and finish][_____].~~

~~2.4.5.2 Message Strips~~

~~[Message strips must be updatable by user.]Message strips must be [sized in accordance with manufacturer's standard] [_____]. Letters and numbers must be provided in accordance with the [drawings] [schedule].~~

~~2.4.6 Electronic Directory System~~

~~Provide [non-interactive][interactive] electronic directory. Electronic directory system must be a complete turnkey system consisting of digital display, hardware, software connected through the local area network (LAN) to a [server][cloud]. Electrical equipment must be UL listed and must comply with NFPA 70. Unit must be [free-standing][wall mounted].~~

~~2.5 METAL PLAQUES~~

~~2.5.1 Cast Metal Plaques~~

~~2.5.1.1 Fabrication~~

~~Cast metal plaques must have the logo, emblem and artwork cast in the [bas relief] [flat relief] [_____] technique. Plaques must be fabricated from [prime aluminum] [bronze] [brass] [_____].~~

~~2.5.1.2 Border~~

~~Border must be [flat band] [plain edge] [bevel] [custom ornamental] [_____].~~

~~2.5.1.3 Finish~~

~~Letter Finish~~	~~[satin] [polished]~~
~~Background Finish~~	~~[[light][dark] aluminum][[dark][ ] bronze]~~
~~Background Texture~~	~~[leather][pebble][smooth][ ]~~

~~2.5.1.4 Mounting~~

~~Mounting must be [concealed] [rosettes and anchors] [rosettes and toggle bolts] [_____].~~

~~2.5.2 Chemically Etched Metal Plaques~~

~~2.5.2.1 Fabrication~~

~~Plaque must be chemically etched one-piece or photochemically engraved metal sheet or plate [aluminum] [brass] [bronze] [zinc] [magnesium] [_____] [_____] mm thick.~~

~~2.5.2.2 Finish~~

~~[Single-etched raised areas must be in [gold-tone] [silver-tone] [bronze-tone] finish and recessed areas must be colorfilled.] [Double-etched raised areas must be [gold-tone] [silver-tone] and recessed textured areas must be [gold-tone] [silver-tone] colorfilled.]~~

~~2.6 DIMENSIONAL BUILDING LETTERS~~

~~2.6.1 Fabrication~~

~~Letters must be [cast][cutout][fabricated channel][molded plastic]. Letters must be [aluminum][bronze][brass][ ]. Package letters for protection until installation.~~

~~2.6.2 Size~~

~~Letter size must be [_____] [as indicated]. Provide letter thickness that is [manufacturer's standard for the size of letter][_____].~~

~~2.6.3 Finish~~

~~Provide [[mill][clear anodized][[light][medium][dark] anodized bronze]] [[polished] bronze with clear coat] [baked enamel] [powder coat][two-component acrylic polyurethane] finish.~~

~~2.6.4 Mounting~~

~~[Threaded studs] [Steel U-bracket, cap screws, and expansion bolts] of number and size recommended by manufacturer, must be supplied for concealed anchorage. Letters which project from the mounting surface must have [stud spacer sleeves] [_____]. Letters, studs, and sleeves must be of the same material. Templates for mounting must be supplied.~~

~~2.7 PRESSURE SENSITIVE LETTERS~~

~~2.7.1 Fabrication~~

~~Ensure that vinyl letter edges and corners of finished letterforms and graphics are true and clean. Do not use letterforms and graphics with rounded positive or negative corners, nicked, cut, or ragged edges.~~

~~2.7.2 Size~~

~~Letter size: [as indicated] [_____].~~

~~2.8 ALUMINUM ALLOY PRODUCTS~~

~~Aluminum extrusions must be at least 3 mm thick, and aluminum plate or sheet must be at least 1.3 mm thick. Extrusions must conform to JIS H 4040; plate and sheet must conform to JIS H 4000. Where anodic coatings are specified, alloy must conform to [_____]. Exposed anodized aluminum finishes must be as shown. Welding for aluminum products must conform to JIS Z 3801, JIS Z 3410, and JIS Z 3841.~~

~~2.9 ANODIC COATING~~

~~Anodized finish must conform to JIS H 8602 as follows:~~

~~a. [Clear (natural).]~~

~~b. [Integral color.]~~

~~c. [Electrolytically deposited color-anodized.]~~

~~2.10 ORGANIC COATING~~

~~Organic coating must conform to JIS K 5906, with total dry film thickness not less than 0.030 mm.~~

~~2.11 FABRICATION AND MANUFACTURE~~

~~2.11.1 Factory Workmanship~~

~~Holes for bolts and screws must be drilled or punched. Drilling and punching must produce clean, true lines and surfaces. Exposed surfaces of work must have a smooth finish and exposed riveting must be flush. Fastenings must be concealed where practicable.~~

~~2.11.2 Dissimilar Materials~~

~~Where dissimilar metals are in contact, the surfaces will be protected to prevent galvanic or corrosive action.~~

2.3 COLOR, FINISH, AND CONTRAST

Color must be ~~[in accordance with Section 09 06 00 SCHEDULES FOR FINISHES]~~ ~~[~~as indicated~~] [_____]~~ in drawings. Finish of all signs must be eggshell, matte, or other non-glare finish as required in handicapped-accessible buildings.

~~2.4 TYPEFACE~~

~~[ADA-ABA compliant font for Room Signs][Helvetica Regular][_____].~~

PART 3 EXECUTION

~~[3.1 PLACEMENT SCHEDULE~~

~~SIGNAGE PLACEMENT SCHEDULE~~				
~~Door/Room Number~~	~~Sign Type~~	~~Text~~	~~Insert(s)~~	~~Symbol/Remarks~~
~~[_______]~~	~~[_______]~~	~~[_______]~~	~~[_______]~~	~~[_______]~~

~~]~~3.1 INSTALLATION

Install signs plumb and true and in accordance with approved manufacturer's instructions at locations shown on the ~~[detail~~ drawings~~] [schedule below] [attachments]. Submit operating instructions outlining the step-by-step procedures required for system operation. The instructions include simplified diagrams for the system as installed, the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Provide each set permanently bound with a hard cover. The following identification must be inscribed on the covers: "OPERATING AND MAINTENANCE INSTRUCTIONS", name and location of the facility, name of the Contractor, and contract number. Submit in accordance with Section 01 78 23 OPERATING AND MAINTENANCE DATA~~. Mounting height and mounting location complies with 36 CFR 1191. Install required blocking. Do not install signs on doors or other surfaces until finishes on such surfaces have been installed. Signs installed on glass surfaces are installed with matching blank back-up plates in accordance with manufacturer's instructions. ~~[Provide illuminated signage in conformance with the requirements of Section 26 51 00 INTERIOR LIGHTING.]~~

Do not install items that show visual evidence of biological growth.

3.1.1 Anchorage

Provide anchorage in accordance with approved manufacturer's instructions. ~~In high humidity interior spaces (for example, bathrooms, locker rooms, pools, trainers) and unconditioned spaces, use corrosion-resistant anchors/fasteners or with approval by the manufacturer, waterproof silicone adhesive.~~ Anchorage not otherwise specified or shown must include slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; ~~toggle bolts and through bolts for masonry;~~ machine carriage bolts for steel~~; lag bolts and screws for wood~~. Provide exposed anchor and fastener materials compatible with metal to which applied with matching color and finish.At interior applications in heavy traffic areas, firmly attach signage to structure walls with tamper-proof fasteners.

a. Signs mounted to painted gypsum board surfaces must be removable for painting maintenance.

~~b. Mount signs mounted to lay-in ceiling grids with clip connections to~~

~~ceiling tees.~~

~~c. Install signs mounted on metal surfaces with magnetic tape.~~

~~d. Install signs mounted on fabric surfaces with hook and loop tape or pin mount.~~

~~e. Install signs to workstation panels with panel clips.~~

3.1.2 Protection and Cleaning

Protect the work against damage during construction. ~~Adjust hardware and electrical equipment for proper operation.~~ Clean glass, frames, and other sign surfaces at completion of signage installation in accordance with the manufacturer's written instructions.

-- End of Section --

SECTION 10 28 13

TOILET ACCESSORIES
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM F2285 (2004; R 2016; E 2016) Standard Consumer Safety Performance Specification for Diaper Changing Tables for Commercial Use

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3864 (2011) Graphical Symbols - Safety Colours and Safety Signs

ISO 7010 (2019) Graphical Symbols - Safety Colours and Safey Signs -

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS R 3220 (2011) Glass in Building - Silvered, Flat-Glass Mirror

JIS Z 2911 (2018) Methods of Test for Fungus Resistance

JIS Z 9103 (2018) Graphic Symbols - Safety Colors and Safety Signs - Range of Chromaticity Coordinates and Measurement Methods for Safety Colors

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-04 Samples

Finishes; G~~[, [_____]]~~

Accessory Items

SD-10 Operation and Maintenance Data

Electric Hand Dryer; G~~[, [_____]]~~

~~[~~1.3 CERTIFICATIONS

1.3.1 ~~Baby~~Diaper Changing Stations

Provide certification that ~~baby~~diaper changing stations meet the performance criteria of ASTM F2285.

Provide certification that ~~baby~~diaper changing stations safety signage meet the requirements of JIS Z 9103, ISO 3864 or ISO 7010.

Provide certification that baby changing stations meet the testing requirements of JIS Z 2911.

~~]~~1.4 DELIVERY, STORAGE, AND HANDLING

Wrap toilet accessories for shipment and storage, then deliver to the jobsite in manufacturer's original packaging, and store in a clean, dry area protected from construction damage and vandalism.

1.5 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a 1 year period.

PART 2 PRODUCTS

2.1 MANUFACTURED UNITS

Provide toilet accessories where indicated in accordance with paragraph SCHEDULE. Porcelain type, tile-wall accessories are specified in Section 09 30 10 CEMENT TILING, QUARRY TILING, AND PAVER TILING. Provide each accessory item complete with the necessary mounting plates of sturdy construction with corrosion resistant surface.

2.1.1 Anchors and Fasteners

Provide anchors and fasteners capable of developing a restraining force commensurate with the strength of the accessory to be mounted and suited for use with the supporting construction. Provide ~~[~~tamperproof design~~] [oval heads]~~ exposed fasteners with finish to match the accessory.

2.1.2 Finishes

Except where noted otherwise, provide the following finishes on metal:

Metal	Finish
Stainless steel	No. 4 satin finish
Carbon steel, copper alloy, and brass	Chromium plated, bright

2.2 ACCESSORY ITEMS

Conform to the requirements for accessory items specified below. Submit fasteners proposed for use for each type of wall construction, mounting, operation, and cleaning instructions and one sample of each other accessory proposed for use. Incorporate approved samples into the finished work, provided they are identified and their locations noted. Submit certificate for each type of accessory specified, attesting that the items meet the specified requirements.

~~2.2.1 Facial Tissue Dispenser (FTD)~~

~~Provide [surface] [recessed] mounted facial tissue dispenser, Type 304 stainless steel face, [satin finish] [bright polished finish]. Secure face of recessed dispenser by friction with suitable spring steel clips. Provide a minimum capacity of [150] [200] [300] two-ply tissues for dispenser.~~

2.2.1 Grab Bar (GB)

Provide an 18 gauge, 32 mm grab bar OD Type 304 stainless steel. Provide form and length for grab bar as indicated. Provide ~~[~~concealed~~]~~ ~~[exposed]~~ mounting flange. Provide grab with ~~[~~satin finish~~] [peened non-slip surface]~~. Furnish installed bars capable of withstanding a 2.225 kN vertical load without coming loose from the fastenings and without obvious permanent deformation. Allow 38 mm space between wall and grab bar.

2.2.2 Medicine Cabinet (MC)

Construct medicine cabinet with cold-rolled carbon steel sheet of minimum 0.76 mm thickness, formed from a single sheet of steel or mechanically formed and spot welded. Provide width, height and depth of cabinet in accordance with paragraph SCHEDULE.

~~2.2.2.1 Sliding Door Cabinet~~

~~Provide [surface mounted vanity] [recessed cabinet] sliding door cabinet assembly with design and lighting arrangement as indicated. Provide a minimum of 2 shelves per cabinet. Provide [a wide] [a narrow] [no] frame mirror.~~

2.2.2.1 Swinging Door Cabinet

Provide swinging door cabinet assembly, including the lighting arrangement, as indicated. Provide ~~[~~surface~~]~~ ~~[recess]~~ mounted assembly. Locate cabinet centrally behind the door with a minimum of two shelves. Provide stainless steel or carbon steel door hinges. Provide permanent type magnets used in door catches. Provide doors ~~[~~with~~]~~ ~~[without]~~ a mirror.

2.2.3 Mirrors, Glass (MG)

Provide clear glass for mirrors conforming to JIS R 3220. Coat glass on one surface with silver coating, copper protective coating, and mirror backing paint. Provide highly adhesive pure silver coating of a thickness which provides reflectivity of 83 percent or more of incident light when viewed through 6 mm thick glass, free of pinholes or other defects. Provide copper protective coating with pure bright reflective copper, homogeneous without sludge, pinholes or other defects, of proper thickness to prevent "adhesion pull" by mirror backing paint. Provide mirror backing paint with two coats of special scratch and abrasion-resistant paint and baked in uniform thickness to provide a protection for silver and copper coatings which will permit normal cutting and edge fabrication.

2.2.4 Mirror, Metal (MM)

Provide a brightly polished stainless steel metal mirror of 0.94 mm minimum thickness, edges turned back 6 mm and recess fitted with tempered hardboard backing, and theft-proof fasteners. Provide size in accordance with paragraph SCHEDULE.

~~2.2.5 Mirror, Tilt (MT)~~

~~Provide surface mounted tilt mirror with full visibility for persons in a wheelchair. Furnish [adjustable] [fixed] tilt mirror, extending at least 100 mm from the wall at the top and tapering to 25 mm at the bottom. Provide size in accordance with [paragraph SCHEDULE] [the drawings] [_____]. Conform to JIS R 3220 and paragraph Glass Mirrors.~~

~~2.2.6 Paper Towel Dispenser (PTD)~~

~~Provide [_____] paper towel dispenser constructed of a minimum [22 gauge carbon steel] [0.7 mm Type 304 stainless steel], [surface] [recessed] mounted. Provide a towel compartment and a [mirror door] [and] [liquid soap dispenser] for each dispenser. Furnish [tumbler key lock] [concealed tumbler key lock] locking mechanism.~~

2.2.5 Combination Paper Towel Dispenser/Waste Receptacle (PTDWR)

Provide ~~[recessed] [~~semi-recessed~~]~~ dispenser/receptacle with a capacity of ~~[400] [~~600~~] [_____]~~ sheets of C-fold, single-fold, or quarter-fold towel. Design waste receptacle to be locked in unit and removable for service. Provide tumbler key locking mechanism. Provide waste receptacle capacity of ~~[45] [68] [_____]~~ L. Fabricate a minimum 0.7 mm stainless steel welded construction unit with all exposed surfaces having a satin finish. Provide waste receptacle that accepts reusable liner standard for unit manufacturer.

2.2.6 Sanitary Napkin Disposer (SND)

Construct a Type 304 stainless steel sanitary napkin disposal with removable leak-proof receptacle for disposable liners. Provide ~~[~~fifty~~] [_____]~~ disposable liners of the type standard with the manufacturer. Retain receptacle in cabinet by tumbler lock. Provide disposer with a door for inserting disposed napkins, ~~[recessed] [~~partition mounted, double access~~] [~~surface mounted~~]~~.

~~2.2.7 Sanitary Napkin and Tampon Dispenser (SNTD)~~

~~Provide sanitary napkin and tampon dispenser [surface mounted] [recessed]. Dispenser, including door of Type 304 stainless steel that dispense both napkins and tampons with a minimum capacity of 20 each. Furnish dispensing mechanism for [complimentary] [coin] operation. Provide coin mechanisms with minimum denominations of [_____] [free]. Hang doors with a full-length corrosion-resistant steel piano hinge and secure with a tumbler lock. Provide keys for coin box different from the door keys.~~

~~2.2.8 Shower Curtain (SC)~~

~~Provide [_____] shower curtain, size to suit conditions. Provide anti-bacterial nylon/vinyl fabric curtain. Furnish [_____] color [as shown in Section 09 06 00 SCHEDULES FOR FINISHES].~~

2.2.7 Shower Curtain Rods (SCR)

Provide Type 304 stainless steel shower curtain rods 32 mm OD by 1.24 mm minimum ~~[~~straight~~] [bent as required]~~ to meet installation conditions.

2.2.8 Soap Dispenser (SD)

Provide soap dispenser ~~[~~surface mounted, liquid type consisting of a vertical Type 304 stainless steel tank with holding capacity of 1.2 L with a corrosion-resistant all-purpose valve that dispenses liquid soaps, lotions, detergents and antiseptic soaps.~~] [surface mounted, powder type constructed of stainless steel or chromium plated zinc die casting, containing a swap feed mechanism and an agitator designed to break up powdered soap, with a minimum capacity of 0.94 L.] [lavatory mounted, liquid type consisting of a polyethylene tank with a minimum 0.94 L holding capacity and a [100 mm] [150 mm] spout length.]~~

2.2.9 Soap Holder (SH)

Provide ~~[~~surface mounted~~] [recessed]~~ Type 304 stainless steel soap holder. Provide stainless steel separate supports.

2.2.10 Shelf, Metal, Heavy Duty (SMHD)

Furnish a minimum 18 gauge stainless steel heavy duty metal shelf with hemmed edges. Provide shelves over 750 mm with intermediate supports. Provide minimum of 16 gauge supports, welded to the shelf, and spaced no more than 750 mm apart.

2.2.11 Shelf, Metal, Light Duty (SMLD)

Support light duty metal shelf between brackets or on brackets. Purpose of brackets is to prevent lateral movement of the shelf. Furnish ~~[450 mm]~~ ~~[~~600 mm~~]~~ long shelf. Provide stainless steel shelf and brackets.

~~2.2.12 Soap and Grab Bar Combination, Recessed (SGR)~~

~~Provide recessed type, Type 304 stainless steel soap and grab bar combination [bright polished finish] [satin finish].~~

~~2.2.13 Hand Sanitizer Dispenser (HSD)~~

~~Provide hand sanitizer dispensers complete with mounting brackets, batteries as recommended by manufacturer, sanitizer solution, and one bottle of refill sanitizer solution for each dispenser installed. Dispenser properties and characteristics:~~

~~a. Wall mounted~~

~~b. Battery operated~~

~~c. Automatic, touchless type that dispenses sanitizer when a hand is placed in proximity of a sensor~~

~~d. Integral tray below the dispensing portal to catch wasted sanitizer~~

~~e. Operated using standard size batteries~~

2.2.12 Towel Bar (TB)

Provide stainless steel towel bar with a minimum thickness of 0.38 mm. Provide minimum 19 mm diameter bar, or 16 mm square. Provide ~~[bright polish] [~~satin~~]~~ finish.

2.2.13 ~~Towel Pin (TP)~~Robe Hook (RH)

Provide ~~towel pin~~robe hook with concealed wall fastenings, and a ~~pin~~hook integral with or permanently fastened to wall flange with maximum projection of 100 mm. Provide ~~[bright polish] [~~satin~~]~~ finish.

2.2.14 Toilet Tissue Dispenser (TTD) and Toilet Paper Holder (TH)

~~Furnish [surface mounted] [recess mounted] toilet tissue holder with two rolls of standard tissue [mounted horizontally] [stacked vertically]. Provide [carbon steel, bright chromium plated] [stainless steel, satin] finish cabinet.~~Furnish surface mounted toilet tissue holder with two rolls of standard tissue stacked vertically with flat top shelf for public toilet (TTD) and single roll with flat top shelf for resident toilet (TH). Provide stainless steel, satin finish cabinet or approved equal.

~~2.2.15 Toilet Tissue Dispenser, Jumbo (TTDJ)~~

~~Provide surface mounted toilet tissue dispenser with 2 rolls of jumbo tissue. Fabricate cabinet of [Type 304, 18 gauge stainless steel with Type 304, 20 gauge stainless steel door] [high-impact plastic body and transparent plastic front cover]. Provide cover with key lock.~~

~~2.2.16 Toothbrush and Tumbler Holder (TTH)~~

~~Provide stainless steel, surface mounted toothbrush and tumbler holder. Furnish holder to hold a minimum of four toothbrushes in a vertical position. Provide 57 plus or minus 3 mm in diameter size of hole for securing tumbler.~~

~~2.2.17 Waste Receptacle (WR)~~

~~Provide Type 304 stainless steel waste receptacle, designed for [recessed] [surface] mounting. Provide reusable liner, of the type standard with the receptacle manufacturer. Provide a minimum [_____] cubic meters capacity. Provide receptacles with push doors and doors for access to the waste compartment with continuous hinges. Furnish [tumbler key lock] [_____] locking mechanism.~~

2.2.15 Toilet Seat Cover Dispenser (TSCD)

Provide Type 304 stainless steel with ~~[recessed mounted] [~~surface mounted~~]~~ toilet seat cover dispensers. Provide dispenser with a minimum capacity of 500 seat covers.

~~2.2.16 Toilet Seat Cover/Tissue Dispenser/Waste Receptacle (TSCTDWR)~~

~~Provide stainless steel and [partition mounted] [recessed mounted]~~

~~[surface mounted] toilet seat cover, tissue dispenser, and waste receptacle combination. Provide a minimum of 500 [seat covers] [seat covers per side] and [2] [4 (2 per side)] standard tissue rolls for each dispenser. Provide a waste receptacle of the reusable liner of type standard with the receptacle manufacturer. Provide receptacle with [_____] cubic meters capacity. Furnish [tumbler key lock] [_____] locking mechanism.~~

2.2.16 Electric Hand Dryer (EHD)

Provide wall ~~mount and electric hand dryer designed to operate at [_____]phase alternating current with a heating element core rating of a maximum [_____]~~mounted electric hand dryer. Provide dryer housing of single piece construction and of ~~[white porcelain enamel] [~~chrome plated steel~~]~~ or ~~[~~baked electrostatically applied epoxy~~] [_____]~~. Submit ~~[~~4~~] [_____]~~ complete copies of maintenance instructions listing routine maintenance procedures and possible breakdowns. Include repair instructions for simplified wiring and control diagrams and other information necessary for unit maintenance.

2.2.17 Diaper Changing Station (DCS)

Provide ~~[recess mount] [~~surface mounted~~]~~ diaper changing station fabricated of high impact plastic with no sharp edges. Provide fold down platform concave to the child's shape, equipped with nylon and hook and loop safety straps and engineered to withstand a minimum static load of~~[~~ 155 kg~~] [113 kg]~~. Provide an integral dispenser for sanitary liners for each unit. Provide pictorial for universal use of safety graphics conforming to JIS Z 2911, ISO 3864 or ISO 7010. Provide stations that comply with these standards: ASTM F2285 Standard Safety Performance Specification for Diaper Changing tables for Diaper Changing Tables for Commercial Use.~~ Furnish color [_____] [as shown in Section 09 06 00 SCHEDULES FOR FINISHES].~~

~~2.2.18 Folding Shower Seat (FSS)~~

~~Folding shower seat must have a frame constructed of type-304 satin finish stainless steel, 16-gauge, 32 mm square tubing, and 18-gauge, 25 mm diameter seamless tubing. Seat must be constructed of one-piece, 13 mm thick water-resistant, ivory colored solid phenolic with black edge. Clearance between back of shower seat and wall must be 38 mm to comply with ADA Accessibility Guidelines (ADAAG). Seat supports must not come into contact with the floor. Seat must be able to lock in upright position when not in use. Seat must be attached to wall by two 75 mm diameter mounting flanges constructed of type-304, 5 mm thick stainless steel with satin finish. Manufacturer's service and parts manual must be provided to building owner/manager upon completion of project.~~

2.2.18 Mop and Broom Holder (MH)

Stainless steel with grip jaw cam mechanism securing ~~[~~3~~][4][5]~~ mop or broom handles. ~~[~~Also includes ~~[~~hooks~~][~~ and ~~][~~storage shelf~~]~~.~~]~~

PART 3 EXECUTION

3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Provide the same finish for the surfaces of fastening devices exposed after installation as the attached accessory. Install accessories at the location and height indicated. Protect exposed surfaces of accessories with strippable plastic or by other means until the installation is accepted. After acceptance of accessories, remove and dispose of strippable plastic protection. Coordinate accessory manufacturer's mounting details with other trades as their work progresses. ~~[~~Use sealants for brackets, plates, anchoring devices and similar items in showers (a silicone or polysulfide sealant) as they are set to provide a watertight installation.~~]~~ After installation, thoroughly clean exposed surfaces and restore damaged work to its original condition or replace with new work.

3.1.1 Recessed Accessories

Fasten accessories with wood screws to studs, blocking or rough frame in wood construction. Set anchors in mortar in masonry construction. Fasten to metal studs or framing with sheet metal screws in metal construction.

3.1.2 Surface Mounted Accessories

Mount on concealed backplates, unless specified otherwise. Conceal fasteners on accessories without backplates. Install accessories with sheet metal screws or wood screws in lead-lined braided jute, PTFE or neoprene sleeves, or lead expansion shields, or with toggle bolts or other approved fasteners as required by the construction. Install backplates in the same manner, or provide with lugs or anchors set in mortar, as required by the construction. Fasten accessories mounted on gypsum board and plaster walls without solid backing into the metal or wood studs or to solid wood blocking secured between wood studs, or to metal backplates secured to metal studs.

3.2 CLEANING

Clean material in accordance with manufacturer's recommendations. Do not use alkaline or abrasive agents. Take precautions to avoid scratching or marring exposed surfaces.

~~3.3 SCHEDULE~~

~~Accessories Required~~						
~~Room or Space~~	~~MG~~	~~PTD~~	~~SMLD~~	~~SD~~	~~SH~~	~~TTD~~
~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~

-- End of Section --

SECTION 10 44 16

FIRE EXTINGUISHERS
11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 101	(2024) Life Safety Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.106	Flammable Liquids
29 CFR 1910.157	(2003) Portable Fire Extinguishers

UL SOLUTIONS (UL)

UL 299	(2024) Dry Chemical Fire Extinguishers

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fire Extinguishers~~[~~; G~~, [_____]]~~

Accessories~~[~~; G~~, [_____]]~~

Cabinets~~[~~; G~~, [_____]]~~

Wall Brackets~~[~~; G~~, [_____]]~~

Schedule~~[~~; G~~, [_____]]~~

SD-03 Product Data

Fire Extinguishers~~[~~; G~~, [_____]]~~

Accessories~~[~~; G~~, [_____]]~~

Cabinets~~[~~; G~~, [_____]]~~

Wall Brackets~~[~~; G~~, [_____]]~~

Replacement Parts List~~[~~; G~~, [_____]]~~

SD-04 Samples

Equipment Samples~~[~~; G~~, [_____]]~~

SD-07 Certificates

Fire Extinguishers Certifications~~[~~; G~~, [_____]]~~

Manufacturer's Warranty with Inspection Tag~~[~~; G~~, [_____]]~~

1.3 DELIVERY, STORAGE, AND HANDLING

Protect materials from weather, soil, and damage during delivery, storage, and construction.

Deliver materials in their original packages, containers, or bundles bearing the brand name and the name and type of the material.

~~[Provide portable fire extinguishers in compliance with NFPA 505 for all ancillary vehicles where Fire Safety Standard for Powered Industrial Trucks, including type designations, special conditions relating to areas of use, conversions, maintenance, or specific operations apply.~~

~~]~~

1.3.1 Samples

Provide the following equipment samples: One of each type of fire extinguisher being installed; one full-sized sample of each type of cabinet being installed; three samples of wall brackets and accessories of each type being used.

Use approved samples for installation, with proper identification and storage.

1.4 WARRANTY

Guarantee that Fire Extinguishers are free of defects in materials, fabrication, finish, and installation and that they will remain so for a period of not less than ~~[_____]~~6 years after completion.

Submit the manufacturer's warranty with inspection tag.

1.5 PROJECT SCHEDULE

For fire extinguishers. Coordinate final fire extinguisher schedule with fire protection cabinet schedule to ensure proper fit and function. Use same designations indicated on Drawings.

PART 2 PRODUCTS

Submit fabrication drawings consisting of fabrication and assembly details performed in the factory and product data for the following items: Fire Extinguishers; Accessories, cabinets, Wall Brackets.

2.1 SYSTEM DESCRIPTION

2.1.1 Types

Submit fire extinguishers certifications showing compliance with local codes and regulations.

Provide fire extinguishers conforming to NFPA 10. Provide quantity and placement in compliance with the applicable sections of ~~NFPA 1,~~ NFPA 101, ~~[NFPA 99], [NFPA 241], [NFPA 303], [NFPA 385], [NFPA 409], [NFPA 418],[~~ 29 CFR 1910.106~~]~~, and 29 CFR 1910.157.

~~[Provide [stored-pressure] [cartridge] [hand-pump] water type fire extinguishers.~~

~~][Provide [foam] type fire extinguishers.~~

~~][Provide carbon-dioxide type fire extinguishers compliant with UL 154.~~

~~][~~Provide dry chemical type fire extinguishers compliant with UL 299. Extinguishers shall be rated 40A:80B:C.

~~][Provide wet chemical type fire extinguishers compliant with UL 8.~~

~~][Provide clean agent type fire extinguishers compliant with UL 2129.~~

~~][Provide dry powder type fire extinguishers.~~

~~][Provide water mist type fire extinguishers compliant with UL 626.~~

~~]~~2.1.2 Material

Provide ~~[~~corrosion-resistant steel~~] [aluminum] [enameled steel] [_____]~~ extinguisher shell.

2.1.3 Size

~~[9.5 liter extinguishers.~~

~~][1.1 kilogram extinguishers.~~

~~][[2.3] [~~4.5~~] [6.8] [9] [13.6]~~ kilogram extinguishers.

~~]~~2.1.4 Accessories

~~[~~ Forged brass valve

~~][Fusible plug~~

~~][Safety release~~

~~][Antifreeze~~

~~][~~ Pressure gage

~~]~~2.2 EQUIPMENT

2.2.1 Cabinets

2.2.1.1 Material

Provide ~~[~~enameled steel~~] [corrosion-resistant steel] [aluminum]~~ cabinets.

2.2.1.2 Type

~~[~~ Provide ~~[~~recessed~~] [trimless] [surface]~~ type cabinets.

~~] [Provide semi-recessed cabinet for a [150] [100] millimeter wall.~~

~~][Provide [recessed] [trimless] [surface] bubble type cabinets.~~

~~][Provide a fire rated cabinet, listed and labeled to comply with ASTM E814 for fire resistance wall rating.]~~

2.2.1.3 Size

Dimension cabinets to accommodate the specified fire extinguishers.

2.2.2 Wall Brackets

Provide~~[running-board][spring-clip][~~ wall-hook~~]~~ fire extinguisher wall brackets.

Provide wall bracket and accessories as approved.

2.2.2.1 Identification

Provide lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. Locate as indicated by the drawings.

Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

Orientation: ~~[~~Vertical~~][~~ or Horizontal~~]~~.

PART 3 EXECUTION

3.1 INSTALLATION

Install Fire Extinguishers where indicated on the drawings. Verify exact locations prior to installation.

Provide extinguishers which are fully charged and ready for operation upon installation. Provide extinguishers complete with Manufacturer's Warranty with Inspection Tag attached.

Install fire extinguishers in locations indicated and in compliance with requirements of authorities having jurisdiction.

Comply with the manufacturer's recommendations for all installations.

3.2 PROTECTION

3.2.1 Repairing

Remove and replace damaged and unacceptable portions of completed work with new work at no additional cost to the Government.

Submit replacement parts list indicating specified items replacement part, replacement cost, and name, address and contact for replacement parts distributor.

3.2.2 Cleaning

Clean all surfaces of the work, and adjacent surfaces which are soiled as a result of the work. Remove from the site all construction equipment, tools, surplus materials and rubbish resulting from the work.

-- End of Section --

SECTION 12 21 00

~~WINDOW BLINDS~~WINDOW TREATMENTS
08/17

PART 1 GENERAL

1.1 SUMMARY

Provide window treatment, conforming to NFPA 701, complete with necessary brackets, fittings, and hardware. Provide each window treatment type as a complete unit in accordance with paragraph WINDOW TREATMENT PLACEMENT SCHEDULE. Mount and operate equipment in accordance with manufacturer's instructions. Completely cover windows to receive a treatment.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

Contractor may substitute compatible JIS / JASS standards for non-Japanese standards, as approved by the Contracting Officer's Representative.

In addition to the American Standards and criteria referenced in the following specification section, the Contractor shall confirm to the requirements of the Japanese Standards as listed in Section 01 42 00 SOURCES FOR REFERENCE PUBLICATIONS, paragraph EQUIVALENT JAPANESE STANDARDS. Where conflicts arise between the American and Japanese Standards, the more stringent criteria shall be used.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~

~~JIS A 1901~~	~~(2015) Determination of the Emission of Volatile Organic Compounds and Aldehydes by Building Products - Small Chamber Test Method~~

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701	(2019) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.][~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES

SD-02 Shop Drawings

Installation

SD-03 Product Data

Window Blinds

SD-04 Samples

Window Blinds; G

Valance; G

SD-06 Test Reports

Window Blinds

SD-07 Certificates

SD-08 Manufacturer's Instructions

Window Blinds

SD-10 Operation and Maintenance Data

Window Blinds~~; G[, [_____]]~~

~~[1.4 CERTIFICATIONS~~

~~1.4.1 Window Blinds~~

~~Provide products certified to meet indoor air quality requirements by JIS A 1901 or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.~~

~~]~~1.4 DELIVERY, STORAGE, AND HANDLING

Deliver components to the jobsite in the manufacturer's original packaging with the brand or company name, item identification, and project reference clearly marked. Store components in a dry location that is adequately ventilated and free from dust, water, or other contaminants and has easy access for inspection and handling. Store materials flat in a clean dry area with temperature maintained above 1 0 degrees C. Do not open containers until needed for installation unless verification inspection is required.

1.5 WARRANTY

Provide manufacturer's standard performance guarantees or warranties that extend beyond a 1 year period.

PART 2 PRODUCTS

2.1 WINDOW BLINDS

Provide each blind, including hardware, accessory items, mounting brackets and fastenings, as a complete unit produced by one manufacturer. Unless otherwise indicated, all parts will be the same color and will match the color of the blind slat. Treat steel features for corrosion resistance. Submit product data and samples of each type and color of window treatment. Provide ~~[~~slat~~][louver]~~ and louver samples 150 mm in length for each color.~~[Window blinds must meet emissions requirements of JIS A 1901 (use the office or classroom requirement, regardless of space type). Provide certification or validation of indoor air quality for window blinds.]~~

2.1.1 Horizontal Blinds

Provide horizontal blinds with~~[50 mm][~~ 25 mm~~]~~ slats. Blind units must be capable of nominally 180 degree partial tilting operation and full-height raising. Blinds must be ~~[~~inside~~][outside]~~ mount. ~~Provide tapes for 50 mm slats with longitudinal reinforced vinyl plastic in 1-piece turn ladder construction.~~ Tapes for 25 mm slats must be braided polyester or nylon.

2.1.1.1 Head Channel and Slats

Provide head channel made of~~[~~ steel or~~]~~ aluminum with corrosion-resistant finish nominal~~[0.46 mm for 50 mm][~~ 0.61 mm for 25 mm~~]~~ slats. Provide slats of aluminum, not less than ~~[0.203][~~0.152~~][0.813]~~ mm thick, and of sufficient strength to prevent sag or bow in the finished blind. Provide a sufficient amount of slats to assure proper control, uniform spacing, and adequate overlap. Enclose all hardware in the headrail.

2.1.1.2 Controls

A transparent tilting wand will be provided to tilt the slats, it will hang vertically by its own weight, and will swivel for easy operation. Provide a tilter control of enclosed construction. Provide moving parts and mechanical drive made of compatible materials which do not require lubrication during normal expected life. The tilter will tilt the slats to any desired angle and hold them at that angle so that any vibration or movement of ladders and slats will not drive the tilter and change the angle of slats. Include a mechanism to prevent over tightening. Provide a wand of sufficient length to reach to within 1500 mm of the floor. ~~[~~ Provide cordless blinds or blinds with cords that are out of reach of children and strangle proof.~~]~~

2.1.1.3 Intermediate Brackets

Provide intermediate brackets for installation, as recommended by the manufacturer, of blinds over ~~[~~1200~~][1500][2100]~~ mm wide.

2.1.1.4 Bottom Rail

Provide bottom rail made of corrosion-resistant steel with factory applied finish. Provide closed oval shaped bottom rail with double-lock seam for maximum strength. Bottom rail and end caps to match slats in color.

2.1.1.5 Braided Ladders

Provide braided ladders of 100 percent polyester yarn, color to match the slat color. Space ladders 15.2 slats per 300 mm of drop in order to provide a uniform overlap of the slats in a closed position.

~~2.1.1.6 Hold-Down Brackets~~

~~Provide universal type hold-down brackets for sill or jamb mount where~~

~~indicated on placement list.~~

~~2.1.2 Light Control and Privacy Blinds~~

~~In addition to requirements for horizontal blinds, provide each unit with a feature that offers hidden slat holes for maximum light control and privacy.~~

2.1.2 Vertical Blinds

Provide vertical blind units capable of nominal 180 degree partial tilting operation and full stackback. Provide blinds that are listed by the manufacturer as designed for heavy duty strength applications including heavy duty hardware. Provide ~~[ceiling][~~wall~~]~~ mounted vertical blinds with ~~[~~outside~~][inside]~~ brackets. Provide blinds that are ~~[sill][~~floor~~]~~ length. Outside mount type installation must provide adequate overlap to control light and privacy.

2.1.2.1 Louvers

Provide louvers ~~[~~which are fire resistant solid vinyl, UV stable, and impact resistant.~~][which are flame retardant fabric having straight, flat, unfrayed edges and flat, without noticeable twists. Provide a weight at the bottom of the louver without the insert discoloring the fabric.] [which are groover extruded from solid vinyl with clear non-yellowing channel lips to accept fabric inserts. Provide fabric inserts that are flame retardant and colorfast.]~~ Louvers that are~~[~~ 90 mm must overlap not less than 10 mm~~][50 mm must overlap not less than mm]~~ and be dimensionally stable.

2.1.2.2 Carriers

Provide carriers to support each louver made of molded plastic to transverse on self-fabricated wheels for smooth, easy operation. The hook of the carrier must have an automatic latch to permit easy installation and removal of the louver, and to securely lock the louver for tilting and traversing.

2.1.2.3 Headrail System

Provide headrail system not less than 1.19 mm thick and made of anodized aluminum alloy or 0.635 mm thick phosphate treated steel with a baked on ivory gloss enamel paint finish. Provide a headrail that extends the full width of the blind and can be closed with an end cap at each end. One cap will contain the traversing and tilting controls. The opposite cap will house the pulley for the traversing cord.

2.1.2.4 Valance

Attach the manufacturer's standard valance to the headrail by metal or plastic holders which grip the top and bottom edge of the valance and accept an insert of the same material as the slats. Provide sufficient clearance behind the valance to permit the louvers to tilt without interference. Extend the headrail cover the full width of the blind.~~[Provide returns].~~

2.1.2.5 Controls

Provide tilting and traversing controls that hang compactly at the side of the blinds and reach within 1500 mm of the floor. Provide ~~[~~tilt/traverse control~~][bead chain tilting control]~~ that tilts all vanes simultaneously to any desired angle and hold them at that angle. Provide louvers that traverse ~~[~~one way to the right~~] [one way to the left] [two-way split]~~ unless indicated otherwise.~~[~~ The traversing control cord will be minimum 1.78 mm~~-~~ in diameter with a minimum breaking strength of 556 N. Anchor the cord to a lead carrier linked to all adjacent carriers.~~]~~ Provide louvers that traverse along the headrail by pulling one side of the looped cord ~~[~~fastened to a cord tension pulley~~][or][a fiberglass wand that tilts the louvers by turning the wand and traverses the louvers by using the wand as a control]~~. Sliding glass doors will have a one way draw with stackback occurring opposite door openings.

2.1.2.6 Connectors and Spacers

The connector must be flexible, smooth and flat to slide unhindered when carriers move independently of each other, and to nest compactly when carriers are stacking. Relate the length of the links to the louver width in order to equally space the traversing louvers, to maintain uniform and adequate overlap of louvers, and to fully cover the width of the opening. Cords must be out of reach of children and strangle-proof.

2.1.2.7 Intermediate Brackets

Provide intermediate installation brackets for blinds over 1575 mm wide.

2.2 COLOR

Provide color, pattern and texture ~~[in accordance with Section 09 06 00 SCHEDULES FOR FINISHES][~~as indicated~~][~~ selected from manufacturer's standard colors~~][[______]~~. Color listed is not intended to limit the selection of equal colors from other manufacturers.~~]~~

2.3 CURTAIN RAILS

Material: Aluminum, Type: Double Bracket Ceiling type or Two rows of Single type.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

3.2 WINDOW TREATMENT PLACEMENT SCHEDULE

~~[All exterior windows include [______].] [Provide window covering as follows:~~As scheduled on the drawings.

~~Room Number/Name~~	~~Window Covering Type~~	~~Vertical Blind Draw Direction~~	~~Window Type~~	~~Quantity~~
~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~

~~]~~

3.3 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Submit drawings showing fabrication and Installation details. Show layout and locations of track, direction of draw, mounting heights, and details. Provide Manufacturer's Instructions and Operation and Maintenance Data. Perform installation of window blinds in accordance with the ~~approved~~ detail drawings and manufacturer's installation instructions. Install units level, plumb, secure, and at proper height and location relative to window units. Provide and install supplementary or miscellaneous items in total, including clips, brackets, or anchorages incidental to or necessary for a sound, secure, and complete installation. Do not start installation until completion of room painting and finishing operations.

3.4 CLEAN-UP

Upon completion of the installation, inspect window treatments for soiling, damage or blemishes; and adjust them for form and appearance and proper operating condition. Repair or replace damaged units as directed by the Contracting Officer. Isolate metal parts from direct contact with concrete, mortar, or dissimilar metals. ~~Ensure blinds installed in recessed pockets can be removable without disturbing the pocket. The entire blind, when retracted, must be contained behind the pocket.~~ For blinds installed outside the jambs and mullions, overlap each jamb and mullion 20 mm or more when the jamb and mullion sizes permit. Include all hardware, brackets, anchors, fasteners, and accessories necessary for a complete, finished installation.

-- End of Section --

SECTION 14 21 23

ELECTRIC TRACTION PASSENGER ELEVATORS
05/16, CHG 1: 05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

Contractor may substitute compatible JIS / JASS standards for non-Japanese standards, as approved by the Contracting Officer's Representative.

In addition to the American Standards and criteria referenced in the following specification section, the Contractor shall confirm to the requirements of the Japanese Standards as listed in Section 01 42 00 SOURCES FOR REFERENCE PUBLICATIONS, paragraph EQUIVALENT JAPANESE STANDARDS. Where conflicts arise between the American and Japanese Standards, the more stringent criteria shall be used.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A17.1/CSA B44 (2021) Safety Code for Elevators and Escalators

ASME A17.2 (2020) Guide for Inspection of Elevators, Escalators, and Moving Walks Includes Inspection Procedures for Electric Traction and Winding Drum Elevators, Hydraulic Elevators, and Escalators and Moving Walks

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020) Structural Welding Code - Steel

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2018) International Building Code

NATIONAL ELEVATOR INDUSTRY, INC. (NEII)

NEII-1 (2000; R thru 2017) Building

Transportation Standards and Guidelines, including the Performance Standards Matrix for New Elevator Installation

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
NFPA 70E	(2021) Standard for Electrical Safety in the Workplace
NFPA 72	(20~~19; TIA 19-1; ERTA 1 2019; TIA 21-1; ERTA 1 2021~~22) National Fire Alarm and Signaling Code
NFPA 101	(2021) Life Safety Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-490-06	(2018, with Change 1, 2021) Elevators
UFC 3-560-01	(2017, with Change 2, 2019) Operations and Maintenance: Electrical Safety

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191	Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

1.2 SUBMITTALS

Government approval is required for submittals with a "G" ~~or "S" classification~~designation;~~.~~ Submittals not having a "G" ~~or "S" classification~~designation; are ~~[for Contractor Quality Control approval.][~~ for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Elevator System; G~~[, [_____]]~~

Elevator Components; G~~[, [_____]]~~

Elevator Machine; G~~[, [_____]]~~

Elevator Controller; G~~[, [_____]]~~

Wiring Diagrams; G~~[, [_____]]~~

SD-03 Product Data

Elevator and Accessories~~[~~; G~~[, [_____]]]~~

Elevator Components~~[~~; G~~[, [_____]]]~~

Data Sheets~~[~~; G~~[, [_____]]]~~

Elevator Microprocessor Controller; G~~[, [_____]]~~

SD-05 Design Data

Emergency Power Systems; G

Heat Loads; G

Reaction Loads; G

SD-07 Certificates

Elevator Parts and Components Price Lists~~; G[, [_____]]~~

Warranty; G

Endorsement Letter; G

Welders' Qualifications; G

Elevator Controller Certification; G~~[, [_____]]~~

SD-10 Operation and Maintenance Data

Elevator, Data Package 4; G~~[, [_____]]~~

Maintenance Control Program (MCP); G~~[, [_____]]~~

Software and Documentation; G~~[, [_____]]~~

Submit in accordance with Sections 01 78 23 OPERATION AND MAINTENANCE DATA~~ and 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI)~~.

1.2.1 Shop Drawing Requirements

Provide assembly and arrangement of elevators, accessories, and elevator components. Show location of elevator machine in elevator machine room (MR) or machinery space (MS). Show location of elevator controller in elevator machine room or elevator control room (CR). Provide details for materials and equipment, including but not limited to operating and signal fixtures, doors, door and car frames, car enclosure, controllers, motors, guide rails and brackets, layout of hoistway in plan and elevation, and other layout information and clearance dimensions.

1.2.2 Product Data Requirements

Provide manufacturers' product data for all elevator components, including but not limited to the following: elevator controller, hoist machine and drive motor, design counterbalance, hoist ropes and shackles, overspeed governor, emergency braking system, car and hall fixture buttons and switches, cab, machine room, control room, and machinery space communication devices, door operator, door protection system, and car and counterweight roller guides and buffers. For data sheets, provide document identification number or bulletin number, published or copyrighted prior to the date of contract bid opening. Provide controller manufacturer's published procedures for performance of each and all testing required by ASME A17.1/CSA B44 and UFC 3-490-06.

1.2.3 Design Data

1.2.3.1 Reaction Loads

Provide calculations by registered professional engineer for reaction loads imposed on building by elevator system. Calculations must comply with ASCE 7-16 and ASME A17.1/CSA B44

1.2.3.2 Heat Loads

Provide calculations from elevator manufacturer, or by registered professional engineer, for total anticipated heat loads generated by all of the elevator equipment.

1.2.3.3 Emergency Power Systems

Where the facility does have an emergency power system, confirm the elevators that will be connected to the emergency power system. Confirm the complete emergency power system and sequence of operation for all elevators, including elevator sequential operation and operation of the elevator lobby manual selection switch. Provide wiring diagrams for building emergency power interface with elevator controls. For elevators not supplied by an emergency power system, provide manufacturers' product data for auxiliary power systems.

1.2.4 Welders' Requirements

Comply with AWS D1.1/D1.1M, Section 5. Include certified copies of field welders' qualifications. List welders' names with corresponding code marks to identify each welder's welding work

1.2.5 Maintenance Control Program (MCP)

For each elevator, prepare and provide a written Maintenance Control Program (MCP) that complies with ASME A17.1/CSA B44 Section 8.6, including written documentation that details the test procedures for each and every test that is required to be performed by ASME A17.1/CSA B44. Assemble all MCP documentation, and supporting technical attachments, in a single MCP package and provide in both electronic and hard copy. Assemble entire hard copy MCP in 3-ring binders. For each elevator provided, the MCP must include only documentation and instruction that apply to the elevator specified.

For each elevator, provide an additional, separate binder that includes all maintenance, repair, replacement, call back, and other records required by ASME A17.1/CSA B44. The records binder must be kept in the elevator mechanical room, maintained by elevator maintenance and service personnel, and be available at all times to authorized personnel.

Provide detailed information regarding emergency service procedures and elevator installation company personnel contact information. Provide a listing of all tools to be provided to the Contracting Officer as components of the elevator system.

1.3 QUALITY ASSURANCE

1.3.1 Qualification

Provide a designed and engineered elevator system by an elevator contractor regularly engaged in the installation of elevator systems. Provide elevator components manufactured by companies regularly engaged in the manufacture of elevator components. Utilize only licensed and certified elevator personnel for the installation, adjusting, testing, and servicing of the elevators.

1.3.1.1 Elevator Contractor's Elevator Technicians

~~For elevator installations in the United States, including United States territories, perform all elevator related work under the direct guidance of a state certified elevator technician with a minimum of three years of experience in the installation of elevator systems of the type and complexity specified in the contract documents. Provide an endorsement letter from the elevator manufacturer, certifying that the elevator specialist is qualified. All elevator technicians must carry a current certification issued by one of the following organizations:~~Perform all elevator related work under the direct guidance of a Japan or US licensed and certified elevator technician with a minimum of three years of experience in the installation of elevator systems of the type and complexity specified in the contract documents. Provide an endorsement letter from the elevator manufacturer, certifying that the elevator specialist is qualified. All elevator technicians must carry a current certification issued by one of the following organizations:

a. National Association of Elevator Contractors (NAEC)

b. National Elevator Industry Education Program (NEIEP)

c. Japan Elevator Association (JEA), and applicable JEAS

d. The Japan Building Equipment and Elevator Center Foundation (JBEECF)

1.3.2 Manufacturers' Technical Support

Provide elevator components from manufacturers that provide factory training and online and live telephone elevator technical support to any elevator installation, service, and maintenance contractor. Provide elevator components from manufacturers that guarantee accessibility to all replacement and repair parts and components to any elevator installation, service, and maintenance contractor. Use only elevator component manufacturers that provide current published price lists for all elevator parts and components.

1.3.3 Operation and Maintenance Data

Assemble all shop drawing and product data material into O&M Data Packages in accordance with Article SUBMITTALS. Provide two complete O&M Data Packages in hard copy and two complete electronic O&M data packages on separate CDs, in PDF format. Provide all O&M Data Packages to Contracting Officer. Include controller diagnostic documentation and software as required under Article CONTROL EQUIPMENT.

1.3.4 Wiring Diagrams

Provide complete wiring diagrams and sequence of operations, which show electrical connections and functions of elevator systems. Provide one set (279 mm by 432 mm minimum size) of wiring diagrams, with individual sheets laminated in plastic and assembled in binder, to be stored in the machine room or control room cabinet. Provide one additional hard copy set and two complete electronic sets on separate CDs, in PDF format. Provide all wiring diagram sets to the Contracting Officer. Coded diagrams are not acceptable unless fully identified.

1.3.5 Machine Room/Control Room Cabinet

For storage of O&M Data Packages and Wiring Diagrams, provide locking metal cabinet with a minimum size of 508 mm W by 305 mm D by 762 mm H. Cabinet must be sized large enough to accommodate all O&M Data and hardware required in paragraphs OPERATION AND MAINTENANCE DATA and WIRING DIAGRAMS. Secure cabinet to machine room or control room wall.

1.4 NEW INSTALLATION SERVICE

Provide elevator warranty service in accordance with the manufacturer's maintenance plan, warranty requirements, and applicable safety codes, for a period of 12 months after the date of acceptance by Contracting Officer. Perform this work during regular working hours. Provide supplies and parts to keep elevator system in operation. Perform service only by factory trained personnel. Provide ~~[~~Monthly~~][Bi-weekly]~~ services to include repairs, adjustments, greasing, oiling, and cleaning. Provide service log in elevator machine room or control room cabinet and update ~~[~~ Monthly~~][Bi-weekly]~~, throughout the one-year warranty period.

Provide 24-hour emergency service, with ~~[~~one hour~~][two hour]~~ on-site response time, during this period without additional cost to the Government.

1.4.1 Periodic Elevator Certification Inspection and Testing

Provide elevator mechanic to support ~~[NAVFAC]~~QEI or JBEECF Certified Elevator Inspector in the periodic six-month and the annual Category 1 elevator certification inspection and testing. Perform Category 1 inspection and testing no greater than 30 days prior to the end of the warranty period. Perform all elevator certification testing in the presence of QEI or JBEECF Certified Elevator Inspector.

In conjunction with the testing noted above, test systems for Emergency Power Operation, Earthquake Emergency Operation, and Hospital Emergency Commandeering Service Operation, as applicable. Schedule so that testing does not interfere with building operations.

1.5 FIRE PROTECTION SYSTEM

Coordinate interface between building fire protection system and elevator controls.

Additional fire protection requirements are located in: ~~[Section 28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE;] [Section 28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE;] [Section 28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE;] [Section 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE;] [Section~~

~~28 31 02.00 20 FIRE ALARM REPORTING SYSTEMS - DIGITAL COMMUNICATORS;] [~~ Section 21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION;~~] [_____]~~ and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

1.5.1 Fire Alarm Initiating Devices

Fire alarm initiating devices are specified in ~~[Section 28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE] [Section 28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE] [Section 28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE] [~~Section 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE~~] [_____],~~ including conduit and wiring from each detector to fire protection addressable modules in elevator machine room or control room.

1.5.2 Fire Sprinklers

Provide fire sprinklers in accordance with all applicable safety codes and with ~~[~~Section 21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION~~][_____]~~. Provide shutoff valve, check valve, and non-adjustable, zero time-delay flow switch, in each sprinkler line immediately outside of each machine room, control room, and hoistway, as applicable. Provide inspectors' test valve for periodic testing of flow switch and shunt trip disconnect.

Pipe sprinkler piping serving these spaces in a series manner with no laterals. Locate inspectors' test connection at the end of pipe runs such that operation of the test connection will purge air from system piping.

1.5.3 Shunt Trip Disconnect

Provide flow switches specified in paragraph FIRE SPRINKLERS to comply with ASME A17.1/CSA B44 and NFPA 72 for shunt trip of the main line power supply. For each elevator, provide control wiring connecting the flow switch to a shunt trip equipped circuit breaker located in the elevator machine room or control room. Upon flow of water, flow switch will instantaneously cause opening of the shunt-trip circuit breaker and remove power from the elevator. Flow switch must also send a signal to fire alarm control panel to indicate water flow condition.

1.5.4 Demolition

Provide complete demolition of existing two (2) Elevators in Camp Zama Tower B743. Demolition scope per contract drawings and includes but is not limited to: Elevator car controllers, elevator hoist machines, elevator hoist ropes, governors, machine bedplates, motor drive systems (inverters and transformers), seismic switch, counterweight frames, sub-weights, buffers, car frames, safeties, car platform, car enclosure, car interior finishes, car/lobby fixtures, hoistway entrance frames, hoistway door panels, car sills, car flooring, rope compensation, deflector sheaves, governor tail weights, door operators, hoistway/car door accessories, buffer channel support, and machine room/hoistway wiring. Dispose per EPA and OSHA requirements and per local laws and ordinances."

PART 2 PRODUCTS

2.1 ELEVATOR DESCRIPTION

Provide elevator system that complies with ASME A17.1/CSA B44 in its entirety, ASME A17.2 in its entirety, and additional requirements specified herein. Provide elevator system that meets or exceeds the NEII-1 Ride Quality Performance Standards Matrix (RQPSM). For elevator speeds of 500 fpm and higher, comply with the RQPSM "High Performance" criteria. For elevator speeds 350 fpm, up to but not including 500 fpm, comply with the RQPSM "Intermediate Performance" criteria.

Provide and install elevators in accordance with 36 CFR 1191 - ABAAS, ICC IBC, IEEE C62.41, NFPA 70 and NFPA 101 requirements.

2.1.1 Elevator Design Parameters

2.1.1.1 Elevator No.~~[_____]~~B - Emergency Medical Service Accessibility (EMSA)

Provide elevator(s) with minimum size and arrangement to accommodate an ambulance stretcher 610 mm by 2134 mm with not less than 127 mm radius corners, in the open, horizontal position.

a. Type: ~~[~~Geared~~] [Gearless]~~

b. Rated load: ~~3500 lb.~~1450 Kg

c. Rated Speed: ~~[200][350][500] fpm~~60 mpm

d. Car Door Type: ~~Single-speed~~Two-speed, side slide.

e. Car Door Opening Width: ~~107 cm minimum, or [_____]~~ 1067 mm +/-.

2.1.1.2 Elevator No.~~[_____]~~A - ~~Larger Capacity (Pallet-Sized) Loading~~ Passenger Elevator

a. Type: ~~[~~Geared~~] [Gearless]~~

b. Rated load: ~~[4000][4500][_____] lb.~~750 Kg

c. Rated Speed: ~~[200][350][500] fpm~~60 mpm

d. Car Door Type: Single-speed, center opening, horizontally sliding.

e. Car Door Opening Width:~~[122 cm][137 cm].~~ 813 mm +/-.

~~2.1.1.3 Elevator No.[_____] - Non-EMSA Elevator~~

~~a. Type: [Geared] [Gearless]~~

~~b. Rated load: 2500 lb.~~

~~c. Rated Speed: [200][350][500] fpm~~

~~d. Car Door Type: Single speed [side slide][center opening], horizontally sliding.~~

~~e. Car Door Opening Width: 107 cm minimum, or [_____].~~

2.1.2 Cab Enclosure and Hoistway Entrance Assemblies

Provide finishes ~~[~~as indicated~~.][as listed below:~~ in the contract drawings.

a. Floor; ~~[carpet][vinyl composition tile][vinyl sheet tile][_____].~~

rubber composition tile, reference contract drawings.

b. Walls; ~~[prefinished steel][laminated plastic] on plywood [stainless steel][______].~~ Provide each cab wall with equally spaced and equally sized wall panels. All wall panel fasteners must be concealed.

Wall trim; ~~[prefinished steel][~~stainless steel~~][______]~~.

Accessories; Provide hand rails on full length of back wall and side walls of elevator cab.

c. Car doors, car door returns, and wall reveals; ~~[prefinished steel panels][~~stainless steel~~][______]~~.

d. Ceilings; ~~[supported][prefinished steel panels][anodized aluminum][egg crate][______].~~supported island stainless steel with LED downlights. Reference contract drawings.

Ceiling frame; ~~[prefinished steel][~~stainless steel~~][anodized aluminum][______]~~.

e. Hoistway Entrance Assembly Material and Finishes; ~~[prefinished steel][~~stainless steel~~][______]~~.~~]~~

2.2 ELEVATOR OPERATION

ASME A17.1/CSA B44, Introduction, Section 3, Definitions.

~~[2.2.1 Single, Two-Stop, Automatic Operation~~

~~Provide Single Two-Stop Automatic Operation.~~

~~][2.2.2 Selective Collective Automatic Operation~~

~~Provide Selective Collective Automatic Operation.~~

~~][~~2.2.1 Duplex Selective Collective Automatic Operation

Provide Duplex Selective Collective Automatic Operation. If a car is taken out of service or fails to respond to a landing call within a predetermined adjustable time limit of approximately 40 to 180 seconds, transfer calls to the other car functioning as a single car Selective Collective elevator until the out-of-service car is returned to the system. Provide Emergency Medical Services Accessibility key switch in each hall station for Elevator B.

~~][2.2.2 Group Automatic Operation~~

~~Provide Group Automatic Operation. If a car is taken out of service, or fails to respond to a landing call within a predetermined adjustable time limit of approximately 40 to 180 seconds, transfer calls to another car until out-of-service car is returned to the system.~~

~~]~~2.3 SPECIAL OPERATION AND CONTROL

Provide the following special operations and control systems.

2.3.1 Keys for Elevator Key Switches

Provide a minimum of twelve keys per unique cylinder used on all key switches for a single elevator. If there is more than one elevator, additional keys will not be required unless there are additional unique lock cylinders. Provide keys with brass or fiberglass tags marked "PROPERTY OF THE U.S. GOVERNMENT" on one side with function of key or approved code number on the other side.

2.3.2 Firefighters' Emergency Operation (FEO)

Provide FEO equipment and signaling devices. The designated level for the FEO Phase I key operated switch is the ~~[~~ground~~][_____]~~ floor. In the FEO Phase I fixture, provide FEO Operating Instructions.

2.3.2.1 Firefighters' Emergency Operation (FEO) Key Box

Provide flush mounted, locking, FEO Key Box of a minimum size of 127 mm W by 229 mm H by 38 mm D. Install at a height of 183 cm above floor level and directly above the FEO Phase I key switch. Provide box equipped with lock that uses the FEO K1 key.

2.3.3 Hoistway Access Operation

Provide hoistway access operation with switches at top and bottom terminal landings. Locate switch 183 cm above floor level, within 305 mm of elevator hoistway entrance frame or with the ferrule exposed when located in the elevator entrance frame.

2.3.4 In-Car Inspection Operation

Provide In-Car Inspection Operation.

2.3.5 Independent Service

Provide exposed key-operated switch in car operating panel to enable independent service and simultaneously disable in-car signals and landing-call responses. Provide indicator lights that automatically illuminate during independent service. For duplex or group operation, if one car is removed from group another car will respond to its hall calls.

~~2.3.6 Selective Door Operation~~

~~For elevator with one or more rear openings at same level as front opening, provide full-selective operation with car and door operating buttons clearly marked for front and rear openings, front and rear car button for each such floor, and front and rear "DOOR OPEN" and "DOOR CLOSE" buttons. Only door for which the button was operated opens or closes.~~

~~[~~2.3.6 Elevator Emergency Power Operation

~~Provide elevator emergency power operation for [all elevators] [elevator 1,2,3...]. Coordinate power supply and control wiring to accomplish initiation and operation of elevators on emergency power.~~Provide elevator emergency power operation for both Elevators A and B. Provide auto-sequencing, and coordinate power supply and control wiring to accomplish initiation and operation of elevators on emergency power. Provide elevator keyed selection switch labeled "A", "B", and "Auto" at first floor. Install as shown in drawings.

~~][2.3.7 Elevator Auxiliary Power Operating System~~

~~Provide elevator auxiliary power operating system for [all elevators] [elevator 1,2,3...].~~

~~][~~2.3.7 Hospital Emergency Commandeering Service (HECS)

~~Provide Hospital Emergency Commandeering Service.~~Provide Hospital Emergency Commandeering Service for Elevator B. Provide keyswitch in each hall station for Elevator B, and provide HECS keyswitch and dedicated operation to remove Elevator B from group service on Elevator B car operating panel.

~~]~~2.4 ELEVATOR DRIVE MACHINE, HOIST MOTOR, AND DRIVE MOTOR

Provide elevator drive machine, hoist motor, and motor drive system that is designed to be installed in an elevator machine room (MR) or an elevator machinery space. The elevator machine, motor, and drive configuration and installation design must be mechanically and electrically interchangeable with a minimum of two other elevator manufacturer's drive machines that are readily available in the elevator industry. Paint or finish ferrous surfaces with a minimum of one coat of manufacturer applied rust- inhibiting paint.

Design the elevator drive system so that the hoist motor amperage does not exceed the motor data tag full load amperage in any operating condition, exclusive of acceleration and deceleration. Provide elevator hoist motor that is designed with Class F insulation and rated for 120 starts/hr. Design the elevator drive system to limit Total Harmonic Distortion to a maximum of 5 percent. No single harmonic may exceed 3 percent.

Provide an elevator drive machine designed for and provided with stranded steel wire rope for elevator suspension and counterbalance. The minimum acceptable diameter of suspension and counterweight ropes is 9.52 mm. Aramid fiber ropes, coated steel ropes, and non-circular coated steel belts may not be used for elevator suspension or counterbalance.

The elevator drive machine must be equipped with machine manufacturer's designed and installed standard means for the manual release of the driving-machine brake. Provide non-regerative motor drive system that does not feedback regenerative power to the building supply or emergency generator. Provide machine room resistor bank to absorb regenerative power during emergency power operation.

2.4.1 Manufacturer's Factory Training and Technical Support

Provide an elevator drive machine from a manufacturer that provides comprehensive factory training and technical support for installation, adjustment, service, and maintenance of the drive system. The training and support must be identified as available to any licensed elevator contractor. Provide verification of an established and documented training schedule, with pricing, for factory training classes that have been provided for a minimum period of one year prior to contract award date.

The elevator drive system must be identified as available for purchase and installation by any licensed elevator contractor. All drive system related components, parts, diagnostic tools, and software must be available for purchase, installation, and use by any licensed elevator contractor; "exchange-only" provisions for the purchase of spare parts are not acceptable.

2.4.2 Ascending Car Overspeed and Unintended Car Movement Protection

Provide elevator Ascending Car Overspeed and Unintended Car Movement Protection means that is designed to act directly upon, and apply a retarding force to, the elevator suspension ropes. In addition to the requirements of ASME A17.1/CSA B44, the means must be designed to detect and stop movement of the elevator suspension ropes that occurs as a result of loss of traction between the suspension ropes and the elevator machine drive sheave. Reference flow chart in contract drawings.

2.5 CONTROL EQUIPMENT

Enclose all elevator control equipment in factory-primed and baked-enamel coated sheet-metal cabinets with ventilation louvers and removable or hinged doors. Mount cabinets at a height of 254 mm above machine room or control room finish floor.

2.5.1 Motor Control Equipment

Provide variable voltage with silicon controlled rectifier (SCR) or Variable-Frequency Variable Voltage (VVVF) alternating current (ac) drive control.

The elevator motor drive system and controller system shall accommodate a voltage dip within a range between 5 percent to 10 percent of the rated voltage. The new emergency generator size will be dependent on the voltage dip limitation. Coordinate elevator manufacturer's voltage dip with the sizing of the emergency generator system and with the generator contractor. Reference SECTION 26 32 15 ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES.

2.5.1.1 Electrical Isolation Protection

Provide individual isolation transformers and individual choke reactors for each individual hoist motor. Provide filtering to maintain harmonic distortion below IEEE C62.41 standards as measured at the elevator machine room or control room disconnect.

2.5.2 Elevator Microprocessor Controller

For each individual elevator controller, and for each group controller, provide a microprocessor controller that complies with the following paragraphs. Provide controller(s) package that includes all hardware and software required for the installation, maintenance, and service of the elevator, in its' entirety. Provide verification of technical support service that the controller manufacturer provides to any licensed elevator installation, service, and maintenance company.

Provide an elevator controller from a manufacturer that provides comprehensive factory training to include controller installation, adjustment, service, and maintenance. The training must be identified as available to any licensed elevator contractor. Provide verification of an established and documented training schedule, with pricing, for factory training classes that manufacturer has provided for a minimum period of one year prior to contract award date.

The elevator controller must be identified as available for purchase and installation by any licensed elevator contractor. All components, parts, diagnostic tools, and software must be available for purchase and installation and use by any licensed elevator contractor; "exchange-only" provisions for the purchase of spare parts are not acceptable. The elevator controller manufacturer must publish an industry competitive price listing for all controller parts, diagnostic tools, and software.

Provide verification of telephone and internet based technical support service that the elevator controller manufacturer provides to any licensed elevator installation, service, and maintenance company at an industry competitive price. The service must include live telephone based technical support for installation, adjustment, maintenance, and troubleshooting of the elevator controller and related elevator components. The service must be available during standard working hours.

Provide an elevator controller that is designed to automatically reestablish normal elevator operation following any temporary loss of power, regardless of duration.

2.5.2.1 Elevator Controller Interface Cabinet

For each individual elevator microprocessor controller, provide a separate elevator control cabinet with an integrated human interface system. For group elevator installations, a single cabinet and interface system with full access to each elevator controller may be utilized. The separate controller interface cabinet must be supplied by the elevator controller manufacturer and include a minimum 305 mm wide keyboard and a minimum 254 mm monitor. The elevator controller interface cabinet must comply with arc-flash protection requirements of NFPA 70E and UFC 3-560-01.

2.5.2.1.1 Elevator Microprocessor Human Interface

The interface system must provide complete elevator controller interface capability and must include the elevator controller manufacturer's comprehensive package of installation and diagnostic software. The microprocessor interface system must provide unrestricted access to all parameters, all levels of adjustment, and all flags necessary for installation, adjustment, maintenance, and troubleshooting of each elevator and for the elevator group. All software programming must be stored in non-volatile memory. The elevator controller fault log must provide non-volatile memory fault log storage of all faults, trouble calls, and fault history for a minimum of one year and the ability to download or print the fault log. The controller interface must also provide the capability to display and diagnose trouble calls, faults, and shutdowns. Expiring software, degrading operation, and "key" access controls are not acceptable.

2.5.2.2 Software and Documentation

Provide three copies of the manufacturer's maintenance and service diagnostic software, with complete software documentation, that will enable the same level of unrestricted access to all controllers of the same make and model, regardless of the installation date or location. Provide signed certification, from the manufacturer's corporate headquarters, that guarantees that the microprocessor software and access system will not terminate the unlimited and unrestricted access at any future date.

2.5.2.3 Elevator Controller Certification

For elevator installations in the United States, including United States territories, provide an elevator microprocessor controller that has a current certificate of safety code compliance~~ issued by the Technical Standards and Safety Authority (TSSA), Toronto, Canada~~.

2.6 OPERATING PANELS, SIGNAL FIXTURES, AND COMMUNICATIONS CABINETS

For all panels and fixtures, provide identical and uniform fixture design, material, finish, and components for all elevators. For all panels and fixtures, legibly and indelibly identify all buttons and all operating positions for each device. Use engraving and backfilling, or photo etching, for button and switch designations. Do not use attached signs. Provide elevator manufacturers' standard grade for all key switches unless otherwise specified. All illuminating panels and fixture components must utilize LED lighting for energy efficiency.

2.6.1 Car and Hall Buttons

For all cab and landing fixture buttons, provide industry-standard, vandal resistant push buttons with positive-stop assembly design. Buttons must be minimum 19 mm diameter, satin-finish stainless steel, with illuminating LED halo.

2.6.2 Passenger Car-Operating Panel

Provide each car with ~~[~~one~~]~~ ~~[two] ~~car operating panel~~[s]~~ that contain~~[~~s~~]~~ operation controls and communication devices. Provide exposed, flush mounted buttons for the controls identified in subparagraph PASSENGER CONTROLS. Provide a lockable service cabinet for the controls listed in subparagraph SERVICE CONTROLS. Use engraving and backfilling or photo etching for button and switch designations. Do not use attached signs.

2.6.2.1 Passenger Controls

In addition to ASME A17.1/CSA B44 requirements, provide the following operating controls, identified as indicated:

a. LED illuminating car-call buttons identified to correspond to landings served by the elevator.

b. "DOOR OPEN" and "DOOR CLOSE" buttons. For front and rear openings at the same floor, include the identification "F" and "R" for each opening.

c. Red, illuminating "ALARM" button.

d. Key-operated "Independent Service" switch.

e. "Help" communication device to include communication between elevator cab and elevator machine room~~ or control room~~, and 24/7 "push-to-talk" telephone device meeting ADA and ABA connected to Base Security.

~~[~~ f. Key-operated "HOSPITAL EMERGENCY COMMANDEERING SERVICE" switch. Elevator B only for car operating panel controls.

2.6.2.2 Service Controls

In addition to ASME A17.1/CSA B44 requirements, provide the following operating controls, identified as indicated:

a. Provide a key-operated, three-position switch for "In car Inspection Operation" and "Hoistway Access". The center switch position will provide normal, automatic operation.

b. "Car Light" switch.

c. "Car Fan" switch with two speed settings identified.

d. 120-volt ac 60 Hz single-phase duplex electrical outlet of ground-fault-circuit-interrupt (GFCI) design.

2.6.2.3 Certificate Window

Provide a minimum 102 mm wide by 152 mm high certificate window for elevator inspection certificate. Locate window in the Service Controls door of the Car Operating Panel.

2.6.2.4 Emergency Signaling Devices

Provide an audible signaling device, operable from the Car Operating Panel button marked "ALARM". The audible signaling device must have a sound pressure rating between 80 and 90 dBA at 3 meters. Provide battery backup power capable of operating the audible signaling device for at least one hour. Provide 24/7 monitored elevator emergency telephone device meeting ADAAG and ABA requirements in each elevator car.

2.6.3 Elevator In-Car Position Indicators

For all elevators, provide illuminating LED position indicator in the Car Operating Panel.

2.6.4 Elevator In-Car Direction Indicators

For 2-stop elevator installations, provide visual direction indicators and audible car arrival signal in the elevator car door jamb, in accordance with ABA Standards. Visual indicators must be visible from the hall call fixture.

2.6.5 Hall Call Landing Fixtures

Provide a hall call fixture adjacent to each elevator. Provide a single push-button for terminal landings and dual push-buttons, up and down, at intermediate landings. Provide LED digital floor indicators and direction arrows in the face of each hall landing fixture to show car position, car direction, and arrival. Two hall call landings fixtures at each floor. Reference contract drawings for further detail and requirements.

2.6.5.1 Designated Landing Hall Call Fixture

2.6.5.1.1 Location of COMMUNICATION MEANS FAILURE (CMF) Visual Signal

When required by ASME A17.1/CSA B44, provide an elevator CMF audible and illuminating signal, and reset switch, in the FEO Designated Landing hall call fixture. Mount the signal and reset switch at a minimum of 178 mm above the "UP" hall call button.

2.6.5.1.2 COMMUNICATION MEANS FAILURE (CMF) Visual and Audible Signal Operation

Provide a CMF visual and audible signal system that conforms to ASME A17.1/CSA B44. Provide continuous verification of operability of the telephone line and immediate activation of audible and visual signals when verification means determines that the telephone line is not functioning. Provide illumination of visual signal at one second intervals. Provide a minimum of 65 dBA audible signal at 30 second intervals.

2.6.5.1.3 Firefighters' Emergency Operation Phase I Switch and Visual Signal

When required by ASME A17.1/CSA B44, provide an elevator Firefighters' Emergency Operation Phase I switch and illuminating visual signal in the FEO Designated Landing hall call fixture. Provide FEO Phase I visual signal that is designed with intermittent, flashing, illumination when actuated by the machine room, control room, or hoistway fire alarm initiating device. Locate FEO Phase I key switch above the CMF visual signal with a minimum of 152 mm vertical between the centerlines of the CMF signal and the FEO Phase I key switch. Locate FEO Phase I visual signal directly above the Phase I switch. In addition, locate Elevator Corridor Call Station Pictograph at top of hall call fixture.

2.6.5.1.4 High Rise Two Way Emergency Communication

Provide emergency communication per ASME A17.1/CSA B44 for elevators exceeding 18.2 meters of vertical travel distance. Two way communication must be provided between each elevator car and the designated entry level elevator lobby.

2.6.5.1.5 Remote Elevator Fire Status Panel

Provide a remote elevator fire status panel is coordination with the Fire Protection work scope. The panel shall minimally include the following items: Digital Indicators showing the location and direction of travel of each elevator, status power indicator showing whether elevators are operational, elevator emergency power status indicators, elevator fire recall switch and operation per ASME A17.1/CSA B44, and elevator emergency power manual selection switch. Locate the panel per Fire Protection contract drawings located in the remote fire control room outside the Tower.

2.6.6 Elevator Car Position and Direction Indicators and Car Arrival Signal

For elevator installations with three or more stops, provide a separate hall landing fixture that includes the visual elevator position indicator, visual direction indicators, and audible car arrival signal, in accordance with ABA Standards. The digital LED position, direction, and arrival indicators shall be located in the face of each hall call station for each elevator at every floor. Reference contract drawings.

2.6.7 Designated Landing Elevator Identification Fixture

For duplex and group elevator installations, provide a separate elevator identification fixture for each elevator, with identification engraved and backfilled with a contrasting color~~. Number elevators from left to right, as seen during primary approach from building main entrance to elevator lobby. For multiple elevator groups, begin numbering with group that is closest to the building main entrance~~ and visible from the area directing in front of each elevator at the first floor level. Number "Elevator B" on the left, and "Elevator A" on the right.

2.6.8 Emergency or Standby Power

When emergency or standby power is provided for elevator operation, provide an elevator emergency power visual indicator that conforms to ASME A17.1/CSA B44. Locate the visual signal in the Firefighters Emergency Operation fixture for each simplex elevator and for each elevator group. When an emergency power selector switch is required, provide switch in ~~a separate, flush mounted fixture located at the designated level, in view of all elevator entrances~~the Fire fighter's Phase I recall fixture as shown on contract drawings.

2.7 CAR DOOR EQUIPMENT

2.7.1 Car Door Operator

Provide elevator door operator equipment and circuitry that is designed and installed as discreet communication. Serial communication must not be used for this system. Door operators shall utilize solid-state closed loop door control system.

2.7.2 Infra-red Curtain Unit

Provide Infra-red Curtain Unit (ICU) with multiple infra-red beams that protect to the full height and width of the door opening. Provide door nudging operation.

2.8 PASSENGER ELEVATOR GUIDES, PLATFORM, AND ENCLOSURE

2.8.1 Roller Guides

Provide coil-spring loaded roller guide assemblies in adjustable mountings on each side of car and counterweight frames in accurate alignment at top and bottom of frames.

2.8.2 Car Enclosure Wall Panels, Return Panels, Doors, Entrance Columns, and Transom

Provide 14 Gauge minimum ~~[prefinished steel][~~Type 304 stainless steel~~]~~ cab wall panels and entrance components. Use same material and finish for all hoistway and car entrance assemblies. Apply sound-deadening material on exterior of all cab wall panels.

2.8.3 Car Enclosure Top

Provide reinforced, 12 gauge minimum steel car enclosure top. Provide hinged emergency exit with lock that complies with the seismic risk zone 2 or greater design requirements of ASME A17.1/CSA B44. Locate emergency exit hinge towards the rear of the elevator cab. Design and configure the elevator cab interior ceiling to provide convenient and unobstructed access to, and use of, emergency exit from inside the elevator cab. Design and configure the elevator cab interior ceiling to provide convenient and unobstructed access to, and use of, emergency exit from inside the elevator cab. Provide car-top safety railing and kick plate, and car-top inspection operating station with LED lighting and GFCI outlet.

2.8.4 Car Door

Provide 16 gauge minimum ~~[prefinished steel][~~Type 304 stainless steel~~]~~ car doors of sandwich construction with flush surfaces on car and landing sides. Provide a minimum of 2 door guide assemblies per door panel, one guide at leading and one at trailing door edge with guides in the sill groove their entire length of travel.

2.8.5 Car Entrance Sill

~~Provide one piece cast nickel silver, stainless steel, or white bronze entrance sill(s). Set sills level and flush with floor finish. Use same material for hoistway and car entrance sills.~~Provide one piece cast nickel silver new car sills. Set sills level and flush with floor finish. Reuse and refurbish existing hoist way sills and sill supports at all floors for both elevators. Adjust sills to meet ADA/ABA requirements for maximum sill-to-sill horizontal gap.

2.8.6 Cab Finish Floor

Provide cab finish floor with top of finish floor flush with the cab sill.

2.8.7 Car Fan

Provide 2-speed fan for car enclosure forced ventilation. Fan must be mounted in the car enclosure top.

2.8.8 Car Lighting

Utilize LED lighting for elevator car interior illumination. Provide a minimum of 10 foot-candles, measured at all areas of the car enclosure floor. Provide automatic car lighting operation that will turn off car lights after 3 minutes of inactivity. Car lights must automatically turn on upon actuation of an elevator car or hall call.

2.8.9 Car Protection Pads and Hooks

Provide fire retardant, hanging car protection pads that provide protection for all car interior wall panels. ~~Provide permanently installed studs in car that are designed for hanging the car protection pads in the car.~~ Provide permanently installed pad support rail headers as indicated on contract drawings on each wall for both elevators A and B.

2.9 PASSENGER ELEVATOR HOISTWAY DOORS AND ENTRANCES

Provide hoistway entrance assemblies with a minimum 1-1/2 hour fire rating. Use same material and finish for all hoistway and car entrance assemblies.

2.9.1 Hoistway Entrance Frames

Provide 14 gage minimum ~~[prefinished carbon sheet steel][~~Type 304 stainless steel~~]~~ hoistway entrance frames. Solidly grout uprights of entrance ways to height of ~~5 feet~~1,524 mm. Reference contract drawings.

2.9.2 Hoistway Entrance Sills

~~Provide one-piece cast nickel silver, stainless steel, or white bronze entrance sills. Set top of landing sill flush with top of finish floor. Solidly grout under full length of sill. Use same material for all hoistway and car entrance sills.~~Reuse and refurbish existing hoistway sills and sill supports for both elevators. Provide repairs or replace existing sills that are overworn or damaged. Provide minor grout fill and support as required.

2.9.3 Hoistway Entrance Doors

Provide ~~[hollow metal][~~Type 304 stainless steel~~]~~ non-vision construction hoistway entrance doors with flush surfaces on car and landing sides. Provide a minimum of 2 door guide assemblies per door panel, one guide at leading edge and one at trailing edge with guides in the sill groove the entire length of door travel. Use same material and finish for all hoistway and car entrance assemblies. Reference contract drawings.

2.9.4 Hoistway Entrance Door Track Dust Covers

Provide sheet metal hoistway door track dust covers at each landing. Dust covers must cover top and hoistway side of door locks and door roller tracks, and extend the full width of the door track and associated hardware. Dust cover sections will not exceed ~~3 feet~~914.4 mm in length.

2.10 HOISTWAY EQUIPMENT

2.10.1 Car and Counterweight Guide Rails and Fastenings

~~Provide T-section type guide rails for car and counterweight. Paint rail shanks with one coat of black enamel.~~Reuse, vertical align/plumb adjust, and refurbish car and counterweight rails. Replace worn or damaged parts with new. Paint shanks with one coat of black enamel.

2.10.2 Pit Equipment and Support Channels

Provide rail-to-rail pit channels to serve as mounting surface for main guide rails and counterweight guide rails. In addition, pit channels will serve as mounting surfaces for car and counterweight buffers. Method of installation of channels, brackets and buffer mounts will be such that pit waterproofing is not punctured. Provide new car and counterweight buffers.

2.10.3 Pit "STOP" Switch

Provide push-to-stop/pull-to-run type pit "STOP" switch.

2.10.4 Traveling Cables

Suspend traveling cables by means of self-tightening webbed devices or internal suspension members.

2.10.5 Hoistway Pit Ladder

Provide continuous horizontal rungs for the full height of the pit ladder.

2.10.6 Car Frame, Safeties, and Governor

Provide car frame, Type B car safeties, Governor, and tail weight assemblies. Design per ASME A17.1/CSA B44 and using Manufacturer's standard systems. Design to fit existing hoist way dimensions while meeting existing car capacity and speed requirements per Code. Governor shall be designed with a bi-directional switch to meet Unintended Car Motion and Ascending Car Over-Speed requirements.

2.10.7 Counterweight Frame, Sub-Weights, and Compensation

Provide manufacturer standard counterweight frame, sub-weights, and dual encapsulated chain compensation to meet manufacturer's engineering requirements. Remain within the Structural design and dimensions of the existing hoist way design. Provide seismic hardware as required per ASME A17.1/CSA B44 or per related Japan standards. Include counterweight displacement hardware, seismic switch, and related items.

2.10.8 Pit Flood Switch

Provide a pit flood detecting float switch with NEMA Type 4 electrical connections and rating connected to elevator controls to prevent elevator car from traveling to the Basement level in the event that the switch is activated. Provide a flood detection alarm in the pit.

PART 3 EXECUTION

3.1 INSTALLATION

Install in accordance with DOD design criteria, contract specifications, manufacturer's instructions, NEII-1 Building Transportation Standards and Guidelines, and all applicable building and safety code requirements.

3.1.1 Structural Members and Finish Materials

Do not cut or alter structural members. Do not alter finish materials from manufacturer's original design. Restore any damaged or defaced work to original condition.

3.1.2 Miscellaneous Requirements

Provide recesses, cutouts, slots, holes, patching, grouting, and refinishing to accommodate elevator installation. Use core drilling to drill all new holes in concrete. Finish work to be straight, level, and plumb. During installation, protect machinery and equipment from dirt, water, or mechanical damage. At completion, clean all work and spot paint.

3.2 FIELD QUALITY CONTROL

The Contractor will provide and utilize a third-party licensed and certified Qualified Elevator Inspector (QEI) to conduct elevator pre-acceptance inspection and testing. The QEI must perform inspections and witness tests to ensure that the installation conforms to all applicable safety codes and contract requirements. The QEI will be directly employed by the Contractor and independent of the elevator contractor.

Upon completion, the QEI must provide written test data for all ASME A17.1/CSA B44 Acceptance Tests and written certification that the elevator is complete and ready for final Acceptance Inspection, Testing, and Commissioning.

3.3 ACCEPTANCE INSPECTION, TESTING AND COMMISSIONING

When elevator system installation is complete and ready for final inspection, notify Contracting Officer that elevator system is ready for Acceptance Inspection, Testing, and Commissioning. Provide QEI certification specified in Article FIELD QUALITY CONTROL.

~~[Contracting Officer will obtain services of Naval Facilities Engineering Command (NAVFAC) QEI Certified Elevator Inspector. NAVFAC QEI will utilize the applicable NAVFAC Elevator Acceptance Inspection Form to record the results of inspection and testing and to identify safety code and contract deficiencies. Specific values must be provided for all tests required by ASME A17.1/CSA B44, ASME A17.2, and contract documents. Upon completion of inspection and testing, the NAVFAC QEI will sign a copy of the completed forms and provide the signed copy to the Contracting Officer or representative. Within 2 weeks of the inspection, the QEI will also prepare a formal inspection report, including all test results and deficiencies. Upon successful completion of inspection and testing, NAVFAC Certified Elevator Inspector will complete, sign and post form NAVFACENGCOM 9-11014/23(Rev.9-2009), Elevator Inspection Certificate.~~

~~][~~Contracting Officer will obtain the services of a third-party QEI Certified Elevator Inspector. The QEI must utilize an Elevator Acceptance Inspection Form to record the results of inspection and all testing and to identify safety code and contract deficiencies. Specific values must be provided for all tests required by ASME A17.1/CSA B44, ASME A17.2, and contract documents. Upon completion of inspection and testing, the QEI must sign a copy of the completed forms and provide to the Contracting Officer. Within 2 weeks of the inspection, the QEI must also prepare a formal inspection report, including all test results and deficiencies. Upon successful completion of inspection and testing, the QEI will complete, sign, and provide a certificate of compliance with ASME A17.1/CSA B44.

~~]~~3.3.1 Acceptance Inspection Support

Prime and Elevator Contractors must provide inspection support and perform all required tests, in order to demonstrate proper operation of each elevator system and to prove that each system complies with contract requirements and all applicable building and safety codes. Inspection procedures in ASME A17.2 form a part of this inspection and acceptance testing. All inspection and testing must be conducted in the presence of the Qualified Elevator Inspector (QEI).

If the elevator does not comply with all contract and safety code requirements on the initial Acceptance Inspection and Test, the Contractor is responsible for all costs involved with re-inspection and re-testing required as a result of contractor delays and discrepancies discovered during inspection and testing.

3.3.2 Testing Materials and Instruments

Provide all testing materials and instruments necessary for Acceptance Inspection, Testing and Commissioning. At a minimum, include calibrated test weights, tachometer, accelerometer, hydraulic pressure gauge, 600-volt megohm meter, volt meter and ammeter, infrared temperature gauge, door pressure gage, dynamometer, and 6 meter tape measure.

3.3.3 Field Tests

3.3.3.1 Endurance Tests

Test each elevator for a period of one hour continuous, automatic operation, with specified rated load in the elevator cab. During the one hour test, stop car at each floor, in both directions of travel, and allow automatic door open and close operation. The requirements for Automatic Operation, Rated Speed, Leveling, Temperature Rise and Motor Amperes must be met throughout the duration of the Endurance Test. Restart the one hour test period from the beginning, following any shutdown or failure.

3.3.3.2 Speed Tests

Determine actual speed of each elevator, in both directions of travel, with rated load and with no load in elevator car. Make Speed tests at the beginning and at the end of the Endurance test. Determine speed by tachometer reading or accelerometer, excluding accelerating and slow-down zones. Under all conditions, minimum acceptable elevator speed is the Rated speed specified. Maximum acceptable elevator speed is 110 percent of Rated speed.

3.3.3.3 Leveling Tests

Test elevator car leveling operation and provide a leveling accuracy equal to or less than 3 mm at each floor with no load in car, balanced load in car, and with rated load in car, in both directions of travel. Determine leveling accuracy at the beginning and at the end of the endurance tests.

3.3.3.4 Temperature Rise Tests

Determine temperature rise of elevator drive machine motor during one-hour full-load test run. Under these conditions, maximum temperature rise must not exceed acceptable temperature rise indicated on manufacturer's data plate. Start test only when equipment is within 5 degrees C of ambient temperature.

3.3.3.5 Balanced Load Test

Place balanced load in the elevator cab, according to the manufacturer's designed counterbalance. Perform electrical and mechanical balanced load tests of car and counterweight.

3.3.3.6 Motor Ampere Tests

At beginning and end of Endurance test, measure and record motor amperage in both directions of travel and in both no-load and rated load conditions.

3.3.3.7 Elevator Performance and Ride Quality Testing

Evaluate elevator performance to ensure compliance with specification requirements related to the NEII-1 Performance Standards Matrix for New Elevator Installations.

-- End of Section --

SECTION 21 12 00

STANDPIPE SYSTEMS
05/11

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide https://www.approvalguide.com/

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13	(2025; TIA 24-3) Standard for the Installation of Sprinkler Systems
NFPA 14	(2024) Standard for the Installation of Standpipes and Hose Systems
NFPA 24	(2025) Standard for the Installation of Private Fire Service Mains and Their Appurtenances

UL SOLUTIONS (UL)

UL Fire Prot Dir	UL Product IQ (updated online) at https://productiq.ulprospector.com/en

~~1.2 RELATED REQUIREMENTS~~

~~Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS applies to this section with additions and modifications specified herein.~~

1.2 SYSTEM DESCRIPTION

Design and provide new~~[~~ manual dry Class I~~][combination][automatic] [wet Class I]~~ standpipe~~[and fire sprinkler]~~ system~~[s]~~ as shown.

1.3 SYSTEM DESCRIPTION

System design and manufacturer's products shall be in accordance with the required and advisory provisions of NFPA 14 except as modified herein.~~[~~ Standpipe system shall be designed by hydraulic calculations.~~][Provide sprinkler portion of system under Section 21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION.]~~ Design each system~~[~~ for earthquakes and~~]~~ to include materials, accessories, and equipment inside and outside the building necessary to provide each system complete and ready for use. Devices and equipment shall be UL Fire Prot Dir listed or FM APP GUIDE approved for fire protection service. In the publications referred to herein, the advisory provisions shall be considered to be mandatory, as though the word "shall" had been substituted for "should" wherever it appears; reference to the "authority having jurisdiction" shall be interpreted to mean the~~[[_____] Division][Engineering Field Activity [_____]], Naval Facilities Engineering Systems Command, Fire Protection Engineer~~USACE Designated Fire Protection Engineer (USACE DFPE).~~[~~ Begin the work at~~[~~ the points indicated~~] [_____]~~.~~]~~

1.3.1 Residual Pressure

The minimum residual pressure at the outlet of the most remote 64 mm hose connection shall be ~~448.2 kPa [~~689.5 kPa~~]~~ while the system is discharging at the required design flow rates.

1.3.2 Friction Losses

Calculate losses in piping in accordance with the Hazen-Williams formula with 'C' value of 120 for steel piping, 150 for copper tubing, and 140 for cement-lined ductile-iron piping.

~~[1.3.3 Water Supply~~

~~Base hydraulic calculations on a static pressure of [_____] kPa (gage) with [_____] L/m available at a residual pressure of [_____] kPa (gage)[at the junction with the existing water distribution piping system.][Base hydraulic calculations on operation of fire pump[s] provided in Section 21 30 00 FIRE PUMPS.]~~

~~]~~1.3.3 Standpipe System Drawings

Prepare in accordance with the requirements for "Plans and Specifications" as specified in NFPA 14. Each drawing shall be A1 841 by 594 mm. Draw plans to a scale not less than 1 mm equals 100 mm Do not commence work until the design of each system and the various components have been approved. Show data essential for proper installation of each system. Show details, plan view, elevations, and sections of the systems supply and piping. Show piping schematic of systems supply, devices, valves, pipe, and fittings.~~[~~ Submit drawings signed by a registered fire protection engineer.~~]~~ Show:

a. Room, space or area layout and include pipe supports and hangers.

b. Field wiring diagrams showing locations of devices and points of connection and terminals used for all electrical field connections in the system, with wiring color code scheme.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Standpipe System; G~~, [_____]~~

SD-03 Product Data

Aboveground Pipe and Fittings; G~~, [_____]~~

Mechanical Couplings; G~~, [_____]~~

Pipe Hangers and Supports; G~~, [_____]~~

Valves, including Gate, Check, and Hose; G~~, [_____]~~

Fire Department Connections; G~~, [_____]~~

~~[~~ ~~Alarm Valves; G, [_____]~~

~~][~~ ~~Water Motor Alarms; G, [_____]~~

~~][~~ ~~Pressure Switch; G, [_____]~~

~~][~~ ~~Waterflow Detector; G, [_____]~~

~~][~~ ~~Fire hose Cabinets; G, [_____]~~

~~]~~ Valve Tamper Switch; G~~, [_____]~~

~~[~~ ~~Backflow Preventer; G, [_____]~~

~~]~~ Buried Pipe and Fittings; G~~, [_____]~~

Data which describes more than one type of item shall be clearly marked to indicate which type the Contractor intends to provide. Submit one original for each item and clear, legible, first-generation photocopies for the remainder of the specified copies. Incomplete or illegible photocopies will not be accepted. Partial submittals will not be accepted.

SD-06 Test Reports

Preliminary Tests; G~~, [_____]~~

Acceptance Tests; G~~, [_____]~~

Submit for all inspections and tests specified under paragraph entitled "Field Quality Control."

SD-07 Certificates

Qualifications of Installer; G~~, [_____]~~

Submit installers qualifications as required under paragraph entitled "Qualifications of Installer."

~~SD-10 Operation and Maintenance Data~~

~~Alarm Valves, Data Package 3; G, [_____]~~

~~Backflow Preventer, Data Package 3; G, [_____]~~

~~Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Furnish one complete package prior to the time that final acceptance tests are performed, and furnish the~~

~~remaining before the contract is completed. Inscribe the following identification on the cover: the words OPERATION AND MAINTENANCE MANUAL, the location of the building, the name of the Contractor, system manufacturer and the contract number. The instructions shall be legible and easily read, with large sheets of drawings folded in. The package shall include: schematic drawings showing piping; circuit drawings; installation instructions; maintenance instructions; safety precautions, diagrams, and illustrations; test procedures; performance data; and parts list.~~

SD-11 Closeout Submittals

System As-Built Drawings; G~~, [_____]~~

1.5 QUALITY ASSURANCE

1.5.1 Qualifications of Installer

Prior to commencing work, submit data showing that the Contractor has successfully installed fire extinguishing standpipe systems of the same type and design as specified herein, or that he has a firm contractual agreement with a subcontractor having the required experience. Include the names and locations of at least two installations where the Contractor, or the subcontractor referred to above, has installed such systems. Indicate the type and design of each system, and certify that the system has performed satisfactorily for a period of at least 18 months.

Qualifications of System Technician: Installation drawings, shop drawing and as-built drawings shall be prepared, by or under the supervision of, an individual who is experienced with the types of works specified herein, and is currently certified by the National Institute for Certification in Engineering Technologies (NICET) as an engineering technician with minimum Level-III certification in Automatic Sprinkler System program. Contractor shall submit data for approval showing the name and certification of all involved individuals with such qualifications at or prior to submittal of drawings.

1.5.2 System As-Built Drawings

Upon completion, and before final acceptance of the work, submit a complete set of as-built drawings of each system. Submit A1 841 by 594 mm reproducible as-built drawings on mylar film with title block similar to full size contract drawings. Furnish as-built(record) working drawings in addition to the as-built drawings required by Division 1, "General Requirements."

1.6 DELIVERY, STORAGE AND HANDLING

Protect stored equipment from weather, humidity and temperature variations, dirt, dust, and other contaminants.

PART 2 PRODUCTS

2.1 ABOVEGROUND PIPING SYSTEMS

Provide fittings for changes in direction of piping and for connections. Make changes in piping sizes through tapered reducing pipe fittings; bushings will not be permitted. Perform welding in the shop; field welding will not be permitted.~~[Conceal piping in areas with suspended ceiling.]~~

2.1.1 Pipe and Fittings

NFPA 14, except as modified herein. Steel piping shall be Schedule 40 for sizes less than 200 mm, and Schedule 30 or 40 for sizes 200 mm and larger. Fittings shall be welded, threaded, or grooved-end type. Plain-end fittings with mechanical couplings and fittings which use steel gripping devices to bite into the pipe when pressure is applied will not be permitted. Rubber gasketed grooved-end pipe and fittings with mechanical couplings shall be permitted in pipe sizes 40 mm and larger. Fittings shall be UL Fire Prot Dir listed or FM APP GUIDE approved for use in~~[~~ dry~~][wet]~~ pipe sprinkler systems. Fittings, mechanical couplings, and rubber gaskets shall be supplied by the same manufacturer. Steel piping with wall thickness less than Schedule 30 shall not be threaded.~~[~~ Side outlet tees using rubber gasketed fittings shall not be permitted.~~]~~ Pipe and fittings shall be metal.

2.1.2 Pipe Hangers and Supports

Provide in accordance with NFPA 14.

2.1.3 Valves

NFPA 14. Provide valves of types approved for fire service. Hose and gate valves shall open by counterclockwise rotation. Provide isolation and check valves as required by NFPA 14. Isolation valves shall be OS&Y type. Check valves shall be flanged clear opening swing-check type with flanged inspection and access cover plate for sizes 100 mm and larger.

2.1.3.1 Hose Valves

Provide bronze~~[pressure regulating type]~~ hose valve with 65 mm National Standard male hose threads, and 65 mm NH female by 40 mm IPT male reducer with cap and chain.~~[Equip valve with a device to regulate pressure at the outlet to a pressure not exceeding [690][_____] kPa[under both flow and no-flow conditions].]~~

2.1.4 Identification Signs

NFPA 14. Attach properly lettered and approved metal signs to each valve and alarm device.

~~[2.1.5 Waterflow Test Connection~~

~~Provide test connections approximately 1.83 m above the floor for each standpipe system or portion of each standpipe system equipped with an alarm device; locate downstream and adjacent to each alarm actuating device. Provide test connection piping to a location where the discharge will be readily visible and where water may be discharged without property damage. Discharge to janitor sinks or similar fixtures shall not be permitted. Provide discharge orifice equivalent to 15 mm sprinkler orifice.[The penetration of the exterior wall shall be no greater than[0.61 meter] [_____] above finished grade.]~~

~~][2.1.6 Main Drains~~

~~Provide separate drain piping to discharge at safe points outside each~~

~~building or to sight cones attached to drains of adequate size to readily receive the full flow from each drain under maximum pressure. Provide auxiliary drains as required by NFPA 13 and NFPA 14.~~

~~]~~2.1.5 Pipe Sleeves

Provide where piping passes entirely through walls, floors, roofs and partitions. Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, floors, roofs and partitions. Provide one inch minimum clearance between exterior of piping and interior of sleeve or core-drilled hole. Firmly pack space with mineral wool insulation. Seal space at both ends of the sleeve or core-drilled hole with plastic waterproof cement which will dry to a firm but pliable mass, or provide a mechanically adjustable segmented elastomeric seal. In fire walls and fire floors, seal both ends of pipe sleeves or core-drilled holes with UL listed fill, void, or cavity material.

2.1.5.1 Sleeves in Masonry and Concrete Walls, Floors, and Roofs

Provide hot-dip galvanized steel, ductile-iron, or cast-iron sleeves. Core drilling of masonry and concrete may be provided in lieu of pipe sleeves when cavities in the core-drilled hole are completely grouted smooth. Extend sleeves in floor slabs 76 mm above finished floors.

2.1.5.2 Sleeves in Partitions

Provide 26 gage galvanized steel sheet.

2.1.6 Escutcheon Plates

Provide one piece or split hinge type metal plates for piping passing through walls, floors, and ceilings in both exposed and concealed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces. Securely anchor plates in place.

2.1.7 Fire Department Connections

Provide connections approximately one meter above finish grade, of the approved two-way type with~~[[65][_____]~~ mm National Standard~~]~~ female hose threads with plug, chain, and identifying fire department connection escutcheon plate.

~~[2.1.8 Alarm Valves~~

~~Provide variable pressure type alarm valve complete with retarding chamber, alarm test valve, alarm shutoff valve, drain valve, pressure gages, accessories, and appurtenances for the proper operation of the system. The alarm shut-off valve in the piping between the alarm valve and the alarm pressure switch shall be a UL listed electrically supervised quarter-turn valve. Connection of switch shall be under Section[28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE.][28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE.][28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE.][28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE.]~~

~~][2.1.9 Water Motor Alarms~~

~~Provide alarms of the approved weatherproof and guarded type, to sound locally on the flow of water in each corresponding standpipe. Mount alarms on the outside of the outer walls of each building. Provide separate drain piping directly to exterior of building.~~

~~][2.1.10 Pressure Switch~~

~~Provide switch with circuit opener or closer[SPDT contacts] for the automatic transmittal of an alarm over the facility fire alarm system. Connect into the building fire alarm system. Alarm actuating device shall have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and shall instantly recycle.~~

~~][2.1.11 Waterflow Detector~~

~~Provide vane-type waterflow detector. Provide detector with adjustable retard feature to prevent false alarms caused by momentary water surges. Connect into the building fire alarm system.[Alarm actuating device shall have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and shall instantly recycle.] Provide detector[at the base of each standpipe riser above main check valve][where indicated] in accordance with manufacturers instructions.~~

~~][2.1.12 Fire Hose Cabinets~~

~~Provide[recessed][semi recessed][surface]-mounted cabinets where indicated. Cabinets shall be prime grade, cold-rolled, reannealed, process-leveled, furniture steel. Fabricate cabinet from 20 gage steel and door and trim from 18 gage steel. Provide fully welded joints ground smooth. On each jamb, provide at least two anchors or reinforcements spaced approximately 610 mm apart for building in or attaching the cabinets to adjacent construction. Doors shall be flush hollow metal type with fully welded joints ground smooth and full glazed opening. Provide door with continuous hinge, latch and pull. Hinge door for 180 degree opening. Glass shall conform to ASTM C1036 and shall be Type II (flat wired glass), Class 1 (clear), Form 1 (wired, polished both sides), Quality q 8 (glazing quality), diamond or square wire mesh, 6.35 mm thick. Factory finish cabinet inside and out with one coat of enamel applied over a primer. Interior finish color shall be white. Exterior finish color shall be [_____].[Fabricate cabinet with sufficient interior space to store one fire extinguisher.]~~

~~][~~2.1.8 Valve Tamper Switch

Provide valve tamper switch(es) to monitor the open position of valve(s) controlling water supply to the standpipe system. Switch contacts shall transfer from the normal (valve open) position to the off-normal (valve closed) position during the first two revolutions of the hand wheel or when the stem of the valve has moved not more than one-fifth of the distance from its normal position. Switch shall be tamper resistant. Removal of the cover shall cause switch to operate into the off-normal position.

~~][2.1.9 Fire Pumps~~

~~Provide as specified in Section 21 30 00 FIRE PUMPS.~~

~~][2.1.10 Backflow Preventer~~

~~Provide[reduced pressure principle][double check] valve assembly backflow preventer with OS&Y gate valve on both ends. Each check valve shall have a drain. Backflow prevention assemblies shall have current "Certificate of Approval from the Foundation for Cross-Connection Control and Hydraulic Research, FCCCHR List. Listing of the specific make, model, design, and size in the FCCCHR List shall be acceptable as the required documentation."~~

~~]~~2.2 BURIED PIPING SYSTEMS

2.2.1 Buried Pipe and Fittings

NFPA 24, outside coated, cement lined, ductile iron pipe and fittings for piping under the building and to a point 1.52 m outside the building walls. Anchor the joints in accordance with NFPA 24 using pipe clamps and steel rods. Minimum pipe size shall be 150 mm. Minimum depth of cover shall be~~ [_____] [~~one meter~~]~~. Piping more than 1.52 m outside the building walls shall be provided under Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

2.2.2 Buried Utility Warning and Identification Tape

Provide detectable tape in accordance with Section 31 00 00 EARTHWORK.

2.3 ELECTRICAL WORK

Provide electrical work associated with this section under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, except for fire alarm wiring. Provide fire alarm wiring and connection to fire alarm systems under Section~~[28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE][28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE][28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE][~~ 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE~~][this section in accordance with NFPA 70 and NFPA 72]~~.

~~[2.3.1 Wiring~~

~~Provide fire alarm wiring and connections to fire alarm systems, under this section and conforming to NFPA 70, and NFPA 72. Wire for 120 volt circuits shall be No. 12 AWG minimum solid conductor. Wire for low voltage DC circuits shall be No. [14][16] AWG minimum solid conductor. All wiring shall be color coded. Wiring, conduit and devices exposed to water or weather shall be weatherproof. Wiring, conduit and devices located in hazardous atmospheres, as defined by NFPA 70[and as shown], shall be explosion proof. All conduit shall be minimum 20 mm size. Identify circuit conductors within each enclosure where a tap, splice or termination is made. Identify conductors by plastic coated self sticking printed markers or by heat-shrink type sleeves. Attach the markers in a manner that will not permit accidental detachment.~~

~~]~~PART 3 EXECUTION

3.1 EXCAVATION, BACKFILLING, AND COMPACTING

Provide under this section as specified in Section 31 00 00 EARTHWORK.

~~3.2 CONNECTIONS TO EXISTING WATER SUPPLY SYSTEMS~~

~~Connections to existing water supply system are specified in Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.~~

3.2 STANDPIPE SYSTEM INSTALLATION

Equipment, materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing shall be in accordance with the NFPA standards referenced herein. Install piping straight and true to bear evenly on hangers and supports.~~[~~ Conceal piping to the maximum extent possible. Piping shall be inspected, tested and approved before being concealed.~~]~~ Provide fittings for changes in direction of piping and for all connections Make changes in piping sizes through standard reducing pipe fittings; do not use bushings. Cut pipe accurately and work into place without springing or forcing. Ream pipe ends and free pipe and fittings from burrs. Clean with solvent to remove all varnish and cutting oil prior to assemble. Make screw joints with PTFE tape applied to male thread only.

~~[3.3 DISINFECTION~~

~~Disinfect new water piping from the point of connection at the water main and existing water piping affected by the Contractor's operation in accordance with AWWA C651. Exercise caution when mixing chlorine disinfectant solutions. Fill piping systems with solution containing minimum of 50 parts per million of free available chlorine and allow solution to stand for a minimum of 24 hours. Flush solution from systems with clean water until maximum residual chlorine content is not greater than 0.2 parts per million. Obtain at least two consecutive satisfactory bacteriological samples from new water piping, analyze by a certified laboratory, and submit results prior to new water piping being placed into service.~~

~~]~~3.3 FIELD PAINTING

Field painting of fire extinguishing standpipe system shall be specified in Section 09 90 00 PAINTS AND COATINGS. Field painting requirements for "Fire Extinguishing Sprinkler Systems" shall apply.

3.3.1 Piping Labels

Provide permanent labels in mechanical rooms, spaced at 6 meters maximum intervals along pipe, indicating "STANDPIPE."

3.4 ELECTRICAL WORK

Provide electrical work associated with this section under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, except for fire alarm wiring. Provide fire alarm wiring and connection to fire alarm systems under Section~~[28 31 60 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE][28 31 66 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, NON-ADDRESSABLE][28 31 70 INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE][~~ 28 31 76 INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE~~][this section in accordance with NFPA 70 and NFPA 72]~~.

~~[3.4.1 Wiring~~

~~Provide fire alarm wiring and connections to fire alarm systems, under~~

~~this section in accordance with NFPA 70, and NFPA 72. Provide wiring in rigid metal conduit or intermediate metal conduit, except electrical metallic tubing may be used in dry locations not enclosed in concrete or where not subject to mechanical damage. Do not run low voltage DC circuits in the same conduit with AC circuits.~~

~~]~~3.5 FLUSHING

Flush the piping system with potable water in accordance with NFPA 14. Continue flushing operation until water is clear, but for not less than 10 minutes.

3.6 FIELD QUALITY CONTROL

Prior to initial operation, inspect equipment and piping systems for compliance with drawings, specifications, and manufacturer's submittals. Perform tests in the presence of the Contracting Officer to determine conformance with the specified requirements.

3.6.1 Preliminary Tests

Each piping system shall be hydrostatically tested at ~~[~~1379~~][_____]~~ kPa (gage) in accordance with NFPA 14 and NFPA 24 and shall show no leakage or reduction in gauge pressure after 2 hours. The Contractor shall conduct complete preliminary tests, which shall encompass all aspects of system operation.~~[Individually test alarms, and all other components and accessories to demonstrate proper functioning. Test water flow alarms by flowing water.]~~ When tests have been completed and all necessary corrections made, submit to the Contracting Officer a signed and dated certificate, similar to that specified in NFPA 13, attesting to the satisfactory completion of all testing and stating that the system is in operating condition. Also include a written request for a formal inspection and test.

3.6.2 Formal Inspection and Tests (Acceptance Tests)

The~~[[_____] Division][Engineering Field Activity [_____]], Naval Facilities Engineering Systems Command, Fire Protection Engineer~~USACE Designated Fire Protection Engineer (USACE DFPE), will witness formal tests and approve all systems before they are accepted. The system shall be considered ready for such testing only after all necessary preliminary tests have been made and all deficiencies found have been corrected to the satisfaction of the Contracting Officer and written certification to this effect is received by the Division Fire Protection Engineer. Submit the request for formal inspection at least 15 working days prior to the date the inspection is to take place. Experienced technicians regularly employed by the Contractor in the installation of both the mechanical and electrical portions of such systems shall be present during the inspection and shall conduct the testing. All instruments, personnel, appliances and equipment for testing shall be furnished by the Contractor.~~[~~ The Government will furnish water for the tests.~~]~~ All necessary tests encompassing all aspects of system operation shall be made including the following, and any deficiency found shall be corrected and the system retested at no cost to the Government.

3.6.2.1 Flow Test

Perform flow tests of each standpipe riser in accordance with NFPA 14. Affix ~~[~~0-1379~~][0-2068]~~ kPa pressure gauges to lowest hose valve and next-to-highest hose valve. Connect lined, 65 mm diameter fire hose with underwriter's playpipe to highest hose valve and flow at least 946 L/m for 5 minutes from standpipe to a safe location outside the building.~~[~~ For dry pipe system, supply system through 65 mm fire hose connected to the nearest fire hydrant which is approximately ~~[_____]~~46 meters away.~~]~~ Furnish hose, nozzles and fittings required for this test.

~~[3.6.2.2 Alarm Testing~~

~~a. Each pressure switch, waterflow detector, and water motor gong shall be activated by flow of water.~~

~~b. Each valve tamper switch shall be activated by partially closing the associated control valve.~~

~~c. Alarm annunciation at the fire alarm control panel shall be verified.~~

~~d. Circuit supervision shall be demonstrated.~~

~~]~~3.6.3 Additional Tests

When deficiencies, defects or malfunctions develop during the tests required, all further testing of the system shall be suspended until proper adjustments, corrections or revisions have been made to assure proper performance of the system. If these revisions require more than a nominal delay, the Contracting Officer shall be notified when the additional work has been completed, to arrange a new inspection and test of the system. All tests required shall be repeated prior to final acceptance, unless directed otherwise.

-- End of Section --

SECTION 21 13 13

WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1	(2015) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.4	(2016) Standard for Gray Iron Threaded Fittings; Classes 125 and 250
ASME B16.21	(2016) Nonmetallic Flat Gaskets for Pipe Flanges

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1013	(2021) Performance Requirements for Reduced Pressure Principle Backflow Prevention Assemblies
ASSE 1015	(2011) Performance Requirements for Double Check Backflow Prevention Assemblies and Double Check Fire Protection Backflow Prevention Assemblies - (ANSI approved 2010)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C104/A21.4	(2016) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water
AWWA C110/A21.10	(2012) Ductile-Iron and Gray-Iron Fittings for Water
AWWA C111/A21.11	(2017) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C203	(2008) Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape - Hot-Applied
AWWA M14	(2015) Manual: Recommended Practice for Backflow Prevention and Cross-Connection Control

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M — (1999; R 2018; E 2018) Standard Specification for Ferritic Malleable Iron Castings

ASTM A53/A53M — (~~2020~~2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A135/A135M — (2009; R2014) Standard Specification for Electric-Resistance-Welded Steel Pipe

ASTM A153/A153M — (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A183 — (2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts

ASTM A536 — (1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings

FM GLOBAL (FM)

FM APP GUIDE — (updated on-line) Approval Guide ~~http~~https://www.appro

INTELLIGENCE COMMUNITY STANDARD (ICS)

ICS 705-1 — (2010) Physical and Technical Security Standard for Sensitive Compartmented Information Facilities

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-71 — (2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13 — (~~2019; ERTA 1 2018; TIA 18-1; TIA 18-2; ; ERTA 2-3 2019; TIA 19-3; TIA 19-4; TIA 19-5; TIA 19-6; ERTA 4 2019; ERTA 5 2020~~ 2025; TIA 24-3) Standard for the Installation of Sprinkler Systems

NFPA 24 — (~~2019; TIA 19-1~~2025) Standard for the Installation of Private Fire Service Mains and Their Appurtenances

NFPA 101 — (~~2021; TIA 21-1~~2024) Life Safety Code

NFPA 291 — (2019) Recommended Practice for Fire Flow Testing and Marking of Hydrants

NFPA 1963 — (2019) Standard for Fire Hose Connections

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET)

NICET 1014-7 (2012) Program Detail Manual for Certification in the Field of Fire Protection Engineering Technology (Field Code 003) Subfield of Automatic Sprinkler System Layout

UNDERWRITERS LABORATORIES (UL)

UL 199 (2020) UL Standard for Safety Automatic Sprinklers for Fire-Protection Service

UL 312 (2010; Reprint Mar 2018) UL Standard for Safety Check Valves for Fire-Protection Service

UL 405 (2013; Bul. 2020) UL Standard for Safety Fire Department Connection Devices

UL 668 (2004; Reprint Jul 2016) UL Standard for Safety Hose Valves for Fire-Protection Service

UL 1626 (2008; Bul. 2018) UL Standard for Safety Residential Sprinklers for Fire-Protection Service

UL Fire Prot Dir ~~(2020) Fire Protection Equipment Directory~~ UL Product IQ (updated online) at https://productiq.ulprospector.com/en

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS B 2301 (2013) Screwed Type Malleable Cast Iron Pipe Fittings

JIS B 2311 (20~~15~~24) Steel Butt-Welding Pipe Fittings for Ordinary Use

JIS G 3454 (2019) Carbon Steel Pipes for Pressure Service (Amendment 1)

1.2 SYSTEM DESCRIPTION

Provide wet pipe ~~[~~sprinkler~~][water spray]~~ system(s) in ~~[~~all areas of the building~~][areas indicated on the drawings] [______]~~. Except as modified herein, the system must meet the requirements of NFPA 13~~[NFPA 13R][and][NFPA 15]~~. Pipe sizes which are not indicated on the Contract drawings must be determined by hydraulic calculations.

1.2.1 Hydraulic Design

1.2.1.1 Basis for Calculations

A waterflow test was performed on ~~(DATE)~~09/09/2024 at ~~(LOCATION)~~B743. The results are as indicated on the design drawings. Perform a fire hydrant flow test prior to shop drawing submittal in accordance with NFPA 291.

Results must include hydrant elevations relative to the building and hydrant number/identifiers for the tested hydrants, including which were flowed, which had a gauge. This information must be presented in a tabular form if multiple hydrants were flowed. The results must be included with the hydraulic calculations. Hydraulic calculations must be based on flow test noted on the design drawing, unless verified by the USACE Designated Fire Protection Engineer (USACE DFPE) approved by Contracting Officer. Hydraulic calculations must be based upon the Hazen-Williams formula with a "C" value noted in NFPA 13 for piping, ~~[~~and ~~[_____]~~140 for existing underground piping~~]~~.~~[~~ Hydraulic calculations must be based on operation of the fire pump(s) provided in Section 21 30 00 FIRE PUMPS.~~]~~ ~~[~~The minimum residual pressure in a service lateral (lead-in) at the ~~[design flow rate][~~150% of the fire pump rated flow~~]~~ must be 138 kPa at ~~[the inlet to the backflow preventer][~~the suction side of the fire pump~~]]~~.

1.2.1.2 Hydraulic Calculations

a. Water supply curves and system requirements must be plotted on semi-logarithmic graph (N^1.85) paper so as to present a summary of the complete hydraulic calculation.

b. Provide a summary sheet listing sprinklers in the design area and their respective hydraulic reference points, elevations, minimum discharge pressures and minimum flows. Elevations of hydraulic reference points (nodes) must be indicated.

c. Documentation must identify each pipe individually and the nodes connected thereto. Indicate the diameter, length, flow, velocity, friction loss, number and type fittings, total friction loss in the pipe, equivalent pipe length and Hazen-Williams coefficient for each pipe.

d. Where the sprinkler system is supplied by interconnected risers, the sprinkler system must be hydraulically calculated using the hydraulically most demanding single riser. The calculations must not assume the simultaneous use of more than one riser.

e. All calculations must include the backflow preventer manufacturer's stated friction loss at the design flow or ~~[83 kPa for reduced pressure][~~55 kPa for double check~~]~~ backflow preventer, whichever is greater.

f. All calculations must be performed back to the actual location of the flow test, taking into account the direction of flow in the service main at the test location.

g. For gridded systems, calculations must show peaking of demand area friction loss to verify that the hydraulically most demanding area is being used. A flow diagram indicating the quantity and direction of flows must be included.

1.2.1.3 Design Criteria

Hydraulically design the system to discharge a minimum density over the hydraulically most demanding floor areaas indicated on the drawings. Hydraulic calculations must be in accordance with the Area/Density Method of NFPA 13. Add an allowance for exterior hose streams shall be as indicated on design drawing at the point of connection to the existing water system.

1.2.2 Sprinkler Coverage

Sprinklers must be uniformly spaced on branch lines. Provide coverage throughout 100 percent of the ~~[~~building~~][area noted on the Contract drawings]~~. This includes, but is not limited to, telephone rooms, electrical equipment rooms (regardless of the fire resistance rating of the enclosure), boiler rooms, switchgear rooms, transformer rooms, attached electrical vaults and other electrical and mechanical spaces. Coverage per sprinkler must be in accordance with NFPA 13. Provide sprinklers below all obstructions in accordance with NFPA 13. Exceptions are as follows:

a. Sprinklers may be omitted from small rooms which are exempted for specific occupancies in accordance with NFPA 101.

~~b. Facilities that are designed in accordance with NFPA 13R.~~

1.2.3 Qualified Fire Protection Engineer (QFPE)

An individual who is a licensed professional engineer (P.E.) who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience. Services of the QFPE must include:

a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Working (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.

b. Provide a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting all outstanding comments.

c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).

d. Witnessing pre-Government ~~[~~and final Government ~~]~~functional performance testing and performing a final installation review.

e. Signing applicable certificates under SD-07.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[~~Contractor Quality Control approval~~][information only]~~. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Partial submittals and submittals not fully complying with NFPA 13 and this specification section must be returned disapproved without review. SD-02, SD-03 and SD-05 must be submitted simultaneously.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE must be returned disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G~~[, [_____]]~~

Sprinkler System Designer; G~~[, [_____]]~~

Sprinkler System Installer; G~~[, [_____]]~~

SD-02 Shop Drawings

Shop Drawing; G~~[, [_____]]~~

SD-03 Product Data

Pipe; G~~[, [_____]]~~

Fittings; G~~[, [_____]]~~

Valves, including gate, check, butterfly, and globe; G~~[, [_____]]~~

~~Alarm Valves; G[, [_____]]~~

Relief Valves; G~~[, [_____]]~~

Sprinklers ; G~~[, [_____]]~~

Pipe Hangers and Supports ; G~~[, [_____]]~~

Sprinkler Alarm Switch; G~~[, [_____]]~~

Valve Supervisory (Tamper) Switch; G~~[, [_____]]~~

Fire Department Connection; G~~[, [_____]]~~

Backflow Prevention Assembly; G~~[, [_____]]~~

Air Vent; G~~[, [_____]]~~

Hose Valve; G~~[, [_____]]~~

~~[~~ Seismic Bracing; G~~[, [_____]]~~

~~]~~ Nameplates; G~~[, [_____]]~~

SD-05 Design Data

~~[~~ Seismic Bracing; G~~[, [_____]]~~

Load calculations for sizing of seismic bracing

~~]~~ Hydraulic Calculations; G~~[, [_____]]~~

SD-06 Test Reports

Test Procedures; G~~[, [_____]]~~

SD-07 Certificates

Verification of Compliant Installation; G~~[, [_____]]~~

Request for Government Final Test; G~~[, [_____]]~~

SD-10 Operation and Maintenance Data

Operating and Maintenance (O&M) Instructions; G~~[, [_____]]~~

Spare Parts Data; G~~[, [_____]]~~

SD-11 Closeout Submittals

As-built drawings

1.4 QUALITY ASSURANCE

1.4.1 Preconstruction Submittals

Within 36 days of Notice to Proceed (NTP) but no less than ~~[~~14 days~~]~~ prior to commencing work on site, the prime Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications will be returned Disapproved Without Review.

1.4.1.1 Shop Drawing

~~[_____]~~3 copies of the shop drawings, no later than 28 days prior to the start of system installation. Working drawings conforming to the requirements prescribed in NFPA 13 and must be no smaller than ~~[ISO A1][ANSI D][~~the Contract Drawings~~]~~. Each set of drawings must include the following:

1. A descriptive index with drawings listed in sequence by number. A legend sheet identifying device symbols, nomenclature, and conventions used in the package.

2. Floor plans drawn to a scale not less than 1:100 equals 1-foot clearly showing locations of devices, equipment, risers, and other details required to clearly describe the proposed arrangement.

3. Actual center-to-center dimensions between sprinklers on branch lines and between branch lines; from end sprinklers to adjacent walls; from walls to branch lines; from sprinkler feed mains, cross mains and branch lines to finished floor and roof or ceiling. A detail must show the dimension from the sprinkler and sprinkler deflector to the ceiling in finished areas.

4. Longitudinal and transverse building sections showing typical branch line and cross main pipe routing, elevation of each typical sprinkler

above finished floor and elevation of "cloud" or false ceilings in relation to the building ceilings.

5. Plan and elevation views which establish that the equipment will fit the allotted spaces with clearance for installation and maintenance.

6. Riser layout drawings drawn to a scale of not less than 1:25 equals 1-foot to show details of each system component, clearances between each other and from other equipment and construction in the room.

7. Details of each type of riser assembly, pipe hanger, ~~[~~sway bracing for earthquake protection, ~~]~~and restraint of underground water main at point-of-entry into the building, and electrical devices and interconnecting wiring. The dimension from the edge of vertical piping to the nearest adjacent wall(s) must be indicated on the drawings when vertical piping is located in stairs or other portions of the means of egress.

8. Details of each type of pipe hanger~~[~~, seismic bracing/restraint~~]~~ and related components.

~~[~~9. Include fire pump curve with shop drawings and hydraulic calculations. ~~]~~

1.4.1.2 Product Data

~~[_____]~~3 copies of annotated catalog data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, options, and other pertinent information, that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

1.4.1.3 Hydraulic Calculations

Calculations must be as outlined in NFPA 13 except that calculations must be performed by computer using software intended specifically for fire protection system design using the design data shown on the drawings.~~[~~ Calculations must include isometric diagram indicating hydraulic nodes and pipe segments.~~][~~ Include fire pump curve with submittal.~~]~~

1.4.1.4 Operating and Maintenance (O&M) Instructions

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA as supplemented and modified by this specification section.

Provide ~~[~~six~~][_____]~~ manuals~~[~~ and one pdf version on electronic media~~]~~. The manuals must include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine maintenance, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment.~~[~~ Each service organization submitted must be capable of providing ~~[~~4~~][_____]~~-hour on-site response to a service call on an emergency basis.~~]~~

Submit spare parts data for each different item of material and equipment specified. The data must include a complete list of parts and supplies, and a list of parts recommended by the manufacturer to be replaced after

1-year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied.

1.4.2 Qualifications

1.4.2.1 Sprinkler System Designer

The sprinkler system designer must be certified as a Level ~~[~~III~~][IV]~~ Technician by National Institute for Certification in Engineering Technologies (NICET) in the Water-Based Systems Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7.

1.4.2.2 Sprinkler System Installer

The sprinkler system installer must be regularly engaged in the installation of the type and complexity of system specified in the contract documents, and must have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

1.4.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of an item or equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation are mandatory requirements.

1.5 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other contaminants. All pipes must be either capped or plugged until installed.

1.6 EXTRA MATERIALS

Spare sprinklers and wrench(es) must be provided as spare parts in accordance with NFPA 13.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

2.1.1 Standard Products

Provide materials, equipment, and devices listed for fire protection service when so required by NFPA 13 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for a classification of material. Material and equipment must be standard products of a manufacturer regularly engaged in the manufacture of the products for at least ~~[~~2~~][_____]~~ years prior to bid.

2.1.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Nameplates must be etched metal or plastic, permanently attached by screws to control units, panels or adjacent walls.

2.1.3 Identification and Marking

Pipe and fitting markings must include name or identifying symbol of manufacturer and nominal size. Pipe must be marked with ASTM designation. Valves and equipment markings must have name or identifying symbol of manufacturer, specific model number, nominal size, name of device, arrow indicating direction of flow, and position of installation (horizontal or vertical), except if valve can be installed in either position. Markings must be included on the body casting or on an etched or stamped metal nameplate permanently on the valve or cover plate.

2.1.4 Pressure Ratings

Valves, fittings, couplings, alarm switches, and similar devices must be rated for the maximum working pressures that can be experienced in the system, but in no case less than ~~[~~1207~~][1724]~~ kPa.

2.2 UNDERGROUND PIPING COMPONENTS

2.2.1 Pipe

Pipe must comply with NFPA 24. Minimum pipe size is ~~[100 mm][~~150 mm~~]~~. Piping more than 1.50 meters outside the building walls must comply with Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING. A continuous section of welded stainless steel fire water service piping from a point outside the building perimeter to a flanged fitting at least 304 mm above the finished floor within the building is acceptable.

2.2.2 Fittings and Gaskets

Fittings must be ductile-iron conforming to AWWA C110/A21.10 with cement mortar lining conforming to AWWA C104/A21.4. Gaskets must be suitable in design and size for the pipe with which such gaskets are to be used. Gaskets for ductile-iron pipe joints must conform to AWWA C111/A21.11.

~~2.2.3 Gate Valve[and Indicator Posts]~~

~~Installation must comply with NFPA 24. Gate valves for use with indicator post must conform to UL 262.[Indicator posts must conform to UL 789. Provide each indicator post with one coat of primer and two coats of red enamel paint.]~~

~~[2.2.4 Valve Boxes~~

~~Except where indicator posts are provided, for each buried valve, provide a cast-iron, ductile-iron, or plastic valve box of a suitable size. Plastic boxes must be constructed of acrylonitrile-butadiene-styrene (ABS) or inorganic fiber-reinforced black polyolefin. Provide cast-iron, ductile-iron, or plastic cover for valve box with the word "WATER" cast on the cover. The minimum box shaft diameter must be 133 mm. Coat cast-iron and ductile-iron boxes with bituminous paint applied to a minimum dry-film thickness of 0.254 mm.~~

2.2.3 Buried Utility Warning and Identification Tape

Provide detectable aluminum foil plastic backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape, 80 mm minimum width, color coded for the utility involved with warning and identification imprinted in bold block letters continuously and repeatedly over the entire tape length. Warning and identification must read "CAUTION BURIED WATER PIPING BELOW" or similar wording. Use permanent code and letter coloring unaffected by moisture and other substances contained in trench backfill material.

2.3 ABOVEGROUND PIPING COMPONENTS

2.3.1 Steel Piping Components

2.3.1.1 Steel Pipe

Except as modified herein, steel pipe must be black as permitted by NFPA 13 and conform to the applicable provisions of ASTM A53/A53M, ASTM A135/A135M , ASTM A153/A153M, or JIS G 3454 - STPG370.

Steel pipe must be Schedule 40 only. Galvanized pipe is not permitted.

2.3.1.2 Fittings

Fittings must be welded, threaded, or grooved-end type. Threaded fittings must be cast-iron conforming to ASME B16.4, malleable-iron conforming to ASME B16.3 or JIS B 2301, ductile-iron conforming to ASTM A536, or steel butt-welded fittings to JIS B 2311. Plain-end fittings with mechanical couplings, fittings that use steel gripping devices to bite into the pipe, steel press fittings and field welded fittings are not permitted. Fittings, mechanical couplings, and rubber gaskets must be supplied by the same manufacturer. Threaded fittings must use Teflon tape or manufacturer's approved joint compound. Saddle tees using rubber gasketed fittings are permitted only when connecting to existing piping for additions or modifications. Saddle tees must use a connection method that completely wraps around the pipe. Reducing couplings are not permitted except as allowed by NFPA 13. Galvanized fittings are not permitted.

2.3.1.3 Grooved Mechanical Joints and Fittings

Joints and fittings must be designed for not less than 1200 kPa service and the product of the same manufacturer. Field welded fittings must not be used. Fitting and coupling housing must be malleable-iron conforming to ASTM A47/A47M, Grade 32510; ductile-iron conforming to ASTM A536, Grade 65-45-12. Rubber gasketed grooved-end pipe and fittings with mechanical couplings are permitted in pipe sizes 50 mm and larger. Gasket must be the flush type that fills the entire cavity between the fitting and the pipe. Nuts and bolts must be heat-treated steel conforming to ASTM A183 and must be cadmium-plated or zinc-electroplated.

2.3.1.4 Flanges

Flanges must conform to NFPA 13 and ASME B16.1. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, 1.6 mm thick, and full face or self-centering flat ring type.

2.3.2 Flexible Sprinkler Hose

The use of flexible hose is not permitted.

2.3.3 Pipe Hangers and Supports

Provide galvanized pipe hangers~~[~~, supports and seismic bracing~~][and supports]~~ in accordance with NFPA 13.~~[~~ Design and install seismic protection in accordance with the requirements of NFPA 13 section titled "Protection of Piping Against Damage Where Subject to Earthquakes for Seismic Design Category ~~[~~"D"~~]["_____"]]~~.

2.3.4 Valves

Provide valves of types approved for fire service. Valves must open by counterclockwise rotation.

2.3.4.1 Control Valve

Manually operated sprinkler control/gate valve must be ~~[~~outside stem and yoke (OS&Y) type~~][or][butterfly type][as indicated on the drawings]~~ and must be listed.

2.3.4.2 Check Valves

Check valves must comply with UL 312. Check valves 100 mm and larger must be of the swing type, have a clear waterway and meet the requirements of MSS SP-71, for Type 3 or 4. Inspection plate must be provided on valves larger than 150 mm.

2.3.4.3 Hose Valve

Valve must comply with UL 668.

2.3.5 ~~[Alarm][~~Riser Check~~]~~ Valves

~~[~~Provide riser check valve, pressure gauges and main drain.~~][Provide variable pressure type alarm check valve, standard trim piping, pressure gauges, bypass, retarding chamber, testing valves, and main drain, and other components as required for a fully operational system. Alarm valves must comply with UL 193.]~~

2.4 ALARM INITIATING AND SUPERVISORY DEVICES

2.4.1 Sprinkler Alarm Switch

Vane or pressure-type flow switch(es). ~~[~~Connection of switch must be by the fire alarm installer~~]~~. ~~[~~Vane type alarm actuating devices must have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and must instantly recycle. ~~][~~Flow switches for elevator power shunt must not have a retard feature.~~]~~

2.4.2 Valve Supervisory (Tamper) Switch

Switch must be integral to the control valve or suitable for mounting to the type of control valve to be supervised open. The switch must be tamper resistant and contain SPDT (Form C) contacts arranged to transfer upon removal of the housing cover or closure of the valve of more than two rotations of the valve stem.

~~[~~2.5 BACKFLOW PREVENTION ASSEMBLY

~~[Reduced-pressure principle][~~Double-check~~]~~ valve assembly backflow preventer complying with ASSE 1013, ASSE 1015 and AWWA M14. Each check valve must have a drain. Backflow prevention assemblies must have current "Certificate of Approval from the Foundation for Cross-Connection Control and Hydraulic Research, FCCCHR List" and be listed for fire protection use. Listing of the specific make, model, design, and size in the FCCCHR List is acceptable as the required documentation.

2.5.1 Backflow Preventer Test Connection

Test connection must consist of a series of listed hose valves with 65-mm National Standard male hose threads with cap and chain.

~~]~~2.6 FIRE DEPARTMENT CONNECTION

Fire department connection must be ~~[freestanding][~~projecting~~][flush]~~ type with cast-brass body, matching ~~[~~wall~~]~~ escutcheon lettered "Auto Spkr" with a ~~[~~polished-brass~~][chromium-plated]~~ finish. ~~[~~The connection must have individual self-closing clappers, caps with drip drains and chains.~~]~~ Female inlets must have ~~[~~65-mm~~][100 mm][125 mm][_____]~~ diameter ~~[~~American National Fire Hose Connection Screw Threads (NH) per NFPA 1963~~] [Storz][_____]~~. Comply with UL 405.

2.7 SPRINKLERS

Sprinklers must comply with UL 199 and NFPA 13. Sprinklers with internal O-rings are not acceptable. Sprinklers in high heat areas including attic spaces or in close proximity to unit heaters must have temperature classification in accordance with NFPA 13. Extended coverage sprinklers are permitted for loading docks, residential occupancies and high-piled storage applications only.

2.7.1 Pendent Sprinkler

Pendent sprinkler must be~~[recessed][~~ quick-response~~][dry pendent]~~ type with nominal K-factor of ~~[~~80~~][~~ or 115~~][160][_____]~~. Pendent sprinklers must have a ~~[polished chrome][stainless steel][~~white polyester~~][_____]~~ finish. Assembly must include an integral escutcheon.

2.7.2 Upright Sprinkler

Upright sprinkler must be ~~[brass][chrome-plated][stainless steel][~~white polyester~~]~~ ~~[~~quick-response type~~][_____]~~ and have a nominal K-factor of ~~[~~80 ~~][~~ or 115~~][160][_____]~~.

2.7.3 Sidewall Sprinkler

Sidewall sprinkler must be the ~~[~~quick-response~~][standard-response]~~ ~~[~~ recessed~~][~~ dry sidewall~~]~~ type. Sidewall sprinkler must have a nominal K-factor of ~~[~~80~~][115][160][_____]~~. Sidewall sprinkler must have a ~~[brass][polished-chrome][stainless steel][~~white polyester~~][_____]~~ finish.

~~[~~2.7.4 Concealed Sprinkler

Concealed sprinkler must be ~~[chrome-plated][stainless steel][~~white polyester~~]~~ ~~[~~quick-response type~~][_____]~~ and have a nominal K-factor of ~~[~~80

~~][115][160][_____]~~. Coverplate must be ~~[chrome][~~white~~][_____]~~.

~~]~~2.7.5 Residential Sprinkler

Residential sprinkler must be ~~[recessed pendent][pendent][sidewall]~~ concealed pendent type with nominal K-factor of ~~[~~60~~][~~ or 80~~]~~. Residential sprinkler must have a ~~[polished chrome][~~white polyester~~][_____]~~ finish. Sprinkler must comply with UL 1626.

~~[~~2.7.6 Corrosion-Resistant Sprinkler

Corrosion-resistant sprinkler must be the ~~[~~upright~~][pendent]~~ type installed in locations as indicated. Corrosion-resistant coatings must be factory-applied by the sprinkler manufacturer.

~~][~~2.7.7 Dry Sprinkler Assembly

Dry sprinkler assembly must be of the~~[~~ pendent~~][sidewall][45-degree]~~or upright quick response type as indicated. Assembly must include an integral escutcheon. Maximum length must not exceed maximum indicated in its listing. Sprinkler must have a ~~[polished chrome][~~corrosion resistant polyester coating~~][white enamel]~~ finish.

~~][2.7.8 Control Mode Specific Application Sprinkler~~

~~Control mode specific application sprinkler must be of the [pendent][upright][dry sidewall] type as indicated. Sprinkler must be specifically listed for high-piled storage only. Sprinkler must have a [polished chrome][rough brass] finish.~~

~~][2.7.9 ESFR Sprinkler~~

~~ESFR sprinkler must be [pendent][upright] and comply with UL 1767. Nominal K-factor must be [_____].~~

~~][2.7.10 Intermediate Level Rack Sprinkler~~

~~Intermediate level rack sprinkler must be of the [upright][pendent] type with nominal K-factor of [80][115]. The sprinkler must be equipped with a deflector plate to shield the fusible element from water discharged above it.~~

~~]~~2.8 ACCESSORIES

2.8.1 Sprinkler Cabinet

Provide spare sprinklers in accordance with NFPA 13 and must be placed in a suitable metal or plastic cabinet of sufficient size to accommodate all the spare sprinklers and wrenches in designated locations. Spare sprinklers must be representative of, and in proportion to, the number of each type and temperature rating of the sprinklers installed as required by NFPA 13. At least one wrench of each type required must be provided.

2.8.2 Pendent Sprinkler Escutcheon

Escutcheon must be one-piece metallic type with a depth of less than 19 mm and suitable for installation on pendent sprinklers. The escutcheon must have a factory finish that matches the pendent sprinkler.

2.8.3 Pipe Escutcheon

Provide split hinge metal plates for piping entering walls, floors, and ceilings in exposed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

2.8.4 Sprinkler Guard

Listed guard must be a steel wire cage designed to encase the sprinkler and protect it from mechanical damage. Guards must be provided on sprinklers located ~~[_____][~~within 2.1 meters of the floor~~][~~ and as indicated~~]~~.

2.8.5 Relief Valve

Relief valves must be listed and installed at there riser in accordance with NFPA 13.

2.8.6 Air Vent

Air vents must be of the automatic type and piped to drain to the building exterior.

2.8.7 Identification Sign

Valve identification sign must be minimum 150 mm wide by 50 mm high with enamel baked finish on minimum 1.214-mm steel or 0.6-mm aluminum with red letters on a white background or white letters on red background. Wording of sign must include, but not be limited to "main drain", "auxiliary drain", "inspector's test", "alarm test", "alarm line", and similar wording as required to identify operational components. Where there is more than one sprinkler system, signage must include specific details as to the respective system.

PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work that is dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative a condition that prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

3.2 INSTALLATION

The installation must be in accordance with the applicable provisions of NFPA 13, NFPA 24 and publications referenced therein.~~[Installation of in-rack sprinklers must comply with applicable provisions of NFPA 13.]~~ Locate sprinklers in a consistent pattern with ceiling grid, lights, and air supply diffusers. Install sprinkler system over and under ducts, piping and platforms when such equipment can negatively affect or disrupt the sprinkler discharge pattern and coverage.

a. Piping offsets, fittings, and other accessories required must be furnished to provide a complete installation and to eliminate interference with other construction.

b. Wherever the contractor's work interconnects with work of other trades the Contractor must coordinate with other Contractors to insure all Contractors have the information necessary so that they may properly install all necessary connections and equipment. Identify all work items needing access (dampers and similar equipment) that are concealed above hung ceilings by permanent color coded pins/tabs in the ceiling directly below the item.

c. Provide required supports and hangers for piping, conduit, and equipment so that loading will not exceed allowable loadings of structure. Submittal of a bid must be a deemed representation that the contractor submitting such bid has ascertained allowable loadings and has included in his estimates the costs associated in furnishing required supports.

3.2.1 Waste Removal

At the conclusion of each day's work, clean up and stockpile on site all waste, debris, and trash which may have accumulated during the day as a result of work by the contractor and of his presence on the job. Sidewalks and streets adjoining the property must be kept broom clean and free of waste, debris, trash and obstructions caused by work of the contractor, which will affect the condition and safety of streets, walks, utilities, and property.

3.3 UNDERGROUND PIPING INSTALLATION

The fire protection water main must be laid, and joints anchored, in accordance with NFPA 24. Minimum depth of cover must be ~~[900][______]~~ mm or the frost line, whichever is deeper. The supply line must terminate inside the building with a flanged piece, the bottom of which must be set not less than 304 mm above the finished floor. A blind flange must be installed temporarily on top of the flanged piece to prevent the entrance of foreign matter into the supply line. A concrete thrust block must be provided at the elbow where the pipe turns up toward the floor. In addition, joints must be anchored in accordance with NFPA 24. Buried steel components must be provided with a corrosion protective coating in accordance with AWWA C203. Piping more than 1500 mm outside the building walls must meet the requirements of Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

3.4 ABOVEGROUND PIPING INSTALLATION

The methods of fabrication and installation of the aboveground piping must fully comply with the requirements and recommended practices of NFPA 13 and this specification section.

3.4.1 Protection of Piping Against Earthquake Damage

Seismic restraint is ~~[not]~~required.

3.4.2 Piping in Exposed Areas

Install exposed piping without diminishing exit access widths, corridors or equipment access. Exposed horizontal piping, including drain piping, must be installed to provide maximum headroom.

3.4.3 Piping in Finished Areas

In areas with suspended or dropped ceilings and in areas with concealed spaces above the ceiling, piping must be concealed above ceilings. Piping must be inspected, hydrostatically tested and approved before being concealed. Risers and similar vertical runs of piping in finished areas must be concealed.

3.4.4 Pendent Sprinklers

a. Drop nipples to pendent sprinklers must consist of minimum 25-mm pipe with a reducing coupling into which the sprinkler must be threaded.

b. Where sprinklers are installed below suspended or dropped ceilings, drop nipples must be cut such that sprinkler ceiling plates or escutcheons are of a uniform depth throughout the finished space. The outlet of the reducing coupling must not extend below the underside of the ceiling.

c. Recessed pendent sprinklers must be installed such that the distance from the sprinkler deflector to the underside of the ceiling must not exceed the manufacturer's listed range and must be of uniform depth throughout the finished area.

d. Pendent sprinklers in suspended ceilings must be located in the center of the tile (+/- 2 inches).~~[~~

~~e. Dry pendent sprinkler assemblies must be such that sprinkler ceiling plates or escutcheons are of the uniform depth throughout the finished space.][~~

e~~f~~. Dry pendent or upright sprinklers must be of the required length to permit the sprinkler to be threaded directly into a branch line tee.~~][~~

f~~g~~. Where the maximum static or flowing pressure, whichever is greater at the sprinkler, applied other than through the fire department connection, exceeds 6.9 bar and a branch line above the ceiling supplies sprinklers in a pendent position below the ceiling, the cumulative horizontal length of an unsupported armover to a sprinkler or sprinkler drop must not exceed 300 mm for steel pipe.~~]~~

3.4.5 Upright Sprinklers

Riser nipples or "sprigs" to upright sprinklers must contain no fittings between the branch line tee and the reducing coupling at the sprinkler.

3.4.6 Pipe Joints

Pipe joints must conform to NFPA 13, except as modified herein. Not more than four threads must show after joint is made up. Welded joints will be permitted, only if welding operations are performed as required by NFPA 13 at the Contractor's fabrication shop, not at the project construction site. Flanged joints must be provided where indicated or required by NFPA 13. Grooved pipe and fittings must be prepared in accordance with the manufacturer's latest published specification according to pipe material, wall thickness and size. Grooved couplings, fittings and grooving tools must be products of the same manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances.

3.4.7 Reducers

Reductions in pipe sizes must be made with one-piece tapered reducing fittings. When standard fittings of the required size are not manufactured, single bushings of the face or hex type will be permitted. Where used, face bushings must be installed with the outer face flush with the face of the fitting opening being reduced. Bushings cannot be used in elbow fittings, in more than one outlet of a tee, in more than two outlets of a cross, or where the reduction in size is less than 13 mm.

3.4.8 Pipe Penetrations

a. Cutting structural members for passage of pipes or for pipe-hanger fastenings will not be permitted. Pipes that must penetrate concrete or masonry walls or concrete floors must be core-drilled and provided with pipe sleeves. Each sleeve must be Schedule 40 galvanized steel, ductile-iron or cast-iron pipe and extend through its respective wall or floor and be cut flush with each wall surface. Sleeves must provide required clearance between the pipe and the sleeve per NFPA 13. The space between the sleeve and the pipe must be firmly packed with mineral wool insulation.

b. Where pipes and sleeves penetrate fire walls, fire partitions, or floors, pipes/sleeves must be firestopped in accordance with Section 07 84 00 FIRESTOPPING.

c. In penetrations that are not fire-rated or not a floor penetration, the space between the sleeve and the pipe must be sealed at both ends with plastic waterproof cement that will dry to a firm but pliable mass or with a mechanically adjustable segmented elastomer seal.

d. All penetrations through the boundary of rooms/areas identified as secure space area must meet ICS 705-1.

3.4.9 Escutcheons

Escutcheons must be provided for pipe penetration in finished areas of ceilings, floors and walls. Escutcheons must be securely fastened to the pipe at surfaces through which piping passes.

3.4.10 Inspector's Test Connection

Unless otherwise indicated, the test connection must consist of 25-mm pipe connected ~~[to the remote branch line][~~at the riser as a combination test and drain valve~~]~~; a test valve located approximately 2 meters above the floor; a smooth bore brass outlet equivalent to the smallest orifice sprinkler used in the system; and a painted metal identification sign affixed to the valve with the words "Inspector's Test". All test connection piping must be inside of the building and penetrate the exterior wall at the location of the discharge orifice only. The discharge orifice must be located outside the building wall no more than 0.6 meters above finished grade, directed so as not to cause damage to adjacent construction or landscaping during full flow discharge, or to the sanitary sewer. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building. Do not discharge to the roof. Discharge to floor drains, janitor sinks or similar fixtures is not permitted.

Provide concrete splash blocks at all drain and inspector's test connection discharge locations if not discharging to a concrete surface. Splash blocks must be large enough to mitigate erosion and not become dislodged during a full flow of the drain. Ensure all discharged water drains away from the facility and does not cause property damage.

3.4.11 Backflow Preventer

Locate within the building or in a heated enclosure in locations subject to freezing. For heated enclosures, provide a low temperature supervisory alarm connected to the facility fire alarm system. Heat trace is not permitted to be used.

Install backflow preventers so that the bottom of the assembly is a minimum of 150 mm above the finished floor/grade. Install horizontal backflow preventers so that the bottom of the assembly is no greater than ~~[_____]~~610 mm above the finished floor/grade. Install vertical backflow preventers so that the upper operating handwheel is no more than ~~[_____]~~ 1.8 meters above the finished floor/grade. Clearance around control valve handles must be minimum 150 mm above grade/finished floor and away from walls.

~~[~~3.4.11.1 Test Connection

Provide downstream of the backflow prevention assembly UL 668 hose valves with 65-mm National Standard male hose threads with cap and chain. Provide one valve for each 946 L/min of system demand or fraction thereof. Provide a permanent sign in accordance with paragraph entitled "Identification Signs" which reads, "Test Valve". Indicate location of test header. If an exterior connection, provide a control valve inside a heated mechanical room to prevent freezing.

~~]~~3.4.12 Drains

a. Main drain piping must be provided to discharge ~~[~~at a safe point outside the building, no more than 0.6 meters above finished grade~~][at the location indicated][to the sanitary sewer]~~. Provide a concrete splash block at drain outlet. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building.

b. Auxiliary drains must be provided as required by NFPA 13. Auxiliary drains are permitted to discharge to a floor drain if the drain is sized to accommodate full flow (min 151 L/min). Discharge to service sinks or similar plumbing fixtures is not permitted.

3.4.13 Installation of Fire Department Connection

Connection must be mounted ~~[~~on the exterior wall approximately 900 mm above finished grade~~][adjacent to and on the sprinkler system side of the backflow preventer]~~. The piping between the connection and the check valve must be provided with an automatic drip in accordance with NFPA 13 and piped to drain to the outside or a floor drain within the same room.

3.4.14 Identification Signs

Signs must be affixed to each control valve, inspector test valve, main drain, auxiliary drain, test valve, and similar valves as appropriate or as required by NFPA 13. Main drain test results must be etched into main drain identification sign. Hydraulic design data must be etched into the nameplates and permanently affixed to each sprinkler riser as specified in NFPA 13. Provide labeling on the surfaces of all feed and cross mains to show the pipe function (e.g., "Sprinkler System", "Fire Department Connection", "Standpipe") and normal valve position (e.g. "Normally Open", "Normally Closed"). For pipe sizes 100 mm and larger provide white painted stenciled letters and arrows, a minimum of 50 mm in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 100 mm, provide white painted stenciled letters and arrows, a minimum of 18 mm in height and visible from the floor.[Provide properly lettered and approved metal sign to elevator flow switch stating the circuits' voltage, and identify the switch as an "Elevator Power Shunt Flow Switch".]

3.5 ELECTRICAL

Except as modified herein, electric equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.[Alarm signal wiring connected to the building fire alarm control system must be by the fire alarm installer.]

3.6 PAINTING

Color code mark piping [red][as specified in Section 09 90 00 PAINTS AND COATINGS].

3.7 FIELD QUALITY CONTROL

3.7.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level [III][or][IV] Fire Sprinkler Technician, and the representative of the installing company,[and reviewed by the QFPE] [60][_____] days prior to performing system tests. Detailed test procedures must list all components of the installed system. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 13–.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government final testing.

a. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.7.2 Pre-Government Testing

3.7.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that equipment is functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" and "System Acceptance" as noted in NFPA 13. The Contractor [and QFPE] must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 13 and the test reports noted below.

a. NFPA 13 Aboveground Material and Test Certificate

b. NFPA 13 Underground Material and Test Certificate

3.7.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the ~~[______]~~~~[~~Designated Fire Protection Engineer (DFPE)~~][Contracting Officers Designated Representative (COR)]~~. Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the ~~[building fire alarm system][~~installation fire alarm reporting system~~]~~ have been completed and tested to confirm communications are fully functional. Submit request for test at least ~~[~~15~~][______]~~ calendar days prior to the requested test date.

3.7.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.7.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

a. The manufacturer's technical representative.

~~[~~b. The contractor's Qualified Fire Protection Engineer (QFPE).~~]~~

c. Marked-up red line drawings of the system as actually installed.

Government Final Tests will be witnessed by the ~~[______], [~~Designated Fire Protection Engineer, ~~][~~Contracting Officer~~][~~, Qualified Fire Protection Engineer (QFPE)~~]~~. At this time, all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.8 MINIMUM SYSTEM TESTS

The system, including the underground water mains, and the aboveground piping and system components, must be tested to ensure that equipment and components function as intended. The underground and aboveground interior piping systems and attached appurtenances subjected to system working pressure must be tested in accordance with NFPA 13 and NFPA 24.

3.8.1 Underground Piping

3.8.1.1 Flushing

Underground piping must be flushed at a minimum of 3 mps in accordance with NFPA 24.

3.8.1.2 Hydrostatic Test

New underground piping must be hydrostatically tested in accordance with NFPA 24.

3.8.2 Aboveground Piping

3.8.2.1 Hydrostatic Test

Aboveground piping must be hydrostatically tested in accordance with NFPA 13. There must be no drop in gauge pressure or visible leakage when the system is subjected to the hydrostatic test. The test pressure must be read from a gauge located at the low elevation point of the system or portion being tested.

3.8.2.2 Backflow Prevention Assembly Forward Flow Test

Each backflow prevention assembly must be tested at system flow demand, including all applicable hose streams, as specified in NFPA 13. The Contractor must provide all equipment and instruments necessary to conduct a complete forward flow test, including 65-mm diameter hoses, playpipe nozzles or flow diffusers, calibrated pressure gauges, and pitot tube gauge. The Contractor must provide all necessary supports to safely secure hoses and nozzles during the test. At the system demand flow, the pressure readings and pressure drop (friction loss) across the assembly must be recorded. A metal placard must be provided on the backflow prevention assembly that lists the pressure readings both upstream and downstream of the assembly, total pressure drop, and the system test flow rate determined during the preliminary testing. The pressure drop must be compared to the manufacturer's data and the readings observed during the final inspections and tests.

3.8.3 Main Drain Flow Test

Following flushing of the underground piping, a main drain test must be made to verify the adequacy of the water supply. Static and residual pressures must be recorded on the certificate specified in paragraph SUBMITTALS.

3.9 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. Submit six sets of detailed as-built drawings. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final acceptance test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the final acceptance test.

a. Provide one set of full size paper as-built drawings and schematics. The drawings must be prepared electronically and sized no less than the contract drawings.~~]~~ ~~[~~Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of ~~[MicroStation][~~AutoCAD, ~~]~~DXF and portable document formats of as-built drawings and schematics.~~]~~

b. Provide operating and maintenance (O&M) instructions.

3.10 ONSITE TRAINING

Conduct a training course for the responding fire department and operating and maintenance personnel as designated by the Contracting Officer. Training must be performed on two separate days (to accommodate different shifts of Fire Department personnel) for a period of ~~[_____][~~4~~]~~ hours of normal working time and must start after the system is functionally complete and after the final acceptance test. The on-site training must cover all of the items contained in the approved Operating and Maintenance Instructions.

-- End of Section --

SECTION 21 30 00

FIRE PUMPS
04/08

PART 1 GENERAL

1.1 SUMMARY

Except as modified in this Section or on the drawings, install fire pumps in conformance with NFPA 20, NFPA 70, and NFPA 72. In the event of a conflict between specific provisions of this specification and applicable NFPA standards, this specification governs. Devices and equipment for fire protection service must be UL Fire Prot Dir listed or FM APP GUIDE approved. Interpret all reference to the authority having jurisdiction to mean the Contracting Officer.

1.2 SEQUENCING

1.2.1 Primary Fire Pump

Primary fire pump shall ~~[~~automatically operate when the pressure drops to ~~[758][_____]~~1048 kPa~~] [automatically upon tripping of the [_____] sprinkler system][~~, ~~[~~and~~][or]~~ manually when the starter is operated~~]~~. ~~[Pump[s] shall continue to run until shut down manually.] [Pump[s] shall automatically shut down after a running time of [_____] minutes unless manually shutdown.]~~ The fire pump shall automatically stop operating when the system pressure reaches ~~[862][_____]~~1151.4 kPa and after the fire pump has operated for the minimum pump run time specified herein.

~~1.2.2 Secondary Fire Pump~~

~~Secondary fire pump shall operate at 69 kPa increments, set below the primary fire pump starting pressure. The fire pump shall automatically stop running at [862][_____] kPa and after the fire pump has operated for the minimum pump run time. Fire pumps shall be prevented from starting simultaneously and shall start sequentially at intervals of 5 to 10 seconds.~~

1.2.2 Pressure Maintenance Pump

Pressure maintenance pump shall operate when the system pressure drops to ~~[793][_____]~~1082.5 kPa. Pump shall automatically stop when the system pressure reaches ~~[862][_____]~~1151.4 kPa and after the pump has operated for the minimum pump run time specified herein.

1.3 FIRE PUMP INSTALLATION RELATED SUBMITTALS

~~[Perform work specified in this section under the supervision of and certified by the Fire Protection Specialist who is certified as a Level [III] [IV] Technician by National Institute for Certification in Engineering Technologies (NICET) in the [Automatic Sprinkler System] [Special Hazards Suppression System] Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7.]~~

~~[~~Perform work specified in this section under the supervision of or prepared by the by fire protection specialist who is the QFPE as stated in paragraph 1.7.1.~~]~~

The Fire Protection Specialist shall prepare a list of the submittals, from the Contract Submittal Register, that relate to the successful installation of the fire pump(s), no later than ~~[~~7~~] [______]~~ days after the approval of the Fire Protection Specialist and the Manufacturer's Representative. The submittals identified on this list shall be accompanied by a letter of approval signed and dated by the Fire Protection Specialist when submitted to the Government.

1.4 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA 10084	(2005) Standard Methods for the Examination of Water and Wastewater
AWWA B300	(2010; Addenda 2011) Hypochlorites
AWWA B301	(2010) Liquid Chlorine
AWWA C110/A21.10	(2012) Ductile-Iron and Gray-Iron Fittings for Water
AWWA C111/A21.11	(2017) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C500	(2009) Metal-Seated Gate Valves for Water Supply Service
AWWA C606	(2015) Grooved and Shouldered Joints

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.18	(2021) Cast Copper Alloy Solder Joint Pressure Fittings
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
ASME B16.26	(2013) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes
ASME B16.39	(2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300
ASME B16.5	(20~~17~~20) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B31.1	(~~2016; Errata 2016~~2024) Power Piping

ASTM INTERNATIONAL (ASTM)

ASTM A183	(2014; R 2020) Standard Specification for

	Carbon Steel Track Bolts and Nuts
ASTM A193/A193M	(2017) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2018) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A449	(2014; R 2020) Standard Specification for Hex Cap Screws, Bolts, and Studs, Steel, Heat Treated, 120/105/90 ksi Minimum Tensile Strength, General Use
ASTM A47/A47M	(1999; R 2018; E 2018) Standard Specification for Ferritic Malleable Iron Castings
ASTM A53/A53M	(~~2020~~2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings
ASTM A795/A795M	(2013) Standard Specification for Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Fire Protection Use
ASTM B135M	(2010) Standard Specification for Seamless Brass Tube (Metric)
ASTM B42	(2020) Standard Specification for Seamless Copper Pipe, Standard Sizes
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B75/B75M	(2020) Standard Specification for Seamless Copper Tube
ASTM B88M	(2020) Standard Specification for Seamless Copper Water Tube (Metric)
ASTM D2000	(2012; R 2017) Standard Classification System for Rubber Products in Automotive Applications
ASTM D3308	(2012; R 20~~17~~22) ~~P~~Standard Specification for PTFE Resin Skived Tape
ASTM F436M	(2011) Hardened Steel Washers (Metric)

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide ~~http~~https://www.appro

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1	(2021) Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1963	(2019) Standard for Fire Hose Connections
NFPA 20	(2016; ERTA 2016) Standard for the Installation of Stationary Pumps for Fire Protection
NFPA 24	(~~2019; TIA 19-1~~2025) Standard for the Installation of Private Fire Service Mains and Their Appurtenances
NFPA 70	~~(2023; ERTA 7 2023; TIA 23-15) National Electrical Code~~(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 72	(~~2022; ERTA 22-1~~2025; TIA 25-4) National Fire Alarm and Signaling Code

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET)

NICET 1014-7	(2012) Program Detail Manual for Certification in the Field of Fire Protection Engineering Technology (Field Code 003) Subfield of Automatic Sprinkler System Layout

UNDERWRITERS LABORATORIES (UL)

UL 1247	(2007; Reprint Apr 2014) Diesel Engines for Driving Stationary Fire Pumps
UL 262	(2004; Reprint Oct 2011) Gate Valves for Fire-Protection Service
UL 448	(2007; Reprint Jan 2016) Centrifugal Stationary Pumps for Fire-Protection Service
UL Fire Prot Dir	~~(2020) Fire Protection Equipment Directory~~ UL Product IQ (updated online) at https://productiq.ulprospector.com/en

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 2301	(2013) Screwed Type Malleable Cast Iron Pipe Fittings
JIS B 2311	(20~~15~~24) Steel Butt-Welding Pipe Fittings

for Ordinary Use

JIS G 3452~~–~~	(20~~16~~19) Carbon Steel Pipes for Ordinary Piping~~ (Amendment 1)~~

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[~~Contractor Quality Control approval.~~] [information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.]~~

~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING. ~~Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Fire Pump Installation Related Submittals

Fire Protection Specialist; G~~[, [_____]]~~

No later than ~~[~~14~~] [_____]~~ days after the Notice to Proceed and prior to the submittal of the fire pump installation drawings

SD-02 Shop Drawings

Installation Drawings; G~~[, [_____]]~~

~~[~~3~~] [_____]~~ copies

As-Built Drawings; G~~[, [_____]]~~

Piping Layout; G~~[, [_____]]~~

Pump Room; G~~[, [_____]]~~

SD-03 Product Data

Catalog Data; G~~[, [_____]]~~

Spare Parts

Preliminary Tests

At least ~~[~~14~~] [_____]~~ days prior to the proposed date and time to begin Preliminary Tests

Field Tests; G~~[, [_____]]~~

At least 2 weeks before starting field tests

Manufacturer's Representative; G~~[, [_____]]~~

Field Training; G~~[, [_____]]~~

Final Acceptance Test

SD-06 Test Reports

Preliminary Tests

~~[~~3~~] [______]~~ copies of the completed Preliminary Tests Reports, no later that ~~[~~7~~] [______]~~ days after the completion of the Preliminary Tests.

SD-07 Certificates

Fire Protection Specialist

No later than ~~[~~14~~] [______]~~ days after the Notice to Proceed and prior to the submittal of the fire pump installation drawings

Qualifications of Welders

Qualifications of Installer

Preliminary Test Certification

Final Test Certification

SD-10 Operation and Maintenance Data

Operating and Maintenance Instructions; G~~[, [_____]]~~

At least ~~[~~14~~] [______]~~ days prior to conducting field training

Flow Meter

Submit Data Package 2 for flow meter and controllers in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6 EXTRA MATERIALS

Submit Spare Parts data for each different item of equipment and material specified. The data shall include a complete list of parts and supplies, with current unit prices and source of supply, and a list of parts recommended by the manufacturer to be replaced after 1 year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied by the Contractor.

1.7 QUALITY ASSURANCE

1.7.1 Quality Control Fire Protection Engineer

A QFPE (Quality Control Fire Protection Engineer)shall be an inidividual who is a registered professional engineering who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveys (NCESES). The role of the QFPE shall be to review and approve the construction drawings, calculations, material data sheets and must monitor the installation of the fire protection systems and certify in writing that the fire protection systems have been constructed and operate as intended in the design plans and specifications.

~~[The QFPE must review the shop drawings, hydraulic calculations and~~

~~material submittals. The shop drawings must bear the Review Stamp of the QFPE prior to submitting the fire extinguishing system shop drawings to the DFPE.]~~[Construction (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations and material data sheets, indicating approval prior to submitting the fire extinguishing system shop drawings to the DFPE.]

1.7.2 Fire Protection Specialist

Work specified in this section shall be performed under the supervision of and certified by the Fire Protection Specialist. Submit the name and documentation of certification of the proposed Fire Protection Specialists. The Fire Protection Specialist shall be an individual who is a registered professional engineer and a Full Member of the Society of Fire Protection Engineers or who is certified as a Level IV Technician by National Institute for Certification in Engineering Technologies (NICET) in the Automatic Sprinkler System Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7. The Fire Protection Specialist shall be regularly engaged in the design and installation of the type and complexity of system specified in the Contract documents, and shall have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

1.7.3 Qualifications of Welders

Submit certificates of each welder's qualifications prior to site welding; certifications shall not be more than one year old.

1.7.4 Qualifications of Installer

Prior to installation, submit data for approval showing that the Contractor has successfully installed fire pumps and associated equipment of the same type and design as specified herein, or that he has a firm contractual agreement with a subcontractor having such required experience. The data shall include the names and locations of at least two installations where the Contractor, or the subcontractor referred to above, has installed such systems. Indicate the type and design of each system and certify that each system has performed satisfactorily in the manner intended for a period of not less than 18 months.

1.7.5 Preliminary Test Certification

When preliminary tests have been completed and corrections made, submit a signed and dated certificate with a request for a formal inspection and tests.

1.7.6 Final Test Certification

Concurrent with the Final Acceptance Test Report, submit certification by the Fire Protection Specialist that the fire pump installation is in accordance with the contract requirements, including signed approval of the Preliminary and Final Acceptance Test Reports.[Submit data for approval showing the name and certification of all involved individuals with such qualifications at or prior to submittal of drawings.]

1.7.7 Manufacturer's Representative

Work specified in this section shall be performed under the supervision of and certified by a representative of the fire pump manufacturer. Submit the name and documentation of certification of the proposed Manufacturer's Representative, concurrent with submittal of the Fire Protection Specialist Qualifications. The Manufacturer's Representative shall be regularly engaged in the installation of the type and complexity of fire pump(s) specified in the Contract documents, and shall have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

1.8 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other contaminants. Additionally, all pipes shall be either capped or plugged until installed.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

a. Materials and equipment shall be standard products of a manufacturer regularly engaged in the manufacture of such products and shall essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening.

b. Submit manufacturer's catalog data included with the Fire Pump Installation Drawings for each separate piece of equipment proposed for use in the system. Catalog data shall indicate the name of the manufacturer of each item of equipment, with data annotated to indicate model to be provided. In addition, a complete equipment list that includes equipment description, model number and quantity shall be provided. Catalog data for material and equipment shall include, but not be limited to, the following:

(1) Fire pumps, drivers and controllers including manufacturer's certified shop test characteristic curve for each pump. Shop test curve may be submitted after approval of catalog data but shall be submitted prior to the final tests.

(2) Pressure maintenance pump and controller.

(3) Piping components.

(4) Valves, including gate, check, globe and relief valves.

(5) Gauges.

(6) Hose valve manifold test header and hose valves.

(7) Flow meter.

(8) Restrictive orifice union.

(9) Associated devices and equipment.

c. All equipment shall have a nameplate that identifies the manufacturer's name, address, type or style, model or serial number, ~~[~~ contract number and accepted date; capacity or size; system in which installed and system which it controls~~]~~ and catalog number. Pumps and motors shall have standard nameplates securely affixed in a conspicuous place and easy to read. Fire pump shall have nameplates and markings in accordance with UL 448. Diesel driver shall have nameplate and markings in accordance with UL 1247. Electric motor nameplates shall provide the minimum information required by NFPA 70, Section 430-7.

2.2 FIRE PUMP

Fire pump shall be ~~[~~electric motor driven~~] [diesel engine driven]~~. Each pump capacity shall be rated at ~~[_____]~~3785 L/second with a rated net pressure of ~~[_____]~~517.1 kPa. Fire pump shall furnish not less than 150 percent of rated flow capacity at not less than 65 percent of rated net pressure. Pump shall be centrifugal ~~[~~horizontal split case~~][water lubricated, vertical shaft turbine][end-suction][in-line]~~ fire pump. Horizontal pump shall be equipped with automatic air release devices. The maximum rated pump speed shall be 2100 rpm when driving the pump at rated capacity. Pump shall be ~~[automatic start and manual stop][manual pushbutton start and stop][~~automatic start and automatic stop~~]~~. Pump shall conform to the requirements of UL 448. Fire pump discharge and suction gauges shall be oil-filled type.

2.3 REQUIREMENTS FOR FIRE PROTECTION SERVICE

2.3.1 General Requirements

Materials and Equipment shall have been tested by Underwriters Laboratories, Inc. and listed in UL Fire Prot Dir or approved by Factory Mutual and listed in FM APP GUIDE. Where the terms "listed" or "approved" appear in this specification, such shall mean listed in UL Fire Prot Dir or FM APP GUIDE.

2.3.2 Alarms

Provide audible and visual alarms as required by NFPA 20 on the controller. Provide remote supervision as required by NFPA 20, in accordance with NFPA 72 under Section ~~[_____]~~28 31 76. Provide remote alarm devices located ~~[at [_____]][~~as indicated~~]~~. Alarm signal shall be activated upon the following conditions: ~~[~~electric motor controller has operated into a pump running condition, loss of electrical power to electric motor starter, and phase reversal on line side of motor starter~~] [engine drive controller has operated into an engine running condition, engine drive controller main switch has been turned to OFF or to MANUAL position, trouble on engine driven controller or engine]~~. Exterior alarm devices shall be weatherproof type. Provide alarm silencing switch and red signal lamp, with signal lamp arranged to come on when switch is placed in OFF position.

2.4 UNDERGROUND PIPING COMPONENTS

2.4.1 Pipe and Fittings

Provide outside-coated, cement mortar-lined, ductile-iron pipe (with a rated working pressure of ~~[1034][~~1207~~][_____]~~ kPa) conforming to NFPA 24 for piping under the building and less than 1.50 m outside of the building walls. Anchor the joints in accordance with NFPA 24; provide concrete thrust block at the elbow where the pipe turns up toward the floor, and restrain the pipe riser with steel rods from the elbow to the flange above the floor. Minimum pipe size shall be 150 mm. Minimum depth of cover shall be as required by NFPA 24, but no less than 1 m. Piping more than 1.50 m outside of the building walls shall be ~~[outside coated, AWWA C104/A21.4cement mortar-lined, AWWA C151/A21.51 ductile-iron pipe, and AWWA C110/A21.10 fittings conforming to NFPA 24][~~provided under Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING~~]~~.

2.4.2 Fittings and Gaskets

Fittings shall be ductile iron conforming to AWWA C110/A21.10. Gaskets shall be suitable in design and size for the pipe with which such gaskets are to be used. Gaskets for ductile iron pipe joints shall conform to AWWA C111/A21.11.

2.4.3 Valves and Valve Boxes

Valves shall be gate valves conforming to AWWA C500 or UL 262. Valves shall have cast-iron body and bronze trim. Valve shall open by counterclockwise rotation. Except for post indicator valves, all underground valves shall be provided with an adjustable cast-iron or ductile iron valve box of a size suitable for the valve on which the box is to be used, but not less than 133 mm in diameter. The box shall be coated with bituminous coating. A cast-iron or ductile-iron cover with the word "WATER" cast on the cover shall be provided for each box.

~~2.4.4 Gate Valve and Indicator Posts~~

~~Gate valves for underground installation shall be of the inside screw type with counterclockwise rotation to open. Where indicating type valves are shown or required, indicating valves shall be gate valves with an approved indicator post of a length to permit the top of the post to be located 900 mm above finished grade. Gate valves and indicator posts shall be provided with one coat of primer and two coats of red enamel paint and shall be listed in UL Fire Prot Dir or FM APP GUIDE.~~

2.4.4 Buried Utility Warning and Identification Tape

Detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping shall be provided for all buried piping. Tape shall be detectable by an electronic detection instrument. Tape shall be provided in rolls, 80 mm minimum width, color-coded for the utility involved and imprinted in bold black letters continuously and repeatedly over the entire tape length. Warning and identification shall be "CAUTION BURIED WATER PIPING BELOW" or similar wording. Code and lettering shall be permanent and unaffected by moisture and other substances contained in the trench backfill material. Tape shall be buried at a depth of 300 mm below the top surface of earth or the top surface of the subgrade under pavement.

2.5 ABOVEGROUND PIPING COMPONENTS

2.5.1 Pipe Sizes 65 mm and Larger

2.5.1.1 Pipe

Piping shall be ~~[~~ASTM A53/A53M, ~~][ASTM A795/A795M]~~, JIS G 3452, Weight Class STD (Standard), Schedule 40 (except for Schedule 30 for pipe sizes

200 mmand greater in diameter), Type E or Type S, Grade A; black steel pipe. Steel pipe shall be joined by means of flanges welded to the pipe or mechanical grooved joints only. Piping shall not be jointed by welding or weld fittings. Suction piping shall be galvanized on the inside in accordance with NFPA 20.

2.5.1.2 Flanges

Flanges shall be ASME B16.5, Class 150 flanges. Flanges shall be provided at valves, connections to equipment, and where indicated.

2.5.1.3 Gaskets

Gaskets shall be AWWA C111/A21.11, cloth inserted red rubber gaskets.

2.5.1.4 Bolts

Bolts shall be ~~[~~ASTM A449, Type ~~[~~1~~][2]][~~ or ASTM A193/A193M, Grade B7~~]~~. Bolts shall extend no less than three full threads beyond the nut with bolts tightened to the required torque.

2.5.1.5 Nuts

Nuts shall be ~~[~~ASTM A194/A194M, Grade 7~~][~~ or ASTM A193/A193M, Grade 5~~][ASTM A563M, Grade [C3][DH3]]~~.

2.5.1.6 Washers

Washers shall meet the requirements of ASTM F436M. Flat circular washers shall be provided under all bolt heads and nuts.

2.5.2 Grooved Mechanical Joints and Fittings

Joints and fittings shall be designed for not less than 1200 kPa service and shall be the product of the same manufacturer. Fitting and coupling houses shall be malleable iron conforming to ASTM A47/A47M, JIS B 2301, JIS B 2311, Grade 32510; ductile iron conforming to ASTM A536, Grade 65-45-12. Gasket shall be the flush type that fills the entire cavity between the fitting and the pipe. Nuts and bolts shall be heat-treated steel conforming to ASTM A183 and shall be cadmium plated or zinc electroplated.Mechanical tees shall not be permitted.

2.5.3 Piping Sizes 50 mm and Smaller

2.5.3.1 Steel Pipe

Steel piping shall be ~~[~~ASTM A795/A795MJIS G 3452, Weight Class STD (Standard), Schedule 40, Type E or Type S, Grade A~~][ASTM A53/A53M, Weight Class XS (Extra Strong)]~~, zinc-coated steel pipe with threaded end connections. Fittings shall be ~~[ASME B16.3][~~ASME B16.39,JIS B 2301, JIS B 2311~~]~~, Class 150, zinc-coated threaded fittings. Unions shall be ASME B16.39, Class 150, zinc-coated unions.

2.5.3.2 Copper Tubing

Copper tubing shall be ASTM B88M, Type L or K, soft annealed. Fittings shall be ASME B16.26, flared joint fittings. Pipe nipples shall be ASTM B42 copper pipe with threaded end connections.

2.5.4 Pipe Hangers and Supports

Pipe hangers and support shall be ~~[MSS SP-58]~~~~[~~UL listed UL Fire Prot Dir or FM approved FM APP GUIDE~~]~~ and shall be the adjustable type. Finish of rods, nuts, washers, hangers, and supports shall be zinc-plated after fabrication.

2.5.5 Valves

Valves shall be UL listed UL Fire Prot Dir or FM approved FM APP GUIDE for fire protection service. Valves shall have flange or threaded end connections.

2.5.5.1 Gate Valves and Control Valves

Gate valves and control valves shall be outside screw and yoke (O.S.&Y.) type which open by counterclockwise rotation. Butterfly-type control valves are not permitted.

2.5.5.2 Tamper Switch

The suction control valves, the discharge control valves, valves to test header and flow meter, and the by-pass control valves shall be equipped with valve tamper switches for monitoring by the fire alarm system.

2.5.5.3 Check Valve

Check valve shall be clear open, swing type check valve with flange or threaded inspection plate.

2.5.5.4 Relief Valve

Relief valve shall be ~~[~~pilot operated~~][~~ or ~~][~~spring operated~~]~~ type conforming to NFPA 20. A means of detecting water motion in the relief lines shall be provided where the discharge is not visible within the pump house.

2.5.5.5 Circulating Relief Valve

An adjustable circulating relief valve shall be provided for each fire pump in accordance with NFPA 20.

2.5.5.6 Suction Pressure Regulating Valve

Suction pressure regulating valve shall be FM approved FM APP GUIDE. Suction pressure shall be monitored through a pressure line to the controlling mechanism of the regulating valve. Valve shall be arranged in accordance with the manufacturer's recommendations.

2.5.6 Hose Valve Manifold Test Header

Construct header of steel pipe. Provide ASME B16.5, Class 150 flanged inlet connection to hose valve manifold assembly. Provide approved bronze hose gate valve with 65 mm National Standard male hose threads with cap and chain; locate one meter above grade in the horizontal position for each test header outlet. Welding shall be metallic arc process in accordance with ASME B31.1.

2.5.7 Pipe Sleeves

A pipe sleeve shall be provided at each location where piping passes entirely through walls, ceilings, roofs, and floors, including pipe entering buildings from the exterior. Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, ceilings, and floors. Provide 25 mm minimum clearance between exterior of piping or pipe insulation, and interior of sleeve or core-drilled hole. Firmly pack space with mineral wool insulation. Seal space at both ends of the sleeve or core-drilled hole with plastic waterproof cement which will dry to a firm but pliable mass, or provide a mechanically adjustable segmented elastomeric seal. In fire walls and fire floors, a fire seal shall be provided between the pipe and the sleeve in accordance with Section 07 84 00 FIRESTOPPING.

a. Sleeves in Masonry and Concrete Walls, Ceilings, Roofs, and Floors: Provide hot-dip galvanized steel, ductile-iron, or cast-iron pipe sleeves. Core drilling of masonry and concrete may be provided in lieu of pipe sleeves provided that cavities in the core-drilled hole be completely grouted smooth.

b. Sleeves in Other Than Masonry and Concrete Walls, Ceilings, Roofs, and Floors: Provide galvanized steel sheet pipe not less than 4.4 kg/square m.

2.5.8 Escutcheon Plates

Provide one-piece or split-hinge metal plates for piping entering floors, walls, and ceilings in exposed areas. Provide polished stainless steel or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on plates in unfinished spaces. Plates shall be secured in place.

2.6 DISINFECTING MATERIALS

2.6.1 Liquid Chlorine

Liquid chlorine shall conform to AWWA B301.

2.6.2 Hypochlorites

Calcium hypochlorite and sodium hypochlorite shall conform to AWWA B300.

2.7 ELECTRIC MOTOR DRIVER

Motors, controllers, contactors, and disconnects shall be provided with their respective pieces of equipment, as specified herein and shall have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Controllers and contactors shall have a maximum of 120-volt control circuits, and auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, the cost of providing additional electrical service and related work shall be included under this section. Motor shall conform to NEMA MG 1 Design B type. Integral size motors shall be the premium efficiency type in accordance with NEMA MG 1. Motor wattage shall be of sufficient size so that the nameplate wattage rating will not be exceeded throughout the entire published pump characteristic curve. The motor and fire pump controller shall be fully compatible.

~~2.8 DIESEL ENGINE DRIVER~~

~~Diesel engine driver shall conform to the requirements of UL 1247 and shall be UL listed UL Fire Prot Dir or FM approved FM APP GUIDE for fire pump service. Driver shall be of the make recommended by the pump manufacturer. The engine shall be closed circuit, liquid-cooled [with raw water heat exchanger][with radiator and engine-driven fan]. Diesel engine shall be electric start type taking current from 2 battery units. Engine shall be equipped with a fuel in-line filter-water separator. Engine conditions shall be monitored with engine instrumentation panel that has a tachometer, hour meter, fuel pressure gauge, lubricating oil pressure gauge, water temperature gauge, and ammeter gauge. Engine shall be connected to horizontal-shaft pump by flexible couplings. For connections to vertical-shaft fire pumps, right-angle gear drives and universal joints shall be used. An engine jacket water heater shall be provided to maintain a temperature of 49 degrees C in accordance with NFPA 20.~~

~~2.8.1 Engine Capacity~~

~~Engine shall have adequate wattage to drive the pump at all conditions of speed and load over the full range of the pump performance curve. The wattage rating of the engine driver shall be as recommended by the pump manufacturer and shall be derated for temperature and elevation in accordance with NFPA 20. Ambient temperature at the pump location shall be [_____] degrees C. Site elevation shall be [_____] meters above mean sea level (MSL).~~

~~2.8.2 Exhaust System External to Engine~~

~~Exhaust system shall comply with the requirements of NFPA 20 and NFPA 37. An exhaust muffler shall be provided for each diesel engine driver to reduce noise levels less than [85][95] dBA. A flexible connector with flange connections shall be provided at the engine. Flexible sections shall be stainless steel suitable for diesel-engines exhaust gas at 538 degrees C.~~

~~2.8.2.1 Steel Pipe and Fittings~~

~~ASTM A53/A53M, [Schedule 40][Weight Class XS (Extra Strong], black steel, welding end connections. ASME B16.9 or ASME B16.11 welding fittings shall be of the same material and weight as the piping.~~

~~2.8.2.2 Flanges~~

~~ASME B16.5, Class [300][150]. Flanges shall be provided at connections to diesel engines, exhaust mufflers, and flexible connections. Gaskets shall be ASME B16.21, composition ring, 1.5875 mm. ASTM A193/A193M, Grade [B8][B7] bolts and ASTM A194/A194M, Grade [8][7] nuts shall be provided.~~

~~2.8.2.3 Piping Insulation~~

~~Comply with EPA requirements in accordance with Section 01 33 29 SUSTAINABILITY REPORTING. Products containing asbestos will not be permitted. Exhaust piping system including the muffler shall be insulated with ASTM C533 calcium silicate insulation, minimum of 75 mm. Insulation shall be secured with not less than 9.525 mm width fibrous glass reinforced waterproof tape or Type 304 stainless steel bands spaced not more than 200 mm on center. An aluminum jacket encasing the insulation shall be provided. The aluminum jacket shall have a minimum thickness of~~

~~0.406 mm, a factory-applied polyethylene and kraft paper moisture barrier on the inside surface. The jacket shall be secured with not less than 13 mm wide stainless steel bands, spaced not less than 200 mm on centers. Longitudinal and circumferential seams of the jacket shall be lapped not less than 75 mm. Jackets on horizontal line shall be installed so that the longitudinal seams are on the bottom side of the pipe. The seams of the jacket for the vertical lines shall be placed on the off-weather side of the pipe. On vertical lines, the circumferential seams of the jacket shall overlap so the lower edge of each jacket overlaps the upper edge of the jacket below.~~

2.8 FIRE PUMP CONTROLLER

Controller shall be the automatic type and UL listed UL Fire Prot Dir or FM approved FM APP GUIDE for fire pump service. Pump shall be arranged for automatic start and stop, and manual push-button stop. Automatic stopping shall be accomplished only after all starting causes have returned to normal and after a minimum pump run time has elapsed. Controllers shall be completely terminally wired, ready for field connections, and mounted in a ~~[NEMA Type 2 drip-proof][~~NEMA Type 4 watertight and dust tight~~]~~ enclosure arranged so that controller current carrying parts will not be less than 300 mm above the floor. Controller shall be provided with voltage surge arresters installed in accordance with NFPA 20. Controller shall be equipped with a bourdon tube pressure switch or a solid state pressure switch with independent high and low adjustments, automatic starting relay actuated from normally closed contacts, visual alarm lamps and supervisory power light. Controller shall be equipped with a thermostat switch with adjustable setting to monitor the pump room temperature and to provide an alarm when temperatures falls below 5 degrees C ~~[Controller shall be equipped with a sequential start timer/relay feature to start multiple fire pumps in sequence.][~~ The controller shall be factory-equipped with a heater operated by thermostat to prevent moisture in the cabinet.~~]~~

2.8.1 Controller for Electric Motor Driven Fire Pump

Controller shall be ~~[~~electronic soft start~~][across the line][auto-transformer][wye-delta, open circuit transition][wye-delta, closed circuit transition] starting type~~. Controller shall be designed ~~[for [_____] kW at [_____] volts][~~as indicated~~]~~. Controller~~[~~ and transfer switch~~]~~ shall have a short circuit rating ~~[of [_____] amps r.m.s. symmetrical at [_____] volts a.c.][~~as indicated~~]~~.~~[~~ An automatic transfer switch (ATS) shall be provided for each fire pump. The ATS shall comply with NFPA 20 and shall be specifically listed for fire pump service. The ATS shall transfer source of power to the alternate source upon loss of normal power.~~]~~ Controller shall monitor pump running, loss of a phase or line power, phase reversal~~[, low reservoir]~~ and pump room temperature. Alarms shall be individually displayed in front of panel by lighting of visual lamps. Each lamp shall be labeled with rigid etched plastic labels. Controller shall be equipped with terminals for remote monitoring of pump running, pump power supply trouble (loss of power or phase and phase reversal), and pump room trouble (pump room temperature~~ [and low reservoir level]~~), and for remote start. Limited service fire pump controllers are not permitted, except for fire pumps driven by electric motors rated less than 11 kW. Controller shall be equipped with a 7-day electric pressure recorder with 24-hour spring wound back-up. The pressure recorder shall provide a readout of the system pressure from 0 to 207 Pa, time, and date. Controller shall require the pumps to run for ten minutes for pumps with driver motors under 149 kW and for 15 minutes for pumps with motors 149 kW and greater, prior to automatic shutdown. The controller shall be equipped with an externally operable isolating switch which manually operates the motor circuit. Means shall be provided in the controller for measuring current for all motor circuit conductors.

~~2.8.2 Controller for Diesel Engine Driven Fire Pump~~

~~Controller shall require the pump to run for 30 minutes prior to automatic shutdown. Controller shall be equipped with two battery chargers; two ammeters; two voltmeters, one for each set of batteries. Controller shall automatically alternate the battery sets for starting the pumps. Controller shall be equipped with the following supervisory alarm functions:~~

~~a. Engine Trouble (individually monitored)~~

~~(1) Engine overspeed~~

~~(2) Low Oil Pressure~~

~~(3) High Water Temperature~~

~~(4) Engine Failure to Start~~

~~(5) Battery~~

~~(6) Battery Charger/AC Power Failure~~

~~b. Main Switch Mis-set~~

~~c. Pump Running~~

~~d. Pump Room Trouble (individually monitored)~~

~~(1) Low Fuel~~

~~(2) Low Pump Room Temperature~~

~~(3) Low Reservoir Level~~

~~Alarms shall be individually displayed in front of panel by lighting of visual lamps, except that individual lamps are not required for pump running and main switch mis-set. Controller shall be equipped with a 7-day electric pressure recorder with 24-hour back-up mounted inside the controller. The pressure recorder shall provide a readout of the system pressure from 0 to 207 Pa, time, and date. The controller shall be equipped with an audible alarm which will activate upon any engine trouble or pump room trouble alarm condition and alarm silence switch. Controller shall be equipped with terminals for field connection of a remote alarm for main switch mis-set, pump running, engine trouble and pump room trouble; and terminals for remote start. When engine emergency overspeed device operates, the controller shall cause the engine to shut down without time delay and lock out until manually reset.~~

~~2.9 BATTERIES~~

~~Batteries for diesel engine driver shall be sealed lead calcium batteries. Batteries shall be mounted in a steel rack with non-corrosive, non-conductive base, not less than 300 mm above the floor.~~

2.9 PRESSURE SENSING LINE

A completely separate pressure sensing line shall be provided for each fire pump and for the jockey pump. The sensing line shall be arranged in accordance with Figure A-7-5.2.1. of NFPA 20. The sensing line shall be 13 mmH58 brass tubing complying with ASTM B135M. The sensing line shall be equipped with two restrictive orifice unions each. Restricted orifice unions shall be ground-face unions with brass restricted diaphragms drilled for a 2.4 mm. Restricted orifice unions shall be mounted in the horizontal position, not less than 1.5 m apart on the sensing line. Two test connections shall be provided for each sensing line. Test connections shall consist of two brass 13 mm globe valves and 8 mm gauge connection tee arranged in accordance with NFPA 20. One of the test connections shall be equipped with a 0 to 2100 kPa water oil-filled gauge. Sensing line shall be connected to the pump discharge piping between the discharge piping control valve and the check valve.

2.10 PRESSURE MAINTENANCE PUMP

2.10.1 General

Pressure maintenance pump shall be electric motor driven,~~[~~ horizontal shaft ~~][~~ or~~][~~ in-line vertical shaft,~~]~~ centrifugal type with a rated discharge of ~~[~~0.63~~][_____]~~ L/second at ~~[862][_____]~~586.1 kPa. Pump shall draft ~~[~~from the suction supply side of the suction pipe gate valve of the fire pump ~~][as indicated]~~ and shall discharge into the system at the downstream side of the pump discharge gate valve. An approved indicating gate valve of the outside screw and yoke (O.S.&Y.) type shall be provided in the maintenance pump discharge and suction piping. Oil-filled water pressure gauge and approved check valve in the maintenance pump discharge piping shall be provided. Check valve shall be swing type with removable inspection plate.

2.10.2 Pressure Maintenance Pump Controller

Pressure maintenance pump controller shall be arranged for automatic and manual starting and stopping and equipped with a "manual-off-automatic" switch. The controller shall be completely prewired, ready for field connections, and wall-mounted in a NEMA Type 2 drip-proof enclosure. The controller shall be equipped with a bourdon tube pressure switch or a solid state pressure switch with independent high and low adjustments for automatic starting and stopping. A sensing line shall be provided connected to the pressure maintenance pump discharge piping between the control valve and the check valve. The sensing line shall conform to paragraph, PRESSURE SENSING LINE. The sensing line shall be completely separate from the fire pump sensing lines. An adjustable run timer shall be provided to prevent frequent starting and stopping of the pump motor. The run timer shall be set for ~~[~~2~~][_____]~~ minutes.

~~2.11 DIESEL FUEL SYSTEM EXTERNAL TO ENGINE~~

~~Fuel system shall be provided that meets all requirements of NFPA 20 and NFPA 37. The fuel tank vent piping shall be equipped with screened weatherproof vent cap. Vents shall be extended to the outside. Each tank shall be equipped with a fuel level gauge. Flexible bronze or stainless steel piping connectors with single braid shall be provided at each piping connection to the diesel engine. Supply, return, and fill piping shall be steel piping, except supply and return piping may be copper tubing. Fuel~~

~~lines shall be protected against mechanical damage. Fill line shall be equipped with 16 mesh removable wire screen. Fill lines shall be extended to the exterior. A weatherproof tank gauge shall be mounted on the exterior wall near each fill line for each tank. The fill cap shall be able to be locked by padlock. The engine supply (suction) connection shall be located on the side of the fuel tank so that 5 percent of the tank volume provides a sump volume not useable by the engine. The elevation of the fuel tank shall be such that the inlet of the fuel supply line is located so that its opening is no lower than the level of the engine fuel transfer pump. The bottom of the tank shall be pitched 21 mm/m to the side opposite the suction inlet connection, and to an accessible 25 mmplugged globe drain valve.~~

~~2.11.1 Fuel Piping~~

~~As specified in NFPA 20.~~

~~2.11.2 Diesel Fuel Tanks~~

~~UL 80 or UL 142 for aboveground tanks.~~

~~2.11.3 Valves~~

~~Provide an indicating and lockable ball valve in the supply line adjacent to the tank suction inlet connection. Provide a check valve in fuel return line. Valves mustl be suitable for oil service. Valves must have union end connections or threaded end connections.~~

~~2.11.3.1 Globe Valve~~

~~MSS SP-80 Class 125~~

~~2.11.3.2 Check Valve~~

~~MSS SP-80, Class 125, swing check~~

~~2.11.3.3 Ball Valve~~

~~Full port design, copper alloy body, 2-position lever handle~~

2.11 JOINTS AND FITTINGS FOR COPPER TUBE

Wrought copper and bronze solder-joint pressure fittings shall conform to ASME B16.22 and ASTM B75/B75M. Cast copper alloy solder-joint pressure fittings shall conform to ASME B16.18. Cast copper alloy fittings for flared copper tube shall conform to ASME B16.26 and ASTM B62. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment. Extracted brazed tee joints produced with an acceptable tool and installed as recommended by the manufacturer may be used. Grooved mechanical joints and fittings shall be designed for not less than 862 kPa service and shall be the product of the same manufacturer. Grooved fitting and mechanical coupling housing shall be ductile iron conforming to ASTM A536. Gaskets for use in grooved joints shall be molded synthetic polymer of pressure responsive design and shall conform to ASTM D2000 for circulating medium up to 110 degrees C. Grooved joints shall conform to AWWA C606 Coupling nuts and bolts for use in grooved joints shall be steel and shall conform to ASTM A183.

2.12 PUMP BASE PLATE AND PAD

Provide a common base plate for each horizontal-shaft fire pump for mounting pump and driver unit. Construct the base plate of cast iron with raised lip tapped for drainage or welded steel shapes with suitable drainage. Provide each base plate for the horizontal fire pumps with a 25 mmgalvanized steel drain line piped to the nearest floor drain. For vertical shaft pumps, pump head shall be provided with a cast-iron base plate and shall serve as the sole plate for mounting the discharge head assembly. Mount pump units and bases on a raised ~~[100][~~150~~]~~ mm reinforced concrete pad that is an integral part of the reinforced concrete floor.

2.13 HOSE VALVE MANIFOLD TEST HEADER

Hose valve test header shall be connected by ASME B16.5, Class 150 flange inlet connection. Hose valves shall be UL listed UL Fire Prot Dir or FM approved FM APP GUIDE bronze hose gate valves with 65 mm American National Fire Hose Connection Screw Standard Threads (NH) in accordance with NFPA 1963. The number of valves shall be in accordance with NFPA 20. Each hose valve shall be equipped with a cap and chain, and located no more than 900 mm and no less than 600 mm above grade.

2.14 FLOW METER

Meter shall be UL listed UL Fire Prot Dir or FM approved FM APP GUIDE as flow meters for fire pump installation with direct flow readout device. Flow meter shall be capable of metering any waterflow quantities between 50 percent and 150 percent of the rated flow of the pumps. Arrange piping to permit flow meter to discharge to pump suction and to discharge through test header. The meter throttle valve and the meter control valves shall be O.S.&Y. valves. Provide automatic air release if flow meter piping between pump discharge and pump suction forms an inverted "U". Meter shall be of the ~~[~~venturi~~][annular probe][orifice plate][_____]~~ type.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

3.2 INSPECTION BY FIRE PROTECTION SPECIALIST

The Fire Protection Specialist and QFPEshall periodically perform a thorough inspection of the fire pump installation, including visual observation of the pump while running, to assure that the installation conforms to the contract requirements. There shall be no excessive vibration, leaks (oil or water), unusual noises, overheating, or other potential problems. Inspection shall include piping and equipment clearance, access, supports, and guards. Any discrepancy shall be brought to the attention of the Contracting Officer in writing, no later than three working days after the discrepancy is discovered. The Fire Protection Specialist and QFPE shall witness the preliminary and final acceptance tests and, after completion of the inspections and a successful final acceptance test, shall sign test results and certify in writing that the installation the fire pump installation is in accordance with the contract requirements.

3.3 INSTALLATION

Equipment, materials, workmanship, fabrication, assembly, erection, installation, examination, inspection and testing shall be in accordance NFPA 20, except as modified herein. In addition, the fire pump and engine shall be installed in accordance with the written instructions of the manufacturer.

3.3.1 Installation Drawings

Submit Fire Pump Installation Drawings consisting of a detailed plan view, detailed elevations and sections of the pump room, equipment and piping, drawn to a scale of not less than 1:20. Drawings shall indicate equipment, piping, and associated pump equipment to scale. Indicate all clearance, such as those between piping and equipment; between equipment and walls, ceiling and floors; and for electrical working distance clearance around all electrical equipment. Include a legend identifying all symbols, nomenclatures, and abbreviations. Indicate a complete piping and equipment layout including elevations and/or section views of the following:

a. Fire pumps, controllers, piping, valves, and associated equipment.

b. Sensing line for each pump including the pressure maintenance pump.

c. Engine fuel system for diesel driven pumps.

d. Engine cooling system for diesel driven pumps.

e. Pipe hangers and sway bracing including support for diesel muffler and exhaust piping.

f. Restraint of underground water main at ~~[entry-point][~~entry-and exit-points~~]~~ to the building including details of pipe clamps, tie rods, mechanical retainer glands, and thrust blocks.

g. A one-line schematic diagram indicating layout and sizes of all piping, devices, valves and fittings.

h. A complete point-to-point connection drawing of the pump power, control and alarm systems, as well as interior wiring schematics of each controller.

3.3.2 Pump Room Configuration

Provide detail plan view of the pump room including elevations and sections showing the fire pumps, associated equipment, and piping. Submit working drawings on sheets not smaller than A1 594 by 841 mm; include data for the proper installation of each system. Show piping schematic of pumps, devices, valves, pipe, and fittings. ~~[~~Provide an isometric drawing of the fire pump and all associated piping~~]~~. Show point to point electrical wiring diagrams. Show piping layout and sensing piping arrangement. Show engine fuel and cooling system. Include:

a. Pumps, drivers, and controllers

b. Hose valve manifold test header

c. Circuit diagrams for pumps

d. Wiring diagrams of each controller

3.3.3 Accessories

Tank supports, piping offsets, fittings, and any other accessories required shall be furnished as specified to provide a complete installation and to eliminate interference with other construction.

3.4 PIPE AND FITTINGS

Piping shall be inspected, tested and approved before burying, covering, or concealing. Fittings shall be provided for changes in direction of piping and for all connections. Changes in piping sizes shall be made using tapered reducing pipe fittings. Bushings shall not be used.~~[~~ Photograph all piping prior to burying, covering, or concealing.~~]~~

3.4.1 Cleaning of Piping

Interior and ends of piping shall be clean and free of any water or foreign material. Piping shall be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of the piping shall be securely closed so that no water or foreign matter will enter the pipes or fittings. Piping shall be inspected before placing in position.

3.4.2 Threaded Connections

Jointing compound for pipe threads shall be ~~[~~polytetrafluoroethylene (PTFE) pipe thread tape conforming to ASTM D3308~~][Teflon pipe thread paste]~~ and shall be applied to male threads only. Exposed ferrous pipe threads shall be provided with one coat of zinc molybdate primer applied to a minimum of dry film thickness of 0.025 mm.

3.4.3 Pipe Hangers and Supports

Additional hangers and supports shall be provided for concentrated loads in aboveground piping, such as for valves and risers.

3.4.3.1 Vertical Piping

Piping shall be supported at each floor, at not more than 3 meters intervals.

3.4.3.2 Horizontal Piping

Horizontal piping supports shall be spaced as follows:

MAXIMUM SPACING (METERS)										
Nominal Pipe Size (mm)	25 and Under	32	40	50	65	80	90	100	125	150+

MAXIMUM SPACING (METERS)										
Copper Tube	1.8	2	2.4							
Steel Pipe	2	2.4	2.7	3	3.3	3.6	3.9	4.2	4.8	5.0

3.4.4 Underground Piping

Installation of underground piping and fittings shall conform to NFPA 24. Joints shall be anchored in accordance with NFPA 24. Concrete thrust block shall be provided at elbow where pipe turns up towards floor, and the pipe riser shall be restrained with steel rods from the elbow to the flange above the floor. After installation in accordance with NFPA 24, rods and nuts shall be thoroughly cleaned and coated with asphalt or other corrosion-retard material approved by the Contracting Officer. Minimum depth of cover shall be 900 mm.

3.4.5 Grooved Mechanical Joint

Grooves shall be prepared according to the coupling manufacturer's instructions. Grooved fittings, couplings, and grooving tools shall be products of the same manufacturer. Pipe and groove dimensions shall comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field shall be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe shall be measured for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances. Grooved joints shall not be used in concealed locations, such as behind solid walls or ceilings, unless an access panel is shown on the drawings for servicing or adjusting the joint.

3.5 ELECTRICAL WORK

Electric motor and controls shall be in accordance with NFPA 20, NFPA 72 and NFPA 70, unless more stringent requirements are specified herein or are indicated on the drawings. Electrical wiring and associated equipment shall be provided in accordance with NFPA 20 and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide wiring in rigid metal conduit or intermediate metal conduit, except electrical metallic tubing conduit may be provided in dry locations not enclosed in concrete or where not subject to mechanical damage.

3.6 PIPE COLOR CODE MARKING

Color code marking of piping as specified in Section 09 90 00 PAINTS AND COATINGS.

3.7 FLUSHING

The fire pump suction and discharge piping shall be flushed at ~~[120][~~150~~]~~ percent of rated capacity of each pump. ~~Where the pump installation consists of more than one pump, the flushing shall be the total quantity of water flowing when all pumps are discharging at [120][150] percent of~~

~~their rated capacities.~~ The new pumps may be used to attain the required flushing volume. No underground piping shall be flushed by using the fire pumps. Flushing operations shall continue until water is clear, but not less than 10 minutes. Submit a signed and dated flushing certificate before requesting field testing.

3.8 FIELD TESTS

Submit system diagrams that show the layout of equipment, piping, and storage units, and typed condensed sequence of operation, wiring and control diagrams, and operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system shall be framed under glass or laminated plastic. After approval, these items shall be posted where directed.

3.8.1 Hydrostatic Test

Piping shall be hydrostatically tested at 1551 kPa for a period of 2-hours, or at least 345 kPa in excess of the maximum pressure, when the maximum pressure in the system is in excess of ~~[~~1207~~][1379]~~ kPa in accordance with NFPA 20.

3.8.2 Preliminary Tests

Submit proposed procedures for Preliminary Tests prior to the proposed date and time to begin Preliminary Tests. The Fire Protection Specialist and QFPE shall take all readings and measurements. The Manufacturer's Representative, a representative of the fire pump controller manufacturer, and a representative of the diesel engine manufacturer (when supplied) shall witness the complete operational testing of the fire pump and drivers. The fire pump controller manufacturer's representative and the diesel engine manufacturer's representative shall each be an experienced technician employed by the respective manufacturers and capable of demonstrating operation of all features of respective components including trouble alarms and operating features. Fire pumps, drivers and equipment shall be thoroughly inspected and tested to insure that the system is correct, complete, and ready for operation. Tests shall ensure that pumps are operating at rated capacity, pressure and speed. Tests shall include manual starting and running to ensure proper operation and to detect leakage or other abnormal conditions, flow testing, automatic start testing, testing of automatic settings, sequence of operation check, test of required accessories; test of pump alarms devices and supervisory signals, test of pump cooling, operational test of relief valves, and test of automatic power transfer, if provided. Pumps shall run without abnormal noise, vibration or heating. If any component or system was found to be defective, inoperative, or not in compliance with the contract requirements during the tests and inspection, the corrections shall be made and the entire preliminary test shall be repeated. Submit Preliminary Tests Reports, to include both the Contractor's Material and Test Certificate for Underground Piping and the Contractor's Material and Test Certificate for Aboveground Piping. All items in the Report shall be signed by the Fire Protection Specialist and the Manufacturer's Representative.

3.8.3 Final Acceptance Test

The Fire Protection Specialist and QFPE shall take all readings and measurements. The Manufacturer's Representative, the fire pump controller manufacturer's representative, and the diesel engine manufacturer's representative (when supplied) shall also witness for the final tests. Repair any damage caused by hose streams or other aspects of the test. Submit proposed date and time to begin Final Acceptance Test, with the Acceptance Procedures. Notification shall be provided at least ~~[~~14~~] [______]~~ days prior to the proposed start of the test. Submit ~~[~~3~~] [______]~~ copies of the completed Final Acceptance Test Reports, no later that ~~[~~7~~] [______]~~ days after the completion of the tests. All items in the reports shall be signed by the Fire Protection Specialist, QFPE and the Manufacturer's Representative. Test reports in booklet form (each copy furnished in a properly labeled three ring binder) showing all field tests and measurements taken during the preliminary and final testing, and documentation that proves compliance with the specified performance criteria, upon completion of the installation and final testing of the installed system. Each test report shall indicate the final position of the controls and pressure switches. The test reports shall include the description of the hydrostatic test conducted on the piping and flushing of the suction and discharge piping. A copy of the manufacturer's certified pump curve for each fire pump shall be included in the report. Notification shall include a copy of the Contractor's Material & Test Certificates. Include the following in the final acceptance test:

3.8.3.1 Flow Tests

Flow tests using the test header, hoses and playpipe nozzles shall be conducted. Flow tests shall be performed at churn (no flow), 75, 100, 125 and 150 percent capacity for each pump and at full capacity of the pump installation. Flow readings shall be taken from each nozzle by means of a calibrated pitot tube with gauge or other approved measuring equipment. Rpm, suction pressure and discharge pressure reading shall be taken as part of each flow test. Voltage and ampere readings shall taken on each phase as part of each flow test for electric-motor driven pumps.

3.8.3.2 Starting Tests

Pumps shall be tested for automatic starting and sequential starting. Setting of the pressure switches shall be tested when pumps are operated by pressure drop. Tests may be performed by operating the test connection on the pressure sensing lines. As a minimum, each pump shall be started automatically 10 times and manually 10 times, in accordance with NFPA 20. Tests of engine-driven pumps shall be divided equally between both set of batteries. The fire pumps shall be operated for a period of a least 10 minutes for each of the starts; except that electric motors over 149 kW shall be operated for at least 15 minutes and shall not be started more than 2 times in 10 hours. Pressure settings that include automatic starting and stopping of the fire pump(s) shall be indicated on an etched plastic placard, attached to the corresponding pump controller.

3.8.3.3 Battery Changeover

Diesel driven fire pumps shall be tested for automatic battery changeover in event of failure of initial battery units.

3.8.3.4 Alarms

All pump alarms, both local and remote, shall be tested. Supervisory alarms for diesel drivers shall be electrically tested for low oil pressure, high engine jacket coolant temperature, shutdown from overspeed, battery failure and battery charger failure.

3.8.3.5 Miscellaneous

Valve tamper switches shall be tested. Pressure recorder operation relief valve settings, valve operations, operation and accuracy of meters and gauges, and other accessory devices shall be verified.

3.8.3.6 Alternate Power Source

On installations with an alternate source of power and an automatic transfer switch, loss of primary power shall be simulated and transfer shall occur while the pump is operating at peak load. Transfer from normal to emergency source and retransfer from emergency to normal source shall not cause opening of overcurrent devices in either line. At least half of the manual and automatic starting operations listed shall be performed with the fire pump connected to the alternate source.

3.8.3.7 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests shall be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.8.3.8 Test Documentation

The Manufacturer's Representative shall supply a copy of the manufacturer's certified curve for each fire pump at the time of the test. The Fire Protection Specialist shall record all test results and plot curve of each pump performance during the test. Complete pump acceptance test data of each fire pump shall be recorded. The pump acceptance test data shall be on forms that give the detail pump information such as that which is indicated in Figure A-11-2.6.3(f) of NFPA 20. All test data records shall be submitted in a three ring binder.

3.8.4 Test Equipment

Provide all equipment and instruments necessary to conduct a complete final test, including 65 mm diameter hoses, playpipe nozzles, pitot tube gauges, portable digital tachometer, voltage and ampere meters, and calibrated oil-filled water pressure gauges. Provide all necessary supports to safely secure hoses and nozzles during the test. The ~~[~~ Government will~~][Contractor shall]~~ furnish water for the tests.

3.9 DISINFECTION

After all system components are installed including pumps, piping, and other associated work, and all hydrostatic tests are successfully completed, thoroughly flush the pumps and all piping to be disinfected with potable water until there is no visible sign of dirt or other residue. and hydrostatic test are successfully completed, each portion of the piping specified in this Section system to be disinfected shall be thoroughly flushed with potable water until all entrained dirt and other foreign materials have been removed before introducing chlorinating material.

3.9.1 Chlorination

The chlorinating material shall be hypochlorites or liquid chlorine. The chlorinating material shall be fed into the sprinkler piping at a constant rate of 50 parts per million (ppm). A properly adjusted hypochlorite solution injected into the system with a hypochlorinator, or liquid chlorine injected into the system through a solution-fed chlorinator and booster pump shall be used. Chlorination application shall continue until the entire system if filled. The water shall remain in the system for a minimum of 24 hours. Each valve in the system shall be opened and closed several times to ensure its proper disinfection. Following the 24-hour period, no less than 25 ppm chlorine residual shall remain in the system.

3.9.2 Flushing

The system shall then be flushed with clean water until the residual chlorine is reduced to less than one part per million. Samples of water in disinfected containers for bacterial examination will be taken from several system locations which are approved by the Contracting Officer.

3.9.3 Sample Testing

Samples shall be tested for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with AWWA 10084. The testing method shall be either the multiple-tube fermentation technique or the membrane-filter technique. The disinfection shall be repeated until tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

3.10 SYSTEM STARTUP

Fully enclose or properly guard coupling, rotating parts, gears, projecting equipment, etc. so as to prevent possible injury to persons that come in close proximity of the equipment. Conduct testing of the fire pumps in a safe manner and ensure that all equipment is safely secured. Hoses and nozzles used to conduct flow tests shall be in excellent condition and shall be safely anchored and secured to prevent any misdirection of the hose streams.

Post operating instructions for pumps, drivers, controllers, and flow meters.

3.11 CLOSEOUT ACTIVITIES

3.11.1 Field Training

The Fire Protection Specialist and the Manufacturer's Representative shall conduct a training course for operating and maintenance personnel as designated by the Contracting Officer. Submit the proposed schedule for field training at least 14 days prior to the start of related training. Training shall be provided for a period of ~~[~~2~~] [8]~~ hours of normal working time and shall start after the fire pump installation is functionally complete and after the Final Acceptance Test. The field instruction shall cover all of the items contained in the approved Operating and Maintenance Instructions. Submit manuals listing step-by-step procedures required for system startup, operation, shutdown, and routine maintenance. The manuals shall include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine maintenance including the name of a local supplier, simplified wiring and controls diagrams, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Data Package 3 shall be submitted for fire pumps and drivers in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. ~~[~~Each service organization submitted shall be capable of providing ~~[~~4~~] [______]~~ hour onsite response to a service call on an emergency basis.~~]~~

3.11.2 As-Built Drawings

Submit As-Built Drawings, no later than ~~[~~14~~][______]~~ days after completion of the Final Tests. Update he Fire Pump Installation Drawings to reflect as-built conditions after all related work is completed and shall be on reproducible full-size mylar film.

3.12 PROTECTION

Carefully remove materials so as not to damage material which is to remain. Replace existing work damaged by the Contractor's operations with new work of the same construction.

-- End of Section --

SECTION 22 00 00

PLUMBING, GENERAL PURPOSE
11/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)~~

~~ANSI C12.1~~	~~(2014; Errata 2016) Electric Meters - Code for Electricity Metering~~
~~ANSI Z21.22/CSA 4.4~~	~~(2015; R 2020) Relief Valves for Hot Water Supply Systems~~

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.19.3/CSA B45.4	(2022) Stainless Steel Plumbing Fixtures
ASME B16.18	(2021) Cast Copper Alloy Solder Joint Pressure Fittings
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings
ASME BPVC SEC VIII D1	(2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

ASTM INTERNATIONAL (ASTM)

ASTM B42	(2020) Standard Specification for Seamless Copper Pipe, Standard Sizes
ASTM B75/B75M	(2020) Standard Specification for Seamless Copper Tube
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B88M	(2020) Standard Specification for Seamless Copper Water Tube (Metric)

~~AMERICAN WATER WORKS ASSOCIATION (AWWA)~~

~~AWWA C700~~	~~(2020) Cold-Water Meters - Displacement Type, Metal Alloy Main Case~~
~~AWWA C701~~	~~(2019) Cold-Water Meters - Turbine Type for Customer Service~~
~~AWWA D100~~	~~(2021) Welded Steel Tanks for Water Storage~~

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

PL 93-523	(1974; A 1999) Safe Drinking Water Act

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 COMM	(2017) Standard and Commentary Accessible and Usable Buildings and Facilities

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A	(20~~18~~24) Standard for the Installation of Air Conditioning and Ventilating Systems

PLASTIC PIPE AND FITTINGS ASSOCIATION (PPFA)

PPFA Fire Man	(2016) Firestopping: Plastic Pipe in Fire Resistive Construction

JAPAN WATER WORKS ASSOCIATION (JWWA)

JWWA K 120	(2008) Water Works Sodium Hypochlorite
JWWA S 101	(2019) Polyethylene Pipe Fitting for Water Distribution

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 4420	(2018) Components for Kitchen Equipments
JIS A 4421	(1991) Drain With Traps for Equipment Units
JIS A 5207	(2014) Sanitary Wares
JIS A 5758	(2022) Sealants for Sealing and Glazing in Buildings
JIS A 6005	(2005) Asphalt Roofing Felts
JIS B 0203	(1999) Taper Pipe Threads
JIS B 2003	(2013) General Rules for Inspection of Valves
JIS B 2004	(1994) General Rules for Marking on Valves
JIS B 2011	(2013) Bronze, Gate, Globe, Angle, and Check Valves (Amendment 2)
JIS B 2031	(2015) Gray Cast Iron Valves (Amendment 1)
JIS B 2061	(2017) Faucets, Ball Taps and Flush Valves
JIS B 2071	(2000) Steel Valves
JIS B 2220	(2012) Steel Pipe Flanges
JIS B 2308	(2013) Stainless Steel Threaded Fittings

JIS B 2309	(2009) Butt-Welding Pipe Fittings for Light Gauge Stainless Steel Tubes for Ordinary Use
JIS B 2404	(2018) Dimensions of Gaskets for Use with Pipe Flanges
JIS B 7414	(2018) Glass Thermometers
JIS B 7505-1	(2017) Aneroid Pressure Gauges -- Part 1: Bourdon Tube Pressure Gauges~~-~~
JIS B 8225	(2012) Safety Valves-Measuring Methods for Coefficient of Discharge
JIS B 8285	(2010) Welding Procedure Qualification Test for Pressure Vessels
JIS B 8325	(2003; R 2008; R 2013) Submersible Motor Pumps For Sump
JIS B 8372-1	(2003) Pneumatic Fluid Power-Compressed Air Pressure Regulators and Filter-Regulators-Part 1: Main Characteristics to be Included in Literature from Suppliers and Product-Making Requirements
JIS B 8410	(2015) Pressure Reducing Valves for Water Works
JIS B 8414	(2011) Relief Valves for Hot Water Appliances (Amendment 3)
JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 9335-2-35	(2019) Household and Similar Electrical Appliances-Safety-Part 2-35: Particular Requirements for Instantaneous Water Heaters
JIS G 3103	(20~~12~~23) Carbon Steel and Molybdenum Alloy Steel Plates for Boilers and Pressure Vessels
JIS G 3118	(2017) Carbon Steel Plates for Pressure Vessels for Intermediate and Moderate Temperature Services
JIS G 3201	(2008) Carbon Steel Forgings for General Use (Amendment 1)
JIS G 4107	~~(2015) Alloy Steel Bolting Materials for High Temperature Service (Amendment 1)~~ (2022) Alloy Steel Bolting Materials for High Temperature Service

JIS G 4303	(2012) Stainless Steel Bars
JIS G 5526	(2014) Ductile Iron Pipes
JIS H 3100	(2018) Copper and Copper Alloy Sheets, Plates and Strips
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS K 6353	(2011) Rubber Goods for Water Works
JIS K 6741	(2016~~; R 2021~~) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes
JIS K 7014	(1997) Fittings and Joints for Fibre Reinforced Plastic Pipes
JIS S 3200-3	(1997) Equipment for Water Supply Service-Test Method of Water Hammer
JIS S 3201	(2017) Testing Methods for Household Water Purifiers
JIS Z 3284-1	(2014) Solder Paste-Part 1: Kinds and Quality Classification
JIS Z 3313	(2009) Flux Cored Wires for Gas Shielded and Self-Shielded Metal Arc Welding of Mild Steel, High Strength Steel and Low Temperature Service Steel
JIS Z 3621	(2014) Recommended Practice for Brazing
JIS Z 9102	(1987) Identification Marking for Piping Systems

~~JAPAN VALVE MANUFACTURERS' ASSOCIATION (JVMA)~~

~~JV-5~~	~~(2008) Pipe end Anticorrosion Screwed Valve~~

JAPAN VALVE MANUFACTURERS' ASSOCIATION (JVMA)

JV-5	(2008) Pipe end Anticorrosion Screwed Valve

JAPAN PIPE FITTINGS ASSOCIATION (JPFA)

JPF-MP-006	(2019) Rool Grooving Dimension Fitting
JPF-DP-001	(2010) Screwed Cast Iron Fitting

JAPAN CAST IRON COVER & WASTE FITTING ASSOCIATION (JCW)

JCW 203	(2012) Floor Clean Out

~~MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)~~

~~MSS SP-44~~	~~(2019) Steel Pipeline Flanges~~
~~MSS SP-67~~	~~(2022) Butterfly Valves~~
~~MSS SP-70~~	~~(2011) Gray Iron Gate Valves, Flanged and Threaded Ends~~
~~MSS SP-71~~	~~(2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends~~
~~MSS SP-72~~	~~(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service~~
~~MSS SP-78~~	~~(2011) Cast Iron Plug Valves, Flanged and Threaded Ends~~
~~MSS SP-80~~	~~(2019) Bronze Gate, Globe, Angle and Check Valves~~
~~MSS SP-83~~	~~(2014) Class 3000 Steel Pipe Unions Socket Welding and Threaded~~
~~MSS SP-85~~	~~(2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends~~
~~MSS SP-110~~	~~(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends~~

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification

NSF INTERNATIONAL (NSF)

NSF 372	(2016) Drinking Water System Components - Lead Content
NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects

~~MINISTRY OF HEALTH, LABOUR AND WELFARE, GOVERNMENT OF JAPAN (MHLW)~~

	~~INDUSTRIAL SAFETY AND HEALTH ACT~~
~~MHLW-177~~	~~(2018) WATER SUPPLY LAW~~
~~CCPV~~	~~(2016) Construction Code of Pressure Vessel~~
~~PSCP~~	~~Performance Standard Compliant Product~~

~~PLUMBING AND DRAINAGE INSTITUTE (PDI)~~

~~PDI WH 201~~ ~~(2010) Water Hammer Arresters Standard~~

~~U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)~~

~~10 CFR 430~~ ~~Energy Conservation Program for Consumer Products~~

PLUMBING AND DRAINAGE INSTITUTE (PDI)

PDI WH 201 (2010) Water Hammer Arresters Standard

UNDERWRITERS LABORATORIES (UL)

UL 499 (2014; Reprint May 2023) UL Standard for Safety Electric Heating Appliances

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[~~for Contractor Quality Control approval.~~][for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

~~[~~ SD-02 Shop Drawings

Plumbing System; G~~[, [_____]]~~

Detail drawings consisting of schedules, performance charts, instructions, diagrams, and other information to illustrate the requirements and operations of systems that are not covered by the Plumbing Code. Detail drawings for the complete plumbing system including piping layouts and locations of connections; dimensions for roughing-in, foundation, and support points; schematic diagrams and wiring diagrams or connection and interconnection diagrams. Detail drawings shall indicate clearances required for maintenance and operation. Where piping and equipment are to be supported other than as indicated, details shall include loadings and proposed support methods. Mechanical drawing plans, elevations, views, and details, shall be drawn to scale.

~~]~~ SD-03 Product Data

Water Heaters

Pumps

~~Backflow Prevention Assemblies~~

~~Swimming Pool [and Spa]Suction Fittings; G[, [_____]]~~

SD-06 Test Reports

Tests, Flushing and Disinfection

Test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, completion and testing of the installed system. Each test report shall indicate the final position of controls.

~~Test of Backflow Prevention Assemblies; G[, [_____]].~~

~~Certification of proper operation shall be as accomplished in accordance with state regulations by an individual certified by the state to perform such tests. If no state requirement exists, the Contractor shall have the manufacturer's representative test the device, to ensure the unit is properly installed and performing as intended. The Contractor shall provide written documentation of the tests performed and signed by the individual performing the tests.~~

SD-10 Operation and Maintenance Data

Plumbing System

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.3 STANDARD PRODUCTS

Specified materials and equipment shall be standard products of a manufacturer regularly engaged in the manufacture of such products. Specified equipment shall essentially duplicate equipment that has performed satisfactorily at least two years prior to bid opening. Standard products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.3.1 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.3.2 Service Support

The equipment items shall be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations shall be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.3.3 Manufacturer's Nameplate

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.3.4 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

1.3.4.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions shall be considered mandatory, the word "should" shall be interpreted as "shall." Reference to the "code official" shall be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" shall be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" shall be interpreted to mean the "lessor." References to the "permit holder" shall be interpreted to mean the "Contractor."

1.3.4.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, shall be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

~~1.5 PERFORMANCE REQUIREMENTS~~

~~1.5.1 Welding~~

~~[Piping shall be welded in accordance with qualified procedures using performance-qualified welders and welding operators. Procedures and welders shall be qualified in accordance with JIS Z 3801. Welding procedures qualified by others, and welders and welding operators qualified by another employer, may be accepted as permitted by JIS B 8285. The Contracting Officer shall be notified 24 hours in advance of tests, and the tests shall be performed at the work site if practicable. Welders or welding operators shall apply their assigned symbols near each weld they make as a permanent record. Structural members shall be welded in accordance with Section 05 05 23.16 STRUCTURAL WELDING.] [Welding and nondestructive testing procedures are specified in Section 40 05 13.96 WELDING PROCESS PIPING.] Structural members shall be welded in accordance with Section 05 05 23.16 STRUCTURAL WELDING.~~

~~1.5.2 Cathodic Protection and Pipe Joint Bonding~~

~~Cathodic protection and pipe joint bonding systems shall be in accordance with [Section 26 42 14.00 10 CATHODIC PROTECTION SYSTEM (SACRIFICIAL ANODE)] [and] [Section 26 42 17.00 10 CATHODIC PROTECTION SYSTEM (IMPRESSED CURRENT)][Section 26 42 13.00 20 CATHODIC PROTECTION BY GALVANIC ANODES] [and] [Section 26 42 19.00 20 CATHODIC PROTECTION BY IMPRESSED CURRENT]Section 26 42 14.00 10 CATHODIC PROTECTION SYSTEM (SACRIFICIAL ANODE) and Section 26 42 19.00 20 CATHODIC PROTECTION BY IMPRESSED CURRENT.~~

1.5 REGULATORY REQUIREMENTS

Unless otherwise required herein, plumbing work shall be in accordance with MLIT-M.

1.6 PROJECT/SITE CONDITIONS

The Contractor shall become familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.7 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors shall be thoroughly familiar with all parts of the installation and shall be trained in operating theory as well as practical operation and maintenance work.

Instruction shall be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished shall be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.8 ACCESSIBILITY OF EQUIPMENT

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

2.1 MATERIALS

Materials for various services shall be in accordance with TABLES I and II. Pipe schedules shall be selected based on service requirements. Pipe fittings shall be compatible with the applicable pipe materials. ~~Plastic pipe, fittings, and solvent cement shall meet NSF/ANSI 14 and shall be NSF~~

~~listed for the service intended. Plastic pipe, fittings, and solvent cement used for potable hot and cold water service shall bear the NSF seal "NSF-PW." Polypropylene pipe and fittings shall conform to dimensional requirements of Schedule 40, Iron Pipe size and shall comply with NSF/ANSI 14, NSF/ANSI 61 and ASTM F2389. Polypropylene piping that will be exposed to UV light shall be provided with a Factory applied UV resistant coating. Pipe threads (except dry seal) shall conform to ASME B1.20.1. Grooved pipe couplings and fittings shall be from the same manufacturer.~~ Material or equipment containing a weighted average of greater than 0.25 percent lead shall not be used in any potable water system intended for human consumption, and shall be certified in accordance with NSF/ANSI 61, Annex G or NSF 372. In line devices such as water meters, building valves, check valves, meter stops, valves, fittings and back flow preventers shall comply with or PL 93-523 and NSF/ANSI 61, Section 8. End point devices such as drinking water fountains, lavatory faucets, kitchen and bar faucets, residential ice makers, supply stops and end point control valves used to dispense water for drinking must meet the requirements of NSF/ANSI 61. Hubless cast-iron soil pipe shall not be installed underground, under concrete floor slabs, or in crawl spaces below kitchen floors. Plastic pipe shall not be installed in air plenums. Plastic pipe shall not be installed in a pressure piping system in buildings greater than three stories including any basement levels.

The NSF requirements of this specification and specification section 01 11 00.0010 Mandatory US Tested Products are independent of the Japanese standards listed elsewhere in this specification.

Thus where Japanese standards are listed for potable water equipment, in line devices, and end point devices, the products must meet both NSF and the Japanese standards. The purpose is to promote the use of Japanese products to the greatest extent possible. Japanese products that are not NSF certified will not be allowed for potable water systems. For the majority of material and equipment that must be NSF certified and where other standards are required both the Japanese and US Standards are referenced in this specification.

A searchable database of NSF certified products and materials is available online at: https://www.nsf.org/certified-products-systems

2.1.1 Pipe Joint Materials

Grooved pipe and hubless cast-iron soil pipe shall not be used underground. Solder containing lead shall not be used with copper pipe.

a. Coupling for Cast-Iron Pipe: for hub and spigot type JIS G 5526, JPF-MP-006. For hubless type: JIS G 5526.

~~b. Coupling for Steel Pipe: JPF-MP-006.~~

~~c. Couplings for Grooved Pipe: [Ductile Iron JIS G 5502] [Malleable Iron JIS G 5705].[Copper JIS G 5502].~~

~~d~~b. Flange Gaskets: Gaskets shall be made of non-asbestos material in accordance with JIS B 2404. Gaskets shall be flat, 1.6 mm thick, and contain Aramid fibers bonded with Styrene Butadiene Rubber (SBR) or Nitro Butadiene Rubber (NBR). Gaskets shall be the full face or self centering flat ring type. Gaskets used for hydrocarbon service shall be bonded with NBR.

~~e~~c. Brazing Material: Brazing material shall conform to JIS Z 3621.

~~f~~d. Brazing Flux: Flux shall be in paste or liquid form appropriate for use with brazing material. Flux shall be as follows: lead-free; have a 100 percent flushable residue; contain slightly acidic reagents; contain potassium borides; and contain fluorides.

~~g~~e. Solder Material: Solder metal shall conform to JIS Z 3284-1.

~~h~~f. Solder Flux: Flux shall be liquid form, non-corrosive, and conform to JIS Z 3313.

~~i~~g. PTFE Tape: PTFE Tape, for use with Threaded Metal or Plastic Pipe.

~~j~~h. Rubber Gaskets for Cast-Iron Soil-Pipe and Fittings (hub and spigot type and hubless type): JIS K 6353.

~~k~~i. Rubber Gaskets for Grooved Pipe: JIS B 2404.

~~l~~j. Flexible Elastomeric Seals: JIS A 5758.

~~m. Bolts and Nuts for Grooved Pipe Couplings: Heat-treated carbon steel, JIS G 4051.~~

~~n~~k. Solvent Cement for Transition Joints between ABS and PVC Nonpressure Piping Components: JWWA S 101.

~~o~~l. Plastic Solvent Cement for ABS Plastic Pipe: JWWA S 101.

~~p~~m. Plastic Solvent Cement for PVC Plastic Pipe: JWWA S 101 and JWWA S 101.

~~q~~n. Plastic Solvent Cement for CPVC Plastic Pipe: JWWA S 101.

~~r~~o. Flanged fittings including, but not limited to, flanges, bolts, nuts and bolt patterns shall be in accordance with JIS B 2220 and shall have the manufacturer's trademark affixed in accordance with JIS B 2004 and JIS Z 9102. Flange material shall conform to JIS G 3201. Blind flange material shall conform to JIS G 3118 and JIS G 3118 cold service and JIS G 3103 and JIS G 3118 for hot service. Bolts shall be high strength or intermediate strength with material conforming to JIS G 4107 and JIS G 4303.

~~s. Plastic Solvent Cement for Styrene Rubber Plastic Pipe: JIS K 6743.~~

~~t. Press fittings for Copper Pipe and Tube: Copper press fittings shall conform to the material and sizing requirements of JIS H 3401. Sealing elements shall be factory installed or an alternative supplied fitting manufacturer. Sealing element shall be selected based on manufacturer's approved application guidelines.~~

~~u~~p. Copper tubing shall conform to JIS H 3300 or ASTM B88M, Type K, L or M.

2.1.2 Miscellaneous Materials

Miscellaneous materials shall conform to the following:

a. Water Hammer Arrester: Performance standards of products conforming to the Japanese Water Supply Law or PDI WH 201.~~[~~ Water hammer arrester shall be ~~[diaphragm][or][~~piston~~]~~ type.~~]~~

b. Copper, Sheet and Strip for Building Construction: JIS H 3100.

c. Asphalt Roof Cement: JIS A 6005.

d. Hose Clamps: MLIT-M.

e. Metallic Cleanouts: JCW 203.

f. Plumbing Fixture Setting Compound: A preformed flexible ring seal molded from hydrocarbon wax material. The seal material shall be nonvolatile nonasphaltic and contain germicide and provide watertight, gastight, odorproof and verminproof properties.

~~g. Coal-Tar Protective Coatings and Linings for Steel Water Pipelines: JWWA K 115.~~

~~h~~g. Hypochlorites: JWWA K 120.

~~i~~h. Liquid Chlorine: JWWA K 120.

~~j~~i. Gauges - Pressure and Vacuum Indicating Dial Type - Elastic Element: JIS B 7505-1.

~~k~~j. Thermometers: JIS B 7414. Mercury shall not be used in thermometers.

2.1.3 Pipe Insulation Material

Insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.2 PIPE HANGERS, INSERTS, AND SUPPORTS

Pipe hangers, inserts, and supports shall conform to MLIT-M.

2.3 VALVES

Valves shall be provided on supplies to equipment and fixtures. Valves 65 mm and smaller shall be bronze with threaded bodies for pipe and solder-type connections for tubing. Valves 80 mm and larger shall have flanged iron bodies and bronze trim. Pressure ratings shall be based upon the application. Grooved end valves may be provided if the manufacturer certifies that the valves meet the performance requirements of applicable MSS standard. Valves shall conform to the following standards:

Description	Standard
Butterfly Valves~~: MSS SP-67 for potable water system. JPF-DP-001 or MSS SP-67 for other than potable water system.~~	JPF-DP-001~~ or MSS SP-67~~

Cast-Iron Gate Valves, Flanged and Threaded Ends~~: MSS SP-70 for potable water system. JIS B 2031 or MSS SP-70 for other than potable water system~~	JIS B 2031~~ or MSS SP-70~~
Cast-Iron Swing Check Valves, Flanged and Threaded Ends~~: MSS SP-71 for potable water system. JIS B 2011 or MSS SP-71 for other than potable water system~~	JIS B 2011~~ or MSS SP-71~~
Ball Valves with Flanged Butt-Welding Ends for General Service~~: MSS SP-72 for potable water system. JV-5 or MSS SP-72 for other than potable water system~~	JV-5~~ or MSS SP-72~~
Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends~~: MSS SP-110 for potable water system. JV-5 or MSS SP-110 for other than potable water system~~	~~MSS SP-110 or ~~JV-5
Bronze Gate, Globe, Angle, and Check Valves~~: MSS SP-80 for potable water system. JV-5 or MSS SP-80 for other than potable water system.~~	JV-5~~ or MSS SP-78~~
Steel Valves, Socket Welding and Threaded Ends~~: ASME B16.34 for potable water system. JIS B 2003 or ASME B16.34 for other than potable water system.~~	JIS B 2003~~ or MSS SP-80~~
Cast-Iron Globe and Angle Valves, Flanged and Threaded Ends~~: MSS SP-85 for potable water system. JIS B 2071 or MSS SP-80 for other than potable water system~~	JIS B 2071~~ or ASME A112.14.1~~
Backwater Valves	
Vacuum Relief Valves~~: "ANSI Z21.22/CSA 4.4 for potable water system. JIS B 8372-1 or ANSI Z21.22/CSA 4.4 for other than potable water system~~	JIS B 8372-1 and JIS B 8225~~ or ANSI Z21.22/CSA 4.4~~
Water Pressure Reducing Valves~~: ASSE 1003 for potable water system. JIS B 8410 or ASSE 1003 for other than potable water system.~~	JIS B 8410~~ or ASSE 1003~~
Water Heater Drain Valves~~: ASME BPVC SEC IV, Part HLW-810; Requirements for Potable-Water Heaters Bottom Drain Valve for potable water system. JIS B 2011 or ASME BPVC SEC IV, Part HLW-810; Requirements for Potable-Water Heaters Bottom Drain Valve for other than potable water system~~	JIS B 2011~~ or ANSI Z21.22/CSA 4.4~~

Trap Seal Primer Valves	JIS B 2061
Temperature and Pressure Relief Valves for Hot Water Supply Systems~~: ANSI Z21.22/CSA 4.4 for potable water system. JIS B 8414 and JIS B 2011 or ANSI Z21.22/CSA 4.4 for other than potable water system.~~	JIS B 8414 and JIS B 2011~~ or ANSI Z21.22/CSA 4.4~~
Temperature and Pressure Relief Valves for Automatically Fired Hot Water Boilers	JIS B 8414

2.3.1 Backwater Valves

Backwater valves shall be either separate from the floor drain or a combination floor drain, P-trap, and backwater valve, as shown. Valves shall have cast-iron bodies with cleanouts large enough to permit removal of interior parts. Valves shall be of the flap type, hinged or pivoted, with revolving disks. Hinge pivots, disks, and seats shall be nonferrous metal. Disks shall be slightly open in a no-flow no-backwater condition. Cleanouts shall extend to finished floor and be fitted with threaded countersunk plugs.

~~2.3.2 Wall Faucets~~

~~Wall faucets with vacuum-breaker backflow preventer shall be brass with 20 mm male inlet threads, hexagon shoulder, and 20 mm hose connection. Faucet handle shall be securely attached to stem.~~

~~2.3.3 Lawn Faucets~~

~~Lawn faucets shall be brass, with either straight or angle bodies, and shall be of the compression type. Body flange shall be provided with internal pipe thread to suit 20 mm pipe. Body shall be suitable for wrench grip. Faucet spout shall have 20 mm exposed hose threads. Faucet handle shall be securely attached to stem.~~

~~2.3.4 Yard Hydrants~~

~~Yard box or post hydrants shall have valve housings located below frost lines. Water from the casing shall be drained after valve is shut off. Hydrant shall be bronze with cast-iron box or casing guard. "T" handle key shall be provided.~~

2.3.2 Relief Valves

Water heaters and hot water storage tanks shall have a combination pressure and temperature (P&T) relief valve. The pressure relief element of a P&T relief valve shall have adequate capacity to prevent excessive pressure buildup in the system when the system is operating at the maximum rate of heat input. The temperature element of a P&T relief valve shall have a relieving capacity which is at least equal to the total input of the heaters when operating at their maximum capacity. Relief valves shall be rated according to JIS B 8225. Relief valves for systems where the maximum rate of heat input is less than 59 kW shall have 20 mm minimum inlets, and 20 mm outlets. Relief valves for systems where the maximum rate of heat input is greater than 59 kW shall have 25 mm minimum inlets, and 25 mm outlets. The discharge pipe from the relief valve shall be the size of the valve outlet.

2.3.3 Thermostatic Mixing Valves

Provide thermostatic mixing valve for lavatory faucets. Mixing valves, thermostatic type, pressure-balanced or combination thermostatic and pressure-balanced shall be line size and shall be constructed with rough or finish bodies either with or without plating. Each valve shall be constructed to control the mixing of hot and cold water and to deliver water at a desired temperature regardless of pressure or input temperature changes. The control element shall be of an approved type. The body shall be of heavy cast bronze, and interior parts shall be brass, bronze, corrosion-resisting steel or copper. The valve shall be equipped with necessary stops, check valves, unions, and sediment strainers on the inlets. Mixing valves shall maintain water temperature within 2 degrees C of any setting.

2.4 FIXTURES

Water closet replacements in major renovations may have a flush valve of up to 6.1 LPF to accommodate existing plumbing capacity. ~~[JIS B 2308 302 stainless steel] [~~Vitreous China~~]~~, nonabsorbent, hard-burned, and vitrified throughout the body shall be provided. Porcelain enameled ware shall have specially selected, clear ~~[white][_____]~~, acid-resisting enamel coating evenly applied on surfaces. No fixture will be accepted that shows cracks, crazes, blisters, thin spots, or other flaws. Fixtures shall be equipped with appurtenances such as traps, faucets, stop valves, and drain fittings. Each fixture and piece of equipment requiring connections to the drainage system~~, except grease interceptors,~~ shall be equipped with a trap. Brass expansion or toggle bolts capped with acorn nuts shall be provided for supports, and polished chromium-plated pipe, valves, and fittings shall be provided where exposed to view. Fixtures with the supply discharge below the rim shall be equipped with backflow preventers. Internal parts of flush valves and flushometer valves, shower mixing valves, shower head face plates, pop-up stoppers of lavatory waste drains, and pop-up stoppers and overflow tees and shoes of bathtub waste drains ~~[may contain acetal resin, fluorocarbon, nylon, acrylonitrile-butadiene-styrene (ABS) or other plastic material, if the material has provided satisfactory service under actual commercial or industrial operating conditions for not less than 2 years][~~shall be copper alloy with all visible surfaces chrome plated~~]~~.~~[Plastic in contact with hot water shall be suitable for 82 degrees C water temperature.]~~

2.4.1 Lavatories

~~[Enameled cast-iron lavatories shall be provided with two cast-iron or steel brackets secured to the underside of the apron and drilled for bolting to the wall in a manner similar to the hanger plate. Exposed brackets shall be porcelain enameled.][~~Vitreous china lavatories shall be provided with two integral molded lugs on the back-underside of the fixture and drilled for bolting to the wall in a manner similar to the hanger plate.~~]~~ Provide faucet with a maximum flow rate of 1.9 L/min at a flowing pressure of 414 kPa. ~~Water volume must be limited to 1.0 L per metering cycle.~~ Provide lavatories with low flow type single lever mixing faucet with satin/brushed stainless steel finish. Provide with thermostatic mixing valve to allow maximum 48.9 degrees C hot water.

~~2.4.2 Automatic Controls~~

~~Provide automatic, sensor operated faucets and flush valves to comply with JIS A 5207 or ASSE 1037 for lavatory faucets, urinals, and water closets. Flushing and faucet systems shall consist of solenoid-activated valves with light beam sensors. Flush valve for water closet shall include an override pushbutton. Flushing devices shall be provided as described in paragraph FIXTURES AND FIXTURE TRIMMINGS.~~

~~2.4.3 Flush Valve Water Closets~~

~~JIS A 5207, [white] [_____] vitreous china, [JIS B 2308 and JIS B 2309 302 Stainless Steel,] siphon jet, elongated bowl, [floor-mounted, floor outlet][wall mounted, wall outlet]. Top of toilet seat height above floor shall be 360 to 380 mm, except 435 to 480 mm for wheelchair water closets. Provide wax bowl ring including plastic sleeve. Provide [white] [_____] solid plastic elongated [open-front seat] [closed-front seat with cover].~~

~~Water flushing volume of the water closet and flush valve combination shall not exceed 4.85 liters per flush.[Provide a dual-flush water closet and flush valve combination that will also provide a second flushing water volume not to exceed 4.8 liters per flush.]~~

~~Provide large diameter flush valve including angle control-stop valve, vacuum breaker, tail pieces, slip nuts, and wall plates; exposed to view components shall be chromium-plated or polished stainless steel. Flush valves shall be nonhold-open type. Mount flush valves not less than 280 mm above the fixture. Mounted height of flush valve shall not interfere with the hand rail in ADA stalls.[Provide solenoid-activated flush valves including electrical-operated light-beam-sensor to energize the solenoid.][Provide piston type, oil operated, flush valve and wall support for salt water service.]~~

~~2.4.4 Flush Valve Urinals~~

~~JIS A 5207, [white] [_____] vitreous china, [JIS B 2308 and JIS B 2309 302 stainless steel],wall-mounted, wall outlet, siphon jet, integral trap, and extended side shields. Provide urinal with the rim 430 mm above the floor. Provide urinal with the rim 610 mm above the floor. Water flushing volume of the urinal and flush valve combination shall not exceed 1.9 liters per flush. Provide large diameter flush valve including angle control-stop valve, vacuum breaker, tail pieces, slip nuts, and wall plates; exposed to view components shall be chromium-plated or polished stainless steel. Flush valves shall be nonhold-open type. Mount flush valves not less than 280 mm above the fixture.[Provide solenoid-activated flush valves including electrical-operated light-beam-sensor to energize the solenoid.] [Provide piston type, oil operated, flush valve and wall support for salt water service.]~~

~~2.4.5 Wheelchair Flush Valve Type Urinals~~

~~JIS A 5207 [white] [_____] vitreous china, [JIS B 2308 and JIS B 2309 302 stainless steel],wall-mounted, wall outlet, blowout action, integral trap, elongated projecting bowl, 510 mm long from wall to front of flare, and JIS B 2061 trim. Provide large diaphragm (not less than 65 mm upper chamber inside diameter at the point where the diaphragm is sealed between the upper and lower chambers), nonhold-open flush valve of chrome plated~~

~~cast brass conforming to JIS B 2061, including vacuum breaker and angle (control-stop) valve with back check. The water flushing volume of the flush valve and urinal combination shall not exceed 1.9 liters per flush. Furnish urinal manufacturer's certification of conformance. Mount urinal with front rim a maximum of 430 mm above floor and flush valve handle a maximum of 1120 mm above floor for use by handicapped on wheelchair.[Provide solenoid-activated flush valves including electrical-operated light-beam-sensor to energize the solenoid.]~~

2.4.2 Flush Tank Water Closets

JIS A 5207 ~~[~~white~~] [_____]~~ vitreous china, ~~[JIS B 2308 and JIS B 2309 302 stainless steel],~~ siphon jet, round bowl, pressure assisted, floor-mounted, floor outlet. Top of toilet seat height above floor shall be 360 to 380 mm, except 435 to 480 mm for wheelchair water closets.~~[Nonfloat swing type flush tank valves are not acceptable.][Gravity tank type water closets are not permitted.]~~ Provide wax bowl ring including plastic sleeve. Water flushing volume of the water closet shall not exceed ~~4.8~~3.6 liters per flush.~~[~~ Provide a dual-flush toilet with a second flushing option that shall not exceed 4.1 liters per flush.~~]~~ Provide ~~[~~ white~~] [_____]~~ solid plastic round closed-front seat with cover.~~[Provide solenoid-activated flush valves including electrical-operated light-beam-sensor to energize the solenoid.]~~

~~2.4.3 Non-Flushing Toilets~~

~~[Provide composting toilets in accordance with manufacturer's recommendations.][Provide vacuum toilet systems in accordance with manufacturer's recommendations.]~~

~~2.4.4 Wall Hung Lavatories~~

~~JIS A 5207 [[white] [_____] vitreous china, JIS B 2308 and JIS B 2309 302 stainless steel],straight back type, minimum dimensions of 480 mm, wide by 430 mm front to rear, with supply openings for use with top mounted centerset faucets, and openings for concealed arm carrier installation.[Provide aerator with faucet.]Provide lavatory faucets and accessories meeting the flow rate and product requirements of the paragraph LAVATORIES. Provide concealed chair carriers with vertical steel pipe supports and concealed arms for the lavatory. Mount lavatory with the front rim 865 mm above floor and with 740 mm minimum clearance from bottom of the front rim to floor.[Provide top mounted washerless centerset lavatory faucets.][Provide top-mounted solenoid-activated lavatory faucets including electrical-operated light-beam-sensor to energize the solenoid.] [Provide filters for chlorine in supply piping to faucets.]~~

~~2.4.5 Countertop Lavatories~~

~~JIS A 5207 [white] [_____] vitreous china, [JIS B 2308 and JIS B 2309 302 stainless steel],self-rimming, minimum dimensions of 480mm wide by 430 mm front to rear, with supply openings for use with top mounted centerset faucets. Furnish template and mounting kit by lavatory manufacturer.[Provide aerator with faucet.]Provide lavatory faucets and accessories meeting the flow rate and product requirements of the paragraph LAVATORIES. [Provide top mounted washerless centerset lavatory faucets.][Provide top-mounted solenoid-activated lavatory faucets including electrical-operated light-beam-sensor to energize the solenoid.] [Provide filters for chlorine in supply piping to faucets.]~~

2.4.3 Kitchen Sinks

JIS B 2308 and JIS B 2309 or ASME A112.19.3/CSA B45.4 20 gage stainless steel with integral mounting rim for flush installation, minimum dimensions of 840 mm wide by 535 mm front to rear, two compartments, with undersides fully sound deadened, with supply openings for use with top mounted washerless sink faucets with hose spray, and with 90 mm drain outlet.~~[~~ Provide aerator with faucet.~~]~~ Water flow rate shall not exceed 8.3 L per minute when measured at a flowing water pressure of 414 kPa. Provide stainless steel drain outlets and stainless steel cup strainers. Provide separate 38 mm P-trap and drain piping to vertical vent piping from each compartment. Provide top mounted washerless sink faucets with hose spray. ~~[Provide filters for chlorine in supply piping to faucets.][~~ Provide JIS A 4420 waste disposer in right compartment.~~][Provide pedal valve for foot-operated flow control.][Provide secondary kitchen sink that drains to graywater system.][Provide sink with disposal chute to compost bucket under sink.]~~ Provide thermostatic mixing valve to allow maximum 48.9 degrees C hot water.

2.4.4 Service Sinks

JIS A 5207, ~~[[~~white~~] [_____]~~ vitreous china JIS B 2308 and JIS B 2309 302 stainless steel~~]~~ with integral back and wall hanger supports, minimum dimensions of 560 mm wide by 510 mm front to rear, with two supply openings in 254 mm high back. Provide floor supported wall outlet cast iron P-trap and stainless steel rim guards as recommended by service sink manufacturer. Provide back mounted washerless service sink faucets with vacuum breaker and 19 mm external hose threads. Provide thermostatic mixing valve to allow maximum 48.9 degrees C hot water.

~~2.4.5 Drinking-Water Coolers~~

~~JIS C 9618 or ANSI/ASHRAE 18 with more than a single thickness of metal between the potable water and the refrigerant in the heat exchanger, wall-hung, bubbler style, air-cooled condensing unit, 5 ml per second minimum capacity, stainless steel splash receptor and basin, [bottle filler] and stainless steel cabinet. Bubblers shall be controlled by push levers or push bars, front mounted or side mounted near the front edge of the cabinet. Bubbler spouts shall be mounted at maximum of 915 mm above floor and at front of unit basin. Spouts shall direct water flow at least 100 mm above unit basin and trajectory parallel or nearly parallel to the front of unit. [Provide filters for chlorine in supply piping to faucets.] Provide concealed steel pipe chair carriers.~~

~~2.4.6 Wheelchair Drinking Water cooler~~

~~JIS C 9618 or ANSI/ASHRAE 18 wall-mounted bubbler style with concealed chair carrier, air-cooled condensing unit, 5 mL per second minimum capacity, stainless steel splash receptor, and all stainless steel cabinet, with 690 mm minimum knee clearance from front bottom of unit to floor and 915 mm maximum spout height above floor [and bottle filler]. Bubblers shall also be controlled by push levers, by push bars, or touch pads one on each side or one on front and both sides of the cabinet. [Provide filters for chlorine in supply piping to faucets.]~~

~~2.4.7 Plastic Bathtub/Shower Units~~

~~JIS A 5207 multi-piece white solid acrylic pressure molded fiberglass reinforced plastic bathtub/shower units. Units shall be scratch~~

~~resistant, waterproof, and reinforced. Provide showerheads meeting the requirements of the paragraph BATHTUB AND SHOWER FAUCETS AND DRAIN FITTI.NGS.[Provide flow restrictor in handshower to flow 6.6 L/min.] [Provide filters for chlorine in supply piping to faucets and showerheads.] Provide recessed type units approximately 1520 mm wide, 760 mm front to rear, 1830 mm high with 380 mm high rim for through-the-floor drain installation with unit bottom or feet firmly supported by a smooth level floor. Provide left or right drain outlet units as required. Units shall have built-in soap dish and minimum of 305 mm long stainless steel horizontal grab bar located on back wall for standing use. Units shall meet performance requirements of JIS A 5207. Install unit in accordance with the manufacturer's written instructions. Finish installation by covering unit attachment flanges with wall board in accordance with unit manufacturer's recommendation. Provide smooth 100 percent silicone rubber [white] [_____] bathtub caulk between the unit and the adjacent walls and floor surfaces.~~

~~2.4.8 Plastic Bathtubs~~

~~JIS A 5207 one piece [white] [_____] solid acrylic pressure molded fiberglass reinforced plastic bathtubs. Bathtubs shall be scratch resistant, waterproof, and reinforced. Provide recessed type bathtubs approximately 1520 mm wide, 760 mm front to rear, 380 mm high rim for through-the-floor drain installation with bathtub bottom or feet firmly supported by a smooth level floor. Provide left or right drain outlet bathtub as required. [Provide filters for chlorine in supply piping to faucets.]Bathtubs shall meet performance requirements of JIS A 5207. Install bathtub in accordance with the manufacturer's written instructions. Finish installation by covering bathtub attachment flanges with dry-wall in accordance with bathtub manufacturer's recommendation. Provide smooth 100 percent silicone rubber [white] [_____] bathtub caulk between the bathtub and the adjacent walls and floor surfaces.~~

~~2.4.9 Plastic Shower Stalls~~

~~JIS A 5207 four piece [white] [_____] solid acrylic pressure molded fiberglass reinforced plastic shower stalls. Shower stalls shall be scratch resistant, waterproof, and reinforced. Provide showerheads meeting the requirements of the paragraph BATHTUB AND SHOWER FAUCETS AND DRAIN FITTINGS.[Provide flow restrictor in handshower to flow 6.6 L/min.] [Provide filters for chlorine in supply piping to showerheads.] Provide recessed type shower stalls approximately 915 mm wide, 915 mm front to rear, 1830 mm high, and 125 high mm high curb with shower stall bottom or feet firmly supported by a smooth level floor. Provide PVC shower floor drains and stainless steel strainers. Shower stalls shall meet performance requirements of JIS A 5207. Install shower stall in accordance with the manufacturer's written instructions. Finish installation by covering shower stall attachment flanges with dry-wall in accordance with shower stall manufacturer's recommendation. Provide smooth 100 percent silicone rubber [white] [_____] bathtub caulk between the top, sides, and bottom of shower stalls and bathroom walls and floors.~~

~~2.4.10 Plastic Bathtub Liners~~

~~JIS A 5532 one piece [white] [_____] plastic bathtub liners. Existing bathtubs shall be identified and measured to insure proper identification in order that each new bathtub liner shall be custom molded to fit the exact contours of the existing bathtubs. Provide left or right drain outlet bathtub liners as required. Bathtub liners shall be inserted over~~

~~and into the existing bathtubs without disturbing the existing ceramic tile wainscot walls and existing floor material. Prepare the existing cast-iron bathtubs, ceramic tile wainscots, and floor to receive the new bathtub liners in accordance with the bathtub liner manufacturer's written instructions. Installation personnel shall be trained by the bathtub liner manufacturer. Seal the bathtub liner to existing bathtub with waterproof adhesive as required to keep moisture out from behind the bathtub liner. Provide smooth [white] [_____] waterproof bathtub sealant between bathtub drains, bathtub, and bathtub liners. Provide replacement chromium-plated overflow cover plates and push-pull bathtub drain stopper assembly. Provide smooth 100 percent silicone rubber [white] [_____] bathtub caulk between the bathtub liner and the adjacent walls and floor surfaces in accordance with the bathtub liners manufacturer's written instructions.~~

~~2.4.11 Plastic Bathtub Wall Surrounds~~

~~JIS A 5207 three piece [white] [_____] sectional pressure molded fiberglass plastic bathtub wall surrounds suitable for installation with existing bathtubs which are approximately 1520 mm wide by 760 mm front to rear. Wall surrounds shall have built-in soap dish and minimum of 305 mm long stainless steel horizontal grab bar located on back wall for standing use. Bathtub wall surrounds shall meet performance requirements of JIS A 5207. for compliance. Install bathtub wall surrounds in accordance with the manufacturers written instructions. Finish installation by covering bathtub wall surround attachment flanges with dry-wall in accordance with bathtub wall surround manufacturer's recommendations. Provide smooth 100 percent silicone rubber [white] [_____] bathtub caulk between the bathtubs and the adjacent walls and floor surfaces.~~

~~2.4.12 Precast Terrazzo Shower Floors~~

~~Terrazzo shall be made of marble chips cast in white portland cement to produce 25 mPa minimum compressive strength 7 days after casting. Provide floor or wall outlet copper alloy body drain cast integral with terrazzo, with polished stainless steel strainers.~~

~~2.4.13 Precast Terrazzo Mop Sinks~~

~~Terrazzo shall be made of marble chips cast in white portland cement to produce 25 mPa minimum compressive strength 7 days after casting. Provide floor or wall outlet copper alloy body drain cast integral with terrazzo, with polished stainless steel strainers.~~

2.4.5 Bathtubs, Cast Iron

JIS A 5207 ~~[~~white~~] [_____]~~ enameled cast iron, recessed type, minimum dimensions of 1520 mm wide by 760 mm front to rear by 410 mm high with drain outlet for above-the-floor drain installation. Provide left or right drain outlet bathtub as indicated. ~~[Provide filters for chlorine in supply piping to faucets.]~~ Provide with overflow drain, low flow type shower head, faucet, and spout. Provide with thermostatic mixing valve to allow maximum 48.9 degrees C hot water.

~~2.4.6 Bathtubs, Porcelain~~

~~JIS A 5207 [white] [_____] porcelain bonded to enameling grade metal, bonded to a structural composite, recessed type, minimum dimensions of 1520 mm wide by 760 mm front to rear by 410 mm high with drain outlet for~~

~~above-the-floor drain installation. Provide left or right drain outlet bathtub as indicated. [Provide filters for chlorine in supply piping to faucets.]~~

~~2.4.7 Emergency Eyewash and Shower~~

~~Ministry of Labour Ordinance (Tokutei-Kagaku-bushitsu-Shogai-Yobou-Kisoku) Article 38 or ANSI/ISEA Z358.1, floor supported free standing unit. Provide deluge shower head, stay-open ball valve operated by pull rod and ring or triangular handle. Provide eyewash and stay-open ball valve operated by foot treadle or push handle.~~

~~2.4.8 Emergency Eye and Face Wash~~

~~Ministry of Labour Ordinance (Tokutei-Kagaku-bushitsu-Shogai-Yobou-Kisoku) Article 38 or ANSI/ISEA Z358.1, wall-mounted self-cleaning, nonclogging eye and face wash with quick opening, full-flow valves, stainless steel eye and face wash receptor. Unit shall deliver 0.19 L/s of aerated water at 207 kPa (gage) flow pressure, with eye and face wash nozzles 840 to 1140 mm above finished floor. Provide copper alloy control valves. Provide an air-gap with the lowest potable eye and face wash water outlet located above the overflow rim by not less than the International Plumbing Code minimum.[Provide a pressure-compensated tempering valve, with leaving water temperature setpoint adjustable throughout the range 15.5 to 35 degrees C.][Provide packaged, alarm system; including an amber strobe lamp, horn with externally adjustable loudness and horn silencing switch, mounting hardware, and waterflow service within enclosures[and for explosion proof service within enclosures].]~~

~~2.5 BACKFLOW PREVENTERS~~

~~Backflow preventers with intermediate atmospheric vent shall conform to JIS S 3200-5,or ASSE 1012. Reduced pressure principle backflow preventers shall be approved product of quality certification center of JWWA or ASSE 1013. Hose connection vacuum breakers shall be approved product of quality certification center of JWWA or ASSE 1011. Pipe applied atmospheric type vacuum breakers shall be approved product of quality certification center of JWWA or ASSE 1001. Pressure vacuum breaker assembly shall be approved product of quality certification center of JWWAor ASSE 1020. Air gaps in plumbing systems shall conform to Ministry of Health Labor and Welfare Ordinance Article 123.~~

2.5 DRAINS

2.5.1 Floor and Shower Drains

Floor and shower drains shall consist of a galvanized body, integral seepage pan, and adjustable perforated or slotted chromium-plated bronze, nickel-bronze, or nickel-brass strainer, consisting of grate and threaded collar. Floor drains shall be cast iron except where metallic waterproofing membrane is installed. Drains shall be of double drainage pattern for embedding in the floor construction. The seepage pan shall have weep holes or channels for drainage to the drainpipe. The strainer shall be adjustable to floor thickness. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or waterproofing membrane shall be provided when required. Drains shall be provided with threaded connection. Between the drain outlet and waste pipe, a neoprene rubber gasket may be installed, provided that the drain is specifically designed for the rubber gasket compression type joint. Floor and shower drains shall conform to JCM 201.

~~2.5.1.1 Metallic Shower Pan Drains~~

~~Where metallic shower pan membrane is installed, polyethylene drain with corrosion-resistant screws securing the clamping device shall be provided. Polyethylene drains shall have fittings to adapt drain to waste piping. Polyethylene for floor drains shall conform to JIS K 7350-3. Drains shall have separate cast-iron "P" trap, circular body, seepage pan, and strainer, unless otherwise indicated.~~

2.5.1.1 Drains and Backwater Valves

Drains and backwater valves installed in connection with waterproofed floors or shower pans shall be equipped with bolted-type device to securely clamp flashing.

2.5.2 Bathtub and Shower Faucets and Drain Fittings

Provide single control pressure equalizing bathtub and shower faucets with body mounted from behind the wall with threaded connections. Provide ball joint self-cleaning shower heads. Provide tubing mounted from behind the wall between bathtub faucets and shower heads and bathtub diverter spouts. Provide separate globe valves or angle valves with union connections in each supply to faucet. Provide trip-lever pop-up drain fittings for above-the-floor drain installations. The top of drain pop-ups, drain outlets, tub overflow outlet, and; control handle for pop-up drain shall be chromium-plated or polished stainless steel. Linkage between drain pop-up and pop-up control handle at bathtub overflow outlet shall be copper alloy or stainless steel. Provide 40 mm copper alloy adjustable tubing with slip nuts and gaskets between bathtub overflow and drain outlet; chromium-plated finish is not required.~~[Provide bathtub and shower valve with ball type control handle.]~~ Provide with thermostatic mixing valve to allow maximum 48.9 degrees C hot water.

~~2.5.3 Area Drains~~

~~Area drains shall be plain pattern with polished stainless steel perforated or slotted grate and bottom outlet. The drain shall be circular or square with a 300 mm nominal overall width or diameter and 250 mm nominal overall depth. Drains shall be cast iron with manufacturer's standard coating. Grate shall be easily lifted out for cleaning. Outlet shall be suitable for inside caulked connection to drain pipe. Drains shall conform to JIS A 5207.~~

2.5.3 Floor Sinks

Floor sinks shall be ~~[circular] [~~square~~]~~, with 300 mm nominal overall width or diameter and 250 mm nominal overall depth. Floor sink shall have an acid-resistant enamel interior finish with cast-iron body, ~~[aluminum][~~ ABS~~]~~ sediment bucket, and perforated grate of cast iron in industrial areas and stainless steel in finished areas. The outlet pipe size shall be as indicated or of the same size as the connecting pipe.

~~2.5.4 Boiler Room Drains~~

~~Boiler room drains shall have combined drain and trap, hinged grate, removable bucket, and threaded brass cleanout with brass backwater valve. The removable galvanized cast-iron sediment bucket shall have rounded~~

~~corners to eliminate fouling and shall be equipped with hand grips. Drain shall have a minimum water seal of 100 mm. The grate area shall be not less than 0.065 square meters.~~

~~2.5.5 Pit Drains~~

~~Pit drains shall consist of a body, integral seepage pan, and nontilting perforated or slotted grate. Drains shall be of double drainage pattern suitable for embedding in the floor construction. The seepage pan shall have weep holes or channels for drainage to the drain pipe. Membrane or flashing clamping device shall be provided when required. Drains shall be cast iron with manufacturer's standard coating. Drains shall be circular and provided with bottom outlet suitable for inside caulked connection, unless otherwise indicated. Drains shall be provided with separate cast-iron "P" traps, unless otherwise indicated.~~

~~2.5.6 Sight Drains~~

~~Sight drains shall consist of body, integral seepage pan, and adjustable strainer with perforated or slotted grate and funnel extension. The strainer shall have a threaded collar to permit adjustment to floor thickness. Drains shall be of double drainage pattern suitable for embedding in the floor construction. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or membrane shall be provided for other than concrete construction. Drains shall have a galvanized heavy cast-iron body and seepage pan and chromium-plated bronze, nickel-bronze, or nickel-brass strainer and funnel combination. Drains shall be provided with threaded connection and with a separate cast-iron "P" trap, unless otherwise indicated. Drains shall be circular, unless otherwise indicated. The funnel shall be securely mounted over an opening in the center of the strainer. Minimum dimensions shall be as follows:~~

~~Area of strainer and collar:~~	~~0.023 square meters~~
~~Height of funnel:~~	~~95 mm~~
~~Diameter of lower portion:~~	~~50 mm of funnel~~
~~Diameter of upper portion:~~	~~100 mm of funnel~~

~~2.5.7 Roof Drains and Expansion Joints~~

~~Roof drains shall conform to MLIT-M, with dome and integral flange, and shall have a device for making a watertight connection between roofing and flashing. The whole assembly shall be galvanized heavy pattern cast iron. For aggregate surface roofing, the drain shall be provided with a gravel stop. On roofs other than concrete construction, roof drains shall be complete with underdeck clamp, sump receiver, and an extension for the insulation thickness where applicable. A clamping device for attaching flashing or waterproofing membrane to the seepage pan without damaging the flashing or membrane shall be provided when required to suit the building construction. Strainer openings shall have a combined area equal to twice that of the drain outlet. The outlet shall be equipped to make a proper connection to threaded pipe of the same size as the downspout. An expansion joint of proper size to receive the conductor pipe shall be provided. The expansion joint shall consist of a heavy cast-iron housing, brass or bronze sleeve, brass or bronze fastening bolts and nuts, and gaskets or packing. The sleeve shall have a nominal thickness of not less~~

~~than 3.5 mm. Gaskets and packing shall be close-cell neoprene, O-ring packing shall be close-cell neoprene of 70 durometer. Packing shall be held in place by a packing gland secured with bolts.~~

~~2.5.8 Swimming Pool [and Spa]Suction Fittings~~

~~Pool water suction fittings in swimming pools [and spas]shall comply with Pool Safety Standard Guidelines. The compliance of the fitting shall include of the associated drain cover, sump, and hardware.~~

~~2.6 SHOWER PAN~~

~~Shower pan may be copper, or nonmetallic material.~~

~~2.6.1 Sheet Copper~~

~~Sheet copper shall be 4.9 kg per square meter weight.~~

~~2.6.2 Plasticized Polyvinyl Chloride Shower Pan Material~~

~~Material shall be sheet form. The material shall be 1.015 mm minimum thickness of plasticized polyvinyl chloride or chlorinated polyethylene and shall be in accordance with JIS K 5981.~~

~~2.6.3 Nonplasticized Polyvinyl Chloride (PVC) Shower Pan Material~~

~~Material shall consist of a plastic waterproofing membrane in sheet form. The material shall be 1.015 mm minimum thickness of nonplasticized PVC and shall have the following minimum properties:~~

~~a. JIS K 7139~~

~~Ultimate Tensile Strength:~~	~~1.79 MPa~~
~~Ultimate Elongation:~~	~~398 percent~~
~~100 Percent Modulus:~~	~~3.07 MPa~~

~~b. JIS K 7128-3~~

~~Tear Strength:~~	~~53 kilonewtons per meter~~

~~c. JIS Z 0208 and JIS K 7129-7:~~

~~Permeance:~~	~~0.46 ng per Pa per second per sq meter~~

~~d. Other Properties:~~

~~Specific Gravity:~~	~~1.29~~
~~PVC Solvent:~~	~~Weldable~~
~~Cold Crack:~~	~~minus 47 degrees C~~
~~Dimensional stability~~	~~100 degrees C~~
~~Hardness, Shore A:~~	~~89~~

2.6 TRAPS

Unless otherwise specified, traps shall be ~~[~~plastic per JIS K 6741 and JIS K 7014~~]~~ ~~[~~or~~]~~ ~~[~~copper-alloy adjustable tube type with slip joint inlet and swivel~~]~~. Traps shall be without a cleanout.~~[~~ Provide traps with removable access panels for easy clean-out at sinks and lavatories. ~~]~~ Tubes shall be copper alloy with walls not less than 0.813 mm thick within commercial tolerances, except on the outside of bends where the thickness may be reduced slightly in manufacture by usual commercial methods. Inlets shall have rubber washer and copper alloy nuts for slip joints above the discharge level. Swivel joints shall be below the discharge level and shall be of metal-to-metal or metal-to-plastic type as required for the application. Nuts shall have flats for wrench grip. Outlets shall have internal pipe thread, except that when required for the application, the outlets shall have sockets for solder-joint connections. The depth of the water seal shall be not less than 50 mm. The interior diameter shall be not more than 3.2 mm over or under the nominal size, and interior surfaces shall be reasonably smooth throughout. A copper alloy "P" trap assembly consisting of an adjustable "P" trap and threaded trap wall nipple with cast brass wall flange shall be provided for lavatories. The assembly shall be a standard manufactured unit and may have a rubber-gasketed swivel joint.

~~2.7 INTERCEPTORS~~

~~2.7.1 Grease Interceptor~~

~~Grease interceptor of the size indicated shall be of reinforced concrete, [or precast concrete construction] [or equivalent capacity commercially available steel grease interceptor] with removable three-section, 9.5 mm checker-plate cover, and shall be installed outside the building. Steel grease interceptor shall be installed in a concrete pit and shall be epoxy-coated to resist corrosion as recommended by the manufacturer. Concrete shall have 21 MPa minimum compressive strength at 28 days. Provide flow control fitting.~~

~~2.7.2 Oil Interceptor~~

~~Cast iron or welded steel, coated inside and outside with white acid resistant epoxy, with internal air relief bypass, bronze cleanout plug, double wall trap seal, removable combination pressure equalizing and flow diffusing baffle and sediment bucket, horizontal baffle, adjustable oil draw-off and vent connections on either side, gas and watertight gasketed nonskid cover, and flow control fitting.~~

~~2.7.3 Sand Interceptors~~

~~Sand interceptor of the size indicated shall be of reinforced concrete, [or precast concrete construction] [or equivalent capacity commercially available steel sand interceptor] with manufacturer's standard checker-plate cover, and shall be installed [outside the building][top flush with the floor][floor mounted]. Steel sand interceptor shall be installed in accordance with manufacturer's recommendations and shall be coated to resist corrosion as recommended by the manufacturer.[Concrete shall have 21 MPa minimum compressive strength at 28 days.]~~

2.7 WATER HEATERS

Water heater types and capacities shall be as indicated. Each water heater shall have replaceable anodes. Each primary water heater shall have controls with an adjustable range that includes 32 to 71 degrees C. ~~Each gas-fired water heater and booster water heater shall have controls with an adjustable range that includes 49 to 82 degrees C.~~ Hot water systems utilizing recirculation systems shall be tied into building off-hour controls. The thermal efficiencies and standby heat losses shall conform to TABLE III in PART 3 of this Section for each type of water heater specified. The only exception is that storage water heaters and hot water storage tanks having more than 2000 liters storage capacity need not meet the standard loss requirement if the tank surface area is insulated to R-12.5 and if a standing light is not used. Plastic materials polyetherimide (PEI) and polyethersulfone (PES) are forbidden to be used for vent piping of combustion gases. A factory pre-charged expansion tank shall be installed on the cold water supply to each water heater. Expansion tanks shall be specifically designed for use on potable water systems and shall be rated for 93 degrees C water temperature and 1034 kPa working pressure. The expansion tank size and acceptance volume shall be ~~[_____] [~~as indicated~~]~~.

2.7.1 Automatic Storage Type

Heaters shall be complete with ~~[control system,] [~~control system, temperature gauge, and pressure gauge,~~]~~ and shall have combination pressure and temperature relief valve.

~~2.7.1.1 Oil-Fired Type~~

~~Oil-fired type water heaters shall conform to JIS S 3024 or UL 732.~~

~~2.7.1.2 Gas-Fired Type~~

~~Gas-fired water heaters shall conform to JIS S 2109 when input is 70 KW per hour or less for simultaneous heater and 42 KW per hour or less for storage type or JIS S 2116 for heaters with input greater than 42 KW per hour.~~

~~2.7.1.3 Electric Type~~

~~Electric type water heaters shall conform to JIS C 9219-1 or UL 174 with dual heating elements. Each element shall be 4.5 KW. The elements shall be wired so that only one element can operate at a time.~~

2.7.1.1 Indirect Heater Type

Steam and high temperature hot water (HTHW) heaters with storage system shall be the assembled product of one manufacturer, and~~-~~ shall be in accordance with MLIT-M or be ASME tested and "U" stamped to code requirements under ASME BPVC SEC VIII D1. The storage tank shall be as specified in paragraph HOT-WATER STORAGE TANKS. The heat exchanger shall be ~~[double wall] [~~single wall~~]~~ type that separates the potable water from the heat transfer medium with a space vented to the atmosphere.

~~a. HTHW Energy Source: The heater element shall have a working pressure of 2758 kPa with water at a temperature of 204 degrees C. The heating surface shall be based on 0.093 square meter of heating surface to heat 76 L or more of water in 1 hour from 4 to 82 degrees C using hot water at a temperature of 178 degrees C. Carbon steel heads shall be used. Tubing shall conform to JIS H 3300 or ASTM B75/B75M. Heating elements shall withstand an internal hydrostatic pressure of 4137 kPa for not less than 15 seconds without leaking or any evidence of damage.~~

~~b~~a. Steam Energy Source: The heater element shall have a working pressure of 1034 kPa per square metergauge~~ (psig)~~ with steam at a temperature of 185 degrees C. The heating surface shall be based on 0.093 square meter of heating surface to heat 76 L or more of water in 1 hour from 4 to 82 degrees C using steam at atmospheric pressure.~~[~~

Cast iron~~] [bronze]~~ heads shall be used. Tubing shall be light-drawn copper tubing conforming to JIS H 3300 or ASTM B75/B75M. Heating elements shall withstand an internal hydrostatic pressure of 1551 kPa for not less than 15 seconds without leaking or any evidence of damage.

~~2.7.2 Instantaneous Water Heater~~

~~Heater shall be crossflow design with service water in the coil and [steam] [hot water] in the shell. An integral internal controller shall be provided, anticipating a change in demand so that the final temperature can be maintained under all normal load conditions when used in conjunction with [pneumatic control system] [pilot-operated temperature control system]. Normal load conditions shall be as specified by the manufacturer for the heater. Unit shall be manufactured in accordance with MLIT-M or ASME BPVC SEC VIII D1, and shall be certified for 1.03 MPa working pressure in the shell and 1.03 MPa working pressure in the coils. Shell shall be carbon steel with copper lining. Heads shall be [cast iron] [bronze] [carbon steel plate with copper lining]. Coils shall be [copper] [copper-nickel]. Shell shall have metal sheathed fiberglass insulation, combination pressure and temperature relief valve, and thermometer. Insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.~~

2.7.2 Electric Instantaneous Water Heaters (Tankless)

JIS C 9335-2-35 or UL 499 flow switch activated, tankless electric instantaneous water heater for wall mounting below sink or lavatory.

2.7.3 Phenolic Resin Coatings for Heater Tubes

The phenolic resin coating system shall be applied at either the coil or coating manufacturer's factory in accordance with manufacturer's standard proven production process. The coating system shall be a product specifically intended for use on the material the water heating tubes/coils are made of and shall be acceptable for use in potable water systems.

~~[~~The entire exterior surface~~]~~ ~~[~~and~~]~~ ~~[~~the first 125 mm to 200 mm inside the tubes~~]~~ of each coil shall be coated with phenolic resin coating system.

2.7.3.1 Standard Product

Provide a phenolic resin coating system that is a standard product of a manufacturer regularly engaged in the manufacturing of products that are of a similar material, design and workmanship.

Standard products are defined as components and equipment that have been in satisfactory commercial or industrial use in similar applications of similar size for at least two years before bid opening.

Prior to this two year period, these standard products were sold on the commercial market using advertisements in manufacturers' catalogs or brochures. These manufacturers' catalogs, or brochures shall have been copyrighted documents or be identified with a manufacturer's document number.

2.8 HOT-WATER STORAGE TANKS

Hot-water storage tanks shall be constructed by one manufacturer, and nameplate shall indicate the working pressure. The tank shall be cement-lined or glass-lined steel type in accordance with MLIT-M ~~or AWWA D100~~. Each tank shall be equipped with a thermometer, style and form as required for the installation, and with 175 mm scale. Thermometer shall have a separable socket suitable for a 20 mm tapped opening. Tanks shall be equipped with a pressure gauge 155 mm minimum diameter face. Insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Storage tank capacity shall be as shown.

2.9 PUMPS

2.9.1 Sump Pumps

Sump pumps shall be of capacities indicated. The pumps shall be of the automatic, electric motor-driven, submerged type, complete with necessary control equipment and with a split or solid cast-iron or steel cover plate. The pumps shall be direct-connected by an approved flexible coupling to a vertical electric motor having a continuous oiling device or packed bearings sealed against dirt and moisture. Motors shall be totally enclosed, fan-cooled of sizes as indicated. The across-the-line magnetic controller shall be equipped with industrial use type enclosure. Integral size motors shall be the premium efficiency type in accordance with JIS B 8325. Each pump shall be fitted with a high-grade thrust bearing mounted above the floor. Each shaft shall have an alignment bearing at each end, and the suction inlet shall be between 75 and 150 mm above the sump bottom. The suction side of each pump shall have a strainer of ample capacity. A float switch assembly, with the switch completely enclosed in an industrial use type enclosure, shall start and stop each motor at predetermined water levels. Duplex pumps shall be equipped with an automatic alternator to change the lead operation from one pump to the other, and for starting the second pump if the flow exceeds the capacity of the first pump. The discharge line from each pump shall be provided with a union or flange, a nonclog swing check valve, and a stop valve in an accessible location near the pump.

2.9.2 Circulating Pumps

Domestic hot water circulating pumps shall be electrically driven, single-stage, centrifugal, with mechanical seals, suitable for the intended service. Pump and motor shall be ~~[integrally mounted on a cast-iron or steel subbase,] [close-coupled with an overhung impeller,] [or] [~~supported by the piping on which it is installed~~]~~. The shaft shall be one-piece, heat-treated, corrosion-resisting steel with impeller and smooth-surfaced housing of bronze.

Motor shall be totally enclosed, fan-cooled and shall have sufficient wattage for the service required. Each pump motor shall be equipped with an across-the-line magnetic controller in an industrial type enclosure with "START-STOP" switch in cover.

Integral size motors shall be premium efficiency type in accordance with JIS C 4212. Pump motors smaller than 746 W shall have integral thermal overload protection in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Guards shall shield exposed moving parts.

2.9.3 Booster Pumps

2.9.3.1 Centrifugal Pumps

Horizontal split-case centrifugal-type booster pumps shall be furnished. The capacities shall be as shown, and the speed shall not exceed 1800 rpm. Pumps shall have a casing of close-grained iron or steel with smooth water passages. A gasket shall be provided between the upper and lower halves of the casing. Suction and discharge connections shall be flanged. Impellers shall be nonoverloading, bronze, balanced to eliminate vibration, and shall be keyed to corrosion-resisting steel shafts. The casings shall be fitted with bronze wearing or sealing rings. Bearings shall be cartridge type, enabling the entire rotating element to be removed without disturbing alignment or exposing the bearings to dirt, water, and other foreign matter. Pumps shall be provided with mechanical seals. Seal boxes shall be machined in the pump casing and at both sides of the pump, and shall be of sufficient depth to include a conventional bronze seal ring and rows of shaft packing. Bedplates shall be close-grain cast iron or steel with ribs and lugs, complete with foundation bolts, and shall have a drip lip with drain hole. Each pump shall be tested at the manufacturer's plant for operating characteristics at the rated capacity and under specified operating conditions. Test curves shall be furnished showing capacity in liters per second, head in meters, efficiency, brake wattage, and operation in parallel with similar pumps. Multiple pump installations shall have pump characteristics compatible for operation in parallel with similar pumps. The electric motor shall be sized for non-overload when operating at any point along the characteristic curve of the pump. Guards shall shield exposed belts and moving parts.

2.9.3.2 Controls

Each pump motor shall be provided with enclosed across-the-line-type magnetic controller complete in an industrial use type enclosure with three position, "HAND-OFF-AUTOMATIC," selector switch in cover. Pumps shall be automatically started and stopped by float or pressure switches, as indicated. The pumps shall start and stop at the levels and pressures indicated. A multiposition sequence selector switch shall be provided so that any two pumps may be operated simultaneously keeping a third pump as a standby.

2.9.4 Flexible Connectors

Flexible connectors shall be provided at the suction and discharge of each pump that is 1 hp or larger. Connectors shall be constructed of neoprene, rubber, or braided bronze, with Class 150 standard flanges. Flexible connectors shall be line size and suitable for the pressure and temperature of the intended service.

~~2.9.5 Sewage Pumps~~

~~Provide single type duplex type with automatic controls to alternate the operation from one pump to the other pump and to start the second pump in the event the first pump cannot handle the incoming flow. Provide high water alarm and check valve.~~

2.10 WATER PRESSURE BOOSTER SYSTEM

~~2.10.1 Constant Speed Pumping System~~

~~Constant speed pumping system with pressure-regulating valves shall employ one lead pump for low flows, and one or more lag pumps for higher flows. Pressure-regulating valves shall be provided with nonslam check feature. The factory prepiped and prewired assembly shall be mounted on a steel frame, complete with pumps, motors, and automatic controls. The system capacity and capacity of individual pumps shall be as indicated. Current sensing relays shall provide staging of the pumps. The pumps shall be protected from thermal buildup, when running at no-flow, by a common thermal relief valve. Pressure gauges shall be mounted on the suction and discharge headers. The control panel shall be industrial use type enclosure. The control panel shall include the following: No-flow shutdown; 7-day time clock; audiovisual alarm; external resets; manual alternation; magnetic motor controllers; time delays; transformer; current relays; "HAND-OFF-AUTOMATIC" switches for each pump; minimum run timers; low suction pressure cutout; and indicating lights for power on, individual motor overload, and low suction pressure. The control circuit shall be interlocked so that the failure of any controller shall energize the succeeding controller.~~

~~2.10.2 Hydro-Pneumatic Water Pressure System~~

~~Hydro-pneumatic water tank shall be constructed in accordance with pressure vessel construction standard (atsuryoku-youki-kouzou-kisoku).~~

~~2.10.3 Variable Speed Pumping System~~

~~Variable speed pumping system shall provide system pressure by varying speed and number of operating pumps. The factory prepiped and prewired assembly shall be mounted on a steel frame complete with pumps, variable speed drives, motors, and controls. The variable speed drives shall be the oil-filled type capable of power transmission throughout their complete speed range without vibration, noise, or shock loading. Each variable speed drive shall be run-tested by the manufacturer for rated performance, and the manufacturer shall furnish written performance certification. System shall have suppressors to prevent noise transmission over electric feed lines. Required electrical control circuitry and system function sensors shall be supplied by the variable speed drive manufacturer. The primary power controls and magnetic motor controllers shall be installed in [the controls supplied by the drive manufacturer] [the motor control center]. The sensors shall be located in the system to control drive speed as a function of [constant pump discharge pressure] [constant system pressure at location indicated]. Connection between the sensors and the variable speed drive controls shall be accomplished with [hydraulic sensing lines] [copper wiring] [telemetry]. Controls shall be in industrial use type enclosures.~~

2.10.1 Domestic Water Storage Tanks

Modular water tank for domestic water storage up to 40‹C consisting of insulated panels made from glass fibre reinforced polyester (GRP), also sometimes referred to as fiberglass reinforced polymer (FRP). Gasket sealed manhole panel having an opening of 600 mm in its center. Manhole opening shall raise 100 mm from the roof panel surface to prevent water and other contaminants from entering the tank. Gaskets shall meet NSF/ANSI 61 requirements. Corrosion-resistant galvanized steel or SUS304 stainless steel assembly bolts. ABS air vents with insect screen and PVC electrode seats and covers. Internal ladder of PVC-U and external ladder of hot-dip galvanized steel. The frame shall meet hot-dip galvanized specification. Tank shall conform to water supply laws, food sanitation laws, and algae growth prevention technical guidelines. Tank shall meet earthquake resistance standards and have a horizontal seismic coefficient of 1.5 G and wind load of 60 m/s. The foundation shall be continuous with a level surface and have adequate strength.

~~2.11 COMPRESSED AIR SYSTEM~~

~~2.11.1 Air Compressors~~

~~Air compressor unit shall be a factory-packaged assembly, including [_____] phase, [_____] volt motor controls, switches, wiring, accessories, and motor controllers, in an industrial use type enclosure. Air compressors shall have manufacturer's name and address, together with trade name, and catalog number on a nameplate securely attached to the equipment. Each compressor shall [start and stop automatically at upper and lower pressure limits of the system] [regulate pressure by constant speed compressor loading and unloading] [have a manual-off-automatic switch that when in the manual position, the compressor loads and unloads to meet the demand and, in the automatic position, a time delay relay shall allow the compressor to operate for an adjustable length of time unloaded, then stop the unit]. Guards shall shield exposed moving parts. Each duplex compressor system shall be provided with [automatic] [manual] alternation system. Each compressor motor shall be provided with an across-the-line-type magnetic controller, complete with low-voltage release. An intake air filter and silencer shall be provided with each compressor. Aftercooler and moisture separator shall be installed between compressors and air receiver to remove moisture and oil condensate before the air enters the receiver. Aftercoolers shall be either air- or water-cooled, as indicated. The air shall pass through a sufficient number of tubes to affect cooling. Tubes shall be sized to give maximum heat transfer. Water to unit shall be controlled by a solenoid or pneumatic valve, which opens when the compressors start and closes when the compressors shut down. Cooling capacity of the aftercooler shall be sized for the total capacity of the compressors. Means shall be provided for draining condensed moisture from the receiver by an automatic float type trap. Capacities of air compressors and receivers shall be as indicated.~~

~~2.11.2 Lubricated Compressors~~

~~Compressors shall be two-stage, V-belt drive, capable of operating continuously against their designed discharge pressure, and shall operate at a speed not in excess of 1800 rpm. Compressors shall have the capacity and discharge pressure indicated. Compressors shall be assembled complete on a common subbase. The compressor main bearings shall be either roller or ball. The discharge passage of the high pressure air shall be piped to the air receiver with a copper pipe or tubing. A pressure gauge calibrated to 1.03 MPa and equipped with a gauge cock and pulsation dampener shall be furnished for installation adjacent to pressure switches.~~

~~2.11.3 Air Receivers~~

~~Receivers shall be designed for 1.38 MPa working pressure. Receivers shall be factory air tested to 1-1/2 times the working pressure. Receivers shall be equipped with safety relief valves and accessories,~~

~~including pressure gauges and automatic and manual drains. The outside of air receivers may be galvanized or supplied with commercial enamel finish. Receivers shall be designed and constructed in accordance with Contruction Code for Pressure Vessel (CCPV) and shall have the design working pressures specified herein.~~

~~2.11.4 Intake Air Supply Filter~~

~~Dry type air filter shall be provided having a collection efficiency of 99 percent of particles larger than 10 microns. Filter body and media shall withstand a maximum 862 kPa, capacity as indicated.~~

~~2.11.5 Pressure Regulators~~

~~The air system shall be provided with the necessary regulator valves to maintain the desired pressure for the installed equipment. Regulators shall be designed for a maximum inlet pressure of 862 kPa and a maximum temperature of 93 degrees C. Regulators shall be single-seated, pilot-operated with valve plug, bronze body and trim or equal, and threaded connections. The regulator valve shall include a pressure gauge and shall be provided with an adjustment screw for adjusting the pressure differential from 0 kPa to 862 kPa. Regulator shall be sized as indicated.~~

~~2.12 DOMESTIC WATER SERVICE METER~~

~~[The requirements for metering and submetering are specified in Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.~~

~~][Cold water meters 50 mm and smaller shall be positive displacement type conforming to JIS B 8570-1 or AWWA C700. Cold water meters 64 mm and larger shall be turbine type conforming to JIS B 8570-1 or AWWA C701. Meter register may be round or straight reading type, [indicating [_____]] [as provided by the local utility]. Meter shall be provided with a pulse generator, remote readout register and all necessary wiring and accessories.~~

~~Meters must be connected to the base wide energy and utility monitoring and control system (if this system exists) using the installation's advanced metering protocols.~~

~~]~~2.11 ELECTRICAL WORK

Provide electrical motor driven equipment specified complete with motors, motor starters, and controls as specified herein and in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide ~~[~~high efficiency type, ~~]~~single-phase, fractional-horsepower alternating-current motors, including motors that are part of a system, corresponding to the applications in accordance with JIS C 4212.~~[~~ In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide polyphase, squirrel-cage medium induction motors with continuous ratings, including motors that are part of a system, that meet the efficiency ratings for premium efficiency motors in accordance with JIS C 4212.~~]~~ Provide motors in accordance with JIS C 4212 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor.

Motors shall be rated for continuous duty with the enclosure specified. Motor duty requirements shall allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque shall be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Motor bearings shall be fitted with grease supply fittings and grease relief to outside of the enclosure.

Controllers and contactors shall have auxiliary contacts for use with the controls provided. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, shall be provided. For packaged equipment, the manufacturer shall provide controllers, including the required monitors and timed restart.

Power wiring and conduit for field installed equipment shall be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.12 MISCELLANEOUS PIPING ITEMS

2.12.1 Escutcheon Plates

Provide one piece or split hinge metal plates for piping entering floors, walls, and ceilings in exposed spaces. Provide chromium-plated on copper alloy plates or polished stainless steel finish in finished spaces. Provide paint finish on plates in unfinished spaces.

2.12.2 Pipe Sleeves

Provide where piping passes entirely through walls, ceilings, roofs, and floors. Sleeves are not required where ~~[supply]~~ drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade, except where penetrating a membrane waterproof floor.

2.12.2.1 Sleeves in Masonry and Concrete

Provide steel pipe sleeves or schedule 40 PVC plastic pipe sleeves. Sleeves are not required where drain, waste, and vent (DWV) piping passes through concrete floor slabs located on grade. Do not core drill masonry or concrete.

2.12.2.2 Sleeves Not in Masonry and Concrete

Provide 26 gage galvanized steel sheet or PVC plastic pipe sleeves.

2.12.3 Pipe Hangers (Supports)

Provide Pipe hangers and supports in accordance with MLIT-M.

2.12.4 Nameplates

Provide 3.2 mm thick melamine laminated plastic nameplates, black matte ~~finish with white center core, for equipment, gages, thermometers, and~~ valves; valves in supplies to faucets will not require nameplates. Accurately align lettering and engrave minimum of 6.4 mm high normal block lettering into the white core. Minimum size of nameplates shall be 25 by 63 mm. Key nameplates to a chart and schedule for each system. Frame charts and schedules under glass and place where directed near each system. Furnish two copies of each chart and schedule.

2.12.5 Remote Fuel Fill Station

Provide a factory assembled wall mounted manual fuel fill station. The cabinet shall be constructed of stainless steel. The fuel fill station shall include all valves and fittings necessary for hose connections from a pumper truck. The station shall be provided with lockable weatherproof spill containment box. Provide 40 mm inlet with quick detachable coupling with check valve, shutoff ball valve. Outlet fitting shall be 40 mm threaded connection.

~~2.12.6 Labels~~

~~Provide labels for sensor operators at flush valves and faucets. Include the following information on each label:~~

~~a. Identification of the sensor and its operation with [graphic] [written] [Braille] description.~~

~~b. Range of the sensor.~~

~~c. Battery replacement schedule.~~

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Piping located in air plenums shall conform to NFPA 90A. Piping located in shafts that constitute air ducts or that enclose air ducts shall be noncombustible in accordance with NFPA 90A. Installation of plastic pipe where in compliance with NFPA may be installed in accordance with PPFA Fire Man. The plumbing system shall be installed complete with necessary fixtures, fittings, traps, valves, and accessories. Water and drainage piping shall be extended 1.5 m outside the building, unless otherwise indicated. A ~~[gate valve] [~~full port ball valve~~] [ball valve]~~ and drain shall be installed on the water service line inside the building approximately 150 mm above the floor from point of entry. Piping shall be connected to the exterior service lines or capped or plugged if the exterior service is not in place. Sewer and water pipes shall be laid in separate trenches, except when otherwise shown. Exterior underground utilities shall be at least 300 mm below the ~~[~~average local frost depth~~] [finish grade]~~ or as indicated on the drawings. If trenches are closed or the pipes are otherwise covered before being connected to the service lines, the location of the end of each plumbing utility shall be marked with a stake or other acceptable means. Valves shall be installed with control no lower than the valve body.

3.1.1 Water Pipe, Fittings, and Connections

3.1.1.1 Utilities

The piping shall be extended to fixtures, outlets, and equipment. The hot-water and cold-water piping system shall be arranged and installed to permit draining. The supply line to each item of equipment or fixture, except faucets, flush valves, or other control valves which are supplied with integral stops, shall be equipped with a shutoff valve to enable isolation of the item for repair and maintenance without interfering with operation of other equipment or fixtures. Supply piping to fixtures, faucets, hydrants, shower heads, and flushing devices shall be anchored to prevent movement.

~~3.1.1.2 Cutting and Repairing~~

~~The work shall be carefully laid out in advance, and unnecessary cutting of construction shall be avoided. Damage to building, piping, wiring, or equipment as a result of cutting shall be repaired by mechanics skilled in the trade involved.~~

3.1.1.2 Protection of Fixtures, Materials, and Equipment

Pipe openings shall be closed with caps or plugs during installation. Fixtures and equipment shall be tightly covered and protected against dirt, water, chemicals, and mechanical injury. Upon completion of the work, the fixtures, materials, and equipment shall be thoroughly cleaned, adjusted, and operated. Safety guards shall be provided for exposed rotating equipment.

3.1.1.3 Mains, Branches, and Runouts

Piping shall be installed as indicated. Pipe shall be accurately cut and worked into place without springing or forcing. Structural portions of the building shall not be weakened. Aboveground piping shall run parallel with the lines of the building, unless otherwise indicated. Branch pipes from service lines may be taken from top, bottom, or side of main, using crossover fittings required by structural or installation conditions. Supply pipes, valves, and fittings shall be kept a sufficient distance from other work and other services to permit not less than 12 mm between finished covering on the different services. Bare and insulated water lines shall not bear directly against building structural elements so as to transmit sound to the structure or to prevent flexible movement of the lines. Water pipe shall not be buried in or under floors unless specifically indicated or approved. Changes in pipe sizes shall be made with reducing fittings. Use of bushings will not be permitted except for use in situations in which standard factory fabricated components are furnished to accommodate specific accepted installation practice. Change in direction shall be made with fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. The center-line radius of bends shall be not less than six diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be acceptable.

3.1.1.4 Pipe Drains

Pipe drains indicated shall consist of 20 mm hose bibb with renewable seat and ~~[gate] [~~full port ball~~] [ball]~~ valve ahead of hose bibb. At other low points, 20 mm brass plugs or caps shall be provided. Disconnection of the supply piping at the fixture is an acceptable drain.

3.1.1.5 Expansion and Contraction of Piping

Allowance shall be made throughout for expansion and contraction of water pipe. Each hot-water and hot-water circulation riser shall have expansion loops or other provisions such as offsets and changes in direction where indicated and required. Risers shall be securely anchored as required or where indicated to force expansion to loops. Branch connections from risers shall be made with ample swing or offset to avoid undue strain on fittings or short pipe lengths. Horizontal runs of pipe over 15 m in length shall be anchored to the wall or the supporting construction about midway on the run to force expansion, evenly divided, toward the ends.

Sufficient flexibility shall be provided on branch runouts from mains and risers to provide for expansion and contraction of piping. Flexibility shall be provided by installing one or more turns in the line so that piping will spring enough to allow for expansion without straining. If mechanical grooved pipe coupling systems are provided, the deviation from design requirements for expansion and contraction may be allowed pending approval of Contracting Officer.

3.1.1.6 Thrust Restraint

Plugs, caps, tees, valves and bends deflecting 11.25 degrees or more, either vertically or horizontally, in waterlines 100 mm in diameter or larger shall be provided with thrust blocks, where indicated, to prevent movement. Thrust blocking shall be concrete of a mix not leaner than: 1 cement, 2-1/2 sand, 5 gravel; and having a compressive strength of not less than 14 MPa after 28 days. Blocking shall be placed between solid ground and the fitting to be anchored. Unless otherwise indicated or directed, the base and thrust bearing sides of the thrust block shall be poured against undisturbed earth. The side of the thrust block not subject to thrust shall be poured against forms. The area of bearing will be as shown. Blocking shall be placed so that the joints of the fitting are accessible for repair. Steel rods and clamps, protected by galvanizing or by coating with bituminous paint, shall be used to anchor vertical down bends into gravity thrust blocks.

3.1.1.7 Commercial-Type Water Hammer Arresters

Commercial-type water hammer arresters shall be provided on hot- and cold-water supplies and shall be located as generally indicated, with precise location and sizing to be in accordance with PDI WH 201 or the Performance Standards Compliant Product (PSCP) of the Water Supply Law. Water hammer arresters, where concealed, shall be accessible by means of access doors or removable panels. Commercial-type water hammer arresters shall conform to JIS S 3200-3~~ or ASSE 1010~~. Vertical capped pipe columns will not be permitted.

~~3.1.2 Compressed Air Piping (Non-Oil Free)~~

~~Compressed air piping shall be installed as specified for water piping and suitable for 862 kPa working pressure. Compressed air piping shall have supply lines and discharge terminals legibly and permanently marked at both ends with the name of the system and the direction of flow.~~

3.1.2 Joints

Installation of pipe and fittings shall be made in accordance with the manufacturer's recommendations. Mitering of joints for elbows and notching of straight runs of pipe for tees will not be permitted. Joints shall be made up with fittings of compatible material and made for the specific purpose intended.

3.1.2.1 Threaded

Threaded joints shall have American Standard taper pipe threads conforming to JIS B 0203. Only male pipe threads shall be coated with graphite or with an approved graphite compound, or with an inert filler and oil, or shall have a polytetrafluoroethylene tape applied.

3.1.2.2 Mechanical Couplings

Mechanical couplings may be used in conjunction with grooved pipe for aboveground, ferrous or non-ferrous, domestic hot and cold water systems, in lieu of unions, brazed, soldered, welded, flanged, or threaded joints.

Mechanical couplings are permitted in accessible locations including behind access plates. Flexible grooved joints will not be permitted, except as vibration isolators adjacent to mechanical equipment. Rigid grooved joints shall incorporate an angle bolt pad design which maintains metal-to-metal contact with equal amount of pad offset of housings upon installation to ensure positive rigid clamping of the pipe.

Designs which can only clamp on the bottom of the groove or which utilize gripping teeth or jaws, or which use misaligned housing bolt holes, or which require a torque wrench or torque specifications will not be permitted.

Grooved fittings and couplings, and grooving tools shall be provided from the same manufacturer. Segmentally welded elbows shall not be used. Grooves shall be prepared in accordance with the coupling manufacturer's latest published standards. Grooving shall be performed by qualified grooving operators having demonstrated proper grooving procedures in accordance with the tool manufacturer's recommendations.

The Contracting Officer shall be notified 24 hours in advance of test to demonstrate operator's capability, and the test shall be performed at the work site, if practical, or at a site agreed upon. The operator shall demonstrate the ability to properly adjust the grooving tool, groove the pipe, and to verify the groove dimensions in accordance with the coupling manufacturer's specifications.

3.1.2.3 Unions and Flanges

Unions, flanges and mechanical couplings shall not be concealed in walls, ceilings, or partitions. Unions shall be used on pipe sizes 65 mm and smaller; flanges shall be used on pipe sizes 80 mm and larger.

~~3.1.2.4 Grooved Mechanical Joints~~

~~Grooves shall be prepared according to the coupling manufacturer's instructions. Grooved fittings, couplings, and grooving tools shall be products of the same manufacturer. Pipe and groove dimensions shall comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field shall be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe shall be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances. Grooved joints shall not be used in concealed locations.~~

~~3.1.2.5 Cast Iron Soil, Waste and Vent Pipe~~

~~Bell and spigot compression and hubless gasketed clamp joints for soil, waste and vent piping shall be installed per the manufacturer's recommendations.~~

3.1.2.4 Copper Tube and Pipe

a. Brazed. Brazed joints shall be made in conformance with JIS Z 3621, JIS H 3300, and JIS H 3300 with flux and are acceptable for all pipe sizes. Copper to copper joints shall include the use of copper-phosphorus or copper-phosphorus-silver brazing metal without flux. Brazing of dissimilar metals (copper to bronze or brass) shall include the use of flux with either a copper-phosphorus, copper-phosphorus-silver or a silver brazing filler metal.

b. Soldered. Soldered joints shall be made with flux and are only acceptable for piping 50 mm and smaller. Soldered joints shall conform to JIS H 3300 and JIS H 3300. Soldered joints shall not be used in compressed air piping between the air compressor and the receiver.

c. Copper Tube Extracted Joint. Mechanically extracted joints shall be made in accordance with MLIT-M.

~~d. Press connection. Copper press connections shall be made in **strict** accordance with the manufacturer's installation instructions for manufactured rated size. The joints shall be pressed using the tool(s) approved by the manufacturer **of that joint**. Minimum distance between fittings shall be in accordance with the manufacturer's requirements.~~

3.1.2.5 Plastic Pipe

Acrylonitrile-Butadiene-Styrene (ABS) pipe shall have joints made with solvent cement. PVC and CPVC pipe shall have joints made with solvent cement elastomeric, threading, (threading of Schedule 80 Pipe is allowed only where required for disconnection and inspection; threading of Schedule 40 Pipe is not allowed), or mated flanged.

~~3.1.2.6 Glass Pipe~~

~~Joints for corrosive waste glass pipe and fittings shall be made with corrosion-resisting steel compression-type couplings with acrylonitrile rubber gaskets lined with polytetrafluoroethylene.~~

3.1.2.6 Corrosive Waste Plastic Pipe

Joints for polyolefin pipe and fittings shall be made by mechanical joint or electrical fusion coil method. Joints for filament-wound reinforced thermosetting resin pipe shall be made in accordance with manufacturer's instructions. Unions or flanges shall be used where required for disconnection and inspection.

~~3.1.2.7 Other Joint Methods~~

3.1.3 Dissimilar Pipe Materials

Connections between ferrous and non-ferrous copper water pipe shall be made with dielectric unions or flange waterways. Dielectric waterways shall have temperature and pressure rating equal to or greater than that specified for the connecting piping. Waterways shall have metal connections on both ends suited to match connecting piping. Dielectric waterways shall be internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Dielectric flanges shall meet the performance requirements described herein for dielectric waterways. Connecting joints between plastic and metallic pipe shall be made with transition fitting for the specific purpose.

~~3.1.4 Corrosion Protection for Buried Pipe and Fittings~~

~~Ductile iron, cast iron, and steel pipe, fittings, and joints shall have a protective coating. Additionally, ductile iron, cast iron, and steel pressure pipe shall have a cathodic protection system and joint bonding. The cathodic protection system, protective coating system, and joint bonding for cathodically protected pipe shall be in accordance with [Section 26 42 14.00 10 CATHODIC PROTECTION SYSTEM (SACRIFICIAL ANODE)] [and] [Section 26 42 17.00 10 CATHODIC PROTECTION SYSTEM (IMPRESSED CURRENT)][Section 26 42 13.00 20 CATHODIC PROTECTION BY GALVANIC ANODES] [and] [Section 26 42 19.00 20 CATHODIC PROTECTION BY IMPRESSED CURRENT] [Section 26 42 14.00 10 CATHODIC PROTECTION SYSTEM (SACRIFICIAL ANODE)][Section 26 42 19.00 20 CATHODIC PROTECTION BY IMPRESSED CURRENT]. and as otherwise specified. The pipe shall be cleaned and the coating system applied prior to pipe tightness testing. Joints and fittings shall be cleaned and the coating system applied after pipe tightness testing. For tape coating systems, the tape shall conform to JWWA K 115 and shall be applied with a 50 percent overlap. Primer utilized with tape type coating systems shall be as recommended by the tape manufacturer.~~

3.1.4 Pipe Sleeves and Flashing

Pipe sleeves shall be furnished and set in their proper and permanent location.

3.1.4.1 Sleeve Requirements

Unless indicated otherwise, provide pipe sleeves meeting the following requirements:

Secure sleeves in position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, ceilings, roofs, and floors.

A modular mechanical type sealing assembly may be installed in lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve. The seals shall consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe and sleeve using galvanized steel bolts, nuts, and pressure plates. The links shall be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tightening of the bolt shall cause the rubber sealing elements to expand and provide a watertight seal between the pipe and the sleeve. Each seal assembly shall be sized as recommended by the manufacturer to fit the pipe and sleeve involved.

Sleeves shall not be installed in structural members, except where indicated or approved. Rectangular and square openings shall be as detailed. Each sleeve shall extend through its respective floor, or roof, and shall be cut flush with each surface, except for special circumstances. Pipe sleeves passing through floors in wet areas such as mechanical equipment rooms, lavatories, kitchens, and other plumbing fixture areas shall extend a minimum of 100 mm above the finished floor.

Unless otherwise indicated, sleeves shall be of a size to provide a minimum of~~[6 mm][~~ 25 mm ~~]~~ clearance between bare pipe or insulation and inside of sleeve or between insulation and inside of sleeve. Sleeves in bearing walls and concrete slab on grade floors shall be steel pipe or cast-iron pipe. Sleeves in nonbearing walls or ceilings may be steel pipe, cast-iron pipe, galvanized sheet metal with lock-type longitudinal seam, or plastic.

Except as otherwise specified, the annular space between pipe and sleeve, or between jacket over insulation and sleeve, shall be sealed as indicated with sealants conforming to JIS A 5758 and with a primer, backstop material and surface preparation as specified in Section 07 92 00 JOINT SEALANTS. The annular space between pipe and sleeve, between bare insulation and sleeve or between jacket over insulation and sleeve shall not be sealed for interior walls which are not designated as fire rated.

Sleeves through below-grade walls in contact with earth shall be recessed 12 mm from wall surfaces on both sides. Annular space between pipe and sleeve shall be filled with backing material and sealants in the joint between the pipe and ~~[~~concrete~~] [masonry]~~ wall as specified above. Sealant selected for the earth side of the wall shall be compatible with dampproofing/waterproofing materials that are to be applied over the joint sealant. Pipe sleeves in fire-rated walls shall conform to the requirements in Section 07 84 00 FIRESTOPPING.

3.1.4.2 Flashing Requirements

Pipes passing through roof shall be installed through a 4.9 kg per square meter copper flashing, each within an integral skirt or flange. Flashing shall be suitably formed, and the skirt or flange shall extend not less than 200 mm from the pipe and shall be set over the roof or floor membrane in a solid coating of bituminous cement. The flashing shall extend up the pipe a minimum of 250 mm. For cleanouts, the flashing shall be turned down into the hub and caulked after placing the ferrule. Pipes passing through pitched roofs shall be flashed, using lead or copper flashing, with an adjustable integral flange of adequate size to extend not less than 200 mm from the pipe in all directions and lapped into the roofing to provide a watertight seal. The annular space between the flashing and the bare pipe or between the flashing and the metal-jacket-covered insulation shall be sealed as indicated. Flashing for dry vents shall be turned down into the pipe to form a waterproof joint. Pipes, up to and including 250 mm in diameter, passing through roof or floor waterproofing membrane may be installed through a cast-iron sleeve with caulking recess, anchor lugs, flashing-clamp device, and pressure ring with brass bolts. Flashing shield shall be fitted into the sleeve clamping device. Pipes passing through wall waterproofing membrane shall be sleeved as described above. A waterproofing clamping flange shall be installed.

3.1.4.3 Waterproofing

Waterproofing at floor-mounted water closets shall be accomplished by forming a flashing guard from soft-tempered sheet copper. The center of the sheet shall be perforated and turned down approximately 40 mm to fit between the outside diameter of the drainpipe and the inside diameter of the cast-iron or steel pipe sleeve. The turned-down portion of the flashing guard shall be embedded in sealant to a depth of approximately 40 mm; then the sealant shall be finished off flush to floor level between the flashing guard and drainpipe. The flashing guard of sheet copper shall extend not less than 200 mm from the drainpipe and shall be lapped between the floor membrane in a solid coating of bituminous cement. If cast-iron water closet floor flanges are used, the space between the pipe sleeve and drainpipe shall be sealed with sealant and the flashing guard shall be upturned approximately 40 mm to fit the outside diameter of the drainpipe and the inside diameter of the water closet floor flange. The upturned portion of the sheet fitted into the floor flange shall be sealed.

3.1.4.4 Optional Counterflashing

Instead of turning the flashing down into a dry vent pipe, or caulking and sealing the annular space between the pipe and flashing or metal-jacket-covered insulation and flashing, counterflashing may be accomplished by utilizing the following:

a. A standard roof coupling for threaded pipe up to 150 mm in diameter.

b. A tack-welded or banded-metal rain shield around the pipe.

3.1.4.5 Pipe Penetrations of Slab on Grade Floors

Where pipes, fixture drains, floor drains, cleanouts or similar items penetrate slab on grade floors, except at penetrations of floors with waterproofing membrane as specified in paragraphs FLASHING REQUIREMENTS and WATERPROOFING, a groove 6 to 13 mm wide by 6 to 10 mm deep shall be formed around the pipe, fitting or drain. The groove shall be filled with a sealant as specified in Section 07 92 00 JOINT SEALANTS.

3.1.4.6 Pipe Penetrations

Provide sealants for all pipe penetrations. All pipe penetrations shall be sealed to prevent infiltration of air, insects, and vermin.

3.1.5 Fire Seal

Where pipes pass through fire walls, fire-partitions, fire-rated pipe chase walls or floors above grade, a fire seal shall be provided as specified in Section 07 84 00 FIRESTOPPING.

3.1.6 Supports

3.1.6.1 General

Hangers used to support piping 50 mm and larger shall be fabricated to permit adequate adjustment after erection while still supporting the load. Pipe guides and anchors shall be installed to keep pipes in accurate alignment, to direct the expansion movement, and to prevent buckling, swaying, and undue strain. Piping subjected to vertical movement when operating temperatures exceed ambient temperatures shall be supported by variable spring hangers and supports or by constant support hangers. In the support of multiple pipe runs on a common base member, a clip or clamp shall be used where each pipe crosses the base support member. Spacing of the base support members shall not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run. Threaded sections of rods shall not be formed or bent.

3.1.6.2 Pipe Supports and Structural Bracing, Seismic Requirements

Piping and attached valves shall be supported and braced to resist seismic loads as specified in Section ~~13 48 00 [SEISMIC] BRACING FOR MISCELLANEOUS~~

~~EQUIPMENT~~13 48 73 SEISMIC CONTROL FOR MISCELLANEOUS EQUIPMENT and ~~[~~Section 23 05 48.19 ~~[~~SEISMIC~~]~~ BRACING FOR HVAC~~][Section 22 05 48.00 20 MECHANICAL SOUND, VIBRATION, AND SEISMIC CONTROL] [as shown]~~. Structural steel required for reinforcement to properly support piping, headers, and equipment, but not shown, shall be provided.~~ Material used for supports shall be as specified in[Section 05 12 00 STRUCTURAL STEEL][Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS][Section 05 51 33 METAL LADDERS][Section 05 52 00 METAL RAILINGS][Section 05 51 00 METAL STAIRS].~~

3.1.6.3 Pipe Hangers, Inserts, and Supports

Installation of pipe hangers, inserts and supports shall conform to MLIT-M except as modified herein.

a. Horizontal pipe supports shall be spaced as specified in MLIT-M and a support shall be installed not over 300 mm from the pipe fitting joint at each change in direction of the piping. Pipe supports shall be spaced not over 1.5 m apart at valves. Operating temperatures in determining hanger spacing for PVC or CPVC pipe shall be 49 degrees C for PVC and 82 degrees C for CPVC. Horizontal pipe runs shall include allowances for expansion and contraction.

b. Vertical pipe shall be supported at each floor, except at slab-on-grade, at intervals of not more than 4.5 m nor more than 2 m from end of risers, and at vent terminations. Vertical pipe risers shall include allowances for expansion and contraction.

3.1.6.4 Structural Attachments

Attachment to building structure concrete and masonry shall be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors shall be applied with a safety factor not less than 5. Supports shall not be attached to metal decking. Supports shall not be attached to the underside of concrete filled floor or concrete roof decks unless approved by the Contracting Officer. Masonry anchors for overhead applications shall be constructed of ferrous materials only.

3.1.7 Welded Installation

Plumbing pipe weldments shall be as indicated. Changes in direction of piping shall be made with welding fittings only; mitering or notching pipe to form elbows and tees or other similar type construction will not be permitted. Branch connection may be made with either welding tees or forged branch outlet fittings. Branch outlet fittings shall be forged, flared for improvement of flow where attached to the run, and reinforced against external strains. Beveling, alignment, heat treatment, and inspection of weld shall conform to JIS B 8285. Weld defects shall be removed and repairs made to the weld, or the weld joints shall be entirely removed and rewelded. After filler metal has been removed from its original package, it shall be protected or stored so that its characteristics or welding properties are not affected. Electrodes that have been wetted or that have lost any of their coating shall not be used.

3.1.8 Pipe Cleanouts

Pipe cleanouts shall be the same size as the pipe except that cleanout plugs larger than 100 mm will not be required. A cleanout installed in connection with cast-iron soil pipe shall consist of a long-sweep 1/4 bend or one or two 1/8 bends extended to the place shown. An extra-heavy cast-brass or cast-iron ferrule with countersunk cast-brass head screw plug shall be caulked into the hub of the fitting and shall be flush with the floor. Cleanouts in connection with other pipe, where indicated, shall be T-pattern, 90-degree branch drainage fittings with cast-brass screw plugs, except plastic plugs shall be installed in plastic pipe. Plugs shall be the same size as the pipe up to and including 100 mm. Cleanout tee branches with screw plug shall be installed at the foot of soil and waste stacks, at the foot of interior downspouts, on each connection to building storm drain where interior downspouts are indicated, and on each building drain outside the building. Cleanout tee branches may be omitted on stacks in single story buildings with slab-on-grade construction or where less than 450 mm of crawl space is provided under the floor. Cleanouts on pipe concealed in partitions shall be provided with chromium plated bronze, nickel bronze, nickel brass or stainless steel flush type access cover plates. Round access covers shall be provided and secured to plugs with securing screw. Square access covers may be provided with matching frames, anchoring lugs and cover screws. Cleanouts in finished walls shall have access covers and frames installed flush with the finished wall. Cleanouts installed in finished floors subject to foot traffic shall be provided with a chrome-plated cast brass, nickel brass, or nickel bronze cover secured to the plug or cover frame and set flush with the finished floor. Heads of fastening screws shall not project above the cover surface. Where cleanouts are provided with adjustable heads, the heads shall be ~~[~~cast iron~~] [or] [plastic]~~.

3.2 WATER HEATERS AND HOT WATER STORAGE TANKS

3.2.1 Relief Valves

No valves shall be installed between a relief valve and its water heater or storage tank. The P&T relief valve shall be installed where the valve actuator comes in contact with the hottest water in the heater. Whenever possible, the relief valve shall be installed directly in a tapping in the tank or heater; otherwise, the P&T valve shall be installed in the hot-water outlet piping. A vacuum relief valve shall be provided on the cold water supply line to the hot-water storage tank or water heater and mounted above and within 150 mm above the top of the tank or water heater.

~~3.2.2 Installation of Gas- and Oil-Fired Water Heater~~

~~Installation shall conform to JIS S 2120, and JIS S 2135, for gas fired and JIS S 3024, JIS S 3021 and JIS S 3030 for oil fired. Storage water heaters that are not equipped with integral heat traps and having vertical pipe risers shall be installed with heat traps directly on both the inlet and outlet. Circulating systems need not have heat traps installed. An acceptable heat trap may be a piping arrangement such as elbows connected so that the inlet and outlet piping make vertically upward runs of not less than 600 mm just before turning downward or directly horizontal into the water heater's inlet and outlet fittings. Commercially available heat traps, specifically designed by the manufacturer for the purpose of effectively restricting the natural tendency of hot water to rise through vertical inlet and outlet piping during standby periods may also be approved.~~

3.2.2 Heat Traps

Piping to and from each water heater and hot water storage tank shall be routed horizontally and downward a minimum of 600 mm before turning in an upward direction.

3.2.3 Connections to Water Heaters

Connections of metallic pipe to water heaters shall be made with dielectric unions or flanges.

3.2.4 Expansion Tank

A pre-charged expansion tank shall be installed on the cold water supply between the water heater inlet and the cold water supply shut-off valve. The Contractor shall adjust the expansion tank air pressure, as recommended by the tank manufacturer, to match incoming water pressure.

~~3.2.5 Direct Fired and Domestic Water Heaters~~

~~Notify the Contracting Officer when any direct fired domestic water heater over 117 kw is operational and ready to be inspected and certified.~~

3.3 FIXTURES AND FIXTURE TRIMMINGS

Polished chromium-plated pipe, valves, and fittings shall be provided where exposed to view. Angle stops, straight stops, stops integral with the faucets, or concealed type of lock-shield, and loose-key pattern stops for supplies with threaded, sweat or solvent weld inlets shall be furnished and installed with fixtures. Where connections between copper tubing and faucets are made by rubber compression fittings, a beading tool shall be used to mechanically deform the tubing above the compression fitting. Exposed traps and supply pipes for fixtures and equipment shall be connected to the rough piping systems at the wall, unless otherwise specified under the item. Floor and wall escutcheons shall be as specified. Drain lines and hot water lines of fixtures for handicapped personnel shall be insulated and do not require polished chrome finish. Plumbing fixtures and accessories shall be installed within the space shown.

3.3.1 Fixture Connections

Where space limitations prohibit standard fittings in conjunction with the cast-iron floor flange, special short-radius fittings shall be provided. Connections between earthenware fixtures and flanges on soil pipe shall be made gastight and watertight with a closet-setting compound or neoprene gasket and seal. Use of natural rubber gaskets or putty will not be permitted. Fixtures with outlet flanges shall be set the proper distance from floor or wall to make a first-class joint with the closet-setting compound or gasket and fixture used.

~~3.3.2 Flushometer Valves~~

~~Flushometer valves shall be secured to prevent movement by anchoring the long finished top spud connecting tube to wall adjacent to valve with approved metal bracket.[Flushometer valves for water closets shall be installed 1 m above the floor, except at water closets intended for use by the physically handicapped where flushometer valves shall be mounted at approximately 760 mm above the floor and arranged to avoid interference with grab bars. In addition, for water closets intended for handicap use, the flush valve handle shall be installed on the wide side of the enclosure.][Bumpers for water closet seats shall be installed on the [wall] [flushometer stop] [flushometer spud].]~~

3.3.2 Height of Fixture Rims Above Floor

Lavatories shall be mounted with rim 775 mm above finished floor. Wall-hung drinking fountains and water coolers shall be installed with rim 1020 mm above floor. Wall-hung service sinks shall be mounted with rim 700 mm above the floor. Installation of fixtures for use by the physically handicapped shall be in accordance with ICC A117.1 COMM.

3.3.3 Shower Bath Outfits

The area around the water supply piping to the mixing valves and behind the escutcheon plate shall be made watertight by caulking or gasketing.

3.3.4 Fixture Supports

Fixture supports for off-the-floor lavatories, urinals, water closets, and other fixtures of similar size, design, and use, shall be of the chair-carrier type. The carrier shall provide the necessary means of mounting the fixture, with a foot or feet to anchor the assembly to the floor slab. Adjustability shall be provided to locate the fixture at the desired height and in proper relation to the wall. Support plates, in lieu of chair carrier, shall be fastened to the wall structure only where it is not possible to anchor a floor-mounted chair carrier to the floor slab.

3.3.4.1 Support for Solid Masonry Construction

Chair carrier shall be anchored to the floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate shall be imbedded in the masonry wall.

3.3.4.2 Support for Concrete-Masonry Wall Construction

Chair carrier shall be anchored to floor slab. Where a floor-anchored chair carrier cannot be used, a suitable wall plate shall be fastened to the concrete wall using through bolts and a back-up plate.

3.3.4.3 Support for Steel Stud Frame Partitions

Chair carrier shall be used. The anchor feet and tubular uprights shall be of the heavy duty design; and feet (bases) shall be steel and welded to a square or rectangular steel tube upright. Wall plates, in lieu of floor-anchored chair carriers, shall be used only if adjoining steel partition studs are suitably reinforced to support a wall plate bolted to these studs.

~~3.3.4.4 Support for Wood Stud Construction~~

~~Where floor is a concrete slab, a floor-anchored chair carrier shall be used. Where entire construction is wood, wood crosspieces shall be installed. Fixture hanger plates, supports, brackets, or mounting lugs shall be fastened with not less than No. 10 wood screws, 6 mm thick minimum steel hanger, or toggle bolts with nut. The wood crosspieces shall extend the full width of the fixture and shall be securely supported.~~

~~3.3.4.5 Wall-Mounted Water Closet Gaskets~~

~~Where wall-mounted water closets are provided, reinforced wax, treated felt, or neoprene gaskets shall be provided. The type of gasket furnished~~

~~shall be as recommended by the chair-carrier manufacturer.~~

3.3.5 Backflow Prevention Devices

Plumbing fixtures, equipment, and pipe connections shall not cross connect or interconnect between a potable water supply and any source of nonpotable water. Backflow preventers shall be installed where indicated and in accordance with MLIT-M at all other locations necessary to preclude a cross-connect or interconnect between a potable water supply and any nonpotable substance. In addition backflow preventers shall be installed at all locations where the potable water outlet is below the flood level of the equipment, or where the potable water outlet will be located below the level of the nonpotable substance. Backflow preventers shall be located so that no part of the device will be submerged. Backflow preventers shall be of sufficient size to allow unrestricted flow of water to the equipment, and preclude the backflow of any nonpotable substance into the potable water system. Bypass piping shall not be provided around backflow preventers. Access shall be provided for maintenance and testing. Each device shall be a standard commercial unit.

3.3.6 Access Panels

Access panels shall be provided for concealed valves and controls, or any item requiring inspection or maintenance. Access panels shall be of sufficient size and located so that the concealed items may be serviced, maintained, or replaced. Access panels shall be as specified in~~[~~ Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS~~][Section 05 51 33 METAL LADDERS][Section 05 52 00 METAL RAILINGS][Section 05 51 00 METAL STAIRS]~~.

~~3.3.7 Sight Drains~~

~~Sight drains shall be installed so that the indirect waste will terminate 50 mm above the flood rim of the funnel to provide an acceptable air gap.~~

3.3.7 Traps

Each trap shall be placed as near the fixture as possible, and no fixture shall be double-trapped. Traps installed on cast-iron soil pipe shall be cast iron. Traps installed on steel pipe or copper tubing shall be recess-drainage pattern, or brass-tube type. Traps installed on plastic pipe may be plastic conforming to JIS A 4421. Traps for acid-resisting waste shall be of the same material as the pipe.

~~3.3.8 Shower Pans~~

~~Before installing shower pan, subfloor shall be free of projections such as nail heads or rough edges of aggregate. Drain shall be a bolt-down, clamping-ring type with weepholes, installed so the lip of the subdrain is flush with subfloor.~~

~~3.3.8.1 General~~

~~The floor of each individual shower, the shower-area portion of combination shower and drying room, and the entire shower and drying room where the two are not separated by curb or partition, shall be made watertight with a shower pan fabricated in place. The shower pan material shall be cut to size and shape of the area indicated, in one piece to the maximum extent practicable, allowing a minimum of 150 mm for turnup on walls or partitions, and shall be folded over the curb with an approximate~~

~~return of 1/4 of curb height. The upstands shall be placed behind any wall or partition finish. Subflooring shall be smooth and clean, with nailheads driven flush with surface, and shall be sloped to drain. Shower pans shall be clamped to drains with the drain clamping ring.~~

~~3.3.8.2 Metal Shower Pans~~

~~When a shower pan of required size cannot be furnished in one piece, metal pieces shall be joined with a flintlock seam and soldered or burned. The corners shall be folded, not cut, and the corner seam shall be soldered or burned. Pans, including upstands, shall be coated on all surfaces with one brush coat of asphalt. Asphalt shall be applied evenly at not less than 1 liter per square meter. A layer of felt covered with building paper shall be placed between shower pans and wood floors. The joining surfaces of metal pan and drain shall be given a brush coat of asphalt after the pan is connected to the drain.~~

~~3.3.8.3 Plasticized Chlorinated Polyethylene Shower Pans~~

~~Corners of plasticized chlorinated polyethylene shower pans shall be folded against the upstand by making a pig-ear fold. Hot-air gun or heat lamp shall be used in making corner folds. Each pig-ear corner fold shall be nailed or stapled 12 mm from the upper edge to hold it in place. Nails shall be galvanized large-head roofing nails. On metal framing or studs, approved duct tape shall be used to secure pig-ear fold and membrane. Where no backing is provided between the studs, the membrane slack shall be taken up by pleating and stapling or nailing to studding 12 mm from upper edge. To adhere the membrane to vertical surfaces, the back of the membrane and the surface to which it will be applied shall be coated with adhesive that becomes dry to the touch in 5 to 10 minutes, after which the membrane shall be pressed into place. Surfaces to be solvent-welded shall be clean. Surfaces to be joined with xylene shall be initially sprayed and vigorously cleaned with a cotton cloth, followed by final coating of xylene and the joining of the surfaces by roller or equivalent means. If ambient or membrane temperatures are below 4 degrees C the membrane and the joint shall be heated prior to application of xylene. Heat may be applied with hot-air gun or heat lamp, taking precautions not to scorch the membrane. Adequate ventilation and wearing of gloves are required when working with xylene. Membrane shall be pressed into position on the drain body, and shall be cut and fit to match so that membrane can be properly clamped and an effective gasket-type seal provided. On wood subflooring, two layers of 0.73 kg per square meter dry felt shall be installed prior to installation of shower pan to ensure a smooth surface for installation.~~

~~3.3.8.4 Nonplasticized Polyvinyl Chloride (PVC) Shower Pans~~

~~Nonplasticized PVC shall be turned up behind walls or wall surfaces a distance of not less than 150 mm in room areas and 75 mm above curb level in curbed spaces with sufficient material to fold over and fasten to outside face of curb. Corners shall be pig-ear type and folded between pan and studs. Only top 25 mm of upstand shall be nailed to hold in place. Nails shall be galvanized large-head roofing type. Approved duct tape shall be used on metal framing or studs to secure pig-ear fold and membrane. Where no backing is provided between studs, the membrane slack shall be taken up by pleating and stapling or nailing to studding at top inch of upstand. To adhere the membrane to vertical surfaces, the back of the membrane and the surface to which it is to be applied shall be coated with adhesive that becomes dry to the touch in 5 to 10 minutes, after~~

~~which the membrane shall be pressed into place. Trim for drain shall be exactly the size of drain opening. Bolt holes shall be pierced to accommodate bolts with a tight fit. Adhesive shall be used between pan and subdrain. Clamping ring shall be bolted firmly. A small amount of gravel or porous materials shall be placed at weepholes so that holes remain clear when setting bed is poured. Membrane shall be solvent welded with PVC solvent cement. Surfaces to be solvent welded shall be clean (free of grease and grime). Sheets shall be laid on a flat surface with an overlap of about 50 mm. Top edge shall be folded back and surface primed with a PVC primer. PVC cement shall be applied and surfaces immediately placed together, while still wet. Joint shall be lightly rolled with a paint roller, then as the joint sets shall be rolled firmly but not so hard as to distort the material. In long lengths, about 600 or 900 mm at a time shall be welded. On wood subflooring, two layers of 0.73 kg per square meter felt shall be installed prior to installation of shower pan to ensure a smooth surface installation.~~

3.4 VIBRATION-ABSORBING FEATURES

Mechanical equipment, including compressors and pumps, shall be isolated from the building structure by approved vibration-absorbing features, unless otherwise shown. Each foundation shall include an adequate number of standard isolation units. Each unit shall consist of machine and floor or foundation fastening, together with intermediate isolation material, and shall be a standard product with printed load rating. Piping connected to mechanical equipment shall be provided with flexible connectors. Isolation unit installation shall limit vibration to ~~[_____]~~20 percent of the lowest equipment rpm.

~~3.4.1 Tank- or Skid-Mounted Compressors~~

~~Floor attachment shall be as recommended by compressor manufacturer. Compressors shall be mounted to resist seismic loads as specified in Section 23 05 48.19 [SEISMIC] BRACING FOR HVAC.~~

~~3.4.2 Foundation-Mounted Compressors~~

~~[Foundation attachment shall be as recommended by the compressor manufacturer.][Foundation shall be as recommended by the compressor manufacturer, except the foundation shall weigh not less than three times the weight of the moving parts.] Compressors shall be mounted to resist seismic loads as specified in Section 23 05 48.19 [SEISMIC] BRACING FOR HVAC.~~

3.5 WATER METER REMOTE READOUT REGISTER

The remote readout register shall be mounted at the location indicated or as directed by the Contracting Officer.

3.6 IDENTIFICATION SYSTEMS

3.6.1 Identification Tags

Identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and valve number shall be installed on valves, except those valves installed on supplies at plumbing fixtures. Tags shall be 35 mm minimum diameter, and marking shall be stamped or engraved. Indentations shall be black, for reading clarity. Tags shall be attached to valves with No. 12 AWG, copper wire, chrome-plated beaded chain, or plastic straps designed for that purpose.

3.6.2 Pipe Color Code Marking

Color code marking of piping shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

~~3.6.3 Color Coding Scheme for Locating Hidden Utility Components~~

~~Scheme shall be provided in buildings having suspended grid ceilings. The color coding scheme shall identify points of access for maintenance and operation of operable components which are not visible from the finished space and installed in the space directly above the suspended grid ceiling. The operable components shall include valves, dampers, switches, linkages and thermostats. The color coding scheme shall consist of a color code board and colored metal disks. Each colored metal disk shall be approximately 12 mm in diameter and secured to removable ceiling panels with fasteners. The fasteners shall be inserted into the ceiling panels so that the fasteners will be concealed from view. The fasteners shall be manually removable without tools and shall not separate from the ceiling panels when panels are dropped from ceiling height. Installation of colored metal disks shall follow completion of the finished surface on which the disks are to be fastened. The color code board shall have the approximate dimensions of 1 m width, 750 mm height, and 12 mm thickness. The board shall be made of wood fiberboard and framed under glass or 1.6 mm transparent plastic cover. Unless otherwise directed, the color code symbols shall be approximately 20 mm in diameter and the related lettering in 12 mm high capital letters. The color code board shall be mounted and located in the mechanical or equipment room. The color code system shall be as indicated below:~~

~~Color~~	~~System~~	~~Item~~	~~Location~~
~~[____]~~	~~[____]~~	~~[____]~~	~~[____]~~

3.7 ESCUTCHEONS

Escutcheons shall be provided at finished surfaces where bare or insulated piping, exposed to view, passes through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Escutcheons shall be fastened securely to pipe or pipe covering and shall be satin-finish, corrosion-resisting steel, polished chromium-plated zinc alloy, or polished chromium-plated copper alloy. Escutcheons shall be either one-piece or split-pattern, held in place by internal spring tension or setscrew.

3.8 PAINTING

Painting of pipes, hangers, supports, and other iron work, either in concealed spaces or exposed spaces, is specified in Section 09 90 00 PAINTS AND COATINGS.

3.8.1 Painting of New Equipment

New equipment painting shall be factory applied or shop applied, and shall be as specified herein, and provided under each individual section.

~~3.8.1.1 Factory Painting Systems~~

~~Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors shall withstand 500 hours in a salt-spray fog test. Salt-spray fog test shall be in accordance with JIS Z 2371 or ASTM B117, and for that test the acceptance criteria shall be as follows: immediately after completion of the test, the paint shall show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shall show no signs of rust creepage beyond 3 mm on either side of the scratch mark.~~

~~The film thickness of the factory painting system applied on the equipment shall not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, the factory painting system shall be designed for the temperature service.~~

3.8.1.1 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 50 degrees C shall be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat shall be aluminum or light gray.

a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C shall receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.

b. Temperatures Between 50 and 205 Degrees C: Metal surfaces subject to temperatures between 50 and 205 degrees C shall receive two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm.

c. Temperatures Greater Than 205 Degrees C: Metal surfaces subject to temperatures greater than 205 degrees C shall receive two coats of 315 degrees C heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm.

3.9 TESTS, FLUSHING AND DISINFECTION

3.9.1 Plumbing System

The following tests shall be performed on the plumbing system in accordance with MLIT-M, except that the drainage and vent system final test shall include the smoke test. The Contractor has the option to perform a peppermint test in lieu of the smoke test. If a peppermint test is chosen, the Contractor must submit a testing procedure and reasons for choosing this option in lieu of the smoke test to the Contracting Officer for approval.

a. Drainage and Vent Systems Test. The final test shall include a smoke test.

b. Building Sewers Tests.

c. Water Supply Systems Tests.

~~3.9.1.1 Test of Backflow Prevention Assemblies~~

~~Backflow prevention assembly shall be tested using gauges specifically designed for the testing of backflow prevention assemblies.~~

~~Backflow prevention assembly test gauges shall be tested annually for accuracy in accordance with the requirements of State or local regulatory agencies. If there is no State or local regulatory agency requirements, gauges shall be tested annually for accuracy in accordance with the requirements of University of Southern California's Foundation of Cross Connection Control and Hydraulic Research or the American Water Works Association Manual of Cross Connection (Manual M-14), or any other approved testing laboratory having equivalent capabilities for both laboratory and field evaluation of backflow prevention assembly test gauges. Report form for each assembly shall include, as a minimum, the following:~~

~~Data on Device~~	~~Data on Testing Firm~~
~~Type of Assembly~~	~~Name~~
~~Manufacturer~~	~~Address~~
~~Model Number~~	~~Certified Tester~~
~~Serial Number~~	~~Certified Tester No.~~
~~Size~~	~~Date of Test~~
~~Location~~	
~~Test Pressure Readings~~	~~Serial Number and Test Data of Gauges~~

~~If the unit fails to meet specified requirements, the unit shall be repaired and retested.~~

~~3.9.1.2 Shower Pans~~

~~After installation of the pan and finished floor, the drain shall be temporarily plugged below the weep holes. The floor area shall be flooded with water to a minimum depth of 25 mm for a period of 24 hours. Any drop in the water level during test, except for evaporation, will be reason for rejection, repair, and retest.~~

~~3.9.1.3 Compressed Air Piping (Nonoil-Free)~~

~~Piping systems shall be filled with oil-free dry air or gaseous nitrogen to~~

~~1.03 MPa and hold this pressure for 2 hours with no drop in pressure.~~

3.9.2 Defective Work

If inspection or test shows defects, such defective work or material shall be replaced or repaired as necessary and inspection and tests shall be repeated. Repairs to piping shall be made with new materials. Caulking of screwed joints or holes will not be acceptable.

3.9.3 System Flushing

3.9.3.1 During Flushing

Before operational tests or disinfection, potable water piping system shall be flushed with ~~[hot]~~ potable water. Sufficient water shall be used to produce a water velocity that is capable of entraining and removing debris in all portions of the piping system. This requires simultaneous operation of all fixtures on a common branch or main in order to produce a flushing velocity of approximately 1.2 meters per second through all portions of the piping system. In the event that this is impossible due to size of system, the Contracting Officer (or the designated representative) shall specify the number of fixtures to be operated during flushing. Contractor shall provide adequate personnel to monitor the flushing operation and to ensure that drain lines are unobstructed in order to prevent flooding of the facility. Contractor shall be responsible for any flood damage resulting from flushing of the system. Flushing shall be continued until entrained dirt and other foreign materials have been removed and until discharge water shows no discoloration. All faucets and drinking water fountains, to include any device considered as an end point device by NSF/ANSI 61, Section 9, shall be flushed a minimum of 1 L per 24 hour period, ten times over a 14 day period.

3.9.3.2 After Flushing

System shall be drained at low points. Strainer screens shall be removed, cleaned, and replaced. After flushing and cleaning, systems shall be prepared for testing by immediately filling water piping with clean, fresh potable water. Any stoppage, discoloration, or other damage to the finish, furnishings, or parts of the building due to the Contractor's failure to properly clean the piping system shall be repaired by the Contractor. When the system flushing is complete, the hot-water system shall be adjusted for uniform circulation. Flushing devices and automatic control systems shall be adjusted for proper operation according to manufacturer's instructions. Flow rates on fixtures must not exceed those stated in PART 2 of this Section. Unless more stringent local requirements exist, lead levels shall not exceed limits established by JIS S 3201. The water supply to the building shall be tested separately to ensure that any lead contamination found during potable water system testing is due to work being performed inside the building.

3.9.4 Operational Test

Upon completion of flushing and prior to disinfection procedures, the Contractor shall subject the plumbing system to operating tests to demonstrate satisfactory installation, connections, adjustments, and functional and operational efficiency. Such operating tests shall cover a period of not less than 8 hours for each system and shall include the following information in a report with conclusion as to the adequacy of the system:

a. Time, date, and duration of test.

b. Water pressures at the most remote and the highest fixtures.

c. Operation of each fixture and fixture trim.

d. Operation of each valve, hydrant, and faucet.

e. Pump suction and discharge pressures.

f. Temperature of each domestic hot-water supply.

g. Operation of each floor and roof drain by flooding with water.

h. Operation of each vacuum breaker and backflow preventer.

i. Complete operation of each water pressure booster system, including pump start pressure and stop pressure.

j. Compressed air readings at each compressor and at each outlet. Each indicating instrument shall be read at 1/2 hour intervals. The report of the test shall be submitted in quadruplicate. The Contractor shall furnish instruments, equipment, and personnel required for the tests; the Government will furnish the necessary water and electricity.

3.9.5 Disinfection

After all system components are provided and operational tests are complete, the entire domestic hot- and cold-water distribution system shall be disinfected. Before introducing disinfecting chlorination material, entire system shall be flushed with potable water until any entrained dirt and other foreign materials have been removed.

Water chlorination procedure shall be in accordance with Japanese Water Supply Law, Article 20 as modified and supplemented by this specification. The chlorinating material shall be hypochlorites or liquid chlorine. The chlorinating material shall be fed into the water piping system at a constant rate at a concentration of at least 50 parts per
million (ppm). Feed a properly adjusted hypochlorite solution injected into the system with a hypochlorinator, or inject liquid chlorine into the system through a solution-feed chlorinator and booster pump until the entire system is completely filled.

Test the chlorine residual level in the water at 6 hour intervals for a continuous period of 24 hours. If at the end of a 6 hour interval, the chlorine residual has dropped to less than 25 ppm, flush the piping including tanks with potable water, and repeat the above chlorination procedures. During the chlorination period, each valve and faucet shall be opened and closed several times.

After the second 24 hour period, verify that no less than 25 ppm chlorine residual remains in the treated system. The 24 hour chlorination procedure must be repeated until no less than 25 ppm chlorine residual remains in the treated system.

Upon the specified verification, the system including tanks shall then be flushed with potable water until the residual chlorine level is reduced to less than one part per million. During the flushing period, each valve and faucet shall be opened and closed several times.

Take additional samples of water in disinfected containers, for bacterial examination, at locations specified by the Contracting Officer
Test these samples for total coliform organisms (coliform bacteria, fecal coliform, streptococcal, and other bacteria) in accordance with MLIT-M. The testing method used shall be EPA approved for drinking water systems and shall comply with applicable local and state requirements.

Disinfection shall be repeated until bacterial tests indicate the absence of coliform organisms (zero mean coliform density per 100 milliliters) in the samples for at least 2 full days. The system will not be accepted until satisfactory bacteriological results have been obtained.

3.10 POSTED INSTRUCTIONS

Framed instructions under glass or in laminated plastic, including wiring and control diagrams showing the complete layout of the entire system, shall be posted where directed. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system shall be prepared in typed form, framed as specified above for the wiring and control diagrams and posted beside the diagrams. The framed instructions shall be posted before acceptance testing of the systems.

3.11 PERFORMANCE OF WATER HEATING EQUIPMENT

~~Standard rating condition terms are as follows:~~

~~EF = Energy factor, minimum overall efficiency.~~

~~ET = Minimum thermal efficiency with 21 degrees C delta T.~~

~~SL = Standby loss is maximum (Btu/h) based on a 38.9 degree C temperature difference between stored water and ambient requirements.~~

~~V = Rated volume in gallons~~

~~Q = Nameplate input rate in kW (Btu/h)~~

~~3.11.1 Storage Water Heaters~~

~~3.11.1.1 Electric~~

~~a. Storage capacity of 227 liters shall have a minimum energy factor (EF) of 0.93 or higher per FEMP requirements.~~

~~b. Storage capacity of 227 liters or more shall have a minimum energy factor (EF) of 0.91 or higher per FEMP requirements.~~

~~3.11.1.2 Gas~~

~~a. Storage capacity of 189 liters or less shall have a minimum energy factor (EF) of 0.67 or higher per FEMP requirements.~~

~~b. Storage capacity of 75.7 liters - or more and input rating of 23 KW W or less:~~

~~c. Rating of less than 22980 W: (75,000 Btu/h) ET shall be 80 percent; maximum SL shall be (Q/800+110x(V^^1/2)), per JIS S 2109~~

~~3.11.1.3 Oil~~

~~a. Storage capacity of 75.7 liters or more and input rating of 31 KW W or less:~~

~~b. Rating of less than 309.75 W/L or input rating more than 31 KW: ET shall be 78 percent; maximum SL shall be (Q/800+100x(V^^1/2)), per JIS S 2109.~~

3.11.1 Unfired Hot Water Storage

All volumes and inputs: shall meet or exceed R-12.5.

~~3.11.2 Instantaneous Water Heater~~

~~3.11.2.1 Gas~~

~~a. Rating of 309.75 W/L and greater and less than 7.57 L with an input greater than 14.66 kW and less than 58.62 kW~~

~~b. Rating of 309.75 W/L and greater and less than 37.85 L with an input of 58.62 kW and greater shall have a minimum thermal efficiency (ET) of 80 percent per JIS S 2109~~

~~c. Rating of 309.75 W/L and greater and 37.85 L and greater with an input of 58.62 kW and greater shall have a minimum thermal efficiency (ET) of 80 percent and the maximum SL shall be Q/800+110x(V^^1/2)) per JIS S 2109~~

~~3.11.2.2 Oil~~

~~a. Rating of 309.75 W/L and greater and less than 7.57 L with an input of 61.55 kW and less~~

~~b. Rating of 309.75 W/Ll and greater and less than 37.85 L with an input greater than 61.55 kW shall have a minimum thermal efficiency (ET) of 80 percent per JIS S 2109~~

~~c. Rating of 309.75 W/L and 37.85 L and greater with an input of greater than 61.55 kW shall have a minimum thermal efficiency (ET) of 78 percent and the maximum SL shall be Q/800+110x(V^^1/2)) per JIS S 2109~~

~~3.11.3 Pool Heaters~~

~~a. Gas/oil fuel, capacities and inputs: ET shall be 78 percent.~~

~~b. Heat Pump, All capacities and inputs shall meet a COP of 4.0.~~

3.12 TABLES

TABLE I								
PIPE AND FITTING MATERIALS FOR DRAINAGE, WASTE, VENT AND CONDENSATE DRAIN PIPING SYSTEMS								
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D	SERVICE E	SERVICE F	SERVICE G
1	Cast iron soil ~~pipe and~~ fittings, hub and spigot, JIS G 5526 with compression gaskets.~~ .~~	X	~~X~~	~~X~~	~~X~~	~~X~~		

SERVICE:

- A - Underground Building Soil, Waste and Storm Drain
- B - Aboveground Soil, Waste, Drain In Buildings
- C - Underground Vent
- D - Aboveground Vent
- E - Interior Rainwater Conductors Aboveground
- F - Corrosive Waste And Vent Above And Belowground
- G - Condensate Drain Aboveground

* - Hard Temper

TABLE II					
PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS					
Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
~~7~~1	Seamless copper pipe, ASTM B42	X	X		X
~~8~~2	Seamless copper water tube, ASTM B88, ASTM B88M	X**	X**	~~X**~~	X***
~~10~~3	Wrought copper and bronze solder-joint pressure fittings, ASME B16.22 for use with Items ~~5, 7~~1 and ~~8~~2	X	X	~~X~~	X
~~11~~4	Cast copper alloy solder-joint pressure fittings, ASME B16.18 for use with Item ~~8~~2	X	X	~~X~~	X

TABLE II

PIPE AND FITTING MATERIALS FOR PRESSURE PIPING SYSTEMS

Item #	Pipe and Fitting Materials	SERVICE A	SERVICE B	SERVICE C	SERVICE D
	SERVICE: A - Cold Water Service Aboveground B - Hot and Cold Water Distribution 82 degrees C Maximum Aboveground C - Compressed Air Lubricated D - Cold Water Service Belowground Indicated types are minimum wall thicknesses. ** - Type L - Hard *** - Type K - Hard temper with brazed joints only or type K-soft temper without joints in or under floors **** - In or under slab floors only brazed joints				

~~TABLE III~~

~~STANDARD RATING CONDITIONS AND MINIMUM PERFORMANCE RATINGS FOR WATER HEATING EQUIPMENT~~

~~FUEL~~	~~STORAGE CAPACITY LITERS~~	~~INPUT RATING~~	~~TEST PROCEDURE~~	~~REQUIRED PERFORMANCE~~
~~A. STORAGE WATER HEATERS~~				
~~Elect.~~	~~227 max~~		~~ISO 14024~~	~~EF = 0.93~~
~~Elect.~~	~~227 min~~		~~ISO 14024~~	~~EF = 0.91~~
~~Elect.~~	~~75.7 min.~~	~~12 kW max.~~	~~ISO 14024~~	~~EF = 0.93-0.00132V minimum~~
~~Elect.~~	~~75.7 min. OR 12 kW min.~~		~~JIS S 2109~~	~~SL = 20+35x(V^^1/2) maximum~~
~~Elect. Heat Pump~~		~~24 Amps or less and 250 Volts or less~~	~~ISO 14024~~	~~EF = 0.93-0.00132V~~
~~Gas~~	~~189 max~~		~~ISO 14024~~	~~EF = 0.67-0.0019V min~~
~~Gas~~	~~75.7 min.~~	~~22 kW max.~~	~~ISO 14024~~	~~EF = 0.80-0.0019V minimum~~

~~TABLE III~~				
~~STANDARD RATING CONDITIONS AND MINIMUM PERFORMANCE RATINGS FOR WATER HEATING EQUIPMENT~~				
~~FUEL~~	~~STORAGE CAPACITY LITERS~~	~~INPUT RATING~~	~~TEST PROCEDURE~~	~~REQUIRED PERFORMANCE~~
~~Gas~~	~~309.75 W/L max.~~	~~22 kW max.~~	~~JIS S 2109~~	~~ET= 80 percent; SL = 1.3+38/V max.~~
~~Oil~~	~~75.7 min.~~	~~30.8 kW max.~~	~~ISO 14024~~	~~EF = 0.59-0.0019V min~~
~~Oil~~	~~309.75 W/L max~~	~~30.8 kW~~	~~JIS S 2109~~	~~ET = 78 percent; SL = (Q/800+110x(V^^1/2)) maximum~~
~~B. Unfired Hot Water Storage, R = 2.2 minimum~~				
~~C. Instantaneous Water Heater~~				
~~Gas~~	~~309.75 W/L min.~~	~~14.66 kW min.~~	~~ISO 14024~~	~~EF = 0.62-0.0019V and 7.57 L max 58.62 kW max.~~
~~Gas~~	~~309.75 W/L min.~~	~~58.62 kW min.~~	~~JIS S 2109~~	~~ET = 80 percent and 37.85 L max 58.62 kW max.~~
~~Gas~~	~~309.75 W/L min.~~	~~58.62 kW min.~~	~~JIS S 2109~~	~~ET = 80 percent and 37.85 L min. SL + (Q/800+110x(V^^1/2)~~
~~Oil~~	~~309.75 W/L min.~~	~~61.552 kW max.~~	~~ISO 14024~~	~~EF = 0.59-0.0019V and 37.85 L max.~~
~~Oil~~	~~309.75 W/L min.~~	~~61.552 kW max.~~	~~JIS S 2109~~	~~ET = 80 percent and 37.85 L min. SL + (Q/800+110x(V^^1/2)~~
~~Oil~~	~~309.75 W/L min.~~	~~61.552 kW max.~~	~~JIS S 2109~~	~~ET = 78 percent and 37.85 L max SL = (Q800+110x(V^^1/2))~~
~~D. Pool Heater~~				
~~Gas or Oil~~	~~All~~	~~All~~		~~ET = 78 percent~~
~~Heat Pump All~~	~~All~~	~~All~~		~~COP = 4.0~~

~~TABLE III~~				
~~STANDARD RATING CONDITIONS AND MINIMUM PERFORMANCE RATINGS FOR WATER HEATING EQUIPMENT~~				
~~FUEL~~	~~STORAGE CAPACITY LITERS~~	~~INPUT RATING~~	~~TEST PROCEDURE~~	~~REQUIRED PERFORMANCE~~
~~TERMS:~~ ~~EF = Energy factor, minimum overall efficiency.~~ ~~ET = Minimum thermal efficiency with 21 degrees C delta T.~~ ~~SL = Standby loss is maximum Watts based on a 38.9 degrees C temperature difference between stored water and ambient requirements.~~ ~~V = Rated storage volume in gallons~~ ~~Q = Nameplate input rate in Watts~~				

-- End of Section --

SECTION 23 05 15

COMMON PIPING FOR HVAC
02/14

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 0203	(2014) Concrete Terminology
JIS A 1108	(2018) Method of Test for Compressive Strength of Concrete
JIS A 1110	(2006) Methods of Test for Density and Water Absorption of Coarse Aggregates
JIS A 5758	(2022) Sealants for Sealing and Glazing in Buildings
JIS A 9504	(20~~21~~24) Man Made Mineral Fibre Thermal Insulation Materials
JIS B 0209-1	(2001) ISO General Purpose Metric Screw Threads-Tolerances-Part 1 : Principles and Basic Data
JIS B 1112	(2019) Cross-Recessed Head Wood Screws
JIS B 1181	(2014) Hexagon Nuts and Hexagon Thin Nuts
JIS B 1220	(2015) Set of Anchor Bolt for Structures
JIS B 2011	(2013) Bronze, Gate, Globe, Angle, and Check Valves (Amendment 2)
JIS B 2031	(2015) Gray Cast Iron Valves (Amendment 1)
JIS B 2032	(2013) Wafer Type Rubber-Seated Butterfly Valves
JIS B 2061	(2017) Faucets, Ball Taps and Flush Valves
JIS B 2220	(2012) Steel Pipe Flanges
JIS B 2239	(2013) Cast Iron Pipe Flanges
JIS B 2240	(2006) Copper Alloy Pipe Flanges
JIS B 2301	(2013) Screwed Type Malleable Cast Iron Pipe Fittings

JIS B 2311	~~(2015) Steel Butt-Welding Pipe Fittings for Ordinary Use~~(2024) Steel Butt-Welding Pipe Fittings for Ordinary Use
JIS B 2312	(2015) Steel Butt-Welding Pipe Fittings
JIS B 7410	(1997) Liquid-In-Glass Thermometers for Testing of Petroleum Product
JIS B 7505-1	~~(2017) Aneroid Pressure Gauges-Part 1: Bourdon Tube Pressure Gauges~~(2017) Aneroid Pressure Gauges -- Part 1: Bourdon Tube Pressure Gauges
JIS B 8267	~~(2015) Construction of Pressure Vessel~~ (2022) Construction of Pressure Vessel
JIS B 8285	(2010) Welding Procedure Qualification Test for Pressure Vessels
JIS F 0602	(1995) Shipbuilding-Non-Asbestos Gaskets to Cargo Piping System-Application Standard
JIS G 3138	(2021) Rolled Steel Bars for Building Structure
JIS G 3201	(2008) Carbon Steel Forgings for General Use (Amendment 1)
JIS G 3202	(2008) Carbon Steel Forgings for Pressure Vessels (Amendment 1)
JIS G 3454	(2019) Carbon Steel Pipes for Pressure Service (Amendment 1)
JIS G 3455	~~(2016) Carbon Steel Pipes for High Pressure Service~~(2024) Carbon Steel Pipes for High Pressure Service
JIS G 3456	(20~~19~~24) Carbon Steel Pipes for High Temperature Service
JIS G 3459	~~(2017) Stainless Steel Pipes (Amendment 1)~~ (2021) Stainless Steel Pipes
JIS G 4051	(2018) Carbon Steels for Machine Structural Use (Amendment 1)
JIS G 4053	(2018) Low-Alloyed Steels for Machine Structural Use (Amendment 1)
JIS G 4107	~~(2015) Alloy Steel Bolting Materials for High Temperature Service (Amendment 1)~~ (2022) Alloy Steel Bolting Materials for High Temperature Service
JIS G 4303	(2012) Stainless Steel Bars

JIS G 5501	~~(2020) Grey Iron Castings~~(1995) Grey Iron Castings
JIS H 3100	(2018) Copper and Copper Alloy Sheets, Plates and Strips
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS K 7311	(1995) Testing Methods for Thermoplastic Polyurethane Elastomers
JIS HB 40-1	(2019) Ferrous Materials & Metallurgy I
JIS HB 40-2	(2019) Ferrous Materials & Metallurgy II

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 834-1	(1999) Fire-Resistance Tests-Elements of Building Construction-Part 1: General Requirements

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification

1.2 GENERAL REQUIREMENTS

~~[~~ Section 23 30 00 AIR SUPPLY, DISTRIBUTION, VENTILATION, AND EXHAUST SYSTEMS applies to work specified in this section~~]~~.

~~[Section 40 17 30.00 40 WELDING GENERAL PIPING applies to work specified in this section.~~

~~]~~ Submit Material, Equipment, and Fixture Lists for pipes, valves and specialties including manufacturer's style or catalog numbers, specification and drawing reference numbers, and warranty information.

Submit Record Drawings for pipes, valves and accessories providing current factual information including deviations and amendments to the drawings, and concealed and visible changes in the work.

Submit Coordination Drawings for pipes, valves and specialties showing coordination of work between different trades and with the structural and architectural elements of work. Detail all drawings sufficiently to show overall dimensions of related items, clearances, and relative locations of work in allotted spaces. Indicate on drawings where conflicts or clearance problems exist between various trades.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[~~for Contractor Quality Control ~~approval.][for~~ information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Material, Equipment, and Fixture Lists~~[~~ G~~]~~

SD-02 Shop Drawings

Record Drawings~~[~~; G~~[, [____]]]~~

Coordination Drawings;~~[~~ G~~]~~

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals~~[ ]~~

1.4 QUALITY ASSURANCE

1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Provide standard products in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use includes applications of equipment and materials under similar circumstances and of similar size. Ensure the product has been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.4.2 Alternative Qualifications

Products having less than a two-year field service record are acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.4.3 Service Support

Ensure the equipment items are supported by service organizations located in Japan. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. Select service organizations that are reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.4.4 Manufacturer's Nameplate

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable.

1.4.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer.

1.4.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions are considered mandatory, the word "should" is interpreted as "shall." Reference to the "code official" is interpreted to mean the "Contracting Officer." For Navy owned property, interpret references to the "owner" to mean the "Contracting Officer." For leased facilities, references to the "owner" is interpreted to mean the "lessor." References to the "permit holder" are interpreted to mean the "Contractor."

1.4.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, are applied as appropriate by the Contracting Officer and as authorized by his administrative cognizance and the FAR.

1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

1.6 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Provide instructors thoroughly familiar with all parts of the installation and trained in operating theory as well as practical operation and maintenance work.

Give instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished is as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.7 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

PART 2 PRODUCTS

~~2.1 ELECTRICAL HEAT TRACING~~

~~Provide heat trace systems for pipes, valves, and fittings. System include all necessary components, including heaters and controls to prevent freezing.~~

~~Provide self-regulating heaters consisting of two 16 AWG tinned-copper bus wires embedded in parallel in a self-regulating polymer core that varies its power output to respond to temperature along its length. Ensure heater is able to be crossed over itself without overheating. Obtain approval before used directly on plastic pipe. Cover heater with a radiation cross-linked modified polyolefin dielectric jacket in accordance with JIS C 3605.~~

~~Provide heater with self-regulating factor of at least [90] percent, in order to provide energy conservation and to prevent overheating.~~

~~Operate heater on line voltages of [120] [208] volts without the use of transformers.~~

~~Size Heater according to the following table:~~

~~Pipe Size (DN)~~

~~(Millimeter Diameter)~~	~~Minus 23 degrees C~~	~~Minus 29 degrees C~~
~~80 or less~~	~~16 watts per meter (wpm)~~	~~16 watts per meter (wpm)~~
~~100~~	~~16 wpm~~	~~26 wpm~~
~~150~~	~~26 wpm~~	~~26 wpm~~
~~200~~	~~2 strips/16 wpm~~	~~2 strips/26 wpm~~
~~300 to 356~~	~~2 strips/26 wpm~~	~~2 strips/26 wpm~~

~~Control systems by an ambient sensing thermostat set at 4 degrees C either directly or through an appropriate contactor.~~

2.1 PIPE AND FITTINGS

Submit equipment and performance data for pipe and fittings consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.1.1 Type BCS, Black Carbon Steel

Ensure pipe DN6 through DN300 is Schedule 40 black carbon steel, conforming to JIS G 3454.

Ensure pipe DN6 through DN250 is Schedule 40 seamless or electric-resistance welded black carbon steel, conforming to JIS G 3454.

Ensure pipe DN300 through DN610 is 9.52 millimeter wall seamless black carbon steel, conforming to JIS G 3454.

Ensure fittings DN50 and under are 1034 kilopascal working steam pressure (wsp) banded black malleable iron screwed, conforming to JIS G 3454 and JIS B 2301.

Ensure unions DN50 and under are 1724 kilopascal female, screwed, black malleable iron with brass-to-iron seat, and ground joint, conforming to JIS B 2301.

Ensure fittings DN65 and over are Steel butt weld, conforming to JIS B 2312 to match pipe wall thickness.

Ensure flanges DN65 and over are 1034 kilopascal forged-steel conforming to JIS B 2220, welding neck to match pipe wall thickness.

2.1.2 Type BCS-125, 862 kilopascal Service

Ensure pipe DN6 through DN40 is Schedule 40 steam, Schedule 80 condensate, furnace butt weld, black carbon steel, conforming to JIS G 3456.

Ensure pipe DN50 through DN250 is Schedule 40 steam, Schedule 80 condensate, seamless or electric-resistance welded black carbon steel, conforming to ~~[~~Grade B (electric-resistance welded)~~]~~ or ~~[~~Type S (seamless) ~~]~~ and JIS G 3456.

Ensure pipe DN300 through DN610 is 9.52 millimeter wall, ~~[~~seamless~~]~~ or ~~[~~ electric-resistance~~]~~ welded black carbon steel, conforming to JIS G 3454.

Ensure fittings DN50 and under are 862 kilopascal wsp, cast iron, screwed end, conforming to JIS G 5501 and JIS G 3454 and JIS B 2301.

Ensure fittings DN50 and under are 1034 kilopascal wsp banded black malleable iron screwed, conforming to JIS G 3455 and JIS B 2301.

Ensure fittings DN25 through DN50 are 14 or 21 megapascal water, oil, or gas (wog) to match pipe wall, forged carbon steel socket weld, conforming to JIS G 3455 and JIS B 2301.

Ensure fittings DN50 and under are 862 kilopascal wsp, cast iron, screwed end, conforming to JIS G 3454 and JIS B 2301.

Ensure fittings DN65 and over are wall thickness to match pipe, long radius butt weld, black carbon steel, conforming to JIS G 4051 and JIS B 2311.

Ensure couplings DN50 and under are commercial standard weight for Schedule 40 pipe and commercial extra heavy weight for Schedule 80 pipe, black carbon steel where threaded, and 14 or 21 megapascal wog forged carbon steel, conforming to JIS G 3455 and JIS B 2301, where welded.

Ensure flanges DN65 and over are 1035 kilopascal, forged carbon-steel welding neck, with raised face or flat face and concentric serrated finish, conforming to JIS G 3455 and JIS B 2301.

~~Conform grooved pipe couplings and fittings in accordance with paragraph GROOVED PIPE COUPLINGS AND FITTINGS.~~

2.1.3 Type CPR, Copper

2.1.3.1 Type CPR-A, Copper Above Ground

Ensure tubing DN50 and under is seamless copper tubing, conforming to JIS H 3300, Type L (hard-drawn for all horizontal and all exposed vertical lines, annealed for concealed vertical lines).

Ensure fittings DN50 and under are 1034 kilopascal wsp wrought-copper solder joint fittings conforming to JIS H 3401.

Ensure unions DN50 and under are 1034 kilopascal wsp wrought-copper solder joint, conforming to JIS H 3401.

2.2 PIPING SPECIALTIES

Submit equipment and performance data for piping specialties consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.2.1 Air Separator

Air separated from converter discharge water is ejected by a reduced-velocity device vented to the compression tank.

Provide a commercially constructed separator, designed and certified to separate not less than 80 percent of entrained air on the first passage of water and not less than 80 percent of residual on each successive pass. Provide shop drawings detailing all piping connections proposed for this work.

Ensure the air separator is carbon steel, designed, fabricated, tested, and stamped in conformance with JIS B 8267 for service pressures not less than 862 kilopascal.

2.2.2 Air Vents

Provide manual air vents using 10 millimeter globe valves.

Provide automatic air vents on pumps, mains, and where indicated using ball-float construction. Ensure the vent inlet is not less than DN20 and the outlet not less than 8 millimeter. Orifice size is 3 millimeter. Provide corrosion-resistant steel trim conforming to ~~[~~JIS G 4303~~]~~ ~~JIS G 4305~~. Fit vent with try-cock. Ensure vent discharges air at any pressure up to 1034 kilopascal–. Ensure outlet is copper tube routed.

2.2.3 Compression Tank

Provide compression tank designed, fabricated, tested, and stamped for a working pressure of not less than 862 kilopascal in accordance with JIS B 8267. Ensure tank is hot-dip galvanized after fabrication to produce not less than 51 grams of zinc coating per square meter foot of single-side surface. Tank accessories include red-lined gage-glass complete with glass protectors and shutoff valves, air charger and drainer, and manual vent.

2.2.4 Dielectric Connections

Electrically insulate dissimlar pipe metals from each other by couplings, unions, or flanges commercially manufactured for that purpose and rated for the service pressure and temperature.

2.2.5 Expansion Vibration Isolation Joints

Construct single or multiple arch-flanged expansion vibration isolation joints of steel-ring reinforced chloroprene-impregnated cloth materials. Design joint to absorb the movement of the pipe sections in which installed with no detrimental effect on the pipe or connected equipment. Back flanges with ferrous-metal backing rings. Provide control rod assemblies to restrict joint movement. Coat all nonmetallic exterior surfaces of the joint with chlorosulphinated polyethylene. Provide grommets in limit bolt hole to absorb noise transmitted through the bolts.

Ensure joints are suitable for continuous-duty working temperature of at least 121 degrees C.

Fill arches with soft chloroprene.

Ensure joint, single-arch, movement limitations and size-related, pressure characteristics conform to MLIT-M.

2.2.6 Flexible Pipe

Construct flexible pipe vibration and pipe-noise eliminators of wire-reinforced, rubber-impregnated cloth and cord materials and be flanged. Back the flanges with ferrous-metal backing rings. Ensure service pressure-rating is a minimum 1.5 times actual service, with surge pressure at 82 degrees C.

Construct flexible pipe vibration and pipe noise eliminators of wire-reinforced chloroprene-impregnated cloth and cord materials. Ensure the pipe is flanged. Provide all flanges backed with ferrous-metal backing rings. Coat nonmetallic exterior surfaces of the flexible pipe with an acid- and oxidation-resistant chlorosulphinated polyethylene. Rate the flexible pipe for continuous duty at 896 kilopascal and 121 degrees C.

Ensure unit pipe lengths, face-to-face, are not less than the following:

INSIDE DIAMETER (DN)	UNIT PIPE LENGTH
To 65, inclusive	305 millimeter

INSIDE DIAMETER (DN)	UNIT PIPE LENGTH
80 to 100, inclusive	450 millimeter
125 to 300, inclusive	600 millimeter
To 80, inclusive	450 millimeter
110 to 250, inclusive	600 millimeter
300 and larger	914 millimeter

2.2.7 Flexible Metallic Pipe

Ensure flexible pipe is the bellows-type with wire braid cover and designed, constructed, and rated in accordance with the applicable requirements of MLIT-M.

Minimum working pressure rating is ~~[345] [~~690~~]~~ kilopascal at 149 degrees C.

Ensure minimum burst pressure is four times working pressure at 149 degrees C. Bellows material is JIS G 3459 corrosion-resistant steel. Ensure braid is JIS G 4053 corrosion-resistant steel wire.

Ensure welded end connections are Schedule 80 carbon steel pipe, conforming to JIS G 3456.

Provide threaded end connections; hex-collared Schedule 40, JIS G 3459 corrosion-resistant steel, conforming to JIS G 3459.

Ensure flanged end connection rating and materials conform to specifications for system primary-pressure rating.

2.2.8 Flexible Metal Steam Hose

Provide a bellows type hose with wire braid cover and designed, constructed, and rated in accordance with the applicable requirements of MLIT-M.

Ensure the working steam pressure rating is 862 kilopascal at 260 degrees C .

Ensure minimum burst pressure is nine times working steam pressure at 149 degrees C.

Ensure bellows material is JIS G 3459 corrosion-resistant steel. Braid is JIS G 4053 corrosion-resistant steel wire.

Provide welded end connections; Schedule 80 carbon steel pressure tube, conforming to JIS G 3456.

Provide threaded end connections; hex-collared Schedule 40, JIS G 3459 corrosion-resistant steel, conforming to JIS G 3459.

Ensure flanged end connection rating and materials conform to specifications for system primary-pressure rating.

2.2.9 Metallic Expansion Joints

Provide metallic-bellows expansion joints conforming to MLIT-M.

Design and construct joints to absorb all of the movements of the pipe sections in which installed, with no detrimental effect on pipe or supporting structure.

Rate, design, and construct joints for pressures to 862 kilopascal and temperatures to 260 degrees C.

Ensure joints have a designed bursting strength in excess of four times their rated pressure.

Provide bellows and internal sleeve material of SUS Type 304, SUS Type 304L, or SUS Type 321.

2.2.10 Hose Faucets

Construct hose faucets with 15 millimeter male inlet threads, hexagon shoulder, and 20 millimeter hose connection, conforming to MLIT-M. Ensure hose-coupling screw threads conform to JIS B 0209-1.

Provide vandal proof, atmospheric-type vacuum breaker on the discharge of all potable water lines.

2.2.11 Pressure Gages

Ensure pressure gages conform to JIS B 7505-1 and to requirements specified herein. Pressure-gage size is 90 millimeter nominal diameter. Ensure case is corrosion-resistant steel, conforming to any of JIS G 4053 series of JIS G 3138. Equip gages with adjustable red marking pointer and damper-screw adjustment in inlet connection. Align service-pressure reading at midpoint of gage range.

Fit steam gages with black steel syphons and steam service pressure-rated gage cocks or valves.

2.2.12 Sight-Flow Indicators

Construct sight-flow indicators for pressure service on 80 millimeter and smaller of bronze with specially treated single- or double-glass sight windows and have a bronze, nylon, or tetrafluoroethylene rotating flow indicator mounted on SUS 304, or SUS 316 corrosion-resistant steel shaft. Body may have screwed or flanged end. Provide pressure- and temperature-rated assembly for the applied service. Flapper flow-type indicators are not acceptable.

2.2.13 Sleeve Couplings

Sleeve couplings for plain-end pipe consist of one steel middle ring, two steel followers, two chloroprene or Buna-N elastomer gaskets, and the necessary steel bolts and nuts.

2.2.14 Thermometers

Ensure thermometers conform to JIS B 7410, except for being filled with a red organic liquid. Provide an industrial pattern armored glass thermometer, (well-threaded and seal-welded). Ensure thermometers installed 1800 millimeter or higher above the floor have an adjustable angle body. Ensure scale is not less than 180 millimeter long and the case face is manufactured from manufacturer's standard polished aluminum JIS G 4053 series polished corrosion-resistant steel. Provide thermometers with nonferrous separable wells. Provide lagging extension to accommodate insulation thickness.

2.2.15 Pump Suction Strainers

Provide a cast iron strainer body, rated for not less than 172 kilopascal at 38 degrees C, with flanges conforming to JIS B 2239. Strainer construction is such that there is a machined surface joint between body and basket that is normal to the centerline of the basket.

Ensure minimum ratio of open area of each basket to pipe area is 3 to 1. Provide a basket with JIS G 4053 series corrosion-resistant steel wire mesh with perforated backing.

Ensure mesh is capable of retaining all particles larger than 1,000 micrometer, with a pressure drop across the strainer body of not more than 5 kilopascal when the basket is two-thirds dirty at maximum system flow rate. Provide reducing fittings from strainer-flange size to pipe size.

Provide a ~~[~~differential-pressure gage~~]~~ fitted with a two-way brass cock across the strainer.

Provide manual air vent cocks in cap of each strainer.

2.2.16 Line Strainers, Water Service

Install Y-type strainers with removable basket. Ensure strainers in sizes DN50 and smaller have screwed ends; in sizes DN65 and larger, strainers have flanged ends. Ensure body working-pressure rating exceeds maximum service pressure of installed system by at least 50 percent. Ensure body has cast-in arrows to indicate direction of flow. Ensure all strainer bodies fitted with screwed screen retainers have straight threads and gasketed with nonferrous metal. For strainer bodies DN65 and larger, fitted with bolted-on screen retainers, provide offset blowdown holes. Fit all strainers larger than DN65 with manufacturer's standard ball-type blowdown valve. Ensure body material is ~~[~~cast bronze conforming to MLIT-M ~~] [cast iron conforming to Class 30]~~. Where system material is nonferrous, use nonferrous metal for the metal strainer body material.

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 1.14 millimeter. Ensure strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is ~~[~~SUS 304, or SUS 316 corrosion-resistant steel~~] [Monel metal]~~.

2.2.17 Line Strainers, Steam Service

Install Type Y strainers with removable strainer element.

Use flanged body end connections for all valves larger than DN50, unless butt weld ends are specified. Use ~~[screwed] [~~socket~~]~~ weld for sizes DN50 and under to suit specified piping system end connection and maintenance requirements~~ [or be welded]~~.

For strainers located in tunnels, trenches, manholes, and valve pits,use welded end connections.

Body working steam pressure rating is the same as the primary valve rating for system in which strainer is installed, except where welded end materials requirements result in higher pressure ratings. Ensure body has integral cast or forged arrows to indicate direction of flow. Provide strainer bodies with blowdown valves that have discharge end plugged with a solid metal plug. Make closure assembly with tetrafluoroethylene tape. Ensure bodies fitted with bolted-on screen retainers have offset blowdown holes.

Body materials are ~~[cast steel conforming to JIS G 5151, Grade WCB] [~~ forged carbon steel conforming to JIS G 3202 or JIS G 3201~~]~~ or ~~[~~ manufacturer's standard metallurgical equivalents for service pressures of 1035 kilopascal wsp and greater, and for lower pressure ratings where welding is required~~]~~ or ~~[~~cast iron conforming to JIS B 2031, Class B, for service pressures 862 kilopascal wsp and less~~]~~.

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 0.51 millimeter or equivalent wire mesh. Strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is SUS 304, or SUS 316 corrosion-resistant steel and fitted with backup screens where necessary to prevent collapse.

2.3 VALVES

Submit equipment and performance data for valves consisting of corrosion resistance and life expectancy. Submit design analysis and calculations consisting of rates of flow, head losses, inlet and outlet design, and pressure calculations. Also include in data, pipe dimensions, as well as temperature ratings, vibration and thrust limitations, minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

2.3.1 Ball and Butterfly Valves

Ensure ball valves conform to JIS B 2061. For valve bodies in sizes DN50 and smaller, use screwed-end connection-type constructed of copper alloy. For valve bodies in sizes DN50~~ DN65~~ and larger, use flanged-end connection type, constructed of material. Balls and stems of valves DN50 and smaller are manufacturer's standard with hard chrome plating finish. Balls and stems of valves DN65 and larger are manufacturer's standard corrosion-resistant steel alloy with hard chrome plating. Balls of valves DN150 and larger may be Brinell hard chrome plating. Ensure valves are suitable for flow from either direction and seal equally tight in either direction. Valves with ball seals held in place by spring washers are not acceptable. Ensure all valves have adjustable packing glands. Seats and seals are fabricated from tetrafluoroethylene.

Ensure butterfly valves conform to JIS B 2032 and are the wafer type for mounting between specified flanges. Ensure valves are rated for 1034 kilopascal shutoff and nonshock working pressure. Select bodies of cast ferrous metal conforming to JIS B 2239 for body wall thickness. Seats and seals are fabricated from resilient elastomer designed for field removal and replacement.

2.3.2 Drain, Vent, and Gage Cocks

Provide ~~[~~T-head~~]~~ or ~~[~~lever handle~~]~~ drain, vent, and gage cocks, ground key type, with washer and screw, constructed of polished JIS B 2240 and rated 862 kilopascal wsp. Ensure end connections are rated for specified service pressure.

Ensure pump vent cocks, and where spray control is required, constructed of manufacturer's standard polished brass. Ensure cocks are 15 millimeter male, end threaded, and rated at not less than 862 kilopascal at 107 degrees C.

2.3.3 Gate Valves (GAV)

Ensure gate valves DN50 and smaller conform to JIS B 2011. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated use union-ring bonnet, screwed-end type. Make packing of non-asbestos type materials. Use rising stem type valves.

Ensure gate valves DN65 and larger, are Type I, (solid wedge disc, tapered seats, steam rated); Class 125 (862 kilopascal steam-working pressure at 178 degrees C saturation); and 1379 kilopascal, wog (nonshock), conforming to JIS B 2031 and to requirements specified herein. Select flanged valves, with bronze trim and outside screw and yoke (OS&Y) construction. Make packing of non-asbestos type materials.

2.3.4 Globe and Angle Valves (GLV-ANV)

Ensure globe and angle valves DN50 and smaller, are 862 kilopascal conforming to JIS B 2011 and to requirements specified herein. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated, use union-ring bonnet, screwed-end type. Ensure disc is free to swivel on the stem in all valve sizes. Composition seating-surface disc construction may be substituted for all metal-disc construction. Make packing of non-asbestos type materials. Ensure disk and packing are suitable for pipe service installed.

Ensure globe and angle valves, DN65 and larger, are cast iron with bronze trim. Ensure valve bodies are cast iron conforming to JIS B 2011. Select flanged valves in conformance with JIS B 2239. Valve construction is outside screw and yoke (OS&Y) type. Make packing of non-asbestos type materials.

2.3.5 Standard Check Valves (SCV)

Ensure standard check valves in sizes DN50 and smaller are 862 kilopascal swing check valves except as otherwise specified. Provide lift checks where indicated. Ensure swing-check pins are nonferrous and suitably hard for the service. Select composition type discs. Ensure the swing-check angle of closure is manufacturer's standard unless a specific angle is needed.

Use cast iron, bronze trim, swing type check valves in sizes DN65 and larger. Ensure valve bodies are cast iron, conforming to~~–~~ JIS B 2031 and valve ends are flanged in conformance with JIS B 2239. Swing-check pin is approved equal corrosion-resistant steel. Angle of closure is manufacturer's standard unless a specific angle is needed. Ensure valves have bolted and gasketed covers.

Provide check valves with external spring-loaded or lever-weighted, positive-closure devices and valve ends are flanged.

2.3.6 Nonslam Check Valves (NSV)

Provide check valves at pump discharges in sizes DN50 and larger with nonslam or silent-check operation conforming to JIS B 2031. Select a valve disc or plate that closes before line flow can reverse to eliminate slam and water-hammer due to check-valve closure. Ensure valve is Class 125 rated for 1379 kilopascal maximum, nonshock pressure at 66 degrees C in sizes to DN300. Use valves that are ~~[~~wafer type to fit between flanges conforming to JIS B 2239~~]~~. Valve body may be cast iron, or equivalent strength ductile iron. Select disks using manufacturer's standard bronze, aluminum bronze, or corrosion-resistant steel. Ensure pins, springs, and miscellaneous trim are manufacturer's standard corrosion-resistant steel.

2.4 MISCELLANEOUS MATERIALS

Submit equipment and performance data for miscellaneous materials consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

2.4.1 Bituminous Coating

Ensure the bituminous coating is a solvent cutback, heavy-bodied material to produce not less than a 0.30 millimeter dry-film thickness in one coat, and is recommended by the manufacturer to be compatible with factory-applied coating and rubber joints.

For previously coal-tar coated and uncoated ferrous surfaces underground, use bituminous coating solvent cutback coal-tar type.

2.4.2 Bolting

Ensure flange and general purpose bolting is hex-head and conforms to JIS G 4107, or above~~ (bolts, for flanged joints in piping systems where one or both flanges are cast iron)~~. Heavy hex-nuts conform to JIS B 1181. Square-head bolts and nuts are not acceptable. Ensure threads are coarse-thread series.

2.4.3 Elastomer Caulk

Use two-component polysulfide- or polyurethane-base elastomer caulking material, conforming to JIS K 7311.

2.4.4 Escutcheons

Manufacture escutcheons from nonferrous metals and chrome-plated except when JIS G 4053 series corrosion-resistant steel is provided. Ensure metals and finish conforms to Japanese standard.

Use one-piece escutcheons where mounted on chrome-plated pipe or tubing, and one-piece of split-pattern type elsewhere. Ensure all escutcheons have provisions consisting of internal spring-tension devices or setscrews for maintaining a fixed position against a surface.

2.4.5 Flashing

Ensure sheetlead conforms to Japanese standard

Ensure sheet copper conforms to JIS H 3100 and be not less than 4.88 kilogram per square meter weight.

2.4.6 Flange Gaskets

Provide compressed non-asbestos sheets, conforming to JIS F 0602, coated on both sides with graphite or similar lubricant, with nitrile composition, binder rated to 399 degrees C.

2.4.7 Grout

Provide shrink-resistant grout as a premixed and packaged metallic-aggregate, mortar-grouting compound conforming to JIS A 0203.

Ensure shrink-resistant grout is a combination of pre-measured and packaged epoxy polyamide or amine resins and selected aggregate mortar grouting compound conforming to the following requirements:

Tensile strength		13.100 Megapascal, minimum
Compressive strength	JIS A 1108 ~~JIS A 1108~~	96.527 Megapascal, minimum
Shrinkage, linear maximum		0.003 mm per millimeter,
Water absorption	JIS A 1110	0.1 percent, maximum
Bond strength to		6.895 Megapascal, minimum steel in shear minimum

2.4.8 Pipe Thread Compounds

Use polytetrafluoroethylene tape not less than 0.05 to 0.08 millimeter thick in potable and process water and in chemical systems for pipe sizes to and including DN25. Use polytetrafluoroethylene dispersions and other suitable compounds for all other applications upon approval by the Contracting Officer; however, do not use lead-containing compounds in potable water systems.

2.5 SUPPORTING ELEMENTS

Submit equipment and performance data for the supporting elements consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

Provide all necessary piping systems and equipment supporting elements, including but not limited to: building structure attachments; supplementary steel; hanger rods, stanchions, and fixtures; vertical pipe attachments; horizontal pipe attachments; anchors; guides; and spring-cushion, variable, or constant supports. Ensure supporting elements are suitable for stresses imposed by systems pressures and temperatures and natural and other external forces normal to this facility without damage to supporting element system or to work being supported.

Ensure supporting elements conform to requirements of MLIT-M, except as noted.

Ensure attachments welded to pipe are made of materials identical to that of pipe or materials accepted as permissible raw materials by referenced code or standard specification.

Ensure supporting elements exposed to weather are hot-dip galvanized or stainless steel. Select materials of such a nature that their apparent and latent-strength characteristics are not reduced due to galvanizing process. Electroplate supporting elements in contact with copper tubing with copper.

Ensure masonry anchor group-, type-, and style-combination designations are in accordance with JIS A 5758 and JIS B 1112. Provide support elements, except for supplementary steel, that are cataloged, load rated, commercially manufactured products.

2.5.1 Building Structure Attachments

2.5.1.1 Anchor Devices, Concrete and Masonry

Ensure anchor devices conform to JIS A 5758 and JIS B 1220.

For cast-in, floor mounted, equipment anchor devices, provide adjustable positions.

Do not use powder-actuated anchoring devices to support any mechanical systems components.

2.5.1.2 Beam Clamps

Ensure beam clamps are center-loading conforming to MLIT-M.

When it is not possible to use center-loading beam clamps, eccentric-loading beam clamps, conforming to MLIT-M may be used for piping sizes DN50 and less and for piping sizes DN50 through DN250 provided two counterbalancing clamps are used per point of pipe support. Where more than one rod is used per point of pipe support, determine rod diameter in accordance with referenced standards.

2.5.1.3 C-Clamps

Do not use C-clamps.

2.5.1.4 Inserts, Concrete

Use concrete conforming to MLIT-M inserts. When applied to piping in sizes DN50 and larger and where otherwise required by imposed loads, insert and wire a 305 millimeter length of 13 millimeter reinforcing rod through wing slots. Submit proprietary-type continuous inserts for approval.

2.5.2 Horizontal Pipe Attachments

2.5.2.1 Single Pipes

Support piping in sizes to and including DN50 by conforming to MLIT-M solid malleable iron pipe rings, except that, use split-band-type rings in sizes up to DN25.

Support piping in sizes through DN200 inclusive by conforming to MLIT-M.

Use MLIT-M assemblies on vapor-sealed insulated piping and have an inside diameter larger than pipe being supported to provide adequate clearance during pipe movement.

Where thermal movement of a point in a piping system DN100 and larger would cause a hanger rod to deflect more than 4 degrees from the vertical or where a horizontal point movement exceeds 13 millimeter, use conforming to MLIT-M.

Support piping in sizes larger than DN200 with conforming to MLIT-M.

Use conforming to MLIT-M shields on all insulated piping. Ensure area of the supporting surface is such that compression deformation of insulated surfaces does not occur. Roll away longitudinal and transverse shield edges from the insulation.

Provide insulated piping without vapor barrier on roll supports with conforming to MLIT-M saddles.

Provide spring supports as indicated.

2.5.2.2 Parallel Pipes

Use trapeze hangers fabricated from structural steel shapes, with U-bolts, in congested areas and where multiple pipe runs occur. Ensure structural steel shapes ~~[~~conform to supplementary steel requirements~~]~~ or ~~[~~be of commercially available, proprietary design, rolled steel~~]~~.

2.5.3 Vertical Pipe Attachments

Ensure vertical pipe attachments are conforming to MLIT-M.

Include complete fabrication and attachment details of any spring supports in shop drawings.

2.5.4 Hanger Rods and Fixtures

Use only circular cross section rod hangers to connect building structure attachments to pipe support devices. Use pipe, straps, or bars of equivalent strength for hangers only where approved by the Contracting Officer.

Provide turnbuckles, swing eyes, and clevises as required by support system to accommodate temperature change, pipe accessibility, and adjustment for load and pitch. Rod couplings are not acceptable.

2.5.5 Supplementary Steel

Where it is necessary to frame structural members between existing members or where structural members are used in lieu of commercially rated supports, design and fabricate such supplementary steel in accordance with MLIT-M.

PART 3 EXECUTION

3.1 PIPE INSTALLATION

Submit certificates for pipes, valves and specialties showing conformance with test requirements as contained in the reference standards contained in this section. Provide certificates verifying Surface Resistance, Shear and Tensile Strengths, Temperature Ratings, Bending Tests, Flattening Tests and Transverse Guided Weld Bend Tests.

Fabricate and install piping systems in accordance with JIS HB 40-1 and JIS HB 40-2.

Ensure connections between steel piping and copper piping are electrically isolated from each other with ~~[dielectric couplings (or unions)] [~~flanged with gaskets~~]~~ rated for the service.

Make final connections to equipment with ~~[unions] [~~flanges~~]~~ provided every 30480 millimeter of straight run. Provide unions in the line downstream of screwed- and welded-end valves.

Ream all pipe ends before joint connections are made.

Make screwed joints with specified joint compound with not more than three threads showing after joint is made up.

Apply joint compounds to the male thread only and exercise care to prevent compound from reaching the unthreaded interior of the pipe.

Provide screwed unions, welded unions, or bolted flanges wherever required to permit convenient removal of equipment, valves, and piping accessories from the piping system for maintenance.

Securely support piping systems with due allowance for thrust forces, thermal expansion and contraction. Do not subject the system to mechanical, chemical, vibrational or other damage as specified in MLIT-M.

Ensure field welded joints conform to the requirements of the JIS HB 40-1 and JIS HB 40-2 and JIS B 8285.

~~[~~ Accomplish preheat and postheat treatment of welds in accordance with JIS B 8285.

~~][~~Take all necessary precautions during installation of flexible pipe and hose including flushing and purging with water, steam, and compressed air to preclude bellows failure due to pipe line debris lodged in bellows. Ensure installation conforms to manufacturer's instructions.

~~]~~3.2 VALVES

Provide valves in piping mains and all branches and at equipment where indicated and as specified.

Provide valves to permit isolation of branch piping and each equipment item from the balance of the system.

Provide riser and downcomer drains above piping shutoff valves in piping DN65 and larger. Tap and fit shutoff valve body with a DN15 plugged globe valve.

Provide valves unavoidably located in furred or other normally inaccessible places with access panels adequately sized for the location and located so that concealed items may be serviced, maintained, or replaced.

3.3 SUPPORTING ELEMENTS INSTALLATION

Provide supporting elements in accordance with the referenced codes and standards.

Support piping from building structure. Do not support piping from roof deck or from other pipe.

Run piping parallel with the lines of the building. Space and install piping and components so that a threaded pipe fitting may be removed between adjacent pipes and so that there is no less than DN15 of clear space between the finished surface and other work and between the finished surface of parallel adjacent piping. Arrange hangars on different adjacent service lines running parallel with each other in line with each other and parallel to the lines of the building.

Install piping support elements at intervals specified hereinafter, at locations not more than 900 millimeter from the ends of each runout, and not over 300 millimeter from each change in direction of piping.

Base load rating for all pipe-hanger supports on insulated weight of lines filled with water and forces imposed. Deflection per span is not exceed slope gradient of pipe. Ensure supports are in accordance with the following minimum rod size and maximum allowable hanger spacing for specified pipe. For concentrated loads such as valves, reduce the allowable span proportionately:

PIPE SIZE (DN) MILLIMETER	ROD SIZE MILLIMETER	STEEL PIPE MILLIMETER	COPPER PIPE MILLIMETER
25 and smaller	10	2500	1850
32 to 40	10	3050	2500
50	10	3050	3050
65 to 90	13	3700	3700
100 to 125	16	5000	4300
150	20	5000	5000
200 to 300	22	6100	6100
356 to 457	25	6100	6100
508 and over	32	6100	6100

Provide vibration isolation supports where needed. Refer to Section 23 05 48.~~00 40~~19 VIBRATION AND SEISMIC CONTROLS FOR HVAC ~~PIPING AND EQUIPMENT~~ where A/C equipment and piping is installed.

Support vertical risers independently of connected horizontal piping, whenever practicable, with fixed or spring supports at the base and at intervals to accommodate system range of thermal conditions. Ensure risers have guides for lateral stability. For risers subject to expansion, provide only one rigid support at a point approximately one-third down from the top. Place clamps under fittings unless otherwise specified. Support carbon-steel pipe at each floor and at not more than 4572 millimeter intervals for pipe DN50 and smaller and at not more than 6096 millimeter intervals for pipe DN65 and larger.

3.4 PENETRATIONS

Provide effective sound stopping and adequate operating clearance to prevent structure contact where piping penetrates walls, floors, or ceilings into occupied spaces adjacent to equipment rooms; where similar penetrations occur between occupied spaces; and where penetrations occur from pipe chases into occupied spaces. Occupied spaces include space above ceilings where no special acoustic treatment of ceiling is provided. Finish penetrations to be compatible with surface being penetrated.

~~[~~ Accomplish sound stopping and vapor-barrier sealing of pipe shafts and large floor and wall openings by packing to high density with properly supported fibrous-glass insulation or, where ambient or surface temperatures do not exceed 49 degrees C, by foaming-in-place with self-extinguishing, 0.9 kilogram density polyurethane foam to a depth not less than 152 millimeter. Finish foam with a rasp. Ensure vapor barrier is not less than 3 millimeter thick vinyl coating applied to visible and accessible surfaces. Where high temperatures and fire stopping are a consideration, use only mineral wool with openings covered by 1.6 millimeter sheet metal.

~~]~~3.5 SLEEVES

Provide sleeves where piping passes through roofs, masonry, concrete walls and floors.

Continuously ~~[weld] [~~braze~~]~~ sleeves passing through steel decks to the deck.

Ensure sleeves that extend through floors, roofs, load bearing walls, and fire barriers are continuous and fabricated from Schedule 40 steel pipe, with welded anchor lugs. Form all other sleeves by molded linear polyethylene liners or similar materials that are removable. Ensure diameter of sleeves is large enough to accommodate pipe, insulation, and jacketing without touching the sleeve and provides a minimum 10 millimeter clearance. Install a sleeve size to accommodate mechanical and thermal motion of pipe precluding transmission of vibration to walls and the generation of noise.

Pack the space between a pipe, bare or insulated, and the inside of a pipe sleeve or a construction surface penetration solid with a mineral fiber conforming to JIS A 9504 to 538 degrees C. Provide this packing wherever the piping passes through firewalls, equipment room walls, floors, and ceilings connected to occupied spaces, and other locations where sleeves or construction-surface penetrations occur between occupied spaces. Where sleeves or construction surface penetrations occur between conditioned and unconditioned spaces, fill the space between a pipe, bare or insulated, and the inside of a pipe sleeve or construction surface penetration with an elastomer caulk to a depth of 13 millimeter. Ensure all caulked surfaces are oil- and grease-free.

Ensure through-penetration fire stop materials and methods are in accordance with ISO 834-1.

Caulk exterior wall sleeves watertight with lead and oakum or mechanically expandable chloroprene inserts with mastic-sealed metal components.

Ensure sleeve height above roof surface is a minimum of 305 and a maximum of 457 millimeter.

3.6 ESCUTCHEONS

Provide escutcheons at all penetrations of piping into finished areas. Where finished areas are separated by partitions through which piping passes, provide escutcheons on both sides of the partition. Where suspended ceilings are installed, provide plates at the underside only of such ceilings. For insulated pipes, select plates large enough to fit around the insulation. Use chrome-plated escutcheons in all occupied spaces and of size sufficient to effectively conceal openings in building construction. Firmly attach escutcheons with setscrews.

3.7 FLASHINGS

Provide flashings at penetrations of building boundaries by mechanical systems and related work.

3.8 UNDERGROUND PIPING INSTALLATION

Prior to being lowered into a trench, clean all piping, visually inspected for apparent defects, and tapped with a hammer to audibly detect hidden defects.

Further inspect suspect cast-ferrous piping by painting with kerosene on external surfaces to reveal cracks.

Distinctly mark defective materials found using a road-traffic quality yellow paint; promptly remove defective material from the site.

After conduit has been inspected, and not less than 48 hours prior to being lowered into a trench, coat all external surfaces of cast ferrous conduit with a compatible bituminous coating for protection against brackish ground water. Apply a single coat, in accordance with the manufacturer's instructions, to result in a dry-film thickness of not less than 0.30 millimeter.

Ensure excavations are dry and clear of extraneous materials when pipe is being laid.

Use wheel cutters for cutting of piping or other machines designed specifically for that purpose. Electric-arc and oxyacetylene cutting is not permitted.

Begin laying of pipe at the low point of a system. When in final acceptance position, ensure it is true to the grades and alignment indicated, with unbroken continuity of invert. Blocking and wedging is not permitted.

Point bell or grooved ends of piping upstream.

Make changes in direction with long sweep fittings.

Provide necessary socket clamping, piers, bases, anchors, and thrust blocking. Protect rods, clamps, and bolting with a coating of bitumen.

Support underground piping below supported or suspended slabs from the slab with a minimum of two supports per length of pipe. Protect supports with a coating of bitumen.

On excavations that occur near and below building footings, provide backfilling material consisting of 13800 kilopascal cured compressive-strength concrete poured or pressure-grouted up to the level of the footing.

Properly support vertical downspouts; soil, waste, and vent stacks; water risers; and similar work on approved piers at the base and provided with approved structural supports attached to building construction.

~~[~~Provide cleanout, flushing, and observation risers.~~]~~

~~3.9 HEAT TRACE CABLE INSTALLATION~~

~~Field apply heater tape and cut to fit as necessary, linearly along the length of pipe after piping has been pressure tested and approved by the Contracting Officer. Secure the heater to piping with [cable ties] [fiberglass tape]. Label thermal insulation on the outside, "Electrical Heat Trace."~~

~~Install power connection, end seals, splice kits and tee kit components provide a complete workable system. Terminate connection to the thermostat and ends of the heat tape in a junction box. Ensure cable and conduit connections are raintight.~~

3.9 DISINFECTION

~~[~~Disinfect water piping, including all valves, fittings, and other devices, with a solution of chlorine and water. Ensure the solution contains not less than 50 parts per million (ppm) of available chlorine. Hold solution for a period of not less than 8 hours, after which the solution contains not less than 10 ppm of available chlorine or redisinfect the piping. After successful sterilization, thoroughly flush the piping before placing into service. Flushing is complete when the flush water contains less than 0.5 ppm of available chlorine. Water for disinfected will be furnished by the Government. Approve disposal of contaminated flush water in accordance with written instructions received from the Environmental authority having jurisdiction through the Contracting Officer and all local, State and Federal Regulations.~~]~~

~~[~~Flush piping with potable water until visible grease, dirt and other contaminants are removed (visual inspection).~~]~~

3.10 OPERATION AND MAINTENANCE

Provide Operation and Maintenance Manuals consistent with manufacturer's standard brochures, schematics, printed instructions, general operating procedures and safety precautions. Submit test data that is clear and readily legible.

3.11 PAINTING OF NEW EQUIPMENT

Factory or shop apply new equipment painting, as specified herein, and provided under each individual section.

3.11.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied withstands 125 hours in a salt-spray fog test, except that equipment located outdoors withstand 500 hours in a salt-spray fog test. Conduct salt-spray fog test is in accordance with Japanese Industry Standards, and for that test the acceptance criteria is as follows: immediately after completion of the test, the inspected paint shows no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shows no signs of rust creepage beyond 3 mm on either side of the scratch mark.

Ensure the film thickness of the factory painting system applied on the equipment is not less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, design the factory painting system for the temperature service.

3.11.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal, surfaces subject to temperatures in excess of 50 degrees C.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Selected color of finish coat is aluminum or light gray.

a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C receives one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.

b. Temperatures Between 50 and 205 Degrees C: Metal surfaces subject to temperatures between 50 and 205 degrees C Receives two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm .

c. Temperatures Greater Than 205 Degrees C: Metal surfaces subject to temperatures greater than 205 degrees C receives two coats of 315 degrees C heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm.

-- End of Section --

SECTION 23 05 48.19

~~[~~SEISMIC~~]~~ BRACING FOR HVAC
05/18, CHG 2: 08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)

FEMA P-414	(January 2004) Installing Seismic Restraints for Duct and Pipe

ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC ES AC193	(2012) Acceptance Criteria for Mechanical Anchors in Concrete Elements

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1048	(2007) Fasteners -- Hot Dip Galvanized Coatings
JIS B 1186	(2013) Sets of High Strength Hexagon Bolt, Hexagon Nut and Plain Washers for Friction Grip Joints
JIS G 3101	~~(2020) Rolled Steels for General Structure~~ (2024) Rolled Steels for General Structure
JIS G 3106	~~(2020) Rolled Steels for Welded Structure~~ (2024) Rolled Steels for Welded Structure
JIS G 3114	(2022) Hot-Rolled Atmospheric Corrosion Resisting Steels for Welded Structure
JIS G 3136	(2022) Rolled Steels for Building Structure
JIS G 3443-3	(2020) Coated Steel Pipes for Water Service -- Part 3: Long-Life External Plastic Coatings
JIS G 3443-4	(2020) Coated Steel Pipes for Water Service -- Part 4: Internal Epoxy Coatings
JIS G 3444	(2021) Carbon Steel Tubes for General Structure
JIS G 3466	(2021) Carbon Steel Square and Rectangular Tubes for General Structure

JIS G 3475 (2021) Carbon Steel Tubes for Building Structure

JIS G 3547 (2015) Zinc-Coated Low Carbon Steel Wires

JIS G 5502 (2024) Spheroidal Graphite Cast Irons (Amendment 1)

JIS G 5527 (2014) Ductile Iron Fittings

JIS K 6741 (2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes

JIS K 6742 (2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes for Water Supply

JIS K 6743 (2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Water Supply

JIS S 3200-7 (2004) Equipment for Water Supply Service -- Test Methods of Effect to Water Quality

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

MFMA-4 (2004) Metal Framing Standards Publication

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM, GOVERNMENT OF JAPAN (MLIT)

MLIT-M (2019) Public Building Construction Standard Specification

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1981 (2008) Seismic Restraint Manual Guidelines for Mechanical Systems, 3rd Edition

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 3~~1~~, 20~~23~~25) Structural Engineering

UFC 4-010-01 (2018; with Change 1, 2020) DoD Minimum Antiterrorism Standards for Buildings

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION (VISCMA)

VISCMA 412 (2014) Installing Seismic Restraints for Mechanical Equipment

1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

Apply the requirements for seismic protection measures described in this section and on the drawings to the mechanical equipment and mechanical systems both inside and outside of the building along with exterior utilities and systems listed below. Where there is a conflict between the specifications and the drawings, the specifications will take precedence. Accomplish resistance to lateral forces induced by earthquakes without consideration of friction resulting from gravity loads.

1.2.2 Mechanical Equipment

Mechanical equipment to be seismically protected must include the following items to the extent required on the drawings or in other sections of these specifications:

~~[~~Equipment/Components with Ip = 1.0~~]~~

~~[Equipment/Components with Ip = 1.5 (Designated Seismic Systems)~~
~~Insert edited list here similar to one above for Ip = 1.0]~~
~~[Non-Mission Critical (NMC) Equipment/Components in Risk Category V~~
~~Insert edited list here similar to one above for Ip = 1.0]~~
~~[Mission Critical Level 1 (MC-1) Equipment/Components in Risk Category V~~
~~Insert edited list here similar to one above for Ip = 1.0]~~
~~[Mission Critical Level 2 (MC-2) Equipment/Components in Risk Category V~~
~~Insert edited list here similar to one above for Ip = 1.0]~~

1.2.3 Mechanical Systems

Mechanical systems to be seismically protected must include the following items to the extent required on the drawings or in this or other sections of these specifications:

~~[~~Mechanical systems with Ip = 1.0~~]~~

a. All Piping and Ducts Inside the Building Except as Specifically Stated Below Under "Items Not Covered By This Section".

b. Chilled Water Distribution Systems Outside of Buildings.

c. Steam, Water, ~~Oil, Gas~~ and Fuel Piping Outside of Buildings.

d. All Water Supply Systems Outside of Buildings.

e. Storm and Sanitary Sewer Systems Outside of Buildings.

f. All Process Piping Outside of Buildings.

g. Heat Distribution Systems (Supply, Return, and Condensate Return) Outside of Buildings.

h. ~~Condenser Water and~~ Refrigerant Piping Outside the Building.

~~i. Pneumatic Tube Distribution System Outside of Buildings.~~

~~j. Cold Storage Refrigeration Systems Outside of Buildings.~~

~~k. Fuel Storage Tanks Outside of Buildings.~~

~~l. Water Storage Tanks Outside of Buildings.~~

~~m~~i. Ductwork Outside of Buildings.

~~n. Stacks.~~

~~o. [_____]~~

~~[Mechanical systems with Ip = 1.5 (Designated Seismic Systems) Insert edited list here similar to one above for Ip = 1.0] [Non-Mission Critical (NMC) Mechanical Systems in Risk Category V Insert edited list here similar to one above for Ip = 1.0] [Mission Critical Level 1 (MC-1) Mechanical Systems in Risk Category V Insert edited list here similar to one above for Ip = 1.0] [Mission Critical Level 2 (MC-2) Mechanical Systems in Risk Category V Insert edited list here similar to one above for Ip = 1.0]~~

1.2.4 Contractor Designed Bracing

Submit copies of the design calculations with the drawings. Calculations must be approved, certified, stamped and signed by a registered Professional Structural Engineer. A non-U.S. citizen may qualify with a 1 Kyu Kenchikushi (1st Class Qualified Architect), in lieu of U.S. licensing qualifications. Calculations must verify the capability of structural members to which bracing is attached for carrying the load from the brace. Design the bracing in accordance with UFC 3-301-01, ~~[UFC 3-301-02],~~ UFC 4-010-01 and additional data furnished by the Contracting Officer. Resistance to lateral forces induced by earthquakes must be accomplished without consideration of friction resulting from gravity loads. UFC 3-301-01 uses parameters for the building, not for the equipment in the building; therefore, corresponding adjustments to the formulas must be required. Loadings determined using UFC 3-301-01 are based on strength design; therefore, ~~AISC 325~~JIS G 3101 and JIS G 3106 Specifications must be used for the design. The bracing for the mechanical equipment designated in paragraph 1.2.2 and systems designated in paragraph 1.2.3 must be developed by the Contractor.

~~[Provide documentation of an independent design review for mission critical (MC) equipment bracing design. Documentation must be signed by the independent reviewer who must also be a registered structural engineer.]~~

1.2.5 Items Not Covered By This Section

1.2.5.1 Fire Protection Systems

Install seismic protection of piping for fire protection systems as specified in Sections 21 30 00 FIRE PUMPS, 21 13 13 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION, and 21 13 16 DRY PIPE SPRINKLER SYSTEMS, FIRE PROTECTION~~, 21 13 18 PREACTION SPRINKLER SYSTEMS, FIRE PROTECTION, and 21 13 24.00 10 AQUEOUS FILM-FORMING FOAM (AFFF) FIRE PROTECTION SYSTEM~~.

1.2.5.2 Items Requiring No Seismic Restraints

Seismic restraints are not required for the following items:

~~a. Gas piping less than 25 mm nominal pipe size.~~

~~b~~a. Piping in boiler and mechanical equipment rooms less than 32 mm nominal pipe size.

~~c~~b. All other piping equal to or less than 80 mm nominal pipe size.

~~d~~c. Rectangular air handling ducts less than 0.56 square meters in cross sectional area.

~~e~~d. Round air handling ducts less than 711 mm in diameter.

~~f~~e. Piping suspended by individual hangers 300 mm or less in length from the top of pipe to the bottom of the supporting structural member where the hanger is attached, except as noted below.

~~g~~f. Ducts suspended by hangers 300 mm or less in length from the top of the duct to the bottom of the supporting structural member, except as noted below.

In exemptions ~~f~~e. and ~~g~~f. all hangers must meet the length requirements. If the length requirement is exceeded by one hanger in the run, brace the entire run. Seismically protect interior piping and ducts not listed above in accordance with the provisions of this specification.

Non-critical items may require seismic restraints if adjacent to critical equipment or systems that must remain operational after an earthquake and could be compromised by impact with non-critical adjacent components.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[~~ for Contractor Quality Control ~~approval.][for~~ information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Coupling and Bracing

Flexible Couplings or Joints

Equipment Restraint

Contractor Designed Bracing; G~~[, [_____]]~~

SD-03 Product Data

Coupling and Bracing; G~~[, [_____]]~~

Flexible Couplings Or Joints; G~~[, [_____]]~~

Equipment Restraint; G~~[, [_____]]~~

Contractor Designed Bracing; G~~[, [_____]]~~

Snubbers

Anchor Bolts

Vibration Isolators

SD-05 Design Data

Design Calculations

SD-06 Test Reports

Anchor Bolts; G~~[, [_____]]~~

PART 2 PRODUCTS

2.1 GENERAL DESIGN REQUIREMENTS

Submit detailed seismic restraint drawings for mechanical equipment, duct systems, piping systems and any other mechanical systems along with calculations, catalog cuts, templates, and erection and installation details, as appropriate, for the items listed below. Indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction. Calculations must be stamped, by a registered structural engineer, and verify the capability of structural members to which bracing is attached for carrying the load from the brace. Include drawing for Mission Critical Equipment indicating the equipment location in the facility sufficient to be used for the installation. Design must be based on actual equipment and system layout. Design must include calculated dead loads, static seismic loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods.

2.2 EQUIPMENT RESTRAINT

Equipment must be rigidly or flexibly mounted as indicated in the specifications and/or drawings depending on vibration isolation requirements as follows below.

Roof mounted equipment such as cooling towers and condensers, both vibration isolated and nonisolated, must have support members designed and anchored to building structural steel or concrete as required for seismic restraint and wind loads.

2.2.1 Rigidly (Base and Suspended) Mounted Equipment

HVAC equipment furnished under this contract must be ~~[rigidly mounted] [~~ rigidly mounted using cast-in-place anchor bolts or post-installed anchors~~]~~ that are qualified for earthquake loading in accordance with ~~ACI 355.2 and ACI 355.4~~MLIT-M. Anchor bolts must conform to ~~ASTM F1554 or~~ JIS B 1186. For any rigid equipment which is rigidly anchored, provide flexible joints for piping, ductwork, electrical conduit, etc., that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Suspended equipment bracing attachments should be located just above the center of gravity to minimize swinging. Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchor bolts for combined shear and overturning.

Roof mounted HVAC equipment roof curbs, framing and attachment to equipment and structure must be designed and braced to withstand seismic loads. ~~[Mission critical base mounted and suspended equipment for Risk Category (RC) V,]~~ Designated Seismic Systems (DSS) assigned to Seismic

Design Category (SDC) C, D, E, or F and Risk Category IV components needed for continued operation after an earthquake must have two nuts provided on each anchor bolt.

2.2.2 Nonrigid or Flexibly-Mounted Equipment

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 6 mm. Equipment flexibly mounted on vibration isolators must have a bumper restraint or snubber in each horizontal direction and vertical restraints must be provided where required to resist overturning. Isolator housing and restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.

Spring vibration isolators must be seismically rated, restrained isolators for equipment subject to load variations and large external forces. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturer's will be a member of VISCMA. Design force, Fp, must be doubled for vibration isolators with an air gap greater than ~~0.25 inches as specified in ASCE 7-16, Chapter 13~~6 mm. Housed springs must not be used for seismic restraint applications because they cannot resist uplift.

2.3 BOLTS AND NUTS

Hex head bolts, and heavy hexagon nuts must be ~~ASTM A325 or~~ JIS B 1186,~~-~~ or ~~ASTM A490 or~~ JIS B 1186 bolts and ~~ASTM A563 or~~ JIS B 1186 ~~-~~nuts. Provide bolts and nuts galvanized in accordance with ~~ASTM A153/A153M~~ JIS B 1048 when used underground or exposed to weather.

2.4 FLEXIBLE JOINTS

Flexible joints must have same pressure and temperature ratings as adjoining pipe. Braided hoses must not be used where there is torsional or axial movement unless manufacturer allows it.

2.4.1 Braided Hose Expansion Joint

Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops are not acceptable. Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops must not be acceptable. Braided hose in a 60 degree flexible V loop arrangement must be used for small diameter pipe connections to coils in ~~variable-air-volume (VAV)terminal units and~~ fan coil units installed in suspended ductwork whether braced or unbraced.

All braided hose expansion joints must be manufactured in accordance with the documented manufacturers weld procedure specifications. The procedure qualification record must be used to document the execution of this procedure and must follow the general "guidelines" of ~~ASME Section IX~~ applicable Japanese standards. Each individual welder must conform to the in-house procedure qualification record and be qualified prior to each production lot. The testing of each individual welder must be documented in a welding procedure qualification record.

2.4.1.1 Corrugated Hose

Corrugated hose must be ~~[~~Type ~~[304] [~~321~~]~~ or ~~[~~316~~]~~ stainless steel~~] [bronze]~~. For fuel oil piping, bronze type shall be used. Braid must be ~~[~~ Type 304 stainless steel for any series 300 stainless steel hose~~]~~ or ~~[~~ bronze for any bronze hose~~]~~. Fittings materials of construction and end fitting type must be consistent with pipe material and equipment/ pipe connection fittings. Copper fittings must not be attached to stainless steel hose.

2.4.1.2 Flexible Hose Expansion Loops

Flexible hose expansion loops must have a factory supplied, hanger / support lug located at the bottom of the 180deg return. ~~[~~Flexible hose expansion loop(s) must be furnished with a plugged FPT to be used for a drain or air release vent.~~]~~ Flexible hose expansion loop(s) must be rated with an operating pressure which is the same as the adjoining pipe. The operating pressure must be based on burst pressure with a 4 to 1 safety factor. ~~[~~For steam service, the operating pressure must be based on burst pressure with a 8 to 1 safety factor.~~]~~

2.4.2 Double Ball Flexible Expansion Joint

Install flexible expansion joints manufactured of ductile iron conforming to the material requirements of ~~ASTM A536 and AWWA C153/A21.53~~JIS G 5502 and JIS G 5527 in the locations indicated on the drawings. Provide foundry certification of material upon request. Each flexible expansion joint must be pressure tested prior to shipment against its own restraint to a minimum of 350 psi (250 psi for flexible expansion joints 2 inch and 30 inches diameter and larger.) A minimum 2:1 safety factor, determined from the published pressure rating, must apply. Factory Mutual Approval for the 3 inch through 12 inch sizes is required. Each flexible expansion joint must consist of an expansion joint designed and cast as an integral part of a ball and socket type flexible joint, having a minimum per ball deflection of: 20°, 2" - 12"; 15°, 14" - 36"; 12°, 42"-48" and 4-inches minimum expansion. Additional expansion sleeves must be available and easily added or removed at the factory or in the field. Both standardized mechanical joint and flange end connections must be available.

2.4.2.1 Internal Surfaces

Line all internal surfaces (wetted parts) with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of ~~AWWA C213~~ JIS G 3443-4. Sealing gaskets must be constructed of EPDM. The coating must meet ~~NSF/ANSI 61~~JIS S 3200-7.

2.4.2.2 Exterior Surfaces

Coat exterior surfaces with a minimum of 6 mils of fusion bonded epoxy conforming to the applicable requirements of ~~AWWA C116/A21.16~~JIS G 3443-3 . Include appropriately sized polyethylene sleeves, meeting ~~AWWA C105/A21.5~~JIS G 3443-3 and other applicable Japanese standards, for direct buried applications.

2.4.3 Double Ball Flexible Expansion Joint Gravity Drain (Non-Pressurized)

Flexible expansion joints gravity drain must be installed in the locations indicated on the drawings and must be manufactured of pvc. All connections whether solvent weld or mechanical must be restrained to allow movement to be transferred to expansion joint. Each ball must allow up to 15 degrees deflection.

End connection outside diameters must be compatible with ~~ASTM D1785, ASTM D2665 and ASTM F891~~JIS K 6741, JIS K 6742, and JIS K 6743 PVC pipe and are to be solvent welded.

2.5 SWAY BRACING MATERIALS

Material used for members listed ~~[~~in this section~~] [and] [on the drawings]~~, must be structural steel conforming with the following:

a. Plates, rods, and rolled shapes, ~~ASTM A36/A36M or~~ JIS G 3101, JIS G 3106, JIS G 3114, JIS G 3136.

b. Wire rope, ~~ASTM A603~~JIS G 3547 pre-stretched. ~~[~~Class B galv coating ~~][Class C galv coating]~~ Ferrule clamps must be qualified by testing for use in seismic applications per VISCMA 412 or equivalent Japanese installation standards. A minimum of two clamps are required on each end of wire rope.

c. Tubes, ~~ASTM A500/A500M, Grade B. or~~ JIS G 3444, JIS G 3475

d. Pipes, ~~ASTM A53/A53M, Grade B or~~ JIS G 3444, JIS G 3466.

e. Angles, ~~ASTM A36/A36M or~~ JIS G 3101, JIS G 3106, JIS G 3114, JIS G 3136.

f. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at random points conforming to MFMA-4.

2.6 MULTIDIRECTIONAL SEISMIC SNUBBERS

Install multidirectional seismic snubbers employing elastomeric pads on ~~[~~ floor- or slab-mounted equipment~~]~~ ~~[~~and~~]~~ ~~[~~large piping~~]~~ as detailed on drawings. These snubbers must provide 6 mm free vertical and horizontal movement from the static deflection point. Snubber medium must consist of multiple pads of cotton duct and neoprene or other suitable materials arranged around a flanged steel trunnion so both horizontal and vertical forces are resisted by the snubber medium.

PART 3 EXECUTION

3.1 COUPLING AND BRACING

a. Submit detail drawings, as specified here and throughout this specification, along with catalog cuts, templates, and erection and installation details, as appropriate, for the items listed. Submittals must be complete in detail; must indicate thickness, type, grade, class of metal, and dimensions; and must show construction details, reinforcement, anchorage, and installation with relation to the building construction.

b. Provide coupling installation conforming to the details shown on the drawings. Provisions of this paragraph apply to all piping within a 1.5 m line around outside of building unless buried in the ground. Piping grouped for support on trapeze-type hangers must be braced at the most frequent interval as determined by applying the requirements of this specification to each piping run on the common support.

c. Size bracing components as required for the total load carried by the common supports. Bracing rigidly attached to pipe flanges, or similar, must not be used where it would interfere with thermal expansion of piping.

d. Adjust isolators and restraints after piping systems has been filled and equipment is at its operating weight, following the manufacturer's written instructions.

e. Install cables at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412.

3.2 BUILDING DRIFT

Provide joints capable of accommodating seismic displacements for vertical piping between floors of the building, where pipes pass through a building seismic or expansion joint, or where rigidly supported pipes connect to equipment with vibration isolators. Provide horizontal piping across expansion joints to accommodate the resultant of the drifts of each building unit in each orthogonal direction. For threaded piping, provide swing joints made of the same piping material. For piping with manufactured ball joints the seismic drift must be ~~[~~0.015~~] [______]~~ meters per meter of height above the base where the seismic separation occurs; this drift value must be used in place of the expansion given in the manufacturer's selection table.

3.3 FLEXIBLE COUPLINGS OR JOINTS

3.3.1 Building Piping

Provide flexible couplings or joints in building piping at bottom of all pipe risers for pipe larger than 90 mm in diameter. Laterally brace flexible couplings or joints without interfering with the action of the flexible coupling or joint. Cast iron waste and vent piping need only comply with these provisions when caulked joints are used. Flexible bell and spigot pipe joints using rubber gaskets may be used at each branch adjacent to tees and elbows for underground waste piping inside of building to satisfy these requirements.

3.3.2 Underground Piping

Install flexible coupling in underground piping and 100 mm or larger conduit, except heat distribution system, where the piping enters the building. Provide couplings that accommodate ~~[______]~~75 mm of relative movement between the pipe and the building in any direction. Provide additional flexible couplings where shown on the drawings.

3.4 PIPE SLEEVES

Size pipe sleeves in interior non-fire rated walls as indicated on the drawings to provide clearances that will permit differential movement of piping without the piping striking the pipe sleeve. Pipe sleeves in fire rated walls must conform to the requirements in Section 07 84 00 FIRESTOPPING.

3.5 SPREADERS

Provide spreaders between adjacent piping runs to prevent contact during seismic activity whenever pipe or insulated pipe surfaces are less than ~~[~~ 100~~][_____]~~ mm apart. Apply spreaders at same interval as sway braces at an equal distance between the sway braces. If rack type hangers are used where the pipes are restrained from contact by mounting to the rack, spreaders are not required for pipes mounted in the rack. Apply spreaders to surface of bare pipe and over insulation on insulated pipes utilizing high-density inserts and pipe protection shields in accordance with the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

3.6 SWAY BRACES FOR PIPING

Provide sway braces to prevent movement of the pipes under seismic loading. Provide braces in both the longitudinal and transverse directions, relative to the axis of the pipe. Provide sufficient braces for equipment to resist a horizontal force as specified in UFC 3-301-01~~[UFC 3-301-02]~~ without exceeding safe working stress of bracing components. Provide bracing that does not interfere with thermal expansion requirements for the pipes as described in other sections of these specifications. For seismic analysis of horizontal pipes, the equivalent static force should be considered to act concurrently with the full dead load of the pipe, including contents.

3.6.1 Transverse Sway Bracing

Provide transverse sway bracing for steel and copper pipe at intervals not to exceed those shown on the drawings. All runs (length of pipe between end joints) must have a minimum of transverse bracing at each end. Provide transverse sway bracing for pipes of materials other than steel and copper at intervals not to exceed the hanger spacing as specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.6.2 Longitudinal Sway Bracing

Provide longitudinal sway bracing at 12 m intervals unless otherwise indicated. All runs (length of pipe between end joints) must have one longitudinal brace minimum. Construct sway braces in accordance with the drawings. Do not use branch lines, walls, or floors as sway braces.

3.6.3 Vertical Runs

Run is defined as length of pipe between end joints. Do not brace vertical runs of piping no more than 3 m vertical intervals. Braces for vertical runs must be above the center of gravity of the segment being braced. Flexible couplings should be provided at the bottoms of risers for pipes larger than 3.5 in. (89 mm) in diameter. Flexible couplings and expansion joints should be braced laterally and longitudinally unless such bracing would interfere with the action of the couplings or joints. When pipes enter buildings, flexible couplings should be provided to allow for relative movement between the soil and building. Construct all sway braces in accordance with the drawings. Attach sway braces to the structural system. Do not connect to branch lines, walls, or floors.

3.6.4 Clamps and Hangers

Apply clamps or hangers on uninsulated pipes directly to pipe. Insulated piping must have clamps or hangers applied over insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

Hanger rod stiffener angle or strut bracing must be securely attached by a series of attachment clamps manufactured from a one piece metal stamping and must include all require attachment hardware and locking nuts. Attachment clamps made from aluminum or cast iron must not be used in seismic applications. Do not weld vertical braces to hanger rods.

3.7 SWAY BRACES FOR DUCTS

3.7.1 Braced Ducts

Provide bracing details and spacing for rectangular and round ducts in accordance with SMACNA 1981 or equivalent Japanese standard. However, the design seismic loadings for these items must not be less than loadings obtained using the procedures in UFC 3-301-01~~[UFC 3-301-02]~~. Bracing must not attach to duct joints. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct.

3.7.2 Unbraced Ducts

Attach hangers for unbraced ducts to the duct within 50 mm of the top of the duct with a minimum of two #10 sheet metal screws in accordance with FEMA P-414. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct. Install unbraced ducts with a 150 mm minimum clearance to vertical ceiling hanger wires.

3.8 EQUIPMENT

3.8.1 General

Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for restraint anchor bolts. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Install neoprene grommet washers or till the gap with epoxy on equipment anchor bolts where clearance between anchor and equipment support hole exceeds 0.125 inches.

3.8.2 Controls

Ensure that controls for critical equipment that must remain operational after an earthquake are certified ~~per paragraph 3.11 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT~~ and are served by emergency power as required.

3.9 ANCHOR BOLTS

3.9.1 Cast-in-Place Anchor Bolts

Use templates to locate cast-in-place bolts accurately and securely in formwork. Anchor bolts must have an embedded straight length equal to at least 12 times nominal diameter of the bolt. Anchor bolts that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or be increased in depth to accommodate bolt lengths.

3.9.2 Drilled-In Anchor Bolts

Drill holes with rotary impact hammer drills Drill bits must be of diameters as specified by the anchor manufacturer. Unless otherwise shown on the Drawings, all holes must be drilled perpendicular to the concrete surface. Where anchors are permitted to be installed in cored holes, use core bits with matched tolerances as specified by the manufacturer. Properly clean cored hole per manufacturer's instructions. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Exercise care in coring or drilling to avoid damaging existing reinforcing or embedded items. Notify the COR if reinforcing steel or other embedded items are encountered during drilling. Take precautions as necessary to avoid damaging prestressing tendons, electrical and telecommunications conduit, and gas lines. Unless otherwise specified, do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Perform anchor installation in accordance with manufacturer instructions.

3.9.2.1 Wedge Anchors, Heavy-Duty Sleeve Anchors, and Undercut Anchors

Protect threads from damage during anchor installation. Heavy-duty sleeve anchors must be installed with sleeve fully engaged in part to be fastened. Set anchors to manufacturer's recommended torque, using a torque wrench. Following attainment of 10% of the specified torque, 100% of the specified torque must be reached within 7 or fewer complete turns of the nut. If the specified torque is not achieved within the required number of turns, the anchor must be removed and replaced unless otherwise directed by the Engineer.

3.9.2.2 Cartridge Injection Adhesive Anchors

Where approved for seismic application, clean all holes per manufacturer instructions to remove loose material and drilling dust prior to installation of adhesive. Inject adhesive into holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive. Follow manufacturer recommendations to ensure proper mixing of adhesive components. Sufficient adhesive must be injected in the hole to ensure that the annular gap is filled to the surface. Remove excess adhesive from the surface. Shim anchors with suitable device to center the anchor in the hole. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

3.9.2.3 Capsule Anchors

Where approved for seismic application, perform drilling and setting operations in accordance with manufacturer instructions. Clean all holes to remove loose material and drilling dust prior to installation of adhesive. Remove water from drilled holes in such a manner as to achieve a surface dry condition. Capsule anchors must be installed with equipment conforming to manufacturer recommendations. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

Observe manufacturer recommendations with respect to installation temperatures for cartridge injection adhesive anchors and capsule anchors.

3.10 ANCHOR BOLT TESTING

Test in place expansion and chemically bonded anchors not more than ~~[~~24 ~~][______]~~ hours after installation of the anchor, conducted by an independent testing agency; testing must be performed on random anchor bolts as described below.

3.10.1 Torque Wrench Testing

Perform torque wrench testing on not less than ~~[~~50~~] [______]~~ percent of the total installed applied torque expansion anchors and at least ~~[~~one anchor~~] [[______] anchors]~~ for every piece of equipment containing more than ~~[~~two~~] [______]~~ anchors. The test torque must equal the minimum required installation torque as required by the bolt manufacturer. Calibrate torque wrenches at the beginning of each day the torque tests are performed. Recalibrate torque wrenches for each bolt diameter whenever tests are run on bolts of various diameters. Apply torque between 20 and 100 percent of wrench capacity. Reach the test torque within one half turn of the nut, except for 9 mm ~~-~~sleeve anchors which must reach their torque by one quarter turn of the nut. If any anchor fails the test, test similar anchors not previously tested until ~~[~~20~~] [______]~~ consecutive anchors pass. Failed anchors must be retightened and retested to the specified torque; if the anchor still fails the test it must be replaced.

3.10.2 Pullout Testing

Test expansion and chemically bonded anchors by applying a pullout load using a hydraulic ram attached to the anchor bolt. Testing must be in accordance with ~~ASTM E488/E488M or~~ ICC ES AC193. At least ~~[~~10~~] [______]~~ percent of each type and size of anchors, but not less than ~~[~~3~~] [______]~~ per day must be tested. Apply the load to the anchor without removing the nut; when that is not possible, the nut must be removed and a threaded coupler must be installed of the same tightness as the original nut. Check the test setup to verify that the anchor is not restrained from withdrawing by the baseplate, the test fixture, or any other fixtures. The support for the testing apparatus must be at least 1.5 times the embedment length away from the bolt being tested. Load each tested anchor to ~~[~~1~~] [______]~~ times the design tension value for the anchor. The anchor must have no observable movement at the test load. If any anchor fails the test, similar type and size anchors not previously tested must be tested until ~~[~~10~~] [______]~~ percent of those type consecutive anchors pass. Remove and replace failed anchors. Fill empty anchor holes and patch failed anchor locations with high-strength non-shrink, nonmetallic grout.

~~3.11 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT~~

~~Equipment and components (including controls) designated as [MC-1 (Mission Critical Level 1] Designated Seismic Systems required to remain operational after an earthquake will be seismic qualified by shake table testing conforming to ICC ES AC156 procedures. The manufacturer is to provide a certification by a fully qualified testing agency for the specific equipment and/or components. Prequalified certifications are acceptable unless noted otherwise.[Seismic component qualification documentation for each piece of equipment must contain the information required in UFC 3-301-02, Section 2-17.2.5 Component Qualification Documentation.]~~

~~Mechanical components that are required to be certified must bear permanent marking or nameplates constructed of a durable heat and water~~

~~resistant material. Nameplates must be mechanically attached to such nonstructural components and placed on each component for clear identification. The nameplate must not be less than 5 inches x 7 inches with red letters 1 inch in height on a white background stating "Certified Equipment." The following statement must be on the nameplate: "This equipment/component is certified. No modifications are allowed unless authorized in advance and documented in the Equipment Certification Documentation file." The nameplate must also contain the component identification number in accordance with the drawings/specifications and the O&M manuals.~~

~~3.12   SPECIAL INSPECTION FOR SEISMIC-RESISTING SYSTEMS AND EQUIPMENT~~

~~Perform special inspections for seismic-resisting mechanical systems, equipment and components [for structures assigned to Risk Category V;] designated mechanical seismic systems and equipment per ICC IBC 1705.12.4; and plumbing and mechanical components per ICC IBC 1705.12.6. Periodic special inspections will be conducted on mechanical equipment as required by Section 1705.12 of the International Building Code and paragraph 2-5.4 of UFC 3-301-01. Provide a Statement of Special Inspections and Final Report in accordance with paragraph 2-2.4.3 of UFC 3-301-01.~~

-- End of Section --

SECTION 23 05 93

TESTING, ADJUSTING, AND BALANCING FOR HVAC
05/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 203	(1990; R 2011) Field Performance Measurements of Fan Systems

ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1	(2016, 7th ed) National Standards for Total System Balance

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB PROCEDURAL STANDARDS	(2015) Procedural Standards for TAB (Testing, Adjusting and Balancing) Environmental Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E	(2024) Standard for Electrical Safety in the Workplace

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 016	(2012) HVAC Air Duct Leakage Test Manual - 2nd Edition
SMACNA 1780	(2023) HVAC Systems - Testing, Adjusting and Balancing, 4th Edition

1.2 DEFINITIONS

In some instances, terminology differs between this Contract and the TAB certifying body or the TAB Standard primarily because this section intends to use the industry standards specified, along with additional requirements listed within this section, to produce optimal results.

a. AABC: Associated Air Balance Council

b. Certifying Body: An independent body responsible for issuing a formal recognition, such as a certificate, proving that the certification meets specific national or international requirements.

c. COTR: Contracting Officer's Technical Representative

d. DALT: Duct Air Leakage Testing (of HVAC duct systems such as supply air (SA), return air (RA), exhaust air (EA), relief air (LA), and outdoor air (OA)).

e. DALT Failure: This phrase means "a leakage rate measured during DALT Field Work and Acceptance Testing which exceeds the allowable leakage rate for the duct test pressure and leakage class indicated."

f. Duct System: When applied to DALT, this phrase means "complete duct system, inclusive of all ductwork, plenums, mains, branches, fittings and duct-mounted components and appurtenances, such as manual balancing dampers, control dampers, access doors, fire dampers, and duct-mounted coils, up to, but excluding flexible ducts and air-handling equipment, such ~~as air-handling units (AHUs),~~ dedicated outdoor air system (DOAS) units, and fan coil units (FCUs)~~, water source heat pumps (WSHPs), and variable-air volume (VAV) terminal units~~."

g. HVAC: Heating, ventilating, and air-conditioning

h. Season 1, Season 2: Depending upon when the project HVAC systems are completed and ready for TAB, Season 1 is defined as Cooling Season or Heating Season, thereby defining Season 2 as the other season.

(1) Cooling Season: A 30-day timeframe centered on the expected date of maximum cooling load or expected warmest day of the year, making the timeframe ~~[~~10-July through 9-August~~][_____]~~.

(2) Heating Season: A 30-day timeframe centered on the expected date of maximum heating load or expected coldest day of the year, making the timeframe ~~[~~20-December through 19-January~~][_____]~~.

i. TAB: Testing, Adjusting, and Balancing (of HVAC and domestic hot-water recirculation systems).

j. TAB Failure: This phrase means "a measurement taken during TAB Field Acceptance Testing which does not fall within the specified Acceptance Tolerance for a specific parameter."

k. TAB Firm: AABC Certified Member Agency, NEBB Certified Firm, TABB Certified Contractor

l. TAB Standard: Standard utilized by TAB Team based on TAB Firm's certifying body:

(1) AABC - National Standards for Total System Balance (AABC MN-1)

(2) NEBB - Procedural Standard for Testing, Adjusting and Balancing of Environmental Systems (NEBB PROCEDURAL STANDARDS)

(3) TABB - HVAC Systems Testing, Adjusting & Balancing Manual (SMACNA 1780)

m. TAB Team Supervisor: AABC Test and Balance Engineer (TBE), NEBB TAB Certified Professional (TAB CP), TABB TAB Certified Supervisor

n. TAB Team: TAB Firm personnel

o. TABB: Testing, Adjusting and Balancing Bureau

1.3 GENERAL REQUIREMENTS

The work includes DALT and TAB of ~~[~~new~~][and][existing]~~ HVAC and domestic hot-water (DHW) recirculation systems, which are located within, on, under, between, and adjacent to buildings.

Perform TAB in accordance with the requirements of the TAB Standard identified by the TAB Firm's certifying body, as supplemented and modified by this specification section. Perform DALT in compliance with the procedures specified in SMACNA 016, as supplemented and modified by this specification section.

1.3.1 Support Personnel

Ensure the TAB Team Supervisor or TAB Team Field Leader is present full-time on the Contract site for overall management and direct observation of the DALT and TAB Field Work. Certified TAB Team Field Technicians may work independently of direct observation by the TAB Team Supervisor or Field Leader.

Ensure the technical personnel, such as factory representatives and HVAC controls contractor technicians, to operate HVAC equipment are present to enable the TAB Team to accomplish the DALT and the TAB measurement work. Provide the personnel to install field designated test ports and test apparatus connections.

1.3.2 Coordination

Coordinate testing schedules with work in other Sections performed on the components or systems to be tested. Ensure support personnel are present at the times required by the TAB Team and cause no delay in the DALT and the TAB Field Work.

1.4 RELATED REQUIREMENTS

~~[Requirements for climate conditions during building pressure testing are specified in Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING.~~

~~][~~Requirements for thermal insulation of ductwork, piping, fittings, and valves are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

~~][~~Requirements for coordination with commissioning are specified in Section 23 08 00 COMMISSIONING OF MECHANICAL AND PLUMBING SYSTEMS.

~~][~~Requirements for calibration and testing of HVAC controls system devices are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

~~][~~Requirements for Detail Drawings, construction and sealing of HVAC air distribution systems are specified in Section 23 30 00 HVAC AIR DISTRIBUTION.

~~]~~

Requirements for price breakdown of HVAC TAB work are specified in Section 01 20 00 PRICE AND PAYMENT PROCEDURES.

~~Requirements for construction scheduling related to HVAC TAB work are specified in Section 01 32 17.00 20 COST LOADED NETWORK ANALYSIS SCHEDULES (NAS).~~

1.5 SUBMITTALS

Unless otherwise specified or directed by the Contracting Officer, provide submittals in electronic format.~~[~~ In addition to the electronic submittal, provide ~~[~~three~~][_____]~~ hard copies of the submittals.~~]~~

Compile the electronic submittal file as a single, complete document, to include the Transmittal Form required by Section ~~01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES~~01 33 00 SUBMITTAL PROCEDURES. Name the electronic submittal file specifically according to its contents and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is clear and legible. Use portable document format (PDF) as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of an ink signature.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES: Submit the following in accordance with Sections 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES and 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES:~~

SD-01 Preconstruction Submittals

TAB Firm And Personnel Qualifications; G~~, [_____]~~

Design Review Report; G~~, [_____]~~

TAB Pre-Field Engineering Plan; G~~, [_____]~~

SD-06 Test Reports

~~[~~ ~~Existing Facility Conditions Report; G, [_____]~~

~~]~~

~~Pre-Final DALT Report; G, [_____]~~

~~Phase [1][_____] Pre-Final DALT Report; G, [_____]~~

~~Phase [2][_____] Pre-Final DALT Report; G, [_____]~~

~~Certified Final DALT Report; G, [_____]~~

Certified TAB Report; G~~, [_____]~~

~~Phase [1][_____] Certified TAB Report; G, [_____]~~

~~Phase [2][_____] Certified TAB Report; G, [_____]~~

Certified TAB Report of Proportional Balancing; G~~, [_____]~~

Certified TAB Report of Season 1; G~~, [_____]~~

~~Phase [1][_____] Certified TAB Report of Proportional Balancing; G, [_____]~~

~~Phase [1][_____] Certified TAB Report of Season 1; G, [_____]~~

~~Phase [2][_____] Certified TAB Report of Proportional Balancing; G, [_____]~~

~~Phase [2][_____] Certified TAB Report of Season 1; G, [_____]~~

SD-11 Closeout Submittals

Certified TAB Report of Season 2; G~~[, [_____]]~~

~~Phase [1][_____] Certified TAB Report of Season 2; G, [_____]~~

~~Phase [2][_____] Certified TAB Report of Season 2; G, [_____]~~

Certified Final TAB Report; G~~, [_____]~~

1.6 TAB FIRM AND PERSONNEL QUALIFICATIONS

Engage the services of a qualified TAB Firm to perform DALT and TAB work for this project. Only approved firm and personnel are allowed to perform DALT and TAB work on this Contract. Any firm or individual that has been the subject of disciplinary action by their certifying body within the 5 years preceding Contract Award is not eligible to perform any duties related to DALT and TAB. Any firm or individual that has been the subject of disciplinary action by their certifying body within the 5 years preceding Notice to Proceed is not eligible to perform any duties related to DALT and TAB. Any firm or individual that has been the subject of disciplinary action by their certifying body within the 5 years preceding Design Acceptance is not eligible to perform any duties related to DALT and TAB. Comply with the following:

a. Certification: Maintain TAB Firm and personnel certifications for the entire duration of duties specified. Renew Certifications which expire prior to completion of the TAB work in a timely manner so there is no lapse in certification. Provide written notice to the Contracting Officer, with a copy of renewed certificates of the TAB Firm and personnel, prior to expiration of the previous certificate(s). If, for any reason, the TAB Firm or its employee loses subject certification during this Contract, immediately notify the Contracting Officer in writing and comply with the requirements for the replacement of TAB Firm and TAB Team Members. An approved successor must validate all work performed for this project by the TAB Firm or TAB Team Member who lost certification.

b. Replacement of TAB Firm and TAB Team Members: Replacement or substitution of firm or individual members may occur if each complies with the applicable qualifications and is approved by the Contracting Officer prior to the performance of its duties on this Contract. Submit replacement TAB Firm and Personnel Qualifications in accordance with this section.

Submit the TAB Firm and Personnel Qualifications within 60 calendar days after the date of Contract Award. Submit the TAB Firm and Personnel Qualifications within 60 calendar days after the date of Notice to Proceed. Submit the TAB Firm and Personnel Qualifications within 60 calendar days after the date of Design Acceptance.

1.6.1 TAB Firm

Provide documentation that the TAB Firm is a first tier subcontractor to the prime Contractor and an independent company, not participating in any other work on this Contract or affiliated to any other company participating in work on this Contract, including, but not limited to, design, furnishing equipment, or construction. No work may be performed by a second tier subcontractor.

Provide a copy of current certificate, with expiration date, to certify that the TAB Firm is certified by AABC, NEBB, or TABB in categories and functions of the DALT and TAB field work specified. Designate the single certifying body to be utilized when the firm is certified by more than one.

1.6.2 TAB Personnel

For each TAB Team role below, provide at least one qualified, full-time employee of the TAB Firm for approval. Include the proposed role(s) of each employee with their qualification documentation. It is acceptable for a TAB Team member to perform duties of another TAB Team role if they meet the qualifications of both.

a. TAB Team Supervisor

Provide documentation, including full name and a copy of current certificate, that the TAB Team Supervisor is a AABC TBE, a NEBB TAB CP, or a TABB TAB Certified Supervisor.

b. TAB Team Field Leader

Provide documentation that the TAB Team Field Leader is a certified TAB Technician by AABC, NEBB, ~~or~~ TABB, or equal Japanese standardized TAB certification and has satisfactorily performed full-time supervision of TAB work of others in the field for a minimum of 3 years immediately preceding Contract Award. ~~Provide documentation that the TAB Team Field Leader is a certified TAB Technician by AABC, NEBB, or TABB, and has satisfactorily performed full-time supervision of TAB work of others in the field for a minimum of 3 years immediately preceding Notice to Proceed. Provide documentation that the TAB Team Field Leader is a certified TAB Technician by AABC, NEBB, or TABB, and has satisfactorily performed full-time supervision of TAB work of others in the field for a minimum of 3 years immediately preceding Design Acceptance.~~ Include full name, a copy of current certificate, and a list of projects, roles performed, and associated dates.

c. TAB Team Field Technician

Provide documentation that each TAB Team Field Technician is a certified TAB Technician by AABC, NEBB, ~~or~~ TABB, or equal Japanese standardized TAB certification or has satisfactorily performed DALT and TAB work in the field for a minimum of 1 year immediately preceding Contract Award.~~Provide documentation that each TAB Team Field Technician is a certified TAB Technician by AABC, NEBB, or TABB, or has satisfactorily performed DALT and TAB work in the field for a minimum of 1 year immediately preceding Notice to Proceed.Provide documentation that each TAB Team Field Technician is a certified TAB Technician by AABC, NEBB, or TABB, or has satisfactorily performed DALT and TAB work in the field for a minimum of 1 year immediately preceding Design Acceptance.~~ Include full name, a copy of current certificate when applicable, and a list of projects, roles performed, and associated dates.

1.7 TAB STANDARD

All recommendations and suggested practices contained in the TAB Standard are mandatory. Maintain a hard copy ~~[and an electronic PDF file copy]~~of the standard on site.

All quality assurance provisions of the TAB Standard, such as performance guarantees, are part of this Contract. For systems or system components not covered in the TAB Standard, TAB procedures and report forms must be developed by the TAB Team Supervisor and included in the TAB Pre-Field Engineering Plan. Where new procedures and requirements, applicable to the Contract requirements have been published or adopted by the certifying body responsible for the TAB Standard, the more stringent of the requirements and recommendations contained in these procedures and requirements are considered mandatory.

1.8 INSTRUMENTATION

It is the responsibility of the TAB Firm to use instrumentation that meets the requirements of the TAB Standard. Select appropriate instruments to achieve the applicable TAB tolerances. Instrumentation used for taking air wet-bulb temperature readings must provide accuracy of a minimum of plus or minus 5 percent at the measured face velocities. Where the instrument manufacturer calibration recommendations are more stringent than those listed in the TAB Standard, adhere to the manufacturer's recommendations. Instrumentation must be in proper operating condition and must be applied in accordance with the instrumentation's manufacturer recommendations.

All instrumentation must bear a valid National Institute of Standards and Technology (NIST) traceable calibration certificate, dated within one year, during field work and Field Acceptance Testing.

1.9 DESIGN REVIEW

The TAB Team Supervisor must review the Contract Plans and Specifications to advise the Contracting Officer of any omissions and deficiencies that would prevent the effective and accurate TAB of the systems in compliance with the requirements of this section~~[, including performing the field work for the reporting of existing facility conditions]~~. Accomplish the following:

a. Verify in the documents the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, flow measuring devices, balancing valves, and manual balancing dampers.

b. Confirm the inclusion of the design values necessary to complete the

DALT and TAB report forms required for the TAB Pre-Field Engineering Plan.

c. Generate a list of submittals, from those in the Contract Submittal Register of Section 01 33 00 SUBMITTAL PROCEDURES, that relate to the successful accomplishment of all HVAC and DHW recirculation systems TAB. Generate a list of submittals, from those in the Contract Submittal Register of Section 01 33 00 SUBMITTAL PROCEDURES, that relate to the successful accomplishment of all HVAC and DHW recirculation systems TAB. Generate a list of submittals, from those in the Contract Submittal Register of Section ~~01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES~~01 33 00 SUBMITTAL PROCEDURES, that relate to the successful accomplishment of all HVAC and DHW recirculation systems TAB.

d. Prepare a report containing a detailed listing of the items identified during the review. Provide the omission and deficiency listing in a tabular format, with each item sequentially numbered and with a reference to the applicable design document. For each omission and deficiency, provide a complete explanation including supporting documentation detailing the design deficiency. If no deficiencies are evident, state so in the report. In addition, the design engineer must respond to the Design Review Report items and the responses, with dated signature, must accompany the report. Include the TAB Related Submittals List in the report.

Submit the Design Review Report to the Contracting Officer for approval within 30 calendar days after approval of the TAB Firm and Personnel Qualifications.

1.10 TAB PRE-FIELD ENGINEERING PLAN

The TAB Team Supervisor must utilize the TAB Standard, Contract Plans, Contract Specifications, approved contract revisions, and contract change orders to prepare a plan for the DALT and TAB field work, including procedures, report forms, diagrams, instrumentation and other supporting documents in compliance with the following:

a. Project Summary: Provide a detailed narrative describing the DALT and TAB Field Work to be performed, including systems and equipment. Clearly distinguish between DALT information and TAB information. ~~Include separate sections for each[construction phase][building].~~

b. Step-by-Step Procedures: Provide a list of the intended procedural steps for DALT and TAB Field Work from start to finish. Include the following:

    (1) The intended procedural steps for TAB work for systems, subsystems, and system components. TAB Standard procedure excerpts may be provided to supplement this requirement.

    (2) A description of how each type of data measurement will be obtained.

    (3) A complete methodology for accomplishing each seasonal TAB Field Work requirement.

    (4) A list of Contractor support personnel required for each step, and tasks they need to perform.

c. Report Forms: Provide the report forms to be used in the DALT and TAB reports to document the DALT and TAB Field Work, with the preliminary information and design values filled in. Utilize contract document design HVAC and DHW systems values. For TAB report forms, use forms inclusive of all data on TAB Standard sample report forms, or generate where no TAB Standard form exists, for all data collection requirements. Ensure report forms contain data fields for each HVAC controls system flow measuring device's calibration, including design flow rate, actual flow rate, controls system reading and factor. For DALT report forms, use forms comparable to the "Air Duct Leakage Test Summary Report Forms" located in SMACNA 016, with an example of design and test data in the plan. Coordinate the key numbering used in the report forms with that used in the TAB Schematic Drawings.

d. TAB Schematic Drawings: Provide the schematic drawings to be used in the required reports, which may include building floor plans, mechanical room plans, equipment schedules, duct system plans, and equipment elevations. Use the contract drawings and, if available, ductwork shop drawings to provide the following:

(1) Air moving equipment and pump schedules.

(2) Thermal energy transfer equipment schedules.

(3) Duct Construction and Leakage Testing Schedule.

(4) Air system plans showing the location and airflow quantities of all supply, return, exhaust, relief, outdoor, and transfer air terminal devices such as registers, grilles, diffusers, louvers, and rooftop hoods and their respective airflow quantities.

(5) A key numbering system on the drawings which identifies each piece of equipment and each air terminal device contained in the air terminal airflow report sheets. It is acceptable to utilize the contract drawing numbering system when provided for equipment.

(6) A unique number or mark for each water flow measurement fitting and balancing valve. It is acceptable to utilize the contract drawing numbering system when provided for valves and fittings.

(7) Intended TAB measurement locations, including where test ports need to be provided by the Contractor.

For Static Pressure Profile and Duct Traverse Equipment Diagrams, provide diagrams, representative of the actual equipment configuration, developed from TAB Standard forms, contract drawings or approved equipment submittals.

For DALT schematic drawings, provide a representative diagram matching the example DALT report form in this plan.

e. Instrumentation: Provide a list of DALT and TAB instruments for this project. Include the instrument name and description, manufacturer, model number, scale range, published accuracy, most recent calibration date, and what the instrument will be used for on this project. Submit instrument manufacturer's literature to document instrument accuracy performance is in compliance with that specified. Provide a copy of the calibration certificate for each calibrated instrument to be used.

f. Pre-DALT Field Work Checklist: Provide a thorough checklist of the work items and inspections that need to be accomplished before DALT Field Work can be performed. Include signature and date lines for the Contractor's quality control representative and mechanical contractor's representative acknowledging completion of the marked checklist items.

g. Pre-TAB Field Work Checklist: Provide a thorough checklist of the work items and inspections that need to be accomplished before the proportional balancing and thermal performance testing TAB Field Work can be performed. Provide a thorough checklist of the work items and inspections that need to be accomplished before the proportional balancing, heating season, and cooling season TAB Field Work can be performed. Include items such as completion of applicable HVAC controls systems installation, cleaning of hydronic system including but not limited to strainers, and replacement of HVAC system air filters. Provide separate sections of checklist items for proportional balancing and thermal performance testing TAB Field Work. Provide separate sections of checklist items for proportional balancing, heating season thermal performance testing, and cooling season thermal performance testing TAB Field Work. Each section must include signature and date lines for the Contractor's quality control representative and mechanical contractor's representative acknowledging completion of the marked checklist items.

Submit the TAB Pre-Field Engineering Plan to the Contracting Officer for approval a minimum of ~~[~~60~~][______]~~ calendar days in advance of commencement of ~~[DALT][~~TAB~~]~~ field work~~[required of Existing Facility Conditions]~~.

1.11 PRE-DALT/TAB MEETING

Meet with the Contracting Officer's Technical Representative (COTR) and the design engineer of the HVAC systems to develop a mutual understanding relative to the details of the DALT and TAB work requirements. Requirements to be discussed include required submittals, work schedules, and field quality control. Ensure that the TAB Team Supervisor and a representative for the sheet metal contractor, mechanical contractor, electrical contractor, and HVAC controls system contractor are present at this meeting. Conduct the Pre-DALT/TAB Meeting a minimum of 30 calendar days in advance of the commencement of ductwork installation.

Provide written notice to the Contracting Officer of the proposed date of the Pre-DALT/TAB Meeting at a minimum of 14 calendar days prior to the proposed date. The notice must include the proposed date and time; required, and optional if any, attendees' names and job titles; a copy of the current submittal register; and a copy of the current project schedule containing the sequencing and durations of the DALT and TAB activities and submittals in this section. The notice must include the proposed date and time; required, and optional if any, attendees' names and job titles; a copy of the current submittal register; and a copy of the current network analysis schedule containing the sequencing and durations of the DALT and TAB activities and submittals in this section.

~~[1.12 EXISTING FACILITY CONDITIONS~~

~~Conduct [DALT][TAB][DALT and TAB]measurements, survey, and generate report(s) in accordance with the procedures in this section,~~

~~[exclusive][inclusive] of Field Acceptance Testing, for the following:~~

~~a. [_____]~~

~~b. [_____]~~

~~c. [_____]~~

~~Perform a field survey, including inspection, by the TAB Team of the existing equipment, ductwork, and associated systems to confirm the location and condition of the equipment, test ports, flow measuring devices, balancing devices, fire and smoke dampers, and other items that impact the [DALT][TAB][DALT and TAB]work required of this Contract. Include a visual inspection of ductwork and duct liner for the presence and condition of duct sealant, material deterioration, and visible duct blockages and restrictions.~~

~~Submit the Existing Facility Conditions Report to the Contracting Officer for approval within [14][_____] calendar days of completion of the associated field work.[Submit an updated TAB Pre-Field Engineering Plan to the Contracting Officer for approval within 30 calendar days of approval of the Existing Facility Conditions Report as a result of this field work.]~~

~~]~~1.12 REPORTS

Format: All submitted documentation must be typed, neat, and organized. Include a dated title page, a certification page, a table of contents identifying the location of each report section, sequentially numbered pages throughout, and all sections bookmarked. Tables, lists, and diagrams must be titled.~~[~~ Hardcopy report forms and report data must be typewritten and bound with a waterproof front and back cover.~~]~~ Handwritten report forms or report data are not acceptable. Submit electronic reports with the additional requirements in accordance with the paragraph SUBMITTALS.

Submit the completed report forms approved in the TAB Pre-Field Engineering Plan, and any other data sheets and supporting documentation necessary, to document the required design data; field measurements data; system configuration and settings during testing; design and installation deficiencies; and, required data evaluations. Reports must be completed by the TAB Team, and reviewed and certified by the TAB Team Supervisor.

~~[1.12.1 Existing Facility Conditions Report~~

~~Provide a consolidated report of the [DALT][TAB][DALT and TAB][_____][and]survey Field Work performed in compliance with the following requirements:~~

~~a. Project Summary: Provide a description of the scope of the Field Work, including a narrative describing the outcome of the Field Work. When applicable, provide notations describing how actual field procedures differed from those in the TAB Pre-Field Engineering Plan.~~

~~b. Survey Results: Provide a narrative describing the items that impact the [DALT][TAB][DALT and TAB]work required of this Contract. For those items considered detrimental to the work or nonfunctional, include explanation of the deficiencies. Define the scheduled need date for repairs of existing nonfunctional equipment and devices.~~

~~c. System Variations: Provide a listing in a tabular format, with each item sequentially numbered, that documents:~~

~~(1) System variations deviating from the [contract documents][As-Built documents]discovered as a result of the Field Work, including any abnormalities and any items for the attention of the Contracting Officer.~~

~~(2) Recommended resolutions developed by the TAB Firm.~~

~~(3) References to the associated Field Work Reports or Supporting Documents, as applicable.~~

~~(4) Scheduled need date for repairs of existing nonfunctional equipment and devices.~~

~~d. Supporting Documents: Provide copies of documentation that clarifies the System Variations such as equipment schedules, product literature, and photographic images. Include this documentation in an appendix to the report with sequential numbering of each separate item for reference in the System Variations list.~~

~~e. Field Work Reports: Provide a copy of the approved reports generated from the [DALT][TAB][DALT and TAB]Field Work in an appendix to the report with sequential numbering of each separate document.~~

~~][~~1.12.1 DALT Reports

Report the data for Pre-Final DALT Reports and the Certified Final DALT Report meeting the following requirements:

a. Project Summary: Provide, if applicable, notations describing how actual field procedures differed from those in the approved TAB Pre-Field Engineering Plan. If no procedures differed, state so in the report.

b. Data from DALT Field Work: After conducting the DALT Field Work, prepare the DALT field data using the report forms approved in the TAB Pre-Field Engineering Plan. Form data must be recorded for each test iteration of each duct selection. Report forms for each test must indicate either "Pass" or "Fail". In addition, submit in the report, a marked duct shop drawing identifying each section of duct tested with assigned node numbers for each section. Node numbers used in the marked duct shop drawings must be included in the completed report forms to identify each duct section.

c. Supporting Documents: Include a copy of all calculations prepared in determining the duct surface area and the allowable leakage for each duct test section. In addition, provide a copy of the calibration curve for each of the DALT orifices used for testing.

d. Instruments: List the instruments used to measure the data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date. Provide a copy of the calibration certificate for each calibrated instrument used.

e. Certification: Include the TAB Team Supervisor's physical or electronic stamp, issued by the certifying body, and dated signature.

The Certified Final DALT Report consists of the approved Pre-Final DALT Reports, organized in chronological order, the DALT-related Project Summary and Step-by-Step Procedures from the approved TAB Pre-Field Engineering Plan with a brief narrative describing any DALT failures and subsequent passing tests, and a separate Certification.

~~1~~1.12.2 TAB Reports

Report the data for the Certified TAB Report of Proportional Balancing, Certified TAB Report of Season 1, and Certified TAB Report of Season 2 in compliance with the following requirements: Report the data for the Certified TAB Report in compliance with the following requirements:

a. Project Summary: Provide a project narrative, including systems and equipment tested, describing the outcome of the testing and balancing and include a list of items documenting system variations deviating from the design tolerances, any abnormalities discovered, and any outstanding items with proposed corrective action. When applicable, provide notations describing how actual field procedures differed from those approved in the TAB Pre-Field Engineering Plan. If no procedures differed, state so in the report.

b. Certification: Include the TAB Team Supervisor's physical or electronic stamp, issued by the certifying body, and dated signature.

c. Data from TAB Field Work: Prepare the TAB field data using the report forms approved in the TAB Pre-Field Engineering Plan and any other data sheets necessary to document the required final system balance and demonstrate accuracy and repeatability during TAB Field Acceptance Testing. For each unique calibration factor utilized, report the traverse airflow, the flow measuring instrument reading (e.g., flow hood airflow), the resulting correction factor, and the location of the traverse taken. Include notation on each form with the date the form's data was collected and with any Supporting Document reference number. Provide a detailed explanation wherever a final measurement did not achieve the required value or fall within the required range of values. Include the following:

(1) System Configuration: Clearly identify system configurations and conditions affecting data for all reported data. Include system operational parameters such as device positioning, system diversity, modes of operation, and setpoints necessary to set up and duplicate the system configuration during the TAB Field Acceptance Testing. Include final hydronic differential pressure setpoint, hydronic system fill pressure, ~~glycol percentage,~~ pump and fan motor frequency in maximum, fan motor frequency in minimum, fan brake horsepower, calibration coefficients and factors, control damper positions, and primary air static pressure setpoint.

(2) Static Pressure Profiles: Provide static pressure profiles in accordance with this section. Illustrate pressures in an Equipment Diagram representative of the actual installation. Include design static pressure and differential pressure values for comparison where provided.

(3) Duct Traverses: Provide duct traverse measurement data in accordance with this section. Include individual velocities from

the duct traverses taken. Illustrate, in an Equipment Diagram, the actual entering and leaving duct traverse airflows for each piece of air-handling equipment exceeding 944 L/S.

(4) Air and Water Systems' Open Paths: Identify air and hydronic system open paths in the report form.

(5) Motor Data: Provide motor current (amperage) and voltage for each phase of motors, including those powered by VFDs. Include current setting for thermal overload protection for motors, including starter or VFD brand, model, enclosure type, installed overload devices, rating(s) or rating range, and setpoints, and revised device ratings and setpoints when applicable.

(6) Seasonal Thermal Performance Data: Include capacity field data and design and calculated capacity, along with ambient outdoor temperature data, on seasonal thermal performance report forms. Include notation to identify system configuration settings that deviate from the design sequences of operation and setpoints when necessary to obtain the design capacity.

(7) Sound Assessment: Provide, for each area, the sound pressure levels measured for each octave band in tabular format and the resultant noise criteria curve and the noise criteria (NC) goals. Include notation indicating those areas with sound levels above 65 dB in octave bands below 63 Hz. Include notation to identify system configuration settings and the design sequences of operation and setpoints utilized to obtain the sound data.

d. Flow Measuring Device Calibration: Provide calibration data at each scheduled design volumetric flow rate.

e. Instruments: Provide an updated list of the instruments used to measure the TAB data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date. Include a copy of the instrument calibration certificate when updated from the copy in the approved TAB Pre-Field Engineering Plan.

f. Performance Curves: Provide copies of the factory pump curves and fan curves for pumps and fans included on the job, and manufacturer equipment data curves or tables correlating pressure drop and water flow rate. Include marks or notation indicating the reported final balance values on each curve.

g. Calibration Charts: Provide a factory calibration chart, table, graph, or diagram, as applicable, for each applicable balancing valve and flow measuring device such as a venturi flow meter and an orifice flow meter.

h. TAB Schematic Drawings: Provide updated drawings and diagrams with the final installed locations of all terminals and devices, any numbering changes, and actual test locations including duct traverse and static pressure measurements. Indicate the air terminal device and water coil of the open paths for those identified.

i. Mechanical Equipment Schedules: Include a copy of the mechanical equipment schedules.

j. Supporting Documents: Provide copies of any request for information

(RFI) with the RFI response; summaries of implemented change order(s); meeting minutes with participants; electronic mail with addresses; and, other documentation substantiating any deviations of the reported design values from the initial contract design documents. Include this documentation in an appendix to the TAB report with sequential numbering of each separate document for reference in the data presentation forms.

k. Guarantee: Provide documentation substantiating the inclusion of the quality assurance program such as a certificate of conformance or performance guarantee issued by the TAB Firm's certifying body.

1.13 SEQUENCING AND SCHEDULING

~~1.13.1 Phased Construction~~

~~This project includes multiple [phases][and][buildings]. DALT and TAB work for each project [phase][and][building] must be scheduled separately and must correspond to each completion milestone in the master schedule. The DALT and TAB work, including all associated submittals, must be completed and approved with each [phase][building], excluding Season 2, prior to turnover.[At completion of the Season 2 TAB Field Acceptance Testing for the final phase in the building, compile all approved TAB reports and submit as one document.][_____].~~

~~The following [phases][buildings] are included:~~

~~a. [Phase [_____]]~~

~~b. [Phase [_____]]~~

~~c. [Phase [_____]]~~

1.13.1 Submittal and Work Schedule

This schedule is structured as though the HVAC and DHW systems construction, and thereby the DALT and TAB work, submittals, and field acceptance testing, is going to be completed in a single phase for the contract, including the seasonal work. ~~All DALT and TAB field work, submittals, and acceptance testing must be planned and completed separately for each[phase][building].~~ This schedule is structured as though the HVAC and DHW systems construction, and thereby the DALT and TAB work, submittals, and field acceptance testing, is going to be completed in a single occurance for the contract, including the seasonal work. All DALT and TAB field work, submittals, and acceptance testing must be planned and completed separately for each occurance. Include the items listed in this schedule in the Project Schedule required by Section 01 32 01.00 10 PROJECT SCHEDULE. ~~Include the items listed in this schedule in the Network Analysis Schedule (NAS) required by Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES.~~ Ensure sufficient time is scheduled to complete each item, including Government review period for submittals. The order of items listed below does not imply a specified sequence:

a. Submission of TAB Firm and Personnel Qualifications

b. Submission of Design Review Report

c. Submission of TAB Pre-Field Engineering Plan

d. Pre-DALT/TAB Meeting

~~[e. Existing Facility Conditions Field Work~~

~~f. Submission of Existing Facility Conditions Report~~

~~]~~ e. DALT Field Work

f. Submission of Pre-Final DALT Reports

g. DALT Field Acceptance Testing

h. Submission of Certified Final DALT Report

i. Proportional Balancing TAB Field Work

~~[j. Sound Assessment Field Work~~

~~]~~ j. Submission of Certified TAB Report

k. TAB Field Acceptance Testing

l. Submission of Certified TAB Report of Proportional Balancing

m. Proportional Balancing TAB Field Acceptance Testing

n. Season 1 TAB Field Work

o. Submission of Certified TAB Report of Season 1

p. Season 1 TAB Field Acceptance Testing

q. Season 2 TAB Field Work

r. Submission of Certified TAB Report of Season 2

s. Season 2 TAB Field Acceptance Testing

1.14 WARRANTY

Furnish workmanship and performance warranty for the ~~[~~DALT and ~~]~~TAB system work performed for a period of ~~[1][~~2~~][3][5][_____]~~ years from the date of Government acceptance of the work, issued directly to the Government. Include provisions that if within the warranty period the system shows evidence of major performance deterioration, or is outside of the tolerances noted within the TAB System Tolerance section, resulting from defective TAB or DALT workmanship, the corrective repair or replacement of the defective materials and correction of the defective workmanship is the responsibility of the TAB Firm. Perform corrective action that becomes necessary because of defective materials and workmanship while system TAB and DALT are under warranty 7 days after notification, unless additional time is approved by the Contracting Officer. Failure to perform repairs within the specified period of time constitutes grounds for having the corrective action and repairs performed by others and the cost billed to the TAB Firm. The Contractor must also provide a ~~[1][~~2~~][3][5][_____]~~ year contractor installation warranty.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 DALT PROCEDURES

3.1.1 Systems Requiring DALT

All duct systems, regardless of duct pressure classification, are subject to DALT including supply, return, exhaust, relief, and outdoor air~~[~~ with exception of transfer air~~]~~. From each duct system indicated as subject to DALT, the Contracting Officer will randomly select sections of each completed duct system for testing such that the duct selections will not exceed 25 percent of the total measured linear footage of each duct system. Include notation within the associated Advance Notice indicating the specific unit(s) when electing the following for any single unit or quantity of similar units:

a. It is acceptable for an entire duct system associated with a single unit (e.g., 100 percent of AHU-1 supply air ductwork) to be leak tested instead of disassembling that system in order to DALT only the 25 percent duct selection specified. Testing an entire duct system for a unit does not reduce the DALT requirements for any other duct system or selection.

b. It is acceptable for the duct systems (e.g., FCU supply air and return air) for 25 percent of the total quantity of similar terminal units and light-duty air-handling equipment (e.g., ~~WSHPs less than 5 tons cooling capacity, VAVs, and~~ FCUs) to be leak tested instead of disassembling 25 percent of each duct system for each of these units.

3.1.2 Prerequisites for DALT Field Work

Complete the following prior to starting DALT Field Work:

a. Submission and approval of the SD-01 Preconstruction Submittals.

b. Installation and sealing in conformance with the contract documents of those duct systems indicated completed and ready for testing.

c. Confirmation ductwork to be air leakage tested is accessible and not concealed or insulated.

d. Work items and inspections indicated in the Pre-DALT Field Work Checklist.

e. Furnish the TAB Team a copy of the ductwork layout Detail Drawings indicating the completed duct systems available for DALT.

3.1.3 Advance Notice for Pre-Final DALT Field Work

Provide the Contracting Officer with written notification including the proposed date for DALT and a copy of the ductwork layout Detail Drawings with indication and dimensions of the duct systems, or portion thereof, completed and ready for DALT. Include with the notice the signed and dated Pre-DALT Field Work Checklist, with all work items certified by the Contractor and mechanical contractor as completed.

Provide the notice after completion of the installation of duct systems indicated as ready for testing.

3.1.4 Ductwork Selections

From each duct system indicated within the Advance Notice of Pre-Final DALT Field Work, the Contracting Officer will select sections of each completed duct system for testing in accordance with this section. Maintain a hard copy of the drawings indicating ductwork selections on site.

Sealing of the ductwork selections is prohibited, with exception of temporary end caps and connection for test apparatus, from the time the Contractor is notified of the selections until DALT measurements are recorded.

Avoid testing duct runs which contain equipment such as dampers, filters, fans, air terminals, or any other device which would have potential for leakage and compromise the validity of the test.

3.1.5 DALT Field Work

Conduct DALT Field work on the ductwork selections in compliance with the procedures specified in SMACNA 016 or equivalent Japanese standard, as supplemented and modified by this section. Use a duct test pressure equivalent to each ductwork selection's pressure classification. Calculate the ductwork selection allowable leakage using the more stringent, lower numerical value, of the duct system leakage class for the duct pressure classification indicated on the drawings or the maximum permitted leakage class defined as follows:

a. Round/Oval Leakage Class = 3 and Rectangular Leakage Class = 4 at duct pressure classification of 1000 Pa and above.

b. Round/Oval Leakage Class = 4 and Rectangular Leakage Class = 8 at duct pressure classification of 500 Pa and 750 Pa.

c. Round/Oval Leakage Class = 8 and Rectangular Leakage Class = 16 at duct pressure classification of 250 Pa.

Complete DALT Field Work on the ductwork selections within 48 hours from receipt of selections, including disconnection, capping, measuring, and testing of ductwork. If any of the duct sections tested is a DALT failure, terminate testing for that system and report the duct selection as failed and the duct system as DALT terminated in the Pre-Final DALT Report.

~~3.1.6 Pre-Final DALT Report~~

~~Upon completion of the DALT Field Work, prepare a Pre-Final DALT Report. Prepare separate Pre-Final DALT Reports when DALT is not completed in a single event[for each phase][for each building] and differentiate these with a sequential numerical suffix such as Pre-Final DALT Report 1.~~

~~Submit the Pre-Final DALT Report to the Contracting Officer for approval within two working days of completion of the related DALT Field Work.~~

~~Submit a Pre-Final DALT Report to the Contracting Officer for approval~~

~~within two working days of completion of the DALT Field Work for each of the following:~~

~~Phase [1][_____] Pre-Final DALT Report~~

~~Phase [2][_____] Pre-Final DALT Report~~

~~[_____]~~

3.1.6 Quality Assurance

3.1.6.1 DALT Field Acceptance Testing

In the presence of the Contracting Officer and TAB Team Field Leader, conduct field acceptance testing of a maximum of 50 percent of the duct sections in the approved Pre-Final DALT Report as selected by the Contracting Officer. If any of the duct sections tested is a DALT failure, terminate data checking for that section; the associated Pre-Final DALT Report data for the given duct selection will be disapproved.

Provide the Contracting Officer with written notification of the proposed date for DALT Field Acceptance Testing within 14 calendar days of approval of the Pre-Final DALT Report and a minimum of 7 calendar days in advance of proposed field acceptance testing date.

3.1.6.2 Retesting and Additional Testing

For any duct section indicated as a DALT failure, make the necessary repairs. Repairs must be applied to similar conditions in all untested duct sections. For each DALT failure, and after repairs are completed, conduct DALT on the untested duct selection(s) of the terminated duct system, on the failed section, and one additional duct section as selected by the Contracting Officer. Prepare and resubmit the Pre-Final DALT Report with the original, retesting, and additional selection test results for approval. Reschedule field acceptance testing of the approved revised report with the Contracting Officer.

~~3.1.7 Certified Final DALT Report~~

~~Prepare the Certified Final DALT Report using data furnished by the TAB Team in the Pre-Final DALT Report(s) for[each phase][each building].~~

~~Submit the Certified Final DALT Report to the Contracting Officer for approval within 14 calendar days of successful completion of all DALT Field Acceptance Testing for[the][each] building.~~

3.2 TAB PROCEDURES

3.2.1 Systems Requiring TAB

All DHW recirculation, air and hydronic systems are subject to TAB. All DHW recirculation, air and hydronic systems are subject to TAB such as the following:

a. Air Systems:

(1) Supply Air (SA)

(2) Return Air (RA)

(3) Outdoor Air (OA)

(4) Exhaust Air (EA)

~~(5) Relief Air (LA)~~

~~(6) [_____]~~

b. Hydronic Systems:

(1) Chilled Water (CHW)

~~(2) Condenser Water (CW)~~

(2) Heating Hot-Water (HHW)

~~(3) [_____]~~

c. Other Systems:

(1) DHW Recirculation

~~(2) [_____]~~

Each system includes the associated equipment, devices, and components such as:

a. Air systems: ~~AHUs,~~ DOAS units, Heating and Ventilating Units (HVUs), ~~VAV terminal units,~~ FCUs, ~~WSHPs,~~ fans, duct systems, energy recovery systems, exhaust hoods, flow measuring and balancing devices, grilles, registers, and diffusers.

b. Hydronic and DHW systems: chillers, ~~boilers, cooling towers,~~ coils, pumps, piping systems, heat exchangers, energy recovery systems, pressure reducing valves, flow-measuring and balancing devices, and pressure and temperature ports.

c. Other systems: ~~DHW heaters, pressure~~Pressure and temperature ports, flow balancing valves, ~~[~~ and~~]~~ thermostatic mixing valves ~~[, _____][, _____][, and _____]~~.

3.2.2 TAB Tolerances

3.2.2.1 System Tolerances

System tolerances apply to the design flow rates as specified or indicated in the contract documents. HVAC systems' tolerance is plus or minus 5 percent, except as indicated below:

a. Air Systems:

~~[~~ (1) OA 23.5 L/s or less: plus 2.35 L/s to minus zero L/s

~~][~~ (2) EA 23.5 L/s or less: plus zero L/s to minus 2.35 L/s

~~][~~ (3) SA and RA of 100 percent recirculating single-zone air-handing equipment: plus or minus 10 percent

][(4) SA and RA 23.5 L/s or less: plus or minus 1.42 L/s

][(5) ~~[_____]~~Chilled water (CHW) at coils: plus or minus 10 percent CW at coils: plus or minus 10 percent

] b. Hydronic Systems:

[(1) Heating Hot-Water (HHW) at coils: plus or minus 10 percentHW at coils: plus or minus 10 percent

][~~(2) [_____]~~

] c. Other Systems:

(1) DHW Recirculation: plus or minus 10 percent of design temperature at the outlet thermostatic mixing valve

[~~(2) [_____]~~

]3.2.2.2 Acceptance Tolerances

Field Acceptance Testing measurements must remain within the associated System Tolerance of the design flow rate.[When the associated System Tolerance is plus or minus 10 percent, Field Acceptance Testing measurements must remain within the associated System Tolerance of the design flow rate and repeat within plus or minus 5 percent of the approved Certified TAB Report data.]

3.2.3 Advance Notice for TAB Field Work

Provide the Contracting Officer with written notification including the proposed dates for TAB Field Work and a list of the systems ready for TAB on the proposed start date. Include with the notice the signed and dated Pre-TAB Field Work Checklist, with the applicable work items certified by the Contractor and mechanical contractor as having been completed and working as designed.

Furnish the Advance Notice for Proportional Balancing TAB Field Work, for Season 1 TAB Field Work, and for Season 2 TAB Field Work TAB Field Work prior to start of the TAB Field Work.Furnish the Advance Notice for TAB Field Work prior to start of the TAB Field Work.

3.2.4 Proportional Balancing TAB Field Work

Conduct TAB Field Work in conformance with the TAB Standard, as supplemented and modified by this section. Test, adjust, and balance the HVAC and DHW recirculation systems until measured flow rates (air and water flow) are within the specified System Tolerances. Do not exceed maximum rated revolutions per minute (RPM) and rated full load amperage for equipment during any mode of operation, including with loaded filters and wet cooling coils. Variable speed drives must be capable of controlling to maximum and minimum flows with a drive operational range not less than 25 Hz. Operation exceeding 60 Hz is permitted only for direct-drive plenum fans rated for operation at higher frequencies. Follow the procedures specified in NFPA 70E during electrical measurements on exposed energized conductors or circuits. Additionally, this work includes the following:

a. Adjustment of flow control and balancing devices such as balancing valves, balancing dampers, and sheaves.

b. Adjustment of variable frequency drives and speed controllers.

c. Calibration of each controls system flow measuring device at each scheduled design volumetric flow rate.

d. Changing out fan sheaves if required to obtain flow rates specified or indicated.

Complete Proportional Balancing TAB Field Work at a minimum of ~~[90][~~120 ~~][_____]~~ calendar days in advance of the scheduled Beneficial Occupancy Date (BOD).

3.2.4.1 Prerequisites for TAB Field Work

Complete the following prior to starting Field Work:

a. DALT Field Work and obtain approval of the Certified Final DALT Report.

b. Installation of pressure and temperature test ports adjacent to inlet and outlet of every pump, heat exchanger, chiller, ~~boiler,~~ and coil, and as required by the TAB Team Supervisor.

c. Work items and inspections indicated in the Pre-TAB Field Work Checklist, including providing new air filters and cleaning screens of hydronic system strainers.

d. Enclosure of the building envelope according to the contract documents with the Air Barrier Pressure Test completed and the Air Leakage Test Reports and Diagnostic Test Reports approved. Hydronic system TAB field work may commence prior to air barrier test completion.

e. Manufacturer's equipment start-up forms for each piece of equipment to be tested.

f. HVAC controls system installation and start-up for each piece of equipment to be tested.

~~[~~3.2.4.2 TAB Air Distribution Systems

Balance systems to minimize throttling losses with at least one open path, with fully open balancing dampers from the fan discharge to an outlet and from fan suction to an inlet, for fans greater than 1 hp.

a. Static Pressure Profile Work

Measure and record static pressure profiles under maximum airflow conditions for air systems of 944 L/S and greater.

Measure static pressure data for supply, return, relief, exhaust, and outdoor air ducts for the systems. The static pressure measurement data must include, in addition to TAB Standard requirements, the following:

(1) Inlet and discharge static pressures for fans.

(2) Static pressure drop across coils and heat reclaim devices

installed in unit cabinetry or the system ductwork.

(3) Static pressure drop across automatic control dampers.

(4) Static pressure drop across air filters, acoustic silencers, moisture eliminators, airflow straighteners, airflow measuring stations, and other pressure drop producing specialty items installed in unit cabinetry or in the system ductwork. Examples of these specialty items are smoke detectors, white sound generators, RF shielding, wave guides, security bars, blast valves, and duct mounted humidifiers.

Do not report static pressure drop across duct fittings provided for the sole purpose of conveying air, such as elbows with or without turning vanes, transitions, offsets, plenums, manual dampers, and branch take-offs.

(5) Report static pressure drop across louvers.

(6) Report static pressure readings in the duct at the point where the ducts connect to air-handling equipment.

b. Duct Traverse Work

Perform duct traverses, in addition to TAB Standard requirements, for main~~[and each branch main]~~ supply, return, exhaust, relief and outdoor air ducts for each piece of air-handling equipment exceeding 472 L/s. Include those ducts which lack 7 1/2 duct diameters upstream and 3 duct diameters downstream of straight duct unobstructed by duct fittings, offsets, and elbows.~~[Perform additional duct traverses as shown on the contract drawings.]~~ Evaluate the suitability of the duct traverse measurement based on satisfying the qualifications for a satisfactory pitot traverse plane as defined by AMCA 203.

~~c. Building Pressurization Work~~

~~Perform measurements upon completion of air system balancing and during climate conditions suitable for a pressure test in accordance with Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING. Record building differential pressure for all sides of the building on each floor where openings allow measurements by instrumentation. Report all system setup parameters affecting building pressure measurement (e.g., exhaust, relief and outdoor airflow, damper positions, and fan speeds) and indicate wind speed during time of building pressure measurements. Measure in maximum and minimum building systems configuration.~~

~~Measurements must occur during building system setup with building exterior doors and windows closed and as follows:~~

~~[(1) Occupied Mode; air-handling equipment and fans in maximum cooling, maximum design airflow.~~

~~][(2) Occupied Mode; air-handling equipment with all zones satisfied such as VAVs at minimum design airflow.~~

~~][(3) Occupied Mode; air-handling equipment in full Economizer Mode.~~

~~][(4) Occupied Mode; air-handling equipment in Demand Controlled~~

~~Ventilation Occupied-Standby Mode.~~

~~][(5) [_____]~~

~~] Measure differential pressure for the following locations: [_____][, _____].~~

~~][~~3.2.4.3 TAB Hydronic and Domestic Hot Water Distribution Systems

Balance systems to minimize throttling losses with at least one path, with fully open balancing valves, from the pump discharge to a terminal device and back to the pump suction, for pumps greater than 1 hp. Measure and record flow rates at all balancing valves and flow measuring devices.

~~a. [_____]~~

~~b. [_____]~~

~~]~~3.2.4.4 Marking of Settings and Test Ports

Upon completing each item of work for TAB Field Work specified and prior to TAB Field Acceptance Testing, the TAB Team is to permanently mark the settings of HVAC adjustment devices including valves, gauges, and dampers so that adjustment can be restored if disturbed at any time. Secure balancing damper handles and set valve memory stops. Label variable frequency drives with final frequency (Hz) and control setpoint. Provide permanent markings clearly indicating the settings on the adjustment devices which result in the data reported on the Certified TAB Report. Permanently and legibly mark and identify the duct test ports. If the ducts have external insulation, make these markings on the exterior side of the insulation. Show the location of test ports on the as-built mechanical drawings with dimensions given where the test port is covered by exterior insulation.

~~[3.2.5 Sound Assessment Field Work~~

~~Conduct sound measurements and assessments in accordance with the TAB Firm certifying body's sound standard, either AABC MN-1, NEBB MASV, or SMACNA 1858 for TABB, as supplemented and modified by this section.~~

~~3.2.5.1 Prerequisites for Sound Field Work~~

~~Complete the following prior to starting Field Work:~~

~~a. Proportional Balancing TAB Field Work.~~

~~b. [_____]~~

~~3.2.5.2 Areas Requiring Sound Assessment~~

~~The following spaces are subject to sound assessment:~~

~~a. HVAC mechanical rooms: Room [_____] [, _____][, _____].~~

~~b. Adjacent spaces sharing a common barrier with each mechanical room: Room [_____] [, _____][, _____].~~

~~[c. AHU-1 System: Rooms: [_____]~~

~~][d. [_____] System: Rooms: [_____]~~

~~][e. [_____] System: Rooms: [_____]~~

~~]3.2.5.3 Procedure~~

~~Conduct sound measurements in each interior room, when unoccupied except for the TAB Team, and each exterior location, when applicable, with all HVAC systems that would cause sound readings in the specified room or location operating in their noisiest mode such as maximum cooling. Measure sound pressure levels in each octave band.~~

~~Attempt to mitigate the sound level and bring the level to within the specified ASHRAE HVAC APP SI HDBK NC goals, if such mitigation is within the TAB Team's control. When no NC goal exists, utilize NC 35. If the sound level cannot be brought into compliance, provide written notice of the deficiency to the Contractor for correction. If the sound level cannot be brought into compliance, provide written notice of the deficiency to the Contracting Officer for resolution.~~

~~3.2.5.4 Timing~~

~~Measure sound levels during the period when the background noise is quietest, unless otherwise specified.~~

~~3.2.5.5 Meters~~

~~Measure sound levels with a sound meter complying with ASA S1.4, Class 1 or 2, and an octave band filter set complying with ASA S1.11 PART 1, Class 1. Calibrate meters as prescribed by the TAB Firm certifying body's sound standard, except that calibrators emitting a sound pressure level tone of 94 dB at 1000 hertz (Hz) are also acceptable.~~

~~3.2.5.6 Background Noise Correction~~

~~Determine background noise component of room sound (noise) levels for each octave band.~~

~~]~~3.2.5 Seasonal Thermal Performance TAB Field Work

Perform operational thermal performance testing of the HVAC systems as soon as all prerequisites are completed, but prior to facility turnover, regardless of the season. In addition, perform thermal performance testing during Season 1 and Season 2 when the outdoor air temperature (OAT) is within the seasonal temperature ranges below for equipment with performance that is OAT dependent. It is acceptable to conduct Season 1 testing prior to the applicable season dates if all prerequisites are complete including OAT within range.

a. Cooling Season: ~~[XX.X to XX.X]~~29 to 34.6 degrees C dry-bulb temperature.

b. Heating Season: ~~[XX.X to XX.X]~~10.8 to -11.3 degrees C dry-bulb temperature.

Complete Season 1 Thermal Performance Testing Field Work a minimum of 60 calendar days in advance of the scheduled BOD.

3.2.5.1 Prerequisites for Season 1 Thermal Performance Field Work

Complete the following prior to starting Field Work:

~~[~~ a. Proportional Balancing TAB Field Work.

~~][b. Sound Assessment Field Work.~~

~~]~~ b. Work items and inspections indicated in the Pre-TAB Field Work Checklist for the applicable season, including confirming clean filters and strainers.

c. Advance Notice for Season 1 TAB Field Work. Advance Notice for Thermal Performance TAB Field Work.

~~d. [_____]~~

3.2.5.2 Thermal Performance Testing

Conduct thermal performance (capacity) tests to verify thermal energy transfer components, devices, and equipment meet the indicated design capacity. This work includes:

a. Verifying, using the same procedures and instruments for proportional balancing TAB field work, flows (air and water) remain within the System Tolerances. If the measurements exceed the specified System Tolerances, report these as deficiencies in accordance with this section.

b. Measuring water temperatures through immersion into fluid stream.

c. For ducted units, measuring airflows via duct traverse.

d. Measuring the range, from minimum to maximum, of the outdoor ambient dry-bulb temperature and corresponding dew-point temperature within which the TAB seasonal data was recorded. Record these temperature ranges on each day of seasonal data measurements.

Report all design data, actual field measurements, and capacity calculations for the components, devices, and equipment below.

3.2.5.3 Units with Coils

Report heating and cooling performance tests for heating and cooling coils such as heating hot-water, chilled-water, direct-expansion, and steam coils:

a. For units with capacities greater than 26,370 Watts cooling, such as factory manufactured units, central built-up units and rooftop units, determine the apparent air-side coil capacity by calculations utilizing direct measurement of airflow, and measurements of entering and leaving wet-bulb and dry-bulb temperatures for cooling capacity and dry-bulb temperatures only for heating capacity. Calculate water-side coil capacity utilizing direct measurements of water flow rate, and entering and leaving water temperatures. Measure and record coil water pressure drop.

Measure and report dry-bulb temperature and relative humidity within the supply air duct downstream of each cooling coil. Report the corresponding dewpoint temperature.

b. For units with capacities of 26,370 Watts or less, such as fan coil units, ~~duct mounted reheat coils associated with VAV terminal units,~~ and unitary units, determine the apparent air-side coil capacity by calculations using direct measurement of airflow, single point measurement centered within the airstream of entering and leaving wet-bulb and dry-bulb temperatures for cooling capacity and dry-bulb temperatures only for heating capacity.

~~3.2.5.4 Units with Heat Recovery Devices~~

~~Report heating and cooling energy recovery performance tests for energy recovery devices such as wheels, coils, and fixed plate exchangers.~~

~~Report total cooling capacity, heating capacity, and energy recovery effectiveness. For outdoor air and exhaust airstreams measure and report airflows and entering and leaving wet-bulb and dry-bulb temperatures across heat recovery device. For water-side calculations, utilize direct measurements of water flow rate and entering and leaving water temperatures. Measure and record water pressure drops.~~

3.2.5.4 Thermal Energy Transfer Equipment

Report heating and cooling performance tests for thermal energy transfer equipment (e.g., boilers, chillers, and cooling towers). Measure and report water flow rates, water-side pressure drops, and entering and leaving water temperatures. Report capacities.

3.2.6 Deficiencies

Strive to meet the intent of this section to maximize the performance of the equipment as designed and installed. However, if deficiencies in equipment or system design or installation prevent TAB work from being accomplished in accordance with this section, the TAB Team Supervisor must submit written notification within five working days from discovery of all such deficiencies for resolution. The TAB Firm must issue notice concurrently to the Contractor for corrective action and to the Contracting Officer for information. Provide a complete explanation, including supporting documentation, detailing the deficiencies and proposed corrective actions in the notification. Where deficiencies are encountered that adversely impact successful completion of:

a. TAB Field Work: Resolve deficiencies prior to submission of the Certified TAB Report.

b. TAB Field Acceptance Testing: The Contractor must, within five working days of the TAB Firm notice, submit written notification directly to the Contracting Officer, with a separate copy to the TAB Firm, of all such deficiencies, the intended or implemented corrective action, and the planned or actual date(s) for completion of each corrective action.

Responsibility for correction of design and installation deficiencies is the Contractor's. Responsibility for correction of installation deficiencies is the Contractor's.

3.2.7 Certified TAB Reports

Upon completing each item of work for TAB Field Work specified, prepare a TAB Report in accordance with this section using data furnished by the TAB Team. In addition, the design engineer must review the Certified TAB Report and the comments, with dated signature, must accompany the report. If there are no comments generated during the Certified TAB Report review process, state so in the report. Prepare separate Certified TAB Reports, identified as indicated, for each~~[~~ construction phase and a final, consolidated Certified Final TAB Report inclusive of Certified TAB Reports for all phases upon completion of all TAB Field Acceptance Testing required for~~[each building of]~~ the project~~][building]~~.

Submit the Certified TAB Report to the Contracting Officer for approval within 14 calendar days of completion of the TAB~~[and Sound Assessment]~~ Field Work.

~~Submit a Certified TAB Report for each phase to the Contracting Officer for approval within 14 calendar days of completion of the TAB[and Sound Assessment] Field Work for each of the following:~~

~~Phase [1][_____] Certified TAB Report~~

~~Phase [2][_____] Certified TAB Report~~

~~[_____]~~

Submit the Certified TAB Report of Proportional Balancing to the Contracting Officer for approval within 14 calendar days of completion of the Proportional Balancing TAB~~[and Sound Assessment]~~ Field Work.

Submit the Certified TAB Report of Season 1 to the Contracting Officer for approval within 14 calendar days of completion of the Season 1 TAB Field Work.

~~Submit a Certified TAB Report of Proportional Balancing to the Contracting Officer for approval within 14 calendar days of completion of the Proportional Balancing TAB[and Sound Assessment] Field Work for each of the following:~~

~~Phase [1][_____] Certified TAB Report of Proportional Balancing~~

~~Phase [2][_____] Certified TAB Report of Proportional Balancing~~

~~[_____]~~

~~Submit a Certified TAB Report of Season 1 to the Contracting Officer for approval within 14 calendar days of completion of the Season 1 TAB Field Work for each of the following:~~

~~Phase [1][_____] Certified TAB Report of Season 1~~

~~Phase [2][_____] Certified TAB Report of Season 1~~

~~[_____]~~

~~[~~ Provide a copy of the approved Certified TAB Report data in an electronic spreadsheet file, along with all supporting graphs, charts, and curves, within 21 calendar days of receipt of TAB Report approval.

~~]~~3.2.8 Quality Assurance

3.2.8.1 TAB Field Acceptance Testing

In the presence of the Contracting Officer and TAB Team Field Leader, conduct field acceptance testing of points and areas in the approved Certified TAB Report as selected by the Contracting Officer. The selections include data types such as water and air volumetric flow rates, velocity, pressure, rotational speed (RPM), temperature, ~~[sound level readings,]~~and electrical current recorded in the TAB Report. Measurement and test procedures are the same as required for TAB Field Work. If any of the measurements is a TAB failure, the Contracting Officer may terminate data verification. The sample group will be disapproved and require retesting and additional testing. Conduct TAB Field Acceptance Testing for the Certified TAB Report of Season 1 and for the Certified TAB Report of Season 2 within the applicable Cooling Season or Heating Season.

Provide the Contracting Officer with written notification of the proposed date for TAB Field Acceptance Testing within 14 calendar days of approval of the Certified TAB Report and a minimum of 7 calendar days in advance of proposed TAB Field Acceptance Testing date. Provide the Contracting Officer with written notification of the proposed date for Proportional Balancing TAB Field Acceptance Testing within 14 calendar days of approval of the Certified TAB Report of Proportional Balancing and a minimum of 7 calendar days in advance of proposed TAB Field Acceptance Testing date. Provide the Contracting Officer with separate written notification of the proposed date for Season 1 and for Season 2 TAB Field Acceptance Testing within 7 calendar days of approval of the associated Certified TAB Report and a minimum of 7 calendar days in advance of proposed TAB Field Acceptance Testing date.

3.2.8.2 Sample Strategy

Field acceptance testing includes verification of selections using the sample strategy indicated for each Category. For each of the following Categories, the Contracting Officer will select a sample group for testing:

Category 1: 100 percent of the air system equipment such as DOAS units, AHUs (rooftop and central stations), return fans, computer room air-conditioning (CRAC) units, and energy recovery units (ERUs) and 25 percent of the associated air terminal devices.

Category 2: 100 percent of the water system equipment such as chillers, ~~boilers, cooling towers, ~~pumps, DHW heaters, and Category 1 equipment's coils.

Category 3: 25 percent of the air terminal units (e.g., ~~VAV terminal units, ~~FCUs), and of the exhaust fans, supply fans, ~~and relief fans ~~and 100 percent of the selected equipment's air terminal devices.

Category 4: 25 percent of the hydronic and DHW recirculation systems' flow measuring devices (e.g., manual balancing valves and flow meters) and Category 3 equipment's hydronic coils.

Quantity is rounded up to a whole number for a sample size and is a minimum of three units or devices when three or more exist.

3.2.8.3 Retesting and Additional Testing

For any data indicated as a TAB Failure and when the data verification is terminated, make the necessary corrections, conduct TAB Field Work, set and secure balancing devices, and mark settings. Prepare and resubmit the applicable TAB Report with the corrected test results for approval. Reschedule TAB Field Acceptance Testing of the approved revised report with the Contracting Officer. For each TAB Failure, and after the revised report is approved, conduct field acceptance testing of the failed data point, one additional selection by the Contracting Officer from the same Category for each failure, and new selections by the Contracting Officer in the same Category using the Category's sample strategy. If there is insufficient quantity of data in the same Category, the Contracting Officer may select data from a different Category.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Season 2 Thermal Performance TAB Field Work

Perform operational thermal performance testing of the HVAC systems during Season 2 when the OAT is within the seasonal temperature range.

3.3.1.1 Prerequisites for Season 2 Thermal Performance Field Work

Complete the following prior to starting Field Work:

a. Proportional Balancing TAB Field Work.

b. Work items and inspections indicated in the Pre-TAB Field Work Checklist for the applicable season, including confirming clean filters and strainers.

c. Advance Notice for Season 2 TAB Field Work.

~~d. [_____]~~

3.3.1.2 Season 2 Thermal Performance Testing

Conduct thermal performance (capacity) tests to verify thermal energy transfer components, devices, and equipment meet the indicated design capacity within the parameters for Season 2. Prepare separate Certified TAB Reports, identified as indicated, for each~~[~~ construction phase and a final, consolidated Certified Final TAB Report inclusive of TAB Field Work for all phases upon completion of all TAB Field Acceptance Testing required for~~[each building of]~~ the project~~][building]~~.

Submit the Certified TAB Report of Season 2 to the Contracting Officer for approval within 14 calendar days of completion of Season 2 Thermal Performance TAB Field Work.

~~Submit a Certified TAB Report of Season 2 to the Contracting Officer for approval within 14 calendar days of completion of Season 2 TAB Field Work for each of the following:~~

~~Phase [1][_____] Certified TAB Report of Season 2~~

~~Phase [2][_____] Certified TAB Report of Season 2~~

~~[______]~~

3.4 PREREQUISITE FOR SYSTEMS ACCEPTANCE

Compliance with field acceptance testing requirements of this section is a prerequisite for the final Contracting Officer acceptance of the HVAC systems.

-- End of Section --

SECTION 23 07 00

THERMAL INSULATION FOR MECHANICAL SYSTEMS
02/13

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. At the discretion of the Government, the manufacturer of any material supplied will be required to furnish test reports pertaining to any of the tests necessary to assure compliance with the standard or standards referenced in this specification.

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide ~~http~~https://www.appro

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 2758 (2014) Paper - Determination of Bursting Strength

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (20~~18~~24) Standard for the Installation of Air Conditioning and Ventilating Systems

NFPA 90B (2018) Standard for the Installation of Warm Air Heating and Air Conditioning Systems

NFPA 96 (2017; TIA 17-1) Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-24179 (1969; Rev A; Am 2 1980; Notice 1 1987) Adhesive, Flexible Unicellular-Plastic Thermal Insulation

MIL-A-3316 (1987; Rev C; Am 2 1990) Adhesives, Fire-Resistant, Thermal Insulation

MIL-PRF-19565 (1988; Rev C) Coating Compounds, Thermal Insulation, Fire- and Water-Resistant, Vapor-Barrier

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1321 (2011) Testing Method for Incombustibility of Internal Finish Material and Procedure of Buildings (Amendment 1)

JIS A 5538	(2018) Adhesives for Wall and Ceiling Boards
JIS A 9504	~~(2021) Man Made Mineral Fibre Thermal Insulation Materials~~(2024) Man Made Mineral Fibre Thermal Insulation Materials
JIS A 9521	(2022) Thermal Insulation Materials for Buildings
JIS G 4304	(2021) Hot-Rolled Stainless Steel Plate, Sheet and Strip
JIS H 4000	(2017) Aluminium and Aluminium Alloy Sheets, Strips and Plates (Amendment 1)
JIS K 7127	(1999) Plastics -- Determination of Tensile Properties -- Part 3: Test Conditions for Films and Sheets
JIS K 7129-1	(2019) Plastics -- Film and Sheeting -- Determination of Water Vapour Transmission Rate -- Part 1: Humidity Detection Sensor Method
JIS K 7129-2	(2019) Plastics -- Film and Sheeting -- Determination of Water Vapour Transmission Rate -- Part 2: Infrared Detection Sensor Method
JIS Z 0208	(2021) Testing Methods for Determination of the Water Vapour Transmission Rate of Moisture-Proof Packaging Materials (Dish Method) (Amendment 1)
JIS Z 2911	(2023) Methods of Tests for Fungus Resistance

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification

UNDERWRITERS LABORATORIES (UL)

UL 723	(2018; Reprint Jun 2025) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

1.2 SYSTEM DESCRIPTION

1.2.1 General

Provide field-applied insulation and accessories on mechanical systems as specified herein; factory-applied insulation is specified under the piping, duct or equipment to be insulated. Insulation of heat distribution systems and chilled water systems outside of buildings shall be as specified in Section ~~33 61 13 PRE-ENGINEERED UNDERGROUND HEAT~~

~~DISTRIBUTION SYSTEM, Section 33 63 13.19 CONCRETE TRENCH HYDRONIC AND STEAM ENERGY DISTRIBUTION, Section 33 60 02 ABOVEGROUND HEAT DISTRIBUTION SYSTEM, and Section 33 61 13.13 PREFABRICATED UNDERGROUND HYDRONIC ENERGY DISTRIBUTION~~33 61 14 EXTERIOR BURIED PREINSULATED WATER PIPING. Field applied insulation materials required for use on Government-furnished items as listed in the SPECIAL CONTRACT REQUIREMENTS shall be furnished and installed by the Contractor.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[~~Contractor Quality Control ~~approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submit the three SD types, SD-02 Shop Drawings, SD-03 Product Data, and SD-08 Manufacturer's Instructions at the same time for each system.

SD-02 Shop Drawings

Pipe Insulation Systems and Associated Accessories

Duct Insulation Systems and Associated Accessories

Equipment Insulation Systems and Associated Accessories

~~1.4 CERTIFICATIONS~~

~~1.4.1 Adhesives and Sealants~~

~~Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein. Include VOC content requirements of SCAQMD Rule 1168, GS-36, or F 4-Star.~~

1.4 QUALITY ASSURANCE

1.4.1 Installer Qualification

Qualified installers shall have successfully completed three or more similar type jobs within the last 5 years.

1.5 DELIVERY, STORAGE, AND HANDLING

Materials shall be delivered in the manufacturer's unopened containers. Materials delivered and placed in storage shall be provided with protection from weather, humidity, dirt, dust and other contaminants. The Contracting Officer may reject insulation material and supplies that become dirty, dusty, wet, or contaminated by some other means. Packages or standard containers of insulation, jacket material, cements, adhesives, and coatings delivered for use, and samples required for approval shall have manufacturer's stamp or label attached giving the name of the manufacturer and brand, and a description of the material, date codes, and approximate shelf life (if applicable). Insulation packages and containers shall be asbestos free.

PART 2 PRODUCTS

2.1 STANDARD PRODUCTS

Provide materials which are the standard products of manufacturers regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Submit a complete list of materials, including manufacturer's descriptive technical literature, performance data, catalog cuts, and installation instructions. The product number, k-value, thickness and furnished accessories including adhesives, sealants and jackets for each mechanical system requiring insulation shall be included. The product data must be copyrighted, have an identifying or publication number, and shall have been published prior to the issuance date of this solicitation. Materials furnished under this section shall be submitted together in a booklet.

2.1.1 Insulation System

Provide insulation systems in accordance with MLIT-M. Provide field-applied insulation for heating, ventilating, and cooling (HVAC) air distribution systems and piping systems that are located within, on, under, and adjacent to buildings; and for plumbing systems.

2.1.2 Surface Burning Characteristics

Unless otherwise specified, insulation must have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321 or UL 723. Flame spread, and smoke developed indexes,shall be determined by ~~ASTM E84~~JIS A 1321 or UL 723. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Prepare and mount test specimens according to ~~ASTM E2231~~JIS A 1321 and JIS A 9521.

2.2 MATERIALS

Insulation exterior shall be cleanable, grease resistant, non-flaking and non-peeling. Materials shall be compatible and shall not contribute to corrosion, soften, or otherwise attack surfaces to which applied in either wet or dry state.

2.2.1 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants must meet limit requirements of "Other" category within SCAQMD Rule 1168 sealants table). Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

2.2.1.1 Acoustical Lining Insulation Adhesive

Adhesive shall be a nonflammable, fire-resistant adhesive conforming to JIS A 5538.

2.2.1.2 Mineral Fiber Insulation Cement

Cement shall be in accordance with JIS A 9504.

2.2.1.3 Lagging Adhesive

Lagging is the material used for thermal insulation, especially around a cylindrical object. This may include the insulation as well as the cloth/material covering the insulation. ~~[~~To resist mold/mildew, lagging adhesive shall meet ~~ASTM D5590 with 0 growth rating~~JIS Z 2911. ~~]~~Lagging adhesives shall be nonflammable and fire-resistant and shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321. Adhesive shall be MIL-A-3316, Class 1, pigmented ~~[~~white~~]~~ or ~~[~~red~~]~~ and be suitable for bonding fibrous glass cloth to faced and unfaced fibrous glass insulation board; for bonding cotton brattice cloth to faced and unfaced fibrous glass insulation board; for sealing edges of and bonding glass tape to joints of fibrous glass board; for bonding lagging cloth to thermal insulation; or Class 2 for attaching fibrous glass insulation to metal surfaces. Lagging adhesives shall be applied in strict accordance with the manufacturer's recommendations for pipe and duct insulation.

2.2.1.4 Contact Adhesive

Adhesives may be any of, but not limited to, the neoprene based, rubber based, or elastomeric type that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321. The adhesive shall not adversely affect, initially or in service, the insulation to which it is applied, nor shall it cause any corrosive effect on metal to which it is applied. Any solvent dispersing medium or volatile component of the adhesive shall have no objectionable odor and shall not contain any benzene or carbon tetrachloride. The dried adhesive shall not emit nauseous, irritating, or toxic volatile matters or aerosols when the adhesive is heated to any temperature up to 100 degrees C. The dried adhesive shall be nonflammable and fire resistant. Flexible Elastomeric Adhesive: Comply with MIL-A-24179, Type II, Class I. Provide product listed in FM APP GUIDE.

2.2.2 Caulking

Caulking shall be in accordance with MLIT-M.

2.2.3 Corner Angles

Nominal 0.406 mm aluminum 25 by 25 mm with factory applied kraft backing. Aluminum shall be JIS H 4000.

2.2.4 Finishing Cement

MLIT-M: Mineral fiber hydraulic-setting thermal insulating and finishing cement. All cements that may come in contact with Austenitic stainless steel must comply with MLIT-M.

2.2.5 Fibrous Glass Cloth and Glass Tape

Fibrous glass cloth, with 20X20 maximum mesh size, and glass tape shall be have maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321. Tape shall be 100 mm wide rolls. Class 3 tape shall be 0.15 kg/square m. Elastomeric Foam Tape: Black vapor-retarder foam tape with acrylic adhesive containing an anti-microbial additive.

2.2.6 Staples

Outward clinching type ~~[monel] [~~JIS G 4304, Type 304 or 316 stainless steel ~~]~~.

2.2.7 Jackets

2.2.7.1 Aluminum Jackets

Aluminum jackets shall be in accordance with MLIT-M, Table 2.3.1 and 2.3.2.

2.2.7.2 Vapor Barrier/Weatherproofing Jacket

Vapor barrier/weatherproofing jacket shall be in accordance with MLIT-M, Table 2.3.1 and 2.3.2.

2.2.8 Vapor Retarder Required

~~ASTM C921~~JIS A 9521 or applicable Japanese standard, Type I, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork, where a minimum puncture resistance of 25 Beach units is acceptable. Minimum tensile strength, 6.1 N/mm width. ~~ASTM C921~~JIS A 9521 or applicable Japanese standard, Type II, minimum puncture resistance 25 Beach units, tensile strength minimum 3.5 N/mm width. Jackets used on insulation exposed in finished areas shall have white finish suitable for painting without sizing. Based on the application, insulation materials that require manufacturer or fabricator applied pipe insulation jackets are cellular glass, when all joints are sealed with a vapor barrier mastic, and mineral fiber. All non-metallic jackets shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321. Flexible elastomerics require (in addition to vapor barrier skin) vapor retarder jacketing for high relative humidity and below ambient temperature applications.

2.2.8.1 Vapor Retarder/Vapor Barrier Mastic Coatings

2.2.8.1.1 Vapor Barrier

The vapor barrier shall be self adhesive (minimum 0.05 mm adhesive, 0.075 mm embossed) greater than 3 plies standard grade, silver, white, black and embossed white jacket for use on hot/cold pipes. Permeability shall be less than 0.02 when tested in accordance with ~~ASTM E96/E96M~~JIS K 7129-1, JIS K 7129-2, JIS Z 0208, or any other approved methods of testing. Products shall meet ~~UL 723 or ASTM E84~~JIS A 1321. Flame and smoke requirements and shall be UV resistant.

2.2.8.1.2 Vapor Retarder

The vapor retarder coating shall be fire and water resistant and appropriately selected for either outdoor or indoor service. Color shall be white. The water vapor permeance of the compound shall be 0.013 perms or less at 1 mm dry film thickness as determined according to ~~procedure B of ASTM E96/E96M or utilizing apparatus described in ASTM E96/E96M~~ JIS K 7129-1, JIS K 7129-2, JIS Z 0208, or any other approved methods of testing. The coating shall be nonflammable, fire resistant type. ~~[~~To resist mold/mildew, coating shall meet ~~ASTM D5590 with 0 growth rating~~ JIS Z 2911. ~~]~~Coating shall meet MIL-PRF-19565 Type II (if selected for indoor service) and be Qualified Products Database listed. All other application and service properties shall be in accordance with ~~ASTM C647~~ JIS A 9504.

2.2.8.2 Laminated Film Vapor Retarder

~~ASTM C1136~~JIS A 9521 or applicable Japanese standard, Type I, maximum moisture vapor transmission 0.02 perms, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork; where Type II, maximum moisture vapor transmission 0.02 perms, a minimum puncture resistance of 25 Beach units is acceptable. Vapor retarder shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321. Flexible Elastomeric exterior foam with factory applied UV Jacket. Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

2.2.8.3 Polyvinylidene Chloride (PVDC) Film Vapor Retarder

The PVDC film vapor retarder shall have a maximum moisture vapor transmission of 0.02 perms, minimum puncture resistance of 150 Beach units, a minimum tensile strength in any direction of 5.3 kN/m when tested in accordance with ~~ASTM D882~~JIS K 7127, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321.

2.2.8.4 Vapor Barrier/Weather Barrier

The vapor barrier shall be greater than 3 ply self adhesive laminate -white vapor barrier jacket- superior performance~~ (less than 0.0000 permeability when tested in accordance with ASTM E96/E96M)~~. Vapor barrier shall meet ~~UL 723 or ASTM E84~~JIS A 1321. 25 flame and 50 smoke to comply with requirements specified in JIS K 7129-1, JIS K 7129-2, JIS Z 0208, or any other approved methods of testing requirements; and UV resistant. Minimum burst strength 1.3 MPa in accordance with ~~[TAPPI T403 OM] [~~ISO 2758 ~~]~~. Tensile strength 0.12 kg/m width (PSTC-1000). Tape shall be as specified for laminated film vapor barrier above.

2.2.9 Vapor Retarder Not Required

~~ASTM C921~~JIS A 9521 or applicable Japanese standard, Type II, Class D, minimum puncture resistance 50 Beach units on all surfaces except ductwork, where maximum moisture vapor transmission 0.10, a minimum puncture resistance of 25 Beach units is acceptable. Jacket shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321.

2.2.10 Wire

Wire shall be in accordance with MLIT-M.

2.2.11 Sealants

Sealants shall be chosen from the butyl polymer type, the styrene-butadiene rubber type, or the butyl type of sealants. Sealants shall ~~have a maximum permeance of 0.02 perms based on Procedure B for ASTM E96/E96M~~comply with JIS K 7129-1, JIS K 7129-2, JIS Z 0208, or any other approved methods of testing, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ~~ASTM E84~~JIS A 1321.

2.3 PIPE INSULATION SYSTEMS

Conform insulation materials to Table 1 and minimum insulation thickness as listed in Table 2 and meet. Limit pipe insulation materials to those listed herein and meeting the following requirements:

2.3.1 Aboveground Cold Pipeline (-34 to 16 deg. C)

Insulation for outdoor, indoor, exposed or concealed applications, shall be as follows:

2.3.1.1 Mineral Fiber Insulation with Integral Wicking Material (MFIWM)

JIS A 9504. Install in accordance with manufacturer's instructions. Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4.

2.3.2 Aboveground Hot Pipeline (Above 16 deg. C)

Insulation for outdoor, indoor, exposed or concealed applications shall meet the following requirements. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

2.3.2.1 Mineral Fiber

JIS A 9504, supply the insulation with manufacturer's recommended factory-applied jacket.

2.3.3 Aboveground Dual Temperature Pipeline

Selection of insulation for use over a dual temperature pipeline system (Outdoor, Indoor - Exposed or Concealed) shall be in accordance with the most limiting/restrictive case. Find an allowable material from paragraph PIPE INSULATION MATERIALS and determine the required thickness from the most restrictive case. Use the thickness listed in paragraphs INSULATION THICKNESS for cold & hot pipe applications.

2.4 DUCT INSULATION SYSTEMS

2.4.1 Field Applied Insulation

"Provide field applied insulation according to manufacturer's installation guidelines. Provide manufacturer's standard reinforced fire-retardant vapor barrier, with identification of installed thermal resistance (R) value and out-of-package R values.

2.4.2 Kitchen Exhaust Ductwork Insulation

Insulation thickness shall be a minimum of 50 mm, blocks or boards, either mineral fiber conforming to JIS A 9504. The enclosure materials and the grease duct enclosure systems shall meet testing requirements for noncombustibility, fire resistance, durability, internal fire, and fire-engulfment with a through-penetration fire stop.

2.4.3 Acoustical Duct Lining

2.4.3.1 General

For ductwork indicated or specified in Section 23 30 00 AIR SUPPLY, DISTRIBUTION, VENTILATION, AND EXHAUST SYSTEM to be acoustically lined, provide external insulation in accordance with this specification section and in addition to the acoustical duct lining. Do not use acoustical lining in place of duct wrap or rigid board insulation (insulation on the exterior of the duct).

2.4.4 Duct Insulation Jackets

2.4.4.1 All-Purpose Jacket

Provide insulation with insulation manufacturer's standard reinforced fire-retardant jacket with or without integral vapor barrier as required by the service. In exposed locations, provide jacket with a white surface suitable for field painting.

2.4.4.2 Metal Jackets

2.4.4.2.1 Aluminum Jackets

JIS H 4000, with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide smooth surface jackets for jacket outside dimension 200 mm and larger. Provide corrugated surface jackets for jacket outside dimension 200 mm and larger. Provide stainless steel bands, minimum width of 13 mm.

2.4.4.2.2 Stainless Steel Jackets

JIS G 4304; Type 304, smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 13 mm.

2.5 EQUIPMENT INSULATION SYSTEMS

Insulate equipment and accessories as specified in accordance with MLIT-M, Chapter 3.

PART 3 EXECUTION

3.1 APPLICATION - GENERAL

Insulation shall only be applied to unheated and uncooled piping and equipment. Flexible elastomeric cellular insulation shall not be compressed at joists, studs, columns, ducts, hangers, etc. The insulation shall not pull apart after a one hour period; any insulation found to pull apart after one hour, shall be replaced.

3.1.1 Display Samples

Submit and display, after approval of materials, actual sections of installed systems, properly insulated in accordance with the specification requirements. Such actual sections must remain accessible to inspection throughout the job and will be reviewed from time to time for controlling the quality of the work throughout the construction site. Each material used shall be identified, by indicating on an attached sheet the specification requirement for the material and the material by each manufacturer intended to meet the requirement. The Contracting Officer will inspect display sample sections at the jobsite. Approved display sample sections shall remain on display at the jobsite during the construction period. Upon completion of construction, the display sample sections will be closed and sealed.

3.1.1.1 Pipe Insulation Display Sections

Display sample sections shall include as a minimum an elbow or tee, a valve, dielectric waterways and flanges, a hanger with protection shield and insulation insert, or dowel as required, at support point, method of fastening and sealing insulation at longitudinal lap, circumferential lap, butt joints at fittings and on pipe runs, and terminating points for each type of pipe insulation used on the job, and for hot pipelines and cold pipelines, both interior and exterior, even when the same type of insulation is used for these services.

3.1.1.2 Duct Insulation Display Sections

Display sample sections for rigid and flexible duct insulation used on the job. Use a temporary covering to enclose and protect display sections for duct insulation exposed to weather

3.1.2 Installation

Except as otherwise specified, material shall be installed in accordance with the manufacturer's written instructions. Insulation materials shall not be applied until ~~[tests] [~~tests and heat tracing~~]~~ specified in other sections of this specification are completed. Material such as rust, scale, dirt and moisture shall be removed from surfaces to receive insulation. Insulation shall be kept clean and dry. Insulation shall not be removed from its shipping containers until the day it is ready to use and shall be returned to like containers or equally protected from dirt and moisture at the end of each workday. Insulation that becomes dirty shall be thoroughly cleaned prior to use. If insulation becomes wet or if cleaning does not restore the surfaces to like new condition, the insulation will be rejected, and shall be immediately removed from the jobsite. Joints shall be staggered on multi layer insulation. Mineral fiber thermal insulating cement shall be mixed with demineralized water when used on stainless steel surfaces. Insulation, jacketing and accessories shall be installed in accordance with plates except where modified herein or on the drawings.

3.1.3 Firestopping

Where pipes and ducts pass through fire walls, fire partitions, above grade floors, and fire rated chase walls, the penetration shall be sealed with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING. The protection of ducts at point of passage through firewalls must be in accordance with NFPA 90A and/or NFPA 90B. All other penetrations, such as piping, conduit, and wiring, through firewalls must be protected with a material or system of the same hourly rating that is listed by MLIT-M.

3.1.4 Painting and Finishing

Painting shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

3.1.5 Welding

No welding shall be done on piping, duct or equipment without written approval of the Contracting Officer. The capacitor discharge welding process may be used for securing metal fasteners to duct.

3.1.6 Pipes/Ducts/Equipment That Require Insulation

Insulation is required on all pipes, ducts, or equipment, except for omitted items as specified.

3.2 PIPE INSULATION SYSTEMS INSTALLATION

Install pipe insulation systems in accordance with the approved plates as supplemented by the manufacturer's published installation instructions.

3.2.1 Pipe Insulation

3.2.1.1 General

Pipe insulation shall be installed on aboveground hot and cold pipeline systems as specified below to form a continuous thermal retarder/barrier, including straight runs, fittings and appurtenances unless specified otherwise. Installation shall be with full length units of insulation and using a single cut piece to complete a run. Cut pieces or scraps abutting each other shall not be used. Pipe insulation shall be omitted on the following:

a. Pipe used solely for fire protection.

b. Chromium plated pipe to plumbing fixtures. However, fixtures for use by the physically handicapped shall have the hot water supply and drain, including the trap, insulated where exposed.

c. Sanitary drain lines.

d. Air chambers.

e. Adjacent insulation.

f. Access plates of fan housings.

g. Cleanouts or handholes.

3.2.1.2 Pipe Insulation Material and Thickness

Pipe insulation materials must be as listed in Table 1.

TABLE 1

Insulation Material for Piping

Service					
	Material	Specification	Type	Class	VR/VB Req'd
Chilled Water (Supply & Return, Dual Temperature Piping, 4.44 C nominal)					
	~~[~~Mineral Fiber with Wicking Material~~][~~. Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4.~~]~~	~~[]~~JIS A 9504	~~[~~I~~]~~		~~[~~Yes~~]~~
Heating Hot Water Supply & Return~~, Heated Oil~~ (Max 121 C)					
	Mineral Fiber	JIS A 9504			
Cold Domestic Water Piping, Makeup Water & Drinking Fountain Drain Piping					
	Mineral Fiber	JIS A 9504			
Hot Domestic Water Supply & Recirculating Piping (Max 93 C)					
	Mineral Fiber	JIS A 9504			
Refrigerant Suction Piping (1.67 degrees C nominal)					
	Shall be pre-insulated by manufacturer				
~~Compressed Air Discharge,~~ Steam and Condensate Return (94 to 121 Degrees C					
	Mineral Fiber	JIS A 9504			
Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel					
	Mineral Fiber	JIS A 9504			
~~Horizontal Roof Drain Leaders (Including Underside of Roof Drain Fittings)~~					
	~~Mineral Fiber~~	~~JIS A 9504~~			
Condensate Drain Located Inside Building					
	Mineral Fiber	JIS A 9504			
~~Medium Temperature Hot Water,~~ Steam and Condensate (122 to 176 Degrees C)					
	Mineral Fiber	JIS A 9504			
~~High Temperature Hot Water &~~ Steam (177 to 371 Degrees C)					
	Mineral Fiber	JIS A 9504			
~~Brine Systems Cryogenics (-34 to -18 Degrees C)~~					

TABLE 1					
Insulation Material for Piping					
Service					
	Material	Specification	Type	Class	VR/VB Req'd
	~~Mineral Fiber~~	~~JIS A 9504~~			
~~Brine Systems Cryogenics (-18 to 1.11 Degrees C)~~					
	~~Mineral Fiber~~	~~JIS A 9504~~			
Note: VR/VB = Vapor Retarder/Vapor Barrier					

TABLE 2						
Piping Insulation Thickness (mm) For flexible cellular foam the thickness should be 13mm instead of 15mm. Economic thickness or prevention of condensation is the basis of these tables. If prevention of condensation is the criterion, the ambient temperature and relative humidity must be stated. Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4.						
Service						
	Material	Tube And Pipe Size (mm)				
		<25	25-<40	40-<100	100-<200	> or = 200
~~[~~Chilled Water (Supply & Return, Dual Temperature Piping, 4.44 Degrees C nominal)~~]~~						
	Mineral Fiber with Wicking Material	25	40	40	50	50
~~[~~Chilled Water (Supply & Return, Dual Temperature Piping, 4.44 Degrees C nominal)~~]~~						
	Mineral Fiber with Wicking Material	25	40	40	50	50
Heating Hot Water Supply & Return~~, Heated Oil~~ (Max 121 C)						
	Mineral Fiber	40	40	50	50	50
Cold Domestic Water Piping, Makeup Water~~ & Drinking Fountain Drain Piping~~						
	Mineral Fiber	40	40	50	50	50

TABLE 2

Piping Insulation Thickness (mm)
For flexible cellular foam the thickness should be 13mm instead of 15mm. Economic thickness or prevention of condensation is the basis of these tables. If prevention of condensation is the criterion, the ambient temperature and relative humidity must be stated. Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4.

Service						
	Material	Tube And Pipe Size (mm)				
		<25	25-<40	40-<100	100-<200	> or = 200
	Flexible Elastomeric Cellular	25	25	25	N/A	N/A
Hot Domestic Water Supply & Recirculating Piping (Max 93 C)						
	Mineral Fiber	25	25	25	40	40
Refrigerant Suction Piping (1.67 degrees C nominal)						
	Shall be pre-insulated by manufacturer					
~~Compressed Air Discharge,~~ Steam and Condensate Return (94 to 121 Degrees C						
	Mineral Fiber	40	40	50	50	50
		40*	50*	65*	80*	90*
Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel						
	Mineral Fiber	40	40	50	50	50
~~Horizontal Roof Drain Leaders (Including Underside of Roof Drain Fittings)~~						
	~~Mineral Fiber~~	~~40~~	~~40~~	~~50~~	~~50~~	~~50~~
Condensate Drain Located Inside Building						
	Mineral Fiber	40	40	50	50	50
~~Medium Temperature Hot Water,~~ Steam and Condensate (122 to 176 Degrees C)						
	Mineral Fiber	40	80	80	100	100
		65*	80*	90*		
~~High Temperature Hot Water &~~ Steam (177 to 371 Degrees C)						

TABLE 2							
Piping Insulation Thickness (mm) For flexible cellular foam the thickness should be 13mm instead of 15mm. Economic thickness or prevention of condensation is the basis of these tables. If prevention of condensation is the criterion, the ambient temperature and relative humidity must be stated. Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4.							
Service							
	Material	Tube And Pipe Size (mm)					
		<25	25-<40	40-<100	100-<200	> or = 200	
	Mineral Fiber	65	80	80	100	100	
~~Brine Systems Cryogenics (-34 to -18 Degrees C)~~							
	~~Mineral Fiber~~	~~40~~	~~40~~	~~50~~	~~50~~	~~50~~	
~~Brine Systems Cryogenics (-18 to 1.11 Degrees C)~~							
	~~Mineral Fiber~~	~~40~~	~~40~~	~~50~~	~~50~~	~~50~~	

3.2.2 Aboveground Cold Pipelines

The following cold pipelines for minus 34 to plus 16 degrees C, shall be insulated in accordance with Table 2 except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted. This includes but is not limited to the following:

a. Make-up water.

b. Horizontal and vertical portions of interior roof drains.

c. Refrigerant suction lines.

d. Chilled water.

e. Dual temperature water, i.e. HVAC hot/chilled water.

f. Air conditioner condensate drains.

~~g. Brine system cryogenics~~

~~h~~g. Exposed lavatory drains and domestic water lines serving plumbing fixtures for handicap persons.

~~i~~h. Domestic cold and chilled drinking water.

3.2.2.1 Insulation Material and Thickness

Insulation thickness for cold pipelines shall be determined using Table 2.

3.2.3 Aboveground Hot Pipelines

3.2.3.1 General Requirements

All hot pipe lines above 16 degrees C, except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted, shall be insulated in accordance with Table 2. This includes but is not limited to the following:

a. Domestic hot water supply & re-circulating system.

b. Steam.

c. Condensate~~ & compressed air discharge~~.

d. Hot water heating.

~~e. Heated oil.~~

~~f. Water defrost lines in refrigerated rooms.~~

Insulation shall be covered, in accordance with manufacturer's recommendations, with a factory applied Type I jacket or field applied aluminum where required or seal welded PVC.

3.2.4 Piping Exposed to Weather

Piping exposed to weather shall be insulated and jacketed in accordance with MLIT-M.

3.3 DUCT INSULATION SYSTEMS INSTALLATION

Install duct insulation systems in accordance with the approved plates as supplemented by the manufacturer's published installation instructions. Duct insulation minimum thickness and insulation level must be as listed in Table 4.

3.3.1 Duct Insulation Minimum Thickness

Duct insulation minimum thickness in accordance with Table 3.

Table 3 - Minimum Duct Insulation (mm)

Cold Air Ducts	50
~~Relief Ducts~~	~~40~~
Fresh Air Intake Ducts	40
~~Warm Air Ducts~~	~~50~~
~~Relief Ducts~~	~~40~~
~~Fresh Air Intake Ducts~~	~~40~~

3.3.2 Insulation and Vapor Retarder/Vapor Barrier for Cold Air Duct

Insulation and vapor retarder/vapor barrier shall be provided for the following cold air ducts and associated equipment.

a. Supply ducts.

b. Return air ducts.

c. Relief ducts.

d. Flexible run-outs (field-insulated).

~~e. Plenums.~~

~~f. Duct-mounted coil casings.~~

~~g~~e. Coil headers and return bends.

~~h~~f. Coil casings.

~~i~~g. Fresh air intake ducts.

~~j~~h. Filter boxes.

~~k~~i. Mixing boxes (field-insulated).

~~l~~j. Supply fans (field-insulated).

~~m~~k. Site-erected air conditioner casings.

~~n~~l. Ducts exposed to weather.

~~o. Combustion air intake ducts.~~

~~3.3.3 Insulation for Warm Air Duct~~

~~Insulation and vapor barrier shall be provided for the following warm air ducts and associated equipment:.~~

~~a. Supply ducts.~~

~~b. Return air ducts.~~

~~c. Relief air ducts~~

~~d. Flexible run-outs (field insulated).~~

~~e. Plenums.~~

~~f. Duct-mounted coil casings.~~

~~g. Coil-headers and return bends.~~

~~h. Coil casings.~~

~~i. Fresh air intake ducts.~~

~~j. Filter boxes.~~

~~k. Mixing boxes.~~

~~l. Supply fans.~~

~~m. Site-erected air conditioner casings.~~

~~n. Ducts exposed to weather.~~

3.3.3 Ducts Handling Air for Dual Purpose

For air handling ducts for dual purpose below and above 16 degrees C, ducts shall be insulated as specified for cold air duct.

3.3.4 Duct Test Holes

After duct systems have been tested, adjusted, and balanced, breaks in the insulation and jacket shall be repaired in accordance with the applicable section of this specification for the type of duct insulation to be repaired.

3.3.5 Duct Exposed to Weather

3.3.5.1 Installation

Ducts exposed to weather shall be insulated and finished as specified in accordance with MLIT-M.

3.3.6 Kitchen Exhaust Duct Insulation

NFPA 96 for ~~[~~ovens~~,] [griddles,] [deep fat fryers,] [steam kettles,] [vegetable steamers,] [high pressure cookers,] [and] [mobile serving units]~~. Provide insulation with 19 mm wide, minimum 4 mm thick galvanized steel bands spaced not over 305 mm o.c.; or 16 gauge galvanized steel wire with corner clips under the wire; or with heavy welded pins spaced not over 305 mm apart each way. Do not use adhesives.

3.4 EQUIPMENT INSULATION SYSTEMS INSTALLATION

Install equipment insulation systems in accordance with MLIT-M.

3.4.1 Insulation for Cold Equipment

Cold equipment below 16 degrees C: Insulation shall be furnished on equipment handling media below 16 degrees C including the following:

a. Pumps.

b. Refrigeration equipment parts that are not factory insulated.

c. Drip pans under chilled equipment.

d. Cold water storage tanks.

e. Water softeners.

~~f. Duct mounted coils.~~

~~g~~f. Cold and chilled water pumps.

~~h. Pneumatic water tanks.~~

~~i~~g. Roof drain bodies.

~~j~~h. Air handling equipment parts that are not factory insulated.

~~k~~i. Expansion and air separation tanks.

3.4.1.1 Insulation Type

Insulation shall be suitable for the temperature encountered. Material and thicknesses shall be as shown in Table 5:

TABLE 5		
Insulation Thickness for Cold Equipment (mm)		
Equipment handling media at indicated temperature		
	Material	Thickness (mm)
	Mineral Fiber	MLIT-M

3.4.2 Insulation for Hot Equipment

Insulation shall be furnished on equipment handling media above 16 degrees C including the following:

a. Converters.

b. Heat exchangers.

c. Hot water generators.

d. Water heaters.

e. Pumps handling media above 54 degrees C.

~~f. Fuel oil heaters.~~

~~g~~f. Hot water storage tanks.

~~h~~g. Air separation tanks.

~~i. Surge tanks.~~

~~j~~h. Flash tanks.

~~k. Feed-water heaters.~~

~~l. Unjacketed boilers or parts of boilers.~~

~~m. Boiler flue gas connection from boiler to stack (if inside).~~

~~n. Induced draft fans.~~

~~o. Fly ash and soot collectors.~~

~~p. Condensate receivers.~~

3.4.2.1 Insulation

Insulation shall be suitable for the temperature encountered. Shell and tube-type heat exchangers shall be insulated for the temperature of the shell medium.

Insulation thickness for hot equipment shall be determined using Table 6:

TABLE 6		
Insulation Thickness for Hot Equipment (mm)		
Equipment handling steam or media at indicated pressure or temperature limit		
	Material	Thickness (mm)
103 kPa or 121 degrees C		
	Rigid Mineral Fiber	50
	Flexible Mineral Fiber	50
1380 kPa or 204 degree C		
	Rigid Mineral Fiber	75
	Flexible Mineral Fiber	75
316 degrees C		
	Rigid Mineral Fiber	125
	Flexible Mineral Fiber	150
316 degrees C: Thickness necessary to limit the external temperature of the insulation to 50 C. Heat transfer calculations shall be submitted to substantiate insulation and thickness selection.		

3.4.2.2 Insulation of Boiler Stack and Diesel Engine Exhaust Pipe

Insulation type and thickness shall be in accordance with the following Table 7.

TABLE 7						
Insulation and Thickness for Boiler Stack and Diesel Engine Exhaust Pipe						
Service & Surface Temperature Range (Degrees C)						
	Material	Outside Diameter (mm)				
		6 - 32	25 - 80	90-125	150 - 250	> or = 280 - 900
Boiler Stack (Up to 204 degrees C)						
	Mineral Fiber JIS A 9504	N/A	N/A	75	90	100
Boiler Stack (205 to 315 degrees C)						
	Mineral Fiber JIS A 9504 ~~ASTM C547 Class 2, ASTM C592 Class 1, or ASTM C612 Class 3~~	N/A	N/A	100	100	125
	Mineral Fiber/Cellular Glass Composite:					
	Mineral Fiber JIS A 9504 ~~ASTM C547 Class 2, ASTM C592 Class 1, or ASTM C612 Class 3~~	25	25	25	25	50
Boiler Stack (316 to 427 degrees C)						
	Mineral Fiber JIS A 9504 ~~ASTM C547 Class 3, ASTM C592 Class 1, or ASTM C612 Class 3 JIS A 9504~~	N/A	N/A	100	100	150
	Mineral Fiber/Cellular Glass Composite:					
	Mineral Fiber JIS A 9504 ~~ASTM C547 Class 2, ASTM C592 Class 1, or ASTM C612 Class 3 JIS A 9504~~	50	50	50	80	80
Diesel Engine Exhaust (Up to 371 degrees C)						

TABLE 7						
Insulation and Thickness for Boiler Stack and Diesel Engine Exhaust Pipe						
Service & Surface Temperature Range (Degrees C)						
	Material	Outside Diameter (mm)				
		6 - 32	25 - 80	90-125	150 - 250	> or = 280 - 900
	Mineral Fiber JIS A 9504 ~~ASTM C547 Class 2, ASTM C592 Class 1, or ASTM C612 Class 3 JIS A 9504~~	50	50	50	80	80

3.4.3 Equipment Handling Dual Temperature Media

Below and above 16 degrees C: equipment handling dual temperature media shall be insulated as specified for cold equipment.

3.4.4 Equipment Exposed to Weather

3.4.4.1 Installation

Equipment exposed to weather shall be insulated and finished in accordance with the requirements for ducts exposed to weather in paragraph DUCT INSULATION INSTALLATION.

-- End of Section --

SECTION 23 08 00

COMMISSIONING OF MECHANICAL~~[~~ AND PLUMBING~~]~~ SYSTEMS
05/23, CHG 1: 08/24

PART 1 GENERAL

Building Commissioning is a systematic, quality-focused process for enhancing the delivery of a project that focuses on verifying and documenting that all of the commissioned systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the project requirements. The purpose is to reduce the cost and performance risks associated with delivering facilities projects, and to increase value to owners, occupants, and users. Comply with specification Section 01 91 00.15 BUILDING COMMISSIONING.

1.1 SEQUENCING AND SCHEDULING

Complete functional performance testing prior to performance verification testing required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Complete the following prior to starting Functional Performance Tests of mechanical systems:

a. All equipment and systems completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications.

b. Final DALT Report submitted and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

c. Pre-final Testing, Adjusting, and Balancing Report submitted in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

d. Performance Verification Tests of the controls systems have been completed and the Performance Verification Test Report has been submitted and approved in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

e. The Certificate of Readiness submitted and approved in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

~~[f. Air Leakage Test Reports and Diagnostic Test Reports submitted and approved in accordance with Section 01 91 19 BUILDING ENCLOSURE COMMISSIONING~~

~~][~~f. Tests, Flushing, and Disinfection in accordance with Section ~~[~~22 00 00 PLUMBING, GENERAL PURPOSE~~][22 00 70 PLUMBING FOR HEALTHCARE FACILITIES]~~.

~~][g. Inspection and Testing in accordance with Section 22 33 30 SOLAR WATER HEATING EQUIPMENT~~

~~]~~1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control ~~approval. Submittals not having a "G" or "S" classification are for~~ information only. When used, a code following the

"G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Test Equipment; G~~, [_____]~~

SD-06 Test Reports

Pipe Flushing, Testing, And Water Treatment Reports; G~~, [_____]~~

Completed Pre-Functional Checklists; G~~, [_____]~~

~~[~~ Seasonal Test Report; G~~, [_____]~~

~~][~~ Full-Load Test Report; G~~, [_____]~~

~~][~~ Post-Construction Trend Log Report; G~~, [_____]~~

~~]~~ SD-07 Certificates

Certificate Of Readiness; G~~, [_____]~~

1.3 ACCESSIBILITY REQUIREMENTS

Equipment, systems, and devices for commissioned systems must be accessible. Make necessary modifications if systems and devices are not accessible for inspections and testing.

Assist commissioning team in testing by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

1.4 COORDINATION

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the Commissioning Provider Lead Commissioning Specialist in accordance with Section 01 91 00.15 and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the Commissioning Provider Lead Commissioning Specialist. Coordinate scheduling of Functional Performance Testing with the commissioning team. Provide plans, reports, notes, and other documentation to the Commissioning Provider Lead Commissioning Specialist as specified in the commissioning plan, as it is completed.

1.5 PIPE FLUSHING, TESTING, AND WATER TREATMENT REPORTS

Test requirements are specified in Division ~~[~~22 and ~~]~~23 piping Sections. Prepare a pipe system cleaning, flushing, and hydrostatic testing log. Provide cleaning, flushing, testing, and water treatment log and final reports.

Include the following in the pipe system cleaning, flushing, and hydrostatic testing log:

a. Minimum flushing water velocity.

b. Water treatment reports.

c. Tracking checklist for managing and ensuring that all pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.

1.6 CERTIFICATE OF READINESS

Submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING for all equipment and systems including start-up reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; Issues Log; HVAC Controls Start-Up Reports; Performance Verification Test Reports;[and the building envelope Air Leakage Test Reports and Diagnostic Test Reports]. Do not schedule Functional Performance Tests for the system until the Certificate of Readiness for that system receives approval by the Government. The CxC, Quality Control Manager, and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness.

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

Provide all testing equipment required to perform testing for the systems to be commissioned, except for equipment specific to and used by TAB as required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Provide a sufficient quantity of two-way radios for each subcontractor. Submit list of Test Equipment and instrumentation to be used for testing including equipment/instrument identification number, equipment application or planned use, manufacturer, make, model, and serial number, and calibration history with certificates. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

2.1.1 Proprietary Equipment

Provide manufacturer's proprietary test equipment and software required by any equipment manufacturer for programming and start-up, whether specified or not. Provide manufacturer test equipment, demonstrate its use, and assist in the commissioning process as needed. Provide data logging equipment and software required to test equipment.

2.1.2 Calibration and Accuracy

Comply with equipment manufacturer's test equipment calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Contracting Officer upon request.

Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Unless otherwise noted, the following minimum requirements apply: Provide temperature sensors and digital thermometers with a certified calibration within the past year to an accuracy of 0.3 degrees C and a resolution of plus or minus 0.1 degrees C. Provide pressure sensors with an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and calibrated within the last year.

PART 3 EXECUTION

3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes to allow the Commissioning Provider Lead Commissioning Specialist~~—~~ to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

3.2 COMMISSIONING CONSTRUCTION OBSERVATION CHECKLISTS

Commissioning construction observation checklists include Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists. Provide commissioning construction observation checklists for the Interim and Final Construction Phase Commissioning Plan in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

~~Download example Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists for Section 01 91 00.15 BUILDING COMMISSIONING at the following location: https://www.wbdg.org/dod/ufgs/forms-graphics-tables. The checklists submitted in the Interim and Final Construction Phase Commissioning Plans must contain the same level of detail shown in the examples. The submitted checklists are not required to match the format of the examples.~~

3.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS. Pre-functional checklists must be tailored to verify the specific installation requirements and details of the construction documents and manufacturer's instructions.

Use the Pre-Functional Checklists prepared by the CxC for physical inspection or testing to demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS.

3.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract~~[and Owner's Project Requirements Document]~~. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

a. Notable system features including information about controls to facilitate understanding of system operation and maintenance

b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements and specifications~~[and Owner's Project Requirements Document]~~. Recommendation must clearly indicate that the system should or should not be approved by the Government.

c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions.

d. Attendees.

e. Identification of the equipment involved in the test.

f. **System feature identification.**

g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station.

h. Actuator operation observations demonstrating actuator responses to commands from the control system.

i. As-found condition of the system operation.

j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.

k. Space for comments for each test item.

Use the Functional Performance Test Checklists prepared by the CxC that list, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING.

3.2.3 Integrated Systems Test Checklists

Integrated Systems Test Checklists must include test procedures that explain, step-by-step, the actions and expected results that will demonstrate that the interactive operations between systems performs in accordance with the contract~~[and Owner's Project Requirements Document]~~. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections in the Integrated Systems Test Checklists:

a. Notable features of the interconnected systems organized by discipline including information to facilitate understanding of system operation

b. Conclusions and recommendations. Conclusions must clearly indicate if the systems do or do not perform in accordance with contract requirements and specifications~~[and the Owner's Project Requirements Document]~~. Recommendation must clearly indicate that the systems should or should not be approved by the Government

c. Test conditions including date and beginning and ending time

d. Identification of the equipment and systems involved in the test

e. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.

f. Space for comments for each test item.

Use the Integrated Systems Test Checklists prepared by the CxC that list, step-by-step, the actions and expected results that will demonstrate that the interactive operations between systems performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING.

3.3 PRE-FUNCTIONAL CHECKS

Pre-Functional Checks are a type of Commissioning Inspection in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Complete one Pre-Functional Checklist for each individual item of equipment or system for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures, and controls. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists. Provide manufacturer's installation manual for each type of unit. Indicate commissioning team member inspection and validation of each Pre-Functional Checklist item by initials. Validation of each Pre-Functional Checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Validation of each item by each team member must occur immediately following completion of their respective duties related to the item. Commissioning Specialist validation of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents~~[and the Owner's Project Requirements Document]~~. Required commissioning team members for Pre-Functional Checks includes the CxC, Mechanical Commissioning Specialist, Quality Control Manager, sub-contractor representative for each trade responsible for construction/installation of the checklist item, and the TAB representative (for items impacting TAB). Submit the initialed and Completed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklists items for each system.

3.4 STARTUP AND INITIAL CHECKOUT

Document start-up and initial testing procedures, and include in the Completed Pre-Functional Checklists submittal, including:

a. Startup tests and factory testing reports.

b. Manufacturer's representative start-up, operating, troubleshooting and maintenance procedures.

~~[~~ c. Additional documentation necessary for third party certification programs.

~~]~~ d. Perform and clearly document system operational checks and quality control checks as they are completed, and providing a copy to the commissioning team.

e. Correct deficiencies and sign the Certificate of Readiness for each system before functional performance testing.

3.5 DUCT AIR LEAKAGE TEST (DALT) REPORT REVIEW AND VERIFICATION

3.5.1 Duct Air Leakage Test (DALT) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final DALT Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for reviewing the pre-final DALT Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to DALT Report approval.

3.5.2 Duct Air Leakage Test (DALT) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the DALT Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for witnessing the DALT Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to DALT Report approval.

3.6 TESTING, ADJUSTING, AND BALANCING REPORT REVIEW AND VERIFICATION

3.6.1 Testing, Adjusting, and Balancing (TAB) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for reviewing the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to TAB Report approval.

3.6.2 Testing, Adjusting, and Balancing (TAB) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for witnessing the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to TAB Report approval.

3.7 HVAC CONTROLS START-UP AND PERFORMANCE VERIFICATION TEST REVIEW

The Mechanical System Technical Commissioning Specialist must review the

Start-Up Testing Report, PVT Procedures and PVT Reports, including endurance testing trend submittals, required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC~~[and Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION]~~. The Mechanical System Technical Commissioning Specialist must review each submittal and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. The Commissioning Specialist is responsible for reviewing the Start-Up Testing Report, PVT Procedures and PVT Reports including endurance testing trend data required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC~~[and Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION]~~ and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to final acceptance.

~~]~~3.8 FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING

Functional Performance and Integrated Systems Testing are a type of Commissioning Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Conduct Functional Performance Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and requirements in this Section. Prior to Functional Performance Testing, complete all prerequisites in accordance with paragraph SEQUENCING AND SCHEDULING. Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents~~[and the Owner's Project Requirements Document]~~.

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

3.8.1 Test Scheduling and Coordination

Schedule and conduct Initial Functional Performance Tests as soon as all contract work is completed, regardless of the season. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests.

Functional Performance Tests and Integrated Systems Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

3.8.2 Preparation

Put equipment and systems into operation and continue operation during each working day of functional performance and integrated systems testing, as required. Verify temperature and pressure taps in accordance with Contract Documents. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

Perform minor adjustments to equipment and systems during Functional Performance Tests as deemed necessary by the commissioning team. Where calibrated DDC sensors cannot be used to record test data, provide measuring instruments, logging devices, and data acquisition equipment to record data for the complete range of test data for the required test period.

3.8.3 Testing Procedures

Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

Follow the Functional Performance Test from the approved Final Construction Phase Commissioning Plan. Perform Functional Performance Tests for each item of equipment and each system required to be commissioned. Verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, and all other performance requirements comply with contract, regardless of the specific items listed within the checklists provided. In general, testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Commissioning Specialists are responsible for determining the order of components and systems to be tested.

Indicate validation of each item of equipment and systems tested by signature of each commissioning team member for each test. The Quality Control Representative, Commissioning Specialists, and Contracting Officer's Representative, if present, must indicate validation after the equipment and systems are free of deficiencies.

3.8.4 Simulating Conditions

Functional performance testing is conducted by simulating conditions at control devices to initiate a control system response. Before testing, calibrate all sensors, transducers and devices. Over-writing control input values through the control system is not acceptable unless approved by the Contracting Officer. Perform each test under conditions that simulate actual conditions as close as is practically possible. Specific examples of simulating conditions are provided below. Do not simulate conditions when damage to the system or building may result.

a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).

b. Dirty filter pressure drops can be simulated by partially blocking filter face.

c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.

d. High outside air temperatures can be simulated with a hair blower.

e. Raising entering cooling coil temperatures by activating a heating/preheat coil can be used to simulate entering cooling coil conditions.

f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer, as noted above, for special cases.

g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 12 degrees C, when the outside air temperature is above 12 degrees C, temporarily change the lockout set point to be minus 17 degrees C above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point to cause damage to the components, systems, or the building structure and contents.

h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition.

i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer.

j. Test dehumidifiers in accordance with the manufacturer's recommendations. Humid conditions can be simulated through an external humidifier or humidifying agent present during testing.

3.8.5 Manufacturer's Representative

Provide a factory trained representative authorized by the equipment manufacturer to perform Functional Performance Testing for the following equipment:

~~[~~ Chillers

~~][~~ ~~Cooling towers and evaporatively cooled condensers~~

~~][~~ ~~Boilers~~

~~][~~ Packaged Direct-Expansion Refrigeration Equipment, including variable refrigerant flow (VRF) systems

~~][~~ ~~Packaged Computer Room [Air Handlers (CRAH)] [Air Conditioners (CRAC)]~~

~~][~~ Booster Pumps

~~][~~ ~~Packaged Air Compressors~~

~~][~~ ~~Water Quality and Chemical Treatment Systems~~

~~][~~ ~~Solar Water Heating Systems~~
Domestic Hot Water System

Fan Coil Units

Exhaust Fans

Heat Exchanger

Dehumidifier

Energy Recovery Ventilator

~~]~~ Ensure the test representative reviews, approves, and signs the completed field test report. Include person's name with signatures.

3.8.6 Integrated Systems Tests

Follow the Integrated Systems Test Checklists from the approved Final Construction Phase Commissioning Plan. Integrated Systems Tests must be performed for the interactive operation between HVAC systems and other systems such as fire protection systems, smoke control systems, or back-up electrical supply. Verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation or correct operation of uninterruptible power supplies or energy generators and connected systems.

3.8.7 Sample Strategy

Perform Functional Performance Tests using the following sample strategy. Perform Functional Performance Tests using the sample strategy identified herein. Test all central plant equipment, primary air handling units, and process cooling or heating equipment. Test all system-level equipment serving multiple zones. ~~[~~Twenty-five~~] [Fifty] [_____]~~ percent sample testing is allowed for large groups of identical equipment with identical controllers serving single zones such as air terminal units, fan coil units, unitary equipment, and plumbing fixtures. Sample size may be no less than three units. Refer to Section 01 91 00.15 BUILDING COMMISSIONING for sample strategy. Complete a Functional Performance Test Checklist for each item of equipment or system to be tested. During testing, Government representatives may select the specific equipment or system to be tested for sample sizes less than 100 percent. Equipment Identifiers are as indicated on the design drawings:

Equipment Identifier	Sample Size (Percent)
~~AHU~~	~~[_____]~~
~~VAV~~	~~[_____]~~
~~CUH~~	~~[_____]~~
~~CWP~~	~~[_____]~~
~~DWH~~	~~[_____]~~
Unit EF	25
DOAS	100
DBF	50

HX	100
ACCU	100
CHWP/HHWP	100
UH	100
General EF	100
DH	50

Perform Integrated Systems Tests and complete an Integrated Systems Test Checklist for for all (100 percent) systems and equipment having interactive operation.

3.8.7.1 100 Percent Sample Procedures

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.8.7.2 Less than 100 Percent Sample Procedures

Randomly test each sample group of identical equipment. If 10 percent of the units in the first sample fail the functional performance tests, test a second sample group, the same size as the first sample group. The second sample must not include any units from the first sample group.

If 10 percent of the units in the second sample fail, test all remaining units. If at any point frequent failures occur the CxC may stop the testing and require the contractor to perform and document a checkout of the remaining units prior to continuing functional testing.

3.9 INTEGRATED SYSTEMS TEST

Perform Integrated Systems Tests only after the Functional Performance Tests for each associated system are completed with all deficiencies resolved and after the related Functional Performance Test Checklists have been signed by each commissioning team member.

3.10 RETESTING REQUIREMENTS

Abort tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test only after all deficiencies identified during the original tests have been corrected.

Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members. Correct deficiencies as identified by the commissioning team and retest the systems to be commissioned.

3.11 SYSTEM ACCEPTANCE

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract and accepted design requirements that can be tested during initial Functional Performance Tests. All test procedures must be successful completed prior to full systems acceptance.

~~[~~3.12 SEASONAL TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover, regardless of the season.

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems during peak heating and cooling seasons during outdoor air condition design extremes. Schedule Seasonal Functional Performance Tests in coordination with the Contracting Officer. Submit Seasonal Test Report within 14 days of test completion. Include seasonal test report in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

Execute seasonal functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.

~~][~~3.13 FULL-LOAD TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover. In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems under full-load conditions.~~[Develop and implement means of artificial loading to demonstrate the ability of the process cooling systems to handle peak process loads.]~~ Schedule Full-Load Functional Performance Tests in coordination with the Contracting Officer. Submit Full-Load Test Report within 14 days of test completion.

Execute full-load functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any full load testing.

~~]~~3.14 TRAINING

The Mechanical Systems Technical Commissioning Specialist must review the training plan required by Section 01 78 00 OPERATION AND MAINTENANCE DATA and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

The Commissioning Provider is responsible for overseeing and approving the training plan required by Section 01 78 00 OPERATION AND MAINTENANCE DATA and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

Coordinate, schedule, and document all required training. At a minimum, include the following items in the training report for commissioned systems:

a. Complete commissioning documentation.

b. Complete O&M data.

c. Complete Training.

d. Purpose of equipment.

e. Principle of how the equipment works.

f. Important parts and assemblies.

g. How the equipment achieves its purpose and necessary operating conditions.

h. Most likely failure modes, causes and corrections.

i. On site demonstration.

j. Provide updates to O&M manuals based on field modifications.

k. Provide training of the post-occupancy operations and maintenance staff.

3.15 COMMISSIONING REPORT

Include all completed Pre-Functional Checklists and Functional Performance and Integrated Systems Test Checklists in the Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING. Include the approved TAB Report.

~~[~~3.16 POST-CONSTRUCTION ENDURANCE TEST

Perform an Endurance Test in accordance with the paragraph ENDURANCE TEST ENDURANCE TESTING in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC once during the peak heating season and once during the peak cooling season during outdoor air condition extremes with the exception that network bandwidth usage measurement and recording is not required.~~[Use the Temporary Trending Hardware, if necessary, in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.]~~

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation and operate as required by the construction contract, the accepted design ~~[, and the Owner's Project Requirements Document]~~. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation, recommendations for correction, and includes a graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist. Provide a Post-Construction Trend Log Report meeting the same requirements as the Endurance Testing Results Format in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following download of the trended data. Include the Post-Construction Trend Log Reports in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING.

~~]~~ -- End of Section --

SECTION 23 09 00

INSTRUMENTATION AND CONTROL FOR HVAC
08/24, CHG 1: 08/25

PART 1 GENERAL

1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, except for the Front End which is specified in Section 25 10 10 UTILITY MONITORING AND CONTROL (UMCS) FRONT END AND INTEGRATION, suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as indicated and shown and in accordance with Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC,~~[Section 23 09 93 SEQUENCES OF OPERATION FOR HVAC CONTROL,]~~ Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS, and other referenced Sections.

~~[1.1.1 Control System Vendor Requirement~~

~~The control system provided under this Section must be [_____]. Configure the equipment as indicated in [attached configuration setting requirements][the configuration settings drawings][_____].~~

~~]~~1.1.1 Proprietary Systems

1.1.1.1 Proprietary Systems Exempted From Open Protocol Requirements

The following systems are specifically exempted from the open protocol requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS:

a. A simple split (DX) system consisting of a single indoor unit and a single outdoor unit from the same manufacturer.

b. Systems in Table I (previously approved by the designer in accordance with UFC 3-410-02).

TABLE I: Systems Approved to Use Proprietary Communications		
System	Type (Multi-Split/VRF or Chiller/Boiler Plant)	Proprietary Multi-Split Engineering Tool Software Required (for Multi-Split/VRF only)

c. A system (not already shown Table I) of multiple boilers or multiple chillers communicating with a proprietary network for which an approved request has been obtained and for which: all units are from the same manufacturer, they are all co-located in the same room, the network connecting them is fully contained in that room, and the units are operating using a common "plant" sequence of operation which stages the units in a manner that requires operational parameters be shared between them and which cannot be accomplished with a single lead-lag command from a third-party controller.

1.1.1.2 Implementation of Proprietary Systems

For proprietary systems exempted from open protocol requirements, a proprietary network and DDC hardware communicating via a proprietary protocol are permitted. For these systems a building control network meeting the requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS must also be provided, along with a gateway or interface to connect the proprietary system to the open building control network.

The proprietary system gateway or interface must provide the required functionality as shown on the points schedule. Scheduling, alarming, trending, overrides, network inputs, network outputs and other protocol related requirements must be met on the open protocol control system as specified in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

1.1.1.3 Proprietary Multi-Split Engineering Tool Software

For each permitted proprietary systems in Table 1 shown as requiring Proprietary Multi-Split Engineering Tool Software, provide the software needed to replace a unit and configure the replacement. Submit hard copies of the software user manuals with the software submittal.

Submit Proprietary Multi-Split Engineering Tool Software on CD-ROM as a Technical Data Package. Submit ~~[_____]~~4 hard copies of the software user manual for each piece of software.

1.1.2 System Requirements

Provide systems meeting the requirements this Section and other Sections referenced by this Section, and which have the following characteristics:

a. The system implements the control sequences of operation ~~[~~shown in the Contract Drawings~~][____]~~ M-800 using DDC hardware to control mechanical and electrical equipment

b. The system meet the requirements of this specification as a stand-alone system and does not require connection to any other system.

c. Control sequences reside in DDC hardware in the building. The building control network is not dependent upon connection to a Utility Monitoring and Control System (UMCS) Front End or to any other system for performance of control sequences. To the greatest extent practical, the hardware performs control sequences without reliance on the building network~~, unless otherwise pre-approved by the Contracting Officer~~.

d. The hardware is installed such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.

e. All necessary documentation, configuration information, programming tools, programs, drivers, and other software are licensed to and otherwise remain with the Government such that the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.

f. Sufficient documentation and data, including rights to documentation and data, are provided such that the Government or their agents can execute work to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.

g. Hardware is installed and configured such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the Contractor, Vendor or Manufacturer.

~~h. All Niagara Framework components have an unrestricted interoperability license with a Niagara Information and Conformance Statement (NICS)which allows open access as defined by Niagara NICS with a value of "ALL" for "Station Compatibility In", "Station Compatibility Out", "Tool Compatibility In" and "Tool Compatibility Out". Note that this will result in the following entries in the license file:~~
~~accept.station.in="*"~~
~~accept.station.out="*"~~
~~accept.wb.in="*"~~
~~accept.wb.out="*"~~

1.1.3 End to End Accuracy

Select products, install and configure the system such that the maximum error of a measured value as read from the DDC Hardware over the network is less than the maximum allowable error specified for the sensor or instrumentation.

1.1.4 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.1.5 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

1.2 RELATED SECTIONS

Related work specified elsewhere:

a. Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

b. Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC.

~~c. Section 23 09 93 SEQUENCES OF OPERATIONS FOR HVAC CONTROLS~~

~~d. Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEMS TESTING~~

~~e. Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS~~

~~[~~

~~f. Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEMS (UMCS) FRONT END AND INTEGRATION~~

~~][~~c. Section 23 81 00 DECENTRALIZED UNITARY HVAC EQUIPMENT.

~~][d. Section 23 81 29 VARIABLE REFRIGERANT FLOW HVAC SYSTEMS~~

~~]~~ d. ~~[~~Section 01 91 00.15 TOTAL BUILDING COMMISSIONING~~][_____]~~.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 511	(2010; R 2016) Certified Ratings Program for Air Control Devices

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135	(2024; INT 1-3 2025; Errata 1 2025; INT 4-6 2025; Addenda CU 2025) BACnet—A Data Communication Protocol for Building Automation and Control Networks
ASHRAE FUN SI	(2021) Fundamentals Handbook, SI Edition

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41	(1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529	(2019) Corrigendum 1 - Amendment 2 - Degrees of Protection Provided by Enclosures (IP Code)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-410-02	(2018; with Change 2, 2021) Direct Digital Control for HVAC and Other Building

UL SOLUTIONS (UL)

UL 5085-3	(2006; Reprint Jan 2022) UL Standard for Safety Low Voltage Transformers - Part 3: Class 2 and Class 3 Transformers

1.4 DEFINITIONS

The following list of definitions includes terms used in Sections referenced by this Section and are included here for completeness. The definitions contained in this Section may disagree with how terms are defined or used in other documents, including documents referenced by this Section. The definitions included here are the authoritative definitions for this Section and all Sections referenced by this Section.

Definitions including "(BACnet)" after the term are definitions derived from ASHRAE 135.

1.4.1 Alarm Generation

Alarm Generation is the monitoring of a value, comparison of the value to alarm conditions and the creation of an alarm when the conditions set for the alarm are met. Note that this does NOT include delivery of the alarm to the final destination (such as a user interface) - see paragraph ALARM ROUTING in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION.

1.4.2 Building Automation and Control Network (BACnet) (BACnet)

The term BACnet is used in two ways. First meaning the BACnet Protocol Standard - the communication requirements as defined by ASHRAE 135 including all annexes and addenda. The second to refer to the overall technology related to the ASHRAE 135 protocol.

1.4.3 BACnet Advanced Application Controller (B-AAC) (BACnet)

A hardware device BTL Listed as a B-AAC, which is required to support BACnet Interoperability Building Blocks (BIBBs) for scheduling and alarming, but is not required to support as many BIBBs as a B-BC.

1.4.4 BACnet Application Specific Controller (B-ASC) (BACnet)

A hardware device BTL Listed as a B-ASC, with fewer BIBB requirements than a B-AAC. It is intended for use in a specific application.

1.4.5 BACnet Building Controller (B-BC) (BACnet)

A hardware device BTL Listed as a B-BC. A general-purpose, field-programmable device capable of carrying out a variety of building automation and control tasks including control and monitoring via direct digital control (DDC) of specific systems and data storage for trend information, time schedules, and alarm data. Like the other BTL Listed controller types (B-AAC, B-ASC etc.) a B-BC device is required to support the server ("B") side of the ReadProperty and WriteProperty services, but unlike the other controller types it is also required to support the client ("A") side of these services. Communication between controllers requires that one of them support the client side and the other support the server side, so a B-BC is often used when communication between controllers is needed.

1.4.6 BACnet Broadcast Management Device (BBMD) (BACnet)

A communications device, typically combined with a BACnet router. A BBMD forwards BACnet broadcast messages to BACnet/IP devices and other BBMDs connected to the same BACnet/IP network. Each IP subnet that is part of a BACnet/IP network must have at least one BBMD. Note there are additional restrictions when multiple BBMDs share an IP subnet.

1.4.7 BACnet/IP (BACnet)

An extension of BACnet, Annex J, defines the use of a reserved UDP socket to transmit BACnet messages over IP networks. A BACnet/IP network is a collection of one or more IP subnets that share the same BACnet network number. See also paragraph BACNET BROADCAST MANAGEMENT DEVICE.

1.4.8 BACnet Internetwork (BACnet)

Two or more BACnet networks, connected with BACnet routers. In a BACnet Internetwork, there exists only one message path between devices.

1.4.9 BACnet Interoperability Building Blocks (BIBBs) (BACnet)

A BIBB is a collection of one or more ASHRAE 135 Services intended to define a higher level of interoperability. BIBBs are combined to build the BACnet functional requirements for a device in a specification. Some BIBBs define additional requirements (beyond requiring support for specific services) in order to achieve a level of interoperability. For example, the BIBB DS-V-A (Data Sharing-View-A), which would typically be used by a front-end, not only requires the client to support the ReadProperty Service, but also provides a list of data types (Object / Properties) which the client must be able to interpret and display for the user.

In the BIBB shorthand notation, -A is the client side and -B is the server side.

The following is a list of some BIBBs used by this or referenced Sections:	
DS-COV-A	Data Sharing-Change of Value (A side)
DS-COV-B	Data Sharing-Change of Value (B side)
NM-RC-B	Network Management-Router Configuration (B side)
DS-RP-A	Data Sharing-Read Property (A side)
DS-RP-B	Data Sharing-Read Property (B side)
DS-RPM-A	Data Sharing-Read Property Multiple (A Side)
DS-RPM-B	Data Sharing-Read Property Multiple (B Side)
DS-WP-A	Data Sharing-Write Property (A Side)

The following is a list of some BIBBs used by this or referenced Sections:	
DM-TS-B	Device Management-Time Synchronization (B Side)
DM-UTC-B	Device Management-UTC Time Synchronization (B Side)
DS-WP-B	Data Sharing-Write Property (B side)
SCHED-E-B	Scheduling-External (B side)
DM-OCD-B	Device Management-Object Creation and Deletion (B side)
AE-N-I-B	Alarm and Event-Notification Internal (B Side)
AE-N-E-B	Alarm and Event-Notification External (B Side)
T-VMT-I-B	Trending-Viewing and Modifying Trends Internal (B Side)
T-VMT-E-B	Trending-Viewing and Modifying Trends External (B Side)

1.4.10 BACnet Network (BACnet)

In BACnet, a portion of the control Internetwork consisting of one or more segments connected by repeaters. Networks are separated by routers.

1.4.11 BACnet Operator Display (B-OD) (BACnet)

A basic operator interface with limited capabilities relative to a B-OWS. It is not intended to perform direct digital control. A B-OD profile could be used for LCD devices, displays affixed to BACnet devices, handheld terminals or other very simple user interfaces.

1.4.12 BACnet Segment (BACnet)

One or more physical segments interconnected by repeaters (ASHRAE 135).

1.4.13 BACnet Smart Actuator (B-SA) (BACnet)

A simple actuator device with limited resources intended for specific applications.

1.4.14 BACnet Smart Sensor (B-SS) (BACnet)

A simple sensing device with limited resources.

1.4.15 BACnet Testing Laboratories (BTL) (BACnet)

Established by BACnet International to support compliance testing and interoperability testing activities and consists of BTL Manager and the BTL Working Group (BTL-WG). BTL also publishes Implementation Guidelines.

1.4.16 BACnet Testing Laboratories (BTL) Listed (BACnet)

A device that has been listed by BACnet Testing Laboratory. Devices may be certified to a specific device profile, in which case the listing indicates that the device supports the required capabilities for that profile, or may be listed as "other".

1.4.17 Binary

A two-state system where an "ON" condition is represented by a high signal level and an "OFF" condition is represented by a low signal level. 'Digital' is sometimes used interchangeably with 'binary'.

1.4.18 Broadcast

Unlike most messages, which are intended for a specific recipient device, a broadcast message is intended for all devices on the network.

1.4.19 Building Control Network (BCN)

The network connecting all DDC Hardware within a building (or specific group of buildings).

1.4.20 Building Point of Connection (BPOC)

A FPOC for a Building Control System. (This term is being phased out of use in preference for FPOC but is still used in some specifications and criteria. When it was used, it typically referred to a piece of control hardware. The current FPOC definition typically refers instead to IT hardware.)

1.4.21 Commandable

See Overridable.

1.4.22 Commandable Objects (BACnet)

Commandable Objects have a Commandable Property, Priority_Array, and Relinquish_Default Property as defined in ASHRAE 135, Clause 19.2, Command Prioritization.

1.4.23 Configurable

A property, setting, or value is configurable if it can be changed via hardware settings on the device, via the use of engineering software or over the control network from the front end, and is retained through (after) loss of power.

a. In a ~~non-Niagara Framework~~ BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:
   (1) via BACnet services (including proprietary BACnet services)
   (2) via hardware settings on the device

~~b. In a Niagara Framework BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:~~
   ~~(1) via BACnet services (including proprietary BACnet services)~~
   ~~(2) via hardware settings on the device~~
   ~~(3) via the Niagara Framework~~

Note this is more stringent than the ASHRAE 135 definition.

1.4.24 Control Logic Diagram

A graphical representation of control logic for multiple processes that make up a system.

1.4.25 Device (BACnet)

A Digital Controller that contains a BACnet Device Object and uses BACnet to communicate with other devices.

1.4.26 Device Object (BACnet)

Every BACnet device requires one Device Object, whose properties represent the network visible properties of that device. Every Device Object requires a unique Object Identifier number on the BACnet Internetwork. This number is often referred to as the device instance or device ID.

1.4.27 Device Profile (BACnet)

A collection of BIBBs determining minimum BACnet capabilities of a device, defined in ASHRAE 135. Standard device profiles include BACnet Advanced Workstations (B-AWS), BACnet Building Controllers (B-BC), BACnet Advanced Application Controllers (B-AAC), BACnet Application Specific Controllers (B-ASC), BACnet Smart Actuator (B-SA), and BACnet Smart Sensor (B-SS).

1.4.28 Digital Controller

An electronic controller, usually with internal programming logic and digital and analog input/output capability, which performs control functions.

1.4.29 Direct Digital Control (DDC)

Digital controllers performing control logic. Usually the controller directly senses physical values, makes control decisions with internal programs, and outputs control signals to directly operate switches, valves, dampers, and motor controllers.

1.4.30 Field Point of Connection (FPOC)

The FPOC is the point of connection between the UMCS IP Network and the field control network (either an IP network, a non-IP network, or a combination of both). The hardware at this location which provides the connection is generally an IT device such as a switch, IP router, or firewall.

In general, the term "FPOC Location" means the place where this connection occurs, and "FPOC Hardware" means the device that provides the connection. Sometimes the term "FPOC" is used to mean either and its actual meaning (i.e. location or hardware) is determined by the context in which it is used.

~~1.4.31 Fox Protocol (Niagara Framework)~~

~~The protocol used for communication between components in the Niagara Framework. By default, Fox uses TCP port 1911.~~

1.4.31 Gateway

A device that translates from one protocol application data format to another. Devices that change only the transport mechanism of the protocol - "translating" from TP/FT-10 to Ethernet/IP or from BACnet MS/TP to BACnet over IP for example - are not gateways as the underlying data format does not change. Gateways are also called Communications Bridges or Protocol Translators. Some devices contain bith gateway and control functions; these devices are considered both gateways and digital controllers and must meet the requirements for both.

~~A Niagara Framework Supervisory Gateway is one type of Gateway.~~

1.4.32 IEEE 802.3 Ethernet

A family of local-area-network technologies providing high-speed networking features over various media, typically Cat 5, 5e or Cat 6 twisted pair copper or fiber optic cable.

1.4.33 Internet Protocol (IP, TCP/IP, UDP/IP)

A communication method, the most common use is the World Wide Web. At the lowest level, it is based on Internet Protocol (IP), a method for conveying and routing packets of information over various LAN media. Two common protocols using IP are User Datagram Protocol (UDP) and Transmission Control Protocol (TCP). UDP conveys information to well-known "sockets" without confirmation of receipt. TCP establishes connections, also known as "sessions", which have end-to-end confirmation and guaranteed sequence of delivery.

1.4.34 Input/Output (I/O)

Physical inputs and outputs to and from a device, although the term sometimes describes network or "virtual" inputs or outputs. See also "Points".

1.4.35 I/O Expansion Unit

An I/O expansion unit provides additional point capacity to a digital controller.

1.4.36 IP subnet

A group of devices which share a defined range IP addresses. Devices on a common IP subnet can share data (including broadcasts) directly without the need for the traffic to traverse an IP router.

~~1.4.37 JACE (Niagara Framework)~~

~~Java Application Control Engine. See paragraph NIAGARA FRAMEWORK SUPERVISORY GATEWAY~~

1.4.37 Local-Area Network (LAN)

A communication network that spans a limited geographic area and uses the same basic communication technology throughout.

1.4.38 Local Display Panels (LDPs)

A DDC Hardware with a display and navigation buttons, and must provide display and adjustment of points as shown on the Points Schedule and as indicated.

1.4.39 MAC Address

Media Access Control address. The physical device address that identifies a device on a Local Area Network.

1.4.40 Multidrop Serial Bus/Token-Passing (MS/TP) (BACnet)

Data link protocol as defined by the BACnet standard. Multiple speeds (data rates) are permitted by the BACnet MS/TP standard.

1.4.41 Monitoring and Control (M&C) Software

The UMCS 'front end' software which performs supervisory functions such as alarm handling, scheduling and data logging and provides a user interface for monitoring the system and configuring these functions.

1.4.42 Network Number (BACnet)

A site-specific number assigned to each network. This network number must be unique throughout the BACnet Internetwork.

~~1.4.43 Niagara Framework (Niagara Framework)~~

~~A set of hardware and software specifications for building and utility control owned by Tridium Inc. and licensed to multiple vendors. The Framework consists of front end (M&C) software, web based clients, field level control hardware, and engineering tools. While the Niagara Framework is not adopted by a recognized standards body and does not use an open licensing model, it is sufficiently well-supported by multiple HVAC vendors to be considered a de-facto Open Standard.~~

~~1.4.44 Niagara Framework Supervisory Gateway (Niagara Framework)~~

~~DDC Hardware component of the Niagara Framework. A typical Niagara architecture has Niagara specific supervisory gateways at the IP level and other (non-Niagara specific) controllers on field networks (TP/FT-10, MS/TP, etc.) beneath the Niagara supervisory gateways. The Niagara specific controllers function as a gateway between the Niagara framework protocol (Fox) and the field network beneath. These supervisory gateways may also be used as general purpose controllers and also have the capability to provide a web-based user interface.~~

~~Note that different vendors refer to this component by different names. The most common name is "JACE"; other names include (but are not limited to)"EC-BOS", "FX-40", "TMN", "SLX" and "UNC".~~

1.4.43 Object (BACnet)

An ASHRAE 135 Object. The concept of organizing BACnet information into standard components with various associated Properties. Examples include Analog Input objects and Binary Output objects.

1.4.44 Object Identifier (BACnet)

A grouping of two Object properties: Object Type (e.g. Analog Value, Schedule, etc.) and Object Instance (in this case, a number). Object Identifiers must be unique within a device.

1.4.45 Object Instance (BACnet)

See paragraph OBJECT IDENTIFIER.

1.4.46 Object Properties (BACnet)

Attributes of an object. Examples include present value and high limit properties of an analog input object. Properties are defined in ASHRAE 135; some are optional and some are required. Objects are controlled by reading from and writing to object properties.

1.4.47 Operator Configurable

Operator configurable values are values that can be changed from a single common front end user interface across multiple vendor systems.

~~For Niagara Framework Systems, a property, setting, or value is Operator Configurable when it is configurable from a Niagara Framework Front End.~~

For non Niagara-based BACnet systems, a property, setting, or value in a device is Operator Configurable when it is Configurable and is either:

a. a Writable Property of a Standard BACnet Object; or

b. a Property of a Standard BACnet Object that is Writable when Out_Of_Service is TRUE and Out_Of_Service is Writable.

1.4.48 Overridable (Point)

A point is Overridable if it can be Overriden (See Override).

1.4.49 Override

Changing the value of a point outside of the normal sequence of operation where the change has priority over the sequence and where there is a mechanism for releasing the change such that the point returns to the normal value. Overrides persist until released or overridden at the same or higher priority but are not required to persist through a loss of power. Overrides are often used by operators to change values, and generally originate at a user interface (workstation or local display panel).

1.4.50 Packaged Equipment

Packaged equipment is a single piece of equipment provided by a manufacturer in a substantially complete and operable condition, where the controls (DDC Hardware) are factory installed, and the equipment is sold and shipped from the manufacturer as a single entity. Disassembly and reassembly of a large piece of equipment for shipping does not prevent it from being packaged equipment. Package units may require field installation of remote sensors. Packaged equipment is also called a "packaged unit".

Note industry may use the term "Packaged System" to mean a collection of equipment that is designed to work together where each piece of equipment is packaged equipment and there is a network that connects the equipment together. A "packaged system" of this type is NOT packaged equipment; it is a collection of packaged equipment, and each piece of equipment must individually meet specification requirements.

1.4.51 Packaged Unit

See packaged equipment.

1.4.52 Performance Verification Test (PVT)

The procedure for determining if the installed BAS meets design criteria prior to final acceptance. The PVT is performed after installation, testing, and balancing of mechanical systems. Typically the PVT is performed by the Contractor in the presence of the Government.

1.4.53 Physical Segment (BACnet)

A single contiguous medium to which BACnet devices are attached (ASHRAE 135).

1.4.54 Polling

A device periodically requesting data from another device.

1.4.55 Points

Physical inputs and outputs, and virtual values read from or written to a controller over the the network. See also paragraph INPUT/OUTPUT (I/O).

1.4.56 Proportional, Integral, and Derivative (PID) Control Loop

A closed-loop control algorithm calculating control output based using the error between a process variable and setpoint. The proportional control component adjusts control output based on the current error, the integral component adjusts based on a sum of errors over time, and the derivative component adjusts based on the rate of change of the errors. Not all components are employed in all implementation of a PID loop, and when a component is not used the corresponding letter is generally dropped from the abbreviation used for that control. For example derivative control is often not required for HVAC systems, leaving "PI" control.

1.4.57 Proprietary (BACnet)

Within the context of BACnet, any extension of or addition to object types, properties, PrivateTransfer services, or enumerations specified in ASHRAE 135. Objects with Object_Type values of 128 and above are Proprietary Objects. Properties with Property_Identifier of 512 and above are proprietary Properties.

1.4.58 Protocol Implementation Conformance Statement (PICS) (BACnet)

A document, created by the manufacturer of a device, which describes which portions of the BACnet standard may be implemented by a given device. ASHRAE 135 requires that all ASHRAE 135 devices have a PICS, and also defines a minimum set of information that must be in it. A device as installed for a specific project may not implement everything in its PICS.

1.4.59 Repeater

A device that connects two control network segments and retransmits all information received on one side onto the other.

1.4.60 Router

Generically, a Router is a device that connects two or more networks and controls traffic between them by retransmitting signals received from one network onto another based on the signal destination.

As used in this Section, a Router connects two or more ASHRAE 135 networks and controls traffic between them by retransmitting signals received from one network onto another based on the signal destination. Routers are used to subdivide a BACnet internetwork and to limit network traffic.

1.4.61 Segment

A 'single' section of a control network that contains no repeaters or routers. There is generally a limit on the number of devices on a segment, and this limit is dependent on the topology/media and device type.

1.4.62 Standard BACnet Objects (BACnet)

Objects with Object_Type values below 128 and specifically enumerated in Clause 21 of ASHRAE 135. Objects which are not proprietary. See paragraph PROPRIETARY.

1.4.63 Standard BACnet Properties (BACnet)

Properties with Property_Identifier values below 512 and specifically enumerated in Clause 21 of ASHRAE 135. Properties which are not proprietary. See Proprietary.

1.4.64 Standard BACnet Services (BACnet)

ASHRAE 135 services other than ConfirmedPrivateTransfer or UnconfirmedPrivateTransfer. See paragraph PROPRIETARY.

1.4.65 UMCS

UMCS stands for Utility Monitoring and Control System. The term refers to all components by which a project site monitors, manages, and controls real-time operation of HVAC and other building systems. These components include the UMCS "front-end" and all field building control systems connected to the front-end. The front-end consists of Monitoring and Control Software (user interface software), browser-based user interfaces and network infrastructure.

The network infrastructure (the "UMCS Network"), is an IP network connecting multiple building or facility control networks to the Monitoring and Control Software.

1.4.66 UMCS Network

The UMCS Network connects multiple building or facility control networks to the Monitoring and Control Software.

1.4.67 Writable Property (BACnet)

A Property is Writable when it can be changed through the use of one or more of the WriteProperty services defined in ASHRAE 135, Clause 15 regardless of the value of any other Property. Note that in the ASHRAE 135 standard, some Properties may be writable when the Out of Service Property is TRUE; for purposes of this Section, Properties that are only writable when the Out of Service Property is TRUE are not considered to be Writable.

1.5 PROJECT SEQUENCING

TABLE II: PROJECT SEQUENCING lists the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column). TABLE II does not specify overall project milestone and completion dates~~[; these dates are specified in the contract documents][_____]~~.

a. Sequencing for Submittals: The sequencing specified for submittals is the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within ~~[~~14~~][_____]~~ days of notification that the submittal has been rejected. Upon resubmittal there will be an additional Government review period. If the submittal is not accepted the process repeats until the submittal is accepted by the Government.

b. Sequencing for Activities: The sequencing specified for activities indicates the earliest the activity may begin.

c. Abbreviations: In TABLE II the abbreviation AAO is used for 'after approval of' and 'ACO' is used for 'after completion of'.

TABLE II. PROJECT SEQUENCING

ITEM #	TYPE	DESCRIPTION	SEQUENCING (START OF ACTIVITY OR DEADLINE FOR SUBMITTAL)
1	S	Existing Conditions Report	
2	S	DDC Contractor Design Drawings	
3	S	Manufacturer's Product Data	
4	S	Pre-construction QC Checklist	
5	E	Install Building Control System	AAO #1 thru #4
6	E	Start-Up and Start-Up Testing	ACO #5
7	S	Post-Construction QC Checklist	~~[[_____] days]~~ACO #6

TABLE II. PROJECT SEQUENCING

ITEM #	TYPE	DESCRIPTION	SEQUENCING (START OF ACTIVITY OR DEADLINE FOR SUBMITTAL)
8	S	Programming Software Configuration Software ~~Niagara Framework Engineering Tool~~ ~~Niagara Framework Wizards~~	~~[[______] days]~~ACO #6
9	S	Draft As-Built Drawings	~~[[______] days]~~ACO #6
10	S	Start-Up Testing Report	~~[[______] days]~~ACO #6
11	S	PVT Procedures	~~[[______]~~14 days ~~]~~before schedule start of #12 and AAO #10
12	~~S,~~E	Execute PVT ~~PVT Testing Activities~~	AAO #9 and #11~~As indicated in PART 3 of this Section~~
13	S	PVT Report	~~[[______] days]~~ACO #12 ~~As indicated in PART 3 of this Section~~
14	S	Controller Application Programs Controller Configuration Settings ~~Niagara Framework Supervisory Gateway Backups~~	~~[[______]~~14 days ~~]~~AAO #13
15	S	Final As-Built Drawings	~~[[______]~~14 days ~~]~~AAO #13
16	S	O&M Instructions	AAO #15
17	S	Training Documentation	AAO #10 and ~~[[______]~~14 days ~~]~~before scheduled start of #18
18	E	Training	AAO #16 and #17
19	S	Closeout QC Checklist	ACO #18

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. ~~Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.~~ Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

DDC Contractor Design Drawings; G~~, [_____]~~

Draft As-Built Drawings; G~~, [_____]~~

Final As-Built Drawings; G~~, [_____]~~

SD-03 Product Data

Programming Software; G~~, [_____]~~

Controller Application Programs; G~~, [_____]~~

Configuration Software; G~~, [_____]~~

Controller Configuration Settings; G~~, [_____]~~

Proprietary Multi-Split Engineering Tool Software; G~~, [_____]~~

Manufacturer's Product Data; G~~, [_____]~~

~~Niagara Framework Supervisory Gateway Backups; G, [_____]~~

~~[~~ ~~Niagara Framework Engineering Tool; G, [_____]]~~

SD-05 Design Data

Boiler Or Chiller Plant Gateway Request

SD-06 Test Reports

~~Existing Conditions Report~~

Pre-Construction Quality Control (QC) Checklist; G~~, [_____]~~

Post-Construction Quality Control (QC) Checklist; G~~, [_____]~~

Start-Up Testing Report; G~~, [_____]~~

~~PVT Procedures; G, [_____]~~

~~PVT Report; G, [_____]~~

~~Control Contractor s Performance Verification Testing Plan; G~~

~~Equipment Supplier s Performance Verification Testing Plan; G~~

~~Endurance Testing Results; G~~

~~Performance Verification Test Report; G~~

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G~~, [_____]~~

Training Documentation; G~~, [_____]~~

SD-11 Closeout Submittals

Enclosure Keys; G~~, [_____]~~

Password Summary Report; G~~, [_____]~~

Closeout Quality Control (QC) Checklist; G~~, [_____]~~

1.7 DATA PACKAGE AND SUBMITTAL REQUIREMENTS

Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

1.8 SOFTWARE FOR DDC HARDWARE AND GATEWAYS

Provide all software related to the programming and configuration of DDC Hardware and Gateways as indicated. License all Software to the project site. The term "controller" as used in these requirements means both DDC Hardware and Gateways.

1.8.1 Configuration Software

For each type of controller, provide the configuration tool software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of the software user manuals for each software with the software submittal.

Submit Configuration Software on CD-ROM as a Technical Data Package. Submit ~~[_____]~~4 hard copies of the software user manual for each piece of software.

1.8.2 Controller Configuration Settings

For each controller, provide copies of the installed configuration settings as source code compatible with the configuration tool software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Configuration Settings on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which files are associated with each device. Submit ~~[2][_____]~~4 copies of the Controller Configuration Settings CD-ROM.

1.8.3 Programming Software

For each type of programmable controller, provide the programming software in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit hard copies of software user manuals for each software with the software submittal.

Submit Programming Software on CD-ROM as a Technical Data Package. Submit ~~[_____]~~4 hard copies of the software user manual for each piece of software.

1.8.4 Controller Application Programs

For each programmable controller, provide copies of the application program as source code compatible with the programming software for that controller in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Controller Application Programs on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which application program is associated with each device. Submit ~~[2][_____]~~4 copies of the Controller Application Programs CD-ROM.

~~1.8.5 Niagara Framework Supervisory Gateway Backups~~

~~For each Niagara Framework Supervisory Gateway, provide a backup of all software within the Niagara Framework Supervisory Gateway, including configuration settings. This backup must be sufficient to allow the restoration of the Niagara Framework Supervisory Gateway or the replacement of the Niagara Framework Supervisory Gateway.~~

~~Submit backups for each Niagara Framework Supervisory Gateway on CD-ROM as a Technical Data Package. Mark each backup indicating clearly the source Niagara Framework Supervisory Gateway.~~

~~[1.8.6 Niagara Framework Engineering Tool(for all Niagara Framework system)~~

~~Provide a Niagara Framework Engineering Tool in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS. Submit software user manuals with the Niagara Framework Engineering Tool submittal.~~

~~Submit the Niagara Framework Engineering Tool on CD-ROM as a Technical Data Package. Submit [_____] hard copies of the software user manual for the Niagara Framework Engineering Tool.~~

~~]~~1.9 BOILER OR CHILLER PLANT GATEWAY REQUEST

If requesting the use of a gateway to a boiler or chiller plant as indicated in paragraph Proprietary Systems Exempted From Open Protocol Requirements, submit a Boiler or Chiller Plant Gateway Request describing the configuration of the boilers or chillers including model numbers for equipment and controllers, the sequence of operation for the units, and a justification for the need to operate the units on a shared non-BACnet network.

1.10 QUALITY CONTROL CHECKLISTS

The QC Checklist for BACnet Systems in APPENDIX A of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

~~The QC Checklist for Niagara Framework Based BACnet Systems in APPENDIX A of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.~~

The QC Representative must verify each item indicated and initial in the space provided to indicate that the requirement has been met. The QC Representative must sign and date the Checklist prior to submission to the Government.

1.10.1 Pre-Construction Quality Control (QC) Checklist

Complete items indicated as Pre-Construction QC Checklist items in the QC Checklist. Submit ~~[~~four~~][_____]~~ copies of the Pre-Construction QC Checklist.

1.10.2 Post-Construction Quality Control (QC) Checklist

Complete items indicated as Post-Construction QC Checklist items in the QC Checklist. Submit ~~[~~four~~][_____]~~ copies of the Post-Construction QC Checklist.

1.10.3 Closeout Quality Control (QC) Checklist

Complete items indicated as Closeout QC Checklist items in the QC Checklist. Submit ~~[~~four~~][_____]~~ copies of the Closeout QC Checklist.

PART 2 PRODUCTS

Provide products meeting the requirements of Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS for BACnet or Niagara BACnet systems, other referenced Sections, and this Section.

2.1 GENERAL PRODUCT REQUIREMENTS

Units of the same type of equipment must be products of a single manufacturer. Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Materials and equipment must be standard products of a manufacturer regularly engaged in the manufacturing of these and similar products. The standard products must have been in a satisfactory commercial or industrial use for two years prior to use on this project. The two year use must include applications of equipment and materials under similar circumstances and of similar size. DDC Hardware not meeting the two-year field service requirement is acceptable provided it has been successfully used by the Contractor in a minimum of two previous projects. The equipment items must be supported by a service organization. Items of the same type and purpose must be identical, including equipment, assemblies, parts and components.

2.2 PRODUCT DATA

Provide manufacturer's product data sheets documenting compliance with product specifications for each product provided under Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, or this Section. Provide product data for all products in a single indexed compendium, organized by product type.

For each manufacturer, model and version (revision) of DDC Hardware provide the Protocol Implementation Conformance Statement (PICS) in accordance with Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

Submit Manufacturer's Product Data on CD-ROM.

2.3 OPERATION ENVIRONMENT

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

a. Pressure: Pressure conditions normally encountered in the installed location.

b. Vibration: Vibration conditions normally encountered in the installed location.

c. Temperature:

   (1) Products installed indoors: Ambient temperatures in the range of 0 to 50 degrees C and temperature conditions outside this range normally encountered at the installed location.

   (2) Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of ~~[~~-37 to +66 degrees C~~] [_____]~~ and temperature conditions outside this range normally encountered at the installed location.

d. Humidity: 10 to 95 percent relative humidity, noncondensing and humidity conditions outside this range normally encountered at the installed location.

2.4 WIRELESS CAPABILITY

Wireless capabilities must comply with Section 25 05 11~~[._____]~~ CYBERSECURITY FOR ~~[~~FACILITY-RELATED CONTROL~~][_____]~~ SYSTEMS

2.5 ENCLOSURES

Enclosures supplied as an integral (pre-packaged) part of another product are acceptable. Provide two Enclosure Keys for each lockable enclosure on a single ring per enclosure with a tag identifying the enclosure the keys operate. Provide enclosures meeting the following minimum requirements:

2.5.1 Outdoors

For enclosures located outdoors, provide enclosures meeting ~~NEMA 250~~ IEC 60529 ~~[~~Type 3~~][Type 4]~~ requirements.

2.5.2 Mechanical and Electrical Rooms

For enclosures located in mechanical or electrical rooms, provide enclosures meeting ~~NEMA 250~~IEC 60529 ~~[~~Type 2~~][Type 4]~~ requirements.

2.5.3 Other Locations

For enclosures in other locations including but not limited to occupied spaces, above ceilings, and in plenum returns, provide enclosures meeting ~~NEMA 250~~IEC 60529 Type 1 requirements.

2.6 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

2.6.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

2.6.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG (1.02 mm diameter) copper or thicker wire rated for 300-volt service.

2.6.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG (1.02 mm diameter) or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

a. minimum 50 mm (2 inch) lay of twist

b. 100 percent shielded pairs

c. at least 300-volt insulation

d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation

e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

2.6.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

2.6.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

PART 3 EXECUTION

~~[3.1 EXISTING CONDITIONS~~

~~3.1.1 Existing Conditions Survey~~

~~Perform a field survey, including testing and inspection of the equipment to be controlled and submit an Existing Conditions Report documenting the current status and its impact on the Contractor's ability to meet this specification. For those items considered nonfunctional, document the deficiency in the report including explanation of the deficiencies and estimated costs to correct the deficiencies. As part of the report, define the scheduled need date for connection to existing equipment. Make written requests and obtain Government approval prior to disconnecting any controls and obtaining equipment downtime.~~

~~Submit [four][_____] copies of the Existing Conditions Report.~~

~~3.1.2 Existing Equipment Downtime~~

~~Make written requests and obtain Government approval prior to disconnecting any controls and obtaining equipment downtime.~~

~~3.1.3 Existing Control System Devices~~

~~Inspect, calibrate, and adjust as necessary to place in proper working order all existing devices which are to be reused.~~

~~]~~3.1 INSTALLATION

Fully install and test the control system in accordance Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC, Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, and this Section.

3.1.1 Dielectric Isolation

Provide dielectric isolation where dissimilar metals are used for connection and support. Install control system in a matter that provides clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Install control system such that it does not interfere with the clearance requirements for mechanical and electrical system maintenance.

3.1.2 Penetrations in Building Exterior

Make all penetrations through and mounting holes in the building exterior watertight.

3.1.3 Device Mounting Criteria

Install devices in accordance with the manufacturer's recommendations and as indicated and shown. Provide a weathershield for all devices installed outdoors. Provide clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Provide clearance for mechanical and electrical system maintenance; do not not interfere with the clearance requirements for mechanical and electrical system maintenance.

3.1.4 Labels and Tags

Key all labels and tags to the unique identifiers shown on the As-Built drawings. For labels exterior to protective enclosures provide engraved plastic labels mechanically attached to the enclosure or DDC Hardware. Labels inside protective enclosures may be attached using adhesive, but must not be hand written. For tags, provide plastic or metal tags mechanically attached directly to each device or attached by a metal chain or wire.

a. Label all Enclosures and DDC Hardware.

b. Tag Airflow measurement arrays (AFMA) with flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient.

c. Tag duct static pressure taps at the location of the pressure tap.

3.1.5 Surge Protection

3.1.5.1 Power-Line Surge Protection

Protect equipment connected to AC circuits to withstand power-line surges in accordance with IEEE C62.41. Do not use fuses for surge protection.

3.1.5.2 Surge Protection for Transmitter and Control Wiring

Protect DDC hardware against or provided DDC hardware capable of withstanding surges induced on control and transmitter wiring installed outdoors and as shown. Protect equipment against the following two waveforms:

a. A waveform with a 10-microsecond rise time, a 1000-microsecond decay time and a peak current of 60 amps.

b. A waveform with an 8-microsecond rise time, a 20-microsecond decay time and a peak current of 500 amperes.

3.1.6 Basic Cybersecurity Requirements

3.1.6.1 Passwords

Password development and the Password Summary Report submittal must comply with Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS.

3.1.6.2 Wireless Capability

Wireless capabilities must comply with Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS.

3.1.6.3 IP Network Physical Security

IP network physical security must comply with Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS.

3.2 DRAWINGS AND CALCULATIONS

Provide drawings in the form and arrangement indicated and shown. Use the same abbreviations, symbols, nomenclature and identifiers shown. Assign a unique identifier as shown to each control system element on a drawing. When packaging drawings, group schedules by system. When space allows, it is permissible to include multiple schedules for the same system on a single sheet. Except for drawings covering all systems, do not put information for different systems on the same sheet.

Submit hardcopy drawings on ~~[~~ISO A1 841 by 594 mm~~][~~ or~~][~~ A3 420 by 297 mm~~]~~ sheets, and electronic drawings in PDF and in ~~[~~AutoCAD ~~][Microstation][Bentley BIM V8][Autodesk Revit 2013]~~ format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

a. Submit DDC Contractor Design Drawings consisting of each drawing indicated with pre-construction information depicting the intended control system design and plans. Submit DDC Contractor Design Drawings as a single complete package: ~~[_____]~~2 hard copies and ~~[_____]~~2 copies on CD-ROM.

b. Submit Draft As-Built Drawings consisting of each drawing indicated updated with as-built data for the system prior to PVT. Submit Draft As-Built Drawings as a single complete package: ~~[_____]~~2 hard copies and ~~[_____]~~2 copies on CD-ROM.

c. Submit Final As-Built Drawings consisting of each drawing indicated updated with all final as-built data. Final As-Built Drawings as a single complete package: ~~[_____]~~2 hard copies and ~~[_____]~~2 copies on CD-ROM.

3.2.1 Sample Drawings

Sample drawings in electronic format are available at the Whole Building Design Guide page for this section:
https://www.wbdg.org/dod/ufgs/ufgs-23-09-00
These drawings may prove useful in demonstrating expected drawing formatting and example content and are provided for illustrative purposes only. Note that these drawings do not meet the content requirements of this Section and must be completed to meet project requirements.

3.2.2 Drawing Index and Legend

Provide an HVAC Control System Drawing Index showing the name and number of the building, military site, State or other similar designation, and Country. In the Drawing Index, list all Contractor Design Drawings, including the drawing number, sheet number, drawing title, and computer filename when used. In the Design Drawing Legend, show and describe all symbols, abbreviations and acronyms used on the Design Drawings. Provide a single Index and Legend for the entire drawing package.

3.2.3 Thermostat and Occupancy Sensor Schedule

Provide a thermostat and occupancy sensor schedule containing each thermostat's unique identifier, room identifier and control features and functions as shown. Provide a single thermostat and occupancy sensor schedule for the entire project.

3.2.4 Valve Schedule

Provide a valve schedule containing each valve's unique identifier, size, flow coefficient Kv (Cv), pressure drop at specified flow rate, spring range, positive positioner range, actuator size, close-off pressure to torque data, dimensions, and access and clearance requirements data. In the valve schedule include actuator selection data supported by calculations of the force required to move and seal the valve, access and clearance requirements. Provide a single valve schedule for the entire project.

3.2.5 Damper Schedule

Provide a damper schedule containing each damper's unique identifier, type (opposed or parallel blade), nominal and actual sizes, orientation of axis and frame, direction of blade rotation, actuator size and spring ranges, operation rate, positive positioner range, location of actuators and damper end switches, arrangement of sections in multi-section dampers, and methods of connecting dampers, actuators, and linkages. Include the AMCA 511 maximum leakage rate at the operating static-pressure differential for each damper in the Damper Schedule. Provide a single damper schedule for the entire project.

3.2.6 Project Summary Equipment Schedule

Provide a project summary equipment schedule containing the manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a single project equipment schedule for the entire project.

3.2.7 Equipment Schedule

Provide system equipment schedules containing the unique identifier, manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a separate equipment schedule for each HVAC system.

3.2.8 Occupancy Schedule

Provide an occupancy schedule drawing containing the same fields as the occupancy schedule Contract Drawing with Contractor updated information. Provide a single occupancy schedule for the entire project.

3.2.9 DDC Hardware Schedule

Provide a single DDC Hardware Schedule for the entire project and including following information for each device.

3.2.9.1 DDC Hardware Identifier

The Unique DDC Hardware Identifier for the device.

3.2.9.2 HVAC System

The system "name" used to identify a specific system (the name used on the system schematic drawing for that system).

3.2.9.3 BACnet Device Information

3.2.9.3.1 Device Object Identifier

The Device Object Identifier: The Object_Identifier of the Device Object

3.2.9.3.2 Network Number

The Network Number for the device.

3.2.9.3.3 MAC Address

The MAC Address for the device

3.2.9.3.4 BTL Listing

The BTL Listing of the device. If the device is listed under multiple BTL Profiles, indicate the profile that matches the use and configuration of the device as installed.

3.2.9.3.5 Proprietary Services Information

If the device uses non-standard ASHRAE 135 services as defined and permitted in Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS, indicate that the device uses non-standard services and include a description of all non-standard services used. Describe usage and content such that a device from another vendor can interoperate with the device using the non-standard service. Provide descriptions with sufficient detail to allow a device from a different manufacturer to be programmed to both read and write the non-standard service request:

a. read: interpret the data contained in the non-standard service and;

b. write: given similar data, generate the appropriate non-standard service request.

3.2.9.3.6 Alarming Information

Indicate whether the device is used for alarm generation, and which types of alarm generation the device implements: intrinsic, local algorithmic, remote algorithmic.

3.2.9.3.7 Scheduling Information

Indicate whether the device is used for scheduling.

3.2.9.3.8 Trending Information

Indicate whether the device is used for trending, and indicate if the device is used to trend local values, remote values, or both.

~~3.2.9.4 Niagara Station ID~~

~~The Niagara Station ID for each Niagara Framework Supervisory Gateway~~

3.2.10 Points Schedule

Provide a Points Schedule in tabular form for each HVAC system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.

b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets ("<" and ">"), such as <___> for a required entry or <value> for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval.

Points Schedule Columns must include:

3.2.10.1 Point Name

The abbreviated name for the point using the indicated naming convention.

3.2.10.2 Description

A brief functional description of the point such as "Supply Air Temperature".

3.2.10.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.

3.2.10.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

3.2.10.5 Range

The range of values, including units, associated with the point, including but not limited to a zone temperature setpoint adjustment range, a sensor measurement range, occupancy values for an occupancy input, or the status of a safety.

3.2.10.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

a. AI: The value comes from a hardware (physical) Analog Input

b. AO: The value is output as a hardware (physical) Analog Output

c. BI: The value comes from a hardware (physical) Binary Input

d. BO: The value is output as a hardware (physical) Binary Output

e. PULSE: The value comes from a hardware (physical) Pulse Accumulator

Input

f. NET-IN: The value is provided from the network (generally from another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.

g. NET-OUT: The value is provided to another controller over the network. Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

3.2.10.7 Object and Property Information

The Object Type and Instance Number for the Object associated with the point. If the value of the point is not in the Present_Value Property, then also provide the Property ID for the Property containing the value of the point. Any point that is displayed at the front end or on an LDP, is trended, is used by another device on the network, or has an alarm condition must be documented here.

~~3.2.10.8 Niagara Station ID~~

~~The Niagara Station ID of the Niagara Framework Supervisory Gateway the point is mapped into.~~

3.2.10.8 Network Data Exchange Information (Gets Data From, Sends Data To)

Provide the DDC Hardware Identifier of other DDC Hardware the point is shared with.

3.2.10.9 Override Information (Object Type and Instance Number)

For each point requiring an Override~~ and not residing in a Niagara Framework Supervisory Gateway~~, indicate if the Object for the point is Commandable or, if the use of a separate Object was specifically approved by the Contracting Officer, provide the Object Type and Instance Number of the Object to be used in overriding the point.

3.2.10.10 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote, the trend Object type and the trend Object instance number. For remote trends provide the DDC Hardware Identifier for the device containing the trend Object in the Points Schedule notes.

3.2.10.11 Alarm Information

Indicate the Alarm Generation Type, Event Enrollment Object Instance Number, and Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have Event Enrollment Objects.)

~~For Niagara BACnet systems: Indicate the Alarm Generation Type and~~

~~Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have a Notification Class Object.)~~

3.2.10.12 Configuration Information

Indicate the means of configuration associated with each point. ~~For points in a Niagara Framework Supervisory Gateway, indicate the point within the Niagara Framework Supervisory Gateway used to configure the value. For other points:~~

a. For Operator Configurable Points indicate BACnet Object and Property information (Name, Type, Identifiers) containing the configurable value. Indicate whether the property is writable always, or only when Out_Of_Service is TRUE.

b. For Configurable Points indicate the BACnet Object and Property information as for Operator Configurable points, or identification of the configurable settings from within the engineering software for the device or identification of the hardware settings on the device.

3.2.11 Riser Diagram

The Riser Diagram of the Building Control Network may be in tabular form, and must show all DDC Hardware and all Network Hardware, including network terminators. For each item, provide the unique identifier, common descriptive name, physical sequential order (previous and next device on the network), room identifier and location within room. A single riser diagram must be submitted for the entire system.

3.2.12 Control System Schematics

Provide control system schematics in the same form as the control system schematic Contract Drawing with Contractor updated information. Provide a control system schematic for each HVAC system.

3.2.13 Sequences of Operation~~[~~ Including Control Logic Diagrams~~]~~

Provide HVAC control system sequence of operation and ~~[~~control logic diagrams~~]~~ in the same format as the Contract Drawings. Within these drawings, refer to devices by their unique identifiers. Submit sequences of operation~~[~~ and control logic diagrams~~]~~ for each HVAC system

3.2.14 Controller, Motor Starter and Relay Wiring Diagram

Provide controller wiring diagrams as functional wiring diagrams which show the interconnection of conductors and cables to each controller and to the identified terminals of input and output devices, starters and package equipment. Show necessary jumpers and ground connections and the labels of all conductors. Identify sources of power required for control systems and for packaged equipment control systems back to the panel board circuit breaker number, controller enclosures, magnetic starter, or packaged equipment control circuit. Show each power supply and transformer not integral to a controller, starter, or packaged equipment. Show the connected volt-ampere load and the power supply volt-ampere rating. Provide wiring diagrams for each HVAC system.

3.3 CONTROLLER TUNING

Tune each controller in a manner consistent with that described in the ASHRAE FUN SI and in the manufacturer's instruction manual. Tuning must consist of adjustment of the proportional, integral, and where applicable, the derivative (PID) settings to provide stable closed-loop control. Each loop must be tuned while the system or plant is operating at a high gain (worst case) condition, where high gain can generally be defined as a low-flow or low-load condition. Upon final adjustment of the PID settings, in response to a change in controller setpoint, the controlled variable must settle out at the new setpoint with no more than two (2) oscillations above and below setpoint. Upon settling out at the new setpoint the controller output must be steady. With the exception of naturally slow processes such as zone temperature control, the controller must settle out at the new setpoint within five (5) minutes. Set the controller to its correct setpoint and record and submit the final PID configuration settings with the O&M Instructions and on the associated Points Schedule.

3.4 START-UP

3.4.1 Start-Up Test

Perform the following startup tests for each control system to ensure that the described control system components are installed and functioning per this specification.

Adjust, calibrate, measure, program, configure, set the time schedules, and otherwise perform all necessary actions to ensure that the systems function as indicated and shown in the sequence of operation and other contract documents.

3.4.1.1 Systems Check

An item-by-item check must be performed for each HVAC system.

3.4.1.1.1 Step 1 - System Inspection

With the system in unoccupied mode and with fan hand-off-auto switches in the OFF position, verify that power and main air are available where required and that all output devices are in their failsafe and normal positions. Inspect each local display panel ~~[and each M&C Client]~~ to verify that all displays indicate shutdown conditions.

3.4.1.1.2 Step 2 - Calibration Accuracy Check

Perform a two-point accuracy check of the calibration of each HVAC control system sensing element and transmitter by comparing the value from the test instrument to the network value provided by the DDC Hardware. Use digital indicating test instruments, such as digital thermometers, motor-driven psychrometers, and tachometers. Use test instruments with accuracy at least twice as accurate as the specified sensor accuracy and with calibration traceable to National Institute of Standards and Technology standards. Check one the first check point in the bottom one-third of the sensor range, and the second in the top one-third of the sensor range. Verify that the sensing element-to-DDC readout accuracies at two points are within the specified product accuracy tolerances, and if not recalibrate or replace the device and repeat the calibration check.

3.4.1.1.3 Step 3 - Actuator Range Check

With the system running, apply a signal to each actuator through the DDC Hardware controller. Verify proper operation of the actuators and positioners for all actuated devices and record the signal levels for the extreme positions of each device. Vary the signal over its full range, and verify that the actuators travel from zero stroke to full stroke within the signal range. Where applicable, verify that all sequenced actuators move from zero stroke to full stroke in the proper direction, and move the connected device in the proper direction from one extreme position to the other. For valve actuators and damper actuators, perform the actuator range check under normal system pressures.

3.4.1.2 Weather Dependent Test

Perform weather dependent test procedures in the appropriate climatic season.

3.4.2 Start-Up Testing Report

Submit ~~[~~4~~] [_____]~~ copies of the Start-Up Testing Report. The report may be submitted as a Technical Data Package documenting the results of the tests performed and certifying that the system is installed and functioning per this specification, and is ready for the Performance Verification Test (PVT).

~~3.5 PERFORMANCE VERIFICATION TEST (PVT)~~

~~3.5.1 PVT Procedures~~

~~Prepare PVT Procedures based on Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING explaining step-by-step, the actions and expected results that will demonstrate that the control system performs in accordance with the sequences of operation, and other contract documents. Submit [4] [_____] copies of the PVT Procedures. The PVT Procedures may be submitted as a Technical Data Package.~~

~~3.5.1.1 Sensor Accuracy Checks~~

~~Include a one-point accuracy check of each sensor in the PVT procedures.~~

~~3.5.1.2 Endurance Test~~

~~Include a [one-week] [_____] endurance test as part of the PVT during which the system is operated continuously.~~

~~Use the building control system BACnet Trend Log or Trend Log Multiple Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.~~

~~Use the building control system Niagara Trend Log Objects to trend all points shown as requiring a trend on the Point Schedule for the entire endurance test. If insufficient buffer capacity exists to trend the entire endurance test, upload trend logs during the course of the endurance test to ensure that no trend data is lost.~~

~~Use the existing trending capabilities as indicated to trend all points~~

~~shown as requiring a trend on the Point Schedule for the entire endurance test.~~

~~3.5.1.3 PVT Equipment List~~

~~Include in the PVT procedures a control system performance verification test equipment list that lists the equipment to be used during performance verification testing. For each piece of equipment, include manufacturer name, model number, equipment function, the date of the latest calibration, and the results of the latest calibration~~

~~3.5.2 PVT Execution~~

~~Demonstrate compliance of the control system with the contract documents. Using test plans and procedures approved by the Government, software capable of reading and writing COV Notification Subscriptions, Notification Class Recipient List Properties, event enrollments, demonstrate all physical and functional requirements of the project. Show, step-by-step, the actions and results demonstrating that the control systems perform in accordance with the sequences of operation. Do not start the performance verification test until after receipt of written permission by the Government, based on Government approval of the PVT Plan and Draft As-Builts and completion of balancing. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. Do not conduct tests during scheduled seasonal off periods of base heating and cooling systems. If the system experiences any failures during the endurance test portion of the PVT, repair the system repeat the endurance test portion of the PVT until the system operates continuously and without failure for the specified endurance test period.~~

~~3.5.3 PVT Report~~

~~Prepare and submit a PVT report documenting all tests performed during the PVT and their results. Include all tests in the PVT procedures and any additional tests performed during PVT. Document test failures and repairs conducted with the test results.~~

~~Submit [four][_____] copies of the PVT Report. The PVT Report may be submitted as a Technical Data Package.~~

~~3.6 PERFORMANCE VERIFICATION TESTING~~

~~3.6.1 General~~

~~PVT testing must demonstrate compliance of controls work with contract document requirements and must be performed by the Controls Contractor and Equipment Suppliers. No less than [14][__] calendar days prior to start of controls system installation, meet with the Contracting Office's technical representative (COTR) [and the designing engineer of the HVAC systems], the Contractor's QA representative, the Contractor's Controls Contractor representative,[and the control system Owner] to develop a mutual understanding relate to the details of the PVT work requirements, including required submittals, work schedule, and field quality control.~~

~~3.6.2 Performance Verification Testing and Commissioning~~

~~PVT testing is a Government quality assurance function that includes systems trending and field tests. Commissioning is a quality control~~

~~function that is the Commissioning Team s responsibility to the extent required by this contract.~~

~~3.6.3 Performance Verification Testing of Equipment with Packaged Controls~~

~~Controls Contractor and Equipment Supplier(s) must share and coordinate PVT testing responsibilities for equipment provided with on-board factory packaged controls such as boiler controllers, dedicated outside air systems (DOAS s), and packaged pumping systems.~~

~~3.6.3.1 Controls Contractor Responsibilities~~

~~The Controls Contractor must provide a PVT Plan separate from Equipment Supplier s performance verification testing plan, perform endurance testing, and perform PVT testing concurrent with Equipment Suppliers testing for equipment provided with on-board factory packaged controls to demonstrate the following:~~

~~a. Equipment enabling and disabling.~~

~~b. Equipment standard and optional control points necessary to accomplish functionality regardless if specified in contract documents or not.~~

~~c. Equipment standard and optional alarms critical to safe operation regardless if specified in contract documents or not.~~

~~d. All control points added by Controls Contractor in addition to onboard factory packaged controls regardless if specified in contract documents or not.~~

~~Refer to paragraphs titled Performance Verification Test Plan and Endurance Testing for additional information.~~

~~3.6.3.2 Equipment Supplier Responsibilities~~

~~Each Equipment Supplier must provide PVT Plans separate from Controls Contractor s plans and perform PVT testing concurrent with Controls Contractor s testing for their equipment provided with on-board factory packaged controls to demonstrate the following:~~

~~a. Equipment standard and optional control features necessary to accomplish functionality regardless if specified in contract documents or not.~~

~~b. Equipment standard and optional operation modes necessary to accomplish functionality regardless if specified in contract documents or not.~~

~~c. Equipment standard and optional alarm conditions for safe operation regardless if specified in contract documents or not.~~

~~Refer to all paragraphs under paragraph titled Performance Verification Testing except for section titled Endurance Testing for additional information.~~

~~3.6.4 Sequencing of Performance Verification Testing Activities~~

~~PVT activities must be sequenced with major activities listed below for Test and Balance (TAB) Contractor, Equipment Suppliers, Commissioning~~

~~Specialists, and others to demonstrate fully functioning systems. Refer to Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS). Complete the items in TABLE III: SEQUENCING OF PVT TESTING ACTIVITIES as schedule activities or milestones.~~

~~TABLE III: SEQUENCING OF PVT TESTING ACTIVITIES~~	
~~SEQUENCE~~	~~ITEM~~
~~1~~	~~Submission, review, and approval of Control Contractors PVT Plans.~~
~~2~~	~~Submission, review, and approval of Equipment Suppliers PVT Plans.~~
~~3~~	~~Submission, review, and approval of certified final Test and Balance Report.~~
~~4~~	~~Conduct commissioning functional performance tests.~~
~~5~~	~~Submission, review, and approval of all of the Commissioning Specialists completed functional performance tests.~~
~~6~~	~~Request Contracting Officer to allow beginning of Government-witnessed PVT testing.~~
~~7~~	~~Contracting Officers approval to begin PVT testing.~~
~~8~~	~~Conduct PVT field work.~~
~~9~~	~~Governments verbal approval of PVT field work for all systems.~~
~~10~~	~~Conduct Test and Balance verification field work.~~
~~11~~	~~Governments written approval of Test and Balance verification field work.~~
~~12~~	~~Submission, review, and approval of endurance testing.~~
~~13~~	~~Governments written approval of PVT field work for all systems.~~
~~14~~	~~Facility acceptance recommendation.~~
~~15~~	~~Submission, review, and approval of Control Contractors PVT Report.~~
~~16~~	~~Submission, review, and approval of Equipment Suppliers PVT Report.~~
~~17~~	~~Conduct applicable re-testing and seasonal testing within 10 months of beneficial occupancy.~~

~~3.6.4.1 PVT Testing for Multi-Phase Construction~~

~~For air moving systems except outside air systems serving multiple phases, all major activities listed in TABLE III through Government s verbal approval of Test and Balance verification field work can be completed by phase if all ductwork construction is completed for that phase.~~

~~For primary systems such as chilled water systems, HVAC heating hot water systems, and outside air systems serving multiple phases, all major~~

~~activities listed listed in TABLE III through Government s verbal approval of Test and Balance verification field work for all air moving systems served by that primary system for that phase must be completed prior to conducting PVT field work for that primary system.~~

~~3.6.5 Control Contractor's Performance Verification Testing Plan~~

~~Submit a detailed PVT Plan of the proposed control systems testing in this contract for approval prior to its use. Develop and use a single PVT Plan for each system with a unique control sequence. Systems sharing an identical control sequence can be tested using copies of the PVT Plan intended for these systems.~~

~~PVT Plans must include system-based, step-by-step test methods demonstrating system performs in accordance with contract document requirements. The Government may provide sample PVT Plans upon request. PVT Plans must include the following:~~

~~a. Control sequences from contract documents segmented such that each control algorithm, operation mode, and alarm condition is immediately followed by numbered test methods required to initiate a response, expected response, space for comments, and pass" or "fail indication for each expected response.~~

~~b. PVT Plans with control sequences from contract documents that are not segmented into parts will not be accepted.~~

~~c. Indication where assisting personnel are required such as Mechanical Contractor.~~

~~d. Signature and date lines for the Contractor's PVT administrator, Contractor's quality assurance representative, and Contracting Officer's representative acknowledging completion of testing.~~

~~3.6.6 Performance Verification Testing Sample Size~~

~~PVT testing sample sizes will be as follows:~~

~~a. 100-Percent of the following systems:~~

~~(1) primary systems including, but not limited to, chilled water and HVAC heating hot water systems~~

~~(2) air handling unit systems including all associated fans except for remote exhaust air fans~~

~~(3) DOAS s including all associated fans except for remote exhaust air fans~~

~~b. 20-Percent of each set of systems with a shared identical control sequence for systems such as:~~

~~(1) air terminal units~~

~~(2) exhaust air fans~~

~~(3) terminal equipment such as fan coil units and unit heaters~~

~~3.6.6.1 Selection of Systems to Test~~

~~For sample sets less than 100-percent, the Government will choose which systems will be tested. The Government may require additional testing if previous testing results are inconsistent or demonstrate improper system control as follows:~~

~~a. An additional 25-percent after five-percent failure rate of first sample set.~~

~~b. 100-percent after any failures occurring in additional sample set.~~

~~3.6.7 Conducting Performance Verification Testing~~

~~At least 15 days prior to preferred test date, request the Contracting Officer to allow the beginning of Government-witnessed PVT testing. Provide an estimated time table required to perform testing of each system. Furnish personnel, equipment, instrumentation, and supplies necessary to perform all aspects of testing. Testing personnel must be regularly employed in the testing and calibration of control systems. After receipt of Contracting Officer s approval to begin testing, perform PVT testing using project's as-built (shop) control system drawings, project's design drawings, and approved PVT Plans.~~

~~During testing, identify deficiencies that do not meet contract document requirements. Deficiencies must be investigated, corrected with corrections documented, and re-tested at a later date following procedures for the initial PVT testing. The Government may require re-testing of any control system components affected by the original failed test.~~

~~3.6.8 Endurance Testing~~

~~3.6.8.1 General~~

~~Conduct endurance testing in conjunction with the PVT to demonstrate control loop stability and accuracy. For all control loops tested, record trend data of the control variables over time, demonstrating that the control loop responds to a sudden change of the control variable set point without excessive overshoot or undershoot. Conduct endurance testing for each system subject to PVT testing. Systems must be operating as normally anticipated during occupancy throughout endurance testing.~~

~~Endurance testing results must clearly demonstrate control loop stability and accuracy. Controlled loop outputs must be stable and accurately maintain each setpoint.~~

~~3.6.8.2 Hardware~~

~~[Use hardware provided in this contract for testing.] [Use Government furnished hardware for testing if available when endurance testing begins. If unavailable, the Contractor must provide suitable hardware for required testing.]~~

~~If insufficient buffer capacity exists to trend the entire endurance test, upload trend data during the course of endurance testing to ensure all trend data is retained. Lost trend data will require retesting of all control points for affected system(s).~~

~~3.6.8.3 Endurance Testing Results Format~~

~~Submit endurance testing results for each tested system in a graphical format complete with clear indication of value(s) for y-axis, value for x-axis, and legend identifying each trended control point. The number of control points contained on a single graph must be such that all control points can be clearly visible. Control points must be logically grouped such that related points appear on a single graph. In addition, submit a separate comma separated value (CSV) file of raw trend data for each trended system. Each trended control point in CSV file must be clearly identified.~~

~~For control points recorded based on change of value, change of value for recording data must be clearly identified for each control point.~~

~~3.6.8.4 Endurance Testing Start, Duration, and Frequency~~

~~Trending of all control points for a given system must start at an identical date and time regardless of the basis of data collection. Duration of all endurance tests must be at least [one-week][_____].~~

~~Unless specified otherwise for control points recorded based on time, frequency of data collection must be [15-minutes] [_____]. Frequency of data collection for specific types of control points is as follows:~~

~~3.6.8.4.1 Points Trended at One Minute Intervals~~

~~a. Temperature for supply air, return air, mixed air, supply water, and return water~~

~~b. Temperature for outside air, supply air, return air and exhaust air entering and leaving energy recovery device~~

~~c. Flow for supply air, return air, outside air, chilled water, and HVAC heating hot water~~

~~d. Flow for exhaust air associated with energy recovery~~

~~e. Relative humidity for outside air and return air~~

~~f. Relative humidity for outside air, supply air, return air and exhaust air entering and leaving energy recovery device~~

~~g. Command and status for control dampers and control valves~~

~~h. Speed for fans and pumps~~

~~i. Pressure for fans and pumps~~

~~3.6.8.4.2 Points Trended at 15 Minute Intervals~~

~~a. Temperature and relative humidity for zones~~

~~b. Temperature and relative humidity for outside air not associated with energy recovery~~

~~c. Command and status for equipment~~

~~d. Pressure relative to the outside for facility~~

~~3.6.8.5 Trended Control Points~~

~~Trended control points for each system must demonstrate each system performs in accordance with contract document requirements. Trended control points must include, but not be limited to, control points listed in contract document points list.~~

~~Minimum control points that are required to be trended for selected systems are listed below. These control points must be trended as applicable to this contract in addition to control points necessary to demonstrate systems perform in accordance with contract document requirements and those listed in contract document s points list.~~

~~[3.6.8.5.1 Air-Cooled Chiller Chilled Water System.~~

~~a. Chiller(s) command and status~~

~~b. Chiller isolation valve(s) command and status~~

~~c. Chilled water pump(s) actual speed~~

~~d. Chilled water pump(s) setpoint and actual differential pressure~~

~~e. Minimum flow bypass control valve command~~

~~f. Minimum system flow setpoint and actual flow~~

~~g. Chilled water supply setpoint and actual temperature~~

~~h. Chilled water return actual temperature~~

~~i. Chilled water actual flow~~

~~j. Outside air actual dry-bulb temperature~~

~~][3.6.8.5.2 HVAC Heating Hot Water System with Boiler.~~

~~a. Boiler(s) command and status~~

~~b. Boiler(s) isolation valve command and status~~

~~c. HVAC heating hot water pump(s) actual speed~~

~~d. HVAC heating hot water pump(s) setpoint and actual differential pressure~~

~~e. Minimum flow bypass control valve command~~

~~f. Minimum system setpoint and actual flow~~

~~g. HVAC heating hot water supply setpoint and actual temperature~~

~~h. HVAC heating hot water return actual temperature~~

~~i. HVAC heating hot water actual flow~~

~~j. Outside air actual dry-bulb temperature~~

~~][3.6.8.5.3 HVAC Heating Hot Water System with Steam-to-Hot Water Heat Exchanger.~~

~~a. Steam control valve(s) command~~

~~b. Heat exchanger isolation valve(s) command and status~~

~~c. HVAC heating hot water pump(s) actual speed~~

~~d. HVAC heating hot water pump(s) setpoint and actual differential pressure~~

~~e. Minimum flow bypass control valve command~~

~~f. Minimum system setpoint and actual flow~~

~~g. HVAC heating hot water supply setpoint and actual temperature~~

~~h. HVAC heating hot water return actual temperature~~

~~i. HVAC heating hot water actual flow~~

~~j. Outside air actual dry-bulb temperature~~

~~][3.6.8.5.4 Air Handling Unit with Relief Air Fan~~

~~a. Outside air actual dry-bulb temperature~~

~~b. Outside air actual relative humidity~~

~~c. Outside air setpoint and actual airflow~~

~~d. Minimum outside air control damper command~~

~~e. Economizer outside air control damper command~~

~~f. Facility setpoint and actual relative pressure~~

~~g. Return air actual dry-bulb temperature~~

~~h. Return air actual relative humidity~~

~~i. Return air control damper command~~

~~j. Relief air control damper command~~

~~k. Relief air fan actual speed~~

~~l. Mixed air setpoint and setpoint and actual temperature~~

~~m. Preheat coil leaving air setpoint and actual temperature~~

~~n. Preheat coil control actuator command~~

~~o. Cooling coil leaving air setpoint and actual temperature~~

~~p. Cooling coil control valve command~~

~~q. Supply air fan actual speed~~

~~r. Discharge air actual temperature~~

~~s. Supply air fan setpoint and actual static pressure~~

~~][3.6.8.5.5 Dedicated Outside Air System (DOAS)~~

~~a. Outside air actual dry-bulb temperature~~

~~b. Outside air actual relative humidity~~

~~c. Outside air isolation damper command and status~~

~~d. Outside air setpoint and actual airflow~~

~~e. Energy recovery wheel command, status, and actual speed~~

~~f. Energy recovery wheel s OA bypass control damper command and status~~

~~g. Energy recovery wheel s defrost cycle command and status~~

~~h. Energy recovery wheel s OA discharge air actual dry-bulb temperature~~

~~i. Energy recovery wheel s OA discharge air actual relative humidity~~

~~j. Preheat coil leaving air setpoint and actual temperature~~

~~k. Preheat coil control actuator command~~

~~l. Cooling coil leaving air setpoint and actual temperature~~

~~m. Cooling coil control valve command~~

~~n. Supply air fan actual speed~~

~~o. Reheat coil control valve command~~

~~p. Discharge air setpoint and actual temperature~~

~~q. Supply air fan setpoint and actual static pressure~~

~~r. Facility setpoint and actual relative pressure~~

~~s. Return air actual dry-bulb temperature~~

~~t. Return air actual relative humidity~~

~~u. Energy recovery wheel s EA bypass control damper command and status~~

~~v. Energy recovery wheel s EA discharge air actual dry-bulb temperature~~

~~w. Energy recovery wheel s EA discharge air actual relative humidity~~

~~x. Exhaust air fan actual speed~~

~~y. Exhaust air isolation damper command and status~~

~~][3.6.8.5.6 Series Fan-Powered Supply Air Terminal Units~~

~~a. Zone setpoint and actual dry-bulb temperature~~

~~b. Zone actual relative humidity~~

~~c. Control damper command~~

~~d. Fan command and status~~

~~e. Heating coil valve command~~

~~f. Airflow actual value~~

~~g. Leaving air actual temperature~~

~~]3.6.8.6 Endurance Testing Sample Size~~

~~Endurance Testing sample sizes ware as follows:~~

~~a. 100-Percent of the following systems:~~

~~(1) primary systems including, but not limited to, chilled water and HVAC heating hot water systems~~

~~(2) air handling unit systems including all associated fans except for remote exhaust air fans~~

~~(3) DOAS s including all associated fans except for remote exhaust air fans~~

~~b. 20-Percent of each set of systems with a shared identical control sequence for systems such as:~~

~~(1) air terminal units~~

~~(2) exhaust air fans~~

~~(3) terminal equipment such as fan coil units and unit heaters~~

~~3.6.8.6.1 Selection of Systems to Test~~

~~For sample sets less than 100-percent, the Government will choose which systems will be tested. The Government may require additional testing if previous testing results are inconsistent or demonstrate improper system control as follows:~~

~~a. An additional 25-percent after five-percent failure rate of first sample set.~~

~~b. 100-percent after any failures occurring in additional sample set.~~

~~3.6.9 Performance Verification Test Report~~

~~Submit a PVT Report after receiving Government s written approval of PVT field work that is intended to document test results and final control system sequences and settings prior to turnover. The PVT Report must contain the following:~~

~~a. Executive summary that briefly discusses results of each system s endurance testing and PVT testing and conclusions for each system.~~

~~b. Endurance testing for each system.~~

~~c. Completed PVT Plan for each system used during testing that includes hand written field notes and participant signatures.~~

~~d. Blank PVT Plan for each system approved prior to testing that is edited to reflect changes occurring during testing. Edits must be typed and must reflect changes to control sequences from contract documents, must reflect changes to numbered test methods required to initiate a response, and must reflect changes to expected response. Only one blank PVT Plan is required for each set of systems sharing an identical control sequence, such as air terminal units, exhaust air fans, fan coil units and unit heaters.~~

~~e. Written certification that the installation and testing of all systems are complete and meet all contract document requirements.~~

3.5 OPERATION AND MAINTENANCE (O&M) INSTRUCTIONS

Provide HVAC control System Operation and Maintenance Instructions which include:

a. "Data Package 3" as indicated in Section 01 78 23 OPERATION AND MAINTENANCE DATA for each piece of control equipment.

b. "Data Package 4" as described in Section 01 78 23 OPERATION AND MAINTENANCE DATA for all air compressors.

c. HVAC control system sequences of operation formatted as indicated.

d. Procedures for the HVAC system start-up, operation and shut-down including the manufacturer's supplied procedures for each piece of equipment, and procedures for the overall HVAC system.

e. As-built HVAC control system detail drawings formatted as indicated.

f. Routine maintenance checklist. Provide the routine maintenance checklist arranged in a columnar format, where the first column lists all installed devices, the second column states the maintenance activity or that no maintenance required, the third column states the frequency of the maintenance activity, and the fourth column is used for additional comments or reference.

g. Qualified service organization list, including at a minimum company name, contact name and phone number.

h. Start-Up Testing Report.

i. Performance Verification Test (PVT) Procedures and Report.

Submit ~~[2] [_____]~~4 copies of the Operation and Maintenance Instructions, indexed and in booklet form. The Operation and Maintenance Instructions may be submitted as a Technical Data Package.

~~[~~3.6 MAINTENANCE AND SERVICE

Provide services, materials and equipment as necessary to maintain the entire system in an operational state as indicated for a period of one year from the date of final acceptance of the project. Minimize impacts on facility operations.

a. The integration of the system specified in this section into a Utility Monitoring and Control System must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period. Integration into a UMCS includes but is not limited to establishing communication between devices in the control system and the front end or devices in another system.

b. The changing of configuration properties must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period.

3.6.1 Description of Work

Provide adjustment and repair of the system including the manufacturer's required sensor and actuator (including transducer) calibration, span and range adjustment.

3.6.2 Personnel

Use only service personnel qualified to accomplish work promptly and satisfactorily. Advise the Government in writing of the name of the designated service representative, and of any changes in personnel.

3.6.3 Scheduled Inspections

Perform two inspections at six-month intervals and provide work required. Perform inspections in ~~[~~June and December~~][_____]~~. During each inspection perform the indicated tasks:

a. Perform visual checks and operational tests of equipment.

b. Clean control system equipment including interior and exterior surfaces.

c. Check and calibrate each field device. Check and calibrate 50 percent of the total analog inputs and outputs during the first inspection. Check and calibrate the remaining 50 percent of the analog inputs and outputs during the second major inspection. Certify analog test instrumentation accuracy to be twice the specified accuracy of the device being calibrated. Randomly check at least 25 percent of all binary inputs and outputs for proper operation during the first inspection. Randomly check at least 25 percent of the remaining binary inputs and outputs during the second inspection. If more than 20 percent of checked inputs or outputs failed the calibration check during any inspection, check and recalibrate all inputs and outputs during that inspection.

d. Run system software diagnostics and correct diagnosed problems.

e. Resolve any previous outstanding problems.

3.6.4 Scheduled Work

This work must be performed ~~[~~during regular working hours, Monday through Friday, excluding Federal holidays~~][______]~~.

3.6.5 Emergency Service

The Government will initiate service calls when the system is not functioning properly. Qualified personnel must be available to provide service to the system. A telephone number where the service supervisor can be reached at all times must be provided. Service personnel must be at the site within 24 hours after receiving a request for service. The control system must be restored to proper operating condition as required per Section 01 78 00 CLOSEOUT SUBMITTALS.

3.6.6 Operation

After performing scheduled adjustments and repairs, verify control system operation as demonstrated by the applicable tests of the performance verification test.

3.6.7 Records and Logs

Keep dated records and logs of each task, with cumulative records for each major component, and for the complete system chronologically. Maintain a continuous log for all devices, including initial analog span and zero calibration values and digital points. Keep complete logs and provide logs for inspection onsite, demonstrating that planned and systematic adjustments and repairs have been accomplished for the control system.

3.6.8 Work Requests

Record each service call request as received and include its location, date and time the call was received, nature of trouble, names of the service personnel assigned to the task, instructions describing what has to be done, the amount and nature of the materials to be used, the time and date work started, and the time and date of completion. Submit a record of the work performed within 5 days after work is accomplished.

3.6.9 System Modifications

Submit recommendations for system modification in writing. Do not make system modifications, including operating parameters and control settings, without prior approval of the Government.

~~]~~3.7 TRAINING

Conduct a training course for ~~[______]~~ operating staff members designated by the Government in the maintenance and operation of the system, including specified hardware and software. Conduct ~~[~~32~~] [______]~~ hours of training at the project site within 30 days after successful completion of the performance verification test. The Government reserves the right to make audio and visual recordings (using Government supplied equipment)of the training sessions for later use. Provide audiovisual equipment and other training materials and supplies required to conduct training. A training day is defined as 8 hours of classroom instruction, including two 15 minute breaks and excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility.

3.7.1 Training Documentation

Prepare training documentation consisting of:

a. Course Attendee List: Develop the list of course attendees in coordination with and signed by the ~~[Controls][~~HVAC~~][Electrical]~~ shop supervisor.

b. Training Manuals: Provide training manuals which include an agenda, defined objectives for each lesson, and a detailed description of the subject matter for each lesson. When presenting portions of the course material by audiovisuals, deliver copies of those audiovisuals as a part of the printed training manuals.

3.7.2 Training Course Content

For guidance in planning the required instruction, assume that attendees will have a high school education, and are familiar with HVAC systems. During the training course, cover all of the material contained in the Operating and Maintenance Instructions, the layout and location of each controller enclosure, the layout of one of each type of equipment and the locations of each, the location of each control device external to the panels, the location of the compressed air station, preventive maintenance, troubleshooting, diagnostics, calibration, adjustment, commissioning, tuning, and repair procedures. Typical systems and similar systems may be treated as a group, with instruction on the physical layout of one such system. Present the results of the performance verification test and the Start-Up Testing Report as benchmarks of HVAC control system performance by which to measure operation and maintenance effectiveness.

3.7.3 Training Documentation Submittal Requirements

Submit hardcopy training manuals and all training materials on CD-ROM. Provide one hardcopy manual for each trainee on the Course Attendee List and ~~[~~2~~][______]~~ additional copies for archive at the project site. Provide ~~[~~2~~][______]~~ copies of the Course Attendee List with the archival copies. Training Documentation may be submitted as a Technical Data Package.

APPENDIX A

QC CHECKLIST FOR BACNET SYSTEMS

This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.

Instructions: Initial each item in the space provided (|____|) verifying that the requirement has been met.

This checklist is for (circle one:)

Pre-Construction QC Checklist Submittal

Post-Construction QC Checklist Submittal

Close-out QC Checklist Submittal

Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:

1	All DDC Hardware is numbered on Control System Schematic Drawings.	|____|
2	Signal lines on Control System Schematic are labeled with the signal type.	|____|
3	Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.	|____|

Items verified for Post-Construction and Closeout QC Checklist Submittals:

4	All sequences are performed as specified using DDC Hardware.	|____|
5	Training schedule and course attendee list has been developed and coordinated with shops and submitted.	|____|

Items verified for Closeout QC Checklist Submittal:

6	Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system.	|____|
7	Programming software has been submitted for all programmable controllers.	|____|
8	All software has been licensed to the Government.	
9	O&M Instructions have been completed and submitted.	|____|
10	Training course has been completed.	|____|

QC CHECKLIST FOR BACNET SYSTEMS

11	All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.	|____|
12	All DDC Hardware is BTL listed.	|____|
13	Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule.	|____|
14	Scheduling, Alarming, and Trending have been implemented using the standard BACnet Objects for these functions.	|____|
15	All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated	|____|

____ __________________

(QC Representative Signature) (Date)

~~**QC CHECKLIST FOR NIAGARA FRAMEWORK BASED BACNET SYSTEMS**~~

~~This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.~~

~~Instructions: Initial each item in the space provided (|____|) verifying that the requirement has been met.~~

~~This checklist is for (circle one:)~~

~~Pre-Construction QC Checklist Submittal~~

~~Post-Construction QC Checklist Submittal~~

~~Close-out QC Checklist Submittal~~

~~Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:~~

~~1~~	~~All DDC Hardware is numbered on Control System Schematic Drawings.~~	~~|____|~~
~~2~~	~~Signal lines on Control System Schematic are labeled with the signal type.~~	~~|____|~~
~~3~~	~~Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.~~	~~|____|~~

	~~QC CHECKLIST FOR NIAGARA FRAMEWORK BASED BACNET SYSTEMS~~	
	~~Items verified for Post-Construction and Closeout QC Checklist Submittals:~~	
~~4~~	~~All sequences are performed as specified using DDC Hardware.~~	~~____~~
~~5~~	~~Training schedule and course attendee list has been developed and coordinated with shops and submitted.~~	~~____~~
	~~Items verified for Closeout QC Checklist Submittal:~~	
~~6~~	~~Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system.~~	~~____~~
~~7~~	~~Programming software has been submitted for all programmable controllers.~~	~~____~~
~~8~~	~~All software has been licensed to the Government.~~	
~~9~~	~~O&M Instructions have been completed and submitted.~~	~~____~~
~~10~~	~~Training course has been completed.~~	~~____~~
~~11~~	~~All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.~~	~~____~~
~~12~~	~~All DDC Hardware is BTL listed.~~	~~____~~
~~13~~	~~Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule.~~	~~____~~
~~14~~	~~Scheduling, Alarming, and Trending have been implemented using Niagara Framework objects and services, and BACnet Instrinsic Alarming as indicated.~~	~~____~~
~~15~~	~~All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated~~	~~____~~
	~~______________________________________ ________________~~	
	~~(QC Representative Signature) (Date)~~	

-- End of Section --

SECTION 23 09 13

INSTRUMENTATION AND CONTROL DEVICES FOR HVAC
11/15, CHG 2: 05/21

PART 1 GENERAL

1.1 SUMMARY

This section provides for the instrumentation control system components excluding direct digital controllers, network controllers, gateways etc. that are necessary for a completely functional automatic control system. When combined with a Direct Digital Control (DDC) system, the Instrumentation and Control Devices covered under this section must be a complete system suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and indicated.

a. Install hardware to perform the control sequences as specified and indicated and to provide control of the equipment as specified and indicated.

b. Install hardware such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.

c. Install and configure hardware such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the installing Contractor.

1.1.1 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

1.1.2 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

1.2 RELATED SECTIONS

Related work specified elsewhere.

~~Section 01 30 00 ADMINISTRATIVE REQUIREMENTS~~

Section 23 30 00 HVAC AIR DISTRIBUTION

Section 23 05 15 COMMON PIPING FOR HVAC

Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-D (2018) Laboratory Methods of Testing Dampers for Rating

AMCA 511 (2010; R 2016) Certified Ratings Program for Air Control Devices

FLUID CONTROLS INSTITUTE (FCI)

FCI 70-2 (2021) Control Valve Seat Leakage

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529 (2019) Corrigendum 1 - Amendment 2 - Degrees of Protection Provided by Enclosures (IP Code)

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 2071 (2000) Steel Valves

JIS B 7505-1 (2017) Aneroid Pressure Gauges -- Part 1: Bourdon Tube Pressure Gauges

JIS G 3459 (2021) Stainless Steel Pipes

JIS G 5502 (2024) Spheroidal Graphite Cast Irons (Amendment 1)

JIS H 3300 (2018) Copper and Copper Alloy Seamless Pipes and Tubes

JIS H 3401 (2001) Pipe Fittings of Copper and Copper Alloys

JIS H 5120 (2016) Copper and Copper Alloy Castings

JIS K 6911 (2006; R 2021) Testing Methods for Thermosetting Plastic (Amendment 1)

JIS K 6922-2 (2018) Plastics -- Polyethylene (PE) Moulding and Extrusion Materials -- Part 2: Preparation of Test Specimens and Determination of Properties

JIS K 7161-1	(2024) Plastics -- Determination of Tensile Properties -- Part 1: General Principles
JIS K 7210-1	(2014) Plastics -- Determination of the Melt Mass-Flow Rate (MFR) and Melt Volume-Flow Rate (MVR) of Thermoplastics -- Part 1: Standard Method
JIS K 7112-1	(2023) Plastics -- Methods for Determining the Density of Non-Cellular Plastics -- Part 1: Immersion Method, Liquid Pycnometer Method and Titration Method
JIS K 7112-2	(2023) Plastics -- Methods for Determining the Density of Non-Cellular Plastics -- Part 2: Density Gradient Column Method
JIS Z 3282	(2017) Soft Solder -- Chemical Compositions and Forms

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems

UL SOLUTIONS (UL)

UL 94	(2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 555	(2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers
UL 555S	(2014; Reprint Oct 2020) UL Standard for Safety Smoke Dampers
UL 1820	(2004; Reprint May 2013) UL Standard for Safety Fire Test of Pneumatic Tubing for Flame and Smoke Characteristics
UL 5085-3	(2006; Reprint Jan 2022) UL Standard for Safety Low Voltage Transformers - Part 3: Class 2 and Class 3 Transformers

1.4 SUBMITTALS

Submittal requirements are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

1.5 DELIVERY AND STORAGE

Store and protect products from the weather, humidity, and temperature variations, dirt and dust, and other contaminants, within the storage condition limits published by the equipment manufacturer.

1.6 INPUT MEASUREMENT ACCURACY

Select, install and configure sensors, transmitters and DDC Hardware such that the maximum error of the measured value at the input of the DDC hardware is less than the maximum allowable error specified for the sensor or instrumentation.

~~1.7 SUBCONTRACTOR SPECIAL REQUIREMENTS~~

~~Perform all work in this section in accordance with the paragraph entitled CONTRACTOR SPECIAL REQUIREMENTS in Section 01 30 00 ADMINISTRATIVE REQUIREMENTS.~~

PART 2 PRODUCTS

2.1 EQUIPMENT

2.1.1 General Requirements

All products used to meet this specification must meet the indicated requirements, but not all products specified here will be required by every project. All products must meet the requirements both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section.

2.1.2 Operation Environment Requirements

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

2.1.2.1 Pressure

Pressure conditions normally encountered in the installed location.

2.1.2.2 Vibration

Vibration conditions normally encountered in the installed location.

2.1.2.3 Temperature

a. Products installed indoors: Ambient temperatures in the range of 0 to 50 degrees C and temperature conditions outside this range normally encountered at the installed location.

b. Products installed outdoors or in unconditioned indoor spaces: Ambient temperatures in the range of ~~[~~-37 to +66 degrees C~~] [_____]~~ and temperature conditions outside this range normally encountered at the installed location.

2.1.2.4 Humidity

10 to 95 percent relative humidity, non-condensing and also humidity conditions outside this range normally encountered at the installed location.

2.2 WEATHERSHIELDS

Provide weathershields constructed of galvanized steel painted white, unpainted aluminum, aluminum painted white, or white PVC.

~~[~~2.3 TUBING

2.3.1 Copper

Provide ~~ASTM B75/B75M or ASTM B88~~JIS H 3300 and JIS H 3401 rated tubing meeting the following requirements:

a. For tubing 9 mm outside diameter and larger provide tubing with minimum wall thickness ~~equal to ASTM B88, Type M~~conforming to JIS H 3300.

b. For tubing less than9 mm outside diameter provide tubing with minimum wall thickness of 0.6 mm.

c. For exposed tubing and tubing for working pressures greater than 207 kPa provide hard copper tubing.

d. Provide fittings which are ~~ASME B16.18 or ASME B16.22~~JIS H 3401 solder type using ~~ASTM B32 95-5~~JIS Z 3282 tin-antimony solder~~, or which are ASME B16.26 compression type~~.

2.3.2 Stainless Steel

For stainless steel tubing provide tubing conforming to ~~ASTM A269/A269M~~ JIS G 3459.

2.3.3 Plastic

Provide plastic tubing with the burning characteristics of linear low-density polyethylene tubing which is self-extinguishing when tested in accordance with ~~ASTM D635~~JIS K 6911, has UL 94 V-2 flammability classification or better, and which withstands stress cracking when tested in accordance with ~~ASTM D1693~~JIS K 6922-2. Provide plastic-tubing bundles with Mylar barrier and flame-retardant polyethylene jacket.

2.3.4 Polyethylene Tubing

Provide flame-resistant, multiple polyethylene tubing in flame-resistant protective sheath with mylar barrier, or unsheathed polyethylene tubing in rigid metal, intermediate metal, or electrical metallic tubing conduit for areas where tubing is exposed. Single, unsheathed, flame-resistant polyethylene tubing may be used where concealed in walls or above ceilings and within control panels. Do not provide polyethylene tubing for ~~[systems indicated as critical and]~~ smoke removal systems, or for systems with working pressures over 206 kPa. Provide compression or brass barbed push-on type fittings. Provide extruded seamless polyethylene tubing conforming to the following:

a. Minimum Burst Pressure Requirements: 690 kPa at 24 degrees C to 172 kPa at 66 degrees C .

b. Stress Crack Resistance: ~~ASTM D1693~~JIS K 6922-2, 200 hours minimum.

c. Tensile Strength (Minimum): ~~ASTM D638~~JIS K 7161-1, 7584 kPa.

d. Flow Rate (Average): ~~ASTM D1238~~JIS K 7210-1, 0.30 decigram per minute.

e. Density (Average): ~~ASTM D792~~JIS K 7112-1 or JIS K 7112-2, 921 kg per cubic meter.

f. Burn rate: ~~ASTM D635~~JIS K 6911.

g. Flame Propagation: UL 1820, less than 1.5 meters~~ ASTM D635~~.

h. Average Optical Density: UL 1820, less than 0.15~~ ASTM D635~~.

2.4 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

2.4.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

2.4.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG (1.02 mm diameter) copper or thicker wire rated for 300-volt service.

2.4.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG (1.02 mm diameter) or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

a. minimum 50 mm (2 inch) lay of twist

b. 100 percent shielded pairs

c. at least 300-volt insulation

d. each pair has a 20 AWG tinned-copper drain wire and individual overall pair insulation

e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

2.4.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

2.4.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

2.5 AUTOMATIC CONTROL VALVES

Provide valves with stainless-steel stems and stuffing boxes with extended necks to clear the piping insulation. Provide valves with bodies meeting ~~ASME B16.34 or ASME B16.15~~JIS B 2071 or JIS H 5120 pressure and temperature class ratings based on the design operating temperature and 150 percent of the system design operating pressure. Unless otherwise specified or indicated, provide valves meeting FCI 70-2 ~~[~~Class III leakage rating~~][Class IV leakage rating]~~. Provide valves rated for modulating or two-position service as indicated, which close against a differential pressure indicated as the Close-Off pressure and which are Normally-Open, Normally-Closed, or Fail-In-Last-Position as indicated.

2.5.1 Valve Type

2.5.1.1 Liquid Service 150 Degrees F or Less

Use either globe valves or ball valves except that butterfly valves may be used for sizes 100 mm and larger.

2.5.1.2 Liquid Service Above 150 Degrees F

a. Two-position valves: Use either globe valves or ball valves except that butterfly valves may be used for sizes 100 mm and larger.

b. Modulating valves: Use globe valves except that butterfly valves may be used for sizes 100 mm and larger.

2.5.1.3 Steam Service

Use globe valves except that butterfly valves may be used for sizes 100 mm and larger.

2.5.2 Valve Flow Coefficient and Flow Characteristic

2.5.2.1 Two-Way Modulating Valves

Provide the valve coefficient (Kv) indicated. Provide equal-percentage flow characteristic for liquid service except for butterfly valves. Provide linear flow characteristic for steam service except for butterfly valves.

2.5.2.2 Three-Way Modulating Valves

Provide the valve coefficient (Kv) indicated. Provide linear flow characteristic with constant total flow throughout full plug travel.

2.5.3 Two-Position Valves

Use full line size full port valves with maximum available (Kv).

2.5.4 Globe Valves

2.5.4.1 Liquid Service Not Exceeding 66 Degrees C

a. Valve body and body connections:

(1) valves 38 mm and smaller: brass or bronze body, with threaded or union ends

(2) valves from 51 mm to 76 mm inclusive: brass, bronze, or iron bodies. 51 millimeters valves with threaded connections; 63 to 76 mm valves with flanged connections

b. Internal valve trim: Brass or bronze.

c. Stems: Stainless steel.

d. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

2.5.4.2 Liquid Service Not Exceeding 121 Degrees C

a. Valve body and body connections:

(a) 1) valves 38 mm and smaller: brass or bronze body, with threaded or union ends

(b) 2) valves from 51 mm to 76 mm inclusive: brass, bronze, or iron bodies. 51 millimeters valves with threaded connections; 63 to 76 mm valves with flanged connections

b. Internal trim: Type 316 stainless steel including seats, seat rings, modulation plugs, valve stems, and springs.

c. Provide valves with non-metallic parts suitable for a minimum continuous operating temperature of 121 degrees C or 28 degrees C above the system design temperature, whichever is higher.

d. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

2.5.4.3 Hot water service 121 Degrees C and above

a. Provide valve bodies conforming to ~~ASME B16.34~~JIS B 2071 Class 300. For valves 25 mm and larger provide valves with bodies which are carbon steel, globe type with welded ends. For valves smaller than 25 mm provide valves with socket-weld ends. Provide valves with virgin polytetrafluoroethylene (PTFE) packing. Provide valve and actuator combinations which are normally closed.

b. Internal trim: Type 316 stainless steel including seats, seat rings, modulation plugs, valve stems, and springs.

2.5.4.4 Steam Service

For steam service, provide valves meeting the following requirements:

a. Valve body and connections:

(a) 1) valves 38 mm and smaller: complete body of brass or bronze, with threaded or union ends

(b) 2) valves from 51 mm to 76 mm inclusive: body of brass, bronze, or carbon steel

(c) 3) valves 100 mm and larger: body of carbon steel. 50 mm valves with threaded connections; valves 63 mm and larger with flanged connections.

b. Internal Trim: Type 316 stainless steel including seats, seat rings, modulation plugs, valve stems, and springs.

c. Valve sizing: sized for ~~[~~103.4 kPa~~] [______]~~ inlet steam pressure with a maximum ~~[~~83 kPa~~] [______]~~ differential through the valve at rated flow, except where indicated otherwise.

2.5.5 Ball Valves

2.5.5.1 Liquid Service Not Exceeding 66 Degrees C

a. Valve body and connections:

(a) 1) valves 38 mm and smaller: bodies of brass or bronze, with threaded or union ends

(b) 2) valves from 51 mm to 76 mm inclusive: bodies of brass, bronze, or iron. 50 mm valves with threaded connections; valves from 63 to 76 mm with flanged connections.

b. Ball: Stainless steel or nickel-plated brass or chrome-plated brass.

c. Seals: Reinforced Teflon seals and EPDM O-rings.

d. Stem: Stainless steel, blow-out proof.

e. Provide valves compatible with a solution of 50 percent ethylene or propylene glycol.

2.5.6 Butterfly Valves

Provide butterfly valves which are threaded lug type suitable for dead-end service and modulation to the fully-closed position, with carbon-steel bodies or with ductile iron bodies in accordance with ~~ASTM A536~~JIS G 5502. Provide butterfly valves with non-corrosive discs, stainless steel shafts supported by bearings, and EPDM seats suitable for temperatures from -28.9 to +121.1 degrees C. Provide valves with rated Kv of the Kv at 70 percent (60 degrees) open position. Provide valves meeting FCI 70-2 Class VI leakage rating.

2.5.7 Pressure Independent Control Valves (PICV)

Provide pressure independent control valves which include a regulator valve which maintains the differential pressure across a flow control valve. Pressure independent control valves must accurately control the flow from 0-100 percent full rated flow regardless of changes in the piping pressure and not vary the flow more than plus or minus 5 percent at any given flow control valve position when the PICV differential pressure lies between the manufacturer's stated minimum and maximum. The rated minimum differential pressure for steady flow must not exceed 34.5 kPa across the PICV. Provide either globe or ball type valves meeting the indicated requirements for globe and ball valves. Provide valves with a flow tag listing full rated flow and minimum required pressure drop. Provide valves with factory installed Pressure/Temperature ports ("Pete's Plugs") to measure the pressure drop to determine the valve flow rate.

~~2.5.8 Duct-Coil and Terminal-Unit-Coil Valves~~

~~For duct or terminal-unit coils provide control valves with either [flare-type][screw type] or solder-type ends. Provide flare nuts for each flare-type end valve.~~

2.6 DAMPERS

2.6.1 Damper Assembly

Provide single damper sections with blades no longer than 1.2 m and which are no higher than 1.8 m and damper blade width of 203 mm or less. When larger sizes are required, combine damper sections. Provide dampers made of steel, or other materials where indicated and with assembly frames constructed of 2.8 mm minimum thickness ~~[~~galvanized~~][~~ or stainless~~]~~ steel channels with mitered and welded corners. Steel channel frames constructed of 1.5 mm minimum thickness are acceptable provided the corners are reinforced.

a. Flat blades must be made rigid by folding the edges. Blade-operating linkages must be within the frame so that blade-connecting devices within the same damper section must not be located directly in the air stream.

b. Damper axles must be 13 mm minimum, plated steel rods supported in the damper frame by stainless steel or bronze bearings. Blades mounted vertically must be supported by thrust bearings.

c. Provide dampers which do not exceed a pressure drop through the damper of 10 Pa at 5 m/s in the wide-open position. Provide dampers with frames not less than 50 mm in width. Provide dampers which have been tested in accordance with AMCA 500-D.

2.6.2 Operating Linkages

For operating links external to dampers, such as crank arms, connecting rods, and line shafting for transmitting motion from damper actuators to dampers, provide links able to withstand a load equal to at least 300 percent of the maximum required damper-operating force without deforming. Rod lengths must be adjustable. Links must be brass, bronze, zinc-coated steel, or stainless steel. Working parts of joints and clevises must be brass, bronze, or stainless steel. Adjustments of crank arms must control the open and closed positions of dampers.

2.6.3 Damper Types

2.6.3.1 Flow Control Dampers

Provide parallel-blade or opposed blade type dampers for outside air, return air, relief air, exhaust, face and bypass dampers as indicated on the Damper Schedule. Blades must have interlocking edges. The channel frames of the dampers must be provided with jamb seals to minimize air leakage. Unless otherwise indicated, dampers must meet minimum AMCA 511 ~~[Class 1A][Class 1][~~Class 2~~]~~ requirements. Outside air damper seals must be suitable for an operating temperature range of -40 to +75 degrees C. Dampers must be rated at not less than 10 m/s air velocity.

2.6.3.2 Mechanical Rooms and Other Utility Space Ventilation Dampers

Provide utility space ventilation dampers as indicated. Unless otherwise indicated provide AMCA 511 class 3 dampers. Provide dampers rated at not less than 7.6 m/s air velocity.

2.6.3.3 Smoke Dampers

Provide smoke-damper and actuator assemblies which meet the current requirements of NFPA 90A, UL 555, and UL 555S. For combination fire and smoke dampers provide dampers rated for 121 degrees C Class II leakage per UL 555S.

2.7 SENSORS AND INSTRUMENTATION

Unless otherwise specified, provide sensors and instrumentation which incorporate an integral transmitter. Sensors and instrumentation, including their transmitters, must meet the specified accuracy and drift requirements at the input of the connected DDC Hardware's analog-to-digital conversion.

2.7.1 Analog and Binary Transmitters

Provide transmitters which match the characteristics of the sensor. Transmitters providing analog values must produce a linear 4-20 mAdc, 0-10 Vdc signal corresponding to the required operating range and must have zero and span adjustment. Transmitters providing binary values must have dry contacts rated at 1A at 24 Volts AC.

2.7.2 Network Transmitters

Sensors and Instrumentation incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS when used in a BACnet network.

2.7.3 Temperature Sensors

Provide the same sensor type throughout the project. Temperature sensors may be provided without transmitters. Where transmitters are used, the range must be the smallest available from the manufacturer and suitable for the application such that the range encompasses the expected range of temperatures to be measured. The end to end accuracy includes the combined effect of sensitivity, hysteresis, linearity and repeatability between the measured variable and the end user interface (graphic presentation) including transmitters if used.

2.7.3.1 Sensor Accuracy and Stability of Control

2.7.3.1.1 Conditioned Space Temperature

Plus or minus 0.3 degrees C over the operating range.

2.7.3.1.2 Unconditioned Space Temperature

a. Plus or minus 0.6 degrees C over the range of -1 to +55 degrees C AND

b. Plus or minus 2 degrees C over the rest of the operating range.

2.7.3.1.3 Duct Temperature

Plus or minus 0.3 degrees C.

2.7.3.1.4 Outside Air Temperature

a. Plus or minus 1 degree C over the range of -35 to +55 degrees C AND

b. Plus or minus 0.6 degrees C over the range of -1 to +40 degrees C.

~~2.7.3.1.5 High Temperature Hot Water~~

~~Plus or minus 2 degrees C.~~

2.7.3.1.5 Chilled Water

Plus or minus 0.4 degrees C over the range of 2 to 18 degrees C.

2.7.3.1.6 Dual Temperature Water

Plus or minus 1 degree C.

2.7.3.1.7 Heating Hot Water

Plus or minus 1 degree C.

~~2.7.3.1.8 Condenser Water~~

~~Plus or minus 1 degree C.~~

2.7.3.2 Transmitter Drift

The maximum allowable transmitter drift: 0.1 degrees C per year.

2.7.3.3 Point Temperature Sensors

Point Sensors must be encapsulated in epoxy, series 300 stainless steel, anodized aluminum, or copper.

2.7.3.4 Temperature Sensor Details

2.7.3.4.1 Room Type

Provide the sensing element components within a decorative protective cover suitable for surrounding decor.

2.7.3.4.2 Duct Probe Type

Ensure the probe is long enough to properly sense the air stream temperature.

2.7.3.4.3 Duct Averaging Type

Continuous averaging sensors must be one foot in length for each 0.1 square m of duct cross-sectional area, and a minimum length of 1.5 m.

2.7.3.4.4 Pipe Immersion Type

For pipes with larger than 7.6 cm diameter, provide minimum 7.6 cm immersion. For pipes with less than 7.6 cm diameter, provide immersion at least half the diameter of the pipe. Provide each sensor with a corresponding pipe-mounted sensor well, unless indicated otherwise. Sensor wells must be stainless steel when used in steel piping, and brass when used in copper piping.

2.7.3.4.5 Outside Air Type

Provide the sensing element rated for outdoor use

2.7.4 Relative Humidity Sensor

Relative humidity sensors must use bulk polymer resistive or thin film capacitive type non-saturating sensing elements capable of withstanding a saturated condition without permanently affecting calibration or sustaining damage. The sensors must include removable protective membrane filters. Where required for exterior installation, sensors must be capable of surviving below freezing temperatures and direct contact with moisture without affecting sensor calibration. When used indoors, the sensor must be capable of being exposed to a condensing air stream (100 percent relative humidity) with no adverse effect to the sensor's calibration or other harm to the instrument. The sensor must be of the wall-mounted or duct-mounted type, as required by the application, and must be provided with any required accessories. Sensors used in duct high-limit applications must have a bulk polymer resistive sensing element. Duct-mounted sensors must be provided with a duct probe designed to protect the sensing element from dust accumulation and mechanical damage. Relative humidity (RH) sensors must measure relative humidity over a range of 0 percent to 100 percent with an accuracy of plus or minus ~~[~~2~~][3]~~ percent. RH sensors must function over a temperature range of 4.4 to 57.2 degrees C and must not drift more than 1 percent per year.

~~2.7.5 Carbon Dioxide (CO2) Sensors~~

~~Provide photometric type CO2 sensors with integral transducers and linear output. Carbon dioxide (CO2) sensors must measure CO2 concentrations between 0 to 2000 parts per million (ppm) using non-dispersible infrared (NDIR) technology with an accuracy of plus or minus 50 ppm and a maximum response time of 1 minute. The sensor must be rated for operation at ambient air temperatures within the range of 0 to 50 degrees C and relative humidity within the range of 20 to 95 percent (non-condensing). The sensor must have a maximum drift of 2 percent per year. The sensor chamber must be manufactured with a non-corrosive material that does not affect carbon dioxide sample concentration. Duct mounted sensors must be provided with a duct probe designed to protect the sensing element from dust accumulation and mechanical damage. The sensor must have a calibration interval no less than 5 years.~~

2.7.5 Differential Pressure Instrumentation

2.7.5.1 Differential Pressure Sensors

Provide Differential Pressure Sensors with ranges as indicated or as required for the application. Pressure sensor ranges must not exceed the high end range indicated on the Points Schedule by more than 50 percent. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The accuracy must be plus or minus 1 percent of full scale. The sensor must have a maximum drift of 2 percent per year

2.7.5.2 Differential Pressure Switch

Provide differential pressure switches with a user-adjustable setpoint which are sized for the application such that the setpoint is between 25 percent and 75 percent of the full range. The over pressure rating must be a minimum of 150 percent of the highest design pressure of either input to the sensor. The switch must have two sets of contacts and each contact must have a rating greater than it's connected load. Contacts must open or close upon rise of pressure above the setpoint or drop of pressure below the setpoint as indicated.

2.7.6 Flow Sensors

2.7.6.1 Airflow Measurement Array (AFMA)

2.7.6.1.1 Airflow Straightener

Provide AFMAs which contain an airflow straightener if required by the AFMA manufacturer's published installation instructions. The straightener must be contained inside a flanged sheet metal casing, with the AFMA located as specified according to the published recommendation of the AFMA manufacturer. In the absence of published documentation, provide airflow straighteners if there is any duct obstruction within 5 duct diameters upstream of the AFMA. Air-flow straighteners, where required, must be constructed of 3 mm aluminum honeycomb and the depth of the straightener must not be less than 40 mm.

2.7.6.1.2 Resistance to Airflow

The resistance to air flow through the AFMA, including the airflow straightener must not exceed 20 Pa at an airflow velocity of 10 m/s. AFMA construction must be suitable for operation at airflow velocities of up to 25 m/s over a temperature range of 4 to 49 degrees C.

2.7.6.1.3 Outside Air Temperature

In outside air measurement or in low-temperature air delivery applications, provide an AFMA certified by the manufacturer to be accurate as specified over a temperature range of ~~[~~-29 to +49 degrees C~~] [_____]~~.

2.7.6.1.4 Pitot Tube AFMA

Each Pitot Tube AFMA must contain an array of velocity sensing elements. The velocity sensing elements must be of the multiple pitot tube type with averaging manifolds. The sensing elements must be distributed across the duct cross section in the quantity and pattern specified or recommended by the published installation instructions of the AFMA manufacturer.

a. Pitot Tube AFMAs for use in airflow velocities over 3.0 m/s must have an accuracy of plus or minus 5 percent over a range of 2.5 to 12.5 m/s.

b. Pitot Tube AFMAs for use in airflow velocities under 3.0 m/s must have an accuracy of plus or minus 5 percent over a range of 0.6 to 12.5 m/s.

2.7.6.1.5 Electronic AFMA

Each electronic AFMA must consist of an array of velocity sensing elements of the resistance temperature detector (RTD) or thermistor type. The sensing elements must be distributed across the duct cross section in the quantity and pattern specified or recommended by the published application data of the AFMA manufacturer. Electronic AFMAs must have an accuracy of plus or minus 5 percent over a range of 0.6 to 12.5 m/s and the output must be temperature compensated over a range of 0 to 100 degrees C.

2.7.6.1.6 Fan Inlet Measurement Devices

Fan inlet measurement devices cannot be used unless indicated on the drawings or schedules.

2.7.6.2 Orifice Plate

Orifice plate must be made of an austenitic stainless steel sheet of 3.0 mm nominal thickness with an accuracy of plus or minus 1 percent of full flow. The orifice plate must be flat within 0.1 mm. The orifice surface roughness must not exceed 0.5 µm. The thickness of the cylindrical face of the orifice must not exceed 2 percent of the pipe inside diameter or 12.5 percent of the orifice diameter, whichever is smaller. The upstream edge of the orifice must be square and sharp. Where orifice plates are used, concentric orifice plates must be used in all applications except steam flow measurement in horizontal pipelines.

2.7.6.3 Flow Nozzle

Flow nozzle must be made of austenitic stainless steel with an accuracy of plus or minus 1 percent of full flow. The inlet nozzle form must be elliptical and the nozzle throat must be the quadrant of an ellipse. The thickness of the nozzle wall and flange must be such that distortion of the nozzle throat from strains caused by the pipeline temperature and pressure, flange bolting, or other methods of installing the nozzle in the pipeline must not cause the accuracy to degrade beyond the specified limit. The outside diameter of the nozzle flange or the design of the flange facing must be such that the nozzle throat must be centered accurately in the pipe.

2.7.6.4 Venturi Tube

Venturi tube must be made of cast iron or cast steel and must have an accuracy of plus or minus 1 percent of full flow. The throat section must be lined with austenitic stainless steel. Thermal expansion characteristics of the lining must be the same as that of the throat casting material. The surface of the throat lining must be machined to a plus or minus 1.2 µm finish, including the short curvature leading from the converging entrance section into the throat.

2.7.6.5 Annular Pitot Tube

Annular pitot tube must be made of austenitic stainless steel with an accuracy of plus or minus 2 percent of full flow and a repeatability of plus or minus 0.5 percent of measured value. The unit must have at least one static port and no less than four total head pressure ports with an averaging manifold.

2.7.6.6 Insertion Turbine Flowmeter

Provide dual axial turbine flowmeter with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Insertion Turbine Flowmeter accuracy must be plus or minus 0.5 percent of rate at calibrated velocity., within plus or minus of rate over a 10:1 turndown and within plus or minus 2 percent of rate over a 50:1 turndown. Repeatability must be plus or minus 0.25 percent of reading. The meter flow sensing element must operate over a range suitable for the installed location with a pressure loss limited to 1 percent of operating pressure at maximum flow rate. The flowmeter ,must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs. The turbine rotor assembly must be constructed of Series 300 stainless steel and use Teflon seals.

2.7.6.7 Vortex Shedding Flowmeter

Vortex Shedding Flowmeter accuracy must be within plus or minus 0.8 percent of the actual reading over the range of the meter. Steam meters must contain density compensation by direct measurement of temperature. Mass flow inferred from specified steam pressure are not acceptable. The flow meter body must be made of austenitic stainless steel and include a weather tight NEMA 4X electronics enclosure. The vortex shedding flowmeter body must not require removal from the piping in order to replace the shedding sensor.

2.7.6.8 Ultrasonic Flow Meter

Provide Ultrasonic Flow Meters complete with matched transducers, self aligning installation hardware and transducer cables. Ultrasonic transducers must be optimized for the specific pipe and process conditions for the application. The flow meter accuracy must plus or minus 1 percent of rate from 0.3 to 12 meters/sec. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc output.

2.7.6.9 Insertion Magnetic Flow Meter

Provide insertion type magnetic flowmeters with all installation hardware necessary to enable insertion and removal of the meter without system shutdown. All parts must meet or exceed the pressure classification of the pipe system it is installed in. Flowmeter accuracy must be no greater than plus or minus 1 percent of rate from 0.6 to 6 meters/sec. Wetted material parts must be 300 series stainless steel. The flowmeter must include either dry contact pulse outputs, 4-20mA, 0-10Vdc or 0-5Vdc outputs.

2.7.6.10 Positive Displacement Flow Meter

The flow meter must be a direct reading, gerotor, nutating disc or vane type displacement device rated for liquid service as indicated. A counter must be mounted on top of the meter, and must consist of a non-resettable mechanical totalizer for local reading, and a pulse transmitter for remote reading. The totalizer must have a six digit register to indicate the volume passed through the meter in ~~[~~liters~~] [gallons]~~, and a sweep-hand dial to indicate down to 1 L. The pulse transmitter must have a hermetically sealed reed switch which is activated by magnets fixed on gears of the counter. The meter must have a bronze body with threaded or flanged connections as required for the application. Output accuracy must be plus or minus 2 percent of the flow range. The maximum pressure drop at full flow must be 34 kPa.

2.7.6.11 Flow Meters, Paddle Type

Sensor must be non-magnetic, with forward curved impeller blades designed for water containing debris. Sensor accuracy must be plus or minus 1 percent of rate of flow, minimum operating flow velocity must be 0.3 meters/second. Sensor repeatability and linearity must be plus or minus 1 percent. Materials which will be wetted must be made from non-corrosive materials and must not contaminate water. The sensor must be rated for installation in pipes of 76 mm to 1 m diameters. The transmitter housing must ~~be a NEMA 250~~IEC 60529 Type 4 enclosure.

2.7.6.12 Flow Switch

Flow switch must have a repetitive accuracy of plus or minus 10 percent of actual flow setting. Switch actuation must be adjustable over the operating flow range, and must be sized for the application such that the setpoint is between 25 percent and 75 percent of the full range.~~.~~ The switch must have Form C snap-action contacts, rated for the application. The flow switch must have non flexible paddle with magnetically actuated contacts and be rated for service at a pressure greater than the installed conditions. Flow switch for use in sewage system must be rated for use in corrosive environments encountered.

~~2.7.6.13 Gas Flow Meter~~

~~Gas flow meter must be diaphragm or bellows type (gas positive displacement meters) for flows up to 19.7 L/sec and axial flow turbine type for flows above 19.7 L/sec, designed specifically for natural gas supply metering, and rated for the pressure, temperature, and flow rates of the installation. Meter must have a minimum turndown ratio of 10 to 1 with an accuracy of plus or minus 1 percent of actual flow rate. The meter index must include a direct reading mechanical totalizing register and electrical impulse dry contact output for remote monitoring. The electrical impulse dry contact output must not require field adjustment or calibration. The electrical impulse dry contact output must have a minimum resolution of 3 cubic meters of gas per pulse and must not exceed 15 pulses per second at the design flow.~~

2.7.7 Electrical Instruments

Provide Electrical Instruments with an input range as indicated or sized for the application. Unless otherwise specified, AC instrumentation must be suitable for 60 Hz operation.

2.7.7.1 Current Transducers

Current transducers must accept an AC current input and must have an accuracy of plus or minus ~~[~~0.5~~] [2]~~ percent of full scale. The device must have a means for calibration. Current transducers for variable frequency applications must be rated for variable frequency operation.

2.7.7.2 Current Sensing Relays (CSRs)

Current sensing relays (CSRs) must provide a normally-open contact with a voltage and amperage rating greater than its connected load. Current sensing relays must be of split-core design. The CSR must be rated for operation at 200 percent of the connected load. Voltage isolation must be a minimum of 600 volts. The CSR must auto-calibrate to the connected load or be adjustable and field calibrated. Current sensors for variable frequency applications must be rated for variable frequency operation.

2.7.7.3 Voltage Transducers

Voltage transducers must accept an AC voltage input and have an accuracy of plus or minus 0.25 percent of full scale. The device must have a means for calibration. Line side fuses for transducer protection must be provided.

2.7.7.4 Energy Metering

~~2.7.7.4.1 Watt or Watthour Transducers~~

~~Watt transducers must measure voltage and current and must output kW or kWh or both kW and kWh as indicated. kW outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1. kWh outputs must have an accuracy of plus or minus 0.5 percent over a power factor range of 0.1 to 1.~~

~~2.7.7.4.2 Watthour Revenue Meter (with and without Demand Register)~~

~~All Watthour revenue meters must measure voltage and current and must be in accordance with ANSI C12.1 with an ANSI C12.20 Accuracy class of [0.5] [0.2] and must have pulse initiators for remote monitoring of Watthour consumption. Pulse initiators must consist of form C contacts with a current rating not to exceed two amperes and voltage not to exceed 500 V, with combinations of VA not to exceed 100 VA, and a life rating of one billion operations. Meter sockets must be in accordance with NEMA/ANSI C12.10. Watthour revenue meters with demand registers must output instantaneous demand in addition to the pulse initiators.~~

2.7.7.4.1 Steam Meters

Steam meters must be the vortex type, with pressure compensation, a minimum turndown ratio of 10 to 1, and an output signal compatible with the DDC system.

2.7.7.4.2 Hydronic BTU Meters

The BTU meter is to be supplied with wall mount hardware and be capable of being installed remote from the flow meter. The BTU meter must include an LCD display for local indication of energy rate and for display of parameters and settings during configuration. Each BTU meter must be factory configured for its specific application and be completely field configurable by the user via a front panel keypad (no special interface device or computer required). The unit must output Energy Rate, Energy Total, Flow Rate, Supply Temperature, and Return Temperature. An integral transmitter is to provide a linear analog or configurable pulse output signal representing the energy rate; and the signal must be compatible with building automation system DDC Hardware to which the output is connected.

~~2.7.8 pH Sensor~~

~~The sensor must be suitable for applications and chemicals encountered in water treatment systems of boilers, chillers and condenser water systems. Construction, wiring, fittings and accessories must be corrosion and chemical resistant with fittings for tank or suspension installation. Housing must be polyvinylidene fluoride with O-rings made of chemical resistant materials which do not corrode or deteriorate with extended exposure to chemicals. The sensor must be encapsulated. Periodic replacement must not be required for continued sensor operation. Sensors must use a ceramic junction and pH sensitive glass membrane capable of withstanding a pressure of 689 kPa at 66 degrees C. The reference cell must be double junction configuration. Sensor range must be 0 to 12 pH, stability 0.05, sensitivity 0.02, and repeatability of plus or minus 0.05 pH value, response of 90 percent of full scale in one second and a linearity of 99 percent of theoretical electrode output measured at 24 degrees C.~~

~~2.7.9 Oxygen Analyzer~~

~~Oxygen analyzer must consist of a zirconium oxide sensor for continuous sampling and an air-powered aspirator to draw flue gas samples. The analyzer must be equipped with filters to remove flue air particles. Sensor probe temperature rating must be 435 degrees C. The sensor assembly must be equipped for flue flange mounting.~~

~~2.7.10 Carbon Monoxide Analyzer~~

~~Carbon monoxide analyzer must consist of an infrared light source in a weather proof steel enclosure for duct or stack mounting. An optical detector/analyzer in a similar enclosure, suitable for duct or stack mounting must be provided. Both assemblies must include internal blower systems to keep optical windows free of dust and ash at all times. The third component of the analyzer must be the electronics cabinet. Automatic flue gas temperature compensation and manual/automatic zeroing devices must be provided. Unit must read parts per million (ppm) of carbon monoxide in the range of [_____] to [_____] ppm and the response time must be less than 3 seconds to 90 percent value. Unit measurement range must not exceed specified range by more that 50 percent. Repeatability must be plus or minus 1 percent of full scale with an accuracy of plus or minus 1 percent of full scale.~~

2.7.8 Occupancy Sensors

Occupancy sensors must have occupancy-sensing sensitivity adjustment and an adjustable off-delay timer with a setpoint of 15 minutes. Adjustments accessible from the face of the unit are preferred. Occupancy sensors must be rated for operation in ambient air temperatures ranging from 5 to 35 degrees C or temperatures normally encountered in the installed location. Sensors integral to wall mount on-off light switches must have an auto-off switch. Wall switch sensors must be decorator style and must fit behind a standard decorator type wall plate. All occupancy sensors, power packs, and slave packs must be UL listed. In addition to any outputs required for lighting control, the occupancy sensor must provide an output for the HVAC control system.

2.7.8.1 Passive Infrared (PIR) Occupancy Sensors

PIR occupancy sensors must have a multi-level, multi-segmented viewing lens and a conical field of view with a viewing angle of 180 degrees and a detection of at least 6 m unless otherwise indicated or specified. PIR Sensors must provide field-adjustable background light-level adjustment with an adjustment range suitable to the light level in the sensed area, room or space. PIR sensors must be immune to false triggering from RFI and EMI.

2.7.8.2 Ultrasonic Occupancy Sensors

Ultrasonic sensors must operate at a minimum frequency 32 kHz and must be designed to not interfere with hearing aids.

2.7.8.3 Dual-Technology Occupancy Sensor (PIR and Ultrasonic)

Dual-Technology Occupancy Sensors must meet the requirements of both PIR and Ultrasonic Occupancy Sensors.

~~2.7.9 Vibration Switch~~

~~Vibration switch must be solid state, enclosed in a NEMA 250 Type 4 or Type 4X housing with sealed wire entry. Unit must have two independent sets of Form C switch contacts with one set to shutdown equipment upon excessive vibration and a second set for monitoring alarm level vibration. The vibration sensing range must be a true rms reading, suitable for the application. The unit must include either displacement response for low speed or velocity response for high speed application. The frequency range must be at least 3 Hz to 500 Hz. Contact time delay must be 3 seconds. The unit must have independent start-up and running delay on each switch contact. Alarm limits must be adjustable and setpoint accuracy must be plus or minus 10 percent of setting with repeatability of plus or minus 2 percent.~~

~~2.7.10 Conductivity Sensor~~

~~Sensor must include local indicating meter and must be suitable for measurement of conductivity of water in boilers, chilled water systems, condenser water systems, distillation systems, or potable water systems as indicated. Sensor must sense from 0 to 10 microSeimens per centimeter (µS/cm) for distillation systems, 0 to 100 µS/cm for boiler, chilled water, and potable water systems and 0 to 1000 µS/cm for condenser water systems. Contractor must field verify the ranges for particular applications and adjust the range as required. The output must be temperature compensated over a range of 0 to 100 degrees C. The accuracy must be plus or minus 2 percent of the full scale reading. Sensor must have automatic zeroing and must require no periodic maintenance or recalibration.~~

~~2.7.11 Compressed Air Dew Point Sensor~~

~~Sensor must be suitable for measurement of dew point from -40 +27 degrees C over a pressure range of 0 to 1 MPa. The transmitter must provide both dry bulb and dew point temperatures on separate outputs. The end to end accuracy of the dew point must be plus or minus 2.8 degrees C and the dry bulb must be plus or minus 0.6 degrees C. Sensor must be automatic zeroing and must require no normal maintenance or periodic recalibration.~~

~~2.7.12 NOx Monitor~~

~~Monitor must continuously monitor and give local indication of boiler stack gas for NOx content. It must be a complete system designed to verify compliance with the Clean Air Act standards for NOx normalized to a 3 percent oxygen basis and must have a range of from 0 to 100 ppm. Sensor must be accurate to plus or minus 5 ppm. Sensor must output NOx and oxygen levels and binary output that changes state when the NOx level is above a locally adjustable NOx setpoint. Sensor must have normal, trouble and alarm lights. Sensor must have heat traced lines if the stack pickup is remote from the sensor. Sensor must be complete with automatic zero and span calibration using a timed calibration gas system, and must not require periodic maintenance or recalibration.~~

~~2.7.13 Turbidity Sensor~~

~~Sensor must include a local indicating meter and must be suitable for measurement of turbidity of water. Sensor must sense from 0 to 1000 Nephelometric Turbidity Units (NTU). Range must be field-verified for the particular application and adjusted as required. The output must be temperature compensated over a range of 0 to 100 degrees C. The accuracy must be plus or minus 5 percent of full scale reading. Sensor must have automatic zeroing and must not require periodic maintenance or recalibration.~~

~~2.7.14 Chlorine Detector~~

~~The detector must measure concentrations of chlorine in water in the range 0 to 20 ppm with a repeatability of plus or minus 1 percent of full scale and an accuracy of plus or minus 2 percent of full scale. The Chlorine Detector transmitter must be housed in a non-corrosive NEMA 250 Type 4X enclosure. Detector must include a local panel with adjustable alarm trip level, local audio and visual alarm with silence function.~~

2.7.9 Floor Mounted Leak Detector

Leak detectors must use electrodes mounted at slab level with a minimum built-in-vertical adjustment of 3 mm. Detector must have a binary output. The indicator must be manual reset type.

2.7.10 Temperature Switch

2.7.10.1 Duct Mount Temperature Low Limit Safety Switch (Freezestat)

Duct mount temperature low limit switches (Freezestats) must be manual reset, low temperature safety switches at least 3 meters long per square meter of coverage which must respond to the coldest 450 mm segment with an accuracy of plus or minus 2 degrees C. The switch must have a field-adjustable setpoint with a range of at least -1 +10 degrees C. The switch must have two sets of contacts, and each contact must have a rating greater than its connected load. Contacts must open or close upon drop of temperature below setpoint as indicated and must remain in this state until reset.

2.7.10.2 Pipe Mount Temperature Limit Switch (Aquastat)

Pipe mount temperature limit switches (aquastats) must have a field adjustable setpoint between 15 and 32 degrees C, an accuracy of plus or minus 2 degrees C and a 5 degrees C fixed deadband. The switch must have two sets of contacts, and each contact must have a rating greater than its connected load. Contacts must open or close upon change of temperature above or below setpoint as indicated.

2.7.11 Damper End Switches

Each end switch must be a hermetically sealed switch with a trip lever and over-travel mechanism. The switch enclosure must be suitable for mounting on the duct exterior and must permit setting the position of the trip lever that actuates the switch. The trip lever must be aligned with the damper blade.

End switches integral to an electric damper actuator are allowed as long as at least one is adjustable over the travel of the actuator.

~~2.7.12 Air Quality Sensors~~

~~Provide full spectrum air quality sensors using a hot wire element based on the Taguchi principle. The sensor must monitor a wide range of gaseous volatile organic components common in indoor air contaminants like paint fumes, solvents, cigarette smoke, and vehicle exhaust. The sensor must automatically compensate for temperature and humidity, have span and calibration potentiometers, operate on 24 VDC power with output of 0-10 VDC, and have a service rating of 0 to 60 degrees C and 5 to 95 percent relative humidity.~~

~~[~~2.8 INDICATING DEVICES

All indicating devices must display readings in ~~[~~metric (SI)~~][English (inch-pound)]~~ units.

2.8.1 Thermometers

Provide bi-metal type thermometers at locations indicated. Thermometers must have either 230 mm long scales or 90 mm diameter dials, with insertion, immersion, or averaging elements. Provide matching thermowells for pipe-mounted installations. Select scale ranges suitable for the intended service, with the normal operating temperature near the scale's midpoint. The thermometer's accuracy must be plus or minus 2 percent of the scale range.

2.8.1.1 Piping System Thermometers

Piping system thermometers must have brass, malleable iron or aluminum alloy case and frame, clear protective face, permanently stabilized glass tube with indicating-fluid column, white face, black numbers, and a 230 mm scale. Piping system thermometers must have an accuracy of plus or minus 1 percent of scale range. Thermometers for piping systems must have rigid stems with straight, angular, or inclined pattern. Thermometer stems must have expansion heads as required to prevent breakage at extreme temperatures. On rigid-stem thermometers, the space between bulb and stem must be filled with a heat-transfer medium.

2.8.1.2 Air-Duct Thermometers

Air-duct thermometers must have perforated stem guards and 45-degree adjustable duct flanges with locking mechanism.

2.8.2 Pressure Gauges

Provide pipe-mounted pressure gauges at the locations indicated. Gauges must conform to ~~ASME B40.100~~JIS B 7505-1 and have a 100 mm diameter dial and shutoff cock.~~ ~~ Select scale ranges suitable for the intended service, with the normal operating pressure near the scale's midpoint. The gauge's accuracy must be plus or minus 2 percent of the scale range.

Gauges must be suitable for field or panel mounting as required, must have black legend on white background, and must have a pointer traveling through a 270-degree arc. Gauge range must be suitable for the application with an upper end of the range not to exceed 150 percent of the design upper limit. Accuracy must be plus or minus 3 percent of scale range. Gauges must meet requirements of ~~ASME B40.100~~JIS B 7505-1.

2.8.3 Low Differential Pressure Gauges

Gauges for low differential pressure measurements must be a minimum of 90 mm (nominal) size with two sets of pressure taps, and must have a diaphragm-actuated pointer, white dial with black figures, and pointer zero adjustment. Gauge range must be suitable for the application with an upper end of the range not to exceed 150 percent of the design upper limit. Accuracy must be plus or minus two percent of scale range.

~~[2.8.4 Pressure Gauges for Pneumatic Controls~~

~~Gauges must [have a 0 to 207 kPa scale][sufficient scale to display the full range of expected pressures] with 5 kPa graduations.~~

~~]]~~2.9 OUTPUT DEVICES

2.9.1 Actuators

Actuators must be electric (electronic)~~ [or pneumatic as indicated]~~. All actuators must be normally open (NO), normally closed (NC) or fail-in-last-position (FILP) as indicated. Normally open and normally closed actuators must be of mechanical spring return type. Electric actuators must have an electronic cut off or other means to provide burnout protection if stalled. Actuators must have a visible position indicator. ~~[~~Electric actuators must provide position feedback to the controller as indicated.~~]~~ Actuators must smoothly and fully open or close the devices to which they are applied. Electric actuators must have a full stroke response time in both directions of 90 seconds or less at rated load. Electric actuators must be of the foot-mounted type with an oil-immersed gear train or the direct-coupled type. Where multiple electric actuators operate from a common signal, the actuators must provide an output signal identical to its input signal to the additional devices. ~~[Pneumatic actuators must be rated for 172 kPa operating pressure except for high-pressure cylinder-type actuators.]~~ All actuators must be rated for their operating environment. Actuators used outdoors must be designed and rated for outdoor use. Actuators under continuous exposure to water, such as those used in sumps, must be submersible.

Actuators incorporating an integral network connection are considered DDC Hardware and must meet the DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

2.9.1.1 Valve Actuators

Valve actuators must provide shutoff pressures and torques as indicated on the Valve Schedule.

2.9.1.2 Damper Actuators

Damper actuators must provide the torque necessary per damper manufacturer's instructions to modulate the dampers smoothly over its full range of operation and torque must be at least 7.3 Nm/square m of damper area for opposed blade dampers and 10.9 Nm/square m of damper area for parallel blade dampers.

~~[2.9.1.3 Positive Positioners~~

~~Positive positioners must be a pneumatic relay with a mechanical position feedback mechanism and an adjustable operating range and starting point.~~

~~]~~2.9.1.3 Electric Actuators

Each actuator must have distinct markings indicating the full-open and full-closed position. Each actuator must deliver the torque required for continuous uniform motion and must have internal end switches to limit the travel, or be capable of withstanding continuous stalling without damage. Actuators must function properly within 85 to 110 percent of rated line voltage. Provide actuators with hardened steel running shafts and gears of steel or copper alloy. Fiber or reinforced nylon gears may be used for torques less than 16 inch-pounds.

a. Two-position actuators must be single direction, spring return, or reversing type. Two position actuator signals may either be the control power voltage or line voltage as needed for torque or appropriate interlock circuits.

b. Modulating actuators must be capable of stopping at any point in the cycle, and starting in either direction from any point. Actuators must be equipped with a switch for reversing direction, and a button to disengage the clutch to allow manual adjustments. Provide the actuator with a hand crank for manual adjustments, as applicable. Modulating actuator input signals can either be a 4 to 20 mAdc or a 0-10 VDC signal.

c. Floating or pulse width modulation actuators are acceptable for non-fail safe applications unless indicated otherwise provided that the floating point control (timed actuation) must have a scheduled re-calibration of span and position no more than once a day and no less than once a week. The schedule for the re-calibration should not affect occupied conditions and be staggered between equipment to prevent falsely loading or unloading central plant equipment.

~~[2.9.1.4 Pneumatic Actuators~~

~~Provide piston or diaphragm type actuators with replaceable diaphragm/piston.~~

~~]2.9.2 Solenoid-Operated Electric to Pneumatic Switch (EPS)~~

~~Solenoid-Operated Electric to Pneumatic Switches (EPS) must accept a voltage input to actuate its air valve. Each valve must have three-port~~

~~operation: common, normally open, and normally closed. Each valve must have an outer cast aluminum body and internal parts of brass, bronze, or stainless steel. The air connection must be a 10 mm NPT threaded connection. Valves must be rated for 345 kPa.~~

~~2.9.3 Electric to Pneumatic Transducers (EP)~~

~~Electric to Pneumatic Transducers (EPs) must convert either a 4-20 mAdc input signal, a 0-10 Vdc input signal to a proportional 0 to 140 kPa pneumatic output. The EP must withstand pressures at least 150 percent of the system supply air pressure (main air). EPs must include independent offset and span adjustment. Steady state air consumption must not be greater than 0.23 L/s. EPs must have a manual adjustable override for the EP pneumatic output. EPs must have sufficient output capacity to provide full range stroke of the actuated device in both directions within [90][_____] seconds.~~

2.9.2 Relays

Relays must have contacts rated for the intended application, indicator light, and dust proof enclosure. The indicator light must be lit when the coil is energized and off when coil is not energized.

Control relay contacts must have utilization category and ratings selected for the application. Each set of contacts must incorporate a normally open (NO), normally closed (NC) and common contact. Relays must be rated for a minimum life of one million operations.

2.10 USER INPUT DEVICES

User Input Devices, including potentiometers, switches and momentary contact push-buttons. Potentiometers must be of the thumb wheel or sliding bar type. Momentary Contact Push-Buttons may include an adjustable timer for their output. User input devices must be labeled for their function.

2.11 MULTIFUNCTION DEVICES

Multifunction devices are products which combine the functions of multiple sensor, user input or output devices into a single product. Unless otherwise specified, the multifunction device must meet all requirements of each component device. Where the requirements for the component devices conflict, the multifunction device must meet the most stringent of the requirements.

2.11.1 Current Sensing Relay Command Switch

The Current Sensing Relay portion must meet all requirements of the Current Sensing Relay input device. The Command Switch portion must meet all requirements of the Relay output device except that it must have at least one normally-open (NO) contact.

Current Sensing Relays used for Variable Frequency Drives must be rated for Variable Frequency applications unless installed on the source side of the drive. If used in this situation, the threshold for showing status must be set to allow for the VFD's control power when the drive is not enabled and provide indication of operation when the drive is enabled at minimum speed.

2.11.2 Space Sensor Module

Space Sensor Modules must be multifunction devices incorporating a temperature sensor and one or more of the following as specified and indicated on the Space Sensor Module Schedule:

a. A temperature indicating device.

b. A User Input Device which must adjust a temperature setpoint output.

c. A User Input Momentary Contact Button and an output to the control system indicating zone occupancy.

d. A three position User Input Switch labeled to indicate heating, cooling and off positions ('HEAT-COOL-OFF' switch) and providing corresponding outputs to the control system.

e. A two position User Input Switch labeled with 'AUTO' and 'ON' positions and providing corresponding output to the control system.

f. A multi-position User Input Switch with 'OFF' and at least two fan speed positions and providing corresponding outputs to the control system.

Space Sensor Modules cannot contain mercury (Hg).

~~[2.12 COMPRESSED AIR STATIONS~~

~~2.12.1 Air Compressor Assembly~~

~~Air compressors for pneumatic control systems must be the tank-mounted, electric motor driven, air cooled, reciprocating type with integral [duplex motors and compressors][single motor and compressor], tank, controller, [alternator switch,]pressure switch, belt guard[s], pressure relief valve, automatic moisture drain valve and must be supported by a steel base mounted on an air storage tank. Compressor piston speeds must not exceed 2.28 meter/second. Provide compressors with a dry-type combination intake air filter and silencer with baked enamel steel housing. The filter must be 99 percent efficient at 10 µm. The pressure switch must start the compressor[s] at 482 kPa and stop the compressor[s] at 620 kPa. The relief valve must be set for 69 to 172 kPa above the control switch cut-off pressure. Provide compressor capacity suitable for not more than a [33] [50] percent run time, at full system control load. Compressors must have a combination type magnetic starter with undervoltage protection and thermal-overload protection for each phase and must automatically restart after a power outage. Motors 0.5 hp and larger must be three-phase.[~~

~~A second (duplex arrangement) compressor of capacity equal to the primary compressor must be provided, with interlocked control to provide automatic changeover upon malfunction or failure of either compressor. A manual selector switch must be provided to index the lead compressor including the automatic changeover.]~~

~~2.12.2 Compressed Air Station Specialties~~

~~2.12.2.1 Refrigerated Air Dryers~~

~~Provide each air compressor tank with a refrigerant air dryer sized for~~

~~continuous operation at full delivery capacity of the compressor. The air must be dried at a pressure of not less than 483 kPa to a temperature not greater than 2 degrees C and an ambient air temperature between 13 and 35 degrees C. The dryer must be provided with an automatic condensate drain trap with manual override feature with an adjustable cycle and drain time. Locate each dryer in the air piping between the tank and the pressure-reducing station. The refrigerant used in the dryer must be one of the fluorocarbon gases and have an Ozone Depletion Potential of not more than 0.05. A five micron pre-filter and coalescing-type 0.03 µm oil removal filter with shut-off valves must be provided in the dryer discharge.~~

~~2.12.2.2 Compressed Air Discharge Filters~~

~~Provide a disposable type in-line filter in the incoming pneumatic main at each pneumatic control panel. The filter must be capable of eliminating 99.99 percent of all liquid or solid contaminants 0.1 micron or larger. Provide the filter with fittings that allow easy removal/replacement. Each filter bowl must be rated for 1034 kPa maximum working pressure. A pressure regulator, with high side and low side pressure gauges, and a safety valve must be provided downstream of the filter.~~

~~2.12.2.3 Air Pressure-Reducing Stations~~

~~Provide air compressors with a pressure-reducing valve (PRV) with a field adjustable range of 0 to 345 kPa discharge pressure, at an inlet pressure of 482 to 620 kPa. Provide a factory-set pressure relief valve downstream of the PRV to relieve over-pressure. Provide a pressure gage upstream of the PRV with range of 0 to 689 kPa and downstream of the PRV with range of 0 to 207 kPa 0 to 30 psig. For two-pressure control systems, provide an additional PRV and downstream pressure gage. Pressure regulators of the relieving type must not be used.~~

~~2.12.2.4 Flexible Pipe Connections~~

~~The flexible pipe connections must be designed for 1034 kPa and 120 degrees C service, and must be constructed of rubber or tetrafluoroethylene resin tubing with a reinforcing protective cover of braided corrosion-resistant steel, bronze, monel, or galvanized steel. The connectors must be suitable for the service intended and must have threaded or soldered ends. The length of the connectors must be as recommended by the manufacturer for the service intended.~~

~~2.12.2.5 Vibration Isolation Units~~

~~The vibration isolation units must be standard products with published loading ratings, and must be single rubber-in-shear, double rubber-in-shear, or spring type.~~

~~2.12.3 Compressed Air Tanks~~

~~The air storage tank must be fabricated for a working pressure of not less than 1380 kPa and constructed and certified in accordance with ASME BPVC SEC VIII D1. The tank must be of sufficient volume so that no more than six compressor starts per hour are required with the starting pressure switch differential set at 140 kPa. The tank must be provided with an automatic condensate drain trap with manual override feature. Provide drain valve and piping routing the drainage to a floor sink or other safe and visible drainage location.~~

~~]~~PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 General Installation Requirements

Perform the installation under the supervision of competent technicians regularly employed in the installation of DDC systems.

3.1.1.1 Device Mounting Criteria

All devices must be installed in accordance with manufacturer's recommendations and as specified and indicated. Control devices to be installed in piping and ductwork must be provided with required gaskets, flanges, thermal compounds, insulation, piping, fittings, and manual valves for shutoff, equalization, purging, and calibration. Strap-on temperature sensing elements must not be used except as specified. Spare thermowells must be installed adjacent to each thermowell containing a sensor and as indicated. Devices located outdoors must have a weathershield.

3.1.1.2 Labels and Tags

Match labels and tags to the unique identifiers indicated on the As-Built drawings. Label all enclosures and instrumentation. Tag all sensors and actuators in mechanical rooms. Tag airflow measurement arrays to show flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient. Tag duct static pressure taps at the location of the pressure tap. Provide plastic or metal tags, mechanically attached directly to each device or attached by a metal chain or wire. Labels exterior to protective enclosures must be engraved plastic and mechanically attached to the enclosure or instrumentation. Labels inside protective enclosures may attached using adhesive, but must not be hand written.

3.1.2 Weathershield

Provide weathershields for sensors located outdoors. Install weathershields such that they prevent the sun from directly striking the sensor and prevent rain from directly striking or dripping onto the sensor. Install weather shields with adequate ventilation so that the sensing element responds to the ambient conditions of the surroundings. When installing weathershields near outside air intake ducts, install them such that normal outside air flow does not cause rainwater to strike the sensor.

3.1.3 Room Instrument Mounting

Mount room instruments, including but not limited to wall mounted non-adjustable space sensor modules and sensors located in occupied spaces, ~~[~~1.5~~][1.2]~~ meters above the floor unless otherwise indicated. Install adjustable devices to be ADA compliant unless otherwise indicated on the Room Sensor Schedule:

a. Space Sensor Modules for Fan Coil Units may be either unit or wall mounted but not mounted on an exterior wall.

b. Wall mount all other Space Sensor Modules.

3.1.4 Indication Devices Installed in Piping and Liquid Systems

Provide snubbers for gauges in piping systems subject to pulsation. For gauges for steam service use pigtail fittings with cock. Install thermometers and temperature sensing elements in liquid systems in thermowells. Provide spare Pressure/Temperature Ports (Pete's Plug) for all temperature and pressure sensing elements installed in liquid systems for calibration/testing.

3.1.5 Occupancy Sensors

Provide a sufficient quantity of occupancy sensors to provide complete coverage of the area (room or space). Occupancy sensors are to be ceiling mounted. Install occupancy sensors in accordance with NFPA 70 requirements and the manufacturer's instructions. Do not locate occupancy sensors within 2 m of HVAC outlets or heating ducts, or where they can "see" beyond any doorway. Installation above doorway(s) is preferred. Do not use ultrasonic sensors in spaces containing ceiling fans. Install sensors to detect motion to within 600 mm of all room entrances and to not trigger due to motion outside the room. Set the off-delay timer to ~~[~~15 ~~][_____]~~ minutes unless otherwise indicated. Adjust sensors prior to beneficial occupancy, but after installation of furniture systems, shelving, partitions, etc. For each controlled area, provide one hundred percent coverage capable of detecting small hand-motion movements, accommodating all occupancy habits of single or multiple occupants at any location within the controlled room.

3.1.6 Switches

3.1.6.1 Temperature Limit Switch

Provide a temperature limit switch (freezestat) to sense the temperature at the location indicated. Provide a sufficient number of temperature limit switches (freezestats) to provide complete coverage of the duct section but no less than 3 m in length per square meter of cross sectional area. Install manual reset limit switches in approved, accessible locations where they can be reset easily. Install temperature limit switch (freezestat) sensing elements in a side-to-side (not top-to-bottom) serpentine pattern with the relay section at the highest point and in accordance with the manufacturer's installation instructions.

3.1.6.2 Hand-Off Auto Switches

Wire safety controls such as smoke detectors and freeze protection thermostats to protect the equipment during both hand and auto operation.

3.1.7 Temperature Sensors

Install temperature sensors in locations that are accessible and provide a good representation of sensed media. Installations in dead spaces are not acceptable. Calibrate and install sensors according to manufacturer’s instructions. Select sensors only for intended application as designated or recommended by manufacturer.

3.1.7.1 Room Temperature Sensors

Mount the sensors on interior walls to sense the average room temperature at the locations indicated. Avoid locations near heat sources such as copy machines or locations by supply air outlet drafts. Mount the center of all user-adjustable sensors ~~[~~1.5 m above the finished floor~~][~~ or 1220 mm above the floor to meet ADA requirements~~][at the height[s] indicated]~~. Non user-adjustable sensors can be mounted as indicated in paragraph ROOM INSTRUMENT MOUNTING.

3.1.7.2 Duct Temperature Sensors

3.1.7.2.1 Probe Type

Place tip of the sensor in the middle of the airstream or in accordance with manufacturer's recommendations or instructions.Provide a gasket between the sensor housing and the duct wall. Seal the duct penetration air tight. When installed in insulated duct, provide enclosure or stand off fitting to accommodate the thickness of duct insulation to allow for maintenance or replacement of the sensor and wiring terminations. Seal the duct insulation penetration vapor tight.

3.1.7.2.2 Averaging Type

Weave the sensing element in a serpentine fashion from side to side perpendicular to the flow, across the duct or air handler cross-section, using durable non-metal supports in accordance with manufacturer's installation instructions. Avoid tight radius bends or kinking of the sensing element. Prevent contact between the sensing element and the duct or air handler internals. Provide a duct access door at the sensor location. The access door must be hinged on the side, factory insulated, have cam type locks, and be as large as the duct will permit, maximum 18 by 18 inches. For sensors inside air handlers, the sensors must be fully accessible through the air handler's access doors without removing any of the air handler's internals.

3.1.7.3 Immersion Temperature Sensors

Provide thermowells for sensors measuring piping, tank, or pressure vessel temperatures. Locate wells to sense continuous flow conditions. Do not install wells using extension couplings. When installed on insulated piping, provide stand enclosure or stand off fitting to accommodate the thickness of the pipe insulation and allow for maintenance or replacement of the sensor or wiring terminations. Where piping diameters are smaller than the length of the wells, provide wells in piping at elbows to sense flow across entire area of well. Wells must not restrict flow area to less than 70 percent of pipe area. Increase piping size as required to avoid restriction. Provide the sensor well with a heat-sensitive transfer agent between the sensor and the well interior ensuring contact between the sensor and the well.

3.1.7.4 Outside Air Temperature Sensors

Provide outside air temperature sensors on the building's north side with a protective weather shade that does not inhibit free air flow across the sensing element, and protects the sensor from snow, ice, and rain. Location must not be near exhaust hoods and other areas such that it is not influenced by radiation or convection sources which may affect the reading. Provide a shield to shade the sensor from direct sunlight.

3.1.8 Air Flow Measurement Arrays (AFMA)

Locate Outside Air AFMAs downstream from the Outside Air filters.

Install AFMAs with the manufacturer's recommended minimum distances between upstream and downstream disturbances. Airflow straighteners may be used to reduce minimum distances as recommended by the AFMA manufacturer.

3.1.9 Duct Static Pressure Sensors

Locate the duct static pressure sensing tap at 75 percent of the distance between the first and last air terminal units ~~[as indicated on the design documents]~~. If the transmitter output is a 0-10Vdc signal, locate the transmitter in the same enclosure as the air handling unit (AHU) controller for the AHU serving the terminal units. If a remote duct static pressure sensor is to be used, run the signal wire back to the controller for the air handling unit.

3.1.10 Relative Humidity Sensors

Install relative humidity sensors in supply air ducts at least 3 m downstream of humidity injection elements.

3.1.11 Meters

3.1.11.1 Flowmeters

Install flowmeters to ensure minimum straight unobstructed piping for at least 10 pipe diameters upstream and at least 5 pipe diameters downstream of the flowmeter, and in accordance with the manufacturer's installation instructions.

3.1.11.2 Energy Meters

Locate energy meters as indicated. Connect each meter output to the DDC system, to measure both instantaneous demand/energy and other variables as indicated.

3.1.12 Dampers

3.1.12.1 Damper Actuators

Provide spring return actuators which fail to a position that protects the served equipment and space on all control dampers related to freeze protection or force protection. For all outside, makeup and relief dampers provide dampers which fail closed. Terminal fan coil units, terminal VAV units, convectors, and unit heaters nay be non-spring return unless indicated otherwise. Do not mount actuators in the air stream. Do not connect multiple actuators to a common drive shaft. Install actuators so that their action seal the damper to the extent required to maintain leakage at or below the specified rate and so that they move the blades smoothly throughout the full range of motion.

3.1.12.2 Damper Installation

Install dampers straight and true, level in all planes, and square in all dimensions. Dampers must move freely without undue stress due to twisting, racking (parallelogramming), bowing, or other installation error. External linkages must operate smoothly over the entire range of motion, without deformation or slipping of any connecting rods, joints or brackets that will prevent a return to it's normal position. Blades must close completely and leakage must not exceed that specified at the rated static pressure. Provide structural support for multi-section dampers. Acceptable methods of structural support include but are not limited to U-channel, angle iron, corner angles and bolts, bent galvanized steel stiffeners, sleeve attachments, braces, and building structure. Where multi-section dampers are installed in ducts or sleeves, they must not sag due to lack of support. Do not use jackshafts to link more than three damper sections. Do not use blade to blade linkages. Install outside and return air dampers such that their blades direct their respective air streams towards each other to provide for maximum mixing of air streams.

3.1.13 Valves

Install the valves in accordance with the manufacturer's instructions.

3.1.13.1 Valve Actuators

Provide spring return actuators on all control valves where freeze protection is required. Spring return actuators for terminal fan coil units, terminal VAV units, convectors, and unit heaters are not required unless indicated otherwise.

3.1.14 Thermometers and Gauges

~~[3.1.14.1 Local Gauges for Actuators~~

~~Provide a pressure gauge at each pneumatic control input and output. Pneumatic actuators must have an accessible and visible pressure gauge installed in the tubing lines at the actuator as indicated.~~

~~]~~3.1.14.1 Thermometers

Mount devices to allow reading while standing on the floor or ground, as applicable.

3.1.15 Wire and Cable

Provide complete electrical wiring for the Control System, including wiring to transformer primaries. Wire and Cable must be installed without splices between control devices and in accordance with NFPA 70 and NFPA 90A. Instrumentation grounding must be installed per the device manufacturer's instructions and as necessary to prevent ground loops, noise, and surges from adversely affecting operation of the system. Test installed ground rods as specified in IEEE 142. Cables and conductor wires must be tagged at both ends, with the identifier indicated on the shop drawings. Electrical work must be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and as indicated. Wiring external to enclosures must be run in raceways~~[, except low-voltage control and low-voltage network wiring may be installed as follows:~~.

~~a. plenum rated cable in suspended ceilings over occupied spaces may be run without raceways~~

~~b. nonmetallic-sheathed cables or metallic-armored cables may be installed as permitted by NFPA 70.]~~

Install control circuit wiring not in raceways in a neat and safe manner. Wiring must not use the suspended ceiling system (including tiles, frames or hangers) for support. Where conduit or raceways are required, control circuit wiring must not run in the same conduit/raceway as power wiring over 50 volts. Run all circuits over 50 volts in conduit, metallic tubing, covered metal raceways, or armored cable.

3.1.16 Copper Tubing

Provide hard-drawn copper tubing in exposed areas and either hard-drawn or annealed copper tubing in concealed areas. Use only tool-made bends. Use only brass or copper solder joint type fittings, except for connections to apparatus. For connections to apparatus use brass compression type fittings.

3.1.17 Plastic Tubing

Install plastic tubing within covered raceways or conduit except when otherwise specified. Do not use plastic tubing for applications where the tubing could be subjected to a temperature exceeding 55 degrees C. For fittings, use brass or acetal resin of the compression or barbed push-on type for instrument service. Except in walls and exposed locations, plastic multitube instrument tubing bundle without conduit or raceway protection may be used where a number of air lines run to the same points, provided the multitube bundle is enclosed in a protective sheath, is run parallel to the building lines and is adequately supported as specified.

~~[3.1.18 Pneumatic Lines~~

~~Run tubing concealed in finished areas, run tubing exposed in unfinished areas like mechanical rooms. For tubing enclosed in concrete, provide rigid metal conduit. Run tubing parallel and perpendicular to building walls. Use1.5 m maximum spacing between tubing supports. With the compressor turned off, test each tubing system pneumatically at 1.5 times the working pressure and prove it air tight, locating and correcting leaks as applicable. Caulking joints is not permitted. Do not run tubing and electrical power conductors in the same conduit.~~

~~a. Install pneumatic lines must such that they are not exposed to outside air temperatures. Conceal pneumatic lines except in mechanical rooms and other areas where other tubing and piping is exposed.~~

~~b. Install all tubes and tube bundles exposed to view in lines parallel to the lines of the building. Route tubing in mechanical/electrical so that the lines are easily traceable.~~

~~c. Purge air lines of dirt, impurities and moisture before connecting to the control equipment. Number-code or color-code air lines and key the coding in the As-Built Drawings for future identification and servicing the control system.~~

~~3.1.18.1 Pneumatic Lines In Mechanical/Electrical Spaces~~

~~In mechanical/electrical spaces, use plastic or copper tubing for pneumatic lines. Install horizontal and vertical runs of plastic tubing or soft copper tubing min raceways or rigid conduit dedicated to tubing. Support dedicated raceways, conduit, and hard copper tubing not installed in raceways every 2 m for horizontal runs and every 2.4 m for vertical runs.~~

~~3.1.18.2 Pneumatic Lines External to Mechanical/Electrical Spaces~~

~~External to mechanical/electrical spaces, use plastic tubing in raceways not containing power wiring or copper tubing with sweat fittings. Support raceways and tubing not in raceways every 2.4 m. For pneumatic lines concealed in walls use hard-drawn copper tubing or plastic tubing in rigid conduit. Plastic tubing in a protective sheath, run parallel to the building lines and supported as specified, may be used above accessible ceilings and in other concealed but accessible locations.~~

~~3.1.18.3 Terminal Single Lines~~

~~For terminal single lines use hard-drawn copper tubing, except when the run is less than 300 mm in length, flexible polyethylene may be used.~~

~~3.1.18.4 Connection to Liquid and Steam Lines~~

~~Use [copper][Series 300 stainless steel] with [brass compression][stainless-steel compression] fittings for connection of sensing elements and transmitters to liquid and steam lines.~~

~~3.1.18.5 Connection to Ductwork~~

~~Use plastic tubing for connections to sensing elements in ductwork.~~

~~3.1.18.6 Tubing in Concrete~~

~~Install tubing in concrete in rigid conduit. Install tubing in walls containing insulation, fill, or other packing materials in raceways dedicated to tubing.~~

~~3.1.18.7 Tubing Connection to Actuators~~

~~For final connections to actuators use plastic tubing no more than 300 mm long and unsupported at the actuator.~~

~~]3.1.19 Compressed Air Stations~~

~~Mount the air compressor assembly on vibration eliminators, in accordance with ASME BPVC SEC VIII D1 for tank clearance. Connect the air line to the tank with a flexible pipe connector. Provide compressed air station specialties with required tubing, including condensate tubing to a floor drain. Compressed air stations must deliver control air meeting the requirements of ISA 7.0.01. Provide foundations and housekeeping pads for the HVAC control system air compressors [in accordance with the air compressor manufacturer's instructions][as specified in Section 23 30 00 HVAC AIR DISTRIBUTION].~~

-- End of Section --

SECTION 23 09 23.02

BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS
08/24

PART 1 GENERAL

1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, except for the front end which is specified in Section 25 10 10 UTILITY MONITORING AND CONTROL (UMCS) FRONT END AND INTEGRATION, suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as specified and shown and in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

1.1.1 System Requirements

Provide a system meeting the requirements of both Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and this Section and with the following characteristics:

a. ~~Except for Gateways, the control system must be an open implementation of BACnet technology using ASHRAE 135 and Fox as the communications protocols. The system must use standard ASHRAE 135 Objects and Properties and the Niagara Framework. The system must use standard ASHRAE 135 Services and the Niagara Framework exclusively for communication over the network. Gateways to packaged units must communicate with other DDC hardware using ASHRAE 135 or the Fox protocol exclusively and may communicate with packaged equipment using other protocols. The control system must be installed such that any two ASHRAE 135 devices on the Internetwork can communicate using standard ASHRAE 135 Services.~~

Except for Gateways, the control system must be an open implementation of BACnet technology using ASHRAE 135 as the communications protocol. The system must use standard ASHRAE 135 Objects and Properties. The system must use standard ASHRAE 135 Services exclusively for communication over the network. Gateways to packaged units must communicate with other DDC hardware using ASHRAE 135 exclusively and may communicate with packaged equipment using other protocols. The control system must be installed such that any two devices on the Internetwork can communicate using standard ASHRAE 135 Services.

b. Install and configure control hardware to provide ASHRAE 135 Objects and Properties ~~or Niagara Framework Objects~~ as indicated and as needed to meet the requirements of this specification.

~~c. Use Niagara Framework hardware and software exclusively for scheduling, trending, and communication with a front end (UMCS). Use Niagara Framework or standard BACnet Objects and services for alarming. Use the Fox protocol for all communication between Niagara Framework Supervisory Gateways; use the ASHRAE 135 protocol for all other building communication. [Niagara Framework Supervisory Gateway must serve web pages as specified.]~~

~~d. Use Niagara Framework Version [4.0 or later][_____]..~~

1.1.2 Verification of Specification Requirements

Review all specifications related to the control system installation and advise the Contracting Officer of any discrepancies before performing any work. If Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC or any other Section referenced in this specification is not included in the project specifications advise the Contracting Officer and either obtain the missing Section or obtain Contracting Officer approval before performing any work.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135	(2024; INT 1-3 2025; Errata 1 2025; INT 4-6 2025; Addenda CU 2025) BACnet—A Data Communication Protocol for Building Automation and Control Networks

BACNET TESTING LABORATORIES (BTL)

BTL Guide	(v.50; 2022) BACnet Testing Laboratory Implementation Guidelines

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.3	(2022; CORR 2024) Ethernet

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-485	(1998a; R 2012) Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15	Radio Frequency Devices (47 CFR 15)

UL SOLUTIONS (UL)

UL 916	(2015; Reprint Oct 2021) UL Standard for Safety Energy Management Equipment

1.3 DEFINITIONS

For definitions related to this section, see Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

1.4 SUBMITTALS

Submittal requirements related to this Section are specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

PART 2 PRODUCTS

All products used to meet this specification must meet the indicated requirements, but not all products specified here will be required by every project. All products must meet the requirements of Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC, Section 25 05 11~~[.______]~~ CYBERSECURITY FOR ~~[~~FACILITY-RELATED CONTROL~~][______]~~ SYSTEMS and this Section.

2.1 NETWORK HARDWARE

2.1.1 BACnet Router

All BACnet Routers must be BACnet/IP Routers and must perform layer 3 routing of ASHRAE 135 packets over an IP network in accordance with ASHRAE 135 Annex J and Clause 6. The router must provide the appropriate connection to the IP network and connections to one or more ASHRAE 135 MS/TP networks.~~-~~ Devices used as BACnet Routers must meet the requirements for DDC Hardware, and ~~except for Niagara Framework Supervisory Gateways, devices used as BACnet routers~~ must support the NM-RC-B BIBB.

2.1.2 BACnet Gateways

In addition to the requirements for DDC Hardware, the BACnet Gateway must ~~be a Niagara Framework Supervisory Gateway or must~~ meet the following requirements:

a. It must perform bi-directional protocol translation from one non-ASHRAE 135 protocol to ASHRAE 135.~~-~~ BACnet Gateways must incorporate a network connection to an ASHRAE 135 network (either BACnet over IP in accordance with Annex J or MS/TP) and a separate connection appropriate for the non-ASHRAE 135 protocol and media.

b. It must retain its configuration after a power loss of an indefinite time, and must automatically return to their pre-power loss state once power is restored.

c. It must allow bi-directional mapping of data between the non-ASHRAE 135 protocol and Standard Objects as defined in ASHRAE 135. It must support the DS-RP-B BIBB for Objects requiring read access and the DS-WP-B BIBB for Objects requiring write access.

d. It must support the DS-COV-B BIBB.

Although Gateways must meet DDC Hardware requirements~~, except for Niagara Framework Supervisory Gateways,~~ they are not DDC Hardware and must not be used when DDC Hardware is required. ~~(Niagara Framework Supervisory Gateways are both Gateways and DDC Hardware.)~~

2.1.3 Ethernet Switch

Ethernet Switches must be as specified in Section 25 05 11~~[.______]~~ CYBERSECURITY FOR ~~[~~FACILITY-RELATED CONTROL~~][______]~~ SYSTEMS

2.2 CONTROL NETWORK WIRING

a. BACnet MS/TP communications wiring must be in accordance with

ASHRAE 135. The wiring must use shielded, three wire (twisted-pair with reference) cable with characteristic impedance between 100 and 120 ohms. Distributed capacitance between conductors must be less than 100 pF per meter.

b. Building Control Network Backbone IP Network must use Ethernet media. Ethernet cables must be CAT-5e at a minimum and meet all requirements of IEEE 802.3~~ [and [_____]]~~.

2.3 DIRECT DIGITAL CONTROL (DDC) HARDWARE

2.3.1 General Requirements

All DDC Hardware must meet the following requirements:

a. It must be locally powered and must incorporate a light to indicate the device is receiving power.

b. It must conform to the BTL Guide.

c. It must be BACnet Testing Laboratory (BTL) Listed.

d. The Manufacturer's Product Data submittal for each piece of DDC Hardware must include the Protocol Implementation Conformance Statement (PICS) for that hardware as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

e. It must communicate and be interoperable in accordance with ASHRAE 135 and have connections for BACnet IP or MS/TP control network wiring.

f. Other than devices controlling terminal units or functioning solely as a BACnet Router, it must support DS-COV-B, DS-RPM-A and DS-RPM-B BIBBs.

g. Devices supporting the DS-RP-A BIBB must also support the DS-COV-A BIBB.

h. Application programs, configuration settings and communication information must be stored in a manner such that they persist through loss of power:

(1) Application programs must persist regardless of the length of time power is lost.

(2) Configured settings must persist for any loss of power less than 2,500 hours.

(3) Communication information, including but not limited to COV subscriptions, event reporting destinations, Notification Class Object settings, and internal communication settings, must persist for any loss of power less than 2,500 hours.

i. Internal Clocks:

(1) Clocks in DDC Hardware incorporating a Clock must continue to function for 120 hours upon loss of power to the DDC Hardware.

(2) DDC Hardware incorporating a Clock must support the DM-TS-B or DM-UTC-B BIBB.

j. It must have all functionality indicated and required to support the application (Sequence of Operation or portion thereof) in which it is used, including but not limited to providing Objects ~~or Niagara Framework Points~~ as specified and as indicated on the Points Schedule.

k. In addition to these general requirements and the DDC Hardware Input-Output (I/O) Function requirements, all DDC Hardware must also meet any additional requirements for the application in which it is used (e.g. scheduling, alarming, trending, etc.).

l. It must meet FCC Part 15 requirements and have UL 916 or equivalent safety listing.

m. ~~Except for Niagara Framework Supervisory Gateways,~~ Device must support Commandable Objects to support Override requirements as detailed in PART 3 EXECUTION.

n. User interfaces which allow for modification of Properties or settings must be password-protected.

o. Devices communicating BACnet MS/TP must meet the following requirements:

(1) Must have a configurable Max_Master Property.

(2) DDC Hardware other than hardware controlling a single terminal unit must have a configurable Max_Info_Frames Property.

(3) Must respond to any valid request within 50 msec with either the appropriate response or with a response of "Reply Postponed".

(4) Must use twisted pair with reference and shield (3-wire media) wiring~~[~~, or twisted pair with shield (2-wire media) wiring and use half-wave rectification~~]~~.

p. Devices communicating BACnet/IP must use UDP Port 0xBAC0. Devices with configurable UDP Ports must default to 0xBAC0.

q. All Device IDs, Network Numbers, and BACnet MAC addresses of devices must be fully configurable without limitation, except MS/TP MAC addresses may be limited by ASHRAE 135 requirements.

r. ~~Except for Niagara Framework Supervisory Gateways,~~ DDC Hardware controlling a single terminal unit must have:

(1) Objects (including the Device Object) with an Object Name Property of at least 8 characters in length.

(2) A configurable Device Object Name.

(3) A configurable Device Object Description Property at least 16 characters in length.

s. Except for Objects in ~~either Niagara Framework Supervisory Gateways or~~ DDC Hardware controlling a single terminal unit, all Objects (including Device Objects) must:

(1) Have a configurable Object Name Property of at least 12 characters in length.

(2) Have a configurable Object Description Property of at least 24 characters in length.

t. For programmable DDC Hardware, provide and license to the project site all programming software required to program the Hardware in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

u. For programmable DDC Hardware, provide copies of the installed application programs (all software that is not common to every controller of the same manufacturer and model) as source code compatible with the supplied programming software in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The submitted application program must be the complete application necessary for controller to function as installed and be sufficient to allow replacement of the installed controller with another controller of the same type.

2.3.2 Hardware Input-Output (I/O) Functions

DDC Hardware incorporating hardware input-output (I/O) functions must meet the following requirements:

2.3.2.1 Analog Inputs

DC Hardware analog inputs (AIs) must be implemented using ASHRAE 135 Analog Input Objects and perform analog to digital (A-to-D) conversion with a minimum resolution of 8 bits plus sign or better as needed to meet the accuracy requirements specified in Section 23 09 00. Signal conditioning including transient rejection must be provided for each analog input. Analog inputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. The AI must incorporate common mode noise rejection of at least 50 dB from 0 to 100 Hz for differential inputs, and normal mode noise rejection of at least 20 dB at 60 Hz from a source impedance of 10,000 ohms.

2.3.2.2 Analog Outputs

DDC Hardware analog outputs (AOs) must be implemented using ASHRAE 135 Analog Output Objects and perform digital to analog (D-to-A) conversion with a minimum resolution of 8 bits plus sign, and output a signal with a range of 4-20 mAdc or 0-10 Vdc. Analog outputs must be capable of being individually calibrated for zero and span. Calibration via software scaling performed as part of point configuration is acceptable. DDC Hardware with Hand-Off-Auto (H-O-A) switches for analog outputs must provide for overriding the output ~~[to 0 percent and to 100 percent][~~ through the range of 0 percent to 100 percent~~]~~.

2.3.2.3 Binary Inputs

DDC Hardware binary inputs (BIs) must be implemented using ASHRAE 135 Binary Input Objects and accept contact closures and must ignore transients of less than 5 milli-second duration. Protection against a transient 50VAC must be provided.

2.3.2.4 Binary Outputs

DDC Hardware binary outputs (BOs) must be implemented using ASHRAE 135

Binary Output Objects and provide relay contact closures or triac outputs for momentary and maintained operation of output devices. DDC Hardware with H-O-A switches for binary outputs must provide for overriding the output open or closed.

2.3.2.4.1 Relay Contact Closures

Closures must have a minimum duration of 0.1 second. Relays must provide at least 180V of isolation. Electromagnetic interference suppression must be provided on all output lines to limit transients to 50 Vac. Minimum contact rating must be 0.5 amperes at 24 Vac.

2.3.2.4.2 Triac Outputs

Triac outputs must provide at least 180 V of isolation. Minimum contact rating must be 0.5 amperes at 24 Vac.

2.3.2.5 Pulse Accumulator

DDC Hardware pulse accumulators must be implemented using either an ASHRAE 135 Accumulator Object or an ASHRAE 135 Analog Value Object where the Present_Value is the totalized pulse count. Pulse accumulators must accept contact closures, ignore transients less than 5 msec duration, protect against
transients of 50 VAC, and accept rates of at least 20 pulses per second.

2.3.2.6 ASHRAE 135 Objects for Hardware Inputs and Outputs

The requirements for use of ASHRAE 135 objects for hardware input and outputs includes devices where the hardware sensor or actuator is integral to the controller (e.g. a VAV box with integral damper actuator, a smart sensor, a VFD, etc.)

~~[2.3.2.7 Integrated H-O-A Switches~~

~~Where integrated H-O-A switches are provided on hardware outputs, controller must provide means of monitoring position or status of H-O-A switch. This feedback may be provided via the Niagara Framework or via any valid BACnet method, including the use of proprietary Objects, Properties, or Services.~~

~~]~~2.3.3 Local Display Panel (LDP)

The Local Display Panels (LDPs) must be DDC Hardware with a display and navigation buttons or a touch screen display, and must provide display and adjustment of ~~Niagara Framework points or~~ ASHRAE 135 Properties as indicated on the Points Schedule and as specified. LDPs must be either BTL Listed as a B-OD, B-OWS, B-AWS, or be an integral part of another piece of DDC Hardware listed as a B-BC. For LDPs listed as B-OWS or B-AWS, the hardware must be BTL listed and the product must come factory installed with all applications necessary for the device to function as an LDP.

The adjustment of values using display and navigation buttons must be password protected.

2.3.4 Expansion Modules and Tethered Hardware

A single piece of DDC Hardware may consist of a base unit and also:

a. An unlimited number of hardware expansion modules, where the individual hardware expansion modules are designed to directly connect, both mechanically and electrically, to the base unit hardware. The expansion modules must be commercially available as an optional add-on to the base unit.

b. A single piece of hardware connected (tethered) to a base unit by a single cable where the cable carries a proprietary protocol between the base unit and tethered hardware. The tethered hardware must not contain control logic and be commercially available as an optional add-on to the base unit as a single package.

   Note that this restriction on tethered hardware does not apply to sensors or actuators using standard binary or analog signals (not a communications protocol); sensors or actuators using standard binary or analog signals are not considered part of the DDC Hardware.

Hardware capable of being installed stand-alone, or without a separate base unit, is DDC Hardware and must not be used as expansion modules or tethered hardware.

2.3.5 Supervisory Control Requirements

2.3.5.1 Scheduling Hardware

DDC Hardware used for scheduling must meet the following requirements:

a. It must be BTL Listed as a B-BC and support the SCHED-E-B BIBB.

b. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Calendar and Schedule Objects, such that a front end BTL listed as a B-AWS may create or delete Calendar and Schedule Objects. It is also preferred but not required that devices supporting the DM-OCD-B BIBB accept any valid value for properties of Calendar and Schedule Objects. Note that there are additional requirements in the EXECUTION Part of this Section for Devices which do not support the DM-OCD-B BIBB as specified.

c. The Date_List property of all Calendar Objects must be writable.

d. The Present_Value Property of Schedule must support the following values: 1, 2, 3, 4.

2.3.5.2 Alarm Generation Hardware

~~Non-Niagara Framework~~ DDC Hardware used for alarm generation must meet the following requirements:

a. Device must support the AE-N-I-B BIBB.

b. The Recipient_List Property must be Writable for all Notification Class Objects used for alarm generation.

c. For all Objects implementing Intrinsic Alarming, the following Properties must be Writable:

(1) Time_Delay
(2) High_Limit
(3) Low_Limit
(4) Deadband
(5) Event_Enable
(6) If the issue date of this project specification is after 1 January 2016, Time_Delay_Normal must be writable.

~~d. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Notification Class Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Notification Class Objects.~~

d. For Event Enrollment Objects used for alarm generation, the following Properties must be Writable:

(1) Event_Parameters
(2) Event_Enable
(3) If the issue date of this project specification is after 1 January 2016, Time_Delay_Normal must be writable.

e. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Notification Class Objects and Event Enrollment Objects, such that a front end BTL listed as a B-AWS may create or delete Notification Class Objects and Event Enrollment Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Notification Class Objects and Event Enrollment Objects. Note that there are additional requirements in the EXECUTION Part of this Section for devices which do not support the DM-OCD-B BIBB as specified.

f. Devices provided to meet the the requirements indicated under "Support for Future Alarm Generation" in the EXECUTION part of this specification must support the AE-N-E-B BIBB.

2.3.5.3 Trending Hardware

DDC Hardware used for collecting trend data must meet the following requirements:

a. Device must support Trend Log or Trend Log Multiple Objects.

b. Device must support the T-VMT-I-B BIBB.

c. Devices provided to meet the EXECUTION requirement for support of Future Trending must support the T-VMT-E-B BIBB.

d. The following properties of all Trend Log or Trend Log Multiple Objects must be present and Writable:

Start_Time
Stop_Time
Log_DeviceObjectProperty
Log Interval Log interval must support an interval of at least 60 minutes duration.

e. Trend Log Objects must support using Intrinsic Reporting to send a BUFFER_FULL event.

f. The device must have a Notification Class Object for the BUFFER_FULL event. The Recipient_List Property must be Writable.

g. Devices must support values of at least 1,000 for Buffer_Size Properties.

h. It is preferred, but not required, that devices support the DM-OCD-B BIBB on all Trend Log Objects, such that a front end BTL listed as a A-AWS may create or delete Trend Log Objects. It is also preferred, but not required that devices supporting the DM-OCD-B BIBB accept any valid value as an initial value for properties of Trend Log Objects. Note that there are additional EXECUTION requirements for devices which do not support the DM-OCD-B BIBB as specified.

~~2.3.6 Niagara Framework Supervisory Gateway~~

~~Any device implementing the Niagara Framework is a Niagara Framework Supervisory Gateway and must meet these requirements. In addition to the general requirements for all DDC Hardware, Niagara Framework Supervisory Gateway Hardware must:~~

~~a. Be direct digital control hardware.~~

~~b. Have an unrestricted interoperability license and its Niagara Information and Conformance Statement (NICS) must allow open access as defined by Niagara NICS.~~

~~c. Manage communications between a field control network and the Niagara Framework Monitoring and Control Software, and between itself and other Niagara Framework Supervisory Gateways. Niagara Framework Supervisory Gateway Hardware must use Fox protocol for communication with other Niagara Framework Components, regardless of the manufacturer of the other components.~~

~~d. Be fully programmable using the Niagara Framework Engineering Tool and must support the following:~~

~~(1) Time synchronization, Calendar, and Scheduling using Niagara Scheduling Objects~~

~~(2) Alarm generation and routing using the Niagara Alarm Service~~

~~(3) Trending using the Niagara History Service and Niagara Trend Log Objects~~

~~(4) Integration of field control networks using the Niagara Framework Engineering Tool~~

~~(5) Configuration of integrated field control system using the Niagara Framework Engineering Tool when supported by the field control system~~

~~e. Meet the following minimum hardware requirements:~~

~~(1) [One][Two] 10/100/1000 Mbps Ethernet Port(s)~~

~~(2) One or more MS/TP ports.[~~

~~(3) Central Processing Unit of 600 Mhz or higher.][~~

~~(4) Embedded operating system.]~~

~~f. Provide access to field control network data and supervisory functions via web interface and support a minimum of 16 simultaneous users. Note: implementation of this capability may not be required on all projects.~~

~~g. Submit a backup of each Niagara Framework Supervisory Gateway as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The backup must be sufficient to restore a Niagara Framework Supervisory Gateway to the final as-built condition such that a new Niagara Framework Supervisory Gateway loaded with the backup is indistinguishable in functionality from the original.~~

~~2.4 NIAGARA FRAMEWORK ENGINEERING TOOL~~

~~The Niagara Framework Engineering Tool must be Niagara Workbench or an equivalent Niagara Framework engineering tool software must:~~

~~a. Have an unrestricted interoperability license and its Niagara Information and Conformance Statement (NICS) must allow open access as defined by Niagara NICS.~~

~~b. Be capable of performing network configuration for Niagara Framework Supervisory Gateways and Niagara Framework Monitoring and Control Software.~~

~~c. Be capable of programming and configuring of Niagara Framework Supervisory Gateways and Niagara Framework Monitoring and Control Software.~~

~~d. Be capable of discovery of Niagara Framework Supervisory Gateways and all points mapped into each Niagara Framework Supervisory Gateway and making these points accessible to Niagara Framework Monitoring and Control Software.~~

~~Monitoring and Control Software is specified in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION.~~

PART 3 EXECUTION

3.1 CONTROL SYSTEM INSTALLATION~~3.1.1 Niagara Framework Engineering Tool~~

~~[The project site currently has the [_____] Niagara Framework Engineering Tool. If this software is not adequate for programming the Niagara Framework Supervisory Gateways provided under this project, provide a Niagara Framework Engineering Tool.][Provide a Niagara Framework Engineering Tool.]~~

3.1.1 Building Control Network (BCN)

Install the Building Control Network (BCN) as a single BACnet Internetwork consisting of a single IP network as the BCN Backbone and zero or more BACnet MS/TP networks. Note that in some cases there may only be a single device on the BCN Backbone.

Except ~~for the IP Network and~~ as permitted for the non-BACnet side of Gateways, use exclusively ASHRAE 135 networks.

3.1.1.1 Building Control Network IP Backbone

Install IP Network Cabling in conduit. Install Ethernet Switches in lockable enclosures. Install the Building Control Network (BCN) IP Backbone such that it is available at the Facility Point of Connection (FPOC) location ~~[~~as indicated~~][_____]~~. When the FPOC location is a room number, provide sufficient additional media to ensure that the Building Control Network (BCN) IP Backbone can be extended to any location in the room.

Use UDP port 0xBAC0 for all BACnet traffic on the IP network. ~~(Note that in a Niagara Framework system there may not be BACnet traffic on the IP Network)~~

3.1.1.2 BACnet MS/TP Networks

When using MS/TP, provide MS/TP networks in accordance with ASHRAE 135 and in accordance with the ASHRAE 135 figure "Mixed Devices on 3-Conductor Cable with Shield" (Figure 9-1.4 in the 2020 version of ASHRAE 135). Ground the shield at the BACnet Router and at no other point. Ground the reference wire at the BACnet Router through a 100 ohm resistor and do not ground it at any other point. In addition:

a. Provide each segment in a doubly terminated bus topology in accordance with TIA-485.

b. Provide each segment with 2 sets of network bias resistors in accordance with ASHRAE 135, with one set of resistors at each end of the MS/TP network.

c. Use 3 wire (twisted pair and reference) with shield media for all MS/TP media installed inside. Use fiber optic isolation in accordance with ASHRAE 135 for all MS/TP media installed outside buildings, or between multiple buildings.

d. For 18 AWG cable, use segments with a maximum length of 1200 m. When using greater distances or different wire gauges comply with the electrical specifications of TIA-485.

e. For each controller that does not use the reference wire provide transient suppression at the network connection of the controller if the controller itself does not incorporate transient suppression.

f. Install no more than 32 devices on each MS/TP segment. Do not use MS/TP to MS/TP routers.

g. Connect each MS/TP network to the BCN backbone via ~~a Niagara Framework Supervisory Gateway configured as~~ a BACnet Router.

h. For BACnet Routers, configure the MS/TP MAC address to 0. Assign MAC Addresses to other devices consecutively beginning at 1, with no gaps.

i. Configure the Max_Master Property of all devices to be 31.

3.1.1.3 Building Control Network (BCN) Installation

Provide a building control network meeting the following requirements:

a. Install all DDC Hardware connected to the Building Control Network.

b. Where multiple pieces of DDC Hardware are used to execute one sequence, install all DDC Hardware executing that sequence on a single MS/TP network dedicated to that sequence.

c. Traffic between BACnet networks must be exclusively via BACnet routers.

~~d. Use the Fox protocol for all traffic both originating and terminating at Niagara Framework components. Use the Fox protocol for all traffic originating or terminating at a Niagara Framework UMCS (including traffic to or from a future UMCS). All other traffic, including traffic between ASHRAE 135 devices and traffic between Niagara Framework Supervisory Gateways and ASHRAE 135 devices must be in accordance with ASHRAE 135.~~

3.1.2 DDC Hardware

Install all DDC Hardware that connects to an IP network in lockable enclosure. Install other DDC Hardware that is not in suspended ceilings in ~~[~~ lockable ~~]~~enclosures. For all DDC hardware with a user interface, coordinate with site to determine proper passwords and configure passwords into device.

a. Except for zone sensors (thermostats), install all Tethered Hardware within 2 m of its base unit.

b. Install and configure all BTL-Listed devices in a manner consistent with their BTL Listing such that the device as provided still meets all requirements necessary for its BTL Listing.

c. Install and configure all BTL-Listed devices in a manner consistent with the BTL Device Implementation Guidelines such that the device as provided meets all those Guidelines.

3.1.2.1 Device Identifiers, Network Addresses, and IP addresses

a. Do not use any Device Identifier or Network Number already used by another BACnet system at the project site.~~ [Coordinate Device IDs and Network Numbers with the installation. The installation POC is [_____]] [Use Device IDs within the range of [_____] to [_____] and Network Numbers in the range of [_____] to [_____]].~~

~~b. [Use IP addresses within the range of [_____] to [_____]] [Coordinate device IP addresses with installation. The installation POC is [_____]].~~

3.1.2.2 ~~ASHRAE 135~~ Object Name Property and Object Description Property

Configure the Object_Names and Object_Descriptions properties of all ~~ASHRAE 135~~ Objects (including Device Objects) as indicated on the Points Schedule (Point Name and Point Description) and as specified. At a minimum:

a. Except for DDC Hardware controlling a single terminal unit, configure the Object_Name and Object_Description properties of all Objects (including Device Objects) as indicated on the Points Schedule and as specified.

b. In DDC Hardware controlling a single terminal unit, configure the Device Object_Name and Device Object_Description as indicated on the Points Schedule and as specified.

When Points Schedule entries exceed the length limitations in the device, notify ~~[_____]~~Contracting Officer and provide recommended alternatives for approval.

~~3.1.2.3 Niagara Framework Point Names and Descriptions~~

~~Configure the names and descriptions of all Points in Niagara Framework Supervisory Gateways as indicated on the Points Schedule and as specified.~~

~~3.1.2.4 Niagara Station IDs~~

~~Ensure that Niagara Station IDs of new Niagara Framework Supervisory Gateways are maintained as unique within UMCS front-end, including ensuring they do not conflict with any existing Niagara Station ID.~~

3.1.2.3 Hand-Off-Auto (H-O-A) Switches

Provide Hand-Off-Auto (H-O-A) switches ~~[for all DDC Hardware analog outputs and binary outputs used for control of systems other than terminal units,]~~as specified and as indicated on the Points Schedule. Provide H-O-A switches that are integral to the controller hardware, an external device co-located with (in the same enclosure as) the controller, integral to the controlled equipment, or an external device co-located with (in the same enclosure as) the controlled equipment.

a. For H-O-A switches integral to DDC Hardware, meet the requirements specified in paragraph DIRECT DIGITAL CONTROL (DDC) HARDWARE.

b. For external H-O-A switches used for binary outputs, provide for overriding the output open or closed.

~~c. For eternal H-O-A switches used for analog outputs, provide for overriding [to 0 percent or 100 percent][through the range of 0 percent to 100 percent].~~

3.1.2.4 Local Display Panels

Provide LDPs to display and override values of ~~points in a Niagara Framework Supervisory Gateway or~~ ASHRAE 135 Object Properties as indicated on the Points Schedule. Install LDPs displaying points for anything other than a terminal unit in the same room as the equipment. Install LDPs displaying points for only terminal units ~~[~~in a mechanical room central to the group of terminal units it serves~~][_____]~~. For LDPs using WriteProperty to commandable objects to implement an override, write values with priority 9.

3.1.2.5 MS/TP Slave Devices

Configure all MS/TP devices as Master devices. Do not configure any devices to act as slave devices.

3.1.2.6 Change of Value (COV) and Read Property

a. To the greatest extent possible, configure all devices to support the SubscribeCOV service (the DS-COV-B BIBB). At a minimum, all devices supporting the DS-RP-B BIBB, other than devices controlling only a single terminal unit, must be configured to support the DS-COV-B BIBB.

b. Whenever supported by the server side, configure client devices to use the DS-COV-A BIBB.

3.1.2.7 Engineering Units

~~[~~ Configure devices to use SI (Metric) units as follows:

a. Temperature in degrees C

b. Air ~~or natural gas~~ flows in Liters per Second (LPS)

c. Water flow in Liters per Second (LPS)

d. Steam flow in kilograms per second (kg/s)

e. Differential Air pressures in Pascals (Pa)

f. Water~~,~~ and steam ~~and natural gas~~ pressures in kiloPascals (kPa)

g. Enthalpy in kiloJoules per kilogram (kJ/kg)

h. Heating and Cooling Energy in kilowatt-hours (kWh)

i. Heating and Cooling load in kilowatts (kW)

j. Electrical Power: kilowatts (kW)

k. Electrical Energy: kilowatt-hours (kWh)~~][Configure devices to use English (Inch-Pound) engineering units as follows:~~

~~a. Temperature in degrees F~~

~~b. Air or natural gas flows in cubic feet per minute (CFM)~~

~~c. Water in gallons per minute (GPM)~~

~~d. Steam flow in pounds per hour (pph)~~

~~e. Differential Air pressures in inches of water column (IWC)~~

~~f. Water, steam, and natural gas pressures in PSI~~

~~g. Enthalpy in BTU/lb~~

~~h. Heating and cooling energy in MBTU (1MBTU = 1,000,000 BTU))~~

~~i. Cooling load in tons (1 ton = 12,000 BTU/hour)~~

~~j. Heating load in MBTU/hour (1MBTU = 1,000,000 BTU)~~

~~k. Electrical Power: kilowatts (kW)~~

~~l. Electrical Energy: kilowatt-hours (kWh)]~~

3.1.2.8 Occupancy Modes

Use the following correspondence between value and occupancy mode whenever an occupancy state or value is required:

a. OCCUPIED mode: a value of one
b. UNOCCUPIED mode: a value of two
c. WARM-UP/COOL-DOWN (PRE-OCCUPANCY) mode: a value of three

Note that elsewhere in this Section the Schedule Object is required to also support a value of four, which is reserved for future use. Also note that the behavior of a system in each of these occupancy modes is indicated in the sequence of operation for the system.

3.1.2.9 Use of BACnet Objects

~~Except as specifically indicated for Niagara Framework Objects,~~ Use only standard non-proprietary ASHRAE 135 Objects and services to accomplish the project scope of work as follows:

a. Use Analog Input or Analog Output Objects for all analog hardware I/O. Do not use Analog Value Object for analog hardware I/O)~~-~~.

b. Use Binary Input or Binary Output Objects for all binary hardware I/O. Do not use Binary Value Objects for binary hardware I/O.

c. Use Analog Value Objects for analog setpoints.

d. Use Accumulator Objects or Analog Value Objects for pulse inputs.

e. For occupancy modes, use Multistate Value Objects and the correspondence between value and occupancy mode specified in paragraph OCCUPANCY MODES.

f. Use Schedule Objects and Calendar Objects for all scheduling. Use Trend Log Objects or Trend Log Multiple Objects for all trending and Notification Class Objects for trend log upload. Use a combination of Event Enrollment Objects,Intrinsic Alarming, and Notification Class Objects for alarm generation.

~~g. Use a combination of Niagara Framework Alarm Extensions and Alarm Services, Intrinsic Alarming, and Notification Class Objects for alarm generation.~~

g. For all other points shown on the Points Schedule as requiring an ASHRAE 135 Object, use the Object type shown on the Points Schedule or, if no Object Type is shown, use a standard Object appropriate to the point.

~~3.1.2.9.1 Niagara Framework Objects~~

~~Points in the Niagara Framework Supervisory Gateway, even if used in a sequence or are shown on the Points Schedule, are not required to be exposed as BACnet Objects unless they are required to be available on the network by another device or sequence of operation (i.e. there is some other reason they are needed).~~

~~Use a Niagara Framework Supervisory Gateway as specified for all scheduling and trending. Use a Niagara Framework Supervisory Gateway as specified for all alarming except for intrinsic alarming.~~

3.1.2.10 Use of Standard BACnet Services

Except as noted in this paragraph, for all DDC Hardware ~~(including Niagara Frameworks Supervisory Gateways when communicating with non-Niagara Framework DDC Hardware)~~ use Standard BACnet Services as defined in this specification (which excludes some ASHRAE 135 services) exclusively for application control functionality and communication.

DDC Hardware that cannot meet this requirement may use non-standard services provided they can provide identical functionality using Standard BACnet Services when communicating with BACnet devices from a different vendor. When implementing non-standard services, document all non-standard services in the DDC Hardware Schedule as specified and as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

3.1.2.11 Device Application Configuration

a. For every property, setting or value shown on the Points Schedule or otherwise indicated as Configurable, provide a value that is retained through loss of power and can be changed via one or more of:

(1) BACnet services (including proprietary services)

(2) Hardware settings on the device

~~(3) The Niagara Framework~~

b. For every property, setting or value ~~in non-Niagara Framework Hardware~~ shown on the Points Schedule or otherwise indicated as Operator Configurable, provide a value that is retained through loss of power and can be changed via one or more of:

(1) A Writable Property of a standard BACnet Object

(2) A Property of a standard BACnet Object that is Writable when Out_Of_Service is TRUE and Out_Of_Service is Writable.

~~(3) Using some other method supported by a Niagara Framework Supervisory Gateway~~

~~c. Configure Niagara Framework Supervisory Gateways such that the property, setting or value is configurable from a Niagara Framework Front End.~~

~~d. For every property, setting or value in a Niagara Framework Supervisory Gateway which is shown on the Points Schedule or otherwise indicated as Operator Configurable, configure the value to be configurable from within the Niagara Framework such that it can be configured from a system graphic page at a Niagara Framework Front End.~~

~~3.1.2.12 Niagara Framework Engineering Tool~~

~~Use the Niagara Framework Engineering Tool to fully discover the field control system and make all field control system information available to the Niagara Framework Supervisory Gateway. Ensure that all points on the points schedule are available to the front end via the Fox protocol.~~

~~[3.1.2.13 Graphics and Web Pages~~

~~Configure Niagara Framework Supervisory Gateways to use web pages to provide a graphical user interface including System Displays[using the project site sample displays], including overrides, as indicated on the Points Schedule and as specified. Label all points on displays with [full English language descriptions][the point name as indicated on the Points Schedule][the point description as indicated on the Points Schedule][_____]. Configure user permissions for access to and executions of action using graphic pages. Coordinate user permissions with [the [Controls] [HVAC] [Electrical] shop supervisor][_____]. Configure the web server to use HTTPS based on the Transport Layer Security (TLS) protocol in accordance with RFC 5246 using a Government furnished certificate.~~

~~]~~3.1.3 Scheduling, Alarming, Trending, and Overrides

3.1.3.1 Scheduling

Configure schedules in BACnet Scheduling Objects to schedule systems as indicated on the Points Schedule and as specified using the indicated correspondence between value and occupancy mode. If no devices supports both the SCHED-E-B and DM-OCD-B BIBBS for Schedule Objects, provide ~~[~~5 ~~][_____]~~ blank Schedule Objects in DDC Hardware BTL listed as B-BCs and supporting the SCHED-E-B BIBB for later use by the site.
~~Configure schedules in Niagara Framework Supervisory Gateway using Niagara Schedule Objects as indicated on the Points Schedule and as specified. When the schedule is controlling occupancy modes in DDC Hardware other than a Niagara Framework Supervisory Gateway use the indicated correspondence between value and occupancy mode.~~

~~Provide a separate schedule for each AHU including it's associated Terminal Units and for each stand-alone Terminal Unit (those not dependent upon AHU service)[or group of stand-alone Terminal Units acting according to a common schedule[as indicated]].~~

~~3.1.3.2 Alarm Configuration~~

~~Configure alarm generation and management as indicated on the Points Schedule and as specified. Configure alarm generation in Niagara Framework Supervisory Gateways using Niagara Framework Alarm Extensions and Alarm Services or in other DDC Hardware (not Niagara Framework Supervisory Gateways) using ASHRAE 135 Intrinsic Alarming. Configure alarm management and routing for all alarms, including those generated via intrinsic alarming in other devices, in the Niagara Framework Supervisory Gateway such that the alarms are able to be accessed from the Niagara Framework Front End.~~

~~Where Intrinsic Alarming is used, configure intrinsic alarming as specified in paragraph "Configuration of ASHRAE 135 Intrinsic Alarm Generation". Configure a Niagara Framework Supervisory Gateway to provide a means to configure the intrinsic alarm parameters such that the Intrinsic Alarm is configurable from the front end via the Niagara Framework.~~

3.1.3.2 Configuration of ~~ASHRAE 135 Intrinsic~~ Alarm Generation

~~Intrinsic alarm generation must meet the following requirements:~~

~~Configure alarm generation as indicated on the Points Schedule and as specified using Intrinsic Alarming in accordance with ASHRAE 135 or Algorithmic Alarming in accordance with ASHRAE 135. Alarm generation must meet the following requirements:~~

a. Send alarm events as Alarms (not Events).

b. Use the ConfirmedNotification Service for alarm events.

c. For alarm generation, support two priority levels for alarms: critical and non-critical. Configure the Priority of Notification Class Objects to use Priority 112 for critical and 224 for non-critical alarms.

d. Number of Notification Class Objects for Alarm Generation:

(1) If the device implements non-critical alarms, or if any Object in the device supports Intrinsic Alarms, then provide a single Notification Class Object specifically for (shared by) all non-critical alarms.

(2) If the device implements critical alarms, provide a single Notification Class Object specifically for (shared by) all critical alarms.

(3) If the device implements both critical and non-critical alarms, provide both Notification Class Objects (one for critical, one for non-critical).

(4) If the device controls equipment other than a single terminal unit, provide both Notification Class Objects (one for critical, one for non-critical) even if no alarm generation is required at time of installation.

e. For all intrinsic alarms configure the Limit_Enable Property to set both HighLimitEnable and LowLimitEnable to TRUE. If the specified alarm conditions are for a single-sided alarm (only High_Limit used or only Low_Limit used) assign a value to the unused limit such that the unused alarm condition will not occur.

f. For all objects supporting intrinsic alarming, even if no alarm generation is required during installation, configure the following Properties as follows:

(1) Notification_Class to point to the non-Critical Notification Class Object in that device.

(2) Limit_Enable to enable both the HighLimitEnable and LowLimitEnable

(3) Notify_Type to Alarm

g. Use of alarm generation types:

(1) Only use algorithmic alarm generation when intrinsic alarm generation is not supported by the device or object, or when the specific alarm conditions cannot be implemented using intrinsic alarm generation.

(2) Only use remote alarm generation when the alarm cannot be

generated using intrinsic or local algorithmic alarm generation on the device containing the referenced property. If remote alarm generation is used, use the same DDC Hardware for all remote alarm generation within a single sequence.

h. ~~Configure the Recipient_List Property of the Notification Class Object to point to the Niagara Framework Supervisory Gateway managing the alarm.~~

3.1.3.3 Support for Future Alarm Generation

For every piece of DDC Hardware, support future alarm generation capabilities by supporting either intrinsic or additional algorithmic alarming. Provide one of the following:

a. Support intrinsic alarming for every Object used by the application in that device.

b. Support additional Event_Enrollment Objects. For DDC hardware controlling a single terminal unit, support at least one additional object. Otherwise, support at least ~~[~~4~~][_____]~~ additional Objects. Support additional Event_Enrollment Objects via one of the following:

(1) Provide unused Event_Enrollment Objects on that device.

(2) Support the DM-OCD-B BIBB and the creation of sufficient Event_Enrollment Objects on that device.

(3) Provide one or more devices in the IP network that support the AE-N-E-B BIBB and have unused Event_Enrollment Objects.

(4) Provide one or more devices on the IP network that support the AE-N-E-B BIBB, the DM-OCD-B BIBB, and the creation of sufficient Event_Enrollment Objects.

The total number of Event_Enrollment Objects required by the project is the sum of the individual device requirements, and the distribution of Event_Enrollment Objects among devices is not further restricted. (Note this allows a single device to contain many Event_Enrollment Objects satisfying the requirements for multiple devices.)

3.1.3.4 Trend Log Configuration

a. Configure trends in Trend Log or Trend Log Multiple Objects as indicated on the Points Schedule and as specified.

b. Configure all trend logs (including any provided to support future trends) to save data on regular intervals using the BUFFER_FULL event to request trend upload from the front end.

c. Configure Trend Log Objects with a minimum Buffer_Size property value of 1,000 and Trend Log Multiple Objects with a minimum Buffer_Size property value of 1,000 per point trended (for example, a Trend Log Multiple Object used to trend 3 points must have a Buffer_Size Property value of at least 3,000).

d. Configure a Notification Class Object in devices doing trending (including devices supporting future trends) to handle the BUFFER_FULL event.

e. When possible, trend each point using an Object in the device containing the point. When it is necessary to trend using a an Object in another device, all trends not on the same Device as the Object being trended must be on a singe device (i.e. all Trend Log and Trend Log Multiple Objects used for remote trending within a sequence must be on the same device).

f. For each trend log, including any trend logs provided to support future trending, configure the following properties as specified:

(1) Logging_Type: Set to Polling

(2) Stop_When_Full: Set to Wrap Around

(3) Buffer_Size: Set to 400 or greater.

(4) Notification_Threshold: Set to 90 percent of full

(5) Notification_Class: Set to the Notification Class Object in that device

(6) Event_Enable: Set to TRUE

(7) Log_Interval: Set to 15 minutes.

g. Future Trending support. Provide support for future trending:

(1) Provide one or more devices on the Building Control Network Backbone IP network which support both the T-VMT-E-B and DM-OCD-B BIBBs for Trend Log Objects. Provide sufficient devices to support the creation of at least ~~[[_____] additional Trend Log Objects][~~one additional Trend Log Object for every terminal unit plus 4 additional Trend Log Objects for every non-terminal unit~~]~~.

(2) Provide ~~[[_____] additional Trend Log Objects][~~one additional Trend Log Object for every terminal unit plus 4 additional Trend Log Objects for every non-terminal unit~~]~~ in one or more devices on the Building Control Network Backbone IP network that support the T-VMT-E-B BIBB for later use by the site.

(3) A combination of these two methods is permitted provided the total required number of Trend Log Objects is met.

~~3.1.3.5 Trending~~

~~Perform all trending using a Niagara Framework Supervisory Gateway using Niagara Framework History Extensions and Niagara Framework History Service exclusively.~~

3.1.3.5 Overrides

Provide an override for each point shown on the Points Schedule as requiring an override. ~~Use the Niagara Framework for all overrides to points in Niagara Framework Supervisory Gateways. For overrides to other points, provide an override to a point in a Niagara Framework Supervisory Gateway via the Niagara Framework where the Niagara Framework Supervisory Gateway overrides the other point as specified.~~

Unless otherwise approved, provide Commandable Objects to support all Overrides~~ in non-Niagara Framework Supervisory Gateway DDC Hardware~~. With specific approval from the Contracting Officer, Overrides for points which are not hardware outputs and which are in DDC hardware controlling a single terminal unit may support overrides via an additional Object provided for the override. No other means of implementing Overrides may be used.

a. Where Commandable Objects are used, ensure that WriteProperty service requests with a Priority of 10 or less take precedence over the SEQUENCE VALUE and that WriteProperty service request with a priority of 11 or more have a lower precedence than the SEQUENCE VALUE.

b. For devices implementing overrides via additional Objects, provide Objects which are NOT Written to as part of the normal Sequence of Operations and are Writable when Out_Of_Service is TRUE and Out_Of_Service is Writable. Use this point as an Override of the normal value when Out_Of_Service is TRUE and the normal value otherwise. Note these Objects may be modified as part of the sequence via local processes, but must not be modified by local processes when Out_Of_Service is TRUE.

3.1.4 BACnet Gateways

The requirements in this paragraph do not themselves permit the installation of hardware not meeting the other requirements of this section. Except for proprietary systems specifically indicated in Section 23 09 00, all control hardware installed under this project must meet the requirements of this specification, including the control hardware providing the network interface for a package unit or split system specified under another section.~~ ~~ Only use gateways to connect to pre-existing control devices, and to proprietary systems specifically permitted by Section 23 09 00.

3.1.4.1 General Gateway Requirements

Provide BACnet Gateways to connect non-BACnet control hardware in accordance with the following:

a. Configure gateways to map writable data points in the controlled equipment to Writable Properties of Standard Objects~~, or to Niagara Framework points,~~ as indicated in the Points Schedule and as specified.

b. Configure gateway to map readable data points in the controlled equipment to Readable Properties of Standard Objects~~, or to Niagara Framework points,~~ as indicated in the Points Schedule and as specified.

c. Configure gateway to support the DS-COV-B BIBB for all points mapped to BACnet Objects.

d. Do not use non-BACnet control hardware for controlling built-up units or any other equipment that was not furnished with factory-installed controls. ~~(Note: A Niagara Framework Supervisory Gateway is BACnet control hardware.)~~

e. Do not use non-BACnet control hardware for system scheduling functions.

f. Each gateway must communicate with and perform protocol translation for non-BACnet control hardware controlling one and only one package unit or a single non-BACnet system specifically permitted by Section 23 09 00.

g. Connect one network port on the gateway to the Building Control Backbone IP Network or to a BACnet MS/TP network and the other port to the single piece of controlled equipment or the non-BACnet system specifically permitted by Section 23 09 00.~~.~~

h. For gateways to existing package units or simple split systems, non-BACnet network wiring connecting the gateway to the package unit must not exceed 3 meters in length and must connect to exactly two devices: the controlled equipment (packaged unit) or split system interface and the gateway.

-- End of Section --

SECTION 23 23 00

REFRIGERANT PIPING
10/07

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1180	(2014) Hexagon Head Bolts and Hexagon Screws
JIS B 2220	(2012) Steel Pipe Flanges
JIS B 2301	(2013) Screwed Type Malleable Cast Iron Pipe Fittings
JIS B 2312	(2015) Steel Butt-Welding Pipe Fittings
JIS B 2316	(2017) Steel Socket-Welding Pipe Fittings
JIS B 2404	(2018) Dimensions of Gaskets for Use with Pipe Flanges
JIS B 8225	(2012) Safety Valves-Measuring Methods for Coefficient of Discharge
JIS B 8471	(2004) Water Pipe Line-Solenoid Valves
JIS B 8605	(2002) Stop Valves for Refrigerants
JIS B 8619	(2018) Thermostatic Refrigerant Expansion Valves-Methods of Testing for Performance
JIS G 3302	(2022) Hot~~–~~-Dip Zinc~~–~~-Coated Steel Sheet and Strip
JIS G 3454	(2019) Carbon Steel Pipes for Pressure Service (Amendment 1)
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS K 7137-2	(2001)Plastics-Polytetrafluoroethylene (PTFE) Semi-Finished Products-

	Part 2 : Preparation of Test Specimens and Determination of Properties
JIS Z 2371	(2015) Methods of Salt Spray Testing
JIS Z 3197	~~(2012) Test Methods for Soldering Fluxes~~ (2021) Test Methods for Soldering Fluxes
JIS Z 3202	(2007) Copper and Copper Alloy Gas Welding Rods
JIS Z 3284-1	(2014) Solder Paste-Part 1: Kinds and Quality Classification
JIS Z 3801	(2018) Standard Qualification Test and Acceptance Requirements for Manual Welding Technique
JIS Z 3821	~~(2018) Standard Qualification Test and Acceptance Requirements for Welding Technique of Stainless Steel~~(2018) Standard Qualification Test and Acceptance Requirements for Welding Technique of Stainless Steel

THE JAPAN WELDING ENGINEERING SOCIETY (JWES)

JWES	Japan Welding Engineering Society

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[~~Contractor Quality Control approval.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Refrigerant Piping System; G~~[, [_____]]~~

SD-06 Test Reports

Refrigerant Piping Tests

SD-10 Operation and Maintenance Data

Maintenance

Operation and Maintenance Manuals

1.3 QUALITY ASSURANCE

1.3.1 Qualifications

Submit ~~[_____]~~4 copies of qualified procedures, and list of names and identification symbols of qualified welders and welding operators, prior to non-factory welding operations.~~[~~ Piping shall be welded in accordance with the qualified procedures using performance qualified welders and welding operators. Procedures and welders shall be qualified in accordance with JIS Z 3801. Notify the Contracting Officer 24 hours in advance of tests to be performed at the work site, if practical. The welder or welding operator shall apply the personally assigned symbol near each weld made, as a permanent record. Structural members shall be welded in accordance with Section ~~[05 12 00 STRUCTURAL STEEL]~~05 50 13 MISCELLANEOUS METAL FABRICATIONS.~~] [Welding and nondestructive testing procedures are specified in Section [40 05 13.96 WELDING PROCESS PIPING] .]~~

1.3.2 Contract Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

1.4 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation is the Contractor's responsibility. Replace any materials found to be damaged at the Contractor's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

1.5 MAINTENANCE

1.5.1 General

Submit Data Package 2 plus operation and maintenance data complying with the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

1.5.2 Extra Materials

Submit spare parts data for each different item of equipment specified, after approval of detail drawings and not later than ~~[_____]~~2 months prior to the date of beneficial occupancy. The data shall include a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

PART 2 PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship and that have been in satisfactory commercial or industrial use for 2 years prior to bid opening.

b. The 2 year use shall include applications of equipment and materials under similar circumstances and of similar size. The 2 years experience shall be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2 year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown.

c. Products shall be supported by a service organization location in Japan. System components shall be environmentally suitable for the indicated locations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. The service organizations shall be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

d. Exposed equipment moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel shall be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired. Welding and cutting safety requirements shall be in accordance with MLIT-M and JIS Z 3801.

e. Manufacturer's standard catalog data, at least ~~[~~5 weeks~~] [______]~~ prior to the purchase or installation of a particular component, highlighted to show material, size, options, performance charts and curves, etc. in adequate detail to demonstrate compliance with contract requirements. Include in the data manufacturer's recommended installation instructions and procedures. Provide data for the following components as a minimum:

(1) Piping and Fittings
(2) Valves
(3) Piping Accessories
(4) Pipe Hangers, Inserts, and Supports

2.2 ELECTRICAL WORK

Electrical equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Field wiring must be in accordance with manufacturer's instructions.

2.3 REFRIGERANT PIPING SYSTEM

Refrigerant piping, valves, fittings, and accessories shall be in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese Standard., except as specified herein. Refrigerant piping, valves, fittings, and accessories shall be compatible with the fluids used and capable of withstanding the pressures and temperatures of the service. Refrigerant piping, valves, and accessories used for refrigerant service shall be cleaned, dehydrated, and sealed (capped or plugged) prior to shipment from the manufacturer's plant. Submit drawings, at least ~~[~~5~~] [______]~~ weeks prior to beginning construction, provided in adequate detail to demonstrate compliance with contract requirements. Drawings shall consist of:

a. Piping layouts which identify all valves and fittings.

b. Plans and elevations which identify clearances required for maintenance and operation.

2.4 PIPE, FITTINGS AND END CONNECTIONS (JOINTS)

2.4.1 Steel Pipe

Steel pipe for refrigerant service shall conform to JIS G 3454 & JIS B 2301, Schedule 40.

2.4.1.1 Welded Fittings and Connections

Butt-welded fittings shall conform to JIS B 2312. Socket-welded fittings shall conform to JIS B 2316. Welded fittings shall be identified with the appropriate grade and marking symbol. Welded valves and pipe connections (both butt-welds and socket-welds types) shall conform to JIS B 2312 or JIS B 2316.

2.4.1.2 Threaded Fittings and Connections

Threaded fitting shall conform to JIS B 2301. Threaded valves and pipe connections shall conform to JIS B 2301.

2.4.1.3 Flanged Fittings and Connections

Flanges shall conform to JIS B 2220. Gaskets shall be non asbestos compressed material in accordance with JIS B 2404, 1.59 mm thickness, full face or self-centering flat ring type. This gaskets shall contain aramid fibers bonded with styrene butadeine rubber (SBR) or nitrile butadeine rubber (NBR). Bolts, nuts, and bolt patterns shall conform to JIS B 2220. Bolts shall be high or intermediate strength material conforming to JIS B 1180.

2.4.2 Copper Tubing

Copper tubing shall conform to JIS H 3300 and JIS H 3401, annealed or hard drawn as required. Copper tubing shall be soft annealed where bending is required and hard drawn where no bending is required. Soft annealed copper tubing shall not be used in sizes larger than 35 mm. Joints shall be brazed except that joints on lines 22 mm and smaller may be flared. Cast copper alloy fittings for flared copper tube shall conform to JIS H 3401. Wrought copper and bronze solder-joint pressure fittings shall conform to JIS H 3401. Joints and fittings for brazed joint shall be wrought-copper or forged-brass sweat fittings. Cast sweat-type joints and fittings shall not be allowed for brazed joints. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment.

2.4.3 Solder

Solder shall conform to JIS Z 3284-1, tin-antimony alloy for service pressures up to 1034 kPa. Solder flux shall be liquid or paste form, non-corrosive and conform to JIS Z 3197.

2.4.4 Brazing Filler Metal

Filler metal shall conform to JIS Z 3202.

2.5 VALVES

Valves shall be designed, manufactured, and tested specifically for refrigerant service. Valve bodies shall be of brass, bronze, steel, or ductile iron construction. Valves 25 mm and smaller shall have brazed or socket welded connections. Valves larger than 25 mm shall have ~~[~~ tongue-and-groove flanged~~]~~ or ~~[~~butt welded~~]~~ end connections. Threaded end connections shall not be used, except in pilot pressure or gauge lines where maintenance disassembly is required and welded flanges cannot be used. Internal parts shall be removable for inspection or replacement without applying heat or breaking pipe connections. Valve stems exposed to the atmosphere shall be stainless steel or corrosion resistant metal plated carbon steel. Direction of flow shall be legibly and permanently indicated on the valve body. Control valve inlets shall be fitted with integral or adapted strainer or filter where recommended or required by the manufacturer. Purge, charge and receiver valves shall be of manufacturer's standard configuration.

2.5.1 Refrigerant Stop Valves

Valve shall be the globe or full-port ball type with a back-seating stem especially packed for refrigerant service. Valve packing shall be replaceable under line pressure. Valve shall be provided with a ~~[~~handwheel ~~]~~ ~~[~~or~~]~~ ~~[~~wrench~~]~~ operator and a seal cap. Valve shall be the straight or angle pattern design as indicated.

2.5.2 Check Valves

Valve shall be the swing or lift type as required to provide positive shutoff at the differential pressure indicated. Valve shall be provide with resilient seat.

2.5.3 Liquid Solenoid Valves

Valves shall comply with JIS B 8471 and be suitable for continuous duty with applied voltages 15 percent under and 5 percent over nominal rated voltage at maximum and minimum encountered pressure and temperature service conditions. Valves shall be direct-acting or pilot-operating type, packless, except that packed stem, seal capped, manual lifting provisions shall be furnished. Solenoid coils shall be moisture-proof, UL approved, totally encapsulated or encapsulated and metal jacketed as required. Valves shall have safe working pressure of 2760 kPa and a maximum operating pressure differential of at least 1375 kPa at 85 percent rated voltage. Valves shall have an operating pressure differential suitable for the refrigerant used.

2.5.4 Expansion Valves

Valve shall conform to JIS B 8619. Valve shall be the diaphragm and spring-loaded type with internal or external equalizers, and bulb and capillary tubing. Valve shall be provided with an external superheat adjustment along with a seal cap. Internal equalizers may be utilized where flowing refrigerant pressure drop between outlet of the valve and inlet to the evaporator coil is negligible and pressure drop across the evaporator is less than the pressure difference corresponding to 1 degree C of saturated suction temperature at evaporator conditions. Bulb charge shall be determined by the manufacturer for the application and such that liquid will remain in the bulb at all operating conditions. Gas limited liquid charged valves and other valve devices for limiting evaporator pressure shall not be used without a distributor or discharge tube or effective means to prevent loss of control when bulb becomes warmer than valve body. Pilot-operated valves shall have a characterized plug to provide required modulating control. A de-energized solenoid valve may be used in the pilot line to close the main valve in lieu of a solenoid valve in the main liquid line. An isolatable pressure gauge shall be provided in the pilot line, at the main valve. Automatic pressure reducing or constant pressure regulating expansion valves may be used only where indicted or for constant evaporator loads.

2.5.5 Safety Relief Valves

Valve shall be the two-way type, unless indicated otherwise. Valve shall conform to JIS B 8225. Valve capacity shall be certified by the National Board of Boiler and Pressure Vessel Inspectors or Japan Boiler Association. Valve shall be of an automatically reseating design after activation.

2.5.6 Evaporator Pressure Regulators, Direct-Acting

Valve shall include a diaphragm/spring assembly, external pressure adjustment with seal cap, and pressure gauge port. Valve shall maintain a constant inlet pressure by balancing inlet pressure on diaphragm against an adjustable spring load. Pressure drop at system design load shall not exceed the pressure difference corresponding to a 1 degree C change in saturated refrigerant temperature at evaporator operating suction temperature. Spring shall be selected for indicated maximum allowable suction pressure range.

2.5.7 Refrigerant Access Valves

Refrigerant access valves and hose connections shall be in accordance with JIS B 8605.

2.6 PIPING ACCESSORIES

2.6.1 Filter Driers

Sizes 15 mm and larger shall be the full flow, replaceable core type. Sizes 13 mm and smaller shall be the sealed type. Cores shall be of suitable desiccant that will not plug, cake, dust, channel, or break down, and shall remove water, acid, and foreign material from the refrigerant. Filter driers shall be constructed so that none of the desiccant will pass into the refrigerant lines. Minimum bursting pressure shall be 10.3 MPa.

2.6.2 Sight Glass and Liquid Level Indicator

2.6.2.1 Assembly and Components

Assembly shall be pressure- and temperature-rated and constructed of materials suitable for the service. Glass shall be borosilicate type. Ferrous components subject to condensation shall be electro-galvanized.

2.6.2.2 Gauge Glass

Gauge glass shall include top and bottom isolation valves fitted with automatic checks, and packing followers; red-line or green-line gauge glass; elastomer or polymer packing to suit the service; and gauge glass guard.

2.6.2.3 Bull's-Eye and Inline Sight Glass Reflex Lens

Bull's-eye and inline sight glass reflex lens shall be provided for dead-end liquid service. For pipe line mounting, two plain lenses in one body suitable for backlighted viewing shall be provided.

2.6.2.4 Moisture Indicator

Indicator shall be a self-reversible action, moisture reactive, color changing media. Indicator shall be furnished with full-color-printing tag containing color, moisture and temperature criteria. Unless otherwise indicated, the moisture indicator shall be an integral part of each corresponding sight glass.

2.6.3 Vibration Dampeners

Dampeners shall be of the all-metallic bellows and woven-wire type.

2.6.4 Flexible Pipe Connectors

Connector shall be a composite of interior corrugated phosphor bronze or stainless steel, as required for fluid service, with exterior reinforcement of bronze, stainless steel or monel wire braid. Assembly shall be constructed with a safety factor of not less than 4 at 150 degrees C. Unless otherwise indicated, the length of a flexible connector shall be as recommended by the manufacturer for the service intended.

2.6.5 Strainers

Strainers used in refrigerant service shall have brass or cast iron body, Y-or angle-pattern, cleanable, not less than 60-mesh noncorroding screen of an area to provide net free area not less than ten times the pipe diameter with pressure rating compatible with the refrigerant service. Screens shall be stainless steel or monel and reinforced spring-loaded where necessary for bypass-proof construction.

2.6.6 Pressure and Vacuum Gauges

Gauges shall conform to MLIT-M and shall be provided with throttling type needle valve or a pulsation dampener and shut-off valve. Gauge shall be a minimum of 85 mm in diameter with a range from 0 kPa to approximately 1.5 times the maximum system working pressure. Each gauge range shall be selected so that at normal operating pressure, the needle is within the middle-third of the range.

2.6.7 Temperature Gauges

Temperature gauges shall be the industrial duty type and be provided for the required temperature range. Gauges shall have Celsius scale in 1 degree graduations scale (black numbers) on a white face. The pointer shall be adjustable. Rigid stem type temperature gauges shall be provided in thermal wells located within 1.5 m of the finished floor. Universal adjustable angle type or remote element type temperature gauges shall be provided in thermal wells located 1.5 to 2.1 m above the finished floor. Remote element type temperature gauges shall be provided in thermal wells located 2.1 m above the finished floor.

2.6.7.1 Stem Cased-Glass

Stem cased-glass case shall be polished stainless steel or cast aluminum, 229 mm long, with clear acrylic lens, and non-mercury filled glass tube with indicating-fluid column.

2.6.7.2 Bimetallic Dial

Bimetallic dial type case shall be not less than 89 mm, stainless steel, and shall be hermetically sealed with clear acrylic lens. Bimetallic element shall be silicone dampened and unit fitted with external calibrator adjustment. Accuracy shall be one percent of dial range.

2.6.7.3 Liquid-, Solid-, and Vapor-Filled Dial

Liquid-, solid-, and vapor-filled dial type cases shall be not less than 89 mm, stainless steel or cast aluminum with clear acrylic lens. Fill shall be nonmercury, suitable for encountered cross-ambients, and connecting capillary tubing shall be double-braided bronze.

2.6.7.4 Thermal Well

Thermal well shall be identical size, 13 or 19 mm NPT connection, brass or stainless steel. Where test wells are indicated, provide captive plug-fitted type 13 mm NPT connection suitable for use with either engraved stem or standard separable socket thermometer or thermostat. Mercury shall not be used in thermometers. Extended neck thermal wells shall be of sufficient length to clear insulation thickness by 25 mm.

2.6.8 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, guides, and supports shall conform to MLIT-M.

2.6.9 Escutcheons

Escutcheons shall be chromium-plated iron or chromium-plated brass, either one piece or split pattern, held in place by internal spring tension or set screws.

2.7 FABRICATION

2.7.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, shall be factory finished with the manufacturer's standard finish, except that items located outside of buildings shall have weather resistant finishes that will withstand ~~[125] [~~500~~]~~ hours exposure to the salt spray test specified in JIS Z 2371 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen shall show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used shall be coated with a zinc-rich coating conforming to JIS H 8641.

2.7.2 Factory Applied Insulation

~~[~~Refrigerant suction lines between the cooler and each compressor ~~[and cold gas inlet connections to gas cooled motors]] [Refrigerant pumps and exposed chilled water lines on absorption chillers]~~ shall be insulated with not less than 19 mm thick unicellular plastic foam. Factory insulated items installed outdoors are not required to be fire-rated.~~ .~~

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, perform a verification of dimensions in the field. Submit a letter, at least ~~[~~2~~] [______]~~ weeks prior to beginning construction, including the date the site was visited, conformation of existing conditions, and any discrepancies found before performing any work.

3.2 INSTALLATION

Pipe and fitting installation shall conform to the requirements of MLIT-M. Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows, doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation are not permitted without written approval. Cut pipe or tubing square, removeremoved by reaming, and permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

3.2.1 Directional Changes

Make changes in direction with fittings, except that bending of pipe 100 mm and smalleris permitted, provided a pipe bender is used and wide weep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees is not permitted. The centerline radius of bends shall not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted.

3.2.2 Functional Requirements

Piping shall be installed 4 mm/m of pipe in the direction of flow to ensure adequate oil drainage. Open ends of refrigerant lines or equipment shall be properly capped or plugged during installation to keep moisture, dirt, or other foreign material out of the system. Piping shall remain capped until installation. Equipment piping shall be in accordance with the equipment manufacturer's recommendations and the contract drawings. Equipment and piping arrangements shall fit into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance.

3.2.3 Fittings and End Connections

3.2.3.1 Threaded Connections

Make threaded connections with tapered threads and make tight with tape complying with JIS K 7137-2 thread-joint compound applied to the male threads only.

3.2.3.2 Brazed Connections

Perform brazing in accordance with MLIT-M, except as modified herein. During brazing, fill the pipe and fittings with a pressure regulated inert gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, clean both the outside of the tube and the inside of the fitting with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux. Remove surplus brazing material at all joints. Make steel tubing joints in accordance with the manufacturer's recommendations. Paint joints in steel tubing with the same material as the baked-on coating within 8 hours after joints are made. Protect tubing against oxidation during brazing by continuous purging of the inside of the piping using nitrogen. Support piping prior to brazing and do not spring or force.

3.2.3.3 Welded Connections

Welded joints in steel refrigerant piping shall be fusion-welded. Branch connections shall be made with welding tees or forged welding branch outlets. Pipe shall be thoroughly cleaned of all scale and foreign matter before the piping is assembled. During welding the pipe and fittings shall be filled with an inert gas, such as nitrogen, to prevent the formation of scale. Beveling, alignment, heat treatment, and inspection of weld shall conform to JIS Z 3821. Weld defects shall be removed and rewelded at no additional cost to the Government. Electrodes shall be stored and dried in accordance with JWES-Welding Technology Education Sheet or as recommended by the manufacturer. Electrodes that have been wetted or that have lost any of their coating shall not be used.

3.2.3.4 Flared Connections

When flared connections are used, a suitable lubricant shall be used between the back of the flare and the nut in order to avoid tearing the flare while tightening the nut.

3.2.3.5 Flanged Connections

When steel refrigerant piping is used, union or flange joints shall be provided in each line immediately preceding the connection to each piece of equipment requiring maintenance, such as compressors, coils, chillers, control valves, and other similar items. Flanged joints shall be assembled square end tight with matched flanges, gaskets, and bolts. Gaskets shall be suitable for use with the refrigerants to be handled.

3.2.4 Valves

3.2.4.1 General

Refrigerant stop valves shall be installed on each side of each piece of equipment such as compressors condensers, evaporators, receivers, and other similar items in multiple-unit installation, to provide partial system isolation as required for maintenance or repair. Stop valves shall be installed with stems horizontal unless otherwise indicated. Ball valves shall be installed with stems positioned to facilitate operation and maintenance. Isolating valves for pressure gauges and switches shall be external to thermal insulation. Safety switches shall not be fitted with isolation valves. Filter dryers having access ports may be considered a point of isolation. Purge valves shall be provided at all points of systems where accumulated noncondensable gases would prevent proper system operation. Valves shall be furnished to match line size, unless otherwise indicated or approved.

3.2.4.2 Expansion Valves

Expansion valves shall be installed with the thermostatic expansion valve bulb located on top of the suction line when the suction line is less than 54 mm in diameter and at the 4 o'clock or 8 o'clock position on lines larger than 54 mm. The bulb shall be securely fastened with two clamps. The bulb shall be insulated. The bulb shall installed in a horizontal portion of the suction line, if possible, with the pigtail on the bottom. If the bulb must be installed in a vertical line, the bulb tubing shall be facing up.

3.2.4.3 Valve Identification

Each system valve, including those which are part of a factory assembly, shall be tagged. Tags shall be in alphanumeric sequence, progressing in direction of fluid flow. Tags shall be embossed, engraved, or stamped plastic or nonferrous metal of various shapes, sized approximately 40 mm diameter, substantially attached to a component or immediately adjacent thereto. Tags shall be attached with nonferrous, heavy duty, bead or link chain, 14 gauge annealed wire, nylon cable bands or as approved. Tag numbers shall be referenced in Operation and Maintenance Manuals and system diagrams.

3.2.5 Vibration Dampers

Vibration damper shall be provided in the suction and discharge lines on spring mounted compressors. Vibration dampers shall be installed parallel with the shaft of the compressor and shall be anchored firmly at the upstream end on the suction line and the downstream end in the discharge line.

3.2.6 Strainers

Strainers shall be provided immediately ahead of solenoid valves and expansion devices. Strainers may be an integral part of an expansion valve.

3.2.7 Filter Dryer

A liquid line filter dryer shall be provided on each refrigerant circuit located such that all liquid refrigerant passes through a filter dryer. Dryers shall be sized in accordance with the manufacturer's recommendations for the system in which it is installed. Dryers shall be installed such that it can be isolated from the system, the isolated portion of the system evacuated, and the filter dryer replaced. Dryers shall be installed in the horizontal position except replaceable core filter dryers may be installed in the vertical position with the access flange on the bottom.

3.2.8 Sight Glass

A moisture indicating sight glass shall be installed in all refrigerant circuits down stream of all filter dryers and where indicated. Site glasses shall be full line size.

3.2.9 Discharge Line Oil Separator

Discharge line oil separator shall be provided in the discharge line from each compressor. Oil return line shall be connected to the compressor as recommended by the compressor manufacturer.

3.2.10 Accumulator

Accumulators shall be provided in the suction line to each compressor.

3.2.11 Flexible Pipe Connectors

Connectors shall be installed perpendicular to line of motion being isolated. Piping for equipment with bidirectional motion shall be fitted with two flexible connectors, in perpendicular planes. Reinforced elastomer flexible connectors shall be installed in accordance with manufacturer's instructions. Piping guides and restraints related to flexible connectors shall be provided as required.

3.2.12 Temperature Gauges

Temperature gauges shall be located specifically on, but not limited to the following: ~~[~~the sensing element of each automatic temperature control device where a thermometer is not an integral part thereof~~]~~ ~~[~~the liquid line leaving a receiver~~]~~ ~~[~~and~~]~~ ~~[~~the suction line at each evaporator or liquid cooler~~]~~. Thermal wells for insertion thermometers and thermostats shall extend beyond thermal insulation surface not less than 25 mm.

3.2.13 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports shall conform to MLIT-M, except as modified herein. Hangers used to support piping 50 mm and larger shall be fabricated to permit adequate adjustment after erection while still supporting the load. Piping subjected to vertical movement, when operating temperatures exceed ambient temperatures, shall be supported by variable spring hangers and supports or by constant support hangers.

3.2.13.1 Horizontal Pipe Supports

Horizontal pipe supports shall be spaced as specified in MLIT-M and a support shall be installed not over 300 mm from the pipe fitting joint at each change in direction of the piping. Pipe supports shall be spaced not over 1.5 m apart at valves. ~~[~~Pipe hanger loads suspended from steel joist with hanger loads between panel points in excess of 23 kg shall have the excess hanger loads suspended from panel points.~~]~~

3.2.13.2 Vertical Pipe Supports

Vertical pipe shall be supported at each floor, except at slab-on-grade, and at intervals of not more than 4.5 m not more than 2.4 m from end of risers, and at vent terminations.

3.2.13.3 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, a clip or clamp shall be used where each pipe crosses the base support member. Spacing of the base support members shall not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.

3.2.13.4 Seismic Requirements

Piping and attached valves shall be supported and braced to resist seismic loads as specified under Building equipment seismic design & construction guidelines 2014 Edition.

3.2.13.5 Structural Attachments

Attachment to building structure concrete and masonry shall be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors shall be applied with a safety factor not less than 5. Supports shall not be attached to metal decking. Masonry anchors for overhead applications shall be constructed of ferrous materials only. Structural steel brackets required to support piping, headers, and equipment, but not shown, shall be provided under this section.

3.2.14 Pipe Alignment Guides

Pipe alignment guides shall be provided where indicated for expansion loops, offsets, and bends and as recommended by the manufacturer for expansion joints, not to exceed 1.5 m on each side of each expansion joint, and in lines 100 mm or smaller not more than 600 mm on each side of the joint.

3.2.15 Pipe Anchors

Anchors shall be provided wherever necessary or indicated to localize expansion or to prevent undue strain on piping. Anchors shall consist of heavy steel collars with lugs and bolts for clamping and attaching anchor braces, unless otherwise indicated. Anchor braces shall be installed in the most effective manner to secure the desired results using turnbuckles where required. Supports, anchors, or stays shall not be attached where they will injure the structure or adjacent construction during installation or by the weight of expansion of the pipeline. Where pipe and conduit penetrations of vapor barrier sealed surfaces occur, these items shall be anchored immediately adjacent to each penetrated surface, to provide essentially zero movement within penetration seal. Detailed drawings of pipe anchors shall be submitted for approval before installation.

3.2.16 Building Surface Penetrations

Sleeves shall not be installed in structural members except where indicated or approved. Sleeves in nonload bearing surfaces shall be galvanized sheet metal, conforming to JIS G 3302, 1.0 mm (20 gauge). Sleeves in load bearing surfaces shall be uncoated carbon steel pipe, conforming to JIS G 3454, ~~[Schedule 30] [Schedule 20] [~~Standard weight~~]~~. Sealants shall be applied to moisture and oil-free surfaces and elastomers to not less than 13 mm depth. Sleeves shall not be installed in structural members.

~~3.2.16.1 Refrigerated Space~~

~~Refrigerated space building surface penetrations shall be fitted with sleeves fabricated from hand-lay-up or helically wound, fibrous glass reinforced polyester or epoxy resin with a minimum thickness equal to equivalent size Schedule 40 steel pipe. Sleeves shall be constructed with integral collar or cold side shall be fitted with a bonded slip-on flange or extended collar. In the case of masonry penetrations where sleeve is~~

~~not cast-in, voids shall be filled with latex mixed mortar cast to shape of sleeve and flange/external collar type sleeve shall be assembled with butyl elastomer vapor barrier sealant through penetration to cold side surface vapor barrier overlap and fastened to surface with masonry anchors. Integral cast-in collar type sleeve shall be flashed [with not less than 100 mm of cold side vapor barrier overlap of sleeve surface.] Normally noninsulated penetrating round surfaces shall be sealed to sleeve bore with mechanically expandable seals in vapor tight manner and remaining warm and cold side sleeve depth shall be insulated with not less than [100] [_____] mm of foamed-in-place rigid polyurethane or foamed-in-place silicone elastomer. Vapor barrier sealant shall be applied to finish warm side insulation surface. Warm side of penetrating surface shall be insulated beyond vapor barrier sealed sleeve insulation for a distance which prevents condensation. Wires in refrigerated space surface penetrating conduit shall be sealed with vapor barrier plugs or compound to prevent moisture migration through conduit and condensation therein.~~

3.2.16.1 General Service Areas

Each sleeve shall extend through its respective wall, floor, or roof, and shall be cut flush with each surface. Pipes passing through concrete or masonry wall or concrete floors or roofs shall be provided with pipe sleeves fitted into place at the time of construction. Sleeves shall be of such size as to provide a minimum of 6 mm all-around clearance between bare pipe and sleeves or between jacketed-insulation and sleeves. Except in pipe chases or interior walls, the annular space between pipe and sleeve or between jacket over-insulation and sleeve shall be sealed in accordance with Section 07 92 00 JOINT SEALANTS.

3.2.16.2 Waterproof Penetrations

Pipes passing through roof or floor waterproofing membrane shall be installed through a 5.17 kg/sq. m copper sleeve, or a 0.81 mm thick aluminum sleeve, each within an integral skirt or flange. Flashing sleeve shall be suitably formed, and skirt or flange shall extend not less than 200 mm from the pipe and be set over the roof or floor membrane in a troweled coating of bituminous cement. The flashing sleeve shall extend up the pipe a minimum of 50 mm above the roof or floor penetration. The annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation shall be sealed as indicated. Penetrations shall be sealed by either one of the following methods.

3.2.16.2.1 Waterproofing Clamping Flange

Pipes up to and including 250 mm in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass bolts. Waterproofing membrane shall be clamped into place and sealant shall be placed in the caulking recess.

3.2.16.2.2 Modular Mechanical Type Sealing Assembly

In lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve or conduit and sleeve, a modular mechanical type sealing assembly may be installed. Seals shall consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. Links shall be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tightening of the bolt shall cause the rubber sealing elements to expand and provide a watertight seal rubber sealing elements to expand and provide a watertight seal between the pipe/conduit seal between the pipe/conduit and the sleeve. Each seal assembly shall be sized as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals shall provide sleeves of the proper diameters.

3.2.16.3 Fire-Rated Penetrations

Penetration of fire-rated walls, partitions, and floors shall be sealed as specified in Section 07 84 00 FIRESTOPPING.

3.2.16.4 Escutcheons

Finished surfaces where exposed piping, bare or insulated, pass through floors, walls, or ceilings, except in boiler, utility, or equipment rooms, shall be provided with escutcheons. Where sleeves project slightly from floors, special deep-type escutcheons shall be used. Escutcheon shall be secured to pipe or pipe covering.

3.2.17 Access Panels

Access panels shall be provided for all concealed valves, vents, controls, and items requiring inspection or maintenance. Access panels shall be of sufficient size and located so that the concealed items may be serviced and maintained or completely removed and replaced. Access panels shall be as specified in Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS and 08 31 00 ACCESS DOORS AND PANELS.

3.2.18 Field Applied Insulation

Field installed insulation shall be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

3.2.19 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

3.2.19.1 Color Coding

Color coding for piping identification is specified in Section 09 90 00 PAINTS AND COATINGS.

3.2.19.2 Color Coding Scheme

A color coding scheme for locating hidden piping shall be in accordance with ~~[~~Section 22 00 00 PLUMBING, GENERAL PURPOSE~~][Section 22 00 70 PLUMBING, HEALTHCARE FACILITIES]~~.

3.2.20 Identification Tags

Provide identification tags made of brass, engraved laminated plastic or engraved anodized aluminum indicating service and item number on all valves and dampers. Tags shall be 35 mm minimum diameter and marking shall be stamped or engraved. Indentations shall be black for reading clarity. Tags shall be attached to valves with No. 12 AWG copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

3.3 CLEANING AND ADJUSTING

Clean uncontaminated system(s) by evacuation and purging procedures currently recommended by refrigerant and refrigerant equipment manufacturers, and as specified herein, to remove small amounts of air and moisture. Systems containing moderate amounts of air, moisture, contaminated refrigerant, or any foreign matter shall be considered contaminated systems. Restoring contaminated systems to clean condition including disassembly, component replacement, evacuation, flushing, purging, and re-charging, shall be performed using currently approved refrigerant and refrigeration manufacturer's procedures. Restoring contaminated systems shall be at no additional cost to the Government as determined by the Contracting Officer. Water shall not be used in any procedure or test.

3.4 TRAINING COURSE

a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training. Conduct a training course for all members of the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.

b. Cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations in the field posted instructions.

c. Submit 4 complete copies of an operation manual in bound 216 by 279 mm booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. Include the manufacturer's name, model number, and parts list in the booklets. Include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features in the manuals.

d. Submit 4 complete copies of maintenance manual in bound 216 by 279 mm booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. Include piping layouts and simplified wiring and control diagrams of the system as installed in the manuals.

3.5 REFRIGERANT PIPING TESTS

After all components of the refrigerant system have been installed and connected, subject the entire refrigeration system to pneumatic, evacuation, and startup tests as described herein. Submit a schedule, at least ~~[~~2~~] [______]~~ weeks prior to the start of related testing, for each test. Identify the proposed date, time, and location for each test. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test. Provide the services of a qualified technician, as required, to perform all tests and procedures indicated herein. Field tests shall be coordinated with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit ~~[6] [_____]~~4 copies of the tests report in bound 216 by 279 mm booklets documenting all phases of the tests performed. The report shall include initial test summaries, all repairs/adjustments made, and the final test results.

3.5.1 Preliminary Procedures

Prior to pneumatic testing, equipment which has been factory tested and refrigerant charged as well as equipment which could be damaged or cause personnel injury by imposed test pressure, positive or negative, shall be isolated from the test pressure or removed from the system. Safety relief valves and rupture discs, where not part of factory sealed systems, shall be removed and openings capped or plugged.

3.5.2 Pneumatic Test

Pressure control and excess pressure protection shall be provided at the source of test pressure. Valves shall be wide open, except those leading to the atmosphere. Test gas shall be dry nitrogen, with minus 55 degrees C dewpoint and less than 5 ppm oil. Test pressure shall be applied in two stages before any refrigerant pipe is insulated or covered. First stage test shall be at 69 kPa with every joint being tested with a thick soap or color indicating solution. Second stage tests shall raise the system to the minimum refrigerant leakage test pressure, with a maximum test pressure 25 percent greater. Pressure above 690 KPa shall be raised in 10 percent increments with a pressure acclimatizing period between increments. The initial test pressure shall be recorded along with the ambient temperature to which the system is exposed. Final test pressures of the second stage shall be maintained on the system for a minimum of 24 hours. At the end of the 24 hour period, the system pressure will be recorded along with the ambient temperature to which the system is exposed. A correction factor of 2 kPa will be allowed for each degree C change between test space initial and final ambient temperature, plus for increase and minus for a decrease. If the corrected system pressure is not exactly equal to the initial system test pressure, then the system shall be investigated for leaking joints. To repair leaks, the joint shall be taken apart, thoroughly cleaned, and reconstructed as a new joint. Joints repaired by caulking, remelting, or back-welding/brazing shall not be acceptable. Following repair, the entire system shall be retested using the pneumatic tests described above. The entire system shall be reassembled once the pneumatic tests are satisfactorily completed.

3.5.3 Evacuation Test

Following satisfactory completion of the pneumatic tests, the pressure shall be relieved and the entire system shall be evacuated to an absolute pressure of 300 micrometers. During evacuation of the system, the ambient temperature shall be higher than 2 degrees C. No more than one system shall be evacuated at one time by one vacuum pump. Once the desired vacuum has been reached, the vacuum line shall be closed and the system shall stand for 1 hour. If the pressure rises over 500 micrometers after the 1 hour period, then the system shall be evacuated again down to 300 micrometers and let set for another 1 hour period. The system shall not be charged until a vacuum of at least 500 micrometers is maintained for a period of 1 hour without the assistance of a vacuum line. If during the testing the pressure continues to rise, check the system for leaks, repair as required, and repeat the evacuation procedure. During evacuation, pressures shall be recorded by a thermocouple-type, electronic-type, or a calibrated-micrometer type gauge.

3.5.4 System Charging and Startup Test

Following satisfactory completion of the evacuation tests, the system shall be charged with the required amount of refrigerant by raising pressure to normal operating pressure and in accordance with manufacturer's procedures. Following charging, the system shall operate with high-side and low-side pressures and corresponding refrigerant temperatures, at design or improved values. The entire system shall be tested for leaks. Fluorocarbon systems shall be tested with halide torch or electronic leak detectors.

3.5.5 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system shall immediately be isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. Under no circumstances shall the refrigerant be discharged into the atmosphere.

3.5.6 Contractor's Responsibility

At all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps shall include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time shall more than 85 g of refrigerant be released to the atmosphere in any one occurrence. Any system leaks within the first year shall be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

-- End of Section --

SECTION 23 30 00

HVAC AIR DISTRIBUTION
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

~~ASHRAE 52.2~~ ~~(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size~~

ASHRAE 52.2 (2017; Addenda B 2020; Errata 1 2020; Addenda C 2022; Errata 2 2024; Addenda D 2025) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A (20~~18~~24) Standard for the Installation of Air Conditioning and Ventilating Systems

NFPA 96 (2017; TIA 17-1) Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1819 (2002) Fire, Smoke and Radiation Damper Installation Guide for HVAC Systems, 5th Edition

UNDERWRITERS LABORATORIES (UL)

UL 555 (2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers

UL 705 (2017; Reprint ~~Oct~~Sep 20~~18~~24) UL Standard for Safety Power Ventilators

UL Electrical Construction (2012) Electrical Construction Equipment Directory

UL Fire Resistance (2014) Fire Resistance Directory

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 4009	(2017) Components of Air Duct
JIS A 9504	(20~~21~~24) Man Made Mineral Fibre Thermal Insulation Materials
JIS A 9511	(20~~21~~24) Preformed Cellular Plastics Thermal Insulation Materials
JIS B 1518	(2013) Rolling Bearings-Dynamic Load Ratings and Rating Life
JIS B 1521	(2012) Rolling Bearings-Deep Groove Ball Bearings
JIS B 1534	(2013) Rolling Bearings-Tapered Roller Bearings
JIS B 8330	(2000) Testing Methods for Turbo-Fans
JIS B 8616	(2015) Package Air Conditioners
JIS B 9908-1	(2019) Test Method of Air Filter Units for Ventilation and Electric Air Cleaners for Ventilation-Part 1: Technical Specifications, Requirements and Classification System Based Upon Particulate Matter Efficiency
JIS C 4203	(2010) Single Phase Induction Motors for General Purpose (Amendment 1)
JIS C 4212	~~(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)~~(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 9603	(1988) Ventilating Fans
JIS C 9803	(1999) Household Electric Direct-Acting Room Heaters-Methods for Measuring Perofrmance
JIS G 3302	(2022) Hot~~–~~-Dip Zinc~~–~~-Coated Steel Sheet and Strip
JIS G 3452	~~(2016) Carbon Steel Pipes for Ordinary Piping (Amendment 1)~~(2019) Carbon Steel Pipes for Ordinary Piping
JIS G 3553	(2011) Crimped Wire Cloth (Amendment 1)
JIS G 4305	~~(2015) Cold-Rolled Stainless Steel Plate, Sheet and Strip (Amendment 1)~~(2021) Cold-Rolled Stainless Steel Plate, Sheet

	and Strip
JIS H 3100	(2018) Copper and Copper Alloy Sheets, Plates and Strips
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 4000	(2017) Aluminium and Aluminium Alloy Sheets, Strips and Plates (Amendment 1)
JIS H 8610	(1999) Electroplated-Coatings of Zinc on Iron or Steel
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS HB 71	(2019) Electrical Safety
JIS K 5600-5-5	(1999) Testing Methods for Paints- Part 5 : Mechanical Property of Film Section 5 : Scratch Hardness (Stylus Method)
JIS K 5600-5-6	(1999) Testing Methods for Paints - Part 5 : Mechanical Property of Film-Section 6 : Adhesion Test (Cross-Cut Test)
JIS K 5600-7-9	(2006) Testing Methods for Paints Part 7 : Determination Of Resistance to Cyclic Corrosion Conditions Section 9 : Salt Fog/Dry/Humidity
JIS Z 2371	(2015) Methods of Salt Spray Testing
JIS Z 4812	(1995) HEPA Filters for Radioactive Aerosols

THE JAPAN REFRIGERATION AND AIR CONDITIONING INDUSTRY ASSOCIATION (JRAIA)

JRA 4036	(2014) Air Handling Unit

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification
JBE-S	(2014) Japanese Building Equipment Seismic Standard Design

1.2 SYSTEM DESCRIPTION

Furnish ductwork, piping offsets, fittings, and accessories as required to provide a complete installation. Coordinate the work of the different trades to avoid interference between piping, equipment, structural, and electrical work. Provide complete, in place, all necessary offsets in piping and ductwork, and all fittings, and other components, required to install the work as indicated and specified.

1.2.1 Mechanical Equipment Identification

The number of charts and diagrams must be equal to or greater than the number of mechanical equipment rooms. Where more than one chart or diagram per space is required, mount these in edge pivoted, swinging leaf, extruded aluminum frame holders which open to 170 degrees.

1.2.1.1 Charts

Provide chart listing of equipment by designation numbers and capacities such as flow rates, pressure and temperature differences, heating and cooling capacities, horsepower, pipe sizes, and voltage and current characteristics.

1.2.1.2 Diagrams

Submit proposed diagrams, at least 2 weeks prior to start of related testing. provide neat mechanical drawings provided with extruded aluminum frame under 3 mm glass or laminated plastic, system diagrams that show the layout of equipment, piping, and ductwork, and typed condensed operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system. After approval, post these items where directed.

1.2.2 Service Labeling

Label equipment, including fans, air handlers, terminal units, etc. with labels made of self-sticking, plastic film designed for permanent installation. Provide labels in accordance with the typical examples below:

SERVICE	LABEL AND TAG DESIGNATION
Air handling unit Number	~~AHU~~DOAS - ~~[~~_____~~]~~
Control and instrument air	CONTROL AND INSTR.
Exhaust Fan Number	EF - ~~[~~_____~~]~~
VAV Box Number	~~VAV~~CH - ~~[~~_____~~]~~
Fan Coil Unit Number	~~FC~~FCU - ~~[~~_____~~]~~
Terminal Box Number	~~TB~~ACCU - ~~[~~_____~~]~~
Unit Ventilator Number	~~UV~~ET - ~~[~~_____~~]~~

Identify similar services with different temperatures or pressures. Where pressures could exceed 860 kilopascal, include the maximum system pressure in the label. Label and arrow piping in accordance with the following:

a. Each point of entry and exit of pipe passing through walls.

b. Each change in direction, i.e., elbows, tees.

c. In congested or hidden areas and at all access panels at each point required to clarify service or indicated hazard.

d. In long straight runs, locate labels at distances within eyesight of each other not to exceed 22 meter. All labels must be visible and legible from the primary service and operating area.

For Bare or Insulated Pipes	
for Outside Diameters of	Lettering
13 thru ~~[_____]~~39 mm	13 mm
40 thru ~~[_____]~~64 mm	~~[_____]~~25 mm
65 mm and larger	~~[_____]~~32 mm

1.2.3 Color Coding

Color coding of all piping systems must be in accordance with MLIT-M or the base standard.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G~~[, [_____]]~~

SD-03 Product Data

Fire Dampers

~~Automatic Smoke-Fire Dampers~~

~~Automatic Smoke Dampers~~

Air Handling Units; G~~[, [_____]]~~

Room Fan-Coil Units; G~~[, [_____]]~~

~~Coil Induction Units; G[, [_____]]~~

~~Constant Volume, Single Duct Terminal Units; G[, [_____]]~~

~~Variable Volume, Single Duct Terminal Units; G[, [_____]]~~

~~Variable Volume, Single Duct, Fan-Powered Terminal Units; G[, [_____]]~~

~~Dual Duct Terminal Units; G[, [_____]]~~

Reheat Units; G~~[, [_____]]~~

Energy Recovery Devices; G~~[, [_____]]~~

Test ~~Procedures~~Diagrams; G~~[, [_____]]~~

SD-06 Test Reports

Performance Tests

Damper Acceptance Test; G~~[, [_____]]~~

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals

Fire Dampers

Manual Balancing Dampers

~~Automatic Smoke-Fire Dampers~~

~~Automatic Smoke Dampers~~

Centrifugal Fans

In-Line Centrifugal Fans

~~Axial Flow Fans~~

~~Centrifugal~~Propeller Type Power Wall Ventilators

Centrifugal Type Power Roof Ventilators

~~Propeller Type Power Roof Ventilators~~

~~Air-Curtain Fans~~

Ceiling Exhaust Fans

Air Handling Units

Room Fan-Coil Units

~~Coil Induction Units~~

~~Constant Volume, Single Duct Terminal Units~~

~~Variable Volume, Single Duct Terminal Units~~

~~Variable Volume, Single Duct, Fan-Powered Terminal Units~~

~~Dual Duct Terminal Units~~

Reheat Units

Unit Ventilators

Energy Recovery Devices

~~Hydronic Modular Panels~~

~~Prefabricated Radiant-Heating Electric Panels~~

1.4 QUALITY ASSURANCE

Except as otherwise specified, approval of materials and equipment is based on manufacturer's published data.

a. Submit a written certificate from any recognized testing agency, adequately equipped and competent to perform such services, stating that the items have been tested and that the units conform to the specified requirements. Outline methods of testing used by the specified agencies.

b. Where materials or equipment are specified to be constructed or tested, or both, in accordance with the standards of the JIS ~~-~~a manufacturer's certificate of compliance of each item is acceptable as proof of compliance.

c. Conformance to such agency requirements does not relieve the item from compliance with other requirements of these specifications.

1.4.1 Prevention of Corrosion

Protect metallic materials against corrosion. Provide rust-inhibiting treatment and standard finish for the equipment enclosures. Do not use aluminum in contact with earth, and where connected to dissimilar metal. Protect aluminum by approved fittings, barrier material, or treatment. Provide hot-dip galvanized ferrous parts such as anchors, bolts, braces, boxes, bodies, clamps, fittings, guards, nuts, pins, rods, shims, thimbles, washers, and miscellaneous parts not of corrosion-resistant steel or nonferrous materials in accordance with JIS H 8641 for exterior locations and cadmium-plated in conformance with JIS H 8610 for interior locations.~~[Provide written certification from the bolt manufacturer that the bolts furnished comply with the requirements of this specification. Include illustrations of product markings, and the number of each type of bolt to be furnished in the certification.]~~

1.4.2 Asbestos Prohibition

Do not use asbestos and asbestos-containing products.

1.4.3 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be Refrigerant Handling Technician (Reibai-Furontou-Toriatsukai-Gijutsusha). Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.4.4 Detail Drawings

Submit detail drawings showing equipment layout, including assembly and installation details and electrical connection diagrams; ductwork layout showing the location of all supports and hangers, typical hanger details, gauge reinforcement, reinforcement spacing rigidity classification, and static pressure and seal classifications. Include any information required to demonstrate that the system has been coordinated and functions properly as a unit on the drawings and show equipment relationship to other parts of the work, including clearances required for operation and maintenance. Submit drawings showing bolt-setting information, and foundation bolts prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Submit function designation of the equipment and any other requirements specified throughout this Section with the shop drawings.

1.4.5 Test Procedures

Submit proposed test procedures and test schedules for the ~~[~~ductwork leak test, and~~]~~ performance tests of systems, at least 2 weeks prior to the start of related testing.

1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored equipment at the jobsite from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Additionally, cap or plug all pipes until installed.

PART 2 PRODUCTS

2.1 STANDARD PRODUCTS

Provide components and equipment that are "standard products" of a manufacturer regularly engaged in the manufacturing of products that are of a similar material, design and workmanship. "Standard products" is defined as being in satisfactory commercial or industrial use for 2 years before bid opening, including applications of components and equipment under similar circumstances and of similar size, satisfactorily completed by a product that is sold on the commercial market through advertisements, manufacturers' catalogs, or brochures. Products having less than a 2-year field service record are acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. Provide equipment items that are supported by a service organization located in Japan.

~~2.2 STANDARD PRODUCTS~~

~~Except for the fabricated duct, plenums and casings specified in paragraphs "Metal Ductwork" and "Plenums and Casings for Field-Fabricated Units",~~
~~provide components and equipment that are standard products of manufacturers regularly engaged in the manufacturing of products that are of a similar material, design and workmanship. This requirement applies to all equipment, including diffusers, registers, fire dampers, and balancing dampers.~~

~~a. Standard products are defined as components and equipment that have been in satisfactory commercial or industrial use in similar applications of similar size for at least two years before bid opening.~~

~~b. Prior to this two year period, these standard products must have been sold on the commercial market using advertisements in manufacturers' catalogs or brochures. These manufacturers' catalogs, or brochures must have been copyrighted documents or have been identified with a manufacturer's document number.~~

~~c. Provide equipment items that are supported by a service organization.~~

2.2 IDENTIFICATION PLATES

In addition to standard manufacturer's identification plates, provide engraved laminated phenolic identification plates for each piece of mechanical equipment. Identification plates are to designate the function of the equipment. Submit designation with the shop drawings. Provide identification plates that are layers, black-white-black, engraved to show white letters on black background. Letters must be upper case. Identification plates 40 mm that are high and smaller must be 1.6 mm thick, with engraved lettering 3 mm high; identification plates larger than 40 mm high must be 3 mm thick, with engraved lettering of suitable height. Identification plates 40 mm high and larger must have beveled edges. Install identification plates using a compatible adhesive.

2.3 EQUIPMENT GUARDS AND ACCESS

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact according to OSHA requirements. Properly guard or cover with insulation of a type specified, high temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard. The requirements for ~~[catwalks,] [~~operating platforms,~~]~~ ~~[~~ladders,~~]~~ ~~[~~and~~]~~ ~~[~~guardrails~~]~~ are specified in Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

2.4 ELECTRICAL WORK

a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00INTERIOR DISTRIBUTION SYSTEM. Provide manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown. For packaged equipment, include manufacturer provided controllers with the required monitors and timed restart.

b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with JIS C 4203.

c. For polyphase motors, provide squirrel-cage medium induction motors, including motors that are part of a system , and that meet the efficiency ratings for premium efficiency motors in accordance with JIS C 4212.

d. Provide motors in accordance with JIS C 4212 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Provide motors rated for continuous duty with the enclosure specified. Provide motor duty that allows for maximum frequency start-stop operation and minimum encountered interval between start and stop. Provide motor torque capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. Fit motor bearings with grease supply fittings and grease relief to outside of the enclosure.

e. ~~[~~Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers are allowed to accomplish the same function. Use solid-state variable-speed controllers for motors rated 7.45 kW or less and adjustable frequency drives for larger motors.~~]~~ ~~[Provide variable frequency drives for motors as specified in Section 26 29 23 VARIABLE FREQUENCY DRIVE SYSTEMS UNDER 600 VOLTS.]~~

2.5 ANCHOR BOLTS

Provide anchor bolts for equipment placed on concrete equipment pads or on concrete slabs. Bolts to be of the size and number recommended by the equipment manufacturer and located by means of suitable templates. Installation of anchor bolts must not degrade the surrounding concrete.

2.6 SEISMIC ANCHORAGE

Anchor equipment in accordance with applicable seismic criteria for the area and as defined in JBE-S.

2.7 PAINTING

Paint equipment units in accordance with approved equipment manufacturer's standards unless specified otherwise. Field retouch only if approved. Otherwise, return equipment to the factory for refinishing.

2.8 DUCT SYSTEMS

2.8.1 Metal Ductwork

Provide metal ductwork construction, including all fittings and components, that complies with MLIT-M.

2.8.1.1 Metallic Flexible Duct

a. Provide duct that conforms to~~-~~ MLIT-M with factory-applied insulation, vapor barrier, and end connections.~~-~~ Provide ducts designed for working pressures of 497 Pa and 373 Pa. Provide flexible round duct length that does not exceed 1525 mm. Secure connections by applying adhesive for 51 mm over rigid duct, apply flexible duct 51 mm over rigid duct, apply metal clamp, and provide minimum of three No. 8 sheet metal screws through clamp and rigid duct.

b. Inner duct core: Provide interlocking spiral or helically corrugated flexible core constructed of zinc-coated steel, aluminum, or stainless steel; or constructed of inner liner of continuous galvanized spring steel wire helix fused to continuous, fire-retardant, flexible vapor barrier film, inner duct core.

c. Insulation: Provide inner duct core that is insulated with mineral fiber blanket type flexible insulation, minimum of 25 mm thick. Provide insulation covered on exterior with manufacturer's standard fire retardant vapor barrier jacket for flexible round duct.

2.8.1.2 Insulated Nonmetallic Flexible Duct Runouts

Use flexible duct runouts only where indicated. Runout length is indicated on the drawings, and is not to exceed 1.5 m. Provide runouts that are preinsulated, factory fabricated. Provide either field or factory applied vapor barrier. Provide not less than 0.60 L glass fabric duct connectors coated on both sides with neoprene. Where coil induction or high velocity units are supplied with vertical air inlets, use a streamlined, vaned and mitered elbow transition piece for connection to the flexible duct or hose. Provide a die-stamped elbow and not a flexible connector as the last elbow to these units other than the vertical air inlet type. Insulated flexible connectors are allowed as runouts. Provide insulated material and vapor barrier that conform to the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Do not expose the insulation material surface to the air stream.

2.8.1.3 General Service Duct Connectors

Provide a flexible duct connector approximately 150 mm in width where sheet metal connections are made to fans or where ducts of dissimilar metals are connected. For round/oval ducts, secure the flexible material by stainless steel or zinc-coated, iron clinch-type draw bands. For rectangular ducts, install the flexible material locked to metal collars using normal duct construction methods.

2.8.1.4 High Temperature Service Duct Connections

Provide material that is approximately 2.38 mm thick, 1.2 to 1.36 kg per square meter weight, plain weave fibrous glass cloth with, nickel/chrome wire reinforcement for service in excess of 650 degrees C.

2.8.1.5 Aluminum Ducts

JIS H 4000, alloy 3003-H14 for aluminum sheet and alloy 6061-T6 or equivalent strength for aluminum connectors and bar stock.

2.8.1.6 Copper Sheets

JIS H 3100, light cold rolled temper.

2.8.1.7 Corrosion Resisting (Stainless) Steel Sheets

JIS G 4305.

2.8.2 Duct Access Doors

Provide hinged access doors conforming to MLIT-M in ductwork and plenums where indicated and at all air flow measuring primaries, automatic dampers, fire dampers, coils, thermostats, and other apparatus requiring service and inspection in the duct system. Provide access doors upstream and downstream of air flow measuring primaries and heating and cooling coils. Provide doors that are a minimum 375 by 450 mm, unless otherwise shown. Where duct size does not accommodate this size door, make the doors as large as practicable. Equip doors 600 by 600 mm or larger with fasteners operable from inside and outside the duct. Use insulated type doors in insulated ducts.

2.8.3 Fire Dampers

Use 1.5 hour rated fire dampers unless otherwise indicated. Provide fire dampers that conform to the requirements of NFPA 90A and UL 555. Perform the fire damper test as outlined in NFPA 90A. Provide a pressure relief door upstream of the fire damper. If the ductwork connected to the fire damper is to be insulated then provide a factory installed pressure relief damper. Provide automatic operating fire dampers with a dynamic rating suitable for the maximum air velocity and pressure differential to which it is subjected. Provide fire dampers approved for the specific application, and install according to their listing. Equip fire dampers with a steel sleeve or adequately sized frame installed in such a manner that disruption of the attached ductwork, if any, does not impair the operation of the damper. Equip sleeves or frames with perimeter mounting angles attached on both sides of the wall or floor opening. Construct ductwork in fire-rated floor-ceiling or roof-ceiling assembly systems with air ducts that pierce the ceiling of the assemblies in conformance with UL Fire Resistance. Provide ~~[~~curtain type with damper blades~~]~~ ~~[~~in the air stream~~] [out of the air stream][or][single blade type][~~ or ~~][~~multi-blade type~~]~~ fire dampers. Install dampers that do not reduce the duct or the air transfer opening cross-sectional area. Install dampers so that the centerline of the damper depth or thickness is located in the centerline of the wall, partition or floor slab depth or thickness. Unless otherwise indicated, comply with the installation details given in SMACNA 1819 and in manufacturer's instructions for fire dampers. Perform acceptance testing of fire dampers according to paragraph Fire Damper Acceptance Test and NFPA 90A.

2.8.4 Manual Balancing Dampers

Furnish manual balancing dampers with accessible operating mechanisms. Use chromium plated operators (with all exposed edges rounded) in finished portions of the building. Provide manual volume control dampers that are operated by locking-type quadrant operators. Install dampers that are 2 gauges heavier than the duct in which installed. Unless otherwise indicated, provide opposed blade type multileaf dampers with maximum blade width of 300 mm. Provide access doors or panels for all concealed damper operators and locking setscrews. Provide stand-off mounting brackets, bases, or adapters not less than the thickness of the insulation when the locking-type quadrant operators for dampers are installed on ducts to be thermally insulated, to provide clearance between the duct surface and the operator. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

2.8.5 Manual Balancing Dampers

a. Furnish manual balancing dampers with accessible operating mechanisms. Use chromium plated operators (with all exposed edges rounded) in finished portions of the building. Provide manual volume control dampers that are operated by locking-type quadrant operators.

b. Unless otherwise indicated, provide opposed blade type multileaf dampers with maximum blade width of 300 mm. Provide access doors or panels for all concealed damper operators and locking setscrews. Provide access doors or panels in hard ceilings, partitions and walls for access to all concealed damper operators and damper locking setscrews. Coordinate location of doors or panels with other affected contractors.

c. Provide stand-off mounting brackets, bases, or adapters not less than the thickness of the insulation when the locking-type quadrant operators for dampers are installed on ducts to be thermally insulated, to provide clearance between the duct surface and the operator. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

2.8.5.1 Square or Rectangular Dampers

2.8.5.1.1 Duct Height 300 mm and Less

2.8.5.1.1.1 Frames

Width	Height	Galvanized Steel Thickness	Length
Maximum 483 mm	Maximum 300 mm	Minimum 0.91 mm	Minimum 75 mm
More than 483 mm	Maximum 300 mm	Minimum 1.6 mm	Minimum 75 mm

2.8.5.1.1.2 Single Leaf Blades

Width	Height	Galvanized Steel Thickness	Length
Maximum 483 mm	Maximum 300 mm	Minimum 0.91 mm	Minimum 75 mm
More than 483 mm	Maximum 300 mm	Minimum 1.6 mm	Minimum 75 mm

2.8.5.1.1.3 Blade Axles

To support the blades of round dampers, provide galvanized steel shafts supporting the blade the entire duct diameter frame-to-frame. Provide axle shafts that extend through standoff bracket and hand quadrant.

Width	Height	Material	Square Shaft
Maximum 483 mm	Maximum 300 mm	Galvanized Steel	Minimum 10 mm
More than 483 mm	Maximum 300 mm	Galvanized Steel	Minimum 13 mm

2.8.5.1.1.4 Axle Bearings

Support the shaft on each end at the frames with shaft bearings. Press fit shaft bearings configuration to provide a tight joint between blade shaft and damper frame.

Width	Height	Material
Maximum 483 mm	Maximum 300 mm	solid nylon, or equivalent solid plastic, or oil-impregnated bronze

Width	Height	Material
More than 483 mm	Maximum 300 mm	oil-impregnated bronze

2.8.5.1.1.5 Control Shaft/Hand Quadrant

Provide dampers with accessible locking-type control shaft/hand quadrant operators.

Provide stand-off mounting brackets, bases, or adapters for the locking-type quadrant operators on dampers installed on ducts to be thermally insulated. Provide a minimum stand-off distance of 50 mm off the metal duct surface. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

2.8.5.1.1.6 Finish

Mill Galvanized

2.8.5.1.2 Duct Height Greater than 300 mm

2.8.5.1.2.1 Dampers

Provide dampers with multi-leaf opposed-type blades.

2.8.5.1.2.2 Frames

Maximum 1200 mm in height; maximum 1200 mm in width; minimum of 1.6 mm galvanized steel,
_minimum of 127 mm long.

2.8.5.1.2.3 Blades

Minimum of 1.6 mm galvanized steel; 150 mm nominal width.

2.8.5.1.2.4 Blade Axles

To support the blades of round dampers, provide galvanized square steel shafts supporting the blade the entire duct diameter frame-to-frame. Provide axle shafts that extend through standoff bracket and hand quadrant.

2.8.5.1.2.5 Axle Bearings

Support the shaft on each end at the frames with shaft bearings constructed of oil-impregnated bronze, or solid nylon, or a solid plastic equivalent to nylon. Press fit shaft bearings configuration to provide a tight joint between blade shaft and damper frame.

2.8.5.1.2.6 Blade Actuator

Minimum 50 mm diameter galvanized steel.

2.8.5.1.2.7 Blade Actuator Linkage

Mill Galvanized steel bar and crank plate with stainless steel pivots.

2.8.5.1.2.8 Control Shaft/Hand Quadrant

Provide dampers with accessible locking-type control shaft/hand quadrant operators.

Provide stand-off mounting brackets, bases, or adapters for the locking-type quadrant operators on dampers installed on ducts to be thermally insulated. Provide a minimum stand-off distance of 50 mm off the metal duct surface. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

2.8.5.1.2.9 Finish

Mill Galvanized.

2.8.5.2 Round Dampers

2.8.5.2.1 Frames

Size	Galvanized Steel Thickness	Length
100 to 500 mm	Minimum 0.91 mm	Minimum 152 mm
550 to 750 mm	Minimum 0.91 mm	Minimum 250 mm
775 to 1000 mm	Minimum 1.6 mm	Minimum 250 mm

2.8.5.2.2 Blades

Size	Galvanized Steel Thickness
100 to 500 mm	Minimum 0.91 mm
550 to 750 mm	Minimum 1.6 mm
775 to 1000 mm	Minimum 3.5 mm

2.8.5.2.3 Blade Axles

To support the blades of round dampers, provide galvanized steel shafts supporting the blade the entire duct diameter frame-to-frame. Provide axle shafts that extend through standoff bracket and hand quadrant.

Size	Shaft Size and Shape
100 to 500 mm	Minimum 10 mm square

Size	Shaft Size and Shape
550 to 750 mm	Minimum 13 mm square
775 to 1000 mm	Minimum 19 mm square

2.8.5.2.4 Axle Bearings

Support the shaft on each end at the frames with shaft bearings constructed of oil-impregnated bronze, or solid nylon, or a solid plastic equivalent to nylon. Press fit shaft bearings configuration to provide a tight joint between blade shaft and damper frame.

Size	Material
100 to 500 mm	solid nylon, or equivalent solid plastic, or oil-impregnated bronze
550 to 750 mm	solid nylon, or equivalent solid plastic, or oil-impregnated bronze
775 to 1000 mm	oil-impregnated bronze, or stainless steel sleeve bearing

2.8.5.2.5 Control Shaft/Hand Quadrant

Provide dampers with accessible locking-type control shaft/hand quadrant operators.

Provide stand-off mounting brackets, bases, or adapters for the locking-type quadrant operators on dampers installed on ducts to be thermally insulated. Provide a minimum stand-off distance of 50 mm off the metal duct surface. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

2.8.5.2.6 Finish

Mill Galvanized.

2.8.6 Automatic Balancing Dampers

Provide dampers as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.

~~2.8.7 Automatic Smoke-Fire Dampers~~

~~Multiple blade type, 82 degrees C fusible fire damper link; smoke damper assembly to include [pneumatically powered][electric] damper operator. UL 555 as a 1.5 hour rated fire damper; further qualified under UL 555S as a leakage rated damper. Provide a leakage rating under UL 555S that is no higher than Class [II][or][III] at an elevated temperature Category B (121 degrees C for 30 minutes). Ensure that pressure drop in the damper open position does not exceed 25 Pa with average duct velocities of 13 m/second.~~

~~2.8.8 Automatic Smoke Dampers~~

~~UL listed multiple blade type, supplied by smoke damper manufacturer, with [pneumatic][electric] damper operator as part of assembly. Qualified under UL 555S with a leakage rating no higher than Class [II][or][III] at an elevated temperature Category B (121 degrees C for 30 minutes). Ensure that pressure drop in the damper open position does not exceed 25 Pa with average duct velocities of 13 m/second.~~

2.8.7 Air Supply And Exhaust Air Dampers

Provide outdoor air supply and exhaust air dampers that have a maximum leakage rate when tested in accordance with JIS A 4009 as required by MLIT-M, including maximum Damper Leakage for:

a. Climate Zones 1,2,6,7,8 the maximum damper leakage at 250 Pa for motorized dampers is 20 L/s per square m of damper area and non-motorized dampers are not allowed.

b. All other Climate Zones the maximum damper leakage at 250 Pa is 50 L/s per square m and for non-motorized dampers is 100 L/s per square m of damper area.

Dampers smaller than 600 mm in either direction may have leakage of 200 L/s per square m.

2.8.8 Air Deflectors (Volume Extractors) and Branch Connections

Provide air deflectors (volume extractors) at all duct mounted supply outlets, at takeoff or extension collars to supply outlets, at duct branch takeoff connections, and at 90 degree elbows, as well as at locations as indicated on the drawings or otherwise specified. Conical branch connections or 45 degree entry connections are allowed in lieu of deflectors for branch connections. Furnish all air deflectors (volume extractors), except those installed in 90 degree elbows, with an approved means of adjustment. Provide easily accessible means for adjustment inside the duct or from an adjustment with sturdy lock on the face of the duct. When installed on ducts to be thermally insulated, provide external adjustments with stand-off mounting brackets, integral with the adjustment device, to provide clearance between the duct surface and the adjustment device not less than the thickness of the thermal insulation. Provide factory-fabricated air deflectors consisting of curved turning vanes or louver blades designed to provide uniform air distribution and change of direction with minimum turbulence or pressure loss. Provide factory or field assembled air deflectors (volume extractors). Make adjustment from the face of the diffuser or by position adjustment and lock external to the duct. Provide stand-off brackets on insulated ducts as described herein. Provide fixed air deflectors (volume extractors), also called turning vanes, in 90 degree elbows.

2.8.9 Plenums and Casings for Field-Fabricated Units

2.8.9.1 Plenum and Casings

Fabricate and erect plenums and casings as shown in MLIT-M, as applicable. Construct system casing of not less than 1.6 mm galvanized sheet steel. Furnish cooling coil drain pans with 25 mm threaded outlet to collect condensation from the cooling coils. Fabricate drain pans from not lighter than 1.6 mm steel, galvanized after fabrication or of 1.3 mm corrosion-resisting sheet steel conforming to JIS G 4305, welded and stiffened. Thermally insulate drain pans exposed to the atmosphere to prevent condensation. Coat insulation with a flame resistant waterproofing material. Provide separate drain pans for each vertical coil section, and a separate drain line for each pan. Size pans to ensure capture of entrained moisture on the downstream-air side of the coil. Seal openings in the casing, such as for piping connections, to prevent air leakage. Size the water seal for the drain to maintain a pressure of at least 500 Pa greater than the maximum negative pressure in the coil space.

2.8.9.2 Casing

Terminate casings at the curb line and bolt each to the curb using galvanized angle, as indicated in MLIT-M.

2.8.9.3 Access Doors

Provide access doors in each section of the casing. Weld doorframes in place, gasket each door with neoprene, hinge with minimum of two brass hinges, and fasten with a minimum of two brass tension fasteners operable from inside and outside of the casing. Where possible, make doors 900 by 450 mm and locate them 450 mm above the floor. Where the space available does not accommodate doors of this size, use doors as large as the space accommodates. Swing doors so that fan suction or pressure holds doors in closed position, airtight. Provide a push-button station, located inside the casing, to stop the supply.

2.8.9.4 Factory-Fabricated Insulated Sheet Metal Panels

Factory-fabricated components are allowed for field-assembled units, provided all requirements specified for field-fabricated plenums and casings are met. Provide panels of modular design, pretested for structural strength, thermal control, condensation control, and acoustical control. Seal and insulate panel joints. Provide and gasket access doors to prevent air leakage. Provide panel construction that is not less than one mm galvanized sheet steel, assembled with fasteners treated against corrosion. Provide standard length panels that deflect not more than 13 mm under operation. Construct details, including joint sealing, not specifically covered, as indicated in MLIT-M. Construct the plenums and casings to withstand the specified internal pressure of the air systems.

2.8.9.5 Duct Liner

Unless otherwise specified, duct liner is not permitted.

2.8.10 Sound Attenuation Equipment

2.8.10.1 Systems with total pressure above 1 kPa

Provide sound attenuators on the discharge duct of each fan operating at a total pressure above 1 kPa, and, when indicated, at the intake of each fan system. Provide sound attenuators elsewhere as indicated. Provide factory fabricated sound attenuators, tested by an independent laboratory for sound and performance characteristics. Provide a net sound reduction as indicated. Maximum permissible pressure drop is not to exceed 157 Pa. Construct traps to be airtight when operating under an internal static pressure of 2.5 kPa. Provide air-side surface capable of withstanding air velocity of 50 m/s. Certify that the equipment can obtain the sound reduction values specified after the equipment is installed in the system and coordinated with the sound information of the system fan to be provided. Provide sound absorbing material conforming to JIS A 9511. Provide sound absorbing material that meets the fire hazard rating requirements for insulation specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. For connection to ductwork, provide a duct transition section. Factory fabricated double-walled internally insulated spiral lock seam and round duct and fittings designed for high pressure air system can be provided if complying with requirements specified for factory fabricated sound attenuators, in lieu of factory fabricated sound attenuators. Construct the double-walled duct and fittings from an outer metal pressure shell of zinc-coated steel sheet, 25 mm thick acoustical blanket insulation, and an internal perforated zinc-coated metal liner. Provide a sufficient length of run to obtain the noise reduction coefficient specified. Certify that the sound reduction value specified can be obtained within the length of duct run provided. Provide welded or spiral lock seams on the outer sheet metal of the double-walled duct to prevent water vapor penetration. Provide duct and fittings with an outer sheet that conforms to the metal thickness of high-pressure spiral and round ducts and fittings shown in MLIT-M. Provide acoustical insulation with a thermal conductivity "k" of not more than 0.0389 W/m-K at 24 degrees C mean temperature. Provide an internal perforated zinc-coated metal liner that is not less than 0.7 mm with perforations not larger than 6.35 mm in diameter providing a net open area not less than 10 percent of the surface.

2.8.10.2 System with total pressure of 1 kPa and Lower

Use sound attenuators only where indicated. Provide factory fabricated sound attenuators that are constructed of galvanized steel sheets. Provide attenuator with outer casing that is not less than 0.85 mm. Provide fibrous glass acoustical fill. Provide net sound reduction indicated. Obtain values on a test unit not less than 600 by 600 mm outside dimensions made by a certified nationally recognized independent acoustical laboratory. Provide air flow capacity as indicated or required. Provide pressure drop through the attenuator that does not exceed the value indicated, or that is not in excess of 15 percent of the total external static pressure of the air handling system, whichever is less. Acoustically test attenuators with metal duct inlet and outlet sections while under the rated air flow conditions. Include with the noise reduction data the effects of flanking paths and vibration transmission. Construct sound attenuators to be airtight when operating at the internal static pressure indicated or specified for the duct system, but in no case less than 500 Pa.

2.8.10.3 Acoustical Duct Liner

Use fibrous glass designed or flexible elastomeric duct liner for lining ductwork and conforming to the requirements of JIS A 9511, Type I and II. Provide uniform density, graduated density, or dual density liner composition, as standard with the manufacturer. Provide not less than 25 mm thick coated lining. Where acoustical duct liner is used, provide the thermal equivalent of the insulation specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS for liner or combination of liner and insulation applied to the exterior of the ductwork. Increase duct sizes shown to compensate for the thickness of the lining used. ~~[In lieu of sheet metal duct with field-applied acoustical lining, provide acoustically equivalent lengths of fibrous glass duct, elastomeric duct liner or factory fabricated double-walled internally insulated duct with~~

~~perforated liner.]~~

2.8.11 Diffusers, Registers, and Grilles

Provide factory-fabricated units of ~~[~~steel~~][corrosion-resistant steel][~~ or ~~][~~aluminum~~]~~ that distribute the specified quantity of air evenly over space intended without causing noticeable drafts, air movement faster than 0.25 m/s in occupied zone, or dead spots anywhere in the conditioned area. Provide outlets for diffusion, spread, throw, and noise level as required for specified performance. Certify performance according to Japanese Industry Standards (JIS). Provide sound rated and certified inlets and outlets according to MLIT-M. Provide sound power level as indicated. Provide diffusers and registers with volume damper with accessible operator, unless otherwise indicated; or if standard with the manufacturer, an automatically controlled device is acceptable. Provide opposed blade type volume dampers for all diffusers and registers, except linear slot diffusers. Provide linear slot diffusers with round or elliptical balancing dampers. Where the inlet and outlet openings are located less than 2 m above the floor, protect them by a grille or screen.

2.8.11.1 Diffusers

Provide diffuser types indicated. Furnish ceiling mounted units with anti-smudge devices, unless the diffuser unit minimizes ceiling smudging through design features. Provide diffusers with air deflectors of the type indicated. Provide air handling troffers or combination light and ceiling diffusers conforming to the requirements of UL Electrical Construction for the interchangeable use as cooled or heated air supply diffusers or return air units. Install ceiling mounted units with rims tight against ceiling. Provide sponge rubber gaskets between ceiling and surface mounted diffusers for air leakage control. Provide suitable trim for flush mounted diffusers. For connecting the duct to diffuser, provide duct collar that is airtight and does not interfere with volume controller. Provide return or exhaust units that are similar to supply diffusers.

~~2.8.11.2 Perforated Plate Diffusers~~

~~Provide adjustable [one-way,] [two-way,] [three-way,][ ][four-way] air pattern controls as indicated. Provide diffuser faceplates that do not sag or deflect when operating under design conditions.~~

~~2.8.11.3 Linear Diffusers~~

~~Make joints between diffuser sections that appear as hairline cracks. Provide alignment slots for insertion of key strips or other concealed means to align exposed butt edges of diffusers.[Equip with plaster frames when mounted in plaster ceiling.] Do not use screws and bolts in exposed face of frames or flanges. Metal-fill and ground smooth frames and flanges exposed below ceiling. Furnish separate pivoted or hinged adjustable air-volume-damper and separate air-deflection blades.~~

~~2.8.11.4 Security Ceiling Diffusers~~

~~Provide diffusers that are steel with faceplate, fixed diffusion louvers, with flat surface margin, and an opposed blade damper. Provide faceplate that is 1.9 mm minimum with 13 by 13 mm holes on 5 mm spacing and a minimum free area of 45 percent.~~

2.8.11.2 Registers and Grilles

Provide units that are four-way directional-control type, except provide return and exhaust registers that are fixed horizontal or vertical louver type similar in appearance to the supply register face. Furnish registers with sponge-rubber gasket between flanges and wall or ceiling. Install wall supply registers at least 150 mm below the ceiling unless otherwise indicated. Locate return and exhaust registers 150 mm above the floor unless otherwise indicated. Achieve four-way directional control by a grille face which can be rotated in 4 positions or by adjustment of horizontal and vertical vanes. Provide grilles as specified for registers, without volume control damper.

~~2.8.11.3 Registers~~

~~Double-deflection supply registers.[Provide manufacturer-furnished volume dampers. Provide volume dampers of the group-operated, opposed-blade type and key adjustable by inserting key through face of register. Operating mechanism must not project through any part of the register face. Automatic volume control devices are acceptable.][Provide exhaust and return registers as specified for supply registers, except provide exhaust and return registers that have a single set of nondirectional face bars or vanes having the same appearance as the supply registers.][Set face bars or vanes at [_____] degrees.]~~

~~2.8.11.4 Security Supply Air Registers Except in Cells~~

~~Provide supply air registers, except in prisoner cells and prisoner holding cells, that are steel with individually adjustable horizontal and vertical vanes, perforated faceplate, flat surface margin and opposed blade damper. Put vertical vanes in front; with 19 mm o.c. vane spacing. Provide a 1.9 mm (minimum) perforated faceplate with 13 by 13 mm holes on 5 mm spacing and a minimum free area of 45 percent.~~

~~2.8.11.5 Security Return and Other Air Registers Except in Cells~~

~~Provide return, exhaust, transfer and relief air registers, except in prisoner cells and prisoner holding cells, that are steel with perforated faceplate, flat surface margin, opposed blade damper, and duct mounting sleeve. Provide 14 gage (minimum) faceplate with 13 by 13 mm holes on 5 mm spacing and a minimum free area of 45 percent.~~

~~2.8.11.6 Security Supply Air Registers in Cells~~

~~Provide supply air registers in prisoner cells and prisoner holding cells that are steel with perforated faceplate, flat surface margin, extension sleeve, opposed blade damper, and back mounting flanges. Provide a 1.9 mm (minimum) faceplate with 13 by 13 mm holes on 5 mm spacing and a minimum free area of 45 percent. Provide a 14 gage (minimum) wall sleeve.~~

~~2.8.11.7 Security Return and Other Type Air Registers in Cells~~

~~Provide steel return, exhaust, transfer and relief air registers in prisoner cells and prisoner holding cells with perforated faceplate, flat surface margin, wall sleeve, opposed blade damper, and back mounting flanges. Provide 1.9 mm (minimum) faceplate with 13 by 13 mm holes on 5 mm spacing and a minimum free area of 45 percent. Provide a 14 gage (minimum)wall sleeve.~~

2.8.12 Louvers

Provide louvers for installation in exterior walls that are associated with the air supply and distribution system as specified in Section ~~[07 60 00 FLASHING AND SHEET METAL] [~~08 91 00 METAL ~~[~~WALL~~][AND][DOOR]~~ LOUVERS~~]~~.

2.8.13 Air Vents, Penthouses, and Goosenecks

Fabricate air vents, penthouses, and goosenecks from galvanized steel ~~[or aluminum]~~ sheets with galvanized~~[or aluminum]~~ structural shapes. Provide sheet metal thickness, reinforcement, and fabrication that conform to MLIT-M. Accurately fit and secure louver blades to frames. Fold or bead edges of louver blades for rigidity and baffle these edges to exclude driving rain. Provide air vents, penthouses, and goosenecks with bird screen.

2.8.14 Bird Screens and Frames

Provide bird screens that conform to JIS G 3553, No. 2 mesh, aluminum or stainless steel. Provide "medium-light" rated aluminum screens. Provide "light" rated stainless steel screens. Provide removable type frames fabricated from either stainless steel or extruded aluminum.

~~2.8.15 Radon Exhaust Ductwork~~

~~Fabricate radon exhaust ductwork installed in or beneath slabs from Schedule 40 PVC pipe that conforms to JIS K 6741. Use solvent cement conforming to JIS K 6741 to make joints. Otherwise provide metal radon exhaust ductwork as specified herein.~~

2.9 AIR SYSTEMS EQUIPMENT

2.9.1 Fans

Test and rate fans according to JIS B 8330. Calculate system effect on air moving devices in accordance with JIS B 8330 where installed ductwork differs from that indicated on drawings. Install air moving devices to minimize fan system effect. Where system effect is unavoidable, determine the most effective way to accommodate the inefficiencies caused by system effect on the installed air moving device. The sound power level of the fans must not exceed 85 dBA when tested according to JIS B 8330 and rated in accordance with JIS B 8330. Connect fans to the motors either directly or indirectly with V-belt drive. Use V-belt drives designed for not less than ~~[~~150~~] [or 140] [or 120]~~ percent of the connected driving capacity. Provide variable pitch motor sheaves for 11 kW and below, and fixed pitch as defined by MLIT-M the fan shaft and the motor shaft. This is a non-adjustable speed. Select variable pitch sheaves to drive the fan at a speed which can produce the specified capacity when set at the approximate midpoint of the sheave adjustment. When fixed pitch sheaves are furnished, provide a replaceable sheave when needed to achieve system air balance. Provide motors for V-belt drives with adjustable rails or bases. Provide removable metal guards for all exposed V-belt drives, and provide speed-test openings at the center of all rotating shafts. Provide fans with personnel screens or guards on both suction and supply ends, except that the screens need not be provided, unless otherwise indicated, where ducts are connected to the fan. Provide fan and motor assemblies with vibration-isolation supports or mountings as indicated. Use vibration-isolation units that are standard products with published loading ratings. Select each fan to produce the capacity required at the fan static pressure indicated. Provide sound power level as indicated. Obtain the sound power level values according to JIS B 8330. Provide standard JIS C 9603, arrangement, rotation, and discharge as indicated.

2.9.1.1 Centrifugal Fans

Provide fully enclosed, single-width single-inlet, or double-width double-inlet centrifugal fans, with JIS B 8330 as required or indicated for the design system pressure. Provide impeller wheels that are rigidly constructed and accurately balanced both statically and dynamically. ~~[~~ Provide forward curved or backward-inclined airfoil design fan blades in wheel sizes up to 750 mm. Provide backward-inclined airfoil design fan blades for wheels over 750 mm in diameter~~]~~. ~~[~~Provide open-wheel radial type booster fans for exhaust dryer systems, and fans suitable for conveying lint and the temperatures encountered. Equip the fan shaft with a heat slinger to dissipate heat buildup along the shaft. Install an access (service) door to facilitate maintenance to these fans.~~]~~ Provide fan wheels over 900 mm in diameter with overhung pulleys and a bearing on each side of the wheel. Provide fan wheels 900 mm or less in diameter that have one or more extra long bearings between the fan wheel and the drive. Provide sleeve type, self-aligning and self-oiling bearings with oil reservoirs, or precision self-aligning roller or ball-type with accessible grease fittings or permanently lubricated type. Connect grease fittings to tubing for serviceability from a single accessible point. Provide L50 rated bearing life at not less than the requirement of JIS B 1518 as defined by JIS B 1521 and JIS B 1534. Provide steel, accurately finished fan shafts, with key seats and keys for impeller hubs and fan pulleys. Provide fan outlets of ample proportions, designed for the attachment of angles and bolts for attaching flexible connections. Provide~~[[manually]~~ ~~[~~automatically~~]~~ operated inlet vanes on suction inlets. Provide ~~[manually] [~~automatically~~]~~ operated outlet dampers.~~]~~ Unless otherwise indicated, provide motors that do not exceed 1800 rpm and have ~~[open] [dripproof] [~~totally enclosed~~] [explosion-proof]~~ enclosures. ~~[~~ Provide ~~[manual] [~~magnetic~~]~~ ~~[~~across-the-line~~] [reduced-voltage-start]~~ type motor starters with ~~[~~general-purpose~~] [weather-resistant] [watertight]~~ enclosure.~~] [Provide remote manual switch with pilot indicating light where indicated.]~~

2.9.1.2 In-Line Centrifugal Fans

Provide in-line fans with centrifugal backward inclined blades, stationary discharge conversion vanes, internal and external belt guards, and adjustable motor mounts. Mount fans in a welded tubular casing. Provide a fan that axially flows the air in and out. Streamline inlets with conversion vanes to eliminate turbulence and provide smooth discharge air flow. Enclose and isolate fan bearings and drive shafts from the air stream. Provide precision, self aligning ball or roller type fan bearings that are sealed against dust and dirt and are permanently lubricated. Provide L50 rated bearing life at not less than the requirement of JIS B 1518 as defined by JIS B 1521 and JIS B 1534.~~[~~ Provide motors with ~~[open][dripproof][~~totally enclosed~~] [explosion-proof]~~ enclosure.~~]~~ ~~[~~Provide ~~[manual] [~~magnetic~~]~~ motor starters across-the-line with ~~[~~general-purpose~~] [weather-resistant] [explosion-proof]~~ enclosures.~~][Provide remote manual switch with pilot indicating light where indicated.]~~

~~2.9.1.3 Axial Flow Fans~~

~~Provide axial flow fans complete with drive components and belt guard,~~

~~with steel housing, cast fan wheel, cast or welded steel diffusers, fan shaft, bearings, and mounting frame as a factory-assembled unit. Provide fan wheels that are dynamically balanced and keyed to the fan shaft, with radially projecting blades of airfoil cross-section. Enclose and isolate fan bearings and drive shafts from the air stream. Permanently lubricate fan bearings or provide them with accessible grease fittings. Provide precision self-aligning ball or roller type fan bearings that are sealed against dust and dirt. Provide fan bearings that have a L50 rated bearing life at not less than the requirement of JIS B 1518 of operation as defined by and JIS B 1534. Provide fan inlets with an aerodynamically shaped bell and an inlet cone. Install diffuser or straightening vanes at the fan discharge to minimize turbulence and provide smooth discharge air flow. Furnish fan unit with [inlet and outlet flanges,] [inlet screen,] [duct equalizer section,] and [manual] [automatic] operation adjustable inlet vanes. Unless otherwise indicated, provide motors that do not exceed 1800 rpm and have [open] [dripproof] [totally enclosed] [explosion-proof] enclosure. [Provide [manual] [magnetic] motor starters across-the-line with [general-purpose] [weather-resistant] [explosion-proof] enclosure.] [Provide remote manual switch with pilot indicating light where indicated.]~~

2.9.1.3 ~~Centrifugal~~Propeller Type Power Wall Ventilators

Provide ~~[~~direct~~][or][V-belt]~~ driven centrifugal type fans with backward inclined, non-overloading wheel. Provide removable and weatherproof motor housing. Provide unit housing that is designed for sealing to building surface and for discharge and condensate drippage away from building surface. Construct housing of heavy gauge aluminum. Equip unit with an ~~[~~ aluminum or plated steel wire discharge bird screen,~~]~~ ~~[~~disconnect switch,~~]~~ ~~[[anodized aluminum][~~stainless steel~~]~~ wall grille,~~]~~ ~~[~~manufacturer's standard ~~[~~gravity~~][motor-operated]~~ damper,~~]~~ an airtight and liquid-tight metallic wall sleeve. Provide ~~[~~totally enclosed fan cooled~~] [dripproof] [explosion-proof]~~ type motor enclosure. Use only lubricated bearings.

2.9.1.4 Centrifugal Type Power Roof Ventilators

Provide ~~[~~direct~~][or][V-belt]~~ driven centrifugal type fans with backward inclined, non-overloading wheel. Provide hinged or removable and weatherproof motor compartment housing, constructed of heavy gauge aluminum. Provide fans with ~~[~~birdscreen,~~]~~ ~~[~~disconnect switch,~~] [[gravity] [~~motorized~~]~~ dampers,~~] [sound curb,]~~ ~~[~~roof curb~~,] and [extended base]~~. Provide ~~[~~dripproof~~] [explosion-proof]~~ type motor enclosure. Provide centrifugal type kitchen exhaust fans according to UL 705, fitted with V-belt drive, round hood, and windband upblast discharge configuration, integral residue trough and collection device, with motor and power transmission components located in outside positively air ventilated compartment. Use only lubricated bearings.

~~2.9.1.5 Propeller Type Power Roof Ventilators~~

~~Provide [direct][or][V-belt] driven fans. Provide hinged or removable weathertight fan housing, fitted with framed rectangular base constructed of aluminum or galvanized steel. Provide [totally enclosed fan cooled] [explosion-proof] type motors. Furnish motors with nonfusible, horsepower rated, manual disconnect mount on unit. Furnish fans with [gravity] [motor operated] dampers, [birdscreen][sound curb][roof curb]. Use only lubricated bearings.~~

~~2.9.1.6 Air-Curtain Fans~~

~~Furnish air curtains with a weatherproof housing constructed of high impact plastic or minimum 1.3 mm rigid welded steel. Provide backward curved, non-overloading, centrifugal type fan wheels, accurately balanced statically and dynamically. Provide motors with totally enclosed fan cooled enclosures. Provide remote manual type motor starters with weather-resistant enclosure actuated when the doorway served is open. Provide air curtains that attain the air velocities specified within 2 seconds following activation. Provide bird screens at air intake and discharge openings. Provide air curtain unit or a multiple unit installation that is at least as wide as the opening to be protected. Provide the air discharge openings to permit outward adjustment of the discharge air. Place installation and adjust according to the manufacturer's written recommendation. Furnish directional controls on air curtains for service windows for easy clean or convenient removal. Design air curtains to prevent the adjustment of the air velocities specified. Make the interior surfaces of the air curtain units accessible for cleaning. Provide certified test data indicating that the fan can provide the air velocities required when fan is mounted as indicated. Provide air curtains designed as fly fans unless otherwise indicated. [Provide air curtains designed for use in service entranceways that develop an air curtain not less than 75 mm thick at the discharge nozzle. Provide air velocity that is not less than 8 m/s across the entire entryway when measured 900 mm above the floor.] [Provide air curtains designed for use on customer entranceways that develop an air curtain not less than 200 mm thick at the discharge opening. Provide velocity that is not less than 3 m/s across the entire entryway when measured 900 mm above the floor. Equip recirculating type air curtains with readily removable filters, or design the filters for in-position cleaning. Provide readily accessible and easily cleanable air capture compartment or design for in-position cleaning.] [Provide air curtains designed for use on service windows that develop an air curtain not less than 200 mm thick at the discharge opening. Provide air velocity that is not less than 3 m/s across the entire opening of the service window measured 900 mm below the air discharge opening.]~~

2.9.1.5 Ceiling Exhaust Fans

Provide centrifugal type, direct driven suspended cabinet-type ceiling exhaust fans. Provide fans with acoustically insulated housing. Provide chatter-proof backdraft damper. Provide egg-crate design or louver design integral face grille. Mount fan motors on vibration isolators. Furnish unit with mounting flange for hanging unit from above.

2.9.2 Coils

Provide fin-and-tube type coils constructed of seamless ~~[~~copper~~][red brass]~~ tubes and ~~[~~aluminum~~][~~ or ~~][~~copper~~]~~ fins mechanically bonded or soldered to the tubes.~~[~~ Provide copper tube wall thickness that is a minimum of ~~[~~0.406~~][0.508][0.6096]~~ mm~~]~~.~~[Provide red brass tube wall thickness that is a minimum of [0.89][1.24] mm].~~ ~~[~~Provide aluminum fins that are ~~[~~0.14 ~~][0.19]~~ mm minimum thickness.~~][~~ Provide copper fins that are 0.114 mm minimum thickness.~~]~~ Provide casing and tube support sheets that are not lighter than 1.6 mm galvanized steel, formed to provide structural strength. When required, provide multiple tube supports to prevent tube sag. Mount coils for counterflow service. Rate and certify coils to meet the requirements of JIS B 8616.

2.9.2.1 Direct-Expansion Coils

Provide suitable direct-expansion coils for the refrigerant involved. Provide refrigerant piping that conforms to JIS H 3300 and clean, dehydrate and seal. Provide seamless copper tubing suction headers or seamless or resistance welded steel tube suction headers with copper connections. Provide supply headers that consist of a distributor which distributes the refrigerant through seamless copper tubing equally to all circuits in the coil. Provide circuited tubes to ensure minimum pressure drop and maximum heat transfer. Provide circuiting that permits refrigerant flow from inlet to suction outlet without causing oil slugging or restricting refrigerant flow in coil. Provide field installed coils which are completely dehydrated and sealed at the factory upon completion of pressure tests. Pressure test coils in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).

2.9.2.2 Water Coils

Install water coils with a pitch of not less than 10 mm/m of the tube length toward the drain end. Use headers constructed of cast iron, welded steel or copper. Furnish each coil with a plugged vent and drain connection extending through the unit casing. Provide removable water coils with drain pans. Pressure test coils in accordance with Japanese Refrigeration Safety Regulations(Nihon-Reitou-Hoan-Kisoku).

~~2.9.2.3 Steam Heating Coils~~

~~Construct steam coils from cast semisteel, welded steel or copper headers, and [red brass][copper] tubes. Construct headers from cast iron, welded steel or copper. Provide fin tube and header section that float within the casing to allow free expansion of tubing for coils subject to high pressure steam service. Provide each coil with a field or factory installed vacuum breaker. Provide single-tube type coils with tubes not less than 13 mm outside diameter, except for steam preheat coils. Provide supply headers that distribute steam evenly to all tubes at the indicated steam pressure. Factory test coils to ensure that, when supplied with a uniform face velocity, temperature across the leaving side is uniform with a maximum variation of no more than 5 percent. Pressure test coils in accordance with in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).~~

~~2.9.2.4 Steam Preheat (Nonfreeze) Coils~~

~~Provide steam-distribution-tube type steam (nonfreeze) coils with condensing tubes not less than 25 mm outside diameter for tube lengths 1.5 m and over and 13 mm outside diameter for tube lengths under 1.5 m. Construct headers from cast iron, welded steel, or copper. Provide distribution tubes that are not less than 15 mm outside diameter for tube lengths 1.5 m and over and 10 mm outside diameter for tube lengths under 1.5 m with orifices to discharge steam to condensing tubes. Install distribution tubes concentric inside of condensing tubes and hold securely in alignment. Limit maximum length of a single coil to 3.66 m. Factory test coils to ensure that, when supplied with a uniform face velocity, temperature across the leaving side is uniform with a maximum variation of no more than 5 percent. Pressure test coils in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).~~

2.9.2.3 Electric Heating Coil

Provide an electric duct heater coil in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). Provide duct- or unit-mounted coil. Provide ~~[nickel chromium resistor, single stage, strip] [~~nickel chromium resistor, single stage, strip or stainless steel, fin tubular~~]~~ type coil. Provide coil with a built-in or surface-mounted high-limit thermostat interlocked electrically so that the coil cannot be energized unless the fan is energized. Provide galvanized steel or aluminum coil casing and support brackets. Mount coil to eliminate noise from expansion and contraction and for complete accessibility for service.

~~2.9.2.4 Eliminators~~

~~Equip each cooling coil having an air velocity of over 2 m/s through the net face area with moisture eliminators, unless the coil manufacturer guarantees, over the signature of a responsible company official, that no moisture can be carried beyond the drip pans under actual conditions of operation. Construct of minimum 24 gage [zinc-coated steel] [copper] [copper nickel] [or] [stainless steel], removable through the nearest access door in the casing or ductwork. Provide eliminators that have not less than two bends at 45 degrees and are spaced not more than 63 mm center-to-center on face. Provide each bend with an integrally formed hook as indicated in the JIS A 4009.~~

~~2.9.2.5 Sprayed Coil Dehumidifiers~~

~~Provide assembly with reinforced, braced, and externally insulated galvanized steel casing, vertical in-line spray pump, bronze self-cleaning spray nozzles, galvanized steel pipe spray headers, adjustable float valve with replaceable neoprene seat, manufacturer's standard cooling coil, and welded black steel drain tank. Provide overflow drain, make-up, and bleed connection.~~

~~2.9.2.6 Corrosion Protection for Coastal Installations~~

2.9.3 Air Filters

List air filters according to requirements of JIS B 9908-1, except list high efficiency particulate air filters of 99.97 percent efficiency by the DOP Test method under the Label Service to meet the requirements of JIS B 9908-1.

2.9.3.1 Extended Surface Pleated Panel Filters

Provide 50 mm depth, sectional, disposable type filters of the size indicated with a MERV of 8 when tested according to JIS B 9908-1. Provide initial resistance at 2.54 m/s that does not exceed 0.09 kPa. Provide UL Class 2 filters, and nonwoven cotton and synthetic fiber mat media. Attach a wire support grid bonded to the media to a moisture resistant fiberboard frame. Bond all four edges of the filter media to the inside of the frame to prevent air bypass and increase rigidity.

2.9.3.2 Extended Surface Nonsupported Pocket Filters

Provide ~~[750][_____]~~ mm depth, sectional, replaceable dry media type filters of the size indicated when tested according to~~–~~ JIS B 9908-1. Provide initial resistance at ~~[~~2.54~~][______]~~ m/s that does not exceed ~~[~~ 0.1125~~][______]~~ kPa. Provide fibrous glass media, supported in the air stream by a wire or non-woven synthetic backing and secured to a galvanized steel metal header. Provide pockets that do not sag or flap at anticipated air flows. Install each filter ~~[with an extended surface pleated panel filter as a prefilter]~~ in a factory preassembled, side access housing or a factory-made sectional frame bank, as indicated.

2.9.3.3 Cartridge Type Filters

Provide 305 mm depth, sectional, replaceable dry media type filters of the size indicated when tested according to JIS B 9908-1. Provide initial resistance at ~~[~~2.54~~][______]~~ m/s that does not exceed ~~[~~0.14~~][______]~~ kPa. Provide JIS Z 4812, and pleated microglass paper media with corrugated aluminum separators, sealed inside the filter cell to form a totally rigid filter assembly. Fluctuations in filter face velocity or turbulent airflow have no effect on filter integrity or performance. Install each filter with an extended surface pleated media panel filter as a prefilter in a factory preassembled side access housing, or a factory-made sectional frame bank, as indicated.

2.9.3.4 Sectional Cleanable Filters

Provide ~~[25][50]~~ mm thick cleanable filters. Provide viscous adhesive in 20 L containers in sufficient quantity for 12 cleaning operations and not less than one L for each filter section. Provide one washing and charging tank for every 100 filter sections or fraction thereof; with each washing and charging unit consisting of a tank and ~~[~~single~~][double]~~ drain rack mounted on legs and drain rack with dividers and partitions to properly support the filters in the draining position.

2.9.3.5 Replaceable Media Filters

Provide the ~~[~~dry-media~~][viscous adhesive]~~ type replaceable media filters, of the size required to suit the application. Provide filtering media that is not less than 50 mm thick fibrous glass media pad supported by a structural wire grid or woven wire mesh. Enclose pad in a holding frame of not less than 1.6 mm galvanized steel, equipped with quick-opening mechanism for changing filter media. Base the air flow capacity of the filter on net filter face velocity not exceeding ~~[~~1.5~~][______]~~ m/s, with initial resistance of ~~[~~32~~][______]~~ Pa.

~~2.9.3.6 Automatic Renewable Media Filters~~

~~Provide the following:~~

~~a. Automatic, renewable media filters consisting of a horizontal or vertical traveling curtain of adhesive-coated bonded fibrous glass supplied in convenient roll form, and filter that does not require water supply, sewer connections, adhesive reservoir, or sprinkler equipment as part of the operation and maintenance requirements.~~

~~b. Basic frame that is fabricated of not less than 2 mm galvanized steel, and sectional design filters with each section of each filter fully factory assembled, requiring no field assembly other than setting in place next to any adjacent sections and the installation of media in roll form.~~

~~c. Each filter complete with initial loading of filter media drive motor adequate to handle the number of sections involved, and [painted steel] [stainless steel] control box containing a warning light to indicate media runout, a runout switch, and a Hand-Off-Auto selector switch.~~

~~d. Media feed across the filter face in [full-face increments] [increments] automatically controlled as determined by [filter pressure differential] [time interval control] [time interval control with pressure override] [photo electric control] to provide substantially constant operating resistance to airflow and varying not more than plus or minus 10 percent. Roll or enclose media in such a way that collected particulates can not re-entrain.~~

~~e. Rolls of clean media, no less than 19.8 m long, rerolled on disposable spools in the rewind section of the filter after the media has accumulated its design dirt load. Equip rewind section with a compression panel to tightly rewind used media for ease of handling. Provide media made of continuous, bonded fibrous glass material, UL Class 2, that does not compress more than 6 mm when subjected to air flow at 2.54 m/s. Factory charge media with an odorless and flame retardant adhesive which does not flow while in storage nor when subjected to temperatures up to 79.4 degrees C. Support media on both the leaving and entering air faces. Clean media must have initial resistance that does not exceed 45 Pa at its rated velocity of 2.54 m/s. Set control so that the resistance to air flow is between 100 and 125 Pa unless otherwise indicated.~~

~~f. Dust holding capacity, of 80 percent average arrestance under these operating conditions, when operating at a steady state with an upper operating resistance of 125 Pa, that is at least 592 (55) grams of ASHRAE Standard Test Dust per square meter of media area, when tested according to the dynamic testing provisions of JIS B 9908-1.~~

~~g. The horizontal type automatic renewable media filters, when used in conjunction with factory fabricated air handling units, that are dimensionally compatible with the connecting air handling units, and horizontal type filter housings with all exposed surfaces factory insulated internally with 25 mm, 24 kg/cubic meter density neoprene coated fibrous glass with thermal conductivity not greater than 0.04 W/m-K of thickness.~~

~~h. Access doors for horizontal filters with double wall construction as specified for plenums and casings for field-fabricated units in paragraph DUCT SYSTEMS.~~

~~2.9.3.7 Electrostatic Filters~~

~~Provide the following:~~

~~a. The combination dry agglomerator/extended surface, nonsupported pocket electrostatic filters or the combination dry agglomerator/automatic renewable, media (roll) type electrostatic filters, as indicated (except as modified). Supply each dry agglomerator electrostatic air filter with the correct quantity of fully housed power packs and equip with silicon rectifiers, manual reset circuit breakers, low voltage safety cutout, relays for field wiring to remote indication of primary and secondary voltages, with lamps mounted in the cover to indicate these functions locally. Equip power pack enclosure with external~~

~~mounting brackets, and low and high voltage terminals fully exposed with access cover removed for ease of installation. Furnish interlock safety switches for each access door and access panel that permits access to either side of the filter, so that the filter is de-energized in the event that a door or panel is opened.~~

~~b. Ozone generation within the filter that does not exceed five parts per one hundred million parts of air. Locate high voltage insulators in a serviceable location outside the moving air stream or on the clean air side of the unit. Fully expose ionizer wire supports and furnish ionizer wires precut to size and with formed loops at each end to facilitate ionizer wire replacement.~~

~~c. Agglomerator cell plates that allow proper air stream entrainment of agglomerates and prevent excessive residual dust build-up, with cells that are open at the top and bottom to prevent accumulation of agglomerates which settle by gravity. Where the dry agglomerator electrostatic filter is indicated to be the automatic renewable media type, provide a storage section that utilizes a horizontal or vertical traveling curtain of adhesive-coated bonded fibrous glass for dry agglomerator storage section service supplied in 19.8 m lengths in convenient roll form. Otherwise, provide section construction and roll media characteristics as specified for automatic renewable media filters. Also a dry agglomerator/renewable media combination with an initial air flow resistance, after installation of clean media, that does not exceed 62.3 Pa at 2.54 m/s face velocity.~~

~~d. Where the dry agglomerator electrostatic filter is indicated to be of the extended surface nonsupported pocket filter type, provide a storage section as specified for extended surface non-supported pocket filters, with sectional holding frames or side access housings as indicated.~~

~~e. A dry agglomerator/extended surface nonsupported pocket filter section combination with initial air flow resistance, after installation of clean filters, that does not exceed 162 Pa at 2.54 m/s face velocity. Furnish front access filters with full height air distribution baffles and upper and lower mounting tracks to permit the baffles to be moved for agglomerator cell inspection and service. When used in conjunction with factory fabricated air handling units, supply side access housings which have dimensional compatibility.~~

~~2.9.3.8 High-Efficiency Particulate Air (HEPA) Filters~~

~~Provide HEPA filters that meet the requirements of JIS Z 4812 and are individually tested and certified to have an efficiency of not less than [95] [99.97] percent in accordance with JIS B 9927, and an initial resistance at [_____] m/s that does not exceed [_____] Pa. Provide filters that are constructed by pleating a continuous sheet of filter medium into closely spaced pleats separated by corrugated aluminum or mineral-fiber inserts, strips of filter medium, or by honeycomb construction of the pleated filter medium. Provide interlocking, dovetailed, molded neoprene rubber gaskets of 5-10 durometer that are cemented to the perimeter of the [upstream] [downstream] face of the filter cell sides. Provide self-extinguishing rubber-base type adhesive or other materials conforming to fire hazard classification specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Provide filter cell sides that are [19 mm thick exterior grade fire-retardant plywood] [cadmium plated steel] [galvanized steel] assembled in a rigid~~

~~manner. Provide overall cell side dimensions that are correct to 2 mm, and squareness that is maintained to within 3.2 mm. Provide holding frames that use spring loaded fasteners or other devices to seal the filter tightly within it and that prevent any bypass leakage around the filter during its installed life. Provide air capacity and the nominal depth of the filter as indicated. Install each filter in a factory preassembled side access housing or a factory-made sectional supporting frame as indicated. Provide prefilters of the type, construction and efficiency indicated.~~

2.9.3.6 Holding Frames

Fabricate frames from not lighter than 1.6 mm sheet steel with rust-inhibitor coating. Equip each holding frame with suitable filter holding devices. Provide gasketed holding frame seats. Make all joints airtight.

2.9.3.7 Filter Gauges

Provide dial type filter gauges, diaphragm actuated draft for all filter stations, including those filters which are furnished as integral parts of factory fabricated air handling units. Provide gauges that are at least 98 mm in diameter, with white dials with black figures, and ~~[graduations] [~~ graduated in 0.0025 kPa,~~]~~ with a minimum range of 0.25 kPa beyond the specified final resistance for the filter bank on which each gauge is applied. Provide each gauge with a screw operated zero adjustment and two static pressure tips with integral compression fittings, two molded plastic vent valves, two 1.5 m minimum lengths of 6.35 mm diameter ~~[~~ aluminum~~] [vinyl]~~ tubing, and all hardware and accessories for gauge mounting.

2.10 AIR HANDLING UNITS

2.10.1 Factory-Fabricated Air Handling Units

Provide ~~[~~single-zone draw-through type~~][or][single-zone blow-through type][or][multizone blow-through type][blow-through double-deck type][blow-through triple deck type]~~ units as indicated. Units must include fans, coils, airtight insulated casing, ~~[~~prefilters,~~] [secondary filter sections,][~~ and ~~][diffuser sections where indicated,] [air blender] adjustable V-belt drives, belt guards for externally mounted motors, access sections where indicated, [mixing box] [combination sectional filter-mixing box,] [[pan][drysteam][spray type] humidifier,]~~ vibration-isolators, and appurtenances required for specified operation. Provide vibration isolators as indicated. Physical dimensions of each air handling unit must be suitable to fit space allotted to the unit with the capacity indicated. Provide air handling unit that is rated in accordance with JRA 4036.

2.10.1.1 Casings

Provide the following:

a. ~~[~~Casing sections ~~[[single] [~~50 mm double~~]~~ wall type~~] [as indicated]~~, constructed of a minimum 1.3 mm galvanized steel, or 1.3 mm corrosion-resisting sheet steel conforming to JIS G 4305.~~][~~ Inner casing of double-wall units that are a minimum one mm solid galvanized steel or corrosion-resisting sheet steel conforming to JIS G 4305.~~]~~ Design and construct casing with an integral insulated structural galvanized steel frame such that exterior panels are non-load bearing.

b. Individually removable exterior panels with standard tools. Removal must not affect the structural integrity of the unit. Furnish casings with access sections, according to paragraph AIR HANDLING UNITS, inspection doors, and access doors, all capable of opening a minimum of 90 degrees, as indicated.

c. Insulated, fully gasketed, double-wall type inspection and access doors, of a minimum 1.3 mm outer and one mm inner panels made of either galvanized steel or corrosion-resisting sheet steel conforming to JIS G 4305. Provide rigid doors with heavy duty hinges and latches. Inspection doors must be a minimum 300 mm wide by 300 mm high. Access doors must be a minimum 600 mm wide, the full height of the unit casing or a minimum of 1800 mm, whichever is less. ~~[Install a minimum 200 by 200 mm sealed glass window suitable for the intended application, in all access doors.]~~

d. Double-wall insulated type drain pan (thickness equal to exterior casing) constructed of 1.4 mm ~~[galvanized steel] [~~corrosion resisting sheet steel conforming to JIS G 4305.~~]~~ Construct drain pans water tight, treated to prevent corrosion, and designed for positive condensate drainage. When 2 or more cooling coils are used, with one stacked above the other, condensate from the upper coils must not flow across the face of lower coils. Provide intermediate drain pans or condensate collection channels and downspouts, as required to carry condensate to the unit drain pan out of the air stream and without moisture carryover. Construct drain pan to allow for easy visual inspection, including underneath the coil without removal of the coil and to allow complete and easy physical cleaning of the pan underneath the coil without removal of the coil. Provide coils that are individually removable from the casing.

e. Insulate single-wall casing sections handling conditioned air with not less than 25 mm thick, 24 kg/cubic meter density coated fibrous glass material having a thermal conductivity not greater than 0.033 W/m-K. Insulate double-wall casing sections handling conditioned air with not less than 50 mm of the same insulation specified for single-wall casings. Foil-faced insulation is not an acceptable substitute for use with double wall casing. Seal double wall insulation completely by inner and outer panels.

f. Factory applied fibrous glass insulation that conforms to JIS A 9511, except that the minimum thickness and density requirements do not apply. Make air handling unit casing insulation uniform over the entire casing. Foil-faced insulation is not an acceptable substitute for use on double-wall access doors and inspections doors ~~[~~and casing sections~~]~~.

g. Duct liner material, coating, and adhesive that conforms to fire-hazard requirements specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Protect exposed insulation edges and joints where insulation panels are butted with a metal nosing strip or coat to meet erosion resistance requirements of JIS A 9511.

h. A latched and hinged inspection door, in the fan and coil sections. Plus additional inspection doors, access doors and access sections ~~[______][~~ where indicated~~]~~.

2.10.1.2 Heating and Cooling Coils

Provide coils as specified in paragraph AIR SYSTEMS EQUIPMENT.

2.10.1.3 Air Filters

Provide air filters as specified in paragraph AIR SYSTEMS EQUIPMENT for types and thickness indicated.

2.10.1.4 Fans

Provide the following:

a. Fans that are double-inlet, centrifugal type with each fan in a separate scroll. Dynamically balance fans and shafts prior to installation into air handling unit, then after it has been installed in the air handling unit, statically and dynamically balance the entire fan assembly. Mount fans on steel shafts, accurately ground and finished.

b. Fan bearings that are sealed against dust and dirt and are precision self-aligning ball or roller type, with L50 rated bearing life at not less than 200,000 hours as defined by JIS B 1521 and JIS B 1534. Provide bearings that are permanently lubricated or lubricated type with lubrication fittings readily accessible at the drive side of the unit. Support bearings by structural shapes, or die formed sheet structural members, or support plates securely attached to the unit casing. Do not fasten bearings directly to the unit sheet metal casing. Furnish fans and scrolls with coating indicated.

c. Fans that are driven by a unit-mounted, or a floor-mounted motor connected to fans by V-belt drive complete with belt guard for externally mounted motors. Furnish belt guards that are the three-sided enclosed type with solid or expanded metal face. Design belt drives for not less than a 1.3 service factor based on motor nameplate rating.

d. ~~[~~Motor sheaves that are variable pitch for 20 kW and below and fixed pitch above 20 kW as defined by MLIT-M.~~]~~ Where fixed sheaves are required, the use of variable pitch sheaves is allowed during air balance, but replace them with an appropriate fixed sheave after air balance is completed. Select variable pitch sheaves to drive the fan at a speed that produces the specified capacity when set at the approximate midpoint of the sheave adjustment. Furnish motors for V-belt drives with adjustable bases, and with ~~[open][splashproof][~~ totally enclosed~~]~~ enclosures.

e. Motor starters of ~~[manual][~~magnetic~~][~~ across-the-line ~~][reduced-voltage-start]~~ type with ~~[~~general-purpose ~~][weather-resistant][watertight]~~ enclosure. Select unit fan or fans to produce the required capacity at the fan static pressure with sound power level as indicated. Obtain the sound power level values according to JIS B 8330, JIS B 8330.

2.10.1.5 Access Sections~~ and Filter/Mixing Boxes~~

Provide access sections where indicated and furnish with access doors as shown. Construct access sections ~~and filter/mixing boxes~~ in a manner identical to the remainder of the unit casing and equip with access doors.

~~Design mixing boxes to minimize air stratification and to promote thorough mixing of the air streams.~~

~~2.10.1.6 Diffuser Sections~~

~~Furnish diffuser sections between the discharge of all housed supply fans [and cooling coils of blow-through single zone units][and][filter sections of those units with high efficiency filters located immediately downstream of the air handling unit fan section]. Provide diffuser sections that are fabricated by the unit manufacturer in a manner identical to the remainder of the unit casing, designed to be airtight under positive static pressures up to [2][_____] kPa and with an access door on each side for inspection purposes. Provide a diffuser section that contains a perforated diffusion plate, fabricated of galvanized steel, Type 316 stainless steel, aluminum, or steel treated for corrosion with manufacturer's standard corrosion-resisting finish, and designed to accomplish uniform air flow across the down-stream [coil][filters] while reducing the higher fan outlet velocity to within plus or minus 5 percent of the required face velocity of the downstream component.~~

2.11 TERMINAL UNITS

2.11.1 Room Fan-Coil Units

Provide base units that include galvanized coil casing, coil assembly drain pan ~~[~~valve and piping package,~~] [outside air damper,] [wall intake box,]~~ air filter, fans, motor, fan drive, motor switch, an enclosure for cabinet models and casing for concealed models, leveling devices integral with the unit for vertical type units, and sound power levels as indicated. Obtain sound power level data or values for these units according to test procedures based on JIS B 8616. Sound power values apply to units provided with factory fabricated cabinet enclosures and standard grilles. Values obtained for the standard cabinet models are acceptable for concealed models without separate test provided there is no variation between models as to the coil configuration, blowers, motor speeds, or relative arrangement of parts. Provide automatic valves and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS. Fasten each unit securely to the building structure. Provide units with capacity indicated. Provide room fan-coil units that are certified as complying with MLIT-M, and meet the requirements of in accordance wiht Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).

2.11.1.1 Enclosures

Fabricate enclosures from not lighter than 1.3 mm steel, reinforced and braced. Provide enclosures with front panels that are removable and have 7 mm closed cell insulation or 13 mm thick dual density foil faced fibrous glass insulation. Make the exposed side of a high density, erosion-proof material suitable for use in air streams with velocities up to 23 m/s. Provide a discharge grille that is ~~[~~adjustable~~] [fixed]~~ and that is of such design as to properly distribute air throughout the conditioned space. Plastic discharge and return grilles are acceptable provided the plastic material is certified by the manufacturer to be classified as flame resistant and the material complies with the heat deflection criteria per in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). Provide galvanized or factory finished ferrous metal surfaces with corrosion resistant enamel, and access doors or removable panels for piping and control compartments, plus easy access for filter replacement. Provide duct discharge collar for concealed models.

2.11.1.2 Fans

Provide steel or aluminum, multiblade, centrifugal type fans. In lieu of metal, fans and scrolls could be of non-metallic materials of suitably reinforced compounds with smooth surfaces. Dynamically and statically balance the fans. Provide accessible assemblies for maintenance. Disassemble and re-assemble by means of mechanical fastening devices and not by epoxies or cements.

2.11.1.3 Coils

Fabricate coils from not less than 10 mm outside diameter seamless copper tubing, with copper or aluminum fins mechanically bonded or soldered to the tubes. Provide coils with not less than 13 mm outside diameter flare or sweat connectors, accessory piping package with thermal connections suitable for connection to the type of control valve supplied, and manual air vent. Test coils hydrostatically at 2000 kPa or under water at 1700 kPa air pressure. Provide coils suitable for 1400 kPa working pressure. Make provisions for coil removal.

2.11.1.4 Drain Pans

Size and locate drain and drip pans to collect all water condensed on and dripping from any item within the unit enclosure or casing. Provide condensate drain pans designed for self-drainage to preclude the buildup of microbial slime and thermally insulated to prevent condensation and constructed of not lighter than 0.9 mm type 304 stainless steel or noncorrosive ABS plastic. Provide insulation with a flame spread rating not over 25 without evidence of continued progressive combustion, a smoke developed rating no higher than 50, and of a waterproof type or coated with a waterproofing material. Design drain pans so as to allow no standing water and pitch to drain. Provide minimum 19 mm NPT or 15 mm OD drain connection in drain pan. Provide plastic or metal auxiliary drain pans to catch drips from control and piping packages, eliminating insulation of the packages; if metal, provide auxiliary pans that comply with the requirements specified above. Extend insulation at control and piping connections 25 mm minimum over the auxiliary drain pan.

2.11.1.5 Manually Operated Outside Air Dampers

Provide manually operated outside air dampers according to the arrangement indicated, and parallel airfoil type dampers of galvanized construction. Provide blades that rotate on stainless steel or nylon sleeve bearings.

2.11.1.6 Filters

Provide disposable type filter that complies with JIS B 9908-1. Provide filters in each unit that are removable without the use of tools.

2.11.1.7 Motors

Provide motors of the permanent split-capacitor type with built-in thermal overload protection, directly connected to unit fans. Provide motor switch with two or three speeds and off, manually operated, and mounted on an identified plate ~~[~~inside the unit below or behind an access door~~][or][adjacent to the room thermostat][as indicated]~~. In lieu of the above fan speed control, a solid-state variable-speed controller having a minimum speed reduction of 50 percent is allowed. Provide motors with permanently-lubricated or oilable sleeve-type or combination ball and sleeve-type bearings with vibration isolating mountings suitable for continuous duty. Provide a motor power consumption, shown in watts, at the fan operating speed selected to meet the specified capacity that does not exceed the following values:

Free Discharge Motors			
Unit Capacity (L/S)	Maximum Power Consumption (Watts)		
	115V	230V	277V
94	70	110	90
142	100	110	110
189	170	150	150
283	180	210	220
378	240	240	230
472	310	250	270
566	440	400	440

High Static Motors	
Unit Capacity (L/S)	Maximum Power Consumption (Watts)
94	145
142	145
189	210
283	320
378	320
472	530
566	530

~~2.11.2 Coil Induction Units~~

~~Provide base unit that includes air plenums, air-discharge nozzles, air discharge grilles, recirculation grilles, water coil assembly, valve and piping package, condensate drain pan, and adjustable air-balancing dampers, plus an enclosure for cabinet models and casing for concealed~~

~~models. Make each unit capable of producing not less than the capacity indicated without exceeding the indicated static pressure. Provide a sound power level as indicated with power level data or values for these units based on tests conducted according to or JIS Z 8734. Sound power values apply to units provided with factory fabricated cabinet enclosures and standard grilles. The values obtained for the standard cabinet models are acceptable for concealed models without separate tests, provided there is no variation between models as to coil configuration, air discharge nozzles, air balancing dampers, or relative arrangement of parts. Provide automatic valves and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS. Secure each unit to the building structure. Provide units with capacity indicated.~~

~~2.11.2.1 Enclosures~~

~~Fabricate enclosures from not lighter than 1.2 mm steel, reinforced and braced. Provide a removable front panel of enclosure and insulate when required acoustically and to prevent condensation. Provide discharge grilles that are [adjustable][integrally stamped] and properly distribute air throughout the conditioned space. Plastic discharge and return grilles are not acceptable. Provide access doors for all piping and control compartments.~~

~~2.11.2.2 Air Plenums~~

~~Fabricate plenums from galvanized steel with interior acoustically baffled and lined with sound absorbing material to attenuate the sound power from the primary air supply to the room. Provide heat-resistant nozzles that are integral with or attached airtight to the plenum. Where coil induction units are supplied with vertical runouts, furnish a streamlined, vaned, mitered elbow transition piece for connection between the unit and ductwork. Provide an adjustable air-balancing damper in each unit.~~

~~2.11.2.3 Coils~~

~~Fabricate coils from not less than 10 mm outside diameter seamless copper tubing, with copper or aluminum fins, mechanically bonded or soldered to the tubes. Furnish coil connections with not less than 13 mm outside diameter flare or sweat connectors, accessory piping package with terminal connections suitable for connection to the type of control valve supplied, and manual air vent. Test coils hydrostatically at 2000 kPa or under water at 1700 kPa air pressure and provide coils suitable for 1400 kPa working pressure.~~

~~2.11.2.4 Screens~~

~~Provide easily accessible lint screens or throwaway filters for each unit.~~

~~2.11.2.5 Drain Pan~~

~~Size and locate drain and drip pans to collect condensed water dripping from any item within the unit enclosure. Provide drain pans constructed of not lighter than 0.9 mm steel, galvanized after fabrication, and thermally insulated to prevent condensation. Provide insulation that has a flame spread rating not over 25 without evidence of continued progressive combustion, a smoke developed rating no higher than 50, and that is a waterproof type or coated with a waterproofing material. In lieu of the above, drain pans constructed of die-formed 0.8 mm steel are allowed, formed from a single sheet and galvanized after fabrication and~~

~~insulated and coated as for the 0.9 mm steel material or of die-formed 0.9 mm type 304 stainless steel insulated as specified above. Pitch drain pans to drain. Provide drain connection when a condensate drain system is indicated. Make connection a minimum 19 mm NPT or 15 mm OD.~~

2.11.2 Variable Air Volume (VAV) and Dual Duct Terminal Units

~~a. Provide VAV and dual duct terminal units that are the type, size, and capacity shown, mounted in the ceiling or wall cavity, plus units that are suitable for single or dual duct system applications. Provide actuators and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS. For each VAV terminal unit, provide a temperature sensor in the unit discharge ductwork.~~

~~b. Provide unit enclosures that are constructed of galvanized steel not lighter than 0.85 mm or aluminum sheet not lighter than 1.3 mm. Provide single or multiple discharge outlets as required. Units with flow limiters are not acceptable. Provide unit air volume that is factory preset and readily field adjustable without special tools. Provide reheat coils as indicated.~~

~~c. Attach a flow chart to each unit. Base acoustic performance of the terminal units upon units tested according to JRA 4036 with the calculations prepared in accordance with JIS B 8616. Provide sound power level as indicated. Show discharge sound power for minimum and [375][_____] Pa inlet static pressure.~~

~~2.11.2.1 Constant Volume, Single Duct Terminal Units~~

~~Provide constant volume, single duct, terminal units that contain within the casing, a constant volume regulator. Provide volume regulators that control air delivery to within plus or minus 5 percent of specified air flow subjected to inlet pressure from 200 to 1500 Pa.~~

~~2.11.2.2 Variable Volume, Single Duct Terminal Units~~

~~Provide variable volume, single duct, terminal units with a calibrated air volume sensing device, air valve or damper, actuator, and accessory relays. Provide units that control air volume to within plus or minus 5 percent of each air set point volume as determined by the thermostat with variations in inlet pressures from 200 to 1500 Pa. Provide units with an internal resistance not exceeding 100 Pa at maximum flow range. Provide external differential pressure taps separate from the control pressure taps for air flow measurement with a 0 to 250 Pa range.~~

~~2.11.2.3 Variable Volume, Single Duct, Fan-Powered Terminal Units~~

~~Provide variable volume, single duct, fan-powered terminal units with a calibrated air volume sensing device, air valve or damper, actuator, fan and motor, and accessory relays. Provide units that control primary air volume to within plus or minus 5 percent of each air set point as determined by the thermostat with variations in inlet pressure from 200 to 1500 Pa. Provide unit fan that is centrifugal, direct-driven, double-inlet type with forward curved blades. Provide either single speed with speed controller or three-speed, permanently lubricated, permanent split-capacitor type fan motor. Isolate fan/motor assembly from the casing to minimize vibration transmission. Provide factory furnished fan control that is wired into the unit control system. Provide a factory-mounted pressure switch to operate the unit fan whenever pressure~~

~~exists at the unit primary air inlet or when the control system fan operates.~~

~~2.11.2.4 Dual Duct Terminal Units~~

~~Provide dual duct terminal units with hot and cold inlet valve or dampers that are controlled in unison by single or dual actuators. Provide actuator as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS. Provide unit that controls delivered air volumes within plus or minus 5 percent with inlet air variations from 250 to 2000 Pa in either duct. Include mixing baffles with the unit casing. Provide cabinet and closed duct leakage that does not exceed 2 percent of maximum rated air volume. Provide units with an internal resistance that does not exceed [_____] Pa at maximum flow range.~~

2.11.2.1 Reheat Units

~~2.11.2.1.1 Hot Water Coils~~

~~Provide fin-and-tube type hot-water coils constructed of seamless copper tubes and copper or aluminum fins mechanically bonded or soldered to the tubes. Provide headers that are constructed of cast iron, welded steel or copper. Provide casing and tube support sheets that are 1.6 mm, galvanized steel, formed to provide structural strength. Provide tubes that are correctly circuited for proper water velocity without excessive pressure drop and are drainable where required or indicated. At the factory, test each coil at not less than 1700 kPa air pressure and provide coils suitable for 1400 kPa working pressure. Install drainable coils in the air handling units with a pitch of not less than 10 mm per m of tube length toward the drain end. Coils must conform to the provisions of JIS B 8616.~~

~~2.11.2.1.2 Steam Coils~~

~~Provide steam coils constructed of cast semisteel, welded steel, or copper headers, red-brass or copper tubes, and copper or aluminum fins mechanically bonded or soldered to the tubes. Roll and bush, braze or weld tubes into headers. Provide coil casings and tube support sheets, with collars of ample width, that are not lighter than 1.6 mm galvanized steel formed to provide structural strength. When required, furnish multiple tube supports to prevent tube sag. Float the fin tube and header section within the casing to allow free expansion of tubing for coils subject to high pressure steam service. Provide coils that are factory pressure tested and capable of withstanding 1700 kPa hydrostatic test pressure or 1400 kPa air pressure, and are for [700] [1400] kPa steam working pressure. Provide steam-distribution tube type preheat coils with condensing tubes having not less than 15 mm outside diameters. Provide distribution tubes that have not less than 10 mm outside diameter, with orifices to discharge steam to condensing tubes. Install distribution tubes concentric inside of condensing tubes held securely in alignment. Limit the maximum length of a single coil to 120 times the diameter of the outside tube. Other heating coils must be single tube type with an outside diameter not less than 13 mm. Provide supply headers that distribute steam evenly to all tubes at the indicated steam pressure. Provide coils that conform to the provisions of JIS B 8616.~~

2.11.2.1.1 Electric Resistance Heaters

Provide the duct-mounting type electric resistance heaters consisting of a nickel-chromium resistor mounted on refractory material and a steel or aluminum frame for attachment to ductwork. Provide electric duct heater that meets the requirement of JIS C 9803 and is provided with a built-in or surface-mounted high-limit thermostat. Interlock electric duct heaters electrically so that they cannot be energized unless the fan is running.

2.11.3 Unit Ventilators

~~Provide unit ventilators that include an enclosure, [galvanized casing,] [cold-rolled steel casing with corrosion resistant coating,] coil assembly, [resistance heating coil assembly,] [valve and piping package,] drain pan, air filters, fan assembly, fan drive, motor, motor controller, dampers, damper operators, and sound power level as indicated. Obtain sound power level data or values for these units according to test procedures based on MLIT-M. Sound power values apply to units provided with factory fabricated cabinet enclosures and standard grilles, when handling standard flow for which the unit air capacity is rated. Secure each unit to the building structure. Provide the unit ventilators with capacity indicated. Provide the year-round classroom type unit ventilator with automatic controls arranged to properly heat, cool, and ventilate the room. Provide automatic valves and controls as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.~~

~~2.11.3.1 Enclosures~~

~~Fabricate enclosures from not lighter than 1.6 mm galvanized steel, reinforced and braced, or all welded framework with panels to provide equivalent strength. Provide casing that is acoustically and thermally insulated internally with not less than 13 mm thick dual density fibrous glass insulation. Make the exposed side a high density, erosion-proof material suitable for use in air streams with velocities up to 246 m/s. Fasten the insulation with waterproof, fire-resistant adhesive. Design front panel for easy removal by one person. Provide discharge grilles that [have adjustable grilles or grilles with adjustable vanes and] properly distribute air throughout the conditioned space. Provide return grilles that are removable where front panel does not provide access to interior components. Plastic discharge or return grilles are not acceptable. Furnish removable panels or access doors for all piping and control compartments. Provide fan switch that is key operated or accessible through a locked access panel. Install gaskets at the back and bottom of the unit for effective air seal, as required.~~

~~2.11.3.2 Electric Resistance Heating Elements~~

~~Provide electric resistance heating elements that are of the sheathed, finned, tubular type, or of the open resistance type designed for direct exposure to the air stream. Provide heating element electrical characteristics as indicated. Where fan motor or control voltage is lower than required for the electric-resistance heating element, install a fused factory mounted and wired transformer.~~

2.11.3.1 Fans

Provide fans that meet the requirements as specified in paragraph AIR SYSTEMS EQUIPMENT. Provide galvanized steel or aluminum, multiblade, centrifugal type fans, dynamically and statically balanced. Equip fan housings with resilient mounted, self-aligning permanently lubricated ball bearings, sleeve bearings, or combination ball and sleeve bearings, capable of not less than 2000 hours of operation on one oiling. Provide direct-connected fans.

2.11.3.2 Coils

Provide coils that are circuited for a maximum water velocity of 2.4 m/s without excessive pressure drop and are otherwise as specified for hot water coils in paragraph TERMINAL UNITS.

2.11.3.3 Drain Pans

Size and locate drain and drip pans to collect all condensed water dripping from any item within the unit enclosure. Provide drain pans constructed of not lighter than 1.2 mm steel, galvanized after fabrication, and thermally insulated to prevent condensation. Provide insulation that is coated with a fire-resistant waterproofing material. In lieu of the above, drain pans constructed of die-formed 1.0 mm steel is allowed, formed from a single sheet and galvanized after fabrication and insulated and coated as for the 1.3 mm steel material, or of die-formed 1.3 mm type 304 stainless steel insulated as specified above. Pitch drain pans to drain. Furnish drain connection unless otherwise indicated. Make the minimum connection 19 mm NDT or 18 mm OD.

2.11.3.4 Filters

Disposable type rated in accordance with ASHRAE 52.2, installed upstream of coil.

2.11.3.5 Dampers

Provide an outside air proportioning damper on each unit. In addition, provide a vane to prevent excessive outside air from entering unit and to prevent blow-through of outside air through the return air grille under high wind pressures. Where outside air and recirculated air proportioning dampers are provided on the unit, an additional vane is not required. Provide face and bypass dampers for each unit to ensure constant air volume at all positions of the dampers. Furnish each unit with a factory installed control cam assembly~~, pneumatic motor,~~ or electric motor to operate the face and bypass dampers and outside air damper or outside air and recirculated air dampers in the sequence as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.

2.11.3.6 Motors

Provide permanent split-capacitor type motors with built-in thermal overload protection and automatic reset. Mount motor on a resilient mounting, isolated from the casing and suitable for operation on electric service available. Provide a manually operated motor switch that provides for 2 or 3 speeds and off, mounted on an identified plate ~~[~~inside the unit below or behind an access door~~][or][adjacent to the room thermostat][as indicated]~~. In lieu of speed control, provide a solid state variable speed controller having minimum speed reduction of 50 percent.

2.11.3.7 Outside Air Intakes

Provide the manufacturer's standard design outside air intakes furnished with 13 mm mesh bird screen or louvers on 13 mm centers.

2.12 ENERGY RECOVERY DEVICES

2.12.1 Rotary Wheel

~~Provide unit that is a factory fabricated and tested assembly for air-to-air energy recovery by transfer of sensible heat from exhaust air to supply air stream, with device performance according to JIS B 8628 and that delivers an energy transfer effectiveness of not less than [70][85][_____] percent with cross-contamination not in excess of [0.1][1.0][_____] percent of exhaust airflow rate at system design differential pressure, including purging sector if provided with wheel. Provide exchange media that is chemically inert, moisture-resistant, fire-retardant, laminated, nonmetallic material which complies with NFPA 90A. Isolate exhaust and supply streams by seals which are static, field adjustable, and replaceable. Equip chain drive mechanisms with ratcheting torque limiter or slip-clutch protective device. Fabricate enclosure from galvanized steel and include provisions for maintenance access. Provide recovery control and rotation failure provisions as indicated.~~

~~2.12.2 Run-Around-Coil~~

~~Provide assembly that is factory fabricated and tested air-to-liquid-to-air energy recovery system for transfer of sensible heat from exhaust air to supply air stream and that delivers an energy transfer effectiveness not less than that indicated without cross-contamination with maximum energy recovery at minimum life cycle cost. Computer optimize components for capacity, effectiveness, number of coil fins per inch, number of coil rows, flow rate, heat transfer rate of [_____] percent by volume of [ethylene][propylene] glycol solution, and frost control. Provide coils that conform to paragraph AIR HANDLING UNITS. Provide related pumps, and piping specialties that conform to requirements of [Section 23 63 00.00 10 COLD STORAGE REFRIGERATION SYSTEMS][Section 23 57 10.00 10 FORCED HOT WATER HEATING SYSTEMS USING WATER AND STEAM HEAT EXCHANGERS][23 69 00.00 20 REFRIGERATION EQUIPMENT FOR COLD STORAGE] [_____].~~

~~2.12.3 Heat Pipe~~

~~Provide a device that is a factory fabricated, assembled and tested, counterflow arrangement, air-to-air heat exchanger for transfer of sensible heat between exhaust and supply streams and that delivers an energy transfer effectiveness not less than that indicated without cross-contamination. Provide heat exchanger tube core that is [15][18][25] mm nominal diameter, seamless aluminum or copper tube with extended surfaces, utilizing wrought aluminum Alloy 3003 or Alloy 5052, temper to suit. Provide maximum fins per unit length and number of tube rows as indicated. Provide tubes that are fitted with internal capillary wick, filled with a refrigerant complying with in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku), selected for system design temperature range, and hermetically sealed. Refrigerants containing chlorofluorocarbons (CFC) are prohibited. Provide heat exchanger frame that is constructed of not less than 1.6 mm galvanized steel and fitted with intermediate tube supports, and flange connections. Provide tube end-covers and a partition of galvanized steel to separate exhaust and supply air streams without cross-contamination and in required area ratio.[Provide a drain pan constructed of welded Type 300 series stainless steel.] Provide heat recovery regulation by [system face and bypass dampers and related control system as indicated][interfacing with~~

~~manufacturer's standard tilt-control mechanism for summer/winter operation, regulating the supply air temperature and frost prevention on weather face of exhaust side at temperature indicated]. Coil must be fitted with pleated flexible connectors.~~

~~2.12.4 Desiccant Wheel~~

~~Provide counterflow supply, regeneration airstreams, a rotary type dehumidifier designed for continuous operation, and extended surface type wheel structure in the axial flow direction with a geometry that allows for laminar flow over the operating range for minimum air pressure differentials. Provide the dehumidifier complete with a drive system utilizing a fractional-horsepower electric motor and speed reducer assembly driving the rotor. Include a slack-side tensioner for automatic take-up for belt-driven wheels. Provide an adsorbing type desiccant material. Apply the desiccant material to the wheel such that the entire surface is active as a desiccant and the desiccant material does not degrade or detach from the surface of the wheel which is fitted with full-face, low-friction contact seals on both sides to prevent cross leakage. Provide rotary structure that has underheat, overheat and rotation fault circuitry. Provide wheel assembly with a warranty for a minimum of five years.~~

2.12.2 Plate Heat Exchanger

Provide energy recovery ventilator unit that is factory-fabricated for indoor installation, consisting of a flat plate cross-flow heat exchanger, cooling coil, supply air fan and motor and exhaust air fan and motor. The casing must be 1 mm G90, galvanized steel, double wall construction with 25 mm insulation. Provide fibrous desiccant cross-flow type heat exchanger core capable of easy removal from the unit.

2.13 FACTORY PAINTING

Factory paint new equipment, which are not of galvanized construction. Paint with a corrosion resisting paint finish according to JIS H 8641 or JIS G 3302. Clean, phosphatize and coat internal and external ferrous metal surfaces with a paint finish which has been tested according to JIS Z 2371 JIS K 5600-7-9, and JIS K 5600-5-5 or JIS K 5600-5-6. Submit evidence of satisfactory paint performance for a minimum of 125 hours for units to be installed indoors and 500 hours for units to be installed outdoors. Provide rating of failure at the scribe mark that is not less than 6, average creepage not greater than 3 mm. Provide rating of the inscribed area that is not less than 10, no failure. On units constructed of galvanized steel that have been welded, provide a final shop docket of zinc-rich protective paint on exterior surfaces of welds or welds that have burned through from the interior according to JIS H 8641.

Field paint factory painting that has been damaged prior to acceptance by the Contracting Officer in compliance with the requirements of paragraph FIELD PAINTING OF MECHANICAL EQUIPMENT.

2.14 SUPPLEMENTAL COMPONENTS/SERVICES

2.14.1 Chilled, Condenser, or Dual Service Water Piping

The requirements for chilled, condenser, or dual service water piping and accessories are specified in Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS

2.14.2 Refrigerant Piping

The requirements for refrigerant piping are specified in Section 23 23 00 REFRIGERANT PIPING.

2.14.3 Water or Steam Heating System Accessories

The requirements for water or steam heating accessories such as expansion tanks and steam traps are specified in Section ~~[23 52 00 HEATING BOILERS][23 21 13.00 20 LOW TEMPERATURE WATER (LTW) HEATING SYSTEM][23 22 26.00 20 STEAM SYSTEM AND TERMINAL UNITS]~~23 57 10.00 10 FORCED HOT WATER HEATING SYSTEMS USING WATER AND STEAM HEAT EXCHANGERS.

2.14.4 Condensate Drain Lines

Provide and install condensate drainage for each item of equipment that generates condensate in accordance with Section ~~[22 00 00 PLUMBING, GENERAL PURPOSE][~~23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS~~]~~ except as modified herein.

2.14.5 Backflow Preventers

The requirements for backflow preventers are specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE.

2.14.6 Insulation

The requirements for shop and field applied insulation are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.14.7 Controls

The requirements for controls are specified in ~~[~~Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS~~][~~ and ~~][~~Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC~~][and][Section 23 09 53.00 20 SPACE TEMPERATURE CONTROL SYSTEMS]~~.

~~2.15 RADIANT PANELS~~

~~2.15.1 Hydronic Modular Panels~~

~~2.15.1.1 Panels~~

~~Modular radiant panels will fit into a standard 600 mm x 600 mm or 600 mm x 1200 mm suspended T-Bar ceiling grid or flush mounted on a drywall ceiling. For flush mounted ceiling applications, the manufacturer will provide a one piece extruded aluminum frame. Panels shall be supported from the T-bar assembly. Panels shall be [14 gauge] or [16 gauge] extruded aluminum or sheet steel.~~

~~2.15.1.2 Heat Sink~~

~~The modular panels shall use extruded aluminum with integrated heat sinks on the back to transfer heat between copper tubes and the panel face.~~

~~2.15.1.3 Water Tubes~~

~~Tubes shall consist of JIS H 3300 [13 mm] [16 mm] O.D. nominal copper tubing. Water connections will be suitable for solder or compression fittings. Heat pads will be used between the soldered fitting and the panel to protect the panel surface. The manufacturer will provide water pressure drop data as well as heating output data derived from tests in accordance with JIS A 1400 (heating). The panels will have the capacity to have multiple passes with connections either on the [same end] or [opposite ends], dependent on the number of passes.~~

~~2.15.1.4 Finish~~

~~All visible components shall be powder coated with highly emissive powder coat polyester paint for optimal radiative properties as well as durability and easy cleaning. Standard finish color shall be white.~~

~~2.15.1.5 Performance~~

~~Manufacturer will provide water pressure drop data as well as heat and cool output data derived from tests in accordance with JIS A 1400 (heating).~~

~~2.15.1.6 Capacity~~

~~Modular radiant panel capacity will be tested and certified by manufacturer in accordance with JIS A 1400 (heating) to meet the required performance. Should any performance rating, chilled or hot water supply temperature, water pressure drop, etc. deviate from the schedule, the manufacturer will submit the updated capacity. [The manufacturer will have factory testing facility available to perform performance test of units in accordance with said standard.]~~

~~2.15.1.7 Water Connections~~

~~Connections will be shipped sealed to limit the introduction of dust and dirt during shipping and construction.~~

~~2.15.1.8 Installation~~

~~Panels will be installed as recommended by the manufacturer.~~

~~2.15.1.9 Accessories~~

~~Stainless steel braded hoses, 300 mm or 450 mm long will be supplied with the panels.~~

~~The top of the heating and cooling panels shall be covered with 38 mm thick 16kg/m3 formaldehyde-free fiber glass insulation with a minimum R = 0.79 m2 deg C/W. The insulation shall be covered with a foil scrim kraft vapor barrier facing.~~

~~2.15.2 Hydronic Linear Panels~~

~~2.15.2.1 Panels~~

~~Linear radiant panels must use extruded aluminum with integrated heat sinks on the back to transfer heat between copper tubes and the panel face. The linear radiant panel is to radiate or absorb heat from or to the~~

~~zone below. Panels must be [14 gauge] or [16 gauge] extruded aluminum.~~

~~2.15.2.2 Heat Sink~~

~~The modular panels must use extruded aluminum with integrated heat sinks on the back to transfer heat between copper tubes and the panel face.~~

~~2.15.2.3 Water Tubes~~

~~Tubes must consist of JIS H 3300 13 mm or 16mm O.D. nominal copper tubing. Water connections will be suitable for solder or compression fittings. The manufacturer will provide water pressure drop data as well as heating output data derived from tests in accordance with JIS A 1400 (heating).~~

~~2.15.2.4 Mounting~~

~~Units must be provided with mounting hardware as required for mounting in T-Bar applications or ceiling flush mounting. The manufacturer's standard hardware for mounting panels abutting each other must be submitted for approval.~~

~~2.15.2.5 Finish~~

~~All visible components must be powder coated with highly emissive powder coat polyester paint for optimal radiative properties as well as durability and easy cleaning. Standard finish color must be white.~~

~~2.15.2.6 Performance~~

~~Manufacturer must provide water pressure drop data as well as heat output data derived from tests in accordance with JIS A 1400(heating).~~

~~2.15.2.7 Capacity~~

~~Modular radiant panel capacity must be tested and certified by manufacturer in accordance with JIS A 1400(heating) to meet the required performance. Should any performance rating, chilled or hot water supply temperature, water pressure drop, etc. deviate from the schedule, the manufacturer must submit the updated capacity. The manufacturer must have factory testing facility available to perform performance test of units in accordance with said standard.~~

~~2.15.2.8 Water Connections~~

~~Connections will be shipped sealed to limit the introduction of dust and dirt during shipping and construction.~~

~~2.15.2.9 Accessories~~

~~Stainless steel braded hoses, 300 mm or 450 mm long will be supplied with the panels.~~

~~The top of the heating and cooling panels must be covered with 38 mm thick 16kg/m3 formaldehyde-free fiber glass insulation with a minimum R = 0.79 m2 deg C/W. The insulation must be covered with a foil scrim kraft vapor barrier facing.~~

~~2.15.3 Prefabricated Radiant-Heating Electric Panels~~

~~2.15.3.1 Description~~

~~Sheet metal enclosed panel with heating element suitable for [lay-in installation flush with T-bar ceiling grid] [or surface mounting] [or recessed mounting].~~

~~2.15.3.2 Panel~~

~~Minimum 0.7 mm thick, galvanized steel sheet back panel riveted to minimum 1.0 mm thick, galvanized steel sheet front panel with fused-on crystalline surface.~~

~~2.15.3.3 Heating Element~~

~~Powdered graphite sandwiched between sheets of electric insulation.~~

~~2.15.3.4 Electrical Connections~~

~~Nonheating, high-temperature, insulated-copper leads, factory connected to heating element.~~

~~2.15.3.5 Exposed-Side Panel Finish~~

~~[Apply silk-screened finish to match appearance of Architect selected acoustical ceiling tiles.] [Baked-enamel finish in color as selected by Architect.]~~

~~2.15.3.6 Surface-Mounting Trim~~

~~Sheet metal with baked-enamel finish in color as selected by Architect.~~

~~2.15.3.7 Wall Thermostat~~

~~Bimetal, sensing elements; with contacts suitable for [low] [line]-voltage circuit, and manually operated on-off switch with contactors, relays, and control transformers.~~

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

3.2 INSTALLATION

a. Install materials and equipment in accordance with the requirements of the contract drawings and approved manufacturer's installation instructions. Accomplish installation by workers skilled in this type of work. Perform installation so that there is no degradation of the designed fire ratings of walls, partitions, ceilings, and floors.

b. No installation is permitted to block or otherwise impede access to any existing machine or system. Install all hinged doors to swing open a minimum of 120 degrees. Provide an area in front of all access doors that clears a minimum of ~~[910][_____]~~ mm. In front of all access doors to electrical circuits, clear the area the minimum distance to energized circuits JIS HB 71 (Electrical-Safety Related work practices)and an additional ~~[910][_____]~~ mm.

c. Except as otherwise indicated, install emergency switches and alarms in conspicuous locations. Mount all indicators, to include gauges, meters, and alarms in order to be easily visible by people in the area.

3.2.1 Condensate Drain Lines

Provide water seals in the condensate drain from all ~~[~~units~~]~~. Provide a depth of each seal of 50 mm plus 0.1 mm for each Pa, of the total static pressure rating of the unit to which the drain is connected. Provide water seals that are constructed of 2 tees and an appropriate U-bend with the open end of each tee plugged. Provide pipe cap or plug cleanouts where indicated. Connect drains indicated to connect to the sanitary waste system using an indirect waste fitting. Insulate air conditioner drain lines as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

3.2.2 Equipment and Installation

Provide frames and supports for tanks, compressors, pumps, valves, air handling units, fans, coils, dampers, and other similar items requiring supports. Floor mount or ceiling hang air handling units as indicated. Anchor and fasten as detailed. Set floor-mounted equipment on not less than 150 mm concrete pads or curbs doweled in place unless otherwise indicated. Make concrete foundations heavy enough to minimize the intensity of the vibrations transmitted to the piping, duct work and the surrounding structure, as recommended in writing by the equipment manufacturer. In lieu of a concrete pad foundation, build a concrete pedestal block with isolators placed between the pedestal block and the floor. Make the concrete foundation or concrete pedestal block a mass not less than three times the weight of the components to be supported. Provide the lines connected to the pump mounted on pedestal blocks with flexible connectors. Submit foundation drawings as specified in paragraph DETAIL DRAWINGS. Provide concrete for foundations as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.2.3 Access Panels

Install access panels for concealed valves, vents, controls, dampers, and items requiring inspection or maintenance of sufficient size, and locate them so that the concealed items are easily serviced and maintained or completely removed and replaced. Provide access panels as specified in Section ~~05 50 13 MISCELLANEOUS METAL FABRICATIONS~~08 31 00 ACCESS DOORS AND PANELS.

3.2.4 Flexible Duct

Install pre-insulated flexible duct in accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint. Provide hangers, when required to suspend the duct, of the type recommended by the duct manufacturer and set at the intervals recommended.

3.2.5 Metal Ductwork

Install according to MLIT-M unless otherwise indicated. Install duct supports for sheet metal ductwork according to MLIT-M, unless otherwise specified. Do not use friction beam clamps indicated in MLIT-M. Anchor risers on high velocity ducts in the center of the vertical run to allow ends of riser to move due to thermal expansion. Erect supports on the risers that allow free vertical movement of the duct. Attach supports only to structural framing members and concrete slabs. Do not anchor supports to metal decking unless a means is provided and approved for preventing the anchor from puncturing the metal decking. Where supports are required between structural framing members, provide suitable intermediate metal framing. Where C-clamps are used, provide retainer clips.

~~3.2.5.1 Underground Ductwork~~

~~Provide PVC plastisol coated galvanized steel underground ductwork with coating on interior and exterior surfaces and watertight joints. Install ductwork as indicated. Maximum burial depth is 2 m.~~

~~3.2.5.2 Radon Exhaust Ductwork~~

~~Perforate subslab suction piping where indicated. Install PVC joints.~~

~~3.2.5.3 Light Duty Corrosive Exhaust Ductwork~~

~~For light duty corrosive exhaust ductwork, use PVC plastisol coated galvanized steel with PVC coating on interior [surfaces][and exterior surfaces][and epoxy wash primer coating on exterior surfaces].~~

~~3.2.6 FRP Ductwork~~

~~Provide fibrous glass reinforced plastic ducting and related structures that conform to JIS A 4009. Provide flanged joints where indicated. Crevice-free butt lay-up joints are acceptable where flanged joints are not indicated. When ambient temperatures are lower than 10 degrees C, heat cure joints by exothermic reaction heat packs.~~

3.2.6 Kitchen Exhaust Ductwork

3.2.6.1 Ducts Conveying Smoke and Grease Laden Vapors

Provide ducts conveying smoke and grease laden vapors that conform to requirements of NFPA 96. Make seams, joints, penetrations, and duct-to-hood collar connections with a liquid tight continuous external weld. Provide duct material that is a ~~[minimum 1.3 mm, Type 304L or 316L, stainless steel] [~~minimum 1.6 mm carbon steel~~]~~. ~~[~~Include with duct construction an external perimeter angle sized in accordance with MLIT-M, except place welded joint reinforcement on maximum of 600 mm centers; continuously welded companion angle bolted flanged joints with flexible ceramic cloth gaskets where indicated; pitched to drain at low points; welded pipe coupling-plug drains at low points; welded fire protection and detergent cleaning penetration; steel framed, stud bolted, and flexible ceramic cloth gasketed cleaning access provisions where indicated. Make angles, pipe couplings, frames, bolts, etc., the same material as that specified for the duct unless indicated otherwise.~~]~~

3.2.6.2 Exposed Ductwork

Provide exposed ductwork that is fabricated from minimum 1.3 mm, Type 304L or 316L, stainless steel with continuously welded joints and seams. Pitch ducts to drain at hoods and low points indicated. Match surface finish to hoods.

3.2.6.3 Concealed Ducts Conveying Moisture Laden Air

Fabricate concealed ducts conveying moisture laden air from minimum ~~[~~1.3 mm, Type 300 series, stainless steel~~] [1.6 mm, galvanized steel] [0.55 mm, tempered copper sheet]~~. Continuously weld, braze, or solder joints to be liquid tight. Pitch ducts to drain at points indicated. Make transitions to other metals liquid tight, companion angle bolted and gasketed.

3.2.6.4 ConstractionConstruction

Ducts shall be constructed of and supported by carbon steel not less than 1.37 mm in thickness or stainless steel not less than 1.09 mm in thickness. All seams, joints, penetrations shall have external weld except duct-to-hood collar connections shall not require a liquidtight continuous external weld as required in NFPA 96.

3.2.6.5 Access Panels

On vertical ductwork where personnel entry is possible, access shall be provided at the top of the vertical riser to accommodate descent. Where personnel entry is not possible, adequate access for cleaning shall be provided on each floor. The exhaust duct shall have a slope downstream of the exhaust air. Access panels shall be of the same material and thickness as the duct. Access panels shall have a gasket or sealant that is rated for 815.6 celsius degrees (1500 F) and shall be greasetight. For hoods with dampers in the exhaust or supply collar, an access panel for cleaning and inspection shall be provided in the duct or the hood within 457 mm of the damper. Access panels shall be provided at the side or at the top of the duct, whichever is more accessible, and at changes of direction. Horizontal duct shall have access panels at 8 feet interval.

3.2.7 Acoustical Duct Lining

Apply lining in cut-to-size pieces attached to the interior of the duct with nonflammable fire resistant adhesive , and manufacture's reconnected material. Provide top and bottom pieces that lap the side pieces and are secured with welded pins, adhered clips of metal, nylon, or high impact plastic, and speed washers or welding cup-head pins installed according to MLIT-M. Provide welded pins, cup-head pins, or adhered clips that do not distort the duct, burn through, nor mar the finish or the surface of the duct. Make pins and washers flush with the surfaces of the duct liner and seal all breaks and punctures of the duct liner coating with the nonflammable, fire resistant adhesive. Coat exposed edges of the liner at the duct ends and at other joints where the lining is subject to erosion with a heavy brush coat of the nonflammable, fire resistant adhesive, to prevent delamination of glass fibers. Apply duct liner to flat sheet metal prior to forming duct through the sheet metal brake. Additionally secure lining at the top and bottom surfaces of the duct by welded pins or adhered clips as specified for cut-to-size pieces. Other methods indicated in MLIT-M to obtain proper installation of duct liners in sheet metal ducts, including adhesives and fasteners, are acceptable.

3.2.8 Dust Control

To prevent the accumulation of dust, debris and foreign material during construction, perform temporary dust control protection. Protect the distribution system (supply and return) with temporary seal-offs at all inlets and outlets at the end of each day's work. Keep temporary protection in place until system is ready for startup.

3.2.9 Insulation

Provide thickness and application of insulation materials for ductwork, piping, and equipment according to Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.~~ Externally insulate outdoor air intake ducts and plenums [up to the point where the outdoor air reaches the conditioning unit][or][up to the point where the outdoor air mixes with the return air stream].~~

3.2.10 Duct Test Holes

Provide holes with closures or threaded holes with plugs in ducts and plenums as indicated or where necessary for the use of pitot tube in balancing the air system. Plug insulated duct at the duct surface, patched over with insulation and then marked to indicate location of test hole if needed for future use.

3.2.11 Power Roof Ventilator Mounting

Provide foamed 13 mm thick, closed-cell, flexible elastomer insulation to cover width of roof curb mounting flange. Where wood nailers are used, predrill holes for fasteners.

3.2.12 Power Transmission Components Adjustment

Test V-belts and sheaves for proper alignment and tension prior to operation and after 72 hours of operation at final speed. Uniformly load belts on drive side to prevent bouncing. Make alignment of direct driven couplings to within 50 percent of manufacturer's maximum allowable range of misalignment.

3.3 EQUIPMENT PADS

Provide equipment pads to the dimensions shown or, if not shown, to conform to the shape of each piece of equipment served with a minimum 75 mm margin around the equipment and supports. Allow equipment bases and foundations, when constructed of concrete or grout, to cure a minimum of ~~[~~ 28~~][14][_____]~~ calendar days before being loaded.

3.4 CUTTING AND PATCHING

Install work in such a manner and at such time that a minimum of cutting and patching of the building structure is required. Make holes in exposed locations, in or through existing floors, by drilling and smooth by sanding. Use of a jackhammer is permitted only where specifically approved. Make holes through masonry walls to accommodate sleeves with an iron pipe masonry core saw.

3.5 CLEANING

Thoroughly clean surfaces of piping and equipment that have become covered with dirt, plaster, or other material during handling and construction before such surfaces are prepared for final finish painting or are enclosed within the building structure. Before final acceptance, clean mechanical equipment, including piping, ducting, and fixtures, and free from dirt, grease, and finger marks. When the work area is in an occupied space such as office, laboratory or warehouse ~~[_____]~~ protect all furniture and equipment from dirt and debris. Incorporate housekeeping for field construction work which leaves all furniture and equipment in the affected area free of construction generated dust and debris; and, all floor surfaces vacuum-swept clean.

3.6 PENETRATIONS

Provide sleeves and prepared openings for duct mains, branches, and other penetrating items, and install during the construction of the surface to be penetrated. Cut sleeves flush with each surface. Place sleeves for round duct 380 mm and smaller. Build framed, prepared openings for round duct larger than 380 mm and square, rectangular or oval ducts. Sleeves and framed openings are also required where grilles, registers, and diffusers are installed at the openings. Provide 25 mm clearance between penetrating and penetrated surfaces except at grilles, registers, and diffusers. Pack spaces between sleeve or opening and duct or duct insulation with mineral fiber conforming with JIS A 9504, Type 1, Class B-2.

3.6.1 Sleeves

Fabricate sleeves, except as otherwise specified or indicated, from 1 mm thick mill galvanized sheet metal. Where sleeves are installed in bearing walls or partitions, provide black steel pipe conforming with JIS G 3452, Schedule 20.

3.6.2 Framed Prepared Openings

Fabricate framed prepared openings from 1 mm galvanized steel, unless otherwise indicated.

3.6.3 Insulation

Provide duct insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS continuous through sleeves and prepared openings except firewall penetrations. Terminate duct insulation at fire dampers and flexible connections. For duct handling air at or below 16 degrees C, provide insulation continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air.

3.6.4 Closure Collars

Provide closure collars of a minimum 100 mm wide, unless otherwise indicated, for exposed ducts and items on each side of penetrated surface, except where equipment is installed. Install collar tight against the surface and fit snugly around the duct or insulation. Grind sharp edges smooth to prevent damage to penetrating surface. Fabricate collars for round ducts 380 mm in diameter or less from 1 mm galvanized steel. Fabricate collars for square and rectangular ducts, or round ducts with minimum dimension over 380 mm from 1.40 mm galvanized steel. Fabricate collars for square and rectangular ducts with a maximum side of 380 mm or less from 1 mm galvanized steel. Install collars with fasteners a maximum of 150 mm on center. Attach to collars a minimum of 4 fasteners where the opening is 300 mm in diameter or less, and a minimum of 8 fasteners where the opening is 500 mm in diameter or less.

3.6.5 Firestopping

Where ducts pass through fire-rated walls, fire partitions, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING.

3.7 FIELD PAINTING OF MECHANICAL EQUIPMENT

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal on metal surfaces subject to temperatures in excess of 50 degrees C. Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Provide aluminum or light gray finish coat.

3.7.1 Temperatures less than 50 degrees C

Immediately after cleaning, apply one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat to metal surfaces subject to temperatures less than 50 degrees C.

3.7.2 Temperatures between 50 and 205 degrees C

Apply two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm to metal surfaces subject to temperatures between 50 and 205 degrees C.

3.7.3 Temperatures greater than 205 degrees C

Apply two coats of heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm to metal surfaces subject to temperatures greater than 205 degrees C.

3.7.4 Finish Painting

The requirements for finish painting of items only primed at the factory, and surfaces not specifically noted otherwise, are specified in Section 09 90 00 PAINTS AND COATINGS.

~~3.7.5 Color Coding Scheme for Locating Hidden Utility Components~~

~~Use scheme in buildings having suspended grid ceilings. Provide color coding scheme that identifies points of access for maintenance and operation of components and equipment that are not visible from the finished space and are accessible from the ceiling grid, consisting of a color code board and colored metal disks. Make each colored metal disk approximately 13 mm diameter and secure to removable ceiling panels with fasteners. Insert each fastener into the ceiling panel so as to be concealed from view. Provide fasteners that are manually removable without the use of tools and that do not separate from the ceiling panels when the panels are dropped from ceiling height. Make installation of colored metal disks follow completion of the finished surface on which the disks are to be fastened. Provide color code board that is approximately 1 m wide, 750 mm high, and 13 mm thick. Make the board of wood fiberboard~~

~~and frame under glass or 1.6 mm transparent plastic cover. Make the color code symbols approximately 19 mm in diameter and the related lettering in 13 mm high capital letters. Mount the color code board [where indicated] [in the mechanical or equipment room]. Make the color code system as indicated below:~~

~~Color~~	~~System~~	~~Item~~	~~Location~~
~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~

3.8 IDENTIFICATION SYSTEMS

Provide identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and item number on all valves and dampers. Provide tags that are 35 mm minimum diameter with stamped or engraved markings. Make indentations black for reading clarity. Attach tags to valves with No. 12 AWG 2 mm diameter corrosion-resistant steel wire, copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

~~3.9 DUCTWORK LEAK TEST~~

~~Perform ductwork leak test for the entire air distribution and exhaust system, including fans, coils, [filters, etc.][filters, etc. designated as static pressure Class 750 Pa through Class 2500 Pa.] Provide test procedure, apparatus, and report that conform to JIS Z 2330. The maximum allowable leakage rate is [_____] L/s. Complete ductwork leak test with satisfactory results prior to applying insulation to ductwork exterior or concealing ductwork.~~

3.9 DUCTWORK LEAK TESTS

The requirements for ductwork leak tests are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC.

3.10 DAMPER ACCEPTANCE TEST

Submit the proposed schedule, at least 2 weeks prior to the start of test. Operate all fire dampers and smoke dampers under normal operating conditions, prior to the occupancy of a building to determine that they function properly. Test each fire damper equipped with fusible link by having the fusible link cut in place. Test dynamic fire dampers with the air handling and distribution system running. Reset all fire dampers with the fusible links replaced after acceptance testing. To ensure optimum operation and performance, install the damper so it is square and free from racking.

3.11 TESTING, ADJUSTING, AND BALANCING

The requirements for testing, adjusting, and balancing are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC. Begin testing, adjusting, and balancing only when the air supply and distribution, including controls, has been completed, with the exception of performance tests.

3.12 PERFORMANCE TESTS

~~After testing, adjusting, and balancing is complete as specified, test each system as a whole to see that all items perform as integral parts of the system and temperatures and conditions are evenly controlled throughout the building. Record the testing during the applicable season. Make corrections and adjustments as necessary to produce the conditions indicated or specified. Conduct capacity tests and general operating tests by an experienced engineer. Provide tests that cover a period of not less than [_____] days for each system and demonstrate that the entire system is functioning according to the specifications. Make coincidental chart recordings at points indicated on the drawings for the duration of the time period and record the temperature at space thermostats or space sensors, the humidity at space humidistats or space sensors and the ambient temperature and humidity in a shaded and weather protected area.~~Conduct performance test as required in section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC and section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

~~Submit test reports for the [ductwork leak test, and] performance tests in booklet form, upon completion of testing. Document phases of tests performed including initial test summary, repairs/adjustments made, and final test results in the reports.~~

3.13 CLEANING AND ADJUSTING

Provide a temporary bypass for water coils to prevent flushing water from passing through coils. Inside of ~~[~~room fan-coil units~~][coil-induction units,] [air terminal units,] [unit ventilators,]~~ thoroughly clean ducts, plenums, and casing of debris and blow free of small particles of rubbish and dust and then vacuum clean before installing outlet faces. Wipe equipment clean, with no traces of oil, dust, dirt, or paint spots. Provide temporary filters prior to startup of all fans that are operated during construction, and provide new filters after all construction dirt has been removed from the building, and the ducts, plenums, casings, and other items specified have been vacuum cleaned. Perform and document that proper "Indoor Air Quality During Construction" procedures have been followed; provide documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. Maintain system in this clean condition until final acceptance. Properly lubricate bearings with oil or grease as recommended by the manufacturer. Tighten belts to proper tension. Adjust control valves and other miscellaneous equipment requiring adjustment to setting indicated or directed. Adjust fans to the speed indicated by the manufacturer to meet specified conditions. Maintain all equipment installed under the contract until close out documentation is received, the project is completed and the building has been documented as beneficially occupied.

~~3.14 RADIANT PANELS~~

~~3.14.1 Installation~~

~~Install radiant panels level and plumb, maintaining sufficient clearance for normal services and maintenance.~~

~~3.14.2 Soldering~~

~~When soldering copper fittings at the panel, a heat pad will be used to protect the panel finish.~~

~~3.14.3 Connections~~

~~Install piping adjacent to radiant panels to allow for service and maintenance.~~

3.14 OPERATION AND MAINTENANCE

3.14.1 Operation and Maintenance Manuals

Submit ~~[six] [_____]~~4 manuals at least 2 weeks prior to field training. Submit data complying with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit Data Package 3 for the items/units listed under SD-10 Operation and Maintenance Data

3.14.2 Operation And Maintenance Training

Conduct a training course for the members of the operating staff as designated by the Contracting Officer. Make the training period consist of a total of ~~[_____]~~8 hours of normal working time and start it after all work specified herein is functionally completed and the Performance Tests have been approved. Conduct field instruction that covers all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations. Submit the proposed On-site Training schedule concurrently with the Operation and Maintenance Manuals and at least 14 days prior to conducting the training course.

-- End of Section --

SECTION 23 57 10.00 10

FORCED HOT WATER HEATING SYSTEMS USING WATER AND STEAM HEAT EXCHANGERS
11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M	(2019) Specification for Filler Metals for Brazing and Braze Welding

ASTM INTERNATIONAL (ASTM)

ASTM D3308	(2012; R 2022) Standard Specification for PTFE Resin Skived Tape

EXPANSION JOINT MANUFACTURERS ASSOCIATION (EJMA)

EJMA Stds	(2015) (10th Ed) EJMA Standards

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 2220	(2012) Steel Pipe Flanges
JIS B 2301	(2013) Screwed Type Malleable Cast Iron Pipe Fittings
JIS B 2302	(2013) Screwed Type Steel Pipe Fittings
JIS B 2311	(2024) Steel Butt-Welding Pipe Fittings for Ordinary Use
JIS B 2316	(2017) Steel Socket-Welding Pipe Fittings
JIS B 2352	(2013) Bellows Type Expansion Joints
JIS B 2404	(2018) Dimensions of Gaskets for Use with Pipe Flanges
JIS B 7505	(2017) Aneroid Pressure Gauges
JIS B 8265	(2024) Construction of Pressure Vessel -- General Principles
JIS C 0920	(2003) Degrees of Protection Provided by Enclosures (IP Code)

JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS G 3302	(2022) Hot-Dip Zinc-Coated Steel Sheet and Strip
JIS G 3452	(2019) Carbon Steel Pipes for Ordinary Piping
JIS G 3455	(2024) Carbon Steel Pipes for High Pressure Service
JIS G 3456	(2024) Carbon Steel Pipes for High Temperature Service
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS H 5120	(2016) Copper and Copper Alloy Castings
JIS H 8615	(2006) Electroplated Coatings of Chromium for Engineering Purposes (Amendment 1)
JIS K 6742	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes for Water Supply
JIS Z 3192	(1999) Methods of Tensile and Shear Tests for Brazed Joint
JIS Z 3197	(2021) Test Methods for Soldering Fluxes
JIS Z 3261	(1998) Silver Brazing Filler Metals
JIS Z 3282	(2017) Soft Solder -- Chemical Compositions and Forms
JIS Z 3801	(2018) Standard Qualification Test and Acceptance Requirements for Manual Welding Technique

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-25	(2018) Standard Marking System for Valves, Fittings, Flanges and Unions
MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
MSS SP-70	(2011) Gray Iron Gate Valves, Flanged and Threaded Ends
MSS SP-71	(2018) Gray Iron Swing Check Valves,

	Flanged and Threaded Ends
MSS SP-80	(2019) Bronze Gate, Globe, Angle, and Check Valves
MSS SP-85	(2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends

PLUMBING-HEATING-COOLING CONTRACTORS ASSOCIATION (PHCC)

NAPHCC NSPC	(2015) National Standard Plumbing Code Illustrated

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2023; with Change 3, 2025) Structural Engineering

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Heating System

SD-03 Product Data

Spare Parts

Welding

Framed Instructions

SD-06 Test Reports

Testing and Cleaning

Water Treatment Testing

SD-07 Certificates

Bolts

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G~~, [_____]~~

1.3 QUALITY ASSURANCE

Procedures and welders must be qualified in accordance with the code under which the welding is specified to be accomplished.

1.4 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and excessive temperature variation; and dirt, dust, or other contaminants.

1.5 EXTRA MATERIALS

Submit spare parts data for each different item of material and equipment specified, after approval of the related submittals and not later than ~~[_____]~~2 months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

2.1.1 Standard Products

Provide materials and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Equipment must be supported by a service organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site.

2.1.2 Nameplates

Place a plate on each major item of equipment having the manufacturer's name, address, type or style, model or serial number, and catalog number secured to the item of equipment.

2.1.3 Equipment Guards and Access

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact in accordance with OSHA requirements. High temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard must be properly guarded or covered with insulation of a type specified. ~~[~~Catwalks, operating platforms, ladders, and guardrails must be provided where shown and must be constructed in accordance with Section ~~[~~08 31 00 ACCESS DOORS AND PANELS~~][~~ and 05 51 33 METAL LADDERS~~]~~.~~]~~

2.1.4 Asbestos Prohibition

Asbestos and asbestos-containing products will not be accepted.

2.1.5 Electrical Work

Provide electrical motor driven equipment specified complete with motors, motor starters, and controls. Electric equipment (including motor efficiencies), and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide integral size motors of the premium efficiency type in accordance with ~~NEMA MG 1~~JIS C 4212. Electrical characteristics must be as specified or indicated. Provide motor starters complete with thermal overload protection and other appurtenances necessary for the motor control specified. Each motor must be of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Manual or automatic control and protective or signal devices required for the operation specified, and any control wiring, conduit, and connection to power required for controls and devices but not shown must be provided.

2.2 PIPING, TUBING, AND FITTINGS

2.2.1 General

Piping, tubing, and fittings must be as follows:

a. Low temperature water piping must be black steel or copper tubing with cast iron, malleable iron or steel, solder-joint, flared-tube or grooved mechanical joint fittings.

b. Steam pipe must be black steel with malleable iron or steel fittings.

c. Condensate return piping must be black steel Schedule 80 with cast iron or malleable iron, Class 250 minimum.

~~d. High temperature water piping must be black steel, Schedule 40.~~

~~e~~d. Vent piping must be black steel, Schedule 40, with black malleable iron fittings.

2.2.2 Steel Pipe

Pipe must conform to ~~ASTM A53/A53M or ASTM A106/A106M~~JIS G 3452 or JIS G 3456, Grade A or B, black steel, Schedule 40, unless otherwise specified. Steel pipe to be bent must be ~~ASTM A53/A53M~~JIS G 3452, Grade A, standard, or Grade B, extra strong weight. Steam pipe must be ~~ASTM A53/A53M~~JIS G 3452 Grade A.

~~2.2.3 High Temperature Water Piping~~

~~Piping must be Type S for 40 mm and smaller, Type S or Type E for pipe 50 mm and larger, schedule 40 steel conforming to ASTM A53/A53M, Grade B; or to ASTM A106/A106M, Grade B.~~

2.2.3 Gauge Piping

Piping must be copper tubing for ~~[steam] [and] [~~low temperature water~~]~~. ~~[Use black steel, ASTM A106/A106M, seamless, Grade A pipe for high temperature.]~~

2.2.4 Copper Tubing

Tubing must conform to ~~ASTM B88, ASTM B88M~~JIS H 3300, Type K or L. Tubing for compressed air tubing must conform to ~~ASTM B251/B251M~~JIS H 3300.

~~2.2.5 High Temperature Water Fittings~~

~~Fittings must be steel welding fittings conforming in physical and chemical properties to ASTM A234/A234M. Buttwelding fittings must conform to ASME B16.9. Socket welded fittings must conform to ASME B16.1. Screwed fittings, when required, must be black forged steel, 2000-pound class, conforming to ASME B16.11. Flanges must be serrated or raised-faced type.~~

2.2.5 Malleable Iron Pipe Fittings

Fittings must conform to ~~ASME B16.3~~JIS B 2301, type required to match adjacent piping.

~~2.2.6 Cast Iron Pipe Fittings~~

~~Fittings must conform to ASME B16.1 or ASME B16.4 type required to match adjacent piping.~~

2.2.6 Steel Pipe Fittings

Fittings must have the manufacturer's trademark affixed in accordance with MSS SP-25 so as to permanently identify the manufacturer.

2.2.6.1 Welded Fittings

Welded fittings must conform to ~~ASTM A234/A234M~~JIS G 3456 with WPA marking. Butt welded fittings must conform to ~~ASME B16.9~~JIS B 2311, and socket welded fittings must conform to ~~ASME B16.11~~JIS B 2316 and JIS B 2302.

~~2.2.6.2 Grooved Mechanical Fittings~~

~~Standard fittings must be of malleable iron conforming to ASTM A47/A47M, Grade 32510, or ductile iron conforming to ASTM A536, Grade 65-45-12. Fittings may also be constructed of steel, conforming to ASTM A106/A106M, Grade B or ASTM A53/A53M.~~

~~2.2.6.3 Grooved Mechanical Pipe Joints~~

~~Pipe joints must conform to AWWA C606. Grooved mechanical joint fittings must be full flow factory manufactured forged steel fittings. Fittings, couplings, gaskets, and pipe grooving tool or grooved end pipe must be products of the same manufacturer. Mechanical pipe couplings must be of the bolted type and must consist of a housing fabricated in two or more parts, a synthetic rubber gasket, and nuts and bolts to secure unit together. Housings must be of malleable iron conforming to ASTM A47/A47M, Grade 32510 or ductile iron conforming to ASTM A536, Grade 65-45-12. Coupling nuts and bolts must be of steel and conform to ASTM A183. Submit written certification that the bolts furnished comply with the requirements of this specification, provided by the bolt manufacturer. The certification must include illustrations of product-required markings, the date of manufacture, and the number of each type of bolt to be furnished based on this certification. Gaskets must be of molded synthetic rubber, Type [EPDM] [Buna-N] with central cavity, pressure responsive configuration and must conform to ASTM D2000.~~

2.2.7 Joints and Fittings for Copper Tubing

Wrought copper and bronze fittings must conform to ~~ASME B16.22~~JIS H 3401 and ~~ASTM B75/B75M~~JIS H 3300. Cast copper alloy fittings must conform to ~~ASME B16.18 and ASTM B828~~JIS H 3401. Flared fittings must conform to ~~ASME B16.26 and ASTM B62~~JIS H 3401 and JIS H 5120. Adaptors may be used for connecting tubing to flanges and threaded ends of valves and equipment. Extracted brazed tee joints produced with an acceptable tool and installed as recommended by the manufacturer may be used. ~~Cast bronze threaded fittings must conform to ASME B16.15. Grooved mechanical joints and fittings must be designed for not less than 862 kPa service and must~~

~~be the product of the same manufacturer. Grooved fitting and mechanical coupling housing must be ductile iron conforming to ASTM A536. Gaskets for use in grooved joints must be molded synthetic polymer of pressure responsive design and must conform to ASTM D2000 for circulating medium up to 110 degrees C. Grooved joints must conform to AWWA C606. Coupling nuts and and bolts for use in grooved joints must be steel and must conform to ASTM A183.~~

~~2.2.8 Steel Flanges~~

~~Flanged fittings including flanges, bolts, nuts, bolt patterns., etc. must be in accordance with ASME B16.5 class 150 and must have the manufacturers trademark affixed in accordance with MSS SP-25. Flange material must conform to ASTM A105/A105M. Flanges for high temperature water systems must be serrated or raised-face type. Blind flange material must conform to ASTM A516/A516M cold service and ASTM A515/A515M for hot service. Bolts must be high strength or intermediate strength with material conforming to ASTM A193/A193M.~~

~~2.2.9 Pipe Threads~~

~~Pipe threads must conform to ASME B1.20.2M.~~

2.2.8 Nipples

Nipples must conform to ~~ASTM A733 or ASTM B687~~JIS B 2302, standard weight.

2.2.9 Unions

Unions must conform to ~~ASME B16.39~~JIS B 2302, type to match adjacent piping.

2.2.10 Adapters

Adapters for copper tubing must be brass or bronze for soldered fittings.

2.2.11 Dielectric Waterways

Dielectric waterways must conform to the tensile strength and dimensional requirements specified in ~~ASME B16.39~~JIS B 2302. Waterways must have metal connections on both ends to match adjacent piping. Metal parts of dielectric waterways must be separated so that the electrical current is below 1 percent of the galvanic current which would exist upon metal-to-metal contact. Dielectric waterways must have temperature and pressure rating equal to or greater than that specified for the connecting piping. Dielectric waterways must be internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Dielectric flanges must meet the performance requirements described herein for dielectric waterways.

~~2.2.12 Grooved Mechanical Joints~~

~~Rigid grooved pipe joints may be provided in lieu of unions, welded, flanges or screwed piping connections at chilled water pumps and allied equipment, and on aboveground pipelines in serviceable locations, if the temperature of the circulating medium does not exceed 110 degrees C. Flexible grooved joints will not be permitted, except as vibration isolators adjacent to mechanical equipment. Rigid grooved joints must incorporate an angle bolt pad design which maintains metal-to-metal~~

~~contact with equal amount of pad offset of housings upon installation to insure positive rigid clamping of the pipe. Designs which can only clamp on the bottom of the groove or which utilize gripping teeth or jaws, or which use misaligned housing bolt holes, or which require a torque wrench or torque specifications, will not be permitted. Rigid grooved pipe couplings must be used with grooved end pipes, fittings, valves and strainers. Rigid couplings must be designed for not less than 862 kPa service and appropriate for static head plus the pumping head, and must provide a water-tight joint. Grooved fittings and couplings, and grooving tools must be provided from the same manufacturer. Segmentally welded elbows must not be used. Grooves must be prepared in accordance with the coupling manufacturer's latest published standards. Grooving must be performed by qualified grooving operators having demonstrated proper grooving procedures in accordance with the tool manufacturer's recommendations. The Contracting Officer must be notified 24 hours in advance of test to demonstrate operator's capability, and the test must be performed at the work site, if practical, or at a site agreed upon. The operator must demonstrate the ability to properly adjust the grooving tool, groove the pipe, and verify the groove dimensions in accordance with the coupling manufacturer's specifications.~~

2.2.12 Flexible Pipe Connectors

Flexible pipe connectors must be designed for 1.034 MPa or 1.034 MPa service as appropriate for the static head plus the system head, and 121 degrees C. Connectors must be installed where indicated. The flexible section must be constructed of rubber, tetrafluoroethylene resin, or corrosion-resisting steel, bronze, monel, or galvanized steel. Materials used and the configuration must be suitable for the pressure, vacuum, temperature, and circulating medium. The flexible section may have threaded, welded, soldered, flanged, grooved, or socket ends. Flanged assemblies must be equipped with limit bolts to restrict maximum travel to the manufacturer's standard limits. Unless otherwise indicated, the length of the flexible connectors must be as recommended by the manufacturer for the service intended. Internal sleeves or liners, compatible with circulating medium, must be provided when recommended by the manufacturer. Provide covers to protect the bellows where indicated.

2.3 MATERIALS AND ACCESSORIES

2.3.1 Iron and Steel Sheets

2.3.1.1 Galvanized Iron and Steel

Galvanized iron and steel must conform to ~~ASTM A653/A653M~~JIS G 3302, with general requirements conforming to ~~ASTM A653/A653M~~JIS G 3302. Gauge numbers specified are Manufacturer's Standard Gauge.

2.3.1.2 Uncoated (Black) Steel

Uncoated (black) steel must conform to ~~ASTM A653/A653M~~JIS G 3302, composition, condition, and finish best suited to the intended use. Gauge numbers specified refer to Manufacturer's Standard Gauge.

2.3.2 Solder

Solder must conform to ~~ASTM B32~~JIS Z 3282. Solder and flux must be lead free. Solder flux must be liquid or paste form, non-corrosive and conform to ~~ASTM B813~~JIS Z 3197 and any additional applicable standards.

2.3.3 Solder, Silver

Silver solder must conform to AWS A5.8/A5.8M or JIS Z 3261 or equal equivalent.

2.3.4 Thermometers

Mercury must not be used in thermometers. Thermometers must have brass, malleable iron, or aluminum alloy case and frame, clear protective face, permanently stabilized glass tube with indicating-fluid column, white face, black numbers, and a 225 mm scale, and thermometers must have rigid stems with straight, angular, or inclined pattern.

2.3.5 Gauges

Provide gauges conforming to ~~ASME B40.100~~JIS B 7505.

2.3.6 Gaskets for Flanges

Composition gaskets must conform to ~~ASME B16.21~~JIS B 2404. Gaskets must be nonasbestos compressed material in accordance with ~~ASME B16.21~~JIS B 2404, 1.6 mm thickness, full face or self-centering flat ring type. Gaskets must contain aramid fibers bonded with styrene butadiene rubber (SBR) or nitrile butadiene rubber (NBR). NBR binder must be used for hydrocarbon service. Gaskets must be suitable for pressure and temperatures of piping system.

2.3.7 Polyethylene Tubing

Low-density virgin polyethylene must conform to ~~ASTM D1248~~JIS K 6742 and other applicable standards, Type I, Category 5, Class B or C.

2.3.8 Bellows-Type Joints

Joints must be flexible, guided expansion joints. Expansion element must be of stainless steel. Bellows-type expansion joints must be in accordance with the applicable requirements of EJMA Stds and ~~ASME B31.1~~ JIS B 2352 with internal liners.

2.3.9 Expansion Joints

Expansion joints must provide for either single or double slip of connected pipes, as required or indicated, and for not less than the traverse indicated. Joints must be designed for hot water working pressure not less than ~~[_____]~~862 kPa and must be in accordance with applicable requirements of EJMA Stds and ~~ASME B31.1~~JIS B 2352. Joints must be designed for packing injection under full line pressure. End connections must be flanged or beveled for welding as indicated. Provide joints with anchor base where required or indicated. Where adjoining pipe is carbon steel, the sliding slip must be seamless steel plated with a minimum of 0.0508 mm of hard chrome conforming to ~~ASTM B650~~JIS H 8615. Joint components must be fabricated from material equivalent to that of the pipeline. Initial settings must be made in accordance with manufacturer's recommendations to compensate for ambient temperature at time of installation. Pipe alignment guides must be installed as recommended by joint manufacturer, but in any case must not be more than 1.5 m from expansion joint except for lines 100 mm or smaller, guides must be installed not more than 600 mm from the joint. Provide service outlets where indicated.

2.3.10 Flexible Ball Joints

Flexible ball joints must be constructed of alloys as appropriate for the service intended. Where so indicated, the ball joint must be designed for packing injection under full line pressure to contain leakage. Joint ends must be threaded (to 50.8 mm only), grooved, flanged or beveled for welding as indicated or required and must be capable of absorbing a minimum of 15-degree angular flex and 360-degree rotation. Balls and sockets must be of equivalent material as the adjoining pipeline. Exterior spherical surface of carbon steel balls must be plated with 0.0508 mm of hard chrome conforming to ~~ASTM B650~~JIS H 8615. Ball type joints must be designed and constructed in accordance with ~~ASME B31.1~~ JIS B 2352 and ~~ASME BPVC SEC VIII D1~~JIS B 8265, where applicable. Flanges where required must conform to ASME B16.5. Gaskets and compression seals must be compatible with the service intended.

2.3.11 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports must conform to MSS SP-58.

2.4 VALVES FOR LOW TEMPERATURE WATER HEATING AND STEAM SYSTEMS

2.4.1 Check Valves

Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 3 or 4, Class 125. Sizes 80 mm through 300 mm, cast iron must conform to MSS SP-71, Type III or IV, Class 125.

2.4.2 Globe Valves

Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 1, 2 or 3, Class 125. Sizes 80 mm through 300 mm, cast iron must conform to MSS SP-85, Type III, Class 125.

2.4.3 Angle Valves

Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 1, 2 or 3, Class 125. Sizes 80 mm through 300 mm, cast iron must conform to MSS SP-85, Type III, Class 125.

2.4.4 Gate Valves

Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 1 or 2, Class 125. Sizes 80 mm through 1200 mm, cast iron must conform to MSS SP-70, Type I, Class 125, Design OT or OF (OS&Y), bronze trim.

2.4.5 Air Vents

Provide air vents at all piping high points in water systems, with block valve in inlet and internal check valve to allow air vent to be isolated for cleaning and inspection. Outlet connection must be piped to nearest open site or suitable drain, or terminated 300 mm above finished grade. Pressure rating of air vent must match pressure rating of piping system. Body and cover must be cast iron or semi-steel with stainless steel or copper float and stainless steel or bronze internal parts. Air vents installed in piping in chase walls or other inaccessible places must be provided with an access panel.

2.4.6 Balancing Valves

Balancing valves must have meter connections with positive shutoff valves. An integral pointer must register degree of valve opening. Valves must be calibrated so that flow in L/minute can be determined when valve opening in degrees and pressure differential across valve is known. Each balancing valve must be constructed with internal seals to prevent leakage and must be supplied with preformed insulation. Valves must be suitable for 121 degrees C temperature and working pressure of the pipe in which installed. Valve bodies must be provided with tapped openings and pipe extensions with shutoff valves outside of pipe insulation. The pipe extensions must be provided with quick connecting hose fittings for a portable meter to measure the pressure differential. One portable differential meter must be furnished. The meter suitable for the operating pressure specified must be complete with hoses, vent, and shutoff valves and carrying case. In lieu of the balancing valve with integral metering connections, a ball valve or plug valve with a separately installed orifice plate or venturi tube may be used for balancing. Provide plug valves and ball valves 200 mm or larger with manual gear operators with position indicators.

2.4.7 Gravity Flow Control Valves

Ends must be soldered, threaded, or flanged type as applicable, and designed for easy cleaning without disconnecting piping. Valves for copper tubing must be bronze. Valves must prevent flow due to gravity when circulators are off.

~~2.4.8 Radiator Valves~~

~~Automatic thermostatic radiator valves must be self-contained [direct sensor] [remote sensor] [wall thermostat] controlled nonelectric temperature control valves. Valve bodies must be constructed of chrome plated brass and must be angle or straight pattern as indicated, with threaded or brazed end connections. Valve disc must be of ethylene propylene or composition material. Thermostatic operators must be a modulating type consisting of a sensing unit counter balanced by a spring setting.~~

~~2.5 VALVES FOR HIGH AND MEDIUM TEMPERATURE WATER SYSTEMS~~

~~2.5.1 Check Valves~~

~~Sizes 65 mm and less, bronze must conform to MSS SP-80, Class 300 Sizes 65 mm and less, bronze must conform to MSS SP-80, Class 300 minimum. Sizes 80 mm through 600 mm, steel must conform to ASME B16.34, Class 300 minimum, flanged ends, swing disc; water, oil gas or steam service to 454 degrees C.~~

~~2.5.2 Globe Valves~~

~~Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 1, 2 or 3, Class 300 minimum. Sizes 80 mm through 600 mm, steel must conform to ASME B16.34, Class 300 minimum, flanged ends; water, oil, gas, or steam service to 454 degrees C.~~

~~2.5.3 Angle Valves~~

~~Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 1, 2 or 3, Class 300 minimum. Sizes 80 mm through 600 mm, steel must conform to ASME B16.34, Class 300 minimum, flanged ends; water, oil, gas, or steam service to 454 degrees C.~~

~~2.5.4 Gate Valves~~

~~Sizes 65 mm and less, bronze must conform to MSS SP-80, Type 1, or 2, Class 300 minimum. Sizes 80 mm through 600 mm, steel must conform to ASME B16.34, Class 300 minimum, flanged ends; water, oil, gas or steam service to 454 degrees C. Gate must be split wedge (double disc) type.~~

2.5 COLD WATER CONNECTIONS

Connections must be provided which include consecutively in line a strainer, backflow prevention device, and water pressure regulator. The backflow prevention device must be provided as indicated and in compliance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

2.5.1 Strainers

Basket or Y-type strainers must be the same size as the pipelines in which they are installed. Strainer bodies must be rated for ~~[~~0.862~~] [1.72]~~ MPa service, with bottoms drilled and plugged. Bodies must have arrows cast on the sides to indicate the direction of flow. Each strainer must be equipped with a removable cover and sediment basket. Basket must not be less than 0.795 mm (22 gauge) and must have perforations to provide a net free area through the basket of at least four times that of the entering pipe.

2.5.2 Pressure Regulating Valve

Valve must be a type that will not stick nor allow pressure to build up on the low side. Valve must be set to maintain a terminal pressure approximately 35 kPa in excess of the static head on the system and must operate within a 138 kPa variation regardless of initial pressure and without objectionable noise under any condition of operation.

2.6 FLASH TANK

Tank must be sized and installed as indicated, and must be of welded construction utilizing black steel sheets not less than 3.175 mm (11 gauge). Provide tank with a handhole and with tapping for the condensate returns, drip lines, vent line, and condensate discharge line to the condensate receiver. Discharge line must be equipped with a float trap. Tank must be ASME rated for ~~[_____]~~862 kPa in accordance with ~~ASME BPVC SEC VIII D1~~ JIS B 8265.

2.7 EXPANSION TANK

Pressurization system must include a replaceable diaphragm-type captive air expansion tank which will accommodate the expanded water of the system generated within the normal operating temperature range, limiting this pressure increase at all components in the system to the maximum allowable pressure at those components. The only air in the system must be the permanent sealed-in air cushion contained in the diaphragm-type tank. Sizes must be as indicated. Expansion tank must be welded steel, constructed, tested and stamped in accordance with ~~ASME BPVC SEC VIII D1~~ JIS B 8265 for a working pressure of ~~[862] [_____]~~ kPa and precharged to the minimum operating pressure. Tank air chamber must be fitted with an air charging valve. Tank must be supported by steel legs or bases for vertical installation or steel saddles for horizontal installations.

2.8 AIR SEPARATOR TANK

External air separation tank must be steel, constructed, tested, and stamped in accordance with ~~ASME BPVC SEC VIII D1~~JIS B 8265 for a working pressure of ~~[862] [_____]~~ kPa. The capacity of the air separation tank indicated is minimum.

2.9 STEAM TRAPS

~~2.9.1 Float Traps~~

~~Capacity, working pressure, and differential pressure of the traps must be as indicated.~~

2.9.1 Float-and-Thermostatic Traps

Traps must be designed for a steam working pressure of approximately 103 kPa, but must operate with a supply pressure of approximately 35 kPa. The capacity of the traps must be as indicated. Trap capacity must be based on a pressure differential of 2 kPa. Provide each float-and-thermostatic trap a hard bronze, monel, or stainless steel valve seat and mechanism and brass float, all of which can be removed easily for inspection or replacement without disturbing the piping connections. Inlet to each trap must have a cast iron strainer, either an integral part of the trap or a separate item of equipment.

~~2.9.2 Bucket Traps~~

~~Traps must be inverted or vertical bucket type with automatic air discharge. Traps must be designed for a working pressure of 1034 kPa, but must operate under a steam supply pressure of approximately 276 to 690 kPa as required. Each trap must have a heavy body and cap of fine-grained, gray cast iron. The bucket must be made of brass; the mechanism of hard bronze; the valve and seat of stainless or monel; or each of equivalent material. Traps must be tested hydrostatically under a pressure of 1.38 MPa. Traps must have capacities as indicated when operating under the specified working conditions. A strainer must be installed in the suction connection of each trap. Impact operated traps, impulse-operated traps, or thermodynamic traps with continuous discharge may be installed in lieu of bucket traps, subject to approval. Thermostatic traps designed for a steam working pressure suitable for the application may be furnished in lieu of the traps specified above. Thermostatic traps must be equipped with valves and seats of stainless steel or monel metal, and must have capacities based on a pressure differential not in excess of the following:~~

~~Steam Working Pressure, kPa (psi)~~	~~Differential, kPa (psi)~~
~~172 - 345 (25-50)~~	~~138 (20)~~
~~621 - 689 (90-100)~~	~~552 (80)~~

2.10 HEAT EXCHANGERS

Heat exchangers must be multiple pass shell and U-tube type or plate and frame type as indicated, to provide low temperature hot water for the heating system when supplied with ~~[~~steam~~] [or] [high temperature hot water] [or] [medium temperature hot water]~~ at the temperatures and pressures indicated. Temperature and pressure for plate and frame exchangers must not exceed 138 degrees C and 1.93 MPa for medium temperature hot water, or 138 degrees C and 241 kPa for steam. Temperature and pressure for shell and U-tube exchangers must not exceed 170 degrees C and 689 kPa for steam or 221 degrees C and 2.76 MPa for high temperature hot water. Exchangers must be constructed in accordance with ~~ASME BPVC SEC VIII D1~~JIS B 8265 and certified with ASME stamp secured to unit. U-tube bundles must be completely removable for cleaning and tube replacement and must be free to expand with shell. Shells must be of seamless steel pipe or welded steel construction and tubes must be seamless tubing as specified below unless otherwise indicated. Tube connections to plates must be leakproof. Provide saddles or cradles to mount shell and U-tube exchangers. Frames of plate and frame type exchangers must be fabricated of carbon steel and finished with baked epoxy enamel. Design fouling factor must be ~~[_____]~~.00025.

2.10.1 Steam Heat Exchangers, Shell and U-Tube Type

Exchangers must operate with steam in shell and low temperature water in tubes. Shell and tube sides must be designed for 1.03 MPa working pressure and factory tested at 2.02 MPa. Steam, water, condensate, and vacuum and pressure relief valve connections must be located in accordance with the manufacturer's standard practice. Connections larger than 80 mm must be ASME 1.03 MPa flanged. Water pressure loss through clean tubes must not exceed 41 kPa and water velocity must not exceed 1.8 m/second unless otherwise indicated. Minimum water velocity in tubes must be not less than 300 mm/second and assure turbulent flow. Tubes must be seamless copper or copper alloy, constructed in accordance with ~~ASTM B75/B75M or ASTM B395/B395M~~JIS H 3300, suitable for the temperatures and pressures specified. Tubes must be not less than 19 mm unless otherwise indicated. Maximum steam inlet nozzle velocity must not exceed 30.5 m/second.

~~2.10.2 High Temperature Water Heat Exchangers, Shell and U-tube Type~~

~~Exchangers must operate with low temperature water in shell and high temperature water in tubes. Shell side must be designed for 1.03 MPa working pressure and factory tested at 2.07 MPa. Tubes must be designed for 2.76 MPa working pressure and an operating temperature of 232 degrees C. High and low temperature water and pressure relief connections must be located in accordance with the manufacturer's standard practice. Water connections larger than 80 mm must be ASME 4.14 MPa flanged for high temperature water, and ASME 4.03 MPa flanged for low temperature water. Water pressure loss through clean tubes must not exceed 41 kPa unless otherwise indicated. Minimum water velocity in tubes must be 300 mm/second and assure turbulent flow. Tubes must be cupronickel or inhibited admiralty, constructed in accordance with ASTM B395/B395M, suitable for the temperatures and pressures specified. Tubes must be not less than 19 mm unless otherwise indicated.~~

~~2.10.3 Steam Heat Exchangers, Plate and Frame Type~~

~~Plates, frames and gaskets must be designed for a working pressure of 2.07 MPa and factory tested at 3.10 MPa. Steam, low temperature water, condensate, and vacuum and pressure relief valve connections must be~~

~~located in accordance with the manufacturer's standard practice. Connections larger than 80 mm must be ASME 4.03 MPa flanged. Water pressure drop through clean plates and headers must not exceed [_____] kPa at the flow rates and temperatures indicated. Plates must be designed to assure turbulent flow at a minimum rate of [_____] L/minute through any 2 plate segment. Plates must be corrugated [Type 304 stainless steel] [Type 316 stainless steel] [nickel-iron-chromium alloy conforming to ASTM B424] [nickel-molybdenum alloy conforming to ASTM B333] [titanium alloy conforming to ASTM B265]. Plate thickness must be not less than [_____] mm.~~

~~2.10.4 Medium Temperature Water Heat Exchangers, Plate and Frame Type~~

~~Plates, frames and gaskets must be designed for a working pressure of 2.07 MPa and factory tested at 31.0 MPa. Medium temperature water, low temperature water, and pressure relief valve connections must be located in accordance with the manufacturer's standard practice. Connections larger than 80 mm must be ASME 2.07 MPa flanged. Water pressure drop through clean plates and headers must not exceed [_____] kPa at the flow rates and temperatures indicated. Plates must be designed to assure turbulent flow at a minimum rate of [_____] L/second through any 2 plate segment. Plates must be corrugated [Type 304 stainless steel] [Type 316 stainless steel] [nickel-iron-chromium alloy conforming to ASTM B424] [nickel-molybdenum alloy conforming to ASTM B333] [titanium alloy conforming to ASTM B265]. Plate thickness must be not less than [_____] mm.~~

2.11 SYSTEM EQUIPMENT AND ACCESSORIES

2.11.1 Circulating Pumps

Pumps for hot water must be of the single-stage centrifugal type, electrically driven. Pumps must be supported ~~[on a concrete foundation] [or] [by the piping on which installed] [~~as indicated~~]~~. Pumps must be either integrally mounted with the motor or direct-connected by means of a flexible-shaft coupling on a cast iron, or steel sub-base. Pump housing must be of close grained cast iron. Shaft must be carbon or alloy steel, turned and ground. Shaft seal must be mechanical-seal or stuffing-box type. Impeller, impeller wearing rings, glands, casing wear rings, and shaft sleeve must be bronze. Bearings must be ball-, roller-, or oil-lubricated, bronze-sleeve type, and must be sealed or isolated to prevent loss of oil or entrance of dirt or water. Motor must be of a type approved by the manufacturer of the pump.

2.11.2 Condensate Pumping Unit

Pump must have a minimum capacity, as indicated, of ~~[_____]~~1.2 L/second when discharging against the specified pressure. The minimum capacity of the tank must be ~~[_____]~~3 liters. Condensate pumping unit must be of the ~~[single] [~~duplex~~]~~, ~~[horizontal-shaft] [~~vertical-shaft~~]~~ type, as indicated. Unit must consist of ~~[one pump] [~~two pumps~~]~~, ~~[one electric motor] [~~two electric motors~~]~~ and a single receiver. Pumps must be centrifugal or turbine type, bronze-fitted throughout with impellers of bronze or other corrosion-resistant metal. Pumps must be free from air-binding when handling condensate with temperatures up to 93 degrees C. Pumps must be connected directly to dripproof enclosed motors. Receiver must be cast iron and must be provided with condensate return, vent, overflow, and pump suction connections, and water level indicator and automatic air vent. Inlet strainer must be provided in the inlet line to the tank. Vent pipe must be galvanized steel, and fittings must be galvanized malleable iron. Vent pipe must be installed as indicated or directed. Vent piping must be flashed as specified. Pump, motor, and receiving tank may be mounted on a single base with the receiver piped to the pumps suctions. Provide a gate valve and check valve in the discharge connection from each pump.

2.11.2.1 Controls

Install enclosed float switches complete with float mechanisms in the head of the receiver. The condensate pump must be controlled automatically by means of the ~~[respective]~~ float switch that will automatically start the motor when the water in the receiving tank reaches the high level and stop the motor when the water reaches the low level. Provide motors with magnetic across-the-line starters equipped with general purpose enclosure and Automatic-Manual-Off selector switch in the cover.

2.11.2.2 Factory Testing

Submit a certificate of compliance from the pump manufacturer covering the actual test of the unit and certifying that the equipment complies with the indicated requirements.

2.11.3 Pressure Gauges and Thermometers

Provide gauges for each heat exchanger and piping as indicated. Provide a thermometer and pressure gauge on the high temperature water supply and return mains. Thermometers must be separable socket type.

2.11.4 Vacuum Relief Valve

Install a vacuum relief valve on the shell of each shell and U-tube steam heat exchanger and on the factory supplied steam inlet nozzle of each plate and frame heat exchanger. On shutoff of steam supply and condensing of steam, the vacuum relief valve must automatically admit air to the heat exchanger.

2.11.5 Pressure Relief Valves

Provide one or more pressure relief valves for each heat exchanger in accordance with ~~ASME BPVC SEC VIII D1~~JIS B 8265. The aggregate relieving capacity of the relief valves must be not less than that required by the above code. Discharge from the valves must be installed as indicated. Pressure relief valves for steam heat exchangers must be located on the low temperature water supply coming from near the heat exchanger as indicated. Relief valves for high temperature water heat exchanger must be installed on the heat exchanger shell.

2.11.6 Drains

Install a drain connection with 19 mm hose bib at the lowest point in the low temperature water return main near the heat exchanger. In addition, install threaded drain connections with threaded cap or plug wherever required for thorough draining of the low temperature water system.

2.11.7 Strainers

Basket or Y-type strainer-body connections must be the same size as the pipe lines in which the connections are installed. The bodies must have arrows clearly cast on the sides to indicate the direction of flow. Each strainer must be equipped with an easily removable cover and sediment basket. The body or bottom opening must be equipped with nipple and gate valve for blowdown. The basket for steam systems must be of not less than 0.635 mm thick stainless steel, or monel with small perforations of sufficient number to provide a net free area through the basket of at least 2.5 times that of the entering pipe. The flow must be into the basket and out through the perforations. ~~[For high temperature water systems, only cast steel bodies must be used.] [~~The strainer bodies for steam systems must be of cast steel or gray cast iron with bottoms drilled and plugged.~~]~~

2.12 INSULATION

Shop and field applied insulation must be as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.13 FACTORY PAINTED EXPOSED SPACE HEATING EQUIPMENT

Radiator and convector enclosures must be coated with the manufacturer's standard rust inhibiting primer for painting in the field as specified in Section 09 90 00 PAINTS AND COATINGS. All other exposed heating equipment must be painted at the factory with the manufacturer's standard primer and enamel finish.

2.14 RADIATORS AND CONVECTORS

The radiator and convector must be the type and size indicated. The supply and return connections must be the same size. Cast iron radiators and nonferrous convectors must be tested hydrostatically at the factory and proved tight under a pressure of not less than ~~[~~207 kPa~~][[______] kPa]~~ or 150 percent of the system operating pressure, whichever is greater. Furnish a certified report of these tests in accordance with paragraph SUBMITTALS.

~~2.14.1 Cast Iron Radiators~~

~~Cast iron radiators must be gray cast iron, free from sandholes and other defects. The sections must be connected with malleable iron nipples not less than 2.286 mm thick at any point. Cast iron radiators must be the legless type mounted on the walls by means of hangers as specified. Adjustable radiator hangers must be secured to the wall and must hold the radiators near both ends, at both top and bottom, in such a manner that the radiators cannot be removed without the use of tools. Not less than two bolts must be used to secure each hanger to the wall. Necessary angles, bolts, bearing plates, toggles, radiator grips, and other parts required for complete installation of the radiators must be provided.~~

~~2.14.2 Extended-Surface, Steel, or Nonferrous Tube-Type Radiators~~

~~Radiators must consist of metal fins permanently bonded to steel or nonferrous pipe cores, with threaded or sweat fittings at each end for connecting to external piping. Radiators must have capacities not less than those indicated, determined in accordance with HYI-005. Radiators must be equipped with [expanded-metal cover grilles fabricated from black steel sheets not less than 1.519 mm (16 gauge), secured either directly to the radiators or to independent brackets.] [solid-front, slotted horizontal-top cover grilles fabricated from black steel sheets not less than 1.214 mm (18 gauge), secured either directly to the radiators or to independent brackets.] [solid-front, slotted sloping-top cover grilles fabricated from black steel sheets not less than 1.519 mm (16 gauge),~~

~~independently secured to masonry with brackets.]~~

~~2.14.3 Convectors~~

~~Convectors must be constructed of cast iron or of nonferrous alloys, and must be installed where indicated. Capacity of convectors must be as indicated. Overall space requirements for convectors must not be greater than the space provided. Convectors must be complete with heating elements and enclosing cabinets having bottom recirculating opening, manual control damper and top supply grille. Convector cabinets must be constructed of black sheet steel not less than 0.912 mm (20 gauge).~~

~~2.14.4 Radiators and Convectors Control~~

~~[The space temperature must be maintained automatically by regulating water flow to the radiators and convectors by the self contained, automatic thermostatic radiator control valves.] [Provide controls as specified in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.]~~

2.15 UNIT HEATERS

Heaters must be as specified below, and must have a heating capacity not in excess of 125 percent of the capacity indicated. ~~[Noise level of each unit heater for areas noted must not exceed the criteria indicated.]~~

~~2.15.1 Propeller Fan Heaters~~

~~Heaters must be designed for suspension and arranged for [horizontal] [vertical] discharge of air as indicated. Casings must be not less than 0.912 mm (20 gauge) black steel and finished with lacquer or enamel. Suitable [stationary] [rotating air] deflectors must be provided to assure proper air and heat penetration capacity at floor level based on established design temperature. Suspension from heating pipes will not be permitted. [Fans for vertical discharge type heaters must operate at speeds not in excess of 1,200 rpm, except that units with 84.4 MJ output capacity or less may operate at speeds up to 1,800 rpm.] [Horizontal discharge type unit heaters must have discharge or face velocities not in excess of the following:~~

~~Unit Capacity, Liters per Second~~	~~Face Velocity, Meters per Second~~
~~Up to 472 (1000)~~	~~4.06 (800)~~
~~473 (1001)~~	~~4.57 (900)~~
~~1417 (3001) and over~~	~~5.08 (1,000)~~

~~]~~

~~2.15.2 Centrifugal Fan Heaters~~

~~Heaters must be arranged for floor or ceiling mounting as indicated. Heating elements and fans must be housed in steel cabinets of sectionalized steel plates or reinforced with angle-iron frames. Cabinets must be constructed of not lighter than 1.27 mm (18 gauge) black steel. Provide each unit heater with a means of diffusing and distributing the air. Fans must be mounted on a common shaft, with one fan to each air outlet. Fan shaft must be equipped with self-aligning ball, roller, or sleeve bearings and accessible means of lubrication. Fan shaft may be~~

~~either directly connected to the driving motor or indirectly connected by adjustable V-belt drive rated at 150 percent of motor capacity. All fans in any one unit heater must be the same size.~~

2.15.1 Heating Elements

~~[Heating coils and radiating fins must be of suitable nonferrous alloy with [threaded] [brazed] fittings at each end for connecting to external piping. The heating elements must be free to expand or contract without developing leaks and must be properly pitched for drainage. The elements must be tested under a hydrostatic pressure of 1.38 MPa and a certified report of the test must be submitted to the Contracting Officer.] [~~ Heating coils must be as specified in Section 23 30 00 HVAC AIR DISTRIBUTION for types indicated.~~]~~ Coils must be suitable for use with water up to 121 degrees C.

2.15.2 Motors

Provide motors with ~~NEMA 250~~JIS C 0920 general purpose enclosure. Motors and motor controls must otherwise be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.15.3 Motor Switches

Provide motors with manual selection switches with~~-~~ "Off," and "Automatic" positions and must be equipped with thermal overload protection.

2.15.4 Controls

Provide controls as specified in 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

2.16 HEATING AND VENTILATING UNITS

Heating and ventilating units must be as specified in Section 23 30 00 HVAC AIR DISTRIBUTION.

2.17 WATER TREATMENT SYSTEM

The water treatment system must be capable of ~~[manually] [~~automatically~~]~~ feeding chemicals into the heating system to prevent corrosion and scale within the heat exchanger and piping system. Submit detail drawings consisting of a complete list of equipment and material, including manufacturer's descriptive and technical literature, performance charts and curves, catalog cuts, and installation instructions. Also show on the drawings complete wiring and schematic diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Show on the drawings proposed layout and anchorage of equipment and appurtenances and equipment relationship to other parts of the work including clearances for maintenance and operation. All water treatment equipment and chemicals must be furnished and installed by a water treatment company regularly engaged in the installation of water treatment equipment and the provision of water treatment chemicals based upon water condition analyses. The water treatment company must provide a water sample analysis taken from the building site, each month for one year.

2.17.1 Chemical Shot Feeder

Provide a shot feeder indicated. Size and capacity of feeder must be based upon local requirements and water analysis. The feeder must be furnished with an air vent, gauge glass, funnel, valves, fittings, and piping. All materials of construction must be compatible with the chemicals being used.

2.17.2 Make Up Water Analysis

The make up water conditions reported ~~as prescribed in ASTM D596~~ are as follows:

Date of Sample	~~[~~_____~~]~~
Temperature	~~[~~_____~~]~~ degrees C
Silica (SiO2)	~~[~~_____~~]~~ ppm (mg/1)
Insoluble	~~[~~_____~~]~~ ppm (mg/1)
Iron and Aluminum Oxides	~~[~~_____~~]~~ ppm (mg/1)
Calcium (Ca)	~~[~~_____~~]~~ ppm (mg/1)
Magnesium (Mg)	~~[~~_____~~]~~ ppm (mg/1)
Sodium and Potassium (Na and K)	~~[~~_____~~]~~ ppm (mg/1)
Carbonate (HCO3)	~~[~~_____~~]~~ ppm (mg/1)
Sulfate (SO4)	~~[~~_____~~]~~ ppm (mg/1)
Chloride (C1)	~~[~~_____~~]~~ ppm (mg/1)
Nitrate (NO3)	~~[~~_____~~]~~ ppm (mg/1)
Turbidity	~~[~~_____~~]~~ unit
pH	~~[~~_____~~]~~
Residual Chlorine	~~[~~_____~~]~~ ppm (mg/1)
Total Alkalinity	~~[~~_____~~]~~ ppm (mg/1)
Noncarbonate Hardness	~~[~~_____~~]~~ epm (mg/1)
Total Hardness	~~[~~_____~~]~~ epm (mg/1)
Dissolved Solids	~~[~~_____~~]~~ ppm (mg/1)
Fluorine	~~[~~_____~~]~~ ppm (mg/1)

Conductivity	~~[_____]~~ microsiemens/cm

2.17.3 Chemicals

The chemical company must provide pretreatment chemicals that will remove and permit flushing of mill scale, oil, grease, and other foreign matter from the water heating system. The chemical company must also provide all treatment chemicals required for the initial fill of the system and for a period of one year of operation. The chemical company must determine the correct chemicals and concentrations required for the water treatment. The chemicals must not be proprietary and must meet required federal, state, and local environmental regulations for the treatment of heating water systems and discharge to the sanitary sewer. The chemicals must remain stable throughout the operating temperature range of the system, and must be compatible with pump seals and other elements of the system.

~~2.17.4 Glycol Solutions~~

~~Provide a [_____] percent concentration by volume of industrial grade [ethylene] [propylene] glycol. The glycol must be tested in accordance with ASTM D1384 with less than 0.013 mm penetration per year for all system metals. The glycol must contain corrosion inhibitors. Silicate based inhibitors must not be used. The solution must be compatible with pump seals, other elements of the system, and all water treatment chemicals used within the system.~~

2.17.4 Test Kits

Provide all required test kits and reagents for determining the proper water conditions.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

3.2 INSTALLATION

Install all work as indicated and in accordance with the manufacturer's diagrams and recommendations.

3.3 COLOR CODE MARKING AND FIELD PAINTING

Color code marking, field painting of exposed pipe, and field painting of factory primed equipment must be as specified in Section 09 90 00 PAINTS AND COATINGS.

3.4 WELDING

Submit ~~[_____]~~4 copies of qualified procedures and list of names and identification symbols of qualified welders and welding operators, prior to welding operations. ~~[~~Piping must be welded in accordance with qualified procedures using performance qualified welders and welding operators. Procedures and welders must be qualified in accordance with

~~ASME BPVC SEC IX~~JIS Z 3801 and any applicable standards. Welding procedures qualified by others, and welders and welding operators qualified by another employer may be accepted as permitted by ~~ASME B31.1~~ JIS Z 3801 and any applicable standards. The Contracting Officer must be notified 24 hours in advance of tests and the tests must be performed at the work site if practical. The welder or welding operator must apply his assigned symbol near each weld he makes as a permanent record.~~]~~ Structural members must be welded in accordance with Section ~~05 05 23.16 STRUCTURAL WELDING~~05 50 13 MISCELLANEOUS METAL FABRICATIONS.~~ [Welding and nondestructive testing procedures for piping must be as specified in Section 40 05 13.96 WELDING, PRESSURE PIPING.]~~

3.5 PIPING

Unless otherwise specified, pipe and fittings installation must conform to the requirements of ~~ASME B31.1~~JIS G 3455 and other applicable piping requirements. Pipe must be cut accurately to measurements established at the job site and worked into place without springing or forcing, completely clearing all windows, doors, and other openings. Cuttings or other weakening of the building structure to facilitate piping installation will not be permitted without written approval. Pipe or tubing must be cut square, must have burrs removed by reaming, and must be so installed as to permit free expansion and contraction without causing damage to building structure, pipe, joints, or hangers. Changes in direction must be made with factory made fittings, except that bending of pipe up to 100 mm will be permitted, provided a pipe bender is used and wide sweep bends are formed. The center line radius of bends must not be less than six diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted. Vent pipes must be installed through the roof as indicated and must be flashed as specified. Horizontal mains must pitch up or down in the direction of flow as indicated. The grade must be not less than 25 mm in 12 m. Reducing fittings must be used for changes in pipe sizes. Open ends of pipelines and equipment must be capped or plugged during installation to keep dirt or other foreign materials out of the systems. Pipe not otherwise specified must be uncoated. Unions and other components for copper pipe or tubing must be brass or bronze. Connections between ferrous and copper piping must be electrically isolated using dielectric unions.

3.5.1 Joints

Except as otherwise specified, joints used on steel pipe must be threaded for fittings 25 mm and smaller; threaded or welded for 32 mm up through 65 mm; and flanged or welded for 80 mm and larger. Joints between sections of copper tubing or copper pipe must be flared or sweated. Pipe and fittings 32 mm and larger installed in inaccessible conduits or trenches beneath concrete floor slabs must be welded. Unless otherwise specified, connections to equipment must be made with black malleable iron unions for pipe 65 mm or smaller in diameter, and with flanges for pipe 80 mm or larger in diameter.

3.5.2 Low Temperature Systems

Piping may have threaded, welded, flanged or flared, sweated, or grooved mechanical joints as applicable and as specified. Reducing fittings must be used for changes in pipe sizes. In horizontal lines, reducing fittings must be the eccentric type to maintain the top of the adjoining pipes at the same level.

3.5.3 Steam Systems

Piping may have threaded, welded, or flanged joints as applicable and as specified. Reducing fittings must be used for changes in pipe sizes. In horizontal steam lines, reducing fittings must be the eccentric type to maintain the bottom of the lines at the same level. Grooved mechanical joints must not be used.

3.5.4 High And Medium Temperature Systems

Temperature systems must have welded joints to the maximum extent practicable, except screwed joints and fittings may be used at connections to equipment and on piping 65 mm and smaller. Equipment connections 80 mm and larger must be flanged. Piping connections 80 mm and larger may be welded or flanged. In horizontal lines, reducing fittings must be the eccentric type to maintain the tops of adjoining pipes at the same level. Grooved mechanical joints must not be used.

3.5.5 Threaded Joints

Threaded joints must be made with tapered threads properly cut, and must be made tight with PTFE tape complying with ASTM D3308 or equal equivalent, or equivalent thread joint compound applied to the male threads only, and in no case to the fittings.

3.5.6 Welded Joints

Joints must be fusion-welded unless otherwise required. Changes in direction of piping must be made with welding fittings only. Branch connection may be made with either welding tees or branch outlet fittings. Branch outlet fittings must be forged, flared for improvement of flow where attached to the run, and reinforced against external strains.

3.5.7 Flanged Joints or Unions

Provide flanged joints or unions in each line immediately preceding the connection to each piece of equipment or material requiring maintenance such as coils, pumps, control valves, and similar items. Flanged joints must be faced true, provided with gaskets, and made square and tight. Full-faced gaskets must be used with cast iron flanges.

3.5.8 Flared and Sweated Pipe and Tubing

Pipe and tubing must be cut square and burrs must be removed. Both inside of fittings and outside of tubing must be cleaned with an abrasive before sweating. Care must be taken to prevent annealing of fittings and hard drawn tubing when making connection. Installation must be made in accordance with the manufacturer's recommendations. Changes in direction of piping must be made with flared or soldered fittings only. Solder and flux must be lead free. Joints for soldered fittings must be made with silver solder or 95:5 tin-antimony solder. Cored solder must not be used. Joints for flared fittings must be of the compression pattern. Provide swing joints or offsets on all branch connections, mains, and risers to provide for expansion and contraction forces without undue stress to the fittings or to short lengths of pipe or tubing.

3.5.9 Mechanical Tee Joint

An extracted mechanical tee joint may be made in copper tube. Joint must be produced with an appropriate tool by drilling a pilot hole and drawing out the tube surface to form a collar having a minimum height of three times the thickness of the tube wall. To prevent the branch tube from being inserted beyond the depth of the extracted joint, provide dimpled depth stops. The branch tube must be notched for proper penetration into fitting to assure a free flow joint. Joints must be brazed in accordance with NAPHCC NSPC or JIS Z 3192. Soldered joints will not be permitted.

3.5.10 Grooved Joints for Copper Tube

Grooves must be prepared according to the coupling manufacturer's instructions. Grooved fittings, couplings, and grooving tools must be products of the same manufacturer. Pipe and groove dimensions must comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances. Grooved joints must not be used in concealed locations.

3.6 CONNECTIONS TO EQUIPMENT

Provide supply and return connections unless otherwise indicated. Valves and traps must be installed in accordance with the manufacturer's recommendations. Unless otherwise indicated, the size of the supply and return pipes to each piece of equipment must be not smaller than the connections on the equipment. Bushed connections are not permitted. Change in sizes must be made with reducers or increasers only.

3.6.1 Low Temperature Water and Steam and Return Connections

Connections, unless otherwise indicated, must be made with malleable iron unions for piping 65 mm or less in diameter and with flanges for pipe 80 mm or more in diameter.

~~3.6.2 High And Medium Temperature Water Connections~~

~~Connections must be made with 13.8 MPa black malleable iron unions for pipe 19 mm or less in diameter and with flanges for pipe 25 mm and larger in diameter.~~

3.7 BRANCH CONNECTIONS

Branches must pitch up or down as indicated, unless otherwise specified. Connection must be made to insure unrestricted circulation, eliminate air pockets, and permit drainage of the system.

3.7.1 Low Temperature Water Branches

Branches taken from mains must pitch with a grade of not less than 25 mm in 3 m. ~~[Special flow fittings must be installed on the mains to bypass portions of water through each radiator. Special flow fittings must be installed as recommended by the manufacturer.]~~

3.7.2 Steam Supply and Condensate Branches

Branches taken from mains must pitch with a grade of not less than 25 mm in 3 m, unless otherwise indicated.

3.7.3 High And Medium Temperature Water Branches

Branches must take off at 45 degrees in the direction of the fluid flow from the supply and return lines and should be branched from the top or upper half of the main line unless otherwise indicated. Abrupt reduction in pipe sizes must be avoided.

3.8 RISERS

The location of risers is approximate. Exact locations of the risers must be as approved.~~ [Steam supply downfeed risers must terminate in a dirt pocket and must be dripped through a trap to the return line.]~~

3.9 SUPPORTS

3.9.1 General

Hangers used to support piping 50 mm and larger must be fabricated to permit adequate adjustment after erection while supporting the load. Pipe guides and anchors must be installed to keep pipes in accurate alignment, to direct the expansion movement, and to prevent buckling, swaying, and undue strain. All piping subjected to vertical movement when operating temperatures exceed ambient temperatures, must be supported by variable spring hangers and supports or by constant support hangers. Where threaded rods are used for support, they must not be formed or bent.

3.9.1.1 Seismic Requirements for Pipe Supports, Standard Bracing

All piping and attached valves must be supported and braced to resist seismic loads as specified under UFC 3-301-01 and Sections 13 48 73 SEISMIC CONTROL FOR ~~NONSTRUCTURAL COMPONENTS~~MISCELLANEOUS EQUIPMENT ~~[~~and 23 05 48.19 SEISMIC BRACING FOR MECHANICAL SYSTEMS~~] [as shown on the drawings]~~. Structural steel required for reinforcement to properly support piping, headers, and equipment but not shown must be provided under this section. Material used for supports must be as specified under Section ~~05 12 00 STRUCTURAL STEEL~~05 50 13 MISCELLANEOUS METAL FABRICATIONS.

3.9.1.2 Structural Attachments

~~Structural steel brackets required to support piping, headers, and equipment, but not shown, must be provided under this section. Material and installation must be as specified under Section 05 12 00 STRUCTURAL STEEL. [~~Pipe hanger loads suspended from steel joist panel points must not exceed 222 N. Loads exceeding 222 N must be suspended from panel points.~~]~~

3.9.1.3 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, a clip or clamp must be used where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for any individual pipe in the multiple pipe run.

3.9.2 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts and supports must conform to MSS SP-58, except as specified as follows:

3.9.2.1 Types 5, 12, and 26

Use of these types is prohibited.

3.9.2.2 Type 3

Type 3 is prohibited on insulated pipe which has a vapor barrier. Type 3 may be used on insulated pipe that does not have a vapor barrier if clamped directly to the pipe and if the clamp bottom does not extend through the insulation and the top clamp attachment does not contact the insulation during pipe movement.

3.9.2.3 Type 18 Inserts

Type 18 inserts must be secured to concrete forms before concrete is placed. Continuous inserts which allow more adjustment may be used if they otherwise meet the requirements for Type 18 inserts.

3.9.2.4 Type 19 and 23 C-Clamps

Type 19 and 23 C-clamps must be torqued in accordance with MSS SP-58 and have both locknuts and retaining devices, furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.

3.9.2.5 Type 20 Attachments

Provide Type 20 attachments used on angles and channels with an added malleable iron heel plate or adapter.

3.9.2.6 Type 24

Type 24 may be used only on trapeze hanger systems or on fabricated frames.

3.9.2.7 Type 39 Saddle or Type 40 Shield

Where Type 39 saddle or Type 40 shield are permitted for a particular pipe attachment application, the Type 39 saddle must be used on all pipe 100 mm and larger.

3.9.2.8 Horizontal Pipe Supports

Space horizontal pipe supports as specified in MSS SP-58 and install a support not over 300 mm from the pipe fitting joint at each change in direction of the piping. Do not space pipe supports over 1.5 m apart at valves.

3.9.2.9 Vertical Pipe Supports

Support vertical pipe at each floor, except at slab-on-grade, and at intervals of not more than 4.5 m, except support pipe not more than 2.4 m from end of risers, and at vent terminations.

3.9.2.10 Type 35 Guides

Provide Type 35 guides using steel, reinforced PTFE or graphite slides where required to allow longitudinal pipe movement. Provide lateral restraints as required. Slide materials must be suitable for the system operating temperatures, atmospheric conditions and bearing loads encountered. Where steel slides do not require provision for restraint or lateral movement, an alternate guide method may be used. On piping 100 mm and larger, a Type 39 saddle may be welded to the pipe and freely rest on a steel plate. On piping under 100 mm, a Type 40 protection shield may be attached to the pipe or insulation and freely rest on a steel slide plate. Where there are high system temperatures and welding to piping is not desirable, then the Type 35 guide must include a pipe cradle, welded to the guide structure and strapped securely to the pipe. The pipe must be separated from the slide material by at least 100 mm or by an amount adequate for the insulation, which ever is greater.

3.9.2.11 Pipe Hanger Size

Except for Type 3, pipe hangers on horizontal insulated pipe must be the size of the outside diameter of the insulation.

3.9.3 Piping in Trenches

Support piping as indicated.

3.10 PIPE SLEEVES

3.10.1 Pipe Passing Through Concrete or Masonry

Provide pipe passing through concrete or masonry walls or concrete floors or roofs with pipe sleeves fitted into place at the time of construction. Sleeves must not be installed in structural members except where indicated or approved. Rectangular and square openings must be as detailed. Each sleeve must extend through its respective wall, floor, or roof, and must be cut flush with each surface. Unless otherwise indicated, sleeves must provide a minimum of 6 mm annular space between bare pipe or insulation surface and sleeves. Sleeves in bearing walls, waterproofing membrane floors, and wet areas must be steel pipe or cast iron pipe. Sleeves in nonbearing walls, floors, or ceilings may be steel pipe, cast iron pipe, or galvanized sheet metal with lock-type longitudinal seam and of the metal thickness indicated. Except in pipe chases or interior walls, the annular space between pipe and sleeve or between jacket over insulation and sleeve in nonfire rated walls and floors must be sealed as indicated and specified in Section 07 92 00 JOINT SEALANTS. Seal penetrations in fire walls and floors in accordance with Section 07 84 00 FIRESTOPPING.

3.10.2 Pipes Passing Through Waterproofing Membranes

Install pipes passing through waterproofing membranes through a 19.5 kg/square meter lead-flashing sleeve, a 4.9 kg/square meter copper sleeve, or a 0.813 mm thick aluminum sleeve, each having an integral skirt or flange. Flashing sleeve must be suitably formed, and the skirt or flange must extend 200 mm or more from the pipe and must be set over the roof or floor membrane in a troweled coating of bituminous cement. The flashing sleeve must extend up the pipe a minimum of 50 mm above the highest flood level of the roof or a minimum of 250 mm above the roof, whichever is greater, or 250 mm above the floor. The annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation must be sealed as indicated. At the Contractor's option, pipes up to and including 250 mm in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass ~~bolts~~bolts. Waterproofing membrane must be clamped into place and sealant must be placed in the caulking recess.

3.10.3 Mechanical Seal Assembly

In lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve or conduit and sleeve, a modular mechanical type sealing assembly may be installed. The seals must consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. The links must be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tightening of the bolts must cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Each seal assembly must be sized as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals must provide sleeves of the proper diameters.

3.10.4 Counterflashing Alternate

As an alternate to caulking and sealing the annular space between the pipe and flashing sleeve or metal-jacket-covered insulation and flashing sleeve, counterflashing may be by standard roof coupling for threaded pipe up to 150 mm in diameter; lead-flashing sleeve for dry vents and turning the sleeve down into the pipe to form a waterproof joint; or tack-welded or banded-metal rain shield round the pipe and sealing as indicated.

3.10.5 Waterproofing Clamping Flange

Pipe passing through wall waterproofing membrane must be sleeved as specified. In addition, a waterproofing clamping flange must be installed as indicated.

3.10.6 Fire Seal

Where pipes pass through fire walls, fire partitions, fire rated pipe chase walls or floors above grade, provide a fire seal as specified in Section 07 84 00 FIRESTOPPING.

3.10.7 Escutcheons

Provide escutcheons at all finished surfaces where exposed piping, bare or covered, passes through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Escutcheons must be fastened securely to pipe sleeves or to extensions of sleeves without any part of sleeves being visible. Where sleeves project slightly from floors, special deep-type escutcheons must be used. Escutcheons must be chromium-plated iron or chromium-plated brass, either one-piece or split pattern, held in place by internal spring tension or setscrew.

3.11 ANCHORS

Provide anchors where necessary or indicated to localize expansion or prevent undue strain on piping. Anchors must consist of heavy steel collars with lugs and bolts for clamping and attaching anchor braces, unless otherwise indicated. Anchor braces must be installed using turnbuckles where required. Supports, anchors, or stays must not be attached in places where construction will be damaged by installation operations or by the weight or expansion of the pipeline.

3.12 PIPE EXPANSION

The expansion of supply and return pipes must be provided for by changes in the direction of the run of pipe, by expansion loops, or by expansion joints as indicated. Low temperature water and steam expansion joints may be one of the types specified.~~ [High] [Medium] temperature water system expansion joints may be one of the joints specified, except slip-tube type.~~

3.12.1 Expansion Loops

Expansion loops must provide adequate expansion of the main straight runs of the system within the stress limits specified in ~~ASME B31.1~~JIS B 8265. The loops must be cold-sprung and installed where indicated. Provide pipe guides as indicated.

3.12.2 Slip-Tube Joints

Slip-tube type expansion joints must be used for steam and low temperature water systems only and must be installed where indicated. The joints must provide for either single or double slip of the connected pipes as indicated and for the traverse indicated. The joints must be designed for a working temperature and pressure suitable for the application and in no case less than ~~[_____]~~862 kPa. The joints must be in accordance with applicable requirements of EJMA Stds and ~~ASME B31.1~~JIS B 8265. End connections must be flanged. Provide anchor bases or support bases must be provided as indicated or required. Initial setting must be made in accordance with the manufacturer's recommendations to allow for ambient temperature at time of installation. Pipe alignment guides must be installed as recommended by the joint manufacturer, but in any case must be not more than 1.5 m from expansion joint, except in lines 100 mm or smaller where guides must be installed not more that 600 mm from the joint.

3.12.3 Bellows-Type Joint

Bellows-type joint design and installation must comply with EJMA Stds standards. The joints must be designed for the working temperature and pressure suitable for the application and must be not less than 1.03 MPa in any case.

3.12.4 Flexible Ball Joints

Flexible ball joints may be threaded (to 50 mm only), flanged, or welded end as required. The ball-type joint must be designed and constructed in accordance with the generally accepted engineering principle stated in ~~ASME B31.1, and ASME BPVC SEC VIII D1~~JIS B 8265, where applicable. Flanges must conform to the diameter and drilling of ~~ASME B16.5~~JIS B 2220. Molded gaskets furnished must be suitable for the service intended.

3.13 VALVES AND EQUIPMENT ACCESSORIES

3.13.1 Valves and Equipment

Install valves at the locations shown or specified, and where required for the proper functioning of the system as directed. Gate valves must be used unless otherwise indicated, specified, or directed. Valves must be installed with their stems horizontal to or above the main body of the valve. Valves used with ferrous piping must have threaded or flanged ends and sweat-type connections for copper tubing.

3.13.2 Gravity Flow-Control Valve

Install the valve to control the flow of water in the supply main near the heat exchanger. The valve must operate so that when the circulating pump starts, the increased pressure within the main will open the valve; when the pump stops, the valve will close. The valve must be constructed with a cast iron body and must be provided with a device whereby the valve can be opened manually to allow gravity circulation. The flow-control valve must be designed for the intended purpose, and must be installed as recommended by the manufacturer.

3.13.3 Thermometer Socket

Provide a thermometer well in each return line for each circuit in multicircuit systems.

3.13.4 Air Vents

Install vents where indicated, and on all high points and piping offsets where air can collect or pocket.

3.13.4.1 Water Air Vents

~~[High] [Medium]~~Provide temperature water air vents must be as indicated. Vent discharge lines must be double-valved with globe valves and must discharge into a funnel drain.

3.13.4.2 Steam Air Vents

Steam air vents must be a quick-acting valve that continuously removes air. Valve must be constructed of corrosion-resisting metal, must be designed to withstand the maximum piping system pressure, and must automatically close tight to prevent escape of steam and condensate. Vent must be provided with a manual isolation valve. Provide a vent on the shell of each steam heat exchanger.

3.14 STEAM TRAPS

Install float Traps in the condensate line as indicated. Other steam traps must be installed where indicated.

3.15 UNIT HEATERS

Install unit heaters as indicated and in accordance with the manufacturer's instructions.

3.16 INSULATION

Thickness of insulation materials for piping and equipment and application must be in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

3.17 MANUFACTURER'S SERVICES

Provide the services of a manufacturer's representative who is experienced in the installation, adjustment, and operation of the equipment specified. The representative must supervise the installation, adjustment, and testing of the equipment.

3.18 TESTING AND CLEANING

Submit performance test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, upon completion and testing of the installed system. Indicate in each test report the final position of controls.

3.18.1 Pressure Testing

Notify the Contracting Officer ~~[_____]~~14 days before the tests are to be conducted. Perform the tests in the presence of the Contracting Officer. Furnish all instruments and personnel required for the tests. Electricity, steam, and water will be furnished by the Government. All test results must be accepted before thermal insulation is installed. The entire low temperature heating system, including heat exchanger, radiators and fittings, must be hydrostatically tested and proved tight under a pressure of 310 kPa for a period of four hours.

3.18.2 Test of Backflow Prevention Assemblies

Test backflow prevention assemblies in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.18.3 Cleaning

After the hydrostatic and backflow prevention tests have been made and prior to the operating tests, the heat exchanger and piping must be thoroughly cleaned by filling the system with a solution of 0.5 kg of caustic soda or 0.5 kg of trisodium phosphate per 200 L of water. Observe the proper safety precautions in the handling and use of these chemicals. Heat the water to approximately 66 degrees C, and circulate the solution in the system for a period of 48 hours, then drain and flush the system thoroughly with fresh water. Wipe clean all equipment , and remove all traces of oil, dust, dirt, or paint spots. The Contractor will be responsible for maintaining the system in a clean condition until final acceptance. Lubricate bearings with oil or grease as recommended by the manufacturer.

3.18.4 Water Treatment Testing

Identify in the water quality test report the chemical composition of the heating water. The report must include a comparison of the condition of the water with the chemical company's recommended conditions. Document any required corrective action within the report. Analyze the heating water ~~[~~prior to the acceptance of the facility~~] [and] [a minimum of once a~~

~~month for a period of one year]~~ by the water treatment company. The analysis must include the following information recorded in accordance with ~~ASTM D596~~the table below.

Date of Sample	~~[~~_____~~]~~
Temperature	~~[~~_____~~]~~ degrees C
Silica (SiO2)	~~[~~_____~~]~~ ppm (mg/1)
Insoluble	~~[~~_____~~]~~ ppm (mg/1)
Iron and Aluminum Oxides	~~[~~_____~~]~~ ppm (mg/1)
Calcium (Ca)	~~[~~_____~~]~~ ppm (mg/1)
Magnesium (Mg)	~~[~~_____~~]~~ ppm (mg/1)
Sodium and Potassium (Na and K)	~~[~~_____~~]~~ ppm (mg/1)
Carbonate (HCO3)	~~[~~_____~~]~~ ppm (mg/1)
Sulfate (SO4)	~~[~~_____~~]~~ ppm (mg/1)
Chloride (C1)	~~[~~_____~~]~~ ppm (mg/1)
Nitrate (NO3)	~~[~~_____~~]~~ ppm (mg/1)
Turbidity	~~[~~_____~~]~~ unit
pH	~~[~~_____~~]~~
Residual Chlorine	~~[~~_____~~]~~ ppm (mg/1)
Total Alkalinity	~~[~~_____~~]~~ ppm (mg/1)
Noncarbonate Hardness	~~[~~_____~~]~~ epm (mg/1)
Total Hardness	~~[~~_____~~]~~ epm (mg/1)
Dissolved Solids	~~[~~_____~~]~~ ppm (mg/1)
Fluorine	~~[~~_____~~]~~ ppm (mg/1)
Conductivity	~~[~~_____~~]~~ microsiemens/cm

3.19 FRAMED INSTRUCTIONS

Submit proposed diagrams, instructions, and other sheets, prior to posting. Show in the instructions wiring and control diagrams and complete layout of the entire system. The instructions must include, in typed form, condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation and procedures for safely starting and stopping the system.

Post framed instructions, containing wiring and control diagrams under glass or in laminated plastic, where directed. Condensed operating instructions, prepared in typed form, must be framed as specified above and posted beside the diagrams. Post the framed instructions before acceptance testing of the system.

3.20 FIELD TRAINING

Provide a field training course for designated operating and maintenance staff members. Provide training for a total period of ~~[_____]~~8 hours of normal working time starting after the system is functionally complete but prior to final acceptance tests. Field training must cover all of the items contained in the approved Operation and Maintenance Manuals. Submit ~~[6] [_____]~~4 copies of operation and ~~[6] [_____]~~4 copies of maintenance manuals for the equipment furnished. One complete set, prior to performance testing and the remainder upon acceptance. Operating manuals must detail the step-by-step procedures required for system startup, operation, and shutdown. Operating manuals must include the manufacturer's name, model number, parts list, and brief description of all equipment and their basic operating features. Maintenance manuals must list routine maintenance procedures, water treatment procedures, possible breakdowns and repairs, and troubleshooting guides. Maintenance manuals must include piping and equipment layout and simplified wiring and control diagrams of the system as installed. Provide manuals prior to the field training course.

3.21 TESTING, ADJUSTING AND BALANCING

Except as specified herein, testing, adjusting, and balancing must be in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS.

-- End of Section --

SECTION 23 64 10

WATER CHILLERS, VAPOR COMPRESSION TYPE
11/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 1518	(2013) Rolling Bearings-Dynamic Load Ratings and Rating Life
JIS B 1521	(2012) Rolling Bearings-Deep Groove Ball Bearings
JIS B 1534	(2013) Rolling Bearings-Tapered Roller Bearings
JIS B 6801	(2003) Manual Blowpipes for Welding, Cutting and Heating
JIS B 8267	(20~~15~~22) Construction of Pressure Vessel
JIS B 8613	(1994) Water Chilling Unit
JIS C 4212	~~(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)~~(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS F 0602	(1995) Shipbuilding-Non-Asbestos Gaskets to Cargo Piping System-Application Standard
JIS G 3101	(20~~20~~24) Rolled Steels for General Structure
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS Z 2371	(2015) Methods of Salt Spray Testing

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification

THE JAPAN REFRIGERATION AND AIR CONDITIONING INDUSTRY ASSOCIATION (JRAIA)

JRA	JRA Standard

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval or for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Water Chiller; G

Water Chiller - Field Acceptance Test Plan

SD-06 Test Reports

Field Acceptance Testing

Water Chiller - Field Acceptance Test Report

System Performance Tests

SD-07 Certificates

Refrigeration System;

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals

1.3 CERTIFICATIONS

1.3.1 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified Technician to meet requirements in Refrigerant Handling Technician (Reibai-Furonrui-Toriatsukai Gijutsusha) under Japan Refrigeration and Air-Conditioning Industry Association. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.4 SAFETY REQUIREMENTS

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with JIS B 6801.

1.5 DELIVERY, STORAGE, AND HANDLING

Stored items must be protected from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation will be the Contractor's responsibility. Any materials found to be damaged must be replaced at the Contractor's expense. During installation, piping and similar openings must be capped to keep out dirt and other foreign matter.

1.6 PROJECT REQUIREMENTS

1.6.1 Verification of Dimensions

The Contractor must become familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

PART 2 PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

Materials and equipment will be standard Commercial cataloged products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. These products must have a two year record of satisfactory field service prior to bid opening. The two year record of service must include applications of equipment and materials under similar circumstances and of similar size. Products having less than a two year record of satisfactory field service will be acceptable if a certified record of satisfactory field service for not less than 6000 hours can be shown. The 6000 hour service record must not include any manufacturer's prototype or factory testing. Satisfactory field service must have been completed by a product that has been, and presently is being sold or offered for sale on the commercial market through the following copyrighted means: advertisements, manufacturer's catalogs, or brochures. Equivalent shall be supported by a product representative and service organization located in Japan

2.2 MANUFACTURER'S STANDARD NAMEPLATES

Major equipment including chillers, compressors, compressor drivers, condensers, water coolers, receivers, refrigerant leak detectors, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life. Plates must be fixed in prominent locations with nonferrous screws or bolts.

Nameplates are required on major components if the manufacturer needs to provide specific engineering and manufacturing information pertaining to the particular component. Should replacement of this component be required, nameplate information will insure correct operation of the unit after replacement of this component.

2.3 ELECTRICAL WORK

a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown, must be provided. For packaged equipment, the manufacturer must provide controllers including the required monitors and timed restart.

b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with JIS C 4212.

c. For polyphase motors, provide squirrel-cage medium induction motors, including motors that are part of a system, and that meet the efficiency ratings for premium efficiency motors in accordance with JIS C 4212.

d. Provide motors in accordance with JIS C 4212 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Motors must be rated for continuous duty with the enclosure specified. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque must be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. ~~[~~Motor bearings must be fitted with grease supply fittings and grease relief to outside of the enclosure.~~]~~ Motor enclosure type may be either TEAO or TEFC.

2.4 SELF-CONTAINED WATER CHILLERS, VAPOR COMPRESSION TYPE

Unless necessary for delivery purposes, units must be assembled, leak-tested, charged (refrigerant and oil), and adjusted at the factory. In lieu of delivery constraints, a chiller may be assembled, leak-tested, charged (refrigerant and oil), and adjusted at the job site by a factory representative. Unit components delivered separately must be sealed and charged with a nitrogen holding charge. Parts weighing 23 kg or more which must be removed for inspection, cleaning, or repair, such as motors, gear boxes, cylinder heads, casing tops, condenser, and cooler heads, must have lifting eyes or lugs. Chiller must be provided with a single point wiring connection for incoming power supply. Chiller's condenser and water cooler must be provided with ~~[~~standard~~]~~ water boxes with ~~[grooved mechanical] [flanged] [~~welded~~]~~ connections.

2.4.1 ~~Scroll, Reciprocating, or~~ Rotary Screw Type

Chiller must be certified for performance per JIS B 8613. The chiller's performance must be rated in accordance with JIS B 8613. Chiller must conform to Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). As a minimum, chiller must include the following components as defined in paragraph CHILLER COMPONENTS.

a. Refrigerant and oil

b. Structural base

c. Chiller refrigerant circuit

d. Controls package

e. Scroll, reciprocating, or rotary screw compressor

f. Compressor driver, ~~[~~electric motor~~] [gas-engine]~~

g. Compressor driver connection

h. Water cooler (evaporator)

i. ~~[~~Air~~][Water]~~-cooled condenser coil

~~j. Heat recovery condenser~~

~~k~~j. Receiver

~~l~~k. Tools

~~2.4.2 Centrifugal or Rotary Screw Type~~

~~Chiller must be certified for performance per JIS B 8613. The chiller's performance must be rated in accordance with JIS B 8613. Chiller must conform to Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). As a minimum, chiller must include the following components as defined in paragraph CHILLER COMPONENTS.~~

~~a. Refrigerant and oil~~

~~b. Structural base~~

~~c. Chiller refrigerant circuit~~

~~d. Controls package~~

~~e. Centrifugal or rotary screw compressor~~

~~f. Compressor driver, [electric motor] [gas-engine] [steam turbine]~~

~~g. Compressor driver connection~~

~~h. Water cooler (evaporator)~~

~~i. [Air][Water]-cooled condenser coil~~

~~j. Heat recovery condenser coil~~

~~k. Receiver~~

~~l. Purge system for chillers which operate below atmospheric pressure~~

~~m. Tools~~

2.5 SPLIT-SYSTEM WATER CHILLER, VAPOR COMPRESSION TYPE

Total chiller system must be certified for performance per JIS B 8613. Individual chiller components must be constructed and rated in accordance with the applicable JRA standards. Chiller system must conform to Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). The construction of chiller must be MLIT-M compliant. The manufacturer must provide certification of compliance. Chiller must be assembled, leak-tested, charged (refrigerant and oil), and adjusted at the job site in strict accordance with manufacturer's recommendations. Unit assembly must be completed in strict accordance with manufacturer's recommendations. Chiller must operate within capacity range and speed recommended by the manufacturer. Parts weighing 23 kg or more which must be removed for inspection, cleaning, or repair, must have lifting eyes or lugs. Chiller must include all customary auxiliaries deemed necessary by the manufacturer for safe, controlled, automatic operation of the equipment. Chiller's water cooler must be provided with ~~[~~standard~~] [marine]~~ water boxes with ~~[grooved mechanical] [flanged] [~~welded~~]~~ connections. Chillers must operate at partial load conditions without increased vibration over normal vibration at full load, and must be capable of continuous operation down to minimum capacity. As a minimum, chiller must include the following components as defined in paragraph CHILLER COMPONENTS.

a. Refrigerant and oil

b. Structural base

c. Chiller refrigerant circuit

d. Controls package

e. Receiver

f. Tools

~~2.5.1 Compressor-Chiller Unit~~

~~As a minimum, the compressor-chiller unit must include the following components as defined in paragraph CHILLER COMPONENTS.~~

~~a. Scroll, reciprocating, or rotary screw compressor~~

~~b. Compressor driver, electric motor~~

~~c. Compressor driver connection~~

~~d. Water cooler (evaporator)~~

~~2.5.2 Condensing Unit~~

~~As a minimum, the condensing unit must include the following components as defined in paragraph CHILLER COMPONENTS.~~

~~a. Scroll, reciprocating, or rotary screw compressor~~

~~b. Compressor driver, electric motor~~

~~c. Compressor driver connection~~

~~d. Air or water cooled condenser~~

~~2.5.3 Remote Air-Cooled Condenser~~

~~Condenser must be a factory-fabricated and assembled unit, consisting of coils, fans, and condenser fan motors. Condenser must be rated in accordance with JIS B 8623. [Unless the condenser coil is completely protected through inherent design, louvered panel coil guards must be provided by the manufacturer to prevent physical damage to the coil.] Manufacturer must certify that the condenser and associated equipment are designed for the submitted condensing temperature. For design conditions, if matched combination catalog ratings matching remote condensers to compressors are not available, the Contractor must furnish a crossplotting~~

~~of the gross heat rejection of the condenser against the gross heat rejection of the compressor, for the design conditions to show the compatibility of the equipment furnished.~~

~~2.5.3.1 Condenser Casing~~

~~Condenser casing must be aluminum not less than [1.016] [2.032] mm or hot-dip galvanized steel not lighter than 18 gauge 1.311 mm or JIS G 3302. [Condensers having horizontal air discharge must be provided with discharge baffle to direct air upward, constructed of the same material and thickness as the casing].~~

~~2.5.3.2 Coil~~

~~[Condenser coil must be of the extended-surface fin-and-tube type and must be constructed of seamless [copper] [or] tubes with compatible [copper] [or] [aluminum] fins. Fins must be soldered or mechanically bonded to the tubes and installed in a metal casing. Coils must be circuited and sized for a minimum of 3 degrees C subcooling and full pumpdown capacity. Coil must be factory leak and pressure tested after assembly in accordance with or in Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese Standard product][The condenser coil must be of the microchannel heat exchanger technology (MCHX) type consisting of a series of flat tubes containing a series of multiple, parallel flow microchannels layered between the refrigerant manifolds in a two-pass arrangement. Provide coils constructed of aluminum alloys for fins, tubes, and manifolds. Coil must be factory leak and pressure tested after assembly in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese product.]~~

~~Coil must be entirely coated in accordance with the requirements of paragraph COIL CORROSION PROTECTION.~~

~~2.5.3.3 Fans~~

~~Provide propeller type fans as best suited for the application. Fans must be direct. Fans must be statically and dynamically balanced.~~

~~2.5.3.4 Condenser Sizing~~

~~Size condensers for full capacity at 16.67 degrees C temperature difference between entering outside air and condensing refrigerant. Subcooling must not be considered in determining compressor and condenser capacities. For design conditions, submit a cross-plot of net refrigeration effect of compressor to establish net refrigeration effect and compatibility or JIS B 8608 of equipment furnished.~~

~~2.5.3.5 Low Ambient Control~~

~~Provide factory mounted head pressure control for operation during low ambient conditions. Head pressure must be controlled by [fan cycling,] [fan speed control,]. Low ambient control must permit compressor operation below[4.4 degrees C][[_____] degrees C].~~

~~2.5.3.6 High Ambient Unloading~~

~~Provide unloading capability to allow operation in high ambient conditions [[_____] degrees C] above design conditions.~~

~~2.5.4 Remote Water-Cooled Condenser~~

~~Condenser must be a factory-fabricated and assembled unit constructed and rated in accordance with JIS B 8623. Condenser may be of either the shell-and-coil or shell-and-tube type design. Condenser's refrigerant side must be designed and factory pressure tested to comply with in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese standard. Condenser's water side must be designed and factory pressure tested. Condensers must be complete with pressure relief valve or rupture disk, water drain connections, refrigerant charging valve, refrigerant valves, liquid-level indicating devices, and stand or saddle. Low pressure refrigerant condenser must be provided with a purge valve located at the highest point in the condenser to purge non-condensibles trapped in the condenser. Condenser shell must be constructed of seamless or welded steel. Coil bundles must be totally removable and arranged to drain completely. Tubes may be either seamless copper, plain, integrally finned with smooth bore or integrally finned with enhanced bore. Each tube must be individually replaceable, except for the coaxial tubes. Tubes must be installed into carbon mild steel tube sheets by rolling. Tube baffles must be properly spaced to provide adequate tube support and cross flow. Condenser performance must be based on water velocities per JIS B 8623 and a fouling factor per JIS B 8613. Water-cooled condensers may be used for refrigerant storage in lieu of a separate liquid receiver, if the condenser storage capacity is 20 percent in excess of the fully charged system for remote water cooled condensers. As a minimum, the condenser must include the following components as defined in paragraph CHILLER COMPONENTS.~~

~~a. Liquid-level indicating devices.~~

~~b. Companion flanges, bolts, and gaskets for flanged water connections.~~

2.6 CHILLER COMPONENTS

2.6.1 Refrigerant and Oil

Refrigerants must be one of the fluorocarbon gases. Refrigerants must have number designations and safety classifications in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese standard. CFC-based refrigerants are prohibited. Refrigerants must have an Ozone Depletion Potential (ODP) no greater than 0.0, with the exception of R-123. Provide SDS sheets for all refrigerants.

2.6.2 Structural Base

Chiller and individual chiller components must be provided with a factory-mounted structural steel base (welded or bolted) or support legs. Chiller and individual chiller components must be isolated from the building structure by means of ~~[molded neoprene isolation pads.] [~~ vibration isolators with published load ratings. Vibration isolators must have isolation characteristics as recommended by the manufacturer for the unit supplied and the service intended.~~]~~

2.6.3 Chiller Refrigerant Circuit

Chiller refrigerant circuit must be completely piped and factory leak tested in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese standard. ~~[For multicompressor units, not less than 2 independent refrigerant circuits must be provided.~~

~~]~~Circuit must include as a minimum a ~~[~~combination filter and drier,~~]~~ combination sight glass and moisture indicator, an electronic or thermostatic expansion valve with external equalizer or float valve, charging ports, compressor service valves for field-serviceable compressors, and superheat adjustment.

2.6.4 Controls Package

Provide chillers with a complete ~~[~~factory-mounted~~] [remote-mounted where indicated]~~, microprocessor based operating and safety control system. Controls package must contain as a minimum a digital display, an on-auto-off switch, ~~[motor starters,] [variable frequency motor controller~~ combination motor and VFD starter,~~]~~ ~~[~~disconnect switches,~~]~~ power wiring, and control wiring. Controls package must provide operating controls, monitoring capabilities, programmable setpoints, safety controls, and ~~[~~BAS~~]~~ ~~[UMCS]~~ interfaces as defined below.

2.6.4.1 Operating Controls

Chiller must be provided with the following adjustable operating controls as a minimum.

a. Leaving chilled water temperature control

b. Adjustable timer or automated controls to prevent a compressor from short cycling

c. Automatic lead/lag controls (adjustable) for multi-compressor units

d. Load limiting

e. System capacity control to adjust the unit capacity in accordance with the system load and the programmable setpoints. Controls must automatically re-cycle the chiller on power interruption.

f. Startup and head pressure controls to allow system operation at all ambient temperatures down to ~~[_____]~~-3.9 degrees C.

~~[~~ g. Fan sequencing for air-cooled condenser

~~]~~2.6.4.2 Monitoring Capabilities

During normal operations, the control system must be capable of monitoring and displaying the following operating parameters. Access and operation of display must not require opening or removing any panels or doors.

a. Entering and leaving chilled water temperatures

b. ~~[Entering and leaving chilled water pressure][Chilled water flow]~~ Entering and leaving chilled water pressure/chilled water flow

~~c. [Entering and leaving condenser water pressure][Condenser water flow]~~

~~d~~c. Self diagnostic

~~e~~d. Operation status

~~f~~e. Operating hours

~~g~~f. Number of starts

~~h~~g. Compressor status (on or off)

~~i~~h. Compressor load (percent)

~~j~~i. Refrigerant discharge and suction pressures

~~k~~j. Magnetic bearing levitation status (if applicable)

~~l~~k. Magnetic bearing temperatures (if applicable)

~~m~~l. Oil pressure

~~n~~m. Condenser water entering and leaving temperatures

~~o~~n. Number of purge cycles over the last 7 days

2.6.4.3 Configurable Setpoints

The control system must be capable of being configured directly at the unit's interface panel. ~~[No parameters may be capable of being changed without first entering a security access code.]~~ The programmable setpoints must include the following as a minimum:

a. Leaving Chilled Water Temperature

~~b. Leaving Condenser Water Temperature~~

~~c~~b. Time Clock/Calendar Date

2.6.4.4 Safety Controls with Manual Reset

Chiller must be provided with the following safety controls which automatically shutdown the chiller and which require manual reset.

a. Low chilled water temperature protection

b. High condenser refrigerant discharge pressure protection

c. Low evaporator pressure protection

d. Chilled water flow detection

e. High motor winding temperature protection

f. Low oil flow protection if applicable

g. Magnetic bearing controller (MBC), Internal fault (if applicable)

h. MBC, High bearing temperature (if applicable)

i. MBC, Communication fault (if applicable)

j. MBC, Power supply fault (if applicable)

k. Motor current overload and phase loss protection

2.6.4.5 Safety Controls with Automatic Reset

Chiller must be provided with the following safety controls which automatically shutdown the chiller and which provide automatic reset.

a. Over/under voltage protection

b. Chilled water flow interlock

c. MBC, Vibration (if applicable)

d. MBC, No levitation (if applicable)

e. Phase reversal protection

2.6.4.6 Remote Alarm

During the initiation of a safety shutdown, a chiller's control system must be capable of activating a remote alarm bell. In coordination with the chiller, the Contractor must provide an alarm circuit (including transformer if applicable) and a minimum 100 mm diameter alarm bell. Alarm circuit must activate bell in the event of machine shutdown due to the chiller's monitoring of safety controls. The alarm bell must not sound for a chiller that uses low-pressure cutout as an operating control.

2.6.4.7 Utility Monitoring and Control System Interface

Provide a Utility Monitoring and Control System (UMCS) interface meeting the requirements of Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and the requirements of ~~[Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS] [or] [~~Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS~~]~~. The interface must provide all system operating conditions, capacity controls, and safety shutdown conditions as network points. In addition, the following points must be overridable via the network interface:

a. Unit Start/Stop

b. Leaving Chilled Water Temperature Setpoint

~~c. Leaving Condenser Water Temperature Setpoint~~

2.6.5 Compressor(s)

~~2.6.5.1 Scroll Compressor(s)~~

~~Compressors must be of the hermetically sealed design. Compressors must be mounted on vibration isolators to minimize vibration and noise. Rotating parts must be statically and dynamically balanced at the factory to minimize vibration. Lubrication system must be centrifugal pump type equipped with a means for determining oil level and an oil charging valve. Crankcase oil heater must be provided.[Provide continuous compressor unloading to [10 percent][15 percent] of full-load capacity by way of variable speed compressor motor controller or variable unloading of the scroll.]~~

2.6.5.1 Rotary Screw Compressor(s)

Compressors must operate stably for indefinite time periods to at least 25 percent capacity reduction without gas bypass external to the compressor. Provision must be made to insure proper lubrication of bearings and shaft seals on shutdown with or without electric power supply. Rotary screw compressors must include:

a. An open or hermetic, positive displacement, oil-injected design directly driven by the compressor driver. Allow access to internal compressor components for repairs, inspection, and replacement of parts.

b. Rotors must be solid steel, possessing sufficient rigidity for proper operation.

c. A maximum rotor operating speed no greater than 3600 RPM. Provide cast iron rotor housing.

d. Casings of cast iron, precision machined for minimal clearance about periphery of rotors with minimal clearance at rotor tops and rotor ends.

e. A lubrication system of the forced-feed type that provides oil at the proper pressure to all parts requiring lubrication.

f. Bearing housing must be conservatively loaded and rated for an L(10) life of not less than the requirement of JIS B 1518. Shaft main bearings of the sleeve type with heavy duty bushings or rolling element type in accordance with JIS B 1521, JIS B 1534.

g. A differential oil pressure or flow cutout to allow the compressor to operate only when the required oil pressure or flow is provided to the bearings.

h. ~~[A temperature- or pressure-initiated, hydraulically actuated, single-slide-valve, capacity-control system to provide minimum automatic capacity modulation from 100 percent to 15 percent.] [~~Use a Variable Frequency Drive (VFD) to modulate capacity modulation from 100 percent to 15 percent.~~]~~

i. An oil separator and oil return system to remove oil entrained in the refrigerant gas and automatically return the oil to the compressor.

j. Crankcase oil heaters must be provided.

~~2.6.5.2 Centrifugal Compressor(s)~~

~~Centrifugal compressors may be either single or multistage, having dynamically balanced impellers, either direct or gear driven by the compressor driver. Impellers must be over-speed tested at 1.2 times the impeller-shaft speed. Impeller shaft must be steel with sufficient rigidity for proper operation at any required operating speed. Compressors must be capable of variable speed operation and may have either oil-free bearing drives or oil-lubricated bearing drives. Centrifugal compressors must include:~~

~~a. Shaft main bearings that are either oil lubricated, oil free ceramic or magnetic levitated. The oil lubricated bearings must be the~~

~~rolling element type in accordance with JIS B 1521 JIS B 1534, journal type with bronze or babbitt liners, or of the aluminum-alloy one-piece insert type. Oil lubricated or oil free ceramic bearings must be rated for an L(10) life of not less than the JIS B 1518 requirement. Magnetic levitated main shaft bearings must be in accordance with ISO 14839-1, ISO 14839-2, ISO 14839-3, ISO 14839-4, and provided with radial and axial magnetic levitated bearings (combination permanent and electro magnets) to levitate the shaft thereby eliminating metal to metal contact and thus eliminating the need for oil. The active magnetic bearings must be equipped with an automatic vibration reduction and balancing system. Each bearing position must be sensed by position sensors and provide real time positioning of the rotor shaft, controlled by on-board digital electronics. In the event of a power failure, the magnetic bearings will remain in operation throughout the compressor coast-down using a reserve power supply. Provide mechanical bearings designed for emergency touchdowns, as a backup to the magnetic bearings.~~

~~b. Casing of cast iron, aluminum, or steel plate with split sections gasketed and bolted or clamped together.~~

~~c. Lubrication system of the forced-feed type that provides oil at the proper pressure to all parts requiring lubrication.~~

~~d. Provisions to ensure proper lubrication of bearings and shaft seals prior to starting and upon stopping with or without electric power supply (if applicable). On units providing forced-feed lubrication prior to starting, a differential oil pressure cutout interlocked with the compressor starting equipment must allow the compressor to operate only when the required oil pressure is provided to the bearings (if applicable).~~

~~e. Oil sump heaters controlled as recommended by the manufacturer.~~

~~f. Temperature-or pressure-actuated prerotation vane, variable geometry diffuser or suction damper to provide automatic capacity modulation from 100 percent capacity to 25 percent capacity. If operation to 25 percent capacity cannot be achieved without providing gas bypass external to the compressor, then the Contractor must indicate in the equipment submittal the load percent at which external hot gas bypass is required to prevent surge and to provide the specified capacity reduction and its impact on performance.~~

2.6.6 Compressor Driver, Electric Motor

Components such as motors, ~~[~~starters~~]~~, ~~[variable speed drives]~~ and wiring must be in accordance with paragraph ELECTRICAL WORK. ~~[Motor starter][Variable frequency drive]~~Combination motor and VFD starter must be ~~[unit mounted] [~~remote mounted~~]~~ as indicated with ~~[~~starter~~][variable frequency drive]~~ type, wiring, and accessories coordinated with the chiller manufacturer.

2.6.7 Compressor Driver Connections

~~[Each compressor must be driven by a V-belt drive or direct connected through a flexible coupling, except that flexible coupling is not required on hermetic units. V-belt drives must be designed for not less than 150 percent of the driving motor capacity. Flexible couplings must be of the type that does not require lubrication.] [~~Each machine driven through speed-increasing gears must be so designed as to assure self-alignment, interchangeable parts, proper lubrication system, and minimum unbalanced forces. Bearings must be of the sleeve or roller type. Gear cases must be oil tight. Shaft extensions must be provided with seals to retain oil and exclude all dust.

~~]~~2.6.8 Water Cooler (Evaporator)

Cooler must be of the shell-and-coil or shell-and-tube type design. Cooler shell must be constructed of seamless or welded steel. Coil bundles must be totally removable and arranged to drain completely. Tubes must be seamless copper, plain, integrally finned with smooth bore or integrally finned with enhanced bore. Each tube must be individually replaceable. Tubes must be installed into carbon mild steel tube sheets by rolling. Tube baffles must be properly spaced to provide adequate tube support and cross flow. Performance must be based on a water velocity not less than0.91 m/s nor more than 3.7 m/s and a fouling factor per JIS B 8613.

Brazed plate heat exchanger must be constructed of 304 or 316 stainless steel, designed to a refrigerant-side working pressure of 3,000 kPa and a waterside working pressure of 1,000 kPa. Evaporator must be factory tested at 1.1 times maximum allowable refrigerant side working pressure and 1.5 times maximum allowable water side working pressure. Provide cooler heaters to protect the evaporator to an ambient of minus 29 degrees C. Provide cooler with factory-installed flow switches. All water connections must use either flanged or grooved-pipe connections. Factory insulate all cold surfaces.

2.6.9 Air-Cooled Condenser Coil

~~[~~Condenser coil must be of the extended-surface fin-and-tube type and must be constructed of seamless ~~[~~copper~~]~~ ~~[~~or~~]~~ ~~[~~aluminum~~]~~ tubes with compatible ~~[~~ copper~~]~~ ~~[~~or~~]~~ ~~[~~aluminum~~]~~ fins. Fins must be soldered or mechanically bonded to the tubes and installed in a metal casing. Coils must be circuited and sized for a minimum of 3 degrees C subcooling and full pumpdown capacity. Coil must be factory leak and pressure tested after assembly in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).~~][The condenser coil must be of the microchannel heat exchanger technology (MCHX) type consisting of a series of flat tubes containing a series of multiple, parallel flow microchannels layered between the refrigerant manifolds in a two-pass arrangement. Provide coils constructed of aluminum alloys for fins, tubes, and manifolds. Coil must be factory leak and pressure tested after assembly in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).]~~

Coil must be entirely coated in accordance with the requirements of paragraph COIL CORROSION PROTECTION.

~~2.6.10 Water-Cooled Condenser Coil~~

~~Condenser must be of the shell-and-coil or shell-and-tube type design. Condenser's refrigerant side must be designed and factory pressure tested to comply with in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). Condenser's water side must be designed and factory pressure tested for not less than [1,000] [1,700] [2000] kPa. Condensers must be complete with refrigerant relief valve/rupture disc assembly, water drain connections, and refrigerant charging valve. Low pressure refrigerant condenser must be provided with~~

~~a purging device to purge non-condensibles trapped in the condenser while keeping refrigerant emissions below requirements of MLIT-M. Purge units must be certified per MLIT-M. Condenser shell must be constructed of seamless or welded steel. Coil bundles must be totally removable and arranged to drain completely. Tubes must be seamless copper, plain, integrally finned with smooth bore or integrally finned with enhanced bore. Each tube must be individually replaceable, except for the coaxial tubes. Tube baffles must be properly spaced to provide adequate tube support and cross flow. Performance must be based on water velocities not less than 0.91 m/s nor more than 3.7 m/s and a fouling factor per JIS B 8613. Water-cooled condensers may be used for refrigerant storage in lieu of a separate liquid receiver, if the condenser storage capacity is 5 percent in excess of the fully charged system for single packaged systems.~~

~~2.6.11 Heat Recovery Condenser Coil~~

~~Condenser must be of the shell-and-coil or shell-and-tube type design and must not be a part of the standard condenser. Condenser must be provided and installed by the chiller manufacturer. Condenser's refrigerant side must be designed and factory pressure tested to comply with in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). Condenser's water side must be designed and factory pressure tested for not less than [1,000] [1,700] [2000] kPa. Condenser must have performance characteristics as indicated on the drawings. Condenser shell must be constructed of seamless or welded steel. Coil bundles must be totally removable and arranged to drain completely. Tubes must be seamless copper, plain, integrally finned with smooth bore or integrally finned with enhanced bore. Each tube must be individually replaceable, except for the coaxial tubes. Tube baffles must be properly spaced to provide adequate tube support and cross flow. Performance must be based on water velocities not less than 0.91 m/s nor more than 3.7 m/s and a fouling factor per JIS B 8613.~~

2.6.10 Receivers

Receiver must bear a stamp certifying compliance with JIS B 8267 and must meet the requirements in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). Inner surfaces must be thoroughly cleaned by sandblasting or other approved means. Each receiver must have a storage capacity not less than 20 percent in excess of that required for the fully-charged system. Each receiver must be equipped with inlet, outlet drop pipe, drain plug, purging valve, in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku), and two bull's eye liquid-level sight glasses. Sight glasses must be in the same vertical plane, 90 degrees apart, perpendicular to the axis of the receiver, and not over 75 mm horizontally from the drop pipe measured along the axis of the receiver. In lieu of bull's eye sight glass, external gauge glass with metal glass guard and automatic closing stop valves may be provided.

2.6.11 Chiller Purge System

Chillers which operate at pressures below atmospheric pressure must be provided with a purge system in accordance with MLIT-M. Purge system must automatically remove air, water vapor, and non-condensible gases from the chiller's refrigerant. Purge system must condense, separate, and return all refrigerant back to the chiller. Purge system must not discharge to occupied areas, or create a potential hazard to personnel. Purge system must include a purge pressure gauge, number of starts counter, and an elapsed time meter. Purge system must include lights or an alarm which indicate excessive purge or an abnormal air leakage into chiller.

2.6.12 Tools

One complete set of special tools, as recommended by the manufacturer for field maintenance of the system, must be provided. Tools must be mounted on a tool board in the equipment room or contained in a toolbox as directed by the Contracting Officer.

2.7 ACCESSORIES

~~2.7.1 Refrigerant Leak Detector~~

~~Detector must be the continuously-operating, halogen-specific type. Detector must be appropriate for the refrigerant in use. Detector must be specifically designed for area monitoring and must include [a single sampling point] [[_____] sampling points] installed where indicated. Detector design and construction must be compatible with the temperature, humidity, barometric pressure and voltage fluctuations of the operating area. Detector must have an adjustable sensitivity such that it can detect refrigerant at or above 3 parts per million (ppm). Detector must be supplied factory-calibrated for the appropriate refrigerant(s). Detector must be provided with an alarm relay output which energizes when the detector detects a refrigerant level at or above the TLV-TWA (or toxicity measurement consistent therewith) for the refrigerant(s) in use. The detector's relay must be capable of initiating corresponding alarms and ventilation systems as indicated on the drawings. Detector must be provided with a failure relay output that energizes when the monitor detects a fault in its operation. [Detector must be compatible with the facility's Building Control Network (BCN). The BCN must be capable of generating an electronic log of the refrigerant level in the operating area, monitoring for detector malfunctions, and monitoring for any refrigerant alarm conditions.]~~

2.7.1 Refrigerant Relief Valve/Rupture Disc Assembly

The assembly must be a combination pressure relief valve and rupture disc designed for refrigerant usage. The assembly must be in accordance with JIS B 8267 and in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku). The assembly must be provided with a pressure gauge assembly which will provide local indication if a rupture disc is broken. Rupture disc must be the non-fragmenting type.

2.7.2 Refrigerant Signs

Refrigerant signs must be a medium-weight aluminum type with a baked enamel finish. Signs must be suitable for indoor or outdoor service. Signs must have a white background with red letters not less than 13 mm in height.

2.7.2.1 Installation Identification

Each new refrigerating system must be provided with a refrigerant sign which indicates the following as a minimum:

a. Contractor's name.

b. Refrigerant number and amount of refrigerant.

c. The lubricant identity and amount.

d. Field test pressure applied.

2.7.2.2 Controls and Piping Identification

Refrigerant systems containing more than 50 kg of refrigerant must be provided with refrigerant signs which designate the following as a minimum:

a. Valves or switches for controlling the refrigerant flow ~~[, the ventilation system,]~~ and the refrigerant compressor(s).

b. Pressure limiting device(s).

~~2.7.3 Automatic Tube Brush Cleaning System~~

~~2.7.3.1 Brush and Basket Sets~~

~~One brush and basket set (one brush and two baskets) must be furnished for each condenser tube. Brushes must be made of nylon bristles, with titanium wire. Baskets must be polypropylene.~~

~~2.7.3.2 Flow-Diverter Valve~~

~~Each system must be equipped with one flow-diverter valve specifically designed for the automatic tube brush cleaning system and have parallel flow connections. The flow-diverter valve must be designed for a working pressure of [1,000][1,700][2000] kPa. End connections must be flanged. Each valve must be provided with an electrically operated air solenoid valve and position indicator.~~

~~2.7.3.3 Control Panel~~

~~The control panel must provide signals to the diverter valve at a preset time interval to reverse water flow to drive the tube brushes down the tubes and then signal the valve to reverse the water flow to drive the brushes back down the tubes to their original position. The controller must have the following features as a minimum:~~

~~a. Timer to initiate the on-load cleaning cycle.~~

~~b. Manual override of preset cleaning cycle.~~

~~c. Power-on indicator.~~

~~d. Diverter-position indicator.~~

~~e. Cleaning-cycle-time adjustment~~

~~f. Flow-switch bypass.~~

2.7.3 Gaskets

Gaskets must conform to JIS F 0602 - classification for compressed sheet with nitrile binder and acrylic fibers.

2.7.4 Bolts and Nuts

Bolts and nuts, except as required for piping applications, must be in accordance with JIS G 3101. The bolt head must be marked to identify the manufacturer and the standard with which the bolt complies in accordance with JIS G 3101.

2.8 FABRICATION

2.8.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand ~~[125] [~~500~~]~~ hours exposure to the salt spray test specified in JIS Z 2371 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 3 mm on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to JIS H 8641.

2.8.2 Factory Applied Insulation

Chiller must be provided with factory installed insulation on surfaces subject to sweating including the water cooler, suction line piping, economizer, and cooling lines. Insulation on heads of coolers may be field applied, however it must be installed to provide easy removal and replacement of heads without damage to the insulation. Where motors are the gas-cooled type, factory installed insulation must be provided on the cold-gas inlet connection to the motor per manufacturer's standard practice. Factory insulated items installed outdoors are not required to be fire-rated. .

2.8.3 Coil Corrosion Protection

Provide coil with a uniformly applied ~~[~~epoxy electrodeposition~~] [phenolic] [vinyl]~~ type coating to all coil surface areas without material bridging between fins. Submit product data on the type coating selected, the application process used, and verification of conformance with the salt spray test requirement. Coating must be applied at either the coil or coating manufacturer's factory. Coating process must ensure complete coil encapsulation. Salt spray test shall be in accordance with JIS Z 2371.

2.9 SUPPLEMENTAL COMPONENTS/SERVICES

2.9.1 Chilled ~~and Condenser~~ Water Piping and Accessories

Chilled and condenser water piping and accessories must be provided and installed in accordance with Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS.

2.9.2 Refrigerant Piping

Refrigerant piping for split-system water chillers must be provided and installed in accordance with Section 23 23 00 REFRIGERANT PIPING.

~~2.9.3 Cooling Tower~~

~~Cooling towers must be provided and installed in accordance with Section 23 65 00 COOLING TOWERS AND REMOTE EVAPORATIVELY-COOLED CONDENSERS.~~

2.9.3 Temperature Controls

Chiller control packages must be fully coordinated with and integrated ~~[~~ into the temperature control system indicated in ~~[~~Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC~~]~~ and ~~[Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS] [or] [~~ Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS~~]] [into the existing air-conditioning system]~~.

PART 3 EXECUTION

3.1 INSTALLATION

Installation of water chiller systems including materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with the manufacturer's written installation instructions, including the following:

(1) Water chiller - installation instructions

3.1.1 Installation Instructions

Provide manufacturer's standard catalog data, at least ~~[~~5~~] [_____]~~ weeks prior to the purchase or installation of a particular component, highlighted to show features such as materials, dimensions, options, performance and efficiency. Data must include manufacturer's recommended installation instructions and procedures. Data must be adequate to demonstrate compliance with contract requirements.

3.1.2 Vibration Isolation

If vibration isolation is specified for a unit, vibration isolator literature must be included containing catalog cuts and certification that the isolation characteristics of the isolators provided meet the manufacturer's recommendations.

3.1.3 Posted Instructions

Provide posted instructions, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.

3.1.4 Verification of Dimensions

Provide a letter including the date the site was visited, conformation of existing conditions, and any discrepancies found.

3.1.5 System Performance Test Schedules

Provide a schedule, at least ~~[~~2~~] [_____]~~ weeks prior to the start of related testing, for the system performance tests. The schedules must identify the proposed date, time, and location for each test.

3.1.6 Certificates

Where the system, components, or equipment are specified to comply with requirements of Japanese Industrial Standards, proof of such compliance must be provided. The label or listing of the specified agency must be acceptable evidence.

3.1.7 Operation and Maintenance Manuals

Provide ~~[Six] [_____]~~4 complete copies of an operation manual in bound 216 by 279 mm booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least ~~[~~4~~] [_____]~~ weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features. ~~[Six] [_____]~~4 complete copies of maintenance manual in bound 216 by 279 booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

3.1.8 Connections to Existing Systems

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Furnish materials required to make connections into existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Furnish labor and tools for making actual connections to existing systems.

3.1.9 Refrigeration System

3.1.9.1 Equipment

Refrigeration equipment and the installation thereof must conform to in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese standard. Necessary supports must be provided for all equipment, appurtenances, and pipe as required, including frames or supports for compressors, pumps, cooling towers, condensers, water coolers, and similar items. Compressors must be isolated from the building structure. If mechanical vibration isolators are not provided, vibration absorbing foundations must be provided. Each foundation must include isolation units consisting of machine and floor or foundation fastenings, together with intermediate isolation material. Other floor-mounted equipment must be set on not less than a 150 mm concrete pad doweled in place. Concrete foundations for floor mounted pumps must have a mass equivalent to three times the weight of the components, pump, base plate, and motor to be supported. In lieu of concrete pad foundation, concrete pedestal block with isolators placed between the pedestal block and the floor may be provided. Concrete pedestal block must be of mass not less than three times the combined pump, motor, and base weights. Isolators must be selected and sized based on load-bearing requirements and the lowest frequency of vibration to be isolated. Isolators must limit vibration to ~~[_____]~~20 percent at lowest equipment rpm. Lines connected to pumps mounted on pedestal blocks must be provided with flexible connectors. Foundation drawings, bolt-setting information, and foundation bolts must be furnished prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Concrete for foundations must be as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE. Equipment must be properly leveled, aligned, and secured in place in accordance with manufacturer's instructions.

3.1.9.2 Field Refrigerant Charging

a. Initial Charge: Upon completion of all the refrigerant pipe tests, the vacuum on the system must be broken by adding the required charge of dry refrigerant for which the system is designed, in accordance with the manufacturer's recommendations. Contractor must provide the complete charge of refrigerant in accordance with manufacturer's recommendations. Upon satisfactory completion of the system performance tests, any refrigerant that has been lost from the system must be replaced. After the system is fully operational, service valve seal caps and blanks over gauge points must be installed and tightened.

b. Refrigerant Leakage: If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must immediately be isolated from the remainder of the system and the refrigerant must be pumped into the system receiver or other suitable container. The refrigerant must not be discharged into the atmosphere.

c. Contractor's Responsibility: The Contractor must, at all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time must more than 85 g of refrigerant be released to the atmosphere in any one occurrence. Any system leaks within the first year must be repaired in accordance with the specified requirements including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

3.1.9.3 Oil Charging

Except for factory sealed units, two complete charges of lubricating oil for each compressor crankcase must be furnished. One charge must be used during the performance testing period, and upon the satisfactory completion of the tests, the oil must be drained and replaced with the second charge.

~~3.1.10 Mechanical Room Ventilation~~

~~Mechanical ventilation systems must be in accordance with Section 23 30 00 AIR SUPPLY, DISTRIBUTION, VENTILATION, AND EXHAUST SYSTEM.~~

3.1.10 Field Applied Insulation

Field installed insulation must be as specified in Section 23 07 00

THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

3.1.11 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

3.2 MANUFACTURER'S FIELD SERVICE

The services of a factory-trained representative must be provided for ~~[_____]~~5 days. The representative shall advise on the following:

a. Hermetic machines:

(1) Testing hermetic water-chilling unit under pressure for refrigerant leaks; evacuation and dehydration of machine to an absolute pressure of not over 300 micrometers.

(2) Charging the machine with refrigerant.

(3) Starting the machine.

b. Open Machines:

(1) Erection, alignment, testing, and dehydrating.

(2) Charging the machine with refrigerant.

(3) Starting the machine.

3.3 CLEANING AND ADJUSTING

Equipment must be wiped clean, with all traces of oil, dust, dirt, or paint spots removed. Provide temporary filters for all fans that are operated during construction. Perform and document that proper Indoor Air Quality During Construction procedures have been followed; this includes providing documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. System must be maintained in this clean condition until final acceptance. Bearings must be properly lubricated with oil or grease as recommended by the manufacturer. Belts must be tightened to proper tension. Control valves and other miscellaneous equipment requiring adjustment must be adjusted to setting indicated or directed. Fans must be adjusted to the speed indicated by the manufacturer to meet specified conditions. At least one week before the official equipment warranty start date, all condenser coils on air-cooled water chillers and split-system water chillers must be cleaned in accordance with the chiller manufacturer's instructions. This work covers two coil cleanings. The condenser coils must be cleaned with an approved coil cleaner by a service technician, factory trained by the chiller manufacturer. The condenser coil cleaner must not have any detrimental affect on the materials or protective coatings on the condenser coils. Testing, adjusting, and balancing must be as specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

3.4 FIELD ACCEPTANCE TESTING

3.4.1 Test Plans

a. Manufacturer's Test Plans: Within ~~[~~120~~][_____]~~ calendar days after contract award, submit the following plans:

(1) Water chiller - Field Acceptance Test Plan

Field acceptance test plans must be developed by the chiller manufacturer detailing recommended field test procedures for that particular type and size of equipment. Field acceptance test plans developed by the installing Contractor, or the equipment sales agency furnishing the equipment, will not be acceptable.

The Contracting Officer will review and approve the field acceptance test plan for each of the listed equipment prior to commencement of field testing of the equipment. The approved field acceptance tests of the chiller and subsequent test reporting.

b. Coordinated testing: Indicate in each field acceptance test plan when work required by this section requires coordination with test work required by other specification sections. Furnish test procedures for the simultaneous or integrated testing of tower system controls which interlock and interface with controls for the equipment provided under ~~[Section 23 09 53.00 20, SPACE TEMPERATURE CONTROL SYSTEMS or~~ Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC~~][Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS] [or] [~~ and Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS~~]~~.

c. Prerequisite testing: Chillers for which performance testing is dependent upon the completion of the work covered by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC must have that work completed as a prerequisite to testing work under this section. Indicate in each field acceptance test plan when such prerequisite work is required.

d. Test procedure: Indicate in each field acceptance test plan each equipment manufacturers published installation, start-up, and field acceptance test procedures. Include in each test plan a detailed step-by-step procedure for testing automatic controls provided by the manufacturer.

Each test plan must include the required test reporting forms to be completed by the Contractor's testing representatives. Procedures must be structured to test the controls through all modes of control to confirm that the controls are performing with the intended sequence of control.

Controller must be verified to be properly calibrated and have the proper set point to provide stable control of their respective equipment.

e. Performance variables: Each test plan must list performance variables that are required to be measured or tested as part of the field test.

Include in the listed variables performance requirements indicated on the equipment schedules on the design drawings. Chiller manufacturer must furnish with each test procedure a description of acceptable results that have been verified.

Chiller manufacturer must identify the acceptable limits or tolerance within which each tested performance variable must acceptably operate.

f. Job specific: Each test plan must be job specific and must address the particular cooling towers and particular conditions which exist in this contract. Generic or general preprinted test procedures are not acceptable.

g. Specialized components: Each test plan must include procedures for field testing and field adjusting specialized components, such as hot gas bypass control valves, or pressure valves.

3.4.2 Testing

a. Each water chiller system must be field acceptance tested in compliance with its approved field acceptance test plan and the resulting following field acceptance test report submitted for approval:

(1) Water chiller - Field Acceptance Test Report

b. Manufacturer's recommended testing: Conduct the manufacturer's recommended field testing in compliance with the approved test plan. Furnish a factory trained field representative authorized by and to represent the equipment manufacturer at the complete execution of the field acceptance testing.

c. Operational test: Conduct a continuous 24 hour operational test for each item of equipment. Equipment shutdown before the test period is completed shall result in the test period being started again and run for the required duration. For the duration of the test period, compile an operational log of each item of equipment. Log required entries every two hours. Use the test report forms for logging the operational variables.

d. Notice of tests: Conduct the manufacturer's recommended tests and the operational tests; record the required data using the approved reporting forms. Notify the Contracting Officer in writing at least 15 calendar days prior to the testing. Within 30 calendar days after acceptable completion of testing, submit each test report for review and approval.

e. Report forms: Type data entries and writing on the test report forms. Completed test report forms for each item of equipment must be reviewed, approved, and signed by the Contractor's test director. The manufacturer's field test representative must review, approve, and sign the report of the manufacturer's recommended test. Signatures must be accompanied by the person's name typed.

f. Deficiency resolution: The test requirements acceptably met; deficiencies identified during the tests must be corrected in compliance with the manufacturer's recommendations and corrections retested in order to verify compliance.

3.5 SYSTEM PERFORMANCE TESTS

~~[Six] [_____]~~4 copies of the report must be provided in bound 216 by 279 mm booklets.

3.5.1 General Requirements

Before each refrigeration system is accepted, tests to demonstrate the general operating characteristics of all equipment must be conducted by the manufacturer's approved start-up representative experienced in system start-up and testing, at such times as directed. Tests must cover a period of not less than 48 hours for each system and must demonstrate that the entire system is functioning in accordance with the drawings and specifications. Corrections and adjustments must be made as necessary and tests must be re-conducted to demonstrate that the entire system is functioning as specified. Prior to acceptance, service valve seal caps and blanks over gauge points must be installed and tightened. Any refrigerant lost during the system startup must be replaced. If tests do not demonstrate satisfactory system performance, deficiencies must be corrected and the system must be retested. Tests must be conducted in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Any material, equipment, instruments, and personnel required for the test must be provided by the Contractor. Field tests must be coordinated with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC.

3.5.2 Test Report

The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system. The report must also include the following information and must be taken at least three different times at outside dry-bulb temperatures that are at least 3 degrees C apart:

a. Date and outside weather conditions.

b. The load on the system based on the following:

(1) The refrigerant used in the system.
(2) Condensing temperature and pressure.
(3) Suction temperature and pressure.
(4) Running current, voltage and proper phase sequence for each phase of all motors.
(5) The actual on-site setting of all operating and safety controls.
(6) Chilled water pressure, flow and temperature in and out of the chiller.
(7) The position of the ~~[~~capacity-reduction gear~~]~~ at machine off, one-third loaded, one-half loaded, two-thirds loaded, and fully loaded.

3.6 DEMONSTRATIONS

Contractor must conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total ~~[_____]~~8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests. The training course must cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations.

Provide a schedule, at least ~~[~~2~~] [_____]~~ weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.

-- End of Section --

SECTION 23 74 33

DEDICATED OUTDOOR AIR SYSTEMS (DOAS)
05/24

PART 1 GENERAL

Submit equipment and performance data for each DOAS consisting of use life, total static pressure, all entering and leaving temperatures of each component, fan curves, provide psychrometric chart with processes shown for both summer and winter conditions, power ratings, capacity ranges, face area classifications, rotational velocities, controls instrument locations ~~[with recommended installation locations], [~~and ~~][~~all information shown on the equipment schedule on the design drawings~~][and][_____]~~.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~[~~ Design and construction must comply with current applicable Host Nation codes and standards. In case of conflicting U.S. and H.N. codes and standards, the most stringent must apply, unless otherwise specified by the applicable Bilateral U.S./H.N. Agreements.

~~]~~ AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 99	(2016) Standards Handbook
AMCA 201	(2002; R 2011) Fans and Systems
AMCA 210	(2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating
AMCA 300	(2014) Reverberant Room Method for Sound Testing of Fans
AMCA 301	(2014) Methods for Calculating Fan Sound Ratings from Laboratory Test Data

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 920 I-P	(2020) Performance Rating of Direct Expansion-Dedicated Outdoor Air System Units
ANSI/AHRI 210/240	(2008; Add 1 2011; Add 2 2012) Performance Rating of Unitary Air-Conditioning & Air-Source Heat Pump Equipment

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 11	(2014; R 2020) Load Ratings and Fatigue Life for Roller Bearings

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2	(2017; Addenda B 2020; Errata 1 2020; Addenda C 2022; Errata 2 2024; Addenda D 2025) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size
ASHRAE 62.1	(2016) Ventilation for Acceptable Indoor Air Quality

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 9511	(2024) Preformed Cellular Plastics Thermal Insulation Materials
JIS B 1514	(2017) Rolling Bearings
JIS B 8616	(2015) Package Air Conditioners
JIS C 4203	(2010) Single Phase Induction Motors for General Purpose (Amendment 1)
JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 8305	(2019) Rigid Steel Condits
JIS G 3302	(2022) Hot-Dip Zinc-Coated Steel Sheet and Strip
JIS G 4305	(2021) Cold-Rolled Stainless Steel Plate, Sheet and Strip
JIS G 5501	(1995) Grey Iron Castings
JIS Z 2371	(2015) Methods of Salt Spray Testing

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-200-01	(2022; with Change 4, 2024) DoD Building Code

UL SOLUTIONS (UL)

UL 586	(2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units
UL 705	(2017; Reprint Sep 2024) UL Standard for Safety Power Ventilators

UL 900	(2015; Reprint Aug 2022) UL Standard for SafetyStandard for Air Filter Units
UL 1995	(2015; Reprint Aug 2022) UL Standard for Safety Heating and Cooling Equipment

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

~~Compressor; G, [_____]~~

Coils; G~~, [_____]~~

Controls; G~~, [_____]~~

Unit Cabinet; G~~, [_____]~~

Casing; G~~, [_____]~~

~~Air-Cooled Condenser; G, [_____]~~

~~Evaporative Condenser; G, [_____]~~

Installation Drawings; G~~, [_____]~~

SD-03 Product Data

Equipment and Performance Data; G~~, [_____]~~

Air-Conditioning Systems; G~~, [_____]~~

~~Compressor; G, [_____]~~

Coils; G~~, [_____]~~

Fans; G~~, [_____]~~

Controls; G~~, [_____]~~

Unit Cabinet; G~~, [_____]~~

Casing; G~~, [_____]~~

Air Filters; G~~, [_____]~~

~~Air-Cooled Condenser; G, [_____]~~

~~Evaporative Condenser; G, [_____]~~

~~Energy Recovery Devices; G, [_____]~~

~~[~~ ~~Humidifier; G, [_____]~~

~~]~~ SD-06 Test Reports

~~Refrigerant Tests, Charging, and Start-Up; G, [_____]~~

System Performance Tests; G~~, [_____]~~

SD-07 Certificates

List of Product Installations

Manufacturer's Warranty; G~~, [_____]~~

~~[~~ Coil Coating Warranty; G~~, [_____]~~

~~]~~ SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions

Operation and Maintenance Training

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals, Data Package 2

1.3 RELATED DOCUMENTS

~~[~~ Section ~~01 45 00~~01 45 00.00 10 QUALITY CONTROL applies to work specified in this section.

~~][~~ Section ~~01 91 00.15~~01 91 00.15 10 TOTAL BUILDING COMMISSIONING applies to work specified in this section.

~~][~~ Section 13 48 73 SEISMIC CONTROL FOR ~~NONSTRUCTURAL COMPONENTS~~ MISCELLANEOUS EQUIPMENT applies to work specified in this section.

~~][~~ Section 23 05 48.19 ~~[~~SEISMIC~~]~~ BRACING FOR HVAC applies to work specified in this section.

~~][~~ Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC applies to work specified in this section.

~~][~~ ~~Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC applies to work specified in this section~~

~~][~~ Section 23 09 13 INSTRUMENTATION AND CONTROL DEVICES FOR HVAC applies to work specified in this section.

~~][~~ ~~Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS applies to work specified in this section.~~

~~][~~ Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS applies to work specified in this section.

~~][~~ Section 23 23 00 REFRIGERANT PIPING applies to work specified in this section.

~~][~~ Section 23 30 00 HVAC AIR DISTRIBUTION applies to work specified in this section.

~~][~~ Section 23 81 00 DECENTRALIZED UNITARY HVAC EQUIPMENT applies to work specified in this section.

~~][~~ ~~Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS applies to work specified in this section.~~

~~]~~1.4 QUALITY CONTROL

Submit a list of product installations of each DOAS showing a minimum of five installed units, similar to those proposed for use, that have been in successful service for a minimum period of 5 years. Provide a list that includes the purchaser, address of installation, service organization, and date of installation.

1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Standard products must have been in satisfactory commercial or industrial use for 5 years prior to bid opening. The 5-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 5 year period.

1.4.2 Service Support

The equipment items must be supported by service organizations. Submit a certified list of a minimum of three qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract within ~~[24][~~48~~][_____]~~ hours.

1.4.3 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

1.4.4 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer. "Authority having jurisdiction" must be in accordance with the applicable year of UFC 1-200-01 DOD BUILDING CODE.

1.4.5 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

1.4.6 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

1.5 DELIVERY, STORAGE AND HANDLING

Handle, store, and protect equipment and materials to prevent damage from the weather, humidity and temperature variations, dirt and dust, or other contaminants before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Additionally, cap or plug all pipes until installed. Replace damaged or defective items.

1.6 ELECTRICAL WORK

a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown. For packaged equipment, include manufacturer provided controllers with the required monitors and timed restart.

b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with JIS C 4203~~NEMA MG 11~~. ~~Provide premium efficiency type integral size motors in accordance with NEMA MG 1.~~

c. For polyphase motors, provide squirrel-cage medium induction motors, including motors that are part of a system , and that meet the efficiency ratings for premium efficiency motors in accordance with ~~NEMA MG 1~~JIS C 4212. Select premium efficiency polyphase motors in accordance with ~~NEMA MG 10~~JIS C 4212.

d. Provide motors in accordance with ~~NEMA MG 1~~JIS C 4212 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Provide motors rated for continuous duty with the enclosure specified. Provide motor duty that allows for maximum frequency start-stop operation and minimum encountered interval between start and stop. Provide motor torque capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. Fit motor bearings with grease supply fittings and grease relief to outside of the enclosure.

~~e. Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers are allowed to accomplish the same function. Use solid-state variable-speed controllers for motors rated 7.45 kW or less and adjustable frequency drives for larger motors. Provide variable frequency drives for motors as specified in Section 26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS.~~

1.7 INSTRUCTION TO GOVERNMENT PERSONNEL

Furnish the services of competent instructors to give full video recorded instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work. Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section.~~[~~ When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.~~]~~

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

1.8 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors. Contractor and manufacturer are responsible for verifying equipment, including any additional parts such as controls devices and other equipment requiring access, fits in the space provided. The Government review does not relieve contractor from delivering the product that meets minimum capacity and mandatory efficiency requirements to meet all codes and standards.

1.9 WARRANTY

Submit the manufacturer's warranty for the unit.

~~[~~Submit the coil coating warranty.~~]~~ Provide unit with the ~~[~~ Manufacturer's Standard Warranty.~~][[1 year][2 year][5 year][10 year][_____ year]manufacturer's warranty.][~~ Provide ~~[~~compressors~~][and][gas heat exchangers]~~ with the~~[~~ Manufacturer's Standard Warranty.~~][[1 year][2 year][5 year][10 year][_____ year]manufacturer's warranty.]]~~

PART 2 PRODUCTS

2.1 TYPES

Provide ~~[~~single-zone draw-through type~~][or][single-zone blow-through~~

~~type][or][multizone blow-through type][blow-through double-deck type][blow-through triple deck type]~~ units as indicated. Units must include fans, coils, airtight insulated casing,~~[energy recovery devices,][~~ required controls devices for temperature and humidity control,~~][~~ prefilters,~~][secondary filter sections,][~~ and ~~][diffuser sections where indicated,][air blender] adjustable V-belt drives, belt guards for externally mounted motors,~~ access sections where indicated,~~[mixing box][combination sectional filter-mixing box,][[pan][drysteam][spray type] humidifier,]~~ vibration-isolators, and appurtenances required for specified operation. ~~Provide vibration isolators as indicated.~~ Physical dimensions of each air handling unit must be suitable to fit space allotted to the unit with the capacity and efficiencies indicated. Provide air handling unit that is rated in accordance with ~~[~~AHRI 920 I-P~~] [AHRI 430] [_____]~~ and AHRI certified for cooling.~~[Protect outdoor air coils from incidental contact to coil fins by a coil guard. Coil guard must be constructed of cross wire welded steel with PVC coating.]~~

2.2 UNIT CONSTRUCTION

2.2.1 Unit Cabinet

Units must have a leakage rate not to exceed 1 percent of supply air volume to ~~[2][2.5][_____]~~ kPa static pressure in either positive or negative pressure tests. Manufacturer must provide test data confirming cabinet construction can meet the requirement. Indoor cabinets must be suitable for the specified indoor service and enclose all unit components. Outdoor cabinets must be suitable for outdoor service with a weathertight, insulated and corrosion-protected structure. Cabinets constructed exclusively for indoor service which have been modified for outdoor service are not acceptable.

2.2.1.1 Class A and Class B Cabinets

Provide an AHU cabinet suitable for the pressure class shown and has leaktight joints, closures, penetrations, and access provisions. Provide a cabinet that does not expand or contract perceptibly when fans are starting or stopping and that does not pulsate during operation. Reinforce cabinet surfaces with deflections in excess of 0.004167 of unsupported span before acceptance. Stiffen pulsating panels, which produce low-frequency noise due to diaphragming of unstable panel walls, to raise the natural frequency to an easily attenuated level. Fabricate the enclosure from continuous hot-dipped-galvanized steel no lighter than 0.91 millimeter thickness, to match the industry standard. Provide mill-galvanized sheet-metal that conforms to ~~ASTM A653/A653M~~JIS G 3302 and that is coated with not less than 0.38 kilogram of zinc per square meter of a two-sided surface. Provide mill-rolled structural-steel that is hot-dip-galvanized or primed and painted. Corrosion-protect cut edges, burns, and scratches in galvanized surfaces. Provide primed and painted black carbon steel cabinet construction that complies with this specification.

Provide removable panels to access the interior of the unit cabinet. Provide seams that are welded, bolted, or gasketed and sealed with a rubber-based mastic. Make entire cabinet floor and ceiling hot-dipped-galvanized steel. Provide removable access doors on both sides of all access, filter, and fan sections for inspection and maintenance.

2.2.1.2 Class C Cabinets

Provide an AHU cabinet that is suitable for the pressure class shown and has leaktight joints, closures, penetrations, and access provisions. Provide a cabinet that does not expand or contract perceptibly when the fans are starting or stopping and that does not pulsate during operation. Reinforce cabinet surfaces with deflections in excess of 0.002778 of unsupported span before acceptance by the Contracting Officer. Stiffen pulsating panels, which produce low-frequency noise due to diaphragming of unstable panel walls, to raise the natural frequency to an easily attenuated level. Provide the enclosure that is fabricated from mill-galvanized or primed and painted carbon sheet steel. Provide mill-galvanized sheet metal that conforms to ~~ASTM A653/A653M~~JIS G 3302 and that is coated with not less than 0.38 kilogram of zinc per square meter of a two-sided surface. Provide mill-rolled structural steel that is hot-dip galvanized or primed and painted. Corrosion-protect edges, burns, and scratches in galvanized surfaces. Provide primed and painted black carbon steel cabinet construction that complies with this specification.

Provide removable panels to access the interior of the unit cabinet. Provide seams that are welded, bolted, or gasketed and sealed with a rubber-based mastic. Make the entire cabinet floor and ceiling hot-dipped galvanized steel. Provide removable access doors on both sides of all access, filter, and fan sections for inspection and maintenance.

2.2.2 Casing

Provide the following:

a. ~~[~~Casing sections~~[[single][~~50 mm double~~]~~ wall type~~][as indicated]~~, constructed of a minimum 1.3 mm galvanized steel, or 1.3 mm corrosion-resisting sheet steel conforming to ~~ASTM A167, Type 304~~ JIS G 4305. ~~][Inner casing of double-wall units that are a minimum one mm solid galvanized steel or corrosion-resisting sheet steel conforming to ASTM A167, Type 304.]~~Design and construct casing with an integral insulated structural galvanized steel frame such that exterior panels are non-load bearing.

b. Individually removable exterior panels with standard tools. Removal must not affect the structural integrity of the unit. Furnish casings with access sections, inspection doors, and access doors, all capable of opening a minimum of 90 degrees, as indicated.

c. Insulated, fully gasketed, double-wall type inspection and access doors, of a minimum 1.3 mm outer and one mm inner panels made of either galvanized steel or corrosion-resisting sheet steel conforming to ~~ASTM A167, Type 304~~JIS G 4305. Provide rigid doors with heavy duty hinges and latches. Inspection doors must be a minimum 300 mm wide by 300 mm high. Access doors must be a minimum 600 mm wide, the full height of the unit casing or a minimum of 1800 mm, whichever is less. ~~[Install a minimum 200 by 200 mm sealed glass window suitable for the intended application, in all access doors.]~~

d. Access doors must be installed to open against the greatest pressure relative to air pressure on each side of access door. Provide 25 mm diameter test ports with screwed caps on casing upstream and downstream of all coils and filters for pressure and temperature measurement across internal components to limit unnecessary or unauthorized access inside the unit.

e. Double-wall insulated type drain pan (thickness equal to exterior casing) constructed of 1.4 mm~~[~~ stainless steel~~][corrosion resisting sheet steel conforming to ASTM A167, Type 304]~~, conforming to ASHRAE 62.1. Galvanized steel pans are not acceptable. Construct drain pans water tight, treated to prevent corrosion, and designed for positive condensate drainage.~~[When two or more cooling coils are used, with one stacked above the other, condensate from the upper coils must not flow across the face of lower coils. Provide intermediate drain pans or condensate collection channels and downspouts, as required to carry condensate to the unit drain pan out of the air stream and without moisture carryover.]~~ Construct drain pan to allow for easy visual inspection, including underneath the coil without removal of the coil and to allow complete and easy physical cleaning of the pan underneath the coil without removal of the coil. Provide coils that are individually removable from the casing. Provide ~~[~~ 40 mm~~][______]~~ minimum drain pan connection~~[as indicated]~~.

f. Casing insulation that conforms to NFPA 90A. Insulate single-wall casing sections handling conditioned air with not less than 25 mm thick, 24 kg/cubic meter density coated fibrous glass material having a thermal conductivity not greater than 0.033 W/m-K. Insulate double-wall casing sections handling conditioned air with not less than 50 mm of the same insulation specified for single-wall casings. Foil-faced insulation is not an acceptable substitute for use with double wall casing. Seal double wall insulation completely by inner and outer panels.

g. Factory applied fibrous glass insulation that conforms to ~~ASTM C1071~~ JIS A 9511, except that the minimum thickness and density requirements do not apply, and that meets the requirements of NFPA 90A. Make air handling unit casing insulation uniform over the entire casing. Foil-faced insulation is not an acceptable substitute for use on double-wall access doors and inspections doors ~~[~~and casing sections~~]~~.

h. Duct liner material, coating, and adhesive that conforms to fire-hazard requirements specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Protect exposed insulation edges and joints where insulation panels are butted with a metal nosing strip or coat to meet erosion resistance requirements of ~~ASTM C1071~~ JIS A 9511.

i. A latched and hinged inspection door, in the fan, filter and coil sections. Plus additional inspection doors, access doors and access sections ~~[______][~~where indicated~~]~~.

2.2.3 Access Sections and Filter/Mixing Boxes

Provide access sections where indicated and furnish with access doors as shown. Construct access sections and filter/mixing boxes in a manner identical to the remainder of the unit casing and equip with access doors. Design mixing boxes to minimize air stratification and to promote thorough mixing of the air streams.

~~2.2.4 Diffuser Sections~~

~~Furnish diffuser sections between the discharge of all housed supply fans [and cooling coils of blow-through single zone units][and][filter sections of those units with high efficiency filters located immediately~~

~~downstream of the air handling unit fan section]. Provide diffuser sections that are fabricated by the unit manufacturer in a manner identical to the remainder of the unit casing, designed to be airtight under positive static pressures up to [2][_____] kPa and with an access door on each side for inspection purposes. Provide a diffuser section that contains a perforated diffusion plate, fabricated of galvanized steel, Type 316 stainless steel, aluminum, or steel treated for corrosion with manufacturer's standard corrosion-resisting finish, and designed to accomplish uniform air flow across the down-stream[coil][filters] while reducing the higher fan outlet velocity to within plus or minus 5 percent of the required face velocity of the downstream component.~~

2.2.4 Roof Curb

Provide a roof curb that mates with the unit to provide support and be completely weather tight. Provide curb with sealing strips to ensure an airtight seal between supply and return openings of the curb and unit. Design curb to allow ductwork to be directly connected to the curb.~~[The roof curb must be provided by the Manufacturer of the equipment.][The roof curb must be a minimum of [_____] mm tall.][Provide an acoustical roof curb to meet noise requirements.]~~

2.2.5 Corrosion Protection

Provide the interior of the unit with corrosion protection which must be capable of withstanding at least 3000 hours, with no visible corrosive effects, when tested in a salt spray and fog atmosphere in accordance with ~~ASTM B117~~JIS Z 2371 test procedure. Air tunnel, fans and dampers must all include the corrosion protection. Provide all coils in contact with outside air a flexible, epoxy polymer e-coat uniformly applied to all coil surface areas without material bridging between fins. Confirm corrosion durability through testing to no less than 3000 hours salt spray per ~~ASTM B117~~JIS Z 2371. Coated coil must receive a spray-applied, UV-resistant polyurethane topcoat to prevent UV degradation of the e-coat. Coating is not required on coils utilizing copper tubes and copper fins.

~~2.2.5.1 Remote Outdoor Condenser Coils~~

~~Epoxy Immersion Coating - Electrically Deposited: The multi-stage corrosion-resistant coating application comprised of cleaning (heated alkaline immersion bath) and reverse-osmosis immersion rinse prior to the start of the coating process. Maintain the coating thickness between 0.6-mil and 1.2-mil. Before the coils are subjected to high-temperature oven cure, treat to permeate immersion rinse and spray. Where the coils are subject to UV exposure, apply UV protection spray treatment comprising of UV-resistant urethane mastic topcoat. Provide complete coating process traceability for each coil and minimum five years of limited warranty. The coating process must be such that uniform coating thickness is maintained at the fin edges. Comply with the applicable ASTM Standards for the following:~~

~~a. Salt Spray Resistance: [500][1,000][3,000][6,000][10,000][_____] hours (ASTM B117)~~

~~b. Humidity Resistance (Minimum 1,000 Hours)~~

~~c. Water Immersion (Minimum 260 Hours)~~

~~d. Cross-Hatch Adhesion (Minimum 4B-5B Rating)~~

~~e. Impact Resistance (Up to 160 Inch/Pound)~~

~~2.2.5.2 Exposed Outdoor Cabinet~~

~~Casing Surfaces (Exterior and Interior): Protect all exposed and accessible metal surfaces with a water-reducible acrylic with stainless steel pigment spray-applied over the manufacturer's standard finish. The spray coating thickness must be 2-4 mils and provide minimum salt-spray resistance of [500][1,000][3,000][6,000][10,000][_____] hours (ASTM B117) and [500][1,000][_____] hours UV resistance (ASTM D4587).~~

~~Provide outdoor cabinets with a roof panels sloped at a minimum pitch of 2 percent. The roof must overhang side and end panels by a minimum of 50 mm.~~

~~2.2.5.3 Corrosion Protection for Coastal Installations~~

~~[_____]~~

2.3 FANS

a. Test and rate fans according to AMCA 210. Calculate system effect on air moving devices in accordance with AMCA 201 where installed ductwork differs from that indicated on drawings. Install air moving devices to minimize fan system effect. Where system effect is unavoidable, determine the most effective way to accommodate the inefficiencies caused by system effect on the installed air moving device. The sound power level of the fans must not exceed 85 dBA when tested according to AMCA 300 and rated in accordance with AMCA 301. Provide all fans with an AMCA seal. ~~Connect fans to the motors either directly or indirectly with V-belt drive. Use V-belt drives designed for not less than [150][140][120] percent of the connected driving capacity. Provide variable pitch motor sheaves for 11 kW and below, and fixed pitch as defined by AHRI Guideline D (A fixed-pitch sheave is provided on both the fan shaft and the motor shaft. This is a non-adjustable speed drive.). Select variable pitch sheaves to drive the fan at a speed which can produce the specified capacity when set at the approximate midpoint of the sheave adjustment. When fixed pitch sheaves are furnished, provide a replaceable sheave when needed to achieve system air balance. Provide motors for V-belt drives with adjustable rails or bases. Provide removable metal guards for all exposed V-belt drives, and provide speed-test openings at the center of all rotating shafts.~~ Provide fans with personnel screens or guards on both suction and supply ends, except that the screens need not be provided, unless otherwise indicated, where ducts are connected to the fan. Provide fan and motor assemblies with vibration-isolation supports or mountings as indicated. Use vibration-isolation units that are standard products with published loading ratings. Select each fan to produce the capacity required at the fan static pressure indicated. Provide sound power level as indicated. Obtain the sound power level values according to AMCA 300. Provide standard AMCA arrangement, rotation, and discharge as indicated. Provide power ventilators that conform to UL 705 and have a UL label.

~~b. Each fan/motor assembly must be factory balanced to AMCA 204, [BV-5, Balance Quality Grade G1.0][BV-4, Balance Quality Grade G2.5][BV-3, Balance Quality Grade G6.3][Balance Quality Grade indicated on Drawings][_____] or better through entire operating speed range from~~

~~minimum speed to maximum speed.~~

b. Mount fan/motor on ~~[aluminum][~~galvanized-steel~~][or][~~powder-coated steel ~~]~~base. Include base and vibration isolators in accordance with requirements indicated. Weld structural members to form a rigid base. Size and design the base construction to withstand the rigors of shipping and rigging. Include the base with lifting lugs or holes.

c. Each fan/motor assembly must be capable of lock-out/tag-out procedure without interrupting operation of other fans in air-handling unit. Fan wheel/motor assembly must pass through the air-handling unit access door servicing fans. Design and incorporate features to permit safe, rapid, and economical maintenance.

d. Include easily removable safety screens where fan inlet and outlet are exposed to maintenance personnel, including walk-in air-handling unit plenums.~~[Safety screens are not required on fan inlets and outlets with backdraft dampers.]~~ Safety screens must be expanded-metal or wire screens, fastened to a flat bar perimeter frame. Screens must comply with OSHA requirements. Screens and frame must be constructed of aluminum, stainless steel, or powder-coated steel. Fasten screens to fan using removable and reusable hardware designed for easy removal by maintenance personnel.

~~2.3.1 Centrifugal Fan Arrays~~

~~Fan arrangement within fan array must produce a uniform airflow and velocity profile across air-handling unit air tunnel when measured[300 mm][_____] upstream of fan inlet and[1200 mm][_____] downstream of fan inlet. Remaining fans in array must continue to operate with one or multiple failed fans. A single mechanical, electrical, and control device failure must not result in a fan array available capacity of less than [33][_____] percent of air-handling unit total scheduled airflow capacity.~~

~~Each fan/motor assembly must be controlled through a variable-frequency controller, except for fans with electronically commutated (EC) motors having integral motor controls.[Include a dedicated variable-frequency controller for each fan/motor assembly in the fan array.][If fan array is served from a single variable-frequency controller, include a redundant variable-frequency controller with automatic switchover in event of primary variable-frequency controller failure.]~~

~~[2.3.1.1 Airflow Measurement~~

~~Each fan within fan array must include airflow measurement indication in L/s. Include airflow totalization of all operating fans in fan array. Airflow measurement instrumentation must not restrict or deflect air travel through fan and must not impact fan air and sound performance. Include digital display of individual fan airflow and total fan array airflow on face of fan control panel. Include a 4- to 20-mA output signal for remote monitoring of total fan array airflow.~~

~~]2.3.1.2 Fan Array Control~~

~~[Include fan control panel with operator interface to control fan array locally through the fan control panel and to switch to control of fan array through a remote-control source. Local control must include on/off operation [and speed adjustment]for entire fan array and each individual fan/motor in fan array.~~

a. ~~The component panel functional intent is manual operation of the fan array which is independent from the building automation system and automated control for the VFD / ECM motor by the building automation system. The component panel must provide all life safety and HVAC equipment safeties AHU shutdown interlocks wired to termination blocks.~~

b. ~~The component panel must be designed using only ladder logic hard wired methods and must not execute any programmed logic specified in the sequence of operation which cannot be submitted in a ladder wiring diagram. The component panel must provide all relevant inputs and outputs necessary for the building automation system to execute the specified sequence of operation.~~

c. ~~Coordinate component panel design with Division 25. Submit design for approval prior to construction of component panel. Laminate final approved as-built ladder wiring diagram and mount inside component panel door.~~

~~][Include fan control panel with control interface for remote control. Fan array on/off operation must be remotely controlled through a single hardwired digital output signal. Fan array speed must be remotely controlled through a single hardwired analog (4- to 20-mA) output signal.~~

~~]2.3.1.3 Fan Array Construction~~

~~Include a precision-spun or die-formed, matched inlet and wheel cone to ensure streamlined airflow into the wheel and full loading of fan blades. Inlet cone must be a smooth hyperbolic shape. Inlet cone must be a single piece, constructed of aluminum or powder-coated steel. Fasten inlet cone to fan panel using bolts, nuts, and washers to provide a positive and secure attachment that can be field removable. Fan blades must be a true hollow airfoil shape, welded to backplate and wheel cone. Construct blades of aluminum, reinforced for AMCA fan class. Design blades to provide smooth airflow over all surfaces of blade. Construct fan hubs of aluminum with integral bracing for extra strength and stiffness. Castings must be sound and free of shrink holes, blow holes, cracks, scale, blisters, or other similar injurious defects. Clean surfaces of castings by blasting, pickling, or any other standard method. Mold-parting fins and remains of gates and risers must be chipped, filed, and ground flush. Design hubs to maintain a high resistance to fatigue and low relative wheel imbalance. Hubs must be keyed and setscrewed to motor shaft for positive attachment. Construct wheel backplates of aluminum. Select entire rotating assembly so first critical speed is at least [30][_____] percent greater than fan design speed and at least [20][_____] percent greater than maximum speed in AMCA fan class.~~

~~Fan must be direct drive, arrangement 4 in accordance with AMCA 99. Adjust wheel width and diameter to match motor speed while providing performance scheduled. Fasten fan wheel directly to motor shaft using a key in motor shaft and setscrew. Construct motor base and pedestal supports of [aluminum][galvanized steel][or][powder-coated steel].~~

~~2.3.1.4 Fan Array Motors~~

~~[See Fan Motors for more information.~~

~~][Provide Electronically Commutated (EC), variable-speed, dc, programmable brushless motor with integral controller/inverter that operates a wound~~

~~stator and senses rotor position to electronically commutate the stator. Controller must control motor speed either through manual adjustment locally at fan array control panel or through a remote 0- to 10-V-dc control signal. Coordinate motor mounting with driven equipment. Motor must be suitable for mounting with motor shaft in either horizontal or vertical position. Motor must be suitable for operation at site altitude and ambient temperature range. Electrical characteristics must be suitable for operation with field power source. Coordinate with electrical installer. Motor efficiency must comply with governing energy codes; [80][_____] percent or higher maintained throughout entire operating range. Power Factor must be [0.9][_____] or higher at full load. Service Factor must be 1.0 or higher.~~

~~a. Provide with zero to 100 percent variable speed control. Controller must be capable of synchronous speed rotation with no slip losses, gradual ramp-up to set point upon receiving a start signal, soft speed change ramps, overcoming reverse rotation without impact, and controlling airflow within 5 percent of set point regardless of static pressure.~~

~~b. Provide with thermal protection to automatically break electrical power to motor when temperature exceeds a safe value and automatically resets and restores power when temperature returns to normal range.~~

~~c. Bearings must be sealed and permanently lubricated ball bearings. Enclosure must be ODP or TEFC. Insulation must be Class B or Class F. Rotor must be permanent magnet with near zero rotor losses that operates independent of motor current.~~

~~d. Enclosure and Frame must be constructed of aluminum, painted steel, or stainless steel. End brackets must be cast aluminum. Shaft must be steel or stainless steel. Motor leads must be Pin or screw terminals. Provide with manufacturer's standard nameplates and paint.~~

~~]2.3.1.5 Fan Array Discharge~~

~~[Include each fan in fan array with integral single-wall enclosure constructed of solid [aluminum][galvanized-steel][or][powder-coated steel]sheet. Enclosure must not increase fan array length beyond size indicated on Drawings. Enclosure must not add static pressure loss. Enclosure must provide a physical separation between operating adjacent fans to prevent negative performance.~~

~~]~~
~~[Include each fan of fan array with integral sound silencer enclosure to reduce the bare fan discharge sound levels by at least [8][15][_____] dBs through octave band frequencies from 125 to 8000 Hz. Enclosure must not increase the fan array length beyond size indicated on Drawings. Silencing enclosure must not add static pressure loss. Double-wall construction consisting of sound-absorbing insulation sandwiched between a solid metal outer skin and perforated metal inner skin. Outer skin material must be[Aluminum][Galvanized steel][Powder-coated steel]. Inner skin material must be [Aluminum][Galvanized steel][Powder-coated steel]. Insulation must be [Mineral fiber][mineral fiber wrapped in a tight woven fiberglass cloth or polymer sheet][or][fiber free]. Enclosure must provide a physical separation between operating adjacent fans to prevent a negative performance.~~

~~]2.3.1.6 Fan Array Accessories~~

~~[Include each fan in the fan array with a backdraft damper at the fan [inlet][or][outlet] to prevent air circulation through a fan that is not operating. Open backdraft damper when fan is operating and close when fan is not operating. Design backdraft damper assembly to operate with little to no static pressure loss with fan operating throughout entire operating range from design to minimum airflow. Add damper pressure loss to fan scheduled total static pressure. If pressure loss requires a change field electrical power, air-handling unit manufacturer must be responsible for associated cost of change. Fasten backdraft damper assembly to fan panel or enclosure using hardware designed for easy removal by maintenance personnel. Dampers must not create measurable additional noise above the sound level of fan. Dampers must not vibrate or rattle. Construct dampers of extruded aluminum, stainless steel, or powder-coated steel.~~

~~]~~
~~[Include[one][two][10 percent of][_____] blank-off panel(s) with each air-handling unit fan array for use by operators in the field to prevent air circulation through any of the fans in fan array that are not operating. Design blank-off panels for attachment to fan panels using easily removable and reusable hardware. Construct blank-offs of aluminum, stainless steel, or powder-coated steel sheets, not less than[1.8 mm][_____] thick. Mount fan blank-off panels in the fan inlet access section for convenient operator access and use in the future.~~

~~]2.3.2 Centrifugal Plenum Fans~~

~~a. Construct fan panel of aluminum or powder-coated steel. Support fan wheel and bearings from a structural aluminum or powder-coated steel framework. Reinforce and brace fan panel to prevent vibration and pulsation. Include stiffeners to form a rigid panel that is free of structural resonance and vibration.~~

~~b. Include a precision-spun or die-formed, matched inlet and wheel cone to ensure streamlined airflow into the wheel and full loading of blades. Inlet and wheel cones must be hyperbolic. Inlet cone must be a single piece, constructed of aluminum or powder-coated carbon steel. Fasten inlet cone to fan panel using bolts, nuts, and washers to provide a positive and secure attachment that can be field removable. Inlet cones that are held in place using retaining clips are unacceptable.~~

~~c. Fan blades must be a true hollow airfoil shape,[continuously] welded to backplate and wheel cone. Construct blades of aluminum, reinforced for AMCA fan class and operating conditions scheduled. Design blades to provide smooth and aerodynamic airflow over all surfaces of blade. Construct fan hubs of cast aluminum or cast iron, ASTM A48/A48M Class 20A and better, with integral bracing for extra strength and stiffness. Castings must be sound and free of shrink holes, blow holes, cracks, scale, blisters, or other similar injurious defects. Clean surfaces of castings by blasting, pickling, or other standard method. Mold-parting fins and remains of gates and risers must be chipped, filed, and ground flush. Design hubs to maintain a high resistance to fatigue and low relative wheel imbalance. Hubs must be keyed and set-screwed to shaft for positive attachment. Construct the wheel backplates of aluminum. Statically and dynamically balance fan wheel before fan is assembled. Select entire rotating assembly so first critical speed is at least [30][_____] percent greater than fan~~

~~design speed and at least [20][_____] percent greater than maximum AMCA class speed.~~

~~d. Fans must be direct drive, arrangement 4 in accordance with AMCA 99 for single-width, single-inlet fans. Adjust wheel width and diameter to match motor speed while providing performance scheduled. Fasten fan wheel directly to motor shaft using a key and setscrew as previously specified. Construct motor base and pedestal of aluminum or powder-coated carbon-steel plate.~~

2.3.1 Housed Centrifugal Fans

Continuously weld housing constructed of ~~[aluminum][~~carbon-steel ~~][~~or ~~][~~ stainless steel ~~]~~sheets, plates, and structural shapes. Support fan housing and shaft bearings from a rigid structural framework. Brace fan housing to prevent vibration and pulsation with external stiffeners to form a rigid housing that is free of operating resonance. Extend fan housing side sheets not more than~~[~~ 13 mm~~][_____]~~ past fan scroll. Fan Cut-off: Designed for pressure distribution required by the application. Fan Blast Area: At least ~~[~~80~~][_____]~~ percent of fan outlet area. Wheel Removal: Construct fan housing for fan wheel(s) removal through the inlet opening when the inlet cone is removed.

2.3.1.1 Housed Centrifugal Fan Construction

~~[~~ a. For fans with wheel diameters nominal~~[~~ 1225 mm~~][_____]~~ and larger, include diagonally flanged, gasketed, and bolted split housings.

~~][~~b. Provide access doors with quick-opening, gasketed, with heavy-duty latches. Conform to housing contour. Locate in 5 o'clock or 7 o'clock position about the end of the shaft and position to gain internal access. Maximize size of square access door up to~~[~~ 600 mm~~][_____]~~.

~~][~~c. Provide~~[~~ DN 25~~][_____]~~ female NPT threaded half coupling welded to lowest point of fan housing. Position the drain connection to accommodate field-installed drain piping. Include each drain coupling with a stainless steel threaded plug.

~~][~~d. Provide welded flanged discharge with a matching companion flange having evenly spaced holes for threaded hardware.

~~][~~e. Provide welded flanged inlet with a matching companion flange having evenly spaced holes for threaded hardware.

~~]~~ f. Include precision-spun or die-formed, matched inlet and wheel cones with hyperbolic shape to ensure streamlined airflow into the wheel and fully load the blades for efficient aerodynamic performance. Inlet cone must be a single piece, constructed of~~[~~ aluminum~~][~~ carbon steel~~][~~ or~~][~~ stainless steel~~]~~. Fasten inlet cone to fan housing using bolts, nuts, and washers to provide a positive and secure attachment that can be removed and replaced in the field.

Provide fan blades with true hollow airfoil shape, continuously welded to backplate or centerplate and wheel cone(s). Construct blades of~~[aluminum][~~ carbon steel~~][~~ or~~][~~ stainless steel~~]~~, reinforced for AMCA fan class. Design blades to provide smooth and aerodynamic airflow over all surfaces of blade. Construct fan hubs of~~[cast aluminum][~~ cast iron, ~~ASTM A48/A48M~~JIS G 5501 ~~Class 20A~~FC150 and better~~][~~ or~~][~~ stainless steel~~]~~, with integral bracing for extra strength and stiffness. Castings must be sound and free of shrink holes, blow holes, cracks, scale, blisters, or other similar injurious defects. Clean surfaces of castings by blasting, pickling, or other standard method. Mold-parting fins and remains of gates and risers must be chipped, filed, and ground flush. Design hubs to maintain a high resistance to fatigue and low relative wheel imbalance. Key and setscrew hubs to shaft for positive attachment. Construct wheel backplates or centerplates of~~[~~ aluminum~~][~~ carbon steel~~][~~ or~~][~~ stainless steel~~]~~. Statically and dynamically balance fan wheel before fan is assembled. Select entire rotating assembly so first critical speed is at least ~~[~~30~~][_____]~~ percent greater than fan design speed and at least ~~[~~20 ~~][_____]~~ percent greater than maximum AMCA class speed.

~~[~~2.3.1.2 Housed Centrifugal Fan Drive

Fan shaft must be one piece, solid ~~[~~carbon~~][~~ or ~~][~~stainless~~]~~ steel, accurately turned, ground, polished, and inspected. Polish shafts at the point of bearing contact to comply with bearing manufacturer's recommended tolerances. Inspect shafts for straightness after the keyways are cut. Coat carbon-steel shafts with a rust-inhibitive coating. Fan bearings must be foot-mounted type, bolted on a rigid welded steel framework that is integral with, or independent of, the housing. Size bearings for L-10 life of at least ~~[~~200,000~~][_____]~~ hours, a DN factor less than ~~[~~200,000 ~~][_____]~~ and a load factor less than ~~[~~2,700,000~~][_____]~~ at the maximum fan class load limit horsepower, including belt pull. Select bearings in accordance with ~~[ABMA 9][and][or][ABMA 11]~~JIS B 1514.

Bearings for Fans with Motor Horsepower up to~~[~~ 7.5 kW~~][_____]~~: Single-row ball or spherical bearings, self-aligning, grease lubricated, and housed in a pillow block housing. Bearings for Larger-Size Fans: Double-row spherical, self-aligning, grease lubricated, and housed in a horizontally split pillow block housing. Extend ~~[~~copper~~][or][plastic]~~ grease lines to an accessible location within sight of bearing for greasing the bearings without removing guards, inlet screens, linkages, and other appurtenances. Terminate bearings and extended grease lines with grease gun fittings.

~~[~~2.3.1.2.1 Direct Drive

Fans must be direct drive Double-Width, Double-Inlet: Arrangement 7 in accordance with AMCA 99. Single-Width, Single-Inlet: Arrangement ~~[~~4~~][~~ or ~~][~~5~~]~~ in accordance with AMCA 99. For AMCA arrangements 4 and 5, fasten fan wheel directly to motor shaft using a key and setscrew indicated. Construct motor base and pedestal for AMCA arrangement 4 and 7 fans of~~[aluminum][~~ carbon-steel~~][~~ or stainless steel~~]~~ plate.

~~][2.3.1.2.2 Belt Drive~~

~~Fans must be Multiple V-Belt Design. Size two belt drives for at least [2.0][_____] times the fan motor horsepower. Size belt drives with more than two belts, for at least [1.5][_____] times the fan motor horsepower. Coordinate, with motor manufacturer, the size and location of motor sheave required to satisfy motor L-10 bearing life using motor manufacturer's written instructions.~~

~~]]~~2.3.1.3 Additional Fan Requirements

~~[a. Sheave and V-belt selections must be in accordance with manufacturer's published data. For constant-Speed Applications, utilize fixed or variable sheaves for applications up to[4 kW][_____] and fixed~~

~~sheaves for applications with larger horsepower. For variable-speed applications, utilize fixed sheaves. Sheave profile must be machined to Mechanical Power Transmission Association Standards. Construct sheaves of high-strength cast iron having a minimum tensile strength of[172.4 MPa][_____]. Sheave rim speeds must not exceed[25.4 m/s][_____]. Sheave side wobble and runout, and eccentricity must be within Mechanical Power Transmission Association and Rubber Manufacturers Association (RMA) tolerances. Balance sheaves to satisfy vibration performance requirements. Mount sheaves on the shaft using a taperlock split and keyed bushing or an integral keyed bushing. V-belts must be oil resistant, non-static conducting, and high quality in accordance with RMA standards. Classic A, B, C, D, and E, non-cogged, cross sections. Multiple belt drives must be a matched set with belt tolerances in accordance with RMA standards. Tension belts in accordance with manufacturer's written instructions.~~

a. Furnish and factory install fan drive guard(s) for bearing assemblies, rotating shafts, sheaves, belts, couplings, and heat slingers. Arrangement 3 fans do not require guards for bearings on the side opposite the drive. Make provision for motor and fan rpm measurement without removing the guard. Construct the guard of flattened expanded aluminum or steel wrapped around a channel frame, suitably braced to prevent vibration. Paint guards using the same coating as fan, except color must be safety yellow. Design attachment to fan for easy removal by maintenance personnel using a clamp and latch.

~~]~~
~~[~~ b. Provide single-width, single-inlet fans with shaft seal(s) where indicated on Drawings. Shaft seal(s) must minimize fan leakage when operating and not operating. Construct shaft seal of an aluminum or split stainless steel plate with non-asbestos material fitted under the split metal plate and seated around the shaft. Secure split plates to housing using threaded hex-head hardware. Shaft seals must be replaceable from outside the housing without disturbing the shaft or bearings. Select shaft seal clearances and materials for operating temperature, pressure, and air quality encountered. Shaft seals, including labyrinth, floating bushing, or close clearance annulus, are acceptable alternatives.

~~][~~c. Piezometer Ring: Mount piezometer ring at fan inlet cone for airflow measurement.

~~]~~2.3.2 Special Fan Construction

Where indicated on Drawings, construct individual fans with the following additional features.

2.3.2.1 Spark-Resistant Construction

Construct fans in accordance with AMCA 99~~[~~ Type A~~][Type B][Type C]~~ or higher rating spark-resistant construction~~[where indicated on Drawings]~~.

~~2.3.2.2 Heat Slinger~~

~~Provide for fans handling air with temperatures exceeding[65 deg C][_____]. Provide a cast aluminum, split-design, internally-finned rotor to create a strong circulation of air over the shaft and inboard bearing, and reduce heat conduction along the shaft to bearings. Secure the split halves together using at least two stainless steel bolts and use a~~

~~setscrew to secure the assembly to fan shaft.~~

2.3.2.2 Corrosion-Resistant Coating

Provide Baked phenolic~~[, equal to Heresite's "P-4403 Brown Baked Phenolic Coating."] <Insert manufacturer's name and product name or designation.>~~. Total Dry Film Thickness: ~~[~~7~~][_____]~~ mils. Application in accordance with manufacturer's written instructions. Mist bonding pass and allow to flash off for several minutes, but not long enough to allow film to completely dry. Not less than three crisscross multipasses maintaining a wet-appearing film. Air dry approximately 45 to 60 minutes with ventilation before applying heat. After air dry period has elapsed, raise temperature in recommended increments of 30 minutes until desired temperature is reached. Bake intermediate coats at approximately half the final temperature for 10 to 20 minutes. Final Bake: ~~[~~ 205 deg C~~][_____]~~ for 1-1/2 hours.

2.3.3 Fan Motors

a. Comply with ~~NEMA MG 1~~JIS C 4212 unless more stringent requirements are indicated. ~~NEMA MG 1~~JIS C 4212~~, [Design B][_____],~~ as required to comply with capacity and torque characteristics; medium-induction motor.

b. Capacity and torque must be sufficient to start, accelerate, and operate connected loads at designated speeds, at installed altitude and environment, with indicated operating sequence, and without exceeding nameplate ratings or considering service factor. Provide ~~NEMA Premium~~ Efficiency rating complying with ~~NEMA MG 1~~JIS C 4212. Motor must operate fan under all conditions indicated without exceeding motor nameplate and without use of motor service factor. Comply with minimum requirements of Class F or Class H insulation, suitable for "inverter-duty" or "drive-duty" applications in accordance with ~~NEMA MG 1~~JIS C 4212. Motor operation through a variable-frequency controller must not adversely affect the motor performance, operation, useful life, and warranty. Motor Service Factor must be ~~[~~1.15~~][_____]~~. Enclosure Type: must be ~~[ODP][or][~~TEFC~~]~~.

c. Provide Shaft grounding system to protect bearings from induced voltage. Shaft grounding system must have low drag (less than 0.05 percent of motor horsepower), and must operate for a minimum of 3 years without periodic maintenance or adjustments. Shaft grounding system must be mounted external ~~[~~or internal~~]~~ to motor enclosure. Provide frames with integrally cast feet unless other requirements of driven equipment require a different arrangement. Frame, front and back end brackets, and front and back end bearing intercaps constructed of cast iron, ~~ASTM A48/A48M~~JIS G 5501, ~~Class 25~~FC150 or better. Fabricate rotor frame from die-cast aluminum, copper, or associated alloys. Key rotors to motor shaft. Rotating assembly must be dynamically balanced to within limits defined in ~~NEMA MG 1~~JIS C 4212. Motors must have the entire rotating assembly between bearing inner caps coated with a corrosion-resistant coating.

d. Copper windings must be spike resistant to withstand 1600 peak V. Entire wound and insulated stator coated with a coating to protect against moisture and corrosion.

e. Solid shaft fabricated of~~[carbon][Type 304 stainless][~~ Type 316 stainless~~][~~ Type 416 stainless~~]~~ steel, accurately turned, ground and polished, and inspected for accuracy. End of shaft with drilled hole for use in field measurements.

f. Grease-lubricated ball or roller bearings. ABMA 11 L-10 motor bearing life of ~~[~~100,000~~][_____]~~ hours. Factory lubricate motor bearings using a premium moisture-resistant polyurea thickened grease with rust inhibitors suitable for extreme operating temperatures encountered. Coordinate special requirements that may impact lubrication and include appropriate lubrication.

g. Equip each bearing housing with an easily accessible grease inlet. Fit grease inlets with a grease fitting and protective fitting cap. Equip inlets with an automatic grease relief fitting to prevent excessive greasing. Equip each bearing housing with grease drain and threaded plug.

h. Conduit box material must be the same as frame. For motor frames 365T and below, furnish conduit boxes sized with internal volumes in accordance with ~~NEMA MG 1~~JIS C 8305. For motor frames larger than 365T, furnish conduit boxes one size larger than ~~NEMA MG 1~~JIS C 8305. Coordinate the location and mounting of conduit box with driven equipment manufacturer. Factory mount conduit box on motor. Provide NRTL-listed clamp-type grounding lug mounted in conduit box.

i. Motor leads must be non-wicking type, Class F temperature rating or better, and permanently numbered over entire length for identification. Lead terminals must be manufacturer's standard.

j. Provide motor with drain holes at the lowest point for drainage of condensate. Each drain hole with a threaded removable plug.

~~[~~ k. TEFC Motor Fans must be corrosion-resistant construction, non-sparking, metallic or non-metallic, bi-directional, and keyed to shaft. Motor Fan Cover must be~~[Steel][~~ Same material as frame~~]~~.

~~]~~ l. Motor hardware must be hex-head, high-strength, zinc-plated carbon steel or stainless steel.

m. Provide lifting eyebolts threaded into frame receptacle and designed to prevent moisture and other foreign material from entering motor cavity when eyebolt is removed.

n. Construct nameplates of aluminum or stainless steel and attach to motor frame with aluminum, stainless steel, or brass drive pins. Engrave or stamp data on the nameplate. At a minimum, include nameplate data in accordance with ~~NEMA MG 1~~JIS C 4212.~~[Also include ABMA bearing designation for the drive and opposite end bearing.]~~

o. Motor must successfully pass~~[500][~~ 1000~~][2000][_____]~~-hour salt spray test for corrosion in accordance with ~~ASTM B117~~JIS Z 2371.

2.4 COILS

Provide fin-and-tube type coils constructed of seamless ~~[~~copper~~][red brass]~~ tubes and ~~[~~aluminum~~][~~ or ~~][~~copper~~]~~ fins mechanically bonded or soldered to the tubes.~~[~~ Provide copper tube wall thickness that is a minimum of ~~[~~ 0.406~~][0.508][0.6096]~~ mm.~~][Provide red brass tube wall thickness that is a minimum of [0.89][1.24] mm.][~~ Provide aluminum fins that are ~~[~~0.14

~~][0.19]~~ mm minimum thickness.~~][~~ Provide copper fins that are 0.114 mm minimum thickness.~~]~~ Provide casing and tube support sheets that are not lighter than 1.6 mm galvanized steel, formed to provide structural strength. When required, provide multiple tube supports to prevent tube sag. Mount coils for counterflow service. Rate and certify coils to meet the requirements of ~~AHRI 410~~JIS B 8616.~~[Provide factory applied phenolic, vinyl or epoxy/electrodeposition coating.]~~

~~2.4.1   Direct-Expansion Coils~~

~~Provide suitable direct-expansion coils for the refrigerant involved. Provide refrigerant piping that conforms to ASTM B280 and clean, dehydrate and seal. Provide seamless copper tubing suction headers or seamless or resistance welded steel tube suction headers with copper connections. Provide supply headers that consist of a distributor which distributes the refrigerant through seamless copper tubing equally to all circuits in the coil. Provide circuited tubes to ensure minimum pressure drop and maximum heat transfer. Provide circuiting that permits refrigerant flow from inlet to suction outlet without causing oil slugging or restricting refrigerant flow in coil. Provide field installed coils which are completely dehydrated and sealed at the factory upon completion of pressure tests. Pressure test coils in accordance with UL 1995.~~

2.4.1   Water Coils

Install water coils with a pitch of not less than 10 mm/m of the tube length toward the drain end. Use headers constructed of cast iron, welded steel or copper. Furnish each coil with a plugged vent and drain connection extending through the unit casing. Provide removable water coils with drain pans. Pressure test coils in accordance with UL 1995.

~~2.4.2   Steam Heating Coils~~

~~Construct steam coils from cast semisteel, welded steel or copper headers, and [red brass][copper] tubes. Construct headers from cast iron, welded steel or copper. Provide fin tube and header section that float within the casing to allow free expansion of tubing for coils subject to high pressure steam service. Provide each coil with a field or factory installed vacuum breaker. Provide single-tube type coils with tubes not less than 13 mm outside diameter, except for steam preheat coils. Provide supply headers that distribute steam evenly to all tubes at the indicated steam pressure. Factory test coils to ensure that, when supplied with a uniform face velocity, temperature across the leaving side is uniform with a maximum variation of no more than 5 percent. Pressure testcoils in accordance with UL 1995.~~

~~2.4.3   Steam Preheat (Nonfreeze) Coils~~

~~Provide steam-distribution-tube type steam (nonfreeze) coils with condensing tubes not less than 25 mm outside diameter for tube lengths 1.5 m and over and 13 mm outside diameter for tube lengths under 1.5 m. Construct headers from cast iron, welded steel, or copper. Provide distribution tubes that are not less than 15 mm outside diameter for tube lengths 1.5 m and over and 10 mm outside diameter for tube lengths under 1.5 m with orifices to discharge steam to condensing tubes. Install distribution tubes concentric inside of condensing tubes and hold securely in alignment. Limit maximum length of a single coil to 3.66 m. Factory test coils to ensure that, when supplied with a uniform face velocity, temperature across the leaving side is uniform with a maximum variation of~~

~~no more than 5 percent. Pressure test coils in accordance with UL 1995.~~

~~2.4.4 Electric Heating Coil~~

~~Provide an electric duct heater coil in accordance with UL 1995 and NFPA 70. Provide duct- or unit-mounted coil. Provide[nickel chromium resistor, single stage, strip][nickel chromium resistor, single stage, strip or stainless steel, fin tubular] type coil. Provide coil with a built-in or surface-mounted high-limit thermostat interlocked electrically so that the coil cannot be energized unless the fan is energized. Provide galvanized steel or aluminum coil casing and support brackets. Mount coil to eliminate noise from expansion and contraction and for complete accessibility for service.~~

~~2.4.5 Eliminators~~

~~Equip each cooling coil having an air velocity of over 2 m/s through the net face area with moisture eliminators, unless the coil manufacturer guarantees, over the signature of a responsible company official, that no moisture can be carried beyond the drip pans under actual conditions of operation. Construct of minimum 24 gage[zinc-coated steel][copper][copper nickel][or][stainless steel], removable through the nearest access door in the casing or ductwork. Provide eliminators that have not less than two bends at 45 degrees and are spaced not more than 63 mm center-to-center on face. Provide each bend with an integrally formed hook as indicated in the SMACNA 1884.~~

~~2.4.6 Sprayed Coil Dehumidifiers~~

~~Provide assembly with reinforced, braced, and externally insulated galvanized steel casing, vertical in-line spray pump, bronze self-cleaning spray nozzles, galvanized steel pipe spray headers, adjustable float valve with replaceable neoprene seat, manufacturer's standard cooling coil, and welded black steel drain tank. Provide overflow drain, make-up, and bleed connection.~~

~~2.5 REFRIGERATION~~

~~2.5.1 Air-to-Refrigerant Coils~~

~~Provide air-to-refrigerant coils with [seamless copper][or][aluminum] tubes of 8 mm minimum diameter with [copper][or][aluminum] fins that are mechanically bonded or soldered to the tubes. Casing must be [galvanized steel][or][aluminum]. Avoid contact of dissimilar metals. Test coils in accordance with ANSI/ASHRAE 15 & 34 at the factory and must be suitable for the working pressure of the installed system. Factory pressure and leak test each coil.~~

~~a. Provide separate expansion devices for each compressor circuit. Condensate drain pans must be removable and double-sloped.~~

~~b. Dual compressor units must have intermingled evaporator coils.~~

~~c. Condensate drain pans must be removable and double-sloped.~~

~~[d. Provide condenser coils with hail protection guards.~~

~~] e. Coat[condenser][evaporator][condenser and evaporator] coil with a uniformly applied[epoxy electrodeposition][phenolic][vinyl][epoxy~~

~~electrodeposition, phenolic, or vinyl] type coating to all coil surface areas without material bridging between fins. Apply coating at either the coil or coating manufacturer's factory. Coating process must ensure complete coil encapsulation. Coating must be capable of withstanding a minimum [500][1000][_____] hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution.~~

~~2.5.2 Compressor~~

~~Provide direct drive, semi-hermetic or hermetic reciprocating, or scroll type compressor capable of operating at partial load conditions. Compressor must be capable of continuous operation down to the lowest step of unloading as specified. Equip compressors of 35 kW and larger with capacity reduction devices to produce automatic capacity reduction of at least 50 percent. If standard with the manufacturer, two or more compressors may be used in lieu of a single compressor with unloading capabilities, in which case the compressors operate in sequence, and each compressor has an independent refrigeration circuit through the condenser and evaporator. Start compressors in the unloaded position. Provide each compressor with vibration isolators, crankcase heater, thermal overloads,[lubrication pump,][high][high and low] pressure safety cutoffs and protection against short cycling.~~

~~2.5.3 Refrigeration Circuit~~

~~Refrigerant containing components must comply with ANSI/ASHRAE 15 & 34 and be factory tested, cleaned, dehydrated, charged, and sealed. Provide refrigerant lines with service pressure tap ports and refrigerant line filter.~~

~~2.5.4 Remote Condenser or Condensing Unit~~

~~Units with capacities 39.5 kW or greater must produce a maximum AHRI sound rating of [85][_____] dB when rated in accordance with ANSI/AHRI 370. Fit each remote condenser coil with a manual isolation valve and an access valve on the coil side. Saturated refrigerant condensing temperature must not exceed 49 degrees C at 40 degrees C ambient. Provide unit with low ambient condenser controls to ensure proper operation in an ambient temperature of [-6][13][_____] degrees C. Provide fan and cabinet construction must be provided as specified in paragraph UNIT CONSTRUCTION. Fan and condenser motors must have[open][dripproof][totally enclosed][explosion proof] enclosures.[Condensing unit must have controls to initiate a refrigerant pump down cycle at system shut down on each refrigerant circuit.][Provide outdoor unit with louvered, weather-proof enclosure for protection of integral elements.]~~

~~2.5.4.1 Air-Cooled Condenser~~

~~Provide unit rated in accordance with ANSI/AHRI 460 and conform to the requirements of UL 1995. Provide factory fabricated, tested, packaged, and self-contained unit. Unit must be complete with casing, propeller or centrifugal type fans, heat rejection coils, connecting piping and wiring, and all necessary appurtenances.~~

~~a. Provide interconnecting refrigeration piping, electrical power, and control wiring between the condensing unit and the indoor unit as required and as indicated. Provide electrical and refrigeration piping terminal connections between[condenser][condensing unit] and~~

~~evaporator units.~~

~~b. Low ambient control for multi-circuited units serving more than one evaporator coil must provide independent condenser pressure controls for each refrigerant circuit. Set controls to produce a minimum of 40 degrees C saturated refrigerant condensing temperature. Provide unit with a liquid subcooling circuit that ensures proper liquid refrigerant flow to the expansion device over the specified application range of the condenser. Unit must be provided with[manufacturer's standard][not less than [4][_____] degrees C] liquid subcooling. Liquid seal the subcooling circuit.~~

~~c. Coils must have[nonferrous][copper or aluminum] tubes of 10 mm minimum diameter with copper or aluminum fins that are mechanically bonded or soldered to the tubes.[Protect coil in accordance with paragraph COILS.] Casing must be galvanized steel or aluminum. Avoid contact of dissimilar metals. Test coils in accordance with ANSI/ASHRAE 15 & 34 at the factory and ensure suitability for the working pressure of the installed system. Dehydrate and seal each coil after testing and prior to evaluation and charging. Provide each unit with a factory operating charge of refrigerant and oil or a holding charge. Field charge unit shipped with a holding charge. Provide separate expansion devices for each compressor circuit.~~

~~d. Provide a complete control system with required accessories for regulating condenser pressure by fan cycling, solid-state variable fan speed, modulating condenser coil or fan dampers, flooding the condenser, or a combination of the above. Construct unit mounted control panels or enclosures in accordance with applicable requirements of NFPA 70 and house in NEMA ICS 6, Class 1 or 3A enclosures. Controls must include[control transformer,][fan motor[starters],][solid-state speed control,][electric heat tracing controls,][time delay start-up,] overload protective devices, interface with local and remote components, and intercomponent wiring to terminal block points.~~

~~2.5.4.2 Evaporative Condenser~~

~~[Provide a counter-flow blow-through design, with single-side air entry.]The unit must have fan assemblies built into the unit base, with all moving parts factory mounted and aligned. Primary construction of the pan section and the cabinet must not be lighter than 1.6 mm steel, protected against corrosion by a zinc coating. Conform the zinc coating ASTM A153/A153M and ASTM A123/A123M, as applicable and have an extra heavy coating of not less than 0.76 kg/square meter of surface. Give cut edges a protective coating of zinc-rich compound. After assembly, apply the manufacturer's standard zinc chromatized aluminum or epoxy paint finish to the exterior of the unit. Unit must be rated in accordance with AHRI 490 I-P and tested in accordance with the requirements of ASHRAE 64.~~

~~a. Provide a watertight pan complete with drain, overflow, and make-up water connections. Provide standard pan accessories to include circular access doors, a lift-out strainer of anti-vortexing design and a brass make-up valve with float ball.~~

~~b. Provide a direct driven, statically and dynamically balanced, [centrifugal][or][propeller] type fan. Do not locate fan and fan motor in the discharge airstream of the unit. Enclose motors in[open][splashproof][totally enclosed] enclosure that is suitable for~~

~~the indicated service. Design the condensing unit design to prevent water from entering into the fan section.~~

~~c. Provide condensing coils with[nonferrous][copper][or][aluminum] tubes of 10 mm minimum diameter without fins.[Protect coil in accordance with paragraph COILS.] Provide [galvanized steel][or][aluminum] casing. Avoid contact of dissimilar metals. Test coils in accordance with ANSI/ASHRAE 15 & 34 at the factory and ensure suitability for the working pressure of the installed system. Dehydrate and seal each coil after testing and prior to evaluation and charging. Provide each unit with [a factory operating charge of refrigerant and oil][or][a holding charge].[Field charge unit shipped with a holding charge with refrigerant and oil.]~~

~~d. Provide a water distribution system that distributes water uniformly over the condensing coil to ensure complete wetting of the coil at all times. Provide[brass,][stainless steel,][or][high-impact plastic] spray nozzles that are the cleanable, non-clogging, removable type. Design nozzles to permit easy disassembly and arrange for easy access.~~

~~e. Provide [a][two] bronze-fitted [centrifugal][or][turbine] type water pump[s] that may be mounted as an integral part of the evaporative condenser or remotely on a separate mounting pad. Pumps must have cast-iron casings. Impellers must be bronze, and shafts stainless steel with bronze casing wearing rings. Use mechanical type shaft seals. Factory coat the pump casing with epoxy paint. Pump motors must have[open][drip proof][totally enclosed][explosion proof] enclosures. Provide a bleed line with a flow valve or fixed orifice in the pump discharge line and extend to the nearest drain for continuous discharge. Fully submerge pump suction and provide with a [galvanized steel][or][monel] screened inlet.~~

~~f. Provide drift eliminators to limit drift loss to not over 0.005 percent of the specified water flow. Construct eliminators of [zinc-coated steel][or][polyvinyl chloride (PVC)]. Eliminators must prevent carry over into the unit's fan section.~~

~~g. Provide the evaporative condenser unit with modulating capacity control dampers mounted in the discharge of the fan housing. On a decrease in refrigerant discharge pressure the dampers must modulate to reduce the airflow through the evaporative condenser. Controls must include a proportional acting pressure controller, a control transformer, motor actuator with linkages and end switches to cycle fan motor on and off. Cycle a fan motor on and off in accordance with the manufacturer's instructions.~~

~~[2.5.4.3 Compressor~~

~~Provide compressor rated in accordance with AHRI 540. Provide direct drive, semi-hermetic or hermetic reciprocating, or scroll type compressor capable of operating at partial load conditions. Compressor must be capable of continuous operation down to the lowest step of unloading as specified. Provide units 35 kW and larger with capacity reduction devices to produce automatic capacity reduction of at least 50 percent. If standard with the manufacturer, two or more compressors may be used in lieu of a single compressor with unloading capabilities, in which case the compressors operate in sequence, and each compressor must have an independent refrigeration circuit through the condenser and evaporator. Each compressor must start in the unloaded position. Provide each~~

~~compressor with vibration isolators, crankcase heater,[lubrication pump,] thermal overloads, and[high][high and low] pressure safety cutoffs and protection against short cycling.~~

~~]2.5.4.4 Fans~~

~~Provide fan wheel shafts supported by either maintenance-accessible grease lubricated antifriction block-type bearings, or permanently lubricated ball bearings. Mount fan motor and fan assembly on a common base to allow consistent belt tension with no relative motion between fan and motor shafts. The entire fan motor and fan assembly must be completely vibrationally isolated from the unit. Select unit fans to produce the airflow required at the fan total pressure. Motor starters, if applicable, must be magnetic across-the-line type with a[n][open drip-proof][totally enclosed][explosion proof] enclosure. Provide [manual][or][automatic-reset] type thermal overload protection. Construct fan wheels of [aluminum][or][galvanized steel]. Provide centrifugal fan wheel housings of galvanized steel, and construct centrifugal fan casings of [aluminum][or][galvanized steel]. Steel elements of fans, except fan shafts, must be [hot-dipped galvanized after fabrication][or][fabricated of mill galvanized steel]. Recoat mill-galvanized steel surfaces and edges damaged or cut during fabrication by forming, punching, drilling, welding, or cutting with an approved zinc-rich compound. Statically and dynamically balance [fan wheels][or][propellers]. Provide double inlet[forward-curved][air foil] type fan wheels. Fan must reach rated rpm before the fan shaft passes through the first critical speed. Fans must be belt-driven with adjustable sheaves. Select the sheave size so that the fan speed at the approximate midpoint of the sheave adjustment produces the specified air quantity. Provide centrifugal scroll-type fans with streamlined orifice inlet and V-belt drive. Each drive must be independent of any other drive. Condenser fans must be propeller type, direct drive, statically balanced with galvanized steel blades and permanently lubricated ball bearings. Protect condenser fan motor drive bearings with water slingers or shields. Fit all belt drives with guards where exposed to contact by personnel.~~

~~2.6 GAS-FIRED HEATING SECTION~~

~~Construct gas-fired heat exchanger and burner of stainless steel suitable for[natural gas][liquid propane gas] fuel supply. Provide burner with[direct spark][pilot] ignition. Heating section must have modulation with a turn down ratio of at least [4][3] to 1. Provide heating section completely assembled and integral to unit. Fire test all units prior to shipment. Valve must include a pressure regulator. Supply combustion air with a centrifugal combustion air blower with built-in thermal over load protection. Safety controls must include a flame sensor and air pressure switch. Mount heater section to eliminate noise from expansion and contraction and completely accessible for service. Gas equipment must bear the AGA label for the type of service involved.[Provide burner in accordance with NFPA 54.]~~

~~2.6.1 Direct Gas-Fired Heating Section~~

~~Direct gas-fired heat module(s) must have a stainless-steel burner with aluminum burner head casting, non-clogging gas ports; spark-ignition intermittent pilot, flame safeguard system with integral flame sensing, air pressure switch and automatic high limit switch set to [85][_____] degrees C. Burner assembly must be mechanically secured to vestibule panels. The burner combustion must be clean and odorless, with combustion~~

~~efficiency limiting the products of combustion to a maximum of 5 ppm carbon monoxide and 0.5 ppm nitrogen dioxide. The burner profile to be equipped with adjustable profile plates.~~

~~2.6.2 Indirect Gas-Fired Heating Section~~

~~Heat module must be listed for outdoor installation with a Category IV venting system. Vent connectors provided must be suitable for connection to commercially available PVC pipe. Module must have [2][4][_____]-pass tubular heat exchangers, constructed of[aluminized steel][type 409 stainless steel]. Heat exchanger tubes must be installed on the vest plate by means of swaged assembly, welded connections are not acceptable. Module must be encased in a weather-tight metal housing with intake air vents.~~

2.5 AIR FILTERS

List air filters according to requirements of UL 900, except list high efficiency particulate air filters of 99.97 percent efficiency by the DOP Test method under the Label Service to meet the requirements of UL 586.

2.5.1 Extended Surface Pleated Panel Filters

Provide 50 mm depth, sectional, disposable type filters of the size indicated with a MERV of 8 when tested according to ASHRAE 52.2. Provide initial resistance at 2.54 m/s that does not exceed 0.09 kPa. Provide UL Class 2 filters, and nonwoven cotton and synthetic fiber mat media. Attach a wire support grid bonded to the media to a moisture resistant fiberboard frame. Bond all four edges of the filter media to the inside of the frame to prevent air bypass and increase rigidity.

2.5.2 Extended Surface Nonsupported Pocket Filters

Provide ~~[~~750~~][_____]~~ mm depth, sectional, replaceable dry media type filters of the size indicated with a MERV of 13 when tested according to ASHRAE 52.2. Provide initial resistance at ~~[~~2.54~~][_____]~~ m/s that does not exceed ~~[~~0.1125~~][_____]~~ kPa. Provide UL Class 1 filters. Provide fibrous glass media, supported in the air stream by a wire or non-woven synthetic backing and secured to a galvanized steel metal header. Provide pockets that do not sag or flap at anticipated air flows. Install each filter ~~[with an extended surface pleated panel filter as a prefilter]~~ in a factory preassembled, side access housing or a factory-made sectional frame bank, as indicated.

2.5.3 Cartridge Type Filters

Provide 305 mm depth, sectional, replaceable dry media type filters of the size indicated with a MERV of 13 when tested according to ASHRAE 52.2. Provide initial resistance at ~~[~~2.54~~][_____]~~ m/s that does not exceed ~~[~~0.14 ~~][_____]~~ kPa. Provide UL class 1 filters, and pleated microglass paper media with corrugated aluminum separators, sealed inside the filter cell to form a totally rigid filter assembly. Fluctuations in filter face velocity or turbulent airflow have no effect on filter integrity or performance. Install each filter~~[~~ with an extended surface pleated media panel filter as a prefilter~~]~~ in a factory preassembled side access housing, or a factory-made sectional frame bank, as indicated.

~~2.5.4 Sectional Cleanable Filters~~

~~Provide [25][50] mm thick cleanable filters. Provide viscous adhesive in 20 L containers in sufficient quantity for 12 cleaning operations and not less than one L for each filter section. Provide one washing and charging tank for every 100 filter sections or fraction thereof; with each washing and charging unit consisting of a tank and[single][double] drain rack mounted on legs and drain rack with dividers and partitions to properly support the filters in the draining position.~~

~~2.5.5 Replaceable Media Filters~~

~~Provide the[dry-media][viscous adhesive] type replaceable media filters, of the size required to suit the application. Provide filtering media that is not less than 50 mm thick fibrous glass media pad supported by a structural wire grid or woven wire mesh. Enclose pad in a holding frame of not less than 1.6 mm galvanized steel, equipped with quick-opening mechanism for changing filter media. Base the air flow capacity of the filter on net filter face velocity not exceeding [1.5][_____] m/s, with initial resistance of [32][_____] Pa. Provide MERV that is not less than [_____] when tested according to ASHRAE 52.2.~~

~~2.5.6 Electrostatic Filters~~

~~Provide the following:~~

a. ~~The combination dry agglomerator/extended surface, nonsupported pocket electrostatic filters or the combination dry agglomerator/automatic renewable, media (roll) type electrostatic filters, as indicated (except as modified). Supply each dry agglomerator electrostatic air filter with the correct quantity of fully housed power packs and equip with silicon rectifiers, manual reset circuit breakers, low voltage safety cutout, relays for field wiring to remote indication of primary and secondary voltages, with lamps mounted in the cover to indicate these functions locally. Equip power pack enclosure with external mounting brackets, and low and high voltage terminals fully exposed with access cover removed for ease of installation. Furnish interlock safety switches for each access door and access panel that permits access to either side of the filter, so that the filter is de-energized in the event that a door or panel is opened.~~

b. ~~Ozone generation within the filter that does not exceed five parts per one hundred million parts of air. Locate high voltage insulators in a serviceable location outside the moving air stream or on the clean air side of the unit. Fully expose ionizer wire supports and furnish ionizer wires precut to size and with formed loops at each end to facilitate ionizer wire replacement.~~

c. ~~Agglomerator cell plates that allow proper air stream entrainment of agglomerates and prevent excessive residual dust build-up, with cells that are open at the top and bottom to prevent accumulation of agglomerates which settle by gravity. Where the dry agglomerator electrostatic filter is indicated to be the automatic renewable media type, provide a storage section that utilizes a horizontal or vertical traveling curtain of adhesive-coated bonded fibrous glass for dry agglomerator storage section service supplied in 19.8 m lengths in convenient roll form. Otherwise, provide section construction and roll media characteristics as specified for automatic renewable media filters. Also a dry agglomerator/renewable media combination with an~~

~~initial air flow resistance, after installation of clean media, that does not exceed 62.3 Pa at 2.54 m/s face velocity.~~

~~d. A MERV of the combination that is not less than 15 when tested according to ASHRAE 52.2 at an average operating resistance of 125 Pa. Where the dry agglomerator electrostatic filter is indicated to be of the extended surface nonsupported pocket filter type, provide a storage section as specified for extended surface non-supported pocket filters, with sectional holding frames or side access housings as indicated.~~

~~e. A dry agglomerator/extended surface nonsupported pocket filter section combination with initial air flow resistance, after installation of clean filters, that does not exceed 162 Pa at 2.54 m/s face velocity, with a MERV of the combination not less than 16 when tested according to ASHRAE 52.2. Furnish front access filters with full height air distribution baffles and upper and lower mounting tracks to permit the baffles to be moved for agglomerator cell inspection and service. When used in conjunction with factory fabricated air handling units, supply side access housings which have dimensional compatibility.~~

~~2.5.7 High-Efficiency Particulate Air (HEPA) Filters~~

~~Provide HEPA filters that meet the requirements of IEST RP-CC-001 and are individually tested and certified to have an efficiency of not less than [95][99.97] percent, and an initial resistance at [_____] m/s that does not exceed [_____] Pa. Provide filters that are constructed by pleating a continuous sheet of filter medium into closely spaced pleats separated by corrugated aluminum or mineral-fiber inserts, strips of filter medium, or by honeycomb construction of the pleated filter medium. Provide interlocking, dovetailed, molded neoprene rubber gaskets of 5-10 durometer that are cemented to the perimeter of the[upstream][downstream] face of the filter cell sides. Provide self-extinguishing rubber-base type adhesive or other materials conforming to fire hazard classification specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Provide filter cell sides that are[19 mm thick exterior grade fire-retardant plywood][cadmium plated steel][galvanized steel] assembled in a rigid manner. Provide overall cell side dimensions that are correct to 2 mm, and squareness that is maintained to within 3.2 mm. Provide holding frames that use spring loaded fasteners or other devices to seal the filter tightly within it and that prevent any bypass leakage around the filter during its installed life. Provide air capacity and the nominal depth of the filter as indicated. Install each filter in a factory preassembled side access housing or a factory-made sectional supporting frame as indicated. Provide prefilters of the type, construction and efficiency indicated.~~

~~2.5.8 Holding Frames~~

~~Fabricate frames from not lighter than 1.6 mm sheet steel with rust-inhibitor coating. Equip each holding frame with suitable filter holding devices. Provide gasketed holding frame seats. Make all joints airtight.~~

~~2.5.9 Filter Gauges~~

~~Provide dial type filter gauges, diaphragm actuated draft for all filter stations, including those filters which are furnished as integral parts of factory fabricated air handling units. Provide gauges that are at least~~

~~98 mm in diameter, with white dials with black figures, and[graduations][graduated in 0.0025 kPa,] with a minimum range of 0.25 kPa beyond the specified final resistance for the filter bank on which each gauge is applied. Provide each gauge with a screw operated zero adjustment and two static pressure tips with integral compression fittings, two molded plastic vent valves, two 1.5 m minimum lengths of 6.35 mm diameter[aluminum][vinyl] tubing, and all hardware and accessories for gauge mounting.~~

~~2.6 ENERGY RECOVERY DEVICES~~

~~2.6.1 Energy Recovery Performance~~

~~Energy recover equipment must comply with[ASHRAE Design Guide for Dedicated Outdoor Air Systems (DOAS)][ANSI/AHRI Guideline V Calculating the Efficiency of Energy Recovery Ventilation and its Effect on Efficiency and Sizing of Building HVAC Systems][and][ANSI/AHRI Guideline W Selecting, Sizing, and Specifying Packaged Air-to-Air Energy Recovery Ventilation Equipment].~~

~~[Energy transfer ratings must be AHRI Certified to AHRI 1060 I-P and bear the AHRI certification seal for AHRI Air-to-Air Energy Recovery Ventilation Equipment Program based on AHRI 1060 I-P.][The manufacturer must provide certified performance data in accordance with ASHRAE 84 and AHRI 1060 I-P. Independent performance test results must be used to rate the product in accordance with the AHRI Air-to-Air Energy Recovery Ventilation Equipment Program.]~~

~~2.6.2 Rotary Wheel~~

~~Provide unit that is a factory fabricated and tested assembly for air-to-air energy recovery by transfer of sensible[and latent] heat from exhaust air to supply air stream, with device performance according to ASHRAE 84 and that delivers an energy transfer effectiveness of not less than [70][85][_____] percent with cross-contamination not in excess of [0.1][1.0][_____] percent of exhaust airflow rate at system design differential pressure, including purging sector if provided with wheel. Provide exchange media that is chemically inert, moisture-resistant, fire-retardant, laminated, nonmetallic material which complies with NFPA 90A. Isolate exhaust and supply streams by seals which are static, field adjustable, and replaceable. Equip chain drive mechanisms with ratcheting torque limiter or slip-clutch protective device. Fabricate enclosure from galvanized steel and include provisions for maintenance access. Provide recovery control and rotation failure provisions as indicated.~~

~~2.6.3 Run-Around-Coil~~

~~Provide assembly that is factory fabricated and tested air-to-liquid-to-air energy recovery system for transfer of sensible heat from exhaust air to supply air stream and that delivers an energy transfer effectiveness not less than that indicated without cross-contamination with maximum energy recovery at minimum life cycle cost. Computer optimize components for capacity, effectiveness, number of coil fins per inch, number of coil rows, flow rate, heat transfer rate of [_____] percent by volume of[ethylene][propylene] glycol solution, and frost control. Provide coils that conform to paragraph COILS. Provide related pumps, and piping specialties that conform to requirements of[Section 23 63 00.00 COLD STORAGE REFRIGERATION SYSTEMS][Section 23 57 10.00 10~~

~~FORCED HOT WATER HEATING SYSTEMS USING WATER AND STEAM HEAT EXCHANGERS][Section23 69 00.00 20 REFRIGERATION EQUIPMENT FOR COLD STORAGE] [_____].~~

~~2.6.4 Heat Pipe~~

~~Provide a device that is a factory fabricated, assembled and tested, counterflow arrangement, air-to-air heat exchanger for transfer of sensible heat[between exhaust and supply streams] and that delivers an energy transfer effectiveness not less than that indicated without cross-contamination. Provide heat exchanger tube core that is [15][18][25] mm nominal diameter, seamless aluminum or copper tube with extended surfaces, utilizing wrought aluminum Alloy 3003 or Alloy 5052, temper to suit. Provide maximum fins per unit length and number of tube rows as indicated. Provide tubes that are fitted with internal capillary wick, filled with a refrigerant complying with ANSI/ASHRAE 15 & 34, selected for system design temperature range, and hermetically sealed. Refrigerants containing chlorofluorocarbons (CFC) are prohibited. Provide heat exchanger frame that is constructed of not less than 1.6 mm galvanized steel and fitted with intermediate tube supports, and flange connections. Provide tube end-covers and a partition of galvanized steel to separate exhaust and supply air streams without cross-contamination and in required area ratio.[Provide a drain pan constructed of welded Type 300 series stainless steel.] Provide heat recovery regulation by[system face and bypass dampers and related control system as indicated][interfacing with manufacturer's standard tilt-control mechanism for summer/winter operation, regulating the supply air temperature and frost prevention on weather face of exhaust side at temperature indicated]. Coil must be fitted with pleated flexible connectors.~~

~~2.6.5 Desiccant Wheel~~

~~Provide counterflow supply, regeneration airstreams, a rotary type dehumidifier designed for continuous operation, and extended surface type wheel structure in the axial flow direction with a geometry that allows for laminar flow over the operating range for minimum air pressure differentials. Provide the dehumidifier complete with a drive system utilizing a fractional-horsepower electric motor and speed reducer assembly driving the rotor. Include a slack-side tensioner for automatic take-up for belt-driven wheels. Provide an adsorbing type desiccant material. Apply the desiccant material to the wheel such that the entire surface is active as a desiccant and the desiccant material does not degrade or detach from the surface of the wheel which is fitted with full-face, low-friction contact seals on both sides to prevent cross leakage. Provide rotary structure that has underheat, overheat and rotation fault circuitry. Provide wheel assembly with a warranty for a minimum of five years.~~

~~2.6.6 Plate Heat Exchanger~~

~~Provide energy recovery ventilator unit that is factory-fabricated for indoor installation, consisting of a flat plate cross-flow heat exchanger, cooling coil, supply air fan and motor and exhaust air fan and motor. The casing must be 1 mm G90, galvanized steel, double wall construction with 25 mm insulation. Provide fibrous desiccant cross-flow type heat exchanger core capable of easy removal from the unit.~~

~~2.6.7 Plate Total Energy Exchanger~~

~~Provide enthalpy plate energy exchanger that must transfer both sensible~~

~~and latent energy between outgoing and incoming air streams in a cross or counter flow arrangement with no moving parts. The enthalpy plate exchanger media must be [coated with hydrophilic resin][or][impregnated with a polymeric desiccant]. The material must allow the exchange water by direct vapor transfer using molecular transport without the need of condensation. The frame supporting enthalpy matrix must be constructed of G90 Galvanized material with end caps constructed of [1.3][1.6] mm galvanized plates. The enthalpy plate exchanger must operate at temperatures between minus 40 degrees C and 60 degrees C.[The enthalpy plate exchanger must have a warranty of at least 5 years against manufacturing defects.][The enthalpy plate exchanger manufacturer must have at least 10 years of experience in the manufacturing of energy recovery components.]~~

~~The enthalpy plate exchanger must bear the AHRI 1060 Certified Product Seal. Sensible, latent and total effectiveness along with pressure drop, exhaust air transfer ratio (EATR) and outdoor air correction factor (OACF) rating must be clearly documented with performance tests conducted in accordance with ASHRAE 84 and per the official AHRI laboratory. The enthalpy plate exchanger must withstand pressure differentials of minimum 1.25 kPa 5 inches water gauge. Following UL 1995, the enthalpy plate exchanger must be a UL Recognized Component and bear the UL Certification Mark (tested under UL723 with success by the UL laboratory). The exchanger must have a flame spread rating not more than 25 and a smoke developed rating not more than 50 when tested in accordance with ASTM E84. The membrane must not promote the growth of mold or bacteria and must have successfully passed ASTM G21 testing. Provide heat exchanger core capable of easy removal from the unit. Enthalpy plate exchangers must be cleanable outside of cabinet.~~

~~[2.7 HUMIDIFIER~~

~~Provide humidifiers that meet the requirements of ANSI/AHRI 640.~~

~~2.7.1 Electrode Canister Type Humidifier~~

~~Provide humidifier of the self-contained steam generating electrode type utilizing a[plastic][disposable] canister with full probes connected to electric power via electrode screw connectors. Construct the electrodes from expanded low carbon steel, zinc plated and dynamically formed for precise current control. The humidifier assembly must include integral fill cup, fill and drain valves and associated piping. Design the canister to collect the mineral deposits in the water and provide clean particle free steam to the air stream. Water chemistry requirements must be provided with humidifier submittal data.~~

~~2.7.2 Ultrasonic Type Humidifier~~

~~Provide self-contained ultrasonic type humidifier operating on the principle of ultrasonic nebulization of water. Make the casing of high-quality stainless steel. The ultrasonic humidifier must not produce any unacceptable noise radiation or frequency interference with communications or other electronic equipment. Water chemistry requirements must be provided with humidifier submittal data.~~

~~2.7.3 Gas-Fired Steam Humidifiers (Stand-Alone)~~

~~Provide a stand-alone gas-fired steam humidifier that includes an enclosed cabinet of[powder coated][baked enamel] [14][_____] gauge steel~~

~~construction with an air gap between cabinet and insulated humidifier tank to ensure safe surface temperatures. Install all tank surfaces insulated with minimum 12 mm thick insulation and enclosed within unit cabinetry.~~

~~Unit must include a drain water cooler to ensure drain water tempering to below 60 degrees C. Humidifier must prevent "back-siphoning" using an internal air gap for supply water and the drain line must include a vacuum breaker to prevent siphon drainage of the tank in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.~~

~~a. Provide a unit that includes heat treated type [316][_____] Stainless Steel combustion chamber(s) and heat exchanger(s).~~

~~b. Each burner, capable of modulation at a [5:1][_____] ratio must provide steam production as indicated on the HUMIDIFIER SCHEDULE. Provide burner in accordance with NFPA 54.~~

~~c. [Control system must seamlessly interface with temperature control system as specified in [Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS][Section 23 09 23.02 BACnet DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS] without requiring gateways or any other interface devices.]Ensure that all controls equipment meets the requirements of UFC 4-010-06.~~

~~2.7.4 Electrically Heated Steam Humidifiers (Stand-Alone)~~

~~Provide a stand-alone electrically heated steam humidifier that includes an enclosed cabinet of[powder coated][baked enamel] [14][_____] gauge steel construction with an air gap between cabinet and insulated humidifier tank to ensure safe surface temperatures. Install all tank surfaces insulated with minimum 12 mm thick insulation and enclosed within unit cabinetry.~~

~~Unit must include a drain water cooler to ensure drain water tempering to below 60 degrees C. Humidifier must prevent "back-siphoning" using an internal air gap for supply water and the drain line must include a vacuum breaker to prevent siphon drainage of the tank in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE.~~

~~a. Provide a unit that includes heat treated type [316][_____] Stainless Steel combustion chamber(s) and heat exchanger(s).~~

~~b. Each humidifier must operate at the voltage and provide steam production as indicated on the HUMIDIFIER SCHEDULE.~~

~~c. [Control system must seamlessly interface with temperature control system as specified in[Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS][Section 23 09 23.02 BACnet DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS] without requiring gateways or any other interface devices.]Ensure that all controls equipment meets the requirements of UFC 4-010-06.~~

~~]~~2.6 CONTROLS

2.6.1 Unit Controls

Provide units internally prewired by manufacturer with a 24 volt electromechanical control circuit powered by an internal transformer. Provide terminal blocks for power wiring and external control wiring. Internally protect unit by ~~[~~fuses~~][~~ or ~~][~~a circuit breaker~~]~~ in accordance with UL 1995. Units with three-phase power must be equipped with phase monitoring protection to protect against problems caused by phase loss, phase imbalance and phase reversal.

a. ~~[~~Provide unit with microprocessor controls to provide all 24V control functions. ~~]~~Control unit by a~~[~~ two stage heating /cooling thermostat ~~][one stage heating/cooling thermostat] with[manual][automatic]~~ changeover.~~[Control unit by a programmable electronic thermostat with heating setback and cooling setup with 7-day programming capability.]~~

b. Provide unit with~~[low voltage electric controls][~~ factory supplied DDC control system~~]~~.

2.6.2 Unit DDC Controller

a. Unit controller must include input, output and self-contained programming as needed for complete control of unit.

b. All program sequences must be stored on board in EEPROM. Batteries cannot be used to retain logic program. Execute all program sequences by controller 10 times per second and must be capable of multiple PID loops for control of multiple devices. Programming of logic controller must be completely modifiable in the field over installed~~[~~ BACnet LANs~~][LonWorks LANs]~~.

c. Temperature Control System Interface: Points must be available from the unit controller for service access and display or control.

d. Provide device, either graphical interface or an additional laptop with communication cabling along with the manufacturer’s diagnostic software, to enable the maintenance/service technicians to log directly into the DOAS stand-alone controller for system diagnostics/maintenance.

2.6.3 Control System Interface

Controls must include a control system interface to a BACnet Control system. The control system interface must meet DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.

~~Controls must include a control system interface to a LonWorks control system. The control system interface must meet DDC Hardware requirements of Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.~~

Controls must include a control system interface to a BACnet or LonWorks control system~~, whichever is used by the control system in the building in which the unit is installed~~. For BACnet, the control system interface must meet DDC Hardware requirements of Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.~~ For LonWorks, the control system interface must meet DDC Hardware requirements of Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS.~~

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment in accordance with the manufacturer's installation instructions and manufacturer's recommendations.

Submit installation drawings for each DOAS in accordance with referenced standards in this section. Contractor and manufacturer are responsible for verifying equipment, including any additional parts such as controls devices and other equipment requiring access, fits in the space provided. The Government review does not relieve contractor from delivering the product that meets minimum capacity and mandatory efficiency requirements to meet all codes and standards.

3.2 TESTING

3.2.1 Quality Control

Test and rate components of the air-conditioning systems as a system in accordance with ANSI/AHRI 210/240.

3.2.2 SYSTEM PERFORMANCE TESTS

Before each system is accepted, conduct tests to demonstrate the general operating characteristics of all equipment by a registered professional engineer or an approved manufacturer's start-up representative experienced in system start-up and testing, at such times as directed. Provide in electronic format.~~[~~ Also provide with ~~[six][______]~~4 copies of the report provided in bound 216 by 279 mm booklets.~~]~~ The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system.

For equipment providing heating and cooling the system performance tests must be performed during the heating and cooling seasons.

a. Submit a schedule, at least ~~[~~2~~][______]~~ weeks prior to the start of related testing, for the system performance tests. The schedules must identify the proposed date, time, and location for each test. Tests must cover a period of not less than ~~[~~48~~][______]~~ hours for each system and must demonstrate that the entire system is functioning in accordance with the drawings and specifications.

b. Make corrections and adjustments, as necessary, tests must be re-conducted to demonstrate that the entire system is functioning as specified. Prior to acceptance, install and tighten service valve seal caps and blanks over gauge points. Replace any refrigerant lost during the system startup.

c. If tests do not demonstrate satisfactory system performance, correct deficiencies and retest the system. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test.

d. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit ~~[six][______]~~4 copies of the report provided in bound 216 by 279 mm booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system. Submit the report including the following information (where values are taken at least three different times at outside dry-bulb temperatures that are at least 3 degrees C apart):

(1) Date and outside weather conditions.

(2) The load on the system based on the following:

~~(a) The refrigerant used in the system.~~

~~(b) Condensing temperature and pressure.~~

~~(c) Suction temperature and pressure.~~

(a) Cold water inlet temperature and pressure.

(b) Ambient~~, condensing~~ and coolant temperatures.

(c) Running current, voltage and proper phase sequence for each phase of all motors.

(3) The actual on-site setting of operating and safety controls.

~~(4) Thermostatic expansion valve superheat - value as determined by field test.~~

~~(5) Subcooling.~~

~~(6) High and low refrigerant temperature switch set-points~~

~~(7) Low oil pressure switch set-point.~~

~~(8) Defrost system timer and thermostat set-points.~~

(4) Moisture content.

(5) Capacity control set-points.

(6) Field data and adjustments which affect unit performance and energy consumption.

(7) Field adjustments and settings which were not permanently marked as an integral part of a device.

~~3.3 REFRIGERANT TESTS, CHARGING, AND START-UP~~

~~Split-system refrigerant piping systems must be tested and charged as specified in Section 23 23 00 REFRIGERANT PIPING. Packaged refrigerant systems which are factory charged must be checked for refrigerant and oil capacity to verify proper refrigerant levels in accordance with manufacturer's recommendations. Following charging, packaged systems must be tested for leaks with a halide torch or an electronic leak detector. Provide in electronic format.[Also provide with [six][_____] copies of each test containing the information described below in bound 216 by 279 mm booklets. Individual reports must be submitted for the refrigerant system tests.]~~

~~a. The date the tests were performed.~~

~~b. A list of equipment used, with calibration certifications.~~

~~c. Initial test summaries.~~

~~d. Repairs/adjustments performed.~~

~~e. Final test results.~~

~~3.3.1 Refrigerant Leakage~~

~~If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must immediately be isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. Under no circumstances must the refrigerant be discharged into the atmosphere.~~

~~3.3.2 Contractor's Responsibility~~

~~Take steps, at all times during the installation and testing of the refrigeration system, to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time must more than 85 g of refrigerant be released to the atmosphere in any one occurrence. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.~~

3.3 CLOSEOUT ACTIVITIES

3.3.1 Operation and Maintenance Manuals, Data Package 2

Submit ~~[six][_____]~~4 manuals at least 2 weeks prior to field training. Submit data complying with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA~~ and 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW)~~. Submit Data Package 3 for the items/units listed under SD-10 Operation and Maintenance Data

3.3.2 Operation and Maintenance Training

Conduct a video recorded training course for the members of the operating staff as designated by the Contracting Officer. Make the training period consist of a total of ~~[16][_____]~~8 hours of normal working time and start it after all work specified herein is functionally completed and the Performance Tests have been approved. Conduct field instruction that covers all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations. Submit the proposed On-site Training schedule concurrently with the Operation and Maintenance Manuals and at least 14 days prior to conducting the training course. Provide a copy of the recorded training to the designated operating staff.

3.3.3 Acceptance

With the warranty, provide a cover letter/sheet clearly marked with the system name, date, and the words "Equipment Warranty" - "Forward to the Systems Engineer/Condition Monitoring Office/Predictive Testing Group for inclusion in the Maintenance Database."

3.4 PAINTING OF NEW EQUIPMENT

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

3.4.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ~~ASTM B117~~JIS Z 2371, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 3 mm on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 50 degrees C, the factory painting system must be designed for the temperature service.

3.4.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 50 degrees C must be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

a. Temperatures Less Than 50 Degrees C: Immediately after cleaning, the metal surfaces subject to temperatures less than 50 degrees C must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.0076 mm, one coat of primer applied to a minimum dry film thickness of 0.0255 mm; and two coats of enamel applied to a minimum dry film thickness of 0.0255 mm per coat.

b. Temperatures Between 50 and 205 Degrees C: Metal surfaces subject to temperatures between 50 and 205 degrees C must receive two coats of 205 degrees C heat-resisting enamel applied to a total minimum thickness of 0.05 mm.

c. Temperatures Greater Than 205 Degrees C: Metal surfaces subject to temperatures greater than 205 degrees C must receive two coats of 315 degrees C heat-resisting paint applied to a total minimum dry film thickness of 0.05 mm.

-- End of Section --

SECTION 23 81 00

DECENTRALIZED UNITARY HVAC EQUIPMENT
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 8607	(2008) Flare Type and Brazing Type Fittings for Refrigerants
JIS B 8615-1	(2013) Non-ducted Air Conditioners and Heat Pumps-Testing and Rating for Performance
JIS B 8616	(2015) Package Air Conditioners
JIS B 9908-1	(2019) Test Method of Air Filter Units for Ventilation and Electric Air Cleaners for Ventilation-Part 1: Technical Specifications, Requirements and Classification System Based Upon Particulate Matter Efficiency
JIS C 4212	~~(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)~~(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 9730-1	(2019) Automatic Electrical Controls-Part 1: General Requirements
JIS C 9730-2-9	(2010) Automatic Electrical Controls for Household and Similar Use-Part 2-9: Particular Requirements for Temperature Sensing Controls
JIS G 3302	(2022) Hot~~—~~-Dip Zinc~~—~~-Coated Steel Sheet and Strip
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS H 8641	(2021) Hot Dip Galvanized Coatings

JIS Z 2371	(2015) Methods of Salt Spray Testing

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-M	(2019) Public Building Construction Standard Specification

THE JAPAN REFRIGERATION AND AIR CONDITIONING INDUSTRY ASSOCIATION (JRAIA)

JRA Standard	Japan Refrigeration and Air Conditioning

THE JAPAN ELECTRICAL MANUFACTURERS' ASSOCIATION (JEMA)

JEM 1038	AC Electromagnetic Contactor
JEM 1167	High Voltage Electromagnetic Contactor

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval or for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Field-Assembled Refrigerant Piping

Control System Wiring Diagrams

SD-03 Product Data

Room Air Conditioners

~~Packaged Terminal Units~~

Heat Pumps, Air to Air

~~Air Conditioners~~

Training; G~~[, [_____]]~~

Posted Instructions

Spare Parts

SD-06 Test Reports

Start-Up and Initial Operational Tests

SD-07 Certificates

Service Organizations

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G~~[, [_____]]~~

Room Air Conditioners, Data Package 3

~~Packaged Terminal Units, Data Package 3~~

Heat Pumps, Air to Air, Data Package 3

~~Air Conditioners, Data Package 3~~

Filters, Data Package 2

Thermostats, Data Package 2

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.3 QUALITY ASSURANCE

1.3.1 Modification of References

Accomplish work in accordance with the referenced publications, except as modified by this section. Consider the advisory or recommended provisions to be mandatory, as though the word "shall" had been substituted for the words "should" or "could" or "may," wherever they appear. Interpret reference to "the Authority having jurisdiction," "the Administrative Authority," "the Owner," or "the Design Engineer" to mean the Contracting Officer.

1.3.2 Detail Drawing

For refrigerant piping, submit piping, including pipe sizes. Submit control system wiring diagrams.

1.3.3 Safety

Design, manufacture, and installation of unitary air conditioning equipment shall conform to Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku).

1.3.4 Sizing

Size equipment based on MLIT-M;_do not oversize.

1.3.5 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be Refrigerant Handling Technician (Reibai-Furontou-Toriatsukai-Gijutsusha) under Japan Refrigeration and Air-Conditioning Industry Association. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

1.4 REFRIGERANTS

Refrigerants must have an Ozone Depletion Potential (ODP) no greater than 0.0, with the exception of R-123. CFC-based refrigerants are prohibited.

HCFCs and Halons shall not be permitted.

1.5 ENVIRONMENTAL REQUIREMENTS

For proper Indoor Environmental Quality, maintain positive pressure within the building. Meet or exceed filter media efficiency as tested in accordance with JIS B 9908-1.

1.6 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Properly protect and care for all material both before and during installation. Submit an inventory of all the stored items. Replace any materials found to be damaged, at no additional cost to the Government. During installation, cap piping and similar openings capped to keep out dirt and other foreign matter.

PART 2 PRODUCTS

2.1 ROOM AIR CONDITIONERS

JIS B 8616. Minimum energy efficiency ratio (EER) shall be in accordance with Room Air Conditioners: Provide units removable from inside the building for servicing without removing the outside cabinet. Construct outside cabinets, including metal grilles to protect condenser coils, of zinc-coated steel or aluminum. Steel and zinc-coated surfaces shall receive at least one coat of primer and manufacturer's standard factory-applied finish Insulate cabinets to prevent condensation and run off of moisture. Provide mounting hardware made of corrosion-resistant material or protected by a corrosion-resistant finish. Provide air filters of the ~~[throw-away] [or] [~~permanent washable~~]~~ type removable without the use of tools and arranged to filter both room and ventilating air. Remove condensate by means of a drain or by evaporation and diffusion. Provide with metal or plastic mounting flanges on each side, top, and bottom of unit. For thru-the-wall installations provide aluminum or shop painted zinc-coated steel flanged telescopic wall sleeves. Design wall sleeves to restrict driving rain. For window mounted units provide shop-painted metal mounting brackets, braces, and sill plates. Mount compressors on vibration isolators. Minimum cooling capacity shall be not less than that indicated.

2.1.1 Units for Operation on 100 Volts

Provide 3-wire cords of manufacturer's standard length. If not existing, provide a receptacle within reach of the standard length cord. Cords shall have a 15- or 20-amp, 3-pole, 125-volt ground type plug to match receptacle.

2.1.2 Units for Operation on 200 Volts

Provide 3-wire cords of manufacturer's standard length. If not existing, provide a receptacle within reach of the standard length cord. Cords shall have a 15-, 20-, or 30-amp, 3-pole, 250-volt or manufacturer's standard ground type plug to match receptacle.

2.1.3 Controls

Mount controls in cabinet. Manual controls shall permit operation of either the fan or the fan and refrigerating equipment. Fan control shall provide two fan speed settings. Automatic controls shall include a thermostat for controlling air temperature. Thermostat shall have an adjustable range, including 18 to 28 degrees C and shall automatically turn the refrigeration system on or off to maintain the preselected temperature within plus or minus 20 degrees C.

~~2.2 PACKAGED TERMINAL UNITS~~

~~2.2.1 Heat Pumps~~

~~JIS B 8616, air-cooled, split type; [Heat pumps shall have a minimum energy efficiency ratio (EER) of [_____], or a minimum Coefficient of Performance (COP) of [_____], and a minimum integrated part load value (IPLV) of [_____].] [Provide supplemental electric resistance heaters integral with unit.] [Provide units suitable for use with minimal ductwork having a total external static resistance up to 25 Pa.]~~

~~2.2.2 Air Conditioners~~

~~JIS B 8616, air-cooled, split type. Provide units with [heating only] [cooling only] [combination heating and cooling] section with indicated capacity. Minimum [seasonal]energy efficiency ratio ([S]EER) shall be [[_____] [S]EER.] [Provide units suitable for use with minimal ductwork having a total external static resistance up to 25 Pa.]~~

~~2.2.3 Indoor Noise Rating~~

~~Rate in accordance with Japanese manufacture's standard ratings. Indoor rating shall not exceed [_____] bels while entire unit is operating at any fan or compressor speed.~~

~~2.2.4 Wall Sleeves and Mounts~~

~~Provide manufacturer's standard wall sleeves and mounts. Wall sleeves shall have seals designed to restrict driving rain and wind. [Provide unit subbase of the same construction and finish as the sleeve to provide for concealed electrical connection, cord storage, and equipped with unit leveling legs.] [Provide subbase with 24-volt remote control circuitry and wall mounted thermostat.]~~

~~2.2.5 Heating Section for Air Conditioners~~

~~a. Electric Coils: Electric resistance heating elements with high temperature-limit safety device, factory-mounted, and wired to chassis.~~

~~b. Hot Water Coils: Serpentine type constructed of seamless copper tubes with aluminum fins mechanically or hydraulically bonded to tubes. Provide factory-furnished tee and manual air vent on return connection. Factory test coils at twice maximum operating pressure.~~

~~c. Steam Coils: Serpentine type constructed of red brass or seamless copper tubes with JIS H 3300 mechanically or hydraulically bonded to tubes. Factory test coils at twice the maximum operating pressure.~~

~~d. Heating unit shall have non-flammable and non-combustible manufacture's standard insulations.~~

~~2.2.6 Refrigeration Sections~~

~~Completely self-contained, slide-in assembly or removable chassis with welded, hermetically sealed, air-cooled refrigeration system, outdoor fan, indoor fan, control box, and ventilation damper. Provide refrigeration sections capable of installation or removal without the use of tools. Refrigeration sections shall include refrigeration circuit tubing, wiring, and safety controls, and shall operate down to 2 degrees C outdoor temperature and 21 degrees C indoor temperature, without compressor short cycling while delivering not less than 100 percent of rated cooling capacity. Units shall have drains to the building exterior to eliminate excess driving rain. Condensate shall not drain onto building exterior or interior.~~

~~a. Compressors: Hermetic type with vibration isolation devices.~~

~~b. Coils: Constructed of seamless copper or aluminum tubing with copper or aluminum fins bonded to tubes. [Coat outdoor air coils with factory applied corrosion resistant treatment. Coils to be coated shall be part of manufacturer's standard product for capacities and ratings indicated and specified. Provide plate type fins.]~~

~~c. Outdoor Fans: Direct connected centrifugal type with aluminum or plastic wheel and forward curved blades or direct connected aluminum propeller type. Design fans so that condensate will evaporate without drip, splash, or spray on building exterior.~~

~~d. Indoor Fans: Direct connected centrifugal type with aluminum, galvanized steel, or plastic wheel and forward curved blades. Provide minimum two-speed motor with built-in overload protection.~~

~~2.2.7 Ventilation Damper Assembly~~

~~Operated by automatic actuator. Dampers shall close on unit shutdown or loss of power and shall open on heating or cooling start-up.~~

~~2.2.8 Air Filters~~

~~Removable without use of tools, and shall filter both recirculated and ventilating air.~~

~~2.2.9 Controls~~

~~Provide controls including, an adjustable thermostat, and switches, to regulate room air temperature through control of refrigerant compressors or heating elements. Controls shall at least have positions for off, high or low fan speed for [heating] [and] [cooling], and fan only operation. [Provide remote mounted night set-back thermostat.]~~

2.2 HEAT PUMPS, AIR TO AIR

Provide factory assembled units complete with accessories, wiring, piping, and controls. Provide units with ~~[outlet grilles.] [supplemental electric heaters.] [humidifiers.] [~~air filters as specified in the paragraph FILTERS.~~]~~

~~2.2.1 Energy Performance~~

~~[Energy performance shall be in accordance with JIS B 8615-1.][Heat~~

~~pumps shall have [a minimum [seasonal] energy efficiency ratio ([S]EER) of [_____],] [a minimum Heating Seasonal Performance Factor (HSPF) of [_____],] [[_____].]].~~

2.2.1 Air Coils

Extended-surface fin and tube type with seamless copper ~~or aluminum~~ tubes with ~~copper or~~ aluminum fins securely bonded to the tubes. ~~On coils with all-aluminum construction, provide tubes of aluminum alloy provide fins of aluminum alloy and provide tube sheets of aluminum alloy.~~ ~~[~~Provide a coating on ~~[~~outdoor air~~] [and] [indoor air]~~ coils as specified in the paragraph COATINGS FOR FINNED TUBE COILS. Coils to be coated shall be part of manufacturer's standard product for capacities and ratings indicated and specified. Provide plate type fins.~~]~~

~~2.2.2 Supplemental Electric Heaters~~

~~Provide electrical resistance heaters [integral with the unit] [for remote installation in ductwork]. Heaters shall have a total capacity as indicated. Provide internal fusing for heaters.~~

2.2.2 Compressors

For compressors above 70 kW, compressor speed shall not exceed 3450 rpm. For equipment over 35 kW, provide automatic capacity reduction of at least 50 percent of rated capacity. Capacity reduction may be accomplished by cylinder unloading, use of multiple, but not more than four compressors, or a combination of the two methods. Units with cylinder unloading shall start with capacity reduction devices in the unloaded position. Units with multiple compressors shall have a means to sequence starting of compressors. Provide compressors with devices to prevent short cycling when shutdown by safety controls. Provide reciprocating compressors with crankcase heaters, and vibration isolators.

2.2.3 Mounting Provisions

Provide units that permit mounting as indicated. ~~[Provide suitable lifting attachment plates to enable equipment to be lifted to normal position.]~~

2.2.4 Temperature Controls

Provide controls as specified in JIS B 8615-1 and as modified herein. Provide indoor ~~thermostats~~thermostats of the adjustable type that conform to applicable requirements of JIS C 9730-1 and JIS C 9730-2-9. Provide manual means for temperature set-back. Provide thermostats capable of controlling supplemental heat as specified in JIS B 8615-1.

2.2.5 Accessories

In addition to accessories specified in JIS B 8615-1, provide the following accessories for heat pump units.

~~a. Protective grille around outside unit coils~~

~~b~~a. Start capacitor kit

~~2.3 AIR CONDITIONERS~~

~~2.3.1 Split-System Type~~

~~Provide separate assemblies designed to be used together. Base ratings on the use of matched assemblies. Units shall have a minimum [SEER] [EER] of [_____] when tested in accordance with JIS B 8616 or JIS B 8615-1 as applicable. Provide capacity, electrical characteristics and operating conditions as indicated. Condensers shall provide not less than 10 degrees F liquid subcooling at standard ratings.~~

~~2.3.2 Single Zone Units~~

~~Provide single zone type units arranged to [draw] [or] [blow] through coil sections. [Air may be blown or drawn through heating section.]~~

~~2.3.3 Multizone Units~~

~~Provide multizone type units arranged to [blow through the cooling and heating sections] [draw through the cooling and heating sections] [blow through the individual cooling and heating coils of each zone].~~

~~2.3.4 Heaters~~

~~Provide as [an integral part of the evaporator-blower unit] [a separate unit for installation in the duct work]. Provide [steam coils] [hot water coils] [gas heaters] [oil heaters] [electric open coils] [electric strip tubular heaters] [electric fin tubular heaters].~~

~~2.3.5 Compressors~~

~~For compressors over 70 kW, compressor speed shall not exceed 3450 rpm. For systems over 35 kW provide automatic capacity reduction of at least 50 percent of rated capacity. Capacity reduction may be accomplished by cylinder unloading, use of multi- or variable speed compressors, use of multiple, but not more than four compressors, or a combination of the two methods. Units with cylinder unloading shall start with capacity reduction devices in the unloaded position. Units with multiple compressors shall have means to sequence starting of compressors. Provide compressors with devices to prevent short cycling when shut down by safety controls. Device shall delay operation of compressor motor for at least 3 minutes but not more than 6 minutes. Provide a pumpdown cycle for units 70 kW and over. Provide reciprocating compressors with crankcase heaters in accordance with the manufacturer's recommendations. If compressors are paralleled, provide not less than two independent circuits.~~

~~2.3.6 Coils~~

~~On coils with all-aluminum construction, provide tubes of aluminum alloy; provide fins of aluminum alloy and provide tube sheets of aluminum alloy. Provide a separate air cooled condenser circuit for each compressor or parallel compressor installation. [Provide a coating on [condenser] [and] [evaporator] coils and fins as specified in the paragraph COATINGS FOR FINNED TUBE COILS. Coils to be coated shall be part of manufacturer's standard product for capacities and ratings indicated and specified. Provide plate type fins.]~~

~~2.3.7 Condenser Controls~~

~~Provide start-up and head pressure controls to allow for system operation at ambient temperatures down to [_____] degrees C.~~

~~2.3.8 Fans~~

~~Provide belt-driven evaporator fans with adjustable pitch pulleys; except for units less than 17 1/2 kW capacity, direct drive with at least two speed taps may be used. Select pulleys at approximately midpoint of the adjustable range.~~

~~2.3.9 Filters~~

~~Provide filters of the type specified in this section.~~

~~2.3.10 Filter Boxes~~

~~Provide when filters are not included integral with air conditioning units. Construct of not less than No. 20 US gage steel with track, hinged access doors with latches, and gaskets between frame and filters. Arrange filters to filter outside and return air. Provide removable filter assemblies, replaceable without the use of tools.~~

~~2.3.11 Mixing Boxes~~

~~Provide of the physical size to match the basic unit and include equal sized flanged openings, sized to individually handle full air flow. Arrange openings as indicated. Provide openings with dampers of parallel or opposed blade type. Provide opposed blade type for modulating dampers and parallel type for two-position dampers. Connect damper shafts together by one continuous linkage bar. Arrange dampers for [automatic] [or] [manual] operation so that when one starts to close from its opened position, the other starts to open from its closed position.~~

~~2.3.12 Thermostats~~

~~Provide adjustable type that conforms to applicable requirements of JIS C 9730-1 and JIS C 9730-2-9. Provide combination heating-cooling type with contacts hermetically sealed against moisture, corrosion, lint, dust, and foreign material. Design to operate on not more than 0.83 degrees C differential and of suitable range calibrated in degrees C. Provide adjustable heat anticipation and fixed cooling anticipation. Provide two independent temperature sensing elements electrically connected to control the compressor and heating equipment, respectively. Accomplish manual switching for system changeover from heating to cooling or cooling to heating and fan operation through the use of a thermostat subbase. Provide system selector switches to provide "COOL" and "OFF" and "HEAT" and fan selector switches to provide "AUTOMATIC" and "ON." Provide relays, contactors, and transformers located in a panel or panels for replacement and service.~~

~~2.3.12.1 Cooling~~

a. ~~When thermostat is in "COOL" position with fan selector switch in "AUTO" position, compressor, evaporator fan, and condenser fan shall cycle together.~~

b. ~~When thermostat is in "COOL" position with fan selector switch in "ON"~~

~~position, compressor, and condenser fan shall cycle together and evaporator fan shall run continuously.~~

~~2.3.12.2 Heating~~

~~a. When thermostat is in "HEAT" position with fan selector switch in "AUTO" position, heater and supply air fan shall cycle together. Provide a separate thermostat to keep the fan running until the heater cools.~~

~~b. When thermostat is in "HEAT" position with fan selector switch in "ON" position, heater shall cycle and supply air fan shall run continuously.~~

~~2.3.12.3 Supply Air Fan~~

~~a. When fan selector switch is in "AUTO" position with thermostat in "OFF" position, fan shall not run.~~

~~b. When fan selector switch is in "ON" position, fan shall run continuously.~~

2.3 FILTERS

Provide filters to filter ~~outside air~~ and return air and locate ~~[as indicated] [~~inside air conditioners~~] [inside filter box] [inside combination air filter mixing box]~~. Provide ~~[replaceable (throw-away)] [high efficiency] [~~cleanable (reusable)~~]~~ type. Filters shall conform to JIS B 9908-1. Polyurethane filters shall not be used on units with multiframe filters.

2.3.1 Replaceable Type Filters

Throw-away frames and media, standard dust holding capacity, 1.79 m/s maximum face velocity, and~~ [~~ 25 mm~~] [50 mm]~~ thick. Filters shall be in accordance with JIS B 9908-1.

~~2.3.2 High Efficiency Filters~~

~~Filters shall have a 99.97% efficiency on 0.30 pm particle when tested in accordance with JIS B 9908-1. Filter assembly shall include; holding frame and fastener assembly, filter cartridge, mounting frame, and retainer assembly. Reinforce filter media with glass fiber mat. Pressure drop across clean filter shall not exceed [_____] Pa gage. Precede high efficiency filters with a replaceable type filter.~~

2.3.2 Cleanable Type Filters

Provide sufficient oil to coat filters six times based on one pint of oil per each 0.93 square meter of filter area. Provide washing and charging tanks for cleaning and coating filters. Filters shall be in accordance with JIS B 9908-1.

2.3.3 Manometers

Provide inclined-type manometers for filter stations of 944 L/s capacity or larger including filters furnished as integral parts of air-handling units and filters installed separately. Provide sufficient length to read at least 250 Pa with 10 major graduations, and equipped with spirit level. Equip manometers with overpressure safety traps to prevent loss of fluid, and two three-way vent valves for checking zero setting. [Mercury shall not be used as the operating fluid.]

2.4 COATINGS FOR FINNED TUBE COILS

Where stipulated in equipment specifications of this section, coat finned tube coils of the affected equipment as specified below. Apply coating at the premises of a company specializing in such work. Degrease and prepare for coating in accordance with the coating applicator's procedures for the type of metals involved. Completed coating shall show no evidence of softening, blistering, cracking, crazing, flaking, loss of adhesion, or "bridging" between the fins.

2.4.1 Phenolic Coating

Provide a resin base thermosetting phenolic coating. Apply coating by immersion dipping of the entire coil. Provide a minimum of two coats. Bake or heat dry coils following immersions. After final immersion and prior to final baking, spray entire coil with particular emphasis given to building up coating on sheared edges. Total dry film thickness shall be 0.064 to 0.076 mm.

2.4.2 Chemical Conversion Coating with Polyelastomer Finish Coat

Dip coils in a chemical conversion solution to molecularly deposit a corrosion resistant coating by electrolysis action. Cure conversion coating at a temperature of 43 to 60 degrees C for a minimum of 3 hours. Coat coil surfaces with a complex polymer primer with a dry film thickness of 0.025 mm. Cure primer coat for a minimum of 1 hour. Using dip tank method, provide three coats of a complex polyelastomer finish coat. After each of the first two finish coats, cure the coils for 1 hour. Following the third coat, spray a fog coat of an inert sealer on the coil surfaces. Total dry film thickness shall be 0.064 to 0.076 mm. Cure finish coat for a minimum of 3 hours. Coating materials shall have 300 percent flexibility, operate in temperatures of minus 46 to plus 104 degrees C, and protect against atmospheres of a pH range of 1 to 14.

2.4.3 Vinyl Coating

Apply coating using an airless fog nozzle. For each coat, make at least two passes with the nozzle. Materials to be applied are as follows:

Total dry film thickness, 0.165 mm maximum.

Vinyl Primer, 24 percent solids by volume: One coat 0.051 mm thick.

Vinyl Copolymer, 30 percent solids by volume: One coat 0.114 mm thick.

2.5 MOTORS AND STARTERS

JIS C 4212 or JEM 1038, and JEM 1167. Variable speed. Motors less than 3/4 kW shall meet JIS C 4212 requirements. Motors 3/4 kW and larger shall meet JIS C 4212 requirements. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters. Provide motors to operate at full capacity with a voltage variation of plus or minus 10 percent of the motor voltage rating. Motor size shall be sufficient for the duty to be performed and shall not exceed its full load nameplate current rating when driven equipment is operated at specified capacity under the most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, the Contractor shall make the necessary adjustments to the wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. ~~[Provide reduced voltage type motor starters.]~~ Provide ~~[general-purpose] [~~weather-resistant~~] [watertight] [explosion proof]~~ type starter enclosures.

2.6 REFRIGERANT PIPING AND ACCESSORIES

Provide accessories as specified in ~~[~~JIS B 8615-1 and~~]~~ this section. Provide suction line accumulators as recommended by equipment manufacturer's installation instructions. ~~[Provide a filter-drier in the liquid line.]~~

2.6.1 Factory Charged Tubing

Provide extra soft, deoxidized, bright annealed copper tubing conforming to JIS H 3300, factory dehydrated and furnished with a balanced charge of refrigerant recommended by manufacturer of equipment being connected. Factory insulate suction line tubing with 9.52 mm minimum thickness of closed cell, foamed plastic conforming to manufacturer's standard close cell foamed plastic material with a permeance rating not to exceed 1.0. Provide quick-connectors with caps or plugs to protect couplings. Include couplings for suction and liquid line connections of the indoor and outdoor sections.

2.6.2 Field-Assembled Refrigerant Piping

Material and dimensional requirements for field-assembled refrigerant piping, valves, fittings, and accessories shall conform to in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese product and JIS H 3300 and JIS B 8607, except as herein specified. Factory clean, dehydrate, and seal piping before delivery to the project location. Provide seamless copper tubing, hard drawn, Type K or L, conforming to JIS H 3300, except that tubing with outside diameters of 6.35 mm and 9.52 mm shall have nominal wall thickness of not less than 7.62 mm and 0.81 mm, respectively. Soft annealed copper tubing conforming to JIS H 3300 may be used where flare connections to equipment are required only in nominal sizes less than one inch outside diameter.

2.6.3 Fittings

JIS H 3401 for solder-joint fittings. Recommended by manufacturer for flared tube fittings.

2.6.4 Pipe Hangers and Supports

Pipe hangers and supports shall be in accordance with MLIT-M.

2.6.5 Pipe Sleeves

Provide sleeves where piping passes through walls, floors, roofs, and partitions. Secure sleeves in proper position and location during construction. Provide sleeves of sufficient length to pass through entire thickness of walls, floors, roofs, and partitions. Provide not less than 6.35 mm space between exterior of piping or pipe insulation and interior of sleeve. Firmly pack space with insulation and caulk at both ends of the sleeve with plastic waterproof cement which will dry to a firm but pliable mass, or provide a segmented elastomeric seal.

2.6.5.1 Sleeves in Masonry and Concrete Walls, Floors, and Roofs

Provide Schedule 40 or Standard Weight zinc-coated steel pipe sleeves. Extend sleeves in floor slabs 80 mm above finished floor.

2.6.5.2 Sleeves in Partitions and Non-Masonry Structures

Provide zinc-coated steel sheet sleeves having a nominal weight of not less than 4.39 kg per square meter, in partitions and other than masonry and concrete walls, floors, and roofs.

2.7 FINISHES

Provide steel surfaces of equipment including packaged terminal units, heat pumps, and air conditioners, that do not have a zinc coating conforming to JIS H 8641, JIS G 3302 or a duplex coating of zinc and paint, with a factory applied coating or paint system. Provide a coating or paint system on actual equipment identical to that on salt-spray test specimens with respect to materials, conditions of application, and dry-film thickness.

2.8 SOURCE QUALITY CONTROL

2.8.1 Salt-Spray Tests

Salt-spray test the factory-applied coating or paint system of equipment including packaged terminal units, heat pumps, and air conditioners in accordance with JIS Z 2371.

2.9 MANUFACTURER'S STANDARD NAMEPLATES

Major equipment including compressors, condensers, receivers, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life. Plates must be fixed in prominent locations.

PART 3 EXECUTION

3.1 EQUIPMENT INSTALLATION

Install equipment and components in a manner to ensure proper and sequential operation of equipment and equipment controls. Install equipment not covered in this section, or in manufacturer's instructions, as recommended by manufacturer's representative. Provide proper foundations for mounting of equipment, accessories, appurtenances, piping and controls including, but not limited to, supports, vibration isolators, stands, guides, anchors, clamps and brackets. Foundations for equipment shall conform to equipment manufacturer's recommendation, unless otherwise indicated. Set anchor bolts and sleeves using templates. Provide anchor bolts of adequate length, and provide with welded-on plates on the head end embedded in the concrete. Level equipment bases, using jacks or steel wedges, and neatly grout-in with a nonshrinking type of grouting mortar. Locate equipment to allow working space for servicing including shaft removal, disassembling compressor cylinders and pistons, replacing or adjusting drives, motors, or shaft seals, access to water heads and valves of shell and tube equipment, tube cleaning or replacement, access to automatic controls, refrigerant charging, lubrication, oil draining and working clearance under overhead lines. Provide electric isolation between dissimilar metals for the purpose of minimizing galvanic corrosion.

3.1.1 Packaged Terminal Air Conditioners and Heat Pumps

Wall sleeve installation shall provide a positive weathertight and airtight seal.

3.1.2 Unitary Air Conditioning System

Install as indicated, in accordance with requirements in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese product, and the manufacturer's installation and operational instructions.

3.1.3 Room Air Conditioners

Install units in accordance with manufacturer's instructions. Provide structural mountings, closures, and seals for weathertight assembly. Pitch unit as recommended by manufacturer to ensure condensate drain to drain pan without overflow.

3.2 PIPING

Brazing, bending, forming and assembly of refrigerant piping shall conform to JIS H 3300 and JIS B 8607.

3.2.1 Pipe Hangers and Supports

Design and fabrication of pipe hangers, supports, and welding attachments shall conform to MLIT-M. Installation of hanger types and supports for bare and covered pipes shall conform to MLIT-M. Unless otherwise indicated, horizontal and vertical piping attachments shall conform to MLIT-M.

3.2.2 Refrigerant Piping

Cut pipe to measurements established at the site and work into place without springing or forcing. Install piping with sufficient flexibility to provide for expansion and contraction due to temperature fluctuation. Where pipe passes through building structure pipe joints shall not be concealed, but shall be located where they may be readily inspected. Install piping to be insulated with sufficient clearance to permit application of insulation. Install piping as indicated and detailed, to avoid interference with other piping, conduit, or equipment. Except where specifically indicated otherwise, run piping plumb and straight and parallel to walls and ceilings. Trapping of lines will not be permitted except where indicated. Provide sleeves of suitable size for lines passing through building structure. Braze refrigerant piping with silver solder. Inside of tubing and fittings shall be free of flux. Clean parts to be jointed with emery cloth and keep hot until solder has penetrated full depth of fitting and extra flux has been expelled. Cool joints in air and remove flame marks and traces of flux. During brazing operation, prevent oxide film from forming on inside of tubing by slowly flowing dry nitrogen through tubing to expel air. Make provisions to automatically return oil on halocarbon systems. Installation of piping shall comply with JIS H 3300 and JIS B 8607.

3.2.3 Returning Oil From Refrigerant System

Install refrigerant lines so that gas velocity in the evaporator suction line is sufficient to move oil along with gas to the compressor. Where equipment location requires vertical risers, line shall be sized to maintain sufficient velocity to lift oil at minimum system loading and corresponding reduction of gas volume. Install a double riser when excess velocity and pressure drop would result from full system loading. Larger riser shall have a trap, of minimum volume, obtained by use of 90- and 45-degree ells. Arrange small riser with inlet close to bottom of horizontal line, and connect to top of upper horizontal line. Do not install valves in risers.

3.2.4 Refrigerant Driers, Sight Glass Indicators, and Strainers

Provide refrigerant driers, sight glass liquid indicators, and strainers in refrigerant piping in accordance with ~~[~~this section~~]~~ when not furnished by the manufacturer as part of the equipment. Install driers in liquid line with service valves and valved bypass line the same size as liquid line in which dryer is installed. Size of driers shall be determined by piping and installation of the unit on location. Install dryers of 820 mL and larger vertically with the cover for removing cartridge at the bottom. Install moisture indicators in the liquid line downstream of the drier. Indicator connections shall be the same size as the liquid line in which it is installed. These devices shall be provided as optional accessories.

3.2.5 Strainer Locations and Installation

Locate strainers close to equipment they are to protect. Provide a strainer in common refrigerant liquid supply to two or more thermal valves in parallel when each thermal valve has a built-in strainer. Install strainers with screen down and in direction of flow as indicated on strainer's body.

3.2.6 Solenoid Valve Installation

Install solenoid valves in horizontal lines with stem vertical and with flow in direction indicated on valve. If not incorporated as integral part of the valve, provide a strainer upstream of the solenoid valve. Provide service valves upstream of the solenoid valve, upstream of the strainer, and downstream of the solenoid valve. Remove the internal parts of the solenoid valve when brazing the valve.

3.3 AUXILIARY DRAIN PANS, DRAIN CONNECTIONS, AND DRAIN LINES

Provide auxiliary drain pans under units located above finished ceilings or over mechanical or electrical equipment where condensate overflow will cause damage to ceilings, piping, and equipment below. Provide separate drain lines for the unit drain and auxiliary drain pans. Trap drain pans from the bottom to ensure complete pan drainage. Provide drain lines full size of drain opening. Traps and piping to drainage disposal points shall conform to Section 22 00 00 PLUMBING, GENERAL PURPOSE.

3.4 ACCESS PANELS

Provide access panels for concealed valves, controls, dampers, and other fittings requiring inspection and maintenance.

3.5 AIR FILTERS

Allow access space for servicing filters. Install filters with suitable sealing to prevent bypassing of air. Perform and document that proper Indoor Air Quality During Construction procedures have been followed; this includes providing documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed.

3.6 IDENTIFICATION TAGS AND PLATES

Provide equipment, gages, thermometers, valves, and controllers with tags numbered and stamped for their use. Provide plates and tags of brass or suitable nonferrous material, securely mounted or attached. Provide minimum letter and numeral size of 3.18 mm high.

3.7 FIELD QUALITY CONTROL

3.7.1 Leak Testing

Upon completion of installation of air conditioning equipment, test factory- and field-installed refrigerant piping with an electronic-type leak detector. Use same type of refrigerant to be provided in the system for leak testing. When nitrogen is used to boost system pressure for testing, ensure that it is eliminated from the system before charging. Minimum refrigerant leak field test pressure shall be as specified in accordance with Japanese Refrigeration Safety Regulations (Nihon-Reitou-Hoan-Kisoku) for Japanese standard. If leaks are detected at time of installation or during warranty period, remove the entire refrigerant charge from the system, correct leaks, and retest system.

3.7.2 Evacuation, Dehydration, and Charging

After field charged refrigerant system is found to be without leaks or after leaks have been repaired on field-charged and factory-charged systems, evacuate the system using a reliable gage and a vacuum pump capable of pulling a vacuum of at least 133 Pa absolute. Evacuate system in accordance with the triple-evacuation and blotter method or in accordance with equipment manufacturer's printed instructions and recharge system.

3.7.3 Start-Up and Initial Operational Tests

Test the air conditioning systems and systems components for proper operation. Adjust safety and automatic control instruments as necessary to ensure proper operation and sequence. Conduct operational tests for not less than 8 hours.

3.7.4 Performance Tests

Upon completion of evacuation, charging, startup, final leak testing, and proper adjustment of controls, test the systems to demonstrate compliance with performance and capacity requirements. Test systems for not less than 8 hours, record readings hourly. At the end of the test period, average the readings, and the average shall be considered to be the system performance.

3.8 TRAINING

Conduct a training course for the operating staff as designated by the

Contracting Officer. The training period must consist of a total ~~[~~8~~]~~ ~~[______]~~ hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.

a. Submit a schedule, at least ~~[~~2~~]~~ ~~[______]~~ weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.

b. Submit the field posted instructions, at least ~~[~~2~~]~~ ~~[______]~~ weeks prior to construction completion, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.

c. The posted instructions must cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations. ~~[~~Submit 4~~[6] [______]~~ complete copies of an operation manual in bound 216 by 279 booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least ~~[~~4~~]~~ ~~[______]~~ weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features.~~]~~

d. Submit 4~~[6] [______]~~ complete copies of maintenance manual in bound 216 by 279 mm booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

3.9 MAINTENANCE

3.9.1 EXTRA MATERIALS

Submit spare parts data for each different item of equipment specified, after approval of detail drawings and not later than ~~[~~2~~]~~ ~~[______]~~ months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

3.9.2 Maintenance Service

Submit a certified list of qualified permanent service organizations, which includes their addresses and qualifications, for support of the equipment. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

-- End of Section --

SECTION 26 05 48

SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT
11/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16	(2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.21.1	(2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)
ASME B18.22M	(1981; R 2017) Metric Plain Washers

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2025) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A500/A500M	(2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A563	(2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts
ASTM A572/A572M	(2021; E 2021) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium

Structural Steel

ASTM A603	(2019) Standard Specification for Zinc-Coated Steel Structural Wire Rope
ASTM A992/A992M	(2022) Standard Specification for Structural Steel Shapes
ASTM B695	(2021) Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM E580/E580M	(2024a) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions
ASTM F436	(2011) Hardened Steel Washers
ASTM F959/F959M	(2017a; R 2023) Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series
ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
ASTM F3125/F3125M	(2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC ES AC156	(2012) Acceptable Criteria for Seismic Certification by Shake-Table Testing of Nonstructural Components

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

MFMA-4	(2004) Metal Framing Standards Publication

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2023; with Change 3, 2025) Structural Engineering
UFC 3-301-02	(2023; with Change 1, 2025) Design of Risk Category V Structures, National Strategic Military Assets

UL SOLUTIONS (UL)

UL 1598 (2021; Reprint Jan 2024) UL Standard for Safety Luminaires

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION (VISCMA)

VISCMA 413 (2014) Installing Seismic Restraints for Electrical Equipment

1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

~~[Design and provide seismic supports and attachments in accordance with UFC 3-301-01 and ASCE 7-16.[Design and provide seismic supports and attachments, and isolation and energy dissipation systems in accordance with UFC 3-301-02.]]~~

~~[~~Provide seismic supports and attachments as indicated and in accordance with UFC 3-301-01 and ASCE 7-16.~~[Provide seismic supports and attachments, and isolation and energy dissipation systems as indicated and in accordance with UFC 3-301-02.]]~~

Components, supports, and attachments must comply with following structural design criteria:

Risk Category: ~~[I] [~~II~~] [III] [IV] [V]~~.
Seismic Design Category: ~~[C] [D] [~~E~~] [F]~~.
Seismic Design Spectral Response Acceleration Parameter at Short Period (SDS): ~~[_____]~~per structural analysis.
Seismic Design Spectral Response Acceleration Parameter at period of 1 second (SD1): ~~[_____]~~per structural analysis.
~~[Seismic relative displacement within structure[, between height [_____] and height [_____]]: [_____ mm .]]~~
~~[Seismic relative displacement between structure [_____] and structure [_____][, at level _____]: [_____ mm .]]~~

Apply the seismic requirements described in this section and on the drawings to the electrical components listed in paragraph ELECTRICAL COMPONENTS.

Electrical components and their supports must be attached or anchored to structure.

1.2.2 Electrical Components

Provide seismic supports and attachments for the following electrical components in accordance with the requirements of this specification:

~~[~~ Components with Importance Factor (Ip) = 1.0:
Battery Systems
Control Panels
Generators
Light Fixtures
Motors
Motor Control Centers
Photovoltaic Panels

Switchgear
Switchboards
Solar Heating Units
Storage Racks
Transformers
Unit Substations
Uninterruptible Power Supply Systems
Distribution Systems ~~[~~Conduit, Cable Tray, Busway, and Raceway~~]~~
Utility and Service Lines
~~[Premanufacturered modular electrical systems: [_____]]~~
~~[Other electrical components: [_____]]~~
~~[_____]]~~

~~[~~ Components with Importance Factor (Ip) = 1.5 ~~[Designated Seismic Systems]~~
Emergency and standby power systems
Fire detection and suppression systems
Emergency exit lighting
~~Utility and service lines in Risk Category IV: [_____]~~
~~[________]]~~

~~[Non-Mission Critical (NMC) Components in Risk Category V Insert edited list here similar to one above for Ip = 1.0]~~

~~[Mission Critical Level 1 (MC-1) Components in Risk Category V Insert edited list here similar to one above for Ip = 1.0]~~

~~[Mission Critical Level 2 (MC-2) Components in Risk Category V Insert edited list here similar to one above for Ip = 1.0]~~

~~[Electrical components classified as nonbuilding structure: [_____]]~~

1.2.3 ~~[Contractor Designed]~~ Supports and Attachments

Provide seismic supports and attachments~~[, and isolation and energy dissipation systems]~~ for electrical components.

~~[~~ Contractor must retain services of a Registered Professional Engineer to design supports and attachments~~[, and isolation and energy dissipation systems]~~ for electrical components.~~[~~ Supports and attachments, and isolation and energy dissipation systems that induce torsion in structural members are not permitted.~~]~~

Submit copies of the Design Calculations and Design Drawings. Calculations and Drawings must be stamped and signed by Contractor's Registered Professional Engineer.~~]~~

~~[~~ Submit Independent Design Review Report, prepared by Contractor's independent Professional Engineer for Risk Category V electrical components, supports and attachments, and isolation and energy dissipation systems in accordance with UFC 3-301-02.~~]~~

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. ~~Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.~~ Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section

01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Equipment Requirements; G, ~~[____]~~RMS

Lighting Fixtures; G, ~~[____]~~AE

SD-03 Product Data

~~[~~ Supports and Attachments; G, ~~[____]~~AE

~~]~~ Equipment Requirements; G, ~~[____]~~RMS

Lighting Fixtures; G, ~~[____]~~RMS

Flexible Fittings; G, ~~[____]~~AE

Anchors; G

~~[~~ ~~Isolation And Energy Dissipation Systems; G~~

~~]~~ SD-05 Design Data

~~[~~ Design Calculations; G, ~~[____]~~DO

~~]~~ Design Drawings; G, ~~[____]~~DO

~~[~~ Independent Design Review Report; G

~~]~~ SD-06 Test Reports

Anchors; G, ~~[____]~~AE

SD-07 Certificates

ICC ES AC156 Shake Table Test; G, ~~[____]~~AE

PART 2 PRODUCTS

2.1 EQUIPMENT REQUIREMENTS

Equipment must be rugged enough to survive design seismic event.~~[~~ Equipment components designated as Designated Seismic Systems or MC-1 (Mission Critical Level 1) must remain operational as required in paragraph SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT.~~]~~

Submit detail drawings of supports and attachments~~[, and isolation and energy dissipation systems]~~ along with~~[~~ calculations,~~]~~ catalog cuts, templates, and erection and installation details, as appropriate, for the items listed in paragraph ELECTRICAL COMPONENTS. Indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction.

~~[~~Submit calculations and drawings that are stamped and signed by Contractor's registered professional engineer. Design must be based on actual equipment and system layout. Design must include calculated loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods. ~~]~~ Include drawing for~~[~~ Mission Critical Equipment and~~]~~ Designated Seismic System Equipment indicating the equipment location in the facility to be used for the installation. Equipment must be rigidly or flexibly mounted as indicated. Roof mounted equipment both vibration isolated and nonisolated, must have support members designed and anchored to building structure.

2.1.1 Rigidly (Base and Suspended) Mounted Equipment

The following specific items of equipment must be constructed and assembled to withstand the seismic forces specified in UFC 3-301-01 and ~~[~~ UFC 3-301-02~~]~~. Entirely locate each item of rigid electrical equipment and rigidly attach on one side only of a building expansion joint. Provide items such as electrical conduit, which cross the building expansion joint with flexible joints that are capable of accommodating displacements equal to the full width of the joint in each orthogonal direction.

Engine-Generators
Substations
Transformers
Switch Boards and Switch Gear
Motor Control Centers
Free Standing Electric Motors
~~[______]~~Battery Racks and Day Tanks Serving Emergency Power Systems

Equipment furnished under this Contract must be ~~[rigidly mounted] [~~rigidly mounted using cast-in-place anchor bolts or post-installed anchors that are qualified for use in cracked concrete. Cast-in-place anchor bolts must conform to ASTM F1554~~]~~. For any rigid equipment which is rigidly anchored, provide flexible joints for electrical conduit, cabletray, busway, and raceway, that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. ~~[~~Mission critical base mounted and suspended equipment for Risk Category (RC) V, and ~~]~~Designated Seismic Systems (DSS) for RC IV buildings assigned to Seismic Design Category (SDC) C, D, E, or F and Risk Category IV components needed for continued operation after an earthquake must have two nuts provided on each bolt.

2.1.2 Nonrigid or Flexibly-Mounted Equipment

The following specific items of equipment must be constructed and assembled to resist seismic loads and meet associated seismic criteria.

Nonrigid or Flexibly-Mounted:
~~[______]~~Ceiling-suspended luminaires and lighting supports.
~~[______]~~Electrical equipment including transformers, panelboards, and switchboards.

~~2.1.3 Premanufactured Modular Electrical Systems~~

~~The following specific items of equipment must be manufactured, constructed, and assembled to resist seismic loads and meet associated seismic criteria.~~

~~Premanufactured modular electrical systems:~~
~~[______]~~
~~[______]~~

2.2 LIGHTING FIXTURES

Provide lighting fixtures and supports and attachments conforming to UL 1598.

2.3 SUPPORTS AND ATTACHMENTS

Material used for members listed ~~[in this section] [and] [~~on the drawings~~]~~, must conform with the following:

a. Plates, rods, and rolled shapes, ASTM A36/A36M, ASTM A572/A572M Grade 50, or ASTM A992/A992M.

b. Wire rope, ASTM A603 pre-stretched, with ~~[~~Class B weight zinc-coated ~~][Class C weight zinc-coated]~~ wires throughout rope. Connect rope at ends using ferrule or saddle-type wire rope clamp systems. Ferrule clamps must be qualified by testing for use in seismic applications per VISCMA 413. Saddle-type clamps must be used with minimum of two clamps at each end of wire rope.

c. Tubes, ASTM A500/A500M, Grade B.

d. Pipes, ASTM A53/A53M, Grade B.

e. Angles, ASTM A36/A36M.

f. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at discrete points conforming to MFMA-4.

g. Fasteners:

(1) High-strength bolts, ASTM F3125/F3125M, Grade A325, heavy hex.

(2) Nuts, ASTM A563. Use heavy hex nuts for high strength bolts and anchor bolts.

(3) Washers, ASME B18.22M and ASME B18.21.1, except use ASTM F436 washers for high strength bolts.

(4) Galvanized coating, ASTM A153/A153M.

(5) Direct Tension Indicator Washers, ASTM F959/F959M.~~[~~ Provide ASTM B695, Class 55, Type 1 galvanizing.~~]~~ Submit product data for direct tension indicator washers.

~~[~~ (6) Standard bolts, ASTM A307.

~~] h. Anchor bolts:~~

~~(1) Cast in anchors: Refer to Section 05 12 00 STRUCTURAL STEEL [05 50 13 MISCELLANEOUS METAL FABRICATIONS] for requirements.~~

~~(2) Post-installed anchors: Refer to Section 05 05 20 POST-INSTALLED CONCRETE AND MASONRY ANCHORS for requirements.~~

~~i. Welding: Refer to Section 05 12 00 STRUCTURAL STEEL[Section 05 50 14 STRUCTURAL METAL FABRICATIONS] for requirements.~~

2.4 FLEXIBLE FITTINGS

Provide flexible fittings to allow conduit, cabletray, busway, and raceway systems to accommodate seismic movement between equipment and conduit, cabletray, busway, and raceway systems and supporting structure. Use specification grade steel fittings conforming to ~~[~~Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM~~][MIL-STD-810]~~. Provide suitable ~~[~~watertight~~]~~ expansion fittings that maintain electrical continuity by bonding jumper or other means.~~[~~ Flexible fittings must accommodate minimum of 100 mm~~[[_____] mm]~~ seismic movement in each orthogonal directions~~]~~.

PART 3 EXECUTION

3.1 SUPPORTS AND ATTACHMENTS

Provide supports and attachments with continuous load path to distribute electrical component seismic loads to structure.

Provide supports and attachments for electrical components that conform~~[to the arrangements shown][~~ to UFC 3-301-01 requirements~~][~~ to UFC 3-301-02 requirements~~]~~. Install vertical diagonal braces at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 413.

Provide bolted and welded connections for supports and attachments in accordance with UFC 3-301-01~~[~~ and UFC 3-301-02~~]~~.

Provide welding in accordance with AWS D1.1/D1.1M.~~[~~ Grind visible welds smooth in the finished installation.~~]~~

3.2 BUILDING DRIFT

Do not attach electrical components to two dissimilar structural elements of a building that may respond differentially during an earthquake unless a flexible joint is provided. Electrical components, supports, and attachments must be capable of accommodating building story drifts, deflections, and relative displacements.

3.3 CONDUIT, CABLETRAY, BUSWAY, AND RACEWAY

Provide supports and attachments for conduit, cabletray, busway, and raceway conforming to the requirements of UFC 3-301-01~~[~~ and UFC 3-301-02~~]~~.

3.4 LIGHTING FIXTURES

Provide lighting fixtures and supports conforming to the following:

3.4.1 Pendant Fixtures

Provide pendant fixtures conforming to the requirements of UFC 3-301-01~~[~~ and UFC 3-301-02~~]~~.

3.4.2 Ceiling Attached Fixtures

3.4.2.1 Recessed Fixtures

Support recessed individual or continuous-row mounted fixtures by a seismic-resistant suspended ceiling support system built in accordance with ASTM E580/E580M. Provide supports and attachments for the fixtures conforming to the requirements of UFC 3-301-01~~[~~ and UFC 3-301-02~~]~~. Recessed lighting fixtures not over 25 kg in weight and not required to be supported separately from the structure, may be supported by and attached directly to the ceiling system runners using screws or bolts, number and size as required by the seismic design. Provide lock or screw attachments for fixture accessories, including diffusers and lenses.

3.4.2.2 Surface-Mounted Fixtures

Attach surface-mounted individual or continuous-row fixtures to a seismic-resistant ceiling support system built in accordance with ~~[~~ ASTM E580/E580M~~][Section 09 51 00 ACOUSTICAL CEILINGS]~~. Provide supports and attachments for the fixtures conforming to the requirements of UFC 3-301-01~~[~~ and UFC 3-301-02~~]~~.

3.4.3 Assembly Mounted on Outlet Box

Design a supporting assembly, that is intended to be mounted on an outlet box, to accommodate mounting features on ~~[100] [75]~~ mm boxes, plaster rings, and fixture studs.

3.4.4 Wall-Mounted Emergency Light Unit

Design and secure attachments for wall-mounted emergency light units for the worst expected seismic disturbance at the site.

3.5 ANCHORS

3.5.1 General

Submit copies of test results to verify the adequacy of the specific anchor and application, as specified. Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for restraint anchor bolts. Identify position of reinforcing steel and other embedded items prior to drilling holes for post-installed anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.~~[~~ Fill annular gap with nonshrink grout at equipment anchor bolts where clearance between anchor and equipment support hole exceeds 3.175 mm.~~]~~

3.5.2 Cast-In-Place Anchors

Use templates to locate cast-in-place bolts accurately and securely in formwork. Provide anchor bolts with an embedded straight length with heavy hex nut and ~~[plate]~~ washer as~~[~~ shown on drawings~~][~~ to provide required strength and ductility~~]~~. Anchor bolts that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or increase depth of the piers or pads to accommodate bolt lengths.

~~3.5.3 Post-Installed Anchors~~

~~Refer to Section 05 05 20 POST-INSTALLED CONCRETE AND MASONRY ANCHORS for requirements.~~

3.6 EQUIPMENT SUPPORT REQUIREMENTS

3.6.1 Suspended Equipment

Provide supports and attachments for components supported from structural systems. Provide supports and attachments that consist of angles, rods, wire rope, bars, channels (struts) or pipes arranged as shown in bracing submittals and secured at both ends with not less than 13 mm bolts. Provide sufficient supports and attachments to resist seismic forces as specified in UFC 3-301-01[and UFC 3-301-02] without exceeding capacity of structural components.[Provide, for approval, specific force calculations in accordance with UFC 3-301-01[and UFC 3-301-02] and for the project.]

Submit details of supports and attachments for acceptance. In lieu of bracing with vertical supports, these items may be supported with hangers inclined at 45 degrees directed up and radially away from equipment and oriented symmetrically in 90-degree intervals on the horizontal plane, bisecting the angles of each corner of the equipment, provided that supporting members are properly sized to support operating weight when hangers are inclined. Where interference is present, the inclined hanger slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 413.

3.6.2 Floor or Pad Mounted Equipment

3.6.2.1 Shear Resistance

Bolt components to floor or pads. Provide bolts to resist seismic forces in accordance with paragraph ANCHORS.

3.6.2.2 Overturning Resistance

Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchor bolts for combined shear and tension. Provide bolts to resist seismic forces in accordance with paragraph ANCHORS.

3.7 SPECIAL TESTING FOR SEISMIC-RESISTING COMPONENTS

Electrical components designated as Designated Seismic Systems [or MC-1 (Mission Critical Level 1)] required to remain operational after an earthquake must be seismic qualified by shake table testing conforming to ICC ES AC156 Shake Table Test procedures. The manufacturer is to provide a certification by a fully qualified testing agency for the specific components. Prequalified certifications are acceptable unless noted otherwise.[Seismic qualification documentation for each component must contain the information required in UFC 3-301-02, Section 2-17.2.5 Component Qualification Documentation.]

Components that are required to be certified must bear permanent marking or nameplates constructed of a durable heat and water resistant material.

Provide component identification nameplates in accordance with UFC 3-301-01 [and UFC 3-301-02].

Mechanically attach nameplates to electrical components.

3.8 SPECIAL INSPECTION FOR COMPONENTS, SUPPORTS, AND ATTACHMENTS

Perform special inspections for seismic-resisting systems, designated seismic systems, components, supports and attachments, and equipment per~~[~~ Section 01 45 35 SPECIAL INSPECTION attachment Schedule of Special Inspections and~~]~~ ICC IBC 1705.13.4; electrical components per ICC IBC 1705.13.6~~[~~, and seismic isolators and energy dissipation systems per UFC 3-301-02~~]~~.

~~[~~ Special Inspector must inspect and test items required in Schedule of Special Inspections and Statement of Special Inspection.

~~][~~Special Inspector must examine Designated Seismic Systems requiring seismic qualification in accordance with ASCE 7-16 and verify the label, anchorage and mounting conform to the certificate of compliance.

~~][~~Contractor must employ an independent Professional Engineer to perform "walk down" inspection of MC-1 electrical components and anchorage in accordance with UFC 3-301-02.

~~]~~ Provide a~~[~~ Statement of Special Inspections and~~]~~ Final Special Inspection Report in accordance with UFC 3-301-01 and ~~[~~UFC 3-301-02~~]~~.

-- End of Section --

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING
11/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. ~~Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.~~ Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Acceptance Tests and Inspections; G, ~~[____]~~RMS

SD-07 Certificates

Qualifications of Organization, and Lead Engineering Technician; G, ~~[____]~~RMS

Acceptance Test and Inspections Procedure; G, ~~[____]~~RMS

1.4 QUALITY ASSURANCE

1.4.1 Qualifications

Engage the services of a qualified testing organization to provide inspection, testing, calibration, and adjustment of the electrical distribution system and generation equipment listed in paragraph ACCEPTANCE TESTS AND INSPECTIONS herein. Organization must be independent of the supplier, manufacturer, and installer of the equipment. The organization must be a first tier subcontractor. No work required by this section of the specification may be performed by a second tier subcontractor.

a. Submit name and qualifications of organization. Organization must have been regularly engaged in the testing of electrical materials, devices, installations, and systems for a minimum of 5 years. The organization must have a calibration program, and test instruments used must be calibrated in accordance with NETA ATS.

b. Submit name and qualifications of the lead engineering technician performing the required testing services. Include a list of three comparable jobs performed by the technician with specific names and telephone numbers for reference. Testing, inspection, calibration, and adjustments must be performed by an engineering technician, certified by NETA (Level III) or the National Institute for Certification in Engineering Technologies (NICET) with a minimum of 5 years' experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices.

1.4.2 Acceptance Tests and Inspections Reports

Submit certified copies of inspection reports and test reports. Include certification of compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate. Type and neatly bind test reports to form a part of the final record. Submit test reports documenting the results of each test not more than 10 days after test is completed.

1.4.3 Acceptance Test and Inspections Procedure

Submit test procedure reports for each item of equipment to be field tested at least 45 days prior to planned testing date. Do not perform testing until after test procedure has been approved.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 ACCEPTANCE TESTS AND INSPECTIONS

Testing organization will perform acceptance tests and inspections. Test methods, procedures, and test values must be performed and evaluated in accordance with NETA ATS, the manufacturer's recommendations, and paragraph FIELD QUALITY CONTROL of each applicable specification section. Tests identified as optional in NETA ATS are not required unless otherwise specified. Place equipment in service only after completion of required tests and evaluation of the test results have been completed. Supply to the testing organization complete sets of shop drawings, settings of adjustable devices, and other information necessary for an accurate test and inspection of the system prior to the performance of any final testing. Notify Contracting Officer at least 14 days in advance of when tests will be conducted by the testing organization. Perform acceptance tests and inspections on applicable equipment and systems specified in the following sections:

a. Section 26 32 15 ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES. Functional engine shutdown tests, vibration base-line test, and load bank test will not be performed by the testing organization, but by the start-up engineer.

~~][b. Section 26 12 19 PAD-MOUNTED, LIQUID-FILLED, MEDIUM-VOLTAGE TRANSFORMERS~~

~~][c. Section 26 12 21 SINGLE-PHASE PAD-MOUNTED TRANSFORMERS~~

~~][d. Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION~~

~~][~~b. Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Medium voltage cables and grounding systems only.

~~][c. Section 26 13 00 SF6/HIGH-FIREPOINT FLUID INSULATED PAD-MOUNTED SWITCHGEAR~~

~~][d. Section 26 11 16 SECONDARY UNIT SUBSTATIONS~~

~~][e. Section 26 11 13.00 20 PRIMARY UNIT SUBSTATION~~

~~][~~c. Section 26 36 23 AUTOMATIC TRANSFER SWITCHES AND BY-PASS/ISOLATION SWITCH

~~][d. Section 26 23 00 LOW VOLTAGE SWITCHGEAR~~

~~][~~d. Section 26 24 13 SWITCHBOARDS

~~][e. Section 26 35 43 400 HERTZ ((HZ) SOLID STATE FREQUENCY CONVERTER. The NETA ATS representative must coordinate with the Contractor and the Converter Manufacturers' representative to witness, document, and validate the Converter Field Quality Control, Inspection and Testing. These tests will be performed by the converter manufacturers' representative, however include the documentation in the overall NETA report as well.~~

~~][f. Section 26 35 44 270 VDC SOLID STATE CONVERTER. The NETA ATS representative must coordinate with the Contractor and the Converter Manufacturers' representative to witness, document, and validate the Converter Field Quality Control, Inspection and Testing. These tests will be performed by the converter manufacturers' representative, however include the documentation in the overall NETA report as well.~~

~~]~~3.2 SYSTEM ACCEPTANCE

Final acceptance of the system is contingent upon satisfactory completion of acceptance tests and inspections.

3.3 PLACING EQUIPMENT IN SERVICE

A representative of the approved testing organization must be present when equipment tested by the organization is initially energized and placed in service.

-- End of Section --

SECTION 26 11 16.00 33

THREE-PHASE, CUBICLE-TYPE PAD-MOUNTED TRANSFORMER
08/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS B 1178	(2015) Foundation Bolts (Amendment 1)
JIS C 0920	(2003) Degrees Of Protection Provided By Enclosures (IP Code)
JIS C 1102-1	(2007)Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 1: Definitions And General Requirements Common To All Parts
JIS C 1102-2	(1997) Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 2: Special Requirements For Ammeters And Voltmeters
JIS C 1102-3	(1997) Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 3: Special Requirements For Wattmeters And Varmeters
JIS C 1102-4	(1997) Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 4: Special Requirements For Frequency Meters
JIS C 1102-5	(1997) Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 5: Special Requirements For Phase Meters, Power Factor Meters And Synchroscopes
JIS C 1102-6	(1997) Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 6: Special Requirements For Ohmmeters (Impedance Meters) And Conductance Meters
JIS C 1102-7	(1997) Direct Acting Indicating Analogue Electrical Measuring Instruments And Their Accessories Part 7: Special Requirements For Multi-Function Instruments
JIS C 1102-8	(1997) Direct Acting Indicating Analogue

Electrical Measuring Instruments And Their Accessories Part 8: Special Requirements For Accessories

JIS C 1103	(1984) Dimensions of Electrical Indicating Instruments for Switchboards
JIS C 1211-1	(2009) Alternating-Current Watt-Hour Meters (For Direct Connection) -- Part 1: General Measuring Instrument
JIS C 1211-2	(2017) Alternating-Current Watt-Hour Meters (For Direct Connection) -- Part 1: General Measuring Instrument
JIS C 1216-1	(2009) Alternating-Current Watt-Hour Meters (For Connection Through Instrument Transformer) -- Part 1: General Measuring Instrument
JIS C 1263-1	(2009) Var-Hour Meters -- Part 1: General Measuring Instrument
JIS C 1281	(1979) Weather-Proof Performance Of Electricity Meters
JIS C 1283-1	(2009) Watt-Hour, Var-Hour And Maximum Demand Indicators For Telemetering -- Part 1: General Measuring Instrument
JIS C 3102	(1984) Annealed Copper Wires for Electrical Purposes
JIS C 3611	(2020) Insulated Wires for Cubicle Type Unit Substation for 6.6 KV Receiving
JIS C 3814	(1999) Indoor Post Insulators
JIS C 3851	(2012) Indoor Post Insulator Of Organic Material
JIS C 4304	(2013) 6 KV Oil-Immersed Distribution Transformers
JIS C 4603	(2019) High Voltage AC circuit breakers
JIS C 4604	(2017) High Voltage Current-Limiting Fuses
JIS C 4605	(2020) AC Load Break Switches for rated voltage above 1kV up to and including 52 kV
JIS C 4606	(R2011) Indoor Use Disconnectors for 3.3 kV or 6.6 kV
JIS C 4607	(1999) Ac Load Break Switches With Tripping Device for 3.3 kV or 6.6 kV
JIS C 4608	(2015)Surge arresters for 6.6 kV Cubicle Type Unit Substation

JIS C 4611	(1999) High-voltage alternating current switch-fuse combinations
JIS C 4620	(2018) Cubicle Type High Voltage Power Receiving Units
JIS C 8105-1	(2021) Luminaires - Part 1: General Requirements For Safety
JIS C 8105-2-2	(2017) Luminaires - Part 2. Luminaires Part 2: Particular requirements for safety - Section 2: Recessed luminaires
JIS C 8106	(2015) Luminaires With Led Light Source Fluorescent Lamp For Commercial, Industrial And Public Lighting
JIS C 8201-2-1	(2021) Low-Voltage Switchgear and Control Gear - Part 2-1: Circuit-Breakers
JIS C 8201-2-2	(2021) Low-Voltage Switchgear And Control Gear - Part 2-2: Circuit-Breakers Incorporating Residual Current Protection
JIS C 8201-7-1	(2016) Low-Voltage Switchgear And Control Gear -- Part 7-1: Ancillary Equipment -- Terminal Blocks For Copper Conductors
JIS C 8303	(2007; R 2022) Plugs And Receptacles For Domestic And Similar General Use
JIS C 8364	(2008; R 2018) Busways
JIS C 8480	(R2020) Box-Type Switchgear Assemblies for Low-Voltage Distribution Purpose
JIS G 3553	(2011) Crimped Wire Cloth (Amendment 1)
JIS G 3555	(2015) Woven Wire Cloth, 4th Edition, Incl Amendments
JIS G 3555 PW-S	(2015) The most basic way of weaving, where vertical and horizontal lines are kept at regular intervals and intersect each other one by one. (Translated Japanese)
JIS H 3140	(2018) Copper Bus Bars
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS Z 8721	(1993) Color Specification - Specification According To Their Three Attributes

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 1226	(2003) Soft Stranded Wire

JAPANESE ELECTROTECHNICAL COMMITTEE (JEC)

JEC 2300	(1998) Vacuum Circuit Breaker
JEC 2310	(2003) Disconnector and Earthing Switch

THE JAPAN ELECTRICAL MANUFACTURERS' ASSOCIATION (JEMA)

JEM 1425	(2011) Metal-enclosed Switch Gear and Control Gear
JEM 1459	(2013) Structure and dimensions of switchboard and control panel

JAPAN POWER CABLE ACCESSORIES ASSOCIATION (JCAA)

JCAA C 3102	(2016) 6600V Cross-linked Polyethylene Insulated Power Cable Rubber Stress Cone Type Cubicle Termination Connection

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT DSKKS	Denki Setsubi Kouji Kanri Shishin (DSKKS) Electrical Construction Supervision Guidelines
MLIT ESS	(2019) MLIT Electrical Standard Specification (ESS)

~~1.2 RELATED EQUIPMENT~~

~~Section 26 00 00.00 20, BASIC ELECTRICAL MATERIAL AND METHOD, applies to this section, with the additions and modifications specified herein.~~

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Cubicle Type Unit Substation Including Concrete Foundation

Distribution panel

~~Cubicle type unit switching station including concrete foundation~~

Switchgear including concrete foundation

SD-03 Product Data

Transformers; G~~[[, [ ]]~~

~~Primary Cutout (PC)G[, [ ]]~~

Disconnecting Switches (DS); G~~[, [ ]]~~

Load Disconnecting Switches (LDS); G~~[, [ ]]~~

Vacuum Circuit Breaker (VCB); G~~, [ ]]~~

Load Break Switches (LBS); G~~[, [ ]]~~

Load Break Switches (LBS) with tripping device; G~~[, [ ]]~~

Load Break Switches (LBS) with fuse; G~~[, [ ]]~~

Circuit breaker; G~~[, [ ]]~~

Circuit Breaker with GFCI; G~~[, [ ]]~~

Automatic Transfer Switch (ATS); G~~[, [ ]]~~

Power Fuses (PF); G FIO~~[, [ ]]~~

Instruments; G~~[, [ ]]~~

Instrument Control Switches; G~~[, [ ]]~~

Buzzer

Test Terminal

Lightning Arrester; G~~[, [ ]]~~

SD-06 Test Reports

Acceptance Checks and Tests

SD-10 Operation and Maintenance Data

Data Transformer(s), Data Package 5; G~~[, [ ]]~~

1.3 QUALITY ASSURANCE

1.3.1 PCB Free Certificate

Submit results of PCB analysis in transformer oil to certify that the transformers installed under this contract are PCB free transformer as specified in Paragraph Also, analysis results shall be required maker name, serial number and other identification data of the transformers which are taken and analyzed oil samples. Analysis results appearing PCB rate shall be submitted to and approved by the Contracting Officer prior to install the transformers. The certificate shall include "PCB analysis report" prepared by the Contractor and "PCB-free certificate" prepared by the transformer manufacturer.

1.3.2 Lead-Containing Paint Material

Use of Lead-containing paint shall not be permitted. Submit the Certification of Lead Free for each enclosure, and field-applied paint in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS.

1.3.3 Cubicle Type Transformer Drawings

Include the following as a minimum:

a. An outline drawing, including front, top, and side views.

b. Nameplate data.

c. Elementary diagrams and wiring diagrams~~[~~ with terminals identified of watthour meter and current transformers~~]~~.

d. One-line diagram, including switch(es)~~[~~, current transformers, meters, and fuses~~]~~.

1.3.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, except when more stringent requirements are specified or indicated, as though the word "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with applicable codes and standards unless more stringent requirements are specified or indicated.

1.3.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship, and:

a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.3.6 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.4 MAINTENANCE

1.4.1 Additions to Operation and Maintenance Data

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein. In addition to requirements of Data Package 5, include the following on the actual transformer(s) provided:

a. An instruction manual with pertinent items and information highlighted

b. An outline drawing, front, top, and side views

c. Prices for spare parts and supply list

d. Routine and field acceptance test reports

e. Fuse curves for primary fuses

~~[~~ f. Information on watthour demand meter, CT's, and fuse block

~~]~~ g. Actual nameplate diagram

h. Date of purchase

PART 2 PRODUCTS

2.1 GENERAL

MLIT ESS. Substation shall be open-type switchgear for secondary distribution with transformation. The substation assembly shall consist of ~~[~~one~~][ ]~~ incoming section, ~~[one][ ]~~2 transformer section and ~~[one][ ]~~4 distribution section. Dimension and feature of the substation shall be as indicated.

2.2 MATERIALS AND EQUIPMENT

All materials, equipment, and devices shall, as a minimum, meet the requirements of JIS where JIS Standards are established for those items, and the requirements of MLIT ESS. All items shall be new unless specified or indicated otherwise.

2.3 ~~[MODIFICATION OF]~~CUBICLE TYPE UNIT SUBSTATION

JIS C 4620 and MLIT ESS. ~~[~~ Substation shall be metal enclosed station type cubicle switchgear for secondary distribution with transformation. The substation assembly shall consist of ~~[~~one~~][]~~ incoming section, ~~[one][]~~2 transformer section and ~~[one][]~~4 distribution section. External doors shall be suitable for handle key. Enclosure of the substation, inside enclosure and oil transformers shall be coated by salt-air proofing paint, heavy-duty type. Dimension and feature of the substation shall be as indicated.~~]~~

2.4 TRANSFORMERS SECTION

2.4.1 Unit Frame

Unit frame shall conform to JEM 1459.

2.4.2 Transformer

Oil immersed transformer shall conform to JIS C 4304. Transformer shall be high-efficiency type.

~~[Molded transformer shall conform to JIS C 4306. Transformer shall be high-efficiency type.]~~

~~[Extra-high voltage type transformer shall conform to JEC 2200.]~~

Voltage tap shall be changed by outside setting.

2.4.2.1 Transformer (Insulation) Oil

The use of PCB containing oil shall not be permitted. Before installation of transformer, the new transformer oil shall be tested in accordance with the method described in the latest Law of the Japanese Government, and submit PCB free certificate with its testing method to the Contracting Officer. Materials containing PCBs (0.5 ppm and above) shall not be permitted to use.

~~2.5 [MODIFICATION OF]CUBICLE TYPE UNIT SWITCHING STATION~~

~~JIS C 4620 and MLIT ESS. [Switching station shall be metal enclosed station-type cubicle switchgear for feeder distribution. The switching station assembly shall consist of [one][ ] incoming section, and [one][ ] distribution section. External doors shall be suitable for handle key. Enclosure of the switching station and inside enclosure shall be coated by salt-air proofing paint, heavy-duty type. Dimension and feature of the switching station shall be as indicated.]~~

2.5 ~~[MODIFICATION OF]~~SWITCHGEAR

~~[~~JIS C 4620,~~]~~ ~~[~~JEM 1425~~]~~ and MLIT ESS. ~~[~~Switchgear shall be metal enclosed ~~[metal-clad]~~ ~~[~~cubicle~~]~~ type switchgear for feeder distribution.~~] The switchgear assembly shall consist of [one] [_____] incoming section, and [one] [_____] distribution section.~~ External doors shall be suitable for handle key. Enclosure of the switchgear and inside enclosure shall be coated by salt-air proofing paint, heavy-duty type. Dimension and feature of the switchgear shall be as indicated.

~~[~~2.5.1 Interrupter Switch for Extra-High-voltage

~~2.5.1.1 Primary Cutout (PC)~~

~~As recommended by the primary cutout manufacturer.~~

2.5.1.1 Disconnecting Switches (DS)

Shall conform to JEC 2310.

2.5.1.2 Vacuum Circuit Breaker (VCB)

Shall conform to JEC 2300.

~~][~~2.5.2 Interrupter Switch for High-Voltage

2.5.2.1 Primary Cutout (PC)

Shall conform to JIS C 4620, Appendix C.

2.5.2.2 Load Disconnecting Switches (LDS)

Shall conform to JIS C 4606.

2.5.2.3 Load Break Switches (Lbs)

Shall conform to JIS C 4605.

LDS means fuse less type of Load Break Switches (LBS).

2.5.2.4 Load Break Switches (LBS) With Tripping Device

Shall conform to JIS C 4607.

2.5.2.5 Load Break Switches (LBS) With Fuse

Shall conform to JIS C 4611. Power fuses (PF) shall confrm to JIS C 4604.

2.5.2.6 Vacuum Circuit Breaker (VCB)

Shall conform to JIS C 4603.

~~]~~2.6 DISTRIBUTION SECTION

Distribution switchboard shall be circuit breaker-equipped unless indicated otherwise. Design shall be such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by JEM. Where "space only" is indicated, make provisions for future installation of breaker sized as indicated. Directories shall be typed to indicate load served by each circuit and mounted in holder behind transparent protective covering.

2.6.1 Unit Frame

Unit frame shall conform to JEM 1459.

2.6.2 Cubicle Type Cabinet

Cabinet shall conform to cubicle-type ~~[~~JIS C 4620~~][~~ and JEM 1425~~]~~. Additional requirements shall be attached at the end of this section as a reference.

2.6.3 Panelboard Type Cabinet

Cabinet shall conform to JIS C 8480 shall have a mounting plate, and shall be provided with wiring gutters of adequate size at top, bottom and sides.

2.6.3.1 Steel Type

Thickness of steel sheet shall be not less than 2.3 mm.

Thickness of sheet steel for the cabinet shall be not less than 2.3 mm for outdoor installation and for indoor installation.

Cabinet located outside the building shall be of hot dip galvanized steel sheet material.

Cabinets shall be painted in accordance with paragraph FACTORY APPLIED FINISH.

2.6.3.2 Stainless Steel Type

Cabinet shall be of Stainless Steel.

2.6.3.3 Weather Proof Type

Cabinet located outside of building and exposure to weather, shall be weather-proofed box.

Weather-proof cabinet shall conform to JIS C 0920 ~~[(IP44)] [(IP54)] [(IPXX and more)] [~~as indicated~~]~~ and exposed screws to weather shall be non-corrosive material.

2.6.4 Interrupter Switch for Low-voltage

2.6.4.1 Circuit Breaker

JIS C 8201-2-1, thermal-magnetic, magnetic or solid-state (electronic) type with interrupting capacity as indicated. Plug-in circuit breakers unacceptable.

Interrupting rating of circuit breakers shall be as indicated. If not shown, do not select circuit breakers less than 10,000A asymmetrical interrupting rating for voltages 240V and under, and 14,000A asymmetrical interrupting rating for 480V and under.

2.6.4.2 Circuit Breaker with GFCI

JIS C 8201-2-2, Plug-in circuit breakers unacceptable. Provide with "push-to-test" button, visible indication of tripped condition, and ability to detect and trip on current imbalance of ~~[15] [30]~~ milliamperes or greater per requirements or as indicated on drawings.

2.6.5 Automatic Transfer Switch (ATS)

As recommended by the auto transfer switch manufacturer.

2.6.6 Power Factor Improvement Equipments

~~[~~Provide as indicated on drawings.~~][_____]~~

2.6.7 Protective Relays, Metering, and Control Devices

Provide protective relays as indicated ~~[on drawings.] [~~per manufacturer's recommendations.~~] [_____]~~

2.6.7.1 Instruments

General of instruments shall conform to JIS C 1102-1, JIS C 1102-8 and JIS C 1103.

Ammeter ~~(~~wide-range type~~)~~, voltmeter ~~(~~wide-range type~~)~~ shall conform to JIS C 1102-2 respectively.

Wattmeters and varmeters shall conform to JIS C 1102-3.

Frequency meters shall conform to JIS C 1102-4.

Phase meters, power factor meters and synchroscopes shall conform to JIS C 1102-5.

Ohmmeters (impedance meters) and conductance meters conform to JIS C 1102-6.

Multi-function instruments shall conform to JIS C 1102-7.

Watt-hour meter with pulse generator shall conform to JIS C 1211-1, JIS C 1211-2, JIS C 1216-1, JIS C 1281 and JIS C 1283-1.

Var-hour meter shall conform to JIS C 1263-1.

[Metering shall be compliant with the current Advanced Meter Reading System (AMRS) Electric Meter Specifications.]

2.6.7.2 Instrument Control Switches

Provide rotary cam-operated type with positive means of indicating contact positions.

2.6.7.3 Buzzer

Shall have a sound output rating of at least 90 decibels at 1 m.

2.6.7.4 Test Terminal

Provide current test terminal and voltage test terminal.

2.6.7.5 Pilot and Indicating Lights

Provide transformer, resistor, or diode type.

2.6.7.6 Lightning Arrester

Shall conform to JIS C 4608.

2.6.7.7 Insulators

Shall conform to JIS C 3814, and JIS C 3851.

2.6.7.8 EMCS Terminal

Provide plywood (600 mm x 600mm x 12mm) with terminal blocks, receptacle outlet and associated wiring as indicated on drawings.

2.6.7.9 Space Heater

As indicated on drawings.

2.6.7.10 Receptacles

Provide receptacle outlet for maintenance.

Shall conform to JIS C 8303, grounding-type and duplex type.

2.6.7.11 LED Lighting Fixtures

Provide LED lighting fixtures inside cabinet.

Shall conform to JIS C 8106, JIS C 8105-1 and JIS C 8105-2-2.

2.6.7.12 Roof Fan

As indicated on drawings.

2.6.7.12.1 Air Intake Fan

Provide Air Intake Fan on side wall of receiving panel. Its cover (outdoor hood) shall have minimum 500mm straight portion duct below lower portion of opening for air intake on side wall. Straight portion duct shall be opened forward downside, and opening of duct shall be covered by crimped wire cloth and woven wire cloth with stainless steel (SUS304) frame. Wire clothes shall be fixed to cover by bolts or screws. Requirements for wire clothes are as follows, and their layer shall be following order from downside (outside);

(1) Crimped wire cloth, JIS G 3553, CR-S (SUS304), Wire dia. 1.6mm, mesh size 9mm
(2) Woven wire cloth, JIS G 3555, PW-S (SUS304), Wire dia. 0.65mm, 8 mesh
(3) Woven wire cloth, JIS G 3555 PW-S (SUS304), Wire dia. 0.65mm, 10 mesh

2.7 WIRE AND CABLES

2.7.1 Current Carrying Section

2.7.1.1 Cable Head

Cable head shall be designed for terminating one single conductor cables per phase and shall be arranged for conduits entering from below. Cable head Shall conform to JCAA C 3102.

2.7.1.2 Bus Duct (Busway)

JIS C 8364.

2.7.1.3 Copper Bus Bars

Shall conform to JIS H 3140.

2.7.1.4 Copper Ground Bus Terminal (Copper Bus Bar for Ground Terminal)

Provide a copper ground bus terminal of sufficient amperage and install. Bus bars shall be JIS H 3140.

2.7.1.5 Terminal Blocks

JIS C 8201-7-1.

~~[2.7.1.6 Extra-High-Voltage Cable (Over 6.6 kV)~~

~~Provide as specified in SECTION 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.~~

~~][~~2.7.1.6 High Voltage Wire (6.6 kV)

High voltage wire for cubicle type unit substation shall conform to JIS C 3611, Type KIP.

~~][~~2.7.1.7 Low-Voltage Cable (600 V)

Provide as specified in SECTION 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

~~]~~2.8 CONDUITS

Provide as specified in SECTION 26 20 00~~,~~ INTERIOR DISTRIBUTION SYSTEM.

2.9 FACTORY APPLIED FINISH

Electrical equipment shall have factory-applied painting systems which shall, as a minimum, meet the requirements of JIS C 0920 corrosion-resistance test and the additional requirements as specified herein. Interior and exterior steel surfaces of equipment enclosures shall be thoroughly cleaned and then receive a rust-inhibitive phosphatizing or equivalent treatment prior to painting. Exterior surfaces shall be free from holes, seams, dents, weld marks, loose scale or other imperfections. Interior surfaces shall receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice. Exterior surfaces shall be primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish. Equipment located indoors shall be ANSI Light Gray,~~[~~ and equipment located outdoors shall be ANSI~~[Light Gray][~~ Dark Gray~~]]~~. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.9.1 Hot Dip Galvanizing

JIS H 8641, Type HDZ35.

2.9.2 Paint of Cabinet

Provide standard factory finishes including rust inhibiting treatment. Unless otherwise specified or indicated finish of outside panels shall be applied factory finish color. Field applied paint shall not be permitted for newly installed panels. The cabinet shall include likely panelboard, power panel, control panel, breaker box, disconnecting switch box, terminal box and steel cabinet for electrical work.

2.9.2.1 Clear Blue

Provide standard factory finishes including rust inhibiting treatment, except that the inside finish of the cabinet shall be vivid orange (2.5YR5/12 of JIS Z 8721) and the outside including exposed parts of trim and door shall be clear blue (2.5PB5/8 of JIS Z 8721).

2.9.2.2 Sand Beige

Provide standard factory finishes including rust inhibiting treatment, except that the inside finish of the cabinet shall be vivid orange (2.5YR5/12 of JIS Z 8721) and the outside including exposed parts of trim and door shall be sand beige (2.5Y8.5/1 of JIS Z 8721).

2.9.2.3 Surrounding

Provide standard factory finishes including rust inhibiting treatment, except that the inside finish of the cabinet shall be vivid orange (2.5YR5/12 of JIS Z 8721) and the outside including exposed parts of trim and door shall match to surrounding wall surface.

2.9.2.4 Fire Red

Provide standard factory finishes including rust inhibiting treatment, except that the inside finish of the cabinet shall be vivid orange (No. 2.5YR5/12 of JIS Z 8721) and the outside including exposed parts of trim and door shall be fire red (No. 7.5R4/14 of JIS Z 8721).

2.9.2.5 Selection of Colors Outside

Selection of color outside cabinet including exposed parts of trim shall be clear blue in the industrial area and mechanical room, unless otherwise specified or indicated. The color of the cabinet located other area shall match to surrounding or sand beige. ~~[~~Directed by the Contracting Officer.~~]~~

~~2.9.3 Nameplate~~

~~Provide as specified in Section 26 00 00.00 20, BASIC ELECTRICAL MATERIALS AND METHODS.~~

2.10 LEAD-IN POCKET FOR TEMPORARY CABLE

As indicated on drawings.

2.11 WARNING SIGN

Warning sign shall be attached at the end of this section.

~~2.12 GROUNDING AND BONDING~~

~~[Provide grounding and bonding as specified in Section 33 71 02.00 20 UNDERGROUND ELECTRICAL DISTRIBUTION.]~~

~~[Provide as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.]~~

2.12 MATERIALS FOR CONCRETE FOUNDATION

Features and dimension of the concrete foundation shall be as indicated.

2.12.1 Concrete Material

Specified in Section 03 30 00~~,~~ CAST-IN-PLACE CONCRETE.

2.12.2 Anchor Bolt

Anchor bolts shall be JIS B 1178, Type L. Bolts shall be stainless steel material conforming to SUS 304.

2.12.3 Anchor Bolt (Expansion Anchor)

Tubular, multi-slit, internal thread, with stud bolt having a head of the expander shape, as indicated on drawing. Do not use plastic material.

2.12.4 Anchor Bolt (Chemical Anchor)

Shall be a two-part system composed of a threaded rod stud and a sealed glass capsule containing premeasured amounts of epoxy acrylic resin, quartz sand, and a hardener contained in a separate vial within the capsule.

2.12.5 Nuts and Washers

Material shall be stainless steel conforming to SUS 304.

2.13 ADDITIONAL REQUIREMENTS OF THE SUBSTATION SWITCHING STATION SWITCHGEAR

Additional requirements of cubicle type unit substation open type unit substation switching station switchgear shall be attached at the end of this section as a reference.

PART 3 EXECUTION

3.1 INSTALLATION

Unless otherwise indicated, installation shall be performed in accordance with MLIT ESS, MLIT DSKKS and to the requirements specified herein.

3.2 GROUNDING

Provide grounding as indicated, in accordance with ~~[~~MLIT ESS and MLIT DSKKS, ~~]~~ ~~[~~except that grounding systems shall have a resistance to solid earth ground not exceeding 5 ohms~~]~~.

3.2.1 Grounding Electrodes

Provide driven ground rods as specified in ~~[Section 33 71 02.00 20 UNDERGROUND ELECTRICAL DISTRIBUTION] [~~Section 26 20 00 Interior Distribution System~~]~~. Connect ground conductors to the upper end of ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.2.2 Transformer Grounding

Provide separate copper grounding conductors and connect them to ~~[~~the ground girdle as indicated~~]~~ and ~~[~~copper ground bus terminal~~]~~. When work in addition to that indicated or specified is required to obtain the specified ground resistance, the provision of the contract covering "Changes" shall apply.

3.2.3 Grounding and Bonding Equipment

Provide separate copper grounding conductors and connect them to copper ground bus terminal. Solid bare copper wire shall be JIS C 3102: Stranded bare copper wire shall be JCS 1226, except as indicated or specified otherwise.

3.2.4 Ground Girdle (Loop Ground)

Provide a 60 sqmm bare copper-ground girdle around substation switching station switchgear. Girdle shall be buried 305 mm (one foot) deep and placed 915 mm (3 feet) laterally from the substation enclosure. Connect girdle to enclosure at two opposite places using 60 sqmm copper.

3.2.5 Connections

Make joints in grounding conductors and ground girdle by exothermic weld or compression connector. Exothermic welds and compression connectors shall be installed as specified in ~~[Section 33 71 02.00 20 UNDERGROUND ELECTRICAL DISTRIBUTION] [~~Section 26 20 00 Interior Distribution System~~]~~.

3.2.6 Resistance

~~[Maximum resistance-to-ground of grounding system shall be as specified in Section 33 71 02.00 20 UNDERGROUND ELECTRICAL DISTRIBUTION.]~~
~~[~~Maximum resistance-to-ground of grounding system shall be as specified in Section 26 20 00 Interior Distribution System.~~]~~

3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

Install and connect cubicle type unit substation open type unit substation switching station switchgear furnished under this section as indicated on project drawings, the approved shop drawings, and as specified herein.

3.4 FIELD APPLIED PAINTING

Field paint shall be specified in Section 09 90 00, PAINTS AND COATINGS.

3.5 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side, but space the signs a maximum of 9 meters apart.

3.6 RESTORATION

Unless otherwise indicated, all existing objects which interfere with new work shall be removed temporary and reinstalled upon completion of new work.

3.7 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

Foundation shall be in accordance with MLIT ESS.

3.7.1 Cast-in-place concrete

Cast-in-place concrete work shall be as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.7.2 Sealing

When the installation is complete, the Contractor shall seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals shall be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

3.8 FIELD QUALITY CONTROL

3.8.1 Testing Methods for Each Field Test

Use design documents and requirements specified in this section to develop test procedures. Procedures shall consist of detailed instructions for a test setup, execution, and evaluation of test results.

Testing shall not proceed until after the Contractor has received written Contracting Officer's approval of the test procedures as specified.

3.8.2 Performance of Acceptance Checks and Tests

First Class Construction Electric Management Engineer (1 Kyu Dekikouji

Sekou Kanrigishi) shall perform acceptance checks and testing in accordance with the manufacturer's recommendations, and include [the following] visual and mechanical inspections and electrical tests, performed in accordance with MLIT ESS, MLIT DSKKS, METI, the Technical Standard for Electrical Equipment TSEE, and JEAC 8001.

a. Protection co-ordination curve line

b. Insulation resistance test

c. Withstand voltage test

d. Protective relays test

e. [Leakage current test of transformer]

f. System test

[Grounding system test are performed in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests.]

3.8.3 Equipment Checks and Tests

3.8.3.1 Interrupter Switches

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Confirm correct application of manufacturer's recommended lubricants.

(4) Verify appropriate anchorage and required area clearances.

(5) Verify appropriate equipment grounding.

(6) Verify correct blade alignment, blade penetration, travel stops, and mechanical operation.

(7) [Verify that fuse sizes and types correspond to approved shop drawings.]

(8) [Verify that each fuse holder has adequate mechanical support.]

(9) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermographic surveying~~[is not]~~[is]required.

(10) Test interlocking systems for correct operation and sequencing.

(11) Verify correct phase barrier materials and installation.

(12) Compare switch blade clearances with industry standards.

(13) Inspect all indicating devices for correct operation

b. Electrical Tests

(1) Perform insulation-resistance tests.

(2) Perform over-potential tests.

(3) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.

(4) Measure closed contact-resistance across each switch blade[and fuse holder].

(5) [Measure fuse resistance.]

(6) Verify heater operation.

3.8.3.2 Interrupter Switchgear (LDS, LBS)

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Confirm correct application of manufacturer's recommended lubricants.

(4) Verify appropriate anchorage and required area clearances.

(5) Verify appropriate equipment grounding.

(6) Verify correct blade alignment, blade penetration, travel stops, and mechanical operation.

(7) [Verify that fuse sizes and types correspond to approved shop drawings.]

(8) [Verify that each fuse holder has adequate mechanical support.]

(9) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

(10) Test interlocking systems for correct operation and sequencing.

(11) Verify correct phase barrier materials and installation.

(12) Compare switch blade clearances with industry standards.

(13) Inspect all indicating devices for correct operation

b. Electrical Tests

(1) Perform insulation-resistance tests.

(2) Perform over-potential tests.

(3) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.

(4) Measure closed contact-resistance across each switch blade~~[~~ and fuse holder~~]~~.

(5) ~~[~~Measure fuse resistance.~~]~~

(6) Verify heater operation.

3.8.3.3 Vacuum Circuit Breaker (VCB)

a. Visual and mechanical inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Confirm correct application of manufacturer's recommended lubricants.

(4) Inspect anchorage, alignment, and grounding.

(5) Perform all mechanical operational tests on both the circuit breaker and its operating mechanism.

(6) Measure critical distances such as contact gap as recommended by manufacturer.

(7) Verify tightness of accessible bolted connections by calibrated torque-wrench method. Thermographic survey~~[is not]~~~~[~~ is ~~]~~ required.

(8) Record as-found and as-left operation counter readings.

b. Electrical Tests

(1) Perform a contact-resistance test.

(2) Verify trip, close, trip-free, and antipump function.

(3) Trip circuit breaker by operation of each protective device.

(4) Perform insulation-resistance tests.

(5) Perform vacuum bottle integrity (overpotential) test across each bottle with the breaker in the open position in strict accordance with manufacturer's instructions. Do not exceed maximum voltage stipulated for this test.

3.8.3.4 Metering and Instrumentation

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify tightness of electrical connections.

b. Electrical Tests

(1) Verify accuracy of meters at 25, 50, 75, and 100 percent of full scale.

(2) Calibrate watthour meters according to manufacturer's published data.

(3) Verify all instrument multipliers.

(4) Verify that current transformer[and voltage transformer] secondary circuits are intact.

3.8.3.5 Switchgear Assemblies

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical, electrical, and mechanical condition.

(3) Confirm correct application of manufacturer's recommended lubricants.

(4) Verify appropriate anchorage, required area clearances, and correct alignment.

(5) Inspect all doors, panels, and sections for paint, dents, scratches, fit, and missing hardware.

(6) Verify that[fuse and] circuit breaker sizes and types correspond to approved shop drawings.

(7) [Verify that current and potential transformer ratios correspond to approved shop drawings.]

(8) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermographic survey~~[is not]~~[is] required.

(9) Confirm correct operation and sequencing of electrical and mechanical interlock systems.

(10) Clean switchgear.

(11) Inspect insulators for evidence of physical damage or contaminated surfaces.

(12) Verify correct barrier[and shutter] installation[and operation].

(13) Exercise all active components.

(14) Inspect all mechanical indicating devices for correct operation.

(15) Verify that vents are clear.

(16) Test operation, alignment, and penetration of instrument transformer withdrawal disconnects.

(17) Inspect control power transformers.

b. Electrical Tests

(1) Perform insulation-resistance tests on each bus section.

(2) Perform overpotential tests.

(3) Perform insulation-resistance test on control wiring; Do not perform this test on wiring connected to solid-state components.

(4) Perform control wiring performance test.

(5) Perform primary current injection tests on the entire current circuit in each section of assembly.

(6) [Perform phasing check on double-ended switchgear to ensure correct bus phasing from each source.]

(7) Verify operation of heaters.

3.8.3.6 Transformers

a. Visual and mechanical inspection

(1) Compare equipment nameplate information with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition. Check for damaged or cracked insulators and leaks.

(3) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

(4) Verify correct liquid level in tanks.

(5) Perform specific inspections and mechanical tests as recommended by manufacturer.

(6) Verify correct equipment grounding.

(7) Verify the presence of transformer surge arresters.

b. Electrical tests

(1) Perform resistance measurements through all bolted

connections with low-resistance ohmmeter.

(2) Verify that the tap-changer is set at specified ratio.

(3) Verify proper secondary voltage phase-to-phase and phase-to-neutral after energization and prior to loading.

(4) Perform transformer test in accordance with JEC 2200 and the transformer manufacture's written instruction.

3.8.3.7 Current Transformers

a. Visual and mechanical inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify correct connection.

(4) Verify that adequate clearances exist between primary circuits and secondary circuit.

(5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

(6) Verify that required grounding and shorting connections provide good contact.

b. Electrical tests

(1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.

(2) Perform insulation-resistance test.

(3) Perform a polarity test.

(4) Perform a ratio-verification test.

3.8.3.8 ~~[~~Kilowatt Demand Meter~~][Watthour Meter]~~

a. Visual and mechanical inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify tightness of electrical connections.

b. Electrical tests

(1) ~~[~~Calibrate kilo watthour meters according to manufacturer's published data.~~]~~

(2) Verify that correct multiplier has been placed on face of meter, where applicable.

(3) Verify that current transformer secondary circuits are intact.

3.8.3.9 Grounding System

a. Visual and mechanical inspection

(1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

(1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete (not exceed 5 ohms). On systems consisting of a single ground rod perform tests before any wire is connected (not exceed 25 ohms). Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground testing megger in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument shall be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.

(2) Submit the measured ground resistance of each ground rod (not exceed 25 ohms) and grounding system (not exceed 5 ohms), indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

3.8.4 Follow-Up Verification

Upon completion of acceptance checks and tests, the Contractor shall show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer shall be given 5 working days advance notice of the dates and times of checking and testing.

INSERT 7 DRAWINGS

-- End of Section --

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM
02/14

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS), Ministry of Land, Infrastructure and Transport (MLIT), Japan Electrical Safety Inspection Associations, Japanese Electrotechnical Committee (JEC) or Japan Wire Industry Association (JCS) standards for non-Japanese standards, as approved by the Contracting Officer's representative.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1	(2022) Electric Meters - Code for Electricity Metering

JAPANESE ELECTROTECHNICAL COMMITTEE (JEC)

JEC 2200	(2015) Transformer

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS C 0365	(2007) Protection Against Electric Shock - Common Aspects for Installation and Equipment
JIS C 0920	(2003) Degrees Of Protection Provided By Enclosures (IP Code)
JIS C 1210	(1979) General Rules for Electricity Meters
JIS C 2336	(2012) Pressure-sensitive polyvinyl chloride tapes for electrical purposes
JIS C 2338	(2012) Polyester adhesive tape for electrical insulation
JIS C 2805	(2010) Crimp terminal for copper wire
JIS C 2806	(2003; R 2018) Bare crimping sleeve for copper wire
JIS C 2810	(1995; R 2021) General rules on non-separable type wire connectors for interior wiring
JIS C 3101	(1994; R 2021) Hard-drawn copper wires for electrical purposes
JIS C 3105	(1994; R 2021) Hard-drawn copper stranded conductors

JIS C 3341	~~(2000; R 2021) Polyvinyl chloride insulated drop service wires~~(2000) Polyvinyl Chloride Insulated Drop Service Wires
JIS C 3401	(2002; R 2022) Control Cables
JIS C 3605	(2022) 600 V Polyethylene Insulated Cables~~, Type CV~~
JIS C 3612	(2002; R 2022) 600V Flame Retardant Polyethylene Insulated Wires
JIS C 4212	(2010; R 2022) Low-Voltage Three-Phase Squirrel-Cage High-Efficiency Induction Motors (Amendment 1)
JIS C 5381-11	(2014; R 2019) Low-voltage surge protective devices -- Part 11: Surge protective devices connected to low-voltage power systems -- Requirements and test methods
JIS C 5381-12	(2021) Low-voltage surge protective devices -- Part 12: Surge protective devices connected to low-voltage power distribution systems -- Selection and application principles
JIS C 6436	(1995; R 2019) Power transformer for electronic equipment
JIS C 8201-2-1	(2021) Low-Voltage Switchgear and Control Gear - Part 2-1: Circuit-Breakers
JIS C 8201-2-2	(2021) Low-Voltage Switchgear And Control Gear - Part 2-2: Circuit-Breakers Incorporating Residual Current Protection
JIS C 8201-3	(2009; R 2018) Low-voltage switchgear and control units-Part 3: Switches,Disconnector, disconnecting switch and fuse assembly unit
JIS C 8201-5-1	(2022) Low-Voltage Switchgear And Control Gear - Part 5-1: Control Circuit Devices And Switching Elements - Electromechanical Control Circuit Devices
JIS C 8201-4-1	(2023) Low-voltage switchgear and controlgear -- Part 4-1: Contactors and motor-starters: Electromechanical contactors and motor-starters
JIS C 8201-4-2	(2010; 2020) Low-voltage switchgear and controlgear -- Part 4-2: Contactors and motor-starters -- AC semiconductor motor controllers and starters

JIS C 8201-7-1	(2016) Low-Voltage Switchgear And Control Gear -- Part 7-1: Ancillary Equipment -- Terminal Blocks For Copper Conductors
JIS C 8222	(2021) Residual current operated circuit-breakers with integral overcurrent protection for household and similar uses (RCBOs)
JIS C 8269-1	(2016) Low-Voltage Fuses -- Part 1: General Requirements
JIS C 8269-2	(2016) Low voltage fuse-Part 2: Additional requirements for expert fuses(Mainly industrial fuses)
JIS C 8281-1	(2019) Plugs and socket-outlets for household and similar purposes -- Part 1: General requirements
JIS C 8303	(2007; R 2022) Plugs And Receptacles For Domestic And Similar General Use
JIS C 8304	(2009; R 2019) Small switches for indoor use
JIS C 8305	(2019) Rigid Steel Conduits
JIS C 8309	(2019) Pliable metal conduits
JIS C 8330	(1999; R 2019) Fittings for rigid metal conduits
JIS C 8340	(1999; R 2019) Boxes And Box Covers For Rigid Metal Conduits
JIS C 8350	(1999; R 2014) Fittings for pliable metal conduits
JIS C 8364	(2008; R 2018) Busways
JIS C 8380	(2009; R 2019) Plastic coated steel pipes for cable-ways
JIS C 8425	(1984) Plastic surface raceways for interior wiring
JIS C 8430	~~(2019) Unplasticized polyvinyl chloride (PVC-U) conduits~~(2019) Unplasticized Polyvinyl Chloride (PVC-U) Conduits
JIS C 8432	(2019) Fittings of unplasticized polyvinyl chloride (PVC-U) conduits
JIS C 8435	(2022) Boxes And Box Covers Of Plastic Conduits
JIS C 8462-1	(2021) Boxes and enclosures for electrical

	accessories for household and similar fixed electrical installations -- Part 1: General requirements
JIS C 8480	(R2020) Box-Type Switchgear Assemblies for Low-Voltage Distribution Purpose
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor
JIS C 60364-6	(2010; R 2019) Low-voltage electrical installations -- Part 6: Verification
JIS C 61000-4-7	(2007; R 2017) Electromagnetic compatibility (EMC) -- Part 4-7: Testing and measurement techniques -- General guide on harmonics and interharmonics measurements and instrumentation
JIS C 61558-1	(2019) Safety of transformers, reactors, power supply units and combinations thereof -- Part 1: General requirements and tests
JIS K 6743	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Water Supply
JIS K 6911	(2006; R 2021) Thermosetting plastic general test method
JIS K 6741	(2016~~; R 2021~~) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes
JIS T 1021	(2019) Hospital grade outlet-sockets and plugs
JIS X 5150	(R2016) Information Technology-Generic Cabling for Customer Premises
JIS Z 9101	(2018) Graphical symbols -- Safety colours and safety signs -- Part 1: Design principles for safety signs and safety markings
JIS Z 9290-1	(2014; R 2019) Protection Against Lightning - Part 1: General Principles

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 1226	(2003) Soft Stranded Wire

ELECTRICAL SAFETY INSPECTION ASSOCIATIONS

Denki Hoan Kyoukai	Japan Standard for Acceptance Testing and Inspections

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT DSKKS	Denki Setsubi Kouji Kanri Shishin (DSKKS) Electrical Construction Supervision Guidelines
MLIT ESS	(2019) MLIT Electrical Standard Specification (ESS)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E	(2024) Standard for Electrical Safety in the Workplace

UNDERWRITERS LABORATORIES (UL)

UL 2043	(2013) Fire Test for Heat and Visible Smoke Release for Discrete Products and Their Accessories Installed in Air-Handling Spaces

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are~~[for Contractor Quality Control approval][~~ for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government~~]~~. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00.

SD-02 Shop Drawings

Panelboards; G~~[, [_____]]~~

Transformers; G~~[, [_____]]~~

Busway; G~~[, [_____]]~~

Cable trays; G~~[, [_____]]~~

Motor control centers; G~~[, [_____]]~~

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Identify circuit terminals on wiring diagrams and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Indicate on the drawings adequate clearance for operation, maintenance, and replacement of operating equipment devices.

Wireways; G~~[, [_____]]~~

~~[Load centers for housing units; G[, [_____]]]~~

Marking strips drawings; G~~[, [_____]]~~

SD-03 Product Data

Receptacles~~;~~

Circuit breakers; G~~[, [_____]]~~

Switches; G~~[, [_____]]~~

Transformers; G~~[, [_____]]~~

Enclosed circuit breakers; G~~[, [_____]]~~

Motor controllers; G~~[, [_____]]~~

~~[Combination motor controllers; G[, [_____]]]~~

~~[Load centers for housing units; G[, [_____]]]~~

Manual motor starters; G~~[, [_____]]~~

~~[~~Residential load centers; G~~[, [_____]]]~~

~~[~~Metering; G~~[, [_____]]]~~

~~[~~Meter base only; G~~[, [_____]]]~~

CATV outlets; G~~[, [_____]]~~

Surge protective devices; G~~[, [_____]]~~

Include performance and characteristic curves.

SD-06 Test Reports

600-volt wiring test~~;~~

Grounding system test~~;~~

Transformer tests~~;~~

Ground-fault receptacle test~~;~~

~~[~~ SD-10 Operation and Maintenance Data

Electrical Systems, Data Package 5~~;~~

Metering, Data Package 5~~;~~

Submit operation and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein.

1.4 QUALITY ASSURANCE

1.4.1 Fuses

Submit coordination data as specified in paragraph, FUSES of this section.

1.4.2 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" or "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with the mandatory and advisory provisions of applicable codes and standards unless more stringent requirements are specified or indicated.

1.4.3 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.4.3.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.4.3.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.5 MAINTENANCE

1.5.1 Electrical Systems

Submit operation and maintenance manuals for electrical systems that provide basic data relating to the design, operation, and maintenance of the electrical distribution system for the building. Include the following:

a. Single line diagram of the "as-built" building electrical system.

b. Schematic diagram of electrical control system (other than HVAC, covered elsewhere).

c. Manufacturers' operating and maintenance manuals on active electrical equipment.

~~]~~1.6 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

~~[~~1.7 SEISMIC REQUIREMENTS

Provide seismic details~~[conforming to[Section 13 48 00 [SEISMIC] BRACING FOR MISCELLANEOUS EQUIPMENT][and to][Section 26 05 48.00 10 SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT]][~~ as indicated~~]~~.

~~]~~PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL or Japanese standards, where UL or applicable Japanese codes and standards are established for those items for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit or Type G Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit or Type G

JIS C 8305, Type G. Diameter of conduit shall be as indicated.

2.2.2 Rigid Nonmetallic Conduit or Unplasticized Polyvinyl Chloride

JIS C 8430 for Type VE or HIVE conduit of diameter of less than 100mm and JIS K 6741 for Type VP or HIVP conduit diameter or 100mm and larger.

2.2.3 Electrical, Zinc-Coated Steel Metallic Tubing (EMT) or Type E Metallic Conduit

JIS C 8305, Type E.

2.2.4 Plastic-Coated Rigid Steel and IMC Conduit or Type G, C or E; LL or LT

JIS C 8380.

2.2.5 Flexible Metal Conduit

JIS C 8309.

2.2.5.1 Liquid-Tight Flexible Metal Conduit, Steel

JIS C 8309.

2.2.6 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

JIS C 8330 and JIS C 8350. Ferrous fittings: cadmium- or zinc-coated in accordance with JIS C 8330 and JIS C 8350.

2.2.6.1 Fittings for Rigid Metal Conduit and IMC or Type G and Type C

Threaded-type. Split couplings unacceptable.

2.2.6.2 Fittings for EMT or Type E Conduit

~~[Die Cast][~~Steel~~][set screw]~~ type.

2.2.7 Fittings for Rigid Nonmetallic Conduit or Unplasticized Polyvinyl Chloride

JIS C 8432 for diameters less than 100mm and JIS K 6743 for diameters 100mm or larger.

2.3 SURFACE RACEWAY

2.3.1 Surface Metal Raceway

Two-piece painted steel, totally enclosed, snap-cover type.~~[~~ Provide multiple outlet-type raceway with grounding-type receptacle where indicated. Provide receptacles as specified herein, spaced a minimum of one every ~~[455] [_____]~~ mm.~~][~~ Wire alternate receptacles on different circuits.~~]~~

2.3.2 Surface Nonmetallic Raceway

JIS C 8425, nonmetallic totally enclosed, snap-cover type.~~[~~ Provide multiple outlet-type raceway with grounding-type receptacle where indicated. Provide receptacles as specified herein, spaced a minimum of one every ~~[455] [_____]~~ mm.~~][~~ Wire alternate receptacles on different circuits.~~]~~

2.4 BUSWAY

JIS C 8364. Provide the following:

a. Buses: ~~[~~ copper~~][or][aluminum]~~.

b. Busways: rated ~~[_____]~~600 volts, ~~[_____]~~as indicated continuous current amperes, three-phase,~~[three-][~~ four-~~]~~wire, and include integral or internal~~[50-percent]~~ ground bus.

c. Short circuit rating: ~~[[_____] root mean square (rms) symmetrical amperes minimum][~~ as indicated~~]~~.

~~[~~ d. Busway systems: suitable for use indoors.

~~]~~ e. Enclosures: ~~[~~ steel~~][aluminum] [metallic]~~.

f. Hardware: plated or otherwise protected to resist corrosion.

g. Joints: one-bolt type with through-bolts, which can be checked for tightness without deenergizing system.

h. Maximum hot spot temperature rise at any point in busway at continuous rated load: do not exceed 55 degrees C above maximum ambient temperature of 40 degrees C in any position.

i. Internal barriers to prevent movement of superheated gases.

j. Coordinate proper voltage phasing of entire bus duct system, for example where busway interfaces with transformers, switchgear, switchboards, motor control centers, and other system components.

2.4.1 Feeder Busways

Provide[ventilated, except that vertical busways within 1830 mm of floors must be unventilated,][unventilated, totally enclosed] low-impedance busway. Provide bus bars fully covered with insulating material, except at stabs. Provide an entirely polarized busway system.

2.4.2 Plug-In Busways

Unventilated type. Provide the following:

a. Plug-in units: [fusible, handle-operated, switch type, horsepower-rated][circuit breaker-type][handle-operated, switch type, equipped with high interrupting-capacity, current-limiting fuses].

b. Bus bars: covered with insulating material throughout, except at joints and other connection points.

[c. A hook stick of suitable length for operating plug-in units from the floor.

]2.5 CABLE TRAYS

Provide the following:

a. Cable trays: form a wireway system, with a nominal[~~[75] [100] [150]~~ mm] depth[as indicated].

b. Cable trays: constructed of~~[aluminum]~~[copper-free aluminum]~~[steel that has been zinc-coated after fabrication]~~.

c. Cable trays: include splice and end plates, dropouts, and miscellaneous hardware.

d. Edges, fittings, and hardware: finished free from burrs and sharp edges.

e. Fittings: ensure not less than load-carrying ability of straight tray sections and have manufacturer's minimum standard radius.

[f. Radius of bends: ~~[305] [610] [915]~~ mm.][Radius of bends: as indicated.

]2.5.1 Basket-Type Cable Trays

Provide[size as indicated][of nominal~~[50,]~~[100~~,~~]~~[150,][200,]~~[and 300~~,][450,][and][600]~~ mm width and ~~[25,]~~[50~~,][and][100]~~ mm depth] with maximum wire mesh spacing of 50 by 100 mm.

2.5.2 Trough-Type Cable Trays

Provide~~[~~ size as indicated~~][of nominal [150] [305] [455] [610] [760] [915] mm width]~~.

2.5.3 Ladder-Type Cable Trays

Provide~~[~~ size as indicated~~][of nominal [150] [305] [455] [610] [760] [915] mm width] with maximum rung spacing of [150] [225] [305] [455] mm~~.

2.5.4 Channel-Type Cable Trays

Provide ~~[~~ size as indicated~~][of nominal [75] [100] [150] mm width]~~. Provide trays with one-piece construction having slots spaced not more than 115 mm on centers.

2.5.5 Solid Bottom-Type Cable Trays

Provide~~[~~ size as indicated~~][of nominal [150] [305] [455] [610] [760] [915] mm width]~~. ~~[~~Provide solid covers.~~][Do not provide solid covers.]~~

~~2.5.6 [Cantilever~~

~~Cantilever-type, center-hung cable trays may be provided at the Contractor's option in lieu of other cable tray types specified.~~

~~][~~2.6 OPEN TELECOMMUNICATIONS CABLE SUPPORT

2.6.1 Open Top Cable Supports

Provide open top cable supports in accordance with UL 2043. Provide ~~[[galvanized][~~ zinc-coated~~][stainless] steel]~~open top cable supports~~[~~ as indicated~~]~~.

2.6.2 Closed Ring Cable Supports

Provide closed ring cable supports in accordance with UL 2043. Provide ~~[[galvanized][~~ zinc-coated~~][stainless] steel]~~closed ring cable supports~~[~~ as indicated~~]~~.

~~]~~2.7 OUTLET BOXES AND COVERS

JIS C 8340, cadmium- or zinc-coated, if ferrous metal. JIS C 8435, if nonmetallic. Rated ~~[~~IP65 for wet locations~~]~~ ~~[~~IP66, 67 or 68 for dust-tight/hazardous locations~~]~~.

2.7.1 Floor Outlet Boxes

Provide the following:

a. Boxes: ~~[~~ adjustable~~][nonadjustable]~~ and concrete tight.

b. Each outlet: consisting of~~[nonmetallic][or][~~ cast-metal~~]~~ body with threaded openings,~~[or sheet-steel body with knockouts]~~ for conduits,~~[adjustable][,][~~ brass flange~~]~~ ring, and cover plate with ~~[19][~~25~~][~~ and 31~~.75][53.92]~~ mm threaded plug.

c. Telecommunications outlets: consisting of~~[surface-mounted, horizontal][~~ flush~~]~~, aluminum ~~or stainless steel~~ housing with a receptacle as specified and~~[25 mm bushed side opening]~~~~[~~ 19 mm top opening~~]~~.

d. Receptacle outlets: consisting of~~[surface-mounted, horizontal]~~~~[~~ flush ~~]~~ aluminum ~~or stainless steel~~ housing with duplex-type receptacle as specified herein.

e. Provide gaskets where necessary to ensure watertight installation.

~~[f. Provide plugs with installation instructions to the Contracting Officer for [5] [_____] percent of outlet boxes for the capping of outlets upon removal of service fittings.~~

~~]~~2.7.2 Outlet Boxes for Telecommunications System

Provide the following:

a. Standard type~~[~~ 100 mm square by 54 mm deep~~]~~~~[120 mm square by 54 mm deep]~~.

~~[~~ b. Outlet boxes for wall-mounted telecommunications outlets: 100 by 54 by 54 mm deep.

~~]~~ c. Depth of boxes: large enough to allow manufacturers' recommended conductor bend radii.

~~[~~ d. Outlet boxes for fiber optic telecommunication outlets: include a minimum 10 mm deep single or two gang plaster ring as shown and installed using a minimum 27 mm conduit system.

~~][~~e. Outlet boxes for handicapped telecommunications station: 100 by 54 by 54 mm deep.

~~][2.7.3 Clock Outlet for Use in Other Than Wired Clock System~~

~~Provide the following:~~

~~a. Outlet box with plastic cover, where required, and single receptacle with clock outlet plate.~~

~~b. Receptacle: recessed sufficiently within box to allow complete insertion of standard cap, flush with plate.~~

~~c. Suitable clip or support for hanging clock: secured to top plate.~~

~~d. Material and finish of plate: as specified in paragraph DEVICE PLATES of this section.~~

~~]~~2.8 CABINETS, JUNCTION BOXES, AND PULL BOXES

Volume greater than 1640 mL, JIS C 8340, hot-dip, zinc-coated, if sheet steel. Rated ~~[~~IP65 for wet locations~~]~~ ~~[~~IP66, 67 or 68 for dust-tight/hazardous locations~~]~~.

2.9 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of Japanese standards for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 12 months prior to date of delivery to site.

2.9.1 Conductors

Provide the following:

a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.

b. Conductors 5.5 sqmm and larger cross sectional area: stranded.

c. Conductors 2.0 mm and smaller diameter: solid.

d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.

~~[~~ e. All conductors: ~~[~~copper.~~][~~ Conductors indicated to be 14 sqmm or smaller diameter: copper. Conductors indicated to be 22 sqmm and larger diameter: copper , unless type of conductor material is specifically indicated, or specified, or required by equipment manufacturer.~~]~~

~~][2.9.1.1 Equipment Manufacturer Requirements~~

~~When manufacturer's equipment requires copper conductors at the terminations or requires copper conductors to be provided between components of equipment, provide copper conductors or splices, splice boxes, and other work required to satisfy manufacturer's requirements.~~

~~]~~2.9.1.1 Minimum Conductor Sizes

Provide minimum conductor size in accordance with the following:

a. Branch circuits: 2.0 mm.

b. Class 1 remote-control and signal circuits: 1.6 mm.

c. Class 2 low-energy, remote-control and signal circuits: 1.2 mm.

d. Class 3 low-energy, remote-control, alarm and signal circuits: 0.65 mm.

2.9.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

2.9.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

a. Grounding conductors: Green.

b. Neutral conductors: White or Gray.

c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.9.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

a. ~~[~~480/277~~][440/254V][420/242V][~~ or 208/120 volt~~]~~, three-phase, four-wire

(1) Phase A - red

(2) Phase B - black

(3) Phase C - blue

b. ~~[~~480 volt~~][440 volt][420 volt][220 volt][210 volt][~~ or 208 volt~~]~~, three-phase, three-wire

(1) Phase A - red

(2) Phase B - black

(3) Phase C - blue

c. ~~[105/210][~~120/240~~]~~ volt, single phase, three-wire:

(1) Phase A - red

(2) Phase B - black

~~[~~ d. On three-phase, four-wire delta system, high leg: orange, as required.

~~]~~2.9.3 Insulation

Unless specified or indicated otherwise, provide power and lighting wires rated for 600-volts, ~~[~~Type EM-IE conforming to JIS C 3612~~][~~ or ~~][~~Type EM-CE, EM-CET or EM-ECEQ conforming to JIS C 3605~~]~~, except that grounding wire may be type TW conforming to JIS C 3612, Type EM-IE; remote-control and signal circuits: Type TW or TF, conforming to JIS C 3401, Type CEV. Where lighting fixtures require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.9.4 Bonding Conductors

JIS C 3101, solid bare copper wire for sizes 8 sqmm and smaller diameter; JIS C 3105 and JCS 1226, Class B, stranded bare copper wire for sizes 14 sqmm and larger diameter, JIS C 3102 Annealed Copper Wire.

2.9.4.1 Telecommunications Bonding Backbone (TBB)

Provide a copper conductor TBB in accordance with JIS C 60364-5-54 with 14 sqmm minimum size, and sized at 3.3 sqmm per linear meter of conductor length up to a maximum size of 1500 sqmm.~~[~~ Provide insulated TBB with insulation as specified in the paragraph INSULATION and meeting the fire ratings of its pathway.~~]~~

2.9.4.2 Bonding Conductor for Telecommunications

Provide a copper conductor Bonding Conductor for Telecommunications between the telecommunications main grounding busbar (TMGB) and the electrical service ground in accordance with JIS C 60364-5-54. Size the bonding conductor for telecommunications the same as the TBB.

~~[~~2.9.5 Service Entrance Cables

Service Entrance (SE) and Underground Service Entrance (USE) Cables, JIS C 3341 and JIS C 3605.

~~][~~2.9.6 EM-EEF Cable

JIS C 3605, Type EM-EEF Cable.

~~][~~2.9.7 Wire and Cable for 400 Hertz (Hz) Circuits

Insulated copper conductors.

~~][2.9.8 Metal-Clad Cable~~

~~Type MC cable.~~

~~][2.9.9 Armored Cable~~

~~Type AC cable.~~

~~][2.9.10 Mineral-Insulated, Metal-Sheathed Cable~~

~~UL listed; Type MI cable. Do not use sheathing containing asbestos fibers.~~

~~][2.9.11 Flat Conductor Cable~~

~~UL listed; Type FCC.~~

~~][2.9.12 Cable Tray Cable or Power Limited Tray Cable~~

~~UL listed; type TC or PLTC.~~

~~][2.9.13 Cord Sets and Power-Supply Cords~~

~~JIS C 8286.~~

~~][~~2.9.8 Polyethylene or Cross-Linked Polyethylene Cable

~~[~~Type ~~[~~CV~~][~~ or CE~~]~~ conforming to JIS C 3605 for 600V.~~][Type [CV][CE], conforming to JIS C 3606 for 6.6kV]~~

~~]~~2.10 SPLICES AND TERMINATION COMPONENTS

JIS C 2805, JIS C 2806 and JIS C 2810 for wire connectors and JIS C 2336, JIS C 2338 for insulating tapes. Connectors for 5.5 sqmm and smaller diameter wires: insulated, pressure-type in accordance with JIS C 2805, JIS C 2806 and JIS C 2810 or JIS C 2806 ~~(~~twist-on splicing connector~~)~~. Provide solderless terminal lugs on stranded conductors. Resin splices, where made, must be made of a rigid tube used to inject polyurethane insulating and sealing resin during a splice installation. Splices shall be performed in accordance with the manufacturer's instructions. The installation must not require heat or any additional materials such as covering.

2.11 DEVICE PLATES

Provide the following:

a. UL listed, one-piece device plates for outlets to suit the devices installed.

b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.

c. For nonmetallic boxes and fittings, other suitable plates may be provided.

~~[~~ d. Plates on finished walls: nylon or lexan, minimum 0.792 mm wall thickness and same color as receptacle or toggle switch with which they are mounted.

~~][e. Plates on finished walls: satin finish stainless steel or brushed-finish aluminum, minimum 0.792 mm thick.~~

~~]~~ f. Screws: machine-type with countersunk heads in color to match finish of plate.

g. Sectional type device plates are not be permitted.

h. Plates installed in wet locations: gasketed and UL listed or JIS listed for "wet locations."

~~[i. Device plates in areas normally accessible to prisoners: brown or ivory finish nylon device plates rated for high abuse. Test device plates for compliance with JIS C 8340 and JIS C 8435 for physical strength. Attach device plates with spanner head bolts.~~

~~]~~2.12 SWITCHES

2.12.1 Toggle Switches

JIS C 8304,~~[~~ single pole~~][~~, double pole~~][~~, three-way~~][~~, and four-way~~]~~, totally enclosed with bodies of thermoplastic or thermoset plastic and mounting strap with grounding screw. Include the following:

a. Handles: ~~[~~ white~~][ivory][brown]~~ thermoplastic.

b. Wiring terminals: screw-type, side-wired~~[~~ or of the solderless pressure type having suitable conductor-release arrangement~~]~~.

c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.

d. Switches: rated quiet-type ac only, ~~[105][120][210][240][~~120/277~~]~~ volts, with current rating and number of poles indicated.

2.12.2 Switch with Red Pilot Handle

Provide the following:

a. Pilot lights that are integrally constructed as a part of the switch's handle.

b. Pilot light color: red and illuminate whenever the switch is closed or "on".

c. Pilot lighted switch: rated 20 amps and ~~[105]~~~~[~~120~~]~~ volts or ~~[210][240][~~277~~]~~ volts as indicated.

d. The circuit's neutral conductor to each switch with a pilot light.

2.12.3 Breakers Used as Switches

For ~~[~~100~~][120]-~~ and ~~[~~200~~][277]~~-Volt fluorescent fixtures, mark breakers "SWD" in accordance with JIS C 8201-2-1. Provide label adjacent to circuit breaker indicating Breaker used as switch.

2.12.4 Disconnect Switches

JIS C 8201-3. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in IP rated enclosure~~[~~ as indicated~~]~~ per JIS C 8462-1.

2.13 FUSES

JIS C 8269-1, JIS C 8269-2. Provide complete set of fuses for each fusible ~~[~~ switch~~][~~ panel~~][~~ and control center~~]~~. Coordinate time-current characteristics curves of fuses serving motors or connected in series with circuit breakers~~[~~ or other circuit protective devices~~]~~ for proper operation. Submit coordination data for approval. Provide fuses with a voltage rating not less than circuit voltage.

2.13.1 Fuseholders

Provide in accordance with JIS C 8269-1 or JIS C 8269-2.

2.13.2 Cartridge Fuses, Current Limiting Type (Class R)

JIS C 8269-1 or JIS C 8269-2~~, [Class[RK-1][RK-5][time-delay type]][Fuse System [A][B][C][D][E][F][G][H][I][J][L]]~~. Provide only Class R associated fuseholders in accordance with JIS C 8269-1 or JIS C 8269-2.

2.14 RECEPTACLES

Provide the following:

a. ~~[~~JIS C 8303, JIS C 8281-1, general purpose specification grade (also designated heavy-duty),~~][~~ JIS C 8303, JIS C 8281-1 and JIS T 1021, hospital grade,~~]~~ grounding-type.

b. Ratings and configurations: as indicated.

c. Bodies: ~~[~~ white~~][ivory][brown]~~.

d. Face and body: thermoplastic supported on a metal mounting strap.

e. Dimensional requirements: per JIS C 8462-1.

f. Screw-type, side-wired wiring terminals or of the solderless pressure type having suitable conductor-release arrangement.

g. Grounding pole connected to mounting strap.

h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.

2.14.1 Switched Duplex Receptacles

Provide separate terminals for each ungrounded pole. Top receptacle: switched when installed.

2.14.2 Weatherproof Receptacles

Provide receptacles, UL listed for use in "wet locations" or weather resistant per JIS C 8303. Include cast metal box with gasketed, hinged, lockable and weatherproof while-in-use, ~~[~~polycarbonate, UV resistant/stabilized~~][~~ die-cast metal/aluminum~~]~~ cover plate.
The cover shall have notches or other openings that allow cords and plugs to be inserted into the receptacle while the cover is closed. (reason: this is an NEC requirement, but some JIS listed products do not have this feature)

2.14.3 Ground-Fault Circuit Interrupter Receptacles

Duplex type for mounting in standard outlet box. Provide device capable of detecting current leak of 6 milliamperes or greater and tripping for Class A ground-fault circuit interrupter devices. Provide screw-type, side-wired wiring terminals or pre-wired (pigtail) leads.

2.14.4 Special Purpose Receptacles

Receptacles serving ~~[_____]~~Tenant Dryers are special purpose.~~[~~ Provide in ratings indicated.~~][[_____] configuration, rated [_____] amperes, [_____] volts.][Furnish one matching plug with each receptacle.]~~

~~2.14.5 [Plugs~~

~~Provide heavy-duty, rubber-covered[three-,][four-,][or][five-]wire cord of required size, install plugs thereon, and attach to equipment. Provide UL listed plugs with receptacles, complete with grounding blades. Where equipment is not available, turn over plugs and cord assemblies to the Government.~~

~~]~~2.14.5 Range Receptacles

~~[ ] ampere~~Ampere receptacle configuration ~~[~~as indicated~~],[flush mounted for housing units,] rated 50 amperes, [105/210][125/250] volts~~.~~[Furnish one matching plug with each receptacle.]~~

2.14.6 Dryer Receptacles

Provide NEMA 14-30R receptacles.

2.14.7 Tamper-Resistant Receptacles

Provide duplex receptacle with mechanical sliding shutters that prevent the insertion of small objects into its contact slots.

2.15 PANELBOARDS

Provide panelboards in accordance with the following:

~~[~~ a. JIS C 8480 and JIS C 8480, JIS C 0920 having a short-circuit current rating~~[~~ as indicated~~][~~ of 10,000 amperes symmetrical minimum~~]~~.

b. Panelboards for use as service disconnecting means: additionally conform to manufacturer's recommendations.

c. Panelboards: circuit breaker-equipped.

d. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.

e. "Specific breaker placement" is required in panelboards to match the breaker placement indicated in the panelboard schedule on the drawings.

f. Use of "Subfeed Breakers" is not acceptable unless specifically indicated otherwise.

g. Main breaker: "separately" mounted~~[~~ "above"~~][~~ or~~][~~ "below"~~]~~ branch breakers.

h. Where "space only" is indicated, make provisions for future installation of breakers.

i. Directories: indicate load served by each circuit in panelboard.

j. Directories: indicate source of service to panelboard (e.g., Panel PA served from Panel MDP).

~~[~~ k. Provide new directories for existing panels modified by this project as indicated.

~~]~~ l. Type directories and mount in holder behind transparent protective covering.

~~[~~ m. Panelboards: listed and labeled for their intended use.

~~]~~ n. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

~~][~~o~~a~~. JIS C 8480 and JIS C 8480, JIS C 0920.

p~~b~~. Panelboards for use as service disconnecting: additionally conform to manufacturer's recommendations.~~.~~

q~~c~~. Panelboards: circuit breaker-equipped.

r~~d~~. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.

s~~e~~. Where "space only" is indicated, make provisions for future installation of breaker sized as indicated.

t~~f~~. Directories: indicate load served by each circuit of panelboard.

u~~g~~. Directories: indicate source of service (upstream panel, switchboard, motor control center, etc.) to panelboard.

v~~h~~. Type directories and mount in holder behind transparent protective covering.

w~~i~~. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

~~]~~2.15.1 Enclosure

Provide panelboard enclosure in accordance with the following:

a. JIS C 0920.

b. Cabinets mounted outdoors or flush-mounted: hot-dipped galvanized after fabrication.

c. Cabinets: painted in accordance with paragraph PAINTING.

d. Outdoor cabinets: Weatherproof rated IP code with~~[conduit hubs welded to the cabinet][~~ a removable steel plate 7 mm thick in the bottom for field drilling for conduit connections~~]~~.

e. Front edges of cabinets: form-flanged or fitted with structural shapes welded or riveted to the sheet steel, for supporting the panelboard front.

f. All cabinets: fabricated such that no part of any surface on the finished cabinet deviates from a true plane by more than 3 mm.

g. Holes: provided in the back of indoor surface-mounted cabinets, with outside spacers and inside stiffeners, for mounting the cabinets with a 15 mm clear space between the back of the cabinet and the wall surface.

h. Flush doors: mounted on hinges that expose only the hinge roll to view when the door is closed.

i. Each door: fitted with a combined catch and lock, except that doors over 600 mm long provided with a three-point latch having a knob with a T-handle, and a cylinder lock.

j. Keys: two provided with each lock, with all locks keyed alike.

k. Finished-head cap screws: provided for mounting the panelboard fronts on the cabinets.

2.15.2 Panelboard Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per JIS C 8480 for connecting grounding conductors; bond to steel cabinet.~~[~~ In addition to equipment grounding bus, provide second "isolated" ground bus, where indicated.~~]~~

~~[~~2.15.2.1 Panelboard Neutrals for Non-Linear Loads

Provide in accordance with the following:.

a. UL listed, with panelboard type specifically UL heat rise tested for use on non-linear loads.

b. Panelboard: heat rise tested in accordance with JIS C 8480, except with the neutral assembly installed and carrying 200 percent of the phase bus current during testing.

c. Verification of the testing procedure: provided upon request.

d. Two neutral assemblies paralleled together with cable is not acceptable.

e. Nameplates for panelboard rated for use on non-linear loads: marked "SUITABLE FOR NON-LINEAR LOADS" and in accordance with paragraph FIELD FABRICATED NAMEPLATES.

f. Provide a neutral label with instructions for wiring the neutral of panelboards rated for use on non-linear loads.

~~]~~2.15.3 Circuit Breakers

JIS C 8201-2-1,~~[~~ thermal magnetic-type~~][~~ solid state-type~~]~~ having an interrupting rating as shown on the drawings. Breaker terminals: UL listed as suitable for type of conductor provided.~~[~~ Where indicated on the drawings, provide circuit breakers with shunt trip devices.~~]~~ Series rated circuit breakers and plug-in circuit breakers are unacceptable.

2.15.3.1 Multipole Breakers

Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any three adjacent breaker poles are connected to Phases A, B, and C, respectively.

2.15.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter or Type RCBO

JIS C 8222 or JIS C 8201-2-2. ~~[~~Provide with "push-to-test" button, visible indication of tripped condition, and ability to detect and trip on current imbalance of 6 milliamperes or greater per requirements of JIS C 8222 or JIS C 8201-2-2 for Class A ground-fault circuit interrupter. ~~][~~ Provide Residual Current Device (RCBO) circuit breaker with a leakage current detection of 15 milliamperes.~~]~~

2.15.3.3 Arc-Fault Circuit Interrupters

JIS C 8201-2-1. Molded case circuit breakers: rated as indicated.~~[Two pole arc-fault circuit-interrupters: rated 120/240 volts. The provision of (two) one pole circuit breakers for shared neutral circuits in lieu of (one) two pole circuit breaker is unacceptable.]~~ Provide with "push-to-test" button.

~~[~~2.15.4 Fusible Switches for Panelboards

JIS C 8201-3, hinged door-type. Provide switches serving as motor disconnect means rated for kilowatt.

~~][2.15.5 400 Hz Panelboard and Breakers~~

~~Provide panelboards and breakers for use on 400 Hz systems rated and labeled "400 Hz."~~

~~[~~2.15.5 Branch Circuit Monitoring Panelboards

Provide a microprocessor-based panelboard monitoring system having the following features:

a. ANSI C12.1 and IEC 62053-21 Class 1 energy revenue metering accuracy.

b. Direct reading metered or calculated values for up to forty-two branch circuits.

c. Monitored values at the branch circuit level for current (A), power (kW), and energy (kWh).

d. Four user-configurable alarm thresholds.

e. Communications with building automation system using Modbus RTU protocol via RS-485 cable connection.

~~][~~2.15.6 Lighting Control Panelboards

Provided a lighting control panelboard having the following features:

a. Minimum sixteen schedules including a 7-day repeating schedule with sixteen daily on/off periods.

b. Minimum sixteen lighting zones grouping branch breakers that are controlled by schedules, manual inputs, or override commands.

c. Electronic clock including real-time, astronomical clock, and leap year and daylight savings time adjustments.

d. Burn-hour tracking.

e. Remote circuit breaker operation.

~~[~~ f. Master Lighting Control Panelboard with controller to control up to ~~[~~8 ~~] [______]~~ control bussed located ~~[individually][~~in slave panelboard~~]~~ up to ~~[~~400~~] [______]~~ feet away from the master panelboard.

~~]~~ g. Communications with building automation system using Modbus RTU protocol via RS-485 cable connection.

~~]][~~2.16 RESIDENTIAL LOAD CENTERS

Provide residential load centers (RLCs) in accordance with the following:

a. JIS C 8480 and JIS C 8480, JIS C 0920.

b. RLCs for use as service disconnecting means: additionally conform to manufacturer's recommendations.

c. Circuit breaker equipped.

d. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.

e. Where "space only" is indicated, make provisions for future installation of breakers sized as indicated.

~~[~~ f. Provide load centers with keyed locks.

~~]~~ g. Provide printed directories.

2.16.1 RLC Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated groundable neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per JIS C 8480 for connecting grounding conductors; bond to steel cabinet.

2.16.2 Circuit Breakers

JIS C 8201-2-1, thermal magnetic-type with interrupting capacity~~[~~ as indicated~~]~~. Breaker terminals: UL listed as suitable for the type of conductor provided.

2.16.2.1 Multipole Breakers

Provide common trip-type with single operating handle. Provide a breaker design such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any two adjacent breaker poles are connected to alternate phases in sequence.

~~[~~2.16.2.2 Circuit Breaker With Ground-Fault Circuit Interrupter or Type RCBO

JIS C 8222. ~~[~~Provide with "push-to-test" button, visible indication of tripped condition, and ability to detect and trip on current imbalance of 6 milliamperes or greater per requirements of JIS C 8222 for Class A ground-fault circuit interrupter devices.~~][~~ Provide Residual Current Device (RCBO) circuit breaker with a leakage current detection of 15 milliamperes or 30 milliamperes to provide ground fault protection.~~]~~

~~]~~2.16.2.3 Arc-Fault Circuit-Interrupters

JIS C 8201-2-1. Molded case circuit breakers: rated as indicated.~~[Two pole arc-fault circuit-interrupters: rated [105/210][120/240] volts. The provision of (two) one pole circuit breakers for shared neutral circuits in lieu of (one) two pole circuit breaker is unacceptable.]~~ Provide with "push-to-test" button.

~~][2.17 LOAD CENTERS FOR HOUSING UNITS~~

~~Provide single-phase panelboards for housing units on this project in~~

~~accordance with the following:~~

~~a. Load center type, circuit breaker equipped, conforming to JIS C 8480 and JIS C 8480 or JIS C 0920.~~

~~b. Panelboards series short-circuit current rating: 22,000 amperes symmetrical minimum for the main breaker and the branch breakers.~~

~~c. Panelboards for use as service disconnecting means: additionally conform to manufacturer's recommendations.~~

~~d. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.~~

~~e. "Specific breaker placement" is required in panelboards to match the breaker placement indicated in the panelboard schedule on the drawings.~~

~~f. Where "space only" is indicated, make provisions for future installation of breakers.~~

~~g. Provide cover with latching door.~~

~~h. Directories: indicate load served by each circuit in panelboard.~~

~~i. Directories: indicate source of service to panelboard (e.g., Panel PA served from panel MDP).~~

~~j. Type directories and mount behind in holder with transparent protective covering on inside of panel door.~~

~~2.17.1 Panelboard Buses~~

~~Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide copper or aluminum bus bars, either tin plated or silver plated. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per JIS C 8480 for connecting grounding conductors; bond to steel cabinet.~~

~~2.17.2 Circuit Breakers~~

~~JIS C 8201-2-1 thermal magnetic type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Half-size and tandem breakers are not acceptable. Provide switch duty rated 15 and 20 ampere breakers. Breakers must not require use of panel trim to secure them to the bus.~~

~~2.17.2.1 Multipole Breakers~~

~~Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any two adjacent breaker poles are connected to Phases A and B respectively.~~

~~2.17.2.2 Arc-Fault Circuit-Interrupters~~

~~JIS C 8201-2-1. Molded case circuit breakers: rated as indicated.[Two pole arc-fault circuit-interrupters: rated [1050/210][120/240] volts. The provision of (two) one pole circuit breakers for shared neutral circuits in lieu of (one) two pole circuit breaker is unacceptable.] Provide with "push-to-test" button.~~

~~]~~2.17 ENCLOSED CIRCUIT BREAKERS

JIS C 8201-2-1. Individual molded case circuit breakers with voltage and continuous current ratings, number of poles, overload trip setting, and short circuit current interrupting rating as indicated. Enclosure type as indicated.~~[Provide solid neutral.]~~

~~[~~2.18 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)

Motor short-circuit protectors, also called motor circuit protectors (MCPs): JIS C 8201-5-1 and JIS C 8201-2-1, and provided as shown. Provide MSCPs that consist of an adjustable instantaneous trip circuit breaker used only in conjunction with a combination motor controller which provides coordinated motor branch-circuit overload and short-circuit protection. Rate MSCPs in accordance with MLIT ESS.

~~]~~2.19 TRANSFORMERS

Provide transformers in accordance with the following:

a. JEC 2200 and JIS C 61558-1, general purpose, dry-type, self-cooled,~~[~~ ventilated~~][unventilated][sealed]~~.

b. Provide transformers in indoor rated or weatherproof IP rated enclosure.

c. Taps for transformers 15 kVA and larger: ~~[~~ Two 2.5 percent taps Full Capacity Above Nominal (FCAN) and four 2.5 percent taps Full Capacity Below Nominal (FCBN)~~] [Two 2.5 percent taps Full Capacity Above Nominal (FCAN) and two 2.5 percent taps Full Capacity Below Nominal (FCBN)] [_____]~~.

d. Transformer insulation system:

(1) 220 degrees C insulation system for transformers 15 kVA and greater, with temperature rise not exceeding~~[150][115][~~ 80~~]~~ degrees C under full-rated load in maximum ambient of 40 degrees C.

(2) 180 degrees C insulation for transformers rated 10 kVA and less, with temperature rise not exceeding~~[150][115][~~ 80~~]~~ degrees C under full-rated load in maximum ambient of 40 degrees C.

~~[e. Transformer of 150 degrees C temperature rise: capable of carrying continuously 100 percent of nameplate kVA without exceeding insulation rating.~~

~~][f. Transformer of 115 degrees C temperature rise: capable of carrying continuously 115 percent of nameplate kVA without exceeding insulation rating.~~

~~][g. Transformer of 80 degrees C temperature rise: capable of carrying continuously 130 percent of nameplate kVA without exceeding insulation rating.~~

~~][h. Transformers: quiet type with maximum sound level at least 3 decibels less than NEMA standard level for transformer ratings indicated.~~

~~]2.19.1 Specified Transformer Efficiency~~

~~Transformers, indicated and specified with: [420][440V][480V] primary, 80 degrees C or 115 degrees C temperature rise, kVA ratings of 37.5 to 100 for single phase or 30 to 500 for three phase, energy efficient type. Minimum efficiency, based on factory test results.~~

2.19.1 ~~[~~Transformers With Non-Linear Loads

Provide transformers for non-linear loads in accordance JIS C 61000-4-7.

~~]~~2.20 MOTORS

Provide motors in accordance with the following:

a. JIS C 4212~~[~~ except provide fire pump motors as specified in Section 21 30 00~~]~~ FIRE PUMPS.

b. Provide the size in terms of kW, or kVA, or full-load current, or a combination of these characteristics, and other characteristics, of each motor as indicated or specified.

c. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters.

d. Rate motors for operation on ~~[~~200-volt, 3-phase circuits with a terminal voltage rating of 200 volts, and those for operation on ~~[~~400 ~~][440]~~ volt, 3 phase circuits with a terminal voltage rating of ~~[~~400 ~~][440]~~ volts.~~][208-volt, 3-phase circuits with a terminal voltage rating of 200 volts, and those for operation on 480-volt, 3-phase circuits with a terminal voltage rating of 460 volts.]~~

e. Use motors designed to operate at full capacity with voltage variation of plus or minus 10 percent of motor voltage rating.

f. Unless otherwise indicated, use continuous duty type motors if rated 745 Watts and above.

h. Where fuse protection is specifically recommended by the equipment manufacturer, provide fused switches in lieu of non-fused switches indicated.

i. Use ~~[~~Inverter-Rated~~] [Inverter-Duty]~~ motors designed to operate with adjustable speed drive (ASD).

2.20.1 High Efficiency Single-Phase Motors

Single-phase fractional-horsepower alternating-current motors: high efficiency types corresponding to the applications listed in JIS C 4212. In exception, for motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.20.2 Premium Efficiency Polyphase Motors

Select polyphase motors based on high efficiency characteristics relative to typical characteristics and applications as listed in JIS C 4212. In addition, continuous rated, polyphase squirrel-cage medium induction motors must meet the requirements for premium efficiency electric motors in accordance with JIS C 4212, including the full load efficiency ratings. In exception, for motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.20.3 Motor Sizes

Provide size for duty to be performed, not exceeding the full-load nameplate current rating when driven equipment is operated at specified capacity under most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, make adjustments to wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. Provide controllers for motors rated 1-hp (746 watts) and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

2.20.4 Wiring and Conduit

Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide power wiring and conduit for field-installed equipment~~[~~, and motor control equipment forming part of motor control centers or switchgear assemblies, the conduit and wiring connecting such centers, assemblies, or other power sources to equipment~~]~~ as specified herein. Power wiring and conduit: conform to the requirements specified herein. Control wiring: provided under, and conform to, the requirements of the section specifying the associated equipment.

2.21 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

a. JIS C 8201-5-1, JIS C 8201-4-1 and JIS C 8201-4-2,~~[~~ except fire pump controllers as specified in Section 21 30 00 FIRE PUMPS~~]~~.

b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.

c. Provide controllers for motors rated 1-hp (746 kilowatt) and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.

d. Provide protection for motors from immediate restart by a time adjustable restart relay.

e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.

f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.

g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.

h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.

~~[~~ i. Provide selector switch with the means for locking in any position.

~~]~~ j. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.

l. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual reset-type pushbutton on outside of motor controller case.

m. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.

~~[n. Minimum short circuit withstand rating of combination motor controller: [_____] rms symmetrical amperes.~~

~~][o. Provide controllers in hazardous locations with classifications as indicated.~~

~~]~~2.21.1 Control Wiring

Provide control wiring in accordance with the following:

a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type CCE/F meeting JIS C 3401, or Type CEE/F meeting JIS C 3401, and passing the flame tests included in those standards.

b. Current transformer secondary leads: not smaller than 5.5 sqmm

c. Control wire minimum size: 1.6mm

d. Power wiring for ~~[~~400~~][440][480]~~-volt circuits and below: the same type as control wiring with 2.0mm minimum size.

e. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

2.21.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

a. JIS C 8201-7-1.

b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.

c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.

d. Terminals: not less than 5.5 sqmm in size with sufficient length and space for connecting at least two indented terminals for 5.5 sqmmconductors to each terminal.

e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four (4) spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.

f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.

g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity, and are constructed to assure positive contact between current-carrying parts.

2.21.2.1 Types of Terminal Blocks

a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.

b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide eEach connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

2.21.3 Control Circuits

~~[~~Control circuits: maximum voltage of ~~[105][~~120~~]~~ volts derived from control transformer in same enclosure. Transformers: conform to JIS C 6436, as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side.~~[Provide fuses in each ungrounded primary feeder].~~ Provide one fused secondary lead with the other lead grounded.~~[~~ For designated systems, as indicated, provide backup power supply, including transformers connected to~~[~~ emergency power source~~][_____]~~. Provide for automatic switchover and alarm upon failure of primary control circuit.~~]]~~

~~[Control circuits: maximum voltage of [105][120] volts derived from a separate control source. Provide terminals and terminal boards. Provide separate control disconnect switch within controller. Provide one fused secondary lead with the other lead grounded.[For designated systems, as indicated, provide backup power supply, including connection to[emergency power source][_____]. Provide for automatic switchover and alarm upon failure of primary control circuit.]]~~

2.21.4 Enclosures for Motor Controllers

JIS C 8462-1.

2.21.5 Multiple-Speed Motor Controllers and Reversible Motor Controllers

Across-the-line-type, electrically and mechanically interlocked. Multiple-speed controllers: include compelling relays and multiple-button, station-type with pilot lights for each speed.

2.21.6 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

2.21.7 Pilot and Indicating Lights

~~[Provide LED cluster lamps.][~~Provide transformer, resistor, or diode type.~~]~~

~~[2.21.8 Reduced-Voltage Controllers~~

~~Provide for polyphase motors [_____] kilowatt and larger. Reduced-voltage starters: single-step, closed transition[autotransformer,][reactor,][primary resistor-type,][solid state-type,] or as indicated, with an adjustable time interval between application of reduced and full voltages to motors.[Wye-delta reduced voltage starter or part winding increment starter having adjustable time delay between application of voltage to first and second winding of motor may be used in lieu of the reduced-voltage starters for starting of[motor-generator sets,][centrifugally operated equipment,][or][reciprocating compressors provided with automatic unloaders].]~~

~~]~~2.22 MANUAL MOTOR STARTERS (MOTOR RATED SWITCHES)

~~[Single][Double][~~Three~~]~~ pole designed for~~[flush][~~ surface~~]~~ mounting with overload protection~~[and pilot lights]~~.

2.22.1 Pilot Lights

~~[~~ Provide yoke-mounted, seven element LED cluster light module. Color:~~[green][red][amber][~~ in accordance with JIS C 8201-4-1 and JIS C 8201-4-2~~]~~. ~~][Provide yoke-mounted, candelabra-base sockets rated 125 volts and fitted with glass or plastic jewels. Provide clear, 6 watt lamp in each pilot switch. Jewels for use with switches controlling motors: green; jewels for other purposes:[white][red][amber].~~

~~]~~2.23 MOTOR CONTROL CENTERS

Provide motor control centers in accordance with the following:

a. JIS C 8201-4-1, JIS C 8201-4-1 and JIS C 8201-4-2.

b. Wiring: in ~~[~~indoor~~][weatherproof][]~~ IP rated enclosure per JIS C 0920.

c. Provide control centers suitable ~~for operation on [_____]-volt, [_____]-phase, [_____]-wire, [_____]~~ Hz system with minimum short-circuit withstand and interrupting rating of~~[100,000][65,000][42,000][25,000][_____]~~22,000 amperes rms symmetrical.

d. Incoming power feeder: ~~[bus duct][~~ cable~~]~~ entering at the~~[~~ top~~][bottom]~~ of enclosure and terminating on~~[terminal lugs][~~ main protective device~~]~~.

~~[~~ e. Main protective device: ~~[~~ molded case circuit breaker~~][low-voltage power circuit breaker][fusible switch]~~ rated at ~~[_____]~~22,000 amperes rms symmetrical interrupting capacity.

~~][f. Arrange busing so that control center can be expanded from both ends.~~

~~] g. Interconnecting wires: copper.~~

~~h. Terminal blocks: plug-in-type so that controllers may be removed without disconnecting individual control wiring.~~

~~2.23.1 Bus Systems~~

~~Provide the following bus systems. Power bus: be braced to withstand fault current of[100,000][65,000][42,000][25,000][_____] amperes rms symmetrical. Wiring troughs: isolated from horizontal and vertical bus bars.~~

~~2.23.1.1 Horizontal and Main Buses~~

~~Horizontal bus: continuous current rating of[600][800][1000][1200][_____] amperes. Main bus: [aluminum, tin-plated][copper, silver-plated] enclosed in isolated compartment at top of each vertical section. Main bus: isolated from wire troughs, starters, and other areas.~~

~~2.23.1.2 Vertical Bus~~

~~Vertical bus: continuous current rating of[300][450][600][_____] amperes, and [aluminum, tin-plated][copper, tin-plated][copper, silver-plated]. Vertical bus: enclosed in flame-retardant, polyester glass "sandwich."~~

~~2.23.1.3 Ground Bus~~

~~Copper ground bus: provided full width of motor control center and equipped with necessary lugs.~~

~~[2.23.1.4 Neutral Bus~~

~~Insulated neutral bus: provided continuous through the motor control~~

~~center; neutral full rated. Provide lugs of appropriate capacity, as required.~~

~~][2.23.2 Combination Motor Controllers~~

~~JIS C 8201-5-1 and other requirements in paragraph, MOTOR CONTROLLERS. Provide in controller a[molded case circuit breaker][fusible switch with clips for [_____]-type fuses for branch circuit protection].[Minimum short circuit withstand rating of combination motor controller: [_____] rms symmetrical amperes.][Circuit breakers for combination controllers: [thermal magnetic][magnetic only].]~~

~~][2.23.3 Space Heaters~~

~~Provide space heaters where indicated on the drawings, controlled using an adjustable 10 to 35 degrees C thermostat, magnetic contactor, and a molded-case circuit breaker[and a [480][400]-[120][100] volt single-phase transformer]. Provide space heaters equipped with 250-watt, [210][240] volt strip elements operated at [105][120] volts and [supplied from the motor control center bus][wired to terminal blocks for connection to [105][120]-volt single-phase power sources located external to the control centers]. Contactors: open type, electrically-held, rated 30 amperes, 2-pole, with 120-volt ac coils.~~

~~]~~2.24 LOCKOUT REQUIREMENTS

Provide disconnecting means capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy in accordance with MLIT ESS. Comply with requirements of Division 23, "Mechanical" for mechanical isolation of machines and other equipment.

2.25 TELECOMMUNICATIONS SYSTEM

Provide system of telecommunications wire-supporting structures (pathway), including: outlet boxes, conduits with pull wires~~[~~ wireways,~~][~~ cable trays,~~]~~ and other accessories for telecommunications outlets and pathway in accordance with the drawings and as specified herein.~~[~~ Additional telecommunications requirements are specified in Section 27 10 00, BUILDING TELECOMMUNICATIONS CABLING SYSTEM.~~]~~

~~[~~2.26 COMMUNITY ANTENNA TELEVISION (CATV) SYSTEM

~~[Additional CATV requirements are specified in[Section 27 54 00.00 20, COMMUNITY ANTENNA TELEVISION (CATV) SYSTEMS.][Section 27 05 14.00 10, CABLE TELEVISION PREMISES DISTRIBUTION SYSTEM.]]~~

~~[~~2.26.1 CATV Outlets

Provide flush mounted, 75-ohm, F-type connector outlet rated from 5 to 1000 MHz in standard electrical outlet boxes~~[with isolation barrier]~~ with mounting frame.

~~][~~2.26.2 CATV Faceplates

Provide modular faceplates for mounting of CATV Outlets.~~[~~ Faceplate: include designation labels and label covers for circuit identification.~~]~~ Faceplate color: match outlet and switch coverplates.

~~][~~2.26.3 Backboards

~~[~~Provide void-free, fire rated interior grade plywood~~, 19 mm thick,[1200 by 2400 mm][~~ as indicated~~]~~. Do not cover the fire stamp on the backboard. ~~][~~ Coordinate CATV backboard requirements with telecommunications backboard requirements as specified in Section 27 10 00, BUILDING TELECOMMUNICATIONS CABLING.~~]~~

~~]]~~2.27 GROUNDING AND BONDING EQUIPMENT

2.27.1 Ground Rods

JIS C 60364-5-54. Ground rods: ~~[~~ copper-clad steel~~][solid copper][stainless steel],~~ with minimum diameter of 14mm and minimum length 1500mm. Sectional ground rods are permitted.

~~[~~2.27.2 Ground Bus

Copper ground bus: provided in the electrical equipment rooms as indicated.

~~]~~2.27.3 Telecommunications ~~[~~and CATV~~]~~ Grounding Busbar

Provide corrosion-resistant grounding busbar suitable for~~[~~ indoor~~][outdoor]~~ installation in accordance with JIS C 60364-5-54. Busbars: plated for reduced contact resistance. If not plated, clean the busbar prior to fastening the conductors to the busbar and apply an anti-oxidant to the contact area to control corrosion and reduce contact resistance. Provide a telecommunications main grounding busbar (TMGB) in the telecommunications entrance facility~~[~~ and a (TGB) in all other telecommunications rooms and equipment rooms~~]~~. The telecommunications main grounding busbar (TMGB)~~[and the telecommunications grounding busbar (TGB)]~~: sized in accordance with the immediate application requirements and with consideration of future growth. Provide telecommunications grounding busbars with the following:

a. Predrilled copper busbar provided with holes for use with standard sized lugs,

b. Minimum dimensions of 6 mm thick by 100 mm wide for the TMGB~~[and 50 mm wide for TGBs]~~ with length as indicated;

c. Listed by a nationally recognized testing laboratory.

~~[2.28 HAZARDOUS LOCATIONS~~

~~Electrical materials, equipment, and devices for installation in hazardous locations, as defined by JIS: specifically approved by Underwriters' Laboratories, Inc., or Factory Mutual for particular "Class," "Division," and "Group" of hazardous locations involved. Boundaries and classifications of hazardous locations: as indicated. Equipment in hazardous locations: comply with JIS C 60079-0 and JIS C 60079-14 for electrical equipment and industrial controls and JIS C 60079-0 and JIS C 60079-14for motors.~~

~~]~~2.28 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.29 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

a. JIS K 6911.

b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.

c. Each nameplate inscription: identify the function and, when applicable, the position.

d. Nameplates: melamine plastic, 3 mm thick, white with ~~[~~black~~] [_____]~~ center core.

~~[e. Provide red laminated plastic label with white center core where indicated.~~

~~]~~ f. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.

g. Minimum size of nameplates: 25 by 65 mm.

h. Lettering size and style: a minimum of 6.35 mm high normal block style.

2.30 WARNING SIGNS

Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer in accordance with NFPA 70E and JIS Z 9101 for switchboards, panelboards, industrial control panels, and motor control centers. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.31 FIRESTOPPING MATERIALS

Provide firestopping around electrical penetrations in accordance with Section 07 84 00, FIRESTOPPING.

2.32 WIREWAYS

Material: steel~~[epoxy painted][~~ galvanized~~]~~ 16 gauge for heights and depths up to 150 by 150 mm, and 14 gauge for heights and depths up to 305 by 305 mm. Provide in length~~[indicated][~~ required for the application~~]~~ with~~[~~ hinged-~~][screw-]~~ cover enclosure per ~~[~~indoor ~~][weatherproof][hazardous]~~ IP rated enclosure per JIS C 8462-1.

~~[~~2.33 METERING

JIS C 1210. Provide a self-contained, socket-mounted, electronic programmable outdoor watthour meter. Meter: either programmed at the factory or programmed in the field. Turn field programming device over to the Contracting Officer at completion of project. Coordinate meter to system requirements. ~~[~~Metering shall be compliant with the current

Advanced Meter Reading System (AMRS) specification.~~]~~

a. Design: Provide watthour meter designed for use on a single-phase, three-wire,~~[210/105][~~ 240/120~~][440/254][420/242][480/240]~~ volt system. Include necessary KYZ pulse initiation hardware for Energy Monitoring and Control System (EMCS).

b. Class: 200; Form: ~~[~~ 2S~~][_____]~~, accuracy: plus or minus 1.0 percent; Finish: Class II.

c. Cover: Polycarbonate and lockable to prevent tampering and unauthorized removal.

d. Kilowatt-hour Register: five digit electronic programmable type.

e. Demand Register:

(1) Provide solid state.

(2) Meter reading multiplier: Indicate multiplier on the meter face.

(3) Demand interval length: programmed for~~[~~ 15~~][30][60]~~ minutes with rolling demand up to six subintervals per interval.

f. Socket: JIS C 1210. Provide ~~[weatherproof][]~~ IP rated, box-mounted socket, ringless, having~~[~~ manual circuit-closing bypass and having~~]~~ jaws compatible with requirements of the meter. Provide manufacturers standard enclosure color unless otherwise indicated.

~~][~~2.34 METER BASE ONLY

JIS C 1210. Provide ~~[~~weatherproof~~][]~~ IP rated, box-mounted socket, ringless, having jaws compatible with requirements of a class: 200 and Form: ~~[~~ 2S~~][_____]~~ self contained watthour meter. Provide gray plastic closing cover and bypass links. Provide manufacturers standard enclosure color unless otherwise indicated.

~~]~~2.35 SURGE PROTECTIVE DEVICES

Provide surge protective devices (SPD) which comply with JIS C 5381-11 and JIS C 5381-12 at the service entrance~~[~~, load centers~~][~~, panelboards~~][~~, MCC~~][and][_____]~~. Provide surge protectors in a ~~[~~indoor~~][weatherproof]~~ IP rated enclosure per JIS C 8462-1. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker.

~~[~~ Provide SPDs per JIS Z 9290-1 for the lightning protection system.

~~]~~2.36 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

a. JIS C 0920 corrosion-resistance test and the additional requirements as specified herein.

b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.

c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.

d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.

e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.

f. Equipment located indoors: Light Gray,~~[~~ and equipment located outdoors: ~~[Light Gray][~~ Dark Gray~~]]~~.

g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.37 SOURCE QUALITY CONTROL

2.37.1 Transformer Factory Tests

Submittal: include routine JEC 2200 and JIS C 61558-1 transformer test results on each transformer and also provide the results of Japanese standard "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

~~[2.38 COORDINATED POWER SYSTEM PROTECTION~~

~~Prepare analyses as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.~~

~~]~~PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of MLIT ESS, JIS C 0365 and to requirements specified herein.

~~[~~3.1.1 Underground Service

Underground service conductors and associated conduit: continuous from service entrance equipment to outdoor power system connection.

~~][3.1.2 Overhead Service~~

~~Overhead service conductors into buildings: terminate at service entrance fittings or weatherhead outside building. Overhead service conductors and support bracket for overhead conductors are included in[Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION.]~~

~~][3.1.3 Hazardous Locations~~

~~Perform work in hazardous locations, as defined by applicable codes and standards, for particular "Class," "Division," and "Group" of hazardous locations involved. Provide conduit and cable seals where required. Provide conduit with tapered threads.~~

3.1.2 Service Entrance Identification

Service entrance disconnect devices, switches, and enclosures: labeled and identified as such.

3.1.2.1 Labels

Wherever work results in service entrance disconnect devices in more than one enclosure, label each enclosure, new and existing, as one of several enclosures containing service entrance disconnect devices. Label, at minimum: indicate number of service disconnect devices housed by enclosure and indicate total number of enclosures that contain service disconnect devices. Provide laminated plastic labels conforming to paragraph FIELD FABRICATED NAMEPLATES. Use lettering of at least 6.35 mm in height, and engrave on black-on-white matte finish. Service entrance disconnect devices in more than one enclosure: provided only as permitted.

3.1.3 Wiring Methods

Provide insulated conductors installed in rigid steel conduit or type G, IMC or type C, rigid nonmetallic conduit or Unplasticized Polyvinyl Chloride, or EMT or Type E, except where specifically indicated or specified otherwise or required to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Shared neutral, or multi-wire branch circuits, are not permitted with arc-fault circuit interrupters. Minimum conduit size: 16 mm in diameter for low voltage lighting and power circuits. Vertical distribution in multiple story buildings: made with metal conduit in fire-rated shafts, with metal conduit extending through shafts for minimum distance of 150 mm. Firestop conduit which penetrates fire-rated walls, fire-rated partitions, or fire-rated floors in accordance with Section 07 84 00, FIRESTOPPING.

3.1.3.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 890-N force tensile strength. Leave minimum 915 mm of slack at each end of pull wire.

3.1.3.2 Metal Clad Cable

Install in accordance with manufacturer's requirements, Type MC cable.

3.1.3.3 Armored Cable

Install in accordance with manufacturer's requirements, Type AC cable.

3.1.3.4 Flat Conductor Cable

Install in accordance with manufacturer's requirements, Type FCC cable.

3.1.4 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Keep conduit minimum 150 mm away from parallel runs of flues and steam or hot water pipes. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project.~~[~~ Run conduits~~[~~ in crawl space~~][~~ under floor slab~~]~~ as if exposed.~~]~~

3.1.4.1 Restrictions Applicable to EMT or Type E

a. Do not install underground.

b. Do not encase in concrete, mortar, grout, or other cementitious materials.

c. Do not use in areas subject to severe physical damage including but not limited to equipment rooms where moving or replacing equipment could physically damage the EMT or Type E conduit.

d. Do not use in hazardous areas.

e. Do not use outdoors.

f. Do not use in fire pump rooms.

g. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

~~[~~3.1.4.2 Restrictions Applicable to Nonmetallic Conduit or Type VE, HIVE, VP, HIVP

a. PVC Schedule 40 and PVC Schedule 80 or Type VE, HIVE, VP, HIVP

(1) Do not use in areas where subject to severe physical damage, including but not limited to, mechanical equipment rooms, electrical equipment rooms, hospitals, power plants, missile magazines, and other such areas.

(2) Do not use in hazardous (classified) areas.

(3) Do not use in fire pump rooms.

(4) Do not use in penetrating fire-rated walls or partitions, or fire-rated floors.

(5) Do not use above grade, except where allowed in this section for rising through floor slab or indicated otherwise.

(6) Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

~~]~~3.1.4.3 Restrictions Applicable to Flexible Conduit

Use only as specified in paragraph FLEXIBLE CONNECTIONS. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.4.4 Underground Conduit

Plastic-coated rigid steel; plastic-coated steel IMC or type C, LL or LT; PVC, Type EPC-40 or Type VE, HIVE, VP, HIVP. Convert nonmetallic conduit, other than PVC Schedule 40 or 80 or Type VE, HIVE, VP, HIVP, to plastic-coated rigid, or IMC, steel conduit before rising through floor slab. Plastic coating: extend minimum 150 mm above floor.

~~[3.1.4.5 Conduit Interior to Buildings for 400 Hz Circuits~~

~~Aluminum or nonmetallic. Where 400-Hz circuit runs underground or through concrete, provide PVC Schedule[40][80] or Type VE, HIVE, VP, HIVP conduit.~~

~~]~~3.1.4.5 Conduit for Circuits Rated Greater Than 600 Volts

Rigid metal conduit or Type G only.

3.1.4.6 Conduit Installed Under Floor Slabs

Conduit run under floor slab: located a minimum of ~~[~~305~~] [_____]~~ mm below the vapor barrier. Seal around conduits at penetrations thru vapor barrier.

3.1.4.7 Conduit Through Floor Slabs

Where conduits rise through floor slabs, do not allow curved portion of bends to be visible above finished slab.

~~[~~3.1.4.8 Conduit Installed in Concrete Floor Slabs

~~[~~Rigid steel or Type G; steel IMC or Type C; fiberglass, or PVC, Type EPC-40~~.][PVC, Type EPC-40,~~ unless indicated otherwise.~~]~~ Locate so as not to adversely affect structural strength of slabs. Install conduit within middle one-third of concrete slab.~~[~~ Do not stack conduits.~~][Do not stack conduits more than two diameters high with minimum vertical separation of [_____] mm.]~~ Space conduits horizontally not closer than three diameters, except at cabinet locations. Curved portions of bends must not be visible above finish slab. Increase slab thickness as necessary to provide minimum 25 mm cover over conduit. Where embedded conduits cross building and/or expansion joints, provide suitable watertight expansion/deflection fittings and bonding jumpers. Expansion/deflection fittings must allow horizontal and vertical movement of raceway. Conduit larger than 27 mm trade size: installed parallel with or at right angles to main reinforcement; when at right angles to reinforcement, install conduit close to one of supports of slab.~~[~~ Where nonmetallic conduit is used, convert raceway to plastic coated rigid steel or Type G or plastic coated steel IMC or Type C; LL or LT before rising above floor, unless specifically indicated.~~]~~

~~]~~3.1.4.9 Stub-Ups

Provide conduits stubbed up through concrete floor for connection to free-standing equipment with adjustable top or coupling threaded inside for plugs, set flush with finished floor. Extend conductors to equipment in rigid steel conduit, except that flexible metal conduit may be used 150 mm above floor. Where no equipment connections are made, install screwdriver-operated threaded flush plugs in conduit end.

3.1.4.10 Conduit Support

Support conduit by pipe straps, wall brackets, threaded rod conduit hangers, or ceiling trapeze. Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to fasteners. Provide vibration resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 40 mm in reinforced concrete beams or to depth of more than 20 mm in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts. Coordinate installationwith above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations.~~[~~ Support exposed risers in wire shafts of multistory buildings by U-clamp hangers at each floor level and at 3050 mm maximum intervals.~~]~~ Where conduit crosses building expansion joints, provide suitable~~[watertight]~~ expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 63 mm inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction. Type E conduit shall be secured and supported at least 1 meter from terminations.

3.1.4.11 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

3.1.4.12 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required.

3.1.4.13 Flexible Connections

Provide flexible steel conduit between 915 and 1830 mm in length for recessed and semirecessed lighting fixtures~~[~~; for equipment subject to vibration, noise transmission, or movement; and for motors~~]~~. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 16 mm diameter. Provide liquidtight flexible~~[~~ nonmetallic~~]~~ conduit in wet and damp locations~~[~~ and in fire pump rooms~~]~~ for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections.

3.1.4.14 Telecommunications and Signal System Pathway

Install telecommunications pathway in accordance with JIS X 5150.

a. Horizontal Pathway: Telecommunications pathways from the work area to the telecommunications room: installed and cabling length requirements in accordance with JIS X 5150. Size conduits[, wireways][, and cable trays] in accordance with JIS X 5150~~[and][as indicated]~~.

b. Backbone Pathway: Telecommunication pathways from the telecommunications entrance facility to telecommunications rooms, and, telecommunications equipment rooms (backbone cabling): installed in accordance with JIS X 5150. Size conduits[, wireways][, and cable trays] for telecommunications risers in accordance with JIS X 5150[and][as indicated].

[3.1.4.15 Community Antenna Television (CATV) System Conduits

Install a system of CATV wire-supporting structures (pathway), including: outlet boxes, conduits with pull wires[wireways,][cable trays,] and other accessories for CATV outlets and pathway in accordance with JIS X 5150. [Provide distribution system with star topology with empty conduit and pullwire from each outlet box to the telecommunications room and empty conduit and pullwire from each telecommunications room to the headend equipment location]~~[Provide distribution system with star topology with empty conduit and pullwire from each outlet to the headend equipment location]~~.

]3.1.5 Busway Installation

Install busways parallel with or at right angles to ceilings, walls, and structural members. Support busways at 1525 mm maximum intervals, and brace to prevent lateral movement. Provide fixed type hinges on risers; spring-type are unacceptable. Provide flanges where busway makes penetrations through walls and floors, and seal to maintain smoke and fire ratings. Provide waterproof curb where busway riser passes through floor. Seal gaps with fire-rated foam and caulk. Provide expansion joints, but only where bus duct crosses building expansion joints. Provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.6 Cable Tray Installation

[Install and ground in accordance with manufacturer's instructions.[In addition, install and ground telecommunications cable tray in accordance with JIS X 5150, and JIS C 60364-5-54]. Install cable trays parallel with or at right angles to ceilings, walls, and structural members. Support[in accordance with manufacturer recommendations but at not more than [1830] [_____] mm intervals][as indicated].[Coat contact surfaces of aluminum connections with an antioxidant compound prior to assembly.] Adjacent cable tray sections: bonded together by connector plates of an identical type as the cable tray sections. For grounding of cable tray system provide 38 sqmm bare copper wire throughout cable tray system, and bond to each section, except use 60 sqmm aluminum wire if cable tray is aluminum. Terminate cable trays 255 mm from both sides of smoke and fire partitions. Install conductors run through smoke and fire partitions in 103 mm rigid steel conduits with grounding bushings, extending 305 mm beyond each side of partitions. Seal conduit on both ends to maintain smoke and fire ratings of partitions. Firestop penetrations in accordance with Section 07 84 00, FIRESTOPPING. Provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

][Install cable trays parallel with or at right angles to ceilings, walls, and structural members. Support[as indicated~~][at maximum [1830] [_____] mm] intervals~~.[In addition, install and ground telecommunications cable tray in accordance with JIS X 5150, and JIS C 60364-5-54][. Coat contact surfaces of aluminum connections with an antioxidant compound prior to assembly.] Ensure edges, fittings, and hardware are finished free from burrs and sharp edges. Provide 38 sqmm AWG bare copper wire throughout cable tray system, and bond to each section. Use 60 sqmm aluminum wire if cable tray is aluminum. Install conductors that run though smoke and fire partitions in 103 mm rigid steel conduits with grounding bushing, extending 305 mm beyond each side of partitions. Seal conduit on both ends to maintain smoke and fire ratings of partitions. Provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

][3.1.7 Telecommunications Cable Support Installation

Install open top and closed ring cable supports on 1.2 m to 1.5 m centers to adequately support and distribute the cable’s weight. Use these types of supports to support a maximum of 50 6.4 mm diameter cables. Install suspended cables with at least 75 mm of clear vertical space above the ceiling tiles and support channels (T-bars). Open top and closed ring cable supports: suspended from or attached to the structural ceiling or walls with hardware or other installation aids specifically designed to support their weight.

]3.1.8 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces,[when surface mounted on interior walls exposed up to 2135 mm above floors and walkways,]~~[or when installed in hazardous areas]~~ and when specifically indicated. Boxes in other locations: sheet steel, except that aluminum boxes may be used with aluminum conduit, and nonmetallic boxes may be used with nonmetallic~~[sheathed cable]~~ conduit system. Provide each box with volume required for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 100 mm square, or octagonal, except that smaller boxes may be installed as required by fixture configurations, as approved. Boxes for use in masonry-block or tile walls: square-cornered, tile-type, or standard boxes having square-cornered, tile-type covers. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixturesfor access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel.~~[Threaded studs driven in by powder charge and provided with lockwashers and nuts[or nail-type nylon anchors] may be used in lieu of wood screws, expansion shields, or machine screws.]~~ In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 610 mm from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

3.1.8.1 Boxes

Boxes for use with raceway systems: minimum 40 mm deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 100 mm square, except that 100 by 50 mm boxes may be used where only one raceway enters outlet. Telecommunications outlets: a minimum of~~[~~ 100 mm square by 54 mm deep~~][120 mm square by 54 mm deep][~~, except for ~~[~~wall mounted telephones~~] [and] [outlet boxes for handicap telephone stations]]~~. Mount outlet boxes flush in finished walls.

3.1.8.2 Pull Boxes

Construct of at least minimum size required~~ [~~ of code-gauge aluminum or galvanized sheet steel,~~][and][compatible with nonmetallic raceway systems,]~~ except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

~~[~~3.1.8.3 Extension Rings

Extension rings are not permitted for new construction. Use only on existing boxes in concealed conduit systems where wall is furred out for new finish.

~~]~~3.1.9 Mounting Heights

Mount panelboards,~~[~~ enclosed~~]~~ circuit breakers,~~[motor controller]~~ and disconnecting switches so height of operating handle at its highest position is maximum 1980 mm above floor. Mount lighting switches~~[~~ and handicapped telecommunications stations~~][~~ 1220 mm above finished floor~~]~~. Mount receptacles~~[~~ and telecommunications outlets~~]~~ 460 mm above finished floor~~[~~, unless otherwise indicated~~]~~.~~[~~ Wall-mounted telecommunications outlets: mounted at height~~[1525 mm above finished floor][~~ indicated~~]~~.~~] [~~ Mount other devices as indicated.~~][~~ Measure mounting heights of wiring devices and outlets~~[in non-hazardous areas]~~to center of device or outlet. ~~][Measure mounting heights of receptacle outlet boxes in the[hazardous area][_____] to the bottom of the outlet box.]~~

~~[~~3.1.10 Nonmetallic Sheathed Cable Installation

Where possible, install cables concealed behind ceiling or wall finish. Thread cables through holes bored on approximate centerline of wood members; notching of end surfaces is not permitted. Provide sleeves through concrete or masonry for threading cables. Install exposed cables parallel to or at right angles to walls or structural members. Protect exposed nonmetallic sheathed cables less than 1220 mm above floors from mechanical injury by installation in conduit or tubing. When cable is used in metal stud construction, insert plastic stud grommets in studs at each point through which cable passes, prior to installation of cable.

~~][~~3.1.11 Mineral Insulated, Metal Sheathed (Type MI) Cable Installation

Mineral-insulated, metal-sheathed cable system, Type MI, may be used in lieu of exposed conduit and wiring. Conductor sizes: not less than those indicated for the conduit installation. Fasten cables within 305 mm of each turn or offset and at 830 mm maximum intervals. Make cable terminations in accordance with cable manufacturer's recommendations. Terminate single-conductor cables of a circuit, having capacities of more than 50 amperes, in a single box or cabinet opening. Color code individual conductors in all outlets and cabinets.

~~]~~3.1.12 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors 14 sqmm and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors 22 sqmm and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and plates; or heat shrink-type sleeves. Identify control circuit terminations in accordance with~~[Section 23 09 53.00 20 SPACE TEMPERATURE CONTROL SYSTEMS.][Section [_____], [_____]][Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC][~~ manufacturer's recommendations~~]~~.~~[~~ Provide telecommunications system conductor identification as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS.~~]~~

3.1.12.1 Marking Strips

Provide marking strips in accordance with the following:

a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.

b. Use permanent ink for the wire numbers

c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.

d. Size marking strips to accommodate the two sets of wire numbers.

e. Assign a device designation in accordance with JIS C 8201-4-1 and JIS C 8201-4-2 to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.

f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external) circuits for the Government's wire designations.

g. Prints of the marking strips drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

3.1.13 Splices

Make splices in accessible locations. Make splices in conductors 5.5 sqmm and smaller diameter with insulated, pressure-type connector. Make splices in conductors 8 sqmm and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

3.1.14 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of 0.58 mm. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

3.1.15 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated walls, partitions, floors, or ceilings in accordance with Section 07 84 00 FIRESTOPPING.

3.1.16 Grounding and Bonding

~~[~~Provide in accordance with JIS Z 9290-1~~]~~. Ground exposed, non-current-carrying metallic parts of electrical equipment,~~[access flooring support system,]~~ metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, ~~[~~ grounding conductor of nonmetallic sheathed cables, ~~]~~and neutral conductor of wiring systems.~~[~~ Make ground connection at main service equipment, and extend grounding conductor to point of entrance of metallic water service. Make connection to water pipe by suitable ground clamp or lug connection to plugged tee. If flanged pipes are encountered, make connection with lug bolted to street side of flanged connection. Supplement metallic water service grounding system with additional made electrode.~~][~~ Make ground connection to driven ground rods on exterior of building.~~][~~ Interconnect all grounding media in or on the structure to provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground metallic piping systems. Make interconnection to the gas line on the customer's side of the meter. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection system.~~]~~ In addition to the requirements specified herein, provide telecommunications grounding in accordance with JIS C 60364-5-54. Where ground fault protection is employed, ensure that connection of ground and neutral does not interfere with correct operation of fault protection. If a splice is made inside a metal box, then the box shall be bonded to the equipment grounding conductor.

3.1.16.1 Ground Rods

Provide cone pointed ground rods. Measure the resistance to ground using the fall-of-potential method described in JIS C 60364-6. Do not exceed 25 ohms under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained with a single rod,~~[_____]~~ additional rods, spaced on center, not less than twice the distance of the length of the rod~~,[or if sectional type rods are used,[_____] additional sections may be coupled and driven with the first rod]~~.~~[~~ In high-ground-resistance, UL listed chemically charged ground rods may be used.~~]~~ If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

3.1.16.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible,~~[~~ excepting specifically those connections for which access for periodic testing is required,~~]~~ by exothermic weld or compression connector.

a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.

b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

3.1.16.3 Ground Bus

Provide a copper ground bus in the electrical equipment rooms as indicated. Noncurrent-carrying metal parts of~~[~~ transformer neutrals and other electrical~~][electrical]~~ equipment: effectively grounded by bonding to the ground bus. Bond the ground bus to both the entrance ground, and to a ground rod or rods as specified above having the upper ends terminating approximately 100 mm above the floor. Make connections and splices of the brazed, welded, bolted, or pressure-connector type, except use pressure connectors or bolted connections for connections to removable equipment.~~[For raised floor equipment rooms in computer and data processing centers, provide a minimum of 4, one at each corner, ground buses connected to the building grounding system. Use bolted connections in lieu of thermoweld, so they can be changed as required by additions and/or alterations.]~~

3.1.16.4 Resistance

Maximum resistance-to-ground of grounding system: do not exceed~~[~~ 5~~][_____]~~ ohms under dry conditions. Where resistance obtained exceeds~~[~~ 5~~][_____]~~ ohms, contact Contracting Officer for further instructions.

3.1.16.5 Telecommunications System

Provide telecommunications grounding in accordance with the following:

a. Telecommunications Grounding Busbars: Provide a telecommunications main grounding busbar (TMGB) in the telecommunications entrance facility. Install the TMGB as close to the electrical service entrance grounding connection as practicable.~~[~~ Provide a telecommunications grounding busbar (TGB) in all other telecommunications rooms and telecommunications equipment rooms. Install the TGB as close to the telecommunications room panelboard as practicable, when equipped. Where a panelboard for telecommunications equipment is not installed in the telecommunications room, locate the TGB near the backbone cabling and associated terminations. In addition, locate the TGB to provide for the shortest and straightest routing of the grounding conductors. Where a panelboard for telecommunications equipment is located within the same room or space as a TGB, bond that panelboard’s alternating current equipment ground (ACEG) bus (when equipped) or the panelboard enclosure to the TGB.~~]~~ Install telecommunications grounding busbars to maintain clearances as required and insulated from its support. A minimum of 50 mm separation from the wall is recommended to allow access to the rear of the busbar and adjust the mounting height to accommodate overhead or underfloor cable routing.

b. Telecommunications Bonding Conductors: Provide main telecommunications service equipment ground consisting of separate bonding conductor for telecommunications, between the TMGB and readily accessible grounding connection of the electrical service. Grounding and bonding conductors should not be placed in ferrous metallic conduit. If it is necessary to place grounding and bonding conductors in ferrous metallic conduit that exceeds 1 m in length, bond the conductors to each end of the conduit using a grounding bushing or a 14 sqmm conductor, minimum.~~[~~ Provide a telecommunications bonding backbone (TBB) that originates at the TMGB extends throughout the building using the telecommunications backbone pathways, and connects to the TGBs in all telecommunications rooms and equipment rooms. Install the TBB conductors such that they are protected from physical and mechanical damage. The TBB conductors should be installed without splices and routed in the shortest possible straight-line path. Make the bonding conductor between a TBB and a TGB continuous. Where splices are necessary, the number of splices should be a minimum. Make the splices accessible and located in telecommunications spaces. Connect joined segments of a TBB using exothermic welding, irreversible compression-type connectors, or equivalent. Install all joints to be adequately supported and protected from damage. Whenever two or more TBBs are used within a multistory building, bond the TBBs together with a grounding equalizer (GE) at the top floor and at a minimum of every third floor in between. Do not connect the TBB and GE to the pathway ground, except at the TMGB or the TGB.~~]~~

c. Telecommunications Grounding Connections: Telecommunications grounding connections to the TMGB~~[or TGB]~~: utilize listed compression two-hole lugs, exothermic welding, suitable and equivalent one hole non-twisting lugs, or other irreversible compression type connections. Bond all metallic pathways, cabinets, and racks for telecommunications cabling and interconnecting hardware located within the same room or space as the TMGB~~[or TGB]~~ to the TMGB~~[or TGB respectively]~~. In a metal frame (structural steel) building, where the steel framework is readily accessible within the room; bond each TMGB~~[and TGB]~~ to the vertical steel metal frame using a minimum 14 sqmm conductor. Where the metal frame is external to the room and readily accessible, bond the metal frame to the TGB or TMGB with a minimum 14 sqmm conductor. When practicable because of shorter distances and, where horizontal steel members are permanently electrically bonded to vertical column members, the TGB may be bonded to these horizontal members in lieu of the vertical column members. All connectors used for bonding to the metal frame of a building must be listed for the intended purpose.

3.1.17 Equipment Connections

Provide power wiring for the connection of motors and control equipment under this section of the specification. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and protective devices within the control circuitry are not included in this section of the specifications and are provided under the section specifying the associated equipment.

3.1.18 Elevator

Provide circuit to line terminals of elevator controller, and disconnect switch on line side of controller, outlet for control power, outlet receptacle and work light at midheight of elevator shaft, and work light and outlet receptacle in elevator pit.

~~[~~3.1.19 Government-Furnished Equipment

Contractor~~[~~ rough-in for Government-furnished equipment~~][make connections to Government-furnished equipment]~~ to make equipment operate as intended, including providing miscellaneous items such as plugs, receptacles, wire, cable, conduit, flexible conduit, and outlet boxes or fittings.

~~]~~3.1.20 Repair of Existing Work

Perform repair of existing work~~[~~, demolition, and modification of existing electrical distribution systems~~]~~ as follows:

3.1.20.1 Workmanship

Lay out work in advance. Exercise care where cutting, channeling, chasing, or drilling of floors, walls, partitions, ceilings, or other surfaces is necessary for proper installation, support, or anchorage of conduit, raceways, or other electrical work. Repair damage to buildings, piping, and equipment using skilled craftsmen of trades involved.

3.1.20.2 Existing Concealed Wiring to be Removed

Disconnect existing concealed wiring to be removed from its source. Remove conductors; cut conduit flush with floor, underside of floor, and through walls; and seal openings.

~~[~~3.1.20.3 Removal of Existing Electrical Distribution System

Removal of existing electrical distribution system equipment includes equipment's associated wiring, including conductors, cables, exposed conduit, surface metal raceways, boxes, and fittings,~~[~~ back to equipment's power source~~]~~ as indicated.

~~][~~3.1.20.4 Continuation of Service

Maintain continuity of existing circuits of equipment to remain. Maintain existing circuits of equipment energized. Restore circuits wiring and power which are to remain but were disturbed during demolition back to original condition.

~~][~~3.1.21 Watthour Meters

JIS C 1210.

~~]~~3.1.22 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible. Maximum allowed lead length is 900 mm.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. ~~[~~ Painting: as specified in Section 09 90 00 PAINTS AND COATINGS.~~][~~ Where field painting of enclosures for panelboards, load centers or the like is specified to match adjacent surfaces, to correct damage to the manufacturer's factory applied coatings, or to meet the indicated or specified safety criteria, provide manufacturer's recommended coatings and apply in accordance to manufacturer's instructions.~~]~~

3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer~~[~~ 5~~][_____]~~ working days notice prior to~~[~~ each~~][_____]~~ test~~[s]~~.

3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring 14 sqmm and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring to provide direct reading of resistance. All existing wiring to be reused shall also be tested.

3.5.3 Transformer Tests

Perform the standard, not optional, tests in accordance with the Inspection and Test Procedures for transformers, dry type, air-cooled, 600 volt and below; as specified in Denki Hoan Kyoukai and MLIT DSKKS. Measure primary and secondary voltages for proper tap settings. Tests need not be performed by a recognized independent testing firm or independent electrical consulting firm.

3.5.4 Ground-Fault Receptacle Test

~~[~~Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed.~~][~~ Test RCBO circuit breakers in accordance with JIS C 8222, Annex G.~~]~~

3.5.5 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

~~[~~3.5.6 Watthour Meter

a. Visual and mechanical inspection

(1) Examine for broken parts, shipping damage, and tightness of connections.

(2) Verify that meter type, scales, and connections are in accordance with approved shop drawings.

b. Electrical tests

(1) Determine accuracy of meter.

(2) Calibrate watthour meters to one-half percent.

(3) Verify that correct multiplier has been placed on face of meter, where applicable.

~~]~~3.5.7 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --

SECTION 26 24 13

SWITCHBOARDS
05/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS), Ministry of Land, Infrastructure and Transport (MLIT), Japan Electrical Safety Inspection Associations or Japanese Architectural Standard Specifications (JASS) for non-Japanese standards, as approved by the Contracting Officer's representative.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2	(2023) National Electrical Safety Code

ARCHITECTURAL INSTITUTE OF JAPAN (AIJ)

JASS 6	(2015) Structural Steelwork Specification for Building Construction

ELECTRICAL SAFETY INSPECTION ASSOCIATIONS

Denki Hoan Kyoukai	Japan Standard for Acceptance Testing and Inspections

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS B 1048	~~(2007) Fasteners - Hot Dip Galvanized Coatings~~(2007) Fasteners -- Hot Dip Galvanized Coatings
JIS C 0365	(2007) Protection Against Electric Shock - Common Aspects for Installation and Equipment
JIS C 1210	(1979) General Rules for Electricity Meters
JIS C 1731-1	(1998) Instrument Transformers for Testing Purpose and Used with General Instrument Part 1: Current Transformer
JIS C 2110-1	(2016) Solid electrical insulation materials-Test methods for strength of dielectric breakdown-Part 1: Tests by applying commercial frequency alternating voltage
JIS C 8201-2-1	(2021) Low-Voltage Switchgear and Control Gear - Part 2-1: Circuit-Breakers
JIS C 8201-2-2	(2021) Low-Voltage Switchgear And Control

	Gear - Part 2-2: Circuit-Breakers Incorporating Residual Current Protection
JIS C 8269-1	(2016) Low-Voltage Fuses -- Part 1: General Requirements
JIS C 8269-2	(2016) Low voltage fuse-Part 2: Additional requirements for expert fuses(Mainly industrial fuses)
JIS C 8462-1	(2021) Boxes and enclosures for electrical accessories for household and similar fixed electrical installations -- Part 1: General requirements
JIS C 8480	(R2020) Box-Type Switchgear Assemblies for Low-Voltage Distribution Purpose
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor
JIS C 60364-6	(2010; R 2019) Low-voltage electrical installations -- Part 6: Verification
JIS C 61558-1	(2019) Safety of transformers, reactors, power supply units and combinations thereof -- Part 1: General requirements and tests
JIS G 3352	(2014) Deck Plate
JIS G 3601	(2012) Stainless clad steel
JIS G 4304	(2021) Hot-Rolled Stainless Steel Plate, Sheet and Strip
JIS H 8641	(2021) Hot Dip Galvanized Coatings
JIS K 6911	(2006; R 2021) Thermosetting plastic general test method
JIS Z 9101	(2018) Graphical symbols -- Safety colours and safety signs -- Part 1: Design principles for safety signs and safety markings

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E	(2024) Standard for Electrical Safety in the Workplace

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor QC approval.][~~ for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29, SUSTAINABITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Switchboard Drawings; G~~[, [_____]]~~

SD-03 Product Data

Switchboard; G~~[, [_____]]~~

SD-06 Test Reports

Switchboard Design Tests~~;~~

SD-10 Operation and Maintenance Data

Switchboard Operation and Maintenance, Data Package 5~~;~~

SD-11 Closeout Submittals

Assembled Operation and Maintenance Manuals~~;~~

Equipment Test Schedule~~;~~

~~[~~ ~~Request for Settings;~~

~~][~~ Required Settings~~;~~
~~]~~
~~[~~ Service Entrance Available Fault Current Label~~;~~

~~]~~1.5 QUALITY ASSURANCE

1.5.1 Product Data

Include manufacturer's information on each submittal for each component, device and accessory provided with the switchboard including:

a. Circuit breaker type, interrupting rating, and trip devices, including available settings.

b. Manufacturer's instruction manuals and published time-current curves (in electronic format) of the main secondary breaker and largest secondary feeder device.

1.5.2 Switchboard Drawings

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Identify circuit terminals on wiring diagrams and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Indicate on the drawings adequate clearance for operation, maintenance, and replacement of operating equipment devices. Include the nameplate data, size, and capacity on submittal. Also include applicable federal, military, industry, and technical society publication references on submittals. Include the following:

a. One-line diagram including breakers~~[~~, fuses~~][~~, current transformers, and meters~~]~~.

b. Outline drawings including front elevation, section views, footprint, and overall dimensions.

c. Bus configuration including dimensions and ampere ratings of bus bars.

d. Markings and rated IP code nameplate data~~[~~, including fuse information (manufacturer's name, catalog number, and ratings)~~]~~.

e. Circuit breaker type, interrupting rating, and trip devices, including available settings.

f. Wiring diagrams and elementary diagrams with terminals identified, and indicating prewired interconnections between items of equipment and the interconnection between the items.

g. Manufacturer's instruction manuals and published time-current curves (in electronic format) of the main secondary breaker and largest secondary feeder device. Use this information (designer of record) to provide breaker settings that ensures protection and coordination are achieved.~~ [For Navy installations, provide electronic format curves using SKM's Power Tools for Windows device library electronic format or EasyPower device library format depending on installation modeling software requirements.]~~

~~[~~ h. Provisions for future expansion by adding switchboard sections.

~~]~~

1.5.3 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" or "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with the mandatory and advisory provisions of applicable codes and standards unless more stringent requirements are specified or indicated.

1.5.4 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship, and:

a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.4.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site are not acceptable.

1.6 MAINTENANCE

1.6.1 Switchboard Operation and Maintenance Data

Submit Operation and Maintenance Manuals in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6.2 Assembled Operation and Maintenance Manuals

Assemble and securely bind manuals in durable, hard covered, water resistant binders. Assemble and index the manuals in the following order with a table of contents:

a. Manufacturer's O&M information required by the paragraph SD-10, OPERATION AND MAINTENANCE DATA.

b. Catalog data required by the paragraph SD-03, PRODUCT DATA.

c. Drawings required by the paragraph SD-02, SHOP DRAWINGS.

d. Prices for spare parts and supply list.

~~[~~ e. Information on metering.

~~]~~ f. Design test reports.

g. Production test reports.

1.7 WARRANTY

Provide equipment items that are supported by service organizations reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

Products and materials not considered to be switchboards and related accessories are specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION, and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.2 SWITCHBOARD

JIS C 8480.

2.2.1 Ratings

Provide equipment with the following ratings:

a. Voltage rating: ~~[480Y/277][~~208Y/120 ~~][440][440Y/254][420Y/242][220][210Y/105][_____]~~ volts AC, ~~[~~50~~][60]~~ hertz, ~~[~~three-phase, ~~[3][~~4~~]~~-wire~~][~~_as indicated~~]~~.

b. Continuous current rating of the main bus: ~~[_____ amperes][~~as indicated~~]~~.

c. Short-circuit current rating: ~~[_____ rms symmetrical amperes][~~as indicated~~]~~.

d. UL listed and labeled~~[~~ as service entrance equipment~~]~~.

2.2.2 Construction

Provide the following:

a. Switchboard: consisting of one or more vertical sections~~[~~ bolted together to form a rigid assembly~~]~~ and ~~[~~rear~~][front and rear]~~ aligned~~[~~ as indicated~~]~~.

b. All circuit breakers: front accessible.

~~[~~ c. Rear aligned switchboards: front accessible load connections.

~~][d. Front and rear aligned switchboards[: rear accessible load connections].~~

~~]~~ ~~e~~c. Where indicated, "space for future" or "space" means to include a vertical bus provided behind a blank front cover. Where indicated, "provision for future" means full hardware provided to mount a breaker suitable for the location.

~~f~~d. Completely factory engineered and assembled, including protective devices and equipment indicated with necessary interconnections, instrumentation, and control wiring.

2.2.2.1 Enclosure

Provide the following:

a. Enclosure: ~~[~~indoor~~][weatherproof]~~ IP rated per JIS C 8462-1~~[~~_as indicated~~][fabricated entirely of 12 gauge, type 304 or 304L~~

~~stainless steel] per JIS G 3601 and JIS G 4304~~.

b. Enclosure: bolted together with removable bolt-on side and~~[~~ hinged~~]~~ rear covers~~[~~, and sloping roof downward toward rear~~]~~.

~~[~~ c. Front~~[~~ and rear~~]~~ doors: provided with~~[~~ stainless steel~~]~~ padlockable vault handles with a three point catch.

~~][~~d. Bases, frames and channels of enclosure: corrosion resistant and fabricated of~~[type 304 or 304L stainless steel][or][~~ galvanized steel ~~]~~_JIS G 3601 and JIS G 4304.

~~]~~ e. Base: includes any part of enclosure that is within 75 mm of concrete pad.

~~[~~ f. Galvanized steel: JIS H 8641 and JIS G 3352 coating, and JIS B 1048 and JIS H 8641, as applicable. Galvanize after fabrication where practicable.

~~]~~ g. Paint color: light gray over rust inhibitor.

~~[~~ h. Paint coating system: comply with~~[~~ JIS C 8480 for galvanized steel~~][and][JIS Z 2371 for stainless steel]~~.

~~]~~2.2.2.2 Bus Bars

Provide the following:

a. Bus bars: ~~[~~copper with silver-plated contact surfaces~~][or][aluminum with tin-plated contact surfaces]~~.

(1) Phase bus bars: ~~[uninsulated][~~insulated with a tape wrap or insulating sleeve providing a minimum breakdown voltage in accordance with JIS C 2110-1~~]~~.

(2) Neutral bus: rated ~~[100][_____] percent of the main bus continuous current rating[~~ as indicated~~]~~.

b. Make bus connections and joints with hardened steel bolts.

c. Main-bus (through bus): rated at the full ampacity of the main throughout the switchboard.

d. Minimum 6.35 mm by 50.8 mm copper ground bus secured to each vertical section along the entire length of the switchboard.

2.2.2.3 Main Section

Provide the main section consisting of~~[~~_a combination section with~~[~~ molded-case circuit breakers~~]~~ for the~~[~~ main and~~]~~ branch devices as indicated~~][main lugs only][an individually mounted [fixed][air power circuit breaker[with current-limiting fuses]][insulated-case circuit breaker][molded-case circuit breaker]][and utility transformer compartment]~~.

~~[~~2.2.2.4 Distribution Sections

Provide the distribution section~~[~~s~~]~~ consisting of~~[[~~individually mounted, ~~]][air power circuit breakers[with current-limiting fuses]][~~

~~insulated-case circuit breakers][~~ molded-case circuit breakers~~][and utility transformer compartments]~~ as indicated.

~~][~~2.2.2.5 Auxiliary Sections

Provide auxiliary sections consisting of indicated~~[~~ instruments,~~][~~ metering equipment,~~][~~ control equipment,~~][~~ transformer,~~][~~ and~~][~~ current transformer compartments~~]~~ as indicated.

~~][~~2.2.2.6 Handles

Provide handles for individually mounted devices of the same design and method of external operation. Label handles prominently to indicate device ampere rating, color coded for device type. Identify ON-OFF indication by handle position and by prominent marking.

~~]~~2.2.3 Protective Device

Provide~~[~~ main and~~]~~ branch protective devices as indicated.

~~[~~2.2.3.1 Power Circuit Breaker

Provide the following:

a. JIS C 8201-2-1 and JIS C 8201-2-2. ~~[~~120 Vac~~][100 Vac][~~ electrically~~][manually]~~ operated~~ [stationary]~~, ~~[unfused][~~fused~~]~~, low-voltage power circuit breaker with a short-circuit current rating~~[of [_____] rms amperes symmetrical][~~ as indicated~~] at [_____] volts~~.

b. Breaker frame size: ~~[~~ ~~~~as indicated~~][[_____] amperes]~~.

~~[~~c. Equip electrically operated breakers with motor-charged, stored-energy closing mechanism to permit rapid and safe closing of the breaker against fault currents within the short time rating of the breaker, independent of the operator's strength or effort in closing the handle.

~~]][~~2.2.3.2 Insulated-Case Breaker

Provide the following:

a. JIS C 8201-2-1. UL listed and labeled,~~[~~ 100 percent rated main breaker ~~][standard rated branch breakers],[electrically]~~ ~~[~~manually~~]~~ operated, low voltage, insulated-case circuit breaker, with a short-circuit current rating~~[of [_____] rms symmetrical amperes][~~ as indicated~~] at [_____] volts~~.

b. Breaker frame size: ~~[[_____] amperes][~~ as indicated~~]~~.

c. Series rated circuit breakers are unacceptable.

~~][~~2.2.3.3 Molded-Case Circuit Breaker

Provide the following:

a. JIS C 8201-2-1. UL listed and labeled,~~[~~ 100 percent rated main breaker ~~][standard rated branch breakers],[electrically][~~ manually~~]~~ operated, low voltage molded-case circuit breaker, with a short-circuit current rating ~~of[[_____] rms symmetrical amperes][~~ as indicated~~] at [_____] volts~~.

b. Breaker frame size: ~~[[_____] amperes][~~as indicated~~]~~.

c. Series rated circuit breakers are unacceptable.

~~][~~2.2.3.4 Fusible Switches

Provide the following:

a. Fusible Switches: quick-make, quick-break, hinged-door type.

~~[~~ b. Switches serving as motor disconnects: horsepower rated.

~~]~~ c. Fuses: current-limiting cartridge type conforming to~~[, Class [RK1][RK5] for 0 to 600 amperes]~~ per JIS C 8269-1 and JIS C 8269-2.

d. Fuseholders: ~~[~~JIS C 8269-1 and JIS C 8269-2~~]~~.

~~][~~2.2.3.5 Integral Combination Breaker and Current-Limiting Fuses

Provide the following:

a. JIS C 8201-2-1.

b. Integral combination molded-case circuit breaker and current-limiting fuses:~~[~~ as indicated~~][rated [_____] amperes]~~ with a minimum short-circuit-current rating equal to the short-circuit-current rating of the switchboard in which the circuit breaker will be mounted.

c. Series rated circuit breakers are unacceptable.

d. Coordination of overcurrent devices of the circuit breaker and current-limiting fuses: for overloads or fault currents of relatively low value, the overcurrent device of the breaker operates to clear the fault. The current-limiting fuses operate to clear the fault for high magnitude short circuits above a predetermined value~~[~~ crossover point~~]~~.

e. Housing for the current-limiting fuses: an individual molding readily removable from the front and located at the load side of the circuit breaker. If the fuse housing is removed, a blown fuse is readily evident by means of a visible indicator.

f. Removal of fuse housing causes the breaker contacts to open, and the breaker contacts can not close with the fuse housing removed. The fuse housing can not be inserted with a blown fuse or with one fuse missing. The blowing of any of the fuses causes the circuit breaker contacts to open.

~~][~~2.2.4 Drawout Breakers

Provide drawout breakers~~[~~ as indicated~~][_____]~~. Equip drawout breakers with disconnecting contacts, wheels, and interlocks for drawout application. Provide main, auxiliary, and control disconnecting contacts with silver-plated, multifinger, positive pressure, self-aligning type. Provide each drawout breaker with four-position operation with each position clearly identified by an indicator on the circuit breaker front panel as follows.

a. Connected Position: Primary and secondary contacts are fully engaged. Breaker must be tripped before racking into or out of position.

b. Test Position: Primary contacts are disconnected but secondary contacts remain fully engaged. This position allows complete test and operation of the breaker without energizing the primary circuit.

c. Disconnected Position: Primary and secondary contacts are disconnected.

d. Withdrawn (Removed) Position: Places breaker completely out of compartment, ready for removal. Removal of the breaker actuates assembly that isolates the primary stabs.

~~]~~2.2.5 Electronic Trip Units

Equip~~[~~ main and~~][~~ distribution~~]~~ breakers~~[~~ as indicated~~]~~ with a solid-state tripping system consisting of three current sensors and a microprocessor-based trip unit that provides true rms sensing adjustable time-current circuit protection. Include the following:

a. Current sensors ampere rating: ~~[~~as indicated~~][[_____] amperes][the same as the breaker frame rating]~~.

b. Trip unit ampere rating: ~~[~~as indicated~~][[_____] amperes]~~.

~~[~~ c. Ground fault protection: ~~[~~as indicated~~][zero sequence sensing][residual type sensing]~~.

~~][~~d. Electronic trip units: provide additional features~~[~~ as indicated~~]~~:

~~]~~

~~[~~ (1) ~~[Indicated]~~Breakers: include long delay pick-up and time settings, and LED indication of cause of circuit breaker trip.

~~][~~ (2) Main breakers: include~~[~~ short delay pick-up and time settings~~][~~ and~~][~~, instantaneous settings~~][~~ and~~][~~ ground fault settings~~][~~ as indicated~~]~~.

~~][~~ (3) Distribution breakers: include~~[~~ short delay pick-up and time settings~~][~~, instantaneous settings~~][~~, and ground fault settings~~][~~ as indicated~~]~~.

~~][~~ (4) ~~[Main]~~Breakers: include a digital display for phase and ground current.

~~][~~ (5) ~~[Main]~~Breakers: include a digital display for watts, vars, VA, kWh, kvarh, and kVAh.

~~][~~ (6) ~~[Main]~~Breakers: include a digital display for phase voltage, and percent THD voltage and current.

~~][~~ ~~(7) [Main]Breakers: include provisions for communication via a network twisted pair cable for remote monitoring and control. Provide the following communications protocol:[DNP3][Modbus][IEC 61850].~~

~~][~~ (8) For electronic trip units that are rated for or can be adjusted to 1,200 amperes or higher, provide arc energy reduction capability with an energy-reducing maintenance switch with local status indicator.

~~][~~2.2.6 Metering

~~[~~2.2.6.1 Digital Meters

~~[JIS C 5381-11 and JIS C 61000-4-5 for surge withstand. [Metering shall be compliant with the current Advanced Meter Reading System (AMRS) specification.] Provide true rms, plus/minus one percent accuracy, programmable, microprocessor-based meter enclosed in a sealed case with the following features.]~~

~~[~~Provide meter(s) and connect the meter(s) to the existing Advanced Meter Infrastructure Data Aquisition System (AMI DAS). The contractor shall use the existing government laptop computers to configure the meter using existing software loaded on the computer. The contractor will not be allowed to modify any software or add any additional software to the computer. Alternatively, the government will configure the meter(s), which must be compatible with the existing system, using existing software. Contract shall insure that the meter(s) will transmit the specified data to the DAS. Meters shall be compatible with the Base AMI DAS per ~~NAVFAC Far East~~Army requirements.~~]~~

a. Display capability:

~~[~~ (1) Multi-Function Meter: Display a selected phase to neutral voltage, phase to phase voltage, percent phase to neutral voltage THD, percent phase to phase voltage THD; a selected phase current, neutral current, percent phase current THD, percent neutral current; selected total PF, kW, KVA, kVAR, FREQ, kVAh, kWh. Detected alarm conditions include over/under current, over/under voltage, over/under KVA, over/under frequency, over/under selected PF/kVAR, voltage phase reversal, voltage imbalance, reverse power, over percent THD. Include a Form C KYZ pulse output relay on the meter.

~~][~~ (2) Power Meter: Display Watts, VARs, and selected KVA/PF. Detected alarm conditions include over/under KVA, over/under PF, over/under VARs, over/under reverse power.

~~][~~ (3) Volt Meter: Provide capability to be selectable between display of the three phases of phase to neutral voltages and display of the three phases of the phase to phase voltages. Detected alarm conditions include over/under voltage, over/under voltage imbalance, over percent THD.

~~][~~ (4) Ammeter: Display phase A, B, and C currents. Detected alarm conditions include over/under current, over percent THD.

~~][~~ (5) Digital Watthour Meter: Provide a single selectable display for watts, total kilowatt hours (kWh) and watt demand (Wd). Include a Form C KYZ pulse output relay on the meter.

~~]~~ b. Design meters to accept~~[~~ input from standard 5A secondary instrument transformers~~][~~ and~~][~~ direct voltage monitoring range to ~~[300][~~600~~]~~ volts, phase to phase~~]~~.

c. Provide programming via a front panel display and a communication interface accessible by a computer.

d. Provide password secured programming stored in non-volatile EEPROM memory.

e. Provide digital communications in a Modbus [RTU] protocol via a [RS232C][RS485] serial port[and an independently addressable [RS232C][RS485] serial port].

f. Provide meter that calculates and stores average max/min demand values with time and date for all readings based on a user selectable sliding window averaging period.

g. Provide meter with programmable hi/low set limits with two dry contact relays when exceeding alarm conditions.

[h. Meter shall have two-way communication with the existing DAS. Provide a communications interface utilizing fiber-optic LC connection.]

][2.2.6.2 Electronic Watthour Meter

[JIS C 1210. Provide a switchboard style electronic programmable watthour meter, semi-flush mounted, as indicated. Meter can be either programmed at the factory or programmed in the field. Turn field programming device over to the Contracting Officer at completion of project. Coordinate meter to system requirements.

a. Design: Provide meter designed for use on a 3-phase, 4-wire, [208Y/120][210Y/105][480Y/277][440Y/254][420Y/242] volt system with 3 current transformers. Include necessary KYZ pulse initiation hardware for Energy Monitoring and Control System (EMCS).

b. Coordination: Provide meter coordinated with ratios of current transformers and transformer secondary voltage.

c. Class: 20. Accuracy: plus or minus 1.0 percent. Finish: Class II.

d. Kilowatt-hour Register: five digit electronic programmable type.

e. Demand Register:

(1) Provide solid state.

(2) Meter reading multiplier: Indicate multiplier on the meter face.

(3) Demand interval length: programmed for [15][30][60] minutes with rolling demand up to six subintervals per interval.

f. Meter fusing: Provide a fuse block mounted in the metering compartment containing one fuse per phase to protect the voltage input to the watthour meter. Size fuses as recommended by the meter manufacturer.

g. Provide meter with a communications port, RS485, with Modbus RTU serial or Ethernet, Modbus-TCP communications.

JIS C 1731-1. Provide single ratio transformers, [60 hertz][50 hertz], [_____] to 5-ampere ratio, [_____]4.0 rating factor, with a metering accuracy class of 0.3 through [_____]B-0.5.

~~[~~ Provide a fuse block mounted in the metering compartment containing one fuse per phase to protect the voltage input to voltage sensing meters. Size fuses as recommended by the meter manufacturer.

~~]]][~~2.2.6.3 Submetering

Provide submetering for ~~[_____] [~~in accordance with drawings~~]~~.

~~]][~~2.2.7 Transformer

Provide transformer section in switchboard in accordance with JIS C 8480 and as indicated. Provide the transformer and section that is suitable for the installation.~~[~~ Test transformers greater than 10 kVA in accordance with JIS C 8480.~~]~~ Provide a transformer conforming to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

~~][~~2.2.8 Heaters

Provide ~~[~~120-volt~~][100-volt]~~ heaters in each switchboard section. Provide heaters of sufficient capacity to control moisture condensation in the section, 250 watts minimum, and controlled by a thermostat~~[and humidistat]~~ located in the section. Provide industrial type thermostat, high limit, to maintain sections within the range of 15 to 32 degrees C.~~[Provide humidistat with a range of 30 to 60 percent relative humidity.]~~ Obtain supply voltage for the heaters from a control power transformer within the switchboard. If heater voltage is different than switchboard voltage, provide transformer rated to carry 125 percent of heater full load rating. Provide transformer with a 220 degrees C insulation system with a temperature rise not exceeding 115 degrees C and conforming to JIS C 61558-1.~~[~~ Energize electric heaters in switchboard assemblies while the equipment is in storage or in place prior to being placed in service. Provide method for easy connection of heater to external power source. Provide temporary, reliable external power source if commercial power at rated voltage is not available on site.~~]~~

~~]~~2.2.9 Terminal Boards

Provide with engraved plastic terminal strips and screw type terminals for external wiring between components and for internal wiring between removable assemblies. Provide short-circuiting type terminal boards associated with current transformer. Terminate conductors for current transformers with ring-tongue lugs. Provide terminal board identification that is identical in similar units. Provide color coded external wiring that is color coded consistently for similar terminal boards.

2.2.10 Wire Marking

Mark control and metering conductors at each end. Provide factory installed, white, plastic tubing, heat stamped with black block type letters on factory-installed wiring. On field-installed wiring, provide white, preprinted, polyvinyl chloride (PVC) sleeves, heat stamped with black block type letters. Provide a single letter or number on each sleeve, elliptically shaped to securely grip the wire, and keyed in such a manner to ensure alignment with adjacent sleeves. Provide specific wire markings using the appropriate combination of individual sleeves. Indicate on each wire marker the device or equipment, including specific terminal number to which the remote end of the wire is attached.

2.3 MANUFACTURER'S NAMEPLATE

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable. This nameplate and method of attachment may be the manufacturer's standard if it contains the required information.

2.4 FIELD FABRICATED NAMEPLATES

JIS K 6911. Provide laminated plastic nameplates for each switchboard, equipment enclosure, relay, switch, and device; as specified in this section or as indicated on the drawings. Identify on each nameplate inscription the function and, when applicable, the position. Provide nameplates of melamine plastic, 3 mm thick, white with ~~[~~black~~][______]~~ center core.~~[~~ Provide red laminated plastic label with white center core where indicated.~~]~~ Provide matte finish surface. Provide square corners. Accurately align lettering and engrave into the core. Provide nameplates with minimum size of 25 by 65 mm. Provide lettering that is a minimum of 6.35 mm high normal block style.

2.5 SOURCE QUALITY CONTROL

2.5.1 Equipment Test Schedule

The Government reserves the right to witness tests. Provide equipment test schedules for tests to be performed at the manufacturer's test facility. Submit required test schedule and location, and notify the Contracting Officer 30 calendar days before scheduled test date. Notify Contracting Officer 15 calendar days in advance of changes to scheduled date.

Provide the following as part of test equipment calibration:

a. Provide a calibration program which assures that all applicable test instruments are maintained within rated accuracy.

b. Accuracy: Traceable to the National Institute of Standards and Technology.

c. Instrument calibration frequency schedule: less than or equal to 12 months for both test floor instruments and leased specialty equipment.

d. Dated calibration labels: visible on all test equipment.

e. Calibrating standard: higher accuracy than that of the instrument tested.

f. Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:

    (1) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.

    (2) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

2.5.2 Switchboard Design Tests

JIS C 8480.

2.5.2.1 Design Tests

Furnish documentation showing the results of design tests on a product of the same series and rating as that provided by this specification.

a. Short-circuit current test.

b. Enclosure tests.

c. Dielectric test.

~~[~~2.5.2.2 Additional Design Tests

In addition to normal design tests, perform the following tests on the actual equipment. Furnish reports which include results of design tests performed on the actual equipment.

a. Temperature rise tests.

b. Continuous current.

~~]~~2.5.3 Switchboard Production Tests

JIS C 8480. Furnish reports which include results of production tests performed on the actual equipment for this project. These tests include:

a. ~~[60-hertz][~~50-hertz~~]~~ dielectric tests.

b. Mechanical operation tests.

c. Electrical operation and control wiring tests.

d. Ground fault sensing equipment test.

~~[2.6 COORDINATED POWER SYSTEM PROTECTION~~

~~Provide a power system study as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.~~

~~]~~2.6 ARC FLASH WARNING LABEL

Provide warning label of potential electrical arc flash hazards for switchboards in accordance with NFPA 70E and JIS Z 9101.~~ .~~

~~[~~2.7 SERVICE ENTRANCE AVAILABLE FAULT CURRENT LABEL

Provide label on exterior of switchboards used as service equipment listing the fault current rating in accordance with NFPA 70E and JIS Z 9101. Locate this self-adhesive warning label on the outside of the switchboard.

]PART 3 EXECUTION

3.1 INSTALLATION

Conform to IEEE C2, JIS C 0365 and to the requirements specified herein. Provide new equipment and materials unless indicated or specified otherwise.

[3.2 GROUNDING

IEEE C2 and JIS C 0365, except that grounds and grounding systems with a resistance to solid earth ground not exceeding [25][_____] ohms.

3.2.1 Grounding Electrodes

Provide driven ground rods as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Connect ground conductors to the upper end of the ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.2.2 Equipment Grounding

Provide bare copper cable not smaller than [100 sqmm] not less than 610 mm below grade connecting to the indicated ground rods. When work in addition to that indicated or specified is directed to obtain the specified ground resistance, the provision of the contract covering "Changes" applies.

3.2.3 Connections

Make joints in grounding conductors and loops by exothermic weld or compression connector. Install exothermic welds and compression connectors as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.2.4 Grounding and Bonding Equipment

JIS C 60364-5-54, except as indicated or specified otherwise.

]3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

Install and connect equipment furnished under this section as indicated on project drawings, the approved shop drawings, and as specified herein.

3.3.1 Switchboard

JIS C 8480.

3.3.2 Meters and Instrument Transformers

JIS C 1210.

3.3.3 Field Applied Painting

Where field painting of enclosures is required to correct damage to the manufacturer's factory applied coatings, provide manufacturer's recommended coatings and apply in accordance with manufacturer's instructions.

3.3.4 Galvanizing Repair

Repair damage to galvanized coatings using JASS 6, zinc rich paint, for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces that repair paint has been applied to.

3.3.5 Field Fabricated Nameplate Mounting

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.4 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

3.4.1 Exterior Location

Mount switchboard on concrete slab as follows:

a. Unless otherwise indicated, provide the slab with dimensions at least 200 mm thick, reinforced with a 150 by 150 mm - MW19 by MW19 (6 by 6 - W2.9 by W2.9) mesh placed uniformly 100 mm from the top of the slab.

b. Place slab on a 150 mm thick, well-compacted gravel base.

c. Install slab such that the top of the concrete slab is approximately 100 mm above the finished grade.

d. Provide edges above grade with 15 mm chamfer.

e. Provide slab of adequate size to project at least 200 mm beyond the equipment.

f. Provide conduit turnups and cable entrance space required by the equipment to be mounted.

g. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant.

h. Cut off and bush conduits 75 mm above slab surface.

i. Provide concrete work as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.4.2 Interior Location

Mount switchboard on concrete slab as follows:

a. Unless otherwise indicated, provide the slab with dimensions at least 100 mm thick.

b. Install slab such that the top of the concrete slab is approximately 100 mm above the finished grade.

c. Provide edges above grade with 15 mm chamfer.

d. Provide slab of adequate size to project at least ~~[~~ 200 mm~~][ ]~~ beyond the equipment.

e. Provide conduit turnups and cable entrance space required by the the equipment to be mounted.

f. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant.

g. Cut off and bush conduits 75 mm above slab surface.

h. Provide concrete work as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.5 FIELD QUALITY CONTROL

~~[Submit request for settings of breakers to the Contracting Officer after approval of switchboard and at least 30 days in advance of their requirement.~~

~~][~~Submit Required Settings of breakers to the Contracting Officer after approval of switchboard and at least 30 days in advance of their requirement.

~~]~~3.5.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with Denki Hoan Kyoukai.

3.5.1.1 Switchboard Assemblies

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical, electrical, and mechanical condition.

(3) Verify appropriate anchorage, required area clearances, and correct alignment.

(4) Clean switchboard and verify shipping bracing, loose parts, and documentation shipped inside cubicles have been removed.

(5) Inspect all doors, panels, and sections for paint, dents, scratches, fit, and missing hardware.

(6) Verify that~~[~~ fuse and~~]~~ circuit breaker sizes and types correspond to approved shop drawings as well as to the circuit breaker’s address for microprocessor-communication packages.

~~[~~ (7) Verify that current transformer ratios correspond to approved shop drawings.

~~]~~ (8) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

(9) Confirm correct operation and sequencing of electrical and mechanical interlock systems.

(10) Confirm correct application of manufacturer's recommended lubricants.

(11) Inspect insulators for evidence of physical damage or contaminated surfaces.

(12) Verify correct barrier installation[and operation].

(13) Exercise all active components.

(14) Inspect all mechanical indicating devices for correct operation.

(15) Verify that filters are in place and vents are clear.

(16) Test operation, alignment, and penetration of instrument transformer withdrawal disconnects.

(17) Inspect control power transformers.

b. Electrical Tests

(1) Perform insulation-resistance tests on each bus section.

(2) Perform dielectric withstand voltage tests.

(3) Perform insulation-resistance test on control wiring; Do not perform this test on wiring connected to solid-state components.

(4) Perform control wiring performance test.

(5) Perform primary current injection tests on the entire current circuit in each section of assembly.

[(6) Perform phasing check on double-ended switchboard to ensure correct bus phasing from each source.

][(7) Verify operation of switchboard heaters.

][3.5.1.2 Circuit Breakers - Low Voltage - Power

a. Visual and Mechanical Inspection

(1) Compare nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Inspect anchorage, alignment, and grounding.

(4) Verify that all maintenance devices are available for servicing and operating the breaker.

(5) Inspect arc chutes.

(6) Inspect moving and stationary contacts for condition, wear, and alignment.

(7) Verify that primary and secondary contact wipe and other dimensions vital to satisfactory operation of the breaker are

correct.

(8) Perform all mechanical operator and contact alignment tests on both the breaker and its operating mechanism.

(9) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

(10) Verify cell fit and element alignment.

(11) Verify racking mechanism.

(12) Confirm correct application of manufacturer's recommended lubricants.

b. Electrical Tests

(1) Perform contact-resistance tests on each breaker.

(2) Perform insulation-resistance tests.

(3) Adjust Breaker(s) for final settings in accordance with Government provided settings.

(4) Determine long-time minimum pickup current by primary current injection.

(5) Determine long-time delay by primary current injection.

[(6) Determine short-time pickup and delay by primary current injection.

][(7) Determine ground-fault pickup and delay by primary current injection.

][(8) Determine instantaneous pickup value by primary current injection.

][(9) Activate auxiliary protective devices, such as ground-fault or undervoltage relays, to ensure operation of shunt trip devices; Check the operation of electrically-operated breakers in their cubicle.

](10) Verify correct operation of any auxiliary features such as trip and pickup indicators, zone interlocking, electrical close and trip operation, trip-free, and antipump function.

(11) Verify operation of charging mechanism.

]3.5.1.3 Circuit Breakers

[Low Voltage - Insulated-Case][and][Low Voltage Molded Case with Solid State Trips]

a. Visual and Mechanical Inspection

(1) Compare nameplate data with specifications and approved shop drawings.

(2) Inspect circuit breaker for correct mounting.

(3) Operate circuit breaker to ensure smooth operation.

(4) Inspect case for cracks or other defects.

(5) Inspect all bolted electrical connections for high resistance using low resistance ohmmeter, verifying tightness of accessible bolted connections and/or cable connections by calibrated torque-wrench method, or performing thermographic survey.

(6) Inspect mechanism contacts and arc chutes in unsealed units.

b. Electrical Tests

(1) Perform contact-resistance tests.

(2) Perform insulation-resistance tests.

(3) Perform Breaker adjustments for final settings in accordance with Government provided settings.

(4) Perform long-time delay time-current characteristic tests

(5) Determine short-time pickup and delay by primary current injection.

(6) Determine ground-fault pickup and time delay by primary current injection.

(7) Determine instantaneous pickup current by primary injection.

(8) Verify correct operation of any auxiliary features such as trip and pickup indicators, zone interlocking, electrical close and trip operation, trip-free, and anti-pump function.

3.5.1.4 Current Transformers

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify correct connection.

(4) Verify that adequate clearances exist between primary and secondary circuit.

(5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

(6) Verify that all required grounding and shorting connections provide good contact.

b. Electrical Tests

(1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.

(2) Perform insulation-resistance tests.

(3) Perform polarity tests.

(4) Perform ratio-verification tests.

3.5.1.5 Metering and Instrumentation

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify tightness of electrical connections.

b. Electrical Tests

(1) Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.

(2) Calibrate watthour meters according to manufacturer's published data.

(3) Verify all instrument multipliers.

(4) Electrically confirm that current transformer and voltage transformer secondary circuits are intact.

3.5.1.6 Grounding System

a. Visual and Mechanical Inspection

(1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

(1) JIS C 60364-6. Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.

(2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

3.5.2 Follow-Up Verification

Upon completion of acceptance checks, settings, and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. Trip circuit breakers by operation of each protective device. Test each item to perform its function not less than three times. As an exception to requirements stated elsewhere in the contract, provide the Contracting Officer 5 working days advance notice of the dates and times for checks, settings, and tests.

-- End of Section --

SECTION 26 27 14.00 20

ELECTRICITY METERING
02/21, CHG 1: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 — (~~2014; Errata 2016~~2022) Electric Meters - Code for Electricity Metering

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP — (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASHRAE 90.1 - SI — (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.90.1 — (2023; ERTA) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

IEEE C57.13 — (2016) Standard Requirements for Instrument Transformers

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS — (2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 62053-22 — (2020) Electricity Metering Equipment (A.C.) - Particular Requirements - Part 22: Static Meters for Active Energy (Classes 0,2 S and 0,5 S)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C12.7 — (2022) Requirements for Watthour Meter Sockets

ANSI C12.18 — (2006; R 2023) Protocol Specification for ANSI Type 2 Optical Port

ANSI C12.20	(2015; E 2018) Electricity Meters - 0.1, 0.2, and 0.5 Accuracy Classes
NEMA C12.19	(2021) Utility Industry End Device Data Tables

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES~~[, the CONTRACT CLAUSES and DD Form 1423]~~:

~~[~~ Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

~~]~~ SD-02 Shop Drawings

Installation Drawings; G~~, [_____]~~

SD-03 Product Data

Electricity Meters; G~~, [_____]~~

~~[~~ The most recent meter product data must be submitted as a Technical Data Package and must be licensed to the project site. Any software must be submitted on CD-ROM and ~~[_____]~~ hard copies of the software user manual must be submitted for each piece of software provided.

~~]~~ Current Transformer; G~~, [_____]~~

~~[~~ Potential Transformer; G~~, [_____]~~

~~]~~ External Communications Devices; G~~, [_____]~~

~~[~~ Configuration Software; G~~, [_____]~~

The most recent version of the configuration software for each type (manufacturer and model) must be submitted as a Technical Data Package and must be licensed to the project site. Software must be submitted on CD-ROM and ~~[_____]~~ hard copies of the software user manual must be submitted for each piece of software provided.

] SD-06 Test Reports

Acceptance Checks and Tests; G~~, [_____]~~

System Functional Verification; G~~, [_____]~~

Building Meter Installation Sheet, per Building; G~~, [_____]~~

Completed Meter Installation Schedule; G~~, [_____]~~

Completed Meter Data Schedule; G~~, [_____]~~

Meter Configuration Template; G~~, [_____]~~

Contractor must fill in the meter configuration template and submit to the Activity for concurrence.

Meter Configuration Report; G~~, [_____]~~

The meter configuration report must be submitted as a Technical Data Package.

SD-10 Operation and Maintenance Data

Electricity Meters and Accessories, Data Package 5; G~~, [_____]~~

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

SD-11 Closeout Submittals

System Functional Verification; G~~, [_____]~~

1.3 QUALITY ASSURANCE

1.3.1 Installation Drawings

Drawings must be provided in hard-copy and ~~[_____]~~ electronic format, and must include but not be limited to the following:

a. Wiring diagrams with terminals identified of [kilowatt] [advanced] meter, [current transformers,] [potential transformers,][protocol modules,][communications interfaces,][Ethernet connections,][telephone lines]. [For each typical meter installation, provide a diagram.]

b. One-line diagram, including meters, [switch(es),][current transformers,][potential transformers,] [protocol modules,][communications interfaces,][Ethernet connections,][telephone outlets,][and fuses]. [For each typical meter installation, provide a diagram.] Provide one-line diagram to the local Public Works department.

1.3.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for one year prior to bid opening. The one-year period must include applications of equipment and materials under similar circumstances and of similar size. The product, or an earlier release of the product, must have been on sale on the commercial market through advertisements, manufacturers catalogs, or brochures during the prior one-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.3.3 Material and Equipment Manufacturing Data

Products manufactured more than 1 year prior to date of delivery to site must not be used, unless specified otherwise.

1.4 MAINTENANCE

1.4.1 Additions to Operation and Maintenance Data

In addition to requirements of Data Package 5, include the following on the actual electricity meters and accessories provided:

a. A condensed description of how the system operates

b. Block diagram indicating major assemblies

c. Troubleshooting information

d. Preventive maintenance

e. Prices for spare parts and supply list

1.5 WARRANTY

The equipment items and software must be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment and software on a regular and emergency basis during the warranty period of the contract.

1.6 SYSTEM DESCRIPTION

1.6.1 System Requirements

Electricity metering, consisting of meters and associated equipment, will be used to record the electricity consumption and other values as described in the requirements that follow and as shown on the drawings. Communication system requirements are contained in a separate specification section as identified in paragraph COMMUNICATIONS INTERFACES.

1.6.2 Selection Criteria

Metering components and software are part of a system that includes the physical meter, data recorder function and communications method. Every building site identified must include sufficient metering components to measure the electrical parameters identified and to store and communicate the values as required.

~~[~~ Contractor must verify that the electricity meter installed on any building site is compatible with the base-wide metering system with respect to the types of meters selected and the method used to program the meters for initial use. Software and meter programming tools are necessary to set up the meters described by this specification. New software tools different from the meter programming methods currently used by base personnel will require an Authority to Operate (ATO) by Command Information Office at the Enterprise level.~~.~~

~~][~~Contractor must verify that the metering system installed on any building site is compatible with the facility-wide or base-wide communication and meter reading protocol system.

~~]~~PART 2 PRODUCTS

2.1 ELECTRICITY METERS AND ACCESSORIES

Provide meter(s) and connect the meter(s) to the existing AMI DAS. The contractor must use the existing government laptop computers to configure the meter using existing software loaded on the computer. The contractor will not be allowed to modify any software or add any additional software to the computer. Alternatively, the government will configure the meter(s), which must be compatible with the existing system, using existing software. Contractor must insure that the meter(s) will transmit the specified data to the DAS.~~ The current meters being used by [_____] are: [ION 8600C] [ION 8650C] [PM 8000] [SHARK 270] [NEXUS 1272] [_____].~~

2.1.1 Physical and Common Requirements

a. Provide metering system components in accordance with the Metering System Schedule shown ~~[~~in this specification~~][on the drawings]~~. Provide Meter configuration template.

b. ~~[~~Replace all existing meter bases. For socket arrangements, use meter and base form of 9S unless installation-specific limitations require the use of a different form type. For panelboards, switchboards, and switchgear, match the existing installation with the new meter base. ~~][~~Existing meter bases can be re-used if they are electrically functional, in physically good condition, and show no signs of corrosion on the electrical contacts. If the existing meter base is usable, the meter base determines meter form factor. If a new meter is being installed, use meter and base form factor of 9S unless installation-specific limitations require the use of a different form type. ~~][~~If use of a socket adaptor arrangement has been approved by the activity, contractor must verify that all clearances are met and doors are able to be properly closed.~~]~~

~~[~~ c. Meter must have NEMA ~~[3R] [~~3R stainless steel~~]~~ enclosure for surface mounting with bottom or rear penetrations.

~~]~~ d. Surge withstand capability must conform to IEEE C37.90.1.

e. Use #12 SIS (XHHW, or equivalent) wiring with ring lugs for all meter connections. Color code and mark the conductors~~[~~ as follows:

(1) Red - Phase A CT - C1
(2) Orange - Phase B CT - C2
(3) Brown - Phase C CT - C3

(4) Gray with white stripe - neutral current return - C0
(5) Black - Phase A voltage - V1
(6) Yellow - Phase B voltage - V2
(7) Blue - Phase C voltage - V3
(8) White - Neutral voltage

~~]~~

2.1.2 Potential Transformer Requirements

a. Meter must be capable of connection to the service voltage phases and magnitude being monitored. If the meter is not rated for the service voltage, provide suitable potential transformers to send an acceptable voltage to the meter.

b. Voltage input must be optically isolated to 2500 volts DC from signal and communications outputs. Components must meet or exceed IEEE C37.90.1.

c. Provide ~~[a pull-out type fuse block containing]~~ one fuse per phase, Class RK type, to protect the voltage input to the meter. Size fuses as recommended by the meter manufacturer. Fusing must either be inside the secondary compartment of the transformer or inside the same enclosure as the CT shorting device.

~~[~~ d. Potential transformers will be used to convert 480 volt inputs to 120 volts for the locations shown on the metering schedule. Potential transformers must be rated indoor or outdoor, as required for the specific application. Voltage rating must provide 120 volts, wye-connected, 3 phase, 4 wire, ~~[60 Hz][~~50 Hz~~]~~, insulation class, 600 volts. Potential transformers BIL must be 10 kV and must have an accuracy class of 0.3 at burdens w, x, and y. Thermal rating must be 500 VA.

~~]~~

~~[~~ e. The Contractor must be responsible for determining the actual voltage ratio of each potential transformer for medium voltage applications. Transformer must conform to IEEE C57.13 and the following requirements.

(1) Type: Dry type, of two-winding construction.

(2) Weather: Outdoor or indoor rated for the application.

(3) Frequency: Nominal ~~[60 Hz][~~50 Hz~~]~~.

(4) Accuracy: Plus or minus 0.3 percent at ~~[60 Hz][~~50 Hz~~]~~.

f. Potential transformers installed inside switchgear and panels must be rated for interior use. Voltage rating must provide 120 volts, wye-connected, 3 phase, 4 wire, ~~[60 Hz][~~50 Hz~~]~~, insulation class, 600 volts. Potential transformers BIL must be a minimum of 10 kV, and have an insulation class and BIL rating that equals or exceeds the ratings of the associated switchgear. Potential transformers must have an accuracy class of 0.15 at burdens w, x, and y. Thermal rating must be 500 VA. Potential transformers must be accessed from the front and mounted in a metering section.

~~]~~2.1.3 Current Transformer Requirements

a. Current transformer must be installed with a rating as shown in the schedule.

b. Current transformers must have an Accuracy Class of 0.15 (with a maximum error of plus/minus 0.3 percent at 5.0 amperes) when operating within the specified rating factor.

c. Current transformers must be solid-core, bracket-mounted for new installations using ring-tongue lugs for electrical connections. Current transformers must be accessible and the associated wiring must be installed in an organized and neat workmanship arrangement. Current transformers that are retrofitted onto existing switchgear busbar can be a busbar split-core design.

d. Current transformers must have:

(1) Insulation Class: All 600 volt and below current transformers must be rated 10 KV BIL.[Current transformers for 2400 and 4160 volt service must be rated 25 KV BIL.]

(2) Frequency: Nominal ~~[60 Hz]~~[50 Hz].

(3) Burden: Burden class must be selected for the load.

(4) Phase Angle Range: 0 to 60 degrees.

e. Meter must accept current input from standard instrument transformers (5A secondary current transformers).

f. Current inputs must have a continuous rating in accordance with IEEE C57.13.

g. Provide one single-ratio current transformer for each phase per power transformer with characteristics listed in the following table.

Single-Ratio Current Transformer Characteristics				
kVA	Sec. Volt	CT Ratio	RF	Meter Acc. Class
[500]	[208Y:120]	[1200:5]	[1.33]	[0.3 thru B0.05]
[750]	[480Y:277]	[800/5]	[1.33]	[0.3 thru B0.05]

2.1.4 Meter Requirements

~~[Notwithstanding any other provision of this contract, meters must be [_____]; no other product will be acceptable. All meters must meet NAVFAC Cyber Security Requirements.~~

~~]~~[Electricity meters must include the following features:

a. Meter must comply with ANSI C12.1, NEMA C12.19, and ANSI C12.20 and must match existing AMI meter system at the installation and be the newest version with ATO.

b. Meter sockets must comply with ANSI C12.7.

[c. Meter must comply with IEC 62053-22, certified by a qualified third party test laboratory.

d. Meter must be certified by a qualified 3rd party test laboratory.

~~]~~ e. Provide socket-mounted or panel mounted meters as indicated on the meter schedule.

~~[~~ (1) Panel-mounted meters must be semi-flush, back-connected, dustproof, draw-out switchboard type. Cases must have window removable covers capable of being sealed against tampering. Meters must be of a type that can be withdrawn through approved sliding contacts from fronts of panels or doors without opening current-transformer secondary circuits, disturbing external circuits, or requiring disconnection of any meter leads. Necessary test devices must be incorporated within each meter and must provide means for testing either from an external source of electric power or from associated instrument transformers or bus voltage.

~~]~~

f. Meter must be a Class 20, transformer rated design.

~~[g. Use Class 200 meters for direct current reading without current transformers for applications with an expected load less than 200 amperes, where indicated.~~

~~] h. Meter must be rated for use at temperature from minus 40 [_____] degrees Centigrade to plus 70 [_____] degrees Centigrade.~~

g. The meters must have an electronic demand recording register and must be secondary reading as indicated. The register must be used to indicate maximum kilowatt demand as well as cumulative or continuously cumulative demand. Demand must be measured on a block-interval basis and must be capable of a 5 to 60 minute interval and initially set to a 15-minute interval. It must have provisions to be programmed to calculate demand on a rolling interval basis. Meter readings must be true RMS.

h. The meter electronic register must be of modular design with non-volatile data storage. Downloading meter stored data must be capable via an ~~[~~optical~~][USB]~~ port. Recording capability of data storage with a minimum capability of 89 days of 15 minute, 2 channel interval data. The meter must be capable of providing at least 2 KYZ pulse outputs (dry contacts). Default initial configuration (unless identified otherwise by base personnel) must meet NAVFAC CIRCUITS Call for Consistency document located on the NAVFAC CIRCUITS Portal and must be:

(1) First channel - kWh
(2) Second channel - kVARh
(3) KYZ output #1 - kWh
(4) KYZ output #2 - kVARh

i. All meters must have identical features available in accordance with this specification. The meter schedule identifies which features must be activated at each meter location.

j. Enable switches for Time of Use (TOU), pulse and load profile measurement module at the factory.

k. Meter must have an optical port on front of meter. Optical device must be compatible with ANSI C12.18.

l. Meters must be 120-480 volts auto ranging.

m. Provide blank tag fixed to the meter faceplate for the addition of the meter multiplier, which will be the product of the current transformer [and potential transformer]ratio and will be filled in by base personnel on the job site. The meter's nameplate must include:

(1) Meter ID number.
(2) Rated voltage.
(3) Current class.
(4) Metering form.
(5) Test amperes.
(6) Frequency.
(7) Catalog number.
(8) Manufacturing date.

n. On switchboard style installations, provide switchboard case with disconnect means for meter removal incorporating short-circuiting of current transformer circuits.

o. Meter covers must be polycarbonate resins with an optical port and reset. Backup battery must be easily accessible for change-out after removing the meter cover.

p. The normal billing data scroll must be fully programmable. The normal billing data scroll requirements provided in the CIRCUITS Call for Consistency Document located on the NAVFAC CIRCUITS Portal. Data scroll display must include the following.

(1) Number of demand resets.
(2) End-of-interval indication.
(3) Maximum demand.
(4) New maximum demand indication.
(5) Cumulative or continuously cumulative.
(6) Time remaining in interval.
(7) Kilowatt hours.

q. The register must incorporate a built-in test mode that allows it to be tested without the loss of any data or parameters. The following quantities must be available for display in the test mode:

(1) Present interval's accumulating demand.
(2) Maximum demand.
(3) Number of impulses being received by the register.

r. Pulse module simple I/O board with programmable ratio selection.

s. Meters must be programmed after installation via an [optical]~~[USB]~~ port. Optical display must show TOU data, peak kWh, semi-peak kWh, off peak kWh, and phase angles.

t. Self-monitoring to provide for:

(1) Unprogrammed register.
(2) RAM checksum error.
(3) ROM checksum error.
(4) Hardware failure.
(5) Memory failure.

(6) EPROM error.
(7) Battery status (fault, condition, or time in service).

u. Liquid crystal alphanumeric displays, 9 digits, blinking squares confirm register operation. Six Large digits for data and smaller digits for display identifier.

v. Display operations, programmable sequence with display identifiers. Display identifiers must be selectable for each item. Continually sequence with time selectable for each item.

w. The meters must support three modes of registers: Normal Mode, Alternate Mode, and Test Mode. The meter also must support a "Toolbox" or "Service Information" (accessible in the field) through an optocom port to a separate computer using the supplied software to allow access to instantaneous service information such as voltage, current, power factor, load demand, and the phase angle for individual phases.

x. Meter must have a standard ~~[~~4~~] [_____]~~-year warranty.

~~]~~2.1.5 Disconnect Method

a. Provide a 10-pole safety disconnect complete with isolation devices for the voltage and current transformer inputs, including a shorting means for the current transformers.

~~[~~ b. Disconnecting wiring blocks must be provided between the current transformer and the meter. A shorting mechanism must be built into the wiring block to allow the current transformer wiring to be changed without removing power to the transformer. The wiring blocks must be located where they are accessible without the necessity of disconnecting power to the transformer.

c. Voltage monitoring circuits must be equipped with disconnect switches to isolate the meter base or socket from the voltage source.

~~]~~2.1.6 Installation Methods

a. Transformer Mounted ("XFMR" in Metering Systems Schedule). Meter base must be located outside on the secondary side of the pad-mounted transformer.

b. Stand Mounted Adjacent to Transformer ("STAND" in Metering Systems Schedule). Meter base must be mounted on a structural steel pole approximately 1.2 meters from the transformer pad. This can be used for multiple meters associated with a single transformers.

c. Building Mounted ("BLDG" in Metering Systems Schedule). Meter base must be mounted on the side of the existing building near the service entrance.

d. Panel Mounted. ("PNL" in Metering Systems Schedule). Meter must be mounted where directed.

e. Commercial meter pedestal ("PED" in Metering Systems Schedule).

2.2 COMMUNICATIONS INTERFACES

Meter must have two-way communication with the existing data acquisition system (DAS). Provide a communications interface~~ utilizing [_____]~~.~~[~~ Refer to ~~Section [_____] for ~~the communication interface requirements for these meters.~~]~~

Provide interfacing software if a meter is used that is different than the existing meters at the Activity to ensure compatibility within the metering system.

Connect to the AMI network~~ utilizing [_____]~~.

~~[Provide [_____].~~

~~][~~2.3 SPARE PARTS

Provide the following spare parts:

a. Power Meter - two for each type used with batteries.

b. Communications interface - one for each type used.

~~]2.4 METERING SYSTEM SCHEDULE~~

~~[_____]~~

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations must conform to ASHRAE 90.1 - IP, ASHRAE 90.1 - SI IEEE C2, NFPA 70 (National Electrical Code), and to the requirements specified herein. Provide new equipment and materials unless indicated or specified otherwise.

~~[~~3.1.1 Existing Condition Survey

The Contractor must perform a field survey, including inspection of all existing equipment, resulting clearances, and new equipment locations intended to be incorporated into the system and furnish an existing conditions report to the Government. The report must identify those items that are non-workable as defined in the contract documents. The Contractor must be held responsible for repairs and modifications necessary to make the system perform as required.

3.1.1.1 Existing Meter Sockets

In some cases, the existing meter sockets will have to be replaced to accommodate the new electrical meters. An existing socket is considered unacceptable for any of the following conditions:

a. It is a non-ANSI form factor meter socket.

b. It is weathered beyond the point of being safe to reuse.

c. It is installed incorrectly, such as a non-weather resistant enclosure installed outdoors.

d. It is not the correct form factor for the existing electrical service.

3.1.1.2 Existing Installations

As part of the existing condition survey, the following applies for installations with existing meters:

a. Replace any meters that do not comply with this section.

b. If CTs are installed, verify that they comply with this section. If they do not comply, replace them with CTs that comply with this section. One CT per phase is required for wye-connected systems.

~~[~~ c. If potential transformers are installed on low-voltage systems, remove the PTs as part of the installation.

~~]~~ d. Install disconnect switches as specified in this section.

~~][~~3.1.2 Scheduling of Work and Outages

The Contract Clauses must govern regarding permission for power outages, scheduling of work, coordination with Government personnel, and special working conditions.~~[_____]~~

~~]~~3.1.3 Configuration Software

The standard meter must include the latest available version of firmware and software. Meter must either be programmed at the factory or must be programmed in the field. Meters must have a password that must be provided to the contracting officer upon project completion. When field programming is performed, turn field programming device over to the Contracting Officer at completion of project. When interfacing software is used for a meter that is different than the existing meters in use at the Activity, turn the software over to the Contracting Officer at completion of the project.

3.2 FIELD QUALITY CONTROL

Perform the following acceptance checks and tests on all installed meters.

3.2.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

a. Meter Assembly

(1) Visual and mechanical inspection.

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical and mechanical condition. Confirm the meter is firmly seated in the socket, the socket is not abnormally heated, the display is visible, and the ring and seal on the cover are intact.

(c) Inspect all electrical connections to ensure they are tight. For Class 200 services, verify tightness of the service conductor terminations for high resistance using low-resistance ohmmeter, or by verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method.

(d) Record model number, serial number, firmware revision, software revision, and rated control voltage.

(e) Verify operation of display and indicating devices.

(f) Record password and user log-in for each meter.

(g) Verify grounding of metering enclosure.

(h) Set all required parameters including instrument transformer ratios, system type, frequency, power demand methods/intervals, and communications requirements. Verify that the CT ratio and the PT ratio are properly included in the meter multiplier or the programming of the meter. Confirm that the multiplier is provided on the meter face or on the meter.

(i) Provide building meter installation sheet, per building for each facility. See example Graphic E-S1.

[(j) Provide the completed meter installation schedule for the installation if multiple meters are to be used. See example Graphic E-S2

][(k) Provide the completed meter data schedule for the installation if multiple meters are to be used. See example Graphic E-S3.

] (2) Electrical tests.

(a) Apply voltage or current as appropriate to each analog input and verify correct measurement and indication.

(b) Confirm correct operation and setting of each auxiliary input/output feature including mechanical relay, digital, and analog.

(c) After initial system energization, confirm measurements and indications are consistent with loads present.

(d) Make note of, and report, any "Error-Code" or "Caution-Code" on the meter's display.

(3) Provide meter configuration report.

b. Current Transformers

(1) Visual and mechanical inspection.

(a) Compare equipment nameplate data with specification and approved shop drawings.

(b) Inspect physical and mechanical condition.

(c) Verify correct connection, including polarity.

(d) Inspect all electrical connections to ensure they are tight.

(e) Verify that required grounding and shorting connections provide good contact.

(2) Electrical Tests.

Verify proper operation by reviewing the meter configuration report.

[c. Potential Transformers

(1) Visual and mechanical inspection.

(a) Verify potential transformers are rigidly mounted.

(b) Verify potential transformers are the correct voltage.

(c) Verify that adequate clearances exist between the primary and secondary circuit.

(2) Electrical Tests.

(a) Verify by the meter configuration report that the polarity and phasing are correct.

]3.2.2 System Functional Verification

Verify that the installed meters are working correctly in accordance with the meter configuration report:

a. The correct meter form is installed.

b. All voltage phases are present.

c. Phase rotation is correct.

d. Phase angles are correct.

e. The new meter accurately measures power magnitude and direction, and can communicate as required by paragraph COMMUNICATIONS INTERFACES.

-- End of Section --

SECTION 26 29 23

ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS
02/20, CHG 1: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)

EN 61800-3	(2017) Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 519	(2022) Standard for Harmonic Control in Electrical Power Systems
IEEE C62.41.1	(2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits
IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61000-3-12	(2012) Electromagnetic Compatibility (EMC) - Part 3-12: Limits - Limits for harmonic currents produced by equipment connected to public low-voltage systems with input current >16 A and </=75 A per phase

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ICS 1	(2022) Standard for Industrial Control and Systems: General Requirements
NEMA ICS 3.1	(2019) Guide for the Application, Handling, Storage, Installation and Maintenance of Medium-Voltage AC Contactors, Controllers and Control Centers
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA ICS 7	(2020) Adjustable-Speed Drives

NEMA ICS 7.2	(2015) Application Guide for AC Adjustable Speed Drive Systems
NEMA ICS 61800-2	(2005) Adjustable Speed Electrical Power Drive Systems Part 2: General Requirements - Rating Specifications for Low Voltage Adjustable Frequency A.C. Power Drive Systems
NEMA MG 1	(2021) Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	~~(2023; ERTA 7 2023; TIA 23-15) National Electrical Code~~(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15	Radio Frequency Devices

UNDERWRITERS LABORATORIES (UL)

UL 489	(~~2016; Rev 2019~~2025) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 61800-5-1	(2016) Adjustable Speed Electrical Power Drive Systems - Part 5-1: Safety Requirements - Electrical, Thermal and Energy

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SYSTEM DESCRIPTION

1.3.1 Performance Requirements

1.3.1.1 Electromagnetic Interference Suppression

Computing devices, as defined by 47 CFR 15 and EN 61800-3 ~~–~~rules and regulations, must be certified to comply with the requirements for class A computing devices and labeled.

1.3.1.2 Electromechanical and Electrical Components

Ensure electrical and electromechanical components of the Adjustable Speed Drive (ASD) do not cause electromagnetic interference to adjacent electrical or electromechanical equipment while in operation.

1.3.2 Electrical Requirements

1.3.2.1 Power Line Surge Protection

IEEE C62.41.1 and IEEE C62.41.2, IEEE 519, IEC 61000-3-12 Control panel must have surge protection, included within the panel to protect the unit from damaging transient voltage surges. Surge protective device must be mounted near the incoming power source and properly wired to all three phases and ground. Fuses must not be used for surge protection.

1.3.2.2 Sensor and Control Wiring Surge Protection

I/O functions as specified must be protected against surges induced on control and sensor wiring installed outdoors and as shown. Test the inputs and outputs in both normal mode and common mode using the following two waveforms:

a. A 10 microsecond by 1000 microsecond waveform with a peak voltage of 1500 volts and a peak current of 60 amperes.

b. An 8 microsecond by 20 microsecond waveform with a peak voltage of 1000 volts and a peak current of 500 amperes.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[~~ for Contractor Quality Control ~~approval.][for~~ information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Schematic Diagrams; G~~[, [_____]]~~

Interconnecting Diagrams; G~~[, [_____]]~~

Installation Drawings; G~~[, [_____]]~~

As-Built Drawings; G~~[, [_____]]~~

SD-03 Product Data

Adjustable Speed Drives; G~~[, [_____]]~~

Wires and Cables

Equipment Schedule

SD-06 Test Reports

ASD Test

Performance Verification Tests

Endurance Test

~~[~~ SD-07 Certificates

Testing Agency's Field Supervisor NETA Certificate; G~~[, [_____]]~~

~~]~~ SD-08 Manufacturer's Instructions

Installation instructions

SD-09 Manufacturer's Field Reports

ASD Test Plan; G~~[, [_____]]~~

Standard Products

SD-10 Operation and Maintenance Data

Adjustable Speed Drives, Data Package 4

1.5 QUALITY ASSURANCE

1.5.1 Schematic Diagrams

Submit diagrams showing circuits and device elements for each replaceable module. Schematic diagrams of printed circuit boards are permitted to group functional assemblies as devices, provided that sufficient information is provided for government maintenance personnel to verify proper operation of the functional assemblies.

1.5.2 Interconnecting Diagrams

Show interconnections between equipment assemblies, and external interfaces, including power and signal conductors. Include for enclosures and external devices.

1.5.3 Installation Drawings

Show floor plan of each site, with ASD's and motors indicated. Indicate ventilation requirements, adequate clearances, and cable routes. Submit drawings for government approval prior to equipment construction or integration. Immediately record modifications to original drawings made during installation for inclusion into the as-built drawings.

1.5.4 Equipment Schedule

Provide schedule of equipment supplied. Schedule must provide a cross reference between manufacturer data and identifiers indicated in shop drawings. Schedule must include the total quantity of each item of equipment supplied and data indicating compatibility with motors being driven. For complete assemblies, such as ASD's, provide the serial numbers of each assembly, and a sub-schedule of components within the assembly. Provide recommended spare parts listing for each assembly or component.

1.5.5 Installation Instructions

Provide installation instructions issued by the manufacturer of the equipment, including notes and recommendations, prior to shipment to the site. Provide operation instructions prior to acceptance testing.

1.5.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6 DELIVERY AND STORAGE

Store delivered equipment to protect from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

1.7 WARRANTY

The complete system must be warranted by the manufacturer for a period of ~~[~~ one year~~] [[_____] years]~~. Repair or replace any component failing to perform its function as specified and documented at no additional cost to the Government. Items repaired or replaced must be warranted for an additional period of at least one year from the date that it becomes functional again, as specified in FAR 52.246-21 Warranty of Construction.

1.8 MAINTENANCE

1.8.1 Spare Parts

Manufacturers provide spare parts in accordance with recommended spare parts list.

~~[Provide one [_____] spare ASD of each model provided for HVAC equipment, fully programmed and ready for back-up operation when connected.~~

~~]~~

1.8.2 Operation and Maintenance Data

Provide in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Provide service and maintenance information including preventive maintenance, assembly, and disassembly procedures. Include electrical drawings from electrical general sections. Provide additional information necessary to provide complete operation, repair, and maintenance information, detailed to the smallest replaceable unit. Include copies of as-built submittals. Provide routine preventative maintenance instructions, and equipment required. Provide instructions on how to modify program settings, and modify the control program. Provide instructions on drive adjustment, trouble-shooting, and configuration. Provide instructions on process tuning and system calibration.

1.8.3 Technical Support

Provide the ASDs with manufacturer's technical telephone support in English, readily available during normal working hours.

PART 2 PRODUCTS

2.1 ADJUSTABLE SPEED DRIVES (ASD)

Provide adjustable speed drive to control the speed of induction motor(s). The ASD must include the following minimum functions, features and ratings.

a. Input circuit breaker per UL 489 with a minimum of 10,000 amps symmetrical interrupting capacity and door interlocked external operator.

b. A converter stage per UL 61800-5-1 must change fixed voltage, fixed frequency, ac line power to a fixed dc voltage. The converter must utilize a full wave bridge design incorporating diode rectifiers. Silicon Controlled Rectifiers (SCR) are not acceptable. The converter must be insensitive to three phase rotation of the ac line and must not cause displacement power factor of less than .95 lagging under any speed and load condition.

c. An inverter stage must change fixed dc voltage to variable frequency, variable ac voltage for application to a standard NEMA MG 1 Part 30 motor designed for use with adjustable frequency power supplies. Switch the inverter to produce a sine coded pulse width modulated (PWM) output waveform.

d. The ASD shall be capable of supplying 110 percent of rated full load current for one minute at maximum ambient temperature.

e. The ASD must be designed to operate from a ~~[_____]~~100 volt, plus or minus 10 percent, three phase, ~~60~~50 Hz supply, and control motors with a corresponding voltage rating.

f. Acceleration and deceleration time must be independently adjustable from one second to 60 seconds.

~~[~~Adjust decelerating time by~~[~~ providing an external dynamic braking resistor designed to meet NEMA ICS 61800-2 to be capable of decelerating six times the motor inertia with no more than 150 percent of rated current with the motor at its base speed.~~][providing an ASD with a regenerative braking designed to return some of braking energy from the motor to the AC power distribution system.][providing each of several ASD used in a process with a common DC bus tie designed to share the regenerative energy between tied in parallel controls.]~~ ~~]~~Required deceleration time may be achieved using not only dynamic braking resistor but with other methods described in NEMA ICS 7.2-2015 paragraph 5.2.5.

g. Adjustable full-time current limiting must limit the current to a preset value which must not exceed 110 percent of the controller rated current. The current limiting action must maintain the V/Hz ratio constant so that variable torque can be maintained. Short time starting override must allow starting current to reach 175 percent of controller rated current to maximum starting torque.

h. The controllers must be capable of producing an output frequency over the range of 3 Hz to 60 Hz (20 to one speed range), without low speed cogging. Over frequency protection must be included such that a failure in the controller electronic circuitry must not cause frequency to exceed 110 percent of the maximum controller output frequency selected.

i. Minimum and maximum output frequency must be adjustable over the following ranges: 1) Minimum frequency 3 Hz to 50 percent of maximum selected frequency; 2) Maximum frequency 40 Hz to 60 Hz.

j. The controller efficiency at any speed must not be less than 96 percent.

k. The controllers must be capable of being restarted into a motor coasting in the forward direction without tripping.

l. Protection of power semiconductor components must be accomplished without the use of fast acting semiconductor output fuses. Subjecting the controllers to any of the following conditions must not result in component failure or the need for fuse replacement:

(1) Short circuit at controller output

(2) Ground fault at controller output

(3) Open circuit at controller output

(4) Input undervoltage

(5) Input overvoltage

(6) Loss of input phase

(7) AC line switching transients

(8) Instantaneous overload

(9) Sustained overload exceeding 115 percent of controller rated current

(10) Over temperature

(11) Phase reversal

m. Solid state motor overload protection must ~~[~~be included such that current exceeding an adjustable threshold must activate a 60 second timing circuit. Should current remain above the threshold continuously for the timing period, the controller will automatically shut down.~~][have~~ Have ~~[~~sensor in each phase,~~][[Class 10] [Class 20] [~~ Class 10/20 selectable~~]~~ tripping characteristic selected to protect motor against voltage and current unbalance and single phasing,~~]~~ ~~[~~ Class II ground-fault protection, with start and run delays to prevent nuisance trip on staring,~~]~~ ~~[~~analog communication module,~~][~~ ~~[~~NC~~] [NO]~~ isolated overload alarm contact,~~]~~ ~~[~~external overload, reset push button ~~]~~.~~]~~

n. Include slip compensation circuit that will sense changing motor load conditions and adjust output frequency to provide speed regulation of NEMA MG 1 Part 30 designed for use with adjustable frequency power supplies motors to within plus or minus 0.5 percent of maximum speed without the necessity of a tachometer generator.

o. The ASD must be factory set for manual restart after the first protective circuit trip for malfunction (overcurrent, undervoltage, overvoltage or overtemperature) or an interruption of power. The ASD must be capable of being set for automatic restart after a selected time delay. If the drive faults again within a specified time period (adjustable 0-60 seconds), a manual restart will be required.~~[Provide Bidirectional Autospeed Search capable of starting the ASD into rotating loads spinning in either direction and returning motor to set speed in proper direction, without causing damage to drive, motor, or load.]~~

p. The ASD must include external fault reset capability. All the necessary logic to accept an external fault reset contact must be included.

q. Provide critical speed lockout circuitry to prevent operating at frequencies with critical harmonics that cause resonant vibrations. The ASD must have a minimum of three user selectable bandwidths.

r. Provide preperly sized ~~[~~NEMA~~][IEC]~~ rated by-pass and isolation contactors to enable operation of motor in the event of ASD failure~~[~~ and for safety transfers motor between power converter output and bypass circuit using a field-selectable automatic and manual bypass mode~~]~~. Install mechanical and electrical interlocks between the by-pass and isolation contactors. Provide a selector switch and transfer delay timer. Motor overload and short circuit protective features must remain in use during the bypass mode.

s. Each individual ASD must meet the following Total Harmonic Distortion (THD) requirements at the input terminals to the factory assembly of the ASD or at the load disconnecting means serving the ASD and filter assembly. These measurements should be taken with the drive set at 90 percent frequency (rpms) and the motor under a minimum of 50 percent demand.

(1) The Voltage THD should not exceed 2.0 percent THD.

(2) The Current THD should not exceed 15.0 percent THD.

(3) If the standard factory ASD does not meet or exceed these requirements the factory must install appropriate equipment (Harmonic Traps, Filters, different Drive technology, etc.) to mitigate the distortion to assure performance of the VFD is within the limits.

(4) These tests should be performed at the Manufacturers Laboratory facilities and submitted as part of the Product Data Submittals, in order to prevent the necessity of adding mitigation equipment in the field. If the requirements listed above are met, IEEE 519 will also be met.

~~[~~ t. Minimum Operating Conditions. Designed and constructed ASD's to operate within the following service conditions:

(1) Ambient Temperature Rating: 0 to 120 degrees F.

(2) Non-condensing relative humidity rating: less than 95 percent.

(3) Ambient rating: Not exceed 1,006 meters.

~~][2.1.1 ASD for Industrial Application~~

~~Provide the following operator control and monitoring devices mounted on the front panel of the ASD:~~

~~a. Manual speed potentiometer.~~

~~b. Hand-Off-Auto (HOA) switch.~~

~~c. Power on light.~~

~~d. Drive run power light.~~

~~e. Local display[capable of including ASD status, frequency, motor RPM, phase current, fault diagnostic in descriptive text, and all programmed parameters].~~

~~][~~2.1.1 ASD for HVAC Application

ASDs must have the following features:

a. A local operator control providing the following functions:

(1) Remote/Local operator selection with password access.

(2) Run/Stop and manual speed commands.

(3) All programming functions.

(4) Scrolling through all display functions.

b. A local operator control panel with the following data displayed:

(1) ASD status.

(2) Frequency.

(3) Motor RPM.

(4) Phase current.

(5) Scrolling through all display functions.

(6) Fault diagnostics in descriptive text.

(7) All programmed parameters.

c. Standard PI loop controller with input terminal for controlled variable and parameter settings.

d. User interface terminals for remote control of ASD speed, speed feedback, and an isolated form C SPDT relay, which energizes on a drive fault condition.

e. An isolated form C SPDT auxiliary relay which energizes on a run command.

f. An adjustable carrier frequency with 16 KHz minimum upper limit.

g. A built-in or external line reactor with 3 percent minimum impedance to protect the DC bus capacitors and rectifier section diodes~~[~~, reduce power line transient voltage, line notching, DC bus over-voltage tripping and improve the inverter over-current and over-voltage conditions~~]~~.

h. Historical logging information and displays:

~~[~~ (1) Real-time clock with current time and date.

~~]~~ (2) Running log of total power versus time.

(3) Total run time.

~~[~~ (4) Fault log, maintaining last ~~[~~four~~][_____]~~ faults with time and data stamp for each.

~~]~~ ~~(5) [_____].~~

~~[~~ i. The ASD must be capable of automatic control by a remote ~~[~~4-20 mA~~][~~ or 0 to 10 VDC~~][_____]~~ signal, by ~~[~~BACnet~~][LONworks][_____]~~ network command, or manually by the ASD control panel.

~~][~~j. ASDs must include the following operator programmable parameters:

(1) Upper and lower limit frequency.

(2) Acceleration and deceleration rate.

(3) Variable torque volts per Hertz curve.

(4) Starting voltage level.

(5) Starting frequency level.

(6) Display speed scaling.

(7) Enable/disable soft stall feature.

(8) Motor overload level.

(9) Motor stall level.

(10) Jump frequency and hysteresis band.

(11) PWM carrier frequency.

~~][~~k. ASD must have the following protective features:

(1) An electronic adjustable inverse time current limit with consideration for additional heating of the motor at frequencies below 45Hz, for the protection of the motor.

(2) An electronic adjustable soft stall feature, allowing the ASD to lower the frequency to a point where the motor will not exceed the full-load amperage when an overload ASD will automatically return to the requested frequency when load conditions permit.

(3) A separate electronic stall at 110 percent ASD rated current, and

a separate hardware trip at 190 percent current.

(4) The ability to shut down if inadvertently started into a rotating load without damaging the ASD or the motor.

(5) The ability to keep a log of a minimum of four previous fault conditions, indicating the fault type and time of occurrence in descriptive text.

(6) The ability to sustain 110 percent rated current for 60 seconds.

(7) The ability to shutdown safely or protect against and record the following fault conditions:

(a) Over current (and an indication if the over current was during acceleration, deceleration, or running).

(b) Over current internal to the drive.

(c) Motor overload at start-up.

(d) Over voltage from utility power.

(e) Motor running overload.

(f) Over voltage during deceleration.

(g) ASD over heat.

(h) Load and ground fault.

(h) Abnormal parameters or data in ASD EEPROM.

~~]]~~2.2 ENCLOSURES

Provide equipment enclosures conforming to NEMA 250, NEMA ICS 7, and NEMA ICS 6, with a heater if located outdoors. An HMCP device shall provide the disconnecting means. The operating handle shall protrude through the door, but the disconnect shall not be mounted on the door. The handle shall indicate ON, OFF, and tripped conditions. The handle shall have provisions to accommodate a minimum of three padlocks in the OFF position. Interlocks shall prevent unauthorized opening or closing of the ASD door with the disconnect handle in the ON position. The door handle interlock should have provisions to be defeated by qualified maintenance personnel.

2.3 WIRES AND CABLES

All wires and cables must conform to NEMA 250, NEMA ICS 7, NFPA 70.

2.4 NAMEPLATES

Nameplates external to NEMA enclosures must conform with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manufacturer's standard, permanent nameplates for internal areas of enclosures.

2.5 SOURCE QUALITY CONTROL

2.5.1 ASD Test Plan

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test plans.

2.5.2 ASD Test Report

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test reports.

PART 3 EXECUTION

3.1 INSTALLATION

Per NEMA ICS 3.1, install equipment in accordance with the approved manufacturer's printed installation drawings, instructions, wiring diagrams, and as indicated on project drawings and the approved shop drawings. A field representative of the drive manufacturer must supervise the installation of all equipment, and wiring.

3.2 GROUNDING

Per NEMA ICS 7.2, ASD must be solidly grounded to the main distribution.

3.3 FIELD QUALITY CONTROL

Specified products must be tested as a system for conformance to specification requirements prior to scheduling the acceptance tests. Conduct performance verification tests in the presence of Government representative, observing and documenting complete compliance of the system to the specifications. Submit a signed copy of the test results, certifying proper system operation before scheduling tests.

3.3.1 ASD Test

A proposed test plan must be submitted to the contracting officer at least 28 calendar days prior to proposed testing for approval. The tests must conform to NEMA ICS 1, NEMA ICS 7, and all manufacturer's safety regulations. The Government reserves the right to witness all tests and review any documentation. Inform the Government at least 14 working days prior to the dates of testing. Perform the ASD test ~~[with the assistance of a factory-authorized service representative][~~engaging a qualified testing agency's field supervisor currently certified by NETA to supervise on-site testing~~]~~.

3.3.2 Performance Verification Tests

"Performance Verification Test" plan must provide the step by step procedure required to establish formal verification of the performance of the ASD. Compliance with the specification requirements must be verified by inspections, review of critical data, demonstrations, and tests. The Government reserves the right to witness all tests, review data, and request other such additional inspections and repeat tests as necessary to ensure that the system and provided services conform to the stated requirements. Inform the Government 14 calendar days prior to the date the test is to be conducted.

3.3.3 Endurance Test

Immediately upon completion of the performance verification test, the endurance test must commence. The system must be operated at varying rates for not less than 192 consecutive hours, at an average effectiveness level of 0.9998, to demonstrate proper functioning of the complete PCS. Continue the test on a day-to-day basis until performance standard is met. The contractor is not allowed in the building during the endurance test. The system must respond as designed.

3.4 DEMONSTRATION

3.4.1 Training

Coordinate training requirements with the Contracting Officer. Provide video tapes, if available, of all training provided to the Government for subsequent use in training new personnel. Provide all training aids, texts, and expendable support material for a self-sufficient presentation shall be provided, the amount of which to be determined by the contracting officer.

3.4.1.1 Instructions to Government Personnel

Provide the services of competent instructors with minimum two-year field experience with the operation and maintenance of similar ASDs who will give full instruction to designated personnel in operation, maintenance, calibration, configuration, and programming of the complete control system. Orient the training specifically to the system installed. Instructors must be thoroughly familiar with the subject matter they are to teach. The number of training days of instruction furnished must be as specified. A training day is defined as eight hours of instruction, including two 15-minute breaks and excluding lunch time; Monday through Friday. Provide a training manual for each student at each training phase which describes in detail the material included in each training program. Provide one additional copy for archiving. Provide equipment and materials required for classroom training. Provide a list of additional related courses, and offers, noting any courses recommended. List each training course individually by name, including duration, approximate cost per person, and location of course. Unused copies of training manuals must be turned over to the Government at the end of last training session.

3.4.1.2 Operating Personnel Training Program

Provide one 2-hour training session at the site at a time and place mutually agreeable between the Contractor and the Government. Provide session to train 4 operation personnel in the functional operations of the system and the procedures that personnel will follow in system operation. This training shall include:

a. System overview

b. General theory of operation

c. System operation

d. Alarm formats

e. Failure recovery procedures

f. Troubleshooting

3.4.1.3 Engineering/Maintenance Personnel Training

Accomplish the training program as specified. Training must be conducted on site at a location designated by the Government. Provide a one-day training session to train ~~four [_____] engineering personnel~~housing maintenance department in the functional operations of the system. This training must include:

a. System overview

b. General theory of operation

c. System operation

d. System configuration

e. Alarm formats

f. Failure recovery procedures

g. Troubleshooting and repair

h. Maintenance and calibration

i. System programming and configuration

-- End of Section --

SECTION 26 32 15

ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.11	(2021) Forged Fittings, Socket-Welding and Threaded
ASME B31.1	(2024) Power Piping
ASME B31.3	(2024) Process Piping
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)

AEIC CS8	(2013) Specification for Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A106/A106M	(2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A181/A181M	(2014; R 2020) Standard Specification for Carbon Steel Forgings, for General-Purpose Piping
ASTM A234/A234M	(2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service

ELECTRICAL GENERATING SYSTEMS ASSOCIATION (EGSA)

EGSA 101P	(1995) Performance Standard for Engine Driven Generator Sets

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1	(2000; R 2011) General Principles for Temperature Limits in the Rating of Electric Equipment and for the Evaluation of Electrical Insulation
IEEE 43	(2013) Recommended Practice for Testing Insulation Resistance of Rotating Machinery
IEEE 48	(2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV
IEEE 81	(2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE 115	(2019) Guide for Test Procedures for Synchronous Machines: Part I Acceptance and Performance Testing; Part II Test Procedures and Parameter Determination for Dynamic Analysis
IEEE 120	(2023) Master Test Guide for Electrical Measurements in Power Circuits
IEEE 404	(2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V
IEEE 484	(2019) Recommended Practice for Installation Design and Implementation of Vented Lead-Acid Batteries for Stationary Applications
IEEE 485	(2020) Recommended Practice for Sizing Lead-Acid Batteries for Stationary Applications
IEEE 519	(2022) Standard for Harmonic Control in Electrical Power Systems
IEEE C2	(2023) National Electrical Safety Code
IEEE C50.12	(2005; R 2010) Standard for Salient Pole 50 HZ and 60 Hz Synchronous Generators and Generation/Motors for Hydraulic Turbine

Applications Rated 5 MVA and above

IEEE C57.13 (2016) Standard Requirements for Instrument Transformers

IEEE C57.13.1 (2006; R 2012) Guide for Field Testing of Relaying Current Transformers

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2024) International Building Code

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60034-2A (1974; ED 1.0) Rotating Electrical Machines Part 2: Methods for Determining Losses and Efficiency of Rotating Electrical Machinery from Tests (Excluding Machines for Traction Vehicles) Measurement of Losses by the Calorimetric Method

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 3046 (2002, 2006, 2009, 2001) Reciprocating Internal Combustion Engines - Performance--Part 1, 3, 4, 5, 6

ISO 8528 (1993; R 2018) Reciprocating Internal Combustion Engine Driven Alternating Current Generator Sets--Part 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58 (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

MSS SP-80 (2019) Bronze Gate, Globe, Angle, and Check Valves

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

NEMA MG 1 (2021) Motors and Generators

NEMA PB 1	(2011) Panelboards
NEMA PB 2	(2011) Deadfront Distribution Switchboards
NEMA WC 74/ICEA S-93-639	(2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy
NEMA/ANSI C12.11	(2024) Instrument Transformers for Revenue Metering, 10 kV BIL through 350 kV BIL (0.6 kV NSV through 69 kV NSV)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30	(2024; ERTA 1 2025) Flammable and Combustible Liquids Code
NFPA 37	(2024; TIA 24-1) Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines
NFPA 54	(2024) National Fuel Gas Code
NFPA 58	(2024; TIA 24-2) Liquefied Petroleum Gas Code
NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 99	(2024, TIA 24-2) Health Care Facilities Code
NFPA 110	(2025) Standard for Emergency and Standby Power Systems

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE ARP892	(1965; R 1994) DC Starter-Generator, Engine
SAE J537	(2023) Storage Batteries

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-5624	(2024; Rev X) Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-16884	(2017; Rev P) Fuel, Naval Distillate
MIL-STD-461	(2015; Rev G) Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
UFC 3-301-01	(2023; with Change 3, 2025) Structural Engineering

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 60	Standards of Performance for New

Stationary Sources

UL SOLUTIONS (UL)

UL 142	(2019; Reprint Jan 2021) UL Standard for Safety Steel Aboveground Tanks for Flammable and Combustible Liquids
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 489	(2025) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 891	(2019; Reprint Mar 2025) UL Standard for Safety Switchboards
UL 1236	(2015; Reprint Feb 2021) UL Standard for Battery Chargers and Charging Engine-Starter Batteries
UL 1437	(2025) Electrical Analog Instruments - Panel Board Types

1.2 RELATED MATERIALS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and Section 26 08 00 APPARATUS INSPECTION AND TESTING apply to this section, except as modified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. ~~Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.~~ Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Engine-Generator Set and Auxiliary Equipment; G~~, [_____]~~

Auxiliary Systems; G~~, [_____]~~

Detailed Drawings; G~~, [_____]~~

Acceptance; G~~, [_____]~~

SD-03 Product Data

Harmonic Requirements; G~~, [_____]~~

Engine-Generator Set Efficiencies; G~~, [_____]~~

~~Emissions; G, [_____]~~

~~filters; G, [_____]~~

special tools; G~~, [_____]~~

Remote Alarm Annunciator; G~~, [_____]~~

Engine-Generator Parameter Schedule

~~Heat Exchanger~~

Generator

~~Manufacturer's Catalog~~

Site Welding

Spare Parts

Onsite Training

Vibration-Isolation

Posted Data and Instructions; G~~, [_____]~~

Instructions; G~~, [_____]~~

Experience

Field Engineer

General Installation

Exciter

SD-05 Design Data

Performance Criteria

~~Sound Limitations; G, [_____]~~

Integral Main Fuel Storage Tank

~~Day Tank~~

Power Factor

~~Heat Exchanger~~

Time-Delay on Alarms

~~Cooling System~~

Vibration Isolation

Battery Charger

Capacity Calculations for Engine-Generator Set; G~~, [_____]~~

Brake Mean Effective Pressure (BMEP) Calculations; G~~, [_____]~~

Torsional Vibration Stress Analysis Computations; G~~, [_____]~~

Capacity Calculations for Batteries; G~~, [_____]~~

Turbocharger Load Calculations; G~~, [_____]~~

SD-06 Test Reports

Performance Tests

Factory Inspection and Tests

Factory Tests

Onsite Inspection and Tests; G~~, [_____]~~

Acceptance Checks and Tests; G~~, [_____]~~

Functional Acceptance Tests; G~~, [_____]~~

Maintenance Procedures; G~~, [_____]~~

Operation and Maintenance Manuals; G~~, [_____]~~

Inspections; G~~, [_____]~~

Functional Acceptance Test Procedure; G~~, [_____]~~

SD-07 Certificates

~~Cooling System~~

Vibration Isolation

Prototype Test

Reliability and Durability

Fuel System Certification; G~~, [_____]~~

Start-Up Engineer; G~~, [_____]~~

Instructor's Qualification Resume; G~~, [_____]~~

Engine Emission Limits; G~~, [_____]~~

~~Sound Limitations~~

Site Visit

Current Balance

Materials and Equipment

Factory Inspection and Tests

SD-09 Manufacturer's Field Reports

Engine Tests; G~~, [_____]~~

Generator Tests; G~~, [_____]~~

Assembled Engine-Generator Set Tests; G~~, [_____]~~

SD-10 Operation and Maintenance Data

Preliminary Assembled Operation and Maintenance Manuals; G~~, [_____]~~

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and the paragraph ASSEMBLED OPERATION AND MAINTENANCE MANUALS.

SD-11 Closeout Submittals

Posted Data and Instructions; G~~, [_____]~~

Training Plan; G~~, [_____]~~

1.4 QUALITY ASSURANCE

1.4.1 Conformance to Codes and Standards

Where equipment is specified to conform to requirements of any code or standard such as UL, NEMA, etc., the design, fabrication and installation must also conform to the code.

1.4.2 Site Welding

~~Weld structural members in accordance with Section 05 05 23.16 STRUCTURAL WELDING.~~ For all other welding, qualify procedures and welders in accordance with ASME BPVC SEC IX.

a. Welding procedures qualified by others, and welders and welding operators qualified by a previously qualified employer may be accepted as permitted by ASME B31.1.

b. Submit a copy of qualifying procedures and a list of names and identification symbols of qualified welders and welding operators.

c. Submit a letter listing the welder qualifying procedures for each welder, complete with supporting data such as test procedures used, what was tested to, and a list of the names of all welders and their identification symbols.

d. Perform welder qualification tests for each welder whose qualifications are not in compliance with the referenced standards. Notify the Contracting Officer 24 hours in advance of qualification tests which must be performed at the work site, if practical.

e. The welder or welding operator must apply the personally assigned symbol near each weld made as a permanent record.

1.4.3 Vibration Limitation

Limit the maximum engine-generator set vibration in the horizontal, vertical, and axial directions to 0.15 mm (peak-peak RMS), with an overall velocity limit of 24 mm/second RMS, at rated speed for all loads through

110 percent of rated speed. ~~[~~Install a vibration isolation system between the floor and the base to limit the maximum vibration transmitted to the floor at all frequencies to a maximum of ~~[_____]~~ (peak force).~~][~~ The engine-generator set must be provided with vibration isolation in accordance with the manufacturer's standard recommendation.~~]~~ Where the vibration isolation system does not secure the base to the structure floor or unit foundation, provide seismic restraints in accordance with the seismic parameters specified.

~~1.4.4 Torsional Analysis~~

~~Submit torsional analysis including prototype testing or calculations which certify and demonstrate that no damaging or dangerous torsional vibrations will occur when the prime mover is connected to the generator, at synchronous speeds, plus/minus 10 percent.~~

1.4.4 Performance Data

Submit vibration isolation system performance data for the range of frequencies generated by the engine-generator set during operation from no load to full load and the maximum vibration transmitted to the floor. Also submit a description of seismic qualification of the engine-generator mounting, base, and vibration isolation.

1.4.5 Seismic Requirements

~~[~~Seismic requirements must be in accordance with UFC 3-301-01 and Sections 13 48 73 SEISMIC CONTROL FOR NONSTRUCTURAL COMPONENTS~~, 23 05 48.19 SEISMIC BRACING FOR MECHANICAL SYSTEMS~~ and 26 05 48 SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT~~]~~ ~~[~~as shown on the drawings~~]~~.

1.4.6 Experience

Each component manufacturer must have a minimum of 3 years' experience in the manufacture, assembly and sale of components used with stationary engine-generator sets for commercial and industrial use. The engine-generator set manufacturer/assembler must have a minimum of 3 years' experience in the manufacture, assembly and sale of stationary engine-generator sets for commercial and industrial use. Submit a statement showing and verifying these requirements.

1.4.7 Field Engineer

The engine-generator set manufacturer or assembler must furnish a qualified field engineer to supervise the complete installation of the engine-generator set, assist in the performance of the onsite tests, and instruct personnel as to the operational and maintenance features of the equipment. The field engineer must have attended the engine generator manufacturer's training courses on installation and operation and maintenance of engine generator sets. Submit a letter listing the qualifications, schools, formal training, and experience of the field engineer.

1.4.8 Detailed Drawings

Submit detailed drawings showing the following:

a. Base-mounted equipment, complete with base and attachments, including anchor bolt template and recommended clearances for maintenance and operation.

b. Starting system.

c. Fuel system.

d. Cooling system.

e. Exhaust system.

f. Electric wiring of relays, breakers, programmable controllers, and switches including single line and wiring diagrams.

g. Lubrication system, including piping, pumps, strainers, filters, ~~[~~heat exchangers for lube oil and turbocharger cooling,~~]~~ ~~[~~electric heater,~~]~~ controls and wiring.

h. Location, type, and description of vibration isolation devices for all applications.

i. The safety system, including wiring schematics.

j. One-line schematic and wiring diagrams of the generator, exciter, regulator, governor, and instrumentation.

k. Panel layouts.

l. Mounting and support for each panel and major piece of electrical equipment.

m. Engine-generator set rigging points and lifting instructions.

1.4.9 Auxiliary Systems Engine-Generator Set and Auxiliary Equipment Drawing Requirements

Submit drawings pertaining to the engine-generator set and auxiliary equipment, including but not limited to the following:

a. Certified outline, general arrangement (setting plan), and anchor bolt details. Show total weight and center of gravity of assembled equipment on the steel sub-base.

b. Detailed elementary, schematic wiring, and interconnection diagrams of the engine starting system, jacket coolant heating system, engine protective devices, engine alarm devices, engine speed governor system, generator and excitation system, and other integral devices.

c. Detailed elementary, schematic wiring; and interconnection diagrams of the fuel system, starting battery system, engine-generator control panel, generator circuit breaker~~[~~, and remote alarm annunciator~~]~~.

d. Dimensional drawings or catalog cuts of exhaust silencers, radiator, fuel day tanks, fuel oil cooler, valves and pumps, intake filters, vibration isolators, and other auxiliary equipment not integral with the engine-generator set.

1.4.10 Auxiliary Systems Drawing Requirements

Submit drawings showing floor plan arrangement of~~[~~ exhaust,~~][~~ air intake,~~][~~

fuel oil cooler,~~][~~ and~~][~~ jacket coolant water~~]~~ systems including arrangement of piping and pipe sizes.

1.4.11 Vibration Isolation System Certification

Submit certification from the manufacturer that the vibration isolation system will reduce the vibration to the limits specified in the paragraph VIBRATION ISOLATION.

1.4.12 Fuel System Certification

When the fuel system requires a fuel oil cooler as described in the paragraph FUEL OIL COOLER, submit certification from the engine manufacturer that the fuel system design is satisfactory.

1.5 DELIVERY, STORAGE, AND HANDLING

Properly protect materials and equipment, in accordance with the manufacturers recommended storage procedures,before, during, and after installation. Protect stored items from the weather and contamination. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

Deliver equipment on pallets or blocking wrapped in heavy-duty plastic, sealed to protect parts and assemblies from moisture and dirt. Protect and prepare batteries for shipment as recommended by the battery manufacturer. Store auxiliary equipment at the site in covered enclosures, protected from atmospheric moisture, dirt, and ground water.

1.6 EXTRA MATERIALS

Provide ~~[~~two~~] [_____]~~ sets of special tools and ~~[~~two~~] [_____]~~ sets of filters required for maintenance. Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts. One handset must be provided for each electronic governor when required to indicate and/or change governor response settings. Furnish 4 liters one gallon of identical paint used on engine-generator set in manufacturer's sealed container with each engine-generator set.

Wrenches and tools specifically designed and required to work on the new equipment, which are not commercially available as standard mechanic's tools, must be furnished to the Contracting Officer.

Provide proposed operating instructions for the engine-generator set and auxiliary equipment laminated between matte-surface thermoplastic sheets and suitable for placement adjacent to corresponding equipment. After approval, install operating instructions where directed.

1.7 MAINTENANCE SERVICES

Submit the operation and maintenance manuals and have them approved prior to commencing onsite tests.

1.7.1 Operation Manual

Provide ~~[~~three~~] [_____]~~ copies of the ~~[~~manufacturers standard operation manual~~] [operation manual in 216 by 279 mm three-ring binders]~~. Sections must be separated by heavy plastic dividers with tabs which identify the material in the section. Fold drawings with the title block visible, and placed in 216 by 279 mm plastic pockets with reinforced holes. The manual must include:

a. Step-by-step procedures for system startup, operation, and shutdown;

b. Drawings, diagrams, and single-line schematics to illustrate and define the electrical, mechanical, and hydraulic systems with their controls, alarms, and safety systems;

c. Procedures for interface and interaction with related systems to include ~~[~~automatic transfer switches~~]~~ ~~[~~fire alarm/suppression systems~~]~~ ~~[~~load shedding systems~~]~~ ~~[~~uninterruptible power supplies~~] [_____]~~.

1.7.2 Maintenance Manual

Provide ~~[~~three~~] [_____]~~ copies of the ~~[~~manufacturers standard maintenance manual~~] [maintenance manual containing the information described below in 216 x 279 mm three-ring binders]~~. Separate each section by a heavy plastic divider with tabs. Fold drawings with the title block visible, and placed in plastic pockets with reinforced holes. The manual must include:

a. ~~[~~Procedures for each routine maintenance item.~~][~~ Procedures for troubleshooting.~~][Factory-service, take-down overhaul, and repair~~ Repair service manuals, with parts lists.~~]~~

b. The manufacturer's recommended maintenance schedule.

c. A component list which includes the manufacturer's name, address, type or style, model or serial number, rating, and catalog number for the major components.

d. A list of spare parts for each piece of equipment and a complete list of materials and supplies needed for operation.

1.7.3 Assembled Operation and Maintenance Manuals

The contents of the assembled operation and maintenance manuals must include the manufacturer's O&M information required by the paragraph SD-10, OPERATION AND MAINTENANCE DATA and the manufacturer's O&M information specified in Section 26 36 23 AUTOMATIC TRANSFER SWITCHES AND BY-PASS/ISOLATION SWITCH.

a. Manuals must be in separate books or volumes, assembled and bound securely in durable, hard covered, water resistant binder, and indexed by major assembly and components in sequential order.

b. A table of contents (index) must be made part of the assembled O&M. The manual must be assembled in the order noted in table of contents.

c. The cover sheet or binder on each volume of the manuals must be identified and marked with the words, "Operation and Maintenance Manual."

1.8 SITE CONDITIONS

Protect the components of the engine-generator set, including cooling system components, pumps, fans, and similar auxiliaries when not operating and provide components capable of the specified outputs in the following environment:

a. Site Location: ~~[_____]~~CAMP ZAMA

b. Site Elevation: ~~[_____]~~40 meters above mean sea level.

c. Ambient Temperatures:

(1) Maximum ~~[_____]~~40 degrees C dry bulb, ~~[_____]~~30 degrees C wet bulb.

(2) Minimum ~~[_____]~~0 degrees C dry bulb.

d. Design Wind Velocity: ~~[_____]~~160 km/h.

e. Prevailing Wind Direction: ~~[_____]~~Variable

f. Seismic Zone: Zone ~~[_____]~~as indicated on drawings as defined by ICC IBC.

PART 2 PRODUCTS

2.1 SYSTEM REQUIREMENTS

a. Provide and install each engine-generator set complete and totally functional, with all necessary ancillary equipment to include: air filtration; starting system; generator controls, protection, and isolation; instrumentation; lubrication; fuel system; cooling system; and engine exhaust system. Each engine-generator set must satisfy the requirements specified in the Engine-Generator Parameter Schedule. Submit certification that the engine-generator set and cooling system function properly in the ambient temperatures specified.

b. Provide each engine-generator set consisting of one engine, one generator, and one exciter mounted, assembled, and aligned on one base; and all other necessary ancillary equipment which may be mounted separately. Assemble sets having a capacity of 750 kW or smaller and attach to the base prior to shipping. Sets over 750 kW capacity may be shipped in sections. Provide set components that are environmentally suitable for the locations shown and that are the manufacturer's standard product offered in catalogs for commercial or industrial use. Provide a generator strip heater for moisture control when the generator is not operating. Identify any nonstandard products or components and the reason for their use.

2.1.1 Engine-Generator Parameter Schedule

Engine-Generator Set and Auxiliary Equipment Capacity Calculations for Engine-Generator Set

ENGINE-GENERATOR PARAMETER SCHEDULE	
Identification	Make/Model
Electrical Characteristics	

ENGINE-GENERATOR PARAMETER SCHEDULE	
Power Rating	~~[Prime][Limited Running Time][~~Emergency Standby~~] [Industrial]~~ Gross bhp rating / Net brake power rating ~~[____]~~200 kW at 0.8 power factor
Governor Type	Type Make / Model ~~[~~Isochronous~~][Droop]~~
Overload Capacity (Prime applications only)	110 percent of Service Load for 1 hour in 12 consecutive hours
Service Load	~~[____]~~365.5 kVA (maximum) ~~[____]~~150.3 kVA (continuous)
Motor Starting kVA (Max.)	~~[____]~~330.5 kVA
Power Factor	~~[~~0.8~~][____]~~ lagging
Engine-Generator Applications	~~[~~stand-alone~~][parallel with infinite bus][parallel with other generators on an isolated bus][parallel with other generators on an infinite bus]~~
Voltage Regulation (No Load to Full Load)(Stand-alone applications)	plus or minus 2 percent (maximum)
Voltage Bandwidth (steady state)	plus or minus ~~[0.5][1][~~2~~]~~ percent
Frequency	~~[~~50~~][60]~~ Hz
Voltage	400Y/230V~~[____] volts~~
Phases	~~[~~3 Phase, Wye~~][3 Phase, Delta][1 Phase]~~
Minimum Generator Sub-transient Reactance	~~[____]~~0.1688 percent Sub-transient
Nonlinear Loads	~~[____]~~22.2 kVA
Max Step Load Increase	~~[~~25~~][50][75] [100]~~ percent of Service Load at ~~[____]~~0.8 PF
Transient Recovery Time with Step Load Increase (Voltage)	~~[____]~~< 3 seconds
Transient Recovery	~~[____]~~< 3 seconds
Time with Step Load Increase (Frequency)	
Maximum Voltage Deviation with Step Load Increase	~~[5][10][~~30~~][____]~~ percent of rated voltage

ENGINE-GENERATOR PARAMETER SCHEDULE					
Maximum Frequency Deviation with Step Load Increase	~~[2.5][~~5~~][______]~~ percent of rated frequency				
Max Step Load Decrease (without shutdown)	100 percent of Service Load at ~~[______]~~0.8 PF				
Frequency Bandwidth (steady state)	plus or minus ~~[______][0.4][~~0.25~~]~~ percent				
Frequency Regulation (droop) (No Load to Full Load)	~~[~~3~~][______]~~ percent (maximum)				
Frequency Bandwidth (steady state)	plus or minus ~~[______][0.4][~~0.25~~]~~ percent				
Reactances	Synchronous reactance, Xd Transient reactance, X'd Sub-transient reactance, X"d Negative sequence reactance, X2 Zero sequence reactance, Xo				
Capacity Calculations					
Calculations must verify that the engine-generator set power rating is adequate for the following load conditions:					
Lighting	~~[______]~~1.50 kW				
Motor Starting Sequence see generator calculations for full motor starting sequence.	Starting Order	Size (HP/kW ~~hp~~)	Locked Rotor NEMA Code	Starting Method	Maximum Voltage Dip
	1.1~~[______]~~	0.18 kW ~~[______]~~	~~[~~F~~] [______]~~	~~[~~Full Voltage~~] [______]~~	~~[25] [______]~~ 30 Percent
	1.2~~[______]~~	7.50 kW ~~[______]~~	~~[~~F~~] [______]~~	~~[~~Full Voltage~~] [______]~~	~~[25] [______]~~ 30 Percent
	2.1~~[______]~~	0.40 kW ~~[______]~~	~~[~~F~~] [______]~~	~~[~~Full Voltage~~] [______]~~	~~[25] [______]~~ 30 Percent
Other Load	kW at 0.8 power factor				
Capacity Calculations for Batteries					
Calculation must verify that the engine starting battery capacity exceeds dc power requirements.					
Mechanical Characteristics					
Engine Description	Strokes/cycle Number of cylinders Bore and Stroke, mm				

ENGINE-GENERATOR PARAMETER SCHEDULE	
Engine Speed	~~[____][900][1200][1800]~~1500 rpm
Piston Speed	~~[____] m/s~~
Heat Exchanger Type	~~[~~fin-tube (radiator)~~][shell-tube]~~
Engine Cooling Type	water/ethylene glycol
Intercooler Type	Air-to-Air / Jacket Water
Induction Method	Naturally Aspirated / Turbocharged
Turbocharger	Make / Model
Max Time to Start and be Ready to Assume Load	~~[~~10~~][____]~~ seconds
Max Summer Indoor Temp (Prior to Engine-generator Operation)	~~[____]~~25 degrees C
Min Winter Indoor Temp (Prior to Engine-generator Operation)	~~[____]~~25 degrees C
Max Allowable Heat Transferred To Engine Generator Space at Rated Output Capacity	~~[____]~~1441 MBTU/hr
Max Summer Outdoor Temp (Ambient)	~~[____]~~25 degrees C
Min Winter Outdoor Temp (Ambient)	~~[____]~~25 degrees C
Installation Elevation	~~[____] above sea level~~
Engine-Generator Set Efficiencies	
Fuel Consumption	At 0.8 power factor, liters / hour for: 1 / 2 load 3 / 4 load Full Load
Generator Efficiency	At 0.8 power factor,(per cent) ~~[~~in accordance with IEC 60034-2A~~]~~ for: 1 / 2 load 3 / 4 load Full Load
Radiator Capacity	Coolant Type L/s coolant L/s air through radiator kW per hr of heat exchange based on optimum coolant temperature to and from engine

ENGINE-GENERATOR PARAMETER SCHEDULE	
Engine-Generator Set Emissions Data	
Exhaust Temperature	Degrees C at full load
Weight of Exhaust Gas	Kg per hr at full load
Weight of Intake Air	Kg per hr at full load
Total Heat Rejected	kW per hr, at full load to: Jacket Coolant System Fuel Oil Cooling System
Emissions	Kg per hr , at full load Total Suspended Particulate Particulate Matter with an average aerodynamic diameter of 10 microns Sulfur Dioxides Nitrogen Oxides (as NO2) Carbon Monoxide Volatile Organic Compounds
Visible Emissions	Percent opacity at full load
Brake Mean Effective Pressure (BMEP) Calculations	
Calculation must verify that the engine meets the specified maximum BMEP, as follows: BMEP kPa = (120,000 X bkW) X (792,000 X bhp) (rpm X liters) Where: bkW bhp = bkW' + bkW" bhp' + bhp" bkW" bhp" is the Brake kW horsepower required by engine driven fan for cooling radiator or motor driven fan for cooling radiator. bkW' bhp' = kW/GEN.EFF. kW/(GEN.EFF. times 0.746) GEN.EFF. = Generator efficiency liters = Total engine piston displacement in liters rpm = Engine revolutions per minute kW = Minimum power rating	
Torsional Vibration Stress Analysis Computations	

ENGINE-GENERATOR PARAMETER SCHEDULE
Torsional vibrational stresses in the crankshaft and generator shaft of assembled engine and driven generator must not exceed 34,450 kPa when engine is driving generator at rated speed while assembled unit is loaded to rated engine-generator set power. Computations must be based on a mathematical model of the assembled generator set provided or based on calculations using measured values from tests on a unit identical to the one provided. Calculations based on models of, or measured data from, the unassembled engine and generator will not be acceptable. Calculations must include: a. A description of the system relating information pertinent to analysis such as operating speed range and identification plate data. b. A mass elastic assembly drawing, showing the arrangement of the units in the generator set and dimensions of shafting, including minimum diameters (or section moduli) of shafting in the system. c. A labeled line diagram of the mass elastic system indicating values of masses, stiffness, equivalent lengths, and equivalent diameters including basic assumptions and definition of terms. d. Sample computations showing procedures used to obtain resulting stress values. e. Computations indicating assembled engine-generator speed of 1800 rpm with assembly loaded to rated generator power and the resulting computed critical torsional stress values in the assembled engine crankshaft and generator shaft.
Turbocharger Load Calculations
NOTE: When the engine-generator set installation includes field installed exhaust system (i.e., the engine-generator set is installed internal to a building in lieu of in a self-contained outdoor enclosure), include the following paragraph. When the proposed exhaust system layout is different from that shown on the contract drawings, submit calculations showing that the external loads from the exhaust system such as weight and thermal expansion do not exceed the engine manufacturer's maximum allowed forces and moments on the turbocharger.

2.1.2 Rated Output Capacity

Provide each engine-generator-set with power equal to the sum of service load plus the machine's efficiency loss and associated ancillary equipment loads. Rated output capacity must also consider engine and/or generator oversizing required to meet requirements in paragraph Engine-Generator Parameter Schedule.

The engine must meet the specified maximum BMEP requirements at rated speed as calculated in accordance with the calculations in the engine-generator parameter schedule. The engine capacity must be based on the following:

a. Engine burning diesel fuel conforming to ~~[~~MIL-DTL-16884~~][ASTM D975, Grade 2-D,]~~ or ~~[~~MIL-DTL-5624, Grade JP-5~~]~~ at an ambient temperature of 29 degrees C . For stationary engines operated in the United States, diesel fuel requirements are found in 40 CFR 60 Subpart IIII.

b. Engine cooled by a radiator fan mechanically driven by the engine or remote with a motor driven fan.

c. Engine cooled by coolant mixture of water and ethylene glycol, 50 percent by volume of each.

Maximum BMEP, kPa			
	Naturally Aspirated	Turbocharged	Turbocharged and Intercooled
Four-cycle engines	~~[_____]~~	~~[_____]~~	~~[_____]~~
Engine speed, rpm:	~~[1800][~~1500~~]~~	~~[_____]~~	~~[_____]~~

2.1.2.1 Engine Emission Limits

Engine must be certified by the manufacturer to meet applicable EPA emission standards found in 40 CFR 60 Subpart IIII. In addition, engine must meet any applicable state or local emission requirements (ex: California SCAQMD).

2.1.2.2 Performance Class

The voltage and frequency behavior of the generator set must be in accordance with ISO 8528 operating limit values for performance Class~~[G1][G2][G3][~~ G4 as follows~~]~~.

2.1.3 Power Ratings

Power ratings must be in accordance with EGSA 101P.

2.1.4 Transient Response

The engine-generator set governor and voltage regulator must cause the engine-generator set to respond to the maximum step load changes such that output voltage and frequency recover to and stabilize within the operational bandwidth within the transient recovery time. The engine-generator set must respond to maximum step load changes such that the maximum voltage and frequency deviations from bandwidth are not exceeded.

2.1.5 Reliability and Durability

~~[Provide prime engine-generator sets that have both an engine and a generator capable of delivering the specified power on a prime basis with an anticipated mean time between overhauls of not less than 10,000 hours operating with a 70 percent load factor. Cite two like engines and two like generators that have performed satisfactorily in a stationary power~~

~~plant, independent from the physical location of the manufacturer's and assembler's facilities. The engine and generators should have been in operation for a minimum of 8000 actual hours at a minimum load of 70 percent of the rated output capacity. During two consecutive years of service, the units should not have experienced any failure resulting in a downtime in excess of 72 hours. Provide engines that are the same model, speed, bore, stroke, number and configuration of cylinders and rated output capacity. Provide generators that are the same model, speed, pitch, cooling, exciter, voltage regulator and rated output capacity.] [~~ Each standby engine-generator set must have both an engine and a generator capable of delivering the specified power on a standby basis with an anticipated mean time between overhauls of no less than 5,000 hours operating with a load factor of 70 percent. Cite two like engines and two like generators that have performed satisfactorily in a stationary power plant, independent and separate from the physical location of the manufacturer's and assembler's facilities, for standby without any failure to start, including all periodic exercise. Provide like engines and generators that have had no failures resulting in downtime for repairs in excess of 72 hours during two consecutive years of service. Provide engines that are the same model, speed, bore, stroke, number and configuration of cylinders, and rated output capacity. Provide generators that are the same model, speed, pitch, cooling, exciter, voltage regulator and rated output capacity.~~]~~

Submit a reliability and durability certification letter from the manufacturer and assembler to prove that existing facilities are and have been successfully utilizing the same components proposed to meet this specification, in similar service. Certification may be based on components, i.e. engines used with different models of generators and generators used with different engines, and does not exclude annual technological improvements made by a manufacturer in the basic standard-model component on which experience was obtained, provided parts interchangeability has not been substantially affected and the current standard model meets the performance requirements specified. Provide a list with the name of the installations, completion dates, and name and telephone number of a point of contact.

~~2.1.6 Parallel Operation~~

~~Configure each engine-generator set specified for parallel operation for [automatic] [manual] parallel operation. Each set must be capable of parallel operation with [a commercial power source on an infinite bus] [one or more sets on an isolated bus] [a commercial power source on an infinite bus and with one or more sets on an isolated bus].~~

~~2.1.7 Load Sharing~~

~~Configure each engine-generator set specified for parallel operation to [manually load share with other sets.] [automatically load share with other sets by proportional loading. Proportional loading must load each set to within 5 percent of its fair share. A set's fair share is its nameplate-rated capacity times the total load, divided by the sum of all nameplate-rated capacities of on-line sets. Incorporate both the real and reactive components of the load.]~~

2.1.6 Engine-Generator Set Enclosure

Provide engine-generator set enclosures that are corrosion resistant and fully weather resistant. The enclosure must contain all set components and provide ventilation to permit operation at Service Load under secured conditions. Provide access doors to controls and equipment requiring periodic maintenance or adjustment. Provide removable panels for access to components requiring periodic replacement. The enclosure must be capable of being removed without disassembly of the engine-generator set or removal of components other than the exhaust system. The enclosure must reduce the noise of the generator set to within the limits specified in the paragraph SOUND LIMITATIONS.

2.1.7 Vibration Isolation

~~[Install a vibration isolation system between the floor and the base. The vibration isolation system must limit the maximum vibration transmitted to the floor at all frequencies to a maximum of [_____] (peak force).] [~~ Provide an engine-generator set with a vibration isolation system in accordance with the manufacturer's standard recommendation.~~]~~ Submit vibration isolation system performance data for the range of frequencies generated by the engine-generator set during operation from no load to full load and the maximum vibration transmitted to the floor plus description of seismic qualification of the engine-generator mounting, base, and vibration isolation. Submit torsional analysis including prototype testing or and calculations which certify and demonstrate that no damaging or dangerous torsional vibrations will occur when the prime mover is connected to the generator, at synchronous speeds, plus 10 percent. Design and qualify vibration isolation systems as an integral part of the base and mounting system in accordance with the seismic parameters specified. Where the vibration isolation system does not secure the base to the structure floor or unit foundation, provide seismic restraints in accordance with the seismic parameters specified.

2.1.8 Harmonic Requirements

Non-linear loads to be served by each engine-generator set are as indicated. The maximum linear load demand (kVA at PF) when non-linear loads will also be in use is as indicated.

2.1.9 Starting Time Requirements

Upon receipt of a signal to start, each engine generator set will start, reach rated frequency and voltage and be ready to assume load within the time specified. For standby sets used in emergency power applications, each engine generator set will start, reach rated frequency and voltage, and power will be supplied to the load terminals of the automatic transfer switch within the starting time specified.

2.2 NAMEPLATES

Provide the manufacturer's name, type or style, model or serial number and rating on a plate secured to the equipment for each major component of this specification. Provide plates and tags sized so that inscription is readily legible to operating or maintenance personnel and securely mounted to or attached in proximity of their identified controls or equipment. Lettering must be normal block lettering, a minimum of 6.4 mm high. As a minimum, provide nameplates for:

Engines Relays

Generators	Transformers (CT & PT)
Regulators	Day tanks
Pumps and pump motors	Governors
Generator Breaker	Air Starting System
Economizers	Heat exchangers (other than base mounted)

Where the following equipment is not provided as a standard component by the engine generator set manufacturer, the nameplate information may be provided in the maintenance manual in lieu of nameplates.

Battery charger	Heaters
Switchboards	Exhaust mufflers
Switchgear	Silencers
Battery	Exciters

2.2.1 Materials

Construct ID plates and tags of 16 gage minimum thickness bronze or stainless steel sheet metal engraved or stamped with inscription. Construct plates and tags not exposed to the weather or high operational temperature of the engine of laminated plastic, 3.2 mm thick, matte white finish with black center core, with lettering accurately aligned and engraved into the core.

2.2.2 Control Devices and Operation Indicators

Provide ID plates or tags for control devices and operation indicators, including valves, off-on switches, visual alarm annunciators, gages and thermometers, that are required for operation and maintenance of provided mechanical systems. Plates or tags must be minimum of 13 mm high and 50 mm long and must indicate component system and component function.

2.2.3 Equipment

Provide ID plates of a minimum size of 75 mm 3 inches high and 130 mm long on provided equipment indicating the following information:

a. Manufacturer's name, address, type and model number, serial number, and certificate of compliance with applicable EPA mission standards;

b. Contract number and accepted date;

c. Capacity or size;

d. System in which installed; and

e. System which it controls.

2.3 SAFETY DEVICES

Exposed moving parts, parts that produce high operating temperatures, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices such that proper operation of the equipment is not impaired.

2.4 MATERIALS AND EQUIPMENT

Submit certification stating that where materials or equipment are specified to comply with requirements of UL, written proof of such compliance has been obtained. The label or listing of the specified agency, or a written certificate from an approved, nationally recognized testing organization equipped to perform such services, stating that the items have been tested and conform to the requirements and testing methods of the specified agency are acceptable as proof.

2.4.1 Circuit Breakers, Low Voltage

UL 489.

2.4.2 Filter Elements

Provide the manufacturer's standard fuel-oil, lubricating-oil, and combustion-air filter elements.

2.4.3 Instrument Transformers

NEMA/ANSI C12.11.

2.4.4 Revenue Metering

IEEE C57.13.

2.4.5 Pipe (Fuel/Lube-Oil, Compressed Air, Coolant, and Exhaust)

ASTM A53/A53M, or ASTM A106/A106M steel pipe. Pipe smaller than 50 mm must be Schedule 80. Pipe 50 mm and larger must be Schedule 40.

2.4.5.1 Flanges and Flanged Fittings

ASTM A181/A181M, Class 60, or ASME B16.5, Grade 1, Class 150.

2.4.5.2 Pipe Welding Fittings

ASTM A234/A234M, Grade WPB or WPC, Class 150 or ASME B16.11, 1360.7 kg.

2.4.5.3 Threaded Fittings

ASME B16.3, Class 150.

2.4.5.4 Valves

MSS SP-80, Class 150.

2.4.5.5 Gaskets

Manufacturer's standard.

2.4.6 Pipe Hangers

MSS SP-58.

2.4.7 Electrical Enclosures

NEMA ICS 6.

2.4.7.1 Switchboards

NEMA PB 2.

2.4.7.2 Panelboards

NEMA PB 1.

2.4.8 Electric Motors

Provide electric motors that conform to the requirements of NEMA MG 1. Motors must have sealed ball bearings and a maximum speed of 1800 rpm. Motors used indoors must have drip-proof frames; enclose those that are used outside. Alternating current motors larger than 373 W must be of the squirrel-cage induction type for operation on 208 volts or higher, ~~[~~5~~0] [60]~~ Hz, and three-phase power. Alternating current motors 373 W or smaller, must be suitable for operation on 120 volts, ~~[~~50~~] [60]~~ Hz, and single-phase power. Direct current motors must be suitable for operation on ~~[~~125~~] [_____]~~ volts.

2.4.9 Motor Controllers

Provide motor controllers and starters that conform to the requirements of NFPA 70 and NEMA ICS 2.

2.5 ENGINE

~~Each engine must operate on [No. 2-D diesel fuel][_____] conforming to [ASTM D975][_____], must be designed for stationary applications and must be complete with ancillaries. The engine must be a standard production model shown in the manufacturer's catalog describing and depicting each engine-generator set and all ancillary equipment in sufficient detail to demonstrate complete specification compliance. The engine must be naturally aspirated, supercharged, or turbocharged. The engine must be 2- or 4-stroke-cycle and compression-ignition type. The engine must be vertical in-line, V- or opposed-piston type, with a solid cast block or individually cast cylinders. The engine must have a minimum of two cylinders. Opposed-piston type engines must have more than four cylinders. Each block must have a coolant drain port. Equip each engine with an over-speed sensor.~~

~~ISO 3046. Diesel engines must be four-cycle naturally aspirated, or turbocharged, or turbocharged and intercooled; vertical in-line or vertical Vee type; designed for stationary service. Engines must be capable of immediate acceleration from rest to normal speed without intermediate idle/warm up period or pre-lubrication to provide essential electrical power. Two-cycle engines are not acceptable.~~

2.5.1 Sub-base Mounting

Mount each engine-generator set on a structural steel sub-base sized to support the engine, generator, and necessary accessories, auxiliaries and control equipment to produce a complete self-contained unit as standard with the manufacturer. Design the structural sub-base to properly support the equipment and maintain proper alignment of the engine-generator set in the specified seismic zone. In addition, provide sub-base with both lifting rings and jacking pads properly located to facilitate shipping and installation of the unit. Factory align engine and generator on the sub-base and securely bolt into place in accordance with the manufacturer's standard practice. Crankshaft must have rigid coupling for connection to the generator.

2.5.2 Assembly

Completely shop assemble each engine-generator set on its structural steel sub-base. Paint entire unit with manufacturer's standard paints and colors. After factory tests and before shipping, thoroughly clean and retouch painting as necessary to provide complete protection.

2.5.3 Turbocharger

If required by the manufacturer to meet the engine-generator set rating, provide turbine type driven by exhaust gas from engine cylinders, and direct connected to the blower supplying air to the engine intake manifold.

2.5.4 Intercooler

Provide manufacturer's standard intercooler for engine size specified.

2.5.5 Crankcase Protection

2.5.6 Miscellaneous Engine Accessories

Provide the following engine accessories where the manufacturer's standard design permits:

a. Piping on engine to inlet and outlet connections, including nonstandard companion flanges.

b. Structural steel sub-base and vibration isolators, foundation bolts, nuts, and pipe sleeves.

c. Level jack screws or shims, as required.

d. Rails, chocks, and shims for installation of sub-base on the foundation.

e. Removable guard, around fan. Support guard, on engine sub-base, to suit manufacturer's standard.

2.5.7 Intercooler

Provide manufacturer's standard intercooler for engine size specified.

2.6 FUEL SYSTEM

Provide fuel system conforming to the requirements of NFPA 30 and NFPA 37 and containing the following elements.

2.6.1 Pumps

Fuel transfer pumps may be mounted on the day tank. Pump~~[~~s~~]~~ must be ~~[~~ duplex,~~]~~ horizontal, positive displacement. Direct-connect pump to motor through a flexible coupling. Equip each pump with a bypass relief valve, if not provided with an internal relief valve. Provide motor and controller in accordance with the paragraphs ELECTRIC MOTORS and MOTOR CONTROLLERS, respectively.

2.6.1.1 Main Pump

Provide engines with an engine driven pump. The pump must supply fuel at a minimum rate sufficient to provide the amount of fuel required to meet the performance indicated within the parameter schedule. Base the fuel flow rate on meeting the load requirements and all necessary recirculation.

2.6.1.2 Auxiliary Fuel Pump

Provide auxiliary fuel pumps to maintain the required engine fuel pressure, if either required by the installation or indicated on the drawings. The auxiliary pump must be driven by a dc electric motor powered by the starting/station batteries. Automatically actuate the auxiliary pump by a pressure-detecting device.

2.6.2 Fuel Filter

Provide a minimum of one full-flow fuel filter for each engine. The filter must be readily accessible and capable of being changed without disconnecting the piping or disturbing other components. Mark the inlet and outlet connections of the filter.

Provide intake filter assemblies for each engine of the oil bath or dry type, as standard with the manufacturer. Filters must be capable of removing a minimum of 92 percent of dirt and abrasive 3 microns and larger from intake air. Size filters to suit engine requirements at 100 percent of rated full load. Design unit for field access for maintenance purposes.

2.6.3 Relief/Bypass Valve

Provide a relief/bypass valve to regulate pressure in the fuel supply line, return excess fuel to a return line and prevent the build-up of excessive pressure in the fuel system.

2.6.4 Integral Main Fuel Storage Tank

Provide each engine with an integral main fuel tank. Each tank must be factory installed and provided as an integral part of the generator manufacturer's product. Provide each tank with connections for fuel supply line, fuel return line, local fuel fill port, gauge, vent line, and float switch assembly. Provide a fuel return line cooler as recommended by the manufacturer and assembler. The temperature of the fuel returning to the tank must be below the flash point of the fuel. Mount the tank within the enclosure for each engine-generator set provided with weatherproof enclosures. The fuel fill line must be accessible without opening the enclosure.

a. All Tanks: UL 142. Provide ~~[integral in skid] [~~free standing~~]~~ double wall (110 percent containment) fuel tanks with a ~~[minimum capacity of [_____] hours of engine-generator set operation at full-rated load] [~~ capacity as indicated~~]~~. Epoxy coat day tanks inside and prime and paint outside. Construct tanks of not less than 4.76 mm steel plate with welded joints and necessary stiffeners on exterior of tank. Provide a braced structural steel framework support. Weld tank top tight. Provide 114 mm square inspection port with a 2 inch NPT fill connection and spill box. Provide proper normal and emergency venting for the primary tank and emergency venting only for the secondary tank / containment basin in accordance with UL 142 requirements.~~[~~ Provide an overflow or return line between the fuel day tank and storage tank in accordance with NFPA 37.~~]~~

b. Float Switches for Day Tanks: Provide tank-top mounted or external float cage, single-pole, single-throw type designed for use on fuel oil tanks. Arrange high level float switches to close on rise of liquid level, and low level float switches to close on fall of liquid level. Mount float cage units with isolating and drain valves. Contacts must be suitable for the station battery voltage.

(1) Critical low level float switch which must activate at 5 percent of normal liquid level must shut engine off.

(2) Low-low level float switch which must activate alarm at 30 percent of normal liquid level.

(3) Low level float switch which must open the fuel oil solenoid valve and start the ~~[~~remote~~]~~ fuel transfer pump at 75 percent of normal liquid level.

(4) High level float switch which must close the fuel oil solenoid valve and stop the ~~[~~remote~~]~~ fuel transfer pump at 90 percent of normal liquid level.

(5) Critical high level float switch which must activate alarm at 95 percent of normal liquid level.

c. Leak Detector Switch for All Tanks: Actuates when fuel is detected in containment basin, stops fuel transfer pump, and closes the fuel oil solenoid valve.

d. Control Panel for All Tanks: Provide NEMA ICS 6, Type ~~[~~1~~] [_____]~~, enclosed control panel for each day tank. Control panel must include the following accessories.

(1) Power available LED (green).

(2) Critical low fuel alarm contacts for shut down of engine.

(3) Low-low level fuel alarm LED.

(4) Low-low level fuel alarm contracts for remote annunciator.

(5) Critical high level fuel alarm LED.

(6) Leak detecting alarm LED.

(7) Alarm horn.

e. Tank Gages for All Tanks: Provide buoyant force type gages for fuel tanks with dial indicator not less than 100 mm in size and arranged for top mounting. Calibrate each reading dial or scale for its specific tank to read from empty to full, with intermediate points of 1/4, 1/2, and 3/4.

f. Integral Base Tanks Used as Primary Tank: Provide a 2 inch opening at the tank fill port, fitted an overfill prevention valve (OPV). Additionally, the fill opening must be perpendicular to the tank in order to allow operation of the OPV. Integral base tank must be sized and configured such that the filling and venting nozzles are outside the generator cabinet for ease of accessibility, inspection, and maintenance. Level gage must be in the line of sight from the fill port.

g. Integral Base Tanks Located Inside Buildings. The tank vents must discharge outside the building in accordance with NFPA 30 and NFPA 37. The fill pipe must terminate outside the building. The fill pipe connection point must be housed in a sealed spill box. High level alarms or level gauges used as overfill protection mechanisms must annunciate at the fill connection point. Provide an overfill prevention valve (OPV) at the tank with a check valve mounted on the fill line in the spill box. The fill connection point must be labeled with tank contents and capacity.

~~h. External tanks (all non-integral base tanks) are specified in Section 33 56 10 FACTORY-FABRICATED FUEL STORAGE TANKS.~~

2.6.4.1 Fuel Transfer Pump~~[~~s~~]~~

Fuel transfer pumps may be mounted on the day tank. Pump~~[~~s~~]~~ must be ~~[~~ duplex,~~]~~ horizontal, positive displacement. Direct-connect pump to motor through a flexible coupling. Equip each pump with a bypass relief valve, if not provided with an internal relief valve. Provide motor and controller in accordance with the paragraphs ELECTRIC MOTORS and MOTOR CONTROLLERS, respectively.

2.6.4.2 Capacity

Each tank must have capacity ~~[~~as shown~~] [to supply fuel to the engine for an uninterrupted [4-hour][_____] period]~~ at 100 percent rated load without being refilled.

2.6.4.3 Local Fuel Fill

Each local fuel fill port on the day tank must have a screw-on cap.

2.6.4.4 Fuel Level Controls

Provide tanks with a float-switch assembly to perform the following functions:

a. Activate the "Low Fuel Level" alarm at 70 percent of the rated tank capacity.

b. Activate the "Overfill Fuel Level" alarm at 95 percent of the rated tank capacity.

2.6.4.5 Arrangement

Integral tanks may allow gravity flow into the engine. Gravity flow tanks and any tank that allows a fuel level above the fuel injectors must have an internal or external factory installed valve located as near as possible to the shell of the tank. The valve must close when the engine is not operating. Provide integral day tanks with any necessary pumps to supply fuel to the engine as recommended by the generator set manufacturer. The fuel supply line from the tank to the manufacturer's standard engine connection must be welded pipe.

~~2.6.5 Day Tank~~

~~Provide engine with [a separate self-supporting] [integral] day tank. Submit calculations for the capacity of each day tank, including allowances for recirculated fuel, usable tank capacity, and duration of fuel supply. Provide connections for fuel supply line, [fuel return line, fuel overflow line, local fuel fill port, gauge, vent line, drain line, and float switch assembly for control for each day tank. Provide a fuel return line cooler as recommended by the manufacturer and assembler. The temperature of the fuel returning to the day tank must be below the flash point of the fuel. Install a temperature sensing device in the fuel supply line], [fuel overflow line, local fuel fill port, gauge, vent line, drain line, and float switch assembly for control]. Mount the day tank within the enclosure for each engine-generator set with weatherproof enclosures. The fuel fill line must be accessible without opening the enclosure.~~

~~2.6.5.1 Capacity, Prime~~

~~Provide day tank with the capacity [as shown] [to supply fuel to the engine for an uninterrupted [8-hour] [_____] period at 100 percent rated load without being refilled, plus any fuel which may be returned to the main fuel storage tank. Submit calculations for the capacity of each day tank, including allowances for recirculated fuel, usable tank capacity, and duration of fuel supply. The calculation of the capacity of each day tank must incorporate the requirement to stop the supply of fuel into the day tank at a "High" level mark of 90 percent of the ultimate volume of the tank].~~

~~2.6.5.2 Capacity, Standby~~

~~Provide day tank with the capacity [as shown] [to supply fuel to the engine for an uninterrupted [4-hour] [_____] period at 100 percent rated load without being refilled, plus any fuel which may be returned to the main fuel storage tank. Submit calculations for the capacity of each day tank, including allowances for recirculated fuel, usable tank capacity, and duration of fuel supply. The calculation of the capacity of each day tank must incorporate the requirement to stop the supply of fuel into the day tank at 90 percent of the ultimate volume of the tank].~~

~~2.6.5.3 Drain Line~~

~~Each day tank drain line must be accessible and equipped with a shutoff valve. Arrange self-supporting day tanks to allow drainage into a 305 mm tall bucket.~~

~~2.6.5.4 Local Fuel Fill~~

~~Each local fuel fill port on the day tank must have a screw-on cap.~~

~~2.6.5.5 Fuel Level Controls~~

~~Provide day tank with a float-switch assembly to perform the following functions:~~

~~a. [When the main storage tank is located higher than the day tank, open the solenoid valve located on the fuel supply line entering the day tank and start the supply of fuel into the day tank] [Start the supply of fuel into the day tank] when the fuel level is at the "Low" level mark, 75 percent of the rated tank capacity.~~

~~b. [When the main storage tank is located higher than the day tank, stop the supply of fuel into the day tank and close the solenoid valve located on the fuel supply line entering the day tank] [Stop the supply of fuel into the day tank] when the fuel level is at 90 percent of the rated tank capacity.~~

~~c. Activate the "Overfill Fuel Level" alarm at 95 percent of the rated tank capacity.~~

~~d. Activate the "Low Fuel Level" alarm at 70 percent of the rated tank capacity.~~

~~e. Activate the automatic fuel supply shut-off valve located on the fill line of the day tank and shut down the fuel pump which supplies fuel to the day tank at 95 percent of the rated tank capacity. Stop the flow of fuel before any fuel can be forced into the fuel overflow line.~~

~~2.6.5.6 Fuel Oil Solenoid Valve~~

~~UL 429. Provide electric solenoid type control valve for each day tank. Solenoid must be rated for starting battery voltage. Valve body must have a minimum working pressure rating of 1033 kPa at 93 degrees C . Valve must be capable of passing 0 to 0.63 L/s of fuel oil. Valves must be two-way, direct acting, normally closed (open when energized, closed when de-energized), with stainless steel body and resilient seat material. Solenoid enclosures must be NEMA ICS 6, Type 1. Body connections must be same size as connecting piping. Valve must be in line before the fuel pump.~~

~~2.6.5.7 Arrangement~~

~~[Integral day tanks may allow gravity flow into the engine. Provide gravity flow tanks with an internal or external valve located as near as possible to the shell of the tank. The valve must close when the engine is not operating. Provide integral day tanks with any necessary pumps to supply fuel to the engine as recommended by the generator set manufacturer. Arrange the overflow connection and the fuel supply line for integral day tanks which do not rely upon gravity flow so that the highest possible fuel level is below the fuel injectors.] [Arrange self-supporting day tanks so that the fuel level in the day tank remains above the suction port of the engine driven fuel pump or be provided with a transfer pump to provide fuel to the engine driven pump. Arrange the overflow connection and fuel supply line so that the highest possible fuel level is below the fuel injectors.] [When the main fuel storage tanks are~~

~~located below the day tank, provide a check valve in the fuel supply line entering the day tank.] [When the main fuel storage tanks are located above the day tank, install a solenoid valve in the fuel supply line entering the day tank. The solenoid valve must be in addition to the automatic fuel shut off valve.] The fuel supply line from the day tank to the manufacturer's standard engine connection must be welded pipe.~~

~~2.6.6 Fuel Supply System~~

~~Provide the fuel supply from the main storage of fuel to the day tank as specified in Section 33 56 10 FACTORY-FABRICATED FUEL STORAGE TANKS.~~

~~2.6.7 Strainer~~

~~[Simplex][Duplex] strainers must comply with Section 33 52 10 SERVICE PIPING, FUEL SYSTEMS.~~

~~2.6.8 Fuel Oil Meters~~

~~Fuel oil meter must comply with Section 33 52 10 SERVICE PIPING, FUEL SYSTEMS.~~

~~2.6.9 Fuel Oil Cooler~~

~~Provide an air cooled fuel oil cooler if the temperature of the fuel returned to the tank from the engine will cause overheating of the tank fuel above the maximum fuel temperature allowed by the engine manufacturer when operating at maximum rated generator power output and low fuel level in the tank. The fuel oil cooler must be furnished by the engine manufacturer for the application and the installation must be complete including piping and power requirements.~~

~~2.7 LUBRICATION~~

~~Provide engine with a separate lube-oil system conforming to NFPA 30 and NFPA 37. Pressurize each system by engine-driven pumps. Regulate system pressure as recommended by the engine manufacturer. Provide a pressure relief valve on the crankcase for closed systems. Vent the crankcase in accordance with the manufacturer's recommendation. Do not vent the crankcase to the engine exhaust system. Crankcase breathers, if provided on engines installed in buildings or enclosures, must be piped to vent to the outside. The system must be readily accessible for service such as draining, refilling, etc. Each system must permit addition of oil and have oil-level indication with the set operating. The system must utilize an oil cooler as recommended by the engine manufacturer.~~

~~2.7.1 Lube-Oil Filter~~

~~Provide one full-flow filter for each pump. The filter must be readily accessible and capable of being changed without disconnecting the piping or disturbing other components. Mark inlet and outlet connections.~~

~~2.7.2 Lube-Oil Sensors~~

~~Equip each engine with lube-oil pressure sensors located downstream of the filters and provide signals for required indication and alarms. Submit two complete sets of filters, required for maintenance, supplied in a suitable storage box. Provide these filters in addition to filters replaced after testing.~~

~~2.7.3 Precirculation Pump~~

~~Provide a motor-driven precirculation pump powered by the station battery, complete with motor starter, if recommended by the engine manufacturer.~~

~~2.8 COOLING SYSTEM~~

~~Provide each engine with its own cooling system to operate automatically while its engine is running. The cooling system coolant must use a combination of water and ethylene-glycol sufficient for freeze protection at the minimum winter outdoor temperature specified. The maximum temperature rise of the coolant across each engine must not exceed that recommended below. Submit a letter which certifies that the engine-generator set and cooling system function properly in the ambient temperature specified, stating the following values:~~

~~a. The maximum allowable inlet temperature of the [coolant fluid][cooling air].~~

~~b. The minimum allowable inlet temperature of the [coolant fluid through the engine][cooling air across the engine].~~

~~c. The maximum allowable temperature rise in the [coolant fluid through the engine][cooling air across the engine].~~

~~d. The minimum allowable inlet fuel temperature.~~

~~2.8.1 Coolant Pumps~~

~~Provide centrifugal coolant pumps. Each engine must have an engine-driven primary pump. Provide secondary pumps that are electric motor driven and have automatic controllers. Control raw-water circulating pump by manual-off-automatic controllers and must be [electric motor] [engine] driven.~~

~~2.8.2 Heat Exchanger~~

~~Provide heat exchanger with the size and capacity to limit the maximum allowable temperature rise in the coolant across the engine to that recommended and submitted for the maximum summer outdoor design temperature and site elevation. Submit manufacturer's data to quantify heat rejected to the space with the engine generator set at rated capacity. Provide heat exchangers that are corrosion resistant, suitable for service in ambient conditions of application.~~

~~2.8.2.1 Fin-Tube-Type Heat Exchanger (Radiator)~~

~~Heat exchanger may be factory coated with corrosion resistant film, provided that corrective measures are taken to restore the heat rejection capability of the radiator to the initial design requirement via over sizing, or other compensating methods. Provide internal surfaces that are compatible with liquid fluid coolant used. Materials and coolant are subject to approval by the Contracting Officer. Provide heat exchangers that are pressure type incorporating a pressure valve, vacuum valve and a cap. Design caps for pressure relief prior to removal. Provide heat exchanger and cooling system that is capable of withstanding a minimum pressure of 48 kPa and protect with a strong grille or screen guard. Provide heat exchanger with at least two tapped holes; equip one tapped~~

~~hole with a drain cock, and plug the rest.~~

~~Provide for each engine-generator set, as standard with the manufacturer.~~

~~a. Design Conditions: Each radiator unit must have ample capacity to remove not less than the total kW per hour of heat rejected by its respective engine at 100 percent full-rated load to the jacket water, fuel oil, and lubricating oil system, and intercooler. Radiator capacity must be rated at optimum temperature of coolant leaving the engine and intercooler as recommended by the engine manufacturer with an ambient dry bulb air temperature outside the enclosure of [_____] degrees C maximum, and [_____] degrees C minimum at the site elevation specified in the paragraph SITE CONDITIONS, and with the coolant mixture specified in the paragraph ENGINE CAPACITY. Pressure drop through the radiator must not exceed 41.34 kPa when circulating the maximum required coolant flow. Radiator air velocity must be a maximum of 7.6 meters per second .~~

~~b. Engine Mounted Radiator Construction: Radiator fan must direct airflow from the engine outward through the radiator. Fan must be V-belt driven directly from the engine crankshaft. Radiator fan must have sufficient capacity to meet design conditions against a static restriction of [_____] Pa of water. Fan static capacity must be adjusted to suit the ductwork furnished. Cooling section must have a tube and fin-type core consisting of copper or copper base alloy tubes with nonferrous fins. Select engine-driven fans for quiet vibration-free operation. Make provision for coolant expansion either by self-contained expansion tanks or separately mounted expansion tanks, as standard with the manufacturer. Provide suitable guards for each fan and drive.~~

~~c. Remote Radiator Construction: Provide radiators as described above, except radiators must be remotely piped and provided with electric motor driven fan. Drive must be multiple V-belt or reduction gears. Expansion tanks must be separately mounted. Air flow must be vertical or horizontal as indicated. Interlock fan with engine operation such that fan must operate when engine operates when recommended by engine manufacturer.[Provide controls and control devices complete which must cycle fan on and off based upon coolant temperature.] Provide motors and controllers in accordance with the paragraphs ELECTRIC MOTORS and MOTOR CONTROLLERS, respectively. Motors, controllers, contactors, and disconnects must conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.~~

~~d. Coolant solution must be a mixture of clean water and ethylene glycol, 50 percent by volume each. Provide an anti-freeze solution tester suitable for the mixture.~~

~~Field installed jacket coolant water piping must conform to the following:~~

~~a. Piping: Provide seamless steel pipe, Schedule 40, conforming to ASTM A53/A53M, Grade A.~~

~~b. Fittings and Flanges: Fittings, 40 mm or smaller, must be malleable iron conforming to ASME B16.3 for Class 300 threaded type. Fittings, 50 mm and larger, must be steel butt welding conforming to ASME B16.9. Utilize either ASME B16.1 or Class A of ASTM A126 for Class 125 cast-iron flanged fittings. Flanges must be Class 150 slip-on forged steel welding flanges in accordance with ASME B16.5, with material in~~

~~accordance with ASTM A181/A181M, Grade I. Provide flat face flanges for connecting to Class 125 standard cast-iron valves, fittings, and equipment connections.~~

~~c. Valves~~

~~(1) Gate Valves: For valves, 40 mm and smaller, provide double disk, rising stem, inside screw, union bonnet type, Class 125 bronze material conforming to MSS SP-80. For valves, 50 mm and larger, provide double-disk, parallel seat type, hydraulic-rated, Class 125, outside screw and yoke type with flanged ends and bronze trim conforming to MSS SP-70. Provide stem packing of material compatible with the system coolant.~~

~~(2) Globe Valves: For valves, 40 mm and smaller, provide rising stem, inside screw, union bonnet type, Class 125 bronze valves conforming to MSS SP-80. For valves, 50 mm and larger, provide Class 125 cast iron, flanged ends, bronze trim globe valves conforming to MSS SP-85. Valves must have renewable composition or cast iron discs compatible with the system coolant.~~

~~(3) Check Valves: MSS SP-71 or MSS SP-80, swing check type.~~

~~d. Hangers and Supports: MSS SP-58.~~

~~e. Piping Sleeves: Provide where piping passes through masonry or concrete walls, floors, roofs, and partitions. Place sleeves during construction. Unless indicated otherwise, pipe sleeves must comply with following requirements: Sleeves in outside walls below and above grade, in floor, or in roof slabs, must be standard weight zinc coated steel pipe. Sleeves in partitions must be zinc coated sheet steel having a nominal weight of not less than 4.4 kg per square meter . Space between piping insulation and the sleeve must be not less than 6 mm . Sleeves must be held securely in proper position and location during construction. Sleeves must be of sufficient length to pass through entire thickness of walls, partitions, or slabs. Sleeves in floor slabs must extend 50 mm above the finished floor. Space between the pipe and the sleeve must be firmly packed with insulation and caulked at both ends of the sleeve with plastic waterproof cement.~~

~~2.8.2.2 Shell and U-Tube Type Heat Exchanger~~

~~Provide multiple pass shell, U-tube type heat exchanger. Exchanger must operate with low temperature water in the shell and high temperature water in the tubes. Provide exchangers that are constructed in accordance with ASME BPVC SEC VIII D1 and certified with ASME stamp secured to the unit. Provide U-tube bundles that are completely removable for cleaning and tube replacement and free to expand with the shell. Construct shells of seamless steel pipe or welded steel. Tubes must be cupronickel or inhibited admiralty, constructed in accordance with ASTM B395/B395M, suitable for the temperatures and pressures specified. Tubes less than 19 mm unless otherwise indicated are not acceptable. Design shell side and tube side for 1.03 MPa working pressure and factory tested at 2.06 MPa. Locate high and low temperature water and pressure relief connections in accordance with the manufacturers standard practice. Water connections larger than 75 mm must be ASME Class 150 flanged. Water pressure loss through clean tubes must be as recommended by the engine manufacturer. Minimum water velocity through tubes must be 300 mm/sec and assure turbulent flow. Provide one or more pressure relief valves for each heat~~

~~exchanger in accordance with ASME BPVC SEC VIII D1. The aggregate relieving capacity of the relief valves must be not less than that required by the above code. Install discharge from the valves indicated. Install the relief valves on the heat exchanger shell. Install a drain connection with 19 mm hose bib at the lowest point in the system near the heat exchanger. Install additional drain connection with threaded cap or plug wherever required for thorough draining of the system.~~

~~2.8.3 Expansion Tank~~

~~The cooling system must include an air expansion tank which will accommodate the expanded water of the system generated within the normal operating temperature range, limiting the pressure increase at all components in the system to the maximum allowable pressure at those components. The tank must be suitable for operating temperature of 121 degrees C and a working pressure of 0.86 MPa. Provide welded steel tank , tested and stamped in accordance with ASME BPVC SEC VIII D1 for the stated working pressure. Do not use a bladder type tank. Support the tank by steel legs or bases for vertical or steel saddles for horizontal installation.~~

~~2.8.4 Thermostatic Control Valve~~

~~Provide a modulating type, thermostatic control valve in the coolant system to maintain the coolant temperature range submitted in paragraph SUBMITTALS.~~

~~2.8.5 Ductwork~~

~~Provide ductwork in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION, except use a flexible connection to connect the duct to the engine radiator. Material for the connection must be wire-reinforced glass. Provide airtight connection.~~

~~2.8.6 Temperature Sensors~~

~~Equip each engine with coolant temperature sensors. Provide temperature sensors with signals for pre-high and high indication and alarms.~~

~~2.9 SOUND LIMITATIONS~~

~~Submit sound power level data for the packaged unit operating at 100 percent load in a free field environment. The data should demonstrate compliance with the sound limitation requirements of this specification. Submit certification from the manufacturer stating that the sound emissions meet the specification. Do not exceed the following sound pressure levels in any of the indicated frequencies when measured in a free field at a radial distance of 22.9 feet 7 meters at 45 degrees apart in all directions when operating at 100 percent load.~~

~~Frequency Band (Hz)~~	~~Maximum Acceptable Sound Level (Decibels)~~
~~31~~	~~[_____]~~
~~63~~	~~[_____]~~

~~Frequency Band (Hz)~~	~~Maximum Acceptable Sound Level (Decibels)~~
~~125~~	~~[_____]~~
~~250~~	~~[_____]~~
~~500~~	~~[_____]~~
~~1,000~~	~~[_____]~~
~~2,000~~	~~[_____]~~
~~4,000~~	~~[_____]~~
~~8,000~~	~~[_____]~~

~~2.10 AIR INTAKE EQUIPMENT~~

~~Locate filters and silencers in locations that are convenient for servicing. Provide high-frequency filter type silencers and locate in the air intake system as recommended by the engine manufacturer. Provide silencer to reduce the noise level at the air intake so that the indicated pressure levels specified in paragraph SOUND LIMITATIONS will not be exceeded. A combined filter-silencer unit meeting requirements for the separate filter and silencer items may be provided. Provide [copper] [rubber] expansion elements in air-intake lines.~~

~~Provide intake filter assemblies for each engine of the oil bath or dry type, as standard with the manufacturer. Filters must be capable of removing a minimum of 92 percent of dirt and abrasive 3 microns and larger from intake air. Size filters to suit engine requirements at 100 percent of rated full load. Design unit for field access for maintenance purposes.~~

~~2.11 EXHAUST SYSTEM~~

~~Provide a separate and complete system for each engine. Support piping to minimize vibration. Where a V-type engine is provided, use a V-type connector, with necessary flexible sections and hardware, to connect the engine exhaust outlets.~~

~~2.11.1 Flexible Sections and Expansion Joints~~

~~Provide a flexible section at each engine and an expansion joint at each muffler. Provide flexible sections and expansion joints that have flanged connections. Provide flexible sections made of convoluted seamless tube without joints or packing. Provide bellows type expansion joints. Provide stainless steel expansion and flexible elements suitable for engine exhaust gas at the maximum exhaust temperature that is specified by the engine manufacturer. Provide expansion and flexible elements that are capable of absorbing vibration from the engine and compensation for thermal expansion and contraction.~~

~~2.11.2 Exhaust Muffler~~

~~Provide a chamber type exhaust muffler. Provide welded steel muffler designed for [outside] [inside] [vertical] [horizontal] mounting. Provide eyebolts, lugs, flanges, or other items as necessary for support in the~~

~~location and position indicated. Do not exceed the engine manufacturer's recommended pressure drop. Outside mufflers must be zinc coated or painted with high temperature 204 degrees C resisting paint. The muffler and exhaust piping together must reduce the noise level to less than the maximum acceptable level listed for sound limitations in paragraph SOUND LIMITATIONS. Provide muffler with a drain valve, nipple, and cap at the low-point of the muffler.~~

~~A[residential class][critical class] silencer must be provided for each engine which will reduce the exhaust sound spectrum by the following listed values at a 23 m radius from the outlet, with generator set loaded to rated capacity and clear weather. Inlet and outlet connections must be flanged.~~

~~Octave Band Center Frequency (Hertz)~~								
~~Minimum Silencer Attenuation Decibels~~	~~63~~	~~125~~	~~250~~	~~500~~	~~1000~~	~~2000~~	~~4000~~	~~8000~~
~~[Residental Class]~~	~~[10]~~	~~[25]~~	~~[32]~~	~~[30]~~	~~[25]~~	~~[25]~~	~~[24]~~	~~[23]~~
~~[Critical Class]~~	~~[15]~~	~~[32]~~	~~[37]~~	~~[36]~~	~~[30]~~	~~[36]~~	~~[37]~~	~~[37]~~

~~2.11.3 Exhaust Piping~~

~~Slope horizontal sections of exhaust piping downward away from the engine to a drip leg for collection of condensate with drain valve and cap. Changes in direction must be long radius. Insulate exhaust piping, mufflers and silencers installed inside any building in accordance with paragraph THERMAL INSULATION and covered to protect personnel. Provide vertical exhaust piping with a hinged, gravity-operated, self-closing, rain cover.~~

~~Field installed exhaust piping must conform to the following:~~

~~a. Exhaust Piping: Provide flanges for connections to engines, exhaust mufflers, and flexible connections. Provide steel pipe conforming to ASTM A53/A53M for each engine complete with necessary fittings, flanges, gaskets, bolts, and nuts. Exhaust piping must be Schedule 40 pipe for 300 mm and smaller, standard weight for sizes 350 mm through 600 mm , and 6 mm wall thickness for sizes larger than 600 mm . Flanges must be Class 150 slip-on forged steel welding flanges in accordance with ASME B16.5, with material in accordance with ASTM A181/A181M, Grade I. Fittings must be butt welding conforming to ASTM A234/A234M, with wall thickness same as adjoining piping. Fittings must be of same material and wall thickness as pipe. Built-up miter welded fittings may be used. Miter angles of each individual section must not exceed 22.5 degrees total and not more than 11.25 degrees relative to the axis of the pipe at any one cut. Gaskets for exhaust piping must be of high temperature asbestos-free material suitable for the service and must be ASME B16.21, composition ring, 1.6 mm thick. Bolting material for exhaust flanges must be alloy-steel bolt-studs conforming to ASTM A193/A193M, Grade B7 bolts and alloy-steel nuts conforming to ASTM A194/A194M, Grade 7. Bolts must be of sufficient length to obtain full bearing on the nuts and must project not more than two full threads beyond the nut. Provide stainless steel counterbalance type rain caps at termination of each exhaust pipe.~~

~~b. Expansion (Flexible) Joints: Provide sections of multiple corrugated stainless steel expansion joints [with liners] in the engine exhaust piping for each engine to absorb expansion strains and vibration transmitted to the piping. Flexible joints must be suitable for operation at 93 degrees C above normal exhaust gas temperature at 100 percent load, 10,000 cycles, minimum. Joints must be flanged and located between engine exhaust manifold and exhaust piping, must be the same size as exhaust piping size, and must be designed and constructed for engine exhaust service.~~

~~c. Hangers and Supports: MSS SP-58.~~

~~d. Piping Sleeves: Provide where piping passes through masonry or concrete walls, floors, roofs, and partitions. Sleeves must be placed during construction. Unless indicated otherwise, pipe sleeves must comply with following requirements: sleeves in outside walls below and above grade, in floor, or in roof slabs, must be standard weight zinc coated steel pipe. Sleeves in partitions must be zinc coated sheet steel having a nominal weight of not less than 4.4 kg per square meter . Space between piping insulation and the sleeve must not be less than 6 mm . Sleeves must be held securely in proper position and location during construction. Sleeves must be sufficient length to pass through entire thickness of walls, partitions, or slabs. Sleeves in floor slabs must extend 50 mm above the finished floor. Space between the pipe and the sleeve must be firmly packed with insulation and caulked at both ends of the sleeve with plastic waterproof cement.~~

~~e. Piping Insulation: Provide exhaust piping insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.~~

~~2.12 PYROMETER~~

~~Provide a pyrometer, [multi-point selector switch, and individual thermocouples] [and thermocouple] with calibrated leads to show the temperature [in each engine cylinder and the combined exhaust] [of the combined exhaust]. For a supercharged engine, provide additional points, thermocouples and leads to show the temperature in the turbocharger exhaust gas outlet and combustion air discharge passages. Graduated scale length less than 150 mm is not acceptable. Provide double pole selector switch with an "off" position, one set of points for each thermocouple, and suitable indicating dial. Calibrate the pyrometer, thermocouples, leads and compensating devices to show true exhaust temperature within plus or minus 1 percent above the highest temperature encountered at 110 percent load conditions.~~

~~2.13 EMISSIONS~~

~~The finished installation must comply with Federal, state, and local regulations and restrictions regarding the limits of emissions, as listed here: [_____]. Submit certification from the engine manufacturer stating that the engine exhaust emissions meet the federal, state, and local regulations and restrictions specified. At a minimum this certification must include emission factors for criteria pollutants including nitrogen oxides, carbon monoxide, particulate matter, sulfur dioxide, non-methane hydrocarbon, and for hazardous air pollutants (HAPs).~~

2.7 STARTING SYSTEM

Provide starting system for ~~[~~standby engine generator sets used in emergency applications in accordance with NFPA 99 and NFPA 110and as follows.~~] [engine generator sets used in non-emergency applications as follows.]~~

2.7.1 Controls

Provide an engine control switch with functions including: run/start (manual), off/reset, and, automatic mode. Provide start-stop logic for adjustable cycle cranking and cool-down operation. Arrange the logic for ~~[~~ manual starting~~]~~ ~~[~~and~~]~~ ~~[~~fully automatic starting in accordance with paragraph AUTOMATIC ENGINE-GENERATOR-SET SYSTEM OPERATION~~]~~. Provide electrical starting systems with an adjustable cranking limit device to limit cranking periods from 1 second up to the maximum duration.

2.7.2 Capacity

Provide starting system with sufficient capacity, at the maximum ~~[outdoor]~~ ~~[~~indoor~~]~~ summer temperature specified to crank the engine without damage or overheating. The system must provide a minimum of three cranking periods with 15 second intervals between cranks. Each cranking period must have a maximum duration of 15 seconds. Starting must be accomplished using an adequately sized dc starter system with a positive shift solenoid to engage the starter motor and to crank the engine continuously for 60 seconds without overheating.

2.7.3 Electrical Starting

Manufacturers recommended dc system, utilizing a negative circuit ground. Starting motors must be in accordance with SAE ARP892.

2.7.3.1 Battery

Provide a starting battery system including the battery, battery rack, intercell connectors, spacers, automatic battery charger with overcurrent protection, metering and relaying. Provide battery in accordance with SAE J537. Size critical system components (rack, protection, etc.) to withstand the seismic acceleration forces specified. Provide ~~[~~lead-acid~~]~~ ~~[nickel-cadmium]~~ battery with sufficient capacity, at the minimum ~~[outdoor]~~ ~~[~~indoor~~]~~ and maximum ~~[outdoor]~~ ~~[~~indoor~~]~~ temperature specified, to provide the specified cranking periods. Valve-regulated lead-acid batteries are not acceptable.

Provide maintenance free, sealed, lead-acid, SAE Type D engine starting batteries.~~[~~ Battery configuration must be two parallel sets of two 12-volt batteries.~~]~~ Batteries must have sufficient capacity to provide 60 seconds of continuous cranking of the engine in an ambient temperature of ~~[_____]~~0 degrees C .

2.7.3.2 Battery Charger

Provide a current-limiting battery charger, conforming to UL 1236, that automatically recharges the batteries. Submit battery charger sizing calculations. The charger must be capable of an equalize charging rate ~~[~~ for recharging fully depleted batteries within ~~[~~24~~] [_____]~~ hours~~]~~ ~~[~~which is manually adjustable in a continuous range~~]~~ and a floating charge rate for maintaining the batteries at fully charged condition. Provide an ammeter to indicate charging rate. Provide a voltmeter to indicate charging voltage. Provide a timer for the equalize charging-rate setting. A battery is considered to be fully depleted when the output voltage falls to a value which will not operate the engine generator set and its components.

Provide ~~[~~120~~] [_____]~~ volt ac, enclosed, automatic equalizing, dual-rate, solid-state, constant voltage type battery charger with automatic ac line compensation. DC output must be voltage regulated and current limited. Charger must have two ranges, float and equalize, and must provide continuous taper charging. The charger must have a continuous output rating of not less than 10 amperes and must be sized to recharge the engine starting batteries in a minimum of 8 hours while providing the control power needs of the engine-generator set. Enclosure must be NEMA ICS 6, Type ~~[~~1~~] [_____]~~. The following accessories must be included:

a. DC ammeter

b. DC voltmeter

c. Equalize light

d. AC on light

e. Low voltage light

f. High voltage light

g. Equalize test button/switch

h. AC circuit breaker

i. Low dc voltage alarm relay

j. High dc voltage alarm relay

k. Current failure relay

l. AC power failure relay

2.7.4 Storage Batteries

Provide storage batteries of suitable rating and capacity to supply and maintain power for the remote alarm annunciator for a period of 90 minutes minimum without the voltage applied falling below 87.5 percent of normal. Provide a ~~[~~120~~] [_____]~~ volt ac automatic battery charger.

2.7.5 Pneumatic

Provide a pneumatic starting system. Provide compressed air system as specified in Section 22 00 00 PLUMBING, GENERAL PURPOSE, for a working pressure of ~~[~~2.07 MPa~~] [1.03 MPa]~~.

2.7.5.1 Air Driven Motors

Provide air driven motors complete with solenoid valve, strainer, and lubricator.

2.7.5.2 Cylinder Injection

Perform starting by admitting compressed air into two or more engine cylinders through a timing valve, or through a distributor into a sufficient number of cylinders to assure successful starting regardless of piston positions.

2.7.6 Starting Aids

Provide one or more of other following methods to assist engine starting.

2.7.6.1 Glow Plugs

Design glow plugs to provide sufficient heat for combustion of fuel within the cylinders to guarantee starting at an ambient temperature of -32 degrees C.

2.7.6.2 Jacket-Coolant Heaters

Mount a thermostatically controlled electric heater in the engine coolant jacketing to automatically maintain the coolant within plus or minus 1.7 degrees C of the control temperature. The heater must operate independently of engine operation so that starting times are minimized. Power for the heaters must be ~~[_____]~~120 volts ac. Include necessary equipment, piping, controls, wiring, and accessories.

2.7.6.2.1 Prime Rated Sets

The control temperature must be the higher of the manufacturer's recommended temperature or the minimum coolant inlet temperature of the engine recommended in paragraph SUBMITTALS.

2.7.6.2.2 Standby Rated Sets

The control temperature must be the temperature recommended by the engine manufacturer to meet the starting time specified at the minimum winter outdoor temperature.

2.7.6.3 Lubricating-Oil Heaters

Mount a thermostatically controlled electric heater in the engine lubricating-oil system to automatically maintain the oil temperature within plus or minus 1.7 degrees C of the control temperature. The heater must operate independently of engine operation so that starting times are minimized. Power for the heaters must be ~~[_____]~~120 volts ac.

2.7.7 Exerciser

Provide exerciser in accordance with Section 26 36 23 AUTOMATIC TRANSFER SWITCH AND BY-PASS/ISOLATION SWITCH.

2.8 GOVERNOR

Provide a forward acting type engine speed governor system. Steady-state frequency band and frequency regulation (droop) must be in accordance with the operating limit values of the performance class specified in the paragraph PERFORMANCE CLASS.

Provide engine with a governor which maintains the frequency within a bandwidth of the rated frequency, over a steady-state load range of zero to 100 percent of rated output capacity. Configure the governor for safe manual adjustment of the speed/frequency during operation of the engine-generator set, without special tools, from 90 to 110 percent of the rated speed/frequency, over a steady state load range of 0 to 100 percent or rated capacity. Submit two complete sets of special tools required for maintenance (except for electronic governor handset). Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts. Provide a suitable tool box for tools. Provide one handset for each electronic governor when required to indicate and/or change governor response settings. ~~[Maintain the midpoint of the frequency bandwidth at the same value for steady-state loads over the range of zero to 100 percent of rated output capacity for isochronous governors.]~~ ~~[~~Maintain the midpoint of the frequency bandwidth linearly for steady-state loads over the range of zero to 100 percent of rated output capacity, ~~[~~with 3 percent droop~~]~~ ~~[~~configured for safe, manual, external adjustment of the droop from zero to ~~[~~7~~] [_____]~~ percent~~]~~ for droop governors.~~]~~

2.9 GENERATOR

Provide synchronous type, one or two bearing, generator conforming to the performance criteria in NEMA MG 1, equipped with winding terminal housings in accordance with NEMA MG 1, equipped with an amortisseur winding, and directly connected to the engine. Submit calculations of the engine and generator output power capability, including efficiency and parasitic load data. Provide ~~[Class H] [~~Class F~~]~~ insulation.

a. Select NEMA MG 1, Part 16, standby duty, and temperature rise of 130 degrees C for engine-generator sets which are expected to operate for less than 300 hours per year. Select NEMA MG 1, Part 22, continuous duty, and temperature rise of 105 degrees C for engine-generator sets expected to operate 300 hours or greater per year or rated 300 kW and above.

b. Select 2/3 pitch design option for engine-generator sets rated 300 kW and above.

c. Select 10-12 lead re-connectable for engine-generator sets rated 300 kW to 800 kW.

d. For applications requiring high SCR loading or in harsh environments laden with salts and chemicals, select vacuum pressure impregnation (VPI) insulated coils. When engine-generator sets are rated 800 kW and larger, also select form wound coils.

e. Provide salient-pole type, ac, brushless-excited, revolving field, air-cooled, self-ventilated, ~~[~~drip-proof guarded,~~]~~ coupled type, synchronous generator conforming to NEMA MG 1, Part ~~[~~16~~] [22]~~, and IEEE C50.12. Generator must be rated for ~~[~~standby~~]~~ ~~[~~continuous~~]~~ duty at 100 percent of the power rating of the engine-generator set as specified in paragraph ENGINE-GENERATOR SET RATINGS AND PERFORMANCE. Temperature rise of each of the various parts of the generator must not exceed~~[130][~~ 105~~]~~ degrees C as measured by resistance, based on a maximum ambient temperature of 40 degrees C. Winding insulation must be Class H.

f. Stator: Stator windings must be ~~[~~2/3 pitch design~~] [~~,~~]~~ ~~[~~10-12 lead re-connectable~~]~~ ~~[~~with VPI insulated ~~[~~and form wound~~]~~ coils~~]~~.

g. Rotor: The rotor must have connected amortisseur windings.

h. Generator Space Heater: Provide ~~[~~120~~] [______]~~ volt ac heaters. Heater capacity must be as recommended by the generator manufacturer to aid in keeping the generator insulation dry.

i. Grounding: Provide non-corrosive steel grounding pads located at two opposite mounting legs.

j. Filters: Provide manufacturer's standard generator cooling air filter assembly.

k. Design generator to protect against mechanical, electrical and thermal damage due to vibration, 25 percent overspeeds, or voltages and temperatures at a rated output capacity of 110 percent for prime applications and 100 percent for standby applications.

l. Provide generator ancillary equipment meeting the short circuit requirements of NEMA MG 1. Select drip-proof guarded option for generators without weatherproof enclosures.

m. Submit manufacturer's standard data for each generator (prototype data at the specified rating or above is acceptable), listing the following information:

(1) Direct-Axis sub-transient reactance (per unit).

(2) The generator kW rating and short circuit current capacity (both symmetric and asymmetric).

2.9.1 Current Balance

At 100 percent rated output capacity, and load impedance equal for each of the 3 phases, the permissible current difference between any 2 phases must not exceed 2 percent of the largest current on either of the 2 phases. Submit certification stating that the flywheel has been statically and dynamically balanced and is capable of being rotated at 125 percent of rated speed without vibration or damage.

2.9.2 Voltage Balance

At any balanced load between 75 and 100 percent of rated output capacity, the difference in line-to-neutral voltage among the 3 phases must not exceed 1 percent of the average line-to-neutral voltage. For a single phase load condition, consisting of 25 percent load at unity power factor placed between any phase and neutral with no load on the other 2 phases, the maximum simultaneous difference in line-to-neutral voltage between the phases must not exceed 3 percent of rated line to neutral voltage. The single-phase load requirement must be valid utilizing normal exciter and regulator control. The interpretation of the 25 percent load for single phase load conditions means 25 percent of rated current at rated phase voltage and unity power factor.

2.9.3 Waveform

The deviation factor of the line-to-line voltage at zero load and at balanced rated output capacity must not exceed 10 percent. The RMS of all harmonics must be less than 5.0 percent and that of any one harmonic less than 3.0 percent of the fundamental at rated output capacity. Design and configure engine-generator to meet the total harmonic distortion limits of IEEE 519.

2.10 EXCITER

Provide brushless generator exciter. Provide semiconductor rectifiers that have a minimum safety factor of 300 percent for peak inverse voltage and forward current ratings for all operating conditions, including 110 percent generator output at 40 degrees C ambient. The exciter and regulator in combination must maintain generator-output voltage within the limits specified.

Provide a brushless excitation system consisting of an exciter and rotating rectifier assembly~~[~~, and permanent magnet generator~~]~~ integral with the generator and a voltage regulator. Insulation class for parts integral with the generator must be as specified in paragraph GENERATOR. System must provide a minimum short circuit of 300 percent rated engine-generator set current for at least 10 seconds. Steady state voltage regulation must be in accordance with the operating limit values of the performance class specified in the paragraph PERFORMANCE CLASS.

a. Exciter and Rotating Rectifier Assembly: Rectifiers must be provided with surge voltage protection.

b. Permanent Magnet Generator: Provide a voltage spike suppression device for permanent magnet generator (PMG) excitation systems.

c. Voltage Regulator: Voltage regulator must be solid state or digital, automatic, three-phase sensing, volts per hertz type regulator. Regulator must receive its input power from a PMG. Voltage variation for any 40 degree C change over the operating temperature range must be less than plus or minus 1.0 percent. Operating temperature must be minus 40 degree C to plus 70 degree C. Voltage adjust range must be plus to minus 5.0 percent of nominal. Inherent regulator features must include over excitation shutdown.

2.10.1 Electromagnetic Interference (EMI) Suppression

Provide as an integral part of the generator and excitation system, EMI suppression complying with MIL-STD-461.

2.11 VOLTAGE REGULATOR

Provide a solid-state voltage regulator, separate from the exciter, for each generator. Maintain the voltage within a bandwidth of the rated voltage, over a steady-state load range of zero to 100 percent of rated output capacity. Configure regulator for safe manual adjustment of the engine-generator voltage output without special tools, during operation, from 90 to 110 percent of the rated voltage over the steady state load range of 0 to 100 percent of rated output capacity. Regulation drift exceeding plus or minus 0.5 percent for an ambient temperature change of 20 degrees C is not acceptable. Reactive droop compensation or reactive differential compensation must load share the reactive load proportionally between sets during parallel operation. Provide voltage regulator with a maximum droop of 2 percent of rated voltage over a load range from 0 to 100 percent of rated output capacity and automatically maintain the generator output voltage within the specified operational bandwidth.

2.12 GENERATOR ISOLATION AND PROTECTION

Provide necessary devices for electrical protection and isolation of each engine-generator set and its ancillary equipment. The generator circuit breaker (IEEE Device 52) ratings must be consistent with the generator rated voltage and frequency, with continuous, short circuit withstand, and interrupting current ratings to match the generator capacity. Provide ~~[manually operated] [electrically operated] [~~operated as indicated~~]~~ generator circuit breaker. Mount a set of surge capacitors at the generator terminals. Provide monitoring and control devices as specified in paragraph GENERATOR PANEL.

The generator circuit breaker must comply with UL 489 requirements for molded case, adjustable thermal magnetic trip type circuit breaker. The circuit breaker continuous current rating must be adequate for the power rating of the engine-generator set and the circuit breaker must be rated to withstand the short circuit current provided by the generator set. Provide circuit breaker in a NEMA ICS 6, Type ~~[~~1~~] [_____]~~ enclosure mounted on the engine-generator set.

2.12.1 Switchboards

Provide free-standing, metal-enclosed, general purpose, 3-phase, 4-wire, ~~[~~ 600~~] [_____]~~ volt rated, with neutral bus and continuous ground bus, switchboards conforming to NEMA PB 2 and UL 891. Neutral bus and ground bus capacity must be ~~[~~as shown~~] [full capacity]~~. Provide panelboards conforming to NEMA PB 1. Provide enclosure designs, construction, materials and coatings ~~[as indicated] [~~suitable for the application and environment~~]~~. Bus continuous current rating must be ~~[~~at least equal to the generator rating and correspond to the UL listed current ratings specified for panelboards and switchboards~~]~~ ~~[~~as indicated~~]~~. Current withstand (short circuit rating) must be ~~[~~equal to the breaker interrupting rating~~]~~ ~~[~~as indicated~~]~~. Provide copper buses.

2.12.2 Devices

Provide switches, circuit breakers, switchgear, fuses, relays, and other protective devices~~ as specified in Section 26 05 73 POWER SYSTEM STUDIES~~.

Furnish with respective pieces of equipment. Motors, controllers, contactors, and disconnects must conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical connections under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide controllers and contactors with maximum of 120-volt control circuits, and auxiliary contacts for use with controls furnished. When motors and equipment furnished are larger than size indicated, the cost of providing additional electrical service and related work must be included under this section.

2.13 SAFETY SYSTEM

Provide and install devices, wiring, remote panels, and local panels, etc., as a complete system to automatically activate the appropriate signals and initiate the appropriate actions. Provide a safety system with a self-test method to verify its operability. Provide alarm signals that have manual acknowledgment and reset devices. The alarm signal systems must reactivate for new signals after acknowledgment is given to any signal. Configure the systems so that loss of any monitoring device will be dealt with as an alarm on that system element.

2.13.1 Audible Signal

Provide audible alarm signal sound at a frequency of ~~[~~70~~] [_____]~~ Hz at a volume of ~~[_____] [~~75~~]~~ dB at 3.1 m. The sound must be continuously activated upon alarm and silenced upon acknowledgment. Locate signal devices as shown.

2.13.2 Visual Signal

The visual alarm signal must be a panel light. The light must be normally off, activated to be blinking upon alarm. The light must change to continuously lit upon acknowledgement. If automatic shutdown occurs, the display must maintain activated status to indicate the cause of failure and must not be reset until cause of alarm has been cleared and/or restored to normal condition. Shutdown alarms must be red; all other alarms must be amber.

2.13.3 Alarms and Action Logic

2.13.3.1 Shutdown

Accomplish simultaneous activation of the audible signal, activation of the visual signal, stopping the engine, and opening the generator main circuit breakers.

2.13.3.2 Problem

Accomplish activation of the visual signal.

2.13.4 Safety Indications and Shutdowns

Provide a local alarm panel with the following shutdown and alarm functions ~~[~~as indicated~~] [in accordance with [NFPA 99] [NFPA 110 level [1] [2]]]~~ mounted either on or adjacent to the engine generator set.

A remote alarm panel is ~~[is not]~~ required for audible alarms, e.g., in the control room.

Indicator Function (at battery voltage)	NFPA 99 Level 1 CV S RA	NFPA 110 Level 1 CV S RA	NFPA 110 Level 2 CV S RA
Overcrank	X X X	X X X	X X O
Low water temperature	X NA X	X NA X	X NA O
High engine temperature pre-alarm	X NA X	X NA X	O NA NA
High engine temperature	X X X	X X X	X X O
Low lube oil pressure pre-alarm	X NA X	NA NA NA	NA NA NA
Low lube oil pressure	X X X	X X X	X X O

Indicator Function (at battery voltage)	NFPA 99 Level 1 CV S RA	NFPA 110 Level 1 CV S RA	NFPA 110 Level 2 CV S RA
Overspeed	X X X	X X X	X X O
Low fuel main tank	X NA X	X NA X	O NA O
Low coolant level	X O X	X O X	X O X
EPS supplying load	X NA NA	X NA NA	O NA NA
Control switch not in automatic position	X NA X	X NA X	X NA X
High battery voltage	X NA NA	X NA NA	O NA NA
Low cranking voltage	X NA X	X NA X	O NA NA
Low voltage in battery	X NA NA	X NA NA	O NA NA
Battery charger ac failure	X NA NA	X NA NA	O NA NA
Lamp test	X NA NA	X NA NA	X NA NA
Contacts for local and remote common alarm	X NA X	X NA X	X NA X
Audible alarm silencing switch	NA NA X	NA NA X	NA NA O
Low starting air pressure	X NA NA	X NA NA	O NA NA
Low starting hydraulic pressure	X NA NA	X NA NA	O NA NA
Air shutdown damper when used	X X X	X X X	X X O
Remote emergency stop	NA X NA	NA X NA	NA X NA

Symbology:
CV: Control panel-mounted visual.
S: Shutdown of EPS indication.
RA: Remote audible.
Symbology:
CV: Control panel-mounted visual.
S: Shutdown of EPS indication.
RA: Remote audible.
X: Required.
O: Optional.
NA: Not applicable.

2.13.5 Time-Delay on Alarms

For startup of the engine-generator set, install time-delay devices bypassing the low lubricating oil pressure alarm during cranking, and the coolant-fluid outlet temperature alarm. Submit the magnitude of monitored values which define alarm or action set points, and the tolerance (plus and/or minus) at which the devices activate the alarm or action for items contained within the alarm panels. The lube-oil time-delay device must return its alarm to normal status after the engine starts. The coolant time-delay device must return its alarm to normal status 5 minutes after the engine starts.

2.14 SYNCHRONIZING PANEL

Provide panel as specified in paragraph PANELS and provide controls, gauges, meters, and displays to include:

a. Frequency meters, dial type, with a range of 90 to 110 percent of rated frequency. Do not use vibrating-reed type meters . One must monitor generator output frequency ("Generator Frequency Meter") and the other must monitor the frequency of the parallel source ("Bus Frequency Meter").

b. Voltmeters, ac, dial type, 3-phase, with 4-position selector switch for the generator output ("Generator Volt Meter") and for the parallel power source ("Bus volt meter").

c. Automatic synchronizer.

d. Manual synchronizing controls.

e. Indicating lights for supplementary indication of synchronization.

f. Synchroscope.

g. Wattmeter, indicating.

2.15 PANELS

Each panel must be of the type and kind necessary to provide specified functions. Mount panels ~~[on the engine-generator set base by vibration/shock absorbing type mountings] [~~as shown~~]~~. Mount instruments flush or semiflush. Provide convenient access to the back of panels to facilitate maintenance. Calibrate instruments using recognized industry calibration standards. Provide a panel identification plate identifying the panel function. Provide a plate identifying the device and its function for each instrument and device on the panel. Provide switch plates identifying the switch-position function.

2.15.1 Enclosures

Design enclosures for the application and environment, conforming to NEMA ICS 6. Locking mechanisms ~~[are optional.] [~~must be keyed alike.~~]~~

Provide for each engine-generator set and fabricate from zinc coated or phosphatized and shop primed 16 gage minimum sheet steel in accordance with the manufacturer's standard design. Provide a complete, weatherproof enclosure for the engine, generator, and auxiliary systems and equipment.

Support exhaust piping and silencer so that the turbocharger is not subjected to exhaust system weight or lateral forces generated in connecting piping that exceed the engine manufacturer's maximum allowed forces and moments. The housing must have sufficient louvered openings to allow entrance of outside air for engine and generator cooling at full load. Design louvered openings to exclude driving rain and snow. Provide properly arranged and sized, hinged panels in the enclosure to allow convenient access to the engine, generator, and control equipment for maintenance and operational procedures. Provide hinged panels with spring type latches which must hold the panels closed securely and will not allow them to vibrate. Brace the housing internally to prevent excessive vibration when the set is in operation

2.15.2 Analog

Provide analog electrical indicating instruments in accordance with UL 1437 with semi-flush mounting. Switchboard, switchgear, and control-room panel-mounted instruments must have 250 degree scales with an accuracy of not less than 99 percent. Unit-mounted instruments must ~~[~~be the manufacturer's standard~~]~~ ~~[~~have 100 degree scales~~]~~ with an accuracy of not less than 98 percent. The instrument's operating temperature range must be minus 20 to plus 65 degrees C. Distorted generator output voltage waveform of a crest factor less than 5 must not affect metering accuracy for phase voltages, hertz and amps.

2.15.3 Electronic

Electronic indicating instruments must be true RMS indicating instruments, 100 percent solid state, state-of-the-art, microprocessor controlled to provide specified functions. Provide control, logic, and function devices that are compatible as a system, sealed, dust and water tight, and that utilize modular components with metal housings and digital instrumentation. Provide an interface module to decode serial link data from the electronic panel and translate alarm, fault and status conditions to set of relay contacts. Instrument accuracy less than 98 percent for unit mounted devices and 99 percent for control room, panel mounted devices, throughout a temperature range of minus 20 to plus 65 degrees C is not acceptable. Provide LED or back lit LCD data display. Additionally, the display must provide indication of cycle programming and diagnostic codes for troubleshooting. Numeral height must be ~~[~~13 mm~~] [______]~~.

2.15.4 Parameter Display

Provide indication or readouts of the tachometer, lubricating-oil pressure, ac voltmeter, ac ammeter, frequency meter, and safety system parameters. Specify a momentary switch for other panels.

2.16 SURGE PROTECTION

Electrical and electronic components must be protected from, or designed to withstand the effects of surges from switching and lightning.

2.17 AUTOMATIC ENGINE-GENERATOR-SET SYSTEM OPERATION

Provide fully automatic operation for the following operations: engine-generator set starting and load transfer upon loss of ~~[~~normal~~] [preferred]~~ source; retransfer upon restoration of the ~~[~~normal~~] [preferred]~~ source; sequential starting; paralleling, and load-sharing for multiple engine-generator sets; and stopping of each engine-generator set after cool-down. Devices must automatically reset after termination of their function.

2.17.1 Automatic Transfer Switch

Provide automatic transfer switches in accordance with Section 26 36 23 AUTOMATIC TRANSFER SWITCH AND BY-PASS/ISOLATION SWITCH.

2.17.2 Monitoring and Transfer

Provide devices to monitor voltage and frequency for the ~~[~~normal~~] [preferred]~~ power source and each engine-generator set, and control transfer from the ~~[normal] [~~preferred~~]~~ source and retransfer upon restoration of the ~~[~~normal~~] [preferred]~~ source. Describe functions, actuation, and time delays as described in Section 26 36 23 AUTOMATIC TRANSFER SWITCH AND BY-PASS/ISOLATION SWITCH.

2.17.3 Automatic Paralleling and Loading of Engine-Generator Sets

Provide an automatic loading system to load and unload engine-generator sets in the sequence indicated. Monitor the system load and cause additional engine-generator sets to start, synchronize, and be connected in parallel with the system bus with increasing load. Actuation of the additional engine-generator set start logic must occur when the load exceeds a percentage set-point of the operating set's rating for a period of approximately 10 seconds. Provide an adjustable set-point range from 50 to 100 percent. When the system load falls below the percentage set-point of the operating set's rating for a period of approximately ~~[_____]~~, the controller must unload and disconnect engine-generator sets from the system, stopping each engine-generator set after cool-down.

2.18 MANUAL ENGINE-GENERATOR-SET SYSTEM OPERATION

Provide complete facilities for manual starting and testing of each set without load, loading and unloading of each set, and synchronization of each set with an energized bus.

2.19 STATION BATTERY SYSTEM

Provide a station battery system including the battery, battery rack, spacers, automatic battery charger and distribution panelboard with overcurrent protection, metering and relaying. Size components to withstand the seismic acceleration forces specified. Provide batteries that have a rated life of 20 years and a manufacturer's 5-year, no cost replacement guarantee.

2.19.1 Battery

Provide ~~[~~lead-acid~~] [nickel-cadmium]~~ battery sized in accordance with IEEE 485 and conforming to the requirements of IEEE 484. Valve-regulated lead-acid batteries are not acceptable. Provide battery environment temperature range between ~~[_____]~~0 and ~~[_____]~~40 degrees. The battery must be rated for at least ~~[_____]~~200 ampere hours at the 8-hour rate.

2.19.2 Battery Capacity

The battery must be rated for at least ~~[_____]~~200 ampere hours at the 8-hour rate, and must have sufficient capacity to serve the following loads without recharging for a period of ~~[_____]~~1 hour~~s~~. At the end of the discharge period, the battery must have the capacity to simultaneously close and trip all the circuit breakers provided, based on a 1-minute load to final voltage of ~~[_____]~~2.05 volts per cell.

a. Diesel-generator safety circuits.

b. Switchgear indicating lights, control relays, protective relays, and other switchgear dc components as required for 24 hours.

c. Voltage regulator (dc power supplies).

d. Emergency-lighting and power load at ~~[_____]~~500 watts for ~~[_____]~~1.5 hours.

2.19.3 Battery Charger

Furnish a current-limiting, ~~[_____]~~24-volt battery charger to automatically recharge the batteries. Provide a charger that is capable of an equalize charging rate ~~[~~for recharging fully depleted batteries within ~~[~~8~~] [_____]~~ hours~~] [which is continuously adjustable]~~ and a floating-charge rate for maintaining the batteries in a fully charged condition. Equip the charger with a low-voltage alarm relay, 0- to 24-hour equalizing timer, an ammeter to indicate charging rate, and necessary circuit breakers. The charger must conform to the requirements of UL 1236. A battery is considered to be fully depleted when the voltage falls to a level incapable of operating the equipment loads served by the battery.

2.20 BASE

Provide a steel base. Design the base to rigidly support the engine-generator set, ensure permanent alignment of rotating parts, be arranged to provide easy access to allow changing of lube-oil, and ensure that alignment is maintained during shipping and normal operation. The base must permit skidding in any direction during installation and must withstand and mitigate the affects of synchronous vibration of the engine and generator. Provide base with ~~[~~suitable holes for anchor bolts~~] [[_____] diameter holes for anchor bolts]~~ and jacking screws for leveling.

2.21 THERMAL INSULATION

Provide thermal insulation as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.22 PAINTING AND FINISHING

Clean, prime and paintthe engine-generator set in accordance with the manufacturer's standard color and practice.

2.23 FACTORY INSPECTION AND TESTS

Submit ~~[~~six~~] [_____]~~ complete reproducible copies of the factory inspection result on the checklist format specified below. Perform the factory tests on each engine-generator set. The component manufacturer's production line test is acceptable as noted. Run each engine-generator set for at least 1 hour at rated output capacity prior to inspections. Complete inspections and make all necessary repairs prior to testing. Use engine generator controls and protective devices that are provided by the generator set manufacturer as part of the standard package for factory tests. When controls and switchgear are not provided as part of the generator set manufacturer's standard package, the actual controls and protective devices provided for the project are not required to be used during the factory test. The Contracting Officer may provide one or more representatives to witness inspections and tests.

2.23.1 Factory Inspection

Perform inspections prior to beginning and after completion of testing of the assembled engine-generator set. Look for leaks, looseness, defects in components, proper assembly, etc. and note any item found to be in need of correction as a necessary repair. Use the following checklist for the inspection:

INSPECTION ITEM	GOOD	BAD	NOTES
Drive belts			
Governor and adjustments			
Engine timing mark			
Starting motor			
Starting aids			
Coolant type and concentration			
Radiator drains			
Block coolant drains			
Coolant fill level			
All coolant line connections			
All coolant hoses			
Combustion air filter			
Combustion air silencer			
Lube oil type			
Lube oil sump drain			
Lube-oil filter			
Lube-oil-level indicator			
Lube-oil-fill level			

INSPECTION ITEM	GOOD	BAD	NOTES
All lube-oil line connections			
All lube-oil lines			
Fuel type and amount			
All fuel-line connections			
All fuel lines			
Fuel filter			
Coupling and shaft alignment			
Voltage regulators			
Battery-charger connections			
All wiring connections			
Instrumentation			
Hazards to personnel			
Base			
Nameplates			
Paint			
Exhaust-heat recovery unit			
Switchboard			
Switchgear			

2.23.2 Factory Tests

Submit a letter giving notice of the proposed dates of factory inspections and tests at least 14 days prior to beginning tests, including:

a. A detailed description of the manufacturer's procedures for factory tests at least ~~[~~14~~] [______]~~ days prior to beginning tests.

b. ~~[~~Six~~] [______]~~ copies of the Factory Test data described below in 216 by 279 mm binders having a minimum of 3 rings from which material may readily be removed and replaced, including a separate section for each test. Separate sections by heavy plastic dividers with tabs. Provide full size (216 by 279 mm minimum) data plots showing grid lines, with full resolution.

(1) A detailed description of the procedures for factory tests.

(2) A list of equipment used, with calibration certifications.

(3) A copy of measurements taken, with required plots and graphs.

(4) The date of testing.

(5) A list of the parameters verified.

(6) The condition specified for the parameter.

(7) The test results, signed and dated.

(8) A description of adjustments made.

On engine-generator set tests where the engine and generator are required to be connected and operated together, the load power factor must be ~~[~~the power factor specified in the engine generator set parameter schedule~~] [[______] power factor]~~. For engine-generator set with dual-fuel operating capability, perform the following tests using ~~[~~the primary fuel type~~] [[______] type fuel]~~. Perform electrical measurements in accordance with IEEE 120. Temperature limits in the rating of electrical equipment and for the evaluation of electrical insulation must be in accordance with IEEE 1. In the following tests where measurements are to be recorded after stabilization of an engine-generator set parameter (voltage, frequency, current, temperature, etc.), stabilization is considered to have occurred when measurements are maintained within the specified bandwidths or tolerances, for a minimum of four consecutive readings. Tests specifically for the generator may be performed utilizing any prime mover.

a. Insulation Resistance for Stator and Exciter Test, IEEE 115 and IEEE 43, to the performance criteria in NEMA MG 1, Part 22. Generator manufacturer's production line test is acceptable.

b. High Potential Test, in accordance with IEEE 115 and NEMA MG 1, test voltage in accordance with NEMA MG 1. Generator manufacturer's production line test is acceptable.

c. Winding Resistance Test, Stator and Exciter, in accordance with IEEE 115. Generator manufacturer's production line test is acceptable.

d. Phase Balance Voltage Test, to the performance criteria specified in paragraph GENERATOR. This test can be performed with any prime mover. Generator manufacturer's production line test results are acceptable.

(1) Start and operate the generator at no load.

(2) Adjust a regulated phase voltage (line-to-neutral) to rated voltage.

(3) Read and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.

(4) Apply 75 percent rated load and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.

(5) Apply rated load and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.

(6) Calculate average line-neutral voltage and percent deviation of individual line-neutral voltages from average for each load condition.

e. Current Balance on Stator Winding Test, by measuring the current on each phase of the winding with the generator operating at 100 percent of Rated Output Capacity, with the load impedance equal for each of the three phases: to the performance criteria specified in paragraph GENERATOR.

f. Voltage Waveform Deviation and Distortion Test in accordance with IEEE 115 to the performance criteria specified in paragraph GENERATOR. Use high-speed recording instruments capable of recording voltage waveform deviation and all distortion, including harmonic distortion. Include appropriate scales to provide a means to measure and interpret results.

g. Voltage and Frequency Droop Test. Verify that the output voltage and frequency are within the specified parameters as follows:

(1) With the generator operating at no load, adjust voltage and frequency to rated voltage and frequency. Record the generator output frequency and line-line and line-neutral voltages.

(2) Increase load to Rated Output Capacity. Record the generator output frequency and line-line and line-neutral voltages.

(3) Calculate the percent droop for voltage and frequency with the following equations:

$$\text{Voltage droop percent} = \frac{\text{(No-Load Volts)} - \text{(Rated Capacity Volts)}}{\text{(Service-Load Volts)}} \times 100$$

$$\text{Frequency droop percent} = \frac{\text{(No-Load Hertz)} - \text{(Rated Capacity Hertz)}}{\text{(Service-Load Hertz)}} \times 100$$

(4) Repeat steps 1 through 3 two additional times without making any adjustments.

h. Frequency and Voltage Stability and Transient Response. Verify that the engine-generator set responds to addition and dropping of blocks of load in accordance with the transient response requirements. Document maximum voltage and frequency variation from bandwidth and verify that voltage and frequency return to and stabilize within the specified bandwidth, within the specified response time period. Document results in tabular form and with high resolution, high speed strip chart recorders or comparable digital recorders, as approved by the Contracting Officer. Include the following tabular data:

(1) Ambient temperature (at 15 minute intervals).

(2) Generator output current (before and after load changes).

(3) Generator output voltage (before and after load changes).

(4) Frequency (before and after load changes).

(5) Generator output power (before and after load changes).

(6) Graphic representations must include the actual instrument trace of voltage and frequency showing: charts marked at start of test; observed steady-state band; mean of observed band; momentary overshoot and undershoot (generator terminal voltage and frequency) and recovery time for each load change together with the voltage and frequency maximum and minimum trace excursions for each steady state load condition prior to and immediately following each load change. Generator terminal voltage and frequency transient recovery time for each step load increase and decrease.

(a) Perform and record engine manufacturer's recommended pre-starting checks and inspections.

(b) Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period and no load. Verify stabilization of voltage and frequency within specified bandwidths.

(c) With the unit at no load, apply the Maximum Step Load Increase.

(d) Apply load in steps equal to the Maximum Step Load Increase until the addition of one more step increase will exceed the Service Load.

(e) Decrease load to the unit such that addition of the Maximum Step Load Increase will load the unit to 100 percent of Service Load.

(f) Apply the Maximum Step Load Increase.

(g) Decrease load to zero percent in steps equal to the Maximum Step Load Decrease.

(h) Repeat steps (c) through (g).

i. Test Voltage Unbalance with Unbalanced Load (Line-to-Neutral) to the performance criteria specified in paragraph GENERATOR. Prototype test data is acceptable in lieu of the actual test. Submit manufacturer's standard certification that prototype tests were performed for the generator model proposed. This test may be performed using any prime mover.

(1) Start and operate the generator set at rate voltage, no load, rated frequency, and under control of the voltage regulator. Read and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.

(2) Apply the specified load between terminals L_1-L_2, L_2-L_0, and L_3-L_0 in turn. Record all instrument readings at each line-neutral condition.

(3) Express the greatest difference between any two of the line-to-line voltages and any two of the line-to-neutral voltages

as a percent of rated voltage.

(4) Compare the largest differences expressed in percent with the maximum allowable difference specified.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the job, perform a Site Visit to verify the information shown on the drawings, before performing any work. Submit a letter stating the date the site was visited and listing discrepancies found. Notify the Contracting Officer in writing of any discrepancies.

3.2 GENERAL INSTALLATION

Provide clear space for operation and maintenance in accordance with NFPA 70 and IEEE C2. Submit a copy of the manufacturer's installation procedures and a detailed description of the manufacturer's recommended break-in procedure. Install pipe, duct, conduit, and ancillary equipment to facilitate easy removal and replacement of major components and parts of the engine-generator set.

3.3 PIPING INSTALLATION

Weld piping. Provide flanged valve connections. Provide flanged connections at equipment. Provide threaded connections to the engine if the manufacturers standard connection is threaded. Except where otherwise specified, use welded flanged fittings to allow for complete dismantling and removal of each piping system from the facility without disconnecting or removing any portion of any other system's equipment or piping. Make connections to equipment with vibration isolation-type flexible connectors. Support and align piping and tubing to prevent stressing of flexible hoses and connectors. Flash pipes extending through the roof. Install piping clear of windows, doors and openings, to permit thermal expansion and contraction without damage to joints or hangers, and install a 13 mm drain valve with cap at each low point.

The installation of gas engines must conform to the requirements of NFPA 37 and its references therein, including NFPA 54, NFPA 58, and ASME B31.3.

3.3.1 Support

Provide hangers, inserts, and supports to accommodate any insulation and conforming to MSS SP-58. Space supports no more than 2.1 m on center for pipes 50 mm in diameter or less, no more than 3.6 m on center for pipes larger than 50 mm but smaller than 100 mm in diameter, and not more than 5.2 m on center for pipes larger than 100 mm in diameter. Provide supports at pipe bends or change of direction.

3.3.1.1 Ceiling and Roof

Support exhaust piping with appropriately sized Type 41 single pipe roll and threaded rods; support all other piping with appropriately sized Type 1 clevis and threaded rods.

3.3.1.2 Wall

Make wall supports for pipe by suspending the pipe from appropriately sized Type 33 brackets with the appropriate ceiling and roof pipe supports.

3.3.2 Flanged Joints

Provide flanges that are Class 125 type, drilled, and of the proper size and configuration to match the equipment and engine connections. Provide gasketed flanged joints that are square and tight.

3.3.3 Cleaning

After fabrication and before assembly, piping interiors must be manually wiped clean of debris.

3.3.4 Pipe Sleeves

Fit pipes passing through construction such as ceilings, floors, or walls with sleeves. Extend each sleeve through and fasten in its respective structure and cut flush with each surface. Build the structure tightly to the sleeve. The inside diameter of each sleeve must be minimum 13 mm, and where pipes pass through combustible materials 25 mm larger than the outside diameter of the passing pipe or pipe insulation/covering.

3.4 ELECTRICAL INSTALLATION

Perform electrical installation in compliance with NFPA 70, IEEE C2, and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. For vibration isolation, provide flexible fittings for conduit, cable trays, and raceways attached to engine-generator sets; provide flexible stranded conductor for metallic conductor cables installed on the engine generator set and from the engine generator set to equipment not mounted on the engine generator set; and provide crimp-type terminals or lugs for terminations of conductors on the engine generator set.

3.5 FIELD PAINTING

Perform field painting as specified in Section 09 90 00 PAINTS AND COATINGS.

3.6 ONSITE INSPECTION AND TESTS

Perform and report on factory tests and inspections prior to shipment. Provide certified copies of manufacturer's test data and results. Test procedures must conform to ASME, IEEE, ~~[~~IEC,~~]~~ and ANSI standards, and to ISO requirements on testing, as appropriate and applicable. The manufacturer performing the tests must provide equipment, labor, and consumables necessary for tests and measuring and indicating devices must be certified to be within calibration. Tests must indicate satisfactory operation and attainment of specified performance. If satisfactory, equipment tested will be given a tentative approval. Equipment must not be shipped before approval of the factory test reports for the following tests.

Submit a letter giving notice of the proposed dates of onsite inspections and tests at least ~~[~~14~~] [_____]~~ days prior to beginning tests.

a. Submit a detailed description of the Contractor's procedures for onsite tests including the test plan and a listing of equipment necessary to perform the tests at least ~~[______]~~30 days prior to beginning tests.

b. Submit ~~[~~six~~] [______]~~ copies of the onsite test data described below in 216 by 279 mm binders having a minimum of 3 rings from which material may readily be removed and replaced, including a separate section for each test. Separate sections by heavy plastic dividers with tabs. Provide full size (216 by 279 mm minimum) data plots showing grid lines, with full resolution.

(1) A detailed description of the procedures for onsite tests.

(2) A list of equipment used, with calibration certifications.

(3) A copy of measurements taken, with required plots and graphs.

(4) The date of testing.

(5) A list of the parameters verified.

(6) The condition specified for the parameter.

(7) The test results, signed and dated.

(8) A description of adjustments made.

3.6.1 Test Conditions

3.6.1.1 Data

Make and record measurements of all parameters necessary to verify that each set meets specified parameters. If the results of any test step are not satisfactory, make adjustments, replacements, or repairs and repeat the step until satisfactory results are obtained. Unless otherwise indicated, record data in 15 minute intervals during engine-generator set operation and include: readings of all engine-generator set meters and gauges for electrical and power parameters; oil pressure; ambient temperature; and engine temperatures available from meters and gauges supplied as permanent equipment on the engine-generator set. Perform electrical measurements in accordance with IEEE 120. Definitions of terms are in accordance with IEEE 100. Provide temperature limits in the rating of electrical equipment and for the evaluation of electrical insulations in accordance with IEEE 1.

3.6.1.2 Power Factor

Submit the generator capability curve showing generator kVA output capability (kW vs. kvar) for both leading and lagging power factors ranging from 0 to 1.0. For all engine-generator set operating tests the load power factor must be ~~[~~the power factor specified in the engine-generator set parameter schedule~~] [[_____] power factor]~~.

3.6.1.3 Contractor Supplied Items

Provide equipment and supplies required for inspections and tests including fuel, test instruments, and loadbanks at the specified power factors.

3.6.1.4 Instruments

Verify readings of panel gauges, meters, displays, and instruments provided as permanent equipment during test runs, using test instruments of greater precision and accuracy. Test instrument accuracy must be within the following: current plus or minus 1.5 percent, voltage plus or minus 1.5 percent, real power plus or minus 1.5 percent, reactive power plus or minus 1.5 percent, power factor plus or minus 3 percent, frequency plus or minus 0.5 percent. Calibrate test instruments by a recognized standards laboratory within 30 days prior to testing.

3.6.1.5 Sequence

Provide the sequence of testing as specified in the approved testing plan unless variance is authorized by the Contracting Officer. Perform field testing in the presence of the Contracting Officer. Schedule and sequence tests in order to optimize run-time periods; however, follow the general order of testing: Construction Tests; Inspections; Pre-operational Tests; Safety Run Tests; Performance Tests; and Final Inspection.

3.6.2 Construction Tests

Perform individual component and equipment functional tests for fuel piping, coolant piping, and lubricating-oil piping, electrical circuit continuity, insulation resistance, circuit protective devices, and equipment not provided by the engine-generator set manufacturer prior to connection to the engine-generator set.

3.6.2.1 Piping Test

a. Flush lube-oil and fuel-oil piping with the same type of fluid intended to flow through the piping, until the outflowing fluid has no obvious sediment or emulsion.

b. Test fuel piping which is external to the engine-generator set in accordance with NFPA 30. Pressure all remaining piping which is external to the engine-generator set with air pressure at 150 percent of the maximum anticipated working pressure, but not less than 1.03 MPa, for a period of 2 hours to prove the piping has no leaks. If piping is to be insulated, perform the test before the insulation is applied.

3.6.2.2 Electrical Equipment Tests

a. Perform low-voltage cable insulation integrity tests for cables connecting the generator breaker to the ~~[~~automatic transfer switch~~] [~~ panelboard~~] [~~main disconnect switch~~] [~~distribution bus~~] [_____]~~. Test low-voltage cable, complete with splices, for insulation resistance after the cables are installed, in their final configuration, ready for connection to the equipment, and prior to energization. Apply a test voltage of 500 volts dc for one minute between each conductor and ground and between all possible combinations conductors in the same trench, duct, or cable, with all other conductors in the same trench, duct, or conduit. Provide the minimum value of insulation as follows:

(1) R in meg-ohms = (rated voltage in kV plus 1) x 304.8/(length of cable in meters)

(2) R in meg-ohms = (rated voltage in kV plus 1) x 1000/(length of cable in feet)

(3) Each cable failing this test must be repaired or replaced. The repair cable must be retested until failures have been eliminated.

b. Perform medium-voltage cable insulation integrity tests for cables connecting the generator breaker to the ~~[generator switchgear] [~~main disconnect switch~~] [distribution bus]~~. After installation and before the operating test or connection to an existing system, perform a high potential test on the medium-voltage cable system. Apply direct-current voltage on each phase conductor of the system by connecting conductors as one terminal and connecting grounds of metallic shields or sheaths of the cable as the other terminal for each test. Prior to making the test, isolate the cables by opening applicable protective devices and disconnecting equipment. Conduct the test with all splices, connectors, and terminations in place. Provide the method, voltage, length of time, and other characteristics of the test for initial installation in accordance with ~~[~~ NEMA WC 74/ICEA S-93-639~~] [_____]~~ for the particular type of cable installed, except provide 28kV and 35kV insulation test voltages in accordance with either AEIC CS8 or AEIC CS8 as applicable, and do not exceed the recommendations of IEEE 404 for cable joints and IEEE 48 for cable terminations unless the cable and accessory manufacturers indicate higher voltages are acceptable for testing. Should any cable fail due to a weakness of conductor insulation or due to defects or injuries incidental to the installation or because of improper installation of cable, cable joints, terminations, or other connections, make necessary repairs or replace cables as directed. Retest repaired or replaced cables.

c. Ground-Resistance Tests. Measure the resistance of ~~[~~each grounding electrode~~] [each grounding electrode system] [the ground mat] [the ground ring]~~ using the fall-of-potential method defined in IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. Take measurements in normally dry weather, not less than 48 hours after rainfall. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance. The combined resistance of separate systems may be used to meet the requirements resistance, but the specified number of electrodes must still be provided as follows:

(1) Single rod electrode - ~~[~~25~~] [_____]~~ ohms.

(2) Multiple rod electrodes - ~~[_____]~~5 ohms.

(3) Ground mat - ~~[_____]~~1 ohms.

d. Examine and test circuit breakers and switchgear in accordance with the manufacturer's published instructions for functional testing.

3.6.3 Inspections

Perform the following inspections jointly by the Contracting Officer and the Contractor, after complete installation of each engine-generator set and its associated equipment, and prior to startup of the engine-generator set. Submit a letter certifying that all facilities are complete and functional; that each system is fully functional; and that each item of equipment is complete, free from damage, adjusted, and ready for beneficial use. Perform checks applicable to the installation. Document and submit the results of those which are physical inspections (I) in accordance with paragraph SUBMITTALS. Present manufacturer's data for the inspections designated (D) at the time of inspection. Verify that equipment type, features, accessibility, installation and condition are in accordance with the contract specification. Provide manufacturer's statements to certify provision of features which cannot be verified visually.

Drive belts	I
Governor type and features	I
Engine timing mark	I
Starting motor	I
Starting aids	I
Coolant type and concentration	D
Radiator drains	I
Block coolant drains	I
Coolant fill level	I
Coolant line connections	I
Coolant hoses	I
Combustion air filter	I
Intake air silencer	I
Lube oil type	D
Lube oil sump drain	I
Lube-oil filter	I
Lube-oil level indicator	I
Lube-oil fill level	I
Lube-oil line connections	I
Lube-oil lines	I
Fuel type	D
Fuel level	I

Fuel-line connections	I
Fuel lines	I
Fuel filter	I
Access for maintenance	I
Voltage regulator	I
Battery-charger connections	I
Wiring and terminations	I
Instrumentation	I
Hazards to personnel	I
Base	I
Nameplates	I
Paint	I
Exhaust-heat system	I
Exhaust muffler	I
Switchboard	I
Switchgear	I
Access provided to controls	I
Enclosure is weather resistant	I
Engine and generator mounting bolts (application)	I

3.6.4 Engine Tests

Perform customary commercial factory tests in accordance with ISO 3046 on each engine and associated engine protective device, including, but not limited to the following:

a. Perform dynamometer test at rated power. Record horsepower at rated speed and nominal characteristics such as lubricating oil pressure, jacket water temperature, and ambient temperature.

b. Test and record the values that the low oil pressure alarm and protective shutdown devices actuate prior to assembly on the engine.

c. Test and record values that the high jacket water temperature alarm and protective shutdown devices actuate prior to assembly on the engine.

3.6.5 Generator Tests

Tests must be performed on the complete factory assembled generator prior to shipment. Conduct tests in accordance with IEEE 115~~[~~, IEC 60034-2A~~]~~, and NEMA MG 1.

3.6.5.1 Routine Tests

Perform the following routine tests on the generators and their exciters:

a. Resistance of armature and field windings.

b. Mechanical balance.

c. Phases sequence.

d. Open circuit saturation curve and phase (voltage) balance test.

e. Insulation resistance of armature and field windings.

f. High potential tes.t

3.6.5.2 Design Tests

Submit the following design tests made on prototype machines that are physically and electrically identical to the generators specified.

a. Temperature rise test

b. Short circuit saturation curve and current balance test

3.6.6 Assembled Engine-Generator Set Tests

~~[Submit the following tests made on prototype machines that are physically and electrically identical to the engine-generator set specified.][~~Perform the following tests on the assembled engine-generator set.~~]~~

3.6.6.1 Initial Stabilization Readings

Operate the engine-generator set and allow the set to stabilize at rated kW at rated power factor, rated voltage, and rated frequency. During this period record instrument readings for output power (kW), terminal voltage, line current, power factor, frequency (rpm) generator (exciter) field voltage and current, lubricating oil pressure, jacket water temperature, and ambient temperature at minimum intervals of 15 minutes. Adjust the load, voltage, and frequency to maintain rated load at rated voltage and frequency. Adjustments to load, voltage, or frequency controls must be recorded on the data sheet at the time of adjustment. Stabilization must be considered to have occurred when four consecutive voltage and current recorded readings of the generator (or exciter) field either remain unchanged or have only minor variations about an equilibrium condition with no evident continued increase or decrease in value after the last adjustment to the load, voltage, or frequency has been made.

3.6.6.2 Regulator Range Test

Remove load and record instrument readings (after transients have subsided). Adjust voltage to the maximum attainable value or to a value just prior to actuation of the overvoltage protection device. Apply rated load and adjust voltage to the minimum attainable value or a value just prior to activation of the under-voltage protection device. The data sheets must indicate the voltage regulation as a percent of rated voltage and the maximum and minimum voltages attainable. Voltage regulation must be defined as follows:

$$\text{Percent Regulation} = \frac{((\text{No-Load Voltage}) - (\text{Rated-Load Voltage})) \times 100}{(\text{Rated-Load Voltage})}$$

3.6.6.3 Frequency Range Test

Adjust the engine-generator set frequency for the maximum attainable frequency at rated load. Record instrument readings. Adjust the engine-generator set frequency for the specified minimum attainable frequency at rated load. Record instrument readings. Reduce the load to zero and adjust the engine-generator set frequency for the maximum attainable frequency. Record instrument readings. Adjust the engine-generator set frequency for the minimum attainable frequency. Record instrument readings. The data sheet must show the maximum and minimum frequencies attained at rated load, and at no load.

3.6.6.4 Transient Response Test

Drop the load to no load and re-apply rated load three times to ensure that the no load and rated load voltage and frequency values are repeatable and that the frequency and voltage regulation is within the limits specified. Record generator terminal voltage and frequency using a high speed strip chart recorder. The data sheet must show the following results:

a. Frequency

(1) Stability bandwidth or deviation in percent of rated frequency.

(2) Recovery time.

(3) Overshoot and undershoot.

b. Voltage

(1) Stability bandwidth or deviation in percent of rated voltage.

(2) Recovery time.

(3) Overshoot and undershoot.

3.6.7 Pre-operational Tests

3.6.7.1 Protective Relays

Visually and mechanically inspect, adjust, test, and calibrate protective relays in accordance with the manufacturer's published instructions. Include pick-up, timing, contact action, restraint, and other aspects necessary to ensure proper calibration and operation. Implement relay settings in accordance with the installation coordination study. Manually or electrically operate relay contacts to verify that the proper breakers and alarms initiate. Field test relaying current transformers in accordance with IEEE C57.13.1.

3.6.7.2 Insulation Test

Test generator and exciter circuits insulation resistance in accordance with IEEE 43. Take stator readings including generator leads to ~~[~~ switchgear~~] [switchboard]~~ at the circuit breaker. Record results of insulation resistance tests. Readings must be within limits specified by the manufacturer. Verify mechanical operation, insulation resistance, protective relay calibration and operation, and wiring continuity of ~~[~~ switchgear~~] [switchboard]~~ assembly. Do not damage generator components during test.

3.6.7.3 Engine-Generator Connection Coupling Test

When the generator provided is a two-bearing machine, inspect and check the engine-generator connection coupling by dial indicator to prove that no misalignment has occurred. Use the dial indicator to measure variation in radial positioning and axial clearance between the coupling halves. Take readings at four points, spaced 90 degrees apart. Align solid couplings and pin-type flexible couplings within a total indicator reading of 0.012 to 0.025 mm for both parallel and angular misalignment. For gear-type or grid-type couplings, 0.05 mm will be acceptable.

3.6.8 Safety Run Test

For the following tests, repeat the associated safety tests if any parts are changed, or adjustments made to the generator set, its controls, or auxiliaries.

a. Perform and record engine manufacturer's recommended prestarting checks and inspections.

b. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.

c. Activate the manual emergency stop switch and verify that the engine stops.

d. Remove the high and pre-high lubricating oil temperature sensing elements from the engine and temporarily install a temperature gauge in their normal locations on the engine (required for safety, not for recorded data). Where necessary provide temporary wiring harness to connect the sensing elements to their permanent electrical leads.

e. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period. Operate the engine-generator set at no load until the output voltage and frequency stabilize. Monitor the temporarily installed temperature gauges. If either temperature reading exceeds the value required for an alarm condition, activate the manual emergency stop switch.

f. Immerse the elements in a vessel containing controlled-temperature hot oil and record the temperature at which the pre-high alarm activates and the temperature at which the engine shuts down. Remove the temporary temperature gauges and reinstall the temperature sensors on the engine.

g. Remove the high and pre-high coolant temperature sensing elements from the engine and temporarily install a temperature gauge in their normal locations on the engine (required for safety, not for recorded data). Where necessary provide temporary wiring harness to connect the sensing elements to their permanent electrical leads.

h. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period. Operate the engine generator-set at no load until the output voltage and frequency stabilize.

i. Immerse the elements in a vessel containing controlled-temperature hot oil and record the temperature at which the pre-high alarm activates and the temperature at which the engine shuts down. Remove the temporary temperature gauges and reinstall the temperature sensors on the engine.

j. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.

k. Operate the engine generator-set for at least 2 hours at 75 percent of Service Load.

l. Verify proper operation and set-points of gauges and instruments.

m. Verify proper operation of ancillary equipment.

n. Manually adjust the governor to increase engine speed past the over-speed limit. Record the RPM at which the engine shuts down.

o. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 75 percent of Service Load.

p. Manually adjust the governor to increase engine speed to within 2 percent of the over-speed trip speed previously determined and operate at that point for 5 minutes. Manually adjust the governor to the rated frequency.

q. Manually fill the day tank to a level above the overfill limit. Record the level at which the overfill alarm sounds. Verify shutdown of the fuel transfer pump. Drain the day tank down below the overfill limit.

r. Shut down the engine. Remove the time-delay low lube oil pressure alarm bypass and try to start the engine.

s. Attach a manifold to the engine oil system (at the oil pressure sensor port) that contains a shutoff valve in series with a connection for the engine's oil pressure sensor followed by an oil pressure gauge ending with a bleed valve. Move the engine's oil pressure sensor from the engine to the manifold. Open the manifold shutoff valve and close the bleed valve.

t. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 75 percent of Service Load.

u. Close the manifold shutoff valve. Slowly allow the pressure in the manifold to bleed off through the bleed valve while watching the pressure gauge. Record the pressure at which the engine shuts down. Catch oil spillage from the bleed valve in a container. Add the oil from the container back to the engine, remove the manifold, and reinstall the engine's oil pressure sensor on the engine.

v. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 100 percent of Service Load. Record the maximum sound level in each frequency band at a distance of ~~[~~22.9~~] [______]~~ m from the end of the exhaust and air intake piping directly along the path of intake and discharge for horizontal piping; or at a radius of ~~[22.9] [~~10.7~~] [______]~~ m from the engine at 45 degrees apart in all directions for vertical piping. ~~[~~If a sound limiting enclosure is provided, modify or replace the enclosure, the muffler, and intake silencer must be modified or replaced as required to meet the sound requirements contained within this specification.~~]~~ ~~[~~If a sound limiting enclosure is not provided, the muffler and air intake silencer as required to meet the sound limitations of this specification. If the sound limitations can not be obtained by modifying or replacing the muffler and air intake silencer, notify the Contracting Officers Representative and provide a recommendation for meeting the sound limitations.~~]~~

w. Manually drain off fuel slowly from the day tank to empty it to below the low fuel level limit and record the level at which the audible alarm sounds. Add fuel back to the day tank to fill it above low level alarm limits.

3.6.9 Performance Tests

In the following tests, where measurements are to be recorded after stabilization of an engine-generator set parameter (voltage, frequency, current, temperature, etc.), stabilization is considered to have occurred when measurements are maintained within the specified bandwidths or tolerances, for a minimum of four consecutive readings. For the following tests, repeat the associated tests if any parts are changed, or adjustments made to the generator set, its controls, or auxiliaries.

3.6.9.1 Continuous Engine Load Run Test

Test the engine-generator set and ancillary systems at service load to demonstrate durability; verify that heat of extended operation does not adversely affect or cause failure in any part of the system; and check all parts of the system. If the engine load run test is interrupted for any reason, repeat the entire test. Accomplish the engine load run test during daylight hours, with an average ambient temperature of ~~[______]~~30 degrees C, during the month of ~~[______]~~August. After each change in load in the following test, measure the vibration at the end bearings (front and back of engine, outboard end of generator) in the horizontal, vertical, and axial directions. Verify that the vibration is within the allowable range. Take data taken at 15 minute intervals and include the following:

Electrical: Output amperes, voltage, real and reactive power, power factor, frequency.

Pressure: Lube-oil.

Temperature: Coolant, Lube-oil, Exhaust, Ambient.

a. Perform and record engine manufacturer's recommended prestarting checks and inspections. Include as a minimum checking of coolant fluid, fuel, and lube-oil levels.

b. Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warmup period.

c. Operate the engine generator-set for 2 hours at 75 percent of Service Load.

d. Increase load to 100 percent of Service Load and operate the engine generator-set for 4 hours.

e. For prime rated units, increase load to 110 percent of Service Load and operate the engine generator-set for 2 hours.

f. Decrease load to 100 percent of Service Load and operate the engine generator-set for 2 hours or until all temperatures have stabilized.

g. Remove load from the engine-generator set.

3.6.9.2 Voltage and Frequency Droop Test

For the following steps, verify that the output voltage and frequency return to and stabilize within the specified bandwidth values following each load change. Record the generator output frequency and line-line and line-neutral voltages following each load change.

a. With the generator operating at no load, adjust voltage and frequency to rated voltage and frequency.

b. Increase load to 100 percent of Rated Output Capacity. Record the generator output frequency and line-line and line-neutral voltages.

c. Calculate the percent droop for voltage and frequency with the following equations.

$$\text{Voltage droop percent} = \frac{\text{No-load volts - rated output capacity volts}}{\text{Rated output capacity volts}} \times 100$$

$$\text{Frequency droop percent} = \frac{\text{No load hertz - rated output capacity hertz}}{\text{Rated output capacity volts}} \times 100$$

d. Repeat steps a. through c. two additional times without making any adjustments.

3.6.9.3 Voltage Regulator Range Test

a. While operating at no load, verify that the voltage regulator adjusts from 90 to 110 percent of rated voltage.

b. Increase load to 100 percent of Rated Output Capacity. Verify that the voltage regulator adjusts from 90 to 110 percent of rated voltage.

3.6.9.4 Governor Adjustment Range Test

a. While operating at no load, verify that the governor adjusts from 90 to 110 percent of rated frequency.

b. Increase load to 100 percent of Rated Output Capacity. Verify that the governor adjusts from 90 to 110 percent of rated frequency.

3.6.9.5 Frequency and Voltage Stability and Transient Response

Verify that the engine-generator set responds to addition and dropping of blocks of load in accordance with the transient response requirements. Document maximum voltage and frequency variation from bandwidth and verify that voltage and frequency return to and stabilize within the specified bandwidth, within the specified response time period. Document results in tabular form and with high resolution, high speed strip chart recorders or comparable digital recorders, as approved by the Contracting Officer. Include the following tabular data:

(1) Ambient temperature (at 15 minute intervals).

(2) Generator output current (before and after load changes).

(3) Generator output voltage (before and after load changes).

(4) Frequency (before and after load changes).

(5) Generator output power (before and after load changes.

(6) Include the actual instrument trace of voltage and frequency in graphic representations showing:

Charts marked at start of test; observed steady-state band; mean of observed band; momentary overshoot and undershoot (generator terminal voltage and frequency) and recovery time for each load change together with the voltage and frequency maximum and minimum trace excursions for each steady state load condition prior to and immediately following each load change. Generator terminal voltage and frequency transient recovery time for each step load increase and decrease.

a. Perform and record engine manufacturer's recommended prestarting checks and inspections.

b. Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period and no load. Verify stabilization of voltage and frequency within specified bandwidths.

c. With the unit at no load, apply the Maximum Step Load Increase.

d. Apply load in steps equal to the Maximum Step Load Increase until the addition of one more step increase will exceed the Service Load.

e. Decrease load to the unit such that addition of the Maximum Step Load Increase will load the unit to 100 percent of Service Load.

f. Apply the Maximum Step Load Increase.

g. Decrease load to zero percent in steps equal to the Maximum Step Load Decrease.

h. Repeat steps c. through g.

~~3.6.10 Parallel Operation Test~~

~~Test the capability of each engine-generator set to parallel and share load with other generator sets, individually and in all combinations. This test must be performed with the voltage regulator and governor adjustment settings used for the Frequency and Voltage Stability and Transient Response test. If settings are changed during the performance of this test, a voltage and frequency stability and transient response test must be performed for each engine generator set using the setting utilized in this test. During operations record load-sharing characteristics of each set in parallel operation. Include the following data:~~

~~(1) Ambient temperature (at 15 minute intervals).~~

~~(2) Generator output current (before and after load changes).~~

~~(3) Generator output voltage (before and after load changes).~~

~~(4) Power division and exchange between generator sets.~~

~~(5) Real power (watts) and reactive power (vars) on each set.~~

~~3.6.10.1 Combinations~~

~~Connect each set, while operating at no load, parallel with one other set in the system, operating at service load, until all possible combinations have been achieved. Verify stabilization of voltage and frequency within specified bandwidths and proportional sharing of real and reactive loads. Document stabilization of voltage and frequency within specified bandwidth, the active power division, active power exchange, reactive power division, and voltage and frequency stability and transient response in the following steps for each combination.~~

~~a. Divide the load proportionally between the sets and operate in parallel for 15 minutes.~~

~~b. Increase the load, in steps equal to the Maximum Step Increase, until each set is loaded to its service load.~~

~~c. Decrease the load, in steps equal to the Maximum Step Decrease, until each set is loaded to approximately 25 percent of its service load.~~

~~d. Increase the load, in steps equal to the Maximum Step Increase, until each set is loaded to approximately 50 percent of its service load. Verify stabilization of voltage and frequency within specified bandwidths and proportional sharing of real and reactive load.~~

~~e. Reduce the sum of the loads on both sets to the output rating of the smaller set.~~

~~f. Transfer a load equal to the output rating of the smaller of the 2 sets to and from each set. Verify stabilization of voltage and~~

~~frequency within specified bandwidths and proportional sharing of real and reactive load.~~

~~g. Document the active power division, active power exchange, reactive power division, and voltage and frequency stability and transient response.~~

~~3.6.10.2 Multiple Combinations~~

~~Connect each set, while operating at no load, parallel with all multiple combinations of all other set in the system, while operating at service load, until all multiple combinations of parallel operations have been achieved.~~

~~3.6.11 Parallel Operation Test (Commercial Source)~~

~~Connect each set parallel with the commercial power source. Operate in parallel for 15 minutes. Verify stabilization of voltage and frequency within specified bandwidths. Record the output voltage, frequency, and loading to demonstrate ability to synchronize with the commercial power source.~~

3.6.10 Automatic Operation Tests

Test the automatic operating system to demonstrate ~~[~~automatic starting,~~] [loading and unloading,] [the response to loss of operating engine-generator sets,]~~ and paralleling of each engine-generator set. Utilize ~~[~~load banks at the indicated power factor~~]~~ ~~[~~and actual loads to be served~~]~~ for this test, and the loading sequence is the indicated sequence. Record load-sharing characteristics during all operations. Perform this test for a minimum of two successive, successful tests. Include the following data:

(1) Ambient temperature (at 15 minute intervals).

(2) Generator output current (before and after load changes).

(3) Generator output voltage (before and after load changes).

(4) Generator output frequency (before and after load changes).

(5) Power division and exchange between generator sets.

(6) Real and reactive power on each set.

a. Initiate loss of the preferred power source and verify the specified sequence of operation.

b. Verify resetting of automatic starting and transfer logic.

~~3.6.11 Automatic Operation Tests for Stand-Alone Operation~~

~~Test the automatic loading system to demonstrate [automatic starting,] [and] [loading and unloading] of each engine-generator set. Utilize the actual loads to be served for this test, and the loading sequence is the indicated sequence. Perform this test for a minimum of two successive, successful tests. Include the following data:~~

~~(1) Ambient temperature (at 15 minute intervals).~~

~~(2) Generator output current (before and after load changes).~~

~~(3) Generator output voltage (before and after load changes).~~

~~(4) Generator output frequency (before and after load changes).~~

~~a. Initiate loss of the primary power source and verify automatic sequence of operation.~~

~~b. Restore the primary power source and verify sequence of operation.~~

~~c. Verify resetting of controls to normal.~~

3.7 GROUNDING

NFPA 70 and IEEE C2, except that grounding systems must have a resistance to solid earth ground not exceeding 5 ohms.

3.7.1 Grounding Electrodes

Provide driven ground rods as specified in ~~[~~Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION~~] [and] [Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION]~~. Connect ground conductors to the upper end of ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.7.2 Engine-Generator Set Grounding

Provide separate copper grounding conductors and connect them to the ground system as indicated. When work in addition to that indicated or specified is required to obtain the specified ground resistance, the provision of the contract covering "Changes" must apply.

3.7.3 Connections

Make joints in grounding conductors by exothermic weld or compression connector. Exothermic welds and compression connectors must be installed as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION paragraph regarding GROUNDING.

3.7.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

3.8 START-UP ENGINEER

Provide the services of a qualified factory trained start-up engineer, regularly employed by the engine-generator set manufacturer. The start-up services must include conducting preliminary operations and functional acceptance tests. The start-up engineer must be present at the engine generator set installation-site, full-time, while preliminary operations and functional acceptance tests are being conducted.

3.9 PREREQUISITES FOR FUNCTIONAL ACCEPTANCE TESTING

Completion of the following requirements is mandatory prior to scheduling functional acceptance tests for the engine-generator set and auxiliary equipment.

~~3.9.1 Piping Tests~~

~~Complete as specified in Section 33 52 10 SERVICE PIPING, FUEL SYSTEMS.~~

3.9.1 Performance of Acceptance Checks and Tests

The acceptance checks and tests must be accomplished by the testing organization as described in Section 26 08 00 APPARATUS INSPECTION AND TESTING.

3.9.2 Generator Sets

Complete as specified in the paragraph ACCEPTANCE CHECKS AND TESTS.

3.9.2.1 Automatic Transfer Switches

Complete acceptance checks and tests as specified in Section 26 36 23 AUTOMATIC TRANSFER SWITCHES AND BY-PASS/ISOLATION SWITCH.

3.9.3 Preliminary Operations

The start-up engineer must conduct manufacturer recommended start-up procedures and tests to verify that the engine-generator set and auxiliary equipment are ready for functional acceptance tests. Give the Contracting Officer 15 days' advance notice that preliminary operations will be conducted. After preliminary operation has been successfully conducted, the start-up engineer will notify the Contracting Officer in writing stating the engine-generator set and auxiliary equipment are ready for functional acceptance tests.

3.9.4 Preliminary Assembled Operation and Maintenance Manuals

Preliminary assembled operation and maintenance manuals must have been submitted to and approved by the Contracting Officer. Manuals must be prepared as specified in the paragraph ASSEMBLED OPERATION AND MAINTENANCE MANUALS.

3.9.5 Functional Acceptance Test Procedure

Test procedure must be prepared by the start-up engineer specifically for the engine-generator set and auxiliary equipment. The test agenda must cover the requirements specified in the paragraph FUNCTIONAL ACCEPTANCE TESTS. The test procedure must indicate in detail how tests are to be conducted. A statement of the tests that are to be performed without indicating how the tests are to be performed is not acceptable. Indicate what work is planned on each workday and identify the calendar dates of the planned workdays. Specify what additional technical support personnel is needed such as factory representatives for major equipment. Specify on which testing workday each technical support personnel is needed. Data recording forms to be used to document test results are to be submitted with the proposed test procedure. A list of test equipment and instruments must also be included in the test procedure.

3.9.6 Test Equipment

Test equipment and instruments must be on hand prior to scheduling field tests or, subject to Contracting Officer approval, evidence must be provided to show that arrangements have been made to have the necessary equipment and instruments on-site prior to field testing.

3.10 FIELD QUALITY CONTROL

Give Contracting Officer ~~[NAVFAC [_____], Code [_____]]~~ 30 days' notice of dates and times scheduled for tests which require the presence of the Contracting Officer. The Contracting Officer will coordinate with the using activity and schedule a time that will eliminate or minimize interruptions and interference with the activity operations. The Contractor must be responsible for costs associated with conducting tests outside of normal working hours and with incorporating special arrangements and procedures, including temporary power conditions. The Contractor must provide labor, equipment, fuel, test load, and consumables required for the specified tests. The test load must be a cataloged product. Calibration of measuring devices and indicating devices must be certified. Refer to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, for requirements for a cataloged product. Perform the following field tests.

3.10.1 Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.10.1.1 Circuit Breakers - Low Voltage Insulated Case/Molded Case

a. Visual and Mechanical Inspection

   (1) Compare nameplate data with specifications and approved shop drawings.

   (2) Inspect circuit breaker for correct mounting.

   (3) Operate circuit breaker to ensure smooth operation.

   (4) Inspect case for cracks or other defects.

   (5) Verify tightness of accessible bolted connections and cable connections by calibrated torque-wrench method. Thermo-graphic survey is not required.

   (6) Inspect mechanism contacts and arc chutes in unsealed units.

b. Electrical Tests

   (1) Perform contact-resistance tests.

   (2) Perform insulation-resistance tests.

   (3) Adjust breaker(s) for final settings in accordance with engine-generator set manufacturer's requirements.

3.10.1.2 Current Transformers

a. Visual and Mechanical Inspection

   (1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify correct connection.

(4) Verify that adequate clearances exist between primary and secondary circuit.

(5) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermo-graphic survey is not required.

(6) Verify that all required grounding and shorting connections provide good contact.

b. Electrical Tests

(1) Perform insulation-resistance tests.

(2) Perform polarity tests.

(3) Perform ratio-verification tests.

3.10.1.3 Metering and Instrumentation

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify tightness of electrical connections.

b. Electrical Tests

(1) Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.

(2) Calibrate watt-hour meters according to manufacturer's published data.

(3) Verify all instrument multipliers.

(4) Electrically confirm that current transformer secondary circuits are intact.

3.10.1.4 Battery Systems

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermo-graphic survey is not required.

(4) Measure electrolyte specific gravity and temperature and visually check fill level.

(5) Verify adequacy of battery support racks, mounting, anchorage, and clearances.

b. Electrical Tests

(1) Set charger float and equalizing voltage levels.

(2) Verify all charger functions and alarms.

(3) Measure each cell voltage and total battery voltage with charger energized and in float mode of operation.

(4) Perform a capacity load test.

3.10.1.5 Engine-Generator Set

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Inspect for correct anchorage and grounding.

b. Electrical and Mechanical Tests

(1) Perform an insulation-resistance test on generator winding with respect to ground. Calculate polarization index.

(2) Perform phase rotation test to determine compatibility with load requirements.

3.10.1.6 Grounding System

a. Visual and Mechanical Inspection

(1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

(1) Perform ground-impedance measurements utilizing the fall-of-potential method defined in IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. Take measurements in normally dry weather, not less than 48 hours after rainfall. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.10.2 Functional Acceptance Tests

The tests must be performed by the start-up engineer. Upon successful test completion, the start-up engineer must provide the Contracting Officer with a written test report within 15 calendar days showing the tests performed and the results of each test. The report must include the completed approved test data forms and certification from the start-up engineer that the test results fall within the manufacturer's recommended limits and meet the specified requirements performance. The report must be dated and signed by the start-up engineer, and submitted for approval by the Contracting Officer. The Contracting Officer ~~[and NAVFAC [_____], Code [_____]]~~ will witness final acceptance tests. Testing must include, but not be limited to, the following:

a. Verify proper functioning of each engine protective shutdown device and pre-shutdown alarm device. Testing of the devices must be accomplished by simulating device actuation and observing proper alarm and engine shutdown operation.

b. Verify proper functioning of the engine over-speed trip device. Testing of the over-speed trip device must be accomplished by raising the speed of the engine-generator set until an over-speed trip is experienced.

c. Verify proper functioning of the crank cycle/terminate relay. Testing of the relay must be accomplished by engaging the starter motor with the engine being prevented from running. Observe the complete crank/rest cycle as described in the paragraph STARTING SYSTEM.

d. Verify proper functioning of the following automatic and manual operations. Testing must include, but not be limited to, the following:

(1) Loss of Utility: Initiate a normal power failure with connected test load of rated kW at 1.0 power factor. Record time delay on start, cranking time until engine starts and runs, time to come up to operating speed, voltage and frequency overshoot, and time to achieve steady state conditions with all switches transferred to emergency position.

(2) Return of Utility: Return normal power and record time delay on retransfer for each automatic transfer switch, and time delay on engine cool-down and shutdown.

(3) Manual starting.

(4) Emergency stop.

e. Operate the engine-generator set at rated current (amperes) until the jacket water temperature stabilizes. Stabilization will be considered to have occurred when three consecutive temperature readings remain unchanged. Continue to operate the generator set for an additional 2 hours. Record instrument readings for terminal voltage, line current, frequency (Hz), engine speed rpm, lubricating oil pressure, jacket water temperature, and ambient temperature at 5 minute intervals for first 15 minutes and at 15 minute intervals thereafter.

~~[~~ f. Emissions Tests. Provide on-site testing by a certified testing organization of each engine-generator set. Testing must be in accordance with an EPA approved method, 40 CFR 60, (Appendix, Method 7, 7A, 7B, 7C, 7D or 7E). Emissions at rated full load must be within the limits specified in the paragraph ENGINE EMISSIONS LIMITS.~~]~~

3.11 DEMONSTRATION

Upon completion of the work and at a time approved by the Contracting Officer, the Contractor must provide instructions by a qualified instructor to the Government personnel in the proper operation and maintenance of the equipment. ~~[______]~~Government personnel must receive training comparable to the equipment manufacturer's factory training. The duration of instruction must be for not less than one 8 hour working day for instruction of operating personnel and not less than one 8 hour working day for instruction of maintenance personnel.

3.11.1 Instructor's Qualification Resume

Instructors must be regular employees of the engine-generator set manufacturer. The instruction personnel provided to satisfy the requirements above must be factory certified by the related equipment manufacturer to provide instruction services. Submit the name and qualification resume of instructor to the Contracting Officer for approval.

3.11.2 Training Plan

Submit training plan 30 calendar days prior to training sessions. Training plan must include scheduling, content, outline, and training material (handouts). Content must include, but not be limited to, the following:

3.11.2.1 Operating Personnel Training

This instruction includes operating the engine-generator set, auxiliary equipment including automatic transfer switches in all modes, and the use of all functions and features specified.

3.11.2.2 Maintenance Personnel Training

Training must include mechanical, hydraulic, electrical, and electronic instructions for the engine-generator set and auxiliary equipment including automatic transfer switches.

a. Mechanical Training: Must include at least the following:

(1) A review of mechanical diagrams and drawings.

(2) Component location and functions.

(3) Troubleshooting procedures and techniques.

(4) Repair procedures.

(5) Assembly/disassembly procedures.

(6) Adjustments (how, when, and where).

(7) Preventive maintenance procedures.

(8) Review of flow diagram.

(9) Valve locations and function.

(10) Valve and hydraulic equipment adjustment and maintenance

procedures.

(11) Hydraulic system maintenance and servicing.

(12) Lubrication points, type, and recommended procedures and frequency.

b. Electrical and Electronic Maintenance Training: Must include at least the following:

(1) A review of electrical and electronic systems including wiring diagrams and drawings.

(2) Troubleshooting procedures for the machine and control systems.

(3) Electrical and electronic equipment servicing and care.

(4) Use of diagnostics to locate the causes of malfunction.

(5) Procedures for adjustments (locating components, adjustments to be made, values to be measured, and equipment required for making adjustments).

(6) Maintenance and troubleshooting procedures for microprocessor or minicomputer where applicable.

(7) Circuit board repair procedures where applicable (with schematics provided).

(8) Use of diagnostic tapes.

(9) Recommended maintenance servicing and repair for motors, switches, relays, solenoids, and other auxiliary equipment and devices.

3.12 ONSITE TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total ~~[_____]~~8 hours of normal working time and must start after the system is functionally completed but prior to final acceptance.

a. Submit a letter giving the date proposed for conducting the onsite training course, the agenda of instruction, a description of the digital video recording to be provided. The course instructions must cover pertinent points involved in operating, starting, stopping, servicing the equipment, as well as major elements of the operation and maintenance manuals. Additionally, the course instructions must demonstrate routine maintenance procedures as described in the operation and maintenance manuals.

b. Submit a digital video recording of the ~~[entire training session] [~~ manufacturers operating and maintenance training course~~]~~.

c. One full size reproducible Mylar ach drawing must accompany the booklets. Mylars must be rolled and placed in a heavy cardboard tube with threaded caps on each end. The manual must include step-by-step procedures for system startup, operation, and shutdown; drawings, diagrams, and single-line schematics to illustrate and define the electrical, mechanical, and hydraulic systems together with their controls, alarms, and safety systems; the manufacturer's name, model number, and a description of equipment in the system. The instructions must include procedures for interface and interaction with related systems to include ~~[~~automatic transfer switches~~]~~ ~~[~~fire alarm/suppression systems~~]~~ ~~[~~load shedding systems~~]~~ ~~[~~uninterruptible power supplies~~] [______]~~. Each booklet must include a CD containing an ASCII file of the procedures.

d. Provide approved operation and maintenance manuals for the training course. Post approved instructions prior to the beginning date of the training course. Coordinate the training course schedule with the using service's work schedule, and submit for approval 14 days prior to beginning date of proposed beginning date of training.

3.13 INSTALLATION

Installation must conform to the applicable requirements of IEEE C2, NFPA 30, NFPA 37, and NFPA 70.

3.14 FINAL TESTING AND INSPECTION

a. Start the engine, record the starting time, make and record all engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.

b. Increase the load in steps no greater than the Maximum Step Load Increase to 100 percent of Service Load, and operate the engine-generator set for at least 30 minutes. Measure the vibration at the end bearings (front and back of engine, outboard end of generator) in the horizontal, vertical, and axial directions. Verify that the vibration is within the same range as previous measurements and is within the required range.

c. Remove load and shut down the engine-generator set after the recommended cool down period.

d. Remove the lube oil filter and have the oil and filter examined by the engine manufacturer for excessive metal, abrasive foreign particles, etc. Verify any corrective action for effectiveness by running the engine for 8 hours at Service Load, then re-examine the oil and filter.

e. Remove the fuel filter and examine the filter for trash, abrasive foreign particles, etc.

f. Visually inspect and check engine and generator mounting bolts for tightness and visible damage.

g. Replace air, oil, and fuel filters with new filters.

3.15 MANUFACTURER'S FIELD SERVICE

The engine generator-set manufacturer must furnish a qualified representative to supervise the installation of the engine generator-set, assist in the performance of the onsite tests, and instruct personnel as to the operational and maintenance features of the equipment.

3.16 POSTED DATA AND INSTRUCTIONS

Post Data and Instructions prior to field acceptance testing of the engine generator set. [Provide two sets of instructions/data, typed and framed under weatherproof laminated plastic, and post side-by-side where directed. Include a one-line diagram, wiring and control diagrams and a complete layout of the system in the first set. Include the condensed operating instructions describing manufacturer's pre-start checklist and precautions; startup procedures for test-mode, manual-start mode, and automatic-start mode (as applicable); running checks, procedures, and precautions; and shutdown procedures, checks, and precautions in the second set. Submit posted data including wiring and control diagrams showing the key mechanical and electrical control elements, and a complete layout of the entire system.

a. Include procedures for interrelated equipment (such as heat recovery systems, co-generation, load-shedding, and automatic transfer switches).] [Provide two sets of typed instructions/data in 216 X 279 mm format, laminated in weatherproof plastic, and placed in three-ring vinyl binders. Place the binders as directed by the Contracting Officer. Provide the instructions prior to acceptance of the engine generator set installation.

b. Include a one-line diagram, wiring and control diagrams and a complete layout of the system in the first set. Include the condensed operating instructions describing manufacturer's pre-start checklist and precautions; startup procedures for test-mode, manual-start mode, and automatic-start mode (as applicable); running checks, procedures, and precautions; and shutdown procedures, checks, and precautions in the second set. Include procedures for interrelated equipment (such as heat recovery systems, co-generation, load-shedding, and automatic transfer switches).]

c. Submit instructions including: the manufacturers pre-start checklist and precautions; startup procedures for test-mode, manual-start mode, and automatic-start mode (as applicable); running checks, procedures, and precautions; and shutdown procedures, checks, and precautions. Include procedures for interrelated equipment (such as heat recovery systems, co-generation, load-shedding, and automatic transfer switches). Provide weatherproof instructions, laminated in plastic, and post where directed.

3.17 ACCEPTANCE

Submit drawings which accurately depict the as-built configuration of the installation, upon acceptance of the engine-generator set installation. Revise layout drawings to reflect the as-built conditions and submit them with the as-built drawings. Final acceptance of the engine-generator set will not be given until the Contractor has successfully completed all tests and all defects in installation material or operation have been corrected.

-- End of Section --

SECTION 26 36 23

AUTOMATIC TRANSFER SWITCHES AND BY-PASS/ISOLATION SWITCH
05/20, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 4 (2015) Application Guideline for Terminal Blocks

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 20 (2025; TIA 25-1) Standard for the Installation of Stationary Pumps for Fire Protection

NFPA 70 (2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code

NFPA 110 (2025) Standard for Emergency and Standby Power Systems

UL SOLUTIONS (UL)

UL 508 (2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment

UL 1008 (2022) UL Standard for Safety Transfer Switch Equipment

UL 1066 (2022) UL Standard for Safety Low-Voltage AC and DC Power Circuit Breakers Used in

Enclosures

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and Section 26 08 00 APPARATUS INSPECTION AND TESTING, applies to this section, with the additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Automatic Transfer Switch Drawings; G~~, [_____]~~

SD-03 Product Data

Automatic Transfer Switches; G~~, [_____]~~

~~[~~ By-Pass/Isolation Switch (BP/IS); G~~, [_____]~~

~~][~~ Remote Annunciator Panel; G~~, [_____]~~

~~][~~ Remote Annunciator and Control System Panel; G~~, [_____]~~

~~]~~ SD-06 Test Reports

~~Acceptance Checks and Tests; G, [_____]~~

Functional Acceptance Tests; G~~, [_____]~~

Factory Testing; G~~, [_____]~~

Factory Test Reports; G~~, [_____]~~

~~[~~ ~~Factory Testing -Medical Facilities; G, [_____]~~

~~]~~ SD-07 Certificates

Proof of Listing; G~~, [_____]~~

SD-10 Operation and Maintenance Data

Operation and Maintenance Manual, Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA, Data Package 5; G~~, [_____]~~

1.4 OPERATION AND MAINTENANCE MANUAL

Assemble and bind manuals in durable, hard-covered, water resistant binders. Assemble and index the manuals per the following table of contents:

a. Manufacturer's O&M per "SD-10 Operation and Maintenance Data".

b. Catalog data required by "SD-03 Product Data"

c. Drawings required by "SD-02 Shop Drawings".

1.4.1 Additions to Operation and Maintenance Manuals

In addition to requirements of SD-10 Data Package 5, include the followings on the actual equipment provided:

a. An outline drawing, front, top, and side views.

b. Prices for spare parts and supply list.

c. Date of Purchase.

d. Corrective maintenance procedures.

e. Operating manual outlining step-by-step procedures for system startup, operation, and shutdown.

f. Include simplified wiring and control diagrams in the manual for system as installed.

g. Provide typical contact voltage drop readings under specified conditions for use during periodic maintenance. Provide instructions for determination of contact integrity.

1.4.2 Spare Parts

Furnish the following the following minimum spare parts and any other spare parts required in one-year operation, of the same material and workmanship, meeting the same requirements, and interchangeable with the corresponding original parts.

a. Fuses: Two of each type and rating.

1.5 QUALITY ASSURANCE

1.5.1 Proof of Listing

Submit proof of listing by UL 1008.

1.5.2 Automatic Transfer Switch Drawings

Include the following as a minimum:

a. An outline drawing, including front, top, and side views.

b. Provide a nameplate of corrosion-resistant material with not less than 3 mm tall characters showing manufacturer's name and equipment ratings. Mount nameplate to front of enclosure and meet the nameplate requirements of NEMA ICS 2.

c. Provide detail drawings that include manufacturer's name and catalog number, electrical ratings, total system transfer statement, reduced normal supply voltage at which transfer to the alternate supply is initiated, transfer delay times, short-circuit current rating, wiring diagram, description of interconnections, testing instructions, acceptable conductor type for terminals, tightening torque for each wire connector, and other required UL 1008 markings.

d. Submit interface equipment connection diagram showing conduit and wiring between ATS and related equipment. Provide diagrams showing interlocking provisions and cautionary notes, if any.

e. Drawings are to indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices.

1.5.3 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated

1.5.4 Standard Product

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship, and:

a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.

c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.4.1 Alternative Qualifications

Products having less than a 2-year field service record are acceptable if the manufacturer has been regularly engaged in the design and production of automatic transfer switches and if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 years prior to date of delivery to site are not acceptable.

1.6 DELIVERY AND STORAGE

Protect equipment placed in storage from humidity and temperature variations, moisture, water intrusion, dirt, dust, or other contaminants. In harsh environments where temperatures exceed non-operational parameters established within this specification, provide an environmentally controlled equipment storage facility to ensure temperature parameters are within equipment specification. Provide documentation of same to the Government when storage is implemented.

1.7 ENVIRONMENTAL CONDITIONS

Provide an ATS that is suitable for prolonged performance under following service conditions:

a. Operating altitude: Sea level to 1,000 meters. (Systems applied at higher altitudes are to be derated in accordance with the manufacturer's instructions).

b. Operating ambient temperature range:-~~[~~-4~~][_____]~~ to ~~[~~40~~][_____]~~ degrees C.

c. Operating relative humidity: 0 to 90 percent, without condensation.

~~[~~1.8 SEISMIC REQUIREMENTS

Provide seismic details~~[~~ conforming to~~[~~ Section 13 48 73, SEISMIC CONTROL FOR NONSTRUCTURAL COMPONENTS~~][~~ and to~~][~~ Section 26 05 48, SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT~~]][~~ as indicated~~]~~.

~~]~~PART 2 PRODUCTS

2.1 AUTOMATIC TRANSFER SWITCHES

Each automatic transfer switch must be rated and marked for total system transfer and have the current and voltage ratings as indicated. Provide a switch operating mechanism that is electrically operated, have quick-make, quick-break, load break contacts, and be mechanically held in both positions. ~~[Switches utilizing circuit breakers are not acceptable.]~~ Provide an ATS that is UL listed. ATS must be manufactured and tested in accordance with applicable requirements of NEMA ICS 2, UL 1008 and UL 1066. ATS must conform to NFPA 110. Provide the ATS with the following characteristics:

a. Voltage: ~~[_____]~~208 volts ~~[~~ac~~][dc]~~.

b. Amperage: ~~[_____]~~400 amps ~~[~~ac~~][dc]~~. Provide an ATS with a continuous load current rating of the switch rating.

c. Number of Phases: ~~[~~Three~~][One]~~.

d. Number of Wires: ~~[~~Four~~][Three][Two]~~.

e. Frequency: ~~[60][~~50~~]~~ Hz.

f. Poles: ~~[Four switched][~~Three switched~~][Two switched]~~. ~~[One of the poles is the neutral.]~~

g. ATS Withstand Current Rating: ATS must be rated to close on and withstand the available RMS symmetrical short circuit current at the ATS terminals. The ATS must be listed in accordance with UL 1008 for 3 ~~[18] [~~30~~]~~ cycle close and withstand ratings. Minimum UL listed close and withstand ratings at 208 VAC ~~[480 VAC]~~ must be 30 ~~[42] [65] [100] [~~200~~]~~ kA.

h. Nonwelding Contacts: Provide contacts that are nonwelding at the available fault current rating. Contacts must be suitable for repetitive power transfer switching. Switches rated 800 amps and above must have segmented, blow-on construction for high withstand and close-on capability and be protected by separate arcing contacts.

i. ~~[~~Phase and Neutral~~] [Phase]~~ Contacts: Provide contacts with silver alloy composition. ~~[~~Provide neutral contacts with the same continuous current rating as main or phase contacts.~~] [Provide neutral contacts with 200 percent the current rating of the phase contacts.]~~

j. Configuration. Provide an ATS for use in ~~[emergency systems][~~legally required standby system~~][optional standby systems][critical operations power systems]~~ described in NFPA 70.~~ [Provide an ATS that is listed for emergency use.]~~

k. ATS Configuration. ~~[~~Provide an open transition ATS.~~][Neutral is to break and make with the phase contacts.][Phase contacts are to break and make, but the neutral is to make before break (overlap).][Provide a closed transition ATS.]~~

~~[~~ l. Service Entrance Rated. Provide an integrated circuit breaker and automatic transfer switch. Provide a separate deadfront compartment for the circuit breaker on switches 600 amp and larger. Provide label indicating that the ATS is the service disconnect. Provide a circuit breaker that is rated for ~~[~~100 percent~~][80 percent]~~ of the switch contact current rating. All components, except as noted herein, are to have a continuous load rating.

~~]~~
~~[m. Viewing Ports. Provide contacts that are viewable from the front of the device when the door is open. Comply with the requirements found in IEEE 602 and NFPA 99~~

~~][~~m. Fire Pump Service. Provide a manual operating means that is externally operable without opening the enclosure on transfer switches for fire pump service. The manual means is to open and close the switch contacts at the same rate of speed as that caused by the automatic operation of the switch. The ATS is to meet the requirements found in NFPA 20.

~~]~~

2.1.1 Undervoltage Sensing - Normal/Preferred Source

Undervoltage Sensing - Normal Source. Provide undervoltage sensing for each phase in the normal/preferred source. Sense low phase-to-ground voltage on each phase. Provide sensing circuit with adjustable dropout, 75-98 percent of nominal value and adjustable pickup, 85-100 percent of nominal value. Factory set dropout value to ~~[~~85~~][90][80][_____]~~ percent. Factory set pickup value to ~~[90][~~95~~][_____]~~ percent.

2.1.2 Adjustable Time Delay - Override Transfer

Adjustable Time Delay - Override Transfer. For override of normal-source voltage sensing to delay transfer~~[~~ and engine starting~~]~~ signals. Engine starting control contacts with adjustable commit-to-start delay circuit, 0.0-6.0 seconds. Factory set at ~~[~~1~~][0.5][_____]~~ second~~[s]~~.

2.1.3 Voltage/Frequency Lockout Relay - Alternate/Emergency Source

Voltage/Frequency Lockout Relay. ~~[Single-][~~Three-~~]~~phase sensing must be provided on the normal and emergency source. Prevent premature transfer to alternate/emergency source. Provide pickup voltage that is adjustable from 85-100 percent of nominal. Factory set for pickup at ~~[90][_____]~~85 percent. Provide pickup frequency that is adjustable from 90-97 percent of nominal. Factor set frequency pickup for ~~[~~95~~][_____]~~ percent.

2.1.4 Adjustable Time Delay - Transfer to Alternate/Emergency Power Source

Adjustable Time Delay - Transfer to Alternate Power Source. Transfer to alternate power source time delay for transfer switches as indicated, adjustable 0-5 minutes. Factory set to ~~[~~0~~][_____]~~ seconds. ATS is to monitor the frequency and voltage of alternate power source and transfer when frequency and voltage are stabilized.

2.1.5 Adjustable Time Delay- Re-transfer to Normal/Preferred Source

Adjustable Time Delay- Transfer to Source. Re-transfer to normal source time delay, adjustable 0-30 minutes. Factory set at ~~[~~10~~][_____]~~ minutes. Time delay is automatically defeated upon loss or sustained undervoltage of alternate power source, provided that normal source has been restored.

~~[~~2.1.6 Engine-Generator Exerciser

Exerciser. Solid-state, programmable-time switch exerciser to allow automatic starting of the generator set, subsequent load transfer, retransfer of load and shuts down engine after a preset cool-down period. Initiates exercise cycle at preset intervals adjustable from on a daily, weekly, bi-weekly or monthly basis.. Running periods are adjustable from 10-30 minutes. Factory settings are for 7-day exercise cycle, 20 minute running period and 5-minute cool-down period. Exerciser features include the following:

a. Exerciser Transfer Selector Switch: Permits selection of exercise with and without load transfer or dual independent exercisers that allow for unloaded and loaded schedule testing.

b. Push-button programming control with digital display of settings.

c. Integral battery operation of time switch when normal control power is not available.

~~][~~2.1.7 Engine Shutdown Time Delay

Engine Shutdown. Provide time delay that is adjustable from ~~[~~0~~][_____]~~ to ~~[~~5~~][_____]~~ minutes and is factory set at ~~[~~5~~][_____]~~ minutes.

~~][~~2.1.8 Engine Starting Contacts

Provide ~~[~~1~~][2][3][4]~~ isolated normally closed and ~~[~~1~~][2][3][4]~~ isolated normally open contact that is rated 5 A at 250 VAC/30 VDC minimum.

~~][~~2.1.9 Controls for Fire Pump Service Automatic Transfer Switch

Provide the following additional controls features:

Phase reversal of the normal source is to initiate transfer to the emergency/alternate source.

~~][~~2.1.10 Delayed Transition With Time Delay Neutral

Provide an adjustable time delay transition for indicated transfer switches to allow safe transfer of highly inductive loads between two non-synchronized sources. This transfer between loads has a programmed neutral position arranged to provide a midpoint between the two working positions, with an intentional time-controlled pause at midpoint during transfer. Pause is adjustable from 1 to 300 seconds. Factory set time delay at ~~[0.5][~~1~~][2][5]~~ second~~s~~. Time delay occurs for both transfer directions. Manufacturer is to provide recommendations for establishing the length of the time delay.

~~][~~2.1.11 Motor Disconnect And Timing Relay

Motor Disconnect and Timing Relay: Controls designate starters so they disconnect motors before transfer and reconnect them selectively at an adjustable time interval after transfer. Control connection to motor starters is through wiring external to automatic transfer switch. Time delay for reconnecting individual motor loads is adjustable between 1 and 60 seconds, and settings are as indicated. Relay contacts handling motor-control circuit inrush and seal currents are rated for actual currents to be encountered.

~~][2.1.12 Make Before Break Neutral~~

~~[Contractor is required to coordinate with UPS manufacturer to determine if the unit being procured requires the neutral to be interconnected. If not required, then break before make neutral contacts are allowed. If required, then provide the ATS with make before break neutral contacts.][Provide the ATS with a make before break neutral. Phase contacts are to break before make.] The neutrals are to make for [50][_____] ms.~~

~~][~~2.1.12 Auxiliary Contact for Uninterruptible Power Supply

Provide a contact that closes when transferred to the alternate power source.

~~]~~2.1.13 Unassigned Auxiliary Contacts

Provide ~~[two][~~three~~][_____]~~ normally open and ~~[two][~~three~~][_____]~~ normally closed, single-pole, double-throw auxiliary contacts for each switch position rated at ~~[~~10~~][15][_____]~~ amperes at ~~[~~240~~][120][480][_____]~~ volts.

2.1.14 Front Panel Devices

Provide devices mounted on cabinet front consisting of:

a. Mode selector switch with the following positions and associated functions. Selector switch can be part of the microprocessor controller consisting of an LCD screen with a graphical interface or as a stand-alone test switch.

    (1) TEST - Simulates loss of normal/preferred source system operation.

    (2) NORMAL - Transfers system to normal/preferred source bypassing re-transfer time delay.

b. Switch position indicating lights or graphical LCD display. Indicate source to which load is connected.

c. Source-Available Monitor. Provide source-available indicating lights or graphical LCD display monitor that is labeled to show when one or both sources of power are available. If indicating lights are used, then the preference is to have Green be normal/preferred power and Red be for alternate/emergency power; however, other color schemes are allowed if clearly marked.

d. Provide a transfer override switch. Provide automatic transfer switch microprocessor based controller, which offers field selectable/adjustable inputs and outputs for transfer switch operation. Override switch must bypass automatic transfer controls so ATS will transfer and remain connected to ~~[alternate][emergency][~~ generator~~][______]~~ power source, regardless of condition of normal/preferred source. Provide an indicating light to show override status. ~~[~~If ~~[~~alternate~~][emergency]~~ source fails and ~~[~~normal~~][preferred]~~ source is available, ATS is to automatically retransfer to ~~[~~normal~~]~~ ~~[preferred]~~ source.~~]~~

e. Lamp test button.

~~[~~2.1.15 Voltage Unbalance

Provide automatic transfer switch controller or control logic to include positive and negative sequence voltage detection to identify a phase loss condition that can adversely effect motor loads.

~~][~~2.1.16 Closed-Transition Transfer Switch

Include the following functions and characteristic for an automatic transfer switch that is to operate in a closed-transition manner.

a. Fully automatic make-before-break operation.

b. Load transfer without interruption, through momentary interconnection of both power sources not exceeding 100 ms, but no less than 50 ms.

c. Initiation of No-Interruption Transfer: Controlled by in-phase monitor and sensors confirming both sources are present and acceptable.

(1) Initiation occurs without active control of generator.

(2) Controls ensure that closed-transition load transfer closure occurs only when the 2 sources are within plus or minus 5 electrical degrees maximum, and plus or minus 5 percent maximum voltage difference.

d. Failure of power source serving load initiates automatic break-before-make transfer.

~~][~~2.1.17 In-Phase Monitor

Provide an in-phase monitor that consists of a factory-wired, internal relay that controls transfer so it occurs only when the two sources are synchronized in phase. Relay compares phase relationship and frequency difference between normal and emergency sources and initiates transfer when both sources are within 5 electrical degrees, and only if transfer can be completed within 60 electrical degrees. Transfer is initiated only if both sources are within 2 Hz of nominal frequency and 70 percent or more of nominal voltage Manufacturer is to provide information regarding what conditions a transfer cannot be accomplished.

][2.2 BY-PASS/ISOLATION SWITCH (BP/IS)

Include non-load-break by-pass/isolation switches for the indicated automatic transfer switches. Designs which disconnect or interrupt the load when bypassing are not acceptable. Include the following features for each combined by-pass/isolation switch and automatic transfer switch:

a. Bypass/isolation switch (BP/IS) and associated ATS are to be made by the same manufacturer and must be completely interconnected and tested at factory and at project site as specified.

b. ATS is to be manufactured, listed and tested in accordance with paragraph AUTOMATIC TRANSFER SWITCH. BP/IS switch current, voltage, closing, and short-circuit withstand closing ratings are to be equal or exceed comparable ratings specified for ATS and have the same phase arrangement and number of poles.

c. Provide externally operated and arranged selector switch or handle so designed and constructed not to stop in an intermediate or neutral position during operation and that one person can safely bypass the ATS. Accomplish isolation of the ATS externally by one person. Bypass and isolation handles must be permanently affixed and operable without opening the enclosure door. Provide interlocks that ensure ATS is disconnected from source and load during isolation. Interlocks prevent ATS operation, except for testing and maintenance, while isolated. BP/IS operation is to be accomplished without disconnecting switch load terminal conductors. Equipment which require separate tools, keys, or other devices to operate the bypass/isolation mechanism which may not be present during an emergency is not acceptable.

d. Provide drawout transfer switch that provides physical separation from bypass switch and live parts and accessibility for testing and maintenance operation.

e. Provide contacts that have the same contact temperature that do not exceed those of the ATS contacts when carrying rated load. Provide contacts as specified for associated ATS, including provisions for inspection of contacts without disassembly of BP/IS or removal of entire contact enclosure.   Provide manufacturer instructions for determining contact integrity in order To facilitate maintenance.

f. The ATS controls remain functional with the ATS isolated or in bypass mode to permit monitoring of the normal power source [and automatic starting of the generator in the event of a loss of the normal power source]. In the isolated mode, the bypass section is capable of functioning as a manual transfer to transfer the load to either power source for maintenane purposes or when automatic control has failed. Equipment that requires automatic controls to be functional to operate the bypass switch is not acceptable.. The ATS can be completely removed from the enclosure, if required for maintenance or repair, while the bypass section continues to power the load.

g. Construct Bypass/isolation switch for convenient removal of parts from front of switch enclosure without removal of other parts or disconnection of external power conductors.

h. Achieve load by-pass to the source with no load interruption. Bypass/isolation equipment that breaks the load is not accpetalbe.

~~[~~ i. Provide drawout bypass switch that provides physical separation from ATS and live parts and accessibility for testing and maintenance operation.~~ [Provide automatic shutters that closed to isolate the bus.]~~

~~][~~j. Provide a means to ensure the switch is transferred to the alternate or emergency power source when normal power source becomes unavailable.

~~]~~2.2.1 Markings

Mark isolation handle positions with engraved plates or other approved means to indicate position or operating condition of associated ATS, as follows:

a. Provide an indication that shows that BP/IS section is providing power to the load.

b. Provide indication of ATS isolation/test position.

c. Provide suitable control labels and instruction signs describing operating instructions.

d. Indicating lamps or LCD screen for indicating that shows the source availability, bypass switch position, transfer switch position, and isolation handle position. If indicating lights are used, provide a lamp test button that turns the indicating lights on, but does not cause any function to take place.

2.2.2 Interconnection

Interconnect BP/IS and associated ATS with suitably sized copper bus bars silver-plated at each connection point, and braced to withstand magnetic and thermal forces created at withstand current rating specified for associated ATS.

~~]~~2.3 ENCLOSURE

Provide an enclosure that meets the following:

a. Provide ATS and accessories in a ~~[free-standing, ~~floor-mounted ~~][wall-mounted]~~, ~~[~~ventilated~~][unventilated]~~ NEMA 250, Type ~~[~~1 ~~][3R][3RX][4][4X][12]~~, smooth sheet metal enclosure constructed in accordance with applicable requirements of NEMA ICS 6, UL 508, UL 1066, and UL 1008. ~~[~~Provide screened and filtered intake vents. Provide screened exhaust vents.~~]~~ ~~[~~Provide door with suitable hinges, locking handle latch, and gasketed jamb.~~]~~ Provide at least No. 14 metal gauge.

b. Factory wiring within enclosure and field wiring terminating within enclosure must comply with NFPA 70. Provide wire that is permanently tagged or marked near terminal at each end with wire number shown on approved detail drawing, when wiring is not color coded. Conform terminal block to NEMA ICS 4. Arrange terminals for entrance of external conductors from ~~[~~top and bottom~~][top][bottom]~~ of enclosure as shown. Main switch terminals, including neutral terminal if used, must be pressure type suitable for termination of external ~~[~~copper~~] ~~

~~[aluminum]~~ conductors shown.

~~[c. Provide thermostatically controlled heater within enclosure to prevent condensation over temperature range stipulated in paragraph SERVICE CONDITIONS.~~

~~]~~2.3.1 Construction

Construct enclosure for ease of removal and replacement of ATS components and control devices from front without disconnection of external power conductors or removal or disassembly of major components.

2.3.2 Cleaning and Painting

Protect both the inside and outside surfaces of an enclosure, including means for fastening against corrosion by enameling, galvanizing, plating, powder coating, or other equivalent means. Protection is not required for metal parts that are inherently resistant to corrosion, bearings, sliding surfaces of hinges, or other parts where such protection is impractical. Provide manufacturer's standard finish material, process, and color that is free from runs, sags, peeling, or other defects. An enclosure marked Type 1, 3R, 4 or 12 is acceptable if there is no visible rust at the conclusion of a salt spray (fog) test using the test method in ASTM B117, employing a 5 percent by weight, salt solution for 24 hours. Type 4X enclosures are acceptable following performance of the above test with an exposure time of 200 hours.

2.3.3 Field Fabricated Nameplates

Nameplate is to comply with ASTM D709. Provide laminated plastic nameplates for each equipment enclosure as specified or as indicated on the drawings. Provide an inscription on each nameplate that identifies the name of the equipment, sources of power, calculated short circuit with date and the location e.g. 'SWB-1 Electrical Room 103'. Provide nameplates that are made of melamine plastic, 3 mm thick, white with ~~[~~black ~~][_____]~~ center core. Provide the nameplate with a surface that is matte finished and that has square corners.. Accurately align lettering and engrave into the core. Provide nameplates that are at least 25 by 65 mm with a minimum lettering size of 6.35 mm high normal block style.

~~[~~2.4 REMOTE ANNUNCIATOR PANEL

~~[~~Provide remote annunciation with LED indicating lights, an audible alarm with silence switch as well as all appropriate labeling~~.][~~ or ~~][P~~provide a remote annunciator panel that utilizes a touchscreen human machine interface (HMI). Minimum screen size is 175 mm.~~]~~ The annunciator is to be configured to handle ~~[1][2][_____]~~3 transfer switches. Provide a surface mounted cabinet. Provide built-in power supply that accepts either 24 VDC or 120VAC~~ or [_____]~~. Provide communications module to support monitoring of ATS. Module must provide status, analog parameters, event logs, equipment settings, and configurations over embedded webpage, open protocol, and automated email while utilizing AES 128-bit encryption. Provide a remote annunciation panel to annunciate the following conditions for the indicated transfer switch(es).

a. Sources available

b. Switch position.

c. Switch in test mode.

d. Failure of communication link.

~~][~~2.5 REMOTE ANNUNCIATOR AND CONTROL SYSTEM PANEL

~~Provide a remote annunciator and control system that utilizes a touchscreen human machine interface (HMI) with the ability to remotely monitor and control multiple transfer switches from a single panel. Minimum screen size is 175 mm. Provide password protection and date/time stamped alarm history. The controller is to have internal battery backup. In the event of a communication link failure, the system is to automatically revert to stand-alone, self-contained operation. Automatic transfer switch sensing, controlling or operating function is not to depend on remote panel for proper operation. Provide a surface mounted cabinet. Communication is to be by a [Modbus] RS-485 connection. The annunciator controller is to be configured to monitor and control [1][2][_____] transfer switches.~~

~~2.5.1 Monitor~~

~~Monitor the following:~~

~~a. Sources available~~

~~b. Switch position.~~

~~c. Switch in test mode.~~

~~d. Overvoltage~~

~~e. Failure of communication link.~~

~~[f. Engine test or exercise.~~
~~]~~
~~2.5.2 Alarm Screen~~

~~Alarm for the following conditions:~~

~~a. Alternate source closed~~

~~b. Undervoltage~~

~~c. Lockout.~~

2.5.1 Control Functions

~~Provide a means to perform the following functions from the controller: an alarm silence button in addition to monitoring the following items:~~

~~a. Control of switch-test initiation.~~

~~b. Control of switch operation in either direction.~~

~~c. Control of time-delay bypass for transfer to normal source.~~

~~d. Control to perform an engine test.~~

~~[~~ a. Provide a means to remotely configure transfer switch controller setpoints. The means to perform these changes must be password protected.

~~]~~

~~[~~ b. Manage up to eight (8) transfer switches from a single remote annunciator and control panel

~~]~~

~~]~~2.6 FACTORY TESTING

Submit a description of proposed field test procedures, including proposed date and steps describing each test, its duration and expected results, not less than ~~[_____]~~2 weeks prior to test date. Submit certified factory and field test reports, within 14 days following completion of tests. Provide reports that are certified and dated and that demonstrate that tests were successfully completed prior to shipment of equipment.

2.6.1 Prototype Factory Testing

A prototype of specified ATS is to be factory tested in accordance with UL 1008. In addition, perform factory tests on each ATS as follows:

a. Insulation resistance test to ensure integrity and continuity of entire system

b. Main switch contact resistance test.

c. Visual inspection to verify that each ATS is as specified.

d. Mechanical test to verify that ATS sections are free of mechanical hindrances.

e. Electrical tests to verify complete system electrical operation and to set up time delays and voltage sensing settings.

2.6.2 Factory Test Reports

Provide three certified copies of factory test reports from the manufacturer.

~~[2.7 FACTORY TESTING -MEDICAL FACILITIES~~

~~The factory tests for ATS and By-Pass/Isolation switches used in medical facilities must be conducted in the following sequence:~~

~~a. General~~

~~b. Normal~~

~~c. Overvoltage~~

~~d. Undervoltage~~

~~e. Overload~~

~~f. Endurance~~

~~g. Temperature Rise~~

~~h. Dielectric Voltage-Withstand~~

~~i. Contact Opening~~

~~j. Dielectric Voltage-Withstand (Repeated)~~

~~k. Withstand~~

~~l. Instrumentation and Calibration of High Capacity~~

~~m. Closing~~

~~n. Dielectric Voltage-Withstand (Repeated)~~

~~o. Strength of Insulating Base and Support~~

~~]~~PART 3 EXECUTION

~~3.1 INSTALLATION~~

~~Installation must conform to the requirements of NFPA 70 and manufacturer's recommendation.~~

~~3.2 PREREQUISITES FOR FUNCTIONAL ACCEPTANCE TESTING~~

~~Completion of the following requirements is mandatory prior to scheduling functional acceptance tests for the automatic transfer switch.~~

~~3.2.1 Performance of Acceptance Checks and tests~~

~~Complete as specified in paragraph entitled "Acceptance Checks and Tests". The Acceptance Checks and Tests are to be accomplished by the Testing organization as described in Section 26 08 00 APPARATUS INSPECTION AND TESTING.~~

~~3.2.2 Manufacturers O&M Information~~

~~The manufacturers O&M information required by the paragraph entitled "SD-10 Operation and Maintenance Data", is to be submitted to and approved by the Contracting Officer.~~

~~3.2.3 Test Equipment~~

~~Ensure all test equipment and instruments is on hand prior to scheduling field tests, or subject to Contracting Officer's approval, evidence must be provided to show that arrangements have been made to have the necessary equipment and instruments on site prior to field testing.~~

3.1 FIELD QUALITY CONTROL

~~Give Contracting Officer 15 days notice of dates and times scheduled for tests which require the presence of the Contracting Officer. The Contracting Officer will coordinate with the using activity and schedule a time that will eliminate or minimize interruptions and interference with the activity operations. The contractor is responsible for costs associated with conducting tests outside of normal working hours and with incorporating special arrangements and procedures, including temporary power conditions. The contractor provides labor, equipment, apparatus, including test load, and consumables required for the specified tests. Calibration of all measuring devices and indicating devices must be certified. Provide the services of a qualified factory-trained~~

~~manufacturer's representative to assist the contractor in installation and start-up of the equipment specified under this section. The manufacturer's representative is to provide technical direction and assistance to the contractor in general assembly of the equipment, connections and adjustments, and testing of the assembly components contained herein. [Provide a test load that is a cataloged product in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.] Perform the following field tests in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.~~

~~3.1.1 Automatic Transfer Switch Acceptance Checks and Tests~~

~~a. Visual and Mechanical Inspection~~

~~(1) Compare equipment nameplate data with specifications and approved shop drawings.~~

~~(2) Inspect physical and mechanical condition.~~

~~(3) Confirm correct application of manufacturer's recommended lubricants.~~

~~(4) Verify that manual transfer warnings are attached and visible.~~

~~(5) Verify tightness of all control connections.~~

~~(6) Verify tightness of accessible bolted connections by calibrated torque-wrench method. Thermographic survey is not required.~~

~~(7) Perform manual transfer operation.~~

~~(8) Verify positive mechanical interlocking between normal and alternate sources.~~

~~b. Electrical Tests~~

~~(1) Measure contact-resistance. Correct values that exceed 500 microhms and values for 1 pole deviating by more than 50 percent from other poles.~~

~~(2) Perform insulation-resistance on each pole, phase-to-phase and phase-to-ground with switch closed, and across each open pole for one minute. Perform tests in both source positions.~~

~~(3) Verify settings and operations of control devices.~~

~~(4) Calibrate and set all relays and timers.~~

~~[~~ ~~(5) Test ground-fault protective device.~~

~~]~~3.1.1 Functional Acceptance Tests

Functional Acceptance Tests must be coordinated with Section 26 32 15 ENGINE-GENERATOR SET STATIONARY 15-2500 KW, WITH AUXILIARIES.~~[~~ Functional Acceptance Test must be coordinated with Section 21 30 00 FIRE PUMPS.~~]~~ Include simulating power failure and demonstrating the following operations for each automatic transfer switch. Demonstrate in service that the automatic transfer switches are in good operating condition, and function not less than five times.

a. Perform automatic transfer tests:

(1) Simulate loss of normal/preferred power.

(2) Return to normal/preferred power.

(3) Simulate loss of emergency/alternate power.

(4) Simulate all forms of single-phase conditions.

b. Verify correct operation and timing of the following functions:

(1) Normal source voltage-sensing relays.

(2) Engine start sequence.

(3) Time delay upon transfer.

(4) Alternate source voltage-sensing relays.

(5) Automatic transfer operation.

(6) Interlocks and limit switch function.

(7) Time delay and retransfer upon normal power restoration.

[(8) By-pass/isolation functional modes and related automatic transfer switch operations.

]3.1.2 Infrared Scanning

After Substantial Completion, but not more than 60 days after Final Acceptance, perform an infrared scan of each switch. Remove all access panels so joints and connections are accessible to portable scanner.

a. Follow-up Infrared Scanning: Perform an additional follow-up infrared scan of each switch 11 months after acceptance.

b. Instrument: Use an infrared scanning device designed to measure temperature or to detect significant deviations from normal values. Provide calibration record for device.

c. Record of Infrared Scanning: Prepare a certified report that identifies switches checked and that describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.2 TRAINING

Provide 4 hours of training to maintenance personnel on the proper operation, maintenance and adjustment of the automatic transfer switch.[Coordinate this training with that of the generator equipment.]

-- End of Section --

SECTION 26 51 00

INTERIOR LIGHTING
05/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS), Japan Luminaires Association (JIL) as approved by the Contracting Officer's representative.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2	(2023) National Electrical Safety Code

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS C 0365	(2007) Protection Against Electric Shock - Common Aspects for Installation and Equipment
JIS C 0920	(2003) Degrees Of Protection Provided By Enclosures (IP Code)
JIS C 8105-1	(2021) Luminaires - Part 1: General Requirements For Safety
JIS C 8105-2-22	(2014) Luminaires - Part 2-22: Particular Requirements - Luminaires For Emergency Lighting
JIS C 8152-2	(2019) Measurement method of white light emitting diode (LED) for lighting-Part 2: LED modules and LED light engines
JIS C 8152-3	(2013) Measurement method of white light emitting diode (LED) for lightingPart 3: Measurement of luminous flux maintenance rate
JIS C 8153	(2015) DC or AC supplied electronic control gear for LED modules -- Performance requirements
JIS C 8154	(2015) LED modules for general lighting -- Safety specifications
JIS C 8155	(2019) LED modules for general lighting -- Performance requirements
JIS C 8201-5-1	(2022) Low-Voltage Switchgear And Control Gear - Part 5-1: Control Circuit Devices And Switching Elements - Electromechanical

	Control Circuit Devices
JIS C 8286	(2021) Electrical accessories -- Cord sets and interconnection cord sets
JIS C 8304	(2009; R 2019) Small switches for indoor use
JIS C 8462-1	(2021) Boxes and enclosures for electrical accessories for household and similar fixed electrical installations -- Part 1: General requirements
JIS C 5402-20-2	(2005) Electromechanical components for electronic equipment -- Basic testing procedures and measuring methods -- Part 20-2: Test 20b -- Flammability tests -- Fireproofness
JIS C 60079-0	(2010; R 2020) Explosive atmospheres -- Part 0: Equipment -- General requirements
JIS C 61000-3-2	(2019) Electromagnetic compatibility (EMC) -- Part 3-2: Limits -- Limits for harmonic current emissions
JIS C 9730-2-7	(2019) Automatic electrical controls -- Part 2-7: Particular requirements for timers and time switches
JIS G 3141	(2021) Cold-reduced carbon steel sheet and strip
JIS G 3302	~~(2022) Hot Dip Zinc Coated Steel Sheet and Strip~~(2022) Hot-Dip Zinc-Coated Steel Sheet and Strip
JIS G 3547	(2015) Cold-rolled stainless steel plate, sheet and strip
JIS G 4309	(2013) Stainless Steel Wires
JIS H 8610	(1999) Electroplated-Coatings of Zinc on Iron or Steel
JIS Z 8113	(1998) Lighting vocabulary
JIS Z 9110	(2010; R 2011) General rules of recommended lighting levels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101	(2021; TIA 21-1) Life Safety Code

~~UNDERWRITERS LABORATORIES (UL)~~UL SOLUTIONS (UL)

UL 2043	(2013) Fire Test for Heat and Visible Smoke Release for Discrete Products and Their Accessories Installed in

	Air-Handling Spaces
UL 924	(2016; Reprint May 2018) UL Standard for Safety Emergency Lighting and Power Equipment
UL 94	~~(2023; Seventh Edition) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances~~(2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires or luminaire accessories are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.~~ Luminaires and accessories mounted on exterior surfaces of buildings are specified in Section 26 56 00 EXTERIOR LIGHTING.~~

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, must be as defined in JIS Z 8113

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the JIS Z 9110 as applicable, for the lighting system specified. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Occupancy/Vacancy Sensor Coverage Layout~~;~~

SD-03 Product Data

Luminaires; G~~[, [_____]]~~

Light Sources; G~~[, [_____]]~~

Drivers, Ballasts and Generators; G~~[, [_____]]~~

LED Luminaire Warranty~~;~~

Vacancy Sensors; G~~[, [_____]]~~

Dimming Controllers (Dimmers); G~~[, [_____]]~~

~~Lighting Contactor; G[, [_____]]~~

Timeswitch; G~~[, [_____]]~~

Power Hook Luminaire Hangers; G~~[, [_____]]~~

Exit Signs; G~~[, [_____]]~~

Emergency Lighting Unit (EBU); G~~[, [_____]]~~

LED Emergency Drivers; G~~[, [_____]]~~

~~Fluorescent Emergency Ballasts; G[, [_____]]~~

Occupancy Sensors; G~~[, [_____]]~~

Ambient Light Level Sensor ; G~~[, [_____]]~~

Lighting Control Panel; G~~[, [_____]]~~

SD-06 Test Reports

Occupancy/Vacancy Sensor Verification Tests;

Energy Efficiency~~;~~

1.5 QUALITY CONTROL

1.5.1 Luminaire Drawings

Include dimensions, accessories, and installation and construction details. Photometric data, including zonal lumen data, average and minimum ratio, aiming diagram, and computerized candlepower distribution data must accompany shop drawings.

1.5.2 Occupancy/Vacancy Sensor Coverage Layout

Provide floor plans showing coverage layouts of all devices using manufacturer's product information.

1.5.3 Luminaire Design Data

Provide long term lumen maintenance projections for each LED luminaire in accordance with JIS C 8152-3. Data used for projections must be obtained from testing in accordance with JIS C 8152-3 Appendix B.

1.5.4 LED Luminaire - Test Report

Submit test report on manufacturer's standard production model luminaire.

1.5.5 LED Light Source - Test Report

Submit test report on manufacturer's standard production LED light source (package, array or module). Include all applicable and required data, as outlined under "8 Report on Test Results" in JIS C 8152-3.

1.5.6 Photometric Plan

1.5.6.1 Computer-generated Photometric Plans

Computer-generated photometric plans for each space are required to verify proposed luminaires and locations meet the required performance criteria of the design using the applicable light loss factor (LLF).

Target illumination levels are provided for each Interior Application. Depending on the application and the recommendations provided by the IES, values are given as one of the following:

a. Minimum: No values anywhere on the calculation grid may be less than this value, within a 10 percent margin of error.

b. Minimum Average: An average, taken over the entire task area for the application, may not be less than this value, within a 10 percent margin of error.

c. Maximum: No values anywhere on the calculation grid may be greater than this value, within a 10 percent margin of error.

d. Maximum Average: An average, taken over the entire task area for the application, may not be greater than this value, within a 10 percent margin of error.

e. Uniformity: Unless otherwise noted, uniformity is calculated as a ratio of the average calculated illuminance over the minimum calculated illuminance of the calculation grid.

1.5.6.2 Schematic Photometric Plan Calculations

Schematic photometric plan calculations must include:

a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every 305 mm across the task area.

b. Average maintained illuminance level.

c. Minimum and maximum maintained illuminance levels.

d. Lighting power density (Watts per square meter).

e. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.6.3 Final Photometric Plan Calculations

Final photometric plan calculations must include:

a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every 305 mm across the task area.

b. Where applicable, vertical illuminance measurements at designated surface, taken at a maximum of every 305 mm across task area.

c. Minimum and maximum maintained illuminance levels.

d. Average maintained illuminance level.

e. Average to minimum and maximum to minimum ratios for horizontal illuminance.

f. Lighting power density (Watts per square meter).

g. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.7 Occupancy/Vacancy Sensor Verification Tests

Submit test report outlining post-installation coverage and operation of sensors.

1.5.8 Test Laboratories

Test laboratories for JIS C 8152-2 and JIS C 8152-3 test reports must be a Japan Accredited Laboratory for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program for both LM-79 and LM-80 or JIS C 8152-2 and JIS C 8152-3 testing

1.5.9 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of applicable codes and standards, unless more stringent requirements are specified or indicated.

1.5.10 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for two years prior to bid opening. The two-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the two-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.10.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.10.2 Material and Equipment Manufacturing Date

Products manufactured more than six months prior to date of delivery to site must not be used, unless specified otherwise.

1.5.10.3 Energy Efficiency

Submit data indicating lumens per watt efficacy and color rendering index of light source.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 LED Luminaire Warranty

a. Provide a written 5 year on-site replacement warranty for material, fixture finish, and workmanship. On-site replacement includes transportation, removal, and installation of new products.

    (1) Include finish warranty to include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

    (2) Material warranty must include:

        (a) All drivers.

        (b) Replacement when more than 10 percent of LED sources in any lightbar or subassembly(s) are defective or non-starting.

b. Warranty period must begin on date of beneficial occupancy. Provide the Contracting Officer with signed warranty certificates prior to final payment.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

Products and materials not considered to be luminaires, luminaire controls, or associated equipment are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. ~~Luminaires, luminaire controls, and associated equipment for exterior applications are specified in Section 26 56 00 EXTERIOR LIGHTING.~~

2.2 LUMINAIRES

JIS C 8105-1, JIS C 61000-3-2 JIS C 8154 and JIS C 8155. Provide luminaires as indicated in luminaire schedule and NL plates or details on project plans. Provide luminaires complete with light sources of quantity, type, and wattage indicated. Provide all luminaires of the same type by the same manufacturer. Luminaires must be specifically designed for use with the driver, ballast or generator and light source provided.

2.2.1 LED Luminaires

Provide luminaires complete with power supplies (drivers) and light sources. Provide design information including lumen output and design life in luminaire schedule on project plans for LED luminaires. LED luminaires must meet the minimum requirements in the following table:

LUMINAIRE TYPE	MINIMUM LUMINAIRE EFFICACY (LE)	MINIMUM COLOR RENDERING INDEX (CRI)
LED TROFFER - 300 x 1200 600 x 600 600 x 1200	90 LPW	80
LED Downlight	50 LPW	90
LED Track or Accent	40 LPW	80
LED Low Bay/High Bay	80 LPW	70
LED Linear Ambient	80 LPW	80

LED luminaires must also meet the following minimum requirements:

a. Luminaires must have a minimum 5 year manufacturer's warranty.

b. Luminaires must have a minimum ~~[______]~~L70 lumen maintenance value of ~~[______]~~50K hours as calculated by JIS C 8152-3, with data obtained per JIS C 8152-3 requirements.

c. Luminaire drive current value must be identical to that provided by test data for luminaire in question.

d. Luminaires must be tested to JIS C 8152-2 and JIS C 8152-3, with the results provided as required in the Submittals paragraph of this specification.

~~2.2.2 Fluorescent Luminaires~~

~~JIS C 8105-1 and JIL 4003. Provide linear and compact fluorescent luminaires complete with housing, ballast and light source. All fluorescent luminaires must be equipped with electronic ballasts.~~

~~2.2.3 Induction Luminaires~~

~~JIS C 8105-1. Provide induction luminaires complete with housing, generator and light source.~~

2.2.2 Luminaires for Hazardous Locations

In addition to requirements stated herein, provide ~~[~~LED, ~~][fluorescent,][HID,][induction]~~ luminaires for hazardous locations which conform to JIS C 60079-0 or which have Factory Mutual certification for the class and division indicated.

2.3 DRIVERS, BALLASTS and GENERATORS

2.3.1 LED Drivers

JIS C 8153, JIS C 8154 and JIS C 8155. LED drivers must be electronic, constant-current type and comply with the following requirements:

a. Output power (watts)and luminous flux (lumens) as shown in luminaire schedule for each luminaire type to meet minimum luminaire efficacy (LE) value provided.

b. Power Factor (PF) greater than or equal to 0.9 over the full dimming range when provided.

c. Current draw Total Harmonic Distortion (THD) of less than 20 percent.

d. Class A sound rating.

e. Operable at input voltage of ~~[120][240][~~120-277~~][105][210]~~ volts at ~~[~~50 ~~][60]~~ hertz.

f. Minimum 5 year manufacturer's warranty.

g. RoHS compliant.

h. Integral thermal protection that reduces or eliminates the output power if case temperature exceeds a value detrimental to the driver.

i. UL listed for dry or damp locations typical of interior installations.

j. ~~[~~Non-dimmable~~]~~, ~~[~~step-dimmable to 50 percent output~~]~~, or fully-dimmable using 0-10V control as indicated in luminaire schedule.

~~2.3.2 Fluorescent Electronic Ballasts~~

~~JIS C 8117. Fluorescent ballasts must not contain any magnetic core and coil components, and must meet the following requirements:~~

~~a. Provide with protection as recommended by JIS C 5381-11 and JIS C 5381-12.~~

~~b. Be designed for the wattage and type of light source provided in the luminaire specified, and have circuit diagrams and light source connection information printed on the exterior of the ballast housing.~~

~~c. Have a full replacement warranty of five years from date of manufacture.~~

~~[d. Provide all fluorescent ballasts as highest-efficiency type.~~
~~]~~

~~2.3.2.1 T8 Programmed[Instant]-Start Fluorescent Ballasts~~

~~Provide programmed[instant]-start T8 electronic fluorescent ballasts with the following characteristics:~~

~~a. Total harmonic distortion (THD): Must be [20 percent][_____ percent] (maximum).~~

~~b. Input wattage at [120/277][105][210] volts.~~

~~[c. Where indicated on project drawings,provide multi-light source luminaires with two or more ballasts to accomplish the switching scenario indicated.~~
~~]~~
~~[d. A single ballast may be used to serve multiple luminaires if they are continuously mounted and factory manufactured for that installation with an integral wireway.~~

~~]2.3.2.2 T5 (long twin tube) and T5HO Fluorescent Ballasts~~

~~a. Total harmonic distortion (THD): Not greater than[25 percent when operating one light source,][15 percent when operating two light sources,][and][20 percent when operating three light sources].~~

~~b. Input wattage shall be per Annex B, JIS C 8117~~

~~[c. Provide three[and four] light source luminaires with two ballasts per luminaire where multilevel switching is indicated.~~
~~]~~
~~[d. A single ballast may be used to serve multiple fixtures if they are continuously mounted and factory manufactured for that installation with an integral wireway.~~

~~]2.3.2.3 Compact Fluorescent Ballasts~~

~~Provide programmed-start ballasts for compact fluorescent luminaires.~~

~~2.3.2.4 Fluorescent Electronic Dimming Ballasts~~

~~Provide fluorescent electronic dimming ballasts with the following characteristics:~~

~~a. Comply with JIS C 8117 and JIL 4003, unless specified otherwise. Provide ballast as recommended by JIS C 5381-11 and JIS C 5381-12. Provide dimming capability range from 100 to 5 percent (minimum range) of light output, flicker free. Ballast must start lamp at any preset light output setting without first having to go to full light output. Provide ballasts designed for the wattage of the light sources used in the indicated application. Provide ballasts designed to operate on the voltage system to which they are connected.~~

~~a. Ballast must be capable of starting and maintaining operation at a minimum of minus 17 degrees C unless otherwise indicated.~~

~~b. Ballasts for T-5 and smaller light sources must have protection circuits as required by JIS C 7617-1 and JIS C 7620-1 as applicable.~~

~~2.3.2.4.1 T-8 Lamp Ballast~~

~~Input wattage shall be per Annex B, JIS C 8117~~

~~2.3.3 Induction Generators~~

~~Generator must be connected, and operate in conjunction with an inductive power coupler or coil(s). Provide solid-state, high-frequency (200 kHz - 2.67 MHz) type, with power factor greater than 0.95, Class A sound rating, maximum input current THD of 15 percent, operating voltage of [105-210][120-240][120-277]V, and a minimum starting temperature of minus 30 degrees F. Provide generator dimmable to a minimum of 50 percent light output.~~

2.4 LIGHT SOURCES

JIS C 8155. Provide type and wattage as indicated in luminaire schedule on project plans.

2.4.1 LED Light Sources

a. Correlated Color Temperature (CCT) of ~~[3000]~~3500~~[4000][_____]~~ degrees K.

b. Minimum Color Rendering Index (CRI) value of 80.

c. High power, white light output utilizing phosphor conversion (PC) process~~[~~ or mixed system of colored LEDs, typically red, green and blue (RGB)~~]~~.

d. RoHS compliant.

~~[~~2.4.1.1 LED Retrofit T8 Tubes

Provide linear T8 tubular LED light sources to replace fluorescent light sources in renovation or energy conservation projects. Provide only where entire luminaires are not being replaced. Light sources must be compatible with existing instant-start or programmed-start ballasts and have the following requirements:

a. Correlated Color Temperature (CCT) of ~~[3000]~~3500~~[4000]~~ degrees K.

b. Total Harmonic Distortion (THD) less than 20 percent, with Power Factor (PF) greater than 90 percent.

c. Minimum lumen per watt efficacy greater than 120.

d. Minimum beam angle of 180 degrees.

e. Minimum 5 year warranty.

f. Minimum Color Rendering Index (CRI) of 80.

~~]2.4.2 Fluorescent Light Sources~~

~~JIS Z 9112. Fluorescent light sources must be low-mercury, energy-savings type and be compliant with the most current TCLP test procedure at the time of manufacture.~~

~~2.4.2.1 Linear Fluorescent Light Sources~~

~~JIS C 7617-1. Provide linear fluorescent light sources with minimum CRI of 85[_____] and CCT of 3500[_____] degrees K.~~

~~2.4.2.1.1 T8 Linear Fluorescent Light Sources~~

~~Provide T8 light sources with medium bi-pin base, rated [_____] watts (maximum), [_____] initial lumens (minimum), and with an average rated life of 30,000[_____] hours using a average three hour burn time and programmed-start ballast.~~

~~2.4.2.1.2 T5HO (High-Output) Linear Fluorescent Light Sources~~

~~Provide T5HO light sources with miniature bi-pin base, rated [_____] watts (maximum), [_____] initial lumens (minimum), and with an average rated life of 30,000[_____] hours using a average three hour burn time and programmed start ballast.~~

~~2.4.2.2 Compact Fluorescent (CFL) Light Sources~~

~~JIS C 7620-1. Provide compact fluorescent (CFL) light sources with minimum CRI of 82[_____] and CCT of 3500[_____] degrees K.~~

2.4.2 Induction Light Sources

Provide induction light sources consisting of an electrodeless, inductively-coupled, phosphor-coated fluorescent envelope, with an average rated life of 100,000 hours minimum rated using three hours operation per start. Light sources must be compliant with the most current TCLP test procedure at the time of manufacture.

2.5 LIGHTING CONTROLS

Provide lighting controls and associated equipment in accordance with drawings and applicable energy codes. ~~[Provide network certification for all networked lighting control systems and devices per requirements of Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS.]~~

2.5.1 Toggle Switches

Provide line-voltage toggle switches as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.5.2 Dimming Controllers (Dimmers)

JIS C 8304. ~~[~~120/277~~][105/210] V[0-10 V]~~ dimmers must provide flicker-free, continuously variable light output throughout the dimming range. Provide radio frequency interference suppression integral to device. Provide dimmers utilizing ~~[~~pulse width modulation (PWM)~~][constant current reduction (CCR) technology]~~. Provide device with a vertical slider, paddle, rotary button, or toggle (with adjacent vertical slider) type control, with finish to match switches and outlets in same area. Provide back box in wall with sufficient depth to accommodate body of switch and wiring. Devices must be capable of operating at their full rated capacity regardless of being single or ganged-mounted, and be compatible with three-way and four-way switching scenarios. Dimmers must be capable of controlling ~~[two-wire][~~three-wire~~][0-10 volt] fluorescent~~ ballasts or LED drivers. Ensure compatibility of dimmer with separate power packs when utilized for lighting control. Dimmers and the ballasts or drivers they control, must be provided from the same manufacturer, or tested and certified as compatible for use together.

2.5.3 Sensors for Lighting Control

UL 94, JIS C 5402-20-2 and JIS C 8201-5-1.

2.5.3.1 Occupancy Sensors

Provide occupancy sensors with coverage patterns as indicated on project plans. Provide no less quantity of sensors as shown on plans, but add additional sensors when required to fulfill coverage requirement for the specific model sensor provided. Sensor must be provided with an adaptive learning function that automatically sets sensor in optimum calibration in a set period of time after installation and a non-volatile memory that saves settings after a power outage. Provide sensors designed for ceiling, wall or wall-box installation as indicated. Operating voltage must be ~~[105][~~120~~][210][240][~~-277~~]~~ volts. ~~[~~ Operating voltage must be 24V in conjunction with a control system or separate power pack which interacts with luminaire being controlled.~~]~~ Provide housing of high-impact, injection-molded thermoplastic with a multi-segmented lens for PIR and dual technology sensors. Sensor operation requires movement to activate luminaires controlled, and turns luminaires off after a set time of inactivity.~~[~~ Provide integral photocell mounted in occupancy sensor housing when indicated.~~]~~

2.5.3.1.1 Passive Infrared (PIR) Sensors

Provide ceiling or wall-mounted PIR sensors meeting the following requirements:

a. Temperature compensated, dual element sensor and a multi-element fresnel lens (Poly IR4 material).

b. Technology to optimize automatic time delay to fit occupant usage patterns.

c. No minimum load requirement for line voltage sensors and be capable of switching from zero to 800 W at ~~[105][~~120~~]~~ VAC, 50~~/60~~ Hz and from zero to 1200 W at ~~[210][240][~~277~~]~~ VAC, 50/60 Hz. Control voltage sensors must not exceed a maximum load requirement of 20 mA at 24VDC.

d. Time delay of five to 30 minutes in increments of five minutes with a walk through and test mode set by DIP switch.

e. LED indicator that remains active during occupancy.

f. Built-in light level sensor that is operational from 0.8 to 18 lux.

g. Coverage pattern tested to applicable standards.

h. Standard five year warranty.

i. No leakage current to load when in the off mode.

2.5.3.1.2 Ultrasonic Sensors

Provide ceiling-mounted ultrasonic sensors meeting the following requirements:

a. Operate at an ultrasonic frequency~~ of [_____]kHz~~.

b. LED on exterior of device to indicate occupant detection.

c. Adjustable time delay period of 15 seconds to 15 minutes~~ [_____]~~.

d. Minimum five year warranty.

~~[~~ e. Provide with isolated relay for integrating control of HVAC or other automated systems.

~~]~~2.5.3.1.3 Dual Technology Sensors

Provide dual technology sensors that meet the requirements for PIR sensors and ultrasonic sensors indicated above. If either the passive infrared or ultrasonic sensing registers occupancy, the luminaires must remain on.

~~2.5.3.1.4 High/Low-Bay Sensors~~

~~Provide occupancy sensors specifically designed for high/low-bay mounting application using passive infrared (PIR) technology, with the following characteristics:~~

~~a. Input voltage of [120][240][120/277][105][210] volts, at [50][60] hertz.~~

~~b. High-impact, injection-molded thermoplastic housing with interchangeable lenses for 360 degree open area coverage or narrow rectangular, warehouse aisle coverage.~~

~~c. Utilize zero-crossing circuitry to prevent damage from high inrush current and to promote long life operation.~~

~~d. Be designed to mount directly to or adjacent to high or low-bay luminaires.~~

2.5.3.1.4 Power Packs for Sensors

UL 2043 Power packs used to provide power to one or more lighting control sensors must meet the following requirements:

a. Input voltage - ~~[105][210][120][240][~~120-277~~]~~ VAC; output voltage - 24 VDC at 225 mA.

b. Plenum-rated, high-impact thermoplastic enclosure.

c. Utilizes zero-crossing circuitry to prevent damage from inrush current.

~~d. Maximum load rating of [_____] amps for electronic[_____] lighting loads.~~

e. RoHS compliant.

2.5.3.2 Vacancy Sensors

Provide vacancy sensors as indicated above under paragraph OCCUPANCY SENSORS, but with requirement of a manual operation to activate luminaires controlled. Provide automatic operation to turn luminaires off after a set period of inactivity.

~~2.5.4 [Lighting Contactor~~

~~JIS C 8201-4-1. Provide an electrically[mechanically]-held lighting contactor housed in a [indoor][weatherproof][ ] rated IP enclosure conforming to JIS C 8462-1. Provide contactor with one[_____] normally-open(NO)[normally closed(NC)], single[double] pole contacts, rated 600 volts, 30 amps. Provide coil operating voltage of [24][120][277][480][105][210][240][400][_____] volts.]~~

~~[~~2.5.4 Timeswitch

JIS C 9730-2-7. ~~[~~Provide electromechanical type timeswitch with a ~~[24 hour][7 day][~~astronomic~~]~~ dial ~~[~~that changes on/off settings according to seasonal variations of sunset and sunrise~~]~~. Provide power to switch from integral synchronous motor with a maximum three watt rating. Rate contacts at 40 amps at ~~[105][210][120][240][~~120-277~~]~~ volts for general purpose loads. Provide contacts in a SPST~~[DPST][SPDT]~~, ~~[~~normally-open (NO) ~~][normally-closed (NC)]~~ configuration. ~~[~~Provide switch with an automatic spring mechanism to maintain accurate time for up to 16 hours~~.] [~~ Provide switch with function that allows automatic control to be skipped on certain selected days of the week.~~][~~ Provide switch with manual bypass~~[~~ remote override~~]~~ control function.~~]]~~

~~[~~Provide electronic type timeswitch with a ~~[24 hour][7 day][~~astronomic~~]~~ programming function ~~[~~that changes on/off settings according to seasonal variations of sunset and sunrise~~]~~, providing a total of 56~~[_____]~~ on/off set points. Provide ~~[12 hour AM/PM][~~24 hour~~]~~ type digital clock display format. Provide power outage back-up for switch for a minimum of ~~[~~seven ~~][_____]~~ days. Provide switch capable of controlling a minimum of ~~[1][~~2 ~~][4][_____]~~ channels or loads. Rate contacts at ~~[~~30~~][_____]~~ amps at ~~[105][210][120][240][~~120/277~~]~~ volts for general purpose loads. Provide contacts in a SPST~~[DPST][SPDT]~~, ~~[~~normally-open (NO)~~][normally-closed (NC)]~~ configuration. ~~[~~Provide switch with ~~[~~function that allows automatic control to be skipped on certain selected days of the week~~][~~ manual bypass or remote override control~~][~~ daylight savings time adjustment~~][~~ additional memory module~~][~~ momentary function for output contacts~~][~~ ability for photosensor input~~]~~.~~]]~~

House timeswitch in a surface-mounted, lockable, ~~[indoor][~~weatherproof~~][ ]~~ rated IP enclosure constructed of painted steel or plastic polymer conforming to JIS C 8462-1.

~~]~~2.5.5 Lighting Control Panel

Provide an electronic, programmable lighting control panel, capable of providing lighting control with input from internal programming, digital switches, time clocks, and other low-voltage control devices.

Enclose panel hardware in a ~~surface[~~flush~~]~~-mounted, ~~[~~indoor~~][weatherproof]~~ rated IP enclosure in accordance with JIS C 0920, painted, steel enclosure, with hinged, lockable access door and ventilation openings. Internal low-voltage compartment must be separated from line-voltage compartment of enclosure with only low-voltage compartment accessible upon opening of door.~~[~~ Provide additional remote cabinets that communicate back to main control panel.~~]~~

Input voltage - ~~[105][210][~~120/277~~]~~ V, ~~[~~50~~][60]~~ Hz, with internal 24 VDC power supply.

Provide ~~8[16][~~32~~][_____]~~ single-pole latching~~[~~ return to close~~]~~ relays rated at ~~[20][~~30~~]~~ amps, ~~[~~120~~][277][105][210]~~ volts.~~[~~ Provide provision for relays to close upon power failure that meets UL 924 and JIS C 8105-2-22.~~]~~

Relay control module must operate at 24 VDC and be rated to control a minimum of ~~8[16][~~32~~][_____]~~ relays.

2.5.6 Local Area Lighting Controller

Provide controller designed for single area or room with the following requirements:

a. ~~[105][~~120~~][210][240][277]~~ volt input, designed for ~~fluorescent or~~ LED lighting loads.

b. 2~~[_____]~~ zone, with 1~~[2][_____]~~ relay~~[s]~~ rated 20 amps~~[each]~~.

c. Provide daylight harvesting capability with full-range dimming control.

d. Inputs for occupancy sensor, photocell, and low-voltage wall switch.

~~[~~ e. Provide capability for receptacle load control.

~~][~~f. Provide full 'OFF' function with input from external time clock input.

~~]~~2.6 EXIT AND EMERGENCY LIGHTING EQUIPMENT

UL 924, JIS C 8105-2-22 and NFPA 101 compliant.

2.6.1 Exit Signs

Provide exit signs consuming a maximum of five watts total.

2.6.1.1 LED Self-Powered Exit Signs

Provide in ~~[~~UV-stable, thermo-plastic~~][~~ painted, die-cast aluminum ~~][painted steel]~~ housing with ~~[UL damp label][~~UL wet label~~][~~ using clear polycarbonate housing~~]~~, configured for ceiling~~[~~ wall~~][~~ end~~]~~ mounting. ~~[~~ Provide edge-lit type with clear acrylic, edge-lit face and aluminum trim having clear aluminum~~[~~ white~~][chrome][brass][_____]~~ finish.~~]~~ Provide 150 mm high, 19 mm stroke red~~[green][_____]~~ lettering on face of sign. Provide chevrons on either side of lettering to indicate direction. Provide single~~[double]~~ face. Equip with automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1 1/2 hours. LEDs must have a minimum rated life of 10 years. ~~[~~Provide self-diagnostic circuitry integral to emergency LED driver.~~]~~

2.6.1.2 LED Remote-Powered Exit Signs

Provide as indicated above for self-powered type, but without battery and charger. Exit sign must contain provision for ~~[105][210][120][240][~~120/277 ~~]~~ VAC or 6-48 VDC input from remote source.

2.6.2 Emergency Lighting Unit (EBU)

Provide in ~~[~~UV-stable, thermo-plastic~~][~~_painted, die-cast aluminum ~~][painted steel]~~ housing with ~~[UL damp label][~~UL wet label~~][~~_IP rated enclosure in accordance with JIS C 0920~~]~~ as indicated. Emergency lighting units must be rated for 12 volts, except units having no remote-mounted lamps and having no more than two unit-mounted light sources may be rated six volts. Equip units with brown-out sensitive circuit to activate battery when input voltage falls to 75 percent of normal. Equip with two ~~[_____]~~ LED, automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free ~~[~~lead-calcium ~~][nickel-cadmium][_____]~~ type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1 1/2 hours. LEDs must have a minimum rated life of 10 years. ~~[~~Provide self-diagnostic circuitry integral to emergency LED driver.~~]~~

2.6.3 LED Emergency Drivers

Provide LED emergency driver with automatic power failure detection, test switch and LED indicator (or combination switch/indicator) located on luminaire exterior, and fully-automatic solid-state charger, battery and inverter integral to a self-contained housing. ~~[~~Provide self-diagnostic function integral to emergency driver.~~]~~ Integral ~~nickel-cadmium[~~ lead-calcium~~][_____]~~ battery is required to supply a minimum of 90 minutes of emergency power at ~~[5][7][~~10~~][_____]~~ watts, ~~[~~10-50~~][_____]~~ VDC~~[~~ compatible with LED forward voltage requirements~~]~~, constant output. Driver must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.~~[~~ Provide central lighting inverter(s) to supply a minimum of 90 minutes of emergency power.~~]~~

~~2.6.4 Fluorescent Emergency Ballasts~~

~~Provide each 'system' with an automatic power failure device, test switch operable from the exterior of the luminaire (or remotely), a pilot light visible from the exterior of the luminaire, and fully automatic solid-state charger, battery, and inverter integral to a self-contained housing. [Provide self-diagnostic function integral to emergency ballast.] Integral [nickel-cadmium][lead-calcium][_____] battery is required to supply a minimum of 90 minutes of emergency power to one[two][_____] light source[s] within luminaire at a minimum of [500][700][1200][_____] lumens output[each]. Provide open-circuit protection and time-delay function to counteract 'end-of-life' circuitry in normal power ballast from interfering with emergency ballast operation. Ballast must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.[Provide central lighting inverter(s) to supply a minimum of 90 minutes of emergency power.]~~

~~[2.6.5 Self-Diagnostic Circuitry for LED and Fluorescent Emergency Drivers/Ballasts~~

~~Provide emergency lighting unit with fully-automatic, integral self-testing/diagnostic electronic circuitry. Circuitry must provide for a one minute diagnostic test every 28 days, and a 30 minute diagnostic test every six months, minimum. Any malfunction of the unit must be indicated by LED(s) visible from the exterior of the luminaire. A manual test switch must also be provided to perform a diagnostic test at any given time.~~

~~]2.6.6 Central Emergency Lighting System~~

~~Provide integrally-housed emergency system rated at [_____] VA/watts, [105][120][210][240][277] volts (input and output), for a minimum period of 90 minutes. Output frequency must be a pure sine wave at [50][60] hertz, with maximum 5 percent total harmonic distortion. Provide system with minimum short circuit rating required for protection against available fault current.~~

~~2.6.6.1 System Operation~~

~~During normal power operation, system charges batteries as needed and allows normal power to pass through to load. Upon loss of normal power, system automatically transfers to emergency mode without interruption of connected loads. Internal batteries provide a minimum of 90 minutes of~~

~~emergency power at this time. Upon normal power being restored, system switches back to normal power mode and fully charges batteries within UL-approved time period.~~

~~2.6.6.2 Battery Charger~~

~~Solid state, monitored, three step float charging type, keeping batteries in a fully charged state. Provide circuitry to prevent deep discharge of batteries in prolonged power outage conditions.~~

~~2.6.6.3 Batteries~~

~~Provide sealed, lead calcium type, designed to operated unattended without maintenance, for a minimum of 10 years.~~

~~2.6.6.4 Enclosure~~

~~Provide system in [indoor][weatherproof][ ] rated IP enclosure in accordance with JIS C 0920 painted steel[aluminum] enclosure with exterior-mounted "push-to-test" button and LED indicator.~~

~~2.6.6.5 Accessories~~

~~Provide [_____] single pole, [_____] ampere output circuit breakers. [Voltmeter and ammmeter for battery[load].]~~

2.7 LUMINAIRE SUPPORT HARDWARE

2.7.1 Wire

JIS G 3547; Galvanized, soft tempered steel, minimum 2.7 mm in diameter, or galvanized, braided steel, minimum 2 mm in diameter.

2.7.2 Wire for Humid Spaces

JIS G 4309; Annealed stainless steel, minimum 2.7 mm in diameter.

Annealed nickel-copper alloy, minimum 2.7 mm in diameter.

2.7.3 Threaded Rods

Threaded steel rods, 4.76 mm diameter, zinc or cadmium coated.

2.7.4 Straps

Galvanized steel, 25 by 4.76 mm, conforming to JIS G 3302, with a light commercial zinc coating or JIS G 3141 with an electrodeposited zinc coating conforming to JIS H 8610.

2.8 POWER HOOK LUMINAIRE HANGERS

JIS C 8105-1. Provide an assembly consisting of through-wired power hook housing, interlocking plug and receptacle, power cord, and luminaire support loop. Power hook housing must be cast aluminum having two 19 mm threaded hubs. Support hook must have safety screw. Fixture support loop must be cast aluminum with provisions for accepting 19 mm threaded stems. Power cord must include 410 mm of 3 conductor 1.2 mm Type SO cord or power cord per JIS C 8286. Assembly must be rated ~~[105][~~120~~]~~ volts or ~~[210][~~277~~]~~ volts, 15 amperes.

2.9 EQUIPMENT IDENTIFICATION

2.9.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.9.2 Labels

Provide labeled luminaires in accordance with JIS C 8105-1 requirements. All luminaires must be clearly marked for operation of specific light sources and ballasts, generators or drivers. Note the following light source characteristics in the format "Use Only _____":

a. Light source diameter code (T-5, T-8), tube configuration (twin, quad, triple), base type, and nominal wattage for fluorescent and compact fluorescent luminaires.

b. Start type (programmed start, instant start) for fluorescent and compact fluorescent luminaires.

c. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.

All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. Ballasts, generators or drivers must have clear markings indicating multi-level outputs and indicate proper terminals for the various outputs.

2.10 FACTORY APPLIED FINISH

Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of JIS C 0920 corrosion-resistance test.

2.11 RECESS- AND FLUSH-MOUNTED LUMINAIRES

Provide access to lamp and ballast from bottom of luminaire. Provide trim ~~[~~and lenses~~]~~ for the exposed surface of flush-mounted luminaires as indicated on project drawings and specifications.

2.12 SUSPENDED LUMINAIRES

Provide hangers capable of supporting twice the combined weight of luminaires supported by hangers. Provide with swivel hangers to ensure a plumb installation. Provide cadmium-plated steel with a swivel-ball tapped for the conduit size indicated. Hangers must allow fixtures to swing within an angle of 0.79 rad. Brace pendants 1219 mm or longer to limit swinging. Single-unit suspended luminaires must have twin-stem hangers. Multiple-unit or continuous row luminaires must have a tubing or stem for wiring at one point and a tubing or rod suspension provided for each unit length of chassis, including one at each end. Provide rods in minimum 4.57 mm diameter.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations must conform to IEEE C2 and JIS C 0365 and to the requirements specified herein. Install luminaires and lighting controls to meet the requirements of applicable codes and standards. To encourage consistency and uniformity, install luminaires of the same manufacture and model number when residing in the same facility or building.

3.1.1 Light Sources

When light sources are not provided as an integral part of the luminaire, deliver light sources of the type, wattage, lumen output, color temperature, color rendering index, and voltage rating indicated to the project site and install just prior to project completion, if not already installed in the luminaires from the factory.

3.1.2 Luminaires

Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Installation must meet requirements of applicable codes and standards. Mounting heights specified or indicated must be to the bottom of the luminaire for ceiling-mounted luminaires and to center of luminaire for wall-mounted luminaires. Obtain approval of the exact mounting height on the job before commencing installation and, where applicable, after coordinating with the type, style, and pattern of the ceiling being installed. Recessed and semi-recessed luminaires must be independently supported from the building structure by a minimum of four wires, straps or rods per luminaire and located near each corner of the luminaire. Ceiling grid clips are not allowed as an alternative to independently supported luminaires. Round luminaires or luminaires smaller in size than the ceiling grid must be independently supported from the building structure by a minimum of four wires, straps or rods per luminaire, spaced approximately equidistant around. Do not support luminaires by acoustical tile ceiling panels. Where luminaires of sizes less than the ceiling grid are indicated to be centered in the acoustical panel, support each independently and provide at least two 19 mm metal channels spanning, and secured to, the ceiling tees for centering and aligning the luminaire. Provide wires, straps, or rods for luminaire support in this section. Luminaires installed in suspended ceilings must also comply with the requirements of Section 09 51 00 ACOUSTICAL CEILINGS.

3.1.3 Suspended Luminaires

Provide suspended luminaires with 0.79 rad swivel hangers so that they hang plumb and level. Locate so that there are no obstructions within the 0.79 rad range in all directions. The stem, canopy and luminaire must be capable of 0.79 rad swing. Pendants, rods, or chains 1.2 meters or longer excluding luminaire must be braced to prevent swaying using three cables at 2.09 rad separation. Suspended luminaires in continuous rows must have internal wireway systems for end to end wiring and must be properly aligned to provide a straight and continuous row without bends, gaps, light leaks or filler pieces. Utilize aligning splines on extruded aluminum luminaires to assure minimal hairline joints. Support steel luminaires to prevent "oil-canning" effects. Luminaire finishes must be free of scratches, nicks, dents, and warps, and must match the color and gloss specified. Match supporting pendants with supported luminaire. Aircraft cable must be stainless steel. Canopies must be finished to match the ceiling and must be low profile unless otherwise shown. Maximum distance between suspension points must be 3.1 meters or as recommended by the manufacturer, whichever is less.

3.1.4 Ballasts, Generators and Power Supplies

Typically, provide ballasts, generators, and power supplies (drivers) integral to luminaire as constructed by the manufacturer.

3.1.5 Exit Signs and Emergency Lighting Units

Wire exit signs and emergency lighting units ahead of the local switch, to the normal lighting circuit located in the same room or area.

~~[3.1.5.1 Exit Signs~~

~~Connect exit signs on separate circuits and serve from [an emergency panel][a separate circuit breaker][a fused disconnect switch]. Provide only one source of control, which would be [the circuit breaker in the emergency panel][the separate circuit breaker][the fused disconnect switch]. Paint source of control red and provide lockout capability.~~

~~][3.1.5.2 Emergency Lighting from Central Emergency System~~

~~Connect emergency lighting from a central emergency system as indicated on the project drawings.~~

~~]~~3.1.6 Photocell Switch Aiming

Aim switch according to manufacturer's recommendations.

3.1.7 Occupancy/Vacancy Sensors

Provide testing of sensor coverage in all spaces where sensors are placed. This should be done only after all furnishings (carpet, furniture, workstations, etc.) have been installed. Provide quantity of sensor units indicated as a minimum. Provide additional units to give full coverage over controlled area. Full coverage must provide hand and arm motion detection for office and administration type areas and walking motion for industrial areas, warehouses, storage rooms and hallways. Locate the sensor(s) as indicated and in accordance with the manufacturer's recommendations to maximize energy savings and to avoid nuisance activation and deactivation due to sudden temperature or airflow changes and usage.

3.1.8 Daylight or Ambient Light Level Sensor

Locate sensor as indicated and in accordance with the manufacturer's recommendations. Adjust sensor for 300 lux or for the indicated light level measured at the work plane for that particular area.

3.2 FIELD APPLIED PAINTING

Paint lighting equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Provide painting as specified in Section 09 90 00 PAINTS AND COATINGS.

-- End of Section --

SECTION 27 05 13.43

TELEVISION DISTRIBUTION SYSTEM
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2	(2023) National Electrical Safety Code
IEEE C62.41.1	(2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits
IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

NATIONAL CABLE AND TELECOMMUNICATIONS ASSOCIATION (NCTA)

NCTA RP	(1989) NCTA Recommended Practices for Measurements on Cable Television Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 76.605	Technical Standards

UL SOLUTIONS (UL)

UL 969	(2017; Reprint May 2023) UL Standard for Safety Marking and Labeling Systems
UL 1581	(2019; Reprint Nov 2023) UL Standard for Safety Reference Standard for Electrical Wires, Cables, and Flexible Cords

1.2 DEFINITIONS

1.2.1 Television Distribution System

The television distribution system system, commonly referred to as cable television, is a network of cables, headend, electronic and passive components that process and amplify television (TV) signals for distribution of adequate signals to each receiver from the headend equipment to the individual television outlets and provides distortion-free signal to TV sets by isolating each receiver from the system an providing the proper amount of signal to each set.

1.2.2 Headend

The connection point between television distribution system equipment and equipment provided by the local television service provider.

1.2.3 Distribution System

Distribution system transports and delivers adequate signals to each receiver. Provides distortion-free signal to TV sets by isolating each receiver from the system and by providing proper amount of signal to each set.

1.2.4 Cable

~~[~~Trunk and feeder cables are low-loss cables used to transport the desired signal from the headend equipment to the telecommunications room in the area to be served. These cables are used to transport signal from the ~~[~~ telecommunications room~~][~~ headend equipment~~]~~ into close proximity to a number of user locations in excess of 60 meters from the ~~[~~ telecommunications room~~][headend equipment]~~. ~~]~~Drop cables are used to transport the desired signal used from the ~~[~~telecommunications closet ~~room][headend equipment]~~ to the wall outlet.

1.3 SYSTEM DESCRIPTION

1.3.1 Headend

Contractor must provide interior equipment up to headend ~~[~~and including the main amplifier~~]~~ located at the interior television distribution system ~~[backboard][~~cabinet~~][rack]~~.

1.3.2 Distribution System

~~[~~ ~~[~~Distribution system must be star topology with each outlet connected to a telecommunications room with a feeder cable or a drop cable and each telecommunications room connected to the headend equipment with a ~~[~~coaxial ~~][fiber optic]~~ trunk cable~~][~~. Distribution system must be star~~ topology with each outlet connected to headend equipment with the drop cable]~~.

~~][Distribution system must be a star topology utilizing a coaxial trunk cable and category [6] UTP drop cables with the exception of low voltage patient television outlets. Low voltage patient television outlets must be connected with coaxial drop cables.~~

~~]~~1.3.3 Cable

~~[~~Provide trunk cables to transport the desired signal from the headend equipment to the telecommunications room in the area to be served. ~~][~~ Provide ~~[trunk] [~~feeder~~]~~ cables to transport signal from the ~~[headend equipment][~~telecommunications room~~]~~ to user locations for cable lengths in excess of 60 meters from the ~~[~~headend equipment~~][telecommunications room]~~. ~~]~~Provide drop cables to transport the desired signal from the ~~[~~ telecommunications ~~room][~~headend equipment~~]~~ to the outlet.

1.3.4 System Components

System must provide high quality TV signals to all outlets~~[with a return~~

~~path for interactive television and cable modem access]~~. Provide any combination of items specified herein to achieve required performance, subject to approvals, limitations, acceptance test, and other requirements specified herein. System must include amplifiers, splitters, combiners, line taps, cables, outlets, tilt compensators and all other parts, components, and equipment necessary to provide a complete and usable system.

1.3.4.1 System Bandwidth

a. Downstream: 50-1000 MHz minimum.

b. Upstream 5-40 MHz minimum.

1.3.5 System Performance

System must be in compliance with 47 CFR 76.605.

1.3.5.1 Receiver Termination Signal Level

Each termination for a TV receiver must have a minimum signal level of 0 decibel millivolts (dBmV) (1000 microvolts) at 55 MHz and of 0 dBmV (1000 microvolts) at 1000 MHz and a maximum signal of 15 dBmV or a level not to overload the receiver for the entire system bandwidth.

1.3.5.2 Distribution System

a. Modulation distortion at power frequencies: 4 percent or less hum distortion;

b. Composite third order distortion for:

    (1) CW carriers: 53 dB.

    (2) Modulated carriers: 59 dB.

c. Subscriber terminal isolation: 18 dB or greater.

d. Carrier to second order beat ratio: 60 dB.

e. Amplitude characteristic:within a range of plus or minus 2 decibels from 0.75 MHz to 5.0 MHz above the lower boundary frequency of the cable television channel, referenced to the average of the highest and lowest amplitudes within these frequency boundaries.

f. Visual, aural carrier level, 24-hour variation: 47 CFR 76.605, subpart (a), rules (4), (5), and (6).

g. Frequency determination: 47 CFR 76.605, subpart (a), rules (1),(2), and (3).

1.3.5.3 System Tolerance

The system must not show a serious loss of carrier to noise when the system levels are lowered 3 dB below normal or a significant distortion when the levels are increased 3 dB above normal, as observed on a TV set located at the far end extremities of the system.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Television Distribution System Wiring Diagrams And Installation Details; G~~, [_____]~~

Television Distribution System Components; G~~, [_____]~~

SD-03 Product Data

Attenuators; G~~, [_____]~~

~~[~~ Amplifiers, including ~~[~~Headend, ~~]~~Trunk, Bridging, and Distribution; G~~, [_____]~~

~~]~~ Cables, including ~~[~~Trunk, Feeder, and ~~]~~Drop; G~~, [_____]~~

Terminators; G~~, [_____]~~

Splitters/Combiners; G~~, [_____]~~

~~[~~ Fiber Optic Couplers; G~~, [_____]~~

~~]~~ Line Taps; G~~, [_____]~~

Coaxial Outlets; G~~, [_____]~~

Coaxial Connectors; G~~, [_____]~~

Tilt Compensator; G~~, [_____]~~

~~[~~ Distribution Hubs; G~~, [_____]~~

~~][~~ Baluns; G~~, [_____]~~

~~][~~ Grounding Block; G~~, [_____]~~

~~]~~ Submittals for each manufactured item must be the current manufacturer's descriptive literature of catalog products, equipment drawings, diagrams, performance and characteristics curves, and catalog cuts.

SD-05 Design Data

Television Distribution System Loss Calculations; G~~, [_____]~~

SD-06 Test Reports

Operational Test Plan; G~~, [_____]~~

Operational Test Procedures; G~~, [_____]~~

System Pretest; G~~, [_____]~~

Acceptance Tests; G~~, [_____]~~

SD-08 Manufacturer's Instructions

Connector Installation; G~~, [_____]~~

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G~~, [_____]~~

Submit Data Package 5 for each component in accordance with requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.5 QUALITY ASSURANCE

1.5.1 Wiring Diagrams and Installation Details

Illustrate how each item of equipment functions in the system and include an overall system schematic indicating the relationship of television distribution system units on one diagram. Drawings must include wiring diagrams and installation details of equipment indicating proposed locations, layout and arrangements, and other items that must be shown to ensure coordinated installation.

1.5.2 Television Distribution System Loss Calculations

Calculations must verify that the system does not exceed the loss values specified in dBmV at the ~~[receiver terminations][~~input of all active devices and the receiver terminations~~]~~. Provide a drawing displaying all distribution network calculations. The drawing should accurately show taps, splitters, outlets, and the type and length of all ~~[~~trunk, feeder, and ~~]~~drop cables. The drawing must show how many taps, splitters, or outlets are served by each tap or splitter.

1.5.3 Operational Test Plan

Test plan must define tests required to ensure that the system meets technical, operational, and performance specifications. Test plan must be based on NCTA RP and be in accordance with FCC proof of performance requirements. Test plan must include plan for testing for signal leakage. Provide test requirements and guidelines.

1.5.4 Operational Test Procedures

Use test plan and design documents to develop test procedures. Procedures must consist of detailed instructions for a test setup, execution, and evaluation of test results.

1.5.5 Connector Installation

Provide manufacturer's instructions for installing connectors.

PART 2 PRODUCTS

2.1 ELECTRONIC EQUIPMENT

Electronic components of similar type must be produced and designed by the same manufacturer as major components of the equipment and must have the manufacturer's name and model permanently attached. Equipment must function properly as a complete integrated system. Equipment must be shielded. The system must be designed to operate within 5 to 1000 MHz bandwidth using 1000 MHz passive devices and a minimum of 1000 MHz active devices.

2.2 HEADEND EQUIPMENT

2.2.1 Headend Amplifiers

Provide broadband distribution amplifiers. Amplifiers must amplify broadband signals from 40 to 1000 MHz~~[~~ and provide an amplified return path for signals from 5 to 40 MHz for 75 ohms impedance~~]~~. Amplifiers must be bidirectional with variable slope and gain control.

2.2.2 Attenuators

Provide attenuators to equalize signal levels, when required. Variable attenuators are not permitted.

2.2.3 Power Supplies

Power supplies must contain a current limiter circuit to protect against short circuits on the radio frequency (RF) line. Provide overvoltage protection to protect solid state equipment from line surges and induced voltages, in accordance with IEEE C62.41.1 and IEEE C62.41.2.

2.3 DISTRIBUTION EQUIPMENT

~~[~~2.3.1 Distribution Amplifiers

Distribution amplifiers must be equipped for 75 ohms input and output impedance. Electronic equipment exposed to weather must be equipped with weatherproof housings. Amplifiers must be bidirectional with variable slope and gain control and must amplify broadband signals from 50 to 1000 MHz~~[and provide an amplified return path for signals from 5 to 40 MHz for 75 ohms impedance]~~.

2.3.1.1 Trunk Amplifiers

Trunk amplifiers must have automatic level and slope control features.

2.3.1.2 Bridging Amplifiers

Bridging amplifiers must be used to connect feeder cables to trunk cables.

~~[~~2.3.1.3 Fiber Optic Amplifiers

Provide fiber optic amplifiers in fiber optic backbone distribution systems as required to provide required optical power to fiber optic receivers.

~~][~~2.3.1.4 Fiber Optic Receiver Distribution Amplifier

Provide fiber optic receiver with integrated broadband distribution amplifier to receive the fiber optic video signal and distribute it to the coaxial cable sub-distribution system.

~~]]~~2.3.2 Cables and Associated Hardware

Cabling must be UL listed for the application and must comply with NFPA 70. Provide a labeling system for cabling as required by UL 969. Cabling manufactured more than 12 months prior to date of installation must not be used.

~~2.3.2.1 Trunk Cable~~

~~[UL 1666. Provide trunk cable with an NFPA 70 rating of CATVR.~~

~~a. Provide RG-11 coaxial cable with the following characteristics:~~

~~(1) 14 AWG copper-clad steel center conductor.~~

~~(2) Gas injected foam polyethylene dielectric with nominal 7.11 mm outer diameter.~~

~~(3) Bonded foil inner-shield and 60 percent aluminum braid[or quad shield].~~

~~(4) 75 ohms impedance.~~

~~(5) 82 to 85 percent nominal velocity of propagation.~~

~~(6) Black PVC jacket~~

~~(7) Maximum attenuation characteristics:~~

~~MHz~~	~~DB/100 m~~
~~5~~	~~1.25~~
~~55~~	~~3.15~~
~~300~~	~~7.38~~
~~350~~	~~7.94~~
~~450~~	~~9.02~~
~~500~~	~~9.51~~
~~600~~	~~10.43~~
~~750~~	~~11.97~~
~~1000~~	~~14.27~~

~~b. Provide 625 Series cable with an NFPA 70 rating of CATVR and the following characteristics:~~

~~(1) Copper-clad aluminum center conductor~~

~~(2) Seamless aluminum tubing shield~~

~~(3) Expanded polyethylene dielectric~~

~~(4) 75 ohms impedance~~

~~(5) Nominal diameter over outer conductor: 15.88 mm.~~

~~(6) Maximum attenuation at 20 degrees C and 1000 MHz: 6.79 dB/100m~~

~~(7) Black medium density polyethylene jacket~~

~~(8) Nominal 87 percent velocity of propagation~~

~~][Provide fiber optic trunk cable as indicated and in accordance with ICEA S-83-596, ANSI/TIA-568.3, UL 1666 and NFPA 70. Provide the number of strands indicated, (but not less than 6 strands between the main telecommunication room and each of the other telecommunication rooms), of single-mode, tight buffered fiber optic cable as recommended by the television distribution system manufacturer.~~

~~][~~2.3.2.1 Feeder Cable

UL 1581, provide RG-11 coaxial trunk cable with an NFPA 70 rating of ~~[~~CATV ~~][CATVP]~~ and the following characteristics:

a. 14 AWG copper-clad steel center conductor.

b. ~~[~~Gas injected foam polyethylene~~][Foam FEP]~~ dielectric with 7.11 mm nominal outer diameter.

c. Bonded foil inner-shield and a minimum of 60 percent aluminum braid~~[or quad shield]~~.

d. 75 ohms impedance.

e. 81 to 84 percent nominal velocity of propagation.

f. ~~[~~Black PVC~~][PVC low smoke polymer or FEP]~~ jacket.

g. Maximum attenuation characteristics:

~~[~~

CATV	
MHz	DB/100 m
50	3.1
100	4.2
200	5.7
400	8.85

CATV	
MHz	DB/100 m
700	11.0
1000	14.26

][

CATVP	
MHz	DB/100 m
50	3.9
100	5.6
200	8.2
400	11.5
700	115.1
900	117.4
1000	18.4

]][2.3.2.2 Drop Cable

UL 1581. Provide RG 6 coaxial cable with an NFPA 70 rating of [CATV][CATVP] and with the following characteristics:

a. No. 18 AWG copper-clad steel center conductor.

b. Bonded foil inner-shield and 90 percent aluminum braid.

c. Characteristic impedance of 75 ohms.

d. [Gas injected foam polyethylene][Foam FEP] dielectric

e. Nominal capacitance, conductor to shield, of 53 pf per 100 m.

f. Maximum operating voltage of 350 V RMS.

g. Maximum attenuation:

[

CATV	
MHz	DB/100 m
10	2.59
50	5.08

CATV	
MHz	DB/100 m
100	7.19
200	10.17
400	14.38
500	15.48
700	19.02
1000	22.74

~~][~~

~~CATVP~~	
~~MHz~~	~~DB/100 m~~
~~10~~	~~2.3~~
~~50~~	~~4.9~~
~~100~~	~~6.9~~
~~200~~	~~10.2~~
~~400~~	~~14.8~~
~~500~~	~~19.7~~
~~700~~	~~22.6~~
~~1000~~	~~23.9~~

~~] h. [Black polyvinyl chloride (PVC)][PVC low smoke polymer or FEP] jacket.~~

~~i. 100 percent sweep testing from 5 MHz to a minimum of 1000 MHz.~~

~~][2.3.2.3 Low Voltage Patient Television Drop Cable~~

~~UL 1581. Provide RG 6 coaxial cable with an NFPA 70 rating of [CATV][CATVP] and with the following characteristics:~~

~~a. No. 18 AWG solid copper center conductor.~~

~~b. Aluminum/Polyesetr/Aluminum tape 100 percent coverage inner-shield and tinned copper outer braid with 60 percent coverage.~~

~~c. Characteristic impedance of 75 ohms.~~

~~d. [Gas injected foam polyethylene][Foam FEP] dielectric~~

~~e. Nominal capacitance, conductor to shield, of 53 pf per m.~~

~~f. Maximum operating voltage of 300 V RMS.~~

~~g. Maximum attenuation:~~

~~[~~

~~CATV~~	
~~MHz~~	~~DB/100 m~~
~~10~~	~~2.3~~
~~50~~	~~4.9~~
~~100~~	~~6.58~~
~~200~~	~~9.2~~
~~400~~	~~13.1~~
~~700~~	~~17.43~~
~~900~~	~~20.07~~
~~1000~~	~~21.3~~
~~1500~~	~~27.3~~

~~][~~

~~CATVP~~	
~~MHz~~	~~DB/100 m~~
~~10~~	~~2.3~~
~~50~~	~~4.9~~
~~100~~	~~6.58~~
~~200~~	~~9.2~~
~~400~~	~~13.1~~
~~700~~	~~17.43~~
~~1000~~	~~21.3~~
~~1500~~	~~27.3~~

~~] h. [Black polyvinyl chloride (PVC)][PVC low smoke polymer or FEP] jacket.~~

~~i. 100 percent sweep testing from 5 MHz to a minimum of 1000 MHz.~~

~~]~~2.3.3 Category 6 Drop Cable

Provide as indicated on the drawings and 27 10 00 BUILDING

TELECOMMUNICATIONS CABLING SYSTEM.

2.3.4 Category 6 Patch Cords

Provide patch cords, as complete assemblies, with matching connectors as specified in 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM. Patch cords must meet minimum performance requirements specified in 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM for cables, cable length and hardware specified.

2.3.5 Terminators

Coaxial terminators must be rated for 75 ohms and 1/4 watt.~~[~~ Category 6 port terminators shawl be plug-in hub type and must have integral 8-pin modular connector.~~]~~

2.3.6 Splitters/Combiners

Use splitters/combiners with characteristics equal to or exceeding the characteristics listed in this paragraph over the entire operating band. All unused outlets must be terminated with 75-ohm terminators.

a. Peak to Valley: Not to exceed 1 dB across bandwidth of device.

b. Return loss: 18 dB minimum.

c. Bandwidth: 5-1000 Mh

~~[~~2.3.7 Fiber Optic Couplers

Use fiber optic couplers to split fiber optic light sources equal to or exceeding the characteristics listed in this paragraph.

a. Number of inputs: 1

b. Wavelength: 1310 and 1550 nm

c. Number of Outputs: ~~[2 with insertion loss of 3.3 dB, uniformity of greater than 0.7 dB, and directivitiy of greater than 50 dB][~~4 with insertion loss of 6.3 dB, uniformity of greater than 1.4 dB, and directivitiy of greater than 50 dB~~][8 with insertion loss of 9.5 dB, uniformity of greater than 2.1 dB, and directivitiy of greater than 50 dB][16 with insertion loss of 12.6 dB, uniformity of greater than 2.8 dB, and directivitiy of greater than 50 dB]~~.

~~]~~2.3.8 Line Taps

Line taps must have 18 dB minimum isolation from each tap to the thru-line. Pressure tapoffs are not permitted. Taps must be rated from 5 to 1000 MHz and must have a peak to valley not to exceed 1 dB to 1 GHz.

2.3.9 Coaxial Outlets

Provide flush mounted, 75-ohm, F-type connector outlets rated from 5 to 1000 MHz in standard electrical outlet boxes~~[~~ with isolation barrier~~]~~.

2.3.10 Coaxial Connectors

Provide one piece connectors. ~~[Trunk and feeder~~Feeder cable connectors must be pin type.]Drop cable connectors must be feed thru type.

2.3.11 Tilt Compensator

Provide tilt compensators as required.

[2.3.12 Distribution Hubs

Rack-mountable passive distribution hubs with 16 front mounted modular 8-pin jacks for connection to UTP 4-pair category 6 cabling, and “F” fitting on rear panel for connection to coaxial trunk cable. Hubs must be tested compliant with FCC Part 15 requirements. Bandwidth: 5 – 860 MHz. Hubs must be supplied with quantities of plug-in terminators to terminate all unused ports. Provide quantities of hubs as indicated.

][2.3.13 Baluns

Provide single-port converters (for use at TV outlet locations). Converter must convert balanced line signal to coaxial unbalanced signal for connection to coaxial inputs of television receiver/monitors. Converter must be equipped with integral 8-pin modular jack on one end and male “F” fitting on the opposite end. Provide quantities as indicated

]2.4 GROUNDING AND BONDING

Provide ground rods and connections in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

[2.4.1 Grounding Block

Provide [corrosion-resistant] grounding block suitable for ~~[outdoor]~~[indoor] installation.

][2.5 BACKBOARDS

Provide void-free, fire rated interior grade plywood, 19 mm thick,[1200 by 2400 mm]~~[as indicated]~~. Backboards must be painted with a gray, nonconductive fire-resistant overcoat. Do not cover the fire stamp on the backboard.

]PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Distribution System

Distribution system must conform to requirements specified herein. Installation must be in accordance with IEEE C2 and NFPA 70.

3.1.1.1 Raceway

Provide cable installed in raceways such as conduit and cable trays in compliance with NFPA 70. Raceway must comply with Section 26 20 00, INTERIOR DISTRIBUTION SYSTEM.[Provide 78 mm, minimum, PVC from interior headend location to exterior television service provider connection location. Coordinate location and requirements with the local cable television service provider.]

~~[~~3.1.1.2 Grounding System

Provide the grounding block ~~[~~at the main television distribution backboard ~~][_____]~~. Ground this device according to the requirements of IEEE C2 and NFPA 70.

~~]~~3.1.1.3 Trunk, Feeder, and Drop Cable

Provide cable to grounding blocks, to line taps, and to outlets.

3.1.1.4 Splitters, Directional Couplers, Attenuators, Amplifiers

Install in accordance with manufacturer's written instructions.

3.2 FIELD QUALITY CONTROL

3.2.1 System Pretest

Upon completing installation of the television distribution system, the Contractor ~~[~~must align and balance the system and ~~]~~must perform complete pretesting. During the system pretest, Contractor, utilizing the approved spectrum analyzer or signal level meter, must verify that the system is fully operational and meets all the system performance requirements of the specification. Contractor must test the signal loss in dBmV at 55~~[~~, 151, 547,~~]~~ and 1000 MHz. The signal levels must be 0 dBmV (1000 microvolts), minimum. The signal must not exceed 15 dBmV over the entire system bandwidth. Any deficiencies found must be corrected and revalidated by follow up testing. Contractor must measure and record the video and audio carrier levels at each of the frequency levels specified at each of the following points in the system:

a. Furthest outlet from~~[each communications room][~~ service entrance point of connection~~]~~.

b. A random sampling of 25 percent of the~~[~~ outlets~~[~~ from each ~~communications room]][~~ housing units~~]~~.

~~[~~ c. At each outlet.

~~][~~d. ~~[Headend and]~~Distribution amplifier inputs and outputs.

~~]~~3.2.2 Acceptance Tests

Contractor must notify the Contracting Officer of system readiness 10 days prior to the date of acceptance testing. Contractor must also coordinate with the local television service provider and allow them to attend witness tests. The television distribution system must be tested in accordance with the approved test plan in the presence of the Contracting Officer's representative to certify acceptable performance. System test must verify that the total system meets all the requirements of the specification and complies with the specified standards. Contractor must verify that no signal leakage exists in conformance with NCTA RP and 47 CFR 76.605. System leakage must also be tested at the headend location with signal applied to system. Deficiencies revealed by the testing must be corrected on the ~~[housing units][~~outlets~~]~~ sampled as well as on the ~~[units][~~outlets~~]~~ not sampled and revalidated by follow-up testing. Contractor must conduct testing at each of the following points in the system:

a. Furthest outlet from ~~[~~each communications room~~][service entrance point of connection]~~.

b. A random sampling of 25 percent of the ~~[~~outlets~~[~~ from each telecommunications room~~]][housing units]~~ as designated by the Contracting Officer.

~~[c. At each outlet.~~

~~][~~c. ~~[~~Headend and ~~]~~Distribution amplifier inputs and outputs.

~~]~~3.3 OPERATION AND MAINTENANCE MANUALS

Submit commercial, off-the-shelf manuals for operation, installation, configuration, and maintenance of products provided as a part of the cable television premises distribution system.

-- End of Section --

SECTION 27 10 00

BUILDING TELECOMMUNICATIONS CABLING SYSTEM
08/11

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. Contractor may substitute compatible Telecommunications Industry Association (TIA) or UL standards, as approved by the Contracting Officer's representative.

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 5507	(2010) LAN Twisted Pair Cable

~~JAPANESE STANDARDS ASSOCIATION (JSA)~~JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS C 3665-1-1	(2007) Tests on electric and optical fibre cables under fire conditions -- Part 1-1: Test for vertical flame propagation for a single insulated wire or cable -- Apparatus
JIS C 5964-4	(2014) Fiber optic connector interfaces -- Part 4: Type SC connector family (F04 Type)
JIS C 5964-4-100	(2018) Fiber optic connector interfaces -- Part 4-100: Type SC connector family -- Simplified receptacle SC-PC connector interfaces (F16 type)
JIS C 5964-18	(2014) Fiber optic connector interfaces -- Part 18: Type MT-RJ connector family (F19 type)
JIS C 5964-20	(2015) Fiber optic connector interfaces -- Part 20: Type LC connector family
JIS C 6011-1	(2015) Mechanical structures for electronic equipment -- Tests for IEC 60917 and IEC 60297 series -- Part 1: Environmental requirements, test set-up and safety aspects for cabinets, racks, subracks and chassis under indoor conditions
JIS C 6012-3-100	(2015) Mechanical structures for electronic equipment -- Dimensions of mechanical structures of the 482.6 mm (19 in) series -- Part 3-100: Basic dimensions of front panels, subracks, chassis, racks and cabinets

JIS C 6185-2	(2014) Optical time-domain reflectometers (OTDR) -- Part 2: Calibration of OTDR for single mode fibers
JIS C 6185-3	(2014) Optical time-domain reflectometers (OTDR) -- Part 3: Calibration of OTDR for multimode fibers
JIS C 6820	~~(2018) General rules of optical fibers~~ (2023) General Rules of Optical Fibers
JIS C 6850	(2006) General rules of optical fiber cables
JIS C 6870-2	(2021) Indoor optical fiber cables -- Part 2: Sectional specification
JIS C 8435	(2022) Boxes And Box Covers Of Plastic Conduits
JIS C 60364-4-42	(2022) Electrical installations of buildings -- Part 4-42: Protection for safety -- Protection against thermal effects
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor
JIS C 61300-2-2	(2011) Fiber optic interconnecting devices and passive components -- Basic test and measurement procedures -- Part 2-2: Tests -- Mating durability
JIS K 6911	(2006; R 2021) Thermosetting plastic general test method
JIS X 5150	(R2016) Information Technology-Generic Cabling for Customer Premises

UNDERWRITERS LABORATORIES (UL)

UL 1666	(2007; Reprint Jun 2012) Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts
UL 723	(2018; Reprint Jun 2025) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and Section 33 82 00 TELECOMMUNICATIONS, OUTSIDE PLANT (OSP), apply to this section with additions and modifications specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification shall be as defined in JIS X 5150, JCS 5507, JIS C 6820, JIS C 6850, and herein.

~~1.3.1 Campus Distributor (CD)~~

~~A distributor from which the campus backbone cabling emanates. (International expression for main cross-connect (MC).)~~

1.3.1 Building Distributor (BD)

A distributor in which the building backbone cables terminate and at which connections to the campus backbone cables may be made. (International expression for intermediate cross-connect (IC).)

1.3.2 Floor Distributor (FD)

A distributor used to connect horizontal cable and cabling subsystems or equipment. (International expression for horizontal cross-connect (HC).)

1.3.3 Telecommunications Room (TR)

An enclosed space for housing telecommunications equipment, cable, terminations, and cross-connects. The room is the recognized cross-connect between the backbone cable and the horizontal cabling.

1.3.4 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service cables (including wireless) including the entrance point at the building wall and continuing to the equipment room.

1.3.5 Equipment Room (ER) (Telecommunications)

An environmentally controlled centralized space for telecommunications equipment that serves the occupants of a building. Equipment housed therein is considered distinct from a telecommunications room because of the nature of its complexity.

1.3.6 Open Cable

Cabling that is not run in a raceway as defined by applicable codes and standards. This refers to cabling that is "open" to the space in which the cable has been installed and is therefore exposed to the environmental conditions associated with that space.

~~1.3.7 Open Office~~

~~A floor space division provided by furniture, moveable partitions, or other means instead of by building walls.~~

1.3.7 Pathway

A physical infrastructure utilized for the placement and routing of telecommunications cable.

1.4 SYSTEM DESCRIPTION

The building telecommunications cabling and pathway system shall include permanently installed backbone and horizontal cabling, horizontal and backbone pathways, service entrance facilities, work area pathways, telecommunications outlet assemblies, conduit, raceway, and hardware for splicing, terminating, and interconnecting cabling necessary to transport telephone and data (including LAN) between equipment items in a building. The horizontal system shall be wired in a star topology from the telecommunications work area to the floor distributor or campus distributor at the center or hub of the star. The backbone cabling and pathway system includes intrabuilding and interbuilding interconnecting cabling, pathway, and terminal hardware. The intrabuilding backbone provides connectivity from the floor distributors to the building distributors or to the campus distributor and from the building distributors to the campus distributor as required. The backbone system shall be wired in a star topology with the campus distributor at the center or hub of the star.~~[The interbuilding backbone system provides connectivity between the campus distributors and is specified in Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP).]~~ Provide telecommunications pathway systems referenced herein as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.~~[~~ The telecommunications contractor must coordinate with the ~~NMCI/COSC/NGEN~~ contractor concerning access to and configuration of telecommunications spaces. The telecommunications contractor may be required to coordinate work effort within the telecommunications spaces with the ~~NMCI/COSC/NGEN~~ contractor.~~]~~

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[~~for Contractor Quality Control approval.~~][for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications drawings; G~~[, [_____]]~~

Telecommunications Space Drawings; G~~[, [_____]]~~

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

SD-03 Product Data

Telecommunications cabling (backbone and horizontal); G~~[, [_____]]~~

Patch panels; G~~[, [_____]]~~

Telecommunications outlet/connector assemblies; G~~[, [_____]]~~

Equipment support frame; G~~[, [_____]]~~

~~[~~Connector blocks; G~~[, [_____]]]~~

SD-06 Test Reports

Telecommunications cabling testing;

SD-10 Operation and Maintenance Data

Telecommunications cabling and pathway system Data Package 5;

SD-11 Closeout Submittals

Record Documentation~~;~~

1.6 QUALITY ASSURANCE

1.6.1 Shop Drawings

In exception to Section 01 33 00 SUBMITTAL PROCEDURES, submitted plan drawings shall be a minimum of 279 by 432 mm in size using a minimum scale of one mm per 100 mm~~[~~, except as specified otherwise~~]~~. Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

1.6.1.1 Telecommunications Drawings

Provide~~[~~ registered communications distribution designer (RCDD) approved,~~]~~ drawings in accordance with applicable codes and standards. The identifier for each termination and cable shall appear on the drawings. Drawings shall depict final telecommunications installed wiring system infrastructure. The drawings should provide details required to prove that the distribution system shall properly support connectivity from the EF telecommunications and ER telecommunications, CD's~~[~~, BD's~~]~~, and FD's to the telecommunications work area outlets.~~[~~ Provide a plastic laminated schematic of the as-installed telecommunications cable system showing cabling, CD's, BD's, FD's, and the EF and ER for telecommunications keyed to floor plans by room number. Mount the laminated schematic in the EF telecommunications space as directed by the Contracting Officer.~~]~~ The following drawings shall be provided as a minimum:

a. T1 - Layout of complete building per floor - Building Area/Serving Zone Boundaries, Backbone Systems, and Horizontal Pathways. Layout of complete building per floor. The drawing indicates location of building areas, serving zones, vertical backbone diagrams, telecommunications rooms, access points, pathways, grounding system, and other systems that need to be viewed from the complete building perspective.

b. T2 - Serving Zones/Building Area Drawings - Drop Locations and Cable Identification (ID'S). Shows a building area or serving zone. These drawings show drop locations, telecommunications rooms, access points and detail call outs for common equipment rooms and other congested areas.

c. T4 - Typical Detail Drawings - Faceplate Labeling, Firestopping, Americans with Disabilities Act (ADA), Safety, Department of Transportation (DOT). Detailed drawings of symbols and typicals such as faceplate labeling, faceplate types, faceplate population installation procedures, detail racking, and raceways.

1.6.1.2 Telecommunications Space Drawings

Provide T3 drawings that include telecommunications rooms plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and ~~[~~cabinet~~][~~, rack~~][~~, backboard~~][~~ and~~]~~ wall elevations. Drawings shall show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks, patch panels and equipment spaces and cabinet/racks. Drawings shall include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings.

1.6.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, and the supervisor (if different from the installer). A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

1.6.2.1 Telecommunications Contractor

The telecommunications contractor shall be a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor shall demonstrate experience in providing successful telecommunications systems within the past 3 years of similar scope and size. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for the telecommunications contractor.

1.6.2.2 Key Personnel

Provide key personnel who are regularly and professionally engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed for this solicitation depending upon how many of the key roles each has successfully provided. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems within the past 3 years.

Supervisors and installers assigned to the installation of this system or any of its components shall be Building Industry Consulting Services International (BICSI) Registered Cabling Installers, Technician Level. Submit documentation of current BICSI certification for each of the key personnel.

In lieu of BICSI certification, supervisors and installers assigned to the installation of this system or any of its components shall have a minimum of [3][______] years experience in the installation of the specified copper and fiber optic cable and components. They shall have factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for each of the key personnel. Documentation for each key person shall include at least two successful system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed for this solicitation. Include specific experience in installing and testing telecommunications systems and provide the names and locations of at least two project installations successfully completed using [optical fiber and] copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience shall have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation. Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work on this project. All key persons shall be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work on this project by the date that the bid was due to the Contracting Officer.

Note that only the key personnel approved by the Contracting Officer in the successful proposal shall do work on this solicitation's telecommunications system. Key personnel shall function in the same roles in this contract, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from The Contracting Officer.

1.6.2.3 Minimum Manufacturer Qualifications

Cabling, equipment and hardware manufacturers shall have a minimum of [3] [______] years experience in the manufacturing, assembly, and factory testing of components which comply with JIS X 5150, JCS 5507 and JIS C 6820 and JIS C 6850.

1.6.3 Test Plan

Provide a complete and detailed test plan for the telecommunications cabling system including a complete list of test equipment for the components and accessories for each cable type specified, [60][______] days prior to the proposed test date. Include procedures for certification, validation, and testing.

1.6.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of applicable codes and standards unless more stringent requirements are specified or indicated.

1.6.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.6.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site shall not be used, unless specified otherwise.

1.7 DELIVERY AND STORAGE

Provide protection from weather, moisture, extreme heat and cold, dirt, dust, and other contaminants for telecommunications cabling and equipment placed in storage.

1.8 ENVIRONMENTAL REQUIREMENTS

Connecting hardware shall be rated for operation under ambient conditions of 0 to 60 degrees C and in the range of 0 to 95 percent relative humidity, noncondensing.

1.9 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.10 MAINTENANCE

1.10.1 Operation and Maintenance Manuals

Commercial off the shelf manuals shall be furnished for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications cabling and pathway system, Data Package

5. Submit operations and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein not later than ~~[~~2~~] [______]~~ months prior to the date of beneficial occupancy. In addition to requirements of Data Package 5, include the requirements of paragraphs TELECOMMUNICATIONS DRAWINGS, TELECOMMUNICATIONS SPACE DRAWINGS, and RECORD DOCUMENTATION. Ensure that these drawings and documents depict the as-built configuration.

1.10.2 Record Documentation

Provide T5 drawings including documentation on cables and termination hardware. T5 drawings shall include schedules to show information for cut-overs and cable plant management, patch panel layouts and cover plate assignments, cross-connect information and connecting terminal layout as a minimum. T5 drawings shall be provided~~[~~ in hard copy format~~][~~ on electronic media using Windows based computer cable management software.~~][~~ A licensed copy of the cable management software including documentation, shall be provided.~~]~~ Provide the following T5 drawing documentation as a minimum:

a. Cables - A record of installed cable shall be provided. The cable records shall ~~[~~include only the required data fields~~][include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility]~~. Include manufacture date of cable with submittal.

b. Termination Hardware - A record of installed patch panels, cross-connect points, distribution frames, terminating block arrangements and type, and outlets shall be provided. Documentation shall include the required data fields~~[~~ as a minimum~~][only]~~.

PART 2 PRODUCTS

2.1 COMPONENTS

Components shall be UL or third party certified. Where equipment or materials are specified to conform to industry and technical society reference standards of the organizations, submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard. Provide a complete system of telecommunications cabling and pathway components using star topology. Provide support structures and pathways, complete with outlets, cables, connecting hardware and telecommunications cabinets/racks. Cabling and interconnecting hardware and components for telecommunications systems shall be listed or third party independent testing laboratory certified, and shall comply with applicable codes and standards and conform to the requirements specified herein.

2.2 TELECOMMUNICATIONS PATHWAY

Provide telecommunications pathways in accordance with JIS X 5150 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.3 TELECOMMUNICATIONS CABLING

Cabling shall be listed for the application and shall comply with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850. Provide a labeling system for cabling as required. Ship cable on reels or in boxes bearing manufacture date for for unshielded twisted pair (UTP) in accordance with JCS 5507~~[~~ and optical fiber cables in accordance with JIS C 6870-2~~]~~ for all cable used on this project. Cabling manufactured more than 12 months prior to date of installation shall not be used.

~~[~~2.3.1 Backbone Cabling

~~[~~2.3.1.1 Backbone Copper

Copper backbone cable shall be solid conductor, 0.2 sqmm, 100 ohm, ~~[~~100~~][_____]~~-pair, Category 3, UTP, in accordance with JCS 5507, JIS X 5150, JCS 5507 and JCS 5000 series, formed into ~~[~~25~~][10]~~ pair binder groups covered with a~~[~~ gray~~][_____]~~ thermoplastic jacket~~[~~ and overall metallic shield~~]~~. Cable shall be imprinted with manufacturers name or identifier, flammability rating, gauge of conductor, transmission performance rating (category designation) at regular length marking intervals in accordance with JCS 5507. Provide plenum (CMP), riser (CMR), or general purpose (CM or CMG)communications rated cabling as required. Substitution of a higher rated cable shall be permitted upon approval prior to installation.

~~][~~2.3.1.2 Backbone Optical Fiber

Provide in accordance with JIS C 6870-2, JIS C 6820, JIS C 6850, UL 1666 and JIS C 3665-1-1. Cable shall be imprinted with fiber count, fiber type and aggregate length at regular intervals not to exceed 1 meter.

Provide the number of strands indicated, (but not less than 12 strands between the main telecommunication room and each of the other telecommunication rooms), of single-mode(OS1), tight buffered fiber optic cable.

~~[~~Provide tight buffered fiber optic multimode, ~~[50/125-um diameter laser optimized(OM3)][50/125-um diameter(OM2)][62.5/125-um diameter(OM1)]~~ cable as indicated.~~]~~

Provide plenum (OFNP), riser (OFNR), or general purpose (OFN or OFNG) rated non-conductive, fiber optic cable as required per application. Substitution of a higher rated cable shall be permitted. The cable cordage jacket, fiber, unit, and group color shall be in accordance with JIS X 5150.

Provide plenum (OFNP) riser (OFNR)~~–~~, or general purpose (OFN or OFNG) rated non-conductive, fiber optic cable as required per application. Substitution of a higher rated cable shall be permitted~~–~~. The cable cordage jacket, fiber, unit, and group color shall be in accordance with JIS X 5150.

~~]]~~2.3.2 Horizontal Cabling

Provide horizontal cable and performance characteristics in accordance with JIS X 5150.

2.3.2.1 Horizontal Copper

Provide horizontal copper cable, UTP, 100 ohm in accordance with JCS 5507, and JIS X 5150. Provide four each individually twisted pair, minimum size 0.2 sqmm conductors, Category 6, with a~~[~~ blue~~][____]~~ thermoplastic jacket. Cable shall be imprinted with manufacturers name or identifier, flammability rating, gauge of conductor, transmission performance rating (category designation) and length marking at regular intervals in accordance with JCS 5507. Provide plenum (CMP), riser (CMR), or general purpose (CM or CMG) communications rated cabling as required per application. Substitution of a higher rated cable shall be permitted . Cables installed in conduit within and under slabs shall be listed and labeled for wet locations. ~~[~~Provide residential Category 6 cabling as required.~~]~~

~~[2.3.2.2 Horizontal Optical Fiber~~

~~Provide optical fiber horizontal cable in accordance with JIS C 6870-2 and JIS C 6820 and JIS C 6850. Cable shall be tight buffered, [multimode, 50/125-um diameter laser optimized, OM3][,][multimode, 50/125-um diameter, OM2][multimode, 62.5.125-um diameter, OM1][single-mode, 8/125-um diameter, OS1]. Cable shall be imprinted with manufacturer, flammability rating and fiber count at regular intervals not to exceed 1 meter.~~

~~Provide plenum (OFNP), riser (OFNR), or general purpose (OFN or OFNG) rated non-conductive, fiber optic cable as required per application. Substitution of a higher rated cable shall be permitted. Cables installed in conduit within and under slabs be listed and labeled for wet locations. The cable jacket shall be of single jacket construction with color coding of cordage jacket, fiber, unit, and group in accordance with JIS X 5150.~~

~~][2.3.3 Work Area Cabling~~

~~2.3.3.1 Work Area Copper~~

~~Provide work area copper cable in accordance with JCS 5507, with a[blue,][____] thermoplastic jacket.~~

~~[2.3.3.2 Work Area Optical Fiber~~

~~Provide optical work area cable in accordance with JIS C 6820 and JIS C 6850.~~

~~]]~~2.4 TELECOMMUNICATIONS SPACES

Provide connecting hardware and termination equipment in the telecommunications entrance facility~~[~~ and telecommunication equipment room~~[~~s~~]]~~ to facilitate installation as shown on design drawings for terminating and cross-connecting permanent cabling. Provide telecommunications interconnecting hardware color coding in accordance with ~~[base][~~facility~~]~~ standards.

2.4.1 Backboards

Provide void-free, interior grade A-C plywood 19 mm thick~~[~~1200 by 2400 mm~~][as indicated]~~. Backboards shall be fire rated by manufacturing process. Fire stamp shall be clearly visible.~~[~~ Paint applied over fire retardant backboard shall be UL 723 and JIS C 60364-4-42 fire retardant paint.

Provide label including paint manufacturer, date painted, UL listing and name of Installer. When painted, paint label and fire stamp shall be clearly visible.] Backboards shall be provided on a minimum of two adjacent walls in the telecommunication spaces.

[2.4.2 Equipment Support Frame

Provide in accordance with JIS C 6011-1 and JIS C 6012-3-100.

[a. Bracket, wall mounted, 8 gauge aluminum. Provide hinged bracket compatible with [482.6 mm][~~584 mm~~] panel mounting.

][b. Racks, floor mounted modular type, [16 gauge steel][or][11 gauge aluminum] construction, minimum, treated to resist corrosion. Provide rack with vertical and horizontal cable management channels, top and bottom cable troughs, grounding lug[and a surge protected power strip with 6 duplex 20 amp receptacles]. Rack shall be compatible with[482.6 mm][~~ 584 mm~~] panel mounting.

][c. Cabinets, freestanding modular type, [16 gauge steel][or][11 gauge aluminum] construction–, minimum, treated to resist corrosion. Cabinet shall have removable and lockable side panels, front and rear doors, and have adjustable feet for leveling. Cabinet shall be vented in the roof and rear door. Cabinet shall have cable access in the roof and base and be compatible with[482.6 mm][~~ 584 mm~~] panel mounting. Provide cabinet with grounding bar[~~,~~][[rack][~~roof~~] mounted 15 cu. m fan with filter–][and][a surge protected power strip with 6 duplex 20 amp receptacles].[All cabinets shall be keyed alike.]

][d. Cabinets, wall-mounted modular type, [16 gauge steel][or][11 gauge aluminum] construction–, minimum, treated to resist corrosion. Cabinet shall have have lockable front[and rear] door[s], louvered side panels,[7 cu. m [~~roof~~][rack] mounted fan,––] ground lug, and top and bottom cable access. Cabinet shall be compatible with[482.6 mm][~~ 584 mm~~] panel mounting.[All cabinets shall be keyed alike.][A [~~duplex AC outlet~~][surge protected power strip with 6 duplex 20 amp receptacles] shall be provided within the cabinet.]

]][2.4.3 Connector Blocks

Provide insulation displacement connector (IDC) Type 110 for Category 6 systems. Provide blocks for the number of horizontal and backbone cables terminated on the block plus 25 percent spare.

][2.4.4 Cable Guides

Provide cable guides specifically manufactured for the purpose of routing cables, wires and patch cords horizontally and vertically on[[482.6][584] mm equipment[racks][cabinets]][and–][telecommunications backboards]. Cable guides of ring or bracket type devices[mounted on [rack][_cabinet] panels][_backboard] for horizontal cable management and individually mounted for vertical cable management. Mount cable guides with screws,[and–][or]nuts and lockwashers.

][2.4.5 Patch Panels

Provide ports for the number of horizontal and backbone cables terminated on the panel plus [25][______] percent spare. Provide pre-connectorized [

optical fiber~~][~~ and ~~][~~copper~~]~~ patch cords for patch panels. Provide patch cords, as complete assemblies, with matching connectors as specified.~~[~~ Provide fiber optic patch cables with crossover orientation in accordance with JIS C 6820 and JIS C 6850~~]~~. Patch cords shall meet minimum performance requirements specified in JIS X 5150, JCS 5507~~ [~~ and JIS C 6820 and JIS C 6850~~]~~ for cables, cable length and hardware specified.

~~2.4.5.1 Modular to 110 Block Patch Panel~~

~~Provide in accordance with JIS X 5150 and JCS 5507. Panels shall be third party verified[and shall comply with EIA/TIACategory 6 requirements]. Panel shall be constructed of 2.2 mm minimum aluminum and shall be [cabinet][rack][wall] mounted and compatible with JIS C 6012-3-100 [482.6 mm][584 mm] equipment[cabinet][rack]. Panel shall provide [48][_____] non-keyed, 8-pin modular ports, wired to [T568A][T568B]. Patch panels shall terminate the building cabling on Type 110 IDCs and shall utilize a printed circuit board interface. The rear of each panel shall have incoming cable strain-relief and routing guides. Panels shall have each port factory numbered and be equipped with laminated plastic nameplates above each port.~~

~~[~~2.4.5.1 Fiber Optic Patch Panel

Provide panel for maintenance and cross-connecting of optical fiber cables. Panel shall be constructed of~~[[~~16~~][18]~~ gauge steel~~][~~ or~~][~~ 11 gauge aluminum~~]~~ minimum and shall be ~~[~~cabinet~~][rack][wall]~~ mounted and compatible with a JIS C 6012-3-100~~[~~ 482.6 mm~~][584 mm]~~ equipment rack. Each panel shall provide ~~[~~12~~][_____]~~ ~~[~~multimode~~][single-mode]~~ adapters as ~~[~~ duplex LC in accordance with JIS C 5964-20 with zirconia ceramic alignment sleeves,~~]~~ ~~[duplex SC in accordance with JIS C 5964-4 and JIS C 5964-4-100 with zirconia ceramic][MT-RJ in accordance with JIS C 5964-18 with thermoplastic][~~ST in accordance with with metallic~~]~~ alignment sleeves. Provide dust cover for unused adapters. The rear of each panel shall have a cable management tray a minimum of 203 mm deep with removable cover, incoming cable strain-relief and routing guides. Panels shall have each adapter factory numbered and be equipped with laminated plastic nameplates above each adapter.

~~]][~~2.4.6 Optical Fiber Distribution Panel

~~[Cabinet][Rack][~~Wall~~]~~ mounted optical fiber distribution panel (OFDP) shall be constructed in accordance with JIS C 6012-3-100 utilizing~~[[~~16 ~~][18]~~ gauge steel~~][~~ or~~][~~ 11 gauge aluminum~~]~~ minimum. Panel shall be divided into two sections, distribution and user. Distribution section shall have strain relief, routing guides, splice tray and shall be lockable, user section shall have a cover for patch cord protection. Each panel shall provide~~[[~~12~~][_____]~~ multimode~~][~~ and~~][[~~12~~][_____]~~ single-mode~~]~~ pigtails and adapters. Provide adapters as~~ [~~ duplex LC with zirconia ceramic~~] [duplex SC with zirconia ceramic][MT-RJ with thermoplastic][~~ ST with metallic~~]~~ alignment sleeves. Provide dust covers for adapters. Provide patch cords as specified in the paragraph PATCH PANELS.

~~]~~2.5 TELECOMMUNICATIONS OUTLET/CONNECTOR ASSEMBLIES

2.5.1 Outlet/Connector Copper

Outlet/connectors shall comply with JIS X 5150 and JCS 5507. UTP outlet/connectors shall be JIS X 5150 listed, non-keyed, 8-pin modular, constructed of high impact rated thermoplastic housing and shall be third party verified~~[~~ and shall comply with JCS 5507 Category 6 requirements.~~]~~ Outlet/connectors provided for UTP cabling shall meet or exceed the requirements for the cable provided. Outlet/connectors shall be terminated using a Type 110 IDC PC board connector, color-coded for both T568A and T568B wiring. Each outlet/connector shall be wired ~~[~~T568A~~][~~ or ~~][T568B][~~as indicated~~]~~. UTP outlet/connectors shall comply with JCS 5507 for ~~[~~200~~] [_____]~~ mating cycles.~~[~~ UTP outlet/connectors installed in outdoor or marine environments shall be jell-filled type containing an anti-corrosive, memory retaining compound.~~]~~

2.5.2 Optical Fiber Adapters(Couplers)

Provide optical fiber adapters suitable for~~[~~ duplex LC in accordance with JIS C 5964-20 with zirconia ceramic alignment sleeves,~~][~~ duplex SC in Accordance with JIS C 5964-4 and JIS C 5964-4-100 with zirconia ceramic alignment sleeves,~~]~~ ~~[~~MT-RJ in accordance with JIS C 5964-18 with thermoplastic alignment sleeves,~~][~~ and~~][~~ ST with metallic alignment sleeves~~]~~ as indicated. Provide dust cover for adapters. Optical fiber adapters shall comply with JIS C 61300-2-2 for ~~[~~500~~][_____]~~ mating cycles.

2.5.3 Optical Fiber Connectors

Provide in accordance with JIS C 61300-2-2.~~[~~ Optical fiber connectors shall be~~[~~ duplex LC in accordance with JIS C 5964-20 with zirconia ceramic alignment sleeves,~~] [~~ duplex SC in accordance with JIS C 5964-4 and JIS C 5964-4-100 with zirconia ceramic~~][~~ MT-RJ in accordance with JIS C 5964-18 with thermoplastic~~][~~ ST in accordance with with metallic~~]~~ ferrule, epoxyless~~[~~ crimp style~~]~~ compatible with~~[[62.5/125][~~50/125~~]~~ multimode~~][8/125 single-mode]~~ fiber. The connectors shall provide a maximum attenuation of 0.3 dB at~~[850][~~ 1300~~][1310][1550]~~ nm with less than a 0.2 dB change after 500 mating cycles.~~]~~

2.5.4 Cover Plates

Telecommunications cover plates shall comply with JIS C 8435, and JIS X 5150, ~~[~~JCS 5507~~]~~, ~~[~~JIS C 6820 and JIS C 6850~~]~~; ~~[~~flush~~][~~ or ~~][~~ oversized~~]~~ design constructed of ~~[~~high impact thermoplastic material ~~[[~~ ivory~~][white][brown]~~ in color~~][~~_to match color of receptacle/switch cover plates specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM~~]][~~_302 stainless material~~][~~ or ~~][~~brass material~~]~~. Provide labeling in accordance with the paragraph LABELING in this section.

~~[2.6 MULTI-USER TELECOMMUNICATIONS OUTLET ASSEMBLY (MUTOA)~~

~~Provide MUTOA(s) in accordance with JIS X 5150.~~

~~][~~2.6 TERMINAL CABINETS

Construct of zinc-coated sheet steel~~,[915 by 610 by 150 mm deep][~~_as indicated~~]~~. Trim shall be fitted with hinged door and locking latch. Doors shall be maximum size openings to box interiors. Boxes shall be provided with 16 mm backboard with two-coat varnish finish. Match trim, hardware, doors, and finishes with panelboards. Provide label and identification systems for telecommunications wiring and components.

~~]~~2.7 GROUNDING AND BONDING PRODUCTS

Provide in accordance with JIS C 60364-5-54. Components shall be identified as required by applicable codes and standards. Provide ground rods, bonding conductors, and grounding busbars as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.8 FIRESTOPPING MATERIAL

Provide as specified in Section 07 84 00 FIRESTOPPING.

2.9 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.10 FIELD FABRICATED NAMEPLATES

JIS K 6911. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 3 mm thick, white with [black] [______] center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be 25 by 65 mm . Lettering shall be a minimum of 6.35 mm high normal block style.

2.11 TESTS, INSPECTIONS, AND VERIFICATIONS

2.11.1 Factory Reel Tests

Provide documentation of the testing and verification actions taken by manufacturer to confirm compliance with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850[, JIS C 6185-2 for single mode optical fiber][, and JIS C 6185-3 for multimode optical fiber] cables.

PART 3 EXECUTION

3.1 INSTALLATION

Install telecommunications cabling and pathway systems, including the horizontal and backbone cable, pathway systems, telecommunications outlet/connector assemblies, and associated hardware in accordance with JIS X 5150, JCS 5507, [JIS C 6820~~ and~~, JIS C 6850,~~],~~ and UL standards as applicable. Provide cabling in a star topology network.[Provide residential cabling in a star wiring architecture from the distribution device as required.] Pathways and outlet boxes shall be installed as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Install telecommunications cabling with copper media in accordance with the following criteria to avoid potential electromagnetic interference between power and telecommunications equipment. The interference ceiling shall not exceed 3.0 volts per meter measured over the usable bandwidth of the telecommunications cabling.[Cabling shall be run with horizontal and vertical cable guides in telecommunications spaces with terminating hardware and interconnection equipment.]

3.1.1 Cabling

Install [UTP,][and][optical fiber] telecommunications cabling system as detailed in JIS X 5150, [JCS 5507,] [JIS C 6820 and JIS C 6850, Appendix A][and applicable codes and standards for residential cabling]. Screw terminals shall not be used except where specifically indicated on plans. Use an approved insulation displacement connection (IDC) tool kit for copper cable terminations. Do not exceed manufacturers' cable pull tensions for copper and optical fiber cables. Provide a device to monitor cable pull tensions. Do not exceed 110 N pull tension for four pair copper cables. Do not chafe or damage outer jacket materials. Use only lubricants approved by cable manufacturer. Do not over cinch cables, or crush cables with staples. For UTP cable, bend radii shall not be less than four times the cable diameter. Cables shall be terminated; no cable shall contain unterminated elements. Cables shall not be spliced. Label cabling in accordance with paragraph LABELING in this section.

~~[3.1.1.1 Open Cable~~

~~Use only where specifically indicated on plans for use in cable trays, or below raised floors. Install in accordance withJIS X 5150, JCS 5507 [[and]JIS C 6820 and JIS C 6850, Appendix A]. Do not exceed cable pull tensions recommended by the manufacturer.[Copper cable not in a wireway or pathway shall be suspended a minimum of [200][_____] mm above ceilings by cable supports no greater than [1.5][_____] m apart. Cable shall not be run through structural members or in contact with pipes, ducts, or other potentially damaging items. Placement of cable parallel to power conductors shall be avoided, if possible; a minimum separation of 300 mm shall be maintained when such placement cannot be avoided.]~~

~~Plenum cable shall be used where open cables are routed through plenum areas. Cable routed exposed under raised floors shall be plenum rated. Plenum cables shall comply with flammability plenum requirements per applicable codes and standards. Install cabling after the flooring system has been installed in raised floor areas.[Cable [1.8][_____] meters long shall be neatly coiled not less than [300][_____] mm in diameter below each feed point in raised floor areas.]~~

~~]~~3.1.1.1 Backbone Cable

a. Copper Backbone Cable. Install intrabuilding backbone copper cable, in indicated pathways, between the campus distributor, located in the telecommunications entrance facility or room, the building distributors and the floor distributors located in telecommunications rooms and telecommunications equipment rooms as indicated on drawings.

b. Optical fiber Backbone Cable. Install intrabuilding backbone optical fiber in indicated pathways. Do not exceed manufacturer's recommended bending radii and pull tension. Prepare cable for pulling by cutting outer jacket 250 mm leaving strength members exposed for approximately 250 mm. Twist strength members together and attach to pulling eye. Vertical cable support intervals shall be in accordance with manufacturer's recommendations.

3.1.1.2 Horizontal Cabling

Install horizontal cabling as indicated on drawings Do not untwist Category 6 UTP cables more than 12 mm from the point of termination to maintain cable geometry. Provide slack cable in the form of a figure eight (not a service loop) on each end of the cable, 3 m in the telecommunications room, and 304 mm in the work area outlet.

3.1.2 Pathway Installations

Provide in accordance with JIS X 5150. Provide building pathway as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

~~[~~3.1.3 Service Entrance Conduit, Overhead

Provide service entrance overhead as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEMS.

~~][~~3.1.4 Service Entrance Conduit, Underground

Provide service entrance underground as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

~~]~~3.1.5 Cable Tray Installation

Install cable tray as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Only ~~[~~CMP~~]~~ ~~[~~and~~]~~ ~~[~~OFNP~~]~~ type cable shall be installed in a plenum.

~~3.1.6 Work Area Outlets~~

~~3.1.6.1 Terminations~~

~~Terminate UTP cable in accordance with JIS X 5150, JCS 5507 and wiring configuration as specified.[Terminate fiber optic cables in accordance with JIS C 6820 and JIS C 6850]~~

~~3.1.6.2 Cover Plates~~

~~As a minimum, each outlet/connector shall be labeled as to its function and a unique number to identify cable link in accordance with the paragraph LABELING in this section.~~

~~3.1.6.3 Cables~~

~~Unshielded twisted pair and fiber optic cables shall have a minimum of 304 mm of slack cable loosely coiled into the telecommunications outlet boxes. Minimum manufacturer's bend radius for each type of cable shall not be exceeded.~~

~~3.1.6.4 Pull Cords~~

~~Pull cords shall be installed in conduit serving telecommunications outlets that do not have cable installed.~~

~~[3.1.6.5 Multi-User Telecommunications Outlet Assembly (MUTOA)~~

~~Run horizontal cable in the ceiling or underneath the floor and terminate each cable on a MUTOA in each individual zone. MUTOAs shall not be located in ceiling spaces, or any obstructed area. MUTOAs shall not be installed in furniture unless that unit of furniture is permanently secured to the building structure. MUTOAs shall be located in an open work area so that each furniture cluster is served by at least one MUTOA. The MUTOA shall be limited to serving a maximum of twelve work areas. Maximum work area cable length requirements shall also be taken into account. MUTOAs must be labeled to include the maximum length of work area cables. MUTOA labeling is in addition to the labeling, or other applicable cabling administration standards. Work area cables extending~~

~~from the MUTOA to the work area device must also be uniquely identified and labeled.~~

~~]~~3.1.6 Telecommunications Space Termination

Install termination hardware required for ~~[~~Category 6~~][~~ and ~~][~~optical fiber ~~]~~ system. An insulation displacement tool shall be used for terminating copper cable to insulation displacement connectors.

~~[~~3.1.6.1 Connector Blocks

Connector blocks shall be ~~[~~cabinet~~][rack][wall]~~ mounted in orderly rows and columns. Adequate vertical and horizontal wire routing areas shall be provided between groups of blocks. Install in accordance with industry standard wire routing guides in accordance with JIS X 5150.

~~]~~3.1.6.2 Patch Panels

Patch panels shall be mounted ~~[~~in equipment ~~[~~cabinets~~]][racks][on the plywood backboard]~~ with sufficient ports to accommodate the installed cable plant plus ~~[~~25~~][______]~~ percent spares.

~~[~~ a. Copper Patch Panel. Copper cable entering a patch panel shall be secured to the panel ~~[with cable ties][~~as recommended by the manufacturer~~]~~ to prevent movement of the cable.

~~][~~b. Fiber Optic Patch Panel. Fiber optic cable loop shall be ~~[~~ ~~[~~900 ~~][_____]~~ mm ~~-~~in length~~][provided as recommended by the manufacturer]~~. The outer jacket of each cable entering a patch panel shall be secured to the panel to prevent movement of the fibers within the panel, using clamps or brackets specifically manufactured for that purpose.

~~][~~3.1.6.3 Equipment Support Frames

Install in accordance with JIS X 5150:

~~[~~a. Bracket, wall mounted. Mount bracket to plywood backboard in accordance with manufacturer's recommendations. Mount rack so height of highest panel does not exceed 1980 mm above floor.~~]~~

~~[~~b. Racks, floor mounted modular type. Permanently anchor rack to the floor in accordance with manufacturer's recommendations.~~]~~

~~[~~c. Cabinets, freestanding modular type. When cabinets are connected together, remove adjoining side panels for cable routing between cabinets.~~[~~ Mount rack mounted fan in ~~[~~roof~~][base]~~ of cabinet.~~]]~~

~~[~~d. Cabinets, wall-mounted modular type. Mount cabinet to plywood backboard in accordance with manufacturer's recommendations. Mount cabinet so height of highest panel does not exceed 1980 mm above floor. ~~]~~

~~]~~3.1.7 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated wall, partitions, floors, or ceilings as specified in Section 07 84 00 FIRESTOPPING.

3.1.8 Grounding and Bonding

Provide in accordance with JIS C 60364-5-54 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.2 LABELING

3.2.1 Labels

Provide labeling in accordance with applicable codes and standards. Handwritten labeling is unacceptable. Stenciled lettering for voice and data circuits shall be provided using~~[~~ thermal ink transfer process~~][~~ laser printer~~] [_____]~~.

3.2.2 Cable

Cables shall be labeled using color labels on both ends with identifiers.

3.2.3 Termination Hardware

Workstation outlets and patch panel connections shall be labeled using color coded labels with identifiers.

3.3 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Painting shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

3.3.1 Painting Backboards

If backboards are required to be painted, then the manufactured fire retardant backboard must be painted with fire retardant paint, so as not to increase flame spread and smoke density and must be appropriately labeled. Label and fire rating stamp must be unpainted.

3.4 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.5 TESTING

3.5.1 Telecommunications Cabling Testing

Perform telecommunications cabling inspection, verification, and performance tests in accordance with JIS X 5150, ~~[~~JCS 5507~~]~~, ~~[~~JIS C 6820 and JIS C 6850~~]~~. Test equipment shall conform to JIS X 5150. Perform optical fiber field inspection tests via attenuation measurements on factory reels and provide results along with manufacturer certification for factory reel tests. Remove failed cable reels from project site upon attenuation test failure.

3.5.1.1 Inspection

Visually inspect UTP and optical fiber jacket materials for UL or third party certification markings. Inspect cabling terminations in telecommunications rooms and at workstations to confirm color code for T568A or T568B pin assignments, and inspect cabling connections to confirm compliance with JIS X 5150, JCS 5507, ~~[~~JIS C 6820 and JIS C 6850~~]~~,~~ [~~ and ~~][~~ applicable codes and standardsfor residential cabling~~]~~. Visually confirm ~~[~~ Category 6,~~]~~ marking of outlets, cover plates, outlet/connectors, and patch panels.

3.5.1.2 Verification Tests

UTP backbone copper cabling shall be tested for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors, and between conductors and shield, if cable has overall shield. Test operation of shorting bars in connection blocks. Test cables after termination but prior to being cross-connected.

~~[[~~For multimode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with JIS C 6820 and JIS C 6850 and JIS C 6850 and JIS C 6185-3 using~~[~~ Method A, Optical Power Meter and Light Source~~][~~ Method B, OTDR~~]~~ for multimode optical fiber.~~][~~ For single-mode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with JIS C 6820 and JIS C 6850 and JIS C 6185-2 using~~[~~ Method A, Optical Power Meter and Light Source~~][~~ Method B, OTDR~~]~~ for single-mode optical fiber.~~]~~ Perform verification acceptance tests.~~]~~

3.5.1.3 Performance Tests

Perform testing for each outlet and MUTOA as follows:

~~[~~ ~~-~~a. Perform Category 6 link tests in accordance with JIS X 5150 and JCS 5507. Tests shall include wire map, length, insertion loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, return loss, propagation delay, and delay skew.

~~][~~b. Optical fiber Links. Perform optical fiber end-to-end link tests in accordance with JIS C 6820 and JIS C 6850.

~~]~~3.5.1.4 Final Verification Tests

Perform verification tests for UTP~~[~~ and optical fiber~~]~~ systems after the complete telecommunications cabling and workstation outlet/connectors are installed.

~~[a. Voice Tests. These tests assume that dial tone service has been installed. Connect to the network interface device at the demarcation point. Go off-hook and listen and receive a dial tone. If a test number is available, make and receive a local, long distance, and DSN telephone call.~~

~~][b. Data Tests. These tests assume the Information Technology Staff has a network installed and are available to assist with testing. Connect to the network interface device at the demarcation point. Log onto the network to ensure proper connection to the network.~~

~~]~~ -- End of Section --

SECTION 28 31 76

INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE
08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S3.2	(2020; R 2025) American National Standard Method for Measuring the Intelligibility of Speech Over Communication Systems

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A17.1/CSA B44	(2022) Safety Code for Elevators and Escalators

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide https://www.approvalguide.com/

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1	(2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits
IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 4	(2024) Standard for Integrated Fire Protection and Life Safety System Testing
NFPA 70	(2023; ERTA 1 2024; TIA 24-1; TIA 25-2) National Electrical Code
NFPA 72	(2025; TIA 25-4) National Fire Alarm and Signaling Code
NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems
NFPA 170	(2024; ERTA 1 2023) Standard for Fire Safety and Emergency Symbols

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-601-02	(2021; with Change 1, 2025) Fire Protection Systems Inspection, Testing, and Maintenance
UFC 4-010-06	(2023) Cybersecurity of Facility-Related Control Systems

UL SOLUTIONS (UL)

UL 228	(2006; Reprint Mar 2022) UL Standard for Safety Door Closers-Holders, With or Without Integral Smoke Detectors
UL 268	(2023) UL Standard for Safety Smoke Detectors for Fire Alarm Systems
UL 268A	(2008; Reprint Aug 2023) Smoke Detectors for Duct Application
UL 464	(2023) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 497A	(2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits
UL 497B	(2004; Reprint Feb 2022) UL Standard for Safety Protectors for Data Communications and Fire Alarm Circuits
UL 864	(2023; Reprint Oct 2024) UL Standard for Safety Control Units and Accessories for Fire Alarm Systems
UL 1283	(2025) UL Standard for Safety Electromagnetic Interference Filters
UL 1449	(2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices
UL 1480	(2023) UL Standard for Safety Speakers for Fire Alarm and Signaling Systems, Including Accessories
UL 1638	(2023) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 1971	(2002; Reprint Feb 2024) UL Standard for Safety Signaling Devices for the Deaf and Hard of Hearing
UL 2017	(2008; Reprint Jan 2024) UL Standard for Safety General-Purpose Signaling Devices and Systems

UL 2572	(2016; Bul. 2018) UL Standard for Safety Mass Notification Systems
UL Fire Prot Dir	UL Product IQ (updated online) at https://productiq.ulprospector.com/en

1.2 RELATED SECTIONS

Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS, applies to this section, with the additions and modifications specified herein. In addition, refer to the following sections for related work and coordination:

~~[~~ Section 21 13 13 WET PIPE SPRINKLER SYSTEM, FIRE PROTECTION
~~][~~ Section 21 30 00 FIRE PUMPS
~~][Section 21 23 00 WET CHEMICAL FIRE EXTINGUISHING SYSTEMS~~
~~][Section 21 13 16 DRY PIPE FIRE SPRINKLER SYSTEMS~~
~~][Section 21 13 18 PREACTION FIRE SPRINKLER SYSTEMS~~
~~][Section 21 13 19 DELUGE FIRE SPRINKLER SYSTEMS~~
~~][~~ Section 23 30 00 HVAC AIR DISTRIBUTION
~~][Section 21 13 24.00 10 AQUEOUS FILM-FORMING FOAM (AFFF) FIRE PROTECTION SYSTEM~~
~~][Section 21 13 25 HIGH-EXPANSION FOAM SYSTEM, FIRE PROTECTION~~
~~][Section 21 21 01 CARBON-DIOXIDE FIRE-EXTINGUISHING SYSTEMS~~
~~][Section 21 23 00 WET CHEMICAL FIRE EXTINGUISHING SYSTEMS~~
~~][Section 08 71 00 DOOR HARDWARE for [door release][door unlocking] and additional work related to finish hardware.~~
~~][~~ Section~~[s] [14 21 13 ELECTRIC TRACTION FREIGHT ELEVATORS][~~14 21 23 ELECTRIC TRACTION PASSENGER ELEVATORS~~] [and] [14 24 13 HYDRAULIC FREIGHT ELEVATORS] [14 24 23 HYDRAULIC PASSENGER ELEVATORS]~~ for additional work related to elevators.
~~][~~ Section 07 84 00 FIRESTOPPING for additional work related to firestopping.

~~]~~1.3 SUMMARY

1.3.1 Scope

a. This work includes designing and ~~[~~providing a new, complete, ~~][modifying the existing]~~ fire alarm and mass notification (MNS) system as described herein and on the contract drawings~~[for the ________________]~~building. Include system wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, initiating devices, notification appliances, supervising station fire alarm transmitters/mass notification transceiver, and other accessories and miscellaneous items required for a complete operational system even though each item is not specifically mentioned or described. Provide system~~[s]~~ complete and ready for operation.~~[Existing interior fire alarm system was manufactured by [_____].]~~ Design and installation must comply with UFC 4-010-06.

b. Provide equipment, materials, installation, workmanship, inspection, and testing in strict accordance with~~-~~ NFPA 72, except as modified herein. ~~[~~The system layout on the drawings show the intent of coverage and suggested locations. Final quantity, system layout, and coordination are the responsibility of the Contractor.~~]~~

~~[~~ c. Each remote fire alarm control unit must be powered from a wiring riser specifically for that use or from a local emergency power panel located on the same floor as the remote fire alarm control unit. Where remote fire control units are provided, equipment for notification appliances may be located in the remote fire alarm control units.

][d. Where a fire pump is provided, the fire alarm and mass notification system must monitor and transmit the fire pump controller signals in accordance with the provisions of NFPA 72.

][e. Where an emergency generator provides standby power supply for life safety system circuits, the generator must be monitored by the FMCU and transmit emergency generator signals in accordance with NFPA 72.

] f. The fire alarm and mass notification system must be independent of the building security, building management, and energy/utility monitoring systems other than for control functions.

1.3.2 Qualified Fire Protection Engineer (QFPE)

Services of the QFPE must include:

a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Construction (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.

b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding comments.

c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).

d. Witnessing pre-Government [and final Government]functional performance testing and performing a final installation review.

e. Signing applicable certificates under SD-07.

1.4 DEFINITIONS

Wherever mentioned in this specification or on the drawings, the equipment, devices, and functions must be defined as follows:

1.4.1 Interface Device

An addressable device that interconnects hard wired systems or devices to an analog/addressable system.

1.4.2 Fire Alarm and Mass Notification Control Unit (FMCU)

A master control unit having the features of a fire alarm control unit (FACU) and an autonomous control unit (ACU) where these units are interconnected to function as a combined fire alarm/mass notification system. The FACU and ACU functions may be contained in a single cabinet or in independent, interconnected, and co-located cabinets.

1.4.3 Remote Fire Alarm and Mass Notification Control Unit

A control unit, physically remote from the fire alarm and mass notification control unit, that receives inputs from automatic and manual fire alarm devices; may supply power to detection devices and interface devices; may provide transfer of power to the notification appliances; may provide transfer of condition to relays or devices connected to the control unit; and reports to and receives signals from the fire alarm and mass notification control unit.

1.4.4 Local Operating Console (LOC)

A unit designed to allow emergency responders or building occupants to operate the MNS including delivery of recorded messages or live voice announcements, initiate visual, textual visual, and audible appliance operation and other relayed functions.

1.4.5 Terminal Cabinet

A steel cabinet with locking, hinge-mounted door where terminal strips are securely mounted inside the cabinet.

1.4.6 Control Module and Relay Module

Terms utilized to describe emergency control function interface devices as defined by NFPA 72.

1.4.7 Designated Fire Protection Engineer (DFPE)

The DoD fire protection engineer that oversees that Area of Responsibility for that project. This is sometimes referred to as the "cognizant" fire protection engineer. Interpret reference to "authority having jurisdiction" and AHJ in referenced standards to mean the Designated Fire Protection Engineer (DFPE). The DFPE may be responsible for review of the contractor submittals having a "G" designation, and for witnessing final inspection and testing.

1.4.8 Qualified Fire Protection Engineer (QFPE)

A QFPE is an individual who is a licensed professional engineer (P.E.), who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the fire alarm designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE will be returned by the Government disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G~~, [_____]~~

Fire alarm system designer; G~~, [_____]~~

Supervisor; G~~, [_____]~~

Technician; G~~, [_____]~~

Installer; G~~, [_____]~~

Test Technician; G~~, [_____]~~

Fire Alarm System Site-Specific Software Acknowledgement; G~~, [_____]~~

SD-02 Shop Drawings

Nameplates; G~~, [_____]~~

Instructions; G~~, [_____]~~

Wiring Diagrams; G~~, [_____]~~

System Layout; G~~, [_____]~~

Notification Appliances; G~~, [_____]~~

Initiating devices; G~~, [_____]~~

Amplifiers; G~~, [_____]~~

Battery Power; G~~, [_____]~~

Voltage Drop Calculations; G~~, [_____]~~

SD-03 Product Data

Fire Alarm and Mass Notification Control Unit (FMCU); G~~, [_____]~~

Local Operating Console (LOC); G~~, [_____]~~

Amplifiers; G~~, [_____]~~

Tone Generators; G~~, [_____]~~

Digitalized voice generators; G~~, [_____]~~

~~[Video Display Unit (VDU); G, [_____]~~

~~]~~ LCD Annunciator; G~~, [_____]~~

Manual Stations; G~~, [_____]~~

Smoke Detectors; G~~, [_____]~~

Duct Smoke Detectors; G~~, [_____]~~

~~[~~ ~~Projected Beam Smoke Detector; G, [_____]~~

~~][~~ ~~Air sampling smoke detectors; G, [_____]~~

~~]~~ ~~Heat Detectors; G, [_____]~~

~~[~~ ~~Flame Detectors; G, [_____]~~

~~][~~ ~~Multi-Criteria Detectors; G, [_____]~~

~~]~~ ~~Carbon monoxide detector; G, [_____]~~

Addressable Interface Devices; G~~, [_____]~~

Addressable Control Modules; G~~, [_____]~~

Isolation Modules; G~~, [_____]~~

Notification Appliances; G~~, [_____]~~

~~Textual Display Sign Control Panel; G, [_____]~~

Textual Display Signs; G~~, [_____]~~

Batteries; G~~, [_____]~~

Battery Chargers; G~~, [_____]~~

~~[~~ Supplemental Notification Appliance Circuit Panels; G~~, [_____]~~

~~][~~ ~~Auxiliary Power Supply Panels; G, [_____]~~

~~]~~ Surge Protective Devices; G~~, [_____]~~

Alarm Wiring; G~~, [_____]~~

~~[~~ Back Boxes and Conduit; G~~, [_____]~~

~~][~~ Ceiling Bridges for Ceiling-Mounted Appliances; G~~, [_____]~~

~~]~~ Terminal Cabinets; G~~, [_____]~~

Digital Alarm Communicator Transmitter (DACT); G~~, [_____]~~

~~[~~ Automatic Fire Alarm Transmitters (including housing); G~~, [_____]~~

~~][~~ ~~Radio Transmitter and Interface Panels; G, [_____]~~

~~][~~ ~~Mass Notification Transceiver; G, [_____]~~

~~]~~ Electromagnetic Door Holders; G~~, [_____]~~

Environmental Enclosures or Guards; G~~, [_____]~~

~~[~~ ~~Firefighter Telephone; G, [_____]~~

~~][~~ ~~Printer; G, [_____]~~

~~]~~ Document Storage Cabinet; G~~, [_____]~~

~~SD-05 Design Data~~

~~[~~ ~~Air Sampling Smoke Detection System Calculations; G, [_____]~~

~~]~~ SD-06 Test Reports

Test Procedures; G~~, [_____]~~

SD-07 Certificates

Verification of Compliant Installation; G~~, [_____]~~

Request for Government Final Test; G~~, [_____]~~

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G~~, [_____]~~

Instruction of Government Employees; G~~, [_____]~~

SD-11 Closeout Submittals

As-Built Drawings

Spare Parts

1.6 SYSTEM OPERATION

Fire alarm system/mass notification system including textual display sign control panel(s), components requiring power, except for the FMCU(s) power supply, must operate on 24 volts DC unless noted otherwise in this section.

The interior fire alarm and mass notification system must be a complete, supervised, noncoded, analog/addressable fire alarm and mass notification system conforming to NFPA 72, UL 864, and UL 2572. Systems meeting UL 2017 only are not acceptable. The system must be activated into the alarm mode by actuation of an alarm initiating device. The system must remain in the alarm mode until the initiating device is reset and the control unit is reset and restored to normal. The system may be placed in the alarm mode by local microphones, LOC, FMCU, or remotely from authorized locations/users.

1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Voice, Textual)

a. Connect alarm initiating devices to initiating device circuits (IDC) ~~[Class "A"][~~Class "B"~~]~~, or to signaling line circuits (SLC) Class ~~["A"][~~"B"~~]["X"]~~ and installed in accordance with NFPA 72.

b. Connect notification appliances to notification appliance circuits (NAC) ~~[Class "A"][~~Class "B"~~]~~.

1.6.2 Functions and Operating Features

The system must provide the following functions and operating features:

a. Power, annunciation, supervision, and control for the system. Addressable systems must be microcomputer (microprocessor or microcontroller) based with a minimum word size of eight bits with sufficient memory to perform as specified.

b. Visual alarm notification appliances must be synchronized as required by NFPA 72.

c. Electrical supervision of the primary power (AC) supply, presence of the battery, battery voltage, and placement of system modules within the control unit.

d. An audible and visual trouble signal to activate upon a single break or open condition, or ground fault~~[(or short circuit for Class "X")]~~. The trouble signal must also operate upon loss of primary power (AC) supply, absence of a battery supply, low battery voltage, or removal of alarm or supervisory control unit modules. After the system returns to normal operating conditions, the trouble signal must again sound until the trouble is acknowledged. A smoke detector in the process of being verified for the actual presence of smoke must not initiate a trouble condition.

e. A trouble signal silence feature that must silence the audible trouble signal, without affecting the visual indicator.

f. Program capability via switches in a locked portion of the FMCU to bypass the automatic notification appliance circuits, ~~[~~fire reporting system~~][~~, air handler shutdown~~][smoke control operation][~~, elevator recall~~][~~, door release~~][door unlocking]~~ features. Operation of this programmed action must indicate on the FMCU display~~[and printer output]~~ as a supervisory or trouble condition.~~[~~ Notification appliance bypass must be selectable by floor.~~]~~

g. Alarm functions must override trouble or supervisory functions. Supervisory functions must override trouble functions.

h. The system must be capable of being programmed from the control unit keyboard. Programmed information must be stored in non-volatile memory.

i. The system must be capable of operating, supervising, and monitoring non-addressable alarm and supervisory devices.

j. There must be no limit, other than maximum system capacity, as to the number of addressable devices that may be in alarm simultaneously.

k. Where the fire alarm/mass notification system is responsible for initiating an action in another emergency control device or system, such as ~~[~~HVAC, ~~][atrium exhaust,][smoke control,][~~elevator recall, ~~][releasing service,]~~the addressable fire alarm relay must be located in the vicinity of the emergency control device.

l. An alarm signal must automatically initiate the following functions:

(1) Transmission of an alarm signal to ~~[~~the fire department~~][a remote supervising station]~~.

(2) Visual indication of the device operated on the FMCU, ~~[Video Display unit (VDU),] [~~and on the ~~[graphic][~~remote ~~]~~annunciator~~]~~. ~~[Indication on the graphic annunciator must be by floor, zone or circuit, and type of device.]~~

~~[~~ ~~(3) Record the event on the system printer.~~

~~]~~ (3) Actuation of alarm notification appliances.

(4) Recording of the event electronically in the history log of the FMCU.

~~[~~ (5) Release of doors held open by electromagnetic devices.

~~][~~ ~~(6) Operation of the [smoke control system][atrium exhaust system].~~

~~][~~ ~~(7) Release of power to electric locks (delayed egress locks) on doors that are part of the means of egress.~~

~~][~~ (6) Elevator recall as described in this section.
~~]~~
~~[~~ ~~(7) Operation of [_____] must release the [_____] fire extinguishing system after a [_____] second time delay.~~

~~][~~ (7) Operation of a sprinkler waterflow switch serving an elevator machinery room or elevator shaft must operate shunt trip circuit breaker(s) to shut down power to the elevators in accordance with ASME A17.1/CSA B44.

~~][~~ ~~(8) Operation of an interface that operates vibrating pagers worn by hearing-impaired occupants.~~

~~]~~ m. A supervisory signal must automatically initiate the following functions:

(1) Visual indication of the device operated on the FMCU~~,~~ ~~[Video Display unit (VDU),] [~~and on the ~~[graphic][~~remote ~~]~~annunciator~~]~~. ~~[Indication on the graphic annunciator must be by floor, zone or circuit, and type of device.]~~

~~[~~ ~~(2) Record the event on the system printer.~~

~~]~~ (2) Transmission of a supervisory signal to ~~[~~the fire department~~][a remote supervising station]~~.

(3) Operation of a duct smoke detector must shut down the appropriate air handler in accordance with NFPA 90A in addition to other requirements of this paragraph and as allowed by NFPA 72.

(4) Recording of the event electronically in the history log of the FMCU.

n. A trouble condition must automatically initiate the following functions:

(1) Visual indication of the device operated on the FMCU, ~~[Video Display unit (VDU),]~~ ~~[~~and on the ~~[graphic][~~remote ~~]~~annunciator~~]~~. ~~[Indication on the graphic annunciator must be by floor, zone or circuit, and type of device.]~~

~~[(2) Record the event on the system printer.~~

~~]~~ (2) Transmission of a trouble signal to ~~[~~the fire department~~][a remote supervising station]~~.

(3) Recording of the event electronically in the history log of the FMCU.

~~[o. Activation of a carbon monoxide alarm initiating device must automatically initiate the following functions:~~

~~(1) Visual indication of the device operated on the FMCU,[LCD, LED Display unit (VDU),][and on the [graphic][remote]annunciator].[Indication on the graphic annunciator must be by floor and room number, device address, and device type.]~~

~~(2) Transmission of a carbon monoxide alarm signal to [the fire department][a remote supervising station].~~

~~(3) Activation of all strobes and the audible carbon monoxide message throughout the building.~~

~~(4) Recording of the event electronically in the history log of the FMCU.~~

~~]~~ o. System control equipment must be programmed to provide a 60-minute to 180-minute delay in transmission of trouble signals resulting from primary power failure.

~~[~~ p. Activation of a LOC pushbutton must activate the audible and visual alarms in the facility. The audible message must be the one associated with the pushbutton activated.

~~][~~1.6.3 Elevator Recall

Provide elevator recall in accordance with ASME A17.1/CSA B44, Section ~~[14 21 13 ELECTRIC TRACTION FREIGHT ELEVATORS][~~14 21 23 ELECTRIC TRACTION PASSENGER ELEVATORS~~][14 24 13 HYDRAULIC FREIGHT ELEVATORS][14 24 23 HYDRAULIC PASSENGER ELEVATORS]~~, and as specified herein. Activation of any smoke detector in an elevator shaft, machine room, or lobby (except at designated recall level) must cause all elevators associated with that shaft, machine room, or lobby to return nonstop to the designated level. Activation of a smoke detector in the lobby or machine room at the designated level must cause all elevators associated with that lobby to return nonstop to the assigned alternate level. Activation of a detector in an elevator shaft, machine room, or lobby must also cause illumination of elevator cab warning signal (fire hat) and complete operation of fire alarm system as specified in paragraph titled "Functions and Operating Features".

~~]~~1.7 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE

Technical data and site-specific software (meaning technical data that relates to computer software) that are specifically identified in this project, and may be required in other specifications, must be delivered, strictly in accordance with the CONTRACT CLAUSES. The fire alarm system manufacturer must submit written confirmation of this contract provision as "Fire Alarm System Site-Specific Software Acknowledgement". Identify data delivered by reference to the specification paragraph against which it is furnished. Data to be submitted must include complete system, equipment, and software descriptions. Descriptions must show how the equipment will operate as a system to meet the performance requirements of this contract. The site-specific software data package must also include the following:

a. Items identified in NFPA 72, titled "Site-Specific Software".

b. Identification of programmable portions of the system equipment and capabilities.

c. Description of system revision and expansion capabilities and methods of implementation detailing both equipment and software requirements.

d. Provision of operational software data on all modes of programmable portions for fire alarm and mass notification.

e. Description of Fire Alarm and Mass Notification Control Unit equipment operation.

f. Description of auxiliary and remote equipment operations.

g. Library of application software.

h. Operation and maintenance manuals.

~~[~~1.8 EXISTING EQUIPMENT

a. Equipment and devices must be compatible and operable with the existing fire alarm/mass notification system and must not impair reliability or operational functions of existing supervising station fire alarm system.~~[The proprietary type supervising station (PSS) is located [in Building [_____].][_____].][~~ The supervising equipment is existing and ~~consists of the following brands and models: [supervising station control unit [_____].][_____][signal reporting components [_____]], [annunciator [_____]][_____]]~~connects to fire alarm systems via direct copper connections.

~~[b. Equipment and devices must be compatible and operable with the existing building fire alarm/mass notification system. Equipment must not impair reliability or operational functions of the existing system. The existing building system control unit is [_____].~~

~~] c. Equipment and devices must be compatible and operable with the existing installation-wide mass notification system and must not impair reliability or operational functions of the existing system. The installation-wide mass notification system utilizes [_____] transceivers.~~

~~]~~1.9 QUALITY ASSURANCE

1.9.1 Submittal Documents

1.9.1.1 Preconstruction Submittals

Within 36 days of contract award but not less than ~~[~~14 days~~][_____]~~ prior to commencing any work on site, the Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the fire alarm subcontractor and QFPE must be returned disapproved without review. All resultant delays must be the sole responsibility of the Contractor.

1.9.1.2 Shop Drawings

Shop drawings must not be smaller than ~~[ISO A1][ANSI D][~~the Contract Drawings~~]~~. Drawings must comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans must be 1/8"=1'.

1.9.1.3 Nameplates

Nameplate illustrations and data to obtain approval by the Contracting Officer before installation.

1.9.1.4 Wiring Diagrams

~~[_____]~~3 copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FMCU and remote FMCU, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams must be job specific and must not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

1.9.1.5 System Layout

~~[_____]~~3 copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Indicate candela rating of each visual notification appliance. Indicate the wattage of each speaker. Clearly identify the locations of isolation modules. Indicate the addresses of all devices, modules, relays, and similar. Show/identify all acoustically similar spaces. Indicate if the environment for the FMCU is within its environmental listing (e.g. temperature/humidity).

Provide a complete description of the system operation in matrix format similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

~~[For air sampling smoke detection systems, provide floor plan layouts indicating location of fire alarm control unit, air sampling piping (lengths of pipe) and sampling ports (sizes and locations). Floor plan~~

~~must also indicate geographic monitor zone boundaries, location of display control unit, bar level annunciation panels if separate, and all other associated equipment that is required to provide a complete operational system.~~

~~]~~1.9.1.6 Notification Appliances

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for notification appliances. Annotate data for each circuit on the drawings.

1.9.1.7 Initiating Devices

Calculations and supporting data on each circuit to indicate that there is at least ~~[~~25~~][_____]~~ percent spare capacity for initiating devices. Annotate data for each circuit on the drawings.

1.9.1.8 Amplifiers

Calculations and supporting data to indicate that amplifiers have sufficient capacity to simultaneously drive all notification speakers at tapped settings plus ~~[~~25~~][_____]~~ percent spare capacity. Annotate data for each circuit on the drawings.

1.9.1.9 Battery Power

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component and each control unit component, and the battery recharging period, must be included on the drawings.

1.9.1.10 Voltage Drop Calculations

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings.

1.9.1.11 Product Data

~~[_____]~~3 copies of annotated descriptive data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, and options that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list must include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted must coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

~~[1.9.1.12 Air Sampling Smoke Detection System Calculations~~

~~Submit air sampling detection system design analysis calculations consisting of battery capacity, loading calculations, and fan speed and~~

~~air flow/transport calculations. Include schematic diagrams showing pipe segments, pipe diameters, lengths of pipe, node numbers, and sample port diameters to verify the requirements are met.~~

~~]~~1.9.1.12 Operation and Maintenance (O&M) Instructions

~~[~~Six~~][_____]~~ copies of the Operation and Maintenance Instructions. The O&M Instructions must be prepared in a single volume or in multiple volumes, with each volume indexed, and may be submitted as a Technical Data Package. Manuals must be approved prior to training. The Interior Fire Alarm And Mass Notification System Operation and Maintenance Instructions must include the following:

a. "Manufacturer Data Package ~~[~~five~~][_____]~~" as specified in ~~[~~Section 01 78 23 OPERATION AND MAINTENANCE DATA~~][_____]~~.

b. Operating manual outlining step-by-step procedures required for system startup, operation, and shutdown. The manual must include the manufacturer's name, model number, service manual, parts list, and preliminary equipment list complete with description of equipment and their basic operating features.

c. Maintenance manual listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. The manuals must include conduit layout, equipment layout and simplified wiring, and control diagrams of the system as installed.

d. Complete procedures for system revision and expansion, detailing both equipment and software requirements.

e. Software submitted for this project on CD/DVD media utilized.

f. Printouts of configuration settings for all devices.

g. Routine maintenance checklist. The routine maintenance checklist must be arranged in a columnar format. The first column must list all installed devices, the second column must state the maintenance activity or state no maintenance required, the third column must state the frequency of the maintenance activity, and the fourth column provided for additional comments or reference. All data (devices, testing frequencies, and similar) must comply with UFC 3-601-02.

h. A final Equipment List must be submitted with the Operating and Maintenance (O&M) manual.

1.9.1.13 As-Built Drawings

The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings must be provided at the time of, or prior to the final Government test.

1.9.2 Qualifications

1.9.2.1 Fire Alarm System Designer

The fire alarm system designer must be certified as a Level ~~[~~III~~][IV]~~ (minimum) Technician by National Institute for Certification in

Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology or meet the qualifications for a QFPE.

1.9.2.2 Supervisor

~~[~~A NICET Level ~~[~~III~~][~~ or ~~][~~IV~~]~~ fire alarm technician must supervise the installation of the fire alarm/mass notification system~~[, including the air sampling smoke detection system]~~.~~][A fire alarm technician with a minimum of eight years of experience must supervise the installation of the fire alarm/mass notification system.]~~ The fire alarm technicians supervising the installation of equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.3 Technician

Fire alarm technicians with a minimum of four years of experience must be utilized to install and terminate fire alarm/mass notification devices, cabinets and control units. The fire alarm technicians installing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings~~[, and must be thoroughly experienced in the installation of air sampling detection systems]~~.

1.9.2.4 Installer

~~[Fire alarm installer with a minimum of two years of experience utilized to assist in the installation of fire alarm/mass notification devices, cabinets and control units] [~~NICET Level II technician to assist in the installation of fire alarm/mass notification devices, cabinets and control units~~]~~. A licensed electrician must be allowed to install wire, cable, conduit and backboxes for the fire alarm system/mass notification system. The fire alarm installer must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.5 Test Technician

Fire alarm technicians with a minimum of eight years of experience and NICET Level ~~[~~III~~][~~ or ~~][~~IV~~]~~ utilized in testing and certification of the installation of the fire alarm/mass notification devices, cabinets and control units. The fire alarm technicians testing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

1.9.2.6 Manufacturer

Components must be of current design and must be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as specified herein.

1.9.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation must be considered as mandatory requirements.

1.10 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

1.11 MAINTENANCE

1.11.1 Spare Parts

Furnish the following spare parts in the manufacturers original unopened containers:

a. ~~[~~Five~~][_____]~~ complete sets of system keys.

b. ~~[~~Two~~][_____]~~ of each type of fuse required by the system.

c. ~~[~~One~~][_____]~~ manual stations.

~~[~~ d. ~~[~~Two~~][_____]~~ of each type of detector installed.

~~]~~ e. ~~[~~Two~~][_____]~~ of each type of detector base and head installed.

~~[~~ f. ~~[~~One~~][_____]~~ electromagnetic door holders.

~~][g. One smoke detector manufacturer's test screen, card or magnet for each ten beam smoke detectors, or fraction thereof, installed in the system.~~

~~][h. Two air sampling smoke detection system filter assemblies.~~

~~]~~ g. ~~[~~Two~~][_____]~~ of each type of audible and visual alarm device installed.

h. ~~[~~One~~][_____]~~ textual visual notification appliance.

i. ~~[~~Two~~][_____]~~ of each type of addressable monitor module installed.

j. ~~[~~Two~~][_____]~~ of each type of addressable control module installed.

k. ~~[~~Two~~][_____]~~ low voltage, ~~[~~one~~][_____]~~ ~~[~~telephone~~][internet][ethernet]~~, and ~~[~~one~~][_____]~~ 120 VAC surge protective device.

~~[l. [Two][_____] spare reams of paper for the system printer, plus sufficient paper for testing.~~

~~][m. [Two][_____] spare printer ribbons.~~

~~]~~1.11.2 Special Tools

Software, connecting cables and proprietary equipment, necessary for the maintenance, testing, and reprogramming of the equipment must be furnished to the Contracting Officer, prior to the instruction of Government employees.

PART 2 PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENT

All fire alarm and mass notification equipment must be listed for use under the applicable reference standards. Interfacing of UL 864 or similar approved industry listing with Mass Notification equipment listed to UL 2572 must be done in a laboratory listed configuration, if the software programming features cannot provide a listed interface control.

2.2 MATERIALS AND EQUIPMENT

2.2.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by NFPA 72 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment must be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least ~~[~~2~~][_____]~~ years prior to bid opening.

2.2.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Major components include, but are not limited to, the following:

a. FMCU

Nameplates must be etched metal or plastic, permanently attached by screws to control units or adjacent walls.

2.2.3 Keys

Keys and locks for equipment, control units and devices must be identical. ~~[~~Master all keys and locks to a single key as required by the ~~[~~ Installation Fire Department~~][_____]]~~. ~~[Keys must be CAT [60][_____].]~~

2.2.4 Instructions

Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. ~~[~~ Install the instructions on the interior of the FMCU.~~][Install the frame in a conspicuous location observable from the FMCU.]~~ The card must show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions must also include procedures for operating live voice microphones. The instructions and their mounting location must be approved by the Contracting Officer before being posted.

2.3 FIRE ALARM AND MASS NOTIFICATION CONTROL UNIT

Provide a complete fire alarm and mass notification control unit (FMCU) fully enclosed in a lockable steel cabinet as specified herein. Operations required for testing or for normal care, maintenance, and use of the system must be performed from the front of the enclosure. If more than a single unit is required at a location to form a complete control unit, the unit cabinets must match exactly.~~[If more than a single unit is required, and is located in the lobby/entrance, notify the Contracting Officer's Designated Representative (COR), prior to installing the equipment.]~~ The system must be capable of defining any module as an alarm module and report alarm trouble, loss of polling, or as a supervisory module, and reporting supervisory short, supervisory open or loss of polling such as waterflow switches, valve supervisory switches, fire pump monitoring, independent smoke detection systems, relays for output function actuation.

a. Each control unit must provide power, supervision, control, and logic for the entire system, utilizing modular components, internally mounted and arranged for easy access. Each control unit must be suitable for operation on a 120 volt, ~~60~~50 hertz, normal building power supply. Provide each control unit with supervisory functions for power failure, internal component placement, and operation.

b. Visual indication of alarm, supervisory, or trouble initiation on the FMCU must be by liquid crystal display or similar means with a minimum of 80 characters. The mass notification control unit must have the capability of temporarily deactivate the fire alarm audible notification appliances while delivering voice messages.

c. Provide secure operator console for initiating recorded messages, strobes and displays; and for delivering live voice messages. Provide capacity for at least eight prerecorded messages. Provide the ability to automatically repeat prerecorded messages. Provide a secure microphone for delivering live messages. Provide adequate discrete outputs to temporarily deactivate fire alarm audible notification, initiate/synchronize strobes and initiate textual visual notification appliances. Provide a complete set of self-diagnostics for controller and appliance network. Provide local diagnostic information display and local diagnostic information and system event log file.

2.3.1 Cabinet

Install control unit components in cabinets large enough to accommodate all components and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and ~~[~~surface~~][semi-recessed]~~ mounting provisions. The enclosure must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must say "Fire Alarm and Mass Notification control unit" and must not be less than 25 mm high. Provide prominent rigid plastic or metal identification plates for lamps, circuits, meters, fuses, and switches.

2.3.2 Silencing Switches

2.3.2.1 Alarm Silencing Switch

Provide an alarm silencing switch at the FMCU that must silence the audible and visual notification appliances. Subsequent activation of initiating devices must cause the notification appliances to re-activate.

2.3.2.2 Supervisory/Trouble Silencing Switch

Provide supervisory and trouble silencing switch(es) that must silence the audible trouble and supervisory signal(s), but not extinguish the visual indicator. This switch must be overridden upon activation of a subsequent supervisory or trouble condition. Audible trouble indication must resound automatically every 24 hours after the silencing feature has been operated if the supervisory or trouble condition still exists.

2.3.3 Non-Interfering

Power and supervise each circuit such that a signal from one device does not prevent the receipt of signals from any other device. Initiating devices must be manually reset by switch from the FMCU after the initiating device or devices have been restored to normal.

2.3.4 Audible Notification System

The Audible Notification System must comply with the requirements of NFPA 72 for Emergency Voice/Alarm Communications System requirements, except as specified herein. The system must be a one-way, multi-channel voice notification system incorporating user selectability of a minimum eight distinct sounds for tone signaling, and the incorporation of a voice module for delivery of recorded messages. Audible appliances must produce a three-pulse temporal pattern for three cycles followed by a voice message that is repeated until the control unit is reset or silenced.~~ For carbon monoxide detector activation, audible appliances must produce a four-pulse temporal pattern for three cycles followed by a voice message that is repeated until the control unit is reset or silenced.~~ Automatic messages must be broadcast through speakers throughout the building/facility but not in stairs or elevator cabs. A live voice message must override the automatic audible output through use of a microphone input at the control unit or the LOC.

a. When using the microphone, live messages must be broadcast ~~[~~throughout a selected floor or floors, ~~][selectable by zone,]~~ or all call. The system must be capable of operating all speakers at the same time.~~ [The Audible Notification System must support Public Address (PA) paging for the facility.][This must be accomplished with the provision of a separate microphone with a head unit that interfaces with the FMCU. The public address paging function must not override any fire alarm or mass notification functions. The microphone must be [desktop][hand-held][_____] style. Hand-held microphones must be housed in a separate protective cabinet. The cabinet must be accessible without the use of a key. The location of the microphone(s) must be approved by the [_____] Designated Fire Protection Engineer (DFPE).] Activation of the public address microphone must not initiate activation of visual notification appliances or LED text displays.~~

b. The microprocessor must actively interrogate circuitry, field wiring, and digital coding necessary for the immediate and accurate rebroadcasting of the stored voice data into the appropriate amplifier input. Loss of operating power, supervisory power, or any other malfunction that could render the digitalized voice module inoperative must automatically cause the three-pulse temporal pattern to take over all functions assigned to the failed unit in the event an alarm is activated.

2.3.4.1 Outputs and Operational Modules

All outputs and operational modules must be fully supervised with on-board diagnostics and trouble reporting circuits. Provide form "C" contacts for system alarm and trouble conditions. Provide circuits for operation of auxiliary appliance during trouble conditions. During a Mass Notification event, the control unit must not generate nor cause any trouble alarms to be generated with the Fire Alarm system.

2.3.4.2 Mass Notification

a. The system must have the capability of utilizing an LOC with redundant controls of the FMCU. Notification Appliance Circuits (NAC) must be provided for the activation of strobe appliances. Audio output must be selectable for line level. A hand-held microphone must be provided and, upon activation, must take priority over any tone signal, recorded message or PA microphone operation in progress, while maintaining the strobe NAC circuit activation.

b. The Mass Notification functions must override the manual or automatic fire alarm notification~~, [and public address (PA) functions]~~. Other fire alarm functions including transmission of a signal(s) to the fire department must remain operational. When a mass notification announcement is disengaged and a fire alarm condition still exists, the audible and visual notification appliances must resume activation for alarm conditions. The fire alarm message must be of lower priority that all other messages (except any "test" messages) and must not override any other messages. If not manually ended, any mass notification announcement will automatically end after 10 minutes.

~~[~~ c. Messages must be recorded professionally utilizing standard industry methods, in a professional ~~[male][~~female~~]~~ voice. Message and tone volumes must both be at the same decibel level. Messages recorded from the system microphone must not be accepted. A 1000 Hz tone (as required by NFPA 72) must precede messages and be similar to the following unless Installation or Facility specific messages are required:

~~(1) "May I have your attention please. May I have your attention please. [Insert installation specific message here.]" (Provide a [2][_____] second pause.) "May I have your attention please, (repeat the tones and message [on a continuous loop][[_____] times])."~~

~~(2) Carbon Monoxide: "May I have your attention please. May I have your attention please. Carbon monoxide has been detected in the building. Please walk to the nearest exit and leave the building." (Provide a [2][_____] second pause.) "May I have your attention please, (repeat the tones and message [on a continuous loop][[_____] times])."~~

(1) Fire: "May I have your attention please. May I have your attention please. A fire emergency has been reported in the building. Please leave the building by the nearest exit~~[~~ or exit stairway~~]~~.~~[~~ Do not use the elevators.~~]~~" (Provide a ~~[~~2~~][_____]~~ second pause.) "May I have your attention please, (repeat the tones and message on a continuous loop)."

(2) Test: "May I have your attention please. May I have your attention please. This is a test of the building mass notification system. Please continue your normal duties. This is only a test." (Provide a ~~[~~2~~][____]~~ second pause.)

(3) All Clear: "May I have your attention please. May I have your attention please. An all clear has been issued, resume normal activities." (Provide a ~~[~~2~~][____]~~ second pause.)

~~]~~ d. Auxiliary Input Module must be designed to be an outboard expansion module to either expand the number of optional LOC's, or allow a telephone interface.

~~[2.3.4.3 Installation-Wide Control~~

~~If an installation-wide control system for mass notification exists on the Base, the autonomous control unit must communicate with the central control unit of the Installation-wide system. The autonomous control unit must receive commands/messages from the central control unit and provide status information.~~

~~]~~2.3.5 Memory

Provide each control unit with non-volatile memory and logic for all functions. The use of long life batteries, capacitors, or other age-dependent devices must not be considered as equal to non-volatile processors, PROMS, or EPROMS.

2.3.6 Field Programmability

Provide control units and control units that are fully field programmable for both input and output of control, initiation, notification, supervisory, and trouble functions. The system program configuration must be menu driven. System changes must be password protected. Any proprietary equipment and proprietary software needed by qualified technicians to implement future changes to the fire alarm system must be provided as part of this contract.

2.3.7 Input/Output Modifications

The FMCU must contain features that allow the bypassing of input devices from the system or the modification of system outputs. These control features must consist of a control unit mounted keypad~~[and a keyboard]~~. Any bypass or modification to the system must indicate a trouble condition on the FMCU.

2.3.8 Resetting

Provide the necessary controls to prevent the resetting of any alarm, supervisory, or trouble signal while the alarm, supervisory or trouble condition on the system still exists.

2.3.9 Walk Test

The FMCU must have a walk test feature. When using this feature, operation of initiating devices must result in limited system outputs, so that the notification appliances operate for only a few seconds and the event is indicated in the history log~~[and on the system printer]~~, but no other outputs occur.

2.3.10 History Logging

The control unit must have the ability to store a minimum of 400 events in a log. These events must be stored in a battery-protected memory and must remain in the memory until the memory is downloaded or cleared manually. Resetting of the control unit must not clear the memory.

2.3.11 Manual Access

An operator at the control unit, having a proper access level, must have the capability to manually access the following information for each initiating device.

a. Primary status.

b. Device type.

c. Present average value.

d. Present sensitivity selected.

e. Detector range (normal, dirty).

~~[2.3.12 Heat Detector Self-Test Routines~~

~~Automatic self-test routines must be performed on each detector that will functionally check detector sensitivity electronics and ensure the accuracy of the value being transmitted. Any detector that fails this test must indicate a trouble condition with the detector location at the control unit.~~

~~]~~2.4 LOCAL OPERATING CONSOLES (LOC)

2.4.1 General

The LOC must consist of a remote microphone station incorporating a push-to-talk (PTT) hand-held microphone and system status indicators. The LOC must have the capability of being utilized to activate prerecorded messages. The unit must incorporate microphone override of any tone generation or recorded messages. The unit must be fully supervised from the FMCU. The housing for the LOC must not be lockable.~~[The LOC must have public address capability with the provision of a separate microphone. The PA paging function must not override any alarm or notification functions. The PA microphone must be [desktop][hand-held][_____] style. Hand-held microphones must be housed in a separate protective cabinet. The cabinet must be accessible without the use of a key. The location of the microphone[s] must be approved by the [_____] Designated Fire Protection Engineer (DFPE). Activation of the PA microphone must not initiate activation of visual notification appliances or LED text displays. The PA paging function must not override any alarm or notification functions.]~~

2.4.2 Multiple LOCs

When an installation has more than one LOC, the LOCs must be programmed to allow only one LOC to be available for paging or messaging at a time. Once one LOC becomes active, all other LOC's will have an indication that the system is busy (Amber Busy Light) and cannot be used at that time.

This is to avoid two messages being given at the same time. It must be possible to override or lockout the LOC's from the FMCU.

2.5 AMPLIFIERS, PREAMPLIFIERS, TONE GENERATORS

Any amplifiers, preamplifiers, tone generators, digitalized voice generators, and other hardware necessary for a complete, operational, textual audible circuit conforming to NFPA 72 must be housed in a remote FMCU, terminal cabinet, or in the FMCU. Individual amplifiers must be 100 watts maximum.

2.5.1 Operation

The system must automatically operate and control all building speakers~~[except those installed in the stairs][and within elevator cabs]~~.~~[The speakers in the stairs[and elevator cabs] must operate only when the microphone is used to deliver live messages.]~~

2.5.2 Construction

Amplifiers must utilize computer grade components and must be provided with output protection devices sufficient to protect the amplifier against any transient up to 10 times the highest rated voltage in the system.

2.5.3 Inputs

Equip each system with separate inputs for the tone generator, digitalized voice driver and control unit mounted microphone~~ [Public Address Paging Function]~~. Microphone inputs must be of the low impedance, balanced line type. Both microphone and tone generator input must be operational on any amplifier.

2.5.4 Tone Generator

The tone generator must produce a three-pulse temporal pattern and must be constantly repeated until interrupted by either the digitalized voice message, the microphone input, or the alarm silence mode as specified. The tone generator must be single channel with an automatic backup generator per channel such that failure of the primary tone generator causes the backup generator to automatically take over the functions of the failed unit and also causes transfer of the common trouble relay. The tone generator must be provided with securely attached labels to identify the component as a tone generator and to identify the specific tone it produces.

2.5.5 Protection Circuits

Each amplifier must be constantly supervised for any condition that could render the amplifier inoperable at its maximum output. Failure of any component must cause illumination of a visual "amplifier trouble" indicator on the control unit, appropriate logging of the condition in the history log~~[and on the system printer]~~, and other actions for trouble conditions as specified.

~~[2.6 VIDEO DISPLAY UNIT (VDU)~~

a. ~~The VDU must be the secondary operator-to-system interface for data retrieval, alarm annunciation, commands, and programming functions. The desk mounted VDU must consist of a LCD monitor and a keyboard.~~

~~The VDU must have a [300][430][_____]-mm minimum [touch]screen, capable of displaying 25 lines of 80 characters each. Communications with the FMCU must be supervised. Faults must be recorded in the history log[and on the printer]. Power required must be 120 VAC, 60 Hz from the same source as the FMCU.~~

~~b. To eliminate confusion during an alarm situation, the screen must have dedicated areas for the following functions:~~

~~(1) Alarm and return to normal.~~

~~(2) Commands, reports, and programming.~~

~~(3) Time, day, and date.~~

~~c. Use full English language throughout to describe system activity and instructions. Full English language descriptors defining system points must be 100 percent field programmable by factory trained personnel, alterable and user definable to accurately describe building areas.~~

~~d. Alarms and other changes of status must be displayed in the screen area reserved for this information. Upon receipt of alarm, an audible alarm must sound and the condition and point type must flash until acknowledged by the operator. Return to normal must also be annunciated and must require operator acknowledgment. The following information must be provided in full English:~~

~~(1) Condition of device (alarm, trouble, or supervisory).~~

~~(2) Type of device (for example, manual pull, waterflow)~~

~~(3) Location of device plus numerical system address.~~

~~e. The system must have multiple levels of priority for displaying alarms to conform with UL 864. Priority levels must be as follows:~~

~~(1) Level 1 - Mass Notification Signals~~

~~(2) Level 2 - Fire Alarm Signals~~

~~(3) Level 3 - Carbon Monoxide Alarm Signals~~

~~(4) Level 4 - Supervisory Signals~~

~~(5) Level 5 - Trouble Signals~~

~~f. Provide the system with memory so that no alarm is lost. A highlighted message must advise the operator when unacknowledged alarms are in the system.~~

~~g. Multiple levels of access must be provided for operators and supervisors via user-defined passwords. Provide the following functions for each level:~~

~~(1) Operator level access functions:~~

~~(a) Display system directory, definable by device.~~

~~(b) Display status of an individual device.~~

~~(c) Manual command (alarm device with an associated command must use the same system address for both functions).~~

~~(d) Report generation, definable by device, output on the VDU[or printer], as desired by the operator.~~

~~(e) Activate building notification appliances.~~

~~(2) Supervisor level access functions:~~

~~(a) Reset time and date.~~

~~(b) Enable or disable event initiated programs[, printouts,] and initiators.~~

~~(c) Enable or disable individual devices and system components.~~

~~h. The above supervisor level functions must not require computer programming skills. Changes to system programs must be [recorded on the printer and]maintained in the control unit as a trouble condition.~~

~~]~~2.6 REMOTE ANNUNCIATOR

~~[~~2.6.1 LCD Annunciator

Provide a ~~[~~semi-recessed~~][flush]~~ mounted annunciator that includes an LCD display. The display must indicate the device in trouble/alarm or any supervisory device. Display the device name, address~~[~~, and actual building location~~]~~. The remote annunciator must duplicate functions of the FMCU for message display, fire alarm, supervisory alarm, and trouble conditions, visual and audible notification, and system reset functions. Remote annunciator must require the use of a key for accessing the reset, control and other functions.

A building floor plan must be provided and mounted (behind Plexiglass or similar protective material) at the annunciator location. The floor plan must indicate all rooms by name and number including the locations of stairs and elevators. The floor plan must show all devices and their programmed address to facilitate identification of their physical location from the LCD display information.

~~][2.6.2 Graphic Annunciator~~

~~Graphic annunciator must be of the [interior][weatherproof] type, [flush][surface][pedestal]-mounted. Annunciator must be provided with the [building][room] floor plan, drawn to scale, with alarm lamps mounted to represent the location of [each concealed detector][each initiating device]. Annunciator graphic must also show the locations of the annunciator and control unit, and must have a "you are here" arrow showing its location. Orient building floor plan on graphic to location of person viewing the graphic(i.e., the direction the viewer is facing must be toward the top of the graphic display). Provide a North arrow.[Principal rooms and areas shown must be labeled with room numbers or titles.] Detectors mounted above ceilings, [on ceilings,]and beneath raised floors and different types of initiating devices must have different symbols or lamps of different colors for identification. Lamps must illuminate upon activation of corresponding device and must remain~~

~~illuminated until the system is reset. Annunciator must have a lamp test switch.~~

~~2.6.2.1 Materials~~

~~Construct the graphic annunciator face plate of [smoked Plexiglas][non-glare matte finish][anodized bronze][anodized aluminum]. The face plate must be backlit with LEDs. Control equipment and wiring must be housed in a [recessed][semi-recessed][surface mounted] back box. The exposed portions of the back box must be [chrome plated][anodized bronze][anodized aluminum] without knockouts.~~

~~2.6.2.2 Programming~~

~~Where programming for the operation of the graphic annunciator is accomplished by a separate software program other than the software for the FMCU, the software program must not require reprogramming after loss of power. The software must be reprogrammable in the field.~~

~~][2.6.3 Printer~~

~~a. Provide a system printer [with no stored memory] to record alarm, supervisory, and trouble conditions without loss of any signal or signals. Printout must be by circuit, device, and function as provided in the FMCU. Printer must operate on a 120 VAC, 60 Hz power supply.~~

~~b. The printer must have at least 80 characters per line and have a 96 ASCII character set. The printer must have a microprocessor-controlled, bi-directional, logic seeking head capable of printing 120 characters per second utilizing a 9 by 7 dot matrix print head. Printer must not contain internal software which is essential for proper operation.~~

~~c. When the FMCU receives a signal, the alarm, supervisory, and trouble condition must be printed. The printout must include the type of signal, the circuit or device reporting, the date, and the time of the occurrence. The printer must differentiate alarm signals from other printed indications. When the system is reset, this condition must also be printed including the same information concerning device, location, date, and time. Provide a means to automatically print a list of existing alarm, supervisory, and trouble conditions in the system. If a printer is off-line when an alarm is received, the system must have a buffer to retain the data and it must be printed when the printer is restored to service. The printer must have an indicator to alert the operator that the paper has run out.~~

~~]~~2.7 MANUAL STATIONS

Provide metal or plastic, ~~[~~semi-flush~~][flush][surface]~~ mounted,double-action, addressable manual stations, that are not subject to operation by jarring or vibration.~~[Provide separately addressed conventional single-action manual pull stations[in hazardous locations,][in wet and damp locations,][and][where indicated on plans].]~~Stations must be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations must be finished in red with molded raised lettering operating instructions of contrasting color. The use of a key must be required to reset the station.

2.8 SMOKE DETECTORS

2.8.1 Spot Type Detectors

Provide addressable ~~[~~photoelectric~~][ionization][laser]~~ smoke detectors as follows:

a. Provide analog/addressable ~~[~~photoelectric smoke detectors utilizing the photoelectric light scattering principle for operation in accordance with UL 268~~][smoke detectors that operate on the ionization principle and are actuated by the presence of visible or invisible products of combustion][laser smoke detectors utilizing laser diode and patented smoke sensing chamber, designed to amplify signals from smoke but diminish stray internal reflections and must, on command from the FMCU, send data to the control unit representing the analog level of smoke density]~~. Smoke detectors must be listed for use with the FMCU.

b. Provide self-restoring type detectors that do not require any readjustment after actuation at the FMCU to restore them to normal operation. The detector must have a visual indicator to show actuation.

c. Vibration must have no effect on the detector's operation. Protect the detection chamber with a fine mesh metallic screen that prevents the entrance of insects or airborne materials. The screen must not inhibit the movement of smoke particles into the chamber.

d. Provide twist lock bases ~~[~~with sounder that produces a minimum of 90 dBA at 3 m~~]~~ with screw terminals for each conductor. The detectors must maintain contact with their bases without the use of springs.

e. The detector address must identify the particular unit, its location within the system~~[~~, and its sensitivity setting~~]~~. Detectors must be of the low voltage type rated for use on a 24 VDC system.

~~[f. Laser smoke detector must be listed for use with the FMCU. Detector must be able to achieve sensitivities from 0.02 percent-per-foot to 2 percent-per-foot obscuration.~~

~~g. Laser smoke detector must provide point identification of the fire location through addressability, must experience no delay in response time due to smoke dilution or smoke transportation time, and must offer complete supervision of wiring and detector.~~

~~][2.8.2 Projected Beam Smoke Detector~~

~~Detectors must consist of [combined transmitter and receiver unit] [separate transmitter and receiver units]. The transmitter unit must emit an infrared beam to the receiver unit [the use of a supplied reflector is required for the combined unit]. When the signal at the receiver falls below a preset threshold, the detector must initiate an alarm. The receiver must contain an LED status indicator that illuminates when an alarm condition exists. Long-term changes to the received signal caused by environmental variations must be automatically compensated. Detectors must incorporate features to assure that they are operational; a trouble signal must be initiated if the beam is obstructed for more than 3 seconds, the limits of the compensation circuit are reached, or the~~

~~housing cover is removed. Detectors must have multiple sensitivity settings in order to meet UL listings for the different distances covered by the beam.~~

~~]~~2.8.2 Duct Smoke Detectors

Duct-mounted addressable photoelectric smoke detectors must consist of a smoke detector, as specified in paragraph Spot Type Detectors, mounted in a special housing fitted with duct sampling tubes. Detector circuitry must be mounted in a metallic or plastic enclosure exterior to the duct.~~[~~ It is not permitted to cut the duct insulation to install the duct detector directly on the duct.~~]~~ Detectors must be listed for operation over the complete range of air velocities, temperature and humidity expected at the detector when the air-handling system is operating. Detectors must be powered from the FMCU.

a. Sampling tubes must run the full width of the duct. The duct detector package must conform to the requirements of NFPA 90A, UL 268A, and must be listed for use in air-handling systems. The control functions, operation, reset, and bypass must be controlled from the FMCU.

b. Lights to indicate the operation and alarm condition must be visible and accessible with the unit installed and the cover in place. Remote indicators must be provided where required by NFPA 72. Remote indicators as well as the affected fan units must be properly identified in etched plastic placards.

c. Detectors must provide for control of auxiliary contacts that provide control, interlock, and shutdown functions specified in Section 23 09 00 to INSTRUMENTATION AND CONTROL FOR HVAC. Auxiliary contacts provide for this function must be located within 1 m of the controlled circuit or appliance. The auxiliary contacts must be supplied by the fire alarm system manufacturer to ensure complete system compatibility.

~~[2.9 AIR SAMPLING SMOKE DETECTION SYSTEM~~

~~The [addressable]air sampling smoke system must consist of a detector assembly housing an integral aspiration fan, filter, laser-based detection chamber and control, output and supervision circuitry.[Each sampling point must be capable of being independently addressable.] The system must consist of a piping or tubing distribution network that runs from the detector assembly(s) to the protected area(s) and is supported by air sampling smoke detection system calculations from a computer-based design modeling tool. The system must include configurable alarm and trouble relay outputs for interface to other systems where required.~~

~~a. System must be complete in all ways. It must include all engineering, and electrical installation, all detection and control equipment, auxiliary devices and controls, alarm interface, functional checkout and testing, training and all other operations necessary for a functional system.~~

~~b. System base detectors and modules must each accommodate up to [40 addressable][_____] microbore sampling tubes where each tube has a sampling point at the end. Additional modules may be used to provide up to [20 addressable][_____] sampling holes per system.~~

~~c. Program alarm thresholds to the following values unless the results of~~

~~the pre-Government system tests indicate a clear need to change them. In the event that such a need is indicated, notify the Contracting Officer and provide complete documentation concerning the need to deviate from these values. Include within the deviation documentation request, information that complies with the paragraph entitled "Sensitivity Verification Test". Ensure initial threshold levels are approved prior to the Government test.~~

~~(1) Alarm Level 1: set ALERT at [_____][0.0250] percent obscuration/foot~~

~~(2) Alarm Level 2: set PRE-ALARM at [_____][0.0500] percent obscuration/foot~~

~~(3) Alarm Level 3: set FIRE 1 at [_____][0.1000] percent obscuration/foot~~

~~(4) Alarm Level 4: set FIRE 2 at [_____][0.2000] percent obscuration/foot~~

~~d. All air sampling smoke detection devices and associated components must be new, standard products or the manufacturer's latest design and suitable to perform the functions intended.~~

~~e. The laser detection chamber must be of the mass light scattering type and capable of detecting a wide range of smoke particle types of varying size. A particle counting method must be employed for the purposes of:~~

~~(1) Preventing large particles from affecting the true smoke reading.~~

~~(2) Monitoring contamination of the filter (for example, dust and dirt) to automatically notify when maintenance is required. The particle counting method must not be used for the purpose of smoke density measurement.~~

~~f. Detector(s) must be self-monitoring for filter contamination and provide indication through system fault when replacement is necessary. Detectors which allow automatic reset of filter status upon removal and re-insertion are not permitted.~~

~~g. Detector(s) must contain relays for alarm and fault conditions. The relays must be software programmable to the required functions.~~

~~h. Detector(s) must permit configuration by programmers that are either integral to the system, portable or PC based.~~

~~i. Detector(s) must allow programming of:~~

~~(1) Smoke threshold alarm levels; ALERT, PRE-ALARM, FIRE 1 and FIRE 2.~~

~~(2) Time delays. Ensure the display control unit contains individual adjustable alarm time delay features for each of the alarm threshold levels. Provide an adjustment range between 0 and 60 seconds. Program the alarm threshold time delays to 30 seconds for alarm levels 1 and 2, and 15 seconds for alarm levels 3 and 4.~~

~~(3) Faults, including airflow, detector, power, filter and network, as well as an indication of the urgency of the fault.~~

~~(4) Configuration of relay outputs for remote indication of alarm and fault conditions.~~

~~(5) General purpose input functionality.~~

~~]2.10 HEAT DETECTORS~~

~~2.10.1 Heat Detectors~~

~~Heat detectors must be analog/addressable and designed for detection of fire by [fixed temperature][combination fixed temperature and rate-of-rise principle][rate-compensating principle] in accordance with UL 521. The alarm condition must be determined by comparing detector value with the stored values.. Detectors located in areas subject to moisture, exterior atmospheric conditions, or hazardous locations [as defined by NFPA 70][and][as indicated], must be types approved for such locations.~~

~~[2.10.1.1 Combination Fixed-Temperature and Rate-of-Rise Detectors~~

~~Detectors must be [surface][semi-flush] mounted in the [vertical][horizontal] orientation and supported independently of wiring connections. Detectors must be self-resetting. Detector must operate at [57.2][90] degrees C. Detector must feature rate compensation. [Detectors rated to operate at 57.2 degrees C must not respond to momentary temperature fluctuations less than 16.7 degrees C per minute between 16 and 38 degrees C].[Detectors rated to operate at 90 degrees C must not respond to momentary temperature fluctuations less than 27.8 degrees C per minute between 16 and 66 degrees C.][The detector assembly must be [weatherproof][and][explosionproof].]~~

~~][2.10.1.2 Rate Compensating Detectors~~

~~Detector backbox must be [surface][flush] mounted in the [vertical][horizontal] orientation and supported independently of wiring connections. Detectors must be self-restoring and hermetically sealed.[The detector assembly must be [weatherproof][and][explosionproof].]~~

~~][2.10.1.3 Line-Type Fixed Temperature Detectors~~

~~Provide [thermostatic][or][thermistor] line-type heat detection cable [with weather-resistant outer covering] where indicated. Cable must be nominally rated for a temperature of [68][88][138] degrees C and must operate on fixed temperature principle only.~~

~~][2.10.1.4 Fixed Temperature Detectors~~

~~Detectors must be [surface][semi-flush] mounted in the [vertical][horizontal] orientation and supported independently of wiring connections. Detectors must be self-restoring. The detectors must have a specific temperature setting [of [57.2][_____] degrees C][as shown].[The detector assembly must be [weatherproof][and][explosionproof].]~~

~~][2.11 FLAME DETECTORS~~

~~Detectors must be sensitive to the micron range best suited for their intended use. Detectors must operate over electrically supervised wiring circuits and the loss of power to the detector must result in a trouble signal. A self-test feature must be provided for each detector to be~~

~~individually tested.~~

~~[2.11.1 Infrared (IR) Single Frequency Flame Detector~~

~~The detector must be sensitive in the range of [_____] to [_____] micrometers only.~~

~~][2.11.2 Infrared (IR) Multi Frequency Flame Detector~~

~~The IR detector must consist of three or more IR sensors, each selected for a different IR frequency. The primary sensor must be sensitive in the range of [_____] to [_____] micrometers only. Secondary sensors are tuned to different IR wavelengths to null out the effect of black body radiation to the primary sensor.~~

~~][2.11.3 Ultraviolet (UV) Flame Detectors~~

~~UV flame detector must be of the narrow band response type which operates on radiated ultraviolet energy and must be sensitive in the range of [_____] to [_____] micrometers only. The cone of vision must be 80 degrees or greater. Each detector must be completely insensitive to light sources in the visible frequency range.~~

~~][2.11.4 Combination UV/IR Flame Detector~~

~~The UV/IR detector must provide discrimination against false alarms by requiring both UV and IR flame detection before an alarm is sent. The UV sensor must be sensitive in the range of 0.185 to 0.265 micrometers only. The IR sensor must be sensitive in the range of [_____] to [_____] micrometers only. Detectors must be completely insensitive to light sources in the visible frequency range.~~

~~]][2.12 MULTI-CRITERIA DETECTORS~~

~~Multi-criteria detectors must contain [fixed temperature [_____] degrees C heat sensor], [rate-of-rise heat sensor], [photoelectric smoke detector], [_____] elements in a single housing.~~

~~]2.13 CARBON MONOXIDE DETECTOR~~

~~Analog/addressable carbon monoxide (CO) detectors must be listed to UL 2075 and set to respond to the sensitivity limits of UL 2034. Carbon monoxide detectors must be listed for use with fire alarm control units. Detectors must be [surface] [semi-flush] mounted in the [vertical][horizontal] orientation and supported independently of wiring connections. Detectors must be self-restoring. For FMCU with no listed compatible addressable CO detectors, provide listed 4-wire detectors.[Do not provide CO detectors with local alarms.] Detector must be provided with an LED status indicator.~~

~~a. Where 4-wire CO detectors are necessary, each 4-wire CO detector must be individually monitored via addressable interface modules for alarm and off normal/trouble conditions (including loss of power to the individual detector). Power circuits for 4-wire CO detectors must be dedicated to powering the CO detectors only. Battery powered and 120 VAC powered detectors are prohibited.~~

~~b. Wiring connections must be made by means of screw terminals and detectors must be equipped with trouble relays. Detectors must be~~

~~able to mount a single-gang electrical box.~~

~~c. A trouble condition at an individual CO detector must not affect any other CO detectors. CO detectors must be powered by the FMCU.~~

~~d. Detectors must be provided with a means to test CO gas entry into the CO sensing cell.~~

2.9 ADDRESSABLE INTERFACE DEVICES

The initiating device being monitored must be configured as a ~~[Class "A"][~~ Class "B"~~]~~ initiating device circuits. The module must be listed as compatible with the control unit. The module must provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. Monitor module must contain an integral LED that flashes each time the monitor module is polled and is visible through the device cover plate. Pull stations with a monitor module in a common backbox are not required to have an LED.~~[Existing fire alarm system initiating device circuits must be connected to a single module to supervise the circuit.]~~ Modules must be listed for the environmental conditions in which they will be installed.

2.10 ADDRESSABLE CONTROL MODULES

The control module must be capable of operating as a relay (dry contact form C) for interfacing the control unit with other systems, and to control door holders or initiate elevator fire service. The module must be listed as compatible with the control unit. The indicating device or the external load being controlled must be configured as ~~[~~Class B~~][Class A]~~ notification appliance circuits. The system must be capable of supervising, audible, visual and dry contact circuits. The control module must have both an input and output address. The supervision must detect a short on the supervised circuit and must prevent power from being applied to the circuit. The control module must provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. The control module must contain an integral LED that flashes each time the control module is polled and is visible through the device cover plate. Control Modules must be listed for the environmental conditions in which they will be installed.

~~[~~2.11 ISOLATION MODULES

a. Provide isolation modules to subdivide each signaling line circuit ~~[into groups of not more than [20 addressable devices][_____]][~~each floor~~][~~ in accordance with NFPA 72~~]~~ between adjacent isolation modules.

b. Isolation modules must provide short circuit isolation for signaling line circuit wiring.

c. Power and communications must be supplied by the SLC and must report faults to the FMCU.

d. After the wiring fault is repaired, the fault isolation modules must test the lines and automatically restore the connection.

~~]~~2.12 NOTIFICATION APPLIANCES

2.12.1 Audible Notification Appliances

Audible appliances must conform to the applicable requirements of UL 464. Appliances must be connected into notification appliance circuits. Surface mounted audible appliances must be painted ~~[red][~~white~~][_____]~~. Recessed audible appliances must be installed with a grill that is painted ~~[red][~~white~~][_____][with a factory finish to match the surface to which it is mounted]~~.

2.12.1.1 Speakers

a. Speakers must conform to the applicable requirements of UL 1480. Speakers must have six different sound output levels and operate with audio line input levels of 70.7 VRMs and 25 VRMs, by means of selectable tap settings. Interior speaker tap settings must include taps of 1/4, 1/2, 1, and 2 watt, at a minimum. Exterior speakers must also be multi-tapped with no more than 15 watt maximum setting. Speakers must incorporate a high efficiency speaker for maximum output at minimum power across a frequency range of 400 Hz to 4,000 Hz, and must have a sealed back construction. Speakers must be capable of installation on standard 100 mm square electrical boxes. Where speakers and strobes are provided in the same location, they may be combined into a single ~~[wall mounted]~~ unit. All inputs must be polarized for compatibility with standard reverse polarity supervision of circuit wiring via the FMCU.

b. Provide speaker mounting plates constructed of cold rolled steel having a minimum thickness of 1.519 mm or molded high impact plastic and equipped with mounting holes and other openings as needed for a complete installation. Fabrication marks and holes must be ground and finished to provide a smooth and neat appearance for each plate. Each plate must be primed and painted.

c. Speakers must utilize screw terminals for termination of all field wiring.

2.12.2 Visual Notification Appliances

Visual notification appliances must conform to the applicable requirements of UL 1638, UL 1971 and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances must have clear high intensity optic lens, xenon flash tubes, or light emitting diode (LED) and be marked "Alert" in letters of contrasting color. The light pattern must be dispersed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate must be 1 flash per second and a minimum of ~~[~~15~~][30][75][_____]~~ candela based on the UL 1971 test unless noted otherwise. Strobe must be ~~[surface][~~semi-flush~~]~~ mounted.

2.12.3 Textual Display Signs

Textual display signs must be ~~[~~LED~~][LCD flat panel][LED scrolling]~~ and must not exceed 400 mm long by 150 mm high by 75 mm deep with a height necessary to meet the requirements of NFPA 72. The text display must spell out the word "EVACUATE" or "ANNOUNCEMENT" ~~[and the remainder of the emergency instructions]~~as appropriate. The design of text display must be such that it cannot be read when not illuminated.

~~[LCD or LED scrolling text displays must meet the following requirements at a minimum:~~

~~a. Two lines of information for high priority messaging.~~

~~b. Minimum of 20 characters per line (40 total) displayed.~~

~~c. Text must be no less than height requirements and color/contrast requirements of NFPA 72.~~

~~d. 32K character memory.~~

~~e. Display must be wall or ceiling mounted.~~

~~f. Mounting brackets for a convenient wall/cubicle mount.~~

~~g. [g. During non-emergency periods, display date and time.]~~

~~h. The system must interface with the textual display sign control panel to activate the proper message.]~~

2.13 ELECTRIC POWER

2.13.1 Primary Power

Power must be ~~[~~120~~][_____]~~ VAC ~~[~~50~~][60]~~ Hz service for the FMCU from the AC service to the building in accordance with NFPA 72.

2.14 SECONDARY POWER SUPPLY

Provide for system operation in the event of primary power source failure. Transfer from normal to auxiliary (secondary) power or restoration from auxiliary to normal power must be automatic and must not cause transmission of a false alarm.

2.14.1 Batteries

Provide sealed, maintenance-free, ~~[~~sealed lead acid~~][lead-calcium][gel cell]~~ batteries as the source for emergency power to the FMCU. Batteries must contain suspended electrolyte. The battery system must be maintained in a fully charged condition by means of a battery charger. Provide an automatic transfer switch to transfer the load to the batteries in the event of the failure of primary power.

2.14.1.1 Capacity

Battery size must be the greater of the following two capacities. This capacity applies to every control unit associated with this system, including supplemental notification appliance circuit panels, ~~auxiliary power supply panels,~~ fire alarm transmitters, and Base-wide mass notification transceivers. When determining the required capacity under alarm condition, visual notification appliances must include both textual and non-textual type appliances.

a. Sufficient capacity to operate the fire alarm system under supervisory and trouble conditions, including audible trouble signal devices for 48 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes.

b. Sufficient capacity to operate the mass notification for 60 minutes after loss of AC power.

2.14.1.2 Battery Power Calculations

a. Verify that battery capacity exceeds supervisory and alarm power requirements for the criteria noted in the paragraph "Capacity" above.

(1) Substantiate the battery calculations for alarm and supervisory power requirements. Include ampere-hour requirements for each system component and each control unit component, and compliance with UL 864.

(2) Provide complete battery calculations for both the alarm and supervisory power requirements. Submit ampere-hour requirements for each system component with the calculations.

(3) Provide voltage drop calculations to indicate that sufficient voltage is available for proper operation of the system and all components. Calculations must be performed using the minimum rated voltage of each component.

b. For battery calculations assume a starting voltage of 24 VDC for starting the calculations to size the batteries. Calculate the required Amp-Hours for the specified standby time, and then calculate the required Amp-Hours for the specified alarm time. Using 20.4 VDC as starting voltage, perform a voltage drop calculation for circuits containing devices or appliances remote from the power sources.

2.14.2 Battery Chargers

Provide a fully automatic, variable charging rate battery charger. The charger must be capable of providing 120 percent of the connected system load and must maintain the batteries at full charge. In the event the batteries are fully discharged (20.4 Volts dc), the charger must recharge the batteries back to 95 percent of full charge within 48 hours after a single discharge cycle as described in paragraph CAPACITY above. Provide pilot light to indicate when batteries are manually placed on a high rate of charge as part of the unit assembly if a high rate switch is provided.

2.15 SURGE PROTECTIVE DEVICES

Surge protective devices must be provided to suppress all voltage transients which might damage fire alarm control unit components. Systems having circuits located outdoors, communications equipment must be protected against surges induced on any signaling line circuit. Cables and conductors, that serve as communications links, must have surge protection circuits installed at each end. The surge protective device must wire in series to the power supply of the protected equipment with screw terminations. Line voltage surge arrestor must be installed directly adjacent to the power panel where the FMCU breaker is located.

a. Surge protective devices for nominal 120 VAC must be UL 1449 listed with a maximum 500 volt suppression level and have a maximum response time of 5 nanoseconds. The surge protective device must also meet IEEE C62.41.1 and IEEE C62.41.2 category B tests for surge capacity. The surge protective device must feature multi-stage construction and be provided with a long-life indicator lamp (either light emitting diode or neon) which extinguishes upon failure of protected components. Any unit fusing must be externally accessible.

b. Surge protective devices for nominal 24 VAC, fire alarm telephone dialer, or ethernet connection must be UL 497B listed, meet IEEE C62.41.1 and have a maximum response time of 1-nanosecond. The surge protective device must feature multi-stage construction and be self-resetting. The surge protective device must be a base and plug style. The base assembly must have screw terminals for fire alarm wiring. The base assembly must accept "plug-in" surge protective module.

c. All surge protective devices (SPD) must be the standard product of a single manufacturer and be equal or better than the following:

(1) For 120 VAC nominal line voltage: UL 1449 and UL 1283 listed, series connected 120 VAC, 20A rated, surge protective device in a NEMA 4x enclosure. Minimum 50,000 amp surge current rating with EMI/RFI filtering and a dry contact circuit for remote monitoring of surge protection status.

(2) For 24-volt nominal line voltage: UL 497B listed, series connected low voltage, 24-volt, 5A rated, loop circuit protector, base and replaceable module.

(3) For alarm telephone dialers: UL 497A listed, series connected, 130-volt, 150 mA rated with self-resetting fuse, dialer circuit protector with modular plug and play.

(4) For IP-DACTS: UL 497B listed, series connected, 6.4-volt, 1.5A rated with 20 kA/pair surge current, data network protector with modular plug and play.

2.16 WIRING

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

2.16.1 Alarm Wiring

IDC and SLC wiring must be ~~[fiber optic][or][~~solid copper~~][or][stranded copper]~~ cable in accordance with the manufacturers requirements. Copper signaling line circuits and initiating device circuit field wiring must be No. ~~[~~14~~][16][_____]~~ AWG size conductors at a minimum. Visual notification appliance circuit conductors, that contain audible alarm appliances, must be copper No. 16 AWG size conductors at a minimum. Speaker circuits must be copper No. ~~[~~16~~][_____]~~ AWG size twisted and shielded conductors at a minimum.~~[~~ Wiring for textual notification appliance circuits must be in accordance with manufacturer's requirements but must be supervised by the FMCU.~~]~~ Wire size must be sufficient to prevent voltage drop problems. Circuits operating at 24 VDC must not operate at less than the listed voltages for the detectors and appliances. Power wiring, operating at 120 VAC minimum, must be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables must comply with NFPA 70.

~~2.17 INTERFACE TO THE BASE-WIDE MASS NOTIFICATION NETWORK~~

~~[2.17.1 Fiber Optic~~

~~The fiber optic transceiver must be fully compatible with EIA standards for RS-232, RS-422 and RS-485 at data rates from 0 (DC) to 2.1 mbps (200 kbps for RS-232) in the low speed mode or from 10 kbps to 10 mbps in the high-speed mode. The fiber optic transceiver must be capable of simplex or full duplex asynchronous transmissions in both point-to-point systems and drop-and-repeat data networks. The fiber optic transceiver must be user configurable for the protocol, speed and mode of operation required. The fiber optic transceiver must be installed as a [stand-alone][card-cage] unit. The fiber optic transceiver must operate on [Multi-mode][Single-mode] fiber optic cable. The fiber optic transceiver must be supplied with [ST][or][FCPC] type optical connectors. Cabling: as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.~~

~~][2.17.2 Radio~~

~~The mass notification transceiver must be bi-direction and meet all the requirements of paragraph, RADIO TRANSMITTER AND INTERFACE PANELS as specified in this specification section. The transceiver utilized in the mass notification system must be capable of the following:~~

~~a. Communication with the central control/monitoring system to provide supervision of communication link and status changes are reported by automatic and manual poll/reply/acknowledge routines.~~

~~b. All monitored points/status changes are transmitted immediately and at programmed intervals until acknowledged by the central control/monitoring system.~~

~~c. Each transceiver must transmits a unique identity code as part of all messages; the code is set by the user at the transceiver.~~

~~][2.17.3 Telephone~~

~~A modem must be provided for communication with the central control/monitoring system. The modem must be 56k, compatible with data mode V.90, utilizing Hayes compatible command codes. The modem must be capable of auto dialing a preset number based on preprogrammed events. The modem must auto answer and provide a secure password protection system. Cabling: as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.~~

~~][2.17.4 Secure Radio System~~

~~2.17.4.1 Communications Network~~

~~The communications network provides two-way signals between central control units and autonomous control units (in individual building systems), and should include redundant (primary and backup) communication links. The system must incorporate technology to prevent easy interruption of the radio traffic for MNS alerting.~~

~~2.17.4.2 Radio Frequency Communications~~

~~Use of radio frequency-type communications systems must comply with~~

~~National Telecommunications and Information Administration (NTIA) requirements. The systems must be designed to minimize the potential for interference, jamming, eavesdropping, and spoofing.~~

~~2.17.4.3 Licensed Radio Frequency Systems~~

~~An approved DD Form 1494 for the system is required prior to operation.~~

~~]~~2.17 AUTOMATIC FIRE ALARM TRANSMITTERS

~~[2.17.1 Radio Transmitter and Interface Panels~~

~~Transmitters must be compatible with proprietary supervising station receiving equipment. Each radio alarm transmitter must be the manufacturer's recognized commercial product, completely assembled, wired, factory tested, and delivered ready for installation and operation. Transmitters must be provided in accordance with applicable portions of NFPA 72, Federal Communications Commission (FCC) 47 CFR 90 and Federal Communications Commission (FCC) 47 CFR 15. Transmitter electronics module must be contained within the physical housing as an integral, removable assembly. The proprietary supervising station receiving equipment is [_____] and the transmitter must be fully compatible with this equipment. At the contractors option, and if listed, the transmitter may be housed in the same control unit as the FMCU. The transmitter must be narrowband radio, with FCC certification for narrowband operation and meets the requirements of the NTIA (National Telecommunications and Information Administration) Manual of Regulations and Procedures for Federal Frequency Management.~~

~~2.17.1.1 Operation~~

~~Operate each transmitter from 120-volt ac power. In the event of 120-volt ac power loss, the transmitter must automatically switch to battery operation. Switchover must be accomplished with no interruption of protective service, and must automatically transmit a trouble message. Upon restoration of ac power, transfer back to normal ac power supply must also be automatic.~~

~~2.17.1.2 Battery Power~~

~~Transmitter standby battery capacity must provide sufficient power to operate the transmitter in a normal standby status for a minimum of 72 hours and be capable of transmitting alarms during that period.~~

~~2.17.1.3 Transmitter Housing~~

~~Use NEMA Type 1 for housing. The housing must contain a lock that is keyed [identical to the fire alarm system for the building][identical to radio alarm transmitter housings on the Installation]. Radio alarm transmitter housing must be factory painted with a suitable priming coat and not less than two coats of a hard, durable weatherproof enamel.~~

~~2.17.1.4 Antenna~~

~~Antenna must be [omnidirectional, coaxial, halfwave dipole antennas][_____] for radio alarm transmitters with a driving point impedance to match transmitter output. The antenna and antenna mounts must be corrosion resistant and designed to withstand wind velocities of 161 km/hour. Do not mount antennas to any portion of the building roofing~~

~~system. Protect the antenna from physical damage.~~

~~][~~2.17.1 Digital Alarm Communicator Transmitter (DACT)

Provide DACT that is compatible with the existing supervising station fire alarm system. Transmitter must have a means to transmit alarm, supervisory, and trouble conditions via a single transmitter. Transmitter must have a source of power for operation that conforms to NFPA 72. Transmitter must be capable of initiating a test signal daily at any selected time. Transmitter must be arranged to seize telephone circuits in accordance with NFPA 72.

~~]~~2.17.2 Signals to Be Transmitted to the Base Receiving Station

The following signals must be sent to the base receiving station:

~~[~~ a. Sprinkler waterflow

~~][~~b. Manual pull stations

~~][~~c. Smoke detectors

~~][~~d. Duct smoke detectors

~~][~~e. Sleeping room smoke detectors

~~][f. Carbon monoxide detectors~~

~~][g. Heat detectors~~

~~][h. Fire extinguishing system~~

~~][~~f. Sprinkler valve supervision

~~][~~g. Fire pump running

~~][~~h. Fire pump supervision

~~][i. Water supply level and temperature~~

~~][j. Combustion engine drive fire pump running~~

~~(1) Selector switch in position than automatic~~

~~(2) Engine over-speed~~

~~(3) Low fuel~~

~~(4) Low battery~~

~~(5) Engine trouble (for example, low oil, over temp)~~

~~]~~2.18 SYSTEM MONITORING

2.18.1 Valves

Each valve affecting the proper operation of a fire protection system, including automatic sprinkler control valves,~~[~~ standpipe control valves,~~]~~ sprinkler service entrance valve,~~[~~ valves at fire pumps,~~]~~ isolating valves for pressure type waterflow or supervision switches, and valves at backflow preventers, whether supplied under this contract or existing, must be electrically monitored to ensure its proper position. Provide each tamper switch with a separate address~~[, unless they are within the same room, then a maximum of five can use the same address]~~.

~~[~~2.18.2 High/Low ~~[Air][~~Nitrogen~~]~~ Supervisory Switches

Provide monitoring of high and low supervisory ~~[air][~~nitrogen~~]~~ for ~~[~~dry ~~pipe][and][preaction]~~standpipe systems. Each air supervisory switch must have a separate address. Switches must be listed extinguishing system attachments. The device must contain double pole, double throw contacts. Operation of the switch must cause a supervisory signal to be transmitted to the FMCU when ~~[air][~~nitrogen~~]~~ pressure in the system monitored sprinkler system increases more than 34.5 kPa above the normal system pressure or drops halfway from the normal pressure to the tripping point.

~~][2.18.3 Room Low Temperature Supervisory Switch~~

~~Provide [monitoring of the]listed supervisory air temperature switch for the [fire pump][sprinkler riser] room[s]. Switch must cause a supervisory signal to be transmitted to the FMCU whenever the temperature in the room drops to below 4.4 degrees C. Device must reset when temperature rises above 4.4 degrees C.~~

~~][~~2.18.3 Electromagnetic Door Holders

Electromagnetic holding devices must operate on ~~[120 VAC][~~24 VDC~~]~~, and require not more than ~~[~~3~~][_____]~~ watts of power to develop 6.9 kPa of holding force. Under normal conditions, the magnets must attract and hold the doors open. Operation must be fail safe with no moving parts. Electromagnetic door hold-open devices must not be required to be held open during building power failure. The device must be listed based on UL 228 tests.

~~]~~2.19 ENVIRONMENTAL ENCLOSURES OR GUARDS

Environmental enclosures must be provided to permit fire alarm/mass notification components to be used in areas that exceed the environmental limits of the listing. The enclosure must be listed for the device or appliance as either a manufactured part number or as a listed compatible accessory for the component is currently listed. Guards required to deter mechanical damage must be either a listed manufactured part or a listed accessory for the category of the initiating device or notification appliance.

~~[2.20 FIREFIGHTER TELEPHONE COMMUNICATION SYSTEM~~

~~2.20.1 General~~

~~Provide a firefighter telephone communication system with complete, common talk, closed circuits. The system must include, but not be limited to, a master control station mounted in the fire alarm control unit, a power supply and standby battery system, and remote telephone stations.~~

~~2.20.2 Features~~

~~The system must include the following features:~~

~~a. A master control station which must provide power, supervision, and control for wiring, components, and circuits. The act of lifting any remote telephone handset from its cradle must cause both a visual and audible signal to annunciate at the master control station. Removing the handset at the master control station and depressing a button at the remote telephone handset must cause the automatic silencing of the audible signal.~~

~~b. Communication between the master control station handset and any/or all remote handsets must require the depressing of a push-to-talk switch located on any/all remote handsets. During the time that the master control handset is removed from its cradle it must be possible to communicate between five remote handsets and the master control station.~~

~~c. handsets must be able to monitor any conversation in progress and join the conversation by pressing the push-to-talk button. It must not be possible to communicate between two or more remote handsets with the master control station handset in its cradle.~~

~~d. The master control station handset must be red in color and equipped with a 1.5 m long strain-relieved coiled cord.~~

~~e. The master control station must monitor wire and connections for any opens, shorts, or grounds which would render the system inoperable or unintelligible.~~

~~f. The master control station must be equipped with a silencing switch and ring-back feature such that any audible trouble signal can be silenced and must be so indicated by the lighting of an amber LED. Once any trouble condition has been corrected, the amber LED must be extinguished.~~

~~g. The master control station must be equipped with a separate, LED annunciated switch for each telephone circuit. In addition, LEDs must provide for the annunciation of operating and supervisory power.~~

~~h. The loss of operating or supervisory power must cause an audible and visual indication at the master control station and must also cause the fire alarm trouble signal to sound on the FMCU.~~

~~i. Switches, LEDs, and controls must be fully labeled.~~

~~2.20.3 Handsets~~

~~Handsets must have the following features:~~

~~a. Provide [surface][flush] mounted remote telephone stations.~~

~~b. Each station must be equipped with a hinged door that is magnetically locked.~~

~~c. Each handset must be permanently wired in place with a coiled cord.~~

~~d. Each handset must be red high-impact cycolac and must be equipped with a push-to-talk switch which, when operated, must signal the master control station and a switch-equipped, storage cradle.~~

~~e. Provide operating and supervising power from the same supply circuit(s) utilized for the FMCU.~~

~~]~~PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

3.2 INSTALLATION

3.2.1 Fire Alarm and Mass Notification Control Unit (FMCU)

Locate the FMCU ~~[~~where indicated on the drawings~~][_____]~~. ~~[Recess][~~Semi-recess~~][Surface mount]~~ the enclosure with the top of the cabinet 2 m above the finished floor or center the cabinet at ~~[~~1.5~~][_____]~~ m, whichever is lower. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the FMCU. Locate the document storage cabinet adjacent to the FMCU unless the Contracting Officer directs otherwise.

3.2.2 Battery Cabinets

When batteries will not fit in the FMCU, locate battery cabinets below or adjacent to the FMCU. Battery cabinets must be installed at an accessible location when standing at floor level. Battery cabinets must not be installed lower than 300 mm above finished floor, measured to the bottom of the cabinet, nor higher than 900 mm above the floor, measured to the top of the cabinet. Installing batteries above drop ceilings or in inaccessible locations is prohibited. Battery cabinets must be large enough to accommodate batteries and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be provided in a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and surface mounting provisions.The cabinet must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must indicate the control unit(s) the batteries power and must not be less than 25 mm high.

3.2.3 Manual Stations

Locate manual stations as required by NFPA 72~~[~~ and as indicated on the drawings~~]~~. Mount stations so they are located no farther than ~~[~~1.5~~][_____]~~ m from the exit door they serve, measured horizontally. Manual stations must be mounted at ~~[1067][~~1117~~][_____]~~ mm measured to the operating handle.

3.2.4 Notification Appliances

a. Locate notification appliance devices~~[~~ as required by NFPA 72~~][~~ where indicated~~][~~ and to meet the intelligibility requirements~~]~~. Where two or more visual notification appliances are located in the same room or corridor or field of view, provide synchronized operation. Devices must use screw terminals for all field wiring.~~[~~ Audible and visual notification appliances mounted in acoustical ceiling tiles must be centered in the tiles plus or minus 50 mm.~~]~~

b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers must be listed weatherproof appliances installed on weatherproof backboxes.

c. Speakers must not be located in close proximity to the FMCU or LOC so as to cause feedback when the microphone is in use.

3.2.5 Smoke ~~and Heat~~ Detectors

Locate detectors~~[~~ as required by NFPA 72 and their listing~~][~~ as indicated on the drawings~~]~~ on a 100 mm mounting box. ~~Install heat detectors not less than 100 mm from a side wall to the near edge. Heat detectors located on the wall must have the top of the detector at least 100 mm below the ceiling, but not more than 300 mm below the ceiling.~~ Smoke detectors are permitted to be on the wall no lower than 300 mm from the ceiling with no minimum distance from the ceiling.~~[In raised floor spaces, install the smoke detectors to protect[21 square meters per detector][_____].]~~ Install smoke detectors no closer than 1 m from air handling supply diffusers. Detectors installed in acoustical ceiling tiles must be centered in the tiles plus or minus 50 mm.

~~3.2.6 Carbon Monoxide Detectors~~

~~Locate detectors[as required by NFPA 72 and their listings][as indicated on the drawings] on a 100-mm mounting box.[Carbon monoxide detectors must be installed separate from smoke and heat detectors.]~~

~~[3.2.7 Air Sampling Smoke Detector~~

~~Locate air sampling smoke detectors in accordance with the manufacturer's instructions. Air sampling smoke detectors must be installed as follows:~~

~~a. Air Sampling Smoke Detector Assembly:~~

~~(1) Detector assembly must be mounted to a wall at a height between 1200 mm to 1800 mm to top of detector measured above the finished floor.~~

~~(2) Mounting must be in a fully accessible and visible location.~~

~~(3) Mounting or attachment to site equipment, cable trays, movable walls, other equipment or equipment supports is not permitted.~~

~~(4) Piping network insertion into the detector inlet must not be glued.~~

~~(5) Air sampling smoke detector assembly must be installed in accordance with this specification section and the manufacturer's installation and instruction manuals.~~

~~(6) Flexible tubing for termination of the sampling pipe network into detector inlet is not permitted unless allowed by its listing.~~

~~(7) Provide red background with white lettering labels that are~~

~~plastic or phenolic type with a minimum of 6.4 mm block lettering to indicate detector and zone. For example: "AIR SAMPLING SOME DETECTOR No. 1-1 No. 5".~~

~~(8) Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the frame in a conspicuous location observable from the ASD panel. The card must show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions must be approved by the Contracting Officer before being posted.~~

~~b. Pipe and Sampling Tube Mounting:~~

~~(1) The pipe and sampling tubing detection network must be mounted as per the design and manufacturer's specification. The hardware used for mounting will depend upon the design and site requirements.~~

~~(2) To minimize flexing, pipes must be secured every 1.5 m.~~

~~(3) Pipes must be suspended between 25 mm and 100 mm below the ceiling. In areas with a suspended ceiling, the pipe network must be installed above the ceiling utilizing the manufacturer's capillary sample port supported by the ceiling.~~

~~(4) The sampling tubes must be of the same length or use the manufacturer's guidelines to run tubes of the required lengths.~~

~~(5) When installing a pipe network in areas subject to high temperature fluctuations allow for the contraction and expansion of pipes.~~

~~(6) Where expansion or contraction of pipes is likely either after installation or on a continuous basis, do not place pipe clips adjacent to couplings and socket unions as these may interfere with the movement of the pipe.~~

~~(7) No bends are permitted within the first 450 mm from the detector inlet.~~

~~(8) The routing of the piping and sample tube network must be coordinated with potential obstructions, including cable trays, grounding bars, and HVAC ductwork.~~

~~(9) All changes in direction must be made with standard elbows or tees.~~

~~(10) All joints must be air-tight and made by using solvent cement, except at the entry to the detector assembly. Refer to ASTM F402.~~

~~(11) All pipes must be supported by mechanical hangers attached to the structure of the building. Not more than 300 mm of pipe must extend beyond the last hanger of each sampling pipe. The final installation must result in no noticeable deflection in the piping network.~~

~~(12) Attachment of air sampling pipes to cable trays, "gray iron", and~~

~~telecommunications equipment is prohibited.~~

~~(13) Clearly label pipe network to distinguish the pipe from other facility pipe work or protective cabling enclosures. For example: "SMOKE DETECTION SAMPLING TUBE - DO NOT DISTURB". In open rooms and exposed areas, provide labels at no greater than 6.1-m intervals. Provide labels every 3 m where piping is installed above suspended ceilings and every 609 mm, centered in the floor panels, where piping is installed within the raised floor cavity.~~

~~(14) Placement of the sampling tube must take into consideration appropriate sampling point locations and spacing.~~

~~c. Air Sampling Points:~~

~~(1) Open area ceiling sampling points must be oriented downward and must be within 25 mm to 100 mm below the underside of the ceiling above where the ceiling is smooth.~~

~~(2) Label all air sampling points with a round red label, each with a center hole to match the diameter of the drilled sampling point. For example: "AIR SAMPLING POINT DIA 3.2 MM". Indicate fractional dimensions in decimal format with a minimum of three decimal places.~~

~~][3.2.8 Graphic Annunciator~~

~~Locate the graphic annunciator as shown on the drawings. Mount the annunciator, with the top 1830 mm above the finished floor or center the annunciator at [1525][_____] mm, whichever is lower.~~

~~]~~3.2.6 LCD REMOTE Annunciator

Locate the LCD annunciator as shown on the drawings. Mount the annunciator, with the top 2 m above the finished floor or center the annunciator at ~~[~~1.5~~][_____]~~ m, whichever is lower.

~~[~~3.2.7 Electromagnetic Door Holder Release

Doors must be held open at a minimum of 90 degrees so as not to impede egress from the space. Mount the armature portion on the door and have an adjusting screw for seating the angle of the contact plate. Wall-mount the electromagnetic release, with a total horizontal projection not exceeding 100 mm. Ensure all doors release to close upon first stage (pre-discharge) alarm. Electrical supervision of wiring external of control unit for magnetic door holding circuits is not required.

~~][3.2.8 Firefighter Telephones~~

~~Mount telephone[handsets][jacks] on the wall in each stair at each floor landing, in each emergency generator room, in each fire pump room, in each elevator machine room, in each elevator lobby, and in each elevator cab 1200 mm above the finished floor.~~

~~]~~3.2.8 Local Operating Console (LOC)

Locate the LOC(s) as required by NFPA 72 and as indicated. Mount the console so that the top message button and microphone is no higher than

1200 mm above the floor and the bottom (lowest) message button and microphone is at least 1-meter above the finished floor.

3.2.9 Ceiling Bridges

Provide ceiling bridges for ceiling-mounted appliances. Ceiling bridges must be as recommended/required by the manufacturer of the ceiling-mounted notification appliance.

3.3 SYSTEM FIELD WIRING

3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and professional manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box must be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FMCU and remote fire alarm/mass notification control units, initiating circuits, switches, relays and terminals.

b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

3.3.2 Terminal Cabinets

Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Terminal size must be appropriate for the size of the wiring to be connected. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the terminal cabinet. Minimum size is 200 mm by 200 mm. Only screw-type terminals are permitted. Provide an identification label, that displays "FIRE ALARM TERMINAL CABINET" with 50 mm lettering, on the front of the terminal cabinet.

3.3.3 Alarm Wiring

a. Voltages must not be mixed in any junction box, housing or device, except those containing power supplies and control relays.

b. Utilize shielded wiring where recommended by the manufacturer. For shielded wiring, ground the shield at only one point, in or adjacent to the FMCU.

c. ~~[~~Pigtail or T-tap connections to signal line circuits, initiating device circuits, supervisory alarm circuits, and notification appliance circuits are prohibited.~~][T-tapping using screw terminal blocks is allowed for Class "B" signaling line circuits.]~~

d. Color coding is required for circuits and must be maintained throughout the circuit. Conductors used for the same functions must be similarly color coded. Conform wiring to NFPA 70.

e. Pull all conductors splice free. The use of wire nuts, crimped connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur must be identified on the as-built drawings. The number and location of splices must be subject to approval by the ~~[_____]~~ Designated Fire Protection Engineer (DFPE).

3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size must be 19 mm in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

a. Galvanized rigid steel (GRS) conduit must be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.

b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.

c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.

d. Flexible metal conduit is permitted for initiating device circuits, notification appliance circuits and signaling line circuits ~~[_____]~~ 1,800 mm in length or less. Use liquid tight flexible metal conduit in damp and wet locations.

e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.

f. Exterior wall penetrations must be weathertight. Conduit must be sealed to prevent the infiltration of moisture.

~~[g. For Class "A" or "X" circuits with conductor lengths of 3 meters or less, the conductors must be permitted to be installed in the same raceway in accordance with NFPA 72.~~

~~]~~3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, FMCU~~[, and remote FMCU]~~ and the LOC must be provided at each conductor connection. Each conductor or cable must have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet, FMCU, and remote FMCU must contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing must be neat, using 12 point lettering minimum size, and mounted within each cabinet, control unit, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

~~[~~3.4 DISCONNECTION AND REMOVAL OF EXISTING SYSTEM

Existing system shall be completely demolished and removed from service prior to installation of the new system.~~Maintain existing fire alarm/mass notification equipment fully operational until the new equipment has been tested and accepted by the Contracting Officer. As new equipment is installed, label it "NOT IN SERVICE" until the new equipment is accepted. Once the new system is completed, tested, and accepted by the Government, it must be placed in service and connected to the supervising station. Remove tags from new equipment and tag the existing equipment "NOT IN SERVICE" until removed from the building.~~

~~a. After acceptance of the new system by the Contracting Officer, remove existing equipment not connected to the new system, remove unused exposed conduit, and restore damaged surfaces. Remove the material from the site and dispose.~~

~~b. Disconnect and remove the existing fire alarm/mass notification and smoke detection systems where indicated and elsewhere in the specification.~~

~~c. Control units and fire alarm devices and appliances disconnected and removed must be turned over to the Contracting Officer.~~

~~d. Properly dispose of fire alarm outlet and junction boxes, wiring, conduit, supports, and other such items.~~

3.5 CONNECTION OF NEW SYSTEM

The following new system connections must be made during the last phase of construction, at the beginning of the pre-Government tests. New system connections must include:

~~a. Connection of new relays to existing magnetic door hold-open devices.~~

~~b. Connection of new elevator recall relays to existing wiring and conduit.~~

a. Connection of new system transmitter to existing installation fire reporting system.

Once these connections are made, system must be left energized. Report immediately to the Contracting Officer, coordination and field problems resulting from the connection of the above components.

~~]~~3.6 FIRESTOPPING

Provide firestopping for holes at conduit penetrations through floor slabs, fire-rated walls, partitions with fire-rated doors, corridor walls, and vertical service shafts in accordance with Section 07 84 00 FIRESTOPPING.

3.7 PAINTING

a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceway, junction boxes and covers red. In lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.

b. In finished areas, paint exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Fire Alarm" and the conduit must have painted red bands 19 mm wide at 3-meter centers and at each side of a floor, wall, or ceiling penetration.

c. Painting must comply with Section 09 90 00 PAINTS AND COATINGS.

3.8 FIELD QUALITY CONTROL

3.8.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level ~~[~~ III~~][~~ or ~~][~~IV~~]~~ Fire Alarm Technician, and the representative of the installing company, ~~[~~and reviewed by the QFPE~~]~~ ~~[~~60~~][~~_____~~]~~ days prior to performing system tests. Detailed test procedures must list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, signaling line devices and circuits, control devices/equipment, batteries, transmitting and receiving equipment, power sources/supply, annunciators, special hazard equipment, emergency communication equipment, interface equipment, and surge protective devices. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government testing. The test data forms must record the test results and must:

a. Identify the NFPA Class of all Initiating Device Circuits (IDC), and Notification Appliance Circuits (NAC), Voice Notification System Circuits (NAC Audio), and Signaling Line Circuits (SLC).

b. Identify each test required by NFPA 72 Test Methods and required test herein to be performed on each component, and describe how these tests must be performed.

c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.

d. Identify all test equipment and personnel required to perform each test including equipment necessary for smoke detector testing. The use of magnets is not permitted.

e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.8.2 Pre-Government Testing

3.8.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The contractor and an authorized representative from each supplier of equipment must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 including all referenced annex sections and the test reports noted below.

a. NFPA 72 Record of Completion.

b. NFPA 72 Record of Inspection and Testing.

c. Fire Alarm and Emergency Communication System Inspection and Testing Form.

d. Audibility test results with marked-up test floor plans.

e. Intelligibility test results with marked-up floor plans.

f. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

3.8.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the ~~[_____][~~ Designated Fire Protection Engineer (DFPE)~~][Contracting Officer's Representative (COR)]~~. Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system~~[and the installation-wide mass notification system have]~~ been completed and tested to confirm communications are fully functional. Submit request for test at least ~~[~~15 ~~][_____]~~ calendar days prior to the requested test date.

3.8.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.8.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

a. The manufacturer's technical representative.

~~[~~ b. The contractor's Qualified Fire Protection Engineer (QFPE).

~~]~~ c. Marked-up red line drawings of the system as actually installed.

d. Loop resistance test results.

e. Complete program printout including input/output addresses.

f. Copy of pre-Government Test Certificate, test procedures and completed test data forms.

g. Audibility test results with marked-up floor plans.

h. Intelligibility test results with marked-up floor plans.

Government Final Tests will be witnessed by the ~~[______], [~~Designated Fire Protection Engineer~~][~~ Contracting Officer's Representative (COR)~~][~~, Qualified Fire Protection Engineer (QFPE)~~]~~. At this time, any and all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.9 MINIMUM SYSTEM TESTS

3.9.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

a. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests must be witnessed by the Contracting Officer and test results recorded for use at the final Government test.

b. Verify the absence of unwanted voltages between circuit conductors and ground. The tests must be accomplished at the pre-Government test with results available at the final system test.

c. Verify that the control unit is in the normal condition as detailed in the manufacturer's O&M manual.

d. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors must be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors must comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision must be tested at each device.

~~e. Carbon Monoxide Detector Tests: Carbon monoxide detectors must be tested in accordance with NFPA 72 and the manufacturer's recommended calibrated test method.~~

e. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.

f. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.

g. Determine that the system is operable under trouble conditions as specified.

h. Visually inspect wiring.

i. Test the battery charger and batteries.

j. Verify that software control and data files have been entered or programmed into the FMCU. Hard copy records of the software must be provided to the Contracting Officer.

k. Verify that red-line drawings are accurate.

l. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.

m. Measure voltage readings for circuits to ensure that voltage drop is not excessive.

n. Disconnect the verification feature for smoke detectors during tests to minimize the amount of smoke needed to activate the sensor. Testing of smoke detectors must be conducted using real smoke or the use of canned smoke which is permitted.

o. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.

p. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

3.9.2 Audibility Tests

Sound pressure levels from audible notification appliances must be a minimum of 15 dBa over ambient with a maximum of 110 dBa in any occupiable area. The provisions for audible notification (audibility and intelligibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

3.9.3 Intelligibility Tests

Intelligibility testing of the System must be accomplished in accordance with NFPA 72 for Voice Evacuation Systems, and ASA S3.2. Following are the specific requirements for intelligibility tests:

a. Intelligibility Requirements: Verify intelligibility by measurement after installation.

b. Ensure that a CIS value greater than the required minimum value is provided in each area where building occupants typically could be found. The minimum required value for CIS is ~~[.7]~~~~[~~.8~~]~~. Rounding of values is permitted.

c. Areas of the building provided with hard wall and ceiling surfaces (such as metal or concrete) that are found to cause excessive sound reflections may be permitted to have a CIS score less than the minimum required value if approved by the DFPE, and if building occupants in these areas can determine that a voice signal is being broadcast and they must walk no more than 10 m to find a location with at least the minimum required CIS value within the same area.

d. Areas of the building where occupants are not expected to be normally present are permitted to have a CIS score less than the minimum required value if personnel can determine that a voice signal is being broadcast and they must walk no more than 15 m to a location with at least the minimum required CIS value within the same area.

e. Take measurements near the head level applicable for most personnel in the space under normal conditions (e.g., standing, sitting, sleeping, as appropriate).

f. The distance the occupant must walk to the location meeting the minimum required CIS value must be measured on the floor or other walking surface as follows:

(1) Along the centerline of the natural path of travel, starting from any point subject to occupancy with less than the minimum required CIS value.

(2) Curving around any corners or obstructions, with a 300 mm clearance there from.

(3) Terminating directly below the location where the minimum required CIS value has been obtained.

Use commercially available test instrumentation to measure intelligibility as specified by NFPA 72 as applicable. Use the mean value of at least three readings to compute the intelligibility score at each test location.

3.10 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the Final Government Test.

a. ~~[~~ The drawings must be prepared electronically and sized no less than the contract drawings.~~][~~ Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of ~~[MicroStation] [~~AutoCAD, ~~]~~DXF and portable document formats of as-built drawings and schematics.~~]~~

b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.

c. Include a riser diagram and drawings showing the as-built location of devices and equipment.

d. Provide Operation and Maintenance (O&M) Instructions.

~~[In existing buildings, the transfer of devices from the existing system~~

~~to the new system and the permission to begin demolition of the old fire alarm system will not be permitted until the as-built drawings and O&M manuals are received.]~~

3.11 INSTRUCTION OF GOVERNMENT EMPLOYEES

3.11.1 Instructor

Provide the services of an instructor, who has received specific training from the manufacturer for the training of other persons regarding the operation, inspection, testing, and maintenance of the system provided. The instructor must train the Government employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the fire alarm system. The instructor must be thoroughly familiar with all parts of this installation. The instructor must be trained in operating theory as well as in practical O&M work. Submit the instructors information and qualifications including the training history.

3.11.2 Required Instruction Time

Provide ~~[~~8~~][16][_____]~~ hours of instruction after final acceptance of the system. The instruction must be given during regular working hours on such dates and times selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer. The training must allow for rescheduling for unforeseen maintenance and fire department responses.

~~[~~3.11.2.1 Technical Training

Equipment manufacturer or a factory representative must provide ~~[~~1 ~~][3][_____]~~ days of on site~~[and 5 days of technical training to the Government at the manufacturing facility]~~. Training must allow for classroom instruction as well as individual hands on programming, troubleshooting and diagnostics exercises.~~[Factory training must occur within [6][12][_____] months of system acceptance.]~~

~~]~~3.11.3 Technical Training Manual

Provide, in manual format, lesson plans, operating instructions, maintenance procedures, and training data for the training courses. The operations training must familiarize designated government personnel with proper operation of the installed system. The maintenance training course must provide the designated government personnel adequate knowledge required to diagnose, repair, maintain, and expand functions inherent to the system.

3.12 EXTRA MATERIALS

3.12.1 Repair Service/Replacement Parts

Repair services and replacement parts for the system must be available for a period of 10 years after the date of final acceptance of this work by the Contracting Officer. During the warranty period, the service technician must be on-site within 24 hours after notification. All repairs must be completed within 24 hours of arrival on-site.

During the warranty period, the installing fire alarm contractor is responsible for conducting all required testing and maintenance in accordance with the requirements and recommended practices of NFPA 72 and the system manufacturer~~[~~s~~]~~. Installing fire alarm contractor is NOT responsible for any damage resulting from abuse, misuse, or neglect of equipment by the end user.

3.12.2 Spare Parts

Spare parts furnished must be directly interchangeable with the corresponding components of the installed system~~[~~s~~]~~. Spare parts must be suitably packaged and identified by nameplate, tagging, or stamping. Spare parts must be delivered to the Contracting Officer at the time of the Government testing and must be accompanied by an inventory list.

3.12.3 Document Storage Cabinet

Upon completion of the project, but prior to project close-out, place in the document storage cabinet copies of the following record documentation:

a. As-built shop drawings

b. Product data sheets

c. Design calculations

d. Site-specific software data package

e. All documentation required by SD-06.

-- End of Section --

SECTION 31 00 00

EARTHWORK
08/08

PART 1 GENERAL

~~1.1 MEASUREMENT PROCEDURES~~

~~1.1.1 Excavation~~

~~The unit of measurement for excavation and borrow will be the cubic meter, computed by the average end area method from cross sections taken before and after the excavation and borrow operations, including the excavation for ditches, gutters, and channel changes, when the material is acceptably utilized or disposed of as herein specified. The measurements will include authorized excavation of rock (except for piping trenches that is covered below), authorized excavation of unsatisfactory subgrade soil, and the volume of loose, scattered rocks and boulders collected within the limits of the work; allowance will be made on the same basis for selected backfill ordered as replacement. The measurement will not include the volume of subgrade material or other material that is scarified or plowed and reused in-place, and will not include the volume excavated without authorization or the volume of any material used for purposes other than directed. The volume of overburden stripped from borrow pits and the volume of excavation for ditches to drain borrow pits, unless used as borrow material, will not be measured for payment. The measurement will not include the volume of any excavation performed prior to the taking of elevations and measurements of the undisturbed grade.~~

~~1.1.2 Piping Trench Excavation~~

~~Measure trench excavation by the number of linear meters along the centerline of the trench and excavate to the depths and widths specified for the particular size of pipe. Replace unstable trench bottoms with a selected granular material. Include the additional width at manholes and similar structures, the furnishing, placing and removal of sheeting and bracing, pumping and bailing, and all incidentals necessary to complete the work required by this section.~~

~~1.1.3 Rock Excavation for Trenches~~

~~Measure and pay for rock excavation by the number of cubic meters of acceptably excavated rock material. Measure the material in place, but base volume on a maximum 750 mm width for pipes 300 mm in diameter or less, and a maximum width of 400 mm greater than the outside diameter of the pipe for pipes over 300 mm in diameter. Provide the measurement to include all authorized overdepth rock excavation as determined by the Contracting Officer. For manholes and other appurtenances, compute volumes of rock excavation on the basis of 300 mm outside of the wall lines of the structures.~~

~~1.1.4 Topsoil Requirements~~

~~Separate excavation, hauling, and spreading or piling of topsoil and related miscellaneous operations will be considered subsidiary obligations of the Contractor, covered under the contract unit price for excavation.~~

~~1.1.5 Overhaul Requirements~~

~~Allow the unit of measurement for overhaul to be the station-meter. The overhaul distance will be the distance in stations between the center of volume of the overhaul material in its original position and the center of volume after placing, minus the free-haul distance in stations. The haul distance will be measured along the shortest route determined by the Contracting Officer as feasible and satisfactory. Do no measure or waste unsatisfactory materials for overhaul where the length of haul for borrow is within the free-haul limits.~~

~~1.1.6 Select Granular Material~~

~~Measure select granular material in place as the actual cubic meters replacing wet or unstable material in trench bottoms [within the limits shown] [in authorized overdepth areas]. Provide unit prices which include furnishing and placing the granular material, excavation and disposal of unsatisfactory material, and additional requirements for sheeting and bracing, pumping, bailing, cleaning, and other incidentals necessary to complete the work.~~

~~1.2 PAYMENT PROCEDURES~~

~~Payment will constitute full compensation for all labor, equipment, tools, supplies, and incidentals necessary to complete the work.~~

~~1.2.1 Classified Excavation~~

~~Classified excavation will be paid for at the contract unit prices per cubic meter for common or rock excavation.~~

~~1.2.2 Piping Trench Excavation~~

~~Payment for trench excavation will constitute full payment for excavation and backfilling, [including specified overdepth] except in rock or unstable trench bottoms.~~

~~1.2.3 Rock Excavation for Trenches~~

~~Payment for rock excavation will be made in addition to the price bid for the trench excavation, and will include all necessary drilling and all incidentals necessary to excavate and dispose of the rock. Select granular material, used as backfill replacing rock excavation, will not be paid for separately, but will be included in the unit price for rock excavation.~~

~~1.2.4 Unclassified Excavation~~

~~Unclassified excavation will be paid for at the contract unit price per cubic meter for unclassified excavation.~~

~~1.2.5 Classified Borrow~~

~~Classified borrow will be paid for at the contract unit prices per cubic meter for common or rock borrow.~~

~~1.2.6 Unclassified Borrow~~

~~Unclassified borrow will be paid for at the contract unit price per cubic meter for unclassified borrow.~~

~~1.2.7 Authorized Overhaul~~

~~The number of station-meters of overhaul to be paid for will be the product of number of cubic meters of overhaul material measured in the original position, multiplied by the overhaul distance measured in stations of 100 meters and will be paid for at the contract unit price per station-meter for overhaul in excess of the free-haul limit as designated in paragraph DEFINITIONS.~~

~~1.2.8 Sheeting and Bracing~~

~~Sheeting and bracing, when shown or authorized by the Contracting Officer to be left in place, will be paid for as follows: [_____].~~

~~1.2.8.1 Timber Sheeting~~

~~Timber sheeting will be paid for as the number of board feet of lumber below finish grade measured in place prior to backfilling. Include in the measurement sheeting wasted when cut off between the finished grade and 300 mm below the finished grade.~~

~~1.2.8.2 Steel Sheeting and Soldier Piles~~

~~Steel sheeting, soldier piles, and steel bracing will be paid for according to the number of pounds of steel calculated. Calculate the steel by multiplying the measured in-place length in meters below finish grade by the unit weight of the section in kg per meter. Obtain unit weight of rolled steel sections from recognized steel manuals. [Included in the measurement sheeting wasted when cut off between the finished grade and a distance of up to [_____] meters below the finished grade.]~~

~~1.3 CRITERIA FOR BIDDING~~

~~Base bids on the following criteria:~~

~~a. Surface elevations are as indicated.~~

~~b. Pipes or other artificial obstructions, except those indicated, will not be encountered.~~

~~c. [Ground water elevations indicated by the boring log were those existing at the time subsurface investigations were made and do not necessarily represent ground water elevation at the time of construction.] [Ground water elevation is [_____] meters below existing surface elevation.]~~

~~d. [Ground water elevation is [_____] meters below existing surface elevation.]~~

~~e. [Material character is indicated by the boring logs.]~~

~~f. [Hard materials [and rock] [will not] [will] be encountered [in [_____] percent of the excavations] [at [_____] meter below existing surface elevations]].~~

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS A 1102	(2014) Method of Test for Sieve Analysis of Aggregates
JIS A 1103	(2014) Method of Test for Amount of Material Passing Test Sieve 75 µm in Aggregates
JIS A 1205	(2020~~09~~) Test Method for Liquid Limit and Plastic Limit of Soils
JIS A 1210	(2020~~09~~) Test Method for Soil Compaction Using a Rammer
JIS A 1211	(2020~~09~~) Test Methods for the California Bearing Ratio (CBR) of Soils in laboratory
JIS A 1214	(2013) Test Method for Soil Density by the Sand Replacement Method

JAPAN MINISTRY OF THE ENVIRONMENT (MOE)

Notification No.46	(2001) Environmental Quality Standards for Soil Contamination

JAPANESE GEOTECHNICAL SOCIETY (JGS)

JGS 0051	(20~~0~~15~~0~~) Method of Classification of Geomaterials for Engineering Purposes
JGS 1614	(2003) Test Method for Soil Density Using Nuclear Gauge

U.S. DEPARTMENT OF DEFENSE HEADQUARTERS, U.S. FORCES JAPAN (USFJ)

JEGS	(2020~~18~~) Japan Environmental Governing Standards

1.2 DEFINITIONS

1.2.1 Satisfactory Materials

Satisfactory materials comprise any materials classified by JGS 0051 as G, G-M, G-C~~, [G-O], [G-V]~~, GM, GC~~, [GO], [GV]~~, S, S-M, S-C~~, [S-O], [S-V]~~, SM, SC~~, [SO], [SV]~~, ML, MH, CL, CH~~, [OL], [OH], [OV], [VH1], [VH2]~~. Satisfactory materials for grading comprise stones less than ~~200~~150 mm, except for fill material for pavements~~ and railroads~~ which comprise stones less than 75 mm in any dimension. Satisfactory materials shall have less than 1% of swell expansion and a minimum CBR of 3 in accordance with JIS A 1211.

1.2.2 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; and material classified as satisfactory which contains root and other organic matter or frozen material. Notify the Contracting Officer when encountering any contaminated materials.

1.2.3 Cohesionless and Cohesive Materials

Cohesionless materials include materials classified in JGS 0051 as G, G-M, G-C, S, S-M, S-C. Cohesive materials include materials classified as GC, SC, ML, CL, MH, CH. Materials classified as GM, SM will be identified as cohesionless only when the fines are nonplastic. Perform testing, required for classifying materials, in accordance with JIS A 1205, JIS A 1102 and JIS A 1103.

1.2.4 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in JIS A 1210 (Methods C,D,E - non repeating method) abbreviated as a percent of laboratory maximum density (Modified Proctor Test).

~~1.2.5 Overhaul~~

~~Overhaul is the authorized transportation of satisfactory excavation or borrow materials in excess of the free-haul limit of [_____] stations. Overhaul is the product of the quantity of materials hauled beyond the free-haul limit, and the distance such materials are hauled beyond the free-haul limit, expressed in station meters.~~

1.2.5 Topsoil

Material suitable for topsoils obtained from ~~[~~offsite areas~~]~~ and ~~[~~ excavations~~]~~ ~~[areas indicated on the drawings]~~ is defined as: ~~N~~natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than 25 mm diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. Amend topsoil pH range to obtain a pH of 5.5 to 7.

1.2.6 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than ~~[_____]~~150 mm in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.2.7 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding ~~[~~0.375~~] [_____]~~ cubic meter in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling that is performed merely to increase production.

1.2.8 Unstable Material

Unstable materials are too wet to properly support the utility pipe, conduit, or appurtenant structure.

1.2.9 ~~Select Granular Material~~Imported Granular Fill

1.2.9.1 General Requirements

~~Select granular material~~Imported granular fill consist of materials classified as ~~[~~G~~]~~, ~~[~~G-M~~]~~, ~~[~~G-C~~]~~, ~~[~~S~~]~~, ~~[~~S-M~~]~~, ~~[~~S-C~~] or [_____]~~ by JGS 0051 where indicated. ~~[~~The liquid limit of such material must not exceed ~~[~~35~~] [_____]~~ percent when tested in accordance with JIS A 1205. The plasticity index must not be greater than ~~[~~12~~] [_____]~~ percent when tested in accordance with JIS A 1205, and not more than ~~[~~35~~] [_____]~~ percent by weight may be finer than 75 micrometers sieve when tested in accordance with JIS A 1103.~~]~~ ~~[Provide a minimum coefficient of permeability of [0.01] [_____] mm per second when tested in accordance with JIS A 1218.]~~

1.2.9.2 California Bearing Ratio Values

~~[~~Bearing Ratio: Provide Design CBR ratio in accordance with JIS A 1211 for a laboratory soaking period of not less than 4 days. ~~[~~Provide ~~[_____~~1~~]~~ percent maximum expansion.~~] [Conform the combined material to the following sieve analysis:]]~~ Imported granular fill shall have a minimum CBR of 20 in accordance with JIS A 1211.

~~Sieve Size~~	~~Percent Passing by Weight~~
~~63 mm~~	~~100~~
~~4.75 mm~~	~~40 - 85~~
~~2.00 mm~~	~~20 - 80~~
~~425 µm~~	~~10 - 60~~
~~75 µm~~	~~5 - 25~~

1.2.9.3 Common Fill

Common fill shall comply with the requirement of satisfactory materials.

1.2.10 Initial Backfill Material

Initial backfill consists of ~~select~~imported granular material or satisfactory materials free from rocks ~~[_____]~~150 mm or larger in any dimension or free from rocks of such size as recommended by the pipe manufacturer, whichever is smaller. For pavements and building structures, initial backfilling is related to zones that are located deeper than 1 meter depth below the top of the subgrade. Subgrade is considered as the zone within 1 meter below the base or subbase courses.

When the pipe is coated or wrapped for corrosion protection, free the initial backfill material of stones larger than ~~[_____]~~75 mm in any dimension or as recommended by the pipe manufacturer, whichever is smaller.

1.2.11 Expansive Soils

Expansive soils are defined as soils that ~~have a plasticity index equal to or greater than [_____] when tested in accordance with JIS A 1205 or~~ have expansion/swelling ratios equal to or greater than ~~[_____]~~1% when tested in accordance with JIS A 1211.

~~1.2.12 Nonfrost Susceptible (NFS) Material~~

~~Nonfrost susceptible material are a uniformly graded washed sand with a maximum particle size of [_____] mm and less than 5 percent passing the 0.075 mm size sieve, and with not more than 3 percent by weight finer than 0.02 mm grain size.~~

~~1.2.13 Pile Supported Structure~~

~~As used herein, a structure where both the foundation and floor slab are pile supported.~~

1.3 SYSTEM DESCRIPTION

~~Subsurface soil boring logs are [shown on the drawings] [appended to the SPECIAL CONTRACT REQUIREMENTS]. The subsoil investigation report and samples of materials taken from subsurface investigations may be examined at [_____]. These data represent the best subsurface information available; however, variations may exist in the subsurface between boring locations.~~

1.3.1 Classification of Excavation

~~[No consideration will be given to the nature of the materials, and all excavation will be designated as unclassified excavation.] [~~Finish the specified excavation on a classified basis, in accordance with the following designations and classifications.~~]~~ All excavations shall meet SAFETY REQUIREMENTS in accordance with EM 385-1-1.

1.3.1.1 Common Excavation

Include common excavation with the satisfactory removal and disposal of all materials not classified as rock excavation.

~~1.3.1.2 Rock Excavation~~

~~Submit notification of encountering rock in the project. Include rock excavation with excavating, grading, disposing of material classified as rock, and the satisfactory removal and disposal of boulders 1/2 cubic meter or more in volume; solid rock; rock material that is in ledges, bedded deposits, and unstratified masses, which cannot be removed without systematic drilling; firmly cemented conglomerate deposits possessing the characteristics of solid rock impossible to remove without systematic drilling; and hard materials (see Definitions). Include the removal of any concrete or masonry structures, except pavements, exceeding 1/2 cubic meter in volume that may be encountered in the work in this classification. If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be~~

~~classified as rock excavation, uncover such material and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation and has taken cross sections as required. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow ample time for classification and cross sectioning of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.~~

1.3.2 Dewatering Work Plan

Temporary dewatering systems may be expected during the construction stage. Fluctuations in groundwater level may occur due to change in seasons and rainfall variations. Submit procedures for accomplishing dewatering operations work.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for ~~[Contractor Quality Control approval.] [~~information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Shoring; G~~[, [_____]]~~

Dewatering Work Plan; G~~[, [_____]]~~

SD-06 Test Reports

Testing

Borrow Site Testing

Within 24 hours of conclusion of physical tests, submit ~~[_____]~~1 cop~~ies~~y of test results, including calibration curves and results of calibration tests.

PART 2 PRODUCTS

2.1 REQUIREMENTS FOR OFFSITE SOILS

Offsite soils shall be satisfactory materials. Test offsite soils brought in for use as backfill for Total Petroleum Hydrocarbons (TPH), Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and full Toxicity Characteristic Leaching Procedure (TCLP) including ignitability, corrosivity and reactivity. Perform tests and verify threshold values in accordance with JAPAN Ministry of the Environment Notification No.46 guidelines and JEGS. Provide Borrow Site Testing for TPH, BTEX and TCLP from a composite sample of material from the borrow site, with at least one test from each borrow site. Do not bring material onsite until tests have been approved by the Contracting Officer.

2.2 BURIED WARNING AND IDENTIFICATION TAPE

Provide ~~[~~polyethylene plastic~~]~~ ~~[and]~~or ~~[~~metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic~~]~~ warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 75 mm minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

Warning Tape Color Codes	
Red	Electric
Yellow	Gas, Oil; Dangerous Materials
Orange	Telephone and Other Communications
Blue	Water Systems
Green	Sewer Systems
White	Steam Systems
Gray	Compressed Air

2.2.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.08 mm and a minimum strength of 10.3 MPa lengthwise, and 8.6 MPa crosswise, with a maximum 350 percent elongation.

2.2.2 Detectable Warning Tape for Non-Metallic Piping

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.10 mm, and a minimum strength of 10.3 MPa lengthwise and 8.6 MPa crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 920 mm deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.3 DETECTION WIRE FOR NON-METALLIC PIPING

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

~~2.4 MATERIAL FOR RIP-RAP~~

~~Provide [Bedding material] [Grout] [Filter fabric] and rock conforming to [these requirements] [[_____] local Standard] for construction indicated.~~

~~2.4.1 Bedding Material~~

~~Provide bedding material consisting of sand, gravel, or crushed rock, well graded, [or poorly graded] with a maximum particle size of 50 mm. Compose material of tough, durable particles. Allow fines passing the 75 micrometers standard sieve with a plasticity index less than six.~~

~~2.4.2 Grout~~

~~Provide durable grout composed of cement, water, an air-entraining admixture, and sand mixed in proportions of one part portland cement to [two] [_____] parts of sand, sufficient water to produce a workable mixture, and an amount of admixture which will entrain sufficient air, as determined by the Contracting Officer. Mix grout in a concrete mixer. Allow a sufficient mixing time to produce a mixture having a consistency permitting gravity flow into the interstices of the rip-rap with limited spading and brooming.~~

~~2.4.3 Rock~~

~~Provide rock fragments sufficiently durable to ensure permanence in the structure and the environment in which it is to be used. Use rock fragments free from cracks, seams, and other defects that would increase the risk of deterioration from natural causes. Provide fragments sized so that no individual fragment exceeds a weight of [68] [_____] kg and that no more than 10 percent of the mixture, by weight, consists of fragments weighing 0.91 kg or less each. Provide rock with a minimum specific gravity of [2.50] [_____]. Do not permit the inclusion of more than trace [1 percent] [_____] quantities of dirt, sand, clay, and rock fines.~~

~~2.5 CAPILLARY WATER BARRIER~~

~~Provide capillary water barrier of clean, poorly graded crushed rock, crushed gravel, or uncrushed gravel placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below. Conform to JIS A 5005 for fine aggregate grading with a maximum of 3 percent by weight passing JIS A 1103, 75 micrometers sieve, [or] [37.5 mm and no more than 2 percent by weight passing the 4.75 mm size sieve] [or coarse aggregate with "S-x" designation (where x is the maximum nominal gravel size) in accordance with JIS A 5001].~~

PART 3 EXECUTION

3.1 STRIPPING OF TOPSOIL

Where indicated or directed, strip topsoil to a depth of ~~[~~100~~] [_____]~~ mm. Spread topsoil on areas already graded and prepared for topsoil, or transported and deposited in stockpiles convenient to areas that are to receive application of the topsoil later, or at locations indicated or specified. Keep topsoil separate from other excavated materials, brush, litter, objectionable weeds, roots, stones larger than 50 mm in diameter, and other materials that would interfere with planting and maintenance operations~~. [Stockpile in locations indicated] [~~. Remove from the site~~]~~ any surplus of topsoil from excavations and gradings.

3.2 GENERAL EXCAVATION

Perform excavation of every type of material encountered within the limits of the project to the lines, grades, and elevations indicated and as specified. Perform the grading in accordance with the typical sections shown and the tolerances specified in paragraph FINISHING. Transport satisfactory excavated materials and place in fill or embankment within the limits of the work. Excavate unsatisfactory materials encountered within the limits of the work below grade and replace with satisfactory materials as directed. Include such excavated material and the satisfactory material ordered as replacement in excavation. Dispose surplus satisfactory excavated material not required for fill or embankment and ~~in areas approved for surplus material storage or designated waste areas. Dispose~~ unsatisfactory excavated material in ~~designated waste or spoil areas~~certified waste facilities. During construction, perform excavation and fill in a manner and sequence that will provide proper drainage at all times. Excavate material required for fill or embankment in excess of that produced by excavation within the grading limits from the borrow areas indicated or from other approved areas selected by the Contractor as specified.

3.2.1 Ditches, Gutters, and Channel Changes

Finish excavation of ditches, gutters, and channel changes by cutting accurately to the cross sections, grades, and elevations shown on ~~Drawing~~ drawing~~ Sheet No. [______]~~. Do not excavate ditches and gutters below grades shown. Backfill the excessive open ditch or gutter excavation with satisfactory, thoroughly compacted, material or with suitable stone or cobble to grades shown. Dispose excavated material as shown or as directed, except in no case allow material be deposited a maximum 1 meter from edge of a ditch. Maintain excavations free from detrimental quantities of leaves, brush, sticks, trash, and other debris until final acceptance of the work.

3.2.2 Drainage Structures

Make excavations to the lines, grades, and elevations shown, or as directed. Provide trenches and foundation pits of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown. Clean rock or other hard foundation material of loose debris and cut to a firm, level, stepped, or serrated surface. Remove loose disintegrated rock and thin strata. Do not disturb the bottom of the excavation when concrete or masonry is to be placed in an excavated area. Do not excavate to the final grade level until just before the concrete or masonry is to be placed. ~~Where pile foundations are to be used, stop the excavation of each pit at an elevation 300 mm above the base of the footing, as specified, before piles are driven. After the pile driving has been completed, remove loose and displaced material and complete excavation, leaving a smooth, solid, undisturbed surface to receive the concrete or masonry.~~

3.2.3 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Completely drain construction site during periods of construction to keep soil materials sufficiently dry. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity ~~[~~and~~] [or]~~ provide temporary ditches, swales, and other drainage features and equipment as required to maintain dry soils. When unsuitable working platforms for equipment operation and unsuitable soil support for subsequent construction features develop, remove unsuitable material and provide new soil material as specified herein. It is the responsibility of the Contractor to assess the soil and ground water conditions presented by the plans and specifications and to employ necessary measures to permit construction to proceed.

3.2.4 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. ~~Do not permit French drains, sumps, ditches or trenches within 0.9 m of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made.~~ Fluctuations in groundwater depth may occur due to change in seasons and rainfall variations. Take control measures by the time the excavation reaches the water level in order to maintain the integrity of the in situ material. While the excavation is open, maintain the water level continuously~~, at least [_____] m below the working level~~. ~~[~~Operate dewatering system continuously until construction work below existing water levels is complete. ~~Submit performance records weekly.] [Measure and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system.] [Relieve hydrostatic head in previous zones below subgrade elevation in layered soils to prevent uplift.]~~

3.2.5 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended in the manufacturer's printed installation manual. Provide vertical trench walls where no manufacturer's printed installation manual is available. Shore vertical trench walls more than ~~[_____]~~1.5 meters high, cut back to a stable slope, or provide with equivalent means of protection for employees who may be exposed to moving ground or cave in. ~~Shore vertical trench walls more than [_____] meters high.~~ Excavate trench walls which are cut back to at least the angle of repose of the soil. Give special attention to slopes which may be adversely affected by weather or moisture content. Do not exceed the trench width below the pipe top of 600 mm plus pipe outside diameter (O.D.) for pipes of less than 600 mm inside diameter, and do not exceed 900 mm plus pipe outside diameter for sizes larger than 600 mm inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures by the Contractor. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.2.5.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of ~~[_____]~~75 mm or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.2.5.2 Removal of Unyielding Material

Where ~~[overdepth is not indicated and]~~ unyielding material is encountered in the bottom of the trench, remove such material ~~[_____]~~100 mm below the required grade and replaced with suitable materials as provided in paragraph BACKFILLING AND COMPACTION.

3.2.5.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with ~~select granular material~~imported granular fill as provided in paragraph BACKFILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

3.2.5.4 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures ~~[sufficient to leave at least 300 mm clear between the outer structure surfaces and the face of the excavation or support members.] [~~of sufficient size to permit the placement and remova~~l~~ of forms for the full length and width of structure footings and foundations as shown.~~]~~ Clean rock or loose debris and cut to a firm surface either level, stepped, or serrated, as shown or as directed. Remove loose disintegrated rock and thin strata. Specify removal of unstable material. When concrete or masonry is to be placed in an excavated area, take special care not to disturb the bottom of the excavation. Do not excavate to the final grade level until just before the concrete or masonry is to be placed.

~~3.2.5.5 Jacking, Boring, and Tunneling~~

~~Unless otherwise indicated, provide excavation by open cut except that sections of a trench may be jacked, bored, or tunneled if, in the opinion of the Contracting Officer, the pipe, cable, or duct can be safely and properly installed and backfill can be properly compacted in such sections.~~

3.2.6 Underground Utilities

The Contractor is responsible for movement of construction machinery and equipment over pipes and utilities during construction. ~~[~~Perform work adjacent to non-Government utilities as indicated in accordance with procedures outlined by utility company.~~]~~ ~~[~~Excavation made with power-driven equipment is not permitted within ~~[600] [_____]~~ mm of known Government-owned utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer.~~]~~ Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

~~3.2.7 Structural Excavation~~

~~Ensure that footing subgrades have been inspected and approved by the Contracting Officer prior to concrete placement. Excavate to bottom of pile cap prior to placing or driving piles, unless authorized otherwise by~~

~~the Contracting Officer. Backfill and compact over excavations and changes in grade due to pile driving operations to 95 percent of JIS A 1210 maximum density.~~

3.3 SELECTION OF BORROW MATERIAL

Select borrow material to meet the requirements and conditions of the particular fill or embankment for which it is to be used. Obtain borrow material from the borrow areas ~~[shown on Drawing Sheet No. [_____]] [~~ within the limits of the project site, selected by the Contractor~~]~~ ~~[~~or~~]~~ ~~[~~ from approved private sources~~]~~. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for hauling from the owners. Borrow material from approved sources on Government-controlled land may be obtained without payment of royalties. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval. Consider necessary clearing, grubbing, and satisfactory drainage of borrow pits and the disposal of debris thereon related operations to the borrow excavation.

3.4 OPENING AND DRAINAGE OF EXCAVATION AND BORROW PITS

Notify the Contracting Officer sufficiently in advance of the opening of any excavation or borrow pit or borrow areas to permit elevations and measurements of the undisturbed ground surface to be taken. Except as otherwise permitted, excavate borrow pits and other excavation areas providing adequate drainage. Transport overburden and other spoil material to designated spoil areas or otherwise dispose of as directed. Provide neatly trimmed and drained borrow pits after the excavation is completed. Ensure that excavation of any area, operation of borrow pits, or dumping of spoil material results in minimum detrimental effects on natural environmental conditions._

3.5 SHORING

3.5.1 General Requirements

Submit a Shoring and Sheeting plan for approval 15 days prior to starting work. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheeting of excavations. Finish shoring, including sheet piling, and install as necessary to protect workmen, banks, adjacent paving, structures, and utilities. Remove shoring, bracing, and sheeting as excavations are backfilled, in a manner to prevent caving.

3.5.2 Geotechnical Engineer

Hire a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for updating the excavation, sheeting and dewatering plans as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly written report, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contracting Officer is responsible for arranging meetings with the Geotechnical

Engineer at any time throughout the contract duration.

3.6 GRADING AREAS

Where indicated, divide work into grading areas within which satisfactory excavated material will be placed in embankments, fills, and required backfills. Do not haul satisfactory material excavated in one grading area to another grading area except when so directed in writing. Place and grade stockpiles of satisfactory ~~[~~and unsatisfactory~~] [and wasted materials]~~ as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

3.7 FINAL GRADE OF SURFACES TO SUPPORT CONCRETE

Do not excavate to final grade until just before concrete is to be placed. ~~[For pile foundations, stop the excavation at an elevation of from 150 to 300 mm above the bottom of the footing before driving piles. After pile driving has been completed, complete the remainder of the excavation to the elevations shown.]~~ Only use excavation methods that will leave the foundation rock in a solid and unshattered condition. Roughen the level surfaces, and cut the sloped surfaces, as indicated, into rough steps or benches to provide a satisfactory bond. Protect shales from slaking and all surfaces from erosion resulting from ponding or water flow.

3.8 GROUND SURFACE PREPARATION

3.8.1 General Requirements

Remove and replace unsatisfactory material with satisfactory materials, as directed by the Contracting Officer, in surfaces to receive fill or in excavated areas. Scarify the surface to a depth of 150 mm before the fill is started. Plow, step, bench, or break up sloped surfaces steeper than 1 vertical to 4 horizontal so that the fill material will bond with the existing material. When subgrades are less than the specified density, break up the ground surface to a minimum depth of 150 mm, pulverizing, and compacting to the specified density. When the subgrade is part fill and part excavation or natural ground, scarify the excavated or natural ground portion to a depth of 300 mm and compact it as specified for the adjacent fill.

~~3.8.2 Frozen Material~~

~~Do not place material on surfaces that are muddy, frozen, or contain frost. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Moisten material as necessary [to plus or minus [_____] percent of optimum moisture] [to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used].~~

3.9 UTILIZATION OF EXCAVATED MATERIALS

Dispose unsatisfactory materials removing from excavations into designated waste disposal or spoil areas. Use satisfactory material removed from excavations, insofar as practicable, in the construction of fills, embankments, subgrades, shoulders, bedding (as backfill), and for similar purposes. Submit procedure and location for disposal of unused satisfactory material. Submit proposed source of borrow material. Do not waste any satisfactory excavated material without specific written authorization. Dispose of satisfactory material, authorized to be wasted, in designated areas approved for surplus material storage or designated waste areas as directed. Clear and grub newly designated waste areas on Government-controlled land before disposal of waste material thereon. Stockpile and use coarse rock from excavations for constructing slopes or embankments adjacent to streams, or sides and bottoms of channels and for protecting against erosion. Do not dispose excavated material to obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. Do not reuse excavated materials for grading purposes at sloped areas. Reuse of excavated materials is only allowed at landscaped areas and for initial backfilling under pavements.

3.10 BURIED TAPE AND DETECTION WIRE

3.10.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape 300 mm below finished grade; under pavements and slabs, bury tape 150 mm below top of subgrade.

3.10.2 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed 300 mm above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of 0.9 m of wire, coiled, remaining accessible in each manhole. Furnish insulated wire over it's entire length. Install wires at manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

3.11 BACKFILLING AND COMPACTION

Place backfill adjacent to any and all types of structures, in successive horizontal layers of loose materia not more than ~~200~~300 mm in depth for initial backfilling zones and not more than 150 mm in depth for other zones. Compact to at least 90 percent laboratory maximum density for cohesive materials or 95 percent laboratory maximum density for cohesionless materials in accordance with JIS A 1210, to prevent wedging action or eccentric loading upon or against the structure. Backfill material must be ~~within the range of -2 to +2 percent of optimum moisture content at the time of compaction.~~properly conditioned to achieve specified grade of compaction. For in-situ clay soils, compact soils at moisture contents corresponding to the wet site of the compaction curve.

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs GROUND SURFACE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.

3.11.1 Trench Backfill

Backfill trenches to the grade shown. ~~[~~Backfill the trench to ~~[_____]~~0.6 meters above the top of pipe prior to performing the required pressure tests. Leave the joints and couplings uncovered during the pressure test.~~]~~ ~~[~~Do not backfill the trench until all specified tests are performed.~~]~~

3.11.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with ~~select granular material~~imported granular fill or initial backfill material.

3.11.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with ~~select granular material~~imported granular fill placed in layers not exceeding 150 mm loose thickness.

3.11.1.3 Bedding and Initial Backfill

~~[~~Provide bedding of the type and thickness shown.~~]~~ Place initial backfill material and compact it with approved tampers to a height of at least 300 mm above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with as shown in design drawings, except as specified herein. Compact backfill to top of pipe to 90 percent of JIS A 1210 maximum density. Provide plastic piping with bedding to spring line of pipe. Provide materials as follows:

3.11.1.3.1 Class I

Angular, 6 to 40 mm, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.

3.11.1.3.2 Class II

Coarse sands and gravels with maximum particle size of 40 mm, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types G, G-M, G-C, S, S-M, S-C are included in this class as specified in JGS 0051.

3.11.1.3.3 Sand

Clean, coarse-grained sand classified as ~~[_____] in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL, [gradation [_____] of the [local Standard] or [~~S~~]~~, ~~[~~S-M~~]~~, ~~[~~S-C~~]~~ by JGS 0051 for ~~[~~bedding~~]~~ ~~[~~and~~]~~ ~~[~~backfill~~]~~ ~~[~~as indicated~~]]~~.

3.11.1.3.4 Gravel and Crushed Stone

Clean, coarsely graded natural gravel, crushed stone or a combination thereof identified as~~ [_____]~~_~~ in accordance with Section 31 23 00.00 20 ~~

~~EXCAVATION AND FILL, [gradation [_____] of the [DOT] [State Standard]] or having a classification of [~~G~~]~~, ~~[~~G-M~~]~~, ~~[~~G-C~~]~~ in accordance with JGS 0051 for ~~[~~bedding~~]~~ ~~[~~and~~]~~ ~~[~~backfill~~]~~ ~~[~~as indicated~~]~~. ~~[~~Do not exceed maximum particle size of ~~[75]~~ ~~[_____]~~ mm.~~]~~

3.11.1.3.5 Imported Granular Fill

Imported granular fill shall be compacted to 95% compaction in accordance with ASTM D1557 or equivalent JIS A 1210 (Modified Proctor). The engineered granular fill shall not exceed 35% and 15% for its limit Limit Liquid (LL) and Plasticity Index (PI), respectively, and shall not have more than 35% fines (soil finer than the No. 200 sieve) and shall have a minimum CBR value of 20.

3.11.1.4 Final Backfill

Fill the remainder of the trench, except for special materials for roadways~~, railroads and airfields,~~ and sidewalks with satisfactory material. Place backfill material and compact as follows:

3.11.1.4.1 Roadways~~, Railroads, and Airfields~~

Place backfill up to the required elevation as specified. Do not permit water flooding or jetting methods of compaction.

3.11.1.4.2 Sidewalks, Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 300 mm loose thickness, and compact it to 85 percent maximum density for cohesive soils and 90 percent maximum density for cohesionless soils. ~~[Allow water flooding or jetting methods of compaction for granular noncohesive backfill material. Do not allow water jetting to penetrate the initial backfill.] [~~Do not permit compaction by water flooding or jetting.~~]~~ Apply this requirement to all other areas not specifically designated above.

3.11.2 Backfill for Appurtenances

After the manhole, catchbasin, inlet, or similar structure has been constructed~~ [and the concrete has been allowed to cure for [_____] days]~~, place backfill in such a manner that the structure is not be damaged by the shock of falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.12 SPECIAL REQUIREMENTS

Special requirements for both excavation and backfill relating to the specific utilities are as follows:

~~3.12.1 Gas Distribution~~

~~Excavate trenches to a depth that will provide a minimum 450 mm of cover in rock excavation and a minimum 600 mm of cover in other excavation.~~

3.12.1 Water Lines

Excavate trenches to a depth that provides a minimum cover of ~~[_____]~~1 meter~~s~~ in traffic areas from the existing ground surface, or from the indicated finished grade, whichever is lower, to the top of the pipe.

~~[For fire protection yard mains or piping, an additional [_____] mm of cover is required.]~~

~~3.12.2 Heat Distribution System~~

~~Free initial backfill material of stones larger than 6.3 mm in any dimension.~~

3.12.2 Electrical Distribution System

Provide a minimum cover of 600 mm from the finished grade to direct burial cable and conduit or duct line, unless otherwise indicated.

~~3.12.3 Sewage Absorption Trenches or Pits~~

~~3.12.3.1 Porous Fill~~

~~Provide backfill material consisting of clean crushed rock or gravel having a gradation [such that 100 percent passes the 50 mm sieve and zero percent passes the 12.5 mm sieve.] [conforming to the requirements of gradation [4.75 mm] [_____] for coarse aggregate in JIS A 5005.]~~

~~3.12.3.2 Cover~~

~~[Filter fabric] [Concrete] [or a layer of straw at least 50 mm thick] as indicated.~~

~~3.12.4 Rip-Rap Construction~~

~~Construct rip-rap [on bedding material] [on filter fabric] [with grout] [in accordance with [_____] local Standard, paragraph [_____]] in the areas indicated. Trim and dress indicated areas to conform to cross sections, lines and grades shown within a tolerance of 30 mm.~~

~~3.12.4.1 Bedding Placement~~

~~Spread [filter fabric] bedding material uniformly to a thickness of at least [75] [_____] mm on prepared subgrade as indicated. [Compaction of bedding is not required. Finish bedding to present even surface free from mounds and windrows.]~~

~~3.12.4.2 Stone Placement~~

~~Place rock for rip-rap on prepared bedding material to produce a well graded mass with the minimum practicable percentage of voids in conformance with lines and grades indicated. Distribute larger rock fragments, with dimensions extending the full depth of the rip-rap throughout the entire mass and eliminate "pockets" of small rock fragments. Rearrange individual pieces by mechanical equipment or by hand as necessary to obtain the distribution of fragment sizes specified above. [For grouted rip-rap, hand-place surface rock with open joints to facilitate grouting and do not fill smaller spaces between surface rock with finer material. Provide at least one "weep hole" through grouted rip-rap for every 4.65 square meters of finished surface. Provide weep holes with columns of bedding material, 100 mm in diameter, extending up to the rip-rap surface without grout.]~~

~~3.12.4.3 Grouting~~

~~[Prior to grouting, wet rip-rap surfaces. Grout rip-rap in successive longitudinal strips, approximately 3 m in width, commencing at the lowest strip and working up the slope. Distribute grout to place of final deposit and work into place between stones with brooms, spades, trowels, or vibrating equipment. Take precautions to prevent grout from penetrating bedding layer. Protect and cure surface for a minimum of 7 days.]~~

~~3.13 EMBANKMENTS~~

~~3.13.1 Earth Embankments~~

~~Construct earth embankments from satisfactory materials free of organic or frozen material and rocks with any dimension greater than 75 mm. Place the material in successive horizontal layers of loose material not more than 200 mm in depth. Spread each layer uniformly on a soil surface that has been moistened or aerated as necessary, and scarified or otherwise broken up so that the fill will bond with the surface on which it is placed. After spreading, plow, disk, or otherwise break up each layer; moisten or aerate as necessary; thoroughly mix; and compact to at least 90 percent laboratory maximum density in accordance with JIS A 1210 for cohesive materials or 95 percent laboratory maximum density for cohesionless materials. Backfill material must be within the range of -2 to +2 percent of optimum moisture content at the time of compaction, when required.~~

~~Compaction requirements for the upper portion of earth embankments forming subgrade for pavements are identical with those requirements specified in paragraph SUBGRADE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.~~

~~3.13.2 Rock Embankments~~

~~Construct rock embankments from material classified as rock excavation, as defined above, placed in successive horizontal layers of loose material not more than [_____] mm in depth. Do not use pieces of rock larger than [_____] mm in the greatest dimension. Spread each layer of material uniformly, completely saturate, and compact to a minimum density of [_____] kg/cubic meter. Adequately bond each successive layer of material to the material on which it is placed. Finish compaction with vibratory compactors weighing at least [_____] metric tons, heavy rubber-tired rollers weighing at least [_____] metric tons, or steel-wheeled rollers weighing at least [_____] metric tons. [Do not use rock excavation as fill material for the construction of pavements.] [In embankments on which pavements are to be constructed, do not use rock above a point [_____] mm below the surface of the pavement.]~~

3.13 SUBGRADE PREPARATION

~~3.13.1 Proof Rolling~~

~~Finish proof rolling on an exposed subgrade free of surface water (wet conditions resulting from rainfall) which would promote degradation of an otherwise acceptable subgrade. [After stripping,] proof roll the existing subgrade of the [_____] with six passes of a [dump truck loaded with 6 cubic meters of soil] [13.6 meter tons, pneumatic-tired roller.] Operate~~

~~the [roller] [truck] in a systematic manner to ensure the number of passes over all areas, and at speeds between 4 to 5.5 km/hour. [When proof rolling, provide one-half of the passes made with the roller in a direction perpendicular to the other passes.] Notify the Contracting Officer a minimum of 3 days prior to proof rolling. Perform proof rolling in the presence of the Contracting Officer. Undercut rutting or pumping of material [as directed by the Contracting Officer] [to a depth of [_____] mm] and replace with [fill and backfill] [select] material. [Prepare bids based on replacing approximately [_____] square meters, with an average depth of [_____] mm at various locations.]~~

3.13.1 Construction

Shape subgrade to line, grade, and cross section, and compact as specified. Include plowing, disking, and any moistening or aerating required to obtain specified compaction for this operation. Remove soft or otherwise unsatisfactory material and replace with satisfactory excavated material or other approved material as directed. Excavate rock encountered in the cut section to a depth of 150 mm below finished grade for the subgrade. Bring up low areas resulting from removal of unsatisfactory material or excavation of rock to required grade with satisfactory materials, and shape the entire subgrade to line, grade, and cross section and compact as specified. ~~[After rolling, the surface of the subgrade for roadways shall not show deviations greater than 13 mm when tested with a 4 m straightedge applied both parallel and at right angles to the centerline of the area.] [After rolling, do not show deviations for the surface of the subgrade for airfields greater than [_____] mm when tested with a [_____] meter straightedge applied both parallel and at right angles to the centerline of the area.]~~ Do not vary the elevation of the finish subgrade more than 15 mm from the established grade and cross section.

3.13.2 Compaction

Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment. Except for paved areas and railroads, compact each layer of the embankment to at least ~~[_____]~~90% for cohesive and 95% for cohesionless materials ~~percent~~ of laboratory maximum density.

~~3.13.2.1 Subgrade for Railroads~~

~~Compact subgrade for railroads to at least 90 percent laboratory maximum density for cohesive materials or 95 percent laboratory maximum density for cohesionless materials in accordance with JIS A 1210.~~

3.13.2.1 Subgrade for Pavements

Compact subgrade for pavements to at least 90% for cohesive and 95% for cohesionless materials of the ~~[_____] percentage laboratory~~ maximum density in accordance with JIS A 1210 ~~for the depth below the surface of the pavement shown~~. When more than one soil classification is present in the subgrade, thoroughly blend, reshape, and compact the top ~~[_____]~~150 mm of subgrade.

~~3.13.2.2 Subgrade for Shoulders~~

~~Compact subgrade for shoulders to at least [_____] percentage laboratory maximum density in accordance with JIS A 1210 for the [depth below the~~

~~surface of shoulder shown] [full depth of the shoulder].~~

~~3.13.2.3 Subgrade for Airfield Pavements~~

~~Compact top 600 mm below finished pavement or top 300 mm of subgrades, whichever is greater, to [100] [_____] percent of JIS A 1210; compact fill and backfill material to [100] [_____] percent of JIS A 1210.~~

~~3.14 SHOULDER CONSTRUCTION~~

~~Construct shoulders of satisfactory excavated or borrow material or as otherwise shown or specified.. Submit advanced notice on shoulder construction for rigid pavements. Construct shoulders immediately after adjacent paving is complete. In the case of rigid pavements, do not construct shoulders until permission of the Contracting Officer has been obtained. Compact the entire shoulder area to at least the percentage of maximum density as specified in paragraph SUBGRADE PREPARATION above, for specific ranges of depth below the surface of the shoulder. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment. Finish shoulder construction in proper sequence in such a manner that adjacent ditches will be drained effectively and that no damage of any kind is done to the adjacent completed pavement. Align the completed shoulders true to grade and shaped to drain in conformity with the cross section shown.~~

3.14 FINISHING

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 30 mm of the grades and elevations indicated except that the degree of finish for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed from settlement or washing to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

3.14.1 Subgrade and Embankments

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not permit the storage or stockpiling of materials on the finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, or frozen subgrade.

~~3.14.2 Capillary Water Barrier~~

~~Place a capillary water barrier under concrete floor and area-way slabs grade directly on the subgrade and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.~~

3.14.2 Grading Around Structures

Construct areas within 1.5 m outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

3.15 PLACING TOPSOIL

On areas to receive topsoil, prepare the compacted subgrade soil to a 50 mm depth for bonding of topsoil with subsoil. Spread topsoil evenly to a thickness of ~~[_____]~~300 mm and grade to the elevations and slopes shown. Do not spread topsoil when frozen or excessively wet or dry. Obtain material required for topsoil in excess of that produced by excavation within the grading limits from ~~[~~offsite areas~~] [areas indicated]~~.

3.16 TESTING

Perform testing by a ~~Corps validated~~Certified commercial testing laboratory or the Contractor's validated testing facility. Submit qualifications of the ~~Corps~~ Certified validated commercial testing laboratory or the Contractor's validated testing facilities. If the Contractor elects to establish testing facilities, do not permit work requiring testing until the Contractor's facilities have been inspected, Corps validated and approved by the Contracting Officer.

a. Determine field in-place density in accordance with ~~[~~JIS A 1214~~]~~ or~~[~~ JGS 1614~~]~~. ~~[~~When JGS 1614 is used, check the calibration curves furnished with the moisture gauges along with density calibration checks as described in JGS 1614; check the calibration of both the density and moisture gauges at the beginning of a job on each different type of material encountered and at intervals as directed by the Contracting Officer. ~~and adjust using only the sand cone method as described in JIS A 1214. results in a wet unit weight of soil in determining the moisture content of the soil when using this method.~~ Since JGS 1614 uses equipment that contains radiactive materials, notify the Base BioEnvironmental and/or Environmental Office and the Contracting Officer if this method is used.

b. ~~Check the calibration curves furnished with the moisture gauges along with density calibration checks as described in JGS 1614; check the calibration of both the density and moisture gauges at the beginning of a job on each different type of material encountered and at intervals as directed by the Contracting Officer.]~~ When test results indicate, as determined by the Contracting Officer, that compaction is not as specified, remove the material, replace and recompact to meet specification requirements.

c. Perform tests on recompacted areas to determine conformance with specification requirements. Appoint a registered professional civil engineer to certify inspections and test results. These certifications shall state that the tests and observations were performed by or under the direct supervision of the engineer and that the results are representative of the materials or conditions being certified by the tests. The following number of tests, if performed at the appropriate time, will be the minimum acceptable for each type operation.

3.16.1 Fill and Backfill Material Gradation

One test per ~~[_____]~~100 cubic meters stockpiled or in-place source material. Determine gradation of fill and backfill material in accordance with ~~[~~JIS A 1102~~]~~ and ~~[~~JIS A 1103~~]~~.

3.16.2 In-Place Densities

a. One test per ~~[_____]~~200 square meters, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines.

b. One test per ~~[_____]~~100~~-~~ square meters, or fraction thereof, of each lift of fill or backfill areas compacted by hand-operated machines.

c. One test per ~~[_____]~~35 linear meters, or fraction thereof, of each lift of embankment or backfill for ~~[~~roads~~] [airfields]~~.

~~d. One test per [_____] linear meters, or fraction thereof, of each lift of embankment or backfill for railroads.~~

3.16.3 Check Tests on In-Place Densities

If JGS 1614 is used, perform one check in-place densities by JIS A 1214 as per each 10 tests conducted by JGS 1614~~ follows:~~.

~~a. One check test per lift for each [_____] square meters, or fraction thereof, of each lift of fill or backfill compacted by other than hand-operated machines.~~

~~b. One check test per lift for each [_____] square meters, of fill or backfill areas compacted by hand-operated machines.~~

~~c. One check test per lift for each [_____] linear meters, or fraction thereof, of embankment or backfill for [roads] [airfields].~~

~~d. One check test per lift for each [_____] linear meters, or fraction thereof, of embankment or backfill for railroads.~~

3.16.4 Moisture Contents

In the stockpile, excavation, or borrow areas, perform tests as required, to meet compaction requirements, per type of material or source of material being placed during stable weather conditions. During unstable weather, perform tests as dictated by local conditions and approved by the Contracting Officer.

3.16.5 Optimum Moisture and Laboratory Maximum Density

Perform tests for each type material or source of material including borrow material to determine the optimum moisture and laboratory maximum density values. One representative test per ~~[_____]~~100 cubic meters of fill and backfill~~fill and backfill~~, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum density.

3.16.6 Tolerance Tests for Subgrades

Perform continuous checks on the degree of finish specified in paragraph SUBGRADE PREPARATION during construction of the subgrades.

3.16.7 Displacement of Sewers

After other required tests have been performed and the trench backfill compacted to ~~[[_____], meters above the top of the pipe] [~~the finished grade surface~~]~~, inspect the pipe to determine whether significant displacement has occurred. Conduct this inspection in the presence of the Contracting Officer. Inspect pipe sizes larger than 900 mm, while inspecting smaller diameter pipe by shining a light or laser between manholes or manhole locations, or by the use of television cameras passed through the pipe. If, in the judgment of the Contracting Officer, the interior of the pipe shows poor alignment or any other defects that would cause improper functioning of the system, replace or repair the defects as directed at no additional cost to the Government.

3.17 DISPOSITION OF SURPLUS MATERIAL

Remove surplus material or other soil material not required or suitable for filling or backfilling, and brush, refuse, stumps, roots, and timber ~~[to a Government disposal area [as indicated][which is located within a haul distance of [_____] km]][from Government property to an approved location] [~~from Government property and delivered to a licensed/permitted facility or to a location approved by the Contracting Officer.~~].~~

-- End of Section --

SECTION 32 11 20

BASE COURSE FOR RIGID AND SUBBASES FOR FLEXIBLE PAVING
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1102	(2014) Method of Test for Sieve Analysis of Aggregates
JIS A 1103	(2014) Method of Test for Amount of Material Passing Test Sieve 75 µm in Aggregates
JIS A 1104	(2006) Methods of Test for Density and Water Absorption of Fine Aggregates
JIS A 1121	(2007) Method of Test for Resistance to Abrasion of Coarse Aggregate by Use of the Los Angeles Machine
JIS A 1201	(2009) Practise for Preparing Disturbed Soil Samples for Soil Testing
JIS A 1205	(2020~~09~~) Test Method for Liquid Limit and Plastic Limit of Soils
JIS A 1210	(2020~~09~~) Test Method for Soil Compaction Using a Rammer
JIS A 1214	(2013) Test Method for Soil Density by the Sand Replacement Method
JIS A 5001	(2008) Crushed Stone for Road Construction
JIS Z 8801	(2006) Test Sieves Part 1: Test Sieves of Metal Wire Cloth

~~JAPANESE GEOTECHNICAL SOCIETY (JGS)~~JAPANESE GEOTECHNICAL SOCIETY (JGS)

JGS 0051	(2015~~00~~) Method of Classification of Geomaterials for Engineering Purposes

1.2 DEGREE OF COMPACTION

Degree of compaction required is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in

JIS A 1210 abbreviated as a percent of laboratory maximum dry density.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Plant, Equipment, and Tools

SD-06 Test Reports

Initial Tests; G

In-Place Tests; G

1.4 EQUIPMENT, TOOLS, AND MACHINES

All plant, equipment, and tools used in the performance of the work will be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform sampling and testing using a laboratory approved in accordance with Section 01 45 00.00 10~~01 45 00.00 2001 45 00.00 40~~ QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with JIS A 1201. When deemed necessary, the sampling will be observed by the Contracting Officer.

1.5.2 Tests

1.5.2.1 Sieve Analysis

Perform sieve analysis in conformance with JIS A 1103 and JIS A 1102 using sieves conforming to JIS Z 8801.

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with JIS A 1205.

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture in accordance with paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with JIS A 1214.

1.5.2.5 Wear Test

Perform wear tests on subbase course and or rigid pavement base course material in conformance with JIS A 1121.

1.5.2.6 Weight of Slag

Determine weight per cubic meter of slag in accordance with JIS A 1104.

1.6 ENVIRONMENTAL REQUIREMENTS

Perform construction when the atmospheric temperature is above 2 degrees C. When the temperature falls below 2 degrees C, protect all completed areas by approved methods against detrimental effects of freezing. Correct completed areas damaged by freezing, rainfall, or other weather conditions to meet specified requirements.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Flexible Paving Subbase Course

Provide aggregates conforming to JIS A 5001, C-30/RC-30 and C-40/RC-40_ consisting of crushed stone or slag, gravel, shell, sand, or other sound, durable, approved materials processed and blended or naturally combined. Provide aggregates which are free from lumps and balls of clay, organic matter, objectionable coatings, and other foreign material. The percentage of loss of material retained on the 4.75 mm sieve must not exceed 50 percent after 500 revolutions when tested in accordance with JIS A 1121. Provide aggregate that is reasonably uniform in density and quality. Provide slag that is an air-cooled, blast-furnace product having a dry weight of not less than 1050 kg/cubic meter. Provide aggregates with a maximum size of 37.5 mm for C-40/RC-40 and 31.5 mm for C-30/RC-30 and within the limits specified as follows:

Maximum Allowable Percentage by Weight
Passing Square-Mesh Sieve

Sieve Designation (mm)	C-30 / RC-30	C-40 /RC-40
53.0	----	100
37.5	100	95-100
31.5	95-100	----
26.5	----	----
19	55-85	50-80
13.2	----	----
4.75	15-45	15-40
2.36	5-30	5-25

The portion of any blended component and of the completed course passing the 0.425 mm must be either nonplastic or have a liquid limit not greater than 25 and a plasticity index not greater than 5.

2.1.2 Rigid Pavement Base Course

Provide aggregates for rigid pavement base course as specified for flexible paving subbase course.

2.2 TESTS, INSPECTIONS, AND VERIFICATIONS

2.2.1 Initial Tests

Perform one of each of the following tests on the proposed material prior to commencing construction to demonstrate that the proposed material meets all specified requirements prior to installation. Complete this testing for each source if materials from more than one source are proposed.

a. Sieve Analysis .

b. Liquid limit and plasticity index.

c. Moisture-density relationship.

d. Wear.

e. Weight per cubic meter of Slag.

Submit certified copies of test results for approval not less than 30 days before material is required for the work.

2.2.2 Approval of Material

Tentative approval of material will be based on initial test results.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated by the Contracting Officer to prevent segregation. Stockpile materials obtained from different sources separately.

3.3 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the subbase or rigid pavement base course. Do not construct subbase or rigid pavement base course on underlying course or subgrade that is frozen. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in JGS 0051, stabilize the surface prior to placement of the overlying course. Stabilize by mixing the overlying course material into the underlying course and compacting by approved methods. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a satisfactory condition until the overlying course is placed.

3.4 GRADE CONTROL

Provide a finished and completed subbase and rigid pavement base courses conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.5 MIXING AND PLACING MATERIALS

Mix and place the materials to obtain uniformity of the material at the water content specified. Make such adjustments in mixing or placing procedures or in equipment as may be directed to obtain the true grades, to minimize segregation and degradation, to reduce or accelerate loss or increase of water, and to insure a satisfactory subbase course.

3.6 LAYER THICKNESS

Compact the completed course to the thickness indicated. No individual layer may be thicker than 150 mm nor be thinner than 75 mm in compacted thickness. Compact the course(s) to a total thickness that is within 13 mm of the thickness indicated. Where the measured thickness is more than 13 mm deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 13 mm thicker than indicated, the course will be considered as conforming to the specified thickness requirements. The average job thickness will be the average of all thickness measurements taken for the job and must be within 6 mm of the thickness indicated. Measure the total thickness of the course(s) at intervals of one measurement for each 500 square meters of completed course. Measure total thickness using 75 mm diameter test holes penetrating the completed course.

3.7 COMPACTION

Compact each layer of the material, as specified, with approved compaction equipment. Maintain water content during the compaction procedure to within plus or minus 2 percent of optimum water content determined from laboratory tests as specified in this Section. Begin rolling at the outside edge of the surface and proceed to the center, overlapping on successive trips at least one-half the width of the roller. Slightly vary the length of alternate trips of the roller. Adjust speed of the roller as needed so that displacement of the aggregate does not occur. Compact mixture with hand-operated power tampers in all places not accessible to the rollers. Continue compaction of the subbase until each layer is compacted through the full depth to at least 100 percent of laboratory maximum density. Continue compaction of the rigid base course until each layer is compacted through the full depth to at least 95 percent of laboratory maximum density. Make such adjustments in compacting or finishing procedures as may be directed by the Contracting Officer to obtain true grades, to minimize segregation and degradation, to reduce or increase water content, and to ensure a satisfactory subbase and rigid pavement base course. Remove any materials that are found to be unsatisfactory and replace with satisfactory material or rework, as directed, to meet the requirements of this specification.

3.8 PROOF ROLLING

In addition to the compaction specified, proof roll subbase course under a flexible airfield pavement in areas designated on the drawings by application of 30 coverages for Class IV runways, 8 coverages for runways that support fighter aircraft only, and 4 coverages to all other paved areas, of a heavy pneumatic-tired roller having four or more tires abreast, each tire loaded to a minimum of 13,600 kg and inflated to a minimum of 862 kPa. A coverage is defined as the application of one tire print over the designated area. In the areas designated, apply proof rolling to the top layer of the completed subbase course. Maintain water content of the top layer of the subbase course as specified in paragraph COMPACTION from start of compaction to completion of proof rolling. Remove any subbase course materials that produce unsatisfactory results by proof rolling and replace with satisfactory materials. Then recompact and proof roll to meet specifications.

3.9 EDGES OF SUBBASE AND RIGID PAVEMENT BASE COURSE

Place approved material along the outer edges of the subbase and rigid pavement base course in sufficient quantity to compact to the thickness of the course being constructed. When the course is being constructed in two or more layers, simultaneously roll and compact at least a 600 mm width of this shoulder material with the rolling and compacting of each layer of the subbase and rigid pavement base course, as directed.

3.10 FINISHING

Finish the surface of the top layer of rigid pavement base course after final compaction and proof rolling by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin layers of material to the top layer of rigid pavement base course to meet grade. If the elevation of the top layer of rigid pavement base course is 13 mm or more below grade, scarify the top layer to a depth of at least 75 mm and blend new material in and compact and proof roll to bring to grade. Make adjustments to rolling and finishing procedures as directed by the Contracting Officer to minimize segregation and degradation, obtain grades, maintain moisture content, and insure an acceptable rigid pavement base course. Should the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the unsatisfactory portion and rework and recompact it or replace as directed.

3.11 SMOOTHNESS TEST

Construct the top layer so that the surface shows no deviations in excess of 10 mm when tested with a 3.00 m straightedge. Take measurements in successive positions parallel to the centerline of the area to be paved. Also take measurements perpendicular to the centerline at 15 meter intervals. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.12 FIELD QUALITY CONTROL

3.12.1 In-Place Tests

Perform one of each of the following tests on samples taken from the placed and compacted subbase and rigid pavement base course. Take samples and test at the rates indicated.

a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 500 square meters, or portion thereof, of completed area.

b. Perform sieve analysis on every lift of material placed and at a frequency of one sieve analysis for every 1,000 square meters, or portion thereof, of material placed.

c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.

d. Measure the thickness of each course at intervals providing at least one measurement for each 500 square meters or part thereof. Measure the thickness using test holes, at least 75 mm in diameter through the course.

3.12.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.13 TRAFFIC

Do not allow traffic on the completed subbase and rigid pavement base coursefor airfield pavements. For roads, do not allow heavy equipment on the completed rigid pavement base course except when necessary for construction. When it is necessary for heavy equipment to travel on the completed rigid pavement base course, protect the area against marring or damage to the completed work.

3.14 MAINTENANCE

Maintain the completed course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact. Retest any course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area that is damaged as necessary to comply with this specification.

3.15 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed as directed . No additional payments will be made for materials that have to be replaced.

-- End of Section --

SECTION 32 11 23

AGGREGATE BASE COURSES
08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1102	(2014) Method of Test for Sieve Analysis of Aggregates
JIS A 1103	(2014) Method of Test for Amount of Material Passing Test Sieve 75 µm in Aggregates
JIS A 1104	(2006) Methods of Test for Density and Water Absorption of Fine Aggregates
JIS A 1109	(2006) Method of Test for Density and Water Absorption of Fine Aggregates
JIS A 1110	(2006) Methods of Test for Density and Water Absorption of Coarse Aggregates
JIS A 1112	(2012) Method of Test for Washing Analysis of Fresh Concrete
JIS A 1121	(2007) Method of Test for Resistance to Abrasion of Coarse Aggregate by Use of the Los Angeles Machine
JIS A 1201	(2009) Practise for Preparing Disturbed Soil Samples for Soil Testing
JIS A 1205	(2020~~09~~) Test Method for Liquid Limit and Plastic Limit of Soils
JIS A 1210	(2020~~09~~) Test Method for Soil Compaction Using a Rammer
JIS A 1214	(2013) Test Method for Soil Density by the Sand Replacement Method
JIS A 5001	(2008) Crushed Stone for Road Construction
JIS Z 8801	(2006) Test Sieves Part 1: Test Sieves of Metal Wire Cloth

~~JAPANESE GEOTECHNICAL SOCIETY (JGS)~~JAPANESE GEOTECHNICAL SOCIETY (JGS)

JGS 0051 (20~~00~~15) Method of Classification of Geomaterials for Engineering Purposes

1.2 DEFINITIONS

For the purposes of this specification, the following definitions apply.

1.2.1 Aggregate Base Course

Aggregate base course (ABC) is well graded, durable aggregate uniformly moistened and mechanically stabilized by compaction.

1.2.2 Graded-Crushed Aggregate Base Course

Graded-crushed aggregate (GCA) base course is well graded, crushed, durable aggregate uniformly moistened and mechanically stabilized by compaction.

1.2.3 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in JIS A 1210 abbreviated as a percent of laboratory maximum dry density.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Plant, Equipment, and Tools

SD-06 Test Reports

Initial Tests; G

In-Place Tests; G

1.4 EQUIPMENT, TOOLS, AND MACHINES

All plant, equipment, and tools used in the performance of the work will be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform sampling and testing using a laboratory approved in accordance with Section 01 45 00.00 10~~01 45 00.00 20~~~~01 45 00.00 40~~ QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with JIS A 1201. When deemed necessary, the sampling will be observed by the Contracting Officer.

1.5.2 Tests

1.5.2.1 Sieve Analysis

Perform sieve analysis in conformance with JIS A 1103 and JIS A 1102 using sieves conforming to JIS Z 8801. ~~.~~

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with JIS A 1205.

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with JIS A 1214.

1.5.2.5 Wear Test

Perform wear tests on ABC and GCA course material in conformance with JIS A 1121.

1.5.2.6 Soundness

Perform soundness tests on GCA in accordance with JIS A 1112.

1.5.2.7 Weight of Slag

Determine weight per cubic meter of slag in accordance with JIS A 1104 on the ABC and GCA course material.

1.6 ENVIRONMENTAL REQUIREMENTS

Perform construction when the atmospheric temperature is above 2 degrees C. When the temperature falls below 2 degrees C, protect all completed areas by approved methods against detrimental effects of freezing. Correct completed areas damaged by freezing, rainfall, or other weather conditions to meet specified requirements.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide Aggregate Base Course (ABC) and Graded-Crushed Aggregate (GCA) base course consisting of clean, sound, durable particles of crushed stone, crushed slag, crushed gravel, crushed recycled concrete, angular sand, or other approved material. Provide ABC that is free of lumps of clay, organic matter, and other objectionable materials or coatings. Provide GCA that is free of silt and clay as defined by JGS 0051, organic matter, and other objectionable materials or coatings. The portion retained on the 4.75 mm sieve is known as coarse aggregate; that portion passing the 4.75 mm sieve is known as fine aggregate. Base course shall conform to JIS A 5001, M-30/RM-30 and M-40/RM-40. When the coarse and fine aggregate is supplied form more than one source, provide aggregate from each source that meets the specified requirements.

2.1.1 Coarse Aggregate

Provide coarse aggregates with angular particles of uniform density. Separately stockpile coarse aggregate supplied from more than one source.

a. Crushed Gravel: Provide crushed gravel that has been manufactured by crushing gravels and that meets all the requirements specified below.

b. Crushed Stone: Provide crushed stone consisting of freshly mined quarry rock, meeting all the requirements specified below.

c. Crushed Recycled Concrete: Provide crushed recycled concrete consisting of previously hardened portland cement concrete or other concrete containing pozzolanic binder material. Provide recycled concrete that is free of all reinforcing steel, bituminous concrete surfacing, and any other foreign material and that has been crushed and processed to meet the required gradations for coarse aggregate. Reject recycled concrete aggregate exceeding this value. Provide crushed recycled concrete that meets all other applicable requirements specified below.

d. Crushed Slag: Provide crushed slag that is an air-cooled blast-furnace product having an air dry unit weight of not less than 1120 kg/cubic meter as determined by JIS A 1104, and meets all the requirements specified below.

2.1.1.1 Aggregate Base Course

The percentage of loss of ABC coarse aggregate must not exceed 50 percent when tested in accordance with JIS A 1121.

2.1.1.2 Graded-Crushed Aggregate Base Course

The percentage of loss of GCA coarse aggregate must not exceed 40 percent loss when tested in accordance with JIS A 1121. Provide GCA coarse aggregate that does not exhibit a loss greater than 18 percent weighted average, at five cycles, when tested for soundness in magnesium sulfate, or 12 percent weighted average, at five cycles, when tested in sodium sulfate in accordance with JIS A 1112.

2.1.2 Fine Aggregate

Provide fine aggregates consisting of angular particles of uniform density.

2.1.2.1 Aggregate Base Course

Provide ABC fine aggregate that consists of screenings, angular sand, crushed recycled concrete fines, or other finely divided mineral matter processed or naturally combined with the coarse aggregate.

2.1.2.2 Graded-Crushed Aggregate Base Course

Provide GCA fine aggregate consisting of angular particles produced by crushing stone, slag, recycled concrete, or gravel that meets the requirements for wear and soundness specified for GCA coarse aggregate. Produce fine aggregate by crushing only particles larger than 4.75 mm sieve in size. Provide fine aggregate that contains at least 90 percent by weight of particles having two or more freshly fractured faces in the portion passing the 4.75 mm sieve and retained on the 2 mm sieve, and in the portion passing the 2 mm sieve and retained on the 0.425 mm sieve. Manufacture fine aggregate from gravel particles 95 percent of which by weight are retained on the 12.5 mm sieve.

2.1.3 Gradation Requirements

Apply the specified gradation requirements to the completed base course. Provide aggregates that are continuously well graded within the limits specified in TABLE 1. Use sieves that conform to JIS Z 8801.

TABLE 1. GRADATION OF AGGREGATES

Percentage by Weight Passing Square-Mesh Sieve

Sieve Designation (mm)	M-30 / RM-30	M-40 / RM-40
53.0	----	100
37.5	100	95-100
31.5	95-100	----
26.5	----	----
19.0	60-90	60-90
13.2	----	----
4.75	30-65	30-65
2.36	20-50	20-50
1.18	----	----
0.425	10-30	10-30
0.075	2-10	2-10

NOTE 1: The values are based on aggregates of uniform specific gravity. If materials from different sources are used for the coarse and fine aggregates, test the materials in accordance with JIS A 1110 and JIS A 1109 to determine their specific gravities. Correct the percentages passing the various sieves as directed by the Contracting Officer if the specific gravities vary by more than 10 percent.

2.2 LIQUID LIMIT AND PLASTICITY INDEX

Apply liquid limit and plasticity index requirements to the completed course and to any component that is blended to meet the required gradation. The portion of any component or of the completed course passing the 0.425 mm sieve must be either nonplastic or have a liquid limit not greater than 25 and a plasticity index not greater than 5.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following tests, on the proposed material prior to commencing construction, to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed.

a. Sieve Analysis.

b. Liquid limit and plasticity index.

c. Moisture-density relationship.

d. Wear.

e. Soundness.

f. Weight per cubic meter of Slag.

Submit certified copies of test results for approval not less than 30 days before material is required for the work.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

When the ABC or GCA is constructed in more than one layer, clean the previously constructed layer of loose and foreign matter by sweeping with power sweepers or power brooms, except that hand brooms may be used in areas where power cleaning is not practicable. Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated by the Contracting Officer to prevent segregation. Stockpile materials obtained from different sources separately.

3.3 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the base course(s). Do not construct base course(s) on underlying course or subgrade that is frozen. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in JGS 0051, stabilize the surface prior to placement of the base course(s). Stabilize by mixing ABC or GCA into the underlying course and compacting by approved methods. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a satisfactory condition until the base course is placed.

3.4 GRADE CONTROL

Provide a finished and completed base course conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.5 MIXING AND PLACING MATERIALS

. Make adjustments in mixing procedures or in equipment, as directed, to obtain true grades, to minimize segregation or degradation, to obtain the required water content, and to insure a satisfactory base course meeting all requirements of this specification. Place the mixed material on the prepared subgrade or subbase in layers of uniform thickness with an approved spreader. Place the layers so that when compacted they will be true to the grades or levels required with the least possible surface disturbance. Where the base course is placed in more than one layer, clean the previously constructed layers of loose and foreign matter by sweeping with power sweepers, power brooms, or hand brooms, as directed. Make adjustments in placing procedures or equipment as may be directed by the Contracting Officer to obtain true grades, to minimize segregation and degradation, to adjust the water content, and to insure an acceptable base course.

3.6 LAYER THICKNESS

Compact the completed base course to the thickness indicated. No individual layer may be thicker than 150 mm nor be thinner than 75 mm in compacted thickness. Compact the base course(s) to a total thickness that is within 13 mm of the thickness indicated. Where the measured thickness is more than 13 mm deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 13 mm thicker than indicated, the course will be considered as conforming to the specified thickness requirements. The average job thickness will be the average of all thickness measurements taken for the job and must be within 6 mm of the thickness indicated. Measure the total thickness of the base course at intervals of one measurement for each 500 square meters of base course. Measure total thickness using 75 mm diameter test holes penetrating the base course.

3.7 COMPACTION

Compact each layer of the base course, as specified, with approved compaction equipment. Maintain water content during the compaction procedure to within plus or minus 2 percent of the optimum water content determined from laboratory tests as specified in this Section. Begin rolling at the outside edge of the surface and proceed to the center, overlapping on successive trips at least one-half the width of the roller. Slightly vary the length of alternate trips of the roller. Adjust speed of the roller as needed so that displacement of the aggregate does not occur. Compact mixture with hand-operated power tampers in all places not accessible to the rollers. Continue compaction until each layer is compacted through the full depth to at least 100 percent of laboratory maximum density. Make such adjustments in compacting or finishing procedures as may be directed by the Contracting Officer to obtain true grades, to minimize segregation and degradation, to reduce or increase water content, and to ensure a satisfactory base course. Remove any materials found to be unsatisfactory and replace with satisfactory material or rework, as directed, to meet the requirements of this specification.

3.8 PROOF ROLLING

In addition to the compaction specified, proof roll base course under a flexible airfield pavement in areas designated on the drawings by application of 30 coverages for Class IV runways, 8 coverages for runways that support fighter aircraft only, and 4 coverages to all other paved areas, of a heavy pneumatic-tired roller having four or more tires abreast, each tire loaded to a minimum of 13,600 kg and inflated to a minimum of 862 kPa. A coverage is defined as the application of one tire print over the designated area. In the areas designated, apply proof rolling to the top of the underlying material on which the base course is laid and to the top of each layer of the completed base course. Maintain water content of the underlying material and each layer of the base course as specified in Paragraph COMPACTION from start of compaction to completion of proof rolling of that layer. Remove any base course materials or any underlying materials that produce unsatisfactory results by proof rolling and replace with satisfactory materials. Then recompact and proof roll to meet these specifications.

3.9 EDGES OF BASE COURSE

Place the base course(s) so that the completed section will be a minimum of 600 mm wider, on all sides, than the next layer that will be placed above it. Place approved material along the outer edges of the base course in sufficient quantity to compact to the thickness of the course being constructed. When the course is being constructed in two or more layers, simultaneously roll and compact at least a 600 mm width of this shoulder material with the rolling and compacting of each layer of the base course, as directed.

3.10 FINISHING

Finish the surface of the top layer of base course after final compaction and proof rolling by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin layers of material to the top layer of base course to meet grade. If the elevation of the top layer of base course is 13 mm or more below grade, scarify the top layer to a depth of at least 75 mm and blend new material in and compact and proof roll to bring to grade. Make adjustments to rolling and finishing procedures as directed by the Contracting Officer to minimize segregation and degradation, obtain grades, maintain moisture content, and insure an acceptable base course. Should the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the unsatisfactory portion and rework and recompact it or replace as directed.

3.11 SMOOTHNESS TEST

Construct the top layer so that the surface shows no deviations in excess of 10 mm when tested with a 3.00 meter straightedge. Take measurements in successive positions parallel to the centerline of the area to be paved. Also take measurements perpendicular to the centerline at 15 meter intervals. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.12 FIELD QUALITY CONTROL

3.12.1 In-Place Tests

Perform each of the following tests on samples taken from the placed and compacted ABC and GCA. Take samples and test at the rates indicated. Perform sampling and testing of recycled concrete aggregate at twice the specified frequency until the material uniformity is established.

a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square meters, or portion thereof, of completed area.

b. Perform sieve analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square meters, or portion thereof, of material placed.

c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.

d. Measure the thickness of the base course at intervals providing at least one measurement for each 500 square meters of base course or part thereof. Measure the thickness using test holes, at least 75 mm in diameter through the base course.

3.12.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.13 TRAFFIC

Do not allow traffic on the completed base course for airfield pavements. For roads, do not allow heavy equipment on the completed base course except when necessary for construction. When it is necessary for heavy equipment to travel on the completed base course, protect the area against marring or damage to the completed work.

3.14 MAINTENANCE

Maintain the base course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact. Retest any base course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area of base course that is damaged as necessary to comply with this specification.

3.15 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed as directed . No additional payments will be made for materials that have to be replaced.

-- End of Section --

SECTION 32 12 13

BITUMINOUS TACK AND PRIME COATS
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS K 2208	(2009) Asphalt Emulsion
JIS K 2251	(2003) Crude Petroleum and Petroleum Products - Sampling

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Sampling and Testing; G

1.3 QUALITY ASSURANCE

Certificates of compliance for asphalt materials delivered will be obtained and checked to ensure that specification requirements are met. Tack coat materials will not be diluted. Prime coat materials when emulsions are used can be diluted on site with potable water up to 1 part emulsion to 1 part water.

1.4 DELIVERY, STORAGE, AND HANDLING

Inspect the materials delivered to the site for contamination and damage. Unload and store the materials with a minimum of handling.

1.5 EQUIPMENT, TOOLS AND MACHINES

1.5.1 General Requirements

Equipment, tools and machines used in the work are subject to approval. Maintain in a satisfactory working condition at all times. Calibrate equipment such as asphalt distributors, scales, batching equipment, spreaders and similar equipment within 12 months of their use. If the calibration expires during project, recalibrate the equipment before work can continue.

1.5.2 Bituminous Distributor

Provide a self propelled distributor with pneumatic tires of such size and number to prevent rutting, shoving or otherwise damaging the surface being sprayed. Design and equip the distributor to spray the bituminous material in a uniform coverage at the specified temperature, at readily determined and controlled total liquid rates from 0.14 to 4.5 L/square meter, with a pressure range of 172.4 to 517.1 kPa and with an allowable variation from the specified rate of not more than plus or minus 5 percent, and at variable widths. Include with the distributor equipment a separate power unit for the bitumen pump, full-circulation spray bars, tachometer, pressure gauges, volume-measuring devices, adequate heaters for heating of materials to the proper application temperature, a thermometer for reading the temperature of tank contents, and a hand hose attachment suitable for applying bituminous material manually to areas inaccessible to the distributor. The distributor will be capable of circulating and agitating the bituminous material during the heating process.

1.5.3 Heating Equipment for Storage Tanks

Use steam, electric, or hot oil heaters for heating the bituminous material. Provide steam heaters consisting of steam coils and equipment for producing steam, so designed that the steam cannot come in contact with the bituminous material. Fix an armored thermometer to the tank with a temperature range from 4.4 to 204.4 degrees C so that the temperature of the bituminous material may be determined at all times.

1.5.4 Power Brooms and Power Blowers

Use power brooms and power blowers suitable for cleaning the surfaces to which the bituminous coat is to be applied.

1.6 ENVIRONMENTAL REQUIREMENTS

Apply bituminous coat only when the surface to receive the bituminous coat is dry. A limited amount of moisture (approximately 0.14 liter/square meter) can be sprayed on the surface of unbound material when prime coat is used to improve coverage and penetration of asphalt material. Apply bituminous coat only when the atmospheric temperature in the shade is 10 degrees C or above and when the temperature has not been below 2 degrees C for the 12 hours prior to application, unless otherwise directed.

PART 2 PRODUCTS

2.1 PRIME COAT

2.1.1 Emulsified Asphalt

Provide emulsified asphalt conforming to JIS K 2208, Type PK-3. . Asphalt emulsion can be diluted up to 1 part water to 1 part emulsion for prime coat use. Do not dilute asphalt emulsion for tack coat use.

2.2 TACK COAT

2.2.1 Emulsified Asphalt

Provide emulsified asphalt conforming to JIS K 2208, Type PK-4. For prime coats the emulsified asphalt can be diluted with up to 1 part emulsion to 1 part water. No dilution is allowed for tack coat applications.

PART 3 EXECUTION

3.1 PREPARATION OF SURFACE

Immediately before applying the bituminous coat, remove all loose material, dirt, clay, or other objectionable material from the surface to be treated by means of a power broom or blower supplemented with hand brooms. Apply treatment only when the surface is dry and clean.

3.2 APPLICATION RATE

The exact quantities within the range specified, which may be varied to suit field conditions, will be determined by the Contracting Officer.

3.2.1 Tack Coat

Apply bituminous material for the tack coat in quantities of not less than 0.30 L nor more than 0.60 L/square meter of residual asphalt onto the pavement surface as approved by the Contracting Officer. Do not dilute asphalt emulsion when used as a tack coat.

3.2.2 Prime Coat

Apply bituminous material for the prime coat in quantities of not less than 1.0 L nor more than 2.0 L/square meter of residual asphalt for asphalt emulsion up to a 1 to 1 dilution rate or for residual asphalt for cutback asphalt.

3.3 APPLICATION TEMPERATURE

3.3.1 Viscosity Relationship

Apply asphalt at a temperature that will provide a viscosity between 10 and 60 seconds, Saybolt Furol, or between 20 and 120 square mm/sec, kinematic. Furnish the temperature viscosity relation to the Contracting Officer.

3.3.2 Temperature Ranges

The viscosity requirements determine the application temperature to be used. The following is a normal range of application temperatures:

Asphalt Emulsion	
All Grades	20-70 degrees C

3.4 APPLICATION

3.4.1 General

Following preparation and subsequent inspection of the surface, apply the bituminous prime or tack coat with the bituminous distributor at the specified rate with uniform distribution over the surface to be treated. Properly treat all areas and spots, not capable of being sprayed with the distributor, with the hand spray. Until the succeeding layer of pavement is placed, maintain the surface by protecting the surface against damage and by repairing deficient areas at no additional cost to the Government. If required, spread clean dry sand to effectively blot up any excess bituminous material. No smoking, fires, or flames other than those from the heaters that are a part of the equipment are permitted within 8 meters of heating, distributing, and transferring operations of cutback materials. Prevent all traffic, except for paving equipment used in constructing the surfacing, from using the underlying material, whether primed or not, until the surfacing is completed. The bituminous coat requirements are described herein.

3.4.2 Prime Coat

The prime coat is required if it will be at least 7 days before the asphalt mixture is constructed on the underlying (base course, etc.) compacted material. The type of liquid asphalt and application rate will be as specified herein. Protect the underlying layer from any damage (water, traffic, etc.) until the surfacing is placed. If the Contractor places the surfacing within seven days, the choice of protection measures or actions to be taken is at the Contractor's option. Repair (recompact or replace) damage to the underlying material caused by lack of, or inadequate, protection by approved methods at no additional cost to the Government. If the Contractor opts to use the prime coat, apply as soon as possible after consolidation of the underlying material. Apply the bituminous material uniformly over the surface to be treated at a pressure range of 172.4 to 517.1 kPa; the rate will be as specified above in paragraph APPLICATION RATE. To obtain uniform application of the prime coat on the surface treated at the junction of previous and subsequent applications, spread building paper on the surface for a sufficient distance back from the ends of each application to start and stop the prime coat on the paper and to ensure that all sprayers will operate at full force on the surface to be treated. Immediately after application remove and destroy the building paper.

3.4.3 Tack Coat

A tack coat should be applied to every bound surface (asphalt or concrete pavement) that is being overlaid with asphalt mixture and at transverse and longitudinal joints. Apply the tack coat when the surface to be treated is clean and dry. Immediately following the preparation of the surface for treatment, apply the bituminous material by means of the bituminous distributor, within the limits of temperature specified herein and at a rate as specified above in paragraph APPLICATION RATE. Apply the bituminous material so that uniform distribution is obtained over the entire surface to be treated. Treat lightly coated areas and spots missed by the distributor by spraying with a hand wand or using other approved method. Following the application of bituminous material, allow the surface to cure without being disturbed for period of time necessary to permit setting of the tack coat. Apply the bituminous tack coat only as far in advance of the placing of the overlying layer as required for that day's operation. Maintain and protect the treated surface from damage until the succeeding course of pavement is placed.

3.5 CURING PERIOD

Following application of the bituminous material and prior to application of the succeeding layer of asphalt mixture allow the bituminous coat to cure and water or volatiles to evaporate prior to overlaying. Maintain the tacked surface in good condition until the succeeding layer of pavement is placed, by protecting the surface against damage and by repairing and recoating deficient areas. Allow the prime coat to cure without being disturbed for a period of at least 48 hours or longer, as may be necessary to attain penetration into the treated course. Furnish and spread enough sand to effectively blot up excess bituminous material.

3.6 FIELD QUALITY CONTROL

Obtain certificates of compliance for all asphalt material delivered to the project. Obtain samples of the bituminous material under the supervision of the Contracting Officer. The sample may be retained and tested by the Government at no cost to the Contractor.

3.7 SAMPLING AND TESTING

Furnish certified copies of the manufacturer's test reports indicating temperature viscosity relationship for cutback asphalt or asphalt cement, compliance with applicable specified requirements, not less than 5 days before the material is required in the work.

3.7.1 Sampling

Unless otherwise specified, sample bituminous material in accordance with JIS K 2251.

3.7.2 Calibration Test

Furnish all equipment, materials, and labor necessary to calibrate the bituminous distributor. Calibrate using the approved job material and prior to applying the bituminous coat material to the prepared surface.

3.7.3 Trial Applications

Before applying the spray application of tack or prime coat, apply three lengths of at least 30 meters for the full width of the distributor bar to evaluate the amount of bituminous material that can be satisfactorily applied.

3.7.3.1 Tack Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous tack coat materials in the amount of 0.23 L/square meter. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.3.2 Prime Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous materials in the amount of 0.66 L/square meter. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.4 Sampling and Testing During Construction

Perform quality control sampling and testing as required in paragraph FIELD QUALITY CONTROL.

3.8 TRAFFIC CONTROLS

Keep traffic off surfaces freshly treated with bituminous material. Provide sufficient warning signs and barricades so that traffic will not travel over freshly treated surfaces.

-- End of Section --

SECTION 32 12 16.16

HOT-MIX ASPHALT (HMA) FOR ROADS
08/09

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1110	(2006) Methods of Test for Density and Water Absorption of Coarse Aggregates
JIS A 1121	(2007) Method of Test for Resistance to Abrasion of Coarse Aggregate by Use of the Los Angeles Machine
JIS A 1122	(2014) Method of Test for Soundness of Aggregate~~-~~ by Use of Sodium Sulfate
JIS A 1137	(2014) Method of Test for Clay Lumps Contained in Aggregates
JIS A 5001	(2008) Crushed Stone for Road Construction
JIS A 5008	(2008) Limestone Filler for Bituminous Paving Mixtures
JIS K 2207	(2006) Petroleum Asphalts

~~JAPAN ROAD ASSOCIATION (JRA)~~JAPAN ROAD ASSOCIATION (JARA)

JRA HAP	(2019) Handbook for Asphalt Pavement

1.2 GENERAL REQUIREMENTS

All materials, equipment, and construction procedures of hot-mix asphalt pavement for this project shall be in accordance with the Japan Road Association standards per publication identified in this specification.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~ Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING. ~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Mix Design; G

Quality Control; G

Material Acceptance; G

SD-06 Test Reports

Aggregates; G

QC Monitoring

1.4 ENVIRONMENTAL REQUIREMENTS

Do not place the hot-mix asphalt upon a wet surface or when the surface temperature of the underlying course is less than specified in Table 3. The temperature requirements may be waived by the Contracting Officer, if requested; however, meet all other requirements, including compaction.

Table 3. Surface Temperature Limitations of Underlying Course

Mat Thickness, mm	Degrees C
75 or greater	4
Less than 75	7

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Perform the work consisting of pavement courses composed of mineral aggregate and asphalt material heated and mixed in a central mixing plant and placed on a prepared course. HMA designed and constructed in accordance with this section shall conform to the lines, grades, thicknesses, and typical cross sections indicated. Construct each course to the depth, section, or elevation required by the drawings and roll, finish, and approve it before the placement of the next course.

2.2 AGGREGATES

Provide aggregates consisting of crushed stone, crushed gravel, crushed slag, screenings, natural sand and mineral filler, as required. The portion of material retained on the 5 mm sieve is coarse aggregate. The portion of material passing the 5 mm sieve and retained on the 0.074 mm sieve is fine aggregate. The portion passing the 0.074 mm sieve is defined as mineral filler. Submit all aggregate test results and samples to the Contracting Officer at least 14 days prior to start of construction.

2.2.1 Coarse Aggregate

Provide coarse aggregate consisting of sound, tough, durable particles, free from films of material that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances and conforming to JIS A 5001. All individual coarse aggregate sources shall meet the following requirements:

	Percent Passing (by Weight)		
Sieve Size (mm)	S-20 (20-13 mm)	S-13 (13-5 mm)	S-5 (5-2.5 mm)
25	100	-	-
20	85 - 100	100	-
13	0 - 15	85 - 100	100
5	-	0 - 15	85 - 100
2.5	-	-	0 - 25

a. At least 75 percent by weight of the aggregate shall have two or more fractured faces in accordance with JRA HAP

b. It shall be of uniform quality, clean, hard and durable, and shall not contain deleterious substances over the maximum percent by weight, such as clay or loam (0.25% maximum), soft stone pieces (5.0% maximum), and flat or elongated stone pieces (10.0% maximum) in accordance with JRA HAP.

c. Specific gravity shall be 2.45 or more when tested with JIS A 1110, water absorption shall be 3.0 percent or less when tested with JIS A 1110, and wear shall be not more than 30 percent when tested with JIS A 1121.

d. Soundness of aggregate shall be not more than 12 percent when tested with JIS A 1122.

2.2.2 Fine Aggregate

Fine aggregate shall consist of natural or manufactured sand, and screenings conforming to JIS A 5001, except as modified herein. Fine aggregate shall meet requirements for wear and soundness specified for coarse aggregate. Since screenings may contain deleterious substances, such as silt and clay, it is necessary to perform sufficient examination before use. Clay lumps content shall be less than 0.25% when tested with JIS A 1137. Gradation of fine aggregate shall be as follows:

Sieve Size (mm)	Percent Passing (by Weight)
5	100
2.5	85 - 100
0.6	25 - 55
0.3	15 - 40
0.15	7 - 28

Sieve Size (mm)	Percent Passing (by Weight)
0.074	0 - 20

2.2.3 Mineral Filler

Mineral filler shall be pulverized limestone or igneous rock that is sufficiently dry and free of lumps and meeting the requirements of JIS A 5008. Moisture content shall be less than 1.0 percent, and specific gravity shall be more than 2.60 percent. Gradation of mineral filler shall be as follows:

Sieve Size (mm)	Percent Passing (by Weight)
0.6	100
0.15	Over 90
0.074	Over 70

2.2.4 Composition of Hot-Mix Asphalt Mixture

2.2.4.1 Aggregate Gradation

The combined aggregate gradation shall conform to gradations specified in Table 1 and shall not vary from the low limit on one sieve to the high limit on the adjacent sieve or vice versa, but grade uniformly from coarse to fine.

Table 1

Total Percent Passing (By Weight)

Sieve Size (mm)	Binder Course (Max. 20 mm)	Wearing Course (Max. 20 mm)	Wearing Course (Max. 13 mm)
25	100	100	-
20	95 - 100	95 - 100	100
13	70 - 90	75 - 90	95 - 100
5	35 - 55	45 - 65	55 -70
2.5	20 - 35	35 - 50	35 -50
0.6	11 - 23	18 - 30	18 - 30

Sieve Size (mm)	Binder Course (Max. 20 mm)	Wearing Course (Max. 20 mm)	Wearing Course (Max. 13 mm)
0.3	5 - 16	10 - 21	10 - 21
0.15	4 - 12	6 - 16	6 - 16
0.074	2 - 7	4 - 8	4 - 8

2.2.4.2 Quantity of Asphalt Cement

Mix asphalt cement with aggregates of corresponding mixes in the following proportions:

Asphalt Cement Percent by Weight of Total Mix	
Binder Course	Wearing Course
4.5 - 6	5 - 7

2.3 ASPHALT CEMENT

Asphalt cement shall conform to JIS K 2207, penetration grade 40-60 (high traffic areas), 60-80 (general traffic conditions), 80-100 (snowy regions), and 100-120 (extremely cold places).

2.4 MIX DESIGN

a. Develop the mix design. The asphalt mix shall be composed of a mixture of well-graded aggregate, mineral filler if required, and asphalt material. The aggregate fractions shall be sized, handled in separate size groups, and combined in such proportions that the resulting mixture meets the grading requirements of the job mix formula (JMF). Submit proposed JMF; do not produce hot-mix asphalt for payment until a JMF has been approved. The hot-mix asphalt shall be designed in accordance with Chapter 5 of the JRA HAP - Handbook for Asphalt Pavement and the criteria shown in Table 2.

2.4.1 JMF Requirements

Submit in writing the job mix formula for approval at least 14 days prior to the start of the test section including as a minimum:

a. Source and proportions, percent by weight, of each ingredient of the mixture.

b. Correct gradation, the percentages passing each size sieve listed in the specification for the mixture to be used, for the aggregate and mineral filler from each separate source and from each different size to be used in the mixture and for the composite mixture.

c. Amount of material passing the JIS 0.074 mm sieve as determined by dry sieving.

d. Number of blows of hammer compaction per side of molded specimen.

e. Temperature viscosity relationship of the asphalt pavement.

f. Stability, flow, percent voids in mineral aggregate, percent air voids, and unit weight.

g. Asphalt absorption by the aggregate.

h. Effective asphalt content as percent by weight of total mix.

i. Temperature of the mixture immediately upon completion of mixing.

j. Asphalt viscosity grade and/or penetration range.

k. Curves for the binder and wearing courses.

2.4.1.1 Marshall Test

Marshall test specimen of hot-mix asphalt mixture shall be prepared in a laboratory to determine the optimum composition of aggregates and the optimum quantity of asphalt cement. Aggregate gradation and asphalt cement content used in the mixtures shall be within the limits specified. Standard value for Marshall test shall be as follows:

Table 2. Mix Design Criteria

Type of Mixture	Binder Course	Wearing Course
Number of Blows (Traffic Classification C or heavier)	75	75
Number of Blows (Traffic Classification B or ligher)	50	50
Percentage of Air Voids (%)	3-7	3-6
Voids Filled with Asphalt (%)	65-85	70-85
Marshall Stability (kgf)	500 or more	750 or more
Flow Value (1/100cm)	20-40	20-40

2.4.2 Adjustments to Field JMF

Keep the Laboratory JMF for each mixture in effect until a new formula is approved in writing by the Contracting Officer. Should a change in sources of any materials be made, perform a new laboratory jmf design and a new JMF approved before the new material is used. The Contractor will be allowed to adjust the Laboratory JMF within the limits specified below to optimize mix volumetric properties with the approval of the Contracting Officer. Adjustments to the Laboratory JMF shall be applied to the field (plant) established JMF and limited to those values as shown. Adjustments shall be targeted to produce or nearly produce 4 percent voids total mix (VTM).

TABLE 3. Field (Plant) Established JMF Tolerances

Sieves, mm	Adjustments (plus or minus), percent
13	3
5	3
2.5	3
0.074	1
Binder Content	0.4

If adjustments are needed that exceed these limits, develop a new mix design. Tolerances given above may permit the aggregate grading to be outside the limits shown in Table 1; while not desirable, this is acceptable, except for the 0.074 mm sieve, which shall remain within the aggregate grading of Table 1.

2.5 RECYCLED HOT MIX ASPHALT

Recycled HMA shall consist of reclaimed asphalt pavement (RAP), coarse aggregate, fine aggregate, mineral filler, and asphalt cement to produce a consistent gradation and asphalt content and properties. When RAP is fed into the plant, the maximum RAP chunk size shall not exceed 50 mm. The amount of RAP shall not exceed 30 percent.

PART 3 EXECUTION

3.1 PREPARATION OF ASPHALT BINDER MATERIAL

Heat the asphalt cement material avoiding local overheating and providing a continuous supply of the asphalt material to the mixer at a uniform temperature. The temperature of unmodified asphalts shall be no more than 160 degrees C when added to the aggregates.

3.2 PREPARATION OF MINERAL AGGREGATE

Heat and dry the aggregate for the mixture prior to mixing. No damage shall occur to the aggregates due to the maximum temperature and rate of heating used. The temperature of the aggregate and mineral filler shall not exceed 175 degrees C when the asphalt cement is added. The temperature shall not be lower than is required to obtain complete coating and uniform distribution on the aggregate particles and to provide a mixture of satisfactory workability.

3.3 PREPARATION OF HOT-MIX ASPHALT MIXTURE

Accurately weigh or gauge the aggregates and the dry mineral filler and convey into the mixer in the proportionate amounts of each aggregate size required to meet the job-mix formula. Introduce required amount of asphalt into the mixer at a temperature at which it can be applied uniformly to the aggregate but not to exceed 163 degrees C. In batch mixing, after the aggregates and mineral filler have been introduced into the mixer and mixed for not less than 15 seconds, add asphalt by spraying or other approved methods, and continue mixing for a period of not less than 20 seconds or as much longer as may be required to obtain a homogeneous mixture. The time required to add or spray the asphalt into the mixer will not be added to the total wet-mixing time provided this operation does not exceed 10 seconds and a homogeneous mixture is obtained. The additional mixing time, when required, will be as directed. The temperature of the mixture at the time of discharge shall not exceed 168 degrees C. The temperature of the aggregate and mineral filler in the mixer shall not exceed 177 degrees C when the asphalt is added. When the mixture is prepared in a twin-pugmill mixer, the volume of the aggregates, mineral filler, and asphalt shall not extend above the tips of the mixer blades when the blades are in a vertical position. Overheated and carbonized mixtures, or mixtures that foam or show indication of free moisture, will be rejected. When free moisture is detected in batch mix plant produced mixture, withdraw the aggregates in the hot bins immediately and return to the respective stockpiles.

3.4 PREPARATION OF THE UNDERLYING SURFACE

Immediately before placing the hot mix asphalt, clean the underlying course of dust and debris. Apply a ~~[~~prime coat~~]~~ ~~[~~and/or~~]~~ ~~[~~tack coat~~]~~ in accordance with the contract specifications.

3.4.1 Raising of Existing Manhole, Handhole, Valve Box and Catch Basin

If there are existing manhole, handhole, valve box and catch basins in existing asphalt concrete pavement to be overlaid, those shall be raised up to flush with the finished surface of new hot-mix asphalt pavement before new asphalt concrete is placed. Existing mortar leveling course under the manhole, handhole and valve box shall be replaced with new up to new level to provide a flush setting.

3.5 TESTING LABORATORY

Submit laboratory certification issued by the local prefectural or central government (Japan Ministry of Land, Infrastructure, Transport and Tourism (MLIT)). Use a laboratory to develop the JMF. The Government will inspect the laboratory equipment and test procedures prior to the start of hot mix operations. The laboratory shall maintain a valid certification for the duration of the project. A statement signed by the manager of the laboratory stating that it meets these requirements or clearly listing all deficiencies shall be submitted to the Contracting Officer prior to the start of construction. The statement shall contain as a minimum:

a. Qualifications of personnel; laboratory manager, supervising technician, and testing technicians.

b. A listing of equipment to be used in developing the job mix.

c. A copy of the laboratory's quality control system.

3.6 TRANSPORTING AND PLACING

3.6.1 Transporting

Transport the hot-mix asphalt from the mixing plant to the site in clean, tight vehicles. Schedule deliveries so that placing and compacting of mixture is uniform with minimum stopping and starting of the paver. Provide adequate artificial lighting for night placements. Hauling over freshly placed material will not be permitted until the material has been compacted as specified, and allowed to cool to 60 degrees C. To deliver mix to the paver, use a material transfer vehicle operated to produce continuous forward motion of the paver.

3.6.2 Placing

Place and compact the mix at a temperature suitable for obtaining density, surface smoothness, and other specified requirements. Upon arrival, place the mixture to the full width by an asphalt paver; it shall be struck off in a uniform layer of such depth that, when the work is completed, it will have the required thickness and conform to the grade and contour indicated. Regulate the speed of the paver to eliminate pulling and tearing of the asphalt mat. Unless otherwise permitted, placement of the mixture shall begin along the centerline of a crowned section or on the high side of areas with a one-way slope. Place the mixture in consecutive adjacent strips having a minimum width of 3 m. The longitudinal joint in one course shall offset the longitudinal joint in the course immediately below by at least 300 mm; however, the joint in the surface course shall be at the centerline of the pavement. Transverse joints in one course shall be offset by at least 3 m from transverse joints in the previous course. Transverse joints in adjacent lanes shall be offset a minimum of 3 m. On isolated areas where irregularities or unavoidable obstacles make the use of mechanical spreading and finishing equipment impractical, the mixture may be spread and luted by hand tools.

3.7 COMPACTION OF MIXTURE

After placing, the mixture shall be thoroughly and uniformly compacted by rolling. Compact the surface as soon as possible without causing displacement, cracking or shoving. The sequence of rolling operations and the type of rollers used shall be at the discretion of the Contractor. The speed of the roller shall, at all times, be sufficiently slow to avoid displacement of the hot mixture and be effective in compaction. Any displacement occurring as a result of reversing the direction of the roller, or from any other cause, shall be corrected at once. Furnish sufficient rollers to handle the output of the plant. Continue rolling until the surface is of uniform texture, true to grade and cross section, and the required field density is obtained. To prevent adhesion of the mixture to the roller, keep the wheels properly moistened but excessive water will not be permitted. In areas not accessible to the roller, the mixture shall be thoroughly compacted with hand tampers. Any mixture that becomes loose and broken, mixed with dirt, contains check-cracking, or is in any way defective shall be removed full depth, replaced with fresh hot mixture and immediately compacted to conform to the surrounding area. This work shall be done at the Contractor's expense. Skin patching will not be allowed.

3.8 JOINTS

The formation of joints shall be performed ensuring a continuous bond between the courses and to obtain the required density. All joints shall have the same texture as other sections of the course and meet the requirements for smoothness and grade.

3.8.1 Transverse Joints

Do not pass the roller over the unprotected end of the freshly laid mixture, except when necessary to form a transverse joint. When necessary to form a transverse joint, it shall be made by means of placing a bulkhead or by tapering the course. The tapered edge shall be cut back to its full depth and width on a straight line to expose a vertical face prior to placing material at the joint. Remove the cutback material from the project. In both methods, all contact surfaces shall be given a light tack coat of asphalt material before placing any fresh mixture against the joint.

3.8.2 Longitudinal Joints

Longitudinal joints which are irregular, damaged, uncompacted, cold (less than 80 degrees C at the time of placing adjacent lanes), or otherwise defective, shall be cut back a maximum of 75 mm from the top of the course with a cutting wheel to expose a clean, sound vertical surface for the full depth of the course. All cutback material shall be removed from the project. All contact surfaces shall be given a light tack coat of asphalt material prior to placing any fresh mixture against the joint. The Contractor will be allowed to use an alternate method if it can be demonstrated that density, smoothness, and texture can be met.

3.9 Finishing at Edge and Limit of Paving

Overlay of new hot-mix asphalt mixture shall be finished evenly in the same thickness indicated. Overlay shall be finished in gentle slope so that the edge of new overlaying pavement shall meet the edge elevation of existing asphalt concrete pavement. And where the drawing indicates "Limit of Paving," overlay shall be finished to provide a smooth transition to existing concrete pavement.

3.10 Compacted Earth Shoulder

Place and compact earth at edges of course for at least 30 cm of the shoulder.

3.11 QUALITY CONTROL

3.11.1 General Quality Control Requirements

Develop and submit an approved Quality Control Plan. Submit aggregate and QC test results. Do not produce hot-mix asphalt for payment until the quality control plan has been approved addressing all elements which affect the quality of the pavement including, but not limited to:

a. Mix Design

b. Aggregate Grading

c. Quality of Materials

d. Stockpile Management

e. Proportioning

f. Mixing and Transportation

g. Mixture Volumetrics

h. Moisture Content of Mixtures

i. Placing and Finishing

j. Joints

k. Compaction

l. Surface Smoothness

3.11.2 Testing Laboratory

Provide a fully equipped asphalt laboratory located at the plant or job site. Laboratory facilities shall be kept clean and all equipment maintained in proper working condition. The Contracting Officer shall be permitted unrestricted access to inspect the Contractor's laboratory facility, to witness quality control activities, and to perform any check testing desired. The Contracting Officer will advise the Contractor in writing of any noted deficiencies concerning the laboratory facility, equipment, supplies, or testing personnel and procedures. When the deficiencies are serious enough to adversely affect test results, the incorporation of the materials into the work shall be suspended immediately and will not be permitted to resume until the deficiencies are corrected.

3.11.3 Quality Control Testing

Perform all quality control tests applicable to these specifications in accordance with the testing criteria and frequency requirements as set forth in Chapter 6 of the JRA HAP - Handbook for Asphalt Pavement. Develop a Quality Control Testing Plan as part of a Quality Control Program. The testing program shall include, but shall not be limited to, tests for the control of asphalt content, aggregate gradation, temperatures, aggregate moisture, moisture in the asphalt mixture, laboratory air voids, stability (, flow , in-place density, grade and smoothness.

3.11.3.1 Additional Testing

Any additional testing, which the Contractor deems necessary to control the process, may be performed at the Contractor's option.

3.11.3.2 QC Monitoring

Submit all QC test results to the Contracting Officer on a daily basis as the tests are performed. The Contracting Officer reserves the right to monitor any of the Contractor's quality control testing and to perform duplicate testing as a check to the Contractor's quality control testing.

3.11.4 Sampling

When directed by the Contracting Officer, sample and test any material which appears inconsistent with similar material being produced, unless such material is voluntarily removed and replaced or deficiencies corrected by the Contractor. All sampling shall be in accordance with standard procedures specified.

3.12 MATERIAL ACCEPTANCE

3.12.1 Grade

The final wearing surface of pavement shall conform to the elevations and cross sections shown and shall vary not more than 15 mm from the plan grade established and approved at site of work. Finished surfaces at juncture with other pavements shall coincide with finished surfaces of abutting pavements. Deviation from the plan elevation will not be permitted in areas of pavements where closer conformance with planned elevation is required for the proper functioning of drainage and other appurtenant structures involved. The grade will be determined by running lines of levels at intervals of 7.6 m, or less, longitudinally and transversely, to determine the elevation of the completed pavement surface. Within 5 working days, after the completion of a particular lot incorporating the final wearing surface, test the final wearing surface of the pavement for conformance with the specified plan grade. Diamond grinding may be used to remove high spots to meet grade requirements. Skin patching for correcting low areas or planing or milling for correcting high areas will not be permitted.

3.12.2 Surface Smoothness

Perform all testing in the presence of the Contracting Officer. Keep detailed notes of the results of the testing and furnish a copy to the Government immediately after each day's testing. Use the profilograph method for all longitudinal testing, except where the runs would be less than 60 m in length and the ends where the straightedge will be used.

3.12.2.1 Smoothness Requirements

3.12.2.1.1 Straightedge Testing

The finished surfaces of the pavements shall have no abrupt change of 6 mm or more, and all pavements shall be within the tolerances of 6 mm in both the longitudinal and transverse directions, when tested with an approved 3 m straightedge.

3.12.2.1.2 Profilograph Testing

The finished surfaces of the pavements shall have no abrupt change of 3 mm or more, and each 0.1 km segment of each pavement lot shall have a Profile Index not greater than 140 mm/km when tested with an approved Japanese construction practice using profilometer. If the extent of the pavement in either direction is less than 60 m, that direction shall be tested by the straightedge method and shall meet requirements specified above.

3.12.2.2 Testing Method

After the final rolling, but not later than 24 hours after placement, test the surface of the pavement in each entire lot in such a manner as to reveal all surface irregularities exceeding the tolerances specified above. Separate testing of individual sublots is not required. If any pavement areas are ground, these areas shall be retested immediately after grinding. Test each lot of the pavement in both a longitudinal and a transverse direction on parallel lines. Set the transverse lines 4.5 m or less apart, as directed. The longitudinal lines shall be at the centerline of each paving lane for lanes less than 6.1 m wide and at the third points for lanes 6.1 m or wider. Also test other areas having obvious deviations. Longitudinal testing lines shall be continuous across all joints.

3.12.2.2.1 Straightedge Testing

Hold the straightedge in contact with the surface and move it ahead one-half the length of the straightedge for each successive measurement. Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length, and measuring the maximum gap between the straightedge and the pavement surface in the area between these two high points.

3.12.2.2.2 Profilograph Testing

Perform profilograph testing using approved equipment and procedures. The equipment shall utilize electronic recording and automatic computerized reduction of data to indicate "must-grind" bumps and the Profile Index for each 0.1 km segment of each pavement lot. Grade breaks on parking lots shall be accommodated by breaking the profile segment into shorter sections and repositioning the blanking band on each segment. The "blanking band" shall be 5 mm wide and the "bump template" shall span 25 mm with an offset of 7.5 mm. Compute the Profile Index for each pass of the profilograph in each 0.1 km segment. The Profile Index for each segment shall be the average of the Profile Indices for each pass in each segment. Furnish a copy of the reduced tapes to the Government at the end of each day's testing.

-- End of Section --

SECTION 32 16 19

CONCRETE CURBS, GUTTERS AND SIDEWALKS
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C309	(2011) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM D1751	(2004; E 2013; R 2013) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1101	(2014) Method of Test for Slump of Concrete
JIS A 1115	(2014) Method of Sampling Fresh Concrete
JIS A 1118	(2017) Method of Test for Air Content of Fresh Concrete by Volumetric Method
JIS A 1128	(2014) Method of Test for Air Content of Fresh Concrete by Pressure Method
JIS A 1132	(2014) Method of Making and Curing Concrete Specimens
JIS A 5308	(2014) Ready-Mixed Concrete
JIS A 5371	(2010) Precast Unreinforced Concrete Product
JIS A 5758	(20~~16~~22) Sealants for Sealing and Glazing in Buildings
JIS G 3112	(2010) Steel Bars for Concrete Reinforcement
JIS G 3551	(2005) Welded Wire Mesh and Rebar Grid
JIS K 6781	(1994) Polyethylene Films for Agriculture

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Concrete; G

SD-06 Test Reports

Field Quality Control

1.3 EQUIPMENT, TOOLS, AND MACHINES

1.3.1 General Requirements

Plant, equipment, machines, and tools used in the work will be subject to approval and must be maintained in a satisfactory working condition at all times. Use equipment capable of producing the required product, meeting grade controls, thickness control and smoothness requirements as specified. Discontinue using equipment that produces unsatisfactory results. Allow the Contracting Officer access at all times to the plant and equipment to ensure proper operation and compliance with specifications.

1.3.2 Slip Form Equipment

Slip form paver or curb forming machines, will be approved based on trial use on the job and must be self-propelled, automatically controlled, crawler mounted, and capable of spreading, consolidating, and shaping the plastic concrete to the desired cross section in one pass.

1.4 ENVIRONMENTAL REQUIREMENTS

1.4.1 Placing During Cold Weather

Do not place concrete when the air temperature reaches 5 degrees C and is falling, or is already below that point. Placement may begin when the air temperature reaches 2 degrees C and is rising, or is already above 5 degrees C. Make provisions to protect the concrete from freezing during the specified curing period. If necessary to place concrete when the temperature of the air, aggregates, or water is below 2 degrees C, placement and protection must be approved in writing. Approval will be contingent upon full conformance with the following provisions. Prepare and protect the underlying material so that it is entirely free of frost when the concrete is deposited. Heat mixing water and aggregates as necessary to result in the temperature of the in-place concrete being between 10 and 30 degrees C. Methods and equipment for heating must be approved. Use only aggregates that are free of ice, snow, and frozen lumps before entering the mixer. Provide covering or other means as needed to maintain the concrete at a temperature of at least 10 degrees C for not less than 72 hours after placing, and at a temperature above freezing for the remainder of the curing period.

1.4.2 Placing During Warm Weather

The temperature of the concrete as placed must not exceed 30 degrees C except where an approved retarder is used. Cool the mixing water and aggregates as necessary to maintain a satisfactory placing temperature. The placing temperature must not exceed 35 degrees C at any time.

PART 2 PRODUCTS

2.1 CONCRETE

Provide concrete conforming to the applicable requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE , JIS A 5308 except as otherwise specified. Concrete must have a minimum compressive strength of 18~~24~~ MPa at 28 days. Size of aggregate must not exceed 37.5 mm. Submit copies of certified delivery tickets for all concrete used in the construction.

2.1.1 Air Content

Use concrete mixtures that have an air content by volume of concrete of 5 to 7 percent, based on measurements made immediately after discharge from the mixer.

2.1.2 Slump

Use concrete with a slump of 75 mm plus or minus 25 mm for hand placed concrete or 25 mm plus or minus 10 mm for slipformed concrete as determined in accordance with JIS A 1101.

2.1.3 Reinforcement Steel

Use reinforcement bars conforming to JIS G 3112. Use wire mesh reinforcement conforming to JIS G 3551.

2.2 CONCRETE CURING MATERIALS

2.2.1 Impervious Sheet Materials

Use impervious sheet materials conforming to JIS K 6781, type optional, except that polyethylene film, if used, must be white opaque.

2.2.2 White Pigmented Membrane-Forming Curing Compound

Use ~~Japanese manufactured~~ white pigmented membrane-forming curing compound conforming to ASTM C309, Type 2.

2.3 CONCRETE PROTECTION MATERIALS

Use concrete protection materials consisting of a linseed oil mixture of equal parts, by volume, of linseed oil and either mineral spirits, naphtha, or turpentine. At the option of the Contractor, commercially prepared linseed oil mixtures, formulated specifically for application to concrete to provide protection against the action of deicing chemicals may be used, except that emulsified mixtures are not acceptable.

2.4 JOINT FILLER STRIPS

2.4.1 Expansion Joint Filler, Premolded

Unless otherwise indicated, use 10 mm thick ~~Japanese manufactured~~ premolded expansion joint filler conforming to ASTM D1751.

2.5 JOINT SEALANTS

Use ~~Japanese manufactured~~ cold-applied joint sealant conforming to JIS A 5758.

2.6 FORM WORK

Design and construct form work to ensure that the finished concrete will conform accurately to the indicated dimensions, lines, and elevations, and within the tolerances specified. Use wood or steel forms that are straight and of sufficient strength to resist springing during depositing and consolidating concrete.

2.6.1 Wood Forms

Use forms that are surfaced plank, 50 mm nominal thickness, straight and free from warp, twist, loose knots, splits or other defects. Use forms with a nominal length of 3 m. Radius bends may be formed with 19 mm boards, laminated to the required thickness.

2.6.2 Steel Forms

Use channel-formed sections with a flat top surface and welded braces at each end and at not less than two intermediate points. Use forms with interlocking and self-aligning ends. Provide flexible forms for radius forming, corner forms, form spreaders, and fillers as needed. Use forms with a nominal length of 3 m and that have a minimum of 3 welded stake pockets per form. Use stake pins consisting of solid steel rods with chamfered heads and pointed tips designed for use with steel forms.

2.6.3 Sidewalk Forms

Use sidewalk forms that are of a height equal to the full depth of the finished sidewalk.

2.6.4 Curb and Gutter Forms

Use curb and gutter outside forms that have a height equal to the full depth of the curb or gutter. Use rigid forms for curb returns, except that benders or thin plank forms may be used for curb or curb returns with a radius of 3 m or more, where grade changes occur in the return, or where the central angle is such that a rigid form with a central angle of 90 degrees cannot be used. Back forms for curb returns may be made of 38 mm benders, for the full height of the curb, cleated together. In lieu of inside forms for curbs, a curb "mule" may be used for forming and finishing this surface, provided the results are approved.

2.6.5 Precast Concrete Curb or Precast Concrete Curb and Gutter

Precast concrete curb or precast concrete curb and gutter, where indicated, shall conform to JIS A 5371.

PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Construct subgrade to the specified grade and cross section prior to concrete placement.

3.1.1 Sidewalk Subgrade

Place and compact the subgrade in accordance with Section 31 00 00 EARTHWORK . Test the subgrade for grade and cross section with a template extending the full width of the sidewalk and supported between side forms.

3.1.2 Curb and Gutter Subgrade

Place and compact the subgrade in accordance with Section 31 00 00 EARTHWORK Section 32 11 23 AGGREGATE BASE COURSES . Test the subgrade for grade and cross section by means of a template extending the full width of the curb and gutter. Use subgrade materials equal in bearing quality to the subgrade under the adjacent pavement.

3.1.3 Maintenance of Subgrade

Maintain subgrade in a smooth, compacted condition in conformity with the required section and established grade until the concrete is placed. The subgrade must be in a moist condition when concrete is placed. Prepare and protect subgrade so that it is free from frost when the concrete is deposited.

3.2 FORM SETTING

Set forms to the indicated alignment, grade and dimensions. Hold forms rigidly in place by a minimum of 3 stakes per form placed at intervals not to exceed 1.2 m. Use additional stakes and braces at corners, deep sections, and radius bends, as required. Use clamps, spreaders, and braces where required to ensure rigidity in the forms. Remove forms in a manner that will not injure the concrete. Do not use bars or heavy tools against the concrete when removing the forms. Promptly and satisfactorily repair concrete found to be defective after form removal. Clean forms and coat with form oil each time before concrete is placed. Wood forms may, instead, be thoroughly wetted with water before concrete is placed, except that with probable freezing temperatures, oiling is mandatory.

3.2.1 Sidewalks

Set forms for sidewalks with the upper edge true to line and grade with an allowable tolerance of 3 mm in any 3 m long section. After forms are set, grade and alignment must be checked with a 3 m straightedge. Sidewalks must have a transverse slope of 20 mm per meter Unless otherwise indicated, construct sidewalks that are located adjacent to curbs with the low side adjacent to the curb. Do not remove side forms less than 12 hours after finishing has been completed.

3.2.2 Curbs and Gutters

Remove forms used along the front of the curb not less than 2 hours nor more than 6 hours after the concrete has been placed. Do not remove forms used along the back of curb until the face and top of the curb have been finished, as specified for concrete finishing. Do not remove gutter forms while the concrete is sufficiently plastic to slump in any direction.

3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING

3.3.1 Formed Sidewalks

Place concrete in the forms in one layer. When consolidated and finished, the sidewalks must be of the thickness indicated. Use a strike-off guided by side forms after concrete has been placed in the forms to bring the surface to proper section to be compacted. Consolidate concrete by tamping and spading or with an approved vibrator. Finish the surface to grade with a strike off.

3.3.2 Concrete Finishing

After straightedging, when most of the water sheen has disappeared, and just before the concrete hardens, finish the surface with a wood or magnesium float or darby to a smooth and uniformly fine granular or sandy texture free of waves, irregularities, or tool marks. Produce a scored surface by brooming with a fiber-bristle brush in a direction transverse to that of the traffic, followed by edging.

3.3.3 Edge and Joint Finishing

Finish all slab edges, including those at formed joints, with an edger having a radius of 3 mm. Edge transverse joints before brooming. Eliminate the flat surface left by the surface face of the edger with brooming. Clean and solidly fill corners and edges which have crumbled and areas which lack sufficient mortar for proper finishing with a properly proportioned mortar mixture and then finish.

3.3.4 Surface and Thickness Tolerances

Finished surfaces must not vary more than 8 mm from the testing edge of a 3 m straightedge. Permissible deficiency in section thickness will be up to 6 mm.

3.4 CURB AND GUTTER CONCRETE PLACEMENT AND FINISHING

3.4.1 Formed Curb and Gutter

Place concrete to the required section in a single lift. Consolidate concrete using approved mechanical vibrators. Curve shaped gutters must be finished with a standard curb "mule".

3.4.2 Curb and Gutter Finishing

Approved slipformed curb and gutter machines may be used in lieu of hand placement.

3.4.3 Concrete Finishing

Float and finish exposed surfaces with a smooth wood float until true to grade and section and uniform in texture. Brush floated surfaces with a fine-hair brush using longitudinal strokes. Round the edges of the gutter and top of the curb with an edging tool to a radius of 13 mm. Immediately after removing the front curb form, rub the face of the curb with a wood or concrete rubbing block and water until blemishes, form marks, and tool marks have been removed. Brush the front curb surface, while still wet, in the same manner as the gutter and curb top. Finish the top surface of gutter and entrance to grade with a wood float.

3.4.4 Joint Finishing

Finish curb edges at formed joints as indicated.

3.4.5 Surface and Thickness Tolerances

Finished surfaces must not vary more than 6 mm from the testing edge of a 3 m straightedge. Permissible deficiency in section thickness will be up to 6 mm.

3.4.6 Precast Concrete Curbs or Precast Concrete Curb and Gutter

Precast concrete curbs or precast concrete curbs and gutters where used shall be installed to line and grade as indicated. Joint width shall be 10 mm. Cement mortar shall be provided to set the curbs and fill the joints. Where only precast concrete curb is used, gutter shall be constructed with cast-in-place concrete. Curbs and gutter shall be accomplished to match existing grade at transition point to existing curbs and gutters.

3.5 SIDEWALK JOINTS

Construct sidewalk joints to divide the surface into rectangular areas. Space transverse contraction joints at a distance equal to the sidewalk width or not to exceed 1.25 times the width, whichever is less, and continuous across the slab. Construct longitudinal contraction joints along the centerline of all sidewalks 3 m or more in width. Construct transverse expansion joints at sidewalk returns and opposite expansion joints in adjoining curbs. Where the sidewalk is not in contact with the curb, install transverse expansion joints as indicated. Form expansion joints around structures and features which project through or into the sidewalk pavement, using joint filler of the type, thickness, and width indicated. Expansion joints are not required between sidewalks and curb that abut the sidewalk longitudinally.

3.5.1 Sidewalk Contraction Joints

Form contraction joints in the fresh concrete by cutting a groove in the top portion of the slab to a depth of at least one-fourth of the sidewalk slab thickness. Unless otherwise approved or indicated, either use a jointer to cut the groove or saw a groove in the hardened concrete with a power-driven saw. Construct sawed joints by sawing a groove in the concrete with a 3 mm blade. Provide an ample supply of saw blades on the jobsite before concrete placement is started. Provide at least one standby sawing unit in good working order at the jobsite at all times during the sawing operations.

3.5.2 Sidewalk Expansion Joints

Form expansion joints using 10 mm joint filler strips. Joint filler in expansion joints surrounding structures and features within the sidewalk may consist of preformed filler material conforming to ASTM D1751. Hold joint filler in place with steel pins or other devices to prevent warping of the filler during floating and finishing. Immediately after finishing operations are completed, round joint edges using an edging tool having a radius of 3 mm. Remove any concrete over the joint filler. At the end of the curing period, clean the top of expansion joints and fill with cold-applied joint sealant. Use joint sealant that is gray or stone in color. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Apply joint sealing material only when the concrete at the joint is surface dry and atmospheric and concrete temperatures are above 10 degrees C. Immediately remove any excess material on exposed surfaces of the concrete and clean the concrete surfaces.

3.5.3 Reinforcement Steel Placement

Accurately and securely fasten reinforcement steel in place with suitable supports and ties before the concrete is placed.

3.6 CURB AND GUTTER JOINTS

Construct curb and gutter joints at right angles to the line of curb and gutter.

3.6.1 Expansion Joints

Form expansion joints by means of preformed expansion joint filler material cut and shaped to the cross section of curb and gutter. Construct expansion joints in curb and gutter directly opposite expansion joints of abutting portland cement concrete pavement using the same type and thickness of joints as joints in the pavement. Where curb and gutter do not abut portland cement concrete pavement, provide expansion joints at least 10 mm in width at intervals not less than 10 meters nor greater than 36 meters. Seal expansion joints immediately following curing of the concrete or as soon thereafter as weather conditions permit. Seal expansion joints and the top 25 mm depth of curb and gutter contraction-joints with joint sealant. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Concrete at the joint must be surface dry and atmospheric and concrete temperatures must be above 10 degrees C at the time of application of joint sealing material. Immediately remove excess material on exposed surfaces of the concrete and clean concrete surfaces. Precast concrete curbs will not require expansion joints.

3.7 CURING AND PROTECTION

3.7.1 General Requirements

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days from the beginning of the curing operation. Protect unhardened concrete from rain and flowing water. All equipment needed for adequate curing and protection of the concrete must be on hand and ready for use before actual concrete placement begins. Protect concrete as necessary to prevent cracking of the pavement due to temperature changes during the curing period.

3.7.1.1 Mat Method

Cover the entire exposed surface with two or more layers of burlap. Overlap mats at least 150 mm. Thoroughly wet the mat with water prior to placing on concrete surface and keep the mat continuously in a saturated condition and in intimate contact with concrete for not less than 7 days.

3.7.1.2 Impervious Sheeting Method

Wet the entire exposed surface with a fine spray of water and then cover with impervious sheeting material. Lay sheets directly on the concrete surface with the light-colored side up and overlapped 300 mm when a continuous sheet is not used. Use sheeting that is not less than 450 mm wider than the concrete surface to be cured. Secure sheeting using heavy wood planks or a bank of moist earth placed along edges and laps in the sheets. Satisfactorily repair or replace sheets that are torn or otherwise damaged during curing. Sheeting must remain on the concrete surface to be cured for not less than 7 days.

3.7.1.3 Membrane Curing Method

Apply a uniform coating of white-pigmented membrane-curing compound to the entire exposed surface of the concrete as soon after finishing as the free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after the removal of forms. Do not allow concrete surface to dry before application of the membrane. If drying has occurred, moisten the surface of the concrete with a fine spray of water and apply the curing compound as soon as the free water disappears. Apply curing compound in two coats by hand-operated pressure sprayers at a coverage of approximately 5 square meters/L for the total of both coats. Apply the second coat in a direction approximately at right angles to the direction of application of the first coat. The compound must form a uniform, continuous, coherent film that will not check, crack, or peel and must be free from pinholes or other imperfections. If pinholes, abrasion, or other discontinuities exist, apply an additional coat to the affected areas within 30 minutes. Respray concrete surfaces that are subjected to heavy rainfall within 3 hours after the curing compound has been applied by the method and at the coverage specified above. Respray areas where the curing compound is damaged by subsequent construction operations within the curing period. Take precautions necessary to ensure that the concrete is properly cured at sawed joints, and that no curing compound enters the joints. Tightly seal the top of the joint opening and the joint groove at exposed edges before the concrete in the region of the joint is resprayed with curing compound. Use a method used for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing period. Provide approved standby facilities for curing concrete pavement at a location accessible to the jobsite for use in the event of mechanical failure of the spraying equipment or other conditions that might prevent correct application of the membrane-curing compound at the proper time. Adequately protect concrete surfaces to which membrane-curing compounds have been applied during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from other possible damage to the continuity of the membrane.

3.7.2 Backfilling

After curing, remove debris and backfill, grade, and compact the area adjoining the concrete to conform to the surrounding area in accordance with lines and grades indicated.

3.7.3 Protection

Protect completed concrete from damage until accepted. Repair damaged concrete and clean concrete discolored during construction. Remove and reconstruct concrete that is damaged for the entire length between regularly scheduled joints. Refinishing the damaged portion will not be acceptable. Dispose of removed material as directed.

3.7.4 Protective Coating

Apply a protective coating of linseed oil mixture to the exposed-to-view concrete surface after the curing period, if concrete will be exposed to de-icing chemicals within 6 weeks after placement. Moist cure concrete to receive a protective coating.

3.7.4.1 Application

Complete curing and backfilling operation prior to applying two coats of protective coating. Concrete must be surface dry and clean before each application. Spray apply at a rate of not more than 11 square meters/L for first application and not more than 15.5 square meters/L for second application, except that the number of applications and coverage for each application for commercially prepared mixture must be in accordance with the manufacturer's instructions. Protect coated surfaces from vehicular and pedestrian traffic until dry.

3.7.4.2 Precautions

Do not heat protective coating by direct application of flame or electrical heaters and protect the coating from exposure to open flame, sparks, and fire adjacent to open containers or applicators. Do not apply material at ambient or material temperatures lower than 10 degrees C.

3.8 FIELD QUALITY CONTROL

Submit copies of all test reports within 24 hours of completion of the test.

3.8.1 General Requirements

Perform the inspection and tests described and meet the specified requirements for inspection details and frequency of testing. Based upon the results of these inspections and tests, take the action and submit reports as required below, and additional tests to ensure that the requirements of these specifications are met.

3.8.2 Concrete Testing

3.8.2.1 Strength Testing

Take concrete samples in accordance with JIS A 1115 not less than once a day nor less than once for every 190 cubic meters of concrete placed. Mold cylinders in accordance with JIS A 1132 for strength testing by an approved laboratory. Each strength test result must be the average of 2 test cylinders from the same concrete sample tested at 28 days, unless otherwise specified or approved. Concrete specified on the basis of compressive strength will be considered satisfactory if the averages of all sets of three consecutive strength test results equal or exceed the specified strength, and no individual strength test result falls below the specified strength by more than 4 MPa.

3.8.2.2 Air Content

Determine air content in accordance with JIS A 1118, or JIS A 1128. Use JIS A 1128 with concretes and mortars made with relatively dense natural aggregates. Make two tests for air content on randomly selected batches of each class of concrete placed during each shift. Make additional tests when excessive variation in concrete workability is reported by the placing foreman or the Government inspector. Notify the placing forman if results are out of tolerance. The placing foreman must take appropriate action to have the air content corrected at the plant. Additional tests for air content will be performed on each truckload of material until such time as the air content is within the tolerance specified.

3.8.2.3 Slump Test

Perform two slump tests on randomly selected batches of each class of concrete for every 190 cubic meters, or fraction thereof, of concrete placed during each shift. Perform additional tests when excessive variation in the workability of the concrete is noted or when excessive crumbling or slumping is noted along the edges of slip-formed concrete.

3.8.3 Thickness Evaluation

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section or by measuring the depth of opening of the extrusion template of the curb forming machine. If a slip form paver is used for sidewalk placement, construct the subgrade true to grade prior to concrete placement. The thickness will be determined by measuring each edge of the completed slab.

3.8.4 Surface Evaluation

Provide finished surfaces for each category of the completed work that are uniform in color and free of blemishes and form or tool marks.

3.9 SURFACE DEFICIENCIES AND CORRECTIONS

3.9.1 Thickness Deficiency

When measurements indicate that the completed concrete section is deficient in thickness by more than 6 mm the deficient section will be removed, between regularly scheduled joints, and replaced.

3.9.2 High Areas

In areas not meeting surface smoothness and plan grade requirements, reduce high areas either by rubbing the freshly finished concrete with carborundum brick and water when the concrete is less than 36 hours old or by grinding the hardened concrete with an approved surface grinding machine after the concrete is 36 hours old or more. The area corrected by grinding the surface of the hardened concrete must not exceed 5 percent of the area of any integral slab, and the depth of grinding must not exceed 6 mm. Remove and replace pavement areas requiring grade or surface smoothness corrections in excess of the limits specified.

3.9.3 Appearance

Exposed surfaces of the finished work will be inspected by the Contracting Officer and deficiencies in appearance will be identified. Remove and replace areas which exhibit excessive cracking, discoloration, form marks, or tool marks or which are otherwise inconsistent with the overall appearances of the work.

-- End of Section --

SECTION 32 17 23

PAVEMENT MARKINGS
08/16, CHG 5: 11/18

PART 1 GENERAL

~~1.1 UNIT PRICES~~

~~1.1.1 Measurement~~

~~1.1.1.1 Surface Preparation~~

~~The unit of measurement for surface preparation (cleaning) is the number of square meters of pavement surface prepared for marking and accepted by the Contracting Officer.~~

~~1.1.1.2 Pavement Striping and Markings~~

~~The unit of measurement for pavement markings is the number of square meters of reflective and/or nonreflective striping or markings actually completed and accepted by the Contracting Officer.~~

~~1.1.1.3 Raised Pavement Markers~~

~~The unit of measurement for raised pavement markers is the number actually placed as specified and approved by the Contracting Officer.~~

~~1.1.1.4 Removal of Pavement Markings on Roads and Automotive Parking Areas~~

~~The unit of measurement for removal of pavement markings is the number of square meters of pavement markings removed as specified and accepted by the Contracting Officer.~~

~~1.1.2 Payment~~

~~The quantities of surface preparation, pavement striping or markings, raised pavement markers, and removal of pavement markings determined as specified in paragraph Measurement, will be paid for at the contract unit price. The payment constitutes full compensation for furnishing all labor, materials, tools, equipment, appliances, and doing all work involved in preparing and marking the pavements as shown on the drawings. Remove and replace any striping or markings which required reflective media, but are placed without it, do not meet the stated minimum retro-reflective requirements, or with other defects, at no cost to the Government. Remove and replace striping or markings which do not conform to the required physical characteristics, alignment or location required at no cost to the Government.~~

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 247	(2013) Standard Specification for Glass Beads Used in Pavement Markings
~~AASHTO M 248~~	~~(1991; R 2012) Standard Specification for Ready-Mixed White and Yellow Traffic Paints~~
~~AASHTO M 249~~	~~(2012; R2016) Standard Specification for White and Yellow Reflective Thermoplastic Striping Material (Solid Form)~~

~~ASTM INTERNATIONAL (ASTM)~~

~~ASTM D471~~	~~(2016a) Standard Test Method for Rubber Property - Effect of Liquids~~
~~ASTM D476~~	~~(2015) Dry Pigmentary Titanium Dioxide Pigments~~
~~ASTM D522/D522M~~	~~(2017) Mandrel Bend Test of Attached Organic Coatings~~
~~ASTM D638~~	~~(2014) Standard Test Method for Tensile Properties of Plastics~~
~~ASTM D695~~	~~(2010) Standard Test Method for Compressive Properties of Rigid Plastics~~
~~ASTM D711~~	~~(2010; R 2015) No-Pick-Up Time of Traffic Paint~~
~~ASTM D823~~	~~(2018) Standard Practices for Producing Films of Uniform Thickness of Paint, Coatings, and Related Products on Test Panels~~
~~ASTM D1652~~	~~(2011; E 2012) Standard Test Method for Epoxy Content of Epoxy Resins~~
~~ASTM D2074~~	~~(2007; R2013) Standard Test Methods for Total, Primary, Secondary, and Tertiary Amine Values of Fatty Amines by Alternative Indicator Method~~
~~ASTM D2240~~	~~(2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness~~
~~ASTM D2621~~	~~(1987; R 2016) Standard Test Method for Infrared Identification of Vehicle Solids from Solvent-Reducible Paints~~
~~ASTM D2697~~	~~(2003; R 2014) Volume Nonvolatile Matter in Clear or Pigmented Coatings~~
~~ASTM D3335~~	~~(1985a; R 2020) Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy~~

~~ASTM D3718~~	~~(1985a; R 2015) Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy~~
~~ASTM D3924~~	~~(2016) Standard Specification for Environment for Conditioning and Testing Paint, Varnish, Lacquer, and Related Materials~~
~~ASTM D3960~~	~~(2005; R 2013) Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings~~
~~ASTM D4060~~	~~(2019) Abrasion Resistance of Organic Coatings by the Taber Abraser~~
~~ASTM D4061~~	~~(2013) Standard Test Method for Retroreflectance of Horizontal Coatings~~
~~ASTM D4280~~	~~(2012) Extended Life Type, Nonplowable, Raised, Retroreflective Pavement Markers~~
~~ASTM D4383~~	~~(2012) Standard Specification for Plowable, Raised Retroreflective Pavement Markers~~
~~ASTM D4505~~	~~(2012; R 2017) Standard Specification for Preformed Retroflective Pavement Marking Tape for Extended Service Life~~
~~ASTM D4541~~	~~(2017) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers~~
~~ASTM D6628~~	~~(2003; R 2015) Standard Specification for Color of Pavement Marking Materials~~
~~ASTM D7234~~	~~(2012) Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers~~
~~ASTM E1347~~	~~(2006; R 2020) Standard Test Method for Color and Color Difference Measurement by Tristimulus (Filter) Colorimetry~~
~~ASTM E1710~~	~~(2011) Standard Test Method for Measurement of Retroreflective Pavement Marking Materials with CEN-Prescribed Geometry Using a Portable Retroreflectometer~~
~~ASTM E2177~~	~~(2011) Standard Test Method for Measuring the Coefficient of Retroreflected Luminance (RL) of Pavement Markings in a Standard Condition of Wetness~~
~~ASTM E2302~~	~~(2003; R 2016) Standard Test Method for Measurement of the Luminance Coefficient~~

	~~Under Diffuse Illumination of Pavement Marking Materials Using a Portable Reflectometer~~
~~ASTM G154~~	~~(2016) Standard Practice for Operating Fluorescent Light Apparatus for UV Exposure of Nonmetallic Materials~~

~~INTERNATIONAL CONCRETE REPAIR INSTITUTE (ICRI)~~

~~ICRI 03732~~	~~(1997) Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, and Polymer Overlays~~

~~MASTER PAINTERS INSTITUTE (MPI)~~

~~MPI 32~~	~~(2012) Traffic Marking Paint, S.B.~~
~~MPI 97~~	~~(2012) Traffic Marking Paint, Latex~~

~~SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)~~

~~SAE AMS-STD-595A~~	~~(2017) Colors used in Government Procurement~~

~~U.S. FEDERAL AVIATION ADMINISTRATION (FAA)~~

~~FAA AC 150/5370-10~~	~~(2018; Rev H; Errata 1 2019) Standard Specifications for Construction of Airports~~

~~U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)~~

~~MUTCD~~	~~(2009; Rev 2012) Manual on Uniform Traffic Control Devices~~

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS TT-B-1325	(Rev D; Notice 1; Notice 2 2017) Beads (Glass Spheres) Retro-Reflective (Metric)
~~FS TT-P-1952~~	~~(2015; Rev F; Notice 1) Paint, Traffic and Airfield Markings, Waterborne~~

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS K 5665	(2016) Traffic Paint

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are ~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Paints for Roads and Streets

Paints for Concrete Curb Markings

~~Surface Preparation Equipment List; G[, [_____]]~~

~~Application Equipment List; G[, [_____]]~~

Exterior Surface Preparation

Safety Data Sheets; G~~[, [_____]]~~

~~Reflective media for airfields; G[, [_____]]~~

~~Reflective media for roads; G[, [_____]]~~

~~Waterborne Paint; G[, [_____]]~~

~~Solventborne Paint; G[, [_____]]~~

Thermoplastic compound; G~~[, [_____]]~~

~~Raised Pavement Markers Primers and Adhesives; G[, [_____]]~~

~~SD-06 Test Reports~~

~~Reflective Media for Airfields; G[, [_____]]~~

~~Reflective Media for Roads; G[, [_____]]~~

~~Waterborne Paint; G[, [_____]]~~

~~Solventborne Paint; G[, [_____]]~~

~~High Build Acrylic Coating (HBAC); G[, [_____]]~~

~~Thermoplastic Compound; G[, [_____]]~~

~~Raised Pavement Markers Primers and Adhesives; G[, [_____]]~~

~~Test Reports~~

SD-07 Certificates

~~Qualifications; G[, [_____]]~~

~~Reflective Media for Airfields~~

~~Reflective Media for Roads~~

~~Waterborne Paint~~

~~Solventborne Paint~~

Volatile Organic Compound~~, (VOC); G[, [_____]]~~

~~Thermoplastic Compound~~

SD-08 Manufacturer's Instructions

~~Waterborne~~ Paint~~; G[, [_____]]~~

~~Solventborne Paint; G[, [_____]]~~

~~Thermoplastic Compound; G[, [_____]]~~

1.3 QUALITY ASSURANCE

1.3.1 Regulatory Requirements

~~Submit certificate stating that the proposed~~Provide pavement marking paint meeting~~s~~ the Volatile Organic Compound, (VOC) regulations of the local Air Pollution Control District having jurisdiction over the geographical area in which the project is located. Submit Safety Data Sheets for each product.

~~1.3.2 Qualifications~~

~~Submit documentation certifying that pertinent personnel are qualified for equipment operation and handling of applicable chemicals. The documentation should include experience on five projects of similar size and scope with references for all personnel.~~

~~1.3.3 Qualifications For Airfield Marking Personnel~~

~~Submit documentation of qualifications in resume format a minimum of [14] [_____] days before pavement marking work is to be performed showing personnel who will be performing the work have experience working on airfields, operating mobile self-powered marking, cleaning, and paint removal equipment and performing these tasks. Include with resume a list of references complete with points of contact and telephone numbers. Provide certification for pavement marking machine operator and Foreman demonstrating experience successfully completing a minimum of two airfield pavement marking projects of similar size and scope. Provide documentation demonstrating personnel have a minimum of [two] [three] [four] years of experience operating similar equipment and performing the same or similar work in similar environments, similar in size and scope of the planned project. The Contracting Officer reserves the right to require additional proof of competency or to reject proposed personnel.~~

1.4 DELIVERY AND STORAGE

Deliver paint materials, thermoplastic compound materials, and reflective media in original sealed containers that plainly show the designated name, specification number, batch number, color, date of manufacture, manufacturer's directions, and name of manufacturer.

Provide storage facilities at the job site~~[~~, only in areas approved by the Contracting Officer,~~]~~ for maintaining materials at temperatures recommended by the manufacturer. ~~[Make available paint stored at the project site or segregated at the source for sampling not less than 30 days prior to date of required approval for use to allow sufficient time for testing. Notify the Contracting Officer when paint is available for sampling.]~~

1.5 PROJECT/SITE CONDITIONS

1.5.1 Environmental Requirements

1.5.1.1 Weather Limitations for Application

Apply pavement markings to clean, dry surfaces, and unless otherwise approved, only when the air and pavement surface temperature is at least 3 degrees C above the dew point and the air and pavement temperatures are within the limits recommended by the pavement marking manufacturer. Allow pavement surfaces to dry after water has been used for cleaning or rainfall has occurred prior to striping or marking. Test the pavement surface for moisture before beginning work each day and after cleaning. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer. ~~Employ the "plastic wrap method" to test the pavement for moisture as specified in paragraph TESTING FOR MOISTURE.~~

1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas

Pavement surface must be free of snow, ice, or slush; with a surface temperature of at least 4 degrees C and rising at the beginning of operations, except those involving shot or sand blasting or grinding. Cease operation during thunderstorms, or during rainfall, except for waterblasting and removal of previously applied chemicals. Cease waterblasting where surface water accumulation alters the effectiveness of material removal.

1.5.2 Traffic Controls

Place warning signs ~~conforming to MUTCD~~ near the beginning of the worksite and well ahead of the worksite for alerting approaching traffic from both directions. Place small markers along newly painted lines or freshly placed raised markers to control traffic and prevent damage to newly painted surfaces or displacement of raised pavement markers. Mark painting equipment with large warning signs indicating slow-moving painting equipment in operation.

When traffic must be rerouted or controlled to accomplish the work, provide necessary warning signs, flag persons, and related equipment for the safe passage of vehicles.

~~1.5.3 Airfield Traffic Control~~

~~Coordinate performance of all work in the controlled zones of the airfield with the Contracting Officer and with the [Flight Operations Officer or Airfield Manager] [control tower]. Neither equipment nor personnel can use any portion of the airfield without permission of these officers unless the runway is closed. Runways will be closed during the following times:~~

~~Day or Date~~	~~Runway Closing Time~~	~~Runway Opening Time~~	~~Important Notes~~
~~[_____]~~	~~[_____]~~	~~[_____]~~	~~[_____]~~

~~1.5.4 Airfield Radio Communication~~

~~No personnel or equipment will be allowed in the controlled zones of the airfield until radio contact has been made with the control tower and permission is granted by the control tower. A radio for this purpose [will be provided by the Government. The Contractor is responsible for the radio and must reimburse the Government for repair or replacement of the radio if it is lost, damaged, or destroyed] [is to be provided by the Contractor as approved by the Contracting Officer]. Maintain contact with the control tower at all times during work in vicinity of the airfield. Notify the control tower when work is completed and all personnel, equipment and materials have been removed from all aircraft operating surfaces.~~

~~1.5.5 Airfield Emergency Landing and Takeoff~~

~~Emergencies take precedence over all operations. Upon notification from the control tower of an emergency landing or imminent takeoff, stop all operations immediately and evacuate all personnel and equipment to an area not utilized for aircraft traffic which is at least 75 m measured perpendicular to and away from the near edge of the runway unless otherwise authorized by the Contracting Officer. Equipment and chemicals or detergents as well as excess water must be able to be removed from the work area within 3 minutes.~~

~~1.5.6 Lighting~~

~~When night operations are necessary, provide all necessary lighting and equipment. [Direct or shade lighting to prevent interference with aircraft, the air traffic control tower, and other base operations. Provide lighting and related equipment capable of being removed from the runway within 15 minutes of notification of an emergency. Night work must be coordinated with the Flight Operations Manager or Airfield Manager and approved in advance by the Contracting Officer.] The Government reserves the right to accept or reject night work on the day following night activities by the Contractor.~~

PART 2 PRODUCTS

2.1 EQUIPMENT

Machines, tools and equipment used in the performance of the work shall be approved by the Contracting Officer and maintained in satisfactory operating condition.

~~2.1.1 Surface Preparation and Paint Removal~~

~~2.1.1.1 Surface Preparation and Paint Removal Equipment for Airfield Pavements~~

~~Prepare all airfield surfaces and remove paint from airfield surfaces in accordance with UFGS 32 01 11.51 Rubber and Paint Removal From Airfield Pavements. Provide submittals in accordance with UFGS 32 01 11.51 Rubber~~

~~and Paint Removal From Airfield Pavements.~~

~~2.1.1.2 Surface Preparation Equipment for Roads and Automotive Parking Areas~~

~~Submit a surface preparation equipment list by serial number, type, model, and manufacturer. Include descriptive data indicating area of coverage per pass, pressure adjustment range, tank and flow capacities, and safety precautions required for the equipment operation. Mobile equipment must allow for removal of markings without damaging the pavement surface or joint sealant. Maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition.~~

~~2.1.1.2.1 Sandblasting Equipment~~

~~Use mobile sandblasting equipment capable of producing a pressurized stream of sand and air that effectively removes paint from the surface without filling voids with debris in asphalt or tar pavements or removing joint sealants in Portland cement concrete pavements. Include with the equipment and air compressor, hoses, and nozzles of adequate size and capacity for removing paint. Equip the compressor with traps and coalescing filters that maintain the compressed air free of oil and water.~~

~~2.1.1.2.2 Waterblasting Equipment~~

~~Use mobile waterblasting equipment capable of producing a pressurized stream of water that effectively removes paint from the pavement surface without significantly damaging the pavement. Provide equipment, tools, and machinery which are safe and in good working order at all times.~~

~~2.1.1.2.3 Shotblasting Equipment~~

~~Use mobile self propelled shotblasting equipment capable of producing an adjustable depth of paint removal and of propelling abrasive particles at high velocities on the paint for effective removal. Ensure each unit is self cleaning and self contained. Use equipment able to confine the abrasive, any dust that is produced, and removed paint and is capable of recycling the abrasive for reuse.~~

~~2.1.1.2.4 Grinding or Scarifying Equipment~~

~~Use equipment capable of removing surface contaminates, paint build-up, or extraneous markings from the pavement surface without leaving any residue. Clean the surface by hydro blast to remove surface contaminates and ash after a weed torch is used to remove paint.~~

~~2.1.1.2.5 Chemical Removal Equipment~~

~~Use chemical equipment capable of applying and removing chemicals and paint from the pavement surface, leaving only non-toxic biodegradable residue without scarring or other damage to the pavement or joints and joint seals.~~

2.1.1 Application Equipment

~~Submit application equipment list appropriate for the material(s) to be used. Include manufacturer's descriptive data and certification for the planned use that indicates area of coverage per pass, pressure adjustment range, tank and flow capacities, and all safety precautions required for~~

~~operating and maintaining the equipment. Provide and maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition, or remove them from the work site. Provide mobile and maneuverable application equipment to the extent that straight lines can be followed and normal curves can be made in a true arc.~~

2.1.1.1 Paint Application Equipment

2.1.1.1.1 Hand-Operated, Push-Type Machines

Provide hand-operated push-type applicator machine of a type commonly used for application of water based paint or two-component, chemically curing paint, thermoplastic, or preformed tape, to pavement surfaces for small marking projects, such as legends and cross-walks, automotive parking areas, or surface painted signs. Provide applicator machine equipped with the necessary tanks and spraying nozzles capable of applying paint uniformly at coverage specified. Hand operated spray guns may be used in areas where push-type machines cannot be used.

2.1.1.1.2 Self-Propelled or Mobile-Drawn Spraying Machines

Provide self-propelled or mobile-drawn spraying machine with suitable arrangements of atomizing nozzles and controls to obtain the specified results. Provide machine having a speed during application capable of applying the stripe widths indicated at the paint coverage rate specified herein and of even uniform thickness with clear-cut edges.

~~2.1.1.1.2.1 Road Marking~~

~~Provide equipment used for marking roads capable of placing the prescribed number of lines at a single pass as solid lines, intermittent lines, or a combination of solid and intermittent lines using a maximum of three different colors of paint as specified.~~

~~2.1.1.1.2.2 Airfield Marking~~

~~Provide self-propelled or mobile-drawn spraying machine for applying the paint for airfield pavements with an arrangement of atomizing nozzles capable of applying the specified line width in a single pass. Provide paint applicator with paint reservoirs or tanks of sufficient capacity and suitable gages to apply paint in accordance with requirements specified. Equip tanks with suitable mechanical agitators. Equip spray mechanism with quick-action valves conveniently located, and include necessary pressure regulators and gages in full view and reach of the operator. Install paint strainers in paint supply lines to ensure freedom from residue and foreign matter that may cause malfunction of the spray guns. The paint applicator must be readily adaptable for attachment of a dispenser for the reflective media approved for use.~~

2.1.1.1.2.1 Hand Application

Provide spray guns for hand application of paint in areas where the mobile paint applicator cannot be used.

~~2.1.1.2 Thermoplastic Application Equipment~~

~~2.1.1.2.1 Thermoplastic Material~~

~~Apply thermoplastic material with equipment that is capable of providing~~

~~continuous uniformity in the dimensions and reflectorization of the marking.~~

~~2.1.1.2.2 Application Equipment~~

a. ~~Provide application equipment capable of continuous mixing and agitation of the material, with conveying parts which prevent accumulation and clogging between the main material reservoir and the extrusion shoe or spray gun. All parts of the equipment which come into contact with the material must be easily accessible and exposed for cleaning and maintenance. All mixing and conveying parts up to and including the extrusion shoes and spray guns must maintain the material at the required temperature with heat-transfer oil or electrical-element-controlled heat.~~

b. ~~Provide application equipment constructed to ensure continuous uniformity in the dimensions of the stripe. Provide an applicator with a means for cleanly cutting off stripe ends squarely and providing a method of applying "skiplines." Provide equipment capable of applying varying widths of traffic markings.~~

c. ~~Provide mobile and maneuverable application equipment allowing straight lines to be followed and normal curves to be made in a true arc. Provide equipment used for the placement of thermoplastic pavement markings of two general types: mobile applicator and portable applicator.~~

d. ~~Equip the applicator with a pressurized or drop-on type bead dispenser capable of uniformly dispensing reflective glass spheres at controlled rates of flow. The bead dispenser must operate automatically to begin flow prior to the flow of binder to assure that the strip is fully reflectorized.~~

~~2.1.1.2.3 Mobile Application Equipment~~

~~Provide a truck-mounted, self-contained pavement marking machine that is capable of hot applying thermoplastic by either the extrusion or spray method.~~

a. ~~Equip the unit to apply the thermoplastic marking material at temperatures according to the manufacturer's instructions, at widths varying from 75 to 300 mm, with an automatic pressurized or drop-on bead dispensing system, capable of operating continuously, and of installing a minimum of 6 lineal km of longitudinal markings in an 8-hour day.~~

b. ~~Equip the mobile unit with a melting kettle which holds a minimum of 2.7 metric tons of molten thermoplastic material; capable of heating the thermoplastic composition to temperatures as recommended by the manufacturer. Use a thermostatically controlled heat transfer liquid. Heating of the composition by direct flame is not allowed. Oil and material temperature gauges must be visible at both ends of the kettle.~~

c. ~~Equip mobile units for application of extruded markings with a minimum of two extrusion shoes; located one on each side of the truck, capable of marking simultaneous edge line and centerline stripes; each being a closed, oil-jacketed unit; holding the molten thermoplastic at a temperature as recommended by the manufacturer; and capable of~~

~~extruding a line of 75 to 200 mm in width; and at a thickness of not less than 3 mm nor more than 5.0 mm , of generally uniform cross section.~~

~~d. Equip mobile units for application of spray markings with a spray gun system capable of marking simultaneous edgeline and centerline stripes. Surround (jacket) the spray system with heating oil to maintain the molten thermoplastic at a temperature of 191 to 218 degrees C, capable of spraying a stripe of 75 to 305 mm in width, and in thicknesses varying from 1.52 mm to 2.49 mm, of generally uniform cross section.~~

~~e. Equip the mobile unit with an electronic programmable line pattern control system, capable of applying skip or solid lines in any sequence, through any and all of the extrusion shoes, or the spray guns, and in programmable cycle lengths. In addition, equip the mobile unit with an automatic counting mechanism capable of recording the number of lineal meters of thermoplastic markings applied to the pavement surface with an accuracy of 0.5 percent.~~

~~2.1.1.2.4 Portable Application Equipment~~

~~Provide portable hand-operated equipment, specifically designed for placing special markings such as crosswalks, stop bars, legends, arrows, and short lengths of lane, edge and centerlines; and capable of applying thermoplastic pavement markings by the extrusion method. Equip the portable applicator with all the necessary components, including a materials storage reservoir, bead dispenser, extrusion shoe, and heating accessories, capable of holding the molten thermoplastic at the temperature recommended by the manufacturer, and of extruding a line of 75 to 305 mm in width, and in thickness of not less than 3 mm nor more than 5 mm and of generally uniform cross section.~~

~~2.1.1.3 Reflective Media Dispenser~~

~~Attach the dispenser for applying the reflective media to the [paint] [thermoplastic] dispenser and designed to operate automatically and simultaneously with the applicator through the same control mechanism. The bead applicator must be capable of adjustment and designed to provide uniform flow of reflective media over the full length and width of the stripe at the rate of coverage specified in paragraph APPLICATION.~~

~~2.1.1.4 Preformed Tape Application Equipment~~

~~Provide and use mechanical application equipment for the placement of preformed marking tape which is a mobile pavement marking machine specifically designed for use in applying pressure-sensitive pavement marking tape of varying widths. Equip the applicator with rollers, or other suitable compaction device to provide initial adhesion of the material with the pavement surface. Use additional tools and devices as needed to properly seat the applied material as recommended by the manufacturer.~~

2.2 MATERIALS

Provide materials conforming to the requirements specified herein.~~[Use waterborne or methacrylate paint for airfield markings.] [Use [waterborne paint] [epoxy paint] [thermoplastic] [high build acrylic] [raised pavement markers] [preformed tape] for roads.] [Use non-reflectorized waterborne~~

~~[or solventborne] paint for automotive parking areas.] The maximum allowable VOC content of pavement markings is [150] [____] grams per liter. Color of markings are indicated on the drawings and must conform to ASTM D6628 for roads and automotive parking areas and SAE AMS-STD-595A for airfields. Provide materials conforming to the requirements specified herein.~~

2.2.1 Paints For Roads And Streets

Shall be hot melt application type, conforming to JIS K 5665, color as indicated.

2.2.2 Paints For Concrete Curb Markings

Shall be normal temperature traffic paint conforming to JIS K 5665, color as indicated.

~~2.2.3 Waterborne Paint~~

~~[FS TT-P-1952, Type [I or II] [III]][MPI 97].~~

~~2.2.4 Solventborne Paint~~

~~[AASHTO M 248][MPI 32].~~

~~2.2.5 Methacrylate Paint~~

~~Formulate methacrylate paint to meet the requirements of FAA AC 150/5370-10, Item P-620.2, Methacrylate.~~

~~2.2.6 Epoxy Paint~~

~~2.2.6.1 Formulation~~

~~Epoxy pavement marking material will be a two component, 100 percent solids, material formulated to provide simple volumetric mixing ratio of two volumes of component A and one volume of component B unless otherwise recommended by the manufacturer.~~

~~2.2.6.2 Composition~~

~~The component A of both white and yellow must be within the following limits:~~

~~TABLE I~~		
	~~White~~	~~Yellow~~
~~Pigments~~	~~Minimum 18 percent by weight Titanium Dioxide (ASTM D476, Type II)~~	~~21-27 percent by weight~~
~~Epoxy Resin~~	~~75-82 percent~~	~~73-79 percent~~

~~The epoxy resin must be free of lead, cadmium, mercury, hexavalent chromium and other toxic heavy metals as defined by the Environmental Protection Agency. Submit a manufacturer's certification of compliance~~

~~with this requirement.~~

~~2.2.6.3 Epoxide Value~~

~~Determine epoxide epoxy number of the epoxy resin in accordance with ASTM D1652 for white and yellow component A on pigment free basis. The epoxide number must be within plus or minus 50 of the published manufacturer's standard.~~

~~2.2.6.4 Total Amine Value~~

~~Determine the amine number on the curing agent (component B) in accordance with ASTM D2074. The amine number must be within plus or minus 50 of the published manufacturer's standard.~~

~~2.2.6.5 Toxicity~~

~~Upon heating to application temperature, the material must not produce fumes which are toxic or injurious to persons or property.~~

~~2.2.6.6 Daylight Directional Reflectance~~

~~Directional reflectance of white and yellow paint (without glass beads) in accordance with ASTM E1347: White 84 percent Yellow 55 percent.~~

~~2.2.6.7 Laboratory Drying Time~~

~~The epoxy pavement marking material must have a maximum no-pick-up time of 30 minutes when tested in accordance with ASTM D711.~~

~~2.2.6.8 Curing~~

~~The epoxy material must be capable of fully curing under a constant surface temperature of 7 Degrees C or above.~~

~~2.2.6.9 Adhesion to Concrete~~

~~The catalyzed epoxy pavement marking material must have a high degree of adhesion to the specified concrete surface (100 percent concrete failure) when tested according to ASTM D7234. The concrete substrate must have a minimum compressive strength of 27.5 MPa. Condition prepared specimens at a temperature of 23.9 plus or minus 1.1 Degrees C for a minimum of 24 hours and a maximum of 72 hours prior to performance of the test.~~

~~2.2.6.10 Hardness~~

~~Epoxy pavement marking materials must have a Shore D Hardness between 75 and 100 when tested in accordance with ASTM D2240. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 72 hours and a maximum of 96 hours prior to performing the tests.~~

~~2.2.6.11 Abrasion Resistance~~

~~The wear index for a catalyzed sample must not exceed 82 when tested in accordance with ASTM D4060 using a 1000 gram load, CS-17 wheels and a test duration of 1000 cycles. Run the test on cured samples of material which have been applied at a film thickness of 15 plus or minus 0.5 mils to code S-16 stainless steel plates. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 48 hours prior to performing the tests.~~

~~2.2.6.12 Tensile Strength~~

~~Epoxy pavement marking materials must have a tensile strength of at least 41,370 kPA when tested in accordance with ASTM D638. Cast the Type IV specimens in a suitable mold and pull at the rate of 6 mm per minute using a suitable dynamic testing machine. Cure the samples at 23.9 plus or minus 1.1 Degrees C for a minimum of 12 hours and a maximum of 48 hours prior to performing the tests.~~

~~2.2.6.13 Compressive Strength~~

~~Catalyzed epoxy pavement marking materials must have a compressive strength of at least 82,700 kPA when tested in accordance with ASTM D695. Condition the cast sample at 23.9 plus or minus 1.1 Degrees C for a minimum of 12 hours and a maximum of 48 hours prior to performing the tests. The rate of compression of these samples must not exceed 6 mm per minute.~~

~~2.2.7 Thermoplastic Compound~~

~~2.2.7.1 Composition Requirements~~

~~Thermoplastic compound must conform to AASHTO M 249. Formulate the binder component as an alkyd resin.~~

~~2.2.7.2 Primer~~

a. ~~Asphalt concrete primer: Provide thermosetting adhesive primer with a solids content of pigment reinforced synthetic rubber and synthetic plastic resin dissolved or dispersed in a volatile organic solvent for asphaltic concrete pavements. The solids content must not be less than 10 percent by weight at 21 degrees C and 60 percent relative humidity. A wet film thickness of 0.13 mm, plus or minus 0.03 mm, must dry to a tack-free condition in less than 5 minutes.~~

b. ~~Portland cement concrete primer: Provide an epoxy resin primer for Portland cement concrete pavements, of the type recommended by the manufacturer of the thermoplastic composition.~~

~~2.2.8 High Build Acrylic Coating (HBAC)~~

~~Formulate High Build Acrylic Coating (HBAC) to meet the requirements of Table II.~~

~~TABLE II - REQUIREMENTS FOR HIGH BUILD ACRYLIC COATINGS (HBAC)~~

~~TEST~~	~~MINIMUM REQUIREMENT (AND MAXIMUM WHERE INDICATED)~~
~~Resin System (ASTM D2621)~~	~~Waterborne 100 percent Acrylic~~
~~Percent Volume Solids (ASTM D2697)~~	~~58 percent~~

~~TABLE II - REQUIREMENTS FOR HIGH BUILD ACRYLIC COATINGS (HBAC)~~	
~~TEST~~	~~MINIMUM REQUIREMENT (AND MAXIMUM WHERE INDICATED)~~
~~Volatile Organic Compound, max. (ASTM D3960)~~	~~150 g/l~~
~~White (SAE AMS-STD-595A)~~	~~37925~~
~~Yellow (SAE AMS-STD-595A)~~	~~33538~~
~~Shore D Hardness (ASTM D2240)~~	~~45~~
~~1/8 inch Mandrel Bend at 5 mils Dry Film Thickness (DFT, one-week cure (ASTM D522/D522M, Method B)~~	~~No visual defects at bend (Conditions at ASTM D3924)~~
~~Adhesion to Concrete and Asphaltic Pavements (ASTM D4541)~~	~~0.97 MPa or 100 percent cohesive failure in pavement~~
~~Accelerated Weathering, Yellow, 2500 Hours UV Exposure (ASTM G154: see note 1)~~	~~Max. color loss to 33655 (SAE AMS-STD-595A)~~
~~Water Absorption at 168 Hours Immersion Tap Water (ASTM D471)~~	~~9.0 percent max. weight increase (conditions at ASTM D3924)~~
~~Application at 1650 microns Wet,One Coat, One-week Cure, (see note 2)~~	~~No visual cracking or curling (conditions at ASTM D3924)~~
~~No Pick-Up at 630 microns (ASTM D711)~~	~~Wet 10 minutes max.~~
~~Lead (ASTM D3335)~~	~~0.06 percent max.~~
~~Cadmium (ASTM D3335)~~	~~0.06 percent max.~~
~~Chromium (ASTM D3718)~~	~~0.00 percent~~
~~Notes:~~	

~~TABLE II - REQUIREMENTS FOR HIGH BUILD ACRYLIC COATINGS (HBAC)~~	
~~TEST~~	~~MINIMUM REQUIREMENT (AND MAXIMUM WHERE INDICATED)~~
~~(1) Properly mix and apply yellow paint at 250 microns plus or minus 50 microns DFT over a suitably sized, clean aluminum substrate (ASTM D823), and cure for a minimum of 48 hours: prepare four individual yellow samples. Expose three samples to continuous Ultraviolet (UV) light for 2500 hours, without cycles condensation, in accordance to ASTM G154: use UVA-340 lamps in the testing apparatus. Following exposure, compare the three exposed samples to the "one" non-exposed sample using SAE AMS-STD-595A colors 33538 and 33655 as visual references: evaluate exposed samples for degree of visual color loss. Yellow paint is rated as passing if each exposed sample appears equivalent to the non-exposed sample, and in addition, displays color loss no greater than SAE AMS-STD-595A color 33655.~~	
~~(2) Using double-stick, foam mounting tape (or equal) with a nominal thickness of 1625 microns, apply a rectangular mold with inner dimensions of 7.6 cm by 25.5 cm to a clean aluminum sample approximately sized at 15 cm by 30 cm by 0.30 cm. Do not remove the tape's plastic backing. Mix and apply excess paint into mold. Remove excess paint, by squeegee or other appropriate draw down technique, to a uniform thickness equal to the tape's height. Perform paint application and draw down within a period of no more than 60 seconds. Approximately one to two minutes following the draw down, remove tape from sample and allow coating to cure for a minimum period of one week ASTM D3924. Using a micrometer or other appropriate device, measure cured coating thickness (less sample thickness) to confirm resulting coating application was at or above 950 microns DFT. Inspect coating for visual signs of cracking and curling. Following a one week cure, the coating is rated as passing if applied greater than 950 microns DFT and visually free of both cracking and curling.~~	

~~2.2.9 Preformed Tape~~

~~Provide adherent reflectorized strip preformed tape in accordance with ASTM D4505 Retroreflectivity Level II, Class 1, 2 or 3, Skid Resistance Level B.~~

~~2.2.10 Raised Pavement Markers Primers and Adhesives~~

~~Use either metallic or nonmetallic markers of the button or prismatic reflector type. Provide permanent color markers as specified for pavement marking, which retain the color and brightness under the action of traffic. Provide button markers with a diameter of not less than 100 mm, spaced not more than 12 m apart on solid longitudinal lines. Make broken centerline marker spacing in segments [of [_____]] [indicated] with gaps [of [_____]] [indicated] between segments. Provide button markers with rounded surfaces presenting a smooth contour to traffic and not projecting~~

~~more than 19 mm above level of pavement. Provide [nonplowable] [plowable] pavement markers and adhesive epoxy conforming to [ASTM D4280] [ASTM D4383].~~

~~2.2.11 Reflective Media~~

~~2.2.11.1 Reflective Media for Airfields~~

~~FS TT-B-1325, [Type I, [Gradation A,] Type III,] [or] [Type IV, Gradation A or B].~~

~~2.2.11.2 Reflective Media for Roads~~

~~[FS TT-B-1325, [Type I, Gradation A] [or] [Type IV, Gradation A or B]] [AASHTO M 247, Type 1].~~

PART 3 EXECUTION

3.1 EXAMINATION

3.1.1 Testing for Moisture

Test the pavement surface for moisture before beginning pavement marking after each period of rainfall, fog, high humidity, or cleaning, or when the ambient temperature has fallen below the dew point. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer or authorized representative.

~~Employ the "plastic wrap method" to test the pavement for moisture as follows: Cover the pavement with a 300 mm by 300 mm section of clear plastic wrap and seal the edges with tape. After 15 minutes, examine the plastic wrap for any visible moisture accumulation inside the plastic. Do not begin marking operations until the test can be performed with no visible moisture accumulation inside the plastic wrap. Re-test surfaces when work has been stopped due to rain.~~

~~3.1.2 Surface Preparation Demonstration~~

~~Prior to surface preparation, demonstrate the proposed procedures and equipment. Prepare areas large enough to determine [cleanliness][, adhesion of remaining coating] and rate of cleaning. [Perform a demonstration removal of pavement marking in an area designated by the Contracting Officer.] [Approved demonstration area establishes the standard for the remainder of the work.]~~

~~3.1.3 Test Stripe Demonstration~~

~~Prior to paint application, demonstrate test stripe application within the work area using the proposed materials and equipment. Apply separate test stripes in each of the line widths and configurations required herein using the proposed equipment. Make the test stripes long enough to determine the proper speed and operating pressures for the vehicle(s) and machinery, but not less than 15 m long.~~

~~3.1.4 Application Rate Demonstration~~

~~During the Test Stripe Demonstration, demonstrate compliance with the application rates specified herein. Document the equipment speed and operating pressures required to meet the specified rates in each~~

~~configuration of the equipment and provide a copy of the documentation to the Contracting Officer prior to proceeding with the work.~~

~~3.1.5 Retroreflective Value Demonstration~~

~~After the test stripes have cured to a "no-track" condition, demonstrate compliance with the average retroreflective values specified herein. Take a minimum of ten readings on each test stripe with a Retroreflectometer with a direct readout in millicandelas per square meter per lux (mcd/m2/lx). Perform testing in accordance with ASTM D4061, ASTM E1710, ASTM E2177, and ASTM E2302.~~

~~3.1.6 Level of Performance Demonstration~~

~~The Contracting Officer will be present at the application demonstrations to observe the results obtained and to validate the operating parameters of the vehicle(s) and equipment. If accepted by the Contracting Officer, the test stripe is the measure of performance required for this project. Do not proceed with the work until the demonstration results are satisfactory to the Contracting Officer.~~

3.2 EXTERIOR SURFACE PREPARATION

Allow new pavement surfaces to cure for a period of not less than ~~[~~30~~]~~ ~~[_____]~~ days before application of marking materials. Thoroughly clean surfaces to be marked before application of the paint. Remove dust, dirt, and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or a combination of these methods as required. Remove ~~[~~rubber deposits,~~]~~ ~~[~~existing paint markings,~~]~~ ~~[~~residual curing compounds,~~]~~ and other coatings adhering to the pavement by ~~[~~water blasting or ~~][~~approved chemical removal method~~][according to the removal requirements and procedures outlined in Section 32 01 11.51]~~. Do not commence painting in any area until the pavement surfaces are dry and clean and have been inspected and approved by the Contracting Officer.

~~a. For Portland Cement Concrete pavement, grinding, light shot blasting, or light scarification, to a resulting profile equal to ICRI 03732 CSP 2, CSP 3, and CSP 4, respectively, can be used in addition to water blasting on most pavements, to either remove existing coatings, or for surface preparation.~~

~~b. Do not use shot blasting on airfield pavements due to the potential of Foreign Object Damage (FOD) to aircraft. Scrub affected areas, where oil or grease is present on old pavements to be marked, with several applications of trisodium phosphate solution or other approved detergent or degreaser and rinse thoroughly after each application. After cleaning oil-soaked areas, seal with shellac or primer recommended by the manufacturer to prevent bleeding through the new paint. Do not commence painting in any area until pavement surfaces are dry and clean.~~

~~3.2.1 Early Painting of Rigid Pavements~~

~~Pretreat rigid pavements that require early painting with an aqueous solution containing 3 percent phosphoric acid and 2 percent zinc chloride. Apply the solution to the areas to be marked.~~

3.2.1 Early Painting of Asphalt Pavements

For asphalt pavement systems requiring painting application at less than 30 days, apply the paint and beads at half the normal application rate, followed by a second application at the normal rate after 30 days.

3.3 APPLICATION

Apply pavement markings to dry pavements only.

3.3.1 Paint~~Paint~~

Apply paint with approved equipment at rate of coverage specified herein. Provide guidelines and templates as necessary to control paint application. Take special precautions in marking numbers, letters, and symbols. Manually paint numbers, letters, and symbols. Sharply outline all edges of markings. The maximum drying time requirements of the paint specifications will be strictly enforced, to prevent undue softening of bitumen, and pickup, displacement, or discoloration by tires of traffic. If there is a deficiency in drying of the markings, painting operations must cease until the cause of the slow drying is determined and corrected. Apply paint at a rate of 2.6 plus or minus 0.1 square meter per liter.

~~3.3.1.1 Waterborne Paint~~

~~3.3.1.1.1 Airfields~~

~~For non-reflectorized [and reflectorized] markings, apply paint conforming to [FS TT-P-1952 [Type I or II at a rate of 3.0 plus or minus 0.15 square meter per liter] [Type III at a rate of 2.7 plus or minus 0.20 square meter per liter] [MPI 97 at a rate of 2.5 plus or minus 0.10 square meter per liter].~~

~~[For reflectorized markings, apply FS TT-B-1325 beads at a rate of 834 plus or minus 60 grams of glass spheres per liter.] [For reflectorized markings, apply paint and glass spheres at the following rates:]~~

~~TABLE III~~

~~Bead Type~~	~~Paint Type~~	~~Paint Application Rate~~	~~Bead Application Rate~~
~~Type I (Gradation A)~~	~~Type I, II, III~~	~~3 plus or minus 0.15 Sq M/Liter~~	~~960 plus or minus 120 g/Liter~~
~~Type III~~	~~Type I, II, III~~	~~3 plus or minus 0.15 Sq M/Liter~~	~~1,200 plus or minus 120 g/Liter~~
~~Type IV (Gradation A)~~	~~Type III~~	~~1.9 plus or minus 0.30 Sq M/Liter~~	~~960 plus or minus 120 g/Liter~~
~~Type IV (Gradation B)~~	~~Type III~~	~~2.4 plus or minus 0.23 Sq M/Liter~~	~~960 plus or minus 120 g/Liter~~

~~3.3.1.1.2 Roads~~

~~Apply paint at a rate of 2.6 plus or minus 0.1 square meter per liter. Apply [FS TT-B-1325 Type I (Gradation A)] [AASHTO M 247 Type 1] beads at a rate of 834 plus or minus 60 grams of glass spheres per liter.~~

~~3.3.1.2 Solventborne Paint~~

~~Apply paint at a minimum wet film thickness of 0.381 mm. Apply [FS TT-B-1325 Type I (Gradation A)] [AASHTO M 247 Type 1] beads at a minimum rate of 715 grams of glass spheres per liter.~~

~~3.3.1.3 Methacrylate Paint~~

~~Apply paint evenly to the pavement surface at a maximum rate of 1.1 square meters per liter. Apply glass spheres conforming to FS TT-B-1325 uniformly to the wet paint on airfield pavement. Use either Type I (Gradation A), Type III, or Type IV (Gradation A or B) beads. Apply Type I (Gradation A) beads at a minimum rate of 1.8 kilograms of glass spheres per liter. Apply Type III beads at a minimum rate of 2.4 kilograms of glass spheres per liter. Apply Type IV (Gradation A or B) beads at a minimum rate of 1.8 kilograms of glass spheres per liter.~~

~~3.3.1.4 Epoxy Paint~~

~~Apply paint evenly to the pavement surface at a wet film thickness of 0.508 mm plus or minus 0.025 mm to cover 2.0 plus or minus 0.1 square meters per liter. Apply glass spheres uniformly to the wet paint on road and street pavement at a rate of 834 plus or minus 60 grams of glass spheres per liter.~~

~~3.3.1.5 High Build Acrylic Coating~~

~~Apply High Build Acrylic Coating (HBAC) at a rate of 1.3 square meters per liter. Apply Type IV (Gradation A) beads at a minimum rate of 1.8 kilograms of glass spheres per liter.~~

3.3.2 Thermoplastic Compound

Place thermoplastic pavement markings, free from dirt or tint, upon dry pavement. The temperature must be a minimum of 4.4 degrees C and rising at the time of installation. Apply all centerline, skipline, edgeline, and other longitudinal type markings with a mobile applicator. Place all special markings, crosswalks, stop bars, legends, arrows, and similar patterns with a portable applicator, using the extrusion method.

3.3.2.1 Primer

After surface preparation has been completed, prime the asphalt or concrete pavement surface with spray equipment. Allow primer materials to "set-up" prior to applying the thermoplastic composition. ~~[~~Allow the asphalt concrete primer to dry to a tack-free condition, usually occurring in less than 10 minutes.~~]~~ ~~[Allow the Portland cement concrete primer to dry in accordance with the thermoplastic manufacturer recommendations. To shorten the curing time of the epoxy resins, an infrared heating device may be used on the concrete primer.]~~ ~~[~~Apply asphalt concrete primer to all asphalt concrete pavements at a wet film thickness of 0.13 mm, plus or minus 0.03 mm (6.5 to 9.82 square meters per liter).~~]~~ ~~[Apply Portland~~

~~cement concrete primer to all concrete pavements (including concrete bridge decks) at a wet film thickness of between 1.0 to 1.3 mm 7.85 to 9.82 square meters per liter.]~~

After the primer has "set-up", apply the thermoplastic at temperatures no lower than 191 degrees C nor higher than 218 degrees C at the point of deposition. Apply all extruded thermoplastic markings at the specified width and at a thickness of not less than 3 mm nor more than 5 mm. Apply all sprayed thermoplastic markings at the specified width and the thickness designated in the contract plans. If the plans do not specify a thickness, apply centerline markings at a wet thickness of 2.3 mm, plus or minus 0.13 mm, and edgeline markings at a wet thickness of 1.5 mm, plus or minus 0.13 mm.

~~[~~Extrude or spray thermoplastic reflectorized pavement marking compound in a molten state onto a primed pavement surface. Following a surface application of glass beads and upon cooling to normal pavement temperatures, the marking must be an adherent reflectorized strip of the specified thickness and width that is capable of resisting deformation by traffic.~~]~~

3.3.2.2 Reflective Media

Immediately after installation of the thermoplastic material, mechanically apply drop-on reflective glass spheres conforming to ~~[~~FS TT-B-1325 Type I (Gradation A)~~]~~ or ~~[~~AASHTO M 247 Type 1~~]~~ at the rate of 0.24 kg per square meter such that the spheres are held by and imbedded in the surface of the molten material. Accomplish drop-on application of the glass spheres to ensure even distribution at the specified rate of coverage. If there is a malfunction of either thermoplastic applicator or reflective media dispenser, discontinue operations until deficiency is corrected.

~~3.3.3 Raised Pavement Markers~~

~~Align prefabricated markers carefully at the spacing indicated on the drawings and permanently fix in place by means of epoxy adhesives. To ensure good bond prior to applying adhesive, thoroughly clean all areas where markers are to be set by water blasting and use of compressed air.~~

~~3.3.4 Preformed Tape~~

~~The pavement surface and ambient air temperature must be a minimum of 15 degrees C and rising. Place the preformed markings in accordance with the manufacturer's written instructions.~~

3.3.3 Cleanup and Waste Disposal

Keep the worksite clean and free of debris and waste from the removal and application operations.~~[Immediately cleanup following removal operations in areas subject to aircraft traffic.]~~ Dispose of debris at approved sites.

3.4 FIELD QUALITY CONTROL

3.4.1 Sampling and Testing

As soon as the paint ~~[~~and thermoplastic~~]~~ materials ~~[and reflective media]~~ are available for sampling, obtain by random selection from the sealed containers, ~~[~~two~~]~~ ~~[four]~~ quart samples of each batch in the presence of the Contracting Officer. ~~[~~Two quarts will be for sampling and testing by the Contractor and two quarts will be for retention by the Government.~~]~~ Accomplish adequate mixing prior to sampling to ensure a uniform, representative sample. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Clearly identify samples by designated name, specification number, batch number, project contract number, intended use, and quantity involved.

~~[~~Test samples by an approved laboratory. If a sample fails to meet specification, replace the material in the area represented by the samples and retest the replacement material as specified above. Submit certified copies of the test reports~~test reports~~, prior to the use of the materials at the jobsite. Include in the report of test results a listing of any specification requirements not verified by the test laboratory.~~]~~ ~~[~~At the discretion of the Contracting Officer, samples provided may be tested by the Government for verification.~~]~~

3.4.2 Material Inspection

Examine material at the job site to determine that it is the material referenced in the report of test results or certificate of compliance. Provide test results substantiating conformance to the specified requirements with each certificate of compliance.

3.4.3 Dimensional Tolerances

Apply all markings in the standard dimensions provide in the drawings. New markings may deviate a maximum of 10 percent larger than the standard dimension. The maximum deviation allowed when painting over an old marking is up to 20 percent larger than the standard dimensions.

3.4.4 Bond Failure Verification

Inspect newly applied markings for signs of bond failure based on visual inspection and comparison to results from Test Stripe Demonstration paragraph.

~~3.4.5 Reflective Media and Coating Application Verification~~

~~Use a wet film thickness gauge to measure the application of wet paint. Use a microscope or magnifying glass to evaluate the embedment of glass beads in the paint. Verify the glass bead embedment with approximately 50 percent of the individual bead spheres embedded and 50 percent of the individual bead spheres exposed.~~

~~3.4.6 Retroreflective Markings~~

~~Collect and record readings for white and yellow retroreflective markings at the rate of one reading per 305 linear m. The minimum acceptable average for white markings is 200 millicandelas per square meter per lux (mcd/m2/lx) (measured with Retroreflectometer). The minimum acceptable average for yellow markings is 175 millicandelas per square meter per lux (mcd/m2/lx). Compute readings by averaging a minimum of 10 readings taken within the area at random locations. Re-mark areas not meeting the retroreflective requirements stated above.~~

~~3.4.7 Material Bond Verification and Operations Area Cleanup for Airfields~~

~~Vacuum sweep the aircraft operating area before it is opened for aircraft~~

~~operations to preclude potential foreign object damaged to aircraft engines. Visually inspect the pavement markings and the material captured by the vacuum. Verify that no significant loss of reflective media has occurred to the pavement marking due to the vacuum cleaning.~~

-- End of Section --

SECTION 32 31 13

CHAIN LINK FENCES AND GATES
11/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5308	(2014) Ready-Mixed Concrete
JIS A 5540	(2008) Turnbuckle for Building
JIS B 1180	~~(2014) Hexagon Head Bolts and Hexagon Head Screws~~(2014) Hexagon Head Bolts and Hexagon Screws
JIS B 1181	(2014) Hexagon Nuts and Hexagon Thin Nuts
JIS B 1256	(2008) Plain Washers
JIS G 3101	~~(2017) Rolled Steels for General Structure (Amendment 1)~~(2020) Rolled Steels for General Structure
JIS G 3444	(20~~16~~21) Carbon Steel Tubes for General Structure
JIS G 3532	(2011) Low Carbon Steel Wires
JIS G 3533	(2008) Barbed Wires
JIS G 3552	(2011) Chain Link Wire Netting
JIS H 8641	(20~~07~~21) Hot Dip Galvanized Coatings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Assembly; G

Location of Gate, Corner, End, and Pull Posts; G

Gate Assembly; G

Gate Hardware and Accessories; G

Erection/Installation Drawings; G

SD-03 Product Data

Fence Assembly; G

Gate Assembly; G

Gate Hardware and Accessories; G

Zinc Coating; G

PVC Coating; G

Fabric; G

Stretcher Bars; G

Barbed Wire; G

~~Precast Concrete Posts; G~~

Padlocks; G

Turnbuckles; G

Truss Rod; G

Tension Wires; G

Wire Ties; G

Concrete; G

SD-07 Certificates

Certificates of Compliance

SD-08 Manufacturer's Instructions

Fence Assembly

Gate Assembly

Hardware Assembly

Accessories

SD-11 Closeout Submittals

Recycled Material Content

1.3 QUALITY CONTROL

1.3.1 Certificates of Compliance

Submit certificates of compliance in accordance with the applicable reference standards and descriptions of this section for the following:

a. Zinc coating

b. PVC coating

c. Fabric

d. Stretcher bars

e. Gate hardware and accessories

f. Concrete

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials off the ground to provide protection against oxidation caused by ground contact.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide fencing and gate materials, as specified.
Submit reports of listing chain-link fencing and accessories regarding weight in grams for zinc coating, thickness of PVC coating.

Submit manufacturer's catalog data for complete fence assembly, gate assembly, hardware assembly and accessories. Provide chain link fence on concrete posts for Okinawa Area and chain link fence on pipe posts for all other regions in Japan.

2.2 COMPONENTS FOR CHAIN LINK FENCE AND GATE

2.2.1 Fabric

Fabric for fence and gates shall be galvanized steel chain link wire netting conforming to JIS G 3552 No. 8, 50 mm standard wire mesh.

2.2.2 Posts , Rails and Braces

~~2.2.2.1 Fence Posts for Okinawa Area~~

~~Shall be the commercial precast concrete posts of the type and size as indicated.~~

2.2.2.1 Fence Posts for Other Regions in Japan

Shall be carbon steel pipe conforming to JIS G 3444, size and thickness as indicated, and galvanized.

2.2.2.2 Fence Rails and Braces, and Pipe Connectors

Shall be carbon steel pipe conforming to JIS G 3444, size and thickness as indicated, and galvanized.

2.2.3 Structural Steel Bars, Plates and Shapes

Shall be JIS G 3101, SS 400, galvanized.

2.2.4 Barbed Wire

Shall be JIS G 3533, #12-gage, two-wire strand, a pitch of 102 mm, 4-pointed barbs and 2 to 7 twists.

2.2.5 Fastening Accessories

Shall be galvanized steel and shall be manufacturer's standard product.

2.2.6 Stretcher Bars

Provide bars that have one-piece lengths equal to the full height of the fabric with a minimum cross section of 5 by 20 millimeter or of size as recommended by the fence manufacturer and conforming to JIS G 3101.

2.2.7 Stretcher Bar Bands

Provide bar bands for securing stretcher bars to posts that are steel, wrought iron, or malleable iron spaced not over 381 millimeter on center or of size as recommended by the fence manufacturer. Bands may also be used in conjunction with special fittings for securing rails to posts. Provide bands with projecting edges chamfered or eased.

2.2.8 Post Tops

Provide tops that are steel, wrought iron, or malleable iron designed as a weathertight closure cap. Provide one cap for each post, unless equal protection is provided by a combination post-cap and wire supporting arm. Provide caps with an opening to permit through passage of the top rail.

2.2.9 Gate Posts

Shall be carbon steel pipe conforming to JIS G 3444, size and thickness as indicated, and galvanized.

2.2.10 Gates

2.2.10.1 Gate Assembly

Shape and size of the gate frame shall be as indicated. Framing and bracing members shall be of steel pipe specified herein.

2.2.10.2 Gate Leaves

For gate leaves, more than 2.44 m wide, provide intermediate members as necessary to provide rigid construction, free from sag or twist. Gate leaves less than 2.44 m wide shall have truss rods or intermediate braces. Provide intermediate braces on all gate frames with an electro-mechanical lock. Attach fabric to the gate frame by method standard with the manufacturer except that welding will not be permitted.

2.2.10.3 Gate Hardware And Accessories

Submit manufacturer's catalog data. Furnish and install latches, hinges, stops, keepers, rollers, and other hardware items as required for the operation of the gate and shall be zinc-coated steel having weight of zinc-coating not less than HDZ 40B, JIS H 8641. Gate latches shall be fork or plunger bar type. Arrange latches for padlocking so that the padlock will be accessible from both sides of the gate. Provide stops for holding the gates in the open position. For high security applications, each end member of gate frames shall be extended sufficiently above the top member to carry three strands of barbed wire in horizontal alignment with barbed wire strands on the fence.

2.2.10.4 Turnbuckles for Gates

JIS A 5540, galvanized, size as indicated.

2.2.10.5 Truss Rod

Shall be JIS G 3101, 10 mm diameter, welded to fence post where indicated.

2.2.10.6 Tension Wires

Provide galvanized, coiled spring wire conforming to JIS G 3532 SWM-G3. Provide Zinc coating that weighs not less than 370 gram per square meter.

2.2.11 Wire Ties

Provide 2.3 millimeter galvanized steel wire conforming to JIS G 3532 for tying fabric to line posts,spaced 300 millimeter on center. For tying fabric to rails and braces, space wire ties 600 millimeter on center. For tying fabric to tension wire, space 2.7 millimeter hog rings 600 millimeter on center.

Manufacturer's standard procedure will be accepted if of equal strength and durability.

Provide wire ties constructed of the same material as the fencing fabric. Provide accessories with polyvinyl (PVC) coatings when PVC-coated fence fabric is required.

2.2.12 Bolts, Nuts and Washers

Steel conforming to JIS B 1180, JIS B 1181, JIS B 1256 respectively.

2.2.13 Padlocks

Provide ~~Japanese manufactured~~ padlocks with keys and chain in conformance with the appropriate specification of the installation agency having jurisdiction.

2.3 OTHER MATERIALS

2.3.1 Zinc Coating

Hot dip galvanization shall be in conformance with JIS H 8641.

2.3.2 Cast-in-Place Concrete for Fence and Gate Posts

Shall be of size as indicated. Concrete in conformance with JIS A 5308 and shall be a minimum compressive strength of 18 MPa at 28 days.

2.3.3 Grout

Provide grout of proportions one part portland cement to three parts clean, well-graded sand and a minimum amount of water to produce a workable mix.

2.4 GROUNDING

Ground the chain link fence and gates as indicated on drawings.

PART 3 EXECUTION

Submit manufacturer's erection/installation drawings and instructions that detail proper assembly and materials in the design for fence, gate, hardware and accessories.

3.1 PREPARATION

Ensure final grading and established elevations are complete prior to commencing fence installation.

3.1.1 Clearing and Grading

Clear fence line of trees, brush, and other obstacles to install fencing . Establish a graded, compacted fence line prior to fencing installation.

3.2 INSTALLATION

3.2.1 Security

Install new chain link fencing, remove existing fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer and cognizant Security Officer.

3.2.2 Fence Installation

Install fence on prepared surfaces to line and grade indicated. Install fence in accordance with fence manufacturer's written installation instructions except as modified herein.

3.2.2.1 Post Spacing

Provide line posts spaced equidistantly apart, not exceeding 3.048 m on center. Provide gate posts spaced as necessary for size of gate openings. Do not exceed 152.4 m on straight runs between braced posts. Provide corner or pull posts, with bracing in both directions, for changes in direction of 0.26 rad or more, or for abrupt changes in grade. Submit drawings showing location of gate, corner, end, and pull posts.

3.2.2.2 Top and Bottom Tension Wire

Install top and bottom tension wires before installing chain-link fabric, and pull wires taut. Place top and bottom tension wires within 203 mm of respective fabric line.

3.2.3 Excavation

Provide excavations for post footings which are drilled holes in virgin or compacted soil, of minimum sizes as indicated.

Space footings for line posts 3048 millimeter on center maximum and at closer intervals when indicated, with bottoms of the holes approximately 75 millimeter below the bottoms of the posts. Set bottom of each post not less than 915 millimeter below finished grade when in firm, undisturbed soil. Set posts deeper, as required, in soft and problem soils and for heavy, lateral loads.

Remove excavated soil from Government property.

When solid rock is encountered near the surface, drill into the rock at least 305 millimeter for line posts and at least 457 millimeter for end, pull, corner, and gate posts. Drill holes at least 25.4 millimeter greater in diameter than the largest dimension of the placed post.

If solid rock is below the soil overburden, drill to the full depth required except that penetration into rock need not exceed the minimum depths specified above.

3.2.4 Setting Posts

Remove loose and foreign materials from holes and moisten the soil prior to placing concrete.

Provide tops of footings that are trowel finished and sloped or domed to shed water away from posts. Set hold-open devices, sleeves, and other accessories in concrete.

Keep exposed concrete moist for at least 7 calendar days after placement or cured with a membrane curing material, as approved.

Grout all posts set into sleeved holes in concrete with an approved grouting material.

Maintain vertical alignment of posts in concrete construction until concrete has set.

3.2.4.1 Earth and Bedrock

Provide concrete bases of dimensions indicated on the manufactures installation drawings, except in bedrock. Compact concrete to eliminate voids, and finish to a dome shape. In bedrock, set posts with a minimum of 25.4 mm of grout around each post. Work grout into hole to eliminate voids, and finish to a dome shape.

3.2.4.2 Concrete Slabs and Walls

Set posts into zinc-coated sleeves, set in concrete slab or wall, to a minimum depth of 305 mm. Fill sleeve joint with lead, nonshrink grout, or other approved material. Set posts for support of removable fence sections into sleeves that provide a tight sliding joint and hold posts aligned and plumb without use of lead or setting material.

3.2.4.3 Bracing

Brace gate, corner, end, and pull posts to nearest post with a horizontal brace used as a compression member, placed at least 305 mm below top of fence, and two diagonal tension rods.

a. Tolerances

Provide posts that are straight and plumb within a vertical tolerance of 6.35 millimeter after the fabric has been stretched. Provide fencing and gates that are true to line with no more than 12.7 millimeter deviation from the established centerline between line posts. Repair defects as directed.

3.2.5 Concrete Strength

Provide concrete that has attained at least 75 percent of its minimum 28-day compressive strength, but in no case sooner than 7 calendar days after placement, before rails, tension wire, or fabric are installed. Do not stretch fabric and wires or hang gates until the concrete has attained its full design strength.

Take samples and test concrete to determine strength as specified.

3.2.6 Supporting Arms (only for Security Chain Link Fence and Gates)

It is the Contractor's option to choose the following type of supporting arms; type (1) or type (2). Type (1): Supporting arm shall be top part of fence and gate post, which shall be one length, seamless pipe through post to arm. Type (2): If the selected manufacturer's standard product uses the sectional type supporting arms, that shall be designed to accomodate the top rail. Install supporting arms as recommended by the manufacturer. In addition to manufacturer's standard connections, securely anchor supporting arms to posts to prevent easy removal with hand tools.

3.2.7 Top Rails

Provide top rails that run continuously through post caps or extension arms, bending to radius for curved runs. Provide expansion couplings as recommended by the fencing manufacturer.

3.2.8 Brace Assembly

Provide bracing assemblies at end and gate posts and at both sides of corner and pull posts, with the horizontal brace located at midheight of the fabric.

Install brace assemblies so posts are plumb when the diagonal rod is under proper tension.

Provide two complete brace assemblies at corner and pull posts where required for stiffness and as indicated.

3.2.9 Tension Wire Installation

Install tension wire by weaving them through the fabric and tying them to each post with not less than 3.9 millimeter galvanized wire or by securing the wire to the fabric with 3.5 millimeter ties or clips spaced 610 millimeter on center.

3.2.10 Fabric Installation

Provide fabric in single lengths between stretch bars with bottom barbs placed approximately 38 millimeter above the ground line. Pull fabric taut and tied to posts, rails, and tension wire with wire ties and bands.

Install fabric on the security side of fence, unless otherwise directed.

Ensure fabric remains under tension after the pulling force is released.

3.2.11 Stretcher Bar Installation

Thread stretcher bars through or clamped to fabric 102 millimeter on center and secured to posts with metal bands spaced 381 millimeter on center.

3.2.12 Barbed Wire (only for Security Chain Link Fence and Gate)

Install barbed wire on supporting arms above fence posts. Extend each end member of gate frames sufficiently above top member to carry three strands of barbed wire in horizontal alignment with barbed wire strands on the fence. Pull each strand taut and securely fasten each strand to each supporting arm and extend member. The method of securing wires shall be as follows: twist tie barbed wire to arm using wire which has been looped through a hole in the supporting arm. Other methods of securing barbed wire are acceptable provided they are equally secure, and are approved in advance by the Contracting Officer.

3.2.13 Gate Installation

Install gates plumb, level, and secure, with full opening without interference. Install ground set items in concrete for anchorage as recommended by the fence manufacturer. Adjust hardware for smooth operation and lubricated where necessary.

3.2.14 Tie Wires

Provide tie wires that are U-shaped to the pipe diameters to which attached. Twist ends of tie wires not less than two full turns and bent so as not to present a hazard.

3.2.15 Fasteners

Install nuts for tension bands and hardware on the side of the fence opposite the fabric side. Peen ends of bolts to prevent removal of nuts.

3.2.16 Zinc-Coating Repair

Clean and repair galvanized surfaces damaged by welding or abrasion, and cut ends of fabric, or other cut sections with specified galvanizing repair material applied in strict conformance with the manufacturer's printed instructions.

3.2.17 Accessories Installation

3.2.17.1 Post Caps

Install post caps as recommended by the manufacturer.

3.2.17.2 Padlocks

Provide padlocks for gate openings and provide chains that are securely attached to gate or gate posts. Provide padlocks keyed alike, and provide two keys for each padlock.

3.2.18 Grounding

Ground all fences crossed by overhead power lines in excess of 600 volts, and all electrical equipment attached to the fence. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 15 m of the fence, and where the fence alignment changes more than 15 degrees. Grounding locations can not exceed 198 m. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by power lines of 600 volts or more at or near the point of crossing and at distances not exceeding 45 m on each side of crossing. Provide ground conductor consisting of No. 6 AWG solid copper wire. Provide copper-clad steel rod grounding electrodes 19 mm by 3.05 m long. Drive electrodes into the earth so that the top of the electrode is at least 152 mm below the grade. Where driving is impracticable, bury electrodes a minimum of 305 mm deep and radially from the fence, with top of the electrode not less than 610 mm or more than 2.4 m from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Total resistance of the fence to ground cannot exceed 25 ohms.

3.3 CLOSEOUT ACTIVITIES

Remove waste fencing materials and other debris from the work site.

Submit manufacturer's data indicating percentage of recycled material content in protective fence materials, including chain link fence, fabric, and gates to verify affirmative procurement compliance.

3.4 RESTORATION

The Contractor shall restore all damaged and disturbed areas caused by this project work to match existing condition as directed.

3.4.1 Surface Course Restoration

Concrete and/or asphalt concrete surface course which has been damaged and disturbed to accomplish this project shall be restored to match existing condition with new concrete and/or new asphalt concrete, including crushed stone base course.

3.4.2 Turf Restoration

Sodded area which has been damaged and disturbed to accomplish this project shall be restored to match existing condition by reusing existing sod or new sod. Water thoroughly immediately after replanting.

-- End of Section --

SECTION 32 92 23

SODDING
04/06, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~ASTM INTERNATIONAL (ASTM)~~

~~ASTM C602~~	~~(2020) Agricultural Liming Materials~~
~~ASTM D4427~~	~~(2018) Standard Classification of Peat Samples by Laboratory Testing~~
~~ASTM D4972~~	~~(2018) Standard Test Methods for pH of Soils~~

~~TURFGRASS PRODUCERS INTERNATIONAL (TPI)~~

~~TPI GSS~~	~~(1995) Guideline Specifications to Turfgrass Sodding~~

~~U.S. DEPARTMENT OF AGRICULTURE (USDA)~~

~~DOA SSIR 42~~	~~(1996) Soil Survey Investigation Report No. 42, Soil Survey Laboratory Methods Manual, Version 3.0~~

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT-PCSS	(2011) Planting Construction Standard Specifications

1.2 DEFINITIONS

1.2.1 Stand of Turf

100 percent ground cover of the established species.

~~1.3 RELATED REQUIREMENTS~~

~~[Section 31 00 00 EARTHWORK], [Section 32 84 24 IRRIGATION SPRINKLER SYSTEMS], [Section 32 96 00 TRANSPLANTING EXTERIOR PLANTS], [Section 32 92 19 SEEDING], [Section 32 92 26 SPRIGGING], [Section 32 93 00 EXTERIOR PLANTS], and Section 32 05 33 LANDSCAPE ESTABLISHMENT applies to this section for pesticide use and plant establishment requirements, with additions and modifications herein.~~

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are

~~[for Contractor Quality Control approval.][~~for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.~~]~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Fertilizer

Include physical characteristics, and recommendations.

SD-06 Test Reports

Topsoil composition tests (reports and recommendations).

SD-07 Certificates

~~[Nursery][~~Sod farm ~~]~~certification for sods. Indicate type of sod in accordance with MLIT-PCSS~~TPI GSS~~.

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery

1.4.1.1 Sod Protection

Protect from drying out and from contamination during delivery, on-site storage, and handling.

1.4.1.2 ~~[~~Fertilizer~~]~~, ~~[~~Gypsum~~]~~, ~~[~~Sulfur~~]~~, ~~[~~Iron~~]~~ ~~[~~and~~]~~ ~~[~~Lime~~]~~ Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to local~~state and federal~~ laws. Instead of containers, ~~[~~fertilizer~~]~~, ~~[~~gypsum~~]~~, ~~[~~sulphur~~]~~, ~~[~~iron~~]~~ ~~[~~and~~]~~ ~~[~~lime~~]~~ may be furnished in bulk with certificate indicating the above information.

1.4.2 Storage

1.4.2.1 Sod Storage

Lightly sprinkle with water, cover with moist burlap, straw, or other approved covering; and protect from exposure to wind and direct sunlight until planted. Provide covering that will allow air to circulate so that internal heat will not develop. Do not store sod longer than 24 hours. Do not store directly on concrete or bituminous surfaces.

1.4.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

1.4.2.3 Handling

Do not drop or dump materials from vehicles.

1.5 TIME RESTRICTIONS AND PLANTING CONDITIONS

1.5.1 Restrictions

Do not plant when the ground is ~~[~~frozen,~~]~~ ~~[~~snow covered,~~]~~ muddy, or when air temperature exceeds ~~[32]~~ ~~[_____]~~ degrees Celsius.

1.6 TIME LIMITATIONS

1.6.1 Sod

Place sod a maximum of thirty six hours after initial harvesting~~, in accordance with TPI GSS as modified herein~~.

PART 2 PRODUCTS

Local manufactured SODDING will be accepted if the product meets the inherent properties as indicated in this specification section and drawings, and conforms to applicable local Standard. Inherent properties include but not limited to: material, type, composition and accessories.

2.1 SODS

2.1.1 Classification

Shall be good quality "Kouraishiba" (Zoysia tennuifolia) sod~~Nursery grown, certified as classified in the TPI GSS~~. Machine cut sod at a uniform thickness of 19 mm within a tolerance of 6 mm, excluding top growth and thatch. Each individual sod piece shall be strong enough to support its own weight when lifted by the ends. Broken pads, irregularly shaped pieces, and torn or uneven ends will be rejected.~~[Wood pegs and wire staples for anchorage shall be as recommended by sod supplier.]~~

2.1.2 Purity

Sod species shall be genetically pure, free of weeds, pests, and disease.

2.1.3 Planting Dates

Lay sod from March~~[_____]~~ to June~~[_____]~~ for warm season spring planting and from August~~[_____]~~ to October~~[_____]~~ for cool season fall planting.

2.1.4 Composition

2.1.4.1 Proportion

See plans.~~Proportion grass species as follows.~~

~~Botanical Name~~	~~Common Name~~	~~Percent~~
~~[_____]~~	~~[_____]~~	~~[_____]~~
~~[_____]~~	~~[_____]~~	~~[_____]~~

~~[2.1.4.2 Sod Farm Overseeding~~

~~At the sod farm provide sod with overseeding of [annual rye grass seed][_____][type recommended by seed producer].~~

~~][2.2 WILDFLOWER SOD~~

~~2.2.1 Classification~~

~~[Certified,] [Field grown] wildflower sod, machine cut at a uniform thickness of [25] [_____] mm within a tolerance of 6 mm, excluding top growth. Top growth shall be a maximum height of [75] [_____] mm. Each individual wildflower sod piece shall be strong enough to support its own weight when lifted by the ends. Broken pads, irregular shaped pieces, and torn or uneven ends will be rejected. [Wood pegs and wire staples for anchorage on slope conditions, three to one or greater, shall be used as recommended by wildflower sod supplier.]~~

~~2.2.2 Composition~~

~~Proportion wildflower species as follows:~~

~~Botanical Name~~	~~Common Name~~	~~Percent~~
~~[_____]~~	~~[_____]~~	~~[_____]~~
~~[_____]~~	~~[_____]~~	~~[_____]~~

~~]~~2.2 TOPSOIL

2.2.1 On-Site Topsoil

Surface soil stripped and stockpiled on site and modified as necessary to meet the requirements specified for topsoil in paragraph entitled "Composition." When available topsoil shall be existing surface soil stripped and stockpiled on-site in accordance with Section ~~[~~31 00 00 EARTHWORK~~][31 23 00.00 20 EXCAVATION AND FILL]~~.

2.2.2 Off-Site Topsoil

Conform to requirements specified in paragraph entitled "Composition." Additional topsoil shall be ~~[~~furnished by the Contractor~~] [obtained from topsoil borrow areas indicated]~~.

2.2.3 Composition

Containing from 5 to 10 percent organic matter as determined by the topsoil composition tests of the Organic Carbon, 6A, Chemical Analysis Method described in MLIT-PCSS~~DOA SSIR 42~~. Maximum particle size, 19 mm, with maximum 3 percent retained on 6 mm screen. The pH shall be tested in accordance with MLIT-PCSS~~ASTM D4972~~. Topsoil shall be free of sticks, stones, roots, and other debris and objectionable materials. Other components shall conform to the following limits: amend topsoil to meet topsoil report recommendations.

~~Silt~~	~~[25-50][7 to 17][_____] percent~~
~~Clay~~	~~[10-30][4 to 12][_____] percent~~

~~Sand~~	~~[20-35][70 to 82][_____] percent~~
~~pH~~	~~[5.5 to 7.0][_____]~~
~~Soluble Salts~~	~~[600] [_____] ppm maximum~~

2.3 SOIL CONDITIONERS

Add conditioners to topsoil as required to bring into compliance with "composition" standard for topsoil as specified herein.

2.3.1 Lime

~~Commercial grade [hydrate] [or] [burnt] limestone containing a calcium carbonate equivalent (C.C.E.) as specified in ASTM C602 of not less than [_____] percent.~~ Lime to meet topsoil test report recommendations.

2.3.2 Aluminum Sulfate

Commercial grade.

2.3.3 Sulfur

100 percent elemental

2.3.4 Iron

100 percent elemental

2.3.5 Peat

Natural product of ~~[~~peat moss~~]~~ derived from a freshwater site and conforming to ~~[ASTM D4427~~MLIT-PCSS~~] [as modified herein]~~. Shred and granulate peat to pass a 12.5 mm mesh screen and condition in storage pile for minimum 6 months after excavation.

2.3.6 Sand

Clean and free of materials harmful to plants.

2.3.7 Perlite

Horticultural grade.

2.3.8 Composted Derivatives

Ground bark, nitrolized sawdust, humus or other green wood waste material free of stones, sticks, and soil stabilized with nitrogen and having the following properties:

2.3.8.1 Particle Size

Minimum percent by weight passing:

4.75 mm screen	95
2.36 mm screen	80

2.3.8.2 Nitrogen Content

Minimum percent based on dry weight:

Fir Sawdust	0.7
Fir or Pine Bark	1.0

2.3.9 Gypsum

Coarsely ground gypsum comprised of calcium sulfate dihydrate 91 percent, calcium 22 percent, sulfur 17 percent; minimum 96 percent passing through 850 micrometers, 100 percent passing thru 970 micrometers screen.

2.3.10 Calcined Clay

Calcined clay shall be granular particles produced from montmorillonite clay calcined to a minimum temperature of 650 degrees C. Gradation: A minimum 90 percent shall pass a 2.36 mm sieve; a minimum 99 percent shall be retained on a 0.250 mm sieve; and a maximum 2 percent shall pass a 0.150 mm sieve. Bulk density: A maximum 640 kilogram per cubic meter.

2.4 FERTILIZER

2.4.1 Granular Fertilizer

Per topsoil test recommendations.~~[Organic][synthetic], granular controlled release fertilizer containing the following minimum percentages, by weight, of plant food nutrients:~~

~~[_____] percent available nitrogen~~
~~[_____] percent available phosphorus~~
~~[_____] percent available potassium~~
~~[_____] percent sulfur~~
~~[[_____] percent iron]~~

2.5 WATER

Source of water shall be approved by Contracting Officer and of suitable quality for irrigation containing no element toxic to plant life.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Extent Of Work

Provide soil preparation (including soil conditioners), fertilizing, and sodding of all newly graded finished earth surfaces, unless indicated otherwise, and at all areas inside or outside the limits of construction that are disturbed by the Contractor's operations.

3.1.2 Soil Preparation

Provide 102 mm of ~~[off-site topsoil][~~on-site topsoil~~]~~ to meet indicated finish grade. After areas have been brought to indicated finish grade, incorporate ~~[~~fertilizer~~]~~, ~~[~~pH adjusters~~]~~, ~~[~~soil conditioners~~]~~ into soil a minimum depth of ~~[~~100~~]~~ ~~[_____]~~ mm by disking, harrowing, tilling or other method approved by the Contracting Officer. Remove debris and stones larger than 19 mm in any dimension remaining on the surface after finish grading. Correct irregularities in finish surfaces to eliminate depressions. Protect finished topsoil areas from damage by vehicular or pedestrian traffic.

~~[~~3.1.2.1 Soil Conditioner Application Rates

Per topsoil test recommendations.~~Apply soil conditioners at rates as determined by laboratory soil analysis of the soils at the job site. For bidding purposes only apply at rates for the following:~~

~~[Lime [[_____] kg per square meter] [[_____] kg per 100 square meters.]~~

~~][Sulfur [[_____] kg per square meter] [[_____] kg per 100 square meters]~~

~~][Iron [[_____] kg per square meter] [[_____] kg per 100 square meters.]~~

~~][Aluminum Sulfate [[_____] kg per square meter] [[_____] kg per 100 square meters.]~~

~~][Peat [[_____] cubic meters per square meter] [[_____] cubic meters per 100 square meters.]~~

~~][Sand [[_____] cubic meters per square meter] [[_____] cubic meters per 100 square meters.]~~

~~][Perlite [[_____] cubic meters per square meter] [[_____] cubic meters per 100 square meters.]~~

~~][Compost Derivatives [[_____] cubic meters per square meter] [[_____] cubic meters per 100 square meters.]~~

~~][Calcined Clay [[_____] cubic meters per square meter] [[_____] cubic meters per 100 square meters.]~~

~~][Gypsum [[_____] cubic meters per square meter] [[_____] cubic meters per 100 square meters.]~~

~~]][~~3.1.2.2 Fertilizer Application Rates

Per topsoil test recommendations.~~Apply fertilizer at rates as determined by laboratory soil analysis of the soils at the job site. For bidding purposes only apply at rates for the following:~~

~~[Organic Granular Fertilizer [[_____] kg per square meter] [[_____] kg per 100 square meters.]~~

~~][Synthetic Granular Fertilizer [[_____] kg per square meter] [[_____] kg per 100 square meters]~~

~~]]~~3.2 SODDING

3.2.1 Finished Grade and Topsoil

Prior to the commencement of the sodding operation, the Contractor shall verify that finished grades are as indicated on drawings; the placing of topsoil, smooth grading, and compaction requirements have been completed in accordance with Section ~~[~~31 00 00 EARTHWORK~~][31 23 00.00 20 EXCAVATION AND FILL]~~.

The prepared surface shall be a maximum 25 mm below the adjoining grade of any surfaced area. New surfaces shall be blended to existing areas. The prepared surface shall be completed with a light raking to remove from the surface debris and stones over a minimum 16 mm in any dimension.

3.2.2 Placing

Place sod a maximum of 36 hours after initial harvesting, in accordance with MLIT-PCSS~~TPI GSS~~ as modified herein.

3.2.3 Sodding Slopes and Ditches

For slopes 2:1 and greater, lay sod with long edge perpendicular to the contour. For V-ditches and flat bottomed ditches, lay sod with long edge perpendicular to flow of water. ~~[Anchor each piece of sod with wood pegs or wire staples maximum 600 mm on center.] [~~On slope areas, start sodding at bottom of the slope.~~]~~

3.2.4 Finishing

After completing sodding, blend edges of sodded area smoothly into surrounding area. Air pockets shall be eliminated and a true and even surface shall be provided. Frayed edges shall be trimmed and holes and missing corners shall be patched with sod.

3.2.5 Rolling

Immediately after sodding, firm entire area except for slopes in excess of 3 to 1 with a roller not exceeding ~~[134] [_____]~~ kg per m for each foot of roller width.

3.2.6 Watering

Start watering areas sodded as required by daily temperature and wind conditions. Apply water at a rate sufficient to ensure thorough wetting of soil to minimum depth of ~~[150] [_____]~~ mm. Run-off, puddling, and wilting shall be prevented. Unless otherwise directed, watering trucks shall not be driven over turf areas. Watering of other adjacent areas or plant material shall be prevented.

3.3 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use.

~~[~~3.4 RENOVATION OF EXISTING TURF AREA

Per topsoil test recommendations.

~~[3.4.1 Aeration~~

~~Upon completion of weed eradication operations and Contracting Officer's approval to proceed, aerate turf areas indicated , by approved device. Core, by pulling soil plugs, to a minimum depth of [_____] mm.[Leave all soil plugs, that are produced, in the turf area.][Remove all debris generated during this operation off site.][After aeration operations are complete, topdress entire area [6.35 mm] [12.70 mm] depth with the following mixture:~~

~~[[_____] percent sand~~

~~][[_____] percent humus~~

~~][[_____] percent gypsum~~

~~][[_____] percent organic fertilizer~~

~~][[_____] percent synthetic fertilizer~~

~~] Blend all parts of topdressing mixture to a uniform consistency throughout.]Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean all soil plugs off of other paving when work is complete.~~

~~][3.4.2 Vertical Mowing~~

~~Upon completion of aerating operation and Contracting Officer's approval to proceed, vertical mow turf areas indicated, by approved device, to a depth of [6 mm] [13 mm] above existing soil level, to reduce thatch build-up, grain, and surface compaction. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work is complete. Remove all debris generated during this operation off site.~~

~~][3.4.3 Dethatching~~

~~Upon completion of aerating operation and Contracting Officer's approval to proceed, dethatch turf areas indicated, by approved device, to a depth of [6 mm] [13 mm] below existing soil level, to reduce thatch build-up, grain, and surface compaction. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work is complete. Remove all debris generated during this operation off site.~~

~~]]~~3.5 RESTORATION

Restore to original condition existing turf areas which have been damaged during turf installation operations. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

-- End of Section --

SECTION 33 11 00

WATER UTILITY DISTRIBUTION PIPING
02/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 24	~~(2016; ERTA 2016) Standard for the Installation of Private Fire Service Mains and Their Appurtenances~~(2019; TIA 19-1) Standard for the Installation of Private Fire Service Mains and Their Appurtenances

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-600-01	~~(2016; with Change 1) Fire Protection Engineering for Facilities~~(2016; with Change 6, 2021) Fire Protection Engineering for Facilities

UNDERWRITERS LABORATORIES (UL)

UL 246	(2011; Reprint Dec 2018) UL Standard for Safety Hydrants for Fire-Protection Service
UL 262	(2004; Reprint Oct 2011) Gate Valves for Fire-Protection Service

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5308	(2014) Ready-Mixed Concrete
JIS A 5314	(2014) Mortar Lining for Ductile Iron Pipes
JIS B 1171	(2015) Cup Head Square Neck Bolts (Amendment 1)
JIS B 1180	~~(2014) Hexagon Head Bolts and Hexagon Head Screws~~(2014) Hexagon Head Bolts and Hexagon Screws
JIS B 1181	(2014) Hexagon Nuts and Hexagon Thin Nuts
JIS B 2011	(2013) Bronze, Gate, Globe, Angle, and Check Valves (Amendment 2)
JIS B 2031	(2015) Gray Cast Iron Valves (Amendment 1)
JIS B 2404	(2018) Dimensions of Gaskets for Use with

	Pipe Flanges
JIS B 7552	(2011) Procedures for Calibration and Testing for Liquid Flowmeter
JIS B 8570-1	(2013) Meters for Cold Water and Hot Water - Part 1: General Specifications
JIS G 5526	(2014) Ductile Iron Pipes
JIS G 5527	(2014) Ductile Iron Fittings
JIS K 1102	(2000) Liquid Chlorine for Industrial Use - Determination of the Chlorine Content.
JIS S 3200-1	(1997) Equipment for Water Supply Use - Test methods of Hydrostatic Pressure

JAPAN WATER WORKS ASSOCIATION (JWWA)

JWWA B 120	(2017) Soft Seal Gate Valve for Water Supply
JWWA B 122	(2013) Ductile Cast Iron Gate Valve for Water Supply
JWWA G 113	(2010) Ductile Iron Pipes for Water Works
JWWA K 156	(2015) Rubber Material for Water Supply Facilities
JWWA K 158	(2017) Polyethylene Sleeve for Ductile Cast Iron Pipe for Water Supply

~~JAPAN DUCTILE IRON PIPE ASSOCIATION (JDPA)~~JAPAN DUCTILE IRON PIPE ASSOCIATION (JDPA)

JDPA T 01	(2017) Ductile Iron Pipe Laying Standard Manual
JDPA Z 2010	(2009) Synthetic Resin Coating for Ductile Iron Pipes and Fittings

JAPAN CAST IRON COVER & WASTE FITTING ASSOCIATION (JCW)

JCW-104	(2012) Valve Box

1.2 DEFINITIONS

1.2.1 Water Transmission Mains

Water transmission mains include water piping having diameters greater than 350 mm, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.2 Water Mains

Water mains include water piping having diameters 100 through 350 mm, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.3 Water Service Lines

Water service lines include water piping from a water main to a building service at a point approximately 1.5 m from building or the point indicated on the drawings, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.2.4 Additional Definitions

For additional definitions refer to the definitions in the applicable referenced standard.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Fire Hydrants;

Meters;

~~Backflow Preventer;~~

Disinfection Procedures; G

SD-06 Test Reports

~~Backflow Preventer Tests; G~~

Bacteriological Samples; G

Hydrostatic Sewer Test; G

Leakage Test; G

Hydrostatic Test; G

SD-07 Certificates

Fire Hydrants

~~Backflow Certificate~~

SD-08 Manufacturer's Instructions

Ductile Iron Piping

~~PVC Piping~~

~~PVCO Piping~~

~~Polyethylene (PE) Pipe~~

~~PVC Piping For Service Lines~~

1.4 QUALITY CONTROL

1.4.1 Regulatory Requirements

Comply with NFPA 24 for materials, installation, and testing of fire main piping and components.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery and Storage

Inspect materials delivered to site for damage. Unload and store with minimum handling and in accordance with manufacturer's instructions. Store materials on site in enclosures or under protective covering. Store plastic piping, jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes, fittings, valves, fire hydrants, and other accessories free of dirt and debris.

1.5.2 Handling

Handle pipe, fittings, valves, fire hydrants, and other accessories in accordance with manufacturer's instructions and in a manner to ensure delivery to the trench in sound undamaged condition. Avoid injury to coatings and linings on pipe and fittings; make repairs if coatings or linings are damaged. Do not place other material, hooks, or pipe inside a pipe or fitting after the coating has been applied. Inspect the pipe for defects before installation. Carry, do not drag pipe to the trench. Use of pinch bars and tongs for aligning or turning pipe will be permitted only on the bare ends of the pipe. Clean the interior of pipe and accessories of foreign matter before being lowered into the trench and keep them clean during laying operations by plugging. Replace defective material without additional expense to the Government. Store rubber gaskets, not immediately installed, under cover or out of direct sunlight.

PART 2 PRODUCTS

2.1 MATERIALS

Provide all materials in accordance with JIS, JWWA and other approved Japanese standards ~~commercial products~~ as indicated herein. Provide valves and fittings with pressure ratings equivalent to the pressure ratings of the pipe. Provide ~~Japanese manufactured~~ materials and components for fire protection service meeting the requirements of NFPA 24 and related UL standards.

2.1.1 Pipe, Fittings, Joints And Couplings

Submit manufacturer's standard drawings or catalog cuts, except submit both drawings and cuts for push-on ~~[~~and rubber-gasketed bell-and-spigot~~]~~

joints. Include information concerning gaskets with submittal for joints and couplings.

2.1.1.1 Ductile-Iron Piping

2.1.1.1.1 Pipe and Fittings

Pipe, except flanged pipe, JIS G 5526 or JWWA G 113~~, Pressure Class [_____] Thickness Class [_____]~~. Flanged pipe, JIS G 5527. Fittings, JIS G 5527 or JWWA G 113; fittings with push-on joint ends are to meet the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design. Provide fittings with pressure ratings equivalent to that of the pipe. Provide compatible pipe ends and fittings for the specified joints. Provide cement-mortar lining, JIS A 5314 , twice the standard thickness on pipe and fittings.

2.1.1.1.2 Joints and Jointing Material

Provide push-on joints or mechanical joints for pipe and fittings unless otherwise indicated. Provide mechanical joints where indicated. Provide flanged joints where indicated. Provide mechanically coupled type joints using a sleeve-type mechanical coupling where indicated. Provide insulating joints where indicated. Sleeve-type mechanical couplings in lieu of push-on joints are acceptable, subject to the limitations specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS.

a. Push-On Joints: Shape of pipe ends and fitting ends, gaskets, and lubricant for joint assembly as recommended in JWWA K 156.

b. Mechanical Joints: Dimensional and material requirements for pipe ends, glands, bolts and nuts, and gaskets as recommended in JWWA K 156.

c. Flanged Joints: Bolts, nuts, and gaskets for flanged connections compatible with JIS G 5527 or JWWA G 113 joints. Provide JIS G 5527 ductile iron flanges. Provide epoxy coated steel set screw flanges. Gasket and lubricants for set screw flanges, in accordance with mechanical-joint gaskets conforming to JWWA K 156 or JIS B 2404.

d. Insulating Joints: Designed to prevent metal-to-metal contact at the joint between adjacent sections of piping. Provide flanged type joint with insulating gasket, insulating bolt sleeves, and insulating washers. Provide full face dielectric type gaskets, for JIS G 5527 fittings. Bolts and nuts, for JIS G 5527 fittings.

e. Sleeve-Type Mechanical Coupled Joints: As specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS.

2.1.1.1.3 Pipe, Joint, Valve, and Fitting Coatings

Provide epoxy resin bonded coating meeting the requirements of JDPA Z 2010. Bonded coating shall have minimum thickness of 0.1 mm for pipe and 0.08 mm for bends.

~~2.1.1.2 Plastic Piping~~

~~2.1.1.2.1 PVC and PVCO Piping~~

~~2.1.1.2.1.1 PVC Piping~~

~~JWWA K 129 or JIS K 6742 plain end or gasket bell end pipe, with a minimum Pressure Class 150 (DR27.5) with ductile iron outside diameter.~~

~~2.1.1.2.1.2 PVCO Piping~~

~~JWWA K 129 plain end or gasket bell end pipe, Pressure Class 165 PVCO pressure pipe, with ductile iron outside diameter.~~

~~2.1.1.2.1.3 Fittings for PVC and PVCO Pipe~~

~~Fittings shall be the same material as the pipe with elastomeric gaskets, in conformance with JWWA K 130 or JIS K 6742.~~

~~2.1.1.2.1.4 Joints and Jointing Material for PVC and PVCO Piping~~

~~a. Push-on joints: Use jointing material as recommended by PVC and PVCO pipe manufacturers between pipes, pipes and metal fittings, valves, and other accessories or compression-type joints/mechanical joints. Provide each joint connection with an elastomeric gasket compatible for the bell or coupling used. Gaskets for push-on joints and compression-type joints/mechanical joints for joint connections between pipe and metal fittings, valves, and other accessories, JIS K 6353 for push-on joints and mechanical joints.~~

~~b. Mechanical Joint: Use mechanically coupled joints having a sleeve-type mechanical coupling, as specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS, as an optional jointing method for plain-end PVC pipe, subject to the limitations specified for mechanically coupled joints using a sleeve-type mechanical coupling as specified for compression-type joints. Provide jointing material in conformance with PVC and PVCO pipe manufacturer's recommendation between pipe and sleeve-type mechanical couplings.~~

~~2.1.1.2.2 PVC Piping for Service Lines~~

~~2.1.1.2.2.1 Pipe and Fittings~~

~~Provide JIS K 6742 pipes and JIS K 6743 fittings.~~

~~2.1.1.2.2.2 Joints and Connections~~

~~Fittings may be joined by the solvent-cement method or threading.~~

~~2.1.1.2.2.3 Solvent Joining~~

~~Provide solvent joints per pipe manufacturer's recommendation.~~

~~2.1.1.2.3 Polyethylene (PE) Pipe~~

~~JWWA K 144 or JIS K 6762 with a minimum Pressure Class 200 (DR11) with ductile iron outside diameter.~~

~~2.1.1.2.3.1 Fittings For PE Pipe~~

~~JWWA K 145 or JWWA B 116.~~

~~2.1.1.2.3.2 Joints and Jointing Materials~~

~~Mechanical Joint: JWWA K 156 Mechanical joint adapter and gaskets for mechanical joints for joint connections between pipe and metal fittings, valves, and other accessories.~~

~~2.1.1.3 Piping Beneath Railroad Right-of-Way~~

~~Piping passing under the right-of-way of a commercial railroad is to conform to the specifications for pipelines conveying nonflammable substances. Provide ductile-iron pipe in lieu of cast-iron pipe. Ductile-iron railroad crossing casing pipe is to conform to and have strength computed in accordance with JIS G 5526.~~

2.1.2 Valves

Provide a JWWA compliant protective interior coating for all valves whose interiors are exposed to sea water or salt water, or where there is a serious corrosion problem other than galvanic corrosion for water having a pH range from 4 to 9.

2.1.2.1 Gate Valves 80 mm Size and Larger on Buried Piping

a. JWWA B 120 or JWWA B 122: nonrising stem type with double-disc gate and mechanical-joint ends or push-on joint ends compatible for the adjoining pipe

b. JWWA B 120 or JWWA B 122: nonrising stem type with mechanical-joint ends or resilient-seated gate valves 80 to 300 mm in size

c. JWWA compliant locally manufactured gate valve meeting the requirements of UL 262 for pipe protection service: inside-screw type with operating nut, double-disc or split-wedge type gate, designed for a hydraulic working pressure of 1200 kPa, and have mechanical-joint ends or push-on joint ends as appropriate for the pipe to which it is joined.

Match materials for gate valves meeting UL 262 to the reference standards specified in JWWA B 120 or JWWA B 122. Gate valves open by counterclockwise rotation of the valve stem. Stuffing boxes have 0-ring stem seals, except for those valves for which gearing is specified, in which case use conventional packing in place of 0-ring seal. Stuffing boxes are bolted and constructed so as to permit easy removal of parts for repair. Use gate valves with special ends for connection to sleeve-type mechanical coupling in lieu of mechanical-joint ends and push-on joint ends. Provide valve ends and gaskets for connection to sleeve-type mechanical couplings that conform to the requirements specified respectively for the joint or coupling. Provide JWWA B 122 _____ mm gate valves with gearing and indicator. Where an indicator post are shown, provide an indicator post flange for JWWA B 122 or locally manufactured gate valves conforming to the requirements of UL 262.Provide all valves from one manufacturer.

2.1.2.2 Gate Valves 75 mm Size and Larger in Valve Pit(s) and Aboveground Locations

a. JWWA B 120 or JWWA B 122 or JIS B 2031: nonrising stem type with double-disc gates and flanged ends

b. JWWA B 120 or JWWA B 122 or JIS B 2031: nonrising stem type with flanged ends

c. JWWA B 120 or JWWA B 122 or JIS B 2031 or JWWA compliant locally manufactured gate valve meeting the requirements of UL 262 for fire protection service: inside-screw type, with double-disc or split-wedge type gate and flanged ends, and designed for a hydraulic working pressure of 1200 kPa

Match materials for gate valves meeting UL 262 to the reference standards specified in JWWA B 120 or JWWA B 122 or JIS B 2031. Provide gate valves with handwheels that open by counterclockwise rotation of the valve stem. Bolt and construct stuffing boxes so as to permit easy removal of parts for repair. Provide all valves from one manufacturer.

~~2.1.2.3 Check Valves~~

~~Provide a JWWA compliant protective interior coating for all valves whose interiors are exposed to sea water or salt water, or where there is a serious corrosion problem other than galvanic corrosion for water having a pH range from 4 to 9.~~

~~a. JWWA B 129 or JIS B 2031: Iron or steel body and cover and flanged ends~~

~~b. JWWA compliant locally manufactured check valve meeting the requirements of UL 312 for fire protection service: Cast iron or steel body and cover, flanged ends, and designed for a minimum working pressure of 1000 kPa.~~

~~Materials for check valves meeting UL 312 are to match the reference standards specified in JWWA B 129. Provide check valves with a clear port opening. Provide all check valves from one manufacturer.~~

~~2.1.2.4 Rubber-Seated Butterfly Valves~~

~~Provide rubber-seated butterfly valves and wafer type valves that match the performance requirements of JWWA B 138. Wafer type valves not meeting laying length requirements are acceptable if supplied and installed with a spacer, providing the specified laying length. Meet all tests required by JWWA B 138. Flanged-end valves are required in a pit. Provide a union or sleeve-type coupling in the pit to permit removal. Direct-bury mechanical-end valves 80 through 250 mm in diameter. Provide a valve box, means for manual operation, and an adjacent pipe joint to facilitate valve removal. Provide valve operators that restrict closing to a rate requiring approximately 60 seconds, from fully open to fully closed.~~

~~2.1.2.5 Pressure Reducing Valves~~

~~Maintain a constant downstream pressure regardless of fluctuations in demand. Using pressure reducing valves capable of providing 1720 kPa operating pressure on the inlet side, with outlet pressure set for 340 kPa.~~

~~Provide hydraulically-operated, pilot controlled, globe or angle type valves that are capable of being actuated either by diaphragm or piston. Provide diaphragm-operated, adjustable, spring-loaded type pilot controls made of lead-free bronze with stainless steel working parts, designed to permit flow when controlling pressure exceeds the spring setting. Construct the bodies of bronze, cast iron or cast steel with lead-free bronze trim; the valve stem of stainless steel; the seat of lead-free bronze; and the valve discs and diaphragms of synthetic rubber. Provide flanged ends.~~

~~2.1.2.6 Air Release, Air/Vacuum, and Combination Air Valves~~

~~Provide JWWA B 137 air release , air vacuum and combination air valves that release air and prevent the formation of a vacuum. Provide valves with an iron body, lead-free bronze trim and stainless steel float that automatically releases air when the lines are being filled with water and admits air into the line when water is being withdrawn in excess of the inflow.~~

2.1.2.3 Water Service Valves

2.1.2.3.1 Gate Valves Smaller than 75 mm in Size on Buried Piping

Gate valves smaller than 75 mm size on Buried Piping JIS B 2011, Class 150, solid wedge, nonrising stem, with flanged or threaded end connections, a union on one side of the valve, and a handwheel operator.

2.1.2.3.2 Gate Valves Smaller Than 75 mm Size in Valve Pits

JIS B 2011, Class 150, solid wedge, inside screw, rising stem. Provide valves with flanged or threaded end connections, a union on one side of the valve, and a handwheel operator.

2.1.2.3.3 Check Valves Smaller than 50 mm in Size

Provide check valves with a minimum working pressure of 1000 kPa or as indicated with a clear waterway equal to the full nominal diameter of the valve. Valves open to permit flow when inlet pressure is greater than the discharge pressure, and close tightly to prevent return flow when discharge pressure exceeds inlet pressure. Cast the size of the valve, working pressure, manufacturer's name, initials, or trademark on the body of each valve.

Provide valves for screwed fittings, made of lead-free bronze and in conformance with JIS B 2011, Class 150, Types compatible for the application.

~~2.1.2.4 Indicator Post~~

~~Provide upright gate valve with indicator post conforming to JWWA compliant local commercial products meeting the requirements of UL 789 and NFPA 24. Construct indicator post body of cast iron, ductile iron or a combination of both, bronze operating nut, cast iron locking wrench with open and shut target window.~~

2.1.2.4 Valve Boxes

Provide a valve box for each gate valve on buried piping, except where indicator post is shown. Construct adjustable valve boxes manufactured from cast iron of a size compatible for the valve on which it is used. Provide cast iron valve boxes conforming to JCW-104. Coat the cast-iron box with a heavy coat of bituminous paint. Provide a round head. Cast the word "WATER" on the lid. The minimum diameter of the shaft of the box is 135 mm or as indicated.

2.1.2.5 Valve Pits

Construct the valve pits at locations indicated or as required above and in accordance with the details shown.

~~2.1.3 Blowoff Valve Assemblies~~

~~Provide blowoff valve assemblies complete with all pipe, fittings, valve, valve box, riser box and lid, riser extension, discharge fitting and other materials required to connect to the water main. Provide blow off valve assemblies 100 mm or larger conforming to JWWA compliant local commercial products.~~

2.1.3 Fire Hydrants

2.1.3.1 Fire Hydrants

Provide fire hydrants where indicated. Paint fire hydrants with at least one coat of primer and two coats of enamel paint. Paint barrel and bonnet colors in accordance with UFC 3-600-01. Stencil fire hydrant number and main size on the fire hydrant barrel using black stencil paint.

Provide a JWWA compliant protective epoxy interior coating on those portions of the fire hydrant continuously in contact with sea water or salt water.

2.1.3.1.1 Dry-Barrel Type and Wet-Barrel Type Fire Hydrants

Provide Dry-barrel type fire hydrants conforming to JWWA compliant local commercial product of aboveground fire hydrants and meeting the requirements of UL 246, "Base Valve" with 150 mm inlet, 135 mm valve opening, one 115 mm pumper connection, and two 65 mm hose connections. Provide Wet-barrel type fire hydrants conforming to JWWA compliant local commercial product of aboveground fire hydrants and meeting the requirements of UL 246, "Wet Barrel" with 150 mm inlet, one 115 mm pumper connection, and two 65 mm hose connections. Individually valve pumper connection and hose connections with independent nozzle gate valves. The locally manufactured and JWWA compliant commercial product of fire hydrants shall be compatible with the 115 mm pumper and two 65 mm hose connection and shall be of the type as applicable to U.S. military construction projects in Japan.

Provide mechanical-joint or push-on joint end inlet , except where flanged end is indicated. Provide fire hydrants with breakable features . Provide fire hydrant with special couplings joining upper and lower sections of fire hydrant barrel and upper and lower sections of fire hydrant stem that break from a force imposed by a moving vehicle.

2.1.4 Meters

Submit certificates certifying all required and recommended tests set forth in the referenced standard and JIS B 8570-1 have been performed and comply with all applicable requirements of the referenced standard and

JIS B 8570-1 within the past three years. Include certification that each meter has been tested for accuracy of registration and that each meter complies with the accuracy and capacity requirements of the referenced standard when tested in accordance with JIS B 7552.

Include a register with all meters whether they are or are not connected to a remote reading system.

~~2.1.4.1 Propeller Type Meters~~

~~Provide Advanced Metering Infrastructure (AMI) and Direct Digital Communication (DDC) compatible meter for waterworks of sizes 50mm to 1800mm, conforming to JWWA compliant local commercial products. Flow tubes or main cases constructed of cast iron or fabricated steel with JWWA compliant protective coating.~~

~~2.1.4.2 Displacement Type Meters~~

~~Provide Advanced Metering Infrastructure (AMI) and Direct Digital Communication (DDC) compatible meter for waterworks of sizes 50 mm or smaller, conforming to JWWA compliant local commercial products. Pressure casings constructed of copper alloy containing not less than 75 percent copper. Provide registers with non-breakable covers and straight-reading registers. Provide non-breakable covers of copper alloy containing not less than 75 percent copper. For meter sizes 13mm through 25 mm provide frost-protection-type design.~~

2.1.4.1 Compound Type Meters

Provide Advanced Metering Infrastructure (AMI) and Direct Digital Communication (DDC) compatible meter with strainers for waterworks of sizes 50 mm through 200 mm, conforming to JWWA compliant local commercial products. Main casing constructed of cast iron or fabricated steel with JWWA compliant protective coating. Equip with tapped bosses near the outlet for field testing purposes.

2.1.4.2 Register

Provide open straight-reading register supplied by the meter manufacturer. Equip register with cubic meters readings. Use encoder type remote register designed in accordance with JWWA compliant local commercial product of water meter.

2.1.4.3 Strainers

Provide strainer recommended and supplied by the local meter manufacturer. Provide strainer of the same material as the meter body (i.e., bronze, ductile, or stainless).

2.1.4.4 Meter Connections

Provide connections compatible with the type of pipe and conditions encountered.

2.1.4.5 Advanced Metering Infrastructure

The Government will supply an Advanced Metering Infrastructure (AMI) compatible water meter(s) for the Contractor to install and connect to the existing AMI Data Acquisition System (DAS). ~~Use the existing Government~~

~~laptop computers to configure the meter using existing software loaded on the computer. Modifications to existing software on the computer or the addition of software to the computer is not allowed. The Contractor must ensure that the meter(s) transmit the metered data to the DAS. The current meters being used by [_____] are: [_____].~~

2.1.4.6 Direct Digital Control System Interface

Provide all meters with the capability of providing pulse output to the DDC system.

2.1.4.7 Meter Setter

For water meter 50 mm or greater, provide a by-pass assembly with the valve located outside the vault. Provide valve box for valve located outside of vault.

2.1.4.8 Meter Boxes Vaults

Provide meter boxes vaults of sufficient size to completely enclose the meter and shutoff valve or service stop and in accordance with the details shown on the drawings. Provide a meter boxes or vaults with a height equal to the distance from invert of the service line to finished grade at the meter location.

~~2.1.4.8.1 Cast Iron~~

~~Provide cast iron meter box and lid. Provide a round lid with precast holes for remote electronic meter reading modules having the word "WATER" cast on the top surface.~~

2.1.4.8.1 Meter Boxes Vaults

2.1.4.8.1.1 Vault Access Door

Provide a single-leaf or double-leaf cast-in aluminum or painted steel diamond-plate access door. ~~with the following dimensions:~~

~~Width: [_____] mm~~

~~Length: [_____] mm~~

Include stainless steel spring or pneumatic lift assist, type 316 stainless steel slam locking latch, automatic hold-open arm with a red release handle, and flush mounted retractable lifting handle. Door must have a minimum load rating 6,800 kg load.

2.1.4.8.1.2 Fittings

Provide flanged fittings for pipe 75 mm and larger.

2.1.4.8.1.3 Vault Valves

Provide ball or outside screw and yoke (OS&Y) or butterfly valves in meter vault.

~~2.1.5 Backflow Preventers~~

~~Provide a JWWA compliant local commercial product reduced pressure principle type backflow preventer meeting the following requirements:~~

~~a. Size: [_____]~~

~~b. Maximum Rated Flow: [_____]~~

~~c. Allowable Pressure Loss: [_____]~~

~~d. Flanged cast iron, mounted gate valve~~

~~e. Strainer of the same material as the backflow preventer~~

~~The particular make, model, and size of backflow preventers to be installed must be included in the latest edition of the List of Approved Backflow Prevention Assemblies and be accompanied by a backflow certificate . Select materials for piping, strainers, and valves used in assembly installation that are galvanically compatible. Materials joined, connected, or otherwise in contact are to have no greater than 0.25 V difference on the Anodic Index, unless separated by a dielectric type union or fitting.~~

~~2.1.5.1 Backflow Preventer Enclosure~~

~~Provide an insulated enclosure where freezing temperature are possible.~~

2.1.5 Disinfection

Chlorinating materials are to conform to: Chlorine, Liquid: JIS K 1102; Hypochlorite, Calcium and Sodium: Approved local commercial product.

2.2 ACCESSORIES

2.2.1 Pipe Restraint

2.2.1.1 Thrust Blocks

Use JIS A 5308 concrete having a minimum compressive strength of 18 MPa at 28 days.

2.2.1.2 Joint Restraint

Provide restrained joints in accordance with NFPA 24, Chapter 10 and in accordance with applicable JWWA standard test method for joint restrain.

Provide mechanical joint restraint restraint devices with gripper wedges incorporated into a follower gland and specifically designed for the pipe material and meeting the requirements of JIS G 5527 or metal harness fabricated by the pipe manufacturer.

2.2.2 Protective Enclosures

Provide Freeze-Protection Enclosures that are insulated and designed to protect aboveground water piping, equipment, or specialties from freezing and damage.

2.2.3 Tapping Sleeves

Provide cast gray, ductile, malleable iron or stainless steel, split-sleeve type tapping sleeves of the sizes indicated for connection to existing main with flanged or grooved outlet, and with bolts, follower rings and gaskets on each end of the sleeve. Utilize similar metals for bolts, nuts, and washers to minimize the possibility of galvanic corrosion. Provide dielectric gaskets where dissimilar metals adjoin. Provide a tapping sleeve assembly with a maximum working pressure of 1000 kPa. Provide bolts with square heads and hexagonal nuts. Longitudinal gaskets and mechanical joints with gaskets as recommended by the manufacturer of the sleeve. When using grooved mechanical tee, utilize an upper housing with full locating collar for rigid positioning which engages a machine-cut hole in pipe, encasing an elastomeric gasket which conforms to the pipe outside diameter around the hole and a lower housing with positioning lugs, secured together during assembly by nuts and bolts as specified, pre-torqued to 67.8 Newton meters.

2.2.4 Sleeve-Type Mechanical Couplings

Use couplings to join plain-end piping by compression of a ring gasket at each end of the adjoining pipe sections. The coupling consists of one middle ring flared or beveled at each end to provide a gasket seat; two follower rings; two resilient tapered rubber gaskets; and bolts and nuts to draw the follower rings toward each other to compress the gaskets. Provide true circular middle ring and the follower rings sections free from irregularities, flat spots, and surface defects; provide for confinement and compression of the gaskets. For ductile iron and PVC pipe, the middle ring is cast-iron; and the follower rings are malleable or ductile iron. Use gaskets for resistance to set after installation and to meet the requirements specified for gaskets for mechanical joint in JIS B 2404. Provide track-head type bolts JIS B 1180, with nuts,JIS B 1181; or round-head square-neck type bolts, JIS B 1171 with hex nuts JIS B 1181. Provide 16 mm diameter bolts. Shape bolt holes in follower rings to hold fast to the necks of the bolts used. Do not use mechanically coupled joints using a sleeve-type mechanical coupling as an optional method of jointing except where pipeline is adequately anchored to resist tension pull across the joint. Provide a tight flexible joint with mechanical couplings under reasonable conditions, such as pipe movements caused by expansion, contraction, slight settling or shifting in the ground, minor variations in trench gradients, and traffic vibrations. Match coupling strength to that of the adjoining pipeline.

2.2.5 Insulating Joints

Provide a rubber-gasketed insulating joint or dielectric coupling between pipe of dissimilar metals which will effectively prevent metal-to-metal contact between adjacent sections of piping.

2.2.6 Bonded Joints

For all ferrous pipe, provide a metallic bond at each joint, including joints made with flexible couplings, caulking, or rubber gaskets, of ferrous metallic piping to effect continuous conductivity. Provide Size 1/0 copper conductor thermal weld type bond wire designed for direct burial and shaped to stand clear of the joint.

2.2.7 Dielectric Fittings

Install dielectric fittings between threaded ferrous and nonferrous metallic pipe, fittings and valves, except where corporation stops join mains to prevent metal-to-metal contact of dissimilar metallic piping elements and compatible with the indicated working pressure.

2.2.8 Tracer Wire for Nonmetallic Piping

Provide a continuous bare copper or aluminum wire not less than 2.5 mm in diameter in sufficient length over each separate run of nonmetallic pipe.

~~2.2.9 Water Service Line Appurtenances~~

~~2.2.9.1 Corporation Stops~~

~~Ground key type; lead-free bronze, compatible with the working pressure of the system and solder-joint, or flared tube compression type joint. Threaded ends for inlet and outlet of corporation stops, coupling nut for connection to flared copper tubing.~~

~~2.2.9.2 Curb or Service Stops~~

~~Ground key, round way, inverted key type; made of lead-free bronze, and compatible with the working pressure of the system. Provide compatible ends for connection to the service piping. Cast an arrow into body of the curb or service stop indicating direction of flow.~~

~~2.2.9.3 Service Clamps~~

~~Provide single or double flattened strap type service clamps used for repairing damaged cast-iron, steel or PVC pipe with a pressure rating not less than that of the pipe being repaired. Provide clamps with a galvanized malleable-iron body with cadmium plated straps and nuts and a rubber gasket cemented to the body.~~

~~2.2.9.4 Goosenecks~~

~~Manufacture goosenecks from Type K copper tubing; provide joint ends for goosenecks compatible with connecting to corporation stop and service line. Where multiple gooseneck connections are required for an individual service, connect goosenecks to the service line through a compatible lead-free brass or bronze branch connection; the total clear area of the branches to be at least equal to the clear area of the service line.~~

~~2.2.9.5 Curb Boxes~~

~~Provide a curb box for each curb or service stop manufactured from cast iron, size capable of containing the stop where it is used. Provide a round head. Cast the word "WATER" on the lid. Factory coat the box with a heavy coat of bituminous paint.~~

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Connections to Existing System

Perform all connections to the existing water system in the presence of the Contracting Officer.

3.1.2 Operation of Existing Valves

Do not operate valves within or directly connected to the existing water system unless expressly directed to do so by the Contracting Officer.

3.1.3 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK.

3.2 INSTALLATION

Install all materials in accordance with the applicable reference standard, manufacturers instructions and as indicated herein.

3.2.1 Piping

3.2.1.1 General Requirements

Install pipe, fittings, joints and couplings in accordance with the applicable referenced standard, the manufacturer's instructions and as specified herein.

3.2.1.1.1 Termination of Water Lines

Terminate the work covered by this section at a point approximately 1.5 m from the building, unless otherwise indicated.

Do not lay water lines in the same trench with gas lines, fuel lines, electric wiring, or any other utility. Do not install copper tubing in the same trench with ferrous piping materials. Where nonferrous metallic pipe (i.e., copper tubing) crosses any ferrous piping, provide a minimum vertical separation of 300 mm between pipes.

3.2.1.1.2 Pipe Laying and Jointing

Remove fins and burrs from pipe and fittings. Before placing in position, clean pipe, fittings, valves, and accessories, and maintain in a clean condition. Provide proper facilities for lowering sections of pipe into trenches. Under no circumstances is it permissible to drop or dump pipe, fittings, valves, or other water line material into trenches. Cut pipe cleanly, squarely, and accurately to the length established at the site and work into place without springing or forcing. Replace a pipe or fitting that does not allow sufficient space for installation of jointing material. Blocking or wedging between bells and spigots is not permitted. Lay bell-and-spigot pipe with the bell end pointing in the direction of laying. Grade the pipeline in straight lines; avoid the formation of dips and low points. Support pipe at the design elevation and grade. Secure firm, uniform support. Wood support blocking is not permitted. Lay pipe so that the full length of each section of pipe and each fitting rests solidly on the pipe bedding; excavate recesses to accommodate bells, joints, and couplings. Provide anchors and supports for fastening work into place. Make provision for expansion and contraction of pipelines. Keep trenches free of water until joints have been assembled. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Do not lay pipe when conditions of trench or weather prevent installation. Provide a minimum of 760 mm depth of cover over top of pipe under non-traffic areas and minimum of 900 mm under traffic areas.~~]~~

3.2.1.1.3 Tracer Wire

Install a continuous length of tracer wire for the full length of each run of nonmetallic pipe. Attach wire to top of pipe in such manner that it will not be displaced during construction operations.

3.2.1.1.4 Connections to Existing Water Lines

Make connections to existing water lines after coordination with the facility and with a minimum interruption of service on the existing line. Make connections to existing lines under pressure in accordance with the recommended procedures of the manufacturer of the pipe being tapped and as indicated.

3.2.1.1.5 Sewer Manholes

No water piping is to pass through or come in contact with any part of a sewer manhole.

3.2.1.1.6 Water Piping Parallel With Sewer Piping

Where the location of the water line is not clearly defined by dimensions on the drawings, do not lay water line closer than 3.0 m, horizontally, from any sewer line.

a. Normal Conditions: Lay water piping at least 3.0 m horizontally from sewer or sewer manhole whenever possible. Measure the distance from outside edge to outside edge of pipe or outside edge of manhole. When local conditions prevent horizontal separation install water piping in a separate trench with the bottom of the water piping at least 450 mm above the top of the sewer piping.

b. Unusual Conditions: When local conditions prevent vertical separation, construct sewer piping of JWWA compliant ductile iron water piping and perform hydrostatic sewer test, without leakage, prior to backfilling. When local conditions prevent vertical separation, test the sewer manhole in place to ensure watertight construction.

3.2.1.1.7 Water Piping Crossing Sewer Piping

Provide at least 450 mm above the top (crown) of the sewer piping and the bottom (invert) of the water piping whenever possible. Measure the distance edge-to-edge. Where water lines cross under gravity sewer lines, construct sewer line of JWWA compliant ductile iron water piping with rubber-gasketed joints and no joint located within 3 m horizontally, of the crossing. Lay water lines which cross sewer force mains and inverted siphons at least 600 mm above these sewer lines; when joints in the sewer line are closer than 900 mm horizontally from the water line relay the sewer line to ensure no joint closer than 900 mm.

a. Normal Conditions: Provide a separation of at least 450 mm between the bottom of the water piping and the top of the sewer piping in cases where water piping crosses above sewer piping.

b. Unusual Conditions: When local conditions prevent a vertical separation described above, construct sewer piping passing over or under water piping of JWWA compliant ductile iron water piping and perform hydrostatic sewer test, without leakage, prior to backfilling. Construct sewer crossing with a minimum 6.1 m length of the JWWA compliant ductile iron water piping, centered at the point of the crossing so that joints are equidistant and as far as possible from the water piping. Protect water piping passing under sewer piping by providing a vertical separation of at least 450 mm between the bottom of the sewer piping and the top of the water piping; adequate structural support for the sewer piping to prevent excessive deflection of the joints and the settling on or damage to the water piping.

3.2.1.1.8 Penetrations

Provide ductile-iron or Schedule 40 steel wall sleeves for pipe passing through walls of valve pits and structures. Fill annular space between walls and sleeves with rich cement mortar. Fill annular space between pipe and sleeves with mastic.

3.2.1.1.9 Flanged Pipe

Only install flanged pipe aboveground or with the flanges in valve pits.

3.2.1.2 Ductile-Iron Piping

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of JDPA T 01 for pipe installation, joint assembly, valve-and-fitting installation, and thrust restraint.

a. Jointing: Make push-on joints with the gaskets and lubricant specified for this type joint; assemble in accordance with the applicable requirements of pipe manufacturer for joint assembly. Make mechanical joints with the gaskets, glands, bolts, and nuts specified for this type joint; assemble in accordance with the applicable requirements of of pipe manufacturer for joint assembly . Make flanged joints with the gaskets, bolts, and nuts specified for this type joint. Make flanged joints up tight; avoid undue strain on flanges, fittings, valves, and other accessories. Align bolt holes for each flanged joint. Use full size bolts for the bolt holes; use of undersized bolts will not be permitted. Do not allow adjoining flange faces to be out of parallel to such degree that the flanged joint cannot be made watertight without overstraining the flange. When flanged pipe or fitting has dimensions that do not allow the making of a flanged joint as specified, replace it. Use set screw flanges to make flanged joints where conditions prevent the use of full-length flanged pipe and assemble in accordance with the recommendations of the set screw flange manufacturer. During installation of set screw gasket provide for confinement and compression of gasket when joint to adjoining flange is made. Assemble joints made with sleeve-type mechanical couplings in accordance with the recommendations of the coupling manufacturer. Make insulating joints with the gaskets, sleeves, washers, bolts, and nuts previously specified for this type joint. Assemble insulating joints as specified for flanged joints, except that bolts with insulating sleeves are to be full size for the bolt holes. Ensure that there is no metal-to-metal contact between dissimilar metals after the joint has been assembled.

b. Allowable Deflection: Follow ductile iron pipe manufacturer's recommendation for the maximum allowable deflection. If the alignment requires deflection in excess of the above limitations, provide special bends or a sufficient number of shorter lengths of pipe to achieve angular deflections within the limit set forth.

c. Exterior Protection: Completely encase buried ductile iron pipelines with polyethylene film, in conformance with JWWA K 158.

~~3.2.1.3 PVC and PVCO Water Main Pipe~~

~~Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS for laying of pipe, joining PVC pipe to fittings and accessories, setting of fire hydrants, valves, and fittings; and with the recommendations for pipe joint assembly and appurtenance per pipe manufacturer's installation instructions.~~

~~a. Jointing: Make push-on joints with the elastomeric gaskets specified for this type joint, using either elastomeric-gasket bell-end pipe or elastomeric-gasket couplings. For pipe-to-pipe push-on joint connections, use only pipe with push-on joint ends having factory-made bevel; for push-on joint connections to metal fittings, valves, and other accessories, cut spigot end of pipe off square and re-bevel pipe end to a bevel approximately the same as that on ductile-iron pipe used for the same type of joint. Use a lubricant recommended by the pipe manufacturer for push-on joints. Assemble push-on joints for pipe-to-pipe joint connections; assemble push-on joints for connection to fittings, valves, and other accessories; make compression-type joints/mechanical joints with the gaskets, glands, bolts, nuts, and internal stiffeners previously specified for this type joint; cut off spigot end of pipe for compression-type joint/mechanical-joint connections and do not re-bevel; assemble joints made with sleeve-type mechanical couplings , all in accordance with pipe manufacturer's installation instructions.~~

~~b. Joint Offset: Construct joint offset . Do not exceed the minimum longitudinal bending as recommended by pipe manufacturer.~~

~~c. Fittings: Install in accordance with PVC and PVCO pipe manufacturer's installation standards.~~

~~3.2.1.4 Polyethylene (PE) Piping~~

~~Install PE pipes in accordance with pipe manufacturer's installation instruction.~~

~~3.2.1.5 Plastic Service Piping~~

~~Install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the applicable requirements, pipe manufacturer's installation instructions , unless otherwise specified. Handle solvent cements used to join plastic piping in accordance with pipe manufacturer's installation instructions.~~

~~3.2.1.5.1 Jointing~~

~~Make solvent-cemented joints for PVC piping using the solvent cement previously specified for this material; assemble joints in accordance with pipe manufacturer's installation instructions. Make plastic pipe joints to other pipe materials in accordance with the recommendations of the plastic pipe manufacturer.~~

~~3.2.1.5.2 Plastic Pipe Connections to Appurtenances~~

~~Connect plastic service lines to corporation stops and gate valves in accordance with the recommendations of the plastic pipe manufacturer.~~

3.2.1.3 Fire Protection Service Lines for Sprinkler Supplies

Connect water service lines used to supply building sprinkler systems for fire protection to the water main in accordance with NFPA 24.

3.2.1.4 Water Service Piping

3.2.1.4.1 Location

Connect water service piping to the building service where the building service has been installed. Where building service has not been installed, terminate water service lines approximately 1.5 m from the building line at the points indicated; close such water service lines with plugs or caps.

3.2.1.4.2 Water Service Line Connections to Water Mains

Connect water service lines to the main by a corporation stop and gooseneck and install a service stop below the frostline. Connect water service lines to ductile-iron water mains in accordance with pipe manufacturer's installation instructions for service taps. Connect water service lines to PVC water mains in accordance with pipe manufacturer's installation instructions.

~~3.2.2 Railroad Right-of-Way~~

~~Install piping passing under the right-of-way of a commercial railroad in accordance with the specifications for pipelines conveying nonflammable substances . For PVC water main pipe, also install in accordance with the recommendations of pipe manufacturer for installation of casings.~~

3.2.2 Meters

Install meters and meter boxes vaults at the locations shown on the drawings. Center meters in the boxes vaults to allow for reading and ease of removal or maintenance. Set top of box or vault at finished grade.

~~3.2.3 Backflow Preventers~~

~~Install backflow preventers of type, size, and capacity indicated a minimum of 300 mm and a maximum of 900 mm above concrete base. Include valves and test cocks. Install according to the manufacturers requirements and the requirements of plumbing and health department and authorities having jurisdiction. Support NPS 63 mm and larger backflow preventers, valves, and piping near floor with 300 mm minimum air gap, and on concrete piers or steel pipe supports. Do not install backflow preventers that have a relief drain in vault or in other spaces subject to flooding. Do not install by-pass piping around backflow preventers.~~

~~3.2.3.1 Backflow Preventer Enclosure~~

~~Install a level concrete base with top of concrete surface approximately 50 mm above grade. Install protective enclosure over valve and~~

~~equipment. Anchor protective enclosure to concrete base.~~

3.2.3 Disinfection

Disinfection of systems supplying non-potable water is not required.

Prior to disinfection, provide disinfection procedures, proposed neutralization and disposal methods of waste water from disinfection as part of the disinfection submittal. Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with the applicable JWWA standards. Ensure a free chlorine residual of not less than 10 mg/L after 24 hour holding period and prior to performing bacteriological tests.

3.2.4 Flushing

Perform bacteriological tests prior to flushing. Flush solution from the systems with domestic water until maximum residual chlorine content is within the range of 0.2 to 0.5 mg/L, the residual chlorine content of the distribution system, or acceptable for domestic use. Use neutralizing chemicals as recommended by JWWA standards.

3.2.5 Pipe Restraint

3.2.5.1 Concrete Thrust Blocks

Install concrete thrust blocks where indicated.

3.2.5.2 Restrained Joints

Install restrained joints in accordance with the manufacturer's instructions or NFPA 24 where indicated. For metal harness use tie rods and clamps as shown in NFPA 24. Provide metal harness fabricated by the pipe manufacturer and furnished with the pipe.

3.2.6 Valves

3.2.6.1 Gate Valves

Install gate valves in accordance with the requirements for valve-and-fitting installation and with the recommendations of the gate valve manufacturer. Install gate valves on PVC and PVCO water mains in accordance with the recommendations of the gate valve manufacturer. Make and assemble joints to gate valves as specified for making and assembling the same type joints between pipe and fittings.

3.2.6.2 Check Valves

Install check valves in accordance with the applicable requirements for valve-and-fitting installation , except as otherwise indicated. Make and assemble joints to check valves as specified for making and assembling the same type joints between pipe and fittings.

3.2.6.3 Air Release, Air/Vacuum, and Combination Air Valves

Install pressure vacuum assemblies of type, size, and capacity indicated. Include valves and test cocks. Install according to the requirements of plumbing and health department and authorities having jurisdiction. Do not install pressure vacuum breaker assemblies in vault or other space subject to flooding.

3.2.7 Blowoff Valve Assemblies

Install blowoff valve assemblies as indicated on the drawings or in accordance with the manufactures recommendations. Install discharge fitting on the end of riser pipe to direct the flow of water so as to minimize damage to surrounding areas.

3.2.8 Fire Hydrants

Install fire hydrants in accordance with NFPA 24 and with the requirements of JDPA T 01 for pipe installation and as indicated. Make and assemble joints as specified for making and assembling the same type joints between pipe and fittings. Provide metal harness as specified under pipe anchorage requirements for the respective pipeline material to which fire hydrant is attached. Install fire hydrants with the 115 mm connections facing the adjacent paved surface. If there are two paved adjacent surfaces, install fire hydrants with the 115 mm connection facing the paved surface where the connecting main is located.

3.3 FIELD QUALITY CONTROL

3.3.1 Tests

Notify the Contracting Officer a minimum of five days in advance of hydrostatic testing. Coordinate the proposed method for disposal of waste water from hydrostatic testing. Perform field tests, and provide labor, equipment, and incidentals required for testing, except that water needed for field tests will be furnished as set forth in paragraph AVAILABILITY AND USE OF UTILITY SERVICES in Section 01 50 00 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS. Provide documentation that all items of work have been constructed in accordance with the Contract documents.

3.3.1.1 Hydrostatic Test

Test the water system in accordance with the applicable JWWA standards. Where water mains provide fire service, test in accordance with the special testing requirements given in the paragraph SPECIAL TESTING REQUIREMENTS FOR FIRE SERVICE. Test ductile-iron water mains in accordance with the requirements of JIS S 3200-1 for hydrostatic testing. The amount of leakage on ductile-iron pipelines with mechanical-joints or push-on joints is not to exceed the amounts given in pipe manufacturer's installation instructions. No leakage will be allowed at joints made by any other methods. Test PVC and PVCO plastic water systems made with PVC pipe for pressure and leakage tests. The amount of leakage on pipelines made of PVC water main pipe is not to exceed the amounts given in pipe manufacturer's installation instructions, except that at joints made with sleeve-type mechanical couplings, no leakage will be allowed. Test water service lines for hydrostatic testing. No leakage will be allowed at copper pipe joints, copper tubing joints (soldered, compression type, brazed), plastic pipe joints, flanged joints, and screwed joints. Do not backfill utility trench or begin testing on any section of a pipeline where concrete thrust blocks have been provided until at least 7 days after placing of the concrete.

3.3.1.2 Hydrostatic Sewer Test

The hydrostatic pressure sewer test will be performed in accordance with the applicable JWWA standard for the piping material with a minimum test pressure of 200 kPa.

3.3.1.3 Leakage Test

For leakage test, use a hydrostatic pressure not less than the maximum working pressure of the system. Leakage test may be performed at the same time and at the same test pressure as the pressure test.

For PE perform leak testing in accordance with the applicable JWWA standards.

3.3.1.4 Bacteriological Testing

Perform bacteriological tests in accordance with the applicable JWWA standards. Analyze samples by a certified laboratory, and submit the results of the bacteriological samples.

~~3.3.1.5 Backflow Preventer Tests~~

~~After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the JIS S 3200-4 or applicable JWWA testing standards.~~

3.3.1.5 Special Testing Requirements for Fire Service

Test water mains and water service lines providing fire service or water and fire service in accordance with NFPA 24. The additional water added to the system must not exceed the limits given in NFPA 24

3.3.1.6 Tracer Wire Continuity Test

Test tracer wire for continuity after service connections have been completed and prior to final pavement or restoration. Verify that tracer wire is locatable with electronic utility locating equipment. Repair breaks or separations and re-test for continuity.

3.4 SYSTEM STARTUP

Water mains and appurtenances must be completely installed, disinfected, flushed, and satisfactory bacteriological sample results received prior to permanent connections being made to the active distribution system. Obtain approval by the Contracting Officer prior to the new water piping being placed into service.

3.5 CLEANUP

Upon completion of the installation of water lines and appurtenances, remove all debris and surplus materials resulting from the work.

-- End of Section --

SECTION 33 30 00

SANITARY SEWERAGE
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5001	(2008) Crushed Stone for Road Construction
JIS A 5005	(2009) Crushed Stone and Manufactured Sand for Concrete
JIS A 5308	(2014) Ready-Mixed Concrete
JIS A 5372	(2016) Precast Reinforced Concrete Products
JIS A 5506	(2008) Manhole Covers for Sewerage Works
JIS G 3101	~~(2017) Rolled Steels for General Structure (Amendment 1)~~(2020) Rolled Steels for General Structure
JIS H 8641	(20~~07~~21) Hot Dip Galvanized Coatings
JIS K 6353	(2011) Rubber Goods for Water Works
JIS K 6739	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Drain
JIS K 6741	~~(2016)~~(2016; R 2021) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes
JIS R 5210	(2009) Portland Cement

~~JAPAN SEWAGE WORKS ASSOCIATION (JSWAS)~~JAPAN SEWAGE WORKS ASSOCIATION (JSWA)

JSWAS K-1	(2010) Rigid PVC Pipe for Sewer

~~THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF JAPAN STANDARD (SHASE-S)~~THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF JAPAN (SHASE)

SHASE-S 209	(2009) Manhole Cover

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation;

submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation Drawings; G

~~SD-03 Product Data~~

~~Pressure Pipe; G~~

~~Sewage Treatment Tanks; G~~

SD-06 Test Reports

Hydrostatic Sewer Test

Infiltration Tests or Exfiltration Tests

~~Tests For Pressure Lines; G~~

Deflection Testing

SD-07 Certificates

~~Gaskets~~

Pre-Installation Inspection Request; G

Post-Installation Inspection; G

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery and Storage

Check upon arrival; identify and segregate as to types, functions, and sizes. Store off the ground in a manner affording easy accessibility and not causing excessive rusting or coating with grease or other objectionable materials.

1.3.1.1 Piping

Inspect materials delivered to site for damage; store with minimum of handling. Store materials on site in enclosures or under protective coverings. Store plastic piping and jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes and fittings free of dirt and debris.

1.3.1.2 Cement, Aggregate, and Reinforcement

As specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

1.3.2 Handling

Handle pipe, fittings, and other accessories in such manner as to ensure delivery to the trench in sound undamaged condition. Take special care not to damage linings of pipe and fittings; if lining is damaged, make satisfactory repairs. Carry, do not drag, pipe to trench. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install the plastic pipe in accordance with the manufacturer's recommendation and discard those materials if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Sanitary Sewer Gravity Pipeline

Provide mains and laterals of concrete pipe or polyvinyl chloride (PVC) plastic pipe. Provide building connections of polyvinyl chloride (PVC) plastic pipe. Provide new and modify existing exterior sanitary gravity sewer piping and appurtenances. Provide each system complete and ready for operation. The exterior sanitary gravity sewer system includes equipment, materials, installation, and workmanship as specified herein more than 1.5 m outside of building walls.

~~2.1.2 Sanitary Sewer Pressure Lines~~

~~Provide pressure lines of ductile iron pressure pipe or polyvinyl chloride (PVC) plastic pressure pipe.~~

2.2 MATERIALS

Provide materials conforming to the respective specifications and other requirements specified below. Submit manufacturer's product specification,standard drawings or catalog cuts.

2.2.1 Gravity Pipe

~~2.2.1.1 Concrete Gravity Sewer Piping~~

~~2.2.1.1.1 Concrete Gravity Pipe~~

~~Provide reinforced concrete pipe conforming to JIS A 5372, external pressure type, Class 1 or Class 2, Type B.~~

~~2.2.1.1.2 Jointing Materials for Concrete Gravity Piping~~

~~Provide gaskets and pipe ends for rubber gasket joint conforming to JIS K 6353. Use gaskets suitable for use with sewage.~~

~~Submit certificates of compliance stating that the fittings or gaskets used for waste drains or lines are oil resistant.~~

2.2.1.1 PVC Gravity Sewer Piping

2.2.1.1.1 Pipe and Fittings

a. Pipe:JIS K 6741, Class VP or VU.

b. Fittings:JIS K 6739

2.2.1.1.2 Joints and Jointing Material

Jointing Materials: Rubber gasket conforming to JIS K 6353. Gaskets shall be suitable for use in sewerage.

2.2.1.1.3 PVC Branch Pipe Connectors

Shall be standard product compatible with the PVC plastic pipe and conforming to JSWAS K-1 and JIS K 6739. Adhesive materials shall be as per branch pipe manufacturer's recommendation.

~~2.2.2 Pressure Pipe~~

~~2.2.2.1 Ductile Iron Pressure Piping~~

~~2.2.2.1.1 Ductile Iron Pressure Pipe and Fittings~~

~~Provide mechanical joint or flanged ductile-iron pipe conforming to JIS G 5526. Provide fittings conforming to JIS G 5527. Use fittings which have a pressure rating at least equivalent to that of the pipe. Pipe and fittings are to have interior cement-mortar lining conforming to JIS A 5314 and exterior pipe coating conforming to JDPA Z 2010.~~

~~2.2.2.1.2 Ductile Iron Pressure Joints and Jointing Materials~~

~~a. Joints, general: Use mechanical joints for pipe and fittings. Use flanged joints where indicated. Joints made with sleeve-type mechanical coupling may be used in lieu of push-on joint.~~

~~b. Mechanical joints: Gaskets are to conform to JIS B 2404.~~

~~c. Flanged joints: Provide bolts, nuts, and gaskets for flanged connections compatible with JIS G 5527 joints.~~

~~d. Joints made with sleeve-type mechanical couplings: Provide bolts conforming to the tensile requirements of JIS B 1180 with nuts conforming to the tensile requirements of JIS B 1181 or round-head square-neck type bolts conforming to JIS B 1171 with hex nuts conforming to JIS B 1181.~~

~~2.2.2.2 PVC Pressure Pipe and Associated Fittings~~

~~2.2.2.2.1 Pipe and Fittings~~

~~Pipe, couplings and fittings are to be manufactured of materials conforming to JIS K 6742 and fittings conforming to JIS K 6743.~~

~~2.2.2.2.2 Solvent Cement Joint~~

~~Provide solvent cement joint per pipe manufacturer's recommendation.~~

~~2.2.3 Piping Beneath Railroad Right-of-Way~~

~~Where pipeline passes under the right-of-way of a commercial railroad, piping is to conform to the specifications for pipelines conveying nonflammable substances. For casing pipe provide ductile-iron pipe in lieu of cast-iron soil pipe. Ductile-iron pipe is to conform to and have strength computed in accordance with JIS G 5526.~~

2.2.2 Portland Cement

Portland cement shall conform to JIS R 5210.

2.2.3 Portland Cement Concrete

Provide portland cement concrete conforming to JIS A 5308, compressive strength of 24 MPa at 28 days, except for concrete cradle and encasement or concrete blocks for manholes. Concrete used for cradle and encasement is to have a compressive strength of 18 MPa minimum at 28 days. Protect concrete in place from freezing and moisture loss for 7 days.

2.2.4 Precast Concrete Manholes

Approved commercial products as shown on drawings and conforming to JIS A 5372 or JSWAS K-1. Joints between precast concrete manhole sections shall be made with flexible watertight, rubber-type gaskets per manhole manufacturer's standards.

2.2.5 Invert Mortar

Mortar for forming manhole inverts shall be composed of cement, sand and water mixed in proportion of 1 part cement to 2 parts of sand, sufficient water to produce a workable mixture. Mortar shall be used in the work within one hour after mixing.

2.2.6 Gaskets and Connectors

Resilient connectors for making joints between manhole and pipes entering manhole are to conform to pipe manufacturer's standards.

~~2.2.7 Sewage Treatment Tanks~~

~~Shall be three functions of primary, secondary, and final sewage treatment and shall meet the requirements of biochemical oxygen demand (BOD) value as indicated by testing in accordance with JIS K 0102.~~

2.2.7 Frames And Covers for Manholes

Shall be local manufacturer's standard product conforming to SHASE-S 209, and shall be of cast iron per JIS A 5506. Size, configuration and loading capacity shall be as indicated on drawings. A letter "S" shall be stamped or cast into covers.

2.2.8 Manhole Steps

Materials shall conform to JIS G 3101, Type SS400, galvanized, and of the size and configuration as shown on drawing. Manhole steps are not required in manholes and inlets less than 1.2 m deep.

2.2.9 Manhole Ladders

Provide a steel ladder conforming to JIS G 3101 where the depth of a manhole exceeds 3.6 m. The ladder is not to be less than 406 mm in width, with 19 mm diameter rungs spaced 305 mm apart. The two stringers are to be a minimum 10 mm thick and 51 mm wide. Galvanize ladders and inserts after fabrication in conformance with JIS H 8641.

2.2.10 Miscellaneous Items

2.2.10.1 Warning Tape

Provide in accordance with requirements as specified in Section 31 00 00 EARTHWORK

2.2.10.2 Sand Fill

Fill around PVC drainage pipes shall be cleaned sand conforming to JIS A 5005.

2.2.10.3 Gravel Base Course

Crushed stone for base course at concrete structures shall be crusher run conforming to JIS A 5001, RC-40.

2.2.10.4 Surface Cleanouts

Surface cleanout shall have cast iron cover and fitting adaptable for connection to the lower PVC pipe. Traffic loading capacity shall be as required for the manhole covers located within the same ground surface condition.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Installation Drawings

Submit Installation Drawings showing complete detail, both plan and side view details with proper layout and elevations.

3.2 INSTALLATION

Backfill after inspection by the Contracting Officer. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer.

3.2.1 Connections to Existing Lines

Obtain approval from the Contracting Officer before making connection to existing line. Conduct work so that there is minimum interruption of service on existing line.

3.2.2 General Requirements for Installation of Pipelines

These general requirements apply except where specific exception is made in the following paragraphs entitled "Special Requirements."

3.2.2.1 Location

Terminate the work covered by this section at a point approximately 1.5 m from the building, unless otherwise indicated. Install pressure sewer lines beneath water lines only, with the top of the sewer line being at least 0.60 m below bottom of water line. When these separation distances can not be met, contact the Contracting Officer for direction.

3.2.2.1.1 Sanitary Piping Installation Parallel with Water Line

3.2.2.1.1.1 Normal Conditions

Install sanitary piping or manholes at least 3 m horizontally from a water line whenever possible. Measure the distance from edge-to-edge.

3.2.2.1.1.2 Unusual Conditions

When local conditions prevent a horizontal separation of 3 m, the sanitary piping or manhole may be laid closer to a water line provided that:

a. The top (crown) of the sanitary piping is to be at least 450 mm below the bottom (invert) of the water main.

b. Where this vertical separation cannot be obtained, construct the sanitary piping with JWWA-approved ductile iron water pipe pressure and conduct a hydrostatic sewer testwithout leakage prior to backfilling.

c. The sewer manhole is to be of watertight construction and tested in place.

3.2.2.1.2 Installation of Sanitary Piping Crossing a Water Line

3.2.2.1.2.1 Normal Conditions

Lay sanitary sewer piping by crossing under water lines to provide a separation of at least 450 mm between the top of the sanitary piping and the bottom of the water line whenever possible.

3.2.2.1.2.2 Unusual Conditions

When local conditions prevent a vertical separation described above, use the following construction:

a. Construct sanitary piping passing over or under water lines with JWWA-approved ductile iron water pressure piping and conduct a hydrostatic sewer test without leakage prior to backfilling.

b. Protect sanitary piping passing over water lines by providing:

    (1) A vertical separation of at least 450 mm between the bottom of the sanitary piping and the top of the water line.

    (2) Adequate structural support for the sanitary piping to prevent excessive deflection of the joints and the settling on and breaking of the water line.

    (3) That the length, minimum 6.1 m, of the sanitary piping be centered at the point of the crossing so that joints are equidistant and as far as possible from the water line.

3.2.2.1.3 Sanitary Sewer Manholes

No water piping shall pass through or come in contact with any part of a sanitary sewer manhole.

3.2.2.2 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK ~~31 23 00.00 20 EXCAVATION AND FILL~~.

3.2.2.3 Pipe Laying and Jointing

Inspect each pipe and fitting before and after installation; replace those found defective and remove from site. Provide proper facilities for lowering sections of pipe into trenches. Lay nonpressure pipe with the bell or groove ends in the upgrade direction. Adjust spigots in bells to give a uniform space all around. Blocking or wedging between bells and spigots will not be permitted. Replace by one of the proper dimensions, pipe or fittings that do not allow sufficient space for installation of joint material. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Provide batterboards not more than 7.50 m apart in trenches for checking and ensuring that pipe invert elevations are as indicated. Laser beam method may be used in lieu of batterboards for the same purpose. Construct branch connections by use of regular fittings or with matching branch pipe connectors from the same pipe manufacturer.

3.2.3 Special Requirements

~~3.2.3.1 Installation of Concrete Gravity Sewer Piping~~

~~Make joints with the gaskets specified for concrete gravity sewer pipe joints. Clean and dry surfaces receiving lubricants, cements, or adhesives. Affix gaskets to pipe not more than 24 hours prior to the installation of the pipe. Protect gaskets from sun, blowing dust, and other deleterious agents at all times. Before installation of the pipe, inspect gaskets and remove and replace loose or improperly affixed gaskets. Align each pipe section with the previously installed pipe section, and pull the joint together. If, while pulling the joint, the gasket becomes loose and can be seen through the exterior joint recess when the pipe is pulled up to within 25 mm of closure, remove the pipe and remake the joint.~~

~~3.2.3.2 Installation of Ductile-Iron Pressure Lines~~

~~a. Make mechanical-joints with the gaskets, glands, bolts, and nuts specified for this type joint. Make flanged joints with gaskets, bolts, and nuts specified for this type joint. Make flanged joints up tight, taking care to avoid undue strain on flanges, fittings, and other accessories. Align bolt holes for each flanged joint. Use full size bolts for the bolt holes; use of undersized bolts to make up for misalignment of bolt holes or for any other purpose will not be permitted. Do not allow adjoining flange faces to be out of parallel to such degree that the flanged joint cannot be made watertight without overstraining the flange. When flanged pipe or fittings have dimensions that do not allow the making of a proper flanged joint as specified, replace it by one of proper dimensions.~~

~~b. Exterior protection: Completely wrap buried ductile iron pipelines with 8 mil (minimum) polyethylene sheet in conformance with JWWA K 158.~~

~~c. Pipe anchorage: Provide concrete thrust blocks (reaction backing) for pipe anchorage. Size and position thrust blocks as indicated. Use~~

~~concrete conforming to JIS A 5308 having a minimum compressive strength of 18 MPa at 28 days.~~

3.2.3.1 Installation of PVC Piping

Make joints to other pipe materials in accordance with the recommendations of the plastic pipe manufacturer.

~~3.2.3.2 Installation of PVC Pressure Pipe~~

~~3.2.3.2.1 Pipe~~

~~Make push-on joints with elastomeric gasket . For pipe-to-pipe push-on joint connections, use only pipe with push-on joint ends having factory-made bevel. For push-on joint connections to fittings, use cut spigot end of pipe off square, marked to match the manufacturer's insertion line and beveled to match factory supplied bevel. . Use an approved lubricant recommended by the pipe manufacturer for push-on joints. Assemble push-on joints for pipe-to-pipe joint connections in accordance with the requirements of the pipe manufacturer for laying the pipe. Assemble push-on joints for connection to fittings in accordance with the requirements of the pipe manufacturer for joining PVC pipe to fittings and accessories.~~

~~3.2.3.2.2 Pipe Anchorage~~

~~Provide concrete thrust blocks (reaction backing) for pipe anchorage. Size and position thrust blocks as indicated. Use concrete conforming to JIS A 5308 having a minimum compressive strength of 18 MPa at 28 days.~~

~~3.2.3.3 Pipeline Installation Beneath Railroad Right-of-Way~~

~~Where pipeline passes under the right-of-way of a commercial railroad, install piping in accordance with the specifications for pipelines conveying nonflammable substances.~~

3.2.4 Concrete Work

Cast-in-place concrete is included in Section 03 30 00 CAST-IN-PLACE CONCRETE. Support the pipe on a concrete cradle, or encased in concrete where indicated or directed.

3.2.5 Manhole Construction

Construct base slab of cast-in-place concrete or use precast concrete base sections. Make inverts in cast-in-place concrete and precast concrete bases with a smooth-surfaced semi-circular bottom conforming to the inside contour of the adjacent sewer sections. For changes in direction of the sewer and entering branches into the manhole, make a circular curve in the manhole invert of as large a radius as manhole size will permit. For cast-in-place concrete construction, either pour bottom slabs and walls integrally or key and bond walls to bottom slab. No parging will be permitted on interior manhole walls. For precast concrete construction, make joints between manhole sections with the gaskets specified for this purpose; install in the manner specified for installing joints in concrete piping. Parging will not be required for precast concrete manholes. Perform cast-in-place concrete work in accordance with the requirements specified under paragraph entitled "Concrete Work" of this section. Make joints between concrete manholes and pipes entering manholes with the resilient connectors specified for this purpose; install in accordance with the recommendations of the connector manufacturer. Where a new manhole is constructed on an existing line, remove existing pipe as necessary to construct the manhole. Cut existing pipe so that pipe ends are approximately flush with the interior face of manhole wall, but not protruding into the manhole. Use resilient connectors as previously specified for pipe connectors to concrete manholes.

3.2.6 Miscellaneous Construction and Installation

3.2.6.1 Connecting to Existing Manholes

Connect pipe to existing manholes such that finish work will conform as nearly as practicable to the applicable requirements specified for new manholes, including all necessary concrete work, cutting, and shaping. Center the connection on the manhole. Holes for the new pipe are be of sufficient diameter to allow packing cement mortar around the entire periphery of the pipe but no larger than 1.5 times the diameter of the pipe. Cut the manhole in a manner that will cause the least damage to the walls.

3.2.6.2 Metal Work

3.2.6.2.1 Workmanship and Finish

Perform metal work so that workmanship and finish will be equal to the best practice in modern structural shops and foundries. Form iron to shape and size with sharp lines and angles. Do shearing and punching so that clean true lines and surfaces are produced. Make castings sound and free from warp, cold shuts, and blow holes that may impair their strength or appearance. Give exposed surfaces a smooth finish with sharp well-defined lines and arises. Provide necessary rabbets, lugs, and brackets wherever necessary for fitting and support.

3.2.6.2.2 Field Painting

After installation, clean cast-iron frames, covers, gratings, and steps not buried in concrete to bare metal, remove mortar, rust, grease, dirt, and other deleterious materials and apply a coat of bituminous paint. Do not paint surfaces subject to abrasion.

3.2.7 Installations of Wye Branches and Branch Connectors

Install wye branches and branch connectorsin an existing sewer using a method which does not damage the integrity of the existing sewer. Do not cutinto piping for connections except when approved by the Contracting Officer. When the connecting pipe cannot be adequately supported on undisturbed earth or tamped backfill, support on a concrete cradle as directed by the Contracting Officer. Provide and install concrete required because of conditions resulting from faulty construction methods or negligence without any additional cost to the Government. Do not damage the existing sewer when installing wye branches in an existing sewer.

~~3.2.8 Construction of Sewage Treatment Tanks~~

~~Construct sewage tank in accordance with approved drawings and manufacturer's catalog data.~~

3.3 FIELD QUALITY CONTROL

The Contracting Officer will conduct field inspections and witness field tests specified in this section. Be able to produce evidence, when required, that each item of work has been constructed in accordance with the drawings and specifications.

3.3.1 Tests

Perform field tests and provide labor, equipment, and incidentals required for testing~~, except that water and electric power needed for field tests will be furnished as set forth in Section [_____]~~.

3.3.1.1 Hydrostatic Sewer Test

When unusual conflicts are encountered between sanitary sewer and waterlines a hydrostatic pressure sewer test will be performed in accordance with the applicable JWWA standard for the piping material with a minimum test pressure of 200 kPa.

3.3.1.2 Leakage Tests for Nonpressure Lines

Test lines for leakage by either infiltration tests or exfiltration tests. Prior to testing for leakage, backfill trench up to at least lower half of pipe. When necessary to prevent pipeline movement during testing, place additional backfill around pipe sufficient to prevent movement, but leaving joints uncovered to permit inspection. When leakage exceeds the allowable amount specified, make satisfactory correction and retest pipeline section in the same manner. When the water table is 60 cm or more above top of pipe at upper end of pipeline section to be tested, measure infiltration using a suitable weir or other acceptable device. When the water table is less than 60 cm above top of pipe at upper end of pipeline section to be tested, make exfiltration test by filling the line to be tested with water so that the head will be at least 1.2 m above top of pipe at upper end of pipeline section being tested. Allow filled pipeline to stand until the pipe has reached its maximum absorption, but not less than 4 hours. After absorption, reestablish the head and measure amount of water needed to maintain this water level during a 2-hour test period. Amount of leakage, as measured by either infiltration or exfiltration test shall not exceed one liter per cm of diameter per hour per 100 m of pipeline. When leakage exceeds the amount specified, make satisfactory correction and retest pipeline section in the same manner as previously specified. Correct all visible leaks regardless of leakage test results.

~~3.3.1.3 Tests for Pressure Lines~~

~~Test pressure lines in accordance with the applicable standards as recommended by pipe manufacturer's installation manual for the respective pressure pipes specified in this specification. For hydrostatic pressure test, use a hydrostatic pressure 345 kPa in excess of the maximum working pressure of the system, but not less than 690 kPa, holding the pressure for a period of not less than one hour. For leakage test, use a hydrostatic pressure not less than the maximum working pressure of the system. Leakage test may be performed at the same time and at the same test pressure as the pressure test.~~

3.3.1.3 Deflection Testing

Perform a deflection test on entire length of installed plastic pipeline on completion of work adjacent to and over the pipeline, including leakage tests, backfilling, placement of fill, grading, paving, concreting, and any other superimposed loads. Deflection of pipe in the installed pipeline under external loads is not to exceed 4.5 percent of the average inside diameter of pipe. Determine whether the allowable deflection has been exceeded by use of a pull-through device or a deflection measuring device.

3.3.1.3.1 Pull-Through Device

This device is to be a spherical, spheroidal, or elliptical ball, a cylinder, or circular sections fused to a common shaft. Space circular sections on the shaft so that the distance from external faces of front and back sections will equal or exceed the diameter of the circular section. Pull-through device may also be of a design recommended by the pipe manufacturer, provided the device meets the applicable requirements specified in this paragraph, including those for diameter of the device, and that the mandrel has a minimum of 9 arms. Ball, cylinder, or circular sections are to conform to the following:

a. A diameter, or minor diameter as applicable, of 95 percent of the average inside diameter of the pipe; tolerance of plus 0.5 percent will be permitted.

b. Homogeneous material throughout, is to have a density greater than 1.0 as related to water at 4 degrees C, and a surface Brinell hardness of not less than 150.

c. Center bored and through-bolted with a 6 mm minimum diameter steel shaft having a yield strength of not less than 483 MPa, with eyes or loops at each end for attaching pulling cables.

d. Suitably Back each eye or loop with a flange or heavy washer such that a pull exerted on opposite end of shaft will produce compression throughout remote end.

3.3.1.3.2 Deflection Measuring Device

Sensitive to 1.0 percent of the diameter of the pipe being tested and be accurate to 1.0 percent of the indicated dimension. Prior approval is required for the deflection measuring device.

3.3.1.3.3 Pull-Through Device Procedure

Pass the pull-through device through each run of pipe, either by pulling it through or flushing it through with water. If the device fails to pass freely through a pipe run, replace pipe which has the excessive deflection and completely retest in same manner and under same conditions.

3.3.1.3.4 Deflection measuring device procedure

Measure deflections through each run of installed pipe. If deflection readings in excess of 4.5 percent of average inside diameter of pipe are obtained, retest pipe by a run from the opposite direction. If retest continues to show a deflection in excess of 4.5 percent of average inside diameter of pipe, replace pipe which has excessive deflection and completely retest in same manner and under same conditions.

3.3.2 Field Tests for Cast-In-Place Concrete

Field testing requirements are covered in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.3.3 Inspection

Check each straight run of pipeline for gross deficiencies by holding a light in a manhole; the light must show a practically full circle of light through the pipeline when viewed from the adjoining end of line.

3.3.3.1 Pre-Installation Inspection

Prior to connecting the new service, perform pre-installation inspection after trenching and layout is complete. Submit pre-installation inspection request for field support at least 14 days in advance. The Installation's Utilities Field Support personnel will perform the pre-installation inspection.

3.3.3.2 Post-Installation Inspection

Perform a post-installation inspection after connection has been made and before the connection is buried. Submit post-installation inspection request for field support at least 14 days in advance. The Installation's Utilities Field Support personnel will perform the post-connection inspection.

-- End of Section --

SECTION 33 40 00

STORM DRAINAGE UTILITIES
02/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

~~JAPANESE INDUSTRIAL STANDARDS (JIS)~~JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 1210 (2020~~09~~) Test Method for Soil Compaction Using a Rammer

JIS A 1214 (2013) Test Method for Soil Density by the Sand Replacement Method

JIS A 5001 (2008) Crushed Stone for Road Construction

JIS A 5005 (2009) Crushed Stone and Manufactured Sand for Concrete

JIS A 5308 (2014) Ready-Mixed Concrete

JIS A 5372 (2016) Precast Reinforced Concrete Products

JIS A 5506 (2008) Manhole Covers for Sewerage Works

JIS G 3101 ~~(2017) Rolled Steels for General Structure (Amendment 1)~~(2020) Rolled Steels for General Structure

JIS H 8641 (20~~07~~21) Hot Dip Galvanized Coatings

JIS K 6739 (2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Drain

JIS K 6741 ~~(2016)~~(2016; R 2021) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes

~~JAPAN SEWAGE WORKS ASSOCIATION (JSWAS)~~JAPAN SEWAGE WORKS ASSOCIATION (JSWA)

JSWAS K-1 (2010) Rigid PVC Pipe for Sewer

~~THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF JAPAN STANDARD (SHASE-S)~~THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF JAPAN (SHASE)

SHASE-S 209 (2009) Manhole Cover

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. ~~Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

~~Crushed Stone for Perforated Pipes~~

~~Geotextile Filter Fabric for Perforated Pipes~~

Leakage Test; G

Determination of Density

Post-Installation Inspection Report; G

SD-08 Manufacturer's Instructions

Placing Pipe

1.3 DELIVERY, STORAGE, AND HANDLING

1.3.1 Delivery and Storage

Materials delivered to site shall be inspected for damage, unloaded, and stored with a minimum of handling. Materials shall not be stored directly on the ground. The inside of pipes and fittings shall be kept free of dirt and debris. Before, during, and after installation, plastic pipe and fittings shall be protected from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer. Solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe shall be stored in accordance with the manufacturer's recommendations and shall be discarded if the storage period exceeds the recommended shelf life. Solvents in use shall be discarded when the recommended pot life is exceeded.

1.3.2 Handling

Materials shall be handled in a manner that ensures delivery to the trench in sound, undamaged condition. Pipe shall be carried to the trench, not dragged.

PART 2 PRODUCTS

2.1 PIPE FOR CULVERTS AND STORM DRAINS

Pipe for culverts and storm drains shall be of the sizes indicated and shall conform to the requirements specified.

~~2.1.1 Concrete Pipe~~

~~Provide reinforced concrete pipe conforming to JIS A 5372, external pressure type, Class 1 or 2, Type B. Provide gaskets and pipe ends for rubber gasket joint conforming to JIS K 6353. Use gaskets suitable for use with sewage.~~

~~2.1.2 Precast Concrete Pipe-Type Gutter~~

~~Japanese manufactured precast concrete product that comes in standard length of 2 m. Locally known as DO pipe or acceptable alternative, and to be used along curb lines to function as gutter with water conveyance underneath. Shall be of the cross section shown on drawing and panels are supplied both with grating and without grating. Traffic loading capacity shall be 25 tons. Jointing materials shall be in accordance with manufacturer's standards.~~

~~2.1.3 Precast Concrete Pipe Culvert Type Ditch~~

~~Japanese manufactured precast concrete product that comes in standard length of 2 m. Locally known as DO pipe or acceptable alternative, and to be used for below grade construction in place of PVC pipe whenever the sand fill layer above the PVC pipe is less than 300 mm. Shall be of the cross section shown on drawing and to be installed where specified on drainage plans. Traffic loading capacity shall be 25 tons. Jointing materials shall be in accordance with manufacturer's standards.~~

2.1.1 Polyvinyl Chloride (PVC) Pipe

a. PVC plastic pipe shall be in conformance with JIS K 6741, Type VP or VU.

b. Fitting for PVC plastic pipe shall be in conformance with JIS K 6739.

c. Jointing material for PVC plastic pipe shall be in conformance with PVC pipe manufacturer's recommendation.

d. PVC branch pipe connectors shall be standard product compatible with the PVC plastic pipe and conforming to JSWAS K-1 and JIS K 6739. Adhesive materials shall be per branch pipe manufacturer's recommendation.

~~2.1.2 Polyethylene (PE) Pipe~~

~~a. Polyethylene pipe shall be in conformance with JIS K 6761.~~

~~b. Jointing material shall conform with polyethylene pipe manufacturer's recommendation.~~

~~2.2 PERFORATED PIPING~~

~~2.2.1 Polyvinyl Chloride (PVC) Pipe~~

~~a. Perforated PVC plastic pipe shall be in conformance with JIS K 6741,Type VP or VU.~~

~~b. Fitting for perforated PVC plastic pipe shall be in conformance with JIS K 6739.~~

~~c. Jointing material for perforated PVC plastic pipe shall be in conformance with PVC pipe manufacturer's recommendation.~~

~~2.2.2 Polyethylene (PE) Pipe~~

~~a. Flexible perforated polyethylene pipe shall be in conformance with JIS K 6761.~~

~~b. Jointing material shall conform with polyethylene pipe manufacturer's recommendation.~~

2.2 OTHER DRAINAGE CONVEYANCE MATERIALS

~~2.2.1 Precast Concrete Free Gradient Ditch~~

~~Manufactured precast concrete product that comes in standard length of 2 m, and to be used as an interceptor trench along curb lines. Typically of inverted U shaped with the flat top functioning as gutter surface with grated inlets while the bottom is open and filled with plain concrete to obtain the desired gradient of the water conveyance underneath. Shall be of the varying cross sections as shown on drawing. Traffic loading capacity shall be 25 tons. Jointing materials shall be in accordance with manufacturer's standards.~~

~~2.2.2 Precast Concrete U-Ditch~~

~~Manufactured precast concrete product conforming to JIS A 5372. Comes in 600 mm panel length and shall be of the cross section shown on drawing. Ditch shall be non-traffic type and with or without cover where specified on drawing.~~

2.2.1 Cast-in-Place Concrete Trench

A reinforced concrete trench of varying cross sections as shown on drawing. Concrete in conformance with JIS A 5308 shall be 24 MPa and shall have frame and grating.

2.3 MISCELLANEOUS MATERIALS

2.3.1 Concrete

Reinforced concrete drainage structures shall use concrete with compressive strength of 24 MPa.

2.3.2 Leveling Mortar

Mortar for setting precast concrete pipe type gutter and culvert pipe ditch (both locally known as DO pipe) shall composed of cement, sand and water mixed in proportion of 1 part cement to 3 parts of sand, sufficient water to produce a workable mixture. Mortar shall be used in the work within one hour after mixing.2.3.3 Leveling Concrete and Concrete Fill

Plain concrete to be used where specified on drainage drawing details shall have a compressive strength of 18 MPa.

2.3.4 Precast Reinforced Concrete Manholes

Approved commercial products as shown on drawings and conforming to JIS A 5372. Joints between precast concrete manhole sections shall be made with flexible watertight, rubber-type gaskets per manhole manufacturer's standards.

2.3.5 Frames, Covers And Grating

2.3.5.1 Manhole Frames and Covers

Shall be ~~Japanese~~ manufacturer's standard product conforming to SHASE-S 209, and shall be of cast iron per JIS A 5506. Size, configuration and loading capacity shall be as indicated on drawings. A letter "D" shall be stamped or cast into covers.

2.3.5.2 Ditch and Inlet Frame and Grating

Shall be ~~Japanese~~ manufacturer's standard product of steel materials and of the size, shape, grating pattern, and loading capacity as indicated on drawings. Grating shall be finished with factory-zinc-coated in accordance with JIS H 8641, and frame shall be finished with factory-bake coated.

2.3.6 Sand Fill And Leveling Sand

Fill around PVC drainage pipes and for setting non-traffic type precast U-ditch shall be cleaned sand conforming to JIS A 5005.

~~2.3.7 Crushed Stone For Perforated Pipes~~

~~Granular aggregates for use in perforated pipes shall be single-sized crushed stone S-20 (#5) of size 13 mm to 20 mm and conforming to JIS A 5001.~~

2.3.7 Gravel Base Course

Crushed stone for base course at concrete structures shall be crusher run conforming to JIS A 5001, RC-40.

~~2.3.8 Flap Gates~~

~~Flap Gates shall be medium or heavy-duty with circular or rectangular opening and double-hinged. Top pivot points shall be adjustable. The seat shall be one-piece cast iron with a raised section around the perimeter of the waterway opening to provide the seating face. The seating face of the seat shall be cast iron or stainless steel. The cover shall be one-piece cast iron with necessary reinforcing rib, lifting eye for manual operation, and bosses to provide a pivot point connection with the links. The seating face of the cover shall be cast iron or stainless steel. Links or hinge arms shall be cast or ductile iron. Holes of pivot points shall be bronze bushed. All fasteners shall be either galvanized steel, bronze or stainless steel.~~

2.4 STEEL LADDER

Steel ladder conforming to JIS G 3101 shall be provided where the depth of the storm drainage structure exceeds 3.66 m. These ladders shall be not less than 406 mm in width, with 19 mm diameter rungs spaced 305 mm apart. The two stringers shall be a minimum 10 mm thick and 63 mm wide. Ladders and inserts shall be galvanized after fabrication in conformance with JIS H 8641.

2.4.1 Manhole Steps

Materials shall conform to JIS G 3101, Type SS 400, galvanized, and of the size and configuration as shown on drawing. Manhole steps are not required in manholes and inlets less than 1.2 m deep.

2.5 RESILIENT CONNECTORS

Flexible, watertight connectors used for connecting pipe to manholes and inlets shall conform to pipe manufacturer's standards.

~~2.6 GEOTEXTILE FILTER FABRIC FOR PERFORATED PIPES~~

~~Submit certification from the manufacturers attesting that the filter fabric meets specification requirements. Provide geotextile that is a nonwoven pervious sheet of polymeric material conforming to JIS L 1908.~~

2.6 Warning Tape

Provide in accordance with requirements as specified in Section 31 00 00 EARTHWORK.

2.7 Cleanouts

Cleanouts installed at vertical PVC downspouts shall use the standard Y-fitting with the compatible PVC cleanout cover conforming to JIS K 6739. Surface cleanout at paved and turfed areas shall be cast iron cover and fitting adaptable for connection to the lower PVC pipe. Traffic loading capacity for surface cleanout shall be as required for manhole covers located within the same ground surface condition.

~~2.8 EROSION CONTROL RIP RAP~~

~~Provide non-erodible rock not exceeding 375 mm in its greatest dimension and choked with sufficient small rocks to provide a dense mass with a minimum thickness of [200 mm] [as indicated].~~

PART 3 EXECUTION

3.1 INSTALLATION OF PIPE CULVERTS, STORM DRAINS, AND DRAINAGE STRUCTURES

Excavation of trenches, and for appurtenances and backfilling for culverts and storm drains, shall be in accordance with the applicable portions of Section 31 00 00 EARTHWORK, ~~31 23 00.00 20 EXCAVATION AND FILL~~ and the requirements specified below.

3.1.1 Trenching

The width of trenches at any point below the top of the pipe shall be not greater than the outside diameter of the pipe plus 500 mm to permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Sheeting and bracing, where required, shall be placed within the trench width as specified, without any overexcavation. Where trench widths are exceeded, redesign with a resultant increase in cost of stronger pipe or special installation procedures will be necessary. Cost of this redesign and increased cost of pipe or installation shall be borne by the Contractor without additional cost to the Government.

3.1.2 Removal of Rock

Rock in either ledge or boulder formation shall be replaced with suitable materials to provide a compacted earth cushion having a thickness between unremoved rock and the pipe of at least 200 mm or 13 mm for each meter of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Where bell-and-spigot pipe is used, the cushion shall be maintained under the bell as well as under the straight portion of the pipe. Rock excavation shall be as specified and defined in Section 31 00 00 EARTHWORK ~~31 23 00.00 20 EXCAVATION AND FILL~~ .

3.1.3 Removal of Unstable Material

Where wet or otherwise unstable soil incapable of properly supporting the pipe, as determined by the Contracting Officer, is unexpectedly encountered in the bottom of a trench, such material shall be removed to the depth required and replaced to the proper grade with select granular material, compacted as provided in paragraph BACKFILLING. When removal of unstable material is due to the fault or neglect of the Contractor while performing shoring and sheeting, water removal, or other specified requirements, such removal and replacement shall be performed at no additional cost to the Government.

3.2 BEDDING

The bedding surface for the pipe shall provide a firm foundation of uniform density throughout the entire length of the pipe.

~~3.2.1 Concrete Pipe Requirements~~

~~Concrete pipe shall be bedded with sand material minimum 200 mm in depth in trenches with soil foundation. Depth of sand bedding in trenches with rock foundation shall be 13 mm in depth per 300 mm of depth of fill, minimum depth of bedding shall be 200 mm up to maximum depth of 600 mm. Where concrete pipes are laid in deep trenches or where traffic loads are expected, the entire pipe line length shall be supported uniformly by concrete foundation or cradle.~~

~~3.2.2 Precast, Concrete Pipe Type Gutter/Ditch and Free-Gradient Ditch~~

~~These type of Precast drainage materials shall be laid using leveling mortar over a concrete foundation base.~~

~~3.2.3 Precast Concrete U-Ditch~~

~~Use leveling sand for non-traffic areas and leveling mortar over concrete foundation for traffic areas.~~

~~3.2.4 Plastic Pipe~~

~~Use clean sand for bedding, haunching and initial backfill for PVC and PE pipes. Minimum depth of bedding shall be 200 mm~~

3.3 PLACING PIPE

Each pipe shall be thoroughly examined before being laid; defective or damaged pipe shall not be used. Plastic pipe shall be protected from exposure to direct sunlight prior to laying, if necessary to maintain adequate pipe stiffness and meet installation deflection requirements. Pipelines shall be laid to the grades and alignment indicated. Proper facilities shall be provided for lowering sections of pipe into trenches. Lifting lugs in vertically elongated pipe shall be placed in the same vertical plane as the major axis of the pipe. Pipe shall not be laid in water, and pipe shall not be laid when trench conditions or weather are unsuitable for such work. Diversion of drainage or dewatering of trenches during construction shall be provided as necessary. Deflection of installed flexible pipe shall not exceed 4.5 percent of the average inside diamater of pipe.

3.3.1 ~~Concrete and~~ Plastic Pipes

Laying shall proceed upgrade with spigot ends of bell-and-spigot pipe pointing in the direction of the flow.

~~3.3.2 Precast Concrete Pipe Type Gutter/Ditch, Free Gradient Ditch and U-Ditch~~

~~Lay the drainage materials in conformance with product manufacturer's instructions.~~

3.3.2 Multiple Culverts

Where multiple lines of pipe are installed, adjacent sides of pipe shall be at least half the nominal pipe diameter or 1 meter apart, whichever is less.

3.4 JOINTING

~~3.4.1 Concrete Pipes and Other Precast Concrete Drainage Conveyance Materials.~~

~~3.4.1.1 Flexible Watertight Joints~~

~~Gaskets and jointing materials shall be as recommended by the particular manufacturer. Cements, or adhesives shall be clean and dry. Gaskets and jointing materials shall be protected from the sun, blowing dust, and other deleterious agents at all times. Gaskets and jointing materials shall be inspected before installing the pipe; any loose or improperly affixed gaskets and jointing materials shall be removed and replaced. The pipe shall be aligned with the previously installed pipe, and the joint pushed home. If, while the joint is being made the gasket becomes visibly dislocated the pipe shall be removed and the joint remade.~~

3.4.1 Plastic Piping

Install pipes and fittings per pipe manufacturer's installation manual.

3.5 DRAINAGE STRUCTURES

3.5.1 Manholes and Inlets

Construction shall be of reinforced concrete, precast reinforced concrete, precast concrete segmental blocks complete with frames and covers or gratings; and with fixed galvanized steel ladders or manhole steps where indicated. Pipe connections to concrete manholes and inlets shall be made with flexible, watertight connectors in conformance with pipe manufacturer.

3.5.2 Walls and Headwalls

Construction shall be as indicated.

3.6 Metal Steps

Individual metal steps shall be adequately anchored to precast manhole walls per manhole or metal step manufacturer's installation standards.

3.7 STEEL LADDER INSTALLATION

Ladder shall be adequately anchored to the wall by means of steel inserts spaced not more than 1.83 m vertically, and shall be installed to provide at least 152 mm of space between the wall and the rungs. The wall along the line of the ladder shall be vertical for its entire length.

3.8 BACKFILLING

3.8.1 Backfilling Pipe in Trenches

After the pipe has been properly bedded clean sand shall be placed along both sides of pipe in layers not exceeding 150 mm in compacted depth. The backfill shall be brought up evenly on both sides of pipe for the full length of pipe. The fill shall be thoroughly compacted under the haunches of the pipe. Each layer shall be thoroughly compacted with mechanical tampers or rammers. This method of filling and compacting shall continue until the fill has reached an elevation of at least 300 mm above the top of the pipe. The remainder of the trench shall be backfilled with select material from excavation or borrow and compacted by spreading and rolling or compacted by mechanical rammers or tampers in layers not exceeding 150 mm. Tests for density shall be made as necessary to ensure conformance to the compaction requirements specified below. Where it is necessary, in the opinion of the Contracting Officer, that sheeting or portions of bracing used be left in place, the contract will be adjusted accordingly. Untreated sheeting shall not be left in place beneath structures or pavements.

3.8.2 Backfilling Pipe in Fill Sections

For pipe placed in fill sections, backfill material and the placement and compaction procedures shall be as specified below. The fill material shall be uniformly spread in layers longitudinally on both sides of the pipe, not exceeding 150 mm in compacted depth, and shall be compacted by rolling parallel with pipe or by mechanical tamping or ramming. Prior to commencing normal filling operations, the crown width of the fill at a height of 300 mm above the top of the pipe shall extend a distance of not less than twice the outside pipe diameter on each side of the pipe or 4 m, whichever is less. After the backfill has reached at least 300 mm above the top of the pipe, the remainder of the fill shall be placed and thoroughly compacted in layers not exceeding 150 mm. Use sand material for this entire region of backfill for flexible and concrete pipe installations.

3.8.3 Movement of Construction Machinery

When compacting by rolling or operating heavy equipment parallel with the pipe, displacement of or injury to the pipe shall be avoided. Movement of construction machinery over a culvert or storm drain at any stage of construction shall be at the Contractor's risk. Any damaged pipe shall be repaired or replaced.

3.8.4 Compaction

3.8.4.1 General Requirements

Cohesionless materials include gravels, gravel-sand mixtures, sands, and gravelly sands. Cohesive materials include clayey and silty gravels, gravel-silt mixtures, clayey and silty sands, sand-clay mixtures, clays, silts, and very fine sands. When results of compaction tests for moisture-density relations are recorded on graphs, cohesionless soils will show straight lines or reverse-shaped moisture-density curves, and cohesive soils will show normal moisture-density curves.

3.8.4.2 Minimum Density

Backfill over and around the pipe and backfill around and adjacent to drainage structures shall be compacted at the approved moisture content to the following applicable minimum density, which will be determined as specified below.

a. Under airfield and heliport pavements, paved roads, streets, parking areas, and similar-use pavements including adjacent shoulder areas, the density shall be not less than 90 percent of maximum density for cohesive material and 95 percent of maximum density for cohesionless material, up to the elevation where requirements for pavement subgrade materials and compaction shall control.

b. Under unpaved or turfed traffic areas, density shall not be less than 90 percent of maximum density for cohesive material and 95 percent of maximum density for cohesionless material.

c. Under nontraffic areas, density shall be not less than that of the surrounding material.

3.9 FIELD PAINTING

3.9.1 Cast-Iron Covers and Frames

After installation, clean cast-iron, not buried in masonry or concrete, of mortar, rust, grease, dirt, and other deleterious materials to bare metal and apply a coat of bituminous paint.

3.10 FIELD QUALITY CONTROL

3.10.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

3.10.1.1 Leakage Test

Lines shall be tested for leakage by exfiltration tests prior to completing backfill. Prior to exfiltration tests, the trench shall be backfilled up to at least the lower half of the pipe. If required, sufficient additional backfill shall be placed to prevent pipe movement during testing, leaving the joints uncovered to permit inspection. Visible leaks encountered shall be corrected regardless of leakage test results. When the water table is 600 mm or more above the top of the pipe at the upper end of the pipeline section to be tested, infiltration shall be measured using a suitable weir or other device acceptable to the Contracting Officer. An exfiltration test shall be made by filling the line to be tested with water so that a head of at least 600 mm is provided above both the water table and the top of the pipe at the upper end of the pipeline to be tested. The filled line shall be allowed to stand until the pipe has reached its maximum absorption, but not less than 4 hours. After absorption, the head shall be reestablished. The amount of water required to maintain this water level during a 2-hour test period shall be measured. Leakage as measured by the exfiltration test shall not exceed 9 mL per mm in diameter per 100 meters of pipeline per hour.

3.10.1.2 Determination Of Density

Testing shall be performed by an approved commercial testing laboratory or by the Contractor subject to approval. Tests shall be performed in sufficient number to ensure that specified density is being obtained. Laboratory tests for moisture-density relations shall be made in accordance withJIS A 1210 except that mechanical tampers may be used provided the results are correlated with those obtained with the specified hand tamper. Field density tests shall be determined in accordance with JIS A 1214.

3.10.1.3 Deflection Testing

Conduct deflection test no sooner than 30 days after completion of final backfill and compaction testing. Clean or flush all lines prior to testing. Perform a deflection test on entire length of installed flexible pipeline upon completion of work adjacent to and over the pipeline, including backfilling, placement of fill, grading, paving, placement of concrete, and any other superimposed loads. Deflection of pipe in the installed pipeline under external loads shall not exceed limits in paragraph PLACING PIPE above as percent of the average inside diameter of pipe. Use a laser profiler or mandrel to determine if allowable deflection has been exceeded.

3.10.1.3.1 Laser Profiler

Inspect pipe interior with laser profiling equipment. Utilize low barrel distortion video equipment for pipe sizes 1.22 m or less. Use a camera with suitable lighting to allow a clear picture of the entire periphery of the pipe interior. Center the camera in the pipe both vertically and horizontally. The camera must be able to pan and tilt to a 90 degree angle with the axis of the pipe rotating 360 degrees. Use equipment to move the camera through the pipe that will not obstruct the camera's view or interfere with proper documentation of the pipe's condition. The video image shall be clear, focused, and relatively free from roll static or other image distortion qualities that would prevent the reviewer from evaluating the condition of the pipe. For initial post installation inspections for pipe sizes larger than 1.22 m, a visual inspection shall be completed of the pipe interior.

3.10.1.3.2 Mandrel

Pass the mandrel through each run of pipe by pulling it by hand. If deflection readings in excess of the allowable deflection of average inside diameter of pipe are obtained, stop and begin test from the opposite direction. The mandrel must meet the Pipe Manufacture's recommendations and the following requirements. Provide a Mandrel that is rigid, nonadjustable, has a minimum of 9 fins, pulling rings at each end, and is engraved with the nominal pipe size and mandrel outside diameter. The mandrel must be 4.5 percent less than the certified-actual pipe diameter for Plastic Pipe. The Government will verify the outside diameter(OD)of the Contractor provided mandrel through the use of Contractor provided proving rings.

3.10.2 Inspection

3.10.2.1 Post-Installation Inspection

Visually inspect each segment of concrete pipe for alignment, settlement, joint separations, soil migration through the joint, cracks, buckling, bulging and deflection. An engineer must evaluate all defects to determine if any remediation or repair is required.

~~3.10.2.1.1 Concrete~~

~~Cracks with a width greater than 0.25 mm. An engineer must evaluate all pipes with cracks with a width greater than 0.25 mm but less than 2.5 mm to determine if any remediation or repair is required.~~

3.10.2.1.1 Flexible Pipe

Check each flexible pipe (PE, PVC)for rips, tears, joint separations, soil migration through the joint, cracks, localized bucking, bulges, settlement and alignment.

3.10.2.1.2 Post-Installation Inspection Report

The deflection results and final post installation inspection report must include: a copy of all video taken, pipe location identification, equipment used for inspection, inspector name, deviation from design, grade, deviation from line, deflection and deformation of flexible pipe, inspector notes, condition of joints, condition of pipe wall (e.g. distress, cracking, wall damage dents, bulges, creases, tears, holes, etc.).

3.10.3 Repair Of Defects

3.10.3.1 Leakage Test

When leakage exceeds the maximum amount specified, correct source of excess leakage by replacing damaged pipe and gaskets and retest.

3.10.3.2 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diameter of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing.

3.10.3.3 Inspection

Replace pipe or repair defects indicated in the Post-Installation Inspection Report.

~~3.10.3.3.1 Concrete~~

~~Replace pipes having cracks with a width greater than 2.5 mm.~~

3.10.3.3.1 Flexible Pipe

Replace pipes having cracks or splits.

3.11 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and external loads during construction.

3.12 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested.

-- End of Section --

SECTION 33 61 14

EXTERIOR BURIED PREINSULATED WATER PIPING
02/10

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

JAPANESE INDUSTRIAL STANDARDS (JIS)

JIS B 2220	(2012) Steel Pipe Flanges
JIS B 2240	(2006) Copper Alloy Pipe Flanges
JIS B 2311	(2024) Steel Butt-Welding Pipe Fittings for Ordinary Use
JIS B 2316	(2017) Steel Socket-Welding Pipe Fittings
JIS G 3452	(2019) Carbon Steel Pipes for Ordinary Piping
JIS G 4107	(2022) Alloy Steel Bolting Materials for High Temperature Service
JIS H 3300	(2018) Copper and Copper Alloy Seamless Pipes and Tubes
JIS H 3401	(2001) Pipe Fittings of Copper and Copper Alloys
JIS K 6380	(2014) Rubber Packing Material -- Classification of Physical Properties
JIS K 6741	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipes
JIS K 7013	(2009) Fibre Reinforced Plastic Pipes (Amendment 1)
JIS Z 3282	(2017) Soft Solder -- Chemical Compositions and Forms
JIS Z 3821	(2018) Standard Qualification Test and Acceptance Requirements for Welding Technique of Stainless Steel

1.2 SYSTEM DESCRIPTION

Provide ~~[~~new ~~and modify existing]~~ exterior buried factory-prefabricated preinsulated water piping system to the first piping connection aboveground or within each building complete and ready for operation. Piping system includes ~~[hot domestic water piping,] [recirculating hot~~

~~domestic water piping,] [~~chilled water piping~~,] [chilled-hot (dual temperature) water piping,] [hot water piping,]~~ and related work~~ [from heat exchanges to each building]~~. ~~[Hot domestic water piping within each building is specified under Section 22 00 00 PLUMBING, GENERAL PURPOSE.] [~~ Chilled water piping, chilled-hot water piping, and hot water piping within each building is specified under~~]~~ ~~[~~Section 23 64 26 CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS~~]~~.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Factory-prefabricated preinsulated water piping system

Preinsulated plastic pipe field joints

Show layout of piping system. Drawings must have Professional Engineer Seal.

SD-03 Product Data

Pipe, fittings, and end connections

Factory-prefabricated preinsulated water piping system

~~Plastic reinforced thermosetting resin (RTR) piping~~

SD-07 Certificates

Certification of welders' qualifications

SD-08 Manufacturer's Instructions

Installation manual for buried factory-prefabricated preinsulated water piping system

1.4 QUALITY ASSURANCE

1.4.1 Certification of Welders' Qualifications

Submit prior to site welding of steel piping; certifications must be not more than one year old.

PART 2 PRODUCTS

2.1 BURIED FACTORY-PREFABRICATED PREINSULATED WATER PIPING SYSTEM

Piping (pipe, fittings, and end connections) system must be suitable for working pressure of 862 kPag at 121 degrees C, except plastic polyvinyl chloride (PVC) chilled water piping must be suitable for working pressure of 862 kPag at 24 degrees C. Piping system must withstand H-20 highway loading with 600 mm of compacted backfill over top of conduit. Mark each section of conduit with fabricator's name, product identification, and publications to which the items conform. Provide each section of carrier pipe including factory-applied insulation and conduit, with waterproof conduit ends at both ends of each section of carrier pipe, except for piping systems which have the field joints insulated and covered with waterproof shrink sleeves.

2.1.1 Factory-Applied Insulation

Polyurethane or polyisocyanate insulation, minimum density of 27.2 kg per cubic meter, rated for not less than 121 degrees C, completely filling space between carrier pipe and conduit.

2.1.2 Factory-Applied Conduit

Conduit material, size, and thickness must be as follows:

Carrier Pipe (mm)	Minimum Conduit Size (mm)	Minimum Conduit Thickness (mm)
50 mm	100 mm	1.5 mm
75 mm	150 mm	1.5 mm
100 mm	200 mm	2.0 mm
150 mm	250 mm	2.5 mm
200 mm	300 mm	3.0 mm
250 mm	350 mm	3.0 mm

a. Plastic PVC pipe conduit: ~~ASTM D1784~~JIS K 6741, Class 12454-B compound extruded seamless PVC plastic pipe.

b. Plastic RTR pipe conduit: ~~ASTM D2996~~JIS K 7013, filament-wound, fiberglass RTR plastic pipe, without liner.

c. Plastic RTR factory lay-up conduit: Conduit must be machine-applied continuous rovings of fiberglass strands saturated with isophthalic polyester or epoxy resin filament wound in helical pattern directly to the outer surface of the pipe insulation. In lieu of minimum conduit size for each size of carrier pipe, provide minimum of 25 mm thick insulation for 50 mm carrier pipe and provide minimum of 38 mm thick insulation for 75 mm and larger carrier pipe.

2.1.3 Factory-Applied End Seals

Provide watertight end seal, or factory lay-up type end seal between carrier pipe and conduit. Provide sufficient surface bonding area between carrier pipe and conduit to ensure permanent watertight end seal suitable for use with temperature limits of carrier pipe.

2.1.4 Factory-Prefabricated Carrier Piping

Pipe, fittings, flanges, and couplings must be marked with manufacturer's name, product identification, and publication to which items conform. Carrier piping must be as specified in this section. Buried carrier pipe connections between straight sections of pipe beyond 1.5 mexterior of buildings may be manufacturer's standard O-ring connections designed to absorb pipe expansion and contraction at working pressure of 862 kPag with no leakage. Connections at elbows and tees must be other than O-ring connections.

2.2 CARRIER PIPING

2.2.1 Copper Tubing

Provide copper tubing for ~~hot domestic water piping, recirculating hot domestic water piping, ~~chilled water piping~~,~~ and chilled-hot water piping~~, and hot water piping~~.

a. Copper tubing: Provide ~~ASTM B88M~~JIS H 3300, Type L or M copper tubing for buried factory-prefabricated preinsulated piping and for aboveground piping. Provide ~~ASME B16.18 or ASME B16.22~~JIS H 3401 solder joint fittings, unions, and flanges; provide adapters as required.

b. Solder for copper tubing: Provide ~~ASTM B32~~JIS Z 3282, 95-5 tin-antimony solder or provide Plumbing Code approved lead-free solder.

c. Flanged connections: Provide ~~ASME B16.24~~JIS B 2240, Class 150, solder joint flat face flanged connections.

d. O-ring connections: Provide between straight sections of pipe beyond 1.5 m of exterior of buildings.

2.2.2 Steel Piping

Provide steel piping for chilled water piping~~,~~ and chilled-hot water piping~~, and hot water piping~~.

a. Steel pipe: Provide ~~ASTM A53/A53M~~JIS G 3452, Type E (electric-resistance welded, Grade A or B), ~~ASTM A53/A53M~~JIS G 3452, Type S (seamless, Grade A or B)~~, or ASTM A106/A106M (seamless, Grade A or B)~~. Provide Weight Class STD (Standard) or Schedule No. 40 black steel pipe for welding end connections. Provide Weight Class XS (Extra Strong) or Schedule No. 80 black steel pipe for threaded end connections.

b. Steel pipe fittings: Provide ~~ASME B16.9~~JIS B 2311 buttwelding fittings of the same material and weight as the piping in which fittings are installed. Provide ~~ASME B16.11~~JIS B 2316 socket welding fittings.

c. Steel pipe flanges: Provide ~~ASME B16.5~~JIS B 2220, Class 150 flanges.

d. O-ring connections: Provide between straight sections of pipe beyond 1.5 m of exterior of buildings.

~~2.2.3 Plastic Reinforced Thermosetting Resin (RTR) Piping~~

~~Provide plastic RTR piping for hot domestic water piping, recirculating hot domestic water piping, chilled water piping, chilled-hot water piping, and hot water piping.~~

~~a. Plastic carrier pipe, fittings, and adhesive: Provide plastic carrier piping conforming to the Federal Agency Approved Brochure. Pipe, fittings, and adhesive must be supplied by same manufacturer. Pipe, fittings, flanges, and couplings must have end connections of the adhesive bell and spigot type. Threaded piping, including pipe, fittings, flanges, and couplings, will not be permitted.~~

~~b. Flanged connections: Provide flat face flanged connections between plastic piping and metal piping. Plastic flanges must be suitable for connecting to ASME Class 150 flanges.~~

~~c. Plastic RTR piping sizes: When piping sizes other than 50, 75, 100, 150, 200 mm are indicated, provide next larger piping size. The connecting system piping must be of the same size or increased to meet next size of RTR piping.~~

~~2.2.4 Plastic PVC Piping~~

~~Provide plastic PVC piping only for chilled water piping.~~

~~a. Plastic PVC carrier pipe, fittings, and cement: ASTM D1785pipe, ASTM D2466 socket type fittings, and ASTM D2564solvent cement must be supplied by the same manufacturer. Pipe, fittings, flanges, and couplings must have solvent cement socket end connections, except piping beyond 1.5 m outside of buildings must have O-ring connections. Plastic PVC piping must be suitable for working pressure of 862 kPag at 24 degrees C.~~

~~b. Flanged connections: Provide flat face flanged connections between plastic piping and metal piping. Plastic flanges must be suitable for connecting to ASME Class 150 flanges.~~

~~c. O-ring connections: Provide between straight sections of pipe beyond 1.5 m of exterior of buildings.~~

2.3 FLANGED CONNECTIONS

Provide ASME Class 150 flat face flanged connections in accordance to JIS.

a. Gaskets: ~~ASTM D1330~~JIS K 6380, except Shore A durometer hardness must be 55 to 65, 3 mm thick ethylene propylene. Provide one piece factory cut full-face gaskets.

b. Bolts: ~~ASTM A193/A193M~~JIS G 4107, Grade B7. Extend minimum of two full threads beyond nut with bolts tightened to required torque.

c. Nuts: ~~ASTM A194/A194M~~JIS G 4107, Grade 7, with Teflon coated threads.

d. Washers: Provide galvanized steel flat circular washers under bolt heads and nuts.

e. Electrically isolating (insulating) gaskets for connections between metal flanges: Provide ~~ASTM D229~~ electrical insulating material of

1000 ohms minimum resistance in accordance to Japanese testing standards. Provide one piece factory cut insulating gaskets between flanges. Provide silicon-coated fiberglass insulating sleeves between bolts and holes in flanges; bolts may have reduced shanks of diameter not less than diameter at root of threads. Provide 3 mm thick high-strength insulating washers next to flanges and provide stainless steel flat circular steel washers over insulating washers and under bolt heads and nuts. Provide bolts 13 mm longer than standard length to compensate for thicker insulating gaskets and washers under bolt heads and nuts.

2.4 BURIED WARNING AND IDENTIFICATION TAPE

Provide detectable aluminum foil plastic backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape in rolls, 75 mm minimum width, color coded for the utility involved with warning and identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning and identification must read "CAUTION BURIED PREINSULATED WATER PIPING BELOW" or similar wording. Use permanent code and letter coloring unaffected by moisture and other substances contained in trench backfill material.

2.5 CONCRETE THRUST BLOCKS

Provide concrete thrust blocks as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE. Concrete must be of 27.6 MPa minimum 28 day compressive strength, air-entrained admixture (0.13 kg per cubic meter) with water-reducing admixture (0.81 kg per cubic meter).

2.6 PIPE SLEEVES

Provide where piping passes entirely through walls and floors. Provide sleeves of sufficient length to pass through entire thickness of walls and floors. Provide 25 mm minimum clearance between exterior of piping or pipe insulation, and interior of sleeve or core-drilled hole. Firmly pack space with mineral wool insulation. Seal space at both ends of sleeve or core-drilled hole with plastic waterproof cement which will dry to a firm but pliable mass, or provide mechanically adjustable segmented elastomeric seal. In fire walls and fire floors, seal both ends of sleeves or core-drilled holes with UL listed fill, void, or cavity material.

a. Sleeves in masonry and concrete walls and floors: Provide hot-dip galvanized steel, ductile-iron, or cast-iron sleeves. Core drilling of masonry and concrete may be provided in lieu of sleeves when cavities in the core-drilled hole are grouted smooth.

b. Sleeves in other than masonry and Concrete walls and floors: Provide 0.5 mm galvanized steel sheet.

2.7 ESCUTCHEON PLATES

Provide split hinge type metal plates for piping entering walls and floors in exposed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

PART 3 EXECUTION

3.1 INSTALLATION

Installation of exterior buried factory-prefabricated preinsulated water piping systems must be in accordance with manufacturer's installation manual. Welding of steel piping including qualification of welders must be in accordance with ~~ASME B31.1~~JIS Z 3821, metallic arc process. Deviations must not be permitted unless authorized in writing by Contracting Officer. Install piping straight and true to bear evenly on sand bedding material. Installation and field assembly of plastic RTR piping must be in accordance with the Federal Agency Approved Brochure.

a. Cleaning of piping: Keep interior and ends of new piping and existing piping affected by the Contractor's operations, cleaned of water and foreign matter during installation by means of plugs or other approved methods. When work is not in progress, securely close open ends of pipe and fittings to prevent entry of water and foreign matter. Inspect piping before placing into position.

b. Demolition: Remove materials so as not to damage materials which are to remain. Replace existing work damaged by the Contractor's operations with new work of the same construction.

3.2 FIELD JOINTS

a. Carrier piping joints without concrete anchor: Pressure test and approve piping joints. Provide joints with polyurethane or polyisocyanate insulation of same type and thickness as insulation on carrier piping. Provide waterproof shrink sleeves to cover insulation and overlap not less than 150 mm of each end of conduit section.

b. Carrier piping joints with concrete anchor: Pressure test and approve piping joints. Provide each elbow and tee with concrete anchors (thrust blocks). Provide waterproof end seals between carrier piping and conduit adjacent to each carrier pipe fitting. Encase carrier pipe fitting and at least 50 mm of each end of conduit with a minimum of 150 mm of concrete.

3.3 BURIED FACTORY-PREFABRICATED PREINSULATED PIPE INSTALLATION

a. Assembly and alignment: Assemble carrier pipe and fittings according to manufacturer's installation manual; assemble plastic RTR piping in accordance with the Federal Agency Approved Brochure. Maintain proper alignment during assembly of joints.

b. Bedding: Accurately grade trench bedding with a minimum of 150 mm of manufactured or natural sand. Backfill sand to a minimum of 150 mm above and below conduit. Lay bedding to firmly support conduit along entire length.

c. Concrete thrust blocks: Encase each elbow and tee of carrier pipe in thrust block with minimum of 0.28 square meter of thrust-bearing surface cast against undisturbed soil, minimum pipe-to-bearing surface single dimension of 250 mm perpendicular to bearing surface, and minimum volume of– 0.25 cubic meter, except as indicated otherwise. Disturbed soil under and around thrust blocks must be compacted.

3.4 FIELD QUALITY CONTROL

Before final acceptance of work, test each system to demonstrate compliance with contract requirements. Thoroughly flush and clean piping before placing in operation. Flush piping at minimum velocity of 2.4 meters per second. Correct defects in the work and repeat tests until work is in compliance with contract requirements. Furnish potable water, electricity, instruments, connecting devices, and personnel for tests.

a. Field tests of carrier piping: Do not cover carrier piping joints with insulation or concrete anchors (thrust blocks), until carrier piping joints pass field tests.

b. Hydrostatic pressure test: Test piping system at 1379 kPag for minimum holding period of 2 hours during which time pressure must not drop more than 28 kPa; test plastic RTR piping in accordance with Federal Agency Approved Brochure. Pressure drop greater than 28 kPag corrected for temperature variation constitutes failure. Valve off piping system and disconnect method of piping system pressurization before starting the 2 hour pressure holding period. During hydrostatic pressure test, examine piping system for leaks. Repair leaking joints, replace damaged and porous pipe and fittings with new materials, and repeat tests.

c. Thrust blocks: If O-ring connections are used, provide temporary thrust blocks prior to hydrostatic pressure testing of piping system. Place bedding and backfill around center portion of piping system, leaving thrust blocks and field joints clear for observation. After successful completion of hydrostatic pressure test, cast concrete thrust blocks.

d. Field inspections: Prior to initial operation, inspect piping system for compliance with drawings, specifications, and manufacturer's submittals.

~~3.5 DISINFECTION~~

~~Disinfect new hot domestic water piping under Section 22 00 00 PLUMBING, GENERAL PURPOSE.~~

-- End of Section --

SECTION 33 71 02

UNDERGROUND ELECTRICAL DISTRIBUTION
02/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS), Ministry of Land, Infrastructure and Transport (MLIT), Japan Electrical Safety Inspection Associations or Japan Power Cable Accessories Association (JCAA) standards for non-Japanese standards, as approved by the Contracting Officer's representative.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2 (2023) National Electrical Safety Code

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS A 5506 (2008) Manhole Covers for Sewerage Works

JIS A 5372 (2016) Precast Reinforced Concrete Products

JIS C 0365 (2007) Protection Against Electric Shock - Common Aspects for Installation and Equipment

JIS C 2336 (2012) Pressure-sensitive polyvinyl chloride tapes for electrical purposes

JIS C 2338 (2012) Polyester adhesive tape for electrical insulation

JIS C 2805 (2010) Crimp terminal for copper wire

JIS C 2806 (2003; R 2018) Bare crimping sleeve for copper wire

JIS C 2810 (1995; R 2021) General rules on non-separable type wire connectors for interior wiring

JIS C 3101 (1994; R 2021) Hard-drawn copper wires for electrical purposes

JIS C 3102 (1984) Annealed Copper Wires for Electrical Purposes

JIS C 3105 (1994; R 2021) Hard-drawn copper stranded conductors

JIS C 3341 ~~(2000; R 2021) Polyvinyl chloride insulated drop service wires~~(2000)

	Polyvinyl Chloride Insulated Drop Service Wires
JIS C 3605	(2022) 600 V Polyethylene Insulated Cables~~, Type CV~~
JIS C 3606	(2022) High-Voltage Cross-Linked Polyethylene Insulated Cables, Type CV or CE
JIS C 8305	(2019) Rigid Steel Conduits
JIS C 8330	(1999; R 2019) Fittings for rigid metal conduits
JIS C 8340	(1999; R 2019) Boxes And Box Covers For Rigid Metal Conduits
JIS C 8350	(1999; R 2014) Fittings for pliable metal conduits
JIS C 8380	(2009; R 2019) Plastic coated steel pipes for cable-ways
JIS C 8432	(2019) Fittings of unplasticized polyvinyl chloride (PVC-U) conduits
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor
JIS C 60364-6	(2010; R 2019) Low-voltage electrical installations -- Part 6: Verification
JIS C 60695-2-4-0	(1995) Fire hazard testing Part 2: Test methods -- Section 4/sheet 0: Diffusion type and premixed type flame test methods
JIS K 6743	(2016) Unplasticized Poly (Vinyl Chloride) (PVC-U) Pipe Fittings for Water Supply

ELECTRICAL SAFETY INSPECTION ASSOCIATIONS

Denki Hoan Kyoukai	Japan Standard for Acceptance Testing and Inspections

JAPAN POWER CABLE ACCESSORIES ASSOCIATION (JCAA)

JCAA A 302	(1992) Indoor termination connection for 6600V cross-linked polyethylene insulated power cable
JCAA A 303	(1992) Outdoor termination connection for 6600V cross-linked polyethylene insulated power cable

JCAA A 305	(1992) Straight connection for 6600 V cross-linked polyethylene insulated power cable

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 0168-4	(2010) Calculation of allowable current for power cables under 33 kV- Part 4: Allowable current for 22 kV, 33 kV crosslinked polyethylene cables
JCS 0501	(2014) Allowable current calculation for power cables above 66kV
JCS 1226	(2003) Soft Stranded Wire

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)

MLIT DSKKS	Denki Setsubi Kouji Kanri Shishin (DSKKS) Electrical Construction Supervision Guidelines
MLIT ESS	(2019) MLIT Electrical Standard Specification (ESS)

THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF JAPAN (SHASE)

SHASE-S 209	(2009) Manhole Cover

~~UNDERWRITERS LABORATORIES (UL)~~UL SOLUTIONS (UL)

UL 94	~~(2023; Seventh Edition) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances~~(2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

UNIFIED FACILITIES CRITERIA (UFC)

UFC 3-310-04	(2013; with Change 1, 2016) Seismic Design of Buildings

~~[~~1.2 SYSTEM DESCRIPTION

Items provided under this section must be specifically suitable for the following service conditions. Seismic details must ~~[~~conform to UFC 3-310-04, "Seismic Design for Buildings"~~ and Sections 13 48 00 [SEISMIC] BRACING FOR MISCELLANEOUS EQUIPMENT and 26 05 48.00 10 SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT] [be as indicated]~~.

a. Fungus Control ~~[_____]~~required

b. Altitude ~~[_____]~~40 m.

c. Ambient Temperature ~~[_____]~~40 degrees C.

d. Frequency ~~[_____]~~50Hz

e. Ventilation ~~[_____]~~Provide natural and/or mechanical ventilation as required to maintain equipment within manufacturer's rated operating limits.

f. Seismic Parameters ~~[_____]~~As indicated in the structural drawings and seismic design criteria in accordance with UFC 3-310-04.

g. Humidity Control ~~[_____]~~Design for relative humidity up to 95 percent, non-condensing

h. Corrosive Areas ~~[_____]~~Yes

~~i. [_____]~~

~~]~~1.3 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.4 DEFINITIONS

a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined.

b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.

c. Japan voltage range categories are defined as follows:
Low Voltage (Voltages less than 1kV)
High Voltage (Voltages 1kV thru 7kV)
Extra High Voltage (Voltages over 7kV)

d. In the text of this section, "~~[~~High~~][Extra-High]~~ voltage cable splices," and "~~[~~High~~][Extra-High]~~ voltage cable joints" are used interchangeably and have the same meaning.

~~[d. Underground structures subject to aircraft loading are indicated on the drawings.~~

~~]~~1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for Contractor Quality Control approval.][~~for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~] Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast underground structures; G~~[, [_____]]~~

SD-03 Product Data

~~[~~High~~][Extra-High]~~ voltage cable; G~~[, [_____]]~~

~~[~~High~~][Extra-High]~~ voltage cable joints; G~~[, [_____]]~~

~~[~~High~~][Extra-High]~~ voltage cable terminations; G~~[, [_____]]~~

~~[~~ ~~Live end caps; G[, [_____]]~~

~~]~~ Precast concrete structures; G~~[, [_____]]~~

Sealing Material

Pulling-In Irons

Manhole frames and covers; G~~[, [_____]]~~

Handhole frames and covers; G~~[, [_____]]~~

~~[~~ ~~Frames and Covers for Airfield Facilities; G[, [_____]]~~

~~][~~ ~~Ductile Iron Frames and Covers for Airfield Facilities; G[, [_____]]~~

~~]~~ Composite/fiberglass handholes; G~~[, [_____]]~~

Cable supports (racks, arms and insulators); G~~[, [_____]]~~

~~[~~ ~~Protective Devices and Coordination Study; G[, [_____]]~~

~~][~~ ~~The study must be submitted with protective device equipment submittals. No time extension or similar contract modifications will be granted for work arising out of the requirements for this study. Approval of protective devices proposed must be based on recommendations of this study. The Government must not be held responsible for any changes to equipment, device ratings, settings, or additional labor for installation of equipment or devices ordered or procured prior to approval of the study.~~

~~]~~ SD-06 Test Reports

Field Acceptance Checks and Tests;

1.6 QUALITY ASSURANCE

1.6.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

a. Material description (i.e., f'c and Fy)

b. Manufacturer's printed assembly and installation instructions

c. Design calculations

d. Reinforcing shop drawings in accordance with Manufacturer's recommendations.

e. Plans and elevations showing opening and pulling-in iron locations and details

1.6.2 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and JIS C 0365 unless more stringent requirements are specified or indicated.

1.6.3 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.3.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.6.3.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable, unless specified otherwise.

PART 2 PRODUCTS

2.1 CONDUIT, DUCTS, AND FITTINGS

2.1.1 Rigid Metal Conduit or Type G

JIS C 8305, Type G. Diameter of conduit shall be as indicated.

2.1.1.1 Rigid Metallic Conduit, PVC Coated or Type G; LL or LT

JIS C 8380, Type G; LL or LT.

2.1.2 Intermediate Metal Conduit or Type C

JIS C 8305, Type C.

2.1.2.1 Intermediate Metal Conduit, PVC Coated or Type C; LL or LT

JIS C 8380, Type C; LL or LT.

2.1.3 Plastic Conduit or Unplasticized Polyvinyl Chloride for Direct Burial and Riser Applications

~~[~~As indicated~~] [JIS C 3653, Type FEP] [or] JIS C 8430 for Type HIVE conduit of diameter of less than 100mm and JIS K 6741 for Type HIVP conduit diameter or 100mm and larger~~.

2.1.4 Plastic Duct or Unplasticized Polyvinyl Chloride for Concrete Encasement

Provide~~[~~ as indicated~~][or JIS C 8430 for Type VE conduit of diameter of less than 100mm and JIS K 6741 for Type VP conduit diameter or 100mm and larger]~~.

2.1.5 Innerduct

Provide corrugated ~~[~~or solid wall~~]~~ polyethylene (PE) or PVC innerducts, or fabric-mesh innerducts, with pullwire. Size as indicated.

2.1.6 Duct Sealant

UL 94 and JIS C 60695-2-4-0. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 2 degrees C to 35 degrees C. Cured sealant must withstand temperature ranges of -29 degrees C to 93 degrees C without loss of function.

2.1.7 Fittings

2.1.7.1 Metal Fittings

JIS C 8330 and JIS C 8350.

2.1.7.2 PVC or Unplasticized Polyvinyl Chloride Conduit Fittings

~~[~~JIS C 8432 for diameters less than 100mm~~][~~ JIS K 6743 for diameters 100mm or larger~~]~~.

~~[~~2.1.7.3 Outlet Boxes for Steel Conduit

Outlet boxes for use with rigid or flexible steel conduit must be cast-metal cadmium or zinc-coated if of ferrous metal with gasketed closures and must conform to JIS C 8340.

~~]~~2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of applicable codes and standards, including listing requirements~~[, or in accordance with JIS C 3362 or JIS C 3367]~~. Wires and cables manufactured more than ~~[24][~~12~~]~~ months prior to date of delivery to the site are not acceptable. Service entrance conductors must conform to JIS C 3341.

2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and ~~[THHN/THWN][~~ EM-CE~~]~~ unless otherwise noted. Conductors 5.5 sqmm and smaller must be solid. Conductors 8 sqmm and larger must be stranded.~~[~~ Conductors 14 sqmm and smaller must be copper. Conductors 22 sqmm and larger may be either copper or aluminum, at the Contractor's option. Do not substitute aluminum for copper if the equivalent aluminum conductor size would exceed 250 sqmm. When the Contractor chooses to use aluminum for conductors 22 sqmm and larger, the Contractor must: increase the conductor size to have the same ampacity as the copper size indicated; increase the conduit and pull box sizes to accommodate the larger size aluminum conductors in accordance with applicable codes and standards; ensure that the pulling tension rating of the aluminum conductor is sufficient; relocate equipment, modify equipment terminations, resize equipment, and resolve to the satisfaction of the Contracting Officer problems that are direct results of the use of aluminum conductors in lieu of copper.~~][All conductors must be copper.]~~

2.2.2 Conductor Material

Unless specified or indicated otherwise or required, wires in conduit, other than service entrance, must be 600-volt,conforming to Type EM-CE, EM-CET or EM-ECEQ conforming to JIS C 3605. Copper conductors must be annealed copper complying with JCS 1226 and JIS C 3102 and JIS C 3105.~~[Aluminum conductors must be Type AA-8000 aluminum conductors complying with JIS C 3108 and JIS C 3109, and must be of an aluminum alloy listed or labeled by UL as "component aluminum-wire stock (conductor material). Type 1350 is not acceptable. Intermixing of copper and aluminum conductors in the same raceway is not permitted.]~~

~~[~~2.2.3 Jackets

Multiconductor cables must have an overall PVC outer jacket.

~~][2.2.4 Direct Buried~~

~~Single-conductor [and multi-conductor]cables must be of a type identified for direct burial.~~

~~][~~2.2.4 In Duct

~~Cables must be single-conductor cable.[Cables in factory-installed, coilable-plastic-duct assemblies where extra physical protection is required.]~~Cables must be as indicated in paragraphs above.

~~]~~2.2.5 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags, or approved equal, in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Conductors must be color coded. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves,or colored electrical tape. Control circuit terminations must be properly identified. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals must be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems must be as follows:

a. Three-phase - Primary

(1) Phase A - red

(2) Phase B - white

(3) Phase C - blue

b. Three-phase - Secondary

(1) Phase A - red

(2) Phase B - black

(3) Phase C - blue

c. 480/277 volt, three phase

(1) Phase A - brown

(2) Phase B - orange

(3) Phase C - yellow

d. Single phase: Black and red

d. On three-phase, four-wire delta system, high leg must be orange, as required.

2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Must provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

a. For use with copper conductors: JIS C 2805, JIS C 2806 and JIS C 2810.

b. For use with aluminum conductors: JIS C 2805, JIS C 2806 and JIS C 2810. For connecting aluminum to copper, connectors must be the circumferentially compressed, metallurgically bonded type.

2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water.

2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor, outdoor, direct-burial or submerged locations.

2.5 ~~[~~HIGH~~][EXTRA-HIGH]~~ VOLTAGE CABLE

Cable (conductor) sizes are designated bymillimeters (mm) and square milimeters (sqmm). Conductor and conduit sizes indicated are for copper conductors unless otherwise noted. Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout cable length. Wires and cables manufactured more than ~~[24][~~12~~]~~ months prior to date of delivery to the site are not acceptable. Provide single conductor type cables unless otherwise indicated.

2.5.1 Cable Configuration

Provide ~~[Type CVT, conforming to JCS 4516 for 3.3kV][~~Type EM-CET, conforming to JIS C 3606 for 6.6kV~~][~~ or ~~][~~Type EM-CET, conforming to JCS 0168-4 for 22kV and 33kV~~][~~ Type CV-CVT, conforming to JCS 0501 for 66kV and 77kV~~]~~.~~[concentric neutral underground distribution cable][metallic armored cables, consisting of three-conductor, multi-conductor cables, with insulation and shielding, as specified, using [a galvanized steel][an aluminum] interlocked tape armor and thermoplastic jacket].~~ Provide cables manufactured for use in~~[~~ duct~~][~~ or~~][direct burial]~~ applications~~[as indicated]~~. Cable must be rated ~~[3.3kV][5 kV][~~6.6kV ~~][15 kV][22kV][25 kV][28 kV][33kV][35 kV][66kV][77kV][as indicated]~~ with ~~[[100][~~133~~]~~ percent insulation level~~][[ ] insulation resistance]~~.

2.5.2 Conductor Material

Provide concentric-lay-stranded, Class B~~[compact round]~~ conductors. Provide~~[aluminum alloy conductors complying with JIS C 3108 and JIS C 3109][aluminum alloy cables, 3/4 hard minimum for regular concentric and compressed stranding or for compacted stranding][~~ soft drawn copper cables complying with JCS 1226 and JIS C 3102 and JIS C 3105 for regular concentric and compressed stranding or JIS C 3606 for compact stranding~~]~~.

2.5.3 Insulation

Provide ~~[~~ethylene-propylene-rubber (EPR) insulation~~][tree-retardant cross-linked thermosetting polyethylene (XLP) insulation]~~.

2.5.4 Shielding

Cables rated for 2 kV and above must have a semiconducting conductor shield, a semiconducting insulation shield, and an overall copper~~[~~ tape~~][~~ or~~][~~ wire~~]~~ shield for each phase.

~~2.5.5 Neutrals~~

~~[Neutral conductors must be [copper][aluminum], employing the same insulation and jacket materials as phase conductors, except that a 600-volt insulation rating is acceptable.][Concentric neutrals conductors must be copper, having a combined ampacity [equal to][1/3 of] the phase conductor ampacity rating.][For high impedance grounded neutral systems, the neutral conductors from the neutral point of the transformer or generator to the connection point at the impedance must utilize [copper][aluminum] conductors, employing the same insulation level and construction as the phase conductors.~~

~~]~~2.5.5 Jackets

Provide cables with a ~~[~~PVC~~][_____]~~ jacket.~~[Direct buried cables must be rated for direct burial.][Provide type UD cables with an overall jacket.]~~ ~~[~~ Provide PVC jackets with a separator that prevents contact with underlying semiconducting insulating shield.~~]~~

2.6 ~~[~~HIGH~~][EXTRA-HIGH]~~ VOLTAGE CABLE TERMINATIONS

Provide indoor terminator/outdoor terminations with skirts. ~~[JCAA A 202 for 3.3kV][~~JCAA A 302 and JCAA A 303 for 6.6kV~~][JCAA A 501 and JCAA A 502 for 22kV and 33kV][JEC 3408 for 66kV and 77kV]~~; of the molded elastomer, prestretched elastomer, or heat-shrinkable elastomer. Acceptable elastomers are track-resistant silicone rubber or track-resistant ethylene propylene compounds, such as ethylene propylene rubber or ethylene propylene diene monomer. Separable insulated connectors may be used for apparatus terminations, when such apparatus is provided with suitable bushings. Terminations, where required, must be provided with mounting brackets suitable for the intended installation and with grounding provisions for the cable shielding, metallic sheath, or armor. Terminations must be provided in a kit, including: skirts, stress control terminator, ground clamp, connectors, lugs, and complete instructions for assembly and installation. Terminations must be the product of one manufacturer, suitable for the type, diameter, insulation class and level, and materials of the cable terminated. Do not use separate parts of copper or copper alloy in contact with aluminum alloy parts in the construction or installation of the terminator.

2.6.1 Cold-Shrink Type

Terminator must be a one-piece design, utilizing the manufacturer's latest technology, where high-dielectric constant (capacitive) stress control is integrated within a skirted insulator made of silicone rubber. Termination must not require heat or flame for installation. Termination kit must contain all necessary materials (except for the lugs). Termination must be designed for installation in low or highly contaminated indoor and outdoor locations and must resist ultraviolet rays and oxidative decomposition.

2.6.2 Heat Shrinkable Type

Terminator must consist of a uniform cross section heat shrinkable polymeric construction stress relief tubing and environmentally sealed outer covering that is nontracking, resists heavy atmospheric contaminants, ultra violet rays and oxidative decomposition. Provide heat shrinkable sheds or skirts of the same material. Termination must be designed for installation in low or highly contaminated indoor or outdoor locations.

~~[~~2.6.3 Separable Insulated Connector Type

Provide connector with steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material. Provide connectors of the loadbreak or deadbreak type as indicated, of suitable construction for the application and the type of cable connected, and that include cable shield adaptors. Provide external clamping points and test points. Separable connectors must not be used in manholes/handholes.

~~[~~ a. 200 Ampere loadbreak connector ratings: Voltage: ~~[~~15 kV, 95 kV BIL ~~][25 kV, 125 kV BIL][35 kV, 150 kV BIL]~~. Short time rating: 10,000 rms symmetrical amperes.
~~]~~
~~[~~ b. 600 Ampere deadbreak connector ratings: Voltage: ~~[15 kV, 95 kV BIL][~~ 25 kV, 125 kV BIL~~][35 kV, 150 kV BIL]~~. Short time rating: 25,000 rms symmetrical amperes.~~[~~ Connectors must have 200 ampere bushing interface~~[for surge arresters][~~ as indicated~~]~~.~~]~~
~~]~~
~~[c. Provide[[one][_____] set[s] of three grounding elbows][and][[one][_____] set[s] of three feed-thru inserts]. Deliver [grounding elbows][and][feed-thru inserts] to the Contracting Officer.~~
~~]~~
~~[~~ ~~d~~c. Install one set of faulted circuit indicators on the test points of each set of separable insulated connectors. Indicators must be self powered; with automatic trip with mechanical flag indication upon overcurrent followed by loss of system voltage, and automatic reset upon restoration of system voltage. Indicators must be compact, sealed corrosion resistant construction with provision for hotstick installation and operation.

~~]]~~2.7 ~~[~~HIGH~~][EXTRA-HIGH]~~ VOLTAGE CABLE JOINTS

Provide joints (splices) in accordance with ~~[JCAA A 203 for 3.3kV][~~ JCAA A 305 for 6.6kV~~][JCAA A 503 for 22kV and 33 kV][JEC 3408 for 66kV and 77kV]~~ suitable for the rated voltage, insulation level, insulation type, and construction of the cable. Joints must be certified by the manufacturer for waterproof, submersible applications. Upon request, supply manufacturer's design qualification test report in accordance with ~~[JCAA A 203 for 3.3kV][~~JCAA A 305 for 6.6kV~~][JCAA A 503 for 22kV and 33 kV][JEC 3408 for 66kV and 77kV]~~. Connectors for joint must be tin-plated electrolytic copper, having ends tapered and having center stops to equalize cable insertion.

2.7.1 Heat-Shrinkable Joint

Consists of a uniform cross-section heat-shrinkable polymeric construction with a linear stress relief system, a high dielectric strength insulating material, and an integrally bonded outer conductor layer for shielding. Replace original cable jacket with a heavy-wall heat-shrinkable sleeve with hot-melt adhesive coating.

2.7.2 Cold-Shrink Rubber-Type Joint

Joint must be of a cold shrink design that does not require any heat source for its installation. Splice insulation and jacket must be of a one-piece factory formed cold shrink sleeve made of black EPDM rubber.

Splice must be packaged three splices per kit, including complete installation instructions.

2.8 TELECOMMUNICATIONS CABLING

Provide telecommunications cabling in accordance with Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP).

~~[2.9 LIVE END CAPS~~

~~Provide live end caps using a "kit" including a heat-shrinkable tube and a high dielectric strength, polymeric plug overlapping the conductor. End cap must conform to applicable portions of [JCAA A 202 for 3.3kV][JCAA A 302 and JCAA A 303 for 6.6kV][JCAA A 501 and JCAA A 502 for 22kV and 33kV][JEC 3408 for 66kV and 77kV].~~

~~]~~2.9 TAPE

2.9.1 Insulating Tape

JIS C 2336 and JIS C 2338, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

2.9.2 Buried Warning and Identification Tape

Provide detectable tape in accordance with Section ~~[31 23 00.00 20 EXCAVATION AND FILL][~~31 00 00 EARTHWORK~~]~~.

2.9.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than 0.762 mm thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

2.10 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of 890 N.

2.11 GROUNDING AND BONDING

2.11.1 Driven Ground Rods

Provide ~~[copper-clad steel ground rods conforming to JIS C 60364-5-54][~~ solid copper ground rods conforming to JIS C 60364-5-54~~][solid stainless steel ground rods]~~ not less than~~[~~ 19 mm~~]~~ in diameter by~~[~~ 3.1 m~~]~~ in length. Sectional type rods may be used for rods 6 meters or longer.

2.11.2 Grounding Conductors

Stranded-bare copper conductors must conform to JIS C 3105 soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to JIS C 3101 for sizes 8 sqmm and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

2.12 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with 20 MPa minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be 30 MPa minimum 28-day compressive strength unless specified otherwise.

2.13 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to JIS A 5372. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

2.13.1 Cast-In-Place Concrete Structures

Concrete must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE.~~[~~ Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.~~][Concrete block must conform to Section 04 20 00, MASONRY.][Concrete block is not allowed in areas subject to aircraft loading.]~~

2.13.2 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures may be provided in lieu of cast-in-place subject to the requirements specified below. Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

2.13.2.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day compressive strength of not less than 30 MPa. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

2.13.2.2 Design for Precast Structures

In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

a. Angle of Internal Friction (phi) = 0.523 rad

b. Unit Weight of Soil (Dry) = 1760 kg/m^3, (Saturated) = 2080 kg/m^3

c. Coefficient of Lateral Earth Pressure (Ka) = 0.33

d. Ground Water Level = 915 mm below ground elevation

e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact. Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading.

f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 26,700 N.

g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for thatparticular component.

h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

2.13.2.3 Construction

Structure top, bottom, and wall must be of a uniform thickness of not less than 150 mm. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 305 mm vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 150 mm from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 13 bars around window openings. Provide drain sumps a minimum of 305 mm in diameter and 100 mm deep for precast structures.

2.13.2.4 Joints

Provide tongue-and-groove joints on mating edges of precast components.

Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

2.13.3 Manhole Frames and Covers

Provide cast iron frames and covers for manholes conforming to SHASE-S 209 and JIS A 5506. Cast the words "ELECTRIC" or "TELECOMMUNICATIONS" in the top face of power and telecommunications manhole covers, respectively.

2.13.4 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Steel covers must be rolled-steel floor plate having an approved antislip surface. Hinges must be of ~~[~~stainless steel with bronze hinge pin~~] [wrought steel]~~, 125 by 125 mm by approximately 4.75 mm thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have nonremovable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised antislip surface, by grinding or other approved method.

~~[2.13.5 Frames and Covers for Airfield Facilities~~

~~Fabricate frames and covers for airfield use of standard commercial grade steel welded by qualified welders. Covers must be of rolled steel floor plate having an approved anti-slip surface. Steel frames and covers must be hot dipped galvanized after fabrication.~~

~~][2.13.6 Ductile Iron Frames and Covers for Airfield Facilities~~

~~At the contractor's option, ductile iron covers and frames designed for a minimum proof load of 45,000 kg may be provided in lieu of the steel frames and covers indicated. Covers must be of the same material as the frames (i.e. ductile iron frame with ductile iron cover, galvanized steel frame with galvanized steel cover). Perform proof loading in accordance with applicable codes and standards. Proof loads must be physically stamped into the cover. Provide the Contracting Officer copies of previous proof load test results performed on the same frames and covers as proposed for this contract. Modify the top of the structure to accept the ductile iron structure in lieu of the steel structure indicated. The finished structure must be level and non-rocking, with the top flush with the surrounding pavement.~~

~~]~~2.13.5 Brick for Manhole Collar

Provide sewer and manhole brick as required per applicable codes and standards.

2.13.6 Composite/Fiberglass Handholes and Covers

Provide handholes and covers of polymer concrete, reinforced with heavy weave fiberglass with a design load (Tier rating) appropriate for or greater than the intended use. All covers are required to have the Tier level rating embossed on the surface and this rating must not exceed the design load of the box.

2.14 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

The metal portion of racks and arms must be zinc-coated after fabrication.

2.14.1 Cable Rack Stanchions

The wall bracket or stanchion must be 100 mm by approximately 38 mm by 4.76 mm channel steel, or 100 mm by approximately 25 mm glass-reinforced nylon with recessed bolt mounting holes, 1220 mm long (minimum) in manholes. Slots for mounting cable rack arms must be spaced at 200 mm intervals.

2.14.2 Rack Arms

Cable rack arms must be steel or malleable iron or glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 200 mm and a maximum of 305 mm.

2.14.3 Insulators

Insulators for metal rack arms must be dry-process glazed porcelain. Insulators are not required for nylon arms.

2.15 CABLE TAGS IN MANHOLES

Provide tags for each power cable located in manholes. The tags must be polyethylene. Do not provide handwritten letters. The first position on the power cable tag must denote the voltage. The second through sixth positions on the tag must identify the circuit. The next to last position must denote the phase of the circuit and include the Greek "phi" symbol. The last position must denote the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

2.15.1 Polyethylene Cable Tags

Provide tags of polyethylene that have an average tensile strength of 22.4 MPa; and that are 2 millimeter thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 77 degrees C. Provide 1.3 mm (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag. Ties must have a minimum loop tensile strength of 778.75 N. The cable tags must have black block letters, numbers, and symbols 25 mm high on a yellow background. Letters, numbers, and symbols must not fall off or change positions regardless of the cable tags' orientation.

2.16 ~~[~~HIGH~~][EXTRA-HIGH]~~ VOLTAGE ABOVE GROUND CABLE TERMINATING CABINETS

Cable terminating cabinets must be hook-stick operable, deadfront construction. Provide cabinets with ~~[~~200 A. loadbreak junctions and elbow-type separable loadbreak connectors, cable parking stands, and grounding lugs~~][600 A. dead-break junctions and elbow-type separable dead-break connectors, cable parking stands, and grounding lugs]~~. Provide cable terminating equipment ~~[~~as indicated~~]~~.

Ratings at ~~[~~50~~][60]~~ Hz must be:

Nominal voltage (kV)	~~[_____]~~6.6
Rated maximum voltage (kV)	~~[[3.3][6.6][15][22][2~~ 7.2
Rated continuous current (A)	~~[[200][~~600~~]]~~
One-second short-time current-carrying capacity (kA)	~~[_____]~~25
BIL (kV)	~~[_____]~~60

2.17 LOW VOLTAGE ABOVE GROUND TERMINATION PEDESTAL

Provide copolymer polypropylene, low voltage above ground termination pedestal manufactured through an injection molding process. Pedestals must resist fertilizers, salt air environments and ultra-violet radiation. Pedestal top must be imprinted with a "WARNING" and "ELECTRIC" identification. Pedestal must contain ~~[three][~~four~~]~~ lay-in six port connectors. Connectors must be dual rated for aluminum or copper, and capable of terminating conductors ranging from ~~[~~5.5 sqmm to 250 sqmm~~]~~. Protect each connector with a clear, hard lexan (plastic) cover. Pedestal must be provided with rust-free material and stainless steel hardware. Pedestal must be lockable.

~~2.18 PROTECTIVE DEVICES AND COORDINATION~~

~~Provide protective devices and coordination as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.~~

2.18 SOURCE QUALITY CONTROL

2.18.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 305 mm long with a 65.5 mm outside diameter, and a 3.175 mm thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. The arc must be directed toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Each sample assembly must be tested at three unrelated points. Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

2.18.2 ~~[~~HIGH~~][EXTRA-HIGH]~~ Voltage Cable Qualification and Production Tests

Results of qualification and production tests as applicable for each type of ~~[~~high~~][extra-high]~~ voltage cable.

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of applicable codes and standards and IEEE C2. In addition to these requirements, install telecommunications in accordance with MLIT ESS.

3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

~~[3.3 CABLE INSTALLATION PLAN AND PROCEDURE~~

~~Obtain from the manufacturer an installation manual or set of instructions which addresses such aspects as cable construction, insulation type, cable diameter, bending radius, cable temperature limits for installation, lubricants, coefficient of friction, conduit cleaning, storage procedures, moisture seals, testing for and purging moisture, maximum allowable pulling tension, and maximum allowable sidewall bearing pressure. [Prepare a checklist of significant requirements][Perform pulling calculations and prepare a pulling plan] and submit along with the manufacturer's instructions in accordance with SUBMITTALS. Install cable strictly in accordance with the cable manufacturer's recommendations and the approved installation plan.~~

~~[Calculations and pulling plan must include:~~

~~a. Site layout drawing with cable pulls identified in numeric order of expected pulling sequence and direction of cable pull.~~

~~b. List of cable installation equipment.~~

~~c. Lubricant manufacturer's application instructions.~~

~~d. Procedure for resealing cable ends to prevent moisture from entering cable.~~

~~e. Cable pulling tension calculations of all cable pulls.~~

~~f. Cable percentage conduit fill.~~

~~g. Cable sidewall bearing pressure.~~

~~h. Cable minimum bend radius and minimum diameter of pulling wheels used.~~

~~i. Cable jam ratio.~~

~~j. Maximum allowable pulling tension on each different type and size of~~

~~conductor.~~

~~k. Maximum allowable pulling tension on pulling device.~~

~~]]~~3.3 UNDERGROUND FEEDERS SUPPLYING BUILDINGS

Terminate underground feeders supplying building at a point 1525 mm outside the building and projections thereof, except that conductors must be continuous to the terminating point indicated. Coordinate connections of the feeders to the service entrance equipment with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide ~~[PVC, Type EPC-40][IMC][RGS][Type G][Type FEP][~~Type VP or ~~][~~Type VE~~][Type HIVP][Type HIVE]~~ conduit from the supply equipment to a point 1525 mm outside the building and projections thereof. Protect ends of underground conduit with plastic plugs until connections are made.

~~[~~ Encase the underground portion of the conduit in a concrete envelope and bury as specified for underground duct with concrete encasement.

~~]~~3.4 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to manufacturer's recommendations. Locate duct entrances and windows in the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Steel and iron must be formed to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

3.4.1 Cast-In-Place Concrete Structures

~~[~~Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.~~][Provide concrete block conforming to Section 04 20 00 MASONRY.][Concrete block is not allowed in areas subject to aircraft loading.]~~

3.4.2 Precast Concrete Construction

Set commercial precast structures on 150 mm of level, 90 percent compacted granular fill, 19 mm to 25 mm size, extending 305 mm beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally conform to the manufacturer's instructions.

3.4.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor the irons must be a minimum of 150 mm from the edge of the sump, and in the walls the irons must be located within 150 mm of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 150 mm of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 150 mm clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 915 mm length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 75 mm from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 100 mm and must be designed to withstand a minimum pulling-in load of 26,700 N. Irons must be hot-dipped galvanized after fabrication.

3.4.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 915 mm apart, and provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 420 mm apart with the end rack being no further than 305 mm from the adjacent wall. Methods of anchoring cable racks must be as follows:

a. Provide a 15 mm diameter by 125 mm long anchor bolt with 75 mm foot cast in structure wall with 50 mm protrusion of threaded portion of bolt into structure. Provide 15 mm steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.

b. Provide concrete channel insert with a minimum load rating of 1192 kg per meter. Insert channel must be steel of the same length as "vertical rack channel;" channel insert must be cast flush in structure wall. Provide 15 mm steel nuts in channel insert to receive 15 mm diameter by 75 mm long steel, square head anchor bolts.

c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 365 kg load rating. Provide 15 mm diameter by 75 mm long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

3.4.5 Field Painting

Cast-iron frames and covers not buried in concrete or masonry must be cleaned of mortar, rust, grease, dirt and other deleterious materials, and given a coat of bituminous paint.

~~[~~3.5 DIRECT BURIAL CABLE SYSTEM

~~Cables must be buried directly in the earth below the frostline [as indicated][to the requirements of IEEE C2 and JIS C 0365, whichever is more stringent].~~

3.5.1 Trenching

Excavate trenches for direct-burial cables to provide a minimum cable cover of 610 mm below finished grade for power conductors operated at 600 volts or less, and 765 mm below finished grade for over 600 volts in accordance with IEEE C2 and JIS C 0365. When rock is encountered, remove to a depth of at least 75 mm below the cable and fill the space with sand or clean earth free from particles larger than 6 mm. Bottoms of trenches must be smooth and free of stones and sharp objects. Where materials in bottoms of trenches are other than sand, a 75 mm 3 inch layer of sand must be laid first and compacted to approximate densities of surrounding firm soil. Trenches must be not less than ~~[150][~~200~~]~~ mm wide, and must be in straight lines between cable markers.~~[~~ Cable plows must not be used.~~]~~ Bends in trenches must have a radius ~~[of not less than 915 mm][~~consistent with the cable manufacturer's published minimum cable bending radius for the cable installed~~]~~.

3.5.2 Cable Installation

Unreel cables along the sides of or in trenches and carefully place on sand or earth bottoms. Pulling cables into direct-burial trenches from a fixed reel position is not permitted, except as required to pull cables through conduits under paving or railroad tracks.

Where two or more cables are laid parallel in the same trench, space cables laterally at not less than 75 mm apart, except that communication cable must be separated from power cable by a minimum distance of 305 mm.

Where direct-burial cables cross under roads or other paving exceeding 1.5 m in width, such cables must be installed in~~[~~ concrete-encased~~]~~ ducts. Where direct-burial cables cross under railroad tracks, such cables must be installed in ~~[~~reinforced concrete-encased ducts~~][ducts installed through rigid galvanized steel sleeves]~~. Ducts must extend at least 1.5 m beyond each edge of any paving and at least 1.5 m beyond each side of any railroad tracks. Cables may be pulled into duct from a fixed reel where suitable rollers are provided in the trench. Where direct burial cable transitions to duct-enclosed cable, direct-burial cables must be centered in duct entrances, and a waterproof nonhardening mastic compound must be used to facilitate such centering. If paving or railroad tracks are in place where cables are to be installed, coated rigid steel conduits driven under the paving or railroad tracks may be used in lieu of concrete-encased ducts. Prevent damage to conduit coatings by providing ferrous pipe jackets or by predrilling. Where cuts are made in any paving, the paving and subbase must be restored to their original condition. Where cable is placed in duct(e.g. under paved areas, roads, or railroads), slope ducts to drain.

3.5.3 Splicing

Provide cables in one piece without splices between connections except where the distance exceeds the lengths in which cables are manufactured.~~[~~ Where splices are required, provide splices designed and rated for direct burial.~~][~~ Where splices are required, install splices only in maintenance manholes/handholes or cabinets/pedestals.~~]~~

3.5.4 Bends

Bends in cables must have an inner radius not less than those specified for the type of cable, per manufacturer's recommendation.

3.5.5 Horizontal Slack

Leave approximately 915 mm of horizontal slack in the ground on each end of cable runs, on each side of connection boxes, and at points where connections are brought above ground. Where cable is brought above ground, leave additional slack to make necessary connections.~~[~~ Enclose splices in lead-sheathed or armored cables in split-type cast-iron splice boxes; after completion of the connection, fill with insulating filler compound and tightly clamp the box.~~]~~

3.5.6 Identification Slabs~~[~~ or Markers~~]~~

Provide a slab at each change of direction of cable, over the ends of ducts or conduits which are installed under paved areas and roadways~~[~~, over the ends of ducts or conduits stubbed out for future use~~][~~, and over each splice~~]~~. Identification slabs must be of concrete, approximately 500 mm square by 150 mm thick and must be set flat in the ground so that top surface projects not less than 20 mm, nor more than 30 mm above ground. Concrete must have a compressive strength of not less than 20 MPa and have a smooth troweled finish on exposed surface. Inscribe an identifying legend such as "electric cable," "telephone cable," "splice," or other applicable designation on the top surface of the slab before concrete hardens. Inscribe circuit identification symbols on slabs as indicated. Letters or figures must be approximately 50 mm high and grooves must be approximately 6 mm in width and depth. Install slabs so that the side nearest the inscription on top must include an arrow indicating the side nearest the cable. Provide color, type and depth of warning tape as specified in Section ~~[31 23 00.00 20 EXCAVATION AND FILL][~~31 00 00 EARTHWORK~~]~~.

~~]~~3.6 UNDERGROUND CONDUIT AND DUCT SYSTEMS

3.6.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 100 sqmm bare copper grounding conductor for ~~[~~high~~][extra-high]~~-voltage distribution duct banks as indicated in drawings. Bond grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of ~~[75 mm][~~100 mm~~]~~ per 30 m. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Provide ducts with end bells whenever duct lines terminate in structures.

Perform changes in ductbank direction as follows:

a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.

b. The minimum manufactured bend radius must be 450 mm for ducts of less than 80 mm diameter, and 900 mm for ducts 80 mm or greater in diameter.

c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of 7.6 m for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections: 30 degrees.

3.6.2 Treatment

Ducts must be kept clean of concrete, dirt, or foreign substances during construction. Field cuts requiring tapers must be made with proper tools and match factory tapers. A coupling recommended by the duct manufacturer must be used whenever an existing duct is connected to a duct of different material or shape. Ducts must be stored to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Ducts must be thoroughly cleaned before being laid. Plastic ducts must be stored on a flat surface and protected from the direct rays of the sun.

3.6.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 75 mm and larger, draw a flexible testing mandrel approximately 305 mm long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 75 mm, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

3.6.4 Jacking and Drilling Under Roads and Structures

Conduits to be installed under existing paved areas which are not to be disturbed, and under roads and railroad tracks, must be zinc-coated, rigid steel, jacked into place. Where ducts are jacked under existing pavement, rigid steel conduit must be installed because of its strength. To protect the corrosion-resistant conduit coating, predrilling or installing conduit inside a larger iron pipe sleeve (jack-and-sleeve) is required. For crossings of existing railroads and airfield pavements greater than 15 m in length, the predrilling method or the jack-and-sleeve method will be used. Separators or spacing blocks must be made of steel, concrete, plastic, or a combination of these materials placed not farther apart than 1.2 m on centers. ~~[~~Hydraulic jet method must not be used.~~]~~

~~[~~3.6.5 Galvanized Conduit Concrete Penetrations

Galvanized conduits which penetrate concrete (slabs, pavement, and walls) in wet locations must be PVC coated and must extend from at least 50 mm within the concrete to the first coupling or fitting outside the concrete (minimum of 150 mm from penetration).

~~]~~3.6.6 Multiple Conduits

Separate multiple conduits by a minimum distance of 75 mm~~[~~, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of ~~[300]~~ mm~~]~~. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 3050 mm of conduit assembly.

3.6.7 Conduit Plugs and Pull Rope

New conduit indicated as being unused or empty must be provided with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 915 mm of slack at each end of unused or empty conduits.

3.6.8 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 610 mm below finished grade. Provide not less than 75 mm clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 75 mm, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 6.25 mm sieve. The first 150 mm layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 75 to 150 mm layers. Provide color, type and depth of warning tape as specified in Section ~~[31 23 00.00 20 EXCAVATION AND FILL][~~31 00 00 EARTHWORK~~]~~.

3.6.8.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 75 mm concrete cover around ducts. Concrete encasement must extend at least 1525 mm beyond the edges of paved areas and roads, and 3660 mm beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 610 mm below finished grade~~[~~, and under railroad tracks not less than 1270 mm below the top of the rails~~]~~.

~~[~~3.6.8.2 Directional Boring

HDPE conduits must be installed below the frostline and as specified herein.

~~[For distribution voltages greater than 1000 volts and less than 34,500 volts, depths to the top of the conduit must not be less than 1220 mm in pavement-covered areas and not less than 3050 mm in non-pavement-covered areas.][~~ For distribution voltages less than 1000 volts, depths to the top of the conduit must not be less than 1220 mm in pavement- or non-pavement-covered areas.~~][~~ For branch circuit wiring less than 600 volts, depths to the top of the conduit must not be less than 610 mm in pavement- or non-pavement-covered areas.~~]~~

~~]~~3.6.9 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 450 mm below finished grade~~[~~, except under roads and pavement, concrete envelope must be not less than 610 mm below finished grade~~][~~, and under railroad tracks not less than 1270 mm below the top of the rails~~]~~. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and must provide at least 75 mm of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 75 mm. Before pouring concrete, anchor duct bank assemblies to prevent the assemblies from floating during concrete pouring. Anchoring must be done by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly.~~[~~ Provide steel reinforcing in the concrete envelope as indicated.~~][Provide color, type and depth of warning tape as specified in Section [31 00 00 EARTHWORK][31 23 00.00 20 EXCAVATION AND FILL.]]~~

3.6.9.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 300 mm in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

3.6.9.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and ~~[~~extend into~~][~~ bend out to tie into the reinforcing of~~]~~ the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

3.6.9.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and ~~[extend into][~~bend out to tie into the reinforcing of~~]~~ the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

3.6.9.4 Connections to Existing Ducts

Where connections to existing duct banks are indicated, excavate the banks to the maximum depth necessary. Cut off the banks and remove loose concrete from the conduits before new concrete-encased ducts are installed. Provide a reinforced concrete collar, poured monolithically with the new duct bank, to take the shear at the joint of the duct banks. ~~[Remove existing cables which constitute interference with the work.][~~ Abandon in place those no longer used ducts and cables which do not interfere with the work.~~]~~

3.6.9.5 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs. Fit concrete envelope of a partially completed duct bank with reinforcing steel extending a minimum of 610 mm back into the envelope and a minimum of 610 mm beyond the end of the envelope. Provide one No. 13 bar in each corner, 75 mm from the edge of the envelope. Secure corner bars with two No. 10 ties, spaced approximately 305 mm apart. Restrain reinforcing assembly from moving during concrete pouring.

[3.6.9.6 Removal of Ducts

Where duct lines are removed from existing underground structures, close the openings to waterproof the structure. Chip out the wall opening to provide a key for the new section of wall.

]3.6.10 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

3.7 CABLE PULLING

[Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables.]Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches, transformers, switchgear, switchboards, and other enclosures. Cable with[tape][or][wire] shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

3.7.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

3.8 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations.

3.8.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

3.9 CONDUCTORS INSTALLED IN PARALLEL

Conductors must be grouped such that each conduit of a parallel run contains 1 Phase A conductor, 1 Phase B conductor, 1 Phase C conductor, and 1 neutral conductor.

3.10 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set.~~[~~ Make splices in underground distribution systems only in accessible locations such as manholes, handholes, or aboveground termination pedestals.~~]~~

~~[~~3.10.1 Terminating Aluminum Conductors

a. Use particular care in making up joints and terminations. Remove surface oxides by cleaning with a wire brush or emery cloth. Apply joint compound to conductors, and use UL-listed solid aluminum connectors for connecting aluminum conductors. When connecting aluminum to copper conductors, use connectors specifically designed for this purpose.

b. Terminate aluminum conductors to copper bus either by: (1) in line splicing a copper pigtail to the aluminum conductor (copper pigtail must have a ampacity at least that of the aluminum conductor); or (2) using a circumferential compression type, aluminum bodied terminal lug UL listed for AL/CU and steel Belleville spring washers, flat washers, bolts, and nuts. Belleville spring washers must be cadmium-plated hardened steel. Install the Belleville spring washers with the crown up toward the nut or bolt head, with the concave side of the Belleville bearing on a heavy-duty, wide series flat washer of larger diameter than the Belleville. Tighten nuts sufficient to flatten Belleville and leave in that position. Lubricate hardware with joint compound prior to making connection. Wire brush and apply joint compound to conductor prior to inserting in lug.

c. Terminate aluminum conductors to aluminum bus by using all-aluminum nuts, bolts, washers, and lugs. Wire brush and apply inhibiting compound to conductor prior to inserting in lug. Lubricate hardware with joint compound prior to making connection; if bus contact surface is unplated, scratch-brush and coat with joint compound (without grit).

~~]~~3.11 ~~[~~HIGH~~][EXTRA-HIGH]~~ VOLTAGE CABLE TERMINATIONS

Make terminations in accordance with the written instruction of the termination kit manufacturer.

3.12 ~~[~~HIGH~~][EXTRA-HIGH]~~ VOLTAGE CABLE JOINTS

Provide power cable joints (splices) suitable for continuous immersion in water. Make joints only in accessible locations in manholes or handholes by using materials and methods in accordance with the written instructions of the joint kit manufacturer.

3.12.1 Joints in Shielded Cables

Cover the joined area with metallic tape, or material like the original cable shield and connect it to the cable shield on each side of the splice. Provide a bare copper ground connection brought out in a watertight manner and grounded to the manhole grounding loop as part of the splice installation. Ground conductors, connections, and rods must be as specified elsewhere in this section. Wire must be trained to the sides of the enclosure to prevent interference with the working area.

~~[3.12.2 Joints in Armored Cables~~

~~Armored cable joints must be enclosed in compound-filled, cast-iron or alloy splice boxes equipped with stuffing boxes and armor clamps of a suitable type and size for the cable being installed.~~

~~]~~3.13 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

~~[3.14 LIVE END CAPS~~

~~Provide live end caps for single conductor [high][extra-high] voltage cables where indicated.~~

~~]~~3.14 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

3.14.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

~~[~~3.14.2 Tape-Wrap

Tape-wrap metallic-sheathed or metallic armored cables without a nonmetallic protective covering over the sheath or armor prior to application of fireproofing. Wrap must be in the form of two tightly applied half-lapped layers of a pressure-sensitive 0.254 mm thick plastic tape, and must extend not less than 25 mm into the duct. Even out irregularities of the cable, such as at splices, with insulation putty before applying tape.

~~]~~3.15 GROUNDING SYSTEMS

IEEE C2 and JIS C 0365, except provide grounding systems with a resistance to solid earth ground not exceeding ~~[~~25~~][_____]~~ ohms.

3.15.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus~~[~~ 150 mm~~][300 mm]~~, installed to provide an earth ground of the appropriate value for the particular equipment being grounded.
If the specified ground resistance is not met, an additional ground rod must be provided (placed not less than 1830 mm from the first rod). Should the resultant (combined) resistance exceed the specified

resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

3.15.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.

b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has been adequately compressed on the ground wire.

3.15.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with 14 sqmm. Ground other noncurrent-carrying metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of 14 sqmm.~~[~~ Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.~~]~~

3.15.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

3.15.5 Manhole Grounding

Loop a 100 sqmm grounding conductor around the interior perimeter, approximately 305 mm above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 914 mm. Connect the conductor to the manhole grounding electrode with 100 sqmm conductor. Connect all incoming 100 sqmm grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 14 sqmm conductor.

~~[3.15.6 Fence Grounding~~

~~[Provide grounding for fences as indicated.][Provide grounding for fences with a ground rod at each fixed gate post and at each corner post.] Drive ground rods until the top is 305 mm below grade. Attach a 22 sqmm copper conductor, by exothermic weld to the ground rods and extend underground to the immediate vicinity of fence post. Lace the conductor vertically into 305 mm of fence mesh and fasten by two approved bronze compression~~

~~fittings, one to bond wire to post and the other to bond wire to fence. Each gate section must be bonded to its gatepost by a 3 by 25 mm flexible braided copper strap and ground post clamps. Clamps must be of the anti-electrolysis type.~~

~~][~~3.15.6 Metal Splice Case Grounding

Metal splice cases for ~~[~~high~~][extra-high]~~-voltage direct-burial cable must be grounded by connection to a driven ground rod located within 600 mm of each splice box using a grounding electrode conductor having a current-carrying capacity of at least 20 percent of the individual phase conductors in the associated splice box, but not less than 14 sqmm.

~~]~~3.16 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with Section ~~[31 23 00.00 20 EXCAVATION AND FILL][~~ 31 00 00 EARTHWORK~~]~~.

3.16.1 Reconditioning of Surfaces

3.16.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct ~~[~~or direct burial cable~~]~~. ~~Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching.[Provide work in accordance with Section 32 92 19 SEEDING and Section 32 93 00 EXTERIOR PLANTS.]~~

3.16.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists ~~[~~, restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified, and to match and tie into the adjacent and surrounding existing surfaces.~~][Make repairs as specified in Section [32 13 13.06 PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES][_____].]~~

3.17 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.17.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 200 mm thick, reinforced with a 150 mm by 150 mm - MW19 by MW19 (6 by 6 - W2.9 by W2.9) mesh, placed uniformly 100 mm from the top of the slab. Slab must be placed on a 150 mm thick, well-compacted gravel base. Top of concrete slab must be approximately 100 mm above finished grade with gradual slope for drainage. Edges above grade must have 15 mm chamfer. Slab must be of adequate size to project at least 200 mm beyond the equipment.

Stub up conduits, with bushings, 50 mm into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

~~[~~3.17.2 Sealing

When the installation is complete, seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

~~]~~3.18 FIELD QUALITY CONTROL

3.18.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with Denki Hoan Kyoukai and MLIT DSKKS.

3.18.1.1 ~~[~~High~~][Extra-High]~~ Voltage Cables

Perform tests after installation of cable, splices, and terminators and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

(1) Inspect exposed cable sections for physical damage.

(2) Verify that cable is supplied and connected in accordance with contract plans and specifications.

(3) Inspect for proper shield grounding, cable support, and cable termination.

(4) Verify that cable bends are not less than ICEA or manufacturer's minimum allowable bending radius.

(5) Inspect for proper fireproofing.

(6) Visually inspect jacket and insulation condition.

(7) Inspect for proper phase identification and arrangement.

b. Electrical Tests

(1) Perform a shield continuity test on each power cable by ohmmeter method. Record ohmic value, resistance values in excess of 10 ohms per 1000 feet of cable must be investigated and justified.

(2) Perform acceptance test on new cables before the new cables are connected to existing cables and placed into service, including terminations and joints. Perform maintenance test on complete cable system after the new cables are connected to existing cables and placed into service, including existing cable, terminations, and joints. Tests must be very low frequency (VLF) alternating voltage withstand tests. VLF test frequency must be 0.05 Hz minimum for a duration of 60 minutes using a sinusoidal waveform. Test voltages must be as follows:

CABLE RATING AC TEST VOLTAGE for ACCEPTANCE TESTING	
3.3kV	9kV (peak)
5 kV	10kV rms(peak)
6.6 kV	17kV rms(peak)
8 kV	13kV rms(peak)
15 kV	20kV rms(peak)
22 kV	44kV rms(peak)
25 kV	31kV rms(peak)
33 kV	63kV (peak)
35 kV	44kV rms(peak)
66 kV	85kV (peak)
77 kV	95kV (peak)

CABLE RATING AC TEST VOLTAGE for MAINTENANCE TESTING	
3.3kV	Per manufacturer requirements.
5 kV	7kV rms(peak)
6.6 kV	Per manufacturer requirements.
8 kV	10kV rms(peak)
15 kV	16kV rms(peak)
22 kV	Per manufacturer requirements.
25 kV	23kV rms(peak)
33 kV	Per manufacturer requirements.
35 kV	33kV rms(peak)
66 kV	Per manufacturer requirements.
77 kV	Per manufacturer requirements.

(3) In lieu of the acceptance testing required in item (2), High

Voltage Direct Current (HVDC) test is an alternative acceptance test on newly installed cables. Maximum applied DC test voltage shall be determined in consultation with the manufacturer of the cable and cable accessories for the type of cable being tested. The maximum test voltage to be maintained for 10 minutes. After reaching the maximum test voltage, the current magnitude should be recorded at least twice: once at approximately 2 min and again at the end of the test period (10 min) Contractor shall submit test procedure and standard use as reference for testing to the government for review and approval.

3.18.1.2 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

(1) Inspect exposed cable sections for physical damage.

(2) Verify that cable is supplied and connected in accordance with contract plans and specifications.

(3) Verify tightness of accessible bolted electrical connections.

(4) Inspect compression-applied connectors for correct cable match and indentation.

(5) Visually inspect jacket and insulation condition.

(6) Inspect for proper phase identification and arrangement.

b. Electrical Tests

(1) Perform insulation resistance tests on wiring 14 sqmm and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.

(2) Perform continuity tests to insure correct cable connection.

3.18.1.3 Grounding System

a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with JIS C 60364-6. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.

Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.18.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

.... -- End of Section --

SECTION 33 82 00

TELECOMMUNICATIONS OUTSIDE PLANT (OSP)
04/06

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only. Contractor may substitute compatible Japan Industrial Standard (JIS) or Japan Wire Industry Association Standard (JCS) standards for non-Japanese standards, as approved by the Contracting Officer's representative.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2	(2023) National Electrical Safety Code

JAPANESE STANDARDS ASSOCIATION (JSA)

JIS C 0365	(2007) Protection Against Electric Shock - Common Aspects for Installation and Equipment
JIS C 2336	(2012) Pressure-sensitive polyvinyl chloride tapes for electrical purposes
JIS C 2338	(2012) Polyester adhesive tape for electrical insulation
JIS C 6823	(2010) Measuring methods for attenuation of optical fibers
JIS C 6835	(2017) Silica glass single-mode optical fibers
JIS C 6837	(2015) All Plastic Multimode Optical Fibers
JIS C 6850	(2006) General rules of optical fiber cables
JIS C 6870-3	(2006) Outdoor optical fiber cables-Part 3: Sectional specification
JIS C 6870-3-10	(2011) Optical fiber cables-Part 3-10: Outdoor cables-Family specification for duct, directly buried and lashed aerial optical telecommunications cables
JIS C 60364-5-54	(2006; R 2015) Building Electrical Equipment-Part 5-54: Selection Of Electrical Equipment and Contruction-Grounding Equipment, Protective Conductor and Protective Bonding Conductor

JIS X 5150	(R2016) Information Technology-Generic Cabling for Customer Premises
JIS X 5151	(2018) Information technology -- Implementation and operation of customer premises cabling -- Part 3: Testing of optical fibre cabling

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)

JCS 5503	(2011) Flame retardant polyolefin sheath LAN twisted pair cable
JCS 5507	(2010) LAN Twisted Pair Cable

1.2 RELATED REQUIREMENTS

~~[~~ Section 27 10 00, BUILDING TELECOMMUNICATIONS CABLING SYSTEM~~,][Section 33 71 01, OVERHEAD TRANSMISSION AND DISTRIBUTION,~~ and~~[~~ Section 33 71 02, UNDERGROUND ELECTRICAL DISTRIBUTION~~]~~ apply to this section with additions and modifications specified herein.

~~]~~1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification shall be as defined in JIS X 5150, JCS 5507, JIS C 6850 and herein.

1.3.1 Campus Distributor (CD)

A distributor from which the campus backbone cabling emanates. (International expression for main cross-connect - (MC).)

1.3.2 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service cables (including antennae) including the entrance point at the building wall and continuing to the entrance room or space.

1.3.3 Entrance Room (ER) (Telecommunications)

A centralized space for telecommunications equipment that serves the occupants of a building. Equipment housed therein is considered distinct from a telecommunications room because of the nature of its complexity.

1.3.4 Building Distributor (BD)

A distributor in which the building backbone cables terminate and at which connections to the campus backbone cables may be made. (International expression for intermediate cross-connect - (IC).)

1.3.5 Pathway

A physical infrastructure utilized for the placement and routing of telecommunications cable.

1.4 SYSTEM DESCRIPTION

The telecommunications outside plant consists of cable, conduit, manholes, poles, etc. required to provide signal paths from the closest point of presence to the new facility, including free standing frames or backboards, interconnecting hardware, terminating cables, lightning and surge protection modules at the entrance facility. The work consists of providing, testing and making operational cabling, interconnecting hardware and lightning and surge protection necessary to form a complete outside plant telecommunications system for continuous use.~~[The telecommunications contractor must coordinate with the NMCI contractor concerning layout and configuration of the EF telecommunications and OSP. The telecommunications contractor may be required to coordinate work effort for access to the EF telecommunications and OSP with the NMCI contractor.]~~

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are ~~[for information only.][~~for Contractor Quality Control approval.~~]~~ When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.~~ Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REPORTING.~~ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications Outside Plant; G~~[, [_____]]~~

Telecommunications Entrance Facility Drawings; G~~[, [_____]]~~

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

SD-03 Product Data

Wire and cable; G~~[, [_____]]~~

Cable splices, and connectors; G~~[, [_____]]~~

Closures; G~~[, [_____]]~~

Building protector assemblies; G~~[, [_____]]~~

Protector modules; G~~[, [_____]]~~

Cross-connect terminal cabinets; G~~[, [_____]]~~

SD-06 Test Reports

Pre-installation tests;

Acceptance tests;

SD-10 Operation and Maintenance Data

Telecommunications Outside plant (OSP), Data Package 5;

Commercial off-the-shelf manuals shall be provided for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications outside plant (OSP). Submit operations and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein not later than~~[~~ 2~~][_____]~~ months prior to the date of beneficial occupancy. In addition to requirements of Data package 5, include the requirements of paragraphs TELECOMMUNICATIONS OUTSIDE PLANT SHOP DRAWINGS and TELECOMMUNICATIONS ENTRANCE FACILITY DRAWINGS.

SD-11 Closeout Submittals

Record Documentation;

In addition to other requirements, provide in accordance with paragraph RECORD DOCUMENTATION.

1.6 QUALITY ASSURANCE

1.6.1 Shop Drawings

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

1.6.1.1 Telecommunications Outside Plant Shop Drawings

Provide Outside Plant Design in accordance with JIS C 6870-3 and JIS C 6870-3-10 for aerial system design, and for underground system design. Provide T0 shop drawings that show the physical and logical connections from the perspective of an entire campus, such as actual building locations, exterior pathways and campus backbone cabling on plan view drawings, major system nodes, and related connections on the logical system drawings. Drawings shall include wiring and schematic diagrams for fiber optic and copper cabling and splices, copper conductor gauge and pair count, fiber pair count and type, pathway duct and innerduct arrangement, associated construction materials, and any details required to demonstrate that cable system has been coordinated and will properly support the switching and transmission system identified in specification and drawings.~~[~~ Provide Registered Communications Distribution Designer (RCDD) approved drawings of the telecommunications outside plant.~~][Update existing telecommunication Outside Plant T0 drawings to include information modified, deleted or added as a result of this installation.]~~ The telecommunications outside plant (OSP) shop drawings shall be included in the operation and maintenance manuals.

1.6.1.2 Telecommunications Entrance Facility Drawings

~~[Provide T3 drawings for EF Telecommunications that include telecommunications entrance facility plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and[cabinet][, rack][, backboard][and] wall elevations. Drawings shall show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks,~~

~~patch panels and equipment spaces and cabinet/racks. Drawings shall include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings.][~~Provide T3 drawings for EF Telecommunications as specified in the paragraph TELECOMMUNICATIONS SPACE DRAWINGS of Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS.~~]~~ The telecommunications entrance facility shop drawings shall be included in the operation and maintenance manuals.

1.6.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, the supervisor (if different from the installer), and the cable splicing and terminating personnel. A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

1.6.3 Outside Plant Test Plan

Prepare and provide a complete and detailed test plan for field tests of the outside plant including a complete list of test equipment for the~~[~~ copper conductor~~][~~ and~~][~~ optical fiber~~]~~ cables, components, and accessories for approval by the Contracting Officer. Include a cut-over plan with procedures and schedules for relocation of facility station numbers without interrupting service to any active location. Submit the plan at least~~[~~ 30~~][_____]~~ days prior to tests for Contracting Officer approval. Provide outside plant testing and performance measurement criteria in accordance with JIS X 5150 and JIS X 5151. Include procedures for certification, validation, and testing that includes fiber optic link performance criteria.

1.6.4 Standard Products

Provide materials and equipment that are standard products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and shall be the manufacturer's latest standard design that has been in satisfactory commercial or industrial use for at least~~[2][~~ 1~~]~~ year~~[~~s~~]~~ prior to bid opening. The~~[2][~~ 1~~]~~-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the~~[2][~~ 1~~]~~ -year period. Products supplied shall be specifically designed and manufactured for use with outside plant telecommunications systems. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.4.1 Alternative Qualifications

Products having less than a~~[2][~~ 1~~]~~-year field service record will be acceptable if a certified record of satisfactory field operation for not less than~~[6000][~~ 3000~~]~~ hours, exclusive of the manufacturers' factory or laboratory tests, is provided.

1.6.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site shall not be used, unless specified otherwise.

1.6.5 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of applicable codes and standards unless more stringent requirements are specified or indicated.

1.6.5.1 Independent Testing Organization Certificate

In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.7 DELIVERY, STORAGE, AND HANDLING

Ship cable on reels in ~~[152][~~305~~][_____]~~ meter length with a minimum overage of 10 percent. Radius of the reel drum shall not be smaller than the minimum bend radius of the cable. Wind cable on the reel so that unwinding can be done without kinking the cable. Two meters of cable at both ends of the cable shall be accessible for testing. Attach permanent label on each reel showing length, cable identification number, cable size, cable type, and date of manufacture. Provide water resistant label and the indelible writing on the labels. Apply end seals to each end of the cables to prevent moisture from entering the cable. Reels with cable shall be suitable for outside storage conditions when temperature ranges from minus 40 degrees C to plus 65 degrees C, with relative humidity from 0 to 100 percent. Equipment, other than cable, delivered and placed in storage shall be stored with protection from weather, humidity and temperature variation, dirt and dust, or other contaminants in accordance with manufacturer's requirements.

1.8 MAINTENANCE

1.8.1 Record Documentation

Provide the activity responsible for telecommunications system maintenance and administration a single complete and accurate set of record documentation for the entire telecommunications system with respect to this project.

~~[Provide T5 drawings including documentation on cables and termination hardware. T5 drawings shall include schedules to show information for cut-overs and cable plant management, patch panel layouts, cross-connect information and connecting terminal layout as a minimum. T5 drawings shall be provided[in hard copy format][on electronic media using Windows~~

~~based computer cable management software.][A licensed copy of the cable management software including documentation, shall be provided.][Update existing record documentation to reflect campus distribution T0 drawings and T3 drawing schedule information modified, deleted or added as a result of this installation.] Provide the following T5 drawing documentation as a minimum:~~

~~a. Cables - A record of installed cable shall be provided. The cable records shall [include only the required data fields][include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility]. Include manufacture date of cable with submittal.~~

~~b. Termination Hardware - Provide a record of installed patch panels, cross-connect points, campus distributor and terminating block arrangements and type. Documentation shall include [only]the required data fields[as a minimum].~~

~~][~~Provide record documentation as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

~~]~~1.9 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Products supplied shall be specifically designed and manufactured for use with outside plant telecommunications systems.

2.2 TELECOMMUNICATIONS ENTRANCE FACILITY

2.2.1 Building Protector Assemblies

Provide self-contained~~[~~ 5 pin~~][~~ screw type~~]~~ unit supplied with a field cable stub factory connected to protector socket blocks to terminate and accept protector modules for ~~[_____]~~ pairs of outside cable. Building protector assembly shall have interconnecting hardware for connection to interior cabling at full capacity. Provide manufacturers instructions for building protector assembly installation. Provide copper cable interconnecting hardware as specified in Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

2.2.2 Protector Modules

Provide~~[~~ three~~][two]~~-electrode gas tube or solid state type~~[~~ 5 pin~~][~~ screw type~~]~~ rated for the application. Provide gas tube protection modules ~~[~~ heavy duty, A>10kA, B>400, C>65A~~][~~ maximum duty, A>20kA, B>1000, C>200A~~]~~ where A is the maximum single impulse discharge current, B is the impulse life and C is the AC discharge current. The gas modules shall shunt high voltage to ground, fail short, and be equipped with an external spark gap and heat coils. Provide the number of surge protection modules equal to the number of pairs of exterior cable of the building protector assembly.

2.2.3 Fiber Optic Terminations

Provide fiber optic cable terminations as specified in 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

2.3 CLOSURES

2.3.1 Copper Conductor Closures

2.3.1.1 Aerial Cable Closures

Provide cable closure assembly consisting of a frame with clamps, a lift-off polyethylene cover, cable nozzles, and drop wire rings. Closure shall be suitable for use on Figure 8 cables. Closures shall be free breathing and suitable for housing~~[straight-through type][branch type][~~ of the type indicated~~]~~ splices of non-pressurized communications cables and shall be sized as indicated. The closure shall be constructed with ultraviolet resistant PVC.

2.3.1.2 Underground Cable Closures

a. Aboveground: Provide aboveground closures constructed of~~[~~ not less than 14 gauge steel~~][ultraviolet resistant PVC]~~ and acceptable for~~[~~ pole~~][stake]~~ mounting. Closures shall be sized and contain a marker as indicated. Covers shall be secured to prevent unauthorized entry.

b. Direct burial: Provide buried closure suitable for enclosing a straight, butt, and branch splice in a container into which can be poured an encapsulating compound. Closure shall have adequate strength to protect the splice and maintain cable shield electrical continuity in the buried environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure. Provide filled splice cases.

c. In vault or manhole: Provide underground closure suitable to house a straight, butt, and branch splice in a protective housing into which can be poured an encapsulating compound. Closure shall be of suitable thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure. Provide filled splice cases.

2.3.2 Fiber Optic Closures

2.3.2.1 Aerial

Provide aerial closure that is free breathing and suitable for housing splice organizer of non -pressurized cables. Closure shall be constructed from heavy PVC with ultraviolet resistance.

2.3.2.2 Direct Burial

Provide buried closure suitable to house splice organizer in protective housing into which can be poured an encapsulating compound. Closure shall have adequate strength to protect the splice and maintain cable shield electrical continuity, when metallic, in buried environment. Encapsulating compound shall be reenterable and shall not alter chemical stability of the closure.

2.3.2.3 In Vault or Manhole

Provide underground closure suitable to house splice organizer in a protective housing into which can be poured an encapsulating compound. Closure shall be of thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound shall be reenterable and shall not alter the chemical stability of the closure.

2.4 PAD MOUNTED CROSS-CONNECT TERMINAL CABINETS

Provide in accordance with the following:

a. Constructed of 14 gauge steel~~ or [_____]~~.

b. Equipped with a double set of hinged doors with closed-cell foam weatherstripping. Doors shall be locked and contain a marker as indicated.

c. Equipped with spool spindle bracket, mounting frames, binding post log, ~~[~~ and~~]~~ jumpering instruction label~~[~~, and load coil mounting provisions~~]~~.

d. Complete with cross connect modules to terminate number of pairs as indicated.

e. Sized as indicated.

2.5 CABLE SPLICES, AND CONNECTORS

2.5.1 Copper Cable Splices

Provide~~[~~ multipair,~~[foldback][~~ in-line~~]][~~ single pair,~~[in-line][butt][~~ box tap~~]]~~ splices of a moisture resistant,~~[two][~~ three~~][_____]~~-wire~~ [~~ insulation displacement~~]~~ connector held rigidly in place to assure maximum continuity. Cables greater than 25 pairs shall be spliced using multipair splicing connectors, which accommodate 25 pairs of conductors at a time. Provide correct connector size to accommodate the cable gauge of the supplied cable.

~~[2.5.2 Copper Cable Splice Connector~~

~~Provide splice connectors with a polycarbonate body and cap and a tin-plated brass contact element. Connector shall accommodate 0.65 to 0.4 mm solid wire with a maximum insulation diameter of 1.65 mm. Fill connector with sealant grease to make a moisture resistant connection.~~

~~][2.5.3 Fiber Optic Cable Splices~~

~~Provide fiber optic cable splices and splicing materials for[fusion][mechanical] methods at locations shown on the construction drawings. The splice insertion loss shall be 0.3 dB maximum when measured in accordance with JIS C 6823 using an Optical Time Domain Reflectometer (OTDR). Splices shall be designed for a return loss of 40.0 db max for single mode fiber when tested in accordance with JIS C 6189. Physically protect each fiber optic splice by a splice kit specially designed for the splice.~~

~~][2.5.4 Fiber Optic Splice Organizer~~

~~Provide splice organizer suitable for housing fiber optic splices in a neat and orderly fashion. Splice organizer shall allow for a minimum of 1 m of fiber for each fiber within the cable to be neatly stored without kinks or twists. Splice organizer shall accommodate individual strain relief for each splice and allow for future maintenance or modification, without damage to the cable or splices. Provide splice organizer hardware, such as splice trays, protective glass shelves, and shield bond connectors in a splice organizer kit.~~

~~]2.5.5 Shield Connectors~~

~~Provide connectors with a stable, low-impedance electrical connection between the cable shield and the bonding conductor.~~

2.6 CONDUIT

Provide conduit as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.7 PLASTIC INSULATING TAPE

JIS C 2336 or JIS C 2338.

2.8 WIRE AND CABLE

2.8.1 Copper Conductor Cable

Solid copper conductors, covered with an extruded solid insulating compound. Insulated conductors shall be twisted into pairs which are then stranded or oscillated to form a cylindrical core. For special high frequency applications, the cable core shall be separated into compartments. Cable shall be completed by the application of a suitable core wrapping material, a corrugated copper or plastic coated aluminum shield, and an overall extruded jacket. Telecommunications contractor shall verify distances between splice points prior to ordering cable in specific cut lengths. Gauge of conductor shall determine the range of numbers of pairs specified; 19 gauge (6 to 400 pairs), 22 gauge (6 to 1200 pairs), 24 gauge (6 to 2100 pairs), and 26 gauge (6 to 3000 pairs). Copper conductor shall conform to the following:

2.8.1.1 Underground

Provide filled cable meeting the requirements of applicable codes and standards.

2.8.1.2 Aerial

Provide filled cable meeting the requirements of JCS 5503.

2.8.1.3 Screen

Provide screen-compartmental core cable filled cable ~~[ ][~~as indicated~~]~~.

2.8.2 Fiber Optic Cable

Provide~~[~~ single-mode, 8/125-um, 0.10 aperture 1310 nm fiber optic cable in accordance with JIS C 6835~~][~~ single-mode, 8/125-um, 0.10 aperture 1550 nm fiber optic cable in accordance with JIS C 6835~~][~~ and~~][~~ multimode 62.5/125-um, 0.275 aperture fiber optic cable in accordance with JIS C 6837 ~~][multimode 50/125-um, 0.275 aperture fiber optic cableJIS C 6837], JIS C 6870-3 and JIS C 6870-3-10~~ including any special requirements made necessary by a specialized design. Provide~~[~~ 12~~][_____]~~ optical fibers~~[~~ as indicated~~]~~. Fiber optic cable shall be specifically designed for outside use with loose buffer construction. Provide fiber optic color code in accordance with JIS C 6870-3-10.

2.8.2.1 Strength Members

Provide~~[~~ central~~][non-central]~~,~~[non-metallic][~~ metallic~~]~~ strength members with sufficient tensile strength for installation and residual rated loads to meet the applicable performance requirements in accordance with JIS C 6870-3 and JIS C 6870-3-10. The strength member is included to serve as a cable core foundation to reduce strain on the fibers, and shall not serve as a pulling strength member.

~~[2.8.2.2 Shielding or Other Metallic Covering~~

~~Provide[copper, copper alloy or copper and steel laminate][copper and stainless steel, coated stainless steel or bare low carbon steel][bare aluminum or coated aluminum],[single][dual] tape covering or shield in accordance with JIS C 6870-3 and JIS C 6870-3-10.~~

~~]2.8.2.3 Performance Requirements~~

~~Provide fiber optic cable with optical and mechanical performance requirements in accordance with JIS C 6870-3 and JIS C 6870-3-10.~~

~~2.8.3 Grounding and Bonding Conductors~~

~~Provide grounding and bonding conductors in accordance with JIS C 60364-5-54, IEEE C2 and JIS C 0365. Solid bare copper wire meeting the requirements of JIS C 3101 for sizes 8 sqmm and smaller and stranded bare copper wire meeting the requirements of JIS C 3105, for sizes 14 sqmm and larger. Insulated conductors shall have 600-volt, Type TW insulation meeting the requirements of JIS C 3307 and JIS C 3401.~~

~~2.9 T-SPAN LINE TREATMENT REPEATERS~~

~~Provide as indicated. Repeaters shall be pedestal mounted with non-pressurized housings, sized as indicated.~~

~~2.10 POLES AND HARDWARE~~

~~Provide poles and hardware as specified in Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION.~~

2.9 CABLE TAGS IN MANHOLES, HANDHOLES, AND VAULTS

Provide tags for each telecommunications cable or wire located in manholes, handholes, and vaults. Cable tags shall be~~[~~ stainless steel~~][~~ or ~~][~~ polyethylene~~]~~ and labeled~~[as indicated]~~. Handwritten labeling is unacceptable.

~~[~~2.9.1 Stainless Steel

Provide stainless steel, cable tags 41.25 mm in diameter 1.58 mm thick minimum, and circular in shape. Tags shall be die stamped with numbers, letters, and symbols not less than 6.35 mm high and approximately 0.38 mm deep in normal block style.

][2.9.2 Polyethylene Cable Tags

Provide tags of polyethylene that have an average tensile strength of 22.4 MPa; and that are two millimeter thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 77 degrees C. Provide 1.3 mm (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag. Ties shall have a minimum loop tensile strength of 778.75 N. The cable tags shall have black block letters, numbers, and symbols 25 mm high on a yellow background. Letters, numbers, and symbols shall not fall off or change positions regardless of the cable tags' orientation.

]2.10 BURIED WARNING AND IDENTIFICATION TAPE

Provide fiber optic media marking and protection. Provide color, type and depth of tape as specified in paragraph BURIED WARNING AND IDENTIFICATION TAPE in Section 31 00 00, EARTHWORK.

2.11 GROUNDING BRAID

Provide grounding braid that provides low electrical impedance connections for dependable shield bonding. Braid shall be made from flat tin-plated copper.

2.12 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.13 FIELD FABRICATED NAMEPLATES

Provide laminated plastic nameplates in accordance with JIS K 6911 for each patch panel, protector assembly, rack, cabinet and other equipment or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 3 mm thick, white with[black][_____] center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be 25 by 65 mm. Lettering shall be a minimum of 6.35 mm high normal block style.

2.14 TESTS, INSPECTIONS, AND VERIFICATIONS

2.14.1 Factory Reel Test Data

Test 100 percent OTDR test of FO media at the factory in accordance with JIS X 5150 and JIS C 6850. Use JIS C 6823 for single mode fiber and for multi mode fiber measurements. Calibrate OTDR to show anomalies of 0.2 dB minimum. Enhanced performance filled OSP copper cables, referred to as Broadband Outside Plant (BBOSP). Enhanced performance air core OSP copper cables shall meet the requirements of JCS 5503. Submit test reports, including manufacture date for each cable reel and receive approval before delivery of cable to the project site.

PART 3 EXECUTION

3.1 INSTALLATION

Install all system components and appurtenances in accordance with manufacturer's instructions IEEE C2, JIS C 0365 and as indicated. Provide all necessary interconnections, services, and adjustments required for a complete and operable telecommunications system.

3.1.1 Contractor Damage

Promptly repair indicated utility lines or systems damaged during site preparation and construction. Damages to lines or systems not indicated, which are caused by Contractor operations, shall be treated as "Changes" under the terms of the Contract Clauses. When Contractor is advised in writing of the location of a nonindicated line or system, such notice shall provide that portion of the line or system with "indicated" status in determining liability for damages. In every event, immediately notify the Contracting Officer of damage.

3.1.2 Cable Inspection and Repair

Handle cable and wire provided in the construction of this project with care. Inspect cable reels for cuts, nicks or other damage. Damaged cable shall be replaced or repaired to the satisfaction of the Contracting Officer. Reel wraps shall remain intact on the reel until the cable is ready for placement.

3.1.3 Direct Burial System

Installation shall be in accordance with JIS C 6870-3 and JIS C 6870-3-10. Under railroad tracks, paved areas, and roadways install cable in conduit encased in concrete. Slope ducts to drain. Excavate trenches by hand or mechanical trenching equipment. Provide a minimum cable cover of 610 mm below finished grade. Trenches shall be not less than 155 mm wide and in straight lines between cable markers. Do not use cable plows. Bends in trenches shall have a radius of not less than ~~[915][______]~~ mm. Where two or more cables are laid parallel in the same trench, space laterally at least 78 mm apart. When rock is encountered, remove it to a depth of at least 78 mm below the cable and fill the space with sand or clean earth free from particles larger than 6 mm. Do not unreel and pull cables into the trench from one end. Cable may be unreeled on grade and lifted into position. Provide color, type and depth of warning tape as specified in paragraph BURIED WARNING AND IDENTIFICATION TAPE in Section 31 00 00 EARTHWORK.

3.1.3.1 Cable Placement

a. Separate cables crossing other cables or metal piping from the other cables or pipe by not less than ~~[78][______]~~ mm of well tamped earth. Do not install circuits for communications under or above traffic signal loops.

b. Cables shall be in one piece without splices between connections except where the distance exceeds the lengths in which the cable is furnished.

c. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.

d. Leave a horizontal slack of approximately 915 mm in the ground on each end of cable runs, on each side of connection boxes, and at points where connections are brought aboveground. Where cable is brought aboveground, leave additional slack to make necessary connections.

3.1.3.2 Identification Slabs~~ [Markers]~~

Provide a marker at each change of direction of the cable, over the ends of ducts or conduits which are installed under paved areas and roadways and over each splice. Identification markers shall be of concrete, approximately 508 mm square by 155 mm thick.

3.1.3.3 Backfill for Rocky Soil

When placing cable in a trench in rocky soil, the cable shall be cushioned by a fill of sand or selected soil at least 53 mm thick on the floor of the trench before placing the cable or wire. The backfill for at least 103 mm above the wire or cable shall be free from stones, rocks, or other hard or sharp materials which might damage the cable or wire. If the buried cable is placed less than 610 mm in depth~~[~~, a protective cover of~~[metal][~~ concrete~~]~~ shall be used~~]~~.

3.1.4 Cable Protection

Provide direct burial cable protection in accordance with applicable codes and standards and as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Galvanized conduits which penetrate concrete (slabs, pavement, and walls) shall be PVC coated and shall extend from the first coupling or fitting outside either side of the concrete minimum of 155 mm per 305 mm burial depth beyond the edge of the surface where cable protection is required; all conduits shall be sealed on each end. Where additional protection is required, cable may be placed in galvanized iron pipe (GIP) sized on a maximum fill of 40 percent of cross-sectional area, or in concrete encased 103 mm PVC pipe. Conduit may be installed by jacking or trenching. Trenches shall be backfilled with earth and mechanically tamped at 155 mm lift so that the earth is restored to the same density, grade and vegetation as adjacent undisturbed material.

3.1.4.1 Cable End Caps

Cable ends shall be sealed at all times with coated heat shrinkable end caps. Cables ends shall be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps shall remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

3.1.5 Underground Duct

Provide underground duct and connections to existing~~[~~ manholes,~~][~~ handholes,~~][~~ concrete pads,~~][~~ and~~][~~ existing ducts~~]~~ as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION with any additional requirements as specified herein.

3.1.6 Reconditioning of Surfaces

Provide reconditioning of surfaces as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.1.7 Penetrations

Caulk and seal cable access penetrations in walls, ceilings and other parts of the building. Seal openings around electrical penetrations through fire resistance-rated wall, partitions, floors, or ceilings in accordance with Section 07 84 00 FIRESTOPPING.

3.1.8 Cable Pulling

Test duct lines with a mandrel and swab out to remove foreign material before the pulling of cables. Avoid damage to cables in setting up pulling apparatus or in placing tools or hardware. Do not step on cables when entering or leaving the manhole. Do not place cables in ducts other than those shown without prior written approval of the Contracting Officer. Roll cable reels in the direction indicated by the arrows painted on the reel flanges. Set up cable reels on the same side of the manhole as the conduit section in which the cable is to be placed. Level the reel and bring into proper alignment with the conduit section so that the cable pays off from the top of the reel in a long smooth bend into the duct without twisting. Under no circumstances shall the cable be paid off from the bottom of a reel. Check the equipment set up prior to beginning the cable pulling to avoid an interruption once pulling has started. Use a cable feeder guide of suitable dimensions between cable reel and face of duct to protect cable and guide cable into the duct as it is paid off the reel. As cable is paid off the reel, lubricate and inspect cable for sheath defects. When defects are noticed, stop pulling operations and notify the Contracting Officer to determine required corrective action. Cable pulling shall also be stopped when reel binds or does not pay off freely. Rectify cause of binding before resuming pulling operations. Provide cable lubricants recommended by the cable manufacturer. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.

3.1.8.1 Cable Tensions

Obtain from the cable manufacturer and provide to the Contracting Officer, the maximum allowable pulling tension. This tension shall not be exceeded.

3.1.8.2 Pulling Eyes

Equip cables 32 mm in diameter and larger with cable manufacturer's factory installed pulling-in eyes. Provide cables with diameter smaller than 32 mm with heat shrinkable type end caps or seals on cable ends when using cable pulling grips. Rings to prevent grip from slipping shall not be beaten into the cable sheath. Use a swivel of 19 mm links between pulling-in eyes or grips and pulling strand.

3.1.8.3 Installation of Cables in Manholes, Handholes, and Vaults

Do not install cables utilizing the shortest route, but route along those walls providing the longest route and the maximum spare cable lengths. ~~[~~ Provide a minimum of 15 meter maintenance loop for all cables at maintenance hole feeding building entrance facilities.~~]~~ Form cables to closely parallel walls, not to interfere with duct entrances, and support cables on brackets and cable insulators at a maximum of 1220 mm. In existing manholes, handholes, and vaults where new ducts are to be terminated, or where new cables are to be installed, modify the existing installation of cables, cable supports, and grounding as required with cables arranged and supported as specified for new cables. Identify each cable with corrosion-resistant embossed metal tags.

~~3.1.9 Aerial Cable Installation~~

~~Pole installation shall be as specified in Section 33 71 01 OVERHEAD TRANSMISSION AND DISTRIBUTION. Where physical obstructions make it necessary to pull distribution wire along the line from a stationary reel, use cable stringing blocks to support wire during placing and tensioning operations. Do not place ladders, cable coils, and other equipment on or against the distribution wire. Wire shall be sagged in accordance with the data shown. Protect cable installed outside of building less than 2.5 meters above finished grade against physical damage.~~

~~3.1.9.1 Figure 8 Distribution Wire~~

~~Perform spiraling of the wire within 24 hours of the tensioning operation. Perform spiraling operations at alternate poles with the approximate length of the spiral being 4575 mm. Do not remove insulation from support members except at bonding and grounding points and at points where ends of support members are terminated in splicing and dead-end devices. Ground support wire at poles to the pole ground.~~

~~3.1.9.2 Suspension Strand~~

~~Place suspension strand as indicated. Tension in accordance with the data indicated. When tensioning strand, loosen cable suspension clamps enough to allow free movement of the strand. Place suspension strand on the road side of the pole line. In tangent construction, point the lip of the suspension strand clamp toward the pole. At angles in the line, point the suspension strand clamp lip away from the load. In level construction place the suspension strand clamp in such a manner that it will hold the strand below the through-bolt. At points where there is an up-pull on the strand, place clamp so that it will support strand above the through-bolt. Make suspension strand electrically continuous throughout its entire length, bond to other bare cables suspension strands and connect to pole ground at each pole.~~

~~3.1.9.3 Aerial Cable~~

~~Keep cable ends sealed at all times using cable end caps. Take cable from reel only as it is placed. During placing operations, do not bend cables in a radius less than 10 times the outside diameter of cable. Place temporary supports sufficiently close together and properly tension the cable where necessary to prevent excessive bending. In those instances where spiraling of cabling is involved, accomplish mounting of enclosures for purposes of loading, splicing, and distribution after the spiraling operation has been completed.~~

3.1.9 Cable Splicing

3.1.9.1 Copper Conductor Splices

Perform splicing in accordance with manufacturer's requirements except that direct buried splices and twisted and soldered splices are not allowed. Exception does not apply for pairs assigned for carrier application.

3.1.9.2 Fiber Optic Splices

Fiber optic splicing shall be in accordance with manufacturer's recommendation and shall exhibit an insertion loss~~[not greater than 0.2 dB for fusion splices][~~ not greater than 0.4 db for mechanical splices~~]~~.

3.1.10 Surge Protection

All cables and conductors, except fiber optic cable, which serve as communication lines through off-premise lines, shall have surge protection installed at each end.

3.1.11 Grounding

Provide grounding and bonding in accordance with JIS C 60364-5-54 and IEEE C2 and JIS C 0365. Ground exposed noncurrent carrying metallic parts of telephone equipment, cable sheaths, cable splices, and terminals.

3.1.11.1 Telecommunications Master Ground Bar (TMGB)

The TMGB is the hub of the basic telecommunications grounding system providing a common point of connection for ground from outside cable, CD, and equipment. Establish a TMGB for connection point for cable stub shields to connector blocks and CD protector assemblies as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.1.11.2 Incoming Cable Shields

Shields shall not be bonded across the splice to the cable stubs. Ground shields of incoming cables in the EF Telecommunications to the TMGB.

3.1.11.3 Campus Distributor Grounding

a. Protection assemblies: Mount CD protector assemblies directly~~[~~ on the telecommunications backboard~~][in the telecommunications [rack][cabinet]]~~. Connect assemblies mounted on each vertical frame with No. 6 AWG copper conductor to provide a low resolution path to TMGB.

~~[~~ b. TMGB connection: Connect TMGB to TGB with copper conductor with a total resistance of less than 0.01 ohms.

~~][~~3.1.12 Cut-Over

All necessary transfers and cut-overs, shall be accomplished by the telecommunications contractor.

~~]~~3.2 LABELING

3.2.1 Labels

Provide labeling for new cabling and termination hardware located within the facility. Handwritten labeling is unacceptable. Stenciled lettering for cable and termination hardware shall be provided using~~[~~ thermal ink transfer process~~][laser printer][_____]~~.

3.2.2 Cable Tag Installation

Install cable tags for each telecommunications cable or wire located in manholes, handholes, and vaults including each splice.[Tag only new wire and cable provided by this contract.][Tag new wire and cable provided under this contract and existing wire and cable which are indicated to have splices and terminations provided by this contract.] The labeling of telecommunications cable tag identifiers shall be[as indicated].[Tag legend shall be as indicated.] Do not provide handwritten letters. Install cable tags so that they are clearly visible without disturbing any cabling or wiring in the manholes, handholes, and vaults.

3.2.3 Termination Hardware

Label patch panels, distribution panels, connector blocks and protection modules using color coded labels with identifiers.

3.3 FIELD APPLIED PAINTING

[Provide ferrous metallic enclosure finishes in accordance with the following procedures. Ensure that surfaces are dry and clean when the coating is applied. Coat joints and crevices. Prior to assembly, paint surfaces which will be concealed or inaccessible after assembly. Apply primer and finish coat in accordance with the manufacturer's recommendations.][Provide ferrous metallic enclosure finishes as specified in Section 09 90 00 PAINTS AND COATINGS.]

[3.3.1 Cleaning

Clean surfaces as required per applicable codes and standards.

][3.3.2 Priming

Prime with a two component polyamide epoxy primer which has a bisphenol-A base, a minimum of 60 percent solids by volume, and an ability to build up a minimum dry film thickness on a vertical surface of 0.127 mm. Apply in two coats to a total dry film thickness of 0.127 to 0.2 mm.

][3.3.3 Finish Coat

Finish with a two component urethane consisting of saturated polyester polyol resin mixed with aliphatic isocyanate which has a minimum of 50 percent solids by volume. Apply to a minimum dry film thickness of 0.05 to 0.076 mm. Color shall be the manufacturer's standard.

]3.4 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.5 FIELD QUALITY CONTROL

Provide the Contracting Officer[10][_____] working days notice prior to[each][_____] test. Provide labor, equipment, and incidentals required for testing. Correct defective material and workmanship disclosed as the results of the tests. Furnish a signed copy of the test results to the Contracting Officer within 3 working days after the tests for each segment of construction are completed. Perform testing as construction progresses and do not wait until all construction is complete before starting field tests.

3.5.1 Pre-Installation Tests

Perform the following tests on cable at the job site before it is removed from the cable reel. For cables with factory installed pulling eyes, these tests shall be performed at the factory and certified test results shall accompany the cable.

3.5.1.1 Cable Capacitance

Perform capacitance tests on at least 10 percent of the pairs within a cable to determine if cable capacitance is within the limits specified.

3.5.1.2 Loop Resistance

Perform DC-loop resistance on at least 10 percent of the pairs within a cable to determine if DC-loop resistance is within the manufacturer's calculated resistance.

3.5.1.3 Pre-Installation Test Results

Provide results of pre-installation tests to the Contracting Officer at least~~[~~ 5~~][______]~~ working days before installation is to start. Results shall indicate reel number of the cable, manufacturer, size of cable, pairs tested, and recorded readings. When pre-installation tests indicate that cable does not meet specifications, remove cable from the job site.

3.5.2 Acceptance Tests

Perform acceptance testing in accordance with JIS X 5150 and JIS X 5151 and as further specified in this section. Provide personnel, equipment, instrumentation, and supplies necessary to perform required testing. Notification of any planned testing shall be given to the Contracting Officer at least~~[~~ 14~~][______]~~ days prior to any test unless specified otherwise. Testing shall not proceed until after the Contractor has received written Contracting Officer's approval of the test plans as specified. Test plans shall define the tests required to ensure that the system meets technical, operational, and performance specifications. The test plans shall define milestones for the tests, equipment, personnel, facilities, and supplies required. The test plans shall identify the capabilities and functions to be tested. Provide test reports in booklet form showing all field tests performed, upon completion and testing of the installed system. Measurements shall be tabulated on a pair by pair or strand by strand basis.

3.5.2.1 Copper Conductor Cable

Perform the following acceptance tests in accordance with JIS C 6870-3 and JIS C 6870-3-10:

a. Wire map (pin to pin continuity)

b. Continuity to remote end

c. Crossed pairs

d. Reversed pairs

e. Split pairs

f. Shorts between two or more conductors

~~3.5.2.2 Fiber Optic Cable~~

~~Test fiber optic cable and as further specified in this section. Two optical tests shall be performed on all optical fibers: Optical Time Domain Reflectometry (OTDR) Test, and Attenuation Test. In addition, a Bandwidth Test shall be performed on all multimode optical fibers. These tests shall be performed on the completed end-to-end spans which include the near-end pre-connectorized single fiber cable assembly, outside plant as specified, and the far-end pre-connectorized single fiber cable assembly.~~

~~a. OTDR Test: The OTDR test shall be used to determine the adequacy of the cable installations by showing any irregularities, such as discontinuities, micro-bendings or improper splices for the cable span under test. Hard copy fiber signature records shall be obtained from the OTDR for each fiber in each span and shall be included in the test results. The OTDR test shall be measured in both directions. A reference length of fiber,[20][____] m minimum, used as the delay line shall be placed before the new end connector and after the far end patch panel connectors for inspection of connector signature. Conduct OTDR test and provide calculation or interpretation of results in accordance with JIS C 6823 for single-mode fiber and for multimode fiber. Splice losses shall not exceed 0.3 db.~~

~~b. Attenuation Test: End-to-end attenuation measurements shall be made on all fibers, in both directions, using a[850][1300][1310][1550] nanometer light source at one end and the optical power meter on the other end to verify that the cable system attenuation requirements are met in accordance with[JIS C 6188 for multimode][and][JIS C 6823 for single-mode] fiber optic cables. The measurement method shall be in accordance with JIS C 6823. Attenuation losses shall not exceed 0.5 db/km at 1310 nm and 1550 nm for single-mode fiber. Attenuation losses shall not exceed 5.0 db/km at 850 nm and 1.5 db/km at 1300 nm for multimode fiber.~~

~~c. Bandwidth Test: The end-to-end bandwidth of all multimode fiber span links shall be measured by the frequency domain method. The bandwidth shall be measured in both directions on all fibers. The bandwidth measurements shall be in accordance with JIS C 6824.~~

3.5.3 Soil Density Tests

~~[a. Determine soil-density relationships for compaction of backfill material.~~

~~][b~~a. Determine soil-density relationships as specified for soil tests in Section 31 00 00 EARTHWORK.

~~]~~ -- End of Section --

SUBMITTAL REGISTER

CONTRACT NO. 27AR0001

TITLE AND LOCATION: ARMY FAMILY HOUSING RENOVATE TOWER B743

CONTRACTOR:

ACTIVITY NO (a)	TRANSMITTAL NO (b)	SPEC SECT (c)	DESCRIPTION ITEM SUBMITTED (d)	PARAGRAPH # (e)	CLASSIFICATION GOVT OR A/E REVWR (f)	CONTRACTOR: SCHEDULE DATES SUBMIT (g)	APPROVAL NEEDED BY (h)	MATERIAL NEEDED BY (i)	CONTRACTOR ACTION ACTION CODE (j)	DATE OF ACTION (k)	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR (l)	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER (m)	DATE RCD FROM OTH REVIEWER (n)	ACTION CODE (o)	DATE OF ACTION (p)	MAILED TO CONTR/ DATE RCD FRM APPR AUTH (q)	REMARKS (r)
		01 11 00.00 10	SD-01 Preconstruction Submittals														
			Initial Site Conditions Photos	1.10													
			Key Personnel Qualifications	1.6	G												
			INDOOR AIR QUALITY (IAQ)	1.9.2	G												
			MANAGEMENT PLAN														
			SD-07 Certificates														
			Monthly Progress Photos	1.10													
			SD-11 Closeout Submittals														
			Construction Completion Photos	1.10													
		01 20 00	SD-07 Certificates														
			Bill Of Lading	3.2	G												
		01 32 01.00 10	SD-01 Preconstruction Submittals														
			Project Scheduler Qualifications	1.3	G												
			Preliminary Project Schedule	3.4.1	G												
			Initial Project Schedule	3.4.2	G												
			Periodic Schedule Update	3.6.2	G												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 13	SD-04 Samples														
			Model Unit	3.1.3													
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G												
			Final IAQ Management Plan	1.16	S												
			Indoor Air Quality (IAQ)	1.16	G												
			Management Plan														
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.
27AR0001

TITLE AND LOCATION
ARMY FAMILY HOUSING RENOVATE TOWER B743

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION: ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY: DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Accident Reports	1.12.2	G													
			LHE Inspection Reports	1.12.3														
			Monthly Exposure Reports	1.5	G													
			SD-07 Certificates															
			Crane Operators/Riggers	1.7.5														
			Activity Hazard Analysis (AHA)	1.9	G													
			Certificate of Compliance	1.12.4														
			Hot Work Permit	1.13														
			Standard Lift Plan	1.12.4	G													
			Certifications	3.10.2														
			Licenses	3.10.2														
		01 45 00.00 10	SD-01 Preconstruction Submittals															
			Contractor Quality Control (CQC)	3.2	G													
			Plan															
			Design Quality Control (DQC)	3.2.2														
			Plan															
			SD-05 Design Data															
			Discipline-Specific Checklists	3.2.2														
			Design Quality Control	3.9.1														
			SD-06 Test Reports															
			Verification Statement	3.9.2														
		01 45 35	SD-01 Preconstruction Submittals															
			SIOR Letter of Acceptance	3.1.1	G													
			Project Manual	3.1.1	G													
			Written Practices	3.1.1														

SUBMITTAL REGISTER

CONTRACT NO. 27AR0001

TITLE AND LOCATION: ARMY FAMILY HOUSING RENOVATE TOWER B743

CONTRACTOR:

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	CLASSIFICATION	GOVT OR A/E REVWR	CONTRACTOR: SCHEDULE DATES SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	CONTRACTOR ACTION: ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY: DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
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		01 45 35	NDT Procedures and Equipment	3.1.1														
			Calibration Records															
			SD-06 Test Reports															
			Daily Reports	3.1.1														
			Biweekly Reports	3.1.1														
			SD-07 Certificates															
			Fabrication Plant	2.1														
			Certificate of Compliance	2.1														
			Special Inspector of Record	1.5.11	G													
			Special Inspector	1.5	G													
			Qualification Records	3.1.1														
			SD-11 Closeout Submittals															
			Interim Final Report	3.1.1														
			Comprehensive Final Report	3.1.3	G													
		01 50 00	SD-01 Preconstruction Submittals															
			Construction Site Plan	1.3	G													
			Traffic Control Plan	3.4.1	G													
			Temporary Utility Connections	3.3.2	G													
			Plan															
		01 57 19	SD-01 Preconstruction Submittals															
			Environmental Protection Plan	1.9	G													
			Environmental Manager	1.8.3	G													
			Qualifications															
			Written Assessment Of Friable	3.11														
			Asbestos Disturbance															
			SD-06 Test Reports															

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		01 57 19	Solid Waste Management Report	3.8.2.1	G												
			Nonhazardous Solid Waste	3.7.2													
			Diversion Report														
			SD-07 Certificates														
			Asbestos Certification	1.7.1													
			Lead Certification	1.7.2													
			SD-11 Closeout Submittals														
			Disposal Documentation for	3.8.3.4	G												
			Hazardous and Regulated Waste														
			Solid Waste Management Report	3.8.2.1	G												
			Hazardous Waste/Debris	3.8.3.1	G												
			Management														
			Environmental Records Binder	1.11													
		01 74 19	SD-01 Preconstruction Submittals														
			Waste Management Plan	1.6	G												
			SD-11 Closeout Submittals														
			Records	1.7	G												
		01 78 00	SD-03 Product Data														
			Warranty Management Plan	1.7.1	G												
			Warranty Tags	1.7.5													
			Final Cleaning	3.8													
			Spare Parts and Spare Parts	1.5													
			Data														
			SD-08 Manufacturer’s Instructions														
			Instructions	1.7.1													

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		01 78 00	SD-10 Operation and Maintenance Data															
			Operation and Maintenance Manuals	3.7	G													
			SD-11 Closeout Submittals															
			As-Built Drawings	3.1	G													
			Record Drawings	3.3	G													
			As-Built Record of Equipment and Materials	1.7.1	G													
			As-Built Record of Equipment and Materials	3.6	G													
			Final Approved Shop Drawings	3.4	G													
			Construction Contract Specifications	3.5	G													
			Interim DD FORM 1354	3.9	G													
			Checklist for DD FORM 1354	3.9	G													
		01 78 23	SD-10 Operation and Maintenance Data															
			O&M Database	1.4	G													
			Training Plan	3.1.1	G													
			Training Outline	3.1.3	G													
			Training Content	3.1.2	G													
			SD-11 Closeout Submittals															
			Training Video Recording	3.1.4	G													
			Validation of Training Completion	3.1.6	G													
		01 91 00.15 10	SD-01 Preconstruction Submittals															

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		01 91 00.15 10	Commissioning Firm	1.7	G													
			Lead Commissioning Specialist	1.7.1														
			Technical Commissioning Specialists	1.7.2														
			Commissioning Firm's Contract	1.7														
			SD-06 Test Reports															
			Interim Construction Phase Commissioning Plan	3.1.2.1	G													
			Final Construction Phase Commissioning Plan	3.1.2.2	G													
			Pre-Functional Checklists	3.1.4.2	G													
			Issues Log	1.8														
			Commissioning Report	3.2	G													
			Post-Construction Trend Log Report	3.3.1	G													
			SD-07 Certificates															
			Certificate of Readiness	1.9	G													
			SD-10 Operation and Maintenance Data															
			Training Plan	3.1.5	G													
			Training Attendance Rosters	3.1.5														
			Systems Manual	3.1.6	G													
			Maintenance and Service Life Plans	3.1.7														
			SD-11 Closeout Submittals															

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		01 91 00.15 10	Construction Phase	3.1.2.1	G													
			Commissioning Plan															
			Final Commissioning Report	3.2	G													
		02 82 00	SD-03 Product Data															
			Amended Water	1.2.2														
			Safety Data Sheets (SDS) for All	1.3.9														
			Materials															
			Encapsulants	2.1														
			Respirators	3.1.2.1														
			Local Exhaust Equipment	3.1.7														
			Pressure Differential Automatic	3.1.7														
			Recording Instrument															
			Vacuums	3.1.8														
			Glovebags	3.1.10														
			SD-06 Test Reports															
			Air Sampling Results	1.5.6														
			Pressure Differential Recordings	1.5.7														
			for Local Exhaust System															
			Clearance Sampling	3.2.13.5														
			Asbestos Disposal Quantity	3.3.3.2														
			Report															
			SD-07 Certificates															
			Employee Training	1.3.4														
			Notifications	1.3.5														
			Respiratory Protection Program	1.3.7														
			Asbestos Hazard Abatement Plan	1.3.10														

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		02 82 00	Testing Laboratory	1.3.11														
			Landfill Approval	1.3.12														
			Delivery Tickets	1.3.12														
			Waste Shipment Records	1.3.12														
			Transporter Certification	1.3.13														
			Medical Certification	1.3.14														
			Private Qualified Person	1.5.2														
			Documentation															
			Competent Person	1.5.3														
			Worker's License	1.5.4														
			Contractor's License	1.5.5														
			Encapsulants	2.1														
			Equipment Used to Contain	3.1														
			Airborne Asbestos Fibers															
			Water Filtration Equipment	3.1.3.3														
			Vacuums	3.1.8														
			Ventilation Systems	3.1.8														
			SD-11 Closeout Submittals															
			Permits	1.3.5														
			Notifications	1.3.5														
			Respirator Program Records	1.3.7.1														
			Rental Equipment	1.7.1														
		02 83 00	SD-01 Preconstruction Submittals															
			Occupational and Environmental	1.5.2.3														
			Assessment Data Report															
			Medical Examinations	1.5.2.4														

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		02 83 00	Lead, Cadmium, Chromium	1.5.1.2	G													
			Compliance Plan															
			Lead, Cadmium, Chromium	1.5.2.2	G													
			Compliance Plan															
			SD-03 Product Data															
			Respirators	1.6.1														
			Vacuum Filters	1.6.4														
			Negative Air Pressure System	1.6.7														
			Materials and Equipment	2.1														
			Expendable Supplies	2.1.1														
			Local Exhaust Equipment	3.1.1.5														
			Pressure Differential Automatic	3.1.1.5														
			Recording Instrument															
			Pressure Differential Log	3.1.1.6														
			SD-06 Test Reports															
			Occupational and Environmental	1.5.2.3														
			Assessment Data Report															
			Pressure Differential Recordings	1.5.3														
			For Local Exhaust System															
			Sampling Results	1.5.2.3	G													
			SD-07 Certificates															
			Notification of the	3.1.1.1														
			Commencement of LBP Hazard															
			Abatement															
			SD-11 Closeout Submittals															

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		02 83 00	Turn-In Documents or Weight Tickets	3.5.2.1														
			Written Evidence Of TSD Approval	3.5.2.1														
		02 84 33	SD-01 Preconstruction Submittals															
			PCB and Lamp Management Plan	1.4.1	G													
			PCB and Lamp Management Plan	1.4.1.1	G													
			PCB and Lamp Management Plan	1.4.3	G													
			PCB and Lamp Management Plan	1.4.4	G													
			PCB and Lamp Management Plan	3.2.1	G													
			PCB and Lamp Management Plan	3.3	G													
			PCB and Lamp Management Plan	3.8	G													
			SD-07 Certificates															
			Qualifications of IH/PQP	1.4.5	G													
			Training certification	1.4.6														
			Copy E of the Japanese Hazardous Waste Manifest	3.10														
			Notification	1.4.7														
			Certification Of Decontamination	3.4.4	G													

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		02 84 33	Post Cleanup Sampling	3.4.5	G													
		03 01 00	SD-01 Preconstruction Submittals															
			Qualifications	1.5.3	G													
			Work Plan	1.5.5	G													
			Work Plan	1.6.1	G													
			Quality Control Plan	1.5.2	G													
			SD-03 Product Data															
			Conventional Concrete	2.5														
			Polymers	2.6														
			Miscellaneous Materials And Equipment	2.7														
			SD-05 Design Data															
			Formwork And Shoring	3.1.3	G													
			Repair Procedures	1.5.1	G													
			Mixture Proportioning	2.8	G													
			SD-06 Test Reports															
			Mixture Proportioning	2.8														
			Quality Control	3.1.5														
			Quality Control	3.3.3														
			Conventional Concrete	2.5														
			SD-07 Certificates															
			Qualifications	1.5.3														
			Reinforcement And Reinforcement Supports	2.4														
			Conventional Concrete	2.5														
		03 30 00	SD-01 Preconstruction Submittals															

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		03 30 00	Concrete Curing Plan	1.6.3.1													
			Quality Control Plan	1.6.5	G												
			Laboratory Accreditation	1.6.8	G												
			SD-02 Shop Drawings														
			Reinforcing Steel	1.6.2.1	G												
			SD-03 Product Data														
			Formwork Materials	2.1													
			Reinforcement	2.6													
			Liquid Chemical Floor Hardeners	2.4.3.1													
			and Sealers														
			Mechanical Reinforcing Bar	2.6.2													
			Connectors														
			Pumping Concrete	1.6.3.2													
			Finishing Plan	1.6.3.3													
			SD-05 Design Data														
			Concrete Mix Design	1.6.1.1	G												
			SD-06 Test Reports														
			Concrete Mix Design	1.6.1.1	G												
			Fly Ash	1.6.4.1													
			Pozzolan	1.6.4.1													
			Slag Cement	1.6.4.2													
			Aggregates	1.6.4.3													
			Tolerance Report	3.9.2.1													
			Compressive Strength Tests	3.13.2.3	G												
			Unit Weight of Structural	3.13.2.5													
			Concrete														

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		03 30 00	Chloride Ion Concentration	3.13.2.6													
			Air Content	3.13.2.4													
			Slump Tests	3.13.2.1													
			Water	2.3.2													
			SD-07 Certificates														
			Reinforcing Bars	2.6.1													
			Welder Qualifications	1.8													
			VOC Content for Form Release	1.6.3.4													
			Agents, Curing Compounds, and														
			Concrete Penetrating Sealers														
			Safety Data Sheets	1.6.3.5													
			Field Testing Technician and	1.6.6.2													
			Testing Agency														
			SD-08 Manufacturer’s Instructions														
			Liquid Chemical Floor Hardeners	2.4.3.1													
			and Sealers														
			Joint Sealants	2.4.5													
			Curing Compound	2.4.1													
		05 12 00	SD-01 Preconstruction Submittals														
			Erection and Erection Bracing	1.4.1.1	G												
			Drawings														
			SD-02 Shop Drawings														
			Fabrication Drawings	1.4.2	G												
			SD-03 Product Data														
			Shop Primer	2.6.2													
			Welding Electrodes and Rods	2.4.1													

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		05 12 00	Direct Tension Indicator Washers	3.2.2														
			Non-Shrink Grout	2.4.2														
			Tension Control Bolts	2.3.3														
			SD-06 Test Reports															
			Weld Inspection Reports	3.7.1.2														
			Bolt Testing Reports	3.7.2.1														
			Embrittlement Test Reports	3.7.3														
			SD-07 Certificates															
			MLIT Structural Steel Fabricator	1.3														
			Quality Certification															
			MLIT Structural Steel Erector	1.3														
			Quality Certification															
			Welding Procedure Specifications	3.4														
			(WPS)															
		05 50 13	SD-02 Shop Drawings															
			Cover Plates and Frames	2.4	G													
			Floor Gratings	2.5														
			Bollards/Pipe Guards	2.6	G													
			Angles and Plates	2.7	G													
			SD-03 Product Data															
			Corner Guards	2.3														
			Cover Plates and Frames	2.4														
			Floor Gratings	2.5														
		05 52 00	SD-02 Shop Drawings															
			Fabrication Drawings	1.2.1	G													
			SD-07 Certificates															

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		05 52 00	Welding Procedures	1.4.1	G												
			Welder Qualification	1.4.2	G												
			SD-08 Manufacturer’s Instructions														
			Installation Instructions	3.2													
		06 20 00	SD-02 Shop Drawings														
			Detail Drawings Indicating All	1.3	G												
			Wood Assemblies														
			SD-03 Product Data														
			Wood Products	2.1													
			Treated Wood Products	1.4													
			Hardware and Accessories	2.3													
			SD-04 Samples														
			Samples	1.5	G												
			SD-07 Certificates														
			Certificates of Grade	1.7.1.1	G												
		06 41 16.00 10	SD-02 Shop Drawings														
			Shop Drawings	2.9													
			Installation	3.1													
			SD-03 Product Data														
			Finish Schedule	1.2													
			SD-04 Samples														
			Plastic Laminates	2.3													
			Cabinet Hardware	2.6													
		06 61 16	SD-02 Shop Drawings														
			Detail Fabrication Drawings	1.4.2	G												
			Installation	3.1	G												

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		06 61 16	SD-03 Product Data															
			Solid Polymer	2.1.1	G													
			Indoor air quality for solid surface	2.2.2														
			seam and sealant products															
			SD-04 Samples															
			Material	2.1	G													
			Counter Tops	2.3.4	G													
			SD-06 Test Reports															
			Test Report Results	2.1.1														
			SD-07 Certificates															
			Qualifications	1.4.1														
			Indoor Air Quality for solid	2.1.1														
			surface fabrication products															
			SD-10 Operation and Maintenance															
			Data															
			Solid Polymer	2.1.1	G													
		07 21 13	SD-03 Product Data															
			Manufacturer's Standard Details	1.4	G													
			Block or Board Insulation	2.1	G													
			Pressure Sensitive Tape	2.2	G													
			Accessories	2.3	G													
			Recycled Content for Block or	2.1.4	S													
			Board Insulation															
			SD-07 Certificates															
			Block or Board Insulation	2.1	G													
			Special Warranties	1.9	G													

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		07 21 13	Special Warranties	1.9	G												
			Indoor Air Quality For Block Or	2.1.5	S												
			Board Insulation														
			SD-08 Manufacturer’s Instructions														
			Block or Board Insulation	2.1													
			Adhesive	2.3.1													
		07 21 16	SD-03 Product Data														
			Blanket Insulation	2.1													
			Accessories	2.2													
			SD-07 Certificates														
			Indoor Air Quality for Adhesives	2.2.1													
			SD-08 Manufacturer’s Instructions														
			Insulation	3.3.1													
		07 56 00	SD-03 Product Data														
			Fluid-Applied Membrane	2.1													
			Membrane Primer	2.3													
			Bond Breaker	2.8													
			SD-07 Certificates														
			Qualifications of Installer	1.9													
			SD-11 Closeout Submittals														
			Warranty	1.7													
			Information Card	3.6													
			Instructions To Government	3.5													
			Personnel														
		07 60 00	SD-02 Shop Drawings														
			Downspouts	3.1.9	G												

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		07 60 00	Base Flashing	3.1.7	G													
			Counterflashing	3.1.8	G													
			Scuppers	3.1.10	G													
			SD-08 Manufacturer's Instructions															
			Quality Control Plan	3.4	G													
		07 84 00	SD-02 Shop Drawings															
			Firestopping System	2.1	G													
			SD-03 Product Data															
			Firestopping Materials	2.2	G													
			SD-06 Test Reports															
			Inspection	3.3	G													
			SD-07 Certificates															
			Inspector Qualifications	1.5.2														
			Firestopping Materials	2.2														
			Installer Qualifications	1.5.1	G													
		08 11 13	SD-02 Shop Drawings															
			Doors	2.1	G													
			Doors	2.1	G													
			Frames	2.4	G													
			Frames	2.4	G													
		08 11 16	SD-02 Shop Drawings															
			Door, Frame, Entrance and	1.5.3	G													
			Window Assembly															
			SD-04 Samples															
			Finish Samples	1.5.4	G													
			SD-05 Design Data															

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		08 11 16	Calculations	1.2.1	G													
			SD-06 Test Reports															
			Air Infiltration	1.2.2	G													
			Water Penetration	1.2.3	G													
			SD-10 Operation and Maintenance															
			Data															
			Adjustments, Cleaning, and	1.5.5	G													
			Maintenance															
		08 14 00	SD-02 Shop Drawings															
			Doors	2.1	G													
			SD-04 Samples															
			Doors	2.1														
			Door Finish Colors	2.3.6.2														
			SD-11 Closeout Submittals															
			Warranty	1.5														
		08 31 00	SD-02 Shop Drawings															
			Access Doors And Panels	1.3	G													
			SD-03 Product Data															
			Access Doors And Panels	1.3	G													
			Hardware	1.3.2	G													
			Accessories	2.1.8	G													
			SD-04 Samples															
			Finishes	2.4	G													
			SD-06 Test Reports															
			Fire-rating(s) of Assemblies	1.3.1	G													
			Acoustical Ratings of Assemblies	1.3.1	G													

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		08 33 23	SD-02 Shop Drawings															
			Overhead Coiling Doors	2.2.1	G													
			Counterbalancing Mechanism	2.2.3	G													
			Electric Door Operators	2.2.4	G													
			Bottom Bars	2.2.1.2	G													
			Guides	2.1.1.1	G													
			Mounting Brackets	2.2.3.1	G													
			Hood	2.2.2.2	G													
			Installation Drawings	2.1.1.1	G													
			SD-03 Product Data															
			Overhead Coiling Doors	2.2.1	G													
			Hardware	2.2.2	G													
			Counterbalancing Mechanism	2.2.3	G													
			Electric Door Operators	2.2.4	G													
			Smoke-Rated Door Assembly	2.2.5	G													
			Recycled content for steel curtain slats	2.2.1.1	S													
			SD-05 Design Data															
			Overhead Coiling Doors	2.2.1	G													
			Hardware	2.2.2	G													
			Counterbalancing Mechanism	2.2.3	G													
			Electric Door Operators	2.2.4	G													
			Smoke-Rated Door	1.3	G													
			SD-10 Operation and Maintenance Data															

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		08 33 23	Operation and Maintenance	1.3.2	G													
			Manuals															
			SD-11 Closeout Submittals															
			Warranty	1.3.1	G													
		08 71 00	SD-02 Shop Drawings															
			Manufacturer's Detail Drawings	1.3	G													
			Verification of Existing Conditions	1.3	G													
			Hardware Schedule	1.5	G													
			Keying System	2.3.8	G													
			SD-08 Manufacturer’s Instructions															
			Installation	3.1														
			SD-10 Operation and Maintenance															
			Data															
			Hardware Schedule	1.5	G													
			SD-11 Closeout Submittals															
			Key Bitting	1.6.1														
		08 81 00	SD-03 Product Data															
			Insulating Glass	1.7.1														
			Sealants	2.4.3.1														
			Joint Backer	2.4.4														
			SD-08 Manufacturer’s Instructions															
			Setting and Sealing Materials	2.4														
			Glass Setting	3.2														
			SD-11 Closeout Submittals															
			Insulated Glass Units	1.7.1														
		08 91 00	SD-02 Shop Drawings															

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		08 91 00	Wall Louvers	1.4													
			SD-03 Product Data														
			Metal Wall Louvers	2.2													
			SD-04 Samples														
			Wall Louver Samples	1.5	G												
		09 22 00	SD-02 Shop Drawings														
			Metal Support Systems	2.1	G												
			SD-03 Product Data														
			Metal Support Systems	2.1													
		09 29 00	SD-08 Manufacturer’s Instructions														
			Safety Data Sheets	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Manufacturer Maintenance	2.1													
			Instructions														
		09 30 10	SD-02 Shop Drawings														
			Detail Drawings	3.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Installation	3.2	G												
			Installation	3.2	G												
		09 51 00	SD-02 Shop Drawings														
			Approved Detail Drawings	2.1	G												
		09 65 00	SD-02 Shop Drawings														
			Resilient Flooring and	2.9	G												
			Accessories														

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		09 65 00	SD-10 Operation and Maintenance														
			Data														
			Resilient Flooring and	2.9	G												
			Accessories														
		09 90 00	SD-02 Shop Drawings														
			Piping Identification	3.12													
			SD-08 Manufacturer’s Instructions														
			Application Instructions	3.4.1													
			Mixing	3.8.2													
			SD-10 Operation and Maintenance														
			Data														
			Coatings	1.3.1.1													
		10 14 00.20	SD-02 Shop Drawings														
			Detail Drawings	1.4.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Approved Manufacturer's	3.1	G												
			Instructions														
			Protection and Cleaning	3.1.2	G												
		10 28 13	SD-04 Samples														
			Finishes	2.1.2	G												
			Accessory Items	2.2													
			SD-10 Operation and Maintenance														
			Data														
			Electric Hand Dryer	2.2.16	G												
		10 44 16	SD-02 Shop Drawings														

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		10 44 16	Fire Extinguishers	2.1.1	G													
			Accessories	Part 2	G													
			Cabinets	Part 2	G													
			Wall Brackets	2.2.2	G													
			Schedule	1.5	G													
			SD-03 Product Data															
			Fire Extinguishers	2.1.1	G													
			Accessories	Part 2	G													
			Cabinets	Part 2	G													
			Wall Brackets	2.2.2	G													
			Replacement Parts List	3.2.1	G													
			SD-04 Samples															
			Equipment Samples	1.3.1	G													
			SD-07 Certificates															
			Fire Extinguishers Certifications	2.1.1	G													
			Manufacturer's Warranty with	1.4	G													
			Inspection Tag															
		12 21 00	SD-02 Shop Drawings															
			Installation	3.3														
			SD-03 Product Data															
			Window Blinds	2.1	G													
			Window Blinds	2.1	G													
			Valance	2.1.2.4	G													
			SD-06 Test Reports															
			Window Blinds	2.1														
			SD-08 Manufacturer’s Instructions															

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		12 21 00	Window Blinds	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Window Blinds	2.1													
		14 21 23	SD-02 Shop Drawings														
			Elevator	2.1	G												
			Elevator Components	1.2.1	G												
			Elevator Components	1.2.2	G												
			Elevator Machine	1.2.1	G												
			Elevator Controller	1.2.1	G												
			Wiring Diagrams	1.3.4	G												
			SD-03 Product Data														
			Elevator	2.1	G												
			Elevator Components	1.2.1	G												
			Elevator Components	1.2.2	G												
			Data Sheets	1.2.2	G												
			Elevator Microprocessor	2.5.2	G												
			Controller														
			SD-05 Design Data														
			Emergency Power Systems	1.2.3.3	G												
			Heat Loads	1.2.3.2	G												
			Reaction Loads	1.2.3.1	G												
			SD-07 Certificates														
			Price Lists	1.3.2													
			Warranty	1.4	G												
			Endorsement Letter	1.3.1.1	G												

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		14 21 23	Welders' Qualifications	1.2.4	G													
			Elevator Controller Certification	2.5.2.3	G													
			SD-10 Operation and Maintenance															
			Data															
			Elevator	2.1	G													
			Maintenance Control Program	1.2.5	G													
			(MCP)															
			Software and Documentation	2.5.2.2	G													
		21 12 00	SD-02 Shop Drawings															
			Standpipe System	1.3.3	G													
			SD-03 Product Data															
			Pipe and Fittings	2.1.1	G													
			Mechanical Couplings	2.1.1	G													
			Pipe Hangers and Supports	2.1.2	G													
			Valves	2.1.3	G													
			Fire Department Connections	2.1.7	G													
			Valve Tamper Switch	2.1.8	G													
			Buried Pipe and Fittings	2.2.1	G													
			SD-06 Test Reports															
			Preliminary Tests	3.6.1	G													
			Acceptance Tests	3.6.2	G													
			SD-07 Certificates															
			Qualifications of Installer	1.5.1	G													
			SD-11 Closeout Submittals															
			System As-Built Drawings	1.5.2	G													
		21 13 13	SD-01 Preconstruction Submittals															

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		21 13 13	Qualified Fire Protection Engineer	1.2.3	G												
			(QFPE)														
			Sprinkler System Designer	1.4.2.1	G												
			Sprinkler System Installer	1.4.2.2	G												
			SD-02 Shop Drawings														
			Shop Drawing	1.2.1.1	G												
			SD-03 Product Data														
			Pipe	2.2.1	G												
			Fittings	2.3.1.2	G												
			Valves	2.3.4	G												
			Relief Valves	2.8.5	G												
			Sprinklers	2.7	G												
			Pipe Hangers and Supports	2.3.3	G												
			Sprinkler Alarm Switch	2.4.1	G												
			Valve Supervisory (Tamper)	2.4.2	G												
			Switch														
			Fire Department Connection	2.6	G												
			Backflow Prevention Assembly	2.5	G												
			Air Vent	2.8.6	G												
			Hose Valve	2.5.1	G												
			Seismic Bracing	2.3.3	G												
			Nameplates	2.1.2	G												
			SD-05 Design Data														
			Seismic Bracing	2.3.3	G												
			Hydraulic Calculations	1.2.1.2	G												
			SD-06 Test Reports														

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		21 13 13	Test Procedures	3.7.1	G													
			SD-07 Certificates															
			Verification of Compliant	3.7.2.1	G													
			Installation															
			Request for Government Final	3.7.2.2	G													
			Test															
			SD-10 Operation and Maintenance															
			Data															
			Operating and Maintenance	3.9	G													
			(O&M) Instructions															
			Spare Parts	1.6	G													
			SD-11 Closeout Submittals															
			As-built drawings	3.9														
		21 30 00	SD-01 Preconstruction Submittals															
			Fire Pump Installation Related	1.3														
			Submittals															
			Fire Protection Specialist	1.7.2	G													
			SD-02 Shop Drawings															
			Installation Drawings	3.3.1	G													
			As-Built Drawings	3.11.2	G													
			Piping Layout	3.3.2	G													
			Pump Room	3.3.2	G													
			SD-03 Product Data															
			Catalog Data	2.1	G													
			Spare Parts	1.6														
			Preliminary Tests	3.8.2														

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		21 30 00	Field Tests	3.8	G												
			Manufacturer's Representative	1.7.7	G												
			Field Training	3.11.1	G												
			Final Acceptance Test	3.8.3													
			SD-06 Test Reports														
			Preliminary Tests	3.8.2													
			SD-07 Certificates														
			Fire Protection Specialist	1.7.2													
			Qualifications of Welders	1.7.3													
			Qualifications of Installer	1.7.4													
			Preliminary Test Certification	1.7.5													
			Final Test Certification	1.7.6													
			SD-10 Operation and Maintenance														
			Data														
			Operating and Maintenance	3.11.1	G												
			Instructions														
			Flow Meter	2.14													
		22 00 00	SD-02 Shop Drawings														
			Plumbing System	3.9.1	G												
			SD-03 Product Data														
			Water Heaters	2.7													
			Pumps	2.9													
			SD-06 Test Reports														
			Tests, Flushing and Disinfection	3.9													
			SD-10 Operation and Maintenance														
			Data														

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		22 00 00	Plumbing System	3.9.1														
		23 05 15	SD-01 Preconstruction Submittals															
			Material, Equipment, and Fixture	1.2	G													
			Lists															
			SD-02 Shop Drawings															
			Record Drawings	1.2	G													
			Coordination Drawings	1.2	G													
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	3.10														
			Manuals															
		23 05 48.19	SD-02 Shop Drawings															
			Coupling and Bracing	3.1														
			Flexible Couplings or Joints	3.3														
			Equipment Restraint	2.2														
			Contractor Designed Bracing	1.2.4	G													
			SD-03 Product Data															
			Coupling and Bracing	3.1	G													
			Flexible Couplings Or Joints	3.3	G													
			Equipment Restraint	2.2	G													
			Contractor Designed Bracing	1.2.4	G													
			Snubbers	2.6														
			Anchor Bolts	3.9														
			Vibration Isolators	2.2.2														
			SD-05 Design Data															
			Design Calculations	1.2.4														

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		23 05 48.19	SD-06 Test Reports															
			Anchor Bolts	3.9	G													
		23 05 93	SD-01 Preconstruction Submittals															
			TAB Firm And Personnel Qualifications	1.6	G													
			TAB Firm And Personnel Qualifications	1.6	G													
			Design Review Report	1.9	G													
			TAB Pre-Field Engineering Plan	1.10	G													
			SD-06 Test Reports															
			Certified TAB Report	3.2.7	G													
			Certified TAB Report of Proportional Balancing	3.2.7	G													
			Certified TAB Report of Season 1	3.2.7	G													
			SD-11 Closeout Submittals															
			Certified TAB Report of Season 2	3.3.1.2	G													
			Certified Final TAB Report	3.2.7	G													
			Certified Final TAB Report	3.3.1.2	G													
		23 07 00	SD-02 Shop Drawings															
			Pipe Insulation Systems	2.3														
			Duct Insulation Systems	3.3														
			Equipment Insulation Systems	3.4														
		23 08 00	SD-03 Product Data															
			Test Equipment	2.1	G													
			SD-06 Test Reports															

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		23 08 00	Pipe Flushing, Testing, And Water Treatment Reports	1.5	G													
			Completed Pre-Functional Checklists	3.3	G													
			Seasonal Test Report	3.12	G													
			Full-Load Test Report	3.13	G													
			Post-Construction Trend Log Report	3.16	G													
			SD-07 Certificates															
			Certificate Of Readiness	1.6	G													
		23 09 00	SD-02 Shop Drawings															
			DDC Contractor Design Drawings	3.2	G													
			Draft As-Built Drawings	3.2	G													
			Final As-Built Drawings	3.2	G													
			SD-03 Product Data															
			Programming Software	1.8.3	G													
			Controller Application Programs	1.8.4	G													
			Configuration Software	1.8.1	G													
			Controller Configuration Settings	1.8.2	G													
			Proprietary Multi-Split Engineering Tool Software	1.1.1.3	G													
			Manufacturer's Product Data	2.2	G													
			SD-05 Design Data															
			Boiler Or Chiller Plant Gateway Request	1.9														
			SD-06 Test Reports															

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		23 09 00	Pre-Construction Quality Control	1.10.1	G												
			(QC) Checklist														
			Post-Construction Quality Control	1.10.2	G												
			(QC) Checklist														
			Start-Up Testing Report	3.4.2	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.5	G												
			(O&M) Instructions														
			Training Documentation	3.7.1	G												
			SD-11 Closeout Submittals														
			Enclosure Keys	2.5	G												
			Password Summary Report	3.1.6.1	G												
			Closeout Quality Control (QC)	1.10.3	G												
			Checklist														
		23 23 00	SD-02 Shop Drawings														
			Refrigerant Piping System	2.3	G												
			SD-06 Test Reports														
			Refrigerant Piping Tests	3.5													
			SD-10 Operation and Maintenance														
			Data														
			Maintenance	1.5													
			Operation and Maintenance	3.2.4.3													
			Manuals														
		23 30 00	SD-02 Shop Drawings														
			Detail Drawings	1.4.4	G												

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		23 30 00	SD-03 Product Data														
			Fire Dampers	2.8.3													
			Air Handling Units	2.10	G												
			Room Fan-Coil Units	2.11.1	G												
			Reheat Units	2.11.2.1	G												
			Energy Recovery Devices	2.12	G												
			Test Procedures	1.4.5													
			Diagrams	1.2.1.2	G												
			SD-06 Test Reports														
			Performance Tests	3.12													
			Damper Acceptance Test	3.10	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.14.1													
			Manuals														
			Fire Dampers	2.8.3													
			Manual Balancing Dampers	2.8.4													
			Centrifugal Fans	2.9.1.1													
			In-Line Centrifugal Fans	2.9.1.2													
			Propeller Type Power Wall	2.9.1.3													
			Ventilators														
			Centrifugal Type Power Roof	2.9.1.4													
			Ventilators														
			Ceiling Exhaust Fans	2.9.1.5													
			Air Handling Units	2.10													
			Room Fan-Coil Units	2.11.1													

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		23 30 00	Reheat Units	2.11.2.1													
			Unit Ventilators	2.11.3													
			Energy Recovery Devices	2.12													
		23 57 10.00 10	SD-02 Shop Drawings														
			Heating System	2.17													
			SD-03 Product Data														
			Spare Parts	1.5													
			Welding	3.4													
			Framed Instructions	3.19													
			SD-06 Test Reports														
			Testing and Cleaning	3.18													
			Water Treatment Testing	3.18.4													
			SD-07 Certificates														
			Bolts	3.10.2													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.20	G												
			Manuals														
		23 64 10	SD-03 Product Data														
			Water Chiller	2.5	G												
			Water Chiller - Field Acceptance	3.4.1													
			Test Plan														
			SD-06 Test Reports														
			Field Acceptance Testing	3.4													
			Water Chiller - Field Acceptance	3.4.2													
			Test Report														

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		23 64 10	System Performance Tests	3.5														
			SD-07 Certificates															
			Refrigeration System	3.1.9														
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	3.6														
			Manuals															
		23 74 33	SD-02 Shop Drawings															
			Coils	2.4	G													
			Controls	2.6														
			Unit Cabinet	2.2.1	G													
			Casing	2.2.2	G													
			Installation Drawings	3.1	G													
			SD-03 Product Data															
			Equipment and Performance	Part 1	G													
			Data															
			Air-Conditioning Systems	3.2.1	G													
			Coils	2.4	G													
			Fans	2.3	G													
			Controls	2.6	G													
			Unit Cabinet	2.2.1	G													
			Casing	2.2.2	G													
			Air Filters	2.5	G													
			SD-06 Test Reports															
			System Performance Tests	3.2.2	G													
			SD-07 Certificates															

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		23 74 33	List of Product Installations	1.4													
			Manufacturer's Warranty	1.9	G												
			Coil Coating Warranty	1.9	G												
			SD-08 Manufacturer’s Instructions														
			Manufacturer's Installation	3.1													
			Instructions														
			Operation and Maintenance	3.3.2													
			Training														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.3.1													
			Manuals, Data Package 2														
		23 81 00	SD-02 Shop Drawings														
			Field-Assembled Refrigerant	2.6.2													
			Piping														
			Control System Wiring Diagrams	1.3.2													
			SD-03 Product Data														
			Room Air Conditioners	2.1													
			Heat Pumps	2.2													
			Training	3.8	G												
			Posted Instructions	3.8													
			Spare Parts	3.9.1													
			SD-06 Test Reports														
			Start-Up and Initial Operational	3.7.3													
			Tests														
			SD-07 Certificates														

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		23 81 00	Service Organizations	3.9.2														
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	3.8	G													
			Manuals															
			Room Air Conditioners	2.1														
			Heat Pumps	2.2														
			Filters	2.3														
			Thermostats	2.2.4														
		26 05 48	SD-02 Shop Drawings															
			Equipment Requirements	2.1	G	RMS												
			Lighting Fixtures	3.4	G	AE												
			SD-03 Product Data															
			Supports and Attachments	1.2.3	G	AE												
			Equipment Requirements	2.1	G	RMS												
			Lighting Fixtures	3.4	G	RMS												
			Flexible Fittings	2.4	G	AE												
			Anchors	3.5	G													
			SD-05 Design Data															
			Design Calculations	1.2.3	G	DO												
			Design Drawings	1.2.3	G	DO												
			Independent Design Review	1.2.3	G													
			Report															
			SD-06 Test Reports															
			Anchors	3.5	G	AE												
			SD-07 Certificates															

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		26 05 48	ICC ES AC156 Shake Table Test	3.7	G AE												
		26 08 00	SD-06 Test Reports														
			Acceptance Tests and	3.1	G RMS												
			Inspections														
			SD-07 Certificates														
			Qualifications	1.4.1	G RMS												
			Acceptance Test and Inspections	1.4.3	G RMS												
			Procedure														
		26 11 16.00 33	SD-02 Shop Drawings														
			Cubicle Type Unit Substation	2.3													
			Distribution	2.6													
			Switchgear	2.5													
			SD-03 Product Data														
			Transformers	2.4	G												
			Transformers	3.8.3.6	G												
			Disconnecting Switches (DS)	2.5.1.1	G												
			Load Disconnecting Switches	2.5.2.2	G												
			(LDS)														
			Vacuum Circuit Breaker (VCB)	2.5.1.2	G												
			Load Break Switches (LBS)	2.5.2.3	G												
			Load Break Switches (LBS) with	2.5.2.4	G												
			tripping device														
			Load Break Switches (LBS) with	2.5.2.5	G												
			fuse														
			Circuit breaker	2.6.4.1	G												
			Circuit Breaker with GFCI	2.6.4.2	G												

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ACTIVITY NO (a)	TRANSMITTAL NO (b)	SPEC SECT (c)	DESCRIPTION ITEM SUBMITTED (d)	PARAGRAPH # (e)	CLASSIFICATION GOVT OR A/E REVWR (f)	CONTRACTOR: SCHEDULE DATES SUBMIT (g)	APPROVAL NEEDED BY (h)	MATERIAL NEEDED BY (i)	CONTRACTOR ACTION ACTION CODE (j)	DATE OF ACTION (k)	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR (l)	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER (m)	DATE RCD FROM OTH REVIEWER (n)	ACTION CODE (o)	DATE OF ACTION (p)	MAILED TO CONTR/ DATE RCD FRM APPR AUTH (q)	REMARKS (r)
		26 11 16.00 33	Automatic Transfer Switch (ATS)	2.6.5	G												
			Power Fuses (PF)	2.5.2.5	G												
			Instruments	2.6.7.1	G												
			Instrument Control Switches	2.6.7.2	G												
			Buzzer	2.6.7.3													
			Test Terminal	2.6.7.4													
			Lightning Arrester	2.6.7.6	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.8.2													
			SD-10 Operation and Maintenance														
			Data														
			Data Package 5	1.4.1	G												
		26 20 00	SD-02 Shop Drawings														
			Panelboards	2.15	G												
			Transformers	2.19	G												
			Busway	2.4	G												
			Cable trays	2.5	G												
			Motor control centers	2.23	G												
			Wireways	2.32	G												
			Marking strips	3.1.12.1	G												
			SD-03 Product Data														
			Receptacles	2.14													
			Circuit breakers	2.15.3	G												
			Switches	2.12	G												
			Transformers	2.19	G												
			Enclosed circuit breakers	2.17	G												

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		26 20 00	Motor controllers	2.21	G												
			Manual motor starters	2.22	G												
			Residential load centers	2.16	G												
			Metering	2.33	G												
			Meter base only	2.34	G												
			CATV outlets	2.26.1	G												
			Surge protective devices	2.35	G												
			SD-06 Test Reports														
			600-volt wiring test	3.5.2													
			Grounding system test	3.5.5													
			Transformer tests	3.5.3													
			Ground-fault receptacle test	3.5.4													
			SD-10 Operation and Maintenance														
			Data														
			Electrical Systems	1.5.1													
			Metering	2.33													
		26 24 13	SD-02 Shop Drawings														
			Switchboard Drawings	1.5.2	G												
			SD-03 Product Data														
			Switchboard	2.2	G												
			SD-06 Test Reports														
			Switchboard Design Tests	2.5.2													
			SD-10 Operation and Maintenance														
			Data														
			Switchboard Operation and	1.6.1													
			Maintenance														

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		26 24 13	SD-11 Closeout Submittals														
			Assembled Operation and	1.6.2													
			Maintenance Manuals														
			Equipment Test Schedule	2.5.1													
			Required Settings	3.5													
			Service Entrance Available Fault	2.7													
			Current Label														
		26 27 14.00 20	SD-02 Shop Drawings														
			Installation Drawings	1.3.1	G												
			SD-03 Product Data														
			Electricity Meters	2.1.4	G												
			Current Transformer	2.1.3	G												
			Potential Transformer	2.1.2	G												
			Communications	2.2	G												
			Configuration Software	3.1.3	G												
			SD-06 Test Reports														
			Acceptance Checks and Tests	3.2.1	G												
			System Functional Verification	3.2.2	G												
			Building Meter Installation Sheet,	3.2.1	G												
			per Building														
			Completed Meter Installation	3.2.1	G												
			Schedule														
			Completed Meter Data Schedule	3.2.1	G												
			Meter Configuration Template	2.1.1	G												
			Meter Configuration Report	3.2.1	G												

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		26 27 14.00 20	SD-10 Operation and Maintenance															
			Data															
			Electricity Meters and	1.4.1	G													
			Accessories															
			SD-11 Closeout Submittals															
			System Functional Verification	3.2.2	G													
		26 29 23	SD-02 Shop Drawings															
			Schematic Diagrams	1.5.1	G													
			Interconnecting Diagrams	1.5.2	G													
			Installation Drawings	1.5.3	G													
			As-Built Drawings	1.5.3	G													
			SD-03 Product Data															
			Adjustable Speed Drives	2.1	G													
			Wires and Cables	2.3														
			Equipment Schedule	1.5.4														
			SD-06 Test Reports															
			ASD Test	3.3.1														
			Performance Verification Tests	3.3.2														
			Endurance Test	3.3.3														
			SD-07 Certificates															
			Testing Agency's Field	3.3.1	G													
			Supervisor															
			SD-08 Manufacturer’s Instructions															
			Installation instructions	1.5.5														
			SD-09 Manufacturer’s Field															
			Reports															

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		26 29 23	ASD Test Plan	2.5.1	G													
			Standard Products	1.5.6														
			SD-10 Operation and Maintenance															
			Data															
			Adjustable Speed Drives	2.1														
		26 32 15	SD-02 Shop Drawings															
			Engine-Generator Set and	2.1.1	G													
			Auxiliary Equipment															
			Auxiliary Systems	1.4.9	G													
			Detailed Drawings	1.4.8	G													
			Acceptance	3.17	G													
			SD-03 Product Data															
			Harmonic Requirements	2.1.8	G													
			Engine-Generator Set	2.1.1	G													
			Efficiencies															
			special tools	2.8	G													
			Remote Alarm Annunciator	2.7.4	G													
			Engine-Generator Parameter	2.1.1														
			Schedule															
			Generator	2.9														
			Site Welding	1.4.2														
			Spare Parts	1.7.2														
			Onsite Training	3.12														
			Vibration-Isolation	2.1.7														
			Posted Data and Instructions	3.16	G													
			Instructions	3.6.7.1	G													

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		26 32 15	Experience	1.4.6													
			Field Engineer	1.4.7													
			General Installation	3.2													
			Exciter	2.10													
			SD-05 Design Data														
			Performance Criteria	2.9													
			Integral Main Fuel Storage Tank	2.6.4													
			Power Factor	3.6.1.2													
			Time-Delay on Alarms	2.13.5													
			Vibration Isolation	2.1.7													
			Battery Charger	2.7.3.2													
			Capacity Calculations for	2.1.1	G												
			Engine-Generator Set														
			Brake Mean Effective Pressure	2.1.1	G												
			(BMEP) Calculations														
			Torsional Vibration Stress	2.1.1	G												
			Analysis Computations														
			Capacity Calculations for	2.1.1	G												
			Batteries														
			Turbocharger Load Calculations	2.1.1	G												
			SD-06 Test Reports														
			Performance Tests	3.6.9													
			Factory Inspection and Tests	2.23													
			Factory Tests	2.23.2													
			Onsite Inspection and Tests	3.6	G												
			Acceptance Checks and Tests	3.9.1	G												

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		26 32 15	Functional Acceptance Tests	3.10.2	G												
			Maintenance Procedures	3.12	G												
			Operation and Maintenance Manuals	3.12	G												
			Inspections	3.6.3	G												
			Functional Acceptance Test Procedure	3.9.5	G												
			SD-07 Certificates														
			Vibration Isolation	2.1.7													
			Prototype Test	2.23.2													
			Reliability and Durability	2.1.5													
			Fuel System Certification	1.4.12	G												
			Start-Up Engineer	3.8	G												
			Instructor's Qualification Resume	3.11.1	G												
			Engine Emission Limits	2.1.2.1	G												
			Site Visit	3.1													
			Current Balance	2.9.1													
			Materials and Equipment	2.4													
			Factory Inspection and Tests	2.23													
			Engine Tests	3.6.4	G												
			Generator Tests	3.6.5	G												
			Assembled Engine-Generator Set Tests	3.6.6	G												
			SD-10 Operation and Maintenance Data														

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		26 32 15	Preliminary Assembled Operation	3.9.4	G													
			and Maintenance Manuals															
			Posted Data and Instructions	3.16	G													
			Training Plan	3.11.2	G													
		26 36 23	SD-02 Shop Drawings															
			Automatic Transfer Switch	1.5.2	G													
			Drawings															
			SD-03 Product Data															
			Automatic Transfer Switches	2.1	G													
			By-Pass/Isolation Switch (BP/IS)	2.2	G													
			Remote Annunciator Panel	2.4	G													
			Remote Annunciator and Control	2.5	G													
			System Panel															
			SD-06 Test Reports															
			Functional Acceptance Tests	3.1.1	G													
			Factory Testing	2.6	G													
			Factory Test Reports	2.6.2	G													
			SD-07 Certificates															
			Proof of Listing	1.5.1	G													
			SD-10 Operation and Maintenance															
			Data															
			Operation and Maintenance	1.4	G													
			Manual															
		26 51 00	SD-02 Shop Drawings															
			Occupancy/Vacancy Sensor	1.5.2														
			Coverage Layout															

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		26 51 00	SD-03 Product Data															
			Luminaires	2.2	G													
			Light Sources	2.4	G													
			Drivers, Ballasts and Generators	2.3	G													
			LED Luminaire Warranty	1.6.1														
			Vacancy Sensors	2.5.3.2	G													
			Dimming Controllers (Dimmers)	2.5.2	G													
			Timeswitch	2.5.4	G													
			Power Hook Luminaire Hangers	2.8	G													
			Exit Signs	2.6.1	G													
			Emergency Lighting Unit (EBU)	2.6.2	G													
			LED Emergency Drivers	2.6.3	G													
			Occupancy Sensors	2.5.3.1	G													
			Ambient Light Level Sensor	3.1.8	G													
			Lighting Control Panel	2.5.5	G													
			SD-06 Test Reports															
			Occupancy/Vacancy Sensor	1.5.7														
			Verification Tests															
			Energy Efficiency	1.5.10.3														
		27 05 13.43	SD-02 Shop Drawings															
			Wiring Diagrams And Installation	1.5.1	G													
			Details															
			System Components	1.3.4	G													
			SD-03 Product Data															
			Attenuators	2.2.2	G													
			Amplifiers	2.3.1	G													

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		27 05 13.43	Cables	2.3.2	G												
			Terminators	2.3.5	G												
			Splitters/Combiners	2.3.6	G												
			Fiber Optic Couplers	2.3.7	G												
			Line Taps	2.3.8	G												
			Coaxial Outlets	2.3.9	G												
			Coaxial Connectors	2.3.10	G												
			Tilt Compensator	2.3.11	G												
			Distribution Hubs	2.3.12	G												
			Baluns	2.3.13	G												
			Grounding Block	2.4.1	G												
			SD-05 Design Data														
			Television Distribution System	1.5.2	G												
			Loss Calculations														
			SD-06 Test Reports														
			Operational Test Plan	1.5.3	G												
			Operational Test Procedures	1.5.4	G												
			System Pretest	3.2.1	G												
			Acceptance Tests	3.2.2	G												
			SD-08 Manufacturer’s Instructions														
			Connector Installation	1.5.5	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.3	G												
			Manuals														
		27 10 00	SD-02 Shop Drawings														

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		27 10 00	Telecommunications drawings	1.6.1.1	G													
			Telecommunications Space	1.6.1.2	G													
			Drawings															
			SD-03 Product Data															
			Telecommunications cabling	2.3	G													
			Patch panels	2.4.5	G													
			Telecommunications	2.5	G													
			outlet/connector assemblies															
			Equipment support frame	2.4.2	G													
			Connector blocks	2.4.3	G													
			SD-06 Test Reports															
			Telecommunications cabling	3.5.1														
			testing															
			SD-10 Operation and Maintenance															
			Data															
			Telecommunications cabling and	1.10.1														
			pathway system															
			SD-11 Closeout Submittals															
			Record Documentation	1.10.2														
		28 31 76	SD-01 Preconstruction Submittals															
			Qualified Fire Protection Engineer	1.3.2	G													
			(QFPE)															
			Fire alarm system designer	1.9.2.1	G													
			Supervisor	1.9.2.2	G													
			Technician	1.9.2.3	G													
			Installer	1.9.2.4	G													

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		28 31 76	Test Technician	1.9.2.5	G													
			Fire Alarm System Site-Specific	1.7	G													
			Software Acknowledgement															
			SD-02 Shop Drawings															
			Nameplates	1.9.1.3	G													
			Instructions	2.2.4	G													
			Wiring Diagrams	1.9.1.4	G													
			System Layout	1.9.1.5	G													
			Notification Appliances	1.9.1.6	G													
			Initiating devices	1.9.1.7	G													
			Amplifiers	1.9.1.8	G													
			Battery Power	1.9.1.9	G													
			Voltage Drop Calculations	1.9.1.10	G													
			SD-03 Product Data															
			Fire Alarm and Mass Notification	2.3	G													
			Control Unit (FMCU)															
			Local Operating Console (LOC)	1.4.4	G													
			Amplifiers	1.9.1.8	G													
			Tone Generators	2.5	G													
			Digitalized voice generators	2.5	G													
			LCD Annunciator	2.6.1	G													
			Manual Stations	2.7	G													
			Smoke Detectors	2.8	G													
			Duct Smoke Detectors	2.8.2	G													
			Addressable Interface Devices	2.9	G													
			Addressable Control Modules	2.10	G													

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		28 31 76	Isolation Modules	2.11	G												
			Notification Appliances	1.9.1.6	G												
			Textual Display Signs	2.12.3	G												
			Batteries	2.14.1	G												
			Battery Chargers	2.14.2	G												
			Supplemental Notification	2.14.1.1	G												
			Appliance Circuit Panels														
			Surge Protective Devices	2.15	G												
			Alarm Wiring	2.15	G												
			Back Boxes and Conduit	3.3.4	G												
			Ceiling Bridges	3.2.9	G												
			Terminal Cabinets	3.3.2	G												
			Digital Alarm Communicator	2.17.1	G												
			Transmitter (DACT)														
			Automatic Fire Alarm	2.17	G												
			Transmitters														
			Electromagnetic Door Holders	2.18.3	G												
			Environmental Enclosures or	2.19	G												
			Guards														
			Document Storage Cabinet	3.12.3	G												
			SD-06 Test Reports														
			Test Procedures	3.8.1	G												
			SD-07 Certificates														
			Verification of Compliant	3.8.2.1	G												
			Installation														

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		28 31 76	Request for Government Final Test	3.8.2.2	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance (O&M) Instructions	3.10	G												
			Instruction of Government Employees	3.11	G												
			SD-11 Closeout Submittals														
			As-Built Drawings	1.9.1.13													
			Spare Parts	1.11.1													
		31 00 00	SD-01 Preconstruction Submittals														
			Shoring	3.5	G												
			Dewatering Work Plan	1.3.2	G												
			SD-06 Test Reports														
			Testing	3.16													
			Borrow Site Testing	2.1													
		32 11 20	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4													
			SD-06 Test Reports														
			Initial Tests	2.2.1	G												
			In-Place Tests	3.12.1	G												
		32 11 23	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4													
			SD-06 Test Reports														
			Initial Tests	2.3.1	G												

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		32 11 23	In-Place Tests	3.12.1	G												
		32 12 13	SD-06 Test Reports														
			Sampling and Testing	3.7	G												
		32 12 16.16	SD-03 Product Data														
			Mix Design	2.4	G												
			Quality Control	3.11	G												
			Material Acceptance	3.12	G												
			SD-06 Test Reports														
			Aggregates	2.2	G												
			QC Monitoring	3.11.3.2													
		32 16 19	SD-03 Product Data														
			Concrete	2.1	G												
			SD-06 Test Reports														
			Field Quality Control	3.8													
		32 17 23	SD-03 Product Data														
			Paints for Roads and Streets	2.2.1													
			Paints for Concrete Curb	2.2.2													
			Markings														
			Exterior Surface Preparation	3.2													
			Safety Data Sheets	1.3.1	G												
			Thermoplastic compound	3.3.2	G												
			SD-07 Certificates														
			Volatile Organic Compound	1.3.1													
			SD-08 Manufacturer’s Instructions														
			Paint	3.3.1													
		32 31 13	SD-02 Shop Drawings														

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		32 31 13	Fence Assembly	2.1	G													
			Location of Gate, Corner, End,	3.2.2.1	G													
			and Pull Posts															
			Gate Assembly	2.2.10.1	G													
			Gate Hardware and Accessories	2.2.10.3	G													
			Erection/Installation Drawings	Part 3	G													
			SD-03 Product Data															
			Fence Assembly	2.1	G													
			Gate Assembly	2.2.10.1	G													
			Gate Hardware and Accessories	2.2.10.3	G													
			Zinc Coating	2.3.1	G													
			PVC Coating	2.1	G													
			Fabric	2.2.1	G													
			Stretcher Bars	2.2.6	G													
			Barbed Wire	2.2.4	G													
			Padlocks	2.2.13	G													
			Turnbuckles	2.2.10.4	G													
			Truss Rod	2.2.10.5	G													
			Tension Wires	2.2.10.6	G													
			Wire Ties	2.2.11	G													
			Concrete	2.3.2	G													
			SD-07 Certificates															
			Certificates of Compliance	1.3.1														
			SD-08 Manufacturer's Instructions															
			Fence Assembly	2.1														
			Gate Assembly	2.2.10.1														

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		32 31 13	Hardware Assembly	2.1														
			Accessories	2.1														
			SD-11 Closeout Submittals															
			Recycled Material Content	3.3														
		32 92 23	SD-03 Product Data															
			Fertilizer	2.4														
			SD-06 Test Reports															
			Topsoil composition tests	2.2.3														
			SD-07 Certificates															
			sods	2.1														
		33 11 00	SD-03 Product Data															
			Fire Hydrants	2.1.3.1														
			Meters	2.1.4														
			Disinfection Procedures	3.2.3	G													
			SD-06 Test Reports															
			Bacteriological Samples	3.3.1.4	G													
			Hydrostatic Sewer Test	3.2.1.1.6	G													
			Leakage Test	3.3.1.3	G													
			Hydrostatic Test	3.3.1.1	G													
			SD-07 Certificates															
			Fire Hydrants	2.1.3.1														
			SD-08 Manufacturer’s Instructions															
			Ductile Iron Piping	2.1.1.1														
		33 30 00	SD-02 Shop Drawings															
			Installation Drawings	3.1.1	G													
			SD-06 Test Reports															

ACTIVITY NO (a)	TRANSMITTAL NO (b)	SPEC SECT (c)	DESCRIPTION ITEM SUBMITTED (d)	PARAGRAPH # (e)	CLASSIFICATION GOVT OR A/E REVWR (f)	CONTRACTOR: SCHEDULE DATES SUBMIT (g)	APPROVAL NEEDED BY (h)	MATERIAL NEEDED BY (i)	CONTRACTOR ACTION ACTION CODE (j)	DATE OF ACTION (k)	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR (l)	APPROVING AUTHORITY DATE FWD TO OTHER REVIEWER (m)	DATE RCD FROM OTH REVIEWER (n)	ACTION CODE (o)	DATE OF ACTION (p)	MAILED TO CONTR/ DATE RCD FRM APPR AUTH (q)	REMARKS (r)
		33 30 00	Hydrostatic Sewer Test	3.3.1.1													
			Infiltration Tests or Exfiltration	3.3.1.2													
			Tests														
			Deflection Testing	3.3.1.3													
			SD-07 Certificates														
			Pre-Installation Inspection	3.3.3.1	G												
			Request														
			Post-Installation Inspection	3.3.3.2	G												
		33 40 00	SD-07 Certificates														
			Leakage Test	3.10.1.1	G												
			Determination of Density	3.10.1.2													
			Post-Installation Inspection	3.10.2.1.2	G												
			Report														
			Placing Pipe	3.3													
		33 61 14	SD-02 Shop Drawings														
			Factory-prefabricated	2.1													
			preinsulated water piping system														
			field joints	3.2													
			SD-03 Product Data														
			Pipe, fittings, and end	2.1													
			connections														
			Factory-prefabricated	2.1													
			preinsulated water piping system														
			SD-07 Certificates														
			Certification of welders'	1.4.1													
			qualifications														

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		33 61 14	SD-08 Manufacturer's Instructions															
			factory-prefabricated preinsulated	2.1														
			water piping system															
		33 71 02	SD-02 Shop Drawings															
			Precast underground structures	1.6.1	G													
			SD-03 Product Data															
			High voltage cable	2.5	G													
			High voltage cable joints	2.7	G													
			High voltage cable terminations	2.6	G													
			Precast concrete structures	2.13.2.1	G													
			Sealing Material	2.13.2.4														
			Pulling-In Irons	3.4.3														
			Manhole frames and covers	2.13.3	G													
			Handhole frames and covers	2.13.4	G													
			Composite/fiberglass handholes	2.13.6	G													
			Cable supports	2.14	G													
			SD-06 Test Reports															
			Field Acceptance Checks and	3.18.1														
			Tests															
		33 82 00	SD-02 Shop Drawings															
			Telecommunications Outside	1.6.1.1	G													
			Plant															
			Telecommunications Entrance	1.6.1.2	G													
			Facility Drawings															
			SD-03 Product Data															
			Wire and cable	2.8	G													

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		33 82 00	Cable splices, and connectors	2.5	G													
			Closures	2.3	G													
			Building protector assemblies	2.2.1	G													
			Protector modules	2.2.2	G													
			Cross-connect terminal cabinets	2.4	G													
			SD-06 Test Reports															
			Pre-installation tests	3.5.1														
			Acceptance tests	3.5.2														
			SD-10 Operation and Maintenance															
			Data															
			Telecommunications Outside	1.6.1.1														
			plant (OSP)															
			SD-11 Closeout Submittals															
			Record Documentation	1.8.1														

TEST REQUIREMENTS LIST

==
This report lists all the test requirements cited in the text.

HINT: With this file opened as a report from the SI Processed Files folder, double-clicking a Section number will open the Section in the Editor.
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SECTION: 02 82 00 **SUBPART:** 3.3.3.1

An area for interim storage of asbestos waste-containing drums or skids will be assigned by the Contracting Officer or his authorized representative.

SECTION: 03 30 00 **SUBPART:** 3.13.1

Collect samples of fresh concrete to perform tests specified.

SECTION: 03 30 00 **SUBPART:** 3.13.2.1

Take concrete samples during concrete placement/discharge.

SECTION: 03 30 00 **SUBPART:** 3.13.2.2

Test the concrete delivered and the concrete in the forms

SECTION: 03 30 00 **SUBPART:** 3.13.2.3

JIS A 1108. Make six 150 mm by 300 mm . Concrete compressive tests must meet the requirements of this section, the Contract Document, and JASS 5 and MLIT-SS Chapter 6. Retest locations represented by erratic core

SECTION: 03 30 00 **SUBPART:** 3.13.2.4

JIS A 1118 or JIS A 1128 for normal weight concrete. Test air-entrained concrete for air content at the same frequency as specified for slump tests.

SECTION: 03 30 00 **SUBPART:** 3.13.2.5

Determine unit weight of normal weight concrete.

SECTION: 03 30 00 **SUBPART:** 3.13.2.6

Determine water soluble ion concentration in accordance with JIS A 1154.

SECTION: 03 30 00 **SUBPART:** 3.13.2.9

When there is evidence that strength of concrete structure in place does not meet specification requirements or there are non-conforming materials, make cores drilled from hardened concrete for compressive strength determination in accordance with JIS A 1107, and as follows:

a. Take at least three representative cores from each member or area of concrete-in-place that is considered potentially deficient. Location of cores will be determined by the Contracting Officer.

b. Test cores after moisture conditioning in accordance with JIS A 1107 if concrete they represent is more than superficially wet under service.

c. Air dry cores, (16 to 27 degrees C with relative humidity less than 60 percent) for 7 days before test and test dry if concrete they represent is dry under service conditions.

d. Strength of cores from each member or area are considered satisfactory if their average is equal to or greater than 85 percent of the 28-day design compressive strength of the class of concrete.

SECTION: 05 12 00 **SUBPART:** 3.7

Perform field tests, and provide labor, equipment, and incidentals required for testing, except that electric power for field tests will be furnished as set forth in Division 1. Notify the Contracting Officer in writing of defective welds, bolts, nuts, and washers within 7 working days of the date of the inspection.

SECTION: 05 12 00 **SUBPART:** 3.7.1.2

If more than 20 percent of welds made by a welder contain defects identified by testing, then all groove welds made by that welder must be tested by ultrasonic testing

SECTION: 05 12 00 **SUBPART:** 3.7.2.1

Test a minimum of 3 bolt, nut, and washer assemblies

SECTION: 05 12 00 **SUBPART:** 3.7.2.3

option to perform nondestructive tests on 5 percent of the installed bolts

SECTION: 05 12 00 **SUBPART:** 3.7.3

ASTM A143/A143M for steel products hot-dip galvanized after fabrication.

SECTION: 05 50 13 **SUBPART:** 3.5

Perform welding, welding inspection, and corrective welding in accordance with JASS 6.

SECTION: 05 52 00 **SUBPART:** 1.4.1

welding procedures testing

SECTION: 05 52 00 **SUBPART:** 1.4.2

Submit certified welder qualification by tests in accordance with JIS Z 3801 or JIS Z 3841, or under an equivalent approved qualification test.

SECTION: 07 56 00 **SUBPART:** 3.4.1

Prior to application of fluid-applied waterproofing, measure moisture content of substrate with a moisture meter in the presence of the Contracting Officer.

SECTION: 07 56 00 **SUBPART:** 3.4.2

Measure wet film thickness every 10 square meters during application by placing flat metal plates on the substrate or using a mil-thickness gage especially manufactured for the purpose.

SECTION: 07 56 00 **SUBPART:** 3.4.3

Test watertightness by measuring water level at beginning and end of the 24 hour period. If water level falls, drain water, allow installation to dry, and inspect.

SECTION: 07 60 00 **SUBPART:** 3.4

Establish and maintain a Quality Control Plan for sheet metal used in conjunction with roofing to assure compliance of the installed sheet metalwork with the contract requirements. Remove work that is not in compliance with the contract and replace or correct. Include quality control, but not be limited to, the following:

a. Observation of environmental conditions; number and skill level of sheet metal workers; condition of substrate.

b. Verification that specified material is provided and installed.

c. Inspection of sheet metalwork, for proper size(s) and thickness(es), fastening and joining, and proper installation.

SECTION: 08 33 23 **SUBPART:** 3.2.1

Test the door opening and closing operation when activated by controls and in accordance with UL 1784 for safety of air leakage tests. Adjust controls and safeties. Replace damaged and malfunctioning controls and equipment. Reset the door-closing mechanism after a successful test.

SECTION: 08 33 23 **SUBPART:** 3.2.1.1

Not more than 90 calendar days after completion and acceptance of the project, examine, lubricate, test, and re-adjust doors as required for proper operation.

SECTION: 08 71 00 **SUBPART:** 1.8

Deliver permanent keys and removable cores to the Contracting Officer, either directly or by certified mail. Deliver construction master keys with the locks.

SECTION: 08 71 00 **SUBPART:** 3.4

Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer.

SECTION: 08 81 00 **SUBPART:** 1.7.1

Warranty insulating glass units against development of material obstruction to vision (such as dust, fogging, or film formation on the inner glass surfaces) caused by failure of the hermetic seal, other than through glass breakage, for a 10-year period following acceptance of the work. Provide new units for any units failing to comply with terms of this warranty within 45 working days after receipt of notice from the Government.

SECTION: 09 65 00 **SUBPART:** 3.3

MOISTURE, ALKALINITY AND BOND TESTS

SECTION: 12 21 00 **SUBPART:** 1.1

Mount and operate equipment in accordance with manufacturer's instructions.

SECTION: 21 12 00 **SUBPART:** 3.6.1

Each piping system shall be hydrostatically tested at 1379 kPa (gage) in accordance with NFPA 14 and NFPA 24 and shall show no leakage or reduction in gauge pressure after 2 hours. The Contractor shall conduct complete preliminary tests, which shall encompass all aspects of system operation. When tests have been completed and all necessary corrections made, submit to the Contracting Officer a signed and dated certificate, similar to that specified in NFPA 13, attesting to the satisfactory completion of all testing and stating that the system is in operating condition. Also include a written request for a formal inspection and test.

SECTION: 21 12 00 **SUBPART:** 3.6.2.1

Perform flow tests of each standpipe riser in accordance with NFPA 14. Affix 0-1379 kPa pressure gauges to lowest hose valve and next-to-highest hose valve. Connect lined, 65 mm diameter fire hose with underwriter's playpipe to highest hose valve and flow at least 946 L/m for 5 minutes from standpipe to a safe location outside the building. For dry pipe system, supply system through 65 mm fire hose connected to the nearest fire hydrant which is approximately 46 meters away. Furnish hose, nozzles and fittings required for this test.

SECTION: 21 12 00 **SUBPART:** 3.6.3

When deficiencies, defects or malfunctions develop during the tests required, all further testing of the system shall be suspended until proper adjustments, corrections or revisions have been made to assure proper performance of the system. If these revisions require more than a nominal delay, the Contracting Officer shall be notified when the additional work has been completed, to arrange a new inspection and test of the system. All tests required shall be repeated prior to final acceptance, unless directed otherwise.

SECTION: 21 13 13 **SUBPART:** 3.8.1.1

Flushing

SECTION: 21 13 13 **SUBPART:** 3.8.1.2

Hydrostatic Test

SECTION: 21 13 13 **SUBPART:** 3.8.2.1

Hydrostatic Test

SECTION: 21 13 13 **SUBPART:** 3.8.2.2

Backflow Prevention Assembly Forward Flow Test

SECTION: 21 13 13 **SUBPART:** 3.8.3

Main Drain Flow Test

SECTION: 22 00 00 **SUBPART:** 3.9.1

Plumbing System

SECTION: 22 00 00 **SUBPART:** 3.9.4

Operational Test

SECTION: 23 05 93 **SUBPART:** 3.1.5

Conduct DALT Field work on the ductwork selections in compliance with the procedures specified in SMACNA 016 or equivalent Japanese standard, as supplemented and modified by this section.

SECTION: 23 05 93 **SUBPART:** 3.2.4

Conduct TAB Field Work in conformance with the TAB Standard, as supplemented and modified by this section.

SECTION: 23 05 93 **SUBPART:** 3.2.5.2

Conduct thermal performance (capacity) tests to verify thermal energy transfer components, devices, and equipment meet the indicated design capacity.

SECTION: 23 05 93 **SUBPART:** 3.3.1.2

Conduct thermal performance (capacity) tests to verify thermal energy transfer components, devices, and equipment meet the indicated design capacity within the parameters for Season 2.

SECTION: 23 57 10.00 10 **SUBPART:** 3.18.1

Pressure Testing

SECTION: 23 74 33 **SUBPART:** 3.2.1

Test and rate components of the air-conditioning systems as a system in accordance with ANSI/AHRI 210/240.

SECTION: 23 81 00 **SUBPART:** 2.8.1

Salt-spray test the factory-applied coating or paint system of equipment including packaged terminal units, heat pumps, and air conditioners in accordance with JIS Z 2371.

SECTION: 23 81 00 **SUBPART:** 3.7.1

Upon completion of installation of air conditioning equipment, test factory- and field-installed refrigerant piping with an electronic-type leak detector.

SECTION: 23 81 00 **SUBPART:** 3.7.3

Test the air conditioning systems and systems components for proper operation. Adjust safety and automatic control instruments as necessary to ensure proper operation and sequence. Conduct operational tests for not less than 8 hours.

SECTION: 23 81 00 **SUBPART:** 3.7.4

Upon completion of evacuation, charging, startup, final leak testing, and proper adjustment of controls, test the systems to demonstrate compliance with performance and capacity requirements. Test systems for not less than 8 hours, record readings hourly.

SECTION: 26 20 00 **SUBPART:** 3.5.1

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

SECTION: 26 20 00 **SUBPART:** 3.5.2

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring 14 sqmm and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring to provide direct reading of resistance. All existing wiring to be reused shall also be tested.

SECTION: 26 20 00 **SUBPART:** 3.5.4

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed.

SECTION: 26 20 00 **SUBPART:** 3.5.4

Test RCBO circuit breakers in accordance with JIS C 8222

SECTION: 26 20 00 **SUBPART:** 3.5.5

Test grounding system to ensure continuity, and that resistance to ground is not excessive.

SECTION: 26 20 00 **SUBPART:** 3.5.7

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter.

SECTION: 26 24 13 **SUBPART:** 3.5.1

Perform in accordance with the manufacturer's recommendations

SECTION: 26 24 13 **SUBPART:** 3.5.2

Upon completion of acceptance checks, settings, and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function.

SECTION: 26 24 13 **SUBPART:** 3.5.2

Test each item to perform its function not less than three times.

SECTION: 26 27 14.00 20 **SUBPART:** 3.2.1

Perform in accordance with the manufacturer's recommendations

SECTION: 26 27 14.00 20 **SUBPART:** 3.2.1

Visual and mechanical inspection

SECTION: 26 27 14.00 20 **SUBPART:** 3.2.1

Electrical tests

SECTION: 26 27 14.00 20 **SUBPART:** 3.2.1

Electrical Tests

SECTION: 26 27 14.00 20 **SUBPART:** 3.2.2

System Functional Verification

SECTION: 26 51 00 **SUBPART:** 3.1.2

Obtain approval of the exact mounting height on the job before commencing installation and, where applicable, after coordinating with the type, style, and pattern of the ceiling being installed.

SECTION: 27 05 13.43 **SUBPART:** 1.3.5.1

Each termination for a TV receiver must have a minimum signal level of 0 decibel millivolts (dBmV) (1000 microvolts) at 55 MHz and of 0 dBmV (1000 microvolts) at 1000 MHz and a maximum signal of 15 dBmV or a level not to overload the receiver for the entire system bandwidth.

SECTION: 27 05 13.43 **SUBPART:** 1.3.5.3

The system must not show a serious loss of carrier to noise when the system levels are lowered 3 dB below normal or a significant distortion when the levels are increased 3 dB above normal, as observed on a TV set located at the far end extremities of the system.

SECTION: 27 05 13.43 **SUBPART:** 2.3.8

Line taps must have 18 dB minimum isolation from each tap to the thru-line.

SECTION: 27 05 13.43 **SUBPART:** 3.2.1

Upon completing installation of the television distribution system, the Contractor must align and balance the system and must perform complete pretesting. During the system pretest, Contractor, utilizing the approved spectrum analyzer or signal level meter, must verify that the system is fully operational and meets all the system performance requirements of the specification. Contractor must test the signal loss in dBmV at 55, 151, 547, and 1000 MHz. The signal levels must be 0 dBmV (1000 microvolts), minimum. The signal must not exceed 15 dBmV over the entire system bandwidth. Any deficiencies found must be corrected and revalidated by follow up testing. Contractor must measure and record the video and audio carrier levels at each of the frequency levels specified at each of the following points in the system:

SECTION: 27 05 13.43 **SUBPART:** 3.2.1

Furthest outlet from service entrance point of connection.

A random sampling of 25 percent of the outlets from each housing units.

SECTION: 27 05 13.43 **SUBPART:** 3.2.2

Contractor must notify the Contracting Officer of system readiness 10 days prior to the date of acceptance testing. Contractor must also coordinate with the local television service provider and allow them to attend witness tests. The television distribution system must be tested in accordance with the approved test plan in the presence of the Contracting Officer's representative to certify acceptable performance. System test must verify that the total system meets all the requirements of the specification and complies with the specified standards. Contractor must verify that no signal leakage exists in conformance with NCTA RP and 47 CFR 76.605. System leakage must also be tested at the headend location with signal applied to system. Deficiencies revealed by the testing must be corrected on the outlets sampled as well as on the outlets not sampled and revalidated by follow-up testing. Contractor must conduct testing at each of the following points in the system:

SECTION: 27 10 00 **SUBPART:** 2.11.1

Provide documentation of the testing and verification actions taken by manufacturer to confirm compliance with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850, JIS C 6185-2 for single mode optical fiber, and JIS C 6185-3 for multimode optical fiber cables.

SECTION: 27 10 00 **SUBPART:** 3.5.1

Perform telecommunications cabling inspection, verification, and performance tests in accordance with JIS X 5150, JCS 5507, JIS C 6820 and JIS C 6850.

SECTION: 27 10 00 **SUBPART:** 3.5.1.2

UTP backbone copper cabling shall be tested for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors, and between conductors and shield, if cable has overall shield. Test operation of shorting bars in connection blocks. Test cables after termination but prior to being cross-connected.

SECTION: 27 10 00 **SUBPART:** 3.5.1.2

For multimode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with JIS C 6820 and JIS C 6850 and JIS C 6850 and JIS C 6185-3 using Method A, Optical Power Meter and Light Source Method B, OTDR for multimode optical fiber. For single-mode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with JIS C 6820 and JIS C 6850 and JIS C 6185-2 using Method A, Optical Power Meter and Light Source Method B, OTDR for single-mode optical fiber. Perform verification acceptance tests.

SECTION: 27 10 00 **SUBPART:** 3.5.1.3

Perform testing for each outlet and MUTOA as follows:

SECTION: 27 10 00 **SUBPART:** 3.5.1.3

a. Perform Category 6 link tests in accordance with JIS X 5150 and JCS 5507. Tests shall include wire map, length, insertion loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, return loss, propagation delay, and delay skew.

SECTION: 27 10 00 **SUBPART:** 3.5.1.3

b. Optical fiber Links. Perform optical fiber end-to-end link tests in accordance with JIS C 6820 and JIS C 6850.

SECTION: 27 10 00 **SUBPART:** 3.5.1.4

Perform verification tests for UTP and optical fiber systems after the complete telecommunications cabling and workstation outlet/connectors are installed.

SECTION: 28 31 76 **SUBPART:** 3.8.2

Pre-Government Testing

SECTION: 28 31 76 **SUBPART:** 3.8.4

Government Final Tests

SECTION: 28 31 76 **SUBPART:** 3.9.1

System Tests

SECTION: 28 31 76 **SUBPART:** 3.9.2

Audibility Tests

SECTION: 28 31 76 **SUBPART:** 3.9.3

Intelligibility Tests

SECTION: 31 00 00 **SUBPART:** 2.1

Borrow Site Testing for TPH, BTEX and TCLP from a composite sample of material from the borrow site

SECTION: 31 00 00 **SUBPART:** 3.16

field in-place density

SECTION: 31 00 00 **SUBPART:** 3.16.1

Fill and Backfill Material Gradation

SECTION: 31 00 00 **SUBPART:** 3.16.2

In-Place Densities

SECTION: 31 00 00 **SUBPART:** 3.16.3

Check Tests on In-Place Densities

SECTION: 31 00 00 **SUBPART:** 3.16.4

Moisture Contents

SECTION: 31 00 00 **SUBPART:** 3.16.5

Optimum Moisture and Laboratory Maximum Density

SECTION: 31 00 00 **SUBPART:** 3.16.6

Tolerance Tests for Subgrades

SECTION: 31 00 00 **SUBPART:** 3.16.7

Displacement of Sewers

SECTION: 32 12 13 **SUBPART:** 3.6

FIELD QUALITY CONTROL

SECTION: 32 12 13 **SUBPART:** 3.7.2

Calibration Test

SECTION: 32 12 13 **SUBPART:** 3.7.3

Trial Applications

SECTION: 32 12 13 **SUBPART:** 3.7.4

Sampling and Testing During Construction

SECTION: 32 16 19 **SUBPART:** 3.3.4

Surface and Thickness Tolerances

SECTION: 32 16 19 **SUBPART:** 3.4.5

Surface and Thickness Tolerances

SECTION: 32 16 19 **SUBPART:** 3.8.2.1

Strength Testing

SECTION: 32 16 19 **SUBPART:** 3.8.2.2

Air Content

SECTION: 32 16 19 **SUBPART:** 3.8.2.3

Slump Test

SECTION: 32 17 23 **SUBPART:** 3.1.1

Test the pavement surface for moisture before beginning pavement marking after each period of rainfall, fog, high humidity, or cleaning, or when the ambient temperature has fallen below the dew point.

SECTION: 32 17 23 **SUBPART:** 3.4.1

Test samples by an approved laboratory. If a sample fails to meet specification, replace the material in the area represented by the samples and retest the replacement material as specified above.

SECTION: 32 31 13 **SUBPART:** 3.2.5

Take samples and test concrete to determine strength as specified

SECTION: 33 11 00 **SUBPART:** 3.3.1

Perform field tests, and provide labor, equipment, and incidentals required for testing, except that water needed for field tests will be furnished as set forth in paragraph AVAILABILITY AND USE OF UTILITY SERVICES in Section 01 50 00 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS.

SECTION: 33 11 00 **SUBPART:** 3.3.1.1

Test the water system in accordance with the applicable JWWA standards. Where water mains provide fire service, test in accordance with the special testing requirements given in the paragraph SPECIAL TESTING REQUIREMENTS FOR FIRE SERVICE. Test ductile-iron water mains in accordance with the requirements of JIS S 3200-1 for hydrostatic testing. The amount of leakage on ductile-iron pipelines with mechanical-joints or push-on joints is not to exceed the amounts given in pipe manufacturer's installation instructions. No leakage will be allowed at joints made by any other methods. Test PVC and PVCO plastic water systems made with PVC pipe for pressure and leakage tests. The amount of leakage on pipelines made of PVC water main pipe is not to exceed the amounts given in pipe manufacturer's installation instructions, except that at joints made with sleeve-type mechanical couplings, no leakage will be allowed. Test water service lines for hydrostatic testing. No leakage will be allowed at copper pipe joints, copper tubing joints (soldered, compression type, brazed), plastic pipe joints, flanged joints, and screwed joints.

SECTION: 33 11 00 **SUBPART:** 3.3.1.2

The hydrostatic pressure sewer test will be performed in accordance with the applicable JWWA standard for the piping material with a minimum test pressure of 200 kPa.

SECTION: 33 11 00 **SUBPART:** 3.3.1.3

For leakage test, use a hydrostatic pressure not less than the maximum working pressure of the system. Leakage test may be performed at the same time and at the same test pressure as the pressure test.

SECTION: 33 11 00 **SUBPART:** 3.3.1.3

For PE perform leak testing in accordance with the applicable JWWA standards

SECTION: 33 11 00 **SUBPART:** 3.3.1.4

Perform bacteriological tests in accordance with the applicable JWWA standards.

SECTION: 33 11 00 **SUBPART:** 3.3.1.5

Test water mains and water service lines providing fire service or water and fire service in accordance with NFPA 24. The additional water added to the system must not exceed the limits given in NFPA 24

SECTION: 33 11 00 **SUBPART:** 3.3.1.6

Test tracer wire for continuity after service connections have been completed and prior to final pavement or restoration. Verify that tracer wire is locatable with electronic utility locating equipment. Repair breaks or separations and re-test for continuity.

SECTION: 33 30 00 **SUBPART:** 3.3.1.1

When unusual conflicts are encountered between sanitary sewer and waterlines a hydrostatic pressure sewer test will be performed in accordance with the applicable JWWA standard for the piping material with a minimum test pressure of 200 kPa.

SECTION: 33 30 00 **SUBPART:** 3.3.1.3

Perform a deflection test on entire length of installed plastic pipeline on completion of work adjacent to and over the pipeline, including leakage tests, backfilling, placement of fill, grading, paving, concreting, and any other superimposed loads

SECTION: 33 40 00 **SUBPART:** 3.10.1.1

Leakage Test

SECTION: 33 40 00 **SUBPART:** 3.10.1.2

Determination Of Density

SECTION: 33 40 00 **SUBPART:** 3.10.1.3

deflection test

SECTION: 33 61 14 **SUBPART:** 3.4

Before final acceptance of work, test each system to demonstrate compliance with contract requirements.

SECTION: 33 61 14 **SUBPART:** 3.4

Do not cover carrier piping joints with insulation or concrete anchors (thrust blocks), until carrier piping joints pass field tests.

SECTION: 33 61 14 **SUBPART:** 3.4

Test piping system at 1379 kPag for minimum holding period of 2 hours during which time pressure must not drop more than 28 kPa; test plastic RTR piping in accordance with Federal Agency Approved Brochure. Pressure drop greater than 28 kPagcorrected for temperature variation constitutes failure. Valve off piping system and disconnect method of piping system pressurization before starting the 2 hour pressure holding period. During hydrostatic pressure test, examine piping system for leaks. Repair leaking joints, replace damaged and porous pipe and fittings with new materials, and repeat tests.

SECTION: 33 71 02 **SUBPART:** 2.18.1

Manufacturer must test one sample assembly consisting of a straight lead tube 305 mm long with a 65.5 mm outside diameter, and a 3.175 mm thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions.

SECTION: 33 71 02 **SUBPART:** 3.7

Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables.

SECTION: 33 71 02 **SUBPART:** 3.18.2

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function.

SECTION: 33 82 00 **SUBPART:** 2.14.1

Test 100 percent OTDR test of FO media at the factory in accordance with JIS X 5150 and JIS C 6850. Use JIS C 6823 for single mode fiber and for multi mode fiber measurements. Calibrate OTDR to show anomalies of 0.2 dB minimum. Enhanced performance filled OSP copper cables, referred to as Broadband Outside Plant (BBOSP). Enhanced performance air core OSP copper cables shall meet the requirements of JCS 5503. Submit test reports, including manufacture date for each cable reel and receive approval before delivery of cable to the project site.

SECTION: 33 82 00 **SUBPART:** 3.5.1

Perform the following tests on cable at the job site before it is removed from the cable reel. For cables with factory installed pulling eyes, these tests shall be performed at the factory and certified test results shall accompany the cable.

SECTION: 33 82 00 **SUBPART:** 3.5.1.1

Perform capacitance tests on at least 10 percent of the pairs within a cable to determine if cable capacitance is within the limits specified.

SECTION: 33 82 00 **SUBPART:** 3.5.1.2

Perform DC-loop resistance on at least 10 percent of the pairs within a cable to determine if DC-loop resistance is within the manufacturer's calculated resistance.

SECTION: 33 82 00 **SUBPART:** 3.5.2.1

Perform the following acceptance tests in accordance with JIS C 6870-3 and JIS C 6870-3-10:

a. Wire map (pin to pin continuity)

b. Continuity to remote end

c. Crossed pairs

d. Reversed pairs

e. Split pairs

f. Shorts between two or more conductors